

WORK ORDER NO.

At the

PREPARED BY:

Architectural:  
Civil:  
Structural:

Mechanical:  
Electrical:

Submitted By:

Date:

APPROVED BY:

For Commander, NAVFAC:  
Date:

## PROJECT TABLE OF CONTENTS

### DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

00 01 15 02/11, CHG 1: 08/14 LIST OF DRAWINGS

### DIVISION 01 - GENERAL REQUIREMENTS

|                |                     |                                                                                                     |
|----------------|---------------------|-----------------------------------------------------------------------------------------------------|
| 01 11 00.00 22 | 05/21               | SUMMARY OF WORK (PWD ME)                                                                            |
| 01 14 00.00 22 | 11/22, CHG 4: 02/25 | WORK RESTRICTIONS (PWD ME)                                                                          |
| 01 20 00.00 22 | 11/20, CHG 5: 02/25 | PRICE AND PAYMENT PROCEDURES (PWD ME)                                                               |
| 01 30 00.00 22 | 11/20, CHG 4: 08/24 | ADMINISTRATIVE REQUIREMENTS (PWD ME)                                                                |
| 01 31 23.13 20 | 08/23, CHG 1: 02/24 | ELECTRONIC CONSTRUCTION AND FACILITY<br>SUPPORT CONTRACT MANAGEMENT SYSTEM<br>(eCMS)                |
| 01 32 17.00 20 | 11/23, CHG 1: 08/24 | COST-LOADED NETWORK ANALYSIS SCHEDULES<br>(NAS)                                                     |
| 01 33 00       | 08/18, CHG 4: 02/21 | SUBMITTAL PROCEDURES                                                                                |
| 01 33 10       | 02/24               | SUBSTITUTIONS                                                                                       |
| 01 35 26.00 22 | 05/24               | GOVERNMENTAL SAFETY REQUIREMENTS (PWD<br>ME)                                                        |
| 01 42 00       | 02/19               | SOURCES FOR REFERENCE PUBLICATIONS                                                                  |
| 01 44 00.00 22 | 04/25               | ARCHAEOLOGICAL MONITORING (PWD ME)                                                                  |
| 01 45 00.00 22 | 08/23, CHG 1: 05/24 | QUALITY CONTROL (PWD ME)                                                                            |
| 01 45 35       | 05/24               | SPECIAL INSPECTIONS                                                                                 |
| 01 50 00.00 22 | 11/20, CHG 3: 08/24 | TEMPORARY CONSTRUCTION FACILITIES AND<br>CONTROLS (PWD ME)                                          |
| 01 57 19.00 22 | 08/22               | TEMPORARY ENVIRONMENTAL CONTROLS -<br>PORTSMOUTH NAVAL SHIPYARD (PWD ME)                            |
| 01 58 00       | 08/19, CHG 5: 08/22 | PROJECT IDENTIFICATION                                                                              |
| 01 74 19       | 02/19, CHG 3:11/21  | CONSTRUCTION WASTE MANAGEMENT AND<br>DISPOSAL                                                       |
| 01 78 00.00 22 | 05/19, CHG 1: 08/21 | CLOSEOUT SUBMITTALS (PWD ME)                                                                        |
| 01 78 23       | 05/23               | OPERATION AND MAINTENANCE DATA                                                                      |
| 01 78 24.00 22 | 05/23, CHG 1: 11/23 | ELECTRONIC OPERATION AND MAINTENANCE<br>SUPPORT INFORMATION (eOMSI) FACILITY<br>DATA WORKBOOK (FDW) |
| 01 91 00.15 20 | 05/23, CHG 3: 08/24 | TOTAL BUILDING COMMISSIONING                                                                        |

### DIVISION 02 - EXISTING CONDITIONS

|                |       |                                                                                                                                           |
|----------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 02 41 00       | 08/22 | DEMOLITION                                                                                                                                |
| 02 42 91       | 11/18 | REMOVAL AND SALVAGE OF HISTORIC<br>CONSTRUCTION MATERIALS                                                                                 |
| 02 81 00       | 11/18 | TRANSPORTATION AND DISPOSAL OF<br>HAZARDOUS MATERIALS                                                                                     |
| 02 82 00.00 22 | 09/21 | ASBESTOS REMEDIATION (PNS PROJECTS)                                                                                                       |
| 02 83 00.00 22 | 3/24  | MANAGEMENT OF LEAD, CADMIUM, AND<br>CHROMIUM (VI) DURING RENOVATION,<br>DEMOLITION, CONSTRUCTION, REMOVAL AND<br>ABATEMENT (PNS PROJECTS) |

### DIVISION 03 - CONCRETE

|                |                     |                                                                 |
|----------------|---------------------|-----------------------------------------------------------------|
| 03 30 00       | 02/19, CHG 2: 05/21 | CAST-IN-PLACE CONCRETE                                          |
| 03 42 13.00 10 | 05/16               | PLANT-PRECAST CONCRETE PRODUCTS FOR<br>BELOW GRADE CONSTRUCTION |

### DIVISION 04 - MASONRY

|          |                     |                                              |
|----------|---------------------|----------------------------------------------|
| 04 03 00 | 11/17               | CONSERVATION TREATMENT FOR PERIOD<br>MASONRY |
| 04 20 00 | 11/15, CHG 2: 05/19 | UNIT MASONRY                                 |

**DIVISION 05 - METALS**

|             |                     |                                                |
|-------------|---------------------|------------------------------------------------|
| 05 05 20    | 05/22               | POST-INSTALLED CONCRETE AND MASONRY<br>ANCHORS |
| 05 05 23.16 | 08/18               | STRUCTURAL WELDING                             |
| 05 12 00    | 08/18, CHG 2: 05/21 | STRUCTURAL STEEL                               |
| 05 40 00    | 05/15, CHG 1: 08/18 | COLD-FORMED METAL FRAMING                      |
| 05 50 13    | 08/24               | MISCELLANEOUS METAL FABRICATIONS               |
| 05 51 00    | 08/23               | METAL STAIRS                                   |
| 05 51 33    | 08/24               | METAL LADDERS                                  |
| 05 52 00    | 08/23               | METAL RAILINGS                                 |

**DIVISION 06 - WOOD, PLASTICS, AND COMPOSITES**

|          |                     |                  |
|----------|---------------------|------------------|
| 06 10 00 | 08/16, CHG 2: 11/18 | ROUGH CARPENTRY  |
| 06 20 00 | 08/24               | FINISH CARPENTRY |

**DIVISION 07 - THERMAL AND MOISTURE PROTECTION**

|                |                     |                                 |
|----------------|---------------------|---------------------------------|
| 07 13 53       | 02/16, CHG 1: 08/17 | ELASTOMERIC SHEET WATERPROOFING |
| 07 19 00       | 08/23               | WATER REPELLENTS                |
| 07 21 13       | 08/23               | BOARD AND BLOCK INSULATION      |
| 07 22 00       | 08/23               | ROOF AND DECK INSULATION        |
| 07 27 10       | 08/23               | BUILDING AIR BARRIER SYSTEM     |
| 07 27 19.01    | 08/23, CHG 1: 08/24 | SELF-ADHERING AIR BARRIERS      |
| 07 60 00       | 08/23               | FLASHING AND SHEET METAL        |
| 07 81 00       | 02/11               | SPRAY-APPLIED FIREPROOFING      |
| 07 84 00.00 22 | 07/24               | FIRESTOPPING                    |
| 07 92 00       | 08/23               | JOINT SEALANTS                  |

**DIVISION 08 - OPENINGS**

|          |                     |                         |
|----------|---------------------|-------------------------|
| 08 11 13 | 08/20               | STEEL DOORS AND FRAMES  |
| 08 31 00 | 05/17, CHG 1: 08/18 | ACCESS DOORS AND PANELS |
| 08 33 23 | 08/20, CHG 1: 02/22 | OVERHEAD COILING DOORS  |
| 08 71 00 | 02/16, CHG 4: 02/22 | DOOR HARDWARE           |
| 08 91 00 | 08/20               | METAL WALL LOUVERS      |

**DIVISION 09 - FINISHES**

|             |       |                                                                 |
|-------------|-------|-----------------------------------------------------------------|
| 09 01 90.50 | 06/25 | PREPARATION OF HISTORIC WOOD AND METAL<br>SURFACES FOR PAINTING |
| 09 29 00    | 08/24 | GYPSUM BOARD                                                    |
| 09 77 23    | 01/25 | PLASTIC WALL PANELS                                             |
| 09 90 00    | 02/21 | PAINTS AND COATINGS                                             |

**DIVISION 10 - SPECIALTIES**

|             |                     |                    |
|-------------|---------------------|--------------------|
| 10 14 00.10 | 08/17, CHG 1: 11/18 | EXTERIOR SIGNAGE   |
| 10 14 00.20 | 08/20               | INTERIOR SIGNAGE   |
| 10 44 16    | 11/19               | FIRE EXTINGUISHERS |

**DIVISION 13 - SPECIAL CONSTRUCTION**

13 34 19 08/20, CHG 1: 02/21 METAL BUILDING SYSTEMS

**DIVISION 21 - FIRE SUPPRESSION**

21 13 13 08/20 WET PIPE SPRINKLER SYSTEMS, FIRE PROTECTION

**DIVISION 22 - PLUMBING**

22 13 29 02/11 SANITARY SEWERAGE PUMPS

**DIVISION 23 - HEATING, VENTILATING, AND AIR CONDITIONING (HVAC)**

23 03 00.00 20 08/10, CHG 3: 08/18 BASIC MECHANICAL MATERIALS AND METHODS  
23 05 15 05/22, CHG 2: 08/24 COMMON PIPING FOR HVAC  
23 05 48.19 05/18, CHG 2: 08/20 SEISMIC BRACING FOR HVAC  
23 05 93 11/15 TESTING, ADJUSTING, AND BALANCING FOR HVAC  
23 07 00 08/24 THERMAL INSULATION FOR MECHANICAL SYSTEMS  
23 08 00 05/23, CHG 1: 08/24 COMMISSIONING OF MECHANICAL SYSTEMS  
23 09 00 08/24 INSTRUMENTATION AND CONTROL FOR HVAC  
23 21 13 12/24 POLYPROPYLENE PRESSURE PIPING  
23 21 23 08/17 HYDRONIC PUMPS  
23 23 00 08/21 REFRIGERANT PIPING  
23 30 00 05/20, CHG 1: 02/22 HVAC AIR DISTRIBUTION  
23 64 26 08/09, CHG 7: 02/24 DESALINATION INTAKE SCREEN SYSTEM  
23 81 00 05/24 DECENTRALIZED UNITARY HVAC EQUIPMENT  
23 83 00.00 20 04/06 ELECTRIC SPACE HEATING EQUIPMENT

**DIVISION 25 - INTEGRATED AUTOMATION**

25 05 11 08/24 CYBERSECURITY FOR DESALINATION SCADA SYSTEM

**DIVISION 26 - ELECTRICAL**

26 05 13.00 10 11/23 MEDIUM-VOLTAGE CABLES  
26 05 19.00 10 11/22 INSULATED WIRE AND CABLE  
26 05 48 11/23 SEISMIC PROTECTION FOR ELECTRICAL EQUIPMENT  
26 05 73 08/23 POWER SYSTEM STUDIES  
26 08 00 11/22 APPARATUS INSPECTION AND TESTING  
26 11 16 11/21 SECONDARY UNIT SUBSTATIONS  
26 20 00 08/23 INTERIOR DISTRIBUTION SYSTEM  
26 24 13 08/21 SWITCHBOARDS  
26 29 23 02/20, CHG 1: 05/21 ADJUSTABLE SPEED DRIVE (ASD) SYSTEMS UNDER 600 VOLTS  
26 51 00 05/20, CHG 2: 11/21 INTERIOR LIGHTING

**DIVISION 27 - COMMUNICATIONS**

27 10 00 08/11 BUILDING TELECOMMUNICATIONS CABLING SYSTEM

**DIVISION 28 - ELECTRONIC SAFETY AND SECURITY**

28 31 60 08/20 INTERIOR FIRE ALARM SYSTEM, NON-ADDRESSABLE



**DIVISION 31 - EARTHWORK**

|                |         |                    |
|----------------|---------|--------------------|
| 31 00 00.00 22 | 08/2024 | EARTHWORK          |
| 31 63 33       | 02/25   | DRILLED MICROPILES |

**DIVISION 32 - EXTERIOR IMPROVEMENTS**

|          |       |                                                         |
|----------|-------|---------------------------------------------------------|
| 32 11 26 | 05/20 | HOT-MIX BITUMINOUS BASE COURSE FOR<br>ROADS AND STREETS |
| 32 12 13 | 05/17 | BITUMINOUS TACK AND PRIME COATS                         |
| 32 31 13 | 11/21 | CHAIN LINK FENCES AND GATES                             |

**DIVISION 33 - UTILITIES**

|             |       |                                                    |
|-------------|-------|----------------------------------------------------|
| 33 11 13.28 | 07/24 | HIGH DENSITY POLYETHYLENE PIPE                     |
| 33 21 00.19 | 11/24 | VERTICAL MULTI-STAGE CENTRIFUGAL PUMPS             |
| 33 61 13.19 | 02/16 | VALVES, PIPING, AND EQUIPMENT IN VALVE<br>MANHOLES |
| 33 71 02    | 08/21 | UNDERGROUND ELECTRICAL DISTRIBUTION                |

**DIVISION 35 - WATERWAY AND MARINE CONSTRUCTION**

|          |                     |                                             |
|----------|---------------------|---------------------------------------------|
| 35 45 01 | 02/21, CHG 3: 08/23 | VERTICAL PUMPS, MIXED-FLOW<br>IMPELLER-TYPE |
|----------|---------------------|---------------------------------------------|

**DIVISION 40 - PROCESS INTERCONNECTIONS**

|                |                     |                                  |
|----------------|---------------------|----------------------------------|
| 40 05 13       | 10/07, CHG 2: 02/20 | PIPELINES, LIQUID PROCESS PIPING |
| 40 05 13.96    | 05/10               | WELDING PROCESS PIPING           |
| 40 17 26.00 20 | 04/06               | WELDING PRESSURE PIPING          |
| 40 60 00       | 05/20               | PROCESS CONTROL                  |

**DIVISION 41 - MATERIAL PROCESSING AND HANDLING EQUIPMENT**

|             |                     |                                                  |
|-------------|---------------------|--------------------------------------------------|
| 41 22 13.14 | 11/19, CHG 1: 02/21 | BRIDGE CRANES, OVERHEAD ELECTRIC, TOP<br>RUNNING |
|-------------|---------------------|--------------------------------------------------|

**DIVISION 43 - PROCESS GAS AND LIQUID HANDLING, PURIFICATION, AND STORAGE EQUIPMENT**

|             |       |                                                         |
|-------------|-------|---------------------------------------------------------|
| 43 21 13.16 | 09/24 | HORIZONTAL END SUCTION PUMPS                            |
| 43 41 45    | 01/25 | FIBERGLASS REINFORCED PLASTIC CHEMICAL<br>STORAGE TANKS |

**DIVISION 46 - WATER AND WASTEWATER EQUIPMENT**

|          |       |                                                                |
|----------|-------|----------------------------------------------------------------|
| 46 30 00 | 08/17 | WATER AND WASTEWATER CHEMICAL FEED<br>SYSTEMS                  |
| 46 64 30 | 11/24 | TEMPORARY ASW PUMP STATION                                     |
| 46 80 00 | 09/24 | BRACKISH WATER REVERSE OSMOSIS UNIT                            |
| 46 81 00 | 09/24 | SEAWATER WATER REVERSE OSMOSIS UNIT                            |
| 46 82 00 | 09/24 | IMMERSED MEMBRANE FILTRATION SYSTEM<br>AND ANCILLARY EQUIPMENT |

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DOCUMENT 00 01 15

LIST OF DRAWINGS  
02/11, CHG 1: 08/14

PART 1 GENERAL

1.1 SUMMARY

This section lists the drawings for the project pursuant to contract clause "DFARS 252.236-7001, Contract Drawings, Maps and Specifications."

1.2 CONTRACT DRAWINGS

Contract drawings are as follows:

| Sheet Number | PWD-MENO   | NAVFAC Dwg No. | Drawing Title                            |
|--------------|------------|----------------|------------------------------------------|
| G-001        | FAC-YR-NUM | #####          | COVER SHEET                              |
| G-002        | FAC-YR-NUM | #####          | DRAWING LIST PART 1 OF 2                 |
| G-003        | FAC-YR-NUM | #####          | DRAWING LIST PART 2 OF 2                 |
| G-004        | FAC-YR-NUM | #####          | CODE SUMMARY & LIFE SAFETY NOTES         |
| G-005        | FAC-YR-NUM | #####          | LIFE SAFETY FLOOR PLAN                   |
| G-006        | FAC-YR-NUM | #####          | OUTAGE SEQUENCING AND MITIGATION PLAN    |
|              |            |                |                                          |
| H-101        | FAC-YR-NUM | #####          | HAZARDOUS MATERIALS PLAN 1 OF 2          |
| H-102        | FAC-YR-NUM | #####          | HAZARDOUS MATERIALS PLAN 2 OF 2          |
|              |            |                |                                          |
| B-001        | FAC-YR-NUM | #####          | GEOTECHNICAL NOTES AND SYMBOLS           |
| B-101        | FAC-YR-NUM | #####          | GEOTECHNICAL BORING LOGS PART 1 OF 3     |
| B-102        | FAC-YR-NUM | #####          | GEOTECHNICAL BORING LOGS PART 2 OF 3     |
| B-103        | FAC-YR-NUM | #####          | GEOTECHNICAL BORING LOGS PART 3 OF 3     |
| B-104        | FAC-YR-NUM | #####          | GEOTECHNICAL BORING LOGS - BID OPTION 13 |
|              |            |                |                                          |
| C-001        | FAC-YR-NUM | #####          | CIVIL GENERAL NOTES                      |

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|       |            |       |                                                             |
|-------|------------|-------|-------------------------------------------------------------|
| C-002 | FAC-YR-NUM | ##### | POTHOLE INFORMATION                                         |
| V-101 | FAC-YR-NUM | ##### | EXISTING SURVEY OVERALL SITE PLAN                           |
| V-102 | FAC-YR-NUM | ##### | EXISTING SURVEY UNDERGROUND UTILITIES PLAN                  |
| V-103 | FAC-YR-NUM | ##### | EXISTING SURVEY ABOVE GROUND UTILITIES PLAN                 |
| V-104 | FAC-YR-NUM | ##### | HISTORICAL QUAY WALLS AND BUILDING FOUNDATION PLANS         |
| GC100 | FAC-YR-NUM | ##### | GENERAL SITE AREAS                                          |
| C-100 | FAC-YR-NUM | ##### | SOIL EROSION AND SEDIMENT CONTROL PLAN                      |
| C-101 | FAC-YR-NUM | ##### | SITE PLAN                                                   |
| C-102 | FAC-YR-NUM | ##### | UTILITY PLAN                                                |
| C-103 | FAC-YR-NUM | ##### | GRADING AND DRAINAGE PLAN                                   |
| C-104 | FAC-YR-NUM | ##### | EXISTING CONDITIONS SITE PLAN - BID OPTIONS                 |
| C-105 | FAC-YR-NUM | ##### | TUNNEL ABANDONMENT PART 1 OF 3 - BID OPTION 4               |
| C-106 | FAC-YR-NUM | ##### | TUNNEL ABANDONMENT PART 2 OF 3 - BID OPTION 5               |
| C-107 | FAC-YR-NUM | ##### | TUNNEL ABANDONMENT PART 3 OF 3 - BID OPTION 6               |
| C-108 | FAC-YR-NUM | ##### | OPTION 13: ACCESSORY STRUCTURE                              |
| C-301 | FAC-YR-NUM | ##### | CIVIL PROFILE - B242 TO B72 AND B72 SEWER                   |
| C-302 | FAC-YR-NUM | ##### | CIVIL PROFILE - QUAY WALL TO BUILDING B72                   |
| C-303 | FAC-YR-NUM | ##### | CIVIL PROFILE - DMH 54-A1 TO DMH 8-61 - BID OPTION 6        |
| C-304 | FAC-YR-NUM | ##### | CIVIL PROFILE - INTAKE TO B242                              |
| CT101 | FAC-YR-NUM | ##### | FIRE TRUCK TURNING MOVEMENT SITE PLAN                       |
| C-501 | FAC-YR-NUM | ##### | CIVIL DETAILS PART 1 OF 4                                   |
| C-502 | FAC-YR-NUM | ##### | CIVIL DETAILS PART 2 OF 4                                   |
| C-503 | FAC-YR-NUM | ##### | CIVIL DETAILS PART 3 OF 4                                   |
| C-504 | FAC-YR-NUM | ##### | CIVIL DETAILS PART 4 OF 4                                   |
|       |            |       |                                                             |
| CG101 | FAC-YR-NUM | ##### | DRY DOCK 2 LAYDOWN PLAN - BID OPTION 9                      |
| CG102 | FAC-YR-NUM | ##### | DRY DOCK 2 TEMPORARY OUTAGE SYSTEM SITE PLAN - BID OPTION 9 |

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|       |            |       |                                                             |
|-------|------------|-------|-------------------------------------------------------------|
| CG501 | FAC-YR-NUM | ##### | DRY DOCK 2 TEMPORARY OUTAGE & CIVIL DETAILS - BID OPTION 9  |
| CG601 | FAC-YR-NUM | ##### | DRY DOCK 2 TEMPORARY PIPING SYSTEM SCHEMATIC - BID OPTION 9 |
|       |            |       |                                                             |
| A-000 | FAC-YR-NUM | ##### | GENERAL PROJECT NOTES                                       |
| AD001 | FAC-YR-NUM | ##### | DEMOLITION NOTES                                            |
| AD101 | FAC-YR-NUM | ##### | DEMOLITION FLOOR PLAN                                       |
| AD102 | FAC-YR-NUM | ##### | DEMOLITION FLOOR PLAN - BID OPTION 8                        |
| A-001 | FAC-YR-NUM | ##### | ABBREVIATIONS AND SYMBOLS                                   |
| A-002 | FAC-YR-NUM | ##### | GENERAL NOTES AND LEGENDS                                   |
| A-003 | FAC-YR-NUM | ##### | PARTITION, CEILING AND ROOF TYPES                           |
| A-101 | FAC-YR-NUM | ##### | PROPOSED FLOOR PLAN - PUMP ROOMS B & S, ROOM C              |
| A-102 | FAC-YR-NUM | ##### | PROPOSED FLOOR PLAN - BID OPTION 8                          |
| A-103 | FAC-YR-NUM | ##### | PROPOSED ROOF PLAN                                          |
| A-200 | FAC-YR-NUM | ##### | EXTERIOR ELEVATIONS                                         |
| A-201 | FAC-YR-NUM | ##### | EXTERIOR ELEVATIONS - BID OPTION 8                          |
| A-231 | FAC-YR-NUM | ##### | AXONONOMETRIC VIEWS                                         |
| A-300 | FAC-YR-NUM | ##### | BUILDING SECTIONS PART 1 OF 2                               |
| A-301 | FAC-YR-NUM | ##### | BUILDING SECTIONS PART 2 OF 2                               |
| A-400 | FAC-YR-NUM | ##### | ENLARGED PLAN -PUMP ROOMS B & S, ROOMS C & M                |
| A-401 | FAC-YR-NUM | ##### | ENLARGED PLAN - BID OPTION 11                               |
| A-402 | FAC-YR-NUM | ##### | PLAN & ELEVATIONS - BID OPTION 13                           |
| A-403 | FAC-YR-NUM | ##### | WALL SECTIONS & DETAILS - BID OPTION 13                     |
| A-410 | FAC-YR-NUM | ##### | ENLARGED RCP -PUMP ROOMS B & S, ROOMS C & M                 |
| A-450 | FAC-YR-NUM | ##### | ENLARGED PLAN DETAILS                                       |
| A-500 | FAC-YR-NUM | ##### | WALL SECTIONS                                               |
| A-501 | FAC-YR-NUM | ##### | DOOR DETAILS                                                |
| A-502 | FAC-YR-NUM | ##### | ROOF DETAILS PART 1 OF 2                                    |

|       |            |       |                                                       |
|-------|------------|-------|-------------------------------------------------------|
| A-503 | FAC-YR-NUM | ##### | ROOF DETAILS PART 2 OF 2                              |
| A-600 | FAC-YR-NUM | ##### | INTERIOR ELEVATIONS PART 1 OF 2                       |
| A-601 | FAC-YR-NUM | ##### | INTERIOR ELEVATIONS PART 2 OF 2                       |
| A-602 | FAC-YR-NUM | ##### | DETAILS                                               |
| A-801 | FAC-YR-NUM | ##### | DOOR DETAILS & SCHEDULE                               |
|       |            |       |                                                       |
| S-001 | FAC-YR-NUM | ##### | GENERAL NOTES PART 1 OF 2                             |
| S-002 | FAC-YR-NUM | ##### | GENERAL NOTES PART 2 OF 2                             |
| S-101 | FAC-YR-NUM | ##### | OVERALL PLAN - B72                                    |
| S-102 | FAC-YR-NUM | ##### | FLOOR PLAN - PUMP ROOM S                              |
| S-103 | FAC-YR-NUM | ##### | FLOOR PLAN - UF TANK AND SUCTION WELL                 |
| S-104 | FAC-YR-NUM | ##### | FOUNDATION PLAN - PUMP ROOM B ADDITION                |
| S-105 | FAC-YR-NUM | ##### | FOUNDATION PLAN - PUMP ROOM B<br>ADDITION, BREAK TANK |
| S-106 | FAC-YR-NUM | ##### | FRAMING PLAN - PUMP ROOM B ADDITION                   |
| S-107 | FAC-YR-NUM | ##### | ROOF FRAMING PLAN - PUMP ROOM S                       |
| S-108 | FAC-YR-NUM | ##### | ROOF FRAMING PLAN - ROOM M                            |
| S-110 | FAC-YR-NUM | ##### | FRAMING PLAN BID OPTIONS - PUMP ROOM S                |
| S-121 | FAC-YR-NUM | ##### | FLOOR PLAN - B242 WET WELL                            |
| S-122 | FAC-YR-NUM | ##### | FLOOR PLAN - PUMP ROOM A                              |
| S-123 | FAC-YR-NUM | ##### | FOUNDATION PLAN - BID OPTION 13                       |
| S-124 | FAC-YR-NUM | ##### | ROOF FRAMING PLAN - PUMP ROOM A                       |
| S-301 | FAC-YR-NUM | ##### | SECTIONS PART 1 OF 10                                 |
| S-302 | FAC-YR-NUM | ##### | SECTIONS PART 2 OF 10                                 |
| S-303 | FAC-YR-NUM | ##### | SECTIONS PART 3 OF 10                                 |
| S-304 | FAC-YR-NUM | ##### | SECTIONS PART 4 OF 10                                 |
| S-305 | FAC-YR-NUM | ##### | SECTIONS PART 5 OF 10                                 |
| S-306 | FAC-YR-NUM | ##### | SECTIONS PART 6 OF 10                                 |

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|       |            |       |                                            |
|-------|------------|-------|--------------------------------------------|
| S-307 | FAC-YR-NUM | ##### | SECTIONS PART 7 OF 10                      |
| S-308 | FAC-YR-NUM | ##### | SECTIONS PART 8 OF 10                      |
| S-309 | FAC-YR-NUM | ##### | SECTIONS PART 9 OF 10                      |
| S-310 | FAC-YR-NUM | ##### | SECTIONS PART 10 OF 10                     |
| S-501 | FAC-YR-NUM | ##### | DETAILS PART 1 OF 6 - B72                  |
| S-502 | FAC-YR-NUM | ##### | DETAILS PART 2 OF 6 - B72                  |
| S-503 | FAC-YR-NUM | ##### | DETAILS PART 3 OF 6 - B242                 |
| S-504 | FAC-YR-NUM | ##### | DETAILS PART 4 OF 6 - B242                 |
| S-505 | FAC-YR-NUM | ##### | DETAILS PART 5 OF 6 - B72                  |
| S-506 | FAC-YR-NUM | ##### | DETAILS PART 6 OF 6 - B72                  |
| SD101 | FAC-YR-NUM | ##### | DEMOLITION FLOOR PLAN - PUMP ROOM S        |
| SD102 | FAC-YR-NUM | ##### | DEMOLITION ROOF FRAMING PLAN - PUMP ROOM S |
| SD103 | FAC-YR-NUM | ##### | DEMOLITION FLOOR PLAN - PUMP ROOM A        |
| SD104 | FAC-YR-NUM | ##### | DEMOLITION ROOF FRAMING PLAN - PUMP ROOM A |
| SD301 | FAC-YR-NUM | ##### | DEMOLITION SECTIONS PART 1 OF 2            |
| SD302 | FAC-YR-NUM | ##### | DEMOLITION SECTIONS PART 2 OF 2            |
|       |            |       |                                            |
| F-001 | FAC-YR-NUM | ##### | FIRE PROTECTION NOTES AND DETAILS          |
| FA101 | FAC-YR-NUM | ##### | FIRE ALARM FLOOR PLAN                      |
| FX101 | FAC-YR-NUM | ##### | FIRE SPRINKLER FLOOR PLAN - PUMP ROOM S    |
| FX102 | FAC-YR-NUM | ##### | FIRE SPRINKLER FLOOR PLAN - PUMP ROOM C    |
| FX500 | FAC-YR-NUM | ##### | FIRE SPRINKLER DETAILS                     |
|       |            |       |                                            |
| M-001 | FAC-YR-NUM | ##### | MECHANICAL SYMBOLS & ABBREVIATIONS         |
| M-002 | FAC-YR-NUM | ##### | MECHANICAL ABBREVIATIONS & NOTES           |
| MD101 | FAC-YR-NUM | ##### | OVERALL DEMOLITION PLAN - B72              |
| MD102 | FAC-YR-NUM | ##### | PUMP ROOM S & ROOM C DEMOLITION PLAN       |

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|       |            |       |                                                        |
|-------|------------|-------|--------------------------------------------------------|
| MD103 | FAC-YR-NUM | ##### | ROOM M DEMOLITION PLAN - BID OPTION 8                  |
| MD301 | FAC-YR-NUM | ##### | PUMP ROOM S DEMOLITION SECTIONS                        |
| M-101 | FAC-YR-NUM | ##### | OVERALL PROPOSED WORK PLAN - B72                       |
| M-102 | FAC-YR-NUM | ##### | PUMP ROOM S & ROOM C PROPOSED WORK PLAN                |
| M-103 | FAC-YR-NUM | ##### | ROOM M & PUMP ROOM B PROPOSED WORK PLAN                |
| M-104 | FAC-YR-NUM | ##### | PUMP ROOM S PROPOSED ROOF PLAN                         |
| MP101 | FAC-YR-NUM | ##### | OVERALL BASEMENT MECHANICAL PROPOSED PIPING PLAN - B72 |
| MP102 | FAC-YR-NUM | ##### | PUMP ROOM S & ROOM C PROPOSED PIPING PLAN              |
| MP400 | FAC-YR-NUM | ##### | ENLARGED PROPOSED PIPING PLANS                         |
| M-501 | FAC-YR-NUM | ##### | MECHANICAL DETAILS PART 1 OF 3                         |
| M-502 | FAC-YR-NUM | ##### | MECHANICAL DETAILS PART 2 OF 3                         |
| M-503 | FAC-YR-NUM | ##### | MECHANICAL DETAILS PART 3 OF 3                         |
| M-600 | FAC-YR-NUM | ##### | MECHANICAL SCHEDULES                                   |
| M-601 | FAC-YR-NUM | ##### | MECHANICAL AIRFLOW DIAGRAM                             |
| M-602 | FAC-YR-NUM | ##### | NORTH TOWER TIE IN DIAGRAM - BID OPTION 2              |
| M-603 | FAC-YR-NUM | ##### | EXISTING SOUTH TOWER TIE IN DIAGRAM - BID OPTION 2     |
| M-700 | FAC-YR-NUM | ##### | MECHANICAL SEQUENCE OF OPERATIONS                      |
|       |            |       |                                                        |
| D-001 | FAC-YR-NUM | ##### | PROCESS NOTES , ABBREVIATIONS, AND LEGEND              |
| DD101 | FAC-YR-NUM | ##### | DEMOLITION PLAN PUMP ROOM A                            |
| D-101 | FAC-YR-NUM | ##### | PUMP ROOM A WET WELL PLAN, SECTION & DETAIL            |
| D-102 | FAC-YR-NUM | ##### | PUMP ROOM A PROCESS FLOOR PLAN                         |
| D-103 | FAC-YR-NUM | ##### | OVERALL PROCESS PLAN - B72                             |
| D-104 | FAC-YR-NUM | ##### | PUMP ROOM S PLAN PART 1 OF 3                           |
| D-105 | FAC-YR-NUM | ##### | PUMP ROOM S PLAN PART 2 OF 3                           |
| D-106 | FAC-YR-NUM | ##### | PUMP ROOM S PLAN PART 3 OF 3                           |
| D-107 | FAC-YR-NUM | ##### | UF BREAK TANK PLAN                                     |

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|       |            |       |                                          |
|-------|------------|-------|------------------------------------------|
| D-108 | FAC-YR-NUM | ##### | PUMP ROOM B PLAN PART 1 OF 2             |
| D-109 | FAC-YR-NUM | ##### | PUMP ROOM B PLAN PART 2 OF 2             |
| D-110 | FAC-YR-NUM | ##### | ROOM C PLAN PART 1 OF 2                  |
| D-111 | FAC-YR-NUM | ##### | ROOM C PLAN PART 2 OF 2                  |
| D-112 | FAC-YR-NUM | ##### | BASEMENT PLAN AND SECTION                |
| D-113 | FAC-YR-NUM | ##### | CONDENSATE TANK PLAN                     |
| D-301 | FAC-YR-NUM | ##### | PUMP ROOM A SECTIONS                     |
| D-401 | FAC-YR-NUM | ##### | INTAKE SCREEN PLAN, SECTION, AND DETAILS |
| D-402 | FAC-YR-NUM | ##### | INTAKE SCREEN FENDER PLAN AND SECTIONS   |
| D-403 | FAC-YR-NUM | ##### | PUMP ROOM S ENLARGED PLAN PART 1 OF 3    |
| D-404 | FAC-YR-NUM | ##### | PUMP ROOM S ENLARGED PLAN PART 2 OF 3    |
| D-405 | FAC-YR-NUM | ##### | PUMP ROOM S ENLARGED PLAN PART 3 OF 3    |
| D-406 | FAC-YR-NUM | ##### | PUMP ROOM S SECTIONS PART 1 OF 6         |
| D-407 | FAC-YR-NUM | ##### | PUMP ROOM S SECTIONS PART 2 OF 6         |
| D-408 | FAC-YR-NUM | ##### | PUMP ROOM S SECTIONS PART 3 OF 6         |
| D-409 | FAC-YR-NUM | ##### | PUMP ROOM S SECTIONS PART 4 OF 6         |
| D-410 | FAC-YR-NUM | ##### | PUMP ROOM S SECTIONS PART 5 OF 6         |
| D-411 | FAC-YR-NUM | ##### | PUMP ROOM S SECTIONS PART 6 OF 6         |
| D-412 | FAC-YR-NUM | ##### | UF BREAK TANK SECTION                    |
| D-413 | FAC-YR-NUM | ##### | PUMP ROOM B ENLARGED PLANS PART 1 OF 2   |
| D-414 | FAC-YR-NUM | ##### | PUMP ROOM B ENLARGED PLANS PART 2 OF 2   |
| D-415 | FAC-YR-NUM | ##### | PUMP ROOM B SECTIONS PART 1 OF 3         |
| D-416 | FAC-YR-NUM | ##### | PUMP ROOM B SECTIONS PART 2 OF 3         |
| D-417 | FAC-YR-NUM | ##### | PUMP ROOM B SECTIONS PART 3 OF 3         |
| D-418 | FAC-YR-NUM | ##### | ROOM C SECTIONS PART 1 OF 2              |
| D-419 | FAC-YR-NUM | ##### | ROOM C SECTIONS PART 2 OF 2              |
| D-420 | FAC-YR-NUM | ##### | OUTFALL AND FENDER PLAN AND SECTIONS     |



|       |            |       |                                                      |
|-------|------------|-------|------------------------------------------------------|
| D-501 | FAC-YR-NUM | ##### | DETAILS PART 1 OF 5                                  |
| D-502 | FAC-YR-NUM | ##### | DETAILS PART 2 OF 5                                  |
| D-503 | FAC-YR-NUM | ##### | DETAILS PART 3 OF 5                                  |
| D-504 | FAC-YR-NUM | ##### | DETAILS PART 4 OF 5                                  |
| D-505 | FAC-YR-NUM | ##### | DETAILS PART 5 OF 5                                  |
| D-601 | FAC-YR-NUM | ##### | PROCESS FLOW DIAGRAM PART 1 OF 2                     |
| D-602 | FAC-YR-NUM | ##### | PROCESS FLOW DIAGRAM PART 2 OF 2                     |
| D-603 | FAC-YR-NUM | ##### | CONTROL SEQUENCE ONE-LINE DIAGRAM                    |
| D-604 | FAC-YR-NUM | ##### | AIR BURST SYSTEM SCHEMATIC DIAGRAM                   |
| D-605 | FAC-YR-NUM | ##### | HYDRAULIC PROFILE                                    |
| D-606 | FAC-YR-NUM | ##### | CHEMICAL FEED SYSTEM SCHEMATICS                      |
| D-607 | FAC-YR-NUM | ##### | PROCESS SCHEDULE AND EQUIPMENT LIST                  |
| DI001 | FAC-YR-NUM | ##### | INSTRUMENTATION LEGEND                               |
| DI100 | FAC-YR-NUM | ##### | CONTROL ROOM AND TELECOMMUNICATIONS<br>PLAN VIEW     |
| DI501 | FAC-YR-NUM | ##### | INSTRUMENTATION DETAILS PART 1 OF 2                  |
| DI502 | FAC-YR-NUM | ##### | INSTRUMENTATION DETAILS PART 2 OF 2                  |
| DI503 | FAC-YR-NUM | ##### | INTAKE SCREEN DETAIL                                 |
| DI504 | FAC-YR-NUM | ##### | TEMPORARY ASW SYSTEM CONTROL DETAILS                 |
| DI601 | FAC-YR-NUM | ##### | NETWORK ARCHITECTURE                                 |
| DI602 | FAC-YR-NUM | ##### | PIPING AND INSTRUMENTATION DIAGRAM -<br>PUMP ROOM A  |
| DI603 | FAC-YR-NUM | ##### | PIPING AND INSTRUMENTATION DIAGRAM -<br>PUMP ROOM S  |
| DI604 | FAC-YR-NUM | ##### | PIPING AND INSTRUMENTATION DIAGRAM -<br>YARD         |
| DI605 | FAC-YR-NUM | ##### | PIPING AND INSTRUMENTATION DIAGRAM -<br>CHEM STORAGE |
| DI701 | FAC-YR-NUM | ##### | CONTROL STRATEGIES - GENERAL                         |
| DI702 | FAC-YR-NUM | ##### | CONTROL STRATEGIES - INTAKE                          |
| DI703 | FAC-YR-NUM | ##### | CONTROL STRATEGIES - PERMEATE                        |
| DI704 | FAC-YR-NUM | ##### | PLC POINTS LISTS                                     |

|       |            |       |                                                         |
|-------|------------|-------|---------------------------------------------------------|
| DI705 | FAC-YR-NUM | ##### | FIELD INSTRUMENT LIST                                   |
|       |            |       |                                                         |
| E-001 | FAC-YR-NUM | ##### | ELECTRICAL SYMBOLS & LEGENDS                            |
| E-002 | FAC-YR-NUM | ##### | ELECTRICAL TYPICAL NOTES AND EQUIPMENT WIRING SCHEDULES |
| E-101 | FAC-YR-NUM | ##### | ELECTRICAL SITE PLAN                                    |
| E-102 | FAC-YR-NUM | ##### | ELECTRICAL SITE GROUNDING PLAN & RISER DIAGRAM - B72    |
| ED101 | FAC-YR-NUM | ##### | ELECTRICAL SITE DEMOLITION PLAN - B72                   |
| ED102 | FAC-YR-NUM | ##### | BASEMENT DEMOLITION PLAN - B72                          |
| ED103 | FAC-YR-NUM | ##### | PUMP ROOM S & ROOM C DEMOLITION PLAN                    |
| EL101 | FAC-YR-NUM | ##### | ROOM M LIGHTING PLAN - BID OPTION 8                     |
| EL102 | FAC-YR-NUM | ##### | PUMP ROOM S & ROOM C LIGHTING PLAN                      |
| EL103 | FAC-YR-NUM | ##### | PUMP ROOM B LIGHTING PLAN                               |
| EP101 | FAC-YR-NUM | ##### | ELECTRICAL SITE & PUMP ROOM B POWER PLAN - B72          |
| EP102 | FAC-YR-NUM | ##### | PUMP ROOM S & ROOM C POWER PLAN                         |
| EP103 | FAC-YR-NUM | ##### | ROOM M POWER PLAN - BID OPTION 8                        |
| EP104 | FAC-YR-NUM | ##### | PUMP ROOM S & PUMP ROOM B ROOF POWER PLAN               |
| EP105 | FAC-YR-NUM | ##### | BASEMENT MV ROUTING PLAN - B72                          |
| EP106 | FAC-YR-NUM | ##### | MIDLEVEL MV ROUTING PLAN - B72                          |
| EP107 | FAC-YR-NUM | ##### | SECOND FLOOR MV ROUTING PLAN - B72                      |
| EP108 | FAC-YR-NUM | ##### | PUMP ROOM A POWER PLAN                                  |
| EP109 | FAC-YR-NUM | ##### | ACCESSORY STRUCTURE SITE PLAN - BID OPTION 13           |
| EP501 | FAC-YR-NUM | ##### | DETAILS PART 1 OF 6                                     |
| EP502 | FAC-YR-NUM | ##### | DETAILS PART 2 OF 6                                     |
| EP503 | FAC-YR-NUM | ##### | DETAILS PART 3 OF 6                                     |
| EP504 | FAC-YR-NUM | ##### | DETAILS PART 4 OF 6                                     |
| EP505 | FAC-YR-NUM | ##### | DETAILS PART 5 OF 6                                     |
| EP506 | FAC-YR-NUM | ##### | DETAILS PART 6 OF 6                                     |

|       |            |       |                                  |
|-------|------------|-------|----------------------------------|
| EP601 | FAC-YR-NUM | ##### | POWER SYSTEM SINGLE LINE DIAGRAM |
| EP701 | FAC-YR-NUM | ##### | SCHEDULES PART 1 OF 2            |
| EP702 | FAC-YR-NUM | ##### | SCHEDULES PART 2 OF 2            |

-- End of Document --

SECTION 01 11 00.00 22

SUMMARY OF WORK (PWD ME)  
05/21

PART 1 GENERAL

This Section applies to Design-Bid-Build projects at the Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Salvage Plan; G

1.2 WORK COVERED BY CONTRACT DOCUMENTS

1.2.1 Project Description

The work includes the construction of a desalination facility within the area of Building 242 and Building 72 and includes a new intake at Building 242 and a new discharge at Berth 4.

This project will repurpose the Pump Room 5 located in Building 72 to be modified and dedicated to the new desalination facility.

The desalination system will employ a multi-step treatment process to provide feed water for boiler and cooling water in Building 72. In the first treatment step of the process, seawater will be pumped through an ultrafiltration (UF) membrane process. This initial unit process is a prefiltration step before Reverse Osmosis (RO) desalination is used. Treated water (permeate) from the UF will be pumped through a two stage, high pressure reverse osmosis membrane process.

The major components of this project include:

- a. A new intake at Building 242.
- b. A new desalination intake pump station within Building 242 that will be installed adjacent to the existing Auxiliary Sea Water (ASW) pump station. The desalination intake pump station will function independently (electrically and hydraulically) of the ASW pump station.
- c. A new piping system that connects Building 242 and Building 72. This piping system will convey raw water from Building 242 and Building 72. New conduits for electrical power and signal wires will also be included.
- d. A new desalination facility primarily placed within:
  - (1) The Pump Room S in Building 72

- (2) A new tank pad adjacent to the Pump Room S (Tank Pad)
- (3) A new Pump house located on the tank pad (BWRO / Room B)
- (4) A new chemical storage room, Room M
- e. A new room located near the condensate tank (Room M)
- f. A new stair system in the Pump Room S.
- g. A new piping system that connects Building 72 to the outfall at Berth 4. This piping system will convey concentrated brine raw water from Building 72 to the outfall. This piping system will also discharge non-saline water used for backwash in the desalination facility operations.
- h. A new outfall at Berth 4.
- i. Partial or complete abandonment of the Saltwater Tunnel (as bid options).
- j. All associated electrical and process control equipment.

#### 1.2.2 Project Sequencing Requirements

The following restrictions apply to the sequencing of the work:

- a. Existing furniture and equipment to remain in construction areas (e.g. office furniture, etc.) must be protected from dust, debris, demolition, and construction activities to ensure the items are not damaged and are useable at project completion. Temporary partitions must be full height. Provide daily cleanup as specified in Section 02 41 00 DEMOLITION and as directed by the Contracting Officer.
- b. The building will be occupied by Government personnel adjacent to and within work areas during the project. Buildings 72 and 242 are 24/7 production facilities. Equipment and personnel must be protected at all times.
- c. Construction sequencing and phasing plans have been developed in coordination with the Government and are provided within the drawing set. The construction sequencing/phasing plans provide proposed construction sequence strategies. Development of an independent construction phasing plan and construction schedule is the responsibility of the Contractor and shall be prepared in coordination with the Contracting Officer. The independent construction phasing plan and construction schedule shall be prepared to avoid impacting production operations.
- d. During the phased work of this project, occupied portions of Buildings 242 and 72 must remain fully operational including, but not limited to, the following utilities: plumbing, heat, ventilation, air conditioning, fire suppression, fire alarm, electrical power, communications systems, and security systems.
- e. Buildings on the Portsmouth Naval Shipyard, including Building 72, are historic facilities and require special coordination to have limited impact on the historic fabric and character defining features of the structure. Work must comply with the SECRETARY OF THE INTERIOR'S

## STANDARDS FOR THE TREATMENT OF HISTORIC PROPERTIES.

- f. The Government may require and will provide escorts for the Contractor while working inside all buildings. Coordinate with the Contracting Officer to confirm which buildings require escorts and to schedule escorts. The project schedule developed in accordance with Section 01 32 17.00 20 COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS) will be used to coordinate required escorts. Notify Contracting Officer weekly of escorts required for the subsequent three weeks of work.

### 1.2.3 Location

The work is located at the Portsmouth Naval Shipyard in Building 72 and Building 242. Note the "Project Site" is the specific area of construction within the limits of work as defined by the contract documents.

### 1.3 OCCUPANCY OF PREMISES

All buildings will be occupied during performance of work under this Contract. At a minimum, identify all building occupancies three weeks in advance on the 3-week look-ahead schedule.

Before work is started, arrange with the Contracting Officer a sequence of procedure, means of access, space for storage of materials and equipment, and use of approaches, corridors, and stairways. While the building is occupied, life safety measures and access to egress components must be maintained. Provide temporary egress lighting, emergency lighting, fire alarm, and fire protection systems along paths of egress.

### 1.4 EXISTING WORK

Remove or alter existing work in such a manner as to prevent injury or damage to any portions of the existing work which remain.

Repair or replace portions of existing work which have been altered during construction operations to match existing or adjoining work, as approved by the Contracting Officer. At the completion of operations, existing work must be in a condition equal to or better than that which existed before new work started.

### 1.5 LOCATION OF UNDERGROUND UTILITIES

Verify location of underground utilities in accordance with Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME).

#### 1.5.1 Notification Prior to Excavation

Notify the Contracting Officer at least fourteen (14) working days prior to starting excavation work. See Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME) and attachments for Dig Safe requirements.

### 1.6 SALVAGE MATERIAL AND EQUIPMENT

Items designated by the Contracting Officer to be salvaged remain the property of the Government. Segregate, itemize, deliver and off-load the salvaged property at the Government designated storage area located on the Portsmouth Naval Shipyard.

Provide a salvage plan, listing material and equipment to be salvaged, and

their storage location. Maintain property control records for material or equipment designated as salvage. Provide a system of property control that is approved by the Contracting Officer. Store and protect salvaged materials and equipment until disposition by the Contracting Officer.

#### 1.7 BENEFICIAL OCCUPANCY REQUIREMENTS

All work elements identified as "Required for Facility Delivery (BOD)" included in Section 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS (PWD ME) Appendix A must be completed and accepted by the Contractor's QC Manager prior to submitting the form to the NAVFAC PWD ME Construction Manager for NAVFAC PWD ME signatures.

The Contractor must submit the completed NRZ Checklist with the required documentation no later than ten (10) working days prior to requesting the Government take Beneficial Occupancy.

Failure of the contractor to verify all work and provide the required documentation will delay the processing of acceptance of Beneficial Occupancy.

#### PART 2 PRODUCTS

Not used.

#### PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 14 00.00 22

WORK RESTRICTIONS (PWD ME)  
**11/22, CHG 4: 02/25**

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at the Portsmouth Naval Shipyard only.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 241 (2022) Standard for Safeguarding  
Construction, Alteration, and Demolition  
Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety -- Safety and Occupational  
Health (SOH) Requirements

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

21 CFR 1040 Performance Standards for Light-Emitting  
Products

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

List of Contact Personnel; G

Flooding Mitigation Strategy Plan

SD-06 Test Reports

PWD ME Contractor Weekend Worklist

CRITICAL INFRASTRUCTURE RISK ANALYSIS (RA) AND RISK MITIGATION  
PLAN (RMP); G

Outage Requests; G

1.3 SPECIAL SCHEDULING REQUIREMENTS (PNSY)

- a. Work requiring interruption (i.e., outage) of any Portsmouth Naval Shipyard service or thoroughfare, including but not limited to



roadways, parking spaces, crane rail, railroads, and utility services, requires coordination around and submission of an outage package prior to commencement of the work. Outage packages must be submitted to the Contracting Officer a minimum of 15 calendar days prior to the desired date of interruption, unless otherwise specified. See paragraph FACILITY/UTILITY/SERVICE OUTAGE REQUIREMENTS herein for additional details and requirements.

- b. Work in certain areas and/or buildings may requiring continuous escorting by a Government employee. See Section 01 11 00.00 22 SUMMARY OF WORK (PWD ME) for further information on Government escorts.
- c. Coordinate the work with the sequencing/phasing requirements outlined in Section 01 11 00.00 22 SUMMARY OF WORK (PWD ME) within paragraph PROJECT SEQUENCING REQUIREMENTS.

#### 1.4 CONTRACTOR ACCESS AND USE OF PREMISES (PNSY)

##### 1.4.1 Portsmouth Naval Shipyard Regulations

Notify the Contracting Officer 21 calendar days in advance for any wide loads. Additional access limitations and coordination with the Contracting Officer may apply.

Ensure that personnel employed on the Portsmouth Naval Shipyard become familiar with and obey Portsmouth Naval Shipyard regulations including safety, fire, traffic, and security regulations. Keep within the limits of the work and avenues of ingress and egress. Wear appropriate personal protective equipment (PPE). Do not enter restricted areas unless required to do so and until cleared for such entry.

Ensure all Contractor equipment, including delivery vehicles, are clearly identified with their company name.

##### 1.4.1.1 Subcontractors and Personnel Contacts

Provide a list of contact personnel of the Contractor and Subcontractors including addresses and telephone numbers for use in the event of an emergency at least 15 days prior to the preconstruction meeting. As changes occur and additional information becomes available, correct and change the information contained in previous lists.

##### 1.4.1.2 Additional Personnel Requirements

##### 1.4.1.1.1 General Construction and Finish Work

General construction and finish work on the Portsmouth Naval Shipyard must be performed by U.S. firms using U.S. citizens. General construction includes construction activities such as building sitework, utilities, foundations, structure, and enclosure or shell, including doors, windows, and facade work. Finish Work includes construction activities such as insulation, floor, partition, and ceiling systems; cabinet work; conveyor systems; specialties; building furnishings, fixtures, and equipment; and mechanical and electrical services and equipment including those specialized for fire protection, security, communication, control, energy conservation, safety, comfort, convenience, and similar purposes.

#### 1.4.1.1.2 Electronic Security Systems Equipment

Electronic Security Systems equipment such as processing control units, workstations, field panels, sensors, high security locks, card readers, cable installation, and system programming, testing and training must be performed by U.S. companies using U.S. citizens who have been subjected to a trustworthiness determination.

#### 1.4.1.2 Installation Access

- a. Obtain access to the Portsmouth Naval Shipyard through participation in the Defense Biometrics Identification System (DBIDS). Requirements for Contractor employee registration, and transition for employees currently under Navy Commercial Access Control System (NCACS), are available at <https://www.cnic.navy.mil/Operations-and-Management/Base-Support/DBIDS/>. No fees are associated with obtaining a DBIDS credential.

##### 1.4.1.2.1 Registration for DBIDS

Registration for DBIDS is available at <https://www.cnic.navy.mil/Operations-and-Management/Base-Support/DBIDS/>. PNS has an online appointment system. Appointments may be made using <https://kiosk.na4.qless.com/kiosk/app/home/191>. Procedure includes:

- a. Present a letter or official award document (i.e. DD Form 1155 or SF 1442) from the Contracting Officer, that provides the purpose for access, to the Portsmouth Naval Shipyard's Pass and Identification/Security Office representative.
- b. Present two (2) forms of valid identification, such as a passport or Real ID Act-compliant state driver's license.
- c. Provide completed SECNAV FORM 5512/1 to the Portsmouth Naval Shipyard's Pass and Identification/Security Office representative to obtain a background check. This form is available for download at <https://www.cnic.navy.mil/Operations-and-Management/Base-Support/DBIDS/>.
- d. Upon successful completion of the background check, the Government will complete the DBIDS enrollment process, which includes Contractor employee photo, fingerprints, base restriction and several other assessments.
- e. Upon successful completion of the enrollment process, the Contractor employee will be issued a DBIDS credential, and will be allowed to proceed to worksite.

All Contractors who possess a Commercial Access Control (CAC) System Card are also required to submit an access request (5512) to support a valid purpose to be on the Portsmouth Naval Shipyard. A second form of valid ID may also be required, if requested, upon arrival at the Portsmouth Naval Shipyard's Entrance Gate. All personnel without CAC cards will need two (2) forms of approved identification for access to the Portsmouth Naval Shipyard.

##### 1.4.1.2.2 DBIDS Eligibility Requirements

Throughout the length of the contract, the Contractor employee must

continue to meet background screen standards. Periodic background screenings are conducted to verify continued DBIDS participation and installation access privileges. DBIDS access privileges will be immediately suspended or revoked if at any time a Contractor employee becomes ineligible.

An adjudication process may be initiated when a background screen failure results in disqualification from participation in the DBIDS, and Contractor employee does not agree with the reason for disqualification. The Government is the final authority.

#### 1.4.1.2.3 DBIDS Notification Requirements

- a. Immediately report instances of lost or stolen badges to the Contracting Officer.
- b. Immediately collect DBIDS credentials and notify the Contracting Officer in writing under the following circumstances:
  - (1) An employee has departed the company without having properly returned or surrendered their DBIDS credential.
  - (2) There is a reasonable basis to conclude that an employee, or former employee, might pose a risk, compromise, or threat to the safety or security to the Portsmouth Naval Shipyard or anyone therein.
- c. Prime contractor is responsible for recovering any badges that are no longer required, including from workers who are no longer employed by prime or subs, and immediately turning badges into the Pass/ID Office. Failure to maintain accountability of issued badges could cause delays in future badge requests. Contracts are warned that badges are checked for currency and any badge not used within 30 days will be disabled.

### 1.5 PORTSMOUTH NAVAL SHIPYARD (PNSY) REGULATIONS

#### 1.5.1 Radiological

##### 1.5.1.1 Radiological Indoctrination (PNSY)

All personnel working at the Portsmouth Naval Shipyard are required to view a 15 minute video briefing on radiological postings and controls in use at the Portsmouth Naval Shipyard. The briefing will be given at the Pass Office prior to issue of security badges and vehicle passes.

Any employee who disregards, alters, moves, or otherwise tampers with a radiological posting, or who disobeys a radiological instruction, may be removed from the Portsmouth Naval Shipyard and denied future access.

##### 1.5.1.2 Yellow Materials (PNSY)

Do not use yellow or orange-yellow colored materials at the Portsmouth Naval Shipyard for the following purposes: Protective clothing, hoods, sheeting, tarpaulins, polyethylene bottles or other containers, tapes, bags, banding, identification marks on tools, boundary markers, ribbons, vent ducts, temporary erosion control devices, survey ribbon, etc. Contact the Contracting Officer for a list of yellow items that have been approved for use on the Portsmouth Naval Shipyard. Dispose of generated yellow colored waste off of the Portsmouth Naval Shipyard. Portsmouth

Naval Shipyard refuse containers must not be used for disposal of yellow colored waste materials. Yellow colored items such as described above are of special significance within the Portsmouth Naval Shipyard and are subject to strict controls. Yellow colored Contract generated debris must be bagged in non-translucent containers, and promptly removed from the Portsmouth Naval Shipyard.

#### 1.5.1.3 Smoke Detectors (PNSY)

Ionization type smoke detectors and duct smoke detectors contain radioactive material and are prohibited from use on the Portsmouth Naval Shipyard. Photoelectric smoke detectors are the only type authorized for use on the Portsmouth Naval Shipyard.

#### 1.5.1.4 Radioactive Sources (PNSY)

All Contracts involving radiation generating devices must conform to the requirements listed in Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME) and U.S. Army Corps of Engineers Safety Manual EM 385-1-1.

#### 1.5.2 Laser Control

Comply with laser safety requirements under 21 CFR 1040 and ANSI Z 136.1-1986 for any work under this Contract utilizing lasers.

#### 1.5.3 Energy Conservation

In cooperation with Government representatives, participate in an active program directed toward the efficient use of energy.

#### 1.5.4 Fire Prevention (PNSY)

Require employees to become familiar with fire prevention regulations within the Portsmouth Naval Shipyard to include the proper method of turning in a fire alarm, storage of flammable and combustible materials, and control of combustible waste and trash. Any "Hot Work" (i.e. welding or cutting) requires a Hot Work Permit prior to commencing such work. Refer to paragraph HOT WORK PERMIT in Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME) for the requirements.

#### 1.5.5 Identification and Control of Seamed (Welded) Pipe and Tubing (PNSY)

Submarine Safety regulations prohibit the use of seamed (welded) pipe or tubing within the Portsmouth Naval Shipyard, unless such pipe or tubing is identified and controlled so as to prevent its inadvertent substitution for seamless pipe or tubing. The following requirements apply and will be strictly enforced:

Any seamed (welded) copper-nickel, carbon steel, carbon-moly steel, stainless steel, nickel-chromium-iron alloy, or nickel-copper pipe or tubing intended for use on the Portsmouth Naval Shipyard must be identified in the following manner PRIOR TO DELIVERY TO THE PORTSMOUTH NAVAL SHIPYARD:

Use a lead-free white paint, to mark a 24-inch long stripe and the word "welded" alternately along the entire length of the pipe or tubing. Apply a one-half inch wide stripe unless the size of the pipe or tubing requires use of a narrower stripe.

Maintain positive control over seamed pipe or tubing until worked into place or removed from the Portsmouth Naval Shipyard.

Seamless pipe or tubing may be substituted for any seamed (welded) pipe or tubing specified in the technical specifications.

The above requirements do not apply to square or rectangular tubing, copper or brass pipe or tubing, or to piping or tubing which has been incorporated into equipment or fixtures prior to delivery to the Portsmouth Naval Shipyard.

#### 1.5.6 Pesticide and Herbicide Control

Do not apply pesticides nor herbicides unless specifically required by this Contract. Where application of pesticides or herbicides is required, provide the submittals required by the specification and obtain written approval prior to any application. The Contracting Officer will be required to review and approve pesticides or herbicides submittals.

#### 1.5.7 Smoking Policy

In accordance with NAVFAC policy, smoking is prohibited within and outside of all buildings on the Portsmouth Naval Shipyard, except in designated smoking areas. This applies to existing buildings, buildings under construction and buildings under renovation. Discarding tobacco materials other than into designated tobacco receptacles is considered littering and subject to fines. Smoking is not permitted within 20 feet of air intakes, doorways, or windows.

#### 1.5.8 Portal Crane Clearance Zone (PNSY)

Ensure there is no construction debris or materials within the Crane Clearance Zone (e.g., between the painted yellow lines on each side of the rail) unless a rail outage/securement has been approved.

### 1.6 WORKING HOURS (PNSY)

Regular working hours will consist of a period between 7 AM and 11 PM, Monday through Friday, and 7 AM Saturday through 11 PM Sunday. Federal holidays will be treated as weekend days, and regular working hours will be between 7 AM and 11 PM. The regular working hours must be confirmed with the Contracting Officer.

Working on Weekends/Federal Holidays: The Contractor must submit the PWD ME Contractor Weekend Worklist to the Contracting Officer by 12 PM on the Wednesday prior to the requested work weekend/Federal Holiday. If the Contractor fails to submit the Weekend Worklist, there are no guarantees that the Contractor will be allowed to work.

#### 1.6.1 Work Outside Regular Hours (PNSY)

Work outside regular working hours requires Contracting Officer approval. Provide written requests fifteen (15) calendar days prior to such work to allow arrangements to be made by the Government for inspecting the work in progress and to allow scheduling of full time escorts in the building(s) if required. During periods of darkness, the different parts of the work must be lighted in a manner approved by the Contracting Officer.

As part of the requests, provide the specific dates, hours, location(s), type(s) of work to be performed, Contract number, and project title. Based on the justification provided, the Contracting Officer may approve work outside regular hours.

#### 1.7 WORK IN OCCUPIED AND EXISTING BUILDING(S)

The Contractor can expect to be working in existing buildings which are occupied. These buildings include, but are not limited to, Buildings 72 and Building 242. The existing buildings and its contents must be kept secure at all times. Provide temporary closures as required to maintain security as directed by the Contracting Officer. Provide dust covers or protective enclosures to protect existing work that remains, and Government material during the construction period. Relocate moveable furniture away from the Contractor's working area(s) as required to perform the work; protect the furniture; and replace the furniture in its original location(s) upon completion of the work. Leave attached equipment in place and protect it against damage, or temporarily disconnect, relocate, protect, and reinstall it at the completion of the work. If determined necessary by the Contracting Officer, the Government will remove and relocate other Government property in the areas of the buildings scheduled to receive work. After providing written notification, allow 15 calendar days for the Government to relocate Government property.

#### 1.8 FACILITY/UTILITY/SERVICE OUTAGE REQUIREMENTS

Outages are required when normal system operation is or could be impacted for any Portsmouth Naval Shipyard service or thoroughfare, including but not limited to roadways, parking spaces, crane rail, railroads, and utility services (i.e., water, sanitary sewer, storm sewer, electric service, fire alarm). This situation can occur when inspecting, repairing, or upgrading utility systems and equipment, buildings, and structures (facilities). These outages may affect one or several buildings, facilities, production processes, and/or geographical locations. Although most outage requirements are addressed in this section, the entire specification must be reviewed for additional outage requirements.

##### 1.8.1 Operation of Portsmouth Naval Shipyard Utilities/Services

The Contractor shall make utility/service cut overs after normal working hours or on weekends/Federal holidays unless otherwise specified/stated on the contract drawings or approved by the Contracting Officer. This cut over time includes time for deactivation and reactivation. Conform to procedures required in paragraph WORK OUTSIDE REGULAR HOURS (PNSY) herein. This includes interruption/outage to water, sanitary sewer, storm sewer, telecommunication services (telephone, internet, radio), electric service, air conditioning, heating, fire alarm, compressed air, and other utility services. Anticipated costs must be included in the bid.

The Contractor shall ensure that new utilities/service are complete, except for the connection, before interruption existing services.

The Contractor must not operate nor disturb the setting of control devices in the Portsmouth Naval Shipyard's utilities/service systems, including water, sewer, electrical, air and steam services, except as otherwise noted herein. The Government will operate 480V and 15KV utility systems for the purposes of securing the systems. The Contractor must notify the Contracting Officer giving 15 calendar days' notice when such operation is

required. The Contractor is responsible for lockout/tagout and grounding; refer to the requirements in Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME). The Contractor shall perform updates to control settings required as per the scope of work.

Connection to Existing Sanitary Sewer Line: Provide positive verification that the existing line conveys sanitary sewer; verify line is not incorrectly connected to a storm drain.

#### 1.8.2 Outage Request Requirements

The outage request process includes development of an outage plan and package, coordination meetings to promote Shipyard and stakeholder understanding and alignment on outages, submittal of the outage request package, and Government approval.

Outages involving Critical Shipyard Infrastructure require a specific Risk Analysis and Risk Mitigation Plan as outlined in paragraph CRITICAL SHIPYARD INFRASTRUCTURE REQUIREMENTS. The Critical Infrastructure Risk Analysis and Risk Mitigation Plan must be submitted to the Contracting Officer for review and approval 15 calendar days prior to the submission of the outage request.

Significant outages required to execute the work may be outlined in the Contract Documents. The Contract Documents do not include all the outages that will be required to support the execution of all work. The Contractor is responsible for identifying all required outages and for all outage request packages.

All outage request packages must be submitted to the Contracting Officer, who will then submit to the Back Shift Representative for final stakeholder concurrence. Submission of an outage request package without prior coordination efforts with the Government, including but not limited to the Outage Coordination Meeting, may result in rejection/denial of the outage request. Any outage requests denied due to incomplete stakeholder alignment or information required to support the requested outages will not be justification for a delay claim. Outage request packages must be submitted at least 15 calendar days in advance of the target outage date.

The Contracting Officer will provide the Outage Request Form and the local Outage Instruction.

The Outage Coordination Meetings referenced below are intended to provide a forum to review the outage plan and information required to support outage reviews.

At a minimum, each outage request package must include the following:

- 1) Specific location/system requiring the outage (i.e., utility, roadway, parking space, crane rail).
- 2) Duration and timing of outage.
- 3) Scope of work required for the outage.
- 4) Outage work plan, including organization responsible for each step to accomplish the outage.
- 5) Adequate sketches/drawings that clearly depict location,

situation, and relevant information/interfaces.

- 6) Scope of outage impact, identifying specific affected systems and stakeholders.
- 7) Hazardous energy control, if applicable.
- 8) For Critical Shipyard Infrastructure (see paragraph CRITICAL SHIPYARD INFRASTRUCTURE REQUIREMENTS) a Critical Infrastructure Risk Analysis and Risk Mitigation plan is required.
- 9) If applicable, include LOTO boundaries and a description of the means to fulfill energy isolation requirements in accordance with EM 385-1-1. Some examples of energy isolation devices and procedures are highlighted in EM 385-1-1. In accordance with EM 385-1-1, where outages involve Government or Utility personnel, coordinate with the Government on all activities involving the control of hazardous energy. These activities include, but are not limited to, a review of Hazardous Energy Control Program (HECP) and Hazardous Energy Control (HEC) procedures, as well as applicable Activity Hazard Analyses (AHAs). In accordance with EM 385-1-1 and NFPA 70E, work on energized electrical circuits for an outage must not be performed without prior Government authorization. Government permission is considered through the outage process and submission of a detailed AHA. Energized work permits are considered only when de-energizing introduces additional or increased hazard or when de-energizing is not feasible.

#### 1.8.2.1 Additional Outage Requirements

The Three Week Look Ahead schedule must include upcoming outages planned for that three week period. Outage planning must start early in the construction process to allow adequate coordination and alignment with affected stakeholders and to integrate timing with critical shipyard operations.

##### a. Outage Coordination Meeting

Outage Coordination Meetings are to be held as necessary, prior to submittal of an outage request package. Refer to Paragraph OUTAGE COORDINATION MEETINGS in Section 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS (PWD ME) for additional information.

##### b. Outage Safety Review Meeting

Outage Safety Review Meetings are to be held prior to any outages that require hazardous energy control (HEC). Refer to Paragraph OUTAGE SAFETY REVIEW MEETINGS in Section 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS (PWD ME) for additional information.

#### 1.9 CRANE AND RAILROAD TRACKAGE INTERRUPTIONS (PNSY)

Crane and railroad trackage are facilities that are subject to strict scheduling and outage approvals. When Contract work is planned within 10' adjacent to ground level trackage (that extends below pavement depth) or on ground level track, submit written requests for Outages & Securements a minimum of 15 calendar days prior to the desired date of interruption to the Contracting Officer. Coordination/alignment on the desired date of



interruption should be accomplished with the crane department (Code 722) before submitting the written request. (See Contracting Officer Representative for Track Outage & Securement Request Forms.)

A. Track Outage is required for:

Any excavation, that is within 10 feet of a rail that will extend below depth of the pavement.

Any work performed that will penetrate a track foundation.

Any work involving trackage system replacement or repair.

Work must not be conducted in affected areas until a written approval to the request for outage in accordance with the requirements specified in Paragraph FACILITY/UTILITY/SERVICE OUTAGE REQUIREMENTS is received.

B. Track Securement:

Track Securement differs from Track Outage in that it does not affect track certification. Track Securement requests must be prepared and submitted to the Contracting Officer, via the outage request process outlined in paragraph OUTAGE SUBMITTAL REQUIREMENTS herein, to obtain scheduling approval from Code 740.2 before work commences that would curtail the use of tracks (e.g., placing equipment or hoses within the crane envelope or within the defined Crane Clearance Zones). Track Securement requests must be prepared using the Request for Track Securement form, which will be provided by the Contracting Officer.

No work must be conducted in affected areas until a written approval to the request for securement is received.

1.10 WORK ADJACENT TO OR IMPACTING CIA SECURITY FENCING (PNSY)

Work adjacent to Portsmouth Naval Shipyard Controlled Industrial Area (CIA) fencing is strictly controlled to ensure security is maintained at all times.

"Breaching the fence" is any repair, alteration, or other work which would allow access into the CIA either over, under, or through an opening in a CIA fence.

Work which will breach CIA fencing (including temporary relocation of CIA fencing) is prohibited unless approval has been obtained from Head of Security Operations (Code 1122) and a Portsmouth Naval Shipyard Police representative is at the worksite during the period that the fence has been breached. This approval must be obtained via the Contracting Officer using the outage request process outlined in paragraph OUTAGE REQUEST REQUIREMENTS herein.

The Contractor must submit the outage request for breaching/relocating the CIA security fence a minimum of 30 calendar days prior to performing the work.

Conditions which breach the fence must be eliminated during all non-work periods to the satisfaction of the Portsmouth Naval Shipyard Police representative. Do not leave the worksite until such conditions are eliminated. All materials used to close openings in fencing and method of installation must be the same type and construction as adjacent,

undisturbed CIA fencing.

The clear zone (10 ft on the inside & outside of the security fence) adjacent to CIA fencing must remain clear of vehicles, materials, equipment or tools. Contractor personnel must be at the location throughout the entire time of any off-loading or work. Under no circumstances shall vehicles, materials, equipment, or tools be left unattended in the clear zone. Any work within the clear zones requires approval from Portsmouth Naval Shipyard Security. Any requests to work in the clear zone must be submitted via the outage request process a minimum of 30 calendar days prior to performing the work.

The only outage exception is for short term (e.g., less than 4 hours) temporary off-loading of materials in the CIA clear zone, which must have prior Shipyard Security approval (Code 1122 verbal or email). During temporary off-loading, Contractor personnel must be at the location throughout the entire time. Under no circumstances shall vehicles, materials, equipment, or tools be left unattended in the clear zone.

#### 1.10.1 Violation of CIA Security Fencing Requirements

If the Contractor violates the security requirements related to the CIA Fencing, Security will issue a Security Deficiency Log (SDL). Upon receiving a SDL, the Contractor must take immediate corrective actions as directed in the SDL including responding to the SDL with documentation of the corrective actions taken and what actions are being taken to prevent recurrence of any further violations. Depending on the security violation, the Contractor may be required to have a Security Stand Down in which all staff including subcontractors will need to be present for full review of the PNSY Security Requirements.

If a second offense occurs, the Contractor will have a Security Stand Down in which all staff including subcontractor will need to be present for full review of the PNSY Security Requirements.

If a third offense occurs, the Project Superintendent is subject to removal by the Contracting Officer for non-compliance with security requirements specified in the Contract. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to Stand Downs, stop work orders, or removal of the Project Superintendent are acceptable as the subject of claim for extension of time for excess costs or damages by the Contractor. Work adjacent to Portsmouth Naval Shipyard Controlled Industrial Area (CIA) fencing is strictly controlled to ensure security is maintained at all times.

#### 1.10.2 CIA Lighting

CIA lighting outages must be approved through Code 1122, and temporary lighting must be provided at the Contractor's expense. Refer to Section 01 50 00.00 22 TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS (PWD ME) paragraph TEMPORARY SECURITY LIGHTING for additional requirements.

#### 1.11 WORK ADJACENT TO AN OVERHEAD CRANE

Provide a minimum vertical clearance of three (3) inches between the highest point of the crane and the lowest overhead obstruction. For buildings where truss sag becomes a factor, increase the clearance as necessary to maintain the minimum required clearance.

The horizontal clearance between the end of the crane and the building columns, knee braces or any other obstructions must not be less than two (2) inches with the crane centered on the runway rails. Pipes, conduits, etc. must not reduce this clearance.

The vertical clearance beneath a bridge crane is to be at least three (3) inches. This clearance is not applicable to the hook block unless it is in its up most position.

For work involving installation, adjustment, or replacement of rail fasteners (e.g. clips/bolts), verify the existing crane envelope and clearance measurements around the rail head prior to beginning work. Notify the Government if the work will reduce the clearance between the bridge crane and rail fasteners and ensure newly installed items will not obstruct bridge crane travel.

Notify the Government to verify that crane clearance has been maintained when the work performed may have changed any physical dimensions of objects or structures adjacent to the crane (e.g., changing or servicing lighting fixtures/pendant assemblies, removal and reinstallation of pipes, conduits, junction boxes, etc.). If the crane is not available (e.g., undergoing maintenance, inspection, etc.), verify crane clearance by taking measurements using reference points (e.g., vertical and horizontal distance from the top of crane rail with respect to the crane envelope, vertical distance from the floor with respect to the crane envelope, etc.).

Where bridge cranes within buildings exist, the contractor must ensure that all installations, modifications or other work included in this contract does not create interferences that fall within the bridge crane clearance zone. The bridge crane clearance zone is 2" on any side of the crane, and 3" above or below the crane. It is the responsibility of the contractor to verify the accuracy of the building and crane dimensions in the field prior to conducting work.

## 1.12 FIRE PROTECTION

### 1.12.1 Compliance (PNSY)

Comply with COE EM 385-1-1, NFPA 241, and Portsmouth Naval Shipyard fire regulations. Obtain approval no later than three (3) working days from the Portsmouth Naval Shipyard Fire Chief via the Contracting Officer prior to commencement of hot work operations.

### 1.12.2 Fire Lanes

#### 1.12.2.1 Fire Lane Width

Fire lanes shall be a minimum of twenty (20) feet in width.

#### 1.12.2.2 Vertical Clearance for Fire Lane

Fire lanes shall have a minimum of thirteen (13) feet six (6) inches nominal vertical clearance, which shall be provided and maintained over the full width of the fire lane.

#### 1.12.2.3 Angle of Approach/Departure

Minimum inside turn radius shall be twenty-five (25) feet and minimum

outside turn radius shall be fifty (50) feet.

#### 1.12.2.4 Parking In or Blocking Fire Lanes

Parking in or blocking fire lanes is expressly prohibited for any reason without prior Fire Department approval. Vehicles left unattended in fire lanes are subject to towing. Contractors will be issued a Non-Compliance Notice for vehicles left unattended in fire lanes.

- a. First Offense: The Contractor will remove the vehicle from the Project Site immediately and the vehicle will not be allowed to return to the Project Site for seven (7) days. The Contractor will perform a safety stand down within three (3) days of the incident and the Fire Department shall be invited to attend the safety stand down.
- b. Second Offense: Contractor and Sub-Contractors vehicles will be permanently barred from the Project Site and a Non-Compliance Notice will be issued. Individuals performing deliveries in support of the Project will be allowed but are expected to be onsite for the shortest duration required to perform the delivery, and the driver/operator shall remain in the immediate vicinity of the vehicle at all times.

#### 1.12.2.5 Approval to Temporarily Block a Roadway/Fire Lane

The Contractor may request approval to temporarily block a roadway/fire lane as part of a Road Outage to complete work associated with the project. Approval may only be given by the Fire Chief or his/her assigned representatives. Arrange through the Contracting Officer with the Fire Chief for scheduling the temporary blockage prior to submitting the outage request package. Final approval will be through the outage process. Refer to Paragraph FACILITY/UTILITY/SERVICE OUTAGE REQUIREMENTS of this section for the outage process. In addition to the requirements outlined within Paragraph OUTAGE REQUEST REQUIREMENTS herein, the Contractor must include the following to any outage packages that include a request to temporarily block a roadway/fire lane:

- a. Alternative analysis that discusses whether a practical alternative to the outage exists. The analysis must address the activity purpose, need, and why the activity cannot be completed by:
  - (1) Utilizing, managing, or expanding the work area to avoid the fire lane.
  - (2) Reducing the size, scope, configuration or density of the activity as proposed (phasing).
- c. Illustration showing the location of Contractor equipment during the proposed outage.
- d. Provisions for emergency access during the outage (e.g. a steel plate shall be kept with 10 feet of the excavation).
- e. Additional provisions may be required by the Fire Department on a site by site basis.

### 1.13 SECURITY REQUIREMENTS

#### 1.13.1 Access to the Portsmouth Naval Shipyard (PNSY)

Contract Clause "FAR 52.204-2, Security Requirements and Alternate II" and

the following apply:

Access to areas designated as restricted NAVSEA spaces will require an escort by a Government Representative. Notify the Contracting Officer at least 14 Calendar Days in advance of the date access is required.

Obtain security badges and vehicle passes to enter the Portsmouth Naval Shipyard at the Portsmouth Naval Shipyard's Pass/Security Office. Furnish proof that employees are U.S. citizens to obtain badges to enter the Portsmouth Naval Shipyard.

Complete Department of Homeland Security Form I-9; Employment Eligibility Verification for each employee and furnish proof that employees are U.S. citizens to obtain badges to enter Portsmouth Naval Shipyard.

#### 1.13.2 Application and Issue of Security Badges

"Temporary" Security Badges will be issued to Contractor personnel requiring access to the Shipyard for less than 90 days. Badge(s) will be issued upon satisfactory proof of U.S. citizenship, in the form of an original or certified birth certificate, passport, or naturalization papers. A picture ID is required in addition to satisfactory proof of citizenship.

"Permanent" (photo) Standard Access Control Badges will be issued to Contractor personnel requiring access to the Shipyard for 90 or more days. Contractor personnel will be required to complete an authorization application form for local record check and a personal information sheet. The forms will be furnished to the Contractor following award of any contract resulting from this solicitation, at the time of the Preconstruction Meeting.

In the event access is required to Contract work areas not permitted by the level of security badge issued, such need must be demonstrated and an escort obtained.

STANDARD ACCESS CONTROL BADGES MUST BE ATTACHED TO THE OUTER GARMENT AND DISPLAYED ABOVE THE WAISTLINE AT ALL TIMES WHILE ON THE PORTSMOUTH NAVAL SHIPYARD.

PERSONNEL MUST NOT ENTER AREAS FOR WHICH THEY HAVE NOT BEEN CLEARED. WHERE A NEED HAS BEEN DEMONSTRATED TO ENTER SUCH AREAS, PERSONNEL WILL BE UNDER CONSTANT ESCORT BY PERSONNEL WHO HAVE BEEN CLEARED. FAILURE TO ADHERE TO POSTED SECURITY REQUIREMENTS MAY RESULT IN REMOVAL OF THE EMPLOYEE FROM THE PORTSMOUTH NAVAL SHIPYARD WITH FUTURE ACCESS DENIED.

#### 1.13.3 Application and Issue of Vehicle Passes for Entry into Portsmouth Naval Shipyard's Controlled Industrial Areas (CIA)

Vehicular access to the CIA must be minimized and all vehicles must comply with the following requirements:

Vehicles must visibly display a CIA vehicle entry pass and inspection pass from the Commercial Vehicle Inspection Station (CVIS), Building 386. CIA passes will only be issued to company owned or leased vehicles or rental vehicles rented in the company name. CIA passes will be issued on weekends and holidays at the Installation Gate. Personnel not possessing the level security badge required for CIA access must be accompanied by a properly badged escort to obtain the CIA vehicle pass.

Vehicles will only be allowed in the CIA for the transportation of tools, parts, and materials to and from the worksite. An exception to this policy, employees may be transported to and from the worksite after full vehicle inspection at CVIS if a specific security plan has been developed and approved by the Portsmouth Naval Shipyard Security Officer.

Parking of company owned vehicles not utilized in the execution of construction activities (e.g. not a working truck or specialty vehicle) within the CIA or in any project laydown space is prohibited.

For Vehicles remaining in the CIA for more than 24 hours: Obtain a CIA Overnight Parking Permit from the PWD ME Construction ET via the Contracting Officer. Vehicles without this pass are subject to being towed. Passes will only be issued to those demonstrating a valid reason as determined by Portsmouth Naval Shipyard Security.

#### 1.13.4 Application and Issue of Crane Passes (PNSY)

Comply with EM 385-1-1. For Crane Passes at the Portsmouth Naval Shipyard to be valid, the Certificate of Compliance for Load Handling Equipment (LHE) and Rigging (ENG Form 6209) must be completed and accepted.

#### 1.13.5 Return of Badges and Vehicle Passes

Ensure all vehicle access permits and personnel badges are returned to the Portsmouth Naval Shipyard Security Officer when the need has ended. Account in writing for each missing pass or badge prior to final payment being made on the Contract. Any vehicle passes issued must be returned to Portsmouth Naval Shipyard Security daily.

#### 1.13.6 Security Responsibilities (PNSY)

Employees must not transport, drink, or have in their possession any alcoholic beverages. Possession of any controlled substances without a physician's prescription is also prohibited. Any employee appearing to be under the influence of intoxicating liquor or narcotics will be apprehended by Portsmouth Naval Shipyard Police, escorted off of the Portsmouth Naval Shipyard, and turned over to the local Police Department.

Any vehicle found to contain controlled substances, including usable residue, may be seized and impounded. Within 24 hours of the work day following any vehicle seizure, the Portsmouth Naval Shipyard Police will have determined whether forfeiture of the vehicle is required. If not, the vehicle will be returned to the owner or authorized agent. If the vehicle is determined to be appropriate for forfeiture, the Portsmouth Naval Shipyard's Legal Officer will notify the Drug Enforcement Administration of such seizure and impoundment, for initiation of forfeiture proceedings pursuant to Title 21, U.S. Code, Section 881. Such actions may be taken regardless of whether the owner/operator of the vehicle had knowledge of the presence of drugs in the vehicle. The Government may pursue criminal or other disciplinary actions pursuant to Title 18, U.S. Code, Section 1382.

Possession of firearms, ammunition, and/or explosives is prohibited. In the event explosives are required for construction work, specific handling requirements and approvals must be obtained from the Portsmouth Naval Shipyard Security Officer via the Contracting Officer.

Cameras, video equipment, or similar photographic equipment must not be introduced into nor removed from the Portsmouth Naval Shipyard. In the event such equipment is required for performance of Contract work, approvals must be obtained from the Portsmouth Naval Shipyard Security Officer via the Contracting Officer.

Weapons (firearms, personal knives with blades 2-1/2 inch long or greater, Mace, Pepper Spray etc.) are not permitted aboard the Portsmouth Naval Shipyard.

At a minimum, all Superintendents, QC Managers, and SSHOs (and anyone acting in this capacity) must have a mobile phone on their persons any time while work is occurring at the Contract jobsite(s). The phone number(s) associated with these phones must be provided during the preconstruction meeting and work will not be allowed to proceed without one.

Laptops and any other computer, regardless of form factor, must not be introduced nor removed from the Portsmouth Naval Shipyard without authorization. Only business use computers are allowed. Personally owned computers are not authorized on the Portsmouth Naval Shipyard. If a computer is required to perform work on the Portsmouth Naval Shipyard, a Visitors Portable Computer Equipment Registration form must be completed and submitted for approval. These forms can be obtained and completed at the pass office upon entrance to the Portsmouth Naval Shipyard or through the Portsmouth Naval Shipyard POC.

Any computing devices that also have or had an installed camera, must follow the same requirements as cell phones. If the computing device is visiting a NAVSEA space, it cannot have a camera. If the device has a camera, the only acceptable remediation is that the camera be professionally removed or drilled out and the resulting void must be filled with an opaque epoxy.

In the event that a camera is required for the performance of the Contracted work, approval must be obtained from the C1120 Security Office. All requests for camera use and deviations must come to C1120 through the Contracting Officer. Authorizations and approved computer registrations must remain with the devices and be presentable upon request. Devices without a documented approval or observed being used in a manner that violates the terms of the Visitor Portable Computer Equipment Registration agreement are subject to search and seizure.

Basic guidance for use of Electronic Devices at Portsmouth Naval Shipyard: Obtain Contracting Officer approval for intent to use any electronic device which has the capability to record, copy, store, export and transmit data, photography, digital images, video or audible information, inside the CIA or any NAVSEA controlled spaces. Provide the manufacturer's data describing the electronic capabilities of the device(s).

Use of wireless networking and communication on the Portsmouth Naval Shipyard requires prior authorization. Communication devices such as cellular hotspots and wireless routers must also be registered using the same Visitors Portable Computer Equipment Registration. This does not include air-cards and straight cellular data. It does include any device that transmits or broadcasts Wi-Fi, including cellular phones with the personal hot-spot feature enabled.

Broadband internet connections must be authorized prior to use on the

Portsmouth Naval Shipyard. All requests for physical cable service from an internet service provider (ISP) must be authorized and coordinated through Code 109. Work through the Contracting Officer to request broadband access.

Driver use of a hand-held cellular phone in a moving vehicle on the Portsmouth Naval Shipyard is prohibited. This prohibition does not include hands-free cellular phone devices. Hands-free devices include console/dash-mounted or otherwise secured cellular phones with integrated features such as voice-activation, speed dial, speakerphone or other similar technology for sending and receiving calls.

Driver use of any portable, personal listening device worn inside the aural canal, around or covering the driver's ear while operating a motor vehicle, is prohibited. Listening devices include wired or wireless earphones and headphones (including blue tooth or similar technology), and do not include hearing aids or devices designed and required for hearing protection.

The use of radar or laser detection devices to indicate the presence of speed recording instruments or to transmit simulated erroneous speeds is prohibited in accordance with OPNAVINST 5100.2H.

Indoctrinate personnel on access limitations to ensure security control is maintained as an integral part of their work pattern and habit.

Unescorted personnel found in security areas requiring a higher level clearance than the level represented by the badge displayed will be removed from the area with possible confiscation of security badges and vehicle passes.

#### 1.13.6.1 Portable Electronic Devices Policy (PNSY)

A PED is a portable electronic device that is easily transportable and has the capability to record, copy, store, export, or transmit data, photography, digital images, video or audible information. Examples of PEDs include, but are not limited to: cell phones, fitness trackers, smart watches, watches with input capability, tablet, laptops, media devices, Bluetooth consumer electronics, cellular transceivers, and portable diagnostic equipment.

PEDs are prohibited within spaces authorized for processing classified information or within spaces where any classified discussion is taking place, except as otherwise authorized. PEDs are permitted in other areas on the Portsmouth Naval Shipyard subject to the usage restrictions as outlined below.

##### 1.13.6.1.1 PEDs in Classified Spaces

These spaces include, but are not limited to, open storage areas, NWAs, Controlled Access Areas (CAA), secure rooms, vaults, occupied dry docks, shipboard, and applicable restricted areas).

PEDs with limited capabilities are permitted.

The following capabilities are prohibited on PEDs in classified spaces:

- a. Cellular and Wi-Fi transceivers.
- b. Photographic, video capture, recording, and transmission, microphone,



- and/or audio capture, recording, and transmission capabilities.
- c. Installed removable media unless otherwise authorized.
- d. Capability to perform radio frequency transmission at greater than 100 milliwatts (mw) Equivalent Isotropically Radiated Power.

#### 1.13.6.1.2 PEDs in U-NNPI Spaces

These spaces include where U-NNPI is processed, transmitted, stored, or discussed (e.g., Security Islands (SIs), CNIAs).

PEDs are permitted as follows:

- a. Contractor owned PEDs are permitted in these areas provided they have their photographic and video capabilities (logically or physically) disabled and they have been marked to identify that the capability is disabled.
  - (1) Contractor owned PEDs, other than cell phones, that do not have markings to identify that the capability is disabled are required to have the PEDs reviewed by Security Services (Code 1123) and Code 109 (if applicable) to ensure the capabilities have been disabled. This will be documented via a laptop registration or camera permit.
  - (2) Personal and Contractor owned cell phones are authorized inside of CNIA spaces if the camera(s) is (are) removed and the void filled with an epoxy or if it meets the requirements of 1.13.6.1.2.a. Additionally, the camera application on the device must indicate that it is disabled.
- b. For SI spaces and inside the CIA, camera capable PEDs (unmodified) are permitted. Although these devices are permitted, use of these devices to take photographs, record video, or record audio is prohibited.

#### 1.13.6.1.3 Violations

Discovery of PEDs in prohibited areas, or discovery of prohibited materials (including Government information or digital photographs that have not been authorized for public release on PEDs) on a personally owned PED, may result in the immediate confiscation of the device by Code 1120 with referral to the appropriate authorities for investigation and, if warranted, personnel action or prosecution. The authority to confiscate a personal PED based on reasonable suspicion for review of images, recordings, or videos is the responsibility of Code 1120 or law enforcement personnel.

Contractors and subcontractors found to be in violation of the Portsmouth Naval Shipyard PED Policy will have the violation reported to the Contracting Officer and may be subject to criminal charges. The penalties for any contractor or subcontractor personnel who are found to be in possession of a PED with prohibited capabilities in violation of this policy will be evaluated on a case by case basis, but at a minimum, the Standard Access Control Badge (SACB) and Defense Biometric Identification System card will be confiscated. Access to the installation will be denied and a Security Deficiency Log (SDL) will be issued. If the violation is more severe (e.g., PED with prohibited capabilities found with pictures inside of a NAVSEA controlled space, etc.) or repeat in nature violations, contractor or subcontractor personnel may be permanently barred from access to NAVSEA controlled spaces.

#### 1.13.7 Access to Unclassified Information

Access to U.S. Navy technical information manuals, documents, drawings, plans, specifications, and other information (e.g., photos, presentations, renderings, papers, etc.) is Government property and restricted to an official need-to-know basis. Handle, control, and safeguard to prevent oral, visual, and documentary disclosure to the public, to foreign sources, and to personnel not having an official need-to-know.

#### 1.13.8 Disclosure of Information

- a. Do not release to anyone outside the Contractor's organization any unclassified information, regardless of medium (e.g., hard copies, computer files, film, tape, document), pertaining to any part of this Contract or any program related to this Contract, unless the Contracting Officer has given prior written approval.
- b. Requests for approval under paragraph (a) must identify the specific information to be released, the medium to be used, and the purpose for the release. Submit the request to the Contracting Officer at least 10 business days before the proposed date for release.
- c. The Contractor agrees to include a similar requirement, including this paragraph (c), in each subcontract under this Contract. Subcontractors must submit requests for authorization to release through the Prime Contractor to the Contracting Officer.

#### 1.13.9 Portsmouth Naval Shipyard Operations Security Statement

During the course of this Contract, in addition to those restrictions, instructions, and guidelines delineated in the Contract Statement of Work, Contract Data Requirements List (CDRL), and/or other references provided, adhere to the following minimum requirements in support of the Portsmouth Naval Shipyard (PNS) OPSEC Program:

Introduction of personal electronic devices, laptops, tablet PCs, cellular phones, cameras, recording devices, and data recording/storage devices into Government spaces is STRICTLY controlled and forbidden in most cases. Company issued equipment required for the performance of work must be approved by the PNS Code 1120 Security Office. Photography and recording is not allowed except for official use and by permit only. Photographs will be reviewed by Code 1123.2 to ensure sensitive information is not revealed.

Do not discuss Government operations in public or over unprotected or unencrypted communications. Do not post to company websites, publications, newsletters or other media, any images, data, or information that reveal sensitive Government operations, personnel, equipment, and/or classified or controlled unclassified information. When in doubt, company press releases related to this Contract must be coordinated through the Contracting Officer's Representative (COR) or Technical Point of Contact, as applicable, and in conjunction with PNS Public Affairs Office (PAO).

Due to observation of events, operations, physical changes, etc. which may reveal National Security Information or Naval Nuclear Propulsion Information (NNPI), specific restrictions are needed to preclude unintentional release of this information to unauthorized parties. (Unauthorized disclosure and transfer of National Security Information is punishable under 18 USC § 793.) Therefore, do not disclose to

unauthorized third parties or post to unofficial sites (including Social Networking sites) any images, data or information, or observed events that reveal sensitive Government operations, personnel, equipment, including, but not limited to:

Tactics, techniques and procedures, production or work schedules, any visible or concealed modifications, upgrades, additions to vessels, weapons or equipment; increases, changes, or decreases in work/deployment frequency or Government personnel, vehicle, vessel movements; specialized equipment orders, deliveries, shipments, etc. Unauthorized disclosures and attempts to solicit this type of information by unauthorized third parties or others not affiliated with this Contract must be reported to PNS Code 1120 Security and/or PNS Police Department. Non-Disclosure requirements remain in effect during the duration of this Contract and indefinitely thereafter.

Government issued badges and identification must be removed and/or concealed from plain sight when off PNS and must not be left in vehicles or unprotected. Badges and passes may not be duplicated, copied, or loaned to others. Lost or stolen identification badges, vehicle passes, etc., must be immediately reported to PNS Code 1120 Security Office and/or PNS Police Department.

It is strongly recommended the Contractor mark and protect related internal production schedules, deliverables, inventories and shortages, and identified vulnerabilities related to production of Government material. Internal company markings (e.g., Business Sensitive, etc.) are appropriate for identifying the aforementioned as sensitive information. All Government information must be destroyed at Contract termination or returned to the Government at the Government's discretion.

#### 1.14 WORK WITHIN PUMPWELLS AND DRY DOCKS (PNSY)

All work completed within the dry docks and pumpwells must have a Flooding Mitigation Strategy Plan. The process for development of this plan must start with a meeting to determine any locations of possible flooding. The plan must cover all penetrations through structures which may be sources of flooding of the pumpwell, the dry dock, or an associated structure. Flooding sources considered include the Piscataqua River as well as surface rain water. The plan must identify the source of flooding, provide a strategy for mitigating flooding from that source and details on when that mitigation will be in place (e.g. pipe that is currently open to the river will have double valve protection when not actively being worked). Provide product data sheets with the plan. The plan must also include an inspection program to verify proper installation and maintenance of all mitigation strategies. The plan must be submitted and approved by the Government prior to commencing the work within the pumpwells and dry docks.

#### 1.15 CONSTRUCTION VEHICLES

Do not utilize any vehicle that will exceed an HS20 wheel load. The use of "off road", "Utility", "ATV", or any vehicles which cannot be legally operated on State roadways or highways is prohibited.

#### 1.16 EMPLOYEE PARKING

Parking on the Portsmouth Naval Shipyard is limited and is not guaranteed. All Contractor and Subcontractor employee privately owned vehicles (POV)

must be parked offsite. Only Contractor company-owned vehicles will be allowed to park on the Portsmouth Naval Shipyard. Refer to Section 01 50 00.00 22 TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS (PWD ME) paragraph EMPLOYEE PARKING PLAN for additional requirements.

Per OPNAV M-5100.23 3604.q.(1)(F), use of motorized (electric, gas, etc.) or human powered scooters, skateboards, roller-skates, roller-blades, and other similar equipment are not authorized in any areas on PNSY.

#### 1.17 PORTSMOUTH NAVAL SHIPYARD 6010 TRAINING

Personnel working inside of or within 10 feet of the coping of the dry dock and its associated structures must have Portsmouth Naval Shipyard 6010 training. Associated structures include pumpwells, service tunnels, altar pits, etc. This training will be provided, at the Portsmouth Naval Shipyard, at no cost and is approximately 4 hours in duration. Coordinate training schedule with the Contracting Officer.

Personnel that complete 6010 training can serve as an escort for 6010 purposes. The escort must be within 10 feet or audio range, whichever is closer, and is responsible for the safety of the personnel they are escorting in an emergency situation.

#### 1.18 CRITICAL SHIPYARD INFRASTRUCTURE REQUIREMENTS

All construction activities (including excavation adjacent to Critical Infrastructure) and outages directly impacting or within three feet of Shipyard Critical Infrastructure are required to have a Critical Infrastructure Risk Analysis and Risk Mitigation Plan. The Contract Documents identify the Critical Infrastructure that could be impacted by the work associated with this project.

The Critical Infrastructure Risk Analysis and Risk Mitigation Plan is the Contractor's responsibility. General information related to project specific outage phasing & sequencing may be provided in the Contract Documents. See Paragraph FACILITY/UTILITY/SERVICE OUTAGE & INTERRUPTIONS PROCESS for Outage Requirements.

The Contractor's proposed means and methods related to any work that may impact Shipyard Critical Infrastructure are subject to Government approval. The anticipated phasing & sequencing for completing this work may be outlined in the Contract Drawings. Any additional work and costs associated with any changes to the phasing & sequencing based on the Contractor's means & methods are the responsibility of the Contractor and will not be a basis for a claim for additional cost and time.

The Contracting Officer will require approval of each CI Risk Mitigation Plan prior to work, and when associated with an outage, it must be submitted along with the Outage Request.

#### DEFINITION:

- a. "Critical Infrastructure" are Shipyard facilities (utilities and structures) that directly affect personnel safety, radiological controls, nuclear and reactor safety, and ship safety and require unhindered reliable operation. This definition is intended to be used in a conservative manner when determining if work impacts Critical Infrastructure, and must be reviewed by the Contracting Officer through appropriate Shipyard personnel in those cases when there is

any question as to the applicability.

- i. Critical structures (and their foundation systems) include: dry dock structures, pump well structures, caissons as well as caisson seat and apron, submarine piers, reactor servicing complexes, crane and rail road rails, and bridges 1 and 2. Also, Radiological Material Storage facilities and Emergency Control Center. All work within three feet of a dry dock pump well and dry dock (when a submarine is docked) must follow Critical Shipyard Infrastructure requirements.
- ii. Critical utilities/services support dry docks and piers with submarines, reactor servicing complexes, Radiological Material Storage facilities, and Emergency Control Center including: dry dock pumping systems (drainage, dewatering, and ASW- both temporary and permanent systems) high voltage electrical systems and low voltage electrical systems, communication systems, high pressure air systems, low pressure air (breathing air), gas lines to the B72, fresh water, firefighting water supply, and high pressure steam.

#### 1.18.1 CRITICAL INFRASTRUCTURE RISK ANALYSIS (RA) AND RISK MITIGATION PLAN (RMP)

The Critical Infrastructure Risk Analysis (RA) and Risk Mitigation Plan (RMP) must include the following:

1. Description of Work: Describe in enough detail to allow someone not involved to have a general understanding of the scope.
2. Risks to Critical Infrastructure:
  - (1) Describe what could go wrong with the work and the impact to Shipyard operations (Air, Water, Electrical Power, Telecom/IT, Rail or Structures), including:
    - (a) Risk Assessment: Evaluate the risk of what could go wrong. (Probability of failure/Severity of Impact. See RISK DETERMINATION TABLE in the Outage Form.)
    - (b) Mitigated Risk Assessment: Evaluate the reduced risk of what could go wrong based on steps in paragraphs WORK CONTROL PLAN and OVERSIGHT outlined below.
  - (1) What could go wrong #2:
    - (a) Risk Assessment #2: Evaluate the risk of what could go wrong. (Probability of failure/Severity of Impact)
    - (b) Mitigated Risk Assessment #2: Evaluate the reduced risk of what could go wrong based on steps in paragraphs WORK CONTROL PLAN and OVERSIGHT outlined below. State why or how the risk is reduced.
3. Work Control Plan:
  - (1) Describe specific steps and actions (how, when, and by whom) to mitigate the risks to Critical Infrastructure that were identified above (Examples only included below).
    - (a) Specific utilities/services in the area will be secured

- prior to work, or, potholing/GPR or other utility/service locating has been performed to identify specific utility/service location and depth, which is included in Preparatory Phase meeting agenda.
- (b) Protective timber matting installed over crane and train rail to prevent damage to rails supporting train and crane movements.
  - (c) Noise/Dust/Vibration. Managed per contract requirements, and to be briefed during Initial Phase meeting.
  - (d) Detailed work plan.
  - (e) Specific selection of equipment/tools for work.
  - (f) Foreman/safety pre-job reviews.
  - (g) Foreman pre-job work brief with crew.

4. Oversight:

- (1) Describe specific oversight work leading up to and for the execution of work (how, when, and by whom) to mitigate the risks to critical infrastructure that were identified above (Examples only included below).
  - (a) Daily Contractor Production Reports.
  - (b) Daily Contractor Quality Control (QC) Reports.
  - (c) QC Manager field visits (include periodicity).
  - (d) Periodic project Coordination Meetings (include parties involved).

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 20 00.00 22

PRICE AND PAYMENT PROCEDURES (PWD ME)  
**11/20, CHG 5: 02/25**

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EP 1110-1-8 (2021) Engineering and Design --  
Construction Equipment Ownership and  
Operating Expense Schedule

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Earned Value Report; G

1.3 EARNED VALUE REPORT

1.3.1 Data Required

This Contract requires the use of a cost-loaded Network Analysis Schedule (NAS). Schedule of Prices must not be used with cost-loaded Network Analysis Schedule (NAS). Use Earned Value Report derived from cost-loaded NAS. Within 30 calendar days from receiving the Notice to Proceed, prepare and deliver to the Contracting Officer an Earned Value Report (construction Contract) as directed by the Contracting Officer. Provide a detailed breakdown of the Contract price, giving quantities for each of the various kinds of work, unit prices, and extended prices. Indirect activities (e.g., meetings, presentations, submittals, surveys, haul road or storage yard maintenance, dust control, security) will not be cost loaded. Evenly distribute the costs across the associated construction activities. Contractor overhead and profit including salaries for field office personnel, if applicable, must be proportionately spread over all pay items and not included as individual pay items.

1.3.2 Payment Schedule Instructions

Payments will not be made until the Earned Value Report from the cost-loaded NAS has been submitted to and accepted by the Contracting Officer.

Additionally, the Earned Value Report must be separated as follows:

a. Primary Facilities Cost Breakdown:

Defined as work on the primary facilities out to the 5 foot line. Work out to the 5 foot line includes construction encompassed within a theoretical line 5 foot from the face of exterior walls and includes attendant construction, such as pad mounted HVAC cooling equipment, cooling towers, and transformers placed beyond the 5 foot line.

b. Supporting Facilities Cost Breakdown:

Defined as site work, including incidental work, outside the 5 foot line.

1.3.3 Schedule Requirements for HVAC TAB

The field work requirements in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC must be broken down in the Earned Value Report from the cost-loaded NAS by separate line items which reflect measurable deliverables. The value for each pay item listed below will be established on a case by case basis for each Contract. The line items are as follows:

- a. Approval of Design Review Report: The TABS Agency is required to conduct a review of the project plans and specifications to identify any feature, or the lack thereof, that would preclude successful testing and balancing of the project HVAC systems. Submit the resulting findings to the Government to allow correction of the design. The progress payment will not be issued until the report is reviewed and approved.
- b. Approval of the pre-field engineering report: The TABS Agency submits a report which outlines the scope of field work. The report must contain details of what systems will be tested, procedures to be used, sample report forms for reporting test results and a quality control checklist of work items that must be completed before TABS field work commences.
- c. Season I field work: Incremental payments are issued as the TABS field work progresses. The TABS Agency mobilizes to the project site and executes the field work as outlined in the pre-field engineering report. The HVAC water and air systems are balanced and operational data must be collected for one seasonal condition (either summer or winter depending on project timing).
- d. Approval of Season I report: On completion of the Season I field work, the data is compiled into a report and submitted to the Government. The report is reviewed, and approved, after ensuring compliance with the pre-field engineering report scope of work.
- e. Completion of Season I field QA check: Contract QC and Government representatives meet the TABS Agency at the jobsite to retest portions of the systems reported in the Season I report. The purpose of these tests are to validate the accuracy and completeness of the previously submitted Season I report.
- f. Approval of Season II report: The TABS Agency completes all Season II field work, which is normally comprised mainly of taking heat transfer



temperature readings, in the season opposite of that under which Season I performance data was compiled. Compile this data into a report and submit to the Government. On completion of submittal review to ensure compliance with the pre-field engineering report scope, progress payment is issued. Progress payment is less than that issued for the Season I report since most of the water and air balancing work effort is completed under Season I.

#### 1.4 CONTRACT MODIFICATIONS

In conjunction with the Contract Clause DFARS 252.236-7000 Modification Proposals-Price Breakdown, and where actual ownership and operating costs of construction equipment cannot be determined from Contractor accounting records, base equipment use rates upon the applicable provisions of the EP 1110-1-8.

#### 1.5 CONTRACTOR'S INVOICE AND CONTRACT PERFORMANCE STATEMENT

##### 1.5.1 Content of Invoice

Requests for payment will be processed in accordance with the Contract Clause FAR 52.232-27 Prompt Payment for Construction Contracts and FAR 52.232-5 Payments Under Fixed-Price Construction Contracts. Invoices not completed in accordance with contract requirements will be returned to the Contractor for correction of the deficiencies. At least ten (10) working days prior to the invoice being uploaded to WAWF, the Contractor's Quality Control (QC) Manager must review the invoice with the Contracting Officer to ensure accuracy. Include the documents listed below in the requests for payment.

- a. The Contractor's invoice, on NAVFAC Form 7300/30 furnished by the Government, showing in summary form, the basis for arriving at the amount of the invoice. Form 7300/30 must include certification by QC Manager as required by the Contract.
- b. The Earned Value Report from the cost-loaded NAS.
- c. Contractor's Monthly Estimate for Voucher and Contractors Certification (NAVFAC Form 4330) with Subcontractor and supplier payment certification. Other documents, including but not limited to, that need to be received prior to processing payment include the following submittals as required. These items are still required monthly even when a pay voucher is not submitted.
- d. Monthly Work-hour report.
- e. Updated Construction Progress Schedule and tabular reports required by the contract.
- f. Contractor Safety Self Evaluation Checklist.
- g. Updated submittal register.
- h. Solid Waste Disposal Report.
- i. Certified payrolls.
- j. Updated testing logs.

- k. Monthly Progress As-Built, in accordance with Section 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME).
- l. Other supporting documents as requested.

#### 1.5.2 Submission of Invoices

If DFARS Clause 252.232-7006 Wide Area WorkFlow Payment Instructions is included in the Contract, provide the documents listed in above paragraph CONTENT OF INVOICE in their entirety as attachments in Wide Area Work Flow (WAWF) for each invoice submitted. The maximum size of each WAWF attachment is two megabytes, but there are no limits on the number of attachments. If a document cannot be attached in WAWF due to system or size restriction, provide it as instructed by the Contracting Officer.

Monthly invoices and supporting forms for work performed through the anniversary award date of the Contract must be submitted to the Contracting Officer within 5 calendar days of the date of invoice. For example, if Contract award date is the 7th of the month, the date of each monthly invoice must be the 7th and the invoice must be submitted by the 12th of the month.

#### 1.5.3 Final Invoice

- a. A final invoice must be accompanied by the certification required by DFARS 252.247.7023 Transportation of Supplies by Sea, and the Contractor's Final Release. If the Contractor is incorporated, the Final Release must contain the corporate seal. An officer of the corporation must sign and the corporate secretary must certify the Final Release.
- b. For final invoices being submitted via WAWF, the original Contractor's Final Release Form and required certification of Transportation of Supplies by Sea must be provided directly to the respective Contracting Officer prior to submission of the final invoice. Once receipt of the original Final Release Form and required certification of Transportation of Supplies by Sea has been confirmed by the Contracting Officer, the Contractor must then submit final invoice and attach a copy of the Final Release Form and required certification of Transportation of Supplies by Sea in WAWF.
- c. Final invoices not accompanied by the Contractor's Final Release and required certification of Transportation of Supplies by Sea will be considered incomplete and will be returned to the Contractor.

#### 1.6 PAYMENTS TO THE CONTRACTOR

Payments will be made on submission of itemized requests by the Contractor which comply with the requirements of this section, and will be subject to reduction for overpayments or increase for underpayments made on previous payments to the Contractor.

##### 1.6.1 Obligation of Government Payments

The obligation of the Government to make payments required under the provisions of this Contract will, at the discretion of the Contracting Officer, be subject to reductions and suspensions permitted under the FAR and agency regulations including the following in accordance with FAR 32.103 Progress Payments Under Construction Contracts:

- a. Reasonable deductions due to defects in material or workmanship;
- b. Claims which the Government may have against the Contractor under or in connection with this Contract;
- c. Unless otherwise adjusted, repayment to the Government upon demand for overpayments made to the Contractor; and
- d. Failure to maintain accurate "as-built" or record drawings in accordance with FAR 52.236.21 and the contract.
- e. Failure to submit accurate monthly schedule updates in accordance with the contract.

#### 1.6.2 Payment for Onsite and Offsite Materials

Progress payments may be made to the Contractor for materials delivered on the site or for materials stored off construction sites under the following conditions:

- a. FAR 52.232-5(b) Payments Under Fixed Price Construction Contracts.
- b. Materials delivered on the site, but not installed, and off-site materials to be considered for progress payment must be major high cost, long lead, special order, or specialty items, not susceptible to deterioration or physical damage in storage or in transit to the construction site. Examples of materials acceptable for payment consideration include, but are not limited to, structural steel, non-magnetic steel, non-magnetic aggregate, equipment, machinery, large pipe and fittings, precast/prestressed concrete products, plastic lumber (e.g., fender piles/curbs), and high-voltage electrical cable. Materials not acceptable for payment include consumable materials such as nails, fasteners, conduits, gypsum board, glass, insulation, and wall coverings.
- c. Provide the Contracting Officer with documentation (e.g., Purchase Orders or Material Contracts) to justify payment for materials in advance of installation. When appropriate, include the Defense Priorities and Allocations System (DFAS) ratings (e.g., DX or DO) on purchase orders or material contracts to ensure timely delivery.
- d. Materials to be considered for progress payment prior to installation must be specifically and separately identified in the Contractor's estimates of work submitted for the Contracting Officer's approval in accordance with Earned Value Report requirement of this Contract. Requests for progress payment consideration for such items must be supported by documents establishing their value (e.g., paid invoices) and that the title requirements of the clause at FAR 52.232-5 Payments Under Fixed-Price Construction Contracts have been met.
- e. Ensure materials are adequately insured and protected from theft and exposure. Provide insurance documentation (e.g., Builder's Risk Insurance) which covers the overall value of the materials to be paid in advance of installation.
- f. Provide a written consent from the surety company with each payment request for offsite materials.

- g. Materials to be considered for progress payments prior to installation must be stored within fifty (50) miles of the Portsmouth Naval Shipyard in Kittery, Maine. Other locations are subject to written approval by the Contracting Officer.
- h. Provide written consent allowing the Government access to the remote storage location for inspection of the stored materials.
- i. Materials shall be separated from other stored materials and be clearly identified with the contract number for which the materials will be used.

Materials in transit to the job site or storage site are not acceptable for payment.

## PART 2 PRODUCTS

Not used.

## PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 30 00.00 22

ADMINISTRATIVE REQUIREMENTS (PWD ME)

11/20, CHG 4: 08/24

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety -- Safety and Occupational Health (SOH) Requirements

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Facility Turnover Planning Meeting Agenda and Red Zone (RZ) Checklist-POAM; G

NAVFAC PWD ME Internal Service Requirements List; G

1.3 MINIMUM INSURANCE REQUIREMENTS

Provide the minimum insurance coverage required by FAR 28.307-2 LIABILITY, during the entire period of performance under this Contract. Provide other insurance coverage as required by State law.

1.4 SUPERVISION

Provide, at a minimum, personnel to meet the requirements of the following supervisory roles for the project, as outlined throughout the Division 01 Specifications:

Project Superintendent  
Project Manager  
Designated Project Scheduler  
Site Safety and Health Officer (SSHO)  
Archaeological Coordinator  
Quality Control (QC) Manager  
Quality Control (QC) Specialists  
Special Inspector (SI)  
Special Inspector of Record (SIOR)  
Environmental Manager

The requirements of each role are enumerated in the following paragraphs. Where the requirements for a role are enumerated in another section, a reference to that section is provided.

#### 1.4.1 Project Superintendent

##### 1.4.1.1 Minimum Communication Requirements

Have at least one qualified superintendent, or competent alternate, capable of reading, writing, and conversing fluently in the English language, on the job-site at all times during the performance of Contract work.

##### 1.4.1.2 Qualifications

Provide Project Superintendent with a minimum of 10 years experience in construction with at least 5 of those years as a superintendent on projects similar in size and complexity. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance. The individual must be capable of interpreting a critical path schedule and construction drawings. The qualification requirements for the alternate superintendent are the same as for the Project Superintendent. The Contracting Officer may request proof of the superintendent's qualifications at any point in the project if the performance of the superintendent is in question.

The Project Superintendent cannot be the Project Manager, nor the Quality Control Manager, nor the Site Safety and Health Officer (SSHO).

The Project Superintendent must be a direct employee of the Prime Contractor, not an employee of a subcontractor.

##### 1.4.1.3 Duties

The Project Superintendent is primarily responsible for managing Subcontractors and coordinating day-to-day production and schedule adherence on the project. The superintendent is required, at a minimum, to attend the Specification Mutual Understanding Meeting, Preconstruction Meeting, Environmental Brief, Outage Coordination Meetings, Outage Safety Review Meetings, Archaeological Coordination Meeting, MEP Coordination Meetings, Red Zone Meetings, partnering meetings, preparatory meetings and Quality Control meetings.

##### 1.4.1.4 Non-Compliance Actions

The Project Superintendent is subject to removal by the Contracting Officer for non-compliance with requirements specified in the Contract and for failure to manage the project to insure timely completion. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders is acceptable as the subject of claim for extension of time for excess costs or damages by the Contractor.

#### 1.4.2 Project Manager

##### 1.4.2.1 Requirements

The Contractor shall assign a qualified Project Manager to the project.

The Project Manager **cannot** be the Project Superintendent, nor the Quality Control Manager, nor the Site Safety and Health Officer (SSHO).

#### 1.4.2.2 Qualifications

Approval of the Project Manager by the Contracting Officer is required prior to start of construction. Provide resume(s) for the proposed Project Manager(s) to the Contracting Officer for approval. The resume(s) shall describe the candidates' experience with references and qualifications. Any substitutions during the construction period shall be approved by the Contracting Officer. The Contracting Officer reserves the right to interview the proposed Project Manager at any time in order to verify the submitted qualifications. Any replacement personnel shall have the same or better qualifications than the previous individual (PM/QCM/SSHO, etc.). The resumes for the personnel must provide a summary of projects of similar size and responsibility. The Project Manager shall complete the course entitled "Construction Quality Management for Contractors" prior to the start of construction.

The Project Manager must be a direct employee of the Prime Contractor, not an employee of a subcontractor.

#### 1.4.2.3 Non-Compliance Actions

The Project Manager is subject to removal by the Contracting Officer for non-compliance with requirements specified in the Contract and for failure to manage the project to insure timely completion. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders is acceptable as the subject of claim for extension of time for excess costs or damages by the Contractor.

#### 1.4.3 Designated Project Scheduler

Provide a Designated Project Scheduler in accordance with the requirements of Section 01 32 17.00 20 COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS).

#### 1.4.4 Site Safety and Health Officer (SSHO)

A Site Safety and Health Officer (SSHO) is a Contractor employee that is responsible for overseeing and ensuring implementation of the Prime Contractor's Safety and Occupational Health (SOH) program according to the Contract, EM 385-1-1, and applicable federal, state, and local requirements.

Provide Site Safety and Health Officers (SSHOs) in accordance with the requirements of Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME).

#### 1.4.5 Archaeological Coordinator

Provide an Archaeological Coordinator in accordance with the requirements of Section 01 44 00.00 22 ARCHAEOLOGICAL MONITORING (PWD ME).

#### 1.4.6 Quality Control (QC) Manager

The Quality Control (QC) Manager is responsible for managing and implementing the QC program on this Contract. Provide a QC Manager in accordance with the requirements of Section 01 45 00.00 22 QUALITY CONTROL

(PWD ME).

#### 1.4.7 Quality Control (QC) Specialists

Provide a separate Quality Control (QC) Specialist for each of the areas as listed in Paragraph QUALITY CONTROL (QC) SPECIALISTS in Section 01 45 00.00 22 QUALITY CONTROL (PWD ME). The QC Specialists must assist and report to the QC Manager and must have no duties other than their assigned quality control duties. The QC Specialists must be provided in accordance with the requirements of Section 01 45 00.00 22 QUALITY CONTROL (PWD ME).

#### 1.4.8 Special Inspector (SI)

Provide a Special Inspector (SI) in accordance with the requirements of Section 01 45 35 SPECIAL INSPECTIONS.

#### 1.4.9 Special Inspector of Record (SIOR)

Provide a Special Inspector of Record (SIOR) in accordance with the requirements of Section 01 45 35 SPECIAL INSPECTIONS.

#### 1.4.10 Environmental Manager

Provide an Environmental Manager in accordance with the requirements of Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME).

### 1.5 SPECIFICATION MUTUAL UNDERSTANDING MEETING

Before the Preconstruction Meeting, and within 21 calendar days from receiving the Notice to Proceed, meet with the Contracting Officer and develop a mutual understanding of the Division 01 specifications and their requirements. This meeting must include the Prime Contractor, the Prime and Subcontractors performing the work, the Contracting Officer, NAVFAC DM/ET, Designer of Record and the NAVFAC PWD ME UEM representative.

At the conclusion of the meeting, the Project Superintendent must certify that the Prime Contractor and its Subcontractors have read, understand, and agree to the requirements of the Division 01 specifications by signing a document which will be prepared by the Contracting Officer.

### 1.6 PRECONSTRUCTION MEETING

**The Preconstruction meeting must take place within 45 calendar days from receiving the Notice to Proceed, and prior to commencement of any work at the site.** Coordinate with the Contracting Officer a time and place to meet for the Preconstruction Meeting. The purpose of this meeting is to discuss and develop a mutual understanding of the administrative requirements of the Contract including but not limited to: daily reporting, invoicing, value engineering, safety, Base-access, outage requests, hot work permits, schedule requirements, quality control, earned value report, shop drawings, submittals, cybersecurity, prosecution of the work, Government acceptance, final inspections, and Contract close-out. Contractor must present and discuss their basic approach to scheduling the construction work and any required phasing.

Prepare and distribute minutes to the attendees within three (3) working days of the preconstruction meeting.



### 1.6.1 Attendees

Contractor attendees must include the Project Manager, Project Superintendent, Site Safety and Health Officer (SSHO), Archaeological Coordinator, Quality Control (QC) Manager, Environmental Manager, and major Subcontractors.

### 1.6.2 Submittals

This section compiles in one location the SD-01 Preconstruction Submittals from the Division 01 specifications that the Contractor is required to submit. The requirements of each submittal are detailed in the specification sections as listed below. This list is not meant to be comprehensive; the Contractor is responsible for confirming the complete list of SD-01 Preconstruction Submittals, including those in Divisions 02 through 46, by reviewing the specifications, including the Submittal Register attached to Section 01 33 00 SUBMITTAL PROCEDURES.

The following SD-01 Preconstruction Submittals must be submitted to the Contracting Officer thirty (30) calendar days prior to the preconstruction meeting:

| Specification  | SD # | SD Description             | Submittal Description                       | Specification Paragraph # |
|----------------|------|----------------------------|---------------------------------------------|---------------------------|
| 01 35 26.00 22 | 01   | Preconstruction Submittals | Accident Prevention Plan; G                 | 1.9                       |
| 01 35 26.00 22 | 01   | Preconstruction Submittals | Indoor Air Quality (IAQ) Management Plan; G | 1.16                      |

The following SD-01 Preconstruction Submittals must be submitted to the Contracting Officer fifteen (15) calendar days prior to the preconstruction meeting:

| Specification  | SD # | SD Description             | Submittal Description             | Specification Paragraph # |
|----------------|------|----------------------------|-----------------------------------|---------------------------|
| 01 14 00.00 22 | 01   | Preconstruction Submittals | List of Contact Personnel; G      | 1.4.1.1                   |
| 01 20 00.00 22 | 01   | Preconstruction Submittals | Earned Value Report; G            | 1.3                       |
| 01 31 23.13 20 | 01   | Preconstruction Submittals | List of Contractor's Personnel; G | 1.4.2                     |
| 01 32 17.00 20 | 01   | Preconstruction Submittals | Baseline NAS; G                   | 1.2.2                     |
| 01 32 17.00 20 | 01   | Preconstruction Submittals | Designated Project Scheduler; G   | 1.9                       |

| Specification  | SD # | SD Description             | Submittal Description                    | Specification Paragraph # |
|----------------|------|----------------------------|------------------------------------------|---------------------------|
| 01 45 00.00 22 | 01   | Preconstruction Submittals | Contractor Quality Control (CQC) Plan; G | 1.5.2                     |
| 01 57 19.00 22 | 01   | Preconstruction Submittals | Preconstruction Survey Report; G         | 1.5.1                     |
| 01 57 19.00 22 | 01   | Preconstruction Submittals | Environmental Management Plan; G         | 3.1                       |
| 01 74 19       | 01   | Preconstruction Submittals | Construction Waste Management Plan; G    | 1.7                       |

The following SD-01 Preconstruction Submittals must be submitted to the Contracting Officer at or before the preconstruction meeting:

| Specification  | SD # | SD Description             | Submittal Description                                                         | Specification Paragraph # |
|----------------|------|----------------------------|-------------------------------------------------------------------------------|---------------------------|
| 01 30 00.00 22 | 01   | Preconstruction Submittals | Facility Turnover Planning Meeting Agenda and Red Zone (RZ) Checklist-POAM; G | 1.14.1                    |
| 01 30 00.00 22 | 01   | Preconstruction Submittals | NAVFAC PWD ME Internal Service Requirements List; G                           | 1.14.3                    |
| 01 32 17.00 20 | 01   | Preconstruction Submittals | Weekly Look Ahead Schedule; G                                                 | 1.3                       |
| 01 33 00       | 01   | Preconstruction Submittals | Submittal Register; G                                                         | 1.8                       |

The following SD-01 Preconstruction and SD-10 Closeout submittals must be submitted to the Contracting Officer prior to the start of construction:

| Specification  | SD # | SD Description             | Submittal Description             | Specification Paragraph # |
|----------------|------|----------------------------|-----------------------------------|---------------------------|
| 01 11 00.00 22 | 01   | Preconstruction Submittals | Salvage Plan; G                   | 1.6                       |
| 01 14 00.00 22 | 01   | Preconstruction Submittals | Flooding Mitigation Strategy Plan | 1.15                      |
| 01 44 00.00 22 | 01   | Preconstruction Submittals | Archaeological Monitoring Plan; G | 1.6                       |

| Specification  | SD # | SD Description             | Submittal Description                                               | Specification Paragraph # |
|----------------|------|----------------------------|---------------------------------------------------------------------|---------------------------|
| 01 44 00.00 22 | 01   | Preconstruction Submittals | Archaeologist Qualifications; G                                     | 1.4                       |
| 01 44 00.00 22 | 01   | Preconstruction Submittals | Archaeological Coordinator Designation Letter; G                    | 1.5                       |
| 01 45 35       | 01   | Preconstruction Submittals | SIOR Letter of Acceptance; G                                        | 3.1.1                     |
| 01 45 35       | 01   | Preconstruction Submittals | Project Manual; G                                                   | 3.1.1                     |
| 01 45 35       | 01   | Preconstruction Submittals | Written NDT Practices; G                                            | 3.1.3                     |
| 01 50 00.00 22 | 01   | Preconstruction Submittals | Construction Site Plan; G                                           | 1.4                       |
| 01 50 00.00 22 | 01   | Preconstruction Submittals | Contractor Computer Cybersecurity Compliance Statements; G          | 1.6.1.4                   |
| 01 50 00.00 22 | 01   | Preconstruction Submittals | Contractor Temporary Network Cybersecurity Compliance Statements; G | 1.6.6                     |
| 01 50 00.00 22 | 01   | Preconstruction Submittals | Employee Parking Plan; G                                            | 1.7                       |
| 01 50 00.00 22 | 01   | Preconstruction Submittals | Traffic Control Plan; G                                             | 3.4.1                     |
| 01 57 19.00 22 | 01   | Preconstruction Submittals | Solid Waste Management Plan; G                                      | 3.4                       |
| 01 57 19.00 22 | 01   | Preconstruction Submittals | Regulatory Notifications; G                                         | 1.5.2                     |
| 01 57 19.00 22 | 01   | Preconstruction Submittals | Dirt and Dust Control Plan; G                                       | 3.14.1                    |
| 01 57 19.00 22 | 10   | Closeout Submittals        | Contractor Hazardous Material Inventory Log; G                      | 3.6                       |
| 01 57 19.00 22 | 01   | Preconstruction Submittals | Storm Water Management/Erosion and Sedimentation Control Plan; G    | 3.1.2.a                   |
| 01 57 19.00 22 | 01   | Preconstruction Submittals | Spill Prevention Control and Countermeasure (SPCC) Plan; G          | 3.1.g(2)                  |

| Specification  | SD # | SD Description             | Submittal Description                   | Specification Paragraph # |
|----------------|------|----------------------------|-----------------------------------------|---------------------------|
| 01 57 19.00 22 | 01   | Preconstruction Submittals | Environmental Manager Qualifications; G | 1.5.4                     |
| 01 57 19.00 22 | 01   | Preconstruction Submittals | Noise Mitigation Plan; G                | 3.16                      |
| 01 57 19.00 22 | 01   | Preconstruction Submittals | Employee Training Records; G            | 1.5.5                     |

Following submission and approval of all SD-01 Preconstruction Submittals, the Contractor shall confirm, in writing, the construction start date with the Contracting Officer at least five (5) working days prior to start date.

#### 1.7 OUTAGE COORDINATION MEETINGS

Outage Coordination Meetings are to be held as necessary, prior to submittal of an outage request package. The meetings are to be scheduled and held by the Contractor, and are intended to develop a mutual understanding of the administrative and technical scope and requirements needed to support development, and ultimate approval, of the outage request packages. Successful execution of the outages within this contract, including receiving outage approval, require the Contractor to adequately inform the Government of the outage, including its scope, impact, and timeline. The Contractor must adequately address any Government concerns in the outage request package. Failure of the Contractor to properly coordinate and plan for an outage, which may result in rejection of an outage request package, are entirely the responsibility of the Contractor and can not be made the subject, or any component of any request for additional time or compensation.

The meeting shall also review any Critical Shipyard Infrastructure that may be impacted by the work and review Paragraph CRITICAL SHIPYARD INFRASTRUCTURE REQUIREMENTS in Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME).

These meetings must include applicable personnel from the Prime Contractor, the Prime and Subcontractors performing the work, the Contracting Officer, the NAVFAC DM/ET, the NAVFAC PWD ME UEM Representatives, and applicable stakeholders. Additional personnel may be included (i.e., the Designer of Record) if requested by the Contracting Officer.

All parties, led by the Contractor, must fully coordinate and be aligned on the activities and dates required to support the outages.

#### 1.8 OUTAGE SAFETY REVIEW MEETINGS

The Outage Safety Review Meetings are to be held prior to any outages that require hazardous energy control (HEC). During these meetings, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the Activity Hazard Analysis (AHA), assurance of trade personnel qualifications, identification of competent persons, and compliance with HEC training in accordance with EM 385-1-1. Clarify when

personal protective equipment is required during switching operations, inspection, and verification.

For utility distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the Subcontractor performing the work must attend the Safety Preparatory Inspection Coordination Meeting, which will also be attended by the Contracting Officer, and is required by EM 385-1-1. The Safety Preparatory Inspection Coordination Meeting will occur immediately preceding the start of work and following the completion of the Outage Coordination Meeting. Both the Safety Preparatory Inspection Coordination Meeting and the Outage Coordination Meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

#### 1.9 ENVIRONMENTAL BRIEF

Attend an Environmental Brief prior to commencing any work on the Portsmouth Naval Shipyard. The Brief will be conducted by the Contracting Officer with PWD-ME EV staff and Shipyard C106 Environmental Representatives. The Brief will be part of the Preconstruction Meeting agenda or held as a separate meeting depending on the size and scope of work and potential environmental impact, as determined by the Government.

The meeting will cover details for all the submittals that are required in Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME) including review of the submittal requirements for the Environmental Management Plan; important notes regarding Installation Restoration (IR) sites, Historic and Cultural Resources, Wetlands, Waste Generation, Concrete Wash Water/Wastewater, and Hazard Materials Management, including sampling, testing and disposal requirements; and other applicable topics.

At the Brief, the Contractor shall provide the following information: types, quantities, and use of hazardous materials that will be brought onto the Portsmouth Naval Shipyard; types and quantities of wastes/wastewater that may be generated during the Contract; types and quantities of oil that will be brought onto the Shipyard; pollution control measures for spill prevention and control; and any bulk oil storage container information including quantity and type of product stored. Discuss the results of the Preconstruction Survey at this time.

The Brief will develop a mutual understanding relative to the details of environmental protection, including measures for protecting natural resources, historic properties, required reports, required permits, specific permit requirements, and other measures to be taken. Identify additional environmental concerns specific to the site (e.g., historic, archaeological and natural resources, IR, erosion and sediment control, spill prevention and control, soil management, testing and disposal requirements, etc.).

Contractor attendees must include the Project Manager, Project Superintendent, Site Safety and Health Officer (SSHO), Archaeological Coordinator, Quality Control (QC) Manager, Environmental Manager, and major Subcontractors.

#### 1.10 ARCHAEOLOGICAL COORDINATION MEETING

Prior to the start of any ground disturbing/excavation work and prior to submission of the Archaeological Monitoring Plan, the Archaeological

Coordinator, project archaeologist, Quality Control (QC) Manager, NAVFAC DM/ET and PWD-ME EV staff must meet to review the proposed approach to the work, establish a protocol for artifact collection, and review the relevant documentation.

For projects with a duration exceeding one year, re-initial archaeological coordination meetings must be held on an annual basis to ensure continued alignment on archaeological protocols. In addition, a re-initial archaeological coordination meeting must be held should there be a change in key personnel (Archaeological Subcontractor, Archaeological Coordinator, QC Manager, Project Superintendent, Project Manager, etc.)

#### 1.11 MEP COORDINATION MEETINGS

The Contractor shall schedule monthly in-person coordination meetings with the mechanical, electrical, piping, and fire protection trades. The intent of the meetings is to review construction progress and coordinate work being performed.

Meeting minutes must be provided within 24 hours of completion of each MEP coordination meeting.

The QC organization will be responsible for leading the MEP Coordination Meetings with other trades encouraged to participate.

#### 1.12 FACILITY TURNOVER PLANNING MEETINGS (Red Zone Meetings)

Meet with the Government to identify strategies to ensure the project is carried to expeditious closure and turnover to the Client. Start the turnover process at the Preconstruction Meeting with a discussion of the Red Zone (RZ) process and convene at regularly scheduled RZ Meetings beginning at approximately 75 percent of construction completion. Refer to Paragraph MEETINGS for additional information on frequency of RZ Meetings. Include the following in the facility Turnover effort:

##### 1.12.1 Red Zone Checklist

The Contracting Officer will provide the Contractor a copy of the Red Zone Checklist template in advance of the Red Zone Facility Turnover Meeting. See Appendix A for an example of the Facility Turnover Planning Meeting Agenda and Red Zone (RZ) Checklist-POAM.

At the initial Red Zone Facility Turnover Meeting, NAVFAC, the Client and Contractor will modify the Red Zone Checklist template by adding or deleting critical activities applicable to the project and assign planned completion dates for each activity. This becomes the Red Zone POAM which will be utilized through to the contract completion.

Items listed on the checklists are required to remain on the checklists if they are part of the project/Contract or required by construction convention. Items not listed on the checklists, but required in the Contract or by construction convention, must be added to the checklists by the Contractor, Client and NAVFAC. The Contracting Officer may request additional activities be added to the Red Zone Checklist at any time as necessary. Checklists are applicable to all Contracts no matter what Category of Work. The Point of Contact and due date must initially be determined during the Facility Turnover Planning Meeting by the NAVFAC, Client and Contractor leads. During execution of the NRZ process, for each item on the entire list, the Construction Manager (CM) must indicate

date completed and initial to indicate completion of the item. If a party fails to complete an item by the due date, this must be noted on the checklist and a new due date established and indicated.

#### 1.12.2 Meetings

- a. Upon Government acceptance of the RZ Checklist-POAM, the Contracting Officer will send out the accepted RZ Checklist-POAM to all attendees. The Project Superintendent is required to lead regular Red Zone Meetings beginning at approximately 75 percent project completion, or three (3) to six (6) months prior to Beneficial Occupancy Date (BOD), whichever comes first.
- b. The Contracting Officer will establish the frequency of the meetings, which is expected to increase as the project completion draws nearer. At the beginning, Red Zone meetings may be every two weeks then increase to weekly towards the final month of the project.
- c. Using RZ Checklist-POAM as a Plan of Action and Milestones (POAM) and basis for discussion, review upcoming critical activities and strategies to ensure work is completed on time.
- d. During the Red Zone Meetings, discuss with the NAVFAC DM/ET any upcoming activities that require Government involvement.
- e. All parties will maintain their copy of the RZ Checklist-POAM documenting the actual completion dates as work is completed and update the RZ Checklist-POAM with revised planned completion dates as necessary to match progress. The CM will maintain the master RZ Checklist-POAM, periodically distributing a scanned copy of the current RZ Checklist-POAM to attendees via email after significant progress is made.

#### 1.12.3 NAVFAC PWD ME Internal Service Requirements List

An initial, pre-edited draft of the NAVFAC PWD ME Internal Service Requirements List will be provided by the Contracting Officer upon request.

Include all information usually listed on manufacturer's name plate. The Internal Service Requirements List must include, as applicable, the following for each piece of equipment installed: description of item, location (by room number), manufacturer, model number, serial number, capacity, floor coverings, wall and ceiling surfaces; types and square footage of coverage, lighting fixtures, bathroom fixtures, windows, and HVAC filters.

Submit a preliminary Internal Service List to the Contracting Officer at the initial Facility Turnover Meeting. Provide the final completed Internal Service List with all required facility system/equipment information to the Contracting Officer at least ninety (90) calendar days prior to the project BOD.

#### 1.13 PARTNERING

To most effectively accomplish this Contract, the Contractor and Government must form a cohesive partnership with the common goal of drawing on the strength of each organization in an effort to achieve a successful project without safety mishaps, conforming to the Contract, within budget, and on schedule. The partnering team must consist of

personnel from both the Government and Contractor including project level and corporate level leadership positions. Key Personnel from the supported command, end user (who will occupy the facility), NAVFAC, PWD Maine Design and Construction team and Subject Matter Experts (SME's), FEAD, Project Manager (PM), Design Manager (DM), Construction Manager (CM), Engineering Technician (ET), Contractor, key Subcontractors, and the Designer of Record are required to participate in the Partnering process.

#### 1.13.1 Facilitated (Formal) Partnering

- a. Within 35 calendar days after award and prior to the start of work, host a Formal Partnering session with key personnel from the project team including both Contractor and Government personnel. All costs associated with the Partnering session including the third-party independent Facilitator Consultant, meeting room and other incidental items are the responsibility of the Contractor.
- b. Before the Facilitated (Formal) Partnering session, coordinate with the Facilitator all requirements for incidental items (such as audio-visual equipment, easels, flipchart paper, colored markers, note pads, pens/pencils, colored flash cards) and have these items available at the Partnering session. Provide copies of any documents required for distribution to all attendees. Participants will bear their own costs for meals, lodging and transportation associated with Partnering.
- c. The Initial Partnering Session must be a duration of one day and be held at a location off base as agreed to by the Contracting Officer. Partnering session may take place concurrently with the Preconstruction Meeting.
- d. Facilitator must be experienced in conducting corporate Partnering sessions and must be a third-party independent facilitating consultant - not an employee of the Contractor. The Facilitator is responsible for leading all aspects of the Partnering session necessary to achieve the Partnering goal.
- e. An outcome of the Partnering session must be an escalation matrix agreed upon by both the Government and Contractor, which identifies key Government and Contractor decision makers by name and anticipated decision durations.
- f. Host follow-on Partnering Sessions at three- to six-month intervals or more frequently if needed and lasting generally a half day or less. Attendees need only be those required to resolve current issues. The same Facilitator used in the Initial Partnering session must lead the follow-on sessions unless an alternative is permitted by the Contractor Officer. All costs associated with follow-on Partnering sessions are the responsibility of the Contractor.

#### 1.14 ELECTRONIC MAIL (E-MAIL) ADDRESS

Establish and maintain electronic mail (e-mail) capability along with the capability to open various electronic attachments as text files, pdf files, and other similar formats. Within 10 days after Contract award, provide the Contracting Officer a single (only one) e-mail address for electronic communications from the Contracting Officer related to this Contract including, but not limited to, Contract documents, invoice information, request for proposals, and other correspondence. The



Contracting Officer may also use email to notify the Contractor of Base access conditions when emergency conditions warrant, such as hurricanes or terrorist threats. Multiple email addresses are not allowed.

It is the Contractor's responsibility to make timely distribution of all Contracting Officer initiated e-mail with its own organization including field office(s). Promptly notify the Contracting Officer, in writing, of any changes to this email address.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

**01 30 00 APPENDIX A****NAVFAC Red Zone  
Facility Turnover Planning Meeting****AGENDA****I. Introduction and Overview – Purpose**

CM

The purpose of the Facility Turnover Planning Meeting is to address elements within the project team's purview – schedule management, assure completed facility complies with contract requirements, and other contractual issues. Each member of the project delivery team (Client, NAVFAC, and the contractor) has critical responsibilities to ensure timely completion and turnover of the new facility and each member should execute the NRZ process to achieve this end. The NRZ process provides a final re-focusing of attention to details of those key elements critical for a successful construction contract completion. In implementing NRZ processes, the NAVFAC/Contractor/Client team take a collective "snapshot" of contract status, identifying remaining actions to be accomplished, and confirm required resources needed for successful contract completion and turnover to the Client.

The Facility Turnover Planning Meeting is a collaborative effort between the Client, NAVFAC, and the contractor and results in a completed "NRZ Checklist/POAM Items" list that identifies the major items (and their due dates) that must be completed by the Contractor, the Client and the NAVFAC team to ensure timely completion of the contract.

**II. Attendees**

NAVFAC Echelon IV (PM); NAVFAC FEAD/ROICC Team (AROICC, CM, ET/QA, Contracting Officer); Client Team (Project Manager, Program Coordinator, User/Tenant); Contractor Team (Project Manager, Project Superintendent, CQC Manager)

**III. Schedule to Completion (POAM)**

Contractor

**IV. Schedule of Final Outfitting and Occupancy (POAM)**

Client

**V. Critical feature(s) of project (POAM)**

CM

**VI. Transfer of Maintenance Responsibility**

CM

**VII. Systems training & O&M Manuals (POAM)**

CQC Manager

**VIII. Other Items to include on NRZ checklists**

All

**IX. Summary of Required Actions and Responsibility**

CM

***Guidelines for conducting Facility Turnover Planning Meeting are as follows:***

- a. Meeting is held at approximately 75% construction contract completion or three to six months prior to BOD. NAVFAC representatives will include the Project Manager, Construction Manager/AROICC (CM) and Design Manager (DM), as appropriate. The contractor representatives include applicable prime contractor staff and decision-makers from major subcontractors. Design-Build contractors will have A-E representatives attending. The Client should include representatives from Public Works Officer (PWO) staff, a Client scope and financial decision maker, a user tenant representative, a facility start-up person, and others such as SPAWAR, NMCI, telephone, and furniture contractor, etc.
- b. The purpose of the meeting is to plan the remaining work, identify critical project features that still need to be completed (such as “soft” construction contract requirements as shown on the NRZ Checklist/POAM Items), and to complete the filling out of the “NRZ Checklist/POAM Items”.
- c. The contractor, client and NAVFAC provide a POC and due date for each item on their checklist. The team fills in the checklists by selecting items applicable to the project, selects due dates on each item, and appoints a person who has responsibility to ensure the item gets completed by the due date. The CM will be responsible to monitor the milestones.

### **NRZ Checklist/POAM Items**

The table below provides typical NRZ checklist items for contractor, Client, and NAVFAC actions. Items listed on the checklists are required to remain on the checklists if they are part of the project/contract or required by construction convention. Items not listed on the checklists, but required in the contract or by construction convention, must be added to the checklists by the contractor, Client and NAVFAC. Checklists are applicable to all contracts no matter what Category of Work.

The Point of Contact and due date shall initially be determined during the Facility Turnover Planning Meeting by the NAVFAC, client and contractor leads. During execution of the NRZ process, for each item on the entire list, the Construction Manager (CM) shall indicate date completed and initial to indicate completion of the item. If a party fails to complete an item by the due date, this should be noted on the checklist and new due date established and indicated. The completed NRZ Checklist/POAM shall be placed in the contract file.

|    | A                  | B | C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | D                                                           | E                       | F               | G                           | H                  | I                                                                                                                                        |
|----|--------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------|-----------------|-----------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Revision - 8/26/24 |   | <p><b>* INFO / DIRECTIONS</b></p> <ul style="list-style-type: none"> <li>- This NRZ Checklist/POAM is a tool to track the status of critical activities required for BOD, to help ensure their timely completion and prevent delays with facility acceptance and turnover.</li> <li>- The critical activities are organized by section according to requirements (e.g. NMCI &amp; BCO acceptance and connections)</li> <li>- Critical items missing from this list should be added as necessary to ensure the list is comprehensive. Likewise, unnecessary items should be deleted.</li> <li>- Any critical items left off the NRZ Checklist that are later identified after initial NRZ meeting is conducted should be added immediately so their progress can be tracked.</li> <li>- A copy of the latest project customized NRZ Checklist/POAM shall be maintained in the Project contract file.</li> </ul> |                                                             |                         |                 |                             |                    |                                                                                                                                          |
| 2  |                    |   | <b>Resp.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Critical Activities</b>                                  | <b>Point of Contact</b> | <b>Due Date</b> | <b>Actual Complete Date</b> | <b>CM Initials</b> | <b>Notes</b>                                                                                                                             |
| 3  |                    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Required for Facility Delivery (BOD)</b>                 |                         |                 |                             |                    |                                                                                                                                          |
| 4  |                    |   | KTR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Electrical Systems Testing                                  |                         |                 |                             |                    |                                                                                                                                          |
| 5  |                    |   | NAVFAC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Transformer Performance Verification                        |                         |                 |                             |                    |                                                                                                                                          |
| 6  |                    |   | KTR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Final Electrical Connections                                |                         |                 |                             |                    |                                                                                                                                          |
| 7  |                    |   | NAVFAC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Coordinate Final Electrical Connections                     |                         |                 |                             |                    | Coordinate w/PW Shops                                                                                                                    |
| 8  |                    |   | NAVFAC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Acceptance Testing of Critical Electrical Systems           |                         |                 |                             |                    | Exceptions must be raised to FEAD Leadership with technical evaluation, for acceptance.                                                  |
| 9  |                    |   | KTR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Plumbing/Backflow Testing                                   |                         |                 |                             |                    |                                                                                                                                          |
| 10 |                    |   | KTR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Critical System Start-up                                    |                         |                 |                             |                    |                                                                                                                                          |
| 11 |                    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                             |                         |                 |                             |                    |                                                                                                                                          |
| 12 |                    |   | KTR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Duct Air Leakage Testing (DALT)                             |                         |                 |                             |                    |                                                                                                                                          |
| 13 |                    |   | NAVFAC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | DALT Field Acceptance                                       |                         |                 |                             |                    |                                                                                                                                          |
| 14 |                    |   | NAVFAC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | All HVAC Commissioning/Testing Complete                     |                         |                 |                             |                    | Exceptions must be raised to FEAD Leadership with technical evaluation, for acceptance.                                                  |
| 15 |                    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                             |                         |                 |                             |                    |                                                                                                                                          |
| 16 |                    |   | KTR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Fire Alarm/Sprinkler Testing                                |                         |                 |                             |                    |                                                                                                                                          |
| 17 |                    |   | NAVFAC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | FA & FP test results to ML & DOR                            |                         |                 |                             |                    | Sent test results to ML FPE 14 days prior to acceptance testing                                                                          |
| 18 |                    |   | NAVFAC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Fire Alarm/Sprinkler Acceptance Test                        |                         |                 |                             |                    | Notify ML FPE 30 days prior to required testing. Exceptions must be raised to FEAD Leadership with technical evaluation, for acceptance. |
| 19 |                    |   | NAVFAC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>All Life Safety Inspections/Testing complete</b>         |                         |                 |                             |                    | Exceptions must be raised up through the DCBL chain so that the Command can take action under our Technical Authority.                   |
| 20 |                    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                             |                         |                 |                             |                    |                                                                                                                                          |
| 21 |                    |   | KTR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NMCI Connections/IT Systems Testing                         |                         |                 |                             |                    |                                                                                                                                          |
| 22 |                    |   | Client                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | NMCI Install or other networks (C109)                       |                         |                 |                             |                    |                                                                                                                                          |
| 23 |                    |   | Client                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Secure Network Installations                                |                         |                 |                             |                    |                                                                                                                                          |
| 24 |                    |   | KTR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | IDS & SCIF Testing                                          |                         |                 |                             |                    |                                                                                                                                          |
| 25 |                    |   | Client                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | IDS & SCIF Security Insp & Acceptance                       |                         |                 |                             |                    |                                                                                                                                          |
| 26 |                    |   | NAVFAC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | IDS & SCIF Insp & Test                                      |                         |                 |                             |                    |                                                                                                                                          |
| 27 |                    |   | KTR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Telecommunications Connections & Test                       |                         |                 |                             |                    |                                                                                                                                          |
| 28 |                    |   | Client                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Telecommunication install (BCO phones)                      |                         |                 |                             |                    |                                                                                                                                          |
| 29 |                    |   | Client                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Mod service contracts for Phone/Utilities/Custodial/Grounds |                         |                 |                             |                    |                                                                                                                                          |
| 30 |                    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                             |                         |                 |                             |                    |                                                                                                                                          |
| 31 |                    |   | KTR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Commissioning Functional Performance Testing (FPT)          |                         |                 |                             |                    |                                                                                                                                          |
| 32 |                    |   | KTR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | TAB (Air and Water Balancing)                               |                         |                 |                             |                    |                                                                                                                                          |
| 33 |                    |   | KTR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | TAB Proportional Balancing Report                           |                         |                 |                             |                    |                                                                                                                                          |

|    | A | B | C      | D                                                                                    | E                | F        | G                    | H           | I                                                                                       |
|----|---|---|--------|--------------------------------------------------------------------------------------|------------------|----------|----------------------|-------------|-----------------------------------------------------------------------------------------|
|    |   |   | Resp.  | Critical Activities                                                                  | Point of Contact | Due Date | Actual Complete Date | CM Initials | Notes                                                                                   |
| 2  |   |   |        |                                                                                      |                  |          |                      |             |                                                                                         |
| 34 |   |   | NAVFAC | Proportional Balanc'g TAB Field Accept                                               |                  |          |                      |             |                                                                                         |
| 35 |   |   | KTR    | TAB Field Acceptance Testing                                                         |                  |          |                      |             |                                                                                         |
| 36 |   |   | KTR    | TAB Season 1 Report*                                                                 |                  |          |                      |             |                                                                                         |
| 37 |   |   | KTR    | TAB Field Acceptance Testing                                                         |                  |          |                      |             |                                                                                         |
| 38 |   |   | NAVFAC | 1st Season TAB Field Acceptance                                                      |                  |          |                      |             |                                                                                         |
| 39 |   |   | KTR    | DDC Trend Logs / Endurance Testing                                                   |                  |          |                      |             |                                                                                         |
| 40 |   |   | KTR    | ACATS (controls) Testing                                                             |                  |          |                      |             |                                                                                         |
| 41 |   |   | NAVFAC | ACATS Performance Verification                                                       |                  |          |                      |             |                                                                                         |
| 42 |   |   | KTR    | Performance Verification Testing (PVT) Controls                                      |                  |          |                      |             |                                                                                         |
| 43 |   |   |        |                                                                                      |                  |          |                      |             |                                                                                         |
| 44 |   |   | NAVFAC | Roofing final acceptance                                                             |                  |          |                      |             | Exceptions must be raised to FEAD Leadership with technical evaluation, for acceptance. |
| 45 |   |   |        |                                                                                      |                  |          |                      |             |                                                                                         |
| 46 |   |   | NAVFAC | Performance Verification Testing (PVT) Controls NAVFAC Acceptance                    |                  |          |                      |             |                                                                                         |
| 47 |   |   |        |                                                                                      |                  |          |                      |             |                                                                                         |
| 48 |   |   | KTR    | Facility-Related Control System Cybersecurity Commissioning                          |                  |          |                      |             |                                                                                         |
| 49 |   |   | NAVFAC | Completed Cyber Hygiene Checklist                                                    |                  |          |                      |             |                                                                                         |
| 50 |   |   | NAVFAC | Completed RMF Step 4 Validation                                                      |                  |          |                      |             |                                                                                         |
| 51 |   |   |        |                                                                                      |                  |          |                      |             |                                                                                         |
| 52 |   |   | Client | GFE status/delivery schedule (GFCI, GFGI)                                            |                  |          |                      |             |                                                                                         |
| 53 |   |   | Client | Client provided equipment KTR installed                                              |                  |          |                      |             |                                                                                         |
| 54 |   |   | Client | Client provided equipment SELF installed                                             |                  |          |                      |             |                                                                                         |
| 55 |   |   |        |                                                                                      |                  |          |                      |             |                                                                                         |
| 56 |   |   | NAVFAC | Training Coordinated/Scheduled with FMS                                              |                  |          |                      |             |                                                                                         |
| 57 |   |   | KTR    | System Training of Navy Personnel                                                    |                  |          |                      |             |                                                                                         |
| 58 |   |   | Client | Attend Training                                                                      |                  |          |                      |             |                                                                                         |
| 59 |   |   |        |                                                                                      |                  |          |                      |             |                                                                                         |
| 60 |   |   | KTR    | Provide As-Built Dwgs & floor plans in CAD                                           |                  |          |                      |             |                                                                                         |
| 61 |   |   | NAVFAC | CM to Submit floor plan As-Built to CAD Dept.                                        |                  |          |                      |             | RQ'd for Fire Bill - RQ'd before BOD(A)                                                 |
| 62 |   |   | KTR    | Submit sample as-built drawings for CAD review                                       |                  |          |                      |             | Submit 3 drawings from ea. discipline for review                                        |
| 63 |   |   | NAVFAC | CM to submit to DM who must review and submit Sample As-Built documents to CAD Dept. |                  |          |                      |             |                                                                                         |
| 64 |   |   |        |                                                                                      |                  |          |                      |             |                                                                                         |
| 65 |   |   | KTR    | Pre-Warranty Conference                                                              |                  |          |                      |             |                                                                                         |
| 66 |   |   | KTR    | Contractor's Pre Final Punch List Complete                                           |                  |          |                      |             |                                                                                         |
| 67 |   |   | NAVFAC | Obtain a copy and ensure QA Inspections are completed.                               |                  |          |                      |             | Include a completed copy for BOD Mangament sign off.                                    |
| 68 |   |   | NAVFAC | Obtain a copy and ensure Special Inspections are completed.                          |                  |          |                      |             | Include a completed copy for BOD Mangament sign off.                                    |
| 69 |   |   | KTR    | Pre Final Inspection                                                                 |                  |          |                      |             |                                                                                         |
| 70 |   |   | NAVFAC | Client walk-thru Inspection                                                          |                  |          |                      |             |                                                                                         |
| 71 |   |   | KTR    | Pre-Final Inspection                                                                 |                  |          |                      |             |                                                                                         |
| 72 |   |   | NAVFAC | Pre-Final Inspection                                                                 |                  |          |                      |             |                                                                                         |

|     | A | B | C      | D                                                                                                                                                                                                      | E                | F        | G                    | H           | I                                                                                                    |
|-----|---|---|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|----------|----------------------|-------------|------------------------------------------------------------------------------------------------------|
|     |   |   | Resp.  | Critical Activities                                                                                                                                                                                    | Point of Contact | Due Date | Actual Complete Date | CM Initials | Notes                                                                                                |
| 2   |   |   |        | Punch List                                                                                                                                                                                             |                  |          |                      |             |                                                                                                      |
| 73  |   |   | KTR    | Final Inspection                                                                                                                                                                                       |                  |          |                      |             |                                                                                                      |
| 74  |   |   | KTR    | Final Inspection                                                                                                                                                                                       |                  |          |                      |             |                                                                                                      |
| 75  |   |   | NAVFAC |                                                                                                                                                                                                        |                  |          |                      |             |                                                                                                      |
| 76  |   |   |        |                                                                                                                                                                                                        |                  |          |                      |             |                                                                                                      |
| 77  |   |   | NAVFAC | BOD/Use and Possession                                                                                                                                                                                 |                  |          |                      |             |                                                                                                      |
| 78  |   |   |        |                                                                                                                                                                                                        |                  |          |                      |             |                                                                                                      |
| 79  |   |   | KTR    | Delivery of O&M Manuals                                                                                                                                                                                |                  |          |                      |             | Must be provided before BOD                                                                          |
|     |   |   |        | CM to submit the O&M's to DM who must review, Identify/Flag if equipment is Critical Shipyard Nuclear Infrastructure, and submit to NAVFAC PWD ME Requirements FMS and CSNI info to UEM as applicable. |                  |          |                      |             | DM is to provide notification to FMS and UEM regarding new Critical Shipyard Nuclear Infrastructure. |
| 80  |   |   | NAVFAC |                                                                                                                                                                                                        |                  |          |                      |             |                                                                                                      |
| 81  |   |   |        |                                                                                                                                                                                                        |                  |          |                      |             |                                                                                                      |
| 82  |   |   | KTR    | Delivery of Product Warranties                                                                                                                                                                         |                  |          |                      |             | Must be provided before BOD                                                                          |
|     |   |   |        | CM to submit Warranties to NAVFAC PWD ME Requirements FMS                                                                                                                                              |                  |          |                      |             |                                                                                                      |
| 83  |   |   | NAVFAC |                                                                                                                                                                                                        |                  |          |                      |             |                                                                                                      |
| 84  |   |   |        |                                                                                                                                                                                                        |                  |          |                      |             |                                                                                                      |
| 85  |   |   | KTR    | As-Built Drawings                                                                                                                                                                                      |                  |          |                      |             | Must be provided 30 cal days before BOD                                                              |
|     |   |   |        | CM to submit the As-Built's to DM who must review and submit documents to CAD Dept.                                                                                                                    |                  |          |                      |             |                                                                                                      |
| 86  |   |   | NAVFAC |                                                                                                                                                                                                        |                  |          |                      |             |                                                                                                      |
| 87  |   |   | KTR    | Submit eOMSI Facility Data Workbook                                                                                                                                                                    |                  |          |                      |             | Must be provided before BOD                                                                          |
|     |   |   |        | CM to submit the eOMSI Workbook to the DM who must review and submit to the NAVFAC PWD ME FOS.                                                                                                         |                  |          |                      |             |                                                                                                      |
| 88  |   |   | NAVFAC |                                                                                                                                                                                                        |                  |          |                      |             |                                                                                                      |
| 89  |   |   | KTR    | Submit the Internal Service Requirement List (ISRL).                                                                                                                                                   |                  |          |                      |             | Must be provided before BOD                                                                          |
|     |   |   |        | CM to submit the ISRL to the DM who must review, Identify/Flag if equipment is Critical Shipyard Nuclear Infrastructure, and submit to the NAVFAC PWD ME FSC. Confirm ISRL submitted to FSC.           |                  |          |                      |             | DM is to provide notification to FSC regarding new Critical Shipyard Nuclear Infrastructure.         |
| 90  |   |   | NAVFAC |                                                                                                                                                                                                        |                  |          |                      |             |                                                                                                      |
| 91  |   |   |        |                                                                                                                                                                                                        |                  |          |                      |             |                                                                                                      |
| 92  |   |   |        | <b>Required for Contract Close-out:</b>                                                                                                                                                                |                  |          |                      |             |                                                                                                      |
| 93  |   |   | NAVFAC | Warranty documentation to FMS                                                                                                                                                                          |                  |          |                      |             | Must be provided before CCD                                                                          |
|     |   |   |        | Site Restoration                                                                                                                                                                                       |                  |          |                      |             |                                                                                                      |
| 94  |   |   | KTR    |                                                                                                                                                                                                        |                  |          |                      |             |                                                                                                      |
|     |   |   |        | Spare Parts, Extra Stock, Special Tools, etc                                                                                                                                                           |                  |          |                      |             |                                                                                                      |
| 95  |   |   | KTR    |                                                                                                                                                                                                        |                  |          |                      |             |                                                                                                      |
| 96  |   |   |        |                                                                                                                                                                                                        |                  |          |                      |             |                                                                                                      |
| 97  |   |   | KTR    | Final Demobilization and Clean-up                                                                                                                                                                      |                  |          |                      |             |                                                                                                      |
| 98  |   |   | KTR    | Temp Construction Fence Removed                                                                                                                                                                        |                  |          |                      |             |                                                                                                      |
| 99  |   |   | KTR    | Project Close-out Meeting                                                                                                                                                                              |                  |          |                      |             |                                                                                                      |
|     |   |   |        | As-Built's to FMS                                                                                                                                                                                      |                  |          |                      |             |                                                                                                      |
| 100 |   |   | NAVFAC |                                                                                                                                                                                                        |                  |          |                      |             |                                                                                                      |
|     |   |   |        | Closeout & submit permits and inspection reports                                                                                                                                                       |                  |          |                      |             |                                                                                                      |
| 101 |   |   | KTR    |                                                                                                                                                                                                        |                  |          |                      |             |                                                                                                      |
|     |   |   |        | Closeout & submit permits and inspection reports                                                                                                                                                       |                  |          |                      |             |                                                                                                      |
| 102 |   |   | NAVFAC |                                                                                                                                                                                                        |                  |          |                      |             |                                                                                                      |
| 103 |   |   | KTR    | Seasonal/Deferred Commissioning FPT                                                                                                                                                                    |                  |          |                      |             |                                                                                                      |
| 104 |   |   | KTR    | 2 <sup>nd</sup> Season TAB Report                                                                                                                                                                      |                  |          |                      |             |                                                                                                      |

NAVFAC Red Zone Facility Turnover Planning Meeting Checklist and POAM

|     | A | B | C      | D                                               | E                | F        | G                    | H           | I     |
|-----|---|---|--------|-------------------------------------------------|------------------|----------|----------------------|-------------|-------|
|     |   |   | Resp.  | Critical Activities                             | Point of Contact | Due Date | Actual Complete Date | CM Initials | Notes |
| 2   |   |   |        |                                                 |                  |          |                      |             |       |
| 105 |   |   | NAVFAC | 2 <sup>nd</sup> Season TAB Field Accept Testing |                  |          |                      |             |       |
| 106 |   |   | NAVFAC | Authority to Operate delivered to client        |                  |          |                      |             |       |
| 107 |   |   | KTR    | Commissioning 10-Month/Warranty Visit           |                  |          |                      |             |       |
| 108 |   |   | NAVFAC | Inform PM of BOD                                |                  |          |                      |             |       |
| 109 |   |   | NAVFAC | BOD (A) entered into eContracts                 |                  |          |                      |             |       |
| 110 |   |   | NAVFAC | BOD Letter to Contractor                        |                  |          |                      |             |       |
| 111 |   |   | NAVFAC | Acceptance Letter to Client                     |                  |          |                      |             |       |
| 112 |   |   | NAVFAC | Contractor Evaluations (CPARS) Complete         |                  |          |                      |             |       |
| 113 |   |   | NAVFAC | Finalize Outstanding Contract Mods              |                  |          |                      |             |       |



SECTION 01 31 23.13 20

ELECTRONIC CONSTRUCTION AND FACILITY SUPPORT CONTRACT MANAGEMENT SYSTEM  
(eCMS)

08/23, CHG 1: 02/24

PART 1 GENERAL

1.1 CONTRACT ADMINISTRATION

Utilize the Naval Facilities Engineering Command's (NAVFAC's) Electronic Construction and Facility Support Contract Management System (eCMS) for the transfer, sharing, and management of electronic technical submittals and documents. The web-based eCMS is the designated means of transferring technical documents between the Contractor and the Government. Paper media or e-mail submission, including originals or copies, of the documents is not permitted unless identified within the contract.

All Government Contracting Specialist/Officer, legal, and command communications will remain the same.

1.2 USER PRIVILEGES

The Contractor's key staff may be provided access to eCMS. Contact the Contracting Officer for eCMS account access. Project roles and system roles will be established to control each user's menu, application, and software privileges, including the ability to create, edit, or delete objects. Additional project roles may be assigned for workflow. The Contracting Officer makes the final decision on roles for the project. User's ability to view and edit documents may be lowered at the discretion of the Contracting Officer.

Only one eCMS user account is required regardless of the user's number of projects. Notify the Contracting Officer within seven calendar days if a Contractor user is no longer associated with the company or project so they can remove them from any open record and inactivate them from the project.

1.2.1 eCMS Subcontractor Users

If the Contractor's user is a Subcontractor, the Subcontractor must be registered under the name of their company and email.

1.2.2 Users with Multiple Roles

Users may have multiple roles associated with their account within eCMS. Roles are used in workflow. When a user is added to the project, they will be assigned the default role from when the user was created. Contact the Contracting Officer to change or add roles to the user for the project.

1.2.3 Loss of Privilege

Users may lose privilege to access eCMS at the discretion of the Contracting Officer. The eCMS is a collaborative system that allows flexibility of use and does not restrict all inappropriate user actions. User activities are logged into eCMS in visible and background data collection. Users found to use eCMS in an inappropriate manner may have their eCMS access revoked. Examples include, but are not limited to,

fraudulent representations, sharing user accounts with others, and changing approved records without the consent of the Contracting Officer. Depending on the severity of the infraction, the users can lose eCMS access for a period of time, permanently for the project, or lose eCMS access for any project. The Contractor may appeal the suspension in writing to the Contracting Officer within 14 calendar days of notice of suspension. The appeal must identify the infraction, supporting information, and steps to ensure the infraction will not happen in the future.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-01 Preconstruction Submittals

##### List of Contractor's Personnel; G

### 1.4 SYSTEM REQUIREMENTS AND CONNECTIVITY

#### 1.4.1 General

NAVFAC eCMS requires a web-browser (platform-neutral) and Internet connection. For best results, recommend using browser in InPrivate/Incognito mode; Internet speeds greater than 40mbps when uploading files, computers with high RAM and Solid State Drives, "White List" eCMS website, Zip or Split files for better uploading. Non-NAVFAC Users are not to use VPN when using eCMS per NAVFAC IT.

The use of eCMS is required by the Contractor and all associated costs and time necessary to utilize eCMS will be borne by the Contractor with no allowance for time extensions and at no additional cost to the government.

Note any computers (including tablet, laptop or other computers) shall not be introduced into nor removed from the Portsmouth Naval Shipyard. In the event such equipment is required for performance of Contract work, approvals must be obtained from the Portsmouth Naval Shipyard Security Officer via the Contracting Officer. Refer to Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME) paragraph SECURITY REQUIREMENTS for additional requirements.

#### 1.4.2 Contractor Personnel List

At least fifteen (15) calendar days prior to the preconstruction meeting,, provide to the Contracting Officer a list of Contractor's personnel who will have the responsibility for the transfer, sharing and management of electronic submittals, RFIs, daily reports, and other files and will require access to the eCMS. Project personnel roles which must be filled as applicable in the eCMS include, at a minimum, the Contractor's Project Manager (KTR-PM), Superintendent (KTSUPT), Quality Control (QC) Manager (KTR-QC), Principal (KTR-PRIN), and Site Safety and Health Officer (KTR-SSHO). Notify the Contracting Officer immediately of any personnel changes to the project. The Contracting Officer reserves the right to perform a security check on all potential users.

Provide the following information:

Company Name

Name (First, Last)

E-mail Address

Project Role (CQM, SSHO, Superintendent, CM, PM, Principal)

Existing or New eCMS User

## 1.5 SECURITY CLASSIFICATION

In accordance with Department of Navy guidance, all military construction contract data are unclassified, unless specified otherwise by a properly designated Original Classification Authority (OCA) and in accordance with an established Security Classification Guide (SCG). Refer to the project's OCA when questions arise about the proper classification of information.

In conformance with the Freedom of Information Act (FOIA), DoD INSTRUCTION 5200.48 CONTROLLED UNCLASSIFIED INFORMATION (CUI), and DoD requirements, any unclassified project documentation uploaded into the eCMS must be designated either "U - UNCLASSIFIED" (U) or "CUI - CONTROLLED UNCLASSIFIED INFORMATION" (CUI). NAVFAC eCMS must only be used for the transaction of unclassified information associated with construction projects. Controlled Unclassified Identification (CUI) documents may be loaded into eCMS with the appropriate markings.

### 1.5.1 Markings on CUI Documents

Contractor's proprietary information, or documents determined by the originator in accordance with CUI guidance, should be marked CUI. Proprietary information not marked CUI can be released under the Freedom of Information Act (FOIA). Apply the appropriation markings before any document is uploaded into eCMS. Markings are not required on Unclassified U documents.

## 1.6 ECMS UTILIZATION

Establish, maintain, and update data and documentation in the eCMS throughout the duration of the contract. Utilize eCMS to transfer all submittals, RFIs, daily reports, and other files required by contract to be forwarded to the government.

Full eCMS use is required. All Submittals/Information to use eCMS Modules including, but not limited to, RFIs, Daily Reports, Meeting Minutes, Communications, Issues, Punch Lists, Checklists, and Flysheets, unless otherwise directed by the Contracting Officer.

### 1.6.1 Restricted Information

Personally Identifiable Information (PII) transmittal such as credit card, driver's license, passport, social security, and payroll number are not permitted in eCMS. Name, address, and email are permitted. Pre-negotiation information such as cost estimates that require formal negotiations are not allowed. For example, proposed changes over the SAP level of \$250k require formal negotiations. Cost estimates for LEAN, ULTRA LEAN, and Design Changes under the SAP level are at the discretion of the

Contracting Officer. The eCMS must only be used for the transaction of unclassified information associated with construction projects. Controlled Unclassified Identification (CUI) documents may be loaded into eCMS with the appropriate markings. Uploading of files directly into the Documents folder is not allowed. All documents must be uploaded using an eCMS module.

#### 1.6.2 Naming Convention for Files

Titles of files uploaded are to be descriptive of the purpose and content of the file. For example RFI\_ROOF\_Leak.doc or for submittals, SUB\_LIGHT\_FIXTURE.pdf. Titles of file to be uploaded must only contain uppercase letters, lowercase letters, numbers, hyphens (-), underscores (\_) and periods (.). Use of any other characters is not allowed and may create an error. When practicable, adding the record number to the title is desired. For example RFI\_XYZ12345\_ROOF\_Leak.doc. When practicable, add the date (day of work) with format YYYYMMDD in the filename. For example DAILY\_REPORT\_20190504. Uploading files with the same title will create a new revision in eCMS. Original revision is Rev 0, the first revision if Rev 1. Uploaded files are to use the default file location regardless of the module used unless directed by the Contracting Officer.

Table 1 also identifies which eCMS application is to be used in the transmittal of data (these are subject to change based on the latest software configuration).

Table 1 - Project Documentation Types

| SUBJECT/NAME                                       | REMARKS                                                                                                                                                                               | ECMS APPLICATION |
|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| As-Built Drawings                                  | Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager | Submittals       |
| Building Information Modeling (BIM)                | Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager | Submittals       |
| Construction Permits                               | Refer to rules of the issuing activity, state or jurisdiction                                                                                                                         | Submittals       |
| Construction Schedules (Activities and Milestones) |                                                                                                                                                                                       | Submittals       |

| SUBJECT/NAME                                                    | REMARKS                                                                                                                                                                                  | ECMS APPLICATION |
|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Construction Schedules                                          |                                                                                                                                                                                          | Submittals       |
| Construction Schedules (3-Week Lookahead)                       | Import the schedule file into the scheduling application, and select "Approve" to establish a new schedule baseline                                                                      | Meeting Minutes  |
| DD 1354 Transfer of Real Property                               | When applicable, required for final billing.                                                                                                                                             | Submittals       |
| Daily Production Reports                                        | Provide weather conditions, crew size, man-hours, equipment, and materials information                                                                                                   | Daily Report     |
| Daily Quality Control (QC) Reports                              | Provide QC Phase, Definable Features of Work<br>Identify visitors                                                                                                                        | Daily Report     |
| Designs and Specifications                                      | 1. Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager | Submittals       |
| Environmental Notice of Violation (NOV), Corrective Action Plan | Refer to rules of the issuing activity, state or jurisdiction                                                                                                                            | Submittals       |
| Environmental Protection Plan (EPP)                             |                                                                                                                                                                                          | Submittals       |
| Invoice (Supporting Documentation)                              | Applies to supporting documentation only. Invoices are submitted in Wide-Area Workflow (WAWF)                                                                                            | Submittals       |
| Jobsite Documentation, Bulletin Board, Labor Laws, SDS          | Redact any PII information when loaded into eCMS                                                                                                                                         | Submittals       |

| SUBJECT/NAME                                                                       | REMARKS                                                                                                                                                                                                                                                 | ECMS APPLICATION              |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|
| Meeting Minutes                                                                    |                                                                                                                                                                                                                                                         | Meeting Minutes               |
| Modification Documents                                                             | Provide final modification documents for the project. Upload into Modifications RFPs folder                                                                                                                                                             | Communications                |
| Operations & Maintenance Support Information (OMSI/eOMSI), Facility Data Worksheet | 1. Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager<br>2. Design reviews will be performed in existing "Dr Checks" | Submittals                    |
| Photographs                                                                        | Subject to base/installation restrictions                                                                                                                                                                                                               | Submittals                    |
| QCM Initial Phase Checklists                                                       |                                                                                                                                                                                                                                                         | Meeting Minutes or Checklists |
| QCM Preparatory Phase Checklists                                                   |                                                                                                                                                                                                                                                         | Meeting Minutes or Checklists |
| Quality Control Plans                                                              |                                                                                                                                                                                                                                                         | Submittals                    |
| QC Certifications                                                                  |                                                                                                                                                                                                                                                         | Submittals                    |
| QC Punch List                                                                      |                                                                                                                                                                                                                                                         | Punch Lists                   |
| Red-Zone Checklist                                                                 |                                                                                                                                                                                                                                                         | Punch List or Checklists      |
| Rework Items List                                                                  |                                                                                                                                                                                                                                                         | Punch Lists                   |
| Request for Information (RFI) Post-Award                                           |                                                                                                                                                                                                                                                         | RFIs                          |
| Safety Plan                                                                        |                                                                                                                                                                                                                                                         | Submittals                    |
| Safety - Activity Hazard Analyses (AHA)                                            |                                                                                                                                                                                                                                                         | Submittals                    |

| SUBJECT/NAME                                                                                  | REMARKS                                                                                                                                                                               | ECMS APPLICATION                              |
|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Safety - Mishap Reports                                                                       |                                                                                                                                                                                       | Daily Report                                  |
| Shop Drawings                                                                                 | Locations of sensitive areas must be labeled as either "Controlled Area" or "Restricted Area" and may be shown on unclassified documents with the approval from Site Security Manager | Submittals                                    |
| Storm Water Pollution Prevention (Notice of Intent - Notice of Termination)                   | Refer to rules of the issuing activity, state or jurisdiction                                                                                                                         | Submittals                                    |
| Submittals and Submittal Register                                                             |                                                                                                                                                                                       | Submittals                                    |
| Testing Plans, Logs, and Reports                                                              |                                                                                                                                                                                       | Submittals                                    |
| Training/Reference Materials                                                                  |                                                                                                                                                                                       | Submittals                                    |
| Training Records (Personnel)                                                                  |                                                                                                                                                                                       | Submittals                                    |
| Utility Outage/Tie-In Request/Approval                                                        |                                                                                                                                                                                       | Submittals                                    |
| Warranties/BOD Letter                                                                         |                                                                                                                                                                                       | Submittals                                    |
| Quality Assurance Reports                                                                     |                                                                                                                                                                                       | Checklists (Government initiated)             |
| Non-Compliance Notices                                                                        |                                                                                                                                                                                       | Non-Compliance Notices (Government initiated) |
| Other Government-prepared documents                                                           |                                                                                                                                                                                       | GOV ONLY                                      |
| Letters to government contracting, claims, REAs, and other contracting officer communications | eCMS is not the primary tool to use in contracting officer communications. eCMS can only store documents or letters after the submission to the contracting officer is made.          | Communications                                |

| SUBJECT/NAME        | REMARKS                                                                                      | ECMS APPLICATION |
|---------------------|----------------------------------------------------------------------------------------------|------------------|
| All Other Documents | Refer to FOIA guidelines and contact the FOIA official to determine whether exemptions exist | As applicable    |

#### 1.6.3 RFIs Module

Create Contractor RFIs using eCMS RFIs module. The Contractor must confirm the numbering convention with the Contracting Officer if different than eCMS default.

If the Government (GOV) Response has "No" Cost or Schedule Impact, this reply is given with the expressed understanding that it does not constitute a basis for any change in the amount or time of subject contract. Information provided in this response does not authorize work not currently included in the contract. If GOV Response is "Yes" or "Potentially" then this response may require a change to the contract. If the Contractor disagrees with either the Government's No Cost determination or No Schedule impact determination, or both, the Contractor has 14 calendar days to notify the Contracting Officer in writing.

#### 1.6.4 Submittals Module

Create Contractor submittals using eCMS Submittals module. The Contractor must confirm the numbering convention with the Contracting Officer if different than eCMS default.

#### 1.6.5 Submittal Packages Module

Create submittal packages using the eCMS Submittal Packages module in lieu of or in addition to Related Objects. Submittal Packages track completion of the packaged submittals and is used in NAVFAC HQ's KPIs.

#### 1.6.6 Communications Module

Create communications using the eCMS Communications module. The Communications module is used to create or document communications that are not a part of other eCMS modules. Use of Communications module will memorialize information into an eCMS record file. The following are Types of Communications:

Email



Memo to File

Face to Face

Telephone

Web Collaboration

Photos

Other Documents

Other

Unless directed by the Contracting Officer, upload documents or files that do not have a corresponding eCMS module. Choose "Photos" Type for Photos and "Other Documents" for all other documents.

#### 1.6.7 Issues Module

Create or respond to issues using the eCMS Issues module. Respond to CPARS issues using the Issues module.

#### 1.6.8 Meeting Minutes Module

Create or respond to Meeting Minutes using the eCMS Meetings module.

Document required contractual meetings. Dates of meetings are used in NAVFAC KPIs. Minimum meetings in eCMS include the following:

Preconstruction (Pre Con)

Initial and Preparatory Three Phases of Control

Quality Control (QC)

#### 1.6.9 Potential Change Items Module

Not used.

#### 1.6.10 Daily Report Module

Create Daily Reports using the eCMS Daily Report Module. The Contractor must confirm the numbering convention with the Contracting Officer if different than eCMS default.

#### 1.6.11 Punchlists Testing Logs (Legacy)

Punchlist Testing Logs is a legacy program that is being replaced by the Punch Lists Module. This module is to be used for reference of past projects. Use the Punch Lists Module for all future work.

#### 1.6.12 Punch Lists Module

The eCMS Punch Lists module is useful for more than just Punchlists. The module includes the capability of batch editing, create items from Optical Character Recognition (OCR) plans, assign tasks and track completion of individual items.

Create the following using the Punch Lists module:

Rework Items List

DFOW List

Punch-Out Inspection

Pre-Final Punchlist Inspection

Final Punchlist Inspection

Testing Logs

#### 1.6.13 FWD UltraLean COAR RFP Module

Not used.

#### 1.6.14 Non-Compliance Notices (NCN) Module

Respond to Non-Compliance Notices listed in the Non-Compliance Notices module.

#### 1.6.15 Checklists

Use Checklist listed in the Contractor's eCMS menu and as directed by the Contracting Officer. Checklists capture data and is used in dashboards and KPIs.

##### 1.6.15.1 Partnering Team Health Survey Checklist

Contractor must use the eCMS checklist to document the partnering team health survey. Partnering Team Health Survey is in accordance with the Partnering Specification of this contract.

#### 1.6.16 Flysheets

Use Flysheets listed in the Contractor's eCMS menu, if available, and as directed by the Contracting Officer. Flysheets allow the Contractor to print out information from other systems and upload into eCMS. The eCMS will use OCR to capture the information as data. Flysheets capture data and used in dashboards and KPIs.

#### 1.6.17 eCMS Outage

In the case where eCMS is unavailable for 8 hours or more, paper or email may be used in the interim to maintain project schedule.

Once the system is operational, all final records are required to be recreated using the appropriate module. Subject/title of the record to include the type of record i.e., RFI/Submittal/Daily Report/Communication/Other, the identification number(s), and the statement "Processed Outside of eCMS". Example, "RFI 001 Processed Outside of eCMS".

#### 1.6.18 User Account Activity

NAVFAC eCMS captures user data and activities that are directly related to the user's account. The user agrees through the use of eCMS, their

account activities will be captured and can be displayed on eCMS printed reports.

#### 1.6.19 Quality Assurance

Requested Government response dates on Transmittals and Submittals must be in accordance with the terms and conditions of the Contract unless previously agreed by the Contracting Officer. Requesting response dates earlier than the required review and response time, without concurrence by the Contracting Officer, may be cause for rejection.

Incomplete submittals will be rejected without further review and must be resubmitted. Required Government response dates for resubmittals must reflect the date of resubmittal, not the original submittal date.

All submittals and associated attachments must be transmitted to the Government via the Contracting Officer. Transmittal forms are no longer required when using eCMS since approval status is tracked on the submittal. Transmittal forms can be attached to submittals if required by the Contracting Officer. Submittals requiring Government approval are "Transmitted For" "Approval". Submittals requiring Contractor approved submittals, including those designated for Contractor Quality Control approval or Information Only, are "\*Transmitted For" "Information Only" in the Submittal Module. Provide and sign the QC Certification or approving statement on the attachment per submittal specification section. When Submittal Packages are required, use eCMS Submittal Packages after creating individual submittals. Importing Submittals from the Submittal Register is optional. Contact the Contracting Officer for the data conversion requirements.

#### PART 2 PRODUCTS

Not Used.

#### PART 3 EXECUTION

Not Used.

-- End of Section --

SECTION 01 32 17.00 20

COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS)

11/23, CHG 1: 08/24

PART 1 GENERAL

1.1 DEFINITIONS

The cost-loaded Network Analysis Schedule (NAS) is a tool to manage the project, both for Contractor and Government activities. The NAS is also used to report progress, evaluate time extensions, and provide the basis for progress payments.

For consistency, when scheduling software terminology is used in this section, the terms in Primavera's scheduling programs are used.

1.2 SCHEDULE REQUIREMENTS PRIOR TO THE START OF WORK

1.2.1 Preliminary Scheduling Meeting

Before preparation of the Project Baseline Schedule, and prior to the start of work, meet with the Contracting Officer to discuss the proposed schedule and the requirements of this section. The Contractor shall propose projected data dates for Monthly Network Analysis Updates for the project and incorporate each monthly update submittal into the submittal register. Discuss required forms, terminology, and submittal requirements of this section and other requirements related to schedule management for this contract. Use the NAVFAC-provided Preliminary Scheduling Meeting Guideline in the completion of the intended mutual understanding.

1.2.2 Project Baseline Schedule

Submit the Baseline NAS within 30 calendar days from receiving the Notice to Proceed. Data date must be set to contract award date and no progress recorded for any activity. Only bonds and the required Preliminary Scheduling Meeting may be recorded as complete prior to acceptance of the Baseline NAS. The acceptance of the Baseline NAS is a condition precedent to:

- a. The Contractor starting demolition work or construction stage(s) of the contract.
- b. Processing Contractor's invoices(s) for any items other than bonds.
- c. Review of any schedule updates. The Contractor is still required to submit monthly network analysis updates. Without an accepted baseline the Government may return schedule update as received for record.

Submittal of the Baseline NAS is the Contractor's certification that the submitted schedule meets the requirements of the Contract Documents and represents the Contractor's plan on how the work will be accomplished. Provide all items listed in paragraph REQUIRED TABULAR REPORTS AND NATIVE P6 XER FILES with Baseline NAS submittal.

### 1.3 THREE-WEEK LOOK AHEAD SCHEDULE

#### 1.3.1 Weekly CQC Coordination and Production Meeting

Submit the first look ahead schedule within 30 days from receiving the Notice to Proceed and continue to submit weekly until contract completion.

Transmit the Three-Week Look Ahead Schedule to the Contracting Officer via eCMS on the first day of the current week that the look ahead schedule covers and at least two working days prior to Quality Control meetings, or as agreed to by the Contracting Officer. Contractor is required to provide all attendees at meetings that require discussion of the schedule.

#### 1.3.2 Look Ahead Schedule Requirements

Prepare and issue a Three-Week Look Ahead schedule to provide a more detailed day-to-day plan of upcoming work identified on the Project Network Analysis Schedule. Requirements include:

- a. For each Look Ahead schedule activity, identify parent NAS activity number(s). The parent NAS activity is the activity in the NAS that would incorporate the Look Ahead schedule activity requirement and scope of work.
- b. Update schedule each week to show the planned work for the current and following two-week period. Also include previous week, as-built work, showing actual start and finish dates.
- c. Include upcoming outages, closures, preparatory meetings, and initial meetings, testing and inspections.
- d. Clearly identify longest path activities on the Three-Week Look Ahead Schedule. Include a key or legend that distinguishes longest path activities. Include all Longest Path activity NAS start/finish dates exceeded and/or occurring during this period.
- e. Identify responsibility for each activity.
- f. The detail work plans are to be bar chart type schedules, derived from but maintained separately from the Project NAS on an electronic spreadsheet program and printed on 11 by 17 inch sheets as directed by the Contracting Officer.
- g. Activities must not exceed 5 working days in duration and have sufficient level of detail to assign crews, tools and equipment required to complete the work.
- h. Lookahead activities are not limited to only onsite work. Include but do not limit to submittal, offsite fabrication, and procurement activities if needed to ensure longest path is maintained. Include design activities and coordination when part of scope.
- i. Upload Lookahead to eCMS as a submittal by the date and time required, as indicated above.

### 1.4 MONTHLY NETWORK ANALYSIS

Submittal of Monthly NAS is the Contractor's certification that the submitted schedule meets the requirements of the Contract Documents and

represents the Contractor's plan on how the work will be accomplished. Provide all items listed in paragraph REQUIRED TABULAR REPORTS AND NATIVE P6 XER FILES with the monthly NAS submittal.

#### 1.4.1 Monthly Network Analysis Updates

- a. Regardless of whether an invoice is being submitted monthly, or no progress in the schedule occurred, an updated schedule (or with new data date) must be submitted monthly to the Government. The Monthly NAS update must be submitted within 10 calendar days of the data date.
- b. Regardless of whether a baseline or other schedule is submitted and/or accepted, an updated schedule must be submitted monthly to the Government. Monthly updates submitted before the baseline being accepted will be considered as (interim and for information only) submittal.
- c. Provide all items listed in paragraph REQUIRED TABULAR REPORTS AND NATIVE P6 XER FILES, with each monthly NAS update submittal.
- d. Meet with Government representative(s) at monthly intervals to review and agree on the information presented in the updated project schedule. The submission of an accepted, updated schedule to the Government is a condition precedent to the processing of the Contractor's invoice.
- e. The Monthly Network Analysis Updates must include identification of all anticipated outages (road, rail, utility, etc.) that are required to complete the work for the coming two-month period, at a minimum. The schedule must include the dates on which the outages are expected to commence, as well as the dates for when the outage request packages are expected to be submitted. Refer to Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME) for outage requirements.
- f. Activity progress must incorporate as-built events as they occurred and correspond to records including but not limited to submittals and daily production and quality control reports.
- g. Software Settings: Handle schedule calculations and Out-of-Sequence progress (if applicable) through Retained Logic, not Progress Override. Show all activity durations and float values in days. Show activity progress using Remaining Duration. Set default activity type to "Task Dependent".
- h. Update schedule must reflect current Contract Completion Date and contract value in accordance with all conformed contract modifications issued prior to data date of NAS update.

#### 1.4.2 As-Built Schedule

As a condition precedent to the release of retention and making final payment, submit an "As-Built Schedule," as the last schedule update showing all activities at 100 percent completion. This schedule must reflect the exact manner in which the project was actually constructed.

#### 1.5 CORRESPONDENCE AND TEST REPORTS

Reference Schedule activity IDs that are being addressed in each correspondence (e.g., letters, Requests for Information (RFIs), e-mails,

meeting minute items, Production and QC Daily Reports, material delivery tickets, photographs) and test report (e.g., concrete, soil compaction, weld, pressure).

#### 1.6 ADDITIONAL SCHEDULING REQUIREMENTS

Other specification sections may include additional scheduling requirements, including systems to be inspected, tested and commissioned, and submittal procedures. Those schedule requirements must be incorporated into the NAS schedule.

#### 1.7 SUBMITTALS

Government approval/acceptance is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-01 Preconstruction Submittals

Baseline NAS; G

Designated Project Scheduler; G

##### SD-07 Certificates

Three-Week Look Ahead Schedule; G

Monthly Network Analysis Updates; G

##### SD-11 Closeout Submittals

As-Built Schedule; G

#### 1.8 SOFTWARE

Prepare and maintain project schedules using Primavera P6 software in a version compatible with Government's current version. Importing data into P6 using data conversion techniques or third party software is cause for rejection of the submitted schedule. Schedules with Performing Organizational Breakdown Structure (POBS) data is cause for rejection.

#### 1.9 DESIGNATED PROJECT SCHEDULER

Within 30 calendar days from receiving the Notice to Proceed, submit to the Contracting Officer for approval an individual who will serve as the Designated Project Scheduler. Include a copy of the candidate's resume with qualifications. The Contracting Officer may remove the Designated Project Scheduler, and require replacement, if the scheduler does not effectively fulfill their duties in accordance with the contract requirements. Payment request will not be processed without an approved Designated Project Scheduler.

##### 1.9.1 Qualifications

The Designated Project Scheduler must have prepared and maintained at least three previous construction schedules, of similar size and complexity to this contract, using Primavera P6.

### 1.9.2 Duties

Duties of the Designated Project Scheduler include to be actively involved and be responsible for the following:

- a. Prepare Baseline NAS.
- b. Prepare Three-Week Look Ahead Schedules.
- c. Prepare Monthly Network Analysis updates.
- d. Prepare tabular reports.
- e. Prepare Time Impact Analysis (TIA) as necessary.
- f. Provide certification that NAS and TIA submittals conform to the contract requirements.
- g. Participate with the Prime Contractor and Government Representative in a monthly meeting at the job site in-person, and scheduled with sufficient time to support the Monthly Network Analysis Updates process, to discuss project status, schedule updates, critical activities including outages, potential delays, and contract modifications impacting the schedule. Have a computer with P6 software available during the meeting.

### 1.10 NETWORK SYSTEM FORMAT

Prepare the schedule in accordance with the following Primavera P6 settings and parameters. Deviation from these settings and parameters, without prior consent of the Contracting Officer, is cause for rejection of schedule submission.

#### 1.10.1 Schedule Activity Properties and Level of Detail

##### 1.10.1.1 Activity Identification and Organization

- a. Identify construction activities planned for the project and other activities that could impact project completion if delayed in the NAS.
- b. Each activity must have a unique name.
- c. Identify administrative type activity/milestones, including all pre-construction submittal and permit requirements prior to demolition or construction stage.
- d. Include times for procurement, Contractor quality control and construction, acceptance testing and training in the schedule.
- e. Include the Government approval time required for the submittals that require Government Approval prior to construction, as indicated in Section 01 33 00 SUBMITTAL PROCEDURES.
- f. Include the Government concurrence period required for outage request package submittals, as indicated in Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME).
- g. Create separate activities for each Phase, Area, Floor Level and Location the activity is occurring.



- h. Do not use construction category activity to represent non-work type reference (e.g. Serial Letter, Request for Information) in NAS. Place Non-work reference within the P6 activity details notebook.

Activity categories included in the schedule are specified below.

#### 1.10.1.2 Activity Logic

- a. With the exception of the Contract Award and Contract Completion Date (CCD) milestone activities, activity must not be open-ended; each activity must have at least one predecessor and at least one successor.
- b. Activities must not have open start or open finish (dangling) logic.
- c. Do not use lead or lag logic without Contracting Officer prior approval.
- d. Minimize redundant logic ties.
- e. Once an activity exists on the schedule it must not be deleted or renamed to change the scope of the activity and must not be removed from the schedule logic without approval from the Contracting Officer.
  - (1) While an activity cannot be deleted, where said activity is no longer applicable to the schedule, but must remain within the logic stream for historical record, change the activity original and remaining duration to zero and clearly label "(NO LONGER REQUIRED)" after the activity name. Actual finish date for activity that falls behind the data date. Redistribute accordingly any remaining budget associated with that activity, to other remaining appropriate activity.
  - (2) Document any such change in the activities' "Notebook," including a date and explanation for the change.
  - (3) The ID number for a "NO LONGER REQUIRED" activity must not be re-used for another activity.

#### 1.10.1.3 Longest Path Activity Baseline Limitation

For P6 settings, critical activities are defined as being on the Longest Path. Longest Path (Critical) Activities must not make up more than 30 percent of all activity within the Construction Baseline Schedule.

#### 1.10.1.4 Assigned Calendars

All NAS activity must be assigned calendars that reflect required and anticipated non-work days.

#### 1.10.1.5 Activity Categories

##### 1.10.1.5.1 Pre-construction Activities

Examples of pre-construction activities include, but are not limited to, bond approval, permits, pre-construction submittals and approvals. Include pre-construction activities that are required to be completed prior to the Contractor starting the demolition or construction stage of work.

#### 1.10.1.5.2 Procurement Activities

Examples of procurement activities include, but are not limited to: Material/equipment submittal preparation, submittal and approval of material/equipment; material/equipment fabrication and delivery, and material/equipment on-site. As a minimum, separate procurement activities must be provided for critical items, long lead items, items requiring Government approval and material/equipment procurement for which payment will be requested in advance of installation. Show each delivery with relationship tie to the Construction Activity specifically for the delivery.

#### 1.10.1.5.3 Government Activities

Government and other agency activities that could impact progress must be clearly identified. Government activities include, but are not limited to; Government approved submittal reviews, Government conducted inspections/tests, environmental permit approvals by State regulators, utility outages, and delivery of Government Furnished Material/Equipment.

#### 1.10.1.5.4 Construction Quality Management (CQM) Activities

The Preparatory and Initial Phase meetings for each Definable Feature of Work identified in the Contractor's Quality Control Plan must be included in the Three-Week Look Ahead Schedule. Preparatory and Initial phase meetings are not required in the NAS, but can be represented by a start milestone linked to successor parent Construction Activity. The Follow-up Phase must be represented by the Construction Activities themselves in the NAS.

#### 1.10.1.5.5 Construction Activities

On-site construction activities that are the responsibility of the Contractor or Contractor's subcontractors must not have a duration in excess of 20 working days without prior approval by the Contracting Officer. Contractor activities must be driven by calendars that reflect Saturdays, Sundays and all Federal Holidays as non-work days, unless otherwise defined in this contract. Federal Holidays are as defined in 5 USC 6103.

#### 1.10.1.5.6 Turnover and Closeout Activities

Include activities or milestones for items on the NAVFAC Red Zone Checklist/POAM that are applicable to this project. As a minimum, include required Contractor testing, required Government acceptance inspections on equipment, Pre-Final Inspection, Punch List Completion, Final Inspection and Acceptance. Add an unconstrained start milestone for the initial NAVFAC Red Zone - Facility Turnover Planning Meeting at approximately 75 percent construction contract completion or six months prior to Contract Completion Date (CCD), whichever is sooner. Include a separate NAVFAC Red Zone - Facility Turnover Planning Meeting Milestone for each phase if the contract requires construction to be completed in phases.

#### 1.10.1.5.7 Testing of HVAC - DALT, TAB, and PVT Activities

Include in the baseline schedule, activities and milestones associated with Government acceptance of Duct Air Leakage Test (DALT), Testing, Adjusting, and Balancing (TAB) and Performance Verification Test (PVT) as required and in accordance with Section 23 05 93 TESTING, ADJUSTING AND

BALANCING FOR HVAC and Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

- a. Identify the general area or location(s) for Government Acceptance Testing of DALT, TAB and PVT.
- b. Incorporate into the baseline schedule, time periods required for advance notification of work, and Government submittal review in accordance with Section 23 05 93 TESTING, ADJUSTING AND BALANCING FOR HVAC, paragraph DALT AND TAB SUBMITTAL AND WORK SCHEDULE.
- c. Include the following as schedule activities or milestones:
  - (1) Pre-DALT/TAB/PVT Meeting
  - (2) TAB Design Review Report, Government review
  - (3) TAB Pre-Field Engineering Report, Government review
  - (4) DALT Field Work
  - (5) DALT Field Acceptance Testing
  - (6) Certified Final DALT Report, Government review
  - (7) Control Contractors PVT Plan, Government review
  - (8) Equipment Suppliers PVT Plan, Government review
  - (9) Season I TAB Field Work
  - (10) Season I Certified Final TAB Report, Government review
  - (11) Endurance Testing, Government review
  - (12) PVT Field Work
  - (13) PVT Report, Government review
  - (14) Season I TAB Field Acceptance Testing
  - (15) Season II TAB Field Work
  - (16) Season II Certified Final TAB Report, Government review
  - (17) Season II TAB Field Acceptance Testing

#### 1.10.1.5.8 Commissioning Activities

Include in the baseline schedule activities and milestones associated with Commissioning.

- a. Identify the general area or location(s) of systems for Commissioning Inspection and Testing
- b. Incorporate into the baseline schedule time periods for Government submittal review

#### 1.10.1.6 Contract Milestones and Constraints

##### 1.10.1.6.1 Project Start Date Milestones

Include as the first activity on the schedule a start milestone titled, "Contract Award", which must have a Mandatory Start constraint equal to the Contract Award Date.

##### 1.10.1.6.2 Preconstruction Meeting Milestone

Include an unconstrained finish milestone on the schedule titled, "Pre construction Meeting". The Preconstruction meeting may be a single day, or it may range over several days. The intent to cover all the Precon topics, including Partnering and DD1354.

##### 1.10.1.6.3 Preconstruction Submittals Finish Milestone

Include an unconstrained finish milestone on the schedule titled, "Preconstruction Submittals". This milestone is complete when all required preconstruction submittals have been reviewed and approved by the Government.

##### 1.10.1.6.4 Contractor Mobilization Finish Milestone

Include an unconstrained finish milestone on the schedule titled, "Contractor Mobilization".

##### 1.10.1.6.5 NAVFAC Red Zone - Facility Turnover Planning Meeting Milestones

See paragraph TURNOVER AND CLOSEOUT ACTIVITIES above.

##### 1.10.1.6.6 Substantial Completion Milestone

Include an unconstrained finish milestone on the schedule titled "Substantial Completion." Substantial Completion is defined as the point in time the Government would consider the project ready for beneficial occupancy wherein by mutual agreement of the Government and Contractor, Government use of the facility is allowed while construction access continues in order to complete remaining items (e.g. punch list and other close out submittals). Include a separate Substantial Completion Milestone for each phase if the contract requires construction to be completed in phases.

##### 1.10.1.6.7 DD-1354 Finish Milestone

Add unconstrained finish milestone, titled "DD-1354" and scheduled 30 calendar days prior to Substantial Completion, whenever a Form DD-1354 is required in accordance with Section 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME).

##### 1.10.1.6.8 Projected Completion Milestone

Include an unconstrained finish milestone on the schedule titled "Projected Completion." Projected Completion is defined as the point in time all contract requirements are complete and verified by the Government with a successful Final Inspection in accordance with Section 01 45 00.00 22 QUALITY CONTROL (PWD ME). This milestone must have the Contract Completion Date (CCD) milestone as its only successor.

#### 1.10.1.6.9 Contract Completion Date (CCD) Milestone

Last schedule entry must be an unconstrained finish milestone titled "Contract Completion (CCD: DD-MM-YY)." DD-MM-YYYY is the current contract completion date at data date, day-month-year corresponding to P6 Must Finish By Date. NAS milestone updates of Project Completion finish date for longest path must reflect calculated float as positive or negative based on CCD. Calculation of schedule updates must be such that if the finish of the "Projected Completion" milestone falls after the contract completion date, then negative float is calculated on the longest path. If the finish of the "Projected Completion" milestone falls before the contract completion date, the float calculation must reflect positive float on the longest path.

#### 1.10.1.6.10 Additional Milestones

Provide up to 5 additional milestones as required by Contracting Officer.

#### 1.10.1.7 Work Breakdown Structure & Activity Code

At a minimum, establish a Work Breakdown Structure (WBS) and provide activity codes identified as follows:

##### 1.10.1.7.1 Work Breakdown Structure (WBS)

Group all activities and milestones within appropriate WBS categories including, at a minimum, the following:

a. Project Milestones:

- (1) Management Milestones
- (2) Project Administrative Meetings
- (3) Permits

b. Pre-Construction Phase:

- (1) Submittals and Reviews
- (2) Procurement
- (3) Mobilization

c. Construction Phase: Create multiple sub-sections in accordance with project specific categories of work including in WBS descending order as follows:

- (1) General Area
  - (a) Type of Work Item
    - 1. Location

d. Project Closeout: Include activity items such as, but not limited to, Punchlist, Demobilization, O&M, As-built Drawings, Training, and As-built NAS.

e. Modifications: Create sub-category of Conformed and Non-Conformed

under Modification WBS. Create multiple sub-sections as the project progresses identified by issue and Fragnet placed in Conformed for modifications issued prior data date, or Non-Conformed for issues not modified to contract prior data date.

- f. Removed Activity: Activity is "removed" by remaining within logic sequence, eliminating duration and adding "(NO LONGER REQUIRED)" after Activity Name in Activity Table.
- g. Commissioning & Testing:
  - (1) Specific area/locations of commissioning
  - (2) Final Testing
  - (3) Training

#### 1.10.1.7.2 Responsibility Code

Use "RESP" for Activity Code Name. All activities in the project schedule must be identified with the resource for completing the task. Activities must not belong to more than one responsible party.

#### 1.10.1.7.3 Activity Category Code

Provide user defined "CAT" codes for Project Level activity codes. Use the following codes:

- a. Assign "PROC" value to Procurement type activity
- b. Assign "PRE-CON" value to Pre-construction activity
- c. Assign "CONS" value to Construction type activity
- d. Assign "TEST" value to dedicated testing type activities
- e. Assign "CX" value to dedicated Commissioning type activities
- f. Assign "CLOS" value to dedicated Close Out type activity
- g. Assign "OTHR" to other activity not otherwise designated

#### 1.10.1.7.4 Drawing Code

Identify all activities in the project schedule with its respective Drawing Code. The Drawing Code is the Sheet Number on the primary project drawing which indicates work to be performed. If an activity does not have an applicable Drawing Code (e.g. Mobilize), the code must be "0000".

#### 1.10.1.7.5 Additional Activity Codes

Provide up to an additional five activity codes as required by the Contracting Officer.

#### 1.10.1.8 Adverse Weather Lost Work Days

Use the National Oceanic and Atmospheric Administration's (NOAA) Summary of Monthly Normals report to obtain the historical average number of days each month with precipitation, using a nominal 30-year, greater than 0.10

inch precipitation amount parameter, as indicated on the Station Report for the NOAA location closest to the project site as the basis for establishing a "Weather Calendar" showing the number of anticipated non-workdays for each month due to adverse weather, in addition to Saturdays, Sundays and all Federal Holidays as non-work days.

Assign the Weather Calendar to any activity that could be impacted by adverse weather. The Contracting Officer will issue a modification in accordance with the contract clauses, giving the Contractor a time only extension for the difference of days between the anticipated and actual adverse weather delay if the number of actual adverse weather delay days exceeds the number of days anticipated for the month in which the delay occurs and the adverse weather delayed activities are on the longest path to contract completion in the period when delay occurred. A lost workday due to weather conditions is defined as a day in which the Contractor cannot work at least 50 percent of the day on the impacted activity. Impacts resulting from adverse weather must be documented in Narrative Report for the month that it occurred.

Make changes to P6 project calendars to reflect as-built conditions where work occurred where originally anticipated as non-work days, and where work did not occur (lost work day).

#### 1.10.1.9 Anticipated Restricted Delays

Unless otherwise noted or defined in Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME), allow in the schedule eight lost workdays for every one year of project duration for instances where base access is not permitted or where work areas are temporarily not accessible for security reasons which causes a delay in the work. Use Anticipated Restricted Delays as basis for establishing a "Security Calendar" showing the number of anticipated non-workdays for each month due to anticipated restrictions, in addition to anticipated adverse weather, Saturdays, Sundays and all Federal Holidays as non-work days. Assign the Security Calendar to any activity that could be impacted by restriction delays. The Contracting Officer will issue a modification in accordance with the contract clauses, giving the Contractor a time extension for the difference of days between the anticipated and actual lost work days if the number of actual restriction delay days exceeds the number of anticipated for the month in which the delay occurs and the restriction delayed activities are critical to contract completion. A lost workday due to restriction delay is defined as a day in which the Contractor cannot work at least 50 percent of the day on the impacted activity.

Impacts resulting from restriction delays must be documented in Narrative Report for the month that it occurred.

Make changes to P6 project calendars to reflect as-built conditions where work occurred where originally anticipated as non-work days, and where work did not occur (lost work day).

#### 1.10.1.10 Cost Loading

The Project Network Analysis Schedule (NAS) must be cost-loaded and will provide the basis for progress payments. Earned Value Reports must be derived from and correspond to cost loaded NAS. Use the Critical Path Method (CPM) and the Precedence Diagram Method (PDM) to satisfy time and cost applications.

Sum of all costs assigned to activities must equal the current contract value as of the data date of the baseline or update schedule. Activity cost loading must be without front-end loading. Provide additional documentation if requested by the Contracting Officer.

#### 1.10.1.10.1 Cost Loading Activities

- a. Cost load construction activities. Define construction activities as on-site work activities that can be traced directly to the construction, alteration, demolition, repair of a project facility, infrastructure, or sitework.
- b. Assigned costs for materials which payment will be requested in advance of installation, including their quantities, to their respective procurement activities that were approved by the Contracting Officer. Assign labor costs, including their quantities, for material and equipment paid for after installation to their respective construction activities.
- c. Costs for mobilization will not be paid as individual pay items with the exception of batch plant set-up, mobilization of dredging equipment or other similar labor-intensive situations. If payment for mobilization costs are approved, assign costs for demobilization activities. The value of demobilization activities must not be less than 40 percent of the total costs for mobilization/demobilization activities. If not approved as a separate line item, evenly disperse over construction activities the costs for mobilization and/or demobilization.
- d. The value of commissioning, testing and closeout WBS section may not be less than 10 percent of the total costs for procurement and construction activities.
- e. ALL activities assigned Government responsibility will have Zero Cost. No contractor cost should be assigned to an activity designated as a Government responsibility.
- f. Do not include field overhead positions as individual cost loaded activities. Evenly disperse indirect costs (e.g., meetings, presentations, submittals, surveys, haul road or storage yard maintenance, dust control, security) and profit to construction activities assigned to the Construction Phase WBS level.

#### 1.10.1.10.2 Partial Payment

Breakdown unit of measure and cost must be defined within P6 Activity Detail Expenses for partial payment of any cost loaded activity. Lump sum cost loaded activity will not be partially paid.

#### 1.10.2 Schedule Software Settings and Restrictions

- a. Activity Constraints: Date/time constraint(s), other than those required by the contract, are not allowed unless accepted by the Contracting Officer. Identify any constraints proposed and provide an explanation for the purpose of the constraint in the Narrative Report as described in paragraph REQUIRED TABULAR REPORTS.
- b. Default Progress Data Disallowed: Actual Start is date work begins on activity with intent to pursue work to substantial completion. Actual



Finish is date work is substantially complete to point where successor activity can begin. Actual dates on the CPM schedule must correspond with activity dates reported on the Contractor Quality Control and Production Reports.

- c. At a minimum, include the following settings and parameters in P6 Schedule preparation:
- (1) General: Define or establish Calendars and Activity Codes at the "Project" level, not the "Global" level.
  - (2) Admin Drop-Down Menu, Admin Preferences, Time Periods Tab:
    - (a) Set time periods for P6 to 8.0 Hours/Day, 40.0 Hours/Week, 172.0 Hours/Month and 2000.0 Hours/Year.
    - (b) Use assigned calendar to specify the number of work hours for each time period: Must be checked.
  - (3) Admin Drop-Down Menu, Admin Preferences, Earned Value Tab:
    - (a) Technique for computing performance percent complete: Use "Activity percent Complete"
    - (b) Technique for computing Estimate to Complete: Use "PF = 1".
    - (c) Earned Value Calculation: Use "Budgeted values with current dates".
  - (4) Project Level, Dates Tab:
    - (a) Set "Must Finish By" date to "Contract Completion Date", and set "Must Finish By" time to 05:00pm.
  - (5) Project Level, Defaults Tab:
    - (a) Duration Type: Set to "Fixed Duration & Units".
    - (b) Percent Complete Type: Set to "Physical".
    - (c) Activity Type: Set to "Task Dependent".
    - (d) Calendar: Set to "Standard 5 Day Workweek". Calendar must reflect Saturday, Sunday and all Federal holidays as non-work days. Alternative calendars may be used with Contracting Officer approval.
  - (6) Project Level, Calculations Tab:
    - (a) Default Price/Unit for activities without resource or role Price/Units: Set to "\$1/h".
    - (b) Activity percent complete based on activity steps: Must be Checked.
    - (c) Link Budget and At Completion for not started activities: Must be Checked.
    - (d) Reset Remaining Duration and Units to Original: Must be

Selected.

(e) Subtract Actual from At Completion: Must be Selected.

(f) Recalculate Actual units and Cost when duration percent complete changes: Must be Checked.

(g) Update units when costs change on resource assignments: Must be Unchecked.

(h) Link Actual to Date and Actual This Period Units and Cost: Must be Checked.

(7) Project Level, Settings Tab:

(a) Define Critical Activities: Check "Longest Path".

(8) Work Breakdown Structure Level, Earned Value Tab:

(a) Technique for Computing Performance Percent Complete: "Activity percent complete" is selected.

(b) Technique for Computing Estimate to Complete (ETC): "PF = 1" is selected.

#### 1.10.3 Required Tabular Reports and Native P6 XER Files

Include the following reports with the Baseline, Monthly Update and any other required schedule submittals:

##### a. Time Scaled Logic Schedule

Provide formatted 11 by 17-inch Time-scaled Logic Schedule in color and landscape-oriented with each schedule submittal. Clearly show activities on the longest path setting Gantt chart longest path activity bars to red. Group activities by WBS and sort by finish date in ascending order. Include the following information in column form for each activity and include accompanying Gantt chart:

(1) Activity ID

(2) Activity Name

(3) Original Duration

(4) Remaining duration

(5) Physical Percent Complete

(6) Start Date

(7) Finish Date

(8) Total Float

(9) Project-level Calendar

##### b. Previous Monthly Update Comparison Time Scaled Logic Schedule (Submit with all Monthly Update Schedule Submittals.)

Provide formatted 11 by 17-inch Time-scaled Logic Schedule in color and landscape-oriented with each monthly update schedule submittal. Clearly show activities on the current month longest path setting Gantt chart longest path activities bars to red. Show previous month activities as yellow bars and previous month milestones in yellow within Gantt chart. Sort by finish date in ascending order. Filter activities for longest path. Maintain and assign the accepted previous month update or the accepted baseline schedule for the first update submittal as the baseline and primary baseline in P6 before printing the schedule. Include the following information in column form for each activity and include accompanying Gantt chart:

- (1) Activity ID
  - (2) Activity Name
  - (3) Original Duration
  - (4) Current Month Remaining Duration
  - (5) Current Month Start Date
  - (6) Previous Month Update Start Date (BL Project Start)
  - (7) Start Date Delta between Current Month and Previous Month  
(Variance - BL Project Start Date)
  - (8) Current Month Finish Date
  - (9) Previous Month Finish Date (BL Project Finish)
  - (10) Finish Date Delta between Current Month and Previous Month  
(Variance - BL Project Start Date)
  - (11) Current Month Total Float
- c. P6 native XER file: Include the back-up native .xer program file compatible with the Government version of Primavera P6. Each native schedule file must have a unique file name to include project name and data date using (yyyy-mm-dd) convention. Each native schedule must have a unique Project ID and Project Name.
- d. Log Report: P6 Scheduling/Leveling Report.
- e. Narrative Report: Identify and justify:
- (1) Provide Project Summary Data in format below:
    - (a) Data Date \_\_\_\_\_
    - (b) Award Date: \_\_\_\_\_
    - (c) Original Project Duration: \_\_\_\_\_ days post Award Date
    - (d) Current Project Duration: \_\_\_\_\_ days post Award Date
    - (e) Time percent elapsed: \_\_\_\_\_ percent at data date

- (f) Original CCD: \_\_\_\_\_
  - (g) Current CCD: \_\_\_\_\_ (thru MOD \_\_\_\_\_)
  - (h) Anticipated CCD: \_\_\_\_\_ (\_\_\_\_ calendar days early/late)
  - (i) Original Contract Value: \$\_\_\_\_\_
  - (j) Current Contract Value: \$\_\_\_\_\_
  - (k) Invoiced Amount: \$\_\_\_\_\_ (\_\_\_\_ percent)
  - (l) Cost Growth: \_\_\_\_ percent
  - (m) Schedule Growth: \_\_\_\_ percent
  - (n) There are a total of \_\_\_\_\_ activities, \_\_\_\_\_ activities complete (\_\_\_\_ percent), \_\_\_\_\_ activities in progress (\_\_\_\_ percent), \_\_\_\_\_ activities not started (\_\_\_\_ percent). Of the in progress and not started activities; \_\_\_\_\_ (\_\_\_\_ percent) are on the longest path. The longest path has duration of \_\_\_\_\_ calendar days from data-date to anticipated project completion.
- (2) Progress made in each area of the project;
  - (3) Longest Path;
  - (4) Date/time constraint(s), other than those required by the contract
  - (5) Listing of all changes made between the previous schedule and current updated schedule include: added or deleted activities, original and remaining durations for activities that have not started, logic (sequence constraint lag/lead), milestones, planned sequence of operations, longest path, calendars or calendar assignments, and cost loading;
  - (6) Any decrease in previously reported activity Earned Amount;
  - (7) Pending items and status thereof, including permits, changes orders, and time extensions;
  - (8) Status of Contract Completion Date and interim milestones;
  - (9) Status of Projected Completion Milestone and account of difference in calendar days between previous update Projected Completion Milestone
  - (10) Current and anticipated delays listing Activity Names and IDs for impacted activities(describe cause of delay and corrective actions(s) and mitigation measures to minimize);
  - (11) Description of current and potential future schedule problem areas.
  - (12) Identification of any weather and restricted lost time as compared to anticipated weather for the month and anticipated restricted days for which the update is submitted. Impacts resulting from adverse weather must be documented in tabular form showing the calendar month (or billing period) with the days on

which construction activity incurred Lost Work Days due to adverse weather. In narrative form, describe the adverse weather cause such as precipitation measurement, temperature, wind or other influencing factors, and why work was impacted. Describe the construction activity(s) that was (were) scheduled, impacted.

Each entry in the narrative report must cite the respective Activity ID and Activity Name, the date and reason for the change, and description of the change.

- f. Earned Value Report: Derive from and correspond to P6 cost loaded schedule. List all activities having a budget amount cost loaded. Compile total earnings on the project from notice to proceed to current progress payment request. Show current budget, previous physical percent complete, to-date physical percent complete, previous earned value, to-date earned value, cost this period and cost to complete on the report for each activity.
- g. Schedule Variance Control (SVC) Diagram: With each schedule submission, provide a SVC diagram showing 1) A Cash Flow Curve indicating planned project cost based on each of projected early and projected late activity finish dates and 2) one curve for Earned Value to-date. Revise Cash Flow Curves when the contract is modified, or as directed by the Contracting Officer Include a legend on report clearly indication 3 curves: early finish, late finish, and earned-value to date.

Use the following settings in Activity Usage Profile Options:

- (1) In the Data section, under Display, the radio box for Cost must be selected.
- (2) In the Data section, under Filter for Bars/Graphs, the checkbox for Total must be checked.
- (3) In the Show Bars/Curves section:
  - (a) Under the By Date column, the checkboxes for Baseline, Actual and Remaining Late must be checked. The checkboxes for Budgeted and Remaining Early must be unchecked.
  - (b) Under the Cumulative column, the checkboxes for Baseline, Actual and Remaining Late must be checked. The checkboxes for Budgeted and Remaining Early must be unchecked.
  - (c) Set the color for Baseline to green.
  - (d) Set the color for Actual to blue.
  - (e) Set the color for Remaining Late to red.
- (4) In the Show Earned Value Curves section, the checkboxes for Planned Value Cost, Earned Value Cost and Estimate at Completion must be unchecked.
- h. Logic Diagram showing timescale from data date to 60 days after data date with filter for longest path. Leave Group By selection blank and sort by finish date in ascending order.

- i. Baseline or Monthly Update Checklist as applicable completed and certified by the Designated Project Scheduler. Baseline Project Schedule and Monthly Update Schedule Checklists can be found on the Whole Building Design Guide website at <https://www.wbdg.org/dod/ufgs/ufgs-forms-graphics-tables>
- j. Screen shot PDF of Primavera P6 Time Periods Settings referenced in paragraph SCHEDULE SOFTWARE SETTINGS AND RESTRICTIONS, list item c.(2): ADMIN DROP-DOWN MENU, ADMIN PREFERENCES, TIME PERIODS TAB
- k. Daily Reported Production Activity: Submit on a monthly basis, in electronic spreadsheet (format provided by the Government), summary of daily reported production activity for the reporting month in the update schedule. Use the following columns for reporting:
  - (1) Date
  - (2) Activity ID
  - (3) Work Description
  - (4) Contractor
  - (5) Billable Hours
- l. HVAC/COMMISSIONING KTR Checklist. See Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC for checklist requirements. Complete checklist listing all items with baseline submittal. Complete checklist as required to complete project, submitting complete checklist in intermediately following monthly update submittal.

#### 1.11 CONTRACT MODIFICATION

##### 1.11.1 Time Impact Analysis (TIA)

Submit a Time Impact Analysis with each cost and time proposal for a proposed change. TIA must illustrate the influence of each change or delay on the Contract Completion Date or milestones. No time extensions will be granted nor delay damages paid unless a delay occurs which consumes all available Project Float, impacts the longest path, and extends the Projected Completion beyond the Contract Completion Date.

- a. Each TIA must be in both narrative and schedule form. The narrative must:
  - 1. Define the scope and conditions of the change;
  - 2. Provide start and finish dates of impact;
  - 3. Successor(s) and predecessor(s) activities to impact period;
  - 4. Responsible party;
  - 5. Describe how the impact originated;
  - 6. How it impacts the schedule's longest path.
- b. The schedule submission must consist of three native XER files:

- (1) Fragnet used to define the scope of the changed condition
  - (2) Most recent accepted schedule update as of the time of the impact start date. Update this schedule to show all activity progress as of the time of the impact start date. The impact start date is identified as the time when an existing activity is impeded for either starting or finishing.
  - (3) The impacted schedule that has the fragnet inserted in the updated schedule and the schedule "run" so that the new completion date is determined.
- c. For claimed as-built project delay, the inserted fragnet TIA method must be modified to account for as-built events known to occur after the data date of schedule update used. Updated schedules for periods following the impact start date will be used to evaluate how the project progressed (as-built) through the finish of impact. Impact to longest path must be determined for each following update period.
- d. All TIAs must include any mitigation, and must determine the apportionment of the overall delay assignable to each individual delay. Apportionment must provide identification of delay type and classification of delay by compensable and non-compensable events. The associated narrative must clearly describe analysis methodology used, and the findings in a chronological listing beginning with the earliest delay event.

(1) Identify and classify types of delay defined as follows:

- (a) Force majeure delay (e.g. weather delay): Any delay event caused by something or someone other than the Government or the Contractor, or the risk of which has not been assigned solely to the Government or the Contractor. If the force majeure delay is on the longest path, in absence of other types of concurrent delays, the Contractor is granted an extension of contract time, classified as a non-compensable event.
  - (b) A Contractor-delay: Any delay event caused by the Contractor, or the risk of which has been assigned solely to the Contractor. If the contractor-delay is on the longest path, in absence of other types of concurrent delays, Contractor is not granted extension of contract time, and classified as a non-compensable event. Where absent other types of delays, and having impact to project completion, Contractor must provide to Contracting Officer a Corrective Action Plan identifying plan to mitigate delay.
  - (c) A Government-delay: Any delay event caused by the Government, or the risk of which has been assigned solely to the Government. If the Government-delay is on the longest path, in absence of other types of concurrent delays, the Contractor is granted an extension of contract time, and classified as a compensable event.
- (2) Functional concurrency must be used to analyze concurrent delays, where: separate delay issues delay project completion, do not necessarily occur at same time, rather occur within same monthly schedule update period at minimum, or within same as-built period under review. If a combination of functionally concurrent delay types occurs, it is considered Concurrent Delay, which is defined

in the following combinations:

(a) Government-delay concurrent with contractor-delay: excusable time extension, classified non-compensable event.

(b) Government-delay concurrent with force majeure delay: excusable time extension, classified non-compensable event.

(c) Contractor-delay concurrent with force majeure delay: excusable time extension, classified non-compensable event.

(3) Pacing delay reacting to another delay (parent delay) equally or more critical than paced activity must be identified prior to pacing. Contracting Officer will notify Contractor prior to pacing. Contractor must notify Contracting Officer prior to pacing. Notification must include identification of parent delay issue, estimated parent delay time period, paced activity(s) identity, and pacing reason(s). Pacing Concurrency is defined as follows:

(a) Government-delay concurrent with contractor-pacing: excusable time extension, classified compensable event.

(b) Contractor-delay concurrent with Government-pacing: inexcusable time extension, classified non-compensable event

e. Submit electronic file containing the narrative and the source schedule files used in the time impact analysis.

#### 1.12 PROJECT FLOAT

Project Float is the length of time between the Contractor's Projected Completion Milestone and the Contract Completion Date. Project Float available in the schedule will not be for the exclusive use of either the Government or the Contractor.

The use of Resource Leveling or other techniques used for the purpose of artificially adjusting activity durations to consume float and influence longest path is prohibited.

#### PART 2 PRODUCTS

Not used.

#### PART 3 EXECUTION

Not used.

-- End of Section --



SECTION 01 33 00

SUBMITTAL PROCEDURES

08/18, CHG 4: 02/21

PART 1 GENERAL

This Section applies only to Design-Bid-Build projects at the Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 DEFINITIONS

1.1.1 Submittal Descriptions (SD)

Submittal requirements are specified in the technical sections. Examples and descriptions of submittals identified by the Submittal Description (SD) numbers and titles follow:

SD-01 Preconstruction Submittals

Submittals that are required prior to or commencing with the start of work on site. Submittals that are required prior to or at the start of construction (work) or the next major phase of the construction on a multiphase contract. No work is authorized without an approved submittal.

Preconstruction Submittals include schedules and a tabular list of locations, features, and other pertinent information regarding products, materials, equipment, or components to be used in the work.

Refer to Section 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS (PWD ME) for a partial listing of required Preconstruction Submittals.

SD-02 Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work.

Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project.

Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

SD-03 Product Data

Catalog cuts, illustrations, schedules, diagrams, performance charts, instructions and brochures illustrating size, physical appearance and other characteristics of materials, systems or equipment for some portion of the work.

Samples of warranty language when the contract requires extended product warranties.

SD-04 Samples

Fabricated or unfabricated physical examples of materials, equipment

or workmanship that illustrate functional and aesthetic characteristics of a material or product and establish standards by which the work can be judged.

Color samples from the manufacturer's standard line (or custom color samples if specified) to be used in selecting or approving colors for the project.

Field samples and mock-ups constructed on the project site establish standards ensuring work can be judged. Includes assemblies or portions of assemblies that are to be incorporated into the project and those that will be removed at conclusion of the work.

#### SD-05 Design Data

Design calculations, mix designs, analyses or other data pertaining to a part of work.

#### SD-06 Test Reports

Report signed by authorized official of testing laboratory that a material, product or system identical to the material, product or system to be provided has been tested in accord with specified requirements. Unless specified in another section, testing must have been within three years of date of contract award for the project.

Report that includes findings of a test required to be performed on an actual portion of the work or prototype prepared for the project before shipment to job site.

Report that includes finding of a test made at the job site or on sample taken from the job site, on portion of work during or after installation.

Investigation reports

Daily logs and checklists

Final acceptance test and operational test procedure

#### SD-07 Certificates

Statements printed on the manufacturer's letterhead and signed by responsible officials of manufacturer of product, system or material attesting that the product, system, or material meets specification requirements. Must be dated after award of project contract and clearly name the project.

Document required of Contractor, or of a manufacturer, supplier, installer or Subcontractor through Contractor. The document purpose is to further promote the orderly progression of a portion of the work by documenting procedures, acceptability of methods, or personnel qualifications.

Confined space entry permits

Text of posted operating instructions

The Contractor shall maintain submitted certifications and licenses

through the duration of the project. Any certification or license that expires prior to or during the execution of the affected Definable Feature of Work (DFOW) shall be resubmitted to the Contracting Officer for approval.

#### SD-08 Manufacturer's Instructions

Preprinted material describing installation of a product, system or material, including special notices and (SDS) concerning impedances, hazards and safety precautions.

#### SD-09 Manufacturer's Field Reports

Documentation of the testing and verification actions taken by manufacturer's representative at the job site, in the vicinity of the job site, or on a sample taken from the job site, on a portion of the work, during or after installation, to confirm compliance with manufacturer's standards or instructions. The documentation must be signed by an authorized official of a testing laboratory or agency and state the test results; and indicate whether the material, product, or system has passed or failed the test.

Factory test reports.

#### SD-10 Operation and Maintenance Data

Data provided by the manufacturer, or the system provider, including manufacturer's help and product line documentation, necessary to maintain and install equipment, for operating and maintenance use by facility personnel.

Data required by operating and maintenance personnel for the safe and efficient operation, maintenance and repair of the item.

Data incorporated in an operations and maintenance manual or control system.

eOMSI submittals per Section 01 78 24.00 22 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI).

#### SD-11 Closeout Submittals

Documentation to record compliance with technical or administrative requirements or to establish an administrative mechanism.

Special requirements necessary to properly close out a construction contract. For example, Record Drawings, as-built drawings, and Red Zone documents. Also, submittal requirements necessary to properly close out a major phase of construction on a multi-phase contract.

##### 1.1.2 Approving Authority

Office or designated person authorized to approve the submittal.

##### 1.1.3 Work

As used in this section, on-site and off-site construction required by contract documents, including labor necessary to produce submittals, construction, materials, products, equipment, and systems incorporated or

to be incorporated in such construction. In exception, excludes work to produce SD-01 submittals.

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-01 Preconstruction Submittals

Submittal Register; G

## 1.3 SUBMITTAL CLASSIFICATION

### 1.3.1 Government Approved (G)

Government approval is required for extensions of design, critical materials, variations, equipment whose compatibility with the entire system must be checked, and other items as designated by the Government.

Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, submittals are considered to be "shop drawings."

Submittal review period for Government Approved (G) submittals is outlined in paragraph SCHEDULING with additional clarifications in paragraph VARIATIONS.

### 1.3.2 For Information Only

Submittals not requiring Government approval will be for information only. Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, they are not considered to be "shop drawings."

## 1.4 FORWARDING SUBMITTALS REQUIRING GOVERNMENT APPROVAL

As soon as practicable after award of contract, and before procurement or fabrication, provide through NAVFAC's Electronic Construction and Facility Support Contract Management System (eCMS) submittals required in the technical sections of this specification, including shop drawings, product data and samples.

### 1.4.1 O&M Data

Submit data specified for a given item within 30 calendar days after the item is delivered to the contract site.

In the event the Contractor fails to deliver O&M data within the time limits specified, the Contracting Officer may withhold from progress payments 50 percent of the price of the items to which such O&M data apply.

### 1.4.2 Submittals Reserved for NAVFAC MIDLANT Approval

As an exception to the standard submittal procedure for Government Approval, submit the following to NAVFAC MIDLANT, unless otherwise noted:

- a. All fire protection and fire alarm system submittals are to be reviewed by:

NAVFAC MIDLANT C145 Fire Protection  
ATTN: NAVFAC FPE  
Bldg Z-140, RM 126  
9742 Maryland Avenue  
Norfolk, VA 23511

- b. Section 01 91 00.15 20 TOTAL BUILDING COMMISSIONING: SD-06  
Commissioning Plan, Certificate of Readiness, and Commissioning Report  
submittals
- c. Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC: All  
submittals

## 1.5 PREPARATION

### 1.5.1 Transmittal Form

Transmittal forms are no longer required when using eCMS since approval status is tracked on the submittal. Transmittal forms can be attached to submittals if required by the Contracting Officer. Refer to paragraph QUALITY ASSURANCE in Section 01 31 23.13 20 ELECTRONIC CONSTRUCTION AND FACILITY SUPPORT CONTRACT MANAGEMENT SYSTEM (eCMS) for additional instructions on proper transmittal of submittals.

### 1.5.2 Identifying Submittals

The Contractor's Quality Control Manager must prepare, review and stamp submittals, including those provided by a subcontractor, before submittal to the Government.

Identify submittals, except sample installations and sample panels, with the following information permanently adhered to or noted on each separate component of each submittal. Mark each copy of each submittal identically, with the following:

- a. Project title and location
- b. Construction contract number
- c. Dates of the drawings and revisions
- d. Name, address, and telephone number of Subcontractor, supplier, manufacturer, and any other Subcontractor associated with the submittal.
- e. Section number of the specification by which submittal is required
- f. Submittal description (SD) number of each component of submittal
- g. For a resubmission, add alphabetic suffix on submittal description, for example, submittal 18 would become 18A, to indicate resubmission
- h. Product identification and location in project.

### 1.5.3 Submittal Format

#### 1.5.3.1 Format of SD-01 Preconstruction Submittals

When the submittal includes a document that is to be used in the project, or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

#### 1.5.3.2 Format for SD-02 Shop Drawings

Provide shop drawings not less than 8 1/2 by 11 inches nor more than 30 by 42 inches, except for full-size patterns or templates. Prepare drawings to accurate size, with scale indicated, unless another form is required. Ensure drawings are suitable for reproduction and of a quality to produce clear, distinct lines and letters, with dark lines on a white background.

- a. Include the nameplate data, size, and capacity on drawings. Also include applicable federal, military, industry, and technical society publication references.
- b. Dimension drawings, except diagrams and schematic drawings. Prepare drawings demonstrating interface with other trades to scale. Use the same unit of measure for shop drawings as indicated on the contract drawings. Identify materials and products for work shown.

Submit an electronic copy of drawings in PDF format and native electronic format.

##### 1.5.3.2.1 Drawing Identification

Include on each drawing the drawing title, number, date, and revision numbers and dates, in addition to information required in paragraph IDENTIFYING SUBMITTALS.

Number drawings in a logical sequence. Each drawing is to bear the number of the submittal in a uniform location next to the title block. Place the Government contract number in the margin, immediately below the title block, for each drawing.

Reserve a blank space, no smaller than 3 by 4 inches on the right-hand side of each sheet for the Government disposition stamp.

#### 1.5.3.3 Format of SD-03 Product Data

Present product data submittals for each section. Include a table of contents, listing the page and catalog item numbers for product data.

Indicate, by prominent notation, each product that is being submitted; indicate the specification section number and paragraph number to which it pertains.

##### 1.5.3.3.1 Product Information

Supplement product data with material prepared for the project to satisfy the submittal requirements where product data does not exist. Identify this material as developed specifically for the project, with information

and format as required for submission of SD-07 Certificates.

Provide product data in units used in the Contract documents. Where product data are included in preprinted catalogs with another unit, submit the dimensions in contract document units, on a separate sheet.

#### 1.5.3.3.2 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

#### 1.5.3.3.3 Data Submission

Collect required data submittals for each specific material, product, unit of work, or system into a single submittal that is marked for choices, options, and portions applicable to the submittal. Mark each copy of the product data identically. Partial submittals will not be accepted for expedition of the construction effort.

Submit the manufacturer's instructions before installation.

#### 1.5.3.4 Format of SD-04 Samples

##### 1.5.3.4.1 Sample Characteristics

Furnish samples in the following sizes, unless otherwise specified or unless the manufacturer has prepackaged samples of approximately the same size as specified:

- a. Sample of Equipment or Device: Full size.
- b. Sample of Materials Less Than 2 by 3 inches: Built up to 8 1/2 by 11 inches.
- c. Sample of Materials Exceeding 8 1/2 by 11 inches: Cut down to 8 1/2 by 11 inches and adequate to indicate color, texture, and material variations.
- d. Sample of Linear Devices or Materials: 10 inch length or length to be supplied, if less than 10 inches. Examples of linear devices or materials are conduit and handrails.
- e. Sample Volume of Nonsolid Materials: Pint. Examples of nonsolid materials are sand and paint.
- f. Color Selection Samples: 2 by 4 inches. Where samples are specified for selection of color, finish, pattern, or texture, submit the full set of available choices for the material or product specified. Sizes

and quantities of samples are to represent their respective standard unit.

g. Sample Panel: 4 by 4 feet.

h. Sample Installation: 100 square feet.

#### 1.5.3.4.2 Sample Incorporation

Reusable Samples: Incorporate returned samples into work only if so specified or indicated. Incorporated samples are to be in undamaged condition at the time of use.

Recording of Sample Installation: Note and preserve the notation of any area constituting a sample installation, but remove the notation at the final clean-up of the project.

#### 1.5.3.4.3 Comparison Sample

Samples Showing Range of Variation: Where variations in color, finish, pattern, or texture are unavoidable due to nature of the materials, submit sets of samples of not less than three units showing extremes and middle of range. Mark each unit to describe its relation to the range of the variation.

When color, texture, or pattern is specified by naming a particular manufacturer and style, include one sample of that manufacturer and style, for comparison.

#### 1.5.3.5 Format of SD-05 Design Data

Provide design data and certificates on 8 1/2 by 11 inch paper.

#### 1.5.3.6 Format of SD-06 Test Reports

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

#### 1.5.3.7 Format of SD-07 Certificates

Provide design data and certificates on 8 1/2 by 11 inch paper.

#### 1.5.3.8 Format of SD-08 Manufacturer's Instructions

Present manufacturer's instructions submittals for each section. Include the manufacturer's name, trade name, place of manufacture, and catalog model or number on product data. Also include applicable federal, military, industry, and technical-society publication references. If supplemental information is needed to clarify the manufacturer's data, submit it as specified for SD-07 Certificates.

Submit the manufacturer's instructions before installation.

##### 1.5.3.8.1 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM),



National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

#### 1.5.3.9 Format of SD-09 Manufacturer's Field Reports

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

#### 1.5.3.10 Format of SD-10 Operation and Maintenance Data (O&M)

Comply with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA for O&M Data format.

#### 1.5.3.11 Format of SD-11 Closeout Submittals

When the submittal includes a document that is to be used in the project or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

### 1.5.4 Source Drawings for Shop Drawings

#### 1.5.4.1 Source Drawings

One (1) set of electronic drawings, in the source format, will be provided by the Government at the Preconstruction Meeting.

#### 1.5.4.2 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction data for the referenced project. Any other use or reuse is at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim, and waives to the fullest extent permitted by law any claim or cause of action of any nature against the Government, its agents, or its subconsultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities, or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic source drawing files are not construction documents. Differences may exist between the source drawing files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic source drawing files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. The Contractor

is responsible for determining if any conflict exists. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished source drawing files, the signed and sealed construction documents govern. Use of these source drawing files does not relieve the Contractor of the duty to fully comply with the contract documents, including and without limitation the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction data related to this contract, remove all previous indication of ownership (seals, logos, signatures, initials and dates).

#### 1.5.5 Electronic File Format

Provide submittals in electronic format, with the exception of material samples required for SD-04 Samples items. Compile the submittal file as a single, complete document, and also separately include the native files which were used to create the PDF. The files should include the original digital files used to create the submittal. Name the electronic submittal file specifically according to its contents; refer to the file naming conventions as outlined in Section 01 31 23.13 20 ELECTRONIC CONSTRUCTION AND FACILITY SUPPORT CONTRACT MANAGEMENT SYSTEM (eCMS). Electronic files must be of sufficient quality that all information is legible. Use PDF as the electronic format, unless otherwise specified or directed by the Contracting Officer. Generate PDF files from original documents with bookmarks so that the text included in the PDF file is searchable and can be copied. If documents are scanned, optical character resolution (OCR) routines are required. Index and bookmark files exceeding 30 pages to allow efficient navigation of the file. When required, the electronic file must include a valid electronic signature or a scan of a signature.

All electronic submittal documents must be transferred, shared, and managed through NAVFAC's Electronic Construction and Facility Support Contract Management System (eCMS). Refer to Section 01 31 23.13 20 ELECTRONIC CONSTRUCTION AND FACILITY SUPPORT CONTRACT MANAGEMENT SYSTEM.

#### 1.6 QUANTITY OF SUBMITTALS

##### 1.6.1 Number of SD-04 Samples

- a. Submit two samples, or two sets of samples showing the range of variation, of each required item. One approved sample or set of samples will be retained by the approving authority and one will be returned to the Contractor.
- b. Submit one sample panel or provide one sample installation where directed. Include components listed in the technical section or as directed.
- c. Submit one sample installation, where directed.
- d. Submit one sample of nonsolid materials.

#### 1.7 INFORMATION ONLY SUBMITTALS

Submittals without a "G" designation must be certified by the QC manager and submitted to the Contracting Officer for information-only. Provide information-only submittals to the Contracting Officer a minimum of 14 calendar days prior to the Preparatory Meeting for the associated

Definable Feature of Work (DFOW). Approval from the Contracting Officer is not required on information only submittals. The Contracting Officer will mark "receipt acknowledged" on submittals for information only and will return only the transmittal cover sheet to the Contractor. Normally, submittals for information only will not be returned. However, the Government reserves the right to return unsatisfactory submittals and require the Contractor to resubmit any item found not to comply with the contract. This does not relieve the Contractor from the obligation to furnish material conforming to the plans and specifications; will not prevent the Contracting Officer from requiring removal and replacement of nonconforming material incorporated in the work; and does not relieve the Contractor of the requirement to furnish samples for testing by the Government laboratory or for check testing by the Government in those instances where the technical specifications so prescribe.

## 1.8 PROJECT SUBMITTAL REGISTER

A sample project Submittal Register showing items of equipment and materials for when submittals are required by the specifications is provided as Appendix A to this specification.

### 1.8.1 Submittal Management

Prepare and maintain the Submittal Register as the work progresses. Do not change data that is output in columns (c), (d), (e), and (f) as delivered by the Government; retain data that is output in columns (a), (g), (h), and (i) as approved. As an attachment, provide a Submittal Register showing items of equipment and materials for which submittals are required by the specifications. This list may not be all-inclusive and additional submittals may be required.

Column (c): Lists specification section in which submittal is required.

Column (d): Lists each submittal description (SD Number. and type, e.g., SD-02 Shop Drawings) required in each specification section.

Column (e): Lists one principal paragraph in each specification section where a material or product is specified. This listing is only to facilitate locating submitted requirements. Do not consider entries in column (e) as limiting the project requirements.

Column (f): Lists the approving authority for each submittal.

Thereafter, the Contractor is to track all submittals by maintaining a complete list, including completion of all data columns and all dates on which submittals are received by and returned by the Government.

### 1.8.2 Preconstruction Use of Submittal Register

Submit the Submittal Register no later than at the Preconstruction Meeting. Verify that all submittals required for the project are listed and add missing submittals. Coordinate and complete the following fields on the register:

Column (a) Activity Number: Activity number from the project schedule.

Column (g) Contractor Submit Date: Scheduled date for the approving authority to receive submittals.

Column (h) Contractor Approval Date: Date that Contractor needs approval of submittal.

Column (i) Contractor Material: Date that Contractor needs material delivered to Contractor control.

#### 1.8.3 Contractor Use of Submittal Register

Update the following fields with each submittal throughout the contract.

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (j) Action Code (k): Date of action used to record Contractor's review when forwarding submittals to QC.

Column (l) Date submittal transmitted.

Column (q) Date approval was received.

#### 1.8.4 Approving Authority Use of Submittal Register

Update the following fields:

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (l) Date submittal was received.

Column (m) through (p) Dates of review actions.

Column (q) Date of return to Contractor.

#### 1.8.5 Action Codes

##### 1.8.5.1 Government Review Action Codes

"A" - "Approved as submitted"

"AN" - "Approved as noted"

"RR" - "Disapproved as submitted"; "Completed"

"NR" - "Not Reviewed"

"RA" - "Receipt Acknowledged"

#### 1.8.6 Delivery of Copies

Submit an updated electronic copy of the Submittal Register to the Contracting Officer with each invoice request. Provide an updated Submittal Register monthly regardless of whether an invoice is submitted.

## 1.9 VARIATIONS

Variations from contract requirements require Contracting Officer approval pursuant to contract Clause FAR 52.236-21 Specifications and Drawings for Construction, and will be considered where advantageous to the Government.

### 1.9.1 Considering Variations

Discussion of variations with the Contracting Officer before submission will help ensure that functional and quality requirements are met and may minimize rejections and resubmittals. For variations that include design changes or some material or product substitutions, the Government may require an evaluation and analysis by a licensed professional engineer hired by the contractor.

Specifically point out variations from contract requirements in a transmittal letter. Failure to point out variations may cause the Government to require rejection and removal of such work at no additional cost to the Government.

### 1.9.2 Warranting that Variations are Compatible

When delivering a variation for approval, the Contractor warrants that this contract has been reviewed to establish that the variation, if incorporated, will be compatible with other elements of work.

### 1.9.3 Review Schedule Extension

In addition to the normal submittal review period, a period of 7 calendar days will be allowed for the Government to consider submittals with variations.

## 1.10 SCHEDULING

Contracting Officer review will be completed within 21 calendar days from the date of submission, except as specified otherwise.

Schedule and submit concurrently product data and shop drawings covering component items forming a system or items that are interrelated. Submit pertinent certifications at the same time. No delay damages or time extensions will be allowed for time lost in late submittals. Allow an additional 21 calendar days for review and approval of submittals for HVAC control systems.

- a. Coordinate scheduling, sequencing, preparing, and processing of submittals with performance of work so that work will not be delayed by submittal processing. The Contractor is responsible for additional time required for Government reviews resulting from required resubmittals. The review period for each resubmittal is the same as for the initial submittal.
- b. Submittals required by the contract documents are listed on the Submittal Register. If a submittal is listed in the Submittal Register but does not pertain to the contract work, the Contractor is to include the submittal in the register and annotate it "N/A" with a brief explanation. Approval by the Contracting Officer does not relieve the Contractor of supplying submittals required by the contract documents but that have been omitted from the register or marked "N/A."

- c. Resubmit the Submittal Register and annotate it monthly with actual submission and approval dates. When all items on the register have been fully approved, no further resubmittal of the Submittal Register is required.
- d. Except as specified otherwise, allow a review period, beginning with receipt by the approving authority, that includes at least 14 calendar days for submittals for QC manager approval and 21 calendar days for submittals where the Contracting Officer is the approving authority. The period of review for submittals with Contracting Officer approval begins when the Government receives the submittal from the QC organization.
- e. For submittals requiring review by a Government fire protection engineer, allow a review period, beginning when the Government receives the submittal from the QC organization, of 30 calendar days for return of the submittal to the Contractor.

#### 1.10.1 Reviewing, Certifying, and Approving Authority

The QC Manager is responsible for reviewing all submittals and certifying that they are in compliance with contract requirements. The approving authority on submittals is the QC Manager unless otherwise specified. At each "Submittal" paragraph in individual specification sections, a notation "G" following a submittal item indicates that the Contracting Officer is the approving authority for that submittal item. Provide an additional copy of the submittal to the Government Approving authority, per the requirements of paragraph FORWARDING SUBMITTALS REQUIRING GOVERNMENT APPROVAL.

#### 1.10.2 Constraints

Conform to provisions of this section, unless explicitly stated otherwise for submittals listed or specified in this contract.

Submit complete submittals for each definable feature of the work. At the same time, submit components of definable features that are interrelated as a system.

When acceptability of a submittal is dependent on conditions, items, or materials included in separate subsequent submittals, the submittal will be returned without review.

Approval of a separate material, product, or component does not imply approval of the assembly in which the item functions.

#### 1.10.3 QC Organization Responsibilities

- a. Review submittals for conformance with project design concepts and compliance with contract documents.
- b. Process submittals based on the approving authority indicated in the Submittal Register.
  - (1) When the QC manager is the approving authority, take appropriate action on the submittal from the possible actions defined in paragraph APPROVED SUBMITTALS.

(2) When the Contracting Officer is the approving authority or when variation has been proposed, forward the submittal to the Government, along with a certifying statement, or return the submittal marked "not reviewed" or "revise and resubmit" as appropriate. The QC organization's review of the submittal determines the appropriate action.

c. Ensure that material is clearly legible.

d. Stamp each sheet of each submittal with a QC certifying statement or an approving statement, except that data submitted in a bound volume or on one sheet printed on two sides may be stamped on the front of the first sheet only.

(1) When the approving authority is the Contracting Officer, the QC organization will certify submittals forwarded to the Contracting Officer with the following certifying statement:

"I hereby certify that the (equipment) (material) (article) shown and marked in this submittal is that proposed to be incorporated with Contract Number \_\_\_\_\_ is in compliance with the contract drawings and specification, can be installed in the allocated spaces, and is submitted for Government approval.

Certified by Submittal Reviewer \_\_\_\_\_, Date \_\_\_\_\_  
(Signature when applicable)

Certified by QC Manager \_\_\_\_\_, Date \_\_\_\_\_"  
(Signature)

(2) When approving authority is the QC manager, the QC manager will use the following approval statement when returning submittals to the Contractor as "Approved" or "Approved as Noted."

"I hereby certify that the (material) (equipment) (article) shown and marked in this submittal and proposed to be incorporated with Contract Number \_\_\_\_\_ is in compliance with the contract drawings and specification, can be installed in the allocated spaces, and is approved for use.

Certified by Submittal Reviewer \_\_\_\_\_, Date \_\_\_\_\_  
(Signature when applicable)

Approved by QC Manager \_\_\_\_\_, Date \_\_\_\_\_"  
(Signature)

e. Sign the certifying statement or approval statement. The QC organization member designated in the approved QC plan is the person signing certifying statements. The use of original ink for signatures is required. Stamped signatures are not acceptable.

f. Update the Submittal Register as submittal actions occur, and maintain the submittal register at the project site until final acceptance of all work by the Contracting Officer.

g. Retain a copy of approved submittals and approved samples at the project site.

h. For "S" submittals, provide a copy of the approved submittal to the

Government Approving authority.

#### 1.11 GOVERNMENT APPROVING AUTHORITY

When the approving authority is the Contracting Officer, the Government will:

- a. Note the date on which the submittal was received from the QC manager.
- b. Review submittals for approval within the scheduling period specified in paragraph SCHEDULING and only for conformance with project design concepts and compliance with contract documents.
- c. Identify returned submittals with one of the actions defined in paragraph REVIEW NOTATIONS and with comments and markings appropriate for the action indicated.

Upon completion of review of submittals requiring Government approval, stamp and date submittals. One copy of the submittal will be retained by the Contracting Officer and one copy of the submittal will be returned to the Contractor.

##### 1.11.1 Review Notations

Submittals will be returned to the Contractor with the following notations:

- a. Submittals marked "approved" or "accepted" authorize proceeding with the work covered.
- b. Submittals marked "approved as noted" or "approved, except as noted, resubmittal not required," authorize proceeding with the work covered provided that the Contractor takes no exception to the corrections.
- c. Submittals marked "not approved," "disapproved," or "revise and resubmit" indicate incomplete submittal or noncompliance with the contract requirements or design concept. Resubmit with appropriate changes. Do not proceed with work for this item until the resubmittal is approved.
- d. Submittals marked "not reviewed" indicate that the submittal has been previously reviewed and approved, is not required, does not have evidence of being reviewed and approved by Contractor, or is not complete. A submittal marked "not reviewed" will be returned with an explanation of the reason it is not reviewed. Resubmit submittals returned for lack of review by Contractor or for being incomplete, with appropriate action, coordination, or change.
- e. Submittals marked "receipt acknowledged" indicate that submittals have been received by the Government. This applies only to "information-only submittals" as previously defined.

#### 1.12 DISAPPROVED SUBMITTALS

Make corrections required by the Contracting Officer. If the Contractor considers any correction or notation on the returned submittals to constitute a change to the contract drawings or specifications, give notice to the Contracting Officer as required under the FAR clause titled CHANGES. The Contractor is responsible for the dimensions and design of connection details and the construction of work. Failure to point out



variations may cause the Government to require rejection and removal of such work at the Contractor's expense.

If changes are necessary to submittals, make such revisions and resubmit in accordance with the procedures above. No item of work requiring a submittal change is to be accomplished until the changed submittals are approved.

All revisions made to a previous submittal shall be clouded and specifically indicated in the transmittal to clearly delineate and identify what has changed. The Government has the right to reject revised submittals that do not cloud or delineate changes from previous submittals.

#### 1.13 APPROVED SUBMITTALS

The Contracting Officer's approval of submittals is not to be construed as a complete check, and indicates only that the general method of construction, materials, detailing, and other information are satisfactory.

Approval or acceptance by the Government for a submittal does not relieve the Contractor of the responsibility for meeting the contract requirements or for any error that may exist, because under the Quality Control (QC) requirements of this contract, the Contractor is responsible for ensuring information contained within each submittal accurately conforms with the requirements of the contract documents.

After submittals have been approved or accepted by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.

#### 1.14 APPROVED SAMPLES

Approval of a sample is only for the characteristics or use named in such approval and is not be construed to change or modify any contract requirements. Before submitting samples, provide assurance that the materials or equipment will be available in quantities required in the project. No change or substitution will be permitted after a sample has been approved.

Match the approved samples for materials and equipment incorporated in the work. If requested, approved samples, including those that may be damaged in testing, will be returned to the Contractor, at its expense, upon completion of the contract. Unapproved samples will also be returned to the Contractor at its expense, if so requested.

Failure of any materials to pass the specified tests will be sufficient cause for refusal to consider, under this contract, any further samples of the same brand or make as that material. The Government reserves the right to disapprove any material or equipment that has previously proved unsatisfactory in service.

Samples of various materials or equipment delivered on the site or in place may be taken by the Contracting Officer for testing. Samples failing to meet contract requirements will automatically void previous approvals. Replace such materials or equipment to meet contract requirements.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

# SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

B72 Resiliency Improvement

CONTRACTOR

| ACTIVITY<br>NO | TRANSMITTAL<br>NO | SPEC<br>SECT   | DESCRIPTION<br>ITEM SUBMITTED     | PARAGRAPH | GOVT<br>CLASSIFICATION | CONTRACTOR:<br>SCHEDULE DATES |                          |                          | CONTRACTOR<br>ACTION |                      | DATE FWD<br>TO APPR<br>AUTH/ | APPROVING AUTHORITY              |                                  |                |                      | MAILED<br>TO<br>CONTR/       | REMARKS |
|----------------|-------------------|----------------|-----------------------------------|-----------|------------------------|-------------------------------|--------------------------|--------------------------|----------------------|----------------------|------------------------------|----------------------------------|----------------------------------|----------------|----------------------|------------------------------|---------|
|                |                   |                |                                   |           |                        | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION | DATE RCD<br>FROM<br>CONTR    | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION | DATE RCD<br>FRM APPR<br>AUTH |         |
| (a)            | (b)               | (c)            | (d)                               | (e)       | (f)                    | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                          | (m)                              | (n)                              | (o)            | (p)                  | (q)                          | (r)     |
|                |                   | 01 11 00.00 22 | SD-01 Preconstruction Submittals  |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Salvage Plan                      | 1.6       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   | 01 14 00.00 22 | SD-01 Preconstruction Submittals  |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | List of Contact Personnel         | 1.4.1.1   | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Flooding Mitigation Strategy Plan | 1.14      |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | SD-06 Test Reports                |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | PWD ME Contractor Weekend         | 1.6       |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Worklist                          |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | CRITICAL INFRASTRUCTURE           | 1.18.1    | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | RISK ANALYSIS (RA) AND RISK       |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | MITIGATION PLAN (RMP)             |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Outage Request                    | 1.8.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   | 01 20 00.00 22 | SD-01 Preconstruction Submittals  |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Earned Value Report               | 1.3       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   | 01 30 00.00 22 | SD-01 Preconstruction Submittals  |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Facility Turnover Planning        | 1.12.1    | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Meeting Agenda and Red Zone       |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | (RZ) Checklist-POAM               |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | NAVFAC PWD ME Internal            | 1.12.3    | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Service Requirements List         |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   | 01 31 23.13 20 | SD-01 Preconstruction Submittals  |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | List of Contractor's Personnel    | 1.4.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   | 01 32 17.00 20 | SD-01 Preconstruction Submittals  |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Baseline NAS                      | 1.2.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Designated Project Scheduler      | 1.9       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | SD-07 Certificates                |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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CONTRACT NO.

TITLE AND LOCATION  
B72 Resiliency Improvement

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| ACTIVITY<br>NO | TRANSMITTAL<br>NO | SPEC<br>SECT   | DESCRIPTION<br>ITEM SUBMITTED                                        | PARAGRAPH | GOVT<br>CLASSIFICATION | CONTRACTOR:<br>SCHEDULE DATES |                          |                          | CONTRACTOR<br>ACTION |                      | DATE FWD<br>TO APPR<br>AUTH/ | APPROVING AUTHORITY              |                                  |                |                      | MAILED<br>TO CONTR/<br><br>DATE RCD<br>FRM APPR<br>AUTH | REMARKS |
|----------------|-------------------|----------------|----------------------------------------------------------------------|-----------|------------------------|-------------------------------|--------------------------|--------------------------|----------------------|----------------------|------------------------------|----------------------------------|----------------------------------|----------------|----------------------|---------------------------------------------------------|---------|
|                |                   |                |                                                                      |           |                        | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION | DATE RCD<br>FROM<br>CONTR    | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                         |         |
| (a)            | (b)               | (c)            | (d)                                                                  | (e)       | (f)                    | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                          | (m)                              | (n)                              | (o)            | (p)                  | (q)                                                     | (r)     |
|                |                   | 01 32 17.00 20 | Three-Week Look Ahead<br>Schedule                                    | 1.3       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Monthly Network Analysis<br>Updates                                  | 1.4.1     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | SD-11 Closeout Submittals<br>As-Built Schedule                       | 1.4.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|                |                   | 01 33 00       | SD-01 Preconstruction Submittals<br>Submittal Register               | 1.8       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|                |                   | 01 35 26.00 22 | SD-01 Preconstruction Submittals<br>Accident Prevention Plan         | 1.9       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Indoor Air Quality (IAQ)<br>Management Plan                          | 1.16      | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | SD-06 Test Reports<br>Monthly Exposure Reports                       | 1.5       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Accident Reports                                                     | 1.13.2    | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | LHE Inspection Reports                                               | 1.13.3    | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | SD-07 Certificates<br>Contractor Safety Self-Evaluation<br>Checklist | 1.6       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Standard Lift Plan                                                   | 1.9.3.1   | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Standard Lift Plan                                                   | 1.13.4    | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Critical Lift Plan                                                   | 1.9.3.2   | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Activity Hazard Analysis (AHA)                                       | 1.10      | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Hot Work Permit                                                      | 1.14      | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Certificate of Compliance                                            | 1.13.4    | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|                |                   | 01 44 00.00 22 | SD-01 Preconstruction Submittals                                     |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |

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| B72 Resiliency Improvement |                   |                |                                  |           |                                 |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
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| (a)                        | (b)               | (c)            | (d)                              | (e)       | (f)                             | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                          | (m)                       | (n)                              | (o)                              | (p)            | (q)                                                     | (r)     |
|                            |                   | 01 44 00.00 22 | Archaeologist Qualifications     | 1.4       | G                               |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Archaeological Coordinator       | 1.5       |                                 |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Archaeological Monitoring Plan   | 1.6       | G                               |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | SD-06 Test Reports               |           |                                 |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Archaeological Resource Report   | 3.1       | G                               |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Archaeological Resource Report   | 3.1       | G                               |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Monthly Progress Reports         | 3.2.2     | G                               |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Archaeological Monitoring Report | 3.2.2     | G                               |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | - Draft                          |           |                                 |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Archaeological Monitoring Report | 3.2.2     | G                               |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | - Final Draft                    |           |                                 |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Archaeological Monitoring Report | 3.2.2     | G                               |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | - Final                          |           |                                 |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Curated Artifacts                | 3.3       | G                               |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   | 01 45 00.00 22 | SD-01 Preconstruction Submittals |           |                                 |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Contractor Quality Control (CQC) | 1.5.2     | G                               |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Plan                             |           |                                 |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | SD-06 Test Reports               |           |                                 |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Verification Statement           | 1.13.3    |                                 |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Contractor Quality Control       | 1.8       |                                 |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Reports                          |           |                                 |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Contractor Production Reports    | 1.8       |                                 |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Preparatory Phase Checklists     | 1.8       |                                 |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Initial Phase Checklists         | 1.8       |                                 |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Testing Plan And Log             | 1.8       |                                 |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |
|                            |                   |                | Rework Items List                | 1.8       |                                 |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                                                         |         |

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|             |                |                |                                             |           |                     | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                              | DATE RCD<br>FROM<br>CONTR | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION   |         |
| (a)         | (b)            | (c)            | (d)                                         | (e)       | (f)                 | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                          | (m)                       | (n)                              | (o)                              | (p)            | (q)                    | (r)     |
|             |                | 01 45 00.00 22 | SD-07 Certificates                          |           |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Certificate Of Readiness                    | 1.6.4.1   |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                | 01 45 35       | SD-01 Preconstruction Submittals            |           |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | SIOR Letter of Acceptance                   | 3.1.1     | G                   |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Project Manual                              | 3.1.1     | G                   |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | SD-06 Test Reports                          |           |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Special Inspection Daily Reports            | 3.1.1     |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Special Inspection Daily Reports            | 3.1.3     |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Special Inspection Biweekly Reports         | 3.1.1     |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | SD-07 Certificates                          |           |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | AISC Certified Steel Fabricator             | 2.1       |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Steel Truss Plant Quality Assurance Program | 2.1       |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Wood Truss Plant Quality Assurance Program  | 2.1       |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | AC472 Accreditation                         | 2.1       |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Certificate of Compliance                   | 2.1       |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Special Inspector of Record                 | 1.5       | G                   |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Special Inspector of Record                 | 1.5.17    | G                   |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Special Inspector                           | 1.5       | G                   |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | SD-11 Closeout Submittals                   |           |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Interim Report                              | 3.1.1     | G                   |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Comprehensive Final Report                  | 3.1.1     | G                   |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Final Report                                |           | G                   |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                | 01 50 00.00 22 | SD-01 Preconstruction Submittals            |           |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |

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|             |                | 01 50 00.00 22 | Construction Site Plan           | 1.3       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Traffic control plan             | 3.4.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Contractor Computer              | 1.6.1.4   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Cybersecurity Compliance         |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Statements                       |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Contractor Temporary Network     | 1.6.6     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Cybersecurity Compliance         |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Statements                       |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Employee Parking Plan            | 1.7       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-03 Product Data               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Backflow Preventers              | 1.4       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-06 Test Reports               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Backflow Preventer Tests         | 2.6       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Backflow Preventer Tests         | 3.5       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-07 Certificates               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Backflow Tester                  | 1.4.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Backflow Preventers              | 1.4       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 01 57 19.00 22 | SD-01 Preconstruction Submittals |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Preconstruction Survey           | 1.5.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Solid Waste Management Plan      | 3.4.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Regulatory Notifications         | 1.5.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Environmental Management Plan    | 3.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | (EMP)                            |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Dirt and Dust Control Plan       | 3.13.1    | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Contractor Hazardous Material    | 3.8.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Inventory Log                    |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|                |                   |                |                                                               |           |                        | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                                                            | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                         |         |
| (a)            | (b)               | (c)            | (d)                                                           | (e)       | (f)                    | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                                                        | (m)                              | (n)                              | (o)            | (p)                  | (q)                                                     | (r)     |
|                |                   | 01 57 19.00 22 | Stormwater Management/Erosion and Sedimentation Control Plan  | 3.2.1     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Spill Prevention, Control, and Countermeasures (SPCC) Plan    | 3.1       | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Environmental Manager Qualifications                          | 1.5.4     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Employee Training Records                                     | 1.5.5     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Noise Mitigation Plan                                         | 3.16      | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | SD-06 Test Reports                                            |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Laboratory Analysis                                           | 3.12.3.2  | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Disposal Requirements                                         | 3.14.2    | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Erosion and Sediment Control                                  | 3.2.3     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Inspection and Corrective Action                              |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Monthly Solid Waste Disposal Report                           | 3.4.2     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | SD-07 Certificates                                            |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | ECATTS Certificate Of Completion                              | 1.4.1     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Erosion And Sediment Control Inspector                        | 1.5.5     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | SD-11 Closeout Submittals                                     |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Stormwater Management and Erosion Control Compliance Notebook | 3.2.1     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Waste Determination Documentation                             | 3.5       | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |



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|                |                   | 01 57 19.00 22 | Disposal Documentation for Hazardous and Regulated Waste | 3.12.3.5  | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |                | Contractor Hazardous Material Inventory Log              | 3.8.1     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |                | Hazardous Waste/Debris Management                        | 3.12.3    | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |                | Regulatory Notifications                                 | 1.5.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |                | Asbestos Free Certification Form                         | 3.12.3.3  | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |                | Assembled Employee Training Records                      | 1.5.5     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |                | Sales Documentation                                      | 3.4.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   | 01 58 00       | SD-02 Shop Drawings                                      |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |                | Preliminary Drawing Indicating Layout And Text Content   | 1.3.1     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   | 01 74 19       | SD-01 Preconstruction Submittals                         |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |                | Construction Waste Management Plan                       | 1.6       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |                | SD-11 Closeout Submittals                                |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |                | Final Construction Waste Diversion Report                | 1.8       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   | 01 78 00.00 22 | SD-03 Product Data                                       |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |                | Warranty Management Plan                                 | 1.6.1     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |                | Warranty Tags                                            | 1.6.4     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |                | Final Cleaning                                           | 3.3       |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |                | Spare Parts Data                                         | 1.5       |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |                | SD-08 Manufacturer's Instructions                        |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |

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|             |                | 01 78 00.00 22 | Instructions                                                | 1.6.1     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |                | SD-10 Operation and Maintenance Data                        |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |                | Operation and Maintenance Manuals                           | 1.6.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |                | SD-11 Closeout Submittals                                   |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |                | As-Built Drawings                                           | 3.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |                | Record Drawings                                             | 3.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |                | Record Drawings                                             | 3.2       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |                | Warranted Equipment And Materials                           | 1.6.1     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |                | Interim DD FORM 1354                                        | 3.4.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |                | List Of Installed Building Equipment Items For DD Form 1354 | 3.4.2     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                | 01 78 23       | SD-10 Operation and Maintenance Data                        |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |                | Training Plan                                               | 3.1.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |                | Training Outline                                            | 3.1.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |                | Training Content                                            | 3.1.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |                | eOMSI Facility Data Workbook                                | 1.4       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |                | Operation And Maintenance Manual, Progress Submittal        | 3.2.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |                | Operation And Maintenance Manual, Prefinal Submittal        | 3.2.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |

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|             |                | 01 78 23       | Operation And Maintenance Manual, Final Submittal | 3.2.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-11 Closeout Submittals                         |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Training Video Recording                          | 3.1.4     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Validation of Training Completion                 | 3.1.6     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Record Drawings And Utility Systems               | 1.6.6.7   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 01 78 24.00 22 | SD-11 Closeout Submittals                         |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | eOMSI Facility Data Workbook, Progress Submittal  | 1.3.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | eOMSI Facility Data Workbook, Prefinal Submittal  | 1.3.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | eOMSI Facility Data Workbook, Final Submittal     | 1.3.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 01 91 00.15 20 | SD-06 Test Reports                                |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Interim Construction Phase                        | 3.4       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Commissioning Plan                                |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Final Construction Phase                          | 3.4       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Commissioning Plan                                |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Initial Commissioning Report                      | 3.8       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Issues Log                                        | 1.12      | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Completed Construction                            | 3.5       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Observation Checklists                            |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-07 Certificates                                |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Commissioning Firm                                | 1.11      | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Certificate Of Readiness                          | 1.13      | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|             |                | 01 91 00.15 20 | SD-10 Operation and Maintenance Data               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Systems Manual                                     | 3.7.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-11 Closeout Submittals                          |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Final Commissioning Report                         | 3.8       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Updated Final Commissioning Report                 | 3.9       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 02 41 00       | SD-01 Preconstruction Submittals                   |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Demolition Plan                                    | 1.2.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Existing Conditions                                | 1.10      |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-07 Certificates                                 |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Notification                                       | 1.7       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-11 Closeout Submittals                          |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Receipts                                           | 3.2.4     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 02 42 91       | SD-01 Preconstruction Submittals                   |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Work Plan                                          | 1.2       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 02 81 00       | SD-03 Product Data                                 |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Packaging Notifications                            | 2.1.1     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Hazardous Waste Management Plan                    | 3.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Onsite Hazardous Waste Management                  | 3.2       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Notices of Non-Compliance and Notices of Violation | 3.3       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-06 Test Reports                                 |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Recordkeeping                                      | 3.7       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|                |                   | 02 81 00       | Exception Report                 | 3.7       | G                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Spill Response                   | 3.8       |                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | SD-07 Certificates               |           |                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Transportation and Disposal      | 1.4.1     | G                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Coordinator                      |           |                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Training                         | 1.4.2     | G                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Certification                    | 1.4.3     |                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Shipping Documents and           | 3.3.3     | G                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Packagings Certification         |           |                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Security Plan                    | 3.3.4     |                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Certificates of Disposal         | 3.3.5     |                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Waste Minimization               | 3.6       |                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   | 02 82 00.00 22 | SD-03 Product Data               |           |                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Amended Water                    | 1.2.2     | G                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Safety Data Sheets (SDS) for All | 1.3.9     | G                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Materials                        |           |                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Respirators                      | 3.1.2.1   | G                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Local Exhaust Equipment          | 3.1.7     | G                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Pressure Differential Automatic  | 3.1.7     | G                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Recording Instrument             |           |                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Vacuums                          | 3.1.8     | G                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Glovebags                        | 3.1.10    | G                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | SD-06 Test Reports               |           |                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Air Sampling Results             | 1.5.5     | G                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Pressure Differential Recordings | 1.5.6     | G                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | for Local Exhaust System         |           |                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |

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|-------------|----------------|----------------|--------------------------------------------------------|-----------|---------------------------------|-------------------------------|--------------------|--------------------|-------------------|----------------|------------------------|----------------------------|----------------------------|-------------|----------------|------------------------|---------|
|             |                |                |                                                        |           |                                 | SUBMIT                        | APPROVAL NEEDED BY | MATERIAL NEEDED BY | ACTION CODE       | DATE OF ACTION | DATE RCD FROM CONTR    | DATE FWD TO OTHER REVIEWER | DATE RCD FROM OTH REVIEWER | ACTION CODE | DATE OF ACTION | DATE RCD FRM APPR AUTH |         |
| (a)         | (b)            | (c)            | (d)                                                    | (e)       | (f)                             | (g)                           | (h)                | (i)                | (j)               | (k)            | (l)                    | (m)                        | (n)                        | (o)         | (p)            | (q)                    | (r)     |
|             |                | 02 82 00.00 22 | Clearance Sampling                                     | 3.2.14.5  | G                               |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Asbestos Disposal Quantity Report                      | 3.3.3.2   | G                               |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | SD-07 Certificates                                     |           |                                 |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Employee Training                                      | 1.3.4     | G                               |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Notifications                                          | 1.3.5     | G                               |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Respiratory Protection Program                         | 1.3.7     | G                               |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Asbestos Hazard Abatement Plan                         | 1.3.10    | G                               |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Asbestos Hazard Abatement Plan Checklist               | 1.3.10    | G                               |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Testing Laboratory                                     | 1.3.11    | G                               |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Medical Certification                                  | 1.3.14    | G                               |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Private Qualified Person Documentation                 | 1.5.1     | G                               |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Competent Person                                       | 1.5.2     | G                               |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Worker's License                                       | 1.5.3     | G                               |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Contractor's License                                   | 1.5.4     | G                               |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Federal, State or Local Citations on Previous Projects | 1.5.7     | G                               |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Equipment Used to Contain Airborne Asbestos Fibers     | 3.1       | G                               |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Water Filtration Equipment                             | 3.1.3.3   | G                               |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Vacuums                                                | 3.1.8     | G                               |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Ventilation Systems                                    | 3.1.8     | G                               |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | SD-11 Closeout Submittals                              |           |                                 |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Permits                                                | 1.3.5     |                                 |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |

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| ACTIVITY<br>NO | TRANSMITTAL<br>NO | SPEC<br>SECT   | DESCRIPTION<br>ITEM SUBMITTED                                | PARAGRAPH | GOVT<br>CLASS<br>/ FIC<br>/ REVIEW<br>/ WORK | CONTRACTOR:<br>SCHEDULE DATES |                          |                          | CONTRACTOR<br>ACTION |                      | DATE FWD<br>TO APPR<br>AUTH/<br><br>DATE RCD<br>FROM<br>CONTR | APPROVING AUTHORITY              |                                  |                |                      | MAILED<br>TO<br>CONTR/<br><br>DATE RCD<br>FRM APPR<br>AUTH | REMARKS |
|----------------|-------------------|----------------|--------------------------------------------------------------|-----------|----------------------------------------------|-------------------------------|--------------------------|--------------------------|----------------------|----------------------|---------------------------------------------------------------|----------------------------------|----------------------------------|----------------|----------------------|------------------------------------------------------------|---------|
|                |                   |                |                                                              |           |                                              | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                                                               | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                            |         |
| (a)            | (b)               | (c)            | (d)                                                          | (e)       | (f)                                          | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                                                           | (m)                              | (n)                              | (o)            | (p)                  | (q)                                                        | (r)     |
|                |                   | 02 82 00.00 22 | Licenses                                                     | 1.3.5     | G                                            |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Notifications                                                | 1.3.5     | G                                            |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Respirator Program Records                                   | 1.3.7.1   | G                                            |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Rental Equipment                                             | 1.7.1     | G                                            |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   | 02 83 00.00 22 | SD-01 Preconstruction Submittals                             |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Statement of Competent Person                                | 1.5.1.1   | G                                            |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Assignment, and Competent<br>Person                          |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Statement of CIH Assignment,<br>and CIH Qualifications       | 1.5.1.1   | G                                            |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Statement Identifying and                                    | 1.5.1.1   | G                                            |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Designating Roles and                                        |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Responsibilities for CP, CIH,<br>SSHO, and all Other Parties |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Involved in Heavy Metals<br>Management                       |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Training Certification                                       | 1.5.1.2   | G                                            |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Occupational and Environmental<br>Assessment Data Report     | 1.5.2.4   | G                                            |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Medical Examinations                                         | 1.5.2.5   | G                                            |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Lead, Cadmium, and Chromium<br>(VI) Waste Management Plan    | 1.5.2.9   | G                                            |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Licenses, Permits and                                        | 1.5.4     | G                                            |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Notifications                                                |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Occupant Protection Plan                                     | 1.5.2.1   | G                                            |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Occupant Protection Plan                                     | 1.5.5     | G                                            |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |

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|             |                |                |                                                            |           |                     | SUBMIT                        | APPROVAL NEEDED BY | MATERIAL NEEDED BY | ACTION CODE       | DATE OF ACTION | DATE RCD FROM CONTR    | DATE FWD TO OTHER REVIEWER | DATE RCD FROM OTH REVIEWER | ACTION CODE | DATE OF ACTION | DATE RCD FRM APPR AUTH |         |
| (a)         | (b)            | (c)            | (d)                                                        | (e)       | (f)                 | (g)                           | (h)                | (i)                | (j)               | (k)            | (l)                    | (m)                        | (n)                        | (o)         | (p)            | (q)                    | (r)     |
|             |                | 02 83 00.00 22 | Lead, Cadmium, and Chromium (VI) Compliance Plan           | 1.5.2.2   | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Lead, Cadmium, and Chromium (VI) Compliance Plan Checklist | 1.5.2.3   | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Waste Characterization Sampling Plan                       | 1.5.2.10  | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Analytical Results                                         | 1.3.3     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | SD-03 Product Data                                         |           |                     |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Respirators                                                | 1.6.1     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Vacuum Filters                                             | 1.6.4     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Negative Air Pressure System                               | 1.6.7     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Materials and Equipment                                    | 2.1       | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Expendable Supplies                                        | 2.1.1     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Local Exhaust Equipment                                    | 3.1.1.5   | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Local Exhaust Equipment                                    | 3.1.1.9   | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Pressure Differential Automatic Recording Instrument       | 3.1.1.5   | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Pressure Differential Automatic Recording Instrument       | 3.1.1.9   | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Pressure Differential Log                                  | 3.1.1.6   | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | SD-06 Test Reports                                         |           |                     |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Sampling and Analysis                                      | 1.3.3     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Occupational and Environmental Assessment Data Report      | 1.5.2.4   | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Sampling Results                                           | 1.5.2.4   | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |



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|-------------|----------------|----------------|--------------------------------------------------------------|-----------|---------------------|-------------------------------|--------------------------|--------------------------|----------------------|----------------------|------------------------------|---------------------------|----------------------------------|----------------------------------|----------------|------------------------|---------|
|             |                |                |                                                              |           |                     | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                              | DATE RCD<br>FROM<br>CONTR | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION   |         |
| (a)         | (b)            | (c)            | (d)                                                          | (e)       | (f)                 | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                          | (m)                       | (n)                              | (o)                              | (p)            | (q)                    | (r)     |
|             |                | 02 83 00.00 22 | Pressure Differential Recordings<br>For Local Exhaust System | 1.5.3     | G                   |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | SD-07 Certificates                                           |           |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Competent Person                                             | 1.5.1.1   |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Testing Laboratory                                           | 1.5.1.3   | G                   |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Occupant Notification                                        | 3.1.1.1   | G                   |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Clearance Certification                                      | 3.5.1.1   | G                   |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | SD-11 Closeout Submittals                                    |           |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Turn-In Documents or Weight<br>Tickets                       | 3.5.2.1   | G                   |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                | 03 30 00       | SD-01 Preconstruction Submittals                             |           |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Concrete Curing Plan                                         | 1.6.3.1   |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Quality Control Plan                                         | 1.6.6     | G                   |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Quality Control Personnel<br>Certifications                  | 1.6.7     | G                   |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Quality Control Organizational<br>Chart                      | 1.6.7     |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Laboratory Accreditation                                     | 1.6.9     | G                   |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Form Removal Schedule                                        | 1.6.2.1   | G                   |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Maturity Method Data                                         | 3.3.10    |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | SD-02 Shop Drawings                                          |           |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Formwork                                                     | 1.6.2.1   |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Reinforcing Steel                                            | 1.6.2.2   | G                   |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | SD-03 Product Data                                           |           |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Joint Sealants                                               | 2.4.5     |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|             |                |                | Joint Filler                                                 | 2.4.4     |                     |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |

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| ACTIVITY<br>NO | TRANSMITTAL<br>NO | SPEC<br>SECT | DESCRIPTION<br>ITEM SUBMITTED      | PARAGRAPH | GOVT<br>CLASSIFICATION<br>OR<br>REVIEW | CONTRACTOR:<br>SCHEDULE DATES |                          |                          | CONTRACTOR<br>ACTION |                      | DATE FWD<br>TO APPR<br>AUTH/ | APPROVING AUTHORITY              |                                  |                |                      | MAILED<br>TO<br>CONTR/       | REMARKS |
|----------------|-------------------|--------------|------------------------------------|-----------|----------------------------------------|-------------------------------|--------------------------|--------------------------|----------------------|----------------------|------------------------------|----------------------------------|----------------------------------|----------------|----------------------|------------------------------|---------|
|                |                   |              |                                    |           |                                        | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION | DATE RCD<br>FROM<br>CONTR    | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION | DATE RCD<br>FRM APPR<br>AUTH |         |
| (a)            | (b)               | (c)          | (d)                                | (e)       | (f)                                    | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                          | (m)                              | (n)                              | (o)            | (p)                  | (q)                          | (r)     |
|                |                   | 03 30 00     | Formwork Materials                 | 2.1       |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Cementitious Materials             | 2.3.1     |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Vapor Retarder and Vapor Barrier   | 2.4.6     |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Concrete Curing Materials          | 2.4.1     |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Reinforcement                      | 2.6       |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Admixtures                         | 2.3.4     |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Waterstops                         | 2.2.2     |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Biodegradable Form Release Agent   | 2.2.3     |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Pumping Concrete                   | 1.6.3.2   |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Finishing Plan                     | 1.6.3.4   |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Nonshrink Grout                    | 2.4.2     |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-05 Design Data                  |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Concrete Mix Design                | 1.6.1.1   | G                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-06 Test Reports                 |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Concrete Mix Design                | 1.6.1.1   | G                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Fly Ash                            | 1.6.4.1   |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Pozzolan                           | 1.6.4.1   |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Slag Cement                        | 1.6.4.2   |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Aggregates                         | 1.6.4.3   |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Tolerance Report                   | 3.10.2.1  |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Compressive Strength Tests         | 3.14.3.3  | G                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Unit Weight of Structural Concrete | 3.14.3.5  |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Air Content                        | 3.14.3.4  |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Slump Tests                        | 3.14.3.1  |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|             |                | 03 30 00       | Water                                                      | 2.3.2     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-07 Certificates                                         |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Reinforcing Bars                                           | 2.6.1     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Silica Fume Manufacturer's Representative                  | 1.6.3.3   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | VOC Content for Form Release                               | 1.6.3.5   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Agents, Curing Compounds, and Concrete Penetrating Sealers |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Safety Data Sheets                                         | 1.6.3.6   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Field Testing Technician and Testing Agency                | 1.6.7.2   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-08 Manufacturer's Instructions                          |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Liquid Chemical Floor Hardeners and Sealers                | 2.4.3.1   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Joint Sealants                                             | 2.4.5     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Curing Compound                                            | 2.4.1     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 03 42 13.00 10 | SD-01 Preconstruction Submittals                           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Quality Control Procedures                                 | 1.3.2.2   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-02 Shop Drawings                                        |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Standard Precast Units                                     | 2.1.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Custom-Made Precast Units                                  | 2.1.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Special Finishes                                           | 3.2.4.3   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-03 Product Data                                         |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Standard Precast Units                                     | 2.1.1     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Proprietary Precast Units                                  | 2.1.3     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Embedded Items                                             | 3.1.3     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|             |                | 03 42 13.00 10 | Accessories                      | 2.2.5     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-05 Design Data                |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Design Calculations              | 2.1.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Concrete Mix Proportions         | 2.1.5.1   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-06 Test Reports               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Test Reports                     | 1.3.2.4   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-07 Certificates               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Quality Control Procedures       | 1.3.2.2   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 04 03 00       | SD-01 Preconstruction Submittals |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Quality Control Plan             | 1.5.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Project Training Program         | 1.5.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Qualifications                   | 1.5.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-02 Shop Drawings              |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Photographic Documentation       | 1.5.5     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Structural Upgrades              | 3.5.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-03 Product Data               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Qualifications                   | 1.5.2     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Cleaning and Restoration         | 1.5.6     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Methods                          |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Cleaning Materials               | 2.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Biocides                         | 2.1.4     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Replacement Mortar And Stucco    | 2.2.2.2   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Mortar Mix                       | 2.4       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Water Repellents Infiltration    | 3.4.8.3   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Stone Consolidants               | 3.4.8.4   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-04 Samples                    |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|                            |                   | 04 03 00     | Mock-ups                                   | 1.5.7.1   | G                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | SD-05 Design Data                          |           |                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Structural Upgrades                        | 3.5.1     | G                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | SD-06 Test Reports                         |           |                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Testing and Matching                       | 2.2.2.1   |                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Existing Sealants for Asbestos<br>and PCBs | 3.5.2     |                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | SD-07 Certificates                         |           |                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Repair Materials                           | 2.2       |                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   | 04 20 00     | SD-02 Shop Drawings                        |           |                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Cut CMU                                    | 3.3.5.1   | G                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Detail Drawings                            | 3.4.1.1   | G                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | SD-03 Product Data                         |           |                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Hot Weather Procedures                     | 1.5.1     | G                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Cold Weather Procedures                    | 1.5.2     | G                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Clay or Shale Brick                        | 2.2.2     | G                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Cement                                     | 2.2.3.2.1 | G                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Cementitious Materials                     | 2.4.1.1   | G                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Insulation                                 | 2.6.8     | G                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | SD-04 Samples                              |           |                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Mock-Up Panel                              | 1.3.1.1   | G                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Clay or Shale Brick                        | 2.2.2     | G                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Concrete Masonry Units (CMU)               | 2.2.3.2   | G                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Dimension Stone Units                      | 2.2.4     | G                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Admixtures for Masonry Mortar              | 2.4.1.4   | G                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              |                                            |           |                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |

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|             |                | 04 20 00  | Anchors, Ties, and Bar Positioners       | 2.6.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Joint Reinforcement                      | 2.6.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Clay Masonry Expansion-Joint Materials   | 2.6.6     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Insulation                               | 2.6.8     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-05 Design Data                        |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Masonry Compressive Strength             | 2.1.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Fire-Rated Concrete Masonry Units        | 2.2.3.3   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Bracing Calculations                     | 3.2.5     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-06 Test Reports                       |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Efflorescence Test                       | 2.2.2.1.3 |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Fire-Rated Concrete Masonry Units        | 2.2.3.3   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Field Testing of Mortar                  | 3.6.1.1   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Field Testing of Grout                   | 3.6.1.2   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Prism Tests                              | 3.6.1.4   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-07 Certificates                       |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Special Masonry Inspector Qualifications | 1.3.2     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Clay or Shale Brick                      | 2.2.2     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Concrete Masonry Units (CMU)             | 2.2.3.2   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Cementitious Materials                   | 2.4.1.1   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Admixtures for Masonry Mortar            | 2.4.1.4   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Admixtures for Grout                     | 2.4.2.2   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |

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|                |                   | 04 20 00     | Anchors, Ties, and Bar Positioners                     | 2.6.2     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Joint Reinforcement                                    | 2.6.3     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Insulation                                             | 2.6.8     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Insulation                                             | 2.6.8     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-08 Manufacturer's Instructions                      |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Admixtures for Masonry Mortar                          | 2.4.1.4   |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Admixtures for Grout                                   | 2.4.2.2   |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-10 Operation and Maintenance Data                   |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Take-Back Program                                      | 3.8       |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   | 05 05 20     | SD-01 Preconstruction Submittals                       |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Installer Qualifications                               | 1.5.1.1   | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Post-Installed Anchor Special Inspector Qualifications | 1.5.1.2   | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-03 Product Data                                     |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Adhesive Anchor System in Concrete                     | 2.1.1.3   | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Non-Shrink, Non-Metallic Grout                         | 3.1.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-06 Test Reports                                     |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Post-Installed Anchor Special Inspections Report       | 3.3.3     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-07 Certificates                                     |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Post-Installed Anchor Certification                    | 2.1.1.1   | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-08 Manufacturer's Instructions                      |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |

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|             |                | 05 05 20    | Manufacturer's Printed Installation Instructions   | 2.1.1.2   | G                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Manufacturer's Material Safety Data Sheets         | 1.5       |                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 05 05 23.16 | SD-01 Preconstruction Submittals                   |           |                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Welding Quality Assurance Plan                     | 3.2       | G                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | SD-03 Product Data                                 |           |                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Welding Procedure Qualifications                   | 1.3       | G                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Welder, Welding Operator, and Tacker Qualification | 1.3.3     |                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Previous Qualifications                            | 1.3.2     |                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Welding Electrodes and Rods                        | 2.2       |                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | SD-06 Test Reports                                 |           |                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Nondestructive Testing                             | 3.3       |                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Weld Inspection Log                                | 3.2       |                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | SD-07 Certificates                                 |           |                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Certified Welding Procedure Specifications (WPS)   | 1.3.1     |                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Certified Brazing Procedure Specifications (BPS)   | 1.3.1     |                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Certified Procedure Qualification Records (PQR)    | 1.3.1     |                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Certified Welder Performance Qualifications (WPQ)  | 1.3.1     |                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Certified Brazer Performance Qualifications (BPQ)  | 1.3.1     |                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |



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|                |                   | 05 05 23.16  | Certified Welding Inspector      | 1.3.4     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Nondestructive Testing Personnel | 1.3.4     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   | 05 12 00     | SD-01 Preconstruction Submittals |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Erection and Erection Bracing    | 1.4.1.1   | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Drawings                         |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-02 Shop Drawings              |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Fabrication Drawings             | 1.4.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-03 Product Data               |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Shop Primer                      | 2.5.2     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Welding Electrodes and Rods      | 2.4.1     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Direct Tension Indicator Washers | 2.3.1.3   |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Recycled Content for Structural  | 2.2.1     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Steel                            |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-05 Design Data                |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Design Calculations for Steel    | 1.4.3     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Connections                      |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-06 Test Reports               |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Class B Coating                  | 2.5.2     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Bolts, Nuts, and Washers         | 2.3       |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Weld Inspection Reports          | 3.7.1.2   |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Direct Tension Indicator Washer  | 3.7.2.1   |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Inspection Reports               |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Bolt Testing Reports             | 3.7.3.1   |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-07 Certificates               |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Steel                            | 2.2       |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Bolts, Nuts, and Washers         | 2.3       |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|             |                | 05 12 00  | AISC Structural Steel Fabricator<br>Quality Certification | 1.3       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | AISC Structural Steel Erector<br>Quality Certification    | 1.3       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Welding Procedures and<br>Qualifications                  | 1.4.4.1   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Welding Electrodes and Rods                               | 2.4.1     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Certified Welding Inspector                               | 3.7.1.1   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | NDT Technician                                            | 3.7.1.2   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Welding Procedure Specifications<br>(WPS)                 | 3.4       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 05 40 00  | SD-02 Shop Drawings                                       |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Framing Components                                        | 1.6.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-03 Product Data                                        |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Studs, Joists                                             | 2.1       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Recycled Content of Steel<br>Products                     | 2.1       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-05 Design Data                                         |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Metal Framing Calculations                                | 1.6.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-07 Certificates                                        |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Load-Bearing Cold-Formed Metal<br>Framing                 | 1.4       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Welds                                                     | 3.2.1     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 05 50 13  | SD-02 Shop Drawings                                       |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Structural Steel Door Frames                              | 2.8       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Cover Plates and Frames                                   | 2.3       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|                |                   | 05 50 13     | Expansion Joint Covers                 | 3.8       | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Installation Drawings                  | 2.7       | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Bollards                               | 2.5       | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Angles and Plates                      | 2.7       | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Roof Hatches                           | 2.9       | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-03 Product Data                     |           |                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Cover Plates and Frames                | 2.3       | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Expansion Joint Covers                 | 3.8       | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Floor Gratings                         | 2.4       | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Structural Steel Door Frames           | 2.8       | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Roof Hatches                           | 2.9       | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Downspout Terminations                 | 2.6       | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-04 Samples                          |           |                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Expansion Joint Covers                 | 3.8       |                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-07 Certificates                     |           |                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Certified Mill                         | 2.1       | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   | 05 51 00     | SD-02 Shop Drawings                    |           |                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Iron and Steel Hardware                | 2.1       | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Steel Shapes, Plates, Bars, and Strips | 2.1       | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Metal Stair System                     | 2.2.1     | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Fabrication Drawings                   | 1.4.1     | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-03 Product Data                     |           |                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Masonry Anchorage Devices              | 2.3.4     | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Protective Coating                     | 2.2.4     | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Steel Pan Stairs                       | 2.2.2     | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |

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|             |                | 05 51 00  | Steel Stairs                                 | 2.3.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Steel Stairs, Circular                       | 2.3.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-05 Design Data                            |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Design Calculations For Steel Connections    | 1.4.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-06 Test Reports                           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Mill Test Reports                            | 2.5       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-07 Certificates                           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Welding Procedures                           | 1.3.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Welder Qualification                         | 1.3.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-08 Manufacturer's Instructions            |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Protective Coating                           | 2.2.4     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Masonry Anchorage Devices                    | 2.3.4     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 05 51 33  | SD-02 Shop Drawings                          |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Ladders                                      | 2.3       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-03 Product Data                           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Ladders                                      | 2.3       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Ladder Safety Devices                        | 2.3.1     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-07 Certificates                           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Fabricator Certification for Ladder Assembly | 1.3       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 05 52 00  | SD-02 Shop Drawings                          |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Fabrication Drawings                         | 1.2.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Fabrication Drawings                         | 1.5.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Iron and Steel Hardware                      | 3.2       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|                |                   | 05 52 00     | Steel Shapes, Plates, Bars and Strips     | 3.2       | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-03 Product Data                        |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Concrete Inserts                          | 2.2.7     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Masonry Anchorage Devices                 | 2.2.8     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Protective Coating                        | 2.1.3     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Steel Railings and Handrails              | 2.2.10    | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Anchorage and Fastening Systems           | 1.2.1     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-05 Design Data                         |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Design Calculations For Steel Connections | 1.5.2     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-06 Test Reports                        |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Welding Inspections                       | 1.4.1     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Mill Test Reports                         | 2.3       | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-07 Certificates                        |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Welder Qualification                      | 1.4.2     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-08 Manufacturer's Instructions         |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Installation Instructions                 | 3.2       |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   | 06 10 00     | SD-02 Shop Drawings                       |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Fabricated Structural Members             | 1.9.1     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Modifications of Structural Members       | 1.9.2     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Nailing Strips                            | 2.2.1     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-03 Product Data                        |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Underlayment                              | 2.4       |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |

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|             |                | 06 10 00  | Plastic Lumber                                 | 2.1.2     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Fire-retardant Treatment                       | 1.8       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Structural-use and OSB Panels                  | 1.4.4     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Structural-use and OSB Panels                  | 2.3.2.2   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Oriented Strand Board                          | 2.3       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Adhesives                                      | 2.5.4     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-05 Design Data                              |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Modifications of Structural Members            | 1.9.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-06 Test Reports                             |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Preservative-treated                           | 1.4.5     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-07 Certificates                             |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Preservative Treatment                         | 1.7       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-10 Operation and Maintenance Data           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Plastic                                        | 1.4.8     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 06 20 00  | SD-02 Shop Drawings                            |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Detail Drawings Indicating All Wood Assemblies | 1.3       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-03 Product Data                             |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Wood Products                                  | 2.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Countertops                                    |           | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Engineered                                     |           | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Treated Wood Products                          | 1.4       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Soffits                                        | 2.2       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Fascias and Trim                               | 2.3       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|                |                   | 06 20 00     | Hardware and Accessories          | 2.7       | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-04 Samples                     |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Samples                           | 1.5       | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-07 Certificates                |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Certificates of Grade             | 1.7.1.1   | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   | 07 13 53     | SD-03 Product Data                |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Manufacturer's Standard Details   | 1.3       | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Elastomeric Waterproofing Sheet   | 2.1       | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Material                          |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Protection Board                  | 2.3       | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Primers, Adhesives, and Mastics   | 1.4       | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Primers, Adhesives, and Mastics   | 2.1       | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-06 Test Reports                |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Elastomeric Waterproofing Sheet   | 2.1       | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Material                          |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Field Quality Control             | 3.6       | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Protective Covering               | 3.7       | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-07 Certificates                |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Elastomeric Waterproofing Sheet   | 2.1       |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Material                          |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Primers, Adhesives, and Mastics   | 1.4       | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Primers, Adhesives, and Mastics   | 2.1       | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Protective Coverings              | 1.4       | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Special Warranties                | 1.8       | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Special Warranties                | 1.8       | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-08 Manufacturer's Instructions |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |

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|                |                   | 07 13 53     | Primers, Adhesives, and Mastics   | 1.4       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Primers, Adhesives, and Mastics   | 2.1       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   | 07 19 00     | SD-03 Product Data                |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Water Repellents                  | 2.2       |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-06 Test Reports                |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Water Absorption                  | 1.3.2     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Accelerated Weathering            | 1.3.2     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Resistance to Chloride Ion        | 1.3.2     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Penetration                       |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Moisture Vapor Transmission       | 1.3.2     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Water Penetration and Leakage     | 1.3.2     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-07 Certificates                |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Manufacturer's Qualifications     | 1.3.1     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Applicator's Qualifications       | 1.3.1     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Evidence of Acceptable Variation  | 1.3.3     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Warranty                          | 1.12      |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-08 Manufacturer's Instructions |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Application                       | 3.4       |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Safety Data Sheets                | 1.7.1     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   | 07 21 13     | SD-02 Shop Drawings               |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Manufacturer's Standard Details   | 1.3       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-03 Product Data                |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Environmental Data for Materials  | 1.4       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Block or Board Insulation         | 2.1       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Vapor Retarder                    | 2.2       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Pressure Sensitive Tape           | 2.3       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |



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|                |                   | 07 21 13     | Protection Board or Coating         | 2.4       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Accessories                         | 2.5       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-07 Certificates                  |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Indoor Air Quality For Block Or     | 2.1.5     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Board Insulation                    |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-08 Manufacturer's Instructions   |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Insulation Installation and         | 3.3.1     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Handling                            |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Protection Board or Coating         | 3.4.5     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Installation                        |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Adhesive                            | 2.5.1     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-11 Closeout Submittals           |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Draft Guarantee                     | 1.7.1     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Final Guarantee                     | 1.7.1     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Draft Warranty                      | 1.7.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Final Warranty                      | 1.7.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   | 07 22 00     | SD-02 Shop Drawings                 |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Insulation Board Layout             | 1.3       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Verification of Existing Conditions | 1.3       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-03 Product Data                  |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Insulation                          | 2.1       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Cover Board                         | 1.4       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Fasteners                           | 2.4       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Moisture Control                    | 2.3       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Recycled Content For Insulation     | 2.1.2     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-05 Design Data                   |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |

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|                |                   | 07 22 00     | Wind Resistance                   | 1.7       | G                                  |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-06 Test Reports                |           |                                    |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Flame Spread Rating               | 1.8.1     | G                                  |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-07 Certificates                |           |                                    |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Installer Qualifications          | 1.6       | G                                  |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Certificates Of Compliance For    | 1.6       | G                                  |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Felt Materials                    |           |                                    |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Indoor Air Quality For Insulation | 2.1.3     |                                    |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Acceptable Foam Blowing Agents    | 2.1.4     |                                    |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-08 Manufacturer's Instructions |           |                                    |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Fasteners                         | 2.4       | G                                  |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Insulation                        | 2.1       | G                                  |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   | 07 27 10     | SD-02 Shop Drawings               |           |                                    |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Air Barrier System Shop           | 2.1       | G                                  |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Drawings                          |           |                                    |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-03 Product Data                |           |                                    |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Air Barrier System Product Data   | 2.1       | G                                  |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-04 Samples                     |           |                                    |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Mock-Up                           | 3.1.2     | G                                  |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Material Samples For Air Barrier  | 2.1       | G                                  |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | System                            |           |                                    |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-05 Design Data                 |           |                                    |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Design Data And Calculations      | 1.7       | G                                  |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | For The Air Barrier System        |           |                                    |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-06 Test Reports                |           |                                    |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Design Review Report              | 1.7       | G                                  |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |

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|             |                | 07 27 10    | Testing and Inspection                  | 3.1.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | SD-07 Certificates                      |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Air Barrier Inspector                   | 1.6       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 07 27 19.01 | SD-01 Preconstruction Submittals        |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Qualifications of Manufacturer          | 1.7.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Qualifications of Installer             | 1.7.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | SD-02 Shop Drawings                     |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Self-adhering Air Barrier Shop Drawings | 1.4.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | SD-03 Product Data                      |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Self-adhering Air Barrier               | 2.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Primers, Adhesives, and Mastics         | 2.2       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Safety Data Sheets                      | 1.4.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | SD-04 Samples                           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Self-adhering Air Barrier System Mockup | 1.4.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | SD-06 Test Reports                      |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Field Peel Adhesion Test                | 1.7.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Flame Propagation of Wall Assemblies    | 1.4.4     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Site Inspections and Testing            | 3.4.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | SD-08 Manufacturer's Instructions       |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Self-adhering Air Barrier               | 1.4.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Manufacturer's Printed Instructions     |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Primers, Adhesives, and Mastics         | 2.2       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 07 60 00    | SD-02 Shop Drawings                     |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|----------------|-------------------|--------------|---------------------------------------------------------|-----------|------------------------|-------------------------------|--------------------------|--------------------------|----------------------|----------------------|------------------------------------------------------------|----------------------------------|----------------------------------|----------------|----------------------|---------------------------------------------------------|---------|
|                |                   |              |                                                         |           |                        | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                                                            | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                         |         |
| (a)            | (b)               | (c)          | (d)                                                     | (e)       | (f)                    | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                                                        | (m)                              | (n)                              | (o)            | (p)                  | (q)                                                     | (r)     |
|                |                   | 07 60 00     | Exposed Sheet Metal                                     | 2.2.1     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Gutters                                                 | 3.1.14    | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Downspouts                                              | 3.1.15    | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Expansion Joints                                        | 3.1.20    | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Gravel Stops and fascia                                 | 2.2.1     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Splash Pans                                             | 3.1.17    | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Flashing for Roof Drains                                | 3.1.16    | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Base Flashing                                           | 3.1.10    | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Counterflashing                                         | 3.1.11    | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Flashing at Roof Penetrations<br>and Equipment Supports | 3.1.21    | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Reglets                                                 | 2.2.16    | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Drip Edges                                              | 3.1.13    | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Recycled Content                                        | 2.1       | S                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-03 Product Data                                      |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Cool Roof                                               | 2.2.13    | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-04 Samples                                           |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Finish Samples                                          | 1.4.2     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-07 Certificates                                      |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Warranty on Finishes                                    | 2.2.12.1  | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-08 Manufacturer's Instructions                       |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Instructions for Installation                           | 1.4.3     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Quality Control Plan                                    | 3.5       | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-10 Operation and Maintenance<br>Data                 |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Cleaning and Maintenance                                | 1.4.3     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |

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|             |                | 07 81 00  | SD-03 Product Data               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Fireproofing Material            | 3.3       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-04 Samples                    |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Spray-Applied Fireproofing       |           | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-06 Test Reports               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Fire Resistance Rating           | 1.2.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Field Tests                      | 3.6       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Evaluation Reports               | 1.2.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-07 Certificates               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Installer Qualifications         | 1.4.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Surface Preparation Report       | 3.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Manufacturer's Inspection Report | 3.5.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 07 92 00  | SD-03 Product Data               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Sealants                         | 2.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Primers                          | 2.2       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Bond Breakers                    | 2.3       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Backstops                        | 2.4       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Cleaning Solvents                | 2.5       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Field Adhesion                   | 3.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-07 Certificates               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Indoor Air Quality For Interior  | 2.1.1     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Sealants                         |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Indoor Air Quality For Interior  | 2.1.3     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Floor Joint Sealants             |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Indoor Air Quality For Interior  | 2.1.4     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Acoustical Sealants              |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|             |                | 08 11 13  | SD-02 Shop Drawings                         |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Doors                                       | 2.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Doors                                       | 2.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Frames                                      | 2.4       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Frames                                      | 2.4       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Accessories                                 | 2.2       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-03 Product Data                          |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Doors                                       | 2.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Recycled Content for Steel Door<br>Product  | 2.1       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Frames                                      | 2.4       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Recycled Content for Steel Frame<br>Product | 2.4       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Accessories                                 | 2.2       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                | 08 31 00  | SD-02 Shop Drawings                         |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Access Doors And Panels                     | 1.3       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-03 Product Data                          |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Access Doors And Panels                     | 1.3       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Hardware                                    | 1.3.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Accessories                                 | 2.2.8     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Recycled Content                            | 2.1       | S                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-04 Samples                               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Finishes                                    | 2.5       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-06 Test Reports                          |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Fire-rating(s) of Assemblies                | 1.3.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Acoustical Ratings of Assemblies            |           | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |

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| (a)            | (b)               | (c)          | (d)                             | (e)       | (f)                                           | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l) | (m)                                                           | (n)                              | (o)                              | (p)            | (q)                                                        | (r)     |
|                |                   | 08 33 23     | SD-02 Shop Drawings             |           |                                               |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Overhead Coiling Doors          | 2.2.1     | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Counterbalancing Mechanism      | 2.2.3     | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Manual Door Operators           | 2.2.4     | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Electric Door Operators         | 2.2.5     | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Bottom Bars                     | 2.2.1.3   | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Guides                          | 2.1.1.1   | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Mounting Brackets               | 2.2.3.1   | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Hood                            | 2.2.2.2   | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Installation Drawings           | 2.1.1.1   | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | SD-03 Product Data              |           |                                               |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Overhead Coiling Doors          | 2.2.1     | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Hardware                        | 2.2.2     | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Counterbalancing Mechanism      | 2.2.3     | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Manual Door Operators           | 2.2.4     | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Electric Door Operators         | 2.2.5     | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | SD-05 Design Data               |           |                                               |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Overhead Coiling Doors          | 2.2.1     | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Hardware                        | 2.2.2     | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Counterbalancing Mechanism      | 2.2.3     | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Manual Door Operators           | 2.2.4     | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Electric Door Operators         | 2.2.5     | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | SD-10 Operation and Maintenance |           |                                               |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Data                            |           |                                               |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Operation and Maintenance       | 1.3.2     | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Manuals                         |           |                                               |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |

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|                |                   | 08 33 23     | SD-11 Closeout Submittals           |           |                                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Warranty                            | 1.3.1     | G                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   | 08 71 00     | SD-02 Shop Drawings                 |           |                                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Manufacturer's Detail Drawings      | 1.3       | G                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Verification of Existing Conditions | 1.3       | G                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Hardware Schedule                   | 1.5       | G                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Keying System                       | 2.3.10    | G                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-03 Product Data                  |           |                                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Hardware Items                      | 2.3       | G                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-08 Manufacturer's Instructions   |           |                                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Installation                        | 3.1       |                                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-10 Operation and Maintenance     |           |                                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Data                                |           |                                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Hardware Schedule                   | 1.5       | G                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-11 Closeout Submittals           |           |                                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Key Bitting                         | 1.6.1     |                                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   | 08 91 00     | SD-02 Shop Drawings                 |           |                                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Wall Louvers                        | 1.4       |                                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-03 Product Data                  |           |                                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Metal Wall Louvers                  | 2.2       |                                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-04 Samples                       |           |                                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Wall Louver Samples                 | 1.5       | G                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Door Louver Samples                 | 1.5       | G                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   | 09 01 90.50  | SD-03 Product Data                  |           |                                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Work Plan                           |           | G                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Materials                           |           |                                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |



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|----------------|-------------------|--------------|------------------------------------|-----------|------------------------|-------------------------------|--------------------------|--------------------------|----------------------|----------------------|------------------------------|----------------------------------|----------------------------------|----------------|----------------------|------------------------------------------------------------|---------|
|                |                   |              |                                    |           |                        | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION | DATE RCD<br>FROM<br>CONTR    | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                            |         |
| (a)            | (b)               | (c)          | (d)                                | (e)       | (f)                    | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                          | (m)                              | (n)                              | (o)            | (p)                  | (q)                                                        | (r)     |
|                |                   | 09 01 90.50  | Qualifications                     |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-07 Certificates                 |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Work Plan                          |           | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   | 09 29 00     | SD-02 Shop Drawings                |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Control Joint Shop Drawings        | 3.2.3.1   |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-03 Product Data                 |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Gypsum Board                       | 2.1.1     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Moisture- and Mold-Resistant       | 2.1.2     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Gypsum Backing Board               |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Trim                               | 2.1.6.1   |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Accessories                        | 2.1.6     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | VOC Content of Joint Compound      | 2.1.3     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-07 Certificates                 |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Asbestos Free Materials            | 2.1       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Indoor Air Quality for Gypsum      | 2.1.1     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Board                              |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Indoor Air Quality for Non-aerosol | 2.1.5     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Adhesives                          |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Indoor Air Quality for Aerosol     | 2.1.5     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Adhesives                          |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-08 Manufacturer's Instructions  |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Safety Data Sheets                 | 2.1       |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-10 Operation and Maintenance    |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Data                               |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Manufacturer Maintenance           | 2.1       |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Instructions                       |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |

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|             |                |           |                                                                                             |           |                     | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION | DATE RCD<br>FROM<br>CONTR    | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                         |         |
| (a)         | (b)            | (c)       | (d)                                                                                         | (e)       | (f)                 | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                          | (m)                              | (n)                              | (o)            | (p)                  | (q)                                                     | (r)     |
|             |                | 09 77 23  | SD-03 Product Data                                                                          |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Product Specifications                                                                      |           | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Physical Properties                                                                         |           | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Product Data Sheets                                                                         |           | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-04 Samples                                                                               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Verification Samples                                                                        |           | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-05 Design Data                                                                           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Performance Data At Minimum,<br>25 Percent, 50 Percent, 75<br>Percent, And 100 Percent Load |           | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-07 Certificates                                                                          |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Certification                                                                               |           | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-08 Manufacturer's Instructions                                                           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Installation                                                                                |           | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-10 Operation and Maintenance<br>Data                                                     |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Maintenance and Cleaning<br>Instructions                                                    |           | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                | 09 90 00  | SD-02 Shop Drawings                                                                         |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Piping Identification                                                                       | 3.9       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-03 Product Data                                                                          |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Coating                                                                                     | 2.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Product Data Sheets                                                                         | 2.1       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-04 Samples                                                                               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Color                                                                                       | 2.3       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Textured Wall Coating System                                                                | 1.6.7     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |

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|                |                   |              |                                           |           |                                                         | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                                                               | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                            |         |
| (a)            | (b)               | (c)          | (d)                                       | (e)       | (f)                                                     | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                                                           | (m)                              | (n)                              | (o)            | (p)                  | (q)                                                        | (r)     |
|                |                   | 09 90 00     | SD-07 Certificates                        |           |                                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Qualification Testing                     | 1.6.6.2   | G                                                       |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Indoor Air Quality for Paints and Primers | 1.6.5     |                                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-08 Manufacturer's Instructions         |           |                                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Application Instructions                  | 3.2.1     |                                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Mixing                                    | 2.1       |                                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Manufacturer's Safety Data Sheets         | 1.8.1     |                                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-10 Operation and Maintenance Data      |           |                                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Coatings                                  | 2.1       | G                                                       |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   | 10 14 00.10  | SD-02 Shop Drawings                       |           |                                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Approved Detail Drawings                  | 3.1       | G                                                       |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-03 Product Data                        |           |                                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Installation                              | 3.1       |                                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Exterior Signage                          | 1.2       | G                                                       |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Wind Load Requirements                    | 1.2.1     |                                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-04 Samples                             |           |                                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Exterior Signage                          | 1.2       | G                                                       |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-10 Operation and Maintenance Data      |           |                                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Protection and Cleaning                   | 3.1.2     | G                                                       |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   | 10 14 00.20  | SD-02 Shop Drawings                       |           |                                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Detail Drawings                           | 1.3.2     | G                                                       |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-03 Product Data                        |           |                                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |

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|                |                   |              |                                                    |           |                                            | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION | DATE RCD<br>FROM<br>CONTR    | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                     |         |
| (a)            | (b)               | (c)          | (d)                                                | (e)       | (f)                                        | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                          | (m)                              | (n)                              | (o)            | (p)                  | (q)                                                 | (r)     |
|                |                   | 10 14 00.20  | Room Identification And Directional Signage System | 2.1       | G                                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Stair Signage                                      | 2.2       | G                                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Exit Door Tactile Sign                             | 2.3       | G                                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Door Tags                                          | 2.4       | G                                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-04 Samples                                      |           |                                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Interior Signage                                   | 1.3.1     | G                                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Room Identification And Directional Signage System | 2.1       | G                                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Stair Signage                                      | 2.2       | G                                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Exit Door Tactile Sign                             | 2.3       | G                                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Door Tags                                          | 2.4       | G                                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-10 Operation and Maintenance Data               |           |                                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Approved Manufacturer's Instructions               | 3.1       | G                                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Protection and Cleaning                            | 3.1.2     | G                                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   | 10 44 16     | SD-02 Shop Drawings                                |           |                                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Fire Extinguishers                                 | 2.1.1     | G                                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Accessories                                        | Part 2    | G                                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Cabinets                                           | Part 2    | G                                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Wall Brackets                                      | 2.2.1     | G                                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Schedule                                           | 1.5       | G                                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-03 Product Data                                 |           |                                            |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Fire Extinguishers                                 | 2.1.1     | G                                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Accessories                                        | Part 2    | G                                          |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |

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|                |                   | 10 44 16     | Cabinets                          | Part 2    | G                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Wall Brackets                     | 2.2.1     | G                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Replacement Parts List            | 3.2.1     | G                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-04 Samples                     |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Equipment Samples                 | 1.3.1     | G                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-07 Certificates                |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Fire Extinguishers Certifications | 2.1.1     | G                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Manufacturer's Warranty with      | 1.4       | G                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Inspection Tag                    |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   | 13 34 19     | SD-01 Preconstruction Submittals  |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Manufacturer's Qualifications     | 1.6.3     | G                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-02 Shop Drawings               |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Detail Drawings                   | 1.6.1     | G                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Erection Plan                     | 1.2.10    | G                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-03 Product Data                |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Manufacturer's Catalog Data       | 1.6.1     | G                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Recycled Content for Structural   | 2.1.1     |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Steel Shapes and Plates           |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Recycled Content for Steel Pipe   | 2.1.2     |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Recycled Content for Aluminum     |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Sheet Materials                   |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Recycled Content for Steel Sheet  | 2.4.1     |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Materials                         |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Recycled Content for Insulation   |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Materials                         |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-04 Samples                     |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|                |                   | 13 34 19     | Coil Stock                                             | 2.1.6     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Roof Panels                                            | 1.6.1     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Wall Panels                                            | 1.6.1     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Fasteners                                              | 2.5.2     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Metal Closure Strips                                   | 2.8.1     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Insulation                                             | 2.4.2     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Vapor Barrier                                          | 1.6.10    | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Manufacturer's Color Charts and<br>Chips               | 2.4.3     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-05 Design Data                                      |           |                                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Manufacturer's Descriptive and<br>Technical Literature | 1.6.1     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Manufacturer's Building Design<br>Analysis             | 1.6.1     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Lateral Force Calculations                             | 1.6.1     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-06 Test Reports                                     |           |                                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Test Reports                                           | 1.6.1     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Coatings and Base Metals                               | 1.6.1     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Factory Color Finish Performance<br>Requirements       | 1.6.1     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-07 Certificates                                     |           |                                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | System Components                                      | 1.6.1     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Coil Stock Certificates                                | 1.6.1     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Aluminized Steel Repair Paint                          | 1.6.1     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Galvanizing Repair Paint                               | 1.6.1     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Enamel Repair Paint                                    | 1.6.1     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |

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|             |                | 13 34 19  | Qualification of Manufacturer             | 1.6.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Qualification of Erector                  | 1.6.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-08 Manufacturer's Instructions         |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Installation of Roof and Wall panels      | 1.6.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Shipping, Handling, and Storage           | 1.7       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-11 Closeout Submittals                 |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Manufacturer's Warranty                   | 3.14.1    | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Contractor's Warranty for Installation    | 3.14.2    | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                | 21 13 13  | SD-01 Preconstruction Submittals          |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Qualified Fire Protection Engineer (QFPE) | 1.2.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Sprinkler System Designer                 | 1.2.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Sprinkler System Designer                 | 1.4.2.1   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Sprinkler System Installer                | 1.4.2.2   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-02 Shop Drawings                       |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Shop Drawing                              | 1.2.1.1   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-03 Product Data                        |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Fittings                                  | 2.2.1.2   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Valves                                    | 2.2.4     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Relief Valves                             | 2.7.4     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Sprinklers                                | 2.6       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Pipe Hangers and Supports                 | 2.2.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Sprinkler Alarm Switch                    | 2.3.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |

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|                |                   | 21 13 13     | Valve Supervisory (Tamper) Switch            | 2.3.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Fire Department Connection                   | 2.5       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Backflow Prevention Assembly                 | 2.4       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Air Vent                                     | 2.7.5     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Hose Valve                                   | 2.4.1     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Nameplates                                   | 2.1.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-05 Design Data                            |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Hydraulic Calculations                       | 1.2.1.2   | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-06 Test Reports                           |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Test Procedures                              | 3.6.1     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-07 Certificates                           |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Verification of Compliant Installation       | 3.6.2.1   | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Request for Government Final Test            | 3.6.2.2   | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-10 Operation and Maintenance Data         |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Operating and Maintenance (O&M) Instructions | 3.8       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Spare Parts                                  | 1.6       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-11 Closeout Submittals                    |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | As-built drawings                            | 3.8       |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   | 22 13 29     | SD-02 Shop Drawings                          |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Equipment Installation                       | 3.2       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-03 Product Data                           |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |



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|                |                   | 22 13 29     | Materials and Equipment                   | 2.1       |                                               |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Framed Instructions                       | 3.4       |                                               |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Spare Parts                               | 1.4       |                                               |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | SD-06 Test Reports                        |           |                                               |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Field Testing and Adjusting<br>Equipment  | 3.5       |                                               |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | SD-10 Operation and Maintenance<br>Data   |           |                                               |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Operating and Maintenance<br>Manuals      | 3.7       | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   | 23 05 15     | SD-01 Preconstruction Submittals          |           |                                               |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Material, Equipment, and Fixture<br>Lists | 1.2       | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | SD-02 Shop Drawings                       |           |                                               |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Record Drawings                           | 1.2       | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Connection Diagrams                       | 1.2       | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Coordination Drawings                     | 1.2       | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Fabrication Drawings                      | 1.2       | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Installation Drawings                     | 3.1       | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | SD-03 Product Data                        |           |                                               |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Pipe and Fittings                         | 2.1       | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Piping Specialties                        | 2.2       | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Emergency Eyewash and Shower<br>Valves    | 2.3       | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Miscellaneous Materials                   | 2.4       | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Supporting Elements                       | 2.5       | G                                             |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |

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|----------------------------|-------------------|--------------|---------------------------------------------------|-----------|----------------------------------------|-------------------------------|--------------------------|--------------------------|----------------------|----------------------|---------------------------------------------------------------|----------------------------------|----------------------------------|----------------|----------------------|------------------------------------------------------------|---------|--|
| B72 Resiliency Improvement |                   |              |                                                   |           |                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
| ACTIVITY<br>NO             | TRANSMITTAL<br>NO | SPEC<br>SECT | DESCRIPTION<br>ITEM SUBMITTED                     | PARAGRAPH | GOVT<br>CLASSIFICATION<br>OR<br>REVIEW | CONTRACTOR:<br>SCHEDULE DATES |                          |                          | CONTRACTOR<br>ACTION |                      |                                                               | APPROVING AUTHORITY              |                                  |                |                      | MAILED<br>TO<br>CONTR/<br><br>DATE RCD<br>FRM APPR<br>AUTH | REMARKS |  |
|                            |                   |              |                                                   |           |                                        | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION | DATE FWD<br>TO APPR<br>AUTH/<br><br>DATE RCD<br>FROM<br>CONTR | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                            |         |  |
| (a)                        | (b)               | (c)          | (d)                                               | (e)       | (f)                                    | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                                                           | (m)                              | (n)                              | (o)            | (p)                  | (q)                                                        | (r)     |  |
|                            |                   | 23 05 15     | Equipment Foundation Data                         | 1.2       | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | SD-04 Samples                                     |           |                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Manufacturer's Standard Color Charts              | 1.2       | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | SD-05 Design Data                                 |           |                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Pipe and Fittings                                 | 2.1       | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Piping Specialties                                | 2.2       | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Emergency Eyewash and Shower Valves               | 2.3       | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | SD-06 Test Reports                                |           |                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Hydrostatic Tests                                 | 3.1       | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Air Tests                                         | 3.1       | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Valve-Operating Tests                             | 3.1       | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Drainage Tests                                    | 3.1       | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Pneumatic Tests                                   | 3.1       | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Non-Destructive Electric Tests                    | 3.1       | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | System Operation Tests                            | 3.1       | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | SD-07 Certificates                                |           |                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Record of Satisfactory Field Operation            | 1.4.2     | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | List of Qualified Permanent Service Organizations | 1.4.3     | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Listing of Product Installations                  | 1.2       | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Records of Existing Conditions                    | 1.2       | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Surface Resistance                                | 3.1       | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |
|                            |                   |              | Shear and Tensile Strengths                       | 3.1       | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |  |

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| ACTIVITY NO | TRANSMITTAL NO | SPEC SECT   | DESCRIPTION<br>ITEM SUBMITTED        | PARAGRAPH | GOVT CLASSIFICATION | CONTRACTOR:<br>SCHEDULE DATES |                          |                          | CONTRACTOR<br>ACTION |                      | DATE FWD<br>TO APPR<br>AUTH/ | APPROVING AUTHORITY              |                                  |                |                      | MAILED<br>TO<br>CONTR/       | REMARKS |
|-------------|----------------|-------------|--------------------------------------|-----------|---------------------|-------------------------------|--------------------------|--------------------------|----------------------|----------------------|------------------------------|----------------------------------|----------------------------------|----------------|----------------------|------------------------------|---------|
|             |                |             |                                      |           |                     | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION | DATE RCD<br>FROM<br>CONTR    | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION | DATE RCD<br>FRM APPR<br>AUTH |         |
| (a)         | (b)            | (c)         | (d)                                  | (e)       | (f)                 | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                          | (m)                              | (n)                              | (o)            | (p)                  | (q)                          | (r)     |
|             |                | 23 05 15    | Temperature Ratings                  | 3.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Bending Tests                        | 3.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Flattening Tests                     | 3.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Transverse Guided Weld Bend Tests    | 3.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | SD-10 Operation and Maintenance Data |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Operation and Maintenance Manuals    | 3.8       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 23 05 48.19 | SD-02 Shop Drawings                  |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Coupling and Bracing                 | 3.1       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Flexible Couplings or Joints         | 3.3       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Equipment Restraint                  | 2.2       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Contractor Designed Bracing          | 1.2.4     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | SD-03 Product Data                   |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Coupling and Bracing                 | 3.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Flexible Couplings Or Joints         | 3.3       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Equipment Restraint                  | 2.2       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Contractor Designed Bracing          | 1.2.4     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Snubbers                             | 2.6       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Anchor Bolts                         | 3.9       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Vibration Isolators                  | 2.2.2     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | SD-05 Design Data                    |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Design Calculations                  | 1.2.4     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | SD-06 Test Reports                   |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Anchor Bolts                         | 3.9       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|----------------|-------------------|--------------|--------------------------------------------------------|-----------|-------------------------------------------------|-------------------------------|--------------------------|--------------------------|----------------------|----------------------|---------------------------------------------------------------|----------------------------------|----------------------------------|----------------|----------------------|------------------------------------------------------------|---------|
|                |                   |              |                                                        |           |                                                 | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                                                               | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                            |         |
| (a)            | (b)               | (c)          | (d)                                                    | (e)       | (f)                                             | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                                                           | (m)                              | (n)                              | (o)            | (p)                  | (q)                                                        | (r)     |
|                |                   | 23 05 93     | SD-01 Preconstruction Submittals                       |           |                                                 |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Records of Existing Conditions                         | 1.3.3     | G                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Independent TAB Agency and<br>Personnel Qualifications | 1.5.1     | G                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | TAB Design Review Report                               | 1.5.3.2   | G                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-02 Shop Drawings                                    |           |                                                 |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | TAB Schematic Drawings and<br>Report Forms             | 1.3.3     | G                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-03 Product Data                                     |           |                                                 |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Equipment and Performance<br>Data                      | 1.3       | G                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | TAB Related HVAC Submittals                            | 1.5.1.3   | G                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-06 Test Reports                                     |           |                                                 |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Prerequisite HVAC Work<br>Checkout List                | 1.5.3.3   | G                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Prerequisite HVAC Work<br>Checkout List                | 1.5.3.3   | G                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Prerequisite HVAC Work<br>Checkout List                | 1.5.3.3   | G                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Proportional Balancing                                 | 3.3.6.1   | G                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Proportional Balancing                                 | 3.6       | G                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Season 1                                               | 3.3.6.2   | G                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Season 1                                               | 3.6       | G                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Season 2                                               | 3.3.6.2   | G                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Season 2                                               | 3.6       | G                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-07 Certificates                                     |           |                                                 |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |

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|                |                   |              |                                                        |           |                                                 | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                                                            | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                         |         |
| (a)            | (b)               | (c)          | (d)                                                    | (e)       | (f)                                             | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                                                        | (m)                              | (n)                              | (o)            | (p)                  | (q)                                                     | (r)     |
|                |                   | 23 05 93     | Independent TAB Agency and<br>Personnel Qualifications | 1.5.1     | G                                               |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | TAB Submittal and Work<br>Schedule                     | 1.5.3.2   | G                                               |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | TAB Pre-Field Engineering<br>Report                    | 1.5.3.3   | G                                               |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Instrument Calibration<br>Certificates                 | 1.5.5     | G                                               |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | TAB Procedures Summary                                 | 3.6       | G                                               |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Proportional Balancing                                 | 3.3.6.1   | G                                               |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Proportional Balancing                                 | 3.6       | G                                               |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Season 1                                               | 3.3.6.2   | G                                               |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Season 1                                               | 3.6       | G                                               |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Season 2                                               | 3.3.6.2   | G                                               |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Season 2                                               | 3.6       | G                                               |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   | 23 07 00     | SD-02 Shop Drawings                                    |           |                                                 |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | MICA Plates                                            | 3.2.2.4   | G AE                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Pipe Insulation Systems                                | 2.3       | G AE                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Pipe Insulation Systems                                | 3.2       | G AE                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Duct Insulation Systems                                | 3.3       | G AE                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Equipment Insulation Systems                           | 3.4       | G AE                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Recycled content for insulation<br>materials           | 2.3.1     | G AE                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-03 Product Data                                     |           |                                                 |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Pipe Insulation Systems                                | 2.3       | G AE                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Pipe Insulation Systems                                | 3.2       | G AE                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |

# SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

B72 Resiliency Improvement

CONTRACTOR

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|-------------|----------------|----------|-----------------------------------|-----------|---------------------|-------------------------------|--------------------------|--------------------------|----------------------|----------------------|------------------------------|----------------------------------|----------------------------------|----------------|----------------------|------------------------------|---------|
|             |                |          |                                   |           |                     | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION | DATE RCD<br>FROM<br>CONTR    | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION | DATE RCD<br>FRM APPR<br>AUTH |         |
| (a)         | (b)            | (c)      | (d)                               | (e)       | (f)                 | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                          | (m)                              | (n)                              | (o)            | (p)                  | (q)                          | (r)     |
|             |                | 23 07 00 | Duct Insulation Systems           | 3.3       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | Equipment Insulation Systems      | 3.4       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | SD-04 Samples                     |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | Thermal Insulation                | 2.2.1.3   | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | Display Samples                   | 3.1.1     | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | SD-07 Certificates                |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | Indoor air quality for adhesives  | 2.2.1     | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | SD-08 Manufacturer's Instructions |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | Pipe Insulation Systems           | 2.3       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | Pipe Insulation Systems           | 3.2       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | Duct Insulation Systems           | 3.3       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | Equipment Insulation Systems      | 3.4       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 23 08 00 | SD-03 Product Data                |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | Test Equipment                    | 2.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | SD-06 Test Reports                |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | Pipe Flushing, Testing, And       | 1.5       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | Water Treatment Reports           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | Completed Pre-Functional          | 3.3       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | Checklists                        |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | Seasonal Test Report              | 3.10      | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | Full-Load Test Report             | 3.11      | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | Post-Construction Trend Log       | 3.14      | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | Report                            |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | SD-07 Certificates                |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |          | Certificate Of Readiness          | 1.6       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 23 09 00 | SD-02 Shop Drawings               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|----------------------------|-------------------|--------------|-----------------------------------|-----------|--------------------------------------------------------------------------|-------------------------------|--------------------------|--------------------------|----------------------|----------------------|------------------------------------------------------------|----------------------------------|----------------------------------|----------------|----------------------|---------------------------------------------------------|---------|
| B72 Resiliency Improvement |                   |              |                                   |           |                                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
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|                            |                   |              |                                   |           |                                                                          | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                                                            | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                         |         |
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|                            |                   | 23 09 00     | DDC Contractor Design Drawings    | 3.2       | G                                                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Draft As-Built Drawings           | 3.2       | G                                                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Final As-Built Drawings           | 3.2       | G                                                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | SD-03 Product Data                |           |                                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Programming Software              | 1.8.3     | G                                                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Controller Application Programs   | 1.8.4     | G                                                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Configuration Software            | 1.8.1     | G                                                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Controller Configuration Settings | 1.8.2     | G                                                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Manufacturer's Product Data       | 2.2       | G                                                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | SD-06 Test Reports                |           |                                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Pre-Construction Quality Control  | 1.9.1     | G                                                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | (QC) Checklist                    |           |                                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Post-Construction Quality Control | 1.9.2     | G                                                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | (QC) Checklist                    |           |                                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Start-Up Testing Report           | 3.4.2     | G                                                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Control Contractor's Performance  | 3.5.5     | G                                                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Verification Testing Plan         |           |                                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Equipment Supplier's              | 3.5.3.1   | G                                                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Performance Verification Testing  |           |                                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Plan                              |           |                                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Endurance Testing Results         | 3.5.8.3   | G                                                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Performance Verification Test     | 3.5.9     | G                                                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Report                            |           |                                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | SD-10 Operation and Maintenance   |           |                                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                            |                   |              | Data                              |           |                                                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |

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|                |                   | 23 09 00     | Operation and Maintenance<br>(O&M) Instructions | 3.6       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Training Documentation                          | 3.8.1     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-11 Closeout Submittals                       |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Enclosure Keys                                  | 2.5       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Closeout Quality Control (QC)<br>Checklist      | 1.9.3     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   | 23 21 23     | SD-02 Shop Drawings                             |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | System Coordination                             | 2.1.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-03 Product Data                              |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Instructions                                    | 2.2.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Equipment Data                                  | 2.2.5     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Training Period                                 | 3.5.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-06 Test Reports                              |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Factory Tests                                   | 2.8       |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Field Quality Control                           | 3.3       |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-07 Certificates                              |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Manufacturer's Representative                   | 1.3.1     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-10 Operation and Maintenance<br>Data         |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Operation and Maintenance<br>Manuals            | 3.5.1     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Training                                        | 3.5.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   | 23 23 00     | SD-02 Shop Drawings                             |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Refrigerant Piping System                       | 2.3       | G AE                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-03 Product Data                              |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |



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|                |                   |              |                                 |           |                        | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                                                            | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                         |         |
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|                |                   | 23 23 00     | Refrigerant Piping System       | 2.3       |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Qualifications                  | 1.3.1     |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Refrigerant Piping Tests        | 3.5       |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Verification of Dimensions      | 3.1       |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-06 Test Reports              |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Refrigerant Piping Tests        | 3.5       |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-07 Certificates              |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Service Organization            | 2.1       |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-10 Operation and Maintenance |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Data                            |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Maintenance                     | 1.5       | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Operation and Maintenance       | 3.4       | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Manuals                         |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Demonstrations                  | 3.4       | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   | 23 30 00     | SD-02 Shop Drawings             |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Detail Drawings                 | 1.4.4     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-03 Product Data              |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Metallic Flexible Duct          | 2.9.1.1   |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Insulated Nonmetallic Flexible  | 2.9.1.2   |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Duct Runouts                    |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Duct Connectors                 | 2.9.1.2   |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Duct Access Doors               | 2.9.2     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Fire Dampers                    | 2.9.3     |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Manual Balancing Dampers        | 2.9.4     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Diffusers                       | 2.9.6.1   |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Louvers                         | 2.9.7     |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |

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| B72 Resiliency Improvement |                |           |                                         |           |                     |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
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|                            |                |           |                                         |           |                     | SUBMIT                        | APPROVAL NEEDED BY | MATERIAL NEEDED BY | ACTION CODE       | DATE OF ACTION |     | DATE FWD TO APPR AUTH/<br><br>DATE RCD FROM CONTR | DATE FWD TO OTHER REVIEWER | DATE RCD FROM OTH REVIEWER | ACTION CODE | DATE OF ACTION                                 |         |
| (a)                        | (b)            | (c)       | (d)                                     | (e)       | (f)                 | (g)                           | (h)                | (i)                | (j)               | (k)            | (l) | (m)                                               | (n)                        | (o)                        | (p)         | (q)                                            | (r)     |
|                            |                | 23 30 00  | Air Vents, Penthouses, and Goosenecks   | 2.9.8     |                     |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | In-Line Centrifugal Fans                | 2.10.1.1  |                     |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | Axial Flow Fans                         | 2.10.1.2  |                     |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | Centrifugal Type Power Roof Ventilators | 2.10.1.3  |                     |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | Unit Ventilators                        | 2.11.1    |                     |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | Indoor Air Quality for Duct Sealants    | 2.9.1     | S                   |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | Diagrams                                | 1.2.1.2   | G                   |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | SD-06 Test Reports                      |           |                     |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | Performance Tests                       | 3.12      | G                   |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | Damper Acceptance Test                  | 3.10      | G                   |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | Test Procedures                         | 1.4.5     | G                   |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | SD-07 Certificates                      |           |                     |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | Ozone Depleting Substances              | 1.4.3     |                     |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | Technician Certification                |           |                     |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | SD-08 Manufacturer's Instructions       |           |                     |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | Manufacturer's Installation             | 3.2       |                     |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | Instructions                            |           |                     |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | Operation and Maintenance               | 3.14.2    |                     |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | Training                                |           |                     |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | SD-10 Operation and Maintenance         |           |                     |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | Data                                    |           |                     |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | Operation and Maintenance               | 3.14.1    | G                   |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |
|                            |                |           | Manuals                                 |           |                     |                               |                    |                    |                   |                |     |                                                   |                            |                            |             |                                                |         |

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|                |                   |              |                                               |           |                                        | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |     | DATE FWD<br>TO APPR<br>AUTH/<br><br>DATE RCD<br>FROM<br>CONTR | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE |                                                            |         |
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|                |                   | 23 30 00     | Fire Dampers                                  | 2.9.3     | G                                      |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Manual Balancing Dampers                      | 2.9.4     | G                                      |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | In-Line Centrifugal Fans                      | 2.10.1.1  | G                                      |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Axial Flow Fans                               | 2.10.1.2  | G                                      |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Centrifugal Type Power Roof Ventilators       | 2.10.1.3  | G                                      |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Unit Ventilators                              | 2.11.1    | G                                      |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | SD-11 Closeout Submittals                     |           |                                        |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Indoor Air Quality During Construction        | 3.13      | S                                      |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   | 23 64 26     | SD-01 Preconstruction Submittals              |           |                                        |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Valves                                        | 2.3       | G AE                                   |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Intake screens and airburst back wash system  | 2.4       | G AE                                   |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Intake Screen Flow Distribution               | 2.4.1     | G AE                                   |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | SD-02 Shop Drawings                           |           |                                        |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Detail Drawings                               | 1.7.1     | G AE                                   |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | SD-03 Product Data                            |           |                                        |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Piping                                        | 2.2       | G AE                                   |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Valves                                        | 2.3       | G AE                                   |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | Intake screens and airburst back wash system; | 2.4       | G AE                                   |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | SD-05 Design Data                             |           |                                        |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |
|                |                   |              | INTAKE SCREENS AND AIRBURST BACK WASH SYSTEM  | 2.4       | G AE                                   |                               |                          |                          |                      |                      |     |                                                               |                                  |                                  |                |                                                            |         |

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|                |                   | 23 64 26     | SD-06 Test Reports                |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Piping Welds NDE Report           | 3.1.1.4   | G AE                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Intake Screen Flow Distribution   | 2.4.1     | G AE                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | INTAKE SCREENS AND                | 2.4       | G AE                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | AIRBURST BACK WASH                |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SYSTEM                            |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-07 Certificates                |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Employer's Record Documents       | 3.1.1.2   | G                                            |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | (For Welding)                     |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Welding Procedures and            | 3.1.1.3   | G                                            |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Qualifications                    |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | INTAKE SCREENS AND                | 2.4       | G AE                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | AIRBURST BACK WASH                |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SYSTEM                            |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-08 Manufacturer's Instructions |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Lesson plan for the Instruction   | 3.5       | G AE                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Course                            |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-10 Operation and Maintenance   |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Data                              |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Valves                            | 2.3       | G AE                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | INTAKE SCREENS AND                | 2.4       | G AE                                         |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | AIRBURST BACK WASH                |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SYSTEM                            |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   | 23 81 00     | SD-03 Product Data                |           |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Spare Parts                       | 3.6.1     |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Posted Instructions               | 3.4       |                                              |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |

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|                |                   |                |                                   |           |                                | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION | DATE RCD<br>FROM<br>CONTR    | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION | DATE RCD<br>FRM APPR<br>AUTH |         |
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|                |                   | 23 81 00       | Coil Corrosion Protection         | 2.6.1     |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | System Performance Tests          | 3.5       |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Training                          | 3.4       | G                              |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Inventory                         | 1.5       |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Supplied Products                 | 2.1       |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Manufacturer's Standard Catalog   | 2.2       |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Data                              |           |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | SD-06 Test Reports                |           |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | System Performance Tests          | 3.5       | G                              |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | SD-07 Certificates                |           |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Service Organizations             | 3.6.2     |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | SD-10 Operation and Maintenance   |           |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Data                              |           |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Operation and Maintenance         | 3.4       | G                              |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Manuals                           |           |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | SD-11 Closeout Submittals         |           |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Ozone Depleting Substances        | 2.2.2.3   | S                              |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   | 23 83 00.00 20 | SD-02 Shop Drawings               |           |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Heater Installation Drawing       | 3.1       |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | SD-03 Product Data                |           |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Electric Unit and Cabinet Heaters | 2.1       |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Thermostat                        | 2.1.6     |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | SD-10 Operation and Maintenance   |           |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Data                              |           |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |                | Electric Unit and Cabinet Heaters | 2.1       |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   | 25 05 11       | SD-01 Preconstruction Submittals  |           |                                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|                |                   |              |                                                                  |           |                        | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                                                            | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                         |         |
| (a)            | (b)               | (c)          | (d)                                                              | (e)       | (f)                    | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                                                        | (m)                              | (n)                              | (o)            | (p)                  | (q)                                                     | (r)     |
|                |                   | 25 05 11     | Device Account Lock Exception Request                            | 3.3.2     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Multiple Ethernet Connection Device Request                      | 3.2.3.2   | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Contractor Computer Cybersecurity Compliance Statements          | 1.9.1.6   | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Contractor Temporary Network Cybersecurity Compliance Statements | 1.9.6     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Cybersecurity Interconnection Schedule                           | 1.7.2     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Proposed STIG and SRG Applicability Report                       | 1.7.1     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Pre-Construction Control System Inventory Report                 | 1.7.4     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-02 Shop Drawings                                              |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Network Communication, Ports, Protocols and Services Report      | 1.7.3     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Cybersecurity Network (Riser) Diagram                            | 1.7.6     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-03 Product Data                                               |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Control System Cybersecurity Documentation                       | 1.7.8     | G                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-06 Test Reports                                               |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              |                                                                  |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |

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|----------------|-------------------|----------------|------------------------------------------------------------------------|-----------|------------------------------------------------------------------------|-------------------------------|--------------------------|--------------------------|----------------------|----------------------|---------------------------------------------------------------|----------------------------------|----------------------------------|----------------|----------------------|------------------------------------------------------------|---------|
|                |                   |                |                                                                        |           |                                                                        | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                                                               | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                            |         |
| (a)            | (b)               | (c)            | (d)                                                                    | (e)       | (f)                                                                    | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                                                           | (m)                              | (n)                              | (o)            | (p)                  | (q)                                                        | (r)     |
|                |                   | 25 05 11       | Control System Cybersecurity Testing Procedures                        | 3.11.1    | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Control System Cybersecurity Testing Report                            | 3.11.3    | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | SD-07 Certificates                                                     |           |                                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Software Licenses                                                      | 1.8       | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | SD-11 Closeout Submittals                                              |           |                                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Confidential Password Report                                           | 3.4.5.3   | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Enclosure Keys                                                         | 3.3.5     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Software and Configuration Backups                                     | 1.7.5     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Auditing Front End Software                                            | 3.5.3     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Device Audit Record Upload                                             | 3.5.4.1   | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Software                                                               |           |                                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | System Maintenance Tool                                                | 3.9       | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Software                                                               |           |                                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Control System Scanning Tools                                          | 3.10.2    | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | STIG, SRG and Vendor Guide                                             | 1.7.7     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Compliance Result Report                                               |           |                                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | Final Control System Inventory Report                                  | 1.7.4     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   | 26 05 13.00 10 | SD-01 Preconstruction Submittals                                       |           |                                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | List of Splices and Terminations to be Installed by Splicer/Terminator | 1.3.1     | G                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |                | SD-02 Shop Drawings                                                    |           |                                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |

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|----------------|-------------------|----------------|-------------------------------------|-----------|---------------------------------------------------------|-------------------------------|--------------------------|--------------------------|----------------------|----------------------|------------------------------|---------------------------|----------------------------------|----------------------------------|----------------|------------------------|---------|
|                |                   |                |                                     |           |                                                         | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                              | DATE RCD<br>FROM<br>CONTR | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION   |         |
| (a)            | (b)               | (c)            | (d)                                 | (e)       | (f)                                                     | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                          | (m)                       | (n)                              | (o)                              | (p)            | (q)                    | (r)     |
|                |                   | 26 05 13.00 10 | Pulling Plan                        | 3.2.2     | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | SD-03 Product Data                  |           |                                                         |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | Single-Conductor Shielded<br>Cables | 2.2.1     | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | Cable Supports and Fittings         | 2.2.5     | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | Polyethylene Cable Tags             | 2.2.6     | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | Polyethylene Cable Tags             | 2.2.6     | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | Fireproof Tape                      | 2.2.7     | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | Splices                             | 2.2.3     | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | Terminations                        | 2.2.4     | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | SD-06 Test Reports                  |           |                                                         |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | Field Testing                       | 3.3       | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | Qualification Test Reports          | 2.4.1     | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | Radiographic Tests                  | 3.3       | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | SD-07 Certificates                  |           |                                                         |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | Splicer/Terminator Certifications   | 1.5.2.1   | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | Cable Installer Qualifications      | 1.5.2.1   | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | SD-08 Manufacturer's Instructions   |           |                                                         |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | Medium-Voltage Power Cables         | 3.4       | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | Terminations                        | 2.2.4     | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | Splices                             | 2.2.3     | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   | 26 05 19.00 10 | SD-01 Preconstruction Submittals    |           |                                                         |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | Cable Installation Plan             | 3.2       | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | SD-03 Product Data                  |           |                                                         |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | Wire and Cable                      | 2.1.1     | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |
|                |                   |                | Conductors                          | 2.1.3.1   | G                                                       |                               |                          |                          |                      |                      |                              |                           |                                  |                                  |                |                        |         |



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|                |                   | 26 05 19.00 10 | Cable Manufacturing Data         | 3.1       |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | SD-06 Test Reports               |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Test Report(s), Inspection       | 3.3       | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Report(s), and Verification      |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Report(s)                        |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   | 26 05 48       | SD-02 Shop Drawings              |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Equipment Requirements           | 2.1       | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | SD-03 Product Data               |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Supports and Attachments         | 1.2.3     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Equipment Requirements           | 2.1       | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Flexible Fittings                | 2.3       | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Anchors                          | 3.4       | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | SD-05 Design Data                |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Design Calculations              | 1.2.3     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Design Drawings                  | 1.2.3     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | SD-06 Test Reports               |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Anchors                          | 3.4       | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | SD-07 Certificates               |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | ICC ES AC156 Shake Table Test    | 3.6       | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   | 26 05 73       | SD-01 Preconstruction Submittals |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Field Examination Plan           | 3.1       | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Arc Flash Label Formats          | 3.2.9     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | SD-06 Test Reports               |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | Field Examination                | 3.1.1     |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | SD-07 Certificates               |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |                | System Analyzer                  | 1.4.1     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |

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|             |                |           |                                              |           |                     | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION | DATE RCD<br>FROM<br>CONTR    | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION | DATE RCD<br>FRM APPR<br>AUTH |         |
| (a)         | (b)            | (c)       | (d)                                          | (e)       | (f)                 | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                          | (m)                              | (n)                              | (o)            | (p)                  | (q)                          | (r)     |
|             |                | 26 05 73  | SD-11 Closeout Submittals                    |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Model Files                                  | 3.3       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Load Flow Study                              | 3.2.4     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Fault Current Study                          | 3.2.5     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | System Coordination Study                    | 3.2.6     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Arc Flash Hazard Study                       | 3.2.7     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 26 08 00  | SD-06 Test Reports                           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Acceptance Tests and<br>Inspections          | 3.1       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-07 Certificates                           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Qualifications                               | 1.4.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Acceptance Test and Inspections<br>Procedure | 1.4.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 26 11 16  | SD-02 Shop Drawings                          |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Unit Substation Drawings                     | 1.5.1.1   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Transformer Drawings                         | 1.5.1.2   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-03 Product Data                           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Fuse Curves                                  | 1.5.1.1   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Secondary Unit Substation                    | 2.2       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Transformer (Liquid-filled)                  | 2.2.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-06 Test Reports                           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Acceptance Checks and Tests                  | 3.6.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-07 Certificates                           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Paint Coating System                         | 1.5.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Transformer Efficiencies                     | 1.5.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|                |                   |              |                                                                       |           |                                                                             | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                                                               | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                            |         |
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|                |                   | 26 11 16     | SD-09 Manufacturer's Field Reports                                    |           |                                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Load Interrupter Switch Production Tests                              | 2.6.2     | G                                                                           |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Transformer Design Tests (Liquid-filled)                              | 2.6.3     | G                                                                           |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Transformer Routine and Other Tests (Liquid-filled)                   | 2.6.4     | G                                                                           |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-10 Operation and Maintenance Data                                  |           |                                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Unit Substations                                                      | 2.2       | G                                                                           |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-11 Closeout Submittals Assembled Operation and Maintenance Manuals | 1.6.1     | G                                                                           |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Equipment Test Schedule                                               | 2.6.1     | G                                                                           |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   | 26 20 00     | SD-02 Shop Drawings                                                   |           |                                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Panelboards                                                           | 2.12      | G                                                                           |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Transformers                                                          | 2.14      | G                                                                           |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Cable Trays                                                           | 2.3       | G                                                                           |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Wireways                                                              | 2.23      | G                                                                           |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Marking Strips                                                        | 3.1.9.1   | G                                                                           |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-03 Product Data                                                    |           |                                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Receptacles                                                           | 2.11      | G                                                                           |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Circuit Breakers                                                      | 2.12.3    | G                                                                           |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Switches                                                              | 2.9       | G                                                                           |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Transformers                                                          | 2.14      | G                                                                           |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |

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|                |                   | 26 20 00     | Enclosed Circuit Breakers       | 2.13      | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Motor Controllers               | 2.15      | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Manual Motor Starters           | 2.16      | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Secondary Bonding Busbar        | 2.18.3    | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Surge Protective Devices        | 2.24      | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Cable Trays                     | 2.3       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Cable Tray Design               | 2.3       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-06 Test Reports              |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | 600-volt Wiring Test            | 3.5.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Grounding System Test           | 3.5.5     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Transformer Tests               | 3.5.3     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Ground-fault Receptacle Test    | 3.5.4     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-07 Certificates              |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Fuses                           | 2.10      | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-09 Manufacturer's Field      |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Reports                         |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Transformer Factory Tests       | 2.26.1    |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-10 Operation and Maintenance |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Data                            |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Electrical Systems              | 1.6.1     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   | 26 24 13     | SD-02 Shop Drawings             |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Switchboard Drawings            | 1.5.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-03 Product Data              |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Switchboard                     | 2.2       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-06 Test Reports              |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Switchboard Design Tests        | 2.5.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|                |                   | 26 24 13     | Switchboard Production Tests    | 2.5.3     | G                                                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Acceptance Checks and Tests     | 3.5.1     | G                                                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-07 Certificates              |           |                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Cybersecurity Installation      | 3.5.1.6   | G                                                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Certification                   |           |                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-10 Operation and Maintenance |           |                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Data                            |           |                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Switchboard Operation and       | 1.6.1     | G                                                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Maintenance                     |           |                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-11 Closeout Submittals       |           |                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Assembled Operation and         | 1.6.2     | G                                                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Maintenance Manuals             |           |                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Equipment Test Schedule         | 2.5.1     | G                                                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Request for Settings            | 3.5       | G                                                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   | 26 29 23     | SD-02 Shop Drawings             |           |                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Schematic Diagrams              | 1.5.1     | G                                                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Interconnecting Diagrams        | 1.5.2     | G                                                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Installation Drawings           | 1.5.3     | G                                                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | As-Built Drawings               | 1.5.3     | G                                                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-03 Product Data              |           |                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Adjustable Speed Drives         | 2.1       | G                                                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Wires and Cables                | 2.3       |                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Equipment Schedule              | 1.5.4     |                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-06 Test Reports              |           |                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | ASD Test                        | 3.3.1     |                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Performance Verification Tests  | 3.3.2     |                                                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |

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|                |                   | 26 29 23     | Endurance Test                       | 3.3.3     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-07 Certificates                   |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Testing Agency's Field Supervisor    | 3.3.1     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-08 Manufacturer's Instructions    |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Installation instructions            | 1.5.5     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-09 Manufacturer's Field Reports   |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | ASD Test Plan                        | 2.5.1     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Standard Products                    | 1.5.6     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-10 Operation and Maintenance Data |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Adjustable Speed Drives              | 2.1       |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   | 26 51 00     | SD-02 Shop Drawings                  |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Luminaire Drawings                   | 1.5.1     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-03 Product Data                   |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Luminaires                           | 2.2       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Light Sources                        | 2.3       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | LED Drivers                          | 2.4       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Luminaire Warranty                   | 1.6.1     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Mini Inverters                       | 2.6.4     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Exit Signs                           | 2.6.1     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Emergency Drivers                    | 2.6.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-04 Samples                        |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Luminaire Samples                    | 2.2.1     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-05 Design Data                    |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |

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|                |                   | 26 51 00     | Luminaire Design Data           | 1.5.2     | G                                  |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Photometric Plan                | 1.5.8     | G                                  |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-06 Test Reports              |           |                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | ANSI/IES LM-79 Test Report      | 1.5.3     | G                                  |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | ANSI/IES LM-80 Test Report      | 1.5.4     | G                                  |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | ANSI/IES TM-21 Test Report      | 1.5.5     | G                                  |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | ANSI/IES TM-30 Test Report      | 1.5.6     | G                                  |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-07 Certificates              |           |                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | LED Driver and Dimming Switch   | 1.5.7     | G                                  |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Compatibility Certificate       |           |                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-10 Operation and Maintenance |           |                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Data                            |           |                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Lighting System                 | 1.7.1     | G                                  |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Maintenance Staff Training Plan | 3.3.2.1   | G                                  |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | End-User Training Plan          | 3.3.2.2   | G                                  |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   | 27 10 00     | SD-02 Shop Drawings             |           |                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Telecommunications Drawings     | 1.6.1.1   | G                                  |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Telecommunications Space        | 1.6.1.2   | G                                  |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Drawings                        |           |                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-03 Product Data              |           |                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Telecommunications Cabling      | 2.3       | G                                  |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Patch Panels                    | 2.4.4     | G                                  |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Telecommunications              | 2.5       | G                                  |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Outlet/Connector Assemblies     |           |                                    |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Equipment Support Frame         | 2.4.1     | G                                  |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Connector Blocks                | 2.4.2     | G                                  |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |

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| (a)         | (b)            | (c)       | (d)                                | (e)       | (f)                 | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                          | (m)                              | (n)                              | (o)            | (p)                  | (q)                                                     | (r)     |
|             |                | 27 10 00  | Spare Parts                        | 1.10.3    | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-06 Test Reports                 |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Telecommunications Cabling         | 3.5.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Testing                            |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-07 Certificates                 |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Telecommunications Contractor      | 1.6.2.1   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Key Personnel                      | 1.6.2.2   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Manufacturer Qualifications        | 1.6.2.3   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Test Plan                          | 1.6.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-09 Manufacturer's Field         |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Reports                            |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Factory Reel Tests                 | 2.10.1    | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-10 Operation and Maintenance    |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Data                               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Telecommunications Cabling and     | 1.10.1    | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Pathway System                     |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-11 Closeout Submittals          |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Record Documentation               | 1.10.2    | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                | 28 31 60  | SD-01 Preconstruction Submittals   |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Qualified Fire Protection Engineer | 1.3.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | (QFPE)                             |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Fire alarm system designer         | 1.8.2.1   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Technician                         | 1.8.2.2   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Installer                          | 1.8.2.3   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Test Technician                    | 1.8.2.4   | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-02 Shop Drawings                |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |



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|                |                   | 28 31 60     | Wiring Diagrams                           | 1.8.1.3   | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | System Layout                             | 1.8.1.4   | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Notification Appliances                   | 1.8.1.5   | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Notification Appliances                   | 2.4       | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Initiating devices                        | 1.8.1.6   | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Battery Power                             | 1.8.1.7   | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Voltage Drop Calculations                 | 1.8.1.8   | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-03 Product Data                        |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Manual Stations                           | 2.3       | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Notification Appliances                   | 1.8.1.5   | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Notification Appliances                   | 2.4       | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Alarm Wiring                              | 2.5.1     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Back Boxes and Conduit                    | 3.3.4     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Terminal Cabinets                         | 3.3.2     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Environmental Enclosures or<br>Guards     | 2.7       | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-06 Test Reports                        |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Test Procedures                           | 3.6.1     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-07 Certificates                        |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Verification of Compliant<br>Installation | 3.6.2.1   | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Request for Government Final<br>Test      | 3.6.2.2   | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-10 Operation and Maintenance<br>Data   |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |

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|             |                |                |                                              |           |                     | SUBMIT                        | APPROVAL NEEDED BY | MATERIAL NEEDED BY | ACTION CODE       | DATE OF ACTION | DATE RCD FROM CONTR    | DATE FWD TO OTHER REVIEWER | DATE RCD FROM OTH REVIEWER | ACTION CODE | DATE OF ACTION | DATE RCD FRM APPR AUTH |         |
| (a)         | (b)            | (c)            | (d)                                          | (e)       | (f)                 | (g)                           | (h)                | (i)                | (j)               | (k)            | (l)                    | (m)                        | (n)                        | (o)         | (p)            | (q)                    | (r)     |
|             |                | 28 31 60       | Operation and Maintenance (O&M) Instructions | 3.8       | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | SD-11 Closeout Submittals                    |           |                     |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | As-Built Drawings; G                         |           |                     |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                | 31 00 00.00 22 | SD-01 Preconstruction Submittals             |           |                     |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Shoring and Sheeting Plan                    | 3.1.1     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Excavation and Trenching Plan                | 1.13.1    | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Dewatering Work Plan                         | 1.13.2    | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Dewatering Work Plan                         | 3.1.4.2   | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Archeologists Qualifications                 | 1.13.5    | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Preconstruction Meeting                      | 1.6.4     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Excess Soil Disposal & Sampling Plan         | 1.12      | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | SD-03 Product Data                           |           |                     |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Flowable Fill Mix Design                     | 2.2       | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Geotextiles                                  | 2.5       |                     |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | SD-06 Test Reports                           |           |                     |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Dewatering Performance Records               | 3.1.4.2   | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Material Test Report                         | 3.10      | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Contractor Sampling Field Log                | 1.4.2     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Book Entry                                   |           |                     |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Laboratory gradation test                    | 3.9.2.1   | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Laboratory compaction tests                  | 3.9.2.1   | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Pipe Inspection Report                       | 3.7.1.6   | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |                | Geotechnical Evaluation Report               | 1.6.1     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                | 31 63 33       | SD-01 Preconstruction Submittals             |           |                     |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |

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|             |                | 31 63 33  | Contractor Experience         | 1.4.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Contractor Experience         | 1.4.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Contractor Experience         | 1.4.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Foreman Operators             | 1.4.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Drill Rig Operators           | 1.4.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Project Reference List        | 1.4.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Personnel List                | 1.4.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Professional Engineer         | 1.4.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Professional Engineer         | 1.4.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Professional Engineer         | 1.4.6     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-02 Shop Drawings           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Working Drawings              | 1.4.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Micropile Structures          | 1.4.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Micropile Structures          | 1.4.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Micropile Structures          | 1.4.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Pile Load Test                | 1.4.5     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-05 Design Data             |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Micropile Structures          | 1.4.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Micropile Structures          | 1.4.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Micropile Structures          | 1.4.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Steel Pipe                    | 1.4.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Reports                       | 1.4.5     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Reports                       | 1.4.6     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Design Criteria and Notes     | 1.4.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Design Criteria and Notes     | 1.4.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Design Calculations           | 1.4.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|                |                   |              |                                   |           |                                            | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                                                            | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                         |         |
| (a)            | (b)               | (c)          | (d)                               | (e)       | (f)                                        | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                                                        | (m)                              | (n)                              | (o)            | (p)                  | (q)                                                     | (r)     |
|                |                   | 31 63 33     | Plans                             | 1.4.3     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Plans                             | 1.4.4     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Plans                             | 1.4.4     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Plans                             | 1.4.5     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Computations                      | 1.4.2     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Computations                      | 1.4.4     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Safety and Strength Factors       | 1.4.2     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-06 Test Reports                |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Verification Load Test Report     | 3.6       | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Proof Load Test Reports           | 3.6       | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-08 Manufacturer's Instructions |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Micropile Structures              | 1.4.2     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Micropile Structures              | 1.4.3     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Micropile Structures              | 1.4.3     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Installation                      | 1.4.4     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Installation                      | 1.4.4     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Installation                      | 1.4.6     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Installation                      | 1.4.6     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   | 32 11 26     | SD-03 Product Data                |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Sources of Aggregates             | 1.4.4     |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Job Mix Formula                   | 2.4.2     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-06 Test Reports                |           |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Sources of Aggregates             | 1.4.4     |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Bituminous Materials              | 1.4.6     |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Bituminous Materials              | 2.2       |                                            |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Test Section                      | 1.6.2     | G                                          |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |

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|                |                   |              |                                               |           |                        | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION | DATE RCD<br>FROM<br>CONTR    | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                     |         |
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|                |                   | 32 11 26     | Service Record                                | 1.4.4     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-09 Manufacturer's Field Reports            |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Batch Tickets                                 | 1.4.3     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   | 32 12 13     | SD-03 Product Data                            |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Waybills and Delivery Tickets                 | 1.1.3     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Local/Regional Materials                      | 2.2.4     |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-06 Test Reports                            |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Sampling and Testing                          | 3.7       |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   | 32 31 13     | SD-02 Shop Drawings                           |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Fence Assembly                                | 2.1       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Location of Gate, Corner, End, and Pull Posts | 3.2.2.1   | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Gate Assembly                                 | 2.1       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Gate Hardware and Accessories                 | 2.3.6     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Erection/Installation Drawings                | Part 3    | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | SD-03 Product Data                            |           |                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Fence Assembly                                | 2.1       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Gate Assembly                                 | 2.1       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Gate Hardware and Accessories                 | 2.3.6     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Zinc Coating                                  | 2.4.1     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | PVC Coating                                   |           | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Aluminum Alloy Coating                        | 2.1       | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Fabric                                        | 2.2.1     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Stretcher Bars                                | 2.2.6     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|                |                   |              | Concrete                                      | 2.4.2     | G                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |

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|                |                   | 32 31 13     | SD-04 Samples                     |           |                                                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Fabric                            | 2.2.1     | G                                                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Posts                             | 2.2.2     | G                                                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Braces                            | 2.2.3     | G                                                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Line Posts                        | 2.2.2.1.1 | G                                                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Sleeves                           | 2.2.5     | G                                                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Top Rail                          | 2.2.3.1   | G                                                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Bottom Rail                       | 2.2.3.3   | G                                                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Tension Wire                      | 2.2.8     | G                                                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Stretcher Bars                    | 2.2.6     | G                                                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Gate Posts                        | 2.3.1     | G                                                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Gate Hardware and Accessories     | 2.3.6     | G                                                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Padlocks                          | 2.3.5     | G                                                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Wire Ties                         | 2.2.4     | G                                                                    |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-07 Certificates                |           |                                                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Certificates of Compliance        |           |                                                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-08 Manufacturer's Instructions |           |                                                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Fence Assembly                    | 2.1       |                                                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Gate Assembly                     | 2.1       |                                                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Hardware Assembly                 | 2.1       |                                                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Accessories                       | 2.1       |                                                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   | 33 21 00.19  | SD-02 Shop Drawings               |           |                                                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | PUMP CONSTRUCTION                 | 2.1       | G AE                                                                 |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-06 Test Reports                |           |                                                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | FIELD AND ACCEPTANCE TESTS        | 3.2       | G AE                                                                 |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |

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|             |                |             |                                 |           |                     | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION | DATE RCD<br>FROM<br>CONTR    | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION | DATE RCD<br>FRM APPR<br>AUTH |         |
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|             |                | 33 21 00.19 | SD-07 Certificates              |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Installation                    | 3.1       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | SD-09 Manufacturer's Field      |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Reports                         |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | FIELD AND ACCEPTANCE            | 3.2       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | TESTS                           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | SD-10 Operation and Maintenance |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Data                            |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Operations and Maintenance      |           | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Manuals                         |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Spare Parts                     | 3.4       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 33 61 13.19 | SD-02 Shop Drawings             |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Detail Drawings                 | 1.3.1     | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | SD-03 Product Data              |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Support of the Equipment        | 2.1       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Piping and Fittings             | 2.5       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Valves                          | 2.6       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Insulating Flanges              | 2.5.4.1   | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Expansion Joints                | 2.12      | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | SD-04 Samples                   |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Insulated Sections              | 1.3.2     | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | SD-10 Operation and Maintenance |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Data                            |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Valve Manholes and Accessories  | 3.7       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |             | Data Package 2                  | 3.7.1     | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 33 71 02    | SD-02 Shop Drawings             |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|             |                | 33 71 02  | Precast Underground Structures        | 1.5.1     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-03 Product Data                    |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Medium Voltage Cable                  | 2.5       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Medium Voltage Cable Joints           | 2.7       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Medium Voltage Cable Terminations     | 2.6       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Precast Concrete Structures           | 2.14.2.1  | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Sealing Material                      | 2.14.2.4  |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Pulling-In Irons                      | 3.4.3     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Manhole Frames and Covers             | 2.14.3    | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Handhole Frames and Covers            | 2.14.4    | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Composite/Fiberglass Handholes        |           | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Cable Supports                        | 2.15      | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-06 Test Reports                    |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Medium Voltage Cable                  | 2.19.2    | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Qualification and Production Tests    |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Existing Cable Assessment Tests       |           | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Field Acceptance Checks and Tests     | 3.18.1    | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Arc-proofing Test                     | 2.19.1    | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Cable Installation Plan and Procedure | 3.3       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-07 Certificates                    |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Cable splicer/terminator              | 1.5.2     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Cable Installer Qualifications        | 1.5.3     | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                | 35 45 01  | SD-01 Preconstruction Submittals      |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |



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|             |                | 35 45 01  | Contractor Selected           | 2.2       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Manufacturers                 |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-02 Shop Drawings           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Detail Drawings               | 1.4.1     | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-03 Product Data            |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Materials                     | 2.2       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Shipping Bills                | 1.5.1     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Pump Curves                   | 2.1.3     | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Installation and Erection     | 3.1.1     | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Instructions Manual           |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Field Tests                   | 3.2       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | In Water Bearing Lubrication  | 2.4.10.1  | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | In Water Bearing Lubrication  | 2.5       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Grout                         | 2.8       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-04 Samples                 |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Materials                     | 2.2       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-05 Design Data             |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Stress-Relieving Procedure    | 2.3.3.3   | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Baseplates and the Anchoring  | 2.8       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Bolts                         |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | G; AE                         |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | SD-06 Test Reports            |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Factory Test                  | 2.9.4     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | G; AE                         |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Balancing                     | 2.4.7.2   | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |
|             |                |           | Field Test Results            | 3.2.1     | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                         |         |

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|             |                | 35 45 01  | SD-07 Certificates              |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Qualified Welders               | 1.4.2     | G                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Examination Procedure And       | 2.3.4     | G                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Qualification Of The Examiner   |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | SD-10 Operation and Maintenance |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Data                            |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Operation and Maintenance       | 3.1.1     | G AE                                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Instructions Manual             |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                | 40 05 13  | SD-02 Shop Drawings             |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Materials and Equipment         | 2.2       |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | SD-03 Product Data              |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Qualifications                  | 1.4       | G AE                                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Welders                         | 1.4.3     | G AE                                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Waste Water Disposal            | 3.17      | G AE                                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Manufacturer's Field Services   | 3.18      | G AE                                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Delivery, Storage and Handling  | 1.5       | G AE                                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Materials and Equipment         | 2.2       | G AE                                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Installation                    | 1.8.2     | G AE                                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Valve Schedule                  | 2.13.2    | G AE                                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Operator Schedule               | 2.13.12.1 | G AE                                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | SD-06 Test Reports              |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Double Containment Piping Leak  | 3.7       | G AE                                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Detection System                |           |                                        |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Pipe Leakage Tests              | 3.15.3    | G AE                                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Hydrostatic Tests               | 3.15.1    | G AE                                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |
|             |                |           | Pneumatic Tests                 | 3.15.2    | G AE                                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                     |         |

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|             |                | 40 05 13       | Valve Testing                    | 3.15.5    | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Disinfection                     | 3.16.3    | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-07 Certificates               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Plastic Piping System            | 2.7       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Plastic Pipe Installation        | 3.3.9     | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-10 Operation and Maintenance  |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Data                             |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Piping and Appurtenances         | 3.3       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 40 05 13.96    | SD-02 Shop Drawings              |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Pressure Piping                  | 1.6       | G                   |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-03 Product Data               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Welding Operations               | 3.1       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-07 Certificates               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Qualifications                   | 1.5       |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 40 17 26.00 20 | SD-02 Shop Drawings              |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Welding Pressure Piping          | 1.5.1     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-07 Certificates               |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Welding Procedures Qualification | 1.5.3     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Nondestructive Examination       | 1.5.2     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | (NDE) Procedures                 |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | NDE personnel Certification      | 1.5.6.2   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Procedures                       |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Inspector Certification          | 1.5.6.1   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | SD-11 Closeout Submittals        |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |                | Weld Identifications             | 1.5.7.1   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                | 40 60 00       | SD-02 Shop Drawings              |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|             |                | 40 60 00  | Contractor Design Drawings                | 3.6.2     | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Draft As-Built Drawings                   | 3.6.2.1   | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-03 Product Data                        |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Control Drawings; G                       |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Sensors; G                                |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Performance Verification Test<br>(PVT); G |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Factory Test Procedure; G                 |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-06 Test Reports                        |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Factory Test Report                       | 2.12.3    |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Testing, Adjusting and<br>Commissioning   | 3.7.1     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Performance Verification<br>Test(PVT)     | 3.7.2     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Endurance Test                            | 3.7.3     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-07 Certificates                        |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Control and Sensor Wiring                 | 3.4.2.2   |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Ground Rods                               | 3.7.1     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Wiring                                    | 3.2.1     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Installation                              | 3.2.1     |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-10 Operation and Maintenance<br>Data   |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Training Manual                           | 3.8       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Control System                            | 2.1       | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | SD-11 Closeout Submittals                 |           |                     |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|             |                |           | Final As-Built Drawings                   | 3.6.2.2   | G AE                |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|                |                   |              |                                     |           |                                        | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |     | DATE FWD<br>TO APPR<br>AUTH/<br><br>DATE RCD<br>FROM CONTR | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE |                                                            |         |
| (a)            | (b)               | (c)          | (d)                                 | (e)       | (f)                                    | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l) | (m)                                                        | (n)                              | (o)                              | (p)            | (q)                                                        | (r)     |
|                |                   | 41 22 13.14  | SD-02 Shop Drawings                 |           |                                        |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | Overhead Electric Traveling Crane   | 1.6.4     | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | Complete Schematic Wiring Diagram   | 3.3.2     | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | Control System and Network Drawings | 1.6.4     | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | Runway Beams and Gantry Assembly    |           | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | SD-03 Product Data                  |           |                                        |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | Gear Reducers                       | 2.3.7.1   | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | Hoist Brakes                        | 2.3.10    | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | Travel Brakes                       | 2.3.9     | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | Couplings                           | 2.3.11    | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | Load Blocks and Hooks               | 2.3.3     | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | Wheels                              | 2.3.8     | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | Hoists                              | 2.3.1     | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | Sheaves                             | 2.3.5     | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | Commercial Hoist and Trolley Units  |           | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | End Trucks                          | 2.2.5     | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | Bridge Rails                        | 2.2.3     | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | End Stops                           | 2.2.7     | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | Bumpers                             | 2.2.7     | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | Variable Frequency Drives           |           | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |
|                |                   |              | Motors                              | 2.4.1     | G                                      |                               |                          |                          |                      |                      |     |                                                            |                                  |                                  |                |                                                            |         |

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|             |                |             |                                     |           |                     | SUBMIT                        | APPROVAL NEEDED BY | MATERIAL NEEDED BY | ACTION CODE       | DATE OF ACTION | DATE RCD FROM CONTR    | DATE FWD TO OTHER REVIEWER | DATE RCD FROM OTH REVIEWER | ACTION CODE | DATE OF ACTION | DATE RCD FRM APPR AUTH |         |
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|             |                | 41 22 13.14 | Runway Conductor System             | 2.4.8.1   | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Bridge Conductor System             | 2.4.8.2   | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Limit Switches                      | 2.4.6     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Radio Control System                | 2.4.7.2   | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Pendant Pushbutton Station          | 2.4.7.1   | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Pendant Conductor System            | 2.4.7.1.1 | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Controls                            | 2.4.2     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Control Parameter Settings          | 3.3.2     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Capacity Overload Protective Device | 2.4.9     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Painting System                     | 2.5       | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Control System and Network          | 2.4.13.1  | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | SD-05 Design Data                   |           |                     |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Load and Sizing Calculations        | 1.6.5     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | SD-06 Test Reports                  |           |                     |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Hook Proof Test                     |           | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Hook Non-destructive Test (NDT)     | 1.6.2.2.1 | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Post-erection Inspection            | 3.4.1     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Operational Tests                   | 3.4.2     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Hook Tram Measurement               | 3.4.4     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Load Tests                          | 3.4.5     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Crane Rail Survey                   |           |                     |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | SD-07 Certificates                  |           |                     |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Wire Rope                           | 1.6.3     |                     |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Breaking Strength Test Report       | 2.3.4     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |
|             |                |             | Crane Runway System                 | 1.6.3     | G                   |                               |                    |                    |                   |                |                        |                            |                            |             |                |                        |         |

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|                |                   | 41 22 13.14  | Hazardous Material               | 1.6.3     | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Loss of Power Test               | 1.6.3     | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Coupling Alignment Verification  | 1.6.3     | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Record                           |           |                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Overload Test                    | 1.6.3     | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Brake Adjustment Record          | 1.6.3     | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Compliance with Listed           | 1.6.3     | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Standards                        |           |                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Public Domain Software           | 1.6.3     | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Software and Services            | 1.6.3     | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-10 Operation and Maintenance  |           |                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Data                             |           |                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Operation and Maintenance        | 3.6       | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Manuals                          |           |                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-11 Closeout Submittals        |           |                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Disabled Ports, Connectors, and  | 2.4.13.2  | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Interfaces                       |           |                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Network-Capable Control          | 2.4.13    | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Devices                          |           |                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Control System Access Control    | 2.4.13.3  | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Control System Account           | 2.4.13.4  | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Management                       |           |                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Patch Management and Updates     | 2.4.13.9  | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Malware Detection and Protection | 2.4.13.10 | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Wireless Technology Provisions   | 2.4.13.12 | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Control System Inventory         | 2.4.13.13 | G                                      |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |

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|                |                   | 41 22 13.14  | Evaluation Status of Hardware and Software    | 2.4.13.2  | G                                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   | 43 41 45     | SD-02 Shop Drawings                           |           |                                                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | CONDITIONS OF SERVICE / STORAGE TANK SCHEDULE | 1.3       | G AE                                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-03 Product Data                            |           |                                                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Installation                                  | 3.2       | G AE                                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Dimensions                                    | 2.4       | G AE                                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-06 Test Reports                            |           |                                                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | INSPECTION AND TESTING                        | 3.3       | G AE                                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-08 Manufacturer's Instructions             |           |                                                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Instructions                                  |           |                                                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Post Curing Requirements                      |           |                                                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-10 Operation and Maintenance Data          |           |                                                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Operation and Maintenance Manual              | 3.4       | G AE                                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-11 Closeout Submittals                     |           |                                                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Warranty                                      | 1.5       | G AE                                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   | 46 30 00     | SD-02 Shop Drawings                           |           |                                                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Detail Drawings                               | 3.2       | G AE                                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-03 Product Data                            |           |                                                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Chemical Feed System                          | 2.1       | G AE                                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Safety Data Sheets                            | 2.3       | G AE                                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Framed Instructions                           | 3.2.3     | G AE                                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |



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|                |                   | 46 30 00     | Auxiliary Equipment and Spare Parts         | 1.3       | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Factory Test Report                         | 2.2.15    | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-06 Test Reports                          |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Operating Tests                             | 3.3.1     | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Tank Testing                                | 3.3.1.1   | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Controlled Volume Pumps - Operational Tests | 3.3.1.2   | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | System Pressure Tests                       | 3.3.1.3   | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Flow Tests                                  | 3.3.1.4   | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Synchronization Tests                       | 3.3.1.5   | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-07 Certificates                          |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Supplied Chemical                           | 2.3       | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Service Organization                        | 2.2.1     | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Certificates                                | 3.3.3     | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-10 Operation and Maintenance Data        |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Operating Instructions                      | 3.4.2     | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Maintenance Instructions                    | 3.4.3     | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-11 Closeout Submittals                   |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Extended Warranty                           | 2.1.4     | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   | 46 64 30     | SD-01 Preconstruction Submittals            |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Sequence of Construction                    | 3.2       | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-02 Shop Drawings                         |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | System Coordination                         |           | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              |                                             |           |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |

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|                |                   |              |                                                                                                                                                             |             |                                                             | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                                                               | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                            |         |
| (a)            | (b)               | (c)          | (d)                                                                                                                                                         | (e)         | (f)                                                         | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                                                           | (m)                              | (n)                              | (o)            | (p)                  | (q)                                                        | (r)     |
|                |                   | 46 64 30     | General Arrangement and<br>Dimensional Drawings for Pumps,<br>Piping Systems, Cooling Water<br>Intake Structure, and associated<br>equipment and components |             | G                                                           |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-03 Product Data                                                                                                                                          |             |                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | TEMPORARY ASW SYSTEM                                                                                                                                        | 2.3         | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-06 Test Reports                                                                                                                                          |             |                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Temporary ASW System<br>Performance                                                                                                                         |             | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Field Quality Control and<br>Commissioning                                                                                                                  | 3.3         | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-07 Certificates                                                                                                                                          |             |                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Manufacturer's Representative                                                                                                                               |             | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Pumps                                                                                                                                                       |             | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Gate Valves                                                                                                                                                 |             | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Butterfly Valves                                                                                                                                            |             | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Check Valves                                                                                                                                                |             | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Air and Vacuum Valves                                                                                                                                       |             | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Pressure Relief Valve                                                                                                                                       |             | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Magnetic Flow Meter                                                                                                                                         |             | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Pressure Transmitter                                                                                                                                        |             | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Piping                                                                                                                                                      |             | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Fittings and Flanges                                                                                                                                        |             | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Bolts, Nuts, Washers and<br>Fasteners                                                                                                                       |             | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |

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|                |                   |              |                                 |           |                                                         | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION | DATE RCD<br>FROM<br>CONTR    | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION | DATE RCD<br>FRM APPR<br>AUTH |         |
| (a)            | (b)               | (c)          | (d)                             | (e)       | (f)                                                     | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                          | (m)                              | (n)                              | (o)            | (p)                  | (q)                          | (r)     |
|                |                   | 46 64 30     | Gaskets                         |           | G AE                                                    |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Union                           |           | G AE                                                    |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   | 46 80 00     | SD-02 Shop Drawings             |           |                                                         |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Equipment Construction Details  |           | G                                                       |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Equipment Installation          |           | G                                                       |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Wiring Diagram                  |           | G                                                       |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-03 Product Data              |           |                                                         |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Pump Curves                     |           | G                                                       |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Pump Certification              |           | G                                                       |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Product Description             |           | G                                                       |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Product Data                    |           | G                                                       |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-06 Test Reports              |           |                                                         |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Feed Pump Motor Test Results    |           | G                                                       |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Motor Test Results              |           | G                                                       |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Factory Acceptance Test         | 3.2       | G                                                       |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Field Test Report               |           | G                                                       |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-07 Certificates              |           |                                                         |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Parts List                      |           | G                                                       |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Spare Parts List                |           | G                                                       |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Installation Certification      |           | G                                                       |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-10 Operation and Maintenance |           |                                                         |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Data                            |           |                                                         |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Operating And Maintenance       |           | G                                                       |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Manual                          |           |                                                         |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | SD-11 Closeout Submittals       |           |                                                         |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |
|                |                   |              | Warranty                        | 1.7       | G                                                       |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                              |         |

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|                |                   |              |                                  |                |                                        | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                                                            | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                         |         |
| (a)            | (b)               | (c)          | (d)                              | (e)            | (f)                                    | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                                                        | (m)                              | (n)                              | (o)            | (p)                  | (q)                                                     | (r)     |
|                |                   | 46 80 00     | Warranty                         | 1.7            | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Warranty                         | 1.7            | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Warranty                         | 1.7            | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   | 46 81 00     | SD-02 Shop Drawings              |                |                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Equipment Construction Details   |                | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Equipment Installation           |                | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Wiring Diagram                   |                | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-03 Product Data               |                |                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Pump Curves                      |                | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Pump Certification               |                | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Product Description              |                | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Feed Pump Motor Test Results     |                | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Motor Test Results               |                | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Factory Acceptance Test          |                | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Field Test Report                |                | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-07 Certificates               |                |                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Parts List                       |                | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | pare Parts List                  |                | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Installation Certification       |                | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-10 Operation and Maintenance  |                |                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Data                             |                |                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Operating And Maintenance        |                | G                                      |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | Manuals                          |                |                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   | 46 82 00     | SD-01 Preconstruction Submittals |                |                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | MEMBRANES                        |                | G AE                                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              |                                  |                |                                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |

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|                |                   |              |                                                                |                |                                                               | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                                                               | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                            |         |
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|                |                   | 46 82 00     | PERMEATE AND BACKWASH<br>ROTARY PUMPS                          |                | G AE                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | ASSOCIATED COMPRESSED<br>AIR EQUIPMENT                         |                | G AE                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | LOW PRESSURE BLOWERS                                           |                | G AE                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | CLEAN-IN-PLACE CIP SYSTEM                                      |                | G AE                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | MFS ELECTRICAL AND<br>CONTROL SYSTEM                           |                | G AE                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | MONITORING EQUIPMENT                                           |                | G AE                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | PIPING AND VALVES                                              |                | G AE                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | INTERNAL TO AND DIRECTLY<br>ABOVE THE MEMBRANE TANKS           |                |                                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | WITH TERMINATION POINTS AT<br>OR NEAR THE EDGE OF THE<br>TANKS |                |                                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | VALVES AND<br>INSTRUMENTATION DEVICES                          |                | G AE                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | DRAIN PUMPS                                                    |                | G AE                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | EQUIPMENT LIST                                                 |                | G AE                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | MECHANICAL BILL OF<br>MATERIALS                                |                | G AE                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | I/O LIST                                                       |                | G AE                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-02 Shop Drawings                                            |                |                                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | DETAIL DRAWINGS                                                |                | G AE                                                          |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              |                                                                |                |                                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              |                                                                |                |                                                               |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |

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|                |                   |              |                               |                |                                                             | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION |                                                               | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                            |         |
| (a)            | (b)               | (c)          | (d)                           | (e)            | (f)                                                         | (g)                           | (h)                      | (i)                      | (j)                  | (k)                  | (l)                                                           | (m)                              | (n)                              | (o)            | (p)                  | (q)                                                        | (r)     |
|                |                   | 46 82 00     | PIPING AND                    |                | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | INSTRUMENTATION DIAGRAMS      |                |                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | (P&IDS)                       |                |                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | PROCESS FLOW DIAGRAM          |                | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | (PFD)                         |                |                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | GENERAL ARRANGEMENT           |                | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | DRAWINGS                      |                |                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | NETWORK ARCHITECTURE          |                | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | DIAGRAM                       |                |                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | ELECTRICAL DRAWINGS           |                | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | PLC AND CONTROL PANEL         |                | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | LAYOUT DRAWINGS               |                |                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | CONTROL LOGIC AND             |                | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SEQUENCE CHARTS               |                |                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-03 Product Data            |                |                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | AUTOMATIC STRAINERS           |                | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | MEMBRANES                     |                | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | PERMEATE PUMPS AND            |                | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | BACKWASH ROTARY PUMPS         |                |                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | ASSOCIATED COMPRESSED         |                | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | AIR EQUIPMENT                 |                |                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | LOW PRESSURE BLOWERS          |                | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | CLEAN-IN-PLACE CIP SYSTEM     |                | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | MEMBRANE SUPPORT              |                | G AE                                                        |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | STRUCTURES AND FRAMES         |                |                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |
|                |                   |              |                               |                |                                                             |                               |                          |                          |                      |                      |                                                               |                                  |                                  |                |                      |                                                            |         |

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

B72 Resiliency Improvement

CONTRACTOR

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|                |                   | 46 82 00     | MFS ELECTRICAL AND<br>CONTROL SYSTEM                           |            | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | MONITORING EQUIPMENT                                           |            | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | PIPING AND VALVES                                              |            | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | INTERNAL TO AND DIRECTLY<br>ABOVE THE MEMBRANE TANKS           |            |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | WITH TERMINATION POINTS AT<br>OR NEAR THE EDGE OF THE<br>TANKS |            |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | VALVES AND<br>INSTRUMENTATION DEVICES                          |            | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | SD-05 Design Data                                              |            |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | AUTOMATIC STRAINERS                                            |            | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | MEMBRANES                                                      |            | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | PERMEATE PUMPS AND<br>BACKWASH ROTARY PUMPS                    |            | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | ASSOCIATED COMPRESSED<br>AIR EQUIPMENT                         |            | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | LOW PRESSURE BLOWERS                                           |            | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | CLEAN-IN-PLACE CIP SYSTEM                                      |            | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | MEMBRANE SUPPORT<br>STRUCTURES AND FRAMES                      |            | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | MFS ELECTRICAL AND<br>CONTROL SYSTEM                           |            | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              | MONITORING EQUIPMENT                                           |            | G AE                   |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |
|                |                   |              |                                                                |            |                        |                               |                          |                          |                      |                      |                                                            |                                  |                                  |                |                      |                                                         |         |

# SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

B72 Resiliency Improvement

CONTRACTOR

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|                |                   |              |                                   |             |                                                           | SUBMIT                        | APPROVAL<br>NEEDED<br>BY | MATERIAL<br>NEEDED<br>BY | ACTION<br>CODE       | DATE<br>OF<br>ACTION | DATE RCD<br>FROM<br>CONTR    | DATE FWD<br>TO OTHER<br>REVIEWER | DATE RCD<br>FROM OTH<br>REVIEWER | ACTION<br>CODE | DATE<br>OF<br>ACTION |                                                            |         |
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|                |                   | 46 82 00     | PIPING AND VALVES                 |             | G AE                                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | INTERNAL TO AND DIRECTLY          |             |                                                           |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | ABOVE THE MEMBRANE TANKS          |             |                                                           |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | WITH TERMINATION POINTS AT        |             |                                                           |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | OR NEAR THE EDGE OF THE           |             |                                                           |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | TANKS                             |             |                                                           |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | VALVES AND                        |             | G AE                                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | INSTRUMENTATION DEVICE            |             |                                                           |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-06 Test Reports                |             |                                                           |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | MEMBRANE AND HOUSING              |             | G AE                                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | TESTS                             |             |                                                           |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-07 Certificates                |             |                                                           |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | EMPLOYER'S RECORD                 |             | G AE                                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | DOCUMENTS (FOR WELDING)           |             |                                                           |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | WELDING PROCEDURES AND            |             | G AE                                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | QUALIFICATIONS                    |             |                                                           |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | CERTIFICATES                      |             | G AE                                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-08 Manufacturer's Instructions |             |                                                           |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | LESSON PLAN FOR THE               |             | G AE                                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | INSTRUCTION COURSE                |             |                                                           |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SD-10 Operation and Maintenance   |             |                                                           |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | Data                              |             |                                                           |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | REQUIREMENTS FOR DATA             |             | G AE                                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | PACKAGES                          |             |                                                           |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | SPARE PARTS DATA                  |             | G AE                                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |
|                |                   |              | AUTOMATIC STRAINERS               |             | G AE                                                      |                               |                          |                          |                      |                      |                              |                                  |                                  |                |                      |                                                            |         |



## SUBMITTAL REGISTER

CONTRACT NO.

### TITLE AND LOCATION

## B72 Resiliency Improvement

CONTRACTOR

[illegible]

SECTION 01 33 10  
SUBSTITUTIONS  
02/24

PART 1 GENERAL

1.1 RELATED DOCUMENTS

Drawings and general provisions of the Contract, including General and Supplementary Conditions and other Division 01 Specification Sections, apply to this Section.

1.2 GENERAL

A general description for each piece of equipment, material and component scheduled on the drawings and described in the specifications include the salient physical, functional, or performance characteristics of the brand name item that an "equal" item must meet to be acceptable for award. The use of brand name manufactures and model number is utilized as the basis of design are utilized in the preparation of the project documents. "Or equals" are acceptable when the salient characteristics of the scheduled or specified equipment, component or device, which are firm requirements, are met without exception.

1.2.1 References

1.2.1.1 FAR 11.104

FAR 11.104 only refers to "or equal" substitutions to the basis of design listed on the schedule or described in the specifications.

Use of Brand Name or Equal Purchase Descriptions:

- a. While the use of performance specifications is preferred to encourage offerers to propose innovative solutions, the use of brand name or equal purchase descriptions may be advantageous under certain circumstances.
- b. Brand name or equal purchase descriptions must include, in addition to the brand name, a general description of those salient physical, functional, or performance characteristics of the brand name item that an "equal" item must meet to be acceptable for award. Use brand name or equal descriptions when the salient characteristics are firm requirements.
- c. Note: a Justification and Approval (J&A) document is NOT required.

1.2.1.2 FAR 11.105

FAR 11.105 is not intended to be used for proprietary or Items Peculiar to One Manufacturer. (Indicated for Reference Only)

Items Peculiar to One Manufacturer: Agency requirements shall not be written so as to require a particular brand name, product, or a feature of a product, peculiar to one manufacturer, thereby precluding consideration of a product manufactured by another company, unless:

- a. The particular brand name, product, or feature is essential to the

Government's requirements, and market research indicates other companies' similar products, or products lacking the particular feature, do not meet, or cannot be modified to meet, the agency's minimum needs;

- i. The authority to contract without providing for full and open competition is supported by the required justifications and approvals (see 6.302-1); or
  - ii. The basis for not providing for maximum practicable competition is documented in the file (see 13.106-1(b)) or justified (see 13.501) when the acquisition is awarded using simplified acquisition procedures.
- b. The documentation or justification is posted for acquisitions over \$ 25,000. (See 5.102(a)(6).)
- iii. (b) For multiple award schedule orders, see 8.405-6.
- c. Note: a Justification and Approval (J&A) document is IS required.

### 1.3 SUMMARY

Section includes administrative and procedural requirements for substitutions.

### 1.4 DEFINITIONS

Substitutions: Changes in products, materials, equipment, and methods of construction from those required by the Contract Documents and proposed by Contractor.

#### 1.4.1 Substitutions for Cause

Changes proposed by Contractor that are required due to changed Project conditions, such as unavailability of product, regulatory changes, or unavailability of required warranty terms.

#### 1.4.2 Substitutions for Convenience

Changes proposed by Contractor or contracting officer that are not required in order to meet other Project requirements but may offer an advantage to Contractor or contracting officer.

#### 1.4.3 Procurement Substitution Requests

Requests for changes in products, materials, equipment, and methods of construction from those indicated in the Procurement and Contracting Documents, submitted prior to receipt of bids.

#### 1.4.4 Substitution Requests

Requests for changes in products, materials, equipment, and methods of construction from those indicated in the Contract Documents, submitted following Contract award.

### 1.5 QUALITY ASSURANCE

Compatibility of Substitutions: Investigate and document compatibility of proposed substitution with related products and materials. Engage a qualified testing agency to perform compatibility tests recommended by

manufacturers.

## 1.6 PROCUREMENT SUBSTITUTIONS

### 1.6.1 General

By submitting a bid, the Bidder represents that its bid is based on materials and equipment described in the Procurement and Contracting Documents, including Addenda. Bidders are encouraged to request approval of qualifying substitute materials and equipment when the Specifications Sections list materials and equipment by product or manufacturer name.

### 1.6.2 Requests

Requests will be received and considered by the contracting officer when the following conditions are satisfied, as determined by Architect; otherwise requests will be returned without action:

- a. Extensive revisions to the Contract Documents are not required.
- b. Proposed changes are in keeping with the general intent of the Contract Documents, including the level of quality of the Work represented by the requirements therein.
- c. The request is fully documented and properly submitted.

### 1.6.3 Requirements

The requested substitution for convenience (not for cause) needs to meet the following requirements:

- a. Substitutions for convenience which provide benefit solely to the contractors shall be at the contractors risk & cost.
- b. Be of the same efficiency and not require additional energy cost to operate the equipment, systems and associated components. The substitution cannot decrease the energy use of the proposed substitution item, but increase the energy of other systems.
- c. Be of roughly the same dimensions, weight, physical size, electrical characteristics, construction, etc. as the basis of design listed in the specifications and identified on the drawing schedules.
- d. The performance of the proposed substitution must be equal to or greater than the basis of design listed in the specifications and identified on the drawing schedules.
- e. The general contractor shall be responsible for all the additional costs of design, engineering, construction, project management, etc., to design, provide, furnish & install the proposed substitution over and above the cost related to the basis of design listed in the specifications and identified on the drawing schedules.
- f. The subcontractor shall coordinate with the general contractor and any other associated subcontractors to address any costs related to the proposed substitution.
- g. Note: there may be conditions or circumstances such as existing buildings, mechanical rooms or confined or limited architectural

spaces and areas where in the event the proposed substitution will not be acceptable as an option to the basis of design listed in the specifications and identified on the drawing schedules.

## 1.7 SUBMITTALS

### Procurement Substitution Request; G

Submit documentation identifying product or fabrication or installation method to be replaced. Include Specification Section number and title and Drawing numbers and titles.

Submit to Construction Manager for the Project. A procurement Substitution Request must be made in writing in compliance with the following requirements:

#### 1.7.1 Requests

Request for substitution of equipment, component and devices may be submitted by each contractor after contract award as part of the submittal process for review and comment by the Architect or Engineer of the project. Request for substitutions for Proprietary Items shall only be accepted if the listed product is no longer available or cannot be procured during the project construction schedule.

The final determination of if the substitution is acceptable is by the Contracting Officer's Representative and by the Designer of Record. All reviews of substitution requests shall be based upon the requirements in this section in terms of the substituted products performance, efficiency and fit within the constraints of the project. The burden is on the contractor to confirm these features and coordinate the product within the physical constraints of the project. The decision shall be final.

#### 1.7.2 Submittal Format

Submit electronically, copies of each written Procurement Substitution Request using the Substitution Request Form, an attachment to this section.

Identify the product or the fabrication or installation method to be replaced in each request. Include related Specifications Sections and drawing numbers. Provide complete documentation on both the product specified and the proposed substitute, including the following information as appropriate:

Point-by-point comparison of specified and proposed substitute product data, fabrication drawings, and installation procedures.

Copies of current, independent third-party test data of salient product or system characteristics.

Samples where applicable or when requested by Architect.

Detailed comparison of significant qualities of the proposed substitute with those of the Work specified. Significant qualities may include attributes such as performance, weight, size, durability, visual effect, sustainable design characteristics, warranties, and specific features and requirements indicated. Indicate deviations, if any, from the Work specified.

Material test reports from a qualified testing agency indicating and interpreting test results for compliance with requirements indicated.

Research reports, where applicable, evidencing compliance with building code in effect for Project.

Coordination information, including a list of changes or modifications needed to other parts of the Work which will become necessary to accommodate the proposed substitute. Cost associated with such changes or modifications will be the responsibility of the Contractor.

- a. Provide certification by manufacturer that the substitute proposed is equal to or superior to that required by the Procurement and Contracting Documents, and that its in-place performance will be equal to or superior to the product or equipment specified in the application indicated.
- b. Bidder, in submitting the Procurement Substitution Request, waives the right to additional payment or an extension of Contract Time because of the failure of the substitute to perform as represented in the Procurement Substitution Request.

#### Substitution Requests; G

Submit three copies of each request for consideration. Identify product or fabrication or installation method to be replaced. Include Specification Section number, title, and Drawing numbers and titles.

1. Substitution Request Form: See attachment to this section.
2. Documentation: Show compliance with requirements for substitutions and the following, as applicable:
  - a. Statement indicating why specified product, fabrication or installation could not be provided, if applicable.
  - b. Coordination information, including a list of changes or revisions needed to other parts of the Work and to construction performed by contracting officer and separate contractors that will be necessary to accommodate proposed substitution.
  - c. Detailed comparison of significant qualities of proposed substitution with those of the Work specified. Include annotated copy of applicable Specification Section. Significant qualities may include attributes such as performance, weight, size, durability, visual effect, sustainable design characteristics, warranties, and specific features and requirements indicated. Indicate deviations, if any, from the Work specified.
  - d. Product Data, including drawings and descriptions of products and fabrication and installation procedures.
  - e. Samples, where applicable or requested.
  - f. Certificates and qualification data, where applicable or requested.
  - g. List of similar installations for completed projects with project names and addresses and names and addresses of architects.

- h. Material test reports from a qualified testing agency indicating and interpreting test results for compliance with requirements indicated.
- i. Research reports evidencing compliance with building code in effect for Project.
- j. Detailed comparison of Contractor's construction schedule using proposed substitution with products specified for the Work, including effect on the overall Contract Time. If specified product or method of construction cannot be provided within the Contract Time, include letter from manufacturer, on manufacturer's letterhead, stating date of receipt of purchase order, lack of availability, or delays in delivery.
- k. Cost information, including a proposal of change, if any, in the Contract Sum.
- l. Contractor's certification that proposed substitution complies with requirements in the Contract Documents except as indicated in substitution request, is compatible with related materials, and is appropriate for applications indicated.
- m. Contractor's waiver of rights to additional payment or time that may subsequently become necessary because of failure of proposed substitution to produce indicated results.

#### 1.7.3 Architect's Action

Architect may request additional information or documentation necessary for evaluation of the Procurement Substitution Request. Architect will notify all bidders of acceptance of the proposed substitute by means of an Addendum to the Procurement and Contracting Documents.

#### 1.7.4 Architect, Engineer or Construction Manager's Action

If necessary, there will be a request additional information or documentation for evaluation within seven days of receipt of a request for substitution. Notification will be provided to Contractor of acceptance or rejection of proposed substitution within 15 days of receipt of request, or seven days of receipt of additional information or documentation, whichever is later.

- a. Forms of Acceptance: =Architect's Supplemental Instructions for minor changes in the Work.
- b. Use product specified if Architect does not issue a decision on use of a proposed substitution within time allocated.

#### 1.7.5 Conditions

The Design Professional will consider Contractor's request for substitution when the following conditions are satisfied. If the following conditions are not satisfied, the Design Professional will return requests without action, except to record noncompliance with these requirements:

- a. Requested substitution offers the government a substantial advantage in cost, time, energy conservation, or other considerations, after

deducting additional responsibilities the government must assume. Contractor's additional responsibilities may include compensation to Architect & Engineer for redesign and evaluation services, increased cost of other construction by the government, and similar considerations.

- b. Requested substitution does not require any revisions to the Contract Documents.
- c. Requested substitution is consistent with the Contract Documents.
- d. Requested substitution will not adversely affect project cost.
- e. Requested substitution will not adversely affect Contractor's construction schedule.
- f. Requested substitution is consistent with the Contract Documents and will produce indicated results.
- g. Requested substitution will provide the specified warranty.
- h. Substitution request is fully documented and properly submitted.
- i. Requested substitution has received necessary approvals of authorities having jurisdiction.
- j. Requested substitution is compatible with other portions of the Work.
- k. Requested substitution has been coordinated with other portions of the Work.
- l. Requested substitution provides specified warranty.
- m. If requested substitution involves more than one contractor, requested substitution has been coordinated with other portions of the Work, is uniform and consistent, is compatible with other products, and is acceptable to all contractors involved.

#### 1.7.6 Approval

Architect, Engineer or Construction Manager's approval of a substitute during bidding does not relieve Contractor of the responsibility to submit required shop drawings and to comply with all other requirements of the Contract Documents.

#### 1.8 QUALITY ASSURANCE

Compatibility of Substitutions: Investigate and document compatibility of proposed substitution with related products and materials. Engage a qualified testing agency to perform compatibility tests recommended by manufacturers.

#### 1.9 PROCEDURES

Coordination: Revise or adjust affected work as necessary to integrate work of the approved substitutions.



## PART 2 PRODUCTS

### 2.1 SUBSTITUTIONS

#### 2.1.1 Substitutions for Cause

Submit requests for substitution immediately on discovery of need for change, but not later than 15 days prior to time required for preparation and review of related submittals.

Conditions: Architect, Engineer or Construction Manager will consider Contractor's request for substitution when the following conditions are satisfied. If the following conditions are not satisfied, Architect will return requests without action, except to record noncompliance with these requirements:

- a. Requested substitution is consistent with the Contract Documents and will produce indicated results.
- b. Substitution request is fully documented and properly submitted.
- c. Requested substitution will not adversely affect Contractor's construction schedule.
- d. Requested substitution has received necessary approvals of authorities. having jurisdiction.
- e. Requested substitution is compatible with other portions of the Work.
- f. Requested substitution has been coordinated with other portions of the Work.
- g. Requested substitution provides specified warranty.
- h. If requested substitution involves more than one contractor, requested substitution has been coordinated with other portions of the Work, is uniform and consistent, is compatible with other products, and is acceptable to all contractors involved.

#### 2.1.2 Substitutions for Convenience

Architect, Engineer or Construction Manager will consider requests for substitution if received within 30 days after the Notice of Award. Requests received after that time may be considered or rejected at discretion of Architect, Engineer or Construction Manager.

Architect, Engineer or Construction Manager will consider Contractor's request for substitution when the following conditions are satisfied. If the following conditions are not satisfied, Architect will return requests without action, except to record noncompliance with these requirements:

- a. Requested substitution offers the government a substantial advantage in cost, time, energy conservation, or other considerations, after deducting additional responsibilities Owner must assume. Owner's additional responsibilities may include compensation to Architect for redesign and evaluation services, increased cost of other construction by the government, and similar considerations.
- b. Requested substitution does not require extensive revisions to the

Contract Documents.

- c. Requested substitution is consistent with the Contract Documents and will produce indicated results.
- d. Substitution request is fully documented and properly submitted.
- e. Requested substitution will not adversely affect Contractor's construction schedule.
- f. Requested substitution has received necessary approvals of authorities having jurisdiction.
- g. Requested substitution is compatible with other portions of the Work.
- h. Requested substitution has been coordinated with other portions of the Work.
- i. Requested substitution provides specified warranty.
- j. If requested substitution involves more than one contractor, requested substitution has been coordinated with other portions of the Work, is uniform and consistent, is compatible with other products, and is acceptable to all contractors involved.

PART 3 EXECUTION - NOT USED

## SUBSTITUTION REQUEST COVER SHEET

This cover sheet is required for all proposed substitutions including "or equals".

Contractor: \_\_\_\_\_

Subcontractor: \_\_\_\_\_

This Request Reference Number : \_\_\_\_\_

Substitution Summary: \_\_\_\_\_

Specification Section and Paragraph Number: \_\_\_\_\_

Contract Drawing and Detail Reference: \_\_\_\_\_

Date of This Request: \_\_\_\_\_

This Request Prepared By: \_\_\_\_\_

Conditions: Indicate all conditions that apply to this proposed substitution:

- ☐ Substitution required because specified item is no longer available.
- ☐ Substitution required because Contractor believes specified item is incorrect, inappropriate, or incompatible.
- ☐ Substitution recommended because it offers the Owner substantial advantage in:
  - ☐ Quality
  - ☐ Time
  - ☐ Cost
  - ☐ This is an "or equal".

Evidence: Indicate evidence attached:

- ☐ Tabulated side by side comparison of specified item and proposed substitution directly comparing each feature, characteristic, and performance.
- ☐ Manufacturer's product data for both specified item and proposed substitution. Origin of all information included on tabulated side by side comparison is highlighted.
- ☐ Details showing how the proposed substitution interfaces with adjacent work.
- ☐ Certification that warranty, if any required, will be provided as required.
- ☐ Certification that this proposed substitution is coordinated with all related and adjacent work.
- ☐ Complete cost change information.
- ☐ Higher cost to Owner as stated in cost change information
- ☐ No change in cost to Owner
- ☐ Lower cost to Owner, credit to Owner as stated in cost change information

Deviations from Contract Documents: Itemize all deviations from Contract Documents if this proposed substitution is approved.

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The undersigned attests that the undersigned has carefully examined this entire submission and that the requirements of the Contract Documents have been met.

Should this alternate material or equipment fail to meet the expectations of the owner within the mandated design liability period, this contractor shall replace the material and or equipment with the original specified items at no additional cost to the owner. This Subcontractor shall be responsible for all collateral impacts to the project for replacement of such deviations and substitutions.

Firm: \_\_\_\_\_

By: \_\_\_\_\_

Signature : \_\_\_\_\_

Date: \_\_\_\_\_

Printed or Typed Name Title:

\_\_\_\_\_

SECTION 01 35 26.00 22

GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME)  
05/24

PART 1 GENERAL

This Section applies to Design-Bid-Build projects at the Portsmouth Naval Shipyard and AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING  
ENGINEERS (ASHRAE)

ASHRAE 52.2 (2017) Method of Testing General  
Ventilation Air-Cleaning Devices for  
Removal Efficiency by Particle Size

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ANSI/ASSP A10.34 (2021) Protection of the Public on or  
Adjacent to Construction Sites

ANSI/ASSP A10.44 (2020) Control of Energy Sources  
(Lockout/Tagout) for Construction and  
Demolition Operations

ASTM INTERNATIONAL (ASTM)

ASTM D6245 (2012) Using Indoor Carbon Dioxide  
Concentrations to Evaluate Indoor Air  
Quality and Ventilation

ASTM D6345 (2010) Standard Guide for Selection of  
Methods for Active, Integrative Sampling  
of Volatile Organic Compounds in Air

ASTM F855 (2020) Standard Specifications for  
Temporary Protective Grounds to Be Used on  
De-energized Electric Power Lines and  
Equipment

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1048 (2016) Guide for Protective Grounding of  
Power Lines

IEEE C2 (2023) National Electrical Safety Code

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z89.1 (2014; R 2019) American National Standard  
for Industrial Head Protection

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA Z535.2 (2011; R 2017) Environmental and Facility  
Safety Signs

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 241 (2022) Standard for Safeguarding  
Construction, Alteration, and Demolition  
Operations

NFPA 306 (2024) Standard for the Control of Gas  
Hazards on Vessels

NFPA 51B (2024) Standard for Fire Prevention During  
Welding, Cutting, and Other Hot Work

NFPA 70 (2023) National Electrical Code

NFPA 70E (2024) Standard for Electrical Safety in  
the Workplace

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION  
(SMACNA)

ANSI/SMACNA 008 (2007) IAQ Guidelines for Occupied  
Buildings Under Construction, 2nd Edition

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety -- Safety and Occupational  
Health (SOH) Requirements

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910 Occupational Safety and Health Standards

29 CFR 1915 Confined and Enclosed Spaces and Other  
Dangerous Atmospheres in Shipyard  
Employment

29 CFR 1926 Safety and Health Regulations for  
Construction

CPL 2.100 (1995) Application of the Permit-Required  
Confined Spaces (PRCS) Standards, 29 CFR  
1910.146

ATTACHMENT A "CONTRACTOR CRANE, LHE, CONSTRUCTION  
EQUIPMENT, AND RIGGING GEAR REQUIREMENTS"

ATTACHMENT B "NAVFAC DIG SAFE UTILITY LOCATE  
REQUEST FORM"

ATTACHMENT C "CONTRACTOR DIG-SAFE CERTIFICATION FORM"

ATTACHMENT D "CONTRACTOR UTILITY LOCATE - PRE-EXCAVATION  
SAFETY CHECKLIST"

|              |                                                          |
|--------------|----------------------------------------------------------|
| ATTACHMENT E | "CONTRACTOR DAILY CHECKLIST FOR<br>TRENCHING/EXCAVATION" |
| ATTACHMENT F | "UNDERGROUND UTILITY STRIKE REVIEW CHECKLIST"            |
| ATTACHMENT G | "DIG SAFE COLOR CODING SYSTEM"                           |

The attachments are included at the end of this Section. If the attachments are missing, notify the Contracting Officer.

## 1.2 DEFINITIONS

The following definitions are for the convenience of the reader. If there is a referenced document in the text of this specification section, that is the document that should define terms for that paragraph. If further clarification is needed, contact the Contracting Officer.

### 1.2.1 Site Safety and Health Officer (SSHO)

A Contractor Employee that is responsible for overseeing and ensuring implementation of the prime Contractor's Safety and Occupational Health (SOH) program according to the Contract, EM 385-1-1, applicable federal, state, and local requirements.

#### 1.2.1.1 Level One SSHO

A designated employee with full-time SOH responsibility that meets and follows the requirements of EM 385-1-1.

#### 1.2.1.2 Level Two SSHO

A designated employee with Level Two SSHO responsibility that meets and follows the requirements of EM 385-1-1. Level Two SSHOs cannot be assigned to projects that have a residual Risk Assessment Code (RAC) of high or extremely high.

#### 1.2.1.3 Level Three SSHO

A designated Qualified Person or Competent Person with SOH responsibility that meets and follows the requirements of EM 385-1-1. Level 3 SSHOs cannot be assigned to projects that have a residual RAC of high or extremely high.

#### 1.2.1.4 Alternate SSHO

An employee that meets the definition of the contract-required level SSHO, but is not the primary SSHO.

### 1.2.2 Competent Person (CP)

The CP is a person designated in writing, who, through training, knowledge, and experience, is capable of identifying, evaluating, and addressing existing and predictable hazards in the working environment or working conditions that are unsanitary, hazardous, or dangerous to personnel, and who has authorization to take prompt corrective measures to eliminate them.

### 1.2.3 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

### 1.3 SUBMITTALS

Government Acceptance or Approval does not remove responsibility from the Contractor and subcontractors for their actions or liability.

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-01 Preconstruction Submittals

Accident Prevention Plan; G

Indoor Air Quality (IAQ) Management Plan; G

#### SD-06 Test Reports

Monthly Exposure Reports; G

Accident Reports; G

LHE Inspection Reports; G

#### SD-07 Certificates

Contractor Safety Self-Evaluation Checklist; G

Standard Lift Plan; G

Critical Lift Plan; G

Activity Hazard Analysis (AHA); G

Hot Work Permit; G

Certificate of Compliance; G

### 1.4 PUBLIC HEALTH EMERGENCIES

In the event of a declared public health emergency, follow safety precautions as required by the Occupational Safety and Health Administration (OSHA) [www.osha.gov](http://www.osha.gov), the Centers for Disease Control and Prevention (CDC) [www.cdc.gov](http://www.cdc.gov), and as required by federal, state and local requirements.

### 1.5 MONTHLY EXPOSURE REPORTS

Provide a Monthly Exposure Report by the fifth of each month. This report is a compilation of employee-hours worked each month for all site workers, both Prime and subcontractor. Failure to submit the report may result in rejection of the progress payment.



## 1.6 CONTRACTOR SAFETY SELF-EVALUATION CHECKLIST

The Contractor must complete the Contractor Safety Self-Evaluation Checklist monthly and submit with each request for payment voucher. This submission is required monthly even when a payment voucher is not requested. An acceptable score of 90 or greater is required. Failure to submit the completed safety self-evaluation checklist or achieve a score of at least 90 may result in rejection of the voucher. The Contractor Safety Self-Evaluation Checklist can be found on the Whole Building Design Guide website at <https://www.wbdg.org/dod/ufgs/ufgs-forms-graphics-tables>

## 1.7 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this Contract, comply with the most recent edition of USACE EM 385-1-1, and applicable Federal, State, and local laws, ordinances, criteria, rules, and regulations at the date of the Solicitation for this Contract. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

### 1.7.1 Subcontractor Safety Requirements

For this Contract, neither Contractor nor any Subcontractor may enter into Contract with any Subcontractor that fails to meet the following requirements. The term Subcontractor in this and the following paragraphs means any entity holding a Contract with the Contractor or with a Subcontractor at any tier.

#### 1.7.1.1 Experience Modification Rate (EMR)

Subcontractors on this Contract must have an effective EMR less than or equal to 1.10, as computed by the National Council on Compensation Insurance (NCCI) or if not available, as computed by the state agency's rating bureau in the state where the Subcontractor is registered, when entering into a subcontract agreement with the Prime Contractor or a Subcontractor at any tier. The Prime Contractor may submit a written request for additional consideration to the Contracting Officer where the specified acceptable EMR range cannot be achieved. Relaxation of the EMR range will only be considered for approval on a case-by-case basis for special conditions and must not be anticipated as tacit approval. Contractor's Site Safety and Health Officer (SSHO) must collect and maintain the certified EMR ratings for all subcontractors on the project and make them available to the Government at the Government's request.

#### 1.7.1.2 OSHA Days Away from Work, Restricted Duty, or Job Transfer (DART) Rate

Subcontractors on this Contract must have a DART rate, calculated from the most recent, complete calendar year, less than or equal to 3.4 when entering into a subcontract agreement with the Prime Contractor or a Subcontractor at any tier. The OSHA Dart Rate is calculated using the following formula:

$$(N/EH) \times 200,000$$

Where:

N = number of injuries and illnesses with days away, restricted work, or job transfer

EH = total hours worked by all employees during most recent, complete calendar year

200,000 = base for 100 full-time equivalent workers (working 40 hours per week, 50 weeks per year)

The Prime Contractor may submit a written request for additional consideration to the Contracting Officer where the specified acceptable OSHA Dart rate range cannot be achieved for a particular subcontractor. Relaxation of the OSHA DART rate range will only be considered for approval on a case-by-case basis for special conditions and must not be anticipated as tacit approval. Contractor's Site Safety and Health Officer (SSHO) must collect and maintain self-certified OSHA DART rates for all subcontractors on the project and make them available to the Government at the Government's request.

## 1.8 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

### 1.8.1 Site Safety and Health Officer (SSHO)

#### 1.8.1.1 Qualifications of SSHO

All SSHOs will have met the training, experience requirements identified in the EM 385-1-1 and this Contract.

#### 1.8.1.2 Duties of SSHO

All SSHOs will carry out the roles and responsibilities as identified in this Contract and the EM 385-1-1. All SSHOs must be designated on an ENG Form 6282, provided by the Contracting Officer. SSHO(s) are subject to dismissal if their required duties are not being effectively carried out. If the SSHO is dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out.

#### 1.8.1.3 Safety Meetings

The SSHO must conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide pertinent Safety and Occupational Health (SOH) training and motivation. The SSHO must conduct meetings at least once a month for all supervisors at the project location. The SSHO, supervisors, or foremen must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the Contracting Officer on request. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

### 1.8.2 Roles and Responsibilities of Prime Contractor and SSHO

The Prime Contractor and SSHO must ensure that the requirements of all applicable OSHA and EM 385-1-1 are met for the project. The Prime

Contractor must ensure an SSHO or an equally qualified Alternate SSHO(s) is at the worksite at all times to implement and administer the Contractor's safety program and Government accepted Accident Prevention Plan. If the required SSHO has to temporarily (that is, up to 24 hours / 1 day) leave the site of work due to unforeseen or emergency situations, a Level One, Two, or Three SSHO may be used in the interim and must be on the site of work at all times when work is being performed.

If the SSHO must be off-site for a period longer than 24 hours / 1 day, a qualified alternate that meets the contract requirements must be onsite.

a. Prime contractor must ensure all SSHOs:

- (1) Are designated on an ENG Form 6282.
- (2) Meet minimum training and experience requirements identified in EM 385-1-1.
- (3) Execute roles and responsibilities identified in EM 385-1-1.

#### 1.8.3 Additional Requirements

The SSHO must not serve as the Quality Control Manager. The SSHO must not serve as the Superintendent.

The SSHO must be a direct employee of the Prime Contractor, not an employee of a subcontractor.

#### 1.8.4 Competent Person for Confined Space Entry

Provide a Confined Space Competent Person (CP) who meets the requirements of EM 385-1-1 and herein. The CP for Confined Space Entry must supervise the entry into each confined space in accordance with EM 385-1-1.

Since this work involves operations that handle combustible or hazardous materials, this person must have the ability to understand and follow through on the air sampling, Personal Protective Equipment (PPE), and instructions of a Marine Chemist, Coast Guard authorized persons, or Certified Industrial Hygienist. Confined space and enclosed space work must comply with NFPA 306, 29 CFR 1915, "Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment," or as applicable in 29 CFR 1910 for general industry, 29 CFR 1926 for construction.

#### 1.8.5 Qualified Trainer Requirements

Individuals qualified to instruct the 40 hour Contract safety awareness course, or portions thereof, must meet the definition of a Competent Person Trainer as defined in the EM 385-1-1, and, at a minimum, possess a working knowledge of the following subject areas: EM 385-1-1, Electrical Standards, Lockout/Tagout, Fall Protection, Confined Space Entry for Construction, Excavation, Trenching and Soil Mechanics, and Scaffolds in accordance with 29 CFR 1926.

Instructors are required to:

- a. Prepare class presentations that cover construction-related safety requirements.
- b. Ensure that all attendees attend all sessions by using a class roster

signed daily by each attendee. Maintain copies of the roster for at least five (5) years. This is a certification class and must be attended 100 percent. In cases of emergency where an attendee cannot make it to a session, the attendee can make it up in another class session for the same subject.

- c. Update training course materials whenever an update of the EM 385-1-1 becomes available.
- d. Provide a written exam of at least 50 questions. Students are required to answer 80 percent correctly to pass.
- e. Request, review, and incorporate student feedback into a continuous course improvement program.

#### 1.8.6 Preconstruction Meeting

For additional information on the Preconstruction Meeting, refer to Section 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS (PWD ME).

- a. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the Preconstruction Meeting. This includes the Project Superintendent, Site Safety and Occupational Health Officer, Quality Control Manager, and any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program, and procedures associated with it).
- b. During the Preconstruction Meeting, discuss the details of the submitted APP to include incorporated plans, programs, procedures, and a listing of anticipated AHAs that will be developed and implemented during the performance of the Contract. This list of proposed AHAs will be reviewed and an agreement will be reached between the Contractor and the Contracting Officer as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays. Refer to paragraph ACCIDENT PREVENTION PLAN (APP) for the required timeline for submission of the APP.

### 1.9 ACCIDENT PREVENTION PLAN

#### 1.9.1 Accident Prevention Plan

Submit the Accident Prevention Plan (APP) for review and acceptance by the Government within 15 days from receiving the Notice to Proceed. A Competent Person must prepare the written job-specific APP. Prepare the APP in accordance with the format and requirements of EM 385-1-1, Eng Form 6293, and herein. The APP must be job-specific and address any unusual or unique aspects of the project or activity for which it is written. The APP must interface with the Contractor's overall safety and health program referenced in the APP in the applicable APP element, and made site-specific. Describe the methods to evaluate past safety performance of potential subcontractors in the selection process. Also, describe innovative methods used to ensure and monitor safe work practices of subcontractors. The Government considers the Prime Contractor to be the "controlling employer" for all worksite safety and health of the subcontractors. Contractors are responsible for informing their subcontractors of the safety provisions under the terms of the Contract

and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed in accordance with the APP and ENG Form 6293 Accident Prevention Plan Worksheet. The SSHO(s) must provide and maintain the APP and a log of signatures by each subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the Prime Contractor must provide an interpreter.

Deficiencies in the submitted APP, identified during the Government's review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to begin until an APP is established that is acceptable to and fully approved by the Government. Once reviewed and accepted by the Contracting Officer, the APP and attachments will be enforced as part of the Contract. Disregarding the provisions of this Contract or the accepted APP is cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. The Contractor shall continuously review and amend the APP, as necessary, throughout the life of the Contract. Changes to the accepted APP must be made with the knowledge and concurrence of the Contracting Officer, project superintendent, SSHO, and Quality Control Manager. Incorporate unusual or high-hazard activities not identified in the original APP as they are discovered. Should any severe hazard exposure (i.e. imminent danger) become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the Contracting Officer within 24 hours of discovery. Eliminate and remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ANSI/ASSP A10.34), and the environment.

#### 1.9.2 Names and Qualifications

Provide plans in accordance with the requirements outlined in EM 385-1-1, including the following:

- a. Names and qualifications (resumes including education, training, experience, and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer(s) and other competent and qualified personnel to be used. Specify the duties of each position.
- b. As a minimum, designate and submit qualifications of Competent Persons (CP) for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use and maintenance. Designate and submit qualifications for additional CPs as applicable to the work performed under this Contract.

#### 1.9.3 Plans

Provide plans in the APP in accordance with the requirements outlined in EM 385-1-1, including, but not limited to, the following:

#### 1.9.3.1 Standard Lift Plan (SLP)

Plan all lifts to avoid situations where the operator cannot maintain safe control of the lift. Prepare a written SLP in accordance with EM 385-1-1 for every lift or series of lifts (if duty cycle or routine lifts are being performed). The SLP must be developed, reviewed, and accepted by all personnel involved in the lift. Keep a copy of the SLP on the Load Handling Equipment (LHE) for the specific approved lift(s). Keep a copy of every SLP for a minimum of three months. Use the ENG Form 6203 (Crane Standard Lift Plan) from EM 385-1-1.

#### 1.9.3.2 Critical Lift Plan

Provide a Critical Lift Plan as required by EM 385-1-1. Before any critical lift, a CP or QP must prepare a Critical Lift Plan and submit it to the Contracting Officer or Contracting Officer's Representative for acceptance. Use the ENG Form 6213 (Load-Handling Equipment Crane Operation Critical Lift Plan) from EM 385-1-1.

#### 1.9.3.3 Fall Protection and Prevention Plan

Provide a Fall Protection and Prevention Plan as required by EM 385-1-1. A Fall Protection and Prevention Plan is required when employees are working at heights or exposed to fall hazards. The plan must be developed by a CP or QP and updated as conditions change, but at least annually.

#### 1.9.3.4 Excavation and Trenching Plan

Identify the safety and health aspects of excavation and provide and prepare the plan in accordance with EM 385-1-1 and Section 31 00 00.00 22 EARTHWORK.

#### 1.9.3.5 Lead, Cadmium, and Chromium Compliance Plan

Identify the safety and health aspects of work involving lead, cadmium, and chromium, and prepare in accordance with Section 02 83 00.00 22 MANAGEMENT OF LEAD, CADMIUM, AND CHROMIUM DURING RENOVATION, DEMOLITION, REMOVAL, AND ABATEMENT (PNS PROJECTS).

#### 1.9.3.6 Asbestos Hazard Abatement Plan

Identify the safety and health aspects of asbestos work, and prepare in accordance with Section 02 82 00.00 22 ASBESTOS REMEDIATION (PNS PROJECTS).

#### 1.9.3.7 Site Demolition Plan

Identify the safety and health aspects, and prepare in accordance with Section 02 41 00 DEMOLITION and referenced sources. Include engineering survey as applicable.

#### 1.10 ACTIVITY HAZARD ANALYSIS (AHA)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, Subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before submitting to the Contracting

Officer. AHAs must be signed by the SSHO, Superintendent, QC Manager, and the subcontractor Foreman performing the work. Format the AHA in accordance with EM 385-1-1 or as directed by the Contracting Officer. Submit the AHA for review at least 15 working days prior to the start of each activity task, or DFO. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel, and the control measures to be implemented.

AHAs must identify competent persons required for phases involving high risk activities, including confined entry, crane and rigging, excavations, trenching, electrical work, fall protection, and scaffolding.

#### 1.10.1 AHA Management

Review the AHA list periodically (at least monthly) at the Contractor held Safety Meeting, and update as necessary when procedures, scheduling, or hazards change. Use the AHA during daily inspections by the SSHO to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

#### 1.10.2 AHA Signature Log

Each employee performing work as part of an activity, task, or DFO must review the AHA for that work and sign a signature log specifically maintained for that AHA prior to starting work on that activity. The SSHO must maintain a signature log on site for every AHA. Provide employees whose primary language is other than English with an interpreter to ensure a clear understanding of the AHA and its contents.

#### 1.11 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

#### 1.12 EMERGENCY MEDICAL TREATMENT

Contractors must arrange for their own emergency medical treatment in accordance with EM 385-1-1. Government has no responsibility to provide emergency medical treatment.

#### 1.13 NOTIFICATIONS AND REPORTS

##### 1.13.1 Mishap Notification

Notify the Contracting Officer in accordance with the EM 385-1-1 Accident Reporting Timeline.

| Table<br>Accident Reporting Required Timeline |                  |                                                                        |
|-----------------------------------------------|------------------|------------------------------------------------------------------------|
| Accident Type                                 | Notify KO or COR | Complete Final Accident Report on<br>ENG 3394 and provide to KO or COR |
|                                               |                  |                                                                        |

| Table<br>Accident Reporting Required Timeline                                                  |                                          |               |
|------------------------------------------------------------------------------------------------|------------------------------------------|---------------|
| Fatality, in-patient hospitalization, amputation, eye loss, or property damage over \$600,000. | Immediately, no later than (NLT) 8 Hours | Within 7 Days |
| All other accidents and near misses                                                            | Immediately no later than (NLT) 24 Hours | Within 7 Days |

Within notification include Contractor name; Contract title; type of Contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any mishap.

#### 1.13.2 Accident Reports

- a. Conduct an accident investigation for recordable injuries and illnesses, property damage, and near misses as defined in EM 385-1-1, to establish the root cause(s) of the accident. Complete the applicable NAVFAC Contractor Significant Incident Report (CSIR), and electronically submit via the NAVFAC Enterprise Safety Applications Management System (ESAMS). Complete and submit an accident investigation report in ESAMS within 7 days for accidents as defined by EM 385-1-1. Complete the investigation report within 30 days. Accidents must include a written report submitted as an attachment in ESAMS using the following outline:

- (1) Summary description to include:

- (a) Process

- (b) Findings

- (c) Outcomes

- (2) Root Cause

- (3) Direct Factors

- (4) Indirect and Contributing Factors

- (5) Corrective Actions

- (6) Recommendations

All accidents are reportable, regardless of whether or not it is recordable.

- b. Near Misses: For Navy projects, complete the applicable documentation



in NAVFAC Contractor Incident Reporting System (CIRS), and electronically submit via the NAVFAC Enterprise Safety Applications Management System (ESAMS). The Contracting Officer will provide the Contractor with the required forms. Near miss reports are considered positive and proactive Contractor safety management actions.

- c. Conduct an accident investigation for any load handling equipment accident (including rigging gear accidents) to establish the root cause(s) of the accident. Complete the LHE Accident Report (Crane and Rigging Accident Report) form and provide the report to the Contracting Officer within 30 calendar days of the accident. Do not proceed with crane operations until cause is determined and corrective actions have been implemented to the satisfaction of the Contracting Officer. The Contracting Officer will provide a blank copy of the accident report form.

#### 1.13.3 LHE Inspection Reports

Submit LHE inspection reports required in accordance with EM 385-1-1 and as specified herein with Daily Reports of Inspections.

#### 1.13.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging

Provide a Certificate of Compliance for LHE entering the Portsmouth Naval Shipyard under this Contract and in accordance with EM 385-1-1. Post certifications on the crane.

Develop a Standard Lift Plan (SLP) in accordance with EM 385-1-1 and using Standard Pre-Lift Crane Plan/Checklist for each lift planned. Submit SLP to the Contracting Officer for approval within 15 calendar days in advance of planned lift.

### 1.14 HOT WORK PERMIT

#### 1.14.1 Permit and Personnel Requirements

Submit and obtain a written permit prior to performing "Hot Work" (i.e. welding or cutting) or operating other flame-producing/spark producing devices, from the Portsmouth Naval Shipyard Fire Department via the Contracting Officer. A permit is required from the Explosives Safety Office for work in and around where explosives are processed, stored, or handled. CONTRACTORS ARE REQUIRED TO MEET ALL CRITERIA BEFORE A PERMIT IS ISSUED. Provide at least two (2) 20 pound 4A:20 BC rated extinguishers for normal "Hot Work". The extinguishers must be current inspection tagged, and contain an approved safety pin and tamper resistant seal. It is also mandatory to have a designated FIRE WATCH for any "Hot Work" performed at the Portsmouth Naval Shipyard. The Fire Watch must be trained in accordance with NFPA 51B and remain on-site for a minimum of 1 hour after completion of the task or as specified on the hot work permit.

When starting work in the facility, require personnel to familiarize themselves with the location of the nearest fire alarm boxes and place in memory the emergency Portsmouth Naval Shipyard Fire Department phone number. REPORT ANY FIRE, NO MATTER HOW SMALL, TO THE RESPONSIBLE PORTSMOUTH NAVAL SHIPYARD FIRE DEPARTMENT AND THE CONTRACTING OFFICER IMMEDIATELY.

a. Duration of Hot Work

At the discretion of the Portsmouth Naval Shipyard Fire Inspectors, a hot work permit may be written for up to one work week, (Monday through Friday), provided work is being performed at one specific location and will not hinder any life safety code.

Hot work permits must be issued for five (5) work days with a time period not to exceed twelve (12) hours. Request for weekend permits must be submitted at least three (3) working days in advance of the work to be performed and will only be valid for the weekend requested.

b. Hot Work Within a Confined Space

Hot work permits within a confined space will be issued on a daily basis with a time period not to exceed a twelve (12) hour shift. All requests for hot work permits within a confined space must be received a minimum of three (3) complete working days prior to entry. A duplicate copy of the air monitoring results will be furnished to the fire inspector at time of issuance.

1.14.2 Work Around Flammable Materials

Obtain permit approval from a NFPA Certified Marine Chemist or Certified Industrial Hygienist for "Hot Work" within or around flammable materials (such as fuel systems or welding/cutting on fuel pipes) or confined spaces (such as sewer wet wells, manholes, or vaults) that have the potential for flammable or explosive atmospheres.

Whenever these materials, except beryllium and chromium (VI), are encountered in indoor operations, local mechanical exhaust ventilation systems that are sufficient to reduce and maintain personal exposures to within acceptable limits must be used and maintained in accordance with manufacturer's instruction and supplemented by exceptions noted in EM 385-1-1.

1.15 CONFINED SPACE ENTRY REQUIREMENTS.

Confined space entry must comply with EM 385-1-1, 29 CFR 1926, 29 CFR 1910, and Directive CPL 2.100. Any potential for a hazard in the confined space requires a permit system to be used. Contractors entering and working in confined spaces while performing shipyard industry work are required to follow the requirements of OSHA 29 CFR 1915.

1.15.1 Rescue Procedures and Coordination with Local Emergency Responders

Develop and implement an on-site rescue and recovery plan and procedures. The rescue plan must not rely on local emergency responders for rescue from a confined space.

1.16 CONSTRUCTION INDOOR AIR QUALITY (IAQ) MANAGEMENT PLAN

Submit an IAQ Management Plan within 15 calendar days from receiving the Notice to Proceed and not less than 10 calendar days before the Preconstruction Meeting. Revise and resubmit plan as required by the Contracting Officer. Make copies of the final plan available to all workers on site. Include provisions in the plan to meet the requirements

specified below and to ensure safe, healthy air for construction workers and building occupants.

#### 1.16.1 Requirements During Construction

Provide for evaluation of indoor Carbon Dioxide concentrations in accordance with ASTM D6245. Provide for evaluation of volatile organic compounds (VOCs) in indoor air in accordance with ASTM D6345. Use filters with a Minimum Efficiency Reporting Value (MERV) of 8 in permanently installed air handlers during construction.

#### 1.16.2 Control Measures

Meet or exceed the requirements of ANSI/SMACNA 008 to help minimize contamination of the building from construction activities. The five requirements of this manual which must be adhered to are described below:

- a. HVAC protection: Isolate return side of HVAC system from surrounding environment to prevent construction dust and debris from entering the duct work and spaces.
- b. Source control: Use low emitting paints and other finishes, sealants, adhesives, and other materials as specified. When available, cleaning products must have a low VOC content and be non-toxic to minimize building contamination. Utilize cleaning techniques that minimize dust generation. Cycle equipment off when not needed. Prohibit idling motor vehicles where emissions could be drawn into building. Designate receiving/storage areas for incoming material that minimize IAQ impacts.
- c. Pathway interruption: When pollutants are generated use strategies such as 100 percent outside air ventilation or erection of physical barriers between work and non-work areas to prevent contamination.
- d. Housekeeping: Clean frequently to remove construction dust and debris. Promptly clean up spills. Remove accumulated water and keep work areas dry to discourage the growth of mold and bacteria. Take extra measures when hazardous materials are involved.
- e. Scheduling: Control the sequence of construction to minimize the absorption of VOCs by other building materials.

##### 1.16.2.1 Moisture Contamination

- a. Remove accumulated water and keep work dry.
- b. Use dehumidification to remove moist, humid air from a work area.
- c. Do not use combustion heaters or generators inside the building.
- d. Protect porous materials from exposure to moisture.
- e. Remove and replace items which remain damp for more than a few hours.

##### 1.16.2.2 Requirements After Construction

After construction ends and prior to occupancy, conduct a building flush-out or test the indoor air contaminant levels. Flush-out must be a minimum 2 weeks with MERV-13 filtration media as determined by ASHRAE 52.2

at 100 percent outside air. Air contamination testing must be consistent with EPA's current Compendium of Methods for the Determination of Air Pollutants in Indoor Air. After building flush-out or testing and prior to occupancy, replace filtration media. Filtration media must have a MERV of 13 as determined by ASHRAE 52.2.

#### 1.17 SEVERE STORM PLAN

In the event of a severe storm warning, the Contractor must comply with the applicable Storm Plan and:

- a. Secure outside equipment and materials and place materials that could be damaged in protected areas.
- b. Check surrounding area, including roof, for loose material, equipment, debris, and other objects that could be blown away or against existing facilities.
- c. Ensure that temporary erosion controls are adequate.

### PART 2 PRODUCTS

#### 2.1 CONFINED SPACE SIGNAGE

Provide permanent signs integral to or securely attached to access covers for new permit-required confined spaces. Signs for confined spaces must comply with NEMA Z535.2. Signs wording: "DANGER--PERMIT-REQUIRED CONFINED SPACE, DO NOT ENTER" in bold letters a minimum of one inch in height and constructed to be clearly legible with all paint removed. The signal word "DANGER" must be red and readable from 5 feet.

### PART 3 EXECUTION

#### 3.1 CONSTRUCTION AND OTHER WORK

Comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHA, Federal and State OSHA regulations, and other related submittals and Portsmouth Naval Shipyard fire and safety regulations. The most stringent standard prevails.

PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Mandatory minimum PPE includes:

- a. Head Protection Hat that meets ANSI/ISEA Z89.1
- b. Long Pants
- c. Appropriate Safety Footwear
- d. Appropriate Class Reflective Vests
- e. Safety Glasses

##### 3.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers

must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones, or other acceptable means). The selected communication must be readily available (easily within the immediate reach) to the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. An employee check-in/check-out communication procedure must be developed to ensure employee safety.

### 3.1.2 Hazardous Material Use

Each hazardous material must receive approval from the Contracting Officer or their designated representative prior to being brought onto the job site or prior to any other use in connection with this Contract. Allow a minimum of 10 working days for processing of the request for use of a hazardous material.

### 3.1.3 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury, polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. Notify the Radiation Safety Officer (RSO) via the Contracting Officer prior to excepted items of radioactive material and devices being brought on Portsmouth Naval Shipyard.

### 3.1.4 Unforeseen Hazardous Material

Contract documents identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e. 29 CFR Part 1910.1000). If material(s) that may be hazardous to human health upon disturbance are encountered during construction operations, stop that portion of work and notify the Contracting Officer immediately. Within 14 calendar days the Government will determine if the material is hazardous. If the material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If the material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification.

## 3.2 UTILITY OUTAGE REQUIREMENTS

Outages are required when normal system operation is or could be impacted for utility services (i.e., water, sanitary sewer, storm sewer, electric services, fire alarm, etc.), roads, parking spaces, and crane rail, among others. For detailed information on the outage process, refer to paragraph FACILITY/UTILITY/SERVICE OUTAGE & INTERRUPTIONS PROCESS in Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME).

## 3.3 OUTAGE SAFETY REVIEW MEETING

Outage Safety Review Meetings are to be held prior to any outages that required hazardous energy control (HEC). Refer to Paragraph OUTAGE SAFETY

REVIEW MEETINGS in Section 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS (PWD ME) for additional information.

### 3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

The Contractor shall provide and operate a Hazardous Energy Control Program (HECP) in accordance with EM 385-1-1, 29 CFR 1910, 29 CFR 1915, ANSI/ASSP A10.44, NFPA 70E.

Contracting Officer will, upon request, apply lockout/tagout tags and take other actions that, because of experience and knowledge, are known to be necessary to make the particular equipment safe to work on.

No person, regardless of position or authority, shall operate any switch, valve, or equipment that has an official lockout/tagout tag attached to it, nor can such tag be removed except as provided in this Section.

No person shall work on any equipment that requires a lockout/tagout tag unless they, their immediate supervisor, project leader, or a subordinate has in their possession the stubs of the required lockout/tagout tags.

When work is to be performed on electrical circuits, only qualified personnel are to perform work on electrical circuits.

A supervisor who is required to enter an area protected by a lockout/tagout tag will be considered a member of the protected group provided he/she notifies the holder of the tag stub each time he/she enters and departs from the protected area.

Portsmouth Naval Shipyard and NAVFAC Personnel use a red lock and a red tag to indicate personnel are working on the systems. Use of a red lock and a red tag is highly encouraged to maintain continuity throughout the Portsmouth Naval Shipyard. The use of another colored locks and tags (blue for Portsmouth Naval Shipyard workers and Yellow for NAVFAC personnel) indicate that the system is out of service for some reason.

Identification markings on building light and power distribution circuits must not be relied on for established safe work conditions.

Before clearance will be given on any equipment other than electrical (generally referred to as mechanical apparatus), the apparatus, valves, or systems must be secured in a passive condition with the appropriate vents, pins, and locks.

Pressurized or vacuum systems must be vented to relieve differential pressure completely.

Vent valves must be tagged open during the course of the work.

Where dangerous gas or fluid systems are involved, or in areas where the environment may be oxygen deficient, system or areas must be purged, ventilated, or otherwise made safe prior to entry.

#### 3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the subcontractor performing the work must attend the Safety Preparatory Inspection C

coordination Meeting, which will also be attended by the Contracting Officer's Representative, and is required by EM 385-1-1. The meeting will occur immediately preceding the start of work and following the completion of the outage safety review meeting. Both the safety preparatory inspection coordination meeting and the outage safety review meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

#### 3.4.2 Lockout/Tagout Isolation

Where the Government or Utility performs equipment isolation and lockout/tagout, the Contractor must place their own locks and tags on each energy-isolating device and proceed in accordance with the HECF. Before any work begins, both the Contractor and the Government or Utility must perform energy isolation verification testing while wearing required PPE detailed in the Contractor's AHA and required by EM 385-1-1. Install personal protective grounds, with tags, to eliminate the potential for induced voltage in accordance with EM 385-1-1.

#### 3.4.3 Tag Placement

Lockout/tagout tags must be completed in accordance with the regulations printed on the back thereof and attached to any device which, if operated, could cause an unsafe condition to exist.

If more than one group is to work on any circuit or equipment, the employee in charge of each group must have a separate set of lockout/tag out tags completed and properly attached.

When it is required that certain equipment be tagged, the Government will review the characteristics of the various systems involved that affect the safety of the operations and the work to be performed; take the necessary actions, including voltage and pressure checks, grounding, and venting, to make the system and equipment safe to work on; and apply such lockout/tag out tags to those switches, valves, vents, or other mechanical devices needed to preserve the safety provided. This operation is referred to as "Providing Safety Clearance."

#### 3.4.4 Tag Removal

Upon completion of work, conduct lockout/tagout removal procedure in accordance with the HECF. In accordance with EM 385-1-1, each lock and tag must be removed from each energy isolating device by the authorized individual or systems operator who applied the device. When any individual or group has completed its part of the work and is clear of the circuits or equipment, the supervisor, project leader, or individual for whom the equipment was tagged must turn in their signed lockout/tagout tag stub to the Contracting Officer. That group or individual's lockout/tag-out tags on equipment may then be removed once formal authorization is provided by the Contracting Officer.

### 3.5 FALL PROTECTION PROGRAM

Establish a fall protection program, for the protection of all employees exposed to fall hazards. Within the program include company policy, identify roles and responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection, storage, care and maintenance of fall protection equipment, and rescue and evacuation procedures in accordance with EM 385-1-1.

### 3.5.1 Fall Protection Equipment and Systems

Enforce use of personal fall protection equipment and systems designated (to include fall arrest, restraint, and positioning) for each specific work activity in the Site Specific Fall Protection and Prevention Plan and AHA at all times when an employee is exposed to a fall hazard. Protect employees from fall hazards as specified in EM 385-1-1.

Provide personal fall protection equipment, systems, subsystems, and components that comply with EM 385-1-1 and 29 CFR 1926.

#### 3.5.1.1 Additional Personal Fall Protection Measures

In addition to the required fall protection systems, other protective measures such as safety skiffs, personal flotation devices, and life rings, are required when working above or next to water in accordance with EM 385-1-1. Personal fall protection systems and equipment are required when working from an articulating or extendable boom, swing stages, or suspended platform. In addition, personal fall protection systems are required when operating other equipment such as scissor lifts. The need for tying-off in such equipment is to prevent ejection of the employee from the equipment during raising, lowering, travel, or while performing work.

#### 3.5.1.2 Personal Fall Protection Equipment

Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest body support device. The use of body belts is not acceptable. Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal D-ring) and specifically designated for attachment to the rest of the system. Snap hooks and carabiners must be self-closing and self-locking, capable of being opened only by at least two consecutive deliberate actions and have a minimum gate strength of 3,600 lbs in all directions. Use webbing, straps, and ropes made of synthetic fiber. The maximum free fall distance when using fall arrest equipment must not exceed 6 feet, unless the proper energy absorbing lanyard is used. Always take into consideration the total fall distance and any swinging of the worker (pendulum-like motion), that can occur during a fall, when attaching a person to a fall arrest system. Equip all full body harnesses with Suspension Trauma Preventers such as stirrups, relief steps, or similar in order to provide short-term relief from the effects of orthostatic intolerance in accordance with EM 385-1-1.

### 3.6 SHIPYARD REQUIREMENTS

All personnel who enter the Controlled Industrial Area (CIA) must wear mandatory personal protective equipment (PPE) at all times and comply with PPE postings of shops both inside and outside the CIA.

### 3.7 EQUIPMENT

#### 3.7.1 Use of Explosives

Use of explosives or blasting is not permitted at Portsmouth Naval Shipyard.



### 3.8 EXCAVATIONS AND UTILITY LOCATING

Soil classification must be performed by a competent person in accordance with 29 CFR 1926 and EM 385-1-1.

#### 3.8.1 Utility Locations

##### A. General

Excavation or ground penetrating work is defined as any operation in which earth, rock or other material below ground is moved or otherwise displaced, by means of power and hand tools, power equipment which includes grading, trenching, digging, boring, auguring, tunneling, scraping and cable or pipe driving except tilling of soil, gardening or displacement of earth, rock or other material for agricultural purposes. Removal of bituminous concrete pavement or concrete is not considered excavation. Ground penetrating work may include but is not limited to installing fence posts, probes, borings, piles, sign posts, stakes or anchor rods of any kind that penetrates the soil more than 3". The "Excavator" is defined as the person directly responsible for performing the excavation or ground penetrating work.

##### B. Underground Utilities Location

The Contractor/Excavator shall fully comply with the State of Maine "DIGSAFE" law (Title 23, MRSA 3360-A). Existing underground utilities shown on the plans are based on the Portsmouth Naval Shipyard Total Map and are shown in their approximate locations only.

The Excavator shall pre-mark the excavation area in white paint only. (Field notes may be done in Pink paint). The Excavator shall notify "DIG SAFE" (1-888-344-7233) at least within 14 calendar days, but no more than 30 calendar days, prior to the commencement of the excavation or ground penetrating activity. The Excavator shall notify Unitil at least within 14 calendar days, but no more than 30 calendar days, prior to the commencement of excavation or ground penetrating activity when the activity is within twenty (20) feet of a natural gas line.

The Prime Contractor / Excavator shall prepare a NAVFAC Dig Safe Utility Locate Request Form (Attachment B) at least within 14 calendar days prior to the commencement of the excavation or ground penetrating activity and submit the form to the NAVFAC DM/ET via the Contracting Officer. (The NAVFAC Dig Safe Utility Locate Request Form is provided as Attachment B.)

The Government will locate and mark the underground utilities within 14 calendar days of receiving the Dig Safe Notification. Unitil will locate and mark the underground natural gas utilities.

Excavation or ground penetrating activities cannot commence until the utilities have been marked in the field and the NAVFAC Dig Safe Utility Locate Request Form has been returned to the Contractor indicating the PWD ME Dig Safe review process has been completed and excavation has been approved by the Contracting Officer and the PWD ME Dig Safe Coordinator.

If the excavation or ground penetrating activities do not commence within 27 days of Dig Safe notification or the excavation work is

expanded outside the location originally specified in the notification, the Excavator shall renotify Dig Safe, the Contracting Officer and the PWD ME Dig Safe Coordinator.

The Contractor shall maintain the utility markings through out the excavation period at the Contractor's expense.

The Contractor shall contact the PWD ME Dig Safe Coordinator (DSC) if there are any questions regarding the underground utilities or the Dig Safe process.

#### C. Third Party Utility Locate

The Contractor must provide the services of a third party, qualified independent utility locating company/person(s) (cannot be the Government's Utility Locating Firm) to positively identify underground utilities in the work area. The third party independent locating firm is in addition to the PWD ME Dig Safe Process.

The Third Party review must be completed in conjunction with the PWD ME Dig Safe marking process. Once the Third Party Locate Company has completed their review of the excavation area and the review of the Government markings are confirmed, the Third Party Locate Company & Contractor must sign the Contractor Dig-Safe Certification Form (Attachment C) and submit the form to the Contracting Officer/DSC prior to commencing excavation. If the Third Party Locate Company finds any discrepancies with the Government's utility markings, the Contractor must notify the Contracting Officer/DSC immediately and work with the DSC to resolve the discrepancy **PRIOR TO THE START OF EXCAVATION.**

#### 3.8.2 Utility Location Verification

Physically verify underground utility locations, including utility depth, by hand digging using wood or fiberglass handled tools when any adjacent construction work is expected to come within three (3) feet of the underground system. When any adjacent construction work is expected to come within twenty (20) feet of a natural gas utility, the Contractor must notify Unitil in addition to notifying "DIG SAFE".

#### 3.8.3 Utilities Within and Under Concrete (roofs, walls, floors, etc.) Bituminous Asphalt, and Other Impervious Surfaces (Including Dry Dock and Berthing Walls)

Utilities located within and under concrete slabs or pier structures, bridges, parking areas, and the like, are extremely difficult to identify. This will require utility locating **both before and after** concrete slab removal when excavation continues below bottom of slab. Whenever Contract work involves chipping, saw cutting, drilling or core drilling through concrete, bituminous asphalt or other impervious surfaces, the existing utility location must be coordinated with Portsmouth Naval Shipyard UEM department in addition to location and depth verification by a third party, independent, private locating company.

The third party, independent, private locating company must locate utility depth by use of Ground Penetrating Radar (GPR), X-ray, bore scope, or ultrasound prior to the start of demolition and construction. Outages to isolate utility systems must be used in circumstances where utilities are unable to be positively identified. The use of historical drawings does

not alleviate the Contractor from meeting this requirement. Any markings made during the utility investigation must be maintained by the Contractor throughout the Contract.

3.8.4 Excavating with Hazardous or Mission Critical Utilities Within the Excavation Area (as identified by DSC or plans)

- a. Refer to Paragraph CRITICAL SHIPYARD INFRASTRUCTURE REQUIREMENTS in Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME) for definition of and additional requirements for Critical Shipyard Infrastructure.
- b. The Contractor must employ supplemental daily Dig Safe procedures to include an additional checkoff on Contractor Daily Activity Plan to confirm all utilities have been clearly marked and reviewed with the SSHO. The SSHO & the Excavator must complete and sign the CONTRACTOR UTILITY LOCATE – PRE-EXCAVATION SAFETY CHECKLIST (Attachment D). This checklist must be reviewed and signed by the NAVFAC DM/ET prior to commencing any excavation/trenching work.
- c. The Contractor must provide a Dig Safe laminated utility color coding system posted in or near all heavy digging equipment for easy reference to type of utility. See Attachment G for an example Color Coding reference.
- d. The SSHO must complete a pre-excavation walk as part of the morning procedure to help ensure all known utilities are identified and markings are refreshed with the appropriate color-coded paint. The Excavation/Trenching Competent person must complete the Contractor Daily Worksite Checklist for Trenching/Excavation Sites included as Attachment E. The Daily Checklist must be completed prior to commencing excavation/trenching work and must be submitted with the NAVFAC DM/ET daily.
- e. Contractor must provide additional danger signage, to mark areas of known live underground utilities. Signage must comply with NEMA Z535.2.
- f. Contractor must ensure a 'spotter' accompanies the equipment operator during excavation work.
- g. Contractor to provide notification to the NAVFAC DM/ET via the Contracting Officer no later than 7 working days prior to holding the preparatory and initial pre-excavation/demo safety review meeting. The NAVFAC DM/ET, as well as the PWD ME DSC, **WILL** attend this meeting to ensure safety and preservation of critical utilities is discussed with adequate focus.
- h. The Contractor must confirm and identify the closest utility isolation points and develop mitigation strategies with the utility owner (Coordinate with the PWD ME DSC) to ensure the safe excavation adjacent to these utilities. Utility Outages to isolate utility systems may need to be considered in circumstances where the excavation work cannot be completed safely.
- i. If there is a utility strike, the Contractor must stop all work and notify the Contracting Officer immediately. The Contractor must complete the Underground Utility Strike Review Checklist (See Attachment F) within 2 working days of the utility strike. See additional requirements in Paragraphs MISHAP NOTIFICATION and ACCIDENT REPORTS herein.

### 3.9 ELECTRICAL

Perform electrical work in accordance with EM 385-1-1.

#### 3.9.1 Electrical Work

As described in EM 385-1-1, electrical work is to be conducted in a de-energized state unless there is no alternative method for accomplishing the work. In those cases obtain an energized work permit from the Contracting Officer. The energized work permit application must be accompanied by the AHA and a summary of why the equipment/circuit needs to be worked energized. Underground electrical spaces must be certified safe for entry before entering to conduct work. Cables that will be cut must be positively identified and de-energized prior to performing each cut. Cables shall be verified as de-energized by using a spiking tool. Attach temporary grounds in accordance with ASTM F855 and IEEE 1048. Perform all medium and high voltage cable cutting remotely using hydraulic cutting tool. When racking in or live switching of circuit breakers, no additional person other than the switch operator is allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear of any energized electrical sources is the preferred method.

When working in energized substations, only qualified electrical workers are permitted to enter. When work requires work near energized circuits as defined by NFPA 70, high voltage personnel must use personal protective equipment that includes, as a minimum, electrical hard hat, safety shoes, insulating gloves, and electrical arc flash protection for personnel as required by NFPA 70E. Insulating blankets, hearing protection, and switching suits may also be required, depending on the specific job and as delineated in the Contractor's AHA. Ensure that each employee is familiar with and complies with these procedures and 29 CFR 1910.

#### 3.9.2 Qualifications

Electrical work must be performed by QP with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National, and Local Certifications or Licenses that a Master or Journeyman Electrician may hold, depending on work being performed, and must be identified in the appropriate AHA. Journeyman/Apprentice ratio must be in accordance with State and Local requirements applicable to where work is being performed.

#### 3.9.3 Arc Flash

Conduct a hazard analysis/arc flash hazard analysis whenever work on or near energized parts greater than 50 volts is necessary, in accordance with NFPA 70E.

All personnel entering the identified arc flash protection boundary must be QPs and properly trained in NFPA 70E requirements and procedures. Unless permitted by NFPA 70E, no Unqualified Person is permitted to approach nearer than the Limited Approach Boundary of energized conductors and circuit parts. Training must be administered by an electrically qualified source and documented.

#### 3.9.4 Grounding

Ground electrical circuits, equipment and enclosures in accordance with NFPA 70 and IEEE C2 to provide a permanent, continuous and effective path to ground unless otherwise noted by EM 385-1-1.

#### 3.9.5 Testing

Temporary electrical distribution systems and devices must be inspected, tested and found acceptable for Ground-Fault Circuit Interrupter (GFCI) protection, polarity, ground continuity, and ground resistance before initial use, before use after modification, and at least monthly. Monthly inspections and tests must be maintained for each temporary electrical distribution system, and signed by the electrical CP or QP.

#### 3.9.6 Portable Extension Cords

Size portable extension cords in accordance with manufacturer ratings for the tool to be powered and protected from damage. Immediately remove from service all damaged extension cords. Portable extension cords must meet the requirements of EM 385-1-1, NFPA 70E, and OSHA electrical standards.

-- End of Section --

## SECTION 01 35 26 – ATTACHMENT A

### CONTRACTOR CRANE, MULTI-PURPOSE MACHINE, FORKLIFT, CONSTRUCTION EQUIPMENT, AND RIGGING GEAR REQUIREMENTS

#### 1 CONTRACTOR CRANE, MULTI-PURPOSE MACHINE, FORKLIFT, CONSTRUCTION EQUIPMENT, AND RIGGING GEAR REQUIREMENTS

1.1 The following is a list of requirements that contractors shall comply with for all contracts that may result in the use of a category 1 or 4 crane, multi-purpose machines, forklifts, construction equipment, and rigging gear when used on Navy property to lift suspended loads. Non-compliance with the requirements of this instruction may result in denial of access, stopping of operations, or removal from Navy property.

#### 1.2 References:

1.2.1 NOT USED

1.2.2 American Society of Mechanical Engineers (ASME) B30.3 (tower cranes), B30.5 (mobile cranes), B30.8 (floating cranes), B30.9 (slings), B30.20 (below the hook lifting devices), B30.22 (articulating booms), B30.26 (rigging hardware); ANSI/ITSDF B56.6 (rough terrain forklifts); Safety Standards for Cableways, Cranes, Derricks, Hoists, Hooks, Jacks, and Slings

1.2.3 CFR, Title 29, Chapter XVII, Part 1917, Marine Terminals.

1.2.4 CFR, Title 29, Chapter XVII, Part 1926, Safety and Health Regulations for Construction

1.2.5 CFR, Title 29, Chapter XVII, Part 1915, Occupational Safety and Health Standards for Shipyard Employment

1.2.6 OPNAVINST 5100.23, Navy Safety and Occupational Health Program Manual

1.2.7 EM 385-1-1, Safety and Health Requirements Manual, U.S. Army Corps of Engineers

1.2.8 NAVFAC Guide Specification NFGS-01525D, Safety Requirements

1.3 These requirements are solely intended to provide for the protection of Government property and personnel and are not intended to, and do not, in any manner whatsoever, relieve the contractor of its responsibility, including, without limitation, its responsibility for the protection of its equipment and personnel.

1.4 Notification Requirement: Contractor shall notify the Contracting Officer 7 calendar days in advance of the intent of bringing a non-Navy owned crane onto Navy property or of any multi-purpose machines, material handling equipment, or construction equipment that may be used in a crane-like application to lift suspended loads. The contractor shall also specify when crane entry onto Navy property is scheduled during back shift, weekend, or holiday hours of operation. All entries shall be through a prearranged entry point. The following documentation shall be provided along with notification: a copy of Form 16-1 and objective evidence of operator qualifications for cranes with rated capacities of 2,000 lbs. or greater. Failure to schedule or provide necessary documentation may result in the crane being denied access to the facility.

1.5 The contractor shall comply with applicable reference 1.2.2 standards (e.g., B30.3 for construction tower cranes, B30.5 for mobile cranes, B30.8 for floating cranes, B30.9 for slings, B30.20 for below the hook lifting devices, and B30.22 for articulating boom cranes), B30.26 for rigging hardware, and ANSI/ITSDF B56.6 for rough terrain forklifts). For barge mounted mobile cranes, require a third party certification from an OSHA accredited organization (or from a state accredited organization for those states with OSHA approved state plans), a load indicating device, a wind-indicating device, and a marine type list and trim indicator readable in one-half degree increments. Third party certification is not required for barge-mounted mobile cranes at naval activities in foreign countries.

## SECTION 01 35 26 – ATTACHMENT A

### CONTRACTOR CRANE, MULTI-PURPOSE MACHINE, FORKLIFT, CONSTRUCTION EQUIPMENT, AND RIGGING GEAR REQUIREMENTS

1.6 Certification of Compliance (reference 1.2.1): The contractor shall complete a certificate of compliance that the crane (or other machine if used to lift suspended loads) and rigging gear meet applicable OSHA and ANSI/ASME regulations (with the contractor citing which OSHA regulations are applicable, e.g., cranes/multi-purpose machines used in cargo transfer shall comply with reference 1.2.3; cranes/multi-purpose machines used in construction, demolition, or maintenance shall comply with reference 1.2.4; cranes/multi-purpose machines used in ship repair shall comply with reference 1.2.5; slings shall comply with ASME B30.9; rigging hardware shall comply with ASME B30.26). For cranes (or other machine if used to lift suspended loads) and rigging equipment at naval activities in foreign countries, the contractor shall certify that the crane and rigging gear conform to the appropriate host country safety standards. The contractor shall also certify that all of its crane (or other machine) operators working on the naval activity have been trained not to bypass safety devices (e.g., anti-two block devices) during lifting operations, and that its operators, riggers, and company officials are aware of the actions required in the event of an accident as specified in the contract. Require that the certifications be posted on the crane. When a crane on Navy property is not authorized for use, the Certification of Compliance shall state, "Operation of this Crane is NOT Authorized."

1.7 The contractor shall certify (reference 1.2.1) that the crane or machine operator is qualified and trained for the operation of the crane to be used. For mobile and commercial truck mounted cranes with OEM rated capacities of greater than 2,000 pounds, the crane operator shall be designated as qualified by a source that qualifies crane operators (i.e., a union, a government agency, or an organization that tests and qualifies crane operators). Proof of current qualification shall be provided.

1.8 For multi-purpose machines, material handling equipment and construction equipment used to lift loads suspended by rigging equipment, the contractor shall have proof or authorization from the machine OEM that the machine is capable of making lifts of loads suspended by rigging equipment. The contractor shall demonstrate that the equipment is properly configured to make such lifts and is equipped with a load chart.

1.9 All hooks used on cranes, hoists, other machines, and rigging gear shall have self-closing latches or the throat opening shall be "moused" (secured with wire, rope, heavy tape, etc.) or otherwise secured to prevent the attached item from coming free of the hook under a slack condition. The following exceptions apply and shall be approved by the contractor's technical organization: items where the hook throat is fully obstructed and not available for manual securing and lifts where securing the hook throat increases the danger to personnel such as forge shop, dip tank, or underwater work.

#### 1.10 Loading Limitations:

CAUTION: Piers and waterfront areas such as along dry docks and quay walls may have load restrictions.

1.10.1 The contractor shall notify the Contracting Officer prior to moving a crane on a pier, dry dock, or other waterfront area. Provide the Contracting Officer with the crane make, model, and configuration in which it is to be used.

1.10.2 The contractor shall comply with crane access routes and load limitations issued with the contract.

1.10.3 Allowable Surface Loads. Loads transferred to soils and pavements shall be minimized to a desired maximum of 3000 pounds per square foot, by placement of cribbing or steel pads under rubber-tired crane outriggers and trailer stanchions/sand shoes, or by placement of mats under treads of crawler cranes. Visually inspect areas adjacent to cribbing or plates and report any unusual bituminous pavement surface conditions, irregularities, or cracking to the Contracting Officer.

1.10.3.1 Outriggers of rubber-tired cranes shall be landed on two layers of timbers of appropriate thickness, oriented at right angles to each other, or landed on properly designed steel pads. Treads of crawler cranes shall run on appropriate mats. Use and design of cribbing, plates and mats shall be in a manner consistent with general construction industry standards.

## SECTION 01 35 26 – ATTACHMENT A

### CONTRACTOR CRANE, MULTI-PURPOSE MACHINE, FORKLIFT, CONSTRUCTION EQUIPMENT, AND RIGGING GEAR REQUIREMENTS

1.10.3.2 Position loads that will remain on trailers detached from tractors to attain a distribution of 65 percent to rear axles and 35 percent to front support stanchions/sand shoes. For example, assuming an 83000 pound maximum gross weight and a soil bearing pressure of 3000 pounds per square foot, the required support under each sand shoe would be 2.5 feet x 2.5 feet. Accordingly, two tiers of timber cribbing at right angles, each 2.5 feet x 2.5 feet x 4 inches, or a properly designed 2.5 feet x 2.5 feet steel pad would be utilized under each trailer stanchion/sand shoe.

1.11 Prior to making any critical lift, the contractor shall provide a critical lift plan for each of the following lifts: lifts over 75 percent of the capacity of the crane, hoist, or other machine (50 percent of the capacity of a barge mounted mobile crane's hoists) at any radius of lift; lifts involving more than one crane, hoist, or other machine; lifts of personnel (lifts of personnel suspended by rigging equipment from multi-purpose machines, material handling equipment, or construction equipment shall not be permitted); lifts made in the vicinity of overhead power lines; erection of cranes; and lifts involving non-routine rigging or operation, sensitive equipment, or unusual safety risks. The plan shall include the following as applicable:

1.11.1 The size and weight of the load to be lifted, including crane (or other machine) and rigging equipment that add to the weight. The OEM's maximum load capacities for the entire range of the lift shall also be provided.

1.11.2 The lift geometry, including the crane (or other machine) position, boom length and angle, height of lift, and radius for the entire range of the lift. Applies to both single and tandem crane/machine lifts.

1.11.3 A rigging plan, showing the lift points, rigging equipment, and rigging procedures.

1.11.4 The environmental conditions under which lift operations are to be stopped.

1.11.5 For lifts of personnel, the plan shall demonstrate compliance with the requirements of reference

1.11.6 For barge mounted mobile cranes, barge stability calculations identifying crane placement/footprint; barge list and trim based on anticipated loading; and load charts based on calculated list and trim specific to the barge the crane is mounted on. The amount of list and trim shall be within the crane manufacturer's requirements.

1.11.7 For lifts in the vicinity of overhead power lines (i.e., if any part of the crane or other machine, including the fully extended boom of a telescoping boom crane or machine, or the load could approach the distances noted in figure 10-3 of reference 1.2.1 during a proposed operation), the plan shall demonstrate compliance to 29 CFR 1926.550(a)(15).

1.12 The following additional documentation is required for contractor provided tower cranes (those cranes defined by ASME B30.3).

1.12.1 Foundation design and requirements

1.12.2 Installation instructions 1.12.4

1.12.3 Assembly and disassembly instructions including climbing/jumping instructions if applicable

1.12.4 Operating manual, limitations, and precautions

1.12.5 Periodic inspection and maintenance requirements

1.13 Crane and Rigging Gear Accident Reporting and Record Keeping: Contractor's operating cranes on Navy property shall report all WHE accidents that occur incidental to an operation, project, or facility as



## SECTION 01 35 26 – ATTACHMENT A

### CONTRACTOR CRANE, MULTI-PURPOSE MACHINE, FORKLIFT, CONSTRUCTION EQUIPMENT, AND RIGGING GEAR REQUIREMENTS

prescribed by paragraphs (1.10.1) through (1.10.3) requirements below. Contractors shall report directly to their respective Contracting Officer. There are two general categories of accidents as defined below. Crane accidents are those that occur during operation of a crane. Rigging gear accidents are those that occur when gear is used by itself in weight handling operation i.e., without a crane.

1.13.1 Crane Accident: For the purpose of this definition, it is assumed there is an "operating envelope" around any crane, and inside the envelope are the following elements:

- The crane
- The operator
- The rigger(s) and crane walker
- Other personnel involved in the operation (supervisor, mechanic, tag line handler, engineer, etc.)
- The rigging gear between the hook and the load
- The load
- The crane's supporting structure (ground, rail, etc.)
- The lift procedure

1.13.1.1 Definition: A crane accident occurs when any one or more of the six elements in the operating envelope fails to perform correctly during operation, including operation during maintenance, or testing resulting in the following:

- Personnel injury or death. Minor injuries that are inherent in any industrial operation, including strains and repetitive motion related injuries, shall be reported by the normal personnel injury reporting process in lieu of these requirements.
- Material or equipment damage
- Dropped load
- Derailment
- Two-blocking
- Overload (This includes load tests when the test load tolerance is exceeded.)
- Collision, including unplanned contact between the load, crane, and/or other objects.

A component failure (e.g., motor burnout, gear tooth failure, bearing failure) is not considered an accident solely due to material or equipment damage unless the component failure results in damage to other components (e.g., dropped boom, dropped load, roll over, etc.). [Bullets] 3, 4, 5, 6, and 7 are considered crane accidents even though no material damage or injury occurs.

Exception: If a crane is used as an anchor point for a portable hoist/rigging gear, rigging gear accident as defined in paragraph 1.10.2 below is not considered a crane accident if the crane is not being operated (no functions are in motion) at the time of the rigging gear accident, unless the accident results in an overload or damage to the crane, in which case it shall be reported as a crane accident.

1.13.2 Rigging Gear Accidents: For the purpose of this definition, it is assumed there is an "operating envelope" around any weight handling operation, and inside the envelope are the following:

- Rigging gear and miscellaneous equipment
- The user of the gear or equipment
- Other personnel involved in the operation (supervisor, mechanic, tag line handler, engineer, etc.)
- The load
- The gear or equipment's supporting structure
- The load's rigging path

## SECTION 01 35 26 – ATTACHMENT A

### CONTRACTOR CRANE, MULTI-PURPOSE MACHINE, FORKLIFT, CONSTRUCTION EQUIPMENT, AND RIGGING GEAR REQUIREMENTS

- The rigging procedure

1.13.2.1 Definition. A rigging gear accident occurs when any one or more of the five elements in the operating envelope fails to perform correctly during weight handling operations resulting in the following:

- Personnel injury or death. Minor injuries that are inherent in any industrial operation, including strains and repetitive motion related injuries, shall be reported by the normal personnel injury reporting process of the activity in lieu of these requirements.
- Material or equipment damage that requires the damaged item to be repaired because it can no longer perform its intended function. This does not include superficial damage such as scratched paint, damaged lagging, or normal wear on rigging gear.
- Dropped load.
- Two-blocking of cranes and powered hoists.
- Overload. (This includes load tests when the test load tolerance is exceeded.)

A component failure (e.g., motor burnout, gear tooth failure, bearing failure) is not considered an accident solely due to material or equipment damage unless the component failure results in damage to other components (e.g., dropped load, damaged load, etc.). [Bullets] 3, 4, and 5 are considered accidents even though no material damage or injury occurs.

1.13.3 The contractor shall notify the Contracting Officer as soon as practical, but not later than four hours, after any WHE accident. The contractor shall secure the accident site and protect evidence until released by the Contracting Officer. The contractor shall conduct an accident investigation to establish the root cause(s) of the accident. Crane operations shall not proceed until cause is determined and corrective actions have been implemented to the satisfaction of the Contracting Officer. The contractor shall provide the Contracting Officer within 30 days of any accident a Crane and Rigging Gear Accident Report using the form provided in reference 1.2.1 consisting of a summary of circumstances, an explanation of causes(s), photographs if available, and corrective actions taken. These notifications and reporting requirements are in addition to those promulgated by reference 1.2.6 and related claimant instructions.

1.14 Each contractor shall perform the following actions prior to conducting crane operations on Navy property:

1.14.1 Inspection Requirements: It shall be the sole responsibility of the contractor to assure the Contracting Officer and/or designated Navy personnel that the crane and associated rigging gear are in good working order and safe for use.

1.14.1.1 Crane Inspection: Perform pre-operational inspection of the crane in the presence of a representative of the Contracting Office of the crane prior to starting work on Navy property. Inspection shall meet all applicable reference 1.2.2, reference 1.2.7 (for NAVFAC construction contracts), and OSHA requirements.

1.14.1.2 Wire Rope Inspection: Perform a Wire Rope Inspection in the presence of a representative of the contracting office to applicable reference 1.2.2, reference 1.2.7 (for NAVFAC construction contracts), and OSHA requirements.

1.14.1.3 Rigging Gear Inspection: Perform a Rigging Gear Inspection in the presence of a representative of the contracting office to applicable reference 1.2.2, reference 1.2.7 (for NAVFAC construction contracts), and OSHA requirements.

SECTION 01 35 26 – ATTACHMENT A  
CONTRACTOR CRANE ENTRY CHECKLIST

|    |                                                                                                                                                                                                                                                                                                                        |  |                      |                                         |                                        |                                         |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------------------------------------|----------------------------------------|-----------------------------------------|
| 1  | Crane Company:                                                                                                                                                                                                                                                                                                         |  | Date of Entry:       |                                         |                                        |                                         |
|    | Crane Manufacturer/Crane Model/Crane Number:                                                                                                                                                                                                                                                                           |  | Time of Entry:       |                                         |                                        |                                         |
| 2  | Date of Annual Inspection Expiration                                                                                                                                                                                                                                                                                   |  |                      |                                         |                                        |                                         |
| 3  | Date of Quadrennial Inspection Expiration                                                                                                                                                                                                                                                                              |  |                      |                                         |                                        |                                         |
| 4  | Name & phone number of Contracting Official (or designated local representative)                                                                                                                                                                                                                                       |  | Contracting Official |                                         |                                        |                                         |
|    |                                                                                                                                                                                                                                                                                                                        |  | Phone Number         |                                         |                                        |                                         |
| 5  | Does the package include a routine or critical lift plan?                                                                                                                                                                                                                                                              |  |                      | <div>Yes</div> <input type="checkbox"/> | <div>No</div> <input type="checkbox"/> |                                         |
| 6  | Location of lift site?                                                                                                                                                                                                                                                                                                 |  |                      |                                         |                                        |                                         |
| 7  | Duration crane will be continuously on the job site (hrs, days, weeks...)                                                                                                                                                                                                                                              |  |                      |                                         |                                        |                                         |
| 8  | Does plan include certification from contractor that the crane complies with ASME B30 standard [B30.5 (mobile cranes), B30.8 (floating cranes), B30.22 (articulating boom cranes), or B30.3 (construction tower cranes)] as applicable?                                                                                |  |                      | <div>Yes</div> <input type="checkbox"/> | <div>No</div> <input type="checkbox"/> |                                         |
| 9  | Does plan include a certificate of compliance?                                                                                                                                                                                                                                                                         |  |                      | <div>Yes</div> <input type="checkbox"/> | <div>No</div> <input type="checkbox"/> |                                         |
| 10 | Which OSHA regulations does the certificate of compliance indicate? (For cranes used in cargo transfer, 29 CFR 1917 applies; for cranes used in construction, demolition, or maintenance, 29 CFR 1926 applies; for cranes used in shipbuilding, ship repair, or ship breaking, 29 CFR 1915 applies).                   |  |                      |                                         |                                        |                                         |
| 11 | Does plan include valid medical certificate and proof of operator qualification from a source that qualifies crane operators (union, governmental agency, or an organization that tests and qualifies crane operators)? Verify qualification for each back-up operator (if provided) on the certificate of compliance. |  |                      | <div>Yes</div> <input type="checkbox"/> | <div>No</div> <input type="checkbox"/> | <div>N/A</div> <input type="checkbox"/> |
| 12 | Does the plan designate a qualified Rigger-in-Charge                                                                                                                                                                                                                                                                   |  |                      | <div>Yes</div> <input type="checkbox"/> | <div>No</div> <input type="checkbox"/> |                                         |
| 13 | What is the weight of the heaviest load to be lifted?                                                                                                                                                                                                                                                                  |  |                      | lbs.                                    |                                        |                                         |
| 14 | What is the weight of the rigging gear?                                                                                                                                                                                                                                                                                |  |                      | lbs.                                    |                                        |                                         |
| 15 | What are the crane components (and their weights) that add to the weight of the load (hook, jib, etc.)?                                                                                                                                                                                                                |  | Main Block           | lbs.                                    |                                        |                                         |
|    |                                                                                                                                                                                                                                                                                                                        |  | Aux. Block           | lbs.                                    |                                        |                                         |
|    |                                                                                                                                                                                                                                                                                                                        |  | Jib (Stowed)         | lbs.                                    |                                        |                                         |
|    |                                                                                                                                                                                                                                                                                                                        |  | Jib (Erected)        | lbs.                                    |                                        |                                         |
|    |                                                                                                                                                                                                                                                                                                                        |  | Other                | lbs.                                    |                                        |                                         |
| 16 | What is the maximum total crane lift (sum of 13, 14 & 15 above)?                                                                                                                                                                                                                                                       |  | TOTAL                | lbs.                                    |                                        |                                         |
| 17 | What is the capacity of the crane as configured                                                                                                                                                                                                                                                                        |  |                      | lbs.                                    |                                        |                                         |
| 18 | What percentage of crane capacity does this lift represent?                                                                                                                                                                                                                                                            |  |                      | %                                       |                                        |                                         |

SECTION 01 35 26 – ATTACHMENT A  
CONTRACTOR CRANE ENTRY CHECKLIST

|    |                                                                                                                                                                                         |            |             |                                        |                                                                                 |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------------|----------------------------------------|---------------------------------------------------------------------------------|
| 19 | What is the main boom length? If a jib will be utilized, indicate the length and offset.                                                                                                |            | <b>MAIN</b> | <b>JIB</b>                             | <b>OFFSET</b>                                                                   |
| 20 | What are the minimum and maximum load radii?                                                                                                                                            | <b>Min</b> |             | <b>Max</b>                             |                                                                                 |
| 21 | Does the plan include the manufacturer's load chart for entire range of lift(s)?                                                                                                        |            |             | <b>Yes</b><br><input type="checkbox"/> | <b>No</b><br><input type="checkbox"/>                                           |
| 22 | Does plan include ground loading and outrigger reaction data to determine cribbing requirements, or a Waterfront Operational Permit?                                                    |            |             | <b>Yes</b><br><input type="checkbox"/> | <b>No</b><br><input type="checkbox"/><br><b>N/A</b><br><input type="checkbox"/> |
| 23 | For crawler crane, does the plan indicate area restrictions for operation?                                                                                                              |            |             | <b>Yes</b><br><input type="checkbox"/> | <b>No</b><br><input type="checkbox"/><br><b>N/A</b><br><input type="checkbox"/> |
| 24 | For floating crane, does plan include maximum allowable list?                                                                                                                           |            |             | <b>Yes</b><br><input type="checkbox"/> | <b>No</b><br><input type="checkbox"/><br><b>N/A</b><br><input type="checkbox"/> |
| 25 | For mobile crane mounted on barge, is crane equipped with load indicating device? Wind indicating device? Marine type list and trim indicator (readable in one-half degree increments)? |            |             | <b>Yes</b><br><input type="checkbox"/> | <b>No</b><br><input type="checkbox"/><br><b>N/A</b><br><input type="checkbox"/> |
| 26 | For mobile crane mounted on barge, does plan include revised load chart?                                                                                                                |            |             | <b>Yes</b><br><input type="checkbox"/> | <b>No</b><br><input type="checkbox"/><br><b>N/A</b><br><input type="checkbox"/> |
| 27 | What are the environmental conditions under which crane operations are to be stopped?                                                                                                   |            |             |                                        |                                                                                 |
| 28 | Will the crane perform critical lifts? <b>(If no, skip items 29 –49.)</b>                                                                                                               |            |             | <b>Yes</b><br><input type="checkbox"/> | <b>No</b><br><input type="checkbox"/>                                           |
| 29 | What circumstances require this lift to be classified as a critical lift? (Blind lift, 75% of chart, non-routine rigging, etc.)                                                         |            |             |                                        |                                                                                 |
| 30 | What are the exact dimensions of the load? (L x W x H)                                                                                                                                  |            |             |                                        |                                                                                 |
| 31 | Does the plan indicate the crane position? (Overhead view)                                                                                                                              |            |             | <b>Yes</b><br><input type="checkbox"/> | <b>No</b><br><input type="checkbox"/>                                           |
| 32 | What is the maximum lift height of the lift?                                                                                                                                            |            |             |                                        |                                                                                 |
| 33 | What is the minimum boom angle?                                                                                                                                                         |            |             |                                        |                                                                                 |
| 34 | What is the maximum boom angle?                                                                                                                                                         |            |             |                                        |                                                                                 |
| 35 | What is the name of the operator?                                                                                                                                                       |            |             |                                        |                                                                                 |
| 36 | Indicate name(s) of backup operator (if required).                                                                                                                                      |            |             |                                        |                                                                                 |
| 37 | Does the plan show lift points?                                                                                                                                                         |            |             | <b>Yes</b><br><input type="checkbox"/> | <b>No</b><br><input type="checkbox"/>                                           |
| 38 | Does the plan describe the rigging procedures?                                                                                                                                          |            |             | <b>Yes</b><br><input type="checkbox"/> | <b>No</b><br><input type="checkbox"/>                                           |
| 39 | Does the plan indicate rigging hardware requirements?                                                                                                                                   |            |             | <b>Yes</b><br><input type="checkbox"/> | <b>No</b><br><input type="checkbox"/>                                           |

Enclosure 2 Cont'd

SECTION 01 35 26 – ATTACHMENT A  
CONTRACTOR CRANE ENTRY CHECKLIST

|    |                                                                                                                                                                         |                                        |                                       |                                        |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|---------------------------------------|----------------------------------------|
| 40 | For personnel lifts, does the plan demonstrate compliance with 29 CFR 1926.550?                                                                                         | <b>Yes</b><br><input type="checkbox"/> | <b>No</b><br><input type="checkbox"/> | <b>N/A</b><br><input type="checkbox"/> |
| 41 | Does EM 385-1-1 govern this lift?                                                                                                                                       | <b>Yes</b><br><input type="checkbox"/> | <b>No</b><br><input type="checkbox"/> | <b>N/A</b><br><input type="checkbox"/> |
| 42 | What are the coordination and communication requirements for the lift (e.g., radio and hand signals)?                                                                   |                                        |                                       |                                        |
| 43 | For tandem or tailing crane lifts, does the plan indicate the make and model of the crane, the line, boom, and swing speeds, and the requirement for an equalizer beam? | <b>Yes</b><br><input type="checkbox"/> | <b>No</b><br><input type="checkbox"/> | <b>N/A</b><br><input type="checkbox"/> |
| 44 | For floating cranes, refer to questions 20-22?                                                                                                                          |                                        |                                       |                                        |
| 45 | What is the name of the lift supervisor?                                                                                                                                |                                        |                                       |                                        |
| 56 | Does the plan indicate the qualifications of the lift supervisor?                                                                                                       | <b>Yes</b><br><input type="checkbox"/> | <b>No</b><br><input type="checkbox"/> |                                        |
| 47 | What are the names of the riggers?                                                                                                                                      |                                        |                                       |                                        |
| 48 | Does the plan indicate the qualifications of the riggers?                                                                                                               | <b>Yes</b><br><input type="checkbox"/> | <b>No</b><br><input type="checkbox"/> |                                        |
| 49 | Did all involved personnel (Operator, Riggers, Lift Supervisor, etc.) sign the critical lift plan?                                                                      | <b>Yes</b><br><input type="checkbox"/> | <b>No</b><br><input type="checkbox"/> |                                        |

| Name                  | Organization | Signature | Date | Phone |
|-----------------------|--------------|-----------|------|-------|
| Contracting Official: |              |           |      |       |
| Wed By                |              |           |      |       |

## Contractor Crane Entry Cover Sheet

### PWD MAINE / OICC PNSY

|            |           |                                                                  |                                           |
|------------|-----------|------------------------------------------------------------------|-------------------------------------------|
| <u>KTR</u> | <u>ET</u> |                                                                  | <b>Duration exceeding 15 days: Y    N</b> |
|            |           | <b>Prime Contractor Name:</b>                                    | <b>Proposed date/time of use:</b>         |
|            |           | <b>Contract Title:</b>                                           |                                           |
|            |           | <b>Crane Company:</b>                                            |                                           |
|            |           | <b>Location of Crane Lift:</b>                                   |                                           |
|            |           | <b><i>Following items to be submitted for NAVFAC Review:</i></b> |                                           |
| <u>KTR</u> | <u>ET</u> | KTR    ET                                                        |                                           |
|            |           | Certificate of compliance Eng Form 6209                          | Crane rigging plan                        |
|            |           | Contractor Crane Entry Checklist                                 | Critical or Standard lift plan            |
|            |           | Crane annual inspection certificate                              | Traffic/Site Setup plan                   |
|            |           | Valid Crane operator's license                                   | Ground loading calculations               |
|            |           | Crane operator medical card                                      | Wire rope inspection certificate          |
|            |           | Rigger/Signal person qualification                               | Scale pan certification                   |
|            |           | Crane load chart                                                 | Spreader bar certification                |

- 1) Prime Contractor SSHO is to meet Contractor Crane at CVIS to review certificate of compliance and verify crane operator license with ET
- 2) Contractor Crane operator to complete standard pre-lift crane plan checklist with signatures and returning to Prime Contractor prior to making crane lift.
- 3) Prime contractor shall submit crane rail outages, parking outages and street outages prior to submitting crane package.
- 4) Prime contractor will coordinate with crane contractor approved site location including traffic control, pedestrian walkways along with swing radius of crane with barrier tape prior to making crane lift.
- 5) Prime contractor will conduct a pre-lift meeting with AHA review to include all parties involved in lift operations and receive signature acknowledge of procedures, verify all rigging gear meets required specified standards.

**ET Concurrence:** \_\_\_\_\_ **Phone#** \_\_\_\_\_ **NAVFAC Crane/**  
**Safety Concurrence:** \_\_\_\_\_ **Navfac use only; Risk Assessment Code;**

Definitions 1 thru 3; **To be assigned by Crane SME**

\_\_\_ Level 1 – Full-Time ET on site to observe crane activities; and completes a daily surveillance sheet.

\_\_\_ Level 2 - ET visits site once daily to observe crane activities; and completes a surveillance sheet for each site visit.

\_\_\_ Level 3 – ET visits site once per week to observe crane activities; and submits a weekly surveillance sheet.

| NAVFAC MIDLANT<br><b>CRITICAL LIFT PLAN</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
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| <b>Date:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       | <b>Qualified Person:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| <b>Location:</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| <i>Contractor is required to fill sections A.thru F. prior to submitting plan.<br/>Sections G.- H to be completed by PWD Maine ET</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| <b>A. TOTAL LOAD</b><br>1. Load Weight <input style="width: 100px;" type="text"/> lbs<br>2. Wt. of Aux. Block <input style="width: 100px;" type="text"/> lbs<br>3. Wt. of Main Block <input style="width: 100px;" type="text"/> lbs<br>4. Wt. of Lifting Beam <input style="width: 100px;" type="text"/> lbs<br>5. Wt. of Sling/Shackles <input style="width: 100px;" type="text"/> lbs<br>6. Wt. of Jib/Ext. (erected/stowed) <input style="width: 100px;" type="text"/> lbs<br>7. Wt. of Hoist Rope <input style="width: 100px;" type="text"/> lbs<br>8. Other: <input style="width: 100px;" type="text"/> lbs<br><br><b>TOTAL WEIGHT</b> <input style="width: 100px;" type="text"/> lbs<br><br><small>Note: Source of load weight (Drawings, Calcs., etc.) must be attached on Page 2.</small>                                                                                                                                                                                                                                                                                                                                                                                    |       | <b>E. CRANE PLACEMENT (Mobile Cranes Only)</b><br>1. Maximum Bearing Pressure <input style="width: 100px;" type="text"/> PSF<br><small>Note: Bearing Pressure Calculations must be attached on Page 3.</small><br>2. Ground Conditions Suitable for Load? <input type="checkbox"/> YES / NO<br><small>Note: Ground Condition Calculations must be attached on Page 3.</small><br>3. High Voltage or Electrical Hazards? <input type="checkbox"/> YES / NO<br><small>Note: If Electrical Hazards are present they must be shown on Page 4.</small><br>4. Obstructions to Lift or Swing? <input type="checkbox"/> YES / NO<br><small>Note: If Obstructions are present they must be shown on Page 4.</small><br>5. Travel with Load Required? <input type="checkbox"/> YES / NO<br>6. Other? <input style="width: 100px;" type="text"/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| <b>B. CRANE</b><br>1. Type of Crane <u>Mobile Hydraulic Truck</u><br>2. Maximum Crane Capacity <input style="width: 100px;" type="text"/> lbs.<br>3. Radius (Maximum) <input style="width: 100px;" type="text"/> ft.<br>4. Radius (Minimum) <input style="width: 100px;" type="text"/> ft.<br>5. Boom Length (Maximum) <input style="width: 100px;" type="text"/> ft.<br>6. Boom Length (Minimum) <input style="width: 100px;" type="text"/> ft.<br>7. Crane Capacity (Max Radius) <input style="width: 100px;" type="text"/> lbs.<br>8. Crane Capacity (Min Radius) <input style="width: 100px;" type="text"/> lbs.<br>9. Boom Angle (Maximum) <input style="width: 100px;" type="text"/> deg.<br>10. Boom Angle (Minimum) <input style="width: 100px;" type="text"/> deg.<br>11. Gross Load of Crane <input style="width: 100px;" type="text"/> lbs.<br>12. Lift is <input style="width: 50px;" type="text"/> % of the Crane's rated capacity<br>13. If Jib/Ext. is to be used:<br>Length <input style="width: 100px;" type="text"/> ft.<br>Offset <input style="width: 100px;" type="text"/> ft.<br>14. Rated Capacity of Jib/Ext. <input style="width: 100px;" type="text"/> lbs |       | <b>F. OPERATOR QUALIFICATIONS</b><br>1. Certified Operator? <input type="checkbox"/> YES / NO<br>2. Option? <input style="width: 100px;" type="text"/><br>3. Certified for Type, Class & Capacity? <input type="checkbox"/> YES / NO<br>4. Designated in writing by emp <input style="width: 100px;" type="text"/><br><br><b>G. COMPLETED IN FIELD:</b> <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th></th> <th style="width: 10%;">(YES)</th> <th style="width: 10%;">NO</th> <th style="width: 10%;">PWD ET</th> </tr> </thead> <tbody> <tr><td>1. Crane Inspected</td><td></td><td></td><td></td></tr> <tr><td>2. Rigging Inspected</td><td></td><td></td><td></td></tr> <tr><td>3. Crane Set-up</td><td></td><td></td><td></td></tr> <tr><td>4. Overhead Hazard Check</td><td></td><td></td><td></td></tr> <tr><td>5. Swing Check</td><td></td><td></td><td></td></tr> <tr><td>6. Counterweight Check</td><td></td><td></td><td></td></tr> <tr><td>7. Operator Qualifications</td><td></td><td></td><td></td></tr> <tr><td>8. Signal Person Qualifications</td><td></td><td></td><td></td></tr> <tr><td>9. Rigger Qualifications</td><td></td><td></td><td></td></tr> <tr><td>10. Load Chart in Crane</td><td></td><td></td><td></td></tr> <tr><td>11. Load Test</td><td></td><td></td><td></td></tr> <tr><td>12. Tag Lines</td><td></td><td></td><td></td></tr> <tr><td>13. Wind Conditions</td><td></td><td></td><td></td></tr> <tr><td>14. Traffic Hazard Check</td><td></td><td></td><td></td></tr> <tr><td>15. Site Control</td><td></td><td></td><td></td></tr> </tbody> </table> |        |       |       | (YES) | NO            | PWD ET | 1. Crane Inspected |  |  |                 | 2. Rigging Inspected |  |  |  | 3. Crane Set-up |  |  |  | 4. Overhead Hazard Check |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  | 5. Swing Check |  |  |  | 6. Counterweight Check |  |  |  | 7. Operator Qualifications |  |  |  | 8. Signal Person Qualifications |  |  |  | 9. Rigger Qualifications |  |  |  | 10. Load Chart in Crane |  |  |  | 11. Load Test |  |  |  | 12. Tag Lines |  |  |  | 13. Wind Conditions |  |  |  | 14. Traffic Hazard Check |  |  |  | 15. Site Control |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | (YES) | NO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | PWD ET |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| 1. Crane Inspected                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| 2. Rigging Inspected                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| 3. Crane Set-up                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| 4. Overhead Hazard Check                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| 5. Swing Check                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| 6. Counterweight Check                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| 7. Operator Qualifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| 8. Signal Person Qualifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| 9. Rigger Qualifications                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| 10. Load Chart in Crane                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| 11. Load Test                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| 12. Tag Lines                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| 13. Wind Conditions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| 14. Traffic Hazard Check                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| 15. Site Control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| <b>C. HOIST ROPE</b> <table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th></th> <th style="width: 15%;">Main</th> <th style="width: 15%;">Aux 1</th> <th style="width: 15%;">Aux 2</th> <th style="width: 15%;"></th> </tr> </thead> <tbody> <tr> <td>1. # of Parts</td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>2. Rope Diamter</td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>3. Capacity</td> <td></td> <td></td> <td></td> <td></td> </tr> </tbody> </table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Main   | Aux 1 | Aux 2 |       | 1. # of Parts |        |                    |  |  | 2. Rope Diamter |                      |  |  |  | 3. Capacity     |  |  |  |                          | <b>H. SIGNATURES IN FIELD:</b><br>1. Crane Operator <input style="width: 100px;" type="text"/><br>2. Rigger <input style="width: 100px;" type="text"/><br>3. Signal Person <input style="width: 100px;" type="text"/><br>4. Lift Supervisor <input style="width: 100px;" type="text"/><br>5. PWD ET <input style="width: 100px;" type="text"/><br>6. Other <input style="width: 100px;" type="text"/> |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Main  | Aux 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Aux 2  |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| 1. # of Parts                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| 2. Rope Diamter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| 3. Capacity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |
| <b>D. RIGGING</b><br>1. Hitch Type(s) <input style="width: 100px;" type="text"/><br>2. No. of Slings: <input style="width: 50px;" type="text"/> Size: <input style="width: 50px;" type="text"/><br>3. Sling Type: <input style="width: 100px;" type="text"/><br>4. Sling Assembly Capacity: <input style="width: 100px;" type="text"/> lbs.<br>5. Shackle Size(s): <input style="width: 100px;" type="text"/><br>6. Shackle Rated Capacity(s) <input style="width: 100px;" type="text"/> lbs.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |       |       |       |               |        |                    |  |  |                 |                      |  |  |  |                 |  |  |  |                          |                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |                |  |  |  |                        |  |  |  |                            |  |  |  |                                 |  |  |  |                          |  |  |  |                         |  |  |  |               |  |  |  |               |  |  |  |                     |  |  |  |                          |  |  |  |                  |  |  |  |

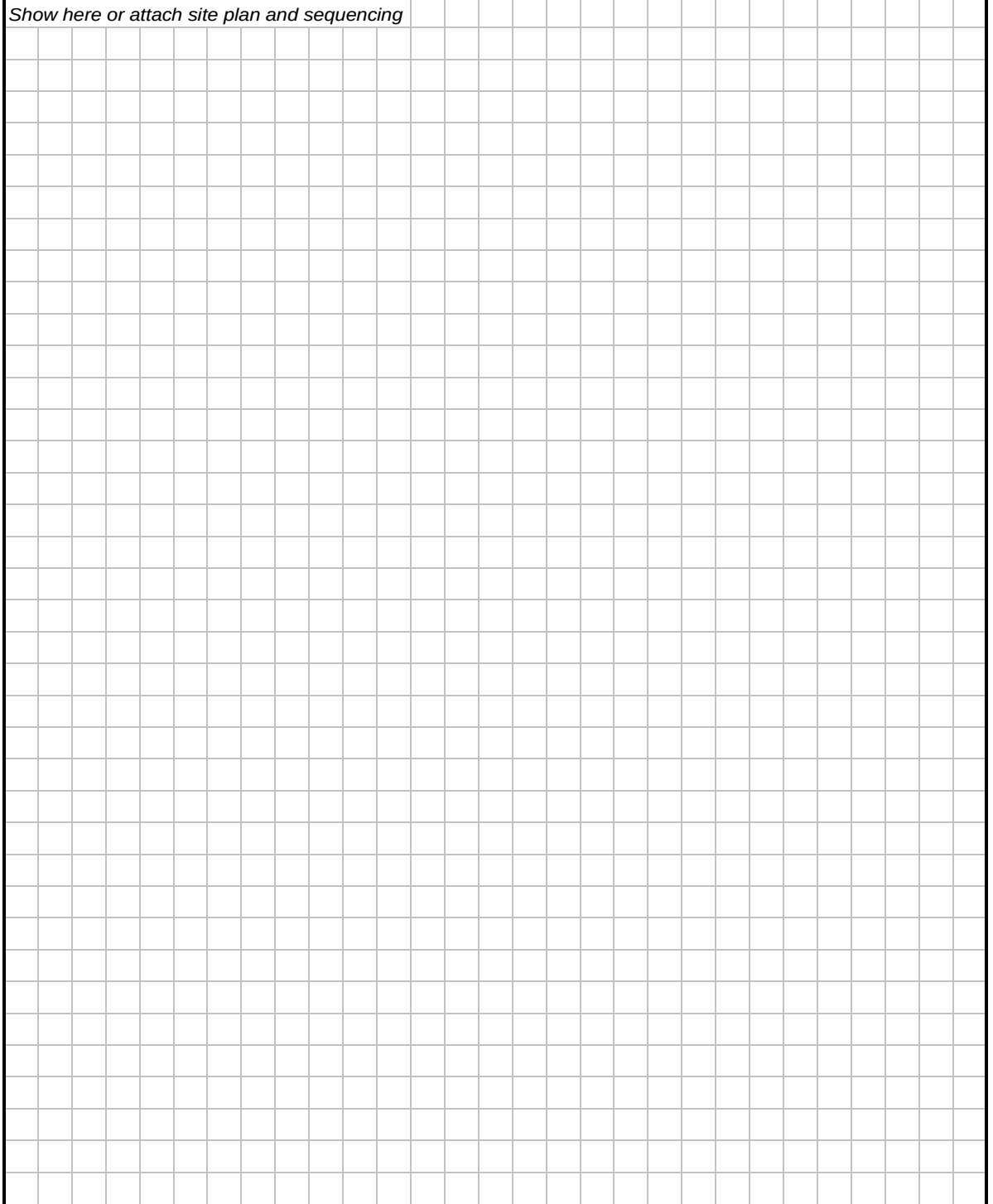
EM 385-1-1  
XX Jul 14

U.S. Army Corps of Engineers  
**CRITICAL LIFT PLAN**

For use of this form, see EM 385-1-1, Section 16. Proponent agency is Crane HHWG.

**SITE PLAN**

*Show here or attach site plan and sequencing*





| P2 CONTRACTOR CRANE / MULTI PURPOSE MACHINE & RIGGING<br>OPERATION CHECKLIST |                                           |
|------------------------------------------------------------------------------|-------------------------------------------|
| Prime Contractor:<br>Phone Number:                                           | Crane Operator's Name:                    |
| Crane Supplier:<br>Phone Number:                                             | Competent Riggers Name:                   |
| Crane Manufacturer:<br>Model No.                                             | Crane Capacity:<br>Crane Serial No:       |
| Inspection Due Date:                                                         | APP Rep:<br>Print _____<br><br>Sign _____ |

| CONTRACTOR CRANE OR RIGGING OPERATION CHECKLIST |                                                                                                                                                                                                                    |     |    |     |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|-----|
|                                                 |                                                                                                                                                                                                                    | YES | NO | N/A |
| 1                                               | Is the Certificate of Compliance, P-1, in the operator's cab (or in the contractor's on-site office for rigging operations) with the current operator's name listed?                                               |     |    |     |
| 2                                               | Is the crane/machine transited to and from the job site correctly? Are the OEM instructions for travel being followed?                                                                                             |     |    |     |
| 3                                               | Does the operator know the weight of the load to be lifted?                                                                                                                                                        |     |    |     |
| 4                                               | Is the load to be lifted within the crane/machine manufacturer's rated capacity in its present configuration?                                                                                                      |     |    |     |
| 5                                               | Are outriggers/stabilizers required and, if so, are they properly extended and down?                                                                                                                               |     |    |     |
| 6                                               | If outrigger/stabilizers are used, and the wheels are not off the ground is this the correct setup in accordance with the OEM?                                                                                     |     |    |     |
| 7                                               | Is the crane/machine level and on firm ground, or if the ground is not firm are adequate supporting materials provided?                                                                                            |     |    |     |
| 8                                               | If supporting materials are provided, is the entire surface of the outrigger/stabilizer pad supported and is the supporting material of sufficient strength to safely support the loaded outrigger/stabilizer pad? |     |    |     |
| 9                                               | If outriggers/stabilizers are not used, is the crane/machine rated for on-rubber lifts by the OEM's load chart?                                                                                                    |     |    |     |
| 10                                              | Is the swing radius of the crane counterweight clear of people and obstructions and are accessible areas within the swing area barricaded to prevent injury or damage?                                             |     |    |     |

Figure P-2 (1 of 3)

|    |                                                                                                                                                                                                                                             | YES | NO | N/A |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|-----|
| 11 | Is the Anti-two block safety device functional? When block is reconfigured has the device been retested to insure correct operation.                                                                                                        |     |    |     |
| 12 | Has the hook been centered over the load in such a manner to minimize swing?                                                                                                                                                                |     |    |     |
| 13 | Is the load well secured and balanced in the sling or lifting device after it is lifted a few inches for verification?                                                                                                                      |     |    |     |
| 14 | Is the lift and rotation path clear of obstructions?                                                                                                                                                                                        |     |    |     |
| 15 | If rotation of the load being lifted is hazardous, is a tagline or other restraint being used?                                                                                                                                              |     |    |     |
| 16 | Are personnel prevented from standing or passing under a suspended load?                                                                                                                                                                    |     |    |     |
| 17 | Is the operator paying full attention to the signal person?                                                                                                                                                                                 |     |    |     |
| 18 | Are proper signals being used? Is the operator responding properly to the signals? Are radios used for blind lifts?                                                                                                                         |     |    |     |
| 19 | Are empty hooks lashed or otherwise secured during travel to prevent swinging?                                                                                                                                                              |     |    |     |
| 20 | Does the operator remain at the controls while the load is suspended?                                                                                                                                                                       |     |    |     |
| 21 | Does the operator ensure that side loading is prohibited?                                                                                                                                                                                   |     |    |     |
| 22 | Are personnel prevented from riding on a load?                                                                                                                                                                                              |     |    |     |
| 23 | Are start and stop motions in a smooth fluid motion (no sudden acceleration or deceleration)?                                                                                                                                               |     |    |     |
| 24 | Is the lift a critical lift?                                                                                                                                                                                                                |     |    |     |
| 25 | If so, is a lift plan provided and understood and check-off sheets initialed and signed off?                                                                                                                                                |     |    |     |
| 26 | If overhead power lines are in the vicinity, is a critical lift plan provided addressing the requirements of 29 CFR 1926.1407-1411?                                                                                                         |     |    |     |
| 27 | If pick and carry operations are allowed and performed, are OEM directions followed (e.g. rotation lock engaged, boom centered over front or rear, etc.)?                                                                                   |     |    |     |
| 28 | When the crane/machine is left unattended, is it in a safe condition?                                                                                                                                                                       |     |    |     |
| 29 | Is rigging gear undamaged and acceptable for the application?                                                                                                                                                                               |     |    |     |
| 30 | Does rigging gear meet applicable ASME or host nation standards (e.g., ASME B30.9 for slings, B30.10 for hooks, B30.26 for rigging hardware such as shackles, safety hoist rings, and eyebolts, B30.20 for below the hook lifting devices)? |     |    |     |

Figure P-2 (2 of 3)



|                                                                            |                                                                                                                                                                                                                 | YES            | NO | N/A |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|----|-----|
| 31                                                                         | Was the rigging gear inspected prior to use?                                                                                                                                                                    |                |    |     |
| 32                                                                         | Is sling protection used to protect slings (especially synthetic slings) and equipment from damage due to abrasion and sharp corners and edges?                                                                 |                |    |     |
| 33                                                                         | Is the rigging gear used in accordance with its working load limit? Is the working load limit marked on the rigging gear?                                                                                       |                |    |     |
| 34                                                                         | Are positive latching devices (or "moussing") used on crane and rigging hooks?                                                                                                                                  |                |    |     |
| 35                                                                         | If a mobile crane is used on a barge, are all rules of 29 CFR 1926.1437(construction) being followed?                                                                                                           |                |    |     |
| 36                                                                         | If a mobile crane is used on a barge are the outriggers/stabilizers blocked or are the crawlers traveling in a defined space as allowed by 29 CFR 1926.1437 (construction)?                                     |                |    |     |
| 37                                                                         | For floating cranes, are rules of 29 CFR 1915 (ship repair) or 29 CFR 1926.1437 (construction) being followed?                                                                                                  |                |    |     |
| 38                                                                         | If a multi-purpose machine, forklift, or construction equipment is being used, is there proof from the OEM (or qualified PE) that the machine is approved for suspended load lifting and is there a load chart? |                |    |     |
| 39                                                                         | If a personnel lift is being performed with a crane or base mounted hoist, are all requirements of NAVFAC P-307, paragraph 11.1.g (5) for a crane or 11.1.m for a base mounted hoist being followed?            |                |    |     |
|                                                                            |                                                                                                                                                                                                                 |                |    |     |
|                                                                            |                                                                                                                                                                                                                 |                |    |     |
| Contractor:                                                                |                                                                                                                                                                                                                 | Subcontractor: |    |     |
| Location:                                                                  |                                                                                                                                                                                                                 | Date:          |    |     |
| Notes:                                                                     |                                                                                                                                                                                                                 |                |    |     |
| Location of Lift: _____<br>Material being lifted and Weight of lift: _____ |                                                                                                                                                                                                                 |                |    |     |
| Signature of Government Representative:                                    |                                                                                                                                                                                                                 |                |    |     |

Figure P-2 (3 of 3)



Form Reset

## Cover Sheet for Contractor Dig Safe Utility Locate Request Form (PWD Maine & OICC Maine & other)

Rev. 4/03/23

|     |    |                                                                                                          |
|-----|----|----------------------------------------------------------------------------------------------------------|
| Ktr |    | <i>Check blocks when action completed.</i>                                                               |
|     |    | <b>Prime Contractor Name:</b>                                                                            |
|     |    | <b>Contract Title:</b>                                                                                   |
|     |    | <b>Requestor Name:</b>                                                                                   |
|     |    | <b>Location of Excavation:</b>                                                                           |
|     |    |                                                                                                          |
| Ktr | ET | <i>Attach Items as Noted for DSC for Review:</i>                                                         |
|     |    | <b>Dig Safe Ticket #:</b>                                                                                |
|     |    | <b>Date of Excavation Start:</b>                                                                         |
|     |    | <b>Renewal Date:</b><br>(If work is planned to exceed 60 days from when State issued Ticket#)            |
|     |    | <b>If job includes digging &lt;20' from a gas line, Unitil has been called by Prime Ktr.</b>             |
|     |    | <b>If Potholing is required by contract, notified DSC with anticipated dates. Potholing is N/A.</b>      |
|     |    | <b>Ensure General Ktr has latest Total Map &amp; Project Plans prior to mapping:</b>                     |
|     |    | <b>Navfac Dig Safe Utility Locate Request Form Attached:</b><br>(Call in data page and Parts #1 thru #4) |
|     |    | <b>Confirm white paint is marked and all items cleared out of the way:</b>                               |
|     |    | <b>Ensure Genral Ktr has latest Total Map &amp; Project Plans prior to mapping:</b>                      |
|     |    | <b>Contractor Dig Safe Certification Form Completed and Attached:</b>                                    |
|     |    | <b>Contractor Utility Locate Pre-Excavation Safety Check List Completed and Attached:</b>                |
|     |    | <b>Anomalies, Dig Safe Notes, Risks, ORM, Mitigations to be discussed at Preparatory Mtg:</b>            |
|     |    | <b>Anomalies, Dig Safe Notes, Risks, ORM, Mitigations to be discussed at Initial Phase Mtg:</b>          |
|     |    | <b>Inside bldgs. for borings:</b><br>(as needed based on design, and for PWD ME shop)                    |

ET Validation all steps are completed. Sign/date\_\_\_\_\_



Public Works Department, Maine

RESET FORM

Form Revision Date 8/14/24

Attachment B

## NAVFAC DIG SAFE UTILITY LOCATE REQUEST FORM

*A Utility Locate Request Form is required for ANY Excavation (i.e. ground penetrating or concrete slab cutting, coring or drilling) either inside or outside of a building, which will penetrate more than 3", on the Shipyard. Submit this Form to NAVFAC PWD-ME DSC at least fourteen (14) calendar days prior to Excavation. See Note on Ticket expiration.*

**PART 1 INFORMATION THE PRIME/A&E/EXCAVATOR WILL NEED TO OBTAIN A DIG SAFE TICKET** (when the Excavator contacts DIG SAFE they will obtain a Dig Safe Ticket as proof of notification.)

**Caller Details:**

Caller Name \_\_\_\_\_ Title \_\_\_\_\_  
Phone # \_\_\_\_\_ Fax # \_\_\_\_\_ Alt # \_\_\_\_\_  
Email address \_\_\_\_\_ Business Hours \_\_\_\_\_ to \_\_\_\_\_  
Company  
Name \_\_\_\_\_  
Address \_\_\_\_\_  
City \_\_\_\_\_ State \_\_\_\_\_ Zip \_\_\_\_\_

**Location Details** (where excavation is planned)

(Check one) State: MA \_\_\_\_\_ ME \_\_\_\_\_ NH \_\_\_\_\_ RI \_\_\_\_\_ VT \_\_\_\_\_  
City/Town \_\_\_\_\_  
(Optional) Latitude \_\_\_\_\_ Longitude \_\_\_\_\_  
Address/Intersection \_\_\_\_\_  
Nearest Cross Street \_\_\_\_\_  
Additional Information \_\_\_\_\_  
Type of Work \_\_\_\_\_  
Excavation Planned Depth (feet) \_\_\_\_\_  
Area (i.e. St to building, in the St., sidewalk area, right side of building)  
\_\_\_\_\_  
Pre-marked? Yes \_\_\_\_\_ No \_\_\_\_\_  
Excavation planned Start Month \_\_\_\_\_ Day \_\_\_\_\_ Year \_\_\_\_\_ Time (military) \_\_\_\_\_  
Excavator Doing Work (if not same as above) \_\_\_\_\_



Form Revision Date 8/14/24

Attachment B cont'd

## **NAVFAC DIG SAFE UTILITY LOCATE REQUEST FORM**

*A Utility Locate Request Form is required for ANY Excavation (i.e. ground penetrating or concrete slab cutting, coring or drilling) either inside or outside of a building, which will penetrate more than 3", on the Shipyard. Submit this Form to PWD-ME DSC at least fourteen (14) calendar days prior to Excavation. See Note on ticket expiration.*

### **Part I – Completed by the Prime Ktr/A&E/Excavator.**

Today's Date: \_\_\_\_/\_\_\_\_/\_\_\_\_ DIG SAFE Ticket #: \_\_\_\_\_

Ticket Expiration Date:(60 days from date obtained): \_\_\_\_/\_\_\_\_/\_\_\_\_

Utility Companies Dig Safe will notify: \_\_\_\_\_

Requested by: \_\_\_\_\_ Phone #: \_\_\_\_\_

Code # / Company: \_\_\_\_\_ E-mail: \_\_\_\_\_

Contract #: \_\_\_\_\_ Project Title: \_\_\_\_\_

Shipyard POC: \_\_\_\_\_ Phone # \_\_\_\_\_

Excavation Location: \_\_\_\_\_ Area Pre-Marked YES\_\_\_\_, NO \_\_\_\_\_

Type of work: \_\_\_\_\_

Depth: (ft): \_\_\_\_\_ Anticipated Excavation Date: \_\_\_\_/\_\_\_\_/\_\_\_\_ Time: (military) \_\_\_\_\_

Attach a marked up PWD-ME Total Utility Map showing excavation/ground penetrating area.

NAVFAC Construction Contractors: Complete Part 1 and Submit Form to NAVFAC CM/ET

AE Firms: Complete Part 1 and Submit Form to PWD-ME DM

Shipyard Personnel: Complete Parts 1 & 2 and Submit Form to PWD-ME DSC

### **Part 2 – To be completed by the NAVFAC CM/ET/DM (If Shipyard work, Part 2 must be completed by Requestor)**

Date: \_\_\_\_/\_\_\_\_/\_\_\_\_ Name: \_\_\_\_\_ Phone # \_\_\_\_\_ Locate

Priority: Routine (3 Business days) \_\_\_\_; Emergency (<3 Business days)\_\_\_\_

\_\_\_\_\_ (DSC Approval Req if <3 Business days)

Part 1 Reviewed and Complete: YES\_\_\_\_, NO \_\_\_\_\_ Initial: \_\_\_\_\_

*Submit Completed Form to DSC. The DSC will review the Request and forward to the PWD-ME FSC Utility Locate Contract Rep who will review and forward to the Utility Locating Company for action.*

### **Part 3 – To be completed by FSC Utility Locating Company**

Approved by FSC Utility Locate Contractor: Initials \_\_\_\_\_

Date Utilities marked in the field: \_\_\_\_/\_\_\_\_/\_\_\_\_ Name: \_\_\_\_\_

Comments: \_\_\_\_\_

Utility Plan Discrepancies Noted: YES\_\_\_\_, NO\_\_\_\_ (PWD-ME FSC PAR & DSC must be Notified if discrepancies are found: Notification made: YES\_\_\_\_ to: \_\_\_\_\_

Comments: \_\_\_\_\_

*Submit to PWD-ME FSC Contract Rep who will forward to the DSC. If Plan Discrepancies are noted, provide a scaled markup plan showing the correct utility locations.*



Form Revision Date 8/14/24

Attachment B cont'd

## **NAVFAC DIG SAFE UTILITY LOCATE REQUEST FORM**

### **Part 4 – To be completed by PWD-ME DSC**

Date \_\_\_/\_\_\_/\_\_\_ Logged Into Database: YES\_\_\_NO\_\_\_ #\_\_\_\_\_

Any Hazardous/Mission Critical Utilities in Excavation Area:

YES\_\_\_ NO\_\_\_ (if YES, ensure Excavator is briefed to include additional controls of Appendix B.)

Comments: \_\_\_\_\_

Any unknown utilities, anomalies (including increased risk of passive detection modes), or discrepancies noted by FSC Utility Locate in Part 3 have been discussed with Project Team and adequate risk mitigations are in place.

DSC or DSC Representative Approval for excavation. (\*See note below).

Signature/Date: \_\_\_\_\_

*DSC return a copy of the completed Form to NAVFAC CM/ET/DM or Shipyard Personnel Requesting the Utility Locate.*

*NAVFAC CM/ET/DM/Requestor ensure the Excavator is aware if any Plan Discrepancies were found, establish risk mitigations, and to use extreme caution when excavating.*

*If Plan Discrepancies are noted, the PWD DSC shall coordinate any Utility Plan Updates with PWD-ME GIS lead.*

*Note:*

*If work has not begun within 30 days from the Dig Safe Ticket approval date or if the excavation area expands beyond the original limits, the Excavator must renew the Dig Safe ticket by notifying DIG SAFE (1-888-344-7233) and the Contracting Officer. If the proposed excavation is not completed within 60 calendar days after obtaining the Dig Safe ticket, the ticket expires. Tickets should be renewed by calling DIG SAFE (as needed) at least 72 hours prior to expiration so there is always an active ticket. The utility locators (not 3rd Party) must be notified to mark out each time a ticket is renewed.*

*\*DSC approval does not waive the construction contractor's responsibility to perform a 3rd Party Locate (if specified by contract) prior to beginning excavation. The excavator is always responsible to ensure utilities are not damaged during excavation. DSC approval does not diminish or change that responsibility.*





Form Revision 4/4/23

Attachment C

Public Works Department, Maine

Reset Form



## CONTRACTOR DIG-SAFE CERTIFICATION FORM

### **References:**

1. Government Safety Requirements (PWD ME) Section 01 35 26.00 22

**Project Title:** \_\_\_\_\_

**Prime Contractor:** \_\_\_\_\_

**Contract #:** \_\_\_\_\_

**Initial Dig-Safe Request #:** \_\_\_\_\_

### **A. Third Party Utility Location Certification**

In accordance with the Specification Section 01 35 26.00 22 Government Safety Requirements para 3.7.1, The Contractor must provide the services of a third party qualified independent utility locating company/person(s) (cannot be the Government's Utility Locating Firm) to positively identify underground utilities in the work area. The utility locating company shall indicate the utility type, size, and depth of all utilities within the proposed area of excavation including, but not limited to, voice/data, electrical, gas, water, air, sanitary sewer, and storm drainage. The third party independent locating firm is in addition to the PWD ME Dig Safe Process.

By signing below, the Contractor certifies that the Third Party Utility Locate Company has complete a review of the excavation area.

**Third Party Utility Locate Company:** \_\_\_\_\_

**Date Utility Locate was completed:** \_\_\_\_\_

**Print Name of the Utility Locator:** \_\_\_\_\_

**Utility Locator Signature/Date:** \_\_\_\_\_

**Any Discrepancies Found with the Marked Utilities (Yes/No):** Yes \_\_\_\_\_ No \_\_\_\_\_

(If Yes, notify the Contracting Officer Immediately)

**Prime Contractor Signature/Date:** \_\_\_\_\_





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## **CONTRACTOR DIG-SAFE CERTIFICATION FORM**

### **B. Prime Contractor Utility Location Certification**

Prime Contractor has performed a review of the following documents to ensure a comprehensive understanding of the sub-surface area to be excavated.

- Government's Dig-Safe report and associated drawings.
- Independent 3rd party Dig-Safe report and associated drawings.
- Project Contract drawings
- Government Total Map of approximate existing utility locations.

Prior to proceeding with excavation the prime contractor has the responsibility to access associated documentation of all items listed above and to mitigate the risk of a potential utility strike.

The final risk assessment shall be identified in the Contractor AHA and reviewed with the Contracting Officers Rep. Also, any hazardous/mission critical utilities identified by the DSC or plans, are covered in the Contractors AHA's and pre-job Safety Briefs prior to commencing excavation.

**By signing below, the Contractor certifies the above documentation has been thoroughly reviewed and evaluated for inconsistencies, anomalies, and stated inadequacies, and the prime contractor has risk mitigation actions planned to ensure excavation does not damage underground utilities.**

**Prime Contractor Signature/date; \_\_\_\_\_**



Reset Form

Public Works Department, Maine

Form Revision Date 4/04/23

Attachment D

**CONTRACTOR UTILITY LOCATE – PRE-EXCAVATION SAFETY  
CHECKLIST**

Project Title: \_\_\_\_\_

Prime Contractor: \_\_\_\_\_

Contract No.: \_\_\_\_\_

Excavator: \_\_\_\_\_

Initial \_\_\_\_\_ Gov't and Utility (if listed in Part 1 Appendix A) Dig Safe Markings Completed & Visible in the field.

Initial \_\_\_\_\_ NAVFAC Dig Safe Utility Locate Request Form returned from DSC and any Noted Discrepancies are Resolved and risks mitigated.

Initial \_\_\_\_\_ Contractor's Independent Third Party Utility Locate Completed and any discrepancies are resolved with NAVFAC (Attach Third Party Certification Form).

**Are Hazardous or Mission Critical Utilities identified in the excavation area (as identified by DSC or plans)?:** Yes \_\_\_\_\_ or No: \_\_\_\_\_  
If Yes complete the following:

\_\_\_\_\_ The Contractor must supplement daily Dig Safe procedures to include an additional checkoff on Contractor Daily Activity Plan ensuring all utilities have been clearly marked and reviewed with the SSHO.

\_\_\_\_\_ The Contractor must have a Dig Safe laminated utility color coding system posted in or near all heavy digging equipment for easy reference to type of utility.

\_\_\_\_\_ The SSHO must complete a pre-excavation walk as part of the morning procedure to ensure all known utilities are identified and markings are refreshed with the appropriate color-coded paint.

\_\_\_\_\_ Contractor must provide additional danger signage, to mark areas of known live underground utilities.

\_\_\_\_\_ Contractor must ensure a 'spotter' accompanies the equipment operator during excavation work.

\_\_\_\_\_ Contractor must provide Construction CM/ET notification no later than 7 working days prior to the preparatory and initial pre-excavation/ demo safety review meeting. (The CM/ET and the DSC will attend this meeting to ensure safety and preservation of Critical Utilities is discussed with adequate focus.)

\_\_\_\_\_ The Contractor must confirm & identify the closest utility isolation points & develop mitigation strategies with the utility owner (Coordinate with the DSC) to ensure the safe excavation adjacent to these utilities. Utility Outages to isolate utility systems may need to be considered in circumstances where the excavation work cannot be completed safely.

**\*NOTE: The Hazardous or Mission Critical Utility requirements listed above are in addition to, and do not replace other requirements. ALL UTILITIES require hand excavation using wood or fiberglass tools when work is expected to come within 3' of the utility, per contract (Div 1 Specs, General Notes, Excavation specs).**

Prime Contractor SSHO Signature/Date: \_\_\_\_\_

Excavator Signature/Date: \_\_\_\_\_

NAVFAC ET Signature/Date: \_\_\_\_\_

## Daily Worksite Checklist for Trenching/Excavation Sites

|               |         |        |                            |            |
|---------------|---------|--------|----------------------------|------------|
| Project:      |         | Date:  | Weather:                   | Soil Type: |
| Trench Depth: | Length: | Width: | Type of Protective System: |            |

|                                       |
|---------------------------------------|
| <b>Signature of Competent Person:</b> |
|---------------------------------------|

| Yes | No | N/A | Excavation                                                                                                                                                                                |
|-----|----|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |    |     | Trench box extends at least 18 inches above the vertical wall of the excavation and to within 2 feet of the bottom of the trench (or less if soil collapsing behind or below trench box). |
|     |    |     | Trench box installed in accordance with manufacturers specific instructions and use limitations.                                                                                          |
|     |    |     | Trench box inspected for damage or defects and pins and spreaders are securely installed.                                                                                                 |
|     |    |     | If other soil protective systems are used, they are installed in accordance with manufacturers Instructions OR are approved by a Registered Professional Engineer.                        |
|     |    |     | All employees at worksite trained in trenching safety procedures.                                                                                                                         |
|     |    |     | Surface encumbrances such as utility poles, heavy equipment supported or removed.                                                                                                         |
|     |    |     | Heavy equipment safety zone at least 1½ times depth of trench for if not supported.                                                                                                       |
|     |    |     | Employees protected from loose rock or soil.                                                                                                                                              |
|     |    |     | Spoils, materials, and equipment set back a minimum of 2' from edge of excavation.                                                                                                        |
|     |    |     | Walkways and bridges over excavations 6' or more in depth are at least 20 inches wide and are equipped with required guardrails.                                                          |
|     |    |     | Ladders placed no more than 25 feet apart.                                                                                                                                                |
|     |    |     | Employees prohibited from working or walking under suspended loads.                                                                                                                       |
|     |    |     | Employees prohibited from working on faces of sloped or benched excavations above other employees.                                                                                        |
|     |    |     | Warning system established and used when mobile equipment is operating near edge of excavation.                                                                                           |
|     |    |     | Barriers provided if trench opening is not readily apparent.                                                                                                                              |
|     |    |     | Barriers, fences available to secure area if left overnight.                                                                                                                              |

| Yes | No | N/A | Personal Protective Equipment                                                                               |
|-----|----|-----|-------------------------------------------------------------------------------------------------------------|
|     |    |     | Hard hats worn by all employees.                                                                            |
|     |    |     | Work boots or safety shoes worn by all employees.                                                           |
|     |    |     | Eye protection worn by all employees (if applicable).                                                       |
|     |    |     | Hearing protection worn by all employees (if applicable).                                                   |
|     |    |     | Warning vests, or other highly visible PPE provided and worn by all employees exposed to vehicular traffic. |

## Daily Worksite Checklist for Trenching/Excavation Sites – pg 2

| Yes | No | N/A | Utilities                                                                           |
|-----|----|-----|-------------------------------------------------------------------------------------|
|     |    |     | Utility companies contacted and/or utilities located.                               |
|     |    |     | Exact location of utilities marked when near excavation.                            |
|     |    |     | Underground installations protected, supported, or removed when excavation is open. |

| Yes | No | N/A | Wet Conditions                                                     |
|-----|----|-----|--------------------------------------------------------------------|
|     |    |     | Precautions taken to protect employees from accumulation of water. |
|     |    |     | Water removal equipment monitored by Competent Person.             |
|     |    |     | Surface water controlled or diverted.                              |
|     |    |     | Inspection made after each rainstorm.                              |

| Yes | No | N/A | Hazardous Atmosphere                                                                                                                                                                                      |
|-----|----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |    |     | Atmosphere tested when there is a possibility of oxygen deficiency or build-up of hazardous gases. If yes, atmosphere will be tested every _____.                                                         |
|     |    |     | Oxygen content is between 19.5% and 21%.                                                                                                                                                                  |
|     |    |     | Flammable gas build-up to 20% of lower explosive limit (LEL).                                                                                                                                             |
|     |    |     | Toxic Levels of gases are below limits set on gas monitor.                                                                                                                                                |
|     |    |     | Ventilation blowing into space and air intake placed away from vehicle exhaust.                                                                                                                           |
|     |    |     | Program Manager contacted if atmosphere is above established limits. Source of contaminant to be determined and eliminated prior to entry or Program Manager will establish special procedures for entry. |

### Emergency Procedures for Trench Cave-Ins

- **GET ALL OTHER EMPLOYEES OUT OF THE TRENCH!!**
- **CALL 438-2333**
- **NOTIFY COMPETENT PERSON**
- **Note time**
- **Note location of trapped worker(s)**
- **Leave all victims hand tools in place**
- **Shut down all heavy equipment**
- **Stop nearby traffic that may cause vibration**
- **Keep everyone back from trench at least 50 feet**
- **Gather information for rescue team**
- **WAIT for rescue team. Do not attempt to rescue.**

*Note-Do not attempt to dig the person out using hand tools or heavy equipment. This could cause the trench to collapse further and could cause further injuries!!!!*



**Public Works Department, Maine**

PWD -ME Dig Safe Coordinator

[gary.r.gagnon4.civ@us.navy.mil](mailto:gary.r.gagnon4.civ@us.navy.mil)

207-994-9208

All fields are mandatory and must be filled in.  
Prime Contractor to complete and submit within 2 working days of the utility strike. For detailed instructions, refer to Dig Safe Strike Checklist Instructions.

**UNDERGROUND UTILITY STRIKE REVIEW CHECKLIST**

**1. General Information:**

- a. Project & Contract#: \_\_\_\_\_
- b. Excavation Site: \_\_\_\_\_
- c. Utility Impacted: \_\_\_\_\_
- d. Depth of Utility Impacted (ft.): \_\_\_\_\_
- e. Date Utility was impacted: \_\_\_\_/\_\_\_\_/\_\_\_\_, Time Utility was impacted: \_\_\_\_\_
- f. Describe how and why damage occurred, what equipment was being used, what went wrong: (attach separate sheet as needed)  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_
- g. Prime Contractor company name: \_\_\_\_\_
- h. Excavator company name: \_\_\_\_\_
- i. Project ET: \_\_\_\_\_
- j. PWD Locate Request Form Revision date used: \_\_\_\_/\_\_\_\_/\_\_\_\_ (Attach a completed copy)
- k. Is this identified as a Hazardous/Mission Critical utility on Part 4 of the Dig Safe Request Form? Yes: \_\_\_\_\_, No: \_\_\_\_\_

**2. Utility Locate Information:**

**a. Dig Safe Locate Ticket;**

- i. Original Dig Safe Ticket #: \_\_\_\_\_ and date pulled: \_\_\_\_\_
- ii. Latest date Dig Safe Ticket was renewed: \_\_\_\_\_ N/A: \_\_\_\_\_
- iii. Date of latest FSC Utility Locate markings in excavation area: \_\_\_\_\_
- iv. Was Prime Contractor maintaining visible markings? Yes: \_\_\_\_\_, No: \_\_\_\_\_
- v. Was the impacted utility currently marked where it was struck? Yes: \_\_\_\_\_
  - 1. If not, describe situation: \_\_\_\_\_  
\_\_\_\_\_
- vi. Did FSC Utility Locate KTR flag anomalies or discrepancies on Field notes that apply to this utility? (Attach FSC Locate sketch and field notes) Yes: \_\_\_\_\_, No: \_\_\_\_\_
  - 1. Was DSC involved in resolution of anomalies or discrepancies? Yes: \_\_\_\_\_, or why not? \_\_\_\_\_
  - 2. Was there a post FSC Locate Tailgate discussion? Yes: \_\_\_\_\_, or why not? \_\_\_\_\_  
\_\_\_\_\_



### Public Works Department, Maine

3. Did Prime Contractor attempt to mitigate utility impact risk from anomalies or discrepancies? Yes: \_\_\_\_\_, No: \_\_\_\_\_  
Describe how, or why not (include separate sheet if needed): \_\_\_\_\_  
\_\_\_\_\_

- vii. Did PWD ME Dig Safe Coordinator approve excavation on Utility Locate Request Form Part 4: Yes: \_\_\_\_\_, or why not? \_\_\_\_\_

**b. 3<sup>rd</sup> Party Utility Locate Information:**

- i. 3rd Party Locate Company: \_\_\_\_\_  
ii. Date of the 3rd Party Locate: \_\_\_\_/\_\_\_\_/\_\_\_\_  
iii. Did the 3rd Party flag anomalies or discrepancies on field notes that apply to this utility? (Attach 3<sup>rd</sup> Party Locate sketch and field notes) Yes: \_\_\_\_\_, No: \_\_\_\_\_  
1. Was DSC involved in resolution of anomalies or discrepancies? Yes: \_\_\_\_\_, or why not? \_\_\_\_\_  
2. Was there a post 3rd Party Tailgate discussion? Yes: \_\_\_\_\_, or why not? \_\_\_\_\_

Did Prime Contractor attempt to mitigate utility impact risk from anomalies or discrepancies? Yes: \_\_\_\_\_, No: \_\_\_\_\_  
Describe how, or why not (include separate sheet if needed): \_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

- iv. Did Prime Contractor complete the Contractor Dig Safe Certification Form prior to excavation? (attach completed copy): Yes: \_\_\_\_\_, or why not? \_\_\_\_\_  
\_\_\_\_\_

**3. Preparatory & Initial Meetings Information:**

- a. Date of the Preparatory Meeting: \_\_\_\_/\_\_\_\_/\_\_\_\_  
b. Did PWD DSC attend the Preparatory meeting? Yes: \_\_\_\_\_ or why not? \_\_\_\_\_  
c. Were Haz/Mission Critical Utilities reviewed at the meeting? Yes: \_\_\_\_\_, No: \_\_\_\_\_  
d. Did PWD UEM personnel attend the meeting? Yes: \_\_\_\_\_, No: \_\_\_\_\_  
e. Were the Utility Markings in the field reviewed? Yes: \_\_\_\_\_, No: \_\_\_\_\_  
f. Were FSC and 3<sup>rd</sup> Party Identified discrepancies or anomalies noted and discussed? Yes: \_\_\_\_\_, If No, why not? \_\_\_\_\_  
g. Date of the Initial Meeting: \_\_\_\_\_  
h. Were Utility Risk Mitigation Strategies reviewed (attach minutes): Yes: \_\_\_\_\_, if No, why not? \_\_\_\_\_  
i. Were Utility Outages part of the Risk Mitigation Plan? Yes: \_\_\_\_\_, No: \_\_\_\_\_  
j. Was excavation performed with expected precaution? Yes: \_\_\_\_\_, No: \_\_\_\_\_



**Public Works Department, Maine**

**4. Utility Mapping Review (DSC input):**

- a. Did the FSC Dig Safe KTR have the latest Total Map? (Explain): Yes: \_\_\_\_\_, No: \_\_\_\_\_

\_\_\_\_\_

- b. Were inaccuracies between the field conditions & the Total Map found? Explain;  
Yes: \_\_\_\_\_, No: \_\_\_\_\_, \_\_\_\_\_

\_\_\_\_\_

- c. If Total Map inaccuracies were identified for this area, has CAD updated for most  
current As-Built Info? Yes: \_\_\_\_\_, No: \_\_\_\_\_

- d. Was digging within 3' of marked utility? Yes: \_\_\_\_\_, No: \_\_\_\_\_

**5. Items to submit along with this Review Checklist:**

- a. \_\_\_\_\_ A copy of the Dig Safe Ticket.  
b. \_\_\_\_\_ Diagram/sketch of the area with specific location of the strike.  
c. \_\_\_\_\_ Appropriate Contractor's Daily Work Notes.  
d. \_\_\_\_\_ FSC Dig Safe Locate Report (sketch and field notes).  
e. \_\_\_\_\_ 3<sup>rd</sup> Party Locate Report (sketch and field notes).  
f. \_\_\_\_\_ Statement from Equipment Operator.  
g. \_\_\_\_\_ Preparatory Meeting Minutes.  
h. \_\_\_\_\_ Initial Meeting Minutes.

**6. Project Specific Comments:**

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

**7. What could be done better?**

a.

b.

c.

**8. Lessons Learned:**

a.

b.

c.



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**ATTACHMENTS FOR DETAILS AND EXPLANATIONS AS NEEDED**



# The Color Coding System

Color coded paint or flags are used to identify the type of underground facilities.

|               |                                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| <b>RED</b>    | <i>Electric Power Lines, Cables, Conduit and Lighting Cables</i>                  |
| <b>YELLOW</b> | <i>Gas, Oil, Steam, Petroleum or Gaseous Materials, Compressed Air</i>            |
| <b>ORANGE</b> | <i>Communication, Alarm or Signal Lines, Cables or Conduit</i>                    |
| <b>BLUE</b>   | <i>Potable Water</i>                                                              |
| <b>GREEN</b>  | <i>Sewers and Drain Lines</i>                                                     |
| <b>PURPLE</b> | <i>Non-Potable Water: Reclaimed Water, Irrigation and Slurry Lines, ASW, etc.</i> |
| <b>PINK</b>   | <i>Temporary Survey Markings/ Unknown Utilities/Anomalies</i>                     |

SECTION 01 42 00

SOURCES FOR REFERENCE PUBLICATIONS

02/19

PART 1 GENERAL

1.1 REFERENCES

Various publications are referenced in other sections of the specifications to establish requirements for the work. These references are identified in each section by document number, date and title. The document number used in the citation is the number assigned by the standards producing organization (e.g., ASTM B564 Standard Specification for Nickel Alloy Forgings). However, when the standards producing organization has not assigned a number to a document, an identifying number has been assigned for reference purposes.

1.2 ORDERING INFORMATION

The addresses of the standards publishing organizations whose documents are referenced in other sections of these specifications are listed below, and if the source of the publications is different from the address of the sponsoring organization, that information is also provided.

AIR BARRIER ASSOCIATION OF AMERICA (ABAA)  
1600 Boston-Providence Hwy  
Walpole, MA 02081  
Ph: 1-866-956-5888  
Fax: 1-866-956-5819  
Internet: <https://www.airbarrier.org/>

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)  
30 West University Drive  
Arlington Heights, IL 60004-1893  
Ph: 847-394-0150  
Fax: 847-253-0088  
E-mail: [communications@amca.org](mailto:communications@amca.org)  
Internet: <http://www.amca.org>

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)  
2111 Wilson Blvd, Suite 400  
Arlington, VA 22201  
Ph: 703-524-8800  
Internet: <http://www.ahrinet.org>

ALUMINUM ASSOCIATION (AA)  
1400 Crystal Drive  
Suite 430  
Arlington, VA 22202  
Ph: 703-358-2960  
E-Mail: [info@aluminum.org](mailto:info@aluminum.org)  
Internet: <https://www.aluminum.org/>

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)  
1900 E Golf Rd, Suite 1250  
Schaumburg, IL 60173  
Ph: 847-303-5664

E-mail: [customerservice@aamanet.org](mailto:customerservice@aamanet.org)  
Internet: <https://aamanet.org/>

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS  
(AASHTO)  
444 North Capital Street, NW, Suite 249  
Washington, DC 20001  
Ph: 202-624-5800  
Fax: 202-624-5806  
E-Mail: [info@aaashto.org](mailto:info@aaashto.org)  
Internet: <https://www.transportation.org/>

AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA)  
330 N. Wabash Ave., Suite 2000  
Chicago, IL 60611  
Ph: 202-367-1155  
E-mail: [info@americanbearings.org](mailto:info@americanbearings.org)  
Internet: <https://www.americanbearings.org/>

AMERICAN CONCRETE INSTITUTE (ACI)  
38800 Country Club Drive  
Farmington Hills, MI 48331-3439  
Ph: 248-848-3700  
Fax: 248-848-3701  
Internet: <https://www.concrete.org/>

AMERICAN CONCRETE PIPE ASSOCIATION (ACPA)  
8445 Freeport Parkway, Suite 350  
Irving, TX 75063-2595  
Ph: 972-506-7216  
Fax: 972-506-7682  
E-mail: [info@concrete-pipe.org](mailto:info@concrete-pipe.org)  
Internet: <https://www.concretepipe.org/>

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)  
1330 Kemper Meadow Drive  
Cincinnati, OH 45240  
Ph: 513-742-2020  
Fax: 513-742-3355  
Internet: <https://www.acgih.org/>

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)  
1001 N. Fairfax Street, Suite 500  
Alexandria, VA 22314-1587  
Ph: 703-684-0211  
Fax: 703-684-0242  
E-mail: [tech@agma.org](mailto:tech@agma.org)  
Internet: <https://www.agma.org/>

AMERICAN HARDBOARD ASSOCIATION (AHA)  
1210 West Northwest Highway  
Palatine, IL 60067  
Ph: 847-934-8800  
Fax: 847-934-8803  
E-mail: [aha@hardboard.org](mailto:aha@hardboard.org)  
Internet: <http://domensino.com/AHA/>

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)  
130 East Randolph, Suite 2000

Chicago, IL 60601  
Ph: 312-670-5444  
Fax: 312-670-5403  
Steel Solutions Center: 866-275-2472  
E-mail: [solutions@aisc.org](mailto:solutions@aisc.org)  
Internet: <https://www.aisc.org/>

AMERICAN INSTITUTE OF TIMBER CONSTRUCTION (AITC)  
7012 South Revere Parkway, Suite 140  
Centennial, CO 80112  
Ph: 503-639-0651  
Fax: 503-684-8928  
E-mail: [mschoen@wclib.org](mailto:mschoen@wclib.org)  
Internet: <http://www.aitc-glulam.org>

AMERICAN IRON AND STEEL INSTITUTE (AISI)  
25 Massachusetts Avenue, NW Suite 800  
Washington, DC 20001  
Ph: 202-452-7100  
Internet: <https://www.steel.org/>

AMERICAN LADDER INSTITUTE (ALI)  
330 N. Wabash, Suite 2000  
Chicago, IL 60611  
Ph: 312-321-6806  
Fax: 312-673-6929  
E-mail: [info@americanladderinstitute.org](mailto:info@americanladderinstitute.org)  
Internet: <https://www.americanladderinstitute.org>

AMERICAN LUMBER STANDARDS COMMITTEE (ALSC)  
7470 New Technology Way, Suite F  
Frederick, MD 21703  
Ph: 301-972-1700  
Fax: 301-540-8004  
E-mail: [alsc@alsc.org](mailto:alsc@alsc.org)  
Internet: <http://www.alsc.org>

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)  
1899 L Street, NW, 11th Floor  
Washington, DC 20036  
Ph: 202-293-8020  
Fax: 202-293-9287  
E-mail: [storemanager@ansi.org](mailto:storemanager@ansi.org)  
Internet: <https://www.ansi.org/>

AMERICAN PETROLEUM INSTITUTE (API)  
1220 L Street, NW  
Washington, DC 20005-4070  
Ph: 202-682-8000  
Internet: <https://www.api.org/>

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)  
P.O. Box 28518  
1711 Arlingate Lane  
Columbus, OH 43228-0518  
Ph: 800-222-2768 or 614-274-6003  
Fax: 614-274-6899  
E-mail: [tjones@asnt.org](mailto:tjones@asnt.org)  
Internet: <https://www.asnt.org/>

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)  
1801 Alexander Bell Drive  
Reston, VA 20191  
Ph: 800-548-2723; 703-295-6300  
Internet: <https://www.asce.org/>

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING  
ENGINEERS (ASHRAE)  
1791 Tullie Circle, NE  
Atlanta, GA 30329  
Ph: 404-636-8400 or 800-527-4723  
Fax: 404-321-5478  
E-mail: [ashrae@ashrae.org](mailto:ashrae@ashrae.org)  
Internet: <https://www.ashrae.org/>

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)  
Two Park Avenue  
New York, NY 10016-5990  
Ph: 800-843-2763  
Fax: 973-882-1717  
E-mail: [customercare@asme.org](mailto:customercare@asme.org)  
Internet: <https://www.asme.org/>

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)  
520 N. Northwest Highway  
Park Ridge, IL 60068  
Ph: 847-699-2929  
E-mail: [customerservice@assp.org](mailto:customerservice@assp.org)  
Internet: <https://www.assp.org/>

AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)  
18927 Hickory Creek Drive, Suite 220  
Mokena, IL 60448  
Ph: 708-995-3019  
Fax: 708-479-6139  
Internet: <http://www.asse-plumbing.org>

AMERICAN WATER WORKS ASSOCIATION (AWWA)  
6666 W. Quincy Avenue  
Denver, CO 80235 USA  
Ph: 303-794-7711 or 800-926-7337  
Fax: 303-347-0804  
Internet: <https://www.awwa.org/>

AMERICAN WELDING SOCIETY (AWS)  
8669 NW 36 Street, #130  
Miami, FL 33166-6672  
Ph: 800-443-9353  
Internet: <https://www.aws.org/>

AMERICAN WOOD COUNCIL (AWC)  
222 Catoctin Circle SE, Suite 201  
Leesburg, VA 20175  
Ph: 800-890-7732  
Fax: 412-741-0609  
E-mail: [publications@awc.org](mailto:publications@awc.org)  
Internet: <https://www.awc.org/>

AMERICAN WOOD PROTECTION ASSOCIATION (AWPA)  
P.O. Box 361784  
Birmingham, AL 35236-1784  
Ph: 205-733-4077  
Fax: 205-733-4075  
Internet: <http://www.awpa.com>

APA - THE ENGINEERED WOOD ASSOCIATION (APA)  
7011 South 19th St.  
Tacoma, WA 98466-5333  
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Fax: 253-565-7265  
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Washington, DC 20301-1400  
Ph: 703-571-3343  
Fax: 215-697-1462  
E-mail: [customerservice@ntis.gov](mailto:customerservice@ntis.gov)  
Internet: <https://www.ntis.gov/>  
Obtain Military Specifications, Standards and Related Publications  
from:  
Acquisition Streamlining and Standardization Information System  
(ASSIST)  
Department of Defense Single Stock Point (DODSSP)  
Document Automation and Production Service (DAPS)  
Building 4/D  
700 Robbins Avenue  
Philadelphia, PA 19111-5094  
Ph: 215-697-6396 - for account/password issues  
Internet: <https://assist.dla.mil/online/start/>; account  
registration required  
Obtain Unified Facilities Criteria (UFC) from:  
Whole Building Design Guide (WBDG)  
National Institute of Building Sciences (NIBS)  
1090 Vermont Avenue NW, Suite 700  
Washington, DC 20005  
Ph: 202-289-7800  
Fax: 202-289-1092  
Internet:  
<https://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc>

U.S. DEPARTMENT OF HOUSING AND URBAN DEVELOPMENT (HUD)  
HUD User  
P.O. Box 23268  
Washington, DC 20026-3268  
Ph: 800-245-2691 or 202-708-3178  
TDD: 800-927-7589  
Fax: 202-708-9981  
E-mail: [helpdesk@huduser.gov](mailto:helpdesk@huduser.gov)  
Internet: <https://www.huduser.gov>

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)  
1200 Pennsylvania Avenue, N.W.  
Washington, DC 20004  
Ph: 202-564-4700  
Internet: <https://www.epa.gov>  
--- Some EPA documents are available only from:

National Technical Information Service (NTIS)  
5301 Shawnee Road  
Alexandria, VA 22312  
Ph: 703-605-6060 or 1-800-363-2068  
Fax: 703-605-6880  
TDD: 703-487-4639  
E-mail: [info@ntis.gov](mailto:info@ntis.gov)  
Internet: <https://www.ntis.gov/>

U.S. FEDERAL AVIATION ADMINISTRATION (FAA)  
Order for sale documents from:  
Superintendent of Documents  
U.S. Government Publishing Office (GPO)  
732 N. Capitol Street, NW  
Washington, DC 20401  
Ph: 202-512-1800 or 866-512-1800  
Bookstore: 202-512-0132  
Internet: <https://www.gpo.gov/>  
Order free documents from:  
U.S. Department of Transportation  
Federal Aviation Administration  
800 Independence Avenue, SW  
Washington, DC 20591  
Ph: 866-835-5322  
Internet: <https://www.faa.gov/>

U.S. FEDERAL COMMUNICATIONS COMMISSION (FCC)  
445 12th Street SW  
Washington, DC 20554  
Ph: 888-225-5322  
TTY: 888-835-5322  
Fax: 866-418-0232  
Internet: <https://www.fcc.gov/>  
Order Publications From:  
Superintendent of Documents  
U.S. Government Publishing Office (GPO)  
732 N. Capitol Street, NW  
Washington, DC 20401  
Ph: 202-512-1800 or 866-512-1800  
Bookstore: 202-512-0132  
Internet: <https://www.gpo.gov/>

U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)  
1200 New Jersey Ave., SE  
Washington, DC 20590  
Ph: 202-366-4000  
E-mail: [ExecSecretariat.FHWA@dot.gov](mailto:ExecSecretariat.FHWA@dot.gov)  
Internet: <https://www.fhwa.dot.gov/>  
Order from:  
Superintendent of Documents  
U.S. Government Publishing Office (GPO)  
732 N. Capitol Street, NW  
Washington, DC 20401  
Ph: 202-512-1800 or 866-512-1800  
Bookstore: 202-512-0132  
Internet: <https://www.gpo.gov/>

U.S. GENERAL SERVICES ADMINISTRATION (GSA)  
General Services Administration

1800 F Street, NW  
Washington, DC 20405  
Ph: 1-844-472-4111  
Internet: <https://www.gsaelibrary.gsa.gov/ElibMain/home.do>  
Obtain documents from:  
Acquisition Streamlining and Standardization Information System  
(ASSIST)  
Internet: <https://assist.dla.mil/online/start/>; account  
registration required

U. S. GREEN BUILDING COUNCIL (USGBC)  
2101 L St NW, Suite 500  
Washington, DC 20037  
Ph: 202-828-7422  
Internet: <https://new.usgbc.org/>

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)  
8601 Adelphi Road  
College Park, MD 20740-6001  
Ph: 866-272-6272  
Internet: <https://www.archives.gov/>  
Order documents from:  
Superintendent of Documents  
U.S. Government Publishing Office (GPO)  
732 N. Capitol Street, NW  
Washington, DC 20401  
Ph: 202-512-1800 or 866-512-1800  
Bookstore: 202-512-0132  
Internet: <https://www.gpo.gov/>

U.S. NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)  
1322 Patterson Ave. SE, Suite 1000  
Washington Navy Yard, DC 20374-5065  
Ph: 202-685-9387  
Internet: <http://www.navfac.navy.mil>

UNDERWRITERS LABORATORIES (UL)  
2600 N.W. Lake Road  
Camas, WA 98607-8542  
Ph: 877-854-3577 or 360-817-5500  
E-mail: [CustomerExperienceCenter@ul.com](mailto:CustomerExperienceCenter@ul.com)  
Internet: <https://www.ul.com/>  
UL Directories available through IHS at <https://ihsmarkit.com/>

VIBRATION ISOLATION AND SEISMIC CONTROL MANUFACTURERS ASSOCIATION  
(VISCMA)  
994 Old Eagle School Road  
Suite 1019  
Wayne, PA 19087-1866  
Ph: 610-971-4850  
E-mail: [info@viscma.com](mailto:info@viscma.com)  
Internet: <http://www.viscma.com>

WEST COAST LUMBER INSPECTION BUREAU (WCLIB)  
6980 S.W. Varns  
Tigard, OR 97223  
Ph: 503-639-0651  
Fax: 503-684-8928  
E-mail: [info@wclib.org](mailto:info@wclib.org)

Internet: <http://www.wclib.org>

WESTERN WOOD PRODUCTS ASSOCIATION (WWPA)  
1500 SW First Ave., Suite 870  
Portland, OR 97201  
Ph: 503-224-3930  
E-mail: [info@wwpa.org](mailto:info@wwpa.org)  
Internet: <http://www.wwpa.org>

WINDOW AND DOOR MANUFACTURERS ASSOCIATION (WDMA)  
2025 M Street, NW, Suite 800  
Washington, DC 20036-3309  
Ph: 202-367-1157  
or  
330 N Wabash Avenue, Suite 2000  
Chicago, IL 60611  
Ph: 312-321-6802  
E-mail: [membersupport@wdma.com](mailto:membersupport@wdma.com)  
Internet: <https://www.wdma.com/>

WOODWORK INSTITUTE (WI)  
3188 Industrial Blvd.  
West Sacramento, CA 95691  
Ph: 916-372-9943  
Fax: 916-372-9950  
E-mail: [info@woodinst.com](mailto:info@woodinst.com)  
Internet: <https://woodworkinstitute.com>

## PART 2 PRODUCTS

Not used

## PART 3 EXECUTION

Not used

-- End of Section --

SECTION 01 44 00.00 22

ARCHAEOLOGICAL MONITORING (PWD ME)  
04/25

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at the Portsmouth Naval Shipyard and at PWD ME AOR Facilities.

1.1 SUMMARY

Perform archaeological monitoring for all forms of soil disturbing activities and excavations within archaeologically sensitive portions of the project area **as identified in the project drawings**. Work must include but is not limited to: background research and review of previous surveys, monitoring during excavation, installation of components, regrading, and other work indicated or specified. Results of monitoring efforts must be summarized in Archaeological Monitoring Reports, including progress reports, a draft monitoring report, a final draft monitoring report, and a final monitoring report. Artifacts collected during monitoring efforts must be catalogued and curated.

The Government will provide available existing documentation pertaining to the project area, which will include previous investigations, plans, photographs, and research.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 79 (2022) Curation of Federally Owned or Administered Archeological Collections

36 CFR 61 (2017) Procedures for State, Tribal, and Local Government Historic Preservation Programs

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. The following shall be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Archaeologist Qualifications; G

Archaeological Coordinator Designation Letter

Archaeological Monitoring Plan; G

## SD-06 Test Reports

Archaeological Resource Report; G

Monthly Progress Reports; G

Archaeological Monitoring Report - Draft; G

Archaeological Monitoring Report - Final Draft; G

Archaeological Monitoring Report - Final; G

Curated Artifacts (if collected); G

### 1.4 QUALITY ASSURANCE

The Contractor must submit Archaeologist Qualifications prior to submission of the Archaeological Monitoring Plan. The Archaeologist Qualifications must be approved by the Government, in accordance with the procedures of Section 01 33 00 SUBMITTAL PROCEDURES, prior to the Archaeological Coordination Meeting and submission of the Archaeological Monitoring Plan. The Archaeologist must meet the Secretary of the Interior's Professional Qualification Standards for Archaeologist as outlined in 36 CFR 61 as required by the National Historic Preservation Act (Section 112, a, 1, A), OPNAVIST 5090.1E (Section 13-3.6), and the Principal Investigator must be an approved Archaeologist on the Maine Historic Preservation Commission's Level II Approved List. Field staff do not need to be named on the Level II list if they are supervised by an archaeologist with those qualifications.

The Archaeologist must specialize in historic archaeology and have experience monitoring construction work in urban or industrial archaeological settings, and have demonstrated experience in recognizing 18th and 19th century burials. The Archaeologist must have a demonstrated track record of success in carrying projects to completion, including the submission of all final project documentation and curated artifacts.

### 1.5 COORDINATION

The Contractor shall designate in writing, with an Archaeological Coordinator Designation Letter, the Archaeological Coordinator responsible for coordination with the archaeological subcontractor. This individual must have a thorough understanding of the project schedule. The Contractor shall notify Contracting Officer of any changes in the Archaeological Coordinator in writing.

The Contractor is responsible for obtaining permission and clearance from the Contracting Officer to enter areas of the required fieldwork.

#### 1.5.1 ARCHAEOLOGICAL COORDINATION MEETING

Prior to the start of any ground disturbing/excavation work and prior to submission of the Archaeological Monitoring Plan, the Archaeological Coordinator, project archaeologist, Contractor's QC Manager, PWD ME CM/ET and PWD-ME Cultural Resources Manager must meet at an Archaeological Coordination Meeting to review the proposed scope of work, establish a protocol for artifact collection, and review the relevant documentation.

The Contractor is responsible for recording minutes of all meetings and

furnishing a copy of the minutes to the Contracting Officer within 3 calendar days of the meeting.

For projects with a duration exceeding one year, re-initial archaeological coordination meetings must be held on an annual basis to ensure continued alignment on archaeological protocols. In addition, a re-initial archaeological coordination meeting must be held should there be a change in key personnel (Archaeological Subcontractor; Archaeological Coordinator; QC Manager, Project Superintendent, Project Manager, etc.)

#### 1.6 ARCHAEOLOGICAL MONITORING PLAN

The Contractor must submit an Archaeological Monitoring Plan prior to the start of construction. The monitoring plan shall be informed by the discussions held at the Archaeological Coordination Meeting and must include:

- a. Identification of the areas to be monitored. Provide supporting graphics.
- b. A description of anticipated or potential resources within the project area, including anticipated locations. Provide supporting graphics.
- c. A description of field methods appropriate to the project including but not limited to mapping/GPS data collection methods, photography, night-work protocols, dewatering protocols, working within Installation Restoration (IR) Sites, human remains discovery protocols, and procedures for encountering potential UXO.
- d. Project-specific discovery protocols as approved through consultation and as discussed at the Archaeological Coordination Meeting. Include protocols for chain of communication.
- e. Artifact collection/discard methods.
- f. Reporting requirements.
- g. Curation requirements and anticipated repository.
- h. Archaeologist Qualifications, including experience related to urban or industrial archaeological settings and in recognizing 18th and 19th century burials. These qualifications must have been approved prior to submission of the Archaeological Monitoring Plan, as outlined in Paragraph QUALITY ASSURANCE.
- i. Points of contact.

#### PART 2 PRODUCTS

Not used.

#### PART 3 EXECUTION

##### 3.1 INSPECTION AND MONITORING

- a. The Archaeologist must observe excavation activities for archaeological resources and shall be authorized to collect archaeological material as warranted for analysis. The Archaeologist must be present on the project site according to a schedule agreed

upon by the Contracting Officer until the Contracting Officer agrees that the project construction activities could have no effects on significant archaeological resources.

- b. The Archaeologist must advise all project Contractors to be on the alert for evidence of the presence of the expected resource(s), of how to identify the evidence of the expected resource(s), and of the appropriate protocol in the event of an archaeological resource discovery. GPS points, photographs, sketch maps, and measured drawings must be provided, as applicable, to document resources encountered during the course of excavations.
- c. If an intact archaeological resource is encountered, all soil disturbing activities in the vicinity of the resource must cease. The Archaeologist must be empowered to halt excavation in the area and must immediately notify the Archaeological Coordinator (who must then notify the Contracting Officer) of the encountered archaeological resource. The Archaeologist must, after making a reasonable effort to assess the identity, integrity, and significance of the encountered archaeological resource, present the findings of this assessment to the Contracting Officer. Following verbal communication of this information, the assessment must be formalized in an Archaeological Resource Report.
- d. If suspected human remains (including but not limited to: bone, bone fragments, teeth, mummified flesh, burials, and cremations) are encountered, stop work in the surrounding area immediately and notify the Contracting Officer and PWD-ME ENV. The remains shall be treated with respect and not be disturbed or photographed. In coordination with the Contracting Officer and PWD-ME ENV, provide the materials and labor necessary to secure the area to prevent damage and protect the discovery.
- e. In the event that potential munitions of explosive concern are identified (i.e., cannonballs, mortars, bullets, etc.), the Archaeologist must immediately halt excavation in the area. The Archaeological Coordinator must immediately notify Security Dispatch, and then notify the Contracting Officer. The Contractor shall vacate the area to a safe distance. The Contractor shall not resume work in the area until directed by the Government.
- f. The Contractor must be responsible for the safety of all work conducted and must ensure that the work area is safe during archaeological investigations. This includes providing appropriate shoring, access, and egress routes, dewatering, and any needed equipment. No one will be expected to enter the excavation until it has been deemed safe. It will be the responsibility of the Contractor to ensure that the archaeological monitors have the ability to enter the excavation as needed.
- g. The Contractor may be performing work at night. When working at night, the Contractor must provide adequate and ample light to allow the monitoring archaeologists the ability to successfully monitor the area for archaeological resources. Lighting provided must be sufficient to illustrate changes in soil type/color, stratigraphy, and to identify/document artifacts and features. If adequate lighting cannot be provided, or if the sensitivity of resources warrant, nighttime work cannot occur within those areas. Refer to Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME) for additional information on where



nighttime work is not permitted.

- h. The Archaeologist must document the resource(s) encountered with photographs (including scale), mapping, measured drawings, collection of GPS points, written description of the resource; and collect artifacts, as appropriate for further analysis.
- i. If a potentially significant resource is encountered, summarize the results of the archaeological investigations in Archaeological Resource Report. The Archaeologist, in coordination with the Contractor, must produce the Resource Report to include a map and written summary of the discovery to support PWD ME coordination with SHPO. The Report must include, at a minimum, a physical description of the resource and its context; the depth at which it was encountered; potential resource identification; a description of the proposed impacts to the resource; and recommended follow-on steps, including options to avoid or minimize impacts on the potential site. The Resource Report must be submitted to the Contracting Officer no later than 2 business days after the potentially significant resource is encountered.

### 3.2 ARCHAEOLOGICAL MONITORING REPORTS

Record findings of the archaeological monitoring efforts in written reports at required intervals, as outlined herein, and with sufficient information. Reports shall always include a combination of narratives and graphics. The Contractor is required to prepare and submit all data, reports, graphics, maps, charts, etc. in an electronic format consistent with the platform currently used by the PWD ME Cultural Resources GIS database. Details and requirements will be provided to the Contractor by the Contracting Officer upon award of this Contract. The report must include a recommendation for the future treatment of archaeological resources encountered during construction that were left in place.

#### 3.2.1 Overall Report Requirements:

Upon the completion of ground disturbing work, the archaeological monitoring effort must be documented in a comprehensive monitoring report. The report must adhere to the industry standards and the Maine SHPO's Archaeological Survey Guidelines and the State Historic Preservation Officer's Standards for Archaeological Work in Maine. The report must summarize the monitoring work, resources encountered, and document their disposition. For resources left in place at the conclusion of construction, the report must include recommendations for future treatment. The report shall include graphics including photographs, measured drawings, and maps (created by GPS points and showing the location(s) of any resources documented, including those resources left in situ).

Contracting Officer review of all monitoring report submissions will be completed within 60 calendar days after the date of submission.

#### 3.2.2 Report Iterations:

- a. Monthly Progress Reports: Provide a brief monthly progress report consisting of a table indicating the dates of archaeological monitoring activities, the names of the archaeological monitors on site each day, and a summary of resources encountered/not encountered. The dates shall be keyed to an accompanying map showing

the areas monitored.

- b. Archaeological Monitoring Report - Draft: Provide one (1) hard copy and one (1) electronic copy of a draft report (99 percent complete) for review by the Government. Once the draft report has been reviewed by the Government, the Contractor must be notified of any recommended changes in a timely manner by the Contracting Officer. Production of the final draft report must be within thirty (30) business days of receipt of the Government comments.
- c. Archaeological Monitoring Report - Final Draft: Provide two (2) hard copies and one (1) electronic copy of the final draft report for review by the Government and Maine SHPO. The final draft report must be completed in accordance with current State of Maine archaeological survey requirements. Draft GIS data must also be provided at this time.
- d. Archaeological Monitoring Report - Final: Once the final draft report has been reviewed by the SHPO, the Contractor must be notified of any recommended changes in a timely manner by the Contracting Officer. Production of three (3) hard copies and three (3) electronic copies of the final reports must be within thirty (40) business days of receipt of the Government comments.

The Contractor must submit all informal documents regarding archaeological field site reconnaissance and archaeological site mapping (field notes, etc.) directly to the Contracting Officer. Upon request or after acceptance of the final report, all field notes, photographs, artifacts, and related data will be turned over to the Contracting Officer.

The Contractor must submit all documentation in editable electronic format (Microsoft Word, Excel, ESRI ArcINFO, etc.) and in PDF upon completion of work. Three (3) hard copies and three (3) electronic copies of the report must also be submitted upon completion and acceptance of work. The electronic copies must be submitted on DVDs or CDs.

### 3.3 ARTIFACT CURATION

Artifacts recovered during the monitoring effort must be cleaned, recorded, and curated in accordance with 36 CFR 79. The Archaeologist must identify a repository for permanent curation and pay initial curation fees. If no suitable repository is identified, curated artifacts must be provided to the Government with the Archaeological Monitoring Report - Final submission.

-- End of Section --

SECTION 01 45 00.00 22

QUALITY CONTROL (PWD ME)  
**08/23, CHG 1: 05/24**

PART 1 GENERAL

This Section applies to only Design-Bid-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

|            |                                                                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C1077 | (2024) Standard Practice for Agencies Testing Concrete and Concrete Aggregates for Use in Construction and Criteria for Testing Agency Evaluation |
| ASTM D3666 | (2016) Standard Specification for Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials                              |
| ASTM D3740 | (2019) Minimum Requirements for Agencies Engaged in the Testing and/or Inspection of Soil and Rock as Used in Engineering Design and Construction |
| ASTM E329  | (2023) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection                                     |
| ASTM E543  | (2021) Standard Specification for Agencies Performing Non-Destructive Testing                                                                     |

U.S. ARMY CORPS OF ENGINEERS (USACE)

|            |                                                                    |
|------------|--------------------------------------------------------------------|
| EM 385-1-1 | (2024) Safety -- Safety and Occupational Health (SOH) Requirements |
|------------|--------------------------------------------------------------------|

1.2 PAYMENT

Separate payment will not be made for providing and maintaining an effective Quality Control program. Include all associated costs in the applicable Schedule item.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Contractor Quality Control (CQC) Plan; G

SD-06 Test Reports

Verification Statement

Contractor Quality Control Reports

Contractor Production Reports

Preparatory Phase Checklists

Initial Phase Checklists

Testing Plan And Log

Rework Items List

SD-07 Certificates

Certificate Of Readiness

1.4 GENERAL REQUIREMENTS

Establish and maintain an effective quality control (QC) system that complies with FAR 52.246-12 Inspection of Construction. QC is comprised of plans, procedures, and organization necessary to produce an end product that complies with the Contract requirements. The QC system covers all construction operations, both onsite and offsite, and must be keyed to the proposed construction sequence. The Quality Control Manager, Superintendent, Project Manager, Site Safety and Health Officer (SSHO), and all on-site supervisors are responsible for the quality of work and are subject to removal by the Contracting Officer for non-compliance with the quality requirements specified in the Contract. The Quality Control Manager must maintain a physical presence at the work site at all times and is the primary individual responsible for all quality control.

1.5 QUALITY CONTROL (QC) PROGRAM REQUIREMENTS

Establish and maintain a QC program as described in this Section. This QC program is a key element in meeting the objectives of NAVFAC Commissioning. The QC program consists of a QC Organization, CQC Plan, a Coordination and Mutual Understanding Meeting, Weekly QC Meetings, three phases of control, submittal review and approval, historic coordination drawings review and approval, testing, completion inspections, QC certifications, independent Special Inspections in accordance with Section 01 45 35 SPECIAL INSPECTIONS, and documentation necessary to provide materials, equipment, workmanship, fabrication, construction and operations which comply with the requirements of this Contract. The QC program must cover on-site and off-site work and be keyed to the work sequence. No construction work or testing may be performed unless the QC Manager is on the work site.

The QC Manager must report to an officer of the firm and not be subordinate to the Project Superintendent, or the Project Manager. The QC Manager must be a direct employee of the Prime Contractor, not an employee of a subcontractor. The QC Manager, Project Superintendent, and Project Manager must work together effectively. Although the QC Manager is the

primary individual responsible for quality control, all individuals will be held responsible for the quality of work on the job.

#### 1.5.1 Meetings

##### 1.5.1.1 Coordination and Mutual Understanding Meeting

Before start of construction, prior to acceptance by the Government of the CQC Plan, and within 21 calendar days from receiving the Notice to Proceed, meet with the Contracting Officer and discuss the Contractor's quality control system. During the meeting, a mutual understanding of the system details must be developed, including the forms for recording the CQC operations, control activities, testing, administration of the system for both onsite and offsite work, and the interrelationship of Contractor's management and control with the Government's Quality Assurance. Minutes of the meeting will be prepared by the QC Manager and signed by the Contractor and the Government. The Contractor shall provide a copy of the signed minutes to all attendees. At a minimum the Coordination and Mutual Understanding Meeting must be repeated when a new QC Manager is appointed. There can be other occasions when subsequent conferences will be called by either party to reconfirm mutual understandings or address deficiencies in the CQC system or procedures which can require corrective action by the Contractor.

##### 1.5.1.1.1 Purpose

The purpose of this meeting is to develop a mutual understanding of the QC details, including documentation, administration for on-site and off-site work, design intent, commissioning in accordance with Section 01 91 00.15 20 TOTAL BUILDING COMMISSIONING, environmental requirements and procedures, coordination of activities to be performed, Special Inspections, and the coordination of the Contractor's management, production, and QC personnel. At the meeting, the Contractor must explain in detail how three phases of control will be implemented for each DFO, as well as how each DFO will be affected by each management plan or requirement as listed below:

- a. Waste Management Plan.
- b. Procedures for noise and acoustics management.
- c. Environmental Management Plan.
- d. Environmental regulatory requirements.
- e. Commissioning Plan requirements in accordance with Section 01 91 00.15 20 TOTAL BUILDING COMMISSIONING.
- f. Special Inspections.
- g. Indoor Air Quality (IAQ) Management Plan

##### 1.5.1.1.2 Coordination of Activities

Coordinate activities included in various sections to assure efficient and orderly installation of each component. Coordinate operations included under different sections that are dependent on each other for proper installation and operation. Schedule construction operations with consideration for indoor air quality as specified in the IAQ Management

Plan. Coordinate prefunctional tests and startup testing with the Commissioning Provider (Cx). Coordinate special inspections.

#### 1.5.1.1.3 Attendees

As a minimum, the Contractor's personnel required to attend include an officer of the firm, the Project Manager, Project Superintendent, QC Manager, Alternate QC Manager, QC Specialists, Special Inspector, Special Inspector of Record, Commissioning Provider (Cx), Environmental Manager, and subcontractor representatives. Each subcontractor who will be assigned QC responsibilities must have a principal of the firm at the meeting.

#### 1.5.1.2 Weekly Quality Control (QC) Meetings

After the start of construction, the QC Manager shall conduct Weekly QC Meetings at the work site with at least the following Contractor personnel in attendance: Project Superintendent, Project Manager, Assistant QC Manager, QC Specialists, Special Inspector, Special Inspector of Record, Cx, and the foremen who are performing the work of the DFOWs. The QC Manager must prepare an agenda and provide a copy to the Contracting Officer no later than two (2) working days before the meeting. The QC Manager must prepare the minutes of the meeting and provide a copy to the Contracting Officer within two (2) working days after the meeting. The Contracting Officer may attend these meetings. At a minimum, accomplish the following at each meeting:

- a. Review the minutes of the previous meeting.
- b. Review the schedule and the status of work and deficiencies/rework. Review the most current approved schedule (in accordance with schedule specification) and the status of work and deficiencies/rework.
- c. Review the status of submittals and Request for Information (RFIs).
- d. Review the work to be accomplished in the next 3 weeks as defined by the schedule section paragraph THREE-WEEK LOOK AHEAD SCHEDULE in Section 01 32 17.00 20 COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS) and all documentation required for that work.
- e. Review Testing Plan and Log including status of tests performed since last QC Meeting.
- f. Resolve QC and production problems. Discuss status of pending change orders.
- g. Address items that may require revising the CQC Plan.
- h. Review Accident Prevention Plan (APP) and effectiveness of the safety program.
- i. Review environmental requirements and procedures.
- j. Review Environmental Management Plan.
- k. Review Waste Management Plan.
- l. Review the status of training completion.

- m. Review Cx Plan and progress. Review Issues Log and resolution.
- n. Review IAQ Management Plan

#### 1.5.2 Contractor Quality Control (CQC) Plan

Submit no later than 30 calendar days from receiving the Notice to Proceed the CQC Plan proposed to implement the requirements FAR 52.246-12 Inspection of Construction. Construction will be permitted to begin only after acceptance of the CQC Plan and other Contract requirements.

##### 1.5.2.1 Content of Contractor Quality Control (CQC) Plan

Provide a CQC Plan, prior to start of construction that includes a table of contents with major sections identified, pages numbered sequentially, and that documents the proposed methods and responsibilities for accomplishing quality control during the construction of the project. The CQC Plan must at a minimum include the following sections:

- a. A description of the quality control organization and acknowledgment that the CQC staff will implement the three phase control system for all aspects of the work specified.
- b. An organizational chart showing the quality control organization with individual names and job titles and lines of authority up to an executive of the company at the home office.
- c. NAMES AND QUALIFICATIONS: Names and qualifications, in resume format, (including position titles and durations for qualifying experiences) for each person in the QC organization. Include the Construction Quality Management (CQM) for Contractors course certifications for the QC personnel as required by the paragraphs entitled CONSTRUCTION QUALITY MANAGEMENT TRAINING.
- d. DUTIES, RESPONSIBILITY AND AUTHORITY OF QC PERSONNEL: Duties, responsibilities, and authorities of each person in the QC organization.
- e. OUTSIDE ORGANIZATIONS: A listing of outside organizations, such as architectural and consulting engineering firms, that will be employed by the Contractor and a description of the services these firms will provide. Example: The fire protection engineer who designs the sprinkler system.
- f. APPOINTMENT LETTERS: Letters signed by an officer of the firm appointing the QC Manager and Alternate QC Manager, CxC, and stating that they are responsible for implementing and managing the QC program as described in this Contract. Include in this letter the responsibility of the QC Manager and Alternate QC Manager to implement and manage the three phases of control, and their authority to stop work which is not in compliance with the Contract. Letters of direction are to be issued by the QC Manager to all other QC Specialists outlining their duties, authorities, and responsibilities. Include copies of the letters in the QC Plan.
- g. SUBMITTAL PROCEDURES AND INITIAL SUBMITTAL REGISTER: Procedures for reviewing, approving, scheduling, and managing submittals, including those of subcontractors, offsite fabricators, suppliers, and purchasing agents. Provide the name(s) of the person(s) in the QC

organization authorized to review and certify submittals prior to approval. Provide the initial submittal of the Submittal Register as specified in Section 01 33 00 SUBMITTAL PROCEDURES.

- h. TESTING LABORATORY INFORMATION: Testing laboratory information required by the paragraph ACCREDITATION REQUIREMENTS, as applicable.
- i. TESTING PLAN AND LOG: A Testing Plan and Log that includes the tests required, associated feature of work required, referenced by the specification paragraph number requiring the test, the frequency, and the person responsible for each test. Use Government forms to log and track tests.
- j. Procedures to complete construction deficiencies from identification through acceptable corrective action. Establish verification procedures that identified deficiencies have been corrected. This phase is performed prior to beginning work on each definable feature of work, after all required plans, documents, materials are approved, and after copies are at the work site.
- k. Reporting procedures, included proposed reporting formats.
- l. Procedures for submitting and reviewing design changes/variations prior to submission to the Contracting Officer.
- m. LIST OF DEFINABLE FEATURES: A Definable Feature of Work (DFOW) is a task that is separate and distinct from other tasks and has control requirements and work crews unique to that task. A DFOW is identified by different trades or disciplines, or it is work by the same trade in a different environment. A DFOW is by definition any item or activity on the construction schedule, and the schedule specification provides direction regarding how the DFOWs are to be structured. Include in the list of DFOWs for all activities on the Construction Schedule. Although each section of the specifications can generally be considered a definable feature of work, there are frequently more than one definable feature under a particular section. Identify the specification section number and schedule activity ID for each DFOW listed. The DFOW list will be reviewed in coordination with the construction schedule and agreed upon during the Coordination of Mutual Understanding Meeting.
- n. PROCEDURES FOR PERFORMING AND TRACKING THE THREE PHASES OF CONTROL: Identify procedures used to ensure the three phases of control to manage the quality on this project. For each Definable Feature of Work (DFOW), a Preparatory and Initial phase checklist must be filled out during the Preparatory and Initial phase meetings. Conduct the Preparatory and Initial Phases and meetings with a view towards obtaining quality construction by planning ahead and identifying potential problems for each DFOW.
- o. Coordinate scheduled work with Special Inspections required by Section 01 45 35 SPECIAL INSPECTIONS, the Statement of Special Inspections, and the Special Inspections Project Manual. Where the applicable code issued by the International Code Council (ICC) calls for inspection by the Building Official, the Contractor must include the inspections in the Quality Control Plan and must perform the inspections required by the applicable ICC. The Contractor must perform these inspections using independent qualified inspectors. Include the Special Inspection Project Manual requirements in the CQC Plan.



p. DAILY CONTRACTOR QUALITY CONTROL REPORT FORM: Template that includes fields for the following (the Contractor may request a template from the Contracting Officer):

- 1) Date.
- 2) Sequential report number.
- 3) Weather and temperature.
- 4) Number of personnel on site by trade or by Subcontract.
- 5) Kind and number of major equipment on site.
- 6) Tests performed and their results, if known.
- 7) Materials and equipment delivered to the site and their conditions.
- 8) Names, affiliations, and positions of visitors to the site with brief explanations of the reasons for visits.
- 9) Brief description of each work activity, noting items that were completed that day.
- 10) Items of work that need attention at a later date, and why.
- 11) Any accidents and injuries.
- 12) Items of concern with respect to maintenance of quality.
- 13) Accounting update of unit price items specified in the Price Schedule utilizing the numbering convention indicated in the Unit Prices Form (e.g., 0001g).
- 14) Any other items of significance.

#### 1.5.3 Acceptance of the Contractor Quality Control (CQC) Plan

The Contracting Officer's acceptance of the CQC Plan is required prior to the start of construction. The Contracting Officer reserves the right to require changes in the QC Plan and operations as necessary, including removal or addition of personnel, to ensure the specified quality of work. The Contracting Officer reserves the right to interview any member of the QC organization at any time in order to verify the submitted qualifications. All QC organization personnel are subject to acceptance by the Contracting Officer. The Contracting Officer may require the removal of any individual for non-compliance with quality requirements specified in the Contract. The removal and replacement of QC organization personnel will not be cause for claim of additional compensation or extensions of the Contract Completion Date (CCD).

#### 1.5.4 Preliminary Construction Work Authorized Prior to Acceptance

The only construction work that is authorized to proceed prior to the acceptance of the CQC Plan is mobilization of storage and office trailers, temporary utilities, and surveying with specific prior approval of the Contracting Officer.

#### 1.5.5 Notification of Changes

Notify the Contracting Officer, in writing, of any proposed changes in the CQC Plan or changes to the QC organization personnel, in accordance with the submittal requirements in Section 01 33 00 SUBMITTAL PROCEDURES. Proposed changes are subject to acceptance by the Contracting Officer.

#### 1.5.6 Special Inspections

Perform all required Special Inspections per Section 01 45 35 SPECIAL INSPECTIONS, the statement of Special Inspections and the Schedule of Special Inspections.

## 1.6 QUALITY CONTROL (QC) ORGANIZATION

### 1.6.1 Quality Control (QC) Manager

#### 1.6.1.1 Duties

Provide a QC Manager at the Project site to implement and manage the QC program. "Project Site" is the specific area of construction within the limits of work as defined by the contract documents. The QC Manager must not perform the duties of Project Superintendent, nor the duties of Project Manager, or SSHO. The only duties and responsibilities of the QC Manager are to manage and implement the QC program on this Contract. The QC Manager must attend the partnering meetings, Coordination and Mutual Understanding Meeting, conduct the Weekly QC Meetings, perform the three phases of control except for those phases of control designated to be performed by QC Specialists, perform submittal review and approval, ensure testing is performed, and provide QC certifications and documentation required in this Contract. The QC Manager is responsible for managing and coordinating the three phases of control and documentation performed by the QC Specialists, testing laboratory personnel, and any other inspection and testing personnel required by this Contract. The QC Manager is the manager of all QC activities and must not be the Special Inspector. The QC manager is responsible for notifying the Special Inspector of activities which require their review. The QC manager is responsible for coordinating the Special Inspection activities, see paragraph CONTRACTOR'S QUALITY CONTROL (QC) MANAGER, in Section 01 45 35 SPECIAL INSPECTIONS. The QC manager is responsible for the quality control for Controlled Area perimeter construction.

#### 1.6.1.2 Qualifications

The QC Manager must be an individual with a minimum of 10 years combined experience in the following positions: Project Superintendent, QC Manager, Project Manager, Project Engineer or Construction Manager on similar size and type construction contracts which included the major trades that are part of this Contract. The individual must have at least two years experience as a QC Manager. The individual must be familiar with the requirements of EM 385-1-1, and have experience in the areas of hazard identification, safety compliance, and sustainability.

The QC Manager and all members of the QC organization must be capable of reading, writing, and conversing fluently in the English language.

#### 1.6.1.3 Construction Quality Management Training

In addition to the above experience and education requirements, the QC Manager and all members of the QC team must have completed the Construction Quality Management (CQM) for Contractors course. If the QC Manager does not have a current certification, obtain the CQM for Contractors course certification within 90 days from receiving the Notice to Proceed. This course is periodically offered by the Naval Facilities Engineering Systems Command and the Army Corps of Engineers. Contact the Contracting Officer for information on the next scheduled class.

The Construction Quality Management Training certificate expires after 5 years. If the QC Manager's certificate has expired, the QC Manager must retake the course to remain current throughout the duration of the Contract.

#### 1.6.2 Organizational Changes

Maintain the QC staff with personnel as required by the specification section at all times. When it is necessary to make changes to the QC staff, revise the CQC Plan to reflect the changes and submit the changes to the Contracting Officer for acceptance.

#### 1.6.3 Alternate Quality Control (QC) Manager Duties and Qualifications

Designate an alternate for the QC Manager at the work site to serve in the event of the designated QC Manager's absence. The period of absence may not exceed 2 weeks at one time, and not more than 30 workdays during a calendar year. The qualification requirements for the Alternate QC Manager must be the same as for the QC Manager.

#### 1.6.4 Commissioning

Commissioning (Cx) is a systematic, quality-focused process for delivery of a project focusing on verifying and documenting all commissioned systems and assemblies are installed, tested, and operating as they were planned and designed to meet the project requirements. The Quality Control requirements outlined in this specification section are key in supporting the objectives of the Cx process, specifically coordinating testing, documenting, and verifying proper system operation. Properly executed the Quality Control support of Cx ensures timely execution of necessary tasks to deliver the fully commissioned and operating systems in coordination with the overall construction and project schedule.

Provide Cx in addition to the quality control requirements of this section and not as a substitute for quality control requirements. The QC Manager is responsible for carrying out the three phases of control while ensuring the functional performance and integrated systems tests are coordinated with the Cx provider as required for each system to be commissioned.

##### 1.6.4.1 Certificate Of Readiness

The QC Manager must issue a Certificate of Readiness for Government approval for each system to be commissioned. Schedule Functional Performance Tests for each system only after the Certificate of Readiness has been approved by the Government for the system. The Certificate of Readiness certifies that all required inspections have been completed and deficiencies that were identified through any prior review, inspection, or test activity have been corrected before the start of Functional Performance Tests. Refer to Cx requirements in Section 01 91 00.15 BUILDING COMMISSIONING for a list of systems to be commissioned and detailed requirements for the Cx provider.

#### 1.6.5 Quality Control (QC) Specialists

Provide a separate QC Specialist at the work site for each of the areas as listed in the Experience Matrix below, who must assist and report to the QC Manager and who must have no duties other than their assigned quality control duties. These individuals or specialized technical companies must be a first tier Subcontractor. QC Specialists must be physically present at the work site with frequency as indicated in the Experience Matrix below, to participate in the QC meetings, perform the three phases of control, including participation in Preparatory and Initial Phase meetings, and to perform and document Follow-up inspections as an extension of the QC Manager for each definable feature of work in their

area of responsibility. QC Specialists must assist and be present for training events, and Critical System Acceptance inspections by the Government. Qualification, experience, Area of Responsibility, and frequency of QC surveillance are provided in the Matrix listed herein.

| <b>Experience Matrix</b>                         |                                                                                                                                                                                   |                                                                                                                                                                                                        |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Area</b>                                   | <b>2-1. Qualification<br/>2-2. Experience</b>                                                                                                                                     | <b>3-1. Area of Responsibility<br/>3-2. Frequency</b>                                                                                                                                                  |
| Testing, Adjusting and Balancing (TAB) Personnel | 2-1 TAB Team Field Leader must be a member of AABC or an experienced technician of the firm certified by the NEBB.<br>2-2. 3 years experience immediately preceding this Contract | 3-1. Area of Responsibility: Testing, Adjusting, and Balancing of HVAC Systems.<br>3-2. Frequency of QC surveillance and inspection will be weekly during installation and including final inspection. |
| Registered Structural Engineer, (P.E.)           | Erection of structural steel, Section 05 12 00 STRUCTURAL STEEL                                                                                                                   | Minimum twice a week                                                                                                                                                                                   |
| Fire Protection QC Specialist (FPQC)             | Note: See paragraph FIRE PROTECTION QC SPECIALIST (FPQC)                                                                                                                          | Note: See paragraph FIRE PROTECTION QC SPECIALIST (FPQC)                                                                                                                                               |
| Roofing Manufacturer's Technical Representative  | 2-1. Installation and testing of roofing systems, Section 07 13 53 ELASTOMERIC SHEET WATERPROOFING<br>2-2. 5 years experience minimum                                             | 3-1. Area of Responsibility: Installation and testing of roofing systems<br>3-2. Frequency of QC surveillance and inspection will be weekly during installation and including final inspection.        |

#### 1.6.6 Fire Protection QC Specialist

Provide a Fire Protection Quality Control Specialist (FPQC) within the QC organization to perform quality control related activities as specified herein on fire protection and life safety systems installed under this Contract.

#### 1.6.6.1 Qualifications

The FPQC must have the following qualifications:

- a. Be a registered Professional Engineer (P.E.) licensed by a Licensing Board in the United States, the District of Columbia, Guam or Puerto Rico, having passed the National Council of Examiners for Engineering and Surveying (NCEES) written examination specifically in the discipline of Fire Protection Engineering.
- b. Have a minimum of 5 years of Fire Protection Engineering experience on projects of similar relevance and complexity to the fire protection work specified under this Contract.
- c. Other than the contractual obligations with the Prime Contractor, the FPQC must have no other business relationship (i.e., employee, owner, partner, operating officer, distributor, salesman, technical representative, family relationship, or financial investment) with the Prime Contractor or subcontractors.
- d. Be employed by an independent engineering firm or company. The firm may identify multiple, to a maximum of five, licensed Fire Protection Engineers for the performance of the duties under the Contractor but must submit the names and qualifications for Government approval for all individuals identified prior to them performing any work under this Contract. These individuals may not be substituted without prior approval from the Contracting Officer.

#### 1.6.6.2 Responsibilities

FPQC duties and responsibilities:

- a. Assist in the development of the CQC Plan including the Testing Plan and Log and executing the three phases of control for work involving the installation and testing of fire protection and life safety systems as an extension of the QC Manager.
- b. Participate in project QC Meetings. Participate in Preparatory and Initial Phase meetings and perform and Follow-up inspections for work involving the installation and testing of fire protection and life safety systems.
- c. Review and certify that all submittals pertaining to fire and life safety systems are complete and accurate prior to submission to the Government. The FPQC specialist is responsible for ensuring submittals are complete and accurate and all corrections have been made prior to submission to the Government. The Government reserves the right to reject any submittal that has not been reviewed first by the FPQC and so marked, in writing, attesting to such review and completeness of the submittal.
- d. The Government reserves the right to reject any submittal or construction that is not in compliance to Contract. Government reviews do not relieve the Contractor responsibility for providing adequate quality control measures and do not constitute or imply acceptance of Contract variation.
- e. Perform construction surveillance in accordance with the Schedule of Fire Protection System Inspections. Construction surveillance includes

but is not limited to performing periodic on-site inspections during construction at specified milestones, performing a pre-final inspection of installed systems and witnessing functional testing; and participating and documenting in an on-site final acceptance inspection of fire protection and life safety systems with the Government FPE.

- f. Document inspection results on a FPQC report prepared each day/night that inspections are performed. The report must include a description of the visual inspection or observation performed, a written summary of findings, a conclusion on compliance with the Contract documents, and signature of the FPQC Specialist. In person inspection must be documented via photo (.jpeg). Photographic documentation must include before and after conditions and physical measurements. Forward the FPQC daily report to the QC Manager who must include the report with the submission of their daily QC Report to the Government each day. Every site visit by the FPQC must be documented on a FPQC daily report.

#### 1.6.6.3 Schedule of Fire Protection System Inspections

A schedule, prepared by the Fire Protection DOR, which lists each of the required visual inspections and observations required by the FPQC. The schedule is included at the end of this section as Attachment A.

#### 1.6.7 Special Inspector and Special Inspector of Record

The Special Inspector (SI) and Special Inspector of Record (SIOR) must be an independent third party hired directly by the Prime Contractor. The SI and SIOR must not be a company employee of the Contractor or any subcontractor performing the work to be inspected. The qualifications of the SI and SIOR are defined in Section 01 45 35 SPECIAL INSPECTIONS.

#### 1.7 SUBMITTAL AND DELIVERABLES REVIEW AND APPROVAL

Procedures for submission, review and approval of submittals are described in Section 01 33 00 SUBMITTAL PROCEDURES. Procedures must include field verification of relevant dimensions and component characteristics by the QC organization prior to submittal being sent to the Contracting Officer. The QC organization is responsible for certifying that all submittals and deliverables are in compliance with the Contract. When Section 01 91 00.15 BUILDING COMMISSIONING are included in the Contract, the submittals required by those sections have to be coordinated with Section 01 33 00 SUBMITTAL PROCEDURES to ensure adequate time is allowed for each type of submittal required.

#### 1.8 OVERVIEW OF REPORT REQUIREMENTS

Prior to commencing work on construction, the Contractor must obtain a set of the current report forms. Contact the Contracting Officer for sample forms. The report forms will consist of the Contractor Quality Control (CQC) Report, CQC Report (Continuation Sheet), Contractor Production Report, Contractor Production Report (Continuation Sheet), Preparatory Phase Checklist, Initial Phase Checklist, Testing Plan and Log, and Rework Items List. Unless otherwise provided by the Contracting Officer, Contractor may use the forms provided as related material located at <https://www.wbdg.org/dod/ufgs/ufgs-01-45-00>.

The Contractor must deliver the following to the Contracting Officer during construction, at the frequencies indicated:

a. Contractor Quality Control Reports

- (1) For work during regular working hours, furnish the original and one copy of the Contractor Quality Control (CQC) Report to the Contracting Officer by 10:00 AM the next working day after the date covered by the report.
- (2) For work outside of regular working hours (e.g., nighttime work), furnish the original and one copy of the CQC Report to the Contracting Officer by 10:00 PM the next working day after the date covered by the report.
- (3) At a minimum, prepare and submit one CQC Report for every 7 days of no work and on the last day of a no work period. All calendar days need to be accounted for throughout the life of the Contract. The first report following a day/night of no work will be for that day/night only.
- (4) Refer to paragraph QUALITY CONTROL ACTIVITIES herein for details on what is to be included in the CQC Reports.

b. Contractor Production Reports

- (1) Contractor Production Reports are required for each day and night that work is performed and must be attached to the CQC Report prepared for that same day or night. Account for each calendar day throughout the life of the Contract.
- (2) Refer to paragraph CONSTRUCTION DOCUMENTATION herein for details on what is to be included in the Contractor Production Reports.

c. Preparatory Phase Checklists

- (1) Submit the Preparatory Phase Checklist for each Preparatory Phase held.
- (2) Refer to paragraph PREPARATORY PHASE herein for details on the Preparatory Phase and requirements of the checklist.

d. Initial Phase Checklists

- (1) Submit the Initial Phase Checklist for each Initial Phase held.
- (2) Refer to paragraph INITIAL PHASE herein for details on the Initial Phase and requirements of the checklist.

e. Testing Plan and Log

- (1) Submit a copy of the updated Testing Plan and Log along with the last daily CQC Report of each month.
- (2) Submit the final Testing Plan and Log to the preparer of the Operation & Maintenance documentation.
- (3) Refer to paragraph TESTING herein for details on testing and what is to be included in the Testing Plan and Log.

f. Rework Items List

- (1) Submit a copy of the Rework Items List along with the last daily CQC Report of each month.
- (2) Refer to paragraph DEFICIENCY/REWORK ITEMS LIST herein for details on what is to be included in the Rework Items List.

## 1.9 THREE PHASES OF CONTROL

QC enables the Contractor to ensure that the construction, including that of subcontractors and suppliers, complies with the requirements of the Contract. At least three phases of control must be conducted by the QC Manager to adequately cover both onsite and offsite work for each definable feature of the construction work as follows:

### 1.9.1 Preparatory Phase

Document the results of the preparatory phase actions by separate minutes prepared by the QC Manager and attached to the CQC Reports. Instruct to applicable workers as to the acceptable level of workmanship required to meet Contract specifications.

Notify the Contracting Officer at least five (5) working days in advance of each preparatory phase.

The Preparatory Phase Meeting must be conducted by the QC Manager and attended by the QC Specialists, the Project Superintendent(s), the SSHO(s), the CxC, the Special Inspector, the Special Inspector of Record, and the foreman responsible for the DFOW or as approved by the Contracting Officer. When the DFOW will be accomplished by a subcontractor, that subcontractor's foreman must attend the Preparatory Phase Meeting. Document the results of the preparatory phase actions in the CQC Report and in the Preparatory Phase Checklist (see Attachment B). This phase is performed prior to beginning work on each DFOW. (Note: Preparatory Phase Meeting shall only be held if the shop drawings and/or submittals have been approved, hard copies are printed for the meeting, the APP and appropriate AHA related to the DFOW have been submitted and the appropriate personnel as stated above are present for the meeting). Perform the following prior to beginning work on each DFOW:

- a. Review each paragraph of the applicable specification sections, reference codes, and standards. Make available during the preparatory inspection a copy of those sections of referenced codes and standards applicable to that portion of the work to be accomplished in the field. Maintain and make available in the field for use by Government personnel until final acceptance of the work.
- b. Review the Contract drawings.
- c. Verify that field measurements are as indicated on construction or shop drawings or both before confirming product orders, to minimize waste due to excessive materials.
- d. Verify that appropriate shop drawings and submittals for materials and equipment have been submitted and approved. Verify receipt of approved factory test results, when required.
- e. Review the testing plan and ensure that provisions have been made to provide the required QC testing.
- f. Examine the work area to ensure that the required preliminary work has been completed and complies with the Contract and ensure any deficiencies/rework items in the preliminary work have been corrected and confirmed by the Contracting Officer
- g. Review coordination of product/material delivery to designated



prepared areas to execute the work.

- h. Examine the required materials, equipment, and sample work to ensure that they are on hand and conform to the approved shop drawings and submitted data and are properly stored.
- i. Check to assure that all materials and equipment have been tested, submitted, and approved.
- j. Review the APP and appropriate Activity Hazard Analysis (AHA) to ensure that applicable safety requirements are met, and that required Safety Data Sheets (SDS) are submitted.
- k. Discuss specific controls to be used, construction methods, construction tolerances, workmanship standards, and the approach that will be used to provide quality construction by planning ahead and identifying potential problems for each DFO. Ensure any portion of the plan requiring separate Contracting Officer acceptance has been approved.
- l. Review the Cx requirements in accordance with Section 01 91 00.15 BUILDING COMMISSIONING and ensure all preliminary work items have been completed and documented.
- m. Review Special Inspections required by Section 01 45 35 SPECIAL INSPECTIONS, the Statement of Special Inspections and the Schedule of Special Inspections.
- n. If work includes demolition work, review demolition work plan to ensure compliance with EM 385-1-1.
- o. If work includes excavation, at a minimum the NAVFAC DM/ET and PWD ME Dig Safe Coordinator (DSC) shall be invited to the Preparatory and Initial Phase Meetings and will review utility locating requirements. (Note: Special attention is required if Hazardous or Mission Critical Utilities are within the excavation area.) If Hazardous and Mission Critical Utilities are located within the excavation area, the NAVFAC DM/ET, PWD ME DSC, PWD ME Design Manager (DM) and the assigned PWD ME Project Civil Engineer must be present at the meeting to review requirements to ensure safety and preservation of critical utilities is discussed with adequate focus. Contractor to provide NAVFAC DM/ET notification no later than 7 working days prior to holding the Preparatory and Initialpre-excavation/demo safety review meeting. The NAVFAC DM/ET, as well as the PWD ME DSC, will attend this meeting to ensure safety and preservation of critical utilities is discussed with adequate focus.
- p. If work includes excavation, review Soil Management Requirements (Note: Special attention is required if the work is within an IR site or Hazardous Soils are within the excavation area. Include Code 106.3 Representatives at this meeting to discuss Soil Management processes and procedures.)
- q. If the work is subject to any environmental (EV) permitting, Archaeological Monitoring, or SHPO approvals, review the terms and conditions of the applicable permits as well as any supplemental controls/ approvals that are required to complete the work. (Note: Include PWD ME EV Representatives at this meeting to review the project specific requirements).

- r. If work includes the removal and disposal of Hazardous materials, review the appropriate work plans to ensure the work plans have been approved and any comments are understood by the work execution team. (Note: Include Code 106.3 Representatives at this meeting to discuss Hazardous Materials handling and disposal processes and procedures.)

#### 1.9.2 Initial Phase

Notify the Contracting Officer at least 2 working days in advance of each initial phase. When construction crews are ready to start work on a DFOW, the QC Manager must conduct the Initial Phase Meeting with the QC Specialists, the Project Superintendent(s), the SSHO(s), the CxC, the Special Inspector, the Special Inspector of Record, and the foreman responsible for that DFOW. Observe the initial segment of the DFOW to ensure that the work complies with Contract requirements. Document the results of the initial phase in the CQC Report and in the Initial Phase Checklist (see Attachment C). Repeat the initial phase for each new crew to work onsite, or when acceptable levels of specified quality are not being met. Indicate the exact location of initial phase for definable feature of work for future references and comparison with follow-up phases. Perform the following for each DFOW:

- a. Check work to ensure that it is in full compliance with the Contract requirements. Review minutes of the preparatory meeting.
- b. Verify adequacy of controls to ensure full Contract compliance. Verify required control inspection and testing comply with the Contract.
- c. Establish level of workmanship and verify that it meets the minimum acceptable workmanship standards. Compare with required sample panels as appropriate.
- d. Resolve any workmanship issues.
- e. Resolve conflicts.
- f. Ensure that testing is performed by the approved laboratory.
- g. Check work procedures for compliance with the APP and the appropriate AHA to ensure that applicable safety requirements are met.
- h. Review project specific work plans (i.e. Cx, HAZMAT Abatement, Stormwater Management) to ensure all preparatory work items have been completed and documented.
- i. Coordinate scheduled work with special inspections required by Section 01 45 35 SPECIAL INSPECTIONS, the Statement of Special Inspections and the Schedule of Special Inspections.

#### 1.9.3 Follow-Up Phase

Perform the following for on-going DFOW daily, or more frequently as necessary, until the completion of each DFOW. The Final Follow-up for any DFOW will clearly note in the daily report the DFOW is completed, and all deficiencies/rework items have been completed in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST. Each DFOW that has completed the Initial Phase and has not completed the Final Follow-up must be included on each daily report. If no work was performed on that DFOW for the period

of that daily report, it must be so noted. Document all Follow-Up activities for DFOWs in the daily CQC Report:

- a. Ensure the work including control testing complies with Contract requirements until completion of that particular work feature. Record checks in the CQC documentation.
- b. Maintain the quality of workmanship required.
- c. Ensure that testing is performed by the approved laboratory.
- d. Ensure that deficiencies/rework items are being corrected. Conduct final follow-up checks and correct all deficiencies prior to the start of additional features of work which may be affected by the deficient work.
- e. Do not build upon nor conceal non-conforming work.
- f. Assure manufacturers' representatives have performed necessary inspections if required and perform safety inspections.
- g. Review the Cx requirements in accordance with Section 01 91 00.15 20 TOTAL BUILDING COMMISSIONING.
- h. Coordinate scheduled work with Special Inspections required by Section 01 45 35 SPECIAL INSPECTIONS, the Statement of Special Inspections and the Schedule of Special Inspections.

#### 1.9.4 Additional Preparatory and Initial Phases

Conduct additional preparatory and initial phases on the same DFOW if the quality of on-going work is unacceptable, if there are changes in the applicable QC organization, if there are changes in the on-site production supervision or work crew, if work on a DFOW has not started within 45 days of the initial preparatory meeting or has resumed after 45 days of inactivity, or if other problems develop.

#### 1.9.5 Notification of Three Phases of Control for Off-Site Work

Notify the Contracting Officer at least 2 weeks prior to the start of the preparatory and initial phases.

#### 1.9.6 Deficiency/Rework Items List

The QC Manager must maintain a list of work that does not comply with the Contract, identifying what items need to be corrected, the activity ID number associated with the item, the date the item was originally discovered, the date the item will be corrected by, and the date the item was corrected.

The QC Manager reviews the list at each weekly QC Meeting:

- a. There is no requirement to report a deficiency/rework item that is corrected the same day it is discovered.
- b. No successor task may be advanced beyond the preparatory phase meeting until all deficiencies/rework items have been cleared by the QC Manager and concurred with by the Contracting Officer. This must be confirmed as part of the Preparatory Phase activities.

- c. Attach a copy of the Rework Items List to the last daily CQC Report of each month.
- d. The Contractor is responsible for including those items identified by the Contracting Officer.
- e. All deficiencies/rework items must be confirmed as corrected by the QC Manager, and concurred by the Contracting Officer, prior to commencement of any completion inspections per paragraph COMPLETION INSPECTIONS unless specifically exempted by the Contracting Officer.
- f. Non-Compliance with these requirements is grounds for removal in accordance with paragraph ACCEPTANCE OF THE QUALITY CONTROL (QC) PLAN.
- g. All delays, concurrent or related to failure to manage, monitor, control, and correct deficiencies/rework items are entirely the responsibility of the Contractor and can not be made the subject, or any component of any request for additional time or compensation.

#### 1.10 TESTING

Perform specified or required tests to verify that control measures are adequate to provide a product which conforms to Contract requirements. Upon request, furnish to the Government duplicate samples of test specimens for possible testing by the Government. Testing includes operation and acceptance tests when specified. Procure the services of an U.S. Army Corps of Engineers approved testing laboratory or establish an approved testing laboratory at the project site or within 10 miles. Perform the following activities and record and provide the following data:

- a. Verify that testing procedures comply with Contract requirements.
- b. Verify that facilities and testing equipment are available and comply with testing standards.
- c. Check test instrument calibration data against certified standards.
- d. Verify that recording forms and test identification control number system, including all test documentation requirements, have been prepared.
- e. Record results of all tests performed, both passing and failing, on the CQC report for the date taken. Include the specification paragraph reference, location where tests were taken, and the sequential control number identifying the test. If approved by the Contracting Officer, the actual test reports may be submitted later, and must include a reference to the test number and date the test was performed. Provide an information copy of tests performed by an offsite or commercial test facility directly to the Contracting Officer. Failure to submit timely test reports as stated results in nonpayment for related work performed and disapproval of the test facility for this Contract.

##### 1.10.1 Accreditation Requirements

Construction materials testing laboratories must be accredited by a laboratory accreditation authority and must submit a copy of the Certificate of Accreditation and Scope of Accreditation to the Contractor

for submission to the Contracting Officer. The laboratory's scope of accreditation must include the appropriate ASTM standards (ASTM E329, ASTM C1077, ASTM D3666, ASTM D3740, ASTM E543) listed in the technical Sections of the specifications. Laboratories engaged in Hazardous Materials Testing must meet the requirements of OSHA and EPA. The policy applies to the specific laboratory performing the actual testing, not just the Corporate Office.

#### 1.10.2 Laboratory Accreditation Authorities

Laboratory Accreditation Authorities include the National Voluntary Laboratory Accreditation Program (NVLAP) administered by the National Institute of Standards and Technology at <https://www.nist.gov/nvlap>, the American Association of State Highway and Transportation Officials (AASHTO) Accreditation Program at <http://www.aashtoresource.org/aap/overview>, International Accreditation Services, Inc. (IAS) at <http://www.iasonline.org>, U. S. Army Corps of Engineers Materials Testing Center (MTC) at <http://www.erdc.usace.army.mil/Media/FactSheets/FactSheetArticleView/tabid/9254/Article/476661/materials-testing-center.aspx>, and the American Association for Laboratory Accreditation (A2LA) program at <http://www.a2la.org/>.

#### 1.10.3 Capability Check

The Contracting Officer retains the right to check laboratory equipment in the proposed laboratory and the laboratory technician's testing procedures, techniques, and other items pertinent to testing, for compliance with the standards set forth in this Contract. Laboratories utilized for testing soils, concrete, asphalt, and steel must meet criteria detailed in ASTM D3740 and ASTM E329.

#### 1.10.4 Test Results

Cite applicable Contract requirements, tests or analytical procedures used. Provide actual results and include a statement that the item tested or analyzed conforms or fails to conform to specified requirements. If the item fails to conform, notify the Contracting Officer immediately. Conspicuously stamp the cover sheet for each report in large red letters "CONFORMS" or "DOES NOT CONFORM" to the specification requirements, whichever is applicable. Test results must be signed by a testing laboratory representative authorized to sign certified test reports. Furnish the signed reports, certifications, and other documentation to the Contracting Officer via the QC Manager. Furnish a summary report of field tests at the end of each month, per the paragraph entitled DOCUMENTATION AND INFORMATION FOR THE CONTRACTING OFFICER.

#### 1.10.5 Test Reports and Monthly Summary Report of Tests

Furnish the signed reports, certifications, and a summary report of field tests at the end of each month to the Contracting Officer. Attach a copy of the summary report to the last daily CQC Report of each month. Provide a copy of the signed test reports and certifications to the Operation and Maintenance Support Information (OMSI) preparer for inclusion into the OMSI documentation, in accordance with Sections 01 78 23 OPERATION AND MAINTENANCE DATA and 01 78 24.00 20 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI).

## 1.11 Quality Control (QC) CERTIFICATIONS

### 1.11.1 Contractor Quality Control (CQC) Report Certification

Contain the following statement within the CQC Report: "On behalf of the Contractor, I certify that this report is complete and correct and equipment and material used, and work performed during this reporting period is in compliance with the Contract drawings and specifications to the best of my knowledge, except as noted in this report."

### 1.11.2 Invoice Certification

Furnish a certificate to the Contracting Officer with each payment request, signed by the QC Manager, attesting that as-built drawings are current, coordinated and attesting that the work for which payment is requested, including stored material, complies with Contract requirements.

### 1.11.3 Completion Certification

Upon completion of work under this Contract, the QC Manager must furnish a certificate to the Contracting Officer attesting that "the work has been completed, inspected, tested and is in compliance with the Contract." Provide a copy of this final QC Certification for completion to the preparer of the Operation & Maintenance (O&M) documentation.

## 1.12 COMPLETION INSPECTIONS

### 1.12.1 Punch-Out Inspection

Near the completion of all work or any increment thereof, established by a completion time stated in the Contract Clause entitled "Commencement, Prosecution, and Completion of Work," or stated elsewhere in the specifications, the QC Manager and the CxC (if applicable) must conduct an inspection of the work and develop a "punch list" of items which do not conform to the approved drawings, specifications, and Contract. Include in the punch list any remaining items on the "Deficiency/Rework Items List", that were not corrected prior to the Punch-Out Inspection as approved by the Contracting Officer in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST. Include within the punch list the estimated date by which the deficiencies will be corrected. Provide a copy of the punch list to the Contracting Officer.

The QC Manager, or staff, must make follow-on inspections to ascertain that all deficiencies have been corrected. All punch list items must be confirmed as corrected by the QC Manager and concurred by the Contracting Officer. Once this is accomplished, notify the Government that the facility is ready for the Government "Pre-Final Inspection".

### 1.12.2 Pre-Final Inspection

The Government and the Contractor's QC Manager will perform this inspection to verify that the facility is complete and ready to be occupied. A Government "Pre-Final Punch List" will be documented by the QC Manager as a result of this inspection. The QC Manager will ensure that all items on this list are corrected and concurred by the Contracting Officer prior to notifying the Government that a "Final" inspection with the Client can be scheduled. All items noted on the "Pre-Final" inspection must be corrected and concurred in a timely manner and be accomplished before the Contract completion date for the work, or any

increment thereof, if the project is divided into increments by separate completion dates unless exceptions are directed by the Contracting Officer.

#### 1.12.3 Final Acceptance Inspection

Notify the Contracting Officer at least 14 calendar days prior to the date a final acceptance inspection can be held. State within the notice that all items previously identified on the pre-final punch list will be corrected and acceptable, along with any other unfinished Contract work, by the date of the final acceptance inspection. The Contractor must be represented by the QC Manager, the Project Superintendent, the CxC, and others deemed necessary. Attendees for the Government will include the Contracting Officer, other Government QA personnel, and personnel representing the Client. Failure of the Contractor to have all Contract work acceptably complete for this inspection will be cause for the Contracting Officer to bill the Contractor for the Government's additional inspection cost in accordance with the Contract Clause entitled "Inspection of Construction."

#### 1.13 DOCUMENTATION AND INFORMATION FOR THE CONTRACTING OFFICER

##### 1.13.1 Construction Documentation

Maintain current and complete records of onsite and offsite QC program operations and activities.

The Project Superintendent and the QC Manager must prepare and sign the Contractor Production and CQC Reports, respectively. Every space on the forms must be filled in. Use N/A if nothing can be reported in one of the spaces. The reporting of work must be identified by terminology consistent with the construction schedule. In the "Remarks" sections of the reports, enter pertinent information including directions received, problems encountered during construction, work progress and delays, conflicts or errors in the drawings or specifications, field changes, safety hazards encountered, instructions given and corrective actions taken, delays encountered, a record of visitors to the work site, quality control problem areas, deviations from the CQC Plan, construction deficiencies encountered, and meetings held. For each entry in the report(s), identify the Schedule Activity No. that is associated with the entered remark.

##### 1.13.2 Quality Control Activities

CQC and Contractor Production reports will be prepared daily/nightly to maintain current records providing factual evidence that required quality control activities and tests have been performed. Include in these records the work of subcontractors and suppliers on an acceptable form that includes, as a minimum, the following information:

- a. The name and area of responsibility of the Contractors and any subcontractors.
- b. Operating plant/equipment with hours worked, idle, or down for repair.
- c. Work performed each day/night, giving location, description, and by whom. When a Network Analysis Schedule (NAS) is used, identify each item of work performed each day by NAS activity number.
- d. Control phase activities performed. Preparatory, and Initial phase

Checklists associated with the DFOV referenced to the construction schedule. Follow-up phase activities identified to the DFOV. If testing or specific QC Specialist activities are associated with the Follow-up phase activities for a specific DFOV note this and include those reports.

- e. Test and control activities performed with results and references to specifications and drawings requirements. Identify the control phase (Preparatory, Initial, Follow-up). List of deficiencies noted, along with corrective action in accordance with the paragraph DEFICIENCY/REWORK ITEMS LIST.
- f. Quantity of materials received at the site with statement as to acceptability, storage, and reference to specifications and drawings requirements.
- g. Submittals and deliverables reviewed, with Contract reference, by whom, and action taken.
- h. Offsite surveillance activities, including actions taken.
- i. Job safety evaluations stating what was checked, results, and instructions or corrective actions.
- j. Instructions given/received and conflicts in plans and specifications.

#### 1.13.3 Verification Statement

Indicate a description of trades working on the project; the number of personnel working; weather conditions encountered; and any delays encountered. Cover both conforming and deficient features and include a statement that equipment and materials incorporated in the work and workmanship comply with the Contract.

Reports need to be signed and dated by the QC Manager. Include copies of test reports and copies of reports prepared by all subordinate quality control personnel within the QC Manager Report.

#### 1.13.4 Reports from the Quality Control (QC) Specialist(s)

Document inspection results on a QC specialist report prepared each day/night work is performed in their area of responsibility. The report must include a description of the visual inspection or observation performed, a written summary of findings, a conclusion on compliance with the Contract documents, and signature of the QC Specialist. Forward all QC Specialist daily reports to the QC Manager, who must include the report(s) with the submission of their QC Report to the Government in accordance with Paragraph OVERVIEW OF REPORT REQUIREMENTS herein. Every site visit by the QC Specialist must be documented on a QC Specialist daily report. In person inspections must be documented with photographs and/or videos. Photographic/video documentation of deficiencies must include before and after conditions and physical measurements, as necessary, and shall be placed on a CD/DVD to be turned over to the Government.

#### 1.13.5 Quality Control Validation

Establish and maintain the following in an electronic folder. Divide folder into a series of tabbed sections as shown below. Ensure folder is updated at each required progress meeting.



- a. CQC Meeting minutes in accordance with paragraph QUALITY CONTROL (QC) MEETINGS.
- b. All completed Preparatory and Initial Phase Checklists, arranged by specification section, further sorted by DFOW referenced to the construction schedule. Submit each individual Phase Checklist the day the phase event occurs as part of the CQC daily report.
- c. All milestone inspections, arranged by Activity Number referenced to the construction schedule.
- d. An up-to-date copy of the Testing Plan and Log with supporting field test reports, arranged by specification section referenced to the DFOW to which individual reports results are associated. Individual field test reports will be submitted within 2 working days after the test is performed in accordance with the paragraph QUALITY CONTROL ACTIVITIES. Monthly Summary Report of Tests: Submit the report as an electronic attachment to the CQC Report at the end of each month.
- e. Copies of all Contract modifications, arranged in numerical order. Also include documentation that modified work was accomplished.
- f. An up-to-date copy of the Rework Items List.
- g. Cx documentation in accordance with Section 01 91 00.15 20 TOTAL BUILDING COMMISSIONING.
- h. Special Inspection reports.
- i. Upon commencement of Completion Inspections of the entire project or any defined portion, maintain up-to-date copies of all punch lists issued by the QC staff to the Contractor and subcontractors and all punch lists issued by the Government in accordance with the paragraph COMPLETION INSPECTIONS.

#### 1.13.6 Testing Plan and Log

As tests are performed, the CxC and the QC Manager will record on the Testing Plan and Log the date the test was performed and the date the test results were forwarded to the Contracting Officer. Attach a copy of the updated Testing Plan and Log to the last QC Report of each month. Provide a copy of the final Testing Plan and Log to the preparer of the Operation & Maintenance (O&M) documentation.

#### 1.13.7 As-Built Drawings

The QC Manager is required to ensure the as-built drawings, required by Section 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME) are kept current on a daily basis and marked to show deviations which have been made from the Contract drawings. The as-built drawings document commences with the QC Manager ensuring all amendments, or changes to the Contract prior to Contract award are accurately noted in the initial document set creating the accurate baseline of the contract prior to any work starting. Ensure each deviation has been identified with the appropriate modifying documentation (e.g., PC No., Modification No., Request for Information No.). The QC Manager or QC Specialist assigned to an area of responsibility must initial each revision. Upon completion of work, the QC Manager will furnish a certificate attesting to the accuracy of the as-built drawings

prior to submission to the Contracting Officer.

#### 1.14 NOTIFICATION ON NON-COMPLIANCE

The Contracting Officer will notify the Contractor of any detected non-compliance with the Contract. Take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, must be deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders will be made the subject of a claim for extension of time for excess costs or damages by the Contractor.

#### 1.15 DELIVERY, STORAGE, AND HANDLING

Designate receiving/storage areas for incoming material to be delivered according to installation schedule and to be placed convenient to work area in order to minimize waste due to excessive materials handling and misapplication. Store and handle materials in a manner as to prevent loss from weather and other damage. Keep materials, products, and accessories covered and off the ground, and store in a dry, secure area. Prevent contact with material that may cause corrosion, discoloration, or staining. Protect all materials and installations from damage by the activities of other trades.

#### PART 2 PRODUCTS

Not used.

#### PART 3 EXECUTION

Not used.

-- End of Section --

# **SCHEDULE OF FIRE PROTECTION SYSTEM INSPECTIONS BY FPQC SPECIALIST**

## **1. INSTRUCTIONS for FPQC Specialist:**

Perform surveillance of fire protection system installation during construction and during pre-final and final acceptance inspections. Surveillance includes either visual inspections or observations performed by the FPQC as indicated in the following Schedule of Fire Protection System Inspections. Check boxes in the schedule indicate the inspections or observations that are applicable to this project, which must be performed and documented by the FPQC.

Surveillance of fire protection system inspections is divided into three phases:

Surveillance During Construction – Includes periodic on-site visits by the FPQC during construction

Pre-Final Acceptance Inspection – FPQC must be onsite during pre-final acceptance inspection

Final Acceptance inspection – FPQC must be onsite during final acceptance inspection

## **2. INSPECTION TYPE**

### **VISUAL INSPECTION:**

Perform on-site visual inspections as indicated in the Schedule of Fire Protection System Inspections as construction progresses to confirm work conforms with the contract requirements including but not limited to verifying proper installation of equipment and materials in accordance with applicable codes. Where work will be covered up by finished walls and ceilings, inspections must be scheduled and performed when work is still visible.

### **OBSERVATION:**

Observe component and system testing in person on-site as indicated in the Schedule of Fire Protection System Inspections. Confirm testing is completed in accordance with applicable codes and that testing was successful.

### **DOCUMENTATION:**

Document all inspections and observations in the FPQC Report in accordance with UFGS 01 45 00. Document any deficiencies and all work observed that has been successfully performed in accordance with the contract documents. Record deficiencies in the rework list.

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# SCHEDULE OF FIRE PROTECTION SYSTEM INSPECTIONS BY FPQC SPECIALIST

## PERFORM SURVEILLANCE DURING CONSTRUCTION

### ~ FIRE SUPPRESSION SYSTEM ~

| PERFORM<br>TASK IF<br>CHECKED       | TASK                                                                                    | INSPECTION<br>TYPE           | DESCRIPTION                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------|-----------------------------------------------------------------------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> | 1. Hydrostatic/leak test for above ground piping prior to ceiling installation          | <b>OBSERVE</b>               | <ul style="list-style-type: none"> <li>✓ Verify proper test method</li> <li>✓ Confirm test was successful</li> </ul>                                                                                                                                                                                                              |
| <input checked="" type="checkbox"/> | 2. Forward flow test for Backflow preventer                                             | <b>OBSERVE</b>               | <ul style="list-style-type: none"> <li>✓ Verify proper test method</li> <li>✓ Confirm test was successful</li> </ul>                                                                                                                                                                                                              |
| <input checked="" type="checkbox"/> | 3. Verify proper installation of sprinkler piping prior to wall or ceiling installation | <b>VISUAL<br/>INSPECTION</b> | <ul style="list-style-type: none"> <li>✓ Inspect pipe hangars, bracing, sprinkler heads</li> <li>✓ Inspect for sprinkler obstructions; damaged, painted or covered heads</li> <li>✓ Verify proper location of control valves, drains, vents, backflow, test header</li> <li>✓ Verify proper component mounting heights</li> </ul> |
| <input checked="" type="checkbox"/> | 4. Document inspections and observations in FPQC Report                                 | <b>DOCUMENT</b>              | <ul style="list-style-type: none"> <li>✓ Submit report for each day inspections or observations occur</li> <li>✓ Annotate deficiencies and satisfactory work observed</li> <li>✓ Record rework items in FPQC report</li> </ul>                                                                                                    |

### ~ END OF SURVEILLANCE DURING CONSTRUCTION SECTION

## PERFORM DURING PRE-FINAL ACCEPTANCE INSPECTION

| PERFORM<br>TASK IF<br>CHECKED       | TASK                                                                                                                   | INSPECTION<br>TYPE           | DESCRIPTION                                                                     |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------------------------------------|
| <input checked="" type="checkbox"/> | 1. Inspect all fire protection systems and all life safety features                                                    | <b>VISUAL<br/>INSPECTION</b> | ✓ Verify proper installation of fire protection and life safety systems         |
| <input checked="" type="checkbox"/> | 2. Witness functional testing of all fire protection and life safety systems                                           | <b>OBSERVE</b>               | ✓ Verify proper testing is performed on fire protection and life safety systems |
| <input checked="" type="checkbox"/> | 3. Identify deficiencies that require correction                                                                       | <b>VISUAL<br/>INSPECTION</b> | ✓ Document deficiencies in FPQC Report and record in rework list                |
| <input checked="" type="checkbox"/> | 4. Verify rework items have been corrected                                                                             | <b>VISUAL<br/>INSPECTION</b> | ✓ Record re-inspection results on FPQC Report                                   |
| <input checked="" type="checkbox"/> | 5. Document results of pre-final inspection and photograph unsatisfactory conditions                                   | <b>DOCUMENT</b>              | ✓ Document results of pre-final inspection in FPQC Report                       |
| <input checked="" type="checkbox"/> | 6. Certify installation and operation of all fire protection and life safety systems conform to the contract documents | <b>DOCUMENT</b>              | ✓ Provide separate letter certifying conformance with contract requirements     |

# SCHEDULE OF FIRE PROTECTION SYSTEM INSPECTIONS BY FPQC SPECIALIST

~ END OF PRE-FINAL INSPECTION SECTION

| PERFORM DURING FINAL ACCEPTANCE INSPECTION |                                                                                                         |                              |                                                                         |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------|
| PERFORM<br>TASK IF<br>CHECKED              | TASK                                                                                                    | INSPECTION<br>TYPE           | DESCRIPTION                                                             |
| <input checked="" type="checkbox"/>        | 1. Verify proper installation and operation of all fire protection systems and all life safety features | <b>VISUAL<br/>INSPECTION</b> | ✓                                                                       |
| <input checked="" type="checkbox"/>        | 2. Note any deficiencies identified by Government FPE on deficiency list                                | <b>DOCUMENT</b>              | ✓ Document deficiencies identified by Government FPE on deficiency list |
| <input checked="" type="checkbox"/>        | 3. Verify all deficiencies have been corrected                                                          | <b>VISUAL<br/>INSPECTION</b> | ✓ Inspect rework to confirm all deficiencies have been corrected        |
| <input checked="" type="checkbox"/>        | 4. Certify that all deficiencies have been corrected                                                    | <b>DOCUMENT</b>              | ✓ Document inspection results on FPQC Report                            |

~ END OF FINAL ACCEPTANCE INSPECTION SECTION

| PREPARATORY PHASE CHECKLIST |                                                                                                                                          | SPEC SECTION              | DATE                   |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|------------------------|
| (CONTINUED ON SECOND PAGE)  |                                                                                                                                          | Enter Spec Section # Here | Enter Date (DD/MMM/YY) |
| CONTRACT NO                 | DEFINABLE FEATURE OF WORK                                                                                                                | SCHEDULE ACT NO.          | INDEX #                |
| Enter Cnt# Here             | Enter DFOV Here                                                                                                                          | Enter Sched Act ID Here   | Enter Index# Here      |
| PERSONNEL PRESENT           | GOVERNMENT REP NOTIFIED _____ HOURS IN ADVANCE: YES <input type="checkbox"/> NO <input type="checkbox"/>                                 |                           |                        |
|                             | NAME                                                                                                                                     | POSITION                  | COMPANY/GOVERNMENT     |
|                             |                                                                                                                                          |                           |                        |
|                             |                                                                                                                                          |                           |                        |
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|                             |                                                                                                                                          |                           |                        |
|                             |                                                                                                                                          |                           |                        |
|                             |                                                                                                                                          |                           |                        |
| SUBMITTALS                  | REVIEW SUBMITTALS AND/OR SUBMITTAL REGISTER. HAVE ALL SUBMITTALS BEEN APPROVED? YES <input type="checkbox"/> NO <input type="checkbox"/> |                           |                        |
|                             | IF NO, WHAT ITEMS HAVE NOT BEEN SUBMITTED? _____                                                                                         |                           |                        |
|                             |                                                                                                                                          |                           |                        |
|                             | ARE ALL MATERIALS ON HAND? YES <input type="checkbox"/> NO <input type="checkbox"/>                                                      |                           |                        |
|                             | IF NO, WHAT ITEMS ARE MISSING? _____                                                                                                     |                           |                        |
|                             |                                                                                                                                          |                           |                        |
| MATERIAL STORAGE            | ARE MATERIALS STORED PROPERLY? YES <input type="checkbox"/> NO <input type="checkbox"/>                                                  |                           |                        |
|                             | IF NO, WHAT ACTION IS TAKEN? _____                                                                                                       |                           |                        |
|                             |                                                                                                                                          |                           |                        |
|                             |                                                                                                                                          |                           |                        |
|                             |                                                                                                                                          |                           |                        |
|                             |                                                                                                                                          |                           |                        |
| SPECIFICATIONS              | REVIEW EACH PARAGRAPH OF SPECIFICATIONS. _____                                                                                           |                           |                        |
|                             |                                                                                                                                          |                           |                        |
|                             |                                                                                                                                          |                           |                        |
|                             | DISCUSS PROCEDURE FOR ACCOMPLISHING THE WORK. _____                                                                                      |                           |                        |
|                             |                                                                                                                                          |                           |                        |
|                             |                                                                                                                                          |                           |                        |
|                             | CLARIFY ANY DIFFERENCES. _____                                                                                                           |                           |                        |
|                             |                                                                                                                                          |                           |                        |
| PRELIMINARY WORK & PERMITS  | ENSURE PRELIMINARY WORK IS CORRECT AND PERMITS ARE ON FILE.                                                                              |                           |                        |
|                             | IF NOT, WHAT ACTION IS TAKEN? _____                                                                                                      |                           |                        |
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|---------------------------------------------------|---------------------------------------------------------------------------------------------|
|                                                   |                                                                                             |
| TESTING                                           | IDENTIFY TEST TO BE PERFORMED, FREQUENCY, AND BY WHOM. _____                                |
|                                                   |                                                                                             |
|                                                   |                                                                                             |
|                                                   | WHEN REQUIRED? _____                                                                        |
|                                                   |                                                                                             |
|                                                   |                                                                                             |
|                                                   | WHERE REQUIRED? _____                                                                       |
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|                                                   |                                                                                             |
|                                                   | REVIEW TESTING PLAN. _____                                                                  |
|                                                   |                                                                                             |
|                                                   |                                                                                             |
|                                                   | HAS TEST FACILITIES BEEN APPROVED? _____                                                    |
|                                                   |                                                                                             |
| SAFETY                                            | ACTIVITY HAZARD ANALYSIS APPROVED? YES <input type="checkbox"/> NO <input type="checkbox"/> |
|                                                   | REVIEW APPLICABLE PORTION OF EM 385-1-1. _____                                              |
|                                                   |                                                                                             |
|                                                   |                                                                                             |
|                                                   |                                                                                             |
| MEETING COMMENTS                                  | NAVY/ROICC COMMENTS DURING MEETING.                                                         |
|                                                   |                                                                                             |
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|                                                   |                                                                                             |
| OTHER ITEMS OR REMARKS                            | OTHER ITEMS OR REMARKS:                                                                     |
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|                                                   |                                                                                             |
| <div>QC MANAGER _____</div> <div>DATE _____</div> |                                                                                             |

**\*\*Note: Preparatory Meeting shall only be held if the shop drawings or submittal have been approved, the APP and appropriate AHA related to the DFOW have been submitted and the appropriate personnel (QC Manager, QC Specialists, Project Sup, CxC, Special Inspector, Special Inspector of Record, Foreman responsible for the DFOW, subcontractor foreman) are present for the meeting.\*\***

## **PREPARATORY PHASE CHECKLIST**

Part 2, DIV 1 Compliance

| Date | Contract No. | Project Title |
|------|--------------|---------------|
|------|--------------|---------------|

- ☐ Identify any changed conditions or modifications that may impact the execution of the work.
- ☐ Examine the work area to ensure that the required preliminary work has been completed.
- ☐ Examine the required materials, equipment, and sample work to ensure that they are on hand and conform to the approved shop drawings and submitted data.
- ☐ Review each paragraph of the applicable specification Sections.
- ☐ Review the Contract drawings
- ☐ Verify that appropriate shop drawings and submittals for materials and equipment have been submitted and approved. Verify receipt of approved factory test results, when required.
- ☐ Discuss specific controls used and construction methods, constructions tolerances, layout/survey controls (horizontal and vertical controls) necessary, workmanship standards, and the approach that will be used to provide quality construction by planning ahead and identifying potential programs for each DFOW.
- ☐ Review the testing plan and ensure that provisions have been made to provide the required QC testing and applicable required A/E (DOR) Quality Assurance Inspections.
- ☐ Review the listing of Government QA inspections that may be implemented as part of the execution of the work.
- ☐ Review any Special Inspections (Section 01 45 35) required as part of the execution of the work.
- ☐ Discuss the QC documentation required to be collected as part of the work. (e.g. Special inspector reports within 24 hours, material testing reports as soon as available.)
- ☐ If work includes demolition work, review demolition work plan to ensure compliance with EM 385-1-1 Section 23 Demolition, Renovation and Re-Occupancy.
- ☐ If work includes excavation, review Utility Locating Requirements. (Note: special attention is required if Hazardous or Mission Critical Utilities are within the excavation area. (If Hazardous and Mission Critical Utilities (as identified in the Scope of Work & Drawings) are located within the excavation area, the PWD ME Dig Safe Coordinator, PWD ME CM/ET, PWD ME Design Manager (DM) and the assigned PWD-ME Project Civil Engineer must be present at the meeting to review requirements to ensure safety and preservation of critical utilities as discussed with adequate focus.)



- ☐ If work includes excavation, review Soil Management Requirements (Note: Special attention is required if the work is within an IR site or hazardous soils are within the excavation area. Include Code 106.3 Representatives at this meeting to discuss Soil Management process and procedures.)
- ☐ If the work is subject to any environmental (EV) permitting, Archaeological Monitoring, or SHPO approvals, review the terms and conditions of the applicable permits as well as any supplemental controls/approvals that are required to complete the work. (Note: Include PWD-ME EV Representatives at this meeting to review the project specific requirements.
- ☐ If work includes the removal and disposal of Hazardous materials, review the appropriate work plans to ensure the work plans have been approved and any comments are understood by the work execution team. (Note: Include Code 106.3 Representatives at this meeting to discuss Hazardous Materials handling and disposal processes and procedures.
- ☐ Review the APP and appropriate Activity Hazard Analysis (AHA) to ensure that applicable safety requirements are met, and that required Safety Data Sheets (SDS) are submitted.
- ☐ Review the processes/strategies that will be implemented to address any issues if the work does not go as planned or if any unforeseen conditions are encountered or if any changes arise that may impact the successful execution of the work.

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QC Manager

| INITIAL PHASE CHECKLIST                                                                           |                                                                                                                       | SPEC SECTION<br>Enter Spec Section # Here   | DATE<br>Enter Date (DD/MMM/YY)                           |
|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------|----------------------------------------------------------|
| CONTRACT NO<br>Enter Cnt# Here                                                                    | DEFINABLE FEATURE OF WORK<br>Enter DFOW Here                                                                          | SCHEDULE ACT NO.<br>Enter Sched Act ID Here | INDEX #<br>Enter Index# Here                             |
| PERSONNEL PRESENT                                                                                 | GOVERNMENT REP NOTIFIED ____ HOURS IN ADVANCE:                                                                        |                                             | YES <input type="checkbox"/> NO <input type="checkbox"/> |
|                                                                                                   | NAME                                                                                                                  | POSITION                                    | COMPANY/GOVERNMENT                                       |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
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|                                                                                                   |                                                                                                                       |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
| PROCEDURE COMPLIANCE                                                                              | IDENTIFY FULL COMPLIANCE WITH PROCEDURES IDENTIFIED AT PREPARATORY. COORDINATE PLANS, SPECIFICATIONS, AND SUBMITTALS. |                                             |                                                          |
|                                                                                                   | COMMENTS: _____                                                                                                       |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
| PRELIMINARY WORK                                                                                  | ENSURE PRELIMINARY WORK IS COMPLETE AND CORRECT. IF NOT, WHAT ACTION IS TAKEN?                                        |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
| WORKMANSHIP                                                                                       | ESTABLISH LEVEL OF WORKMANSHIP.                                                                                       |                                             |                                                          |
|                                                                                                   | WHERE IS WORK LOCATED? _____                                                                                          |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
|                                                                                                   | IS SAMPLE PANEL REQUIRED? YES <input type="checkbox"/> NO <input type="checkbox"/>                                    |                                             |                                                          |
|                                                                                                   | WILL THE INITIAL WORK BE CONSIDERED AS A SAMPLE? YES <input type="checkbox"/> NO <input type="checkbox"/>             |                                             |                                                          |
| (IF YES, MAINTAIN IN PRESENT CONDITION AS LONG AS POSSIBLE AND DESCRIBE LOCATION OF SAMPLE) _____ |                                                                                                                       |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
| RESOLUTION                                                                                        | RESOLVE ANY DIFFERENCES.                                                                                              |                                             |                                                          |
|                                                                                                   | COMMENTS: _____                                                                                                       |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
| CHECK SAFETY                                                                                      | REVIEW JOB CONDITIONS USING EM 385-1-1 AND JOB HAZARD ANALYSIS                                                        |                                             |                                                          |
|                                                                                                   | COMMENTS: _____                                                                                                       |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
| OTHER                                                                                             | OTHER ITEMS OR REMARKS                                                                                                |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |
|                                                                                                   |                                                                                                                       |                                             |                                                          |

QC MANAGER

DATE

SECTION 01 45 35

SPECIAL INSPECTIONS  
05/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-22 (2022; Supp 1 2023; Supp 2 2023) Minimum  
Design Loads and Associated Criteria for  
Buildings and Other Structures

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2021) International Building Code

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01 (2023; with Change 2, 2024) Structural  
Engineering

UFC 3-301-02 (2023) Design of Risk Category V  
Structures, National Strategic Military  
Assets

1.2 GENERAL REQUIREMENTS

Perform Special Inspections in accordance with the Statement of Special Inspections, Schedule of Special Inspections, Chapter 17 of ICC IBC, and UFC 3-301-01. The Statement of Special Inspections and Schedule of Special Inspections are included as an attachment to this specification. Special Inspections are to be performed by an independent third party and are intended to ensure that the work of the Prime Contractor is in accordance with the Contract Documents and applicable building codes. Special inspections do not take the place of the three phases of control inspections performed by the Contractor's QC Manager or any testing and inspections required by other sections of the specifications.

Structural observations will be performed by the Structural Engineer of Record (SER). The Contractor must provide notification to the Contracting Officer and Structural Engineer of Record (SER) 14 days prior to the following points of construction to allow for structural observation:

- a. Tank pad at Building 72
- b. Concrete demolition and cast-in-place in Building 242
- c. Ultrafilter concrete cast-in-place
- d. Foundation work for tank pad

- e. Masonry construction
- f. Cast-in-place concrete

### 1.3 DEFINITIONS

#### 1.3.1 Continuous Special Inspections

Continuous Special Inspections is the constant monitoring of specific tasks by a special inspector. These inspections must be carried out continuously over the duration of the particular tasks.

#### 1.3.2 Perform

Perform these tasks for each weld, fastener or bolted connection, and noted required verification.

#### 1.3.3 Observe

Observe these items randomly during the course of each work day to insure that applicable requirements are being met. Observe these Special Inspections items on a periodic daily basis. Operations need not be delayed pending these inspections.

#### 1.3.4 Special Inspector (SI)

A qualified person retained by the Contractor and approved by the Contracting Officer as having the competence necessary to inspect a particular type of construction requiring Special Inspections. The SI must be an independent third party hired directly by the Prime Contractor.

#### 1.3.5 Associate Special Inspector (ASI)

A qualified person who assists the SI in performing Special Inspections but who must perform inspection under the direct supervision of the SI and who cannot perform inspections without the SI on site.

#### 1.3.6 Third Party

A Special inspector must not be an employee of the Contractor or of any Sub-Contractor performing the work to be inspected.

#### 1.3.7 Special Inspector of Record (SIOR)

A licensed professional engineer in responsible charge of supervision of all special inspectors for the project and approved by the Contracting Officer. The SIOR must be an independent third-party entity hired directly by the Prime Contractor.

#### 1.3.8 Contracting Officer

The Government official having overall authority for administrative contracting actions. Certain contracting actions may be delegated to the Contracting Officer's Representative (COR).

#### 1.3.9 Quality Control (QC) Manager

An individual retained by the Prime Contractor and qualified in accordance with the Section 01 45 00.00 22 QUALITY CONTROL (PWD ME) having the

overall responsibility for the Contractor's QC organization.

#### 1.3.10 Structural Engineer of Record (SER)

A registered design professional employed by the Government responsible for the overall design and review of submittal documents prepared by others. The SER is registered or licensed to practice their respective design profession as defined by the statutory requirements of the professional registration laws in the state in which the design professional works. The SER is also referred to as the Engineer of Record (EOR) in design documents.

#### 1.3.11 Statement of Special Inspections (SSI)

A document developed by the SER identifying the materials, systems, components and work required to have Special Inspections. This statement is included at the end of this specification as Attachment A.

#### 1.3.12 Schedule of Special Inspections (SSI)

A schedule which lists each of the required Special Inspections, the extent to which each Special Inspection is to be performed, and the required frequency for each in accordance with ICC IBC Chapter 17. This schedule is included at the end of this specification as Attachment B.

#### 1.3.13 Designated Seismic Systems (DSS)

Those nonstructural components that require design in accordance with ASCE 7-22 Chapter 13 and for which the component importance factor,  $I_p$ , is greater than 1.0. This designation applies to systems that are required to be operational following the Design Earthquake for RC I - IV structures and following the MCER for RC V structures. All systems in RC V facilities designated as MC-1 in accordance with UFC 3-301-02 are considered part of the Designated Seismic Systems.

#### 1.3.14 Definable Feature of Work (DFOW)

An inspection group that is separate and distinct from other inspection groups, having inspection requirements or inspectors that are unique.

### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-01 Preconstruction Submittals

SIOR Letter of Acceptance; G

Special Inspections Project Manual; G

#### SD-06 Test Reports

Special Inspection Daily Reports

Special Inspection Biweekly Reports

SD-07 Certificates

AISC Certified Steel Fabricator  
Steel Truss Plant Quality Assurance Program  
Wood Truss Plant Quality Assurance Program  
AC472 Accreditation  
Certificate of Compliance  
Special Inspector of Record Qualifications; G  
Special Inspector Qualifications; G

SD-11 Closeout Submittals

Interim Report of Special Inspections from each DFO; G  
Comprehensive Final Report of Special Inspections; G  
Final Report of Structural Observations where required; G

1.5 SPECIAL INSPECTOR QUALIFICATIONS

Submit qualifications for each special inspector and the special inspector of record according to the following certification requirements. The Government will consider alternate equivalent certification upon request. Without written acceptance deviations to the following qualifications are not approved.

1.5.1 Steel Construction and High Strength Bolting

1.5.1.1 Special Inspector

- a. ICC Structural Steel and Bolting Special Inspector certificate with one year of related experience, or
- b. Registered Professional Engineer with 3 years of related experience

1.5.1.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.2 Welding Structural Steel

1.5.2.1 Special Inspector

- a. ICC Structural Welding Special Inspector certificate with one year of related experience, or
- b. AWS Certified Welding Inspector

1.5.2.2 Associate Special Inspector

AWS Certified Associate Welding Inspector

1.5.3 Nondestructive Testing of Welds

1.5.3.1 Special Inspector

NDT Level III Certificate

1.5.3.2 Associate Special Inspector

NDT Level II Certificate plus one year of related experience

1.5.4 Cold Formed Steel Framing

1.5.4.1 Special Inspector

- a. ICC Structural Steel and Bolting Special Inspector certificate with one year of related experience, or
- b. ICC Commercial Building Inspector with one year of experience, or
- c. Registered Professional Engineer with 3 years related experience

1.5.4.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.5 Concrete Construction

1.5.5.1 Special Inspector

- a. ICC Reinforced Concrete Special Inspector Certificate with one year of related experience, or
- b. ACI Concrete Construction Special Inspector, or
- c. Registered Professional Engineer with 3 years of related experience

1.5.5.2 Associate Special Inspector

- a. ACI Concrete Construction Special Inspector-in-Training, or
- b. Engineer-In-Training with one year of related experience

1.5.6 Prestressed Concrete Construction

1.5.6.1 Special Inspector

- a. ICC Prestressed Concrete Special Inspector Certificate with one year of related experience, or
- b. PCI Quality Control Technician/Inspector Level II Certificate with one year of related experience, or
- c. Registered Professional Engineer with 3 years of related experience

1.5.6.2 Associate Special Inspector

- a. PCI Quality Control Technician/Inspector Level I Certificate with one year of related experience, or



- b. Engineer-In-Training with one year of related experience

#### 1.5.7 Post-tensioned Concrete Construction

##### 1.5.7.1 Special Inspector

- a. PTI Level 2 Unbonded PT Inspector Certificate, or
- b. Registered Professional Engineer with 3 years of related experience

##### 1.5.7.2 Associate Special Inspector

- a. PTI Level 1 Unbonded PT Inspector Certificate with one year of related experience, or
- b. Engineer-In-Training with one year of related experience

#### 1.5.8 Masonry Construction

##### 1.5.8.1 Special Inspector

- a. ICC Structural Masonry Special Inspector Certificate with one year of related experience, or
- b. Registered Professional Engineer with 3 years of related experience

##### 1.5.8.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

#### 1.5.9 Wood

##### 1.5.9.1 Special Inspector

- a. ICC Commercial Building Inspector Certificate with one year of related experience, or
- b. ICC Residential Building Inspector with one year of experience, or
- c. Registered Professional Engineer with 3 years of related experience

##### 1.5.9.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

#### 1.5.10 Verification of Site Soil Condition, Fill Placement and Load-Bearing Requirements

##### 1.5.10.1 Special Inspector

- a. ICC Soils Special Inspector Certificate with one year of related experience, or
- b. National Institute for Certification in Engineering Technologies (NICET) Soils Technician Level II Certificate in Construction Material Testing, or
- c. Geologist-In-Training with 3 years of related experience, or

- d. Registered Professional Engineer with 3 years of related experience

1.5.10.2 Associate Special Inspector

- a. NICET Soils Technician Level I Certificate in Construction Material Testing with one year of related experience, or
- b. Engineer-In-Training with one year of related experience

1.5.11 Deep Foundations

1.5.11.1 Special Inspector

- a. NICET Soils Technician Level II Certificate in Construction Material Testing, or
- b. Geologist-In-Training with 3 years of related experience, or
- c. Registered Professional Engineer with 3 years of related experience

1.5.11.2 Associate Special Inspector

- a. NICET Soils Technician Level I Certificate in Construction Material Testing with one year of related experience, or
- b. NICET Geotechnical Engineering Technician Level I Construction or Generalist Certificate with one year of related experience, or
- c. Engineer-In-Training with one year of related experience

1.5.12 Sprayed Fire-Resistive Material

1.5.12.1 Special Inspector

- a. ICC Spray-applied Fireproofing Special Inspector Certificate, or
- b. ICC Fire Inspector I Certificate with one year of related experience, or
- c. Registered Professional Engineer or Architect with related experience

1.5.12.2 Associate Special Inspector

Engineer-In-Training with one year of related experience

1.5.13 Intumescent Fire-Resistive Coatings

1.5.13.1 Special Inspector

- a. ICC Spray-applied Fireproofing Special Inspector Certificate, or
- b. ICC Fire Inspector I Certificate with one year of related experience, or
- c. Registered Professional Engineer or Architect with related experience

1.5.13.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.14 Exterior Insulation and Finish System (EIFS)

1.5.14.1 Special Inspector

- a. Association of the Wall and Ceiling Industry (AWCI) EIFS Inspector Certificate, or
- b. Exterior Design Institute Certificate, or
- c. Registered Professional Engineer or Architect with related experience

1.5.14.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.15 Fire-Resistant Penetrations and Joints

1.5.15.1 Special Inspector

- a. Passed the UL Firestop Exam with one year of related experience, or
- b. Passed the FM Firestop Exam with one year of related experience, or
- c. Passed the International Firestop Council Exam with one year of related experience, or
- d. Registered Professional Engineer with related experience

1.5.15.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.16 Smoke Control

1.5.16.1 Special Inspector

- a. Associated Air Balance Council (AABC) Technician Certification with one year of related experience, or
- b. Registered Professional Engineer with related experience

1.5.16.2 Associate Special Inspector

Engineer-In-Training with one year of related experience.

1.5.17 Special Inspector of Record (SIOR)

Registered Professional Engineer with 5 years of related experience.

PART 2 PRODUCTS

2.1 SPECIAL INSPECTION OF FABRICATED ITEMS

Special Inspection of fabricator's work performed in the fabricator's shop is required to be inspected in accordance with the Statement of Special Inspections and the Schedule of Special Inspections unless the fabricator is certified by the approved agency to perform such work without Special Inspections. Submit the following certifications to the Contracting

Officer for information to allow work performed in the fabricator's shop to not be subjected to Special Inspections.

AISC Certified Steel Fabricator.

Truss Plate Institute (TPI) steel truss plant quality assurance program certification.

Truss Plate Institute (TPI) wood truss plant quality assurance program certification.

International Accreditation Service, AC472 Accreditation

Precast Concrete Institute (PCI) Certified Plant, Group C - Commercial (Structural) Products

At the completion of fabrication, submit a certificate of compliance, to be included with the comprehensive final report of Special Inspections, stating that the materials supplied and work performed by the fabricator are in accordance with the construction documents.

### PART 3 EXECUTION

#### 3.1 RESPONSIBILITIES

##### 3.1.1 Special Inspector of Record

- a. Supervise all Special Inspectors required by the Contract Documents and the IBC.
- b. Submit an SIOR Letter of Acceptance to the Contracting Officer attesting to acceptance of the duties of SIOR, signed and sealed by the SIOR.
- c. Verify the qualifications of all of the Special Inspectors.
- d. Verify the qualifications of fabricators.
- e. Submit Special Inspection agency's written NDT practices for the monitoring and control of the agency's operations to include the following:
  - (1) The agency's procedures for the selection and administration of inspection personnel, describing the training, experience and examination requirements for qualification and certification of inspection personnel.
  - (2) The agency's inspection procedures, including general inspection, material controls, and visual welding inspection.
- f. Submit Qualification Records for Nondestructive Testing Technicians for nondestructive testing (NDT) technicians designated for the project.
- g. Submit NDT procedures and equipment calibration records for NDT to be performed and equipment to be used for the project.
- h. Prepare a Special Inspections Project Manual, which must cover the

following:

- (1) Roles and responsibilities of the following individuals during Special Inspections: SIOR, SI, ASI, General Contractor's QC Manager and SER.
- (2) Organizational chart or communication plan, indicating lines of communication.
- (3) Contractor's internal plan for scheduling inspections. Address items such as timeliness of inspection requests, whom to contact for inspection requests, and availability of alternate inspectors.
- (4) Indicate the Government reporting requirements.
- (5) Propose forms or templates to be used by SI and SIOR to document inspections.
- (6) Indicate procedures for tracking nonconforming work and verification that corrective work is complete.
- (7) Indicate how the SIOR and SI will participate in weekly QC meetings.
- (8) Indicate how Special Inspections of shop fabricated items will be handled when the fabricator's shop is not certified in accordance with paragraph SPECIAL INSPECTION OF FABRICATED ITEMS.
- (9) Include a section in the manual that covers each specific item requiring Special Inspections that is indicated on the Schedule of Special Inspections. Provide names and qualifications of each special inspector who will be performing the Special Inspections for each specific item. Provide details on how the Special Inspections are to be carried out for each item so that the expectations are clear for the General Contractor and the Subcontractor performing the work.

Make a copy of the Special Inspections Project Manual available on the job site during construction. Submit a copy of the Special Inspections Project Manual for approval.

- i. Attend coordination and mutual understanding meeting where the information in the Special Inspections Project Manual will be reviewed to verify that all parties have a clear understanding of the Special Inspection provisions and the individual duties and responsibilities of each party.
- j. Maintain a 3-ring binder for the Special Inspector's daily and special inspection biweekly reports and the Special Inspections Project Manual. This binder must be located in a conspicuous place in the project trailer/office to allow review by the Contracting Officer and the SER.
- k. Submit a copy of the Special Inspector's special inspection daily reports to the QC Manager. The QC Manager must submit copies of the Special Inspector's daily reports to the Contracting Officer with the Contractor's Daily Quality Control Reports.
- l. Discrepancies that are observed during Special Inspections must be

reported to the QC Manager for correction. If discrepancies are not corrected before the Special Inspector leaves the site, the observed discrepancies must be documented in the daily report.

- m. Submit a biweekly Special Inspections report until all work requiring Special Inspections is complete. A report is required for each biweekly period in which Special Inspection activity occurs; the report must include the following:
    - (1) A brief summary of the work performed during the reporting time frame.
    - (2) Changes from and discrepancies with the drawings, specifications and mechanical or electrical component certification that were observed during the reporting period.
    - (3) Discrepancies which were resolved or corrected.
    - (4) A list of nonconforming items requiring resolution.
    - (5) All applicable test results including nondestructive testing reports.
  - n. At the completion of each Definable Feature of Work (DFOW) requiring Special Inspections, submit an interim report that documents the Special Inspections completed for that DFOW, including corrections of all discrepancies noted in the daily reports. Interim reports of Special Inspections must be signed and dated by the SIOR.
  - o. At the completion of the project submit a comprehensive final report of Special Inspections that documents the Special Inspections completed for the project including corrections of all discrepancies noted in the daily reports. The comprehensive final report of Special Inspections must be signed, dated and sealed by the SIOR.
- 3.1.2 Quality Control Manager
- a. Maintain a rework items list that includes discrepancies noted on the Special Inspector's daily report.
- 3.1.3 Special Inspectors
- a. Inspect all elements of the project which the special inspector is qualified to inspect and which are identified in the Schedule of Special Inspections.
  - b. Attend preparatory phase meetings related to the Definable Feature of Work (DFOW) which the Special Inspector is qualified to inspect.
  - c. Submit special inspection daily reports to the SIOR.

3.2 DEFECTIVE WORK

Check work as it progresses, but failure to detect any defective work or materials must in no way prevent later rejection if defective work or materials are discovered, nor obligate the Contracting Officer to accept the defective work or materials.

-- End of Section --

# SCHEDULE OF SPECIAL INSPECTIONS

Reference UFGS 01 45 35 for all requirements not noted as part of this schedule.

**INSPECTION DEFINITIONS:**

- PERFORM:** Perform these tasks for each weld, fastener or bolted connection, and required verification.
- OBSERVE:** Observe these items randomly during the course of each work day to insure that applicable requirements are being met. Operations need not be delayed pending these inspections at contractor’s risk.
- DOCUMENT:** Document, with a report, that the work has been performed in accordance with the contract documents. This is in addition to any other reports required in the Special Inspections guide specification.
- CONTINUOUS:** Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.
- PERIODIC:** Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.

The Seismic Design Category for this project is: ☐ A, ☐ B, ☐ C, ☐ D, ☐ E, ☐ F (check appropriate box)

**DESIGNER NOTES (delete this box after reviewing):**

1. This schedule contains minimum requirements. Do not delete applicable inspection tasks unless notes in blue indicate it is acceptable to do so.
2. Blue text = designers notes. The designer must review and edit all blue text in this schedule prior to inserting this schedule into the special inspections spec (UFGS 01 45 35).
3. Check section boxes with ANY inspection tasks applicable to your project. You may choose to delete unchecked sections or leave them in the schedule unchecked.
4. Individual rows/tasks that are not applicable to the project may be left in the section, as the inspector can determine whether they occur/apply (e.g. metal trusses in the light gauge framing section for example).
5. Design discipline sections are color coded for easier reference by designers. This schedule does NOT need to be printed in color.
6. When finished editing, delete this note box and save this schedule as a PDF and insert into the project specifications (special inspections section).



**A. STRUCTURAL STEEL – WELDING****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐**

| STEEL INSPECTION <u>PRIOR</u> TO WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22: Table N5.4-1 and Table C-N5.4-1 |                              |                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                                                                   | INSPECTION TYPE <sup>1</sup> | DESCRIPTION                                                                                                                                                                                                                                                                                                    |
| 1. Verify that the welding procedures specification (WPS) is available                                                                                 | <b>PERFORM</b>               |                                                                                                                                                                                                                                                                                                                |
| 2. Verify manufacturer certifications for welding consumables are available                                                                            | <b>PERFORM</b>               |                                                                                                                                                                                                                                                                                                                |
| 3. Verify material identification                                                                                                                      | <b>PERFORM</b>               | Type and grade.                                                                                                                                                                                                                                                                                                |
| 4. Verify use of qualified welders                                                                                                                     | <b>PERFORM</b>               | Welding by welders, welding operators, and tack welders who are qualified in conformance with requirements.                                                                                                                                                                                                    |
| 5. Welder Identification System                                                                                                                        | OBSERVE                      | <ul style="list-style-type: none"> <li>✓ Fabricator or erector, as applicable, shall maintain a system by which a welder who has welded a joint or member can be identified.</li> <li>✓ Die stamping of members subjected to fatigue shall be prohibited unless approved by the engineer of record.</li> </ul> |
| 6. Fit-up of groove welds (including joint geometry)                                                                                                   | OBSERVE                      | <ul style="list-style-type: none"> <li>✓ Joint preparation</li> <li>✓ Dimensions (alignment, root opening, root face, bevel)</li> <li>✓ Cleanliness (condition of steel surfaces)</li> <li>✓ Tacking (tack weld quality and location)</li> <li>✓ Backing type and fit (if applicable)</li> </ul>               |
| 7. Fit-up of CJP groove welds of HSS T-, Y-, and K-connections without backing (including joint geometry)                                              | <b>PERFORM</b>               | <ul style="list-style-type: none"> <li>✓ Joint preparations</li> <li>✓ Dimensions (alignment, root opening, root face, bevel)</li> <li>✓ Cleanliness (condition of steel surfaces)</li> <li>✓ Tacking (tack weld quality and location)</li> </ul>                                                              |
| 8. Configuration and finish of access holes                                                                                                            | OBSERVE                      |                                                                                                                                                                                                                                                                                                                |
| 9. Fit-up of fillet welds                                                                                                                              | OBSERVE                      | <ul style="list-style-type: none"> <li>✓ Dimensions (alignment, gaps at root)</li> <li>✓ Cleanliness (condition of steel surfaces)</li> <li>✓ Tacking (tack weld quality and location)</li> </ul>                                                                                                              |
| STEEL INSPECTION <u>DURING</u> WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22: Table N5.4-2 and Table C-N5.4-2   |                              |                                                                                                                                                                                                                                                                                                                |
| TASK                                                                                                                                                   | INSPECTION TYPE              | DESCRIPTION                                                                                                                                                                                                                                                                                                    |
| 10. Control and handling of welding consumables                                                                                                        | OBSERVE                      | <ul style="list-style-type: none"> <li>✓ Packaging</li> <li>✓ Electrode atmospheric exposure control</li> </ul>                                                                                                                                                                                                |
| 11. No welding over cracked tack welds                                                                                                                 | OBSERVE                      |                                                                                                                                                                                                                                                                                                                |
| 12. Environmental conditions                                                                                                                           | OBSERVE                      | <ul style="list-style-type: none"> <li>✓ Wind speed within limits</li> <li>✓ Precipitation and temperature</li> </ul>                                                                                                                                                                                          |

<sup>1</sup> **PERFORM:** Perform these tasks for each weld, fastener, or bolted connection, and required verification.

**OBSERVE:** Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

| STEEL INSPECTION <u>DURING</u> WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22: Table N5.4-2 and Table C-N5.4-2 |                              |                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                                                                 | INSPECTION TYPE              | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                      |
| 13. Welding Procedures Specification followed                                                                                                        | OBSERVE                      | <ul style="list-style-type: none"> <li>✓ Settings on welding equipment</li> <li>✓ Travel speed</li> <li>✓ Selected welding materials</li> <li>✓ Shielding gas type/flow rate</li> <li>✓ Preheat applied</li> <li>✓ Interpass temperature maintained (min./max.)</li> <li>✓ Proper position (F, V, H, OH)</li> <li>✓ Intermix of filler metals avoided</li> </ul> |
| 14. Welding techniques                                                                                                                               | OBSERVE                      | <ul style="list-style-type: none"> <li>✓ Interpass and final cleaning</li> <li>✓ Each pass within profile limitations</li> <li>✓ Each pass meets quality requirements</li> </ul>                                                                                                                                                                                 |
| STEEL INSPECTION <u>AFTER</u> WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22: Table N5.4-3 and Table C-N5.4-3  |                              |                                                                                                                                                                                                                                                                                                                                                                  |
| TASK                                                                                                                                                 | INSPECTION TYPE <sup>1</sup> | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                      |
| 15. Welds cleaned                                                                                                                                    | OBSERVE                      | ✓                                                                                                                                                                                                                                                                                                                                                                |
| 16. Verify size, length, and location of all welds                                                                                                   | <b>PERFORM</b>               | ✓ Size, length, and location of all welds conform to the requirements of the construction drawings.                                                                                                                                                                                                                                                              |
| 17. Verify welds meet visual acceptance criteria                                                                                                     | <b>PERFORM AND DOCUMENT</b>  | <ul style="list-style-type: none"> <li>✓ Crack prohibition</li> <li>✓ Weld/base-metal fusion</li> <li>✓ Crater cross section</li> <li>✓ Weld profiles</li> <li>✓ Weld size</li> <li>✓ Undercut</li> <li>✓ Porosity</li> </ul>                                                                                                                                    |
| 18. Check for arc strikes                                                                                                                            | <b>PERFORM</b>               |                                                                                                                                                                                                                                                                                                                                                                  |
| 19. Visually inspect k-area                                                                                                                          | <b>PERFORM</b>               | When welding of doubler plates, continuity plates or stiffeners has been performed in the k-area, visually inspect the web k-area for cracks within 3 in. (75 mm) of the weld                                                                                                                                                                                    |
| 20. Verify backing removed, weld tabs removed and finished, and fillet welds added where required                                                    | <b>PERFORM</b>               |                                                                                                                                                                                                                                                                                                                                                                  |
| 21. Repair activities                                                                                                                                | <b>PERFORM AND DOCUMENT</b>  |                                                                                                                                                                                                                                                                                                                                                                  |
| 22. Acceptance or rejection of welded joint or member                                                                                                | <b>PERFORM AND DOCUMENT</b>  |                                                                                                                                                                                                                                                                                                                                                                  |
| 23. No prohibited welds have been added without the approval of the engineer of record                                                               | OBSERVE                      |                                                                                                                                                                                                                                                                                                                                                                  |

**END SECTION**

<sup>1</sup> **PERFORM:** Perform these tasks for each weld, fastener, or bolted connection, and required verification.  
**OBSERVE:** Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.  
**DOCUMENT:** Document in a report that the work has been performed as required. This is in addition to all other required reports.

**B. STRUCTURAL STEEL – BOLTING****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| STEEL INSPECTION TASKS <u>PRIOR</u> TO BOLTING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22: Table N5.6-1 and Table C-N5.6-1           |                              |                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                                                                                   | INSPECTION TYPE <sup>1</sup> | DESCRIPTION                                                                                                                                    |
| 1. Verify manufacture’s certifications available for fastener materials                                                                                                | <b>PERFORM</b>               |                                                                                                                                                |
| 2. Fasteners marked in accordance with ASTM requirements                                                                                                               | OBSERVE                      |                                                                                                                                                |
| 3. Proper fasteners selected for joint detail (grade, type, bolt length if threads are to be excluded from shear plane)                                                | OBSERVE                      |                                                                                                                                                |
| 4. Proper bolting procedure selected for joint detail                                                                                                                  | OBSERVE                      |                                                                                                                                                |
| 5. Connecting elements, including appropriate faying surface condition and hole preparation, if specified, meet applicable requirements                                | OBSERVE                      |                                                                                                                                                |
| 6. Pre-installation verification testing for each fastener assembly performed by installation personnel, carefully observing and documenting the installation methods. | OBSERVE                      | Check if the fastener assemblies and the installation methods are capable of achieving 105%, or more, of the minimum required bolt pretension. |
| 7. Proper storage provided for bolts, nuts, washers, and other fastener components                                                                                     | OBSERVE                      |                                                                                                                                                |
| STEEL INSPECTION TASKS <u>DURING</u> BOLTING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22: Table N5.6-2 and Table C-N5.6-2             |                              |                                                                                                                                                |
| TASK                                                                                                                                                                   | INSPECTION TYPE              | DESCRIPTION                                                                                                                                    |
| 8. Fastener assemblies of suitable condition, placed in all holes and washers (if required) are positioned as required                                                 | OBSERVE                      |                                                                                                                                                |
| 9. Joint brought to the snug-tight condition prior to pretensioning operation                                                                                          | OBSERVE                      |                                                                                                                                                |
| 10. Fastener component not turned by the wrench prevented from rotating                                                                                                | OBSERVE                      |                                                                                                                                                |
| 11. Bolts are pretensioned in accordance with RCSC <i>Specification</i> , progressing systematically from the most rigid point toward the free edges                   | OBSERVE                      |                                                                                                                                                |
| STEEL INSPECTION TASKS <u>AFTER</u> BOLTING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22: Table N5.6.3 and Table C-N5.6-3              |                              |                                                                                                                                                |
| TASK                                                                                                                                                                   | INSPECTION TYPE              | DESCRIPTION                                                                                                                                    |
| 12. Acceptance or rejection of all bolted connections                                                                                                                  | <b>DOCUMENT</b>              |                                                                                                                                                |

**END SECTION**

<sup>1</sup> **PERFORM:** Perform these tasks for each weld, fastener, or bolted connection, and required verification.

**OBSERVE:** Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor’s risk.

**DOCUMENT:** Document in a report that the work has been performed as required. This is in addition to all other required reports.

C. STRUCTURAL STEEL - NONDESTRUCTIVE TESTING

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

| NONDESTRUCTIVE TESTING OF WELDED JOINTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22: Section N5.5 |                              |                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                                               | INSPECTION TYPE <sup>1</sup> | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                     |
| 1. Verify use of qualified nondestructive testing personnel                                                                        | PERFORM                      | Visual weld inspection and nondestructive testing (NDT) shall be conducted by personnel qualified in accordance with AWS D1.8 Clause 7.2                                                                                                                                                                                                                        |
| 2. CJP groove welds                                                                                                                | OBSERVE                      | <b>[NOTE: DOR <u>must</u> delete this row if Section D (SEISMIC PROVISIONS SECTION) is checked]</b><br>Dye penetrant testing (DT) and ultrasonic testing (UT) shall be performed on 20% of CJP groove welds for materials greater than 5/16” (8 mm) thick. Testing rate must be increased to 100% if greater than 5% of welds tested have unacceptable defects. |
| 3. Welded joints subject to fatigue                                                                                                | OBSERVE                      | Dye penetrant testing (DT) and Ultrasonic testing (UT) shall be performed on 100% of welded joints identified on contract drawings as being subject to fatigue. Table A-3.1 in AISC 360-22 provides Fatigue Design Parameters for welded joints that must be checked.                                                                                           |

END SECTION

<sup>1</sup> **PERFORM:** Perform these tasks for each weld, fastener, or bolted connection, and required verification.  
**OBSERVE:** Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor’s risk.

**D. STRUCTURAL STEEL – AISC 341 REQUIREMENTS (SEISMIC PROVISIONS)****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| NONDESTRUCTIVE TESTING OF WELDED JOINTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 341-22: Section J7.2                                                                     |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                                                                                                                   | INSPECTION TYPE <sup>1</sup> | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>[NOTE: DOR may uncheck this section for projects NOT designed in accordance with AISC 341 (Seismic Provisions) or for projects designed according to AISC 341, but using an R value equal to 3]</b> |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 4. CJP groove welds                                                                                                                                                                                    | OBSERVE                      | <p>Ultrasonic testing (UT) shall be performed on 100% of complete-joint-penetration (CJP) groove welds in materials 5/16 in. (8 mm) thick or greater. UT in materials less than 5/16 in. (8 mm) thick is not required. Welds shall be inspected by UT in compliance with AWS D1.8/D1.8M.</p> <p>Magnetic particle testing (MT) shall be performed on 25% of all beam-to-column CJP groove welds. Welds shall be inspected by MT in compliance with AWS D1.8/ D1.8M.</p> <p>For ordinary moment frames in structures in risk categories I or II, UT and MT of CJP groove welds shall be required only for demand critical welds.</p> |
| 5. Beam cope and access hole.                                                                                                                                                                          | OBSERVE                      | At welded splices and connections, thermally cut surfaces of beam copes and access holes shall be tested using magnetic particle testing (MT) or dye penetrant testing (DT), when the flange thickness exceeds 1 1/2 in. for rolled shapes, or when the web thickness exceeds 1 1/2 in. for built-up shapes.                                                                                                                                                                                                                                                                                                                        |
| 6. <i>k</i> -area NDT (AISC 341 Table J7.1 and AWS D1.8 Clause 7.4)                                                                                                                                    | <b>PERFORM AND DOCUMENT</b>  | Where welding of doubler plates, continuity plates or stiffeners has been performed in the <i>k</i> -area, the web shall be tested for cracks using magnetic particle testing (MT). The MT inspection area shall include the <i>k</i> -area base metal within 3 in. of the weld. The MT shall be performed no sooner than 48 hours following completion of the welding.                                                                                                                                                                                                                                                             |
| 7. Placement of reinforcing or contouring fillet welds (Table J7.1)                                                                                                                                    | <b>PERFORM AND DOCUMENT</b>  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 8. Weld tab removal sites                                                                                                                                                                              | OBSERVE                      | At the end of welds where weld tabs have been removed, magnetic particle testing shall be performed on the same beam-to-column joints receiving UT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

**END SECTION**

<sup>1</sup> **PERFORM:** Perform these tasks for each weld, fastener, or bolted connection, and required verification.  
**OBSERVE:** Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.  
**DOCUMENT:** Document in a report that the work has been performed as required. This is in addition to all other required reports

**E. STRUCTURAL STEEL - COMPOSITE CONSTRUCTION<sup>1</sup>****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| COMPOSITE CONSTRUCTION <u>PRIOR TO</u> PLACING CONCRETE – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22: Table N5.4-2, AISC 341-22: Table J10.1 |                 |             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------|
| TASK                                                                                                                                                                         | INSPECTION TYPE | DESCRIPTION |
| 1. Placement and installation of steel headed stud anchors                                                                                                                   | <b>PERFORM</b>  |             |
| 2. Material identification of reinforcing steel (Type/Grade)                                                                                                                 | OBSERVE         |             |
| 3. Determination of carbon equivalent for reinforcing steel other than ASTM A706                                                                                             | OBSERVE         |             |
| 4. Proper reinforcing steel size, spacing, clearances, support, and orientation                                                                                              | OBSERVE         |             |
| 5. Reinforcing steel has not been rebent in the field                                                                                                                        | OBSERVE         |             |
| 6. Required reinforcing steel clearances have been provided                                                                                                                  | OBSERVE         |             |
| 7. Reinforcing steel has been tied and supported as required                                                                                                                 | OBSERVE         |             |
| 8. Composite member has required size                                                                                                                                        | OBSERVE         |             |

**END SECTION****F. STRUCTURAL STEEL - OTHER INSPECTIONS****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| OTHER STEEL INSPECTIONS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.1, AISC 360-22 Section N5.8, AISC 341-22: Tables J9.1 & J11.1 |                              |                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                                                                | INSPECTION TYPE <sup>2</sup> | DESCRIPTION                                                                                                                                                                      |
| 1. Verify anchor rods and other embedments supporting structural steel                                                                              | <b>PERFORM</b>               | Verify the diameter, grade, type, and length of the anchor rod or embedded item, and the extent or depth of embedment prior to placement of concrete.                            |
| 2. Fabricated steel or erected steel frame                                                                                                          | OBSERVE                      | Verify compliance with the details shown on the construction documents, such as braces, stiffeners, member locations and proper application of joint details at each connection. |
| 3. Check reduced beam sections (RBS) where/if they occur                                                                                            | <b>PERFORM AND DOCUMENT</b>  | ✓ Contour and finish<br>✓ Dimensional tolerances                                                                                                                                 |
| 4. Check protected zones                                                                                                                            | <b>PERFORM AND DOCUMENT</b>  | No holes or unapproved attachments made by fabricator or erector                                                                                                                 |
| 5. Check H-piles where/if they occur                                                                                                                | <b>PERFORM AND DOCUMENT</b>  | No holes or unapproved attachments made by the responsible contractor                                                                                                            |

**END SECTION****G. STRUCTURAL COLD-FORMED METAL DECK PLACEMENT**<sup>1</sup> See Concrete Construction Section for all concrete related inspection of composite steel construction.<sup>2</sup> **PERFORM:** Perform these tasks for each weld, fastener, or bolted connection, and required verification.**OBSERVE:** Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.**DOCUMENT:** Document in a report that the work has been performed as required. This is in addition to all other required reports.

**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐**

**METAL DECK INSPECTION PRIOR TO DECK PLACEMENT – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.3, SDI QA/QC-2022, Appendix 1, Table 1.1**

| TASK                                                                                                                                                             | INSPECTION TYPE <sup>1</sup> | DESCRIPTION |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------|
| 1. Verify compliance of materials (deck and all deck accessories) with construction documents, including profiles, material properties, and base metal thickness | <b>PERFORM</b>               |             |
| 2. Acceptance or rejection of deck and deck accessories                                                                                                          | <b>DOCUMENT</b>              |             |

**METAL DECK INSPECTION AFTER DECK PLACEMENT – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.3, SDI QA/QC-2022, Appendix 1, Table 1.2**

| TASK                                                                                                            | INSPECTION TYPE | DESCRIPTION |
|-----------------------------------------------------------------------------------------------------------------|-----------------|-------------|
| 3. Verify compliance of deck and all deck accessory installation with construction documents                    | <b>PERFORM</b>  |             |
| 4. Verify deck materials are represented by the mill certifications that comply with the construction documents | <b>PERFORM</b>  |             |
| 5. Acceptance or rejection of installation of deck and deck accessories                                         | <b>DOCUMENT</b> |             |

**METAL DECK INSPECTION PRIOR TO WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.3, SDI QA/QC-2022, Appendix 1, Table 1.3**

| TASK                                                                          | INSPECTION TYPE | DESCRIPTION |
|-------------------------------------------------------------------------------|-----------------|-------------|
| 6. Verify welding procedure specification (WPS) is available                  | <b>PERFORM</b>  |             |
| 7. Verify manufacturer's certifications for welding consumables are available | <b>PERFORM</b>  |             |
| 8. Material identification (type/grade)                                       | <b>OBSERVE</b>  |             |
| 9. Check welding equipment                                                    | <b>PERFORM</b>  |             |

**END SECTION**

<sup>1</sup>**PERFORM:** Perform these tasks for each weld, fastener, or bolted connection, and required verification.

**OBSERVE:** Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

**DOCUMENT:** Document in a report that the work has been performed as required. This is in addition to all other required reports.

H. STRUCTURAL COLD-FORMED METAL DECK – WELDING

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

| METAL DECK INSPECTION <u>DURING</u> WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.3, SDI QA/QC-2022, Appendix 1, Table 1.4 |                              |             |
|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------|
| TASK                                                                                                                                               | INSPECTION TYPE <sup>1</sup> | DESCRIPTION |
| 1. Use of qualified welders                                                                                                                        | OBSERVE                      |             |
| 2. Control and handling of welding consumables                                                                                                     | OBSERVE                      |             |
| 3. Environmental conditions (wind speed, moisture, temperature)                                                                                    | OBSERVE                      |             |
| 4. WPS followed                                                                                                                                    | OBSERVE                      |             |
| METAL DECK INSPECTION <u>AFTER</u> WELDING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.3, SDI QA/QC-2022, Appendix 1, Table 1.5  |                              |             |
| TASK                                                                                                                                               | INSPECTION TYPE              | DESCRIPTION |
| 5. Verify size and location of welds, including support, sidelap, and perimeter welds.                                                             | PERFORM                      |             |
| 6. Verify welds meet visual acceptance criteria                                                                                                    | PERFORM                      |             |
| 7. Verify repair activities                                                                                                                        | PERFORM                      |             |
| 8. Acceptance or rejection of welds                                                                                                                | DOCUMENT                     |             |

END SECTION

<sup>1</sup>**PERFORM:** Perform these tasks for each weld, fastener, or bolted connection, and required verification.  
**OBSERVE:** Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor’s risk.  
**DOCUMENT:** Document in a report that the work has been performed as required. This is in addition to all other required reports.



**I. STRUCTURAL COLD-FORMED METAL DECK – FASTENING****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| METAL DECK INSPECTION <u>BEFORE</u> MECHANICAL FASTENING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.3, SDI QA/QC-2022, Appendix 1, Table 1.6 |                              |                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                                                                            | INSPECTION TYPE              | DESCRIPTION                                                                                                                                                                                                              |
| 1. Manufacturer installation instructions available for mechanical fasteners                                                                                    | OBSERVE                      |                                                                                                                                                                                                                          |
| 2. Proper tools available for fastener installation                                                                                                             | OBSERVE                      |                                                                                                                                                                                                                          |
| 3. Proper storage for mechanical fasteners                                                                                                                      | OBSERVE                      | ✓ Fasteners are located away from moisture and in a protected location, such as off the ground and away from excessive handling.<br>✓ Uncoated fasteners are stored in airtight containers to reduce corrosion and rust. |
| METAL DECK INSPECTION <u>DURING</u> MECHANICAL FASTENING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.3, SDI QA/QC-2022, Appendix 1, Table 1.7 |                              |                                                                                                                                                                                                                          |
| TASK                                                                                                                                                            | INSPECTION TYPE              | DESCRIPTION                                                                                                                                                                                                              |
| 4. Fasteners are positioned as required                                                                                                                         | OBSERVE                      |                                                                                                                                                                                                                          |
| 5. Fasteners are installed in accordance with manufacturer's instructions                                                                                       | OBSERVE                      |                                                                                                                                                                                                                          |
| METAL DECK INSPECTION <u>AFTER</u> MECHANICAL FASTENING – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.3, SDI QA/QC-2022, Appendix 1, Table 1.8  |                              |                                                                                                                                                                                                                          |
| TASK                                                                                                                                                            | INSPECTION TYPE <sup>1</sup> | DESCRIPTION                                                                                                                                                                                                              |
| 6. Check spacing, type, and installation of support fasteners                                                                                                   | <b>PERFORM</b>               |                                                                                                                                                                                                                          |
| 7. Check spacing, type, and installation of side-lap fasteners                                                                                                  | <b>PERFORM</b>               |                                                                                                                                                                                                                          |
| 8. Check spacing, type, and installation of perimeter fasteners                                                                                                 | <b>PERFORM</b>               |                                                                                                                                                                                                                          |
| 9. Verify repair activities                                                                                                                                     | <b>PERFORM</b>               |                                                                                                                                                                                                                          |
| 10. Acceptance or rejection of mechanical fasteners                                                                                                             | <b>DOCUMENT</b>              |                                                                                                                                                                                                                          |

**END SECTION**

<sup>1</sup> **PERFORM:** Perform these tasks for each weld, fastener, or bolted connection, and required verification.  
**OBSERVE:** Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.  
**DOCUMENT:** Document in a report that the work has been performed as required. This is in addition to all other required reports.

**J. STRUCTURAL COLD-FORMED STEEL FRAMING AND/OR COLD-FORMED TRUSSES****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| COLD-FORMED STEEL CONSTRUCTION AND CONNECTIONS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.2.5, 1705.12.2, 1705.12.3, 1705.13.3, 1705.13.9, UFC 4 023 03 |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                                                                                     | INSPECTION TYPE <sup>1</sup> | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 1. Verify compliance of trusses spanning 60-feet or greater where/if applicable                                                                                          | <b>PERFORM</b>               | Verify that temporary and permanent truss restraint/bracing is installed in accordance with approved truss submittal package.                                                                                                                                                                                                                                                                                                                                                                  |
| 2. Welded connections (seismic and/or wind resisting system)                                                                                                             | OBSERVE                      | Visually inspect all welds that are part of the main wind or seismic force resisting system, including shear walls, braces, collectors (drag struts), and hold-downs.<br><b>[NOTE: DOR must identify critical wind and/or seismic force resisting welds in the contract drawings so that the special inspector can confirm compliance.]</b>                                                                                                                                                    |
| 3. Connections (seismic and/or wind resisting system)                                                                                                                    | OBSERVE                      | Visually inspect all screw attachment, bolting, anchoring and other fastening of components within the main wind or seismic force resisting system, including roof deck, roof framing, exterior wall covering, wall to roof/floor connections, braces, collectors (drag struts) and hold-downs.<br><b>[NOTE: DOR must identify critical wind and/or seismic force resisting connection/fastener components in the contract drawings so that the special inspector can confirm compliance.]</b> |
| 4. Cold-formed steel (progressive collapse resisting system where/if applicable)                                                                                         | OBSERVE                      | Verify proper welding operations, screw attachment, bolting, anchoring and other fastening of components within the progressive collapse resisting system, including horizontal tie force elements, vertical tie force elements and bridging elements (UFC 4 023 03).<br><b>[NOTE: DOR must identify critical progressive collapse resisting connection/fastener components in the contract drawings so that the special inspector can confirm compliance.]</b>                                |

**END SECTION****K. STRUCTURAL OPEN-WEB STEEL JOISTS****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| OPEN-WEB STEEL JOISTS AND JOIST GIRDERS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC TABLE 1705.2.4, 2024 IBC 2207.1, SJI 100 OR 200 |                              |                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                | INSPECTION TYPE <sup>1</sup> | DESCRIPTION                                                                                                                                                                                                |
| 1. Installation of open-web steel joists and joist girders                                                                                     | OBSERVE                      | <ul style="list-style-type: none"> <li>✓ End connections – welded or bolted for compliance with SJI 100 or 200</li> <li>✓ Bridging – horizontal and diagonal for compliance with SJI 100 or 200</li> </ul> |

**END SECTION**

<sup>1</sup> **PERFORM:** Perform these tasks for each weld, fastener, or bolted connection, and required verification.

**OBSERVE:** Observe these items on a random sampling basis daily to ensure that applicable requirements are met. Operations need not be delayed pending these inspections at contractor's risk.

**L. STRUCTURAL CONCRETE CONSTRUCTION****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐**

| CONCRETE CONSTRUCTION, INCLUDING COMPOSITE DECK – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC TABLE 1705.3 (ACI 318-19 REFERENCES NOTED IN IBC TABLE) |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                                                                           | INSPECTION TYPE <sup>1</sup> | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1. Inspect reinforcement, including prestressing tendons, and verify placement.                                                                                | PERIODIC                     | Verify prior to placing concrete that reinforcement is of specified type, grade and size; that it is free of oil, dirt and unacceptable rust; that it is located and spaced properly; that specified cover has been provided; that hooks, bends, ties, stirrups and supplemental reinforcement are placed correctly; that specified lap lengths, stagger and offsets are provided; and that all mechanical connections are installed per the manufacturer's instructions and/or evaluation report. |
| 2. Inspect reinforcing bar welding                                                                                                                             | PERIODIC                     | <ul style="list-style-type: none"> <li>✓ Verify weldability of reinforcing bars other than ASTM A 706</li> <li>✓ Inspect single-pass fillet welds, maximum 5/16" in accordance with AWS D1.4</li> </ul>                                                                                                                                                                                                                                                                                            |
|                                                                                                                                                                | CONTINUOUS                   | <ul style="list-style-type: none"> <li>✓ Inspect welding of reinforcement for special moment frame, boundary elements of special shear walls, and coupling beams.</li> <li>✓ Inspect welded reinforcement splices.</li> <li>✓ Inspect welding of primary tension reinforcement in corbels.</li> </ul>                                                                                                                                                                                              |
|                                                                                                                                                                | PERIODIC                     | <ul style="list-style-type: none"> <li>✓ Inspect all other welds in accordance with AWS D1.4</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                            |
| 3. Inspect cast-in-place anchors, post-installed drilled anchors, and adhesive anchors other than those in item 4 below                                        | PERIODIC                     | <ul style="list-style-type: none"> <li>✓ Verify prior to placing concrete that cast in place anchors and post installed drilled anchors have proper embedment, spacing and edge distance.</li> <li>✓ Inspection requirements for adhesive anchors are different from those for other post-installed anchors and require assessment and qualification under the provisions of ACI 355.4 which may require proof loading.</li> </ul>                                                                 |
| 4. Inspect post-installed adhesive anchors in horizontal or upward inclined orientations and                                                                   | CONTINUOUS AND DOCUMENT      | <ul style="list-style-type: none"> <li>✓ Inspect as required per approved ICC-ES report</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                 |

<sup>1</sup> **PERIODIC:** Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.

**DOCUMENT:** Document in a report that the work has been performed as required. This is in addition to all other required reports.

**CONTINUOUS:** Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

| CONCRETE CONSTRUCTION, INCLUDING COMPOSITE DECK – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC TABLE 1705.3 (ACI 318-19 REFERENCES NOTED IN IBC TABLE)                                                  |                              |                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                                                                                                                            | INSPECTION TYPE <sup>1</sup> | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                  |
| subject to sustained tension loading.                                                                                                                                                                           |                              | <ul style="list-style-type: none"> <li>✓ Verify that installer is certified for installation of horizontal and overhead installation applications</li> <li>✓ Verify proof loading as required by the contract documents</li> </ul>                                                                                                                           |
| 5. Verify use of required mix design                                                                                                                                                                            | PERIODIC                     | Verify that all mixes used comply with the approved construction documents                                                                                                                                                                                                                                                                                   |
| 6. Prior to concrete placement, fabricate specimens for strength tests, perform slump and air content tests, and determine the temperature of the concrete                                                      | CONTINUOUS                   | At the time fresh concrete is sampled to fabricate specimens for strength test, verify these tests are performed by qualified technicians.                                                                                                                                                                                                                   |
| 7. Inspect concrete and/or shotcrete placement for proper application techniques                                                                                                                                | CONTINUOUS                   | Verify proper application techniques are used during concrete conveyance and that depositing avoids segregation or contamination. Verify that concrete is properly consolidated.                                                                                                                                                                             |
| 8. Verify maintenance of specified curing temperature and technique                                                                                                                                             | PERIODIC                     | Inspect curing and cold weather, or hot weather protection procedures.                                                                                                                                                                                                                                                                                       |
| 9. Inspect prestressed concrete                                                                                                                                                                                 | CONTINUOUS                   | Verify application of prestressing forces and grouting of bonded prestressing tendons.                                                                                                                                                                                                                                                                       |
| 10. Inspect erection of precast concrete members                                                                                                                                                                | PERIODIC                     | <ul style="list-style-type: none"> <li>✓ Verify dimensional tolerances for precast members and interfacing members.</li> <li>✓ Verify details of lifting devices, embedments, and related reinforcement required to resist temporary loads from handling, storage, transportation, and erection, if designed by the licensed design professional.</li> </ul> |
| 11. Inspect precast concrete diaphragm connections or reinforcement at joints classified as moderate or high deformability element (MDE or HDE) in structures assigned to Seismic Design Category C, D, E, or F | CONTINUOUS                   | Inspect such connections and reinforcement in the field for: <ul style="list-style-type: none"> <li>✓ Installation of the embedded parts</li> <li>✓ Completion of the continuity of reinforcement across joints.</li> <li>✓ Completion of connections in the field</li> </ul>                                                                                |
| 12. Inspect installation tolerances of precast concrete diaphragm connections for compliance with ACI 550.5                                                                                                     | PERIODIC                     |                                                                                                                                                                                                                                                                                                                                                              |

| CONCRETE CONSTRUCTION, INCLUDING COMPOSITE DECK – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC TABLE 1705.3 (ACI 318-19 REFERENCES NOTED IN IBC TABLE)           |                              |                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                                                                                     | INSPECTION TYPE <sup>1</sup> | DESCRIPTION                                                                                                                                                                    |
| 13. Verify in-situ concrete strength, prior to stressing of tendons in post-tensioned concrete and prior to removal of shores and forms from beams and structural slabs. | PERIODIC                     | See ACI 318 26.11.2                                                                                                                                                            |
| 14. Inspect formwork for shape, location and dimensions of the concrete member being formed.                                                                             | PERIODIC                     | Formwork fabrication and installation shall result in final structure that conforms to shapes, lines, and dimensions of the members as required by the construction documents. |

END SECTION

**M. STRUCTURAL - MASONRY CONSTRUCTION (ALL RISK CATEGORIES)****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐**

| MASONRY CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH <u>AT START</u> OF CONSTRUCTION<br>2024 IBC 1705.4 (TMS 402/602-22 TABLE 3 & 4)                                             |                              |                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                                                                                                           | INSPECTION TYPE <sup>1</sup> | DESCRIPTION                                                                                                                                      |
| 1. Verify compliance with approved submittals prior to start                                                                                                                                   | PERIODIC                     |                                                                                                                                                  |
| 2. Verify proportions of site-mixed mortar.                                                                                                                                                    | PERIODIC                     |                                                                                                                                                  |
| 3. Verify grade, type and size of reinforcement, connectors, and anchor bolts                                                                                                                  | PERIODIC                     |                                                                                                                                                  |
| 4. Verify sample panel construction                                                                                                                                                            | PERIODIC<br>CONTINUOUS       | [NOTE: DOR must either delete 'PERIODIC' for Risk Category IV/V, or delete 'CONTINUOUS' for Risk Categories I/II/ III]                           |
| MASONRY CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH <u>PRIOR TO</u> GROUTING<br>IBC 1705.4 (TMS 402/602-22 TABLE 4)                                                             |                              |                                                                                                                                                  |
| TASK                                                                                                                                                                                           | INSPECTION TYPE              | DESCRIPTION                                                                                                                                      |
| 5. Verify grout space                                                                                                                                                                          | PERIODIC<br>CONTINUOUS       | [NOTE: DOR must either delete 'PERIODIC' for Risk Category IV/V, or delete 'CONTINUOUS' for Risk Categories I/II/ III]                           |
| 6. Verify placement of reinforcement, connectors, and anchor bolts                                                                                                                             | PERIODIC<br>CONTINUOUS       | [NOTE: DOR must either delete 'PERIODIC' for Risk Category IV/V, or delete 'CONTINUOUS' for Risk Categories I/II/ III]                           |
| 7. Verify proportions of site-prepared grout                                                                                                                                                   | PERIODIC                     |                                                                                                                                                  |
| MASONRY CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH <u>DURING</u> CONSTRUCTION<br>IBC 1705.4 (ACI 530-16 TABLE 4)                                                               |                              |                                                                                                                                                  |
| TASK                                                                                                                                                                                           | INSPECTION TYPE              | DESCRIPTION                                                                                                                                      |
| 8. Verify compliance of materials and procedures with the approved submittals                                                                                                                  | PERIODIC                     |                                                                                                                                                  |
| 9. Verify placement of masonry units and mortar joint construction                                                                                                                             | PERIODIC                     |                                                                                                                                                  |
| 10. Verify size and location of structural elements                                                                                                                                            | PERIODIC                     |                                                                                                                                                  |
| 11. Verify type, size and placement of reinforcement, connectors, anchor bolts, and anchorages, including details of anchorage of masonry to structural members, frames, or other construction | PERIODIC<br>CONTINUOUS       | [NOTE: DOR must either delete 'PERIODIC' for Risk Category IV/V, or delete 'CONTINUOUS' for Risk Categories I/II/III]                            |
| 12. Verify type, size and location of veneer ties and movement joints                                                                                                                          | PERIODIC                     | Periodic inspection of veneers is required when the height of the veneer exceeds 60 ft (18.3 m) above grade plane. See Commentary in Appendix A. |

<sup>1</sup> **PERIODIC:** Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.

**CONTINUOUS:** Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

| MASONRY CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH <u>DURING</u> CONSTRUCTION<br>IBC 1705.4 (ACI 530-16 TABLE 4)                              |                               |                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                                                                          | INSPECTION TYPE               | DESCRIPTION                                                                                                                  |
| 13. Verify installation of adhered veneer                                                                                                                     | PERIODIC                      | Periodic inspection of veneers is required when the height of the veneer exceeds 60 ft (18.3 m) above grade plane            |
| 14. Verify welding of reinforcement                                                                                                                           | <b>CONTINUOUS</b>             |                                                                                                                              |
| 15. Verify preparation, construction, and protection of masonry during cold weather (temperature below 40°F (4.4°C) or hot weather (temp above 90°F (32.2°C)) | PERIODIC                      |                                                                                                                              |
| 16. Verify placement of grout                                                                                                                                 | <b>CONTINUOUS</b>             |                                                                                                                              |
| 17. Observe preparation of grout specimens, mortar specimens, and/or prisms                                                                                   | PERIODIC<br><b>CONTINUOUS</b> | <b>[NOTE: DOR must either delete 'PERIODIC' for Risk Category IV/V, or delete 'CONTINUOUS' for Risk Categories I/II/III]</b> |

**END SECTION**

**N. STRUCTURAL – WOOD CONSTRUCTION – SPECIALTY ITEMS****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐**

| WOOD CONSTRUCTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.5                                                |                              |                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                                           | INSPECTION TYPE <sup>1</sup> | DESCRIPTION                                                                                                                                                                                                                                                       |
| 1. High-load diaphragms where applicable                                                                                       | PERIODIC                     | Verify thickness and grade of sheathing, size of framing members at panel edges, nail diameters and length, and the number of fastener lines and that fastener spacing is per approved contract documents.                                                        |
| 2. Metal-plate connected wood trusses spanning 60 feet or greater                                                              | PERIODIC                     | Verify that the temporary installation restraint/bracing and the permanent individual truss member restraint/bracing are installed in accordance with the approved truss submittal package                                                                        |
| Mass timber elements in Type IV-A, IV-B, and IV-C construction. Refer to 2024 IBC 602.4 for details about Type IV construction |                              |                                                                                                                                                                                                                                                                   |
| 3. Inspect anchorage and connections of mass timber construction to timber deep foundation systems                             | PERIODIC                     |                                                                                                                                                                                                                                                                   |
| 4. Inspect erection of mass timber construction                                                                                | PERIODIC                     |                                                                                                                                                                                                                                                                   |
| 5. Inspect connections where installation methods are required to meet design loads                                            |                              |                                                                                                                                                                                                                                                                   |
| a) Threaded fasteners                                                                                                          | PERIODIC                     | <ul style="list-style-type: none"> <li>✓ Verify use of proper installation equipment</li> <li>✓ Verify use of pre-drilled holes where required</li> <li>✓ Inspect screws, including diameter, length, head type, spacing, installation angle and depth</li> </ul> |
| b) Adhesive anchors connections installed in horizontal or upwardly inclined orientation to resist sustained tension loads.    | CONTINUOUS                   |                                                                                                                                                                                                                                                                   |
| c) Adhesive anchors not defined in preceding cell                                                                              | PERIODIC                     |                                                                                                                                                                                                                                                                   |
| d) Bolted connections                                                                                                          | PERIODIC                     |                                                                                                                                                                                                                                                                   |
| e) Concealed connections                                                                                                       | PERIODIC                     |                                                                                                                                                                                                                                                                   |

**END SECTION**

<sup>1</sup> **PERIODIC:** Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.

**CONTINUOUS:** Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.



O. STRUCTURAL WOOD CONSTRUCTION - SEISMIC & WIND

THIS SECTION IS APPLICABLE IF BOX IS CHECKED: ☐

| WOOD CONSTRUCTION SEISMIC AND WIND – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.12 & 1705.13.2                                                                                                                                                                                                                                           |                              |                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                                                                                                                                                                                                                                                                    | INSPECTION TYPE <sup>1</sup> | DESCRIPTION                                                                                                                                                              |
| <b>[NOTE: DOR may uncheck this section where the lateral resistance is provided by structural sheathing, the sheathing nailing/fasteners (shear walls, shear panels and diaphragms) are consistently greater than 4” on center, or if the design wind speed (ASD) is less than 140 mph (62.6 meters/sec) AND the seismic design category is A or B]</b> |                              |                                                                                                                                                                          |
| 1. Nailing, bolting, anchoring and other fastening of elements of the main wind/seismic force-resisting system                                                                                                                                                                                                                                          | PERIODIC                     | Includes connectors for: shear wall sheathing, roof/floor sheathing, drag struts/collectors (double top plates), braces, hold downs, roof connections to exterior walls. |
| 2. Field gluing operations of elements of the main wind/seismic force-resisting system                                                                                                                                                                                                                                                                  | CONTINUOUS                   |                                                                                                                                                                          |

END SECTION

P. STRUCTURAL ISOLATION AND ENERGY DISSIPATION SYSTEMS

ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED: ☐

| ISOLATION AND ENERGY DISSIPATION SYSTEMS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.13.8, 1705.14.4             |                              |                                                                                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>[NOTE: This section is <u>not</u> applicable to Seismic Design Category A. Uncheck this section if this category applies]</b> |                              |                                                                                                                                                                         |
| TASK                                                                                                                             | INSPECTION TYPE <sup>2</sup> | DESCRIPTION                                                                                                                                                             |
| 1. Verify fabrication and installation of isolator units and energy dissipation devices                                          | PERFORM                      | Verify that fabrication and installation of isolator units and energy dissipation devices conform to manufacturer’s recommendations and approved construction documents |
| 2. Test of seismic isolation systems in seismically isolated structures                                                          | PERFORM                      | Seismic isolation systems in seismically isolated structures shall be tested in accordance with ASCE 7, Section 17.8                                                    |

END SECTION

<sup>1</sup> **PERIODIC:** Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.

**CONTINUOUS:** Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

<sup>2</sup> **PERFORM:** Perform these tasks for each weld, fastener, or bolted connection, and required verification.

**Q. GEOTECHNICAL - SOILS INSPECTION****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| SOILS INSPECTION – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.6 and TABLE 1705.6                                                                                                                                        |                              |             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------|
| TASK                                                                                                                                                                                                                                   | INSPECTION TYPE <sup>1</sup> | DESCRIPTION |
| 1. Verify materials below shallow foundations are adequate to achieve the design bearing capacity.                                                                                                                                     | PERIODIC                     |             |
| 2. Verify excavations are extended to proper depth and have reached proper material                                                                                                                                                    | PERIODIC                     |             |
| 3. Perform classification and testing of compacted fill materials                                                                                                                                                                      | PERIODIC                     |             |
| 4. Prior to placement of compacted fill, inspect subgrade and verify that site has been prepared properly                                                                                                                              | PERIODIC                     |             |
| 5. During fill placement, verify use of proper materials and procedures in accordance with the provisions of the approved geotechnical report. Verify densities and lift thicknesses during placement and compaction of compacted fill | <b>CONTINUOUS</b>            |             |

**END SECTION****R. GEOTECHNICAL - DRIVEN DEEP FOUNDATION ELEMENTS****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| DRIVEN DEEP FOUNDATION ELEMENTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.7 and TABLE 1705.7                                                                                                                                                         |                   |                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------------------------------------------|
| TASK                                                                                                                                                                                                                                                                   | INSPECTION TYPE   | DESCRIPTION                                         |
| 1. Verify element materials, sizes and lengths comply with requirements                                                                                                                                                                                                | <b>CONTINUOUS</b> |                                                     |
| 2. Inspect driving operations and maintain complete and accurate records for each element                                                                                                                                                                              | <b>CONTINUOUS</b> |                                                     |
| 3. Verify placement locations and plumbness, confirm type and size of hammer, record number of blows per foot of penetration, determine required penetrations to achieve design capacity, record tip and butt elevations and document any damage to foundation element | <b>CONTINUOUS</b> |                                                     |
| 4. Determine capacities of test elements and conduct additional load tests if required.                                                                                                                                                                                | <b>CONTINUOUS</b> |                                                     |
| 5. For steel, concrete, concrete-filled, perform additional special inspections in accordance with the steel and concrete sections in this schedule                                                                                                                    |                   |                                                     |
| 6. For specialty elements, perform additional inspections as determined by the licensed design professional in responsible charge                                                                                                                                      |                   | In accordance with Statement of Special Inspections |

**END SECTION**

<sup>1</sup> **PERIODIC:** Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.

**CONTINUOUS:** Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

**S. GEOTECHNICAL - HELICAL PILE FOUNDATIONS****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| HELICAL PILE FOUNDATIONS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.9                                                                                                                                                                                                                                                 |                   |             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------|
| TASK                                                                                                                                                                                                                                                                                                                                   | INSPECTION TYPE   | DESCRIPTION |
| 1. Record installation equipment used, pile dimensions, tip elevations, final depth, final installation torque and other pertinent installation data as required. The approved geotechnical report and the contract documents prepared by the licensed design professional in responsible charge shall be used to determine compliance | <b>CONTINUOUS</b> |             |
| 2. Determine capacities of test elements and conduct additional load tests if required.                                                                                                                                                                                                                                                | <b>CONTINUOUS</b> |             |

**END SECTION****T. GEOTECHNICAL – CAST-IN-PLACE CONCRETE DEEP FOUNDATION ELEMENTS****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| CAST-IN-PLACE CONCRETE DEEP FOUNDATION ELEMENTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.8                                                                                                                      |                              |             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------|
| TASK                                                                                                                                                                                                                               | INSPECTION TYPE <sup>1</sup> | DESCRIPTION |
| 1. Inspect drilling operations and maintain complete and accurate records for each element                                                                                                                                         | <b>CONTINUOUS</b>            |             |
| 2. Verify placement locations and plumbness, confirm element diameters, bell diameters (if applicable), lengths, embedment into bedrock (if applicable) and adequate end-bearing strata capacity. Record concrete or grout volumes | <b>CONTINUOUS</b>            |             |
| 3. For concrete elements, perform tests and additional special inspections in accordance with Concrete section of this schedule                                                                                                    | <b>CONTINUOUS</b>            |             |
| 4. Determine capacities of test elements and conduct additional load tests if required.                                                                                                                                            | <b>CONTINUOUS</b>            |             |

**END SECTION**

<sup>1</sup> **CONTINUOUS:** Constant monitoring of identified tasks by a special inspector over the duration of performance of said tasks.

**U. FIRE PROTECTION - SPRAYED FIRE-RESISTANT MATERIALS****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| SPRAYED FIRE RESISTANT MATERIALS (SFRM) – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.15 |                              |                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                   | INSPECTION TYPE <sup>1</sup> | DESCRIPTION                                                                                                                                                                                            |
| 1. Check substrate condition                                                                           | PERIODIC                     | Prior to application, confirm that surfaces have been prepared according to the approved fire-resistant design and manufacturer's instructions. See 2024 IBC 1705.15.2 and 1705.15.3 for more details. |
| 2. Verify material thickness                                                                           | PERIODIC                     | Verify SFRM thickness according to 2024 IBC 1705.15.4                                                                                                                                                  |
| 3. Verify material density                                                                             | PERIODIC                     | Verify SFRM density according to 2024 IBC 1705.15.5                                                                                                                                                    |
| 4. Verify bond strength                                                                                | PERIODIC                     | Verify bond strength of cured SFRM according to 2024 IBC 1705.15.6                                                                                                                                     |
| 5. Inspect the condition of finished application                                                       | PERIODIC                     |                                                                                                                                                                                                        |

**END SECTION****V. FIRE PROTECTION - INTUMESCENT FIRE-RESISTIVE MATERIALS****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| INTUMESCENT FIRE-RESISTIVE MATERIALS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.16 |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                | INSPECTION TYPE | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 1. Inspect according to AWCI 12-B and the contract documents                                        | PERIODIC        | <p>Inspections shall be performed in accordance with AWCI 12-B, Standard Practice for the Testing and Inspection of Field Applied Thin Film Intumescent Fire-Resistive Materials.</p> <p>Inspections shall be performed during construction. Additional visual inspection shall be performed after rough installation and where applicable, prior to the concealment of electrical, automatic sprinkler, mechanical and plumbing systems.</p> |

**END SECTION****W. FIRE PROTECTION – FIRE-RESISTANT PENETRATIONS AND JOINTS****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| FIRE-RESISTANT PENETRATIONS AND JOINTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.18 |                 |                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                  | INSPECTION TYPE | DESCRIPTION                                                                                                                                                                                                                                                                                                                                               |
| 1. Inspect of penetration firestop systems conducted in accordance with ASTM E2174.                   | PERIODIC        | <p><b>[NOTE: This section applies to Risk Category III, IV, &amp; V structures, or in fire areas containing Group R occupancies with an occupant load greater than 250. DOR may choose to uncheck this section where project is assigned to Risk Category I or II. Confirm Risk Category with licensed design professional in responsible charge]</b></p> |
| 2. Inspect of fire-resistant joint systems conducted in accordance with ASTM E2393                    | PERIODIC        |                                                                                                                                                                                                                                                                                                                                                           |

**END SECTION**

<sup>1</sup> **PERIODIC:** Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.

**X. FIRE PROTECTION – SMOKE CONTROL**

**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| SMOKE CONTROL – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.19                     |                              |                                                              |
|--------------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------|
| TASK                                                                                             | INSPECTION TYPE <sup>1</sup> | DESCRIPTION                                                  |
| 1. Verify device locations and perform leakage testing                                           | PERIODIC                     | Perform during erection of ductwork and prior to concealment |
| 2. Perform pressure difference testing, flow measurements and detection and control verification | PERIODIC                     | Perform prior to occupancy and after sufficient completion   |

**END SECTION**

**Y. SEALANT – SEALING OF MASS TIMBER**

**ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| SEALING OF MASS TIMBER – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.20 |                 |                                                                                         |
|---------------------------------------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------------|
| TASK                                                                                  | INSPECTION TYPE | DESCRIPTION                                                                             |
| 1. Inspect sealants or adhesives                                                      | PERIODIC        | Inspect sealants or adhesives where sealant and adhesive are required by 2024 IBC 703.7 |

**END SECTION**

<sup>1</sup> **PERIODIC:** Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.

**Z. ARCHITECTURAL - EXTERIOR INSULATION AND FINISH SYSTEMS****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| EXTERIOR INSULATION AND FINISH SYSTEMS (EIFS) – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.17 |                              |                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                         | INSPECTION TYPE <sup>1</sup> | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                     |
| 1. Water-resistive barrier coating applied over a sheathing substrate.                                       | PERIODIC                     | <p>Verify that water-resistive barrier coating complies with ASTM E2570.</p> <p><b>[NOTE: Not required for EIFS applications installed over:</b></p> <ul style="list-style-type: none"> <li>a) a water-resistive barrier with a means of draining moisture to the exterior.</li> <li>b) masonry or concrete walls</li> </ul> <p><b>Uncheck this section in those cases]</b></p> |

**END SECTION****AA.Architectural – Architectural Components****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| ARCHITECTURAL COMPONENTS – VERIFY THE FOLLOWING ARE IN COMPLIANCE WITH 2024 IBC 1705.13.5, 1705.13.7 and TABLE 1705.13.7                                                                                                                        |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TASK                                                                                                                                                                                                                                            | INSPECTION TYPE | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>[NOTE: This section is not applicable to Seismic Design Categories A, B, &amp; C. Uncheck this section if one of those categories applies. Confirm Seismic Design Category with the licensed design professional in responsible charge.]</b> |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1. Erection and fastening of exterior cladding and interior and exterior veneer.                                                                                                                                                                | PERIODIC        | Verify appropriate materials, fasteners and attachment at commencement of work and at completion. <b>[NOTE: Inspection not required if height is less than 30 feet or weight is less than 5psf]</b>                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2. Interior and exterior non-load-bearing walls                                                                                                                                                                                                 | PERIODIC        | Verify appropriate materials, fasteners and attachment at commencement of work and at completion. <b>[NOTE: Inspection not required if interior non-load-bearing walls weigh less than 15psf]</b>                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 3. Access floors                                                                                                                                                                                                                                | PERIODIC        | Verify that anchorage complies with approved construction documents.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4. Storage racks                                                                                                                                                                                                                                | PERIODIC        | <p>Steel storage racks and steel cantilevered storage racks that are 8 feet in height or greater shall be inspected for:</p> <ul style="list-style-type: none"> <li>✓ Materials used, to verify compliance with one or more of the material test reports in accordance with the approved construction documents</li> <li>✓ Fabricated storage rack elements</li> <li>✓ Storage rack anchorage installation</li> <li>✓ Completion of storage rack system</li> </ul> <p>Inspection of post-installed anchors shall comply with approved ICC-ES report.</p> <p><b>[NOTE: Not required for racks less than 8 feet in height]</b></p> |

**END SECTION**

<sup>1</sup> **PERIODIC:** Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.

**BB. PLUMBING/MECHANICAL/ELECTRICAL - DESIGNATED SEISMIC SYSTEMS****ALL OR PORTIONS OF THIS SECTION ARE APPLICABLE IF BOX IS CHECKED:** ☐

| PLUMBING, MECHANICAL AND ELECTRICAL – DESIGNATED SEISMIC SYSTEMS<br>2024 IBC 1705.13.6                                                                                                                                               |                              |                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------------------------------------------------------------------------------------------|
| TASK                                                                                                                                                                                                                                 | INSPECTION TYPE <sup>1</sup> | DESCRIPTION                                                                                         |
| <b>[NOTE: This section is not applicable to Seismic Design Categories A or B. Uncheck this section if one of those categories applies. Confirm Seismic Design Category with licensed design professional in responsible charge.]</b> |                              |                                                                                                     |
| 1. Check anchorage of electrical equipment for emergency and standby power systems                                                                                                                                                   | PERIODIC                     | ✓ Check for general conformance                                                                     |
| 2. Check anchorage of all other electrical equipment in <a href="#">Seismic Design Categories E and F only</a> (See first page of this schedule for Seismic Design Category)                                                         | PERIODIC                     | ✓ Check for general conformance                                                                     |
| 3. Check installation and anchorage of piping designed to carry hazardous materials and their associated mechanical units.                                                                                                           | PERIODIC                     | ✓ Check for general conformance                                                                     |
| 4. Check installation and anchorage of vibration isolation systems where the construction documents require a nominal clearance of ¼" or less between support framing and restraint.                                                 | PERIODIC                     | ✓ Check for general conformance                                                                     |
| 5. Verify clearance between fire sprinkler piping and surrounding mechanical and electrical equipment, including ductwork, piping and their structural supports.                                                                     | PERIODIC                     | ✓ Check for minimum clearances noted in ASCE 7 13.2.3 or a nominal clearance of not less than 3 in. |

**END SECTION**

<sup>1</sup> **PERIODIC:** Special inspection by a special inspector who is intermittently present where the work to be inspected has been or is being performed.

## APPENDIX A

### **Commentary on M12**

Table 4 of TMS 602-22 does not have any Level 3 inspection requirements (for Risk Category IV structures). The leadership of the standard committee was contacted. They indicated that there are no requirements because veneers are covered under Part 4: Prescriptive Design Methods, and for prescriptive design there is never any Level 3 inspection requirement according to Table 3.1 of TMS 402-22. Although veneer is covered in Part 4, Chapter 13 of TMS 402-22, in Section 13.2.3, there is Engineered Design of Anchored Masonry Veneer and in Section 13.3.3, there is Engineered Design of Adhered Masonry Veneer. Engineered Design is covered in Part 3 of the standard, and for Part 3, Engineered Design, there are Level 3 inspection requirements in Table 3.1 of TMS 402-22. Upon further discussion, the leadership of the standard committee suggested to keep inspection Periodic for all Risk Category structures.



Project: P234 - Maintenance Hangar  
 Location: Anytown, VA  
 Project #: 12345678  
 Date: 3/1/2020



## **STATEMENT OF SPECIAL INSPECTIONS**

Seismic Design Category: E

Risk Category: III

Number of Stories above Grade: 1

Height of Highest Occupied Roof or Floor: 75

Hazardous Occupancy or attached to such? No Group H Occupancies

### **Special Inspector of Record (SIOR)**

A Special Inspector of Record (SIOR) IS required (per UFGS 01 45 35, Section 1.3.7)

SIOR Name (Registered Professional): John Doe P.E.

Professional Registration Number: C222222

Consulting Firm Name (if any): ABC Structural Consulting

SIOR Office AND Mobile Phone Number: 1234567895 (Mobile), 123456789 (Office)

### **Lateral Force Resisting System (LFRS)**

2024 IBC 1704.3.2 and 1704.3.3

Following is a listing of critical elements of the main wind/seismic force-resisting system for this structure. Carefully inspect these elements as part of the roles and responsibilities of the Special Inspector (reference the Schedule of Special Inspections for inspection checklists).

| <b>Vertical LFRS Elements</b>                         | <b>Notes</b>                                      |
|-------------------------------------------------------|---------------------------------------------------|
| Steel Ordinary Concentrically Braced Frames           | North - South Direction Only, See Plan            |
| Steel Special Concentrically Braced Frames            | Lettered Gridline only, see plan for call out     |
| Steel Ordinary Moment Frames (SMF)                    | Both orthogonal Directions, See plan for call out |
| Ordinary Reinforced Concrete Shear Walls              | See Key Notes                                     |
| Concrete Grade Beams Resisting Lateral Loads          | Office Portion Only, see plan key notes           |
| Ordinary Reinforced Masonry Shear Walls               | High Bay, Numbered Gridlines only, see plan       |
| OSB Sheathed Shear Walls (nailing, sill bolting, Etc) | See schedule on plan                              |
| Wood Shear Wall Hold Downs                            | Identified on Plan & in Detail Sheet X.XX         |
| <b>Horizontal LFRS Elements</b>                       | <b>Notes</b>                                      |
| Continuous Roof Ties                                  | See Key Notes on Roof Plan                        |
| Collector Elements                                    | Identified on plan with key notes                 |
| Concrete topping over metal deck                      | 2nd floor and roof                                |
| Metal Roof Deck & Related Fastening System            | See Roof Plan                                     |
| Out-of-Plane Wall Connections                         | See Structural Details XX & XX                    |
| Diaphragm Cross Rod Bracings                          | See Roof Plan for locations                       |
| Cast-in-Place Concrete Floor and Roof Diaphragms      | See sheet XX for details and rebar schedule       |

|                                    |                            |
|------------------------------------|----------------------------|
| Special Force Transfer Connections | See Detail X on Sheet X.XX |
|------------------------------------|----------------------------|

Statement of Special Inspections Page 1 of 2

Project: P234 - Maintenance Hangar

Location: Anytown, VA

Project #: 12345678

Date: 3/1/2020

## Designated Seismic Systems (DSS)

(2024 IBC 1705.13.4) (ASCE 7-22, 13.2.3, C13.2.2) (UFC 3-301-1, 2-5.3)

### ELECTRICAL Components that are part of the Designated Seismic System (DSS) , which require a Certificate of Compliance

|    |                                       |
|----|---------------------------------------|
| 1. | DSS Emergency or Standby Power System |
| 2. | DSS Component XX                      |
| 3. | DSS Component XX                      |
| 4. | DSS Component XX                      |
| 5. | DSS Component XX                      |

If additional space is required, append an additional sheet listing the remaining DSS

### MECHANICAL/PLUMBING Components that are part of the Designated Seismic System (DSS), which require a Certificate of Compliance

|    |                                                                          |
|----|--------------------------------------------------------------------------|
| 1. | DSS Gas lines and associated fittings, anchorage, & flexible Connections |
| 2. | DSS Component XX                                                         |
| 3. | DSS Component XX                                                         |
| 4. | DSS Component XX                                                         |
| 5. | DSS Component XX                                                         |
| 6. | DSS Component XX                                                         |

If additional space is required, append an additional sheet listing the remaining DSS

### OTHER Designated Seismic Systems (DSS) requiring a Certificate of Compliance

|    |                                     |
|----|-------------------------------------|
| 1. | DSS Building egress stair systems   |
| 2. | DSS Building fire sprinkler systems |
| 3. | DSS Component XX                    |
| 4. | DSS Component XX                    |
| 5. | DSS Component XX                    |
| 6. | DSS Component XX                    |

## Final Walk-Down Inspection and Report

(UFC 3 301 01 SECTION 2-5.4)

Designated Seismic Systems shall receive a final walk-down inspection by the Registered Design Professional in Responsible Charge

Final Walk-Down Report, Prepared by the Registered Design Professional in Responsible Charge, must Include:

1. Record observations of Final Walk-Down Inspection
2. Document that Inspections were performed in accordance with the Schedule of Special Inspections
3. Document that all Designated Seismic Systems are installed according to construction/manufacture document requirements, and that Compliance Certificates have been collected (UFC 03 301 01, 3-5.2, 13.2.3.2).

SECTION 01 50 00.00 22

TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS (PWD ME)  
**11/20, CHG 3: 08/24**

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C511 (2017; R 2021) Reduced-Pressure Principle  
Backflow Prevention Assembly

FOUNDATION FOR CROSS-CONNECTION CONTROL AND HYDRAULIC RESEARCH  
(FCCCHR)

FCCCHR List (continuously updated) List of Approved  
Backflow Prevention Assemblies

FCCCHR Manual (10th Edition) Manual of Cross-Connection  
Control

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 241 (2022) Standard for Safeguarding  
Construction, Alteration, and Demolition  
Operations

NFPA 70 (2023) National Electrical Code

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety -- Safety and Occupational  
Health (SOH) Requirements

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-530-01 (2023; with Change 1, 2023) Interior and  
Exterior Lighting Systems

U.S. FEDERAL AVIATION ADMINISTRATION (FAA)

FAA AC 70/7460-1 (2016; Rev L; Change 2) Obstruction  
Marking and Lighting

U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)

MUTCD (2009; Rev 2012) Manual on Uniform Traffic  
Control Devices

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

### SD-01 Preconstruction Submittals

Construction Site Plan; G

Traffic control plan; G

Contractor Computer Cybersecurity Compliance Statements; G

Contractor Temporary Network Cybersecurity Compliance Statements; G

Employee Parking Plan; G

### SD-03 Product Data

Backflow Preventers; G

### SD-06 Test Reports

Backflow Preventer Tests; G

### SD-07 Certificates

Backflow Tester Certification; G

Backflow Preventers Certificate of Full Approval; G

## 1.3 CONSTRUCTION SITE PLAN

Prior to the start of work, submit for Government approval a site plan showing the locations and dimensions of temporary facilities (including layouts and details, equipment and material storage areas (onsite and offsite), access and haul routes, avenues of ingress/egress to the fenced area and details of fence installations).

Identify any areas which may have to be graveled to prevent the tracking of mud. Indicate if the use of a supplemental or other staging area is desired. Show locations of safety and construction fences, site trailers, construction entrances, trash dumpsters, temporary sanitary facilities, and worker parking areas.

Prior to the start of Construction, furnish and erect temporary project safety fencing at the work site in accordance with the plans and the following requirements:

- a. Temporary project fencing (or a substitute acceptable to the Contracting Officer (GDA) and delineated in the Accident Prevention Plan(APP)) must be provided on all projects. See also EM 385-1-1.
- b. Fencing must extend from grade to a minimum of 4 feet above grade and must have a maximum mesh size of 2 inches. Fencing must remain rigid/taut with a minimum of 200 pounds of force exerted on it from

any direction with less than 4 inches of deflection.

- c. Signs warning of the presence of construction hazards and requiring unauthorized persons to keep out of the construction area must be posted on the fencing. At a minimum, signs must be posted every 150 feet. Fenced sides of projects that are less than 150 feet must, at minimum, have at least one warning sign.
- d. Depending upon the nature and location of the project site, the Contractor may request not to install temporary fencing in some sections of the project site. The SSHO must submit a risk analysis (AHA) to the Contracting Officer for review. This must be based on a risk analysis of public exposure and other project specific considerations, and must be included in the applicable AHA.
- e. Access gates to the job site shall remain closed when not actively in use to prevent none essential personnel from entering the construction area.

If the Contracting Officer approves the request and has determined fencing is not required, install signs and other acceptable barrier systems warning of construction hazards. The signs must be conspicuously posted.

If at any time it is determined the risk to the public changes based on the work, take immediate action to address any risk to the public.

#### 1.4 BACKFLOW PREVENTERS CERTIFICATE

##### 1.4.1 Backflow Tester Certificate

Prior to testing, submit to the Contracting Officer certification issued by the State or local regulatory agency attesting that the backflow tester has successfully completed a certification course sponsored by the regulatory agency. Tester must not be affiliated with any company participating in other phases of this Contract.

##### 1.4.2 Backflow Prevention Training Certificate

Submit a certificate recognized by the State or local authority that states the Contractor has completed at least 10 hours of training in backflow preventer installations. The certificate must be current.

#### 1.5 DOD CONDITION OF READINESS (COR)

DOD will set the Condition of Readiness (COR) based on the weather forecast for sustained winds 50 knots (58 mph) or greater. Contact the Contracting Officer for the current COR setting.

Monitor weather conditions a minimum of twice a day and take appropriate actions according to the approved Emergency Plan in the accepted Accident Prevention Plan, EM 385-1-1 Emergency Plan requirements and the instructions below.

Unless otherwise directed by the Contracting Officer, comply with:

- a. Condition FOUR (Sustained winds of 58 mph or greater expected within 72 hours): Normal daily jobsite cleanup and good housekeeping practices. Collect and store in piles or containers scrap lumber, waste material, and rubbish for removal and disposal at the close of

each work day. Maintain the construction site including storage areas, free of accumulation of debris. Stack form lumber in neat piles less than 3 feet high. Remove all debris, trash, or objects that could become missile hazards. Review requirements pertaining to "Condition THREE" and continue action as necessary to attain "Condition FOUR" readiness. Contact Contracting Officer for weather and COR updates and completion of required actions.

- b. Condition THREE (Sustained winds of 58 mph or greater expected within 48 hours): Maintain "Condition FOUR" requirements and commence securing operations necessary for "Condition ONE" which cannot be completed within 18 hours. Cease all routine activities which might interfere with securing operations. Commence securing and stow all gear and portable equipment. Make preparations for securing buildings. Reinforce or remove formwork and scaffolding. Secure machinery, tools, equipment, materials, or remove from the jobsite. Expend every effort to clear all missile hazards and loose equipment from general Portsmouth Naval Shipyard areas. Contact Contracting Officer for weather and COR updates and completion of required actions. Review requirements pertaining to "Condition TWO" and continue action as necessary to attain "Condition THREE" readiness.
- c. Condition TWO (Sustained winds of 58 mph or greater expected within 24 hours): Secure the jobsite, and leave Government premises.
- d. Condition ONE (Sustained winds of 58 mph or greater expected within 12 hours): Contractor access to the jobsite and Government premises is prohibited.

## 1.6 CYBERSECURITY DURING CONSTRUCTION

The Contractor must meet the following requirements throughout the construction process.

### 1.6.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. When used, Contractor computers must meet the following requirements along with the requirements of paragraph SECURITY RESPONSIBILITIES (PNSY) of Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME).

Note: Any Computers (including tablet, laptop, or other computers) must not be introduced into nor removed from the Portsmouth Naval Shipyard. Obtain approvals from the PNSY Security Officer via the Contracting Officer. Cameras, video equipment, or similar photographic equipment installed in computers must be disabled. Proof of computer Portsmouth Naval Shipyard Security Officer approval must be with the tablet, laptop, or other computers at all times.

#### 1.6.1.1 Operating System

The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

#### 1.6.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software

manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. All computers used on this project must be scanned using the installed software at least once per day.

#### 1.6.1.3 Passwords and Passphrases

The passwords and passphrases for all computers must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number, and one special character.

#### 1.6.1.4 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using Contractor owned computers. Contractor Computer Cybersecurity Compliance Statements must use the template published at <http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-01-50-00>. Each Statement must be signed by a cybersecurity representative for the relevant company.

#### 1.6.2 Temporary IP Networks

Temporary Contractor-installed IP networks may be used during construction. When used, temporary Contractor-installed IP networks must meet the following requirements:

##### 1.6.2.1 Network Boundaries and Connections

The network must not extend outside the project site and must not connect to any IP network other than IP networks provided under this project or Government furnished IP networks provided for this purpose. Any and all network access from outside the project site is prohibited.

#### 1.6.3 Government Access to Network

Government personnel, as defined, prescribed, and identified by the Contracting Officer, must be allowed to have complete and immediate access to the network at any time in order to verify compliance with this specification. If there is a Government agency responsible for network access, identify that agency.

#### 1.6.4 Temporary Wireless IP Networks

##### 1.6.4.1 Contractor Provided Wi-Fi Network

Network access must be available throughout the jobsite and for the entire Contract period. The Government will use the Wi-Fi service for coordination in the field and related activities. Do not attach or connect the temporary Wi-Fi network Government networks or systems unrelated to construction activities.

Password information must adhere to requirements outlined in Paragraph PASSWORDS AND PASSPHRASES. Provide password and Wi-Fi network name (SSID) to the Contracting Officer for Government personnel access. Wi-Fi performance must be capable of handling bandwidth to support Government and Contractor activities, including drawing and model access onsite. Provide a point of contact for reporting Wi-Fi outages.

#### 1.6.4.2 Temporary Wireless IP Networks Security

In addition to the other requirements on temporary IP networks, temporary wireless IP (WiFi) networks must not interfere with existing wireless network and must use WPA2 security. Network names (SSID) for wireless networks must be changed from their default values.

#### 1.6.5 Passwords and Passphrases

The passwords and passphrases for all network devices and network access must be changed from their default values. Passwords must be a minimum of 8 characters with a minimum of one uppercase letter, one lowercase letter, one number, and one special character.

#### 1.6.6 Contractor Temporary Network Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Temporary Network Cybersecurity Compliance Statements for each company implementing a temporary IP network. Contractor Temporary Network Cybersecurity Compliance Statements must use the template published at <http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-01-50-00>. Each Statement must be signed by a cybersecurity representative for the relevant company. If no temporary IP networks will be used, provide a single copy of the Statement indicating this.

#### 1.7 Employee Parking Plan

Parking on the Portsmouth Naval Shipyard is limited and is not guaranteed. All Contractor and Subcontractor employee privately owned vehicles (POV) must be parked offsite. Parking of Contractor and Subcontractor privately owned vehicles is not allowed on the Portsmouth Naval Shipyard. The Employee Parking Plan must include the anticipated number of Contractor and Subcontractor employees with the anticipated number of POV parking that will be required to support the construction effort. The plan must include the proposed off-yard parking and shuttling plans that will be required. If a POV is found parked on the Portsmouth Naval Shipyard or found violating any Shipyard or Town of Kittery parking regulations, the vehicle will be subject to towing and a non-compliance notice will be issued.

Only Contractor company-owned vehicles will be allowed to park on the Portsmouth Naval Shipyard. The Contractor must park company-owned vehicles in the Jamaica Island Parking Lot only while not in active use. If a company-owned vehicle is found parked outside the Jamaica Island Parking Lot and is not in active use, or is found violating any Shipyard parking regulations, the vehicle will be subject to towing and a non-compliance notice will be issued. Parking of company-owned vehicles must not interfere with existing and established parking requirements of the Portsmouth Naval Shipyard. There may not be sufficient on-yard parking available for all company-owned vehicles. The Contractor may need to make arrangement for off-shipyard parking of company-owned vehicles. Limited parking will not be the basis for a claim for additional time or costs. The Employee Parking Plan must include the anticipated number of Contractor and Subcontractor company-owned vehicles that will be required to support the construction effort. The plan must identify how the Contractor will ensure company-owned vehicles will park at the Jamaica Island Parking Lot only when not in active use, and include proposed off-yard parking that may be required for company-owned vehicles.



## PART 2 PRODUCTS

### 2.1 TEMPORARY SIGNAGE

#### 2.1.1 Bulletin Board

Prior to the commencement of work activities, provide a clear weatherproof covered bulletin board not less than 36 by 48 inches in size for displaying the Equal Employment Opportunity poster, a copy of the wage decision contained in the Contract, Wage Rate Information poster, Safety and Health Information as required by EM 385-1-1 and other information approved by the Contracting Officer. Coordinate requirements herein with 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME). Locate the bulletin board at the project site in a conspicuous place easily accessible to all employees, and in location as approved by the Contracting Officer.

#### 2.1.2 Project Identification Signs

The requirements for the signs, their content, and location are as specified in Section 01 58 00 PROJECT IDENTIFICATION. Erect signs within 15 days from receiving the Notice to Proceed. Correct the data required by the safety sign daily, with light colored metallic or non-metallic numerals.

#### 2.1.3 Warning Signs

Post temporary signs, tags, and labels to give workers and the public adequate warning and caution of construction hazards according to EM 385-1-1. Attach signs to the perimeter fencing every 150 feet warning the public of the presence of construction hazards. Signs must require unauthorized persons to keep out of the construction site. Correct the data required by safety signs daily. Post signs at all points of entry designating the construction site as a hard hat area.

### 2.2 TEMPORARY TRAFFIC CONTROL

#### 2.2.1 Barricades

Erect and maintain temporary barricades to limit public access to hazardous areas. Barricades are required whenever safe public access to paved areas such as roads, parking areas, or sidewalks is prevented by construction activities or as otherwise necessary to ensure the safety of both pedestrian and vehicular traffic. Securely place barricades clearly visible with adequate illumination to provide sufficient visual warning of the hazard during both day and night.

### 2.3 FENCING

Provide fencing along the construction site and at all open excavations and tunnels to control access by unauthorized personnel. Safety fencing must be highly visible to be seen by pedestrians and vehicular traffic. All fencing must meet the requirements of EM 385-1-1. Remove the fence upon completion and acceptance of the work. Fencing must be installed to be able to restrain a force of at least 250 pounds against it.

### 2.3.1 Chain Link Panel Fencing

Temporary panel fencing must be galvanized steel chain link panels 6 feet high. Multiple fencing panels may be linked together at the bases to form long spans as needed. Each panel base must be weighted down using sand bags or other suitable materials in order for the fencing to withstand anticipated winds while remaining upright. Fencing must remain rigid and taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection. Completely remove fencing and posts at the completion of construction and restore surfaces disturbed or damaged to its original condition. Equip fence with a lockable gate. Gate must remain locked when construction personnel are not present.

### 2.3.2 Polyethylene Mesh Safety Fencing

Temporary safety fencing must be a high visibility orange colored, high density polyethylene grid, a minimum of 48 inches high and maximum mesh size of 2 inches. Fencing must extend from the grade to a minimum of 48 inches above the grade and be tightly secured to T-posts spaced as necessary to maintain a rigid and taut fence. Fencing must remain rigid and taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection.

### 2.3.3 Post-Driven Chain Link Fencing

Temporary post-driven fencing must be galvanized chain link fencing 6 feet high supported by a tightly secured to galvanized steel posts driven below grade. Fence posts must be located on minimum 10 foot centers. Posts may be set in various surfaces such as sand, soil, asphalt or concrete as necessary. Chain link fencing must remain rigid and taut with a minimum of 200 pounds of force exerted on it from any direction with less than 4 inches of deflection. Completely remove fencing and posts at the completion of construction and restore surfaces disturbed or damaged to its original condition. Locate and identify underground utilities prior to setting fence posts. Equip fence with lockable gate. Gate must remain locked when construction personnel are not present.

## 2.4 TEMPORARY SECURITY LIGHTING

Provide temporary security lighting if any existing permanent security lighting is disturbed during the execution of the work. Any CIA lighting outages must be approved through Code 1122 and temporary lighting must be provided that is equal to or greater than the current footcandle rating of the area. Refer to UFC 3-530-01, Table 6-2, for minimum lighting criteria footcandle ratings. The clear zone at the Portsmouth Naval Shipyard is 10 feet. Allow a minimum of 5 working days for review of proposed temporary lighting by Code 1122.

### 2.4.1 Temporary Pedestrian Lighting

Provide temporary pedestrian lighting if any existing permanent pedestrian lighting along sidewalk, walkways, or at crosswalks is disturbed during the execution of the work.

## 2.5 TEMPORARY WIRING

Provide temporary wiring in accordance with EM 385-1-1, NFPA 241, and NFPA 70. Include monthly inspection and testing of all equipment and

apparatus.

## 2.6 BACKFLOW PREVENTERS

Certificate of Full Approval from FCCCHR List, University of Southern California, attesting that the design, size, and make of each backflow preventer has satisfactorily passed the complete sequence of performance testing and evaluation for the respective level of approval. Certificate of Provisional Approval is not acceptable.

Reduced pressure principle type conforming to the applicable requirements of AWWA C511. Provide backflow preventers complete with 150 pound flanged cast iron mounted gate valve, stainless steel or bronze internal parts.

Backflow Preventers must be reduced pressure principle type conforming to the applicable requirements of AWWA C511. The particular make, model/design, and size of backflow preventers to be installed must be included in the latest edition of the List of Approved Backflow Prevention Assemblies issued by the FCCCHR List and be accompanied by a Certificate of Full Approval from FCCCHR List. After installation conduct Backflow Preventer Tests and provide test reports verifying that the installation meets the FCCCHR Manual Standards.

## PART 3 EXECUTION

### 3.1 EMPLOYEE PARKING

Parking on the Portsmouth Naval Shipyard is limited and is not guaranteed. All Contractor and Subcontractor employee privately owned vehicles (POV) must be parked offsite. Only Contractor company-owned vehicles will be allowed to park on the Portsmouth Naval Shipyard. Refer to paragraph EMPLOYEE PARKING PLAN for additional requirements.

### 3.2 AVAILABILITY AND USE OF UTILITY SERVICES

#### 3.2.1 Temporary Utilities

Provide temporary utilities required for construction. Materials may be new or used, must be adequate for the required usage, not create unsafe conditions, and not violate applicable codes and standards.

#### 3.2.2 Government Provided Utility Services

- a. Reasonable amounts of the following utilities will be made available without charge.

| Utility Services |
|------------------|
| Electricity      |
| Potable Water    |
| Salt Water       |
| Compressed Air   |
| Steam            |

|                  |
|------------------|
| Utility Services |
| Sanitary Sewer   |
|                  |

- b. The Contractor must coordinate directly with Unitil, the natural gas provider for the Portsmouth Naval Shipyard, for use of natural gas.
- c. The point at which the Government will deliver such utilities and the quantity available must be coordinated with the Contracting Officer. Pay all costs incurred in connecting, converting, and transferring the utilities to the work. Make connections, including providing backflow-preventing devices on connections to domestic water lines; providing meters; providing transformers; and make disconnections. Provide the backflow devices as specified above and NAVFAC PWD ME shop personnel (at the Portsmouth Naval Shipyard only) will install the backflow preventer. Do not operate any Portsmouth Naval Shipyard water system valves. Notify the Contracting Officer a minimum of 15 calendar days prior to the desired date of the installation of the backflow device. Under no circumstances will taps to Shipyard fire hydrants be allowed for obtaining domestic water.

3.2.3 Sanitation

Provide and maintain within the construction area minimum field-type sanitary facilities approved in accordance with EM 385-1-1. Locate the facilities behind the construction fence or out of the public view. Clean units and empty wastes at least once a week or more frequently into a municipal, district, or Portsmouth Naval Shipyard sanitary sewage system, or remove waste to a commercial facility. Obtain approval from the system owner prior to discharge into a municipal, district, or commercial sanitary sewer system. Penalties or fines associated with improper discharge will be the responsibility of the Contractor. Coordinate with the Contracting Officer and follow Portsmouth Naval Shipyard regulations and procedures when discharging into the Portsmouth Naval Shipyard sanitary sewer system. Maintain these conveniences at all times without nuisance. Include provisions for pest control and elimination of odors. Government toilet facilities will not be available to Contractor's personnel.

3.2.4 Telephone

Make arrangements and pay all costs for telephone facilities desired.

3.2.5 Obstruction Lighting of Cranes

Provide a minimum of two (2) aviation red or high intensity white obstruction lights on temporary structures (including cranes) over 100 feet above ground level. Light construction and installation must comply with FAA AC 70/7460-1. Lights must be operational during periods of reduced visibility, darkness, and as directed by the Contracting Officer.

3.2.6 Fire Protection

Provide temporary fire protection equipment for the protection of personnel and property during construction. Remove debris and flammable materials daily to minimize potential hazards.

### 3.3 STATION OPERATION AFFECT ON CONTRACTOR OPERATIONS

#### 3.3.1 Restricted Access Areas

The Government will monitor work in areas indicated. Notify the Contracting Officer at least 14 calendar days prior to starting work in these areas.

### 3.4 TRAFFIC PROVISIONS

#### 3.4.1 Maintenance of Traffic

- a. Conduct operations in a manner that will not close any thoroughfare or interfere in any way with traffic on railways or roadways except with written permission of the Contracting Officer at least 15 calendar days prior to the proposed modification date, and provide to the Contracting Officer a Traffic Control Plan for Government approval detailing the proposed controls to traffic movement for approval. The plan must be in accordance with State and local regulations and the MUTCD, Part VI. Contractor may move oversized and slow-moving vehicles to the worksite provided requirements of the State of Maine Department of Transportation have been met.
- b. Contractor may move oversized and slow-moving vehicles to the worksite provided requirements of the State of Maine Department of Transportation and local jurisdictions have been met.
- c. Conduct work so as to minimize obstruction of traffic and pedestrians on the facility and maintain at least one lane of traffic at all times. Obtain approval from the Contracting Officer prior to starting any activity that will obstruct vehicle or pedestrian traffic.
- d. Provide, erect, and maintain, at no expense to the Government, lights, barriers, signals, passageways, detours, and other items, that may be required by the Life Safety Signage and overhead protection authority having jurisdiction.
- e. Provide cones, signs, barricades, lights, or other traffic control devices and personnel required to control traffic. Do not use foil-backed material for temporary pavement marking because of its potential to conduct electricity during accidents involving downed power lines.
- f. The area adjacent to the project site will be an active Shipyard work site. All work shall be coordinated to avoid impacting Shipyard operations to the extent practical. Work shall not impact Shipyard Fire Department access in any way unless otherwise approved by the Contracting Officer, Government and Shipyard Fire Department via the outage request process as detailed in Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME).

#### 3.4.2 Protection of Traffic

Maintain and protect traffic on all affected roads during the construction period as noted on the Contractor's approved Traffic Control Plan, except as otherwise specifically directed by the Contracting Officer. Measures for the protection and diversion of traffic, including the provision of watchmen and flagmen, erection of barricades, placing of lights around and in front of equipment and the work, and the erection and maintenance of

adequate warning, danger, and direction signs, will be as required by the State and local authorities having jurisdiction. Provide self-illuminated (lighted) barricades during hours of darkness. Brightly-colored (orange) vests are required for all personnel working in roadways. Protect the traveling public from damage to person and property. Minimize the interference with public traffic on roads selected for hauling material to and from the site. Investigate the adequacy of existing roads and their allowable load limit. The Contractor is responsible for the repair of damage to roads caused by construction operations.

#### 3.4.3 Rush Hour Restrictions

Do not interfere with the peak traffic flows preceding and during normal operations, unless shown on the Contractor's approved Traffic Control Plan or without notification to and approval by the Contracting Officer. Peak traffic flows on the Portsmouth Naval Shipyard are between the hours of 0500-0700 and 1500-1800, Monday through Friday.

#### 3.4.4 Dust Control

Dust control methods and procedures must be approved by the Contracting Officer. Coordinate dust control methods with Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME).

#### 3.5 REDUCED PRESSURE BACKFLOW PREVENTERS

Provide an approved reduced pressure backflow prevention assembly at each location where the Contractor taps into the Government potable water supply.

Perform backflow preventer tests using test equipment, procedures, and certification forms conforming to those outlined in the latest edition of the Manual of Cross-Connection Control published by the FCCCHR Manual. Test and tag each reduced pressure backflow preventer upon initial installation (prior to continued water use) and monthly thereafter. Tag must contain the following information: make, model, serial number, dates of tests, results, maintenance performed, and signature of tester. Record test results on certification forms conforming to requirements cited earlier in this paragraph.

#### 3.6 CONTRACTOR'S TEMPORARY FACILITIES

Contractor is responsible for security of their property. Provide adequate outside security lighting at the temporary facilities. Trailers must be anchored to resist high winds and meet applicable State or local standards for anchoring mobile trailers. Coordinate anchoring with EM 385-1-1. The Contract Clause entitled "FAR 52.236-10, Operations and Storage Areas" and the following apply:

##### 3.6.1 Safety Systems

Protect the integrity of all installed safety systems or personnel safety devices. Obtain prior approval from the Contracting Officer if entrance into systems serving safety devices is required. If it is temporarily necessary to remove or disable personnel safety devices in order to accomplish Contract requirements, provide alternative means of protection prior to removing or disabling any permanently installed safety devices or equipment and obtain approval from the Contracting Officer.

### 3.6.2 Administrative Field Offices

Provide and maintain administrative field office facilities within the construction area at the designated site. Government office and warehouse facilities will not be available to Contractor personnel.

In the event a new building is constructed for the temporary project field office, it shall be approximately 8 feet in width, 32 feet in length and have a minimum of 7 feet headroom. Final dimensions of the temporary project field office must be confirmed with the Contracting Officer prior to installation. Equip the building with approved electrical wiring, at least one double convenience outlet and the required switches and fuses to provide 110-120 volt power. Provide a work table with stool, desk with chair, two additional chairs, and one legal size file cabinet that can be locked. The building must be waterproof, supplied with a heater, have a minimum of two doors, electric lights, a telephone, a battery-operated smoke detector alarm, a sufficient number of adjustable windows for adequate light and ventilation, and a supply of approved drinking water. Provide approved sanitary facilities. Screen the windows and doors and provide the doors with deadbolt type locking devices or a padlock and heavy-duty hasp bolted to the door. Door hinge pins must be non-removable. Arrange the windows to open and to be securely fastened from the inside. Protect glass panels in windows by bars or heavy mesh screens to prevent easy access. In warm weather, provide air conditioning capable of maintaining the office at 50 percent relative humidity and a room temperature 20 degrees F below the outside temperature when the outside temperature is 95 degrees F. Unless otherwise directed by the Contracting Officer, remove the building from the site upon completion and acceptance of the work.

### 3.6.3 Storage Area

Construct a temporary 6 foot high chain link fence around trailers and materials. Include plastic strip inserts, colored green, so that visibility through the fence is obstructed. Fence posts may be driven, in lieu of concrete bases, where soil conditions permit. Do not place or store trailers, materials, or equipment outside the fenced area unless such trailers, materials, or equipment are assigned a separate and distinct storage area by the Contracting Officer away from the vicinity of the construction site but within the boundaries of the Portsmouth Naval Shipyard. Trailers, equipment, or materials must not be open to public view with the exception of those items which are in support of ongoing work on the current day. Do not stockpile materials outside the fence in preparation for the next day's work. Park mobile equipment, such as tractors, wheeled lifting equipment, cranes, trucks, and like equipment within the fenced area at the end of each work day.

Keep fencing in a state of good repair and proper alignment. If the Contractor elects to traverse grassed or unpaved areas which are not established roadways with construction equipment or other vehicles, cover the grassed or unpaved areas with a layer of gravel as necessary to prevent rutting and the tracking of mud onto paved or established roadways; gravel gradation must be at the Contractor's discretion. Mow and maintain grass located within the boundaries of the construction site for the duration of the project. Grass and vegetation along fences, structures, under trailers, and in areas not accessible to mowers must be edged or trimmed neatly.

#### 3.6.4 Supplemental Storage Area

Upon request, and pending availability, the Contracting Officer will designate another or supplemental area for the use and storage of trailers, equipment, and materials. This area may not be in close proximity of the construction site but will be within the Portsmouth Naval Shipyard boundaries. Maintain the area in a clean and orderly fashion and secured if needed to protect supplies and equipment. Utilities will not be provided to this area by the Government.

#### 3.6.5 Appearance of Trailers

- a. Trailers must be roadworthy and comply with all appropriate State and local vehicle requirements. Trailers which are rusted, have peeling paint or are otherwise in need of repair will not be allowed on Portsmouth Naval Shipyard property. Trailers must present a clean and neat exterior appearance and be in a state of good repair.
- b. Maintain the temporary facilities. Failure to do so will be sufficient reason to require their removal at the Contractor's expense.

#### 3.6.6 Trailers or Storage Buildings

- a. Trailers or storage buildings will be permitted, where space is available, subject to the approval of the Contracting Officer.
- b. Mount a sign not smaller than 13.78 by 13.78 inches on the trailer or building that shows the company name, business phone number, emergency phone number and conforms to the following requirements and sketch:

|                                 |                                        |
|---------------------------------|----------------------------------------|
| Graphic panel                   | Aluminum, painted blue                 |
| Copy                            | Screen painted or vinyl die-cut, white |
| Typeface                        | Univers 65 u/lc                        |
| See Sketch No. 01500 (graphic). |                                        |

#### 3.6.7 Laydown Space

Parking and laydown space on the site is limited to the areas shown on the plans. Manage the onsite work including equipment, storage trailers, material, material deliveries to allow the work to be completed within the specified Contract duration. This may require the Contractor to locate suitable storage offsite and multiple equipment mobilizations to allow the work to be completed. Equipment or materials not used to complete the work must be removed from the site. If additional offsite storage, additional mobilization or demobilizations are required, all these costs must be included in the base bid.

Failure to plan the work based on the space limitations must not be the basis for any claim nor an equitable price or Contract time adjustment.

#### 3.6.8 Weather Protection of Temporary Facilities and Stored Materials

Take necessary precautions to ensure that roof openings and other critical openings in the temporary facilities are monitored carefully. Take



immediate actions required to seal off such openings when rain or other detrimental weather is imminent, and at the end of each workday. Ensure that the openings are completely sealed off to protect materials and equipment in the temporary facilities from damage.

### 3.6.9 Building and Site Storm Protection

When a warning of gale force winds is issued, take precautions to minimize danger to persons, and protect the work and nearby Government property. Precautions must include, but are not limited to, closing openings; removing loose materials, tools, and equipment from exposed locations; and removing or securing scaffolding and other temporary work. Close openings in the work when storms of lesser intensity pose a threat to the work or any nearby Government property.

### 3.6.10 Temporary Partitions

Provide "No Dust" temporary partitions of wood or metal frame, heavy duty plastic sheathing and negative pressure HEPA filtered ventilation. Provide access door and vestibules as required to prevent dust from escaping the enclosure. Refer to Section 02 41 00 DEMOLITION for additional information and requirements if included as part of the specifications.

## 3.7 PLANT COMMUNICATION

Whenever the individual elements of the plant are located so that operation by normal voice between these elements is not satisfactory, install a satisfactory means of communication, such as telephone or other suitable devices and make available for use by Government personnel.

## 3.8 TEMPORARY PROJECT SAFETY FENCING

As soon as practicable, but not later than 15 calendar days after the date established for commencement of work, furnish and erect temporary project safety fencing at the work site. Maintain the safety fencing during the life of the Contract and, upon completion and acceptance of the work, remove from the work site.

## 3.9 CLEANUP

Remove construction debris, waste materials, packaging material, and the like from the work site daily. Any dirt or mud which is tracked onto paved or surfaced roadways must be cleaned away. Store all salvageable materials resulting from demolition activities within the fenced area described above or at the supplemental storage area. Neatly stack stored materials not in trailers, whether new or salvaged.

## 3.10 RESTORATION OF STORAGE AREA

Upon completion of the project remove the bulletin board, signs, barricades, haul roads, and all other temporary products from the site. After removal of trailers, materials, and equipment from within the fenced area, remove the fence. Restore areas used during the performance of the Contract to the original or better condition. Remove gravel used to traverse grassed areas and restore the areas to their original condition, including top soil and seeding as necessary.

-- End of Section --



SECTION 01 57 19.00 22

TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME)  
08/22

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at Portsmouth Naval Shipyard only.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only. Note: This is not an all-inclusive list of publications and other references may be applicable.

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA SW-846 (Third Edition; Update IV) Test Methods  
for Evaluating Solid Waste:  
Physical/Chemical Methods

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|                 |                                                                                                     |
|-----------------|-----------------------------------------------------------------------------------------------------|
| 29 CFR 1910.120 | Hazardous Waste Operations and Emergency Response                                                   |
| 40 CFR 50       | National Primary and Secondary Ambient Air Quality Standards                                        |
| 40 CFR 82       | Protection of Stratospheric Ozone                                                                   |
| 40 CFR 112      | Oil Pollution Prevention                                                                            |
| 40 CFR 241      | Guidelines for Disposal of Solid Waste                                                              |
| 40 CFR 243      | Guidelines for the Storage and Collection of Residential, Commercial, and Institutional Solid Waste |
| 40 CFR 258      | Subtitle D Landfill Requirements                                                                    |
| 40 CFR 260      | Hazardous Waste Management System: General                                                          |
| 40 CFR 261      | Identification and Listing of Hazardous Waste                                                       |
| 40 CFR 261.7    | Residues of Hazardous Waste in Empty Containers                                                     |
| 40 CFR 262      | Standards Applicable to Generators of Hazardous Waste                                               |
| 40 CFR 263      | Standards Applicable to Transporters of Hazardous Waste                                             |
| 40 CFR 264      | Standards for Owners and Operators of                                                               |

Hazardous Waste Treatment, Storage, and  
Disposal Facilities

|                      |                                                                                                                                              |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 40 CFR 265           | Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities                             |
| 40 CFR 266           | Standards for the Management of Specific Hazardous Wastes and Specific Types of Hazardous Waste Management Facilities                        |
| 40 CFR 268           | Land Disposal Restrictions                                                                                                                   |
| 40 CFR 270           | EPA Administered Permit Programs: The Hazardous Waste Permit Program                                                                         |
| 40 CFR 272           | Approved State Hazardous Waste Management Programs                                                                                           |
| 40 CFR 273           | Standards for Universal Waste Management                                                                                                     |
| 40 CFR 279           | Standards for the Management of Used Oil                                                                                                     |
| 40 CFR 280           | Technical Standards and Corrective Action Requirements for Owners and Operators of Underground Storage Tanks (UST)                           |
| 40 CFR 300           | National Oil and Hazardous Substances Pollution Contingency Plan                                                                             |
| 40 CFR 300.125       | National Oil and Hazardous Substances Pollution Contingency Plan - Notification and Communications                                           |
| 40 CFR 355           | Emergency Planning and Notification                                                                                                          |
| 40 CFR 372-SUBPART D | Specific Toxic Chemical Listings                                                                                                             |
| 40 CFR 761           | Polychlorinated Biphenyls (PCBs) Manufacturing, Processing, Distribution in Commerce, and Use Prohibitions                                   |
| 49 CFR 171           | General Information, Regulations, and Definitions                                                                                            |
| 49 CFR 172           | Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements |
| 49 CFR 173           | Shippers - General Requirements for Shipments and Packagings                                                                                 |
| 49 CFR 178           | Specifications for Packagings                                                                                                                |

**STATE OF MAINE REGULATIONS**

The following STATE OF MAINE REGULATIONS are available on the Internet at:  
<http://www.maine.gov/dep/permits.htm>

STATE OF MAINE Statutes are available on the internet at  
<https://legislature.maine.gov/statutes/38/title38ch0sec0.html>

STATE OF MAINE RULES are available on the internet at  
[https://www.maine.gov/sos/rulemaking/agency-rules/  
departement-environmental-protection-rules](https://www.maine.gov/sos/rulemaking/agency-rules/departement-environmental-protection-rules)

MAINE DEP AIR BUREAU CHAPTER 101 Visible Emissions Regulations;  
<http://www.maine.gov/dep/air/rules/index.html>

MAINE DEP AIR BUREAU CHAPTER 151 Architectural and Industrial Maintenance  
(AIM) COATINGS; <http://www.maine.gov/dep/air/rules/index.html>

MAINE DEP AIR BUREAU CHAPTER 159 Control of Volatile Organic Compounds  
from Adhesives and Sealants; <http://www.maine.gov/dep/air/rules/index.html>

MAINE DEP 38 MSRA 420-C Erosion and Sedimentation Control Law and Rules

MAINE DEP 38 MSRA 420-D Stormwater Management

MAINE 38 MRSA 439-B Contractors Certified in Erosion Control  
(Latest Edition)

MAINE DEP MSRA 481-490 Site Location of Development

MAINE 38 MSRA 850 Identification of Hazardous Waste

MAINE 38 MSRA 851 Standards for Generators of Hazardous Waste

MAINE 38 MSRA 852 Land Disposal Restrictions

MAINE 38 MSRA 853 Licensing of Transporters of Hazardous Waste

MAINE 38 MSRA 857 Hazardous Waste Manifest Requirements

MAINE 38 MSRA 858 Universal Waste Rules

MAINE 38 MSRA 860 Waste Oil Management Rules

MAINE 88 MRSA 480A-480Z Natural Resources Protection Act

CODE OF MAINE RULE 06-096 Chapter 500 Stormwater Management

CODE OF MAINE RULE 06-096 Chapter 573 Snow Dumps: Best Management  
Practices for Pollution Prevention

MAINE DEP Maine Erosion and Sediment Control Practices  
Field Guide for Contractors, latest edition.

MEPDES Permit #MER042000 General Permit for the Discharge of  
Stormwater from State or Federally Owned  
Municipal Separate Storm Sewer Systems

## 1.2 DEFINITIONS

### 1.2.1 Class I and II Ozone Depleting Substance (ODS)

Class I ODS is defined in Section 602(a) of The Clean Air Act. A list of

Class I ODS can be found on the EPA website at the following weblink.  
<https://www.epa.gov/ozone-layer-protection/ozone-depleting-substances>.

Class II ODS is defined in Section 602(s) of The Clean Air Act. A list of Class II ODS can be found on the EPA website at the following weblink.  
<https://www.epa.gov/ozone-layer-protection/ozone-depleting-substances>.

#### 1.2.2 Contractor Generated Hazardous Waste

Contractor generated hazardous waste is materials that, if expired, abandoned or disposed of, may meet the definition of a hazardous waste. These waste streams would typically consist of material brought on site by the Contractor to execute work, but are not fully consumed during the course of construction. Examples include, but are not limited to, excess paint thinners (i.e., methyl ethyl ketone, toluene), waste thinners, excess paints, excess solvents, waste solvents, excess pesticides, and contaminated pesticide equipment rinse water.

#### 1.2.3 Electronics Waste

Electronics waste is discarded electronic devices intended for salvage, recycling, and disposal.

#### 1.2.4 Environmental Pollution and Damage

Environmental pollution and damage is the presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances of importance to human life; affect other species of importance to humankind; or degrade the environment aesthetically, culturally or historically.

#### 1.2.5 Environmental Protection

Environmental protection is the prevention/control of pollution and habitat disruption that may occur to the environment during construction. The control of environmental pollution and damage requires consideration of land, water, and air; biological and cultural resources; and includes management of visual aesthetics; noise; solid, chemical, gaseous, and liquid waste; radiant energy and radioactive material as well as other pollutants.

#### 1.2.6 Hazardous Debris

As defined in paragraph SOLID WASTE, debris that contains listed hazardous waste (either on the debris surface, or in its interstices, such as pore structure) in accordance with 40 CFR 261. Hazardous debris also includes debris that exhibits a characteristic in accordance with 40 CFR 261.

#### 1.2.7 Hazardous Materials

Hazardous material is any material that: Is defined in 49 CFR 171, and listed in 49 CFR 172, regulated as a hazardous material in accordance with 49 CFR 173, or requires a Safety Data Sheet (SDS) per 29 CFR 1910.120; or during end use, treatment, handling, packaging, storage, transportation, or disposal meets or has components that meet or have potential to meet the definition of a hazardous waste as defined by 40 CFR 261 Subparts A, B, C, or D.

Designation of a material by this definition, when separately regulated or controlled by other sections or directives, does not eliminate the need for adherence to that hazard-specific guidance which takes precedence over this instruction for "control" purposes. Such material includes ammunition, weapons, explosive actuated devices, propellants, pyrotechnics, chemical and biological warfare materials, medical and pharmaceutical supplies, medical waste and infectious materials, bulk fuels, radioactive materials, and other materials such as asbestos, mercury, and polychlorinated biphenyls (PCBs).

#### 1.2.8 Hazardous Waste

Hazardous Waste is any material that meets the definition of a solid waste and exhibits a hazardous characteristic (ignitability, corrosivity, reactivity, or toxicity) as specified in 40 CFR 261, Subpart C, or contains a listed hazardous waste as identified in 40 CFR 261, Subpart D or as identified by Maine Department of Environmental Protection (MEDEP) MAINE 38 MSRA 850-860. Hazardous waste controls also apply to Universal Wastes.

#### 1.2.9 Land Application

Land Application means spreading or spraying discharge water at a rate that allows the water to percolate into the soil. No sheeting action, soil erosion, discharge into storm sewers, discharge into defined drainage areas, or discharge into the "waters of the United States" must occur. Comply with federal, state, and local laws and regulations.

#### 1.2.10 Municipal Separate Storm Sewer System (MS4) Permit

MS4 permits are those held by municipalities or installations to obtain NPDES permit coverage for their stormwater discharges.

#### 1.2.11 National Pollutant Discharge Elimination System (NPDES)

The NPDES permit program controls water pollution by regulating point sources that discharge pollutants into waters of the United States.

#### 1.2.12 Waste Oil/Used Oil and Oily Waste/Oil Contaminated Waste

Waste oil or used oil is a petroleum-base or other alternative oil (i.e., vegetable oil, environmental friendly oil) which, through use or handling, has become unsuitable for its original purpose due to the presence of impurities or loss of original properties. It must have sufficient liquid content to be free flowing. Waste oil includes, but is not limited to, automotive crankcase and lubricating oils; industrial lube and non-lube oils; oil recovered from oil tank cleaning operations; and oil spilled to land or water.

Oily wastes or oil contaminated wastes are those materials which are, or were, mixed with Petroleum, Oils, and Lubricants (POLs) and have become separated from that POLs. Oily wastes also means debris generated from the clean-up of oil spills. Oily waste are not considered waste oil. Oily wastes must be turned into Building 357 for disposal, but do not need to be turned in by the end of the shift generated. Oily wastes are solid wastes and must be disposed of appropriately in accordance with the requirements of this specification section

All waste oil must be collected and turned in to Building 357 for

disposal. A hazardous waste determination form may be required prior to disposal. All waste characterization sampling must be conducted in accordance with Paragraph 3.13.4.2 herein entitled SAMPLING AND ANALYSIS OF HAZARDOUS WASTE. Waste oil must be minimized through good housekeeping practices and employee education.

#### 1.2.13 Regulated Waste

Those solid wastes that have specific additional Federal, State, or local controls for handling, storage, or disposal.

#### 1.2.14 Sediment

Sediment is soil and other debris that have eroded and have been transported by runoff water or wind.

#### 1.2.15 Solid Waste

Solid waste is a solid, liquid, semi-solid, or contained gaseous waste. A solid waste can be a hazardous waste, non-hazardous waste, or non-Resource Conservation and Recovery Act (RCRA) regulated waste. Types of solid waste typically generated at construction sites may include:

##### 1.2.15.1 Debris

Debris is non-hazardous solid material generated during the construction, demolition, or renovation of a structure which exceeds 2.5 inch particle size that is: a manufactured object; plant or animal matter; or natural geologic material (for example, cobbles and boulders), broken or removed concrete, masonry, and rock asphalt paving; ceramics; roofing paper and shingles. Inert materials may be reinforced with or contain ferrous wire, rods, accessories, and weldments. A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if the mixture is comprised primarily of debris by volume, based on visual inspection.

##### 1.2.15.2 Green Waste

Green waste is the vegetative matter from landscaping, land clearing and grubbing, including, but not limited to, grass, bushes, scrubs, small trees and saplings, tree stumps, and plant roots. Marketable trees, grasses, and plants that are indicated to remain, be re-located, or be re-used are not included.

##### 1.2.15.3 Material Not Regulated as Solid Waste

Material not regulated as solid waste is nuclear source or byproduct materials regulated under the Federal Atomic Energy Act of 1954 as amended; suspended or dissolved materials in domestic sewage effluent or irrigation return flows, or other regulated point source discharges; regulated air emissions; and fluids or wastes associated with natural gas or crude oil exploration or production.

##### 1.2.15.4 Non-Hazardous Waste

Non-hazardous waste is waste that is excluded from, or does not meet, hazardous waste criteria in accordance with 40 CFR 261.



#### 1.2.15.5 Recyclables

Recyclables are materials, equipment, and assemblies such as doors, windows, door and window frames, plumbing fixtures, glazing, and mirrors that are recovered and sold as recyclable, and structural components. It also includes commercial-grade refrigeration equipment with Freon removed, household appliances where the basic material content is metal, clean polyethylene terephthalate bottles, cooking oil, used fuel oil, textiles, high-grade paper products and corrugated cardboard, stackable pallets in good condition, clean crating material, and clean rubber/vehicle tires. Paint cans that meet the definition of empty containers in accordance with 40 CFR 261.7 may be included as recyclable if sold to a scrap metal company.

#### 1.2.15.6 Surplus Soil

Surplus soil is existing soil that is in excess of what is required for this work, including aggregates intended, but not used, for on-site mixing of concrete, mortars, and paving. Contaminated soil meeting the definition of hazardous material or hazardous waste is not included and must be managed in accordance with paragraph HAZARDOUS MATERIAL MANAGEMENT.

#### 1.2.15.7 Scrap Metal

This includes scrap and excess ferrous and non-ferrous metals such as reinforcing steel, structural shapes, pipe, and wire that are recovered or collected and disposed of as scrap. Scrap metal meeting the definition of hazardous material or hazardous waste is not included. Scrap metal coated with PCB contaminated paint > or = to 50 ppm is considered a hazardous waste and requires Code 106.3 coordination for recycling/disposal.

#### 1.2.15.8 Wood

Wood is dimension and non-dimension lumber, plywood, chipboard, and hardboard. Treated and/or painted wood that meets the definition of lead contaminated or lead based contaminated paint is not included. Treated wood includes, but is not limited to, lumber, utility poles, crossties, and other wood products with chemical treatment.

#### 1.2.15.9 Paint Cans

Metal cans that are RCRA empty of paints, solvents, thinners, and adhesives. NOTE: Aerosol (paint) cans are Hazardous Wastes and must not be disposed of as solid waste or be considered in any definition of "empty", "paint", or "metal" cans.

#### 1.2.15.10 Special Waste

Special Waste: "Special waste" means any solid waste generated by sources other than household and typical commercial establishments that do not meet the definition of a hazardous waste but exists in such an unusual quantity or in such a chemical or physical state, or any combination thereof, that may disrupt or impair effective waste management or threaten the public health, human safety, or the environment and requires special handling, transportation, and disposal procedures. Special waste includes, but is not limited to:

- (1) Ash;

- (2) Industrial and industrial process waste;
- (3) Sludge and dewatered septage;
- (4) Debris from nonhazardous chemical spills and cleanup of those spills;
- (5) Contaminated soils and dredge materials;
- (6) Asbestos and asbestos-containing waste;
- (7) Sand blast grit and non-liquid paint waste;
- (8) PCB and PCB contaminated waste;
- (9) High and low pH waste;
- (10) Spent filter media residue;
- (11) Shredder residue; and
- (12) Railroad ties with or without creosote.

#### 1.2.16 Surface Discharge

Surface discharge means discharge of water into drainage ditches, storm sewers, or creeks meeting the definition of "waters of the United States". Surface discharges from construction sites are discrete, identifiable sources and require a permit from the governing agency. Comply with federal, state, and local laws and regulations.

#### 1.2.17 Wastewater

Wastewater is the used water and solids that flow through a sanitary sewer to a treatment plant.

#### 1.2.18 Stormwater

Stormwater is any precipitation in an urban or suburban area that does not evaporate or soak into the ground, but instead collects and flows into storm drains, rivers, and streams.

#### 1.2.19 Waters of the United States

Waters of the United States means Federally jurisdictional waters, including wetlands, that are subject to regulation under Section 404 of the Clean Water Act or navigable waters, as defined under the Rivers and Harbors Act.

#### 1.2.20 Wetlands

Wetlands are those areas that are inundated or saturated by surface or groundwater at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions.

#### 1.2.21 Universal Waste

The universal waste regulations streamline collection requirements for

certain hazardous wastes in the following categories: batteries, pesticides, and mercury-containing equipment (e.g., thermostats) and lamps (e.g., fluorescent bulbs). The rule is designed to reduce hazardous waste in the municipal solid waste (MSW) stream by making it easier for universal waste handlers to collect these items and send them for recycling or proper disposal. These regulations can be found at 40 CFR 273.

#### 1.2.22 Waste Hazardous Material (WHM)

Any waste material which because of its quantity, concentration, or physical, chemical, or infectious characteristics may pose a substantial hazard to human health or the environment and which has been so designated. Used oil not containing any hazardous waste, as defined above, falls under this definition.

#### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-01 Preconstruction Submittals

- Preconstruction Survey Report; G
- Solid Waste Management Plan; G
- Regulatory Notifications; G
- Environmental Management Plan (EMP); G
- Dirt and Dust Control Plan; G
- Contractor Hazardous Material Inventory Log; G
- Stormwater Management/Erosion and Sedimentation Control Plan; G
- Spill Prevention, Control, and Countermeasures (SPCC) Plan; G
- Environmental Manager Qualifications; G
- Employee Training Records; G
- Noise Mitigation Plan; G

##### SD-06 Test Reports

- Laboratory Analysis; G
- Disposal Requirements; G
- Erosion and Sediment Control Inspection and Corrective Action; G
- Monthly Solid Waste Disposal Report; G

##### SD-07 Certificates

- ECATTS Certificate Of Completion; G

Erosion And Sediment Control Inspector; G

#### SD-11 Closeout Submittals

Some of the records listed below are also required as part of other submittals. For the "Records" submittal, maintain onsite a separate three-ring Environmental Records binder and submit at the completion of the project. Make separate parts to the binder corresponding to each of the applicable sub items listed below.

Stormwater Management and Erosion Control Compliance Notebook; G

Waste Determination Documentation; G

Disposal Documentation for Hazardous and Regulated Waste; G

Contractor Hazardous Material Inventory Log; G

Hazardous Waste/Debris Management; G

Regulatory Notifications; G

Asbestos Free Certification Form; G

Assembled Employee Training Records; G

Sales Documentation; G

#### 1.4 ENVIRONMENTAL PROTECTION REQUIREMENTS

Provide and maintain, during the life of the Contract, environmental protection as defined. Plan for and provide environmental protective measures to control pollution that develops during construction practice. Plan for and provide environmental protective measures required to correct conditions that develop during the construction of permanent or temporary environmental features associated with the project. Protect the environmental features within the project boundaries and those affected outside the limits of permanent work during the entire duration of this Contract. Comply with Federal, State, and local regulations pertaining to the environment, including water, air, solid waste, hazardous waste and substances, oily substances, and noise pollution.

Tests and procedures assessing whether construction operations comply with Applicable Environmental Laws may be required. Analytical work must be performed by qualified laboratories; and where required by law, the laboratories must be certified.

##### 1.4.1 Training in Environmental Compliance Assessment Training and Tracking System (ECATTS)

The Environmental Manager is responsible for environmental compliance on projects. The Environmental Manager must complete ECATTS training modules (installation specific or general) prior to starting respective portions of onsite work under this Contract. If personnel changes occur for any of these positions after starting work, replacement personnel must complete ECATTS training within 14 calendar days of assignment to the project.

Submit an ECATTS certificate of completion for personnel who have

completed the required "Environmental Compliance Assessment Training and Tracking System (ECATTS)" training. This training is web-based and can be accessed from any computer with Internet access using the following instructions.

Register for NAVFAC Environmental Compliance Assessment, Training and Tracking System by logging on to <https://environmentaltraining.ecatts.com>. Obtain the password for registration from the Contracting Officer.

This training has been structured to allow Contractor personnel to receive credit under this contract and also to carry forward credit to future contracts. Ensure the Environmental Manager review their training plans for new modules or updated training requirements prior to beginning work. Some training modules are tailored for specific State regulatory requirements; therefore, working in multiple states will require personnel to retake modules tailored to the State where the Contract work is being performed.

ECATTS is available for use by all personnel associated with this project. These other personnel are encouraged (but not required) to take the training and may do so at their discretion.

#### 1.4.2 Conformance with the Environmental Management System

Perform work under this contract consistent with the policy and objectives identified in the installation's Environmental Management System (EMS). Perform work in a manner that conforms to objectives and targets of the environmental programs and operational controls identified by the EMS. Support Government personnel when environmental compliance and EMS audits are conducted by escorting auditors at the Project site, answering questions, and providing proof of records being maintained. Provide monitoring and measurement information as necessary to address environmental performance relative to environmental, energy, and transportation management goals. In the event an EMS nonconformance or environmental noncompliance associated with the contracted services, tasks, or actions occurs, take corrective and preventative actions. In addition, employees must be aware of their roles and responsibilities under the installation EMS and of how these EMS roles and responsibilities affect work performed under the contract.

Coordinate with the installation's EMS coordinator to identify training needs associated with environmental aspects and the EMS, and arrange training or take other action to meet these needs. Provide training documentation to the Contracting Officer. The Installation Environmental Office will retain associated environmental compliance records. Make EMS Awareness training completion certificates available to Government auditors during EMS audits and include the certificates in the Employee Training Records. See paragraph EMPLOYEE TRAINING RECORDS.

### 1.5 QUALITY ASSURANCE

#### 1.5.1 Preconstruction Survey and Protection of Features

This paragraph supplements the Contract Clause PROTECTION OF EXISTING VEGETATION, STRUCTURES, EQUIPMENT, UTILITIES, AND IMPROVEMENTS. Prior to start of any onsite construction activities, perform a Preconstruction Survey of the project site with the Contracting Officer, and take photographs showing existing environmental conditions in and adjacent to the site. Submit a report for the record with a copy provided to the

Contracting Officer. Obtain a camera pass from Portsmouth Naval Shipyard Security (via Contracting Officer) prior to use of a camera at Portsmouth Naval Shipyard. Cameras may only be used in accordance with the procedures and requirements outlined in Section 01 14 00.00 22 WORK RESTRICTIONS (PWD ME). Include in the report a plan describing the features requiring protection under the provisions of the Contract Clauses that are not specifically identified on the drawings as environmental features requiring protection along with the condition of trees, shrubs and grassed areas immediately adjacent to the site of work and adjacent to the Contractor's assigned storage area and access route(s), as applicable. The Contractor and Contracting Officer will sign this Preconstruction Survey Report upon mutual agreement regarding its accuracy and completeness. Protect those environmental features included in the Preconstruction Survey Report and any indicated on the drawings, regardless of interference that their preservation may cause to the work under the Contract. Submit the Preconstruction Survey Report within 30 calendar days from receiving the Notice to Proceed, or at least 15 calendar days prior to the Environmental Brief, whichever comes first in the project schedule.

#### 1.5.2 Regulatory Notifications

Provide regulatory notification requirements in accordance with Federal, State, and local regulations. In cases where the Government will also provide public notification (such as stormwater permitting), coordinate with the Contracting Officer. Submit copies of all regulatory notifications to the Contracting Officer at least 30 calendar days prior to commencement of work activities. Typically, regulatory notifications must be provided for the following (this listing is not all inclusive): demolition, renovation, NPDES defined site work, construction, removal or use of a permitted air emissions source, and remediation of controlled substances (asbestos, hazardous waste, lead paint).

#### 1.5.3 Environmental Brief

Attend an Environmental Brief prior to commencing any work on the Portsmouth Naval Shipyard. Refer to Section 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS (PWD ME) paragraph ENVIRONMENTAL BRIEF for details on this meeting.

#### 1.5.4 Environmental Manager

Appoint in writing an Environmental Manager for the project site. The Environmental Manager is directly responsible for coordinating Contractor compliance with Federal, State, local, and Portsmouth Naval Shipyard requirements. The Environmental Manager cannot perform the duties of the Project Superintendent, nor the SSHO, nor the Quality Control Manager, nor the Project Manager. The Environmental Manager must ensure compliance with Hazardous Waste Program requirements (including hazardous waste handling, storage, daily turn-in, manifesting, and disposal as applicable); implement the Environmental Management Plan (EMP); ensure environmental permits are obtained, maintained, and closed out; ensure compliance with Stormwater Program Management requirements; ensure compliance with Hazardous Materials (storage, handling, and reporting) requirements; and coordinate any remediation of regulated substances (lead, asbestos, PCB containing building materials, PCB Universal Waste/e-Waste, etc.). The person in this position must be trained to adequately accomplish the following duties: ensure waste segregation and storage compatibility requirements are met; inspect and manage Satellite Accumulation areas; ensure only authorized personnel add wastes to

containers; ensure all personnel are trained in 40 CFR requirements in accordance with their position requirements; coordinate removal or turn-in of waste containers; implement, inspect, and maintain erosion and sediment controls as required by State law; maintain the Environmental Records binder and required documentation; and ensure compliance with all Spill Prevention, Control, and Countermeasures (SPCC) requirements, including but not limited to, the proper storage of tanks and containers and their secondary containment, inspections, spill procedures, etc. including environmental permits compliance and close-out. Submit Environmental Manager Qualifications to the Contracting Officer.

#### 1.5.5 Employee Training Records

Prepare and maintain employee training records throughout the term of the contract meeting applicable 40 CFR requirements. Provide Employee Training Records in the Environmental Records Binder. Ensure every employee completes a program of classroom instruction or on-the-job training that teaches them to perform their duties in a way that ensures compliance with Federal, State, and local regulatory requirements for RCRA Large Quantity Generator. Provide a Position Description for each employee, by subcontractor, based on the Davis-Bacon Wage Rate designation or other equivalent method, evaluating the employee's association with hazardous and regulated wastes. This Position Description will include training requirements as defined in 40 CFR 265 for a Large Quantity Generator facility. Submit these Assembled Employee Training Records to the Contracting Officer at the conclusion of the project, unless otherwise directed.

Train personnel to meet EPA and state requirements. Conduct environmental protection/pollution control meetings for personnel prior to commencing construction activities. Conduct additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda: methods of detecting and avoiding pollution; familiarization with statutory and contractual pollution standards; installation and care of devices, vegetative covers, and instruments required for monitoring purposes to ensure adequate and continuous environmental protection/pollution control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of archaeological sites, artifacts, waters of the United States; and endangered species and their habitat that are known to be in the area. Provide a copy of the Erosion and Sediment Control Inspector Certification as required by Maine.

#### 1.5.6 Non-Compliance Notifications

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with Federal, State or local environmental laws or regulations, permits, and other elements of the Contractor's EMP. After receipt of such notice, the Contractor shall inform the Contracting Officer of the proposed corrective action and take such action when approved by the Contracting Officer. The Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. FAR 52.242-14 Suspension of Work provides that a suspension, delay, or interruption of work due to the fault or negligence of the Contractor allows for no adjustments to the contract for time extensions or equitable adjustments. In addition to a suspension of work, the Contracting Officer may use additional authorities under the contract or law.

#### 1.6 SITE WASTE REMOVAL (SWR) MEETING

A SWR meeting must be held with OICC PNSY, PWD-ME EV staff, and Code 106.3 fourteen (14) calendar days prior to the removal of wastes from a project site to ensure all pertinent requirements of the Standard Operating Procedure for Site Waste Removal (SWR) have been met. An example Site Waste Removal Form is included as Attachment A. No transport of wastes will be allowed until full concurrence is provided by OICC PNSY, PWD-ME EV staff, and Code 106.3 following the satisfactory completion of the SWR.

#### 1.7 ENVIRONMENTAL RECORDS BINDER

Maintain onsite a separate three-ring Environmental Records Binder and submit at the completion of the project. Make separate parts within the binder that correspond to each submittal listed under paragraph CLOSEOUT SUBMITTALS in this section.

#### 1.8 FACILITY HAZARDOUS WASTE GENERATOR STATUS

Portsmouth Naval Shipyard is designated as a Large Quantity Generator. Meet the regulatory requirements of this generator designation for any work conducted within the boundaries of the Portsmouth Naval Shipyard. Comply with provisions of federal, state, and local regulatory requirements applicable to this generator status regarding training and storage, handling, and disposal of construction derived wastes.

### PART 2 PRODUCTS

Not Used

### PART 3 EXECUTION

#### 3.1 ENVIRONMENTAL MANAGEMENT PLAN (EMP)

Prior to initiating any work on site, the Contractor will meet with the Contracting Officer, Shipyard Code 106.3 Representatives, and PWD-ME EV Staff at the Environmental Brief to discuss the proposed Environmental Management Plan (EMP) and develop a mutual understanding relative to the details of environmental protection required to be addressed in the EMP, including measures for protecting natural resources and other measures to be taken. See paragraph ENVIRONMENTAL BRIEF for additional details on the meeting.

The EMP must be submitted in the following format and include all sections specified below, unless a separate plan submittal is accomplished for the EMP section as identified in a.(1)(b). All sections not applicable to the work shall be marked N/A.

##### a. Description of the Environmental Management Plan

###### (1) General overview and purpose

(a) A brief description of the work to be performed to include each type of work activity that will potentially impact the environment (e.g. soil excavation, hazardous waste, abatement (PCB, Lead, Asbestos), concrete slurry/wastewater, emergency generator installation, paint blasting, 55 gallon and greater oil/fuel storage etc.).



(b) A list of other plans required by environmental permit or being submitted separately (e.g. Solid Waste Management Plan, Hazardous Waste Management Plan, Storm Water Pollution Prevention Plan (SWPPP), Spill Prevention Controls and Countermeasures (SPCC) plan, etc).

(c) A copy of any standard or project specific operating procedures that will be used to effectively manage and protect the environment on the project site.

(d) Communication and training procedures that will be used to convey environmental management requirements to employees and Subcontractors.

(e) Emergency contact information for senior construction project contractor personnel and those responsible for environmental management and spill response (office phone number, cell phone number, and e-mail address).

(2) General site information including a site plan showing haul routes, construction material laydown and storage areas, tracking pads, construction trailer locations, sanitary facilities, excavation and soil stockpile locations, dewatering and infiltration areas, location map showing selected erosion and sediment controls, and all other construction facilities required for the work.

(3) A letter signed by an officer of the firm appointing the Environmental Manager and stating that person is responsible for managing and implementing the Environmental Program as described in this contract. Include in this letter the Environmental Manager's authority to direct the removal and replacement of non-conforming work.

b. Management of Natural Resources

(1) Land resources

(2) Tree protection

(3) Replacement of damaged landscape features

(4) Temporary construction

(5) Stream crossings

(6) Fish and wildlife resources

(7) Wetland areas

c. Protection of Historical and Archaeological Resources

(1) Objectives

(2) Methods

(a) Identify any historic properties affected by the project and areas of archaeological monitoring, treatment of unanticipated archaeological resources, and outline mitigation requirements per

this contract.

d. Dirt and Dust Control

Additional information/requirements for dirt and dust control, including those for the Dirt and Dust Control Plan, are found in paragraph DUST CONTROL of this specification.

- (1) Describe how dust will be controlled in construction zones, along haul routes, and at construction entrances/exits.
- (2) List of control measures, operational practices, and Best Management Practices (BMPs) to be employed for dirt and dust control. Selected BMPs must adhere to the Maine Erosion and Sediment Control Practices Field Guide for Contractors, latest edition. Examples include construction entrance/exit design, dust suppressants, stabilization areas, using covers on dump trucks, and sweeping and vacuuming.

e. Stormwater Management and Control

Additional information/requirements for stormwater management and control are found in paragraphs EROSION AND SEDIMENT CONTROL MEASURES and SOLID WASTE MANAGEMENT PLAN of this specification and may be necessary based on the work to be performed. This plan section must include:

- (1) A statement that dumpsters and rollofs will be equipped with a secure lid or covering. Watertight dumpsters and rollofs are preferred. All dumpsters and rollofs must be kept closed at all times, except when being loaded with trash or construction debris.
- (2) Erodible soil areas.
- (3) Temporary measures for runoff control.
- (4) BMPs selected for use from the Maine Erosion and Sediment Control Practices Field Guide for Contractors, latest edition.
- (5) A location map of selected erosion and sediment controls and manufacturer data sheets of products to be used (e.g. erosion control sock, silt sacks, filter fabric etc.).
- (6) A statement that erosion and sediment control inspections will be performed for all projects with ground disturbances greater than or equal to 1 cubic yard and will comply with erosion and sediment control practices within a shoreland zone. Inspections must be completed at least once per week as well as BEFORE and AFTER a storm event that has the potential to create runoff but at a minimum for every storm resulting in 0.25 inches of precipitation or greater.
- (7) The Erosion and Sediment Control Inspector(s) name(s) and current certificate for Maine Certification in Erosion Control Practices.
- (8) A statement that a certified individual will be responsible for management of erosion and sediment control practices at the site each day earth-moving activities occur.

- (9) Erosion and sediment control BMP maintenance and corrective action procedures, see paragraph EROSION AND SEDIMENT CONTROL INSPECTION AND CORRECTIVE ACTION.
  - (10) Include a draft (or blank) inspection form that will be used for Erosion and Sediment Control Inspections.
  - (11) Concrete wastewater/slurry containment and turn-in procedures.  
All concrete wastewater/slurry for disposal must be turned in to Building 357 by the end of the shift generated.
- f. Protection of the Environment from Waste Derived from Contract Operations

- (1) Control and disposal of solid and sanitary waste.
- (2) Control and disposal procedures for hazardous materials and hazardous waste management. Additional information for requirements below are found in paragraph HAZARDOUS WASTE/DEBRIS MANAGEMENT of this specification and may be necessary based on the work to be performed.

All hazardous waste generation must comply with the procedures as outlined in this specification as well as in accordance with Portsmouth Naval Shipyard standards.

At a minimum, this section must include the following:

- (a) A list and projected amount of potentially hazardous waste, universal waste or otherwise regulated waste streams that may be generated. Hazardous waste, universal waste, and other regulated waste (e.g., waste/used oil, concrete wastewater/slurry, paint and paint related debris, used aerosol cans, etc.) must be turned in to Building 357, Hazardous Waste Storage Facility (HWSF) by the end of the shift generated, unless otherwise approved by Code 106.3. The end of the shift generated is the end of the Building 357 shift, which run from 0600 - 1400 Monday through Friday;
- (b) Procedures to be employed to complete a written waste determination for wastes which are to be generated. Written waste determinations and procedures will be as requested by the Government;
- (c) Sampling/analysis plan, including laboratory method(s) that will be used for waste determinations and copies of relevant laboratory certifications (See paragraph CONTROL AND MANAGEMENT OF HAZARDOUS WASTES of this specification for details);
- (d) Management procedures for daily accumulation, storage (if approved by Code 106.3), labeling, transportation, turn-in and disposal of hazardous materials and hazardous waste. All waste turned in to Building 357 must be in containers provided by HWSF personnel and have the contents clearly identified. Disposal or treatment of hazardous waste by the Contractor is not allowed unless specifically noted in contract documents or approved by Code 106.3;
- (e) Management procedures and regulatory documentation ensuring disposal of hazardous waste complies with Land Disposal

Restrictions, as applicable (40 CFR 268);

(f) Management procedures for recyclable hazardous materials such as lead-acid batteries, used oil, and similar;

(g) Waste/used oil management procedures in accordance with 40 CFR 279, this specification, and Portsmouth Naval Shipyard standards. All waste/used oil must be turned in to Building 357 by the end of the shift generated. The end of the shift generated is the end of the Building 357 shift, which run from 0600 - 1400 Monday through Friday;

(h) Pollution prevention/hazardous waste minimization procedures;

(i) Plans for the disposal of hazardous and other regulated waste by permitted facilities (as approved by Code 106.3); and

(j) Procedures to be employed to ensure all required employee training records are maintained.

g. Prevention of Releases to the Environment

This plan section must include:

(1) Procedures to prevent releases to the environment and emergency response procedures and notifications to make in the event of a release to the environment. See paragraph RELEASES/SPILLS OF OIL AND HAZARDOUS SUBSTANCES of this specification for additional detail on items to include in this plan section.

(2) A Spill Prevention, Control, and Countermeasures (SPCC) Plan is required if work is anticipated to extend beyond 6 months, AND will use bulk oil storage containers 55 gallons or greater, in accordance with 40 CFR 112. All SPCC plans must be approved by Code 106.3. Plans need not be certified by a Professional Engineer but must clearly demonstrate proper management of all tanks and containers on site. The SPCC plan must include:

(a) Type of tank or container, quantity stored, type of product stored, and location.

(b) Secondary containment required for tanks/containers 55 gallons or greater; double-wall tanks preferred.

(c) Tank inspection forms (industry standard forms are acceptable, but the use of Portsmouth Naval Shipyard inspection forms is preferred). Records must be kept for 3 years or for the duration of the project. Tanks must be inspected monthly.

i) Bulk storage containers, to include those on equipment, 55 gallons or greater require monthly inspection.

ii) Inspection procedures and an inspection sheet for the release of retained stormwater from secondary containment used for bulk storage tanks and containers.

(d) Where spill kits are located and a description of the spill kit contents for the type of spill anticipated.

(e) If transferring fuel: how often, what type of fuel, and where. Coordinate with the Contracting Officer and Code 106.3 prior to transferring fuel over water.

(f) Who to notify in case of ANY spill (Portsmouth Naval Shipyard Fire Department: 207-438-2333, Contracting Officer, NRC, MEDEP, etc.). All regulatory notification must be coordinated with Code 106.3.

(g) How to clean up a spill safely and a statement that all spill cleanup waste will be brought to Building 357 by the end of the shift generated. The end of the shift generated is the end of the Building 357 shift, which run from 0600 - 1400 Monday through Friday.

#### h. Air Emissions/Air Permitting

(1) Installation of a permanent emergency generator for use by the government requires coordination with Code 106.3 for air permitting determination prior to the installation of any portion of the generator or generator support infrastructure.

(2) Provide details of painting and paint blasting operations to include the following, as applicable:

(a) Spray painting and blasting must be performed within containment.

(b) Tarps used to contain the material must prevent visible emissions from exceeding 20% opacity.

(c) Unless otherwise approved, debris collection systems must be suspended from the structure and extend a minimum of 10 feet wider in all directions than the area in which work is done. Collection systems must be positioned in a manner such that no loss of paint residue occurs, emptied at least once a day and must not remain in place overnight without being emptied.

(d) Painting and Blasting operations must not be performed when the direction and velocity of the wind is such that residue falls outside the limits of the collector.

(e) Spray painting dry fall paint will not be allowed if wind speeds exceed 30 feet/s (20mph), unless approved by Code 106.3.

#### i. Regulatory Notification and Permits

List what notifications must be made and what permits must be obtained for the work. Demonstrate that those permits were obtained or applied for by including copies of all applicable environmental permits. The EMP will not be approved until the permits have been obtained.

##### 3.1.1 Environmental Management Plan Review

The Contractor must submit a draft Environmental Management Plan to the Government at least 15 calendar days prior to the Preconstruction Meeting or Environmental Brief, whichever comes first in the project schedule. This will allow for sufficient review and comment prior to important preconstruction meetings, and provide the opportunity for constructive

feedback and collaboration on the important submittal.

Commencement of work must not begin until the Environmental Management Plan has been approved by the Contracting Officer, Code 106.3, and PWD-ME EV Staff.

### 3.1.2 Licenses, State and Federal Permits

Copies of the approved permit(s) are available from the Contracting Officer. Maintain copies of all permits at the project site. Comply with all terms and conditions of the approved permits.

Where required by the State regulatory authority, the inspections and certifications will be provided through the services of a Professional Engineer (PE), registered (licensed) in the State of Maine. Where a PE is not required, the individual must be otherwise qualified by other current State licensure, specific training, and prior experience (minimum 5 years). As a part of the Contractor Quality Control Plan, which is required to be submitted for approval by Section 01 45 00.00 22 QUALITY CONTROL (PWD ME), provide a sub item containing the name, appropriate professional registration or license number, address, and telephone number of the professionals or other qualified persons who will be performing the inspections and certifications for each permit.

### 3.2 PROTECTION OF NATURAL RESOURCES

Preserve the natural resources within the project boundaries and outside the limits of permanent work and as specified in the permits issued for the work. Restore to an equivalent or improved condition upon completion of work. Confine construction activities to within the limits of the work indicated or specified.

Do not disturb fish and wildlife. Do not alter water flows or otherwise significantly disturb the native habitat adjacent to the project and critical to the survival of fish and wildlife, except as indicated or specified.

Except in areas to be cleared, do not remove, cut, deface, injure, or destroy trees or shrubs without the Contracting Officer's permission. Do not fasten or attach ropes, cables, or guys to existing nearby trees for anchorages unless authorized by the Contracting Officer. Where such use of attached ropes, cables, or guys is authorized, the Contractor will be responsible for any resultant damage.

Protect existing trees which are to remain to ensure they are not injured, bruised, defaced, or otherwise damaged by construction operations. Remove displaced rocks from uncleared areas. By approved excavation, remove trees with 30 percent or more of their root systems destroyed. Remove trees and other landscape features scarred or damaged by equipment operations, and replace with equivalent, undamaged trees and landscape features. Obtain Contracting Officer's approval before replacement. Tree wound paint must not be used for tree cuts or stumps.

In the event that tree protection is no longer feasible, or it is identified in the field that further tree removal is required the Contractor shall reach out to the PWD-ME EV Natural Resources Manager, via the Contracting Officer, prior to any tree or vegetation clearing. Additional tree clearing without approval is not permitted. Further mitigation and instruction is provided in the TREE AND SHRUBBERY

PRESERVATION AND REPLACEMENT POLICY FOR NAVFAC PWD-ME INSTALLATIONS  
(Attachment B).

3.2.1 Erosion and Sediment Control Measures

The State of Maine Erosion and Sediment Control Law requires persons undertaking activity involving filling, displacing, or exposing soil or other earthen materials to take measures to prevent unreasonable erosion of soil or sediment beyond the project site or into a protected natural resource.

At the Portsmouth Naval Shipyard, the Piscataqua River, Upper Meade Pond, and Lower Meade Pond are protected natural resources under State Law. Erosion control measures must be in place before the activity begins, be maintained, and must remain in place and functional until the site is permanently stabilized.

Temporary and permanent erosion control measures must meet, at a minimum, the Best Management Practices (BMPs) presented in the Maine Erosion and Sediment Control Practices Field Guide for Contractors, latest edition. Other techniques may be employed if approved by Code 106.3 and PWD-ME EV and demonstrated to the Contracting Officer that the practice will achieve the required result of no release of sediment per State law.

Site work including any filling, excavation, landscaping, and/or other earthwork in excess of one cubic yard of disturbance must comply with State of Maine requirements for certification in erosion and sediment control practices within a shoreland zone. A certified individual must be responsible for management of erosion and sediment control practices at the site each day earth moving activities occur. A certified individual is required to visit the site every day to ensure proper erosion and sediment control practices are followed. As an alternative, the Contractor may choose to contract with a certified individual to supervise the work in shoreland areas.

The shipyard is entirely within the federal coastal zone as administered under the Coastal Zone Management Act. The shoreland zone is any area within 250 feet of mean high water.

a. Stormwater Management/Erosion and Sedimentation Control Plan

- (1) Submit a Stormwater Management/Erosion and Sedimentation Control Plan to the Contracting Officer, for review and approval by Code 106.3 and PWD-ME Environmental, for all earthwork in excess of one cubic yard of disturbance. The Plan must demonstrate effective selection, implementation and maintenance of BMPs demonstrating compliance with the Portsmouth Naval Shipyard's Maine Pollutant Discharge Elimination System Municipal Storm-Separate Sewer System Permit (MS4).

CMR 06-096 Chapter 500 Stormwater Management and terms and conditions specified in other approved permits (e.g., Maine Construction General Permit (MCGP), U.S. Army Corps of Engineers (ACOE)) issued for the work.

Provide details of chosen temporary erosion and sediment controls to be employed specific to the work site to include a site plan showing the locations of controls. Ensure proposed controls comply with MEDEP approved plans and State regulations.

Submit Stormwater Management and Erosion Control Compliance Notebook at project completion or as directed by the Contracting Officer.

The Plan must:

- (a) Identify potential sources of pollution which may be reasonably expected to affect the quality of stormwater discharge from the site.
- (b) Describe and ensure implementation of practices which will be used to reduce the pollutants in stormwater discharge at the manufacturing, storage and lay down, and construction sites.
- (c) Describe and ensure full compliance with all MEDEP General Permits (MS4/MCGP) and any other regulatory permits (e.g., ACOE, USEPA) specific to the project.
- (d) Describe and ensure compliance with MEDEP over winter stabilization and construction requirements, as applicable.
- (e) Identify inspections and maintenance schedules for BMPs demonstrating compliance with the State of Maine standards. Maintenance procedures must address regular cleaning of drainage structures and repair of temporary erosion control structures, as well as a final cleaning of all drainage structures and removal and reclamation of temporary erosion and sediment control BMPs upon completion of the project.
- (f) Select applicable BMPs from the Maine Erosion and Sediment Control Practices Field Guide for Contractors, latest edition. Submit a site plan showing locations of controls and provide manufacturer's product sheets for each control to be used.
- (g) Include documentation that the individual responsible for management of erosion and sediment control practices at the site is certified in accordance with the State of Maine DEP regulations.
- (h) Control of Manufactured Concrete Product Waste Plan. At a minimum, must include a description of concrete manufacturing activities and the management of concrete wastewater/wash water as described in paragraph herein entitled CONCRETE WASH WATER.

### 3.2.2 Stockpiles

Assumed Non-Hazardous soil stockpiles require erosion and sediment controls around the downhill leading edge of the stockpile, or if a leading edge cannot be found, around the entire stockpile perimeter; a MEDEP approved erosion control cover on top of the stockpile when inactive 7 or more days, when stockpile operations are complete and while awaiting soil characterization test results; and while awaiting test results, placarded as Pending Characterization and based on those sample results placarded as either Non-Hazardous or Hazardous. Immediately cover with 6 mil plastic sheeting if soil characterization test results return as Hazardous.

Known Hazardous (IR Soil) or suspect contaminated soil stockpiles require 6 mil plastic sheeting underneath; erosion control sock around the entire



stockpile perimeter; a 6 mil plastic sheeting cover unless actively being worked or while obtaining soil characterization test samples; and at all times placarded as Hazardous for IR soil stockpiles. Suspect contaminated soil stockpiles must be placarded as Pending Characterization while awaiting soil characterization test results and based on those results placarded as either Non-Hazardous or Hazardous. IR soils stockpiles must be located within the boundaries of the IR site from which the soil originated. When working in or near IR sites, the contractor shall clearly pre-mark the IR site boundary to ensure soil remains segregated within their respective zones.

### 3.2.3 Erosion and Sediment Control Inspection and Corrective Action

Inspection reports must be kept on file at the project site and submitted electronically to the Contracting Officer upon request. The State of Maine requires inspections of disturbed and impervious areas, erosion and sediment control measures, areas used for storage that are exposed to precipitation, and locations where vehicles enter or exit the site.

Inspections must be performed at least once per week as well as BEFORE and AFTER a storm event. A storm event is any precipitation event with the potential to create runoff but at a minimum for every storm resulting in 0.25 inches of precipitation or greater. Inspection reports must document compliance with State of Maine requirements. If erosion and sediment control BMPs need to be repaired, the repair work should be initiated upon discovery of the problem but no later than the end of the next workday. If additional BMPs or significant repair of BMPs are necessary, implementation must be completed within 7 calendar days and prior to any storm event (rainfall). For each BMP requiring maintenance, BMP needing replacement, and location needing additional BMPs, note in the inspection report the corrective action taken and when it was taken.

### 3.2.4 Burnoff

Burnoff of the ground cover is not permitted.

## 3.3 PROTECTION OF CULTURAL RESOURCES

### 3.3.1 Archaeological Resources

Existing archaeological resources within the work area are shown on the drawings. Protect these resources and be responsible for their preservation during the life of the Contract. Refer to Section 01 44 00.00 22 ARCHAEOLOGICAL MONITORING (PWD ME) for requirements on archaeological monitoring, inspection, and reporting. The Government retains ownership over archaeological resources.

### 3.3.2 Historical Resources

Existing historical resources within the work area are shown on the drawings. Protect these resources and be responsible for their preservation during the life of the Contract.

## 3.4 SOLID WASTE MANAGEMENT

### 3.4.1 SOLID WASTE MANAGEMENT PLAN

Provide a written Solid Waste Management Plan (SWMP) to the Contracting Officer, of intended licensed disposal sites for Government approval and

for submission to State regulatory agencies. At a minimum, the SWMP must contain, but not be limited to, the following wastes: stumps and grubbings, excess soil, construction debris, demolition debris, household solid waste, special waste, and industrial solid waste. The submission must contain the name of the disposal facility, address, facility phone number, and the waste type and quantity to be disposed of at the facility.

If waste from the site is taken to a transfer station, identify the facility or facilities at which the waste is ultimately disposed. Government approval for the facility must be obtained prior to transporting wastes off Government property.

Provide the Contracting Officer with written notification of the quantity of anticipated solid waste or debris that is anticipated or estimated to be generated by construction. Include in the report the locations where various types of waste will be disposed or recycled. Include letters of acceptance from the receiving location or as applicable; submit one copy of the receiving location State and local Solid Waste Management Permit or license showing such agency's approval of the disposal plan before transporting wastes off Government property.

#### 3.4.2 Monthly Solid Waste Disposal Report

Monthly, submit a solid waste disposal report to the Contracting Officer. For each waste, the report will state the classification (using the definitions provided in this Section), amount, location, and name of the business receiving the solid waste.

Provide copies of the waste handling facilities' weight tickets, receipts, bills of sale, and other sales documentation. In lieu of sales documentation, a statement indicating the disposal location for the solid waste that is signed by an employee authorized to legally obligate or bind the firm may be submitted. The sales documentation must include the receiver's tax identification number and business, EPA, or State registration number, along with the receiver's delivery and business addresses and telephone numbers. For each solid waste retained for the Contractor's own use, submit the information previously described in this paragraph on the solid waste disposal report. Prices paid or received do not have to be reported to the Contracting Officer unless required by other provisions or specifications of this Contract or public law.

#### 3.4.3 Control and Management of Solid Wastes

Pick up solid wastes, and place into containers that are regularly emptied. Containers must be kept covered except when being loaded with trash and debris. Watertight covered dumpsters and roll-offs are preferred. Do not prepare or cook food on the project site. Prevent contamination of the site or other areas when handling and disposing of wastes. At project completion, leave the areas clean. Recycling is encouraged and can be coordinated with the Contracting Officer and the Portsmouth Naval Shipyard Recycling Coordinator. Remove all solid waste (including non-hazardous debris) from Government property and dispose off-site at an approved landfill. Solid waste disposal off-site must comply with the most stringent local, State, and Federal requirements including 40 CFR 241, 40 CFR 243, and 40 CFR 258. Discharges from dumpsters and roll-offs are prohibited.

Manage hazardous material used in construction including, but not limited to, aerosol cans, waste paint, cleaning solvents, contaminated brushes,

and used rags, as per environmental law and Portsmouth Naval Shipyard requirements. Note: Scrap metal coated with PCB contaminated paint > or = to 50 ppm is considered a hazardous waste and requires Code 106.3 coordination for recycling/disposal.

#### 3.4.3.1 Dumpsters and Roll-offs

Dry weather discharges from dumpsters and roll-offs are prohibited. Equip dumpsters and roll-offs with a secure lid or covering. Keep covers closed at all times, except when being loaded with trash and debris, or provide secondary containment to ensure that discharges have a control. Locate dumpsters and roll-offs behind the construction fence or out of the public view and away from storm drains. Empty site dumpsters and roll-offs at least once a week or as needed to keep the site free of debris and trash. If necessary, provide 55-gallon trash containers to collect debris in the construction site area. Locate the trash containers behind the construction fence or out of the public view. Empty trash containers at least once a day. For large demolitions, dumpsters and roll-offs must use a cover and watertight containers are preferred; debris must not be higher than the sides before emptying. Water must not be collected in watertight roll-offs or dumpsters. Any water collected in roll-offs, dumpsters, or secondary containment could become contaminated with oil or other contaminants and may require waste characterization by the Contractor prior to release and/or removal from the Portsmouth Naval Shipyard.

#### 3.5 WASTE DETERMINATION DOCUMENTATION

Complete a Waste Determination form (provided by the Contracting Officer at the preconstruction meeting) for Contractor-derived wastes to be generated, as required by Code 106.3. Base the waste determination upon either a constituent listing from the manufacturer used in conjunction with consideration of the process by which the waste was generated, EPA approved analytical data, and/or laboratory analysis (Safety Data Sheets (SDS) by themselves are not adequate). Attach support documentation to the Waste Determination form. A Waste Determination form may be requested for the following types of wastes (this listing is not all inclusive): oil- and latex-based painting and caulking products, solvents, adhesives, aerosols, petroleum products, and containers of the original materials.

#### 3.6 POLLUTION PREVENTION/HAZARDOUS WASTE MINIMIZATION

Minimize the use of hazardous materials and the generation of waste. Include procedures for pollution prevention/ hazardous waste minimization in the Hazardous Waste Management Section of the Environmental Management Plan. Obtain a copy of the Portsmouth Naval Shipyard's Pollution Prevention/Hazardous Waste Minimization Plan for reference material when preparing this part of the EMP. If no written plan exists, obtain information by contacting the Contracting Officer. Describe the anticipated types of the hazardous materials to be used in the construction when requesting information.

#### 3.7 WHM/HW MATERIALS PROHIBITION

Do not improperly dispose of hazardous materials/hazardous waste on Government property. No hazardous material will be brought onto Government property that does not directly relate to requirements for the performance of this Contract.

Incidental materials used to support the contract including, but not

limited to, aerosol cans, waste paint, cleaning solvents, contaminated brushes, rags, clothing, empty containers, etc. may be considered hazardous wastes and must be turned in to Building 357 for disposal by the Government as described in the paragraph entitled HAZARDOUS MATERIAL MANAGEMENT of this Section. The list is illustrative rather than inclusive. Universal wastes must be managed with controls similar to those for hazardous waste.

Do not discharge any materials to sanitary sewer, storm drain, or to the Piscataqua River or conduct waste treatment or disposal on Government property without written approval by the Contracting Officer and Code 106.3.

### 3.8 HAZARDOUS MATERIAL MANAGEMENT

Hazardous materials for disposal may be considered hazardous wastes and must be turned in to Building 357 for hazardous waste determination by the end of the shift generated. The end of the shift generated is the end of the Building 357 shift, which run from 0600 - 1400 Monday through Friday.

Include hazardous material control procedures in the Environmental Management Plan as described in paragraph ENVIRONMENTAL MANAGEMENT PLAN (EMP) herein and as applicable in the Accident Prevention Plan (APP) in accordance with Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME). Address procedures and proper handling of hazardous materials, including the appropriate transportation requirements. All hazardous material must be in active use or properly stored. Hazardous materials must not be left unattended or uncontrolled while on Portsmouth Naval Shipyard.

Submit an SDS and estimated quantities to be used for each hazardous material to the Contracting Officer prior to bringing the material on the Portsmouth Naval Shipyard. Typical materials requiring SDS and quantity reporting include, but are not limited to, oil- and latex-based painting and caulking products, solvents, adhesives, aerosol, and petroleum products. Use hazardous materials in a manner that minimizes the amount of hazardous waste generated. Containers of hazardous materials must have National Fire Protection Association labels or their equivalent. Certify that hazardous materials removed from the site are hazardous materials and do not meet the definition of hazardous waste in accordance with 40 CFR 261 and state and Portsmouth Naval Shipyard requirements.

#### 3.8.1 Contractor Hazardous Material Inventory Log

Submit the "Contractor Hazardous Material Inventory Log" (found at: <https://www.wbdg.org/dod/ufgs/ufgs-forms-graphics-tables>), which provides information required by EPCRA Sections 312 and 313 along with corresponding Safety Data Sheets (SDS), to the Contracting Officer at the start and at the end of construction (30 days from final acceptance), and update no later than January 31 of each calendar year during the life of the Contract. Keep copies of the SDSs for hazardous materials onsite. At the end of the project, provide the Contracting Officer with copies of the SDSs, and the maximum quantity of each material that was present at the site at any one time, the dates the material was present, the amount of each material that was used during the project, and how the material was used. The Contracting Officer may request documentation for any spills or releases, environmental reports, or off-site transfers.

### 3.9 PETROLEUM PRODUCTS AND REFUELING

Conduct the fueling and lubricating of equipment and motor vehicles in a manner that protects against spills and evaporation, away from storm drains and surface waters. Manage waste/used oil generated on site in accordance with 40 CFR 279. Determine if any waste/used oil generated while onsite exhibits a characteristic of hazardous waste. All waste/used oil will be turned in to Building 357 and disposed of by the Government.

#### 3.9.1 Oil Storage Including Fuel Tanks

Provide secondary containment and overfill protection for oil and hazardous substance storage tanks. Prevent oil or hazardous substances from entering the ground, drainage areas, storm drains, or navigable waters. In accordance with 40 CFR 112, surround all temporary fuel oil or petroleum storage tanks with a temporary berm or containment of sufficient size and strength to contain the contents of the tanks, plus 10 percent freeboard for precipitation. Construct the berm to be impervious to oil for 72 hours and so that no discharge will permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Use drip pans during oil transfer operations; adequate absorbent material must be onsite to clean up any spills and prevent releases to the environment. Cover tanks and drip pans during inclement weather. Provide procedures and equipment to prevent overfilling of tanks.

Monitor and remove any rainwater that accumulates in open containment dikes, berms, or spill pallets. Inspect the accumulated rainwater prior to draining from a containment dike, berm or spill pallet to the environment, to determine there is no oil sheen present.

#### 3.9.2 Inadvertent Discovery of Petroleum Contaminated Soil or Hazardous Wastes

If petroleum contaminated soil, or suspected hazardous waste is found during construction that was not identified in the Contract Documents, immediately notify the Contracting Officer. Do not disturb this material until authorized by the Contracting Officer.

### 3.10 FUEL TANKS

Petroleum products and lubricants required to sustain up to 30 days of construction activity may be kept onsite. Storage and refilling practices must comply with 40 CFR 112. Secondary containment must be provided and be no less than 110 percent of the tank volume plus five inches of free-board. If a secondary berm is used for containment then the berm must be impervious to oil for 72 hours and be constructed so that any discharge will not permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Drip pans are required. Cover tanks and drip pans during inclement weather to prevent spills, as necessary.

### 3.11 RELEASES/SPILLS OF OIL AND HAZARDOUS SUBSTANCES

Exercise due diligence to prevent, contain, and respond to **ALL** spills of hazardous material, hazardous substances, hazardous waste, sewage, regulated gas, petroleum, lubrication oil, and other substances regulated in accordance with 40 CFR 300. In the event of a spill, take prompt, effective action to stop, contain, curtail, or otherwise limit the amount, duration, and severity of the spill/release. In the event of **ANY** releases of oil and hazardous substances, chemicals, or gases; immediately (within

15 minutes) notify the Portsmouth Naval Shipyard Fire Department 207-438-2333, the Portsmouth Naval Shipyard's Command Duty Officer, Code 106.3: 207-438-4477, and the Contracting Officer. If the response is inadequate, the Navy may respond. If this should occur, reimbursement to the Government for spill response assistance and analysis will be required. Contractor will provide SDSs, Waste Profile Sheets, and/or other technical data on the material spilled to emergency response personnel.

The Contractor is responsible for verbal and written notifications as required by the Federal 40 CFR 300.125 and 40 CFR 355, State, and local regulations and Navy Instructions. These notifications will be done in coordination with Code 106.3. Within 2 hours of notifying off-Shipyard officials, the Contractor must communicate with the Contracting Officer. Spill response must be in accordance with 40 CFR 300 and applicable State and local regulations. Contain and clean up these spills without cost to the Government. If Government assistance is requested or required, reimburse the Government for such assistance. Within 5 working days, provide copies of the written notification and documentation that a verbal notification was made within 20 days.

Maintain spill cleanup equipment and materials at the work site. Clean up all hazardous and non-hazardous (WHM) waste spills. Reimburse the Government for all material, equipment, and clothing generated during any spill cleanup. Reimburse the Government for all costs incurred including sample analysis materials, equipment, and labor if the Government must initiate its own spill cleanup procedures, for spills, when:

- a. Spill cleanup procedure has not begun within one hour of spill discovery/occurrence, or
- b. If, in the Government's judgment, the spill cleanup is not adequately abating a life-threatening situation and/or is a threat to any body of water or environmentally sensitive areas.

All discharged material and other spill cleanup wastes (e.g., absorbent, disposable clothing, brushes, rags, brooms, containers, decontamination water etc.), must be collected, containerized, properly marked and brought to Building 357 by the end of the shift generated, which run from 0600 - 1400 Monday through Friday. Decontamination water must be collected and turned in separately from other generated wastes or as directed by Code 106.3.

### 3.12 CONTROL AND MANAGEMENT OF HAZARDOUS WASTES

At the time of the preconstruction meeting, the Contractor will be briefed and provided written information regarding hazardous waste management. The Government will provide technical and oversight guidance in all aspects of hazardous waste management.

#### 3.12.1 General

All hazardous wastes generated within the confines of the Portsmouth Naval Shipyard must be disposed of by the Government. Accordingly, all hazardous wastes generated to accomplish the requirements of this Contract must be considered Government-generated, and must be disposed of by the Government. Do not bring hazardous wastes onto Government property. Hazardous wastes must be handled in compliance with 40 CFR 260-268, 273, 279 and State of Maine MEDEP Regulations Chapter 850 to 860. For hazardous waste spills, immediately notify the Portsmouth Naval Shipyard

Fire Department 207-438-2333, the Portsmouth Naval Shipyard's Command Duty Officer, Code 106.3: 207-438-4477, and the Contracting Officer.

### 3.12.2 Containers

Use only Government-furnished, Government-labeled containers for the packaging of hazardous soils and hazardous wastes. Containers are requested and picked up at Building 357 following approval of the Management Plan required above.

- a. Segregate hazardous and non-hazardous soils/wastes. Hazardous soils/wastes must be placed into containers provided by the Government. All containers of hazardous waste must have a Code 106.3 tracking number affixed to the container, as applicable.
- b. Hazardous soils must be turned over to the Government, at Building 357, as coordinated with Code 106.3. All other containers of waste hazardous materials and hazardous waste, full or partially full, must be turned in to Building 357 by the end of the shift generated. The end of the shift generated is the end of the Building 357 shift, which run from 0600 - 1400 Monday through Friday.

While in control of hazardous soils and hazardous wastes, the soils/wastes must be handled in accordance with Portsmouth Naval Shipyard requirements.

- c. Prior to Government acceptance of the containers, provide the certification required by the "Submittals" paragraph of this Section, and such additional information regarding contents of the containers. Submittal of a Waste Determination form may be required for proper waste characterization as requested by Code 106.3.

### 3.12.3 Hazardous Waste/Debris Management

Identify construction activities that will generate hazardous waste or debris and universal wastes. Provide a documented waste determination for resultant waste streams as requested by Code 106.3 or other Government personnel. Hazardous waste or debris must be identified, labeled, handled, stored, and disposed of in accordance with all Federal, State, and local regulations including 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, 40 CFR 265, 40 CFR 266, 40 CFR 268, Maine Department of Environmental Protection (MEDEP) MAINE 38 MSRA 850-860, this specification, and Portsmouth Naval Shipyard standards.

Manage hazardous wastes and universal wastes in accordance with the approved Hazardous Waste Management Section of the Environmental Management Plan. Store hazardous wastes and universal wastes in Government-provided containers in accordance with 49 CFR 173 and 49 CFR 178. Hazardous waste generated within the confines of Government facilities is identified as being generated by the Government.

Prior to removal of any hazardous waste from Government property, hazardous waste manifests must be signed by Portsmouth Naval Shipyard personnel from Code 106.3. Provide the Contracting Officer with a copy of waste determination documentation for any solid waste streams that have any potential to be hazardous waste or contain any chemical constituents listed in 40 CFR 372-SUBPART D.

### 3.12.3.1 Waste Storage/Satellite Accumulation/90 Day Storage Areas

If the work requires the temporary storage/collection of regulated or hazardous wastes, request the establishment of a Regulated Waste Storage Area, a Satellite Accumulation Area, or a 90 Day Storage Area at the point of generation. Regulated waste and Hazardous Waste Accumulation Areas (HWAAs) are approved by Code 106.3.

Submit a request in writing to the Contracting Officer providing the following information (Attach Site Plan to the Request):

|                                             |       |                        |       |
|---------------------------------------------|-------|------------------------|-------|
| <u>Contract Number</u>                      | _____ | <u>Contractor</u>      | _____ |
| <u>Haz/Waste or<br/>Regulated Waste POC</u> | _____ | <u>Phone Number</u>    | _____ |
| <u>Type of Waste</u>                        | _____ | <u>Source of Waste</u> | _____ |
| <u>Emergency POC</u>                        | _____ | <u>Phone Number</u>    | _____ |
| <u>Location of the Site:</u>                | _____ |                        |       |

Attach a Waste Determination form for the expected waste streams. Allow ten (10) working days for processing this request. Additional compliance requirements (e.g., training and contingency planning) that may be required are the responsibility of the Contractor. Barricade the designated area where waste is being stored and post a sign identifying as follows:

"DANGER - UNAUTHORIZED PERSONNEL KEEP OUT"

### 3.12.3.2 Sampling and Analysis of Hazardous Waste

#### a. Waste Sampling

Sample waste in accordance with EPA SW-846 , this specification, and Portsmouth Naval Shipyard standards. Each sampled drum or container must be clearly marked with the Contractor's identification number and cross referenced to the chemical analysis performed. All sampling events will require a Code 106.3 reviewed and approved sampling plan and availability for Code 106.3 oversight.

#### b. Laboratory Analysis

Follow the analytical procedure and methods in accordance with 40 CFR 261. Provide all analytical results and reports performed to the Contracting Officer and Code 106.3 Environmental Sampling Project Manager for review and approval.

All laboratory analysis for hazardous waste identification must be performed by a laboratory compliant with this specification and Portsmouth Naval Shipyard standards. Proof of compliance must be made available upon request. All analyses provided by laboratories that are not compliant with the stated requirements will be rejected.

#### c. Analysis Type

Identify waste material/hazardous waste by analyzing for properties



that are reasonably suspected of the waste. Soil and other materials may require specific analysis for acceptance to a disposal facility - contact Code 106.3 personnel at the Hazardous Waste Storage Facility, Building 357, before choosing parameters.

d. Typical Hazardous Waste and PCB Sampling Plan Requirements

- i. Confirmation that the sampling plan will be reviewed and approved by the Navy (Code 106.3) prior to conducting any sampling. Code 106.3 review period is fourteen (14) calendar days from receipt of the plan.
- ii. The timing of waste characterization sampling and analysis. PCB sampling must be conducted prior to the accomplishment of any work. To the extent feasible, all other waste sampling and analysis should be completed prior to the start of work. Waste sampling after the start of work requires Code 106.3 approval.
- iii. A project summary with a brief description of the project, a description of the anticipated waste and how the waste will be generated, a map or diagram with sample locations and type, an estimate of the amount of waste to be generated, and the name and location of the disposal facility the waste will go to if waste is to be disposed of offsite, as approved by Code 106.3.
- iv. A statement in the plan that the sample collection, management, and analysis will be performed per all applicable local, state, and federal statutes and regulations, project specifications, Navy and PNSY requirements to include Portsmouth Naval Shipyard standards.
- v. The proposed date(s) for sampling and expected duration of the sampling event(s). All hazardous waste and PCB sampling requires Code 106.3 oversight with seven (7) calendar day notification prior to a sampling event.
- vi. A statement that all materials used in the collection of the samples (gloves, wipes, trash, etc.) and hazardous debris generated by the sampling process will be containerized in a bag and delivered to Building 357 for disposal by the end of the shift generated.
- vii. A narrative of how the sample(s) will be taken (e.g. discrete, composite, grab), a description of the decontamination process for reusable equipment between consecutive samples, documentation that the Environmental Field Technician (EFT) conducting the waste sampling has training in proper collection and decontamination procedures, and a statement giving rationale for why the sample(s) will be representative.
- viii. A list of all equipment to be used for sampling. Examples include:
  1. Individual sample jars and vials;
  2. Field log;
  3. Site diagram;
  4. Chain-of-Custody (COC) forms;
  5. Nitrile gloves;
  6. Stainless steel or plastic soil scoops;

7. Metal chisels of assorted sizes;
  8. Razor knives and replacement blades;
  9. Water and spray bottle;
  10. Trash bags;
  11. Disposable cleaning wipes;
  12. Paper towels;
  13. Duct tape;
  14. Flashlight;
  15. Personnel lifts; and
  16. Other equipment as necessary.
- ix. A listing of the constituents to be analyzed for, identifying the analytical tests methods to be performed with the corresponding EPA test methods numbers.
- x. The name and location of the laboratory the samples will be sent to and confirmation that the selected lab has Maine Department of Environmental Protection certification, National Environmental Laboratories Accreditation Conference (NELAC) certification and certifications for all corresponding EPA test methods, as applicable.
- xi. Confirmation that the samples will be submitted, under strict chain-of-custody (COC) requirements, to the certified laboratory.
- xii. Confirmation that the sampling results report will be transmitted to the Contracting Officer and Code 106.3 for each material sampled. The report must include quality control data, be reported down to the Method Detection Limit (MDL), and include:
1. Laboratory analytical results;
  2. Field log book entries;
  3. Copy of the Chain-of-Custody (COC)
  4. Pertinent diagrams and maps; and
  5. All other supporting data

### 3.12.3.3 Asbestos Certification

Items, components, or materials disturbed by or included in work under this Contract may involve asbestos. Other materials in the general area around where work will be performed may contain asbestos. All thermal insulation, in all work areas, should be considered to be asbestos unless positively identified by conspicuous tags or previous laboratory analysis certifying them as asbestos free.

Inadvertent discovery of non-disclosed asbestos that will result in an abatement action requires a change in scope before proceeding. Upon discovery of asbestos containing material not identified in the Contract documents, immediately stop all work that would generate further damage to the material, evacuate the asbestos exposed area, and notify the Contracting Officer for resolution of the situation prior to resuming normal work activities in the affected area. Do not remove or perform work on any asbestos containing materials without the prior approval of the Contracting Officer. Do not engage in any activity, which would remove or damage such materials or cause the generation of fibers from such materials.

Asbestos containing waste must be managed and disposed of in accordance with applicable environmental law. Asbestos containing waste must be

manifested and the manifest provided to the Contracting Officer. Disposal of asbestos-containing waste must be coordinated with PWD-ME EV and Code 106.3.

Provide the Asbestos Free Certification Form prior to the Government taking beneficial occupancy certifying that all materials, including those supplied by third parties, are asbestos free. The Contracting Officer will provide the form upon request.

#### 3.12.3.4 Polychlorinated Biphenyls (PCBs)

Polychlorinated Biphenyls (PCBs) were manufactured between 1950 and 1979 and widely used in building construction and renovation materials (e.g., paints, varnishes, caulks, rubber/felt gaskets, adhesives, siding, mastics, roofing, tar paper, fireproofing etc.). Work may include the disturbance of items, components, or materials containing PCBs as identified in project documents. Suspect PCB containing materials must be analytically tested prior to disturbance and managed per 40 CFR 761 and applicable environmental law. If adequate PCB testing for industrial hygiene and waste disposal is not provided in the contract documents, the Contractor shall be responsible for PCB suspect materials testing prior to the start of work.

- a. Sampling of suspect PCB material requires a Code 106.3 reviewed and approved sampling plan and Code 106.3 oversight.
- b. Disposal of all PCB containing waste(s)  $\geq 1$  ppm must be coordinated with B357/Code 106.3 personnel.

Inadvertent discovery of undisclosed PCBs resulting in a new or expanded abatement action requires a change in scope review before proceeding. Upon discovery of PCB containing material not identified in the Contract documents, immediately stop all work that could impact PCB containing materials and notify the Contracting Officer for resolution of the situation prior to resuming normal work activities in the affected area.

#### 3.12.3.5 Hazardous Waste Disposal

Control of stored waste, packaging, sampling, analysis, and disposal will be determined by the details in the Contract. The requirements for jobs in the following paragraphs will be used as the guidelines for disposal of any hazardous waste generated and disposed of by the Contractor. All hazardous waste disposed of by the Contractor must be approved by and coordinated with Code 106.3 prior to transporting wastes off Portsmouth Naval Shipyard property.

##### a. Responsibilities for Disposal

Responsibilities include any generation of WHM/HW requiring disposal of solid waste or liquid.

- (1) Agree to provide all service necessary for the final treatment/disposal of the hazardous material/waste in accordance with all local, State, and Federal laws and regulations, and the terms and conditions of the Contract within sixty (60) days after the materials have been generated. These services must include all necessary personnel, labor, transportation, packaging, detailed analysis (if required for disposal, and/or

transportation, including manifesting or completing waste profile sheets, equipment, and the compilation of all documentation is required).

- (2) Contain all waste in accordance with 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, 40 CFR 265, 40 CFR 266, 40 CFR 268, 40 CFR 270, 40 CFR 272, 40 CFR 273, 40 CFR 279, 40 CFR 280, and 40 CFR 761.
- (3) Obtaining a representative sample of the material generated for each job done to provide waste stream determination.
- (4) Analyzing for each sample taken and providing analytical results to the Contracting Officer. Provide two copies of the results.
- (5) Determine the DOT proper shipping names for all waste (each container requiring disposal) and will demonstrate how this determination is developed and supported by the sampling and analysis requirements contained herein to the Contracting Officer.

#### Disposal Turn-In Requirements

For any waste hazardous materials or hazardous waste generated which requires disposal, the following conditions must be met in order to be acceptable for disposal:

- a. Drums compatible with waste contents and drums meet DOT requirements for 49 CFR 173 for transportation of materials.
- b. Drums banded to wooden pallets. No more than three (3) 55-gallon drums to a pallet, or two (2) 85-gallon over packs.
- c. Band using 1-1/4 inch minimum band on upper third of drum.
- d. Contents of drum identified on the outside of the drum, as well as the volume of material, name of material manufacturer, and other vendor information as available.
- e. Always have three (3) to five (5) inches of empty space above volume of material. This space is called 'outage'.
- f. Provide disposal documentation for hazardous and regulated waste.

#### 3.12.4 Class I ODS Prohibition

Class I ODS must not be used in the performance of this Contract, nor be provided as part of the equipment. This prohibition will be considered to prevail over any other provision, specification, drawing, or referenced documents. Class I and II ODSs are Government property and must be returned to the Government for appropriate management. Coordinate with the Installation Environmental Office to determine the appropriate location for turn in of all reclaimed refrigerant. Regulations related to the protection of stratosphere ozone may be found in 40 CFR 82.

Heating and air conditioning technicians must be certified through an EPA-approved program. Copies of certifications must be maintained at the employees' place of business and be carried as a wallet card by the technician, as provided by environmental law. Accidental venting of a refrigerant is a release and must be reported to the Contracting Officer.

### 3.12.5 Universal Waste/e-Waste Management

Universal waste including, but not limited to, some mercury containing building products such as florescent lamps, mercury vapor lamps, high pressure sodium lamps, CRTs, batteries, aerosol paint containers, electrical equipment containing PCBs, and consumed electronic devices, must be managed in accordance with 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 273, MAINE DEP 38 MSRA 858, and applicable environmental law.

### 3.13 DUST CONTROL

Dust control must meet the requirements of the BMPs in the Maine Erosion and Sediment Control Practices Field Guide for Contractors, latest edition. Keep dust down at all times, including during nonworking periods. Maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and other work areas within or outside the project boundaries free from particulates that would exceed 40 CFR 50, state, and local air pollution standards or that would cause a hazard or a nuisance. Sprinkle or treat, with dust suppressants, the soil at the site, haul roads, and other areas disturbed by operations. Dry power brooming will not be permitted. Instead, use vacuuming or sweeping. Air blowing will be permitted only for cleaning nonparticulate debris such as steel reinforcing bars. Only wet cutting will be permitted for cutting concrete blocks, concrete, and bituminous concrete. Runoff from cutting water is prohibited from entering storm drains or leaving the work site. Do not unnecessarily shake bags of cement, concrete mortar, or plaster.

When temporary dust control measures are employed, repetitive treatment must be applied as needed to accomplish control.

Visible emissions from a fugitive emission source (including stockpiles and roadways) must not exceed an opacity of 20 percent, except for no more than five (5) minutes in any 1-hour period.

Dust suppression during demolition work may include using a manned water hose. Runoff from the site is prohibited.

#### 3.13.1 Dirt and Dust Control Plan

Submit truck and material haul routes along with a plan for controlling dirt, debris, and dust on Portsmouth Naval Shipyard roadways. As a minimum, identify in the plan the Subcontractor and equipment for cleaning along the haul route and measures to reduce the dirt, dust, and debris from roadways.

### 3.14 ABRASIVE AND/OR WET BLASTING AND WATER CUTTING

#### 3.14.1 Blasting Operations

##### a. Abrasive Blasting

The use of silica sand is prohibited in sandblasting.

Provide tarpaulin drop cloths and windscreens to enclose abrasive blasting operations to confine and collect dust, abrasive, agent, paint chips, and other debris.

Abrasive blasting must take place in containments with emissions vented through bag house filters and emissions must be limited to 10% opacity on a six minute block average. The bag houses must be used to control PM emission and operated properly at all times abrasive blasting is being performed.

b. Wet Blasting and Water Cutting

The use of wet blasting requires the capture and proper disposal of all wastes, including the blasting water, associated with the process. Provide enclosed containments to confine and collect wastewater and debris.

3.14.2 Disposal Requirements

Submit analytical results of the wastes and/or debris generated from blasting operations per paragraph entitled LABORATORY ANALYSIS of this Section. Hazardous waste generated from blasting operations must be managed in accordance with paragraph entitled HAZARDOUS WASTE/DEBRIS MANAGEMENT of this Section and the approved EMP. Concrete wash water and oily waste generated from blasting operations must be disposed of in accordance with the policy outlined in these specifications.

3.15 SPRAY PAINTING OPERATIONS

Spray painting operations must take place in containment. Emissions from spray painting must vent through air filters and are limited to 10% opacity on a six-minute block average. The air filters are used to control particulate emissions.

3.16 NOISE

Make the maximum use of low-noise emission products, as certified by the EPA and/or sound deadening enclosures to limit noise within the limits of work. Blasting or the use of explosives will not be permitted. Confine any operations that may generate excessive noise to the period between 7 a.m. and 5 p.m., Monday through Friday, exclusive of holidays, unless otherwise specified or approved by the Contracting Officer.

The maximum permissible sound pressure levels, as measured at the limits of the Navy Property boundary, must not exceed the maximum noise levels as specified in the Town of Kittery's Ordinance and all applicable OSHA Regulations. The Kittery Ordinance specifies the maximum permissible sound level limit at the Navy Property Boundary to be 65 dBA from 0700 to 2100 and 60 dBA from 2100 to 0700.

For any construction or demolition operations that may generate noise levels in excess of 80 dBA in adjacent facility work areas or abutting Non-Government owned properties adjacent to the work area, the Contractor must provide a Noise Mitigation Plan to the Contractor Officer at least 15 calendar days prior to commencing any work that would create noise the would exceed the specified maximum noise levels. The Noise Mitigation Plan may include the use low-noise emission equipment, sound deadening enclosures or other mitigation measures to ensure compliance with the Kittery Noise Ordinance levels or the adjacent work areas are not impacted.

Provide all noise monitoring equipment necessary to record and document noise levels associated with any operations generating excessive noise to

ensure these noise levels specified are not exceeded. Any work activities that generates noise that exceeds the specified maximum noise levels must stop immediately. The Contractor must take corrective actions to ensure the noise levels are mitigated. Any delays in the work, are the Contractors responsibility and will not be justification for a time extension.

### 3.17 MERCURY MATERIALS

Mercury is prohibited in the work of this Contract, unless specified otherwise, and with the exception of mercury vapor lamps and fluorescent lamps. Dumping of mercury-containing materials and devices such as mercury vapor lamps, fluorescent lamps, and mercury switches in rubbish containers is prohibited. Remove without breaking, pack to prevent breakage, and transport out of the activity in an unbroken condition for disposal as directed. Immediately (within 15 minutes) notify instances of breakage or mercury spillage to the Portsmouth Naval Shipyard Fire Department 207-438-2333, the Portsmouth Naval Shipyard's Command Duty Officer, Code 106.3: 207-438-4477, and the Contracting Officer. Clean the mercury spill area(s) to the satisfaction of the Contracting Officer. Cleanup of a mercury spill must not be recycled and must be managed as a hazardous waste for disposal.

### 3.18 CONCRETE WASH WATER

Concrete wash water is water, pressure washing water, or storm water that has come into contact with cement, uncured concrete, concrete dust, or other material of a similar nature generated during construction activities including, but not limited to, washing down ready-mix trucks, mixers, wheelbarrows, pre casting equipment, forms, manufactured cast concrete sections, tools, concrete areas; masonry cutting operations; cleaning up of split mortar or block fill; and hosing away excess materials.

Water or stormwater that has come into contact with pre casting equipment, forms, tools, etc. which have been subjected to oil based form release agents will be considered an oily waste if a visual inspection indicates any signs of oily residue. Oily wastes must be collected and disposed of in accordance with Portsmouth Naval Shipyard policy.

Concrete wash water must be collected and either placed in a concrete wash out structure for settling solids and evaporating liquid or turned-in as wastewater to Building 357 by the end of the shift generated. As approved by Code 106.3, large volumes of concrete wastewater may be disposed of at a facility designed and approved for the disposal or recycling of concrete wastewater. Under no circumstances will clean water be added to concrete wash water/wastewater for dilution purposes or any other reason. Containment wash out structures must be watertight, designed to promote evaporation, and provide adequate freeboard to contain the wash water, solids, and rainfall to prevent overflow. Washout must only occur in designated areas that have been approved by the Contracting Officer.

Inspect all concrete wash out structures daily to determine filled capacity. Remove all materials from containment wash out structures when 75% fill capacity has been reached. Concrete wash out structure operations must be conducted to reduce the amount of concrete wastewater generated for disposal. Remove liquids or cover structures before predicted rainstorms to prevent overflows and infiltration of rainwater. Inspect structures for holes and tears daily and repair to maintain

watertight conditions.

Hardened solids can be removed from containment structures and recycled, reused, or disposed of per regulatory requirements. Liquids remaining in the containment structure must be vacuumed and turned-in to Building 357 or, as approved by Code 106.3, disposed of at a facility designated for disposal/recycling of concrete wastewater.

#### 3.18.1 Pollution Prevention

Store dry and wet concrete supplies under cover away from drainage areas. Concrete wash water must not be released to the storm drain system, sewer system, roadways, other uncontained impervious surfaces, or to natural waterways including the Piscataqua River and its tributaries. Take all precautions necessary to prevent rainwater or stormwater runoff from coming into contact with concrete wash water. Divert clean stormwater and roof runoff from contact with concrete wash water. Take all measures necessary to minimize the volume of concrete wash water generated. Protect all waterways, catch basins, and storm drain structures from potential discharges of concrete wash water. Collect and control concrete wash water separately from wastewater determined to be oily waste.

#### 3.18.2 On-Shipyard Disposal

ALL CONCRETE WASTEWATER MUST BE COLLECTED AND TURNED-IN TO BUILDING 357 BY THE END OF THE SHIFT GENERATED. CONCRETE WASH WATER THAT IS COLLECTED AND DEPOSITED IN WASH OUT STRUCTURES MUST BE TURNED IN TO BUILDING 357 AT OR BEFORE THE CONCRETE WASH OUT STRUCTURE REACHES 75% CAPACITY.

#### 3.18.3 Off-Shipyard Disposal

Concrete wastewater may be disposed of off-site as approved by Code 106.3. The Contractor must submit for approval the disposal facility designated for concrete wash water/wastewater and provide careful oversight to prevent improper dumping of concrete wash water. Any clean up resulting from improper control of concrete wash water will be at no expense to the Government.

### 3.19 DISPOSAL OF CHLORINATED WATER AND DECHLORINATION REQUIREMENTS

Chlorinated water created during disinfection procedures must not be directly discharged to storm drains or sanitary sewers without prior dechlorination and approval by Code 106.3. Chlorinated water must be neutralized by the controlled addition of a reducing chemical such as sodium thiosulfate, sodium bisulfate, sodium sulfite, sulfur dioxide, or ascorbic acid (commonly known as Vitamin C), as approved by Code 106.3. Dechlorination must be sufficiently effective to reduce total residual chlorine concentration to 1 mg/L or below.

### 3.20 SNOW STORAGE AND DISPOSAL

Site and operate snow storage and disposal areas to prevent the discharge of snow directly into surface waters and minimize discharges of pollutants from snow maintenance activities. As necessary, the Contractor must employ applicable snow dump BMPs from CMR 06-096 Chapter 573 Snow Dumps: Best Management Practices for Pollution Prevention for snow storage areas.

-- End of Section --





☐ INITIAL SITE WASTE REMOVAL (SWR) ☐ DAILY REMOVAL RECORD (DRR)

REVISION 3.0, 09DEC2021

| WASTE ACTIVITY DESCRIPTION                                                                                                                                                                      |      |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------------|
| 1. Contract No:                                                                                                                                                                                 |      |            |
| 2. Contract title:                                                                                                                                                                              |      |            |
| 3. Prime contractor company name:                                                                                                                                                               |      |            |
| 4. Hauling contractor company name:                                                                                                                                                             |      |            |
| 5. Anticipated hauling date range:                                                                                                                                                              |      |            |
| 6. Waste type: <input type="checkbox"/> Non-hazardous soils<br><input type="checkbox"/> Hazardous soils<br><input type="checkbox"/> Special waste _____<br><input type="checkbox"/> Other _____ |      |            |
| 7a. Waste source:<br><br>WMI Profile #:<br><br>7b. Current waste location:                                                                                                                      |      |            |
| 8. Anticipated total hauling amount (tons):                                                                                                                                                     |      |            |
| 9. PNSY approved waste disposal facility:                                                                                                                                                       |      |            |
| <b>WASTE REMOVAL APPROVALS</b> – The signatures below verify that an onsite visual inspection of the material has been performed, and that this form is a record of the facts agreed upon.      |      |            |
| 10. Prime contractor signature:                                                                                                                                                                 |      |            |
| 11. NAVFAC CM/ET signature:                                                                                                                                                                     |      |            |
| 12. Code 106.3 signature:                                                                                                                                                                       |      |            |
|                                                                                                                                                                                                 |      |            |
| DAILY REMOVAL RECORD - to be filled each morning of waste removal                                                                                                                               |      |            |
| 13. Hauling date:                                                                                                                                                                               |      |            |
| 14. Anticipated hauling amount:                                                                                                                                                                 | tons | truckloads |
| 15. Hauling contractor truck ID number(s):                                                                                                                                                      |      |            |
| 16. Prime contractor signature:                                                                                                                                                                 |      |            |
| 17. NAVFAC CM/ET signature:                                                                                                                                                                     |      |            |

## ☐ INITIAL SITE WASTE REMOVAL (SWR) ☐ DAILY REMOVAL RECORD (DRR)

### WASTE REMOVAL PASS ATTRIBUTE DESCRIPTIONS

1. *Contract No.* The unique NAVFAC contract identification number applied to the project requesting waste removal.
2. *Contract title.* The unique contract title/description applied to the project requesting waste removal.
3. *Prime contractor company name.* The project's prime contractor company name.
4. *Hauling contractor company name.* The subcontractor hired to haul waste from the project site to the PNSY approved waste disposal company.
5. *Anticipated hauling date(s).* Hauling date range agreed upon at SWR meeting for duration of waste removal in the approved phase of removal.
6. *Waste description.* Description of waste in terms of hazardous/non-hazardous designation and any other characteristic determined at the SWR meeting.
- 7a. *Waste source.* Detailed written description of the source of the waste to be removed. Include removal areas, buildings, streets, land marks, coordinates, and other information as available.
- 7b. *Current waste location.* Detailed written description of the waste location if displaced from the original source during removal. Include removal areas, bin designations, stockpile locations, buildings, streets, land marks, coordinates and other information as available.
8. *Anticipated total hauling amount (tons).* Total amount of waste anticipated for removal in the approved phase of removal.
9. *PNSY approved waste disposal facility.* Name of the PNSY approved waste disposal facility.
10. *Prime contractor signature.* Written signature of project's prime contractor.
11. *NAVFAC CM/ET signature.* Written signature of NAVFAC CM or ET in attendance at the SWR meeting.
12. *Code 106.3 signature.* Written signature of Code 106.3 representative in attendance at the SWR meeting.
13. *Hauling date.* Date of the waste removal.
14. *Anticipated hauling amount.* Entire hauling amount (total tons) anticipated for said hauling date.
15. *Hauling contractor truck ID number(s).* The identification number of the contractor's truck.
16. *Prime contractor signature.* Written signature of prime contractor.
17. *NAVFAC CM/ET signature.* Written signature of project's NAVFAC CM or ET overseeing daily removal of waste from the site.

**NAVFAC PWD-ME EV NATURAL RESOURCE PROGRAM  
TREE AND SHRUBBERY PRESERVATION AND REPLACEMENT POLICY FOR  
NAVFAC PWD-ME INSTALLATIONS**

Updated: 2/20/2025

- References:**
- (a) Executive Order 13148, Greening the Government through Leadership in Environmental Management of 22 April 00;
  - (b) OPNAV M-5090.1, Chapter 12, Natural Resources Conservation dated 25 June 2021;
  - (c) NAVFAC MIDLANT Tree Preservation and Protection Plan – Integrated Natural Resource Management Plan Appendix;
  - (d) NAVSHIPYD PTSMHINST 11041.3C, Maintenance of Trees and Shrubbery
  - (e) Portsmouth Naval Shipyard Natural Resource Process Document;
  - (f) Integrated Natural Resource Management Plan, Naval Support Activity Cutler, Cutler, Maine;
  - (g) Integrated Natural Resource Management Plan, Great Pond Outdoor Adventure Center, Great Pond, Maine;
  - (h) Integrated Natural Resource Management Plan for the Survival, Evasion, Resistance, and Escape School, Redington, Maine;
  - (i) Portsmouth Naval Shipyard Tree Inventory Report;
  - (j) Guide for Plant Appraisal, International Society of Arboriculture, 9th edition;
  - (k) Navy Timber Payback Clause dated July 2015;
  - (l) Memorandum of Agreement (MOA) Between the United States Department of Agriculture and the United States Department of Defense for the Conduct of Forest Insect and Disease Suppression on Lands Administered by the U.S. Department of Defense, 11 Dec 90;
  - (m) E.O. 13112, Invasive Species, as amended by E.O. 13751, Safeguarding the Nation from the Impacts of Invasive Species;
  - (n) Public Law 104–332, National Invasive Species Act of 1996;
  - (o) DoD Instruction 4715.03 of 18 March 2011, Natural Resources Conservation Program;

### 1. Purpose

- a. This policy establishes a program for forest conservation and tree and shrubbery protection at Naval Installations throughout the NAVFAC PWD-ME AOR. The hierarchy for protection initiatives begins with preservation of existing trees and green space wherever practical, and ends with mitigation at specified ratios to compensate for unavoidable loss.
- b. The policy parallels the intent of references (a)-(0). It has been determined that the protection and preservation of trees is not only desirable but essential to sustaining military missions throughout the PWD-ME AOR.

### 2. Policy

- a. It is the intent of this policy to prevent the unauthorized destruction or disfigurement of existing trees and shrubbery. It is further intended to perpetuate vegetation growth, encourage preservation and to provide means to sustain the functions and values currently provided by trees, shrubbery and forested areas.
- b. Where tree preservation is not practicable, mitigation for tree loss may be required. The Natural Resources Manager (NRM) will assist with preparation of tree inventories at potential development areas. Such inventories and determinations will be made during the preliminary siting phase of a project. References (e)-(i) contain forest inventory data at some NAVFAC PWD-ME Installations. Inventories have not been completed for all installations.
- c. Potential mitigation includes replacement planting and/or reimbursement for removal of wood products from federal lands per references (b) and (k). Payment for all wood products shall be made to the Navy's Forestry Program in accordance with reference (b) and (k). Project/activity proponents should consider costs for mitigation (including long-term maintenance) early in the planning process.
- d. Mitigation exempt projects and/or actions (as identified in Section 4) require NRM review and approval to ensure compliance with the intent of this policy and other federal and State natural resource regulations.

### 3. Application

- a. The terms and provisions of this policy shall apply to:
  - i. New site development, including roadway, utilities, training areas, and other infrastructure.
  - ii. Expansion of existing facilities, including roadway, utilities, training areas, and other infrastructure.
  - iii. Maintenance of clear zones.
  - iv. Grounds maintenance activities.

### 4. Mitigation Exempt

- a. Hazard trees. Hazard trees are considered trees that pose an eminent threat to personnel, property and/or operations.
- b. Commercial forestry operations, conducted under an approved Integrated Natural Resources Management Plan or Forest Management Plan.
- c. Tree removal necessary to meet critical military operational requirements.
- d. Mitigation exempt projects are still required to adhere to reference (k), Navy Timber Payback Clause for removal of wood product from federal lands. The NRM in

conjunction with the NAVFAC MIDLANT region forester are responsible for determining value of timber, if any.

5. Endangered Species Act Compliance

- a. All projects and actions involving the removal of trees that are 3-inches or greater in diameter at breast height (including projects exempt from tree mitigation or the Timber Payback Clause) will require Endangered Species Act (ESA), Section 7 consultation with the U.S. Fish and Wildlife Service for the northern long-eared bat (*Myotis septentrionalis*). ESA consultation actions shall be coordinated through the installation NRM. No tree cutting shall occur until written authorization is received from the NRM and/or the NAVFAC Public Works Officer (PWO). Time of year restrictions may apply to tree removal actions.

6. Tree Preservation Plans and Tree Protection

- a. Proponents of all projects and activities, which may affect existing trees, shall coordinate with the Natural Resources Manager to identify all trees in the affected area and to develop a project/activity-specific tree preservation plan in accordance with this policy. All trees designated to be preserved shall be identified on applicable project drawings, and shall be marked in the field prior to the start of work. Existing trees designated for retention shall be protected in accordance with **Attachment A**, Tree Preservation and Protection Plan.
- b. Certain trees and forests are considered priority areas for protection, and shall be left in an undisturbed condition unless no practicable alternative is identified. The following areas are designated as priority areas for tree protection:
  - i. Trees in wetlands, floodplains or riparian buffers except for constructed drainage and stormwater management structures.
  - ii. Contiguous forests - forested corridors that connect with other forested tracts.
  - iii. Critical habitat - protected areas for rare, threatened or endangered species.
  - iv. Culturally significant trees - associated with historic sites and/or landscapes.
  - v. Specimen trees - trees 30 inches in diameter or larger.
  - vi. Rare and/or species of special concern.

7. Mitigation for Unavoidable Losses

- a. Where tree replacement is required, replacement ratios shall be based on the size of the individual trees scheduled for removal.
- b. The ratio for determining the number of required replacement trees shall be one replacement tree for every 6" increment in the diameter breast height (dbh) of the tree to be removed. A minimum 1:1 replacement ratio shall be accomplished. As an example, the following replacement ratios shall be implemented:
  - i. Removal of 3" to 6" dbh tree: 1:1 replacement
  - ii. Removal of 6.1" to 12" dbh tree: 2:1 replacement
  - iii. Removal of 12.1" to 18" dbh tree: 3:1 replacement, and so on...
- c. Replacement tree species and planting locations must be approved by the installation NRM. Replacement planting shall be conducted in accordance with the requirements set forth in Attachment (A) and use native species (Attachment B). If there is a conflict

between the methods described in Attachment A and the nursery/suppliers methods, the supplier's methods shall be followed. Failure to follow supplier instruction will void any warranty provided by the supplier. The following guidelines shall be used in developing appropriate tree replacement procedures for each project:

- i. Size. At the time of planting, replacement trees shall be a minimum of two (2) inches caliper.
- ii. Siting. Tree locations shall be based on sound urban forestry practices, and shall ensure adequate distance from buildings, sidewalks, roads, utilities, and other development to preclude the need for future tree removal. The location of replacement trees shall favor the benefits provided by trees in an urban setting, including noise attenuation, shading of cooling units and buildings, storm water management benefits, and sensible placement with respect to turf management areas.
- iii. Reforestation. In instances where space or conditions at the project site are undesirable for tree replacement, the Natural Resources Manager will assist in identifying priority reforestation locations on NAVFAC PWD-ME installations. Priority planting areas include wetland and other riparian buffers, corridors that connect existing forests, buffers between differing land uses, plantings to achieve energy conservation, expansion of existing forests, and habitat improvement projects.
- iv. Planting Seasons. Trees shall be planted during the proper planting season to benefit survival rates. For most shade tree species this is the dormant season or leaf-off period and runs generally from September through December, except when the ground is frozen.

#### 8. Unauthorized Destruction of Trees

- a. Instances where trees have been willfully damaged or are found in physically or structurally poor condition as a result of improper protection shall be adjudged as destruction of government property.
- b. In instances where compensation for destruction cannot be agreed upon based on the replacement criteria outlined in Section 7, the NRM may use procedures detailed by reference (j) and other relevant sources to calculate tree value. The appraisal process includes valuation of the tree species, size, condition and location. Tree replacement and reimbursement will be the responsibility of the person or persons who caused the destruction. Reimbursements will be made to the Navy Forestry Account in accordance with reference (k).

#### 9. Removal of Wood Products

- a. Federal regulations, Navy policy and instructions, and NAVFAC guidelines prohibit forest products from being given to the contractor, traded, or destroyed without proper compensation to the Navy Forestry Account. This includes tree removal associated with land clearing activities for construction, or the establishment, expansion, or maintenance of clear zones. Financial reimbursement must be turned over to the servicing Navy accounting and finance officer and deposited in the Navy Forestry Account. The project proponent shall work with the NRM to determine the value of

timber to be removed. The project proponent is responsible for providing financial reimbursement to the Navy for the assessed value of the timber.

- b. Contractors have 30 days from the date of notice to proceed to pay the appraised amount for the timber. Payment should be made and deposited into the Navy's forestry account with a copy of the transactions sent to NAVFAC HQ Forestry PM in accordance with reference (k). Coordination for payment and notification can be facilitated through the NRM.
- c. Trees managed primarily for landscaping and aesthetic purposes within the administrative, housing, or recreation areas of our installations fall into the category of "urban forestry," and some bases have more stringent, specific guidelines for their mitigation. The trimming, care, or removal of these urban trees is a base function. Their designation as urban trees is a product of their location in the base environment. Whether or not these trees pose a safety concern is a safety issue rather than a forestry issue. Urban tree issues shall be coordinated with the NRM who will work with the NAVFAC forester to determine if mitigation or payment is justified for their removal.
- d. Contractors, foresters, and natural resources managers must cooperate fully in the planning, coordination, and execution of field operations to prevent and suppress damaging forest insect and disease outbreaks, consistent with the terms of reference (l) and whenever it is determined to be necessary by either the regional U.S. Forest Service (USFS) supervisor or cooperating State forestry department or commission. Forestry management objectives should also include control and prevention of invasive species (references (m), (n), and (o)) and abide by all State and federal wood product quarantine laws.



**Attachment A: NAVFAC MIDLANT URBAN FORESTRY BMPs &  
TREE PRESERVATION AND PROTECTION PLAN**

# **Urban Forestry Best Management Practices**

## **Beneficial Landscaping**

The 1994 Presidential Memorandum on Environmentally and Economically Beneficial Landscape Practices on Federal Landscaped Grounds (60 Federal Register 40837) provides the overriding guidance for landscaping requirements on federal properties. EO 13148, Greening the Government through Leadership in Environmental Management, required federal agencies to incorporate beneficial landscaping into landscaping programs, policies, and practices by April 2001. The term beneficial landscaping describes practices that integrate native vegetation and wildlife habitat into the landscape and minimize the adverse effects that landscaping has on the natural environment. In accordance with specific directives of the presidential memorandum and to the extent practicable, federal landscaping projects should comply with the following guidelines:

- ☐ Use regionally native plants;
- ☐ Use construction practices that minimize adverse effects on the natural habitat;
- ☐ Reduce fertilizer and pesticide use;
- ☐ Use water-efficient practices; and,
- ☐ Create outdoor demonstrations to promote awareness of the environmental and economic benefits of beneficial landscaping.

The use of regionally native plant species, which are generally better suited for local site conditions than nonnative species, reduces the need for intensive maintenance and the use of fertilizers and pesticides. Native plant species are also less likely to become invasive pests than nonnative species and serve as better sources of food and cover for native wildlife.

Effective use of native trees and shrubs in landscaping also can provide economic and environmental benefits. When properly placed around buildings, trees and shrubs reduce energy consumption by moderating the effects of the sun and wind. Planting deciduous trees on the east- and west-facing sides of buildings provides summer shade, and planting evergreens on the north-facing side blocks cold winter winds. Other benefits provided by landscape plants include water conservation and improvement of water quality. These benefits are achieved through reducing the impact of precipitation, reducing flow velocities, and capturing and storing excess runoff. In addition, landscaping with a variety of trees and shrubs provides habitat that attracts wildlife to the urban environment, which benefits both the wildlife and their human observers.

Selecting species that are suitable for a site requires knowledge of plant characteristics such as the species mature size, longevity, tolerance to soil compaction and pollution, and susceptibility to disease and insect pests.

## **Selection of Plant Materials for Landscaping**

The size of plants selected depends on budget, site conditions, planting season, available labor, and desired results. Container-grown stock is more expensive, but is less susceptible to drying and is better able to compete with surrounding vegetation. Sizes of containers vary from 6-inch tube grown seedlings (tublings) to large pots or balled and burlapped saplings. Two- to three-gallon container-grown stock is widely available from private nurseries, survives transplanting better than bare-root, and is appropriate for use on a wide range of sites. Areas up to several acres in size can be planted economically with this size planting stock. Large balled and burlapped stock also has a good survival rate after transplanting in poor or compacted urban soils, but is more costly per plant and is more labor intensive to transport and install than smaller stock. Balled and burlapped stock is most suitable for planting around buildings, along streets, and in high visibility areas that are required to look good quickly. Planting a mixture of sizes of woody plants is an option that creates more diversity and a more naturalistic appearance.

## **Planting**

The planning process should allow for planting during a suitable season. The type of planting stock used, in part, determines the appropriate time for planting. Bare-root seedlings should be planted in the spring before the emergence of new leaves. Larger woody material is best planted in the late fall after leaves have dropped. At this time transpiration is minimal and root growth increases. Roots are often damaged in the transplanting process; therefore, planting during the fall allows additional time for root development before the summer months when transpiration peaks. Groundcovers can be planted at any time, as long as there is adequate rainfall or available supplemental watering.

Proper tree planting is another vital element of a healthy urban forest. Using correct planting methods can increase a tree's ability to establish quickly and improve its health and longevity. Planting techniques differ somewhat with the type of material planted, though the goal of each is to provide an environment that encourages root growth. Guidelines that apply to most types of planting stock are that the planting hole should be three to five times greater in diameter than the root ball of the material to be planted and only as deep as the root ball. It is important not to bury the roots too deeply or they will not be able to get enough oxygen. Appropriate planting guidelines for various plant materials are presented in Appendix "X". Soil amendments should not be added directly to the planting holes for trees and shrubs. These amendments cause problems with soil moisture and root growth. If fertilizers are applied, it is important to use a slow-release product with low solubility so nutrients are not easily leached away. To ensure the greatest chance of survival, urban tree and shrub planting should be performed by trained Station personnel or qualified tree care professionals.

## **Tree and Shrub Care**

The care that newly planted materials receive after planting is critical to their health and longevity. Ensuring adequate soil moisture immediately after planting and during the first two years of establishment is the key factor in planting success. Over watering can deprive the tree of air and should also be avoided.

Preventing damage from mowers and string trimmers is a significant problem for landscape managers. Wounds in a tree's bark make the tree more susceptible to disease and pest infestations and reduce its chance of survival. Mulch can be an effective method of protecting trees from mower damage, when used properly. Mulch protects trees by reducing weed growth around the plant's base, which reduces the need to mow near the plant. Mulch should be applied to a weed-free area around the root mat in a layer about three to four inches thick. Mulch should not be applied too close to the tree trunk or too deeply as this creates an environment that promotes fungal growth and decay.

## Urban Forest Guidance - Pruning Trees

Pruning is the most common tree maintenance procedure. Although forest trees grow quite well with only nature's pruning, landscape trees require a higher level of care to maintain their safety and aesthetics. Pruning should be done with an understanding of how the tree responds to each cut. Improper pruning can cause damage that will last for the life of the tree, or worse, it will shorten the tree's life.

### Reasons for Pruning

Since each cut has the potential to change the growth of the tree, no branch should be removed without a reason. Common reasons for pruning are to remove dead branches, to remove crowded or rubbing limbs, and to eliminate hazards. Trees may also be pruned to increase light and air penetration to the inside of the tree's crown or to the landscape below. In most cases, mature trees are pruned as a corrective or preventative measure.

### When to Prune

Most routine pruning to remove weak, diseased or dead limbs can be accomplished at any time during the year with little effect on the tree. As a rule, growth is maximized and wound closure is fastest if pruning takes place before the spring growth flush. Some trees, such as maples and birches, tend to "bleed" if pruned early in the spring. This may be unsightly, but is of little consequence to the tree.

A few tree diseases, such as oak wilt, can be spread when pruning wounds allow spores access into the tree. Susceptible trees should not be pruned during active transmission periods.

Heavy pruning just after the spring growth flush should be avoided. This is when trees have just expended a great deal of energy to produce foliage and early shoot growth. Removal of a large percentage of foliage at this time can stress the tree.

### Making Proper Pruning Cuts to Mature Trees

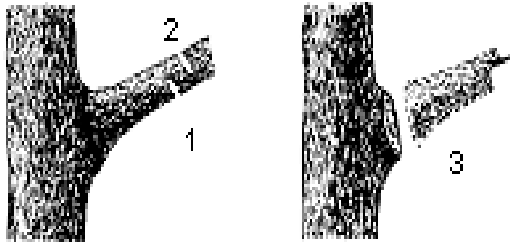
Pruning cuts should be made just outside the branch collar. The branch collar contains trunk or parent branch tissue and should not be damaged or removed. If trunk collar has grown out on a dead limb to be removed, make the cut just beyond the collar. Do not cut the collar (see figure).

**On a dead branch that has a collar of live wood, the final cut should be made just beyond the outer edge of the collar.**



If a large limb is to be removed, its weight should first be reduced. This is done by making an undercut about 12-18 inches from the limb's point of attachment. A second cut is made from the top, directly above or a few inches further out on the limb. This removes the limb leaving the 12-18 inch stub. The stub is removed by cutting back to the branch collar. This technique reduces the possibility of tearing the bark

**Use the 3-cut method to remove a large limb.**



## **How Much Should be Pruned?**

The amount of live tissue that should be removed depends on the tree size, species, and age, as well as the pruning objectives. Younger trees will tolerate the removal of a higher percentage of living tissue than mature trees. A common mistake is to remove too much inner foliage and small branches. It is important to maintain an even distribution of foliage along large limbs and in the lower portion of the crown. A widely accepted rule of thumb is never to remove more than one fourth of a tree's leaf bearing crown. In a mature tree, pruning even that much could have negative effects. Removing even a single, large-diameter limb can create a wound that the tree may not be able to close. The older and larger a tree becomes, the less energy it has in reserve to close wounds and defend against decay or insect attack. The pruning of large, mature trees is usually limited to the removal of dead or potentially hazardous limbs.

## **Wound Dressings**

Wound dressings were once thought to accelerate wound closure, protect against insects and diseases, and reduce decay. However, research has shown that dressings do not reduce decay or speed closure, and rarely prevent insect or disease infestations. Most experts recommend that wound dressings not be used. If a dressing must be used for cosmetic purposes, then only a thin coating of a non-toxic material should be applied.

## **Newly Planted Trees**

Pruning of newly planted trees should be limited to corrective pruning. Remove torn or broken branches. Save other pruning measures for the second or third year. The belief that trees should be pruned when planted to compensate for root loss is misguided. Trees need their leaves and shoot tips to provide food and the substances, which stimulate new root production. Unpruned trees establish faster, with a stronger root system than trees pruned at the time of planting.

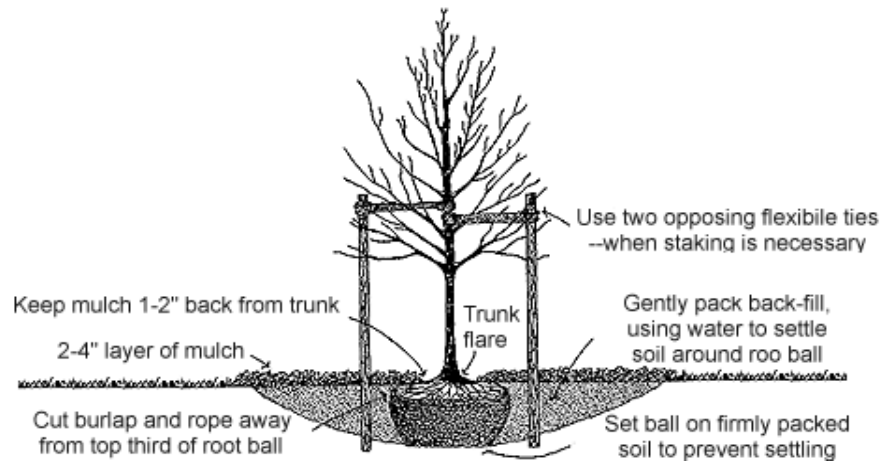
(From International Society Arboriculture at <http://www.isa-arbor.com/consumer/pruning.html>)

## New Tree Planting

The ideal time to plant trees and shrubs is during the dormant season, in the fall after leafdrop or early spring before bud-break. Weather conditions are cool and allow plants to establish roots in the new location before spring rains and summer heat stimulate new top growth. However, trees properly cared for in the nursery or garden center, and given the appropriate care during transport to prevent damage, can be planted throughout the growing season. In either situation, proper handling during planting is essential to ensure a healthy future for new trees and shrubs. *Before you begin planting your tree, be sure you have had all underground utilities located prior to digging.*

If the tree you are planting is balled and burlapped, or bare rooted, it is important to understand that the tree's root system has been reduced by 90-95% of its original size during transplanting. As a result of the trauma caused by the digging process, trees will commonly exhibit what is known as **transplant shock**. Transplant shock is indicated by slow growth and reduced vigor following transplanting. Proper site preparation before and during planting, coupled with good follow up care will reduce the amount of time the plant experiences transplant shock and will allow the tree to quickly establish in its new location. Carefully follow eight simple steps and you can significantly reduce the stress placed on the plant at the time of planting.

1. **Dig a shallow, broad planting hole.** Make the hole wide, as much as three times the diameter of the root ball, but only as deep as the root ball. It is important to make the hole wide because the tree roots on the newly establishing tree must push through surrounding soil to establish. On most planting sites in new developments, the existing soils have been compacted and are unsuitable for healthy root growth. Breaking up the soil in a large area around the tree provides the newly emerging roots room to expand into loose soil to hasten establishment.
2. **Identify the trunk flare.** The trunk flare is where the roots spread at the base of the tree. This point should be partially visible after the tree has been planted (see diagram). If the trunk flare is not partially visible, you may have to remove some soil from the top of the root ball. Find it so you can determine how deep the hole needs to be for proper planting.
3. **Place the tree at the proper height.** Before placing the tree in the hole, check to see that the hole has been dug to the proper depth, and no more. The majority of the roots on the newly planted tree will develop in the top 12" of soil. If the tree is planted too deep, new roots will have difficulty developing due to a lack of oxygen. It is better to plant the tree a little high, 1-2" above the base of the trunk flare, than to plant it at or below the original growing level. This will allow for some settling (see diagram). To avoid damage when setting the tree in the hole, always lift the tree by the root ball, and never by the trunk.



1. **Straighten the tree in the hole.** Before you begin backfilling have someone view the tree from several directions to confirm the tree is straight. Once you begin backfilling it is difficult to reposition.
2. **Fill the hole, gently but firmly.** Fill the hole about 1/3 full and gently but firmly pack the soil around the base of the root ball. Then, if the tree is balled and burlapped, cut and remove the string and wire from around the trunk and top 1/3 of the root ball (see diagram). Be careful not to damage the trunk or roots in the process. Fill the remainder of the hole, taking care to firmly pack soil to eliminate air pockets that may cause roots to dry out. To avoid this problem, add the soil a few inches at a time and settle with water. Continue this process until the hole is filled and the tree is firmly planted. It is not recommended to apply fertilizer at the time of planting.
3. **Stake the tree, if necessary.** If the tree is grown and dug properly at the nursery, staking for support is not necessary in most home landscape situations. Studies have shown that trees will establish more quickly and develop stronger trunk and root systems if they are not staked at the time of planting. However, protective staking may be required on sites where lawn mower damage, vandalism or windy conditions are concerns. If staking is necessary for support, two stakes used in conjunction with a wide flexible tie material will hold the tree upright, provide flexibility, and minimize injury to the trunk (see diagram). Remove support staking and ties after the first year of growth. Leave protective staking in place as long as necessary.
4. **Mulch the base of the tree.** Mulch is simply organic matter applied to the area at the base of the tree. It acts as a blanket to hold moisture, protect against harsh soil temperatures, both hot and cold, and reduces competition from grass and weeds. Some good choices are leaf litter, pine straw, shredded bark, peat moss, or wood chips. A two to four inch layer is ideal. More than four inches may cause a problem with gas exchange. When placing mulch, care should be taken so that the actual trunk of the tree is not covered. This may cause decay of the living bark at the base of the tree. A mulch-free area, one to two inches wide at the base of the tree, is sufficient to avoid moist bark conditions and prevent decay.

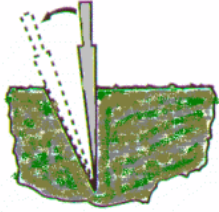


8. **Follow-up care.** Keep the soil moist but not soaked; overwatering will cause leaves to turn yellow or fall off. Water trees at least once a week, barring rain, and more frequently during hot weather. When the soil is dry below the surface of the mulch, it is time to water. Continue until mid-fall, tapering off for lower temperatures that require less frequent watering. Other follow-up care may include minor pruning of branches damaged during the planting process. Prune sparingly immediately after planting, and wait to begin necessary corrective pruning until after a full season of growth in the new location.

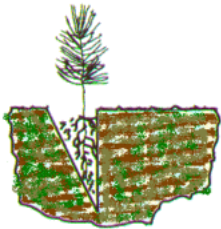
(Taken from the International Society of Arboriculture at <http://www.isa-arbor.com/consumer/planting.htm>)

## How to Plant with a Dibble Bar

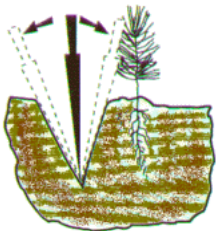
1. Push the blade vertically into the soil then pull the handle toward you to open the hole.



2. Set the seedling 1 to 3 inches deeper than the nursery depth with the roots straight.



3. Push the blade into the soil just behind the planting hole then pull the handle back to close the bottom of the hole. Push the handle forward to close the top of the hole.



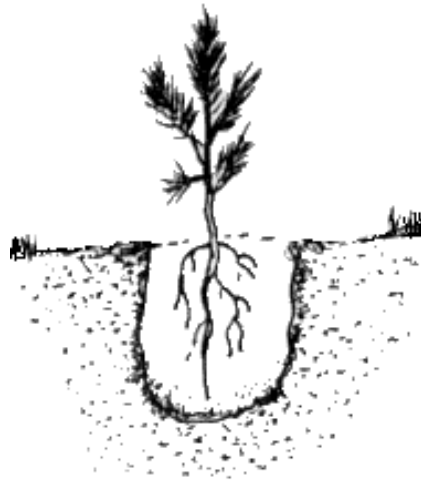
4. Pack the soil firmly with your heel.



(From South Carolina Forestry Commission at [www.state.sc.us/forest/refplant.htm](http://www.state.sc.us/forest/refplant.htm))

## How to Plant Bare-root Trees

1. It is best to plant bare-root trees immediately, in order to keep the fragile roots from drying out. If you can't plant because of weather or soil conditions, store the trees in a cool place and keep the roots moist.
2. Unpack tree and soak in water 3 to 6 hours. Do not plant with packing materials attached to roots, and do not allow roots to dry out.
3. Dig a hole, wider than seems necessary, so the roots can spread without crowding. Remove any grass within a three-foot circular area. To aid root growth, turn soil in an area up to 3 feet in diameter.
4. Plant the tree at the same depth it stood in the nursery, without crowding the roots. Partially fill the hole, firming the soil around the lower roots. Do not add soil amendments.



5. Shovel in the remaining soil. It should be firmly, but not tightly packed with your heel. Construct a water-holding basin around the tree. Give the tree plenty of water.
6. After the water has soaked in, place a 2-inch deep protective mulch area 3 feet in diameter around the base of the tree (but not touching the trunk).
7. Water the tree generously every week or 10 days during the first year of establishment.

## Recognizing Hazardous Trees

**Hazardous Trees & Utility Lines.** Trees that fall into utility lines have additional serious consequences. Not only can they injure people or property near the line, but hitting a line may cause power outages, surges, fires and other damage. Downed lines still conducting electricity are especially dangerous. A tree with a potential to fall into a utility line is a very serious situation.

### Tree Hazard Checklist

Consider these questions . . .

1. Are there large dead branches in the tree?
2. Are there detached branches hanging in the tree?
3. Does the tree have cavities or rotten wood along the trunk or in major branches?
4. Are mushrooms present at the base of the tree?
5. Are there cracks or splits in the trunk or where branches are attached?
6. Have any branches fallen from the tree?
7. Have adjacent trees fallen over or died?
8. Has the trunk developed a strong lean?
9. Do many of the major branches arise from one point on the trunk?
10. Have the roots been broken off, injured or damaged by lowering the soil level, installing pavement, repairing sidewalks or digging trenches?
11. Has the site recently been changed by construction, raising the soil level or installing lawns?
12. Have the leaves prematurely developed an unusual color or size?
13. Have trees in adjacent wooded areas been removed?
14. Has the tree been topped or otherwise heavily pruned?

### Managing Tree Hazards

One of these treatments may help make your tree safer. Reducing the risk associated with hazardous trees can take many forms.

1. **Prune the tree.** Remove the defective branches of the tree. Since in appropriate pruning may also weaken a tree,
2. **Provide routine care.** Mature trees need routine care in the form of water, fertilizer (In some cases), mulch and pruning as dictated by the season and their structure.

A number of treatments are best done by a Certified Arborist

1. **Cable and brace the tree.** Provide physical support for weak branches and stems to increase their strength and stability.
2. **Remove the tree.** Some hazardous trees are best removed. If possible, plant a new tree in an appropriate place as a replacement. (From International Society of Arboriculture at <http://www.isa-arbor.com/consumer/hazards.html>)

## **Root Pruning Guidelines**

Trenching and digging in the soil near trees can cut roots, and this can damage the tree resulting in tree decline or the tree falling over. This can cause liability and safety concerns. Root pruning is more injurious to old mature trees than it is for younger more vigorous trees. Cutting roots greater than about one inch diameter during trenching and digging can mean problems for the tree. In some cases roots of one to three inches diameter represent the major structural roots holding the tree upright.

The impact from pruning roots depends on several factors. Damage typically increases with more cuts, bigger cuts, and cuts made closer to the trunk. Root pruning, trenching, and other construction activities close to the trunk result in more injury on shallow, compacted soils or on soils that drain poorly than on well drained soils. This is due to the shallow roots common on sites with shallow soils or high water table. Trees that are leaning are poor candidates for root pruning. Prune roots only with sharp tools to avoid tearing behind the cuts. Root prune no closer to the trunk than a distance equal to 3 times the trunk diameter, preferably 5 times the trunk diameter.

**Attachment B – NAVFAC PWD-ME Natural Resource Program**  
**Native Plant Species Approved for Planting**

### Native Trees:

| Species               | Scientific name               | Sun | Moisture | Max Height | Notes                                              |
|-----------------------|-------------------------------|-----|----------|------------|----------------------------------------------------|
| Balsam Fir            | <i>Abies balsamea</i>         | F-P | A        | 75'        | Open growth in hot dry locations, evergreen        |
| Red Maple             | <i>Acer rubrum</i>            | F-P | A        | 60'        | Excellent fall color; tolerates wet spring soil    |
| Sugar Maple           | <i>Acer saccharum</i>         | F-P | A        | 75'        | Excellent fall color, beautiful large shade tree   |
| Mountain Maple        | <i>Acer spicatum</i>          | F-P | W-A      | 30'        | Useful in naturalizing                             |
| Yellow Birch          | <i>Betula alleghaniensis</i>  | F-P | A        | 100'       | Does best in cool soils, beautiful bark            |
| Paper Birch           | <i>Betula papyrifera</i>      | F   | A        | 70'        | White bark year-round, tolerates poor, dry soil    |
| Gray Birch            | <i>Betula populifolia</i>     | F   | A        | 40'        | Does well in poor soils; good for naturalizing     |
| American Hornbeam     | <i>Carpinus caroliniana</i>   | F   | A        | 30'        | Use for naturalizing; tolerates periodic flooding  |
| Pagoda Dogwood        | <i>Cornus alternifolia</i>    | F-P | A        | 25'        | Moist soil is important, white flowers in June     |
| (Champlain) Hawthorne | <i>Crataegus submollis</i>    | F   | A        | 30'        | Red fruits in fall attract birds                   |
| White Ash             | <i>Fraxinus americana</i>     | F   | A        | 80'        | Good fall color; tolerates alkaline soil           |
| Green Ash             | <i>Fraxinus pennsylvanica</i> | F-P | A        | 60'        | Grows fast; tolerates salt, drought, alkaline soil |
| Tamarack              | <i>Larix laricina</i>         | F   | W-A      | 80'        | Good for moist-wet sites & well drained sites      |
| Black Gum             | <i>Nyssa sylvatica</i>        | F-P | A        | 50'        | Excellent yellow-orange fall leaf color            |
| American hophornbeam  | <i>Ostrya virginiana</i>      | F-P | A-D      | 40'        | Slow to establish after transplanting              |
| White Spruce          | <i>Picea glauca</i>           | F-P | A        | 60'        | Good specimen or windbreak; evergreen              |
| Black Spruce          | <i>Picea mariana</i>          | F-P | A        | 40'        | Tolerates wet sites; evergreen                     |
| Jack Pine             | <i>Pinus banksiana</i>        | F   | D        | 50'        | Useful for windbreaks in sandy soil; evergreen     |
| Red Pine              | <i>Pinus resinosa</i>         | F   | D        | 80'        | Good windbreak; tolerates dry soils well           |
| White Pine            | <i>Pinus strobus</i>          | F   | A-D      | 80'        | Handsome specimen; not tolerant of salt            |
| Bigtooth Aspen        | <i>Populus grandidentata</i>  | F   | A-D      | 70'        | Fast growing; short lived; good yellow fall color  |
| Quaking Aspen         | <i>Populus tremuloides</i>    | F   | A        | 50'        | Fast growing, short lived; good yellow fall color  |
| Pin Cherry            | <i>Prunus pensylvanica</i>    | F   | A        | 35'        | Adaptable; fast growing; tolerates poor soil       |
| Black Cherry          | <i>Prunus serotina</i>        | F   | A        | 60'        | White flowers (spring); wildlife food source       |
| White Oak             | <i>Quercus alba</i>           | F   | A        | 80'        | Large tree; transplant when young                  |
| Red Oak               | <i>Quercus rubra</i>          | F   | A        | 75'        | Transplants readily; good fall red leaf color      |
| Black Willow          | <i>Salix nigra</i>            | F   | W-A      | 35'        | Tolerates wet soils; twigs can cause lawn litter   |
| American Mountainash  | <i>Sorbus americana</i>       | F   | A        | 30'        | Fruits good in wildlife landscape                  |
| White Cedar           | <i>Thuja occidentalis</i>     | F-P | A        | 60'        | Useful hedge plant; tolerates alkaline soil        |
| Basswood              | <i>Tilia americana</i>        | F-P | A        | 80'        | Tolerates alkaline soil; use in urban landscape    |
| Eastern Hemlock       | <i>Tsuga canadensis</i>       | F-S | A        | 70'        | Does not tolerate drought or windy sites           |

#### Sun Exposure

F = Full Sun

P = Part Sun

S = Shade

#### Soil Moisture

W = Wet

A = Average

D = Dry

### Native Shrubs:

| Species                | Scientific name                  | Sun | Moisture | Max Height | Notes                                              |
|------------------------|----------------------------------|-----|----------|------------|----------------------------------------------------|
| Downy Serviceberry     | <i>Amelanchier arborea</i>       | F-P | A        | 25'        | Edible/wildlife landscapes; orange fall color      |
| Shadblow Serviceberry  | <i>Amelanchier canadensis</i>    | F-P | A        | 20'        | Edible/wildlife landscapes; yellow/gold fall color |
| Alleghany Serviceberry | <i>Amelanchier laevis</i>        | F-P | A        | 25'        | Edible/wildlife landscapes; spring leaves bronze   |
| Bog Rosemary           | <i>Andromeda polifolia</i>       | F-P | W        | 1'         | Evergreen leaves; requires moist acidic soil       |
| Black Chokeberry       | <i>Aronia melanocarpa</i>        | F-P | W-A      | 6'         | Tolerates wet or dry soils; wine-red fall color    |
| Buttonbush             | <i>Cephalanthus occidentalis</i> | F   | W        | 6'         | Good for wetland plantings                         |
| Sweetfern              | <i>Comptonia peregrina</i>       | F-P | A-D      | 3'         | Aromatic foliage; good in dry and sandy soil       |
| Gray Dogwood           | <i>Cornus racemosa</i>           | F-S | A        | 15'        | Suckering; white fall fruits eaten by many birds   |
| Redosier Dogwood       | <i>Cornus sericea</i>            | F   | W-A      | 6'         | Red stems attractive in winter; tolerates wet soil |
| American Filbert       | <i>Corylus americana</i>         | F-P | A        | 15'        | Wildlife eats fruit; tolerates alkaline soil       |
| Leatherwood            | <i>Dirca palustris</i>           | S   | A        | 4'         | Yellow fall color; thrives in moist, shady sites   |
| Common Witchhazel      | <i>Hamamelis virginiana</i>      | F-P | A        | 15'        | Avoid drought; yellow flowers/leaves in fall       |
| Winterberry            | <i>Ilex verticillata</i>         | F-P | W-A      | 10'        | Bright red fruits persist into midwinter           |
| Common Juniper         | <i>Juniperus communis</i>        | F   | A-D      | 3'         | Takes drought, wind, sterile or alkaline soils     |
| Lambkill               | <i>Kalmia angustifolia</i>       | F-P | A        | 3'         | Adaptable to many soils; best in acidic soil       |
| Labrador Tea           | <i>Ledum groenlandicum</i>       | F-P | W-A      | 3'         | Transplants well; good for moist-to-wet sites      |
| Sweetgale              | <i>Myrica gale</i>               | F   | A-D      | 4'         | Bushy plant; dark green, aromatic foliage          |
| Northern Bayberry      | <i>Myrica pensylvanica</i>       | F-P | A-D      | 6'         | Good for massing; useful in poor soil sites        |
| Bush Cinquefoil        | <i>Potentilla fruticosa</i>      | F   | A-D      | 4'         | Summer-flowering shrub; tolerates alkaline soil    |
| Beach Plum             | <i>Prunus maritima</i>           | F   | A        | 6'         | Good for edible landscape; salt-tolerant           |
| Chokecherry            | <i>Prunus virginiana</i>         | F   | A        | 30'        | White flowers in spring; wildlife food source      |
| Rhodora                | <i>Rhododendron canadense</i>    | F-P | W-A      | 3'         | Magenta flowers in spring; best in acidic soil     |
| Meadow Rose            | <i>Rosa blanda</i>               | F   | A        | 5'         | Single light pink flowers; suckering plant         |
| Carolina Rose          | <i>Rosa carolina</i>             | F   | A        | 5'         | Pink single flowers midsummer                      |
| Virginia Rose          | <i>Rosa virginiana</i>           | F   | A        | 5'         | Good in dry & seaside landscape; good barrier      |
| American Elder         | <i>Sambucus canadensis</i>       | F   | A        | 12'        | Edible landscapes; tolerates alkaline soil         |
| Scarlet Elder          | <i>Sambucus pubens</i>           | F   | A        | 20'        | Flowers in July; red fruit mid-summer              |
| Highbush Blueberry     | <i>Vaccinium corymbosum</i>      | F   | A        | 8'         | Edible/wildlife landscapes; best in acidic soil    |
| Mapleleaf Viburnum     | <i>Viburnum acerifolium</i>      | P-S | A        | 6'         | Good for mass plantings in shady sites             |
| Hobblebush             | <i>Viburnum alnifolium</i>       | P-S | A        | 8'         | Open shrub; good for naturalized landscape         |
| Arrowwood Viburnum     | <i>Viburnum dentatum</i>         | F-P | A        | 15'        | Durable; good hedge; tolerates alkaline soil       |
| Nannyberry             | <i>Viburnum lentago</i>          | F-P | A        | 15'        | Good for wildlife and naturalized landscapes       |
| Highbush Cranberry     | <i>V. opulus var. trilobum</i>   | F-P | A        | 12'        | Use for screening; good for wildlife landscapes    |

#### Sun Exposure

F = Full Sun

P = Part Sun

S = Shade

#### Soil Moisture

W = Wet

A = Average

D = Dry



### Native Ground Covers and Vines:

| Species              | Scientific name                    | Sun | Moisture | Max Height | Notes                                            |
|----------------------|------------------------------------|-----|----------|------------|--------------------------------------------------|
| Running Serviceberry | <i>Amelanchier stolonifera</i>     | F-P | A        | 2'         | Stoloniferous groundcover; forms thickets        |
| Bearberry            | <i>Arctostaphylos uva-ursi</i>     | F-P | A-D      | 6"         | Best in poor, sandy, acidic soils; salt-tolerant |
| Virginbower          | <i>Clematis virginiana</i>         | F   | A        | --         | Climbing vine; best in alkaline soil             |
| Bunchberry           | <i>Cornus canadensis</i>           | P-S | A        | 6"         | Groundcover; white flowers & red berries (birds) |
| Checkerberry         | <i>Gaultheria procumbens</i>       | P-S | A        | 6"         | Evergreen groundcover; leaves reddish in fall    |
| Creeping Juniper     | <i>Juniperus horizontalis</i>      | F   | A-D      | 1'         | Tolerates hot, dry, alkaline soil; evergreen     |
| Partridgeberry       | <i>Mitchella repens</i>            | S   | A        | 2"         | Delicate plant; red fruits persist into winter   |
| Woodbine             | <i>Parthenocissus quinquefolia</i> | F-S | A-D      | --         | Hardy vine; maroon in fall; takes alkaline soil  |
| Lowbush Blueberry    | <i>Vaccinium angustifolium</i>     | F   | A        | 2'         | Edible/wildlife landscapes; acidic soil          |
| Cranberry            | <i>Vaccinium macrocarpon</i>       | F   | W-A      | 6"         | Edible/wildlife landscapes; acidic soil          |
| Fox Grape            | <i>Vitis labruscas</i>             | F   | A        | --         | Handsome foliage; good vine for arbors           |

#### Sun Exposure

F = Full Sun

P = Part Sun

S = Shade

#### Soil Moisture

W = Wet

A = Average

D = Dry

## Native Flowering Perennials

| Species              | Scientific name                   | Sun | Moisture | Max Height | Notes                                          |
|----------------------|-----------------------------------|-----|----------|------------|------------------------------------------------|
| Baneberry            | <i>Actaea pachypoda</i>           | P-S | A        | 24"        | Attractive, but poisonous fruit                |
| Columbine            | <i>Aquilegia canadensis</i>       | F-S | A-D      | 12"        | Early spring flowers                           |
| Spikenard            | <i>Aralia racemosa</i>            | P-S | A-D      | 36"        | Good for back of border                        |
| Jack-in-the-Pulpit   | <i>Arisaema triphyllum</i>        | P-S | W-A      | 12"        | Flower green and brown; bright red fruit       |
| Milkweed             | <i>Asclepias syriaca</i>          | F   | D        | 36"        | Attracts butterflies                           |
| New England Aster    | <i>Aster novae-angliae</i>        | F-P | A-D      | 24"        | Fall flowering; deep purple                    |
| New York Aster       | <i>Aster novi-belgii</i>          | F-P | A-D      | 24"        | Fall flowering; purple                         |
| Marsh Marigold       | <i>Caltha palustris</i>           | F-P | W-A      | 12"        | Showy yellow flowers in early spring           |
| Harebell             | <i>Campanula rotundifolia</i>     | F-P | A-D      | 12"        | Delicate blue-purple flowers                   |
| Blue Cohosh          | <i>Caulophyllum thalictroides</i> | P-S | A-D      | 36"        | Blue fruits; back of border                    |
| Turtlehead           | <i>Chelone glabra</i>             | P-S | W-A      | 24"        | Interesting white flowers in midsummer         |
| Bluebead Lily        | <i>Clintonia borealis</i>         | P-S | A        | 12"        | Yellow ball-shaped flowers, poison fruit       |
| Trout-lily           | <i>Eupatorium americanum</i>      | P-S | A        | 6"         | Flowers early-spring                           |
| Joe-pye Weed         | <i>Eupatorium maculatum</i>       | F   | W-A      | 48"        | Purple flowers in fall; attracts butterflies   |
| Boneset              | <i>Eupatorium perfoliatum</i>     | F   | W-A      | 24"        | Green flowers; good for drying                 |
| Blue Flag            | <i>Iris versicolor</i>            | F-P | W-A      | 24"        | Elegant form; blue-purple flowers              |
| Indian Cucumber-root | <i>Medeola virginica</i>          | P-S | A        | 12"        | Interesting magenta floral bracts              |
| Obedient Plant       | <i>Physostegia virginiana</i>     | F-S | A-D      | 24"        | flowers pink; leaves dark green; cut flowers   |
| Solomon's Seal       | <i>Polygonatum pubescens</i>      | P-S | A        | 18"        | Arching stems w/ white flowers (spring)        |
| Silvery Cinquefoil   | <i>Potentilla anserina</i>        | F   | D        | 6"         | Yellow flowers, silvery leaves                 |
| Bloodroot            | <i>Sanguinaria canadensis</i>     | P-S | A        | 12"        | Showy white flowers in early spring            |
| Foam Flower          | <i>Tiarella cordifolia</i>        | P   | A        | 6"         | Delicate white flowers in early spring         |
| Wild-oats            | <i>Uvularia sessilifolia</i>      | P-S | A        | 6"         | Cream, bell-shaped flowers in early spring     |
| Sweet White Violet   | <i>Viola blanda</i>               | P-S | A        | 2"         | Small white flowers, ground cover, self-sows   |
| Marsh Blue Violet    | <i>Viola cucullata</i>            | P-S | A        | 6"         | Blue flowers, spreading; self-sows             |
| Labrador Violet      | <i>Viola labradorica</i>          | F-P | A-D      | 6"         | Magenta flowers; reddish foliage; self-sows    |
| Downy Yellow Violet  | <i>Viola pubescens</i>            | P-S | A-D      | 6"         | Yellow flowers                                 |
| Round-leaved Violet  | <i>Viola rotundifolia</i>         | P   | A-D      | 3"         | Yellow flowers, leaves large, round & leathery |

### Sun Exposure

F = Full Sun

P = Part Sun

S = Shade

### Soil Moisture

W = Wet

A = Average

D = Dry

### Native Perennial Ferns:

| Species             | Scientific name                   | Sun | Moisture | Max Height | Notes                                            |
|---------------------|-----------------------------------|-----|----------|------------|--------------------------------------------------|
| Maidenhair Fern     | <i>Adiantum pedatum</i>           | P-S | A        | 18"        | Delicate black stem, fronds almost circular      |
| Lady Fern           | <i>Athyrium filix-femina</i>      | P-S | W-A      | 18"        | Lacey Fronds; reddish in spring                  |
| Hay-scented Fern    | <i>Dennstaedtia punctilobula</i>  | F-P | A-D      | 12"        | Fragrant lacey fronds; tolerates hot, dry sites  |
| Spinulose Wood Fern | <i>Dryopteris carthusiana</i>     | P-S | A        | 24"        | Lacey fronds; reddish in spring                  |
| Marginal Wood Fern  | <i>Dryopteris marginalis</i>      | F-P | A-D      | 24"        | Fronds blue-green; tolerates rocky sites         |
| Ostrich Fern        | <i>Mattuccia struthiopteris</i>   | P-S | A        | 36"        | Beautiful fronds; fiddleheads edible             |
| Sensitive Fern      | <i>Onoclea sensibilis</i>         | F-P | W-A      | 12"        | Easy to grow; spreads                            |
| Cinnamon Fern       | <i>Osmunda cinnamomea</i>         | P-S | W-A      | 36"        | Attractive cinnamon color fertile frond (spring) |
| Interrupted Fern    | <i>Osmunda claytoniana</i>        | F-S | W-D      | 36"        | Spreads well; luxuriant spring growth            |
| Royal Fern          | <i>Osmunda regalis</i>            | F-S | W-A      | 36"        | Vase-shaped; sterile fronds finely dissected     |
| Long Beech Fern     | <i>Phegopteris connectilis</i>    | P-S | A        | 6"         | Smaller size fern, low growing, spreads well     |
| Christmas Fern      | <i>Polystichum acrostichoides</i> | P-S | A        | 12"        | Leathery, evergreen fronds                       |

#### Sun Exposure

F = Full Sun

P = Part Sun

S = Shade

#### Soil Moisture

W = Wet

A = Average

D = Dry

### Native Grass Species:

Riverbank Wild Rye (*Elymus riparius*), Creeping Red Fescue (*Festuca rubra*), Little Bluestem (*Schizachyrium scoparium*), Big Bluestem (*Andropogon gerardii*), Switch Grass (*Panicum virgatum*), Upland Bentgrass (*Agrostis perennans*), Virginia Wild Rye (*Elymus virginicus*), Red Fescue (*Festuca rubra*), Indian Grass (*Sorghastrum nutans*), Upland Bentgrass (*Agrostis perennans*), Canada Wild Rye (*Elymus canadensis*), Narrowleafed Blue Eyed Grass (*Sisyrinchium angustifolium*), Perennial Ryegrass (*Lolium perenne*), Annual Ryegrass (*Lolium multiflorum*), Smooth Panic Grass (*Panicum dichotomiflorum*), Side Oats Grama (*Bouteloua curtipendula*), Sand Dropseed (*Sporobolus cryptandrus*), Sand Lovegrass (*Eragrostis trichodes*).

### Salt Tolerant Grass Species:

Canada Wild Rye (*Elymus Canadensis*), Side Oats Grama (*Bouteloua curtipendula*), Creeping Red Fescue (*Festuca rubra*), Big Bluestem (*Andropogon gerardii*), Indian Grass (*Sorghastrum nutans*), Switch Grass (*Panicum virgatum*), Sand Dropseed (*Sporobolus cryptandrus*), Sand Lovegrass (*Eragrostis trichodes*).

SECTION 01 58 00

PROJECT IDENTIFICATION  
08/19, CHG 5: 08/22

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects as applicable at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WOOD PROTECTION ASSOCIATION (AWPA)

AWPA U1 (2024) Use Category System: User  
Specification for Treated Wood

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Preliminary Drawing Indicating Layout And Text Content; G

1.3 PROJECT IDENTIFICATION SIGN

Prior to initiating any work on site, provide one project identification sign at the location indicated designated by the Contracting Officer. Construct the sign in accordance with project sign detail, which can be downloaded at:  
<http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>. Maintain sign throughout the life of the project. Upon completion of the project, remove the sign from the site.

1.3.1 Project Identification Signboard

Provide a project identification signboard in accordance with attached Plates 1USN, 3, and 4. Provide a preliminary drawing indicating layout and text content. Erect a signboard at a conspicuous location on the job site where directed by the Contracting Officer.

- a. The field of the sign consists of a 4 by 8 foot sheet of grade B-B medium density overlaid exterior plywood.
- b. Lumber is B or better Southern pine, pressure-preservative treated in accordance with AWPA U1. Nails are aluminum or galvanized steel.
- c. Give one coat of exterior alkyd primer and two coats of exterior alkyd enamel paint to the entire signboard and supports. Perform the lettering and sign work by a skilled sign painter using paint known in

the trade as bulletin colors. The colors, lettering sizes, and lettering styles are as indicated. Where preservative-treated lumber is required, utilize only cured pressure-treated wood which has had the chemicals leached from the surface of the wood prior to painting.

- d. Use spray applied automotive quality high gloss acrylic white enamel paint as background for the NAVFAC logo. NAVFAC logo is an applied 2 mil film sticker/decal with either transparent or white background or paint the logo by stencil onto the sign. The weather resistant sticker/decal film is rated for a minimum of 2-year exterior vertical exposure. Mount the self-adhering sticker to the sign with pressure sensitive, permanent acrylic adhesive. Shop cut sticker/decal to rectangular shape and provide pull-off backing sheet on adhesive side of design sticker for shipping.
- e. Sign paint colors (manufacturer's numbers/types listed below for color identification only)
  - (1) Blue = To match dark blue color in the NAVFAC logo.
  - (2) White = To match Brilliant White color in the NAVFAC logo.
- f. NAVFAC logo must retain proportions and design integrity. NAVFAC logos in electronic format may be obtained from WBDG at the following link:  
<http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>. Use the following to choose color values for the paint to be used:
  - (1) Dark Blue = equivalent to CMYK values 100, 72, 0, 8.
  - (2) Light Blue = equivalent to CMYK values 69, 34, 0, 0.
  - (3) Cyan = equivalent to CMYK values 100, 9, 0, 6.
  - (4) Yellow = equivalent to CMYK values 0.9, 94, 0.

## PART 2 PRODUCTS

Not used.

## PART 3 EXECUTION

Not used.

-- End of Section --

SECTION 01 74 19

CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL  
**02/19, CHG 3:11/21**

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 DEFINITIONS

1.1.1 Co-mingle

The practice of placing unrelated materials together in a single container, usually for benefits of convenience and speed.

1.1.2 Construction Waste

Waste generated by construction activities, such as scrap materials, damaged or spoiled materials, temporary and expendable construction materials, and other waste generated by the workforce during construction activities.

1.1.3 Demolition Debris/Waste

Waste generated from demolition activities, including minor incidental demolition waste materials generated as a result of Intentional dismantling of all or portions of a building, to include clearing of building contents that have been destroyed or damaged.

1.1.4 Disposal

Depositing waste in a solid waste disposal facility, usually a managed landfill, regulated in the US under the Resource Conservation and Recovery Act (RCRA).

1.1.5 Diversion

The practice of diverting waste from disposal in a landfill, by means of eliminating or minimizing waste, or reuse of materials.

1.1.6 Final Construction Waste Diversion Report

A written assertion by a material recovery facility operator identifying constituent materials diverted from disposal, usually including summary tabulations of materials, weight in short-ton.

1.1.7 Recycling

The series of activities, including collection, separation, and processing, by which products or other materials are diverted from the solid waste stream for use in the form of raw materials in the manufacture of new products sold or distributed in commerce, or the reuse of such materials as substitutes for goods made of virgin materials, other than fuel.

#### 1.1.8 Reuse

The use of a product or materials again for the same purpose, in its original form or with little enhancement or change.

#### 1.1.9 Salvage

Usable, salable items derived from buildings undergoing demolition or deconstruction, parts from vehicles, machinery, other equipment, or other components.

#### 1.1.10 Source Separation

The practice of administering and implementing a management strategy to identify and segregate unrelated waste at the first opportunity.

### 1.2 CONSTRUCTION WASTE (INCLUDES DEMOLITION DEBRIS/WASTE)

Divert a minimum of 60 percent by weight of the project construction waste and demolition debris/waste from the landfill or incinerator. Follow applicable industry standards in the management of waste. Apply sound environmental principles in the management of waste. (1) Practice efficient waste management when sizing, cutting, and installing products and materials and (2) use all reasonable means to divert construction waste and demolition debris/waste from landfills and incinerators and to facilitate the recycling or reuse of excess construction materials.

### 1.3 CONSTRUCTION WASTE MANAGEMENT

Implement a Construction Waste Management Program for the project. Take a pro-active, responsible role in the management of construction waste, recycling process, disposal of demolition debris/waste, and require all Subcontractors, vendors, and suppliers to participate in the Construction Waste Management Program. Establish a process for clear tracking, and documentation of construction waste and demolition debris/waste.

#### 1.3.1 Implementation of Construction Waste Management Program

Develop and document how the Construction Waste Management Program will be implemented in a Construction Waste Management Plan. Submit a Construction Waste Management Plan to the Contracting Officer for approval. Construction waste and demolition debris/waste materials include un-used construction materials not incorporated in the final work, as well as demolition debris/waste materials from demolition activities or deconstruction activities. In the management of waste, consider the availability of viable markets, the condition of materials, the ability to provide material in suitable condition and in a quantity acceptable to available markets, and time constraints imposed by internal project completion mandates.

#### 1.3.2 Oversight

The Environmental Manager, as specified in Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME), is responsible for overseeing and documenting results from executing the Construction Waste Management Plan for the project and is responsible for all waste being disposed of in accordance with the contract documents and state & Federal regulations. If any waste materials are removed from the Portsmouth Naval Shipyard not in compliance with the requirements stated

in the contract documents, the person assigned as the Environmental Manager is subject to removal by the Contracting Officer for non-compliance with requirements specified in the Contract. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders is acceptable as the subject of claim for extension of time for excess costs or damages by the Contractor. The Contractor will be responsible to take all required actions to address and correct any non-compliance as well paying any fines as result of improper handling or disposal. This will reflect poorly on the Prime Contractor's performance rating (CPARS) and will be noted in their final performance evaluation.

#### 1.3.3 Special Programs

Implement special programs involving rebates or similar incentives related to recycling of construction waste and demolition debris/waste materials. Retain revenue or savings from salvaged or recycling, unless otherwise directed. Ensure firms and facilities used for recycling, reuse, and disposal are permitted for the intended use to the extent required by Federal, State, and local regulations.

#### 1.3.4 Special Instructions

Provide on-site instruction of appropriate separation, handling, recycling, salvage, reuse, and return methods to be used by all parties at the appropriate stages of the projects. Designation of single source separating or commingling will be clearly marked on the containers.

#### 1.3.5 Waste Streams

Delineate waste streams and characterization, including estimated material types and quantities of waste, in the Construction Waste Management Plan. Manage all waste streams associated with the project. Typical waste streams are listed below. Include additional waste streams not listed:

- a. Land Clearing Debris
- b. Asphalt
- c. Masonry and CMU
- d. Concrete
- e. Metals (includes, but is not limited to, banding, stud trim, ductwork, piping, rebar, roofing, other trim, steel, iron, galvanized, stainless steel, aluminum, copper, zinc, bronze)
- f. Wood (nails and staples allowed)
- g. Glass
- h. Paper
- i. Plastics (PET, HDPE, PVC, LDPE, PP, PS, Other)
- j. Gypsum
- k. Non-hazardous paint and paint cans



- l. Carpet
- m. Ceiling Tiles
- n. Insulation
- o. Beverage Containers

#### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-01 Preconstruction Submittals

Construction Waste Management Plan; G

##### SD-11 Closeout Submittals

Final Construction Waste Diversion Report; G

#### 1.5 MEETINGS

Conduct Construction Waste Management meetings. After award of the Contract and prior to commencement of work, schedule and conduct a meeting with the Contracting Officer to discuss the proposed Construction Waste Management Plan and to develop a mutual understanding relative to the management of the Construction Waste Management Program and how waste diversion requirements will be met.

At a minimum, discuss and document waste management goals at following meetings:

- a. Preconstruction meeting.
- b. Weekly Quality Control meetings.
- c. Work safety meeting (if applicable).

#### 1.6 CONSTRUCTION WASTE MANAGEMENT PLAN

Submit the Construction Waste Management Plan at least 15 calendar days prior to the Preconstruction Meeting or Environmental Brief, whichever comes first. This will allow for sufficient review and comment prior to important preconstruction meeting, and provide the opportunity for constructive feedback and collaboration on the important submittal. Revise and resubmit the Construction Waste Management Plan as necessary, in order for construction to begin. Execute demolition or deconstruction activities in accordance with Section 02 41 00 DEMOLITION. Manage demolition debris/waste or deconstruction materials in accordance with the approved construction waste management plan.

An approved Construction Waste Management Plan will not relieve the Contractor of responsibility for compliance with applicable environmental regulations or meeting project cumulative waste diversion requirement. Ensure all Subcontractors receive a copy of the approved Construction

Waste Management Plan. The plan demonstrates how to meet the project waste diversion requirement. Also, include the following in the plan:

- a. Identify the names of individuals responsible for waste management and waste management tracking, along with roles and responsibilities on the project.
- b. Actions that will be taken to reduce solid waste generation, including coordination with Subcontractors to ensure awareness and participation.
- c. Description of the regular meetings to be held to address waste management.
- d. Description of the specific approaches to be used in recycling/reuse of the various materials generated, including the areas on site and equipment to be used for processing, sorting, and temporary storage of materials.
- e. Name of landfill and/or incinerator to be used.
- f. Identification of local and regional re-use programs, including non-profit organizations such as schools, local housing agencies, and organization that accept used materials such as material exchange networks and resale stores. Include the name, location, phone number for each re-use facility identified, and provide a copy of the permit or license for each facility.
- g. List of specific materials, by type and quantity, that will be salvaged for resale, salvaged and reused on the current project, salvaged and stored for reuse on a future project, or recycled. Identify the recycling facilities by name, address, and phone number.
- h. Identification of materials that cannot be recycled or reused with an explanation or justification, to be approved by the Contracting Officer.
- i. Description of the means by which any materials identified in item (g) above will be protected from contamination.
- j. Description of the means of transportation of the recyclable materials (whether materials will be site-separated and self-hauled to designated centers, or whether mixed materials will be collected by a waste hauler and removed from the site).
- k. Copy of training plan for Subcontractors and other services to prevent contamination by co-mingling materials identified for diversion and waste materials.

Distribute copies of the waste management plan to each Subcontractor, Environmental Manager, and the Contracting Officer.

## 1.7 RECORDS (DOCUMENTATION)

### 1.7.1 General

Maintain records to document the types and quantities of waste generated and diverted through re-use, recycling and/or sale to third parties; through disposal to a landfill or incinerator facility. Provide explanations for any materials not recycled, reused or sold. Collect and

retain manifests, weight tickets, sales receipts, and invoices specifically identifying diverted project waste materials or disposed materials.

#### 1.7.2 Accumulated

Maintain a running record of materials generated and diverted from landfill disposal, including accumulated diversion rates for the project. Make records available to the Contracting Officer during construction or incidental demolition activities. Provide a copy of the diversion records to the Contracting Officer upon completion of the construction, incidental demolitions, or minor deconstruction activities.

#### 1.8 FINAL CONSTRUCTION WASTE DIVERSION REPORT

A Final Construction Waste Diversion Report is required at the end of the project. Provide the Contracting Officer the Final Construction Waste Diversion Report 60 days prior to the Beneficial Occupancy Date (BOD).

#### 1.9 COLLECTION

Collect, store, protect, and handle reusable and recyclable materials at the site in a manner which prevents contamination, and provides protection from the elements to preserve their usefulness and monetary value. Provide receptacles and storage areas designated specifically for recyclable and reusable materials and label them clearly and appropriately to prevent contamination from other waste materials. Keep receptacles or storage areas neat and clean.

Train Subcontractors and other service providers to either separate waste streams or use the co-mingling method as described in the construction waste management plan. Handle hazardous waste and hazardous materials in accordance with applicable regulations and coordinate with Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME) and Section 02 81 00 TRANSPORTATION AND DISPOSAL OF HAZARDOUS MATERIALS. Separate materials by one of the following methods described herein:

##### 1.9.1 Source Separation Method

Separate waste products and materials that are recyclable from trash and sort as described below into appropriately marked separate containers and then transport to the respective recycling facility for further processing. Deliver materials in accordance with recycling or reuse facility requirements (e.g., free of dirt, adhesives, solvents, petroleum contamination, and other substances deleterious to the recycling process). Separate materials into the category types as defined in the Construction Waste Management Plan.

##### 1.9.2 Co-Mingled Method

Place waste products and recyclable materials into a single container and then transport to an authorized recycling facility, which meets all applicable requirements to accept and dispose of recyclable materials in accordance with all applicable local, state and federal regulations. The Co-mingled materials must be sorted and processed in accordance with the approved Construction Waste Management Plan.

### 1.9.3 Other Methods

Other methods proposed by the Contractor may be used when approved by the Contracting Officer.

### 1.10 DISPOSAL

Control accumulation of waste materials and trash. Recycle or dispose of collected materials offsite at intervals approved by the Contracting Officer and in compliance with waste management procedures as described in the waste management plan. Except as otherwise specified in other Sections of the specifications, dispose of in accordance with the following:

#### 1.10.1 Reuse

Give first consideration to reusing construction and demolition materials as a disposition strategy. Recover for reuse materials, products, and components as described in the approved construction waste management plan. Coordinate with the Contracting Officer to identify onsite reuse opportunities or material sales or donation available through Government resale or donation programs. Sale of recovered materials is not allowed on the Portsmouth Naval Shipyard. Consider the use of surplus industrial supply broker services, who match entities with reusable or repurpose industrial materials with entities in need of such materials.

#### 1.10.2 Recycle

Recycle non-hazardous construction and demolition/debris materials that are not suitable for reuse. Track rejection of contaminated recyclable materials by the recycling facility. Rejected recyclables materials will not be counted as a percentage of diversion calculation. Recycle all fluorescent lamps, HID lamps, mercury (Hg) -containing thermostats and ampoules, and PCBs-containing ballasts and electrical components as directed by the Contracting Officer. Do not crush lamps on site as this creates a hazardous waste stream with additional handling requirements.

#### 1.10.3 Waste

Dispose by landfill or incineration only those waste materials with no practical use, economic benefit, or recycling opportunity.

### PART 2 PRODUCTS

Not used.

### PART 3 EXECUTION

Not used.

-- End of Section --

| CONSTRUCTION AND DEMOLITION SOLID WASTE REPORT                |             |                         |             |         |
|---------------------------------------------------------------|-------------|-------------------------|-------------|---------|
| <b>SITE:</b> _____                                            |             | <b>Month:</b> _____     |             |         |
| <b>Contractor's Company Name:</b> _____                       |             | <b>Contract #</b> _____ |             |         |
| <b>Contractor's POC and Telephone or Email Address:</b> _____ |             |                         |             |         |
| <b>Project Description:</b> _____                             |             |                         |             |         |
|                                                               |             |                         |             |         |
| SECTION 1                                                     | Tons        | Cost                    | Revenue     | Remarks |
| <b>Recycled (tons)</b>                                        |             |                         |             |         |
| Concrete(incl: brick & block)                                 |             |                         |             |         |
| Wood                                                          |             |                         |             |         |
| Metal                                                         |             |                         |             |         |
| Asphalt                                                       |             |                         |             |         |
| Green waste(clearing debris)                                  |             |                         |             |         |
| Dirt                                                          |             |                         |             |         |
| Sand                                                          |             |                         |             |         |
| Gravel/Rock                                                   |             |                         |             |         |
| Mixed                                                         |             |                         |             |         |
| Misc                                                          |             |                         |             |         |
| <b>Subtotal - Recycled</b>                                    | <b>0.00</b> | <b>\$ -</b>             | <b>\$ -</b> |         |
| SECTION 2                                                     |             |                         |             |         |
| <b>Landfilled (tons)</b>                                      |             |                         |             |         |
| Concrete(incl: brick & block)                                 |             |                         |             |         |
| Wood                                                          |             |                         |             |         |
| Metal                                                         |             |                         |             |         |
| Asphalt                                                       |             |                         |             |         |
| Green Waste(clearing debris)                                  |             |                         |             |         |
| General C&D                                                   |             |                         |             |         |
| Dirt                                                          |             |                         |             |         |
| Sand                                                          |             |                         |             |         |
| Gravel/Rock                                                   |             |                         |             |         |
| Mixed                                                         |             |                         |             |         |
| Misc                                                          |             |                         |             |         |
| <b>Subtotal - Landfilled</b>                                  | <b>0.00</b> | <b>\$ -</b>             | <b>\$ -</b> |         |
| <b>Solid Waste (tons)</b>                                     |             |                         |             |         |
| <b>Total Solid Waste</b>                                      | <b>0.00</b> | <b>\$ -</b>             | <b>\$ -</b> |         |

|                                                                  |
|------------------------------------------------------------------|
| <b>REPORTING DEADLINE IS NO LATER THAN THE 3RD OF EACH MONTH</b> |
|------------------------------------------------------------------|

SECTION 01 78 00.00 22

CLOSEOUT SUBMITTALS (PWD ME)  
05/19, CHG 1: 08/21

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E1971 (2005; R 2011) Standard Guide for  
Stewardship for the Cleaning of Commercial  
and Institutional Buildings

GREEN SEAL (GS)

GS-37 (2017) Cleaning Products for Industrial  
and Institutional Use

U.S. DEPARTMENT OF DEFENSE (DOD)

FC 1-300-09N (2024) Navy and Marine Corps Design

UFC 1-300-08 (2023) Criteria for Transfer and  
Acceptance of DoD Real Property

1.2 DEFINITIONS

1.2.1 As-Built Drawings

As-built drawings are the marked-up drawings, maintained by the Contractor onsite, that depict actual conditions and deviations from the Contract Documents. These deviations and additions may result from coordination required by, but not limited to: Contract modifications; official responses to submitted Requests for Information (RFIs); direction from the Contracting Officer; design that is the responsibility of the Contractor; and differing site conditions. Maintain the as-builts throughout construction as red-lined hard copies on site. These files serve as the basis for the creation of the record drawings.

1.2.2 Record Drawings

The record drawings are the final compilation of actual conditions reflected in the as-built drawings.

1.3 SOURCE DRAWING FILES

One (1) set of electronic drawings, in the source format, will be provided by the Government at the Preconstruction Meeting for Record Drawing

preparation.

#### 1.3.1 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction data for the referenced project. Any other use or reuse is at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim and waives to the fullest extent permitted by law, any claim or clause of action of any nature against the Government, its agents or sub consultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic CAD drawing files are not construction documents. Differences may exist between the CAD files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic CAD files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished Source Drawing files, the signed and sealed construction documents govern. The Contractor is responsible for determining if any conflict exists. Use of these Source Drawing files does not relieve the Contractor of duty to fully comply with the contract documents, including and without limitation, the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic Source Drawing files for use in producing construction drawings and data related to this contract, remove all previous indicia ownership (seals, logos, signatures, initials and dates).

#### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-03 Product Data

Warranty Management Plan

Warranty Tags

Final Cleaning

Spare Parts Data

##### SD-08 Manufacturer's Instructions

Posted Instructions

##### SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

Comply with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA for O&M Data format. Refer to Section 01 78 24.00 22 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI) for additional requirements.

SD-11 Closeout Submittals

As-Built Drawings; G

Record Drawings; G

Warranted Equipment And Materials

Interim DD FORM 1354; G

List Of Installed Building Equipment Items For DD Form 1354

1.5 SPARE PARTS DATA

Submit two copies of the Spare Parts Data list.

- a. Indicate manufacturer's name, part number, and stock level required for test and balance, pre-commissioning, maintenance and repair activities. List those items that may be standard to the normal maintenance of the system.

1.6 WARRANTY MANAGEMENT

1.6.1 Warranty Management Plan

Develop a Warranty Management Plan which contains information relevant to FAR 52.246-21 Warranty of Construction.

At least 30 days before the planned Pre-Warranty Conference, submit one set of the Warranty Management Plan. Include within the Warranty Management Plan all required actions and documents to assure that the Government receives all warranties to which it is entitled. The plan narrative must contain sufficient detail to render it suitable for use by future maintenance and repair personnel, whether tradesmen, or of engineering background, not necessarily familiar with this Contract. The term "status" as indicated below must include due date and whether item has been submitted or was accomplished. Submit warranty information, made available during the construction phase, to the Contracting Officer for approval prior to each monthly pay estimate. Assemble approved information in a binder and turn over to the Government upon acceptance of the work. The construction warranty period must begin on the date of acceptance and continue for the full product warranty period. Conduct a joint 4 month and 9 month warranty inspection, measured from time of acceptance; with the Contractor, Contracting Officer and the Customer Representative. The Warranty Management Plan must include, but is not limited to, the following:

- a. Roles and responsibilities of personnel associated with the warranty process, including points of contact and telephone numbers within the organizations of the Contractors, Subcontractors, manufacturers or suppliers involved.



- b. For each warranty, the name, address, telephone number, and e-mail of each of the guarantor's representatives nearest to the project location.
- c. A list and status of delivery of Certificates of Warranty for extended warranty items, including roofs, HVAC balancing, pumps, motors, transformers, and for commissioned systems, such as fire protection and alarm systems, sprinkler systems, and lightning protection systems.
- d. Warranted equipment and materials list for each warranted equipment, item, feature of construction or system indicating:
  - (1) Name of item.
  - (2) Model and serial numbers.
  - (3) Location where installed.
  - (4) Name and phone numbers of manufacturers or suppliers.
  - (5) Names, addresses and telephone numbers of sources of spare parts.
  - (6) Warranties and terms of warranty. Include one-year overall warranty of construction, including the starting date of warranty of construction. Items which have warranties longer than one year must be indicated with separate warranty expiration dates.
  - (7) Cross-reference to warranty certificates as applicable.
  - (8) Starting point and duration of warranty period.
  - (9) Summary of maintenance procedures required to continue the warranty in force.
  - (10) Cross-reference to specific pertinent Operation and Maintenance manuals.
  - (11) Organization, names and phone numbers of persons to call for warranty service.
  - (12) Typical response time and repair time expected for various warranted equipment.
- e. The plans for attendance at the 4 and 9 month post-construction warranty inspections conducted by the Government.
- f. Procedure and status of tagging of equipment covered by warranties longer than one year.
- g. Copies of instructions to be posted near selected pieces of equipment where operation is critical for warranty or safety reasons.

#### 1.6.2 Performance Bond

The Performance Bond must remain effective throughout the construction and warranty period.

- a. In the event the Contractor fails to commence and diligently pursue any construction warranty work required, the Contracting Officer will have the work performed by others, and after completion of the work, will charge the remaining construction warranty funds of expenses incurred by the Government while performing the work, including, but not limited to administrative expenses.
- b. In the event sufficient funds are not available to cover the construction warranty work performed by the Government at the Contractor's expense, the Contracting Officer will have the right to recoup expenses from the bonding company.
- c. Following oral or written notification of required construction

warranty repair work, respond in a timely manner. Written verification will follow oral instructions. Failure to respond will be cause for the Contracting Officer to proceed against the Contractor.

#### 1.6.3 Pre-Warranty Conference

Prior to Contract completion, and at a time designated by the Contracting Officer, meet with the Contracting Officer to develop a mutual understanding with respect to the requirements of this Section. At this meeting, establish and review communication procedures for Contractor notification of construction warranty defects, priorities with respect to the type of defect, reasonable time required for Contractor response, and other details deemed necessary by the Contracting Officer for the execution of the construction warranty. In connection with these requirements and at the time of the Contractor's quality control completion inspection, furnish the name, telephone number and address of a licensed and bonded company which is authorized to initiate and pursue construction warranty work action on behalf of the Contractor. This point of contact must be located within the local service area of the warranted construction, be continuously available, and be responsive to Government inquiry on warranty work action and status. This requirement does not relieve the Contractor of any of its responsibilities in connection with other portions of this provision.

#### 1.6.4 Warranty Tags

At the time of installation, tag each warranted item with a durable, oil and water resistant tag approved by the Contracting Officer. Attach each tag with a copper wire and spray with a silicone waterproof coating. Also, submit two record copies of the warranty tags showing the layout and design. The date of acceptance and the QC signature must remain blank until the project is accepted for beneficial occupancy. Show the following information on the tag.

|                                         |  |
|-----------------------------------------|--|
| Type of product/material                |  |
| Model number                            |  |
| Serial number                           |  |
| Contract number                         |  |
| Warranty period from/to                 |  |
| Inspector's signature                   |  |
| Construction Contractor                 |  |
| Address                                 |  |
| Telephone number                        |  |
| Warranty contact                        |  |
| Address                                 |  |
| Telephone number                        |  |
| Warranty response time<br>priority code |  |

WARNING - PROJECT PERSONNEL TO PERFORM ONLY OPERATIONAL MAINTENANCE DURING THE WARRANTY PERIOD.

## PART 2 PRODUCTS

Not used.

## PART 3 EXECUTION

### 3.1 AS-BUILT DRAWINGS

Provide and maintain two black line print copies of the PDF Contract Drawings for As-Built Drawings. Maintain the as-builts throughout construction as red-lined hard copies onsite.

Provide Monthly Progress As-Built Drawings. Provide one (1) set of working as-built drawings, electronically in the specified software and format, to the Contracting Officer with the Monthly Progress Invoice. The Contracting Officer will also review the red-lined hard copies on site monthly, during the review of the Monthly Progress Invoice. The Contractor shall coordinate with the Contracting Officer to arrange for monthly in-person review of the red lined hard-copies onsite.

Failure to submit the working as-built drawings in electronic format and/or to maintain the working as-built drawings in hard copy format, as specified herein, may result in rejection of the Monthly Progress Invoice.

At project completion, after approval of the complete set of Record Drawings, provide the two (2) sets of the red-lined hard copies of the Contract Drawings to the Contracting Officer.

#### 3.1.1 Markup Guidelines

Make comments and markup the drawings complete without reference to letters, memos, or materials that are not part of the As-Built drawing. Show what was changed, how it was changed, where item(s) were relocated and change related details. These working as-built markup prints must be neat, legible and accurate as follows:

- a. Use base colors of red, green, and blue. Color code for changes as follows:
  - (1) Special (Blue) - Items requiring special information, coordination, or special detailing or detailing notes.
  - (2) Deletions (Red) - Over-strike deleted graphic items (lines), lettering in notes and leaders.
  - (3) Additions (Green) - Added items, lettering in notes and leaders.
- b. Provide a legend if colors other than the "base" colors of red, green, and blue are used.

- c. Add and denote any additional equipment or material facilities, service lines, incorporated under As-Built Revisions if not already shown in legend.
- d. Use frequent written explanations on markup drawings to describe changes. Do not totally rely on graphic means to convey the revision.
- e. Use legible lettering and precise and clear digital values when marking prints. Clarify ambiguities concerning the nature and application of change involved.
- f. Wherever a revision is made, also make changes to related section views, details, legend, profiles, plans and elevation views, schedules, notes and call out designations, and mark accordingly to avoid conflicting data on all other sheets.
- g. For deletions, cross out all features, data and captions that relate to that revision.
- h. For changes on small-scale drawings and in restricted areas, provide large-scale inserts, with leaders to the applicable location.
- i. Indicate one of the following when attaching a print or sketch to a markup print:
  - 1) Add an entire drawing to Contract Drawings.
  - 2) Change the Contract Drawing to show changes on the drawing.
  - 3) Provided for reference only to further detail the initial design.
- j. Incorporate all shop and fabrication drawings into the markup drawings.

### 3.1.2 As-Built Drawings Content

Show on the as-built drawings, but not limited to, the following information:

- a. The actual location (horizontal and vertical position based on the datum established for the contract drawings), kinds and sizes of all sub-surface utility lines. In order that the location of these lines and appurtenances may be determined in the event the surface openings or indicators become covered over or obscured, show by offset dimensions to two permanently fixed surface features the end of each run including each change in direction on the record drawings. Locate valves, manholes, utility structures, splice boxes and similar appurtenances by dimensioning along the utility run from a reference point. Also record the average depth below the surface of each run.
- b. The location and dimensions of any changes within the building structure.
- c. Layout and schematic drawings of electrical circuits and piping.
- d. Correct grade, elevations, cross section, or alignment of roads, earthwork, structures or utilities if any changes were made from Contract plans.

- e. Changes in details of design or additional information obtained from working drawings specified to be prepared or furnished by the Contractor; including but not limited to shop drawings, fabrication, erection, installation plans and placing details, pipe sizes, insulation material, dimensions of equipment, and foundations.
- f. The topography, invert elevations and grades of drainage structures and pipes installed or affected as part of the project construction.
- g. Changes or Revisions which result from the final inspection.
- h. Where Contract Drawings or Specifications present options, show only the option selected for construction on the working as-built markup drawings.
- i. If borrow material for this project is from sources on Government property, or if Government property is used as a spoil area, furnish a contour map of the final borrow pit/spoil area elevations.
- j. Systems designed or enhanced by the Contractor, such as HVAC controls, fire alarm, fire sprinkler, and irrigation systems.
- k. Changes in location of equipment and architectural features.
- l. Modifications and compliance with FC 1-300-09N procedures.
- m. Actual location of anchors, construction and control joints, etc., in concrete.
- n. Unusual or uncharted obstructions that are encountered in the Contract work area during construction.
- o. Location, extent, thickness, and size of stone protection particularly where it will be normally submerged by water.

### 3.2 RECORD DRAWINGS

Prepare and provide Record Drawings and Source Documents in accordance with FC 1-300-09N.

Within 30 calendar days of the BOD, provide four copies of Record Drawings and Documents for that DFOW on separate CDs or DVDs.

Submit one (1) set of electronic files on compact disc, read-only memory CD-ROM, one (1) set of mylars and one set of the approved working record drawings. They must be complete in all details and identical in form and function to the Contract Drawing files supplied by the Government. Any transactions or adjustments necessary to accomplish this is the responsibility of the Contractor. The Government reserves the right to reject any drawing files it deems incompatible with the customer's CADD system. Paper prints, drawing files, and storage media submitted will become the property of the Government upon final approval. Failure to submit final record drawing files and marked prints as specified will be cause for withholding any payment due under this Contract. Approval and acceptance of final record drawings must be accomplished before final payment is made.

### 3.2.1 Computer Aided Design and Drafting (CADD) Requirements

Only employ personnel proficient in the preparation of CADD drawings to modify the Contract Drawings or prepare any additional drawings sheets required. Modifications to the Record Drawings must be equal in quality and detail to that of the original Contract Drawings and per PWD ME CADD Standards and As-Built CADD Standards. These standards are included as Attachments A and B.

Line colors, line weights, lettering, layering conventions, and symbols must remain consistent throughout the record drawing set, regardless of either as-built or record drawing. Modify the original Contract Drawing files to reflect the Construction Contract as-built conditions reviewed and accepted by **the Contracting Officer**. Each as-built condition added to a drawing file must be encapsulated by a closed polygon or "revision cloud." A revision symbol must be placed outside the "revision cloud" with the appropriate letter designating the revision sequence. Annotate in the "revision block" of each drawing file modified as to the type of revisions made to the drawing file. The Contract Drawings are to be edited to reflect the as-built conditions only. No part of the original drawings must be deleted, erased or rendered illegible. Parts of the Contract Drawing found to be in error or modified during construction, must be over struck using methods described not to obscure the original drawing, and annotations must be added adjacent that clearly explain the modification, including accurate dimensions locating the feature. If additional drawings are required, the drawings must be prepared using the specified electronic file format applying, the same graphic standards specified for the original drawings. The title block and drawing border to be used for any new final record drawings must be identical to that used to create the Contract Drawings. Modifications, additions, and corrections to the Contract Drawings must be made to the electronic AutoCAD file(s). The original Contract Drawing files in the AutoCAD software format currently in use by PWD-ME will be furnished on a compact disc (CD). Provide all computer software and hardware necessary to prepare the final record drawing set. The Contracting Officer will review the final record drawing set for accuracy and return them for required corrections, changes, additions, and deletions.

a. Provide Record Drawings (CADD) in the following format:

- (1) As-built Layering; follow original drawing layer naming conventions followed by "-AB".
- (2) Deletions (Cyan) - Over-strike deleted graphic items (lines), lettering in notes and leaders.
- (3) Additions (Cyan) - Added items, lettering in notes and leaders.
- (4) Special (Cyan) - Items requiring special information, coordination, or special detailing or detailing notes.
- (5) Furnish the Contract record drawing files in the AutoCAD software format currently in use by PWD-ME.

b. Drawing files modified for as-built condition must be renamed by adding an underscore and the letters "AB" to the end of the existing file name. Drawing files where no modifications were required must be renamed by adding an underscore and the letters "RD" to the end of the existing file name.

- c. When final revisions have been completed to the record drawing set, add the wording "RECORD DRAWINGS / AS-BUILT CONDITIONS" followed by the name of the Contractor in letters at least 3/16 inch high in the lower left hand corner of the cover sheet drawing. Mark all other Contract Drawings in the same location and manner as either "Record Drawing" denoting no revisions on the sheet or "As built Drawing" denoting modifications, additions, or corrections have been made to the drawing sheet. Modify the revision block to reflect either "record drawing", for no changes or "as built drawing", for changes and date for submittal.

### 3.3 CLEANUP

Provide final cleaning in accordance with ASTM E1971 and submit two copies of the listing of completed final clean-up items. Leave site and surrounding premises "broom clean." Comply with GS-37 for general purpose cleaning and bathroom cleaning. Use only nonhazardous cleaning materials, including natural cleaning materials, in the final cleanup. Clean interior and exterior glass surfaces exposed to view; remove temporary labels, stains and foreign substances; polish transparent and glossy surfaces; vacuum carpeted and soft surfaces. Clean equipment and fixtures to a sanitary condition. Replace filters of operating equipment and comply with the Indoor Air Quality (IAQ) Management Plan. Clean debris from roofs, gutters, downspouts and drainage systems. Sweep paved areas and rake clean landscaped areas. Remove waste and surplus materials, rubbish and construction facilities from the site. Recycle, salvage, and return construction and demolition waste from project in accordance with Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME) and 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL.

### 3.4 REAL PROPERTY RECORD

Refer to UFC 1-300-08 for instruction on completing the DD FORM 1354. Contact the Contracting Officer for any project specific information necessary to complete the DD FORM 1354.

#### 3.4.1 Interim DD FORM 1354

Near the completion of the Project, but a minimum of 60 calendar days prior to final acceptance of the work, complete and submit an accounting of all installed property with Interim DD FORM 1354. The Interim DD FORM 1354 need only include the installed property associated with that partial BOD. Include any additional assets, improvements, and alterations from the Draft DD FORM 1354.

#### 3.4.2 Completed DD FORM 1354

For convenience, a blank fillable PDF DD FORM 1354 may be obtained at the following link:

[www.esd.whs.mil/Portals/54/Documents/DD/forms/dd/dd1354.pdf](http://www.esd.whs.mil/Portals/54/Documents/DD/forms/dd/dd1354.pdf)

Submit the completed List of Installed Building Equipment items for DD FORM 1354. Attach this list to the updated DD FORM 1354.

### 3.5 Certificate to Occupy/Operate

The Contractor shall provide the necessary information to the Contracting Officer, upon request, to support the Government's completion of the

Certificate to Occupy/Operate (CTO) Form. CTO is applicable to any new work or renovation that affects life safety. This may include but is not limited to fire protection systems, structural systems, means of egress, electrical safety, interior finishes, and change of occupancy/use. The Government must submit the CTO Form 30 calendar days prior to BOD. The Contracting Officer will provide an example of the CTO Form to the Contractor, for reference, upon request.

-- End of Section --



# **NAVFAC MID-ATLANTIC**

**Public Works Department  
Maine**

## **PORTSMOUTH NAVAL SHIPYARD**

### **AS-BUILT CADD STANDARDS**

**June 2014**



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# STANDARDS FOR EDITING EXISTING CONTRACT DRAWINGS TO REFLECT AS-BUILT CONDITIONS

Following are the *Standards* for editing existing contract drawings to reflect the construction as-built conditions for **NAVFAC MID-ATLANTIC; Public Works Department Maine (PWD-ME)**. The purpose of these *Standards* is to provide a uniform system of drawing formats. This system will be used in retrieving information from the drawings in the future. Also, these *Standards* were established to eliminate various recurring problems encountered when transferring electronic files from non-government sources. All drawings modifications found to be generated outside of these standards will be returned to the drawing provider for correction at no additional cost to the **GOVERNMENT**.

## **As-built Drawings:**

### **Drawing Sets:**

The contractor shall maintain at the jobsite one set of full-size hard copy prints of the contract drawings, accurately marked in red with adequate dimensions, to show all variations between the construction actually provided and that indicated or specified in the contract documents, including buried or concealed construction. **Special attention shall be given to recording the horizontal and vertical location of all buried utilities that differ from the contract drawings.** Existing utility lines and features revealed during the course of construction shall also be accurately located and dimensioned, using permanent existing features and station coordinates as a reference. Acceptable features include; building corners, centers of utility manhole covers, fire hydrants, etc. Existing topographic features that differ from those shown on the contract drawings shall also be accurately located and recorded. Where a choice of materials or methods is permitted herein or where variations in scope or character of work from that of the original contract are authorized, the drawings shall be marked to define the construction provided. The representations of such changes shall conform to standard drafting practices and shall include such supplementary notes, legends, and details as necessary to clearly portray the as-built construction. These drawings shall be available for review by the Contracting Officer at all times. Upon completion of the work, the marked up prints shall be certified as correct, signed by the Contractor, and delivered to the Contracting Officer for review by **PWD-ME**. After acceptance of the as-builts, the contractor will add the as-built information to the original contract drawing files and plot new Mylar reflecting the as-built conditions.

Requests for partial payments will not be approved if the marked prints are not kept current. Request for final payment will not be approved until Contracting Officer receives delivery of the original electronic contract drawing CADD files, modified to reflect the as-built conditions and new Mylar plotted to reflect the as-built conditions.

## **Computer Aided Drafting Design (CADD) As-built Drawings:**

### **File formats:**

Although all methods of CADD drawing are permitted, the final product of all computer-aided drawings shall be made compatible with the **AutoCAD** currently in use by **PWD-ME** Drawings

created using non-AutoCAD programs that do not support AutoCAD's DWG file format, shall be transferred to the Drawing Interchange Format (DXF).

If the drawing provider's CADD software does not support the DXF format, it is the responsibility of the drawing provider to contact the Contracting Officer who will in turn contact the **PWD-ME** to make special arrangements for file transfer.

Each drawing file must not depend on any other drawing file to completely represent the finished product (*Mylar*). All drawings found to use dependent external reference drawings, **will be returned to the drawing provider for correction.**

Each drawing file must not depend on any **"THIRD PARTY"** utility or software to represent the finished Mylar. Before using any third party software to create a finished product, the drawing provider shall verify that no additional cost or effort is required by the **PWD-ME** to completely represent the finished product (*Mylar*).

### **Drawing Sheets:**

Electronic files of the contract drawings shall be provided to the Contractor following award of the contract. The contractor will add the accepted as-built conditions to the original electronic file of the drawing sheet. If additional sheets need to be added to the contract drawing package because of insufficient sheet space, it is the responsibility of the Contractor to ensure the PWD-ME CADD Drawing Standards are obtained and followed. Additional drawing numbers will be assigned by **PWD-ME** personnel prior to submittal of as-built drawings.

### **Basic Drawing Standards:**

#### **Electronic As-builts:**

The contractor shall modify the original contract drawing files to reflect the construction contract as-built conditions reviewed and accepted by **the Contracting Officer**. Each as-built condition added to a drawing file shall be encapsulated by a closed polygon or "revision cloud. A revision symbol shall be placed outside the "revision cloud" with the appropriate letter designating the revision sequence. The contractor shall annotate in the "revision block" of each drawing file modified as to the type of revisions made to the drawing file. The contract drawings are to be edited to reflect the as-built conditions only. No part of the original drawings shall be deleted, erased or rendered illegible. Parts of the contract drawing found to be in error or modified during construction, shall be over struck using methods described not to obscure the original drawing, and annotations will be added adjacent that clearly explain the modification, including accurate dimensions locating the feature.

#### **Drawing Sheet Management:**

Where construction projects encompass multiple facilities (buildings) or systems (utilities), all plans, details, legends, notes etc., for an individual facility or system must appear on the drawing sheets for each facility or system. Standardized details must appear on the drawing sheets for each facility or system where they apply. Cross-referencing details between facilities or systems is not permitted.

## Drawing Units:

All drawings of buildings, structures, floor plans, etc. are to be drawn full size, with one drawing unit equal to one inch. All site plans, location plans, etc. are to be drawn full size, with one drawing unit equal to one foot. The original drawing's origin shall not be modified.

## Entity Management:

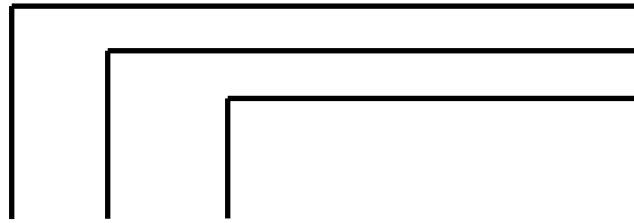
All additions to the original contract drawings will be made on additional layers as designated below. The minimum text height for the "D" size sheet is 1/8in plotted and 3/32in plotted for the "B" size sheet. All as-built modification layers are to be **CYAN** in, including all annotations, dimensions, leaders and callouts.

Entities discovered during construction but not represented on the original construction drawings shall be assigned to layers as described below. This information shall be added to the drawing in such a way not to obscure any original drawing entity when plotted.

## Level/layer Management:

All drawing entities shall be grouped together by layer. Drawing entities shall be assigned to layers by discipline code for that entity, see table page D4. This includes non-text entities such as strikeouts, lines, circles, symbols, etc. See Table-1 for typical layer-naming.

Table-1

|                                                                                     |                                                      |
|-------------------------------------------------------------------------------------|------------------------------------------------------|
|  | As-built Code (Always "Z")                           |
|                                                                                     | Discipline Code (Non-text entities)                  |
|                                                                                     | Annotations (Text describing the As-built Condition) |
|                                                                                     |                                                      |

Identify entities by discipline. For instance: the example below shows layering configuration for all architectural as-built entities **EXCLUDING TEXT** that deviate from the original design:

**Z-ARCH.**

Annotations, dimensions, leaders and callouts for those architectural entities shall be assigned to layer:

**Z-ARCH-TEXT.**

All annotations shall be assigned to layers by discipline and designated as **TEXT**.

Each layer name will follow the discipline code for that entity. See Table-2 for Discipline Descriptions:

Table-2

| <b><i>Discipline Code:</i></b> | <b><i>Discipline:</i></b> | <b><i>Description:</i></b>                                                                                  |
|--------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------|
| ARCH                           | Architectural             | Architectural design, building's interior/exterior, walls, doors, windows etc.                              |
| CIVIL                          | Civil                     | Site work, external to buildings and structures. Typically surface work.                                    |
| ELEC                           | Electrical                | Electrical work pertaining to buildings/structures internal/external and distribution.                      |
| FP                             | Fire Protection           | Fire Protection systems pertaining to buildings/structures internal/external                                |
| GEN                            | General                   | Typically drawing information, general notes, symbols, etc....                                              |
| MECH                           | Mechanical                | Mechanical work, HVAC components, compressed air, etc. pertaining to buildings/structures internal/external |
| PLUMB                          | Plumbing                  | Plumbing, piping and fixtures, etc.. pertaining to buildings/structures internal/external                   |
| UTIL                           | Utilities                 | Utility distribution systems, duct work, manholes and piping/cabling underground/direct buried              |
| STRUC                          | Structural                | All structural work, building framing, conc. duct reinforcement, misc. steel work, etc.                     |
| TELE                           | Telecommunications        | Phone lines, cable TV, computer data lines, etc. pertaining to buildings/structures internal/external       |

### **Electronic Deliverables:**

Submit all as-built documentation and drawing files to the Contracting Officer via recordable CD in the file formats discussed in this and other sections. Electronic transfer of files via E-mail or other methods is not permitted.

### **Standards Deviation:**

Contact the Contracting Officer if questions arise about these standards or these drawing standards cannot be followed. Otherwise, all drawing files will be returned to the drawing provider for correction at no additional cost to the **PWD-ME**.

Last updated June 18, 2014

# **NAVFAC MID-ATLANTIC**

**Public Works Department  
Maine**

## **PORTSMOUTH NAVAL SHIPYARD**

### **CADD STANDARDS**

**June 2014**



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## 1. CADD STANDARDS

All design documents and deliverables prepared by In-House forces and/or contracted A/E firms including IDIQ, Design Bid-Build and Design-Build process shall conform to the following NAVFAC MIDLANT and PWD-ME CADD standards areas not addressed on this standard shall follow:

- UFC 1-300-09N

U.S. DEPARTMENT OF DEFENSE (DOD) [UFCs available on <http://dod.wbdg.org/>]

These are NAVFAC MIDLANT PWD-MAINE specific guidelines that supersede all others for contract documents utilized for execution. Refer to the U.S. National CAD Standard Version 3.0 for any items not covered in this or the referenced NAVFAC UFCs.

### 1. A. General: (for both Design-Build and Design-Bid-Build)

For drawing templates, pen tables, sample symbology, and instructions on sheet arrangement & procedures, see attached NAVFAC MIDLANT PWD-MAINE; Sheet Border Use and Procedures. Provide \*.PDF format of all drawing sheets at each submittal phase. In addition, provide \*.PDF and \*.DWG format of all drawing sheets at the final design submission. CADD files shall be made available upon request, in addition to the full size, stamped Mylar and half-size hard copies or as directed by the specific contract specifications.

### 1. B. Design Drawings

Prepare, organize, and present design drawings in accordance with the requirements of UFC 1-300-09N.

All drawings and their associated PDFs will maintain a “PRELIMINARY Not for Construction” stamp across the signature areas of the title block, until the actual final design submittal. The stamp shall be in translucent lettering across the Project Title area of the drawing title block and shall be displayed on layer “G-ANNO-TTLB-PRLM”. The “G-ANNO-TTLB-PRLM” shall be off for final submittal.

Design revisions shall be tracked by number and date in the revision block of each drawing sheet during the design process. Design revisions shall be assigned to layer “G-ANNO-TTLB-PRLM” and shall be frozen for final submittal.

### 1. C. Drafting Conventions

#### 1. C.1. Fonts

The standard text height for a plotted D-size drawing shall be **1/8”** for typical text, and **1/4”** for titles, and **1”** maximum for project titles on cover sheets. The standard text height for a plotted B-size drawing shall be **3/32”** for typical text, and **1/8”** for titles, and **1/2”** maximum for project titles on cover sheets. For typical text, use the “**RomanS**” font. For titles, use the “**Swis 721**”

**BT”** font. Both fonts shall have a width ratio of no less than **0.8**.

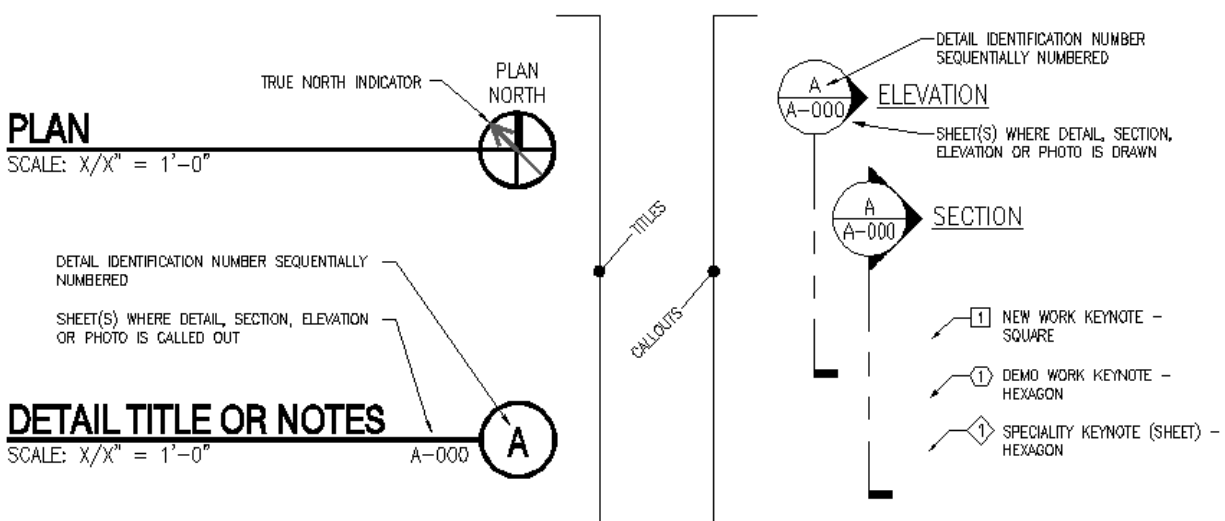
**ROMANS – 1/8” TALL**

**Swis 721 BT** – 1/4” TALL (up to 1”)

These are the only fonts that should be used within the drawing area, with one exception: on civil site plans, the Design Manager (EIC/AIC) may use text (“RomanS” font, 1/8” height {Leroy 100}, 0.8 width factor, with an oblique angle of 12 degrees) to annotate existing site features only. Typical (or Normal) text as defined above must be used for all other annotations.

### 1. C. 2. Symbols

Approved symbology in use by PWD-ME is provided on the PWD-ME title sheet. On the PWD-ME title and standard sheet, borders and plot styles shall delivered at each project design kickoff MTG.



**Figure 3-1 PWD-ME Approved Symbols**

Note that detail, section, elevation and photograph callouts utilize two-part bubbles which indicate a detail identification number and identify sheet where the detail, section, elevation or photo is shown. Additional sheets(s) on which the detail is called out is (are) displayed to the bottom left of the bubble.

### 1. C. 3. Line Weights / Pen Tables

NAVFAC’s Electronic Documents and Deliverables Working Group (EDDWG) developed a comprehensive pen table that utilized the National Cad Standard 255-pen table as a basis, but added much needed thinner lines and additional greyscales (shaded or screened pens) that had not made the transition from the original US Coast Guard standard. The NAVFAC-specific pen

table is provided as filename **NavFacStd.ctb**.

The pen tables can be thought of in groups of 20. After pen 19, pens 20 thru 39 are in the “Rust” range when displayed on screen. The first 10 (pens 20 – 29) look “Rust” and print shades of “Rust”; while the second 10 (pens 30 – 39) look “Rust” but print Black. Typically, design drawings for contract documents utilize the pens that print Black (currently colored pens are used for renderings and some other instances, but not normally on contract drawings).

Since these pen tables are established to be legible when printed at full-size (22”x34”), the corresponding text height is 1/8”. To be consistent, the related B-size (11”x17”) & A-size sheet (8-1/2”x11”)—normally used for sketches, uses fonts and line weights that are ½ the size of the large format (D-size) documents—otherwise the fonts are too big and the line weights are too bold. The associated pen table for B-size & A-size documents is **NavFacStd-Sketch.ctb**

## **1. D. NAVFAC Logo**

A NAVFAC Publications/Logo Guide will be published at a later date, but in the interim, all deliverables shall have the NAVFAC Brand Logo on the title block area of the drawing border (as displayed in the Templates). The logo should not be removed, deleted, or turned off and should appear on all PDFs to be utilized for the electronic bid solicitations. If contract drawings are produced by an A/E firm and that firm desires to have their logo on the drawings then the A/E must apply that logo to all sheets consistently in the same location. Typically that is in the lower- or upper-right hand corner of the drawform, adjacent to the title block.

## **1. E. Units of Measure**

Unless otherwise directed, for all projects the default shall be English units of measurement (U.S. Customary System of units). All drawings, specifications, cost estimates, and design calculations shall reference this system of units.

## **1. F. Datums and Coordinate Systems**

**VERTICAL DATUM:** All elevations, grades, or profiles shall be represented in U.S. feet and referenced to NAVD88 vertical datum and will be provided as required.

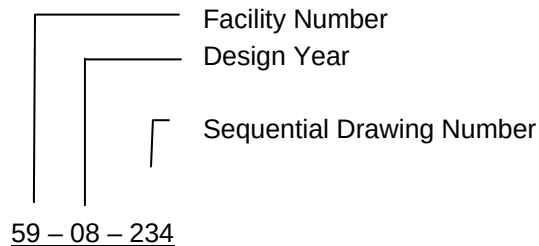
**HORIZONTAL DATUM:** All Portsmouth Naval Shipyard mapping, utility information, site drawings and surveys are represented in U.S. feet and referenced to the state plane coordinate system, NAD83 Maine west, zone 1802. All electronic mapping and utility information shall be maintained to this system.

## **1. G. Drawing Numbers/File Naming**

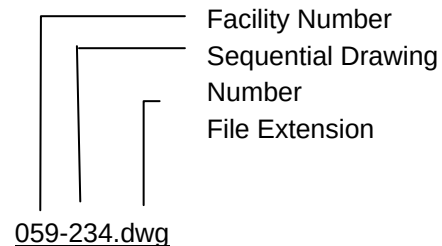
In addition to NAVFAC drawings numbers discussed in UFC 1-300-09N, Chapter 4-3, “Drawing Numbers” the Government Project Manager will provide PWD (Public Works Department) Drawing Numbers for each drawing sheet, similar to NAVFAC drawing numbers. The PWD Drawing Number to be requested by the drawing provider at the 100% design stage. The CADD

files shall be named by the PWD drawing number in accordance with the following guidance: The CADD file for each ANSI D (22" x 34") size drawing sheet shall be named by the corresponding PWD Number, minus the design year designator. Where the assigned PWD Number is: 59-08-501 the corresponding CADD file shall be named 059-501.DWG. Where a single drawing file is used to create multiple drawing sheets via layout tabs, the drawing file shall be named by the PWD Number of the first drawing sheet of the drawing set. Drawing sets shall be arranged by discipline.

#### PW Number



#### CADD File Name



**Figure 7-1 DRAWING NUMBERS/FILE NAMING**

Files for sketches, drawings other than ANSI D (22" x 34") size shall be named by construction contract number and sheet number of the first sheet of the sheet set.

#### **1. G. 1. File Naming Convention for PDF Files:**

ADOBE PDF file naming shall follow CADD file naming methods.

#### **1. H. Electronic Submittals**

Submit all CADD files for the final drawings in Autodesk Drawing (\*.DWG) compatible with current PWD-ME CADD format. Drawing files shall be full files, uncompressed and unzipped. All submitted CADD files shall maintain their original and complete data structure as produced in their native Autodesk application (i.e. Architectural Desktop, Land Development Desktop, Civil 3D, Building Systems...). All files utilizing AutoCAD's "Xref" feature shall be delivered with all external files "bound" to the drawing file. There shall be no external references in the final drawing file.

Multiple layout tabs are allowed, the tabs shall be annotated by PWD number for each sheet as described in section PWD-ME 7.0 Drawing Numbers/File Naming. The electronic file(s) shall be named by the PWD number of the first sheet of the sheet set. When required, the drawing file shall be split by sheet discipline. Corresponding PDF files shall be grouped into one file and named by the project title.

#### **1. I. Record Documents ...1.10.1 Record Drawings (As-Built drawings, G)**

The Quality Control Manager shall deliver the marked-up As-Built drawings to the [Contractor's](#) Designer of Record who shall incorporate all as-built modifications.

The as-built modifications shall be accomplished by electronic drafting methods on the original CADD (\*.DWG) design drawing files to create a complete set of record drawings. For each Record drawing, provide a CADD drawing identical to signed [A/E- or] Contractor-originated \*.PDF drawings, that incorporates modifications to the as-built conditions. In addition, copy initials and dates from the Contracting Officer approved \*.PDF documents to the title block of the Record CADD (\*.DWG) drawings. The Record electronic files shall use the file name of the original signed CADD drawing file name with the suffix “-RD” before the file extension, “.DWG” (example; 059-501-RD.dwg – see Drawing File Naming). The original design, RFP, reference, or definitive drawings are not required for inclusion in the Record set of drawings.

After all as-built conditions are recorded on the CADD (\*.DWG) files, produce a PDF file of each individual record drawing in conformance with UFC 1-300-09N. Generate PDF drawing files using a PDF page size that corresponds to the original document sheet size (NAVFAC utilizes an ANSI D-size, 22”x34” frame and a 0” border all around) and a PDF print resolution that results in clear detail of all drawing features.

Provide one set of signed and stamped record drawings, plotted on Mylar that fully represent the PDF file. Provide all project final submittals including as-builts in accordance with UFC 1-300-09N Chapter 7-5.2. Each CD-ROM shall be marked with Project Name, Construction Contract Number, Project Number, Specification Number, and Record Drawing date.

### **1. I. 1. Source Documents**

In addition to the drawings provide the specifications, design analysis, reports, survey data, calculations, and any other contract documents utilized in creating the design package (drawings, specifications, and cost estimate) on the CD-ROM disk(s) as specified in preceding paragraph.

### **1. J. Forwarding Submittals**

The [A/E- or] Contractor shall provide all submittals, that lend themselves to paper format (i.e. 8-1/2”x11” documents, 11”x17” or tabloid sheets, or 22”x34” drawings), in electronic \*.PDF (Portable Document Format) to the Contracting Officer. The appropriate forwarding e-mail address(es) will be provided at the Pre-Construction meeting. Provision of electronic submittals with appropriate electronic transmittal may alleviate some of the paper copies required herein but must be verified at Pre-Construction meeting. In such instances a minimum of one hard copy shall be forwarded to the Contracting Officer.

SECTION 01 78 23

OPERATION AND MAINTENANCE DATA  
05/23

PART 1 GENERAL

This Section applies to Design-Bid-Build and Design-Build projects at Portsmouth Naval Shipyard and PWD ME AOR Facilities.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING  
ENGINEERS (ASHRAE)

ASHRAE GUIDELINE 1.4 (2019) Preparing Systems Manuals for  
Facilities

ASTM INTERNATIONAL (ASTM)

ASTM E1971 (2005; R 2011) Standard Guide for  
Stewardship for the Cleaning of Commercial  
and Institutional Buildings

ASTM E2166 (2016; R 2023) Standard Practice for  
Organizing and Managing Building Data

U.S. DEPARTMENT OF DEFENSE (DOD)

FC 1-300-09N (2024) Navy and Marine Corps Design

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-10 Operation and Maintenance Data

Training Plan; G

Training Outline; G

Training Content; G

eOMSI Facility Data Workbook; G

Operation And Maintenance Manual, Progress Submittal; G

Operation And Maintenance Manual, Prefinal Submittal; G

Operation And Maintenance Manual, Final Submittal; G

## SD-11 Closeout Submittals

Training Video Recording; G

Validation of Training Completion; G

Record Drawings And Utility Systems; G

### 1.3 MEETINGS

To assure that Operation and Maintenance Manual and eOMSI Facility Data Workbook requirements are being met through the duration of the project, organize the following meetings and discuss the subsequent topics:

#### 1.3.1 Preconstruction Meeting

At a minimum, discuss the following:

- a. The requirement for Operation and Maintenance Manuals and eOMSI Facility Data deliverables under this contract including coordination meetings and multiple submissions related to partial Beneficial Occupancy Dates (BODs);
- b. Processes and method of gathering Facility Data information during construction;
- c. Primary roles and responsibilities associated with the development and delivery of the Operation and Maintenance Manuals and eOMSI Facility Data deliverables; and
- d. Identify and agree upon a date and attendance list for the meetings described below.

#### 1.3.2 Operation and Maintenance Manual and eOMSI Facility Data Workbook Coordination Meeting

Facilitate a meeting after the Preconstruction Meeting and prior to the submission of the first Operation and Maintenance Manual Progress Submittal. Meeting attendance must include the Contractor's Operation and Maintenance Manual and eOMSI Facility Data Workbook Preparer, Quality Control Manager, the Commissioning Authority (CxA), the Government's Design Manager (DM), the Contracting Officer's Representative, and Government's facility data reviewer. Include any Mechanical, Electrical, and Fire Protection Sub-Contractors.

The purpose of this meeting is to reach a mutual understanding of the scope of work concerning the contract requirements for Operation and Maintenance Manual and coordinate the efforts necessary by both the Government and Contractor to ensure an accurate collection, preparation and timely Government review of the Operation and Maintenance Manual.

#### 1.3.3 Submittal Coordination Meeting

Facilitate a meeting following submission and Government review of each design or progress submittal of the Operation and Maintenance Manual and eOMSI Facility Data Workbook.

- a. Include personnel from the Coordination meeting and any additional personnel identified.

- b. The purpose of this meeting is to demonstrate ongoing compliance with the requirements identified in this specification. Discuss Government review comments and unresolved items preventing completion and Government approval of the Operation and Maintenance Manuals and eOMSI Facility Data Workbook (FDW).
- c. The applicable deliverables, along with Government remarks associated with review of these submittals serve as the primary guide and agenda for this meeting.

#### 1.3.4 Facility Turnover Meeting

Include Operation and Maintenance Manual in NAVFAC Red Zone (NRZ) facility turnover meetings as specified in Section 01 30 00.00 22 ADMINISTRATIVE REQUIREMENTS (PWD ME).

#### 1.4 eOMSI FACILITY DATA WORKBOOK

Develop an editable, electronic spreadsheet based on the equipment in the Operation and Maintenance Manuals that contains the information required to start a preventive maintenance program as outlined in 01 78 24.00 22 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI).

#### 1.5 OPERATION AND MAINTENANCE MANUAL MANUAL MEDIA

Assemble the Operation and Maintenance Manual into an electronically bookmarked file using the most current version of Adobe Acrobat or similar software capable of producing PDF file format. Provide compact disks (CD) or data digital versatile disk (DVD) as appropriate, so that each one contains operation, maintenance and record files, project record documents, and training videos. Include a complete bookmarked operation and maintenance directory.

##### 1.5.1 CD or DVD Label and Disk Holder or Case

Provide the following information on the disk label and disk holder or case:

- a. Building Number
- b. Project Title
- c. Activity and Location
- d. Construction Contract Number
- e. Prepared For: (Contracting Agency)
- f. Prepared By: (Name, title, phone number and email address)
- g. Include the disk content on the disk label
- h. Date
- i. Virus scanning program used



#### 1.5.2 O&M Manual Tabbed Hard Copy

Provide a hard copy of the O&M manual upon completion of the project. Provide tabs for each section and subsection for ease of navigation by the user.

#### 1.6 O&M MANUAL CONTENT

Organize the bookmarked Operation and Maintenance Manual into the following Parts in accordance with ASHRAE GUIDELINE 1.4, and as modified and detailed below. Word template for O&M Manual is available at: <https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-01-78-23>.

##### 1.6.1 Part 1: Executive Summary

Provide a summary of the information found in the O&M manual including the purpose of the manual and a description of the manual's organization.

##### 1.6.2 Part 2: Facility Design and Construction

###### 1.6.2.1 General Facility and Systems Description

Provide an overview of the intent for design and use of the facility. Provide a PDF of the Record Drawings prepared in accordance with FC 1-300-09N and 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME) and bookmarked using the sheet title and sheet number. Include uncluttered 11 by 17 inch floor plans with room numbers, type or function of space, and overall facility dimensions on the floor plans. Do not include items such as construction instructions, references, or frame numbers.

Detail the overall dimensions of the facility, number of floors, foundation type, expected number of occupants, and facility Category Code list and generally describe all the facility systems and any special building features (for example, HVAC Controls, Sprinkler Systems, Cranes, Elevators, and Generators). Include photographs marked up and labeled to show key operating components and the overall facility appearance.

###### 1.6.2.2 Contract Documents, RFP, Amendments, and Modifications

Provide the contract construction documents complete, to include specifications, drawings, Request for Proposal, amendments, and modifications.

###### 1.6.2.3 Room Inventory of Real Property and Finishes

Provide a list of installed equipment furnished under this contract. Include all information usually listed on manufacturer's name plate. Include, as applicable, the following information for each piece of equipment installed: description of item, all dimensions, location by room number, model number, serial number, capacity, name and address of manufacturer, name and address of equipment supplier, condition, spare parts list, manufacturer's catalog, and warranty. Real property includes, but is not limited to, floor coverings, wall surfaces, ceiling surfaces, windows, roofing, HVAC filters, plumbing fixtures, and lighting fixtures. Submit the final list 30 days after transfer of the completed facility.

Include spatial data defining actual net square footage and data of each room. Also include the room finish schedule including room names and

numbers. Include schedules in the construction drawings in the room inventory. Add a column to each schedule to record what was provided by the contractor during construction. Provide a PDF of room inventory. Key the designations to the related area depicted on the contract drawings. List the following data:

| RECORD OF DESIGNATED EQUIPMENT AND MATERIALS DATA |                       |                                                    |                      |            |
|---------------------------------------------------|-----------------------|----------------------------------------------------|----------------------|------------|
| Description                                       | Specification Section | Manufacturer and Catalog, Model, and Serial Number | Composition and Size | Where Used |
|                                                   |                       |                                                    |                      |            |

### 1.6.3 Part 3: Facilities, Systems, and Assemblies Information

#### 1.6.3.1 Organization

Bookmark information in this section using the current version of ASTM E2166 Uniformat II, UFGS numbers, and document type as outlined in the example below. Bookmark/tab each item to the third level for easy navigation of the manual.

Example as shown in Table below:

| PARTS AND SUBPART NUMBERING                                       |
|-------------------------------------------------------------------|
| 3.1 B20 EXTERIOR CLOSURE (System)                                 |
| 3.1.1 B2030 EXTERIOR DOORS (Subsystem)                            |
| 3.1.1.1 B2030110 GLAZED DOORS (Component)                         |
| 3.1.1.1.1 Applicable specifications List in UFGS Format           |
| 3.1.1.1.2 Manufacturer's Operations and Maintenance Data          |
| 3.1.1.1.3 Approved Submittal                                      |
| 3.1.1.1.4 Coordination/Shop Drawings                              |
| 3.1.1.1.5 Sequence of Operation for Operating Equipment           |
| 3.1.1.1.6 Testing Equipment Information and Performance Data      |
| 3.1.1.1.7 Routine Maintenance Requirements                        |
| 3.1.1.1.8 Repair Procedures                                       |
| 3.1.1.1.9 Emergency Procedures & Locations of Applicable Controls |
| 3.1.1.1.10 Warranties                                             |
| 3.1.1.1.11 Record Drawings and Utility Systems                    |

| PARTS AND SUBPART NUMBERING                                      |
|------------------------------------------------------------------|
| 3.1.1.1.12 Contractor / Supplies Listing and Contact Information |

#### 1.6.3.2 Related Specifications

Reference each specification related to the subsystem in this section, and locate the actual specification section in Part 2 of the O&M Manual. List specifications in table format as shown in the below example.

| UFGS Number | Specification Title | Page Spec Begins in Part 2 |
|-------------|---------------------|----------------------------|
|             |                     |                            |
|             |                     |                            |
|             |                     |                            |

#### 1.6.3.3 Manufacturer's Operations and Maintenance Data

Provide a copy of all manufacturer specifications and cutsheets. Provide text-searchable, high-quality document files from the manufacturer's online or electronic documentation. Color documents are preferred. Provide documents specific to the product(s) installed under this Contract. Provide identification and coverage for the parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Provide Uniformat II Level 3 identification for D20, D30, D40 installed equipment. When possible, do not submit document files containing multiple product catalogs from the same manufacturer, or product data from multiple manufacturers in the same files. Provide documents directly from the manufacturer whenever possible. Do not provide scanned copies of hardcopy documents. Provide identification and coverage for the parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Include special hardware requirements, such as requirement to use high-strength bolts and nuts. Identify parts by make, model, serial number, and source of supply to allow reordering without further identification. Provide clear and legible illustrations, drawings, and exploded views to enable easy identification of the items. When illustrations omit the part numbers and description, both the illustrations and separate listing must show the index, reference, or key number that will cross-reference the illustrated part to the listed part. Group the parts shown in the listings by components, assemblies, and subassemblies in accordance with the manufacturer's standard practice. Parts data may cover more than one model or series of equipment, components, assemblies, subassemblies, attachments, or accessories, such as typically shown in a master part catalog.

#### 1.6.3.4 Approved Submittals and Certifications

Provide a copy of all submittals documented with the required approval as applicable for each UFGS specification listed in the table outlined in applicable specifications. Include copies of SD-07 Certificates submittals documented with the required approval, SD-08 Manufacturer's

Instructions submittals documented with the required approval, and SD-10 Operation and Maintenance Data submittals documents with the required approval.

#### 1.6.3.5 Approved Coordination/Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work. Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project. Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

#### 1.6.4 Sequence of Operation for Operating Equipment

Provide record one-line diagrams for each floor, delineating mechanical equipment location within the building. Provide specific instructions, procedures, and illustrations for the following phases of operation for the installed model and features of each system:

##### 1.6.4.1 Safety Precautions and Hazards

List personnel hazards and equipment or product safety precautions for operating conditions. List all residual hazards identified in the Activity Hazard Analysis provided under Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME). Provide recommended safeguards for each identified hazard. Specify if any certifications or licenses are required to operate the equipment.

##### 1.6.4.2 Operator Prestart

Provide procedures required to install, set up, and prepare each system for use.

##### 1.6.4.3 Startup, Shutdown, and Post-Shutdown Procedures

Provide narrative description for Startup, Shutdown and Post-shutdown operating procedures including the control sequence for each procedure.

##### 1.6.4.4 Normal Operations

Provide Control Diagrams with data to explain operation and control of systems and specific equipment. Provide narrative description of Normal Operating Procedures.

##### 1.6.4.5 Emergency Operations

Provide Emergency Procedures for equipment malfunctions to permit a short period of continued operation or to shut down the equipment to prevent further damage to systems and equipment. Provide Emergency Shutdown Instructions for fire, explosion, spills, or other foreseeable contingencies. Provide guidance and procedures for emergency operation of utility systems including required valve positions, valve locations and zones or portions of systems controlled.

##### 1.6.4.6 Operator Service Requirements

Provide instructions for services to be performed by the operator such as lubrication, adjustment, inspection, and recording gauge readings.

#### 1.6.4.7 Environmental Conditions

Provide a list of Environmental Conditions (temperature, humidity, and other relevant data) that are best suited for the operation of each product, component or system. Describe conditions under which the item equipment should not be allowed to run.

#### 1.6.4.8 Operating Log

Provide forms, sample logs, and instructions for maintaining necessary operating records.

#### 1.6.4.9 Additional Requirements for Equipment Control Systems

Provide Data Package 5 and the following for all control systems:

- a. Provide a narrative description on how to perform and apply functions, features, modes, and other operations, including unoccupied operation, seasonal changeover, manual operation, and alarms. Include detailed technical manual for programming and customizing control loops and algorithms.
- b. Submit complete controls equipment schedules, full as-built sequence of operations, wiring and logic diagrams, Input/Output Tables, equipment schedules, copies of checkout tests and calibrations performed by the Contractor (not Cx tests), and all associated information.
- c. Full points list. Provide a listing of rooms with the following information for each room:
  - (1) Floor
  - (2) Room number
  - (3) Room name
  - (4) Air handler unit ID
  - (5) Reference drawing number
  - (6) Air terminal unit tag ID
  - (7) Heating or cooling valve tag ID
  - (8) Minimum cfm
  - (9) Maximum cfm
- e. Full print out of all schedules and set points after testing and acceptance of the system.
- f. Full as-built print out of software program.
- g. Marking of system sensors and thermostats on the as-built floor plan and mechanical drawings with their control system designations.

#### 1.6.4.10 Testing Equipment Information and Performance Data

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components. Provide final set points.

Include completed prefunctional checklists, functional performance test forms, and monitoring reports. Include recommended schedule for retesting and blank test forms. Provide final set points.

#### 1.6.5 Routine Maintenance Requirements

##### 1.6.5.1 Preventive Maintenance Plan, Schedule, and Procedures

Provide manufacturer's schedule for routine preventive maintenance, inspections, condition monitoring (predictive tests) and adjustments required to ensure proper and economical operation and to minimize repairs. Provide instructions stating when the systems should be retested. Provide manufacturer's projection of preventive maintenance work-hours on a daily, weekly, monthly, and annual basis including craft requirements by type of craft. For periodic calibrations, provide manufacturer's specified frequency and procedures for each separate operation.

- a. Define the anticipated time required to perform each test (work-hours), test apparatus, number of personnel identified by responsibility, and a testing validation procedure permitting the record operation capability requirements within the schedule. Provide a remarks column for the testing validation procedure referencing operating limits of time, pressure, temperature, volume, voltage, current, acceleration, velocity, alignment, calibration, adjustments, cleaning, or special system notes. Delineate procedures for preventive maintenance, inspection, adjustment, lubrication and cleaning necessary to minimize repairs.
- b. Repair requirements must inform operators how to check out, troubleshoot, repair, and replace components of the system. Include electrical and mechanical schematics and diagrams and diagnostic techniques necessary to enable operation and troubleshooting of the system after acceptance.

##### 1.6.5.2 Lubrication Data

Include the following preventive maintenance lubrication data, in addition to instructions for lubrication required under paragraph OPERATOR SERVICE REQUIREMENTS:

- a. A table showing recommended lubricants for specific temperature ranges and applications.
- b. Charts with a schematic diagram of the equipment showing lubrication points, recommended types and grades of lubricants, and capacities. Provide procedural instructions for Oil Sampling for all equipment.
- c. A Lubrication Schedule showing service interval frequency.

#### 1.6.6 Repair Procedures

Provide instructions and a list of tools required to repair or restore the

product or equipment to proper condition or operating standards. Provide manufacturer's recommended procedures and instructions for correcting problems and making repairs for the installed model and features of each system. Include potential environmental and indoor air quality impacts of recommended maintenance procedures and materials. Specify if any certifications or licenses are required to repair the equipment.

#### 1.6.6.1 Troubleshooting Guides and Diagnostic Techniques

Provide step-by-step procedures to promptly isolate the cause of typical malfunctions. Describe clearly why the checkout is performed and what conditions are to be sought. Identify tests or inspections and test equipment required to determine whether parts and equipment may be reused or require replacement.

#### 1.6.6.2 Wiring Diagrams and Control Diagrams

Provide point-to-point drawings of wiring and control circuits including factory-field interfaces. Provide a complete and accurate depiction of the actual job specific wiring and control work. On diagrams, number electrical and electronic wiring and pneumatic control tubing and the terminals for each type, identically to actual installation configuration and numbering.

#### 1.6.6.3 Removal and Replacement Instructions

Provide step-by-step procedures and a list of required tools and supplies for removal, replacement, disassembly, and assembly of components, assemblies, subassemblies, accessories, and attachments. Provide tolerances, dimensions, settings and adjustments required. Use a combination of text and illustrations.

#### 1.6.6.4 Repair Work-Hours

Provide manufacturer's projection of repair work-hours including requirements by type of craft. Identify, and tabulate separately, repair that requires the equipment manufacturer to complete or to participate.

#### 1.6.6.5 Warranty Information

List and explain the various warranties and clearly identify the servicing and technical precautions prescribed by the manufacturers or contract documents in order to keep warranties in force. Identify if replacement of a subassembly, attachment or accessory requires the entire assembly to be replaced. Include warranty information for primary components of the system. Provide copies of warranties required by Section 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME).

#### 1.6.6.6 Extended Warranty Information

List all warranties for products, equipment, components, and sub-components whose duration exceeds one year. For each warranty listed, indicate the applicable specification section, duration, start date, end date, and the point of contact for warranty fulfillment. Also, list or reference the specific operation and maintenance procedures that must be performed to keep the warranty valid. Provide copies of warranties required by Section 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME).

#### 1.6.6.7 Record Drawings And Utility Systems

The record drawings are the final compilation of actual conditions reflected in the as-built drawings. Provide record drawings as outlined in 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME).

Using Record Source Drawings, show and document details of the actual installation of the utility systems, annotate and highlight the Operation and Maintenance information. Provide the drawings as outlined in paragraph UTILITY RECORD DRAWINGS at a large enough scale to differentiate designated isolation units from surrounding valves and switches.

#### 1.6.6.8 Personnel Training Requirements

Provide information available from the manufacturers that is needed for use in training designated personnel to properly operate and maintain the equipment and systems.

#### 1.6.6.9 Contractor / Supplier Listing and Contact Information

Provide a list that includes the name, address, telephone number, email and website of the General Contractor and each Subcontractor who installed the product or equipment, or system. For each item, also provide the name address and telephone number of the manufacturer's representative and service organization that can provide replacements most convenient to the project site. Provide the name, address, and telephone number of the product, equipment, and system manufacturers.

### 1.6.7 Part 4: Facility Operations

#### 1.6.7.1 Completed Facility Operating Plan

Provide a plan that documents that procedures for the operation of systems and assemblies in the facility. The systems that should be included in the Operating Plan include, but are not limited to:

- a. Electrical systems and equipment
- b. Mechanical systems and equipment
- c. Fire Protection systems and equipment
- d. Control Systems and equipment
- e. Architectural and Structural systems, fixtures, structures, and equipment
- f. Vertical transportation such as elevators and escalators

#### 1.6.7.2 Testing Equipment and Special Tool Information

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components. Provide final set points.

#### 1.6.7.3 Testing and Performance Data

Include completed prefunctional checklists, functional performance test forms, and monitoring reports. Include recommended schedule for retesting



and blank test forms. Provide final set points.

#### 1.6.7.4 Approved Field Test Reports and Manufacturer's Field Reports

Compile and provide approved Field Test Reports (SD-06) and Manufacturer's Field Reports (SD-09) submittals.

#### 1.6.7.5 Maintenance Plans, Procedures, Checklists, Records, and Spare Parts Inventory

##### 1.6.7.5.1 Maintenance Schedules

Include recommended maintenance schedules for systems and equipment.

##### 1.6.7.5.2 Ongoing Commissioning Operational and Maintenance Record Keeping

Include ongoing commissioning and optimization procedures and documentation to monitor and improve the performance of facility systems.

##### 1.6.7.5.3 Janitorial and Cleaning Plans and Procedures

Include a copy of facility cleaning and janitorial plan with procedures and intended chemicals and equipment.

Provide environmentally friendly cleaning recommendations in accordance with ASTM E1971.

#### 1.6.7.6 Utility Record Drawings

##### 1.6.7.6.1 Utility Schematic Diagrams

Provide a one-line schematic diagram for each utility system such as power, water, wastewater, and gas/fuel. Schematic diagram must show from the point where the utility line is connected to the mainline up to the 5 foot connection point to the facility. Indicate location or area designation for route of transmission or distribution lines; locations of duct banks, manholes/handholes or poles; isolation units such as valves and switches; and utility facilities such as pump stations, lift stations, and substations.

##### 1.6.7.6.2 Enlarged Connection and Cutoff Plans

Provide enlarged floor plans and provide information between the 5 foot utilities connection point and where utilities connect to facility distribution. Enlarge floor plans/elevations of the rooms where the utility enters the building and indicate on these plans the locations of the main interiors and exterior connection and cutoff points for the utilities. Also enlarge floor plans/elevations of the rooms where equipment is located. Include enough information to enable someone unfamiliar with the facility to locate the connection and cutoff points. Indicate designations such as room number, panel number, circuit breaker, or valve number of each utility and equipment connection and cutoff point, and what that connection and cutoff point controls.

##### 1.6.7.6.2.1 Description of Utility Metering and Monitoring Systems

Provide in narrative format a description of the utility metering and monitoring systems. Include locations, function, and related systems.

#### 1.6.7.6.2.2 Procedures for Tracking Utility Use and Reporting

Procedures for usage reporting and tracking in support of establishing and monitoring utility budgets and costs, and in developing annual energy reports.

#### 1.6.7.6.2.3 One-Line Diagrams and Meter Location of Systems

Provide one-line diagrams and design drawings that highlight meter locations on the site.

#### 1.6.7.6.3 Spare Parts and Supply Lists

Provide lists of spare parts and supplies required for repair to ensure continued service or operation without unreasonable delays. Special consideration is required for facilities at remote locations. List spare parts and supplies that have a long lead-time to obtain.

#### 1.6.8 Part 5: Training

Provide a copy of training plans used for each type of equipment along with training materials used, arranged in specification sequence. Provide a copy of training records, sign-in sheets, and agendas. Include training and documentation on the updating and continued use of the O&M Manual.

#### 1.6.9 Part 6: Cx Project Report and TAB Report

Provide the final Cx Plan and complete Cx reports with evaluation and testing forms and records for each building system. Include relevant commissioned system assemblies test reports including installers checklists of assemblies. Provide all Cx Progress Reports, issues and resolutions logs with resolution or status of each item, and a list of any open items and seasonal or additional testing required.

#### 1.6.10 Part 7: Regulatory Requirements

Provide information describing regulatory and policies compliance requirements or provide a reference to where it is stored.

#### 1.6.11 Part 8: Permits

Provide information requiring frequently asked questions and associated answers or provide a reference to where it is stored.

#### 1.6.12 Part 9: Operations and Maintenance Manual Approval

Provide a signed document stating that the project O&M Manual has been reviewed and confirming agreement with the approach it presents. Include contact information for the signer for coordination of any future changes.

#### 1.7 SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES

Provide the O&M data packages specified in individual technical sections. O&M Data Packages are one of the components of the O&M Manual. The information required in each type of data package follows:

##### 1.7.1 Package Quality

Documents must be fully legible. Operation and Maintenance data must be

consistent with the manufacturer's standard brochures, schematics, printed instructions, general operating procedures, and safety precautions.

1.7.2 Data Package 1

- a. Safety precautions and hazards
- b. Cleaning recommendations
- c. Maintenance and repair procedures
- d. Warranty information
- e. Extended warranty information
- f. Contractor information
- g. Spare parts and supply list

1.7.3 Data Package 2

- a. Safety precautions and hazards
- b. Normal operations
- c. Environmental conditions
- d. Lubrication data
- e. Preventive maintenance plan, schedule, and procedures
- f. Cleaning recommendations
- g. Maintenance and repair procedures
- h. Removal and replacement instructions
- i. Spare parts and supply list
- j. Parts identification
- k. Warranty information
- l. Extended warranty information
- m. Contractor information

1.7.4 Data Package 3

- a. Safety precautions and hazards
- b. Operator prestart
- c. Startup, shutdown, and post-shutdown procedures
- d. Normal operations
- e. Emergency operations

- f. Environmental conditions
- g. Operating log
- h. Lubrication data
- i. Preventive maintenance plan, schedule, and procedures
- j. Cleaning recommendations
- k. Troubleshooting guides and diagnostic techniques
- l. Wiring diagrams and control diagrams
- m. Maintenance and repair procedures
- n. Removal and replacement instructions
- o. Spare parts and supply list
- p. Product submittal data
- q. O&M submittal data
- r. Parts identification
- s. Warranty information
- t. Extended warranty information
- u. Testing equipment and special tool information
- v. Testing and performance data
- w. Contractor information
- x. Field test reports

1.7.5 Data Package 4

- a. Safety precautions and hazards
- b. Operator prestart
- c. Startup, shutdown, and post-shutdown procedures
- d. Normal operations
- e. Emergency operations
- f. Operator service requirements
- g. Environmental conditions
- h. Operating log
- i. Lubrication data
- j. Preventive maintenance plan, schedule, and procedures

- k. Cleaning recommendations
- l. Troubleshooting guides and diagnostic techniques
- m. Wiring diagrams and control diagrams
- n. Repair procedures
- o. Removal and replacement instructions
- p. Spare parts and supply list
- q. Repair work-hours
- r. Product submittal data
- s. O&M submittal data
- t. Parts identification
- u. Warranty information
- v. Extended warranty information
- w. Personnel training requirements
- x. Testing equipment and special tool information
- y. Testing and performance data
- z. Contractor information
- aa. Field test reports

1.7.6 Data Package 5

- a. Safety precautions and hazards
- b. Operator prestart
- c. Start-up, shutdown, and post-shutdown procedures
- d. Normal operations
- e. Environmental conditions
- f. Preventive maintenance plan, schedule, and procedures
- g. Troubleshooting guides and diagnostic techniques
- h. Wiring and control diagrams
- i. Maintenance and repair procedures
- j. Removal and replacement instructions
- k. Spare parts and supply list

- l. Product submittal data
- m. Manufacturer's instructions
- n. O&M submittal data
- o. Parts identification
- p. Testing equipment and special tool information
- q. Warranty information
- r. Extended warranty information
- s. Testing and performance data
- t. Contractor information
- u. Field test reports
- v. Additional requirements for HVAC control systems

#### 1.7.7 Changes to Submittals

Provide manufacturer-originated changes or revisions to submitted data if a component of an item is so affected subsequent to acceptance of the O&M Data. Submit changes, additions, or revisions required by the Contracting Officer for final acceptance of submitted data within 30 calendar days of the notification of this change requirement.

## PART 2 PRODUCTS

Not Used

## PART 3 EXECUTION

### 3.1 TRAINING

Prior to acceptance of the facility by the Contracting Officer for Beneficial Occupancy, provide comprehensive training for the systems and equipment specified in the technical specifications. The training must be targeted for the Facilities Management Specialist, building maintenance personnel, and applicable building occupants. Instructors must be well-versed in the particular systems that they are presenting. Address aspects of the Operation and Maintenance Manual submitted in accordance with Section 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME). Training must include classroom or field lectures based on the system operating requirements. The location of classroom training requires approval by the Contracting Officer.

#### 3.1.1 Training Plan

Submit a written training plan to the Contracting Officer for approval at least 60 calendar days prior to the scheduled training. The training plan must be approved by the Quality Control Manager (QC) prior to forwarding to the Contracting Officer. Also, coordinate the training schedule with the Contracting Officer and QC Manager. Include within the plan the following elements:

- a. Equipment included in training
- b. Intended audience
- c. Location of training
- d. Dates of training
- e. Objectives
- f. Outline of the information to be presented and subjects covered including description
- g. Start and finish times and duration of training on each subject
- h. Methods (e.g. classroom lecture, video, site walk-through, actual operational demonstrations, written handouts)
- i. Instructor names and instructor qualifications for each subject
- j. List of texts and other materials to be furnished by the Contractor that are required to support training
- k. Description of proposed software to be used for video recording of training sessions.

#### 3.1.2 Training Content

The core of this training must be based on manufacturer's recommendations and the operation and maintenance information. The QC Manager is responsible for overseeing and approving the content and adequacy of the training. Spend 95 percent of the instruction time during the presentation on the OPERATION AND MAINTENANCE DATA. Include the following for each system training presentation:

- a. Start-up, normal operation, shutdown, unoccupied operation, seasonal changeover, manual operation, controls set-up and programming, troubleshooting, and alarms.
- b. Relevant health and safety issues.
- c. Discussion of how the feature or system is environmentally responsive. Advise adjustments and optimizing methods for energy conservation.
- d. Design intent.
- e. Use of O&M Manual Files.
- f. Review of control drawings and schematics.
- g. Interactions with other systems.
- h. Special maintenance and replacement sources.
- i. Tenant interaction issues.

### 3.1.3 Training Outline

Provide the Operation and Maintenance Manual Files (Bookmarked PDF), and a written course outline listing the major and minor topics to be discussed by the instructor on each day of the course to each trainee in the course. Provide the course outline 14 calendar days prior to the training.

### 3.1.4 Training Video Recording

Record classroom training session(s) on video. Provide to the Contracting Officer two copies of the training session(s) in DVD video recording format. Capture within the recording, in video and audio, the instructors' training presentations including question and answer periods with the attendees. The recording camera(s) must be attended by a person during the recording sessions to assure proper size of exhibits and projections during the recording are visible and readable when viewed as training.

### 3.1.5 Unresolved Questions from Attendees

If, at the end of the training course, there are questions from attendees that remain unresolved, the instructor must send the answers, in writing, to the Contracting Officer for transmittal to the attendees, and the training video must be modified to include the appropriate clarifications.

### 3.1.6 Validation of Training Completion

Ensure that each attendee at each training session signs a class roster daily to confirm Government participation in the training. At the completion of training, submit a signed validation letter that includes a sample record of training for reporting what systems were included in the training, who provided the training, when and where the training was performed, and copies of the signed class rosters. Provide two copies of the validation to the Contracting Officer, and one copy to the Operation and Maintenance Manual Preparer for inclusion into the Manual's documentation.

### 3.1.7 Quality Control Coordination

Coordinate this training with the QC in accordance with Section 01 45 00.00 22 QUALITY CONTROL (PWD ME).

## 3.2 SUBMITTAL SCHEDULING

### 3.2.1 Operation And Maintenance Manual, Progress Submittal

Submit the Progress submittal when construction is approximately 50 percent complete to the Contracting Officer for approval. Provide Operation and Maintenance Manual Files (Bookmarked PDF). Include the elements and portions of system construction completed up to this point. The purpose of this submittal is to verify progress is in accordance with contract requirements as discussed during the Operation and Maintenance Manual Coordination Meeting.

### 3.2.2 Operation And Maintenance Manual, Prefinal Submittal

Submit the 100 percent submittal of the Operation and Maintenance Prefinal Submittal to the Contracting Officer for approval within 60 calendar days of the Beneficial Occupancy Date (BOD). This submittal must provide a



complete, working document that can be used to operate and maintain the facility. Any portion of the submittal that is incomplete or inaccurate requires the entire submittal to be returned for correction. Any discrepancies discovered during the Government's review of Operation and Maintenance Progress submittal must be corrected prior to the Prefinal submission. The Prefinal Submittal must include Operation and Maintenance Manual Files (Bookmarked PDF).

### 3.2.3 Operation And Maintenance Manual, Final Submittal

Submit completed Operation and Maintenance Manual Files (Bookmarked PDF) and one hard copy of the Operation and Maintenance Manuals. The Final submittal is due at BOD. Any discrepancies discovered during the Government's review of the Prefinal submittal, including the Field Verification, must be corrected prior to the Final submission.

-- End of Section --

SECTION 01 78 24.00 20

ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI) FACILITY  
DATA WORKBOOK (FDW)  
**05/23, CHG 1: 11/23**

PART 1 GENERAL

1.1 DEFINITIONS AND ABBREVIATIONS

1.1.1 eOMSI

Electronic Operation and Maintenance Support Information

1.1.2 eOMSI Facility Data Workbook (FDW)

A Microsoft Excel file containing required facility information populated by the Contractor.

1.1.3 Systems

The words "system", "systems", and "equipment", when used in this document refer to as-built systems and equipment.

1.1.4 KTR

An abbreviation for "Contractor."

1.2 OPERATION AND MAINTENANCE MANUAL AND eOMSI FACILITY DATA WORKBOOK MEETINGS

Meeting requirements are identified in Section 01 78 23 OPERATION AND MAINTENANCE DATA. Refer to paragraph MEETINGS.

1.3 SUBMITTAL SCHEDULING

1.3.1 eOMSI Facility Data Workbook, Progress Submittal

Submit the Progress submittal when construction is approximately 50 percent complete, to the Contracting Officer for approval. Provide eOMSI Facility Data Workbook (Excel). Include the elements and portions of system construction completed up to this point.

The purpose of this submittal is to verify progress is in accordance with contract requirements as discussed during the Operation and Maintenance Manual and eOMSI Facility Data Workbook Meetings. Field verify a portion of the eOMSI information in accordance with paragraph FIELD VERIFICATION.

1.3.2 eOMSI Facility Data Workbook, Prefinal Submittal

Submit the 100 percent submittal of the eOMSI FDW Prefinal Submittal to the Contracting Officer for approval within 60 calendar days of the Beneficial Occupancy Date (BOD). This submittal must provide a complete, working document that can be used to operate and maintain the facility. Any portion of the submittal that is incomplete or inaccurate requires the entire submittal to be returned for correction. Any discrepancies discovered during the Government's review of eOMSI FDW Progress submittal must be corrected prior to the Prefinal submission.

### 1.3.3 eOMSI Facility Data Workbook, Final Submittal

Submit completed eOMSI Facility Data Workbook (Excel). The Final submittal is due at BOD. Any discrepancies discovered during the Government's review of the Prefinal eOMSI submittal, including the Field Verification, must be corrected prior to the Final eOMSI submission.

### 1.4 UNITS OF MEASURE

Provide eOMSI FDW utilizing the units of measure used in the Government generated contract documents.

### 1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-11 Closeout Submittals

eOMSI Facility Data Workbook, Progress Submittal; G

eOMSI Facility Data Workbook, Prefinal Submittal; G

eOMSI Facility Data Workbook, Final Submittal; G

## PART 2 PRODUCTS

### 2.1 eOMSI FILES FORMAT

Provide eOMSI Facility Data Workbook in (EXCEL) format via eCMS, in accordance with the requirements of Section 01 31 23.13 20 ELECTRONIC CONSTRUCTION AND FACILITY SUPPORT CONTRACT MANAGEMENT SYSTEM (eCMS). Scan eOMSI Facility Data Workbook for viruses, malware, and spyware using a commercially available scanning program that is routinely updated to identify and remove current virus threats.

### 2.2 eOMSI FACILITY DATA WORKBOOK

An initial, pre-edited draft of the Model & Facility Data Matrix tab within the Facility Data Workbook is attached to this section as Attachment A. The Government will provide this Facility Data Workbook electronically to the Contractor upon award. Add, delete, and update Mastersystems, Systems, and Subsystems that may have changed during construction, or any items that may have been omitted or missed during design, at no additional cost to the Government. Complete the KTR Facility Data File tab based on the selection of Mastersystems, Systems, and Subsystems installed. The following tabs are included in the eOMSI Facility Data File Workbook and serve the purpose stated:

- a. Instructions Tab: Instructions for completing Model & Facility Data Matrix Tab and KTR Facility Data File Tab. If a discrepancy exists between what is required in this section and the Workbook, the instructions within the workbook take precedence.
- b. Model & Facility Data Matrix Tab: - The Matrix lists Required Facility

Asset Fields for each SYSTEM and SUBSYSTEM. The Designer of Record selects SYSTEMS and SUBSYSTEMS that are within the project scope, which the Contractor needs to include and populate in KTR Facility Data File tab. The "Required Facility Asset Field Position Numbers," one through seventeen, are pre-populated, and are not editable.

- c. Required Facility Asset Fields Tab: Defines the 35 Required Facility Asset Field Position Numbers used in Model and Facility Data Matrix and KTR Facility Data File tabs.
- d. KTR Sample Facility Data File Tab: Sample KTR eOMSI facility data file. This tab provides an example of the mandatory fields of equipment installed by the Contractor, and populated in the KTR eOMSI Facility Data File Tab, along with their descriptions.
- e. KTR Facility Data File Tab: Required eOMSI facility data file deliverable provided to the Government. Provide a separate and unique new row for each facility component or piece of equipment installed. Coordinate with the Government's Contracting Officer for specific facility component naming convention. Equipment list shall be inclusive of contractor supplied equipment and vendor provided equipment packages broken down to component level detail in accordance with the Model & Facility Data Matrix Tab.

## PART 3 EXECUTION

### 3.1 FIELD VERIFICATION

Field verify eOMSI Facility Data Workbook information with Contractor and Government personnel. Include the following personnel in this meeting: Contractor's eOMSI Manual and Facility Data Workbook Preparer and Quality Control Manager, Commissioning Authority, and the Government's Contracting Officer. The Government may elect to invite additional representatives to this meeting. Request, and provide, an eOMSI Field Verification Meeting no sooner than 14 calendar days after submission of the Progress eOMSI submittal, and another, no sooner than 14 calendar days after submission of the Prefinal eOMSI submittal. During this meeting, the Government and Contractor will verify that the eOMSI Facility Data Workbook is complete and accurate.

Field verify that at least 5 Subsystems under each of the Mastersystems are accurate, for a total of 25 Subsystems. For each of these items, verify that the required facility asset field, as defined in the "Model & Facility Data Matrix" tab, contains the specified data and it is accurate (i.e. item description, manufacturer, model no., serial no.). 100 percent accuracy of eOMSI information is required for successful field verification. If data discrepancies are discovered amongst the 25 Subsystems verified, resubmit an updated eOMSI FDW, and request a make-up Field Verification Meeting. At the make-up field verification meeting 25 new Subsystems and their associated required facility asset fields will be field verified; the 25 new Subsystems must be 100% accurate. Any discrepancies discovered must be corrected prior to next eOMSI Facility Data Workbook Submittal.

- (1) A10 - FOUNDATIONS
- (2) B20 - EXTERIOR ENCLOSURE
- (3) D30 - HVAC
- (4) D40 - FIRE PROTECTION
- (5) D50 - ELECTRICAL

### 3.2 eOMSI TRAINING

Provide training on eOMSI Facility Data Workbook in accordance with  
Section 01 78 23 OPERATION AND MAINTENANCE DATA.

-- End of Section --

| MODEL AND FACILITY DATA MATRIX                                                                                |                                           |              |     |                         |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------|-----|-------------------------|
| STEP 1: Is This a BIM Project?                                                                                |                                           |              |     | No                      |
| STEP 2: Select Yes or No in Column E for each MASTERSYSTEM, SYSTEM and SUBSYSTEM that is In the Project Scope |                                           |              |     | In Scope<br>(Yes or No) |
| CLASSIFICATION ID                                                                                             | MASTERSYSTEM / SYSTEM / SUBSYSTEM Name    | System Type  | UOM |                         |
| A10                                                                                                           | A10 - FOUNDATIONS                         | MASTERSYSTEM | ●   | Yes                     |
| A1010999                                                                                                      | A1010999 - OTHER                          | SUBSYSTEM    | EA  | Yes                     |
| A1020                                                                                                         | A1020 - SPECIAL FOUNDATIONS               | SYSTEM       | ●   | Yes                     |
| A1020130                                                                                                      | A1020130 - STEEL PIPE PILES               | SUBSYSTEM    | EA  | Yes                     |
| A1020210                                                                                                      | A1020210 - GRADE BEAMS                    | SUBSYSTEM    | LF  | Yes                     |
| A1030999                                                                                                      | A1030999 - OTHER                          | SUBSYSTEM    | EA  | Yes                     |
| A20                                                                                                           | A20 - BASEMENT CONSTRUCTION               | MASTERSYSTEM | ●   | Yes                     |
| A2010                                                                                                         | A2010 - BASEMENT EXCAVATION               | SYSTEM       | ●   | Yes                     |
| A2010999                                                                                                      | A2010999 - OTHER BASEMENT EXCAVATION      | SUBSYSTEM    | EA  | Yes                     |
| A2020                                                                                                         | A2020 - BASEMENT WALLS                    | SYSTEM       | ●   | Yes                     |
| A2020110                                                                                                      | A2020110 - CIP CONCRETE                   | SUBSYSTEM    | LF  | Yes                     |
| B10                                                                                                           | B10 - SUPERSTRUCTURE                      | MASTERSYSTEM | ●   | Yes                     |
| B1010                                                                                                         | B1010 - FLOOR CONSTRUCTION                | SYSTEM       | ●   | Yes                     |
| B1010220                                                                                                      | B1010220 - CIP CONCRETE BEAM AND SLAB     | SUBSYSTEM    | SF  | Yes                     |
| B1010244                                                                                                      | B1010244 - LIGHT GAUGE STEEL FLOOR SYSTEM | SUBSYSTEM    | SF  | Yes                     |
| B1010999                                                                                                      | B1010999 - OTHER                          | SUBSYSTEM    | SF  | Yes                     |
| B1020                                                                                                         | B1020 - ROOF CONSTRUCTION                 | SYSTEM       | ●   | Yes                     |
| B1020137                                                                                                      | B1020137 - AWNING/CANOPY                  | SUBSYSTEM    | SF  | Yes                     |
| B1030                                                                                                         | B1030 - ROOF CONSTRUCTION                 | SYSTEM       | ●   | Yes                     |
| B103001                                                                                                       | B103001 - METAL STRUCTURAL FRAME          | SUBSYSTEM    | LF  | Yes                     |
| B20                                                                                                           | B20 - EXTERIOR ENCLOSURE                  | MASTERSYSTEM | ●   | Yes                     |
| B2010109                                                                                                      | B2010109 - CONCRETE BLOCK                 | SUBSYSTEM    | SF  | Yes                     |
| B2010999                                                                                                      | B2010999 - OTHER                          | SUBSYSTEM    | SF  | Yes                     |
| B2030                                                                                                         | B2030 - EXTERIOR DOORS                    | SYSTEM       | ●   | Yes                     |
| B2030220                                                                                                      | B2030220 - STEEL DOORS                    | SUBSYSTEM    | EA  | Yes                     |
| B2030410                                                                                                      | B2030410 - OVERHEAD DOORS                 | SUBSYSTEM    | EA  | Yes                     |
| B30                                                                                                           | B30 - ROOFING                             | MASTERSYSTEM | ●   | Yes                     |
| B3010120                                                                                                      | B3010120 - SINGLE PLY MEMBRANE            | SUBSYSTEM    | SF  | Yes                     |
| B3010999                                                                                                      | B3010999 - OTHER                          | SUBSYSTEM    |     | Yes                     |
| B3020                                                                                                         | B3020 - ROOF OPENING                      | SYSTEM       | ●   | Yes                     |
| B3020210                                                                                                      | B3020210 - HATCHES                        | SUBSYSTEM    | EA  | Yes                     |
| C10                                                                                                           | C10 - INTERIOR CONSTRUCTION               | MASTERSYSTEM | ●   | Yes                     |

| MODEL AND FACILITY DATA MATRIX                                                                                |                                             |              |        |                         |
|---------------------------------------------------------------------------------------------------------------|---------------------------------------------|--------------|--------|-------------------------|
| STEP 1: Is This a BIM Project?                                                                                |                                             |              |        | No                      |
| STEP 2: Select Yes or No in Column E for each MASTERSYSTEM, SYSTEM and SUBSYSTEM that is In the Project Scope |                                             |              |        | In Scope<br>(Yes or No) |
| CLASSIFICATION ID                                                                                             | MASTERSYSTEM / SYSTEM / SUBSYSTEM Name      | System Type  | UOM    |                         |
| C1010                                                                                                         | C1010 - PARTITIONS                          | SYSTEM       | ●      | Yes                     |
| C1010999                                                                                                      | C1010999 - OTHER                            | SUBSYSTEM    | EA     | Yes                     |
| C1020                                                                                                         | C1020 - INTERIOR DOORS                      | SYSTEM       | ●      | Yes                     |
| C1020114                                                                                                      | C1020114 - METAL DOOR                       | SUBSYSTEM    | EA     | Yes                     |
| C1030                                                                                                         | C1030 - FITTINGS                            | SYSTEM       | ●      | Yes                     |
| C1030831                                                                                                      | C1030831 - COUNTERTOP                       | SUBSYSTEM    | LF     | Yes                     |
| C1030999                                                                                                      | C1030999 - OTHER                            | SUBSYSTEM    | LF     | Yes                     |
| C20                                                                                                           | C20 - STAIRS                                | MASTERSYSTEM | ●      | Yes                     |
| C2010110                                                                                                      | C2010110 - STAIRS                           | SUBSYSTEM    | FLIGHT | Yes                     |
| C2020                                                                                                         | C2020 - STAIR FINISHES                      | SYSTEM       | ●      | Yes                     |
| C2020999                                                                                                      | C2020999 - OTHER STAIR FINISHES             | SUBSYSTEM    | EA     | Yes                     |
| D20                                                                                                           | D20 - PLUMBING                              | MASTERSYSTEM | ●      | Yes                     |
| D2010410                                                                                                      | D2010410 - SINK                             | SUBSYSTEM    | EA     | Yes                     |
| D2010901                                                                                                      | D2010901 - EMERGENCY SHOWER                 | SUBSYSTEM    | EA     | Yes                     |
| D2010902                                                                                                      | D2010902 - EMERGENCY EYE WASH               | SUBSYSTEM    | EA     | Yes                     |
| D2020                                                                                                         | D2020 - DOMESTIC WATER DISTRIBUTION         | SYSTEM       | ●      | Yes                     |
| D2020903                                                                                                      | D2020903 - BACKFLOW PREVENTER               | SUBSYSTEM    | EA     | Yes                     |
| D2030                                                                                                         | D2030 - SANITARY WASTE                      | SYSTEM       | ●      | Yes                     |
| D2030999                                                                                                      | D2030999 - OTHER                            | SUBSYSTEM    | EA     | Yes                     |
| D30                                                                                                           | D30 - HVAC                                  | MASTERSYSTEM | ●      | Yes                     |
| D3050                                                                                                         | D3050 - TERMINAL & PACKAGE UNITS            | SYSTEM       | ●      | Yes                     |
| D3050120                                                                                                      | D3050120 - UNIT HEATERS, GAS                | SUBSYSTEM    | EA     | Yes                     |
| D3060                                                                                                         | D3060 - CONTROLS & INSTRUMENTATION          | SYSTEM       | ●      | Yes                     |
| D3060901                                                                                                      | D3060901 - HVAC CONTROLS, ELECTRIC, DIGITAL | SUBSYSTEM    | EA     | Yes                     |
| D50                                                                                                           | D50 - ELECTRICAL                            | MASTERSYSTEM | ●      | Yes                     |
| D5010                                                                                                         | D5010 - ELECTRICAL SERVICE & DISTRIBUTION   | SYSTEM       | ●      | Yes                     |
| D5010120                                                                                                      | D5010120 - ELECTRICAL SERVICE               | SUBSYSTEM    | EA     | Yes                     |
| D5010240                                                                                                      | D5010240 - SWITCHGEAR                       | SUBSYSTEM    | EA     | Yes                     |
| D5010902                                                                                                      | D5010902 - TRANSFORMER                      | SUBSYSTEM    | EA     | Yes                     |
| D5010906                                                                                                      | D5010906 - PANELBOARDS                      | SUBSYSTEM    | EA     | Yes                     |
| D5020                                                                                                         | D5020 - LIGHTING & BRANCH WIRING            | SYSTEM       | ●      | Yes                     |
| D5020248                                                                                                      | D5020248 - INTERIOR LIGHTING, FLUORECENT    | SUBSYSTEM    | EA     | Yes                     |

| MODEL AND FACILITY DATA MATRIX                                                                                |                                               |              |     |                         |
|---------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------|-----|-------------------------|
| STEP 1: Is This a BIM Project?                                                                                |                                               |              |     | No                      |
| STEP 2: Select Yes or No in Column E for each MASTERSYSTEM, SYSTEM and SUBSYSTEM that is In the Project Scope |                                               |              |     | In Scope<br>(Yes or No) |
| CLASSIFICATION ID                                                                                             | MASTERSYSTEM / SYSTEM / SUBSYSTEM Name        | System Type  | UOM |                         |
| D5020901                                                                                                      | D5020901 - EXIT LIGHTING                      | SUBSYSTEM    | EA  | Yes                     |
| E10                                                                                                           | E10 - EQUIPMENT                               | MASTERSYSTEM | ●   | Yes                     |
| E1040                                                                                                         | E1040 - INDUSTRIAL PROCESS EQUIPMENT          | SYSTEM       | ●   | Yes                     |
| E1040999                                                                                                      | E1040999 - OTHER INDUSTRIAL PROCESS EQUIPMENT | SUBSYSTEM    | EA  | Yes                     |
| E1090                                                                                                         | E1090 - OTHER EQUIPMENT                       | SYSTEM       | ●   | Yes                     |
| E1090999                                                                                                      | E1090999 - OTHER EQUIPMENT                    | SUBSYSTEM    | EA  | Yes                     |
| F10                                                                                                           | F10 - SPECIAL CONSTRUCTION                    | MASTERSYSTEM | ●   | Yes                     |
| F1040                                                                                                         | F1040 - SPECIAL FACILITIES                    | SYSTEM       | ●   | Yes                     |
| F104002                                                                                                       | F104002 - LIQUID AND GAS STORAGE TANKS        | SUBSYSTEM    | EA  | Yes                     |
| F1050                                                                                                         | F1050 - SPECIAL CONTROLS AND INSTRUMENTATION  | SYSTEM       | ●   | Yes                     |
| F105001                                                                                                       | F105001 - RECORDING INSTRUMENTATION           | SUBSYSTEM    | EA  | Yes                     |
| G20                                                                                                           | G20 - SITE IMPROVEMENTS                       | MASTERSYSTEM | ●   | Yes                     |
| G2010                                                                                                         | G2010 - PAVEMENTS                             | SYSTEM       | ●   | Yes                     |
| G201003.01                                                                                                    | G201003.01 - FLEXIBLE ASPHALT PAVEMENTS       | SUBSYSTEM    | SF  | Yes                     |
| G2020                                                                                                         | G2020 - A PARKING LOT /PARKING AREA PAVEMENT  | SYSTEM       | ●   | Yes                     |
| G2020061                                                                                                      | G2020061 - ASHPALT RESURFACING                | SUBSYSTEM    | EA  | Yes                     |
| G2040                                                                                                         | G2040 - SITE IMPROVEMENTS & AMMENITIES        | SYSTEM       | ●   | Yes                     |
| G90                                                                                                           | G90 - OTHER SITE CONSTRUCTION                 | MASTERSYSTEM | ●   | Yes                     |
| G9010                                                                                                         | G9010 - SERVICE AND PEDESTRIAN TUNNELS        | SYSTEM       | ●   | Yes                     |
| G901001.01                                                                                                    | G901001.01 - SERVICE TUNNELS                  | SUBSYSTEM    | LF  | Yes                     |
| G901001.07                                                                                                    | G901001.07 - TRENCH BOXES                     | SUBSYSTEM    | CY  | Yes                     |
| H10                                                                                                           | H10 - WATERFRONT STRUCTURES                   | MASTERSYSTEM | ●   | Yes                     |
| H1040                                                                                                         | H1040 - FENDER SYSTEMS                        | SYSTEM       | ●   | Yes                     |
| H104099                                                                                                       | H104099 - OTHER FENDER SYSTEMS                | SUBSYSTEM    | EA  | Yes                     |
| H1099                                                                                                         | H1099 - OTHER WATERFRONT STRUCTURES           | SYSTEM       | ●   | Yes                     |
| H109999                                                                                                       | H109999 - OTHER WATERFRONT STRUCTURES         | SUBSYSTEM    | EA  | Yes                     |
| H2020                                                                                                         | H2020 - PUMP HOUSE                            | SYSTEM       | ●   | Yes                     |
| H202001                                                                                                       | H202001 - PUMP HOUSE BASIC STRUCTURES         | SUBSYSTEM    | EA  | Yes                     |
| H202002                                                                                                       | H202002 - PUMP HOUSE ELECTRIC EQUIPMENT       | SUBSYSTEM    | EA  | Yes                     |
| H202003                                                                                                       | H202003 - PUMP HOUSE MISC. SYSTEMS/EQUIPMENT  | SUBSYSTEM    | EA  | Yes                     |



| MODEL AND FACILITY DATA MATRIX                                                                                |                                        |              |     |                         |
|---------------------------------------------------------------------------------------------------------------|----------------------------------------|--------------|-----|-------------------------|
| STEP 1: Is This a BIM Project?                                                                                |                                        |              |     | No                      |
| STEP 2: Select Yes or No in Column E for each MASTERSYSTEM, SYSTEM and SUBSYSTEM that is In the Project Scope |                                        |              |     | In Scope<br>(Yes or No) |
| CLASSIFICATION ID                                                                                             | MASTERSYSTEM / SYSTEM / SUBSYSTEM Name | System Type  | UOM |                         |
| K20                                                                                                           | K20 - Non-Potable Water Utilities      | MASTERSYSTEM | ●   |                         |
| K2010                                                                                                         | K2010 - Production                     | SYSTEM       | ●   |                         |
| K2010130                                                                                                      | K2010130 - Pump Stations               | SUBSYSTEM    | EA  |                         |
| K2010140                                                                                                      | K2010140 - Storage                     | SUBSYSTEM    | EA  |                         |
| K2010150                                                                                                      | K2010150 - Plant Compressed Air        | SUBSYSTEM    | EA  |                         |
| K2010160                                                                                                      | K2010160 - Plant Controls              | SUBSYSTEM    | EA  |                         |
| K2010180                                                                                                      | K2010180 - Plant Electrical            | SUBSYSTEM    | EA  |                         |
| P10                                                                                                           | P10 - Compressed Air Utilities         | MASTERSYSTEM | ●   |                         |
| P1010                                                                                                         | P1010 - Production                     | SYSTEM       | ●   |                         |
| P1010130                                                                                                      | P1010130 - Compressed Air              | SUBSYSTEM    | EA  | Yes                     |

SECTION 01 91 00.15 20

TOTAL BUILDING COMMISSIONING  
**05/23, CHG 3: 08/24**

PART 1 GENERAL

Total Building Commissioning is a systematic, quality-focused process for enhancing the delivery of a project that focuses on verifying and documenting that all of the commissioned systems and assemblies are planned, designed, installed, tested, operated, and maintained to meet the project requirements. The purpose is to reduce the cost and performance risks associated with delivering facilities projects, and to increase value to owners, occupants, and users.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING  
ENGINEERS (ASHRAE)

ASHRAE 90.1 - IP (2019) Energy Standard for Buildings  
Except Low-Rise Residential Buildings

ASHRAE 202 (2018) Commissioning Process for Buildings  
and Systems

ASSOCIATED AIR BALANCE COUNCIL (AABC)

ACG Commissioning Guideline (2005) Commissioning Guideline

NATIONAL ENVIRONMENTAL BALANCING BUREAU (NEBB)

NEBB S1110 (2019) Whole Building Technical  
Commissioning of New Construction; 2nd  
Edition

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION  
(SMACNA)

ANSI/SMACNA 014 (2013) HVAC Systems Commissioning Manual,  
2nd Edition

1.2 DEFINITIONS

Commissioning Process (Cx) - a quality-focused process for enhancing the delivery of a project. Refer to ASHRAE 202 for a comprehensive description of the commissioning process.

Commissioning Provider (CxP) - The entity who leads, plans, and coordinates the Commissioning Team. The terms Commissioning Provider, Commissioning Firm, Lead Commissioning Specialist, Commissioning Specialist, and Commissioning Authority (CA or CxA) when used by sustainable Third Party Certification (TPC) programs, are interchangeable.

Commissioning Authority - The Government retains the authority for oversight and assurance of the entire commissioning process, and final approval of all commissioning deliverables.

Government Acceptance Testing Representatives - Government Acceptance Testing Representatives perform the inherently Governmental function of technical oversight and quality assurance for critical systems, and is distinctly separate from the commissioning process. Government Acceptance Testing Representatives witness final testing of critical systems and report systems' acceptance to the Contracting Officer. Submittals to be surveilled and approved by Government Acceptance Testing Representatives are identified in Section 01 33 00 SUBMITTAL PROCEDURES. Testing required to be witnessed by Government Acceptance Testing Representatives are identified in system level sections.

### 1.3 COMMUNICATION WITH THE GOVERNMENT

The Commissioning Provider (CxC) must submit all plans, schedules, reports, and documentation directly to the Contracting Officer's Representative concurrent with submission to the Quality Control (QC) Manager.

The Commissioning Provider must have direct communication with the Contracting Officer's Representative regarding all elements of the commissioning process; however, the Government has no direct contract authority with the Commissioning Provider.

### 1.4 COMMUNICATION WITH GOVERNMENT ACCEPTANCE TESTING REPRESENTATIVES

The QC Manager must communicate directly with the Government Acceptance Testing Representatives and Contracting Officer's Representative regarding Government acceptance testing activities. Inform the Contracting Officer's Representative when systems are ready for testing to be witnessed by Government Acceptance Testing Representatives, and allow access to the construction site and system(s) to be tested.

### 1.5 SYSTEMS TO BE COMMISSIONED

Coordinate commissioning and quality control activities for the following systems, equipment, and associated controls. System-specific requirements are located in the associated specification sections. Commission the following systems, equipment, and associated controls in accordance with this section and the inspection, testing, and quality control requirements of their respective sections:

- Ventilating and associated controls (HVAC)

- Lighting systems: interior and exterior

- Desalinization Process Equipment (RO Membranes, Pumps, Filters, etc)

- Salt Water Pump Station

- CWIS System (Air Compressor, Screens, valves, etc)

- Existing North and South Cooling Towers System

- Existing Condensate Pump System

#### 1.5.1 Desalination Commissioning

Please see attached guidance titled Startup, Performance Testing and Demonstration.

#### 1.6 RELATED SECTIONS

Refer to the following technical sections for additional commissioning requirements for respective systems:

23 08 00 COMMISSIONING OF MECHANICAL SYSTEMS

26 08 00 APPARATUS INSPECTION AND TESTING

26 51 00 INTERIOR LIGHTING

#### 1.7 COMMISSIONING TEAM

The Commissioning team will include, but is not limited to the following team members.

Ensure all Construction Activities for systems to be commissioned are coordinated with the appropriate commissioning team members.

- a. Commissioning Provider (CxC)
- b. Quality Control (QC) Manager
- c. Sub-Contractor Representatives for each trade responsible for construction/installation of systems to be commissioned
- d. Construction Manager (CM)
- e. Designer of Record (DOR)
- f. Technical Commissioning Specialists for each system to be commissioned
- g. TAB Representative
- h. Equipment manufacturer representatives
- i. Government Contracting Officer
- j. Government Representatives
- k. Government Acceptance Testing Representatives
- l. Installation Maintenance Representative
- m. Facility End User

#### 1.8 PROJECT SCHEDULE

Include the following tasks in the project schedule required by Section 01 32 17.00 20 COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS) and in the construction phase commissioning plan. Ensure sufficient time is scheduled to complete each item. The order of items listed below is not intended to imply a specified sequence:

- a. Submission and approval of the Commissioning Firm Qualifications
- b. Submission and approval of the Interim and Final Construction Phase Commissioning Plans
- c. Commissioning Kickoff Meeting
- d. Regular Commissioning Coordination Meetings
- e. Installation of permanent utilities (gas, water, electric, steam & condensate, compressed air)
- f. Manufacturer's Equipment Start-Up for each of the systems to be commissioned
- g. Submission and approval of the Completed Commissioning Observation Checklists
- h. Submission and approval of Certificate of Readiness for each system to be commissioned
- i. Commissioning Testing, including Functional Performance Testing, for each system to be commissioned
- j. Integrated Systems Tests
- k. Post-test deficiency correction for each system to be commissioned
- l. Re-Testing
- m. Training for each of the systems to be commissioned
- n. Submission and approval of the Initial and Final Commissioning Reports
- o. Final testing required to be witnessed by Government Acceptance Testing Representatives, as identified in system level sections.

#### 1.9 PHASING

This project includes multiple buildings Commissioning activities for each project building must be scheduled separately and must correspond to each completion milestone in the master schedule.

#### 1.10 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-06 Test Reports

Interim Construction Phase Commissioning Plan; G

Final Construction Phase Commissioning Plan; G; S

Initial Commissioning Report; G

Issues Log; G

Completed Construction Observation Checklists; G

SD-07 Certificates

Commissioning Firm; G

Certificate Of Readiness; G

SD-10 Operation and Maintenance Data

Systems Manual; G

SD-11 Closeout Submittals

Final Commissioning Report; G

Updated Final Commissioning Report; G

1.11 COMMISSIONING FIRM

Employ the services of a Commissioning Firm and all Commissioning Specialists required to perform work for this project. The Commissioning Firm must be a first-tier subcontractor that is financially and corporately independent from prime contractor and all other subcontractors and the Designer of Record and that is not participating in any other work on this Contract, including design, furnishing equipment, construction, or testing, adjusting, and balancing.

- a. Submit the Commissioning Firm's and Commissioning Specialists' qualifications, including the name of the firm and each CxC and each certification, no later than 30 calendar days after Notice to Proceed.
- b. If, for any reason, a specialist loses a certification during this period, immediately notify the Contracting Officer and submit another Commissioning Specialist for approval. Validate all work performed for this project by the CxC who lost a certification by an approved successor.

1.11.1 Commissioning Specialists (CxC)

Assign Lead Commissioning Specialist and other appropriate Commissioning Specialists for the systems to be commissioned.

1.11.1.1 Commissioning Provider (CxC)

Commissioning Provider (CxC) coordinates all aspects of the commissioning process. Duties include leading and overseeing the commissioning work, and acting as the primary point of contact for the commissioning work. CxC may serve as a systems Specialist if all requirements for both designations are met. CxC must have a minimum of five years of commissioning experience, including two projects of similar size and complexity to this project.

CxC must be certified in one of the following:

NEBB qualified Systems Commissioning Administrator (SCA)

ACG Certified Commissioning Authority (CxA)

ICB/TABB Certified Commissioning Supervisor

BCA Certified Commissioning Professional (CCP)

AEE Certified Building Commissioning Professional (CBCP)

University of Wisconsin-Madison Qualified Commissioning Process  
Provider (QCxP)

ASHRAE Building Commissioning Professional (BCxP).

#### 1.11.1.2 Commissioning Specialists

Refer to the related technical commissioning specification for additional qualifications for each Commissioning Specialist associated with each system. Include all Commissioning Specialist qualifications with the Commissioning Firm submittal:

- a. Mechanical Commissioning Specialist: The technical work associated with mechanical systems to be commissioned must be performed by a Commissioning Specialist certified by NEBB, ACG, ICB/TABB, AEE, University of Wisconsin-Madison, ASHRAE, or BCA in the commissioning of HVAC systems with five years of experience in the commissioning of HVAC systems.
- b. Electrical Commissioning Specialist: The technical work associated with electrical systems to be commissioned must be performed by an engineering technician with five years of experience inspecting, testing, and calibrating electrical distribution and generation equipment, systems, and devices. The lead engineering technician specified in specification Section 26 08 00 APPARATUS INSPECTION AND TESTING may serve as the Electrical Commissioning Specialist.

#### 1.11.2 Commissioning Standard

Comply with ASHRAE 202 and the commissioning standards under which the Commissioning Firm and Specialists qualifications are approved, which may include ACG Commissioning Guideline, NEBB S1110, or ANSI/SMACNA 014. Comply with the requirements of the commissioning standard under which the Commissioning Firm and Specialists qualifications are approved. When the firm and specialists are certified by BCA, AEE, ASHRAE, or the University of Wisconsin-Madison, comply with the requirements of one of these acceptable standards: ACG Commissioning Guideline, NEBB S1110, ANSI/SMACNA 014, or ASHRAE 202. In a conflict, the most stringent requirements apply. In addition, comply with ASHRAE 90.1 - IP commissioning requirements for all systems. Refer to related technical commissioning specification sections for additional standards requirements. Comply with applicable International Electrical Testing Association (NETA) testing standards for electrical systems. The following requirements apply to all project commissioning and test standards:

- a. Implement all recommendations and suggested practices contained in the Commissioning Standard and test standards.
- b. Use the Commissioning Standard for all aspects of Commissioning, including calibration of instruments.
- c. Where the instrument manufacturer calibration recommendations are more

stringent than those listed in the Commissioning Standard, adhere to the manufacturer calibration recommendations.

- d. All quality assurance provisions of the Commissioning Standard such as performance guarantees are part of this contract.
- e. The Commissioning Specialists must develop commissioning procedures for any systems or system components not covered in the Commissioning Standard.
- f. Use any new requirements, recommendations, and procedures published or adopted by the body responsible for the applicable Commissioning Standards at the time of project award.
- g. If there is a conflict between the requirements of the contract documents and the commissioning standard used, the contract documents take precedent.

#### 1.12 ISSUES LOG

The Commissioning Specialist develops and maintain an Issues Log for the systems to be commissioned. The issues log documents and tracks resolution of deficiencies identified during submittal reviews, inspection, and testing. At any point during construction, any commissioning team member finding deficiencies may communicate those deficiencies in writing to the Commissioning Specialist for inclusion into the Issues Log. For each issue, the Issues Log includes, but is not limited to, a unique reference number, description of the issue with contract requirement referenced, location of or equipment name/tags exhibiting the issue, the initials of the individual's name whom reported the issue, the date of first observation, the proposed resolution of the issue and date proposed, the date of any subsequent observations with applicable additional information, and the date of implementation of the final resolution of the issue as confirmed by the Commissioning Specialist and Contracting Officer. Issues must not be deleted from the issues log.

CxC must submit the Issues Log monthly and within three working days from changes to the Issue Log. The CxC is responsible for distributing the Issues Log to the Commissioning Team. QC Manager must track construction deficiencies identified in the Issues Log using a quality management system in accordance with Section 01 45 00.00 22 QUALITY CONTROL (PWD ME). The QC manager is responsible for notifying the CxC and Contracting Officer of outstanding deficiencies and tracking them to resolution in accordance with Section 01 45 00.00 22 QUALITY CONTROL (PWD ME), "Quality Control Plan".

#### 1.13 CERTIFICATE OF READINESS

Prior to scheduling Commissioning Tests, the Quality Control Manager must issue a Certificate of Readiness for each system, certifying that inspections have been completed, open issues have been resolved, and the system is ready for Commissioning Tests. Refer to each related technical commissioning specification section for additional requirements.

Submit the Certificate of Readiness for each system 20 working days prior to Commissioning Tests of that system. Do not schedule Commissioning Tests for a system until the Certificate of Readiness is approved by the Government.



## PART 2 PRODUCTS

Not used.

## PART 3 EXECUTION

### 3.1 CONSTRUCTION SUBMITTAL REVIEWS

Coordinate construction submittal document reviews for commissioned systems and assemblies with the CxC. The commissioning submittal review does not replace the Government submittal review, in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

The CxC must identify construction submittals to be provided by the contractor for the commissioned systems. The CxC must evaluate construction submittals for compliance with the contract documents. Include a copy of the submittal document review transmittal and response in the Commissioning Report.

### 3.2 COMMISSIONING KICKOFF MEETING

Conduct a Commissioning Kickoff Meeting, led by the CxC, after approval of the Commissioning Firm and Commissioning Specialists, and no later than 30 days following approval of the Commissioning Firm and Specialists. Discuss the commissioning process including contract requirements, lines of communication, roles and responsibilities, schedules, documentation requirements, inspection and test procedures, and logistics as specified in this section.

The Construction Manager, Quality Control team, Designer of Record, and the Government Acceptance Testing Representatives and other Government team members must attend this meeting. Invite the Facility End User to attend this meeting.

### 3.3 REGULAR COMMISSIONING COORDINATION MEETINGS

The Quality Control team, Designer of Record, and the Government Acceptance Testing Representatives and other Government team members must attend this meeting. Sub-Contractor representatives for each trade responsible for construction/installation of systems to be commissioned must attend this meeting as requested by the CxC. Invite the Facility End User to attend this meeting.

CxC must conduct monthly commissioning coordination meetings when installation of commissioned systems begins. Provide status of commissioned systems, open issues log items, outstanding submittals, and upcoming commissioning activities. Conduct bi-weekly commissioning coordination meetings within 30 days of the scheduled date for commissioning testing.

### 3.4 CONSTRUCTION PHASE COMMISSIONING PLANS

The Interim Construction Phase Commissioning Plan identifies the commissioning and testing standards and outline the overall commissioning process, the commissioning schedule, the commissioning team members and responsibilities, lines of communication, and documentation requirements for the construction phase of the project. Submit the Interim Construction Phase Commissioning Plan, prepared by the CxC, 14 calendar days after the Construction Commissioning Coordination Meeting and 14 days

prior to the start of construction of the building enclosure. Include the following items in the commissioning schedule in addition to the items in paragraph PROJECT SCHEDULE: Commissioning Site Visits, Seasonal Testing, Warranty Phase Site Visit, Post-Construction Endurance Testing, and Updated Commissioning Report.

The Final Construction Phase Commissioning Plan includes the information provided in the Interim Construction Phase Commissioning Plan as well as commissioning construction observation checklists and test procedures such as the Pre-Functional Checklists, Integrated Systems Test Checklists, and Functional Performance Test Checklists and other Commissioning Test Checklists for each building, for each system required to be commissioned, and for each component for inclusion in the Final Construction Phase Commissioning Plan. Refer to the related technical commissioning specification section for additional requirements for checklists. Submit the Final Construction Phase Commissioning Plan, prepared by the CxC, no later than 90 calendar days prior to the start of Commissioning Inspections.

#### 3.4.1 Construction Observation Checklists

Construction Observation Checklists must include items for physical inspection or testing that demonstrate that installation and start-up of equipment and systems is complete. Refer to paragraph COMMISSIONING INSPECTIONS. Construction observation checklists must be tailored to verify the specific installation requirements and details of the construction documents and manufacturer's instructions.

#### 3.4.2 Test Procedures and Checklists

Test procedures and checklists must include procedures that explain, step-by-step, the actions and expected results that will demonstrate that the system performs in accordance with the contract. Refer to paragraph COMMISSIONING AND INTEGRATED SYSTEMS TESTS. Include the following sections and details appropriate to the systems being tested in the test procedures and checklists:

- a. Notable system features including information about controls to facilitate understanding of system operation
- b. Conclusions and recommendations. Conclusions must clearly indicate if system does or does not perform in accordance with contract requirements. Recommendation must clearly indicate that the system should or should not be approved by the Government.
- c. Test conditions including date, beginning and ending time, and beginning and ending outdoor air conditions
- d. Attendees
- e. Identification of the equipment involved in the test
- f. Control system feature identification
- g. Point-to-point observations including demonstrating system flow meters and sensors have been calibrated and are correctly displayed on the Operator work station
- h. Actuator operation observations demonstrating actuator responses to commands from the control system

- i. As-found condition of the system operation
- j. List of test items with step numbers along with the corresponding feature or control operation, intended test procedure, expected system response, and pass/fail indication.
- k. Space for comments for each test item.

#### 3.4.3 Integrated Systems Test Checklists

Integrated Systems Test Checklists must include test procedures that explain, step-by-step, the actions and expected results that will demonstrate that the interactive operations between systems performs in accordance with the contract. Refer to paragraph COMMISSIONING AND INTEGRATED SYSTEMS TESTS. Include the following sections in the Integrated Systems Test Checklists:

- a. Notable features of the interconnected systems organized by discipline including information to facilitate understanding of system operation
- b. Conclusions and recommendations. Conclusions must clearly indicate if the systems do or do not perform in accordance with contract requirements. Recommendation must clearly indicate that the systems should or should not be approved by the Government
- c. Test conditions including date and beginning and ending time
- d. Identification of the equipment and systems involved in the test
- e. List of test items with step numbers along with the corresponding feature or control operation, intended test procedure, expected system response, and pass/fail indication.
- f. Space for comments for each test item.

#### 3.5 COMMISSIONING INSPECTIONS

Complete inspections using Construction Observation Checklists for each individual item of equipment or system for each system required to be commissioned in accordance with the commissioning plan. Indicate commissioning team member inspection and validation of each checklist item by initials. Validation of each checklist item by each team member indicates that item conforms to the contract documents and validated design in their area of responsibility. Commissioning Specialist validation of each checklist item indicates that each item has been installed correctly and in accordance with contract documents. Submit the initialed and Completed Construction Observation Checklists no later than 7 calendar days after completion of inspection of all checklist items for each system. Include manufacturer start-up checklists associated with equipment with the submission of the checklists.

#### 3.6 COMMISSIONING AND INTEGRATED SYSTEMS TESTS

Demonstrate that all system, equipment, and components have been installed correctly, that the systems operate and perform, including interactive operation between systems, in accordance with contract documents. Perform tests as specified in related technical commissioning specifications. Provide all materials, services, and labor required to perform Integrated

Systems Tests, and Commissioning Tests.

Commissioning Specialist's duties include leading and documenting all tests for the systems to be commissioned with appropriate sub-contractors performing the Tests. The representatives listed in the paragraph COMMISSIONING TEAM must attend the tests with the exception of the Construction Manager.

Perform Integrated Systems Tests only after the Commissioning Tests for each associated system are completed with all deficiencies resolved and after the related Commissioning Test Checklists have been signed by each commissioning team member.

#### 3.6.1 Test Scheduling and Coordination

Schedule Commissioning Tests for each system only after the Certificate of Readiness has been approved by the Government for the system. Correct all deficiencies identified through any prior review, inspection, or test activity before the start of Commissioning Tests.

Commissioning Tests and Integrated Systems Tests must be performed with the CxC present. Government reserves the right to witness all tests. Coordinate test schedule with Government representatives.

#### 3.6.2 Testing Procedures

Commissioning tests include tests such as functional performance and integrated systems tests. For electrical systems testing, include testing of sensor calibrations, control responses, safetis, interlocks, operating modes, capacity, lighting levels, and verification of all other electrical system contract performance requirements. Perform test procedures in accordance with the commissioning standards specified.

#### 3.6.3 Integrated Systems Tests

Follow the Integrated Systems Test Checklists from the approved Final Construction Phase Commissioning Plan. Integrated Systems Tests must be performed for the interactive operation between systems such as HVAC systems, fire protection systems, back-up electrical supply, energy generation systems, and other systems, and verify correct interactive operation, acceptable speed of response, and other contract requirements for both normal and failure modes. Examples of Integrated Systems Tests include the correct operation of HVAC systems during emergency system activation, correct operation of uninterruptible power supplies or energy generators and connected systems, or lighting system operation during power outage or emergency system activation.

#### 3.6.4 Sample Strategy

Refer to the sections identified in paragraph RELATED SECTIONS for sample strategy.

For electrical systems, test 100 percent of all equipment and systems except for the following. Test at the sample rate shown:

##### a. Lighting Controls

### 3.6.5 Aborted Tests and Re-Testing

Abort any test if any deficiency prevents successful completion of the test or if any required commissioning team member is not present for the test. Re-test after all deficiencies identified during the original test have been corrected. Contracting Officer may withhold payment equivalent to lost time, re-testing, and aborted tests. These costs may include salary, travel costs, and per diem for Government team members.

### 3.7 TRAINING PLAN

CxC must review the training plan for training associated with the equipment and systems to be commissioned, checking that each plan has the trainer name, trainer contract information, training schedule and location. Submit review at least 30 days prior to the first training event. Incorporate CxC review comments prior to submitting training plan in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA and 01 78 24.00 22 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI). Update and resubmit the training plan based on any corrective action taken.

Document training attendance using training attendance rosters and submit completed attendance rosters no later than 7 calendar days following the completion of training for each system to be commissioned.

#### 3.7.1 SYSTEMS MANUAL

The Systems Manual includes system single line diagrams, as-built sequences of operation and controls drawings, as-built control setpoints, recommended schedule for sensor and actuator calibration, recommended schedule of maintenance when not in the O&M manuals, recommended re-testing schedule with proposed testing forms, and full equipment warranty information for all commissioned systems. Incorporate CxC review comments prior to submitting Systems Manual in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA and 01 78 24.00 22 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI). Update and resubmit the system manual information based on any corrective action taken during the warranty period.

### 3.8 COMMISSIONING REPORT

Submit an Initial Commissioning Report no later than 14 calendar days following commissioning team validates all Commissioning Tests, including Functional Performance Tests, and Integrated Systems Tests, with the exception of Seasonal Tests. Submit a Final Commissioning Report upon completion of training and trend log reviews. **The Final Commissioning Report must be submitted to the Contracting Officer prior to BOD. BOD will not be accepted if there are any outstanding deficiencies.** Include the following information in the Final Commissioning Report:

- a. An executive summary describing the overall commissioning process, the results of the commissioning process, outstanding deficiencies and recommended resolutions, and seasonal testing that must be scheduled for a later date. Indicate, in the executive summary, whether the systems meet the requirements of the contract documents.
- b. A list of deficiencies discovered during the commissioning process and the corrective actions taken in the report.

- c. Completed Building Envelope Inspection Checklists, Pre-Functional Checklists and other Commissioning Observation Checklists, Commissioning Test Checklists such as Functional Performance Test Checklists and Integrated Systems Test Checklists, the Final Construction Phase Commissioning Plan, the Issues Log, Training Attendance Rosters, the Design Review Reports, Submittal Review Report, and any other documents as specified by related technical commissioning specification sections.

### 3.9 WARRANTY PHASE SITE VISIT

The Commissioning Provider must visit the building site concurrent with the 9 month warranty inspection to inspect building system equipment and review building operation with the building operating/maintenance staff, and identify any deficiency of the building systems to operate in accordance with the contract documents. The Commissioning Specialist must notify the Contracting Officer of any identified deficiencies and the proposed corrective action. Submit Updated Final Commissioning Report and Systems Manuals, documenting the results of the warranty phase inspection. Include other warranty or post-construction phase activities as specified in related technical commissioning specification sections, such as Seasonal testing results.

-- End of Section --

## STARTUP, PERFORMANCE TESTING AND DEMONSTRATION

### PART 1 GENERAL

#### 1.0 SYSTEM DESCRIPTION

##### A. Definition

1. A system is defined as an arrangement of items of equipment, components, piping, wiring, materials, or incidentals so related or connected as to form a functional and operational unit.

##### B. Project Classified Systems

1. For this project, system classifications must include but not necessarily be limited to the following:

- a) Water Quality Analyzers
- b) Intake Screen
- c) Air Burst Cleaning System
- d) Raw Water, Intermediate, and Final Pumping Systems
- e) Automatic Backwashing Filters and Associated Systems
- f) Static Mixer
- g) Chemical Feed Systems
- h) Ultrafiltration System and Blower System
- i) UF Drain System
- j) Reverse Osmosis Systems (Seawater and Brackish Water)
- k) Clean in Place (CIP) Systems and Associated Heater.
- l) Sample Pumps and Associated Waste System
- m) Instrumentation and Process Control (SCADA)
- n) Site Electrical Including Heat Tape System
- o) Individual Building Electrical
- p) Individual Building Heating, Air Conditioning and Ventilating as well as Plumbing
- q) Meter Vault
- r) Storage Tank Systems

s) Waste Disposal Systems

2.0 EQUIPMENT START-UP AND PERFORMANCE TEST PROTOCOL FOR THE DESALINATION PLANT

A. General

1. The Contractor must pay for all chemicals used during equipment startup. The price carried in the bid will be the value from the Northeast/Merrimac Valley Chemical Consortium for the year of the bid plus 3% increase for each year until the project is started up. Values for the chemical quantities can be found here; <http://nemvcc.com/>. In the case of proprietary chemicals recommended by the RO vendor that are not on the consortium bid, the contractor shall carry the cost of those chemicals at the time of the bid plus 3% increase for each year until the project is started up.
  - a) The difference between the unit prices and the actual price paid by the Contractor must be used for reimbursement purposes for chemicals used by the Contractor.
  - b) Once the demonstration is complete, the contractor shall fill the tanks to full prior to transition to the Owner.
2. The Owner will pay for all electrical power, pneumatic air for the system, and municipal water. The Contractor will provide temporary pumping, water de-chlorination and for the Contractor's and/or equipment suppliers' staff for the Desalination Plant start up and demonstration.
3. The Contractor must be responsible for disposal of all water treated during demonstration periods at the Desalination Plant, including all de-chlorination and temporary pumping required. Treated water and or waste water during the demonstration will be diverted to the river. The contractor shall refer to the sequence of operations in Attachment of Specification 46 82 00 for details on the cleaning and neutralization process.
4. The seller's representative must be on-site one day a week during the demonstration test period. The seller's representative must be on site for three days during the first week.
5. After the Desalination Plant and all the independent systems have been successfully passed the pre-equipment started up check, all process components must be disinfected. Following the disinfection procedures, the Contractor must de-chlorinate and dispose of all heavily chlorinated water and drain all the chemical storage tanks.
6. The Contractor must start the Desalination Plant, establish proper chemistry, and then operate the new Desalination Plant for 48 days (twenty-four hours per day) as part of the demonstration.
7. Demonstration shall begin no later than thirty (30) days after completion of start-up and commissioning, or three (3) months from the date of receipt of all membranes



components to the project site, whichever occurs first. The Contractor must coordinate with the Engineer and Owner on the mutually agreed time to begin the testing.

**B. Definition of Successful Demonstration**

1. Successful completion will be defined as in Table 1. Details are outlined in the following specifications as well as raw water quality ranges.
  - a) 46 80 00 Brackish Water Reverse Osmosis Unit
  - b) 46 81 00 Seawater Water Reverse Osmosis Unit
  - c) 46 82 00 Immersed Membrane Filtration and Ancillary Equipment.

*Table 1: Performance Testing Requirements*

| Item                                                       | Target Quality                 | Frequency            | Notes                                                     |
|------------------------------------------------------------|--------------------------------|----------------------|-----------------------------------------------------------|
| <b>Ultrafiltration Unit</b>                                |                                |                      |                                                           |
| Turbidity, NTU                                             | <0.1 NTU                       | 95% of measurements  | Continuous permeate from each UF membrane turbidity meter |
| Turbidity, NTU                                             | <0.5 NTU                       | 100% of measurements | Continuous permeate from each UF membrane turbidity meter |
| Conductivity                                               | <100 umhos                     | 95% of measurements  | 1 Time a Day                                              |
| Total Iron                                                 | <0.1 mg/L                      | 95% of measurements  | 1 Time a Week                                             |
| Hardness                                                   | <0.1 mg/L as CaCO <sub>3</sub> | 95% of measurements  | 1 Time a Week                                             |
| Alkalinity                                                 | <150 mg/L as CaCO <sub>3</sub> | 95% of measurements  | 1 Time a Week                                             |
| Salt Density Index                                         | <5                             | 100% of measurements | 2 Times a day (a.m. and p.m.)                             |
| Hydraulic Efficiency at permeate flow of 777 gpm or better | 90.0%                          | 100% of measurements | Calculated Daily                                          |
| <b>Seawater R.O.</b>                                       |                                |                      |                                                           |
| Permeate Total Dissolved Solids at 25°C                    | 135 mg/L                       | 95% of measurements  | 1 Time a day                                              |

|                                                            |           |                      |                  |
|------------------------------------------------------------|-----------|----------------------|------------------|
| Permeate Total Dissolved Solids at 1°C                     | 43 mg/L   | 95% of measurements  | 1 Time a day     |
| Permeate Conductivity at 25°C                              | 282 µS/cm | 95% of measurements  | 1 Time a day     |
| Permeate Conductivity at 1°C                               | 92 µS/cm  | 95% of measurements  | 1 Time a day     |
| Hydraulic Efficiency at permeate flow of 312 gpm or better | 47.4%     | 100% of measurements | Calculated Daily |
| <b>Brackish water R.O.</b>                                 |           |                      |                  |
| Permeate Total Dissolved Solids at 25°C                    | 3 mg/L    | 95% of measurements  | 1 Time a day     |
| Permeate Total Dissolved Solids at 1°C                     | 1 mg/L    | 95% of measurements  | 1 Time a day     |
| Permeate Conductivity at 25°C                              | 5 µS/cm   | 95% of measurements  | 1 Time a day     |
| Permeate Conductivity at 1°C                               | 2 µS/cm   | 95% of measurements  | 1 Time a day     |
| Hydraulic Efficiency at permeate flow of 278 gpm or better | 89.0%     | 100% of measurements | Calculated Daily |
| <b>Final Water Production</b>                              |           |                      |                  |
| Flow, gpm                                                  | ≥278 gpm  | 100% of measurements | Calculated Daily |

Note for temperatures between 1°C and 25°C the performance values should be between those listed and as anticipated by the RO manufacturer. Contractor to provide this information prior to start of the test.

2. The Contractor will provide a report for the demonstration period outlining the demonstration period results prior to or with the acceptance test request letter by the Contractor. The Engineer will review the report as part of the request for acceptance and allow to asked questions and additional information to help aid in assessing the performance. The report shall have the following sections to it:

- a) INTRODUCTION

- (1) 1.1 Study Objectives
  - (2) 1.2 Applicable Standards and Goals
- b) METHODS AND MATERIALS
  - (1) 2.1 Method of Treatment
  - (2) 2.2 Raw Water Supply
  - (3) 2.3 Testing Equipment
  - (4) 2.4 System Configuration
  - (5) 2.5 Termination Criteria
  - (6) 2.6 Chemical Pretreatment
  - (7) 2.7 Field Analytical Methods
- c) RESULTS
  - (1) 3.1 Raw Water Quality
  - (2) 3.2 Finished Water Quality vs. Performance Criteria
  - (3) 3.3 Wash Water Characteristics
  - (4) 3.4 Pressure Losses
  - (5) 3.5 UF and RO Runtimes
  - (6) 3.6 Issues Encountered and How Addressed
- d) CONCLUSIONS
- e) APPENDICES
  - (1) 5.1 Field sampling results and notes
  - (2) 5.2 Certified laboratory results
- 3. Any repairs required in order for the Desalination Plant to meet successful completion must be by the Contractor.
- 4. The Desalination Plant demonstration must be conducted during a consecutive period of forty-eight (48) days total at 24 hours a day. The Desalination Plant must produce acceptable finish water quality and production volumes during this period.
  - a) Periods of acceptable down times are as follows. In these cases, the demonstration period will continue once the event is addressed or passed.
    - (1) Power interruption in the main power supply.
    - (2) Programming issues that are addressed within 4 hours.

- (3) Any influent or operating parameter outside the accepted operating ranges defined in these specifications.
- (4) Errors/Omissions in the design that are not discovered until startup.
- (5) Equipment mechanical or electrical failures that are deemed by the Engineer to be within the manufacturer's or contractor's control will not be used as a reason to allow the test to continue without resetting the demonstration schedule. Items deemed outside of the manufacturer's or contractor's control will be allowed for the demonstration test to continue.

- b) If the Desalination Plant does not properly treat the raw water for the demonstration period, the demonstration must be deemed unacceptable by the Engineer, and the Contractor must coordinate with the Owner and Engineer, and re-schedule another demonstration period. If the Engineer determines that the first demonstration fails as well as any subsequent failures, the Contractor must be responsible for all costs and expenses of the Owner and Engineer, for subsequent demonstrations. The retest must be at a time that is mutually acceptable to the Contractor and Owner.

C. Start-up must consist of the following items, in the order shown:

- 1. Equipment Pre-Start-up Check
- 2. Disinfection of the Desalination Plant
- 3. Manufacturer's Equipment Training
- 4. Equipment Start-up and Performance Testing
- 5. Performance Testing
- 6. Plant Process Start-up and Demonstration

D. Pre-Equipment Start-up Check

- 1. Contractor must check out each item of equipment in the presence of the Engineer, to show that it is properly installed, and is functioning and ready for equipment start-up and performance testing.
- 2. As a minimum, the Contractor must do the following:
  - a) Assure that the equipment is properly installed, painted, leveled, wired and/or insulated.
  - b) Assure that all piping and plumbing is properly installed and that all valving is properly set.
  - c) Assure that all pipeline and plumbing is cleaned and pressure tested, and/or chlorinated as required.
  - d) Assure that all equipment is properly lubricated.

- e) Assure that all safety related accessories are properly installed.
  - f) Bump or momentarily jog equipment to establish operation and proper rotation.
3. Contractor must arrange for equipment manufacturers to be present, or verify that these procedures may be done without the manufacturer's representatives being present.
4. Prior to startup, the Contractor must undertake the following procedures, in the order listed:
- a) Ensure that all required written statements and/or guarantees from manufacturers by individual Specification Sections comply with Contract Documents. Contractor must use a checklist to identify all such requirements, by Specification Section, and must submit a copy of the completed checklist to the Engineer. All such written statements and/or guarantees from manufacturers must be submitted to the Engineer.
  - b) Ensure work is complete before starting up of any unit or system. Certify to Owner that specifically required services or respective equipment manufacturer's representative by individual sections of specifications have been performed in accordance with Contract Documents.
  - c) Ensure systems are tested hydraulically, mechanically, and electrically. Ensure systems which require calibration, commissioning, and balancing are fully Certified as complete in performance, in accordance with Contract Documents. Ensure required tagging, identification, and/or stenciling is complete.
  - d) Provide all labor, supervision, utilities, chemicals, equipment, tools, materials, vehicles or any other items necessary to start up, operate, and demonstrate the system.
  - e) Gages, meters, recorders and monitors must be provided by the Contractor to supplement or augment the instrumentation system provided under this Contract to properly demonstrate that all equipment fully satisfies the requirements of the Contract Documents. All devices employed for the purpose of measuring the performance of the plant's equipment and systems must be specially selected to provide a level of certainty consistent with the variables to be monitored. All instruments must be recently calibrated, and the Contractor must be prepared at all times to demonstrate, through recalibration, the certainty of all instruments employed for testing purposes. Calibration procedures must be in accordance with applicable standards of ASTM, ISA and IEEE. The adequacy of all gages, meters, recorders and monitors must be subject to review of the Engineer.
  - f) The Contractor must provide sign off forms for all installed and operation testing to be accomplished under this contract. Sign off forms must be provided for each item of mechanical, electrical and instrumentation equipment provided or installed under this contract and must contain provisions for

recording relevant performance data for original testing, and not less than three retests. Separate sections on the form must be provided to record values for the pre-operation checkout, initials of representatives of the equipment manufacturers, the Contractor and the Engineer.

- g) The Contractor must maintain a master file of all equipment signoff forms, which must be available for inspection by the Engineer. Upon completion of all equipment testing, the Contractor must furnish the Engineer with the original and two copies of the signoff forms for each equipment item.

#### E. Disinfection of the Desalination Plant

1. The Contractor must disinfect all treatment process units, starting with the raw water piping, and completing with the disinfection of the finish water piping, permeate tank and pumping systems to the cooling systems. Items, such as the membrane systems which are sensitive to chlorine shall be disinfected according to the manufacturer's guidance. Tanks used for the UF membranes must be disinfected prior to the installation of the membranes. The Contractor shall be responsible for neutralizing and legally disposal of any storage solutions used for membrane storage during shipment. The Contractor must plan on carrying the cost of this solution to be disposed off site. The Contractor can request from the Owner to dispose of the neutralized solution into the sewer but the Contractor should carry the cost of disposal in their bid in the event that said solution is not acceptable to the receiving authority.
2. The Contractor must disinfect all new piping (process, potable water, and chemical feed), in accordance with AWWA C651 (Disinfecting Water Mains). The entire treatment system including but not limited to the water mains, static mixer, basins, chemical storage tanks, chemical feeders, pumping systems, etc. must be disinfected in accordance with AWWA C652 (Disinfection of Water Storage Facilities) or AWWA C653 (Disinfection of Water Treatment Plants). The exception to this will be the membrane systems will be disinfected according the manufacturer guidelines.
3. The Contractor must collect all samples under the direction of the Engineer. The Contractor must pay for and submit all the samples for bacteriological analyses, as well as any re-sampling and re-disinfection that may be required.
4. The Contractor must be responsible for providing all equipment necessary to disinfect the desalination plant including any temporary chemical feed systems and/or pumping equipment.
5. The Owner will supply a water tap to the municipally supplied water to provide for disinfection purposes. Contractor shall carry the cost for a water meter and backflow preventor to meter the water as well as any costs incurred for the purchased water from the Public Water System.
6. The Contractor must be responsible for de-chlorinating and disposal of all chlorinated water. All chlorinated water must be dechlorinated and neutralized prior to discharge.

#### F. Manufacturer's Training

1. The contractor must refer to specifications for requirements of manufacturer's training
2. The Contractor must note the following:
  - a) No training can be conducted until the Owner has received approved Manufacturers' Operation & Maintenance Manuals.
  - b) Owner will not accept any item of equipment prior to receiving approved manufacturer training for the equipment. This applies only to equipment requiring manufacturer's training in accordance with the specifications.
  - c) Engineer must approve the completeness of all training and verify its completion.
  - d) Training must be conducted in addition to, and exclusive of start-up and performance testing.
  - e) Training must be completed prior to demonstration testing.

G. Equipment Start-up

1. Prior to testing any equipment, the contractor must obtain written certification from the manufacturer, that the equipment is properly installed, calibrated and ready for safe and efficient operation as intended by the Engineer and manufacturer.
2. Prior to start-up and testing of any item of equipment, the equipment and all associated piping, channels, basins, wetwells, etc. must be thoroughly cleaned, disinfected and flushed.
3. The Contractor, with assistance from the manufacturer's factory trained service engineer, and in the presence of the Engineer, must start-up and operate each item of equipment to show that it is performing according to the requirements of the specifications. With this step, the Contractor must assure that the equipment is ready for plant and process start-up.
4. During startup of equipment, the Contractor must provide knowledgeable and experienced person(s) to instruct Owner's designated personnel on the Operation and Maintenance of each system. This service must be in addition to services provided by equipment manufacturer's authorized representative(s) prescribed by individual Sections of this specification. Instructions during the startup period must be in the form of a comprehensive "overview", and not simply a repeat of previous Operation and Maintenance instructions. Provide a minimum of 4 hours of Operation and Maintenance instructions on each system. For the membrane systems a minimum of 8 hours total for all three membrane system.

H. Performance Tests

1. General
  - a) Full tests must be made at the site on each item of equipment after it has been properly installed, started and certified ready for operation. These tests must

demonstrate that each item of equipment will operate properly by itself and in conjunction with all other treatment plants equipment, in accordance with the performance specification.

- b) The Contractor must furnish all necessary labor, tools, equipment, chemicals to perform field tests to determine that the supplied equipment, including controls and alarms meet hydraulic, electric, mechanical and performance requirements in accordance with the specifications.
  - c) All incomplete and/or unsuccessful tests must be done over, to the satisfaction of the Engineer.
2. Vertical Turbine /Centrifugal/Rotary Lobe Pumps
- a) Assure that pumpage falls on pump curves by testing three points sufficiently spaced to characterize the curve.
  - b) Record the following readings on each pump:
    - (1) Test Flow Rate and TDH during Test
    - (2) Amperes per phase, at the Test Flow Rate and TDH
    - (3) Voltage each phase, at the Test Flow Rate and TDH
    - (4) HP input to motor, at the Test Flow Rate and TDH
  - c) Determine the efficiency of the pumping units at test and design conditions.
  - d) Record maximum ampere and voltage readings of each motor and determine that they compare to the nameplate data.
  - e) Record unusual noise and/or vibration and correct based on manufacturer's recommendations.
  - f) Verify proper operation of flow control valves.
  - g) Verify pump speed.
3. Chemical Feed Systems
- a) Calibrate each chemical feed pump. For centrifugal pumps the characteristic curve should be developed for the site head conditions over the flow range of the pump. For peristaltic or positive displacement pumps the characteristic curve should be developed over the speed and where applicable stroke adjustment full range.
  - b) Ensure that feed pumps pace accurately with either flow signals and/or other designated 4-20 mA signals or pulse signals.
  - c) Check all piping, valves, and connections for leaks.
  - d) Check and verify proper operation of all auxiliary components such as back



pressure valves, anti-siphon valves (vacuum breakers), pressure relief valves, pulsation dampers, pressure gauges, etc.

- e) Check and verify proper operation of all control systems, alarms, timers, etc.
4. Screen Airburst System
- a) Confirmation of pressure build up in supply tank.
  - b) Confirmation of cleaning of the screens to initial headloss across the screen.
  - c) Confirmation of headloss recording across the screen as well as calculation of estimated flow velocities across the screen.
  - d) Confirmation of communication with the SCADA system.
5. Automatic Backwashing System.
- a) Confirmation of headloss build up and successful backwashing.
  - b) Confirmation of alarms and communication with SCADA.
  - c) Confirmation ABW waste pumps supply pump waste to outfall.
  - d) Adjust pressure reducing valve as necessary.
6. Sample Water Pumps and Sump Pumps
- a) Record maximum ampere and voltage readings of each motor and determine that they compare to the nameplate data.
  - b) Record unusual noise and/or vibration and correct according to manufacturer's instructions.
  - c) Determine maximum pumpage.
7. Process Equipment, Instrumentation, and Controls
- a) Calibrate and/or verify calibration and proper operation and function of all process and analytical instruments, and document results, for the following:
    - (1) Level monitors and controls
    - (2) Flow meters and associated transmitters
    - (3) Indicating Controllers
    - (4) Indicators
      - a. Totalizers
    - (5) Electrical Relays and current trips
    - (6) Current-to-current converters
    - (7) Annunciator and alarm horn

- (8) PLCs and transmitting and receiving equipment
- (9) Lightening and surge protection equipment
- (10) Power supplies and UPS systems
- (11) Analyzers (AII)
- (12) Differential Pressure cells
- (13) Selector switches
- (14) SCADA system Hardware and Software
- (15) Membrane Process Equipment
- (16) Electrically and Pneumatically actuated process control valves. The control valves for the R.O. systems must operate to setpoint within the timeframe specified by the R.O. manufacturers.
- (17) Misc. Equipment

8. Electrical

- a) Verify proper operation and function of all electrical equipment and instruments, and document results, for the following:
  - (1) Motor Control Centers
  - (2) Lighting Panels
  - (3) Alarm systems
  - (4) Control Panels

9. Heating and Ventilating as well as Plumbing

- a) Calibrate and/or verify proper operation and function of all HVAC equipment, and document results.
- b) Confirm all drains are functioning.

I. Plant Process Start-Up and Demonstration

1. General

- a) Plant process start-up must demonstrate and verify that each item of equipment operates properly, one in conjunction with another, can achieve the specified process and/or hydraulic capacity under the specified range of raw water quality conditions, and that the entire plant operates in an acceptable fashion. During the "Plant Process Start-Up", each item of equipment must be started and used to perform its design function, and design capacity, within the process train. Water must be chemically treated and filtered during this process. Water quality tests must be made to determine whether the plant is performing

according to design requirements, and is acceptable to Owner as a finished product.

- b) All start-up equipment and training requirements must be complete prior to "Plant Process Start-up".
- c) The Contractor must check and verify proper operation of all equipment as the individual components are required to operate individually, and collectively, in conjunction with one another.
  - (1) The Contractor must demonstrate efficient and proper operation of the entire plant, such that each individual component and the total plant operates in an acceptable fashion to the Engineer, Owner, and any other State or Federal Authority having legal jurisdiction.
  - (2) The Contractor must verify with the appropriate equipment manufacturers that the plants process start-up may be done without the manufacturer being present for each piece of equipment.
- d) Demonstration testing shall be conducted during a period of forty-eight (48) days total. For the duration of the Demonstration test the Owner's operation staff will be responsible for the full operation of the Desalination Plant. During the forty-eight (48) day Demonstration test the membrane manufacturing representative will be present on-site for the first three (3) days to provide assistance.
- e) Upon successful completion of the Acceptance Test of all process trains, the Contractor must provide a written report outlining the performance of the demonstration test with a notice of substantial completion. The Contractor must provide requested information from the Engineer during the review with sufficient time (1 week) to review the results. Within four (4) weeks immediately following receipt of such notice, the Owner must notify the Contractor in writing that it accepts the test results, or it does not accept the test results, or, alternatively a mutually agreeable time in writing when such notice will be given. In the case of non-acceptance, the Owner must state the specific reason for non-acceptance. In the absence of such a reply from the Owner within the four (4) week period, the Contractor shall deem the Demonstration test was accepted.

END OF SECTION

SECTION 02 41 00

DEMOLITION  
08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI Guideline K (2009) Guideline for Containers for Recovered Non-Flammable Fluorocarbon Refrigerants

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO M 145 (1991; R 2012) Standard Specification for Classification of Soils and Soil-Aggregate Mixtures for Highway Construction Purposes

AASHTO T 180 (2017) Standard Method of Test for Moisture-Density Relations of Soils Using a 4.54-kg (10-lb) Rammer and a 457-mm (18-in.) Drop

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.6 (2006) Safety & Health Program Requirements for Demolition Operations - American National Standard for Construction and Demolition Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety -- Safety and Occupational Health (SOH) Requirements

U.S. DEFENSE LOGISTICS AGENCY (DLA)

DLA 4145.25 (Jun 2000; Reaffirmed Oct 2010) Storage and Handling of Liquefied and Gaseous Compressed Gases and Their Full and Empty Cylinders;  
<https://www.dla.mil/Portals/104/Documents/Disposition%20of%20Hazardous%20Materials/4145.25%20Storage%20and%20Handling%20of%20Liquefied%20and%20Gaseous%20Compressed%20Gases%20and%20Their%20Full%20and%20Empty%20Cylinders.pdf>

U.S. DEPARTMENT OF DEFENSE (DOD)

DOD 4000.25-1-M (2006) MILSTRIP - Military Standard Requisitioning and Issue Procedures

MIL-STD-129 (2014; Rev R; Change 1 2018; Change 2

2019; Change 3 2023) Military Marking for  
Shipment and Storage

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 61                      National Emission Standards for Hazardous  
Air Pollutants

40 CFR 82                      Protection of Stratospheric Ozone

1.2 PROJECT DESCRIPTION

1.2.1 Definitions

1.2.1.1 Demolition

Demolition is the process of tearing apart and removing any feature of a facility together with any related handling and disposal operations.

1.2.1.2 Deconstruction

Deconstruction is the process of taking apart a facility with the primary goal of preserving the value of all useful building materials.

1.2.1.3 Demolition Plan

Demolition Plan is the planned steps and processes for managing demolition activities and identifying the required sequencing activities and disposal mechanisms.

1.2.1.4 Deconstruction Plan

Deconstruction Plan is the planned steps and processes for dismantling all or portions of a structure or assembly, to include managing sequencing activities, storage, re-installation activities, salvage and disposal mechanisms.

1.2.2 Demolition/Deconstruction Plan

Prepare a Demolition Plan and submit proposed salvage, demolition, and removal procedures for approval before work is started. Include in the plan procedures for careful removal and disposition of materials specified to be salvaged, coordination with other work in progress. Identify components and materials to be salvaged for reuse or recycling with reference to paragraph Existing Facilities to be Removed. Append tracking forms for all removed materials indicating type, quantities, condition, destination, and end use. Coordinate with Waste Management Plan in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Provide procedures for safe conduct of the work in accordance with EM 385-1-1. Plan must be approved by Contracting Officer prior to work beginning.

1.2.3 General Requirements

Do not begin demolition or deconstruction until authorization is received from the Contracting Officer. The work of this section is to be performed in a manner that maximizes the value derived from the salvage and recycling of materials. The work includes demolition, , salvage of identified items and materials, and removal of resulting rubbish and

debris. Remove rubbish and debris from Government property daily, unless otherwise directed. Store materials that cannot be removed daily in areas specified by the Contracting Officer. In the interest of occupational safety and health, perform the work in accordance with EM 385-1-1, Section 17, Demolition, and other applicable Sections.

### 1.3 ITEMS TO REMAIN IN PLACE

Comply with FAR 52.236-9 to protect existing vegetation, structures, equipment, utilities, and improvements. Coordinate the work of this section with all other work indicated. Construct and maintain shoring, bracing, and supports as required. Ensure that structural elements are not overloaded. Increase structural supports or add new supports as may be required as a result of any cutting, removal, deconstruction, or demolition work performed under this contract. Do not overload structural elements or pavements to remain. Provide new supports and reinforcement for existing construction weakened by demolition, deconstruction, or removal work. Repairs, reinforcement, or structural replacement require approval by the Contracting Officer prior to performing such work.

#### 1.3.1 Existing Construction Limits and Protection

Do not disturb existing construction beyond the extent indicated or necessary for installation of new construction. Provide temporary shoring and bracing for support of building components to prevent settlement or other movement. Provide protective measures to control accumulation and migration of dust and dirt in all work areas. Remove snow, dust, dirt, and debris from work areas daily.

#### 1.3.2 Weather Protection

For portions of the building to remain, protect building interior and materials and equipment from the weather at all times. Where removal of existing roofing is necessary to accomplish work, have materials and workmen ready to provide adequate and temporary covering of exposed areas.

#### 1.3.3 Utility Service

Maintain existing utilities indicated to stay in service and protect against damage during demolition and deconstruction operations. Prior to start of work, utilities serving each area of alteration or removal will be shut off by the Government and disconnected and sealed by the Contractor.

#### 1.3.4 Facilities

Protect electrical and mechanical services and utilities. Where removal of existing utilities and pavement is specified or indicated, provide approved barricades, temporary covering of exposed areas, and temporary services or connections for electrical and mechanical utilities. Floors, roofs, walls, columns, pilasters, and other structural components that are designed and constructed to stand without lateral support or shoring, and are determined to be in stable condition, must remain standing without additional bracing, shoring, or lateral support until demolished or deconstructed, unless directed otherwise by the Contracting Officer. Ensure that no elements determined to be unstable are left unsupported and place and secure bracing, shoring, or lateral supports as may be required as a result of any cutting, removal, deconstruction, or demolition work performed under this contract.

#### 1.4 BURNING

The use of burning at the project site for the disposal of refuse and debris will not be permitted .

#### 1.5 AVAILABILITY OF WORK AREAS

Areas in which the work is to be accomplished will be available at contract award.

#### 1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-01 Preconstruction Submittals

Demolition Plan; G

Existing Conditions

##### SD-07 Certificates

Notification; G

##### SD-11 Closeout Submittals

Receipts

#### 1.7 QUALITY ASSURANCE

Submit timely notification of demolition and renovation projects to Federal, State, regional, and local authorities in accordance with 40 CFR 61, Subpart M. Notify the Regional Office of the United States Environmental Protection Agency (USEPA) and the Contracting Officer in writing 10 working days prior to the commencement of work in accordance with 40 CFR 61, Subpart M. Comply with federal, state, and local hauling and disposal regulations. In addition to the requirements of the "Contract Clauses," conform to the safety requirements contained in ASSP A10.6. Comply with the Environmental Protection Agency requirements specified. Use of explosives will not be permitted.

##### 1.7.1 Dust and Debris Control

Prevent the spread of dust and debris to occupied portions of the building and avoid the creation of a nuisance or hazard in the surrounding area. Do not use water if it results in hazardous or objectionable conditions such as, but not limited to, ice, flooding, or pollution. Vacuum and dust the work area daily . Sweep pavements as often as necessary to control the spread of debris that may result in foreign object damage potential to aircraft.

## 1.8 PROTECTION

### 1.8.1 Traffic Control Signs

- a. Where pedestrian and driver safety is endangered in the area of removal work, use traffic barricades with flashing lights. Notify the Contracting Officer prior to beginning such work.

### 1.8.2 Protection of Personnel

Before, during and after the demolition work continuously evaluate the condition of the site specific features structure being demolished and take immediate action to protect all personnel working in and around the project site. No area, section, or component of floors, roofs, walls, columns, pilasters, or other structural element will be allowed to be left standing without sufficient bracing, shoring, or lateral support to prevent collapse or failure while workmen remove debris or perform other work in the immediate area.

## 1.9 RELOCATIONS

Perform the removal and reinstallation of relocated items as indicated with workmen skilled in the trades involved. Repair or replace items to be relocated which are damaged by the Contractor with new undamaged items as approved by the Contracting Officer.

### 1.10 EXISTING CONDITIONS

Before beginning any demolition or deconstruction work, survey the site and examine the drawings and specifications to determine the extent of the work. Record existing conditions in the presence of the Contracting Officer or the Contracting Officer's Representative showing the condition of structures and other facilities adjacent to areas of alteration or removal. Photographs or electronic images with a minimum resolution of 3072 x 2304 pixels, capable of a print resolution of 300 dpi, will be acceptable as a record of existing conditions. Include in the record the elevation of the top of foundation walls, finish floor elevations, possible conflicting electrical conduits, plumbing lines, alarms systems, the location and extent of existing cracks and other damage and description of surface conditions that exist prior to starting work. It is the Contractor's responsibility to verify and document all required outages which will be required during the course of work, and to note these outages on the record document. Submit survey results to the Contracting Officer or the Contracting Officer's Representative.

## PART 2 PRODUCTS

### 2.1 FILL MATERIAL

- a. Comply with excavating, backfilling, and compacting procedures for soils used as backfill material to fill basements, voids, depressions or excavations resulting from demolition or deconstruction of structures. Provide fill material consisting of waste products from demolition or deconstruction until all waste appropriate for this purpose is consumed.
- b. Provide fill material conforming to the definition of satisfactory soil material as defined in AASHTO M 145, Soil Classification Groups A-1, A-2-4, A-2-5 and A-3. In addition, fill material must be free



from roots and other organic matter, trash, debris, frozen materials, and stones larger than 2 inches in any dimension.

- c. Proposed fill material must be sampled and tested by an approved soil testing laboratory, as follows:

|                            |                             |
|----------------------------|-----------------------------|
| Soil classification        | AASHTO M 145                |
| Moisture-density relations | AASHTO T 180, Method B or D |

## PART 3 EXECUTION

### 3.1 CONCURRENT EARTH-MOVING OPERATIONS

Do not begin excavation, filling, and other earth-moving operations that are sequential to demolition work in areas occupied by structures to be demolished until all demolition in the area has been completed and debris removed. Fill holes and other hazardous openings.

### 3.2 DISPOSITION OF MATERIAL

#### 3.2.1 Title to Materials

Except for salvaged items specified in related Sections, and for materials or equipment scheduled for salvage, all materials and equipment removed and not reused or salvaged, become the property of the Contractor and must be removed from Government property. Materials approved for storage by the Contracting Officer must be removed before completion of the contract. Title to materials resulting from demolition and deconstruction, and materials and equipment to be removed, is vested in the Contractor upon approval by the Contracting Officer. The Government will not be responsible for the condition or loss of, or damage to, such property after contract award. Showing for sale or selling materials and equipment on site is prohibited.

#### 3.2.2 Reuse of Materials and Equipment

Remove and store materials and equipment listed in the Demolition Plan to be reused or relocated to prevent damage, and reinstall as the work progresses. Coordinate the re-use of materials and equipment with the re-use requirements in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Capture re-use of materials in the diversion calculations for the project.

#### 3.2.3 Salvaged Materials and Equipment

Remove materials and equipment that are listed in the Demolition Plan to be removed by the Contractor and that are to remain the property of the Government, and deliver to a storage site, as directed by the Contracting Officer.

- a. Salvage items and material to the maximum extent possible.
- b. Store all materials salvaged for the Contractor as approved by the Contracting Officer and remove from Government property before completion of the contract. Coordinate the salvaged materials with tracking requirements in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Capture salvaged materials in the

diversion calculations for the project.

- c. Remove salvaged items to remain the property of the Government in a manner to prevent damage, and packed or crated to protect the items from damage while in storage or during shipment. Items damaged during removal or storage must be repaired or replaced to match existing items. Properly identify the contents of containers. Deliver the following items reserved as property of the Government to the areas designated.
- e. Remove historical items in accordance with 02 42 91 REMOVAL AND SALVAGE OF HISTORIC CONSTRUCTION MATERIALS
- f. Remove and capture all Class I ODS refrigerants in accordance with the Clean Air Act Amendment of 1990, and turn in to the Navy as directed by the Commanding Officer.

#### 3.2.4 Disposal of Ozone Depleting Substance (ODS)

Class I and Class II ODS are defined in Section, 602(a) and (b), of The Clean Air Act. Prevent discharge of Class I and Class II ODS to the atmosphere. Place recovered ODS in cylinders meeting AHRI Guideline K suitable for the type ODS (filled to no more than 80 percent capacity) and provide appropriate labeling. Recovered ODS must be Turn over recovered ODS to the Contracting Officer. Dispose products, equipment and appliances containing ODS in a sealed, self-contained system (e.g. residential refrigerators and window air conditioners) in accordance with 40 CFR 82. Submit Receipts or bills of lading, as specified. Submit a shipping receipt or bill of lading for all containers of ozone depleting substance (ODS) shipped to the Defense Depot, Richmond, Virginia.

##### 3.2.4.1 Special Instructions

No more than one type of ODS is permitted in each container. Apply a warning/hazardous label to the containers in accordance with Department of Transportation regulations. Provide a tag with the following information on all cylinders including but not limited to fire extinguishers, spheres, or canisters containing an ODS:

- a. Activity name and unit identification code
- b. Activity point of contact and phone number
- c. Type of ODS and pounds of ODS contained
- d. Date of shipment
- e. National stock number (for information, call (804) 279-4525).

##### 3.2.5 Transportation Guidance

Ship all ODS containers in accordance with MIL-STD-129, DLA 4145.25 (also referenced one of the following: Naval Supply Instruction 4440.128C and DOD 4000.25-1-M.

### 3.2.6 Unsalvageable and Non-Recyclable Material

Dispose of unsalvageable and non-recyclable noncombustible material in the disposal area located by the Contracting Officer.

### 3.3 CLEANUP

Remove debris and rubbish from project site and similar excavations. Remove and transport the debris in a manner that prevents spillage on streets or adjacent areas. Apply local regulations regarding hauling and disposal.

### 3.4 DISPOSAL OF REMOVED MATERIALS

#### 3.4.1 Regulation of Removed Materials

Dispose of debris, rubbish, scrap, and other nonsalvageable materials resulting from removal operations with all applicable federal, state and local regulations as contractually specified in the Waste Management Plan . Storage of removed materials on the project site is prohibited.

#### 3.4.2 Burning on Government Property

Burning of materials will not be permitted on Government property .

#### 3.4.3 Removal to Spoil Areas on Government Property

Transport noncombustible materials removed from demolition and deconstruction structures to designated spoil areas on Government property.

#### 3.4.4 Removal from Government Property

Transport waste materials removed from demolished and deconstructed structures, except waste soil, from Government property for legal disposal. Dispose of waste soil per Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME).

### 3.5 REUSE OF SALVAGED ITEMS

Recondition salvaged materials and equipment designated for reuse before installation. Replace items damaged during removal and salvage operations or restore them as necessary to usable condition.

-- End of Section --

SECTION 02 42 91

REMOVAL AND SALVAGE OF HISTORIC CONSTRUCTION MATERIALS  
**11/18**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|                  |                                               |
|------------------|-----------------------------------------------|
| 29 CFR 1910.1000 | Air Contaminants                              |
| 29 CFR 1926.55   | Gases, Vapors, Fumes, Dusts, and Mists        |
| 40 CFR 261       | Identification and Listing of Hazardous Waste |

1.2 PROJECT DESCRIPTION

The work includes removal and salvage of identified historic items and materials, and removal of resulting rubbish and debris. General demolition of non-historic materials and removal of resulting rubbish and debris must comply with the requirements of Section 02 41 00 DEMOLITION. Store salvaged or recycled materials daily in areas and in a manner specified by the Contracting Officer. In the interest of conservation, pursue salvage and recycling to the maximum extent possible. Submit a Work Plan that includes procedures proposed for the accomplishment of the work. The Work Plan procedures must provide for safe conduct of the work, careful removal and disposition of materials specified to be salvaged or recycled, dust control, protection of property which is to remain undisturbed, coordination with other work in progress, and timely disconnection of utility services. Include a detailed description of the methods and equipment to be used for each operation, and the sequence of operations in the Work Plan.

1.2.1 Dust Control

Control the amount of dust resulting from removal, salvage and demolition operations to prevent the spread of dust to occupied portions of the construction site, to avoid creation of a nuisance in the surrounding area and to minimize occupational exposures. Occupational exposures cannot exceed the requirements in 29 CFR 1910.1000 and 29 CFR 1926.55. Use of water to control dust will not be permitted when it will result in, or create, damage to existing building materials and hazardous or objectionable conditions such as ice, flooding and pollution.

1.2.2 Protection

1.2.2.1 Protection of Existing Historic Property

Survey the site and examine the drawings and specifications to determine the extent of work before beginning any removal, salvage or demolition work. Take necessary precautions to avoid damage to existing historic

items that are to remain in place, to be reused, or to remain the property of the Government. Repair or restore items damaged by the Contractor to original condition, or replace, as approved by the Contracting Officer. Coordinate the work of this section with all other work and construct and maintain shoring, bracing and supports, as required. Ensure that structural elements are not overloaded and provide additional supports as may be required as a result of any cutting, removal, or demolition work performed under this contract.

#### 1.2.2.2 Protection From the Weather

Protect the interior of buildings to remain and salvageable materials from the weather at all times. Store salvaged historic materials off the ground and under weathertight covering.

#### 1.2.2.3 Environmental Protection

Comply with the requirements of Section 02 82 00.00 22 ASBESTOS REMEDIATION 02 83 00.00 22 MANAGEMENT OF LEAD, CADMIUM, AND CHROMIUM (VI) DURING RENOVATION, DEMOLITION, CONSTRUCTION, REMOVAL AND ABATEMENT (PNS PROJECTS). Ensure a thorough inspection has been performed for hazardous materials prior to beginning work.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Work Plan; G

### 1.4 QUALIFICATIONS

Provide qualified workers trained and experienced in whole-building recycling, including removal and salvage of historic materials. Submit documentation of five consecutive years of work of this type with a list of similar projects identifying when, where, and for whom the work was done. Provide a current point-of-contact for identified references.

## PART 2 PRODUCTS

Not used.

## PART 3 EXECUTION

### 3.1 HAZARDOUS MATERIALS

Unforeseen hazardous materials may exist in wall cavities, beneath floors, in chases, inside various components, as well as other building materials. Exercise extreme care when performing demolition and salvage operations to ensure unexpected hazardous materials area not inadvertently disturbed creating a potential exposure concern. If suspect hazardous materials are observed or encountered, stop work and notify the Contracting Officer immediately.

### 3.2 SALVAGED ITEMS

Salvage items to the maximum extent possible. Remove historic items to be salvaged from the structure prior to demolition work. Remove salvageable items by hand labor to the maximum extent possible. Do not damage historic portions of the structure to remain or items identified for salvage. Remove furnishings, equipment, and materials not scheduled for salvage or recycling prior to any salvaging procedures. Keep a complete recording of all salvaged materials including the condition of such materials before, and after, salvage operations.

#### 3.2.1 Site Work

Remove intact and salvage the following site items: fences and gates, site furnishings, , site ornaments, and site lighting fixtures.

#### 3.2.2 Masonry

Remove intact and salvage the following masonry items: brick, stone, terra cotta, cast stone, clay tile trim elements, cornerstones .

#### 3.2.3 Metals

Remove intact and salvage the following metal items: cast architectural ornament, architectural metalwork, ornamental ironwork, and gratings and metal walkways.

#### 3.2.4 Wood

Remove intact and salvage the following materials: architectural woodwork, and wood timbers ( 5 x 5 inch or larger).

#### 3.2.5 Thermal and Moisture Protection

Remove intact and salvage the following materials: slate tiles, clay tiles, gutters, leaders, and downspouts, roof accessories and , exterior siding.

#### 3.2.6 Doors and Windows

Remove doors and windows with associated hardware and operating mechanisms intact (including glass) and salvage in accordance with the schedule.

#### 3.2.7 Finishes

Protect the following special or historic finishes : stenciling, decorative tile, molded ornament, decorative ceiling materials, wood flooring, and textured wall coverings.

#### 3.2.8 Equipment and Specialty Items

Remove intact and salvage the following equipment and specialty items: plaques and contents of cornerstones.

#### 3.2.9 Mechanical Equipment

Remove intact and salvage the following mechanical equipment: registers and grilles.

### 3.2.10 Electrical Equipment

Remove intact and salvage the following electrical fixtures and equipment: telecommunications equipment.

### 3.3 RECYCLED MATERIALS

Recycle materials to the maximum extent possible. Use hand labor wherever possible to remove recyclable materials. Do not damage historic portions of the structure indicated to remain and items identified for salvage while removing materials for recycling. Recycle the following materials: dimension lumber, scrap wood, paper and cardboard, gypsum wallboard, asphalt, rubble, glass, and metals.

### 3.4 DISPOSITION OF MATERIALS

The Contractor, upon receipt of notice to proceed, is vested with the title to materials and equipment to be demolished, except Government and using service salvage and historical items. The Government will not be responsible for the condition, loss or damage to such property after notice to proceed.

#### 3.4.1 Material Salvaged for the Contractor

Temporarily store salvaged material as approved by the Contracting Officer and remove from Government property before completion of the contract. Sale of salvaged material on the site is prohibited.

#### 3.4.2 Items Salvaged for the Government

Remove salvaged items to remain the property of the Government in a manner to prevent damage, packed or crated to protect the items from damage, or as directed by the Contracting Officer. Repair or replace items damaged during removal or storage to match existing items. Properly label and identify containers as to contents.

### 3.5 CLEAN-UP

Upon completion of the work, clean portions of structure to remain and adjacent areas and structures of dust, dirt, and debris caused by salvage and demolition operations. Verify that debris and rubbish created by the work is non-hazardous. If any debris and rubbish is suspect hazardous waste characterize it in accordance with 40 CFR 261. If the debris and rubbish is determined to be hazardous materials notify the Contracting Officer. Remove and transport non-hazardous debris and rubbish in a manner that prevents spillage on streets or adjacent areas. Transport and dispose of all material in accordance with all local, state and Federal regulations. Provide copies of all disposal manifests to the the Contracting Officer.

-- End of Section --

SECTION 02 81 00

TRANSPORTATION AND DISPOSAL OF HAZARDOUS MATERIALS  
**11/18**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INTERNATIONAL AIR TRANSPORT ASSOCIATION (IATA)

IATA DGR (2018) Dangerous Goods Regulations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EP 415-1-266 (2000) Construction -- Resident Engineer Management Guide (REMG) for Hazardous, Toxic, and Radioactive Waste (HTRW) Projects

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|            |                                                                                                                       |
|------------|-----------------------------------------------------------------------------------------------------------------------|
| 40 CFR 61  | National Emission Standards for Hazardous Air Pollutants                                                              |
| 40 CFR 261 | Identification and Listing of Hazardous Waste                                                                         |
| 40 CFR 262 | Standards Applicable to Generators of Hazardous Waste                                                                 |
| 40 CFR 263 | Standards Applicable to Transporters of Hazardous Waste                                                               |
| 40 CFR 264 | Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities                     |
| 40 CFR 265 | Interim Status Standards for Owners and Operators of Hazardous Waste Treatment, Storage, and Disposal Facilities      |
| 40 CFR 266 | Standards for the Management of Specific Hazardous Wastes and Specific Types of Hazardous Waste Management Facilities |
| 40 CFR 268 | Land Disposal Restrictions                                                                                            |
| 40 CFR 270 | EPA Administered Permit Programs: The Hazardous Waste Permit Program                                                  |
| 40 CFR 279 | Standards for the Management of Used Oil                                                                              |
| 40 CFR 300 | National Oil and Hazardous Substances                                                                                 |



Pollution Contingency Plan

|            |                                                                                                                                              |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 40 CFR 302 | Designation, Reportable Quantities, and Notification                                                                                         |
| 40 CFR 761 | Polychlorinated Biphenyls (PCBs) Manufacturing, Processing, Distribution in Commerce, and Use Prohibitions                                   |
| 49 CFR 107 | Hazardous Materials Program Procedures                                                                                                       |
| 49 CFR 172 | Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements |
| 49 CFR 173 | Shippers - General Requirements for Shipments and Packagings                                                                                 |
| 49 CFR 178 | Specifications for Packagings                                                                                                                |

1.2 DEFINITIONS

1.2.1 Hazardous Material

A substance or material which has been determined by the Secretary of Transportation to be capable of posing an unreasonable risk to health, safety, and property when transported in commerce, and which has been so designated pursuant to the Hazardous Materials Transportation Act, 49 U.S.C. Appendix Section 1801 et seq. The term includes materials designated as hazardous materials under the provisions of 49 CFR 172, Sections .101 and .102 and materials which meet the defining criteria for hazard classes and divisions in 49 CFR 173. EPA designated hazardous wastes are also hazardous materials.

1.2.2 Hazardous Waste

A waste which meets criteria established in RCRA or specified by the EPA in 40 CFR 261 or which has been designated as hazardous by a RCRA authorized state program.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Packaging Notifications

Hazardous Waste Management Plan; G

Onsite Hazardous Waste Management; G

Notices of Non-Compliance and Notices of Violation

SD-06 Test Reports

Recordkeeping; G

Exception Report; G

Spill Response

#### SD-07 Certificates

Transportation and Disposal Coordinator; G

Training; G

Certification

Shipping Documents and Packagings Certification; G

Security Plan

Certificates of Disposal

Waste Minimization

### 1.4 QUALITY ASSURANCE

#### 1.4.1 Transportation and Disposal Coordinator

Designate, by position and title, one person to act as the Transportation and Disposal Coordinator (TDC) for this contract. The TDC must serve as the single point of contact for all environmental regulatory matters and have overall responsibility for total environmental compliance at the site including, but not limited to, accurate identification and classification of hazardous waste and hazardous materials; determination of proper shipping names; identification of marking, labeling, packaging and placarding requirements; completion of waste profiles, hazardous waste manifests, asbestos waste shipment records, PCB manifests, bill of lading, exception and discrepancy reports; and all other environmental documentation. The TDC must have, at a minimum, one year of specialized experience in the management and transportation of hazardous waste and have been Department of Transportation certified under 49 CFR 172, Subpart H.

#### 1.4.2 Training

Hazardous materials employees must be trained, tested, and certified to safely and effectively carry out their assigned duties in accordance with Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME). Employees transporting hazardous materials or preparing hazardous materials for transportation, including samples, must be trained, tested, and certified in accordance with 49 CFR 172, Subpart H, including security awareness and any applicable security plans. Hazardous material employees must also be trained in accordance with IATA DGR when shipping hazardous materials by air. Employees must be trained, tested, and certified in accordance with 49 CFR 172, Subpart H to determine that shipments do not constitute DOT regulated hazardous materials.

#### 1.4.3 Certification

The hazardous materials transporter must possess a current certificate of

registration issued by the Research and Special Programs Administration (RSPA), U.S. Department of Transportation, when required by 49 CFR 107, Subpart G. Submit copies of the certificates or written statements certifying exemption from these requirements.

#### 1.4.4 Laws and Regulations Requirements

Comply with Federal, state, and local laws and regulations which are applicable. These requirements are amended frequently and compliance with amendments is required as they become effective. Notify the Contracting Officer immediately if compliance exceeds the scope of work or conflicts with specific requirements of the contract.

## PART 2 PRODUCTS

### 2.1 MATERIALS

Provide all the materials required for the packaging, labeling, marking, placarding and transportation of hazardous wastes and hazardous materials in conformance with Department of Transportation standards and IATA DGR and EP 415-1-266. Details in this specification must not be construed as establishing the limits of the Contractor's responsibility.

#### 2.1.1 Packagings

Provide bulk and non-bulk containers for packaging hazardous materials/wastes consistent with the authorizations referenced in the Hazardous Materials Table in 49 CFR 172, Section .101, Column 8. Bulk and non-bulk packaging must meet the corresponding specifications in 49 CFR 173 referenced in the Hazardous Materials Table, 49 CFR 172, Section .101. Packaging must conform to the general packaging requirements of Subpart B of 49 CFR 173, to the requirements of 49 CFR 178 at the specified packing group performance level, to the requirements of special provisions of column 7 of the Hazardous Materials Table in 49 CFR 172, Section .101, and be compatible with the material to be packaged as required by 40 CFR 262. Also provide other packaging related materials such as materials used to cushion or fill voids in overpacked containers. The hazardous materials being packaged must not react dangerously with, decompose or ignite the sorbent packaging materials. Additionally, sorbents used to treat free liquids to be disposed of in landfills must be non-biodegradable as specified in 40 CFR 264, Section .314. In addition, packaging notifications will be provided to the Government in accordance with 49 CFR 172, Section .178.2(c) regarding type and dimensions of closures, including gaskets, needed to satisfy performance test requirements.

#### 2.1.2 Markings

Provide markings for each hazardous material/waste package, freight container, and transport vehicle consistent with the requirements of 49 CFR 172, Subpart D and 40 CFR 262, Section .32 (for hazardous waste) 40 CFR 761, Section .45 (for PCBs) 40 CFR 61, Section .149(d) (for asbestos). Markings must withstand 180 day exposure to conditions reasonably expected to be encountered during container storage and transportation, without deterioration or substantial color change.

#### 2.1.3 Labeling

Provide primary and subsidiary labels for hazardous materials/wastes consistent with the requirements in the Hazardous Materials Table in

49 CFR 172, Section .101, Column 6. Labels must meet design specifications required by 49 CFR 172, Subpart E including size, shape, color, printing, and symbol requirements. Labels must be durable weather resistant and withstanding a 180 day exposure to conditions reasonably expected to be encountered during container storage and transportation, without deterioration or substantial color change.

#### 2.1.4 Placards

For each offsite shipment of hazardous material/waste, provide primary and subsidiary placards consistent with the requirements of 49 CFR 172, Subpart F. Provide placards for each side and each end of bulk packaging, freight containers, transport vehicles, and rail cars requiring such placarding. Placards may be plastic, metal, or other material capable of withstanding, without deterioration, a 30 day exposure to open weather conditions and must meet design requirements specified in 49 CFR 172, Subpart F.

#### 2.1.5 Spill Response Materials

Provide spill response materials including, but not limited to, containers, adsorbent, shovels, and personal protective equipment. Spill response materials must be available at all times when hazardous materials/wastes are being handled or transported. Spill response materials must be compatible with the type of material being handled.

### 2.2 EQUIPMENT AND TOOLS

Provide miscellaneous equipment and tools necessary to handle hazardous materials and hazardous wastes in a safe and environmentally sound manner.

## PART 3 EXECUTION

### 3.1 HAZARDOUS WASTE MANAGEMENT PLAN

Prepare a Hazardous Waste Management Plan detailing the manner in which hazardous wastes will be managed and describing the types and volumes of hazardous wastes anticipated to be managed. The plan must address both onsite and offsite hazardous waste management. Describe the methods to be used to ensure accurate piece counts or weights of shipments; describe waste minimization methods; identify and describe facilities to be used for treatment, storage, and disposal (TSD); identify areas onsite where hazardous wastes are to be handled; and identify whether transfer facilities are to be used; and if so, how the wastes will be tracked to ultimate disposal. Submit the plan to the Contracting Officer for approval prior to start of work. Submit written documentation of weekly hazardous waste inspections on a monthly basis.

### 3.2 ONSITE HAZARDOUS WASTE MANAGEMENT

Coordinate the onsite management of all hazardous materials and waste with the installation environmental function and the Contracting Officer. These paragraphs apply to Government owned waste only. The Contractor is responsible for ensuring compliance with Federal, state, and local hazardous waste laws and regulations and verifying those requirements when preparing reports, waste shipment records, hazardous waste manifests, or other documents. Identify hazardous wastes using criteria set forth in 40 CFR 261 or applicable state and local laws, regulations, and ordinances. Comply with generator requirements in 40 CFR 262 when

accumulating hazardous waste onsite. Onsite accumulation times must be restricted to applicable time frames referenced in 40 CFR 262, Section .34. Accumulation start dates commence when waste container is transferred into a 90 day accumulation site or permitted storage facility. Only use containers in good condition and compatible with the waste to be stored. Ensure containers are closed except when adding or removing waste, and immediately mark all hazardous waste containers with the words "hazardous waste" and other information required by 40 CFR 262, Section .32 as soon as the waste is containerized. An additional marking must be placed on containers of "unknowns" designating the date sampled, and the suspected hazard. Inspect containers for signs of deterioration and for responding to any spills or leaks. Inspect all hazardous waste areas weekly and provide written documentation of the inspection. Include date and time of inspection, name of individual conducting the inspection, problems noted, and corrective actions taken on the inspection logs.

### 3.2.1 Hazardous Waste Classification

Identify, in consultation with the Contracting Officer, all waste codes applicable to each hazardous waste stream based on requirements in 40 CFR 261 or applicable state or local law or regulation. Also identify applicable treatment standards in 40 CFR 268 and state land disposal restrictions and make a determination as to whether or not the waste meets or exceeds the standards. Submit waste profiles, analyses, classification and treatment standards information to Contracting Officer for review and approval.

### 3.3 OFFSITE HAZARDOUS WASTE MANAGEMENT

Coordinate the off site transfer of all hazardous materials and waste with the installation environmental function and the Contracting Officer. Use RCRA Subtitle C permitted facilities which meet the requirements of 40 CFR 264 or facilities operating under interim status which meet the requirements of 40 CFR 265. Do not use offsite treatment, storage, and disposal facilities with significant RCRA violations or compliance problems (such as facilities known to be releasing hazardous constituents into ground water, surface water, soil, or air). Submit Notices of Non-Compliance and Notices of Violation by a Federal, state, or local regulatory agency issued to the Contractor in relation to any work performed under this contract. Immediately provide copies of such notices to the Contracting Officer. Also furnish relevant documents regarding the incident and any information requested by the Contracting Officer, and coordinate its response to the notice with the Contracting Officer or the designated representative prior to submission to the notifying authority. Also furnish a copy to the Contracting Officer of all documents submitted to the regulatory authority, including the final reply to the notice, and all other materials, until the matter is resolved.

#### 3.3.1 Treatment, Storage, and Disposal Facility and Transporter

Provide the Contracting Officer with EPA ID numbers, names, locations, and telephone numbers of TSD facilities and transporters. This information must be contained in the Hazardous Waste Management Plan and be approved by the Contracting Officer prior to waste disposal.

#### 3.3.2 Facility Status Information

Facilities receiving hazardous waste must be permitted in accordance with 40 CFR 270 or operating under interim status in accordance with 40 CFR 265

requirements, or permitted by a state authorized by the Environmental Protection Agency to administer the RCRA permit program. Additionally, prior to using a TSD Facility, contact the EPA Regional Offsite Coordinator specified in 40 CFR 300, Section .440, to determine the facility's status, and document all information necessary to satisfy the requirements of the EPA Offsite policy and submit this information to the Contracting Officer in the Hazardous Waste Management Plan.

### 3.3.3 Shipping Documents and Packagings Certification

Prior to shipment of any hazardous material offsite and a minimum of 14 days prior to anticipated pickup, provide for review written certification to the Contracting Officer that hazardous materials have been properly packaged, labeled, and marked in accordance with Department of Transportation and EPA requirements. Furnish designated disposal facility packaging assurances not later than 35 days after acceptance of the shipment. The Contractor's TDC must also provide written certification regarding waste minimization efforts documenting that efforts have been taken to reduce the volume and toxicity of waste to the degree economically practicable and that the method of treatment, storage, or disposal selected minimizes threats to human health and the environment.

### 3.3.4 Transportation

Prior to conducting hazardous materials activities, the Contractor responsible for pre-transportation activities must either certify to the Government that a Security Plan is in place which meets the requirements of 49 CFR 172, Subpart I or in the event that the types or amounts of hazardous materials are excluded from the security planning requirements, a written statement to that effect detailing the basis for the exception. Use manifests for transporting hazardous wastes as required by 40 CFR 263 or applicable state or local law or regulation. Transportation must comply with all requirements in the Department of Transportation referenced regulations in the 49 CFR series. Prepare hazardous waste manifests for each shipment of hazardous waste shipped offsite. Complete manifests using instructions in 40 CFR 262, Subpart B and applicable state or local law or regulation. Submit manifests and waste profiles to Contracting Officer for review and approval. Prepare land disposal restriction notifications as required by 40 CFR 268 or applicable state or local law or regulation for each shipment of hazardous waste. Submit notifications with the manifest to the Contracting Officer for review and approval. .

### 3.3.5 Treatment and Disposal of Hazardous Wastes

Coordinate any off site shipments of hazardous materials or hazardous wastes with the installation environmental function. Initial, or satellite hazardous waste accumulation is limited to 55 gallons (or 1 quart of acutely hazardous waste). Once a waste stream exceeds 55 gallons, it must be transferred to an on-site 90 day (180 day small quantity generator) accumulation area, or a permitted hazardous waste treatment, storage or disposal facility within three days. Ship hazardous wastes only to facilities which are properly permitted to accept the hazardous waste or operating under interim status. Ensure wastes are treated to meet land disposal treatment standards in 40 CFR 268 prior to land disposal. Propose TSD facilities via submission of the Hazardous Waste Management Plan, subject to the approval of the Contracting Officer. Submit Certificates of Disposal documenting the ultimate disposal, destruction or placement of hazardous wastes, polychlorinated

biphenyls (PCBs), and asbestos within 180 days of initial shipment. Receipt of these certificates will be required for final payment.

### 3.4 RADIOACTIVE MATERIALS MANAGEMENT

Consult with the Contracting Officer, to evaluate, prior to shipment of any material offsite, whether the material is regulated as a hazardous waste in addition to being regulated as a radioactive material. Perform the evaluation to determine proper shipping descriptions, marking requirements, and other criteria, as described below.

#### 3.4.1 Identification of Proper Shipping Names

Use 49 CFR 172, Section .101 to identify proper shipping names for each hazardous material (including hazardous wastes) to be shipped offsite. Submit proper shipping names to the Contracting Officer in the form of draft shipping documents for review and approval.

#### 3.4.2 Packaging, Labeling, and Marking

Package, label, and mark hazardous materials/wastes using the specified materials and in accordance with the referenced authorizations. Mark each container of hazardous waste of 110 gallons or less with the following:

"HAZARDOUS WASTE - Federal Law Prohibits Improper Disposal.

If found, contact the nearest police or public safety authority or the U.S. Environmental Protection Agency.

Generator's name \_\_\_\_\_  
Manifest Document Number \_\_\_\_\_".

#### 3.4.3 Shipping Documents

Ensure that each shipment of hazardous material sent offsite is accompanied by properly completed shipping documents. This includes shipments of samples that may potentially meet the definition of a Department of Transportation regulated hazardous material.

##### 3.4.3.1 PCB Waste Shipment Documents

Prepare hazardous waste manifests for each shipment of PCB waste shipped offsite. Complete manifests using instructions in 40 CFR 761, Sections .207 and .208 and other applicable requirements. Submit documents to Contracting Officer for review and approval.

##### 3.4.3.2 Asbestos Waste Shipment Documents

Prepare waste shipment records, as required by 40 CFR 61, for shipments of asbestos. Submit waste shipment records to the Contracting Officer for review and approval. Waste shipment records must be signed by the Contractor.

##### 3.4.3.3 Other Hazardous Material Shipment Documents

Prepare a bill of lading for each shipment of hazardous material which is not accompanied by a hazardous waste manifest or asbestos waste shipment record which fulfills the shipping paper requirements. The bill of lading must satisfy the requirements of 49 CFR 172, Subpart C, and 40 CFR 279 if shipping used oil and applicable state or local law or regulation, and must be submitted to the Contracting Officer for review and approval. For

laboratory samples and treatability study samples, prepare bills of lading and other documentation as necessary to satisfy conditions of the sample exclusions in 40 CFR 261, Section .4(d) and (e) and any applicable state or local law or regulation. Bill of ladings requiring shipper's certifications must be signed by the Contractor.

### 3.5 SPECIAL REQUIREMENTS FOR ASBESTOS WASTES

If work involves asbestos containing wastes, manage these wastes in accordance with specification Section 02 82 00.00 22 ASBESTOS REMEDIATION.

### 3.6 WASTE MINIMIZATION

Minimize the generation of hazardous waste to the maximum extent practicable and take all necessary precautions to avoid mixing clean and contaminated wastes. Identify and evaluate recycling and reclamation options as alternatives to land disposal. Requirements of 40 CFR 266 apply to: hazardous wastes recycled in a manner constituting disposal; hazardous waste burned for energy recovery; lead-acid battery recycling; and hazardous wastes with economically recoverable precious metals. Submit written certification that waste minimization efforts have been undertaken to reduce the volume and toxicity of waste to the degree economically practicable and that the method of treatment, storage, or disposal selected minimizes threats to human health and the environment.

### 3.7 RECORDKEEPING

Maintain adequate records to support information provided to the Contracting Officer regarding exception reports, annual reports, and biennial reports; maintain asbestos waste shipment records for a minimum of 3 years from the date of shipment or any longer period required by applicable law or regulation or other provision of this contract; and maintain bill of ladings for a minimum of 375 days from the date of shipment or longer period required by applicable law or regulation or other provision of this contract. Submit information necessary to file state annual or EPA biennial reports for hazardous waste transported, treated, stored, or disposed of under this contract. Do not forward these data directly to the regulatory agency but to the Contracting Officer at the specified time. Submit the information necessary for filing of the formal reports in the form and format required by the governing Federal or state regulatory agency. A cover letter must accompany the data to include the contract number, Contractor name, and project location. In the events that a manifest copy documenting receipt of hazardous waste at the treatment storage and disposal facility is not received within 35 days of shipment initiation, or that a manifest copy documenting receipt of PCB waste at the designated facility is not received within 35 days of shipment initiation, prepare and submit an exception report to the Contracting Officer within 37 days of shipment initiation.

### 3.8 SPILL RESPONSE

In the event of a spill or release of a hazardous substance (as designated in 40 CFR 302), or pollutant or contaminant, or oil (as governed by the Oil Pollution Act (OPA), 33 U.S.C. 2701 et seq.), notify the Contracting Officer immediately. Direction from the Contracting Officer concerning a spill or release is not considered a change under the contract. If the spill exceeds a reporting threshold, follow the pre-established procedures for immediate reporting to the Contracting Officer. Comply with applicable requirements of Federal, state, or local laws or regulations



regarding any spill incident.

### 3.9 EMERGENCY CONTACTS

Comply with the emergency contact provisions in 49 CFR 172, Section .604. Whenever the Contractor ships hazardous materials, provide a 24 hr emergency response contact and phone number of a person knowledgeable about the hazardous materials being shipped and who has comprehensive emergency response and incident mitigation information for that material, or has immediate access to a person who possesses such knowledge and information. Monitor the phone on a 24 hour basis at all times when the hazardous materials are in transportation, including during storage incidental to transportation. Ensure that information regarding this emergency contact and phone number are placed on all hazardous material shipping documents. Designate an emergency coordinator and post the following information at areas in which hazardous wastes are managed:

- a. The name of the emergency coordinator.
- b. Phone number through which the emergency coordinator can be contacted on a 24 hour basis.
- c. The telephone number of the local fire department.
- d. The location of fire extinguishers and spill control materials.

| Attachment A<br>SAMPLE OFF-SITE POLICY CERTIFICATION MEMO                                                                                                                       |                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| Project/Contract #:                                                                                                                                                             |                              |
| Waste Stream:                                                                                                                                                                   |                              |
| Primary TSD Facility,<br>EPA ID # and Location:                                                                                                                                 |                              |
| Alter. TSD Facility,<br>EPA ID # and Location:                                                                                                                                  |                              |
| EPA Region                                                                                                                                                                      | Contact                      |
| I                                                                                                                                                                               | 888-372-7341                 |
| II                                                                                                                                                                              | 212-673-4040                 |
| III                                                                                                                                                                             | 800-438-2474 or 215-814-5000 |
| IV                                                                                                                                                                              | 800-241-1754 or 404-562-9900 |
| V                                                                                                                                                                               | 312-353-2000                 |
| VI                                                                                                                                                                              | 800-887-6063 or 214-665-2210 |
| VII                                                                                                                                                                             | 800-223-0425                 |
| VIII                                                                                                                                                                            | 800-424-8802                 |
| IX                                                                                                                                                                              | 415-947-8713                 |
| X                                                                                                                                                                               | 800-424-4372 or 206-553-4973 |
| EPA representative<br>contacted:                                                                                                                                                |                              |
| EPA representative<br>phone number:                                                                                                                                             |                              |
| Date contacted:                                                                                                                                                                 |                              |
| Comment:                                                                                                                                                                        |                              |
| The above EPA representative was contacted on _____. As of that date<br>the above sites were considered acceptable in accordance with the Off-Site<br>Policy in 40 CFR 300.440. |                              |
| Date:                                                                                                                                                                           | Signature:                   |
| Phone number:                                                                                                                                                                   |                              |

-- End of Section --

SECTION 02 82 00.00 22

ASBESTOS REMEDIATION  
**(PNS PROJECTS)**  
09/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP Z9.2 (2018) Fundamentals Governing the Design and Operation of Local Exhaust Ventilation Systems

ASTM INTERNATIONAL (ASTM)

ASTM D4397 (2016) Standard Specification for Polyethylene Sheeting for Construction, Industrial, and Agricultural Applications

ASTM E1368 (2014) Visual Inspection of Asbestos Abatement Projects

COMPRESSED GAS ASSOCIATION (CGA)

CGA G-7 (2014) Compressed Air for Human Respiration; 6th Edition

INTERNATIONAL SAFETY EQUIPMENT ASSOCIATION (ISEA)

ANSI/ISEA Z87.1 (2015) Occupational and Educational Personal Eye and Face Protection Devices

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 701 (2023; ERTA 1 2023) Standard Methods of Fire Tests for Flame Propagation of Textiles and Films

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH)

NIOSH NMAM (2016; 5th Ed) NIOSH Manual of Analytical Methods

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety -- Safety and Occupational Health (SOH) Requirements

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA 340/1-90/018 (1990) Asbestos/NESHAP Regulated Asbestos

Containing Materials Guidance

EPA 560/5-85-024

(1985) Guidance for Controlling  
Asbestos-Containing Materials in Buildings  
(Purple Book)

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|                     |                                                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 29 CFR 1910.147     | The Control of Hazardous Energy (Lock Out/Tag Out)                                                                                           |
| 29 CFR 1926.103     | Respiratory Protection                                                                                                                       |
| 29 CFR 1926.1101    | Asbestos                                                                                                                                     |
| 29 CFR 1926.200     | Accident Prevention Signs and Tags                                                                                                           |
| 29 CFR 1926.51      | Sanitation                                                                                                                                   |
| 29 CFR 1926.59      | Hazard Communication                                                                                                                         |
| 40 CFR 61-SUBPART A | General Provisions                                                                                                                           |
| 40 CFR 61-SUBPART M | National Emission Standard for Asbestos                                                                                                      |
| 40 CFR 763          | Asbestos                                                                                                                                     |
| 42 CFR 84           | Approval of Respiratory Protective Devices                                                                                                   |
| 49 CFR 107          | Hazardous Materials Program Procedures                                                                                                       |
| 49 CFR 171          | General Information, Regulations, and Definitions                                                                                            |
| 49 CFR 172          | Hazardous Materials Table, Special Provisions, Hazardous Materials Communications, Emergency Response Information, and Training Requirements |

U.S. NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)

|                      |                                                                           |
|----------------------|---------------------------------------------------------------------------|
| NAVFAC P-502         | (2017) Asbestos Program Management                                        |
| ND OPNAVINST 5100.23 | (2005; Rev G) Navy Occupational Safety and Health (NAVOSH) Program Manual |

UNDERWRITERS LABORATORIES (UL)

|        |                                                                                               |
|--------|-----------------------------------------------------------------------------------------------|
| UL 586 | (2009; Reprint Sep 2022) UL Standard for Safety High-Efficiency Particulate, Air Filter Units |
|--------|-----------------------------------------------------------------------------------------------|

STATE OF MAINE, DEPARTMENT OF ENVIRONMENTAL PROTECTION (DEP)

|                |                                                                                                                                                                                     |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ME L&C PROGRAM | State of Maine Licensing and Certification Program<br><a href="http://www.mainelegislature.org/legis/statutes/38/tit13">http://www.mainelegislature.org/legis/statutes/38/tit13</a> |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

06 096 CMR

425 Asbestos Management Regulations  
<http://www.maine.gov/sos/cec/rules/06/096/096c425.doc>

## 1.2 DEFINITIONS

### 1.2.1 ACM

Asbestos Containing Materials.

### 1.2.2 Amended Water

Water containing a wetting agent or surfactant with a maximum surface tension of 0.00042 psi.

### 1.2.3 Area Sampling

Sampling of asbestos fiber concentrations which approximates the concentrations of asbestos in the theoretical breathing zone but is not actually collected in the breathing zone of an employee.

### 1.2.4 Asbestos

The term asbestos includes chrysotile, amosite, crocidolite, tremolite asbestos, anthophyllite asbestos, and actinolite asbestos and any of these minerals that has been chemically treated or altered. Materials are considered to contain asbestos if the asbestos content of the material is determined to be equal to or greater than one percent.

### 1.2.5 Asbestos Control Area

That area where asbestos removal operations are performed which is isolated by physical boundaries which assist in the prevention of the uncontrolled release of asbestos dust, fibers, or debris.

### 1.2.6 Asbestos Fibers

Those fibers having an aspect ratio of at least 3:1 and longer than 5 micrometers as determined by National Institute for Occupational Safety and Health (NIOSH) Method 7400.

### 1.2.7 Asbestos Permissible Exposure Limit

0.1 fibers per cubic centimeter of air as an 8-hour time weighted average measured in the breathing zone as defined by 29 CFR 1926.1101 or other Federal legislation having legal jurisdiction for the protection of workers health.

### 1.2.8 Authorized Person

Any person authorized by the Contractor and required by work duties to be present in the regulated areas.

### 1.2.9 Background

The ambient airborne asbestos concentration in an uncontaminated area as measured prior to any asbestos hazard abatement efforts. Background concentrations for other (contaminated) areas are measured in similar but asbestos free locations.

#### 1.2.10 Competent Person (CP)

A person meeting the requirements for competent person as specified in 29 CFR 1926.1101 including a person capable of identifying existing asbestos hazards in the workplace and selecting the appropriate control strategy for asbestos exposure, who has the authority to take prompt corrective measures to eliminate them, and is specifically trained in a training course which meet the criteria of EPA's Model Accreditation Plan (40 CFR 763) for project designer or supervisor, or its equivalent. The competent person must have a current State of Maine asbestos contractor's or supervisor's license.

#### 1.2.11 Contractor

The Contractor is that individual, or entity under contract to perform the herein listed work.

#### 1.2.12 Disposal Bag

A 6 mil thick, leak-tight plastic bag, pre-labeled in accordance with 29 CFR 1926.1101, used for transporting asbestos waste from containment to disposal site.

#### 1.2.13 Disturbance

Activities that disrupt the matrix of ACM, crumble or pulverize ACM, or generate visible debris from ACM. Disturbance includes cutting away small amounts of ACM, no greater than the amount which can be contained in one standard sized glovebag or waste bag, not larger than 60 inches in length and width in order to access a building component.

#### 1.2.14 Friable Asbestos Material

A term defined in 40 CFR 61-SUBPART M and EPA 340/1-90/018 meaning any material which contains equal to or greater than 1 percent asbestos, as determined using the method specified in 40 CFR 763, Polarized Light Microscopy (PLM), that when dry, can be crumbled, pulverized, or reduced to powder by hand pressure.

#### 1.2.15 Glovebag Technique

Those asbestos removal and control techniques put forth in 29 CFR 1926.1101.

#### 1.2.16 Government Consultant (GC)

That qualified person employed directly by the Government to monitor, sample, inspect the work or in some other way advise the Contracting Officer. The GC is normally a private consultant, but can be an employee of the Government.

#### 1.2.17 HEPA Filter Equipment

High efficiency particulate air (HEPA) filtered vacuum and exhaust ventilation equipment with a filter system capable of collecting and retaining asbestos fibers. Filters must retain 99.97 percent of particles 0.3 microns or larger as indicated in UL 586.

1.2.18 Model Accreditation Plan (MAP)

USEPA training accreditation requirements for persons who work with asbestos as specified in 40 CFR 763.

1.2.19 Negative Pressure Enclosure (NPE)

That engineering control technique described as a negative pressure enclosure in 29 CFR 1926.1101.

1.2.20 NESHAP

National Emission Standards for Hazardous Air Pollutants. The USEPA NESHAP regulation for asbestos is at 40 CFR 61-SUBPART M.

1.2.21 Nonfriable Asbestos Material

Material that contains asbestos in which the fibers have been immobilized by a bonding agent, coating, binder, or other material so that the asbestos is well bound and will not normally release asbestos fibers during any appropriate use, handling, storage or transportation. It is understood that asbestos fibers may be released under other conditions such as demolition, removal, or mishap.

1.2.22 Permissible Exposure Limits (PELs)

1.2.22.1 PEL-Time Weighted Average (TWA)

Concentration of asbestos not in excess of 0.1 fibers per cubic centimeter of air (f/cc) as an 8-hour time weighted average (TWA).

1.2.22.2 PEL-Excursion Limit

An airborne concentration of asbestos not in excess of 1.0 f/cc of air as averaged over a sampling period of 30 minutes.

1.2.23 Personal Sampling

Air sampling which is performed to determine asbestos fiber concentrations within the breathing zone of a specific employee, as performed in accordance with 29 CFR 1926.1101.

1.2.24 Private Qualified Person (PQP)

That qualified person hired by the Contractor to perform the herein listed tasks.

1.2.25 Qualified Person (QP)

A Registered Architect, Professional Engineer, Certified Industrial Hygienist, consultant or other qualified person who has successfully completed training and is therefore accredited under a legitimate State Model Accreditation Plan as described in 40 CFR 763 as a Building Inspector, Contractor/Supervisor Abatement Worker, and Asbestos Project Designer; and has successfully completed the National Institute of Occupational Safety and Health (NIOSH) 582 course "Sampling and Evaluating Airborne Asbestos Dust" or equivalent. The QP must be qualified to perform visual inspections as indicated in ASTM E1368. The QP must be appropriately licensed in the State of Maine.

#### 1.2.26 TEM

Refers to Transmission Electron Microscopy.

#### 1.2.27 Time Weighted Average (TWA)

The TWA is an 8-hour time weighted average airborne concentration of asbestos fibers.

#### 1.2.28 Transite

A generic name for asbestos cement wallboard and pipe.

#### 1.2.29 Wetting Agent

A chemical added to water to reduce the water's surface tension thereby increasing the water's ability to soak into the material to which it is applied. An equivalent wetting agent must have a surface tension of at most 0.00042 psi.

#### 1.2.30 Worker

Individual (not designated as the Competent Person or a supervisor) who performs asbestos work and has completed asbestos worker training required by 29 CFR 1926.1101, to include EPA Model Accreditation Plan (MAP) "Worker" training; accreditation, if required by the OSHA Class of work to be performed or by the state where the work is to be performed. The worker must be appropriately licensed in the State of Maine.

### 1.3 REQUIREMENTS

#### 1.3.1 Description of Work

The work covered by this Section includes the handling and control of asbestos containing materials and describes some of the resultant procedures and equipment required to protect workers, the environment and occupants of the building or area, or both, from contact with airborne asbestos fibers. The work also includes the disposal of any asbestos containing materials generated by the work. More specific operational procedures must be outlined in the Asbestos Hazard Abatement Plan called for elsewhere in this specification. The asbestos work includes the demolition and removal and as indicated on the Drawings, which is governed by MEDEP Chapter 425: Asbestos Management Regulations, 40 CFR 763, NAVFAC P-502, and other applicable Federal, State, and local requirements. Refer to the report located at the end of this Section. Under normal conditions non-friable or chemically bound materials containing asbestos would not be considered hazardous; however, this material may release airborne asbestos fibers during disturbance, demolition, and/or removal and therefore must be handled in accordance with the removal and disposal procedures as specified herein. Provide negative pressure enclosure and glovebag techniques as outlined in this specification. The building(s) work area(s) will be evacuated during the asbestos abatement work. A competent person must supervise asbestos removal work as specified herein.

##### 1.3.1.1 Wallboard/Joint Compound

Both composite samples of the wallboard and discrete samples of the components (wallboard and joint compound) have been tested and results are



attached.

#### 1.3.2 Unexpected Discovery of Asbestos

Notify the Contracting Officer if any previously untested building components suspected to contain asbestos are impacted by the work.

#### 1.3.3 Medical Requirements

Provide medical requirements including but not limited to medical surveillance and medical record keeping as listed in 29 CFR 1926.1101.

##### 1.3.3.1 Medical Examinations

Before exposure to airborne asbestos fibers, provide workers with a comprehensive medical examination as required by 29 CFR 1926.1101 or other pertinent State or local directives. This requirement must have been satisfied within the 12 months prior to the start of work on this Contract. The same medical examination must be given on an annual basis to employees engaged in an occupation involving asbestos and within 30 calendar days before or after the termination of employment in such occupation. Specifically identify x-ray films of asbestos workers to the consulting radiologist and mark medical record jackets with the word "ASBESTOS."

##### 1.3.3.2 Medical Records

Maintain complete and accurate records of employees' medical examinations, medical records, and exposure data for a period of indefinite time after termination of employment and make records of the required medical examinations and exposure data available for inspection and copying to: The Assistant Secretary of Labor for Occupational Safety and Health (OSHA), or authorized representatives of them, and an employee's physician upon the request of the employee or former employee.

#### 1.3.4 Employee Training

Submit certificates, prior to the start of work but after the main abatement submittal, signed by each employee indicating that the employee has received training in the proper handling of materials and wastes that contain asbestos in accordance with 40 CFR 763 and 49 CFR 172; understands the health implications and risks involved, including the illnesses possible from exposure to airborne asbestos fibers; understands the use and limits of the respiratory equipment to be used; and understands the results of monitoring of airborne quantities of asbestos as related to health and respiratory equipment as indicated in 29 CFR 1926.1101 on an initial and annual basis. Organize certificates by individual worker, not grouped by type of certification. Post appropriate evidence of compliance with the training requirements of 40 CFR 763. Train personnel involved in the asbestos control work in accordance with United States Environmental Protection Agency (USEPA) Asbestos Hazard Emergency Response Act (AHERA) training criteria or State training criteria whichever is more stringent. Document the training by providing: dates of training, training entity, course outline, names of instructors, and qualifications of instructors upon request by the Contracting Officer. Furnish each employee with respirator training and fit testing administered by the PQP as required by 29 CFR 1926.1101 and 29 CFR 1926.103. Fully cover engineering and other hazard control techniques and procedures. Asbestos workers must have a current State of Maine asbestos worker's license. Comply with the

ME L&C PROGRAM.

#### 1.3.5 Permits, Licenses and Notifications

Prior to the start of work, obtain necessary permits and licenses in conjunction with asbestos removal, encapsulation, hauling, and disposition, and furnish notification of such actions required by Federal, State, regional, and local authorities. Notify the Maine Department of Environmental Protection, the NAVFAC Asbestos Program Manager, and the Contracting Officer in writing 20 working days prior to commencement of work in accordance with MEDEP Chapter 425: Asbestos Management Regulations and 40 CFR 61-SUBPART M and for approval by NAVFAC NPT. Notify the Contracting Officer and other appropriate Government agencies in writing 20 working days prior to the start of asbestos work as indicated in applicable laws, ordinances, criteria, rules, and regulations. Submit copies of all Notifications to the Contracting Officer. Notify the Shipyard fire department 3 days prior to removing fire-proofing material from the building including notice that the material contains asbestos.

#### 1.3.6 Environment, Safety and Health Compliance

In addition to detailed requirements of this specification, comply with those applicable laws, ordinances, criteria, rules, and regulations of Federal, State, regional, and local authorities regarding handling, storing, transporting, and disposing of asbestos waste materials. Comply with the applicable requirements of the current issue of EM 385-1-1, 29 CFR 1926.1101, 40 CFR 61-SUBPART A, 40 CFR 61-SUBPART M, 40 CFR 763, 49 CFR 171, and ND OPNAVINST 5100.23. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting the work. Where the requirements of this specification, applicable laws, rules, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirement as defined by the Government apply. The following laws, ordinances, criteria, rules and regulations regarding removal, handling, storing, transporting and disposing of asbestos materials apply:

- a. U.S. EPA, Title 40, CFR Part 61, National Standards for Hazardous Air Pollutants (NESHAP), Subparts A and M.
- b. OSHA, 29 CFR, Part 1926.1101, Asbestos in Construction.
- c. State of Maine, Department of Environmental Protection (ME DEP), 06 096 CMR Asbestos Management Regulations, Chapter 425, effective date 4-3-2011.

#### 1.3.7 Respiratory Protection Program

Establish and implement a respirator program as required by 29 CFR 1926.1101, and 29 CFR 1926.103. Submit a written description of the program to the Contracting Officer. Submit a written program manual or operating procedure including methods of compliance with regulatory statutes.

##### 1.3.7.1 Respirator Program Records

Submit records of the respirator program as required by 29 CFR 1926.103, and 29 CFR 1926.1101.

#### 1.3.7.2 Respirator Fit Testing

The Contractor's PQP must conduct a qualitative or quantitative fit test conforming to 29 CFR 1926.103 for each worker required to wear a respirator, and any authorized visitors who enter a regulated area where respirators are required to be worn. A respirator fit test must be performed prior to initially wearing a respirator and every 12 months thereafter. If physical changes develop that will affect the fit, a new fit test must be performed. Functional fit checks must be performed each time a respirator is put on and in accordance with the manufacturer's recommendation.

#### 1.3.7.3 Respirator Selection and Use Requirements

Provide respirators, and ensure that they are used as required by 29 CFR 1926.1101 and in accordance with CGA G-7 and the manufacturer's recommendations. Respirators must be approved by the National Institute for Occupational Safety and Health NIOSH, under the provisions of 42 CFR 84, for use in environments containing airborne asbestos fibers. For air-purifying respirators, the particulate filter must be high-efficiency particulate air (HEPA)/(N-,R-,P-100). The initial respirator selection and the decisions regarding the upgrading or downgrading of respirator type must be made by the Contractor's Designated IH based on the measured or anticipated airborne asbestos fiber concentrations to be encountered.

#### 1.3.8 Asbestos Hazard Control Supervisor

The Contractor must be represented on site by a supervisor, trained using the model Contractor accreditation plan as indicated in the Federal statutes for all portions of the herein listed work.

#### 1.3.9 Hazard Communication

Adhere to all parts of 29 CFR 1926.59 and provide the Contracting Officer with a copy of the Safety Data Sheets (SDS) for all materials brought to the site.

#### 1.3.10 Asbestos Hazard Abatement Plan

Submit a detailed plan of the safety precautions such as lockout, tagout, tryout, fall protection, and confined space entry procedures and equipment and work procedures to be used in the removal and demolition of materials containing asbestos. The plan, not to be combined with other hazard abatement plans, must be prepared, signed, and sealed by the PQP. Provide a Table of Contents for each abatement submittal, which follows the sequence of requirements in the Contract. The plan must include but not be limited to the precise personal protective equipment to be used including, but not limited to, respiratory protection, type of whole-body protection and if reusable coveralls are to be employed decontamination methods (operations and quality control plan), the location of asbestos control areas including clean and dirty areas, buffer zones, showers, storage areas, change rooms, removal method, interface of trades involved in the construction, sequencing of asbestos related work, disposal plan, type of wetting agent and asbestos sealer to be used, locations of local exhaust equipment, planned air monitoring strategies, and a detailed description of the method to be employed in order to control environmental pollution. The plan must also include (both fire and medical emergency) response plans and an Activity Hazard Analyses (AHAs) in accordance with EM 385-1-1. The Asbestos Hazard Abatement Plan must be approved in

writing prior to starting any asbestos work. The Contractor, Asbestos Hazard Control Supervisor, CP, and PQP must meet with the Contracting Officer prior to beginning work, to discuss in detail the Asbestos Hazard Abatement Plan, including work procedures and safety precautions. Once approved by the Contracting Officer, the plan will be enforced as if an addition to the specification. Any changes required in the specification as a result of the plan must be identified specifically in the plan to allow for free discussion and approval by the Contracting Officer prior to starting work.

The Plan must be accompanied by a completed Asbestos Hazard Abatement Plan Checklist included as Attachment A at the end of this specification and referenced in Paragraph entitled SUBMITTALS herein. The Checklist must be prepared and signed by the PQP.

#### 1.3.11 Testing Laboratory

Submit the name, address, and telephone number of each testing laboratory selected for the sampling, analysis, and reporting of airborne concentrations of asbestos fibers along with evidence that each laboratory selected holds the appropriate State license and permits and certification that each laboratory is American Industrial Hygiene Association (AIHA) accredited and that persons counting the samples have been judged proficient by current inclusion on the AIHA Asbestos Analysis Registry (AAR) and successful participation of the laboratory in the Proficiency Analytical Testing (PAT) Program. Where analysis to determine asbestos content in bulk materials or transmission electron microscopy is required, submit evidence that the laboratory is accredited by the National Institute of Science and Technology (NIST) under National Voluntary Laboratory Accreditation Program (NVLAP) for asbestos analysis. The testing laboratory firm must be independent of the asbestos contractor and must have no employee or employer relationship which could constitute a conflict of interest.

#### 1.3.12 Landfill Approval

Landfill approval is not required as Portsmouth Naval Shipyard takes responsibility for disposal of all asbestos waste generated on the Shipyard.

#### 1.3.13 Transporter Certification

Transporter certification is not required as Portsmouth Naval Shipyard takes responsibility for transport and disposal of all asbestos waste generated on the Shipyard.

#### 1.3.14 Medical Certification

Provide a written certification for each worker and supervisor, signed by a licensed physician indicating that the worker and supervisor has met or exceeded all of the medical prerequisites listed herein and in 29 CFR 1926.1101 and 29 CFR 1926.103 as prescribed by law. Submit certificates prior to the start of work but after the main abatement submittal.

#### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control

approval. Submit the following in accordance with Section 01 33 00  
SUBMITTAL PROCEDURES:

SD-03 Product Data

Amended Water; G

Safety Data Sheets (SDS) for All Materials; G

Respirators; G

Local Exhaust Equipment; G

Pressure Differential Automatic Recording Instrument; G

Vacuums; G

Glovebags; G

SD-06 Test Reports

Air Sampling Results; G

Pressure Differential Recordings for Local Exhaust System; G

Clearance Sampling; G

Asbestos Disposal Quantity Report; G

SD-07 Certificates

Employee Training; G

Notifications; G

Respiratory Protection Program; G

Asbestos Hazard Abatement Plan; G

Asbestos Hazard Abatement Plan Checklist; G

Testing Laboratory; G

Medical Certification; G

Private Qualified Person Documentation; G

Competent Person; G

Worker's License; G

Contractor's License; G

Federal, State or Local Citations on Previous Projects; G

Equipment Used to Contain Airborne Asbestos Fibers; G

Water Filtration Equipment; G

Vacuums; G

Ventilation Systems; G

#### SD-11 Closeout Submittals

Permits and Licenses; G

Notifications; G

Respirator Program Records; G

Rental Equipment; G

### 1.5 QUALITY ASSURANCE

#### 1.5.1 Private Qualified Person Documentation

Submit the name, address, and telephone number of the Private Qualified Person (PQP) selected to prepare the Asbestos Hazard Abatement Plan, direct monitoring and training, and documented evidence that the PQP has successfully completed training in and is accredited and where required is certified as, a Building Inspector, Contractor/Supervisor Abatement Worker, and Asbestos Project Designer as described by 40 CFR 763 and has successfully completed the National Institute of Occupational Safety and Health (NIOSH) 582 course "Sampling and Evaluating Airborne Asbestos Dust" or equivalent. The PQP must be appropriately licensed in the State of Maine as a Project Monitor. The PQP and the asbestos contractor must not have an employee/employer relationship or financial relationship which could constitute a conflict of interest. The PQP must be a first tier subcontractor.

#### 1.5.2 Competent Person Documentation

The Competent Person must be experienced in the administration and supervision of asbestos abatement projects including exposure assessment and monitoring, work practices, abatement methods, protective measures for personnel, setting up and inspecting asbestos abatement work areas, evaluating the integrity of containment barriers, placement and operation of local exhaust systems, ACM generated waste containment and disposal procedures, decontamination units installation and maintenance requirements, site safety and health requirements, and notification of other employees onsite. The Competent Person must be on-site at all times when asbestos abatement activities are underway. Submit training certification and a current State of Maine Asbestos Contractor's and Supervisor's License. Submit evidence that the Competent Person has a minimum of 2 years of on-the-job asbestos abatement experience relevant to OSHA competent person requirements.

#### 1.5.3 Worker's License

Submit documentation that workers meet the requirements of 29 CFR 1926.1101, 40 CFR 61-SUBPART M and have a current State of Maine Asbestos Workers License.

#### 1.5.4 Contractor's License

Submit a copy of the asbestos contractor's license issued by the State of Maine. Submit the following certification along with the license: "I certify that the personnel I am responsible for during the course of this project fully understand the contents of 29 CFR 1926.1101, 40 CFR 61-SUBPART M, EM 385-1-1, and the Federal, State and local requirements for those asbestos abatement activities that they will be involved in." This certification statement must be signed by the Company's President or Chief Executive.

#### 1.5.5 Air Sampling Results

Complete fiber counting and provide results to the PQP and GC for review within 16 hours of the "time off" of the sample pump. Notify the Contracting Officer and Asbestos Program Manager immediately of any airborne levels of asbestos fibers in excess of the acceptable limits. Submit sampling results to the Contracting Officer and Asbestos Program Manager and the affected Contractor employees where required by law within three working days, signed by the testing laboratory employee performing air sampling, the employee that analyzed the sample, and the PQP and GC. Notify the Contractor, the Contracting Officer and Asbestos Program Manager immediately of any variance in the pressure differential which could cause adjacent unsealed areas to have asbestos fiber concentrations in excess of 0.010 fibers per cubic centimeter or background whichever is higher. In no circumstance must levels exceed 0.10 fibers per cubic centimeter. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

#### 1.5.6 Pressure Differential Recordings for Local Exhaust System

Provide a local exhaust system that creates a negative pressure of at least 0.02 inches of water relative to the pressure external to the enclosure and operate it continuously, 24-hours a day, until the temporary enclosure of the asbestos control area is removed. Submit pressure differential recordings for each work day to the PQP and GC for review and to the Contracting Officer and Asbestos Program Manager within 24-hours from the end of each work day. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

#### 1.5.7 Federal, State or Local Citations on Previous Projects

Submit a statement, signed by an officer of the company, containing a record of any citations issued by Federal, State or local regulatory agencies relating to asbestos activities within the last 5 years (including projects, dates, and resolutions); a list of penalties incurred through non-compliance with asbestos project specifications, including liquidated damages, overruns in scheduled time limitations and resolutions; and situations in which an asbestos-related contract has been terminated (including projects, dates, and reasons for terminations). If there are none, a negative declaration signed by an officer of the company must be provided.

#### 1.5.8 Preconstruction Conference

Conduct a safety preconstruction conference to discuss the details of the Asbestos Hazard Abatement Plan, Accident Prevention Plan (APP) including

the AHAs required in specification Section 01 35 26.00 22 GOVERNMENTAL SAFETY REQUIREMENTS (PWD ME). The safety preconstruction conference must include the Contractor and their Designated Competent Person, NAVFAC Asbestos Program Manager, Designated IH and Project Supervisor and the Contracting Officer. Deficiencies in the APP will be discussed. Onsite work must not begin until the APP has been accepted.

## 1.6 SECURITY

A log book must be kept documenting entry into and out of the regulated area. Entry into regulated areas must only be by personnel authorized by the Contractor and the Contracting Officer. Personnel authorized to enter regulated areas must be trained, medically evaluated, and wear the required personal protective equipment.

## 1.7 EQUIPMENT

### 1.7.1 Rental Equipment

Provide a copy of the written notification to the rental company concerning the intended use of the equipment and the possibility of asbestos contamination of the equipment.

## PART 2 PRODUCTS

### 2.1 DUCT TAPE

Industrial grade duct tape of appropriate widths suitable for bonding sheet plastic and disposal container.

### 2.2 DISPOSAL CONTAINERS

Leak-tight (defined as solids, liquids, or dust that cannot escape or spill out) disposal containers must be provided for ACM wastes as required by 29 CFR 1926.1101. Disposal containers can be in the form of:

- a. Disposal Bags

### 2.3 SHEET PLASTIC

Sheet plastic must be polyethylene of 6 mil minimum thickness and must be provided in the largest sheet size necessary to minimize seams. Film must be clear and conform to ASTM D4397, except as specified below.

#### 2.3.1 Flame Resistant

Where a potential for fire exists, flame-resistant sheets must be provided. Film must be frosted and must conform to the requirements of NFPA 701.

#### 2.3.2 Reinforced

Reinforced sheets must be provided where high skin strength is required, such as where it constitutes the only barrier between the regulated area and the outdoor environment. The sheet stock must consist of translucent, nylon-reinforced or woven-polyethylene thread laminated between 2 layers of polyethylene film. Film must meet flame resistant standards of NFPA 701.



## 2.4 MASTIC REMOVING SOLVENT

Mastic removing solvent must be nonflammable and must not contain methylene chloride, glycol ether, or halogenated hydrocarbons. Solvents used onsite must have a flash point greater than 140 degrees F.

## 2.5 LEAK-TIGHT WRAPPING

At least two layers of 6 mil minimum thick polyethylene sheet stock must be used for the containment of removed asbestos-containing components or materials such as large tanks, boilers, insulated pipe segments and other materials. Upon placement of the ACM component or material, each layer must be individually leak-tight sealed with duct tape.

## 2.6 VIEWING INSPECTION WINDOW

Where feasible, a minimum of one clear, 1/8 inch thick, acrylic sheet, 18 by 24 inches, must be installed as a viewing inspection window at eye level on a wall in each containment enclosure. The windows must be sealed leak-tight with industrial grade duct tape.

## 2.7 WETTING AGENTS

Removal encapsulant (a penetrating encapsulant) must be provided when conducting removal abatement activities that require a longer removal time or are subject to rapid evaporation of amended water. The removal encapsulant must be capable of wetting the ACM and retarding fiber release during disturbance of the ACM greater than or equal to that provided by amended water. Performance requirements for penetrating encapsulants are specified in paragraph ENCAPSULANTS above.

# PART 3 EXECUTION

## 3.1 EQUIPMENT

Provide the Contracting Officer or the Contracting Officer's Representative, with at least two complete sets of personal protective equipment as required for entry to and inspection of the asbestos control area. Provide equivalent training to the Contracting Officer or a designated representative as provided to Contractor employees in the use of the required personal protective equipment. Provide manufacturer's certificate of compliance for all equipment used to contain airborne asbestos fibers.

### 3.1.1 Air Monitoring Equipment

The Contractor's PQP must approve air monitoring equipment. The equipment must include, but must not be limited to:

- a. High-volume sampling pumps that can be calibrated and operated at a constant airflow up to 16 liters per minute.
- b. Low-volume, battery powered, body-attachable, portable personal pumps that can be calibrated to a constant airflow up to approximately 3.5 liters per minute, and a self-contained rechargeable power pack capable of sustaining the calibrated flow rate for a minimum of 10 hours. The pumps must also be equipped with an automatic flow control unit which must maintain a constant flow, even as filter resistance increases due to accumulation of fiber and debris on the filter

surface.

- c. Single use standard 25 mm diameter cassette, open face, 0.8 micron pore size, mixed cellulose ester membrane filters and cassettes with 50 mm electrically conductive extension cowl, and shrink bands for personal air sampling.

### 3.1.2 Respirators

Select respirators from those approved by the National Institute for Occupational Safety and Health (NIOSH), Department of Health and Human Services.

#### 3.1.2.1 Respirators for Handling Asbestos

Provide personnel engaged in pre-cleaning, cleanup, handling, encapsulation, removal and demolition of asbestos materials with respiratory protection as indicated in 29 CFR 1926.1101 and 29 CFR 1926.103. Breathing air must comply with CGA G-7.

### 3.1.3 Exterior Whole Body Protection

#### 3.1.3.1 Outer Protective Clothing

Provide personnel exposed to asbestos with disposable "non-breathable," or reusable "non-breathable" whole body outer protective clothing, head coverings, gloves, and foot coverings. Provide disposable plastic or rubber gloves to protect hands. Cloth gloves may be worn inside the plastic or rubber gloves for comfort, but must not be used alone. Make sleeves secure at the wrists, make foot coverings secure at the ankles, and make clothing secure at the neck by the use of tape.

#### 3.1.3.2 Work Clothing

Provide cloth work clothes for wear under the outer protective clothing and foot coverings and either dispose of or properly decontaminate them as recommended by the PQP after each use.

#### 3.1.3.3 Personal Decontamination Unit

Provide a temporary, negative pressure unit with a separate decontamination locker room and clean locker room with a shower that complies with 29 CFR 1926.51(f)(4)(ii) through (V) in between for personnel required to wear whole body protective clothing. Provide two separate lockers for each asbestos worker, one in each locker room. Keep street clothing and street shoes in the clean locker. HEPA vacuum and remove asbestos contaminated disposable protective clothing while still wearing respirators at the boundary of the asbestos work area and seal in impermeable bags or containers for disposal. HEPA vacuum and remove asbestos contaminated reusable protective clothing while still wearing respirators at the boundary of the asbestos work area, seal in two impermeable bags, label outer bag as asbestos contaminated waste, and transport for decontamination. Do not wear work clothing between home and work. Locate showers between the decontamination locker room and the clean locker room and require that all employees shower before changing into street clothes. Collect used shower water and filter with approved water filtration equipment to remove asbestos contamination. Wastewater filters must be installed in series with the first stage pore size 20 microns and the second stage pore size of 5 microns. Dispose of filters

and residue as asbestos waste. Discharge clean water to the sanitary system. Dispose of asbestos contaminated work clothing as asbestos contaminated waste or properly decontaminate as specified in the Contractor's Asbestos Hazard Abatement Plan. Keep the floor of the decontamination unit's clean room dry and clean at all times. Proper housekeeping and hygiene requirements must be maintained. Provide soap and towels for showering, washing and drying. Cloth towels provided must be disposed of as ACM waste or must be laundered in accordance with 29 CFR 1926.1101. Physically attach the decontamination units to the asbestos control area. Construct both a personnel decontamination unit and an equipment decontamination unit onto and integral with each asbestos control area.

#### 3.1.3.4 Eye Protection

Provide eye protection that complies with ANSI/ISEA Z87.1 when operations present a potential eye injury hazard. Provide goggles to personnel engaged in asbestos abatement operations when the use of a full face respirator is not required.

#### 3.1.4 Regulated Areas

All Class I, II, and III asbestos work must be conducted within regulated areas. The regulated area must be demarcated to minimize the number of persons within the area and to protect persons outside the area from exposure to airborne asbestos. Control access to regulated areas, ensure that only authorized personnel enter, and verify that Contractor required medical surveillance, training and respiratory protection program requirements are met prior to allowing entrance.

#### 3.1.5 Load-out Unit

Provide a temporary load-out unit that is adjacent and connected to the regulated area and access tunnel. Attach the load-out unit in a leak-tight manner to each regulated area.

#### 3.1.6 Warning Signs and Labels

Provide warning signs printed in English and at all approaches to asbestos control areas. Locate signs at such a distance that personnel may read the sign and take the necessary protective steps required before entering the area. Provide labels and affix to all asbestos materials, scrap, waste, debris, and other products contaminated with asbestos. Containers with preprinted warning labels conforming to the requirements are acceptable.

##### 3.1.6.1 Warning Sign

Provide vertical format conforming to 29 CFR 1926.200, and 29 CFR 1926.1101 minimum 20 by 14 inches displaying the following legend in the lower panel:

| <u>Legend</u> | <u>Notation</u>                     |
|---------------|-------------------------------------|
| DANGER        | one inch Sans Serif Gothic or Block |

| Legend                                                           | Notation                            |
|------------------------------------------------------------------|-------------------------------------|
| ASBESTOS                                                         | one inch Sans Serif Gothic or Block |
| MAY CAUSE CANCER                                                 | one inch Sans Serif Gothic or Block |
| CAUSES DAMAGE TO LUNGS                                           | 1/4 inch Sans Serif Gothic or Block |
| AUTHORIZED PERSONNEL ONLY                                        | 1/4 inch Sans Serif Gothic or Block |
| WEAR RESPIRATORY PROTECTION AND PROTECTIVE CLOTHING IN THIS AREA | 1/4 inch Sans Serif Gothic or Block |

Spacing between lines must be at least equal to the height of the upper of any two lines.

### 3.1.6.2 Warning Labels

Provide labels conforming to 29 CFR 1926.1101 of sufficient size to be clearly legible, displaying the following legend:

|                                         |
|-----------------------------------------|
| DANGER                                  |
| CONTAINS ASBESTOS FIBERS                |
| MAY CAUSE CANCER                        |
| CAUSES DAMAGE TO LUNGS                  |
| DO NOT BREATHE DUST AVOID CREATING DUST |

### 3.1.7 Local Exhaust System

Provide a local exhaust system in the asbestos control area in accordance with ASSP Z9.2 and 29 CFR 1926.1101 that will provide at least four air changes per hour inside of the negative pressure enclosure. Local exhaust equipment must be operated 24-hours per day, until the asbestos control area is removed and must be leak proof to the filter and equipped with HEPA filters. Maintain a minimum pressure differential in the control area of minus 0.02 inch of water column relative to adjacent, unsealed areas. Provide continuous 24-hour per day monitoring of the pressure differential with a pressure differential automatic recording instrument. The building ventilation system must not be used as the local exhaust system for the asbestos control area. Filters on exhaust equipment must conform to ASSP Z9.2 and UL 586. Terminate the local exhaust system out of doors and remote from any public access or ventilation system intakes.

### 3.1.8 Tools

Vacuums must be leak proof to the filter and equipped with HEPA filters. Filters on vacuums must conform to ASSP Z9.2 and UL 586. Do not use power tools to remove asbestos containing materials unless the tool is equipped with effective, integral HEPA filtered exhaust ventilation systems. Remove all residual asbestos from reusable tools prior to storage or reuse. Reusable tools must be thoroughly decontaminated prior to being removed from the regulated areas.

### 3.1.9 Rental Equipment

If rental equipment is to be used, furnish written notification to the rental agency concerning the intended use of the equipment and the possibility of asbestos contamination of the equipment.

### 3.1.10 Glovebags

Glovebag operations must be performed in accordance with the requirements of the State of Maine Department of Environmental Protection. Submit written manufacturers proof that glovebags will not break down under expected temperatures and conditions.

### 3.1.11 Decontamination Area Exit Procedures

Note: Whenever possible, decontamination area must be contiguous with, and connected to, the negative pressure enclosure.

Ensure that the following procedures are followed:

- a. Before leaving the regulated area, remove all gross contamination and debris from work clothing using a HEPA vacuum.
- b. Employees must remove their protective clothing in the equipment room and deposit the clothing in labeled impermeable bags or containers for disposal or laundering.
- c. Employees must not remove their respirators until showering.
- d. Employees must shower prior to entering the clean room. If a shower has not been located between the equipment room and the clean room or the work is performed outdoors, ensure that employees engaged in Class I asbestos jobs: a) Remove asbestos contamination from their work suits in the equipment room or decontamination area using a HEPA vacuum before proceeding to a shower that is not adjacent to the work area; or b) Remove their contaminated work suits in the equipment room, without cleaning worksuits, and proceed to a shower that is not adjacent to the work area.

## 3.2 WORK PROCEDURE

Perform asbestos related work in accordance with MEDEP Chapter 425: Asbestos Management Regulations, 29 CFR 1926.1101, 40 CFR 61-SUBPART M, 49 CFR 107, 49 CFR 172, NAVFAC P-502 and as specified herein. Use wet removal procedures and negative pressure enclosure techniques. Wear and utilize protective clothing and equipment as specified herein. No eating, smoking, drinking, chewing gum, tobacco, or applying cosmetics is permitted in the asbestos work or control areas. Personnel of other trades not engaged in the encapsulation removal and demolition of asbestos containing material must not be exposed at any time to airborne concentrations of asbestos unless all the personnel protection and training provisions of this specification are complied with by the trade personnel. Seal all rooftop penetrations, except plumbing vents, prior to asbestos roofing work. Shut down the building heating, ventilating, and air conditioning system, cap the openings to the system, and provide temporary heating and ventilation, prior to the commencement of asbestos work. Power to the regulated area must be locked-out and tagged in accordance with 29 CFR 1910.147. Disconnect electrical service when encapsulation wet removal is performed and provide temporary electrical

service with verifiable ground fault circuit interrupter (GFCI) protection prior to the use of any water encapsulant. All electrical work must be performed by a licensed electrician. Stop abatement work in the regulated area immediately when the airborne total fiber concentration: (1) equals or exceeds 0.010 f/cc, or the pre-abatement concentration, whichever is greater, outside the regulated area; or (2) equals or exceeds 1.0 f/cc inside the regulated area. Correct the condition to the satisfaction of the Contracting Officer, including visual inspection and air sampling. Work must resume only upon notification by the Contracting Officer. Corrective actions must be documented. If an asbestos fiber release or spill occurs, stop work immediately, correct the condition to the satisfaction of the Contracting Officer including clearance sampling, prior to resumption of work.

### 3.2.1 Building Ventilation System and Critical Barriers

Building ventilation system supply and return air ducts in a regulated area must be shut down and isolated by lockable switch or other positive means in accordance with 29 CFR 1910.147. The airtight seals must consist of air-tight rigid covers for building ventilation supply and exhaust grills where the ventilation system is required to remain in service during abatement. Edges to wall, ceiling and floor surfaces must be sealed with industrial grade duct tape.

- a. A Competent Person must supervise the work.
- b. For indoor work, critical barriers must be placed over all openings to the regulated area.
- c. Impermeable dropcloths must be placed on surfaces beneath all removal activity.

### 3.2.2 Protection of Existing Work to Remain

Perform work without damage or contamination of adjacent work. Where such work is damaged or contaminated as verified by the Contracting Officer using visual inspection or sample analysis, it must be restored to its original condition or decontaminated by the Contractor at no expense to the Government as deemed appropriate by the Contracting Officer. This includes inadvertent spill of dirt, dust, or debris in which it is reasonable to conclude that asbestos may exist. When these spills occur, stop work immediately. Then clean up the spill. When satisfactory visual inspection and air sampling results are obtained from the PQP work may proceed at the discretion of the Contracting Officer.

### 3.2.3 Furnishings

Furniture and equipment will remain in the building. Cover and seal furnishings and equipment with 6-mil plastic sheet or remove from the work area and store in a location on site approved by the Contracting Officer.

### 3.2.4 Precleaning

Wet wipe and HEPA vacuum all surfaces potentially contaminated with asbestos prior to establishment of an enclosure.

### 3.2.5 Asbestos Control Area Requirements

#### 3.2.5.1 Negative Pressure Enclosure

Removal of Asbestos Contaminated Materials (ACM) require the use of a negative pressure enclosure. Block and seal openings in areas where the release of airborne asbestos fibers can be expected. Establish an asbestos negative pressure enclosure with the use of curtains, portable partitions, or other enclosures in order to prevent the escape of asbestos fibers from the contaminated asbestos work area. Negative pressure enclosure development must include protective covering of uncontaminated walls, and ceilings with a continuous membrane of two layers of minimum 6-mil plastic sheet sealed with tape to prevent water or other damage. Provide two layers of 6-mil plastic sheet over floors and extend a minimum of 12 inches up walls. Seal all joints with tape. Provide local exhaust system in the asbestos control area. Openings will be allowed in enclosures of asbestos control areas for personnel and equipment entry and exit, the supply and exhaust of air for the local exhaust system and the removal of properly containerized asbestos containing materials. Replace local exhaust system filters as required to maintain the efficiency of the system.

#### 3.2.5.2 Glovebag

If the construction of a negative pressure enclosure is infeasible for the removal of ACM, use alternate techniques as indicated in MEDEP Chapter 425: Asbestos Management Regulations and 29 CFR 1926.1101. Establish designated limits for the asbestos regulated area with the use of rope or other continuous barriers, and maintain all other requirements for asbestos control areas. The PQP must conduct personal samples of each worker engaged in asbestos handling (removal, disposal, transport and other associated work) throughout the duration of the project. If the quantity of airborne asbestos fibers monitored at the breathing zone of the workers at any time exceeds background or 0.010 fibers per cubic centimeter whichever is greater, stop work, evacuate personnel in adjacent areas or provide personnel with approved protective equipment at the discretion of the Contracting Officer. This sampling may be duplicated by the Government at the discretion of the Contracting Officer. If the air sampling results obtained by the Government differ from those obtained by the Contractor, the Government will determine which results predominate. If adjacent areas are contaminated as determined by the Contracting Officer, clean the contaminated areas, monitor, and visually inspect the area as specified herein.

#### 3.2.6 Removal Procedures

Wet asbestos material with a fine spray of amended water during removal, cutting, or other handling so as to reduce the emission of airborne fibers. Remove material and immediately place in 6 mil plastic disposal bags. Remove asbestos containing material in a gradual manner, with continuous application of the amended water or wetting agent in such a manner that no asbestos material is disturbed prior to being adequately wetted. Where unusual circumstances prohibit the use of 6 mil plastic bags, submit an alternate proposal for containment of asbestos fibers to the Contracting Officer for approval. For example, in the case where both piping and insulation are to be removed, the Contractor may elect to wet the insulation, wrap the pipes and insulation in plastic and remove the pipe by sections. Containerize asbestos containing material while wet. Do not allow asbestos material to accumulate or become dry. Lower and

otherwise handle asbestos containing material as indicated in MEDEP Chapter 425: Asbestos Management Regulations and 40 CFR 61-SUBPART M.

#### 3.2.6.1 Sealing Contaminated Items Designated for Disposal

Remove contaminated architectural, mechanical, and electrical appurtenances such as venetian blinds, full-height partitions, carpeting, duct work, pipes and fittings, radiators, light fixtures, conduit, panels, and other contaminated items designated for removal by completely coating the items with an asbestos lock-down encapsulant at the demolition site before removing the items from the asbestos control area. These items need not be vacuumed. The asbestos lock-down encapsulant must be tinted a contrasting color and spray-applied by airless method. Thoroughness of sealing operation must be visually gauged by the extent of colored coating on exposed surfaces. Lock-down encapsulants must comply with the performance requirements specified herein.

#### 3.2.6.2 Exposed Pipe Insulation Edges

Contain edges of asbestos insulation to remain that are exposed by a removal operation. Wet and cut the rough ends true and square with sharp tools and then encapsulate the edges with a 1/4 inch thick layer of non-asbestos containing insulating cement troweled to a smooth hard finish. When cement is dry, lag the end with a layer of non-asbestos lagging cloth, overlapping the existing ends by at least 4 inches. When insulating cement and cloth is an impractical method of sealing a raw edge of asbestos, take appropriate steps to seal the raw edges as approved by the Contracting Officer.

#### 3.2.7 Methods of Compliance

##### 3.2.7.1 Mandated Practices

The specific abatement techniques and items identified must be detailed in the Contractor's AHAP. Use the following engineering controls and work practices in all operations, regardless of the levels of exposure:

- a. Vacuum cleaners equipped with HEPA filters.
- b. Wet methods or wetting agents except where it can be demonstrated that the use of wet methods is unfeasible due to the creation of electrical hazards, equipment malfunction, and in roofing.
- c. Prompt clean-up and disposal.
- d. Inspection and repair of polyethylene.
- e. Cleaning of equipment and surfaces of containers prior to removing them from the equipment room or area.

##### 3.2.7.2 Control Methods

Use the following control methods:

- a. Local exhaust ventilation equipped with HEPA filter;
- b. Enclosure or isolation of processes producing asbestos dust;
- c. Where the feasible engineering and work practice controls are not



sufficient to reduce employee exposure to or below the PELs, use them to reduce employee exposure to the lowest levels attainable and must supplement them by the use of respiratory protection.

#### 3.2.7.3 Unacceptable Practices

The following work practices must not be used:

- a. High-speed abrasive disc saws that are not equipped with point of cut ventilator or enclosures with HEPA filtered exhaust air.
- b. Compressed air used to remove asbestos containing materials, unless the compressed air is used in conjunction with an enclosed ventilation system designed to capture the dust cloud created by the compressed air.
- c. Dry sweeping, shoveling, or other dry clean up.
- d. Employee rotation as a means of reducing employee exposure to asbestos.

#### 3.2.8 Class I Work Procedures

In addition to requirements of paragraphs MANDATED PRACTICES and CONTROL METHODS, the following engineering controls and work practices must be used:

- a. A Competent Person must supervise the installation and operation of the control methods.
- b. Place critical barriers over all openings to the regulated area.
- c. HVAC systems must be isolated in the regulated area by sealing with a double layer of plastic or air-tight rigid covers.
- d. Impermeable dropcloths (6 mil or greater thickness) must be placed on surfaces beneath all removal activity.
- e. Where a negative exposure assessment has not been provided or where exposure monitoring shows the PEL was exceeded, the regulated area must be ventilated with a HEPA unit and employees must use PPE.

#### 3.2.9 Specific Control Methods for Class I Work

Use Class I work procedures, control methods and removal methods for the following ACM:

- a. Spray Applied Fireproofing
- b. Gypsum Wallboard and Joint Compound
- c. Thermal System Insulation and Mudded Pipe Fittings
- d. Plaster and Textured Ceilings and Walls
- e. Vermiculite

##### 3.2.9.1 Negative Pressure Enclosure (NPE) System

The system must provide at least four air changes per hour inside the

containment. The local exhaust unit equipment must be operated 24-hours per day until the containment is removed. The NPE must be smoke tested for leaks at the beginning of each shift and be sufficient to maintain a minimum pressure differential of minus 0.02 inch of water column relative to adjacent, unsealed areas. Pressure differential must be monitored continuously, 24-hours per day, with an automatic manometric recording instrument and Records must be provided daily on the same day collected to the Contracting Officer and Asbestos Program Manager. The Contracting Officer and Asbestos Program Manager must be notified immediately if the pressure differential falls below the prescribed minimum. The building ventilation system must not be used as the local exhaust system for the regulated area. The NPE must terminate outdoors unless an alternate arrangement is allowed by the Contracting Officer and Asbestos Program Manager. All filters used must be new at the beginning of the project and must be periodically changed as necessary and disposed of as ACM waste. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

#### 3.2.9.2 Glovebag Systems

Glovebags must be used without modification, smoke-tested for leaks, and completely cover the circumference of pipe or other structures where the work is to be done. Glovebags must be used only once and must not be moved. Glovebags must not be used on surfaces that have temperatures exceeding 150 degrees F. Prior to disposal, glovebags must be collapsed using a HEPA vacuum. Before beginning the operation, loose and friable material adjacent to the glovebag operation must be wrapped and sealed in 2 layers of plastic or otherwise rendered intact. At least two persons must perform glovebag removal. Asbestos regulated work areas must be established for glovebag abatement. Designated boundary limits for the asbestos work must be established with rope or other continuous barriers and all other requirements for asbestos control areas must be maintained, including area signage and boundary warning tape.

- a. Attach HEPA vacuum systems to the bag to prevent collapse during removal of ACM.
- b. The negative pressure glove boxes must be fitted with gloved apertures and a bagging outlet and constructed with rigid sides from metal or other material which can withstand the weight of the ACM and water used during removal. A negative pressure must be created in the system using a HEPA filtration system. The box must be smoke tested for leaks prior to each use.

#### 3.2.9.3 Mini-Enclosure

Single bulkhead containment, double bulkhead containment, or mini-containment (small walk-in enclosure) to accommodate no more than two persons, may be used if the disturbance or removal can be completely contained by the enclosure. The mini-enclosure must be inspected for leaks and smoke tested before each use. Air movement must be directed away from the employee's breathing zone within the mini-enclosure.

#### 3.2.9.4 Wrap and Cut Operation

Prior to cutting pipe, the asbestos-containing insulation must be wrapped with polyethylene and securely sealed with duct tape to prevent asbestos becoming airborne as a result of the cutting process. The following steps

must be taken: install glovebag, strip back sections to be cut 6 inches from point of cut, and cut pipe into manageable sections.

#### 3.2.9.5 Class I Removal Method

Class I ACM must be removed using a control method described above. Prepare work area as previously specified. Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area. Spread one layer of 6-mil seamless plastic sheeting on the floor below the work area. Using the following techniques are applicable:

- a. Remove asbestos containing spray applied fireproofing using a scraper and wet methods and immediately place into 6-mil thickness disposal bag. After removal of the material use a wire brush to clean the exposed substrate to remove residual material. Continue wet cleaning until surfaces are free of visible debris.
- b. Cut manageable sections of gypsum wallboard and joint compound and immediately place into a 6-mil minimum thickness disposal bag or other approved container. Make every effort to keep the material from falling to the floor of the work area. Use a wire brush and wet clean to remove residual material from studs. Continue wet cleaning until the surface is clean of visible material and encapsulate stud walls.
- c. Remove ACM thermal system insulation and mudded pipe fittings using mechanical means and wet methods and immediately place into 6-mil thickness disposal bag. Continue wet cleaning until surfaces are free of visible debris.
- d. Remove ACM plaster ceilings or walls using mechanical means and adequately wet methods and immediately place into 6-mil thickness disposal bag. Make every effort to keep the material from falling to the floor of the work area. Continue wet cleaning until surfaces are free of visible debris.
- e. Remove ACM textured ceiling finish using a scraper and wet methods and immediately place into 6-mil thickness disposal bag. Floors are considered contaminated from fallen textured ceiling finish. Clean up debris on floor and dispose of carpet as asbestos contaminated material. After removal of the material use a wire brush to clean the exposed concrete ceiling to remove residual material. Continue wet cleaning until surfaces are free of visible debris.
- f. Remove ACM vermiculite using mechanical means and adequately wet methods and immediately place into 6-mil thickness disposal bag. Make every effort to keep the material from falling to the floor of the work area. Continue wet cleaning until surfaces are free of visible debris.

Bag all asbestos debris which has fallen to the floor as asbestos-containing debris. Place all debris in plastic disposal bags of 6-mil minimum thickness. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 6-mil minimum thickness disposal bag. Containerize asbestos containing waste while wet. Lower and otherwise handle asbestos containing materials as indicated in 40 CFR 61-SUBPART M. Conduct area monitoring of airborne

fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the quantity of airborne asbestos fibers monitored at the breathing zone of the workers or the designated limits at any time exceeds background or 0.010 fibers per cubic centimeter, whichever is greater, stop work, and immediately correct the situation.

### 3.2.10 Class II Work Procedures

In addition to the requirements of paragraphs MANDATED PRACTICES and CONTROL METHODS, the following engineering controls and work practices must be used:

- a. A Competent Person must supervise the work.
- b. For indoor work, critical barriers must be placed over all openings to the regulated area.
- c. Impermeable dropcloths must be placed on surfaces beneath all removal activity.

### 3.2.11 Specific Control Methods for Class II Work

#### 3.2.11.1 Suspect Fire Doors

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. A detached decontamination system may be used. Spread 6-mil plastic sheeting on the ground beneath the work area and around the perimeter of the work area extending out in all directions. Remove door intact from hinges and wrap with 6-mil plastic sheeting. Inspect the interior areas of the door to determine if ACM is present. If ACM is not present the door may be disposed of as general construction debris. If ACM is present place whole door in enclosed container for disposal. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.010 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

#### 3.2.11.2 Roofing Materials

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. When removing roofing materials which contain ACM as described in 29 CFR 1926.1101(g)(8)(ii), use the following practices. Roofing material must be removed in an intact state. Wet methods must be used to remove roofing materials that are not intact, or that will be rendered not intact during removal, unless such wet methods are not feasible or will create safety hazards. When removing built-up roofs, with asbestos-containing roofing felts and an aggregate surface, using a power roof cutter, all dust resulting from the cutting operations must be collected by a HEPA dust collector, or must be HEPA vacuumed by vacuuming along the cut line. Asbestos-containing roofing material must not be dropped or thrown to the ground, but must be lowered to the ground via covered, dust-tight chute, crane, hoist or other method approved by the Contracting Officer. Any ACM

that is not intact must be lowered to the ground as soon as practicable, but not later than the end of the work shift. While the material remains on the roof it must be kept wet or placed in an impermeable waste bag or wrapped in plastic sheeting. Intact ACM must be lowered to the ground as soon as practicable, but not later than the end of the work shift. Unwrapped material must be transferred to a closed receptacle. Critical barriers must be placed over roof level heating and ventilation air intakes. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.010 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

### 3.2.11.3 Cementitious Siding and Shingles or Transite Panels

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. When removing cementitious asbestos-containing siding, shingles or Transite panels use the following work practices. Intentionally cutting, abrading or breaking is prohibited. Each panel or shingle must be sprayed with amended water prior to removal. Nails must be cut with flat, sharp instruments. Unwrapped or unbagged panels or shingles must be immediately lowered to the ground via covered dust-tight chute, crane or hoist, or placed in an impervious waste bag or wrapped in plastic sheeting and lowered to the ground no later than the end of the work shift. Place debris into a 6-mil minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 6-mil minimum thickness disposal bag. Containerize asbestos containing waste while wet. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.010 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

### 3.2.11.4 Gaskets

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. Gaskets must be thoroughly wetted with amended water prior to removal and immediately placed in a disposal container. If a gasket is visibly deteriorated and unlikely to be removed intact, removal must be undertaken within a glovebag. Any scraping to remove residue must be performed wet. Place debris into a 6-mil minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 6-mil minimum thickness disposal bag. Containerize asbestos containing waste while wet. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal

samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.010 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

#### 3.2.12 NOT USED

#### 3.2.13 Abatement of Asbestos Contaminated Soil

Establish designated limits for the asbestos regulated work area with the use of red barrier tape, critical barriers, signs, and maintain all other requirements for asbestos control area except local exhaust. Asbestos contaminated soil must be removed from areas to a minimum depth of 2 inches. Soil must be thoroughly dampened with amended water and then removed by manual shoveling into labeled containers. Place debris into a 6-mil minimum thickness disposal bag or other approved container. Once the material is in the disposal bag, apply additional water as needed to achieve "adequately wet" conditions for NESHAP compliance. Place bagged asbestos waste under negative pressure with the use of a HEPA vacuum, goose neck and duck tape to seal the bag, wash to remove any visible contamination and place into a second 6-mil minimum thickness disposal bag. Containerize asbestos containing waste while wet. Conduct area monitoring of airborne fibers during the work shift at the designated limits of the asbestos work area and conduct personal samples of each worker engaged in the work. If the airborne fiber concentration of the workers or designated limits at any time exceeds background or 0.010 fibers per cubic centimeter, whichever is greater, stop work immediately and correct the situation.

#### 3.2.14 Air Sampling

Perform sampling of airborne concentrations of asbestos fibers in accordance with 29 CFR 1926.1101, the Contractor's air monitoring plan and as specified herein. Sampling performed in accordance with 29 CFR 1926.1101 must be performed by the PQP. Sampling performed for environmental and quality control reasons must be performed by the PQP. Unless otherwise specified, use NIOSH Method 7400 for sampling and analysis. Monitoring may be duplicated by the Government at the discretion of the Contracting Officer and Asbestos Program Manager. If the air sampling results obtained by the Government differ from those results obtained by the Contractor, the Government will determine which results predominate. Results of breathing zone samples must be posted at the job site and made available to the Contracting Officer and Asbestos Program Manager. Submit all documentation regarding initial exposure assessments, negative exposure assessments, and air-monitoring results. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

##### 3.2.14.1 Sampling Prior to Asbestos Work

Provide area air sampling and establish the baseline one day prior to the masking and sealing operations for each demolition or removal site. Establish the background by performing area sampling in similar but uncontaminated sites in the building.

##### 3.2.14.2 Sampling During Asbestos Work

The PQP must provide personal sampling as indicated in 29 CFR 1926.1101.

Breathing zone samples must be taken for at least 25 percent of the workers in each shift, or a minimum of two, whichever is greater. Air sample fiber counting must be completed and results provided within 24-hours (breathing zone samples), and 8 hours (environmental/clearance monitoring) after completion of a sampling period. At the same time, the GC will provide area sampling close to the work inside the enclosure, outside the clean room entrance to the enclosure, at the waste load-out, and at the exhaust opening of the local exhaust system. If sampling outside the enclosure shows airborne levels have exceeded background or 0.010 fibers per cubic centimeter, whichever is greater, stop all work, correct the condition(s) causing the increase, and notify the Contracting Officer and Asbestos Program Manager immediately. Where alternate methods are used, perform personal and area air sampling at locations and frequencies that will accurately characterize the evolving airborne asbestos levels. The written results must be signed by testing laboratory analyst, testing laboratory principal and the Contractor's PQP. The air sampling results must be documented on a Contractor's daily air monitoring log. Copies of the air monitoring results must also be submitted directly to the Asbestos Program Manager. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements.

#### 3.2.14.3 Final Clearance Requirements, NIOSH PCM Method

For PCM sampling and analysis using NIOSH NMAM Method 7400, the fiber concentration inside the abated regulated area, for each airborne sample, must be less than 0.010 f/cc. The abatement inside the regulated area is considered complete when every PCM final clearance sample is below the clearance limit. If any sample result is greater than 0.010 total f/cc, the asbestos fiber concentration (asbestos f/cc) must be confirmed from that same filter using NIOSH NMAM Method 7402 (TEM) at Contractor's expense. If any confirmation sample result is greater than 0.010 asbestos f/cc, abatement is incomplete and cleaning must be repeated at the Contractor's expense. Upon completion of any required recleaning, resampling with results to meet the above clearance criteria must be done at the Contractor's expense.

#### 3.2.14.4 Final Clearance Requirements, EPA TEM Method

For EPA TEM sampling and analysis, using the EPA Method specified in 40 CFR 763, abatement inside the regulated area is considered complete when the arithmetic mean asbestos concentration of the five inside samples is less than or equal to 70 structures per square millimeter (70 S/mm). When the arithmetic mean is greater than 70 S/mm, the three blank samples must be analyzed. If the three blank samples are greater than 70 S/mm, resampling must be done. If less than 70 S/mm, the five outside samples must be analyzed and a Z-test analysis performed. When the Z-test results are less than 1.65, the decontamination must be considered complete. If the Z-test results are more than 1.65, the abatement is incomplete and cleaning must be repeated. Upon completion of any required recleaning, resampling with results to meet the above clearance criteria must be done at the Contractor's expense.

#### 3.2.14.5 Sampling After Final Clean-Up (Clearance Sampling)

Provide area sampling of asbestos fibers using aggressive air sampling techniques as defined in the EPA 560/5-85-024 and establish an airborne asbestos concentration of less than 0.010 fibers per cubic centimeter after final clean-up but before removal of the enclosure or the asbestos

work control area. After final cleanup and the asbestos control area is dry but prior to clearance sampling, the PQP and GC must perform a visual inspection in accordance with ASTM E1368 to ensure that the asbestos control and work area is free of any accumulations of dirt, dust, or debris. Prepare a written report signed and dated by the PQP documenting that the asbestos control area is free of dust, dirt, and debris and all waste has been removed. Collect the appropriate number of samples in accordance with Maine DEP regulations. Use transmission electron microscopy (TEM) to analyze clearance samples and report the results in accordance with current NIOSH criteria. The asbestos fiber counts from these samples must be less than 0.010 fibers per cubic centimeter or be not greater than the background, whichever is greater. Should any of the final samples indicate a higher value take appropriate actions to re-clean the area and repeat the sampling and TEM analysis at the Contractor's expense.

#### 3.2.14.6 Air Clearance Failure

If clearance sampling results fail to meet the final clearance requirements, pay all costs associated with the required recleaning, resampling, and analysis, until final clearance requirements are met.

#### 3.2.15 Lock-Down

Prior to removal of plastic barriers and after pre-clearance clean up of gross contamination, the PQP must conduct a visual inspection of all areas affected by the removal in accordance with ASTM E1368. Inspect for any visible fibers. Spray apply a post removal (lock-down) encapsulant to ceiling, walls, floors and other areas exposed in the removal area. The exposed area includes but is not limited to plastic barriers, furnishings and articles to be discarded as well as dirty change room, air locks for bag removal and decontamination chambers.

#### 3.2.16 Site Inspection

While performing asbestos engineering control work, the Contractor must be subject to on-site inspection by the Contracting Officer who may be assisted by or represented by safety or industrial hygiene personnel. If the work is found to be in violation of this specification, the Contracting Officer or his representative will issue a stop work order to be in effect immediately and until the violation is resolved. All related costs including standby time required to resolve the violation must be at the Contractor's expense.

### 3.3 CLEAN-UP AND DISPOSAL

#### 3.3.1 Housekeeping

Essential parts of asbestos dust control are housekeeping and clean-up procedures. Maintain surfaces of the asbestos control area free of accumulations of asbestos fibers. Give meticulous attention to restricting the spread of dust and debris; keep waste from being distributed over the general area. Use HEPA filtered vacuum cleaners. DO NOT BLOW DOWN THE SPACE WITH COMPRESSED AIR. When asbestos removal is complete, all asbestos waste is removed from the work-site, and final clean-up is completed, the Contracting Officer will attest that the area is safe before the signs can be removed. After final clean-up and acceptable airborne concentrations are attained but before the HEPA unit is turned off and the enclosure removed, remove all pre-filters on the



building HVAC system and provide new pre-filters. Dispose of filters as asbestos contaminated materials. Reestablish HVAC mechanical, and electrical systems in proper working order. The PQP and CP will visually inspect all surfaces within the enclosure for residual material or accumulated dust or debris. The Contractor must re-clean all areas showing dust or residual materials. If re-cleaning is required, air sample and establish an acceptable asbestos airborne concentration after re-cleaning. The Contracting Officer must agree that the area is safe in writing before unrestricted entry will be permitted. The Government must have the option to perform monitoring to determine if the areas are safe before entry is permitted.

### 3.3.2 Title to Materials

All waste materials, except as specified otherwise, become the property of the Contractor and must be disposed of as specified in applicable local, State, and Federal regulations and herein.

The Portsmouth Naval Shipyard Hazardous Waste Facility takes title and is responsible for all asbestos waste disposal.

### 3.3.3 Disposal of Asbestos

#### 3.3.3.1 Procedure for Disposal

Coordinate all waste disposal with the Contracting Officer and the Portsmouth Naval Shipyard Hazardous Waste Facility (Code 106, Building 357). Collect asbestos waste, contaminated waste water filters, asbestos contaminated water, scrap, debris, bags, containers, equipment, and asbestos contaminated clothing which may produce airborne concentrations of asbestos fibers and place in sealed fiber-proof, waterproof, non-returnable containers (e.g. double plastic bags 6 mils thick, cartons, drums or cans). Wastes within the containers must be adequately wet in accordance with MEDEP Chapter 425: Asbestos Management Regulations and 40 CFR 61-SUBPART M. All waste materials, including, but not limited to, piping, must be less than 5 feet in length. Affix a warning and Department of Transportation (DOT) label to each container including the bags or use at least 6 mils thick bags with the approved warnings and DOT labeling preprinted on the bag. Clearly indicate on the outside of each container the name of the waste generator and the location at which the waste was generated. Prevent contamination of the transport vehicle (especially if the transport vehicle is a rented truck likely to be used in the future for non-asbestos purposes). These precautions include lining the vehicle cargo area with plastic sheeting (similar to work area enclosure) and thorough cleaning of the cargo area after transport and unloading of asbestos debris is complete.

The Contractor must carefully remove material and immediately place it in approved asbestos disposal bags/containers. If the bags/containers are not pre-marked, a Contractor furnished caution label must be permanently attached to the bags/containers. The Contractor must also furnish and affix a label permanently attached to the bags which provides the following information:

- \* Abatement Contractor Name
- \* Construction Contract Number
- \* Building Number
- \* Portsmouth Naval Shipyard, Kittery, Maine
- \* Date

Contractor must ensure all asbestos materials are properly packaged prior to placement into government-furnished dumpsters.

#### 3.3.3.2 Asbestos Disposal Quantity Report

Direct the PQP to record and report, to the Contracting Officer and Asbestos Program Manager, the amount of asbestos containing material removed and released for disposal. Deliver the report for the previous day at the beginning of each day shift with amounts of material removed during the previous day reported in linear feet or square feet as described initially in this specification and in cubic feet for the amount of asbestos containing material released for disposal.

Allow the GC to inspect, record and report the amount of asbestos containing material removed and released for disposal on a daily basis.

-- End of Section --

SECTION 02 83 00.00 22

MANAGEMENT OF LEAD, CADMIUM, AND CHROMIUM (VI) DURING RENOVATION,  
DEMOLITION, CONSTRUCTION, REMOVAL AND ABATEMENT  
**(PNS PROJECTS)**  
**3/24**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP/SAFE Z9.2 (2018) Fundamentals Governing the Design  
and Operation of Local Exhaust Ventilation  
Systems

ASTM INTERNATIONAL (ASTM)

ASTM E1613 (2012) Standard Test Method for  
Determination of Lead by Inductively  
Coupled Plasma Atomic Emission  
Spectrometry (ICP-AES), Flame Atomic  
Absorption Spectrometry (FAAS), or  
Graphite Furnace Atomic Absorption  
Spectrometry (GFAAS) Techniques

ASTM E1644 (2017) Standard Practice for Hot Plate  
Digestion of Dust Wipe Samples for the  
Determination of Lead

ASTM E1726 (2001; R 2009) Preparation of Soil Samples  
by Hotplate Digestion for Subsequent Lead  
Analysis

ASTM E1727 (2016) Standard Practice for Field  
Collection of Soil Samples for Subsequent  
Lead Determination

ASTM E1728 (2016) Collection of Settled Dust Samples  
Using Wipe Sampling Methods for Subsequent  
Lead Determination

ASTM E1792 (2003; R 2016) Standard Specification for  
Wipe Sampling Materials for Lead in  
Surface Dust

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 701 (2023; ERTA 1 2023) Standard Methods of  
Fire Tests for Flame Propagation of  
Textiles and Films

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety -- Safety and Occupational  
Health (SOH) Requirements

U.S. DEPARTMENT OF HOUSING AND URBAN DEVELOPMENT (HUD)

HUD 6780 (1995; Errata Aug 1996; Rev Ch. 7 - 1997)  
Guidelines for the Evaluation and Control  
of Lead-Based Paint Hazards in Housing

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|                  |                                                                                                                                                       |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 29 CFR 1926.103  | Respiratory Protection                                                                                                                                |
| 29 CFR 1926.1126 | Chromium (VI)                                                                                                                                         |
| 29 CFR 1926.1127 | Cadmium                                                                                                                                               |
| 29 CFR 1926.21   | Safety Training and Education                                                                                                                         |
| 29 CFR 1926.33   | Access to Employee Exposure and Medical<br>Records                                                                                                    |
| 29 CFR 1926.55   | Gases, Vapors, Fumes, Dusts, and Mists                                                                                                                |
| 29 CFR 1926.59   | Hazard Communication                                                                                                                                  |
| 29 CFR 1926.62   | Lead                                                                                                                                                  |
| 29 CFR 1926.65   | Hazardous Waste Operations and Emergency<br>Response                                                                                                  |
| 40 CFR 260       | Hazardous Waste Management System: General                                                                                                            |
| 40 CFR 261       | Identification and Listing of Hazardous<br>Waste                                                                                                      |
| 40 CFR 262       | Standards Applicable to Generators of<br>Hazardous Waste                                                                                              |
| 40 CFR 263       | Standards Applicable to Transporters of<br>Hazardous Waste                                                                                            |
| 40 CFR 264       | Standards for Owners and Operators of<br>Hazardous Waste Treatment, Storage, and<br>Disposal Facilities                                               |
| 40 CFR 265       | Interim Status Standards for Owners and<br>Operators of Hazardous Waste Treatment,<br>Storage, and Disposal Facilities                                |
| 40 CFR 745       | Lead-Based Paint Poisoning Prevention in<br>Certain Residential Structures                                                                            |
| 49 CFR 172       | Hazardous Materials Table, Special<br>Provisions, Hazardous Materials<br>Communications, Emergency Response<br>Information, and Training Requirements |

49 CFR 178

Specifications for Packagings

U.S. NAVAL FACILITIES ENGINEERING COMMAND (NAVFAC)

ND OPNAVINST 5100.23

(2005; Rev G) Navy Occupational Safety and  
Health (NAVOSH) Program Manual

UNDERWRITERS LABORATORIES (UL)

UL 586

(2009; Reprint Sep 2022) UL Standard for  
Safety High-Efficiency Particulate, Air  
Filter Units

## 1.2 DEFINITIONS

### 1.2.1 Abatement

Measures defined in 40 CFR 745, Section 223, designed to permanently eliminate lead-based paint hazards.

### 1.2.2 Action Level

Employee exposure, without regard to use of respirators, to an airborne concentration of lead of 30 micrograms per cubic meter of air averaged over an 8-hour period; to an airborne concentration of cadmium of 2.5 micrograms per cubic meter of air averaged over an 8-hour period; to an airborne concentration of chromium (VI) of 2.5 micrograms per cubic meter of air averaged over an 8-hour period.

### 1.2.3 Area Sampling

Sampling of lead, cadmium, and chromium (VI) concentrations within the lead, cadmium, and chromium (VI) control area and inside the physical boundaries which is representative of the airborne lead, cadmium, and chromium (VI) concentrations but is not collected in the breathing zone of personnel (approximately 5 to 6 feet above the floor).

### 1.2.4 Cadmium Permissible Exposure Limit (PEL)

Five micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.1127. If an employee is exposed for more than 8-hours in a work day, determine the PEL by the following formula:

$$\text{PEL (micrograms/cubic meter of air)} = 40/\text{No. hrs worked per day}$$

### 1.2.5 Certified Industrial Hygienist (CIH)

As used in this section refers to a person retained by the Contractor who is certified as an industrial hygienist and who is trained in the recognition and control of lead, cadmium and chromium (VI) hazards in accordance with current federal, State, and local regulations. CIH must be certified for comprehensive practice by the American Board of Industrial Hygiene. The Certified Industrial Hygienist must be independent of the Contractor and must have no employee or employer relationship which could constitute a conflict of interest.

#### 1.2.6 Child-Occupied Facility

Real property which is a building or portion of a building constructed prior to 1978 visited regularly by the same child, six-years of age or under, on at least two different days within any week (Sunday through Saturday period), provided that each day's visit lasts at least 3 hours, and the combined annual visits last at least 60 hours. Child-occupied facilities include but are not limited to, day-care centers, preschools and kindergarten classrooms.

#### 1.2.7 Chromium (VI) Permissible Exposure Limit (PEL)

Five micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.1126. If an employee is exposed for more than 8 hours in a work day, determine the PEL by the following formula:

$$\text{PEL (micrograms/cubic meter of air)} = 40/\text{No. hrs worked per day}$$

#### 1.2.8 Competent Person (CP)

As used in this section, refers to a person employed by the Contractor who is trained in the recognition and control of lead, cadmium and chromium (VI) hazards in accordance with current federal, State, and local regulations and has the authority to take prompt corrective actions to control the lead, cadmium and chromium (VI) hazard. The Contractor may provide more than one CP as required to supervise and monitor the work. The CP must be a Certified Industrial Hygienist (CIH) certified by the American Board of Industrial Hygiene.

#### 1.2.9 Contaminated Room

Refers to a room for removal of contaminated personal protective equipment (PPE).

#### 1.2.10 Decontamination Shower Facility

That facility that encompasses a clean clothing storage room, and a contaminated clothing storage and disposal rooms, with a shower facility in between.

#### 1.2.11 Deleading

Activities conducted by a person who offers to eliminate lead-based paint or lead-based paint hazards or paints containing cadmium/chromium (VI) or to plan such activities in commercial buildings, bridges or other structures.

#### 1.2.12 Eight-Hour Time Weighted Average (TWA)

Airborne concentration of lead, cadmium, and chromium (VI) to which an employee is exposed, averaged over an 8-hour workday as indicated in 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

#### 1.2.13 High Efficiency Particulate Air (HEPA) Filter Equipment

HEPA filtered vacuuming equipment with a UL 586 filter system capable of collecting and retaining lead, cadmium, and chromium (VI)-contaminated particulate. A high efficiency particulate filter demonstrates at least 99.97 percent efficiency against 0.3 micron or larger size particles.

#### 1.2.14 Lead

Metallic lead, inorganic lead compounds, and organic lead soaps. The use of the term Lead in this section also refers to paints which contain detectable concentrations of Cadmium and Chromium (VI). For the purposes of the section lead-based paint (LBP) and paint with lead (PWL) also contains cadmium and chromium (VI).

#### 1.2.15 Lead-Based Paint (LBP)

Paint or other surface coating that contains lead in excess of 1.0 milligrams per centimeter squared or 0.5 percent by weight.

#### 1.2.16 Lead-Based Paint Activities

In the case of target housing or child occupied facilities, lead-based paint activities include; a lead-based paint inspection, a risk assessment, or abatement of lead-based paint hazards.

#### 1.2.17 Lead-Based Paint Hazards

Paint-lead hazard, dust-lead hazard or soil-lead hazard as identified in 40 CFR 745, Section 65. Any condition that causes exposure to lead from lead-contaminated dust, lead-contaminated soil, lead-based paint that is deteriorated or present in accessible surfaces, friction surfaces, or impact surfaces that would result in adverse human health effects.

#### 1.2.18 Lead, Cadmium, and Chromium (VI) Control Area

A system of control methods to prevent the spread of lead, cadmium, and chromium (VI) dust, paint chips debris, and/or any other associated materials to adjacent areas that may include temporary containment, floor or ground cover protection, physical boundaries, and warning signs to prevent unauthorized entry of personnel. HEPA filtered local exhaust equipment may be used as engineering controls to further reduce personnel exposures or building/outdoor environmental contamination.

#### 1.2.19 Lead Permissible Exposure Limit (PEL)

Fifty micrograms per cubic meter of air as an 8-hour time weighted average as determined by 29 CFR 1926.62. If an employee is exposed for more than 8 hours in a work day, determine the PEL by the following formula:

$$\text{PEL (micrograms/cubic meter of air)} = 400/\text{No. hrs worked per day}$$

#### 1.2.20 Material Containing Lead/Paint with Lead (MCL/PWL)

Any material, including paint, which contains lead as determined by the testing laboratory using a valid test method. The requirements of this section do not apply if no detectable concentrations of lead are found using an appropriate quantitative method for analyzing PCL or MCL using laboratory instruments with specified limits of detection. An X-Ray Fluorescence (XRF) instrument is not considered a valid test method.

#### 1.2.21 Personal Sampling

Sampling of airborne lead, cadmium, and chromium (VI) concentrations within the breathing zone of an employee to determine the 8-hour time

weighted average concentration in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Samples must be representative of the employees' work tasks. Breathing zone must be considered an area within a hemisphere, forward of the shoulders, with a radius of 6 to 9 inches and centered at the nose or mouth of an employee.

#### 1.2.22 Physical Boundary

Area physically roped or partitioned off around lead, cadmium, and chromium (VI) control area to limit unauthorized entry of personnel.

#### 1.2.23 Target Housing

Residential real property which is housing constructed prior to 1978, except housing for the elderly or persons with disabilities (unless any one or more children age 6 years or under resides or is expected to reside in such housing for the elderly or persons with disabilities) or any zero bedroom dwelling.

### 1.3 DESCRIPTION

This specification covers all project-related activities, including, but not limited to, renovation, construction, demolition, removal, and/or abatement, impacting PWL and/or material containing lead, cadmium, and chromium (VI). The work covered by this section includes work tasks and the precautions specified in this section for the protection of workers, building occupants, the building interior and exterior, furniture and equipment, and the environment during and after the performance of all project-related activities in accordance with state and federal regulations and as specified herein.

#### 1.3.1 Protection of Existing Areas To Remain

Any and all work performed under these specifications must be performed without damaging or impacting adjacent work and areas. Where such work or areas are damaged or impacted, restore work and areas to the original condition.

#### 1.3.2 Coordination with Other Work

Coordinate with work being performed in adjacent areas to ensure there are no exposure issues. Explain coordination procedures in the Lead, Cadmium, and Chromium (VI) Compliance Plan and describe how the Contractor will prevent lead, cadmium and chromium (VI) exposure to other contractors and Government personnel performing work unrelated to lead, cadmium and chromium (VI) activities.

#### 1.3.3 Sampling and Analysis

Submit all analytical results from sampling conducted during any and all work performed under these specifications, within the time-frames mandated by these specifications. Keep the log of results current with project activities and brief the results to the Contracting Officer as analytical results are reported.

Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements. The Government will pursue any and all remedies for non-compliance including, but not limited to, stopping work, nonpayment for related work completed, and contractor removal. All delays, concurrent



or related to failure to comply with these specifications are entirely the responsibility of the Contractor and shall not be made the subject , or any component of any request for additional time or compensation.

#### 1.3.3.1 Dust Wipe Materials, Sampling and Analysis

Sampling must conform to ASTM E1728 and ASTM E1792. Analysis must conform to ASTM E1613 and ASTM E1644.

#### 1.3.3.2 Soil Sampling and Analysis

Sampling must conform to ASTM E1727. Analysis must conform to ASTM E1613 and ASTM E1726.

#### 1.3.3.3 Clearance Monitoring

Clearance sampling via air, wipe, and soil sampling must be performed in accordance with the requirements of Section entitled FIELD QUALITY CONTROL.

#### 1.3.4 Clearance Requirements

Clearance requirements for air, wipe, and soil sampling are included in Section entitled in Section entitled FIELD QUALITY CONTROL.

### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-01 Preconstruction Submittals

Statement of Competent Person Assignment, and Competent Person Qualifications; G

Statement of CIH Assignment, and CIH Qualifications; G

Statement Identifying and Designating Roles and Responsibilities for CP, CIH, SSHO, and all Other Parties Involved in Heavy Metals Management; G

Training Certification; G

Occupational and Environmental Assessment Data Report; G

Medical Examinations; G

Lead, Cadmium, and Chromium (VI) Waste Management Plan; G

Licenses, Permits and Notifications; G

Occupant Protection Plan; G

Lead, Cadmium, and Chromium (VI) Compliance Plan; G

Lead, Cadmium, and Chromium (VI) Compliance Plan Checklist; G

Waste Characterization Sampling Plan; G

Analytical Results; G

SD-03 Product Data

Respirators; G

Vacuum Filters; G

Negative Air Pressure System; G

Materials and Equipment; G

Expendable Supplies; G

Local Exhaust Equipment; G

Pressure Differential Automatic Recording Instrument; G

Pressure Differential Log; G

SD-06 Test Reports

Sampling and Analysis; G

Occupational and Environmental Assessment Data Report; G

Sampling Results; G

Pressure Differential Recordings For Local Exhaust System; G

SD-07 Certificates

Competent Person

Testing Laboratory; G

Occupant Notification; G

Clearance Certification; G

SD-11 Closeout Submittals

Turn-In Documents or Weight Tickets; G

1.5 QUALITY ASSURANCE

1.5.1 Qualifications

1.5.1.1 Competent Person (CP)

Submit name, address, and telephone number of the CP selected to perform responsibilities specified in Section entitled COMPETENT PERSON (CP) RESPONSIBILITIES. Provide documented construction project-related experience with implementation of OSHA's Lead in Construction standard (29 CFR 1926.62), Chromium (VI) standard (29 CFR 1926.1126), and Cadmium standard (29 CFR 1926.1127) which shows ability to assess occupational and environmental exposure to lead, cadmium, and chromium (VI); experience with the use of respirators, personal protective equipment and other exposure reduction methods to protect employee health. Demonstrate a minimum of 5 years experience implementing OSHA's Lead in Construction

standard

(29 CFR 1926.62), Chromium (VI) standard (29 CFR 1926.1126), and Cadmium standard (29 CFR 1926.1127). Submit proper documentation that the CP is trained and licensed and certified in accordance with federal, State and local laws. The competent person must possess the qualifications detailed in Section entitled COMPETENT PERSON herein. Contractor must provide the following:

1. Statement of Competent Person Assignment, and Competent Person
2. Statement of CIH Assignment, and CIH Qualifications
3. Statement Identifying and Designating Roles and Responsibilities for CP, CIH, SSHO, and all Other Parties Involved in Heavy Metals Management

#### 1.5.1.2 Training Certification

Submit a current certificate for each worker and supervisor, signed and dated by the accredited training provider, stating that the employee has received the required lead, cadmium and chromium (VI) training specified in 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 and 40 CFR 745 and is certified to perform or supervise any and all work performed under these specifications.

#### 1.5.1.3 Testing Laboratory

Submit the name, address, and telephone number of the testing laboratory selected to perform the air, soil, and wipe, analysis, testing, and reporting of airborne concentrations of lead, cadmium and chromium (VI). Use a laboratory participating in the EPA National Lead Laboratory Accreditation Program (NLLAP) by being accredited by either the American Association for Laboratory Accreditation (A2LA) or the American Industrial Hygiene Association (AIHA) and that is successfully participating in the Environmental Lead Proficiency Analytical Testing (ELPAT) program to perform sample analysis. Laboratories selected to perform blood lead analysis must be OSHA approved. Maine Department of Environmental Protection certification, National Environmental Laboratories Accreditation Conference (NELAC) certification, and certifications for all corresponding EPA test methods are required for waste determination testing, as applicable.

#### 1.5.2 Requirements

##### 1.5.2.1 Competent Person (CP) Responsibilities

- a. Verify training meets all federal, State, and local requirements.
- b. Review and approve Lead, Cadmium, and Chromium (VI) Compliance Plan for conformance to the applicable referenced standards.
- c. Continuously inspect LBP/PWL or MCL work for conformance with the applicable specifications and approved plan. Direct Contractor to immediately correct any non-conformance; Direct Contractor to Stop Work if non-conformance continues. Notify Contractor and Contracting Officer if non-conforming work is not corrected and/or stopped.
- d. Perform air sampling. Recommend upgrades or downgrades (whichever is appropriate based on exposure) on the use of PPE (respirators included) and engineering controls.

- e. Ensure work is performed in strict accordance with specifications at all times. Direct Contractor to immediately correct any non-conformance; Direct Contractor to Stop Work if non-conformance continues. Notify Contractor and Contracting Officer if non-conforming work is not corrected and/or stopped.
- f. Control work to prevent hazardous exposure to human beings, to adjacent areas, and to the environment at all times.
- g. Supervise final cleaning of the lead, cadmium, and chromium (VI) control area, take clearance wipe samples; review clearance sample results with respect to established clearance thresholds, determine requirements for additional cleaning, and advise the Contractor and Contracting Officer of the results and need for cleaning within the time-frames established in these specifications. Direct the Contractor to Stop Work in subject area pending achievement of clearance thresholds.
- h. Certify the conditions of the work as called for elsewhere in this specification.
- i. The CP must be certified pursuant to 40 CFR 745, Section 226 and is responsible for development and implementation of the occupant protection plan, all summary and status reports, and supervision of any and all work performed under these specifications.

#### 1.5.2.2 Lead, Cadmium, and Chromium (VI) Compliance Plan

Submit a detailed job-specific plan of the work procedures to be used in the disturbance of lead, cadmium and chromium (VI), LBP/PWL or MCL. Include in the plan a sketch showing the location, size, and details of lead, cadmium, and chromium (VI) control areas, critical barriers, physical boundaries, location and details of decontamination facilities, viewing ports, and mechanical ventilation system. Include a description of equipment and materials, work practices, controls and job responsibilities for each activity of any and all work performed under these specifications. Include in the plan, eating, drinking, smoking, hygiene facilities and sanitary procedures, interface of trades, sequencing of lead, cadmium, and chromium (VI) related work, collected waste water and dust containing lead, cadmium, and chromium (VI) and debris, air sampling, respirators, personal protective equipment, and a detailed description of the method of containment of the operation to ensure that lead, cadmium, and chromium (VI) is not released outside of the lead, cadmium, and chromium (VI) control area. Include site preparation, cleanup and clearance procedures. Include occupational and environmental sampling, training and strategy, sampling and analysis strategy and methodology, frequency of sampling, duration of sampling, and qualifications of sampling personnel in the sampling portion of the plan. Include a description of arrangements made among contractors on multicontractor worksite's to inform affected employees and to clarify responsibilities to control exposures. If work will occur on a historic building, then include in the plan a description of how the proposed activities will comply with other relevant specifications to avoid or minimize impacts to historic fabric (for example, but not limited to, specification Sections related to historic masonry).

The plan must be developed and signed by a CIH or, as appropriate, a certified Lead Project Designer in the State of Maine. The plan must include the name and certification number of the person signing the plan.

In occupied buildings, the plan must also include an occupant protection program that describes the measures that will be taken during the work to notify and protect the building occupants.

#### 1.5.2.3 Lead, Cadmium, and Chromium (VI) Compliance Plan Checklist

The Contractor must complete the Lead, Cadmium, and Chromium (VI) Compliance Plan Checklist included as Attachment A at the end of this specification. The Checklist must be signed by the properly-certified person preparing the Plan and must be submitted to the Government for review and approval with the Lead, Cadmium, and Chromium (VI) Compliance Plan.

#### 1.5.2.4 Occupational and Environmental Assessment Data Report

If initial monitoring is necessary, submit occupational and environmental sampling results to the Contracting Officer and PD&C HAZMAT POC within three working days of collection, signed by the testing laboratory employee performing the analysis, the employee that performed the sampling, and the CP.

In order to reduce the full implementation of 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 the Contractor must provide documentation. Submit a report that supports the determination to reduce full implementation of the requirements of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and supporting the Lead, Cadmium, and Chromium (VI) Compliance Plan.

- a. The initial monitoring must represent each job classification, or if working conditions are similar to previous jobs by the same employer, provide previously collected exposure data that can be used to estimate worker exposures per 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. The data must represent the worker's regular daily exposure to lead, cadmium, and chromium (VI) for stated work. The initial monitoring must represent each task and/or type of activity. The initial monitoring must represent each/every PWL and/or material containing lead, cadmium, and chromium (VI).
- b. Submit worker exposure data gathered during the task based trigger operations of 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 with a complete process description. This includes, but is not limited to, manual demolition, manual scraping, manual sanding, heat gun, power tool cleaning, rivet busting, cleanup of dry expendable abrasives, abrasive blast enclosure removal, abrasive blasting, welding, cutting and torch burning where lead, cadmium and chromium (VI) containing coatings are present.
- c. The initial assessment must determine the requirement for further monitoring and the need to fully implement the control and protective requirements including the lead, cadmium, and chromium (VI) compliance plan per 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements. The Government will pursue any and all remedies for non-compliance including, but not limited to, stopping work, nonpayment for related work completed, and contractor removal. All delays, concurrent or related to failure to comply with these specifications are entirely the responsibility of the Contractor and shall not be made the subject, or any

component of any request for additional time or compensation.

#### 1.5.2.5 Medical Examinations

Submit a signed letter from an officer of the company confirming each worker engaged in heavy metals related activities in the contract has current medical surveillance as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Where exposure is determined to be above a lead, cadmium, or chromium (VI) action level the contractor must provide pre-work and post-work blood levels as requested by the Government. All medical records submitted shall not contain any Protected Health Information (PHI) (e.g. home address, social security number, physicians name, medical record number, etc.).

Initial medical surveillance as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 must be made available to all employees exposed to lead, cadmium, and chromium (VI) at any time (one day) above the action level. Full medical surveillance must be made available to all employees on an annual basis who are or may be exposed to lead, cadmium and chromium (VI) in excess of the action level for more than 30 days a year or as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Adequate records must show that employees meet the medical surveillance requirements of 29 CFR 1926.33, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 29 CFR 1926.103. Provide medical surveillance to all personnel exposed to lead, cadmium, and chromium (VI) as indicated in 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Maintain complete and accurate medical records of employees for the duration of employment plus 30 years.

#### 1.5.2.6 Training

Train each employee performing any and all work performed under these specifications prior to the time of initial job assignment and annually thereafter, in accordance with 29 CFR 1926.21, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, 40 CFR 745, 49 CFR 172, and State of Maine and local regulations where appropriate.

#### 1.5.2.7 Respiratory Protection Program

- a. Provide each employee required to wear a respirator a respirator fit test at the time of initial fitting and at least annually thereafter as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.
- b. Establish and implement a respiratory protection program as required by 29 CFR 1926.103, 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, and 29 CFR 1926.55.\
- c. Provide training and fit test certificates for each employee performing LBP/ MCL/PWL work. Records submitted shall not contain any Protected Health Information (PHI) (e.g. home address, social security number, physicians name, medical record number, etc.).

#### 1.5.2.8 Hazard Communication Program

Establish and implement a Hazard Communication Program as required by 29 CFR 1926.59.

#### 1.5.2.9 Lead, Cadmium, and Chromium (VI) Waste Management

The Lead, Cadmium, and Chromium (VI) Waste Management Plan must comply with applicable requirements of federal, State, and local hazardous and

non-hazardous waste regulations and address:

- a. Identification and classification of wastes associated with the work.
- b. Estimated quantities of wastes to be generated and disposed of.
- c. Names and qualifications (experience and training) of personnel who will be working on-site with waste materials.
- d. List of waste handling equipment to be used in performing the work, to include cleaning, volume reduction, and transport equipment.
- e. Spill prevention, containment, and cleanup contingency measures including a health and safety plan to be implemented in accordance with 29 CFR 1926.65.
- f. Work plan and schedule for waste characterization, containment and management. Proper containment of the waste includes using acceptable waste containers, as well as proper marking/labeling of the containers. Clean up and containerize wastes daily.
- g. Include any process that may alter or treat waste. Note: altering or treatment of a hazardous waste is not allowed, including any process that dilutes the waste or renders a hazardous waste non-hazardous waste.
- h. All hazardous waste must be placed in a Shipyard-permitted Hazardous Waste Accumulation Area or Contractors must turn hazardous waste in to a Hazardous Waste Accumulation Area not later than the end of the shift on which it is generated. Responsibility for compliance is upon the Contractor. All hazardous wastes generated within the confines of the Shipyard are disposed of by the Government. Accordingly, all hazardous wastes generated by the Contractor to accomplish requirements of this contract will be considered Government-generated, and disposed of by the Government. Contractor shall not bring hazardous wastes onto Government property.

#### 1.5.2.10 Waste Characterization Sampling Plan

A sampling plan must be developed that details practices to be followed for the collection, management, and analysis of waste characterization samples to determine if waste materials generated by the referenced project will be classified as hazardous or non-hazardous to ensure proper disposal. The plan must propose sampling that meets the requirements of all applicable local, state, and federal statutes and regulations, Navy and PNSY requirements, and project specifications. At a minimum, the sampling plan must include:

- a. Confirmation that the sampling plan will be reviewed and approved by the Navy (Code 106.3) prior to conducting any sampling. Code 106.3 review period is fourteen days from receipt of the plan.
- b. The timing of waste characterization sampling and analysis. To the extent feasible, waste sampling and analysis must be completed prior to the accomplishment of any project work to properly characterize waste and ensure hazardous waste is not commingled with non-hazardous waste.
- c. A project summary with a brief description of the project, a

description of the anticipated waste and how the waste will be generated, a map or diagram with sample locations and type, an estimate of the amount of waste to be generated, and the name and location of the disposal facility the waste will go to if waste is to be disposed of off-site, as approved by Code 106.3.

- d. A statement in the plan that the sample collection, management, and analysis will be performed per all applicable local, state, and federal statutes and regulations, project specifications, Navy and PNSY requirements to include NAVSHIPYD PTSMHINST 5090.8 (latest revision) and other PNSY guidance as applicable.
- e. The proposed date(s) for sampling and expected duration of the sampling event(s). All waste sampling requires Code 106.3 oversight with seven (7) calendar day notification prior to a sampling event.
- f. A narrative of how the sample(s) will be taken (e.g. discrete, composite, grab), a list of all equipment to be used for sampling including PPE, sampling instruments, containers; a description of the decontamination process for reusable equipment between consecutive samples, documentation that the Environmental Field Technician (EFT) conducting the waste sampling has training in proper collection and decontamination procedures, and a statement giving rationale for why the sample(s) will be representative.
- g. A listing of the constituents to be analyzed for, identifying the analytical tests methods to be performed with the corresponding EPA test methods numbers. At a minimum, the waste characterization must include laboratory analysis for RCRA 8 Metals via the Toxicity Characteristic Leaching Procedure (TCLP). Depending on the methods used in project implementation, additional waste characterization analysis may be required (for example, waste characterization analysis for corrosivity may be required if high pH solutions are used for paint removal).
- h. The name and location of the laboratory the samples will be sent to and confirmation that the selected lab has Maine Department of Environmental Protection certification, National Environmental Laboratories Accreditation Conference (NELAC) certification and certifications for all corresponding EPA test methods.
- i. Confirmation that the samples will be submitted, under strict chain-of-custody (COC) requirements, to the certified laboratory.
  - 1. Laboratory analytical results.
  - 2. Field log book entries.
  - 3. Copy of the Chain of Custody (COC).
  - 4. Pertinent diagrams and maps.
  - 5. All other supporting data.
- j. A statement that all materials used in the collection of the samples (gloves, wipes, trash, etc.) and hazardous debris generated by the sampling process will be containerized in a bag and delivered to Building 357 for disposal by the end of the shift generated.



- k. Confirmation that the sampling procedure will include, but not be limited to, the following:
- l. The Contractor will notify the Contracting Officer's Representative and PNSY Code 106.3 Sampling Program Manager seven (7) calendar days prior to each sampling event to ensure the area and materials to be sampled are witnessed by all parties.
1. A narrative of how the sample(s) will be taken (e.g. discrete, composite, grab), a list of all equipment to be used for sampling including PPE, sampling instruments, containers; a description of the decontamination process for reusable equipment between consecutive samples, documentation that the Environmental Field Technician (EFT) conducting the waste sampling has training in proper collection and decontamination procedures, and a statement giving rationale for why the sample(s) will be representative.
  2. The EFT taking the samples will inspect the work area/waste with Code 106.3 to visually and tactilely assess the materials to be sampled.
  3. Using a field log, the EFT will record the locations, types, quantities and other salient details related to materials to be sampled.
  4. The EFT will collect sufficient sample size of the materials from the project work area to complete all required analytical testing. Samples must be representative of all materials that comprise the waste for disposal.
  5. The EFT will place the sample in a clean jar or vial, secure the jar or vial closed and label it with a unique sample number.
  6. The EFT will record the sample number on the COC and note the location of the sample in the field log and on the site drawing.
  7. The EFT will verify that the sample number(s) on the jar/vial are accurate and consistent with the COC.
  8. The EFT will place the secured and labeled jar/vial into a cooler and arrange for prompt delivery of bulk samples to the analytical laboratory.
  9. The EFT will clean up any debris generated by the sampling process.
  10. The EFT will decontaminate the collection tool using wet wipes, water and paper towels after each sample is collected, or other approved method. Razor blades will be replaced on a periodic basis as determined by the EFT.
  11. All samples will be collected in a manner to eliminate cross-contamination by other samples.
  12. All materials used in the collection of the samples (gloves, wipes, trash, etc.) and hazardous debris generated by the sampling process will be containerized in a bag and delivered to Building 357 for disposal by the end of the shift generated.
  13. The EFT will obtain the following sampling equipment to perform

the bulk sampling.

- (a) Individual sample jars and vials;
- (b) Field log;
- (c) Site diagram;
- (d) Chain-of-Custody (COC) forms;
- (e) Nitrile gloves;
- (f) Stainless steel or plastic soil scoops;
- (g) Metal chisels of assorted sizes;
- (h) Razor knives and replacement blades;
- (i) Water and spray bottle;
- (j) Trash bags;
- (k) Disposable cleaning wipes;
- (l) Paper towels;
- (m) Duct tape;
- (n) Flashlight;
- (o) Personnel lifts; and
- (p) Other equipment as necessary.

#### 1.5.2.11 Environmental, Safety and Health Compliance

In addition to the detailed requirements of this specification, comply with laws, ordinances, rules, and regulations of federal, State, and local authorities regarding lead, cadmium and chromium. Comply with the applicable requirements of the current issue of 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127, EM 385-1-1, and ND OPNAVINST 5100.23. Submit matters regarding interpretation of standards to the Contracting Officer for resolution before starting work. Where specification requirements and the referenced documents vary, the most stringent requirements apply.

#### 1.5.3 Pressure Differential Recordings for Local Exhaust System

Provide a local exhaust system that creates a negative pressure of at least 0.02 inches of water relative to the pressure external to the enclosure and operate it continuously, 24-hours a day, until the temporary enclosure of the lead, cadmium, and chromium (VI) control area is removed. Submit pressure differential recordings for each work day to the Contracting Officer and PD&C HAZMAT POC within 24-hours from the end of each work day. Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements. The Government will pursue any and all remedies for non-compliance including, but not limited to, stopping work, nonpayment for related work completed, and contractor removal. All delays, concurrent

or related to failure to comply with these specifications are entirely the responsibility of the Contractor and shall not be made the subject , or any component of any request for additional time or compensation.

#### 1.5.4 Licenses, Permits and Notifications

Certify and submit in writing to the Maine Department of Environmental Protection and the Contracting Officer at least 10 days prior to the commencement of work that licenses, permits, and notifications have been obtained. All associated fees or costs incurred in obtaining the licenses, permits and notifications are included in the Contract price.

#### 1.5.5 Occupant Protection Plan

The CP must develop and implement an Occupant Protection Plan describing the measures and management procedures to be taken during any and all work performed under these specifications to protect the building occupants/building facilities and the outside environment from exposure to any lead, cadmium and chromium (VI) impacts while any and all work is performed under these specifications.

#### 1.5.6 Pre-Construction Conference

Along with the CP, meet with the Contracting Officer to discuss in detail the Lead, Cadmium, and Chromium (VI) Waste Management Plan and the Lead, Cadmium, and Chromium (VI) Compliance Plan, including procedures and precautions for the work, roles and responsibilities, monitoring and sampling requirements, and Stop Work conditions.

### 1.6 EQUIPMENT

#### 1.6.1 Respirators

Furnish appropriate respirators approved by the National Institute for Occupational Safety and Health (NIOSH), Department of Health and Human Services, for use in atmospheres containing lead, cadmium and chromium (VI) dust, fume and mist. Respirators must comply with the requirements of 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

#### 1.6.2 Special Protective Clothing

Personnel exposed to lead, cadmium, and chromium (VI) must wear proper disposable protective whole body clothing, head covering, gloves, eye, and foot coverings as required by 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127. Furnish proper disposable plastic or rubber gloves to protect hands. Reduce the level of protection only after obtaining approval from the CP.

#### 1.6.3 Rental Equipment Notification

If rental equipment is to be used during any and all work performed under these specifications, notify the rental agency in writing concerning the intended use of the equipment.

#### 1.6.4 Vacuum Filters

Utilize UL-Labeled HEPA filters for any and all work performed under these specifications.

#### 1.6.5 Equipment for Government Personnel

Furnish the Contracting Officer with two complete sets of personal protective equipment (PPE) daily, as required herein, for entry into and inspection of any and all performed under these specifications work within the lead, cadmium and chromium (VI) controlled area. Personal protective equipment must include disposable whole body covering, including appropriate foot, head, eye, and hand protection. PPE remains the property of the Contractor. The Government will provide respiratory protection for the Contracting Officer.

#### 1.6.6 Abrasive Removal Equipment

The use of powered machines for vibrating, sanding, grinding, or abrasive blasting is prohibited unless equipped with local exhaust ventilation systems equipped with high efficiency particulate air (HEPA) filters. The use of abrasive removal equipment is prohibited unless expressly approved by the NAVFAC Cultural Resources (CR) Managers and the Contracting Officer. Documentation of CR Manager approval must be included in the 'Lead, Cadmium, and Chromium (VI) Compliance Plan.

#### 1.6.7 Negative Air Pressure System

##### 1.6.7.1 Minimum Requirements

Do not proceed with work in the area until containment is set up and HEPA filtration systems are in place. The negative air pressure system must meet the requirements of ASSP/SAFE Z9.2 including approved HEPA filters in accordance with UL 586. Negative air pressure equipment must be equipped with new HEPA filters, and be sufficient to maintain a minimum pressure differential of negative 0.02 inch of water column relative to adjacent, unsealed areas. Negative air pressure system minimum requirements are listed as follows:

- a. The unit must be capable of delivering its rated volume of air with a clean first stage filter, an intermediate filter and a primary HEPA filter in place.
- b. The HEPA filter must be certified as being capable of trapping and retaining mono-disperse particles as small as 0.3 micrometers at a minimum efficiency of 99.97 percent.
- c. The unit must be capable of continuing to deliver no less than 70 percent of rated capacity when the HEPA filter is 70 percent full or measures 2.5 inches of water static pressure differential on a magnehelic gauge.
- d. Equip the unit with a manometer-type negative pressure differential monitor with minor scale division of 0.02 inch of water and accuracy within plus or minus 1.0 percent. The manometer must be calibrated daily as recommended by the manufacturer.
- e. Equip the unit with a means for the operator to easily interpret the readings in terms of the volumetric flow rate of air per minute moving through the machine at any given moment.
- f. Equip the unit with an electronic mechanism that automatically shuts the machine off in the event of a filter breach or absence of a filter.

- g. Equip the unit with an audible horn that sounds an alarm when the machine has shut itself off.
- h. Equip the unit with an automatic safety mechanism that prevents a worker from improperly inserting the main HEPA filter.

#### 1.6.7.2 Auxiliary Generator

Provide an auxiliary generator with capacity to power a minimum of 50 percent of the negative air machines at any time during the work. When power fails, the generator controls must automatically start the generator and switch the negative air pressure system machines to generator power. The generator must not present a carbon monoxide hazard to workers.

#### 1.6.8 Vacuum Systems

Vacuum systems must be suitably sized for the project, and filters must be capable of trapping and retaining all mono-disperse particles as small as 0.3 micrometers (mean aerodynamic diameter) at a minimum efficiency of 99.97 percent. Properly dispose of used filters that are being replaced.

#### 1.6.9 Heat Blower Guns

Heat blower guns must be flameless, electrical, paint-softener type with controls to limit temperature to 1,100 degrees F. Heat blower must be (grounded) 120 volts ac, and must be equipped with cone, fan, glass protector and spoon reflector nozzles.

### 1.7 PROJECT/SITE CONDITIONS

#### 1.7.1 Protection of Existing Work to Remain

Perform work without damage or contamination of adjacent areas and/or areas within work area that were required to be protected (i.e. HVAC systems, furniture, equipment, etc.). Where such areas are damaged or impacted, restore work to its original condition or better as determined by the Contracting Officer. Achievement of restoration must be documented via air, wipe and/or soil sampling, as required under the Section entitled FIELD QUALITY CONTROL.

## PART 2 PRODUCTS

### 2.1 MATERIALS AND EQUIPMENT

Keep materials and equipment needed to complete the project available and on the site. Submit a description of the materials and equipment required; including Safety Data Sheets (SDSs) for material brought onsite to perform the work.

#### 2.1.1 Expendable Supplies

Submit a description of the expendable supplies required.

##### 2.1.1.1 Polyethylene Bags

Disposable bags must be polyethylene plastic and be a minimum of 6 mils thick (4 mils thick if double bags are used) or any other thick plastic material shown to demonstrate at least equivalent performance; and capable of being made leak-tight. Leak-tight means that solids, liquids or dust

cannot escape or spill out.

#### 2.1.1.2 Polyethylene Leak-tight Wrapping

Wrapping used to wrap lead, cadmium, and chromium (VI) contaminated debris must be polyethylene plastic that is a minimum of 6 mils thick or any other thick plastic material shown to demonstrate at least equivalent performance.

#### 2.1.1.3 Polyethylene Sheeting

Sheeting must be polyethylene plastic with a minimum thickness of 6 mil, or any other thick plastic material shown to demonstrate at least equivalent performance; and be provided in the largest sheet size reasonably accommodated by the project to minimize the number of seams. Where the project location constitutes an out-of-the-ordinary potential for fire, or where unusual fire hazards cannot be eliminated, provide flame-resistant polyethylene sheets which conform to the requirements of NFPA 701.

#### 2.1.1.4 Tape and Adhesive Spray

Tape and adhesive must be capable of sealing joints between polyethylene sheets and for attachment of polyethylene sheets to adjacent surfaces. After dry application, tape or adhesive must retain adhesion when exposed to wet conditions, including amended water. Tape must be minimum 2 inches wide, industrial strength.

#### 2.1.1.5 Containers

When used, containers must be leak-tight and be labeled in accordance with EPA, DOT and OSHA standards. For containerization of hazardous waste, the Contractor must use containers provided by B357.

#### 2.1.1.6 Chemical Paint Strippers

SDS for all chemicals must be provided to the Government as part of the prework submittals.

Chemical paint strippers must not contain methylene chloride and be formulated to prevent stain, discoloration, or raising of the substrate materials.

#### 2.1.1.7 Chemical Paint Stripper Neutralizer

SDS for all chemicals must be provided to the Government as part of the prework submittals.

Neutralizers for paint strippers must be compatible with the substrate and suitable for use with the chemical stripper that has been applied to the surface.

#### 2.1.1.8 Detergents and Cleaners

SDS for all chemicals must be provided to the Government as part of the prework submittals.

Detergents or cleaning agents must not contain trisodium phosphate and have demonstrated effectiveness in lead, cadmium and chromium (VI) control

work using cleaning techniques specified by HUD 6780 guidelines.

### PART 3 EXECUTION

#### 3.1 PREPARATION

##### 3.1.1 Protection

###### 3.1.1.1 Notification

- a. Notify the Contracting Officer 20 days prior to the start of any and all work performed under these specifications.

- b. Occupant Notification

Provide building occupants with copy of Occupant Protection Plan prepared under Section entitled OCCUPANT PROTECTION PLAN. Where appropriate, submit occupant written acknowledgment of the delivery of lead hazard information pamphlet (EPA 747-K-99-001 "Protect Your Family From Lead in Your Home") prior to commencing the renovation work for each affected unit using language provided in 40 CFR 745 Subpart E.

###### 3.1.1.2 Lead, Cadmium, and Chromium (VI) Control Area

- a. Physical Boundary - Provide physical boundaries around the lead, cadmium, and chromium (VI) control area by roping off the area designated in the work plan or providing curtains, portable partitions or other enclosures to ensure that lead, cadmium and chromium (VI) will not escape outside of the lead, cadmium and chromium (VI) control area. Prohibit all personnel, except those specifically authorized to enter, from accessing the lead, cadmium, and chromium (VI) control areas.
- b. Warning Signs - Provide warning signs at approaches to lead, cadmium, and chromium (VI) control areas. Locate signs at such a distance that personnel may read the sign and take the necessary precautions before entering the area. Signs must comply with the requirements of 29 CFR 1926.62.

###### 3.1.1.3 Furniture and Equipment :

Furniture and equipment will remain in the lead, cadmium, and chromium (VI) control area. Protect and cover furnishings and equipment or remove furnishings and equipment from the work area and store in a location approved by the Contracting Officer.

Furniture and equipment will be removed by the government from the lead, cadmium, and chromium (VI) control area.

Existing furniture and equipment is lead, cadmium and chromium (VI) contaminated, decontaminate and dispose of as lead, cadmium, and chromium (VI) contaminated waste.

###### 3.1.1.4 Heating, Ventilating and Air Conditioning (HVAC) Systems

Shut down, lock out, and isolate HVAC systems that supply, exhaust, or pass through the lead, cadmium, and chromium (VI) control areas. Seal intake and exhaust vents in the lead, cadmium, and chromium (VI) control

area with 6 mil plastic sheet and tape. Seal seams in HVAC components that pass through the lead, cadmium, and chromium (VI) control area. Provide temporary HVAC system for areas in which HVAC has been shut down outside the lead, cadmium, and chromium (VI) control area.

#### 3.1.1.5 Local Exhaust System

Provide a local exhaust system in the lead, cadmium, and chromium (VI) control area in accordance with ASSP/SAFE Z9.2, 29 CFR 1926.62, 29 CFR 1926.1126 and 29 CFR 1926.1127 that will provide at least 4 air changes per hour inside of the negative pressure enclosure. Local exhaust equipment must be operated 24-hours per day, until all wipe testing clearance thresholds have been achieved and the lead, cadmium, and chromium (VI) control area is removed. The enclosure must be leak proof and equipped with HEPA filters. Maintain a minimum pressure differential in the lead, cadmium, and chromium (VI) control area of negative 0.02 inch of water column relative to adjacent, unsealed areas. Provide continuous 24-hour per day monitoring of the pressure differential with a pressure differential automatic recording instrument. The building ventilation system must not be used as the local exhaust system for the lead, cadmium, and chromium (VI) control area. Filters on exhaust equipment must conform to ASSP/SAFE Z9.2 and UL 586. Terminate the local exhaust system out of doors and remote from any public access or ventilation system intakes.

#### 3.1.1.6 Negative Air Pressure System Containment

- a. Operate the negative air pressure systems to provide at least 4 air changes per hour inside the containment. Operate the local exhaust unit equipment continuously until all wipe testing clearance thresholds have been achieved and the containment is removed. Smoke test the negative air pressure system for leaks at the beginning of each shift. The certified supervisor, under the oversight of the CP, is responsible to continuously monitor and keep a pressure differential log with an automatic manometric recording instrument. Stop work and notify the Contracting Officer immediately if the pressure differential falls below the prescribed minimum. Submit the continuously monitored pressure differential log, as specified. Do not use the building ventilation system as the local exhaust system. Terminate the local exhaust system out of doors unless the Contracting Officer allows an alternate arrangement. All filters must be new at the beginning of the project and be periodically changed as necessary to maintain specified pressure differential and disposed of as lead, cadmium and chromium (VI) contaminated waste.
- b. Discontinuing Negative Air Pressure System. Operate the negative air pressure system continuously during work activities unless otherwise authorized by the Contracting Officer. At the completion of the project, units must be run until full cleanup has been completed and final clearance testing requirements have been met. Protocols for dismantling of the negative air pressure systems must be as presented in the Lead, Cadmium, and Chromium (VI) Compliance Plan. Seal the HEPA filter machine intakes with polyethylene to prevent environmental contamination.

Failure to submit the required data within the time frame detailed in the specifications will be considered non-compliance with project Contract requirements. The Government will pursue any and all remedies for non-compliance including, but not limited to, stopping work, nonpayment for related work completed, and contractor removal. All delays, concurrent or



related to failure to comply with these specifications are entirely the responsibility of the Contractor and shall not be made the subject , or any component of any request for additional time or compensation.

#### 3.1.1.7 Decontamination Shower Facility

If the potential for exceedance of the PEL for lead, cadmium and/or chromium (VI) exists during work performed under these specifications, the decontamination facility must include a shower. The shower must be located within the entry and exit of the containment area, where necessary.

Provide clean and contaminated change rooms and shower facilities in accordance with this specification and 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127.

#### 3.1.1.8 Eye Wash Station

Provide suitable facilities within the work area for quick drenching or flushing of the eyes where eyes may be exposed to injurious corrosive materials.

#### 3.1.1.9 Mechanical Ventilation System

Provide a local exhaust system in the lead, cadmium, and chromium (VI) control area in accordance with ASSP/SAFE Z9.2, 29 CFR 1926.62, 29 CFR 1926.1126 and 29 CFR 1926.1127 that will provide at least 4 air changes per hour inside of the negative pressure enclosure. Local exhaust equipment must be operated 24-hours per day, until all wipe testing clearance thresholds have been achieved and the lead, cadmium, and chromium (VI) control area is removed. The enclosure must be leak proof and equipped with HEPA filters. Maintain a minimum pressure differential in the lead, cadmium, and chromium (VI) control area of negative 0.02 inch of water column relative to adjacent, unsealed areas. Provide continuous 24-hour per day monitoring of the pressure differential with a pressure differential automatic recording instrument. The building ventilation system must not be used as the local exhaust system for the lead, cadmium, and chromium (VI) control area. Filters on exhaust equipment must conform to ASSP/SAFE Z9.2 and UL 586. Terminate the local exhaust system out of doors and remote from any public access or ventilation system intakes.

#### 3.1.1.10 Personnel Protection

Personnel must wear and use protective clothing and equipment as specified herein. Eating, smoking, or drinking or application of cosmetics is not permitted in the lead, cadmium, and chromium (VI) control area. No one will be permitted in the lead, cadmium, and chromium (VI) control area unless they have been appropriately trained and provided with protective equipment.

### 3.2 ERECTION

#### 3.2.1 Lead, Cadmium, and Chromium (VI) Control Area Requirements

Establish a lead, cadmium, and chromium (VI) control area by completely establishing barriers and physical boundaries around the area or structure where any and all work under these specifications will be performed that will fully control paint chips, dust and any other materials containing lead, cadmium, and chromium (VI).

Full containment - Contain operations by the use of critical barriers and

HEPA filtered exhaust or a negative pressure enclosure system with decontamination facilities and with HEPA filtered exhaust where work practices will create airborne dust containing lead, cadmium, chromium (VI) (such as sanding, sawing, grinding, thermal cutting, or digging or demolition activities) as required by the CP. For containment areas larger than 1,000 square feet install a minimum of two 18-inch square viewing ports. Locate ports to provide a view of the required work from the exterior of the enclosed contaminated area. Glaze ports with laminated safety glass.

### 3.3 APPLICATION

#### 3.3.1 Lead, Cadmium, and Chromium (VI) Work

Perform any and all work under these specifications in accordance with approved Lead, Cadmium, and Chromium (VI) Compliance Plan. Use procedures and equipment required to limit occupational exposure, impacts to building interiors/exterior, and environmental contamination with lead, cadmium, and chromium (VI) when the work is performed in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 or 40 CFR 745 and as specified herein. Dispose of all PWL or MCL and associated waste in compliance with these specifications, Navy and PNSY requirements, and federal, State, and local requirements.

#### 3.3.2 Removal of Materials with Lead, Cadmium, and Chromium (VI) or Material Containing Lead, Cadmium, and Chromium (VI)

Manual or power sanding or grinding of lead, cadmium, and chromium (VI) surfaces or materials is not permitted unless using wet methods and/or tools equipped with HEPA attachments. The dry sanding or grinding of surfaces that contain lead, cadmium, and chromium (VI) is prohibited. The use of abrasive removal equipment is prohibited unless expressly approved by the NAVFAC Cultural Resources (CR) Managers and the Contracting Officer. Documentation of CR Manager approval must be included in the Lead, Cadmium, and Chromium (VI) Compliance Plan. Provide methodology for removing lead, cadmium, and chromium (VI) in the Lead, Cadmium, and Chromium (VI) Compliance Plan. Select lead, cadmium, and chromium (VI) removal processes to prevent contamination of work areas outside the control area and/or areas that are required to be protected (i.e. HVAC systems, furniture, equipment, etc.), with lead, cadmium, and chromium (VI) contaminated dust or other lead, cadmium, and chromium (VI) contaminated debris or waste and to ensure that unprotected personnel are not exposed to concentrations of lead, cadmium, and chromium (VI) exceeding the applicable Action Levels. Describe this removal process in the Lead, Cadmium, and Chromium (VI) Compliance Plan.

##### 3.3.2.1 Paint with Lead, Cadmium, and Chromium (VI) or Material Containing Lead, Cadmium, and Chromium (VI) - Indoor Removal

Perform manual or mechanical removal and thermal cutting in the lead, cadmium, and chromium (VI) control areas using enclosures, barriers or containments and powered locally exhausted tools equipped with HEPA filters. Collect residue and debris for disposal in accordance with federal, State, and local requirements.

##### 3.3.2.2 Paint with Lead, Cadmium, and Chromium (VI) or Material Containing Lead, Cadmium, and Chromium (VI) - Outdoor Removal

Perform outdoor removal as indicated in Federal, State, and local

regulations and in the Lead, Cadmium, and Chromium (VI) Compliance Plan. The worksite preparation (barriers or containments) must be job dependent and presented in the Lead, Cadmium, and Chromium (VI) Compliance Plan.

### 3.3.3 Personnel Exiting Procedures

Whenever personnel exit the lead, cadmium, and chromium (VI) controlled area, they must perform the following procedures and must not leave the work place wearing any clothing or equipment worn in the control area:

- a. Vacuum all clothing before entering the contaminated change room.
- b. Remove protective clothing in the contaminated change room, and place them in an approved impermeable disposal bag.
- c. Shower.
- d. Change to clean clothes prior to leaving the clean clothes storage area.

## 3.4 FIELD QUALITY CONTROL

### 3.4.1 Tests

#### 3.4.1.1 Air, Wipe and Soil Sampling Before and During Work

Conduct sampling for lead, cadmium, and chromium (VI) in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, and 29 CFR 1926.1127 and as specified herein. Air, wipe and soil sampling must be directed or performed by the CP.

- a. The CP must be on the job site directing and/or performing the air, soil and wipe sampling and inspecting the project work to ensure that the requirements of the contract, local, state and federal statutes and regulations, Navy and PNSY requirements, and project specifications have been satisfied during all work performed under these specifications.
- b. Collect personal air samples on employees who are anticipated to have the greatest risk of exposure as determined by the CP. In addition, collect air samples on at least twenty-five percent of the work crew or a minimum of two employees, whichever is greater, during each work shift.
- c. Conduct area air sampling daily, on each shift in which any and all work under these specifications are performed, in areas immediately adjacent to the lead, cadmium and chromium (VI) control area at exhaust points for negative air machines, at entrances to the negative pressure enclosure (NPE), and at other locations determined appropriate for the project and site conditions. Conduct sufficient area monitoring to ensure unprotected personnel are not exposed at or above the applicable Action Levels of 30 micrograms of lead per cubic meter of air or 2.5 micrograms of cadmium/chromium (VI) per cubic meter of air. If 30 micrograms of lead per cubic meter of air or 2.5 micrograms of cadmium/chromium (VI) per cubic meter of air is reached or exceeded, stop work, correct the conditions(s) causing the increased levels. Notify the Contracting Officer and PD&C HAZMAT POC immediately. Determine if condition(s) require any further change in work methods. Work may not resume until the CP and the Contracting Officer give approval.

- d. Submit the analytical results for the personal and area air monitoring samples, signed by the CP, to the Contracting Officer within 72-hours after the air samples are collected.
- e. Before any work begins, the CP must collect and analyze baseline wipe, air, and soil (if bare soil is present during exterior remediation operations) samples in accordance with methods defined by federal, State, and local standards inside and outside of the physical boundary to establish baseline conditions in the facility prior to performance of any and all work under these specifications. Provide Initial Sample Results to the Contracting Officer before work begins.
- f. Surface Wipe Samples- If at any time lead, cadmium and/or chromium (VI) concentrations in area air monitoring samples exceed Action Levels, stop work and collect surface wipe samples exterior to the work area. Samples must be collected at a sufficient frequency to properly evaluate area conditions. Surface wipe sample results must meet criteria in Section entitled CLEARANCE CERTIFICATION. If sample results exceed clearance criteria, then the area must be cleaned until the criteria is met. Work may recommence when the criteria are met.

Failure to submit the required data within the time frames detailed in the specifications will be considered non-compliance with project Contract requirements. The Government will pursue any and all remedies for non-compliance including, but not limited to, stopping work, nonpayment for related work completed, and contractor removal. All delays, concurrent or related to failure to comply with these specifications are entirely the responsibility of the Contractor and shall not be made the subject, or any component of any request for additional time or compensation.

#### 3.4.1.2 Wipe and Soil Sampling After Work

The clearance thresholds for required wipe sampling are as follows:

Target clearance levels- Option 1

- (1) Lead- 400 ug/sq. ft. on floors or horizontal surfaces, or below concentrations detected in wipe samples collected before commencement of work- whichever is lower.
- (2) Chromium (VI)- Below concentrations detected in wipe samples collected before commencement of work.
- (3) Cadmium- Below concentrations detected in wipe samples collected before commencement of work.

Target clearance levels- Option 2

- (1)Lead- 40 ug/sq.ft. on floors; 250 ug/sq.ft. on interior window sills; 400 ug/sq. ft. on window troughs; or below concentrations detected in wipe samples collected before commencement of work- whichever is lower.
- (2)Chromium (VI)- Below concentrations detected in wipe samples collected before commencement of work.
- (3)Cadmium- Below concentrations detected in wipe samples collected before commencement of work.

Target clearance levels- Option 3

(1) Lead- 10 ug/sq.ft. on floors; 100 ug/sq.ft. on interior window sills; 400 ug/sq.ft. on window troughs; or below concentrations detected in wipe samples collected before commencement of work- whichever is lower.

(2) Chromium (VI)- Below concentrations detected in wipe samples collected before commencement of work.

(3) Cadmium- Below concentrations detected in wipe samples collected before commencement of work.

Target clearance levels- Option 4

(1) Lead- Below concentrations detected in wipe samples collected before commencement of work.

(2) Chromium (VI)- Below concentrations detected in wipe samples collected before commencement of work.

(3) Cadmium- Below concentrations detected in wipe samples collected before commencement of work.

The CP must perform surface wipe sampling for clearance certification pursuant to Section entitled SAMPLING AND ANALYSIS herein. Subsequent project activities may not commence in the subject area until clearance thresholds have been achieved. The CP must certify surface wipe sample results collected inside and outside the work area are less than the thresholds detailed in Section entitled CLEARANCE REQUIREMENTS herein.

For exterior work, soil samples taken at the exterior of the work site must be used to determine if soil lead, cadmium, and chromium (VI) concentrations have increased at a statistically significant concentration (significant at the 95 percent confidence limit) from the soil lead, cadmium, and chromium (VI) concentrations prior to the operation. If soil lead, cadmium, and chromium (VI) concentrations either show a statistically significant increase above soil lead, cadmium, and chromium (VI) concentrations prior to work or soil lead, cadmium, and chromium (VI) concentrations above any applicable federal or state standard for lead, cadmium, and chromium (VI) in soil, the soil must be remediated.

For any and all work performed under these specifications, surface wipe and soil sampling must be conducted and clearance determinations made according to the requirements detailed in Section entitled SAMPLING AND ANALYSIS herein and to the work practice standards presented in 40 CFR 745.227.

#### 3.4.1.3 Testing of Material Containing Lead, Cadmium, and Chromium (VI) Residue

Test residue in accordance with 40 CFR 261 for hazardous waste.

### 3.5 CLEANING AND DISPOSAL

#### 3.5.1 Cleanup

Maintain surfaces of the lead, cadmium, and chromium (VI) control area free of accumulations of dust and debris. Restrict the spread of dust and debris; keep waste from being distributed over the work area. Do not dry sweep or use pressurized air to clean up the area. At the end of each

shift and when the the work performed under these specifications has been completed, clean the controlled area of all visible contamination by vacuuming with a HEPA filtered vacuum cleaner, wet mopping the area and wet wiping the area in accordance with these specifications and the approved Lead, Cadmium, and Chromium (VI) Compliance Plan. Reclean areas showing dust or debris. After visible dust and debris is removed, wet wipe and HEPA vacuum all surfaces in the controlled area. If adjacent areas become impacted at any time during the work, clean, visually inspect, and then wipe sample all impacted areas. The CP must then certify in writing that the area has been cleaned of lead, cadmium and chromium (VI) impacts before clearance testing.

#### 3.5.1.1 Clearance Certification

The CP must certify in writing to the Contracting Officer and the PD&C HAZMAT POC that air samples collected outside the lead, cadmium, and chromium(VI) control area during performance of any and all work under these specifications are less than the applicable Action Levels of 30 micrograms of lead per cubic meter of air and less than 2.5 micrograms of cadmium/chromium (VI) per cubic meter of air; the respiratory protection used for the employees was adequate; the work procedures were performed in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127; that there were no visible accumulations of material and dust containing lead, cadmium, and chromium (VI) left in the work site, and all applicable wipe and soil clearance sample analytical results are below applicable clearance thresholds. Do not remove the lead, cadmium, and chromium (VI) control area or roped off boundary and warning signs prior to the Contracting Officer's acknowledgement of receipt and approval of the CP certification. Clearance sampling results must be submitted to NAVFAC within 72 hours after the samples are taken.

#### 3.5.2 Disposal

- a. Dispose of material, whether hazardous or non-hazardous in accordance with local, state, and federal statutes and regulations, Navy and PNSY requirements, and project specifications. Ensure all waste is properly characterized in accordance with the Waste Characterization Sampling Plan developed under Section entitled WASTE CHARACTERIZATION SAMPLING PLAN here. The result of each waste characterization will dictate disposal requirements.
- b. Contractor is responsible for segregation of waste. Collect lead, cadmium, and chromium (VI) contaminated waste, scrap, debris, bags, containers, equipment, and lead, cadmium, and chromium (VI) -contaminated clothing. Label the containers in accordance with 29 CFR 1926.62, 29 CFR 1926.1126, 29 CFR 1926.1127 and 40 CFR 261, 40 CFR 262 and corresponding state regulations.
- c. Accumulate waste materials in U.S. Department of Transportation (49 CFR 178) approved 55 gallon drums or appropriately sized container for smaller volumes. Properly label each drum to identify the type of hazardous material (49 CFR 172). For hazardous waste, the collection container requires marking/labeling in accordance with 40 CFR 262 and corresponding state regulations.
- d. Manage lead, cadmium, and chromium (VI) or lead, cadmium, and chromium (VI) contaminated waste in accordance with 40 CFR 260, 40 CFR 261, 40 CFR 262, 40 CFR 263, 40 CFR 264, and 40 CFR 265.

- e. All lead, cadmium, and chromium (VI) waste generation, management, and disposal will be coordinated with the Shipyard environmental function (Code 106, B357) and the Contracting Officer.

All hazardous waste must be turned in to B357 not later than the end of the shift on which it is generated. Responsibility for compliance is upon the Contractor. All hazardous wastes generated within the confines of the Shipyard are disposed of by the Government. Accordingly, all hazardous wastes generated by the Contractor to accomplish requirements of this contract will be considered Government-generated, and disposed of by the Government. Contractor shall not bring hazardous wastes onto Government property.

Containers positioned within the work area boundaries shall have covers in place whenever containers are not in use.

#### 3.5.2.1 Disposal Documentation

For non-hazardous waste, provide a certificate that the waste was accepted by the disposal facility. Provide turn-in documents or weight tickets for non-hazardous waste disposal.

#### 3.5.2.2 Payment for Non-Hazardous Waste

Payment for disposal of non-hazardous waste will not be made until a signed copy of the manifest from the disposal facility is received and approved by the Contracting Officer. The manifest must detail and certify the amount of non-hazardous waste delivered to the disposal facility.

-- End of Section --

SECTION 03 30 00

CAST-IN-PLACE CONCRETE  
**02/19, CHG 2: 05/21**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

|            |                                                                                                                        |
|------------|------------------------------------------------------------------------------------------------------------------------|
| ACI 117    | (2010; Errata 2011) Specifications for Tolerances for Concrete Construction and Materials and Commentary               |
| ACI 121R   | (2008) Guide for Concrete Construction Quality Systems in Conformance with ISO 9001                                    |
| ACI 301    | (2016) Specifications for Structural Concrete                                                                          |
| ACI 302.1R | (2015) Guide for Concrete Floor and Slab Construction                                                                  |
| ACI 304.2R | (2017) Guide to Placing Concrete by Pumping Methods                                                                    |
| ACI 304R   | (2000; R 2009) Guide for Measuring, Mixing, Transporting, and Placing Concrete                                         |
| ACI 305R   | (2020) Guide to Hot Weather Concreting                                                                                 |
| ACI 306R   | (2016) Guide to Cold Weather Concreting                                                                                |
| ACI 308.1  | (2011) Specification for Curing Concrete                                                                               |
| ACI SP-2   | (2007; Abstract: 10th Edition) ACI Manual of Concrete Inspection                                                       |
| ACI SP-15  | (2011) Field Reference Manual: Standard Specifications for Structural Concrete ACI 301-05 with Selected ACI References |

AMERICAN HARDBOARD ASSOCIATION (AHA)

|            |                                |
|------------|--------------------------------|
| AHA A135.4 | (1995; R 2004) Basic Hardboard |
|------------|--------------------------------|

AMERICAN WELDING SOCIETY (AWS)

|                |                                                    |
|----------------|----------------------------------------------------|
| AWS D1.4/D1.4M | (2011) Structural Welding Code - Reinforcing Steel |
|----------------|----------------------------------------------------|



ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                            |
|-----------------|------------------------------------------------------------------------------------------------------------|
| ASTM A53/A53M   | (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless      |
| ASTM A615/A615M | (2024) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement          |
| ASTM A706/A706M | (2016) Standard Specification for Low-Alloy Steel Deformed and Plain Bars for Concrete Reinforcement       |
| ASTM A775/A775M | (2017) Standard Specification for Epoxy-Coated Steel Reinforcing Bars                                      |
| ASTM A934/A934M | (2016) Standard Specification for Epoxy-Coated Prefabricated Steel Reinforcing Bars                        |
| ASTM C31/C31M   | (2021a) Standard Practice for Making and Curing Concrete Test Specimens in the Field                       |
| ASTM C33/C33M   | (2018) Standard Specification for Concrete Aggregates                                                      |
| ASTM C39/C39M   | (2021) Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens                     |
| ASTM C78/C78M   | (2021) Standard Test Method for Flexural Strength of Concrete (Using Simple Beam with Third-Point Loading) |
| ASTM C94/C94M   | (2023) Standard Specification for Ready-Mixed Concrete                                                     |
| ASTM C136/C136M | (2019) Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates                               |
| ASTM C138/C138M | (2017a) Standard Test Method for Density (Unit Weight), Yield, and Air Content (Gravimetric) of Concrete   |
| ASTM C143/C143M | (2020) Standard Test Method for Slump of Hydraulic-Cement Concrete                                         |
| ASTM C150/C150M | (2024) Standard Specification for Portland Cement                                                          |
| ASTM C172/C172M | (2017) Standard Practice for Sampling Freshly Mixed Concrete                                               |
| ASTM C173/C173M | (2016) Standard Test Method for Air Content of Freshly Mixed Concrete by the Volumetric Method             |
| ASTM C231/C231M | (2017a) Standard Test Method for Air                                                                       |

Content of Freshly Mixed Concrete by the  
Pressure Method

|                   |                                                                                                                        |
|-------------------|------------------------------------------------------------------------------------------------------------------------|
| ASTM C260/C260M   | (2010a; R 2016) Standard Specification for Air-Entraining Admixtures for Concrete                                      |
| ASTM C311/C311M   | (2018) Standard Test Methods for Sampling and Testing Fly Ash or Natural Pozzolans for Use in Portland-Cement Concrete |
| ASTM C330/C330M   | (2017a) Standard Specification for Lightweight Aggregates for Structural Concrete                                      |
| ASTM C494/C494M   | (2024) Standard Specification for Chemical Admixtures for Concrete                                                     |
| ASTM C567/C567M   | (2019) Determining Density of Structural Lightweight Concrete                                                          |
| ASTM C595/C595M   | (2020) Standard Specification for Blended Hydraulic Cements                                                            |
| ASTM C618         | (2023; E 2023) Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete        |
| ASTM C803/C803M   | (2018) Standard Test Method for Penetration Resistance of Hardened Concrete                                            |
| ASTM C845/C845M   | (2018) Standard Specification for Expansive Hydraulic Cement                                                           |
| ASTM C873/C873M   | (2015) Standard Test Method for Compressive Strength of Concrete Cylinders Cast in Place in Cylindrical Molds          |
| ASTM C900         | (2015) Standard Test Method for Pullout Strength of Hardened Concrete                                                  |
| ASTM C920         | (2018) Standard Specification for Elastomeric Joint Sealants                                                           |
| ASTM C989/C989M   | (2022) Standard Specification for Slag Cement for Use in Concrete and Mortars                                          |
| ASTM C1012/C1012M | (2018b) Standard Test Method for Length Change of Hydraulic-Cement Mortars Exposed to a Sulfate Solution               |
| ASTM C1017/C1017M | (2013; E 2015) Standard Specification for Chemical Admixtures for Use in Producing Flowing Concrete                    |
| ASTM C1074        | (2019) Standard Practice for Estimating Concrete Strength by the Maturity Method                                       |
| ASTM C1077        | (2024) Standard Practice for Agencies Testing Concrete and Concrete Aggregates                                         |

for Use in Construction and Criteria for  
Testing Agency Evaluation

|                   |                                                                                                                                                                                  |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C1107/C1107M | (2020) Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)                                                                                               |
| ASTM C1157/C1157M | (2020a) Standard Performance Specification for Hydraulic Cement                                                                                                                  |
| ASTM C1218/C1218M | (2020c) Standard Test Method for Water-Soluble Chloride in Mortar and Concrete                                                                                                   |
| ASTM C1260        | (2021) Standard Test Method for Potential Alkali Reactivity of Aggregates (Mortar-Bar Method)                                                                                    |
| ASTM C1293        | (2008; R 2015) Standard Test Method for Determination of Length Change of Concrete Due to Alkali-Silica Reaction                                                                 |
| ASTM C1567        | (2021) Standard Test Method for Potential Alkali-Silica Reactivity of Combinations of Cementitious Materials and Aggregate (Accelerated Mortar-Bar Method)                       |
| ASTM C1602/C1602M | (2018) Standard Specification for Mixing Water Used in Production of Hydraulic Cement Concrete                                                                                   |
| ASTM C1778        | (2016) Standard Guide for Reducing the Risk of Deleterious Alkali-Aggregate Reaction in Concrete                                                                                 |
| ASTM D412         | (2016; R 2021) Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension                                                                                |
| ASTM D471         | (2016a) Standard Test Method for Rubber Property - Effect of Liquids                                                                                                             |
| ASTM D1751        | (2004; E 2013; R 2013) Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Nonextruding and Resilient Bituminous Types) |
| ASTM D1752        | (2018) Standard Specification for Preformed Sponge Rubber, Cork and Recycled PVC Expansion Joint Fillers for Concrete Paving and Structural Construction                         |
| ASTM D2628        | (1991; R 2016) Standard Specification for Preformed Polychloroprene Elastomeric Joint Seals for Concrete Pavements                                                               |
| ASTM D2835        | (1989; R 2017) Standard Specification for Lubricant for Installation of Preformed Compression Seals in Concrete Pavements                                                        |

|                   |                                                                                                                                                                         |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D5759        | (2012; R 2020) Characterization of Coal Fly Ash and Clean Coal Combustion Fly Ash for Potential Uses                                                                    |
| ASTM D6690        | (2015) Standard Specification for Joint and Crack Sealants, Hot Applied, for Concrete and Asphalt Pavements                                                             |
| ASTM E96/E96M     | (2024) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials                                                                |
| ASTM E329         | (2023) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection                                                           |
| ASTM E1155        | (2020) Standard Test Method for Determining Floor Flatness and Floor Levelness Numbers                                                                                  |
| ASTM E1643        | (2018a) Standard Practice for Selection, Design, Installation, and Inspection of Water Vapor Retarders Used in Contact with Earth or Granular Fill Under Concrete Slabs |
| ASTM E1745        | (2017) Standard Specification for Water Vapor Retarders Used in Contact with Soil or Granular Fill under Concrete Slabs                                                 |
| ASTM E1993/E1993M | (1998; R 2020) Standard Specification for Bituminous Water Vapor Retarders Used in Contact with Soil or Granular Fill Under Concrete Slabs                              |

CONCRETE REINFORCING STEEL INSTITUTE (CRSI)

|            |                                                    |
|------------|----------------------------------------------------|
| CRSI 10MSP | (2018) Manual of Standard Practice                 |
| CRSI RB4.1 | (2016) Supports for Reinforcement Used in Concrete |

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

|           |                                                                   |
|-----------|-------------------------------------------------------------------|
| NIST PS 1 | (2009) DOC Voluntary Product Standard PS 1-07, Structural Plywood |
|-----------|-------------------------------------------------------------------|

U.S. ARMY CORPS OF ENGINEERS (USACE)

|               |                                                                           |
|---------------|---------------------------------------------------------------------------|
| COE CRD-C 104 | (1980) Method of Calculation of the Fineness Modulus of Aggregate         |
| COE CRD-C 513 | (1974) Corps of Engineers Specifications for Rubber Waterstops            |
| COE CRD-C 572 | (1974) Corps of Engineers Specifications for Polyvinylchloride Waterstops |

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FS SS-S-200

(Rev E; Am 1; Notice 1) Sealant, Joint,  
Two-Component, Jet-Blast-Resistant,  
Cold-Applied, for Portland Cement Concrete  
Pavement

U.S. GREEN BUILDING COUNCIL (USGBC)

LEED NC

(2009) Leadership in Energy and  
Environmental Design(tm) New Construction  
Rating System

1.2 DEFINITIONS

- a. "Cementitious material" as used herein must include all portland cement, pozzolan, fly ash, slag cement, and silica fume.
- b. "Exposed to public view" means situated so that it can be seen from eye level from a public location after completion of the building. A public location is accessible to persons not responsible for operation or maintenance of the building.
- c. "Chemical admixtures" are materials in the form of powder or fluids that are added to the concrete to give it certain characteristics not obtainable with plain concrete mixes.
- d. "Supplementary cementing materials" (SCM) include coal fly ash, silica fume, slag cement, natural or calcined pozzolans, and ultra-fine coal ash when used in such proportions to replace the portland cement that result in improvement to sustainability and durability and reduced cost.
- e. "Design strength" (f'c) is the specified compressive strength of concrete at time(s) specified in this section to meet structural design criteria.
- f. "Mass Concrete" is any concrete system that approaches a maximum temperature of 158 degrees F within the first 72 hours of placement. In addition, it includes all concrete elements with a section thickness of 3 feet or more regardless of temperature.
- g. "Mixture proportioning" is the process of designing concrete mixture proportions to enable it to meet the strength, service life and constructability requirements of the project while minimizing the initial and life-cycle cost.
- h. "Mixture proportions" are the masses or volumes of individual ingredients used to make a unit measure (cubic meter or cubic yard) of concrete.
- i. "Pozzolan" is a siliceous or siliceous and aluminous material, which in itself possesses little or no cementitious value but will, in finely divided form and in the presence of moisture, chemically react with calcium hydroxide at ordinary temperatures to form compounds possessing cementitious properties.
- j. "Workability (or consistence)" is the ability of a fresh (plastic) concrete mix to fill the form/mould properly with the desired work

(vibration) and without reducing the concrete's quality. Workability depends on water content, chemical admixtures, aggregate (shape and size distribution), cementitious content and age (level of hydration).

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-01 Preconstruction Submittals

- Concrete Curing Plan
- Quality Control Plan; G
- Quality Control Personnel Certifications; G
- Quality Control Organizational Chart
- Laboratory Accreditation; G
- Form Removal Schedule; G
- Maturity Method Datah

#### SD-02 Shop Drawings

- Formwork
- Reinforcing Steel; G

#### SD-03 Product Data

- Joint Sealants; (LEED NC)
- Joint Filler; (LEED NC)
- Formwork Materials
- Cementitious Materials; (LEED NC)
- Vapor Retarder and Vapor Barrier
- Concrete Curing Materials
- Reinforcement; (LEED NC)
- Admixtures

Waterstops

Biodegradable Form Release Agent

Pumping Concrete

Finishing PlanNonshrink Grout

SD-05 Design Data

Concrete Mix Design; G

SD-06 Test Reports

Concrete Mix Design; G

Fly Ash

Pozzolan

Slag Cement

Aggregates

Tolerance ReportCompressive Strength Tests; G

Unit Weight of Structural Concrete

Air ContentSlump Tests

Water

SD-07 Certificates

Reinforcing Bars

Silica Fume Manufacturer's Representative

VOC Content for Form Release Agents, Curing Compounds, and  
Concrete Penetrating SealersSafety Data Sheets

Field Testing Technician and Testing Agency

SD-08 Manufacturer's Instructions

Liquid Chemical Floor Hardeners and Sealers

Joint Sealants; (LEED NC)

Curing Compound

#### 1.4 MODIFICATION OF REFERENCES

Accomplish work in accordance with ACI publications except as modified herein. Consider the advisory or recommended provisions to be mandatory. Interpret reference to the "Building Official," the "Structural Engineer," and the "Architect/Engineer" to mean the Contracting Officer.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

Follow ACI 301, ACI 304R and ASTM A934/A934M requirements and recommendations. Do not deliver concrete until forms, reinforcement, embedded items, and chamfer strips are in place and ready for concrete placement. Do not store concrete curing compounds with materials that have a high capacity to adsorb volatile organic compound (VOC) emissions. Do not store concrete curing compounds or sealers in occupied spaces.

##### 1.5.1 Reinforcement

Store reinforcement of different sizes and shapes in separate piles or racks raised above the ground to avoid excessive rusting. Protect from contaminants such as grease, oil, and dirt. Ensure bar sizes can be accurately identified after bundles are broken and tags removed.

#### 1.6 QUALITY ASSURANCE

##### 1.6.1 Design Data

###### 1.6.1.1 Concrete Mix Design

Sixty days minimum prior to concrete placement, submit a mix design for each strength and type of concrete. Submit a complete list of materials including type; brand; source and amount of cement, supplementary cementitious materials, and admixtures; and applicable reference specifications. Submit mill test and all other test for cement, supplementary cementitious materials, aggregates, and admixtures. Provide documentation of maximum nominal aggregate size, gradation analysis, percentage retained and passing sieve, and a graph of percentage retained versus sieve size. Provide mix proportion data using at least three different water-cementitious material ratios for each type of mixture, which produce a range of strength encompassing those required for each type of concrete required. If source material changes, resubmit mix proportion data using revised source material. Provide only materials that have been proven by trial mix studies to meet the requirements of this specification, unless otherwise approved in writing by the Contracting Officer. Indicate clearly in the submittal where each mix design is used when more than one mix design is submitted. Resubmit data on concrete components if the qualities or source of components changes. For previously approved concrete mix designs used within the past twelve months, the previous mix design may be re-submitted without further trial batch testing if accompanied by material test data conducted within the last six months. Obtain mix design approval from the contracting officer prior to concrete placement.

##### 1.6.2 Shop Drawings

###### 1.6.2.1 Formwork

Drawings showing details of formwork including, but not limited to; joints, supports, studding and shoring, and sequence of form and shoring



removal. Indicate placement schedule, construction, location and method of forming control joints. Include locations of inserts, conduit, sleeves and other embedded items. Reproductions of contract drawings are unacceptable. Submit form removal schedule indicating element and minimum length of time for form removal.

Design, fabricate, erect, support, brace, and maintain formwork so that it is able to support, without failure, all vertical and lateral loads that may reasonably be anticipated to be applied to the formwork.1.6.2.2  
Reinforcing Steel

Indicate bending diagrams, assembly diagrams, splicing and laps of bars, shapes, dimensions, and details of bar reinforcing, accessories, and concrete cover. Do not scale dimensions from structural drawings to determine lengths of reinforcing bars. Reproductions of contract drawings are unacceptable.

### 1.6.3 Control Submittals

#### 1.6.3.1 Concrete Curing Plan

Submit proposed materials, methods and duration for curing concrete elements in accordance with ACI 308.1.

#### 1.6.3.2 Pumping Concrete

Submit proposed materials and methods for pumping concrete. Submittal must include mix designs, pumping equipment including type of pump and size and material for pipe, and maximum length and height concrete is to be pumped.1.6.3.3 Silica Fume Manufacturer's Representative

The manufacturer's representative must be present at mix plant to ensure proper mix, including high range water reducer, and batching methods during the first 3 days of concrete mix preparation and placement. After which the manufacturer's representative must designate a representative at the concrete producer's plant to ensure the concrete mix procedures meet the silica fume manufacturer's recommendations. Representative to attend and advise at finishing of sample slab.1.6.3.4 Finishing Plan

Submit proposed material and procedures to be used in obtaining the finish for the floors. Include qualification of person to be used for obtaining floor tolerance measurement, description of measuring equipment to be used, and a sketch showing lines and locations the measuring equipment will follow.1.6.3.5 VOC Content for form release agents, curing compounds, and concrete penetrating sealers

Submit certification for the form release agent, curing compounds, and concrete penetrating sealers that indicate the VOC content of each product.1.6.3.6 Safety Data Sheets

Submit Safety Data Sheets (SDS) for all materials that are regulated for hazardous health effects. SDS must be readily accessible during each work shift to employees when they are at the construction site.

### 1.6.4 Test Reports

#### 1.6.4.1 Fly Ash and Pozzolan

Submit test results in accordance with ASTM C618 for fly ash and

pozzolan. Submit test results performed within 6 months of submittal date.

#### 1.6.4.2 Slag Cement

Submit test results in accordance with ASTM C989/C989M for slag cement. Submit test results performed within 6 months of submittal date.

#### 1.6.4.3 Aggregates

Submit test results in accordance with ASTM C33/C33M, or ASTM C330/C330M for lightweight aggregate, and ASTM C1293 or ASTM C1567 as required in the paragraph titled ALKALI-AGGREGATE REACTION.

#### 1.6.5 Field Samples

#### 1.6.6 Quality Control Plan

Develop and submit for approval a concrete quality control program in accordance with the guidelines of ACI 121R and as specified herein. The plan must include approved laboratories. Provide direct oversight for the concrete qualification program inclusive of associated sampling and testing. All quality control reports must be provided to the Contracting Officer, Quality Manager and Concrete Supplier. Maintain a copy of ACI SP-15 and CRSI 10MSP at project site.

#### 1.6.7 Quality Control Personnel Certifications

The Contractor must submit for approval the responsibilities of the various quality control personnel, including the names and qualifications of the individuals in those positions and a quality control organizational chart defining the quality control hierarchy and the responsibility of the various positions. Quality control personnel must be employed by the Contractor.

Submit American Concrete Institute certification for the following:

- a. CQC personnel responsible for inspection of concrete operations.
- b. Lead Foreman or Journeyman of the Concrete Placing, Finishing, and Curing Crews.
- c. Field Testing Technicians: ACI Concrete Field Testing Technician, Grade I.

##### 1.6.7.1 Quality Manager Qualifications

The quality manager must hold a current license as a professional engineer in a U.S. state or territory with experience on at least five similar projects. Evidence of extraordinary proven experience may be considered by the Contracting Officer as sufficient to act as the Quality Manager.

##### 1.6.7.2 Field Testing Technician and Testing Agency

Submit data on qualifications of proposed testing agency and technicians for approval by the Contracting Officer prior to performing testing on concrete.

- a. Work on concrete under this contract must be performed by an ACI Concrete Field Testing Technician Grade 1 qualified in accordance with

ACI SP-2 or equivalent. Equivalent certification programs must include requirements for written and performance examinations as stipulated in ACI SP-2.

- b. Testing agencies that perform testing services on reinforcing steel must meet the requirements of ASTM E329.
- c. Testing agencies that perform testing services on concrete materials must meet the requirements of ASTM C1077.

#### 1.6.8 Laboratory Qualifications for Concrete Qualification Testing

The concrete testing laboratory must have the necessary equipment and experience to accomplish required testing. The laboratory must meet the requirements of ASTM C1077 and be Cement and Concrete Reference Laboratory (CCRL) inspected.

#### 1.6.9 Laboratory Accreditation

Laboratory and testing facilities must be provided by and at the expense of the Contractor. The laboratories performing the tests must be accredited in accordance with ASTM C1077, including ASTM C78/C78M and ASTM C1260. The accreditation must be current and must include the required test methods, as specified. Furthermore, the testing must comply with the following requirements:

- a. Aggregate Testing and Mix Proportioning: Aggregate testing and mixture proportioning studies must be performed by an accredited laboratory and under the direction of a registered professional engineer in a U.S. state or territory competent in concrete materials who is competent in concrete materials and must sign all reports and designs.
- b. Acceptance Testing: Furnish all materials, labor, and facilities required for molding, curing, testing, and protecting test specimens at the site and in the laboratory. Furnish and maintain boxes or other facilities suitable for storing and curing the specimens at the site while in the mold within the temperature range stipulated by ASTM C31/C31M.
- c. Contractor Quality Control: All sampling and testing must be performed by an approved, onsite, independent, accredited laboratory.

### 1.7 ENVIRONMENTAL REQUIREMENTS

#### 1.7.1 Submittals for Environmental Performance

- a. Provide data indication the percentage of post-industrial pozzolan (fly ash, slag cement) cement substitution as a percentage of the full product composite by weight.
- b. Provide data indicating the percentage of post-industrial and post-consumer recycled content aggregate.
- c. Provide product data indicating the percentage of post-consumer recycled steel content in each type of steel reinforcement as a percentage of the full product composite by weight.
- d. Provide product data stating the location where all products were

manufactured

- e. For projects using FSC certified formwork, provide chain-of-custody documentation for all certified wood products.
- f. For projects using reusable formwork, provide data showing how formwork is reused.
- g. Provide SDS product information data showing that form release agents meet any environmental performance goals such as using vegetable and soy based products.
- h. Provide SDS product information data showing that concrete adhesives meet any environmental performance goals including low emitting, low volatile organic compound products.

## PART 2 PRODUCTS

### 2.1 FORMWORK MATERIALS

- a. Form-facing material in contact with concrete must be lumber, plywood, or treated paper that creates specified appearance and texture of concrete surface. Submit product information on proposed form-facing materials if different from that specified herein.
- b. Design formwork, shores, reshores, and backshores to support loads transmitted to them and to comply with applicable building code requirements.
- c. Design formwork and shoring for load redistribution resulting from stressing of post-tensioned reinforcement. Ensure that formwork allows movement resulting from application of prestressing force.
- d. Design formwork to withstand pressure resulting from placement and vibration of concrete and to maintain specified tolerances.
- e. Design formwork to accommodate waterstop materials in joints at locations indicated in Contract Documents.
- f. Provide temporary openings in formwork if needed to facilitate cleaning and inspection.
- g. Design formwork joints to inhibit leakage of mortar.
- h. Limit deflection of facing materials for concrete surfaces exposed to view to 1/240 of center-to-center spacing of facing supports.
- i. Do not use earth cuts as forms for vertical or sloping surfaces.
- j. Submit product information on proposed form-facing materials if different from that specified herein.
- k. Submit shop drawings for formwork, shoring, reshoring, and backshoring. Shop drawings must be signed and sealed by a licensed design engineer.
- l. Submit design calculations for formwork, shoring, reshoring, and backshoring. Design calculations must be signed and sealed by a licensed design engineer.

- m. Submit procedure for reshoring and backshoring, including drawings signed and sealed by a licensed design engineer. Include on shop drawings the formwork removal procedure and magnitude of construction loads used for design of reshoring or backshoring system. Indicate in procedure the magnitude of live and dead loads assumed for required capacity of the structure at time of reshoring or backshoring.
- n. Submit manufacturer's product data on form liner proposed for use with each formed surface.2.1.1 Wood Forms

Use lumber as specified in Section 06 10 00 ROUGH CARPENTRY and as follows. Provide lumber that is square edged or tongue-and-groove boards, free of raised grain, knotholes, or other surface defects. Provide plywood that complies with NIST PS 1, B-B concrete form panels or better or AHA A135.4, hardboard for smooth form lining. Submit data verifying that composite wood products contain no urea formaldehyde resins.2.1.1.1 Concrete Form Plywood (Standard Rough)

Provide plywood that conforms to NIST PS 1, B-B, concrete form, not less than 5/8-inch thick.2.1.1.2 Overlaid Concrete Form Plywood (Standard Smooth)

Provide plywood that conforms to NIST PS 1, B-B, high density form overlay, not less than 5/8-inch thick.2.2 FORMWORK ACCESSORIES

- a. Use commercially manufactured formwork accessories, including ties and hangers.
- b. Form ties and accessories must not reduce the effective cover of the reinforcement.

#### 2.2.1 Form Ties

- a. Use form ties with ends or end fasteners that can be removed without damage to concrete.
- b. Where indicated in Contract Documents, use form ties with integral water barrier plates or other acceptable positive water barriers in walls.
- c. The breakback distance for ferrous ties must be at least 3/4 in. for Surface Finish-2.0 or Surface Finish-3.0, as defined in ACI 301.
- d. If the breakback distance is less than 3/4 in., use coated or corrosion-resistant ties.
- e. Submit manufacturer's data sheet on form ties.

#### 2.2.2 Waterstops

Submit manufacturer's data sheet on waterstop materials and splices.

##### 2.2.2.1 PVC Waterstop

Polyvinylchloride waterstops must conform to COE CRD-C 572.

#### 2.2.2.2 Rubber Waterstop

Rubber waterstops must conform to COE CRD-C 513.

#### 2.2.2.3 Thermoplastic Elastomeric Rubber Waterstop

Thermoplastic elastomeric rubber waterstops must conform to ASTM D471.

#### 2.2.2.4 Hydrophilic Waterstop

Swellable strip type compound of polymer modified chloroprene rubber that swells upon contact with water must conform to the following requirements when tested in accordance to ASTM D412: Tensile strength 420 psi minimum; ultimate elongation 600 percent minimum. Hardness must be 50 minimum on the type A durometer and the volumetric expansion ratio in distilled water at 70 degrees F must be 3 to 1 minimum.

#### 2.2.3 Biodegradable Form Release Agent

- a. Provide form release agent that is colorless, biodegradable, and water-based, with a low (maximum of 55 grams/liter (g/l)) VOC content.
- b. Provide product that does not bond with, stain, or adversely affect concrete surfaces and does not impair subsequent treatments of concrete surfaces.
- c. Provide form release agent that reduces formwork moisture absorption, and does not contain diesel fuel, petroleum-based lubricating oils, waxes, or kerosene. Submit documentation indicating type of biobased material in product and biobased content. Indicate relative dollar value of biobased content products to total dollar value of products included in project.
- d. Submit manufacturer's product data on formwork release agent for use on each form-facing material.

#### 2.2.4 Chamfer Materials

Use lumber materials with dimensions of 3/4 x 3/4 in.

#### 2.2.5 Construction and movement joints

- a. Submit details and locations of construction joints in accordance with the requirements herein.
- b. Locate construction joints within middle one-third of spans of slabs, beams, and girders. If a beam intersects a girder within the middle one-third of girder span, the distance between the construction joint in the girder and the edge of the beam must be at least twice the width of the larger member.
- c. For members with post-tensioning tendons, locate construction joints where tendons pass through centroid of concrete section.
- d. Locate construction joints in walls and columns at underside of slabs, beams, or girders and at tops of footings or slabs.
- e. Make construction joints perpendicular to main reinforcement.

- f. Provide movement joints where indicated in Contract Documents or in accepted alternate locations.
- g. Submit location and detail of movement joints if different from those indicated in Contract Documents.
- h. Submit manufacturer's data sheet on expansion joint materials.
- i. Provide keyways where indicated in Contract Documents.

#### 2.2.6 Other Embedded items

Use sleeves, inserts, anchors, and other embedded items of material and design indicated in Contract Documents.

### 2.3 CONCRETE MATERIALS

#### 2.3.1 Cementitious Materials

##### 2.3.1.1 Portland Cement

- a. Unless otherwise specified, provide cement that conforms to ASTM C150/C150M Type Either I II .
- b. Use one brand and type of cement for formed concrete having exposed-to-view finished surfaces.
- c. Supplier must certify that no hazardous waste is used in the fuel mix or raw materials.
- d. Submit information along with evidence demonstrating compliance with referenced standards. Submittals must include types of cementitious materials, manufacturing locations, shipping locations, and certificates showing compliance.
- e. Cementitious materials must be stored and kept dry and free from contaminants.

##### 2.3.1.2 Blended Cements

- a. Blended cements must conform to ASTM C595/C595M Type Either IL or ASTM C1157/C1157M Type GU .
- b. Slag cement added to the Type IS blend must meet ASTM C989/C989M.
- c. The pozzolan added to the Type IP blend must be ASTM C618 Class F fly ash and must be interground with the cement clinker. The manufacturer must state in writing that the amount of pozzolan in the finished cement will not vary more than plus or minus 5 mass percent of the finished cement from lot-to-lot or within a lot. The percentage and type of pozzolan used in the blend must not change from that submitted for the aggregate evaluation and mixture proportioning.

##### 2.3.1.3 Fly Ash

- a. ASTM C618, Class F, except that the maximum allowable loss on ignition must not exceed 3 percent.

- b. If fly ash is used it shall range from 15 to 35 percent by weight of cementitious material, provided the fly ash does not reduce the amount of cement in the concrete mix below the minimum requirements of local building codes. Where the use of fly ash cannot meet the minimum level, it shall not be used. Report the chemical analysis of the fly ash in accordance with ASTM C311/C311M. Evaluate and classify fly ash in accordance with ASTM D5759.

#### 2.3.1.4 Slag Cement

ASTM C989/C989M, Grade Either 100 or 120.

#### 2.3.1.5 Silica Fume

#### 2.3.1.6 Other Supplementary Cementitious Materials

Natural pozzolan must be raw or calcined and conform to ASTM C618, Class N, including the optional requirements for uniformity and effectiveness in controlling ASR and must have an ignition loss not exceeding 3 percent. Class N pozzolan for use in mitigating ASR must have a Calcium Oxide (CaO) content of less than 13 percent and total equivalent alkali content less than 3 percent.

Ultra Fine Fly Ash (UFFA) and Ultra Fine Pozzolan (UFP) must conform to ASTM C618, Class F or N, and the following additional requirements:

- a. The strength activity index at 28 days of age must be at least 95 percent of the control specimens.
- b. The average particle size must not exceed 6 microns.
- c. The sum of SiO<sub>2</sub> + Al<sub>2</sub>O<sub>3</sub> + Fe<sub>2</sub>O<sub>3</sub> must be greater than 77 percent.

#### 2.3.2 Water

- a. Water or ice must comply with the requirements of ASTM C1602/C1602M.
- b. Minimize the amount of water in the mix. Improve workability by adjusting the grading of the aggregate and using admixture rather than by adding water.
- c. Water must be potable free from injurious amounts of oils, acids, alkalis, salts, organic materials, or other substances deleterious to concrete.
- d. Protect mixing water and ice from contamination during storage and delivery.
- e. Submit test report showing water complies with ASTM C1602/C1602M.
- f. When nonpotable source is proposed for use, submit documentation on effects of water on strength and setting time in compliance with ASTM C1602/C1602M.

#### 2.3.3 Aggregate

##### 2.3.3.1 Normal-Weight Aggregate

- a. Aggregates must conform to ASTM C33/C33M unless otherwise specified in



the Contract Documents or approved by the contracting officer.

- b. Aggregates used in concrete must be obtained from the same sources and have the same size range as aggregates used in concrete represented by submitted field test records or used in trial mixtures.
- d. Store and handle aggregate in a manner that will avoid segregation and prevents contamination by other materials or other sizes of aggregates. Store aggregates in locations that will permit them to drain freely. Do not use aggregates that contain frozen lumps.
- e. Submit types, pit or quarry locations, producers' names, aggregate supplier statement of compliance with ASTM C33/C33M, and ASTM C1293 expansion data not more than 18 months old.

#### 2.3.4 Admixtures

- a. Chemical admixtures must conform to ASTM C494/C494M.
- b. Air-entraining admixtures must conform to ASTM C260/C260M.
- c. Chemical admixtures for use in producing flowing concrete must conform to ASTM C1017/C1017M.
- d. Do not use calcium chloride admixtures.
- e. Use a corrosion-inhibiting admixture for concrete classified under exposure category C2.
- f. Admixtures used in concrete must be the same as those used in the concrete represented by submitted field test records or used in trial mixtures.
- g. Protect stored admixtures against contamination, evaporation, or damage.
- h. To ensure uniform distribution of constituents, provide agitating equipment for admixtures used in the form of suspensions or unstable solutions. Protect liquid admixtures from freezing and from temperature changes that would adversely affect their characteristics.
- i. Submit types, brand names, producers' names, manufacturer's technical data sheets, and certificates showing compliance with standards required herein.

#### 2.4 MISCELLANEOUS MATERIALS

##### 2.4.1 Concrete Curing Materials

Provide concrete curing material in accordance with ACI 301 Section 5 and ACI 308.1 Section 2. Submit product data for concrete curing compounds. Submit manufactures instructions for placement of curing compound.

##### 2.4.2 Nonshrink Grout

Nonshrink grout in accordance with ASTM C1107/C1107M.

### 2.4.3 Floor Finish Materials

#### 2.4.3.1 Liquid Chemical Floor Hardeners and Sealers

- a. Hardener must be a colorless aqueous solution containing a blend of inorganic silicate or silicate material and proprietary components combined with a wetting agent; that penetrates, hardens, and densifies concrete surfaces. Submit manufacturer's instructions for placement of liquid chemical floor hardener.
- b. Use concrete penetrating sealers with a low (maximum 100 grams/liter, less water and less exempt compounds) VOC content. Submit manufacturer's instructions for placement of sealers.

#### 2.4.3.2 Abrasive Aggregate for Nonslip Aggregate Finish

Aggregate must be packaged, factory-graded fused aluminum oxide grits, or it may be crushed emery containing not less than 40-percent aluminum oxide and not less than 25-percent ferric oxide. Aggregate must be rust proof and nonglazing and must be unaffected by freezing, moisture, and cleaning materials.

Aggregate must be packaged, factory-graded, silicon carbide grits. Aggregate must be rust proof and must be unaffected by freezing, moisture, and cleaning materials.

Aggregate must be well-graded in size from particles retained on No. 30 sieve 0.0236 inch to particles passing No. 8 sieve 0.0929 inch.

#### 2.4.3.3 Dry Materials for Colored Wear-Resistant Finish

Provide materials that are packaged, dry, and a combination of materials formulated for producing colored and wear-resistant monolithic surface treatments; they must include portland cement, graded-quartz aggregate, coloring pigments, and dispersing agents. Provide coloring pigments that are finely ground, nonfacing mineral oxides prepared especially for the purpose and interground with the cement.

#### 2.4.3.4 Aggregate for Heavy-Duty Wear-Resistant Finish

Provide aggregate that is traprock or emery, as follows:

Traprock must be packaged, crushed, natural, fine-to-medium-grained, igneous rock, such as diabase, basalt, or black granite. Traprock aggregate must be well-graded in size from particles retained on No. 4 sieve 0.187 inch to particles passing 3/8-inch sieve.

Emery must be packaged, factory-graded, crushed, natural-emery ore, cubical or polyhedral in form, containing not less than 35-percent aluminum oxide and not less than 24-percent ferric oxide. Emery aggregate must be well graded in size from particles retained on No. 50 sieve 0.0118 inch to particles passing No. 8 sieve 0.0929 inch.

Provide iron aggregate, as follows:

Iron must be packaged, ground and graded cubicle iron particles with dispersing agents, formulated to blend with portland cement for producing wear-resistant monolithic surface treatments. Provide aggregate that is free of nonferrous metals, oil, grease, soluble alkaline compounds, rust,

and impurities and must be well-graded in size from particles retained on No. 50 sieve 0.0118 inch to particles passing No. 8 sieve 0.0929 inch.

#### 2.4.3.5 Aggregate for Heavy-Duty Floor Topping

Provide emery (or may be traprock or traprock-screenings) fine aggregates, as specified.

Provide emery that is packaged, factory-graded, crushed natural emery ore containing not less than 35-percent aluminum oxide and not less than 24-percent ferric oxide. Provide aggregate that is cubical or polyhedral in form and does not change its physical or chemical nature in the presence of moisture. Grade aggregate to a fineness modulus of 3.9 to 4.0, with 100 percent passing 3/8-inch sieve and not less than 95 percent retained on No. 100 sieve. Deliver emery in moisture-resistant bags.

Provide traprock that is packaged, crushed, natural, fine- to medium-grained igneous rock such as diabase, basalt, or black granite. Uniformly grade coarse aggregate with 100 percent passing 1/2-inch sieve, 30 to 50 percent passing 3/8-inch sieve, 0 to 15 percent passing No. 4 sieve, and 0 to 5 percent passing No. 8 sieve.

Provide fine aggregate using traprock that conforms to ASTM C33/C33M, except gradation. Grade fine aggregate within the following limits:

| <u>SIEVE</u> | <u>PERCENT PASSING</u> |
|--------------|------------------------|
| 3/8 in.      | 100                    |
| No. 4        | 95 to 100              |
| No. 8        | 65 to 80               |
| No. 16       | 45 to 65               |
| No. 30       | 25 to 45               |
| No. 50       | 5 to 15                |
| No. 100      | 0 to 5                 |

Deliver traprock coarse aggregate and fine aggregate in moisture-resistant bags.

2.4.4 Expansion/Contraction Joint FillerASTM D1751 or ASTM D1752 Type I or Type II. Material must be 1/2 inch thick, unless otherwise indicated.

#### 2.4.5 Joint Sealants

Submit manufacturer's product data, indicating VOC content.

##### 2.4.5.1 Horizontal Surfaces, 3 Percent Slope, Maximum

ASTM D6690 or ASTM C920, Type M, Class 25, Use T.

##### 2.4.5.2 Vertical Surfaces Greater Than 3 Percent Slope

ASTM C920, Type M, Grade NS, Class 25, Use T NT. FS SS-S-200, no sag.

#### 2.4.5.3 Preformed Polychloroprene Elastomeric Type

ASTM D2628.

#### 2.4.5.4 Lubricant for Preformed Compression Seals

ASTM D2835.

#### 2.4.6 Vapor Retarder and Vapor Barrier

ASTM E1745 Class C A polyethylene sheeting, minimum 10 mil 15 mil thickness or other equivalent material with a maximum permeance rating of 0.04 perms per ASTM E96/E96M.

ASTM E1745 Class C A polyethylene sheeting, minimum 15 mil thickness or ASTM E1993/E1993M bituminous membrane or other equivalent material with a maximum permeance rating of 0.01 perms per ASTM E96/E96M.

Consider plastic vapor retarders and adhesives with a high recycled content, low toxicity low VOC (Volatile Organic Compounds) levels.

#### 2.4.7 Dovetail Anchor Slot

Preformed metal slot approximately 1 inch by 1 inch of not less than 22 gage galvanized steel cast in concrete. Coordinate actual size and throat opening with dovetail anchors and provide with removable filler material.

### 2.5 CONCRETE MIX DESIGN

#### 2.5.1 Properties and Requirements

- a. Use materials and material combinations listed in this section and the contract documents.
- b. Cementitious material content must be adequate for concrete to satisfy the specified requirements for strength, w/cm, durability, and finishability described in this section and the contract documents.

The minimum cementitious material content for concrete used in floors must meet the following requirements:

| Nominal maximum size of aggregate, in. | Minimum cementitious material content, pounds per cubic yard |
|----------------------------------------|--------------------------------------------------------------|
| 1-1/2                                  | 470                                                          |
| 1                                      | 520                                                          |
| 3/4                                    | 540                                                          |
| 3/8                                    | 610                                                          |

- c. Selected target slump must meet the requirements this section, the contract documents, and must not exceed 9 in. Concrete must not show visible signs of segregation.
- d. The target slump must be enforced for the duration of the project. Determine the slump by ASTM C143/C143M. Slump tolerances must meet

the requirements of ACI 117.

- e. The nominal maximum size of coarse aggregate for a mixture must not exceed three-fourths of the minimum clear spacing between reinforcement, one-fifth of the narrowest dimension between sides of forms, or one-third of the thickness of slabs or toppings.
- f. Concrete must be air entrained for members assigned to Exposure Class F1, F2, or F3. The total air content must be in accordance with the requirements of the paragraph titled DURABILITY.
- g. Measure air content at the point of delivery in accordance with ASTM C173/C173M or ASTM C231/C231M.
- h. Concrete for slabs to receive a hard-troweled finish must not contain an air-entraining admixture or have a total air content greater than 3 percent.
- i. Concrete properties and requirements for each portion of the structure are specified in the table below. Refer to the paragraph titled DURABILITY for more details on exposure categories and their requirements.

|                        | Minimum $f'_c$ psi     | Exposure Categories <sup>^</sup> | Miscellaneous Requirements |
|------------------------|------------------------|----------------------------------|----------------------------|
| <b>Slabs-on-ground</b> | 5000<br><br>at 28 days | S2 ;<br>C1 ;<br>W1;              |                            |

## 2.5.2 Durability

### 2.5.2.1 Alkali-Aggregate Reaction

Do not use any aggregate susceptible to alkali-carbonate reaction (ACR). Use one of the three options below for qualifying concrete mixtures to reduce the potential of alkali-silica reaction (ASR):

- a. For each aggregate used in concrete, the expansion result determined in accordance with ASTM C1293 must not exceed 0.04 percent at one year.
- b. For each aggregate used in concrete, the expansion result of the aggregate and cementitious materials combination determined in accordance with ASTM C1567 must not exceed 0.10 percent at an age of 16 days.
- c. Alkali content in concrete (LBA) must not exceed 4 pounds per cubic

yard for moderately reactive aggregate or 3 pounds per cubic yard for highly reactive aggregate. Reactivity must be determined by testing in accordance with ASTM C1293 and categorized in accordance with ASTM C1778. Alkali content is calculated as follows:  
LBA = (cement content, pounds per cubic yard) × (equivalent alkali content of portland cement in percent/100 percent)

#### 2.5.2.2 Freezing and Thawing Resistance

- a. Provide concrete meeting the following requirements based on exposure class assigned to members for freezing-and-thawing exposure in Contract Documents:

| Exposure class | Maximum $w/cm^*$ | Minimum $f'c$ , psi | Air content               | Additional Requirements                             |
|----------------|------------------|---------------------|---------------------------|-----------------------------------------------------|
| F2             | 0.45             | 4500                | Depends on aggregate size | See limits on maximum cementitious material by mass |

\*The maximum  $w/cm$  limits do not apply to lightweight concrete.

- b. Concrete must be air entrained for members assigned to Exposure Class F1, F2, or F3. The total air content must meet the requirements of the following table:

| Nominal maximum aggregate size, in. | Total air content, percent <sup>^</sup> |                   |
|-------------------------------------|-----------------------------------------|-------------------|
|                                     | Exposure Class F2 and F3                | Exposure Class F1 |
| 3/8                                 | 7.5                                     | 6.0               |
| 1/2                                 | 7.0                                     | 5.5               |
| 3/4                                 | 6.0                                     | 5.0               |
| 1                                   | 6.0                                     | 4.5               |
| 1-1/2                               | 5.5                                     | 4.5               |
| 2                                   | 5.0                                     | 4.0               |
| 3                                   | 5.5                                     | 3.5               |

\*Tolerance on air content as delivered must be plus/minus 1.5 percent.

<sup>^</sup>For  $f'c$  greater than 5000 psi, reducing air content by 1.0 percentage point is acceptable.

- c. Submit documentation verifying compliance with specified requirements.

- d. For sections of the structure that are assigned Exposure Class F3, submit certification on cement composition verifying that concrete mixture meets the requirements of the following table:

| <b>Cementitious material</b>                         | <b>Maximum percent of total cementitious material by mass*</b> |
|------------------------------------------------------|----------------------------------------------------------------|
| Fly ash or other pozzolans conforming to ASTM C618   | 25                                                             |
| Slag cement conforming to ASTM C989/C989M            | 50                                                             |
| Total of fly ash or other pozzolans, and slag cement | 50 <sup>^</sup>                                                |
| Total of fly ash or other pozzolans                  | 35 <sup>^</sup>                                                |

\*Total cementitious material also includes ASTM C150/C150M, ASTM C595/C595M, ASTM C845/C845M, and ASTM C1157/C1157M cement. The maximum percentages above must include:

- i. Fly ash or other pozzolans present in ASTM C1157/C1157M or ASTM C595/C595M Type IP blended cement.
- ii. Slag cement present in ASTM C1157/C1157M or ASTM C595/C595M Type IS blended cement.

<sup>^</sup>Fly ash or other pozzolans and silica fume must constitute no more than 25 percent and 10 percent, respectively, of the total mass of the cementitious materials.

#### 2.5.2.3 Corrosion and Chloride Content

- a. Provide concrete meeting the requirements of the following table based on the exposure class assigned to members requiring protection against reinforcement corrosion in Contract Documents.
- b. Submit documentation verifying compliance with specified requirements.
- c. Water-soluble chloride ion content contributed from constituents including water, aggregates, cementitious materials, and admixtures must be determined for the concrete mixture by ASTM C1218/C1218M at age between 28 and 42 days.
- d. The maximum water-soluble chloride ion (Cl-) content in concrete, percent by mass of cement is as follows:

| <b>Exposure class</b>      | <b>Maximum w/cm*</b> | <b>Minimum f'c, psi</b> | <b>Maximum water-soluble chloride ion (Cl-) content in concrete, percent by mass of cement</b> |
|----------------------------|----------------------|-------------------------|------------------------------------------------------------------------------------------------|
| <b>Reinforced concrete</b> |                      |                         |                                                                                                |

| Exposure class | Maximum w/cm* | Minimum f'c, psi | Maximum water-soluble chloride ion (CL-) content in concrete, percent by mass of cement |
|----------------|---------------|------------------|-----------------------------------------------------------------------------------------|
| C1             | N/A           | 2500             | 0.30                                                                                    |

\*The maximum w/cm limits do not apply to lightweight concrete.

#### 2.5.2.4 Sulfate Resistance

- a. Provide concrete meeting the requirements of the following table based on the exposure class assigned to members for sulfate exposure.

| Exposure class | Maximum w/cm | Minimum f'c, psi | Required cementitious materials-types |                                   |                   | Calcium chloride admixture |
|----------------|--------------|------------------|---------------------------------------|-----------------------------------|-------------------|----------------------------|
|                |              |                  | ASTM C150/C150M                       | ASTM C595/C595M                   | ASTM C1157/C1157M |                            |
| S2             | 0.45         | 4500             | IV^                                   | IP(HS);<br>IS(<70)(HS);<br>IT(HS) | HS                | Not permitted              |

\* For seawater exposure, other types of portland cements with tricalcium aluminate (C3A) contents up to 10 percent are acceptable if the w/cm does not exceed 0.40.

\*\* The amount of the specific source of the pozzolan or slag cement to be used shall be at least the amount determined by test or service record to improve sulfate resistance when used in concrete containing Type V cement. Alternatively, the amount of the specific source of the pozzolan or slag used shall not be less than the amount tested in accordance with ASTM C1012/C1012M and meeting the requirements maximum expansion requirements listed herein.

^ Other available types of cement, such as Type III or Type I, are acceptable in exposure classes S1 or S2 if the C3A contents are less than 8 or 5 percent, respectively.

- b. The maximum w/cm limits for sulfate exposure do not apply to lightweight concrete.
- c. Alternative combinations of cementitious materials of those listed in this paragraph are acceptable if they meet the maximum expansion requirements listed in the following table:

| Exposure class | Maximum expansion when tested using ASTM C1012/C1012M |               |              |
|----------------|-------------------------------------------------------|---------------|--------------|
|                | At 6 months                                           | At 12 months  | At 18 months |
| S2             | 0.05 percent                                          | 0.10 percent^ | N/A          |

^The 12-month expansion limit applies only when the measured expansion exceeds the 6-month maximum expansion limit.

#### 2.5.2.5 Concrete Temperature

The temperature of concrete as delivered must not exceed 95°F.



#### 2.5.2.6 Concrete permeability

- a. Provide concrete meeting the requirements of the following table based on exposure class assigned to members requiring low permeability in the Contract Documents.

| Exposure class | Maximum w/cm* | Minimum f'c, psi | Additional minimum requirements |
|----------------|---------------|------------------|---------------------------------|
| W1             | 0.5           | 4000             | None                            |

\*The maximum w/cm limits do not apply to lightweight concrete.

- b. Submit documentation verifying compliance with specified requirements.

#### 2.5.3 Trial Mixtures

Trial mixtures must be in accordance to ACI 301.

#### 2.5.4 Ready-Mix Concrete

Provide concrete that meets the requirements of ASTM C94/C94M.

Ready-mixed concrete manufacturer must provide duplicate delivery tickets with each load of concrete delivered. Provide delivery tickets with the following information in addition to that required by ASTM C94/C94M:

- Type and brand cement
- Cement and supplementary cementitious materials content in 94-pound bags per cubic yard of concrete
- Maximum size of aggregate
- Amount and brand name of admixtures
- Total water content expressed by water cementitious material ratio

#### 2.6 REINFORCEMENT

- Bend reinforcement cold. Fabricate reinforcement in accordance with fabricating tolerances of ACI 117.
- Submit manufacturer's certified test report for reinforcement.
- Submit placing drawings showing fabrication dimensions and placement locations of reinforcement and reinforcement supports. Placing drawings must indicate locations of splices, lengths of lap splices, and details of mechanical and welded splices.
- Submit request with locations and details of splices not indicated in Contract Documents.

- h. Submit request for field cutting, including location and type of bar to be cut and reason field cutting is required.

#### 2.6.1 Reinforcing Bars

- a. Reinforcing bars must be deformed, except spirals, load-transfer dowels, and welded wire reinforcement, which may be plain.
- b. ASTM A615/A615M with the bars marked S, Grade 60
- d. Submit mill certificates for reinforcing bars.

##### 2.6.1.1 Epoxy-Coated Reinforcing Bars

- a. Provide epoxy-coated reinforcing bars that conform to ASTM A775/A775M, Grade 60.
- b. Coatings must be applied in plants that are certified in accordance with Concrete Reinforcing Steel Institute (CRSI) Epoxy Coating Plant Certification Program or an equivalent program acceptable to the contracting officer.
- c. Coating damage incurred during shipment, storage, handling, and placing of epoxy-coated reinforcing bars must be repaired. Repair damaged coating areas with patching material conforming to ASTM A775/A775M or ASTM A934/A934M as applicable and in accordance with material manufacturer's written recommendations. Damaged coating area must not exceed 2 percent of surface area in each linear foot of each bar or bar must not be used. The 2 percent limit on damaged coating area must include repaired areas damaged before shipment as required by ASTM A775/A775M or ASTM A934/A934M as applicable. Fading of coating color shall not be cause for rejection of epoxy-coated reinforcing bars.
- d. Submit concrete Reinforcing Steel Institute (CRSI) Epoxy Coating Plant Certification inspection and quality-control program of plant applying epoxy coating if proposed plant is not certified in accordance with CRSI Epoxy Coating Plant Certification Program.

#### 2.6.2 Reinforcing Bar Supports

- a. Provide reinforcement support types within structure as required by Contract Documents. Reinforcement supports must conform to CRSI RB4.1. Submit description of reinforcement supports and materials for fastening coated reinforcement if not in conformance with CRSI RB4.1.

#### 2.6.3 Dowels for Load Transfer in Floors

Provide greased dowels for load transfer in floors of the type, design, weight, and dimensions indicated. Provide dowel bars that are plain-billet steel conforming to ASTM A615/A615M, Grade 40. Provide dowel pipe that is steel conforming to ASTM A53/A53M.

#### 2.6.4 Welding

- a. Provide weldable reinforcing bars that conform to ASTM A706/A706M and ASTM A615/A615M and Supplement S1, Grade 60, except that the maximum carbon content must be 0.55 percent.
- b. Comply with AWS D1.4/D1.4M unless otherwise specified. Do not tack weld reinforcing bars.
- c. Welded assemblies of steel reinforcement produced under factory conditions, such as welded wire reinforcement, bar mats, and deformed bar anchors, are allowed.
- d. After completing welds on zinc-coated (galvanized), epoxy-coated, or zinc and epoxy dual-coated reinforcement, coat welds and repair coating damage as previously specified.

#### PART 3 EXECUTION

#### 3.1 EXAMINATION

- a. Do not begin installation until substrates have been properly constructed; verify that substrates are level.
- b. If substrate preparation is the responsibility of another installer, notify Contracting Officer of unsatisfactory preparation before processing.
- c. Check field dimensions before beginning installation. If dimensions vary too much from design dimensions for proper installation, notify Contracting Officer and wait for instructions before beginning installation.

#### 3.2 PREPARATION

Determine quantity of concrete needed and minimize the production of excess concrete. Designate locations or uses for potential excess concrete before the concrete is poured.

##### 3.2.1 General

- a. Surfaces against which concrete is to be placed must be free of debris, loose material, standing water, snow, ice, and other deleterious substances before start of concrete placing.
- b. Remove standing water without washing over freshly deposited concrete. Divert flow of water through side drains provided for such purpose.

##### 3.2.2 Reinforcement and Other Embedded Items

- a. Secure reinforcement, joint materials, and other embedded materials in position, inspected, and approved before start of concrete placing.
- b. When concrete is placed, reinforcement must be free of materials deleterious to bond. Reinforcement with rust, mill scale, or a combination of both will be considered satisfactory, provided minimum nominal dimensions, nominal weight, and minimum average height of deformations of a hand-wire-brushed test specimen are not less than applicable ASTM specification requirements.

### 3.3 FORMS

- a. Provide forms, shoring, and scaffolding for concrete placement. Set forms mortar-tight and true to line and grade.
- b. Chamfer above grade exposed joints, edges, and external corners of concrete 0.75 inch. Place chamfer strips in corners of formwork to produce beveled edges on permanently exposed surfaces. Do not bevel reentrant corners or edges of formed joints of concrete.
- c. Provide formwork with clean-out openings to permit inspection and removal of debris.
- d. Inspect formwork and remove foreign material before concrete is placed.
- e. At construction joints, lap form-facing materials over the concrete of previous placement. Ensure formwork is placed against hardened concrete so offsets at construction joints conform to specified tolerances.
- f. Provide positive means of adjustment (such as wedges or jacks) of shores and struts. Do not make adjustments in formwork after concrete has reached initial setting. Brace formwork to resist lateral deflection and lateral instability.
- g. Fasten form wedges in place after final adjustment of forms and before concrete placement.
- h. Provide anchoring and bracing to control upward and lateral movement of formwork system.
- i. Construct formwork for openings to facilitate removal and to produce opening dimensions as specified and within tolerances.
- j. Provide runways for moving equipment. Support runways directly on formwork or structural members. Do not support runways on reinforcement. Loading applied by runways must not exceed capacity of formwork or structural members.
- k. Position and support expansion joint materials, waterstops, and other embedded items to prevent displacement. Fill voids in sleeves, inserts, and anchor slots temporarily with removable material to prevent concrete entry into voids.
- l. Clean surfaces of formwork and embedded materials of mortar, grout, and foreign materials before concrete placement.

#### 3.3.1 Coating

- a. Cover formwork surfaces with an acceptable material that inhibits bond with concrete.
- b. If formwork release agent is used, apply to formwork surfaces in accordance with manufacturer's recommendations before placing reinforcement. Remove excess release agent on formwork prior to concrete placement.
- c. Do not allow formwork release agent to contact reinforcement or hardened concrete against which fresh concrete is to be placed.

### 3.3.2 Reshoring

- a. Do not allow structural members to be loaded with combined dead and construction loads in excess of loads indicated in the accepted procedure.
- b. Install and remove reshores or backshores in accordance with accepted procedure.
- c. For floors supporting shores under newly placed concrete, either leave original supporting shores in place, or install reshores or backshores. Shoring system and supporting slabs must resist anticipated loads. Locate reshores and backshores directly under a shore position or as indicated on formwork shop drawings.
- d. In multistory buildings, place reshoring or backshoring over a sufficient number of stories to distribute weight of newly placed concrete, forms, and construction live loads.

### 3.3.3 Reuse

- a. Reuse forms providing the structural integrity of concrete and the aesthetics of exposed concrete are not compromised.
- b. Wood forms must not be clogged with paste and must be capable of absorbing high water-cementitious material ratio paste.
- c. Remove leaked mortar from formwork joints before reuse.

### 3.3.4 Forms for Standard Rough Form Finish

Provide formwork in accordance with ACI 301 Section 5 with a surface finish, SF-1.0, for formed surfaces that are to be concealed by other construction.

### 3.3.5 Forms for Standard Smooth Form Finish

Provide formwork in accordance with ACI 301 Section 5 with a surface finish, SF-3.0, for formed surfaces that are exposed to view. Do not provide mockup of concrete surface appearance and texture.

### 3.3.6 Form Ties

- a. For post-tensioned structures, do not remove formwork supports until stressing records have been accepted by the Contracting Officer.
- b. After ends or end fasteners of form ties have been removed, repair tie holes in accordance with ACI 301 Section 5 requirements.

### 3.3.7 Forms for Concrete Pan Joist Construction

Pan-form units for one-way or two-way concrete joist and slab construction must be factory-fabricated units of the approximate section indicated. Units must consist of steel or molded fiberglass concrete form pans. Closure units must be furnished as required.

### 3.3.8 Tolerances for Form Construction

- a. Construct formwork so concrete surfaces conform to tolerances in ACI 117.
- b. Position and secure sleeves, inserts, anchors, and other embedded items such that embedded items are positioned within ACI 117 tolerances.
- c. To maintain specified elevation and thickness within tolerances, install formwork to compensate for deflection and anticipated settlement in formwork during concrete placement. Set formwork and intermediate screed strips for slabs to produce designated elevation, camber, and contour of finished surface before formwork removal. If specified finish requires use of vibrating screeds or roller pipe screeds, ensure that edge forms and screed strips are strong enough to support such equipment.

### 3.3.9 Removal of Forms and Supports

- a. If vertical formed surfaces require finishing, remove forms as soon as removal operations will not damage concrete.
- b. Remove top forms on sloping surfaces of concrete as soon as removal will not allow concrete to sag. Perform repairs and finishing operations required. If forms are removed before end of specified curing period, provide curing and protection.
- c. Do not damage concrete during removal of vertical formwork for columns, walls, and sides of beams. Perform needed repair and finishing operations required on vertical surfaces. If forms are removed before end of specified curing period, provide curing and protection.
- d. Leave formwork and shoring in place to support construction loads and weight of concrete in beams, slabs, and other structural members until in-place required strength of concrete is reached.
- e. Form-facing material and horizontal facing support members may be removed before in-place concrete reaches specified compressive strength if shores and other supports are designed to allow facing removal without deflection of supported slab or member.

### 3.3.10 Strength of Concrete Required for Removal of Formwork

If removal of formwork, reshoring, or backshoring is based on concrete reaching a specified in-place strength, mold and field-cure cylinders in accordance with ASTM C31/C31M. Test cylinders in accordance with ASTM C39/C39M. Alternatively, use one or more of the methods listed herein to evaluate in-place concrete strength for formwork removal.

- a. Tests of cast-in-place cylinders in accordance with ASTM C873/C873M. This option is limited to slabs with concrete depths from 5 to 12 in.
- b. Penetration resistance in accordance with ASTM C803/C803M.
- c. Pullout strength in accordance with ASTM C900.
- d. Maturity method in accordance with ASTM C1074. Submit maturity method

data using project materials and concrete mix proportions used on the project to demonstrate the correlation between maturity and compressive strength of laboratory cured test specimens to the Contracting Officer.

### 3.4 WATERSTOP INSTALLATION AND SPLICES

- a. Provide waterstops in construction joints as indicated.
- b. Install formwork to accommodate waterstop materials. Locate waterstops in joints where indicated in Contract Documents. Minimize number of splices in waterstop. Splice waterstops in accordance with manufacturer's written instructions. Install factory-manufactured premolded mitered corners.
- c. Install waterstops to form a continuous diaphragm in each joint. Make adequate provisions to support and protect waterstops during progress of work. Protect waterstops protruding from joints from damage.

#### 3.4.1 PVC Waterstop

Make splices by heat sealing the adjacent waterstop edges together using a thermoplastic splicing iron utilizing a non-stick surface specifically designed for waterstop welding. Reform waterstops at splices with a remolding iron with ribs or corrugations to match the pattern of the waterstop. The spliced area, when cooled, must show no signs of separation, holes, or other imperfections when bent by hand in as sharp an angle as possible.

#### 3.4.2 Rubber Waterstop

Rubber waterstops must be spliced using cold bond adhesive as recommended by the manufacturer.

#### 3.4.3 Thermoplastic Elastomeric Rubber Waterstop

Fittings must be shop made using a machine specifically designed to mechanically weld the waterstop. A portable power saw must be used to miter or straight cut the ends to be joined to ensure good alignment and contact between joined surfaces. Maintain continuity of the characteristic features of the cross section of the waterstop (for example ribs, tabular center axis, and protrusions) across the splice.

#### 3.4.4 Hydrophilic Waterstop

Miter cut ends to be joined with sharp knife or shears. The ends must be adhered with adhesive.

### 3.5 PLACING REINFORCEMENT AND MISCELLANEOUS MATERIALS

- a. Unless otherwise specified, placing reinforcement and miscellaneous materials must be in accordance to ACI 301. Provide bars, welded wire reinforcement, wire ties, supports, and other devices necessary to install and secure reinforcement.
- b. Reinforcement must not have rust, scale, oil, grease, clay, or foreign substances that would reduce the bond. Rusting of reinforcement is a basis of rejection if the effective cross-sectional area or the nominal weight per unit length has been reduced. Remove loose rust prior to placing steel. Tack welding is prohibited.

- c. Nonprestressed cast-in-place concrete members must have concrete cover for reinforcement given in the following table:

| Concrete Exposure                                | Member                   | Reinforcement          | Specified cover, in. |
|--------------------------------------------------|--------------------------|------------------------|----------------------|
| Not exposed to weather or in contact with ground | Slabs, joists, and walls | No. 11 bar and smaller | 3/4                  |
|                                                  |                          |                        |                      |
|                                                  |                          |                        |                      |

#### 3.5.1 General

Provide details of reinforcement that are in accordance with the Contract Documents.

#### 3.5.2 Vapor Retarder and Vapor Barrier

- Install in accordance with ASTM E1643. Provide beneath the on-grade concrete floor slab. Use the greatest widths and lengths practicable to eliminate joints wherever possible. Lap joints a minimum of 12 inches and tape.
- Remove torn, punctured, or damaged vapor retarder and vapor barrier material and provide with new vapor retarder and vapor barrier prior to placing concrete. Concrete placement must not damage vapor retarder and vapor barrier material. Place vapor barrier directly on underlying subgrade, base course, or capillary water barrier, unless it consists of crushed material or large granular material which could puncture the vapor barrier. In this case, a thin layer of approximately 1/2 inch of fine graded material should be rolled or compacted over the fill before installation of the vapor barrier to reduce the possibility of puncture. Control concrete placement so as to prevent damage to the vapor barrier.

#### 3.5.3 Reinforcement Supports

Provide reinforcement support in accordance with CRSI RB4.1 and ACI 301 Section 3 requirements. Supports for coated or galvanized bars must also be coated with electrically compatible material for a distance of at least 2 inches beyond the point of contact with the bars.



### 3.5.4 Epoxy Coated Reinforcing

Epoxy Coated Reinforcing must meet the requirements of ASTM A934/A934M including Appendix X2, "Guidelines for Job Site Practices" except as otherwise specified herein.

#### 3.5.4.1 Epoxy Coated Reinforcing Steel Placement and Coating Repair

Carefully handle and install bars to minimize job site patching. Use the same precautions as described in the paragraph titled EPOXY-COATED REINFORCING BARS. Do not drag bars over other bars or over abrasive surfaces. Keep bar free of dirt and grit. When possible, assemble reinforcement as tied cages prior to final placement into the forms. Support assembled cages on padded supports. It is not expected that coated bars, when in final position ready for concrete placement, are completely free of damaged areas; however, excessive nicks and scrapes which expose steel is cause for rejection. Criteria for defects which require repair and for those that do not require repair are as indicated. Inspect for defects and provide required repairs prior to assembly. After assembly, reinspect and provide final repairs.

- a. Immediately prior to application of the patching material, manually remove any rust and debonded coating from the reinforcement by suitable techniques employing devices such as wire brushes and emery paper. Exercise care during this surface preparation so that the damaged areas are not enlarged more than necessary to accomplish the repair. Clean damaged areas of dirt, debris, oil, and similar materials prior to application of the patching material.
- b. Do repair and patching in accordance with the patching material manufacturer's recommendations. These recommendations, including cure times, must be available at the job site at all times.
- c. Allow adequate time for the patching materials to cure in accordance with the manufacturer's recommendation prior to concrete placement.
- d. Rinse placed reinforcing bars with fresh water to remove chloride contamination prior to placing concrete.

#### 3.5.5 Splicing

As indicated in the Contract Documents. For splices not indicated follow ACI 301. Do not splice at points of maximum stress. Overlap welded wire reinforcement the spacing of the cross wires, plus 2 inches.

#### 3.5.6 Fabrication

Shop fabricate reinforcing bars to conform to shapes and dimensions indicated for reinforcement, and as follows:

- a. Provide fabrication tolerances that are in accordance with ACI 117.
- b. Provide hooks and bends that are in accordance with the Contract Documents.

Reinforcement must be bent cold to shapes as indicated. Bending must be done in the shop. Rebending of a reinforcing bar that has been bent incorrectly is not be permitted. Bending must be in accordance with standard approved practice and by approved machine methods.

Deliver reinforcing bars bundled, tagged, and marked. Tags must be metal

with bar size, length, mark, and other information pressed in by machine. Marks must correspond with those used on the placing drawings.

Do not use reinforcement that has any of the following defects:

- a. Bar lengths, depths, and bends beyond specified fabrication tolerances
- b. Bends or kinks not indicated on drawings or approved shop drawings
- c. Bars with reduced cross-section due to rusting or other cause

Replace defective reinforcement with new reinforcement having required shape, form, and cross-section area.

### 3.5.7 Placing Reinforcement

Place reinforcement in accordance with ACI 301.

For slabs other than on grade, supports for which any portion is less than 1 inch from concrete surfaces that are exposed to view or to be painted must be of precast concrete units, plastic-coated steel, or stainless steel protected bar supports. Precast concrete units must be wedge shaped, not larger than 3-1/2 by 3-1/2 inches, and of thickness equal to that indicated for concrete protection of reinforcement. Provide precast units that have cast-in galvanized tie wire hooked for anchorage and blend with concrete surfaces after finishing is completed.

Provide reinforcement that is supported and secured together to prevent displacement by construction loads or by placing of wet concrete, and as follows:

- a. Provide supports for reinforcing bars that are sufficient in number and have sufficient strength to carry the reinforcement they support, and in accordance with ACI 301 and CRSI 10MSP. Do not use supports to support runways for concrete conveying equipment and similar construction loads.
- d. Secure reinforcements to supports by means of tie wire. Wire must be black, soft iron wire, not less than 16 gage.
- e. Reinforcement must be accurately placed, securely tied at intersections, and held in position during placing of concrete by spacers, chairs, or other approved supports. Point wire-tie ends away from the form. Unless otherwise indicated, numbers, type, and spacing of supports must conform to the Contract Documents.

### 3.5.8 Spacing of Reinforcing Bars

- a. Spacing must be as indicated in the Contract Documents.
- b. Reinforcing bars may be relocated to avoid interference with other

reinforcement, or with conduit, pipe, or other embedded items. If any reinforcing bar is moved a distance exceeding one bar diameter or specified placing tolerance, resulting rearrangement of reinforcement is subject to preapproval by the Contracting Officer.

#### 3.5.9 Concrete Protection for Reinforcement

Additional concrete protection must be in accordance with the Contract Documents.

#### 3.5.10 Welding

Welding must be in accordance with AWS D1.4/D1.4M.3.6 BATCHING, MEASURING, MIXING, AND TRANSPORTING CONCRETE

In accordance with ASTM C94/C94M, ACI 301, ACI 302.1R and ACI 304R, except as modified herein. Batching equipment must be such that the concrete ingredients are consistently measured within the following tolerances: 1 percent for cement and water, 2 percent for aggregate, and 3 percent for admixtures. Furnish mandatory batch ticket information for each load of ready mix concrete.

#### 3.6.1 Measuring

Make measurements at intervals as specified in paragraphs SAMPLING and TESTING.

#### 3.6.2 Mixing

- a. Mix concrete in accordance with ASTM C94/C94M, ACI 301 and ACI 304R.
- b. Machine mix concrete. Begin mixing within 30 minutes after the cement has been added to the aggregates. Place concrete within 90 minutes of either addition of mixing water to cement and aggregates or addition of cement to aggregates if the concrete temperature is less than 84 degrees F.
- c. Reduce mixing time and place concrete within 60 minutes if the concrete temperature is greater than 84 degrees F except as follows: if set retarding admixture is used and slump requirements can be met, limit for placing concrete may remain at 90 minutes. Additional water may be added, provided that both the specified maximum slump and submitted water-cementitious material ratio are not exceeded and the required concrete strength is still met. When additional water is added, an additional 30 revolutions of the mixer at mixing speed is required.
- d. If the entrained air content falls below the specified limit, add a sufficient quantity of admixture, within the manufacturer's recommended dosage, to bring the entrained air content within the specified limits. Dissolve admixtures in the mixing water and mix in the drum to uniformly distribute the admixture throughout the batch. Do not reconstitute concrete that has begun to solidify.

#### 3.6.3 Transporting

Transport concrete from the mixer to the forms as rapidly as practicable.

Prevent segregation or loss of ingredients. Clean transporting equipment thoroughly before each batch. Do not use aluminum pipe or chutes. Remove concrete which has segregated in transporting and dispose of as directed.

### 3.7 PLACING CONCRETE

Place concrete in accordance with ACI 301 Section 5. Concrete shall be placed within 15 minutes of discharge into non-agitating equipment.

#### 3.7.1 Footing Placement

Concrete for footings may be placed in excavations without forms upon inspection and approval by the Contracting Officer. Excavation width must be a minimum of 4 inches greater than indicated.

#### 3.7.2 Pumping

ACI 304R and ACI 304.2R. Pumping must not result in separation or loss of materials nor cause interruptions sufficient to permit loss of plasticity between successive increments. Loss of slump in pumping equipment must not exceed 2 inches at discharge/placement. Do not convey concrete through pipe made of aluminum or aluminum alloy. Avoid rapid changes in pipe sizes. Limit maximum size of coarse aggregate to 33 percent of the diameter of the pipe. Limit maximum size of well-rounded aggregate to 40 percent of the pipe diameter. Take samples for testing at both the point of delivery to the pump and at the discharge end.

#### 3.7.3 Cold Weather

Cold weather concrete must meet the requirements of ACI 301 unless otherwise specified. Do not allow concrete temperature to decrease below 50 degrees F. Obtain approval prior to placing concrete when the ambient temperature is below 40 degrees F or when concrete is likely to be subjected to freezing temperatures within 24 hours. Cover concrete and provide sufficient heat to maintain 50 degrees F minimum adjacent to both the formwork and the structure while curing. Limit the rate of cooling to 37 degrees F in any 1 hour and 50 degrees F per 24 hours after heat application.

#### 3.7.4 Hot Weather

Hot weather concrete must meet the requirements of ACI 301 unless otherwise specified. Maintain required concrete temperature using Figure 4.2 in ACI 305R to prevent the evaporation rate from exceeding 0.2 pound of water per square foot of exposed concrete per hour. Cool ingredients before mixing or use other suitable means to control concrete temperature and prevent rapid drying of newly placed concrete. Shade the fresh concrete as soon as possible after placing. Start curing when the surface of the fresh concrete is sufficiently hard to permit curing without damage. Provide water hoses, pipes, spraying equipment, and water hauling equipment, where job site is remote to water source, to maintain a moist concrete surface throughout the curing period. Provide burlap cover or other suitable, permeable material with fog spray or continuous wetting of the concrete when weather conditions prevent the use of either liquid membrane curing compound or impervious sheets. For vertical surfaces, protect forms from direct sunlight and add water to top of structure once concrete is set.

### 3.7.5 Bonding

Surfaces of set concrete at joints, must be roughened and cleaned of laitance, coatings, loose particles, and foreign matter. Roughen surfaces in a manner that exposes the aggregate uniformly and does not leave laitance, loosened particles of aggregate, nor damaged concrete at the surface.

Obtain bonding of fresh concrete that has set as follows:

- a. At joints between footings and walls or columns, between walls or columns and the beams or slabs they support, and elsewhere unless otherwise specified; roughened and cleaned surface of set concrete must be dampened, but not saturated, immediately prior to placing of fresh concrete.
- b. At joints in exposed-to-view work; at vertical joints in walls; at joints near midpoint of span in girders, beams, supported slabs, other structural members; in work designed to contain liquids; the roughened and cleaned surface of set concrete must be dampened but not saturated and covered with a cement grout coating.
- c. Provide cement grout that consists of equal parts of portland cement and fine aggregate by weight with not more than 6 gallons of water per sack of cement. Apply cement grout with a stiff broom or brush to a minimum thickness of 1/16 inch. Deposit fresh concrete before cement grout has attained its initial set.

### 3.8 WASTE MANAGEMENT

Provide as specified in the Waste Management Plan and as follows.

#### 3.8.1 Mixing Equipment

Before concrete pours, designate Contractor-owned site meeting environmental standards for cleaning out concrete mixing trucks. Minimize water used to wash equipment.

#### 3.8.2 Reinforcing Steel

Collect reinforcing steel and place in designated area for recycling.

#### 3.8.3 Other Waste

Identify concrete manufacturer's or supplier's policy for collection or return of construction waste, unused material, deconstruction waste, and/or packaging material. Return excess cement to supplier.

### 3.9 SURFACE FINISHES EXCEPT FLOOR, SLAB, AND PAVEMENT FINISHES

#### 3.9.1 Defects

Repair surface defects in accordance with ACI 301 Section 5.

#### 3.9.2 Not Against Forms (Top of Walls)

Surfaces not otherwise specified must be finished with wood floats to even surfaces. Finish must match adjacent finishes.

### 3.9.3 Formed Surfaces

#### 3.9.3.1 Tolerances

Tolerances in accordance with ACI 117 and as indicated.

#### 3.9.3.2 As-Cast Rough Form

Provide for surfaces not exposed to public view a surface finish SF-1.0. Patch holes and defects in accordance with ACI 301.

#### 3.9.3.3 Standard Smooth Finish

Provide for surfaces exposed to public view a surface finish SF-3.0. Patch holes and defects in accordance with ACI 301.

### 3.9.4 Smooth-Rubbed Finish

Provide a smooth-rubbed finish per ACI 301 Section 5 in the locations indicated. 3.10 FLOOR, SLAB, AND PAVEMENT FINISHES AND MISCELLANEOUS CONSTRUCTION

In accordance with ACI 301 and ACI 302.1R, unless otherwise specified. Slope floors uniformly to drains where drains are provided.

### 3.10.1 Finish

Place, consolidate, and immediately strike off concrete to obtain proper contour, grade, and elevation before bleedwater appears. Permit concrete to attain a set sufficient for floating and supporting the weight of the finisher and equipment. If bleedwater is present prior to floating the surface, drag the excess water off or remove by absorption with porous materials. Do not use dry cement to absorb bleedwater. Grate tampers ("jitterbugs") shall not be used.

#### 3.10.1.1 Steel Troweled

Use for floors intended as walking surfaces and for reception of floor coverings. Finish concrete in accordance with ACI 301 Section 5 for a steel troweled finish.

#### 3.10.2 Flat Floor Finishes

ACI 302.1R. Construct in accordance with one of the methods recommended in Table 10.15.3a, "Slab-on-ground flatness/levelness construction guide" or Table 10.15.3b, "Suspended slab flatness/levelness construction guide" appropriate for the type of construction. ACI 117 for tolerance tested by ASTM E1155.

##### a. Specified Conventional Value:

Floor Flatness (Ff) 20 minimum  
Floor Levelness (FL) 15 minimum

##### b. Specified Industrial:

Floor Flatness (Ff) 30 minimum  
Floor Levelness (FL) 20 minimum

#### 3.10.2.1 Measurement of Floor Tolerances

Test slab within 24 hours of the final troweling. Provide tests to Contracting Officer within 12 hours after collecting the data. Floor flatness inspector is required to provide a tolerance report which must include:

- a. Key plan showing location of data collected.
- b. Results required by ASTM E1155.

#### 3.10.2.2 Remedies for Out of Tolerance Work

Contractor is required to repair and retest any floors not meeting specified tolerances. Prior to repair, Contractor must submit and receive approval for the proposed repair, including product data from any materials proposed. Repairs must not result in damage to structural integrity of the floor. For floors exposed to public view, repairs must prevent any uneven or unusual coloring of the surface.

#### 3.10.3 Curbs and Gutters

Provide contraction joints spaced every 10 feet maximum unless otherwise indicated. Cut contraction joints 3/4 inch deep with a jointing tool after the surface has been finished. Provide expansion joints 1/2 inch thick and spaced every 100 feet maximum unless otherwise indicated. Perform pavement finish.

#### 3.11.1 Construction Joints

Make and locate joints not indicated so as not to impair strength and appearance of the structure, as approved. Joints must be perpendicular to main reinforcement. Reinforcement must be continued and developed across construction joints. Locate construction joints as follows:

##### 3.11.1.1 Construction Joints for Constructability Purposes

- a. In walls, at top of footing; at top of slabs on ground; at top and bottom of door and window openings or where required to conform to architectural details; and at underside of deepest beam or girder framing into wall.
- b. In columns or piers, at top of footing; at top of slabs on ground; and at underside of deepest beam or girder framing into column or pier.
- c. Near midpoint of spans for supported slabs, beams, and girders unless a beam intersects a girder at the center, in which case construction joints in girder must offset a distance equal to twice the width of the beam. Make transfer of shear through construction joint by use of inclined reinforcement.

Provide keyways at least 1-1/2-inches deep in construction joints in walls and slabs and between walls and footings; approved bulkheads may be used for slabs.

#### 3.11.2 Isolation Joints in Slabs on Ground

- a. Provide joints at points of contact between slabs on ground and vertical surfaces, such as column pedestals, foundation walls, grade

beams, and elsewhere as indicated.

- b. Fill joints with premolded joint filler strips 1/2 inch thick, extending full slab depth. Install filler strips at proper level below finish floor elevation with a slightly tapered, dress-and-oiled wood strip temporarily secured to top of filler strip to form a groove not less than 3/4 inch in depth where joint is sealed with sealing compound and not less than 1/4 inch in depth where joint sealing is not required. Remove wood strip after concrete has set. Contractor must clean groove of foreign matter and loose particles after surface has dried.

### 3.11.3 Contraction Joints in Slabs on Ground

- a. Provide joints to form panels as indicated.
- b. Under and on exact line of each control joint, cut 50 percent of welded wire reinforcement before placing concrete.
- c. Sawcut contraction joints into slab on ground in accordance with ACI 301 Section 5.
- d. Joints must be 1/8-inch wide by 1/5 to 1/4 of slab depth and formed by inserting hand-pressed fiberboard strip into fresh concrete until top surface of strip is flush with slab surface. After concrete has cured for at least 7 days, the Contractor must remove inserts and clean groove of foreign matter and loose particles.
- e. Sawcutting will be limited to within 12 hours after set and at 1/4 slab depth.

### 3.11.4 Sealing Joints in Slabs on Ground

- a. Contraction and control joints which are to receive finish flooring material must be sealed with joint sealing compound after concrete curing period. Slightly underfill groove with joint sealing compound to prevent extrusion of compound. Remove excess material as soon after sealing as possible.
- b. Sealed groove must be left ready to receive filling material that is provided as part of finish floor covering work.

## 3.12 CONCRETE FLOOR TOPPING

### 3.12.1 Standard Floor Topping

Provide topping for treads and platforms of metal steel stairs and elsewhere as indicated.

#### 3.12.1.1 Preparations Prior to Placing

- a. When topping is placed on a green concrete base slab, screed surface of base slab to a level not more than 1-1/2 inches nor less than 1 inch below required finish surface. Remove water and laitance from surface of base slab before placing topping mixture. As soon as water ceases to rise to surface of base slab, place topping.
- b. When topping is placed on a hardened concrete base slab, remove dirt, loose material, oil, grease, asphalt, paint, and other contaminants from base slab surface, leaving a clean surface. Prior to placing



topping mixture, 2-1/2-inches minimum, slab surface must be dampened and left free of standing water. Immediately before topping mixture is placed, broom a coat of neat cement grout onto surface of slab. Do not allow cement grout to set or dry before topping is placed.

- c. When topping is placed on a metal surface, such as metal pans for steel stairs, remove dirt, loose material, oil, grease, asphalt, paint, and other contaminants from metal surface.

#### 3.12.1.2 Placing

Spread standard topping mixture evenly on previously prepared base slab or metal surface, brought to correct level with a straightedge, and struck off. Topping must be consolidated, floated, checked for trueness of surface, and refloated as specified for float finish.

#### 3.12.1.3 Finishing

Give trowel finish standard floor topping surfaces.

Give other finishes standard floor topping surfaces as indicated.

#### 3.12.2 Heavy-Duty Floor Topping

Provide topping where indicated.

##### 3.12.2.1 Heavy-duty Topping Mixture

Provide mixture that consists of 1 part portland cement and 2-1/2 parts emery aggregate or 1 part fine aggregate and 1-1/2 parts traprock coarse aggregate, by volume. Exact proportions of mixture must conform to recommendations of aggregate manufacturer. Mixing water must not exceed 3-1/4 gallons per 94-pound sack of cement including unabsorbed moisture in aggregate. Maximum slump must be 1 inch.

##### 3.12.2.2 Base Slab

- a. Screed surface of slab to a level no more than 1-1/2 inches nor less than 1 inch below grade of finished floor.
- b. Give slab a scratch finish as specified.
- c. Preparations prior to placing.

Remove dirt, loose material, oil, grease, asphalt, paint and other contaminants from base slab surface. Prior to placing topping mixture, dampen slab surface and leave free of standing water. Immediately before topping mixture is placed, broom a coat of neat cement grout onto surface of slab. Allow cement grout to set or dry before topping mixture is placed.

##### 3.12.2.3 Placing

Spread heavy-duty topping mixture evenly on previously prepared base slab, and bring to correct level with a straightedge, and strike off. Provide topping that is consolidated, floated, and checked for trueness of surface as specified for float finish, except that power-driven floats is the impact type.

#### 3.12.2.4 Finishing

Give trowel finish heavy-duty floor topping surfaces. Provide trowel finish as specified, except that additional troweling after first power troweling must be not less than three hand-troweling operations.

### 3.13 CURING AND PROTECTION

Curing and protection in accordance with ACI 301 Section 5, unless otherwise specified. Begin curing immediately following form removal. Avoid damage to concrete from vibration created by blasting, pile driving, movement of equipment in the vicinity, disturbance of formwork or protruding reinforcement, and any other activity resulting in ground vibrations. Protect concrete from injurious action by sun, rain, flowing water, frost, mechanical injury, tire marks, and oil stains. Do not allow concrete to dry out from time of placement until the expiration of the specified curing period. Do not use membrane-forming compound on surfaces where appearance would be objectionable, on any surface to be painted, where coverings are to be bonded to the concrete, or on concrete to which other concrete is to be bonded. If forms are removed prior to the expiration of the curing period, provide another curing procedure specified herein for the remaining portion of the curing period. Provide moist curing for those areas receiving liquid chemical sealer, hardener, or epoxy coating. Allow curing compound/sealer installations to cure prior to the installation of materials that adsorb VOCs. .

#### 3.13.1 Curing Periods

ACI 301 Section 5, except 10 days for retaining walls, pavement or chimneys. Begin curing immediately after placement. Protect concrete from premature drying, excessively hot temperatures, and mechanical injury; and maintain minimal moisture loss at a relatively constant temperature for the period necessary for hydration of the cement and hardening of the concrete. The materials and methods of curing are subject to approval by the Contracting Officer.

#### 3.13.2 Curing Formed Surfaces

Accomplish curing of formed surfaces, including undersurfaces of girders, beams, supported slabs, and other similar surfaces by moist curing with forms in place for full curing period or until forms are removed. If forms are removed before end of curing period, accomplish final curing of formed surfaces by any of the curing methods specified above, as applicable.

#### 3.13.3 Curing Unformed Surfaces

- a. Accomplish initial curing of unformed surfaces, such as monolithic slabs, floor topping, and other flat surfaces, by membrane curing.

#### 3.13.4 Temperature of Concrete During Curing

When temperature of atmosphere is 41 degrees F and below, maintain temperature of concrete at not less than 55 degrees F throughout concrete curing period or 45 degrees F when the curing period is measured by

maturity. When necessary, make arrangements before start of concrete placing for heating, covering, insulation, or housing as required to maintain specified temperature and moisture conditions for concrete during curing period.

When the temperature of atmosphere is 80 degrees F and above or during other climatic conditions which cause too rapid drying of concrete, make arrangements before start of concrete placing for installation of wind breaks, of shading, and for fog spraying, wet sprinkling, or moisture-retaining covering of light color as required to protect concrete during curing period.

Changes in temperature of concrete must be uniform and not exceed 37 degrees F in any 1 hour nor 80 degrees F in any 24-hour period.

### 3.13.5 Protection from Mechanical Injury

During curing period, protect concrete from damaging mechanical disturbances, particularly load stresses, heavy shock, and excessive vibration and from damage caused by rain or running water.

### 3.13.6 Protection After Curing

Protect finished concrete surfaces from damage by construction operations.

## 3.14 FIELD QUALITY CONTROL

### 3.14.1 Aggregate Testing

#### 3.14.1.1 Fine Aggregate

At least once during each shift when the concrete plant is operating, there shall be one sieve analysis and fineness modulus determination in accordance with ASTM C136/C136M and COE CRD-C 104 for the fine aggregate or for each fine aggregate if it is batched in more than one size or classification. The location at which samples are taken may be selected by the Contractor as the most advantageous for control. However, the Contractor is responsible for delivering fine aggregate to the mixer within specification limits. When the amount passing on any sieve is outside the specification limits, the fine aggregate shall be immediately resampled and retested. If there is another failure on any sieve, the fact shall be immediately reported to the Contracting Officer, concreting shall be stopped, and immediate steps taken to correct the grading.

#### 3.14.1.2 Coarse Aggregate

At least once during each shift in which the concrete plant is operating, there shall be a sieve analysis in accordance with ASTM C136/C136M for each size of coarse aggregate. The location at which samples are taken may be selected by the Contractor as the most advantageous for production control. However, the Contractor shall be responsible for delivering the aggregate to the mixer within specification limits. A test record of samples of aggregate taken at the same locations shall show the results of the current test as well as the average results of the five most recent tests including the current test. The Contractor may adopt limits for control coarser than the specification limits for samples taken other than as delivered to the mixer to allow for degradation during handling. When the amount passing any sieve is outside the specification limits, the coarse aggregate shall be immediately resampled and retested. If the

second sample fails on any sieve, that fact shall be reported to the Contracting Officer. Where two consecutive averages of 5 tests are outside specification limits, the operation shall be considered out of control and reported to the Contracting Officer. Concreting shall be stopped and immediate steps shall be taken to correct the grading.

### 3.14.2 Concrete Sampling

ASTM C172/C172M. Collect samples of fresh concrete to perform tests specified. ASTM C31/C31M for making test specimens.

### 3.14.3 Concrete Testing

#### 3.14.3.1 Slump Tests

ASTM C143/C143M. Take concrete samples during concrete placement/discharge. The maximum slump may be increased as specified with the addition of an approved admixture provided that the water-cementitious material ratio is not exceeded. Perform tests at commencement of concrete placement, when test cylinders are made, and for each batch (minimum) or every 20 cubic yards (maximum) of concrete.

#### 3.14.3.2 Temperature Tests

Test the concrete delivered and the concrete in the forms. Perform tests in hot or cold weather conditions (below 50 degrees F and above 80 degrees F) for each batch (minimum) or every 20 cubic yards (maximum) of concrete, until the specified temperature is obtained, and whenever test cylinders and slump tests are made.

#### 3.14.3.3 Compressive Strength Tests

ASTM C39/C39M. Make eight 4 inch by 8 inch test cylinders for each set of tests in accordance with ASTM C31/C31M, ASTM C172/C172M and applicable requirements of ACI 305R and ACI 306R. Take precautions to prevent evaporation and loss of water from the specimen. Test two cylinders at 7 days, two cylinders at 28 days, two cylinders at 56 days and hold two cylinder in reserve. Take samples for strength tests of each mix design of concrete placed each day not less than once a day, nor less than once for each 100 cubic yards of concrete for the first 500 cubic yards, then every 500 cubic yards thereafter, nor less than once for each 5400 square feet of surface area for slabs or walls. For the entire project, take no less than five sets of samples and perform strength tests for each mix design of concrete placed. Each strength test result must be the average of two cylinders from the same concrete sample tested at 28 days 56 days . Concrete compressive tests must meet the requirements of this section, the Contract Document, and ACI 301. Retest locations represented by erratic core strengths. Where retest does not meet concrete compressive strength requirements submit a mitigation or remediation plan for review and approval by the contracting officer. Repair core holes with nonshrink grout. Match color and finish of adjacent concrete.

#### 3.14.3.4 Air Content

ASTM C173/C173M or ASTM C231/C231M for normal weight concrete . Test air-entrained concrete for air content at the same frequency as specified for slump tests.

#### 3.14.3.5 Unit Weight of Structural Concrete

ASTM C567/C567M and ASTM C138/C138M. Determine unit weight of lightweight and normal weight concrete. Perform test for every 20 cubic yards maximum.

#### 3.14.3.6 Strength of Concrete Structure

The strength of the concrete structure will be considered to be deficient if any of the following conditions are identified:

- a. Failure to meet compressive strength tests as evaluated.
- b. Reinforcement not conforming to requirements specified.
- c. Concrete which differs from required dimensions or location in such a manner as to reduce strength.
- d. Concrete curing and protection of concrete against extremes of temperature during curing, not conforming to requirements specified.
- e. Concrete subjected to damaging mechanical disturbances, particularly load stresses, heavy shock, and excessive vibration.
- f. Poor workmanship likely to result in deficient strength.

Where the strength of the concrete structure is considered deficient submit a mitigation or remediation plan for review and approval by the contracting officer.

#### 3.14.3.7 Non-Conforming Materials

Factors that indicate that there are non-conforming materials include (but not limited to) excessive compressive strength, inadequate compressive strength, excessive slump, excessive voids and honeycombing, concrete delivery records that indicate excessive time between mixing and placement, or excessive water was added to the mixture during delivery and placement. Any of these indicators alone are sufficient reason for the Contracting Officer to request additional sampling and testing.

Investigations into non-conforming materials must be conducted at the Contractor's expense. The Contractor must be responsible for the investigation and must make written recommendations to adequately mitigate or remediate the non-conforming material. The Contracting Officer may accept, accept with reduced payment, require mitigation, or require removal and replacement of non-conforming material at no additional cost to the Government.

#### 3.15 REPAIR, REHABILITATION AND REMOVAL

Before the Contracting Officer accepts the structure the Contractor must inspect the structure for cracks, damage and substandard concrete placements that may adversely affect the service life of the structure. A report documenting these defects must be prepared which includes recommendations for repair, removal or remediation must be submitted to the Contracting Officer for approval before any corrective work is accomplished.

### 3.15.1 Crack Repair

Prior to final acceptance, all cracks in excess of 0.02 inches wide must be documented and repaired. The proposed method and materials to repair the cracks must be submitted to the Contracting Officer for approval. The proposal must address the amount of movement expected in the crack due to temperature changes and loading.

### 3.15.2 Repair of Weak Surfaces

Weak surfaces are defined as mortar-rich, rain-damaged, uncured, or containing exposed voids or deleterious materials. Concrete surfaces with weak surfaces less than 1/4 inch thick must be diamond ground to remove the weak surface. Surfaces containing weak surfaces greater than 1/4 inch thick must be removed and replaced or mitigated in a manner acceptable to the Contracting Officer.

### 3.15.3 Failure of Quality Assurance Test Results

Proposed mitigation efforts by the Contractor must be approved by the Contracting Officer prior to proceeding.

-- End of Section --

SECTION 03 42 13.00 10

PLANT-PRECAST CONCRETE PRODUCTS FOR BELOW GRADE CONSTRUCTION  
**05/16**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

|           |                                                                                                             |
|-----------|-------------------------------------------------------------------------------------------------------------|
| ACI 211.1 | (1991; R 2009) Standard Practice for Selecting Proportions for Normal, Heavyweight and Mass Concrete        |
| ACI 211.2 | (1998; R 2004) Standard Practice for Selecting Proportions for Structural Lightweight Concrete              |
| ACI 305R  | (2020) Guide to Hot Weather Concreting                                                                      |
| ACI 306.1 | (1990; R 2002) Standard Specification for Cold Weather Concreting                                           |
| ACI 318   | (2019; R 2022) Building Code Requirements for Structural Concrete (ACI 318-19) and Commentary (ACI 318R-19) |

AMERICAN CONCRETE PIPE ASSOCIATION (ACPA)

|             |                                                                       |
|-------------|-----------------------------------------------------------------------|
| ACPA 01-102 | (2000) Concrete Pipe Handbook                                         |
| ACPA 01-110 | (1984) Design Manual for Sulfide and Corrosion Prediction and Control |
| ACPA QPC    | (2021) QCast Plant Certification Manual                               |

AMERICAN WELDING SOCIETY (AWS)

|                |                                                       |
|----------------|-------------------------------------------------------|
| AWS D1.1/D1.1M | (2020; Errata 1 2021) Structural Welding Code - Steel |
| AWS D1.4/D1.4M | (2018) Structural Welding Code - Reinforcing Steel    |

ASTM INTERNATIONAL (ASTM)

|                 |                                                                                     |
|-----------------|-------------------------------------------------------------------------------------|
| ASTM A36/A36M   | (2019) Standard Specification for Carbon Structural Steel                           |
| ASTM A153/A153M | (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware |

|                   |                                                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------------------------------|
| ASTM A615/A615M   | (2024) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement                   |
| ASTM A706/A706M   | (2024) Standard Specification for Low-Alloy Steel Deformed and Plain Bars for Concrete Reinforcement                |
| ASTM A767/A767M   | (2019) Standard Specification for Zinc-Coated (Galvanized) Steel Bars for Concrete Reinforcement                    |
| ASTM A775/A775M   | (2022) Standard Specification for Epoxy-Coated Steel Reinforcing Bars                                               |
| ASTM A884/A884M   | (2019; Errata 1) Standard Specification for Epoxy-Coated Steel Wire and Welded Wire Reinforcement                   |
| ASTM A1064/A1064M | (2024) Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete |
| ASTM C31/C31M     | (2024c) Standard Practice for Making and Curing Concrete Test Specimens in the Field                                |
| ASTM C39/C39M     | (2024) Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens                              |
| ASTM C138/C138M   | (2024a) Standard Test Method for Density (Unit Weight), Yield, and Air Content (Gravimetric) of Concrete            |
| ASTM C143/C143M   | (2020) Standard Test Method for Slump of Hydraulic-Cement Concrete                                                  |
| ASTM C171         | (2020) Standard Specification for Sheet Materials for Curing Concrete                                               |
| ASTM C173/C173M   | (2024a) Standard Test Method for Air Content of Freshly Mixed Concrete by the Volumetric Method                     |
| ASTM C192/C192M   | (2024) Standard Practice for Making and Curing Concrete Test Specimens in the Laboratory                            |
| ASTM C231/C231M   | (2024) Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method                        |
| ASTM C309         | (2019) Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete                             |
| ASTM C443         | (2021) Standard Specification for Joints for Concrete Pipe and Manholes, Using                                      |



Rubber Gaskets

|                   |                                                                                                                                                |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C857         | (2016) Standard Practice for Minimum Structural Design Loading for Underground Precast Concrete Utility Structures                             |
| ASTM C858         | (2010; E 2012) Standard Specification for Underground Precast Concrete Utility Structures                                                      |
| ASTM C877         | (2021) Standard Specification for External Sealing Bands for Concrete Pipe, Manholes, and Precast Box Sections                                 |
| ASTM C891         | (2020) Standard Practice for Installation of Underground Precast Concrete Utility Structures                                                   |
| ASTM C920         | (2018; R 2024) Standard Specification for Elastomeric Joint Sealants                                                                           |
| ASTM C923         | (2008; R 2013; E 2016) Standard Specification for Resilient Connectors Between Reinforced Concrete Manhole Structures, Pipes and Laterals      |
| ASTM C990         | (2009; R 2019) Standard Specification for Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants |
| ASTM C1064/C1064M | (2023) Standard Test Method for Temperature of Freshly Mixed Hydraulic-Cement Concrete                                                         |
| ASTM C1107/C1107M | (2020) Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)                                                             |
| ASTM C1116/C1116M | (2023) Standard Specification for Fiber-Reinforced Concrete                                                                                    |
| ASTM C1244        | (2020) Standard Test Method for Concrete Sewer Manholes by the Negative Air Pressure (Vacuum) Test Prior to Backfill                           |
| ASTM C1478        | (2019) Standard Specification for Storm Drain Resilient Connectors Between Reinforced Concrete Storm Sewer Structures, Pipes and Laterals      |

CSA GROUP (CSA)

|           |                                                              |
|-----------|--------------------------------------------------------------|
| CSA A23.4 | (2016; R 2021) Precast Concrete - Materials and Construction |
|-----------|--------------------------------------------------------------|

NATIONAL PRECAST CONCRETE ASSOCIATION (NPCA)

|                |                                                                           |
|----------------|---------------------------------------------------------------------------|
| NPCA QC Manual | (2017) Quality Control Manual for Precast and Prestressed Concrete Plants |
|----------------|---------------------------------------------------------------------------|

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-01 Preconstruction Submittals

Quality Control Procedures

### SD-02 Shop Drawings

Standard Precast Units; G

Custom-Made Precast Units; G

Special Finishes

### SD-03 Product Data

Standard Precast Units

Proprietary Precast Units

Embedded Items

Accessories

### SD-05 Design Data

Design Calculations; G

Concrete Mix Proportions

### SD-06 Test Reports

Test Reports

### SD-07 Certificates

Quality Control Procedures

## 1.3 QUALITY ASSURANCE

Demonstrate adherence to the standards set forth in NPCA QC Manual or ACPA QPC. Meet requirements written in the subparagraphs below.

### 1.3.1 NPCA and ACPA Plant Certification

The precast concrete producer must be certified by the National Precast Concrete Association's or the American Concrete Pipe Association's Plant Certification Program prior to and during production of the products for this project.

### 1.3.2 Qualifications, Quality Control and Inspection

#### 1.3.2.1 Qualifications

Select a precast concrete producer that has been in the business of producing precast concrete units similar to those specified for a minimum of 3 years. The precast concrete producer must maintain a permanent quality control department or retain an independent testing agency on a continuing basis.

#### 1.3.2.2 Quality Control Procedures

Submit quality control procedures established by the precast manufacturer in accordance with NPCA QC Manual and ACPA QPC. Show that the following QC tests are performed as required and in accordance with the ASTM standards indicated.

##### 1.3.2.2.1 Slump

Perform a slump test for each 150 cu yd of concrete produced, or once a day, whichever comes first. Perform slump tests in accordance with ASTM C143/C143M.

##### 1.3.2.2.2 Temperature

Measure the temperature of fresh concrete when slump or air content tests are made and when compressive test specimens are made in accordance with ASTM C1064/C1064M.

##### 1.3.2.2.3 Compressive Strength

Make at least four compressive strength specimens for each 150 cubic yards of concrete of each mix in accordance with the following Standards: ASTM C31/C31M, ASTM C192/C192M, ASTM C39/C39M.

##### 1.3.2.2.4 Air Content

Perform tests for air content on air-entrained, wet-cast concrete for each 150 cu yd of concrete, but not less often than once each day when air-entrained concrete is used. Determine the air content in accordance with either ASTM C231/C231M or ASTM C173/C173M for normal weight aggregates and ASTM C173/C173M for lightweight aggregates.

##### 1.3.2.2.5 Unit Weight

Perform tests for unit weight a minimum of once per week to verify the yield of batch mixes. Perform unit weight tests for each 100 cu yd of lightweight concrete in accordance with ASTM C138/C138M.

#### 1.3.2.3 Inspection

The Contracting Officer may place a Government Quality Assurance inspector in the plant when the units covered by this specification are being manufactured. The precast concrete producer must give notice 14 days prior to the time production of the units will commence, as well as 14 days prior to when the completed units will be available for plant inspection. Neither the exercise nor waiver of inspection at the plant will affect the Government's right to enforce contractual provisions after units are transported or erected.

#### 1.3.2.4 Test Reports

Submit the following:

##### 1.3.2.4.1 Material Certifications or Laboratory Test Reports

Include mill tests and all other test data, for portland cement, blended cement, pozzolans, ground granulated blast furnace slag, silica fume, aggregate, admixtures, and curing compound proposed for use on this project.

##### 1.3.2.4.2 Mix Test

Submit reports showing that the mix has been successfully tested to produce concrete with the properties specified and will be suitable for the job conditions. Such tests may include compressive strength, flexural strength, plastic or hardened air content, freeze thaw durability, abrasion and absorption. Clearly detail in the specifications special tests for precast concrete or cast-in items.

##### 1.3.2.4.3 Self-Consolidating Concrete

Submit sufficient documentation, when the use of self-consolidating concrete (SCC) is proposed, showing a minimum of 30-days production track records demonstrating that SCC is appropriate for casting of the product.

##### 1.3.2.4.4 In-Plant QA/QC Inspection Reports

Submit inspection reports upon the request of the Contracting Officer.

#### 1.4 DELIVERY, STORAGE, AND HANDLING

##### 1.4.1 Delivery

Deliver precast units to the site in accordance with the delivery schedule to avoid excessive build-up of units in storage at the site. Upon delivery to the jobsite, all precast concrete units will be inspected by the Contracting Officer for quality and final acceptance.

##### 1.4.2 Storage

Store units off the ground or in a manner that minimizes potential damage.

##### 1.4.3 Handling

Handle, transport, and store products in a manner to minimize damage. Lifting devices or holes must be consistent with industry standards. Perform lifting with methods or devices intended for this purpose as indicated on shop drawings.

#### PART 2 PRODUCTS

##### 2.1 SYSTEM DESCRIPTION

Furnish precast concrete units designed and fabricated by an experienced and acceptable precast concrete manufacturer who has been, for at least three years, regularly and continuously engaged in the manufacture of precast concrete work similar to that indicated on the drawings.

Coordinate precast work with the work of other trades. Below grade structures must comply with ASTM C858.

#### 2.1.1 Standard Precast Units

Design standard precast concrete units to withstand indicated design load conditions in accordance with applicable industry design standards ACI 318, ASTM C857. Design must also consider stresses induced during handling, shipping and installation as to avoid product cracking or other handling damage. Indicate design loads for precast concrete units on the shop drawings. Submit drawings for standard precast concrete units furnished by the precast concrete producer for approval by the Contracting Officer. These drawings must demonstrate that the applicable industry design standards have been met. Include installation and construction information on shop drawings. Include details of steel reinforcement size and placement as well as supporting design calculations, if appropriate. Produce precast concrete units in accordance with the approved drawings. Submit cut sheets, for standard precast concrete units, showing conformance to project drawings and requirements, and to applicable industry design standards listed in this specification.

#### 2.1.2 Custom-Made Precast Units

Submit design calculations for custom-made precast units, prepared and sealed by a registered professional engineer, for approval prior to fabrication. Include in the calculations the analysis of units for lifting stresses and the sizing of lifting devices. Submit drawings furnished by the precast concrete producer for approval by the Contracting Officer. Show on these drawings complete design, installation, and construction information in such detail as to enable the Contracting Officer to determine the adequacy of the proposed units for the intended purpose. Include details of steel reinforcement size and placement as well as supporting design calculations, if appropriate. Produce precast concrete units in accordance with the approved drawings.

#### 2.1.3 Proprietary Precast Units

Products manufactured under franchise arrangements must conform to all the requirements specified by the franchiser. Items not included in the franchise specification, but included in this specification, must conform to the requirements in this specification. Submit standard plans or informative literature, for proprietary precast concrete units. Make available supporting calculations and design details upon request. Provide sufficient information as to demonstrate that such products will perform the intended task.

#### 2.1.4 Joints and Sealants

Provide joints and sealants between adjacent units of the type and configuration indicated on shop drawings meeting specified design and performance requirements.

#### 2.1.5 Concrete Mix Design

##### 2.1.5.1 Concrete Mix Proportions

Base selection of proportions for concrete on the methodology presented in ACI 211.1 for normal weight concrete and ACI 211.2 for lightweight concrete. Develop the concrete proportions using the same type and brand

of cement, the same type and brand of pozzolan, the same type and gradation of aggregates, and the same type and brand of admixture that will be used in the manufacture of precast concrete units for the project. Do not use calcium chloride in precast concrete containing reinforcing steel or other embedded metal items. At a minimum of thirty days prior to precast concrete unit manufacturing, the precast concrete producer will submit a mix design and proportions for each strength and type of concrete that will be used. Furnish a complete list of materials, including quantity, type, brand and applicable data sheets for all mix design constituents as well as applicable reference specifications. The use of self-consolidating concrete is permitted, provided that mix design proportions and constituents meet the requirements of this specification.

#### 2.1.5.2 Concrete Strength

Provide precast concrete units with a 28-day compressive strength ( $f'_c$ ) of 5,000 psi.

#### 2.1.5.3 Water-to-Cement Ratio

Where exposed to freezing and thawing, furnish concrete containing entrained air and with a water-cementitious ratio of 0.45 or less. Where not exposed to freezing, but required to have a low permeability, furnish concrete with a water-cementitious ratio of 0.48 or less. Where exposed to deicer salts, brackish water, or seawater, furnish concrete with a water-cementitious ratio of 0.40 or less, for corrosion protection.

#### 2.1.5.4 Air Content

The air content of concrete that will be exposed to freezing conditions must be within the limits given below.

| NOMINAL MAXIMUM AGGREGATE SIZE                                                                        | AIR CONTENT PERCENT |                            |
|-------------------------------------------------------------------------------------------------------|---------------------|----------------------------|
|                                                                                                       | EXPOSURE CLASS F1   | EXPOSURE CLASSES F2 and F3 |
| 3/8 inch                                                                                              | 6.0                 | 7.5                        |
| 1/2 inch                                                                                              | 5.5                 | 7.0                        |
| 3/4 inch                                                                                              | 5.0                 | 6.0                        |
| 1.0 inch                                                                                              | 4.5                 | 6.0                        |
| 1.5 inch                                                                                              | 4.5                 | 5.5                        |
| Note: For specified compressive strengths greater than 5000 psi, air content may be reduced 1 percent |                     |                            |

#### 2.1.5.5 Corrosion Control for Sanitary Sewer Systems

Follow design recommendations outlined in Chapter 7 of ACPA 01-102 or the ACPA 01-110 when hydrogen sulfide is indicated as a potential problem.

## 2.2 MATERIALS

Except as otherwise specified in the following paragraphs, conform material to Section 03 30 00 CAST-IN-PLACE CONCRETE.

### 2.2.1 Pigments

Non-fading and lime-resistant

### 2.2.2 Reinforcement

#### 2.2.2.1 Reinforcing Bars

- a. Deformed Billet-steel: ASTM A615/A615M
- b. Deformed Low-alloy steel: ASTM A706/A706M

#### 2.2.2.2 Reinforcing Wire

- a. Plain Wire: ASTM A1064/A1064M
- b. Deformed Wire: ASTM A1064/A1064M

#### 2.2.2.3 Welded Wire Reinforcement

- a. Plain Wire: ASTM A1064/A1064M
- b. Deformed Wire: ASTM A1064/A1064M

#### 2.2.2.4 Epoxy Coated Reinforcement

- a. Reinforcing Bars: ASTM A775/A775M
- b. Wires and Welded Wire: ASTM A884/A884M

#### 2.2.2.5 Galvanized Reinforcement

Provide galvanized reinforcement conforming to ASTM A767/A767M.

### 2.2.3 Synthetic Fiber Reinforcement

Provide fiber reinforced concrete in accordance with ASTM C1116/C1116M Type III, synthetic fiber reinforced concrete, and as follows. Synthetic reinforcing fibers must be monofilament polypropylene fibers. Provide fibers that have a specific gravity of 0.9, a minimum tensile strength of 70 ksi, graded per manufacturer, and specifically manufactured to an optimum gradation for use as concrete secondary reinforcement.

### 2.2.4 Inserts and Embedded Metal

All items embedded in concrete must be of the type required for the intended task, and meet the following standards.

- a. Structural Steel Plates, Angles, etc.: ASTM A36/A36M
- b. Hot-dipped Galvanized: ASTM A153/A153M
- c. Proprietary Items: In accordance with manufacturers published literature

#### 2.2.5 Accessories

Submit proper installation instructions and relevant product data for items including, but not limited to, sealants, gaskets, connectors, steps, cable racks and other items installed before or after delivery.

- a. Rubber Gaskets for Circular Concrete Sewer Pipe and Culvert Pipe: ASTM C443.
- b. External Sealing Bands for Noncircular Sewer, Storm Drain and Culvert Pipe: ASTM C877.
- c. Preformed Flexible Joint Sealants for Concrete Pipe, Manholes, and Manufactured Box Sections: ASTM C990.
- d. Elastomeric Joint Sealants: ASTM C920

#### 2.2.6 Pipe Entry Connectors

Pipe entry connectors must conform to ASTM C923 or ASTM C1478.

#### 2.2.7 Grout

Nonshrink Grout must conform to ASTM C1107/C1107M. Cementitious grout must be a mixture of portland cement, sand, and water. Proportion one part cement to approximately 2.5 parts sand, with the amount of water based on placement method.

### PART 3 EXECUTION

#### 3.1 FABRICATION AND PLACEMENT

Perform fabrication in accordance with NPCA QC Manual or ACPA QPC unless specified otherwise.

##### 3.1.1 Forms

Use forms, for manufacturing precast concrete products, of the type and design consistent with industry standards and practices. They should be capable of consistently providing uniform products and dimensions. Construct forms so that the forces and vibrations to which the forms will be subjected can cause no product damage. Clean forms of concrete build-up after each use. Apply form release agents according to the manufacturers recommendations and do not allow to build up on the form casting surfaces.

##### 3.1.2 Reinforcement

Follow applicable ASTM Standard or ACI 318 for placement and splicing. Fabricate cages of reinforcement either by tying the bars, wires or welded wire reinforcement into rigid assemblies or by welding, where permissible, in accordance with AWS D1.4/D1.4M. Position reinforcing as specified by the design and so that the concrete cover conforms to requirements. The tolerance on concrete cover must be one-third of that specified but not more than 1/2 inch. Provide concrete cover not less than 1/2 inch. Take positive means to assure that the reinforcement does not move significantly during the casting operations.



### 3.1.3 Embedded Items

Position embedded items at locations specified in the design documents. Perform welding in accordance with AWS D1.1/D1.1M when necessary. Hold rigidly in place inserts, plates, weldments, lifting devices and other items to be imbedded in precast concrete products so that they do not move significantly during casting operations. Submit product data sheets and proper installation instruction for anchors, lifting inserts and other devices. Clearly indicate the products dimensions and safe working load.

### 3.1.4 Synthetic Fiber Reinforced Concrete

Add fiber reinforcement to the concrete mix at the batch plant in accordance with the applicable sections of ASTM C1116/C1116M and the recommendations of the manufacturer. Use a minimum of 1.5 pounds of fibers per cubic yard of concrete.

## 3.2 CONCRETE

### 3.2.1 Concrete Mixing

Mixing operations must produce batch-to-batch uniformity of strength, consistency, and appearance.

### 3.2.2 Concrete Placing

Deposit concrete into forms as near to its final location as practical. Keep the free fall of the concrete to a minimum. Consolidate concrete in such a manner that segregation of the concrete is minimized and honeycombed areas are kept to a minimum. Use vibrators to consolidate concrete with frequencies and amplitudes sufficient to produce well consolidated concrete.

#### 3.2.2.1 Cold Weather Concreting

Perform cold weather concreting in accordance with ACI 306.1.

- a. Provide adequate equipment for heating concrete materials and protecting concrete during freezing or near-freezing weather.
- b. All concrete materials, reinforcement, forms, fillers, and ground with which concrete is to come in contact must be free from frost.
- c. Do not use frozen materials or materials containing ice.
- d. In cold weather the temperature of concrete at the time of placing must not be below 45 degrees F. Discard concrete that freezes before its compressive strength reaches 500 psi.

#### 3.2.2.2 Hot Weather Concreting

Follow recommendations for hot weather concreting in ACI 305R. During hot weather, give proper attention to constituents, production methods, handling, placing, protection, and curing to prevent excessive concrete temperatures or water evaporation that could impair required strength or serviceability of the member or structure. The temperature of concrete at the time of placing must not exceed 90 degrees F.

### 3.2.3 Concrete Curing

Commence curing immediately following the initial set and completion of surface finishing.

#### 3.2.3.1 Curing by Moisture Retention

Prevent moisture evaporation from exposed surfaces until adequate strength for stripping is reached by one of the following methods:

- a. Cover with polyethylene sheets a minimum of 6 mils thick in accordance with ASTM C171.
- b. Cover with burlap or other absorptive material and keep continually moist.
- c. Use a membrane-curing compound, conforming to ASTM C309 and applied at a rate not less than 200 square ft/gallon, or in accordance with manufacturers' recommendations.

#### 3.2.3.2 Curing with Heat and Moisture

Do not subject concrete to steam or hot air until after the concrete has attained its initial set. Apply steam, if used, within a suitable enclosure, which permits free circulation of the steam in accordance with CSA A23.4. If hot air is used for curing, take precautions to prevent moisture loss from the concrete. The temperature of the concrete must not be permitted to exceed 150 degrees F. These requirements do not apply to products cured with steam under pressure in an autoclave.

### 3.2.4 Surface Finish

Finish unformed surfaces of wet-cast precast concrete products as specified. If no finishing procedure is specified, finish such surfaces using a strike-off to level the concrete with the top of the form.

#### 3.2.4.1 Formed Non-Architectural Surfaces

Cast surfaces against approved forms following industry practices in cleaning forms, designing concrete mixes, placing and curing concrete. Normal color variations, form joint marks, small surface holes caused by air bubbles, and minor chips and spalls will be accepted but no major imperfections, honeycombs or other major defects will be permitted.

#### 3.2.4.2 Unformed Surfaces

Finish unformed surfaces with a vibrating screed, or by hand with a float. Normal color variations, minor indentations, minor chips and spalls will be accepted. Major imperfections, honeycombs, or other major defects are not permitted.

#### 3.2.4.3 Special Finishes

Troweled, broom or other finishes must be according to the requirements of project documents and performed in accordance with industry standards or supplier specifications. Submit finishes for approval when required by the project documents. The sample finishes must be approved prior to the start of production.

### 3.2.5 Stripping Products from Forms

Do not remove products from the forms until the concrete reaches the compressive strength for stripping required by the design. If no such requirement exists, products may be removed from the forms after the final set of concrete provided that stripping damage is minimal.

### 3.2.6 Patching and Repair

No repair is required to formed surfaces that are relatively free of air voids and honeycombed areas, unless the surfaces are required by the design to be finished.

#### 3.2.6.1 Repairing Minor Defects

Defects that will not impair the functional use or expected life of a precast concrete product may be repaired by any method that does not impair the product.

#### 3.2.6.2 Repairing Honeycombed Areas

When honeycombed areas are to be repaired, remove all loose material and cut back the areas into essentially horizontal or vertical planes to a depth at which coarse aggregate particles break under chipping rather than being dislodged. Use proprietary repair materials in accordance with the manufacturer's instructions. If a proprietary repair material is not used, saturate the area with water. Immediately prior to repair, the area should be damp, but free of excess water. Apply a cement-sand grout or an approved bonding agent to the chipped surfaces, followed immediately by consolidating an appropriate repair material into the cavity.

#### 3.2.6.3 Repairing Major Defects

Evaluate, by qualified personnel, defects in precast concrete products which impair the functional use or the expected life of products to determine if repairs are feasible and, if so, to establish the repair procedure.

### 3.2.7 Shipping Products

Do not ship products until they are at least five days old, unless it can be shown that the concrete strength has reached at least 75 percent of the specified 28-day strength, or that damage will not result, impairing the performance of the product.

## 3.3 INSTALLATION

### 3.3.1 Site Access

It is the Contractor's responsibility to provide adequate access to the site to facilitate hauling, storage and proper handling of the precast concrete products.

### 3.3.2 General Requirements

- a. Install precast concrete products to the lines and grades shown in the contract documents or otherwise specified.
- b. Lift products by suitable lifting devices at points provided by the

precast concrete producer.

- c. Install products in accordance with the precast concrete producer's instructions. In the absence of such instructions, install underground utility structures in accordance with ASTM C891. Install pipe and manhole sections in accordance with the procedures outlined by the American Concrete Pipe Association.
- d. Field modifications to the product will relieve the precast producer of liability even if such modifications result in the failure of the product.

### 3.3.3 Water Tightness

Where water tightness is a necessary performance characteristic of the precast concrete product's end use, watertight joints, connectors and inserts should be used to ensure the integrity of the entire system.

## 3.4 FIELD QUALITY CONTROL

### 3.4.1 Site Tests

When water tightness testing is required for an underground product, use one of the following methods:

### 3.4.2 Vacuum Testing

Prior to backfill vacuum test system according to ASTM C1244.

### 3.4.3 Water Testing

Perform water testing according to the contract documents and precast concrete producer's recommendations.

-- End of Section --

SECTION 04 03 00

CONSERVATION TREATMENT FOR PERIOD MASONRY  
**11/17**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)

ACGIH 0100 (2017; Suppl 2020) Documentation of the  
Threshold Limit Values and Biological  
Exposure Indices

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon  
Structural Steel

ASTM C34 (2023) Standard Specification for  
Structural Clay Loadbearing Wall Tile

ASTM C67/C67M (2023a) Standard Test Methods for Sampling  
and Testing Brick and Structural Clay Tile

ASTM C170/C170M (2023) Standard Test Method for  
Compressive Strength of Dimension Stone

ASTM C216 (2023) Standard Specification for Facing  
Brick (Solid Masonry Units Made from Clay  
or Shale)

ASTM C926 (2023a) Standard Specification for  
Application of Portland Cement-Based  
Plaster

ASTM C1196 (2014) Standard Test Method for In Situ  
Compressive Stress Within Solid Unit  
Masonry Estimated Using Flatjack  
Measurements

ASTM C1197 (2014) Standard Test Method for In Situ  
Measurement of Masonry Deformability  
Properties Using the Flatjack Method

ASTM C1324 (2020) Standard Test Method for  
Examination and Analysis of Hardened  
Masonry Mortar

ASTM C1364 (2023) Standard Specification for  
Architectural Cast Stone

ASTM C1531 (2015) Standard Test Methods for In Situ

Measurement of Masonry Mortar Joint Shear  
Strength Index

|               |                                                                                                          |
|---------------|----------------------------------------------------------------------------------------------------------|
| ASTM C1601    | (2014) Standard Test Method for Field Measurement of Water Penetration of Masonry Wall Surfaces          |
| ASTM C1713    | (2012) Standard Specification for Mortars for the Repair of Historic Masonry                             |
| ASTM E96/E96M | (2024) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials |
| ASTM E2260    | (2012) Standard Guide for Repointing (Tuckpointing) Historic Masonry                                     |
| ASTM E2659    | (2018) Standard Practice for Certificate Programs                                                        |

NATIONAL PARK SERVICE (NPS)

|                 |                                                                                                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NPS Hist Prop   | (2017) National Standards for the Treatment of Historic Properties with Guidelines for Preserving, Rehabilitating, Restoring, and Reconstructing Historic Buildings |
| NPS TPS Brief 1 | (2000) Assessing Cleaning and Water-Repellent Treatments for Historic Masonry Buildings                                                                             |
| NPS TPS Brief 2 | (1998) Repointing Mortar Joints in Historic Masonry Buildings                                                                                                       |

THE MASONRY SOCIETY (TMS)

|          |                                                                                                                                                                                                                      |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TMS MSJC | (2016) Masonry Standard Joint Committee's (MSJC) Book - Building Code Requirements and Specification for Masonry Structures, Containing TMS 402/ACI 530/ASCE 5, TMS 602/ACI 530.1/ASCE 6, and Companion Commentaries |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

|                   |                                                          |
|-------------------|----------------------------------------------------------|
| GSA HPTP 07656-01 | Installing Lead Stone Flashing to Protect Masonry Joints |
|-------------------|----------------------------------------------------------|

1.2 DEFINITIONS

Terms are defined below as applicable to this project.

1.2.1 Aggregates

The sand component of mortar.

#### 1.2.2 Biocides

A chemical treatment that inhibits, deters, or controls organic growth. Such growth is typically removed by cleaning following biocide treatment..

#### 1.2.3 Binder

The component of mortar that binds together the aggregate particles into a cohesive material.

#### 1.2.4 Dispersed Lime Crack Injection

A repair method in which dispersed lime material is injected into small cracks ranging in width from hairline to 1/8 in by use of needle or syringe.

#### 1.2.5 Consolidant

A chemical product meant to strengthen loose or deteriorated stone.

#### 1.2.6 Dutchman

A repair method in which deteriorated stone is removed in part and replaced with salvaged, harvested or new stone to make a seamless patch.

#### 1.2.7 Harvested

Units removed from inconspicuous areas of the building.

#### 1.2.8 In situ

A term referencing a repair procedure in which the masonry units and mortar remain in place and are repaired without removal from the wall system

#### 1.2.9 Joint Sealant

A flexible, chemical product that is used to create a weather-tight seal at the boundary of masonry units with other units or dissimilar materials.

#### 1.2.10 Lead Flashing

An extruded lead material that is inserted into joints to assist in precluding water entry into the masonry.

#### 1.2.11 Lime Wash

A protective surface treatment comprised of calcium hydroxide particles in suspension in water, along with small amounts of calcium carbonate, silica particles and other minerals.

#### 1.2.12 Mockup

Specific area on the building approved by Contracting Officer to demonstrate the ability to apply, match and install specified materials.

#### 1.2.13 Mortar

A mixture of binders, aggregates, and pigments used for reconstruction,

repointing or stucco applications.

#### 1.2.14 New Elements

New, non-historic materials added to masonry structures to aid in their ability to resist loads (typically seismic) or to resist water infiltration.

#### 1.2.15 Patch

The use of substitute repair materials to treat damaged or deteriorated masonry units in situ.

#### 1.2.16 Remediate

An intervention of a historic masonry structure and its component materials with the intent to maintain the original fabric to the greatest extent possible.

#### 1.2.17 Remove

Specifically for historic masonry materials, the term means to detach an item from existing construction to the limits indicated.

#### 1.2.18 Replace

To reinstall an item in its original position (or where indicated) after remedial treatment, or to duplicate and reinstall an entire item with new material; with the original item serving as the pattern for creating the duplicate.

#### 1.2.19 Repoint

To remove existing mortar joints to the specified depth and replace with a mortar that matches in color, texture, and performance with water vapor transmission, bond, hardness, and flexibility compatible with original mortar, as assessed in accordance with ASTM C1713.

#### 1.2.20 Retool

A repair method in which a chisel is used to re-create the surrounding stone texture finish by removing loose pieces of stone.

#### 1.2.21 Stucco

A mixture of binders and aggregates, sometimes including animal hair or fibers used for the repair treatment of existing stucco.

#### 1.2.22 Surface Treatment

The application of traditional materials or contemporary chemical products to the surface of masonry to provide protection to the masonry units and mortar and/or reduce water infiltration.

#### 1.2.23 Test Panel

Specific area on the building approved by the Contracting Officer to demonstrate individual applicator competency and workmanship proficiency prior to the start of restoration work.



#### 1.2.24 Tuckpointing

Often called skim-coating, an American practice of surface repairing mortar joints without the required removal of existing deteriorated mortar beneath. This practice is not recommended for mortar joint repair work on historic masonry. There is also an acceptable British form of tuckpointing practice that involves careful thin penciling of smaller joints within larger ones to give the wall the appearance of an ashlar finish.

#### 1.2.25 Water Repellent

A surface-applied chemical intended to reduce liquid water entry into a masonry wall without significantly affecting the vapor transmission properties of the original material.

#### 1.2.26 Wall System

A term used to address the fact that masonry structures are comprised of different materials but function holistically, requiring that all restoration and cleaning process take into account the implications of the treatment to the adjacent materials and the building as a whole.

#### 1.2.27 Masonry Treatment Requirement (MTR)

Defined treatments that are required by the specification (contract) documents for project specific repairs to masonry.

#### 1.2.28 Saturated Surface Dry (SSD)

Condition of the wall surface after water has been applied sufficient to saturate more than the surface, then allowed to dry until the surface is dry but the body of the masonry still has moisture.

### 1.3 ADMINISTRATIVE REQUIREMENTS

#### 1.3.1 Pre-Installation Meeting

Prior to beginning the work of this Section, convene a meeting with the Contracting Officer's Representative(s) to review the requirements of the Quality Control Plan, Project Training Program, installation procedures, location of required mockup areas, and all job conditions and processes. All subcontracting firms involved with this work must participate in this meeting.

### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-01 Preconstruction Submittals

Quality Control Plan; G

Project Training Program; G

Qualifications; G

SD-02 Shop Drawings; G

Photographic Documentation

Structural Upgrades; G

SD-03 Product Data

Qualifications

Cleaning and Restoration Methods; G

Cleaning Materials; G

Biocides

Replacement Mortar And Stucco; G

Mortar Mix; G

Water Repellents Infiltration; G

Stone Consolidants; G

SD-04 Samples

Mock-ups; G

SD-05 Design Data

Calculations for Structural Upgrades; G

SD-06 Test Reports

Testing and Matching

Existing Sealants for Asbestos and PCBs

SD-07 Certificates

Repair Materials

1.5 QUALITY CONTROL

1.5.1 Quality Control Plan

Prior to beginning restoration and cleaning work, submit a written Quality Control Plan. Include a separate section in the overall project Quality Control Plan specifically addressing this restoration and cleaning work. Do not proceed without written approval of the Quality Control Plan. At a minimum, include the following items in the Quality Control Plan:

- a. Describe methods of dust containment during the work specific to the work of this section.
- b. Describe the methods of protecting surrounding masonry, windows, doors, roof, and building trim as well as surrounding landscape. Provide drawings of protection when requested.

- c. Describe the work procedures, materials, and proposed tools to use for each Masonry Treatment Requirement (MTR) specified.
- d. Describe the sequence of each MTR.
- e. Describe how each MTR sequence and the overall construction schedule changes with weather variations and how completed work will be protected.
- f. Describe the methods for surveying original layout and collecting datum points and plumb lines for rebuilding masonry.
- g. Describe the methods for shoring and providing a safe working environment.
- h. Describe the methods for select deconstruction of individual masonry units and tools/methods for cleaning the masonry for reuse.
- i. Describe the method and approach to mortar joint removal.
- j. Describe the method and approach for assuring repair material compatibility with original materials.
- k. Describe the method and approach to cleaning mortar, coating, smears and old patching materials from the masonry surfaces.
- l. Describe, in detail, the procedures relating to techniques and tools proposed for masonry matching.
- m. Describe the complete masonry removal and matching procedures; include equipment, approach, length of time the masonry will be out of the wall, documentation on mapping the location, and where (on-site or in shop) the masonry units will be repaired.
- n. Describe the procedure for matching of different colors at different locations.
- o. Describe the procedure for mixing and matching of repair materials.
- p. Describe the methods and system by which the use of reclaimed masonry units can be utilized.
- q. Describe the methods for setting masonry back into its original position and maintaining the original bond patterns and joint width.
- r. Describe the methods of transition points where replacement/preservation work will meet the original historic work.
- s. Describe the on-site project training program. Provide the opportunity for workers to be trained in each masonry treatment requirement (MTR) as work proceeds.

#### 1.5.2 Qualifications

##### 1.5.2.1 Historic Masonry Consultant

- a. Secure the services of a historic masonry consultant with a minimum of 10 years experience applying NPS Hist Prop as they relate to the work

in this section.

- b. Submit a resume that describes five relevant projects within that period and include how NPS Hist Prop was applied to the work of similar scope and scale and what jurisdiction or agency was involved in approving the work..
- c. The consultant's services include:
  - (1) Investigating the condition of the masonry materials and mortar.
  - (2) Arranging for material analysis in the laboratory
  - (3) Recommending appropriate cleaning methods and materials
  - (4) Recommending restoration options.
  - (5) Providing project specific specifications.
  - (6) Providing an on-site training program.
  - (7) Providing quality control services during construction.
  - (8) Recommending appropriate repair and restoration materials.

#### 1.5.2.2 Masonry Firm

- a. The firm performing the masonry work must have a minimum of five years experience on relevant projects.
- b. The firm must have completed work similar in material, design, and extent to that indicated for this Project and demonstrate a record of successful in-service performance.
- c. Proven implementation of NPS Hist Prop and related Preservation Briefs are required.
- d. Submit a resume that describes the required experience.

#### 1.5.2.3 Field Supervisor

Retain an experienced full-time supervisor on the project site at all times when masonry restoration is in progress. A single individual must be responsible for supervising the historic masonry restoration work throughout the duration of the project.

Submit a resume that describes the required experience.

#### 1.5.2.4 Masonry Applicator

- a. Employ craftspeople who are experienced with and specialize in restoration work of the types they will be performing.
- b. All masonry restoration treatments must be performed by a craftsperson that is familiar with historic masonry construction and has worked on historic masonry projects for at least five years.
- c. Only skilled technicians who are familiar and experienced with the materials and methods specified may be used.

- d. Submit resumes for all historic masonry applicators, demonstrating the required experience.

#### 1.5.3 Project Training Definition and Use

In addition to five years demonstrable experience on masonry restoration projects, offer workers project training certificate(s) within the framework of ASTM E2659. Project training certificates are earned by individual workers and issued with the understanding that they are for limited time use, enforceable only to this specific project and for a specific MTR. It is not necessary, nor a requirement of this specification, that all restoration workers obtain all project training certificates offered. Rather it is desirable that workers be trained for each project specific task they will perform to ensure the highest quality results from the cleaning and restoration program.

#### 1.5.4 Mortar Analyst

Laboratory mortar analysis equipment should be operated by and results analyzed by trained personnel experienced with analysis of historic masonry mortar.

#### 1.5.5 Documentation

Submit digital photographic documentation of the all phases of masonry restoration, including prior to the start of restoration work.

Provide thorough photo documentation of the project and project details and targeted areas.

#### 1.5.6 Cleaning and Restoration Methods

##### 1.5.6.1 General Procedure

- a. Submit the cleaning and restoration methods, and materials selected for a specific structure for approval before work starts.
- b. Take into account the total construction system of the building to be worked upon, including different masonry and mortar materials, as well as non-masonry elements which may be affected by the work.
- c. Utilize mockups to identify the appropriate cleaning and restoration treatment and materials and set the standard for each project task.
- d. Demonstrate the correct execution of the approved cleaning and restoration methods and materials during the on-site workmanship training program within the framework of ASTM E2659.

##### 1.5.6.2 Cleaning Products and Procedures

###### 1.5.6.2.1 General Cleaning Requirements

- a. Establish cleaning products and procedures during the mockup process.
- b. Select the least aggressive method used to achieve the desired level of cleanliness.
- c. Where chemical products are selected for cleaning, use them in

accordance with the manufacturer's instructions.

#### 1.5.6.2.2 Cleaning Mock-Ups

- a. Demonstrate the materials, equipment, and methods to be used in cleaning in a test section approximately 3 feet by 3 feet.
- b. Locate test patches in inconspicuous areas of the building. The areas tested are subject to approval by the Contracting Officer. The areas tested must exhibit soiling characteristics representative of those larger areas to be cleaned.
- c. Adjust the cleaning process as required and the test section rerun until an acceptable process is obtained.
- d. Conduct tests on areas to be stripped of paint.
- e. Allow tested areas to dry before a determination is made on the effectiveness of a particular treatment.

#### 1.5.7 Masonry Restoration Products and Procedures

##### 1.5.7.1 General Restoration Requirements

- a. Do not use masonry or mortar in the work until the mock-ups and the represented material and workmanship have been submitted and approved.
- b. Demonstrate the methods and quality of workmanship to be performed in each masonry treatment requirement (MTR). Provide a mock-up for each MTR indicated.

##### 1.5.7.2 General Restoration Mock-Up Requirements

- a. Throughout restoration, retain approved mock-up panels in undisturbed condition, suitably marked, as a standard for judging completed work.
- b. Review manufacturer's product data sheets to determine suitability of each product for each surface.
- c. Apply products using manufacturer-approved application methods, determining actual requirements for application.
- d. Obtain approval as to the preservation treatment approach, design, and workmanship to include, but not limited to the verification of all material applications and finishes as specified to the requirements of color, texture, profiles, and finishes before proceeding with work.

##### 1.5.7.2.1 Mock-ups

May be performed on inconspicuous sections of actual construction under the same weather conditions expected the remainder of the work.

- a. Location and number as directed.
- b. Size: 3 feet by 3 feet or as appropriate for the repair specified.
- c. Repair unacceptable work.

### 1.5.7.3 Restoration Mock-Ups

#### 1.5.7.3.1 Repointing

Repoint mortar joints, minimum acceptable mock up dimensions: twelve feet in length - 2/3 horizontal joints and 1/3 vertical joints. Demonstrate method for cutting out mortar joints, preparing wall for repointing, mixing mortar, installing mortar and curing the mortar. Prepare and place repointing mortar in accordance with NPS TPS Brief 2 and in compliance with NPS Hist Prop.

#### 1.5.7.3.2 Retooling Stone Masonry In Situ

Demonstrate treatment technique and methods to retool three deteriorated stone faces in situ in all known historic profile textures identified.

#### 1.5.7.3.3 Masonry Removal and Replacement

Fully remove masonry and replace to specified dimensions and texture. Select size of masonry units representing typical conditions. Return one masonry unit to same location, set to surrounding profile joint width and bond pattern. Set masonry unit using specified mortar. Confirm with Contracting Officer's Representative that the replacement masonry units meet specification requirements for matching and that sufficient quantity required for the work have been identified. Leave one stone dry-set into opening set on wood shims for evaluation and approval of preparation conditions.

#### 1.5.7.3.4 Repair Material

##### 1.5.7.3.4.1 Patching

Apply repair material on at least two masonry units for repair. Include one masonry unit on which to demonstrate proficiency in removing previous patching material and repairing with new substitute repair material. Include the removal of metal anchors at two locations and fill in the holes with repair material on the second masonry unit (where applicable).

##### 1.5.7.3.4.2 Dutchman

Undertake dutchman repairs in two locations, including one that is only cut and prepared for application. Demonstrate the quality of the stone insert, as well as the workmanship and techniques to be performed in the dutchman repairs. Do not proceed with other dutchman repairs until the technique has been approved.

#### 1.5.7.3.5 Crack Repair

Repair one crack, 2 feet in length, using mortar. Repair one crack, 2 feet in length, using dispersed hydrated lime injection technique with appropriate repair material.

#### 1.5.7.3.6 Surface Treatments

Install a minimum 1.5 square meter 16 square foot mockup for each surface treatment on each substrate to be treated. For stucco, demonstrate the means for installing each coat; including mechanical support systems such as wood or metal lath. For water repellents and/or consolidants, demonstrate the equipment and installation procedure. Allow 48 hours for

limewash applications to dry to their final color and appearance.

#### 1.5.7.3.7 New Masonry Elements

Install new components in a manner demonstrating their final installation on the structure.

### 1.6 DELIVERY, STORAGE, AND HANDLING

- a. Furnish cement in suitable bags used for packaging cements.
- b. Provide packages with labeling that clearly defines contents, manufacturer, and batch identification.
- c. Provide detergents, masonry cleaners, paint removers, solvents, epoxies and other chemicals used for masonry cleaning in sealed containers that legibly show the designated name, formula or specification number, quantity, date of manufacture, manufacturer's formulation number, manufacturer's directions including any warnings and special precautions, and name of manufacturer.
- d. Store materials in weathertight structures which will protect all materials from moisture and contaminants.
- e. Store accessories to avoid contamination and deterioration.
- f. Do not use admixtures which have been in storage onsite for six months or longer, or which have been subjected to freezing, unless retested and proven to meet the specified requirements.

### 1.7 PROJECT/SITE CONDITIONS

#### 1.7.1 Environmental Requirements

- a. Do not place materials when weather conditions adversely affect the quality of the finished product.
- b. Do not place masonry or mortar when the air or surface temperature is below 40 degrees F in the shade and will remain so for at least 48 hours after completion of the work. Heated enclosures may be used to overcome ambient weather restrictions, where such enclosures are feasible.
- c. Do not place masonry or mortar when air or surface temperature is above 90 degrees F with a wind speed above 8 miles per hour and will remain so for at least 48 hours after completion of the work.
- d. Do not place masonry or mortar when air or surface temperature is above 100 degrees F with or without wind and will remain so for at least 48 hours after completion of the work.
- e. Do not produce or place materials during periods of rain or other precipitation. Stop material placements, and protect all in-place material from exposure, during periods of rain or other precipitation.
- f. Clean masonry surfaces when air temperatures are above 40 degrees F and will remain so until masonry has dried out, but for not less than 7 days after completion of the work.



- g. Do not perform work in wind conditions that may blow materials onto surfaces not intended to be treated.

#### 1.7.2 Masonry Installation Requirements

- a. Phase work during hot weather by performing work on the shady side(s) of the building during daylight hours and on the daylight side(s) of the building during cooler evening hours to prevent premature evaporation of the water from the mortar.
- b. Do not use frozen materials or materials mixed or coated with ice or frost. Do not apply materials to frozen surfaces; allow complete thawing prior to installation.
- c. Do not lower the freezing point of mortar by the use of admixtures or anti-freeze agents. Do not add chlorides or admixtures containing greater than 0.2 percent chlorides to the mortar, per TMS MSJC.
- d. Prevent mortar from staining the face of the masonry or other exposed surfaces. Immediately remove mortar that comes in contact with such surfaces. Cover partially completed work when work is not in progress. Protect sills, ledges and projections from mortar droppings. Building damage resulting from work of this Section is the Contractor's responsibility. Restore damaged areas to the satisfaction of the Owner at no expense to the Owner. Do not apply products under conditions outside product manufacturer's requirements.

### 1.8 WARRANTY

#### 1.8.1 Cleaning Warranty

Warrant cleaning procedures for a period of two years against harm to substrate (masonry and mortar) or to adjacent materials including, but not limited to discoloration of substrate from improper procedures or usage, chemical damage from inadequate rinse procedures, and abrasive damage from improper procedures.

#### 1.8.2 Repair Warranty

Warrant repair procedures, including repointing, for a period of two years against: discoloration or mismatch of new mortar to adjacent original historic mortar, discoloration or damage to masonry from improper mortar clean-up, loss of bond between masonry and mortar, fracturing of masonry edges from improper mortar joint preparation procedures or improper mortar formulation, and occurrence of efflorescence from improper repair procedures.

## PART 2 PRODUCTS

### 2.1 CLEANING MATERIALS

#### 2.1.1 General Requirements

Selection of appropriate cleaning products requires a clear understanding of the masonry materials to be cleaned, a rationale for the cleaning, and an understanding of the anticipated level of cleanliness expected from the cleaning program. Overly aggressive cleaning methods and materials can cause subtle, long-term damage to masonry units. Use products that have a minimum 5 year performance record on relevant projects. Select the

products predicated on long-term negative effects to the masonry rather than current level of cleanliness of the comparable structure.

#### 2.1.2 Paint Removers

- a. Provide chemical paint removers that are water soluble, low toxicity products, effective for removal of paint on masonry without altering, damaging, or discoloring the masonry surface.
- b. Provide commercially available poulticing materials designed to adhere to and peel off paint without damaging the underlying masonry or project specific mixtures that include absorbent materials and cleaning solutions which can be demonstrated to do no harm to the masonry.

#### 2.1.3 Chemical Cleaners

- a. Provide commercially available products that have a proven record of cleaning masonry without altering, damaging or discoloring the masonry units, mortar or surrounding materials.
- b. Provide the associated pre and post treatment material to neutralize the long term effects of the chemicals.

#### 2.1.4 Biocides

Provide commercially available biocides with accompanying product literature containing information on the product as well as the expected service life of the material and any detrimental effects it may have on the masonry or mortar.

#### 2.1.5 Liquid Strippable Masking Agent

Provide manufacturer's standard liquid, film-forming, strippable masking material for protecting glass, metal, and polished stone surfaces from the damaging effect of acidic and alkaline masonry cleaners.

#### 2.1.6 Cleaning Implements

Furnish brushes that contain natural or nylon fiber bristles only. Do not use metallic wire brushes. Use scrapers and application paddles made of wood with rounded edges. Metallic tools are not permitted.

#### 2.1.7 Water

Obtain potable water from a local source.

### 2.2 REPAIR MATERIALS

#### 2.2.1 General

Use repair materials of one type and from one source, when used in repair treatments that will have surfaces exposed in the finished structure.

## 2.2.2 Mortar and Stucco

### 2.2.2.1 Testing and Matching

- a. Take test specimens of existing mortar and stucco from a sound and intact representative portion of the structure, at locations indicated by the Contracting Officer's Representative and assess in accordance with ASTM C1713 and ASTM C1324.
- b. Subject a part of the historic mortar sample to petrographic examination and differential thermal analysis, or X-ray diffraction, or analytical chemistry to determine the binder components.
- c. Aggregate Analysis
  - (1) Separate aggregate of the mortar sample from the binder by taking the crushed mortar sample and either gently blowing away the fine binder material, placing the crushed sample in a centrifuge, or chemically separating the aggregate from the binder.
  - (2) Rinse the separated aggregate clean with water and dry. Examine the aggregate with a magnifying glass, and record the component materials as to range of materials, sizes, colors, as well as the presence of other materials.
  - (3) Perform sand analysis using a sieve analysis of the aggregate as part of the ASTM C1324 process.
- d. Match the replacement mortar and stucco to the original existing material in color, texture and tooling.

### 2.2.2.2 Replacement Mortar and Stucco

Provide replacement mortar and stucco that will:

- a. Coexist with the old in a sympathetic, supportive and, if necessary, sacrificial capacity.
- b. Have greater vapor permeability and be softer (measured in compressive strength) than the masonry units.
  - (1) Measure water vapor transmission in accordance with ASTM E96/E96M.
  - (2) Prepare ASTM E96/E96M water vapor transmission specimens with thickness similar to that expected in service, or a maximum of 1/2 inch, whichever is thinner.
- c. Be as vapor permeable, and as soft, or softer, (measured in compressive strength) than the existing historic mortar or stucco.

### 2.2.2.3 Binder Content

Provide binder type or mixture of mortar (and stucco) with a cement, lime, or combination thereof consistent with the original existing mortar (and stucco) content in order to provide uniform durability, weathering characteristics, and the same, or better, life-cycle performance expectations.

#### 2.2.2.4 Repointing Mortar

Repointing mortar may be site mixed, Type S, Type N, Type L, Type O and Type K.

#### 2.2.2.5 Admixtures

Do not use admixtures in the mortar or stucco unless specifically approved in writing by the Contracting Officer.

#### 2.2.3 Crack Injection

- a. Comply with the dispersed hydrated lime manufacturer's written instructions.
- b. Inject cracks that are no greater than 1/8 inch in width and masonry is soundly bonded but cracked.
- c. Inject the full length of the cracks unless specifically instructed otherwise.

#### 2.2.4 Replacement Masonry Materials

##### 2.2.4.1 Clay Brick

- a. Provide replacement brick matching color, shape, size, texture, appearance, and thermal expansion properties of the existing historic brick.
- b. Test brick in comparison to the original existing historic brick using ASTM C67/C67M.
- c. Do not use reclaimed brick unless approved by Contracting Officer.
- d. Provide brick meeting the requirements of ASTM C216 Grade SW, including a rating of "not effloresced", unless otherwise specified.

##### 2.2.4.2 Stone

- a. Provide replacement stone matching type, color, shape, size, texture, finish-profile, and compressive strength of the existing historic stone units.
- b. Test replacement stone in comparison to the existing historic stone using ASTM C170/C170M.

##### 2.2.4.3 Terra Cotta

- a. Provide replacement terra cotta matching color, shape, size, texture and finish-profile of the existing historic terra cotta units.
- b. Test replacement terra cotta in comparison to the existing historic terra cotta using ASTM C34.

##### 2.2.4.4 Architectural Precast Stone

- a. Provide replacement architectural precast stone matching color, shape, size, texture and finish-profile of the existing historic architectural precast stone units.

- b. Test replacement architectural precast stone in comparison to the existing historic architectural precast stone using ASTM C1364.

## 2.2.5 Surface Treatments

### 2.2.5.1 General

Provide commercially available coatings with water vapor permeability of 0.98 or greater, as measured in accordance with ASTM E96/E96M, including silanes and siloxanes.

### 2.2.5.2 Consolidants

Provide commercially available consolidants designed to strengthen loose or deteriorated stone without damaging intact stone or reducing water vapor permeability below 0.98, as measured in accordance with ASTM E96/E96M.

### 2.2.5.3 Water Repellents

Provide commercially available water repellents designed to preclude water droplet entry into the masonry walls without reducing water vapor permeability below 0.98, as measured in accordance with ASTM E96/E96M.

## 2.2.6 Miscellaneous Materials

### 2.2.6.1 Cementitious Grout

Use cementitious grout, recommended by the manufacturer for the application, to bond steel anchors to masonry.

### 2.2.6.2 Metal Attachments

- a. Provide threaded or deformed stainless steel anchors for spall repairs, size as indicated.
- b. Provide other plates, angles, anchors, and embedments conforming to ASTM A36/A36M, prime painted with inorganic zinc primer.

### 2.2.6.3 Lead Flashing

Provide commercially available lead flashing conforming to GSA HPTP 07656-01.

## 2.3 EQUIPMENT

### 2.3.1 Cleaning Equipment

Provide cleaning equipment that does not cause staining, erosion, marring, or other damage or changes in the appearance of the surfaces to be cleaned.

#### 2.3.1.1 Sandblasting

Use of sandblasting equipment is not allowed for cleaning masonry surfaces.

#### 2.3.1.2 Water Blasting

- a. Provide water blasting equipment including a trailer-mounted water tank, pumps, high-pressure hose, wand with safety release cutoff

control, nozzle, and auxiliary water re-supply equipment.

- b. Do not operate the equipment at a pressure which will cause etching or other damage to the masonry surface or mortar joints. Operate the equipment at a discharge capacity of 55 to 400 psi and 2.5 to 3 gpm for general surface cleaning operations.
- c. Provide water tank and auxiliary re-supply equipment of sufficient capacity to permit continuous operations.
- d. Provide protective covers and barriers as required to prevent over-spray onto adjacent surfaces.

#### 2.3.1.3 Alternative Blasting Equipment

- a. Alternative blasting methods require equipment designed to discharge sponges, walnut shells, ice, soda and other friable materials.
- b. Operate equipment in accordance with manufacturer's recommendations and maintain in good working order.
- c. Do not operate equipment at a pressure which will cause etching or other damage to the masonry surface or mortar joints.
- d. Determine discharge capacity on a case by case basis during the mockup test panel demonstration and approval process.
- e. Provide protective covers and barriers as required to prevent over-spray onto adjacent surfaces.

#### 2.3.2 Spray Equipment

- a. Provide spray equipment for chemical cleaners with low-pressure tanks or chemical pumps suitable for chemical cleaner indicated, and equipped with stainless steel, cone-shaped spray-tip.
- b. Disperse water through a fan-shaped spray tip at an angle of not less than 15 degrees.
- c. Deliver water at a pressure not greater than 400 psi and at a volume between 2.5 and 3 gpm.
- d. Deliver heated water at flow rates indicated maintaining between 140 and 180 degrees F.

#### 2.3.3 Drilling Equipment

- a. Use standard small, powered, handheld masonry drills, commonly used for drilling small holes in concrete and masonry to drill holes in masonry for patch anchors and other applications.
- b. Use drills in rotary mode only. Do not use impact type drills.

#### 2.3.4 Compressed Air Supplies

- a. Use compressed air equipment that delivers clean, oil and moisture free compressed air at the surface to be cleaned. Use a minimum of two in-line air filters to remove oil and moisture from the air supply.

- b. Test the compressed air supply during each shift for the presence of oil and moisture.

#### 2.3.5 Material Handling and Associated Equipment

##### 2.3.5.1 Mixing, Transporting, and Placing Job Materials

- a. Provide equipment used for mixing, transporting, placing, and confining masonry and mortar placements capable of satisfactorily mixing material and supporting uninterrupted placement operations.
- b. Provide equipment used for mixing, conveying, and placing of materials that is clean, free of old materials and contaminants, and in conformance with material manufacturer's recommendations.

##### 2.3.5.2 Associated Equipment

Provide associated equipment, such as mixer timing equipment, valves, pressure gauges, pressure hoses, other hardware, and tools, as required to ensure a continuous supply of material and operation control.

#### 2.4 Mortar Mix

##### 2.4.1 General

- a. Proportion materials appropriately with regard to the effect of moisture content on the individual components (cement, sand and lime).
- b. Batch materials using volumetric measurement devices and consistently consolidate the material in these devices to ensure the uniformity of the mortar. Do not batch by shovel counts.

##### 2.4.2 Batching

- a. Utilize a calibrated measuring device for batching Portland cement.
- b. Utilize a calibrated measuring device for batching hydrated lime or lime putty.
- c. Utilize a calibrated measuring device for batching the sand.

##### 2.4.3 Cement and Lime Proportions

- a. Fill the measuring device with portland cement, hydrated lime or lime putty.
- b. Briskly strike the bottom of the measuring device against the ground a minimum of ten times and then strike the top flush.
- c. For dry hydrate lime, fill the measuring device using a minimum of three lifts, strike the bottom of the measuring device against the ground a minimum of ten times for each lift and then strike the top flush. Mix dry hydrate lime to a wet paste that is 40 to 42 percent solid.
- d. For lime putty briskly strike the bottom of the measuring device against the ground a minimum of ten times and then strike the top flush. No additional lime is required when measuring from putty.

#### 2.4.4 Sand Proportions

- a. Proportion sand when the sand is in saturated surface dry (SSD), loose damp condition.
- b. Proportion the sand by filling a measuring device using a minimum of three lifts, striking the sides a minimum of ten times, and then striking the top flush.

### PART 3 EXECUTION

#### 3.1 EXAMINATION

- a. Undertake masonry renovation only after complete evaluation and analysis of the areas to be repaired are completed, including sampling and testing of the existing mortar to determine its composition and qualities. Do not start repair work until conditions that have caused masonry deterioration have been identified and corrected.
- b. Use the gentlest means to perform the work and take the greatest of care to ensure that the historic materials are not damaged in the process of the work, as established by mock-ups and testing.
- c. In addition to requirements in this Section, comply with NPS Hist Prop.

##### 3.1.1 Field (In Situ) Mortar Examination

- a. Detect cracks, degradation and de-bonding from the surrounding masonry.
- b. Determine previous surface coating treatments that may be contributing to the current conditions.
- c. Compare the bedding mortar with the pointing mortar and determine the cross-sectional characteristics of the wall.
- d. Determine the level of moisture movement in the in situ mortar, and if the mortar or masonry units are handling the brunt of the water movement through the wall.
- e. Evaluate in situ mortar joint shear strength in accordance with ASTM C1531.

##### 3.1.2 Taking and Preparation of Samples

- a. Take and analyze samples of unweathered original historic mortar and different types of mortar in the structure in order to match the new mortar to be used for repointing.
- b. Remove three or four samples of each type of mortar to be matched with a hand chisel from several locations on the building. Mortar samples to be intact pieces with a minimum size of 1 ounce.
- c. Set aside the largest sample for comparison with the repointing mortar.
- d. Place the remaining samples in labeled, sealed sample bags for transport to the laboratory for evaluation per Part 2 of this Specification.



### 3.2 PREPARATION

#### 3.2.1 Protection

- a. Protect persons, motor vehicles, adjacent surfaces, surrounding buildings, equipment, and landscape materials from chemicals used and runoff from cleaning and paint removal operations.
- b. Erect temporary protection covers, which will remain in operation during the course of the work, over pedestrian walkways and at personnel and vehicular points of entrance and exit.
- c. Protect the interior of buildings from the weather, cleaning, and repair operations at all times.
- d. Do not expose workers to chemical substances in excess of the limits established by ACGIH 0100. Comply with more stringent regulations where applicable.

#### 3.2.2 Surface Preparation

- a. Do not proceed with cleaning until mock-ups have been approved.
- b. Do not proceed with repointing or stucco until existing mortar and stucco have been analyzed and suitable repair materials have been determined.
- c. Do not proceed with restoration work until the cause of observed distresses have been identified and corrected.
- d. Do not proceed with surface treatments until all other restoration work has been completed.

#### 3.2.3 Equipment and Techniques Demonstration

- a. Demonstrate equipment and techniques of operation in an approved location.
- b. Assemble dependable and sufficient equipment, appropriate and adequate to accomplish the work specified, at the work site with sufficient lead time before the start of the work to permit inspection, calibration of weighing and measuring devices, adjustment of parts, and the making of any repairs that may be required.
- c. Maintain the equipment in good working condition throughout the project.

### 3.3 MASONRY CLEANING

#### 3.3.1 General

- a. Exercise caution against over-cleaning of surfaces, which may be detrimental, and which may remove desirable historic surface details or patinas. For example, if cleaning reveals unexpected joint painting or historic signage; suspend the cleaning action, protect the exposed area and notify the Contracting Officer.
- b. Do not damage or mar historic materials in the process of cleaning.

- c. Perform cleaning per NPS TPS Brief 1.
- d. Protect open joints to prevent water and cleaner intrusion into the interior of the structure.
- e. Protect non-masonry materials and severely deteriorated masonry by approved methods prior to initiation of cleaning operations.
- f. Remove all organic and inorganic contaminants from the surface and pores of the substrate, without causing any short or long-term negative consequences.
- g. Clean surfaces evenly with no evidence of streaking or bleaching.
- h. Do not affect the density, porosity, or color of the existing masonry or mortar.
- i. Maintain a neutral pH on surface of cleaned masonry units.
- j. Use the gentlest methods possible for cleaning historic masonry to achieve the desired results.
- k. Proceed with cleaning in an orderly manner, working from top to bottom of each scaffold width and from one end of each elevation to the other.
- l. Perform cleaning in a manner which results in uniform coverage of all surfaces, including corners, moldings, interstices and which produces an even effect without streaking or damage to masonry.
- m. Use the following sequence of methods to determine the least aggressive, effective cleaning method:
  - (1) Water with non-metallic brushes (cold water).
  - (2) Water with mild soap
  - (3) Water with stronger soap
  - (4) Water with stronger soap plus ammonia
  - (5) Water with stronger soap plus vinegar (but not on calcareous masonry)
  - (6) Stronger chemical cleaners, only when above methods are determined to be ineffective by the Contracting Officer

### 3.3.2 Chemical Cleaners

- a. Do not use chemical cleaners without approval from the Contracting Officer.
- b. Do not use acidic chemical cleaners on limestone, marble, concrete and other calcareous (calcium containing) masonry materials. If chemical cleaners are used on such materials, use alkaline based cleaners with neutralizing afterwashes.

### 3.3.3 Paint Removal

- a. Prior to removal, test existing paint for lead in accordance with Section 02 83 00.00 22 MANAGEMENT OF LEAD, CADMIUM, AND CHROMIUM (VI) DURING RENOVATION, DEMOLITION, CONSTRUCTION, REMOVAL AND ABATEMENT (PNS PROJECTS).
- b. Clean areas where paint is to be removed with water and detergent

solution to remove surface dirt. Rinse and allow to dry.

- c. Remove paint and other coatings from masonry surfaces in areas indicated prior to general cleaning.
- d. Do not damage or mar masonry in the process of paint removal.
- e. Apply chemical paint removers in accordance with manufacturer's instructions.
- f. Protect surrounding painted surfaces from exposure to chemical paint removers to avoid damage.
- g. Remove paint containing lead in accordance with Section 02 83 00.00 22 MANAGEMENT OF LEAD, CADMIUM, AND CHROMIUM (VI) DURING RENOVATION, DEMOLITION, CONSTRUCTION, REMOVAL AND ABATEMENT (PNS PROJECTS)

### 3.3.4 Water Cleaning

#### 3.3.4.1 Pressure Spraying

- a. Spray apply water to masonry surfaces to comply with requirements indicated by test patches for location, purpose, water temperature, pressure, volume, and equipment.
- b. Unless otherwise indicated, wash the surface with clean, low pressure water (pressure of less than 55 psi and 2.5 to 3 gpm discharge) and hold spray nozzle not less than 12 inches from surface of masonry.
- c. Apply water side to side and top to bottom in overlapping bands to produce uniform coverage.

#### 3.3.4.2 Hand Scrubbing

- a. Scrub surfaces to be cleaned to remove surface contaminants.
- b. Pre-wet surfaces and use hand-held natural bristle or nylon brushes.
- c. Do not use wire brushes.

#### 3.3.4.3 Rinsing

- a. Rinse scrubbed surfaces clean of all contaminants and cleaning solutions with water in a low-to-moderate pressure spray, working from top to bottom of each treated area.
- b. Remove all traces of contaminants and cleaning solutions.

### 3.3.5 Chemical Cleaning

#### 3.3.5.1 General

- a. Chemical cleaning is the use of any product in addition to water, including detergents, ammonia, vinegar, and bleach.
- b. Use gentlest means possible to achieve the desired result as determined by test patches.
- c. Proceed in an orderly manner, working from top to bottom of each

scaffold width and from one end of each elevation to the other.

- d. Provide uniform coverage of all surfaces, including corners, moldings, interstices and produce an even effect without streaking or damage to masonry.
- e. Do not apply chemical cleaners to the same masonry surfaces more than twice.

#### 3.3.5.2 Surface Prewetting

- a. Wet masonry surfaces to be cleaned with chemical cleaners with water using a low pressure spray before application of any cleaner.
- b. Prewet walls working from top to bottom, except work bottom to top on one-story walls.
- c. Do not prewet masonry surface prior to applying biocides.

#### 3.3.5.3 Acidic Chemical Cleaning

- a. Apply acidic chemical cleaners according to manufacturer's instructions.
- b. Do not apply acidic chemical cleaners to masonry with high calcium content (e.g. marble, limestone).
- c. Apply acidic cleaners to masonry surfaces by low pressure spray 50 psi max., roller, or brush.
- d. Leave cleaner on on masonry surface for the time period recommended by the manufacturer.
- e. Employ manual scrubbing by brushes as indicated by test patches for the specific location.
- f. Rinse cleaned surfaces with a low-to-moderate pressure spray of water to remove all traces of chemical cleaner.

#### 3.3.5.4 Alkaline Chemical Cleaning

##### 3.3.5.4.1 Prewash Phase

- a. Apply alkaline chemical cleaners to masonry surfaces according to manufacturer's instructions, by low pressure spray 50 psi max., roller, or brush.
- b. Leave cleaner on masonry surface for the time period recommended by the manufacturer.
- c. Employ manual scrubbing by brushes as indicated by test patches for the specific location.
- d. Rinse cleaned surfaces with a low-to-moderate pressure spray of water.

##### 3.3.5.4.2 Afterwash Phase

- a. Immediately after rinsing of alkaline cleaned surfaces, apply a neutralizing afterwash to the cleaned masonry areas.

- b. Apply neutralizing afterwash according to manufacturer's instructions, by low pressure spray 50 psi max., roller, or brush.
- c. Leave afterwash on masonry surface for the time period recommended by manufacturer.
- d. Rinse cleaned surfaces with a low-to-moderate pressure spray of water to remove all traces of chemical cleaners.

#### 3.3.5.5 Rinsing and pH Testing

- a. Determine the pH of masonry surfaces that have been chemically cleaned using pH monitoring pencils or papers.
- b. Rinse chemically cleaned masonry, using a low pressure spray, until a neutral pH (7) reading is obtained from the masonry unit surface.

### 3.4 MASONRY REPAIR

#### 3.4.1 General

- a. Match repaired surfaces with adjacent existing surfaces in all respects.
- b. Demonstrate the materials, methods and equipment proposed for use in the repair work in mock-ups, as specified in PART 2.
- c. Use products in accordance with the manufacturer's instructions.
- d. Proceed with masonry repair only after the cause of deterioration has been corrected.
- e. Assist Historic Masonry Consultant with performing field investigation to determine the causes and extent of degradation. Utilize the following techniques.
  - (1) Employ a field microscope to closely assess the conditions at the surface of the mortar and masonry units. Detect cracks and assess for degradation and debonding from the surrounding masonry. Detect previous surface coating treatments on the mortar and masonry that may be contributing to the current conditions.
  - (2) Employ a boroscope to examine mortar deeper in the joint. Compare the bedding mortar with the pointing mortar and ascertain the cross-sectional characteristics of the wall.
  - (3) Employ moisture meters to determine the level of moisture in the mortar and masonry, and if the mortar or masonry units are handling the brunt of the water movement through the wall. Infrared thermography, employed by a trained investigator, can provide additional information on the moisture conditions.
  - (4) Employ RILEM tubes using the method of RILEM II.4 or water penetration testing in accordance with ASTM C1601 to determine the rate of water uptake into the masonry.
  - (5) To access the physical characteristics of hard mortar, use a spring loaded or pendulum impact device to determine surface

hardness as an indicator of relative compressive strength. For evaluating softer mortars, mortar integrity deeper in the wall, and the condition of the masonry units, use a drill resistance tool by an experienced consultant.

- (6) Utilize technologies such as ground penetrating radar or metal detection equipment to map metal reinforcement and embedments in the wall.
- (7) Use flat jack or jacks and rams to gather information on in situ compressive stress (ASTM C1196, masonry compressive response ASTM C1197, and mortar joint shear strength ASTM C1531.

### 3.4.2 Repointing Masonry

Repoint masonry in accordance with NPS TPS Brief 2, using ASTM E2260 as a reference guide.

#### 3.4.2.1 Wall Preparation

- a. Remove old caulking, grout, or non-original mortar from previously repaired joints to a minimum depth of 2.5 times the width of the joint. Cut all joints (unless otherwise noted) back to sound, solid, back up material. Leave a clean, square face at the back of the joint to provide for maximum contact of repointing mortar.
- b. Shallow or feather edging is not permitted. Remove loose particles from joints. Clean joints, followed by blowing with filtered, dry, compressed air or vacuum.
- c. Cut out existing horizontal mortar joints (bed joints) that are filled with a hard Portland mortar using a diamond blade that is narrower than the joint width. Cut out the middle one-third of the mortar joint using a rotary power saw. Remove the remaining mortar from the masonry joints by hand using masonry chisels or pneumatic carving tools.
- d. Do not use rotary power saws to cut out vertical joints (head joints) . Remove all vertical head joints by hand using a pneumatic carving tool, or hammer and chisel.
- e. Remove existing historic lime-based mortar using only small-headed chisels that are no wider than half the width of the existing masonry joints. Pneumatic air carving chisels are permitted as are specially designed mortar removal reciprocating tools (i.e. Arbortech Saw).
- f. Do not widen the existing masonry joints. Do not chip or spall the surrounding masonry edges in the process of mortar removal. Damage to surrounding masonry units resulting from rotary blade over running is not permitted. Damages to adjacent materials exceeding 1/8 inch in size are the responsibility of the contractor and must be repaired by removal and replacement of damaged materials.
- g. Permit applicators to be trained at the project site in this masonry treatment requirement.

#### 3.4.2.2 Presoaking Masonry / Mortar Consistency / Lifts

- a. Use the same mortar as the repointing mortar for setting the

replacement masonry.

- b. Soak exposed surfaces of historic masonry adjacent to joint with water prior to repointing.
- c. Allow time for excess water to run off and evaporate prior to repointing. Joint surfaces must be damp but free from standing water.
- d. Maintain a water sprayer on site at all times during the repointing process.
- e. The mortar material must resemble the consistency of brown sugar during installation. This drier consistency enables the material to be tightly packed into the joint, allows for cleaner work, and prevents shrinkage cracks as the mortar cures.
- f. Allow mixed repointing mortar to stand for not less than one-half hour and not more than one and one-half hours for pre-hydration to reduce post-curing shrinkage. After this time, water can be added to small batches by hand to bring the mortar to a stiff yet workable consistency. Use repointing mortar within two and one-half hours after initial mixing and within one hour after adding water to bring the mortar to a working consistency. Retempering of the mortar to replace evaporated water is permitted within these time frames.
- g. Point joints in layers or "lifts" where the joints are deeper than 1-1/4 inch. Apply in layers not less than 1/2 the depth but not more than 1-1/4 inch or until a uniform depth is formed.

#### 3.4.2.3 Compression / Joint Finish / Curing

- a. Compress each layer thoroughly.
- b. When mortar is thumbprint hard at the surface of the wall, finish the joints to match the original historic joint profile.
- c. For Type L mortar:
  - (1) Allow water evaporation from the freshly repointed walls in order to initiate the carbonation process in high lime content mortars. The carbonation of lime mortar initially requires wet-and-dry cycles, which can be created by water misting the joints after the mortar application when dry weather conditions prevail. Finish the joint profile before these cycles are started.
  - (2) Depending on the environmental conditions (temperature and humidity), carry out water misting until a full nine alternating wet-and-dry cycles are completed.
  - (3) Adjust curing methods to ensure that the repointing mortar is damp without eroding the surface of the mortar.

#### 3.4.2.4 Protection

- a. Keep the mortar from drying out too quickly or from becoming too wet.
- b. Protect mortar from direct sun and high winds for the first 72 hours after installation or from driving rain for the first 24 hours, using plastic sheeting if necessary. Do not create a greenhouse effect by

sealing off air movement in an attempt to protect the wall with plastic. Allow for air circulation to facilitate the carbonation process.

#### 3.4.3 Retooling Stone Masonry In situ

- a. Scale off all loose pieces of original stone from masonry intended to remain in place, including surface material in powder or granular form and detachments of planer elements, spalls and chips.
- b. Assess all stone on building by sounding (tapping with a small hammer) or by using impact echo (for massive stones), surface penetrating radar, or infrared thermography in order to distinguish fully intact stone from those in which delamination may be hidden or pieces of unstable material may not be immediately visible.
- c. Remove and replace stone units that are designated for retooling in situ, but develop a solid stone substrate that is no longer in plane or plumb with the surrounding stone masonry surfaces after chiseling is complete.

#### 3.4.4 Masonry Removal and Replacement

- a. Before removing any deteriorated masonry units, establish bonding patterns, levels and coursings. Remove masonry that has deteriorated or is damaged beyond repair, as determined through investigation and evaluation. Carefully demolish or remove entire units from joint to joint, without damaging surrounding units in a manner that permits replacement with full-size units. Support and protect remaining masonry work that surrounds removal area. Maintain flashing, reinforcement, lintels, and adjoining construction in an undamaged condition. Notify Contracting Officer of unforeseen detrimental conditions including voids, cracks, bulges, and loose masonry units in existing masonry backup, rotted wood, rusted metal, and other deteriorated items. Remove as many whole masonry units as possible without damage.
- b. Remove mortar, loose particles, and soil from masonry by cleaning with hand chisels, non-metallic brushes, and water.
- c. Remove sealants by cutting close to masonry units with utility knife and cleaning with solvents. Clean surrounding masonry areas by removing mortar, dust, and loose particles in preparation for replacement.
- d. Replace removed masonry with masonry units removed from inconspicuous areas of the building, where possible, or with new masonry units matching the existing units. Butter vertical joints for full width before setting and set units in full bed of mortar, unless otherwise indicated. Remove mortar used for laying/setting masonry units before mortar sets to the repointing depth of the surrounding area. Repoint new mortar joints in repaired area to comply with requirements for repointing at existing masonry units.
- e. If a few isolated masonry units are to be replaced, remove each without disturbing the surrounding masonry. Remove deteriorated masonry units and mortar requiring replacement by hand chiseling. Do not damage adjoining masonry units during the removal of deteriorated units and mortar.



- f. Test the new element for fitting into its space without mortar. Use wedges made from non-expanding, non-corrosive material such as plastic to support and align the new unit, cover them with at least 1-1/2 inches of mortar when pointing is complete.
- g. Cover the four sides of the space with sufficient mortar to ensure that there will be no air spaces when the new unit is set. Fill the back of the space with mortar only if it matches existing construction.
- h. Line up and set the new unit by tapping it into place with a wooden or rubber mallet. Align the face of new unit with that of existing masonry.
- i. Repoint joints to match the rest of the wall after new units have been properly installed and adjusted.
- j. Clean replacement areas with a non-metallic brush and water to remove excess mortar.

#### 3.4.5 Material Repair

Repair or replace original historic masonry materials only if surfaces are extensively deteriorated (surface missing to a depth of 4 inches or more) or are threatening the safety of the structure or individuals. If additional damage is found, notify the Contracting Officer. Repairs and replacements must match the materials, colors, and finish of the existing historic masonry as closely as possible.

##### 3.4.5.1 Selective Demolition

- a. Remove unsound, weak, or damaged masonry and mortar in areas as indicated.
- b. Remove loose particles, laitance, spalling, cracked, or debonded masonry and mortar and foreign materials with hand tools unless otherwise noted.
- c. Clean surfaces prepared for repair free of dust, dirt, masonry chips, oil or other contaminants, rinsed with water, and dried before repair work is begun.
- d. Protect surfaces of the structure, and surfaces adjacent to the work area from damage which may result from removal, cleaning, and repair operations.

##### 3.4.5.2 Application of Substitute Repair Materials

- a. Place repair materials to rebuild spalled or damaged areas to match the original surface finish, level, texture, bonding patterns, color and porosity. Match the finished appearance of the substitute repair material patch with the adjacent existing surface. Apply samples to the masonry units in situ.
- b. Do not install repair material in thicknesses exceeding 2 inches. Utilize a Dutchman repair approach or replacement unit for masonry repairs in excess of 2 inches.

- c. Remove loose mortar and masonry prior to installation of the repair material. "Sound" the masonry with a hammer to verify its integrity. If necessary, cut away an additional 1/2 inch of the masonry substrate to ensure the surface to be repaired is solid and stable.
- d. Remove all deteriorated stone, mortar, sealant residue, and previous repair materials back to sound substrate using hammer and chisel or power equipment. Finish edges square to a minimum depth of 1/2 inch. Do not feather edges. Roughen substrate surface to achieve surface roughness required by manufacturer for good bond, but do not overly damage the substrate surface.
- e. Remove sealant residue. Cut out used anchors, threaded rod anchors and/or dowels within the damaged masonry area. Any anchors that are free of rust, solidly embedded, and do not project beyond the solid masonry surface may remain.
- f. Using clean water and a non-metallic scrub brush, clean dust from surface and pores of the substrate.
- g. Pre-wet the substrate with water prior to the application of the repair material to prevent the substrate from drawing out the moisture too quickly. Re-wet the surface with water again immediately before applying the repair material. Use methods approved by the repair material manufacturer to deliver the substitute repair work as demonstrated.
- h. Follow manufacturers' instructions pertaining to the placement of materials. If the manufacturer requires that installers of a specified product be trained, provide this documentation to the Contracting Officer. Training certificates previously issued by product companies for the application of specified products cannot be substituted for the Project Training "Substitute Repair Material Certificate" on this project.
- i. Masonry and Material Repair Finishes and Color
  - (1) Match the exposed surfaces of masonry and substitute material repair finish, color, texture, and surface detail with the original surface. Mechanical finishing and texturing may be required to produce the required finish and appearance.
  - (2) Conceal bond lines between the repaired area and adjacent surfaces.
  - (3) Replicate all surface details, including tooling and machine marks.
  - (4) Use low-impact energy type equipment in finishing and texturing, which will not weaken the patch or damage the patch bond and the adjacent masonry.

#### 3.4.5.3 Patch Anchors

- a. Provide patch anchors to ensure that the patch is tied to the existing masonry structure at a frequency of at least one patch anchor per 4 square inches of patch plan surface area; specific locations for patch anchors must be as indicated.
- b. Use small handheld, low-speed rotary masonry drills to produce holes in the existing masonry, within the limits for the patch anchor

installation.

- (1) Drill holes into the existing substrate material of the masonry using rotary (non-hammer) drills making holes with a diameter of 1/8 inch larger than the anchor diameter and a depth of 4 inches, except as otherwise indicated or directed.
- (2) Drill holes must not penetrate completely through the masonry, and must provide at least 1 inch of cover around the drill hole.
- (3) Clean holes by water blasting to remove drill dust and other debris and then blow dry with filtered, dry, compressed air.
- (4) Condition drill holes in accordance with the epoxy adhesive manufacturer's recommendations.

c. Clean anchors to remove all contaminants which may hinder epoxy bond.

d. Pressure inject adhesive into the back of the drilled holes.

- (1) Fill holes without spilling excess grout when the anchors are inserted.
- (2) Insert anchors immediately into the holes.
- (3) Set back anchors from the exterior face at least 1 inch.
- (4) Install anchors without breaking or chipping the exposed masonry surface.
- (5) Use socked or screen tube anchors where voids exist in the masonry units or between the wythes.

#### 3.4.5.4 Cleanup

- a. Protect masonry surfaces from excess grout adhesive and spills.
- b. Leave the surface of the masonry in a clean and uncontaminated condition.

#### 3.4.6 Dutchman Repairs

- a. Select stone for Dutchman repairs from the following three sources listed in order of priority:
  - (1) Stone harvested from the same elevation and stone type.
  - (2) Approved salvaged stone.
  - (3) New stone made from a similar stone type.
- b. Fit the new piece into place with tolerances of no more than plus or minus 1/16-inch.
- c. Provide supporting rods of stainless steel as necessary for the extent of the repair and the location.
- d. Closely blend repairs in with the surrounding original materials.

### 3.4.7 Crack Injection with Dispersed Hydrated Lime (DHL)

#### 3.4.7.1 General

- a. Notify the Contracting Officer as to when and where the installation will occur at least 48 hours prior to start.
- b. Provide samples to the Government representative from the dispenser during the course of the injection.
- c. Apply in accordance with the manufacturer's instructions.

#### 3.4.7.2 Application of DHL

- a. Drill 1/8-inch diameter, downward-sloping injection holes. For transverse cracks less than 1/8 inch wide, drill holes through center of crack at 1 to 1.5 inches on center.
- b. Clean out drill holes and cracks with compressed air and potable water. Remove dirt and organic matter, loose material, sealants, and failed crack repair materials.
- c. Inject Dispersed Hydrated Lime using hypodermic needles or pressure ports through holes sequentially, beginning at one end of area and working to opposite end. Do not exceed 10 psi injection pressure. Where possible begin at lower end of injection area and work upward. Inject Dispersed Hydrated Lime until it extrudes from adjacent holes. After Dispersed Hydrated Lime has set, remove excess material and patch injection holes and surface of cracks with appropriate surface treatment.

#### 3.4.7.3 Tools and Equipment

Do not use tools and equipment that have not been cleaned of set dispersed hydrated lime.

### 3.4.8 Surface Treatments

#### 3.4.8.1 Stucco

- a. Apply stucco on a clean surface in accordance with ASTM C926 at a thickness matching surrounding historic surfaces.
- b. Soak the substrate with water to saturated surface dry (SSD) condition prior to application of scratch-coat.
- c. Apply the scratch-coat and allow to partially-set on the wall surface.
- d. Use a scratch rake to create the keys into the scratch coat for acceptance of the finish coat.
- e. Apply the finish coat approximately 24 hours after the scratch coat application.
- f. Soak the scratch coat with water to SSD condition prior to the application of the finish coat.
- g. Apply the textured finish and profile to match the surrounding historic surfaces.

#### 3.4.8.2 Limewashes

- a. Apply limewash using fiber brushes in three thin coats on saturated surface dry (SSD) raw masonry surfaces.
- b. Do not allow the material to dry out before it has had a chance to absorb into the masonry surfaces.
- c. Work from top to bottom of the wall working from the dry-edge.
- d. Allow six hours drying time between coats
- e. Where colors are desired, use natural earth pigments.
- f. Verify all applications, materials and colors through mock ups panels applied to the substrate prior to the start of the work.

#### 3.4.8.3 Water Repellents Infiltration

Application of water proofing is not allowed.

Application of water repellents may be performed upon Contracting Officer approval of the recommendation and justification, by the historic masonry consultant, that no other means will control water infiltration. Apply water repellents per manufacturer's instructions.

#### 3.4.8.4 Stone Consolidants

Use of stone consolidants requires Contracting Officer approval of the historic masonry consultants recommendation, including justifying data. Apply stone consolidants per manufacturer's instructions.

### 3.5 INSTALLATION OF NEW ELEMENTS

Evaluate new materials and components for both functional and aesthetic impacts on historic structures.

#### 3.5.1 Structural Upgrades

For mechanical anchors used to reinforce masonry structures, provide design by a registered professional structural engineer. Strengthening measures must take into account the current loads and stresses in the structure and the nature in which the building has historically managed thermal and other environmental changes or cycles.

Submit manufacturers literature, design analysis and detail drawings for the proposed additional materials.

#### 3.5.2 Joint Sealant and Lead Flashing

- a. Test existing sealants for asbestos and PCBs before performing demolition.
- b. Provide joint sealing as specified in Section 07 92 00 JOINT SEALANTS.
  - (1) Augmentation with lead flashing is allowed for upward facing joints exposed to weather.

- (2) Install sealants and lead flashing in accordance with manufacturer's recommendations.

### 3.6 FINAL CLEANING

- a. No sooner than 72 hours after completion of the repair work and after joints are sealed, wash down faces and other exposed surfaces of masonry with water applied with a soft bristle brush, then rinse with clean water.
- b. Discolorations that cannot be removed by these procedures, are considered defective work.
- c. Perform cleaning work when temperature and humidity conditions allow the surfaces to dry rapidly.
- d. Protect adjacent surfaces from damage during cleaning operations.

### 3.7 PROTECTION OF WORK

Protect work against damage from subsequent operations.

### 3.8 DEFECTIVE WORK

Repair or replace defective work as directed by Contracting Officer, using approved procedures.

### 3.9 FINAL INSPECTION

Following completion of the work, inspect the structure for damage, staining, and other distresses. Inspect the patches for cracking, crazing, delamination, unsoundness, staining and other defects. Inspect the finish, texture, color and shade, and surface tolerances of the patches to verify that all requirements have been met. Repair surfaces exhibiting defects as directed.

- a. Following completion of the work, inspect the structure for damage, staining, and other distresses.
  - (1) Inspect patches for cracking, crazing, delamination, unsoundness, staining and other defects.
  - (2) Inspect finish, texture, color and shade, and surface tolerances of the patches to verify that all requirements have been met.
- b. Repair surfaces exhibiting defects as directed by Contracting Officer.

-- End of Section --

SECTION 04 20 00

UNIT MASONRY  
**11/15, CHG 2: 05/19**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 216.1 (2014) Code Requirements for Determining  
Fire Resistance of Concrete and Masonry  
Construction Assemblies

ACI SP-66 (2004) ACI Detailing Manual

ASTM INTERNATIONAL (ASTM)

ASTM A153/A153M (2023) Standard Specification for Zinc  
Coating (Hot-Dip) on Iron and Steel  
Hardware

ASTM A167 (2011) Standard Specification for  
Stainless and Heat-Resisting  
Chromium-Nickel Steel Plate, Sheet, and  
Strip

ASTM A185/A185M (2007) Standard Specification for Steel  
Welded Wire Reinforcement, Plain, for  
Concrete

ASTM A615/A615M (2024) Standard Specification for Deformed  
and Plain Carbon-Steel Bars for Concrete  
Reinforcement

ASTM A641/A641M (2019) Standard Specification for  
Zinc-Coated (Galvanized) Carbon Steel Wire

ASTM A653/A653M (2023) Standard Specification for Steel  
Sheet, Zinc-Coated (Galvanized) or  
Zinc-Iron Alloy-Coated (Galvannealed) by  
the Hot-Dip Process

ASTM A951/A951M (2011) Standard Specification for Steel  
Wire for Masonry Joint Reinforcement

ASTM A996/A996M (2016) Standard Specification for  
Rail-Steel and Axle-Steel Deformed Bars  
for Concrete Reinforcement

ASTM A1008/A1008M (2024) Standard Specification for Steel,  
Sheet, Cold-Rolled, Carbon, Structural,  
High-Strength Low-Alloy, High-Strength

Low-Alloy with Improved Formability,  
Solution Hardened, and Bake Hardenable

|                   |                                                                                                                              |
|-------------------|------------------------------------------------------------------------------------------------------------------------------|
| ASTM A1064/A1064M | (2024) Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete          |
| ASTM B370         | (2022) Standard Specification for Copper Sheet and Strip for Building Construction                                           |
| ASTM C67/C67M     | (2023a) Standard Test Methods for Sampling and Testing Brick and Structural Clay Tile                                        |
| ASTM C90          | (2024) Standard Specification for Loadbearing Concrete Masonry Units                                                         |
| ASTM C129         | (2023) Standard Specification for Nonloadbearing Concrete Masonry Units                                                      |
| ASTM C207         | (2024) Standard Specification for Hydrated Lime for Masonry Purposes                                                         |
| ASTM C216         | (2023) Standard Specification for Facing Brick (Solid Masonry Units Made from Clay or Shale)                                 |
| ASTM C270         | (2019a; E 2019) Standard Specification for Mortar for Unit Masonry                                                           |
| ASTM C476         | (2023) Standard Specification for Grout for Masonry                                                                          |
| ASTM C494/C494M   | (2024) Standard Specification for Chemical Admixtures for Concrete                                                           |
| ASTM C586         | (2011) Standard Test Method for Potential Alkali Reactivity of Carbonate Rocks as Concrete Aggregates (Rock-Cylinder Method) |
| ASTM C616/C615M   | (2011) Standard Specification for Granite Dimension Stone                                                                    |
| ASTM C616/C616M   | (2010) Standard Specification for Quartz-Based Dimension Stone                                                               |
| ASTM C641         | (2023) Standard Test Method for Iron Staining Materials in Lightweight Concrete Aggregates                                   |
| ASTM C780         | (2023) Standard Test Method for Preconstruction and Construction Evaluation of Mortars for Plain and Reinforced Unit Masonry |
| ASTM C979/C979M   | (2016) Standard Specification for Pigments for Integrally Colored Concrete                                                   |
| ASTM C1019        | (2024) Standard Test Method for Sampling                                                                                     |



and Testing Grout

|                   |                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------|
| ASTM C1314        | (2014) Standard Test Method for<br>Compressive Strength of Masonry Prisms               |
| ASTM C1384        | (2012a) Standard Specification for<br>Admixtures for Masonry Mortars                    |
| ASTM C1611/C1611M | (2021) Standard Test Method for Slump Flow<br>of Self-Consolidating Concrete            |
| ASTM D2000        | (2018) Standard Classification System for<br>Rubber Products in Automotive Applications |
| ASTM D2287        | (2019) Nonrigid Vinyl Chloride Polymer and<br>Copolymer Molding and Extrusion Compounds |

THE MASONRY SOCIETY (TMS)

|          |                                                                                                                                                                                                                                     |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TMS MSJC | (2016) Masonry Standard Joint Committee's<br>(MSJC) Book - Building Code Requirements<br>and Specification for Masonry Structures,<br>Containing TMS 402/ACI 530/ASCE 5, TMS<br>602/ACI 530.1/ASCE 6, and Companion<br>Commentaries |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Cut CMU Drawings; G

Reinforcement Detail Drawings; G

SD-03 Product Data

Hot Weather Procedures; G

Cold Weather Procedures; G

Clay or Shale Brick; G

Cement; G

Cementitious Materials; G

Insulation; G

SD-04 Samples

Mock-Up Panel; G

Clay or Shale Brick; G

Concrete Masonry Units (CMU); G

Dimension Stone Units; G

Admixtures for Masonry Mortar; G

Anchors, Ties, and Bar Positioners; G

Joint Reinforcement; G

Clay Masonry Expansion-Joint Materials; G

Insulation; G

SD-05 Design Data

Masonry Compressive Strength; G

Fire-Rated Concrete Masonry Units

Bracing Calculations; G

SD-06 Test Reports

Efflorescence Test

Fire-Rated Concrete Masonry Units

Field Testing of Mortar

Field Testing of Grout

Prism Tests

SD-07 Certificates

Special Masonry Inspector Qualifications

Clay or Shale Brick

Concrete Masonry Units (CMU)

Cementitious Materials

Admixtures for Masonry Mortar

Admixtures for Grout

Anchors, Ties, and Bar Positioners

Joint Reinforcement

Insulation

Insulation

SD-08 Manufacturer's Instructions

Admixtures for Masonry Mortar

Admixtures for Grout

SD-10 Operation and Maintenance Data

Take-Back Program

### 1.3 QUALITY ASSURANCE

#### 1.3.1 Masonry Mock-Up Panels

##### 1.3.1.1 Mock-Up Panel Location

After material samples are approved and prior to starting masonry work, construct a mock-up panel for each type and color of masonry required. At least 48 hours prior to constructing the panel or panels, submit written notification to the Contracting Officer. Do not build-in mock-up panels as part of the structure; locate mock-up panels where directed. Construct portable mock-up panels or locate in an area where they will not be disrupted during construction.

##### 1.3.1.2 Mock-Up Panel Configuration

Construct mock-up panels L-shaped or otherwise configured to represent all of the wall elements. Construct panels of the size necessary to demonstrate the acceptable level of workmanship for each type of masonry represented on the project. Provide a straight panel or a leg of an L-shaped panel of minimum size 8 feet long by 4 feet high.

##### 1.3.1.3 Mock-Up Panel Composition

Show full color range, texture, and bond pattern of the masonry work. Demonstrate mortar joint tooling; grouting of reinforced vertical cores, collar joints, bond beams, and lintels; positioning, securing, and lapping of reinforcing steel; positioning and lapping of joint reinforcement

(including prefabricated corners); and cleaning of masonry work during the construction of the panels. Also include installation or application procedures for anchors, wall ties, CMU control joints, brick expansion joints, insulation, flashing, brick soldier, row lock courses and weeps. Include a a masonry bonded corner . When the panel represents reinforced masonry, include a 2 by 2 foot opening placed at least 2 feet above the panel base and 2 feet away from all free edges, corners, and control joints. Provide required reinforcing around this opening as well as at wall corners and control joints.

#### 1.3.1.4 Mock-Up Panel Construction Method

Where anchored veneer walls or cavity walls are required, demonstrate and receive approval for the method of construction; i.e., either bring up the two wythes together or separately, with the insulation and appropriate ties placed within the specified tolerances across the cavity. Demonstrate provisions to preclude mortar or grout droppings in the cavity and to provide a clear open air space of the dimensions shown on the drawings. Where masonry is to be grouted, demonstrate and receive approval on the method that will be used to bring up the masonry wythes; support the reinforcing bars; and grout cells, bond beams, lintels, and collar joints using the requirements specified herein. When water-repellent is specified to be applied to the masonry, apply the approved product to the mock-up panel. Construct panels on a properly designed concrete foundation.

#### 1.3.1.5 Mock-Up Panel Purpose

The completed panels is used as the standard of workmanship for the type of masonry represented. Do not commence masonry work until the mock-up panel for that type of masonry construction has been completed and approved. Protect panels from the weather and construction operations until the masonry work has been completed and approved. Perform cleaning procedures on the mockup and obtain approval of the Contracting Officer prior to cleaning the building. After completion of the work, completely remove the mock-up panels, including all foundation concrete, from the construction site.

#### 1.3.2 Special Masonry Inspector Qualifications

Refer to Section 01 45 35 SPECIAL INSPECTIONS for qualifications and responsibilities of the masonry special inspector.

### 1.4 DELIVERY, STORAGE, AND HANDLING

Deliver, store, handle, and protect material to avoid chipping, breakage, and contact with soil or contaminating material. Store and prepare materials in already disturbed areas to minimize project site disturbance and size of project site.

#### 1.4.1 Masonry Units

Cover and protect masonry units from precipitation. Conform to handling and storage requirements of TMS MSJC.

- a. Pack glazed brick, glazed structural clay tile, and prefaced concrete masonry units in the manufacturer's standard paper cartons, trays, or shrink wrapped pallets with a divider between each unit. Do not stack pallets. Do not remove units from cartons until cartons are placed on

scaffolds or in the location where units are to be laid.

- b. Mark prefabricated lintels on top sides to show either the lintel schedule number or the number and size of top and bottom bars.

#### 1.4.2 Reinforcement, Anchors, and Ties

Store steel reinforcing bars, coated anchors, ties, and joint reinforcement above the ground. Maintain steel reinforcing bars and uncoated ties free of loose mill scale and loose rust.

#### 1.4.3 Cementitious Materials, Sand and Aggregates

Deliver cementitious and other packaged materials in unopened containers, plainly marked and labeled with manufacturers' names and brands. Store cementitious material in dry, weathertight enclosures or completely cover. Handle cementitious materials in a manner that will prevent the inclusion of foreign materials and damage by water or dampness. Store sand and aggregates in a manner to prevent contamination and segregation.

### 1.5 PROJECT/SITE CONDITIONS

Conform to TMS MSJC for hot and cold weather masonry erection.

#### 1.5.1 Hot Weather Procedures

When ambient air temperature exceeds 100 degrees F, or exceeds 90 degrees F and the wind velocity is greater than 8 mph, comply with TMS MSJC Article 1.8 D for: preparation prior to conducting masonry work; construction while masonry work is in progress; and protection for newly completed masonry.

#### 1.5.2 Cold Weather Procedures

When ambient temperature is below 40 degrees F, comply with TMS MSJC Article 1.8 C for: preparation prior to conducting masonry work; construction while masonry work is in progress; and protection for newly completed masonry.

## PART 2 PRODUCTS

### 2.1 SYSTEM DESCRIPTION

#### 2.1.1 Design - Specified Compressive Strength of Masonry

The specified compressive strength of masonry,  $f'_m$ , is as indicated for each type of masonry .

#### 2.1.2 Performance - Verify Masonry Compressive Strength

Verify specified compressive strength of masonry using the "Unit Strength Method" of TMS MSJC. Submit calculations and certifications of unit and mortar strength.

Verify specified compressive strength of masonry using the "Prism Test Method" of TMS MSJC when the "Unit Strength Method" cannot be used. Submit test results.

## 2.2 MANUFACTURED UNITS

### 2.2.1 General Requirements

Do not change the source of materials, which will affect the appearance of the finished work, after the work has started except with Contracting Officer's approval. Submit test reports from an approved independent laboratory. Certify test reports on a previously tested material as the same materials as that proposed for use in this project. Submit certificates of compliance stating that the materials meet the specified requirements.

### 2.2.2 Clay or Shale Brick

#### 2.2.2.1 General

##### 2.2.2.1.1 Sample Submittal

Submit brick samples as specified, showing the color range and texture of clay or shale brick. Limit units used on the project to those that conform to the approved sample. Submit sample of colored mortar with applicable masonry unit and color samples of three stretcher units and one unit for each type of special shape.

##### 2.2.2.1.2 Uniformity

Manufacture bricks at one time and from the same run. Deliver clay or shale brick units factory-blended to provide a uniform appearance and color range in the completed wall.

##### 2.2.2.1.3 Efflorescence Test

Test clay brick that will be exposed to weathering for efflorescence in accordance with ASTM C67/C67M. Schedule tests far enough in advance of starting masonry work to permit retesting if necessary. Units meeting the definition of "effloresced" are subject to rejection.

### 2.2.2.2 Solid Clay or Shale Brick

Provide solid clay or shale brick that conforms to ASTM C216, Type FBS. Provide brick with minimum compressive strength of 3,350 psi. Where brick cores, recesses, or deformation would be exposed to view, provide 100 percent solid units. Provide brick with texture and color tange to match the brick to match existing adjacent building..

Provide brick with specified sizes.

- a. Modular size, 3-5/8 inches thick, 2-1/4 inches high, and 7-5/8 inches long.

### 2.2.3 Concrete Units

#### 2.2.3.1 Aggregates

Test lightweight aggregates, and blends of lightweight and heavier aggregates in proportions used in producing the units, for stain-producing iron compounds in accordance with ASTM C641, visual classification method. Do not incorporate aggregates for which the iron stain deposited on the

filter paper exceeds the "light stain" classification.

Use industrial waste by-products (air-cooled slag, cinders, or bottom ash), ground waste glass and concrete, granulated slag, and expanded slag in aggregates.

#### 2.2.3.2 Concrete Masonry Units (CMU)

##### 2.2.3.2.1 Cement

Use only cement that has a low alkali content and is of one brand.

##### 2.2.3.2.2 Size

Provide units with specified nominal dimension of 8 inches wide, 8 inches high, and 16 inches long.

##### 2.2.3.2.3 Weather Exposure

Provide concrete masonry units with water-repellant admixture added during manufacture where units will be exposed to weather.

##### 2.2.3.2.4 Unit Types

- a. Hollow Load-Bearing Units: ASTM C90, normal weight. Provide load-bearing units for exterior walls, foundation walls, load-bearing walls, and shear walls.
- b. Hollow Non-Load-Bearing Units: ASTM C129, lightweight or normal weight. Load-bearing units may be provided in lieu of non-load-bearing units.
- c. Solid Load-Bearing Units: ASTM C90, lightweight or normal weight units. Provide solid units as indicated.

##### 2.2.3.2.5 Jamb Units

Provide jamb units of the shapes and sizes to conform with wall units. Solid units may be incorporated in the masonry work where necessary to fill out at corners, gable slopes, and elsewhere as approved.

Provide sash jamb units with a 3/4 by 3/4 inch groove near the center at end of each unit.

#### 2.2.3.3 Fire-Rated Concrete Masonry Units

For indicated fire-rated construction, provide concrete masonry units of minimum equivalent thickness for the fire rating indicated and the corresponding type of aggregates indicated in TABLE I. Units containing more than one of the aggregates listed in TABLE I will be rated by linear interpolation based on the percent by dry-rodded volume of each aggregate used in manufacturing the units.

| TABLE I<br>FIRE-RATED CONCRETE MASONRY UNITS          |                                                                  |          |        |            |         |         |         |
|-------------------------------------------------------|------------------------------------------------------------------|----------|--------|------------|---------|---------|---------|
| Aggregate Type                                        | Minimum Equivalent Thickness for Fire-Resistance Rating,<br>inch |          |        |            |         |         |         |
|                                                       | 1/2 hour                                                         | 3/4 hour | 1 hour | 1-1/2 hour | 2 hours | 3 hours | 4 hours |
| Calcareous or siliceous gravel (other than limestone) | 2.0                                                              | 2.4      | 2.8    | 3.6        | 4.2     | 5.3     | 6.2     |
| Limestone, cinders, or air-cooled slag                | 1.9                                                              | 2.3      | 2.7    | 3.4        | 4.0     | 5.0     | 5.9     |
| Expanded clay, expanded shale, or expanded slate      | 1.8                                                              | 2.2      | 2.6    | 3.3        | 3.6     | 4.4     | 5.1     |
| Expanded slag or pumice                               | 1.5                                                              | 1.9      | 2.1    | 2.7        | 3.2     | 4.0     | 4.7     |

Determine equivalent thickness in accordance with ACI 216.1. Where walls are to receive plaster or be faced with brick, or otherwise form an assembly; include the thickness of plaster or brick or other material in the assembly in determining the equivalent thickness. Submit calculation results.

#### 2.2.4 DIMENSION STONE UNITS

Provide dimension stone for trim, sills, lintels, and copings cut to the design shown and conforming to:

|           |                 |                                                                                                        |
|-----------|-----------------|--------------------------------------------------------------------------------------------------------|
| Limestone | ASTM C586       | Standard buff color with a smooth machine finish free from tool marks                                  |
| Sandstone | ASTM C616/C616M | Standard grade, buff, gray, or buff brown, with a smooth finish free from clay pits and tool marks     |
| Granite   | ASTM C616/C615M | Commercial grade of medium or moderately coarse grain, with a light or medium gray or light pink color |

Provide a smooth machine finish on washes, 4-cut finish on treads, and 6-cut or equivalent machine finish on other exposed surfaces. Except when supported by a steel member, provide lintels 4 inches or more in thickness from face to back edge and of the depth required to support the masonry over the opening. Fabricate stone with beds and joints at right angles to the face, and with sharp, true arises. Provide copings and sills with washes, and where overhanging the walls, with drips cut on the underside. Submit samples as specified.



## 2.3 EQUIPMENT

### 2.3.1 Vibrators

Maintain at least one spare vibrator on site at all times.

### 2.3.2 Grout Pumps

Pumping through aluminum tubes is not permitted.

## 2.4 MATERIALS

### 2.4.1 Mortar Materials

#### 2.4.1.1 Cementitious Materials

Provide cementitious materials that conform to those permitted by ASTM C270.

#### 2.4.1.2 Hydrated Lime and Alternates

Provide lime that conforms to one of the materials permitted by ASTM C207 for use in combination with portland cement, hydraulic cement, and blended hydraulic cement. Do not use lime in combination with masonry cement or mortar cement.

#### 2.4.1.3 Colored Mortar

Use mortar pigment that conforms to ASTM C979/C979M. Add pigment to mortar to produce a uniform color matching existing adjacent building.. Furnish pigments in accurately pre-measured and packaged units that can be added to a measured amount of cementitious materials or supply pigments via preblended cementitious materials or dry mortar mix.

- a. In masonry cement or mortar cement, do not exceed 5 percent of cement weight for mineral oxide pigment; do not exceed 1 percent of cement weight for carbon black pigment.
- b. In cement-lime mortar mix, do not exceed 10 percent of cementitious materials' weight for mineral oxide pigment; do not exceed 2 percent of cementitious materials' weight for carbon black pigment.

#### 2.4.1.4 Admixtures for Masonry Mortar

In cold weather, use a non-chloride based accelerating admixture that conforms to ASTM C1384, unless Type III portland cement is used in the mortar.

In showers and kitchens, use mortar that contains a water-repellent admixture that conforms to ASTM C1384. Provide a water-repellent admixture, conforming to ASTM C1384 and of the same brand and manufacturer as the block's integral water-repellent, in the mortar used to place concrete masonry units that have an integral water-repellent admixture.

#### 2.4.1.5 Aggregate and Water

Provide aggregate (sand) and water that conform to materials permitted by ASTM C270.

#### 2.4.2 Grout and Ready-Mix Grout Materials

##### 2.4.2.1 Cementitious Materials for Grout

Provide cementitious materials that conform to those permitted by ASTM C476.

##### 2.4.2.2 Admixtures for Grout

Water-reducing admixtures that conform to ASTM C494/C494M Type F or G and viscosity-modifying admixtures that conform to ASTM C494/C494M Type S are permitted for use in grout. Other admixtures require approval by the Contracting Officer.

In cold weather, a non-chloride based accelerating admixture may be used subject to approval by the Contracting Officer; use accelerating admixture that is non-corrosive and conforms to ASTM C494/C494M, Type C.

##### 2.4.2.3 Aggregate and Water

Provide fine and coarse aggregates and water that conform to materials permitted by ASTM C476.

#### 2.5 MORTAR AND GROUT MIXES

##### 2.5.1 Mortar Mix

- a. Provide mortar Type S unless specified otherwise herein.
- b. Use ASTM C270 Type S cement-lime mortar or mortar cement mortar for seismic-force-resisting elements indicated.
- d. Provide Type N or S mortar for non-load-bearing, non-shear-wall interior masonry.
- e. For field-batched mortar, measure component materials by volume. Use measuring boxes for materials that do not come in packages, such as sand, for consistent batching. Mix cementitious materials and aggregates between 3 and 5 minutes in a mechanical batch mixer with a sufficient amount of water to produce a workable consistency. Do not hand mix mortar unless approved by the Contracting Officer. Maintain workability of mortar by remixing or retempering. Discard mortar that has begun to stiffen or is not used within 2-1/2 hours after initial mixing.
- f. For preblended mortar, follow manufacturer's mixing instructions.

##### 2.5.2 Grout and Ready Mix Grout Mix

Use grout that conforms to ASTM C476, fine or coarse, depending upon cmu size.. Use conventional grout with a slump between 8 and 11 inches. Use self-consolidating grout with slump flow of 24 to 30 inches and a visual stability index (VSI) not greater than 1. Provide minimum grout strength of 2000 psi in 28 days, as tested in accordance with ASTM C1019. Do not change proportions and do not use materials with different physical or chemical characteristics in grout for the work unless additional evidence

is furnished that grout meets the specified requirements. Use ready-mixed grout that conforms to ASTM C476.

## 2.6 ACCESSORIES

### 2.6.1 Grout Barriers

Grout barriers for vertical cores that consist of fine mesh wire, fiberglass, or expanded metal.

### 2.6.2 Anchors, Ties, and Bar Positioners

#### 2.6.2.1 General

- a. Fabricate anchors and ties without drips or crimps. Size anchors and ties to provide a minimum of 5/8 inch mortar cover from each face of masonry.
- b. Fabricate steel wire anchors and ties from wire conforming to ASTM A1064/A1064M and hot-dip galvanize in accordance with ASTM A153/A153M.
- c. Fabricate joint reinforcement in conformance with ASTM A951/A951M. Hot dip galvanize joint reinforcement in exterior walls and in interior walls exposed to moist environment in conformance with ASTM A153/A153M. Galvanize joint reinforcement in other interior walls in conformance with ASTM A641/A641M; coordinate with paragraph JOINT REINFORCEMENT below.
- d. Fabricate sheet metal anchors and ties in conformance with ASTM A1008/A1008M. Hot dip galvanize sheet metal anchors and ties in exterior walls and in interior walls exposed to moist environment in compliance with ASTM A153/A153M Class B. Galvanize sheet metal anchors and ties in other interior walls in compliance with ASTM A653/A653M, Coating Designation G60.
- e. Submit two anchors, ties and bar positioners of each type used, as samples.

#### 2.6.2.2 Wire Mesh Anchors

Provide wire mesh anchors of 1/4 inch mesh galvanized hardware cloth, conforming to ASTM A185/A185M, with length not less than 12 inches, at intersections of interior non-bearing masonry walls.

#### 2.6.2.3 Wall Ties for Multi-Wythe Masonry Construction

Provide adjustable type wall ties, if approved for use, that consist of two essentially U-shaped elements fabricated of minimum W2.8 diameter steel wire or pintle type ties that are inserted to eyes of horizontal joint reinforcement, hot-dip galvanized. Provide adjustable ties with double pintle legs and allows a maximum offset of 1-1/4 inch between each element of the tie and maximum distance between connecting parts no more than 1/16 inch. Form the pintle and eye elements so that both can be in the same plane. Wall ties may also be of a continuous type conforming to paragraph JOINT REINFORCEMENT.

#### 2.6.2.4 Dovetail Anchors

Provide dovetail anchors of 3/16 inch diameter steel wire, triangular shaped, and attached to a 12 gauge or heavier steel dovetail section. Use these anchors to connect the exterior masonry wythe as it passes over the face of concrete columns, beams, or walls. Fill cells immediately above and below these anchors unless solid units are used. Furnish dovetail slots, which are specified to be installed by others, in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE.

#### 2.6.2.5 Adjustable Anchors

#### 2.6.2.6 Bar Positioners

Factory-fabricate bar positioners, used to prevent displacement of reinforcing bars during the course of construction, from 9 gauge steel wire or equivalent, and hot-dip galvanized. Bar positioners must be suitable for intended use and be corrosion resistant steel. Bar positioners not fully contained within the wythe must be hot-dip galvanized.

#### 2.6.3 Joint Reinforcement

Factory fabricate joint reinforcement in conformance with ASTM A951/A951M, welded construction. Provide ladder type joint reinforcement, having one longitudinal wire in the mortar bed of each face shell for hollow units and one wire for solid units and with all wires a minimum of 9 gauge. Size joint reinforcement to provide a minimum of 5/8 inch cover from each face. Space crosswires not more than 16 inches. Provide joint reinforcement for straight runs in flat sections not less than 10 feet long. Provide joint reinforcement with factory formed corners and intersections. If approved for use, joint reinforcement may be furnished with adjustable wall tie features. Submit one piece of each type used, including corner and wall intersection pieces, showing at least two cross wires.

#### 2.6.4 Reinforcing Steel Bars

Provide reinforcing steel bars and rods conforming to ASTM A615/A615M or ASTM A996/A996M, Grade 60.

#### 2.6.5 Concrete Masonry Control Joint Keys

Provide control joint keys of a factory fabricated solid section of natural or synthetic rubber (or combination thereof) conforming to ASTM D2000 M2AA-805 with a minimum durometer hardness of 80 or polyvinyl chloride conforming to ASTM D2287 Type PVC 654-4 with a minimum durometer hardness of 85. Form the control joint key with a solid shear section not less than 5/8 inch thick and 3/8 inch thick flanges, with a tolerance of plus or minus 1/16 inch, to fit neatly, but without forcing, in masonry unit jamb sash grooves.

#### 2.6.6 Clay Masonry Expansion-Joint Materials

Provide backer rod and sealant, adequate to accommodate joint compression and extension equal to 50 percent of the width of the joint. Provide the backer rod of compressible rod stock of closed cell polyethylene foam,

polyurethane foam, butyl rubber foam, or other flexible, nonabsorptive material as recommended by the sealant manufacturer. Provide sealant in conformance with Section 07 92 00 JOINT SEALANTS.

Submit one piece of each type of material used.

#### 2.6.7 Through Wall Flashing and Weeps

##### 2.6.7.1 General

Provide coated copper, copper or stainless steel sheet, self-adhesive rubberized sheet, or reinforced membrane sheet flashing except that flashing indicated to terminate in reglets must be metal or coated-metal flashing and except that the material must be one which is not adversely affected by dampproofing material.

##### 2.6.7.2 Coated-Copper Flashing

Provide 7 ounce, electrolytic copper sheet, uniformly coated on both sides with acidproof, alkaliproof, asphalt impregnated kraft paper or polyethylene sheets.

##### 2.6.7.3 Copper or Stainless Steel Flashing

Provide copper sheet, complying with ASTM B370, minimum 16 ounce weight; or stainless steel, ASTM A167, Type 304 or 316, 0.015 inch thick, No. 2D finish. Where indicated, provide with factory-fabricated deformations that mechanically bond flashing against horizontal movement in all directions, where deformations consist of dimples, diagonal corrugations, or a combination of dimples and transverse corrugations.

##### 2.6.7.4 Rubberized Flashing

Provide self-adhesive rubberized asphalt sheet flashing consisting of 32-mil thick pliable and highly adhesive rubberized asphalt compound bonded completely and integrally to 8-mil thick, high density, cross-laminated polyethylene film to produce an overall thickness of 40 mils. Provide rubberized, asphalt-based mastic and surface conditioner that are each approved by flashing manufacturer for use with flashing material.

##### 2.6.7.5 Weep Ventilators

Provide weep ventilators that are prefabricated from stainless steel or plastic. Provide inserts with grill or louver-type openings designed to allow the passage of moisture from cavities and to prevent the entrance of insects, and with a rectangular closure strip to prevent mortar droppings from clogging the opening. Provide ventilators with compressible flanges to fit in a standard 3/8 inch wide mortar joint and with height equal to the nominal height of the unit.

##### 2.6.7.6 Single-Wythe Exterior Wall CMU Flashing System

In single-wythe exterior CMU walls, provide a system of CMU cell flashing pans and interlocking CMU web covers made from UV-resistant, high-density polyethylene. For exterior CMU walls, provide a flashing/weep system in open cores that do not receive grout. Cell flashing pans are to have integral weep spouts built into mortar bed joints that extend into the cell to prevent clogging with mortar.

#### 2.6.7.7 Metal Drip Edge

Provide stainless steel drip edge, 15-mil thick, hemmed edges, with down-turned drip at the outside edge and upturned dam at the inside edge for use with membrane flashings.

#### 2.6.8 RIGID BOARD-TYPE INSULATION

Provide rigid board-type insulation as specified in Section 07 21 13 BOARD AND BLOCK INSULATION.

### PART 3 EXECUTION

#### 3.1 EXAMINATION

Prior to start of work, verify the applicable conditions as set forth in TMS MSJC, inspection.

#### 3.2 PREPARATION

##### 3.2.1 Stains

Protect exposed surfaces from mortar and other stains. When mortar joints are tooled, remove mortar from exposed surfaces with fiber brushes and wooden paddles. Protect base of walls from splash stains by covering adjacent ground with sand, sawdust, or polyethylene.

##### 3.2.2 Loads

Do not apply uniform loads for at least 12 hours or concentrated loads for at least 72 hours after masonry is constructed. Provide temporary bracing as required.

##### 3.2.3 Concrete Surfaces

Where masonry is to be placed, clean concrete of laitance, dust, dirt, oil, organic matter, or other foreign materials and slightly roughen to provide a surface texture with a depth of at least 1/8 inch. Sandblast, if necessary, to remove laitance from pores and to expose the aggregate.

##### 3.2.4 Shelf Angles

Adjust shelf angles as required to keep the masonry level and at the proper elevation.

##### 3.2.5 Bracing

Provide bracing and scaffolding necessary for masonry work. Design bracing to resist wind pressure as required by OSHA and local codes and submit bracing calculations, sealed by a registered professional engineer. Do not remove bracing in less than 10 days.

#### 3.3 ERECTION

##### 3.3.1 General

- a. Coordinate masonry work with the work of other trades to accommodate built-in items and to avoid cutting and patching. Lay masonry units

in running bond pattern. Lay facing courses level with back-up courses, unless the use of adjustable ties has been approved in which case the tolerances is plus or minus 1/2 inch. Adjust each unit to its final position while mortar is still soft and has plastic consistency.

- b. Remove and clean units that have been disturbed after the mortar has stiffened, and relay with fresh mortar. Keep air spaces, cavities, chases, expansion joints, and spaces to be grouted free from mortar and other debris. Select units to be used in exposed masonry surfaces from those having the least amount of chipped edges or other imperfections detracting from the appearance of the finished work.
- c. When necessary to temporarily discontinue the work, step (rack) back the masonry for joining when work resumes. Toothing may be used only when specifically approved by the Contracting Officer. Before resuming work, remove loose mortar and thoroughly clean the exposed joint. Cover the top of walls subjected to rain or snow with nonstaining waterproof covering or membrane when work is not in process. Extend the covering a minimum of 610 mm 2 feet down on each side of the wall and hold securely in place.
- d. Ensure that units being laid and surfaces to receive units are free of water film and frost. Lay solid units in a nonfurrowed full bed of mortar. Bevel mortar for veneer wythes and slope down toward the cavity side. Shove units into place so that the vertical joints are tight. Completely fill vertical joints between solid units with mortar, except where indicated at control, expansion, and isolation joints. Place hollow units so that mortar extends to the depth of the face shell at heads and beds, unless otherwise indicated. Mortar will be permitted to protrude up to 1/2 inch into the space or cells to be grouted. Provide means to prevent mortar from dropping into the space below or clean grout spaces prior to grouting.
- e. In multi-wythe construction with collar joints no more than 3/4 inch wide, bring up the inner wythe not more than 16 inches ahead of the outer wythe. Fill collar joints with mortar during the laying of the facing wythe, and do not lag the laying of the facing wythe by back-buttering each unit as it is laid.

#### 3.3.1.1 Jointing

Tool mortar joints when the mortar is thumbprint hard. Tool horizontal joints after tooling vertical joints. Brush mortar joints to remove loose and excess mortar.

##### 3.3.1.1.1 Tooled Joints

Tool mortar joints in exposed exterior and interior masonry surfaces concave, using a jointer that is slightly larger than the joint width so that complete contact is made along the edges of the unit. Perform tooling so that the mortar is compressed and the joint surface is sealed. Use a jointer of sufficient length to obtain a straight and true mortar joint. No exterior joints are to be left un-tooled.

##### 3.3.1.1.2 Flush Joints

Flush cut mortar joints in concealed masonry surfaces and joints at electrical outlet boxes in wet areas. Finish flush cut joints by cutting

off the mortar flush with the face of the wall. Point joints in unparged masonry walls below grade tight. For architectural units, such as fluted units, completely fill both the head and bed joints and flush cut.

#### 3.3.1.1.3 Door and Window Frame Joints

On the exposed interior side of exterior frames, rake joints between frames and abutting masonry walls to a depth of 3/8 inch. On the exterior side of exterior frames, rake joints between frames and abutting masonry walls to a depth of 3/8 inch.

#### 3.3.1.1.4 Joint Widths

- a. Construct brick masonry with mortar joint widths equal to the difference between the specified and nominal dimensions of the unit, within tolerances permitted by TMS MSJC.
- b. Provide 3/8 inch wide mortar joints in concrete masonry, except for prefaced concrete masonry units.
- c. Provide 3/8 inch wide mortar joints on unfaced side of prefaced concrete masonry units and not less than 3/16 inch nor more than 1/4 inch wide on prefaced side.
- d. Maintain mortar joint widths within tolerances permitted by TMS MSJC

#### 3.3.1.2 Cutting and Fitting

Use full units of the proper size wherever possible, in lieu of cut units. Locate cut units where they would have the least impact on the architectural aesthetic goals of the facility. Perform cutting and fitting, including that required to accommodate the work of others, by masonry mechanics using power masonry saws. Concrete masonry units may be wet or dry cut. Before being placed in the work, dry wet-cut units to the same surface-dry appearance as uncut units being laid in the wall. Provide cut edges that are clean, true and sharp.

- a. Carefully make openings in the masonry so that wall plates, cover plates or escutcheons required by the installation will completely conceal the openings and will have bottoms parallel with the masonry bed joints. Provide reinforced masonry lintels above openings over 12 inches wide for pipes, ducts, cable trays, and other wall penetrations, unless steel sleeves are used.
- b. Do not reduce masonry units in size by more than one-third in height and one-half in length. Do not locate cut products at ends of walls, corners, and other openings.

#### 3.3.1.3 Unfinished Work

Rack back unfinished work for joining with new work. Tooothing may be resorted to only when specifically approved by the Contracting Officer. Remove loose mortar and thoroughly clean the exposed joints before laying new work.

#### 3.3.1.4 Clay Masonry Expansion Joints

Provide clay masonry expansion joints as indicated. Construct by leaving a gap . Ensure that no mortar or other noncompressible materials are within



the joint. Install backer rod and sealant in accordance with Section 07 92 00 JOINT SEALANTS.

#### 3.3.1.5 Control Joints

Provide control joints in concrete masonry as indicated. Construct by raking out mortar within the head joint in accordance with the details shown on the Drawings. Form a continuous vertical joint at control joint locations, including through bond beams, by utilizing half blocks in alternating courses on each side of the joint. Interrupt the control joint key in courses containing continuous bond beam reinforcement. Do not interrupt the horizontal reinforcement and grout at the control joint.

Where mortar was placed in the joint, rake both faces of the control joints to a depth of 3/4 inch. Install backer rod and sealant on both faces in accordance with Section 07 92 00 JOINT SEALANTS.

#### 3.3.1.6 Decorative Architectural Units

Place decorative masonry units with the patterned face shell properly aligned in the completed wall.

### 3.3.2 Clay or Shale Brick Masonry

#### 3.3.2.1 Brick Placement

Blend all brick at the jobsite from several cubes to produce a uniform appearance when installed. An observable "banding" or "layering" of colors or textures caused by improperly mixed brick is unacceptable. Lay brick facing with the better face exposed. Lay brick in running bond with each course bonded at corners, unless otherwise indicated. Lay molded brick with the frog side down. Do not lay brick that is cored, recessed, or has other deformations in a manner that allows those deformations to be exposed to view; lay 100 percent solid units in these areas. Completely fill head and bed joints of solid units with mortar. Lay hollow units with mortar joints as specified for concrete masonry units.

Place exterior face of salvaged bricks towards the exterior.

#### 3.3.2.2 Wetting of Units

Wet clay, shale brick, or hollow brick units having an initial rate of absorption of more than 1 gram per minute per square inch of bed surface in conformance with ASTM C67/C67M. Ensure that each unit is nearly saturated when wetted but surface dry when laid.

Test clay or shale brick daily on the job, prior to laying, as follows: Using a wax pencil, draw a circle the size of a quarter on five randomly selected bricks. Apply 20 drops of water with a medicine dropper to the surface within the circle on each brick. If the average time that the water is completely absorbed in the five bricks is less than 1-1/2 minutes, wet bricks represented by the five bricks tested.

### 3.3.3 Anchored Veneer Construction

- a. Construct exterior masonry wythes to the thickness indicated on the drawings. Provide a minimum 1.5 inch air space behind the masonry veneer. Provide means to ensure that the cavity space and flashings

are kept clean of mortar droppings and other loose debris. Maintain chases and raked-out joints free from mortar and debris.

- b. Place masonry in running bond pattern.
- c. For veneer over stud framing, do not install veneer until the exterior sheathing, moisture barrier, veneer anchors and flashing have been installed on the backing. Take extreme care to avoid damage to the moisture barrier and flashing during construction of the masonry veneer. Repair or replace portions of the moisture barrier and flashing that are damaged prior to completion of the veneer. Provide a continuous cavity as indicated.
- d. For veneer with a masonry backup wythe, lay up both the inner and the outer wythes together except when adjustable joint reinforcement assemblies are approved for use. When both wythes are not brought up together, install through-wall flashings with the exterior wythe, securing the top edge of the flashing with a termination bar and sealant, or protect flashings that are installed with the interior wythe from damage until they are fully enclosed in the wall.
- e. Provide anchors (ties) to connect the veneer to its backing in sufficient quantity to comply with the following requirements: and maximum vertical spacing of 16-inches, and maximum horizontal spacing of 32-inches. Provide additional anchors around openings larger than 16 inch in either direction. Space anchors around perimeter of opening at a maximum of 24 inches on center. Place anchors within 12 inches of openings. Anchors with drips are not permitted.
- f. With solid units, embed anchors in mortar joint and extend into the veneer a minimum of 1-1/2 inch, with at least 5/8 inch mortar cover to the outside face.
- g. With hollow units, embed anchors in mortar or grout and extend into the veneer a minimum of 1-1/2 inch, with at least 5/8 inch mortar or grout cover to outside face.

#### 3.3.4 Composite Walls

Tie masonry wythes together with joint reinforcement or with unit wall ties. Embed wall ties at least 1-1/2 inch into mortar of solid units and at least 1/2 inch into the mortar of the outer face shell of hollow units. Provide at least one tie every 2.67 square feet for wire size W1.7 and at least one tie every 4.50 square feet for wire size W2.8. Space ties at a maximum of 36 inches horizontally and 24 inches vertically. Do not cross expansion joints or control joints with ties. Fill collar joints between masonry facing and masonry backup solidly with grout.

#### 3.3.5 Reinforced, Single Wythe Concrete Masonry Units Walls

##### 3.3.5.1 Concrete Masonry Unit Placement

- a. Fully bed units used to form piers, pilasters, columns, starting courses on footings, solid foundation walls, lintels, and beams, and where cells are to be filled with grout in mortar under both face shells and webs. Provide mortar beds under both face shells for other units. Mortar head joints for a distance in from the face of the unit not less than the thickness of the face shell.

- b. Solidly grout foundation walls below grade.
- c. Stiffen double walls at wall-mounted plumbing fixtures by use of strap anchors, two above each fixture and two below each fixture, located to avoid pipe runs, and extending from center to center of each wall within the double wall. Adequately reinforce walls and partitions for support of wall-hung plumbing fixtures when chair carriers are not specified.
- d. Submit drawings showing elevations of walls exposed to view and indicating the location of all cut CMU products.

#### 3.3.5.2 Preparation for Reinforcement

Lay units in such a manner as to preserve the unobstructed vertical continuity of cores to be grouted. Remove mortar protrusions extending 1/2 inch or more into cells before placing grout. Position reinforcing bars accurately as indicated before placing grout. Where vertical reinforcement occurs, fill cores solid with grout in accordance with paragraph PLACING GROUT in this Section.

#### 3.3.6 Cavity Walls (Multi-Wythe Noncomposite Walls)

Provide a continuous cavity as indicated. Bevel mortar beds away from cavity to prevent projection into cavity when bricks are shoved in place. Keep cavities clear and clean of mortar droppings. At the bottom of cavity walls, in the course immediately above the through-wall flashing, temporarily omit one brick every 4 feet. Clean mortar droppings and debris out of the cavity through the temporary openings at least once each day masonry is laid, and more often when required to keep the cavities clean. Fill in the openings with bricks and mortar after the wall is complete and the cavity has been inspected and found clean.

Securely tie the two wythes together with horizontal joint reinforcement, or provide ties to connect the masonry wythes in sufficient quantity to comply with the following requirements: maximum vertical spacing of 16-inches, and maximum horizontal spacing of 32-inches. Provide additional ties around openings larger than 16 inches in either direction. Space ties around perimeter of opening at a maximum of 3 feet on center. Place ties within 12 inches of openings. Ties with drips are not permitted.

#### 3.3.7 ANCHORAGE

##### 3.3.7.1 Anchorage to Concrete

Anchor masonry to the face of concrete columns, beams, or walls with dovetail anchors spaced not over 16 inches on centers vertically and 24 inches on center horizontally.

##### 3.3.7.2 Anchorage to Structural Steel

Anchor masonry to vertical structural steel framing with adjustable steel wire anchors spaced not over 16 inches on centers vertically, and if applicable, not over 24 inches on centers horizontally.

##### 3.3.7.3 Anchorage at Intersecting Walls

Provide wire mesh anchors at maximum 16 inches spacing at intersections of interior non-bearing masonry walls.

### 3.3.8 Lintels

#### 3.3.8.1 Masonry Lintels

Construct masonry lintels with lintel units filled solid with grout in all courses and reinforced with a minimum of two No. 4 bars in the bottom course unless otherwise indicated. Extend lintel reinforcement beyond each side of masonry opening 40 bar diameters or 24 inches, whichever is greater. Support reinforcing bars in place prior to grouting and locate 1/2 inch above the bottom inside surface of the lintel unit.

#### 3.3.8.2 Precast Concrete and Steel Lintels

Provide precast concrete and steel lintels as shown on the Drawings. Set lintels in a full bed of mortar with faces plumb and true. Provide steel and precast lintels with a minimum bearing length of 8 inches unless otherwise indicated. In partially grouted masonry, provide fully grouted units under the full lintel bearing length, unless otherwise indicated.

### 3.3.9 Sills and Copings

Set sills and copings in a full bed of mortar with faces plumb and true. Slope sills and copings to drain water. Mechanically anchor copings and sills longer than 4 feet as indicated.

## 3.4 INSTALLATION

### 3.4.1 Bar Reinforcement Installation

#### 3.4.1.1 Preparation

Submit detail drawings showing bar splice locations. Identify bent bars on a bending diagram and reference and locate such bars on the drawings. Show wall dimensions, bar clearances, and wall openings. Utilize bending details that conform to the requirements of ACI SP-66. No approval will be given to the shop drawings until the Contractor certifies that all openings, including those for mechanical and electrical service, are shown. If, during construction, additional masonry openings are required, resubmit the approved shop drawings with the additional openings shown along with the proposed changes. Clearly highlight location of these additional openings. Provide wall elevation drawings with minimum scale of 1/4 inch per foot. Submit drawings including plans, elevations, and details of wall reinforcement; details of reinforcing bars at corners and wall intersections; offsets; tops, bottoms, and ends of walls; control and expansion joints; lintels; and wall openings.

Clean reinforcement of loose, flaky rust, scale, grease, mortar, grout, and other coatings that might destroy or reduce its bond prior to placing grout. Do not use bars with kinks or bends not shown on the approved shop drawings. Place reinforcement prior to grouting. Unless otherwise indicated, extend vertical wall reinforcement to within 2 inches of tops of walls.

#### 3.4.1.2 Positioning Bars

- a. Accurately place vertical bars within the cells at the positions

indicated on the drawings. Maintain a minimum clearance of 1/2 inch between the bars and masonry units. Provide minimum clearance between parallel bars of 1/2 inch between the bars and masonry units for coarse grout and a minimum clearance of 1/4 inch between the bars and masonry units for fine grout. Provide minimum clearance between parallel bars of 1 inch or one diameter of the reinforcement, whichever is greater. Vertical reinforcement may be held in place using bar positioners located near the ends of each bar and at intermediate intervals of not more than 192 diameters of the reinforcement or by other means to prevent displacement beyond permitted tolerances. As masonry work progresses, secure vertical reinforcement to prevent displacement beyond allowable tolerances.

- b. Wire column and pilaster lateral ties in position around the vertical reinforcing bars. Place lateral ties in contact with the vertical reinforcement and do not place in horizontal mortar bed joints.
- c. Position horizontal reinforcing bars as indicated. Stagger splices in adjacent horizontal bars, unless otherwise indicated.
- d. Form splices by lapping bars as indicated. Do not cut, bend or eliminate reinforcing bars. Foundation dowel bars may be field-bent when permitted by TMS MSJC.

#### 3.4.1.3 Splices of Bar Reinforcement

Lap splice reinforcing bars as indicated. When used, provide welded or mechanical connections that develop at least 125 percent of the specified yield strength of the reinforcement.

#### 3.4.2 Placing Grout

##### 3.4.2.1 General

Fill cells containing reinforcing bars with grout. Solidly grout hollow masonry units in walls or partitions supporting plumbing, heating, or other mechanical fixtures, voids at door and window jambs, and other indicated spaces. Solidly grout cells under lintel bearings on each side of openings for full height of openings. Solidly grout walls below grade, lintels, and bond beams. Units other than open end units may require grouting each course to preclude voids in the units.

Discard site-mixed grout that is not placed within 1-1/2 hours after water is first added to the batch or when the specified slump is not met without adding water after initial mixing. Discard ready-mixed grout that does not meet the specified slump without adding water other than water that was added at the time of initial discharge. Allow sufficient time between grout lifts to preclude displacement or cracking of face shells of masonry units. Provide a grout shear key between lifts when grouting is delayed and the lower lift loses plasticity. If blowouts, flowouts, misalignment, or cracking of face shells should occur during construction, tear down the wall and rebuild.

##### 3.4.2.2 Vertical Grout Barriers for Multi-Wythe Composite Walls

In multi-wythe composite walls, provide grout barriers in the collar joint not more than 30 feet apart, or as required, to limit the horizontal flow of grout for each pour.

#### 3.4.2.3 Horizontal Grout Barriers

Embed horizontal grout barriers in mortar below cells of hollow units receiving grout.

#### 3.4.2.4 Grout Holes and Cleanouts

##### 3.4.2.4.1 Grout Holes

Provide grouting holes in slabs, spandrel beams, and other in-place overhead construction. Locate holes over vertical reinforcing bars or as required to facilitate grout fill in bond beams. Provide additional openings spaced not more than 16 inches on centers where grouting of hollow unit masonry is indicated. Form such openings not less than 4 inches in diameter or 3 by 4 inches in horizontal dimensions. Upon completion of grouting operations, plug and finish grouting holes to match surrounding surfaces.

##### 3.4.2.4.2 Cleanouts for Hollow Unit Masonry Construction

For hollow masonry units, provide cleanout holes at the bottom of every grout pour in cores containing vertical reinforcement when the height of the grout pour exceeds 5 feet 4 inches. Where all cells are to be grouted, construct cleanout courses using bond beam units in an inverted position to permit cleaning of all cells. Provide cleanout holes at a maximum spacing of 32 inches where all cells are to be filled with grout.

Establish a new series of cleanouts if grouting operations are stopped for more than 4 hours. Provide cleanouts not less than 3 by 3 inch by cutting openings in one face shell. Manufacturer's standard cutout units may be used at the Contractor's option. Do not cleanout holes until masonry work, reinforcement, and final cleaning of the grout spaces have been completed and inspected. For walls which will be exposed to view, close cleanout holes in an approved manner to match surrounding masonry.

##### 3.4.2.4.3 Cleanouts for Multi-Wythe Composite Masonry Construction

Provide cleanouts for construction of walls that incorporate a grout filled cavity between solid masonry wythes, provide cleanouts at the bottom of every pour by omitting every other masonry unit from one wythe. Establish a new series of cleanouts if grouting operations are stopped for more than 4 hours. Do not plug cleanout holes until masonry work, reinforcement, and final cleaning of the grout spaces have been completed and inspected. For walls which will be exposed to view, close cleanout holes in an approved manner to match surrounding masonry.

#### 3.4.2.5 Grout Placement

A grout pour is the total height of masonry to be grouted prior to erection of additional masonry. A grout lift is an increment of grout placement within a grout pour. A grout pour is filled by one or more lifts of grout.

- a. Lay masonry to the top of a pour permitted by TMS MSJC Table 7, based on the size of the grout space and the type of grout. Prior to grouting, remove masonry protrusions that extend 1/2 inch or more into cells or spaces to be grouted. Provide grout holes and cleanouts in accordance with paragraph GROUT HOLES AND CLEANOUTS above when the grout pour height exceeds 5 feet 4 inches. Hold reinforcement, bolts,

and embedded connections rigidly in position before grouting is started. Do not prewet concrete masonry units.

- b. Place grout using a hand bucket, concrete hopper, or grout pump to fill the grout space without segregation of aggregate. Operate grout pumps to produce a continuous stream of grout without air pockets, segregation, or contamination.
- c. If the masonry has cured at least 4 hours, grout slump is maintained between 10 to 11 inches, and no intermediate reinforced bond beams are placed between the top and bottom of the pour height, place conventional grout in lifts not exceeding 12 feet 8 inches. For the same curing and slump conditions but with intermediate bond beams, limit conventional grout lift to the bottom of the lowest bond beam that is more than 5 feet 4 inches above the bottom of the lift, but do not exceed 12 feet 8 inches. If masonry has not cured at least 4 hours or grout slump is not maintained between 10 to 11 inches, place conventional grout in lifts not exceeding 5 feet 4 inches.
- d. Consolidate conventional grout lift and reconsolidate after initial settlement before placing next lift. For grout pours that are 12 inches or less in height, consolidate and reconsolidate grout by mechanical vibration or puddling. For grout pours that are greater than 12 inches in height, consolidate and reconsolidate grout by mechanical vibration. Apply vibrators at uniformly spaced points not further apart than the visible effectiveness of the machine. Limit duration of vibration to time necessary to produce satisfactory consolidation without causing segregation. If previous lift is not permitted to set, dip vibrator into previous lift. Do not insert vibrators into lower lifts that are in a semi-solidified state. If lower lift sets prior to placement of subsequent lift, form a grout key by terminating grout a minimum of 1-1/2 inch below a mortar joint. Vibrate each vertical cell containing reinforcement in partially grouted masonry. Do not form grout keys within beams.
- e. If the masonry has cured 4 hours, place self-consolidating grout (SCG) in lifts not exceeding the pour height. If masonry has not cured for at least 4 hours, place SCG in lifts not exceeding 5 feet 4 inches. Do not mechanically consolidate self-consolidating grout. Place self-consolidating grout in accordance with manufacturer's recommendations.
- f. Upon completion of each day's grouting, remove waste materials and debris from the equipment, and dispose of outside the masonry.

#### 3.4.3 Joint Reinforcement Installation

Install joint reinforcement at 16 inches on center unless otherwise indicated. Lap joint reinforcement not less than 6 inches. Install prefabricated sections at corners and wall intersections. Place the longitudinal wires of joint reinforcement in mortar beds to provide not less than 5/8 inch cover to either face of the unit.

#### 3.4.4 Bond Beams

Reinforce and grout bond beams as indicated and as described in paragraphs above. Install grout barriers under bond beam units to retain the grout as required, unless wall is fully grouted or solid bottom units are used. For high lift grouting in partially grouted masonry, provide grout

retaining material on the top of bond beams to prevent upward flow of grout. Ensure that reinforcement is continuous, including around corners, except through control joints or expansion joints, unless otherwise indicated.

#### 3.4.5 Flashing and Weeps

- a. Install through-wall flashing at obstructions in the cavity and where indicated on Drawings. Ensure continuity of the flashing at laps and inside and outside corners by splicing in a manner approved by the flashing manufacturer. Ensure that the top edge of the flashing is sealed by attaching a termination bar and applying compatible sealant at the top edge of the termination bar. Terminate the horizontal leg of the flashing by extending the sheet metal 1/2 inch beyond the outside face of masonry and turning downward with a stainless steel hemmed drip. Provide sealant below the drip edge of through-wall flashing.
- b. Wherever through-wall flashing occurs, provide weep holes to drain flashing to exterior at acceptable locations as indicated. Provide weeps of weep ventilators. Locate weeps not more than 24 inches on centers in mortar joints of the exterior wythe directly on the horizontal leg of through-wall flashing over foundations, bond beams, and any other horizontal interruptions of the cavity. Place weep holes perfectly horizontal or slightly canted downward to encourage water drainage outward and not inward. Other methods may be used for providing weeps when spacing is reduced to 16 inches on center and approved by the Contracting Officer. Maintain weeps free of mortar and other obstructions.

### 3.5 APPLICATION

#### 3.5.1 Insulation

Insulate cavity walls (multi-wythe noncomposite masonry walls), where shown, by installing board-type insulation on the cavity side of the inner wythe. Apply board type insulation directly to the masonry or thru-wall flashing with adhesive. Neatly fit insulation between obstructions without impaling insulation on ties or anchors. Apply insulation in parallel courses with vertical joints breaking midway over the course below and in moderate contact with adjoining units without forcing. Cut to fit neatly against adjoining surfaces. Tape or seal the joints between the boards.

#### 3.5.2 Interface with Other Products

##### 3.5.2.1 Built-In Items

Fill spaces around built-in items with mortar. Point openings around flush-mount electrical outlet boxes in wet locations with mortar. Embed anchors, ties, wall plugs, accessories, flashing, pipe sleeves and other items required to be built-in as the masonry work progresses. Fully embed anchors, ties and joint reinforcement in the mortar. Fill cells receiving anchor bolts and cells of the first course below bearing plates with grout, unless otherwise indicated.



#### 3.5.2.2 Door and Window Frame Joints

On the exposed interior and exterior sides of exterior frames, rake joints between frames and abutting masonry walls to a depth of 3/8 inch.

#### 3.5.2.3 Bearing Plates

Set bearing plates for beams, joists, joist girders and similar structural members to the proper line and elevation with damp-pack bedding mortar, except where non-shrink grout is indicated. Provide bedding mortar and non-shrink grout as specified in Section 03 30 00 CAST-IN-PLACE CONCRETE.

#### 3.5.3 Tolerances

Lay masonry plumb, true to line, with courses level within the tolerances of TMS MSJC, Article 3.3 F.

### 3.6 FIELD QUALITY CONTROL

#### 3.6.1 Tests

##### 3.6.1.1 Field Testing of Mortar

Perform mortar testing at the following frequency: 25 times per 1000. For each required mortar test, provide a minimum of three mortar samples. Perform initial mortar testing prior to construction for comparison purposes during construction.

Prepare and test mortar samples for mortar aggregate ratio in accordance with ASTM C780 Appendix A4.

##### 3.6.1.2 Field Testing of Grout

- a. Perform grout testing at the following frequency: 2 times per month. For each required grout property to be evaluated, provide a minimum of three specimens.
- b. Sample and test conventional and self-consolidating grout for compressive strength and temperature in accordance with ASTM C1019.
- c. Evaluate slump in conventional grout in accordance with ASTM C1019.
- d. Evaluate slump flow and visual stability index of self-consolidating grout in accordance with ASTM C1611/C1611M.

##### 3.6.1.3 Clay Brick Efflorescence Test

Test clay brick that will be exposed to weathering for efflorescence in accordance with ASTM C67/C67M. Schedule tests far enough in advance of starting masonry work to permit retesting if necessary. Units meeting the definition of "effloresced" are subject to rejection.

##### 3.6.1.4 Prism Tests

Perform at least one prism test sample for each 5,000 square feet of wall but not less than three such tests for any building. Evaluate three prisms in each test. Fabricate, store, handle, and test prisms in accordance with ASTM C1314.

Seven-day tests may be used provided the relationship between the 7- and 28-day strengths of the masonry is established by the tests of the materials used. If the compressive strength of any prism falls below the specified value by more than 500 psi, take steps to assure that the load-carrying capacity of the structure is not jeopardized. If the likelihood of low-strength masonry is confirmed and computations indicate that the load-carrying capacity may have been significantly reduced, tests of cores drilled, or prisms sawed, from the area in question may be required. In such case, take three specimens for each prism test more than 500 psi below the specified value. Masonry in the area in question will be considered structurally adequate if the average compressive strength of three specimens is equal to or exceeds the specified value. Additional testing of specimens extracted from locations represented by erratic core or prism strength test results will be permitted.

### 3.6.2 Special Inspection

Perform special inspections and testing in accordance with Section 01 45 35 SPECIAL INSPECTIONS.

## 3.7 POINTING AND CLEANING

After mortar joints have attained their initial set, but prior to hardening, completely remove mortar and grout daubs and splashings from masonry-unit surfaces that will be exposed or painted. Before completion of the work, rake out defects in joints of masonry to be exposed or painted, fill with mortar, and tool to match existing joints. Immediately after grout work is completed, remove scum and stains that have percolated through the masonry work using a low pressure stream of water and a stiff bristled brush. Do not clean masonry surfaces, other than removing excess surface mortar, until mortar in joints has hardened. Leave masonry surfaces clean, free of mortar daubs, dirt, stain, and discoloration, including scum from cleaning operations, and with tight mortar joints throughout. Do not use metal tools and metal brushes for cleaning.

### 3.7.1 Dry-Brushing Concrete Masonry

Dry brush exposed concrete masonry surfaces at the end of each day's work and after any required pointing, using stiff-fiber bristled brushes.

### 3.7.2 Clay Brick Surfaces

Clean exposed clay brick masonry surfaces to obtain surfaces free of stain, dirt, mortar and grout daubs, efflorescence, and discoloration or scum from cleaning operations. Perform cleaning in accordance with the approved cleaning procedure demonstrated on the mockup.

After cleaning, examine the sample panel of similar material for discoloration or stain as a result of cleaning. If the sample panel is discolored or stained, change the method of cleaning to ensure that the masonry surfaces in the structure will not be adversely affected. Water-soak exposed masonry surfaces and then clean with a proprietary masonry cleaning agent specifically recommended for the color and texture by the clay brick manufacturer and manufacturer of the cleaning product. Apply the solution with stiff fiber brushes, followed immediately by thorough rinsing with clean water. Use proprietary cleaning agents in conformance with the cleaning product manufacturer's printed recommendations. Remove efflorescence in conformance with the brick manufacturer's recommendations.

### 3.8 CLOSE-OUT TAKE-BACK PROGRAM

Collect information from manufacturer for take-back program options. Set aside masonry units, full and partial to be returned to manufacturer for recycling into new product. When such a service is not available, seek local recyclers to reclaim the materials. Submit documentation that includes contact information, summary of procedures, and the limitations and conditions applicable to the project. Indicate manufacturer's commitment to reclaim materials for recycling and/or reuse.

### 3.9 PROTECTION

Protect facing materials against staining. Cover top of walls with nonstaining waterproof covering or membrane to protect from moisture intrusion when work is not in progress. Continue covering the top of the unfinished walls until the wall is waterproofed with a complete roof or parapet system. Extend covering a minimum of 2 feet down on each side of the wall and hold securely in place. Before starting or resuming work, clean top surface of masonry in place of loose mortar and foreign material.

-- End of Section --

SECTION 05 05 20

POST-INSTALLED CONCRETE AND MASONRY ANCHORS  
**05/22**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

- |           |                                                                                                |
|-----------|------------------------------------------------------------------------------------------------|
| ACI 355.2 | (2007) Qualification of Post-Installed Mechanical Anchors in Concrete and Commentary           |
| ACI 355.4 | (2011) Qualification of Post-Installed Adhesive Anchors in Concrete (ACI 355.4) and Commentary |

ASTM INTERNATIONAL (ASTM)

- |                 |                                                                                                                                                         |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A36/A36M   | (2019) Standard Specification for Carbon Structural Steel                                                                                               |
| ASTM A153/A153M | (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware                                                                     |
| ASTM A193/A193M | (2024) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications |
| ASTM A615/A615M | (2024) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement                                                       |
| ASTM E488/E488M | (2022) Standard Test Methods for Strength of Anchors in Concrete Elements                                                                               |

INTERNATIONAL CODE COUNCIL (ICC)

- |         |                                    |
|---------|------------------------------------|
| ICC IBC | (2021) International Building Code |
|---------|------------------------------------|

U.S. ARMY CORPS OF ENGINEERS (USACE)

- |            |                                                                    |
|------------|--------------------------------------------------------------------|
| EM 385-1-1 | (2024) Safety -- Safety and Occupational Health (SOH) Requirements |
|------------|--------------------------------------------------------------------|

1.2 APPLICABILITY

This guide specification covers the requirements for all anchors that are post-installed into hardened concrete, concrete masonry, or brick. This

guide specification does not cover through bolts, powder or pneumatic actuated nails, or cast in anchors. Refer to Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS for requirements of through bolts, powder or pneumatic actuated nails, or cast in anchors.

### 1.3 DEFINITIONS

#### 1.3.1 Anchor

"Anchor" includes steel elements post-installed into hardened concrete, concrete masonry, or brick and used to transmit applied loads.

#### 1.3.2 Periodic Special Inspections

"Periodic Special Inspection" as used herein means that, as a minimum, the Post-Installed Anchor Special Inspector must perform inspections in accordance with this specification.

#### 1.3.3 Continuous Special Inspections

"Continuous Special Inspection" as used herein means that the Post-Installed Anchor Special Inspector observes the drilling and cleaning of holes, the injection of adhesive into the holes, and the insertion of anchors into the holes. When applicable for the type of installation, or as indicated in the project drawings, "Continuous Special Inspection" also includes observation of measures to secure the anchor during the adhesive curing period.

### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-01 Preconstruction Submittals

Installer Qualifications; G  
Post-Installed Anchor Special Inspector Qualifications; G

#### SD-03 Product Data

Adhesive Anchor System in Concrete; G  
Non-Shrink, Non-Metallic Grout; G

#### SD-06 Test Reports

Post-Installed Anchor Special Inspections Report; G

#### SD-07 Certificates

Post-Installed Anchor Certification; G

#### SD-08 Manufacturer's Instructions

Manufacturer's Printed Installation Instructions; G

Manufacturer's Material Safety Data Sheets

## 1.5 QUALITY ASSURANCE

Perform all work in accordance with EM 385-1-1 and all manufacturer's instructions and recommendations. To protect personnel from overexposure to toxic materials, conform to the applicable manufacturer's Material Safety Data Sheets (MSDS) or local regulation. Submit the MSDS for epoxies and other potentially hazardous materials.

### 1.5.1 Qualifications

The submittals must identify individuals who will be working on this contract and their relevant experience and training. Do not make changes in approved personnel without prior approval of the Contracting Officer.

#### 1.5.1.1 Installer Qualifications

Each worker engaged in the installation of post-installed anchors must have satisfactorily completed an applicable certification program or equivalent instruction program through the manufacturer or manufacturer's representative for all anchoring products they will install. A manufacturer's representative must train all installers per the installation instructions as listed in the ICC-ES Evaluation Report for the anchor being installed. Training must consist of a review and performance test of the complete installation process, including but not limited to:

- (1) Hole drilling procedure
- (2) Hole preparation & cleaning technique
- (3) Adhesive injection technique & dispenser training / maintenance
- (4) Rebar dowel preparation and installation
- (5) Proof loading/torquing
- (6) Installation in horizontal and upward orientations

Submit certification for each worker showing that they have completed the above training within three years prior to onsite work. Certification must include organization or manufacturer's name, instructor's name and qualifications, trainee's name, list of instruction received, date of instruction, and confirmation of successful performance tests.

#### 1.5.1.2 Post-Installed Anchor Special Inspector Qualifications

The Contractor must retain the services of a third party Special Inspector independent of the installing contractor and manufacturer. The individual(s) who perform special inspections for post-installed anchors must meet all Installer Qualification requirements and have a minimum of 3 years of experience as a Special Inspector on previous projects involving similar scope of work. Submit resumes, pertinent information, past experience, and training.

## 1.6 DELIVERY, STORAGE, AND HANDLING

### 1.6.1 Packing, Shipping, Handling, and Unloading

Deliver products to job site in manufacturer's or distributor's original

packaging undamaged, complete with installation instructions. Inspect materials delivered to site for damage. Unload and store with minimal handling.

#### 1.6.2 Storage

Protect, store, and handle materials in accordance with manufacturer's recommendations to prevent damage or deterioration. Do not allow chemical materials to freeze. Remove materials that have not be stored in accordance with the manufacturer's recommendations, including expired materials, from the job site.

### PART 2 PRODUCTS

#### 2.1 MATERIALS

##### 2.1.1 Post-Installed Anchors

Provide anchors of the type, effective embedment, and diameter indicated on contract drawings. Minimum spacing and concrete edge distances must be as shown on contract drawings. Design values listed must be as tested according to ASTM E488/E488M for the substrate type, substrate moisture condition, concrete aggregate type (normal weight or lightweight concrete), and concrete/masonry strength. Minimum allowable strength tension and shear values must be as indicated on contract drawings. If more than one type of anchor is to be used on a project, clearly indicate on the submittal where each type of anchor will be used.

##### 2.1.1.1 Post-Installed Anchor Certification

Submit product information with recommended design values and physical characteristics for each type anchor shown on the drawings.

Provide certified test reports showing compliance with specified performance characteristics and physical properties. Anchors must have one of the following certifications:

- (1) ICC-ES Evaluation Report indicating conformance with current applicable ICC ES Acceptance Criteria
- (2) Third party Evaluation Report in conformance with ACI 355.2 or ACI 355.4, as applicable. Third party must be accredited under ISO/IEC 17025 by a recognized accreditation body conforming to the requirements of ISO/IEC 17011 in accordance with ACI 355.4."

##### 2.1.1.2 Manufacturer's Printed Installation Instructions

Submit manufacturer's instructions for each anchor type shown on the drawings.

##### 2.1.1.3 Adhesive Anchor System in Concrete

Use an adhesive to bond steel anchors to concrete. The adhesive must be a moisture insensitive, structural adhesive. Anchors must have been tested and qualified for performance in cracked and uncracked concrete, horizontal and overhead applications, and long term creep in accordance with ACI 355.4.

Threaded rod anchors must meet the requirements of ASTM A36/A36M.

Threaded rods must be galvanized in accordance with ASTM A153/A153M or stainless steel in accordance with ASTM A193/A193M unless otherwise indicated. Reinforcing bars must meet the requirements of ASTM A615/A615M Grade 60.

Adhesive anchors must have the below characteristic bond strengths for uncracked and cracked concrete in 4500 psi concrete with maximum short term temperatures of 130 degrees F and maximum long term temperatures of 110 degrees F:

| ANCHOR             | Tau,uncr<br>(characteristic bond<br>strength, uncracked<br>concrete) | Tau,cr (characteristic<br>bond strength, cracked<br>concrete) |
|--------------------|----------------------------------------------------------------------|---------------------------------------------------------------|
|                    | (psi)                                                                | (psi)                                                         |
|                    |                                                                      |                                                               |
| #5 reinforcing bar | 1654                                                                 | 1140                                                          |

## 2.2 EQUIPMENT

Assemble at the site of the work, sufficient equipment that is dependable, appropriate and adequate to accomplish the work specified. Maintain the equipment in good working condition.

## PART 3 EXECUTION

### 3.1 ANCHORING AND REINFORCING

Install anchors in accordance with the spacing and edge clearances indicated on the drawings. Anchor capacity is also highly dependent on proper installation. Follow all manufacturer and Evaluation Report installation instructions.

#### 3.1.1 Drilling and Installing Adhesive Anchors

Drill holes for anchors using drilling equipment and bits suitable for the intended purpose, in accordance with Manufacturer's Printed Installation Instructions and Evaluation Report installation instructions. Diameter of holes must be as recommended by the anchor manufacturer. Unless otherwise shown on the drawings, all holes must be drilled perpendicular to the concrete surface. Deviations more than 10 degrees from perpendicular are not acceptable. Unless otherwise specified, do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength. Adhesive anchors must not be installed in concrete that is less than 21 days old.

Clean holes, place grout, and install anchors in accordance with anchor manufacturer's recommendations. Remove excess adhesive after the anchor has been set in place. Remove spills on adjacent surfaces. Protect threads and anchor from damage during anchor installation. Ensure proper embedment and placement in accordance with contract documents and all other work. Do not disturb or load anchors before manufacturer specified cure time has elapsed.

Adhesives must be stored at temperatures prescribed by the manufacturer and must not be used beyond the expiration date.



### 3.1.2 Unused or Repairs to Drilled Holes

Any holes made for anchors that are not used must be filled with non-shrink, non metallic grout suitable for the orientation and size of hole and have a minimum compressive strength of 4000 psi. Repair must completely fill hole and be flush with existing concrete or masonry. Place in accordance with manufacturer's recommended instructions. Final anchor positions must not be within 1 inch of repair patches.

### 3.2 EMBEDDED ITEMS

Existing reinforcing bars or other embedded items in the structure may conflict with specified anchor locations. Existing reinforcing and embedded items must not be damaged during installation of post-installed anchors.

The contractor must review the as-built drawings and must use Radar detection systems (such as Hilti Ferroskan), X-Ray, or other appropriate means to accurately locate the position of existing reinforcing bars and embedded items at the locations of the anchors in the field. Scanning method must provide enough accuracy and precision to locate the space between rebar. Structural integrity of existing concrete or masonry must not be impaired by investigating method.

Create a template at each anchor connection location prior to fabricating holes in connection plates. Template must be made by locating existing reinforcing with an approved reinforcement detection system.

### 3.3 TESTS AND INSPECTIONS

#### 3.3.1 Adhesive Anchors

For adhesive anchors, periodic special inspection are required as a minimum. Where adhesive anchors are used to resist sustained tension in horizontal or upwardly inclined orientations, or where the findings of the Evaluation Report for the adhesive anchor product require it, continuous special inspection is needed. Inspections must be in accordance with ICC IBC and the Evaluation Report.

Adhesive anchors must be inspected during installation, to verify anchor type, anchor dimensions, base material type, base material age, drill bit, hole dimensions, hole cleaning procedures, anchor spacing, edge distances, base material thickness, effective embedment, tightening torque, adhesive mixing, filling of the hole with adhesive, adherence to the manufacturer's printed installation instructions, and any additional items recommended in the Evaluation Report.

The Special Inspector must perform proof loading on the first three anchors of each type and size, for each installer, and a minimum of 5 percent of randomly selected anchors. Anchor selection will be determined by the Post-Installed Anchor Special Inspector. Perform confined tension proof load testing in accordance with ASTM E488/E488M. Use incremental loading for tensile test. Maintain proof load for a minimum of 10 seconds. Consider anchors to have failed if displacement exceeds 0.1 inch D/10, where D is the nominal anchor diameter, or if any of the failure modes listed in ASTM E488/E488M occur.

Adhesive anchors and capsule anchors must not be torque tested.

Proof loads must be the minimum of the value shown on the Drawings, the values shown in the table below, or 50 percent of the tension capacity of the anchor:

| ANCHOR                              | EFFECTIVE EMBEDMENT | CONFINED TENSION PROOF LOAD |
|-------------------------------------|---------------------|-----------------------------|
|                                     | (inches)            | (pounds)                    |
|                                     |                     |                             |
| Adhesive anchor, #5 reinforcing bar | 2-3/4" to 10"       | 6935                        |

### 3.3.2 Action Required from Failed Tests/Inspections

Immediately report failed anchor locations and test results to the Contracting Officer. Anchors that fail to meet proof/torque load or installation requirements must be regarded as malfunctioning. Do not re-use holes unless specifically allowed by manufacturer's published instructions and approved by the Post-Installed Anchor Special Inspector.

If any of the tested anchors fail to achieve the specified torque or proof load within the limits of the contract documents, test a minimum of two adjacent anchors for each anchor that fails.

Continuously special inspect and proof load/torque test any replacement anchors.

Fill unused anchor holes and patch failed anchor locations in accordance with this specification. Prior to performing the repair, the Contractor must submit to the Contracting Officer for approval, the proposed fill and patch materials.

Additional tests, repairs, delays, or modification of work to accommodate failed tests will be at no cost to the Government.

### 3.3.3 Post-Installed Anchor Special Inspections Report

Report the results of all inspections weekly. Submit report as an electronic PDF file to the Contracting Officer for review by the Engineer of Record. The report must include the following:

- (1) Exact locations of the inspected and tested work
- (2) Inspector's name
- (3) Date of inspection
- (4) Summary of work completed during the inspection period
- (5) Test results
- (6) Statement by the Special Inspector that clearly identifies the tested anchors as being acceptable or rejected.
- (7) Statement by the Special Inspector confirming that the materials and installation procedures conform with the approved contract

documents and the manufacturer's published installation instructions.

#### 3.4 DUST CONTROL

Control dust resulting from demolition to prevent the spread of dust and avoid creation of a nuisance in the surrounding area. Do not use water when it will result in, or create, hazardous or objectionable conditions such as ice, flooding, or pollution.

-- End of Section --

SECTION 05 05 23.16

STRUCTURAL WELDING  
**08/18**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 360 (2016) Specification for Structural Steel Buildings

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)

ANSI/ASNT CP-189 (2020) ASNT Standard for Qualification and Certification of Nondestructive Testing Personnel

AMERICAN WELDING SOCIETY (AWS)

AWS A2.4 (2012) Standard Symbols for Welding, Brazing and Nondestructive Examination

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding Code - Steel

AWS QC1 (2016) Specification for AWS Certification of Welding Inspectors

AWS Z49.1 (2021) Safety in Welding and Cutting and Allied Processes

ASTM INTERNATIONAL (ASTM)

ASTM E165/E165M (2023) Standard Practice for Liquid Penetrant Examination for General Industry

ASTM E709 (2021) Standard Guide for Magnetic Particle Testing

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Welding Quality Assurance Plan; G

SD-03 Product Data

Welding Procedure Qualifications; G

Welder, Welding Operator, and Tacker Qualification

Previous Qualifications

Welding Electrodes and Rods

SD-06 Test Reports

Nondestructive Testing

Weld Inspection Log

SD-07 Certificates

Certified Welding Procedure Specifications (WPS)

Certified Brazing Procedure Specifications (BPS)

Certified Procedure Qualification Records (PQR)

Certified Welder Performance Qualifications (WPQ)

Certified Brazer Performance Qualifications (BPQ)

Certified Welding Inspector

Nondestructive Testing Personnel

### 1.3 QUALITY ASSURANCE

Except for pre-qualified (in accordance with AWS D1.1/D1.1M) and previously qualified procedures, each Contractor performing welding must record in detail and qualify the welding procedure specification for any welding procedure followed in the fabrication of weldments. Conform welding procedure qualifications to AWS D1.1/D1.1M and to the specifications in this section. Submit for approval copies of the welding procedure specification and the procedure qualification records for each type of welding being performed. Submission of the welder, welding operator, or tacker qualification test records is also required. Approval of any procedure, however, does not relieve the Contractor of the sole responsibility for producing a finished structure meeting all the specified requirements. Submit this information on the forms in Annex M of AWS D1.1/D1.1M. Individually identify and clearly reference on the detail drawings and erection drawings all welding procedure specifications, or suitably key them to the contract drawings. In case of conflict between this specification and AWS D1.1/D1.1M, this specification governs.

#### 1.3.1 General Requirements

Fabricate work in an AISC Certified Fabrication Plant, Category BU. Erect work by an AISC Certified Erector, Category CSE.

a. For Structural Projects, provide documentation of the following:

- (1) Component Thickness 1/8 inch and greater: Qualification documents (WPS, PQR, and WPQ) in accordance with AWS D1.1/D1.1M.
- (2) Component Thickness Less than 1/8 inch: Qualification documents (WPS, PQR, and WPQ) in accordance with AWS D1.3/D1.3M.
- (3) Reinforcing Steel: Qualification documents (WPS, PQR, and WPQ) in accordance with AWS D1.4/D1.4M.

b. For other applications, provide documentation of the following:

- (1) Submit two copies of the Certified Welding Procedure Specifications (WPS), Certified Brazing Procedure Specifications (BPS) and Certified Procedure Qualification Records (PQR) to the Contracting Officer for approval.
- (2) Submit two copies of the Certified Welder Performance Qualifications (WPQ) and Certified Brazer Performance Qualifications (BPQ) to the Contracting Officer for approval within fifteen calendar days prior to any employee welding on the project material.
- (3) Machinery: Qualification documents (WPS, PQR, and WPQ) in accordance with AWS D14.4/D14.4M.

#### 1.3.2 Previous Qualifications

Welding procedures previously qualified by test in accordance with AWS D1.1/D1.1M, may be accepted for this contract without re-qualification, upon receipt of the test results, if the following conditions are met:

- a. Testing was performed by an approved testing laboratory, technical consultant, or the Contractor's approved quality control organization.
- b. The qualified welding procedure conforms to the requirements of this specification and is applicable to welding conditions encountered under this contract.
- c. The welder, welding operator, and tacker qualification tests conform to the requirements of this specification and are applicable to welding conditions encountered under this contract.

#### 1.3.3 Welder, Welding Operator, and Tacker Qualification

Each welder, welding operator, and tacker assigned to work on this contract must be qualified in accordance with the applicable requirements of AWS D1.1/D1.1M and as specified in this section. Welders, welding operators, and tackers who make acceptable procedure qualification test welds will be considered qualified for the welding procedure used within the applicable essential variables for welder qualification.

##### 1.3.3.1 Previous Personnel Qualifications

At the discretion of the Contracting Officer, welders, welding operators, and tackers qualified by test within the previous 6 months may be accepted for this contract without re-qualification if all the following conditions are met:

- a. Copies of the welding procedure specifications, the procedure qualification test records, and the welder, welding operator, and tacker qualification test records are submitted and approved in accordance with the specified requirements for detail drawings.
- b. Testing was performed by an approved testing laboratory, technical consultant, or the Contractor's approved quality control organization.
- c. The welder, welding operator, and tacker qualification tests conform to the requirements of this specification and are applicable to welding conditions encountered under this contract.

#### 1.3.3.2 Certificates

Before assigning any welder, welding operator, or tacker to work under this contract, submit the names and certification that each individual is qualified as specified. State in the certification the type of welding and positions for which the welder, welding operator, or tacker is qualified, the code and procedure under which the individual is qualified, the date qualified, and the name of the firm and person certifying the qualification tests. Keep the certification current, on file, and furnish 3 copies.

#### 1.3.3.3 Renewal of Qualification

Re-qualification of a welder or welding operator is required under any of the following conditions:

- a. It has been more than 6 months since the welder or welding operator has used the specific welding process for which he is qualified.
- b. There is specific reason to question the welder or welding operator's ability to make welds that meet the requirements of these specifications.
- c. The welder or welding operator was qualified by an employer other than those firms performing work under this contract, and a qualification test has not been taken within the past 12 months. Submit as evidence of conformance all records showing periods of employment, name of employer where welder, or welding operator, was last employed, and the process for which qualified.
- d. A tacker who passes the qualification test is considered eligible to perform tack welding indefinitely in the positions and with the processes for which he/she is qualified, unless there is some specific reason to question the tacker's ability or there has been a gap greater than 6 months since he/she last used the process. In such a case, the tacker is required to pass the prescribed tack welding test.

#### 1.3.4 Inspector Qualification

Submit certificates indicating that certified welding inspectors meet the requirements of AWS QC1. Submit qualifications for nondestructive testing personnel in accordance with the requirements of ANSI/ASNT CP-189 for Levels I or II in the applicable nondestructive testing method. Level I inspectors must have direct supervision of a Level II inspector.

### 1.3.5 Symbols and Safety

Use symbols in accordance with AWS A2.4, unless otherwise indicated. Follow safe welding practices and safety precautions during welding in conformance with AWS Z49.1.

## PART 2 PRODUCTS

### 2.1 SYSTEM DESCRIPTION

Conform the design of welded connections to AISC 360, unless otherwise indicated or specified. Material with welds will not be accepted unless the welding is specified or indicated on the drawings or otherwise approved. Perform welding as specified in this section, except where additional requirements are shown on the drawings or are specified in other sections. Do not commence welding until welding procedures, inspectors, nondestructive testing personnel, welders, welding operators, and tackers have been qualified and the submittals approved by the Contracting Officer. Perform all testing at or near the work site. Maintain records of the test results obtained in welding procedure, welder, welding operator, and tacker performance qualifications.

#### 2.1.1 Pre-erection Conference

Hold a pre-erection conference prior to the start of the field welding, to bring all affected parties together and to gain a naturally clear understanding of the project and the Welding Procedure Specifications (WPS) (submitted for all welding, including welding done using pre-qualified procedures). Mandatory attendance is required by all Contractor's welding production and inspection personnel and appropriate Government personnel. Include as items for discussion: responsibilities of various parties; welding procedures and processes to be followed; welding sequence (both within a joint and joint sequence within the building); inspection requirements and procedures, both visual and nondestructive testing; welding schedule; and other items deemed necessary by the attendees.

### 2.2 WELDING EQUIPMENT AND MATERIALS

Provide all welding equipment, welding electrodes and rods, welding wire, and fluxes capable of producing satisfactory welds when used by a qualified welder or welding operator. Provide welding equipment and materials that comply with the applicable requirements of AWS D1.1/D1.1M. Submit product data on welding electrodes and rods.

## PART 3 EXECUTION

### 3.1 WELDING OPERATIONS

#### 3.1.1 Requirements

Conform workmanship and techniques for welded construction to the requirements of AWS D1.1/D1.1M and AISC 360. When AWS D1.1/D1.1M and the AISC 360 specification conflict, the requirements of AWS D1.1/D1.1M govern.

#### 3.1.2 Identification

Identify all welds in one of the following ways:



- a. Submit written records to indicate the location of welds made by each welder, welding operator, or tacker.
- b. Identify all work performed by each welder, welding operator, or tacker with an assigned number, letter, or symbol to identify welds made by that individual. The Contracting Officer may require welders, welding operators, and tackers to apply their symbol next to the weld by means of rubber stamp, felt-tipped marker with waterproof ink, or other methods that do not cause an indentation in the metal. Place the identification mark for seam welds adjacent to the weld at 3 foot intervals. Identification with die stamps or electric etchers is not allowed.

### 3.2 QUALITY CONTROL

Perform testing using an approved inspection or testing laboratory or technical consultant; or if approved, the Contractor's inspection and testing personnel may be used instead of the commercial inspection or testing laboratory or technical consultant. A Certified Welding Inspector must perform visual inspection on 100 percent of all welds. Document this inspection in the Visual Weld Inspection Log. Test 50% of CJP welds using ultrasonic testing per Table 6.2 of AWS D1.1/D1.1M. Randomly test 50% of all PJP and fillet welds or as indicated by magnetic particle or dye penetrant testing. Verify the welds conform to paragraph STANDARDS OF ACCEPTANCE. Conform procedures and techniques for inspection with applicable requirements of AWS D1.1/D1.1M, ASTM E165/E165M, and ASTM E709. Submit a Welding Quality Assurance Plan and records of tests and inspections.

### 3.3 STANDARDS OF ACCEPTANCE

Conform dimensional tolerances for welded construction, details of welds, and quality of welds with the applicable requirements of AWS D1.1/D1.1M and the contract drawings. Submit all records of nondestructive testing.

#### 3.3.1 Nondestructive Testing

The welding is subject to inspection and tests in the mill, shop, and field. Inspection and tests in the mill or shop do not relieve the Contractor of the responsibility to furnish weldments of satisfactory quality. When materials or workmanship do not conform to the specification requirements, the Government reserves the right to reject material or workmanship or both at any time before final acceptance of the structure containing the weldment. Any indication of a defect is regarded as a defect, unless re-evaluation by nondestructive methods or by surface conditioning shows that no unacceptable defect is present. Submit all records of nondestructive testing in accordance with paragraph STANDARDS OF ACCEPTANCE.

#### 3.3.2 Destructive Tests

Make all repairs when metallographic specimens are removed from any part of a structure. Employ only qualified welders or welding operators, and use the proper joints and welding procedures, including peening or heat treatment if required, to develop the full strength of the members and joints cut and to relieve residual stress.

### 3.4 GOVERNMENT INSPECTION AND TESTING

In addition to the inspection and tests performed by the Contractor for quality control, the Government will perform inspection and testing for acceptance to the extent determined by the Contracting Officer. The work may be performed by the Government's own forces or under a separate contract for inspection and testing. The Government reserves the right to perform supplemental nondestructive and destructive tests to determine compliance with paragraph STANDARDS OF ACCEPTANCE.

### 3.5 CORRECTIONS AND REPAIRS

If inspection or testing indicates defects in the weld joints, repair defective welds using a qualified welder or welding operator as applicable. Conduct corrections in accordance with the requirements of AWS D1.1/D1.1M and the specifications. Repair all defects in accordance with the approved procedures. Repair defects discovered between passes before additional weld material is deposited. Wherever a defect is removed and repair by welding is not required, blend the affected area into the surrounding surface to eliminate sharp notches, crevices, or corners. After a defect is thought to have been removed, and before re-welding, examine the area by suitable methods to ensure that the defect has been eliminated. Repaired welds must meet the inspection requirements for the original welds.

-- End of Section --

SECTION 05 12 00

STRUCTURAL STEEL  
08/18, CHG 2: 05/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

|                      |                                                                                                                 |
|----------------------|-----------------------------------------------------------------------------------------------------------------|
| AISC 207             | (2016; R 2017) Certification Standard for Steel Fabrication and Erection, and Manufacturing of Metal Components |
| AISC 303             | (2016) Code of Standard Practice for Steel Buildings and Bridges                                                |
| AISC 325             | (2017) Steel Construction Manual                                                                                |
| AISC 326             | (2009) Detailing for Steel Construction                                                                         |
| AISC 360             | (2016) Specification for Structural Steel Buildings                                                             |
| AISC DESIGN GUIDE 10 | (1997) Erection Bracing of Low-Rise Structural Steel Buildings                                                  |

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)

|                  |                                                                                              |
|------------------|----------------------------------------------------------------------------------------------|
| ANSI/ASNT CP-189 | (2020) ASNT Standard for Qualification and Certification of Nondestructive Testing Personnel |
|------------------|----------------------------------------------------------------------------------------------|

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|            |                                                             |
|------------|-------------------------------------------------------------|
| ASME B46.1 | (2020) Surface Texture, Surface Roughness, Waviness and Lay |
|------------|-------------------------------------------------------------|

AMERICAN WELDING SOCIETY (AWS)

|                |                                                                             |
|----------------|-----------------------------------------------------------------------------|
| AWS A2.4       | (2012) Standard Symbols for Welding, Brazing and Nondestructive Examination |
| AWS D1.1/D1.1M | (2020; Errata 1 2021) Structural Welding Code - Steel                       |
| AWS QC1        | (2016) Specification for AWS Certification of Welding Inspectors            |

ASTM INTERNATIONAL (ASTM)

|             |                                                                                     |
|-------------|-------------------------------------------------------------------------------------|
| ASTM A6/A6M | (2024b) Standard Specification for General Requirements for Rolled Structural Steel |
|-------------|-------------------------------------------------------------------------------------|

Bars, Plates, Shapes, and Sheet Piling

|                   |                                                                                                                                                                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A36/A36M     | (2019) Standard Specification for Carbon Structural Steel                                                                                                                                                                                               |
| ASTM A563         | (2021; E 2022a) Standard Specification for Carbon and Alloy Steel Nuts                                                                                                                                                                                  |
| ASTM A780/A780M   | (2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings                                                                                                                                                        |
| ASTM A992/A992M   | (2022) Standard Specification for Structural Steel Shapes                                                                                                                                                                                               |
| ASTM F436/F436M   | (2019) Standard Specification for Hardened Steel Washers Inch and Metric Dimensions                                                                                                                                                                     |
| ASTM F959/F959M   | (2017a; R 2023) Standard Specification for Compressible-Washer-Type Direct Tension Indicators for Use with Structural Fasteners, Inch and Metric Series                                                                                                 |
| ASTM F3125/F3125M | (2019) Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength |

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

|                     |                                                                    |
|---------------------|--------------------------------------------------------------------|
| SSPC PA 1           | (2024) Shop, Field, and Maintenance Coating of Metals              |
| SSPC Paint 20       | (2019) Zinc-Rich Primers (Type I, Inorganic, and Type II, Organic) |
| SSPC Paint 29       | (2002; E 2004) Zinc Dust Sacrificial Primer, Performance-Based     |
| SSPC SP 3           | (2018) Power Tool Cleaning                                         |
| SSPC SP 6/NACE No.3 | (2007) Commercial Blast Cleaning                                   |

U.S. DEPARTMENT OF DEFENSE (DOD)

|              |                                                    |
|--------------|----------------------------------------------------|
| UFC 3-301-01 | (2023; with Change 2, 2024) Structural Engineering |
|--------------|----------------------------------------------------|

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|                             |                |
|-----------------------------|----------------|
| 29 CFR Part 1926, Subpart R | Steel Erection |
|-----------------------------|----------------|

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for

Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Erection and Erection Bracing Drawings; G

SD-02 Shop Drawings

Fabrication Drawings Including Details of Connections; G

SD-03 Product Data

Shop Primer

Welding Electrodes and Rods

Direct Tension Indicator Washers

Recycled Content for Structural Steel

SD-05 Design Data

Design Calculations for Steel Connections; G

SD-06 Test Reports

Class B Coating

Bolts, Nuts, and Washers

Weld Inspection Reports

Direct Tension Indicator Washer Inspection Reports

Bolt Testing Reports

SD-07 Certificates

Steel

Bolts, Nuts, and Washers

AISC Structural Steel Fabricator Quality Certification

AISC Structural Steel Erector Quality Certification

Welding Procedures and Qualifications

Welding Electrodes and Rods

Certified Welding Inspector

NDT Technician

## Welding Procedure Specifications (WPS)

### 1.3 AISC QUALITY CERTIFICATION

Work must be fabricated by an AISC Certified Structural Steel Fabricator, in accordance with AISC 207, Category BU. Submit AISC Structural Steel Fabricator quality certification.

Work must be erected by an AISC Structural Steel Certified Erector, in accordance with AISC 207, Category CSE. Submit AISC Structural Steel erector quality certification.

### 1.4 QUALITY ASSURANCE

#### 1.4.1 Preconstruction Submittals

##### 1.4.1.1 Erection and Erection Bracing Drawings

Submit for record purposes. Indicate the sequence of erection, temporary shoring and bracing. The erection drawings must conform to AISC 303. Erection drawings must be reviewed, stamped and sealed by a registered professional engineer.

##### 1.4.2 Fabrication Drawing Requirements

Submit fabrication drawings for approval prior to fabrication. Prepare in accordance with AISC 303, AISC 326 and AISC 325. Fabrication drawings must not be reproductions of contract drawings. Sign and seal fabrication drawings by a registered professional engineer. Include complete information for the fabrication and erection of the structure's components, including the location, type, and size of bolts, welds, member sizes and lengths, connection details, blocks, copes, and cuts. Use AWS A2.4 standard welding symbols. Clearly highlight any deviations from the details shown on the contract drawings highlighted on the fabrication drawings. Explain the reasons for any deviations from the contract drawings.

##### 1.4.3 Delegated Connection Design

Design structural steel connection indicated in the contract documents per AISC 303, Option 3, using the connection loads indicated. Submit design calculations for steel connections signed and sealed by a registered professional engineer.

##### 1.4.4 Certifications

##### 1.4.4.1 Welding Procedures and Qualifications

Prior to welding, submit certification for each welder stating the type of welding and positions qualified for, the code and procedure qualified under, date qualified, and the firm and individual certifying the qualification tests. If the qualification date of the welder or welding operator is more than 6 months old, the welding operator's qualification certificate must be accompanied by a current certificate by the welder attesting to the fact that he has been engaged in welding since the date of certification, with no break in welding service greater than 6 months.

Conform to all requirements specified in AWS D1.1/D1.1M.

## PART 2 PRODUCTS

### 2.1 SYSTEM DESCRIPTION

Provide the structural steel system, including shop primer, complete and ready for use. Provide structural steel systems including design, materials, installation, workmanship, fabrication, assembly, erection, inspection, quality control, and testing in accordance with AISC 303, AISC 360, and UFC 3-301-01 except as modified in this contract.

### 2.2 STEEL

#### 2.2.1 Structural Steel

Wide flange and WT shapes, ASTM A992/A992M. Angles, Channels and Plates, ASTM A36/A36M. Provide structural steel containing a minimum of 80 percent recycled content. Submit data identifying percentage of recycled content for structural steel.

### 2.3 BOLTS, NUTS, AND WASHERS

Submit the certified manufacturer's mill reports which clearly show the applicable ASTM mechanical and chemical requirements together with the actual test results for the supplied fasteners.

#### 2.3.1 High-Strength Bolts

High strength bolts and nuts must be shipped together in the same shipping container. Fasteners indicated to be galvanized shall be tested by the supplier to show that the galvanized nut with the supplied lubricant provided may be rotated from the snug tight condition well in excess of the rotation required for pretensioned installation without stripping. The supplier shall supply nuts that have been lubricated and tested with the supplied bolts.

##### 2.3.1.1 Bolts

ASTM F3125/F3125M, Grade A325M A325, Type 1 Heavy Hex Head Style, plain finish hot dipped zinc coating.

##### 2.3.1.2 Nuts

ASTM A563, Grade and Style as specified in the applicable ASTM bolt standard.

##### 2.3.1.3 Direct Tension Indicator Washers

ASTM F959/F959M. Submit product data for direct tension indicator washers.

##### 2.3.1.4 Washers

ASTM F436/F436M, plain carbon steel.

### 2.4 STRUCTURAL STEEL ACCESSORIES

#### 2.4.1 Welding Electrodes and Rods

AWS D1.1/D1.1M. Submit product data for welding electrodes and rods.

## 2.5 FABRICATION

Fabrication must be in accordance with the applicable provisions of AISC 325. Fabrication and assembly must be done in the shop to the greatest extent possible. Punch, subpunch and ream, or drill bolt holes perpendicular to the surface of the member.

Compression joints depending on contact bearing must have a surface roughness not in excess of 500 micro inch as determined by ASME B46.1, and ends must be square within the tolerances for milled ends specified in ASTM A6/A6M.

Shop splices of members between field splices will be permitted only where indicated on the Contract Drawings. Splices not indicated require the approval of the Contracting Officer.

### 2.5.1 Markings

Prior to erection, identify members by a painted erection mark. Connecting parts assembled in the shop for reaming holes in field connections must be match marked with scratch and notch marks. Do not locate erection markings on areas to be welded. Do not locate match markings in areas that will decrease member strength or cause stress concentrations.

### 2.5.2 Shop Primer

SSPC Paint 20 or SSPC Paint 29, (zinc rich primer). Shop prime structural steel, except as modified herein, in accordance with SSPC PA 1. Do not prime steel surfaces embedded in concrete, galvanized surfaces, or surfaces within 0.5 inch of the toe of the welds prior to welding (except surfaces on which metal decking and shear studs are to be welded). If flash rusting occurs, re-clean the surface prior to application of primer. Apply primer to a minimum dry film thickness of 2.0 mil. Submit shop primer product data.

Prime slip critical surfaces with a Class B coating in accordance with AISC 325. Submit test report for Class B coating.

Prior to assembly, prime surfaces which will be concealed or inaccessible after assembly. Do not apply primer in foggy or rainy weather; when the ambient temperature is below 45 degrees F or over 95 degrees F; or when the primer may be exposed to temperatures below 40 degrees F within 48 hours after application, unless approved otherwise by the Contracting Officer. Repair damaged primed surfaces with an additional coat of primer.

#### 2.5.2.1 Cleaning

SSPC SP 6/NACE No.3, except steel exposed in spaces above ceilings, attic spaces, furred spaces, and chases that will be hidden to view in finished construction may be cleaned to SSPC SP 3 when recommended by the shop primer manufacturer. Maintain steel surfaces free from rust, dirt, oil, grease, and other contaminants through final assembly.

## 2.6 DRAINAGE HOLES

Drill adequate drainage holes to eliminate water traps. Hole diameter must be 1/2 inch and location indicated on the detail drawings. Hole size



and locations must not affect the structural integrity.

### PART 3 EXECUTION

#### 3.1 ERECTION

- a. Erection of structural steel, except as indicated in item b. below, must be in accordance with the applicable provisions of AISC 325, AISC 303 and 29 CFR Part 1926, Subpart R.
- b. For low-rise structural steel buildings ( 60 feet tall or less and a maximum of 2 stories), erect the structure in accordance with AISC DESIGN GUIDE 10.

After final positioning of steel members, provide full bearing under base plates and bearing plates using nonshrink grout. Place nonshrink grout in accordance with the manufacturer's instructions.

##### 3.1.1 STORAGE

Store the material out of contact with the ground in such manner and location as to minimize deterioration.

#### 3.2 CONNECTIONS

Except as modified in this section, design connections indicated in accordance with AISC 360. Build connections into existing work. Do not tighten anchor bolts set in concrete with impact torque wrenches. Holes must not be cut or enlarged by burning. Bolts, nuts, and washers must be clean of dirt and rust, and lubricated immediately prior to installation.

##### 3.2.1 High-Strength Bolts

Provide direct tension indicator washers in all ASTM F3125/F3125M, Grade A325 and Grade A490 bolted connections. Bolts must be installed in connection holes and initially brought to a snug tight fit. After the initial tightening procedure, fully tension bolts, progressing from the most rigid part of a connection to the free edges.

Fastener components shall be protected from dirt and moisture in closed containers at the site of the installation. Fastener components that are not incorporated into the work shall be returned to protected storage at the end of the work shift.

##### 3.2.1.1 Installation of Direct Tension Indicator Washers (DTIW)

Where possible, install the DTIW under the bolt head and tighten the nut. If the DTIW is installed adjacent to the turned element, provide a flat washer between the DTIW and nut when the nut is turned for tightening, and between the DTIW and bolt head when the bolt head is turned for tightening. In addition to the DTIW, provide flat washers under both the bolt head and nut when ASTM F3125/F3125M, Grade A490 bolts are used.

#### 3.3 GAS CUTTING

Use of gas-cutting torch in the field for correcting fabrication errors is not permitted on any major member in the structural framing. Use of a gas cutting torch will be permitted on minor members not under stress only after approval has been obtained from the Contracting Officer.

### 3.4 WELDING

Welding must be in accordance with AWS D1.1/D1.1M. Provide AWS D1.1/D1.1M qualified welders, welding operators, and tackers.

Develop and submit the Welding Procedure Specifications (WPS) for all welding, including welding done using prequalified procedures. Submit for approval all WPS, whether prequalified or qualified by testing.

#### 3.4.1 Removal of Temporary Welds, Run-Off Plates, and Backing Strips

Remove only from finished areas.

### 3.5 SHOP PRIMER REPAIR

Repair shop primer in accordance with the paint manufacturer's recommendation for surfaces damaged by handling, transporting, cutting, welding, or bolting.

#### 3.5.1 Field Priming

Field prime steel exposed to the weather, or located in building areas without HVAC for control of relative humidity. After erection, the field bolt heads and nuts, field welds, and any abrasions in the shop coat must be cleaned and primed with paint of the same quality as that used for the shop coat.

### 3.6 GALVANIZING REPAIR

Repair damage to galvanized coatings using ASTM A780/A780M zinc rich paint for galvanizing damaged by handling, transporting, cutting, welding, or bolting. Do not heat surfaces to which repair paint has been applied.

### 3.7 FIELD QUALITY CONTROL

Perform field tests, and provide labor, equipment, and incidentals required for testing, except that electric power for field tests will be furnished as set forth in Division 1. Notify the Contracting Officer in writing of defective welds, bolts, nuts, and washers within 7 working days of the date of the inspection.

#### 3.7.1 Welds

##### 3.7.1.1 Visual Inspection

AWS D1.1/D1.1M. Furnish the services of AWS-certified welding inspectors for fabrication and erection inspection and testing and verification inspections. A Certified Welding Inspector must perform visual inspection on 100 percent of all welds. Document this inspection in the Visual Weld Inspection Log. Submit certificates indicating that certified welding inspectors meet the requirements of AWS QC1.

Inspect proper preparation, size, gaging location, and acceptability of all welds; identification marking; operation and current characteristics of welding sets in use.

##### 3.7.1.2 Nondestructive Testing

Nondestructive testing must be in accordance with AWS D1.1/D1.1M.

Ultrasonic testing must be performed in accordance with Table 6.2 of AWS D1.1/D1.1M. Test locations must be selected by the Contracting Officer. All personnel performing NDT must be certified in accordance with ANSI/ASNT CP-189 in the method of testing being performed. Submit certificates showing compliance with ANSI/ASNT CP-189 for all NDT technicians. If more than 20 percent of welds made by a welder contain defects identified by testing, then all groove welds made by that welder must be tested by ultrasonic testing, and all fillet welds made by that welder must be inspected by magnetic particle testing (MT) or dye penetrant testing (PT) as approved by the Contracting Officer. When groove welds made by an individual welder are required to be tested, magnetic particle or dye penetrant testing may be used only in areas inaccessible to ultrasonic testing. Retest all repaired areas. Submit weld inspection reports.

Testing frequency: Provide the following types and number of tests:

| Test Type         | Number of Tests                    |
|-------------------|------------------------------------|
|                   |                                    |
| Magnetic Particle | 50 percent of PJP and Fillet Welds |
| Dye Penetrant     | 50 percent of PJP and Fillet Welds |
|                   |                                    |

### 3.7.2 Direct Tension Indicator Washers

#### 3.7.2.1 Direct Tension Indicator Washer Compression

Test direct tension indicator washers in place to verify that they have been compressed sufficiently to provide the 0.015 inch gap, as required by ASTM F959/F959M. Submit direct tension indicator washer inspection reports.

### 3.7.3 High-Strength Bolts

#### 3.7.3.1 Testing Bolt, Nut, and Washer Assemblies

Test a minimum of 3 bolt, nut, and washer assemblies from each mill certificate batch in a tension measuring device at the job site prior to the beginning of bolting start-up. Demonstrate that the bolts and nuts, when used together, can develop tension not less than the provisions specified in AISC 360, depending on bolt size and grade. The bolt tension must be developed by tightening the nut. A representative of the manufacturer or supplier must be present to ensure that the fasteners are properly used, and to demonstrate that the fastener assemblies supplied satisfy the specified requirements. Submit bolt testing reports.

#### 3.7.3.2 Inspection

Inspection procedures must be in accordance with AISC 360. Confirm and report to the Contracting Officer that the materials meet the project specification and that they are properly stored. Confirm that the faying surfaces have been properly prepared before the connections are assembled. Observe the specified job site testing and calibration, and confirm that the procedure to be used provides the required tension. Monitor the work to ensure the testing procedures are routinely followed

on joints that are specified to be fully tensioned.

Inspect calibration of torque wrenches for high-strength bolts.

#### 3.7.3.3 Testing

The Government has the option to perform nondestructive tests on 5 percent of the installed bolts to verify compliance with pre-load bolt tension requirements. Provide the required access for the Government to perform the tests. The nondestructive testing will be done in-place using an ultrasonic measuring device or any other device capable of determining in-place pre-load bolt tension. The test locations must be selected by the Contracting Officer. If more than 10 percent of the bolts tested contain defects identified by testing, then all bolts used from the batch from which the tested bolts were taken, must be tested at the Contractor's expense. Retest new bolts after installation at the Contractor's expense.

-- End of Section --

SECTION 05 40 00

COLD-FORMED METAL FRAMING  
**05/15, CHG 1: 08/18**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 318 (2019; R 2022) Building Code Requirements for Structural Concrete (ACI 318-19) and Commentary (ACI 318R-19)

AMERICAN IRON AND STEEL INSTITUTE (AISI)

AISI S100 (2012) North American Specification for the Design of Cold-Formed Steel Structural Members

AISI S110 (2007; Suppl 1; Reaffirmed 2012) Standard for Seismic Design of Cold-Formed Steel Structural Systems - Special Bolted Moment Frames

AISI S200 (2007) North American Standard for Cold-Formed Steel Framing - General Provision

AISI S201 (2007) North American Standard for Cold-Formed Steel Framing - Product Data

AISI S202 (2011) Code of Standard Practice for Cold-formed Steel Structural Framing

AISI S211 (2007) North American Standard for Cold-Formed Steel Framing - Wall Stud Design

AISI S212 (2007) North American Standard for Cold-Formed Steel Framing - Header Design

AISI S213 (2007; Suppl 1 2009) North American Standard for Cold-Formed Steel Framing - Lateral Design

AISI S214 (2012) North American Standard for Cold-Formed Steel Framing - Truss Design

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding Code - Steel

AWS D1.3/D1.3M (2018) Structural Welding Code - Sheet Steel

ASTM INTERNATIONAL (ASTM)

ASTM A123/A123M (2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A153/A153M (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM A307 (2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength

ASTM A370 (2024) Standard Test Methods and Definitions for Mechanical Testing of Steel Products

ASTM A653/A653M (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM A1003/A1003M (2023) Standard Specification for Steel Sheet, Carbon, Metallic- and Nonmetallic-Coated for Cold-Formed Framing Members

ASTM C955 (2017) Standard Specification for Cold-Formed Steel Structural Framing Members

ASTM C1007 (2020) Standard Specification for Installation of Load Bearing (Transverse and Axial) Steel Studs and Related Accessories

ASTM C1513 (2018) Standard Specification for Steel Tapping Screws for Cold-Formed Steel Framing Connections

ASTM E119 (2024) Standard Test Methods for Fire Tests of Building Construction and Materials

ASTM E329 (2023) Standard Specification for Agencies Engaged in Construction Inspection, Testing, or Special Inspection

ASTM E488/E488M (2022) Standard Test Methods for Strength of Anchors in Concrete Elements

ASTM F1554 (2020) Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength

ASTM F1941 (2010) Standard Specification for  
Electrodeposited Coatings on Threaded  
Fasteners (Unified Inch Screw Threads  
(UN/UNR))

ASTM F2329/F2329M (2015; R 2023) Standard Specification for  
Zinc Coating, Hot-Dip, Requirements for  
Application to Carbon and Alloy Steel  
Bolts, Screws, Washers, Nuts, and Special  
Threaded Fasteners

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2021) International Building Code

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01 (2023; with Change 2, 2024) Structural  
Engineering

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Framing Components; G

SD-03 Product Data

Steel Studs, Joists, Tracks, Bracing, Bridging and Accessories

Recycled Content of Steel Products

SD-05 Design Data

Metal Framing Calculations; G

SD-07 Certificates

Load-Bearing Cold-Formed Metal Framing

Welds

## 1.3 DELIVERY, STORAGE, AND HANDLING

Steel framing and related accessories shall be stored and handled in accordance with the AISI S202, "Code of Standard Practice for Cold-Formed Steel Structural Framing".

## 1.4 LOAD-BEARING COLD-FORMED METAL FRAMING

Include top and bottom tracks, bracing, fastenings, and other accessories necessary for complete installation. Framing members shall have the structural properties indicated. Where physical structural properties are

not indicated, they shall be as necessary to withstand all imposed loads.

Submit mill certificates or test reports from independent testing agency, qualified in accordance with ASTM E329, showing that the steel sheet used in the manufacture of each cold-formed component complies with the minimum yield strengths and uncoated steel thickness specified. Test reports shall be based on the results of three coupon tests in accordance with ASTM A370.

#### 1.5 MAXIMUM DEFLECTION

Deflections of structural members shall not exceed the more restrictive of the limitations of ICC IBC and UFC 3-301-01.

#### 1.6 QUALITY ASSURANCE

- a. Engineering Responsibility: Preparation of Shop Drawings, design calculations, and other structural data by a registered professional engineer.
- b. Testing Agency Qualifications: An independent testing agency, acceptable to authorities having jurisdiction, qualified according to ASTM E329 for testing indicated.
- c. Product Tests: Mill certificates or data from a qualified independent testing agency indicating steel sheet complies with requirements, including base-metal thickness, yield strength, tensile strength, total elongation, chemical requirements, and metallic-coating thickness.
- d. Welding Qualifications: Qualify procedures and personnel according to the following:
  - (1) AWS D1.1/D1.1M, "Structural Welding Code - Steel".
  - (2) AWS D1.3/D1.3M, "Structural Welding Code - Sheet Steel".
- e. Fire-Test-Response Characteristics: Where indicated, provide cold-formed metal framing identical to that of assemblies tested for fire resistance per ASTM E119 by, and displaying a classification label from, a testing and inspecting agency acceptable to authorities having jurisdiction.
- f. AISI Specifications and Standards: Comply with:
  - (1) AISI S100, "North American Specification for the Design of Cold-Formed Steel Structural Members".
  - (2) AISI S110, "Standard for Seismic Design of Cold-Formed Steel Structural Systems - Special Bolted Moment Frames".
  - (3) AISI S200, "North American Standard for Cold-Formed Steel Framing - General Provision".
  - (4) AISI S201, "North American Standard for Cold-Formed Steel Framing - Product Data".
  - (5) AISI S202, "Code of Standard Practice for Cold-Formed Steel Structural Framing".



- (6) AISI S211, "North American Standard for Cold-Formed Steel Framing - Wall Stud Design".
- (7) AISI S212, "North American Standard for Cold-Formed Steel Framing - Header Design".
- (8) AISI S213, "North American Standard for Cold-Formed Steel Framing - Lateral Design".
- (9) AISI S214, "North American Standard for Cold-Formed Steel Framing - Truss Design".

#### 1.6.1 Drawing Requirements

Submit framing components to show sizes, thicknesses, layout, material designations, methods of installation, and accessories including the following:

- a. Cross sections, plans, and/or elevations showing component types and locations for each framing application; including shop coatings and material thicknesses for each framing component.
- b. Connection details showing fastener type, quantity, location, and other information to assure proper installation.
- c. Drawings depicting panel configuration, dimensions, components, locations, and construction sequence if the Contractor elects to install prefabricated/prefinished frames.

Sign and seal fabrication drawings by a registered professional engineer.

#### 1.6.2 Design Data Required

Submit metal framing calculations with design criteria and structural loading to verify sizes, thickness, and spacing of members and connections signed and sealed by a registered professional engineer. Show methods and practices used in installation.

### PART 2 PRODUCTS

#### 2.1 STEEL STUDS, JOISTS, TRACKS, BRACING, BRIDGING AND ACCESSORIES

Framing components shall comply with ASTM C955 and the following.

- a. Provide products with an average recycled content of steel products so postconsumer recycled content plus one half of preconsumer recycled content not less than 25 percent.
- b. Steel Sheet: ASTM A1003/A1003M, Structural Grade, Type H, metallic coated, of grade and coating weight as follows:
  - (1) Grade: As required by structural performance.
  - (2) Coating: G90 (Z275).
- c. Steel Studs: Manufacturer's standard C-shaped steel studs, of web depths indicated, punched, with stiffened flanges, and as follows:

(1) Minimum Base-Metal Thickness: 0.0677 inch.

(2) Flange Width: 1-5/8 inches.

d. Steel Track: Manufacturer's standard U-shaped steel track, of web depths indicated, unpunched, with straight flanges, and as follows:

(1) Minimum Base-Metal Thickness: 0.0677 inch.

(2) Flange Width: 1-1/4 inches.

#### 2.1.1 Studs and Joists of 54 mils (0.054 Inch) and Heavier

Galvanized steel, ASTM A653/A653M and ASTM A1003/A1003M, SS Grade 50, G90.

#### 2.1.2 Studs and Joists of 43 mils (0.043 Inch) and Lighter

Studs and Joists of 43 mils (0.043 Inch) and Lighter, Track, and Accessories (All thicknesses): Galvanized steel, ASTM A653/A653M and ASTM A1003/A1003M, SS, Grade 33 33,000 psi G60.

#### 2.1.3 Sizes, Thickness, Section Modulus, and Other Structural Properties

Size and thickness as required.

### 2.2 MARKINGS

Studs and track shall have product markings stamped on the web of the section. The markings shall be repeated throughout the length of the member at a maximum spacing of 4 feet on center and shall be legible and easily read. The product marking shall include the following:

- a. An ICC number.
- b. Manufacturer's identification.
- c. Minimum delivered uncoated steel thickness.
- d. Protective coating designator.
- e. Minimum yield strength.

### 2.3 CONNECTIONS

#### 2.3.1 Steel-To-Concrete Connections

- a. Anchor Rods: ASTM F1554, Grade 55; galvanized per ASTM A153/A153M.
- b. Post-Installed Concrete Anchors: Adhesive or expansion anchors fabricated from corrosion-resistant materials with allowable load capacities in accordance with ICC-ES AC193 and ACI 318 greater than or equal to the design load as determined by testing per ASTM E488/E488M conducted by a qualified testing agency.
- c. Power-Actuated Fasteners: Fabricated from corrosion-resistant materials with allowable load capacities in accordance with ICC-ES AC 70 greater than or equal to the design load as determined by testing per ASTM E1190 conducted by a qualified testing agency.

### 2.3.2 Steel-To-Steel Connections

- a. Screws: ASTM C1513, corrosion-resistant-coated, self-drilling, self-tapping steel screws of the type and size indicated. Provide low-profile head beneath sheathing and manufacturer's standard elsewhere. Electroplated to a minimum of 5 micron zinc coating per ASTM F1941 or hot-dipped galvanized per ASTM A123/A123M or ASTM A153/A153M.
- b. Bolts: ASTM A307 coated by hot-dip process per ASTM F2329/F2329M or zinc-coated by mechanical-deposition process per ASTM B695, Class 55.
- c. Welding Electrodes: Comply with AWS standards.

### 2.4 PLASTIC GROMMETS

Supply plastic grommets for stud webs as recommended by stud manufacturer, to protect electrical wires and plumbing piping. Prevent metal-to-metal contact between wiring/piping and studs.

### 2.5 SEALER GASKET

Closed-cell neoprene foam, 1/4-inch thick, selected from manufacturer's standard widths to match width of bottom track on concrete slab or foundation.

## PART 3 EXECUTION

### 3.1 TRUSS FABRICATION

- a. Fabricate cold-formed steel trusses and accessories plumb, square, and true to line, and with connections securely fastened, according to referenced AISI's specifications and standards, manufacturer's written instructions, and requirements in this Section.
- b. Truss must be fabricated either on site or off site prior to erection.
- c. Fabricate trusses using jigs or templates.
- d. Splices can only occur at joints.
- e. Cut truss members by sawing or shearing: do not torch cut.
- f. Fasten cold-formed steel truss members by welding, screw fastening, clinch fastening, pneumatic pin fastening, or riveting as standard with fabricator.
- g. Fasten other materials to cold-formed steel trusses by welding, bolting, pneumatic pin fastening, or screw fastening, according to Shop Drawings.
- h. Reinforce, stiffen, and brace trusses to withstand handling, delivery, and erection stresses. Lift fabricated trusses to prevent damage or permanent distortion.

### 3.2 FASTENING

Fasten framing members together by welding or by using self-drilling, self-tapping screws. Electrodes and screw connections shall be as

required and indicated in the design calculations.

#### 3.2.1 Welds

All welding shall be performed in accordance with AWS D1.3/D1.3M, as modified by AISI S100. All welders, welding operations, and welding procedures shall be qualified according to AWS D1.3/D1.3M. Submit certified copies of welder qualifications test records showing qualification in accordance with AWS D1.3/D1.3M. All welds shall be cleaned and coated with rust inhibitive galvanizing paint. Do not field weld materials lighter than 43 mils.

#### 3.2.2 Screws

Screws shall be of the self-drilling self-tapping type, size, and location as required. Screw penetration through joined materials shall not be less than three exposed threads. Minimum spacings and edge distances for screws shall be as specified in AISI S100. Screws covered by sheathing materials shall have low profile heads.

#### 3.2.3 Anchors

Anchors shall be of the type, size, and location as required.

#### 3.2.4 Powder-Actuated Fasteners

Powder-actuated fasteners shall be of the type, size, and location as required.

### 3.3 INSTALLATION

Install cold-formed framing in accordance with ASTM C1007 and AISI S200.

Install cold-formed steel framing according to AISI S202 and to manufacturer's written instructions unless more stringent requirements are indicated.

#### 3.3.1 Tracks

Provide accurately aligned runners at top and bottom of studs. Install sealer gasket under bottom of track on concrete slab or foundation. Anchor tracks as indicated in design calculations. Butt weld joints in tracks or splice with stud inserts. Fasteners shall be at least 3 inches from the edge of concrete slabs.

#### 3.3.2 Studs

Cut studs square and set with firm bearing against webs of top and bottom tracks. Position studs vertically in tracks and space as indicated in design. Do not splice studs. Provide at least two studs at jambs of doors and other openings 2 feet wide or larger. Provide jack studs over openings, as necessary, to maintain indicated stud spacing. Provide tripled studs at corners, positioned to receive interior and exterior finishes. Fasten studs to top and bottom tracks by welding or screwing both flanges to the tracks. Framed wall openings shall include headers and supporting components as shown on the drawings. Headers shall be installed in all openings that are larger than the stud spacing in a wall. In curtain wall construction, provide for vertical movement where studs connect to the structural frame. Provide horizontal bracing in

accordance with the design calculations and AISI S100. Bracing shall be not less than the following:

| <u>LOAD</u>    | <u>HEIGHT</u> | <u>BRACING</u>          |
|----------------|---------------|-------------------------|
| Wind load only | Up to 10 feet | One row at mid-height   |
|                | Over 10 feet  | Rows 5'-0" o.c. maximum |
| Axial load     | Up to 10 feet | Two rows at 1/3 points  |
|                | Over 10 feet  | Rows 3'-4" o.c. maximum |

### 3.3.3 Joists and Trusses

- a. Provide a stud directly under each joist or truss. The maximum spacing of studs as indicated shall be maintained.
- b. Install, bridge, and brace cold-formed steel trusses according to AISI S200, AISI S214, AISI's "Code of Standard Practice for Cold-Formed Steel Structural Framing," and manufacturer's written instructions unless more stringent requirements are indicated.
- c. Install temporary bracing and supports. Maintain braces and supports in place, undisturbed, until entire integrated supporting structure has been completed and permanent connections to framing are secured.
- d. Do not alter, cut, or remove framing members or connections of trusses.

### 3.3.4 Erection Tolerances

- a. Framing members which will be covered by finishes such as wallboard, plaster, or ceramic tile set in a mortar setting bed, shall be within the following limits:
  - (1) Layout of walls and partitions: 1/4 inch from intended position;
  - (2) Plates and runners: 1/4 inch in 8 feet from a straight line;
  - (3) Studs: 1/4 inch in 8 feet out of plumb, not cumulative; and
  - (4) Face of framing members: 1/4 inch in 8 feet from a true plane.
- b. Framing members which will be covered by ceramic tile set in dry-set mortar, latex-portland cement mortar, or organic adhesive shall be within the following limits:
  - (1) Layout of walls and partitions: 1/4 inch from intended position;
  - (2) Plates and runners: 1/8 inch in 8 feet from a straight line;

(3) Studs: 1/8 inch in 8 feet out of plumb, not cumulative; and

(4) Face of framing members: 1/8 inch in 8 feet from a true plane.

-- End of Section --

SECTION 05 50 13

MISCELLANEOUS METAL FABRICATIONS  
**08/24**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ALUMINUM ASSOCIATION (AA)

AA DAF45 (2003; Reaffirmed 2009) Designation System  
for Aluminum Finishes

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

ANSI/AISC 303 (2022) Code of Standard Practice for  
Structural Steel Buildings and Bridges

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.2.1 (2012; R 2021) Square, Hex, Heavy Hex, and  
Askew Head Bolts and Hex, Heavy Hex, Hex  
Flange, Lobed Head, and Lag Screws (Inch  
Series)

ASME B18.2.2 (2022) Nuts for General Applications:  
Machine Screw Nuts, and Hex, Square, Hex  
Flange, and Coupling Nuts (Inch Series)

ASME B18.6.2 (2020) Square Head Set Screws and Slotted  
Headless Set Screws (Inch Series)

ASME B18.6.3 (2024) Machine Screws, Tapping Screws, and  
Metallic Drive Screws (Inch Series)

ASME B18.21.1 (2009; R 2016) Washers: Helical  
Spring-Lock, Tooth Lock, and Plain Washers  
(Inch Series)

ASME B18.21.2M (1999; R 2014) Lock Washers (Metric Series)

ASME B18.22M (1981; R 2017) Metric Plain Washers

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.3 (2020) Safety Requirements for  
Powder-Actuated Fastening Systems American  
National Standard for Construction and  
Demolition Operations

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding

Code - Steel

AWS D1.2/D1.2M

(2014; Errata 1 2014; Errata 2 2020)  
Structural Welding Code - Aluminum

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M

(2019) Standard Specification for Carbon  
Structural Steel

ASTM A47/A47M

(1999; R 2022; E 2022) Standard  
Specification for Ferritic Malleable Iron  
Castings

ASTM A48/A48M

(2022) Standard Specification for Gray  
Iron Castings

ASTM A53/A53M

(2024) Standard Specification for Pipe,  
Steel, Black and Hot-Dipped, Zinc-Coated,  
Welded and Seamless

ASTM A123/A123M

(2024) Standard Specification for Zinc  
(Hot-Dip Galvanized) Coatings on Iron and  
Steel Products

ASTM A153/A153M

(2023) Standard Specification for Zinc  
Coating (Hot-Dip) on Iron and Steel  
Hardware

ASTM A276/A276M

(2024) Standard Specification for  
Stainless Steel Bars and Shapes

ASTM A283/A283M

(2024) Standard Specification for Low and  
Intermediate Tensile Strength Carbon Steel  
Plates

ASTM A307

(2021) Standard Specification for Carbon  
Steel Bolts, Studs, and Threaded Rod 60  
000 PSI Tensile Strength

ASTM A500/A500M

(2023) Standard Specification for  
Cold-Formed Welded and Seamless Carbon  
Steel Structural Tubing in Rounds and  
Shapes

ASTM A653/A653M

(2023) Standard Specification for Steel  
Sheet, Zinc-Coated (Galvanized) or  
Zinc-Iron Alloy-Coated (Galvannealed) by  
the Hot-Dip Process

ASTM A780/A780M

(2020) Standard Practice for Repair of  
Damaged and Uncoated Areas of Hot-Dip  
Galvanized Coatings

ASTM A786/A786M

(2015; R 2021) Standard Specification for  
Hot-Rolled Carbon, Low-Alloy,  
High-Strength Low-Alloy, and Alloy Steel  
Floor Plates



|                   |                                                                                                                       |
|-------------------|-----------------------------------------------------------------------------------------------------------------------|
| ASTM A924/A924M   | (2022a) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process       |
| ASTM B26/B26M     | (2018; E 2018) Standard Specification for Aluminum-Alloy Sand Castings                                                |
| ASTM B108/B108M   | (2019) Standard Specification for Aluminum-Alloy Permanent Mold Castings                                              |
| ASTM B209/B209M   | (2021a) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate                                        |
| ASTM B221         | (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes          |
| ASTM B221M        | (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric) |
| ASTM C1513        | (2018) Standard Specification for Steel Tapping Screws for Cold-Formed Steel Framing Connections                      |
| ASTM D1187/D1187M | (1997; R 2018) Standard Specification for Asphalt-Base Emulsions for Use as Protective Coatings for Metal             |
| ASTM F1554        | (2020) Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength                             |

MASTER PAINTERS INSTITUTE (MPI)

|        |                                                |
|--------|------------------------------------------------|
| MPI 79 | (2016) Primer, Alkyd, Anti-Corrosive for Metal |
|--------|------------------------------------------------|

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

|               |                                            |
|---------------|--------------------------------------------|
| NAAMM MBG 531 | (2017) Metal Bar Grating Manual            |
| NAAMM MBG 532 | (2019) Heavy Duty Metal Bar Grating Manual |

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

|                     |                                                       |
|---------------------|-------------------------------------------------------|
| SSPC PA 1           | (2024) Shop, Field, and Maintenance Coating of Metals |
| SSPC SP 3           | (2018) Power Tool Cleaning                            |
| SSPC SP 6/NACE No.3 | (2007) Commercial Blast Cleaning                      |

U.S. ARMY CORPS OF ENGINEERS (USACE)

|            |                                                                    |
|------------|--------------------------------------------------------------------|
| EM 385-1-1 | (2024) Safety -- Safety and Occupational Health (SOH) Requirements |
|------------|--------------------------------------------------------------------|

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

Structural Steel Door Frames, Fabrication Drawings; G  
Cover Plates and Frames, Installation Drawings; G  
Expansion Joint Covers, Installation Drawings; G  
Sill Angle, Installation Drawings; G  
Bollards; G  
Embedded Angles and Plates, Installation Drawings; G  
Roof Hatches, Installation Drawings; G

### SD-03 Product Data

Cover Plates and Frames; G  
Expansion Joint Covers; G  
Floor Gratings; G  
Structural Steel Door Frames; G  
Roof Hatches; G  
Each Downspout Terminations Type; G

### SD-04 Samples

Expansion Joint Covers

### SD-07 Certificates

Certified Mill Test Reports for Chemistry and Mechanical Properties; G

## 1.3 QUALIFICATION OF WELDERS

Qualify procedures and personnel according to the following:

- a. AWS D1.1/D1.1M Structural Welding Code - Steel.
- b. AWS D1.2/D1.2M Structural Welding Code - Aluminum.

#### 1.4 DELIVERY, STORAGE, AND PROTECTION

Protect from corrosion, deformation, and other types of damage. Store items in an enclosed area free from contact with soil and weather. Remove and replace damaged items with new items.

#### 1.5 MISCELLANEOUS REQUIREMENTS

##### 1.5.1 Fabrication Drawings

Submit fabrication drawings showing layout(s), connections to structural system, and anchoring details as specified in ANSI/AISC 303.

##### 1.5.2 Installation Drawings

Submit templates, erection, and installation drawings indicating thickness, type, grade, class of metal, and dimensions. Show construction details, reinforcement, anchorage, and installation in relation to the building construction.

### PART 2 PRODUCTS

#### 2.1 MATERIALS

Provide exposed fastenings of compatible materials (avoid contact of dissimilar metals). Coordinate color and finish with the material to which fastenings are applied. Submit the manufacturer's certified mill reports which clearly show the applicable ASTM mechanical and chemical requirements together with the actual test results for the supplied materials.

##### 2.1.1 Structural Carbon Steel

Provide in accordance with ASTM A36/A36M.

##### 2.1.2 Stainless Steel Bars and Shapes

ASTM A276/A276M, Type 316L.

##### 2.1.3 Structural Tubing

Provide in accordance with ASTM A500/A500M.

##### 2.1.4 Steel Pipe

Provide in accordance with ASTM A53/A53M, Type E or S, Grade B.

##### 2.1.5 Fittings for Steel Pipe

Provide standard malleable iron fittings in accordance with ASTM A47/A47M.

##### 2.1.6 Gratings

- a. Provide gray cast iron in accordance with ASTM A48/A48M, Class 40.
- b. Provide metal plank grating, non-slip requirement, steel in accordance with ASTM A653/A653M, Z275 G90.
- c. Provide metal bar type grating in accordance with NAAMM MBG 531 and

NAAMM MBG 532.

#### 2.1.7 Floor Plates

- a. Provide patterned floor plate in accordance with ASTM A786/A786M. Provide steel plate not less than 14 gauge.
- b. Provide abrasive-surface floor plate: Steel plate with abrasive material metallurgically bonded to steel.

#### 2.1.8 Anchor Bolts

Provide in accordance with ASTM F1554. Where exposed, provide anchor bolts of the same material, color, and finish as the metal to which they are applied.

##### 2.1.8.1 Lag Screws and Bolts

Provide in accordance with ASME B18.2.1, type and grade best suited for the purpose.

##### 2.1.8.2 Toggle Bolts

Provide in accordance with ASME B18.2.1.

##### 2.1.8.3 Bolts, Nuts, Studs and Rivets

Provide in accordance with ASME B18.2.2 or ASTM A307.

##### 2.1.8.4 Powder Actuated Fasteners

Follow safety provisions in accordance with ASSP A10.3.

##### 2.1.8.5 Screws

Provide in accordance with ASME B18.2.1, ASME B18.6.2, ASME B18.6.3 and ASTM C1513.

##### 2.1.8.6 Washers

Provide plain washers in accordance with ASME B18.22M, ASME B18.21.1. Provide beveled washers for American Standard beams and channels, square or rectangular, tapered in thickness, and smooth. Provide lock washers in accordance with ASME B18.21.2M, ASME B18.21.1.

#### 2.1.9 Aluminum Alloy Products

Provide in accordance with ASTM B209/B209M for sheet plate, ASTM B221M, ASTM B221M, ASTM B221 for extrusions, and ASTM B26/B26M or ASTM B108/B108M for castings. Provide aluminum extrusions at least 1/8 inch thick and aluminum plate or sheet at least 0.050 inch thick.

## 2.2 FABRICATION FINISHES

### 2.2.1 Galvanizing

Hot-dip galvanize items specified to be zinc-coated, after fabrication where practicable. Provide galvanizing in accordance with ASTM A123/A123M, ASTM A153/A153M, ASTM A653/A653M, or ASTM A924/A924M, Z275 G90.

### 2.2.2 Galvanize

Anchor bolts, grating fasteners, washers, and parts or devices necessary for proper installation, unless indicated otherwise.

### 2.2.3 Repair of Zinc-Coated Surfaces

Repair damaged surfaces with galvanizing repair method and paint in accordance with ASTM A780/A780M or by application of stick or thick paste material specifically designed for repair of galvanizing, as approved by Contracting Officer. Clean areas to be repaired and remove slag from welds. Heat, with a torch, surfaces to which stick or paste material will be applied. Heat to a temperature sufficient to melt the metals in the stick or paste. Spread molten material uniformly over surfaces to be coated and wipe off excess material.

### 2.2.4 Shop Cleaning and Painting

#### 2.2.4.1 Surface Preparation

Blast clean surfaces in accordance with SSPC SP 6/NACE No.3. Surfaces that will be exposed in spaces above ceiling or in attic spaces, crawl spaces, furred spaces, and chases may be cleaned in accordance with SSPC SP 3 in lieu of being blast cleaned. Wash cleaned surfaces which become contaminated with rust, dirt, oil, grease, or other contaminants with solvents until thoroughly clean. Steel to be embedded in concrete must be free of dirt and grease prior to embed. Do not paint or galvanize bearing surfaces, including contact surfaces within slip critical joints. Shop coat these surfaces with rust prevention.

#### 2.2.4.2 Pretreatment, Priming and Painting

Apply pre-treatment, primer, and paint in accordance with manufacturer's printed instructions. On surfaces concealed in the finished construction or not accessible for finish painting, apply an additional prime coat to a minimum dry film thickness of 1.0 mil. Tint additional prime coat with a small amount of tinting pigment.

### 2.2.5 Nonferrous Metal Surfaces

Protect by plating, anodic, or organic coatings.

### 2.2.6 Aluminum Surfaces

#### 2.2.6.1 Surface Condition

Before finishes are applied, remove roll marks, scratches, rolled-in scratches, kinks, stains, pits, orange peel, die marks, structural streaks, and other defects which will affect uniform appearance of finished surfaces.

#### 2.2.6.2 Aluminum Finishes

Unexposed sheet, plate and extrusions may have mill finish as fabricated. Sandblast castings' finish, medium, AA DAF45. Unless otherwise specified, provide all other aluminum items with a standard mill finish. Provide a coating thickness not less than that specified for protective and decorative type finishes for items used in interior locations or

architectural Class I type finish for items used in exterior locations. Provide in accordance with AA DAF45. Provide a polished satin finish on items to be anodized.

### 2.3 COVER PLATES AND FRAMES

Where indicated, fabricate cover plates of 1/4 inch thick rolled steel weighing not more than 100 pounds per plate with a slip-resistant, carbon steel in accordance with ASTM A283/A283M. Provide aluminum oxide or silicon carbide on wearing surfaces. Provide galvanized plate. Reinforce to sustain a live load of 250 pounds per square foot. Provide structural steel shapes and plates for frames, with bent steel bars or headed anchors welded to frame for anchoring to concrete. Miter and weld all corners. Butt joint straight runs. Allow for expansion on straight runs over 15 feet. Provide flush drop handles for removal where indicated; form from 1/4 inch round stock. Remove sharp edges and burrs from cover plates and exposed edges of frames. Weld all connections and grind top surface smooth. Weld bar stops every 6 inches. Provide 1/8 inch clearance at edges and between cover plate and frames after fabrication.

### 2.4 FLOOR GRATINGS

Design steel grating in accordance with NAAMM MBG 531 and NAAMM MBG 532 for bar type gratings, or in accordance with manufacturer's charts for plank grating. Galvanize steel floor gratings.

- a. Design floor gratings to support a stress live load of 250 pounds per square foot for the spans indicated, with maximum deflection of  $L/240$ .
- b. In accordance with NAAMM MBG 531, NAAMM MBG 532, band edges of grating with bars of the same size as the bearing bars. Weld banding in accordance with the manufacturer's standard for trim. Design tops of bearing bars, cross or intermediate bars to be in the same plane and to match grating finish.
- b. NAAMM MBG 531, band ends of gratings with bars of the same or greater thickness than the metal used for grating. Weld banding bars to bearing bars or channels at least every fourth bar or channel and in every corner. Tack weld intervening bars or channels. Band diagonal or round cuts by welding bars of the same or greater thickness as the grating and in accordance with the manufacturer's standard for trim.
- c. Attach gratings to structural members with welded-on anchors.
- d. Provide slip resistant surface finishes.

### 2.5 BOLLARDS

Provide 6 inch galvanized extra strong weight steel pipe in accordance with ASTM A53/A53M. Anchor posts in concrete as indicated and fill solidly with concrete.

### 2.6 DOWNSPOUT TERMINATIONS

Provide 4 by 4 inch aluminum downspout tile adapter with manufacturer's standard powder coated finish. Weld all seams for units.

Provide 5 by 4 inch cast iron downspout boot with cleanout access and manufacturer's standard cast iron strap.

## 2.7 MISCELLANEOUS PLATES AND SHAPES

Submit installation drawings for and provide items that do not form a part of the structural steel framework, such as lintels, sill angles, and miscellaneous mountings and frames. Provide lintels fabricated from structural steel shapes over openings in masonry walls and partitions as indicated and as required to support wall loads over openings. Provide with connections and fasteners and/or welds. Construct to have at least 8 in bearing on masonry at each end.

Provide angles and plates in accordance with ASTM A36/A36M, for embedment as indicated. Galvanize embedded items exposed to the elements in accordance with ASTM A123/A123M.

## 2.8 STRUCTURAL STEEL DOOR FRAMES

- a. Provide frames as indicated. Unless otherwise indicated, construct frames of structural shapes, or shape and plate composite, to form a full depth channel shape with at least 1-1/2 inch outstanding legs. For single swing doors, provide continuous 5/8 by 1-1/2 inch bar stock stops at head and jambs.
- b. Provide support where track, guides, hoods, hangers, operators, and other accessories are required.
- c. Provide jamb anchors near top, bottom, and at not more than 24 inch intervals. Provide the bottom of each jamb member with a clip angle welded in place with two 1/2 inch diameter floor bolts for adjustment.
- d. Provide spreaders between bottoms of floor jamb members. When floor construction permits, spreaders may be left in place and concealed in the floor.
- e. Provide frames of rolled shapes as indicated. Miter and weld heads to jambs, or provide riveted clip angle connections concealed in the finished work. Provide frames for swinging doors with 5/8 by 1-1/2 inch solid bar stops secured to the frame by welding or by 1/4 inch diameter countersunk machine screws spaced not more than 12 inches on centers. Stiffen head openings greater than 3 feet as necessary to limit deflection to not more than 1/16 inch. Secure frames to masonry with zinc-coated metal anchors spaced not more than 30 inches on centers. Where necessary to engage the threads of machine screws for fastening hardware, back frames on inside faces with steel plates of suitable thickness. Tap frames and reinforcing plates as necessary for the installation of hardware and other work. Countersink rivets and screw heads where they will be exposed in the finished work. Grind welds smooth.

## 2.9 ROOF HATCHES (SCUTTLES)

Provide aluminum sheets not less than 14 gauge with 3 inch beaded flange, welded and ground at corners. Provide a minimum clear opening of 30 by 36 inches. Insulate cover and curb with 1 inch thick rigid fiberboard insulation, covered and protected by aluminum sheet of not less than 26 gauge. Provide with 12 inches high curb, formed with 3 inch mounting flanges with holes for securing to the roof deck.

## PART 3 EXECUTION

### 3.1 GENERAL INSTALLATION REQUIREMENTS

Install items at locations indicated in accordance with manufacturer's instructions. Verify all field dimensions prior to fabrication. Include materials and parts necessary to complete each assembly, whether indicated or not. Miss-alignment and miss-sizing of holes for fasteners is cause for rejection. Conceal fastenings where practicable. Joints exposed to weather must be watertight.

### 3.2 WORKMANSHIP

Provide miscellaneous metalwork that is true and accurate in shape, size, and profile. Make angles and lines continuous and straight. Make curves consistent, smooth and unfaceted. Provide continuous welding along the entire area of contact except where tack welding is permitted. Do not tack weld exposed connections. Unless otherwise indicated and approved, provide a smooth finish on exposed surfaces. Provide countersunk rivets where exposed. Provide coped and mitered corner joints aligned flush and without gaps. Do not weld, cut or abrade surfaces of exterior units that have been hot-dip galvanized after fabrication and are bolted or screwed field connections.

### 3.3 ANCHORAGE, FASTENINGS, AND CONNECTIONS

Provide anchorage as necessary, whether indicated or not, for fastening miscellaneous metal items securely in place. Include slotted inserts, expansion shields, powder-driven fasteners, toggle bolts (when approved for concrete), through bolts for masonry, headed shear studs, machine and carriage bolts for steel, through bolts, lag bolts, and screws for wood. Do not use wood plugs. Provide non-ferrous attachments for non-ferrous metal. Provide exposed fastenings of compatible materials (avoid contact of dissimilar metals), that generally match in color and finish the surfaces to which they are applied. Conceal fastenings where practicable. Provide all fasteners flush with the surfaces they fasten, unless indicated otherwise. Test a minimum of two bolt, nut, and washer assemblies from each certified mill batch in a tension measuring device at the job site prior to the beginning of bolting start-up.

### 3.4 BUILT-IN WORK

Where necessary and not otherwise indicated, form built-in metal work for anchorage with concrete or masonry. Provide built-in metal work in ample time for securing in place as the work progresses.

### 3.5 WELDING

Perform welding, welding inspection, and corrective welding in accordance with AWS D1.1/D1.1M. Use continuous welds on all exposed connections. Grind visible welds smooth in the finished installation. Provide welded headed shear studs in accordance with AWS D1.1/D1.1M, Clause 7, except as otherwise specified. Provide in accordance with the safety requirements of EM 385-1-1.

### 3.6 DISSIMILAR METALS

Where dissimilar metals are in contact, protect surfaces with a coating in accordance with MPI 79 to prevent galvanic or corrosive action. Where



aluminum is in contact with concrete, plaster, mortar, masonry, wood, or absorptive materials subject to wetting, protect in accordance with ASTM D1187/D1187M, asphalt-base emulsion. Clean surfaces with metal shavings from installation at the end of each work day.

### 3.7 PREPARATION

#### 3.7.1 Material Coatings and Surfaces

Remove rust preventive coating just prior to field erection, using a remover approved by the metal manufacturer. Surfaces, when assembled, must be free of rust, grease, dirt and other foreign matter.

#### 3.7.2 Environmental Conditions

Do not clean or paint surfaces when damp or exposed to foggy or rainy weather, when metallic surface temperature is less than minus 5 degrees F above the dew point of the surrounding air, or when surface temperature is below 45 degrees F or over 95 degrees F, unless approved by the Contracting Officer. Metal surfaces to be painted must be dry for a minimum of 48 hours prior to the application of primer or paint.

### 3.8 EXPANSION JOINT COVERS

Provide in accordance with manufacturer's written instructions. Verify installation allows specified movement prior to completion of work

### 3.9 COVER PLATES AND FRAMES

Provide tops of cover plates and frames flush with finished surface. Test for trip hazards and adjust for any encountered lippage.

### 3.10 ROOF HATCH (SCUTTLES)

Construction and accessories as follows:

- a. Provide insulated cover and curb with mounting flanges for securing to roof deck. Provide curbs with integral metal cap flashing of the same gauge and metal as the curb, fully welded and ground at corners for weather tightness.
- b. Provide hatches completely assembled, with pintle hinges, compression spring operators enclosed in telescopic tubes, positive snap latches with turn handles on inside and outside, and neoprene draft seals. Provide fasteners for padlocking from the inside. Provide covers with automatic hold-open arms complete with grip handle to permit one hand release. Cover action must be smooth through its entire range of motion with an operating pressure of approximately 30 pounds.

### 3.11 DOOR GUARD FRAME

Mount door guard frames over glazed openings using 1/4 inch lag bolts on the interiors of wood doors or tamperproof through bolts on the interiors of metal doors.

### 3.12 INSTALLATION OF BOLLARDS

Set bollards/pipe guards vertically in concrete piers. Fill hollow cores with concrete having a compressive strength of 3000 psi.

### 3.13 INSTALLATION OF DOWNSPOUT TERMINATIONS

Secure downspouts terminations to downspouts and substrate per manufacturer's instructions.

### 3.14 STRUCTURAL STEEL DOOR FRAMES

Secure door frames to the floor slab by means of angle clips and expansion bolts. Provide any necessary reinforcements and drill and tap frames as required for hardware. Clean metal shavings from finished surfaces at the end of each work day.

For freight elevator hoistway entrances, include a non-skid metal sill installed in accordance with the elevator manufacturer's written installation instructions.

### 3.15 INSTALLATION MISCELLANEOUS PLATES AND SHAPES

Provide lintels fabricated from structural steel shapes over openings in masonry walls and partitions as indicated and as required to support wall loads over openings. Provide with connections and fasteners and/or welds. Construct to have at least 8 inches bearing on masonry at each end.

### 3.16 REPAIRS

#### a. Touchup Painting:

- (1) Immediately after erection, clean field welds, bolted connections, and abraded areas. Paint uncoated and abraded areas with same material as used for shop painting to comply with SSPC PA 1 for touching up shop-painted surfaces.

#### b. Galvanized Surfaces:

- (1) Clean field welds, bolted connections, and abraded areas and repair galvanizing to comply with ASTM A780/A780M.

-- End of Section --

SECTION 05 51 00

METAL STAIRS  
**08/23**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS  
(AASHTO)

AASHTO M 314 (1990; R 2013) Standard Specification for  
Steel Anchor Bolts

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 325 (2017) Steel Construction Manual

AISC 326 (2009) Detailing for Steel Construction

AISC 360 (2016) Specification for Structural Steel  
Buildings

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-22 (2022; Supp 1 2023; Supp 2 2023) Minimum  
Design Loads and Associated Criteria for  
Buildings and Other Structures

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.2.1 (2012; R 2021) Square, Hex, Heavy Hex, and  
Askew Head Bolts and Hex, Heavy Hex, Hex  
Flange, Lobed Head, and Lag Screws (Inch  
Series)

ASME B18.6.1 (2016) Wood Screws (Inch Series)

ASME B18.6.3 (2024) Machine Screws, Tapping Screws, and  
Metallic Drive Screws (Inch Series)

ASME B18.21.1 (2009; R 2016) Washers: Helical  
Spring-Lock, Tooth Lock, and Plain Washers  
(Inch Series)

AMERICAN WELDING SOCIETY (AWS)

AWS A2.4 (2012) Standard Symbols for Welding,  
Brazing and Nondestructive Examination

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding  
Code - Steel

ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A6/A6M     | (2024b) Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes, and Sheet Piling                            |
| ASTM A29/A29M   | (2023) Standard Specification for General Requirements for Steel Bars, Carbon and Alloy, Hot-Wrought                                                  |
| ASTM A36/A36M   | (2019) Standard Specification for Carbon Structural Steel                                                                                             |
| ASTM A47/A47M   | (1999; R 2022; E 2022) Standard Specification for Ferritic Malleable Iron Castings                                                                    |
| ASTM A48/A48M   | (2022) Standard Specification for Gray Iron Castings                                                                                                  |
| ASTM A53/A53M   | (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless                                                 |
| ASTM A108       | (2024) Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished                                                                          |
| ASTM A123/A123M | (2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products                                                       |
| ASTM A153/A153M | (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware                                                                   |
| ASTM A283/A283M | (2024) Standard Specification for Low and Intermediate Tensile Strength Carbon Steel Plates                                                           |
| ASTM A307       | (2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength                                             |
| ASTM A449       | (2014; R 2020) Standard Specification for Hex Cap Screws, Bolts, and Studs, Steel, Heat Treated, 120/105/90 ksi Minimum Tensile Strength, General Use |
| ASTM A500/A500M | (2023) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes                                 |
| ASTM A512       | (2006; R 2012) Standard Specification for Cold-Drawn Buttweld Carbon Steel Mechanical Tubing                                                          |
| ASTM A568/A568M | (2019a) Standard Specification for Steel, Sheet, Carbon, Structural, and                                                                              |

High-Strength, Low-Alloy, Hot-Rolled and  
Cold-Rolled, General Requirements for

|                   |                                                                                                                                                                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A575         | (2020) Standard Specification for Steel Bars, Carbon, Merchant Quality, M-Grades                                                                                                                                                                        |
| ASTM A653/A653M   | (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process                                                                                                                 |
| ASTM A924/A924M   | (2022a) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process                                                                                                                                         |
| ASTM A1008/A1008M | (2024) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable                                                     |
| ASTM A1011/A1011M | (2023) Standard Specification for Steel Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength                                                            |
| ASTM B209         | (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate                                                                                                                                                                           |
| ASTM C514         | (2004; R 2020) Standard Specification for Nails for the Application of Gypsum Board                                                                                                                                                                     |
| ASTM C636/C636M   | (2013) Standard Practice for Installation of Metal Ceiling Suspension Systems for Acoustical Tile and Lay-In Panels                                                                                                                                     |
| ASTM E303         | (2022) Standard Test Method for Measuring Surface Frictional Properties Using the British Pendulum Tester                                                                                                                                               |
| ASTM E488/E488M   | (2022) Standard Test Methods for Strength of Anchors in Concrete Elements                                                                                                                                                                               |
| ASTM F3125/F3125M | (2019) Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength |

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

|               |                                 |
|---------------|---------------------------------|
| NAAMM MBG 531 | (2017) Metal Bar Grating Manual |
|---------------|---------------------------------|

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S"

classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

- Iron and Steel Hardware; G
- Steel Shapes, Plates, Bars, and Strips; G
- Metal Stair System; G
- Fabrication Drawings; G

SD-03 Product Data

- Masonry Anchorage Devices; G
- Protective Coating; G
- Steel Pan Stairs; G
- Steel Stairs; G
- Steel Stairs, Circular; G

SD-05 Design Data

- Design Calculations For Steel Connections; G

SD-06 Test Reports

- Mill Test Reports; G

SD-07 Certificates

- Welding Procedures; G
- Welder Qualification; G

SD-08 Manufacturer's Instructions

- Protective Coating; G
- Masonry Anchorage Devices; G

1.3 QUALITY CONTROL

1.3.1 Qualifications for Welding Work

Submit welding procedures in accordance with AWS D1.1/D1.1M. Make test specimens in the presence of the Contracting Officer, and have the specimens tested by an approved testing laboratory at the Contractor's expense.

Certify welder qualification by tests in accordance with AWS D1.1/D1.1M, or under an equivalent approved qualification test. In addition, perform

tests on test pieces in positions and with clearances equivalent to those actually encountered. If a test weld fails to meet requirements, ensure that two test welds are retested immediately and that each test weld is made and passes. Failure in the immediate retest requires that the welder be retested after further practice or training and a complete set of test welds be made.

#### 1.4 QUALITY ASSURANCE

##### 1.4.1 Fabrication Drawing Requirements

Submit fabrication drawings for approval prior to fabrication. Prepare in accordance with AISC 325, AISC 326, ASCE 7-22, and State of Maine Building Code. Fabrication drawings will not be reproductions of contract drawings.

Sign and seal fabrication drawings by a registered professional engineer.

Include complete information for the fabrication and erection of the structure's components, including the location, type, and size of bolts, welds, member sizes and lengths, connection details, blocks, copes, and cuts. Use AWS A2.4 standard welding symbols. Clearly highlight any deviations from the details shown on the contract drawings highlighted on the fabrication drawings. Explain the reasons for any deviations from the contract drawings.

##### 1.4.2 Delegated Connection Design

Design structural steel connection indicated in the contract documents per AISC 360. Submit design calculations for steel connections signed and sealed by a registered professional engineer.

#### PART 2 PRODUCTS

##### 2.1 SYSTEM DESCRIPTION

Submit complete and detailed fabrication drawings for all iron and steel hardware, and for all steel shapes, plates, bars, and strips used in accordance with the design specifications referenced in this section.

##### 2.2 FABRICATION

Preassemble items in the shop to the greatest extent possible. Disassemble units only to the extent necessary for shipping and handling. Clearly mark units for reassembly and coordinated installation.

For the fabrication of work exposed to view, use only materials that are smooth and free of surface blemishes, including pitting, seam marks, roller marks, rolled trade names, and roughness. Remove blemishes by grinding, or by welding and grinding, before cleaning and treating surfaces and applying surface finishes, including zinc coatings.

##### 2.2.1 General Fabrication

Prepare and submit metal stair system shop drawings with detailed plans and elevations at scales not less than 1 inch to 1 foot and with details of sections and connections at scales not less than 3 inches to 1 foot. Also detail the placement drawings, diagrams, and templates for installation of anchorages, including concrete inserts, anchor bolts, and miscellaneous metal items having integral anchorage devices.

Use materials of size and thicknesses indicated or, if not indicated, of

the size and thickness necessary to produce a finished product that is strong enough and durable enough for its intended use. Work the materials to the dimensions indicated on approved detail drawings, using proven methods of fabrication and support. Use the type of materials indicated or specified for the various components of work.

Form exposed work true to line and level, with accurate angles and surfaces and with straight sharp edges. Ease exposed edges to a radius of approximately 1/32 inch, and bend metal corners to the smallest radius possible without causing grain separation or otherwise impairing the work.

Continuously weld corners and seams in accordance with the recommendations of AWS D1.1/D1.1M. Grind exposed welds smooth and flush to match and blend with adjoining surfaces.

Form exposed connections with hairline joints that are flush and smooth, using concealed fasteners wherever possible. Use exposed fasteners of the type indicated or, if not indicated, use Phillips flat-head (countersunk) screws or bolts.

Provide and coordinate anchorage of the type indicated for the supporting structure. Fabricate anchoring devices, and space them as indicated and as necessary to provide adequate support for the intended use of the work.

Use hot-rolled steel bars for work fabricated from bar stock unless work is indicated or specified as fabricated from cold-finished or cold-rolled stock.

## 2.2.2 Steel Pan Stairs

### 2.2.2.1 General

Joining pieces by welding. Fabricate units so that bolts and other fastenings do not appear on finished surfaces. Make joints true and tight, and connections between parts lighttight. Grind continuous welds smooth where exposed.

Construct metal stair units to sizes and arrangements indicated to support a minimum live load of 100 pounds per square foot. Provide framing, hangers, columns, struts, clips, brackets, bearing plates, and other components as required for the support of stairs and platforms.

### 2.2.2.2 Stair Framing

Fabricate stringers of structural steel channels, or plates, or a combination thereof as indicated. Provide closures for exposed ends of strings.

Construct platforms of structural steel channel headers and miscellaneous framing members as indicated. Bolt headers to stringers and newels, and bolt framing members to stringers and headers.

### 2.2.2.3 Riser, Subtread, and Subplatform Metal Pans

Form metal pans of 0.1084-inch (12-gage) galvanized structural steel sheets, conforming to ASTM A653/A653M, Grade A, with zinc coating conforming to ASTM A653/A653M and ASTM A924/A924M. Shape the pans to the configuration indicated.



Construct risers and subread metal pans with steel angle supporting brackets, of the size indicated, welded to stringers. Secure metal pans to brackets with rivets or welds. Secure subplatform metal pans to platform frames with welds.

#### 2.2.2.4 Metal Safety Nosings

Between stringers, provide abrasive cast metal safety nosings, 4 inches wide by the full length of the step. Fabricate nosings to the thickness, profile, and surface pattern indicated. Equip each nosing with integral anchors for embedding in the pan fill material, and space the anchors not more than 4 inches from each end and not more than 15 inches on center.

#### 2.2.2.5 Steel Floor Plate Treads and Platforms

Provide raised-pattern steel floor plate fabricated from steel complying with ASTM A36/A36M. Provide the pattern indicated or, if not indicated, as selected from the manufacturer's standard patterns.

Form treads of 1/4-inch thick steel floor plate with integral nosing and back-edge stiffener. Weld steel supporting brackets to strings, and weld treads to brackets.

Fabricate platforms of steel floor plate to the thickness indicated. Provide nosing that match treads at landings. Secure floor plates to platform framing members with welds.

#### 2.2.2.6 Safety Nosings for Concrete Treads

Provide safety nosings of cast aluminum with cross-hatched abrasive surfaces, or extruded aluminum with abrasive inserts, at least 4 inches wide and 1/4 inch thick for metal-pan cement-filled treads extending the full length of the tread for stairs and for platforms and landings. Provide safety nosings with anchors embedded a minimum of 3/4 inch in the concrete and with tops flush with the top of the traffic surface.

#### 2.2.2.7 Safety Treads

Plank grating ASTM A653/A653M, G-90 2.2.2.8 Steel Framing for Concrete Stairs

When necessary, modify fabricated units to fit actual dimensions of the supporting structure. Join steel components by welding. Provide 14-gage steel risers unless otherwise indicated. Arrange components to receive finish materials as indicated.

#### 2.2.3 Floor Grating Treads and Platforms

Provide floor grating treads and platforms conforming to ASTM A6/A6M, ASTM A29/A29M and NAAMM MBG 531, "Metal Bar Grating Manual." Provide the pattern, spacing, and bar sizes as indicated:

- a. Galvanized finish, conforming to ASTM A123/A123M.

Fabricate grating treads with steel plate nosings on one edge and with steel angle or steel plate carriers at each end for string connections. Secure treads to strings with bolts.

Match the nosings of grating platforms with the nosing of grating treads

at landings. Provide toeplates where the open-sided edges of floor grating meet platform framing members.

#### 2.2.4 Protective Coating

Hot-dip galvanize steelwork as indicated in accordance with ASTM A123/A123M. Touch up abraded surfaces and cut ends of galvanized members with zinc-dust, zinc-oxide primer, or an approved galvanizing repair compound.

### 2.3 COMPONENTS

#### 2.3.1 Steel Stairs

Provide steel stairs complete with stringers, grating treads, landings, columns, handrails, and necessary bolts and other fastenings. Hot-dip-galvanize steel stairs and accessories.

##### 2.3.1.1 Design Loads

Design stairs to sustain a uniform load of not less than 100 pounds per square foot, or a concentrated load of 300 pounds per 4 square inches applied where it is most critical. Limit deflection to  $L/360$  or  $1/4$  inch, whichever is less. Except for a commercial product, design and fabricate steel stairs to conform to AISC 360.

##### 2.3.1.2 Materials

Provide steel stairs of welded construction except that bolts may be used where welding is not practicable. Do not use screw or screw-type connections.

- a. Structural Steel: ASTM A36/A36M.
- b. Gratings for Treads and Landings: NAAMM MBG 531 or Plank grating; ASTM A653/A653M, G-90 for steel; ASTM B209 for aluminum.
- c. Support steel grating on angle cleats welded to stringers or treads with integral cleats, welded or bolted to the stringer. Close exposed ends.
- e. Before fabrication, obtain necessary field measurements and verify drawing dimensions.
- f. Clean metal surfaces free of mill scale, flake rust, and rust pitting before shop finishing. Weld permanent connections. Finish welds flush and smooth on surfaces that will be exposed after installation.

#### 2.3.2 Steel Stairs, Circular

Provide standard open riser constructed of steel, with a minimum outside diameter of 6 feet and with 12 treads to the circle. Construct the center pole from one continuous length of circular, cold-drawn, seamless tube with a minimum outside diameter of  $3\frac{1}{2}$  inches and with caps at the top plate and base plate having countersunk machine screws and expansion shields for fastening to the concrete floor slab. Construct treads and platforms from steel grating conforming to NAAMM MBG 531. Design slip-resistant gratings to exceed a static coefficient of friction of 0.5 as tested in accordance with ASTM E303.

### 2.3.3 Soffit Clips

Provide clips with holes for attaching metal furring for plastered soffits. Space the clips not more than 12 inches on center, and weld them to stair treads and platforms as required.

### 2.3.4 Masonry Anchorage Devices

Provide masonry anchorage devices consisting of expansion shields complying with AASHTO M 314, ASTM E488/E488M and ASTM C514 as follows:

- d. Bolt anchor expansion shields for bolts; closed-end bottom-bearing class, Group II, Type 2, Class 1.

Use toggle bolts of the tumble-wing type, conforming to ASTM A449, ASTM C636/C636M, and ASTM F3125/F3125M, type, class, and style as required.

### 2.3.5 Fasteners

Select galvanized zinc-coated fasteners conforming to ASTM A153/A153M for exterior applications or where the fasteners are built into exterior walls or floor systems. Select the fasteners for the type, grade, and class required for the installation of steel stair items:

- a. Standard/regular hexagon-head bolts and nuts, conforming to ASTM A307, Grade A.
- b. Square-head lag bolts conforming to ASME B18.2.1.
- c. Cadmium-plated steel machine screws, conforming to ASME B18.6.3.
- d. Flat-head carbon steel wood screws, conforming to ASME B18.6.1.
- e. Plain, round, general-assembly-grade, carbon steel washers, conforming to ASME B18.21.1.
- f. Helical-spring, carbon steel lockwashers, conforming to ASME B18.2.1.

## 2.4 MATERIALS

### 2.4.1 Structural Steel Plates, Shapes and Bars

Structural size shapes and plates, conforming to ASTM A36/A36M, unless otherwise noted, except bent or cold-formed plates.

Steel plates - bent or cold-formed, conforming to ASTM A283/A283M, Grade C.

Steel bars and bar-size shapes, conforming to ASTM A36/A36M, unless otherwise noted for steel bars and bar-size shapes.

### 2.4.2 Structural Steel Tubing

Provide the following:

- a. Structural steel tubing, hot-formed, welded or seamless, conforming to ASTM A500/A500M, Grade B, unless otherwise noted.

#### 2.4.3 Hot-Rolled Carbon Steel Bars

Provide the following:

- a. Hot-rolled carbon steel bars and bar-size shapes, conforming to ASTM A575, grade as selected by the fabricator.

#### 2.4.4 Cold-Finished Steel Bars

Provide the following:

- a. Cold-finished steel bars conforming to ASTM A108, grade as selected by the fabricator.

#### 2.4.5 Hot-Rolled Carbon Steel Sheets and Strips

Provide the following:

- a. Hot-rolled carbon sheets and strips conforming to ASTM A568/A568M and ASTM A1011/A1011M, pickled and oiled.

#### 2.4.6 Cold-Rolled Carbon Steel Sheets

Provide the following:

- a. Cold-rolled carbon steel sheets conforming to ASTM A1008/A1008M.

#### 2.4.7 Galvanized Carbon Steel Sheets

Provide the following:

- a. Galvanized carbon steel sheets conforming to ASTM A653/A653M, with galvanizing conforming to ASTM A653/A653M and ASTM A924/A924M.

#### 2.4.8 Cold-Drawn Steel Tubing

Provide the following:

- a. Cold-drawn steel tubing conforming to ASTM A512, sunk drawn, butt-welded, cold-finished, and stress-relieved.

#### 2.4.9 Gray Iron Castings

Provide the following:

- a. Gray iron castings conforming to ASTM A48/A48M, Class 30.

#### 2.4.10 Malleable Iron Castings

Provide the following:

- a. Malleable iron castings conforming to ASTM A47/A47M, grade as selected.

#### 2.4.11 Steel Pipe

Provide the following:

- a. Steel pipe conforming to ASTM A53/A53M, type as selected, Grade B; primed finish, unless galvanizing is required; standard weight

(Schedule 40).

## 2.5 Mill Test Reports

Provide mill test reports for . Mill test report will include but is not limited to material heat number, material grade, edition year and type of specification met, material dimension, mechanical properties, chemical analysis, heat treatment (if applicable), and certified inspector signature.

## PART 3 EXECUTION

### 3.1 PREPARATION

Clean surfaces thoroughly before installation. Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions. Examine materials upon arrival at site. Notify the carrier and manufacturer of any damage.

Protect installed products until completion of project. Touch up, repair or replace, damaged products before substantial completion

### 3.2 INSTALLATION

Install in accordance with the manufacturer's instructions and approved submittals. Install in proper relationship with adjacent construction.

Install items at locations indicated, according to the manufacturer's instructions. Verify all measurements and take all field measurements necessary before fabrication. Ensure that exposed fastenings are compatible with generally match the color and finish of, and harmonize with the material to which they are applied. Include materials and parts necessary to complete each item, even though such work is not definitely shown or specified. Poor matching of holes for fasteners is cause for rejection. Conceal fastenings where practicable. Select thickness of metal and details of assembly and supports that adequately strengthen and stiffen the construction. Form joints exposed to the weather to exclude water.

#### 3.2.1 Field Preparation

Remove rust-preventive coating just before field erection, using a remover approved by the coating manufacturer. Provide surfaces, when assembled, free of rust, grease, dirt and other foreign matter.

#### 3.2.2 Field Welding

Comply with AWS D1.1/D1.1M in executing manual shielded-metal arc welding, (for appearance and quality of new welds) and in correcting existing welding.

#### 3.2.3 Safety Nosings

Completely embed nosing in concrete before the initial set of the concrete occurs and finish flush with the top of the concrete surface.

-- End of Section --

SECTION 05 51 33

METAL LADDERS  
**08/24**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN LADDER INSTITUTE (ALI)

ALI A14.3 (2008; R 2018) Ladders - Fixed - Safety Requirements

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP Z359.16 (2016) Safety Requirements for Climbing Ladder Fall Arrest Systems

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A47/A47M (1999; R 2022; E 2022) Standard Specification for Ferritic Malleable Iron Castings

ASTM A53/A53M (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A123/A123M (2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A153/A153M (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM A500/A500M (2023) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes

ASTM A653/A653M (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

|                 |                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------|
| ASTM A780/A780M | (2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings                |
| ASTM A924/A924M | (2022a) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process |

MASTER PAINTERS INSTITUTE (MPI)

|        |                                                |
|--------|------------------------------------------------|
| MPI 79 | (2016) Primer, Alkyd, Anti-Corrosive for Metal |
|--------|------------------------------------------------|

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|                |                                                                                          |
|----------------|------------------------------------------------------------------------------------------|
| 29 CFR 1910.23 | (Nov 2016) Ladders                                                                       |
| 29 CFR 1910.29 | (Nov 2016) Fall Protection System and Falling Object Protection - Criteria and Practices |

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Ladders, Installation Drawings

SD-03 Product Data

Ladders

Ladder Safety Devices (Climbing Ladder Fall Arrest Systems)

SD-07 Certificates

Fabricator Certification for Ladder Assembly

## 1.3 CERTIFICATES

Provide fabricator certification for ladder assembly stating that the ladder and associated components have been fabricated according to the requirements of 29 CFR 1910.23.

## 1.4 QUALIFICATION OF WELDERS

Qualify welders in accordance with AWS D1.1/D1.1M. Use procedures, materials, and equipment of the type required for the work.

## 1.5 DELIVERY, STORAGE, AND PROTECTION

Protect from corrosion, deformation, and other types of damage. Store items in an enclosed area free from contact with soil and weather. Remove and replace damaged items with new items.

## PART 2 PRODUCTS

### 2.1 MATERIALS

#### 2.1.1 Structural Carbon Steel

ASTM A36/A36M.

#### 2.1.2 Structural Tubing

ASTM A500/A500M.

#### 2.1.3 Steel Pipe

ASTM A53/A53M, Type E or S, Grade B.

#### 2.1.4 Fittings for Steel Pipe

Standard malleable iron fittings ASTM A47/A47M.

### 2.2 FABRICATION FINISHES

#### 2.2.1 Galvanizing

Hot-dip galvanize items specified to be zinc-coated, after fabrication where practicable. Galvanizing: ASTM A123/A123M, ASTM A153/A153M, ASTM A653/A653M or ASTM A924/A924M, G90, as applicable.

#### 2.2.2 Galvanize

Anchor bolts, washers, and parts or devices necessary for proper installation, unless indicated otherwise.

#### 2.2.3 Repair of Zinc-Coated Surfaces

Repair damaged surfaces with galvanizing repair method and paint conforming to ASTM A780/A780M or by application of stick or thick paste material specifically designed for repair of galvanizing, as approved by Contracting Officer. Clean areas to be repaired and remove slag from welds. Heat surfaces to which stick or paste material is applied, with a torch to a temperature sufficient to melt the metallics in stick or paste; spread molten material uniformly over surfaces to be coated and wipe off excess material.

#### 2.2.4 Nonferrous Metal Surfaces

Protect by plating, anodic, or organic coatings.

### 2.3 LADDERS

Furnish ladders for the permeate tank (05-07-002), CIP tank (05-07-003) and UF Break Tank (05-07-001). Fabricate vertical ladders conforming to 29 CFR 1910.23 and Section 5 of ALI A14.3. Design assembly, including rung connections and methods of attachment, to support a live load of 1300 N 300 pounds per rung, or 1000 pounds point load, or 100 pounds per square foot load applied downward. Ladders requiring a "Safety Device" must have stringers and rungs that meet the minimum strength requirements of the Safety Device anchor points dynamic loading. Ladder rungs, step and



cleats must be spaced not less than 10 inches and not more than 16 inches wide (measured before installation of ladder safety system), spaced no more than 14 inches apart, plug welded or shouldered and headed into stringers. Install ladders so that the maximum perpendicular distance from the centerline of the steps or rungs, or grab bars, or both, to the vessel. Provide heavy clip angles riveted or bolted to the stringer and drilled as indicated. Provide intermediate clip angles not over 48 inches on centers. The top rung of the ladder must be level with the top of the access level, parapet or landing served by the ladder. Extend the side rails of through or side step ladders 42 inches above the access level. The drawings must indicate ladder locations and details of critical dimensions and materials.

#### 2.3.1 Ladder Safety Devices (Climbing Ladder Fall Arrest Systems)

Conform to 29 CFR 1910.29, Section 7 of ALI A14.3 and ASSP Z359.16. Install ladder safety devices on ladders over 20 feet long or more. The 6,000 mm 20 foot distance shall be measured from the top ladder rung to the top of the finished floor or outdoor pavement. If vessel is installed on a housekeeping pad, the 6,000 mm 20 foot distance shall be measured from the top ladder rung to the bottom of the housekeeping pad. The ladder safety systems must meet the design requirement of the ladders which they serve. The ladder safety system must be capable of sustaining a minimum static load of four personnel that may be attached to the fall safety device and climb the ladder at one time. The ladder must have ladder members or anchor points that meet the minimum loading requirements as defined by the safety device and the ladder must be designed to handle the dynamic load being applied in the event of a fall. Each ladder safety system must allow the worker to climb up and down using both hands and does not require the employee continuously, hold, push, or pull any part of the system while climbing. The connection between the carrier or lifeline and the point of attachment to the body harness does not exceed 9 inches. The ladder safety system consists of a rigid or flexible carrier. Mountings for the rigid carriers are attached at each end of the carrier, with intermediate mountings spaced as necessary, along the entire length of the carrier. Mountings for flexible carrier are attached at each end of the carrier and cable guides for flexible carriers are installed at least 25 feet apart but not more than 40 feet apart along the entire length of the carrier. The design and installation of mountings and cable guides does not reduce the design strength of the ladder.

#### 2.3.2 Ladder Security Gate

Provide ladders with either a ladder security guard door or ladder cage security guard gate. Ladder guard doors and gates are to be a minimum height of 6 feet 0 inches and constructed of 16 gauge galvanized steel. Provide a minimum of two hinges that are 4 inches by 4 inches by 1/8 inch zinc-coated plates steel with a 1/4 inch minimum diameter pin.

### PART 3 EXECUTION

#### 3.1 GENERAL INSTALLATION REQUIREMENTS

Install items at locations indicated, according to manufacturer's instructions. Verify all measurements and take all field measurements necessary before fabrication. Provide exposed fastenings of compatible materials, generally matching in color and finish, and harmonize with the material to which fastenings are applied. Include materials and parts necessary to complete each item, even though such work is not definitely

shown or specified. Poor matching of holes for fasteners will be cause for rejection. Conceal fastenings where practicable. Thickness of metal and details of assembly and supports must provide strength and stiffness. Formed joints exposed to the weather to exclude water. Items listed below require additional procedures.

### 3.2 WORKMANSHIP

Metalwork must be well formed to shape and size, with sharp lines and angles and true curves. Drilling and punching must produce clean true lines and surfaces. Continuously weld along the entire area of contact. Do not tack weld exposed connections of work in place. Grind exposed welds smooth. Provide smooth finish on exposed surfaces of work in place, unless otherwise approved. Where tight fits are required, mill joints. Cope or miter corner joints, well formed, and in true alignment. Install in accordance with manufacturer's installation instructions and approved drawings, cuts, and details.

### 3.3 ANCHORAGE, FASTENINGS, AND CONNECTIONS

Provide anchorage where necessary for fastening metal items securely in place. Include for anchorage not otherwise specified or indicated slotted inserts, expansion anchors, and powder-actuated fasteners, when approved for concrete; toggle bolts and through bolts for masonry; machine bolts, carriage bolts and powder-actuated threaded studs for steel; through bolts, lag bolts, and screws for wood. Do not use wood plugs in any material. Provide non-ferrous attachments for non-ferrous metal. Make exposed fastenings of compatible materials, generally matching in color and finish, to which fastenings are applied. Conceal fastenings where practicable.

### 3.4 WELDING

Perform welding, welding inspection, and corrective welding, in accordance with AWS D1.1/D1.1M. Use continuous welds on all exposed connections. Grind visible welds smooth in the finished installation.

### 3.5 FINISHES

#### 3.5.1 Dissimilar Materials

Where dissimilar metals are in contact, protect surfaces with a coat conforming to MPI 79 to prevent galvanic or corrosive action.

### 3.6 LADDERS

Secure to the adjacent construction with the clip angles attached to the stringer. Install intermediate clip angles not over 48 inches on center. Vessel vendor shall provide pre-fabricated anchor points. Coordinate with vessel manufacturer to identify the required locations of the anchor points to ensure the ladder is properly secured and supported. Do not drill directly into the vessel. Ensure ladder is installed such that the bottom ladder rung is no less than 610 MM 24 inches above the closest flat surface including concrete step, housekeeping pad, finished floor, or outdoor pavement.

-- End of Section --

SECTION 05 52 00

METAL RAILINGS  
**08/23**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS  
(AASHTO)

AASHTO M 314 (1990; R 2013) Standard Specification for  
Steel Anchor Bolts

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 325 (2017) Steel Construction Manual

AISC 326 (2009) Detailing for Steel Construction

AISC 360 (2016) Specification for Structural Steel  
Buildings

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-22 (2022; Supp 1 2023; Supp 2 2023) Minimum  
Design Loads and Associated Criteria for  
Buildings and Other Structures

AMERICAN WELDING SOCIETY (AWS)

AWS A2.4 (2012) Standard Symbols for Welding,  
Brazing and Nondestructive Examination

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding  
Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A27/A27M (2020) Standard Specification for Steel  
Castings, Carbon, for General Application

ASTM A36/A36M (2019) Standard Specification for Carbon  
Structural Steel

ASTM A47/A47M (1999; R 2022; E 2022) Standard  
Specification for Ferritic Malleable Iron  
Castings

ASTM A53/A53M (2024) Standard Specification for Pipe,  
Steel, Black and Hot-Dipped, Zinc-Coated,  
Welded and Seamless

|                 |                                                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------|
| ASTM A108       | (2024) Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished                                          |
| ASTM A123/A123M | (2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products                       |
| ASTM A153/A153M | (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware                                   |
| ASTM A283/A283M | (2024) Standard Specification for Low and Intermediate Tensile Strength Carbon Steel Plates                           |
| ASTM A307       | (2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength             |
| ASTM A467/A467M | (2020) Standard Specification for Machine Coil Chain                                                                  |
| ASTM A500/A500M | (2023) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes |
| ASTM A512       | (2006; R 2012) Standard Specification for Cold-Drawn Buttweld Carbon Steel Mechanical Tubing                          |
| ASTM A575       | (2020) Standard Specification for Steel Bars, Carbon, Merchant Quality, M-Grades                                      |
| ASTM B26/B26M   | (2018; E 2018) Standard Specification for Aluminum-Alloy Sand Castings                                                |
| ASTM C514       | (2004; R 2020) Standard Specification for Nails for the Application of Gypsum Board                                   |
| ASTM E488/E488M | (2022) Standard Test Methods for Strength of Anchors in Concrete Elements                                             |

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

|               |                                            |
|---------------|--------------------------------------------|
| NAAMM AMP 521 | (2001; R 2012) Pipe Railing Systems Manual |
|---------------|--------------------------------------------|

1.2 ADMINISTRATIVE REQUIREMENTS

1.2.1 Preinstallation Meetings

Within 30 days of contract award, submit fabrication drawings to the Contracting Officer for the following items:

- b. Steel shapes, plates, bars and strips
- c. Steel railings and handrails
- e. Anchorage and fastening systems

Submit manufacturer's catalog data, including two copies of manufacturers specifications, load tables, dimension diagrams, and anchor details for the following items:

- a. Structural-steel plates, shapes, and bars
- b. Structural-steel tubing
- c. Cold-finished steel bars
- d. Hot-rolled carbon steel bars
- e. Cold-drawn steel tubing
- f. Concrete inserts
- g. Masonry anchorage devices
- h. Protective coating
- i. Steel railings and handrails
- k. Anchorage and fastening systems

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

Fabrication Drawings; G

Iron and Steel Hardware; G

Steel Shapes, Plates, Bars and Strips; G

#### SD-03 Product Data

Concrete Inserts; G

Masonry Anchorage Devices; G

Protective Coating; G

Steel Railings and Handrails; G

Anchorage and Fastening Systems; G

#### SD-05 Design Data

Design Calculations For Steel Connections; G

#### SD-06 Test Reports

Welding Inspections; G

Mill Test Reports; G

SD-07 Certificates

Welder Qualification; G

SD-08 Manufacturer's Instructions

Installation Instructions

#### 1.4 QUALITY CONTROL

##### 1.4.1 Welding Procedures

Submit results of welding inspections testing in accordance with AWS D1.1/D1.1M by an approved testing laboratory at the Contractor's expense. Provide laboratory welding inspection results to Contracting Officer.

##### 1.4.2 Welder Qualification

Submit certified welder qualification by tests in accordance with AWS D1.1/D1.1M, or under an equivalent approved qualification test. In addition, perform tests on test pieces in positions and with clearances equivalent to those actually encountered. If a test weld fails to meet requirements, conduct an immediate retest of two test welds and ensure that each test weld passes. Failure in the immediate retest will require that the welder be retested after further practice or training and make a complete set of test welds.

#### 1.5 QUALITY ASSURANCE

##### 1.5.1 Fabrication Drawing Requirements

Submit fabrication drawings for approval prior to fabrication. Prepare in accordance with AISC 325, AISC 326, ASCE 7-22, and State of Maine Building Code. Fabrication drawings will not be reproductions of contract drawings.

Sign and seal fabrication drawings by a registered professional engineer.

Include complete information for the fabrication and erection of the structure's components, including the location, type, and size of bolts, welds, member sizes and lengths, connection details, blocks, copes, and cuts. Use AWS A2.4 standard welding symbols. Clearly highlight any deviations from the details shown on the contract drawings highlighted on the fabrication drawings. Explain the reasons for any deviations from the contract drawings.

##### 1.5.2 Delegated Connection Design

Design structural steel connection indicated in the contract documents per AISC 360. Submit design calculations for steel connections signed and sealed by a registered professional engineer.

## PART 2 PRODUCTS

### 2.1 FABRICATION

Preassemble items in the shop to the greatest extent possible. Disassemble units only to the extent necessary for shipping and handling. Clearly mark units for reassembly and coordinated installation.

For the fabrication of work exposed to view, use only materials that are smooth and free of surface blemishes, including pitting, seam marks, roller marks, rolled trade names, and roughness. Remove blemishes by grinding, or by welding and grinding, before cleaning, treating, and applying surface finishes, including zinc coatings.

Provide railing and handrail detail plans and elevations at not less than 1 inch to 1 foot. Provide details of sections and connections at not less than 3 inches to 1 foot. Also detail setting drawings, diagrams, templates for installation of anchorages, including concrete inserts, anchor bolts, and miscellaneous metal items having integral anchors.

Use materials of size and thicknesses indicated or, if not indicated, of the size and thickness necessary to produce adequate strength and durability in the finished product for its intended use. Work the materials to the dimensions indicated on approved detail drawings, using proven details of fabrication and support. Use the type of materials indicated or specified for the various components of work.

Form exposed work true to line and level, with accurate angles and surfaces and straight sharp edges. Ensure that all exposed edges are eased to a radius of approximately 1/32 inch. Bend metal corners to the smallest radius possible without causing grain separation or otherwise impairing the work.

Weld corners and seams continuously and in accordance with the recommendations of AWS D1.1/D1.1M. Grind exposed welds smooth and flush to match and blend with adjoining surfaces.

Form the exposed connections with hairline joints that are flush and smooth, using concealed fasteners wherever possible. Use exposed fasteners of the type indicated or, if not indicated, use countersunk Phillips flathead screws or bolts.

Provide anchorage of the type indicated and coordinated with the supporting structure. Fabricate anchoring devices and space as indicated and as required to provide adequate support for the intended use of the work.

Use hot-rolled steel bars for work fabricated from bar stock unless work is indicated or specified to be fabricated from cold-finished or cold-rolled stock.

#### 2.1.1 Aluminum Railings

Fabrication: Provide fabrication jointing by one of the following methods:

- a. Use flush-type rail fittings, welded and ground smooth with splice locks secured with 3/8 inch recessed-head set screws.
- b. Ensure that mitered and welded joints made by fitting; post to top

rail; intermediate rail to post; and corners, are groove welded and ground smooth. Where allowed by the Contracting Officer, provide butt splices reinforced by a tight-fitting dowel or sleeve not less than 6 inches in length. Tack-weld or epoxy-cement the dowel or sleeve to one side of the splice.

- c. Assemble railings using slip-on aluminum-magnesium alloy fittings for joints. Fasten fittings to pipe or tube with 1/4 or 3/8 inch stainless-steel recessed-head setscrews. Provide assembled railings with fittings only at vertical supports or at rail terminations attached to walls. Provide expansion joints at the midpoint of panels. Provide a setscrew in only one side of the slip-on sleeve. Provide alloy fittings to conform to ASTM B26/B26M.

#### 2.1.2 Steel Handrails

Fabricate joint posts, rail, and corners by one of the following methods:

- a. Flush-type rail fittings of commercial standard, welded and ground smooth, with railing splice locks secured with 3/8 inch hexagonal-recessed-head setscrews.
- b. Mitered and welded joints made by fitting post to top rail and intermediate rail to post, mitering corners, groove-welding joints, and grinding smooth. Butt railing splices and reinforce them by a tight-fitting interior sleeve not less than 6 inches long.
- c. Railings may be bent at corners in lieu of jointing, provided that bends are made in suitable jigs and the pipe is not crushed.

#### 2.1.3 Protective Coating

Provide hot-dipped galvanized steelwork as indicated in accordance with ASTM A123/A123M. Touch up abraded surfaces and cut ends of galvanized members with zinc-dust, zinc-oxide primer, or an approved galvanizing repair compound.

### 2.2 COMPONENTS

#### 2.2.1 Structural Steel Plates, Shapes and Bars

Provide structural-size shapes and plates, except plates to be bent or cold-formed, conforming to ASTM A36/A36M, unless otherwise noted.

Provide steel plates, to be bent or cold-formed, conforming to ASTM A283/A283M, Grade C.

Provide steel bars and bar-size shapes conforming to ASTM A36/A36M, unless otherwise noted.

#### 2.2.2 Structural-Steel Tubing

Provide structural-steel tubing, hot-formed, welded or seamless, conforming to ASTM A500/A500M, Grade B, unless otherwise noted.

#### 2.2.3 Hot-Rolled Carbon Steel Bars

Provide bars and bar-size shapes conforming to ASTM A575, grade as selected by the fabricator.



#### 2.2.4 Cold-Finished Steel Bars

Provide cold-finished steel bars conforming to ASTM A108, grade as selected by the fabricator.

#### 2.2.5 Cold-Drawn Steel Tubing

Provide tubing conforming to ASTM A512, sunk-drawn, butt-welded, cold-finished, and stress-relieved.

#### 2.2.6 Steel Pipe

Provide pipe conforming to ASTM A53/A53M, type as selected, Grade B; primed finish, unless galvanizing is required; standard weight (Schedule 40).

#### 2.2.7 Concrete Inserts

Provide threaded-type concrete inserts consisting of galvanized ferrous castings, internally threaded to receive 3/4 inch diameter machine bolts; either malleable iron conforming to ASTM A47/A47M or cast steel conforming to ASTM A27/A27M, hot-dip galvanized in accordance with ASTM A153/A153M.

#### 2.2.8 Masonry Anchorage Devices

Provide masonry anchorage devices consisting of expansion shields complying with AASHTO M 314, ASTM E488/E488M and ASTM C514 as follows:

Provide bolt anchor expansion shields for bolts; closed-end bottom-bearing class, Group II, Type 2, Class 1.

#### 2.2.9 Fasteners

Provide galvanized zinc-coated fasteners in accordance with ASTM A153/A153M used for exterior applications or where built into exterior walls or floor systems. Select fasteners for the type, grade, and class required for the installation of steel stair items.

Provide standard hexagon-head bolts, conforming to ASTM A307, Grade A.

#### 2.2.10 Steel Railings And Handrails

Design handrails to resist a concentrated load of 200 lb in any direction at any point of the top of the rail or 50 lb per foot applied horizontally to the top of the rail, whichever is more severe. NAAMM AMP 521, provide the same size rail and post. Provide pipe collars of the same material and finish as the handrail and posts.

##### 2.2.10.1 Steel Handrails

Provide steel handrails, including inserts in concrete, steel pipe conforming to ASTM A53/A53M or structural tubing conforming to ASTM A500/A500M, Grade A or B of equivalent strength. Provide steel railings of 2 inch nominal size, hot-dip galvanized .

Provide kickplates between railing posts where indicated, and consisting of 1/8 inch steel flat bars not less than 6 inches high. Secure kickplates as indicated.

Provide galvanized railings, including pipe, fittings, brackets, fasteners, and other ferrous metal components.

#### 2.2.11 Safety Chains And Guardrails

Provide safety chains of galvanized steel, straight-link type, 3/16 inch diameter, with at least 12 links per foot, and with snap hooks on each end. Test safety chain in accordance with ASTM A467/A467M, Class CS. Provide snap hooks of boat type. Provide galvanized 3/8 inch bolt with 3/4 inch eye diameter for attachment of chain, anchored as indicated. Supply two chains, 4 inches longer than the anchorage spacing, for each guarded area. Locate safety chain where indicated. Mount the top chain feet 6 inches above the floor and mount the lower chain 2 feet above the floor .

#### 2.3 Mill Test Reports

Provide mill test reports for . Mill test report will include but is not limited to material heat number, material grade, edition year and type of specification met, material dimension, mechanical properties, chemical analysis, heat treatment (if applicable), and certified inspector signature.

### PART 3 EXECUTION

#### 3.1 PREPARATION

Adjust stair railings and handrails before securing in place in order to ensure proper matching at butting joints and correct alignment throughout their length. Space posts not more than 4 feet on center. Plumb posts in each direction. Secure posts and rail ends to building construction as follows:

- a. Anchor posts in concrete by means of pipe sleeves set and anchored into concrete. Provide sleeves of galvanized, standard-weight, steel pipe, not less than 6 inches long, and having an inside diameter not less than 1/2 inch greater than the outside diameter of the inserted pipe post. Provide steel plate closure secured to the bottom of the sleeve, with closure width and length not less than 1 inch greater than the outside diameter of the sleeve. After posts have been inserted into sleeves, fill the annular space between the post and sleeve with nonshrink grout or a quick-setting hydraulic cement. Cover anchorage joint with a round steel flange welded to the post.
- b. Anchor posts to steel with oval steel flanges, angle type or floor type as required by conditions, welded to posts and bolted to the steel supporting members.

Secure handrails to walls by means of wall brackets and wall return fitting at handrail ends. Provide brackets of malleable iron castings, with not less than 3 inch projection from the finished wall surface to the center of the pipe, drilled to receive one 3/8 inch bolt. Locate brackets not more than 60 inches on center. Provide wall return fittings of cast iron castings, flush type, with the same projection as that specified for wall brackets. Secure wall brackets and wall return fittings to building construction as follows:

- a. For concrete and solid masonry anchorage, use bolt anchor expansion

shields and lag bolts.

- b. For hollow masonry and stud partition anchorage, use toggle bolts having square heads.

Install toe boards and brackets where indicated. Make splices, where required, at expansion joints. Install removable sections as indicated.

### 3.2 INSTALLATION

Submit manufacturer's installation instructions for the following products to be used in the fabrication of steel stair railing and hand rail work:

- a. Structural-steel plates, shapes, and bars
- b. Structural-steel tubing
- c. Cold-finished steel bars
- d. Hot-rolled carbon steel bars
- e. Cold-drawn steel tubing
- f. Protective coating
- g. Masonry anchorage devices
- h. Steel railings and handrails
- j. Anchorage and fastening systems

Provide complete, detailed fabrication and installation drawings for all iron and steel hardware, and for all steel shapes, plates, bars, and strips used in accordance with the design specifications cited in this section.

#### 3.2.1 Steel Handrail

Install handrail in pipe sleeves embedded in concrete and filled with nonshrink grout or quick-setting anchoring cement with anchorage covered with standard pipe collar pinned to post by means of masonry with expansion shields and bolts or toggle bolts. Secure rail ends by steel pipe flanges anchored by expansion shields and bolts.

### 3.3 FIELD QUALITY CONTROL

#### 3.3.1 Field Welding

Ensure that procedures of manual shielded metal arc welding, appearance and quality of welds made, and methods used in correcting welding work comply with AWS D1.1/D1.1M.

-- End of Section --

SECTION 06 10 00

ROUGH CARPENTRY  
**08/16, CHG 2: 11/18**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN INSTITUTE OF TIMBER CONSTRUCTION (AITC)

|                  |                                                                                                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------|
| AITC 111         | (2005) Recommended Practice for Protection of Structural Glued Laminated Timber During Transit, Storage and Erection |
| AITC TCM         | (2012) Timber Construction Manual, 5th Edition                                                                       |
| ANSI/AITC A190.1 | (2007) American National Standard, Structural Glued Laminated Timber                                                 |

AMERICAN LUMBER STANDARDS COMMITTEE (ALSC)

|            |                                          |
|------------|------------------------------------------|
| ALSC PS 20 | (2021) American Softwood Lumber Standard |
|------------|------------------------------------------|

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|                 |                                                                                                                                      |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------|
| ASME B18.2.1    | (2012; R 2021) Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series) |
| ASME B18.2.2    | (2022) Nuts for General Applications: Machine Screw Nuts, and Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)               |
| ASME B18.5.2.1M | (2006; R 2011) Metric Round Head Short Square Neck Bolts                                                                             |
| ASME B18.5.2.2M | (1982; R 2010) Metric Round Head Square Neck Bolts                                                                                   |
| ASME B18.6.1    | (2016) Wood Screws (Inch Series)                                                                                                     |

AMERICAN WOOD COUNCIL (AWC)

|          |                                                                         |
|----------|-------------------------------------------------------------------------|
| AWC NDS  | (2018) National Design Specification (NDS) for Wood Construction        |
| AWC WFCM | (2012) Wood Frame Construction Manual for One- and Two-Family Dwellings |

AMERICAN WOOD PROTECTION ASSOCIATION (AWPA)

|           |                                                                                             |
|-----------|---------------------------------------------------------------------------------------------|
| AWPA BOOK | (2015) AWP Book of Standards                                                                |
| AWPA M2   | (2019) Standard for the Inspection of Preservative Treated Wood Products for Industrial Use |
| AWPA M6   | (2024) Brands Used on Preservative Treated Materials                                        |
| AWPA P5   | (2015) Standard for Waterborne Preservatives                                                |
| AWPA P18  | (2014) Nonpressure Preservatives                                                            |
| AWPA P49  | (2015; R 2021) Standard for Fire Retardant FR-1                                             |
| AWPA T1   | (2024) Use Category System: Processing and Treatment Standard                               |
| AWPA U1   | (2024) Use Category System: User Specification for Treated Wood                             |

APA - THE ENGINEERED WOOD ASSOCIATION (APA)

|              |                                                                                               |
|--------------|-----------------------------------------------------------------------------------------------|
| APA E30      | (2016) Engineered Wood Construction Guide                                                     |
| APA E445     | (2002) Performance Standards and Qualification Policy for Structural-Use Panels (APA PRP-108) |
| APA EWS R540 | (2013) Builder Tips: Proper Storage and Handling of Glulam Beams                              |
| APA EWS T300 | (2007) Technical Note: Glulam Connection Details                                              |
| APA F405     | (19) Product Guide: Performance Rated Panels                                                  |
| APA L870     | (2010) Voluntary Product Standard, PS 1-09, Structural Plywood                                |
| APA S350     | (2019) PS 2-18, Performance Standard for Wood-Based Structural-Use Panels                     |

ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                           |
|-----------------|-----------------------------------------------------------------------------------------------------------|
| ASTM A153/A153M | (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware                       |
| ASTM A307       | (2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength |
| ASTM A653/A653M | (2023) Standard Specification for Steel                                                                   |

Sheet, Zinc-Coated (Galvanized) or  
Zinc-Iron Alloy-Coated (Galvannealed) by  
the Hot-Dip Process

|                   |                                                                                                                                                                           |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C1136        | (2023) Standard Specification for<br>Flexible, Low Permeance Vapor Retarders<br>for Thermal Insulation                                                                    |
| ASTM D198         | (2015) Standard Test Methods of Static<br>Tests of Lumber in Structural Sizes                                                                                             |
| ASTM D696         | (2016) Standard Test Method for<br>Coefficient of Linear Thermal Expansion of<br>Plastics Between -30 degrees C and 30<br>degrees C With a Vitreous Silica<br>Dilatometer |
| ASTM D1435        | (2013) Standard Practice for Outdoor<br>Weathering of Plastics                                                                                                            |
| ASTM D1972        | (1997; R 2005) Standard Practice for<br>Generic Marking of Plastic Products                                                                                               |
| ASTM D2344/D2344M | (2016) Standard Test Method for Short-Beam<br>Strength of Polymer Matrix Composite<br>Materials and Their Laminates                                                       |
| ASTM D2898        | (2010; R 2017) Standard Practice for<br>Accelerated Weathering of<br>Fire-Retardant-Treated Wood for Fire<br>Testing                                                      |
| ASTM D6108        | (2013) Standard Test Method for<br>Compressive Properties of Plastic Lumber<br>and Shapes                                                                                 |
| ASTM D6109        | (2013) Standard Test Methods for Flexural<br>Properties of Unreinforced and Reinforced<br>Plastic Lumber and Related Products                                             |
| ASTM D6111        | (2013a) Standard Test Method for Bulk<br>Density and Specific Gravity of Plastic<br>Lumber and Shapes by Displacement                                                     |
| ASTM D6112        | (2013) Compressive and Flexural Creep and<br>Creep-Rupture of Plastic Lumber and Shapes                                                                                   |
| ASTM D6117        | (2016) Standard Test Methods for<br>Mechanical Fasteners in Plastic Lumber and<br>Shapes                                                                                  |
| ASTM E96/E96M     | (2024) Standard Test Methods for<br>Gravimetric Determination of Water Vapor<br>Transmission Rate of Materials                                                            |
| ASTM F547         | (2017) Standard Terminology of Nails for<br>Use with Wood and Wood-Base Materials                                                                                         |
| ASTM F1667/F1667M | (2021a) Standard Specification for Driven                                                                                                                                 |

Fasteners: Nails, Spikes, and Staples

CALIFORNIA AIR RESOURCES BOARD (CARB)

CARB 93120 (2007) Airborne Toxic Control Measure  
(ATCM) to Reduce Formaldehyde Emissions  
from Composite Wood Products

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350 (2017; Version 1.2) Standard Method for  
the Testing and Evaluation of Volatile  
Organic Chemical Emissions from Indoor  
Sources using Environmental Chambers

COMPOSITE PANEL ASSOCIATION (CPA)

ANSI/CPA A135.4 (2012; R2020) Basic Hardboard

GREEN SEAL (GS)

GS-36 (2013) Adhesives for Commercial Use

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2021) International Building Code

NATIONAL HARDWOOD LUMBER ASSOCIATION (NHLA)

NHLA Rules (2023) Rules for the Measurement &  
Inspection of Hardwood & Cypress

NORTHEASTERN LUMBER MANUFACTURERS ASSOCIATION (NELMA)

NELMA Grading Rules (2021) Standard Grading Rules for  
Northeastern Lumber

REDWOOD INSPECTION SERVICE (RIS) OF THE WESTERN WOOD PRODUCTS  
ASSOCIATION (WWPA)

RIS Grade Use (2019) California Redwood Lumber Grades  
and Uses

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168 (2022) Adhesive and Sealant Applications

SOUTHERN CYPRESS MANUFACTURERS ASSOCIATION (SCMA)

SCMA Spec (1986; Supple. No. 1, Aug 1993) Standard  
Specifications for Grades of Southern  
Cypress

SOUTHERN PINE INSPECTION BUREAU (SPIB)

SPIB 1003 (2021) Standard Grading Rules for Southern  
Pine Lumber

TRUSS PLATE INSTITUTE (TPI)

TPI 1 (2022) National Design Standard for Metal  
Plate Connected Wood Truss Construction

U.S. DEPARTMENT OF COMMERCE (DOC)

DOC/NIST PS58 (1973) Basic Hardboard (ANSI A135.4)

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

CID A-A-1923 (Rev A; Notice 1; Notice 2; Notice 3;  
Notice 4) Shield, Expansion (Lag, Machine  
and Externally Threaded Wedge Bolt Anchors)

CID A-A-1924 (Rev A; Notice 3) Shield, Expansion (Self  
Drilling Tubular Expansion Shell Bolt  
Anchors)

CID A-A-1925 (Rev A; Notice 1; Notice 2; Notice 3;  
Notice 4) Shield Expansion (Nail Anchors)

FS UU-B-790 (Rev A; Notice 2) Building Paper Vegetable  
Fiber: (Kraft, Waterproofed, Water  
Repellent and Fire Resistant)

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 770 Formaldehyde Standards for Composite Wood  
Products

UNDERWRITERS LABORATORIES (UL)

UL 2818 (2022) GREENGUARD Certification Program  
For Chemical Emissions For Building  
Materials, Finishes And Furnishings

WEST COAST LUMBER INSPECTION BUREAU (WCLIB)

WCLIB 17 (2015; R 2018) Standard Grading Rules for  
West Coast Lumber

WESTERN WOOD PRODUCTS ASSOCIATION (WWPA)

WWPA G-5 (2021) Western Lumber Grading Rules

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S"  
classification. Submittals not having a "G" or "S" classification are for  
Contractor Quality Control approval. Submit the following in accordance  
with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Fabricated Structural Members; G



Modifications of Structural Members; G

Drawings of structural laminated members, fabricated wood trusses, engineered wood joists and rafters, and other fabricated structural members indicating materials, shop fabrication, and field erection details; including methods of fastening.

Nailers and Nailing Strips; G

Drawings of field erection details, including materials and methods of fastening nailers in conformance with Factory Mutual wind uplift rated systems specified in other Sections of these specifications.

SD-03 Product Data

Underlayment

Plastic Lumber

Fire-retardant Treatment

Structural-use and OSB Panels

Oriented Strand Board

Adhesives

SD-05 Design Data

Modifications of Structural Members; G

Design analysis and calculations showing design criteria used to accomplish the applicable analysis.

SD-06 Test Reports

Preservative-treated Lumber and Plywood

SD-07 Certificates

Preservative Treatment

SD-10 Operation and Maintenance Data

Plastic

When not labeled, identify types in Operation and Maintenance Manual.

### 1.3 DELIVERY AND STORAGE

Deliver materials to the site in an undamaged condition. Store, protect, handle, and install prefabricated structural elements in accordance with manufacturer's instructions and as specified. Store materials off the ground to provide proper ventilation, with drainage to avoid standing

water, and protection against ground moisture and dampness. Store materials with a moisture barrier at both the ground level and as a cover forming a well ventilated enclosure. Store wood I-beams and glue-laminated beams and joists on edge. Adhere to requirements for stacking, lifting, bracing, cutting, notching, and special fastening requirements. Handle and store laminated timber in accordance with AITC 111 or APA EWS R540. Do not use materials that have visible moisture or biological growth. Remove defective and damaged materials and provide new materials. Store separated reusable wood waste convenient to cutting station and area of work.

#### 1.4 GRADING AND MARKING

##### 1.4.1 Lumber

Mark each piece of framing and board lumber or each bundle of small pieces of lumber with the grade mark of a recognized association or independent inspection agency. Such association or agency must be certified by the Board of Review, American Lumber Standards Committee, to grade the species used. Surfaces that are to be exposed to view must not bear grademarks, stamps, or any type of identifying mark. Hammer marking will be permitted on timbers when all surfaces will be exposed to view.

##### 1.4.2 Structural Glued Laminated Timber

Mark each member with the mark of a recognized association or independent inspection agency that maintains continuing control over the quality of structural glued laminated timber products. The marking must indicate compliance with ANSI/AITC A190.1 and must include all identification information required by ANSI/AITC A190.1.

##### 1.4.3 Plywood

Mark each sheet with the mark of a recognized association or independent inspection agency that maintains continuing control over the quality of the plywood. The mark must identify the plywood by species group or span rating, exposure durability classification, grade, and compliance with APA L870. Surfaces that are to be exposed to view must not bear grademarks or other types of identifying marks.

##### 1.4.4 Structural-Use and OSB Panels

Mark each panel with the mark of a recognized association or independent inspection agency that maintains continuing control over the quality of the panel. The mark must indicate end use, span rating, and exposure durability classification. Oriented Strand Board (OSB), APA F405.

##### 1.4.5 Preservative-Treated Lumber and Plywood

The Contractor is responsible for the quality of treated wood products. Each treated piece must be inspected in accordance with AWP A M2 and permanently marked or branded, by the producer, in accordance with AWP A M6. The Contractor must provide Contracting Officer's Representative (COR) with the inspection report of an approved independent inspection agency that offered products comply with applicable AWP A Standards. The appropriate Quality Mark on each piece will be accepted, in lieu of inspection reports, as evidence of compliance with applicable AWP A treatment standards.

#### 1.4.6 Fire-Retardant Treated Lumber

Mark each piece in accordance with AWP A M6, except pieces that are to be natural or transparent finished. In addition, exterior fire-retardant lumber must be distinguished by a permanent penetrating blue stain. Labels of a nationally recognized independent testing agency will be accepted as evidence of conformance to the fire-retardant requirements of AWP A M6.

#### 1.4.7 Hardboard, Gypsum Board, and Fiberboard

Mark each sheet or bundle to identify the standard under which the material is produced and the producer.

#### 1.4.8 Plastic Lumber

Label plastic products to be incorporated into the project in accordance with ASTM D1972, or provide product data indicating polymeric information in the Operation and Maintenance Manual.

- a. Type 1: Polyethylene Terephthalate (PET, PETE).
- b. Type 2: High Density Polyethylene (HDPE).
- c. Type 3: Vinyl (Polyvinyl Chloride or PVC).
- d. Type 4: Low Density Polyethylene (LDPE).
- e. Type 5: Polypropylene (PP).
- f. Type 6: Polystyrene (PS).
- g. Type 7: Other. Use of this code indicates that the package in question. is made with a resin other than the six listed above, or is made of more than one resin listed above, and used in a multi-layer combination.

### 1.5 SIZES AND SURFACING

ALSC PS 20 for dressed sizes of yard and structural lumber. Lumber must be surfaced four sides. Size references, unless otherwise specified, are nominal sizes, and actual sizes must be within manufacturing tolerances allowed by the standard under which the product is produced. Other measurements are IP or SI standard.

### 1.6 MOISTURE CONTENT

Air-dry or kiln-dry lumber. Kiln-dry treated lumber after treatment. Maximum moisture content of wood products must be as follows at the time of delivery to the job site:

- a. Framing lumber and board, 19 percent maximum
- b. Timbers 5 inches and thicker, 25 percent maximum
- d. Materials other than lumber; moisture content must be in accordance with standard under which the product is produced

### 1.7 PRESERVATIVE TREATMENT

Treat wood products with waterborne wood preservatives conforming to AWP A P5. Pressure treatment of wood products must conform to the requirements of AWP A BOOK Use Category System Standards U1 and T1. Pressure-treated wood products must not contain arsenic, chromium, or other agents classified as carcinogenic, probably carcinogenic, or

possibly carcinogenic to humans (compounds in Groups 1, 2A, or 2B) by the International Agency for Research on Cancer (IARC), Lyon, France. Pressure-treated wood products must not exceed the limits of the U.S. EPA's Toxic Characteristic Leaching Procedure (TCLP), and must not be classified as hazardous waste. Submit certification from treating plant stating chemicals and process used and net amount of preservatives retained are in conformance with specified standards. In accordance with AWP A U1 provide non-copper preservative treatment such as EL2, PTI or SBX, DOT for products in direct contact with sheet metal.

- a. 0.25 pcf intended for above ground use.
- b. 0.40 pcf intended for ground contact and fresh water use. 0.60 pcf intended for Ammoniacal Copper Quaternary Compound (ACQ)-treated foundations. 0.80 to 1.00 pcf intended for ACQ-treated pilings. All wood must be air or kiln dried after treatment. Specific treatments must be verified by the report of an approved independent inspection agency, or the AWP A Quality Mark on each piece. Do not incise surfaces of lumber that will be exposed. Minimize cutting and avoid breathing sawdust. Brush coat areas that are cut or drilled after treatment with either the same preservative used in the treatment or with a 2 percent copper naphthenate solution. All lumber and woodwork must be preservative treated. Plastic lumber must not be preservative treated. The following items must be preservative treated:
  - (1) Wood framing, woodwork, and plywood up to and including the subflooring at the first-floor level of structures having crawl spaces when the bottoms of such items are 24 inches or less from the earth underneath.
  - (2) Wood members that are in contact with water.
  - (3) Exterior wood steps, platforms, and railings; and all wood framing of open, roofed structures.
  - (4) Wood sills, soles, plates, furring, and sleepers that are less than 24 inches from the ground, furring and nailers that are set into or in contact with concrete or masonry.
  - (5) Nailers, edge strips, crickets, curbs, and cants for roof decks.

#### 1.7.1 Existing Structures

Use borate, permethrin, or a sodium silicate wood mineralization process to treat wood. Use borate for interior applications only.

#### 1.7.2 New Construction

Use a boron-based preservative conforming to AWP A P18, sodium silicate wood mineralization process, or Ammoniacal Copper Quaternary Compound to treat wood. Use boron-based preservatives for above-ground applications only.

#### 1.8 FIRE-RETARDANT TREATMENT

Fire-retardant treated wood must be pressure treated with fire retardants conforming to AWP A P49. Fire retardant treatment of wood products must conform to the requirements of AWP A U1, Commodity Specification H and AWP A T1, Section H. Treatment and performance inspection must be by an

independent and qualified testing agency that establishes performance ratings. Each piece or bundle of treated material must bear identification of the testing agency to indicate performance in accordance with such rating. Treated materials to be exposed to rain wetting must be subjected to an accelerated weathering technique in accordance with ASTM D2898 prior to being tested. Such items which will not be inside a building, and such items which will be exposed to heat or high humidity, must receive exterior fire-retardant treatment. Fire-retardant-treated wood products must be free of halogens, sulfates, ammonium phosphate, and formaldehyde. Items to be treated include the following:

## 1.9 QUALITY ASSURANCE

### 1.9.1 Drawing Requirements

For fabricated structural members, trusses, glulam members, indicate materials, details of construction, methods of fastening, and erection details. Include reference to design criteria used and manufacturers design calculations. Submit drawings for all proposed modifications of structural members. Do not proceed with modifications until the submittal has been approved.

### 1.9.2 Data Required

Submit calculations and drawings for all proposed modifications of structural members. Do not proceed with modifications until the submittal has been approved.

### 1.9.3 Humidity Requirements

Sequence work to minimize use of temporary HVAC to dry out building and control humidity.

### 1.9.4 Plastic Lumber Performance

Plastic lumber intended for use in exterior applications must have no fading or discoloration and no change in dimensional stability as tested in accordance with ASTM D1435 for a period of 5 years.

## 1.10 ENVIRONMENTAL REQUIREMENTS

During and immediately after installation of treated wood, engineered wood products, and laminated wood products at interior spaces, provide temporary ventilation.

## 1.11 CERTIFICATIONS

### 1.11.1 Indoor Air Quality Certifications

Submit required indoor air quality certifications in one submittal package.

#### 1.11.1.1 Adhesives and Sealants

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party programs that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor

air quality product requirements cited herein.

#### 1.11.1.2 Composite Wood, Wood Structural Panel and Agrifiber Products

For purposes of this specification, composite wood and agrifiber products include particleboard, medium density fiberboard (MDF), strawboard, panel substrates, and door cores. Provide products certified to meet requirements of both 40 CFR 770 and CARB 93120. Provide current product certification documentation from certification body.

### PART 2 PRODUCTS

#### 2.1 MATERIALS

##### 2.1.1 Virgin Lumber

Lumber fabricated from old growth timber is not permitted. Avoid companies who buy, sell, or use old growth timber in their operations, when possible.

##### 2.1.2 Plastic Lumber

HDPE lumber must contain a minimum of 90 percent total recycled content. Mixed plastics and cellulose lumber must contain a minimum of 100 percent total recovered materials content, with a minimum of 50 percent post-consumer recycled content. HDPE/fiberglass lumber must contain a minimum of 95 percent total recovered materials content with a minimum of 75 percent post-consumer recycled content. Other mixed resin lumber must contain a minimum of 95 percent total recovered materials content with a minimum of 50 percent post-consumer recycled content.

##### 2.1.2.1 Shear Parallel to Length

Maximum 1,000 psi in accordance with ASTM D2344/D2344M.

##### 2.1.2.2 Density

ASTM D6111.

##### 2.1.2.3 Compressive Strength

- a. Secant Modulus: Minimum 70,000 psi in accordance with ASTM D6108.
- b. Stress at 3 percent strain: Minimum 1,500 psi in accordance with ASTM D6108.
- c. Compression Parallel to Grain: Minimum 3,000 psi in accordance with ASTM D6112.
- d. Compression Perpendicular to Grain: Minimum 1,000 psi in accordance with ASTM D6112.

##### 2.1.2.4 Flexural Strength

Minimum 2,000 psi in accordance with ASTM D6109.

2.1.2.5 Tensile Strength

Minimum 1,250 psi in accordance with ASTM D198.

2.1.2.6 Coefficient of Thermal Expansion

Maximum 0.000080 in/in/degree F in accordance with ASTM D696.

2.1.2.7 Screw Withdrawal

350 lbs in accordance with ASTM D6117.

2.1.2.8 Nail Withdrawal

150 lbs in accordance with ASTM D6117.

2.2 LUMBER

2.2.1 Framing Lumber

Framing lumber such as studs, plates, caps, collar beams, cant strips, bucks, sleepers, nailing strips, and nailers and board lumber such as subflooring and wall and roof sheathing must be one of the species listed in the table below. Minimum grade of species must be as listed.

| Table of Grades for Framing and Board Lumber |                                                                                                                                                                                                                                                                                                                             |                                                                                                                              |                           |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Grading Rules                                | Species                                                                                                                                                                                                                                                                                                                     | Framing                                                                                                                      | Board Lumber              |
| WWPA G-5 standard grading rules              | Aspen, Douglas Fir-Larch, Douglas Fir South, Engelmann Spruce-Lodgepole Pine, Engelmann Spruce, Hem-Fir, Idaho White Pine, Lodgepole Pine, Mountain Hemlock, Mountain Hemlock-Hem-Fir, Ponderosa Pine-Sugar Pine, Ponderosa Pine-Lodgepole Pine, Subalpine Fir, White Woods, Western Woods, Western Cedars, Western Hemlock | All Species: Standard Light Framing or No. 3 Structural Light Framing (Stud Grade for 2x4 nominal size, 10 feet and shorter) | All Species: No. 3 Common |
| WCLIB 17 standard grading rules              | Douglas Fir-Larch, Hem-Fir, Mountain Hemlock, Sitka Spruce, Western Cedars, Western Hemlock                                                                                                                                                                                                                                 | All Species: Standard Light Framing or No. 3 Structural Light Framing (Stud Grade for 2x4 nominal size, 10 feet and shorter) | All Species: Standard     |



| <u>Table of Grades for Framing and Board Lumber</u> |                                                                                                              |                                                                                                                                 |                                                                                  |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <u>Grading Rules</u>                                | <u>Species</u>                                                                                               | <u>Framing</u>                                                                                                                  | <u>Board Lumber</u>                                                              |
| SPIB 1003 standard grading rules                    | Southern Pine                                                                                                | All Species:<br>Standard Light Framing or No. 3 Structural Light Framing (Stud Grade for 2x4 nominal size, 10 feet and shorter) | No. 2 Boards                                                                     |
| SCMA Spec standard specifications                   | Cypress                                                                                                      | No. 2 Common                                                                                                                    | No. 2 Common                                                                     |
| NELMA Grading Rules standard grading rules          | Balsam Fir, Eastern Hemlock-Tamarack, Eastern Spruce, Eastern White Pine, Northern Pine, Northern Pine-Cedar | All Species:<br>Standard Light Framing or No. 3 Structural Light Framing (Stud Grade for 2x4 nominal size, 10 feet and shorter) | All Species:<br>No. 3 Common except Standard for Eastern White and Northern Pine |
| RIS Grade Use standard specifications               | Redwood                                                                                                      | All Species:<br>Standard Light Framing or No. 3 Structural Light Framing (Stud Grade for 2x4 nominal size, 10 feet and shorter) | Construction Heart                                                               |

| Table of Grades for Framing and Board Lumber                                       |         |                 |              |
|------------------------------------------------------------------------------------|---------|-----------------|--------------|
| Grading Rules                                                                      | Species | Framing         | Board Lumber |
| NHLA Rules rules for the measurement and inspection of hardwood and cypress lumber | Cypress | No. 2 Dimension | No. 2 Common |

## 2.3 PLYWOOD, STRUCTURAL-USE, AND ORIENTED STRAND BOARD (OSB) PANELS

APA L870, APA S350, APA E445, and APA F405 respectively.

### 2.3.1 Subflooring

#### 2.3.1.1 Plywood

C-D Grade, Exposure 1 durability classification, Span rating of 24/16 or greater.

#### 2.3.1.2 Structural-Use and OSB Panels

Sheathing grade with durability equivalent to Exposure 1, Span Rating of 32/16 or greater. OSB, APA E445, Rated Sturd-I-Floor.

### 2.3.2 Wall Sheathing

#### 2.3.2.1 Plywood

C-D Grade, Exposure 1, and a minimum thickness of 1/2 inch, except where indicated to have greater thickness. Provide exterior grade material with phenol resin for interior and exterior applications.

#### 2.3.2.2 Structural-Use and OSB Panels

Sheathing grade with durability equivalent to Exposure 1, Span Rating of 16/0 or greater. OSB, APA Rated Sheathing. OSB must be a phenolic-glued board.

### 2.3.3 Roof Sheathing

#### 2.3.3.1 Plywood

C-D Grade, Exposure 1, with an Identification Index of not less than 24/0. Provide exterior grade material with phenol resin for all applications.

### 2.3.4 Other Uses

#### 2.3.4.1 Plywood

Plywood for electrical panel backing. B-C Grade, Exposure 1, Fire Retardant Treated on 2x wood sleepers attached to wall.

## 2.4 UNDERLAYMENT

Underlayment must conform to one of the following:

### 2.4.1 Hardboard

ANSI/CPA A135.4 service class, sanded one side, 1/4 inch thick, 4 feet wide.

### 2.4.2 Plywood

Plywood must conform to APA L870, underlayment grade with exterior glue, or C-C (Plugged) exterior grade 11/32 inch thick, 4 feet wide.

### 2.4.3 Oriented Strand Board

OSB underlayment grade 0.225 inch.

## 2.5 OTHER MATERIALS

### 2.5.1 Hardboard Underlayment

DOC/NIST PS58, service class, sanded on one side, 1/4 inch thick 4 feet wide.

### 2.5.2 Building Paper

FS UU-B-790, Type I, Grade D, Style 1.

### 2.5.3 Miscellaneous Wood Members

#### 2.5.3.1 Nonstress Graded Members

Members must include bridging, corner bracing, furring, grounds, and nailing strips. Members must be in accordance with TABLE I for the species used. Sizes must be as follows unless otherwise shown:

| Member         | Size inch                                                                                                    |
|----------------|--------------------------------------------------------------------------------------------------------------|
| Bridging       | 1 x 3 or 1 x 4 for use between members 2 x 12 and smaller; 2 x 4 for use between members larger than 2 x 12. |
| Corner bracing | 1 x 4.                                                                                                       |
| Furring        | 1 x 3                                                                                                        |
| Grounds        | Plaster thickness by 38.                                                                                     |

| Member         | Size inch                                                                            |
|----------------|--------------------------------------------------------------------------------------|
| Nailing strips | 1 x 3 or 1 x 4 when used as shingle base or interior finish, otherwise 2 inch stock. |

#### 2.5.3.2 Sill Plates

Sill plates must be standard or number 2 grade.

#### 2.5.3.3 Blocking

Blocking must be standard or number 2 grade.

#### 2.5.3.4 Rough Bucks and Frames

Rough bucks and frames must be straight standard or number 2 grade.

#### 2.5.4 Adhesives

Comply with applicable regulations regarding toxic and hazardous materials and as specified. Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide aerosol adhesives used on the interior of the building meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of GS-36.

### 2.6 ROUGH HARDWARE

Unless otherwise indicated or specified, rough hardware must be of the type and size necessary for the project requirements. Sizes, types, and spacing of fastenings of manufactured building materials must be as recommended by the product manufacturer unless otherwise indicated or specified. Rough hardware exposed to the weather or embedded in or in contact with preservative treated wood, exterior masonry, or concrete walls or slabs must be hot-dip zinc-coated in accordance with ASTM A153/A153M. Nails and fastenings for fire-retardant treated lumber and woodwork exposed to the weather must be copper alloy or hot-dipped galvanized fasteners as recommended by the treated wood manufacturer.

#### 2.6.1 Bolts, Nuts, Studs, and Rivets

ASME B18.2.1, ASME B18.5.2.1M, ASME B18.5.2.2M and ASME B18.2.2.

#### 2.6.2 Anchor Bolts

ASTM A307, size as indicated, complete with nuts and washers.

#### 2.6.3 Expansion Shields

CID A-A-1923, CID A-A-1924, and CID A-A-1925. Except as shown otherwise, maximum size of devices must be 3/8 inch.

#### 2.6.4 Lag Screws and Lag Bolts

ASME B18.2.1.

#### 2.6.5 Wood Screws

ASME B18.6.1.

#### 2.6.6 Nails and Staples

ASTM F547, size and type best suited for purpose; staples must be as recommended by the manufacturer of the materials to be joined. For sheathing and subflooring, length of nails must be sufficient to extend 1 inch into supports. In general, 8-penny or larger nails must be used for nailing through 1 inch thick lumber and for toe nailing 2 inch thick lumber; 16-penny or larger nails must be used for nailing through 2 inch thick lumber. Nails used with treated lumber and sheathing must be hot-dipped galvanized in accordance with ASTM A153/A153M. Nailing must be in accordance with the recommended nailing schedule contained in AWC WFCM. Where detailed nailing requirements are not specified, nail size and spacing must be sufficient to develop an adequate strength for the connection. The connection's strength must be verified against the nail capacity tables in AWC NDS. Reasonable judgment backed by experience must ensure that the designed connection will not cause the wood to split. If a load situation exceeds a reasonable limit for nails, a specialized connector must be used.

#### 2.6.7 Wire Nails

ASTM F1667/F1667M.

#### 2.6.8 Timber Connectors

Unless otherwise specified, timber connectors must be in accordance with TPI 1, APA EWS T300 or AITC TCM.

#### 2.6.9 Clip Angles

Steel, 3/16 inch thick, size best suited for intended use; or zinc-coated steel or iron commercial clips designed for connecting wood members.

#### 2.6.10 Joist Hangers

Steel or iron, zinc coated, sized to fit the supported member, of sufficient strength to develop the full strength of the supported member in accordance with ICC IBC, and furnished complete with any special nails required.

#### 2.6.11 Tie Straps

For joists supported by the lower flange of steel beams, provide 1/8 by 1-1/2 inch steel strap, 2 feet long, except as indicated otherwise.

#### 2.6.12 Joist Anchors

For joists supported by masonry walls, provide anchors 3/16 by 1 1/2 inch steel tee or strap, bent and of length to provide 4 inches embedment into wall and 12 inches along joist except as indicated otherwise. For joists

parallel to masonry or concrete walls, provide anchors 1/4 by 1-1/4 inch minimum cross-sectional area, steel strap, length as necessary to extend over top of first three joists and into wall 4 inches, and with wall end of bend or pin type , except as indicated otherwise.

#### 2.6.13 Door Buck Anchors

Metal anchors, 1/8 by 1-1/4 inch steel, 12 inches long, with ends bent 2 inches , except as indicated otherwise. Anchors must be screwed to the backs of bucks and built into masonry or concrete. Locate 8 inches above sills and below heads and not more than 24 inches intermediately between. Anchorage of bucks to steel framing must be as necessary to suit the conditions.

#### 2.6.14 Toothed Rings and Shear Plates

AWC NDS.

#### 2.6.15 Beam Anchors

Steel U-shaped strap anchors 1/4 inch thick by 1-1/2 inches wide , except as indicated otherwise.

#### 2.6.16 Metal Framing Anchors

Construct anchors to the configuration shown using hot dip zinc-coated steel conforming to ASTM A653/A653M, G90. Except where otherwise shown, Steel must be not lighter than 18 gage. Special nails supplied by the manufacturer must be used for all nailing.

#### 2.6.17 Panel Edge Clips

Extruded aluminum or galvanized steel, H-shaped clips to prevent differential deflection of roof sheathing.

### 2.7 AIR INFILTRATION BARRIER

Air infiltration barrier must be building paper meeting the requirements of ASTM C1136, Type IV, style optional or a tear and puncture resistant olefin building wrap (polyethylene or polypropylene) with a moisture vapor transmission rate of 163 g per square meter per 24 hours in accordance with ASTM E96/E96M, Method A.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Do not install building construction materials that show visual evidence of biological growth.

Conform to AWC WFCM and install in accordance with the National Association of Home Builders (NAHB) Advanced Framing Techniques: Optimum Value Engineering, unless otherwise indicated or specified. Select lumber sizes to minimize waste. Fit framing lumber and other rough carpentry, set accurately to the required lines and levels, and secure in place in a rigid manner. Space plastic lumber boards as necessary to allow for lengthwise expansion and contraction. Do not splice framing members between bearing points. Set joists, rafters, and purlins with their crown edge up. Frame members for the passage of pipes, conduits, and ducts.

Provide adequate support as appropriate to the application, climate, and modulus of elasticity of the product. Do not cut or bore structural members for the passage of ducts or pipes without approval. Reinforce all members damaged by such cutting or boring by means of specially formed and approved sheet metal or bar steel shapes, or remove and provide new, as approved. Provide as necessary for the proper completion of the work all framing members not indicated or specified. Spiking and nailing not indicated or specified otherwise must be in accordance with the Nailing Schedule contained in ICC IBC; perform bolting in an approved manner. Spikes, nails, and bolts must be drawn up tight. Install plastic lumber with screws or bolts; if nails are used, use ring shank or spiral shank nails. Use slate or steel shims when leveling joists, beams, and girders on masonry or concrete. Do not use shimming on wood or metal bearings. When joists, beams, and girders are placed on masonry or concrete, a wood base plate must be positioned and leveled with grout. The joist, beam, or girder must then be placed on the plate. When joists, beams, and girders are set into masonry or concrete, a pocket must be formed into the wall. The joist, beam, or girder must then be placed into the pocket and leveled with a steel shim.

### 3.1.1 Sills

Set sills level and square and wedge with steel or slate shims; point or grout with non-shrinking cement mortar to provide continuous and solid bearing. Anchor sills to the foundations as indicated. Where sizes and spacing of anchor bolts are not indicated, provide not less than 5/8 inch diameter bolts at all corners and splices and space at a maximum of 6 feet o.c. between corner bolts. Provide at least two bolts for each sill member. Lap and splice sills at corners and bolt through the laps or butt the ends and through-bolt not more than 6 inches from the ends. Provide bolts with plate washers and nuts. Bolts in exterior walls must be zinc-coated.

#### 3.1.1.1 Anchors in Masonry

Except where indicated otherwise, Embed anchor bolts not less than 15 inches in masonry unit walls and provide each with a nut and a 2 inch diameter washer at bottom end. Fully grout bolts with mortar.

#### 3.1.1.2 Anchors in Concrete

Except where indicated otherwise, Embed anchor bolts not less than 8 inches in poured concrete walls and provide each with a nut and a 2 inch diameter washer at bottom end. A bent end may be substituted for the nut and washer; bend must be not less than 90 degrees. Powder-actuated fasteners spaced 3 feet o.c. may be provided in lieu of bolts for single thickness plates on concrete.

### 3.1.2 Wall Sheathing

#### 3.1.2.1 Plywood, Structural-Use, and OSB Panel Wall Sheathing

Apply horizontally or vertically. Extend sheathing over and nail to sill and top plate. Abut sheathing edges over centerlines of supports. Allow 1/8 inch spacing between panels and 1/8 inch at windows and doors. If sheathing is applied horizontally, stagger vertical end joints. Nail panels with 6-penny nails spaced 6 inches o.c. along edges of the panel and 12 inches o.c. over intermediate supports. Keep nails 3/8 inches away from panel ledges. Provide 2 by 4 blocking for horizontal edges not

otherwise supported.

### 3.1.3 Wood Sheathing

Sheathing end joints must be made over framing members and so alternated that there will be at least two boards between joints on the same support. Each board must bear on at least three supports. Boards must be nailed at each support using two nails for boards 6 inches and less in width and three nails for boards more than 6 inches in width. Roof sheathing must not be installed where roof decking is installed.

### 3.1.4 Building Paper

Provide building paper on wood board sheathing for all types of exterior siding. Apply paper shingle fashion, horizontally, beginning at the bottom of the wall. Lap edges 4 inches, and nail with one inch, zinc-coated roofing nails, spaced 12 inches o.c. and driven through tin discs.

### 3.1.5 Ceiling Joists

Size as indicated and set accurately and in alignment. Toe-nail joists to all plates with not less than three 10-penny nails. Frame openings in ceilings with headers and trimmers.

### 3.1.6 Plywood Roof Sheathing

Install with the grain of the outer plies or long dimension at right angles to supports. Stagger end joints and locate over the centerlines of supports. Allow 1/8 inch spacing at panel ends and 1/4 inch at panel edges. Nail panels with 8-penny common nails or 6-penny annular rings or screw-type nails spaced 6 inches o.c. at supported edges and 12 inches o.c. at intermediate bearings. Do not use staples in roof sheathing. Where the support spacing exceeds the maximum span for an unsupported edge, provide adequate blocking, tongue-and-groove edges, or panel edge clips, in accordance with APA E30.

### 3.1.7 Plastic Lumber

In conjunction with above requirements, follow manufacturer's recommendations for plastic lumber installation, including requirements for structural support, thermal movement, working, fastening, and finishing. Use standard woodworking tools, including carbide tips, coarse saw blades, and routers with aggressive cutters. Follow manufacturer's recommendations for repair by melting.

## 3.2 MISCELLANEOUS

### 3.2.1 Rough Wood Bucks

2 inch nominal thickness. Set wood bucks true and plumb. Anchor bucks to concrete or masonry with steel straps extending into the wall 8 inches minimum. Place anchors near the top and bottom of the buck and space uniformly at 2 foot maximum intervals.

### 3.2.2 Wood Blocking

Provide proper sizes and shapes at proper locations for the installation and attachment of wood and other finish materials, fixtures, equipment,



and items indicated or specified.

### 3.2.3 Wood Grounds

Provide for fastening wood trim, finish materials, and other items to plastered walls and ceilings. Install grounds in proper alignment and true with an 8 foot straightedge.

### 3.2.4 Wood Furring

Provide where shown and as necessary for facing materials specified. Except as shown otherwise, furring strips must be nominal one by 3, continuous, and spaced 16 inches o.c. Erect furring vertically or horizontally as necessary. Nail furring strips to masonry. Do not use wood plugs. Provide furring strips around openings, behind bases, and at angles and corners. Furring must be plumb, rigid, and level and must be shimmed as necessary to provide a true, even plane with surfaces suitable to receive the finish required. Form furring for offsets and breaks in walls or ceilings on 1 by 4 wood strips spaced 16 inches o.c.

### 3.2.5 Temporary Closures

Provide with hinged doors and padlocks and install during construction at exterior doorways and other ground level openings that are not otherwise closed. Cover windows and other unprotected openings with polyethylene or other approved material, stretched on wood frames. Provide dustproof barrier partitions to isolate areas as directed.

### 3.2.6 Wood Sleepers

Run wood sleepers in lengths as long as practicable and stagger end joints in adjacent rows.

### 3.2.7 Sill Plates

Sill plates must be set level and square and anchor bolted at not more than 6 feet on centers and not more than 12 inches from end of each piece. A minimum of two anchors must be used for each piece.

## 3.3 ERECTION TOLERANCES

- a. Framing members which will be covered by finishes such as wallboard, plaster, or ceramic tile set in a mortar setting bed, must be within the following limits:
  - (1) Layout of walls and partitions: 1/4 inch from intended position;
  - (2) Plates and runners: 1/4 inch in 8 feet from a straight line;
  - (3) Studs: 1/4 inch in 8 feet out of plumb, not cumulative; and
  - (4) Face of framing members: 1/4 inch in 8 feet from a true plane.
- b. Framing members which will be covered by ceramic tile set in dry-set mortar, latex-portland cement mortar, or organic adhesive must be within the following limits:
  - (1) Layout of walls and partitions: 1/4 inch from intended position;

- (2) Plates and runners: 1/8 inch in 8 feet from a straight line;
- (3) Studs: 1/8 inch in 8 feet out of plumb, not cumulative; and
- (4) Face of framing members: 1/8 in 8 feet from a true plane.

### 3.4 WASTE MANAGEMENT OF WOOD PRODUCTS

In accordance with the Waste Management Plan and as specified. Clearly separate damaged wood and other scrap lumber for acceptable alternative uses on site, including bracing, blocking, cripples, ties, and shims.

Separate treated, stained, painted, and contaminated wood and place in designated area for hazardous materials. Dispose of according to local regulations. Do not leave any wood, shavings, sawdust, or other wood waste buried in fill or on the ground. Prevent sawdust and wood shavings from entering the storm drainage system. Do not burn scrap lumber that has been pressure treated, or lumber that is less than one year old.

### 3.5 SCHEDULE

Some metric measurements in this section are based on mathematical conversion of inch-pound measurements. Typical conversion is as shown:

| <u>PRODUCTS</u> | <u>INCH-POUND Nominal</u> | <u>METRIC Conversion</u> |
|-----------------|---------------------------|--------------------------|
| Sawn lumber     | 2 by 4                    | 38 by 89 mm              |
|                 | 1 by                      | 19 mm by                 |
| Stud spacing    | 16 inches                 | 400 mm                   |
|                 | If not 48 inches panel    | 406 mm                   |
| Plywood         | 48 by 96 inches           | 1200 mm by 2400 mm       |

-- End of Section --

SECTION 06 20 00

FINISH CARPENTRY  
**08/24**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN LUMBER STANDARDS COMMITTEE (ALSC)

ALSC PS 20 (2021) American Softwood Lumber Standard

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.2.1 (2012; R 2021) Square, Hex, Heavy Hex, and Askew Head Bolts and Hex, Heavy Hex, Hex Flange, Lobed Head, and Lag Screws (Inch Series)

ASME B18.2.2 (2022) Nuts for General Applications: Machine Screw Nuts, and Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)

ASME B18.6.1 (2016) Wood Screws (Inch Series)

AMERICAN WOOD PROTECTION ASSOCIATION (AWPA)

AWPA M4 (2023) Standard for the Care of Preservative-Treated Wood Products

AWPA U1 (2024) Use Category System: User Specification for Treated Wood

APA - THE ENGINEERED WOOD ASSOCIATION (APA)

APA L870 (2010) Voluntary Product Standard, PS 1-09, Structural Plywood

ARCHITECTURAL WOODWORK INSTITUTE (AWI)

ANSI/AWI 0400 (2022) Factory Finishing Standard

ANSI/AWI 0620 (2018) Finish Carpentry/Installation Standard

AWI 100 (2018) Architectural Woodwork Institute Submittal Standards

AWI 200 (2018) Architectural Woodwork Institute Care & Storage Standards

ASTM INTERNATIONAL (ASTM)

ASTM D2898 (2010; R 2017) Standard Practice for Accelerated Weathering of Fire-Retardant-Treated Wood for Fire Testing

ASTM F547 (2017) Standard Terminology of Nails for Use with Wood and Wood-Base Materials

HARDWOOD PLYWOOD AND VENEER ASSOCIATION (HPVA)

ANSI/HPVA HP-1 (2020) American National Standard for Hardwood and Decorative Plywood

MOULDING AND MILLWORK PRODUCERS ASSOCIATION (MMPA)

WMMPA WM 6 Quality Industry Standards Booklet

NATIONAL HARDWOOD LUMBER ASSOCIATION (NHLA)

NHLA Rules (2023) Rules for the Measurement & Inspection of Hardwood & Cypress

NORTHEASTERN LUMBER MANUFACTURERS ASSOCIATION (NELMA)

NELMA Grading Rules (2021) Standard Grading Rules for Northeastern Lumber

REDWOOD INSPECTION SERVICE (RIS) OF THE WESTERN WOOD PRODUCTS ASSOCIATION (WWPA)

RIS Grade Use (2019) California Redwood Lumber Grades and Uses

SOUTHERN PINE INSPECTION BUREAU (SPIB)

SPIB 1003 (2021) Standard Grading Rules for Southern Pine Lumber

WEST COAST LUMBER INSPECTION BUREAU (WCLIB)

WCLIB 17 (2015; R 2018) Standard Grading Rules for West Coast Lumber

WESTERN WOOD PRODUCTS ASSOCIATION (WWPA)

WWPA G-5 (2021) Western Lumber Grading Rules

WINDOW AND DOOR MANUFACTURERS ASSOCIATION (WDMA)

WDMA I.S.4 (2019) Preservative Treatment for Millwork

WOODWORK INSTITUTE (WI)

NAAWS 4.0 (2021 Errata Edition) North American Architectural Woodwork Standards

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

Submittals to follow AWI 100 2018 standards.

### SD-02 Shop Drawings

Detail Drawings Indicating All Wood Assemblies; G

### SD-03 Product Data

Wood Products; G

Countertops; G

Engineered Wood Products; G

Treated Wood Products; G

Soffits; G

Fascias and Trim; G

Hardware and Accessories; G

### SD-04 Samples

Samples; G

### SD-07 Certificates

Certificates of Grade; G

## 1.3 DETAIL DRAWINGS

Submit detail drawings indicating all wood assemblies proposed for use in the project. Indicate materials, species, grade, density, grain, finish details of construction, location of use in the project, finishes, types, method and arrangement of fasteners, and installation details. This includes all fabricated assemblies.

## 1.4 PRODUCT DATA

Submit Manufacturers printed data including proposed species, grade, density grain, and finish as applicable; sufficient to demonstrate compliance with this specification for each type of wood product specified. For treated wood products also provide documentation of environmentally safe preservatives for each type of wood product specified.

Provide Manufacturers printed data for hardware and all wood accessories including but not limited to edge banding, adhesives, and sealers.

## 1.5 SAMPLES

Samples indicating proposed species, grade, density grain, and finish for

each type of wood product specified. Provide samples of sufficient size to show pattern and color ranges of proposed products.

#### 1.6 DELIVERY, STORAGE, AND HANDLING

Deliver wood products to the jobsite in an undamaged condition. Stack materials to ensure ventilation and drainage. Protect against dampness before and after delivery. Store materials under cover in a well ventilated enclosure and protect against extreme changes in temperature and humidity. Keep materials wrapped and separated from off-gassing materials (such as drying paints and adhesives). Do not use materials that have visible moisture or biological growth. Do not store products in building until wet trade materials are dry and humidity of the space is within wood manufacturer's tolerance limits for storage. Follow AWI 200 for care and storage.

#### 1.7 QUALITY ASSURANCE

##### 1.7.1 Certifications

###### 1.7.1.1 Certified Wood Grades

Provide certificates of grade from the grading agency on graded but unmarked lumber or plywood attesting that materials meet the grade requirements specified herein.

##### 1.7.2 Lumber

Identify each piece or each bundle of lumber, millwork, and trim by the grade mark of a recognized association or independent inspection agency certified by the Board of Review of the ALSC to grade the species.

##### 1.7.3 Plywood

Provide each sheet of plywood with the mark of a recognized association or independent inspection agency that maintains continuing control over the quality of the plywood. Provide marks that identify plywood by species group or span rating, exposure durability classification, grade, and compliance with APA L870.

##### 1.7.4 Hardboard

Provide materials marks or written documentation identifying the producer and the applicable standard.

##### 1.7.5 Pressure Treated Lumber and Plywood

Inspect each treated piece in accordance with AWWA U1.

##### 1.7.6 Non-Pressure Treated Woodwork and Millwork

Mark, stamp, or label to indicate compliance with WDMA I.S.4.

##### 1.7.7 Fire-Retardant Treated Lumber

Confirm each piece has an Underwriters Laboratories fire resistance label or comparable label of another nationally recognized independent fire retardant materials testing laboratory.

## PART 2 PRODUCTS

### 2.1 WOOD PRODUCTS

#### 2.1.1 Sizes and Patterns of Wood Products

Provide yard and board lumber sizes in accordance with ALSC PS 20. Provide shaped lumber and millwork in the patterns indicated and in standard patterns of the association covering the species. Size references, unless otherwise specified, are nominal sizes. Provide actual sizes within manufacturing tolerances allowed by the applicable standard.

#### 2.1.2 Species and Grades

Provide in accordance with AWWA U1 Use Category System Tables unless otherwise specified herein.

#### 2.1.3 Trim, Finish, and Frames

Provide species and grades listed in the table below for all painted wood materials. Provide materials one grade higher than those listed in the table below for all stained materials with a natural or transparent finish. Provide trim, except window stools and aprons with hollow backs.

| TABLE OF GRADES FOR WOOD TO RECEIVE PAINT FINISH |                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                 |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grading Rules                                    | Species                                                                                                                                                                                                                                                                                                          | Exterior and Interior Trim, Finish, and Frames                                                                                                                                                  |
| WWPA G-5 standard grading rules                  | Aspen, Douglas Fir-Larch, Douglas Fir South, Engelmann Spruce-Lodgepole Pine, Engelmann Spruce, Hem-Fir, Idaho White Pine, Lodgepole Pine, Mountain Hemlock, Mountain Hemlock-Hem-Fir, Ponderosa Pine-Sugar Pine, (Ponderosa Pine-Lodgepole Pine,) White Woods, (Western Woods,) Western Cedars, Western Hemlock | All Species: C & BTR. Select (Choice & BTR Idaho White Pine) or Superior Finish. Western Red Cedar can be graded C & BTR. Select or A & BTR in accordance with Special Western Red Cedar Rules. |
| WCLIB 17 standard grading rules                  | Douglas Fir-Larch, Hem-Fir, Mountain Hemlock, Sitka Spruce, Western Cedars, Western Hemlock                                                                                                                                                                                                                      | All Species: C & BTR VG, except A for Western Red Cedar                                                                                                                                         |
| SPIB 1003 standard grading rules                 | Southern Pine                                                                                                                                                                                                                                                                                                    | C & BTR                                                                                                                                                                                         |

| TABLE OF GRADES FOR WOOD TO RECEIVE PAINT FINISH |                                                                                                                              |                                                                             |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Grading Rules                                    | Species                                                                                                                      | Exterior and Interior Trim, Finish, and Frames                              |
| NHLA Rules                                       | Cypress                                                                                                                      | C-Select                                                                    |
| NELMA Grading Rules standard grading rules **    | Balsam Fir, Eastern Hemlock-Tamarack, Eastern Spruce, Eastern White Pine, Northern Pine, Northern Pine, Northern White Cedar | All Species: C-Select except C & BTR for Eastern White Pine and Norway Pine |
| RIS Grade Use standard specifications            | Redwood                                                                                                                      | Clear, Clear All Heart                                                      |
| NHLA Rules                                       | Cypress                                                                                                                      | B Finish                                                                    |
|                                                  | Red Gum, Soft Elm, Birch                                                                                                     | Select or BTR (for interior use only)                                       |

Note: \*\* <https://www.nelma.org/the-grade-rule-book/>

#### 2.1.4 Utility Shelving

Provide utility shelving in a suitable species equal to or exceeding the requirements of No. 3 common white fir under WPA G-5, 1 inch thick; or plywood, interior type, Grade A-B, 1/2 inch thick, any species group.

#### 2.1.5 Hardwood Plywood

ANSI/HPVA HP-1, Type I (Exterior) Common Veneer Species Grade A-C,; Decorative Softwood Grade D,; Lumber Cores Clear Grade, Regular Grade, of thickness indicated, to be fire rated treated wood (FRTW).

### 2.2 SOFFITS

### 2.3 FASCIAS AND TRIM

#### 2.3.1 Wood

Provide species and grades for all fascia and trim, including exterior door and window casings, in accordance with AWP U1 Use Category System Tables. Provide sizes indicated. Metal corners can be provided in lieu of wood corner boards for horizontal siding. If metal corners are used, provide galvanized steel or aluminum, completely coated with primer compatible for the specific metal substrate.

### 2.4 MOISTURE CONTENT OF WOOD PRODUCTS

Air dry or kiln dry lumber. Kiln dry treated lumber after treatment. All wood products at the time of delivery to the jobsite, and when installed to meet the maximum moisture contents listed below.



- a. Interior Paneling: 12 percent.
- b. Interior Finish Lumber, Trim, and Millwork: 1-1/4 Inches Nominal or Less in Thickness: 12 percent on 85 percent of the pieces and 15 percent on remainder.

## 2.5 PRESERVATIVE TREATMENT OF WOOD PRODUCTS

### 2.5.1 Non-Pressure Treatment

Treat woodwork and millwork, such as cabinets, exterior trim, door trim, and window trim, in accordance with WDMA I.S.4, with either 2 percent copper naphthenate, 3 percent zinc naphthenate, or 1.8 percent copper-8-quinolinolate. Provide a liberal brush coat of preservative treatment to field cuts and holes.

### 2.5.2 Pressure Treatment

Treat lumber and plywood used on the exterior of buildings or in contact with masonry or concrete with a waterborne preservative listed in AWP A U1 (P series is included therein by reference) as applicable, and inspected in accordance with AWP A U1. Identify treatment on each piece of material by the quality mark of an agency accredited by the Board of Review of the American Lumber Standards Committee. Provide treated plywood to a reflection level as follows:

Provide all exterior wood moulding and millwork within 18 inches of soil or in contact with water or concrete with preservative treatment in accordance with WMP A WM 6. Provide a field treatment in accordance with AWP A M4 of exposed areas of treated wood that have been cut or drilled. Items of all-heart material of cedar, cypress, or redwood do not require preservative treatment except when in direct contact with soil.

## 2.6 FIRE-RETARDANT TREATMENT

### 2.6.1 Wood Products

Provide a qualified independent testing agency that establishes performance ratings to inspect treatment and performance. Pressure treat fire-retardant treated lumber and plywood in accordance with AWP A U1. Comply with material use as defined in AWP A U1 for Interior Type A and B and Exterior Type. Subject treated materials that will be exposed to rain wetting to an accelerated weathering technique in accordance with ASTM D2898, Method A, prior to being tested for compliance with AWP A U1. For each piece or bundle of treated material, provide rating identification of the testing agency to indicate performance.

## 2.7 HARDWARE AND ACCESSORIES

Provide sizes, types, and spacings of hardware and accessories as recommended in writing by the wood product manufacturer, except as otherwise specified.

### 2.7.1 Wood Screws

ASME B18.6.1.

## 2.7.2 Bolts, Nuts, Lag Screws, and Studs

ASME B18.2.1 and ASME B18.2.2.

## 2.7.3 Nails

Use nails of a size and type best suited for each application and in accordance with ASTM F547. Use hot-dipped galvanized or aluminum nails for exterior applications. For siding, provide nails of sufficient length to extend 1-1/2 inches into supports, including wood sheathing over framing. Where nailing is impractical, provide screws of a size and type best suited for each application.

## 2.8 FABRICATION

### 2.8.1 Quality Standards (QS)

#### 2.8.1.1 Grades

The term "Premium" NAAWS 4.0. Provide items not otherwise specified in a specific grade as "Custom" grade.

#### 2.8.1.2 Adhesives

Select adhesives for durability and permanent bonding. Address expansion and contraction, bond strength, fire rating, moisture resistance, and manufacturer's recommendations factors for all bonded materials.

### 2.8.2 Cabinets

Unless specified otherwise, provide wall and base cabinets of the same construction, materials, and finishes as countertops. Fabricate cabinets with solid ends and frame fronts, or with frames all around. Provide frames of solid hardwood not less than 3/4 by 1-1/2 inches. Provide ends, bottoms, backs, partitions, and doors as hardwood plywood. Mortise and tenon, dovetail, or dowel and glue joints to produce a rigid unit. Cover exposed edges of plywood with hardwood strips. Provide cabinet doors, frames, and solid exposed ends 3/4 inch thick minimum. Provide cabinet bottoms, partitions, and framed ends to be 1/2 inch minimum. Provide shelves to be 5/8 inch thick minimum. Provide cabinet backs 1/4 inch thick minimum.

#### 2.8.2.1 Finish

Provide a clear factory finish in accordance with ANSI/AWI 0400 on wood surfaces after fabrication. Provide fabricator's standard natural finish equivalent to one coat of sealer, one coat of varnish on all surfaces and a second coat of varnish on surfaces exposed to view. Provide spar varnish in exterior or wet area applications. Sand lightly and wipe clean between coats.

## PART 3 EXECUTION

Do not install building construction materials that show visual evidence of biological growth. Follow ANSI/AWI 0620 for installation of wood trim, paneling, casework, integrated door systems, countertops and other related interior finishes.

### 3.1 FINISH WORK

Apply primer to finish work before installing. Where practicable, shop assemble and finish millwork items. Construct joints tight and in a manner to conceal shrinkage but to avoid cupping, twisting and warping after installation. Miter trim and mouldings at exterior angles; cope at interior angles and at returns. Provide millwork and trim in maximum practical lengths. Fasten finish work with finish nails. Provide blind nailing where practicable. Set face nails for putty stopping.

#### 3.1.1 Exterior Finish Work

Machine sand exposed flat members and square edges. Machine finish semi-exposed surfaces. Construct joints to exclude water. In addition to nailing, glue joints with waterproof glue as necessary for weather resistant construction. Evenly distribute end joints in built-up members. Provide shoulder joints in flat work. Reinforce backs of wide-faced miters with metal rings and waterproof glue. Unless otherwise indicated, provide fascia and other flat members 3/4 inch thick minimum. Provide door and window trim in single lengths. Provide braced, blocked, and rigidly anchored cornices for support and protection of vertical joints. Provide soffits in largest practical size. Align joints of plywood over centerlines of supports. Fasten soffits with aluminum or stainless steel nails. Back prime all concealed surfaces of exterior trim.

#### 3.1.2 Interior Finish Work

After installation, sand exposed surfaces smooth. Provide window and door trim in single lengths.

#### 3.1.3 Door Frames

Set plumb and square. Provide solid blocking at not more than 16 inches on center for each jamb. Position blocking to occur behind hinges and lock strikes. Double wedge frames and fasten with finish nails. Set nails for putty stopping.

### 3.2 SHELVING

Support 1 inch nominal thick wood shelf material or 3/4 or 23/32 inch thick plywood shelf material with end and intermediate supports arranged to prevent buckling and sagging. Provide hook strips 1 by 4 inches nominal and cleats 1 by 2 inches nominal. Provide cleats except where hook strips are specified or indicated. Where adjustable shelving is indicated, provide standards and brackets or shelf rests for each shelf. Anchor standards to wall at not more than 2 feet on center.

#### 3.2.1 Storage Rooms

Unless otherwise indicated, provide storage rooms with shelves 11-1/4 inches wide, bottom shelf 18 inches above the floor, top shelf 18 inches below the ceiling, and intermediate shelves approximately 18 inches apart.

#### 3.2.2 Room Closets

Provide two shelves 11-1/4 inches wide. Support lower shelf by hook strips at back and ends, and provide full length wood or metal clothes hanger rods unless indicated otherwise.

### 3.3 FASCIA AND EXTERIOR TRIM

Construct, caulk, and machine sand exposed surfaces and edges to exclude water. In addition to nailing, glue joints as necessary for weather resistance. Evenly distribute end joints in built-up members. Shoulder joints in flat work. Reinforce backs of wide-faced miters with metal rings and glue. Provide fascia and other flat members in maximum practicable lengths. Braced, block, and rigidly anchor cornices for support and protection of vertical joints.

### 3.4 MOULDING AND INTERIOR TRIM

Install mouldings and interior trim straight, plumb, level and with closely fitted joints. Provide exposed surfaces machine sanded at the shop. Cope returns and interior angles at moulded items and miter external corners. Shoulder intersections of flatwork to ease any inherent changes in plane. Provide window and door trim in single lengths. Blind nail to the extent practicable. Set and stop face nailing with a nonstaining putty to match the applied finish. Use screws for attachment to metal; set and stop screws in accordance with the same quality requirements for nails.

-- End of Section --

SECTION 07 13 53

ELASTOMERIC SHEET WATERPROOFING  
**02/16, CHG 1: 08/17**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D297       | (2015; R 2019) Rubber Products - Chemical Analysis                                                                                            |
| ASTM D412       | (2016; R 2021) Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension                                             |
| ASTM D471       | (2016a) Standard Test Method for Rubber Property - Effect of Liquids                                                                          |
| ASTM D573       | (2004; R 2019) Standard Test Method for Rubber - Deterioration in an Air Oven                                                                 |
| ASTM D624       | (2000; R 2020) Standard Test Method for Tear Strength of Conventional Vulcanized Rubber and Thermoplastic Elastomers                          |
| ASTM D746       | (2014) Standard Test Method for Brittleness Temperature of Plastics and Elastomers by Impact                                                  |
| ASTM D1149      | (2018) Standard Test Method for Rubber Deterioration - Cracking in an Ozone Controlled Environment                                            |
| ASTM D2240      | (2015; E 2017) Standard Test Method for Rubber Property - Durometer Hardness                                                                  |
| ASTM E96/E96M   | (2024) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials                                      |
| ASTM E154/E154M | (2008a; R 2019) Standard Test Methods for Water Vapor Retarders Used in Contact with Earth Under Concrete Slabs, on Walls, or as Ground Cover |

INTERNATIONAL CODE COUNCIL (ICC)

|         |                                    |
|---------|------------------------------------|
| ICC IBC | (2021) International Building Code |
|---------|------------------------------------|

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-03 Product Data

Manufacturer's Standard Details; G  
Elastomeric Waterproofing Sheet Material; G  
Protection Board; G  
Primers, Adhesives, and Mastics; G

### SD-06 Test Reports

Elastomeric Waterproofing Sheet Material; G  
Field Quality Control documentation; G  
Protective Covering; G

### SD-07 Certificates

Elastomeric Waterproofing Sheet Material; G  
Primers, Adhesives, and Mastics; G  
Protective Coverings; G  
Draft Special Warranties; G  
Final Special Warranties; G

### SD-08 Manufacturer's Instructions

Primers, Adhesives, and Mastics; G

## 1.3 MANUFACTURER'S DETAILS

Submit manufacturer's standard details indicating methods of attachment and spacing, transition and termination details, and installation details. Include verification of existing conditions.

## 1.4 PRODUCT DATA

Include data for material descriptions, recommendations for product shelf life, requirements for protective coverings, and manufacturer's Safety Data Sheets (SDS) for primers, adhesives, and mastics.

## 1.5 CODE REQUIREMENTS

Provide membrane waterproofing system in accordance with ICC IBC Section 1805 Dampproofing and Waterproofing.

## 1.6 DELIVERY, STORAGE, HANDLING, IDENTIFICATION

Deliver and store materials in accordance with manufacturer's printed instructions, out of the weather, in manufacturer's original packaging with brand name and product identification clearly marked. Keep materials wrapped and separated from off-gassing materials (such as drying paints and adhesives). Do not use materials that have visible moisture or biological growth. Do not permit unidentified materials in the work area or in the project.

## 1.7 ENVIRONMENTAL CONDITIONS

Do not apply waterproofing during inclement weather or when there is ice, frost, surface moisture, or visible dampness on the surface to receive waterproofing for when ambient and surface temperatures are 40 degrees F or below. The restriction on the application of waterproofing materials when ambient and surface temperatures are below 40 degrees F will be waived if the Contractor devises a means, approved by the Contracting Officer in writing, of maintaining the surface and ambient temperatures above 40 degrees F.

## 1.8 SPECIAL WARRANTIES

### 1.8.1 Guarantee

Guarantee waterproofing membrane installation against failure due to leaks for a period of two years from the date of Beneficial Occupancy. Submit draft and final guarantees in accordance with Sections 01 78 00.00 22 CLOSEOUT SUBMITTALS and 01 78 23 OPERATION AND MAINTENANCE DATA.

### 1.8.2 Warranty

Provide manufacturer's material warranty for all system components for a period of ten years from the date of Beneficial Occupancy. Submit draft and final warranties in accordance with Sections 01 78 00.00 22 CLOSEOUT SUBMITTALS and 01 78 23 OPERATION AND MAINTENANCE DATA.

## PART 2 PRODUCTS

### 2.1 MATERIALS

Provide one of the types of elastomeric waterproofing sheet material and related primers, adhesives, and mastics as specified herein. Ensure compatibility of waterproofing materials with each other, and with materials on which they are applied. Provide materials that comply with applicable requirements cited below when tested in accordance with the referenced ASTM publications.

### 2.2 BUTYL RUBBER SHEETING

Not less than 60 mils minimum thickness.

#### 2.2.1 Butyl Rubber Sheeting Performance Requirements

- a. Thickness Tolerance, ASTM D412: Plus or minus 10 percent.
- b. Specific Gravity, ASTM D297: 1.20, plus or minus 0.05.

- c. Tensile Strength, ASTM D412: 1200 psi minimum.
- d. Tensile Stress at 300 percent elongation, ASTM D412: 600 psi minimum.
- e. Elongation, ASTM D412: 300 percent minimum.
- f. Tear Resistance, Die C, ASTM D624: 150 pound force per inch (lbf/inch) minimum.
- g. Shore A Hardness, ASTM D2240: 5-second interval before reading; 60 plus or minus 10.
- h. Ozone Resistance, ASTM D1149: No cracks, 7 days - 50 pphm - 100 degrees F, 20 percent elongation.
- i. Heating Aging-Accelerated, ASTM D573: Tensile retention, 60 percent of minimum original elongation retention; 60 percent of minimum original requirement; 7 days, 240 degrees F.
- j. Butyl Identification, ASTM D471 REV A, Tricresyl Phosphate Immersion: Maximum volume swell 10 percent, 70 hrs, 212 degrees F.
- k. Low Temperature Flexibility, ASTM D746: No failure at minus 40 degrees F.
- l. Water Absorption, ASTM D471 REV A: Plus 1 percent maximum. 7 days, 158 degrees F.
- m. Exposure to Fungi and Bacteria in Soil, ASTM E154/E154M REV A, Minimum 16 Weeks: Unaffected.
- n. Water Vapor Transmission, 80 degrees F Permeance, ASTM E96/E96M, Procedure B or BW: 0.15 perms maximum.

#### 2.2.2 Adhesive, Cement, and Tape for Use with Butyl Rubber

As recommended by the butyl rubber waterproofing membrane manufacturer.

#### 2.3 Protection Board

Provide protection board that is compatible with the waterproofing membrane. Use a minimum 1/2 inch thick fir bitumen impregnated board 1 inch for polystyrene 1/8 inch thick for vertical and 1/4 inch for horizontal premolded bituminous protection board as recommended by the manufacturer.

### PART 3 EXECUTION

#### 3.1 VERIFICATION OF CONDITIONS

Before starting the work, verify surfaces that must be waterproofed are in satisfactory condition. Notify the Contracting Officer of defects or conditions anticipated to prevent a satisfactory application. Do not start application until defects and conditions have been corrected.

#### 3.2 SURFACE PREPARATION

Ensure surfaces to receive treatment are clean, dry, smooth, and free from deleterious materials and projections. Thoroughly wet holes, joints, cracks, and voids in concrete with water and fill with Portland cement



mortar, strike flush, and permit to dry. Cut off high spots or grind smooth. Finish top surfaces of projecting masonry or concrete ledges below grade, except footings, to a steep bevel with Portland cement mortar. Sweep surfaces to receive covering before applying waterproofing to remove dust and foreign matter. Cure concrete by a method compatible with the waterproofing system.

### 3.3 APPLICATION

#### 3.3.1 Building Envelope Requirements

Provide a continuous waterproofing system at all material and building transitions. Lap, wrap, fasten and seal products in accordance with manufacturer's printed instructions. Envelope assembly variations are not permitted without written approval from the Contracting Officer's Representative.

#### 3.3.2 General Installation Requirements

Provide sheet waterproofing in accordance with manufacturer's printed installation instructions. Ensure the surface to receive membrane is clean, smooth and dry without surface irregularities; correct deficiencies prior to installation. When using solvent welding liquid, avoid prolonged contact with skin and breathing of vapor and provide adequate ventilation. Carry waterproofing of horizontal surfaces up abutting vertical surfaces and adhere solid to the substrate. Avoid wrinkles and buckles in applying membrane and joint reinforcement.

##### 3.3.2.1 Non-Self-Adhering Membrane

Unroll membrane and allow to remain flat for at least one-half hour before application. Apply an asphalt concrete primer prior to application of asphaltic adhesive. Where solvent adhesive is applied, allow major portion of solvent to evaporate so that bonding adhesive does not stick to a dry finger touching it. Apply elastomeric waterproofing membrane in a full bed of adhesive at a uniform coverage rate in accordance with the membrane manufacturer's printed instructions. Where membrane on horizontal surfaces are to receive concrete fill, apply adhesive in 4 inch wide strips at 2 feet on center. Pull membrane tight without stretching. As soon as adhesive is fully set and dry, recheck lap splices. Where openings or fishmouths appear, reseal and reroll lap splices.

##### 3.3.2.2 Protection

Protect membrane over horizontal surfaces from traffic during installation. Use only equipment with rubber tires. Provide walkway protection where heavy traffic from other trades is expected. Do not store material on membrane.

#### 3.3.3 Butyl Rubber

Lap sheets at sides and ends a minimum of 6 inches over the preceding sheet. Apply lap splicing cement over entire 6 inches splice area prior to application of sealant. Make sealant continuous along the entire length of the splice. Maintain a continuous bead of sealant at all membrane splices or as required by the manufacturer. Provide a tongue and groove cemented splice a minimum of 6 inches wide with factory made heat vulcanized seam of not less than 2 inches or as required by the manufacturer, when membrane is below water table.

### 3.3.4 Thermoplastic Membrane (PVC)

Consult with membrane manufacturer prior to grid application. Install 12 inches wide sub-membrane containment grid as required by manufacturer. Provide the containment grid at intervals across the width and length of the substrate, at the base of all transitions, walls, curbs, penetrations, and at the perimeter of each deck/substrate section. Fully adhere strips to the deck in a full bedding of two-part urethane adhesive. Weld adjacent sheets in accordance with manufacturer's instructions. Hot-air weld all side and end lap joints. Provide lap area a minimum of 3 inch wide when machine welding, and a minimum of 4 inch wide when hand welding but not less than recommended by the manufacturer. Orient overlaps with the direction of flow of water.

### 3.4 COMPOSITE, SELF-ADHERING MEMBRANE

Lap sheets at edges and ends a minimum of 2-1/2 inches over the preceding sheet. Provide all side laps a minimum 2-1/2 inches and end laps 5 inches. Provide self-adhesive, mastic laps in accordance with manufacturer's recommendation. Roll or firmly press to adhere membrane to substrate. Cover corners and joints with two layers of reinforcement by first applying a 12 inch width of membrane centered along the axis. Flash drains and projections with a second ply of membrane for a distance of 6 inches from the drain or projection. Finish exposed, terminated edges of membrane on horizontal or vertical surfaces with a troweled bead of mastic. Apply mastic around edges of membrane, and drains and projections. Apply mastic at end of each work day.

### 3.5 FLASHING

Flash penetrations through membrane. Seal all penetrations where reinforcing bars penetrate a waterproofing membrane with the appropriate sealant or mastic flashing component. Embed elastomeric membrane in a heavy coat of adhesive, except for self-adhering membrane. Position continuous metal reglets horizontally on footing and vertically on intersecting and connecting walls, and as specified in Section 07 60 00 FLASHING AND SHEET METAL. Metal reglets are to receive exposed edges of membrane waterproofing. Secure membrane into reglets by lead wedges and fill with cement as recommended in writing by manufacturer of waterproofing materials. Counterflash upper edge of membrane waterproofing and protective covering as specified in Section 07 60 00 FLASHING AND SHEET METAL.

### 3.6 FIELD QUALITY CONTROL

Notify the Contracting Officer 5 working days prior to date of performing tests. Before concealment, cover elastomeric waterproofing on horizontal surfaces over finished spaces with 4 inches of ponded water for 24 hours. Do not add water after start of 24 hour period. Accurately measure water level at beginning and end of 24 hour period. If water level falls, remove water and inspect waterproofing membrane. Make repairs or replacement as directed, and repeat test. Do not proceed with work that conceals membrane waterproofing before receiving approval and acceptance of the Contracting Officer.

### 3.7 PROTECTIVE COVERING

After installation has been inspected and approved by the Contracting

Officer, apply a protective covering to the membrane waterproofing prior to backfilling. Protect vertical membrane waterproofing with a 1/2 inch minimum thickness of asphalt plank; 1/2 inch minimum thickness of fiberboard; or 1/8 inch minimum thickness of compatible water resistant bitumen type protection board with edges abutting adjacent edges and exposed surfaces covered by a taping system recommended by manufacturer of protection board. Cover horizontal membrane waterproofing with similar protection board and Portland cement mortar not less than 3/4 inch thick; place uniformly and allow to set before installing subsequent construction.

-- End of Section --

SECTION 07 19 00

WATER REPELLENTS  
**08/23**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 501.2 (2015) Quality Assurance and Diagnostic Water Leakage Field Check of Installed Storefronts, Curtain Walls and Sloped Glazing Systems

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO T 259 (2002; R 2017) Standard Method of Test for Resistance of Concrete to Chloride Ion Penetration

AASHTO T 260 (2021) Standard Method of Test for Sampling and Testing for Chloride Ion in Concrete and Concrete Raw Materials

ASTM INTERNATIONAL (ASTM)

ASTM C140/C140M (2023a) Standard Test Methods for Sampling and Testing Concrete Masonry Units and Related Units

ASTM C642 (2021) Standard Test Method for Density, Absorption, and Voids in Hardened Concrete

ASTM D1653 (2013; R 2021) Standard Test Methods for Water Vapor Transmission of Organic Coating Films

ASTM D2369 (2020) Standard Test Method for Volatile Content of Coatings

ASTM D3278 (2021) Standard Test Methods for Flash Point of Liquids by Small Scale Closed-Cup Apparatus

ASTM E96/E96M (2024) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials

ASTM E514/E514M (2020) Standard Test Method for Water Penetration and Leakage Through Masonry

ASTM G154

(2023) Standard Practice for Operating  
Fluorescent Ultraviolet (UV) Lamp  
Apparatus for Exposure of Materials

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.1000

Air Contaminants

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-03 Product Data

Water Repellents

### SD-06 Test Reports

Water Absorption

Accelerated Weathering

Resistance to Chloride Ion Penetration

Moisture Vapor Transmission

Water Penetration and Leakage

### SD-07 Certificates

Manufacturer's Qualifications

Applicator's Qualifications

Evidence of Acceptable Variation

Warranty

### SD-08 Manufacturer's Instructions

Application Instructions

Manufacturer's Safety Data Sheets

## 1.3 QUALITY ASSURANCE

### 1.3.1 Qualifications

- a. Manufacturer's qualifications: Minimum five years record of successful in-service experience of water repellent treatments manufactured for concrete, concrete masonry, application.
- b. Applicator's qualifications: Minimum five years successful experience in projects of similar scope using specified or similar treatment materials and manufacturer's approval for application.

### 1.3.2 Performance Requirements

- a. Water absorption: ASTM C140/C140M. Comparison of treated and untreated specimens.
- b. Moisture vapor transmission: ASTM E96/E96M. Comparison of treated and untreated specimens.
- c. Water penetration and leakage through masonry: ASTM E514/E514M.
- d. Resistance to chloride ion penetration: AASHTO T 259 and AASHTO T 260.
- e. Resistance to Accelerated weathering: ASTM G154

### 1.3.3 Evidence of Acceptable Variation

If a product proposed for use does not conform to requirements of the referenced specification, submit for approval to the Contracting Officer, evidence that the proposed product is either equal to or better than the product specified. Include the following:

- a. Identification of the proposed substitution;
- b. Reason why the substitution is necessary;
- c. A comparative analysis of the specified product and the proposed substitution, including tabulations of the composition of pigment and vehicle;
- d. The difference between the specified product and the proposed substitution; and
- e. Other information necessary for an accurate comparison of the proposed substitution and the specified product.

## 1.4 SAMPLE TEST PANEL

The approved Sample Test Panel will serve as the standard of quality for all other water repellent coating work. Do not proceed with application until the sample panel has been approved by the Contracting Officer.

### 1.4.1 Preparation of Sample Test Panel

Prior to commencing work, including bulk purchase and delivery of material, apply water repellent treatment to a minimum 4 feet high by 4 feet long concrete, concrete masonry, test-panel specified in 04 20 00, MASONRY.. Provide a full height expansion joint at mid-panel length. Prepare and seal joint with materials approved for project use.

#### 1.4.1.1 Testing

Similar to AAMA 501.2 hose testing, provide field water testing of water repellent treated surfaces in the presence of the Contracting Officer and the water repellent treatment manufacturer's representative as follows.

- a. Apply water repellent to left side of mock-up and allow to cure prior to application of treatment to right side.
- b. Twenty days after completion of application of treatment, test mock-up

with 5/8 inch garden hose, with spray nozzle, located 10 feet from wall and aimed upward so water strikes wall at 45-degree downward angle. After water has run continuously for three hours observe back side of mock-up for water penetration and leakage. If leakage is detected, make changes as needed and retest.

- c. Coordinate testing procedures and modify project treatment application as required to pass mock-up tests for water penetration and leakage resistance.

#### 1.4.1.2 Approval

Proceed with water repellent treatment work only after completion of field test application and approval of mock-up and tests by the Contracting Officer.

#### 1.4.2 Pre-Installation Meeting

- a. Attend pre-installation meeting required prior to commencement of concrete, concrete masonry, installation.
- b. Review procedures and coordination required between water repellent treatment work and work of other trades which could affect work to be performed under this section of the work.
- c. Convene additional pre-installation meeting prior to water repellent treatment application for coordination with work not previously coordinated including joint sealants.

### 1.5 REGULATORY REQUIREMENTS

#### 1.5.1 Environmental Protection

In addition to requirements specified in Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME) for environmental protection, provide coating materials that conform to the restrictions of the . Notify the Contracting Officer of water repellent coating specified herein which fails to conform to the local Air Quality Management District Rules at the location of the Project. In localities where the specified coating is prohibited, the Contracting Officer may direct the substitution of an acceptable coating.

### 1.6 DELIVERY, STORAGE, AND HANDLING

Deliver materials in original sealed containers, clearly marked with the manufacturer's name, brand name, type of material, batch number, percent solids by weight and volume, and date of manufacturer. Store materials off the ground, in a dry area where the temperature will be not less than 50 degrees F nor more than 85 degrees F.

### 1.7 SAFETY METHODS

Apply coating materials using safety methods and equipment in accordance with Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME), and the following:

#### 1.7.1 Toxic Materials

To protect personnel from overexposure to toxic materials, conform to the

most stringent guidance of:

- a. The coating manufacturer when using solvents or other chemicals. Use impermeable gloves, chemical goggles or face shield, and other recommended protective clothing and equipment to avoid exposure of skin, eyes, and respiratory system. Conduct work in a manner to minimize exposure of building occupants and the general public.
- b. 29 CFR 1910.1000.
- c. Threshold Limit Values (R) of the American Conference of Governmental Industrial Hygienists.
- d. Manufacturer's Safety Data Sheets.

## 1.8 ENVIRONMENTAL CONDITIONS

### 1.8.1 Weather and Substrate Conditions

Do not proceed with application of water repellents under any of the following conditions, except with written recommendations of manufacturer.

- a. Ambient temperature is less than 40 degrees F.
- b. Substrate faces have cured less than one month.
- c. Rain or temperature below 40 degrees F are predicted for a period of 24 hours before or after treatment.
- d. Earlier than three days after surfaces are wet.
- e. Substrate is frozen or surface temperature is less than 40 degrees F and falling.

### 1.8.2 Moisture Condition

Determine moisture content of substrate meets manufacturer's requirements prior to application of water repellent material.

## 1.9 SEQUENCING AND SCHEDULING

### 1.9.1 Masonry Surfaces

Do not start water repellent coating until all joint tooling, pointing and masonry cleaning operations have been completed. Allow masonry to cure for at least 60 days under normal weather conditions before applying water repellent.

### 1.9.2 Plaster Surfaces

Do not start water repellent coating until all shrinkage and stress cracks are repaired and sound, all surfaces are free of defects and cleaning operations have been completed. Allow plaster to cure for at least 30 days under normal weather conditions before applying water repellent.

### 1.9.3 Concrete Surfaces

Do not start water repellent coating until all patching, pointing and cleaning operations have been completed and concrete has cured a minimum



of 30 days under normal weather conditions.

#### 1.9.4 Sealants

Do not apply water repellents until the sealants for joints adjacent to surfaces receiving water repellent treatment have been installed and cured.

- a. Water repellent work may precede sealant application only if sealant adhesion and compatibility have been tested and verified using substrate, water repellent, and sealant materials identical to those used in the work.
- b. Provide manufacturers' test results of compatibility.

#### 1.10 INSPECTIONS

Notify the manufacturer's representative a minimum of 72 hours prior to scheduled application of water repellents for field inspection. Inspect surfaces and obtain approval in writing from the manufacturer's representative prior to application of water repellent coating.

#### 1.11 SURFACES TO BE COATED

Coat all exterior concrete, masonry, surfaces. This includes back faces of parapets, top of walls, edges and returns adjacent to windows and door frames and free-standing walls.

#### 1.12 WARRANTY

Provide a warranty, issued jointly by the manufacturer and the applicator of the water repellent treatment against moisture penetration through the treated structurally sound surface for a period of five years. Warranty to provide the material, labor, and equipment necessary to remedy the problem. At the satisfactory completion of the work, complete the warranty sign, notarize, and submit to the Contracting Officer.

### PART 2 PRODUCTS

#### 2.1 MATERIALS

Provide a clear, non-yellowing, deep-penetrating, VOC compliant water repellent solution. Material must not stain or discolor and must produce a mechanical and chemical interlocking bond with the substrate to the depth of the penetration.

#### 2.2 WATER REPELLENTS

##### 2.2.1 Silane, 20 Percent Solids

Penetrating water repellent. A monomeric compound containing approximately 20 percent alkyltrialkoxysilanes with alcohol, mineral spirits, water, and other proprietary solvent carrier.

- a. Composition: Modified alkylalkoxysilane.
- b. Active alkylalkoxysilane content: ASTM D2369 20 percent by weight, plus or minus 1 percent.
- c. Appearance: White, milky liquid.

- d. Average depth of penetration: Up to 3/8 inch depending on substrate.
- e. VOC content: Less than 350 grams per liter.
- f. Flash point, ASTM D3278.
- g. Specific gravity, at 78 degrees F: 0.96 to 0.98.
- h. Density: .0 to 8.2 pounds per gallon.

#### 2.2.2 Silane, 40 Percent Solids

Penetrating water repellent. A monomeric compound containing approximately 40 percent alkyltrialkoxysilanes with alcohol, mineral spirits, or water.

- a. Composition: Modified alkylalkoxysilane.
- b. Active alkylalkoxysilane content: ASTM D2369 40 percent by weight, plus or minus 1.5 percent.
- c. Appearance: White, milky liquid.
- d. Average depth of penetration: Up to 3/8 inch depending on substrate.
- e. VOC content: Less than 350 grams per liter.
- f. Flash point, ASTM D3278.
- g. Specific gravity, at 78 degrees F: 0.94 to 0.97.
- h. Density: 7.8 to 8.1 pounds per gallon.

#### 2.2.3 Silane, 85 Percent Solids or Greater

Penetrating water repellent. A monomeric compound containing 85 percent or greater alkyltrialkoxysilanes with alcohol, mineral spirits, or water.

- a. Composition: Modified alkylalkoxysilane.
- b. Active alkylalkoxysilane content: ASTM D2369 85 percent by weight, plus or minus 1 percent.
- c. Appearance: White, milky liquid.
- d. Average depth of penetration: Up to 3/8 inch depending on substrate.
- e. VOC content: Less than 350 grams per liter.
- f. Flash point, ASTM D3278.
- g. Specific gravity, at 78 degrees F: 0.96 to 0.98.
- h. Density: 8.0 to 8.2 pounds per gallon.

#### 2.2.4 Siloxanes

Penetrating water repellent. Alkylalkoxysiloxanes that are oligomeric

with alcohol, ethanol, mineral spirits, or water.

- a. Solids by weight: ASTM D2369, 7.5 to 16.0 percent.
- b. Volatile Organic Content (VOC) after blending: Less than 175 grams per liter.
- c. Density, activated: 8.4 pounds per gallon, plus or minus one percent.
- d. Flash point, ASTM D3278: Greater than 212 degrees F.

#### 2.2.5 Low-Solids Acrylic

Water-clear, breathing coating of acrylic resins, water-based, solvent-based, or acrylic emulsions solution containing less than 15 percent solids by volume.

#### 2.2.6 High-Solids Acrylic

Water-clear, breathing coating of acrylic resins, water-based, solvent-based, or acrylic emulsions solution containing 15 percent solids or more by volume.

#### 2.2.7 VOC-Complying Water Repellents

Products certified by the manufacturer that they comply with local regulations controlling use of volatile organic compounds (VOC's).

### 2.3 PERFORMANCE CRITERIA

#### 2.3.1 Silane, 20 Percent Solids

- a. Water absorption test: ASTM C642 and ASTM E514/E514M.
- b. Moisture vapor transmission: ASTM D1653, 28.33 perms or 51.61 percent maximum compared to untreated surfaces.
- c. Resistance to chloride ion penetration: AASHTO T 259 and AASHTO T 260.
- d. Water penetration and leakage through masonry, ASTM E514/E514M percentage reduction of leakage: 97 percent minimum.
- e. Resistance to accelerated weathering, ASTM G154 testing 2,500 hours: No loss in repellency.
- f. Drying time under normal conditions: Four hours per 75 degrees F.

#### 2.3.2 Silane, 40 Percent Solids

- a. Average depth of penetration: 3/8 inches depending on substrate
- b. Resistance to chloride ion penetration, AASHTO T 259 and AASHTO T 260.
- c. Water absorption test, ASTM E514/E514M: 0.42 percent per 48 hours; 1.2 percent per 50 days.
- d. Moisture vapor transmission: ASTM D1653, 28.33 perms or 51.61 percent maximum compared to untreated surfaces.

- e. Resistance to accelerated weathering, ASTM G154. Testing 2,500 hours: No loss in repellency.
- f. Drying time under normal conditions: Four hours per 75 degrees F.

#### 2.3.3 Silane, 85 Percent Solids or Greater

- a. Average depth of penetration: 3/8 inches depending on substrate.
- b. Resistance to chloride ion penetration, AASHTO T 259 and AASHTO T 260.
- c. Water absorption test, ASTM E514/E514M: 0.42 percent per 48 hours; 1.2 percent per 50 days.
- d. Moisture vapor transmission: ASTM D1653, 28.33 perms or 51.61 percent maximum compared to untreated surfaces.
- e. Resistance to accelerated weathering, ASTM G154. Testing 2,500 hours: No loss in repellency.
- f. Drying time under normal conditions: Four hours per 75 degrees F.

#### 2.3.4 Siloxanes

- a. Dry time for recoat, if necessary: One to two hours depending on weather conditions.
- b. Penetration: 3/8 inch, depending on substrate.
- c. Water penetration and leakage through masonry, ASTM E514/E514M, percentage reduction of leakage: 97.0 percent minimum.
- d. Moisture vapor transmission, ASTM E96/E96M: 47.5 perms or 82 percent maximum compared to untreated sample.
- e. Resistance to accelerated weathering, ASTM G154. Testing 2,500 hours: No loss in repellency.
- f. Resistance to chloride ion penetration, AASHTO T 259 and AASHTO T 260.

### PART 3 EXECUTION

#### 3.1 EXAMINATION

Examine concrete, , or masonry surfaces to be treated to ensure that:

- a. All visible cracks, voids or holes have been repaired.
- b. All mortar joints in masonry are tight and sound, have not been re-set or misaligned and show no cracks or spalling.
- c. Moisture contents of walls does not exceed 15 percent when measured on an electronic moisture register, calibrated for the appropriate substrate.
- d. Concrete surfaces are free of form release agents, curing compounds and other compounds that would prevent full penetration of the water repellent material.

Do not start water repellent treatment work until all deficiencies have been corrected, examined, and found acceptable to the Contracting Officer and the water repellent treatment manufacturer. Do not apply treatment to damp, dirty, dusty, or otherwise unsuitable surfaces. Comply with the manufacturer's recommendations for suitability of surface.

### 3.2 PREPARATION

#### 3.2.1 Surface Preparation

Prepare substrates in accordance with water repellent treatment manufacturer's recommendation. Clean surfaces of dust, dirt, efflorescence, alkaline, and foreign matter detrimental to proper application of water repellent treatment.

#### 3.2.2 Protection

Provide masking or protective covering for materials which could be damaged by water repellent treatment.

- a. Protect glass, glazed products, and prefinished products from contact with water repellent treatment.
- b. Protect landscape materials with breathing-type drop cloths; plastic covers are not acceptable.

#### 3.2.3 Compatibility

- a. Confirm treatment compatibility with each type of joint sealer within or adjacent to surfaces receiving water repellent treatment in accordance with manufacturer's recommendations.
- b. When recommended by joint sealer manufacturer, apply treatment after application and cure of joint sealers. Coordinate treatment with joint sealers.
- c. Mask surfaces indicated to receive joint sealers which would be adversely affected by water repellent treatment where treatment must be applied prior to application of joint sealers.

### 3.3 MIXING

Mix water repellent material thoroughly in accordance with the manufacturer's recommendations. Mix, in quantities required for that day's work, all containers prior to application. Mix each container the same length of time.

### 3.4 APPLICATION

In strict accordance with the manufacturers written requirements. Do not start application without the manufacturer's representative being present or their written acceptance of the surface to be treated. Submit manufacturer's instructions including preparation, application, recommended equipment to be used, safety measures, and protection of completed application.

### 3.4.1 Water Repellent Treatment

#### 3.4.1.1 Spray Application

Spray apply water repellent material to exterior concrete, and masonry surfaces using low-pressure airless spray equipment in strict accordance with manufacturer's printed application, instructions, and precautions. Maintain copies at the job site. Apply flood coat in an overlapping pattern allowing approximately 8 to 10 inch rundown on the vertical surface. Maintain a wet edge at all overlaps, both vertical and horizontal. Hold gun maximum 18 inches from wall.

#### 3.4.1.2 Brush or Roller Application

Brush or roller apply water repellent material only at locations where overspray would affect adjacent materials and where not practical for spray applications.

#### 3.4.1.3 Covered Surfaces

Coat all exterior concrete, or masonry surfaces including back faces of parapets, tops of walls, edges and returns adjacent to window and door frames, window sills, and free-standing walls.

#### 3.4.1.4 Rate of Application

Apply materials to exterior surfaces at the coverages recommended by the manufacturer and as determined from sample panel test. Increase or decrease application rates depending upon the surface texture and porosity of the substrate to achieve even appearance and total water repellency.

#### 3.4.1.5 Number of Coats

The results of the sample panel test determines the number of coats required to achieve full coverage and protection.

#### 3.4.1.6 Appearance

If unevenness in appearance, lines of work termination or scaffold lines exist, or detectable changes from the approved sample panel occur, the Contracting Officer may require additional treatment at no additional cost to the Government. Apply required additional treatment to a natural break off point.

### 3.5 CLEANING

Clean all runs, drips, and overspray from adjacent surfaces while the water repellent treatment is still wet in a manner recommended by the manufacturer.

### 3.6 FIELD QUALITY CONTROL

Do not remove drums containing water repellent material from the job site until completion of all water repellent treatment and until so authorized by the Contracting Officer.

#### 3.6.1 Field Testing

At a time not less than twenty days after completion of the water

repellent coating application, subject a representative wall area of the building to the Navy Hose Stream Field Test similar to AAMA 501.2 hose test to simulate rainfall for a period of three hours. Use a minimum 5/8 inch diameter hose and a fixed lawn sprinkler spray head which will direct a full flow of water against the wall. Place the sprinkler head so that the water will strike the wall downward at a 45-degree angle to the wall. If the inside of the wall shows traces of moisture during or following the test, apply another coat of water repellent, at the manufacturer's recommended coverage rate to the entire building. Repeat testing and re-coating process until no moisture shows on the inside wall face. Accomplish required work retesting and re-coating at no additional cost to the Government.

### 3.6.2 Site Inspection

Inspect treatment in progress by manufacturer's representative to verify compliance with manufacturer instructions and recommendations.

-- End of Section --

SECTION 07 21 13

BOARD AND BLOCK INSULATION  
**08/23**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

|                   |                                                                                                                                               |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C272/C272M   | (2018) Standard Test Method for Water Absorption of Core Materials for Sandwich Constructions                                                 |
| ASTM C578         | (2023) Standard Specification for Rigid, Cellular Polystyrene Thermal Insulation                                                              |
| ASTM C612         | (2014; R 2019) Standard Specification for Mineral Fiber Block and Board Thermal Insulation                                                    |
| ASTM C930         | (2019) Standard Classification of Potential Health and Safety Concerns Associated with Thermal Insulation Materials and Accessories           |
| ASTM D1621        | (2016) Standard Test Method for Compressive Properties of Rigid Cellular Plastics                                                             |
| ASTM D3833/D3833M | (1996; R 2019) Standard Test Method for Water Vapor Transmission of Pressure-Sensitive Tapes                                                  |
| ASTM D4397        | (2016) Standard Specification for Polyethylene Sheeting for Construction, Industrial, and Agricultural Applications                           |
| ASTM E84          | (2023) Standard Test Method for Surface Burning Characteristics of Building Materials                                                         |
| ASTM E96/E96M     | (2024) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials                                      |
| ASTM E154/E154M   | (2008a; R 2019) Standard Test Methods for Water Vapor Retarders Used in Contact with Earth Under Concrete Slabs, on Walls, or as Ground Cover |



INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2021) International Building Code

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

NFPA 211 (2019) Standard for Chimneys, Fireplaces,  
Vents, and Solid Fuel-Burning Appliances

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.134 Respiratory Protection

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Manufacturer's Standard Details; G

SD-03 Product Data

Environmental Data for Materials; G

Block or Board Insulation; G

Vapor Retarder; G

Pressure Sensitive Tape; G

Protection Board or Coating; G

Accessories including sealants; G

SD-07 Certificates

Indoor Air Quality For Block Or Board Insulation

SD-08 Manufacturer's Instructions

Insulation Installation and Handling

Protection Board or Coating Installation

Adhesive

SD-11 Closeout Submittals

Draft Guarantee; G

Final Guarantee; G

Draft Warranty; G

Final Warranty; G

### 1.3 MANUFACTURER'S DETAILS

Submit manufacturer's standard details indicating methods of attachment and spacing, transition and termination details, and installation details. Include verification of existing conditions.

### 1.4 PRODUCT DATA

Include environmental data for materials descriptions, recommendations for product shelf life, requirements for protection board or coatings, and precautions for flammability and toxicity. Include data to verify compatibility of sealants with insulation.

### 1.5 DELIVERY, STORAGE, AND HANDLING

#### 1.5.1 Delivery

Deliver materials to the site in original sealed wrapping bearing manufacturer's name and brand designation, specification number, type, grade, R-value, and class. Store and handle to protect from damage. Do not allow insulation materials to become wet, soiled, crushed, or covered with ice or snow. Comply with manufacturer's recommendations for handling, storing, and protecting of materials before and during installation.

#### 1.5.2 Storage

Inspect materials delivered to the site for damage and store out of weather in manufacturer's original packaging. Store only in dry locations, not subject to open flames or sparks, and easily accessible for inspection and handling. Keep materials wrapped and separated from off-gassing materials (such as drying paints and adhesives). Do not use materials that have visible moisture or biological growth. Comply with manufacturer's recommendations for handling, storage, and protection of materials before and during installation.

### 1.6 SAFETY PRECAUTIONS

#### 1.6.1 Respirators

Provide installers with dust/mist respirators, training in their use, and protective clothing, all approved by the National Institute for Occupational Safety and Health (NIOSH)/Mine Safety and Health Administration (MSHA) and in accordance with 29 CFR 1910.134.

#### 1.6.2 Other Safety Considerations

Comply with the safety requirements of ASTM C930.

### 1.7 SPECIAL WARRANTIES

#### 1.7.1 Guarantee

Guarantee insulation installation against failure due to ultraviolet light

exposure for a period of three years from the date of Beneficial Occupancy or Substantial Completion. Submit draft guarantee and final guarantee in accordance with Sections 01 78 00.00 22 CLOSEOUT SUBMITTALS .

#### 1.7.2 Warranty

Provide manufacturer's material warranty for all system components for a period of three years from the date of Beneficial Occupancy or Substantial Completion. Submit draft warranty and final warranty in accordance with Sections 01 78 00.00 22 CLOSEOUT SUBMITTALS .

### PART 2 PRODUCTS

#### 2.1 BLOCK OR BOARD INSULATION

Provide thermal insulating materials as recommended by manufacturer for each type of application indicated. Provide insulation with the following physical properties and in accordance with the following standards:

- a. Extruded Preformed Cellular Polystyrene (XPS): ASTM C578 REV A, TypeIV, 25 PSI
- b. Mineral Fiber Block and Board: ASTM C612, Type IVA, without membrane facing.

##### 2.1.1 Thermal Resistance

Provide R-values as indicated on the drawings.

##### 2.1.2 Fire Protection Requirements

- a. Flame spread index of 75 or less when tested in accordance with ASTM E84.
- b. Smoke developed index of 450 or less when tested in accordance with ASTM E84.
- c. Provide insulated assemblies in accordance ICC IBC Chapter Fire and Smoke Protection Features.

##### 2.1.3 Other Material Properties

Provide thermal insulating materials with the following properties:

- a. Rigid cellular plastics: Compressive Resistance at Yield: Not less than 25 pounds per square inch (psi) when measured according to ASTM D1621.
- c. Water Vapor Permeance: Not more than 1.1 perms or less when measured according to ASTM E96/E96M, desiccant method, in the thickness required to provide the specified thermal resistance, including facings, if any.
- e. Water Absorption: Not more than 2 percent by total immersion, by volume, when measured according to ASTM C272/C272M.

##### 2.1.4 Premolded Concrete Masonry Insert

Provide in accordance with ASTM C578 REV A. Provide inserts in concrete

masonry units that are installed at the masonry unit manufacturing plant. Provide insert with thickness of not less than 1 1/4 inches.

#### 2.1.5 Indoor Air Quality

Provide certification of indoor air quality for block or board insulation.

#### 2.1.6 Prohibited Materials

Do not provide materials containing asbestos.

### 2.2 VAPOR RETARDER

#### 2.2.1 Vapor Retarder in Framed Walls and Roofs

- a. 6 mil thick polyethylene sheeting conforming to ASTM D4397 and having a water vapor permeance of one Perm or less when tested in accordance with ASTM E96/E96M.

#### 2.2.2 Vapor Retarder under Floor Slab

- a. Water vapor permeance: 0.2 Perm or less when tested in accordance with ASTM E96/E96M.
- b. Puncture resistance: Maximum load no less than 40 pounds when tested according to ASTM E154/E154M REV A.
- c. Minimum thickness, not less than 20 mils.

### 2.3 PRESSURE SENSITIVE TAPE

As recommended by manufacturer of vapor retarder(s). Match water vapor permeance rating for each vapor retarder specified. Provide tape in accordance with ASTM D3833/D3833M.

### 2.4 PROTECTION BOARD OR COATING

As recommended by insulation manufacturer.

### 2.5 ACCESSORIES

#### 2.5.1 Adhesive

As recommended by insulation manufacturer.

#### 2.5.2 Mechanical Fasteners

Corrosion resistant fasteners as recommended by the insulation manufacturer.

## PART 3 EXECUTION

### 3.1 EXISTING CONDITIONS

Prior to installation, ensure all areas that will be in contact with the insulation are dry and free of projections that could cause voids, compressed insulation, or punctured vapor retarders. For foundation perimeter or under slab applications, check that subsurface fill is flat, smooth, dry, and well tamped. Do not proceed with installation if

moisture or other conditions are present and notify the Contracting Officer of such conditions. Do not proceed with the work until conditions have been corrected and verified to be dry.

### 3.2 PREPARATION

#### 3.2.1 Blocking Around Heat Producing Devices

Provide noncombustible blocking at all spaces between heat producing devices and the floors, ceilings and roofs through which they pass. Provide in accordance with ICC IBC Section 2111.13 Fireplace Fireblocking and with the following clearances:

- a. Recessed lighting fixtures, including wiring compartments, ballasts, and other heat producing devices, unless certified for installation surrounded by insulation: 3 inches from outside face of fixtures and devices or as required by NFPA 70 and, if insulation is placed above fixture or device, 24 inches above fixture.
- b. Vents and vent connectors used for venting products of combustion, flues, and chimneys other than masonry chimneys: Minimum clearances as required by NFPA 211.

Blocking is not required if chimneys or flues are certified in writing by the chimney or flue manufacturer for use in contact with specific insulating materials.

### 3.3 INSTALLATION

#### 3.3.1 Insulation Installation and Handling

Provide insulation in accordance with the manufacturer's printed installation instructions. Keep material dry and free of extraneous materials.

#### 3.3.2 Electrical Wiring

Do not install insulation in a manner that would enclose electrical conductors, that are not installed in conduit, between two layers of insulation.

#### 3.3.3 Cold Climate Requirement

Place insulation on the outside of pipes.

#### 3.3.4 Continuity of Insulation

Butt tightly against adjoining boards, studs, rafters, joists, sill plates, headers and obstructions. Provide continuity and integrity of insulation at corners, wall to ceiling joint, roof, and floor. Avoid creating thermal bridges and voids. Provide and verify continuity of insulative barrier throughout the building enclosure.

#### 3.3.5 Coordination

Verify final installed insulation thicknesses comply with thicknesses indicated and with the approved insulation submittal(s).

### 3.4 INSTALLATION ON WALLS

#### 3.4.1 Installation using Furring Strips

Install insulation between members as recommended by insulation manufacturer.

#### 3.4.2 Installation on Masonry Walls

Apply board directly to masonry with adhesive or fasteners as recommended by the insulation manufacturer. Fit between obstructions without impaling board on ties or anchors. Apply in parallel courses with joints breaking midway over course below. Place boards in moderate contact with adjoining insulation without forcing and without gaps. Cut and shape as required to fit around wall penetrations, projections or openings to accommodate conduit or other utilities. Seal around cutouts with sealant. Install insulation in wall cavities so that it leaves at least a nominal one inch air space outside of the insulation to allow for cavity drainage.

#### 3.4.3 Adhesive Attachment to Concrete and Masonry Walls

Apply adhesive to wall and completely cover wall with insulation.

- c. As recommended by the insulation manufacturer.
- d. Use only full back method for pieces of one square foot or less.
- e. Butt all edges of insulation and seal edges with tape.

#### 3.4.4 Mechanical Attachment on Concrete and Masonry Walls

Cut insulation to cover walls. Apply adhesive to wall and set clip or other mechanical fastener in adhesive as recommended by manufacturer. After curing of adhesive, install insulation over fasteners and bend split prongs to provide a flush condition with the insulation. Butt all edges of insulation and seal with tape.

#### 3.4.5 Protection Board or Coating Installation

Install protection board or coating in accordance with manufacturer's printed instructions. Install protection over all exterior exposed insulation and to one foot below grade.

### 3.5 PERIMETER SLAB INSULATION

Install perimeter thermal insulation where heated spaces are adjacent to exterior walls, slab edges in slab-on-grade, or floating slab construction.

#### 3.5.1 Manufacturer's Instructions

Layout insulation, tape edges, provide vapor retarder and other required accessories to protect against vermin, insects, and damage in accordance with manufacturer's printed instructions.

#### 3.5.2 Insulation on Vertical Surfaces

Provide thermal insulation on exterior of foundation walls below grade. Fasten insulation with adhesive or mechanical fasteners.

### 3.6 VAPOR RETARDER

Apply vapor retarder continuous across all surfaces. Overlap all joints at least 6 inches and seal with pressure sensitive tape. Seal at sills, header, windows, doors and utility penetrations. Repair punctures or tears with pressure sensitive tape.

### 3.7 ACCESS PANELS AND DOORS

Attach insulation to all access panels greater than one square foot and all access doors in insulated floors and ceilings. Use insulation with same R-Value as that for the floor or ceiling in which each panel occurs.

-- End of Section --

SECTION 07 22 00

ROOF AND DECK INSULATION  
**08/23**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7 (2017) Minimum Design Loads for Buildings and Other Structures

ASTM INTERNATIONAL (ASTM)

ASTM C1289 (2023a) Standard Specification for Faced Rigid Cellular Polyisocyanurate Thermal Insulation Board

ASTM D41/D41M (2011; R 2016) Standard Specification for Asphalt Primer Used in Roofing, Dampproofing, and Waterproofing

ASTM D226/D226M (2017) Standard Specification for Asphalt-Saturated Organic Felt Used in Roofing and Waterproofing

ASTM D4601/D4601M (2004; R 2020) Standard Specification for Asphalt-Coated Glass Fiber Base Sheet Used in Roofing

ASTM E84 (2023) Standard Test Method for Surface Burning Characteristics of Building Materials

FM GLOBAL (FM)

FM 4450 (1989) Approval Standard for Class 1 Insulated Steel Deck Roofs

FM 4470 (2022) Single-Ply, Polymer-Modified Bitumen Sheet, Built-up Roof (BUR), and Liquid Applied Roof Assemblies for Use in Class 1 and Noncombustible Roof Deck Construction

FM APP GUIDE (updated on-line) Approval Guide  
<https://www.approvalguide.com/>

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2021) International Building Code



SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

SCS SCS Global Services (SCS) Indoor Advantage

UNDERWRITERS LABORATORIES (UL)

UL 1256 (2023) Fire Test of Roof Deck Constructions

UL 2818 (2022) GREENGUARD Certification Program  
For Chemical Emissions For Building  
Materials, Finishes And Furnishings

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Insulation Board Layout and Attachment; G

Verification of Existing Conditions; G

SD-03 Product Data

Insulation; G

Cover Board; G

Fasteners; G

Moisture Control; G

Recycled Content For Insulation

SD-05 Design Data

Wind Resistance; G

SD-06 Test Reports

Flame Spread Rating; G

SD-07 Certificates

Installer Qualifications; G

Certificates Of Compliance For Felt Materials; G

Indoor Air Quality For Insulation

Acceptable Foam Blowing Agents

## SD-08 Manufacturer's Instructions

Nails and Fasteners; G

Roof Insulation; G

### 1.3 SHOP DRAWINGS

Submit insulation board layout and attachment indicating methods of attachment and spacing of fasteners on each board, transitions, tapered components, thicknesses of materials, and closure and termination conditions. Show locations of ridges, valleys, crickets, interface with, and slope to, roof drains. Base shop drawings on verified field measurements and include verification of existing conditions..

### 1.4 PRODUCT DATA

Include data for material descriptions, recommendations for product shelf life, requirements for cover board or coatings, and precautions for flammability and toxicity. Include data to verify compatibility of sealants with insulation.

### 1.5 MANUFACTURER'S INSTRUCTIONS

Include field of roof, perimeter, and corner attachment requirements.

Provide a complete description of installation sequencing for each phase of the roofing system. Include weatherproofing procedures.

### 1.6 QUALITY CONTROL

Provide certification of installer qualifications from the insulation manufacturer confirming the specific installer has the required qualifications for installing the specific roof insulation system(s) indicated.

Provide certificates of compliance for felt materials.

### 1.7 WIND RESISTANCE REQUIREMENTS

The complete roof system assembly must be rated and installed to resist wind loads calculated in accordance with ASCE 7 and validated by uplift resistance testing in accordance with Factory Mutual (FM) test procedures. Coordinate with roof covering attachment requirements and submit wind resistance test certification, attachment patterns for field, perimeter, and corner roof areas along with perimeter and corner boundary dimensions.

### 1.8 FIRE PERFORMANCE REQUIREMENTS

#### 1.8.1 Insulation in Roof Systems

Comply with the requirements of ICC IBC or UL 1256 or FM 4450 or FM 4470. Roof insulation must have a flame spread rating of 75 or less when tested in accordance with ASTM E84. Additional documentation of compliance with flame spread rating is not required when insulation of the type used for this project as part of the specific roof assembly is listed and labeled as FM Class 1 approved.

#### 1.8.2 Thermal Barrier Requirements

Separate polyurethane insulation from a steel deck with a thermal barrier of glass mat gypsum roof board or other approved barrier material in accordance with the requirements of the ICC IBC or FM 4450 or FM 4470 or UL 1256.

#### 1.8.3 Fire Resistance Ratings for Roofs

Provide in accordance with ICC IBC Chapter 7 and Table 721.1(3) Min Fire and Smoke Protection For Floor and Roof Systems.

#### 1.9 CERTIFICATIONS

Provide products certified to meet indoor air quality requirements by UL 2818(Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification by other third-party programs. Provide current product certification documentation from certification body.

#### 1.10 DELIVERY, STORAGE, AND HANDLING

##### 1.10.1 Delivery

Deliver materials to the project site in manufacturer's unopened and undamaged standard commercial containers bearing the following legible information:

- a. Name of manufacturer
- b. Brand designation
- c. Specification number, type, and class, as applicable, where materials are covered by a referenced specification

Deliver materials in sufficient quantity to allow continuity of the work.

##### 1.10.2 Storage and Handling

Store and handle materials in accordance with manufacturer's printed instructions. Protect from damage, exposure to open flame or other ignition sources, wetting, condensation, and moisture absorption. Keep materials wrapped and separated from off-gassing materials (such as drying paints and adhesives). Do not use materials that have visible moisture or biological growth. Store in an enclosed building or trailer that provides a dry, adequately ventilated environment. Store felt rolls on ends. For the 24 hours immediately before application of felts, store felts in an area maintained at a temperature no lower than 50 degrees F above grade and having ventilation on all sides. Replace damaged material with new material.

#### 1.11 ENVIRONMENTAL CONDITIONS

As per manufacturer's recommendations or Government's instructions, do not install roof insulation during inclement weather or when air temperature is below 40 degrees F and interior humidity is 45 percent or greater, or when there is visible ice, frost, or moisture on the roof deck.

1.12 PROTECTION

1.12.1 Completed Work

Cover completed work with cover board for the duration of construction. Avoid traffic on completed work particularly when ambient temperature is above 80 degrees F. Replace crushed or damaged insulation prior to roof surface installation.

PART 2 PRODUCTS

2.1 INSULATION

2.1.1 Insulation Types

Provide roof insulation that is compatible with attachment methods for the specified insulation and roof membrane.

- a. Polyisocyanurate Board: Provide in accordance with ASTM C1289 REV A Type II, fibrous felt or glass mat membrane both sides, except minimum compressive strength of 20 pounds per square inch (psi).

2.1.2 Recycled Materials

Provide thermal insulation materials containing recycled content. Unless specified otherwise, the minimum required recycled content for listed materials are:

|                                |                              |
|--------------------------------|------------------------------|
| Polyisocyanurate/polyurethane: | 9 percent recovered material |
|--------------------------------|------------------------------|

Provide data identifying percentage of recycled content for insulation.

2.1.3 Indoor Air Quality

Provide certification of indoor air quality for insulation.

2.1.4 Prohibited Materials

Products that contain high ozone depleting or high Global Warming Potential (GWP) blowing agents are prohibited. For a list of acceptable substitute foam blowing agents for the type of insulation used see <https://www.epa.gov/snap/foam-blowing-agents>. Provide validation of acceptable foam blowing agents that no prohibited materials are used.

2.1.5 Insulation Thickness

As necessary to provide the thermal resistance (R-value) indicated. Base calculation on the R-value for aged insulation. For insulation over steel decks, satisfy both specified R-value and minimum thickness for width of rib opening recommended in insulation manufacturer's published literature.

2.2 BITUMENS

2.2.1 Asphalt Primer

Provide in accordance with ASTM D41/D41M.

## 2.3 MOISTURE CONTROL

### 2.3.1 Vapor Retarder

#### 2.3.1.1 Asphalt Saturated Felt Base Sheet for Single Layer Application

Provide in accordance with ASTM D4601/D4601M, weighing not less than 35 pounds per 100 square feet.

### 2.3.2 Organic Roofing

Provide in accordance with ASTM D226/D226M, Type I.

## 2.4 FASTENERS

Provide flush-driven fasteners through flat round or hexagonal steel or plastic plates. Provide zinc-coated steel plates, flat round not less than 1 3/8 inch diameter, hexagonal not less than 28 gage. Provide high-density plastic plates, molded thermoplastic with smooth top surface, reinforcing ribs and not less than 3 inches in diameter. Fully recess fastener head into plastic plate after it is driven. Form plates to prevent dishing. Do not use bell or cup shaped plates. Provide fasteners in accordance with insulation manufacturer's recommendations for holding power when driven, or a minimum of 120 pounds each in steel deck, whichever is the higher minimum. Provide fasteners for steel or concrete decks in accordance with FM APP GUIDE (<https://www.approvalguide.com/>) for Class I roof deck construction, and spaced to withstand uplift pressure of as indicated on structural drawings.

### 2.4.1 Fasteners for Steel Decks

Approved hardened penetrating fasteners or screws in accordance with FM 4450 and listed in FM APP GUIDE for Class I roof deck construction. Quantity and placement to withstand a minimum uplift pressure as indicated on structural drawings. in accordance with FM APP GUIDE.

## 2.5 FOAM ADHESIVE

Foam adhesive as recommended by the insulation manufacturer for adhering the insulation to concrete surfaces or underlying insulation layers and resist the required wind uplift pressure. Product must be compatible with insulation and air and water barrier materials, and with demonstrated capability to bond insulation securely to substrates without damaging insulation and substrates.

## PART 3 EXECUTION

### 3.1 EXAMINATION AND PREPARATION

#### 3.1.1 Surface Inspection

Ensure surfaces are clean, smooth, and dry prior to application. Ensure surfaces receiving vapor retarder are free of projections that might puncture the vapor retarder. Check roof deck surfaces, including surfaces sloped to roof drains and outlets, for defects before starting work.

The Contracting Officer will inspect and approve the surfaces immediately before starting installation. Prior to installing insulation, perform the following:

- b. Examine steel decks to ensure that panels are properly secured to structural members and to each other and that surfaces of top flanges are flat or slightly convex.

### 3.1.2 Surface Preparation

To correct defects and inaccuracies in roof deck surface to eliminate poor drainage from hollow or low spots, perform the following:

- a. Provide wood nailers of the same thickness as the insulation at eaves, edges, curbs, walls, and roof openings for securing of cant strips, gravel stops, flashing flanges, and curbs.
- d. Cover steel decks with a layer of insulation board of sufficient width to span the width of a deck rib opening, and in accordance with fire safety requirements. Secure with piercing or self-drilling, self-tapping fasteners of quantity and placement in accordance with FM APP GUIDE. Locate insulation joints parallel to ribs of deck on solid bearing surfaces only, not over open ribs.

### 3.2 INSTALLATION OF VAPOR RETARDER

Install vapor retarder in direct contact with roof deck surface. Unless otherwise specified, vapor retarder must consist of either two plies of No. 15 asphalt-saturated felt, two plies of asphalt-coated glass felt. Lay vapor retarder at right angles to direction of slope. Install first ply of felt or base sheet as specified herein for the specific deck. Apply second ply of 2-ply vapor retarder system using asphalt at rate of 20 to 35 lbs per 100 square feet, applied within plus or minus 25 degrees F of EVT. Do not heat asphalt above asphalt's FBT or 525 degrees F, whichever is less. Use thermometers to check temperatures during heating and application. Completely seal side and end laps. Asphalt must be visible beyond all edges of each ply as it is being installed. Lay plies free of wrinkles, buckles, creases or fishmouths. Do not walk on mopped surfaces while asphalt is sticky. Press out air bubbles to obtain complete adhesion between surfaces. At walls, eaves, rakes, and other vertical surfaces, extend vapor retarder organic felts or separate plies 9 inches, with not less than 9 inches on the substrate, and the extended portion turned back and mopped in over the top of the insulation. At roof penetrations other than walls, eaves and rakes, and vertical surfaces, extend vapor retarder or separate plies 9 inches to form a lap folded back over the edge of the insulation. Provide asphalt roof cement under the vapor retarder for at least 9 inches from walls, eaves, rakes and other penetrations.

#### 3.2.1 Vapor Retarder on Steel Decks

Even mop the mechanically secured insulation surface with asphalt before installing vapor retarder. For a two-ply vapor retarder, install each sheet lapping 19 inches over the preceding sheet. Lap ends not less than 4 inches. Stagger the laps a minimum of 12 inches. Cement felts together with solid mopping of asphalt. Apply asphalt moppings at rate of 20 to 35 lbs per 100 square feet.

### 3.3 INSULATION INSTALLATION

Apply insulation in two layers with staggered joints when total required

thickness of insulation exceeds 1/2 inch. Lay insulation so that continuous longitudinal joints are perpendicular to direction of roofing, as specified in Section 07 61 15.00 20 Aluminum Standing Seam Roofing, and end joints of each course are staggered with those of adjoining courses. When using multiple layers of insulation, provide joints of each succeeding layer that are parallel and offset in both directions with respect to the layer below. Keep insulation 1/2 inch clear of vertical surfaces penetrating and projecting from roof surface. Verify required slopes to each roof drain.

### 3.3.1 Installation Using Mechanical Fasteners and Foam Adhesive

Secure first layer of insulation and thermal barrier to deck with piercing or self-drilling, self-tapping fasteners. Engage fasteners by driving them through insulation into top flange of steel deck. Use driving method prescribed by fastener manufacturer. Locate insulation joints parallel to ribs of deck on solid bearing surfaces only, not over open ribs. Secure succeeding layers with foam adhesive using installation procedures as recommended by the insulation manufacturer for adhering the insulation and resisting the required wind uplift pressure. Installation must bond insulation securely to substrates without damaging insulation and substrates.

### 3.3.2 Special Precautions for Installation of Foam Insulation

#### 3.3.2.1 Polyisocyanurate Insulation

Where polyisocyanurate foam board insulation is provided, install cover board over top surface of foam board insulation. Stagger joints of insulation with respect to foam board insulation below.

### 3.3.3 Cant Strips

Where indicated, provide cant strips at intersections of roof with walls, parapets, and curbs extending above roof. Wood cant strips must bear on and be anchored to wood blocking. Fit cant strips flush to vertical surfaces. Where possible, nail cant strips to adjoining surfaces. Where cant strips are installed against non-nailable materials, install in .

### 3.3.4 Tapered Edge Strips

Where indicated, provide edge strips in the right angle formed by the juncture of roof and wood nailing strips that extend above the level of the roof. Install edge strips flush to vertical surfaces of wood nailing strips. Where possible, nail edge strips to adjoining surfaces. Where installed against non-nailable materials, install in .

## 3.4 PROTECTION

### 3.4.1 Protection of Applied Insulation

Completely cover each day's installation of insulation with finished roofing specified in 07 61 15.00 20 Aluminum Standing Seam Roofing on same day. Phased construction is not permitted. Protect open spaces between insulation and parapets or other walls and spaces at curbs, scuttles, and expansion joints, until permanent roofing and flashing are applied. Storing, walking, wheeling, or trucking directly on insulation or on roofed surfaces is not permitted. Provide smooth, clean board or plank walkways, runways, and platforms near supports, as necessary, to

distribute weight in accordance with indicated live load limits of roof construction. Protect exposed edges of insulation with cutoffs at the end of each workday or whenever precipitation is imminent. Cutoffs must be two layers of bituminous-saturated felt set in plastic bituminous cement set in roof cement. Fill all profile voids in cutoffs to prevent trapping moisture below the membrane. Remove cutoffs when work resumes.

#### 3.4.2 Damaged Work and Materials

Restore work and materials that become damaged during construction to original condition or replace with new materials.

#### 3.5 INSPECTION

Establish and maintain inspection procedures to assure compliance of the installed roof insulation with Contract requirements. Remove, replace, correct in an approved manner, any work found not in compliance. Quality control must include, but is not limited to, the following:

- a. Observation of environmental conditions; number and skill level of insulation workers; start and end time of work.
- b. Verification of certification, listing or label compliance with FM Data Sheets.
- c. Verification of proper storage and handling of insulation and vapor retarder materials before, during, and after installation.
- d. Inspection of vapor retarder application, including edge envelopes and mechanical fastening.
- e. Inspection of mechanical fasteners; type, number, length, and spacing.
- f. Coordination with other materials, cants, sleepers, and nailing strips.
- g. Inspection of insulation joint orientation and laps between layers, joint width and bearing of edges of insulation on deck.
- h. Installation of cutoffs and proper joining of work on subsequent days.
- i. Continuation of complete roofing system installation to cover insulation installed same day.
- j. Verification of required slope to each roof drain.

-- End of Section --



SECTION 07 27 10

BUILDING AIR BARRIER SYSTEM

08/23

PART 1 GENERAL

1.1 SUMMARY

This Section specifies the construction and quality control of the installation of an air barrier system. Construct the air barrier system indicated, taking responsibility for the means, methods, and workmanship of the installation of the air barrier system. The air barrier must be contiguous and connected across all surfaces of the enclosed air barrier envelope indicated. The maximum leakage requirements of individual air barrier components and materials are specified in the other specification sections covering these items.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referenced within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

|               |                                                                                                                      |
|---------------|----------------------------------------------------------------------------------------------------------------------|
| ASTM D4541    | (2022) Standard Test Method for Pull-Off Strength of Coatings Using Portable Adhesion Testers                        |
| ASTM E96/E96M | (2024) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials             |
| ASTM E2178    | (2021a) Standard Test Method for Determining Air Leakage Rate and Calculation of Air Permeance of Building Materials |
| ASTM E2357    | (2018) Standard Test Method for Determining Air Leakage Rate of Air Barrier Assemblies                               |

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                                                                                                                   |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| NFPA 285 | (2023) Standard Fire Test Method for Evaluation of Fire Propagation Characteristics of Exterior Wall Assemblies Containing Combustible Components |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|

1.3 DEFINITIONS

The following terms as they apply to this section:

1.3.1 Air Barrier Accessory

Products designated to maintain air tightness between air barrier

materials, air barrier assemblies and air barrier components, to fasten them to the structure of the building, or both (e.g., sealants, tapes, backer rods, transition membranes, fasteners, strapping, primers).

#### 1.3.2 Air Barrier Assembly

The combination of air barrier materials and air barrier accessories that are designated and designed within the environmental separator to act as a continuous barrier to the movement of air through the environmental separator.

#### 1.3.3 Air Barrier Component

Pre-manufactured elements such as windows, doors, dampers and service elements that are installed in the environmental separator.

#### 1.3.4 Air Barrier Envelope

The combination of air barrier assemblies and air barrier components, connected by air barrier accessories that are designed to provide a continuous barrier to the movement of air through an environmental separator. There may be more than one air barrier envelope in a single building. Also known as Air Barrier System.

#### 1.3.5 Air Barrier Material

A building material that is designed, tested or produced to provide the primary resistance to airflow through an air barrier assembly of a building enclosure.

#### 1.3.6 Air Barrier System

Same as AIR BARRIER ENVELOPE.

#### 1.3.7 Air Leakage Rate

The rate of airflow (CFM) driven through a unit surface area (sq.ft.) of an assembly or system by a unit static pressure difference (Pa) across the assembly. (example: 0.25 CFM/sq.ft. @ 75 Pa)

#### 1.3.8 Air Leakage

The total airflow (CFM) driven through the air barrier system by a unit static pressure difference (Pa) across the air barrier envelope. (example: 6500 CFM @ 75 Pa)

#### 1.3.9 Air Permeance

The tested rate of airflow (CFM) through a unit area (sq.ft.) of a material driven by unit static pressure difference (Pa) across the material (example: 0.004 CFM/sq.ft. @ 75 Pa) as established by ASTM E2178.

#### 1.3.10 Environmental Separator

The parts of a building that separate the controlled interior environment from the uncontrolled exterior environment, or that separate spaces within a building that have dissimilar environments. Also known as the Control Layer.

#### 1.3.11 Vapor Permeance

Vapor permeance is separated into three classes based on the water vapor permeance of a material as tested via ASTM E96/E96M

Class I Vapor Barrier/Retarder 0.1 perm or less

Class II Vapor Barrier/Retarder 0.1 perm to 1.0 perm

Class III Vapor Barrier/Retarder 1.0 perm to 10 perm

#### 1.4 PREPARATORY PHASE OR PRECONSTRUCTION CONFERENCE

Organize pre-construction conferences between the air barrier inspector and the sub-contractors involved in the construction of, or penetration of, the air barrier system to discuss where the work of each sub-contractor begins and ends, the sequence of installation, and each sub-contractor's responsibility to ensure airtight joints, junctures, penetrations and transitions between materials. Discuss the products, and assemblies of products specified in the different sections to be installed by the different sub-contractors.

#### 1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-02 Shop Drawings

Air Barrier System Shop Drawings; G, Manufacturer produced warranted air barrier system

##### SD-03 Product Data

Air Barrier System Product Data; G

##### SD-04 Samples

Mock-Up; G

Material Samples For Air Barrier System; G

##### SD-05 Design Data

Design Data And Calculations For The Air Barrier System; G

##### SD-06 Test Reports

Design Review Report; G

Testing and Inspection; G

##### SD-07 Certificates

Air Barrier Inspector; G

## 1.6 AIR BARRIER INSPECTOR

Employ a designated Air Barrier Inspector on this project. The Air Barrier Inspector performs a Design Review, oversees quality control testing specified in these specifications, performs quality control air barrier inspection as specified, interfaces with the designer and product manufacturer's representatives to assure installation requirements are met, and verifies that the constructed work is in accordance with both the manufacturer's recommendations for products used, the content of this specification and other Contract drawings or documents. Qualifications for the Air Barrier Inspector are as follows:

- a. Training and certification as an Air Barrier Auditor from the Air Barrier Association of America (ABAA) or other third-party air barrier association.
- b. Or, provide documentation in resume format that demonstrates that the individual proposed has the experience, knowledge, skills and abilities to fulfill the above stated duties as the air barrier inspector.
- c. It is acceptable that this individual be employed by the firm performing the building pressurization test or another independent third-party entity, provided they meet the above requirements but must not be a member of the installing Contractor or firm.

Provide copies of Air Barrier Inspector qualifications 30 days after Notice to Proceed.

## 1.7 DESIGN REVIEW

Review the Contract Plans and Specifications and advise the Contracting Officer of any deficiencies that would prevent the construction of an effective air barrier system. Provide a Design Review Report individually listing each deficiency and the corresponding proposed corrective action necessary for a proper air barrier system. Provide copies of the Design Review Report not later than 14 days after approval of the Air Barrier Inspector Qualifications. Submit design data and calculations for the Air Barrier System for a manufacturer produced and warranted air barrier system.

## PART 2 PRODUCTS

### 2.1 AIR BARRIER

Provide air barrier system of compatible parts from one or several manufacturers coordinated by the Contractor or provide a single warranted system provided by a primary manufacturer. The air barrier system as part of a tested exterior wall assembly must meet the conditions of acceptance as tested in accordance with NFPA 285. Materials used for roof assembly air barrier must conform to the appropriate UL and FM wind and fire requirements for the specified roof assemblies.

If a complete air barrier system from a single manufacturer is utilized, whether warranted or not warranted, the air barrier system must conform to ASTM E2357.

Materials in the following categories, as used in the air barrier system or assembly of the exterior wall system, are tested and are required to

conform to ASTM E2178: Self-adhered sheet membranes, fluid applied membranes, spray polyurethane foam, mechanically fastened commercial building wrap, factory bonded membranes to sheathing, and adhesive backed commercial building wrap and accessory products.

Other materials used as an air barrier such as concrete, glass, wood, metal or gypsum board may or may not conform to ASTM E2178 but are acceptable provided that when integrated into the air barrier system or assemblies they are not subject to material or environmental induced degradation in their final produced state once incorporated in the permanent construction.

All materials used must be identifiable through manufacturer testing data or literature to be compatible with all the attached or adjoining materials or substrates used in the system.

Provide Air Barrier System Shop Drawings, Material Samples for Air Barrier System and Air Barrier System Product Data.

### PART 3 EXECUTION

#### 3.1 QUALITY CONTROL

##### 3.1.1 Documentation and Reporting

Document the entire installation process on daily job site reports. These reports include information on the Installer, substrates, substrate preparation, products used, ambient and substrate temperature, the location of the air barrier installation, the results of the quality control procedures, and testing results.

##### 3.1.2 Construction Mock-Up

Build mock-up prior to building envelope construction.

- a. Prepare a construction mock-up to demonstrate proper installation of the air barrier assemblies and components. Include air barrier system connections between floor and wall, wall and window, and wall and roof. Also, include the sealing method between membrane joints at transitions from one material or component to another, at pipe or conduit penetrations of the wall and roof, and at duct penetration of the wall and roof. Work must not begin until the mock-up is satisfactory to the Contracting Officer.
- b. Size the mock-up to approximately 8 feet long by 8 feet high. The mock-up must be representative of primary exterior wall assemblies and glazing components including backup wall and typical penetrations as acceptable to the Contracting Officer. A corner of the actual building may be used as the mock-up.
- c. Mock-Up Tests for Adhesion: Test the mock-up of materials for adhesion in accordance with manufacturer's recommendations. Perform the test after the curing period recommended by the manufacturer. Record the mode of failure and the area which failed in accordance with ASTM D4541. When the air barrier material manufacturer has established a minimum adhesion level for the product on the particular substrate, the inspection report must indicate whether this requirement has been met. Where the manufacturer has not declared a minimum adhesion value for their product/substrate combination, simply record the value.

### 3.1.3 Quality Control Testing And Inspection

Conduct the following tests and inspections as applicable in the presence of the Contracting Officer during installation of the air barrier system, and submit quality control reports as indicated below.

- a. Provide a Daily Report of Observations with a copy to the Contracting Officer.
- b. Inspect to assure continuity of the air barrier system throughout the building enclosure and that all gaps are covered, the covering is structurally sound, and all penetrations are sealed allowing for no infiltration or exfiltration through the air barrier system.
- c. Inspect to assure structural support of the air barrier system to withstand design air pressures.
- d. Inspect to assure masonry surfaces receiving air barrier materials are smooth, clean, and free of cavities, protrusions and mortar droppings, with mortar joints struck flush or as required by the manufacturer of the air barrier material.
- e. Inspect and test to assure site conditions for application temperature, and dryness of substrates are within guidelines.
- f. Inspect to assure substrate surfaces are properly primed, if applicable, and in accordance with manufacturer's instructions. Priming must extend at least 2 inches beyond the air barrier material to make it obvious that the primer was applied to the substrate before the air barrier material.
- g. Inspect to assure laps in materials are at least a 2-inch minimum, shingled in the correct direction or mastic applied in accordance with manufacturer's recommendations, and with no fishmouths.
- h. Inspect to assure that a roller has been used to enhance adhesion. Identify any defects such as fishmouths, wrinkles, areas of lost adhesion, and improper curing. Note the intended remedy for the deficiencies.
- i. Measure application thickness of liquid applied materials to assure that manufacturer's specifications for the specific substrate are met.
- j. Inspect to assure that the correct materials are installed for compatibility.
- k. Inspect to assure proper transitions for change in direction and structural support at gaps.
- l. Inspect to assure proper connection between assemblies (membrane and sealants) for cleaning, preparation and priming of surfaces, structural support, integrity and continuity of seal.
- m. Perform adhesion tests for fluid-applied and self-adhered air barrier membranes to assure that the manufacturer's specified adhesion strength properties are met. Determine the bond strength of coatings to substrate in accordance with ASTM D4541.

- o. Provide written test reports of all tests performed.

### 3.2 REPAIR AND PROTECTION

Upon completion of inspection, testing, sample removal and similar services, repair damaged construction and restore substrates, coatings and finishes. Protect construction exposed by or for quality control service activities, and protect repaired construction.

-- End of Section --

SECTION 07 27 19.01

SELF-ADHERING AIR BARRIERS  
**08/23, CHG 1: 08/24**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR BARRIER ASSOCIATION OF AMERICA (ABAA)

ABAA Accreditation

Accreditation

ABAA QAP

Quality Assurance Program

ASTM INTERNATIONAL (ASTM)

ASTM D146/D146M

(2004; E 2012; R 2012) Sampling and Testing Bitumen-Saturated Felts and Woven Fabrics for Roofing and Waterproofing

ASTM D412

(2016; R 2021) Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers - Tension

ASTM D570

(1998; E 2010; R 2010) Standard Test Method for Water Absorption of Plastics

ASTM D903

(1998; R 2017) Standard Test Method for Peel or Stripping Strength of Adhesive Bonds

ASTM D1876

(2008; R 2015; E 2015) Standard Test Method for Peel Resistance of Adhesives (T-Peel Test)

ASTM D4263

(1983; R 2018) Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method

ASTM D4541

(2022) Standard Test Method for Pull-Off Strength of Coatings Using Portable Adhesion Testers

ASTM E96/E96M

(2024) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials

ASTM E154/E154M

(2008a; R 2019) Standard Test Methods for Water Vapor Retarders Used in Contact with Earth Under Concrete Slabs, on Walls, or as Ground Cover

ASTM E283/E283M

(2019) Standard Test Method for



Determining Rate of Air Leakage Through  
Exterior Windows, Skylights, Curtain  
Walls, and Doors Under Specified Pressure  
Differences Across the Specimen

ASTM E331 (2000; R 2023) Standard Test Method for  
Water Penetration of Exterior Windows,  
Skylights, Doors, and Curtain Walls by  
Uniform Static Air Pressure Difference

ASTM E2178 (2021a) Standard Test Method for  
Determining Air Leakage Rate and  
Calculation of Air Permeance of Building  
Materials

ASTM E2357 (2018) Standard Test Method for  
Determining Air Leakage Rate of Air  
Barrier Assemblies

#### NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 285 (2023) Standard Fire Test Method for  
Evaluation of Fire Propagation  
Characteristics of Exterior Wall  
Assemblies Containing Combustible  
Components

### 1.2 RELATED REQUIREMENTS

Coordinate the requirements of Section 07 27 10 BUILDING AIR BARRIER SYSTEM and other building enclosure sections to provide a complete building air barrier system. Submit all materials, components, and assemblies of the air barrier system together as one complete submittal package.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-01 Preconstruction Submittals

Qualifications of Manufacturer; G

Qualifications of Installer; G

#### SD-02 Shop Drawings

Self-adhering Air Barrier Shop Drawings; G

#### SD-03 Product Data

Self-adhering Air Barrier; G

Primers, Adhesives, and Mastics; G

Safety Data Sheets; G

SD-04 Samples

Self-adhering Air Barrier System Mockup; G

SD-06 Test Reports

Field Peel Adhesion Test; G

Flame Propagation of Wall Assemblies; G

Site Inspections and Testing; G

SD-08 Manufacturer's Instructions

Self-adhering Air Barrier Manufacturer's Printed Instructions; G

Primers, Adhesives, and Mastics; G

1.4 MISCELLANEOUS REQUIREMENTS

For self-adhering air barrier provide the following:

1.4.1 Shop Drawings

Submit self-adhering air barrier shop drawings showing locations and extent of air barrier assemblies and details of all typical conditions, intersections with other building enclosure assemblies and materials, and membrane counterflashings. Show details for bridging of gaps in construction, treatment of inside and outside corners, expansion joints, methods of attachment of materials covering the self-adhered barrier without compromising the barrier. Indicate how miscellaneous penetrations such as conduit, pipes, electric boxes, brick ties, and similar items will be sealed.

1.4.2 Product Data

Submit manufacturer's technical data indicating compliance with performance and environmental requirements in Part 2 of this specification. Also submit self-adhering air barrier manufacturer's printed instructions for evaluating, preparing, and treating substrates, temperature and other limitations of installation conditions, safety requirements for installation, and Safety Data Sheets. Indicate flame and smoke spread ratings for all products.

1.4.3 Mockup

Provide a self-adhering air barrier system mockup. Apply product in an area designated by the Contracting Officer. Apply an area of not less than 54 square feet. Include all components specified as representative of the complete system. Notify the Contracting Officer a minimum of 48 hours prior to the test application. Select a test area representative of conditions to be covered including window or door openings, wall to ceiling transitions, flashings, and penetrations, as applicable.

1.4.4 Test Reports

Submit test reports indicating that field peel-adhesion tests on all materials have been performed and the changes made, if required, in order to achieve successful and lasting adhesion. Submit test reports for flame

propagation of wall assemblies tested in accordance with NFPA 285.

## 1.5 DELIVERY, STORAGE, AND HANDLING

### 1.5.1 Delivery

Deliver and store materials in sufficient quantity to allow for uninterrupted flow of work. Inspect materials delivered to the site for damage and store out of weather. Deliver materials to the jobsite in their original unopened packages, clearly marked with the manufacturer's name, brand designation, description of contents, and shelf life of containerized materials. Store and handle to protect from damage.

### 1.5.2 Storage

Inspect materials delivered to the site for damage; unload and store out of weather in manufacturer's original packaging. Store only in dry locations, not subject to open flames or sparks, and easily accessible for inspection and handling. Protect stored materials from direct sunlight. Keep materials sealed and separated from absorptive materials, such as wood and insulation.

## 1.6 AIR BARRIER TESTING

Perform air barrier testing in accordance with Section 07 27 10 BUILDING AIR BARRIER SYSTEM.

## 1.7 QUALITY ASSURANCE

### 1.7.1 Qualifications of Manufacturer

Submit documentation verifying that the manufacturer of the self-adhering air barrier is currently accredited by Air Barrier Association of America (ABAA Accreditation <https://www.airbarrier.org/>).

### 1.7.2 Qualifications of Installer

Submit documentation verifying that installers of the self-adhering air barrier are currently certified in accordance with the ABAA QAP Quality Assurance Program (<https://www.airbarrier.org/qap/>).

### 1.7.3 Field Peel Adhesion Test

Perform a field peel-adhesion test on the construction mockup. Test the self-adhering air barrier for adhesion in accordance with ASTM D4541 using a Type II pull tester except use a disk that is 4 inches in diameter and cut through the membrane to separate the material attached to the dish from the surrounding material. Perform test after curing period in accordance with manufacturer's written recommendations. Record mode of failure and area which failed in accordance with ASTM D4541. Compare adhesion values with the manufacturer's established minimum values for the particular combination of material and substrate. Indicate on the inspection report whether the manufacturer's requirement has been met. Where the manufacturer has not declared a minimum adhesion value for their product and substrate combination, the inspector must record actual values.

## 1.8 PRECONSTRUCTION MEETING

Conduct a preconstruction meeting a minimum of two weeks prior to

commencing work specified in this specification Section. Agenda must include, at a minimum, construction and testing of mockup, sequence of construction, coordination with substrate preparation, materials approved for use, compatibility of materials, coordination with installation of adjacent and covering materials, and details of construction. Attendance is required by representatives of related trades including covering materials, substrate materials, adjacent materials, and materials and components of the air barrier system.

## 1.9 ENVIRONMENTAL CONDITIONS

### 1.9.1 Temperature

Install air barrier within the range of ambient and substrate temperatures as recommended in writing by the air barrier manufacturer. Verify that the surface to receive self-adhering air barrier is dry for a minimum of 48 hours prior to the installation of the barrier. Do not apply air barrier to damp or wet substrates. Do not apply during inclement weather or when ice, frost, surface moisture, or visible dampness is present on surfaces to be covered, or when precipitation is imminent.

### 1.9.2 Exposure to Weather and Ultraviolet Light

Protect air barrier products from direct exposure to rain, snow, sunlight, mist, and other extreme weather conditions. Replace, at no additional cost to the Government, barrier products that have been exposed to ultraviolet (sun)light longer than allowed by manufacturer's written requirements.

## PART 2 PRODUCTS

### 2.1 SELF-ADHERING AIR BARRIER

Provide minimum 0.040 inch thick self-adhering, vapor retarding, air barrier membrane consisting of a cross-laminated high density polyethylene (HDPE) film, fully coated with rubberized butyl-based or acrylic-based adhesive. Provide membrane in rolls of various widths interleaved with disposable silicone release paper. Self-adhering air barrier must exhibit no visible water leakage when tested in accordance with ASTM E331 and must perform as a liquid water drainage plane flashed to discharge to the exterior incidental condensation or water penetration. Use regular or low temperature formulation depending on site conditions, within temperature ranges specified by manufacturer.

#### 2.1.1 Physical Properties

- a. Air Permeance (ASTM E2178): In accordance with Section 07 27 10 BUILDING AIR BARRIER SYSTEM Less than 0.004 CFM per sf at 1.57 psf.
- b. Assembly Air Leakage (ASTM E2357, ASTM E283/E283M): In accordance with Section 07 27 10 BUILDING AIR BARRIER SYSTEM less than 0.04 CFM per sf at 1.57 psf at one inch.
- c. Tensile Strength (ASTM D412 die C modified): Not less than 400 psi.
- d. Tensile Elongation (ASTM D412 die C modified): Not less than 200 percent.
- e. Puncture Resistance (ASTM E154/E154M): Not less than 40 lbs.

- f. Pliability (ASTM D146/D146M): Unaffected at minus 25 degrees F, 0.063 inch mandrel.
- g. Lap Adhesion (ASTM D1876 modified): Not less than 4.0 lbs per inch.
- h. Peel Adhesion (ASTM D903): Not less than 5.0 lbs per inch.
- i. Water Vapor Permeance (Vapor Impermeable Air Barrier) (ASTM E96/E96M, desiccant method A): 0.1 perms or less.
- j. Water Absorption (ASTM D570): Not to exceed 0.12 percent by weight.
- k. Flame propagation of wall assemblies (NFPA 285): Pass

## 2.2 PRIMERS, ADHESIVES, AND MASTICS

Provide primers, adhesives, mastics and other accessory materials as recommended in writing by the manufacturer of the self-adhering air barrier for adequate bonding to each type of substrate. Adhesive must be butyl- or acrylic-based.

## 2.3 SHEET METAL FLASHING

Provide as specified in Section 07 60 00 FLASHING AND SHEET METAL.

## 2.4 JOINT SEALANTS

Provide as specified in Section 07 92 00 JOINT SEALANTS. Verify compatibility with adjacent products that are or will be in contact with one another.

# PART 3 EXECUTION

## 3.1 EXAMINATION

Before installing air barrier, examine substrates, areas, and conditions under which air barrier assemblies will be applied, with Installer present, for compliance with requirements. Ensure the following conditions are met:

- a. Surfaces are sound, dry, even, and free of oil, grease, dirt, excess mortar or other contaminants.
- b. Concrete surfaces are cured and dry, smooth without large voids, spalled areas or sharp protrusions.
- c. Verify substrate is visibly dry and free of moisture. Test for capillary moisture by plastic sheet method in accordance with ASTM D4263 and take suitable measures until substrate passes moisture test.
- d. Verify sealants used in sheathing are compatible with membrane proposed for use. Perform field peel adhesion test on materials to which sealants are adhered.

## 3.2 PREPARATION

Clean, prepare, and treat substrate in accordance with manufacturer's

written instructions. Ensure clean, dust-free, and dry substrate for air barrier application.

- a. Prime masonry and concrete substrates with conditioning primer.
- b. Prime gypsum sheathing an adequate number of coats to achieve required bond, with adequate drying time between coats.
- c. Prime wood, metal, and painted substrates with primer.
- d. Prepare, treat, and seal vertical and horizontal surfaces at terminations and penetrations through air barrier and at protrusions.

### 3.3 INSTALLATION

#### 3.3.1 Installation of Self-adhering Air Barrier

Install materials in accordance with manufacturer's recommendations and the following:

- a. Apply primer at rate recommended by manufacturer prior to membrane installation. Allow primer to dry completely before membrane application. Apply as many coats as necessary for proper adhesion.
- b. When membrane is properly positioned, press into place and roll membrane with roller immediately after placement.
- c. Apply membrane sheets to shed water naturally without interception by a sheet edge, unless that edge is sealed with permanently flexible termination mastic.
- d. Position subsequent sheets of membrane applied above so that membrane overlaps the membrane sheet below by a minimum of 2-1/2 inches, unless greater overlap is recommended by manufacturer. Roll into place with roller.
- e. Make side laps a minimum of 2-1/2 inches and end laps a minimum of 5 inches, unless greater overlap is recommended by manufacturer. Roll seams with roller.
- f. Roll membrane to adhere to substrate. Cover corners and joints with two layers of reinforcement by first applying a 12 inch width of membrane centered along the axis. Flash drains and projections with a second ply of membrane for a distance of 6 inches from the drain or projection.
- g. Seal around all penetrations through the air barrier resulting from pipes, vents, conduit, electrical fixtures, structural members, or other construction passing through it. Seal with termination mastic, extruded silicone sealant, membrane counterflashing or other sealing methods in accordance with manufacturer's written recommendations.
- h. Continuously connect the air barrier between walls, roof, floor and below grade assemblies to form a continuous integrated air barrier system around the entire building enclosure. Extend the air barrier membrane into rough openings such as doors, windows, louvers, and other exterior penetrations. Seal edges of barrier at junctures with rough openings.

- i. At changes in substrate plane, provide transition material (e.g., bead of sealant, mastic, extruded silicone sealant, membrane counterflashing or other material recommended by manufacturer) under membrane to eliminate all sharp 90-degree inside corners and to make a smooth transition from one plane to another.
- j. Provide mechanically fastened non-corrosive metal sheet to span gaps in substrate plane and to make a smooth transition from one plane to the other. Continuously support membrane with substrate.
- k. At deflection and control joints, provide backup for the membrane to accommodate anticipated movement.
- l. At expansion and seismic joints provide transition to the joint assemblies.
- m. Apply a bead or trowel coat of mastic along membrane seams at reverse lapped seams, rough cuts, and as recommended by the manufacturer.
- n. At end of each working day, seal top edge of membrane to substrate with termination mastic.
- o. Do not allow materials to come in contact with chemically incompatible materials.
- p. Counterflash upper edge of thru-wall flashing and air barrier. Counter flashing is specified in Section 07 60 00 FLASHING AND SHEET METAL. See applicable wall-type Sections for thru-wall flashing requirements.

### 3.4 FIELD QUALITY CONTROL

#### 3.4.1 Site Inspections and Testing

Provide site inspections and testing in accordance with the manufacturer's instructions, Section 07 27 10 BUILDING AIR BARRIER SYSTEM and this section.

- a. Conduct inspections and testing at 5, 50, and 95 percent completion of this scope of work. Forward written site inspections and testing reports to the Contracting Officer within five working days of the inspection and test being performed.
- b. If inspections reveal defects, promptly remove and replace defective work at no additional expense to the Government.

#### 3.5 FIELD PEEL ADHESION TEST

Conduct in accordance with test protocol indicated in Part 1, paragraph FIELD PEEL ADHESION TEST.

### 3.6 PROTECTION AND CLEANING

#### 3.6.1 Protection

##### 3.6.1.1 Adjacent Surfaces

Protect exposed adjacent surfaces that could be damaged by primers and adhesives associated with air barrier membrane. Provide protection during

application and the remainder of construction in accordance with manufacturer's written instructions.

#### 3.6.1.2 The Air Barrier Assembly

Protect finished portions of the air barrier assembly from damage during ongoing application and throughout the remainder of the construction period in accordance with manufacturer's written instructions. Coordinate timing of installation of materials that will cover the air barrier membrane to ensure the exposure period does not exceed that recommended by the air barrier manufacturer's written installation instructions. Remove and replace, at no additional cost to the Government, membrane products that exceed the manufacturer's allowed exposure limits.

#### 3.6.2 Cleaning

Clean spillage and soiling from adjacent construction using cleaning agents and procedures recommended by manufacturer of affected construction and as acceptable to the primary material manufacturer.

-- End of Section --



SECTION 07 60 00

FLASHING AND SHEET METAL  
**08/23**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING  
ENGINEERS (ASHRAE)

ASHRAE 90.1 - IP (2019) Energy Standard for Buildings  
Except Low-Rise Residential Buildings

AMERICAN WELDING SOCIETY (AWS)

AWS D1.2/D1.2M (2014; Errata 1 2014; Errata 2 2020)  
Structural Welding Code - Aluminum

ASTM INTERNATIONAL (ASTM)

ASTM A308/A308M (2010) Standard Specification for Steel  
Sheet, Terne (Lead-Tin Alloy) Coated by  
the Hot Dip Process

ASTM A480/A480M (2024) Standard Specification for General  
Requirements for Flat-Rolled Stainless and  
Heat-Resisting Steel Plate, Sheet, and  
Strip

ASTM A653/A653M (2023) Standard Specification for Steel  
Sheet, Zinc-Coated (Galvanized) or  
Zinc-Iron Alloy-Coated (Galvannealed) by  
the Hot-Dip Process

ASTM A792/A792M (2022) Standard Specification for Steel  
Sheet, 55% Aluminum-Zinc Alloy-Coated by  
the Hot-Dip Process

ASTM B32 (2020) Standard Specification for Solder  
Metal

ASTM B69 (2021) Standard Specification for Rolled  
Zinc

ASTM B101 (2022) Standard Specification for  
Lead-Coated Copper Sheet and Strip for  
Building Construction

ASTM B209/B209M (2021a) Standard Specification for  
Aluminum and Aluminum-Alloy Sheet and Plate

ASTM B221 (2021) Standard Specification for Aluminum

and Aluminum-Alloy Extruded Bars, Rods,  
Wire, Profiles, and Tubes

ASTM B370 (2022) Standard Specification for Copper  
Sheet and Strip for Building Construction

ASTM D41/D41M (2011; R 2016) Standard Specification for  
Asphalt Primer Used in Roofing,  
Dampproofing, and Waterproofing

ASTM D226/D226M (2017) Standard Specification for  
Asphalt-Saturated Organic Felt Used in  
Roofing and Waterproofing

ASTM D1784 (2020) Standard Specification for Rigid  
Poly(Vinyl Chloride) (PVC) Compounds and  
Chlorinated Poly(Vinyl Chloride) (CPVC)  
Compounds

ASTM D2244 (2021) Standard Practice for Calculation  
of Color Tolerances and Color Differences  
from Instrumentally Measured Color  
Coordinates

ASTM D4214 (2007; R 2015) Standard Test Method for  
Evaluating the Degree of Chalking of  
Exterior Paint Films

ASTM D4586/D4586M (2007; R 2018) Asphalt Roof Cement,  
Asbestos-Free

ASTM E1980 (2011; R 2019) Standard Practice for  
Calculating Solar Reflectance Index of  
Horizontal and Low-Sloped Opaque Surfaces

ASTM E2112 (2023) Standard Practice for Installation  
of Exterior Windows, Doors and Skylights

#### COOL ROOF RATING COUNCIL (CRRC)

ANSI/CRRC S100 (2016) Standard Test Methods for  
Determining Radiative Properties of  
Materials

#### NATIONAL ROOFING CONTRACTORS ASSOCIATION (NRCA)

NRCA 0429 (2022) The NRCA Roofing Manual:  
Architectural Metal Flashing, Condensation  
and Air Leakage Control and Reroofing

#### SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

SMACNA 1793 (2012) Architectural Sheet Metal Manual,  
7th Edition

#### SINGLE PLY ROOFING INDUSTRY (SPRI)

ANSI/SPRI RD-1 (2019) Performance Standard for Retrofit

## Drains

ANSI/SPRI/FM 4435/ES-1

(2017) Test Standard for Edge Systems Used  
with Low Slope Roofing Systems

### 1.2 GENERAL REQUIREMENTS

Finished sheet metal assemblies must form a weathertight enclosure without waves, warps, buckles, fastening stresses or distortion, while allowing for expansion and contraction without damage to the system. The sheet metal installer is responsible for cutting, fitting, drilling, and other operations in connection with sheet metal modifications required to accommodate the work of other trades. Coordinate installation of sheet metal items used in conjunction with roofing with roofing work to permit continuous, uninterrupted roofing operations.

#### 1.2.1 General Material Requirements

All materials specified in this Section installed in conjunction with the roofing system must be provided by the roofing system manufacturer, or by a manufacturer approved by the roofing system manufacturer for use in the roofing system, and must form a part of the Warranty as required by the applicable roofing system Section.

#### 1.2.2 Additional Wind Uplift Requirements

All flashing and edge materials specified in this Section must be installed in conjunction with the requirements of the roofing system to meet the requirements of ES-1 wind uplift. Details must include but not limited to ANSI/SPRI/FM 4435/ES-1 (ES-1), "Wind Test Design Standard for Edge Systems" used with Low Slope Roofing Systems, Test Methods: RE-1, RE-2 and RE-3 using uplift roofing details of the NRCA 0429.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

Exposed Sheet Metal Coverings; G

Gutters; G

Downspouts; G

Expansion Joints; G

Gravel Stops and fascia; G

Splash Pans; G

Flashing for Roof Drains; G

Base Flashing; G

Counterflashing; G

Flashing at Roof Penetrations and Equipment Supports; G

Reglets; G

Drip Edges; G

Recycled Content; S

SD-03 Product Data

Cool Roof Data; G

SD-04 Samples

Finish Samples; G

SD-07 Certificates

Warranty on Finishes; G

SD-08 Manufacturer's Instructions

Instructions for Installation; G

Quality Control Plan; G

SD-10 Operation and Maintenance Data

Cleaning and Maintenance; G

1.4 MISCELLANEOUS REQUIREMENTS

1.4.1 Product Data

Indicate thicknesses, dimensions, fastenings, anchoring methods, expansion joints, and other provisions necessary for thermal expansion and contraction. Scaled manufacturer's catalog data may be submitted for factory fabricated items.

1.4.2 Finish Samples

Submit two color charts and two finish sample chips from manufacturer's standard color and finish options for each type of finish indicated.

1.4.3 Operation and Maintenance Data

Submit detailed instructions for installation and quality control during installation, cleaning and maintenance, for each type of assembly indicated.

1.5 DELIVERY, HANDLING, AND STORAGE

Package and protect materials during shipment. Uncrate and inspect materials for damage, dampness, and wet-storage stains upon delivery to the job site. Remove from the site and replace damaged materials that cannot be restored to like-new condition. Handle sheet metal items to avoid damage to surfaces, edges, and ends. Store materials in dry, weather-tight, ventilated areas until installation.

## PART 2 PRODUCTS

### 2.1 RECYCLED CONTENT

Provide products with recycled content. Provide data for each product with recycled content, identifying percentage of recycled content.

### 2.2 MATERIALS

Use any metal listed by NRCA 0429 or SMACNA 1793 for a particular item, unless otherwise indicated. Provide materials and configurations in accordance with NRCA 0429 or SMACNA 1793 for each material, while also meeting the minimum thickness requirements specified in this Section. Different items need not be of the same metal, except that contact between dissimilar metals must be avoided.

Provide sheet metal items in 8 to 10 foot lengths. Single pieces less than 8 feet long may be used to connect to factory-fabricated inside and outside corners, and at ends of runs. Factory fabricate corner pieces with minimum 12 inch legs. Provide accessories and other items essential to complete the sheet metal installation. Provide accessories made of the same or compatible materials as the items to which they are applied. Fabricate sheet metal items of the materials specified below and to the gage, thickness, or weight shown in Table I at the end of this section. Provide sheet metal items with mill finish unless specified otherwise. Where more than one material is listed for a particular item in Table I, each is acceptable and may be used, except as follows:

#### 2.2.1 Exposed Sheet Metal Items

Provide exposed sheet metal items of the same material. Consider the following as exposed sheet metal: gutters, including hangers; downspouts; gravel stops and fascia; cap, valley, steeped, base, and eave flashings and related accessories.

#### 2.2.2 Drainage

Do not use copper for an exposed item if drainage from that item will pass over exposed masonry, stonework or other metal surfaces. In addition to the metals listed in Table I, lead-coated copper may be used for such items.

#### 2.2.3 Copper, Sheet and Strip

Provide in accordance with ASTM B370, cold-rolled temper, H 00 (standard).

#### 2.2.4 Lead-Coated Copper Sheet

Provide in accordance with ASTM B101.

#### 2.2.5 Lead Sheet

Provide in a minimum weight of 4 pounds per square foot.

#### 2.2.6 Steel Sheet, Zinc-Coated (Galvanized)

Provide in accordance with ASTM A653/A653M, a minimum of 24 gauge, and a minimum zinc coat weighting of 90 ounces per square feet.

#### 2.2.7 Zinc Sheet and Strip

Provide in accordance with ASTM B69, Type I, a minimum of 0.024 inch thick.

#### 2.2.8 Steel Sheet, Aluminum Zinc-Coated

Provide in accordance with ASTM A792/A792M, a minimum of 24 gauge.

#### 2.2.9 Stainless Steel

Provide in accordance with ASTM A480/A480M, Type 302 or 304, 2D Finish, fully annealed, dead-soft temper, a minimum of 24 gauge.

#### 2.2.10 Terne-Coated Steel

Provide in accordance with ASTM A308/A308M, a minimum of 14 by 20 inch with minimum of 40 pound coating per double base box. ASTM A308/A308M.

#### 2.2.11 Aluminum Alloy Sheet and Plate

Provide in accordance with ASTM B209/B209M anodized color form alloy, and temper appropriate for use. Provide material not less than 0.032-in in thickness.

#### 2.2.12 Finishes

Provide exposed exterior sheet metal and aluminum with a baked on, factory applied coil fluoropolymer color coating or approved equal siliconized polyester coating. Dry film thickness of coatings must be 0.8 to 1.3 mils. Color to be selected from manufacturer's full range of color choices. Field applications of color coatings are prohibited and will be rejected.

##### 2.2.12.1 Warranty on Finishes

Provide a manufacturer's warranty to repair, or replace, sheet metal flashing and trim that shows evidence of deterioration of factory-applied finishes within 20 years from date of project substantial completion. Deterioration includes the following:

- a. Color fading more than 5 Delta E units when tested in accordance with ASTM D2244.
- b. Chalking in excess of a No. 8 rating when tested in accordance with ASTM D4214.
- c. Cracking, checking, peeling, or failure of paint to adhere to bare metal.

#### 2.2.13 Cool Roof Finishes

Provide roof finishes having a minimum 3-year aged solar reflectance of 0.55, and a minimum 3-year aged thermal emittance of 0.75 when tested in accordance with ANSI/CRRC S100, or, a minimum 3-year aged Solar Reflectance Index of 64 when determined in accordance with the Solar Reflectance Index method in ASTM E1980 using a convection coefficient of 2.1 BTU per h ft<sup>2</sup>, to comply with ASHRAE 90.1 - IP.

2.2.14 Aluminum Alloy, Extruded Bars, Rods, Shapes, and Tubes

ASTM B221.

2.2.15 Solder

Provide in accordance with ASTM B32, 95-5 tin-antimony.

2.2.16 Reglets

2.2.16.1 Polyvinyl Chloride Reglets

Provide in accordance with ASTM D1784, Type II, Grade 1, Class 14333-D, 0.075 inch minimum thickness.

2.2.16.2 Metal Reglets

Provide factory fabricated caulked type or friction type reglets with a minimum opening of 1/4 inch and a depth of 1-1/4 inch, as approved.

2.2.16.2.1 Caulked Reglets

Provide with rounded edges, temporary reinforcing cores, and accessories as required for securing to adjacent construction. Provide built-up mitered corner pieces for inside and outside corners.

2.2.16.2.2 Friction Reglets

Provide with flashing receiving slots not less than 5/8 inch deep, one inch jointing tongues, and upper and lower anchoring flanges installed at 24 inch maximum snap-lock type receiver.

2.2.17 Scuppers

Line interiors of scupper openings with sheet metal. Provide a drip edge at bottom edges with returns of not less than one inch against the face of the outside wall at the top and sides. Provide the perimeter of the lining approximately 1/2 inch less than the perimeter of the scupper.

2.2.18 Conductor Heads

Provide conductor heads and screens in the same material as downspouts. Provide outlet tubes not less than 4 inches long.

2.2.19 Splash Pans

Provide splash pans where downspouts discharge onto roof surfaces and at locations indicated. Unless otherwise indicated, provide pans not less than 24 inches long by 18 inches wide with metal ribs across bottoms of pans. Provide sides of pans with vertical baffles not less than one inch high in the front, and 4 inches high in the back.

2.2.20 Copings

Unless otherwise indicated, provide copings in copper sheets, 8 or 10 feet long, joined by a 3/4 inch locked and soldered seam.

#### 2.2.21 Bituminous Plastic Cement

Provide in accordance with ASTM D4586/D4586M, Type I.

#### 2.2.22 Roofing Felt

Provide in accordance with ASTM D226/D226M Type I.

#### 2.2.23 Asphalt Primer

Provide in accordance with ASTM D41/D41M.

#### 2.2.24 Fasteners

Use the same metal as, or a metal compatible with the item fastened. Use stainless steel fasteners to fasten. Confirm compatibility of fasteners and items to be fastened to avoid galvanic corrosion due to dissimilar materials.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

##### 3.1.1 Workmanship

Make lines and angles sharp and true. Free exposed surfaces from visible wave, warp, buckle, and tool marks. Fold back exposed edges neatly to form a 1/2 inch hem on the concealed side. Make sheet metal exposed to the weather watertight with provisions for expansion and contraction.

Make surfaces to receive sheet metal plumb and true, clean, even, smooth, dry, and free of defects and projections. For installation of items not shown in detail or not covered by specifications conform to the applicable requirements of SMACNA 1793, Architectural Sheet Metal Manual. Provide sheet metal flashing in the angles formed where roof decks abut walls, curbs, ventilators, pipes, or other vertical surfaces and wherever indicated and necessary to make the work watertight. Join sheet metal items together as shown in Table II.

##### 3.1.2 Nailing

Confine nailing of sheet metal generally to sheet metal having a maximum width of 18 inches. Confine nailing of flashing to one edge only. Space nails evenly not over 3 inch on center and approximately 1/2 inch from edge unless otherwise specified or indicated. Face nailing is not permitted. Where sheet metal is applied to other than wood surfaces, include in shop drawings, the locations for sleepers and nailing strips required to secure the work. Secure flashing at one-half the normal interval to ensure a wind-resistant installation.

##### 3.1.3 Cleats

Provide cleats for sheet metal 18 inches and over in width. Space cleats evenly not over 12 inches on center unless otherwise specified or indicated. Unless otherwise specified, provide cleats of 2 inches wide by 3 inches long and of the same material and thickness as the sheet metal being installed. Secure one end of the cleat with two nails and the cleat folded back over the nailheads. Lock the other end into the seam. Pre-tin cleats for soldered seams.



#### 3.1.4 Bolts, Rivets, and Screws

Install bolts, rivets, and screws where indicated or required. Provide compatible washers where required to protect surface of sheet metal and to provide a watertight connection. Provide mechanically formed joints in aluminum sheets 0.040 inches or less in thickness.

#### 3.1.5 Seams

Straight and uniform in width and height with no solder showing on the face.

##### 3.1.5.1 Flat-lock Seams

Finish not less than 3/4 inch wide.

##### 3.1.5.2 Lap Seams

Finish soldered seams not less than one inch wide. Overlap seams not soldered, not less than 3 inches.

##### 3.1.5.3 Loose-Lock Expansion Seams

Not less than 3 inches wide; provide minimum one inch movement within the joint. Completely fill the joints with the specified sealant, applied at not less than 1/8 inch thick bed.

##### 3.1.5.4 Standing Seams

Not less than one inch high, double locked without solder.

##### 3.1.5.5 Flat Seams

Make seams in the direction of the flow.

#### 3.1.6 Soldering

Where soldering is specified, apply to copper, terne-coated stainless steel, zinc-coated steel, and stainless steel items. Pre-tin edges of sheet metal before soldering is begun. Seal the joints in aluminum sheets of 0.040 inch or less in thickness with specified sealants. Do not solder aluminum.

##### 3.1.6.1 Edges

Scrape or wire-brush the edges of lead-coated material to be soldered to produce a bright surface. Flux brush the seams in before soldering. Treat with soldering acid flux the edges of stainless steel to be pre-tinned. Seal the joints in aluminum sheets of 0.040 inch or less in thickness with specified sealants. Do not solder aluminum.

#### 3.1.7 Welding and Mechanical Fastening

Use welding for aluminum of thickness greater than 0.040 inch. Aluminum 0.040 inch or less in thickness must be butted and the space backed with formed flashing plate; or lock joined, mechanically fastened, and filled with sealant as recommended by the aluminum manufacturer.

#### 3.1.7.1 Welding of Aluminum

Use welding of the inert gas, shield-arc type. For procedures, appearance and quality of welds, and the methods used in correcting welding work, conform to AWS D1.2/D1.2M.

#### 3.1.7.2 Mechanical Fastening of Aluminum

Use No. 12, aluminum alloy, sheet metal screws or other suitable aluminum alloy or stainless steel fasteners. Drive fasteners in holes made with a No. 26 drill in securing side laps, end laps, and flashings. Space fasteners 12 inches maximum on center. Where end lap fasteners are required to improve closure, locate the end lap fasteners not more than 2 inches from the end of the overlapping sheet.

#### 3.1.8 Protection from Contact with Dissimilar Materials

##### 3.1.8.1 Copper or Copper-bearing Alloys

Paint with heavy-bodied bituminous paint surfaces in contact with dissimilar metal, or separate the surfaces by means of moistureproof building felts.

##### 3.1.8.2 Aluminum

Do not allow aluminum surfaces in direct contact with other metals except stainless steel, zinc, or zinc coating. Where aluminum contacts another metal, paint the dissimilar metal with a primer followed by two coats of aluminum paint. Where drainage from a dissimilar metal passes over aluminum, paint the dissimilar metal with a non-lead pigmented paint. Aluminum may be used over concrete construction, provided that required reglets are of stainless steel and aluminum surface in contact with concrete or masonry is coated with bituminous paint or zinc chromate primer.

##### 3.1.8.3 Metal Surfaces

Paint surfaces in contact with mortar, concrete, or other masonry materials with alkali-resistant coatings such as heavy-bodied bituminous paint.

##### 3.1.8.4 Wood or Other Absorptive Materials

Paint surfaces that may become repeatedly wet and in contact with metal with two coats of aluminum paint or a coat of heavy-bodied bituminous paint.

#### 3.1.9 Expansion and Contraction

Provide expansion and contraction joints at not more than 32 foot intervals for aluminum and at not more than 40 foot intervals for other metals. Provide an additional joint where the distance between the last expansion joint and the end of the continuous run is more than half the required interval. Space joints evenly. Join extruded aluminum gravel stops and fascia by expansion and contraction joints spaced not more than 12 feet apart.

### 3.1.10 Base Flashing

Extend up vertical surfaces of the flashing not less than 8 inches and not less than 4 inches under the roof covering. Where finish wall coverings form a counterflashing, extend the vertical leg of the flashing up behind the applied wall covering not less than 6 inches. Overlap the flashing strips with the previously laid flashing not less than 3 inches. Fasten the strips at their upper edge to the deck. Horizontal flashing at vertical surfaces must extend vertically above the roof surface and fastened at their upper edge to the deck a minimum of 6 inches on center with hex headed, galvanized shielded screws a minimum of 2 inch lap of any surface. Solder end laps and provide for expansion and contraction. Extend the metal flashing over crickets at the up-slope side of and similar vertical surfaces extending through sloping roofs, the metal flashings. Extend the metal flashings onto the roof covering not less than 4.5 inches at the lower side of and similar vertical surfaces extending through the roof decks. Install and fit the flashings so as to be completely weathertight. Provide factory-fabricated base flashing for interior and exterior corners. Do not use metal base flashing on built-up roofing.

### 3.1.11 Counterflashing

Except where indicated or specified otherwise, insert counterflashing in reglets located from 9 to 10 inches above roof decks, extend down vertical surfaces over upturned vertical leg of base flashings not less than 3 inches. Fold the exposed edges of counterflashings 1/2 inch. Where stepped counterflashings are required, they may be installed in short lengths a minimum 8 inches by 8 inches or may be of the preformed single piece type. Provide end laps in counterflashings not less than 3 inches and make it weathertight with plastic cement. Do not make lengths of metal counterflashings exceed 10 feet. Form flashings to the required shapes before installation. Factory form corners not less than 12 inches from the angle. Secure the flashings in the reglets with lead wedges and space not more than 18 inches apart; on short runs, place wedges closer together. Fill caulked-type reglets or raked joints which receive counterflashing with caulking compound. Turn up the concealed edge of counterflashings built into masonry or concrete walls not less than 1/4 inch and extend not less than 2 inches into the walls. Install counterflashing to provide a spring action against base flashing. Utilize two piece counterflashing components to minimize damage of counterflashing and reglets during future reroofing activities.

### 3.1.12 Metal Reglets

Keep temporary cores in place during installation. Ensure factory fabricated caulked type or friction type, reglets have a minimum opening of 1/4 inch and a minimum depth of 1-1/4 inch, when installed.

#### 3.1.12.1 Caulked Reglets

Wedge flashing in reglets with lead wedges every 18 inches, caulked full and solid with an approved compound.

#### 3.1.12.2 Friction Reglets

Install flashing snap lock receivers at 24 inches on center maximum. When flashing has been inserted the full depth of the slot, caulk the slot, lock with wedges, and fill with sealant.

### 3.1.13 Metal Drip Edges

Provide a metal drip edge, designed to allow water run-off to drip free of underlying construction, at eaves and rakes prior to the application of roofing shingles. Apply directly on the wood deck at the eaves and over the underlay along the rakes. Extend back from the edge of the deck not more than 3 inches and secure with compatible nails spaced not more than 10 inches on center along upper edge.

### 3.1.14 Gutters

The hung type of shape indicated and supported on underside by brackets that permit free thermal movement of the gutter. Provide gutters in sizes indicated complete with mitered corners, end caps, outlets, brackets, and other accessories necessary for installation. Bead with hemmed edge or reinforce the outer edge of gutter with a stiffening bar not less than 3/4 by 3/16 inch of material compatible with gutter. Fabricate gutters in sections not less than 8 feet. Lap the sections a minimum of one inch in the direction of flow or provide with concealed splice plate 6 inches minimum. Join the gutters, other than aluminum, by riveted and soldered joints. Join aluminum gutters with riveted sealed joints. Provide expansion-type slip joints midway between outlets. Install gutters below slope line of the roof so that snow and ice can slide clear. Support gutters on adjustable hangers spaced not more than 30 inches on center. Adjust gutters to slope uniformly to outlets, with high points occurring midway between outlets. Fabricate hangers and fastenings from compatible metals.

### 3.1.15 Downspouts

Space supports for downspouts according to the manufacturer's recommendation for the masonry or substrate. Types, shapes and sizes are indicated. Provide complete including elbows and offsets. Provide downspouts in approximately 10 foot lengths. Provide end joints to telescope not less than 1/2 inch and lock longitudinal joints. Provide gutter outlets with wire ball strainers for each outlet. Provide strainers to fit tightly into outlets and be of the same material used for gutters. Keep downspouts not less than one inch away from walls. Fasten to the walls at top, bottom, and at an intermediate point not to exceed 5 feet on center with leader straps or concealed rack-and-pin type fasteners. Form straps and fasteners of metal compatible with the downspouts.

#### 3.1.15.1 Terminations

Neatly fit into the drainage connection the downspouts terminating in drainage lines and fill the joints with a portland cement mortar cap sloped away from the downspout. Provide downspouts terminating in splash blocks with elbow-type fittings. Provide splash pans as specified.

### 3.1.16 Flashing for Roof Drains

Provide a 30 inches square sheet indicated. Taper insulation to drain from 24 inches out. Set flashing on finished felts in a full bed of asphalt roof cement, ASTM D4586/D4586M. Heavily coat the drain flashing ring with asphalt roof cement. Clamp the roof membrane, flashing sheet, and stripping felt in the drain clamping ring. Secure clamps so that felts and drain flashing are free of wrinkles and folds. Retrofit roof drains

must conform to ANSI/SPRI RD-1.

#### 3.1.17 Splash Pans

Install splash pans lapped with horizontal roof flanges not less than 4 inches wide to form a continuous surface. Bend the rear flange of the pan to contour of can't strip and extend up 6 inches under the side wall covering or to height of base flashing under counterflashing. Bed the pans and roof flanges in plastic bituminous cement and strip-flash as specified.

#### 3.1.18 Flashing at Wall Openings

Install pan flashing in the rough opening sill at all penetrations in the exterior wall assemblies, such as windows, louvers, storefronts and curtain walls. Pan sill flashing must have end dams at both jambs a minimum of 2 in high and a rear dam of 2 in high. Flashing must comply with ASTM E2112 and SMACNA 1793.

#### 3.1.19 Sheet Metal Covering on Flat, Sloped, or Curved Surfaces

Except as specified or indicated otherwise, cover and flash all minor flat, sloped, or curved surfaces such as crickets, bulkheads, dormers and small decks with metal sheets of the material used for flashing; maximum size of sheets, 16 by 18 inches. Fasten sheets to sheathing with metal cleats. Lock seams and solder. Lock aluminum seams as recommended by aluminum manufacturer. Provide an underlayment of roofing felt for all sheet metal covering.

#### 3.1.20 Expansion Joints

Provide expansion joints for roofs, walls, and floors as indicated. Provide expansion joints in continuous sheet metal at and at 32 foot intervals for aluminum,. Provide evenly spaced joints. Provide an additional joint where the distance between the last expansion joint and the end of the continuous run is more than half the required interval spacing. Conform to the requirements of Table I.

##### 3.1.20.1 Roof Expansion Joints

Consist of curb with wood nailing members on each side of joint, bituminous base flashing, metal counterflashing, and metal joint cover. Bituminous base flashing is specified in Roofing Section. Provide counterflashing as specified in paragraph COUNTERFLASHING, except as follows: Provide counterflashing with vertical leg of suitable depth to enable forming into a horizontal continuous cleat. Secure the inner edge to the nailing member. Make the outer edge projection not less than one inch for flashing on one side of the expansion joint and be less than the width of the expansion joint plus one inch for flashing on the other side of the joint. Hook the expansion joint cover over the projecting outer edges of counterflashing. Provide roof joint with a joint cover of the width indicated. Hook and lock one edge of the joint cover over the shorter projecting flange of the continuous cleat, and the other edge hooked over and loose locked with the longer projecting flange. Joints are specified in Table II.

##### 3.1.20.2 Floor and Wall Expansion Joints

Provide U-shape with extended flanges for expansion joints in concrete and

masonry walls and in floor slabs.

#### 3.1.21 Flashing at Roof Penetrations and Equipment Supports

Provide metal flashing for all pipes, ducts, and conduits projecting through the roof surface and for equipment supports, guy wire anchors, and similar items supported by or attached to the roof deck. Goose-necks, rain hoods, power roof ventilators, and vent stacks are specified in 07 61 15.00 20, ALUMINUM STANDING SEAM ROOFING.

#### 3.1.22 Single Pipe Vents

See Table I, footnote (d). Set flange of sleeve in bituminous plastic cement and nail 3 inches on center. Bend the top of sleeve over and extend down into the vent pipe a minimum of 2 inches. For long runs or long rises above the deck, where it is impractical to cover the vent pipe with lead, use a two-piece formed metal housing. Set metal housing with a metal sleeve having a 4 inches roof flange in bituminous plastic cement and nailed 3 inches on center. Extend sleeve a minimum of 8 inches above the roof deck and lapped a minimum of 3 inches by a metal hood secured to the vent pipe by a draw band. Seal the area of hood in contact with vent pipe with an approved sealant.

#### 3.1.23 Stepped Flashing

Provide stepped flashing where sloping roofs surfaced with shingles abut vertical surfaces. Place separate pieces of base flashing in alternate shingle courses.

#### 3.1.24 Through Wall Flashing

Provide through wall flashing as required in the applicable wall system Sections.

### 3.2 PAINTING

Touch ups in the field may be applied only after metal substrates have been cleaned and pretreated in accordance with manufacturer's written instructions and products.

Field-paint dissimilar sheet metals in contact to separate and deter galvanic interactions.

#### 3.2.1 Aluminum Surfaces

Clean with solvent and apply one coat of zinc-molybdate primer and one coat of aluminum paint.

### 3.3 CLEANING

Clean exposed sheet metal work at completion of installation. Remove grease and oil films, handling marks, contamination from steel wool, fittings and drilling debris, and scrub-clean. Free the exposed metal surfaces of dents, creases, waves, scratch marks, and solder or weld marks.

### 3.4 REPAIRS TO FINISH

Scratches, abrasions, and minor surface defects of finish may be repaired in accordance with the manufacturer's printed instructions and as

approved. Repair damaged surfaces caused by scratches, blemishes, and variations of color and surface texture. Replace items which cannot be repaired.

### 3.5 FIELD QUALITY CONTROL

Establish and maintain a Quality Control Plan for sheet metal used in conjunction with roofing to assure compliance of the installed sheet metalwork with the Contract requirements. Remove work that is not in compliance with the Contract and replace or correct. Include quality control, but not be limited to, the following:

- a. Observation of environmental conditions; number and skill level of sheet metal workers; condition of substrate.
- b. Verification that specified material is provided and installed.
- c. Inspection of sheet metalwork, for proper size(s) and thickness(es), fastening and joining, and proper installation.

#### 3.5.1 Procedure

Submit for approval prior to start of roofing work. Include a checklist of points to be observed. Document the actual quality control observations and inspections. Provide a copy of the documentation to the Contracting Officer at the end of each day.

| TABLE I. SHEET METAL WEIGHTS, THICKNESSES, AND GAGES |                          |                       |                                   |
|------------------------------------------------------|--------------------------|-----------------------|-----------------------------------|
| Sheet Metal Items                                    | Aluminum, inch           | Stainless Steel, inch | Zinc-Coated Steel, U.S. Std. Gage |
| Building Expansion Joints                            |                          |                       |                                   |
| Cover                                                | .032                     | .015                  | 24                                |
| Covering on minor flat, pitched or curved surfaces   | .040                     | .018                  | -                                 |
| Downspouts and leaders                               | .032                     | .015                  | 24                                |
| Downspout clips and anchors                          | .040 clip<br>.125 anchor | -                     | -                                 |
| Downspout straps, 2-inch                             | .060                     | .050                  | -                                 |

| TABLE I. SHEET METAL WEIGHTS, THICKNESSES, AND GAGES |                         |                       |                                   |
|------------------------------------------------------|-------------------------|-----------------------|-----------------------------------|
| Sheet Metal Items                                    | Aluminum, inch          | Stainless Steel, inch | Zinc-Coated Steel, U.S. Std. Gage |
| Strainers, wire diameter or gage                     | .144 diameter           | .109 diameter         |                                   |
| Flashings:                                           |                         |                       |                                   |
| Base                                                 | .040                    | .018                  | 24                                |
| Cap (Counter-flashing)                               | .032                    | .015                  | 26                                |
| Pipe vent sleeve (d)                                 |                         |                       |                                   |
| Coping                                               | -                       | -                     | -                                 |
| Gravel stops and fascia:                             |                         |                       |                                   |
| Extrusions                                           | .075                    | -                     | -                                 |
| Sheets, corrugated                                   | .032                    | .015                  | -                                 |
| Sheets, smooth                                       | .050                    | .018                  | 24                                |
| Edge strip                                           | .050                    | .025                  | -                                 |
| Gutters:                                             |                         |                       |                                   |
| Gutter section                                       | .032                    | .015                  | 24                                |
| Continuous cleat                                     | .032                    | .015                  | 24                                |
| Hangers, dimensions                                  | 1 inch by .080 inch (c) | 1 inch by .037 inch   | -                                 |
| Joint Cover plates (See Table II)                    | .032                    | .015                  | 24                                |
| Reglets (c)                                          | -                       | .010                  | -                                 |
| Splash pans                                          | .040                    | .018                  | -                                 |
| (a) Brass.                                           |                         |                       |                                   |
| (b) May be lead weighing 4 pounds per square foot.   |                         |                       |                                   |
| (c) May be polyvinyl chloride.                       |                         |                       |                                   |



| TABLE I. SHEET METAL WEIGHTS, THICKNESSES, AND GAGES                                                                                                |                |                       |                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------------|-----------------------------------|
| Sheet Metal Items                                                                                                                                   | Aluminum, inch | Stainless Steel, inch | Zinc-Coated Steel, U.S. Std. Gage |
| (d) 2.5 pound minimum lead sleeve with 4 inch flange. Where lead sleeve is impractical, refer to paragraph SINGLE PIPE VENTS for optional material. |                |                       |                                   |

| TABLE II. SHEET METAL JOINTS                                 |                                                                             |                                                                        |                                                                                                                                                     |
|--------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| TYPE OF JOINT                                                |                                                                             |                                                                        |                                                                                                                                                     |
| Item Designation                                             | Copper, Terne-Coated Stainless Steel, Zinc-Coated Steel and Stainless Steel | Aluminum                                                               | Remarks                                                                                                                                             |
| Joint cap for building expansion seam, cleated joint at roof | 1.25 inch single lock, standing seam, cleated                               | 1.25 inch single lock, standing                                        | --                                                                                                                                                  |
| Flashings                                                    |                                                                             |                                                                        |                                                                                                                                                     |
| Base                                                         | One inch 3 inch lap for expansion joint                                     | One inch flat locked, soldered; sealed; 3 inch lap for expansion joint | Aluminum manufacturer's recommended hard setting sealant for locked aluminum joints. Fill each metal expansion joint with a joint sealing compound. |
| Cap-in reglet                                                | 3 inch lap                                                                  | 3 inch lap                                                             | Seal groove with joint sealing compound.                                                                                                            |
| Reglets                                                      | Butt joint                                                                  | --                                                                     | Seal reglet groove with joint sealing compound.                                                                                                     |

| TABLE II. SHEET METAL JOINTS                                                           |                                                                                        |                                                                                               |                                                                                 |
|----------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| TYPE OF JOINT                                                                          |                                                                                        |                                                                                               |                                                                                 |
| Item Designation                                                                       | Copper, Terne-Coated Stainless Steel, Zinc-Coated Steel and Stainless Steel            | Aluminum                                                                                      | Remarks                                                                         |
| Eave                                                                                   | One inch flat locked, cleated. One inch loose locked, sealed expansion joint, cleated. | One inch flat locked, locked, cleated one inch loose locked, sealed expansion joints, cleated | Same as base flashing.                                                          |
| Stepped                                                                                | 3 inch lap                                                                             | 3 inch lap                                                                                    | --                                                                              |
| Valley                                                                                 | 6 inch lap cleated                                                                     | 6 inch lap cleated                                                                            | --                                                                              |
| Edge strip                                                                             | Butt                                                                                   | Butt                                                                                          | --                                                                              |
| Gravel stops:                                                                          |                                                                                        |                                                                                               |                                                                                 |
| Extrusions                                                                             | --                                                                                     | Butt with 1/2 inch space                                                                      | Use sheet flashing beneath and a cover plate                                    |
| Sheet, smooth                                                                          | Butt with 1/4 inch space                                                               | Butt with 1/4 inch space                                                                      | Use sheet flashing backup plate.                                                |
| Sheet, corrugated                                                                      | Butt with 1/4 inch space                                                               | Butt with 1/4 inch space                                                                      | Use sheet flashing beneath and a cover plate or a combination unit              |
| Gutters                                                                                | 1.5 inch lap, riveted and soldered                                                     | One inch flat locked riveted and sealed                                                       | Aluminum producers recommended hard setting sealant for locked aluminum joints. |
| (a) Provide a 3 inch lap elastomeric flashing with manufacturer's recommended sealant. |                                                                                        |                                                                                               |                                                                                 |
| (b) Seal Polyvinyl chloride reglet with manufacturer's recommended sealant.            |                                                                                        |                                                                                               |                                                                                 |

-- End of Section --

SECTION 07 81 00

SPRAY-APPLIED FIREPROOFING  
**02/11**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASSOCIATION OF THE WALL AND CEILING INDUSTRY (AWCI)

AWCI TM 12-A (1997; 3rd Ed) Standard Practice for the Testing and Inspection of Field Applied Sprayed Fire-Resistive Materials; An Annotated Guide

ASTM INTERNATIONAL (ASTM)

ASTM E119 (2024) Standard Test Methods for Fire Tests of Building Construction and Materials

ASTM E605/E605M (2019; R 2023) Standard Test Methods for Thickness and Density of Sprayed Fire-Resistive Material (SFRM) Applied to Structural Members

ASTM E736 (2000; R 2011) Cohesion/Adhesion of Sprayed Fire-Resistive Materials Applied to Structural Members

ICC EVALUATION SERVICE, INC. (ICC-ES)

ICC-ES AC23 (2012; R 2016) Acceptance Criteria for Sprayed Fire-resistant Materials (SFRMs), Intumescent Fire-resistant Coatings and Mastic Fire-resistant Coatings Used to Protect Structural Steel Members

UL SOLUTIONS (UL)

UL 263 (2011; Reprint Aug 2021) UL Standard for Safety Fire Tests of Building Construction and Materials

UL Fire Resistance (2014) Fire Resistance Directory

1.2 SYSTEM DESCRIPTION

1.2.1 General Requirements

Protect all structural steel, undersides of steel floors (if required) and steel roof decks (if required) with spray-applied fireproofing to a fire resistance hour-rating as indicated below, unless otherwise indicated.

### 1.2.2 Fire Resistance Rating

Fire resistance ratings must be in accordance with the fire rated assemblies listed in UL Fire Resistance. Proposed materials not listed in UL Fire Resistance must have fire resistance ratings at least equal to the UL Fire Resistance ratings as determined by an approved independent testing laboratory, based on tests specified in UL 263 or ASTM E119. Submit reports and test records, attesting that the fireproofing material conforms to the specified requirements. Each test report must conform to the report requirements specified by the test method. For the underside of the decking use metal lath installed prior to the fireproofing material or Rigid Board Fireproofing Material as outlined in the UL Fire Resistance Directory Volume 1. Apply fireproofing to structural steel members, with the following hourly fire resistance rating and in accordance with the following UL design or approved equivalent. Use unrestrained fire resistance ratings, unless the architect/engineer has specified that the degree of thermal restraint of the construction meets or exceeds the degree of thermal restraint of the tested assembly. Performance tests must be in accordance with ASTM E119.

| Fire Rating Schedule         |               |                     |
|------------------------------|---------------|---------------------|
| Element                      | Hourly Rating | UL Design Reference |
| Columns supporting equipment | 2             | Y615, Y616          |
| Roof decks                   | 1             | D902                |

### 1.2.3 Evaluation Reports - ICC-ES Reports

Evaluate materials in accordance with ICC-ES AC23. Include ICC-ES Reports as part of the Submittals below. The reports will identify the product as code compliant and having met the physical performance requirements outlined in paragraphs "Dry Density and Cohesion/Adhesion" through "Air Erosion".

## 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-03 Product Data

Fireproofing Material; G

#### SD-04 Samples

Spray-Applied Fireproofing; G

#### SD-06 Test Reports

Fire Resistance Rating; G

Field Tests; G

Evaluation Reports; G

SD-07 Certificates

Installer Qualifications; G

Surface Preparation Report; G

Manufacturer's Inspection Report; G

#### 1.4 QUALITY ASSURANCE

##### 1.4.1 Installer Qualifications

Engage an experienced installer that is certified, licensed, or otherwise qualified by the spray-on fireproofing manufacturer as having the necessary experience, staff, and training to install the manufacturer's products in accordance with specified requirements. Submit manufacturer's certification that each listed installer is qualified and trained to install the specified fireproofing. Show evidence that each fireproofing installer has had a minimum of 3 years experience in installing the specified type of fireproofing. Each installer of fireproofing material must be trained, have a minimum of 3 years experience and a minimum of three installations using fireproofing of the type specified. A manufacturer's willingness to sell its products to the Contractor or installer does not infer qualification of the buyer.

##### 1.4.2 Pre-Installation Meeting

Hold a meeting with the installer, field testing agency, the manufacturer, subcontractors (whose employees come into contact with the fireproofing), and the Contracting Officer prior to the installation of any fireproofing material to review the substrates for acceptability, method of application, applied thickness, patching, repair, inspection and testing procedures.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

Deliver packaged material in the original unopened containers, marked to show the brand name, the manufacturer, and the UL markings. Keep fireproofing material dry until ready to be used, and store off the ground, under cover and away from damp surfaces. Damaged or opened containers will be rejected. Apply material with shelf-life prior to expiration of the shelf-life.

#### 1.6 PROJECT/SITE CONDITIONS

##### 1.6.1 Temperature

Maintain substrate and ambient air temperatures above 40 degrees F during application and for 24 hours before and after application. Maintain relative humidity within the limits recommended by the fireproofing manufacturer.

##### 1.6.2 Ventilation

Provide adequate ventilation to properly dry the fireproofing after application. In enclosed areas, provide a minimum of 4 air exchanges per

hour by forced air circulation.

## PART 2 PRODUCTS

### 2.1 SPRAY-APPLIED INTUMESCENT EPOXY COATING SYSTEM

Provide a two-component epoxy based intumescent fire protective coating that meets the following requirements.

- a. On curing it forms a flexible and tough epoxy barrier which transforms into a ceramic-like, insulating char to provide thermal protection of the substrate in the event of a fire.
- b. The coating system includes the manufacturer's required surface preparation, primer, and fire protective layer, and topcoat.
- c. The coating system protects the substrate from corrosion and retain its fire protection properties under aggressive chemical environments.
- d. Resistant to solvents, acids, alkalis, salts and abrasion while retaining its fire protective properties.

Provide a system that exhibits the following properties:

#### 2.1.1 Percent Solids by Weight

100 percent

#### 2.1.2 In Service Temperature Restrictions

Up to 150 degrees F

#### 2.1.3 Application Method

Air spray or specialized plural component airless equipment approved by the manufacture. Troweling can be used for small areas or touch-up work.

#### 2.1.4 Drying Time

Approximately 24 hours to achieve a Shore D hardness of 25.

#### 2.1.5 Shelf Life

Minimum shelf life under proper storage condition is 1 Year from date of manufacture.

#### 2.1.6 Pot Life

Approximately 40 minutes at 77 degrees F and 50 percent relative humidity. Pot life is not a factor when using specialized plural component airless spray equipment.

#### 2.1.7 Flash Point

Greater than 212 degrees F Pensky-Martens for each component.

## PART 3 EXECUTION

### 3.1 SURFACE PREPARATION

Thoroughly clean surfaces to be fireproofed of dirt, grease, oil, paint, primers, loose rust, rolling lubricant, mill scale or other contaminants that will interfere with the proper bonding of the sprayed fireproofing to the substrate. Test painted/primed steel substrates in accordance with ASTM E736, with specified sprayed fireproofing material, to provide the required fire-resistance rating; painted or primed steel surfaces may require a fireproofing bond test to determine if the paint formulation will impair proper adhesion. Certify the acceptability of surfaces to receive sprayed-applied fireproofing by inspection and submit a Surface Preparation Report accordingly. List the structural members and the areas that have been inspected and certified. Clear overhead areas to be fireproofed of all obstructions interfering with the uniform application of the spray-applied fireproofing. Install hardware such as support sleeves, inserts, clips, hanger attachment devices and the like prior to the application of the fireproofing. Condition of the surfaces must be acceptable to the manufacturer prior to application of spray-applied fireproofing. Applications listed for use on primed surfaces must be in accordance with the manufacturer's recommendations and standards, and detailed in submittal item SD-03 Product Data.

### 3.2 PROTECTION

Cover surfaces not to receive spray-applied fireproofing to prevent contamination by splatter, rebound and overspray. Cover exterior openings in areas to receive spray-applied fireproofing prior to and during application of fireproofing with tarpaulins or other approved material. Clean surfaces not to receive fireproofing of fireproofing and sealer.

### 3.3 FIREPROOFING MATERIAL

Mix fireproofing material in accordance with the manufacturer's recommendations. Submit data identifying performance characteristics of fireproofing material. Data includes recommended application requirements and indicate thickness of fireproofing to be applied to achieve each required fire rating.

### 3.4 APPLICATION

#### 3.4.1 Sequence

Prior to application of fireproofing on each floor, inspect and approve application equipment, water supply and pressure, and the application procedures. If fireproofing is required to be applied to underside of steel roof deck and steel floor assemblies, apply it only after respective roof or floor construction is complete. Do not allow roof or floor traffic during application. Apply fireproofing material prior to the installation of ductwork, piping and conduits which would interfere with uniform application of the fireproofing.

#### 3.4.2 Application Technique

Maintain water pressure and volume to manufacturer's recommendations throughout the fireproofing application. Apply fireproofing material to the thickness and density established for the specified fire resistance rating, in accordance with the procedure recommended by the manufacturer,

and to a uniform density and texture. Do not tamp fireproofing material to achieve the desired density.

#### 3.4.3 Sealer Application

If sealer is required by the product used, apply it after field testing has been conducted and after corrective measures and repairs, if required, have been completed.

#### 3.4.4 Applied Thickness

The minimum average thickness must be no less than 0.375 inches. Thicknesses must not be less than required to achieve designated fire resistance ratings. If the specified thickness is greater than or equal to 1 inch, any individual measurement must not be less than the specified thickness minus 0.25 inches. If the specified thickness is less than 1 inch, any individual measurement must not be less than the specified thickness minus 25 percent.

#### 3.4.5 Application of Spray-Applied Intumescent Epoxy Coating System

Prepare surfaces and apply the spray-applied Intumescent epoxy coating system in accordance with the manufacturer's written recommendations.

### 3.5 MANUFACTURER'S SERVICES

#### 3.5.1 General

The manufacturer, or its representative, must be onsite prior to, periodically during, and at completion of the application, to provide the specified inspections and certifications; and to ensure that preparations are adequate and that the material is applied according to manufacturer's recommendations and the contract requirements.

#### 3.5.2 Manufacturer's Inspection

Inspect the fireproofing work after the work is completed on each floor or area, including testing, repair and clean-up, and certify that the work complies with the manufacturer's criteria and recommendations. Before the sprayed material is covered, and after all of the fireproofing work is completed, including repair, testing, and clean-up; and after mechanical, electrical and other work in contact with fireproofing material has been completed, re-inspect the work and certify that the entire project complies with the manufacturer's criteria and recommendations. Obtain and submit the Manufacturer's Inspection Report and certifications of approval stating that the spray-applied fireproofing in the entire project complies with the manufacturer's criteria and recommendations.

### 3.6 FIELD TESTS

Test the applied fireproofing by an approved independent testing laboratory to be selected by the A/E and paid for by the Contractor. Submit test reports documenting results of tests on the applied material in the project. Include defects identified, repair procedures, and results of the retests when required. Perform the tests in approved locations: for density in accordance with ASTM E736, cohesion/adhesion in accordance with ASTM E736, and for thickness in accordance with ASTM E605/E605M. Determine densities in accordance with ASTM E605/E605M or Appendix A, "Alternate Method for Density Determination" of AWCI TM 12-A.



Take density determinations at the flat portion of deck, beam bottom flange, beam web, column, and an equivalent area from the top of the lower beam flange. Areas showing a density less than specified will be rejected. Locate a test sample every 10,000 square feet of floor area or two for each floor, whichever produces the greatest number of test areas. Correct any area showing less than minimum requirements. Proposed corrective measures, in writing, must be approved before starting the corrective action. Retest corrected work.

#### 3.6.1 Structural Components

Test each structural component type at floor and roof decks, beams, columns, joists, and trusses. Minimum average thickness must be as required by UL Fire Resistance. Density and cohesion/adhesion must be as specified.

#### 3.6.2 Repair

Additional fireproofing material may be added to provide proper thickness. Correct rejected areas of fireproofing to meet specified requirements by adding fireproofing material to provide the proper thickness, or by removing defects and respraying with new fireproofing material. Use same type of fireproofing material for repairs as originally applied or use patching materials recommended by the manufacturer. Retest and reinspect repaired areas. Apply fireproofing material to voids or damaged areas by hand-trowel, or by respraying.

#### 3.6.3 Visual Inspections

Inspections must be made by the certified independent laboratory prior to closure of concealed areas. These inspections may be phased, but must not occur less than 5 working days prior to the enclosure of the fireproofing. Sprayed areas must receive a final inspection. Inspect fireproofed surfaces after mechanical, electrical, and other work in contact with fireproofing material has been completed and before sprayed material is covered. Patch any locations missing fireproofing in accordance with the manufacturer's requirements.

#### 3.6.4 Patching

Patch and repair damaged fireproofing. The patching material must be the same as that specified for that area.

#### 3.7 CLEANUP

Thoroughly clean surfaces not indicated to receive fireproofing of sprayed material within a 24 hour period after application.

-- End of Section --

SECTION 07 84 00.00 22

FIRESTOPPING  
07/24

PART 1 GENERAL

1.1 SUMMARY

Furnish and install tested and listed firestopping systems, combination of materials, or devices to form an effective barrier against the spread of flame, smoke and gases, and maintain the integrity of fire resistance rated walls, barriers, partitions, floors, and ceiling-floor assemblies, including through-penetrations and construction joints and gaps.

- a. Through-penetrations include the annular space around pipes, tubes, conduit, wires, cables and vents.
- b. Construction joints include those used to accommodate expansion, contraction, wind, or seismic movement. Firestopping material must not interfere with the required movement of the joint.

Gaps requiring firestopping include gaps between the curtain wall and the floor slab and between the top of the fire-rated walls and the roof or floor deck above and at the intersection of shaft assemblies and adjoining fire resistance rated assemblies.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

|                   |                                                                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM E84          | (2023) Standard Test Method for Surface Burning Characteristics of Building Materials                                                     |
| ASTM E119         | (2022) Standard Test Methods for Fire Tests of Building Construction and Materials                                                        |
| ASTM E814         | (2013a; R 2017) Standard Test Method for Fire Tests of Penetration Firestop Systems                                                       |
| ASTM E1399/E1399M | (1997; R 2022) Standard Test Method for Cyclic Movement and Measuring the Minimum and Maximum Joint Widths of Architectural Joint Systems |
| ASTM E1966        | (2015; R 2019) Standard Test Method for Fire-Resistive Joint Systems                                                                      |

FM GLOBAL (FM)

|         |                                         |
|---------|-----------------------------------------|
| FM 4991 | (2013) Approval of Firestop Contractors |
|---------|-----------------------------------------|

FM APP GUIDE

(updated on-line) Approval Guide  
<http://www.approvalguide.com/>

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2021) International Building Code

UNDERWRITERS LABORATORIES (UL)

UL 723 (2020) UL Standard for Safety Test for  
Surface Burning Characteristics of  
Building Materials

UL 1479 (2015; Reprint May 2021) Fire Tests of  
Through-Penetration Firestops

UL 2079 (2015; Reprint Jul 2020) Tests for Fire  
Resistance of Building Joint Systems

UL Fire Resistance (2014) Fire Resistance Directory

### 1.3 SEQUENCING

Coordinate the specified work with other trades. Apply firestopping materials at penetrations of pipes and ducts, prior to insulating, unless insulation meets requirements specified for firestopping. Apply firestopping materials at building joints and construction gaps, prior to completion of enclosing walls or assemblies. Cast-in-place firestop devices must be located and installed in place before concrete placement. Pipe, conduit or cable bundles must be installed through cast-in-place device after concrete placement but before area is concealed or made inaccessible. Firestop material must be inspected and approved prior to final completion and enclosing of any assemblies that may conceal installed firestop.

### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submittals with an "S" are for inclusion in the Sustainability eNotebook, in conformance to Section 01 33 29 SUSTAINABILITY REQUIREMENTS AND REPORTING. Shop drawings (SD-02), product data (SD-03) and Certificates (SD-07) must be prepared by the firestopping contractor, combined and submitted as one complete package. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Firestopping System Shop Drawings; G

Include Manufacturer's installation instructions with the UL Listing details and Engineering Judgements.

SD-02 Product Data

Firestopping Materials; G

SD-06 Test Reports

Inspection Forms; G  
Inspection Report; G

SD-07 Certificates

Firestopping Inspector Qualifications; G  
Firestopping Installer Qualifications; G

1.5 QUALITY ASSURANCE

1.5.1 Firestopping Installer Qualifications

The Firestopping Installer must meet the requirements of item a or b below:

a. FM Research approved in accordance with FM 4991, operating as a UL Certified Firestop Contractor.

b. The Firestopping Installer must:

1) Be Certified, licensed, or otherwise qualified by the firestopping manufacturer as having the necessary training; and

2) Have a minimum of 3 years' experience in the installation of the manufacturer's products in accordance with specified requirements. Submit documentation of this experience. The documentation must include at least three projects (one per year) with the types of firestopping systems installed, and identify manufacturer of the products used. The Installer must have been certified during performance of those installations; and

3) Trained by a direct representative of the manufacturer (not distributor or agent) in the proper selection and installation procedures of the manufacturer's products. The Firestopping Installer must obtain from the manufacturer and submit written certification of training, and retain proof of certification for duration of firestopping installation. The credentials must reflect that the individual is trained and certified for installing products. Basic training in firestopping systems, materials or selection does not satisfy these requirements. A "wallet card" provided by the manufacturer after 1-2 hours of training does not establish qualifications or credentials. A manufacturer's willingness to sell its firestopping products to the Contractor or to an installer engaged by the Contractor does not in itself confer Firestopping Installer Qualifications on the buyer.

There must be a single qualified Installer for the entire project. Multiple qualified Installers employed by the same organization will be permitted. Each Installer's qualifications must be approved by the Government. When multiple Installers are submitted, identify the systems by trade or discipline for which each Installer will be responsible.

1.5.2 Firestopping Inspector Qualifications

The Firestopping Inspector must meet all of the following:

a. Be completely independent of, and divested from, the Firestopping Installer, the manufacturer, and the supplier of any material or item being inspected; and

- b. Not be a competitor of the Firestopping Installer, the contractor, the manufacturer, or supplier of any material or item being inspected; and
- d. Have a minimum of two years' experience in construction field inspections of firestopping systems, products, and assemblies. Provide proof of experience.

In addition, the Firestopping Inspector must meet one of the following::

- a. Passed the UL Firestop exam within the last three years; or
- b. Passed the FM Firestop exam within the last three years; or
- c. Passed the International Firestop Council (IFC) exam within the last three years; or
- d. Completed the International Code Council (ICC) Firestopping Credential of Learning Achievement within the last three years; or
- e. Be a registered Professional Engineer who has passed the National Council of Examiners for Engineering and Surveying (NCEES) Fire Protection Engineer exam. Submit documentation demonstrating experience in construction field inspections of firestopping systems, products, and assemblies for a minimum of 3 relevant projects. Include a description of the project and date of construction completion.

Qualifications must be valid throughout the entirety of the project inspections. Include in the qualifications submittal a signed statement asserting compliance with the requirements stated in this paragraph. If meeting the requirements outlined herein, the project Fire Protection Quality Control Specialist (FPQC) provided in Section 01 45 00 Quality Control, may serve as the Firestopping Inspector.

## 1.6 DELIVERY, STORAGE, AND HANDLING

Deliver materials in the original unopened packages or containers showing name of the manufacturer and the brand name. Store materials off the ground, protected from damage and exposure to elements and temperatures in accordance with manufacturer requirements. Remove damaged or deteriorated materials from the site. Use materials within their indicated shelf life.

## PART 2 PRODUCTS

### 2.1 FIRESTOPPING SYSTEM SHOP DRAWINGS

Submit detail drawings including manufacturer's descriptive data, typical details conforming to UL Fire Resistance or other details certified by another nationally recognized testing laboratory, installation instructions or UL listing details for a firestopping assembly in lieu of fire-test data or report. For those firestop applications for which no UL tested system is available through a manufacturer, submit a manufacturer's engineering judgement, derived from similar UL system designs or other tests for review and approval prior to installation. Submittal must indicate the firestopping material to be provided for each type of application. When more than a total of 5 penetrations and/or construction joints are to receive firestopping, provide Shop Drawings that indicate

location of each penetration and joint, its "F" "T" and "L" ratings, and type of firestopping system.

Submit floor plans indicating locations of joint systems and types of penetrations, and types of firestopping used at each location. Record type by UL list printed numbers or Engineering Judgement (EJ) reference number. Separate plans may be required to depict joint systems and penetrations for clarity, depending on the complexity of the project. Penetration system drawings must show penetrants of rated assemblies by all trades, including but not limited to, structural, mechanical, controls, plumbing, fire alarm and mass notification, electrical, telecommunications, audiovisual, security, industrial/process systems, and compressed air. Joint system drawings must include all head of wall, base of wall and vertical joints systems for rated assemblies.

## 2.2 FIRESTOPPING MATERIALS

Provide firestopping materials, supplied from a single domestic manufacturer, consisting of commercially manufactured, asbestos-free, nontoxic products FM APP GUIDE approved, or UL listed, for use with applicable construction and penetrating items, complying with the following minimum requirements:

### 2.2.1 Fire Hazard Classification

Provide material that has a flame spread of 25 or less, and a smoke developed rating of 50 or less, when tested in accordance with ASTM E84 or UL 723. Material must be an approved firestopping material as listed in UL Fire Resistance or by a nationally recognized testing laboratory.

### 2.2.2 Toxicity

Materials must have a Health Rating of 0 or 1, as indicated in the product Material Safety Data Sheet. Products must not contain hazardous chemicals or require harmful chemicals to clean material or equipment.

### 2.2.3 Fire Resistance Rating

Firestop systems must be UL Fire Resistance listed or FM APP GUIDE approved with "F" rating at least equal to fire-rating of fire wall or floor in which penetrated openings are to be protected. Where required, firestop systems must also have "T" rating at least equal to the fire-rated floor in which the openings are to be protected.

#### 2.2.3.1 Penetrations

Firestopping materials for through-penetrations, as described in paragraph SUMMARY, must provide "F", "T" and "L" fire resistance ratings in accordance with ASTM E814 or UL 1479. Provide fire resistance ratings as follows:

##### 2.2.3.1.1 Penetrations of Fire Resistance Rated Walls, Barriers and Partitions.

F Rating = Rating of wall, barrier, or partition being penetrated.

#### 2.2.3.1.2 Penetrations of Fire Resistance Rated Floors, Floor-Ceiling Assemblies and the Ceiling Membrane of Roof-Ceiling Assemblies.

Rating of wall, barrier, or assembly being penetratedWhere the penetrating item is outside of a wall cavity the F rating must be equal to the fire resistance rating of the floor penetrated, and the T rating must be in accordance with the requirements of ICC IBC.

#### 2.2.3.1.3 Penetrations of Fire and Smoke Resistance Rated Walls, Barriers, Partitions, Floors, Floor-Ceiling Assemblies, and the ceiling membrane of Roof-Ceiling Assemblies.

#### 2.2.3.2 Construction Joints and Gaps

Fire resistance ratings of construction joints, as described in paragraph SUMMARY, and gaps such as those between floor slabs and curtain walls must be the same as the construction in which they occur. Construction joints and gaps must be provided with firestopping materials and systems that have been tested in accordance with ASTM E119, ASTM E1966 or UL 2079 to meet the required fire resistance rating. Curtain wall joints must be provided with firestopping materials and systems that have been tested in accordance with ASTM E2307 to meet the required fire resistance rating. Systems installed at construction joints must meet the cycling requirements of ASTM E1399/E1399M or UL 2079. All joints at the intersection of the top of a fire resistance rated wall and the underside of a fire-rated floor, floor ceiling, or roof ceiling assembly must provide a minimum Class II movement capability as defined and tested in accordance with ASTM E1399/E1399M.

#### 2.2.4 Material Certification

Submit certificates attesting that firestopping material complies with the specified requirements. For all intumescent firestop materials used in through penetration systems, manufacturer must provide certification of compliance with UL 1479.

All materials must bear a Listing Label. Manufacturer's container labels must include the manufacturer's name, product name and product description. Other components of the firestop must also be identifiable by labeling or other method by the Designated Fire Protection Engineer (DFPE).

### PART 3 EXECUTION

#### 3.1 PREPARATION

The installer must notify the inspector of the arrival of the materials. Prior to installation, the Inspector must verify that all materials received for the installation meet the requirements of the approved Firestopping Materials.

Areas to receive firestopping must be free of dirt, grease, oil, or loose materials which may affect the fitting or fire resistance of the firestopping system. For cast-in-place firestop devices, formwork or metal deck to receive device prior to concrete placement must be sound and capable of supporting device. Prepare surfaces as recommended by the manufacturer.

### 3.2 INSTALLATION

Firestopping systems for filling floor voids 4 inches or more in any direction must be capable of supporting the same load as the floor is designed to support or be protected by a permanent barrier to prevent loading or traffic in the firestopped area. Install firestopping in accordance with manufacturer's written instructions. Provide tested and listed firestop systems in the following locations, except in floor slabs on grade:

- a. Penetrations of duct, conduit, tubing, cable and pipe through floors and through fire-resistance rated walls, barriers, partitions, and ceiling-floor assemblies.
- b. Penetrations of vertical shafts such as pipe chases, elevator shafts, and utility chutes.
- c. Gaps at the intersection of floor slabs and curtain walls, including inside of hollow curtain walls at the floor slab.
- d. Gaps at perimeter of fire-resistance rated walls, barriers, and partitions, such as the bottom of walls and floor slabs and between the top of the walls and the bottom of roof decks.
- e. Construction joints in floors and fire rated walls and partitions.
- f. Other locations where required to maintain fire resistance rating of the construction.

#### 3.2.1 Insulated Pipes and Ducts

Thermal insulation must be cut and removed where pipes or ducts pass through firestopping, unless insulation meets requirements specified for firestopping. Replace thermal insulation with a material having equal thermal insulating and firestopping characteristics.

#### 3.2.2 Fire Dampers

Install and firestop fire dampers in accordance with Section 23 30 00 HVAC AIR DISTRIBUTION. Firestop installed with fire damper must be tested and approved for use in fire damper system.

#### 3.2.3 Data and Communication Cabling

Cabling for data and communication applications must be sealed with re-enterable firestopping products devices.

##### 3.2.3.1 Re-Enterable Devices

Provide firestopping devices that are pre-manufactured modular devices, containing built-in self-sealing intumescent inserts. Allow for cable moves, additions or changes without the need to remove or replace any firestop materials. Devices must be capable of maintaining the fire resistance rating of the penetrated membrane at 0 percent to 100 percent visual fill of penetrants; while maintaining "L" rating of <10 cfm/sf measured at ambient temperature and 400 degrees F at 0 percent to 100 percent visual fill.



### 3.2.3.2 Re-Sealable Products

Provide firestopping pre-manufactured modular products, containing self-sealing intumescent inserts. Firestopping products must allow for cable moves, additions or changes. Devices must be capable of maintaining the fire resistance rating of the penetrated membrane at 0 percent to 100 percent visual fill of penetrants.

### 3.3 INSPECTION

Install one of each type of penetration and have it inspected and accepted prior to the installation of the remainder of the penetrations by one of the following:

- a. The Division, Naval Facilities Engineering Command, Designated Fire Protection Engineer (DFPE).
- b. The Firestopping Inspector, witnessed by a designated Government representative.
- c. The Firestopping Inspector, with a detailed Inspection Form submitted to the Division, Naval Facilities Engineering Command, Designated Fire Protection Engineer (DFPE) for review and approval within two days of the inspection. The Inspection Form must include:
  - 1) General project identification information.
  - 2) Name of Firestopping Inspector.
  - 3) Location of system with a partial floor plan to highlight location.
  - 4) UL system or Engineering Judgement and rating.
  - 5) Description of rated assembly being penetrated and type of penetrant.
  - 6) Photographs of destructive samples taken or photographs of steps taken during the installation.
  - 7) Measurement of cut samples must be shown in photographs with a ruler. Provide photographs of the joint or penetration prior to firestopping, after firestopping, and intermediate steps. Photographs may be provided in electronic (.jpeg or .pdf) format. Photographs must provide sufficient angles and detail to show conformance with requirements of the system.
  - 8) General observations.
  - 9) Statement of compliance or non-compliance.

The means of inspection must be coordinated with the DFPE prior to the inspection. The manufacturer's technical representative of the firestopping material must be present at the inspection. The manufacturer's technical representative may be the Firestopping Installer.

For all projects, do not cover or enclose the remainder of the firestopped areas until inspection is complete and approved by the Contracting Officer. The Firestopping Inspector must inspect the applications

initially to ensure adequate preparations (clean surfaces suitable for application, etc.) and periodically during the work to assure work has been accomplished according to the manufacturer's written instructions and the specified requirements. Submit written reports indicating locations and types of penetrations and types of firestopping used at each location. Record type by UL listed printed numbers.

### 3.3.1 Inspection Standards

3.3.1.2 The Firestopping Inspector must verify materials and systems used for firestopping and fire resistive joint systems are listed and labeled for the use intended.

3.3.1.3 The Firestopping Inspector must verify that the required firestop systems and fire resistive joint systems have been installed.

3.3.1.4 The Firestopping Inspector must verify compliance of the firestop systems and fire resistive joint systems by observing the installation process and by taking and recording measurements of the substrates and materials being installed or by destructive examination of completed installations.

3.3.1.5 Inspection frequency of firestop systems and fire resistive joint systems will depend on the method of inspection and the scope of the project.

3.3.1.5.1 The method of inspection for firestop systems must be one of the following:

- a. The Firestopping Inspector must be on site during installation and randomly witness a minimum of 10% of each type of firestop system being installed; or
- b. The Firestopping Inspector must conduct a post installation inspection, which requires destructive type verification of the firestop system and repair of the firestop system. A minimum of 2%, but not less than one, of each type of firestop system must be inspected per floor or for each area of a floor when a floor is larger than 10,000 SF. An area consists of 10,000 SF or less.

3.3.1.5.2 The method of inspection for fire resistive joint systems must be one of the following:

- a. The Firestopping Inspector must be on site during installation and randomly witness a minimum of 5% of total lineal feet of each type of fire resistive joint system being installed; or
- b. The Firestopping Inspector must conduct a post installation inspection in accordance with one of the following:
  - 1) Destructive type verification of the fire resistive joint system and repair of the joint system; or
  - 2) Disassembly and verification of the components and re-installation of the joint system; or
  - 3) Visual inspection and verification of the component or entire joint system, where a visual inspection establishes conformance; or

- 4) Other appropriate methods showing compliance with the approval process or the manufacturers' instructions.

Inspection must consist of a minimum of one sampling per type of joint system per 500 lineal feet.

3.3.1.6 Prior to construction, the Firestopping Inspector must coordinate with the DFPE to determine the types of firestop systems and subsequently the number of each type that is to be inspected. The determination of a "type" will typically be a function of a unique combination of parameters, including penetrant type (for example, metal pipe, plastic pipe, cabling), firestop material or device (for example, intumescent caulk, collar, sealant), penetrated substrate (for example, gypsum wall, concrete floor, composite floor deck), and rating of penetrated item.

3.3.1.7 Any type of firestop system that does not comply with this specification will require repair or replacement and re-inspection of that firestop system plus one full additional inspection, of the number specified herein, of that type firestop system. If non-compliance occurs on 10% or more of the quantity of firestop products or firestop systems, then inspection of those particular type firestop systems must cease. The Firestopping Installer must inspect their own work, and repair or replace those like firestops within the area prior to re-commencement of inspections by the Firestopping Inspector.

3.3.1.8 The Firestopping Inspector must maintain inspection forms. All observed deficiencies must be documented and marked on the inspection forms. The Firestopping Inspector must identify the location where a required firestop system has been omitted or where the inspection results indicate non-compliance. The Inspection Forms must include the following:

- a. General project identification information
- b. Name of Firestopping Inspector
- c. Location of system with a partial floor plan to highlight location
- d. UL system or Engineering Judgement and rating
- e. Description of rated assembly being penetrated and type of penetrant
- f. Photographs of destructive samples taken or photographs of steps taken during the installation
- g. Measurement of cut samples must be shown in photographs with a ruler. Provide photographs of the joint or penetration prior to firestopping, after firestopping, and intermediate steps. Photographs may be provided in electronic (.jpeg or .pdf) format. Photographs must provide sufficient angles and detail to show conformance with requirements of the system.
- h. General observations
- i. Statement of compliance or non-compliance

Submit the inspection forms to the DFPE for surveillance within two days of the inspection.

### 3.3.2 Inspection Report

Submit inspection report stating that firestopping work has been inspected and found to be applied according to the manufacturer's recommendations and the specified requirements.

3.3.2.1 The report must contain a cover page with the following:

- a. The project name, location, and contract number; and
- b. The name and address of the Firestopping Inspector; and
- c. The name and address of the Firestopping Installer(s), as well as the prime contractor, if different; and
- d. Report date.

3.3.2.2 The report must include drawings indicating locations and types of all penetrations and fire resistive joint systems and types of firestopping used at each location. Type must be identified by UL listed numbers. Annotate inspected systems with a unique designator that correlates with the Inspection Forms. The drawings must include smoke barriers and partitions and indicate the methods and materials used to protect penetrations.

3.3.2.3 Include UL system details or Engineering Judgements, annotated to identify specific ratings and assembly components.

3.3.2.4 The report must contain a summary page with the following:

- a. Types and quantity of each firestop system and fire resistive joint system on the project.
- b. The verification method used to ascertain compliance.
- c. The quantity of each firestop system and fire resistive joint system inspected on the project and a written statement by the Firestopping Inspector that the number of firestop systems and fire resistive joint systems inspected comply with this specification.
- d. Percentages of deficiencies for each type of firestop system and fire resistive joint system referenced.
- e. A total number of deficiencies, expressed as a percentage of the total number of firestop systems and fire resistive joint systems inspected.
- f. Narrative of inspection observations and description of means and methods used to correct deficiencies. Include a description of deviations from the design documents.
- g. Additional information, as determined by the Firestopping Inspector, required to adequately convey compliance.

3.3.2.5 The report must contain copies of all Inspection Forms generated during the inspection process, arranged chronologically.

3.3.2.6 Submit the Inspection Report at least 14 days prior to Government acceptance testing for review and approval.

### 3.3.3 Acceptance

The Government may require additional destructive testing at the time of Government acceptance testing. Systems that are subject to destructive testing must be replaced, at no cost to the Government, by the project's qualified Firestopping Installer. Confirmation that the system replaced complies must be provided by the Firestopping Inspector.

-- End of Section --

SECTION 07 92 00

JOINT SEALANTS  
**08/23**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

|            |                                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------|
| ASTM C734  | (2015; R 2019) Standard Test Method for Low-Temperature Flexibility of Latex Sealants After Artificial Weathering |
| ASTM C919  | (2022) Standard Practice for Use of Sealants in Acoustical Applications                                           |
| ASTM C920  | (2018) Standard Specification for Elastomeric Joint Sealants                                                      |
| ASTM C1193 | (2016) Standard Guide for Use of Joint Sealants                                                                   |
| ASTM C1521 | (2019; R 2020) Standard Practice for Evaluating Adhesion of Installed Weatherproofing Sealant Joints              |
| ASTM D217  | (2021a) Standard Test Methods for Cone Penetration of Lubricating Grease                                          |
| ASTM D1056 | (2020) Standard Specification for Flexible Cellular Materials - Sponge or Expanded Rubber                         |
| ASTM E84   | (2023) Standard Test Method for Surface Burning Characteristics of Building Materials                             |

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

|                    |                                                                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CDPH SECTION 01350 | (2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

|                  |                                          |
|------------------|------------------------------------------|
| SCAQMD Rule 1168 | (2022) Adhesive and Sealant Applications |
|------------------|------------------------------------------|

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for

Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Sealants; G

Primers; G

Bond Breakers; G

Backstops; G

Cleaning Solvents; G

SD-06 Test Reports

Field Adhesion; G

SD-07 Certificates

Indoor Air Quality For Interior Sealants

Indoor Air Quality For Interior Floor Joint Sealants

Indoor Air Quality For Interior Acoustical Sealants

1.3 PRODUCT DATA

Include storage requirements, shelf life, curing time, instructions for mixing and application, and accessories. Provide manufacturer's Safety Data Sheets (SDS) for each solvent, primer and sealant material proposed.

1.4 ENVIRONMENTAL CONDITIONS

Apply sealant when the ambient temperature is between 40 and 90 degrees F.

1.5 DELIVERY AND STORAGE

Deliver materials to the jobsite in unopened manufacturers' sealed shipping containers, with brand name, date of manufacture, color, and material designation clearly marked thereon. Label elastomeric sealant containers to identify type, class, grade, and use. Handle and store materials in accordance with manufacturer's printed instructions. Prevent exposure to foreign materials or subjection to sustained temperatures exceeding 90 degrees F or lower than 0 degrees F. Keep materials and containers closed and separated from absorptive materials such as wood and insulation.

1.6 QUALITY ASSURANCE

1.6.1 Compatibility with Substrate

Verify that each sealant is compatible for use with each joint substrate in accordance with sealant manufacturer's printed recommendations for each application.

### 1.6.2 Joint Tolerance

Provide joint tolerances in accordance with manufacturer's printed instructions.

### 1.6.3 Mock-Up

Provide a mock-up of each type of sealant using materials, colors, and techniques approved for use on the project. Approved mock-ups may be incorporated into the Work.

### 1.6.4 Adhesion

Provide in accordance with ASTM C1193 or ASTM C1521.

## PART 2 PRODUCTS

### 2.1 SEALANTS

Provide sealant products that have been tested, found suitable, and documented as such by the manufacturer for the particular substrates to which they will be applied.

In areas with ambient temperatures that exceed 110 degrees F, do not use polybutene, bituminous, acrylic-latex, polyvinyl acetate latex sealants, polychloroprene (neoprene), polyvinyl chloride (PVC), and polyurethane foams, and neoprene, PVC, and styrene butadiene rubber extruded seals and closure strips due to these materials having maximum recommended surface temperature ranges from 130 to 180 degrees F.

#### 2.1.1 Interior Sealants

Provide ASTM C920, Type S or M, Grade NS, Class 12.5, Use NT. Provide sealant products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide certification or validation of indoor air quality for interior sealants. Location(s) and color(s) of sealant for the following (Note, color "as selected" refers to manufacturer's full range of color options):

| LOCATION                                                                                                                                                                 | COLOR       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| a. Small voids between walls or partitions and adjacent lockers, casework, shelving, door frames, built-in or surface mounted equipment and fixtures, and similar items. | As selected |
| b. Perimeter of frames at doors, windows, and access panels which adjoin exposed interior concrete and masonry surfaces.                                                 | As Selected |
| c. Joints of interior masonry walls and partitions which adjoin columns, pilasters, concrete walls, and exterior walls unless otherwise detailed.                        | As Selected |



| LOCATION                                                                                                                        | COLOR       |
|---------------------------------------------------------------------------------------------------------------------------------|-------------|
| d. Joints between edge members for acoustical tile and adjoining vertical surfaces.                                             | As Selected |
| e. Interior locations, not otherwise indicated or specified, where small voids exist between materials specified to be painted. | As Selected |

### 2.1.2 Exterior Sealants

For joints in vertical surfaces, provide ASTM C920, Type S or M, Grade NS, Class 25, Use NT. For joints in horizontal surfaces, provide ASTM C920, Type S or M, Grade P, Class 25, Use T. Provide location(s) and color(s) of sealant as follows (Note, color "as selected" refers to manufacturer's full range of color options):

| LOCATION                                                                                                                                                                                                                   | COLOR                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| a. Joints and recesses formed where frames and subsills of windows, doors, louvers, and vents adjoin masonry, concrete, or metal frames. Use sealant at both exterior and interior surfaces of exterior wall penetrations. | Match adjacent surface color |
| b. Joints between new and existing exterior masonry walls.                                                                                                                                                                 | Match adjacent surface color |
| c. Masonry joints where shelf angles occur.                                                                                                                                                                                | Match adjacent surface color |
| e. Expansion and control joints.                                                                                                                                                                                           | As Selected                  |
| f. Interior face of expansion joints in exterior concrete or masonry walls where metal expansion joint covers are not required.                                                                                            | Match adjacent surface color |
| g. Voids where items pass through exterior walls.                                                                                                                                                                          | Match adjacent surface color |
| h. Metal reglets, where flashing is inserted into masonry joints, and where flashing is penetrated by coping dowels.                                                                                                       | Match adjacent surface color |
| i. Metal-to-metal joints where sealant is indicated or specified.                                                                                                                                                          | Match adjacent surface color |
| j. Joints between ends of gravel stops, fascia, copings, and adjacent walls.                                                                                                                                               | Match adjacent surface color |

### 2.1.3 Floor Joint Sealants

ASTM C920, Type S or M, Grade P, Class 25, Use T. Provide sealant products used on the interior of the building (defined as inside of the

weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide certification or validation of indoor air quality for interior floor joint sealants. Provide location(s) and color(s) of sealant as follows (Note, color "as selected" refers to manufacturer's full range of color options):

| LOCATION                                                                      | COLOR       |
|-------------------------------------------------------------------------------|-------------|
| a. Seats of metal thresholds for exterior doors.                              | As selected |
| b. Control and expansion joints in floors, slabs, ceramic tile, and walkways. | As Selected |

#### 2.1.4 Acoustical Sealants

Rubber or polymer based acoustical sealant in accordance with ASTM C919 to have a flame spread of 25 or less and a smoke developed rating of 50 or less when tested in accordance with ASTM E84. Provide non-staining acoustical sealant with a consistency of 250 to 310 when tested in accordance with ASTM D217. Acoustical sealant must remain flexible and adhesive after 500 hours of accelerated weathering as specified in ASTM C734. Provide sealant products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide certification or validation of indoor air quality for interior acoustical sealants.

#### 2.2 PRIMERS

Non-staining, quick drying type and consistency as recommended by the sealant manufacturer for the particular application. Provide primers for interior applications that meet the indoor air quality requirements of the paragraph INTERIOR SEALANTS above.

#### 2.3 BOND BREAKERS

Type and consistency as recommended by the sealant manufacturer to prevent adhesion of the sealant to the backing or to the bottom of the joint. Provide bond breakers for interior applications that meet the indoor air quality requirements of the paragraph INTERIOR SEALANTS above.

#### 2.4 BACKSTOPS

Provide glass fiber roving, neoprene, butyl, polyurethane, or polyethylene foams free from oil or other staining elements as recommended by sealant manufacturer. Provide 25 to 33 percent oversized backing for closed cell and 40 to 50 percent oversized backing for open cell material, unless otherwise indicated. Provide backstop material that is compatible with sealant. Do not use oakum or other types of absorptive materials as backstops.

#### 2.4.1 Neoprene

Provide in accordance with ASTM D1056, closed cell expanded neoprene cord Type 2, Class C, neoprene backing.

#### 2.5 CLEANING SOLVENTS

Provide type(s) recommended by the sealant manufacturer and in accordance with environmental requirements herein. Protect adjacent aluminum and bronze surfaces from solvents. Provide solvents for interior applications that meet the indoor air quality requirements of the paragraph INTERIOR SEALANTS above.

### PART 3 EXECUTION

#### 3.1 FIELD QUALITY CONTROL

Perform a field adhesion test in accordance with manufacturer's instructions and ASTM C1193, Method A or ASTM C1521, Method A, Tail Procedure. Remove sealants that fail adhesion testing; clean substrates, reapply sealants, and re-test. Test sealants adjacent to failed sealants. Submit field adhesion test report indicating tests, locations, dates, results, and remedial actions taken.

#### 3.2 SURFACE PREPARATION

Prepare surfaces according to manufacturer's printed installation instructions. Clean surfaces from dirt, frost, moisture, grease, oil, wax, lacquer, paint, or other foreign matter that would destroy or impair adhesion. Remove oil and grease with solvent; thoroughly remove solvents prior to sealant installation. Wipe surfaces dry with clean cloths. When resealing an existing joint, remove existing caulk or sealant prior to applying new sealant. For surface types not listed below, provide in accordance with sealant manufacturer's printed instructions for each specific surface.

##### 3.2.1 Steel Surfaces

Remove loose mill scale by sandblasting or, if sandblasting is impractical or would damage finished work, scraping and wire brushing. Remove protective coatings by sandblasting or using a residue free solvent. Remove resulting debris and solvent residue prior to sealant installation.

##### 3.2.2 Aluminum or Bronze Surfaces

Remove temporary protective coatings from surfaces that will be in contact with sealant. When masking tape is used as a protective coating, remove tape and residual adhesive prior to sealant application. For removing protective coatings and final cleaning, use non-staining solvents recommended by the manufacturer of the item(s) containing aluminum or bronze surfaces.

##### 3.2.3 Concrete and Masonry Surfaces

Where surfaces have been treated with curing compounds, oil, or other such materials, remove materials by sandblasting or wire brushing. Remove laitance, efflorescence and loose mortar from the joint cavity. Remove resulting debris prior to sealant installation.

### 3.3 SEALANT PREPARATION

Do not add liquids, solvents, or powders to sealants. Mix multicomponent elastomeric sealants in accordance with manufacturer's printed instructions.

### 3.4 APPLICATION

#### 3.4.1 Joint Width-To-Depth Ratios

Acceptable Ratios:

| JOINT WIDTH                                    | JOINT DEPTH  |                |
|------------------------------------------------|--------------|----------------|
|                                                | Minimum      | Maximum        |
| For metal, glass, or other nonporous surfaces: |              |                |
| 1/4 inch (minimum)                             | 1/4 inch     | 1/4 inch       |
| over 1/4 inch                                  | 1/2 of width | Equal to width |
| For concrete or masonry                        |              |                |
| 1/4 inch (minimum)                             | 1/4 inch     | 1/4 inch       |
| over 1/4 inch to 1/2 inch                      | 1/4 inch     | Equal to width |
| over 1/2 inch to 1 inch                        | 1/2 inch     | 5/8 inch       |
| Over 1 inch                                    | prohibited   |                |

Unacceptable Ratios: Where joints of acceptable width-to-depth ratios have not been provided, clean out joints to acceptable depths and grind or cut to acceptable widths without damage to the adjoining work. Grinding is prohibited at metal surfaces.

#### 3.4.2 Unacceptable Sealant Use

Do not install sealants in lieu of other required building enclosure weatherproofing components such as flashing, drainage components, and joint closure accessories, or to close gaps between walls, floors, roofs, windows, and doors, that exceed acceptable installation tolerances. Remove sealants that have been used in an unacceptable manner and correct building enclosure deficiencies to comply with contract documents requirements.

#### 3.4.3 Masking Tape

Place masking tape on the finished surface on one or both sides of joint cavities to protect adjacent finished surfaces from primer or sealant smears. Remove masking tape within 10 minutes of joint filling and tooling.

#### 3.4.4 Backstops

Provide backstops dry and free of tears or holes. Tightly pack the back

or bottom of joint cavities with backstop material to provide joints in specified depths. Provide backstops where indicated and where backstops are not indicated but joint cavities exceed the acceptable maximum depths specified in JOINT WIDTH-TO-DEPTH RATIOS Table.

#### 3.4.5 Primer

Clean out loose particles from joints immediately prior to application of. Apply primer to joints in concrete masonry units, wood, and other porous surfaces in accordance with sealant manufacturer's printed instructions. Do not apply primer to exposed finished surfaces.

#### 3.4.6 Bond Breaker

Provide bond breakers to surfaces not intended to bond in accordance with, sealant manufacturer's printed instructions for each type of surface and sealant combination specified.

#### 3.4.7 Sealants

Provide sealants compatible with the material(s) to which they are applied. Do not use a sealant that has exceeded its shelf life or has jelled and cannot be discharged in a continuous flow from the sealant gun. Apply sealants in accordance with the manufacturer's printed instructions with a gun having a nozzle that fits the joint width. Work sealant into joints so as to fill the joints solidly without air pockets. Tool sealant after application to ensure adhesion. Apply sealant uniformly smooth and free of wrinkles. Upon completion of sealant application, roughen partially filled or unfilled joints, apply additional sealant, and tool smooth as specified. Apply sealer over sealants in accordance with the sealant manufacturer's printed instructions.

### 3.5 PROTECTION AND CLEANING

#### 3.5.1 Protection

Protect areas adjacent to joints from sealant smears. Masking tape may be used for this purpose if removed 5 to 10 minutes after the joint is filled and no residual tape marks remain.

#### 3.5.2 Final Cleaning

Upon completion of sealant application, remove remaining smears and stains and leave the work in a clean and neat condition.

- a. Masonry and Other Porous Surfaces: Immediately remove fresh sealant that has been smeared on adjacent masonry, rub clean with a solvent, and remove solvent residue, in accordance with sealant manufacturer's printed instructions. Allow excess sealant to cure for 24 hours then remove by wire brushing or sanding. Remove resulting debris.
- b. Metal and Other Non-Porous Surfaces: Remove excess sealant with a solvent moistened cloth. Remove solvent residue in accordance with solvent manufacturer's printed instructions.

-- End of Section --

SECTION 08 11 13

STEEL DOORS AND FRAMES

**08/20**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding  
Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM C578 (2023) Standard Specification for Rigid,  
Cellular Polystyrene Thermal Insulation

ASTM C591 (2022) Standard Specification for Unfaced  
Preformed Rigid Cellular Polyisocyanurate  
Thermal Insulation

ASTM C612 (2014; R 2019) Standard Specification for  
Mineral Fiber Block and Board Thermal  
Insulation

ASTM D2863 (2019) Standard Test Method for Measuring  
the Minimum Oxygen Concentration to  
Support Candle-Like Combustion of Plastics  
(Oxygen Index)

ASTM E1300 (2016) Standard Practice for Determining  
Load Resistance of Glass in Buildings

ASTM F2248 (2012) Standard Practice for Specifying an  
Equivalent 3-Second Duration Design  
Loading for Blast Resistant Glazing  
Fabricated with Laminated Glass

BUILDERS HARDWARE MANUFACTURERS ASSOCIATION (BHMA)

ANSI/BHMA A156.115 (2016) Hardware Preparation in Steel Doors  
and Steel Frames

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 80 (2022) Standard for Fire Doors and Other  
Opening Protectives

NFPA 105 (2022) Standard for Smoke Door Assemblies  
and Other Opening Protectives

NFPA 252 (2022) Standard Methods of Fire Tests of

Door Assemblies

STEEL DOOR INSTITUTE (SDI/DOOR)

|                  |                                                                                                                    |
|------------------|--------------------------------------------------------------------------------------------------------------------|
| SDI/DOOR 111     | (2009) Recommended Details for Standard Steel Doors, Frames, and Accessories and Related Components                |
| SDI/DOOR 113     | (2023) Standard Practice for Determining the Steady-State Thermal Transmittance of Steel Door and Frame Assemblies |
| SDI/DOOR A250.6  | (2020) Recommended Practice for Hardware Reinforcing on Standard Steel Doors and Frames                            |
| SDI/DOOR A250.8  | (2023) Specifications for Standard Steel Doors and Frames                                                          |
| SDI/DOOR A250.11 | (2022) Recommended Erection Instructions for Steel Frames                                                          |

UNDERWRITERS LABORATORIES (UL)

|        |                                                                                                 |
|--------|-------------------------------------------------------------------------------------------------|
| UL 10C | (2016; Reprint May 2021) UL Standard for Safety Positive Pressure Fire Tests of Door Assemblies |
|--------|-------------------------------------------------------------------------------------------------|

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Doors; G

Frames; G

Accessories

Schedule of Doors; G

Schedule of Frames; G

SD-03 Product Data

Doors; G

Recycled Content for Steel Door Product

Frames; G

Recycled Content for Steel Frame Product

## Accessories

### 1.3 DELIVERY, STORAGE, AND HANDLING

Deliver doors, frames, and accessories undamaged and with protective wrappings or packaging. Provide temporary steel spreaders securely fastened to the bottom of each welded frame. Store doors and frames on platforms under cover in clean, dry, ventilated, and accessible locations, with 1/4 inch airspace between doors. Remove damp or wet packaging immediately and wipe affected surfaces dry. Replace damaged materials with new.

## PART 2 PRODUCTS

### 2.1 STANDARD STEEL DOORS

SDI/DOOR A250.8, except as specified otherwise. Prepare doors to receive door hardware as specified in Section 08 71 00 DOOR HARDWARE. Undercut where indicated. Provide exterior doors with top edge closed flush and sealed to prevent water intrusion. Provide doors at 1-3/4 inch thick, unless otherwise indicated. Provide door material that uses a minimum of 25 percent recycled content. Provide data indicating percentage of recycled content for steel door product. Provide exterior glazing in accordance with ASTM F2248 and ASTM E1300.

#### 2.1.1 Classification - Level, Performance, Model

##### 2.1.1.1 Heavy Duty Doors

SDI/DOOR A250.8, Level 2, physical performance Level B, Model 2, with core construction as required by the manufacturer for interior doors and for exterior doors, of size(s) and design(s) indicated. Where vertical stiffener cores are required, the space between the stiffeners must be filled with board insulation.

### 2.2 ACCESSORIES

#### 2.2.1 Astragals

For pairs of exterior steel doors which will not have aluminum astragals or removable mullions, as specified in Section 08 71 00 DOOR HARDWARE provide overlapping steel astragals with the doors.

#### 2.2.2 Moldings

Provide moldings around glass of interior and exterior doors and louvers of interior doors. Provide nonremovable moldings on outside of exterior doors and on corridor side of interior doors. Other moldings may be stationary or removable. Secure inside moldings to stationary moldings, or provide snap-on moldings.

### 2.3 INSULATION CORES

Provide insulating cores at all exterior doors, and provide an apparent U-factor of .48 in accordance with SDI/DOOR 113 and conforming to:

- a. Rigid Cellular Polyisocyanurate Foam: ASTM C591, Type I or II, foamed-in-place or in board form, with oxygen index of not less than 22 percent when tested in accordance with ASTM D2863; or



- b. Rigid Polystyrene Foam Board: ASTM C578, Type I or II; or
- c. Mineral board: ASTM C612, Type I.

## 2.4 STANDARD STEEL FRAMES

SDI/DOOR A250.8, Level 2, except as otherwise specified. Form frames to sizes and shapes indicated, with welded corners. Provide steel frames for doors, transoms, sidelights, unless otherwise indicated. Provide frame product that uses a minimum of 25 percent recycled content. Provide data indicating percentage of recycled content for steel frame product.

### 2.4.1 Welded Frames

Continuously weld frame faces at corner joints. Mechanically interlock or continuously weld stops and rabbets. Grind welds smooth.

Weld frames in accordance with the recommended practice of the Structural Welding Code Sections 1 through 6, AWS D1.1/D1.1M and in accordance with the practice specified by the producer of the metal being welded.

### 2.4.2 Mullions and Transom Bars

Provide mullions and transom bars of closed or tubular construction with heads and jambs butt-welded together. Bottom of door mullions must have adjustable floor anchors and spreader connections.

### 2.4.3 Stops and Beads

Provide form and loose stops and beads from 20 gage steel. Provide for glazed and other openings in standard steel frames. Secure beads to frames with oval-head, countersunk Phillips self-tapping sheet metal screws or concealed clips and fasteners. Space fasteners approximately 12 to 16 inch on center. Miter molded shapes at corners. Butt or miter square or rectangular beads at corners.

### 2.4.4 Terminated Stops

Where indicated, terminate interior door frame stops 6 inch above floor. Do not terminate stops of frames for doors.

### 2.4.5 Anchors

Provide anchors to secure the frame to adjoining construction. Provide steel anchors, zinc-coated not lighter than 18 gage.

#### 2.4.5.1 Wall Anchors

Provide at least three anchors for each jamb. For frames which are more than 7.5 feet in height, provide one additional anchor for each jamb for each additional 2.5 feet or fraction thereof.

- a. Masonry: Provide anchors of corrugated or perforated steel straps or 3/16 inch diameter steel wire, adjustable or T-shaped;
- b. Stud partitions: Weld or otherwise securely fasten anchors to backs of frames. Design anchors to be fastened ;

- c. Completed openings: Secure frames to previously placed concrete or masonry with expansion bolts in accordance with SDI/DOOR 111; and
- d. Solid plaster partitions: Secure anchors solidly to back of frames and tie into the lath. Provide adjustable top strut anchors on each side of frame for fastening to structural members or ceiling construction above. Provide size and type of strut anchors as recommended by the frame manufacturer.

#### 2.4.5.2 Floor Anchors

Provide floor anchors drilled for 3/8 inch anchor bolts at bottom of each jamb member. Where floor fill occurs, terminate bottom of frames at the indicated finished floor levels and support by adjustable extension clips resting on and anchored to the structural slabs.

### 2.5 FIRE AND SMOKE DOORS AND FRAMES

Provide fire and smoke doors and frames in accordance with NFPA 80 and NFPA 105 and this specification. Include insulated core materials in fire doors where indicated in the door schedule.

#### 2.5.1 Labels

Provide fire doors and frames bearing the label of Underwriters Laboratories (UL), Factory Mutual Engineering and Research (FM), or Warnock Hersey International (WHI) attesting to the rating required. Testing must be in accordance with NFPA 252 or UL 10C. Provide labels that are metal with raised letters, bearing the name or file number of the door and frame manufacturer. Labels must be permanently affixed at the factory to frames and to the hinge edge of the door. Do not paint door and labels.

#### 2.5.2 Oversized Doors

For fire doors and frames which exceed the size for which testing and labeling are available, furnish certificates stating that the doors and frames are identical in design, materials, and construction to a door which has been tested and meets the requirements for the class indicated.

#### 2.5.3 Astragal on Fire and Smoke Doors

On pairs of labeled fire doors, conform to NFPA 80 and UL requirements.  
On smoke control doors, conform to NFPA 105.

### 2.6 EXTERIOR FRAMES

Provide thermal insulation in all exterior frames. Provide frames of a minimum Level 4, with frames of a minimum thickness of 0.067 inch, 14 gage.

### 2.7 HARDWARE PREPARATION

Drill and tap doors and frames to receive finish hardware. Prepare doors and frames for hardware in accordance with the applicable requirements of SDI/DOOR A250.8 and SDI/DOOR A250.6. For additional requirements refer to ANSI/BHMA A156.115. Drill and tap for surface-applied hardware at the project site. Build additional reinforcing for surface-applied hardware into the door at the factory. Punch door frames, with the exception of

frames that will have weatherstripping or lightproof or soundproof gasketing, to receive a minimum of two rubber or vinyl door silencers on lock side of single doors and one silencer for each leaf at heads of double doors. Set lock strikes out to provide clearance for silencers.

## 2.8 FINISHES

### 2.8.1 Factory-Primed Finish

Thoroughly clean all surfaces of doors and frames then chemically treat and factory prime with a rust inhibiting coating as specified in SDI/DOOR A250.8

## 2.9 FABRICATION AND WORKMANSHIP

Provide finished doors and frames that are strong and rigid, neat in appearance, and free from defects, waves, scratches, cuts, dents, ridges, holes, warp, and buckle. Provide molded members that are clean cut, straight, and true, with joints coped or mitered, well formed, and in true alignment. Dress exposed welded and soldered joints smooth. Design door frame sections for use with the wall construction indicated. Corner joints must be well formed and in true alignment. Conceal fastenings where practicable. On wraparound frames for masonry partitions, provide a throat opening 1/8 inch larger than the actual masonry thickness. Design other frames in exposed masonry walls or partitions to allow sufficient space between the inside back of trim and masonry to receive caulking compound.

### 2.10 PROVISIONS FOR GLAZING

Materials are specified in Section 08 81 00, GLAZING.

## PART 3 EXECUTION

### 3.1 INSTALLATION

#### 3.1.1 Frames

Set frames in accordance with SDI/DOOR A250.11. Plumb, align, and brace securely until permanent anchors are set. Anchor bottoms of frames with expansion bolts or powder-actuated fasteners. Build in or secure wall anchors to adjoining construction.

#### 3.1.2 Doors

Hang doors in accordance with clearances specified in SDI/DOOR A250.8. After erection and glazing, clean and adjust hardware.

#### 3.1.3 Fire and Smoke Doors and Frames

Install fire doors and frames, including hardware, in accordance with NFPA 80. Install fire rated smoke doors and frames in accordance with NFPA 80 and NFPA 105.

### 3.2 PROTECTION

Protect doors and frames from damage. Repair damaged doors and frames prior to completion and acceptance of the project or replace with new, as directed. Wire brush rusted frames until rust is removed. Clean

thoroughly. Apply an all-over coat of rust-inhibitive paint of the same type used for shop coat.

### 3.3 CLEANING

Upon completion, clean exposed surfaces of doors and frames thoroughly. Remove mastic smears and other unsightly marks.

-- End of Section --

SECTION 08 31 00

ACCESS DOORS AND PANELS

05/17, CHG 1: 08/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding  
Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon  
Structural Steel

ASTM A653/A653M (2023) Standard Specification for Steel  
Sheet, Zinc-Coated (Galvanized) or  
Zinc-Iron Alloy-Coated (Galvannealed) by  
the Hot-Dip Process

ASTM A666 (2023) Standard Specification for Annealed  
or Cold-Worked Austenitic Stainless Steel  
Sheet, Strip, Plate and Flat Bar

ASTM A1008/A1008M (2023) Standard Specification for Steel,  
Sheet, Cold-Rolled, Carbon, Structural,  
High-Strength Low-Alloy, High-Strength  
Low-Alloy with Improved Formability,  
Solution Hardened, and Bake Hardenable

ASTM E119 (2022) Standard Test Methods for Fire  
Tests of Building Construction and  
Materials

MASTER PAINTERS INSTITUTE (MPI)

MPI 79 (2016) Primer, Alkyd, Anti-Corrosive for  
Metal

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 252 (2022) Standard Methods of Fire Tests of  
Door Assemblies

NFPA 288 (2017) Standard Methods of Fire Tests of  
Horizontal Fire Door Assemblies Installed  
in Horizontal Fire Resistance-Rated  
Assemblies

UNDERWRITERS LABORATORIES (UL)

|        |                                                                                                   |
|--------|---------------------------------------------------------------------------------------------------|
| UL 10B | (2008; Reprint May 2020) Fire Tests of Door Assemblies                                            |
| UL 263 | (2011; Reprint Aug 2021) UL Standard for Safety Fire Tests of Building Construction and Materials |

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Access Doors And Panels; G

SD-03 Product Data

Access Doors And Panels; G

Hardware Including Locks and Keys; G

Accessories; G Recycled Content; S

SD-04 Samples

Finishes; G

SD-06 Test Reports

Fire-rating(s) of Assemblies; G

Acoustical Ratings of Assemblies; G

1.3 MISCELLANEOUS REQUIREMENTS

For access doors and panels provide the following:

1.3.1 Shop Drawings

For field assembled access doors and panels, provide plans, elevations, sections, and details for each type of access door and panel assembly. Indicate frame, surface and edge construction, materials, and accessories. Indicate types of finished surfaces and details for panel edge conditions. Provide a door schedule with a unique number for each access door and panel, specific location in the project, location of hinges and hardware for each door. Indicate fire-rating(s) of assemblies

1.3.2 Product Data

For shop assembled access doors and panels, provide literature indicating sizes, types, frame and edge construction, finishes, hardware, accessories such as gaskets, seals and weatherstripping, and location of each door and panel in the project. Indicate fire-ratings of assemblies. Provide details of adjoining work for each condition indicated.

### 1.3.3 Finish Samples

Submit two color charts from manufacturer's standard color and finish options for each type of frame and panel assembly finish indicated.

### 1.3.4 Test Reports

Provide test reports for fire-rated assemblies when tested in accordance with NFPA 252 or UL 10B for fire-rated access door assemblies installed vertically and NFPA 288 for fire-rated access door assemblies installed horizontally.

## 1.4 PERFORMANCE REQUIREMENTS

### 1.4.1 Structural Requirements

Provide floor access assemblies to support live loads indicated for floors. Deflection must not exceed 1/180 of span.

### 1.4.2 Fire-Rating Requirements

Provide access panels with a minimum fire-rating of as indicated on the Drawings.

## 1.5 DELIVERY, STORAGE, AND PROTECTION

Protect from corrosion, deformation, and other types of damage. Store items in an enclosed area free from contact with soil and weather. Remove and replace damaged items with new items.

## PART 2 PRODUCTS

### 2.1 RECYCLED CONTENT

Provide products with recycled content. Provide data for each product with recycled content, identifying percentage of recycled content.

### 2.2 MATERIALS

#### 2.2.1 Steel Plates, Shapes, and Bars

Provide in accordance with ASTM A36/A36M.

#### 2.2.2 Sheet Steel

Provide cold rolled steel sheet substrate in accordance with ASTM A1008/A1008M, Commercial Steel (CS), exposed.

#### 2.2.3 Stainless Steel

Provide in accordance with ASTM A666, type 302 or 304.

#### 2.2.4 Metallic Coated Steel Sheet

Provide in accordance with ASTM A653/A653M, Commercial Steel (CS), Type B; with minimum G60 (Z180) or A60 (ZF180) metallic coating.

#### 2.2.5 Hardware

Provide automatic closing devices. Provide latch releases operable from insides of doors. Provide anchors in accordance with applicable fire test parameters.

#### 2.2.6 Hinges

Provide concealed spring hinges, 175 degrees of opening, with removable hinge pins to allow removal of door panel from frame. Provide hinges of same steel as door and frame or in accordance with manufacturer's written recommendations. If providing non-continuous hinges, provide in numbers required to maintain alignment of door panel with frame. Provide coatings as necessary to permanently protect dissimilar metals from contact with one another; see Part 3 herein for more information.

#### 2.2.7 Locks

Unless otherwise indicated, provide flush screwdriver operated cam lock. Provide plastic sleeve or stainless steel bushings to protect holes in surface finishes for screwdriver to access lock.

#### 2.2.8 Accessories

Provide anchors in size, number and location on four sides to secure access door to substrate. Provide anchors in types as recommended by manufacturer's written installation instructions for each substrate indicated. Provide shims, bushings, clips, gaskets, and other devices as necessary for a complete installation.

### 2.3 FABRICATION

#### 2.3.1 Thickness, Size, Edges

Fabricate frames for access doors of steel not lighter than 16 gage with welded joints and anchorage for securing to adjacent construction. Provide doors a minimum of 24 by 24 inches and of not lighter than 16 gage steel, with stiffened edges and welded attachments. Provide with eased (lightly rounded) edges, without burrs, snags or sharpness and exposed welds ground smooth.

#### 2.3.2 Welding

Provide in accordance with AWS D1.1/D1.1M.

### 2.4 ACCESS ASSEMBLY TYPES

Unless indicated otherwise, provide flush-face steel access doors and panels with steel frames and flanges.

#### 2.4.1 Fire-rated Doors

##### 2.4.1.1 Door Construction

Provide ceiling access door construction in accordance with ASTM E119 or UL 263. Provide wall access doors in accordance with NFPA 252 or UL 10B.



#### 2.4.1.2 Labels

Provide class B opening according to UL 10B or test by another nationally recognized laboratory, approved by the Contracting Officer. Provide fire-rating as indicated herein, with a maximum temperature rise of 216 degrees F.

#### 2.4.1.3 Door Panel and Frame

Steel sheet, with mineral fiber insulation core, insulated sandwich type construction.

### 2.5 FINISHES

Provide steel frames and panel surfaces with a powder coated finish. Provide manufacturer's standard two coat finish system consisting of one coat primer and one thermoset topcoat. Provide dry film thickness in 2 mils minimum. Provide exposed fastenings that approximately match the color and finish of the each material to which fastenings are applied.

## PART 3 EXECUTION

### 3.1 PREPARATION

Field verify all measurements prior to fabrication. Verify access door locations and sizes provide required maintenance access to installed building services components. Protect existing construction and completed work from damage during installation.

### 3.2 GENERAL INSTALLATION REQUIREMENTS

Install items at locations indicated, in accordance with manufacturer's written instructions. Include materials and parts as necessary for a complete installation of each item. Conceal fastenings where practicable. Poor matching of holes to fasteners is cause for rejection of the work.

### 3.3 ACCESS LOCATIONS

Install removable access panels directly below each valve, flow indicator, damper, air splitter or other utility requiring access that is located above ceilings, other than at acoustical panel ceilings, and that would otherwise not be accessible. Install access doors and panels permitting access to service valves, traps, dampers, cleanouts, and other mechanical, electrical and conveyor control items concealed in walls and partitions.

### 3.4 FIELD PAINTING

Field painting primed access doors in accordance with the requirements of Section 09 90 00 PAINTS AND COATINGS.

### 3.5 DISSIMILAR MATERIALS

Where dissimilar metals are in contact, protect surfaces with a coating in accordance with MPI 79 to prevent galvanic or corrosive action.

### 3.6 ADJUSTMENT

Adjust hardware so that door panel opens freely. Adjust door when closed

center door panel in frame.

### 3.7 ENVIRONMENTAL CONDITIONS

Do not paint surfaces when damp or exposed to weather, when surface temperature is below 45 degrees F or over 95 degrees F, unless approved by the Contracting Officer.

-- End of Section --

SECTION 08 33 23

OVERHEAD COILING DOORS

08/20, CHG 1: 02/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE FUN IP (2021) Fundamentals Handbook, I-P Edition

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B29.400 (2001; (R 2008) (R 2013) (R 2018) (R 2023)) Combination, "H" Type Mill Chains, and Sprockets

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A47/A47M (1999; R 2022; E 2022) Standard Specification for Ferritic Malleable Iron Castings

ASTM A48/A48M (2022) Standard Specification for Gray Iron Castings

ASTM A53/A53M (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A123/A123M (2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A153/A153M (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM A307 (2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength

|                 |                                                                                                                                                                  |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A653/A653M | (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process                          |
| ASTM A666       | (2023) Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate and Flat Bar                                            |
| ASTM A780/A780M | (2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings                                                                 |
| ASTM A924/A924M | (2022a) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process                                                  |
| ASTM B209       | (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate                                                                                    |
| ASTM B221       | (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes                                                     |
| ASTM E84        | (2023) Standard Test Method for Surface Burning Characteristics of Building Materials                                                                            |
| ASTM E330/E330M | (2014; R 2021) Standard Test Method for Structural Performance of Exterior Windows, Doors, Skylights and Curtain Walls by Uniform Static Air Pressure Difference |
| ASTM F568M      | (2007) Standard Specification for Carbon and Alloy Steel Externally Threaded Metric Fasteners                                                                    |

DOOR AND ACCESS SYSTEM MANUFACTURERS ASSOCIATION (DASMA)

|                |                                                                                                                                                                                   |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ANSI/DASMA 108 | (2017) Standard Method for Testing Sectional Garage Doors, Rolling Doors and Flexible Doors: Determination of Structural Performance Under Uniform Static Air Pressure Difference |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

|            |                                                                                                        |
|------------|--------------------------------------------------------------------------------------------------------|
| NEMA ICS 1 | (2022) Standard for Industrial Control and Systems: General Requirements                               |
| NEMA ICS 2 | (2000; R 2020) Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 V |
| NEMA ICS 6 | (1993; R 2016) Industrial Control and Systems: Enclosures                                              |
| NEMA MG 1  | (2021) Motors and Generators                                                                           |

NEMA ST 1 (1988; R 1994; R 1997) Specialty  
Transformers (Except General Purpose Type)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

UNDERWRITERS LABORATORIES (UL)

UL 325 (2017; Reprint Feb 2020) UL Standard for  
Safety Door, Drapery, Gate, Louver, and  
Window Operators and Systems

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

Overhead Coiling Doors; G  
Counterbalancing Mechanism; G  
Manual Door Operators; G  
Electric Door Operators; G  
Bottom Bars; G  
Guides; G  
Mounting Brackets; G  
Hood; G  
Installation Drawings; G

### SD-03 Product Data

Overhead Coiling Doors; G  
Hardware; G  
Counterbalancing Mechanism; G  
Manual Door Operators; G  
Electric Door Operators; G

### SD-05 Design Data

Overhead Coiling Doors; G

Hardware; G

Counterbalancing Mechanism; G

Manual Door Operators; G

Electric Door Operators; G

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals, Data Package 2; G

SD-11 Closeout Submittals

Warranty; G

### 1.3 QUALITY CONTROL

#### 1.3.1 Warranty

Furnish a written guarantee that the helical spring and counterbalance mechanism are free from defects in material and workmanship for not less than two years after completion and acceptance of the project.

Warrant that upon notification by the Government, any defects in material, workmanship, and door operation are immediately correct within the same time period covered by the guarantee, at no cost to the Government.

#### 1.3.2 Operation And Maintenance Submittals

Submit 6 copies of the operation and maintenance manuals 30 calendar days prior to testing the Overhead Coiling Door Assemblies. Update and resubmit data for final approval no later than 30 calendar days prior to cContract completion.

Submit Operation and Maintenance Manuals for Overhead Coiling Door Assemblies, including the following items:

Manual Door Operators

Electric Door Operators

Counterbalancing Mechanism

Painting

Provide operation and maintenance manuals which are consistent with manufacturer's standard brochures, schematics, printed instructions, operating procedures, and safety precautions.

### 1.4 DELIVERY, STORAGE, AND HANDLING

Deliver doors to the jobsite wrapped in a protective covering with the brands and names clearly marked thereon. Store doors in an adequately ventilated dry location that is free from dirt and dust, water, or other contaminants. Store in a manner that permits easy access for inspection

and handling. Handle doors carefully to prevent damage. Remove damaged items that cannot be restored to like-new condition and provide new items.

## PART 2 PRODUCTS

### 2.1 SYSTEM DESCRIPTION

Provide overhead coiling doors with interlocking slats, complete with anchoring and door hardware, guides, hood, and operating mechanisms, and designed for use on openings as indicated. Doors must be spring counterbalanced, rolling type, and designed for use on exterior openings, as indicated. Doors must be operated by electric-power with auxiliary hand chain operation. Doors to be surface-mounted type with guides at jambs set back a sufficient distance to provide a clear opening when door is in open position. Mount exterior doors as indicated .

#### 2.1.1 Design Requirements

##### 2.1.1.1 Door Detail Shop Drawings

Provide installation drawings for door assemblies which show: elevations of each door type, shape and thickness of materials, finishes, details of joints and connections, details of guides and fittings, rough opening dimensions, location and description of hardware, anchorage locations, and counterbalancing mechanism and door operator details. Show locations of replaceable fusible links on wiring diagrams for power, signal and controls. For motor-operated doors include supporting brackets for motors, location, type, and ratings of motors, and safety devices. Include a schedule showing the location of each door with the drawings.

#### 2.1.2 Performance Requirements

##### 2.1.2.1 Wind Loading

Design and fabricate door assembly to withstand the wind loading pressure of at least 20 pounds per square foot in accordance with ANSI/DASMA 108. Provide test data showing compliance with ASTM E330/E330M. Sound engineering principles may be used to interpolate or extrapolate test results to door sizes not specifically tested. Ensure that the complete assembly meets or exceeds the requirements of ASCE 7-16.

##### 2.1.2.2 Operational Cycle Life

Design all portions of the door, hardware and operating mechanism that are subject to movement, wear, or stress fatigue to operate through a minimum number of 10 cycles per day. One complete cycle of door operation is defined as when the door is in the closed position, moves to the fully open position, and returns to the closed position.

## 2.2 COMPONENTS

### 2.2.1 Overhead Coiling Doors

#### 2.2.1.1 Curtain Materials and Construction

Provide curtain slats fabricated from aluminum sheets conforming to ASTM B209, or ASTM B221 extrusions, alloy and tempering standard from the manufacturer for type of use and finish indicated; with a thickness as

required by the size of the door to meet the required windload.

Fabricate doors from interlocking cold-rolled slats, with section profiles as specified, designed to withstand the specified wind loading. Ensure the provided slats are continuous without splices for the width of the door.

Provide slats filled with manufacturer's standard thermal insulation, complying with the maximum flame-spread and smoke-developed indexes of 75 and 450, respectively, according to ASTM E84. Enclose the insulation completely within the slat faces on the interior surface of the slats.

#### 2.2.1.2 Insulated Curtains

Form curtains from manufacturer's standard shapes of interlocking slats. Supply a slat system with a minimum R-value of 5.0 when calculated in accordance with ASHRAE FUN IP. Slat to consist of a urethane core not less than 11/16 inch thick, completely enclosed within metal facings. Slat steel thickness as required by the size of the door to meet specified performance requirements. The insulated slat assembly requires a flame spread rating of not more than 25 and a smoke development factor of not more than 50 when tested in accordance with ASTM E84.

#### 2.2.1.3 Curtain Bottom Bar

Install curtain bottom bars as pairs of angles or using extrusions from the manufacturer's standard steel, stainless and aluminum extrusions not less than 2.0 by 2.0 inches by 0.188 inch. Do not use aluminum on doors more than 16 feet wide. Ensure steel extrusions conform to ASTM A36/A36M. Stainless steel extrusions conforming to ASTM A666, Type 304. Aluminum extrusions conforming to ASTM B221. Galvanize angles and fasteners in accordance with ASTM A653/A653M and ASTM A924/A924M. Coat welds and abrasions with paint conforming to ASTM A780/A780M.

Provide two minimum 2 inch by 2 inch by 1/8 inch structural steel angles.

#### 2.2.1.4 Endlocks (and Windlocks)

Provide endlocks of Grade B cast steel conforming to ASTM A47/A47M, galvanized in accordance with ASTM A153/A153M. Secure locks at every other curtain slat. In addition to endlocks, exterior doors which are more than 16 feet wide or which have a design wind load of more than 20 pounds per square foot, must have windlocks of manufacturer's standard design. Windlocks must prevent curtain from leaving guide because of deflection from wind pressure or other forces.

#### 2.2.1.5 Weather Stripping

Provide a hood baffle inside the hood that is a minimum 1/16 inch thick sheet of vinyl, neoprene rubber or equivalent. Provide guide weather stripping that is a minimum 1/16 inch thick sheet of vinyl, neoprene rubber, or equivalent.

Provide bottom bar weather-stripping that is a minimum 1/16 inch thick sheet of vinyl, neoprene rubber, or equivalent.

#### 2.2.1.6 Locking Devices

Ensure that the slide bolt engages through slots in tracks for locking by



padlock, located on both left and right jamb sides, operable from coil side.

Provide a locking device assembly which includes cylinder lock, operating handle, cam plate, and adjustable locking bars to engage through slots in tracks.

Provide a chain lock keeper suitable for a standard padlock.

#### 2.2.1.7 Safety Interlock

Equip power-operated doors with a safety interlock switch to disengage power supply when the door is locked, or provide an operator with an internal lock sensing device to prevent the door opening when the door is locked.

#### 2.2.2 Hardware

Ensure that all hardware conforms to ASTM A153/A153M, ASTM A307, and ASTM F568M.

##### 2.2.2.1 Guides

Fabricate curtain jamb guides from the manufacturer's standard angles or channels of same material and finish as curtain slats unless otherwise indicated. Provide guides with sufficient depth or incorporate a steel locking bar to retain the curtain in place under the wind pressure specified. Ensure curtain operates smoothly. Slot bolt holes for track adjustment. Securely attach guides to adjoining construction with not less than 3/8 inch diameter bolts, spaced near each end and not over 30 inches apart.

Ensure guides are roll-formed steel channel bolted to angle or structural grade, three angle assembly of steel to form a slot of sufficient depth to retain curtains in guides to achieve 20 psf windload standard. Guides may be provided with integral windlock bars and removable bottom bar stops.

##### 2.2.2.2 Hood

Provide a hood with a minimum 24-gauge sheet metal, flanged at top for attachment to header and flanged at bottom to provide longitudinal stiffness. The hood encloses the curtain coil and counterbalance mechanism.

Hoods for openings more than 12 feet in width must have intermediate support brackets to prevent excessive sag. 2.2.3 Counterbalancing Mechanism

Counterbalance doors by means of manufacturer's standard mechanism with an adjustable-tension, steel helical torsion spring mounted, around a steel shaft and contained in a spring barrel connected to top of curtain with barrel rings. Use grease-sealed or self-lubricating bearings for rotating members.

##### 2.2.3.1 Brackets

Provide the manufacturer's standard mounting brackets with one located at each end of the counterbalance barrel conforming to ASTM A36/A36M. Provide brackets of hot-rolled steel.

#### 2.2.3.2 Counterbalance Barrels

Curtain must roll up on a barrel supported at head of opening on brackets and be balanced by a torsion spring system in the barrel. Fabricate spring barrel of manufacturer's standard hot-formed, structural-quality, welded or seamless carbon-steel pipe, conforming to ASTM A53/A53M or equivalent. Ensure the barrel is of sufficient diameter and wall thickness to support rolled-up curtain without distortion of slats. Limit barrel deflection to not more than 0.03 inch per foot of span under full load.

##### a. Barrel

Provide steel pipe capable of supporting curtain load with maximum deflection of 0.03 inches per foot of width.

##### b. Spring Balance

Provide an oil-tempered, heat-treated steel helical torsion spring assembly designed for proper balance of door. Ensure that effort to operate manually operated units does not exceed 25 lbs. At least 80 percent of the door weight must be counterbalanced at any position. Provide wheel for applying and adjusting spring torque.

#### 2.2.3.3 Spring Balance

Install one or more oil-tempered, heat-treated steel helical torsion springs within the barrel, capable of producing sufficient torque to assure easy operation of the door curtain. Provide and size springs to counterbalance weight of curtain, with uniform adjustment accessible from outside barrel. Secure ends of springs to barrel and shaft with cast-steel barrel plugs.

#### 2.2.3.4 Torsion Rod for Counter Balance

Fabricate rod from the manufacturer's standard cold-rolled steel, sized to hold fixed spring ends and carry torsional load.

#### 2.2.3.5 Counterbalance Shaft Assembly

##### b. Spring Balance

Provide an oil-tempered, heat-treated steel helical torsion spring assembly designed for proper balance of door. Ensure that maximum effort to operate does not exceed 25 pounds. Provide wheel for applying and adjusting spring torque.

#### 2.2.4 Manual Door Operators

##### 2.2.4.1 Manual Chain-Hoist Door Operators

Provide door operators which consist of an endless steel hand chain, chain-pocket wheel, guard, and a geared reduction unit of at least a 3 to 1 ratio. Required pull for operation cannot exceed 35 pounds. Chain must extend to within 3 feet of floor.

Provide chain hoists with a mechanism allowing the curtain to be stopped

at any point in its upward or downward travel and to remain in that position until moved to the fully open or closed position. Provide hand chains of galvanized steel. Ensure that the yield point of the chain is at least three times the required hand-chain pull.

Provide chain sprocket wheels of cast iron conforming to ASTM A48/A48M.

#### 2.2.5 Electric Door Operators

Provide electrical wiring and door operating controls conforming to the applicable requirements of NFPA 70 and UL 325. The door manufacturer must furnish automatic control and safety devices, including extra flexible type SO cable and spring-loaded automatic takeup reel or equivalent device, as required for proper operation of the doors. Conduit, wiring, and mounting of controls are specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

Electrical materials, equipment, and devices for installation in hazardous locations, as defined by NFPA 70, must be specifically approved by Underwriters Laboratories or an independent testing agency using equivalent standards, for the particular chemical group and the class and division of hazardous location involved.

Electric door-operator assemblies need to be the sizes and capacities recommended and provided by the door manufacturer for specified doors. Furnish complete assemblies with electric motors and factory-prewired motor controls, starter, gear reduction units, solenoid-operated brakes, clutch, remote-control stations, manual or automatic control devices, and accessories as required for proper operation of the doors.

Design the operators so that motors may be removed without disturbing the limit-switch adjustment and affecting the emergency auxiliary operators.

Provide a manual operator of crank-gear or chain-gear mechanisms with a release clutch to permit manual operation of doors in case of power failure. Arrange the emergency manual operator so that it may be put into and out of operation from floor level, and its use does not affect the adjustment of the limit switches. Provide an electrical or mechanical device that automatically disconnects the motor from the operating mechanism when the emergency manual operating mechanism is engaged.

##### 2.2.5.1 Door-Operator Types

Provide an operator mounted to the right or left door head plate with the operator on top of the door-hood assembly and connected to the door drive shaft with drive chain and sprockets. Headroom is required for this type of mounting.

##### 2.2.5.2 Electric Motors

Provide motors which are the high-starting-torque, reversible, constant-duty electrical type with overload protection of sufficient torque and horsepower to move the door in either direction from any position. Ensure they produce a door-travel speed of not less than 8 nor more than 12 inches per second without exceeding the horsepower rating.

Provide motors which conform to NEMA MG 1 designation, temperature rating,

service factor, enclosure type, and efficiency to the requirements specified. Motors must be suitable for operation on current of the characteristics indicated. Motor enclosures must be the drip-proof type or NEMA TEFC and TENV type. Install motors in approved locations.

#### 2.2.5.3 Motor Bearings

Select bearings with bronze-sleeve or heavy-duty ball or roller antifriction type with full provisions for the type of thrust imposed by the specific duty load.

Pre-lubricate and factory seal bearings in motors less than 1/2 horsepower.

Equip motors coupled to worm-gear reduction units with either ball or roller bearings.

Equip bearings in motors 1/2 horsepower or larger with lubrication service fittings. Fit lubrication fittings with color-coded plastic or metal dust caps.

In any motor, bearings that are lubricated at the factory for extended duty periods do not need to be lubricated for a given number of operating hours. Display this information on an appropriate tag or label on the motor with instructions for lubrication cycle maintenance.

#### 2.2.5.4 Motor Starters, Controls, and Enclosures

Provide each door motor with: a factory-wired, unfused, disconnect switch; a reversing, across-the-line magnetic starter with thermal overload protection; 24-volt operating coils with a control transformer limit switch; and a safety interlock assembled in a NEMA ICS 6 type enclosure as specified herein. Ensure control equipment conforms to NEMA ICS 1 and NEMA ICS 2.

Provide adjustable switches, electrically interlocked with the motor controls and set to stop the door automatically at the fully open and fully closed position.

#### 2.2.5.5 Control Enclosures

Provide control enclosures that conform to NEMA ICS 6 for NEMA Type 4, oil-tight and dust-tight NEMA Type 12.

#### 2.2.5.6 Transformer

Provide starters with 230/460 to 115 volt control transformers with one secondary fuse when required to reduce the voltage on control circuits to 24volts or less. Provide a transformer conforming to NEMA ST 1.

#### 2.2.5.7 Sensing-Edge Device

Provide each door with a pneumatic or electric sensing device that meets UL 325, extends the full width of the door, and is located within a U-section neoprene or rubber astragal, mounted on the bottom rail of the bottom door section. Device needs to immediately stop and reverse the door upon contact with an obstruction in the door opening or upon failure of the device or any component of the control system and cause the door to return to its user-defined open position. Any momentary door-closing circuit must be automatically locked out and the door must be operable

manually or with constant pressure controls until the failure or damage has been corrected. A sensing device is not a substitute for a limit switch.

Connect sensing device to the control circuit through a retracting cord and reel.

#### 2.2.5.8 Remote-Control Stations

Provide exterior control stations that are full-guarded, momentary-contact three-button standard-duty, surface-mounted, weatherproof type, NEMA ICS 6, Type 4 enclosures, key-operated, with the same operating functions as specified herein for interior remote-control stations.

#### 2.2.5.9 Speed-Reduction Units

Provide speed-reduction units consisting of hardened-steel worm and bronze worm gear assemblies or planetary gear reducers running in oil or grease and inside a sealed casing, coupled to the motor through a flexible coupling. Drive shafts need to rotate on ball- or roller-bearing assemblies that are integral with the unit.

Provide minimum ratings of speed reduction units in accordance with AGMA provisions for class of service.

Ground worm gears to provide accurate thread form; machine teeth for all other types of gearing. Surface harden all gears.

Provide antifriction type bearings equipped with oil seals.

#### 2.2.5.10 Chain Drives

Provide roller chains that are a power-transmission series steel roller type conforming to ASME B29.400, with a minimum safety factor of 10 times the design load.

Heat-treat or otherwise harden roller-chain side bars, rollers, pins, and bushings.

Provide high-carbon steel chain sprockets with machine-cut hardened teeth, finished bore and keyseat, and hollow-head setscrews.

#### 2.2.5.11 Brakes

Provide 360-degree shoe brakes or shoe and drum brakes. Ensure the brakes are solenoid-operated and electrically interlocked to the control circuit to set automatically when power is interrupted.

#### 2.2.5.12 Clutches

Ensure clutches are friction type or adjustable centrifugal type.

#### 2.2.5.13 Weather/Smoke Seal Sensing Edge

Provide automatic stop control by an automatic sensing switch within neoprene astragal extending the full width of door bottom bar.

Provide an electric sensing edge device. Ensure the door immediately stops downward travel when contact occurs before door fully closes.

Provide a self-monitoring sensing edge connection to the motor operator.

#### 2.2.6 Surface Finishing

Comply with NAAMM's "Metal Finishes Manual for Architectural and Metal Products" for recommendations for applying and designating finishes. Noticeable variations in the same metal component are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved samples and are assembled or installed to minimize contrast.

##### 2.2.6.1 Galvanized and Shop-Primed Finish

Surfaces specified must have a zinc coating, a phosphate treatment, and a shop prime coat of rust-inhibitive paint. The galvanized coating must conform to ASTM A653/A653M, coating designation Z275 (G90), for steel sheets, and ASTM A123/A123M for iron and steel products. The weight of coatings for products must be as designated in Table I of ASTM A123/A123M for the thickness of base metal to be coated. The prime coat must be a type especially developed for materials treated by phosphates and adapted to application by dipping or spraying. Repair damaged zinc-coated surfaces by the materials and methods conforming to ASTM A780/A780M and spot prime. At the option of the Contractor, a two-part system including bonderizing, baked-on epoxy primer, and baked-on enamel top coat may be applied to slats and hoods before forming, in lieu of prime coat specified.

##### 2.2.6.2 Baked-Enamel or Powder-Coat Finish

Manufacturer's standard baked-on finish consisting of prime coat and thermosetting topcoat. Comply with the coating manufacturer's written instructions for cleaning, pretreatment, application, and minimum dry film thickness.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

Install overhead coiling door assembly, anchors and inserts for guides, brackets, motors, switches, hardware, and other accessories in accordance with approved detail drawings and manufacturer's written instructions. Upon completion of installation, ensure doors are free from all distortion.

Install overhead coiling doors, motors, hoods, and operators at the mounting locations as indicated for each door in the Contract Documents and as required by the manufacturer.

Install overhead coiling doors, switches, and controls along accessible routes in compliance with regulatory requirements for accessibility and as required by the manufacturer.

##### 3.1.1 Field Painted Finish

Ensure field painted steel doors and frames are in accordance with Section 09 90 00 PAINTS AND COATINGS and the manufacturer's written instructions. Protect the weather stripping from paint. Ensure that the finishes are free of scratches or other blemishes.

### 3.2 ADJUSTING AND CLEANING

#### 3.2.1 Acceptance Provisions

After installation, adjust the hardware and moving parts. Lubricate bearings and sliding parts as recommended by manufacturer to provide smooth operating functions for ease movement, free of warping, twisting, or distortion of the door assembly.

Adjust seals to provide a weather-tight fit around entire perimeter.

Engage a factory-authorized service representative to perform startup service and checks according to the manufacturer's written instructions.

Test the door opening and closing operation when activated by controls system. Adjust controls and safeties. Replace damaged and malfunctioning controls and equipment. Reset the door-closing mechanism after a successful test.

Test and make final adjustment of new doors at no additional cost to the Government.

##### 3.2.1.1 Maintenance and Adjustment

Not more than 90 calendar days after completion and acceptance of the project, examine, lubricate, test, and re-adjust doors as required for proper operation.

##### 3.2.1.2 Cleaning

Clean doors in accordance with manufacturer's approved instructions.

-- End of Section --

SECTION 08 71 00

DOOR HARDWARE  
02/16, CHG 4: 02/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

|           |                                                                                                                                                                                 |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM E283 | (2019) Standard Test Method for Determining the Rate of Air Leakage Through Exterior Windows, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

BUILDERS HARDWARE MANUFACTURERS ASSOCIATION (BHMA)

|                   |                                                         |
|-------------------|---------------------------------------------------------|
| ANSI/BHMA A156.1  | (2021) Butts and Hinges                                 |
| ANSI/BHMA A156.2  | (2022) Bored and Preassembled Locks and Latches         |
| ANSI/BHMA A156.3  | (2020) Exit Devices                                     |
| ANSI/BHMA A156.4  | (2013) Door Controls - Closers                          |
| ANSI/BHMA A156.5  | (2020) Cylinder and Input Devices for Locks             |
| ANSI/BHMA A156.6  | (2021) Architectural Door Trim                          |
| ANSI/BHMA A156.7  | (2016) Template Hinge Dimensions                        |
| ANSI/BHMA A156.8  | (2021) Door Controls - Overhead Stops and Holders       |
| ANSI/BHMA A156.13 | (2022) Mortise Locks & Latches Series 1000              |
| ANSI/BHMA A156.16 | (2023) Auxiliary Hardware                               |
| ANSI/BHMA A156.17 | (2019) Self Closing Hinges & Pivots                     |
| ANSI/BHMA A156.18 | (2020) Materials and Finishes                           |
| ANSI/BHMA A156.21 | (2019) Thresholds                                       |
| ANSI/BHMA A156.22 | (2021) Gasketing                                        |
| ANSI/BHMA A156.26 | (2012) Continuous Hinges                                |
| ANSI/BHMA A156.29 | (2012) Exit Locks, Exit Alarms, Alarms for Exit Devices |



ANSI/BHMA A156.36 (2010) Auxiliary Locks

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 72 (2022; ERTA 22-1) National Fire Alarm and Signaling Code

NFPA 80 (2022) Standard for Fire Doors and Other Opening Protectives

NFPA 101 (2021; TIA 21-1) Life Safety Code

NFPA 252 (2022) Standard Methods of Fire Tests of Door Assemblies

STEEL DOOR INSTITUTE (SDI/DOOR)

SDI/DOOR A250.8 (2023) Specifications for Standard Steel Doors and Frames

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 1191 Americans with Disabilities Act (ADA) Accessibility Guidelines for Buildings and Facilities; Architectural Barriers Act (ABA) Accessibility Guidelines

UNDERWRITERS LABORATORIES (UL)

UL Bld Mat Dir (updated continuously online) Building Materials Directory

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

Manufacturer's Detail Drawings; G

Verification of Existing Conditions; G

Hardware Schedule; G

Keying System; G

### SD-03 Product Data

Hardware Items; G

### SD-08 Manufacturer's Instructions

Installation

### SD-10 Operation and Maintenance Data

Hardware Schedule Items, Data Package 1; G

SD-11 Closeout Submittals

Key Bitting

1.3 SHOP DRAWINGS

Submit manufacturer's detail drawings indicating all hardware assembly components and interface with adjacent construction. Base shop drawings on verified field measurements and include verification of existing conditions.

1.4 PRODUCT DATA

Indicate fire-ratings at applicable components. Provide documentation of ABA/ADA accessibility compliance of applicable components, as required by 36 CFR 1191 Appendix D - Technical.

1.5 HARDWARE SCHEDULE

Provide Hardware Item List and Hardware Schedule containing the following information, and additional information as needed to identify the complete make up of each hardware set and its application to each opening:

1.5.1 Hardware Item List:

- a. Hardware Type
- b. Item Number
- c. Quantity
- d. Size(s)
- e. Reference Publication / Type Number
- f. Manufacturer's Name / Catalog Number
- g. Key Control Symbols
- h. UL Mark (If fire rated and listed)
- i. BHMA Finish(es)
- j. Remarks

1.5.2 Hardware Schedule

- a. Hardware Set Number
- b. Opening Number(s)
- c. Opening Description (single/double leaf, hand, size, door/frame material)
- d. Fire Rating
- e. Sound Rating

- f. Hardware Items
- g. Quantity
- h. Size
- i. BHMA Finish
- j. Remarks

In addition, submit hardware schedule data package 1 in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

## 1.6 KEY BITTING CHART REQUIREMENTS

### 1.6.1 Requirements

Submit key bitting charts to the Contracting Officer prior to completion of the work. Include:

- a. Complete listing of all keys (e.g. AA1 and AA2).
- b. Complete listing of all key cuts (AA1-123456, AA2-123458).
- c. Tabulation showing which key fits which door.
- d. Copy of floor plan showing doors and door numbers.
- e. Listing of 20 percent more key cuts than are presently required in each master system.

## 1.7 QUALITY ASSURANCE

### 1.7.1 Hardware Manufacturers and Modifications

Provide, as far as feasible, locks, hinges, pivots, and closers of one lock, hinge, pivot, or closer manufacturer's make. Modify hardware as necessary to provide features indicated or specified.

### 1.7.2 Key Shop Drawings Coordination Meeting

Prior to the submission of the key shop drawing, the Contracting Officer, Contractor, Door Hardware Subcontractor, using Activity and Base Locksmith must meet to discuss and coordinate key requirements for the facility.

## 1.8 DELIVERY, STORAGE, AND HANDLING

Deliver hardware in original individual containers, complete with necessary appurtenances including fasteners and instructions. Mark each individual container with item number as shown on hardware schedule. Deliver permanent keys to the Contracting Officer, either directly or by certified mail. Deliver construction master keys with the locks.

## PART 2 PRODUCTS

### 2.1 TEMPLATE HARDWARE

Hardware applied to metal or to prefinished doors must be manufactured

using a template. Provide templates to door and frame manufacturers in accordance with ANSI/BHMA A156.7 for template hinges. Coordinate hardware items to prevent interference with other hardware.

## 2.2 HARDWARE FOR FIRE DOORS AND EXIT DOORS

Provide all hardware necessary to meet the requirements of NFPA 72 for door alarms, NFPA 80 for fire doors, NFPA 101 for exit doors, NFPA 252 for fire tests of door assemblies, ABA/ADA accessibility requirements, and all other requirements indicated, even if such hardware is not specifically mentioned in paragraph HARDWARE SCHEDULE. Provide Underwriters Laboratories, Inc. labels for such hardware in accordance with UL Bld Mat Dir or equivalent labels in accordance with another testing laboratory approved in writing by the Contracting Officer.

## 2.3 HARDWARE ITEMS

Clearly and permanently mark with the manufacturer's name or trademark, hinges, pivots, locks, latches, exit devices, bolts and closers where the identifying mark is visible after the item is installed. For closers with covers, the name or trademark may be beneath the cover. Coordinate electrified door hardware components with corresponding components specified in Division 28 ELECTRONIC SECURITY SYSTEMS (ESS).

### 2.3.1 Hinges

Provide in accordance with ANSI/BHMA A156.1. Provide hinges that are 4-1/2 by 4-1/2 inch unless otherwise indicated. Construct loose pin hinges for interior doors and reverse-bevel exterior doors so that pins are non-removable when door is closed. Other anti-friction bearing hinges may be provided in lieu of ball bearing hinges.

#### 2.3.2 Continuous Hinges

Where continuous hinges are required, provide in accordance with ANSI/BHMA A156.26.

#### 2.3.3 Pivots

Provide in accordance with ANSI/BHMA A156.17.

#### 2.3.4 Spring Hinges

Provide in accordance with ANSI/BHMA A156.17.

#### 2.3.5 Locks and Latches

- a. At exterior locations provide locksets of full stainless steel type 302 or 304 construction including fronts, strike, escutcheons, knobs, bolts and all interior working parts. Marine Grade I, fully non-ferrous.
- b. In non-air-conditioned interior environments or humid interior environments, provide interior locksets on the same Marine Grade I, fully non-ferrous as exterior locksets.

##### 2.3.5.1 Mortise Locks and Latches

Provide in accordance with ANSI/BHMA A156.13, Series 1000, Operational

Grade 1, Security Grade 2. Provide mortise locks with escutcheons not less than 7 by 2-1/4 inch with a bushing at least 1/4 inch long. Cut escutcheons to fit cylinders and provide trim items with straight, beveled, or smoothly rounded sides, corners, and edges. Provide knobs and roses of mortise locks with screwless shanks and no exposed screws.

#### 2.3.5.2 Bored Locks and Latches

Provide in accordance with ANSI/BHMA A156.2, Series 4000, Grade 1. Provide factory installed lead lining in locks for lead -shielded doors.

#### 2.3.5.3 Auxiliary Locks

Provide in accordance with ANSI/BHMA A156.36, Grade 1.

#### 2.3.6 Exit Devices

Provide in accordance with ANSI/BHMA A156.3, Grade 1. Provide adjustable strikes for rim type and vertical rod devices. Provide open back strikes for pairs of doors with mortise and vertical rod devices. Provide escutcheons not less than 7 by 2-1/4 inch.

#### 2.3.7 Exit Locks With Alarm

Provide in accordance with ANSI/BHMA A156.3 and ANSI/BHMA A156.29, Type E0431 (with full width horizontal actuating bar) for single doors; Type E0431 (with actuating bar) or E0471 (with actuating bar and top and bottom bolts, both leaves active) for pairs of doors, unless otherwise specified. Provide door alarms integrated with the fire alarm system in accordance with NFPA 72.

#### 2.3.8 Cylinders and Cores

Provide cylinders and cores for new locks, including locks provided under other sections of this specification. Provide cylinders and cores with seven pin tumblers. Provide cylinders from the products of one manufacturer, and provide cores from the products of one manufacturer.

#### 2.3.9 Push Button Mechanisms

Provide in accordance with ANSI/BHMA A156.5, Grade 1.

#### 2.3.10 Keying System

Provide

The Government will provide permanent cylinders with cores and keys for mortise locksets, auxiliary locks, and exit devices.

#### 2.3.11 Lock Trim

Provide cast, forged, or heavy wrought construction and commercial plain design for lock trim.

##### 2.3.11.1 Knobs and Roses

Provide in accordance with ANSI/BHMA A156.2 and ANSI/BHMA A156.13 for knobs, roses, and escutcheons. For unreinforced knobs, roses, and escutcheons, provide a 0.050 inch thickness. For reinforced knobs, roses,

and escutcheons, provide an outer shell thickness of 0.035 inch and a combined total thickness of 0.070 inch, except at knob shanks. Provide knob shanks 0.060 inch thick.

#### 2.3.11.2 Lever Handles

Provide lever handles . Provide in accordance with ANSI/BHMA A156.3 for mortise locks of lever handles for exit devices. Provide lever handle locks with a breakaway feature (such as a weakened spindle or a shear key) to prevent irreparable damage to the lock when force in excess of that specified in ANSI/BHMA A156.13 is applied to the lever handle. Provide lever handles return to within 1/2 inch of the door face.

#### 2.3.12 Door Bolts

Provide in accordance with ANSI/BHMA A156.16. Provide dustproof strikes for bottom bolts, except at doors having metal thresholds. Provide automatic latching flush bolts in accordance with ANSI/BHMA A156.3, Type 25.

#### 2.3.13 Closers

Provide in accordance with ANSI/BHMA A156.4, Series C02000, Grade 1, with PT 4C. Provide with brackets, arms, mounting devices, fasteners, pivots, and other features necessary for the particular application. Size closers in accordance with manufacturer's printed recommendations, or provide multi-size closers, Sizes 1 through 6, and list sizes in the Hardware Schedule. Provide manufacturer's 10 year warranty.

Use stainless steel inside bracketed or door mounted closers on exterior doors. Non-ferrous closers, such as aluminum or cast bronze, are permissible where door utilization is minimal. On interior doors use closers of 302 or 304 stainless steel or non-ferrous materials. On surface-mounted closers use or apply rust inhibiting finish on all ferrous parts. Also apply this finish on concealed closers.

##### 2.3.13.1 Identification Marking

Engrave each closer with manufacturer's name or trademark, date of manufacture, and manufacturer's size designation in locations that will be visible after installation.

#### 2.3.14 Overhead Holders

Provide in accordance with ANSI/BHMA A156.8.

#### 2.3.15 Door Protection Plates

Provide in accordance with ANSI/BHMA A156.6.

##### 2.3.15.1 Sizes of Kick Plates

2 inch less than door width for single doors; 1 inch less than door width for pairs of doors. Provide 10 inch kick plates for flush doors.

#### 2.3.16 Door Stops and Silencers

Provide in accordance with ANSI/BHMA A156.16. Silencers Type L03011. Provide three silencers for each single door, two for each pair.

### 2.3.17 Thresholds

Provide in accordance with ANSI/BHMA A156.21. Use J35100, with vinyl or silicone rubber insert in face of stop, for exterior doors opening out, unless specified otherwise.

### 2.3.18 Weatherstripping Gasketing

Provide in accordance with ANSI/BHMA A156.22. Provide the type and function designation where specified in paragraph HARDWARE SCHEDULE. Provide a set to include head and jamb seals, sweep strips, and, for pairs of doors, astragals. Air leakage of weatherstripped doors not to exceed 0.5 cubic feet per minute of air per square foot of door area when tested in accordance with ASTM E283. Provide weatherstripping with one of the following:

#### 2.3.18.1 Interlocking Type

Zinc or bronze not less than 0.018 inch thick.

#### 2.3.18.2 Spring Tension Type

Spring bronze or stainless steel not less than 0.008 inch thick.

### 2.3.19 Lightproofing and Soundproofing Gasketing

Provide in accordance with ANSI/BHMA A156.22. Provide adjustable doorstops at heads, jams and automatic door bottoms in accordance with the hardware set, of extruded aluminum, anodized, surface applied, with vinyl fin seals between plunger and housing. Provide doorstops with solid neoprene tube, silicone rubber, or closed cell sponge gasket. Provide door bottoms with adjustable operating rod and silicone rubber or closed cell sponge neoprene gasket. Provide doorstops that are mitered at corners. Provide type and function designation where specified in paragraph HARDWARE SETS.

### 2.3.20 Rain Drips

Provide in accordance with ANSI/BHMA A156.22. Provide extruded aluminum rain drips, not less than 0.08 inch thick, factory painted finish. Provide the manufacturer's full range of color choices to the Contracting Officer for color selection. Provide rain drips with a 4 inch overlap on each side of each exterior door that is not protected by an awning, roof, eave or other horizontal projection. Set drips in sealant and fasten with stainless steel screws.

#### 2.3.20.1 Door Rain Drips

Approximately 1-1/2 inch high by 5/8 inch projection. Align bottom with bottom edge of door.

#### 2.3.20.2 Overhead Rain Drips

Approximately 1-1/2 inch high by 2-1/2 inch projection. Align bottom with door frame rabbet.

#### 2.3.21 Auxiliary Hardware (Other than locks)

Provide in accordance with ANSI/BHMA A156.16, Grade 1.

#### 2.3.22 Special Tools

Provide special tools, such as spanner and socket wrenches and dogging keys, as required to service and adjust hardware items.

### 2.4 FASTENERS

Provide fasteners of type, quality, size, and quantity appropriate to the specific application. Fastener finish to match hardware. Provide stainless steel or nonferrous metal fasteners in locations exposed to weather. Verify metals in contact with one another are compatible and will avoid galvanic corrosion when exposed to weather.

### 2.5 FINISHES

Provide in accordance with ANSI/BHMA A156.18. Provide hardware in BHMA 630 finish (satin stainless steel), unless specified otherwise. Provide items not manufactured in stainless steel in BHMA 626 finish (satin chromium plated) over brass or bronze, except aluminum paint finish for surface door closers, and except BHMA 652 finish (satin chromium plated) for steel hinges. Provide hinges for exterior doors in stainless steel with BHMA 630 finish or chromium plated brass or bronze with BHMA 626 finish. Furnish exit devices in BHMA 626 finish in lieu of BHMA 630 finish. Match exposed parts of concealed closers to lock and door trim. Match hardware finish for aluminum doors to the doors.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Provide hardware in accordance with manufacturers' printed installation instructions. Fasten hardware to wood surfaces with full-threaded wood screws or sheet metal screws. Provide machine screws set in expansion shields for fastening hardware to solid concrete and masonry surfaces. Provide toggle bolts where required for fastening to hollow core construction. Provide through bolts where necessary for satisfactory installation.

#### 3.1.1 Weatherstripping Installation

Provide full contact, weathertight seals that allow operation of doors without binding the weatherstripping.

##### 3.1.1.1 Stop Applied Weatherstripping

Fasten in place with color matched sheet metal screws not more than 9 inch on center after doors and frames have been finish painted.

##### 3.1.1.2 Interlocking Type Weatherstripping

Provide interlocking, self adjusting type on heads and jambs and flexible hook type at sills. Nail weatherstripping to door 1 inch on center and to heads and jambs at 4 inch on center.



#### 3.1.1.3 Spring Tension Type Weatherstripping

Provide spring tension type on heads and jambs. Provide bronze nails with bronze. Provide stainless steel nails with stainless steel. Space nails not more than 1-1/2 inch on center.

#### 3.1.2 Lightproofing and Soundproofing Installation

Provide as specified for stop applied weatherstripping.

#### 3.1.3 Threshold Installation

Extend thresholds the full width of the opening and notch end for jamb stops. Set thresholds in a full bed of sealant and anchor to floor with cadmium-plated, countersunk, steel screws. For aluminum thresholds placed on top of concrete surfaces, coat the underside surfaces that are in contact with the concrete with fluid applied waterproofing as a separation measure prior to placement.

### 3.2 FIRE DOORS AND EXIT DOORS

Provide hardware in accordance with NFPA 72 for door alarms, NFPA 80 for fire doors, NFPA 101 for exit doors, and NFPA 252 for fire tests of door assemblies. .

### 3.3 HARDWARE LOCATIONS

Provide in accordance with SDI/DOOR A250.8, unless indicated or specified otherwise.

- a. Kick and Armor Plates: Push side of single-acting doors. Both sides of double-acting doors.
- b. Mop Plates: Bottom flush with bottom of door.

### 3.4 FIELD QUALITY CONTROL

After installation, protect hardware from paint, stains, blemishes, and other damage until acceptance of work. Submit notice of testing 15 days before scheduled, so that testing can be witnessed by the Contracting Officer. Adjust hinges, locks, latches, bolts, holders, closers, and other items to operate properly. Demonstrate that permanent keys operate respective locks, and give keys to the Contracting Officer. Correct, repair, and finish, errors in cutting and fitting and damage to adjoining work.

-- End of Section --

SECTION 08 91 00

METAL WALL LOUVERS  
**08/20**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 500-L (2015) Laboratory Methods of Testing  
Louvers for Rating

AMCA 511 (2010; R 2016) Certified Ratings Program  
for Air Control Devices

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 2605 (2020) Voluntary Specification,  
Performance Requirements and Test  
Procedures for Superior Performing Organic  
Coatings on Aluminum Extrusions and Panels

ASTM INTERNATIONAL (ASTM)

ASTM A123/A123M (2024) Standard Specification for Zinc  
(Hot-Dip Galvanized) Coatings on Iron and  
Steel Products

ASTM A653/A653M (2023) Standard Specification for Steel  
Sheet, Zinc-Coated (Galvanized) or  
Zinc-Iron Alloy-Coated (Galvannealed) by  
the Hot-Dip Process

ASTM A780/A780M (2020) Standard Practice for Repair of  
Damaged and Uncoated Areas of Hot-Dip  
Galvanized Coatings

ASTM A1008/A1008M (2024) Standard Specification for Steel,  
Sheet, Cold-Rolled, Carbon, Structural,  
High-Strength Low-Alloy, High-Strength  
Low-Alloy with Improved Formability,  
Solution Hardened, and Bake Hardenable

ASTM B209 (2014) Standard Specification for Aluminum  
and Aluminum-Alloy Sheet and Plate

ASTM B221 (2021) Standard Specification for Aluminum  
and Aluminum-Alloy Extruded Bars, Rods,  
Wire, Profiles, and Tubes

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Wall Louvers

SD-03 Product Data

Metal Wall Louvers

SD-04 Samples

Wall Louver Samples; G

Door Louver Samples; G

## 1.3 DELIVERY, STORAGE, AND PROTECTION

Deliver materials to the site in an undamaged condition. Carefully store materials off the ground to provide proper ventilation, drainage, and protection against dampness. Louvers must be free from nicks, scratches, and blemishes. Replace defective or damaged materials with new.

## 1.4 DETAIL DRAWINGS

Show all information necessary for fabrication and installation of wall louvers. Indicate materials, sizes, thicknesses, fastenings, and profiles.

## 1.5 COLOR SAMPLES

Colors of finishes for wall louver samples and door louver samples must closely approximate colors indicated. Where color is not indicated, submit the manufacturer's standard colors to the Contracting Officer for selection.

## PART 2 PRODUCTS

### 2.1 MATERIALS

#### 2.1.1 Galvanized Steel Sheet

ASTM A653/A653M, coating designation G90.

#### 2.1.2 Aluminum Sheet

ASTM B209, alloy 3003 or 5005 with temper as required for forming.

#### 2.1.3 Extruded Aluminum

ASTM B221, alloy 6063-T5 or -T52.

#### 2.1.4 Cold Rolled Steel Sheet

ASTM A1008/A1008M, Class 1, with matte finish. Use for interior louvers only.

### 2.2 METAL WALL LOUVERS

WeatherWind driven rain resistant type, with bird screens and made to withstand a wind load of not less than 30 pounds per square foot. Wall louvers must bear the AMCA certified ratings program seal for air performance and water penetration in accordance with AMCA 500-L and AMCA 511. The rating must show a water penetration of 0.20 or less ounce per square foot of free area at a free velocity of 800 feet per minute.

#### 2.2.1 Extruded Aluminum Louvers

Fabricated of extruded 6063-T5 or -T52 aluminum with a wall thickness of not less than 0.081 inch.

#### 2.2.2 Formed Metal Louvers

Formed of stainless steel sheet not thinner than 16 U.S. gage, or aluminum sheet not less than 0.08 inch thick.

#### 2.2.3 Mullions and Mullion Covers

Same material and finish as louvers. Provide mullions for all louvers more than 5 feet in width at not more than 5 feet on centers. Provide mullion covers on both faces of joints between louvers.

#### 2.2.4 Screens and Frames

For aluminum louvers, provide 1/2 inch square mesh, 14 or 16 gage aluminum or 1/4 inch square mesh, 16 gage aluminum bird screening. For steel louvers, provide 1/2 inch square mesh, 12 or 16 gage zinc-coated steel; 1/2 inch square mesh, 16 gage copper; or 1/4 inch square mesh, 16 gage zinc-coated steel or copper bird screening. Mount screens in removable, rewirable frames of same material and finish as the louvers.

### 2.3 FASTENERS AND ACCESSORIES

Provide stainless steel screws and fasteners for aluminum louvers and zinc-coated or stainless steel screws and fasteners for steel louvers. Provide other accessories as required for complete and proper installation.

### 2.4 FINISHES

#### 2.4.1 Aluminum

Exposed aluminum surfaces must be factory finished with an organic coating. Color will be selected from manufacturer's full range of colors. Louvers must have the same finish.

##### 2.4.1.1 Organic Coating

Clean and prime exposed aluminum surfaces. Provide a superior performance finish in accordance with AAMA 2605 with total dry film thickness of not less than 1.2 mil.

#### 2.4.2 Steel

Surfaces specified must have a zinc coating, a phosphate treatment, and a shop prime coat of rust-inhibitive paint. The galvanized coating must conform to ASTM A653/A653M, coating designation Z275 (G90). The weight of zinc coatings must be as designated in Table I of ASTM A123/A123M for the thickness of base metal to be coated. The prime coat must be a type especially developed for materials treated by phosphates and adapted to application by dipping or spraying. Repair damaged zinc-coated surfaces by the materials and methods conforming to ASTM A780/A780M and spot prime. At the option of the Contractor, a two-part system including bonderizing, baked-on epoxy primer, and baked-on enamel top coat may be applied before forming, in lieu of prime coat specified.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

##### 3.1.1 Wall Louvers

Install using stops or moldings, flanges, strap anchors, or jamb fasteners as appropriate for the wall construction and in accordance with manufacturer's recommendations.

##### 3.1.2 Screens and Frames

Attach frames to louvers with screws or bolts.

#### 3.2 PROTECTION FROM CONTACT OF DISSIMILAR MATERIALS

##### 3.2.1 Copper or Copper-Bearing Alloys

Paint copper or copper-bearing alloys in contact with dissimilar metal with heavy-bodied bituminous paint or separate with inert membrane.

##### 3.2.2 Aluminum

Where aluminum contacts metal other than zinc, paint the dissimilar metal with a primer and two coats of aluminum paint.

##### 3.2.3 Metal

Paint metal in contact with mortar, concrete, or other masonry materials with alkali-resistant coatings such as heavy-bodied bituminous paint.

##### 3.2.4 Wood

Paint wood or other absorptive materials that may become repeatedly wet and in contact with metal with two coats of aluminum paint or a coat of heavy-bodied bituminous paint.

-- End of Section --

SECTION 09 01 90.50

PREPARATION OF HISTORIC WOOD AND METAL SURFACES FOR PAINTING  
**06/25**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)

ACGIH 0100 (2017; Suppl 2020) Documentation of the  
Threshold Limit Values and Biological  
Exposure Indices

ASTM INTERNATIONAL (ASTM)

ASTM D173/D173M (2003; R 2011; E 2012) Bitumen-Saturated  
Cotton Fabrics Used in Roofing and  
Waterproofing

ASTM D3274 (2009; R 2017) Standard Test Method for  
Evaluating Degree of Surface Disfigurement  
of Paint Films by Fungal or Algal Growth,  
or Soil and Dirt Accumulation

ASTM D3359 (2017) Standard Test Methods for Rating  
Adhesion by Tape Test

ASTM D4214 (2007; R 2015) Standard Test Method for  
Evaluating the Degree of Chalking of  
Exterior Paint Films

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC SP 1 (2015) Solvent Cleaning

SSPC SP 2 (2024) Hand Tool Cleaning

SSPC SP 3 (2024) Power Tool Cleaning

SSPC SP 6/NACE No.3 (2007) Commercial Blast Cleaning

1.2 SUMMARY

The procedures proposed for the accomplishment of the work shall provide for safe conduct of the work, careful removal and disposition of materials specified to be salvaged, protection of property which is to remain undisturbed, and coordination with other work in progress. Submit the names, quantity represented, and intended use for proprietary brands of materials proposed to be substituted for the specified materials when the required quantity of a particular batch is 50 gallons or less. Submit manufacturer's current printed product description, safety data sheets (SDS) and technical data sheets for each product. Detailed mixing,

thinning and application instructions, minimum and maximum application temperature, and curing and drying times shall be provided for each product submitted. Include in the work plan a Safety and Health plan describing procedures for handling monitoring, and disposition of VOCs and other hazardous and toxic materials. Submit one copy of the Work Plan and a certificate stating that products proposed for use meet the VOC regulations of the local Air Pollution Control Districts having jurisdiction over the geographical area in which the project is located. The procedures shall include a detailed description of the methods and equipment to be used for each operation, and the sequence of operations. Test the materials designated by the Contracting Officer.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-03 Product Data

Work Plan; G

Materials

Qualifications

#### SD-07 Certificates

Work Plan; G

### 1.4 QUALITY ASSURANCE

Work shall comply with the ACCIDENT PREVENTION PLAN, including the Activity Hazard Analysis as specified in the CONTRACT CLAUSES. The Activity Hazard Analysis shall include analyses of the potential impact of surface preparation operations on personnel and on others involved in and adjacent to the work zone.

#### 1.4.1 Worker Exposures

Exposure of workers to chemical substances shall not exceed limits as established by ACGIH 0100.

#### 1.4.2 Training

Inform workers, having access to an affected work area, of the contents of the applicable SDS and of potential health and safety hazard and protective controls associated with materials used on the project. An affected work area is one which may receive dust, mists, and odors from the surface preparation operations. Workers involved in surface preparation and clean-up must be trained in the safe handling and application, and the exposure limit, for each material which the worker will use in the project. Instruct personnel having a need to use respirators and masks in the use and maintenance of such equipment.

#### 1.4.3 Coordination

Coordinate work to minimize exposure of building occupants, other

Contractor personnel, and visitors to mists and odors from surface preparation and cleaning operations.

1.4.4 Qualifications

Provide qualified workers trained and experienced in the preparation for painting of wood and metal surfaces in historic structures, submit documentation of 5 consecutive years of work of this type and a statement certified by the Contractor attesting that the experience and qualifications of the workers (journeymen) comply with the specifications. Provide a list of similar jobs identifying when, where, and for whom the work was done and a current point-of-contact for identified references.

1.5 DELIVERY, STORAGE, AND HANDLING

Deliver paint removers, solvents, and other chemicals, used for surface preparation, in sealed containers that legibly show the designated name, formula or specification number, quantity, date of manufacture, manufacturer's formulation number, manufacturer's directions including any warnings and special precautions, and name of manufacturer. Furnish such materials in containers not larger than 5 gallons; store them in accordance with the manufacturer's written directions; and, as a minimum, store them off the ground, under cover, with sufficient ventilation to prevent the buildup of flammable vapors and at temperatures between 40 and 95 degrees F.

1.6 ENVIRONMENTAL REQUIREMENTS

Unless otherwise recommended by the product manufacturer, the ambient temperature shall be between 45 and 95 degrees F when applying paint removers, solvents, or other preparation materials.

PART 2 PRODUCTS

2.1 PAINT REMOVERS

Chemical paint removers shall be a commercial item specifically manufactured for the type of paint to be removed.

2.2 EPOXY CONSOLIDANTS

2.2.1 Liquid Consolidant

Provide liquid wood consolidant consisting of a 2-part, low-viscosity liquid epoxy that meets the criteria of Table 1.

2.2.2 Epoxy Paste

Provide epoxy paste consisting of a 2-part, thixotropic paste that meets the criteria of Table 1.

| TABLE 1 |                    |             |
|---------|--------------------|-------------|
|         | LIQUID CONSOLIDANT | EPOXY PASTE |



| TABLE 1                      |                      |                             |
|------------------------------|----------------------|-----------------------------|
| Properties                   | Low-Viscosity Liquid | No-slump, Thixotropic Paste |
| Toxicity                     | Low                  | Very Low                    |
| Toxicity Cured               | Non-Toxic            | Non-Toxic                   |
| Ratios                       | 1:1 by Volume        | 1:1 by Volume               |
| Pot Life @ Room Temp.        | 30 minutes min.      | 50 minutes min.             |
| Hardening @ Room Temp.       | 1 hr. or longer      | 1 hr. or longer             |
| Hardening @ 140 deg. F       | 16 min. or less      | 18 min. or less             |
| Viscosity Poises @ 72 deg. F | 4.7 max.             | Thixotropic paste           |
| Solids                       | 95 percent min.      | 98 percent min.             |
| Tensile Strength             | 4000 psi             | 2500 psi                    |
| Elongation                   | 50 percent           | 4 percent                   |
| Compressive Strength         |                      |                             |
| Failure                      | 19,000 psi           | ---                         |
| Yield                        | 3500 psi             | 5500 psi                    |

## PART 3 EXECUTION

### 3.1 GENERAL REQUIREMENTS

Use methods for preparation of historic wood and metal surfaces for painting which are the gentlest possible to achieve the desired results. Historic substrate materials shall not be damaged or marred in the process of surface preparations. Collect and analyze samples of the existing paint finishes for the purpose of documentation or matching, if so directed by the Contracting Officer and Cultural Resources Manager - PWD Maine. Material and application requirements for paints are covered in Section 09 90 00 PAINTS AND COATINGS.

### 3.2 VENTILATION

Ventilate interior work zones, having a volume of 10,000 cubic feet or less, at a minimum of 2 air exchanges per hour. Maintain ventilation in larger work zones by means of mechanical exhaust. Solvent vapors shall be exhausted outdoors, away from air intakes and workers. Temporarily seal return air inlets in the work zone before start of work until the prepared surfaces have dried. Operators and personnel in the vicinity of paint removal processes involving chemicals or mechanical action (sanding or blasting) shall wear respirators.

### 3.3 PROTECTION OF AREAS NOT TO BE PAINTED

Remove or protect items not to be painted, which are in contact with or adjacent to painted surfaces, prior to surface preparation and painting operations. Replace items removed prior to painting when painting is completed. Following completion of painting, workers skilled in the trades involved shall reinstall removed items. Surfaces contaminated by preparation materials shall be restored to original condition.

### 3.4 CLEANING OF SURFACES

Surfaces to be painted shall be clean and free of grease, dirt, dust and other foreign matter before application of paint or surface treatments. After cleaning, surfaces shall exhibit a surface disfigurement rating of 7 or greater when evaluated in accordance with ASTM D3274. Dirt and surface contaminants shall be cleaned by brush with solutions of water and detergent or trisodium phosphate, then rinsed clean with water and let dry. Surfaces on which mildew or other microbiological growth is present shall be cleaned with a detergent solution containing household bleach. Oil and grease shall be removed with clean cloths and cleaning solvents prior to mechanical cleaning. Cleaning solvents shall be of low toxicity with a flashpoint in excess of 100 degrees F. Cleaning shall be programmed so that dust and other contaminants will not fall on newly prepared or newly painted surfaces.

### 3.5 EXISTING PAINT

Existing paint shall be tested for adhesion to substrate in accordance with ASTM D3359, Test Method A and shall obtain a rating of 4 or better in order to be considered sound. Existing paint meeting this requirement may be considered a satisfactory base for repainting.

### 3.6 PAINT REMOVAL

Remove flaking, cracking, blistering, peeling or otherwise deteriorated paint by scraping with hand scrapers. After scraping, removal of large areas of paint or paint on architectural details shall be accomplished using sanders, heat guns or heat plates, or chemical paint removers. Paint shall be removed to bare substrate or first sound paint layer. Open flame heat devices shall not be used. Mechanical paint removal shall not damage or mar the substrate material.

#### 3.6.1 Chemical Paint Removers

Use chemical paint removers in accordance with manufacturer's recommendations. If chemical strippers are used, substrate shall be neutralized after stripping to a pH of 5 to 8.5.

#### 3.6.2 Lead Paint

In preparation of lead-based painted surfaces for repainting, follow the procedures described in Section 02 83 00.00 22 MANAGEMENT OF LEAD, CADMIUM, AND CHROMIUM (VI) DURING RENOVATION, DEMOLITION, CONSTRUCTION, REMOVAL AND ABATEMENT (PNS PROJECTS).

### 3.7 SURFACE PREPARATION

After cleaning and removal of deteriorated paint, edges of remaining chipped paint shall be feather-edged and sanded smooth. Repair damaged

areas such as, but not limited to, nail holes, cracks, chips, and spalls with suitable material to match adjacent undamaged areas. Slick surfaces shall be roughened. Clean rusty metal surfaces in accordance with SSPC SP 3. Remove chalk so that when tested in accordance with ASTM D4214, the chalk resistance rating is no less than 8. New, proposed coatings shall be compatible with existing coatings. If existing surfaces are glossy, the gloss shall be reduced.

### 3.8 WOOD SURFACES

Wood surfaces shall be cleaned of foreign matter. Wood surfaces adjacent to surfaces to receive water-thinned paints shall be primed and/or touched up before applying water-thinned paints. Small, dry seasoned knots shall be scraped, cleaned, and given a thin coat of commercial knot sealer before application of the priming coat. Pitch on large, open, unseasoned knots and all other beads or streaks of pitch shall be scraped off, or, if it is still soft, removed with mineral spirits or turpentine, and the resinous area shall be thinly coated with knot sealer.

#### 3.8.1 Interior Wood Surfaces

Interior wood surfaces to receive stain shall be sanded. Oak and other open-grain wood to receive stain shall be given a coat of wood filler recommended by the finish manufacturer not less than 8 hours before the application of stain; excess filler shall be removed and the surface sanded smooth. Moisture content of the wood shall not exceed 12 percent as measured by a moisture meter, unless otherwise authorized.

#### 3.8.2 Wood Repair

Remove and repair badly decayed areas. Replace areas and pieces decayed beyond repair with new pieces that match originals in all respects. Moderately decayed areas, weathered, or gouged wood shall be patched with approved patching compounds, and shall be sanded smooth. The source or cause of wood decay shall be identified and corrected prior to application of patching materials. Wet wood shall be completely dried to a moisture content not exceeding 12 percent, as measured by a moisture meter, to its full depth before patching, unless otherwise authorized. Wood that is to be patched shall be clean of dust, grease, and loose paint.

##### 3.8.2.1 Epoxy Wood Repair

Epoxy wood repair materials shall be applied in accordance with manufacturer's written instructions. Health and safety instructions shall be followed in accordance with the manufacturer's instructions. Clean mixing equipment shall be used to avoid contamination. Mix and proportions shall be as directed by the manufacturer. Batches shall be only large enough to complete the specific job intended. Patching materials shall be completely cured before painting or reinstallation of patched pieces.

##### 3.8.2.2 Epoxy Consolidant and Epoxy Paste

Epoxy liquid wood consolidant shall be used: 1) to penetrate and impregnate deteriorated wood sections in order to reinforce wood fibers that have become softened or absorbent. 2) as a primer for areas that are to receive epoxy paste filler. Epoxy paste shall be used to fill areas where portions of wood are missing such as holes, cracks, gaps, gouges, and other voids.

### 3.8.3 Epoxy Exposed Ferrous Metals

Exposed ferrous metals such as nail heads on or in contact with wood surfaces to be painted with water-thinned paints, shall be spot-primed with a suitable corrosion-inhibitive primer capable of preventing flash rusting and compatible with the coating specified for the adjacent areas.

### 3.8.4 Finishing Nails

Finishing nails shall be set, and all holes and surface imperfections shall be primed. After priming, holes and imperfections in finish surfaces shall be filled with putty or plastic wood filler, colored to match the finish coat if natural finish is required, allowed to dry, and sanded smooth. Putty or wood filler shall be compatible with subsequent coatings.

### 3.8.5 Wood Preservative

Areas of bare wood in exterior locations prone to excessive moisture or standing water shall be treated with a commercial, fungicide, paintable water repellant/preservative. Water repellant/preservatives shall not be used on interior surfaces.

## 3.9 METAL SURFACES

Metal surfaces shall be cleaned of foreign matter. Programs for preparation of metal shall be in accordance with SSPC PA Guide 5. Grease, oil, and other soluble contaminants shall be removed by solvent cleaning in accordance with SSPC SP 1. Surfaces shall be free from soils and corrosion; e.g. grease, oil, solder flux, welding flux, weld spatter, sand, rust, scale, and other contaminants that might interfere with the application of the new finish. Cleaning methods shall be the gentlest possible to achieve the desired result. Metals which are soft, thin, or exhibit fine detail shall not be abrasively cleaned. Evidence of corrosion or contamination on a previously cleaned surface shall be cause for recleaning prior to painting.

### 3.9.1 Ferrous Surfaces

Ferrous surfaces that contain loose rust, loose mill scale, and other foreign substances shall be cleaned mechanically with hand tools according to SSPC SP 2, power tools according to SSPC SP 3 or by blast cleaning according to SSPC SP 6/NACE No.3. Shop-coated ferrous surfaces shall be protected from corrosion by treating and touching up corroded areas immediately upon detection.

### 3.9.2 Nonferrous Metallic Surfaces

Galvanized, aluminum and aluminum-alloy, lead, copper, and other nonferrous metal surfaces shall be solvent-cleaned in accordance with SSPC SP 1.

#### 3.9.2.1 Aluminum

Aluminum surfaces shall be treated in accordance with ASTM D173/D173M or ASTM D173/D173M. Steel wool, steel brushes and uninhibited caustic etching solutions, such as sodium hydroxide, shall not be used on aluminum.

#### 3.9.2.2 Zinc

Zinc surfaces including zinc-coated substrates, shall be cleaned prior to painting as follows: degrease, soak in a mild and inhibited alkaline cleaner, rinse with clean overflowing water, clean anodically in an acid (e.g. 0.25 to 0.75 percent sulfuric acid), and rinse with clean overflowing water.

#### 3.10 TIMING

Surfaces that have been cleaned, pretreated, and otherwise prepared for painting shall be given a coat of the specified first coat as soon as practical after such pretreatment has been completed, but prior to any deterioration of the prepared surface. Unless otherwise directed, the first coat primer shall be applied within 48 hours of surface preparation.

#### 3.11 SURFACES TO BE PREPARED FOR PAINTING

Surfaces shall be prepared as specified and as shown in the painting schedule in Section 09 90 00 PAINTS AND COATINGS.

#### 3.12 CLEANING

Place cloths, cotton waste and other debris, that might constitute a fire hazard, in closed metal containers for removal at the end of each day. Containers shall be removed from the site or destroyed in an approved manner. Preparation materials and other deposits on adjacent surfaces shall be removed and the entire job left clean and ready for painting.

-- End of Section --

SECTION 09 29 00

GYPSUM BOARD  
**08/24**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

|                   |                                                                                                                                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM B221         | (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes                                                                                         |
| ASTM C475/C475M   | (2017; R 2022) Standard Specification for Joint Compound and Joint Tape for Finishing Gypsum Board                                                                                                   |
| ASTM C514         | (2004; R 2020) Standard Specification for Nails for the Application of Gypsum Board                                                                                                                  |
| ASTM C557         | (2003; R 2017) Standard Specification for Adhesives for Fastening Gypsum Wallboard to Wood Framing                                                                                                   |
| ASTM C840         | (2023) Standard Specification for Application and Finishing of Gypsum Board                                                                                                                          |
| ASTM C954         | (2022) Standard Specification for Steel Drill Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Steel Studs from 0.033 in. (0.84 mm) to 0.112 in. (2.84 mm) in Thickness |
| ASTM C1002        | (2022) Standard Specification for Steel Self-Piercing Tapping Screws for the Application of Gypsum Panel Products or Metal Plaster Bases to Wood Studs or Steel Studs                                |
| ASTM C1047        | (2019) Standard Specification for Accessories for Gypsum Wallboard and Gypsum Veneer Base                                                                                                            |
| ASTM C1063        | (2023) Standard Specification for Installation of Lathing and Furring to Receive Interior and Exterior Portland Cement-Based Plaster                                                                 |
| ASTM C1396/C1396M | (2017) Standard Specification for Gypsum Board                                                                                                                                                       |
| ASTM D1784        | (2020) Standard Specification for Rigid                                                                                                                                                              |

Poly(Vinyl Chloride) (PVC) Compounds and  
Chlorinated Poly(Vinyl Chloride) (CPVC)  
Compounds

ASTM D3273

(2021) Standard Test Method for Resistance  
to Growth of Mold on the Surface of  
Interior Coatings in an Environmental  
Chamber

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350

(2017; Version 1.2) Standard Method for  
the Testing and Evaluation of Volatile  
Organic Chemical Emissions from Indoor  
Sources using Environmental Chambers

FM GLOBAL (FM)

FM APP GUIDE

(updated on-line) Approval Guide  
<https://www.approvalguide.com/>

GREEN SEAL (GS)

GS-36

(2013) Adhesives for Commercial Use

GYPSUM ASSOCIATION (GA)

GA 214

(2010) Recommended Levels of Gypsum Board  
Finish

GA 216

(2016) Application and Finishing of Gypsum  
Panel Products

GA 600

(2021) Fire Resistance and Sound Control  
Design Manual

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168

(2022) Adhesive and Sealant Applications

UNDERWRITERS LABORATORIES (UL)

UL 2818

(2022) GREENGUARD Certification Program  
For Chemical Emissions For Building  
Materials, Finishes And Furnishings

UL Fire Resistance

(2014) Fire Resistance Directory

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S"  
classification. Submittals not having a "G" or "S" classification are for  
Contractor Quality Control approval. Submit the following in accordance  
with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Control Joint Shop Drawings

SD-03 Product Data

Gypsum Board

Moisture- and Mold-Resistant Gypsum Backing Board

Trim

Accessories

VOC Content of Joint Compound

SD-07 Certificates

Asbestos Free Materials; G

Indoor Air Quality for Gypsum Board

Indoor Air Quality for Non-aerosol Adhesives

Indoor Air Quality for Aerosol Adhesives

SD-08 Manufacturer's Instructions

Safety Data Sheets

SD-10 Operation and Maintenance Data

Manufacturer Maintenance Instructions

1.3 CERTIFICATIONS

1.3.1 Indoor Air Quality Certifications

Submit required indoor air quality certifications in one submittal package.

1.3.1.1 Adhesives and Sealants

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party program that products meet the requirements of this Section. Provide adhesive products certified to meet ASTM C557. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.



#### 1.4 DELIVERY, STORAGE, AND HANDLING

##### 1.4.1 Delivery

Deliver materials in the original packages, containers, or bundles with each bearing the brand name, applicable standard designation, and name of manufacturer, or supplier.

##### 1.4.2 Storage

Keep materials dry by storing inside a sheltered building. Where necessary to store gypsum board, glass mat faced gypsum board, and cementitious backer units outside, store flat and off the ground, properly supported on a level platform, and protected from direct exposure to rain, snow, sunlight, and other extreme weather conditions. Provide adequate ventilation to prevent condensation. Remove plastic wrapping that may inhibit ventilation. Store per manufacturer's recommendations for allowable temperature and humidity range. Do not store panels where temperatures may exceed 125 deg F. Do not store gypsum wallboard with materials which have high emissions of volatile organic compounds (VOCs) or other contaminants. Do not store panels near materials that may offgas or emit harmful fumes, such as kerosene heaters, fresh paint, or adhesives. Do not use materials that have visible moisture or biological growth. Do not stack panels higher than manufacturer's recommendations, and no higher than 17 feet.

##### 1.4.3 Handling

Neatly stack gypsum board, glass mat faced gypsum board, and cementitious backer units flat to prevent sagging or damage to the edges, ends, and surfaces. Handle per manufacturer's recommendations.

#### 1.5 QUALIFICATIONS

Furnish type of gypsum board work specialized by the installer with a minimum of 3 years of documented successful experience.

#### 1.6 SCHEDULING

The gypsum wallboard must be taped, finished and primed before the installation of the highly-emitting materials.

Commence application only after the area scheduled for gypsum board work is completely weathertight. The heating, ventilating, and air-conditioning systems must be complete and in operation prior to application of the gypsum board. If the mechanical system cannot be activated before gypsum board is begun, the gypsum board work may proceed in accordance with an approved plan to maintain the environmental conditions specified below. Apply gypsum board prior to the installation of finish flooring and acoustic ceiling.

#### 1.7 ENVIRONMENTAL REQUIREMENTS

Do not expose the gypsum board to excessive sunlight prior to gypsum board application. Maintain a continuous uniform temperature of not less than 50 degrees F and not more than 80 degrees F for at least one week prior to the application of gypsum board work, while the gypsum board application is being done, and for at least one week after the gypsum board is set. Shield air supply and distribution devices to prevent any uneven flow of

air across the plastered surfaces. Provide ventilation to exhaust moist air to the outside during gypsum board application, set, and until gypsum board jointing is dry. In glazed areas, keep windows open top and bottom or side to side 3 to 4 inches. Reduce openings in cold weather to prevent freezing of joint compound when applied. For enclosed areas lacking natural ventilation, provide temporary mechanical means for ventilation. In unglazed areas subjected to hot, dry winds or temperature differentials from day to night of 20 degrees F or more, screen openings with cheesecloth or similar materials. Avoid rapid drying. During periods of low indoor humidity, provide minimum air circulation following gypsum boarding and until gypsum board jointing complete and is dry.

## 1.8 FIRE RESISTIVE CONSTRUCTION

Comply with specified fire-rated assemblies for design numbers indicated per UL Fire Resistance or FM APP GUIDE.

## PART 2 PRODUCTS

### 2.1 MATERIALS

Conform to specifications, standards and requirements specified. Provide gypsum board types, gypsum backing board types, cementitious backing units, and joint treating materials manufactured from asbestos free materials only. Submit Safety Data Sheets and manufacturer maintenance instructions for gypsum materials including adhesives.

#### 2.1.1 Gypsum Board

ASTM C1396/C1396M. Gypsum board must contain a minimum of 10 percent post-consumer recycled content, or a minimum of 20 percent post-industrial recycled content. Gypsum Board must not contain asbestos. Provide gypsum wall board and panels meeting the emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type). Provide certification or validation of indoor air quality for gypsum board.

##### 2.1.1.1 Regular

48 inch wide, 5/8 inch thick, tapered edges.

##### 2.1.1.2 Type X (Special Fire-Resistant)

48 inch wide, 5/8 inch thick, tapered edges.

##### 2.1.1.3 Type C (Special Fire-Resistant)

48 inch wide, 5/8 inch thick, tapered edges.

##### 2.1.1.4 Mold Resistant / Anti-Microbial Gypsum

ASTM D3273. 48 inch wide, 5/8 inch thick, tapered edges.

#### 2.1.2 Regular Moisture- and Mold-Resistant Gypsum Backing Board

ASTM C1396/C1396M

2.1.2.1 Regular

48 inch wide, 5/8 inch thick, tapered edges.

2.1.2.2 Type X (Special Fire-Resistant)

48 inch wide, 5/8 inch thick, tapered edges.

2.1.2.3 Type C (Special Fire-Resistant)

48 inch wide, 5/8 inch thick, tapered edges.

2.1.3 Joint Treatment Materials

ASTM C475/C475M. Product must be low emitting VOC types with VOC limits not exceeding 50 g/L. Joint materials must not contain asbestos. Provide data identifying VOC content of joint compound. Use all purpose joint and texturing compound containing inert fillers and natural binders, including lime compound. Pre-mixed compounds must be free of antifreeze, vinyl adhesives, preservatives, biocides and other slow releasing compounds.

2.1.3.1 Embedding Compound

Specifically formulated and manufactured for use in embedding tape at gypsum board joints and compatible with tape, substrate and fasteners.

2.1.3.2 Finishing or Topping Compound

Specifically formulated and manufactured for use as a finishing compound.

2.1.3.3 All-Purpose Compound

Specifically formulated and manufactured to serve as both a taping and a finishing compound and compatible with tape, substrate, and fasteners.

2.1.3.4 Setting or Hardening Type Compound

Specifically formulated and manufactured for use with fiber glass mesh tape.

2.1.3.5 Joint Tape

Use cross-laminated, tapered edge, reinforced paper, or fiber glass mesh tape recommended by the manufacturer.

2.1.4 Fasteners

2.1.4.1 Nails

ASTM C514.

2.1.4.2 Screws

ASTM C1002, Type "G", Type "S", or Type "W" steel drill screws for fastening gypsum board to gypsum board, wood framing members, and steel framing members less than 0.033 inch thick. ASTM C954 steel drill screws for fastening gypsum board to steel framing members 0.033 to 0.112 inch thick. Provide cementitious backer unit screws with a polymer coating.

#### 2.1.4.3 Staples

No. 16 USS gage flattened galvanized wire staples with 7/16 inch wide crown outside measurement and divergent point for base ply of two-ply gypsum board application. Use as follows:

| <u>Length of Legs</u> | <u>Thickness of Gypsum Board</u> |
|-----------------------|----------------------------------|
| 1-1/8 inches          | 1/2 inch                         |
| 1-1/4 inches          | 5/8 inch                         |

#### 2.1.5 Adhesives

Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168. Provide aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) meeting either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of GS-36. Provide certification or validation of indoor air quality for non-aerosol adhesives applied on the interior of the building (inside of the weatherproofing system). Provide certification or validation of indoor air quality for aerosol adhesives used on the interior of the building (inside of the weatherproofing system).

##### 2.1.5.1 Adhesive for Fastening Gypsum Board to Metal Framing

Type recommended by gypsum board manufacturer.

#### 2.1.6 Accessories

ASTM C1047. Fabricate from primed aluminum or plastic designed for intended use. Accessories manufactured with paper flanges are not acceptable. Flanges must be free of dirt, grease, and other materials that may adversely affect bond of joint treatment. Provide prefinished or job decorated materials.

##### 2.1.6.1 Trim

Provide trim for plaster including, as required:

- a. Cornerbeads.
- b. Bullnose Beads.
- c. LC Beads.
- d. L-Beads.
- e. U-Beads.
- f. Control Joints.
- g. Curved-edge Cornerbeads.

h. Base-of-wall Trim.

i. Trim required to complete the work.

#### 2.1.6.2 Materials

Aluminum: ASTM B221 6063-T5 alloy aluminum extrusions with a factory applied corrosion inhibiting primer and stainless steel fasteners.

Plastic: ASTM D1784 Extruded PVC with ASTM C1063 galvanized fasteners.

#### 2.1.7 Water

Provide clean, fresh, and potable water.

### PART 3 EXECUTION

#### 3.1 EXAMINATION

##### 3.1.1 Framing and Furring

Verify that framing and furring are securely attached and of sizes and spacing to provide a suitable substrate to receive gypsum board and cementitious backer units. Verify that all blocking, headers and supports are in place to support plumbing fixtures and to receive soap dishes, grab bars, towel racks, and similar items. Do not proceed with work until framing and furring are acceptable for application of gypsum board and cementitious backer units.

##### 3.1.2 Gypsum Board and Framing

Verify that surfaces of gypsum board and framing to be bonded with an adhesive are free of dust, dirt, grease, and any other foreign matter. Do not proceed with work until surfaces are acceptable for application of gypsum board with adhesive.

##### 3.1.3 Building Construction Materials

Do not install building construction materials that show visual evidence of biological growth.

#### 3.2 APPLICATION OF GYPSUM BOARD

Apply gypsum board to framing and furring members in accordance with ASTM C840 or GA 216 and the requirements specified. Apply gypsum board with separate panels in moderate contact; do not force in place. Stagger end joints of adjoining panels. Neatly fit abutting end and edge joints. Use gypsum board of maximum practical length; select panel sizes to minimize waste. Cut out gypsum board to make neat, close, and tight joints around openings. In vertical application of gypsum board, provide panels in lengths required to reach full height of vertical surfaces in one continuous piece. Lay out panels to minimize waste; reuse cutoffs whenever feasible. Surfaces of gypsum board and substrate members may be bonded together with an adhesive, except where prohibited by fire rating(s). Treat edges of cutouts for plumbing pipes, screwheads, and joints with moisture-resistant compound as recommended by the gypsum board manufacturer. Minimize framing by floating corners with single studs and drywall clips. Install gypsum 1/2 inch ceiling board over framing at 24

inch on center. Provide type of gypsum board for use in each system specified herein as indicated.

### 3.2.1 Solid Gypsum Board Partitions

Provide in accordance with ASTM C840, System V or GA 216.

### 3.2.2 Application of Gypsum Board to Steel Framing and Furring

Apply in accordance with ASTM C840, System VIII or GA 216.

### 3.2.3 Control Joints

Install expansion and contraction joints in ceilings and walls in accordance with ASTM C840, System XIII or GA 216. Joints in fire-rated and sound-rated ceiling and walls to maintain the fire and sound rating. Fill control joints between studs in fire-rated construction with firestopping insulation to match the fire-rating of construction.

#### 3.2.3.1 Control Joint Shop Drawings

- a. Submit control joint shop drawings for approval.
- b. Install control joints where a partition, wall, or ceiling traverses a construction joint in the base building structure.
- c. Install control joints where a wall or partition runs in an uninterrupted straight plane exceeding 30 feet.
- d. Install control joints in interior ceilings with a perimeter relief so that linear dimensions between control joints do not exceed 50 feet.
- e. Install control joints in interior ceilings without perimeter relief so that linear dimensions between control joints do not exceed 30 feet.
- f. Install control joints in exterior ceilings and soffits so that linear dimensions between control joints do not exceed 30 feet.
- g. Install control joints or intermediate blocking where ceiling framing members change direction.
- h. In addition to the above, install control joints where indicated on the drawings as a design accent or architectural feature.
- i. Do not install control joints where building or structural expansion or seismic joints are located. Install suitable joint expansion or seismic joint assemblies as indicated on the drawings.

### 3.2.4 Application of Abuse Resistant Gypsum Board

Apply in accordance with applicable system of ASTM C840 as specified or GA 216. Follow manufacturers written instructions on how to cut, drill, and attach board.

## 3.3 FINISHING OF GYPSUM BOARD

Tape and finish gypsum board in accordance with ASTM C840, GA 214, and GA 216. Finish plenum areas above ceilings to Level 1 in accordance with GA 214. Finish and paint walls required to be sound rated assembly field

tested from true floor to true ceiling to Level 5 in accordance with GA 214. Finish moisture-resistant gypsum backing board, ASTM C1396/C1396M, to receive ceramic tile to Level 2 in accordance with GA 214. Finish walls and ceilings to receive a heavy-grade wall covering or heave textured finish before painting to Level 3 in accordance with GA 214. Finish walls and ceilings without critical lighting to receive flat paints, light textures, or wall coverings to Level 4 in accordance with GA 214. Unless otherwise specified, finish all gypsum board walls, partitions, and ceilings to Level 5 in accordance with GA 214. Provide joint, fastener depression, and corner treatment. Tool joints as smoothly as possible to minimize sanding and dust. Do not use self-adhering fiber glass mesh tape with conventional drying type joint compounds; use setting or hardening type compounds only. Provide treatment for moisture-resistant gypsum board as recommended by the gypsum board manufacturer. Protect workers, building occupants, and HVAC systems from gypsum dust.

#### 3.3.1 Uniform Surface

Wherever gypsum board is to receive eggshell, semigloss, or gloss paint finish, or where severe, up or down lighting conditions occur, finish gypsum wall surface in accordance to GA 214 Level 5. In accordance with GA 214 Level 5, apply a thin skim coat of joint compound to the entire gypsum board surface, after the two-coat joint and fastener treatment is complete and dry.

#### 3.4 SEALING

Seal openings around pipes, fixtures, and other items projecting through gypsum board and cementitious backer units as specified in Section 07 92 00 JOINT SEALANTS. Apply material with exposed surface flush with gypsum board or cementitious backer units.

#### 3.5 FIRE-RESISTANT ASSEMBLIES

Wherever fire-rated construction is indicated, provide materials and application methods, including types and spacing of fasteners, wall and ceiling framing in accordance with the specifications contained in UL Fire Resistance for the Design Number(s) indicated or GA 600 for the File Number(s) indicated. Joints of fire-rated gypsum board enclosures must be closed and sealed in accordance with UL test requirements or GA requirements. Seal penetrations through rated partitions and ceilings tight in accordance with tested systems.

#### 3.6 PATCHING

Patch surface defects in gypsum board to a smooth, uniform appearance, ready to receive finishes.

#### 3.7 SHAFTWALL FRAMING

Install the shaftwall system in accordance with the system manufacturer's published instructions. Coordinate bucks, anchors, blocking and other items placed in or behind shaftwall framing with electrical and mechanical work. Patch or replace fireproofing materials which are damaged or removed during shaftwall construction.

-- End of Section --

SECTION 09 77 23

PLASTIC WALL PANELS  
**01/25**

PART 1 GENERAL

1.1 SUMMARY

Furnish and install the following:

- a. Decorative wall panels consisting of corrugated polypropylene substrate with smooth finish.
  - (1) Mounting hardware, adhesives, furring, accessories and trims.
  - (2) Silicone sealant for all joints between panels and moldings, and between panel system and abutting materials.

1.2 RELATED REQUIREMENTS

- a. 06 10 00 ROUGH CARPENTRY

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Product Specifications; G

Physical Properties; G

Product Data Sheets; G

SD-04 Samples

Verification Samples; G

SD-05 Design Data

Performance Data At Minimum, 25 Percent, 50 Percent, 75 Percent,  
And 100 Percent Load; G

SD-07 Certificates

Certification; G

Manufacturer's written certification stating that panel system and all related components to be furnished hereunder, meet or exceed the requirements specified under this Section that all fire-resistive requirements for the indicated requirements have been met.

SD-08 Manufacturer's Instructions



Installation; G

SD-10 Operation and Maintenance Data

Maintenance and Cleaning Instructions; G

#### 1.4 EXTRA MATERIALS

Deliver extra materials to Owner. Furnish extra materials described below matching products installed, packaged with protective covering for storage and identified with appropriate labels:

Wall Panels: Furnish quantity of full-size units equal to 3 percent of the amount installed.

#### 1.5 QUALITY ASSURANCE

Installer qualifications: Installer shall have a minimum of 5 years experience with composite wall panel work similar in scope and size to this project.

#### 1.6 DELIVERY, STORAGE, AND HANDLING

- a. Delivery: Deliver materials to site in manufacturer's original, unopened containers and packaging, with labels clearly identifying product name and manufacturer
- b. Handling: Protect materials during handling and installation to prevent damage.
- c. Storage:
  - (1) Store materials in clean, dry area indoors.
  - (2) Store wall panels flat.

#### 1.7 ENVIRONMENTAL CONDITIONS

##### 1.7.1 Installing Wall Panels

- a. Cold Temperatures: When installing wall panels in temperatures below 40 degrees F, warm to a minimum of 60 degrees F overnight and leave space between panels to allow for expansion in accordance with manufacturer's instructions.
- b. Warm Temperatures: When installing wall panels in temperatures above 70 degrees F, warm panels to a minimum of 60 degrees F in accordance with manufacturer's instructions.

##### 1.7.2 Cutting Panels

Before field cutting wall panels in temperatures below 40 degrees F, warm panels to a minimum of 60 degrees F overnight.

#### 1.8 WARRANTY

- a. General: Submit warranties under provisions of 01 78 00.00 22 CLOSEOUT SUBMITTALS (PWD ME).

b. Manufacturer Warranty:

- (1) In addition to the specific guaranteed requirements of the GENERAL CONDITIONS and SUPPLEMENTAL GENERAL CONDITIONS, the Contractor shall obtain in the Owner's name the standard written manufacturer's guarantee of all materials furnished under this Section where such guarantees are offered in the manufacturer's published product data. All these guarantees shall be in addition to, and not in lieu of, other liabilities which the Contractor may have by law or other provisions of the Contract Documents.

c. Warranty Period for Wall Panels: 2 years

PART 2 PRODUCTS

2.1 MANUFACTURERS

- a. Specified Manufacturer for decorative panels: To establish a standard of quality, design and function desired, specifications have been based on Nudo, Springfield, IL, product: "UtiLite Wall Panel".
- b. Acceptable Manufacturers: Subject to compliance with the requirements specified herein, manufacturers offering products which may be incorporated in the work include the following, or approved equal:

- (1) Nudo, Springfield, IL.
- (2) Extrutech Plastics, Inc., Manitowoc, WI.
- (3) Parkland Plastics, Middlebury, IN.

c. Materials:

- (1) Panels: Tongue and groove, rib reinforced, non-porous, waterproof, 100 percent virgin polyvinyl chloride panels with a smooth finish.
- (2) Outside surface: Flat.
- (3) Thickness: 0.040 inch.
- (4) Width: 16 inches.
- (5) USDA and CFIA acceptance.
- (6) Color: White, glossy finish.
- (7) Size: As indicated on the Drawings.
- (8) Surface Burning Characteristics, Class A per ASTM E84.
- (9) Trim: 100 percent virgin polyvinyl chloride in color to match panels.

2.2 ACCESSORIES

- a. Blocking: Provide solid 5/4 inch thick PVC blocking for wall-mounted equipment.

- b. 1/2 inch PVC furring.
- c. Sealant: One part acetoxysilicone rubber sealant, USDA approved as recommended by panel manufacturer.
- d. Fasteners as recommended manufacturer.

### PART 3 EXECUTION

#### 3.1 EXAMINATION

- a. Notify the Engineer of conditions that would adversely affect installation or subsequent use.
- b. Do not begin preparation or installation until unacceptable conditions are corrected.

#### 3.2 PREPARATION

Ensure walls are dry and clean.

#### 3.3 INSTALLATION

- a. Prior to installing panels, remove packaging and allow panels to acclimate to room temperature and humidity for at least 48 hours.
  - (1) Wall substrate must be dry and free from dirt, dust, grease and other contaminants. Walls must be flat and even. Remove high spots and fill low spots with material acceptable to panel manufacturer.
  - (2) Painted walls must be prepared to adhesive manufacturer specifications for proper adhesion.
- b. General:
  - (1) Inspect panels for any defects. Do not install panels of unacceptable quality.
  - (2) Field cutting of all wall systems should be accomplished using a circular saw with fine tooth carbide blade.
    - (a) Position panel so that the saw blade enters the finished HPL side first to avoid chipping or damage. Protect decorative laminate face of panel by covering work area, do not remove protective will until after installation.
- c. For installation to studs or furring straps, use mechanical fasteners spaced 24 inches on center. Furring straps shall be no less than 18 gauge, 3-1/2 inches wide.
  - (1) Install panels using the following method.
    - (a) Aluminum moldings.
- d. Installation Using Aluminum Moldings:
  - (1) Start in the corner. Mark plumb line 48 1/8 inches from corner.

- (2) Apply adhesive directly to entire back of composite wall panel using correct trowel with 100% adhesive coverage using crosshatch pattern. Apply adhesive to within 1/2 inch of all edges of panel.
- (3) Slide panel into molding and withdraw 1/16 inch for aluminum moldings to provide appropriate gap. Align with plumb line.
- (4) Begin in top corner nearest molding with laminate roller, rolling down and out toward the edge without molding.
- (5) Continue rolling down and out working across panel away from previously installed panel or initial molding. Remove all trapped air.
- (6) Install one-piece division bar and caps or next molding by sliding onto panel.
- (7) Repeat process, working in one direction around room.
- (8) Immediately remove all adhesive residue. To remove, clean with nonabrasive cotton cloth and warm water. If necessary, use a mild nonabrasive detergent. For cleanup with solvent based adhesives, use mineral spirits or acetone to remove residue.

### 3.4 CLEANING

- a. Clean panel surfaces in compliance with manufacturer's recommendations.
  - (1) Use a clean, damp, nonabrasive cotton cloth and a mild liquid detergent or household cleaner.
  - (2) Rinse with clean water using a clean, nonabrasive cotton cloth.
  - (3) Dry panels with a soft, clean nonabrasive cotton cloth.
  - (4) Do not use cleaners containing acid, alkali or sodium hypochlorite.
- b. Daily clean work areas by sweeping and disposing of debris and scraps.
- c. Completely clean all panel surfaces, clean all surfaces of adjacent surfaces which have been marked or soiled by the work of this Section, removing all excess sealants and adhesives with solvents which will not damage the surfaces in any way.
- d. Upon completion of the work of this Section, remove tools, equipment and all rubbish and debris from the work area; leave area in broom-clean condition.

### 3.5 PROTECTION

During the operation of sealant work, protect the work of other trades against undue soilage and damage by the exercise of reasonable care and precautions. Repair or replace any work so damaged and soiled.

-- End of Section --

SECTION 09 90 00

PAINTS AND COATINGS  
02/21

PART 1 GENERAL

1.1 RELATED REQUIREMENTS

1.1.1 Painting Included

Where a space or surface is indicated to be painted, include the following unless indicated otherwise.

- a. Surfaces behind portable objects and surface mounted articles readily detachable by removal of fasteners, such as screws and bolts.
- b. New factory finished surfaces that require identification or color coding and factory finished surfaces that are damaged during performance of the work.
- c. Existing coated surfaces that are damaged during performance of the work.

1.1.1.1 Interior Painting

Includes new surfaces, existing uncoated surfaces, and existing coated surfaces of the building and appurtenances as indicated and existing coated surfaces made bare by cleaning operations. Where a space or surface is indicated to be painted, include the following items, unless indicated otherwise.

- a. Exposed columns, girders, beams, joists, and metal deck; and
- b. Other contiguous surfaces.

1.1.2 Painting Excluded

Do not paint the following unless indicated otherwise.

- a. Surfaces concealed and made inaccessible by panelboards, fixed ductwork, machinery, and equipment fixed in place.
- b. Surfaces in concealed spaces. Concealed spaces are defined as enclosed spaces above suspended ceilings, furred spaces, attic spaces, crawl spaces, elevator shafts and chases.
- c. Steel to be embedded in concrete.
- d. Copper, stainless steel, aluminum, anodized aluminum, brass, and lead except existing coated surfaces.
- e. Hardware, fittings, and other factory finished items.

1.1.3 Mechanical and Electrical Painting

Includes field coating of interior new and existing surfaces.

- a. Where a space or surface is indicated to be painted, include the following items unless indicated otherwise.
  - (1) Exposed piping, conduit, and ductwork;
  - (2) Supports, hangers, air grilles, and registers;
  - (3) Miscellaneous metalwork and insulation coverings.
- b. Do not paint the following, unless indicated otherwise:
  - (1) New zinc-coated, aluminum, and copper surfaces under insulation
  - (2) New aluminum jacket on piping
  - (3) New interior ferrous piping under insulation.

#### 1.1.3.1 Fire Extinguishing Sprinkler Systems

Clean, pretreat, prime, and paint new fire extinguishing sprinkler systems including valves, piping, conduit, hangers, supports, miscellaneous metalwork, and accessories. Apply coatings to clean, dry surfaces, using clean brushes.

#### 1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

##### AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS (ACGIH)

|            |                                                                                                |
|------------|------------------------------------------------------------------------------------------------|
| ACGIH 0100 | (2017; Suppl 2020) Documentation of the Threshold Limit Values and Biological Exposure Indices |
|------------|------------------------------------------------------------------------------------------------|

##### ASTM INTERNATIONAL (ASTM)

|            |                                                                                                                 |
|------------|-----------------------------------------------------------------------------------------------------------------|
| ASTM D235  | (2022) Standard Specification for Mineral Spirits (Petroleum Spirits) (Hydrocarbon Dry Cleaning Solvent)        |
| ASTM D523  | (2014; R 2018) Standard Test Method for Specular Gloss                                                          |
| ASTM D4214 | (2007; R 2015) Standard Test Method for Evaluating the Degree of Chalking of Exterior Paint Films               |
| ASTM D4263 | (1983; R 2018) Standard Test Method for Indicating Moisture in Concrete by the Plastic Sheet Method             |
| ASTM D4444 | (2013; R 2018) Standard Test Method for Laboratory Standardization and Calibration of Hand-Held Moisture Meters |
| ASTM D6386 | (2022) Standard Practice for Preparation of Zinc (Hot-Dip Galvanized) Coated Iron                               |

and Steel Product and Hardware Surfaces  
for Painting

ASTM F1869

(2023) Standard Test Method for Measuring  
Moisture Vapor Emission Rate of Concrete  
Subfloor Using Anhydrous Calcium Chloride

CENTERS FOR DISEASE CONTROL AND PREVENTION (CDC)

Intelligence Bulletin 65

(2013) Occupational Exposure to Carbon  
Nanotubes and Nanofibers

MASTER PAINTERS INSTITUTE (MPI)

|        |                                                                       |
|--------|-----------------------------------------------------------------------|
| MPI 2  | (2012) Aluminum Heat Resistant Enamel (up to 427 C and 800 F          |
| MPI 3  | (2016) Primer, Alkali Resistant, Water Based                          |
| MPI 4  | (2016) Interior/Exterior Latex Block Filler                           |
| MPI 9  | (2016) Alkyd, Exterior Gloss (MPI Gloss Level 6)                      |
| MPI 17 | (2016) Primer, Bonding, Water Based                                   |
| MPI 19 | (2012) Primer, Zinc Rich, Inorganic                                   |
| MPI 21 | (2012) Heat Resistant Coating, (Up to 205°C/402°F), MPI Gloss Level 6 |
| MPI 22 | (2012) Aluminum Paint, High Heat (up to 590° C/1100° F)               |
| MPI 23 | (2015) Primer, Metal, Surface Tolerant                                |
| MPI 42 | (2012) Textured Coating, Latex, Flat                                  |
| MPI 44 | (2016) Latex, Interior, (MPI Gloss Level 2)                           |
| MPI 47 | (2016) Alkyd, Interior, Semi-Gloss (MPI Gloss Level 5)                |
| MPI 48 | (2016) Alkyd, Interior, Gloss (MPI Gloss Level 6-7)                   |
| MPI 49 | (2015) Alkyd, Interior, Flat (MPI Gloss Level 1)                      |
| MPI 50 | (2015) Primer Sealer, Latex, Interior                                 |
| MPI 51 | (2016) Alkyd, Interior, (MPI Gloss Level 3)2                          |
| MPI 52 | (2016) Latex, Interior, (MPI Gloss Level 3)                           |
| MPI 54 | (2016) Latex, Interior, Semi-Gloss (MPI Gloss Level 5)                |

|              |                                                                                        |
|--------------|----------------------------------------------------------------------------------------|
| MPI 72       | (2016) Polyurethane, Two-Component, Pigmented, Gloss (MPI Gloss Level 6-7)             |
| MPI 76       | (2016) Primer, Alkyd, Quick Dry, for Metal                                             |
| MPI 77       | (2015) Epoxy, Gloss                                                                    |
| MPI 95       | (2015) Primer, Quick Dry, for Aluminum                                                 |
| MPI 101      | (2016) Primer, Epoxy, Anti-Corrosive, for Metal                                        |
| MPI 107      | (2016) Primer, Rust-Inhibitive, Water Based                                            |
| MPI 116      | (2012) Block Filler, Epoxy                                                             |
| MPI 138      | (2016) Latex, Interior, High Performance Architectural, (MPI Gloss Level 2)            |
| MPI 139      | (2016) Latex, Interior, High Performance Architectural, (MPI Gloss Level 3)            |
| MPI 140      | (2016) Latex, Interior, High Performance Architectural, (MPI Gloss Level 4)            |
| MPI 141      | (2016) Latex, Interior, High Performance Architectural, Semi-Gloss (MPI Gloss Level 5) |
| MPI 144      | (2016) Latex, Interior, Institutional Low Odor/VOC, (MPI Gloss Level 2)                |
| MPI 145      | (2016) Latex, Interior, Institutional Low Odor/VOC, ( MPI Gloss Level 3)               |
| MPI 146      | (2016) Latex, Interior, Institutional Low Odor/VOC, (MPI Gloss Level 4)                |
| MPI 147      | (May 2016) Latex, Interior, Institutional Low Odor/VOC, Semi-Gloss (MPI Gloss Level 5) |
| MPI 149      | (2016) Primer Sealer, Interior, Institutional Low Odor/VOC                             |
| MPI 151      | (2016) Light Industrial Coating, Interior, Water Based (MPI Gloss Level 3)             |
| MPI 153      | (2016) Light Industrial Coating, Interior, Water Based, Semi-Gloss (MPI Gloss Level 5) |
| MPI 154      | (2016) Light Industrial Coating, Interior, Water Based, Gloss (MPI Gloss Level 6)      |
| MPI ASM      | (2019) Architectural Painting Specification Manual                                     |
| MPI GPS-1-14 | (2014) Green Performance Standard GPS-1-14                                             |



|                                        |                                                                                                                                                                                             |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MPI GPS-2-14                           | (2014) Green Performance Standard GPS-2-14                                                                                                                                                  |
| MPI MRM                                | (2015) Maintenance Repainting Manual                                                                                                                                                        |
| SOCIETY FOR PROTECTIVE COATINGS (SSPC) |                                                                                                                                                                                             |
| SSPC 7/NACE No.4                       | (2007) Brush-Off Blast Cleaning                                                                                                                                                             |
| SSPC Glossary                          | (2011) SSPC Protective Coatings Glossary                                                                                                                                                    |
| SSPC Guide 6                           | (2021) Guide for Containing Surface Preparation Debris Generated During Paint Removal Operations                                                                                            |
| SSPC Guide 7                           | (2015) Guide to the Disposal of Lead-Contaminated Surface Preparation Debris                                                                                                                |
| SSPC PA 1                              | (2024) Shop, Field, and Maintenance Coating of Metals                                                                                                                                       |
| SSPC QP 1                              | (2019) Standard Procedure for Evaluating the Qualifications of Industrial/Marine Painting Contractors (Field Application to Complex Industrial Steel Structures and Other Metal Components) |
| SSPC SP 1                              | (2015) Solvent Cleaning                                                                                                                                                                     |
| SSPC SP 2                              | (2018) Hand Tool Cleaning                                                                                                                                                                   |
| SSPC SP 3                              | (2018) Power Tool Cleaning                                                                                                                                                                  |
| SSPC SP 6/NACE No.3                    | (2007) Commercial Blast Cleaning                                                                                                                                                            |
| SSPC SP 10/NACE No. 2                  | (2015) Near-White Blast Cleaning                                                                                                                                                            |
| SSPC VIS 1                             | (2002; E 2004) Guide and Reference Photographs for Steel Surfaces Prepared by Dry Abrasive Blast Cleaning                                                                                   |
| SSPC VIS 3                             | (2004) Guide and Reference Photographs for Steel Surfaces Prepared by Hand and Power Tool Cleaning                                                                                          |
| SSPC VIS 4/NACE VIS 7                  | (1998; E 2000; E 2004) Guide and Reference Photographs for Steel Surfaces Prepared by Waterjetting                                                                                          |
| SSPC-SP WJ-1/NACE WJ-1                 | (2012) Clean to Bare Substrate, Waterjet Cleaning of Metals                                                                                                                                 |
| SSPC-SP WJ-2/NACE WJ-2                 | (2012) Very Thorough Cleaning, Waterjet Cleaning of Metals                                                                                                                                  |
| SSPC-SP WJ-3/NACE WJ-3                 | (2012) Thorough Cleaning, Waterjet Cleaning of Metals                                                                                                                                       |

SSPC-SP WJ-4/NACE WJ-4 (2012) Light Cleaning, Waterjet Cleaning  
of Metals

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE AMS-STD-595A (2017) Colors used in Government  
Procurement

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety -- Safety and Occupational  
Health (SOH) Requirements

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-STD-101 (2014; Rev C) Color Code for Pipelines and  
for Compressed Gas Cylinders

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA Method 24 (2000) Determination of Volatile Matter  
Content, Water Content, Density, Volume  
Solids, and Weight Solids of Surface  
Coatings

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FED-STD-313 (2018) Material Safety Data,  
Transportation Data and Disposal Data for  
Hazardous Materials Furnished to  
Government Activities

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.1000 Air Contaminants

29 CFR 1910.1001 Asbestos

29 CFR 1910.1025 Lead

29 CFR 1926.62 Lead

### 1.3 DEFINITIONS

#### 1.3.1 Qualification Testing

Qualification testing is the performance of all test requirements listed in the product specification. This testing is accomplished by MPI to qualify each product for the MPI Approved Product List, and may also be accomplished by Contractor's third-party testing lab if an alternative to Batch Quality Conformance Testing by MPI is desired.

#### 1.3.2 Batch Quality Conformance Testing

Batch quality conformance testing determines that the product provided is the same as the product qualified to the appropriate product specification. This testing must be accomplished by an MPI testing lab.

### 1.3.3 Coating

SSPC Glossary; (1) A liquid, liquefiable, or mastic composition that is converted to a solid protective, decorative, or functional adherent film after application as a thin layer; (2) Generic term for paint, lacquer, enamel.

### 1.3.4 DFT or dft

Dry film thickness, the film thickness of the fully cured, dry paint or coating.

### 1.3.5 DSD

Degree of Surface Degradation, the MPI system of defining degree of surface degradation. Five levels are generically defined under the Assessment sections in the MPI MRM, MPI Maintenance Repainting Manual.

### 1.3.6 INT

MPI short term designation for an interior coating system.

### 1.3.7 Loose Paint

Paint or coating that can be removed with a dull putty knife.

### 1.3.8 mil / mils

The English measurement for 0.001 in or one one-thousandth of an inch.

### 1.3.9 MPI Gloss Levels

MPI system of defining gloss. Seven gloss levels (G1 to G7) are generically defined under the Evaluation sections of the MPI Manuals. Traditionally, Flat refers to G1/G2, Eggshell refers to G3, Semigloss refers to G5, and Gloss refers to G6.

Gloss levels are defined by MPI as follows:

| Gloss Level | Description   | Units at 60 degree angle | Units at 80 degree angle |
|-------------|---------------|--------------------------|--------------------------|
| G1          | Matte or Flat | 0 to 5                   | 10 max                   |
| G2          | Velvet        | 0 to 10                  | 10 to 35                 |
| G3          | Eggshell      | 10 to 25                 | 10 to 35                 |
| G4          | Satin         | 20 to 35                 | 35 min                   |
| G5          | Semi-Gloss    | 35 to 70                 |                          |
| G6          | Gloss         | 70 to 85                 |                          |
| G7          | High Gloss    |                          |                          |

Gloss is tested in accordance with ASTM D523. Historically, the Government has used Flat (G1 / G2), Eggshell (G3), Semi-Gloss (G5), and Gloss (G6).

#### 1.3.10 MPI System Number

The MPI coating system number in each MPI Division found in either the MPI Architectural Painting Specification Manual or the Maintenance Repainting Manual and defined as an exterior (EXT/REX) or interior system (INT/RIN).

#### 1.3.11 Paint

SSPC Glossary; (1) Any pigmented liquid, liquefiable, or mastic composition designed for application to a substrate in a thin layer that is converted to an opaque solid film after application. Used for protection, decoration, identification, or to serve some other functional purposes; (2) Application of a coating material.

#### 1.3.12 RIN

MPI short term designation for an interior coating system used in repainting projects or over existing coating systems.

#### 1.4 SCHEDULING

Allow paint, polyurethane, varnish, and wood stain installations to cure prior to the installation of materials that adsorb VOCs, including unprimed gypsum wallboard, .

#### 1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

Samples of specified materials may be taken and tested for compliance with specification requirements.

##### SD-02 Shop Drawings

Piping Identification

##### SD-03 Product Data

Coating; G

Product Data Sheets

##### SD-04 Samples

Color; G

Textured Wall Coating System; G

##### SD-07 Certificates

Qualification Testing laboratory for coatings; G

Indoor Air Quality for Paints and Primers

##### SD-08 Manufacturer's Instructions

Application Instructions

Mixing

Manufacturer's Safety Data Sheets

SD-10 Operation and Maintenance Data

Coatings, Data Package 1; G

## 1.6 QUALITY ASSURANCE

### 1.6.1 Regulatory Requirements

#### 1.6.1.1 Environmental Protection

In addition to requirements specified elsewhere for environmental protection, provide coating materials that conform to the restrictions of the local Air Pollution Control District and regional jurisdiction. Notify Contracting Officer of any paint specified herein which fails to conform.

#### 1.6.1.2 Lead Content

Do not use coatings having a lead content over 0.06 percent by weight of nonvolatile content.

#### 1.6.1.3 Chromate Content

Do not use coatings containing zinc-chromate or strontium-chromate.

#### 1.6.1.4 Asbestos Content

Provide asbestos-free materials.

#### 1.6.1.5 Mercury Content

Provide materials free of mercury or mercury compounds.

#### 1.6.1.6 Silica

Provide abrasive blast media containing no free crystalline silica.

#### 1.6.1.7 Human Carcinogens

Provide materials that do not contain ACGIH 0100 confirmed human carcinogens (A1) or suspected human carcinogens (A2).

#### 1.6.1.8 Carbon Based Fibers / Tubes

Materials must not contain carbon based fibers such as carbon nanotubes or carbon nanofibers. Intelligence Bulletin 65 ranks toxicity of carbon nanotubes on a par with asbestos.

### 1.6.2 Coating Contractor's Qualification

Submit the name, address, telephone number, and e-mail address of the

Contractor that will be performing all surface preparation and coating application. Submit evidence that key personnel have successfully performed surface preparation and application of coatings on NAVFAC Projects on a minimum of three similar projects within the past three years. List information by individual and include the following:

a. Name of individual and proposed position for this work.

b. Information about each previous assignment including:

Position or responsibility

Employer (if other than the Contractor)

Name of facility owner

Mailing address and telephone number of facility owner

Name of individual in facility owner's organization who can be contacted as a reference

Location, size and description of structure

Dates work was carried out

Description of work carried out on structure

#### 1.6.3 SSPC QP 1 Certification

Contractors that perform surface preparation or coating application on steel substrates must be certified by the Society for Protective Coatings (formerly Steel Structures Painting Council) (SSPC) to the requirements of SSPC QP 1 prior to Contract award, and must remain certified while accomplishing any surface preparation or coating application. If a Contractor's certification expires, the firm will not be allowed to perform any work until the certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered. Notify the Contracting Officer of any change in Contractor certification status. Notify the Contracting Officer of all scheduled and unannounced on-site audits from SSPC and furnish a copy of all audit reports.

#### 1.6.4 Approved Products List

The current MPI, "Approved Product List" which lists paint by brand, label, product name and product code as of the date of Contract award, will be used to determine compliance with the submittal requirements of this specification. The Contractor may choose to use a subsequent MPI "Approved Product List", however, only one list may be used for the entire Contract and each coating system is to be from a single manufacturer. Provide all coats on a particular substrate from a single manufacturer. No variation from the MPI Approved Products List is acceptable.

#### 1.6.5 Paints and Coatings Indoor Air Quality Certifications

Provide paint and coating products certified to meet indoor air quality requirements by MPI GPS-1-14, MPI GPS-2-14 or provide certification by other third-party programs. Provide current product certification documentation from certification body.

Provide certification of Indoor Air Quality for Paints and Primers.  
Submit required indoor air quality certifications in one submittal package.

#### 1.6.6 Field Samples and Tests

The Contracting Officer may choose up to two coatings that have been delivered to the site to be tested at no cost to the Government. Take samples of each chosen product as specified in the paragraph SAMPLING PROCEDURE. Test each chosen product as specified in the paragraph TESTING PROCEDURE. Remove products from the job site which do not conform, and replace with new products that conform to the referenced specification. Test replacement products that failed initial testing as specified in the paragraph TESTING PROCEDURE at no cost to the Government.

##### 1.6.6.1 Sampling Procedure

Select paint at random from the products that have been delivered to the job site for sample testing. The Contractor must provide one quart samples of the selected paint materials. Take samples in the presence of the Contracting Officer, and label, and identify each sample. Provide labels in accordance with the paragraph PACKAGING, LABELING, AND STORAGE.

##### 1.6.6.2 Testing Procedure

Provide Batch Quality Conformance Testing for specified products, as defined by and performed by MPI. As an alternative to Batch Quality Conformance Testing, the Contractor may provide Qualification Testing for specified products above to the appropriate MPI product specification, using the third-party laboratory approved under the paragraph QUALIFICATION TESTING laboratory for coatings. Include the backup data and summary of the test results within the qualification testing lab report. Provide a summary listing of all the reference specification requirements and the result of each test. Clearly indicate in the summary whether the tested paint meets each test requirement. Note that Qualification Testing may take 4 to 6 weeks to perform, due to the extent of testing required.

Submit name, address, telephone number, FAX number, and e-mail address of the independent third party laboratory selected to perform testing of coating samples for compliance with specification requirements. Submit documentation that laboratory is regularly engaged in testing of paint samples for conformance with specifications, and that employees performing testing are qualified. If MPI is chosen to perform the Batch Quality Conformance testing, the above submittal information is not required, only a letter is required from the Contractor stating that MPI will perform the testing.

#### 1.6.7 Textured Wall Coating System

Three complete samples of each indicated type, pattern, and color of textured wall coating system applied to a panel of the same material as that on which the coating system will be applied in the work. Provide samples of wall coating systems minimum 5 by 7 inches and of sufficient size to show pattern repeat and texture.

#### 1.7 PACKAGING, LABELING, AND STORAGE

Provide paints in sealed containers that legibly show the Contract

specification number, designation name, formula or specification number, batch number, color, quantity, date of manufacture, manufacturer's formulation number, manufacturer's directions including any warnings and special precautions, and name and address of manufacturer. Furnish pigmented paints in containers not larger than 5 gallons. Store paints and thinners in accordance with the manufacturer's written directions, and as a minimum, stored off the ground, under cover, with sufficient ventilation to prevent the buildup of flammable vapors, and at temperatures between 40 to 95 degrees F. Do not store paint, polyurethane, varnish, or wood stain products with materials that have a high capacity to absorb VOC emissions. Do not store paint, polyurethane, varnish, or wood stain products in occupied spaces.

## 1.8 SAFETY AND HEALTH

Comply with applicable Federal, State, and local laws and regulations, and with the ACCIDENT PREVENTION PLAN, including the Activity Hazard Analysis as specified in Section 01 35 26.00 20 GOVERNMENTAL SAFETY REQUIREMENTS and in Appendix A of EM 385-1-1. Include in the Activity Hazard Analysis the potential impact of painting operations on painting personnel and on others involved in and adjacent to the work zone.

### 1.8.1 Toxic Materials

To protect personnel from overexposure to toxic materials, conform to the most stringent guidance of:

- a. The applicable manufacturer's Safety Data Sheets (SDS) or local regulation.
- b. 29 CFR 1910.1000.
- c. ACGIH 0100, threshold limit values.
- d. The appropriate OSHA standard in 29 CFR 1910.1025 and 29 CFR 1926.62 for surface preparation on painted surfaces containing lead. Removal and disposal of coatings which contain lead is specified in Section 02 83 00.00 22 MANAGEMENT OF LEAD, CADMIUM, AND CHROMIUM (VI) DURING RENOVATION, DEMOLITION, CONSTRUCTION, REMOVAL AND ABATEMENT. Additional guidance is given in SSPC Guide 6 and SSPC Guide 7. Refer to drawings for list of hazardous materials located on this project. Coordinate paint preparation activities with this specification section.
- e. The appropriate OSHA standards in 29 CFR 1910.1001 for surface preparation of painted surfaces containing asbestos. Removal and disposal of coatings which contain asbestos materials is specified in Section 02 82 00.00 22 ASBESTOS REMEDIATION. Refer to drawings for list of hazardous materials located on this project. Coordinate paint preparation activities with this specification section.

Submit manufacturer's Safety Data Sheets for coatings, solvents, and other potentially hazardous materials, as defined in FED-STD-313.

## 1.9 ENVIRONMENTAL REQUIREMENTS

Comply, at minimum, with manufacturer recommendations for space ventilation during and after installation. Isolate area of application from rest of building when applying high-emission paints or coatings.



### 1.9.1 Coatings

Do not apply coating when air or substrate conditions are:

- a. Less than 5 degrees F above dew point;
- b. Below 50 degrees F or over 95 degrees F, unless specifically pre-approved by the Contracting Officer and the product manufacturer. Do not, under any circumstances, violate the manufacturer's application recommendations.

### 1.9.2 Post-Application

Vacate space for as long as possible after application. Wait a minimum of 48 hours before occupying freshly painted rooms. Maintain one of the following ventilation conditions during the curing period, or for 72 hours after application:

- a. Supply 100 percent outside air 24 hours a day.
- b. Supply airflow at a rate of 6 air changes per hour, when outside temperatures are between 55 degrees F and 85 degrees F and humidity is between 30 percent and 60 percent.
- c. Supply airflow at a rate of 1.5 air changes per hour, when outside air conditions are not within the range stipulated above.

## PART 2 PRODUCTS

### 2.1 MATERIALS

Conform to the coating specifications and standards referenced in PART 3. Submit Product Data Sheets for specified coatings and solvents. Provide preprinted cleaning and maintenance instructions for all coating systems. Submit Manufacturer's Instructions on Mixing: Detailed mixing instructions, minimum and maximum application temperature and humidity, pot life, and curing and drying times between coats.

### 2.2 COLOR CODING FOR SHORE-TO SHIP UTILITY CONNECTIONS

Color Coding For Shore-To-Ship Utility Connections: Paint hose connection fittings and shut-off valves the designated color. In addition to color coding provide 2 inch high stenciled letters using black stencil paint, clearly designating service for each connection.

| Color Coding for Shore-to-Ship       |                    |                      |
|--------------------------------------|--------------------|----------------------|
| Utility Connections                  |                    |                      |
| Service                              | Color              | SAE AMS-STD-595A No. |
| Potable Water*                       | Blue               | 15044                |
| Water Provided for Fire Protection** | Red                | 11105                |
| Chilled Water                        | Striped Blue/White | 15044 / 17886        |

| Color Coding for Shore-to-Ship                                                                                                                                                                                                  |                      |               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------|
| Oily Waste Water                                                                                                                                                                                                                | Striped Yellow/Black | 13528 / 17038 |
| Sewer                                                                                                                                                                                                                           | Gold                 | 17043         |
| Steam                                                                                                                                                                                                                           | White                | 17886         |
| High Pressure Air                                                                                                                                                                                                               | Gray                 | 16081         |
| Low Pressure Air                                                                                                                                                                                                                | Tan                  | 10324         |
| Fuels                                                                                                                                                                                                                           | Yellow               | 13655         |
| * This includes connections serving domestic functions.                                                                                                                                                                         |                      |               |
| ** This includes non-potable salt water or, at some locations, fresh water connections provided for fire protection (may also include flushing and cooling requirements). Note: This does not include waterfront fire hydrants. |                      |               |

### 2.3 COLOR SELECTION OF FINISH COATS

Provide colors of finish coats as indicated or specified. Allow Contracting Officer to select colors not indicated or specified. Manufacturers' names and color identification are used for the purpose of color identification only. Named products are acceptable for use only if they conform to specified requirements. Products of other manufacturers are acceptable if the colors are approximately the colors indicated and the product conforms to specified requirements.

Provide color, texture, and pattern of wall coating systems as indicated on Contract Drawings. Submit manufacturer's samples of paint colors. Cross reference color samples to color scheme as indicated. Submit color stencil codes. Tint each coat progressively darker to enable confirmation of the number of coats.

## PART 3 EXECUTION

### 3.1 PROTECTION OF AREAS AND SPACES NOT TO BE PAINTED

Prior to surface preparation and coating applications, remove, mask, or otherwise protect hardware, hardware accessories, machined surfaces, radiator covers, plates, lighting fixtures, public and private property, and other such items not to be coated that are in contact with surfaces to be coated. Following completion of painting, reinstall removed items by workmen skilled in the trades. Restore surfaces contaminated by coating materials, to original condition and repair damaged items.

### 3.2 SURFACE PREPARATION

Remove dirt, splinters, loose particles, grease, oil, disintegrated coatings, and other foreign matter and substances deleterious to coating performance as specified for each substrate before application of paint or surface treatments. Remove oil and grease prior to mechanical cleaning. Schedule cleaning so that dust and other contaminants will not fall on wet, newly painted surfaces. Spot-prime exposed ferrous metals such as

nail heads on or in contact with surfaces to be painted with water-thinned paints, with a suitable corrosion-inhibitive primer capable of preventing flash rusting and compatible with the coating specified for the adjacent areas. Refer to MPI ASM and MPI MRM for additional more specific substrate preparation requirements.

### 3.2.1 Additional Requirements for Preparation of Surfaces With Existing Coatings

Before application of coatings, perform the following on surfaces covered by soundly-adhered coatings, defined as those which cannot be removed with a putty knife:

- a. Test existing finishes for lead before sanding, scraping, or removing. If lead is present, refer to paragraph Toxic Materials.
- b. Wipe previously painted surfaces to receive solvent-based coatings, except stucco and similarly rough surfaces clean with a clean, dry cloth saturated with mineral spirits, ASTM D235 or as specified in MPI MRM. Wipe the surfaces dry with a clean, dry, lint free cloth. Wipe immediately preceding the application of the first coat of any coating, unless specified otherwise.
- c. Sand existing glossy surfaces to be painted to reduce gloss. Brush, and wipe clean with a damp cloth to remove dust.
- d. The requirements specified are minimum. Comply also with the application instructions of the paint manufacturer and specific surface preparation requirements as outlined in MPI MRM Exterior Surface Preparation and Interior Surface Preparation.
- e. Thoroughly clean previously painted surfaces specified to be repainted damaged during construction of all grease, dirt, dust or other foreign matter.
- f. Remove blistering, cracking, flaking and peeling or otherwise deteriorated coatings.
- g. Remove chalk so that when tested in accordance with ASTM D4214, the chalk resistance rating is no less than 8.
- h. Roughen slick surfaces. Repair damaged areas such as, but not limited to, nail holes, cracks, chips, and spalls with suitable material to match adjacent undamaged areas.
- i. Feather and sand smooth edges of chipped paint.
- j. Clean rusty metal surfaces in accordance with SSPC requirements. Use solvent, mechanical, or chemical cleaning methods to provide surfaces suitable for painting.
- k. Provide new, proposed coatings that are compatible with existing coatings.

### 3.2.2 Existing Coated Surfaces with Minor Defects

Sand, spackle, and treat minor defects to render them smooth. Minor defects are defined as scratches, nicks, cracks, gouges, spalls, alligatoring, chalking, and irregularities due to partial peeling of

previous coatings. Remove chalking by sanding or blasting so that when tested in accordance with ASTM D4214, the chalk rating is not less than 8.

### 3.2.3 Removal of Existing Coatings

Remove existing coatings from the following surfaces:

- a. Surfaces containing large areas of minor defects;
- b. Surfaces containing more than 20 percent peeling area; and
- c. Surfaces designated by the Contracting Officer, such as surfaces where rust shows through existing coatings.

### 3.2.4 Substrate Repair

- a. Repair substrate surface damaged during coating removal;
- b. Sand edges of adjacent soundly-adhered existing coatings so they are tapered as smooth as practical to areas involved with coating removal; and
- c. Clean and prime the substrate as specified.

## 3.3 PREPARATION OF METAL SURFACES

### 3.3.1 Existing and New Ferrous Surfaces

- a. Ferrous Surfaces including Shop-coated Surfaces and Small Areas That Contain Rust, Mill Scale and Other Foreign Substances: Solvent clean or detergent wash in accordance with SSPC SP 1 to remove oil and grease. Where shop coat is missing or damaged, clean according to SSPC SP 6/NACE No.3, or SSPC SP 10/NACE No. 2.; Protect shop-coated ferrous surfaces from corrosion by treating and touching up corroded areas immediately upon detection.
- b. Surfaces With More Than 20 Percent Rust, Mill Scale, and Other Foreign Substances: Clean entire surface in accordance with SSPC SP 6/NACE No.3 / SSPC-SP WJ-3/NACE WJ-3.
- c. Metal Floor Surfaces to Receive Nonslip Coating: Clean in accordance with SSPC SP 10/NACE No. 2.

### 3.3.2 Final Ferrous Surface Condition:

#### 3.3.2.1 Tool Cleaned Surfaces

Comply with SSPC SP 2 and SSPC SP 3. Use as a visual reference, photographs in SSPC VIS 3 for the appearance of cleaned surfaces.

#### 3.3.2.2 Abrasive Blast Cleaned Surfaces

Comply with SSPC 7/NACE No.4, SSPC SP 6/NACE No.3, and SSPC SP 10/NACE No. 2. Use as a visual reference, photographs in SSPC VIS 1 for the appearance of cleaned surfaces.

#### 3.3.2.3 Waterjet Cleaned Surfaces

Comply with SSPC-SP WJ-1/NACE WJ-1, SSPC-SP WJ-2/NACE WJ-2,

SSPC-SP WJ-3/NACE WJ-3 or SSPC-SP WJ-4/NACE WJ-4. Use as a visual reference, photographs in SSPC VIS 4/NACE VIS 7 for the appearance of cleaned surfaces.

### 3.3.3 Galvanized Surfaces

- a. New or Existing Galvanized Surfaces With Only Dirt and Zinc Oxidation Products: Clean with solvent, steam, or non-alkaline detergent solution in accordance with SSPC SP 1. Completely remove coating by brush-off abrasive blast if the galvanized metal has been passivated or stabilized. Do not "passivate" or "stabilize" new galvanized steel to be coated. If the absence of hexavalent stain inhibitors is not documented, test as described in ASTM D6386, Appendix X2, and remove by one of the methods described therein.
- b. Galvanized with Slight Coating Deterioration or with Little or No Rusting: Water jetting to SSPC-SP WJ-3/NACE WJ-3 to remove loose coating from surfaces with less than 20 percent coating deterioration and no blistering, peeling, or cracking. Use inhibitor as recommended by the coating manufacturer to prevent rusting.
- c. Galvanized With Severe Deteriorated Coating or Severe Rusting: Spot abrasive blast rusted areas as described for steel in SSPC SP 6/NACE No.3, and waterjet to SSPC-SP WJ-3/NACE WJ-3 to remove existing coating.

### 3.3.4 Non-Ferrous Metallic Surfaces

Aluminum and aluminum-alloy, lead, copper, and other nonferrous metal surfaces.

Surface Cleaning: Solvent clean in accordance with SSPC SP 1 and wash with mild non-alkaline detergent to remove dirt and water soluble contaminants.

### 3.3.5 Terne-Coated Metal Surfaces

Solvent clean surfaces with mineral spirits, ASTM D235. Wipe dry with clean, dry cloths.

### 3.3.6 Existing Surfaces with a Bituminous or Mastic-Type Coating

Remove chalk, mildew, and other loose material by washing with a solution of 1/2 cup trisodium phosphate, 1/4 cup household detergent, one quart 5 percent sodium hypochlorite solution and 3 quarts of warm water.

## 3.4 PREPARATION OF CONCRETE AND CEMENTITIOUS SURFACE

### 3.4.1 Concrete and Masonry

- a. Curing: Allow concrete, stucco and masonry surfaces to cure at least 30 days before painting, and concrete slab on grade to cure at least 90 days before painting.
- b. Surface Cleaning: Remove the following deleterious substances.
  - (1) Dirt, Chalking, Grease, and Oil: Wash new and existing uncoated surfaces with a solution composed of 1/2 cup trisodium phosphate, 1/4 cup household detergent, and 4 quarts of warm water. Then

rinse thoroughly with fresh water. Wash existing coated surfaces with a suitable detergent and rinse thoroughly. For large areas, water blasting may be used.

(2) Fungus and Mold: Wash new, existing coated, and existing uncoated surfaces with a solution composed of 1/2 cup trisodium phosphate, 1/4 cup household detergent, one quart 5 percent sodium hypochlorite solution and 3 quarts of warm water. Rinse thoroughly with fresh water.

(3) Paint and Loose Particles: Remove by wire brushing.

(4) Efflorescence: Remove by scraping or wire brushing followed by washing with a 5 to 10 percent by weight aqueous solution of hydrochloric (muriatic) acid. Do not allow acid to remain on the surface for more than five minutes before rinsing with fresh water. Do not acid clean more than 4 square feet of surface, per workman, at one time.

(5) Removal of Existing Coatings: For surfaces to receive textured coating MPI 42, remove existing coatings including soundly adhered coatings if recommended by textured coating manufacturer.

c. Cosmetic Repair of Minor Defects: Repair or fill mortar joints and minor defects, including but not limited to spalls, in accordance with manufacturer's recommendations and prior to coating application.

d. Allowable Moisture Content: Latex coatings may be applied to damp surfaces, but not to surfaces with droplets of water. Do not apply epoxies to damp vertical surfaces as determined by ASTM D4263 or horizontal surfaces that exceed 3 lbs of moisture per 1000 square feet in 24 hours as determined by ASTM F1869. In all cases follow manufacturer's recommendations. Allow surfaces to cure a minimum of 30 days before painting.

### 3.4.2 Gypsum Board, Plaster, and Stucco

#### 3.4.2.1 Surface Cleaning

Verify that plaster and stucco surfaces are free from loose matter and that gypsum board is dry. Remove loose dirt and dust by brushing with a soft brush, rubbing with a dry cloth, or vacuum-cleaning prior to application of the first coat material. A damp cloth or sponge may be used if paint is water-based.

#### 3.4.2.2 Repair of Minor Defects

Prior to painting, repair joints, cracks, holes, surface irregularities, and other minor defects with patching plaster or spackling compound and sand smooth.

#### 3.4.2.3 Allowable Moisture Content

Latex coatings may be applied to damp surfaces, but not surfaces with droplets of water. Do not apply epoxies to damp surfaces as determined by ASTM D4263. Verify that new plaster to be coated has a maximum moisture content of 8 percent, when measured in accordance with ASTM D4444, Method A, unless otherwise authorized. In addition to moisture content requirements, allow new plaster to age a minimum of 30 days before

preparation for painting.

### 3.5 APPLICATION

#### 3.5.1 Coating Application

- a. Comply with applicable federal, state and local laws enacted to ensure compliance with Federal Clean Air Standards. Apply coating materials in accordance with SSPC PA 1. SSPC PA 1 methods are applicable to all substrates, except as modified herein.
- b. At the time of application, paint must show no signs of deterioration. Maintain uniform suspension of pigments during application.
- c. Unless otherwise specified or recommended by the paint manufacturer, paint may be applied by brush, roller, or spray. Use trigger operated spray nozzles for water hoses. Use rollers for applying paints and enamels of a type designed for the coating to be applied and the surface to be coated. Wear protective clothing and respirators when applying oil-based paints or using spray equipment with any paints.
- d. Only apply paints, except water-thinned types, to surfaces that are completely free of moisture as determined by sight or touch.
- e. Thoroughly work coating materials into joints, crevices, and open spaces. Pay special attention to ensure that all edges, corners, crevices, welds, and rivets receive a film thickness equal to that of adjacent painted surfaces.
- f. Apply each coat of paint so that dry film is of uniform thickness and free from runs, drops, ridges, waves, pinholes or other voids, laps, brush marks, and variations in color, texture, and finish. Completely hide all blemishes.
- g. Touch up damaged coatings before applying subsequent coats. Broom clean and clear dust from interior areas before and during the application of coating material.
- h. Apply paint to new fire extinguishing sprinkler systems including valves, piping, conduit, hangers, supports, miscellaneous metal work, and accessories. Shield sprinkler heads with protective coverings while painting is in progress. Remove sprinkler heads which have been painted and replace with new sprinkler heads. Unfinished spaces include attic spaces, spaces above suspended ceilings, crawl spaces, pipe chases, mechanical equipment room, and space where walls or ceiling are not painted or not constructed of a prefinished material. Upon completion of painting, remove protective covering from sprinkler heads.
- i. Piping in Unfinished Areas: Provide primed surfaces with one coat of red alkyd gloss enamel (MPI 9) applied to a minimum dry film thickness of 1.0 mil in attic spaces, spaces above suspended ceilings, crawl spaces, pipe chases, mechanical equipment room, and spaces where walls or ceiling are not painted or not constructed of a prefinished material.
- j. Piping in Finished Areas: Provide primed surfaces with two coats of paint to match adjacent surfaces, except provide valves and operating

accessories with one coat of red alkyd gloss enamel (MPI 9) applied to a minimum dry film thickness of 1.0 mil or two component gloss polyurethane (MPI 72) in exterior applications.

- k. Provide labeling on the surfaces of all feed and cross mains to show the pipe function such as "Sprinkler System", "Fire Department Connection", "Standpipe". For pipe sizes 4-inch and larger provide white painted stenciled letters and arrows, a minimum of 2 in in height and visible from at least two sides when viewed from the floor. For pipe sizes less than 4-inch, provide white painted stenciled letters and arrows, a minimum of 0.75 in in height and visible from the floor.
- l. All fire suppression system valves must be marked with permanent tags indicating normally open or normally closed.
- m. Drying Time: Allow time between coats, as recommended by the coating manufacturer, to permit thorough drying, but not to present topcoat adhesion problems. Provide each coat in specified condition to receive next coat.
- n. Primers, and Intermediate Coats: Do not allow primers or intermediate coats to dry more than 30 days, or longer than recommended by manufacturer, before applying subsequent coats. Follow manufacturer's recommendations for surface preparation if primers or intermediate coats are allowed to dry longer than recommended by manufacturers of subsequent coatings. Cover each preceding coat or surface completely by ensuring visually perceptible difference in shades of successive coats.
- o. Finished Surfaces: Provide finished surfaces free from runs, drops, ridges, waves, laps, brush marks, and variations in colors.
- p. Thermosetting Paints: Apply topcoats over thermosetting paints (epoxies and urethanes) within the overcoat window recommended by the manufacturer.

### 3.5.2 Mixing and Thinning of Paints

Reduce paints to proper consistency by adding fresh paint, except when thinning is mandatory to suit surface, temperature, weather conditions, application methods, or for the type of paint being used. Obtain written permission from the Contracting Officer to use thinners. Verify that the written permission includes quantities and types of thinners to use.

When thinning is allowed, thin paints immediately prior to application with not more than one pint of suitable thinner per gallon. The use of thinner does not relieve the Contractor from obtaining complete hiding, full film thickness, or required gloss. Thinning cannot cause the paint to exceed limits on volatile organic compounds. Do not mix paints of different manufacturers.

### 3.5.3 Two-Component Systems

Mix two-component systems in accordance with manufacturer's instructions. Follow recommendation by the manufacturer for any thinning of the first coat to ensure proper penetration and sealing for each type of substrate.



### 3.5.4 Coating Systems

- a. Systems by Substrates: Apply coatings that conform to the respective specifications listed in the following Tables:

| Table for Interior Applications |                                                               |
|---------------------------------|---------------------------------------------------------------|
| MPI Division                    | Substrate Application                                         |
| MPI Division 3                  | Interior Concrete Paint Table                                 |
| MPI Division 4                  | Interior Concrete Masonry Units Paint Table                   |
| MPI Division 5                  | Interior Metal, Ferrous and Non-Ferrous Paint Table           |
| MPI Division 6                  | Interior Wood Paint Table                                     |
| MPI Division 9                  | Interior Plaster, Gypsum Board, Textured Surfaces Paint Table |

- b. Minimum Dry Film Thickness (DFT): Apply paints, primers, varnishes, enamels, undercoats, and other coatings to a minimum dry film thickness of 1.5 mil each coat unless specified otherwise in the Tables. Coating thickness, where specified, refers to the minimum dry film thickness.
- c. Coatings for Surfaces Not Specified Otherwise: Coat unspecified surfaces the same as surfaces having similar conditions of exposure.
- d. Existing Surfaces Damaged During Performance of the Work, Including New Patches In Existing Surfaces: Coat surfaces with the following:
- (1) One coat of primer.
  - (2) One coat of undercoat or intermediate coat.
  - (3) One topcoat to match adjacent surfaces.
- e. Existing Coated Surfaces To Be Painted: Apply coatings conforming to the respective specifications listed in the Tables herein, except that pretreatments, sealers and fillers need not be provided on surfaces where existing coatings are soundly adhered and in good condition. Do not omit undercoats or primers.

### 3.6 COATING SYSTEMS FOR METAL

Apply coatings of Tables in MPI Division 5 for Exterior and Interior.

- a. Apply specified ferrous metal primer to steel surfaces on the same day that surface is cleaned, to surfaces that meet all specified surface preparation requirements at time of application.
- b. Inaccessible Surfaces: Prior to erection, use one coat of specified primer on metal surfaces that will be inaccessible after erection.
- c. Shop-primed Surfaces: Touch up exposed substrates and damaged coatings to protect from rusting prior to applying field primer.

- d. Surface Previously Coated with Epoxy or Urethane: Apply MPI 101, 1.5 mils DFT immediately prior to application of epoxy or urethane coatings.
- e. Pipes and Tubing: The semitransparent film applied to some pipes and tubing at the mill is not to be considered a shop coat. Overcoat these items with the specified ferrous-metal primer prior to application of finish coats.
- f. Exposed Nails, Screws, Fasteners, and Miscellaneous Ferrous Surfaces. On surfaces to be coated with water thinned coatings, spot prime exposed nails and other ferrous metal with latex primer MPI 107.

### 3.7 COATING SYSTEMS FOR CONCRETE AND CEMENTITIOUS SUBSTRATES

Apply coatings of Tables in MPI Division 3, 4 and 9 for Exterior and Interior.

### 3.8 COATING SYSTEMS FOR WOOD AND PLYWOOD

- a. Apply coatings of Tables in MPI Division 6 for Exterior and Interior.
- b. Prior to erection, apply two coats of specified primer to treat and prime wood surfaces which will be inaccessible after erection.
- c. Apply stains in accordance with manufacturer's printed instructions.

### 3.9 PIPING IDENTIFICATION

Piping Identification, Including Surfaces In Concealed Spaces: Provide in accordance with MIL-STD-101. Place stenciling in clearly visible locations. On piping not covered by MIL-STD-101, stencil approved names or code letters, in letters a minimum of 1/2 inch high for piping and a minimum of 2 inches high elsewhere. Stencil arrow-shaped markings on piping to indicate direction of flow using black stencil paint.

### 3.10 INSPECTION AND ACCEPTANCE

In addition to meeting previously specified requirements, demonstrate mobility of moving components, including swinging and sliding doors, cabinets, and windows with operable sash, for inspection by the Contracting Officer. Perform this demonstration after appropriate curing and drying times of coatings have elapsed and prior to invoicing for final payment.

### 3.11 WASTE MANAGEMENT

As specified in the Waste Management Plan and as follows. Do not use kerosene or any such organic solvents to clean up water based paints. Properly dispose of paints or solvents in designated containers. Close and seal partially used containers of paint to maintain quality as necessary for reuse. Store in protected, well-ventilated, fire-safe area at moderate temperature. Place materials defined as hazardous or toxic waste in designated containers.

### 3.12 PAINT TABLES

All DFT's are minimum values. Use only materials with a MPI GPS-1-14

green check mark having a minimum MPI "Environmentally Friendly" E1 rating based on VOC (EPA Method 24) content levels. Acceptable products are listed in the MPI Green Approved Products List, available at <https://www.mpi.net/APL/index.asp>.

### 3.12.1 Interior Paint Tables

#### 3.12.1.1 MPI Division 3: Interior Concrete Paint Table

A. New and uncoated existing and Existing, previously painted Concrete, vertical surfaces, not specified otherwise

| Latex                                        |                                 |        |              |         |            |
|----------------------------------------------|---------------------------------|--------|--------------|---------|------------|
| New, uncoated<br>Existing                    | Existing,<br>previously painted | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 3.1A-G2<br>(Flat)                    | MPI RIN 3.1A-G2<br>(Flat)       | MPI 3  | MPI 44       | MPI 44  | 4 mils     |
| MPI INT 3.1A-G3<br>(Eggshell)                | MPI RIN 3.1A-G3<br>(Eggshell)   | MPI 3  | MPI 52       | MPI 52  | 4 mils     |
| MPI INT 3.1A-G5                              | MPI RIN 3.1A-G5<br>(Semigloss)  | MPI 3  | MPI 54       | MPI 54  | 4 mils     |
| Topcoat: Coating to match adjacent surfaces. |                                 |        |              |         |            |

| High Performance Architectural Latex         |                                 |        |              |         |            |
|----------------------------------------------|---------------------------------|--------|--------------|---------|------------|
| New, uncoated<br>Existing                    | Existing,<br>previously painted | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 3.1C-G2<br>(Flat)                    | MPI RIN 3.1J-G2<br>(Flat)       | MPI 3  | MPI 138      | MPI 138 | 4 mils     |
| MPI INT 3.1C-G3<br>(Eggshell)                | MPI RIN 3.1J-G3<br>(Eggshell)   | MPI 3  | MPI 139      | MPI 139 | 4 mils     |
| MPI INT 3.1C-G4<br>(satin)                   | MPI RIN 3.1J-G4                 | MPI 3  | MPI 140      | MPI 140 | 4 mils     |
| MPI INT 3.1C-G5<br>(Semigloss)               | MPI RIN 3.1J-G5<br>(Semigloss)  | MPI 3  | MPI 141      | MPI 141 | 4 mils     |
| Topcoat: Coating to match adjacent surfaces. |                                 |        |              |         |            |

| Institutional Low Odor / Low VOC Latex |
|----------------------------------------|
|----------------------------------------|

| New, uncoated<br>Existing                    | Existing,<br>previously painted | Primer  | Intermediate | Topcoat | System DFT |
|----------------------------------------------|---------------------------------|---------|--------------|---------|------------|
| MPI INT 3.1M-G2<br>(Flat)                    | MPI RIN 3.1L-G2<br>(Flat)       | MPI 149 | MPI 144      | MPI 144 | 4 mils     |
| MPI INT 3.1M-G3<br>(Eggshell)                | MPI RIN 3.1L-G3<br>(Eggshell)   | MPI 149 | MPI 145      | MPI 145 | 4 mils     |
| MPI INT 3.1M-G4<br>(satin)                   | MPI RIN 3.1L-G4                 | MPI 149 | MPI 146      | MPI 146 | 4 mils     |
| MPI INT 3.1M-G5<br>(Semigloss)               | MPI RIN 3.1L-G5<br>(Semigloss)  | MPI 149 | MPI 147      | MPI 147 | 4 mils     |
| Topcoat: Coating to match adjacent surfaces. |                                 |         |              |         |            |

| Waterborne Light Industrial Coating          |                                 |        |              |         |            |
|----------------------------------------------|---------------------------------|--------|--------------|---------|------------|
| New, uncoated<br>Existing                    | Existing,<br>previously painted | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 3.1L-G3(Eggshell)                    | MPI RIN 3.1C-G3(Eggshell)       | MPI 3  | MPI 151      | MPI 151 | 4.8 mils   |
| MPI INT 3.1L-G5(Semigloss)                   | MPI RIN 3.1C-G5(Semigloss)      | MPI 3  | MPI 153      | MPI 153 | 4.8 mils   |
| MPI INT 3.1L-G6(Gloss)                       | MPI RIN 3.1C-G6(Gloss)          | MPI 3  | MPI 154      | MPI 154 | 4.8 mils   |
| Topcoat: Coating to match adjacent surfaces. |                                 |        |              |         |            |

| Alkyd                                        |                                 |        |              |         |            |
|----------------------------------------------|---------------------------------|--------|--------------|---------|------------|
| New, uncoated<br>Existing                    | Existing,<br>previously painted | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 3.1D-G3<br>(Eggshell)                | MPI RIN 3.1D-G3<br>(Eggshell)   | MPI 3  | MPI 51       | MPI 51  | 4.5 mils   |
| MPI INT 3.1D-G5<br>(Semigloss)               | MPI RIN 3.1D-G5<br>(Semigloss)  | MPI 3  | MPI 47       | MPI 47  | 4.5 mils   |
| MPI INT 3.1D-G6<br>(Gloss)                   | MPI RIN 3.1D-G6<br>(Gloss)      | MPI 3  | MPI 48       | MPI 48  | 4.5 mils   |
| Topcoat: Coating to match adjacent surfaces. |                                 |        |              |         |            |

| Epoxy |  |  |  |  |  |
|-------|--|--|--|--|--|
|-------|--|--|--|--|--|

| New, uncoated<br>Existing                                                    | Existing,<br>previously painted | Primer | Intermediate | Topcoat | System DFT |
|------------------------------------------------------------------------------|---------------------------------|--------|--------------|---------|------------|
| MPI INT 3.1F-G6<br>(Gloss)                                                   | MPI RIN 3.1E-G6<br>(Gloss)      | MPI 77 | MPI 77       | MPI 77  | 4 mils     |
| Note: Primer may be reduced for penetration per manufacturer's instructions. |                                 |        |              |         |            |

| Epoxy                                                                        |                                 |        |              |         |            |
|------------------------------------------------------------------------------|---------------------------------|--------|--------------|---------|------------|
| New, uncoated<br>Existing                                                    | Existing,<br>previously painted | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 3.1F                                                                 | MPI RIN 3.1E                    | MPI 77 | MPI 77       | MPI 77  | 4 mils     |
| Note: Primer may be reduced for penetration per manufacturer's instructions. |                                 |        |              |         |            |

### 3.12.1.2 MPI Division 4: Interior Concrete Masonry Units Paint Table

#### A. New and uncoated Existing Concrete Masonry

| High Performance Architectural Latex |        |        |              |         |            |
|--------------------------------------|--------|--------|--------------|---------|------------|
| New, uncoated<br>Existing            | Filler | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 4.2D-G2<br>(Flat)            | MPI 4  | N/A    | MPI 139      | MPI 138 | 11 mils    |
| MPI INT 4.2D-G3<br>(Eggshell)        | MPI 4  | N/A    | MPI 139      | MPI 139 | 11 mils    |
| MPI INT 4.2D-G4<br>(Satin)           | MPI 4  | N/A    | MPI 140      | MPI 140 | 11 mils    |
| MPI INT 4.2D-G5<br>(Semigloss)       | MPI 4  | N/A    | MPI 141      | MPI 141 | 11 mils    |
| Fill all holes in masonry surface    |        |        |              |         |            |

| Institutional Low Odor / Low VOC Latex |        |        |              |         |            |
|----------------------------------------|--------|--------|--------------|---------|------------|
| New, uncoated<br>Existing              | Filler | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 4.2E-G2<br>(Flat)              | MPI 4  | N/A    | MPI 144      | MPI 144 | 4 mils     |
| MPI INT 4.2E-G3<br>(Eggshell)          | MPI 4  | N/A    | MPI 145      | MPI 145 | 4 mils     |

|                                   |       |     |         |         |        |
|-----------------------------------|-------|-----|---------|---------|--------|
| MPI INT 4.2E-G4<br>(Satin)        | MPI 4 | N/A | MPI 146 | MPI 146 | 4 mils |
| MPI INT 4.2E-G5<br>(Semigloss)    | MPI 4 | N/A | MPI 147 | MPI 147 | 4 mils |
| Fill all holes in masonry surface |       |     |         |         |        |

B. Existing, Previously Painted Concrete Masonry

| High Performance Architectural Latex |        |         |              |         |            |
|--------------------------------------|--------|---------|--------------|---------|------------|
| Existing,<br>previously<br>painted   | Filler | Primer  | Intermediate | Topcoat | System DFT |
| MPI RIN 4.2K-G2<br>(Flat)            | N/A    | MPI 138 | MPI 138      | MPI 138 | 4.5 mils   |
| MPI RIN 4.2K-G3<br>(Eggshell)        | N/A    | MPI 139 | MPI 139      | MPI 139 | 4.5 mils   |
| MPI RIN 4.2K-G4                      | N/A    | MPI 140 | MPI 140      | MPI 140 | 4.5 mils   |
| MPI RIN 4.2K-G5<br>(Semigloss)       | N/A    | MPI 141 | MPI 141      | MPI 141 | 4.5 mils   |

| Institutional Low Odor / Low VOC Latex |        |         |              |         |            |
|----------------------------------------|--------|---------|--------------|---------|------------|
| Existing,<br>previously painted        | Filler | Primer  | Intermediate | Topcoat | System DFT |
| MPI RIN 4.2L-G2<br>(Flat)              | N/A    | MPI 144 | MPI 144      | MPI 144 | 4 mils     |
| MPI RIN 4.2L-G3<br>(Eggshell)          | N/A    | MPI 145 | MPI 145      | MPI 145 | 4 mils     |
| MPI RIN 4.2L-G4<br>(Satin)             | N/A    | MPI 146 | MPI 146      | MPI 146 | 4 mils     |
| MPI RIN 4.2L-G5<br>(Semigloss)         | N/A    | MPI 147 | MPI 147      | MPI 147 | 4 mils     |

C. New and uncoated Existing Concrete masonry units in unless otherwise specified

| Waterborne Light Industrial Coating |        |        |              |         |            |
|-------------------------------------|--------|--------|--------------|---------|------------|
| New, uncoated<br>Existing           | Filler | Primer | Intermediate | Topcoat | System DFT |

|                                   |       |     |         |         |         |
|-----------------------------------|-------|-----|---------|---------|---------|
| MPI INT<br>4.2K-G3(Eggshell)      | MPI 4 | N/A | MPI 151 | MPI 151 | 11 mils |
| MPI INT<br>4.2K-G5(Semigloss)     | MPI 4 | N/A | MPI 153 | MPI 153 | 11 mils |
| MPI INT<br>4.2K-G6(Gloss)         | MPI 4 | N/A | MPI 154 | MPI 154 | 11 mils |
| Fill all holes in masonry surface |       |     |         |         |         |

| Alkyd                             |        |        |              |         |            |
|-----------------------------------|--------|--------|--------------|---------|------------|
| New, uncoated<br>Existing         | Filler | Primer | Intermediate | Topcoat | System DFT |
| MPI INT<br>4.2K-G3(Eggshell)      | MPI 4  | MPI 50 | MPI 51       | MPI 51  | 12 mils    |
| MPI INT<br>4.2K-G5(Semigloss)     | MPI 4  | MPI 50 | MPI 47       | MPI 47  | 12 mils    |
| MPI INT<br>4.2K-G6(Gloss)         | MPI 4  | MPI 50 | MPI 48       | MPI 48  | 12 mils    |
| Fill all holes in masonry surface |        |        |              |         |            |

| Epoxy                             |         |        |              |         |            |
|-----------------------------------|---------|--------|--------------|---------|------------|
| New, uncoated<br>Existing         | Filler  | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 4.2G-G6<br>(Gloss)        | MPI 116 | N/A    | MPI 77       | MPI 77  | 10 mils    |
| Fill all holes in masonry surface |         |        |              |         |            |

D. Existing, previously painted, concrete masonry units in unless otherwise specified

| Waterborne Light Industrial Coating |        |         |              |         |            |
|-------------------------------------|--------|---------|--------------|---------|------------|
| Existing,<br>previously painted     | Filler | Primer  | Intermediate | Topcoat | System DFT |
| MPI RIN<br>4.2G-G3(Eggshell)        | N/A    | MPI 151 | MPI 151      | MPI 151 | 4.5 mils   |
| MPI RIN<br>4.2G-G5(Semigloss)       | N/A    | MPI 153 | MPI 153      | MPI 153 | 4.5 mils   |

|                        |     |         |         |         |          |
|------------------------|-----|---------|---------|---------|----------|
| MPI RIN 4.2G-G6(Gloss) | N/A | MPI 154 | MPI 154 | MPI 154 | 4.5 mils |
|------------------------|-----|---------|---------|---------|----------|

| Alkyd                        |        |        |              |         |            |
|------------------------------|--------|--------|--------------|---------|------------|
| Existing, previously painted | Filler | Primer | Intermediate | Topcoat | System DFT |
| MPI RIN 4.2C-G3 (Eggshell)   | N/A    | MPI 17 | MPI 51       | MPI 51  | 4.5 mils   |
| MPI RIN 4.2C-G5 (Semigloss)  | N/A    | MPI 17 | MPI 47       | MPI 47  | 4.5 mils   |
| MPI RIN 4.2C-G6 (Gloss)      | N/A    | MPI 17 | MPI 48       | MPI 48  | 4.5 mils   |

| Epoxy                        |        |        |              |         |            |
|------------------------------|--------|--------|--------------|---------|------------|
| Existing, previously painted | Filler | Primer | Intermediate | Topcoat | System DFT |
| MPI RIN 4.2D-G6 (Gloss)      | N/A    | MPI 77 | MPI 77       | MPI 77  | 5 mils     |

### 3.12.1.3 MPI Division 5: Interior Metal, Ferrous and Non-Ferrous Paint Table

#### A. Interior Steel / Ferrous Surfaces

(1) Metal, not otherwise specified except floors, hot metal surfaces, and new prefinished equipment

| High Performance Architectural Latex         |        |              |         |            |
|----------------------------------------------|--------|--------------|---------|------------|
| New, uncoated Existing                       | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 5.1R-G2 (Flat)                       | MPI 76 | MPI 138      | MPI 138 | 5 mils     |
| MPI INT 5.1R-G3 (Eggshell)                   | MPI 76 | MPI 139      | MPI 139 | 5 mils     |
| MPI INT 5.1R-G5 (Semigloss)                  | MPI 76 | MPI 141      | MPI 141 | 5 mils     |
| Topcoat: Coating to match adjacent surfaces. |        |              |         |            |

| Alkyd                  |        |              |         |            |
|------------------------|--------|--------------|---------|------------|
| New, uncoated Existing | Primer | Intermediate | Topcoat | System DFT |



|                                              |        |        |        |           |
|----------------------------------------------|--------|--------|--------|-----------|
| MPI INT 5.1E-G2 (Flat)                       | MPI 76 | MPI 49 | MPI 49 | 5.25 mils |
| MPI INT 5.1E-G3 (Eggshell)                   | MPI 76 | MPI 51 | MPI 51 | 5.25 mils |
| MPI INT 5.1E-G5 (Semigloss)                  | MPI 76 | MPI 47 | MPI 47 | 5.25 mils |
| MPI INT 5.1E-G6 (Gloss)                      | MPI 76 | MPI 48 | MPI 48 | 5.25 mils |
| Topcoat: Coating to match adjacent surfaces. |        |        |        |           |

(2) Metal floors (non-shop-primed surfaces or non-slip deck surfaces) with non-skid additive (NSA), load at manufacturer's recommendations

| Alkyd (over q.d. Alkyd Primer)               |        |              |         |            |
|----------------------------------------------|--------|--------------|---------|------------|
| New, uncoated Existing                       | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 5.1E-G5 (Semi-Gloss)                 | MPI 76 | MPI 47       | MPI 47  | 5.25 mils  |
| Topcoat: Coating to match adjacent surfaces. |        |              |         |            |

| Epoxy                                        |         |              |         |            |
|----------------------------------------------|---------|--------------|---------|------------|
| New, uncoated Existing                       | Primer  | Intermediate | Topcoat | System DFT |
| MPI INT 5.1L-G6 (Gloss)                      | MPI 101 | MPI 101      | MPI 101 | 5.25 mils  |
| Topcoat: Coating to match adjacent surfaces. |         |              |         |            |

(3) Metal in not otherwise specified except floors, hot metal surfaces, and new prefinished equipment

| Alkyd                       |        |              |         |            |
|-----------------------------|--------|--------------|---------|------------|
| New, uncoated Existing      | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 5.1E-G3 (Eggshell)  | MPI 76 | MPI 51       | MPI 51  | 5.25 mils  |
| MPI INT 5.1E-G5 (Semigloss) | MPI 76 | MPI 47       | MPI 47  | 5.25 mils  |
| MPI INT 5.1E-G6 (Gloss)     | MPI 76 | MPI 48       | MPI 48  | 5.25 mils  |

Topcoat: Coating to match adjacent surfaces.

| Alkyd; For Hand Tool Cleaning                |        |              |         |            |
|----------------------------------------------|--------|--------------|---------|------------|
| New, uncoated Existing                       | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 5.1T-G3 (Eggshell)                   | MPI 23 | MPI 51       | MPI 51  | 5.25 mils  |
| MPI INT 5.1T-G5 (Semigloss)                  | MPI 23 | MPI 47       | MPI 47  | 5.25 mils  |
| MPI INT 5.1T-G6 (Gloss)                      | MPI 23 | MPI 48       | MPI 48  | 5.25 mils  |
| Topcoat: Coating to match adjacent surfaces. |        |              |         |            |

(5) Miscellaneous non-ferrous metal items not otherwise specified except floors, hot metal surfaces, and new prefinished equipment. Match surrounding finish

| High Performance Architectural Latex         |        |              |         |            |
|----------------------------------------------|--------|--------------|---------|------------|
| New, uncoated Existing                       | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 5.4F-G2 (Flat)                       | MPI 95 | MPI 138      | MPI 138 | 5 mils     |
| MPI INT 5.4F-G3 (Eggshell)                   | MPI 95 | MPI 139      | MPI 139 | 5 mils     |
| MPI INT 5.4F-G4 (Satin)                      | MPI 95 | MPI 140      | MPI 140 | 5 mils     |
| MPI INT 5.4F-G5 (Semigloss)                  | MPI 95 | MPI 141      | MPI 141 | 5 mils     |
| Topcoat: Coating to match adjacent surfaces. |        |              |         |            |

| Alkyd                      |        |              |         |            |
|----------------------------|--------|--------------|---------|------------|
| New, uncoated Existing     | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 5.4J-G2 (Flat)     | MPI 95 | MPI 49       | MPI 49  | 5 mils     |
| MPI INT 5.4J-G3 (Eggshell) | MPI 95 | MPI 51       | MPI 51  | 5 mils     |

|                                              |        |        |        |        |
|----------------------------------------------|--------|--------|--------|--------|
| MPI INT 5.4J-G5<br>(Semigloss)               | MPI 95 | MPI 47 | MPI 47 | 5 mils |
| MPI INT 5.4J-G6 (Gloss)                      | MPI 95 | MPI 48 | MPI 48 | 5 mils |
| Topcoat: Coating to match adjacent surfaces. |        |        |        |        |

#### B. Hot Surfaces

(1) Hot metal surfaces including smokestacks subject to temperatures up to 400 degrees F

| Heat Resistant Enamel                                                    |        |              |         |                  |
|--------------------------------------------------------------------------|--------|--------------|---------|------------------|
| New                                                                      | N/A    | Intermediate | Topcoat | System DFT       |
| MPI INT 5.2A                                                             | MPI 21 | N/A          | N/A     | Per Manufacturer |
| Surface preparation and number of coats per manufacturer's instructions. |        |              |         |                  |

(2) Ferrous metal subject to high temperature, up to 750 degrees F

| Inorganic Zinc Rich Coating                                              |        |              |         |                  |
|--------------------------------------------------------------------------|--------|--------------|---------|------------------|
| New                                                                      | N/A    | Intermediate | Topcoat | System DFT       |
| MPI INT 5.2C                                                             | MPI 19 | N/A          | N/A     | Per Manufacturer |
| Surface preparation and number of coats per manufacturer's instructions. |        |              |         |                  |

| Heat Resistant Aluminum Enamel                                           |       |              |         |                  |
|--------------------------------------------------------------------------|-------|--------------|---------|------------------|
| New                                                                      | N/A   | Intermediate | Topcoat | System DFT       |
| MPI INT 5.2B<br>(Aluminum Finish)                                        | MPI 2 | N/A          | N/A     | Per Manufacturer |
| Surface preparation and number of coats per manufacturer's instructions. |       |              |         |                  |

(3) New and Existing Surfaces made bare subject to temperatures up to 1100 degrees F

(1) New surfaces and Existing surfaces made bare cleaning to SSPC SP 10/NACE No. 2 subject to temperatures up to 1100 degrees F:

| Heat Resistant Coating                                                   |              |        |              |         |                  |
|--------------------------------------------------------------------------|--------------|--------|--------------|---------|------------------|
| New                                                                      | Existing     | N/A    | Intermediate | Topcoat | System DFT       |
| MPI INT 5.2D                                                             | MPI RIN 5.2D | MPI 22 | N/A          | N/A     | Per Manufacturer |
| Surface preparation and number of coats per manufacturer's instructions. |              |        |              |         |                  |

### 3.12.1.4 MPI Division 9: Interior Plaster, Gypsum Board, Textured Surfaces Paint Table

A. Interior New and Existing, previously painted Plaster and Wallboard not otherwise specified

| Latex                                        |                              |        |              |         |            |
|----------------------------------------------|------------------------------|--------|--------------|---------|------------|
| New                                          | Existing, previously painted | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 9.2A-G2 (Flat)                       | RIN 9.2A-G2 (Flat)           | MPI 50 | MPI 44       | MPI 44  | 4 mils     |
| MPI INT 9.2A-G3 (Eggshell)                   | RIN 9.2A-G3 (Eggshell)       | MPI 50 | MPI 52       | MPI 52  | 4 mils     |
| MPI INT 9.2A-G5 (Semigloss)                  | RIN 9.2A-G5 (Semigloss)      | MPI 50 | MPI 54       | MPI 54  | 4 mils     |
| Topcoat: Coating to match adjacent surfaces. |                              |        |              |         |            |

| High Performance Architectural Latex - High Traffic Areas |                              |        |              |         |            |
|-----------------------------------------------------------|------------------------------|--------|--------------|---------|------------|
| New                                                       | Existing, previously painted | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 9.2B-G2 (Flat)                                    | MPI RIN 9.2B-G2 (Flat)       | MPI 50 | MPI 138      | MPI 138 | 4 mils     |
| MPI INT 9.2B-G3 (Eggshell)                                | MPI RIN 9.2B-G3 (Eggshell)   | MPI 50 | MPI 139      | MPI 139 | 4 mils     |
| MPI INT 9.2B-G5 (Semigloss)                               | MPI RIN 9.2B-G5 (Semigloss)  | MPI 50 | MPI 141      | MPI 141 | 4 mils     |
| Topcoat: Coating to match adjacent surfaces.              |                              |        |              |         |            |

Institutional Low Odor / Low VOC Latex, New

| Institutional Low Odor / Low VOC Latex |        |              |         |            |
|----------------------------------------|--------|--------------|---------|------------|
| New                                    | Primer | Intermediate | Topcoat | System DFT |

|                                              |         |         |         |        |
|----------------------------------------------|---------|---------|---------|--------|
| MPI INT 9.2M-G2 (Flat)                       | MPI 149 | MPI 144 | MPI 144 | 4 mils |
| MPI INT 9.2M-G3 (Eggshell)                   | MPI 149 | MPI 145 | MPI 145 | 4 mils |
| MPI INT 9.2M-G4 (Satin)                      | MPI 149 | MPI 146 | MPI 146 | 4 mils |
| MPI INT 9.2M-G5 (Semigloss)                  | MPI 149 | MPI 147 | MPI 147 | 4 mils |
| Topcoat: Coating to match adjacent surfaces. |         |         |         |        |

Institutional Low Odor / Low VOC Latex, Existing, previously painted

| Institutional Low Odor / Low VOC Latex       |         |              |         |            |
|----------------------------------------------|---------|--------------|---------|------------|
| Existing, previously painted                 | Primer  | Intermediate | Topcoat | System DFT |
| MPI RIN 9.2M-G2 (Flat)                       | MPI 144 | MPI 144      | MPI 144 | 4 mils     |
| MPI RIN 9.2M-G3 (Eggshell)                   | MPI 144 | MPI 145      | MPI 145 | 4 mils     |
| MPI RIN 9.2M-G4 (Satin)                      | MPI 144 | MPI 146      | MPI 146 | 4 mils     |
| MPI RIN 9.2M-G5 (Semigloss)                  | MPI 144 | MPI 147      | MPI 147 | 4 mils     |
| Topcoat: Coating to match adjacent surfaces. |         |              |         |            |

B. Interior New and Existing, previously painted Plaster and Wallboard in and other high humidity areas not otherwise specified

| Waterborne Light Industrial Coating          |                                    |        |              |         |            |
|----------------------------------------------|------------------------------------|--------|--------------|---------|------------|
| New, uncoated<br>Existing                    | Existing,<br>previously<br>painted | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 9.2L-G5 (Semigloss)                  | MPI RIN 9.2L-G5 (Semigloss)        | MPI 50 | MPI 153      | MPI 153 | 4 mils     |
| Topcoat: Coating to match adjacent surfaces. |                                    |        |              |         |            |

| Alkyd                     |                                    |        |              |         |            |
|---------------------------|------------------------------------|--------|--------------|---------|------------|
| New, uncoated<br>Existing | Existing,<br>previously<br>painted | Primer | Intermediate | Topcoat | System DFT |
|                           |                                    |        |              |         |            |

|                                              |                                |        |        |        |        |
|----------------------------------------------|--------------------------------|--------|--------|--------|--------|
| MPI INT 9.2C-G5<br>(Semigloss)               | MPI RIN 9.2C-G5<br>(Semigloss) | MPI 50 | MPI 47 | MPI 47 | 4 mils |
| Topcoat: Coating to match adjacent surfaces. |                                |        |        |        |        |

Epoxy, New, uncoated Existing

| Epoxy                                        |        |              |         |            |
|----------------------------------------------|--------|--------------|---------|------------|
| New, uncoated Existing                       | Primer | Intermediate | Topcoat | System DFT |
| MPI INT 9.2E-G6 (Gloss)                      | MPI 50 | MPI 77       | MPI 77  | 4 mils     |
| Topcoat: Coating to match adjacent surfaces. |        |              |         |            |

Epoxy, Existing, previously painted

| Epoxy                                        |        |              |         |            |
|----------------------------------------------|--------|--------------|---------|------------|
| Existing, previously painted                 | Primer | Intermediate | Topcoat | System DFT |
| MPI RIN 9.2D-G6 (Gloss)                      | MPI 17 | MPI 77       | MPI 77  | 4 mils     |
| Topcoat: Coating to match adjacent surfaces. |        |              |         |            |

-- End of Section --

SECTION 10 14 00.10

EXTERIOR SIGNAGE  
**08/17, CHG 1: 11/18**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WELDING SOCIETY (AWS)

|                |                                                       |
|----------------|-------------------------------------------------------|
| AWS C1.1M/C1.1 | (2019) Recommended Practices for Resistance Welding   |
| AWS D1.1/D1.1M | (2020; Errata 1 2021) Structural Welding Code - Steel |

ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A123/A123M | (2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products                                         |
| ASTM A653/A653M | (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process |
| ASTM A924/A924M | (2022a) Standard Specification for General Requirements for Steel Sheet, Metallic-Coated by the Hot-Dip Process                         |
| ASTM B26/B26M   | (2018; E 2018) Standard Specification for Aluminum-Alloy Sand Castings                                                                  |
| ASTM B108/B108M | (2019) Standard Specification for Aluminum-Alloy Permanent Mold Castings                                                                |
| ASTM B209       | (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate                                                           |
| ASTM B221       | (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes                            |

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

|               |                              |
|---------------|------------------------------|
| NAAMM AMP 500 | (2006) Metal Finishes Manual |
|---------------|------------------------------|

1.2 GENERAL REQUIREMENTS

All exterior signage must be provided by a single manufacturer. Exterior signage must be of the design, detail, sizes, types, and message content shown on the drawings, must conform to the requirements specified, and

must be provided at the locations indicated. Submit exterior signage schedule in electronic media with spread sheet format. Spread sheet must include sign location, sign type, and message. Signs must be complete with lettering, framing as detailed, and related components for a complete installation. Each sample must consist of a complete sign panel with letters and symbols. Samples may be installed in the work, provided each sample is identified and location recorded. Submit three color samples for each material requiring color and 12 inch square sample of sign face color sample.

#### 1.2.1 Wind Load Requirements

Exterior signage must be designed to withstand windload as indicated on structural drawings. Submit design analysis and supporting calculations performed in support of specified signage.

#### 1.2.2 Character Proportions and Heights

Letters and numbers on indicated signs for handicapped-accessible buildings must have a width-to-height ratio between 3:5 and 1:1 and a stroke-width-to-height ratio between 1:5 and 1:10. Characters and numbers on indicated signs must be sized according to the viewing distance from which they are to be read. The minimum height is measured using an upper case letter "X". Lower case characters are permitted.

#### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Approved Detail Drawings; G

SD-03 Product Data

Installation

Exterior Signage; G

Wind Load Requirements

SD-04 Samples

Exterior Signage; G

SD-10 Operation and Maintenance Data

Protection and Cleaning; G

#### 1.4 QUALIFICATIONS

Signs, plaques, and dimensional letters must be the standard product of a



manufacturer regularly engaged in the manufacture of the products. Items of equipment must essentially duplicate equipment that has been in satisfactory use at least 2 years prior to bid opening.

#### 1.5 DELIVERY AND STORAGE

Materials must be wrapped for shipment and storage, delivered to the jobsite in manufacturer's original packaging, and stored in a clean, dry area in accordance with manufacturer's instructions.

#### 1.6 WARRANTY

Manufacturer's standard performance guarantees or warranties that extend beyond a one year period must be provided.

### PART 2 PRODUCTS

#### 2.1 DIMENSIONAL BUILDING LETTERS

##### 2.1.1 Fabrication

Letters must be fabricated from cast aluminum. Letters must be cleaned by chemical etching or cleaned ultrasonically in a special degreasing bath. Letters must be packaged for protection until installation.

##### 2.1.2 Typeface

Typeface must be helvetica medium.

##### 2.1.3 Size

Letter size must be as indicated.

##### 2.1.4 Finish

Baked enamel or two-component acrylic polyurethane finish must be provided.

##### 2.1.5 Mounting

Threaded studs of number and size as recommended by manufacturer, must be used for concealed anchorage. Letters which project from the building line must have stud spacer sleeves. Letters, studs, and sleeves must be of the same material. Supply templates for mounting.

#### 2.2 ALUMINUM ALLOY PRODUCTS

Aluminum alloy products must conform to ASTM B209 for sheet or plate, ASTM B221 for extrusions and ASTM B26/B26M or ASTM B108/B108M for castings. Aluminum extrusions must be provided at least 1/8 inch thick and aluminum plate or sheet at least 16 gauge thick. Welding for aluminum products must conform to AWS C1.1M/C1.1.

#### 2.3 ORGANIC COATING

Clean, prime and give surfaces a semi-gloss baked enamel or two-component acrylic polyurethane finish in accordance with NAAMM AMP 500, AMP 505, with total dry film thickness not less than 1.2 mils.

## 2.4 ANCHORS AND FASTENERS

Exposed anchor and fastener materials must be compatible with metal to which applied and must match in color and finish and must be non-rusting, non-corroding, and non-staining. Exposed fasteners must be tamper-proof.

## 2.5 SHOP FABRICATION AND MANUFACTURE

### 2.5.1 Factory Workmanship

Work must be assembled in the shop, as far as practical, ready for installation at the site. Work that cannot be shop assembled must be given a trial fit in the shop to ensure proper field assembly. Holes for bolts and screws must be drilled or punched. Drilling and punching must produce clean, true lines and surfaces. Welding to or on structural steel must be in accordance with AWS D1.1/D1.1M. Welding must be continuous along the entire area of contact. Exposed welds must be ground smooth. Exposed surfaces of work must have a smooth finish and exposed riveting must be flush. Fastenings must be concealed where practical. Items specified to be galvanized must be by hot-dip process after fabrication if practical. Galvanization must be in accordance with ASTM A123/A123M and ASTM A653/A653M, as applicable. Other metallic coatings of steel sheet must be in accordance with ASTM A924/A924M. Joints exposed to the weather must be formed to exclude water. Drainage and weep holes must be included as required to prevent condensation buildup.

### 2.5.2 Dissimilar Materials

Where dissimilar metals are in contact, or where aluminum is in contact with concrete, mortar, masonry, wet or pressure-treated wood, or absorptive materials subject to wetting, the surfaces must be protected with a coat of asphalt varnish or a coat of zinc-molybdate primer to prevent galvanic or corrosive action.

### 2.5.3 Shop Painting

Surfaces of miscellaneous metal work, except nonferrous metal, corrosion resisting steel, and zinc-coated work, must be given one coat of zinc-molybdate primer or an approved rust-resisting treatment and metallic primer in accordance with manufacturer's standard practice. Surfaces of items to be embedded in concrete must not be painted. Upon completion of work, damaged surfaces must be recoated.

## 2.6 COLOR, FINISH, AND CONTRAST

Color must be selected from manufacturers standard colors. For buildings required to be handicapped-accessible, the characters and background of signs must be eggshell, matte, or other non-glare finish. Characters and symbols must contrast with their background - either light characters on a dark background or dark characters on a light background.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Signs, plaques, or dimensional letters must be installed in accordance with approved manufacturer's instructions at locations shown on the approved detail drawings; submit drawings showing elevations of each type of sign; dimensions, details, and methods of mounting or anchoring; shape

and thickness of materials; and details of construction. A schedule showing the location, each sign type, and message must be included. Circuits installed underground must conform to the requirements of Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION. Steel conduits installed underground and illuminated signage mounted directly on buildings must be in conformance with the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Signs must be installed plumb and true at mounting heights indicated, and by method shown or specified. Signs mounted on other surfaces must not be installed until finishes on such surfaces have been completed. Submit manufacturer's installation instructions and cleaning instructions.

#### 3.1.1 Anchorage

Anchorage and fastener materials must be in accordance with approved manufacturer's instructions for the indicated substrate. Anchorage not otherwise specified or indicated must include slotted inserts, expansion shields, and powder-driven fasteners when approved for concrete; toggle bolts and through bolts for masonry; machine carriage bolts for steel; lag bolts and screws for wood.

#### 3.1.2 Protection and Cleaning

The work must be protected against damage during construction. Hardware and electrical equipment must be adjusted for proper operation. Glass, frames, and other sign surfaces must be cleaned in accordance with manufacturer's instructions. After signs are completed and inspected, cover all project identification, directional, and other signs which may mislead the public. Covering must be maintained until instructed to be removed by the Contracting Officer or until the facility is to be opened for business. Submit six copies of maintenance instructions listing routine maintenance procedures, possible breakdowns and repairs, and troubleshooting guides. The instructions must include simplified diagrams for the equipment as installed. Signs must be cleaned, as required, at time of cover removal.

#### 3.2 FIELD PAINTED FINISH

Miscellaneous metals and frames must be field painted in accordance with Section 09 90 00 PAINTS AND COATINGS. Anodized metals, masonry, and glass must be protected from paint. Finish must be free of scratches or other blemishes.

-- End of Section --

SECTION 10 14 00.20

INTERIOR SIGNAGE

08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

ALUMINUM ASSOCIATION (AA)

AA DAF45 (2003; Reaffirmed 2009) Designation System  
for Aluminum Finishes

AA PK-1 (2015) Pink Sheets: Designations and  
Chemical Composition Limits for Aluminum  
Alloys in the Form of Castings & Ingot

AMERICAN ARCHITECTURAL MANUFACTURERS ASSOCIATION (AAMA)

AAMA 2604 (2017a) Voluntary Specification,  
Performance Requirements and Test  
Procedures for High Performance Organic  
Coatings on Aluminum Extrusions and Panels

AMERICAN WELDING SOCIETY (AWS)

AWS D1.2/D1.2M (2014; Errata 1 2014; Errata 2 2020)  
Structural Welding Code - Aluminum

ASTM INTERNATIONAL (ASTM)

ASTM B209 (2014) Standard Specification for Aluminum  
and Aluminum-Alloy Sheet and Plate

ASTM B221 (2021) Standard Specification for Aluminum  
and Aluminum-Alloy Extruded Bars, Rods,  
Wire, Profiles, and Tubes

ASTM D635 (2018) Standard Test Method for Rate of  
Burning and/or Extent and Time of Burning  
of Plastics in a Horizontal Position

INTERNATIONAL CODE COUNCIL (ICC)

ICC A117.1 (2017) Standard And Commentary Accessible  
and Usable Buildings and Facilities

ICC/ANSI A117.1 (2009) Accessible and Usable Buildings and  
Facilities

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA LD 3 (2005) Standard for High-Pressure

Decorative Laminates

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 101 (2024) Life Safety Code

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST SP 800-82 (2015; Rev 2) Guide to Industrial Control  
Systems (ICS) Security

U.S. DEPARTMENT OF DEFENSE (DOD)

DOD 8510.01 (2022) Risk Management Framework (RMF) for  
DoD Systems

DODI 8500.01 (2014) Cybersecurity

UFC 4-010-06 (2023) Cybersecurity of Facility-Related  
Control Systems

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

36 CFR 1191 Americans with Disabilities Act (ADA)  
Accessibility Guidelines for Buildings and  
Facilities; Architectural Barriers Act  
(ABA) Accessibility Guidelines

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Detail Drawings; G

SD-03 Product Data

Room Identification And Directional Signage System; G

Stair Signage; G

Exit Door Tactile Sign; G

Door Tags; G

SD-04 Samples

Interior Signage; G

Room Identification And Directional Signage System; G

Stair Signage; G

Exit Door Tactile Sign; G

Door Tags; G

SD-10 Operation and Maintenance Data

Approved Manufacturer's Instructions; G

Protection and Cleaning; G

### 1.3 QUALITY ASSURANCE

#### 1.3.1 Samples

Submit interior signage samples of each of the following sign types showing typical quality, workmanship and color: all sign types included in project , Stair Signage, Exit Door Tactile Sign, Door Tags, , , , . Approved samples may be installed in the work, provided each sample is identified and location recorded.

#### 1.3.2 Detail Drawings

Submit detail drawings showing elevations of each type of sign, dimensions, details and methods of mounting or anchoring, mounting height, shape and thickness of materials, and details of construction. Include a schedule showing the location, each sign type, and message.

#### 1.3.3 Sign Fabricator

Sign Fabricator to follow room number strategies created by designer. The room numbering system to be reviewed and approved by the Contracting Officer and command end users during the shop drawing phase, and prior to fabrication.

#### 1.3.4 Cybersecurity

- a. The Risk Management Framework (RMF) is the process by which information systems are accredited for operation by a designated official from the Using Military Department. It is the standard process under which all DoD information systems achieve and maintain their Authority To Operate. The cybersecurity process is documented in DOD 8510.01 and NIST SP 800-82. Refer to UFC 4-010-06 and DODI 8500.01 for additional requirements.
- b. All systems that are IP addressable or interface with the Assured Network required certification to operate. Coordinate with the Government to initiate and complete the accreditation process.
- c. Cybersecurity requires input from the system vendor or provider and support from the local IMD. The local IMD-IA office is the point of contact for all Cyber Security requirements. The local CMIO is the

point of contact for all clinical and functional system requirements.

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Package materials to prevent damage and deterioration during shipment, handling, storage and installation. Deliver products to the jobsite in manufacturer's original packaging and store in a clean, dry area in accordance with manufacturer's instructions.

#### 1.5 WARRANTY

Provide manufacturer's warranty to repair or replace defective interior signage materials and workmanship for a period of 2 years from date of final acceptance of the work.

### PART 2 PRODUCTS

#### 2.1 ROOM IDENTIFICATION AND DIRECTIONAL SIGNAGE SYSTEM

Provide signs, plaques, directories, and dimensional building letters that are standard products of manufacturers regularly engaged in the manufacture of such products that essentially duplicate signs that have been in satisfactory use at least 2 years prior to bid opening. Obtain signage from a single manufacturer with edges and corners of finished letter forms and graphics true and clean.

##### 2.1.1 Panel Sign Systems

Provide decorative laminate face with applied tactile graphics, sign is fabricated of a balanced core sandwiched between 0.035 inch standard grade high pressure laminated faces. Provide signs that can accept images .

##### 2.1.2 Standard Room Signs

Provide signs that include tactile letters, symbols and Braille for interior rooms or spaces where the sign is not likely to change over time. Tactile text descriptions are required for pictograms that are provided to identify a permanent room. Examples include interior signs that label restrooms, stairs, room numbers or letters, and room names. These permanent room signs can include paper inserts for updatable information.

##### 2.1.2.1 Tactile Letters, Symbols and Braille

Provide ADA compliant material per 36 CFR 1191 which is raised 1/32 inch from the first surface, has a minimum 5/8 inch in height and is an ADA acceptable font. The color of the tactile letters is required to contrast with the sign face color per ADA standards. The ADA required Braille has a minimum durometer reading of 90. All raised letters, numbers and symbols are to comply.

##### 2.1.3 Directional Signs

Directional signs provide arrows with messages which point to critical destinations such as departments, offices, or other pertinent destinations. These can be a panel sign system with a series of permanently attached messages or a modular system with updatable inserts. Directional signs have header panels with applied or direct print messages.

#### 2.1.4 Message Inserts

Provide updatable message inserts covered with a clear matte 0.015 inch vinyl protective overlay. The insert is direct printed clear acetate over large format color print on white photo paper. Manufacturer is required to offer online ordering capabilities to facilitate and expedite ordering packages of replacement, color-coated paper inserts.

#### 2.1.5 Type of Mounting for Signs

Mount framed plaques with manufacturer's standard (1/6 inch) 1.59 mm thick closed cell vinyl foam with adhesive backing. Adhesive must be transparent, long aging, high tech formulation on two sides of the vinyl foam. Double-faced tape consisting of acrylic adhesive on polyurethane foam used in conjunction with silicone adhesive.

#### 2.1.6 Character Proportions and Heights

Letters and numbers on signs conform to 36 CFR 1191.

### 2.2 STAIR SIGNAGE

Provide signs on stairs serving three or more stories with special signage within the enclosure at each floor landing conforming to NFPA 101. Indicate the floor level, the terminus of the top and bottom of the stair enclosure, and the identification of the stair enclosure. Also, state the floor level of, and the direction to, exit discharge. Locate the signage inside the enclosure in a position that is visible when the door is in the open or closed position and install in conformance with 36 CFR 1191. Provide tactile for floor level designation in accordance with ICC A117.1.

### 2.3 EXIT DOOR TACTILE SIGN

Provide tactile sign with the message EXIT at each exit door that requires an exit sign to conform with NFPA 101. Sign tactile message is to comply with ICC/ANSI A117.1.

### 2.4 DOOR TAGS

Provide one door tag plate for each room entry door. Provide room number as determined by Contracting Officer.

#### 2.4.1 Engraved Copy

Machine engrave letters, numbers, symbols, and other graphics into panel sign on face to produce precisely formed copy and sharp images, incised to uniform depth. Melamine plastic engraving stock used for ADA compliant graphic is three-ply lamination contrasting color core meeting ASTM D635.

### 2.5 MATERIALS

#### 2.5.1 Aluminum Alloy Products

Aluminum extrusions are at least 1/8 inch thick, and aluminum plate or sheet are at least 0.0508 inch thick. Extrusions conform to ASTM B221; plate and sheet conforms to ASTM B209. Where anodic coatings are specified, alloy conforms to AA PK-1 alloy designation 514.0. Exposed anodized aluminum finishes are as shown. Welding for aluminum products conforms to AWS D1.2/D1.2M.



#### 2.5.2 Anodic Coating

Anodized finish conforms to AA DAF45 as follows:

- a.
- b. Integral color anodized designation AA-M10-C22-A32, Architectural Class 0.4 to 0.7 mil.
- c.

#### 2.5.3 Organic Coating

Organic coating conforms to AAMA 2604, with total dry film thickness not less than 1.2 mils.

#### 2.5.4 Plastic Laminate Sheet

ANSI/NEMA LD 3, general purpose HGS grade, 0.048 inch nominal thickness.

#### 2.5.5 Fabrication and Manufacture

##### 2.5.5.1 Factory Workmanship

Holes for bolts and screws are drilled or punched. Drilling and punching produces clean, true lines and surfaces. Exposed surfaces of work have a smooth finish; exposed riveting is flush. Conceal fastenings where practicable.

##### 2.5.5.2 Dissimilar Materials

Where dissimilar metals are in contact, protect surfaces prevent galvanic or corrosive action.

#### 2.5.6 Typeface

ADA-ABA compliant font for Room Signs.

### 2.6 GRAPHICS

Provide signage graphics for modular signs to the following:

#### 2.6.1 Photopolymer

Integral graphics and Braille achieved by photomechanical stratification processes. Provide photopolymer used for ADA compliant graphics of the type that has a minimum durometer reading of 90. Tactile graphics are raised 1/32 inch from the first surface of plaque by photomechanical stratification process.

### 2.7 COLOR, FINISH, AND CONTRAST

Provide color for review from full range of colors from manufacturers. Finish of eggshell, matte, or other non-glare finish for all signs as required in handicapped-accessible buildings.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Install signs plumb and true and in accordance with approved manufacturer's instructions at locations shown on the detail drawings. Submit operating instructions outlining the step-by-step procedures required for system operation. The instructions include simplified diagrams for the system as installed, the manufacturer's name, model number, service manual, parts list, and brief description of all equipment and their basic operating features. Provide each set permanently bound with a hard cover. The following identification must be inscribed on the covers: "OPERATING AND MAINTENANCE INSTRUCTIONS", name and location of the facility, name of the Contractor, and contract number. Submit in accordance with Section 01 78 23 OPERATING AND MAINTENANCE DATA. Mounting height and mounting location complies with 36 CFR 1191. Install required blocking. Do not install signs on doors or other surfaces until finishes on such surfaces have been installed. Signs installed on glass surfaces are installed with matching blank back-up plates in accordance with manufacturer's instructions.

Do not install items that show visual evidence of biological growth.

#### 3.1.1 Anchorage

Provide anchorage in accordance with approved manufacturer's instructions. Anchorage not otherwise specified or shown includes slotted inserts, expansion shields, and powder-driven fasteners when approved for concrete; toggle bolts and through bolts for masonry; machine carriage bolts for steel; lag bolts and screws for wood. Provide exposed anchor and fastener materials compatible with metal to which applied with matching color and finish.

- a. Signs mounted to painted gypsum board surfaces must be removable for painting maintenance.

#### 3.1.2 Protection and Cleaning

Protect the work against damage during construction. Adjust hardware and electrical equipment for proper operation. Clean glass, frames, and other sign surfaces at completion of signage installation in accordance with the manufacturer's written instructions.

-- End of Section --

SECTION 10 44 16

FIRE EXTINGUISHERS  
**11/19**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

- |          |                                                              |
|----------|--------------------------------------------------------------|
| NFPA 1   | (2021) Fire Code                                             |
| NFPA 10  | (2022; ERTA 1 2021) Standard for Portable Fire Extinguishers |
| NFPA 101 | (2024) Life Safety Code                                      |

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

- |                 |                                    |
|-----------------|------------------------------------|
| 29 CFR 1910.106 | Flammable Liquids                  |
| 29 CFR 1910.157 | (2003) Portable Fire Extinguishers |

UNDERWRITERS LABORATORIES (UL)

- |        |                                                  |
|--------|--------------------------------------------------|
| UL 299 | (2012; May 2021) Dry Chemical Fire Extinguishers |
|--------|--------------------------------------------------|

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

- Fire Extinguishers; G
- Accessories; G
- Cabinets; G
- Wall Brackets; G
- Schedule; G

SD-03 Product Data

- Fire Extinguishers; G
- Accessories; G

Cabinets; G

Wall Brackets; G

Replacement Parts List; G

SD-04 Samples

Equipment Samples; G

SD-07 Certificates

Fire Extinguishers Certifications; G

Manufacturer's Warranty with Inspection Tag; G

### 1.3 DELIVERY, STORAGE, AND HANDLING

Protect materials from weather, soil, and damage during delivery, storage, and construction.

Deliver materials in their original packages, containers, or bundles bearing the brand name and the name and type of the material.

#### 1.3.1 Samples

Provide the following equipment samples: One of each type of fire extinguisher being installed; one full-sized sample of each type of cabinet being installed; three samples of wall brackets and accessories of each type being used.

Use approved samples for installation, with proper identification and storage.

### 1.4 WARRANTY

Guarantee that Fire Extinguishers are free of defects in materials, fabrication, finish, and installation and that they will remain so for a period of not less than 2 years after completion.

Submit the manufacturer's warranty with inspection tag.

### 1.5 PROJECT SCHEDULE

For fire extinguishers. Coordinate final fire extinguisher schedule with fire protection cabinet schedule to ensure proper fit and function. Use same designations indicated on Drawings.

## PART 2 PRODUCTS

Submit fabrication drawings consisting of fabrication and assembly details performed in the factory and product data for the following items: Fire Extinguishers; Accessories, cabinets, Wall Brackets.

### 2.1 SYSTEM DESCRIPTION

#### 2.1.1 Types

Submit fire extinguishers certifications showing compliance with local

codes and regulations.

Provide fire extinguishers conforming to NFPA 10. Provide quantity and placement in compliance with the applicable sections of NFPA 1, NFPA 101 29 CFR 1910.106 and 29 CFR 1910.157.

Provide dry chemical type fire extinguishers compliant with UL 299.

#### 2.1.2 Material

Provide enameled steel extinguisher shell.

#### 2.1.3 Size

10 pounds extinguishers.

### 2.2 EQUIPMENT

#### 2.2.1 Wall Brackets

Provide wall-hook fire extinguisher wall brackets.

Provide wall bracket and accessories as approved.

##### 2.2.1.1 Identification

Provide lettering complying with authorities having jurisdiction for letter style, size, spacing, and location. The preferred method of identification is to use wall mounted tent style plastic or metal signs. Locate as indicated by the drawings.

Identify bracket-mounted fire extinguishers with the words "FIRE EXTINGUISHER" in red letter decals applied to mounting surface.

Orientation: Vertical.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Install Fire Extinguishers where indicated on the drawings. Verify exact locations prior to installation.

Provide extinguishers which are fully charged and ready for operation upon installation. Provide extinguishers complete with Manufacturer's Warranty with Inspection Tag attached.

Install fire extinguishers in locations indicated and in compliance with requirements of authorities having jurisdiction.

Comply with the manufacturer's recommendations for all installations.

### 3.2 PROTECTION

#### 3.2.1 Repairing

Remove and replace damaged and unacceptable portions of completed work with new work at no additional cost to the Government.

Submit replacement parts list indicating specified items replacement part, replacement cost, and name, address and contact for replacement parts distributor.

### 3.2.2 Cleaning

Clean all surfaces of the work, and adjacent surfaces which are soiled as a result of the work. Remove from the site all construction equipment, tools, surplus materials and rubbish resulting from the work.

-- End of Section --

SECTION 13 34 19

METAL BUILDING SYSTEMS

08/20, CHG 1: 02/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 325 (2017) Steel Construction Manual

AISC 360 (2016) Specification for Structural Steel Buildings

AMERICAN IRON AND STEEL INSTITUTE (AISI)

AISC/AISI 121 (2007) Standard Definitions for Use in the Design of Steel Structures

AISI D100 (2017) Cold-Formed Steel Design Manual

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 90.1 - IP (2019) Energy Standard for Buildings Except Low-Rise Residential Buildings

ASHRAE 90.1 - SI (2019) Energy Standard for Buildings Except Low-Rise Residential Buildings

AMERICAN WELDING SOCIETY (AWS)

AWS A5.1/A5.1M (2012) Specification for Carbon Steel Electrodes for Shielded Metal Arc Welding

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding Code - Steel

AWS D1.3/D1.3M (2018) Structural Welding Code - Sheet Steel

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

|                 |                                                                                                                                                                         |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A53/A53M   | (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless                                                                   |
| ASTM A123/A123M | (2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products                                                                         |
| ASTM A153/A153M | (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware                                                                                     |
| ASTM A307       | (2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength                                                               |
| ASTM A500/A500M | (2023) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes                                                   |
| ASTM A501/A501M | (2021) Standard Specification for Hot-Formed Welded and Seamless Carbon Steel Structural Tubing                                                                         |
| ASTM A529/A529M | (2019) Standard Specification for High-Strength Carbon-Manganese Steel of Structural Quality                                                                            |
| ASTM A563       | (2021; E 2022a) Standard Specification for Carbon and Alloy Steel Nuts                                                                                                  |
| ASTM A563M      | (2007; R 2013) Standard Specification for Carbon and Alloy Steel Nuts (Metric)                                                                                          |
| ASTM A572/A572M | (2021; E 2021) Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Structural Steel                                                                   |
| ASTM A606/A606M | (2023) Standard Specification for Steel Sheet and Strip, High-Strength, Low-Alloy, Hot-Rolled and Cold-Rolled, with Improved Atmospheric Corrosion Resistance           |
| ASTM A653/A653M | (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process                                 |
| ASTM A755/A755M | (2018) Standard Specification for Steel Sheet, Metallic Coated by the Hot-Dip Process and Prepainted by the Coil-Coating Process for Exterior Exposed Building Products |
| ASTM A780/A780M | (2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings                                                                        |



|                   |                                                                                                                                                                                                     |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A992/A992M   | (2022) Standard Specification for Structural Steel Shapes                                                                                                                                           |
| ASTM A1008/A1008M | (2023) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable |
| ASTM A1011/A1011M | (2023) Standard Specification for Steel Sheet and Strip, Hot-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, and Ultra-High Strength        |
| ASTM B117         | (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus                                                                                                                                   |
| ASTM B209         | (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate                                                                                                                       |
| ASTM B221         | (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes                                                                                        |
| ASTM B221M        | (2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes (Metric)                                                                               |
| ASTM B695         | (2021) Standard Specification for Coatings of Zinc Mechanically Deposited on Iron and Steel                                                                                                         |
| ASTM C273/C273M   | (2020) Standard Test Method for Shear Properties of Sandwich Core Materials                                                                                                                         |
| ASTM C518         | (2021) Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter Apparatus                                                                              |
| ASTM C920         | (2018) Standard Specification for Elastomeric Joint Sealants                                                                                                                                        |
| ASTM C1289        | (2023a) Standard Specification for Faced Rigid Cellular Polyisocyanurate Thermal Insulation Board                                                                                                   |
| ASTM C1363        | (2024) Standard Test Method for Thermal Performance of Building Materials and Envelope Assemblies by Means of a Hot Box Apparatus                                                                   |
| ASTM D522/D522M   | (2017) Mandrel Bend Test of Attached Organic Coatings                                                                                                                                               |
| ASTM D523         | (2014; R 2018) Standard Test Method for Specular Gloss                                                                                                                                              |

|                   |                                                                                                                                   |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| ASTM D714         | (2002; R 2017) Standard Test Method for Evaluating Degree of Blistering of Paints                                                 |
| ASTM D822         | (2013; R 2018) Filtered Open-Flame Carbon-Arc Exposures of Paint and Related Coatings                                             |
| ASTM D968         | (2022) Standard Test Methods for Abrasion Resistance of Organic Coatings by Falling Abrasive                                      |
| ASTM D1056        | (2020) Standard Specification for Flexible Cellular Materials - Sponge or Expanded Rubber                                         |
| ASTM D1308        | (2002; R 2013) Effect of Household Chemicals on Clear and Pigmented Organic Finishes                                              |
| ASTM D1621        | (2016) Standard Test Method for Compressive Properties of Rigid Cellular Plastics                                                 |
| ASTM D1622/D1622M | (2014) Apparent Density of Rigid Cellular Plastics                                                                                |
| ASTM D1667        | (2022) Standard Specification for Flexible Cellular Materials - Poly (Vinyl Chloride) Foam (Closed-Cell)                          |
| ASTM D2244        | (2021) Standard Practice for Calculation of Color Tolerances and Color Differences from Instrumentally Measured Color Coordinates |
| ASTM D2247        | (2015; R 2020) Standard Practice for Testing Water Resistance of Coatings in 100% Relative Humidity                               |
| ASTM D2794        | (1993; R 2019) Standard Test Method for Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact)               |
| ASTM D3363        | (2005; E 2011; R 2011; E 2012) Film Hardness by Pencil Test                                                                       |
| ASTM D4214        | (2007; R 2015) Standard Test Method for Evaluating the Degree of Chalking of Exterior Paint Films                                 |
| ASTM D6226        | (2021) Standard Test Method for Open Cell Content of Rigid Cellular Plastics                                                      |
| ASTM DEFONLINE    | (2008) ASTM Online Dictionary of Engineering Science and Technology                                                               |
| ASTM E84          | (2023) Standard Test Method for Surface Burning Characteristics of Building Materials                                             |

|                   |                                                                                                                                                                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM E96/E96M     | (2024) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials                                                                                                                                                |
| ASTM E119         | (2022) Standard Test Methods for Fire Tests of Building Construction and Materials                                                                                                                                                                      |
| ASTM E136         | (2024a) Standard Test Method for Assessing Combustibility of Materials Using a Vertical Tube Furnace at 750 Degrees C                                                                                                                                   |
| ASTM E283         | (2019) Standard Test Method for Determining the Rate of Air Leakage Through Exterior Windows, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen                                                                         |
| ASTM E331         | (2000; R 2023) Standard Test Method for Water Penetration of Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform Static Air Pressure Difference                                                                                            |
| ASTM E1592        | (2017) Standard Test Method for Structural Performance of Sheet Metal Roof and Siding Systems by Uniform Static Air Pressure Difference                                                                                                                 |
| ASTM E1646        | (1995; R 2018) Standard Test Method for Water Penetration of Exterior Metal Roof Panel Systems by Uniform Air Pressure Difference                                                                                                                       |
| ASTM E1680        | (2016) Standard Test Method for Rate of Air Leakage Through Exterior Metal Roof Panel Systems                                                                                                                                                           |
| ASTM F436/F436M   | (2019) Standard Specification for Hardened Steel Washers Inch and Metric Dimensions                                                                                                                                                                     |
| ASTM F844         | (2019) Standard Specification for Washers, Steel, Plain (Flat), Unhardened for General Use                                                                                                                                                              |
| ASTM F1852        | (2014) Standard Specification for "Twist Off" Type Tension Control Structural Bolt/Nut/Washer Assemblies, Steel, Heat Treated, 120/105 ksi Minimum Tensile Strength                                                                                     |
| ASTM F3125/F3125M | (2019) Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength |

ASTM G152 (2013; R 2021) Standard Practice for  
Operating Open Flame Carbon Arc Light  
Apparatus for Exposure of Nonmetallic  
Materials

ASTM G153 (2013; R 2021) Standard Practice for  
Operating Enclosed Carbon Arc Light  
Apparatus for Exposure of Nonmetallic  
Materials

METAL BUILDING MANUFACTURERS ASSOCIATION (MBMA)

MBMA MBSM (2018) Metal Building Systems Manual

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM AMP 500 (2006) Metal Finishes Manual

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 80 (2022) Standard for Fire Doors and Other  
Opening Protectives

NFPA 252 (2022) Standard Methods of Fire Tests of  
Door Assemblies

NATIONAL ROOFING CONTRACTORS ASSOCIATION (NRCA)

NRCA RoofMan (2020) The NRCA Roofing Manual

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION  
(SMACNA)

SMACNA 1793 (2012) Architectural Sheet Metal Manual,  
7th Edition

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC Paint 15 (1999; E 2004) Steel Joist Shop  
Primer/Metal Building Primer

SSPC Painting Manual (2002) Good Painting Practice, Steel  
Structures Painting Manual, Volume 1

SSPC SP 2 (2018) Hand Tool Cleaning

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety -- Safety and Health  
Requirements Manual

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01 (2023; with Change 1, 2023) Structural  
Engineering

UFC 3-301-02 (2023) Design of Risk Category V  
Structures, National Strategic Military  
Assets

UNDERWRITERS LABORATORIES (UL)

UL Bld Mat Dir

(updated continuously online) Building  
Materials Directory

1.2 GENERAL REQUIREMENTS

1.2.1 Design Parameters

Design and construct pre-engineered metal buildings of size, shape, height, fenestration, siting, and configuration indicated. Coordinate site utility services, accessibility requirements, vehicular and pedestrian access, mechanical, electrical, plumbing and fire protection requirements, interior construction and finishes, and such other items as may be necessary for a complete, functional building.

1.2.2 Structural Performance

Provide metal building systems capable of withstanding the effects of gravity loads and the following loads and stresses within the limits and conditions indicated.

1.2.2.1 Engineering

Design metal building systems conforming to procedures described in MBMA MBSM.

1.2.2.2 Design Loads

Design and construct to the requirements of UFC 3-301-01, Structural Engineering.

1.2.3 Thermal Performance

Provide insulated metal panel assemblies with the following maximum U-factors when assemblies are tested or calculated according to ASHRAE 90.1 - SI ASHRAE 90.1 - IP Appendix A, and minimum R-values for opaque elements when tested according to ASTM C1363 or ASTM C518.

1.2.3.1 Metal Roof Panel Assemblies

b. R-Value: 38

1.2.3.2 Metal Wall Panel Assemblies

b. R-Value: 32

1.2.4 Air Infiltration for Metal Roof Panels

Air leakage through assembly must not exceed 0.04 cfm/sq.ft. of roof area when lab tested according to ASTM E1680 at negative test-pressure difference of 1.57 lbf/sq.ft..

1.2.5 Air Infiltration for Metal Wall Panels

Air leakage through assembly of not more than 0.04 cfm/sq.ft. of wall area when labtested according to ASTM E283 at static-air-pressure difference of 1.57 lbf/sq.ft..

#### 1.2.6 Water Penetration for Metal Roof Panels

No water penetration when tested according to ASTM E1646 at test-pressure difference of 2.86 lbf/sq.ft..

#### 1.2.7 Water Penetration for Metal Wall Panels

No water penetration when tested according to ASTM E331 at a minimum differential pressure of 20 percent of inward-acting, wind-load design pressure of not less than 6.24 lbf/sq.ft. and not more than 12 lbf/sq. ft.

#### 1.2.8 Specular Gloss

Finished roof surfaces to have a specular gloss value of 30 plus or minus 5 at an angle of 60 degrees when measured in accordance with ASTM D523.

#### 1.2.9 Wind-Uplift Resistance

Design for wind-uplift resistance in accordance with UFC 3-301-01.

#### 1.2.10 Erection Plan

Provide plans and a written erection/lifting procedure with required plans clearly showing the intended sequence and method of erection in accordance with EM 385-1-1 "Safety - Safety and Health Requirements". Indicate required crane lifting requirements, temporary support structures, member size and locations of braced or guyed temporary supports, and locations of bracing or guys anchor points. Clearly define the required framing sequence and conditions necessary to ensure the structure is maintained in a properly braced and stable condition throughout the complete erection process.

### 1.3 DEFINITIONS

- a. Bay: Dimension between main frames measured normal to frame (at centerline of frame) for interior bays, and dimension from centerline of first interior main frame measured normal to end wall (outside face of end-wall girt) for end bays.
- b. Clear Span: Distance between supports of beams, girders, or trusses (measured from lowest level of connecting area of a column and a rafter frame or knee).
- c. Eave Height: Vertical dimension from finished floor to eave (the line along the sidewall formed by intersection of the planes of the roof and wall).
- d. Terminology Standard: Refer to MBMA "Metal Building Systems Manual" for definitions of terms for metal building system construction not otherwise defined in this Section or in referenced standards.

### 1.4 SYSTEM DESCRIPTION

General: Provide a complete, integrated set of metal building system manufacturer's standard mutually dependent components and assemblies that form a metal building system capable of withstanding structural and other loads, thermally induced movement, and exposure to weather without failure or infiltration of water into building interior. Include primary and

secondary framing, metal roof panels, metal wall panels, and accessories complying with requirements indicated.

Provide metal building system of size and with spacing, slopes, and spans indicated.

#### 1.4.1 Primary Frame Type

- a. Lean To: Solid- or truss-member, structural-framing system without interior columns, designed to be partially supported by another structure.

#### 1.4.2 Fixed End-Wall Framing

Provide manufacturer's standard fixed end wall, for buildings not required to be expandable, consisting of primary frame, capable of supporting one-half of a bay design load, and end-wall columns.

#### 1.4.3 Secondary Frame Type

Provide manufacturer's standard purlins and joists and girts.

#### 1.4.4 Eave Height

Eave height is noted on drawings.

#### 1.4.5 Bay Spacing

Bay Spacing noted on drawings.

#### 1.4.6 Roof Slope

Roof slope must be 3/4 inch per 12 inches.

#### 1.4.7 Roof System

Provide manufacturer's standard vertical-rib, standing-seam with insulation.

#### 1.4.8 Exterior Wall System

Provide factory-assembled, insulated metal wall panels complete with vapor barrier conforming to ASTM E96/E96M

### 1.5 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-01 Preconstruction Submittals

Manufacturer's Qualifications; G

#### SD-02 Shop Drawings

Detail Drawings; G

Erection Plan; G

SD-03 Product Data

Manufacturer's Catalog Data; G  
Recycled Content for Structural Steel Shapes and Plates;  
Recycled Content for Steel Pipe;  
Recycled Content for Aluminum Sheet Materials;  
Recycled Content for Steel Sheet Materials;  
Recycled Content for Insulation Materials;

SD-04 Samples

Coil Stock, 12 inches long by the actual panel width; G  
Roof Panels, 12 inches long by actual panel width; G  
Wall Panels, 12 inches long by actual panel width; G  
Fasteners; G  
Metal Closure Strips 10 inches long of each type; G  
Insulation, approximately 8 by 11 inches; G  
Vapor Barrier; G  
Manufacturer's Color Charts and Chips, 4 by 4 inches; G

SD-05 Design Data

Manufacturer's Descriptive and Technical Literature; G  
Manufacturer's Building Design Analysis; G  
Lateral Force Calculations; G

SD-06 Test Reports

Test Reports; G  
Coatings and Base Metals; G  
Factory Color Finish Performance Requirements; G

SD-07 Certificates

System Components; G  
Coil Stock Certificates; G  
Aluminized Steel Repair Paint; G  
Galvanizing Repair Paint; G



Enamel Repair Paint; G

Qualification of Manufacturer; G

Qualification of Erector; G

SD-08 Manufacturer's Instructions

Installation of Roof and Wall panels; G

Shipping, Handling, and Storage; G

SD-11 Closeout Submittals

Manufacturer's Warranty; G

Contractor's Warranty for Installation; G

1.6 QUALITY ASSURANCE

1.6.1 Pre-Erection Conference

After submittals are received and approved but before metal building system work, including associated work, is performed, the Contracting Officer will hold a pre-erection conference to review the following:

- a. The detail drawings, specifications, and manufacturer's descriptive and technical literature.
- b. Finalize construction schedule and verify availability of materials, erector's personnel, equipment, and facilities needed to make progress and avoid delays.
- c. Methods and procedures related to metal building system erection, including, but not limited to: qualification of manufacturer, qualification of erector, manufacturer's catalog data, manufacturer's building design analysis, lateral force calculations, written instructions and test reports. Lateral force calculations must include all analysis and confirmation of system components required to transfer lateral forces to the foundation.
- d. Support conditions for compliance with requirements, including alignment between and erection of structural members.
- e. Flashing, special roofing and siding details, roof and wall penetrations, openings, and condition of other construction that will affect the metal building system, including coatings and base metals, factory color finish performance requirements, system components, and coil stock certificates.
- f. Governing regulations and requirements for, certificates, insurance, tests and inspections if applicable.
- g. Temporary protection requirements for metal panel assembly during and after installation.
- h. Samples of roof panels, wall panels, aluminized steel repair paint, galvanizing repair paint, and enamel repair paint.

#### 1.6.1.1 Pre-Roofing and Siding Installation Conference

After structural framing system erection and approval but before roofing, siding, insulation and vapor barrier work, including associated work, is performed; the Contracting Officer will hold a pre-roofing and siding conference to review the following:

- a. Examine purlins, sub-girts and formed shapes conditions for compliance with requirements, including flatness and attachment to structural members.
- b. Review structural limitations of purlins, sub-girts and formed shapes during construction and after roofing and siding.
- c. Review flashings, special roof and wall details, roof drainage, roof and wall penetrations, roof equipment curbs, and condition of other construction that will affect the metal building system.
- d. Review temporary protection requirements for metal roof and wall panels' assembly during and after installation.
- e. Review roof and wall observation and repair procedures after metal building system erection.

#### 1.6.2 Manufacturer's Technical Representative

The representative must have authorization from manufacturer to approve field changes and be thoroughly familiar with the products, erection of structural framing and installation of roof and wall panels in the geographical area where construction will take place.

#### 1.6.3 Manufacturer's Qualifications

Metal building system manufacturer must have a minimum of five years experience as a qualified manufacturer and a member of MBMA of metal building systems and accessory products.

Provide engineering services by an authorized currently licensed engineer in the geographical area where construction will take place, having a minimum of four years experience as an engineer knowledgeable in building design analysis, protocols and procedures for the "Metal Building Systems Manual" (MBMA MBSM); ASCE 7-16, the building code in the geographic area where the construction will take place and ASTM E1592. Provide certified engineering calculations using the products submitted for:

- a. Roof and Wall Wind Loads with basic wind speed, exposure category, co-efficient, importance factor, designate type of facility, negative pressures for each zone, methods and requirements of attachment.
- b. Roof Dead and Live Loads
- c. Collateral Loads
- d. Foundation Loads
- e. Roof Snow Load
- f. Seismic Loads

#### 1.6.4 Qualification of Erection Contractor

An experienced erector who has specialized in erecting and installing work similar in material, design, and extent to that indicated for this Project and must be approved and certified by the metal building system manufacturer.

#### 1.6.5 Single Source

Obtain primary and secondary components and structural framing members, each type of metal roof, wall and liner panel assemblies, clips, closures and other accessories from the standard products of the single source from a single manufacturer to operate as a complete system for the intended use.

#### 1.6.6 Welding

Qualify procedures and personnel according to AWS A5.1/A5.1M, AWS D1.1/D1.1M, and AWS D1.3/D1.3M.

#### 1.6.7 Structural Steel

Comply with AISC 325, AISC 360, for design requirements and allowable stresses.

#### 1.6.8 Cold-Formed Steel

Comply with AISC/AISI 121 and AISI D100 for design requirements and allowable stresses.

#### 1.6.9 Fire-Resistance Ratings

Where indicated, provide metal panels identical to those of assemblies tested for fire resistance in accordance with ASTM E119, as certified by a qualified testing agency. Identify products with appropriate markings of applicable testing agency.

Indicate design designations from UL Bld Mat Dir or from the listings of another qualified testing agency. Combustion Characteristics must conform to ASTM E136.

#### 1.6.10 Surface-Burning Characteristics

Provide metal panels having insulation and vapor barrier material with the following surface-burning characteristics as determined by testing identical products according to ASTM E84 by a qualified testing agency. Identify products with appropriate markings of applicable testing agency showing:

- a. Flame-Spread Index: 25 or less.
- b. Smoke-Developed Index: 50 or less.

#### 1.6.11 Fabrication

Fabricate and finish metal panels and accessories at the factory to greatest extent possible, by manufacturer's standard procedures and processes and as necessary to fulfill indicated performance requirements. Comply with indicated profiles with dimensional and structural requirements. Provide metal panel profile, including major ribs and

intermediate stiffening ribs, if any, for full length of panel. Aluminum and aluminum-alloy sheet and plate must conform to ASTM B209. Fabricate metal panel side laps with factory-installed captive gaskets or separator strips that provide a tight seal and prevent metal-to-metal contact, in a manner that will seal weather-tight and minimize noise from movements within panel assembly.

Sheet Metal Accessories: Fabricate flashing and trim to comply with recommendations in SMACNA 1793 that apply to the design, dimensions, metal, and other characteristics of item indicated:

- a. Form exposed sheet metal accessories that are without excessive oil canning, buckling, and tool marks and that are true to line and levels indicated, with exposed edges folded back to form hems.
- b. End Seams: Fabricate nonmoving seams with flat-lock seams. Form seams and seal with epoxy seam sealer. Rivet joints for additional strength.
- c. Sealed Joints: Form non-expansion but movable joints in metal to accommodate elastomeric sealant to comply with SMACNA standards.
- d. Conceal fasteners and expansion provisions where possible. Exposed fasteners are not allowed on faces of accessories exposed to view.
- e. Fabricate cleats and attachment devices of size and metal thickness recommended by SMACNA or by metal building system manufacturer for application, but not less than thickness of metal being secured.

#### 1.6.12 Finishes

Comply with NAAMM AMP 500 for recommendations for applying and designating finishes.

Appearance of Finished Work: Noticeable variations in same piece are not acceptable. Variations in appearance of adjoining components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

### 1.7 SHIPPING, HANDLING AND STORAGE

#### 1.7.1 Delivery

Package and deliver components, sheets, metal panels, and other manufactured items so as not to be damaged or deformed and protected during transportation and handling.

#### 1.7.2 Storage

Stack and store metal panels horizontally on platforms or pallets, covered with suitable weather-tight and ventilated covering to ensure dryness, with positive slope for drainage of water. Store in a manner to prevent bending, warping, twisting, and surface damage. Do not store metal wall panels in contact with other materials that might cause staining, denting, or other surface damage. Retain strippable protective covering on metal panel for entire period up to metal panel installation.

### 1.7.3 Protection of Materials

Protect foam-plastic insulation as follows:

- a. Do not expose to sunlight, except to extent necessary for period of installation and concealment.
- b. Protect against ignition at all times. Do not deliver foam-plastic insulation materials to project site before installation time.

Complete installation and concealment of plastic materials as rapidly as possible in each area of construction to minimize ultraviolet exposure.

## 1.8 PROJECT CONDITIONS

### 1.8.1 Weather Limitations

Proceed with installation preparation only when existing and forecasted weather conditions permit work to proceed without water entering into existing panel system or building.

### 1.8.2 Field Measurements

#### 1.8.2.1 Established Dimensions for Foundations

Comply with established dimensions on approved anchor-bolt plans, established foundation dimensions, and proceed with fabricating structural framing. Do not proceed without verifying field measurements. Coordinate anchor-bolt installation to ensure that actual anchorage dimensions correspond to established dimensions.

#### 1.8.2.2 Established Dimensions for Metal Panels

Where field measurements cannot be made without delaying the work, either establish framing and opening dimensions and proceed with fabricating metal panels without field measurements, or allow for field trimming metal panels. Coordinate construction to ensure that actual building dimensions, locations of structural members, and openings correspond to established dimensions.

#### 1.8.2.3 Verification Record

Verify locations of all framing and opening dimensions by field measurements before metal panel fabrication and indicate measurements on Shop Drawings.

## 1.9 COORDINATION

Coordinate final design and placement of foundation between structural engineer of record, geotechnical engineer, MBMA and Contractor. Coordinate size and location of concrete foundations and casting of anchor-bolt inserts into foundation walls and footings. Concrete, reinforcement, and formwork requirements are specified in section on CAST-IN-PLACE CONCRETE.

Coordinate metal panel assemblies with rain drainage work, flashing, trim, and construction of supports and other adjoining work to provide a leak-proof, secure, and non-corrosive installation.

## 1.10 WARRANTY

### 1.10.1 Building System Warranty

Furnish manufacturer's no-dollar-limit warranty for the metal building system. The warranty period is to be no less than 20 years from the date of acceptance of the work and be issued directly to the Government. The warranty must provide that if within the warranty period, the metal building system shows evidence of deterioration resulting from defective materials or workmanship, correcting of any defects is the responsibility of the metal building system manufacturer. Repairs that become necessary because of defective materials and workmanship while metal building system is under warranty are to be performed within 32 hours after notification, unless additional time is approved by the Contracting Officer. Failure to perform repairs within 32 hours of notification will constitute grounds for having emergency repairs performed by others and will not void the warranty.

### 1.10.2 Roof System Weather-Tightness Warranty

Furnish manufacturer's no-dollar-limit warranty for the metal panel system. The warranty period is to be no less than 20 years from the date of acceptance of the work and be issued directly to the Government.

The warranty is to provide that if within the warranty period the roof panel system shows evidence of corrosion, perforation, rupture, lost of weather-tightness or excess weathering due to deterioration of the panel system resulting from defective materials and correction of the defective workmanship is to be the responsibility of the metal building system manufacturer.

Repairs that become necessary because of defective materials and workmanship while roof panel system is under warranty are to be performed within 24 hours after notification, unless additional time is approved by the Contracting Officer. Failure to perform temporary repairs within 24 hours of notification will constitute grounds for having emergency repairs performed by others and not void the warranty. Immediate follow-up and completion of permanent repairs must be performed within 10 days from date of notification.

### 1.10.3 Roof and Wall Panel Finish Warranty

Furnish manufacturer's no-dollar-limit warranty for the metal panel system. The warranty period is to be no less than 20 years from the date of acceptance of the work and be issued directly to the Government.

The warranty is to provide that if within the warranty period the metal panel system shows evidence of checking, delaminating cracking, peeling, chalk in excess of a numerical rating of eight, as determined by ASTM D4214 test procedures; or change colors in excess of five CIE or Hunter units in accordance with ASTM D2244 or excess weathering due to deterioration of the panel system resulting from defective materials and finish or correction of the defective workmanship is to be the responsibility of the metal building system manufacturer.

Liability under this warranty is exclusively limited to replacing the defective coated materials.

Repairs that become necessary because of defective materials and

workmanship while roof and wall panel system is under warranty are to be performed within 32 hours after notification, unless additional time is approved by the Contracting Officer. Failure to perform repairs within 32 hours of notification will constitute grounds for having emergency repairs performed by others and not void the warranty.

## PART 2 PRODUCTS

### 2.1 STRUCTURAL FRAMING MATERIALS

#### 2.1.1 Steel Shapes and Plates

Wide flange and WT shapes: ASTM A992/A992M; ASTM A572/A572M or ASTM A529/A529M. Angles, Channels and Plates: ASTM A36/A36M, ASTM A572/A572M or ASTM A529/A529M. Provide structural steel shapes and plates containing a minimum of 80 percent recycled content. Submit data identifying percentage of recycled content for structural steel shapes and plates.

#### 2.1.2 Steel Pipe

ASTM A36/A36M, ASTM A53/A53M, ASTM A572/A572M or ASTM A529/A529M. Provide steel pipe containing a minimum of 50 percent recycled content. Submit data identifying percentage of recycled content for steel pipe.

#### 2.1.3 Cold-Formed and Hot Formed Hollow Structural Sections

Cold formed: ASTM A500/A500M or ASTM B221, ASTM B221M. Hot-formed: ASTM A501/A501M.

#### 2.1.4 Structural-Steel Sheet

Hot-rolled, ASTM A1011/A1011M or cold-rolled, ASTM A1008/A1008M.

#### 2.1.5 Metallic-Coated Steel Sheet

ASTM A653/A653M, ASTM A606/A606M.

#### 2.1.6 Metallic-Coated Steel Sheet Pre-painted with Coil Stock Coating

Steel sheet metallic coated by the hot-dip process and pre-painted by the coil-coating process to comply with ASTM A755/A755M.

- a. Zinc-Coated (Galvanized) Steel Sheet: ASTM A653/A653M, and ASTM A123/A123M.

#### 2.1.7 High-Strength Bolts, Nuts, and Washers

ASTM F3125/F3125M, heavy hex steel structural bolts; ASTM A563 heavy hex carbon-steel nuts; and ASTM F436/F436M hardened carbon-steel washers.

Finish: Hot-dip zinc coating, ASTM A153/A153M.

Tension-Control, High-Strength Bolt-Nut-Washer Assemblies: ASTM F1852, heavy-hex-head steel structural bolts with spline.

Finish: Mechanically deposited zinc coating, ASTM B695.

#### 2.1.8 Non-High-Strength Bolts, Nuts, and Washers

ASTM A307, ASTM A563, and ASTM F844.

Finish: ASTM A153/A153M.

#### 2.1.9 Anchor Rods

ASTM A36/A36M.

- a. Configuration: Straight.
- b. Nuts: ASTM A563 heavy hex carbon steel.
- c. Plate Washers: ASTM A36/A36M carbon steel.
- d. Washers: ASTM F436/F436M hardened carbon steel.
- e. Finish: Hot-dip zinc coating, ASTM A153/A153M.

#### 2.1.10 Threaded Rods

ASTM A36/A36M.

- a. Nuts: ASTM A563MASTM A563 hex carbon steel.
- b. Washers: ASTM A36/A36M carbon steel.
- c. Finish: Hot-dip zinc coating, ASTM A153/A153M.

#### 2.1.11 Primer

SSPC-Paint 15, Type I, red oxide.

### 2.2 FABRICATION

#### 2.2.1 General

Comply with MBMA MBSM - "Metal Building Systems Manual": Chapter IV, Section 9, "Fabrication and Erection Tolerances."

### 2.3 STRUCTURAL FRAMING

#### 2.3.1 General

Clean all framing members to remove loose rust and mill scale. Provide 1 shop coat of primer to an average dry film thickness of 1 mil according to SSPC SP 2. Balance of painting and coating procedures must conform to SSPC Paint 15 and SSPC Painting Manual.

#### 2.3.2 Primary Framing

Manufacturer's standard structural primary framing system includes transverse and lean-to frames; rafter, rakes, and canopy beams; sidewall, intermediate, end-wall, and corner columns; and wind bracing designed to withstand required loads and specified requirements. Provide frames with attachment plates, bearing plates, and splice members. Provide frame span and spacing indicated.



Shop fabricate framing components by welding or by using high-strength bolts to the indicated size and section with base-plates, bearing plates, stiffeners, and other items required. Cut, form, punch, drill, and weld framing for bolted field erection.

- a. Rigid Clear-Span Frames: I-shaped frame sections fabricated from shop-welded, built-up steel plates or structural-steel shapes. Interior columns are not permitted.
- c. Frame Configuration: Lean to, with high side connected to, and supported by, another structure.
- d. Exterior Column Type: Uniform depth
- e. Rafter Type: Uniform depth.

#### 2.3.3 Secondary Framing

Manufacturer's standard secondary framing members, including purlins, girts, eave struts, flange bracing, base members, gable angles, clips, headers, jambs, and other miscellaneous structural members. Fabricate framing from cold-formed, structural-steel sheet or roll-formed, metallic-coated steel sheet pre-painted with coil coating, unless otherwise indicated.

Shop fabricate framing components by roll-forming or break-forming to the indicated size and section with base-plates, bearing plates, stiffeners, and other plates required for erection. Cut, form, punch, drill, and weld secondary framing for bolted field connections to primary framing.

- a. Purlins: C or Z-shaped sections; fabricated from steel sheet, built-up steel plates, or structural-steel shapes; minimum depth as required to comply with system performance requirements.
- b. Girts: C or Z-shaped sections; fabricated from steel sheet, built-up steel plates, or structural-steel shapes. Form ends of Z-sections with stiffening lips angled 40 to 50 degrees to flange minimum depth as required to comply with system performance requirements.
- c. Eave Struts: Unequal-flange, C-shaped sections; fabricated from steel sheet, built-up steel plates, or structural-steel shapes; to provide adequate backup for metal panels.
- d. Flange Bracing: Structural-steel angles or cold-formed structural tubing to stiffen primary frame flanges.
- e. Sag Bracing: Structural-steel angles.
- f. Base or Sill Angles: Zinc-coated (galvanized) steel sheet.
- g. Purlin and Girt Clips: Steel sheet. Provide galvanized clips where clips are connected to galvanized framing members.
- h. Secondary End-Wall Framing: Manufacturer's standard sections fabricated from zinc-coated (galvanized) steel sheet.
- i. Framing for Openings: Channel shapes; fabricated cold-formed, structural-steel sheet or structural-steel shapes. Frame head and jamb of door openings, and head, jamb, and sill of other openings.

- j. Miscellaneous Structural Members: Manufacturer's standard sections fabricated from cold-formed, structural-steel sheet; built-up steel plates; or zinc-coated (galvanized) steel sheet; designed to withstand required loads.

#### 2.3.4 Bracing

Provide adjustable wind bracing as follows:

- c. Angles: Fabricated from structural-steel shapes to match primary framing, of size required to withstand design loads.
- d. Rigid Portal Frames: Fabricate from shop-welded, built-up steel plates or structural-steel shapes to match primary framing; of size required to withstand design loads.
- e. Fixed-Base Columns: Fabricate from shop-welded, built-up steel plates or structural-steel shapes to match primary framing; of size required to withstand design loads.
- f. Diaphragm Action of Metal Panels: Design metal building to resist wind forces through diaphragm action of metal panels.
- g. Bracing: Provide wind bracing using any method specified above, at manufacturer's option.

#### 2.4 PANEL MATERIALS

##### 2.4.1 Steel Sheet

Roll-form steel roof wall panels to the specified profile, with  $f_y = 22$  gauge and depth as indicated. Steel sheets must contain insulated wall panel: exterior face 24g min and interior face: 26g min a minimum recycled content of 25 percent. Provide data identifying percentage of recycled content for steel sheet materials. Material must be plumb and true, and within the tolerances listed:

- a. Galvanized Steel Sheet conforming to ASTM A653/A653M and AISI D100.
- c. Individual panels to have continuous length to cover the entire length of any unbroken roof slope and wall area with no joints or seams and formed without warping, waviness, or ripples that are not part of the panel profile and free of damage to the finish coating system.
- d. Provide panels with thermal expansion and contraction consistent with the type of system specified;

profile and coverage to be a minimum height and width from manufacturer's standard for the indicated roof slope wall area.

profile to be a 2-1/2 inch high standing seam, 16 inch coverage, with mechanical crimping or snap-together seams with concealed clips and fasteners.

##### 2.4.2 Foam-Insulation Core Wall Panel

Provide factory-formed steel roof wall panel assembly fabricated from two sheets of metal with modified polyisocyanurate or polyurethane foam

insulation core foamed-in-place during fabrication with joints between panels designed to form weather-tight seals. Include accessories required for weather-tight installation.

- a. Closed-Cell Content: 90 percent when tested according to ASTM D6226, ASTM C1289.
- b. Density: 2.0 to 2.6 lb/cu. ft. when tested according to ASTM D1622/D1622M.
- c. Compressive Strength: Minimum 20 psi when tested according to ASTM D1621.
- d. Shear Strength: 26 psi when tested according to ASTM C273/C273M.

#### 2.4.3 Finish

All panels are to receive a factory-applied polyvinylidene fluoride of Kynar 500/Hylar 5000 finish consisting of a baked-on top-coat with a manufacturer's recommended prime coat conforming to the following:

- a. Metal Preparation: All metal is to have the surfaces carefully prepared for painting on a continuous process coil coating line by alkali cleaning, hot water rinsing, application of chemical conversion coating, cold water rinsing, sealing with acid rinse, and thorough drying.
- b. Prime Coating: A base coat of epoxy paint, specifically formulated to interact with the top-coat, is to be applied to the prepared surfaces by roll coating to a dry film thickness of 0.20 plus 0.05 mils. This prime coat must be oven cured prior to application of finish coat.
- c. Exterior Finish Coating: Apply the finish coating over the primer by roll coating to dry film thickness of 0.80 plus 0.05 mils for a total dry film thickness of 1.00 plus 0.10 mils. This finish coat must be oven-cured.
- d. Interior Finish Coating: Apply a wash-coat on the reverse side over the primer by roll coating to a dry film thickness of 0.30 plus 0.05 mils for a total dry film thickness of 0.50 plus 0.10 mils. The wash-coat must be oven-cured.
- e. Color: The exterior finish chosen from the manufacturer's color charts and chips.
- f. Physical Properties: Coating must conform to the industry and manufacturer's standard performance criteria as listed by the following certified test reports:

Chalking: ASTM DEF01N1  
Color Change and Conformity: ASTM D2244  
Weatherometer: ASTM G152, ASTM G153 and ASTM D822  
Humidity: ASTM D2247 and ASTM D714  
Salt Spray: ASTM B117  
Chemical Pollution: ASTM D1308  
Gloss at 60 degrees: ASTM D523  
Pencil Hardness: ASTM D3363  
Reverse Impact: ASTM D2794  
Flexibility: ASTM D522/D522M

Abrasion: ASTM D968  
Flame Spread: ASTM E84

#### 2.4.4 Repair Of Finish Protection

Repair paint for enameled metal panel must be compatible paint of the same formula and color as the specified finish furnished by the metal panel manufacturer, conforming to ASTM A780/A780M.

### 2.5 MISCELLANEOUS METAL FRAMING

#### 2.5.1 General

Cold-formed metallic-coated steel sheet conforming to ASTM A653/A653M and specified in Section 05 40 00 COLD-FORMED METAL FRAMING unless otherwise indicated.

#### 2.5.2 Fasteners for Miscellaneous Metal Framing

Refer to the following paragraph FASTENERS.

### 2.6 FASTENERS

#### 2.6.1 General

Type, material, corrosion resistance, size and sufficient length to penetrate the supporting member a minimum of 1 inch with other properties required to fasten miscellaneous metal framing members to substrates in accordance with the metal panel manufacturer's and ASCE 7-16 requirements.

#### 2.6.2 Exposed Fasteners

Fasteners for metal panels to be corrosion resistant coated steel, aluminum, stainless steel, or nylon capped steel compatible with the sheet panel or flashing and of a type and size recommended by the manufacturer to meet the performance requirements and design loads. Fasteners for accessories to be the manufacturer's standard. Provide an integral metal washer matching the color of attached material with compressible sealing EPDM gasket approximately 3/32 inch thick.

#### 2.6.3 Screws

Screws to be corrosion resistant coated steel, aluminum or stainless steel being the type and size recommended by the manufacturer to meet the performance requirements.

#### 2.6.4 Rivets

Rivets to be closed-end type, corrosion resistant coated steel, aluminum or stainless steel where watertight connections are required.

#### 2.6.5 Attachment Clips

Fabricate clips from steel hot-dipped galvanized in accordance with ASTM A653/A653M or Series 300 stainless steel. Size, shape, thickness and capacity as required meeting the insulation thickness and design load criteria specified.

## 2.7 FRAMES AND MATERIALS FOR OPENINGS

### 2.7.1 Doors

Fire-Rated and Non-Fire-Rated Door Assemblies conforming with NFPA 80 and based on testing according to NFPA 252 as specified in Division 08 - OPENINGS unless otherwise indicated.

## 2.8 ACCESSORIES

### 2.8.1 General

All accessories to be compatible with the metal panels; sheet metal flashing, trim, metal closure strips, caps and similar metal accessories must not be less than the minimum thickness specified for the metal panels. Exposed metal accessories/finishes to match the panels, except as otherwise indicated. Molded foam rib, ridge and other closure strips to be non-absorbent closed-cell or solid-cell synthetic rubber or pre-molded neoprene to match configuration of the panels.

### 2.8.2 Roof and Wall Accessories and Specialties

Aluminum miscellaneous roof and wall equipment or penetrations conforming to AAMA, ASTM, and UL as specified in Division 07 unless otherwise indicated.

### 2.8.3 Rubber Closure Strips

Closed-cell, expanded cellular rubber conforming to ASTM D1056 and ASTM D1667; extruded or molded to the configuration of the specified metal panel and in lengths supplied by the metal panel manufacturer.

### 2.8.4 Metal Closure Strips

Factory fabricated closure strips to be the same material, thickness, color, finish and profile of the specified roof and/or wall panel.

### 2.8.5 2.6.6 Joint Sealants

#### 2.8.5.1 Sealants

Sealants are to be an approved gun type for use in hand or air-pressure caulking guns at temperatures above 40 degrees F (or frost-free application at temperatures above 10 degrees F with minimum solid content of 85 percent of the total volume. Sealant is to dry with a tough, durable surface skin which permits it to remain soft and pliable underneath, providing a weather-tight joint. No migratory staining is permitted on painted or unpainted metal, stone, glass, vinyl, or wood.

Prime all joints to receive sealants with a compatible one-component or two-component primer as recommended by the metal panel manufacturer.

#### 2.8.5.2 Shop-Applied

Sealant for shop-applied caulking must be an approved gun grade, non-sag one component polysulfide or silicone conforming to ASTM C920, Type II, and with a curing time to ensure the sealant's plasticity at the time of field erection.

#### 2.8.5.3 Field-Applied

Sealant for field-applied caulking must be an approved gun grade, non-sag one component polysulfide or two-component polyurethane with an initial maximum Shore A durometer hardness of 25, and conforming to ASTM C920, Type II. Color to match panel colors.

#### 2.8.5.4 Tape Sealant

Pressure sensitive, 100 percent solid with a release paper backing; permanently elastic, non-sagging, non-toxic and non-staining as approved by the metal panel manufacturer.

### 2.9 SHEET METAL FLASHING AND TRIM

#### 2.9.1 Fabrication

Shop fabricate sheet metal flashing and trim where practicable to comply with recommendations in SMACNA 1793 that apply to design, dimensions, metal, and other characteristics of item indicated. Obtain field measurements for accurate fit before shop fabrication.

Fabricate sheet metal flashing and trim without excessive oil canning, buckling, and tool marks and true to line and levels indicated, with exposed edges folded back to form hems.

### 2.10 FINISHES

#### 2.10.1 General

Comply with NAAMM AMP 500 for recommendations for applying and designating finishes.

#### 2.10.2 Appearance of Finished Work

Variations in appearance of abutting or adjacent pieces are acceptable if they are within one-half of the range of approved Samples. Noticeable variations in the same piece are not acceptable. Variations in appearance of other components are acceptable if they are within the range of approved Samples and are assembled or installed to minimize contrast.

## PART 3 EXECUTION

### 3.1 EXAMINATION

Before erection proceeds, examine with the erector present, the concrete foundation dimensions, concrete and masonry bearing surfaces, anchor bolt size and placement, survey slab elevation, locations of bearing plates, and other embedments to receive structural framing with the metal building manufacturer's templates and drawings before erecting any steel components for compliance with requirements for installation tolerances and other conditions affecting performance of the Work.

Examine primary and secondary framing to verify that rafters, purlins, angles, channels, and other structural and metal panel support members and anchorages have been installed within alignment tolerances required by metal building manufacturer, UL, ASTM, ASCE 7-16 and as required by the building code for the geographical area where construction will take place.

Examine roughing-in for components and systems penetrating metal roof or wall panels to verify actual locations of penetrations relative to seam locations of metal panels before metal roof or wall panel installation.

Submit to the Contracting Officer a written report, endorsed by Erector, listing conditions detrimental to performance of the Work.

Proceed with erection only after unsatisfactory conditions have been corrected.

### 3.2 PREPARATION

Provide temporary shoring, guys, braces, and other supports during erection to keep the structural framing secure, plumb, and in alignment against temporary construction loading or loads equal in intensity of the building design loads. Remove temporary support systems when permanent structural framing, connections, and bracing are in place, unless otherwise indicated.

Clean substrates of substances harmful to insulation, including removing projections capable of interfering with insulation attachment and performance.

Miscellaneous Framing: Install sub-purlins, girts, angles, furring, and other miscellaneous support members or anchorage for the metal roof or wall panels, doors, windows, roof curbs, ventilators and louvers according to metal building manufacturer's written instructions.

### 3.3 ERECTION OF STRUCTURAL FRAMING

Erect metal building system according to manufacturer's written erection instructions, approved shop drawings and other erection documents in accordance with MBMA MBSM - "Metal Building Systems Manual".

Do not field cut, drill, or alter structural members without written approval from metal building system manufacturer's professional engineer and the Contracting Officer.

Set structural framing accurately in locations and to elevations indicated and according to AISC 325 specifications. Maintain structural stability of frame during erection.

Clean and roughen concrete and masonry bearing surfaces prior to setting plates. Clean bottom surface of plates.

Align and adjust structural framing before permanent bolt-up and connections. Perform necessary adjustments and alignment to compensate for changes or discrepancies in elevations.

Maintain erection tolerances of structural framing in accordance with AISC 360.

### 3.4 METAL WALL PANEL INSTALLATION

Provide metal wall panels of full length from sill to eave as indicated, unless otherwise indicated or restricted by shipping limitations. Anchor metal wall panels and other components of the Work securely in place, in accordance with MBMA MBSM.

Erect wall panel system in accordance with the approved erection drawings, the printed instructions and safety precautions of the metal building manufacturer.

Sheets are not to be subjected to overloading, abuse, or undue impact. Do not install bent, chipped, or defective sheets.

Sheets must be erected true and plumb and in exact alignment with the horizontal and vertical edges of the building, securely anchored, and with the indicated eave, and sill.

Work is to allow for thermal movement of the wall panel, movement of the building structure, and to provide permanent freedom from noise due to wind pressure.

Field cutting metal wall panels by torch is not permitted.

### 3.5 ROOF PANEL INSTALLATION

Provide metal roof panels of full length from eave to ridge or eave to wall as indicated, unless otherwise indicated or restricted by shipping limitations. Anchor metal roof panels and other components of the Work securely in place in accordance with NRCA RoofMan and MBMA MBSM.

Erect roofing system in accordance with the approved erection drawings, the printed instructions and safety precautions of the metal building manufacturer.

Sheets are not to be subjected to overloading, abuse, or undue impact. Do not install bent, chipped, or defective sheets.

Sheets must be erected true and plumb and in exact alignment with the horizontal and vertical edges of the building, securely anchored, and with the indicated rake and eave overhang.

Work must allow for thermal movement of the roofing, movement of the building structure, and provide permanent freedom from noise due to wind pressure.

Field cutting metal roof panels by torch is not permitted.

Roofing sheets must be laid with corrugations in the direction of the roof slope. End laps of exterior roofing must not be less than 8 inches; the side laps of standard exterior corrugated sheets must be not less than 2-1/2 corrugations.

Do not permit storage, walking, wheeling, or trucking directly on applied roofing materials. Provide temporary walkways, runways, and platforms of smooth clean boards or planks as necessary to avoid damage to the installed roofing materials, and to distribute weight to conform to the indicated live load limits of roof construction.

### 3.6 METAL PANEL FASTENER INSTALLATION

Anchor metal panels and other components of the Work securely in place, using manufacturer's approved fasteners according to manufacturers' written instructions.



### 3.7 FLASHING, TRIM AND CLOSURE INSTALLATION

- a. Comply with performance requirements, manufacturer's written installation instructions, and SMACNA 1793. Provide concealed fasteners where possible, and set units true to line and level as indicated. Install work with laps, joints, and seams that will be permanently watertight and weather resistant.
- b. Sheet metalwork is to be accomplished to form weather-tight construction without waves, warps, buckles, fastening stresses or distortion, and allow for expansion and contraction. Cutting, fitting, drilling, and other operations in connection with sheet metal required to accommodate the work of other trades is to be performed by sheet metal mechanics.

### 3.8 DOOR AND FRAME INSTALLATION

Install doors and frames plumb, rigid, properly aligned, and securely fastened in place according to manufacturer's written instructions. Coordinate installation with metal panel flashings and other components. Caulk and seal perimeter of each door frame with elastomeric sealant compatible with metal panels. Comply with installation requirements in Division 08 - OPENINGS.

### 3.9 LOUVER INSTALLATION

Install louvers plumb, rigid, properly aligned, without warp or rack of frames, and securely fastened in place according to manufacturer's written instructions. Coordinate installation with metal panel flashings and other components. Caulk and seal perimeter of each louver frame with elastomeric sealant compatible with for metal panels. Comply with installation requirements in Division 08 - OPENINGS.

### 3.10 ACCESSORY INSTALLATION

#### 3.10.1 General

Install accessories with positive anchorage to building and weather-tight mounting, and provide for thermal expansion. Coordinate installation with flashings and other components.

#### 3.10.2 Dissimilar Metals

Where dissimilar metals contact one another or corrosive substrates are present, protect against galvanic action by painting dissimilar metal surfaces with bituminous coating, by applying rubberized-asphalt underlayment to each surface, or by other permanent separation techniques as recommended by the metal building manufacturer.

#### 3.10.3 Gutters and Downspouts

Comply with performance requirements, manufacturer's written installation instructions, and install sheet metal roof drainage items to produce complete roof drainage system according to SMACNA 1793 recommendations and as indicated. Coordinate installation of roof perimeter flashing with installation of roof drainage system.

#### 3.10.4 Insulation

Comply with performance requirements and manufacturer's written installation instructions. Install insulation concurrently with metal panel installation, in thickness indicated to cover entire roof and wall area, as specified in Division 07 - THERMAL AND MOISTURE PROTECTION.

#### 3.10.5 Roof and Wall Accessories and Specialties

Install roof and wall accessories and specialties complete with necessary hardware, anchors, dampers, weather guards, rain caps, and equipment supports as specified in Division 07 - THERMAL AND MOISTURE PROTECTION, unless otherwise indicated.

### 3.11 CLEAN-UP AND PROTECTION

#### 3.11.1 Structural Framing

Clean all exposed structural framing at completion of installation. Remove metal shavings, filings, bolts, and wires from work area. Remove grease and oil films, excess sealants, handling marks, contamination from steel wool, fittings and drilling debris and scrub the work clean. Exposed metal surfaces to be free of dents, creases, waves, scratch marks, solder or weld marks, and damage to the finish coating.

#### 3.11.2 Metal Panels

Clean all exposed sheet metal work at completion of installation. Remove metal shavings, filings, nails, bolts, and wires from work area. Remove protective coverings/films, grease and oil films, excess sealants, handling marks, contamination from steel wool, fittings and drilling debris and scrub the work clean. Exposed metal surfaces to be free of dents, creases, waves, scratch marks, solder or weld marks, and damage to the finish coating.

#### 3.11.3 Touch-Up Painting

After erection, promptly clean, prepare, and prime or re-prime field connections, rust spots, and abraded surfaces of prime-painted structural framing and accessories. Clean and touch-up paint with manufacturer's touch-up paint.

### 3.12 WASTE MANAGEMENT

Dispose of construction waste in accordance with the requirements of Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL.

### 3.13 SPECIAL INSPECTION AND TESTING FOR SEISMIC-RESISTING SYSTEMS

Perform special inspections and testing for seismic-resisting systems and components in accordance with Section 01 45 35 SPECIAL INSPECTIONS. When buildings are classified as Risk Category V, perform special inspections and testing in accordance with UFC 3-301-02.

### 3.14 WARRANTY

#### 3.14.1 Manufacturer's Warranty

Submit all manufacturers' signed warranties to Contracting Officer prior

to final commissioning and acceptance.

3.14.2 Contractor's Warranty For Installation

Submit warranty for installation to the Contracting Officer prior to final commissioning and acceptance.

3.14.3 Contractor's Five Year No Penal Sum Warranty

CONTRACTOR'S FIVE YEAR NO PENAL SUM WARRANTY  
FOR  
METAL BUILDING SYSTEM

FACILITY DESCRIPTION: \_\_\_\_\_

BUILDING NUMBER: \_\_\_\_\_

CORPS OF ENGINEERS CONTRACT NUMBER: \_\_\_\_\_

CONTRACTOR

CONTRACTOR: \_\_\_\_\_

ADDRESS: \_\_\_\_\_

POINT OF CONTACT: \_\_\_\_\_

TELEPHONE NUMBER: \_\_\_\_\_

OWNER

OWNER: \_\_\_\_\_

ADDRESS: \_\_\_\_\_

POINT OF CONTACT: \_\_\_\_\_

TELEPHONE NUMBER: \_\_\_\_\_

CONSTRUCTION AGENT

CONSTRUCTION AGENT: \_\_\_\_\_

ADDRESS: \_\_\_\_\_

POINT OF CONTACT: \_\_\_\_\_

TELEPHONE NUMBER: \_\_\_\_\_

CONTRACTOR'S TWENTY YEAR NO PENAL SUM WARRANTY  
FOR  
METAL BUILDING SYSTEM  
(continued)

THE METAL BUILDING SYSTEM INSTALLED ON THE ABOVE NAMED BUILDING IS WARRANTED BY FOR A PERIOD OF TWENTY (20) YEARS AGAINST WORKMANSHIP AND MATERIAL DEFICIENCIES, WIND DAMAGE AND STRUCTURAL FAILURE WITHIN PROJECT SPECIFIED DESIGN LOADS, AND LEAKAGE. THE METAL BUILDING SYSTEM COVERED UNDER THIS WARRANTY INCLUDES, BUT IS NOT LIMITED TO, THE FOLLOWING:

FRAMING AND STRUCTURAL MEMBERS, ROOFING AND SIDING PANELS AND SEAMS, INTERIOR OR EXTERIOR GUTTERS AND DOWNSPOUTS, ACCESSORIES, TRIM, FLASHINGS AND MISCELLANEOUS BUILDING CLOSURE ITEMS SUCH AS DOORS AND WINDOWS (WHEN FURNISHED BY THE MANUFACTURER), CONNECTORS, COMPONENTS, AND FASTENERS, AND OTHER SYSTEM COMPONENTS AND ASSEMBLIES INSTALLED TO PROVIDE A WEATHERTIGHT SYSTEM; AND ITEMS SPECIFIED IN OTHER SECTIONS OF THESE SPECIFICATIONS THAT BECOME PART OF THE METAL BUILDING SYSTEM.

ALL MATERIAL AND WORKMANSHIP DEFICIENCIES, SYSTEM DETERIORATION CAUSED BY EXPOSURE TO THE ELEMENTS OR INADEQUATE RESISTANCE TO SPECIFIED SERVICE DESIGN LOADS, WATER LEAKS AND WIND UPLIFT DAMAGE MUST BE REPAIRED AS APPROVED BY THE CONTRACTING OFFICER.

ALL MATERIAL DEFICIENCIES, WIND DAMAGE, STRUCTURAL FAILURE AND LEAKAGE ASSOCIATED WITH THE METAL BUILDING SYSTEM COVERED UNDER THIS WARRANTY MUST BE REPAIRED AS APPROVED BY THE CONTRACTING OFFICER.

THIS WARRANTY COVERS THE ENTIRE COST OF REPAIR OR REPLACEMENT, INCLUDING ALL MATERIAL, LABOR, AND RELATED MARKUPS. THE ABOVE REFERENCED WARRANTY COMMENCED ON THE DATE OF FINAL ACCEPTANCE ON SUBSTANTIAL COMPLETION AND WILL REMAIN IN EFFECT FOR STATED DURATION FROM THIS DATE.

SIGNED, DATED, AND NOTARIZED (BY COMPANY PRESIDENT)

---

(Company President)

---

(Date)

CONTRACTOR'S TWENTY YEAR NO PENAL SUM WARRANTY  
FOR  
METAL BUILDING SYSTEM  
(continued)

THE CONTRACTOR HEREBY SUPPLEMENTS THIS WARRANTY WITH WRITTEN WARRANTIES FROM THE MANUFACTURER AND/OR INSTALLER OF THE METAL BUILDING SYSTEM, WHICH IS SUBMITTED ALONG WITH THE CONTRACTOR'S WARRANTY. HOWEVER, THE CONTRACTOR IS ULTIMATELY RESPONSIBLE FOR THIS WARRANTY AS OUTLINED IN THE SPECIFICATIONS AND AS INDICATED IN THIS WARRANTY.

EXCLUSIONS FROM COVERAGE

1. NATURAL DISASTERS, ACTS OF GOD (LIGHTNING, FIRE, EXPLOSIONS, SUSTAINED WIND FORCES IN EXCESS OF THE DESIGN CRITERIA, EARTHQUAKES, AND HAIL).
2. ACTS OF NEGLIGENCE OR ABUSE OR MISUSE BY GOVERNMENT OR OTHER PERSONNEL, INCLUDING ACCIDENTS, VANDALISM, CIVIL DISOBEDIENCE, WAR, OR DAMAGE CAUSED BY FALLING OBJECTS.
3. DAMAGE BY STRUCTURAL FAILURE, SETTLEMENT, MOVEMENT, DISTORTION, WARPAGE, OR DISPLACEMENT OF THE BUILDING STRUCTURE OR ALTERATIONS MADE TO THE BUILDING.
4. CORROSION CAUSED BY EXPOSURE TO CORROSIVE CHEMICALS, ASH OR FUMES GENERATED OR RELEASED INSIDE OR OUTSIDE THE BUILDING FROM CHEMICAL PLANTS, FOUNDRIES, PLATING WORKS, KILNS, FERTILIZER FACTORIES, PAPER PLANTS, AND THE LIKE.
5. FAILURE OF ANY PART OF THE BUILDING SYSTEM DUE TO ACTIONS BY THE OWNER WHICH INHIBIT FREE DRAINAGE FROM THE ROOF, GUTTERS AND DOWNSPOUTS; OR CONDITIONS WHICH CREATE PONDING WATER ON THE ROOF OR AGAINST THE BUILDING SIDING.
6. THIS WARRANTY APPLIES TO THE METAL BUILDING SYSTEM. IT DOES NOT INCLUDE ANY CONSEQUENTIAL DAMAGE TO THE BUILDING INTERIOR OR CONTENTS WHICH IS COVERED BY THE WARRANTY OF CONSTRUCTION CLAUSE INCLUDED IN THIS CONTRACT.
7. THIS WARRANTY CANNOT BE TRANSFERRED TO ANOTHER OWNER WITHOUT WRITTEN CONSENT OF THE CONTRACTOR AND THIS WARRANTY AND THE CONTRACT PROVISIONS TAKE PRECEDENCE OVER ANY CONFLICTS WITH STATE STATUTES. REPORTS OF LEAKS AND BUILDING SYSTEM DEFICIENCIES MUST BE RESPONDED TO WITHIN 48 HOURS OF RECEIPT OF NOTICE BY TELEPHONE OR IN WRITING FROM EITHER THE OWNER, OR CONTRACTING OFFICER. EMERGENCY REPAIRS, TO PREVENT FURTHER ROOF LEAKS, MUST BE INITIATED IMMEDIATELY; A WRITTEN PLAN MUST BE SUBMITTED FOR APPROVAL TO REPAIR OR REPLACE THIS SSSMR SYSTEM WITHIN SEVEN CALENDAR DAYS. ACTUAL WORK FOR PERMANENT REPAIRS OR REPLACEMENT MUST BE STARTED WITHIN 30 DAYS AFTER RECEIPT OF NOTICE, AND COMPLETED WITHIN A REASONABLE TIME FRAME. IF THE CONTRACTOR FAILS TO ADEQUATELY RESPOND TO THE WARRANTY PROVISIONS, AS STATED

CONTRACTOR'S TWENTY YEAR NO PENAL SUM WARRANTY  
FOR  
METAL BUILDING SYSTEM  
(Exclusions from Coverage Continued)

POST A FRAMED COPY OF THIS WARRANTY IN THE MECHANICAL ROOM OR OTHER APPROVED  
LOCATION DURING THE ENTIRE WARRANTY PERIOD.

-- End of Section --

SECTION 21 13 13

WET PIPE SPRINKLER SYSTEMS, FIRE PROTECTION  
**08/20**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|             |                                                                             |
|-------------|-----------------------------------------------------------------------------|
| ASME B16.1  | (2020) Gray Iron Pipe Flanges and Flanged Fittings Classes 25, 125, and 250 |
| ASME B16.3  | (2021) Malleable Iron Threaded Fittings, Classes 150 and 300                |
| ASME B16.4  | (2021) Gray Iron Threaded Fittings; Classes 125 and 250                     |
| ASME B16.21 | (2021) Nonmetallic Flat Gaskets for Pipe Flanges                            |

AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)

|           |                                                                                               |
|-----------|-----------------------------------------------------------------------------------------------|
| ASSE 1013 | (2021) Performance Requirements for Reduced Pressure Principle Backflow Prevention Assemblies |
| ASSE 1015 | (2021) Performance Requirements for Double Check Backflow Prevention Assemblies               |

AMERICAN WATER WORKS ASSOCIATION (AWWA)

|          |                                                                                          |
|----------|------------------------------------------------------------------------------------------|
| AWWA M14 | (2024) Manual: Recommended Practice for Backflow Prevention and Cross-Connection Control |
|----------|------------------------------------------------------------------------------------------|

ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------|
| ASTM A47/A47M   | (1999; R 2022; E 2022) Standard Specification for Ferritic Malleable Iron Castings                    |
| ASTM A53/A53M   | (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless |
| ASTM A135/A135M | (2021) Standard Specification for Electric-Resistance-Welded Steel Pipe                               |
| ASTM A153/A153M | (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware                   |



ASTM A183 (2014; R 2020) Standard Specification for Carbon Steel Track Bolts and Nuts

ASTM A536 (2024) Standard Specification for Ductile Iron Castings

FM GLOBAL (FM)

FM APP GUIDE (updated on-line) Approval Guide  
<https://www.approvalguide.com/>

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS  
INDUSTRY (MSS)

MSS SP-71 (2018) Gray Iron Swing Check Valves,  
Flanged and Threaded Ends

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 13 (2024) Standard for the Installation of Sprinkler Systems

NFPA 24 (2025) Standard for the Installation of Private Fire Service Mains and Their Appurtenances

NFPA 291 (2025) Recommended Practice for Fire Flow Testing and Marking of Hydrants

NATIONAL INSTITUTE FOR CERTIFICATION IN ENGINEERING TECHNOLOGIES  
(NICET)

NICET 1014-7 (2012) Program Detail Manual for Certification in the Field of Fire Protection Engineering Technology (Field Code 003) Subfield of Automatic Sprinkler System Layout

UL SOLUTIONS (UL)

UL 199 (2020) UL Standard for Safety Automatic Sprinklers for Fire-Protection Service

UL 312 (2022) UL Standard for Safety Check Valves for Fire-Protection Service

UL 668 (2004; Reprint Oct 2021) UL Standard for Safety Hose Valves for Fire-Protection Service

UL Fire Prot Dir UL Product IQ (updated online) at  
<https://productiq.ulpropsector.com/en>

## 1.2 SYSTEM DESCRIPTION

Provide wet pipe sprinkler system(s) in areas indicated on the drawings. Except as modified herein, the system must meet the requirements of NFPA 13. Pipe sizes which are not indicated on the Contract drawings must be

determined by hydraulic calculations.

### 1.2.1 Hydraulic Design

#### 1.2.1.1 Basis for Calculations

A waterflow test was performed on 19 November 2024 at Building 72 and resulted in a static pressure of 64 psi with a residual pressure of 62 psi while flowing 2350 gpm. The results include an adjustment for the lowest level of water within the tank. The contractor must also include this adjustment for the lowest water level within the tank. Perform a fire hydrant flow test prior to shop drawing submittal in accordance with NFPA 291. Results must include hydrant elevations relative to the building and hydrant number/identifiers for the tested hydrants, including which were flowed, which had a gauge. This information must be presented in a tabular form if multiple hydrants were flowed. The results must be included with the hydraulic calculations. Hydraulic calculations must be based on flow test noted in this paragraph, unless verified by the NAVFAC Fire Protection Engineer and approved by Contracting Officer. Hydraulic calculations must be based upon the Hazen-Williams formula with a "C" value noted in NFPA 13 for piping. .

#### 1.2.1.2 Hydraulic Calculations

- a. Water supply curves and system requirements must be plotted on semi-logarithmic graph ( $N^{1.85}$ ) paper so as to present a summary of the complete hydraulic calculation.
- b. Provide a summary sheet listing sprinklers in the design area and their respective hydraulic reference points, elevations, minimum discharge pressures and minimum flows. Elevations of hydraulic reference points (nodes) must be indicated.
- c. Documentation must identify each pipe individually and the nodes connected thereto. Indicate the diameter, length, flow, velocity, friction loss, number and type fittings, total friction loss in the pipe, equivalent pipe length and Hazen-Williams coefficient for each pipe.
- d. Where the sprinkler system is supplied by interconnected risers, the sprinkler system must be hydraulically calculated using the hydraulically most demanding single riser. The calculations must not assume the simultaneous use of more than one riser.
- e. All calculations must include the backflow preventer manufacturer's stated friction loss at the design flow or 8 psi for double check backflow preventer, whichever is greater.
- f. All calculations must be performed back to the actual location of the flow test, taking into account the direction of flow in the service main at the test location.
- g. For gridded systems, calculations must show peaking of demand area friction loss to verify that the hydraulically most demanding area is being used. A flow diagram indicating the quantity and direction of flows must be included.

#### 1.2.1.3 Design Criteria

Hydraulically design the system to discharge a minimum density of 0.2 gpm/square foot over the hydraulically most demanding 2500 square feet of floor area as indicated on the drawings. Hydraulic calculations must be in accordance with the Area/Density Method of NFPA 13. Add an allowance for exterior hose streams of 250 gpm to the sprinkler system demand at the fire hydrant shown on the drawings closest to the point where the water service enters the building .

#### 1.2.2 Sprinkler Coverage

Sprinklers must be uniformly spaced on branch lines. Provide coverage throughout 100 percent of the area noted on the Contract drawings. Coverage per sprinkler must be in accordance with NFPA 13. Provide sprinklers below all obstructions in accordance with NFPA 13. Exceptions are as follows:

#### 1.2.3 Qualified Fire Protection Engineer (QFPE)

The QFPE in this section is the same as the FPQC in Section 01 45 00.00 22. Wherever "Qualified Fire Protection Engineer (QFPE)" or (QFPE) appears in this section, substitute with "Fire Protection Quality Control Specialist (FPQC)". The Fire Protection Designer of Record can also be the the FPQC. Services of the QFPE must include:

- a. Reviewing SD-02, SD-03, and SD-05 submittal packages for completeness and compliance with the provisions of this specification. Working (shop) drawings and calculations must be prepared and signed by the Sprinkler System Designer.
- b. Provide a letter documenting that the SD-02, SD-03, and SD-05 submittal package has been reviewed and noting all outstanding comments. The QFPE will affix their professional engineering stamp with signature to this letter indicating approval prior to submitting to the DFPE.
- c. Performing in-progress construction surveillance prior to installation of ceilings (rough-in inspection).
- d. Witnessing pre-Government and final Government functional performance testing and performing a final installation review.
- e. Signing applicable certificates under SD-07.

#### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Partial submittals and submittals not fully complying with NFPA 13 and this specification section must be returned disapproved without review. SD-02, SD-03 and SD-05 must be submitted simultaneously.

Shop drawings (SD-02), product data (SD-03) and calculations (SD-05) must be prepared by the Fire Sprinkler Designer and combined and submitted as one complete package. The QFPE must review the SD-02/SD-03/SD-05 submittal package for completeness and compliance with the Contract

provisions prior to submission to the Government. The QFPE must provide a Letter of Confirmation that they have reviewed the submittal package for compliance with the contract provisions. This letter must include their professional engineer stamp and signature. Partial submittals and submittals not reviewed by the QFPE must be returned disapproved without review.

Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Qualified Fire Protection Engineer (QFPE); G

Sprinkler System Designer; G

Sprinkler System Installer; G

SD-02 Shop Drawings

Shop Drawing; G

SD-03 Product Data

Steel Pipe; G

Fittings; G

Valves, including gate, check, butterfly, and globe; G

Relief Valves; G

Sprinklers ; G

Pipe Hangers and Supports ; G

Sprinkler Alarm Switch; G

Valve Supervisory (Tamper) Switch; G

Fire Department Connection; G

Backflow Prevention Assembly; G

Air Vent; G

Hose Valve; G

Nameplates; G

SD-05 Design Data

Hydraulic Calculations; G

SD-06 Test Reports

Test Procedures; G

SD-07 Certificates

Verification of Compliant Installation; G

Request for Government Final Test; G

SD-10 Operation and Maintenance Data

Operating and Maintenance (O&M) Instructions; G

Spare Parts Data; G

SD-11 Closeout Submittals

As-built drawings

1.4 QUALITY ASSURANCE

1.4.1 Preconstruction Submittals

Within 36 days of contract award but no less than 14 days prior to commencing work on site, the prime Contractor must submit the following for review and approval. SD-02, SD-03 and SD-05 submittals received prior to the review and approval of the qualifications will be returned Disapproved Without Review.

1.4.1.1 Shop Drawing

Electronic copies of the shop drawings, no later than 28 days prior to the start of system installation. Working drawings conforming to the requirements prescribed in NFPA 13 and must be no smaller than the Contract Drawings. Each set of drawings must include the following:

- a. A descriptive index with drawings listed in sequence by number. A legend sheet identifying device symbols, nomenclature, and conventions used in the package.
- b. Floor plans drawn to a scale not less than 1/8-inch equals 1-foot clearly showing locations of devices, equipment, risers, and other details required to clearly describe the proposed arrangement.
- c. Actual center-to-center dimensions between sprinklers on branch lines and between branch lines; from end sprinklers to adjacent walls; from walls to branch lines; from sprinkler feed mains, cross mains and branch lines to finished floor and roof or ceiling. A detail must show the dimension from the sprinkler and sprinkler deflector to the ceiling in finished areas.
- d.
- e. Plan and elevation views which establish that the equipment will fit the allotted spaces with clearance for installation and maintenance.
- f. Riser layout drawings drawn to a scale of not less than 1/2-inch equals 1-foot to show details of each system component, clearances between each other and from other equipment and construction in the room.
- g. Details of each type of riser assembly, pipe hanger, and restraint of

underground water main at point-of-entry into the building, and electrical devices and interconnecting wiring. The dimension from the edge of vertical piping to the nearest adjacent wall(s) must be indicated on the drawings when vertical piping is located in stairs or other portions of the means of egress.

h. Details of each type of pipe hanger and related components.

#### 1.4.1.2 Product Data

Electronic copies of annotated catalog data to show the specific model, type, and size of each item. Catalog cuts must also indicate the NRTL listing. The data must be highlighted to show model, size, options, and other pertinent information, that are intended for consideration. Data must be adequate to demonstrate compliance with all contract requirements. Product data for all equipment must be combined into a single submittal.

#### 1.4.1.3 Hydraulic Calculations

Calculations must be as outlined in NFPA 13 except that calculations must be performed by computer using software intended specifically for fire protection system design using the design data shown on the drawings.

#### 1.4.1.4 Operating and Maintenance (O&M) Instructions

Submit in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA as supplemented and modified by this specification section.

Provide two manuals and one pdf version on electronic media. The manuals must include the manufacturer's name, model number, parts list, list of parts and tools that should be kept in stock by the owner for routine maintenance, troubleshooting guide, and recommended service organization (including address and telephone number) for each item of equipment.

Submit spare parts data for each different item of material and equipment specified. The data must include a complete list of parts and supplies, and a list of parts recommended by the manufacturer to be replaced after 1-year and 3 years of service. Include a list of special tools and test equipment required for maintenance and testing of the products supplied.

#### 1.4.2 Qualifications

##### 1.4.2.1 Sprinkler System Designer

The sprinkler system designer must be certified as a Level III Technician by National Institute for Certification in Engineering Technologies (NICET) in the Water-Based Systems Layout subfield of Fire Protection Engineering Technology in accordance with NICET 1014-7.

##### 1.4.2.2 Sprinkler System Installer

The sprinkler system installer must be regularly engaged in the installation of the type and complexity of system specified in the contract documents, and must have served in a similar capacity for at least three systems that have performed in the manner intended for a period of not less than 6 months.

#### 1.4.3 Regulatory Requirements

Equipment and material must be listed or approved. Listed or approved, as used in this Section, means listed, labeled or approved by a Nationally Recognized Testing Laboratory (NRTL) such as UL Fire Prot Dir or FM APP GUIDE. The omission of these terms under the description of an item or equipment described must not be construed as waiving this requirement. All listings or approvals by testing laboratories must be from an existing ANSI or UL published standard. The recommended practices stated in the manufacturer's literature or documentation are mandatory requirements.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

Protect all equipment delivered and placed in storage from the weather, excessive humidity and temperature variations, dirt and dust, or other contaminants. All pipes must be either capped or plugged until installed.

#### 1.6 EXTRA MATERIALS

Spare sprinklers and wrench(es) must be provided as spare parts in accordance with NFPA 13.

### PART 2 PRODUCTS

#### 2.1 MATERIALS AND EQUIPMENT

##### 2.1.1 Standard Products

Provide materials, equipment, and devices listed for fire protection service when so required by NFPA 13 or this specification. Select material from one manufacturer, where possible, and not a combination of manufacturers, for a classification of material. Material and equipment must be standard products of a manufacturer regularly engaged in the manufacture of the products for at least 2 years prior to bid.

##### 2.1.2 Nameplates

Major components of equipment must have the manufacturer's name, address, type or style, model or serial number, catalog number, date of installation, installing Contractor's name and address, and the contract number provided on a new name plate permanently affixed to the item or equipment. Nameplates must be etched metal or plastic, permanently attached by screws to control units, panels or adjacent walls.

##### 2.1.3 Identification and Marking

Pipe and fitting markings must include name or identifying symbol of manufacturer and nominal size. Pipe must be marked with ASTM designation. Valves and equipment markings must have name or identifying symbol of manufacturer, specific model number, nominal size, name of device, arrow indicating direction of flow, and position of installation (horizontal or vertical), except if valve can be installed in either position. Markings must be included on the body casting or on an etched or stamped metal nameplate permanently on the valve or cover plate.

##### 2.1.4 Pressure Ratings

Valves, fittings, couplings, alarm switches, and similar devices must be

rated for the maximum working pressures that can be experienced in the system, but in no case less than 175 psi.

## 2.2 ABOVEGROUND PIPING COMPONENTS

### 2.2.1 Steel Piping Components

#### 2.2.1.1 Steel Pipe

Except as modified herein, steel pipe must be black as permitted by NFPA 13 and conform to the applicable provisions of ASTM A53/A53M, ASTM A135/A135M or ASTM A153/A153M.

Steel pipe must be minimum Schedule 40 for sizes 2 inches and less; and minimum Schedule 10 for sizes larger than 2 inches. Steel pipe must be Schedule 40 for all size pipe in Room C. Steel piping with wall thickness less than Schedule 40 must not be threaded.

#### 2.2.1.2 Fittings

Fittings must be welded, threaded, or grooved-end type. Threaded fittings must be cast-iron conforming to ASME B16.4, malleable-iron conforming to ASME B16.3 or ductile-iron conforming to ASTM A536. Plain-end fittings with mechanical couplings, fittings that use steel gripping devices to bite into the pipe, steel press fittings and field welded fittings are not permitted. Fittings, mechanical couplings, and rubber gaskets must be supplied by the same manufacturer. Threaded fittings must use Teflon tape or manufacturer's approved joint compound. Reducing couplings are not permitted except as allowed by NFPA 13.

#### 2.2.1.3 Grooved Mechanical Joints and Fittings

Joints and fittings must be designed for not less than 175 psi service and the product of the same manufacturer. Field welded fittings must not be used. Fitting and coupling housing must be malleable-iron conforming to ASTM A47/A47M, Grade 32510; ductile-iron conforming to ASTM A536, Grade 65-45-12. Rubber gasketed grooved-end pipe and fittings with mechanical couplings are permitted in pipe sizes 2 inches and larger. Gasket must be the flush type that fills the entire cavity between the fitting and the pipe. Nuts and bolts must be heat-treated steel conforming to ASTM A183 and must be cadmium-plated or zinc-electroplated.

#### 2.2.1.4 Flanges

Flanges must conform to NFPA 13 and ASME B16.1. Gaskets must be non-asbestos compressed material in accordance with ASME B16.21, 1/16-inch thick, and full face or self-centering flat ring type.

### 2.2.2 Flexible Sprinkler Hose

The use of flexible hose is not permitted.

### 2.2.3 Pipe Hangers and Supports

Provide galvanized pipe hangers and supports in accordance with NFPA 13..

### 2.2.4 Valves

Provide valves of types approved for fire service. Valves must open by



counterclockwise rotation.

#### 2.2.4.1 Control Valve

Manually operated sprinkler control/gate valve must be outside stem and yoke (OS&Y) type or butterfly type and must be listed.

#### 2.2.4.2 Check Valves

Check valves must comply with UL 312. Check valves 4 inches and larger must be of the swing type, have a clear waterway and meet the requirements of MSS SP-71, for Type 3 or 4. Inspection plate must be provided on valves larger than 6 inches.

#### 2.2.4.3 Hose Valve

Valve must comply with UL 668.

### 2.3 ALARM INITIATING AND SUPERVISORY DEVICES

#### 2.3.1 Sprinkler Alarm Switch

Vane or pressure-type flow switch(es). Connection of switch must be by the fire alarm installer. Vane type alarm actuating devices must have mechanical diaphragm controlled retard device adjustable from 10 to 60 seconds and must instantly recycle.

#### 2.3.2 Valve Supervisory (Tamper) Switch

Switch must be integral to the control valve or suitable for mounting to the type of control valve to be supervised open. The switch must be tamper resistant and contain SPDT (Form C) contacts arranged to transfer upon removal of the housing cover or closure of the valve of more than two rotations of the valve stem.

### 2.4 BACKFLOW PREVENTION ASSEMBLY

Double-check valve assembly backflow preventer complying with ASSE 1013, ASSE 1015 and AWWA M14. Each check valve must have a drain. Backflow prevention assemblies must have current "Certificate of Approval from the Foundation for Cross-Connection Control and Hydraulic Research, FCCCHR List" and be listed for fire protection use. Listing of the specific make, model, design, and size in the FCCCHR List is acceptable as the required documentation.

#### 2.4.1 Backflow Preventer Test Connection

Test connection must consist of a series of listed hose valves with 2 1/2-inch National Standard male hose threads with cap and chain.

### 2.5 FIRE DEPARTMENT CONNECTION

Fire department connection for the fire sprinkler system must be flush type with cast-brass body, matching wall escutcheon lettered "Auto Sprk" with a polished-brass finish. Female inlets must have 4-inch diameter Storz.

The fire department connections for the tank suppression must be flush type with cast-brass body, matching wall escutcheon lettered "TANK SUPPRESSION"

with a polished-brass finish. Female inlets must have 2 1/2-inch diameter American National Fire Hose Connection Screw Threads (NH).

## 2.6 SPRINKLERS

Sprinklers must comply with UL 199 and NFPA 13. Sprinklers with internal O-rings are not acceptable. Sprinklers in high heat areas including attic spaces or in close proximity to unit heaters must have temperature classification in accordance with NFPA 13. Extended coverage sprinklers are permitted for loading docks, residential occupancies and high-piled storage applications only.

### 2.6.1 Pendent Sprinkler

Pendent sprinkler must be quick-response type with nominal K-factor as indicated on the drawings.

### 2.6.2 Upright Sprinkler

Upright sprinkler must be brass quick-response type and have a nominal K-factor as indicate on the drawings.

### 2.6.3 Corrosion-Resistant Sprinkler

Corrosion-resistant sprinkler must be the upright type installed in locations as indicated. Corrosion-resistant coatings must be factory-applied by the sprinkler manufacturer.

## 2.7 ACCESSORIES

### 2.7.1 Sprinkler Cabinet

Provide spare sprinklers in accordance with NFPA 13 and must be placed in a suitable metal or plastic cabinet of sufficient size to accommodate all the spare sprinklers and wrenches in designated locations. Spare sprinklers must be representative of, and in proportion to, the number of each type and temperature rating of the sprinklers installed as required by NFPA 13. At least one wrench of each type required must be provided.

### 2.7.2 Pipe Escutcheon

Provide split hinge metal plates for piping entering walls, floors, and ceilings in exposed spaces. Provide polished stainless steel plates or chromium-plated finish on copper alloy plates in finished spaces. Provide paint finish on metal plates in unfinished spaces.

### 2.7.3 Sprinkler Guard

Listed guard must be a steel wire cage designed to encase the sprinkler and protect it from mechanical damage. Guards must be provided on sprinklers located within 7 feet of the floor.

### 2.7.4 Relief Valve

Relief valves must be listed and installed at the riser in accordance with NFPA 13.

#### 2.7.5 Air Vent

Air vents must be of the automatic type.

#### 2.7.6 Identification Sign

Valve identification sign must be minimum 6 inches wide by 2 inches high with enamel baked finish on minimum 18 gage steel or 0.024-inch aluminum with red letters on a white background or white letters on red background. Wording of sign must include, but not be limited to "main drain", "auxiliary drain", "inspector's test", "alarm test", "alarm line", and similar wording as required to identify operational components. Where there is more than one sprinkler system, signage must include specific details as to the respective system.

### PART 3 EXECUTION

#### 3.1 VERIFYING ACTUAL FIELD CONDITIONS

Before commencing work, examine all adjoining work on which the contractor's work that is dependent for perfect workmanship according to the intent of this specification section, and report to the Contracting Officer's Representative a condition that prevents performance of first class work. No "waiver of responsibility" for incomplete, inadequate or defective adjoining work will be considered unless notice has been filed before submittal of a proposal.

#### 3.2 INSTALLATION

The installation must be in accordance with the applicable provisions of NFPA 13, NFPA 24 and publications referenced therein. Locate sprinklers in a consistent pattern with ceiling grid, lights, and air supply diffusers. Install sprinkler system over and under ducts, piping and platforms when such equipment can negatively affect or disrupt the sprinkler discharge pattern and coverage.

- a. Piping offsets, fittings, and other accessories required must be furnished to provide a complete installation and to eliminate interference with other construction.
- b. Wherever the contractor's work interconnects with work of other trades the Contractor must coordinate with other Contractors to insure all Contractors have the information necessary so that they may properly install all necessary connections and equipment. Identify all work items needing access (dampers and similar equipment) that are concealed above hung ceilings by permanent color coded pins/tabs in the ceiling directly below the item.
- c. Provide required supports and hangers for piping, conduit, and equipment so that loading will not exceed allowable loadings of structure. Submittal of a bid must be a deemed representation that the contractor submitting such bid has ascertained allowable loadings and has included in his estimates the costs associated in furnishing required supports.

##### 3.2.1 Waste Removal

At the conclusion of each day's work, clean up and stockpile on site all waste, debris, and trash which may have accumulated during the day as a

result of work by the contractor and of his presence on the job. Sidewalks and streets adjoining the property must be kept broom clean and free of waste, debris, trash and obstructions caused by work of the contractor, which will affect the condition and safety of streets, walks, utilities, and property.

### 3.3 ABOVEGROUND PIPING INSTALLATION

The methods of fabrication and installation of the aboveground piping must fully comply with the requirements and recommended practices of NFPA 13 and this specification section.

#### 3.3.1 Piping in Exposed Areas

Install exposed piping without diminishing exit access widths, corridors or equipment access. Exposed horizontal piping, including drain piping, must be installed to provide maximum headroom.

#### 3.3.2 Piping in Finished Areas

In areas with suspended or dropped ceilings and in areas with concealed spaces above the ceiling, piping must be concealed above ceilings. Piping must be inspected, hydrostatically tested and approved before being concealed. Risers and similar vertical runs of piping in finished areas must be concealed.

#### 3.3.3 Pendent Sprinklers

- a. Drop nipples to pendent sprinklers must consist of minimum 1-inch pipe with a reducing coupling into which the sprinkler must be threaded.
- b. Where sprinklers are installed below suspended or dropped ceilings, drop nipples must be cut such that sprinkler ceiling plates or escutcheons are of a uniform depth throughout the finished space. The outlet of the reducing coupling must not extend below the underside of the ceiling.

#### 3.3.4 Upright Sprinklers

Riser nipples or "sprigs" to upright sprinklers must contain no fittings between the branch line tee and the reducing coupling at the sprinkler.

#### 3.3.5 Pipe Joints

Pipe joints must conform to NFPA 13, except as modified herein. Not more than four threads must show after joint is made up. Welded joints will be permitted, only if welding operations are performed as required by NFPA 13 at the Contractor's fabrication shop, not at the project construction site. Flanged joints must be provided where indicated or required by NFPA 13. Grooved pipe and fittings must be prepared in accordance with the manufacturer's latest published specification according to pipe material, wall thickness and size. Grooved couplings, fittings and grooving tools must be products of the same manufacturer. For copper tubing, pipe and groove dimensions must comply with the tolerances specified by the coupling manufacturer. The diameter of grooves made in the field must be measured using a "go/no-go" gauge, vernier or dial caliper, narrow-land micrometer, or other method specifically approved by

the coupling manufacturer for the intended application. Groove width and dimension of groove from end of pipe must be measured and recorded for each change in grooving tool setup to verify compliance with coupling manufacturer's tolerances.

### 3.3.6 Reducers

Reductions in pipe sizes must be made with one-piece tapered reducing fittings. When standard fittings of the required size are not manufactured, single bushings of the face or hex type will be permitted. Where used, face bushings must be installed with the outer face flush with the face of the fitting opening being reduced. Bushings cannot be used in elbow fittings, in more than one outlet of a tee, in more than two outlets of a cross, or where the reduction in size is less than 1/2-inch.

### 3.3.7 Pipe Penetrations

- a. Cutting structural members for passage of pipes or for pipe-hanger fastenings will not be permitted. Pipes that must penetrate concrete or masonry walls or concrete floors must be core-drilled and provided with pipe sleeves. Each sleeve must be Schedule 40 galvanized steel, ductile-iron or cast-iron pipe and extend through its respective wall or floor and be cut flush with each wall surface. Sleeves must provide required clearance between the pipe and the sleeve per NFPA 13. The space between the sleeve and the pipe must be firmly packed with mineral wool insulation.
- b. Where pipes and sleeves penetrate fire walls, fire partitions, or floors, pipes/sleeves must be firestopped in accordance with Section 07 84 00 FIRESTOPPING.
- c. In penetrations that are not fire-rated or not a floor penetration, the space between the sleeve and the pipe must be sealed at both ends with plastic waterproof cement that will dry to a firm but pliable mass or with a mechanically adjustable segmented elastomer seal.

### 3.3.8 Escutcheons

Escutcheons must be provided for pipe penetration in finished areas of ceilings, floors and walls. Escutcheons must be securely fastened to the pipe at surfaces through which piping passes.

### 3.3.9 Inspector's Test Connection

Unless otherwise indicated, the test connection must consist of 1-inch pipe connected at the riser as a combination test and drain valve; a test valve located approximately 7 feet above the floor; a smooth bore brass outlet equivalent to the smallest orifice sprinkler used in the system; and a painted metal identification sign affixed to the valve with the words "Inspector's Test". All test connection piping must be inside of the building and penetrate the exterior wall at the location of the discharge orifice only. The discharge orifice must be located outside the building wall no more than 2 feet above finished grade, directed so as not to cause damage to adjacent construction or landscaping during full flow discharge, or to the sanitary sewer. Discharge to the exterior must not interfere with exiting from the facility. Water discharge or runoff must not cross the path of egress from the building. Do not discharge to the

roof. Discharge to floor drains, janitor sinks or similar fixtures is not permitted.

Provide concrete splash blocks at all drain and inspector's test connection discharge locations if not discharging to a concrete surface. Splash blocks must be large enough to mitigate erosion and not become dislodged during a full flow of the drain. Ensure all discharged water drains away from the facility and does not cause property damage.

### 3.3.10 Backflow Preventer

Locate within the building or in a heated enclosure in locations subject to freezing. For heated enclosures, provide a low temperature supervisory alarm connected to the facility fire alarm system. Heat trace is not permitted to be used.

Install backflow preventers so that the bottom of the assembly is a minimum of 6 inches above the finished floor/grade. Install horizontal backflow preventers so that the bottom of the assembly is no greater than 24 inches above the finished floor/grade. Install vertical backflow preventers so that the upper operating handwheel is no more than 6 feet above the finished floor/grade. Clearance around control valve handles must be minimum 6 inches above grade/finished floor and away from walls.

#### 3.3.10.1 Test Connection

Provide downstream of the backflow prevention assembly UL 668 hose valves with 2.5-inch National Standard male hose threads with cap and chain. Provide one valve for each 250 gpm of system demand or fraction thereof. Provide a permanent sign in accordance with paragraph entitled "Identification Signs" which reads, "Test Valve". Indicate location of test header. If an exterior connection, provide a control valve inside a heated mechanical room to prevent freezing.

### 3.3.11 Drains

- a. Main drain piping must be provided to discharge at a safe point outside the building, no more than 2 feet above finished grade. Provide a concrete splash block at drain outlet. Discharge to the exterior must not interfere with exiting from the facility. Water discharge or runoff must not cross the path of egress from the building.
- b. Auxiliary drains must be provided as required by NFPA 13. Auxiliary drains are permitted to discharge to a floor drain if the drain is sized to accommodate full flow (min 40 gpm). Discharge to service sinks or similar plumbing fixtures is not permitted.

### 3.3.12 Installation of Fire Department Connection

Connection must be mounted on the exterior wall approximately 3 feet above finished grade. The piping between the connection and the check valve must be provided with an automatic drip in accordance with NFPA 13 and piped to drain to the outside or a floor drain within the same room.

### 3.3.13 Identification Signs

Signs must be affixed to each control valve, inspector test valve, main drain, auxiliary drain, test valve, and similar valves as appropriate or

as required by NFPA 13. Main drain test results must be etched into main drain identification sign. Hydraulic design data must be etched into the nameplates and permanently affixed to each sprinkler riser as specified in NFPA 13. Provide labeling on the surfaces of all feed and cross mains to show the pipe function (e.g., "Sprinkler System", "Fire Department Connection", "Standpipe") and normal valve position (e.g. "Normally Open", "Normally Closed"). For pipe sizes 4-inch and larger provide white painted stenciled letters and arrows, a minimum of 2 inches in height and visible from at least two sides when viewed from the floor. For pipe sizes less than 4-inch, provide white painted stenciled letters and arrows, a minimum of 0.75-inch in height and visible from the floor.

### 3.4 ELECTRICAL

Except as modified herein, electric equipment and wiring must be in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Alarm signal wiring connected to the building fire alarm control system must be by the fire alarm installer.

### 3.5 PAINTING

Color code mark piping red.

### 3.6 FIELD QUALITY CONTROL

#### 3.6.1 Test Procedures

Submit detailed test procedures, prepared and signed by the NICET Level III Fire Sprinkler Designer, and the representative of the installing company, 60 days prior to performing system tests. Detailed test procedures must list all components of the installed system. Test procedures must include sequence of testing, time estimate for each test, and sample test data forms. The test data forms must be in a check-off format (pass/fail with space to add applicable test data; similar to the forms in NFPA 13). The test procedures and accompanying test data forms must be used for the pre-Government testing and the Government final testing.

- a. Provide space to identify the date and time of each test. Provide space to identify the names and signatures of the individuals conducting and witnessing each test.

#### 3.6.2 Pre-Government Testing

##### 3.6.2.1 Verification of Compliant Installation

Conduct inspections and tests to ensure that equipment is functioning properly. Tests must meet the requirements of paragraph entitled "Minimum System Tests" and "System Acceptance" as noted in NFPA 13. The Contractor and QFPE must be in attendance at the pre-Government testing to make necessary adjustments. After inspection and testing is complete, provide a signed Verification of Compliant Installation letter by the QFPE that the installation is complete, compliant with the specification and fully operable. The letter must include the names and titles of the witnesses to the pre-Government tests. Provide all completion documentation as required by NFPA 13 and the test reports noted below.

- a. NFPA 13 Aboveground Material and Test Certificate

### 3.6.2.2 Request for Government Final Test

When the verification of compliant installation has been completed, submit a formal request for Government final test to the Designated Fire Protection Engineer (DFPE). Government final testing will not be scheduled until the DFPE has received copies of the request for Government final testing and Verification of Compliant Installation letter with all required reports. Government final testing will not be performed until after the connections to the building fire alarm system have been completed and tested to confirm communications are fully functional. Submit request for test at least 15 calendar days prior to the requested test date.

### 3.6.3 Correction of Deficiencies

If equipment was found to be defective or non-compliant with contract requirements, perform corrective actions and repeat the tests. Tests must be conducted and repeated if necessary until the system has been demonstrated to comply with all contract requirements.

### 3.6.4 Government Final Tests

The tests must be performed in accordance with the approved test procedures in the presence of the DFPE. Furnish instruments and personnel required for the tests. The following must be provided at the job site for Government Final Testing:

- a. The manufacturer's technical representative.
- b. b. The contractor's Qualified Fire Protection Engineer (QFPE).
- c. Marked-up red line drawings of the system as actually installed.

Government Final Tests will be witnessed by the Designated Fire Protection Engineer, Qualified Fire Protection Engineer (QFPE). At this time, all required tests noted in the paragraph "Minimum System Tests" must be repeated at their discretion.

## 3.7 MINIMUM SYSTEM TESTS

The system, including the underground water mains, and the aboveground piping and system components, must be tested to ensure that equipment and components function as intended. The underground and aboveground interior piping systems and attached appurtenances subjected to system working pressure must be tested in accordance with NFPA 13 and NFPA 24.

### 3.7.1 Aboveground Piping

#### 3.7.1.1 Hydrostatic Test

Aboveground piping must be hydrostatically tested in accordance with NFPA 13. There must be no drop in gauge pressure or visible leakage when the system is subjected to the hydrostatic test. The test pressure must be read from a gauge located at the low elevation point of the system or portion being tested.



#### 3.7.1.2 Backflow Prevention Assembly Forward Flow Test

Each backflow prevention assembly must be tested at system flow demand, including all applicable hose streams, as specified in NFPA 13. The Contractor must provide all equipment and instruments necessary to conduct a complete forward flow test, including 2.5-inch diameter hoses, playpipe nozzles or flow diffusers, calibrated pressure gauges, and pitot tube gauge. The Contractor must provide all necessary supports to safely secure hoses and nozzles during the test. At the system demand flow, the pressure readings and pressure drop (friction loss) across the assembly must be recorded. A metal placard must be provided on the backflow prevention assembly that lists the pressure readings both upstream and downstream of the assembly, total pressure drop, and the system test flow rate determined during the preliminary testing. The pressure drop must be compared to the manufacturer's data and the readings observed during the final inspections and tests.

#### 3.7.2 Main Drain Flow Test

Following flushing of the underground piping, a main drain test must be made to verify the adequacy of the water supply. Static and residual pressures must be recorded on the certificate specified in paragraph SUBMITTALS.

### 3.8 SYSTEM ACCEPTANCE

Following acceptance of the system, as-built drawings and O&M manuals must be delivered to the Contracting Officer for review and acceptance. Submit six sets of detailed as-built drawings. The drawings must show the system as installed, including deviations from both the project drawings and the approved shop drawings. These drawings must be submitted within two weeks after the final acceptance test of the system. At least one set of as-built (marked-up) drawings must be provided at the time of, or prior to the final acceptance test.

- a. Provide one set of full size paper as-built drawings and schematics. The drawings must be prepared electronically and sized no less than the contract drawings. Furnish one set of CDs or DVDs containing software back-up and CAD based drawings in latest version of AutoCAD, DXF and portable document formats of as-built drawings and schematics.
- b. Provide operating and maintenance (O&M) instructions.

### 3.9 ONSITE TRAINING

Conduct a training course for the responding fire department and operating and maintenance personnel as designated by the Contracting Officer. Training must be performed on two separate days (to accommodate different shifts of Fire Department personnel) for a period of 4 hours of normal working time and must start after the system is functionally complete and after the final acceptance test. The on-site training must cover all of the items contained in the approved Operating and Maintenance Instructions.

-- End of Section --

SECTION 22 13 29

SANITARY SEWERAGE PUMPS  
**02/11**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B40.100 (2022) Pressure Gauges and Gauge Attachments

ASTM INTERNATIONAL (ASTM)

ASTM A153/A153M (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 1 (2022) Standard for Industrial Control and Systems: General Requirements

NEMA MG 1 (2021) Motors and Generators

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Equipment Installation; G

SD-03 Product Data

Materials and Equipment

Framed Instructions

Spare Parts

SD-06 Test Reports

Field Testing and Adjusting Equipment

SD-10 Operation and Maintenance Data

Operating and Maintenance Manuals; G

1.3 DELIVERY, STORAGE, AND HANDLING

Protect from the weather, excessive humidity and excessive temperature variation; and dirt, dust, or other contaminants all equipment delivered and placed in storage.

1.4 EXTRA MATERIALS

Submit spare parts data for each different item of material and equipment specified, after approval of the related submittals, and not later than 3 months prior to the date of beneficial occupancy. Include in the data a complete list of parts and supplies, with current unit prices and source of supply

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

Provide materials and equipment which are the standard products of a manufacturer regularly engaged in the manufacture of such products and that essentially duplicate items that have been in satisfactory use for at least 2 years prior to bid opening. Provide equipment supported by a service organization that is, in the opinion of the Contracting Officer, reasonably convenient to the site.

Pump material and all equipment shall be suitable for the intended application and compatible with the seawater environment.

2.1.1 Nameplates

Provide each major item of equipment with the manufacturer's name, address, type or style, model or serial number, and catalog number on a plate secured to the item of equipment.

2.1.2 Equipment Guards

Enclose or guard belts, pulleys, chains, gears, projecting setscrews, keys, and other rotating parts so located that any person may come in close proximity thereto.

2.1.3 Special Tools

Provide one set of special tools, calibration devices, and instruments required for operation, calibration, and maintenance of the equipment.

2.1.4 Electric Motors

Provide motors conforming to NEMA MG 1.

2.1.5 Motor Controls

Provide controls conforming to NEMA ICS 1.

2.1.6 Bolts, Nuts, Anchors, and Washers

Provide steel bolts, nuts, anchors, and washers; galvanized in accordance with ASTM A153/A153M.

### 2.1.7 Pressure Gauges

Provide compound gauges on the suction side of pumps and standard pressure gauges on the discharge side of pumps complying with ASME B40.100. Use gauge ranges as appropriate for the particular installation.

## 2.2 SUBMERSIBLE CENTRIFUGAL PUMPS

Two UF drain submersible pumps will be installed inside the wet well to manage the backwash drainage from the UF tanks and to handle any overflow scenarios. The pumps will operate based on level control and will be driven by variable frequency drives (VFDs). The basis of design is the Flygt NP 3127 HT 3 Adaptive 488 model, equipped with a 10 HP motor rated at 3-phase, 480 volts. Each pump is designed to deliver 558 GPM at 18 PSI.

- a. Pump Manufacturers - Flygt Model NP 3127 H3 or approved equal.
- b. Impeller Material - Material shall be certified for seawater application.
- c. The pumps shall be certified for seawater application to ensure their suitability and durability in marine environments.
- d. Pump minimum water depth shall be sufficient to completely cover the pump motor at all time.
- e. Piping within the wetwell shall be PVC or HDPE. Fitting within the wetwell shall be PVC or HDPE.
- f. The contractor shall install check valves and isolation valves on the discharge pipe, to be located in a separate vault as shown in the drawings.
- g. The contractor shall install the following components to ensure a complete and fully operational the wet well pump station: upper guide rail bracket, guide rail, pump lifting chain, cable holder, float switch, high-level alarm, and level sensor.
- h. All installed equipment shall be suitable for seawater applications. The UF drain pumps, along with the associated equipment and level measurement devices, shall be supplied by the UF manufacturer.

## 2.3 ELECTRICAL WORK

Provide electrical motor driven equipment specified complete with motors, motor starters, controls and wiring in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Specify or indicate electrical characteristics. Provide motor starters complete with thermal overload protection and other appurtenances necessary for the motor control specified. Provide manual or automatic control and protective or signal devices required for the operation specified, and any control wiring required for controls and devices but are not shown.

## PART 3 EXECUTION

### 3.1 EXAMINATION

After becoming familiar with all details of the work, verify all

dimensions in the field, and advise the Contracting Officer of any discrepancy before performing the work.

### 3.2 EQUIPMENT INSTALLATION

Submit Drawings containing complete wiring and schematic diagrams and any other details required to demonstrate that the system has been coordinated and will properly function as a unit. Show on the Drawings proposed layout and anchorage of equipment and appurtenances, and equipment relationship to other parts of the work including clearances for maintenance and operation.

#### 3.2.1 Pump Installation

Install pumping equipment and appurtenances in the position indicated and in accordance with the manufacturer's written instructions. Provide all appurtenances required for a complete and operating pumping system, including such items as piping, conduit, valves, wall sleeves, wall pipes, concrete foundations, anchors, grouting, pumps, drivers, power supply, seal water units, and controls.

#### 3.2.2 Concrete

Provide concrete conforming to Section 03 30 00 CAST-IN-PLACE CONCRETE.

### 3.3 PAINTING

Pumps and motors must be thoroughly cleaned, primed, and given two finish coats of paint at the factory in accordance with the recommendations of the manufacturer. Field painting required for ferrous surfaces not finished at the factory is specified in Section 09 90 00 PAINTS AND COATINGS.

### 3.4 FRAMED INSTRUCTIONS

Post, where directed, framed instructions containing wiring and control diagrams under glass or in laminated plastic. Frame condensed operating instructions, prepared in typed form, as specified above and posted beside the diagrams. Post the framed instructions before acceptance testing of the system. Submit pump characteristic curves showing capacity in gpm, net positive suction head (NPSH), head, efficiency, and pumping horsepower from 0 gpm to 110 percent (100 percent for positive displacement pumps) of design capacity. Submit a complete list of equipment and material, including manufacturer's descriptive data and technical literature, performance charts and curves, catalog cuts, and installation instructions. Diagrams, instructions, and other sheets proposed for posting.

### 3.5 FIELD TESTING AND ADJUSTING EQUIPMENT

#### 3.5.1 Operational Test

Prior to acceptance, perform an operational test of all pumps, drivers, and control systems to determine if the installed equipment meets the purpose and intent of the specifications. Demonstrate that the equipment is not electrically, mechanically, structurally, or otherwise defective; is in safe and satisfactory operating condition; and conforms with the specified operating characteristics. Prior to applying electrical power to any motor driven equipment, rotate the drive train by hand to

demonstrate free operation of all mechanical parts. Include checks for excessive vibration, leaks in all piping and seals, correct operation of control systems and equipment, proper alignment, excessive noise levels, and power consumption.

### 3.5.2 Retesting

If any deficiencies are revealed during any test, correct such deficiencies and reconduct the tests.

### 3.5.3 Performance Test Reports

Submit performance test reports in booklet form showing all field tests performed to adjust each component and all field tests performed to prove compliance with the specified performance criteria, upon completion and testing of the installed system. In each test report indicate the final position of controls.

## 3.6 MANUFACTURER'S SERVICES

Provide the services of a manufacturer's representative who is experienced in the installation, adjustment, and operation of the equipment specified. Supervise the installation, adjustment, and testing of the equipment.

## 3.7 FIELD TRAINING

Provide a field training course for designated operating and maintenance staff members. Provide training for a total period of 8 hours of normal working time and start after the system is functionally complete but prior to final acceptance tests. Cover all of the items contained in the operating and maintenance manuals. Submit six copies of operation and six copies of maintenance manuals for the equipment furnished. One complete set prior to performance testing and the remainder upon acceptance. Detail the step-by-step procedures required for system startup, operation, and shutdown. Include in the operation manuals the manufacturer's name, model number, parts list, and brief description of all equipment and their basic operating features. List in the maintenance manuals routine maintenance procedures, possible breakdowns and repairs, and troubleshooting guides. Include piping and equipment layout and simplified wiring and control diagrams of the system as installed in the maintenance manuals. Manuals must be approved prior to the field training course.

-- End of Section --

SECTION 23 03 00.00 20

BASIC MECHANICAL MATERIALS AND METHODS

08/10, CHG 3: 08/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM B117 (2019) Standard Practice for Operating  
Salt Spray (Fog) Apparatus

1.2 RELATED REQUIREMENTS

This section applies to all sections of Divisions: 21, FIRE SUPPRESSION; 22, PLUMBING; and 23, HEATING, VENTILATING, AND AIR CONDITIONING of this project specification, unless specified otherwise in the individual section.

1.3 QUALITY ASSURANCE

1.3.1 Material and Equipment Qualifications

Provide materials and equipment that are standard products of manufacturers regularly engaged in the manufacture of such products, which are of a similar material, design and workmanship. Standard products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year use must include applications of equipment and materials under similar circumstances and of similar size. The product must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2 year period.

1.3.2 Alternative Qualifications

Products having less than a two-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturer's factory or laboratory tests, can be shown.

1.3.3 Service Support

The equipment items must be supported by service organizations. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. These service organizations must be reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.3.4 Manufacturer's Nameplate

For each item of equipment, provide a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a

conspicuous place; the nameplate of the distributing agent will not be acceptable.

#### 1.3.5 Modification of References

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction", or words of similar meaning, to mean the Contracting Officer.

##### 1.3.5.1 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions must be considered mandatory, the word "should" is interpreted as "must." Reference to the "code official" must be interpreted to mean the "Contracting Officer." For Navy owned property, references to the "owner" must be interpreted to mean the "Contracting Officer." For leased facilities, references to the "owner" must be interpreted to mean the "lessor." References to the "permit holder" must be interpreted to mean the "Contractor."

##### 1.3.5.2 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, must be applied appropriately by the Contracting Officer as authorized by his administrative cognizance and the FAR.

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

#### 1.5 ELECTRICAL REQUIREMENTS

Furnish motors, controllers, disconnects and contactors with their respective pieces of equipment. Motors, controllers, disconnects and contactors must conform to and have electrical connections provided under Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Furnish internal wiring for components of packaged equipment as an integral part of the equipment. Extended voltage range motors will not be permitted. Controllers and contactors shall have a maximum of 120 volt control circuits, and must have auxiliary contacts for use with the controls furnished. When motors and equipment furnished are larger than sizes indicated, the cost of additional electrical service and related work must be included under the section that specified that motor or equipment. Power wiring and conduit for field installed equipment must be provided under and conform to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

#### 1.6 INSTRUCTION TO GOVERNMENT PERSONNEL

When specified in other sections, furnish the services of competent



instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the specified equipment or system. Instructors must be thoroughly familiar with all parts of the installation and must be trained in operating theory as well as practical operation and maintenance work.

Instruction must be given during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The number of man-days (8 hours per day) of instruction furnished must be as specified in the individual section. When more than 4 man-days of instruction are specified, use approximately half of the time for classroom instruction. Use other time for instruction with the equipment or system.

When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

#### 1.7 ACCESSIBILITY

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors.

### PART 2 PRODUCTS

Not used.

### PART 3 EXECUTION

#### 3.1 PAINTING OF NEW EQUIPMENT

New equipment painting must be factory applied or shop applied, and must be as specified herein, and provided under each individual section.

##### 3.1.1 Factory Painting Systems

Manufacturer's standard factory painting systems may be provided subject to certification that the factory painting system applied will withstand 125 hours in a salt-spray fog test, except that equipment located outdoors must withstand 500 hours in a salt-spray fog test. Salt-spray fog test must be in accordance with ASTM B117, and for that test the acceptance criteria must be as follows: immediately after completion of the test, the paint must show no signs of blistering, wrinkling, or cracking, and no loss of adhesion; and the specimen must show no signs of rust creepage beyond 0.125 inch on either side of the scratch mark.

The film thickness of the factory painting system applied on the equipment must not be less than the film thickness used on the test specimen. If manufacturer's standard factory painting system is being proposed for use on surfaces subject to temperatures above 120 degrees F, the factory painting system must be designed for the temperature service.

##### 3.1.2 Shop Painting Systems for Metal Surfaces

Clean, pretreat, prime and paint metal surfaces; except aluminum surfaces need not be painted. Apply coatings to clean dry surfaces. Clean the

surfaces to remove dust, dirt, rust, oil and grease by wire brushing and solvent degreasing prior to application of paint, except metal surfaces subject to temperatures in excess of 120 degrees F must be cleaned to bare metal.

Where more than one coat of paint is specified, apply the second coat after the preceding coat is thoroughly dry. Lightly sand damaged painting and retouch before applying the succeeding coat. Color of finish coat must be aluminum or light gray.

- a. Temperatures Less Than 120 Degrees F: Immediately after cleaning, the metal surfaces subject to temperatures less than 120 degrees F must receive one coat of pretreatment primer applied to a minimum dry film thickness of 0.3 mil, one coat of primer applied to a minimum dry film thickness of 1 mil; and two coats of enamel applied to a minimum dry film thickness of 1 mil per coat.
- b. Temperatures Between 120 and 400 Degrees F: Metal surfaces subject to temperatures between 120 and 400 degrees F must receive two coats of 400 degrees F heat-resisting enamel applied to a total minimum thickness of 2 mils.
- c. Temperatures Greater Than 400 Degrees F: Metal surfaces subject to temperatures greater than 400 degrees F must receive two coats of 600 degrees F heat-resisting paint applied to a total minimum dry film thickness of 2 mils.

-- End of Section --

SECTION 23 05 15

COMMON PIPING FOR HVAC  
**05/22, CHG 2: 08/24**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 325 (2017) Steel Construction Manual

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|                           |                                                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------|
| ASME A112.18.1/CSA B125.1 | (2018) Plumbing Supply Fittings                                                                                          |
| ASME A112.19.2/CSA B45.1  | (2018; ERTA 2018) Standard for Vitreous China Plumbing Fixtures and Hydraulic Requirements for Water Closets and Urinals |
| ASME B1.20.7              | (1991; R 2013) Standard for Hose Coupling Screw Threads (Inch)                                                           |
| ASME B16.1                | (2020) Gray Iron Pipe Flanges and Flanged Fittings Classes 25, 125, and 250                                              |
| ASME B16.3                | (2021) Malleable Iron Threaded Fittings, Classes 150 and 300                                                             |
| ASME B16.5                | (2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard                                     |
| ASME B16.9                | (2018) Factory-Made Wrought Buttwelding Fittings                                                                         |
| ASME B16.22               | (2021) Wrought Copper and Copper Alloy Solder Joint Pressure Fittings                                                    |
| ASME B16.39               | (2020) Standard for Malleable Iron Threaded Pipe Unions; Classes 150, 250, and 300                                       |
| ASME B31.3                | (2022; Errata 2023) Process Piping                                                                                       |
| ASME B40.100              | (2022) Pressure Gauges and Gauge Attachments                                                                             |
| ASME BPVC SEC IX          | (2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications                                           |

AMERICAN WELDING SOCIETY (AWS)

|                |                                                                      |
|----------------|----------------------------------------------------------------------|
| AWS A5.8/A5.8M | (2019) Specification for Filler Metals for Brazing and Braze Welding |
| AWS WHB-2.9    | (2004) Welding Handbook; Volume 2, Welding Processes, Part 1         |

ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                                 |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A6/A6M     | (2024b) Standard Specification for General Requirements for Rolled Structural Steel Bars, Plates, Shapes, and Sheet Piling                      |
| ASTM A53/A53M   | (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless                                           |
| ASTM A126       | (2004; R 2023) Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings                                             |
| ASTM A197/A197M | (2020; R 2023; E 2023) Standard Specification for Cupola Malleable Iron                                                                         |
| ASTM A234/A234M | (2024) Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service             |
| ASTM A276/A276M | (2024) Standard Specification for Stainless Steel Bars and Shapes                                                                               |
| ASTM A278/A278M | (2001; R 2020) Standard Specification for Gray Iron Castings for Pressure-Containing Parts for Temperatures Up to 650 degrees F (350 degrees C) |
| ASTM A307       | (2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength                                       |
| ASTM A480/A480M | (2024) Standard Specification for General Requirements for Flat-Rolled Stainless and Heat-Resisting Steel Plate, Sheet, and Strip               |
| ASTM A563       | (2021; E 2022a) Standard Specification for Carbon and Alloy Steel Nuts                                                                          |
| ASTM B32        | (2020) Standard Specification for Solder Metal                                                                                                  |
| ASTM B62        | (2017) Standard Specification for Composition Bronze or Ounce Metal Castings                                                                    |
| ASTM B88        | (2022) Standard Specification for Seamless Copper Water Tube                                                                                    |

|                 |                                                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------|
| ASTM B117       | (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus                                                             |
| ASTM B370       | (2022) Standard Specification for Copper Sheet and Strip for Building Construction                                            |
| ASTM B749       | (2020) Standard Specification for Lead and Lead Alloy Strip, Sheet and Plate Products                                         |
| ASTM C67/C67M   | (2023a) Standard Test Methods for Sampling and Testing Brick and Structural Clay Tile                                         |
| ASTM C109/C109M | (2023) Standard Test Method for Compressive Strength of Hydraulic Cement Mortars (Using 2-in. or (50-mm) Cube Specimens)      |
| ASTM C404       | (2024) Standard Specification for Aggregates for Masonry Grout                                                                |
| ASTM C476       | (2023) Standard Specification for Grout for Masonry                                                                           |
| ASTM C553       | (2013; R 2019) Standard Specification for Mineral Fiber Blanket Thermal Insulation for Commercial and Industrial Applications |
| ASTM C920       | (2018) Standard Specification for Elastomeric Joint Sealants                                                                  |
| ASTM E1         | (2014) Standard Specification for ASTM Liquid-in-Glass Thermometers                                                           |
| ASTM E814       | (2024) Standard Test Method for Fire Tests of Penetration Firestop Systems                                                    |
| ASTM F104       | (2011; R 2020) Standard Classification System for Nonmetallic Gasket Materials                                                |

FLUID SEALING ASSOCIATION (FSA)

|          |                                                                                                     |
|----------|-----------------------------------------------------------------------------------------------------|
| FSA-0017 | (1995e6) Standard for Non-Metallic Expansion Joints and Flexible Pipe Connectors Technical Handbook |
|----------|-----------------------------------------------------------------------------------------------------|

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

|           |                                                                                                                |
|-----------|----------------------------------------------------------------------------------------------------------------|
| MSS SP-58 | (2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation |
| MSS SP-67 | (2022) Butterfly Valves                                                                                        |
| MSS SP-70 | (2011) Gray Iron Gate Valves, Flanged and Threaded Ends                                                        |
| MSS SP-72 | (2010a) Ball Valves with Flanged or Butt-Welding Ends for General Service                                      |

|            |                                                                                      |
|------------|--------------------------------------------------------------------------------------|
| MSS SP-80  | (2019) Bronze Gate, Globe, Angle and Check Valves                                    |
| MSS SP-110 | (2010) Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends   |
| MSS SP-125 | (2010) Gray Iron and Ductile Iron In-Line, Spring-Loaded, Center-Guided Check Valves |

U.S. DEPARTMENT OF DEFENSE (DOD)

|             |                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------|
| MIL-C-18480 | (1982; Rev B; Notice 2 2009; Notice 3 2022) Coating Compound, Bituminous, Solvent, Coal-Tar Base |
|-------------|--------------------------------------------------------------------------------------------------|

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| CID A-A-1922  | (Rev A; Notice 1; Notice 2; Notice 3; Notice 4) Shield, Expansion (Caulking Anchors, Single Lead)                           |
| CID A-A-1923  | (Rev A; Notice 1; Notice 2; Notice 3; Notice 4) Shield, Expansion (Lag, Machine and Externally Threaded Wedge Bolt Anchors) |
| CID A-A-1924  | (Rev A; Notice 3) Shield, Expansion (Self Drilling Tubular Expansion Shell Bolt Anchors)                                    |
| CID A-A-1925  | (Rev A; Notice 1; Notice 2; Notice 3; Notice 4) Shield Expansion (Nail Anchors)                                             |
| CID A-A-55614 | (Basic; Notice 2) Shield, Expansion (Non-Drilling Expansion Anchors)                                                        |
| CID A-A-55615 | (Basic; Notice 1; Notice 2; Notice 3; Notice 4) Shield, Expansion (Wood Screw and Lag Bolt Self-Threading Anchors)          |

UL SOLUTIONS (UL)

|         |                                                                      |
|---------|----------------------------------------------------------------------|
| UL 1479 | (2015; Reprint Apr 2024) Fire Tests of Through-Penetration Firestops |
|---------|----------------------------------------------------------------------|

1.2 GENERAL REQUIREMENTS

- a. Section 23 30 00 HVAC AIR DISTRIBUTION applies to work specified in this section
- b. Section 23 05 48.19 SEISMIC BRACING FOR HVAC applies to work specified in this section.
- c. Section 40 05 13.96 WELDING PROCESS PIPING applies to work specified in this section.
- d. Submit Records of Existing Conditions consisting of the results of Contractor's survey of work area conditions and features of existing

structures and facilities within and adjacent to the jobsite.  
Commencement of work constitutes acceptance of the existing conditions.

- e. Include with Equipment Foundation Data for piping systems all plan dimensions of foundations and relative elevations, equipment weight and operating loads, horizontal and vertical loads, horizontal and vertical clearances for installation, and size and location of anchor bolts.
- f. Submit Fabrication Drawings for pipes, valves and specialties consisting of fabrication and assembly details to be performed in the factory.
- g. Submit Material, Equipment, and Fixture Lists for pipes, valves and specialties including manufacturer's style or catalog numbers, specification and drawing reference numbers, warranty information, and fabrication site information. Provide a complete list of construction equipment to be used.
- h. Submit Manufacturer's Standard Color Charts for pipes, valves and specialties showing the manufacturer's recommended color and finish selections.
- i. Include with Listing of Product Installations for piping systems identification of at least 5 units, similar to those proposed for use, that have been in successful service for a minimum period of 5 years. Include in the list purchaser, address of installation, service organization, and date of installation.
- j. Submit Record Drawings for pipes, valves and accessories providing current factual information including deviations and amendments to the drawings, and concealed and visible changes in the work.
- k. Submit Connection Diagrams for pipes, valves and specialties indicating the relations and connections of devices and apparatus by showing the general physical layout of all controls, the interconnection of one system (or portion of system) with another, and internal tubing, wiring, and other devices.
- l. Submit Coordination Drawings for pipes, valves and specialties showing coordination of work between different trades and with the structural and architectural elements of work. Detail all drawings sufficiently to show overall dimensions of related items, clearances, and relative locations of work in allotted spaces. Indicate on drawings where conflicts or clearance problems exist between various trades.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Material, Equipment, and Fixture Lists; G

SD-02 Shop Drawings

Record Drawings; G  
Connection Diagrams; G  
Coordination Drawings; G  
Fabrication Drawings; G  
Installation Drawings; G

SD-03 Product Data

Pipe and Fittings; G  
Piping Specialties; G  
Emergency Eyewash and Shower; G  
Valves; G  
Miscellaneous Materials; G  
Supporting Elements; G  
Equipment Foundation Data; G

SD-04 Samples

Manufacturer's Standard Color Charts; G

SD-05 Design Data

Pipe and Fittings; G  
Piping Specialties; G  
Emergency Eyewash and Shower; G  
Valves; G

SD-06 Test Reports

Hydrostatic Tests; G  
Air Tests; G  
Valve-Operating Tests; G  
Drainage Tests; G  
Pneumatic Tests; G  
Non-Destructive Electric Tests; G  
System Operation Tests; G

SD-07 Certificates

Record of Satisfactory Field Operation; G



List of Qualified Permanent Service Organizations; G

Listing of Product Installations; G

Records of Existing Conditions; G

Surface Resistance; G

Shear and Tensile Strengths; G

Temperature Ratings; G

Bending Tests; G

Flattening Tests; G

Transverse Guided Weld Bend Tests; G

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

#### 1.4 QUALITY ASSURANCE

##### 1.4.1 Material and Equipment Qualifications

Provide materials and equipment that are standard products of manufacturers regularly engaged in the manufacture of such products, which are of a similar material, design and workmanship. Provide standard products in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year use includes applications of equipment and materials under similar circumstances and of similar size. Ensure the product has been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2 year period.

##### 1.4.2 Alternative Qualifications

Products having less than a two-year field service record are acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturer's factory or laboratory tests, can be shown.

##### 1.4.3 Service Support

Ensure the equipment items are supported by service organizations. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. Select service organizations that are reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

##### 1.4.4 Manufacturer's Nameplate

Provide a nameplate on each item of equipment bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent is not acceptable.

#### 1.4.5 Modification of References

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer.

##### 1.4.5.1 Definitions

For the International Code Council (ICC) Codes referenced in the contract documents, advisory provisions are considered mandatory, the word "should" is interpreted as "shall." Reference to the "code official" is interpreted to mean the "Contracting Officer." For Navy owned property, interpret references to the "owner" to mean the "Contracting Officer." For leased facilities, references to the "owner" is interpreted to mean the "lessor." References to the "permit holder" are interpreted to mean the "Contractor."

##### 1.4.5.2 Administrative Interpretations

For ICC Codes referenced in the contract documents, the provisions of Chapter 1, "Administrator," do not apply. These administrative requirements are covered by the applicable Federal Acquisition Regulations (FAR) included in this contract and by the authority granted to the Officer in Charge of Construction to administer the construction of this project. References in the ICC Codes to sections of Chapter 1, are applied as appropriate by the Contracting Officer and as authorized by his administrative cognizance and the FAR.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

Handle, store, and protect equipment and materials to prevent damage before and during installation in accordance with the manufacturer's recommendations, and as approved by the Contracting Officer. Replace damaged or defective items.

#### 1.6 ELECTRICAL REQUIREMENTS

Furnish motors, controllers, disconnects and contactors with their respective pieces of equipment. Ensure motors, controllers, disconnects and contactors conform to and have electrical connections provided under Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Furnish internal wiring for components of packaged equipment as an integral part of the equipment. Extended voltage range motors is not permitted. Provide controllers and contactors with a maximum of 120 volt control circuits, and auxiliary contacts for use with the controls furnished. When motors and equipment furnished are larger than sizes indicated, include the cost of additional electrical service and related work under the section that specified that motor or equipment. Provide power wiring and conduit for field installed equipment under and conform to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

#### 1.7 INSTRUCTION TO GOVERNMENT PERSONNEL

When specified in other sections, furnish the services of competent instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including

pertinent safety requirements, of the specified equipment or system. Provide instructors thoroughly familiar with all parts of the installation and trained in operating theory as well as practical operation and maintenance work.

Give instruction during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The number of man-days (8 hours per day) of instruction furnished is as specified in the individual section. When more than four man-days of instruction are specified, use approximately half of the time for classroom instruction. Use other time for instruction with the equipment or system.

When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

## 1.8 ACCESSIBILITY

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible. Install concealed valves, expansion joints, controls, dampers, and equipment requiring access, in locations freely accessible through access doors.

## PART 2 PRODUCTS

### 2.1 PIPE AND FITTINGS

Submit equipment and performance data for pipe and fittings consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis. Submit design analysis and calculations consisting of surface resistance, rates of flow, head losses, inlet and outlet design, required radius of bend, and pressure calculations. Also include in data pipe size, shape, and dimensions, as well as temperature ratings, vibration and thrust limitations minimum burst pressures, shut-off and non-shock pressures and weld characteristics.

#### 2.1.1 Type BCS, Black Carbon Steel

Ensure pipe 1/8 through 12 inches is Schedule 40 black carbon steel, conforming to ASTM A53/A53M.

Ensure pipe 1/8 through 10 inches is Schedule 40 seamless or electric-resistance welded black carbon steel, conforming to ASTM A53/A53M, Type E, Grade B (electric-resistance welded). Grade A should be used for permissible field bending, in both cases.

Ensure pipe 12 through 24 inches is 0.375-inch wall black carbon steel, conforming to ASTM A53/A53M, Type E, Grade B (electric-resistance welded).

Ensure fittings 2 inches and under are 150-pounds per square inch, gage (psig) working steam pressure (wsp) banded black malleable iron screwed, conforming to ASTM A197/A197M and ASME B16.3.

Ensure unions 2 inches and under are 250 pounds per square inch, wsp female, screwed, black malleable iron with brass-to-iron seat, and ground joint, conforming to ASME B16.39.

Ensure fittings 2-1/2 inches and over are Steel butt weld, conforming to

ASTM A234/A234M and ASME B16.9 to match pipe wall thickness.

Ensure flanges 2-1/2 inches and over are 150-pound forged-steel conforming to ASME B16.5, welding neck to match pipe wall thickness.

## 2.1.2 Type CPR, Copper

### 2.1.2.1 Type CPR-A, Copper Above Ground

Ensure tubing 2 inches and under is seamless copper tubing, conforming to ASTM B88, Type L (hard-drawn for all horizontal and all exposed vertical lines, annealed for concealed vertical lines).

Ensure fittings 2 inches and under are 150-psig wsp wrought-copper solder joint fittings conforming to ASME B16.22.

Ensure unions 2 inches and under are 150-psig wsp wrought-copper solder joint, conforming to ASME B16.22.

Provide brazing rod with Classification BCuP-5, conforming to AWS A5.8/A5.8M.

Use solder, alloy Sb-5, conforming to ASTM B32.

## 2.2 PIPING SPECIALTIES

Submit equipment and performance data for piping specialties consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis. Submit design analysis and calculations consisting of surface resistance, rates of flow, head losses, inlet and outlet design, required radius of bend, and pressure calculations. Also include in data pipe size, shape, and dimensions, as well as temperature ratings, vibration and thrust limitations minimum burst pressures, shut-off and non-shock pressures and weld characteristics.

### 2.2.1 Air Vents

Provide manual air vents using 3/8-inch globe valves.

Provide automatic air vents on pumps, mains, and where indicated using ball-float construction. Ensure the vent inlet is not less than 3/4-inch ips and the outlet not less than 1/4-inch ips. Orifice size is 1/8 inch. Provide corrosion-resistant steel trim conforming to ASTM A276/A276M or ASTM A480/A480M. Fit vent with try-cock. Ensure vent discharges air at any pressure up to 150 psi. Ensure outlet is copper tube routed.

### 2.2.2 Dielectric Connections

Electrically isolate dissimilar pipe metals from each other by couplings, unions, or flanges commercially manufactured for that purpose and rated for the service pressure and temperature.

### 2.2.3 Expansion Vibration Isolation Joints

Construct single or multiple arch-flanged expansion vibration isolation joints of steel-ring reinforced chloroprene-impregnated cloth materials. Design joint to absorb the movement of the pipe sections in which installed with no detrimental effect on the pipe or connected equipment.

Back flanges with ferrous-metal backing rings. Provide control rod assemblies to restrict joint movement. Coat all nonmetallic exterior surfaces of the joint with chlorosulphonated polyethylene. Provide grommets in limit bolt hole to absorb noise transmitted through the bolts.

Ensure joints are suitable for continuous-duty working temperature of at least 250 degrees F.

Ensure joint, single-arch, movement limitations and size-related, pressure characteristics conform to FSA-0017.

#### 2.2.4 Hose Faucets

Construct hose faucets with 1/2 inch male inlet threads, hexagon shoulder, and 3/4 inch hose connection, conforming to ASME A112.18.1/CSA B125.1. Ensure hose-coupling screw threads conform to ASME B1.20.7.

Provide vandal proof, atmospheric-type vacuum breaker on the discharge of all potable water lines.

#### 2.2.5 Pressure Gages

Ensure pressure gages conform to ASME B40.100 and to requirements specified herein. Pressure-gage size is 3-1/2 inches nominal diameter. Ensure case is corrosion-resistant steel, conforming to any of the AISI 300 series of ASTM A6/A6M, with an ASM No. 4 standard commercial polish or better. Equip gages with adjustable red marking pointer and damper-screw adjustment in inlet connection. Align service-pressure reading at midpoint of gage range. Ensure all gages are Grade B or better and be equipped with gage isolators.

#### 2.2.6 Sight-Flow Indicators

Construct sight-flow indicators for pressure service on 3-inch ips and smaller of bronze with specially treated single- or double-glass sight windows and have a bronze, nylon, or tetrafluoroethylene rotating flow indicator mounted on an AISI Type 304 or 316 corrosion-resistant steel shaft. Body may have screwed or flanged end. Provide pressure- and temperature-rated assembly for the applied service. Flapper flow-type indicators are not acceptable.

#### 2.2.7 Sleeve Couplings

Sleeve couplings for plain-end pipe consist of one steel middle ring, two steel followers, two chloroprene or Buna-N elastomer gaskets, and the necessary steel bolts and nuts.

#### 2.2.8 Thermometers

Ensure thermometers conform to ASTM E1, except for being filled with a red organic liquid. Provide an industrial pattern armored glass thermometer, (well-threaded and seal-welded). Ensure thermometers installed 6 feet or higher above the floor have an adjustable angle body. Ensure scale is not less than 7 inches long and the case face is manufactured from manufacturer's standard polished aluminum or AISI 300 series polished corrosion-resistant steel. Thermometer range is 0 - 200. Provide thermometers with nonferrous separable wells. Provide lagging extension

to accommodate insulation thickness.

#### 2.2.9 Pump Suction Strainers

Provide a cast iron strainer body, rated for not less than 25 psig at 100 degrees F, with flanges conforming to ASME B16.1, Class 125. Strainer construction is such that there is a machined surface joint between body and basket that is normal to the centerline of the basket.

Ensure minimum ratio of open area of each basket to pipe area is 3 to 1. Provide a basket with AISI 300 series corrosion-resistant steel wire mesh with perforated backing.

Ensure mesh is capable of retaining all particles larger than 1,000 micrometer, with a pressure drop across the strainer body of not more than 0.5 psi when the basket is two-thirds dirty at maximum system flow rate. Provide reducing fittings from strainer-flange size to pipe size.

Provide a differential-pressure gage fitted with a two-way brass cock across the strainer.

Provide manual air vent cocks in cap of each strainer.

#### 2.2.10 Line Strainers, Water Service

Provide Y-type strainers with removable basket. Ensure strainers in sizes 2-inch ips and smaller have screwed ends; in sizes 2-1/2-inch ips and larger, strainers have flanged ends. Ensure body working-pressure rating exceeds maximum service pressure of installed system by at least 50 percent. Ensure body has cast-in arrows to indicate direction of flow. Ensure all strainer bodies fitted with screwed screen retainers have straight threads and gasketed with nonferrous metal. For strainer bodies 2-1/2-inches and larger, fitted with bolted-on screen retainers, provide offset blowdown holes. Fit all strainers larger than 2-1/2-inches with manufacturer's standard ball-type blowdown valve. Ensure body material is cast iron conforming to Class 30 ASTM A278/A278M. Where system material is nonferrous, use nonferrous metal for the metal strainer body material.

Ensure minimum free-hole area of strainer element is equal to not less than 3.4 times the internal area of connecting piping. Strainer screens perforation size is not to exceed 0.045-inch. Ensure strainer screens have finished ends fitted to machined screen chamber surfaces to preclude bypass flow. Strainer element material is Monel metal.

#### 2.2.11 Emergency Eyewash and Shower

ANSI/ISEA Z358.1, floor supported free standing unit with privacy curtain. Provide deluge shower head, stay-open ball valve operated by pull rod and ring or triangular handle. Provide eyewash dust caps and stay-open ball valve operated by foot treadle or push handle. Unit must deliver a minimum of 20 gpm of water at 30 psig flow pressure for 15 minutes. Eye wash unit must have a minimum of 0.4 gpm of water at 30 psig flow pressure for 15 minutes. Position eye wash nozzles 33 inches to 45 inches above finished floor. Provide copper alloy control valves. Provide an air-gap with the lowest potable eye wash water outlet located above the overflow rim by not less than the International Plumbing Code minimum. Provide a pressure-compensated tempering valve, with leaving water temperature setpoint adjustable throughout the range 60 degrees F to 95 degrees F. Provide packaged, UL listed, alarm system; including an amber

strobe lamp, horn with externally adjustable loudness and horn silencing switch, mounting hardware, and waterflow service within NEMA Type 3 or 4 enclosures.

## 2.3 VALVES

Submit equipment and performance data for valves consisting of corrosion resistance and life expectancy. Submit design analysis and calculations consisting of rates of flow, head losses, inlet and outlet design, and pressure calculations. Also include in data, pipe dimensions, as well as temperature ratings, vibration and thrust limitations, minimum burst pressures, shut-off and non-shock pressures and weld characteristics.

### 2.3.1 Ball and Butterfly Valves

Ensure ball valves conform to MSS SP-72 for flanged valves and MSS SP-110 for screwed-end valves for Figure 1B vertically split body, 1C top entry three piece body and are rated for service at not less than 175 psig at 200 degrees F. For valve bodies in sizes 2 inches and smaller, use screwed-end connection-type constructed of Class A copper alloy. For valve bodies in sizes 2-1/2 inches and larger, use flanged-end connection type, constructed of Class D, E, or F material. Balls and stems of valves 2 inches and smaller are manufacturer's standard with hard chrome plating finish. Balls and stems of valves 2-1/2 inches and larger are manufacturer's standard Class C corrosion-resistant steel alloy with hard chrome plating. Balls of valves 6 inches and larger may be Class D with 900 Brinell hard chrome plating. Ensure valves are suitable for flow from either direction and seal equally tight in either direction. Valves with ball seals held in place by spring washers are not acceptable. Ensure all valves have adjustable packing glands. Seats and seals are fabricated from tetrafluoroethylene.

Ensure butterfly valves conform to MSS SP-67 and are the lug type for mounting between specified flanges. Ensure valves are rated for 150-psig shutoff and nonshock working pressure. Select bodies of cast ferrous metal conforming to ASTM A126, Class B, and to ASME B16.1 for body wall thickness. Seats and seals are fabricated from resilient elastomer designed for field removal and replacement.

### 2.3.2 Drain, Vent, and Gage Cocks

Provide lever handle drain, vent, and gage cocks, ground key type, with washer and screw, constructed of polished ASTM B62 bronze, and rated 125-psi wsp. Ensure end connections are rated for specified service pressure.

Ensure pump vent cocks, and where spray control is required, are UL umbrella-hood type, constructed of manufacturer's standard polished brass. Ensure cocks are 1/2-inch ips male, end threaded, and rated at not less than 125 psi at 225 degrees F.

### 2.3.3 Gate Valves (GAV)

Ensure gate valves 2 inches and smaller conform to MSS SP-80. For valves located in tunnels, equipment rooms, factory-assembled equipment, and where indicated use union-ring bonnet, screwed-end type. Make packing of non-asbestos type materials. Use rising stem type valves.

Ensure gate valves 2-1/2 inches and larger, are Type I, (solid wedge disc, tapered seats, steam rated); Class 125 (125-psig steam-working pressure at 353 degrees F saturation); and 200-psig, wog (nonshock), conforming to MSS SP-70 and to requirements specified herein. Select flanged valves, with bronze trim and outside screw and yoke (OS&Y) construction. Make packing of non-asbestos type materials.

For gate valves 8-inches and larger, provide gate valve with integral 3/4-inch globe valve bypass. Bypass globe valve shall be at least the same pressure class as the main gate valve.

#### 2.3.4 Globe and Angle Valves (GLV-ANV)

Ensure globe and angle valves 2 inches and smaller, are 125-pound, 125-psi conforming to MSS SP-80 and to requirements specified herein. For valves located in tunnels, equipment rooms, factory-assembled equipment, and where indicated, use union-ring bonnet, screwed-end type. Ensure disc is free to swivel on the stem in all valve sizes. Composition seating-surface disc construction may be substituted for all metal-disc construction. Make packing of non-asbestos type materials. Ensure disk and packing are suitable for pipe service installed.

Ensure globe and angle valves, 2-1/2 inches and larger, are cast iron with bronze trim. Ensure valve bodies are cast iron conforming to ASTM A126, Class A, as specified for Class 1 valves under MSS SP-80. Select flanged valves in conformance with ASME B16.1. Valve construction is outside screw and yoke (OS&Y) type. Make packing of non-asbestos type materials.

#### 2.3.5 Standard Check Valves (SCV)

Ensure standard check valves in sizes 2 inches and smaller are 125-psi swing check valves except as otherwise specified. Provide lift checks where indicated. Ensure swing-check pins are nonferrous and suitably hard for the service. Select composition type discs. Ensure the swing-check angle of closure is manufacturer's standard unless a specific angle is needed.

Use cast iron, bronze trim, swing type check valves in sizes 2-1/2 inches and larger. Ensure valve bodies are cast iron, conforming to ASTM A126, Class A and valve ends are flanged in conformance with ASME B16.1. Swing-check pin is AISI Type or approved equal corrosion-resistant steel. Angle of closure is manufacturer's standard unless a specific angle is needed. Ensure valves have bolted and gasketed covers.

#### 2.3.6 Nonslam Check Valves (NSV)

Provide check valves at pump discharges in sizes 2 inches and larger with nonslam or silent-check operation conforming to MSS SP-125. Select a valve disc or plate that closes before line flow can reverse to eliminate slam and water-hammer due to check-valve closure. Ensure valve is Class 125 rated for 200-psi maximum, nonshock pressure at 150 degrees F in sizes to 12 inches. Use valves that are center-guided, globe-body type, fitted with flanges conforming to ASME B16.1. Valve body may be cast iron, or equivalent strength ductile iron. Select disks using manufacturer's standard bronze, aluminum bronze, or corrosion-resistant steel. Ensure pins, springs, and miscellaneous trim are manufacturer's standard



corrosion-resistant steel. Disk and shaft seals are Buna-N elastomer tetrafluoroethylene.

## 2.4 MISCELLANEOUS MATERIALS

Submit equipment and performance data for miscellaneous materials consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis.

### 2.4.1 Bituminous Coating

Ensure the bituminous coating is a solvent cutback, heavy-bodied material to produce not less than a 12-mil dry-film thickness in one coat, and is recommended by the manufacturer to be compatible with factory-applied coating and rubber joints.

For previously coal-tar coated and uncoated ferrous surfaces underground, use bituminous coating solvent cutback coal-tar type, conforming to MIL-C-18480.

### 2.4.2 Bolting

Ensure flange and general purpose bolting is hex-head and conforms to ASTM A307, Grade B (bolts, for flanged joints in piping systems where one or both flanges are cast iron). Heavy hex-nuts conform to ASTM A563. Square-head bolts and nuts are not acceptable. Ensure threads are coarse-thread series.

### 2.4.3 Elastomer Caulk

Use two-component polysulfide- or polyurethane-base elastomer caulking material, conforming to ASTM C920.

### 2.4.4 Escutcheons

Manufacture escutcheons from nonferrous metals and chrome-plated except when AISI 300 series corrosion-resistant steel is provided. Ensure metals and finish conforms to ASME A112.19.2/CSA B45.1.

Use one-piece escutcheons where mounted on chrome-plated pipe or tubing, and one-piece of split-pattern type elsewhere. Ensure all escutcheons have provisions consisting of internal spring-tension devices or setscrews for maintaining a fixed position against a surface.

### 2.4.5 Flashing

Ensure sheetlead conforms to ASTM B749, UNS Alloy Number L50049 (intended for use in laboratories and shops in general application).

Ensure sheet copper conforms to ASTM B370 and be not less than 16 ounces per square foot weight.

### 2.4.6 Flange Gaskets

Provide compressed non-asbestos sheets, conforming to ASTM F104, coated on both sides with graphite or similar lubricant, with nitrile composition, binder rated to 750 degrees F.

#### 2.4.7 Grout

Provide shrink-resistant grout as a premixed and packaged metallic-aggregate, mortar-grouting compound conforming to ASTM C404 and ASTM C476.

Ensure shrink-resistant grout is a combination of pre-measured and packaged epoxy polyamide or amine resins and selected aggregate mortar grouting compound conforming to the following requirements:

|                      |                 |                                              |
|----------------------|-----------------|----------------------------------------------|
| Tensile strength     |                 | 1,900 psi, minimum                           |
| Compressive strength | ASTM C109/C109M | 14,000 psi, minimum                          |
| Shrinkage, linear    |                 | 0.00012 inch per inch, maximum               |
| Water absorption     | ASTM C67/C67M   | 0.1 percent, maximum                         |
| Bond strength to     |                 | 1,000 psi, minimum steel in shear<br>minimum |

#### 2.4.8 Pipe Thread Compounds

Use polytetrafluoroethylene tape not less than 2 to 3 mils thick in potable and process water and in chemical systems for pipe sizes to and including 1-inch ips. Use polytetrafluoroethylene dispersions and other suitable compounds for all other applications upon approval by the Contracting Officer; however, do not use lead-containing compounds in potable water systems.

#### 2.5 SUPPORTING ELEMENTS

Submit equipment and performance data for the supporting elements consisting of corrosion resistance, life expectancy, gage tolerances, and grade line analysis.

Provide all necessary piping systems and equipment supporting elements, including but not limited to: building structure attachments; supplementary steel; hanger rods, stanchions, and fixtures; vertical pipe attachments; horizontal pipe attachments; anchors; guides; and spring-cushion, variable, or constant supports. Ensure supporting elements are suitable for stresses imposed by systems pressures and temperatures and natural and other external forces normal to this facility without damage to supporting element system or to work being supported.

Ensure supporting elements conform to requirements of ASME B31.3, and MSS SP-58, except as noted.

Ensure attachments welded to pipe are made of materials identical to that of pipe or materials accepted as permissible raw materials by referenced code or standard specification.

Ensure supporting elements exposed to weather are hot-dip galvanized or stainless steel. Select materials of such a nature that their apparent and latent-strength characteristics are not reduced due to galvanizing process. Electroplate supporting elements in contact with copper tubing with copper.

Type designations specified herein are based on MSS SP-58. Ensure masonry

anchor group-, type-, and style-combination designations are in accordance with CID A-A-1922, CID A-A-1923, CID A-A-1924, CID A-A-1925 , CID A-A-55614, and CID A-A-55615. Provide support elements, except for supplementary steel, that are cataloged, load rated, commercially manufactured products.

#### 2.5.1 Building Structure Attachments

##### 2.5.1.1 Anchor Devices, Concrete and Masonry

Ensure anchor devices conform to CID A-A-1922, CID A-A-1923, CID A-A-1924, CID A-A-1925, CID A-A-55614, and CID A-A-55615

For cast-in, floor mounted, equipment anchor devices, provide adjustable positions.

Do not use powder-actuated anchoring devices to support any mechanical systems components.

##### 2.5.1.2 Beam Clamps

Ensure beam clamps are center-loading MSS SP-58 Type 21, 28, 29, or 30.

When it is not possible to use center-loading beam clamps, eccentric-loading beam clamps, MSS SP-58 Type 25 or 27 may be used for piping sizes 2 inches and less and for piping sizes 2 through 10 inches provided two counterbalancing clamps are used per point of pipe support. Where more than one rod is used per point of pipe support, determine rod diameter in accordance with referenced standards.

##### 2.5.1.3 Type 19 and 23 C-Clamps

Torque Type 19 and 23 C-clamps in accordance with MSS SP-58 and have both locknuts and retaining devices furnished by the manufacturer. Field fabricated C-clamp bodies or retaining devices are not acceptable.

##### 2.5.1.4 Inserts, Concrete

Use concrete MSS SP-58 Type 18 inserts. When applied to piping in sizes 2 inches ips and larger and where otherwise required by imposed loads, insert and wire a 1-foot length of 1/2-inch reinforcing rod through wing slots. Submit proprietary-type continuous inserts for approval.

#### 2.5.2 Horizontal Pipe Attachments

##### 2.5.2.1 Single Pipes

Support piping in sizes to and including 2-inch ips by MSS SP-58 Type 6 solid malleable iron pipe rings, except that, use split-band-type rings in sizes up to 1-inch ips.

Support piping in sizes through 8-inch ips inclusive by MSS SP-58 Type 1, 3 or 4 attachments.

Use MSS SP-58 Type 1 and Type 6 assemblies on vapor-sealed insulated piping and have an inside diameter larger than pipe being supported to provide adequate clearance during pipe movement.

Where thermal movement of a point in a piping system 4 inches and larger would cause a hanger rod to deflect more than 4 degrees from the vertical

or where a horizontal point movement exceeds 1/2 inch, use MSS SP-58 Type 41, 44 through 46 or 49 pipe rolls.

Support piping in sizes larger than 8-inch ips with MSS SP-58 Type 41, 44 through 46 or 49 pipe rolls.

Use MSS SP-58 Type 40 shields on all insulated piping. Ensure area of the supporting surface is such that compression deformation of insulated surfaces does not occur. Roll away longitudinal and transverse shield edges from the insulation.

Provide insulated piping without vapor barrier on roll supports with MSS SP-58 Type 39 saddles.

Provide spring supports as indicated.

#### 2.5.2.2 Parallel Pipes

Use trapeze hangers fabricated from structural steel shapes, with U-bolts, in congested areas and where multiple pipe runs occur. Ensure structural steel shapes conform to supplementary steel requirements.

#### 2.5.3 Vertical Pipe Attachments

Ensure vertical pipe attachments are MSS SP-58 Type 8.

Include complete fabrication and attachment details of any spring supports in shop drawings.

#### 2.5.4 Hanger Rods and Fixtures

Use only circular cross section rod hangers to connect building structure attachments to pipe support devices. Use pipe, straps, or bars of equivalent strength for hangers only where approved by the Contracting Officer.

Provide turnbuckles, swing eyes, and clevises as required by support system to accommodate temperature change, pipe accessibility, and adjustment for load and pitch. Rod couplings are not acceptable.

#### 2.5.5 Supplementary Steel

Where it is necessary to frame structural members between existing members or where structural members are used in lieu of commercially rated supports, design and fabricate such supplementary steel in accordance with AISC 325.

### PART 3 EXECUTION

#### 3.1 PIPE INSTALLATION

Submit certificates for pipes, valves and specialties showing conformance with test requirements as contained in the reference standards contained in this section. Provide certificates verifying Surface Resistance, Shear and Tensile Strengths, Temperature Ratings, Bending Tests, Flattening Tests and Transverse Guided Weld Bend Tests.

Provide test reports for Hydrostatic Tests, Air Tests, Valve-Operating Tests, Drainage Tests, Pneumatic Tests, Non-Destructive Electric Tests and

System Operation Tests, in compliance with referenced standards contained within this section.

Fabricate and install piping systems in accordance with ASME B31.3, MSS SP-58, and AWS WHB-2.9.

Submit Installation Drawings for pipes, valves and specialties. Drawings include the manufacturer's design and construction calculations, forces required to obtain rated axial, lateral, or angular movements, installation criteria, anchor and guide requirements for equipment, and equipment room layout and design. Ensure drawings specifically advise on procedures to be followed and provisions required to protect expansion joints during specified hydrostatic testing operations.

Ensure connections between steel piping and copper piping are electrically isolated from each other. Dielectric pipe unions shall be installed to prevent galvanic corrosion. The dielectric unions shall have metal connections on both ends. The ends shall be threaded, flanged, or brazed to match adjacent piping. The metal parts of the union shall be separated so that the electrical current is below one percent of the galvanic current which would exist upon metal-to-metal contact. Gaskets, flanges, and unions shall be installed in accordance with manufacturer's recommendations.

Make final connections to equipment with unions or flanges installed every 100 feet of straight run. Install unions in the line downstream of screwed- and welded-end valves.

Ream all pipe ends before joint connections are made.

Make screwed joints with specified joint compound with not more than three threads showing after joint is made up.

Apply joint compounds to the male thread only and exercise care to prevent compound from reaching the unthreaded interior of the pipe.

Install screwed unions, welded unions, or bolted flanges wherever required to permit convenient removal of equipment, valves, and piping accessories from the piping system for maintenance.

Securely support piping systems with due allowance for thrust forces, thermal expansion and contraction. Do not subject the system to mechanical, chemical, vibrational or other damage as specified in ASME B31.3.

Ensure field welded joints conform to the requirements of the AWS WHB-2.9, ASME B31.3, and ASME BPVC SEC IX.

Accomplish preheat and postheat treatment of welds in accordance with ASME BPVC SEC IX and ASME B31.3.

### 3.2 VALVES

Install valves in piping mains and all branches and at equipment where indicated and as specified.

Install valves to permit isolation of branch piping and each equipment

item from the balance of the system.

Install riser and downcomer drains above piping shutoff valves in piping 2-1/2 inches and larger. Tap and fit shutoff valve body with a 1/2-inch plugged globe valve.

Install valves unavoidably located in furred or other normally inaccessible places with access panels adequately sized for the location and located so that concealed items may be serviced, maintained, or replaced.

### 3.3 SUPPORTING ELEMENTS INSTALLATION

Install supporting elements in accordance with the referenced codes and standards.

Support piping from building structure. Do not support piping from roof deck or from other pipe.

Run piping parallel with the lines of the building. Space and install piping and components so that a threaded pipe fitting may be removed between adjacent pipes and so that there is no less than 1/2 inch of clear space between the finished surface and other work and between the finished surface of parallel adjacent piping. Arrange hangers on different adjacent service lines running parallel with each other in line with each other and parallel to the lines of the building.

Install piping support elements at intervals specified hereinafter, at locations not more than 3 feet from the ends of each runout, and not over 1 foot from each change in direction of piping.

Base load rating for all pipe-hanger supports on insulated weight of lines filled with water and forces imposed. Deflection per span is not exceed slope gradient of pipe. Ensure supports are in accordance with the following minimum rod size and maximum allowable hanger spacing for specified pipe. For concentrated loads such as valves, reduce the allowable span proportionately:

| <u>PIPE SIZE</u><br><u>INCHES</u> | <u>ROD SIZE</u><br><u>INCHES</u> | <u>STEEL PIPE</u><br><u>FEET</u> | <u>COPPER PIPE</u><br><u>FEET</u> |
|-----------------------------------|----------------------------------|----------------------------------|-----------------------------------|
| 1 and smaller                     | 3/8                              | 8                                | 6                                 |
| 1-1/4 to 1-1/2                    | 3/8                              | 10                               | 8                                 |
| 2                                 | 3/8                              | 10                               | 8                                 |
| 2-1/2 to 3-1/2                    | 1/2                              | 12                               | 12                                |
| 4 to 5                            | 5/8                              | 16                               | 14                                |
| 6                                 | 3/4                              | 16                               | 16                                |
| 8 to 12                           | 7/8                              | 20                               | 20                                |
| 14 to 18                          | 1                                | 20                               | 20                                |
| 20 and over                       | 1-1/4                            | 20                               | 20                                |

Install vibration isolation supports where needed. Refer to Section 23 05 48.19 SEISMIC BRACING FOR HVAC where A/C equipment and piping is installed.

Support vertical risers independently of connected horizontal piping, whenever practicable, with fixed or spring supports at the base and at intervals to accommodate system range of thermal conditions. Ensure risers have guides for lateral stability. For risers subject to expansion, install only one rigid support at a point approximately one-third down from the top. Place clamps under fittings unless otherwise specified. Support carbon-steel pipe at each floor and at not more than 15-foot intervals for pipe 2 inches and smaller and at not more than 20-foot intervals for pipe 2-1/2 inches and larger.

### 3.4 PENETRATIONS

Install effective sound stopping and adequate operating clearance to prevent structure contact where piping penetrates walls, floors, or ceilings into occupied spaces adjacent to equipment rooms; where similar penetrations occur between occupied spaces; and where penetrations occur from pipe chases into occupied spaces. Occupied spaces include space above ceilings where no special acoustic treatment of ceiling is provided. Finish penetrations to be compatible with surface being penetrated.

### 3.5 SLEEVES

Install sleeves where piping passes through roofs, masonry, concrete walls and floors.

Continuously weld sleeves passing through steel decks to the deck.

Ensure sleeves that extend through floors, roofs, load bearing walls, and fire barriers are continuous and fabricated from Schedule 40 steel pipe, with welded anchor lugs. Form all other sleeves by molded linear polyethylene liners or similar materials that are removable. Ensure diameter of sleeves is large enough to accommodate pipe, insulation, and jacketing without touching the sleeve and provides a minimum 3/8-inch clearance. Install a sleeve size to accommodate mechanical and thermal motion of pipe precluding transmission of vibration to walls and the generation of noise.

Pack the space between a pipe, bare or insulated, and the inside of a pipe sleeve or a construction surface penetration solid with a mineral fiber conforming to ASTM C553 Type V (flexible blanket), to 1,000 degrees F. Install this packing wherever the piping passes through firewalls, equipment room walls, floors, and ceilings connected to occupied spaces, and other locations where sleeves or construction-surface penetrations occur between occupied spaces. Where sleeves or construction surface penetrations occur between conditioned and unconditioned spaces, fill the space between a pipe, bare or insulated, and the inside of a pipe sleeve or construction surface penetration with an elastomer caulk to a depth of 1/2 inch. Ensure all caulked surfaces are oil- and grease-free.

Ensure through-penetration fire stop materials and methods are in accordance with ASTM E814 and UL 1479.

Caulk exterior wall sleeves watertight with lead and oakum or mechanically expandable chloroprene inserts with mastic-sealed metal components.

Ensure sleeve height above roof surface is a minimum of 12 and a maximum of 18-inches.

### 3.6 ESCUTCHEONS

Install escutcheons at all penetrations of piping into finished areas. Where finished areas are separated by partitions through which piping passes, install escutcheons on both sides of the partition. Where suspended ceilings are installed, install plates at the underside only of such ceilings. For insulated pipes, select plates large enough to fit around the insulation. Use chrome-plated escutcheons in all occupied spaces and of size sufficient to effectively conceal openings in building construction. Firmly attach escutcheons with setscrews.

### 3.7 FLASHINGS

Install flashings at penetrations of building boundaries by mechanical systems and related work.

### 3.8 OPERATION AND MAINTENANCE

Provide Operation and Maintenance Manuals consistent with manufacturer's standard brochures, schematics, printed instructions, general operating procedures and safety precautions. Submit test data that is clear and readily legible.

### 3.9 PAINTING OF NEW EQUIPMENT

Factory or shop apply new equipment painting, as specified herein, and provided under each individual section.

#### 3.9.1 Factory Painting Systems

Manufacturer's standard factory painting systems may be provided subject to certification that the factory painting system applied withstands 125 hours in a salt-spray fog test, except that equipment located outdoors withstand 500 hours in a salt-spray fog test. Conduct salt-spray fog test is in accordance with ASTM B117, and for that test the acceptance criteria is as follows: immediately after completion of the test, the inspected paint shows no signs of blistering, wrinkling, or cracking, and no loss of adhesion; and the specimen shows no signs of rust creepage beyond 0.125 inch on either side of the scratch mark.

Ensure the film thickness of the factory painting system applied on the equipment is not less than the film thickness used on the test specimen. If manufacturer's standard factory painting system is being proposed for use on surfaces subject to temperatures above 120 degrees F, design the factory painting system for the temperature service.

#### 3.9.2 Shop Painting Systems for Metal Surfaces

Clean, pretreat, prime and paint metal surfaces; except aluminum surfaces need not be painted. Apply coatings to clean dry surfaces. Clean the surfaces to remove dust, dirt, rust, oil and grease by wire brushing and solvent degreasing prior to application of paint, except clean to bare metal, surfaces subject to temperatures in excess of 120 degrees F.

Where more than one coat of paint is specified, apply the second coat after the preceding coat is thoroughly dry. Lightly sand damaged painting



and retouch before applying the succeeding coat. Selected color of finish coat is aluminum or light gray.

- a. Temperatures Less Than 120 Degrees F: Immediately after cleaning, the metal surfaces subject to temperatures less than 120 degrees F receives one coat of pretreatment primer applied to a minimum dry film thickness of 0.3 mil, one coat of primer applied to a minimum dry film thickness of one mil; and two coats of enamel applied to a minimum dry film thickness of one mil per coat.
- b. Temperatures Between 120 and 400 Degrees F: Metal surfaces subject to temperatures between 120 and 400 degrees F Receives two coats of 400 degrees F heat-resisting enamel applied to a total minimum thickness of 2 mils.
- c. Temperatures Greater Than 400 Degrees F: Metal surfaces subject to temperatures greater than 400 degrees F receives two coats of 600 degrees F heat-resisting paint applied to a total minimum dry film thickness of 2 mils.

-- End of Section --

SECTION 23 05 48.19

SEISMIC BRACING FOR HVAC  
**05/18, CHG 2: 08/20**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 355.2 (2007) Qualification of Post-Installed Mechanical Anchors in Concrete and Commentary

ACI 355.4 (2011) Qualification of Post-Installed Adhesive Anchors in Concrete (ACI 355.4) and Commentary

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 325 (2017) Steel Construction Manual

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C105/A21.5 (2018) Polyethylene Encasement for Ductile-Iron Pipe Systems

AWWA C116/A21.16 (2015) Protective Fusion-Bonded Coatings for the Interior and Exterior Surfaces of Ductile-Iron and Gray Iron Fittings

AWWA C153/A21.53 (2019) Ductile-Iron Compact Fittings for Water Service

AWWA C213 (2022) Fusion-Bonded Epoxy Coatings and Linings for Steel Water Pipe and Fittings

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A53/A53M (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A153/A153M (2023) Standard Specification for Zinc

Coating (Hot-Dip) on Iron and Steel  
Hardware

|                 |                                                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------|
| ASTM A325       | (2014) Standard Specification for Structural Bolts, Steel, Heat Treated, 120/105 ksi Minimum Tensile Strength         |
| ASTM A490       | (2014a) Standard Specification for Structural Bolts, Alloy Steel, Heat Treated, 150 ksi Minimum Tensile Strength      |
| ASTM A500/A500M | (2023) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes |
| ASTM A536       | (1984; R 2019; E 2019) Standard Specification for Ductile Iron Castings                                               |
| ASTM A563       | (2021; E 2022a) Standard Specification for Carbon and Alloy Steel Nuts                                                |
| ASTM A603       | (2019) Standard Specification for Zinc-Coated Steel Structural Wire Rope                                              |
| ASTM D1785      | (2021) Standard Specification for Poly(Vinyl Chloride) (PVC), Plastic Pipe, Schedules 40, 80, and 120                 |
| ASTM D2665      | (2014) Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Drain, Waste, and Vent Pipe and Fittings         |
| ASTM E488/E488M | (2022) Standard Test Methods for Strength of Anchors in Concrete Elements                                             |
| ASTM F891       | (2024) Standard Specification for Coextruded Poly (Vinyl Chloride) (PVC) Plastic Pipe with a Cellular Core            |
| ASTM F1554      | (2020) Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength                             |

FEDERAL EMERGENCY MANAGEMENT AGENCY (FEMA)

|            |                                                                |
|------------|----------------------------------------------------------------|
| FEMA P-414 | (January 2004) Installing Seismic Restraints for Duct and Pipe |
|------------|----------------------------------------------------------------|

ICC EVALUATION SERVICE, INC. (ICC-ES)

|              |                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------|
| ICC ES AC156 | (2012) Acceptable Criteria for Seismic Certification by Shake-Table Testing of Nonstructural Components |
| ICC ES AC193 | (2012) Acceptance Criteria for Mechanical Anchors in Concrete Elements                                  |

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC (2021) International Building Code

METAL FRAMING MANUFACTURERS ASSOCIATION (MFMA)

MFMA-4 (2004) Metal Framing Standards Publication

NSF INTERNATIONAL (NSF)

NSF/ANSI/CAN 61 (2022) Drinking Water System Components -  
Health Effects

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION  
(SMACNA)

SMACNA 1981 (2008) Seismic Restraint Manual Guidelines  
for Mechanical Systems, 3rd Edition

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01 (2023; with Change 2, 2024) Structural  
Engineering

UFC 3-301-02 (2023) Design of Risk Category V  
Structures, National Strategic Military  
Assets

UFC 4-010-01 (2018; with Change 3, 2024) DoD Minimum  
Antiterrorism Standards for Buildings

VIBRATION ISOLATION AND SEISMIC CONTROL MANUFACTURERS ASSOCIATION  
(VISCMA)

VISCMA 412 (2014) Installing Seismic Restraints for  
Mechanical Equipment

## 1.2 SYSTEM DESCRIPTION

### 1.2.1 General Requirements

Apply the requirements for seismic protection measures described in this section and on the drawings to the mechanical equipment and mechanical systems both inside and outside of the building along with exterior utilities and systems listed below. Where there is a conflict between the specifications and the drawings, the specifications will take precedence. Accomplish resistance to lateral forces induced by earthquakes without consideration of friction resulting from gravity loads.

### 1.2.2 Mechanical Equipment

Mechanical equipment to be seismically protected must include the following items to the extent required on the drawings or in other sections of these specifications:

Equipment/Components with  $I_p = 1.0$

|                    |                                      |
|--------------------|--------------------------------------|
|                    |                                      |
|                    |                                      |
|                    | Valves and Fittings for Piping       |
|                    |                                      |
|                    |                                      |
|                    |                                      |
| Air Conditioners   |                                      |
| Pumps with Motors  |                                      |
|                    |                                      |
|                    |                                      |
|                    | Unit Heaters                         |
|                    | Exhaust, Return and Misc. Fans       |
|                    |                                      |
|                    | Pumps                                |
| Air Terminal Units | Unitary HVAC Systems                 |
|                    |                                      |
|                    | Instrumentation and Control for HVAC |
| Duct Mounted Coils |                                      |

### 1.2.3 Mechanical Systems

Mechanical systems to be seismically protected must include the following items to the extent required on the drawings or in this or other sections of these specifications:

Mechanical systems with  $I_p = 1.0$

- a. All Piping and Ducts Inside the Building Except as Specifically Stated Below Under "Items Not Covered By This Section".
- d. All Water Supply Systems Outside of Buildings.
- f. All Process Piping Outside of Buildings.

h. Condenser Water and Refrigerant Piping Outside the Building.

m. Ductwork Outside of Buildings.

#### 1.2.4 Contractor Designed Bracing

Submit copies of the design calculations with the drawings. Calculations must be approved, certified, stamped and signed by a registered Professional Structural Engineer. Calculations must verify the capability of structural members to which bracing is attached for carrying the load from the brace. Design the bracing in accordance with UFC 3-301-01, UFC 3-301-02, UFC 4-010-01 and additional data furnished by the Contracting Officer. Resistance to lateral forces induced by earthquakes must be accomplished without consideration of friction resulting from gravity loads. UFC 3-301-01 uses parameters for the building, not for the equipment in the building; therefore, corresponding adjustments to the formulas must be required. Loadings determined using UFC 3-301-01 are based on strength design; therefore, AISC 325 Specifications must be used for the design. The bracing for the mechanical equipment designated in paragraph 1.2.2 and systems designated in paragraph 1.2.3 must be developed by the Contractor.

Provide documentation of an independent design review for mission critical (MC) equipment bracing design. Documentation must be signed by the independent reviewer who must also be a registered structural engineer.

#### 1.2.5 Items Not Covered By This Section

##### 1.2.5.1 Items Requiring No Seismic Restraints

Seismic restraints are not required for the following items:

- a. Gas piping less than 1 inch nominal pipe size.
- b. Piping in boiler and mechanical equipment rooms less than 1-1/4 inches nominal pipe size.
- c. All other piping equal to or less than 3 inches nominal pipe size.
- d. Rectangular air handling ducts less than 6 square feet in cross sectional area.
- e. Round air handling ducts less than 28 inches in diameter.
- f. Piping suspended by individual hangers 12 inches or less in length from the top of pipe to the bottom of the supporting structural member where the hanger is attached, except as noted below.
- g. Ducts suspended by hangers 12 inches or less in length from the top of the duct to the bottom of the supporting structural member, except as

noted below.

In exemptions f. and g. all hangers must meet the length requirements. If the length requirement is exceeded by one hanger in the run, brace the entire run. Seismically protect interior piping and ducts not listed above in accordance with the provisions of this specification.

Non-critical items may require seismic restraints if adjacent to critical equipment or systems that must remain operational after an earthquake and could be compromised by impact with non-critical adjacent components.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

- Coupling and Bracing
- Flexible Couplings or Joints
- Equipment Restraint
- Contractor Designed Bracing; G

#### SD-03 Product Data

- Coupling and Bracing; G
- Flexible Couplings Or Joints; G
- Equipment Restraint; G
- Contractor Designed Bracing; G
- Snubbers
- Anchor Bolts
- Vibration Isolators

#### SD-05 Design Data

- Design Calculations

#### SD-06 Test Reports

- Anchor Bolts; G

## PART 2 PRODUCTS

### 2.1 GENERAL DESIGN REQUIREMENTS

Submit detailed seismic restraint drawings for mechanical equipment, duct systems, piping systems and any other mechanical systems along with

calculations, catalog cuts, templates, and erection and installation details, as appropriate, for the items listed below. Indicate thickness, type, grade, class of metal, and dimensions; and show construction details, reinforcement, anchorage, and installation with relation to the building construction. Calculations must be stamped, by a registered structural engineer, and verify the capability of structural members to which bracing is attached for carrying the load from the brace. Include drawing for Mission Critical Equipment indicating the equipment location in the facility sufficient to be used for the installation. Design must be based on actual equipment and system layout. Design must include calculated dead loads, static seismic loads and capacity of materials utilized for the connection of the equipment or system to the structure. Analysis must detail anchoring methods.

## 2.2 EQUIPMENT RESTRAINT

Equipment must be rigidly or flexibly mounted as indicated in the specifications and/or drawings depending on vibration isolation requirements as follows below.

Roof mounted equipment such as cooling towers and condensers, both vibration isolated and nonisolated, must have support members designed and anchored to building structural steel or concrete as required for seismic restraint and wind loads.

### 2.2.1 Rigidly (Base and Suspended) Mounted Equipment

HVAC equipment furnished under this contract must be rigidly mounted that are qualified for earthquake loading in accordance with ACI 355.2 and ACI 355.4. Anchor bolts must conform to ASTM F1554. For any rigid equipment which is rigidly anchored, provide flexible joints for piping, ductwork, electrical conduit, etc., that are capable of accommodating displacements equal to the full width of the joint in both orthogonal directions. Suspended equipment bracing attachments should be located just above the center of gravity to minimize swinging. Use the ratio of the overturning moment from seismic forces to the resisting moment due to gravity loads to determine if overturning forces need to be considered in the sizing of anchor bolts. Provide calculations to verify the adequacy of the anchor bolts for combined shear and overturning.

Roof mounted HVAC equipment roof curbs, framing and attachment to equipment and structure must be designed and braced to withstand seismic loads. Designated Seismic Systems (DSS) assigned to Seismic Design Category (SDC) C, D, E, or F and Risk Category IV components needed for continued operation after an earthquake must have two nuts provided on each anchor bolt.

### 2.2.2 Nonrigid or Flexibly-Mounted Equipment

Select vibration isolation devices so that the maximum movement of equipment from the static deflection point is 1/4 inch. Equipment flexibly mounted on vibration isolators must have a bumper restraint or snubber in each horizontal direction and vertical restraints must be provided where required to resist overturning. Isolator housing and restraints must be constructed of ductile materials. A viscoelastic pad or similar material of appropriate thickness must be used between the bumper and components to limit the impact load. Restraints must be designed to resist the calculated horizontal lateral and vertical forces.



Spring vibration isolators must be seismically rated, restrained isolators for equipment subject to load variations and large external forces. The seismically rated housing must be sized to meet or exceed the force requirements applicable to the project and meet the required isolation criteria. Spring vibration isolator manufacturer's will be a member of VISCMA. Design force,  $F_p$ , must be doubled for vibration isolators with an air gap greater than 0.25 inches as specified in ASCE 7-16, Chapter 13. Housed springs must not be used for seismic restraint applications because they cannot resist uplift.

## 2.3 BOLTS AND NUTS

Hex head bolts, and heavy hexagon nuts must be ASTM A325 or ASTM A490 bolts and ASTM A563 nuts. Provide bolts and nuts galvanized in accordance with ASTM A153/A153M when used underground or exposed to weather.

## 2.4 FLEXIBLE JOINTS

Flexible joints must have same pressure and temperature ratings as adjoining pipe. Braided hoses must not be used where there is torsional or axial movement unless manufacturer allows it.

### 2.4.1 Braided Hose Expansion Joint

Braided hose expansion joint(s) must be installed in the locations indicated on the drawings and as required to accommodate any thermal expansion, contraction or seismic movement of the piping system. Joints must consist of two parallel sections of corrugated metal hose, compatible braid, and 180 degree return bend with inlet and outlet connections. Field fabricated loops are not acceptable. Braided hose expansion joint(s) must be installed in the locations indicated on the drawings and as required to accommodate any thermal expansion, contraction or seismic movement of the piping system. Joints must consist of two parallel sections of corrugated metal hose, compatible braid, and 180 degree return bend with inlet and outlet connections. Field fabricated loops must not be acceptable. Braided hose in a 60 degree flexible V loop arrangement must be used for small diameter pipe connections to coils in variable-air-volume (VAV) terminal units and fan coil units installed in suspended ductwork whether braced or unbraced.

All braided hose expansion joints must be manufactured in accordance with the documented manufacturers weld procedure specifications. The procedure qualification record must be used to document the execution of this procedure and must follow the general "guidelines" of ASME Section IX. Each individual welder must conform to the in-house procedure qualification record and be qualified prior to each production lot. The testing of each individual welder must be documented in a welding procedure qualification record.

#### 2.4.1.1 Corrugated Hose

Corrugated hose must be Type 304 stainless steel. Braid must be Type 304 stainless steel for any series 300 stainless steel hose. Fittings materials of construction and end fitting type must be consistent with pipe material and equipment/ pipe connection fittings. Copper fittings must not be attached to stainless steel hose.

#### 2.4.1.2 Flexible Hose Expansion Loops

Flexible hose expansion loops must have a factory supplied, hanger / support lug located at the bottom of the 180deg return. Flexible hose expansion loop(s) must be furnished with a plugged FPT to be used for a drain or air release vent. Flexible hose expansion loop(s) must be rated with an operating pressure which is the same as the adjoining pipe. The operating pressure must be based on burst pressure with a 4 to 1 safety factor.

#### 2.4.2 Double Ball Flexible Expansion Joint

Install flexible expansion joints manufactured of ductile iron conforming to the material requirements of ASTM A536 and AWWA C153/A21.53 in the locations indicated on the drawings. Provide foundry certification of material upon request. Each flexible expansion joint must be pressure tested prior to shipment against its own restraint to a minimum of 350 psi (250 psi for flexible expansion joints 2 inch and 30 inches diameter and larger.) A minimum 2:1 safety factor, determined from the published pressure rating, must apply. Factory Mutual Approval for the 3 inch through 12 inch sizes is required. Each flexible expansion joint must consist of an expansion joint designed and cast as an integral part of a ball and socket type flexible joint, having a minimum per ball deflection of: 20°, 2" - 12"; 15°, 14" - 36"; 12°, 42"-48" and 4-inches minimum expansion. Additional expansion sleeves must be available and easily added or removed at the factory or in the field. Both standardized mechanical joint and flange end connections must be available.

##### 2.4.2.1 Internal Surfaces

Line all internal surfaces (wetted parts) with a minimum of 15 mils of fusion bonded epoxy conforming to the applicable requirements of AWWA C213. Sealing gaskets must be constructed of EPDM. The coating must meet NSF/ANSI/CAN 61.

##### 2.4.2.2 Exterior Surfaces

Coat exterior surfaces with a minimum of 6 mils of fusion bonded epoxy conforming to the applicable requirements of AWWA C116/A21.16. Include appropriately sized polyethylene sleeves, meeting AWWA C105/A21.5, for direct buried applications.

#### 2.4.3 Double Ball Flexible Expansion Joint Gravity Drain (Non-Pressurized)

Flexible expansion joints gravity drain must be installed in the locations indicated on the drawings and must be manufactured of pvc. All connections whether solvent weld or mechanical must be restrained to allow movement to be transferred to expansion joint. Each ball must allow up to 15 degrees deflection.

End connection outside diameters must be compatible with ASTM D1785, ASTM D2665 and ASTM F891 PVC pipe and are to be solvent welded.

#### 2.5 SWAY BRACING MATERIALS

Material used for members listed in this section and on the drawings, must be structural steel conforming with the following:

- a. Plates, rods, and rolled shapes, ASTM A36/A36M.

- b. Wire rope, ASTM A603 pre-stretched. Class C galv coating Ferrule clamps must be qualified by testing for use in seismic applications per VISCMA 412. A minimum of two clamps are required on each end of wire rope.
- c. Tubes, ASTM A500/A500M, Grade B.
- d. Pipes, ASTM A53/A53M, Grade B.
- e. Angles, ASTM A36/A36M.
- f. Channels (Struts) with in-turned lips and associated hardware for fastening to channels at random points conforming to MFMA-4

## 2.6 MULTIDIRECTIONAL SEISMIC SNUBBERS

Install multidirectional seismic snubbers employing elastomeric pads on floor- or slab-mounted equipment and large piping as detailed on drawings. These snubbers must provide 1/4 inch free vertical and horizontal movement from the static deflection point. Snubber medium must consist of multiple pads of cotton duct and neoprene or other suitable materials arranged around a flanged steel trunnion so both horizontal and vertical forces are resisted by the snubber medium.

## PART 3 EXECUTION

### 3.1 COUPLING AND BRACING

- a. Submit detail drawings, as specified here and throughout this specification, along with catalog cuts, templates, and erection and installation details, as appropriate, for the items listed. Submittals must be complete in detail; must indicate thickness, type, grade, class of metal, and dimensions; and must show construction details, reinforcement, anchorage, and installation with relation to the building construction.
- b. Provide coupling installation conforming to the details shown on the drawings. Provisions of this paragraph apply to all piping within a 5 foot line around outside of building unless buried in the ground. Piping grouped for support on trapeze-type hangers must be braced at the most frequent interval as determined by applying the requirements of this specification to each piping run on the common support.
- c. Size bracing components as required for the total load carried by the common supports. Bracing rigidly attached to pipe flanges, or similar, must not be used where it would interfere with thermal expansion of piping.
- d. Adjust isolators and restraints after piping systems has been filled and equipment is at its operating weight, following the manufacturer's written instructions.
- e. Install cables at a 45-degree slope. Where interference is present, the slope may be minimum of 30 degrees or a maximum of 60 degrees per VISCMA 412.

### 3.2 BUILDING DRIFT

Provide joints capable of accommodating seismic displacements for vertical piping between floors of the building, where pipes pass through a building seismic or expansion joint, or where rigidly supported pipes connect to equipment with vibration isolators. Provide horizontal piping across expansion joints to accommodate the resultant of the drifts of each building unit in each orthogonal direction. For threaded piping, provide swing joints made of the same piping material. For piping with manufactured ball joints the seismic drift must be 0.015 feet per foot of height above the base where the seismic separation occurs; this drift value must be used in place of the expansion given in the manufacturer's selection table.

### 3.3 FLEXIBLE COUPLINGS OR JOINTS

#### 3.3.1 Building Piping

Provide flexible couplings or joints in building piping at bottom of all pipe risers for pipe larger than 3-1/2 inches in diameter. Laterally brace flexible couplings or joints without interfering with the action of the flexible coupling or joint. Cast iron waste and vent piping need only comply with these provisions when caulked joints are used. Flexible bell and spigot pipe joints using rubber gaskets may be used at each branch adjacent to tees and elbows for underground waste piping inside of building to satisfy these requirements.

### 3.4 PIPE SLEEVES

Size pipe sleeves in interior non-fire rated walls as indicated on the drawings to provide clearances that will permit differential movement of piping without the piping striking the pipe sleeve. Pipe sleeves in fire rated walls must conform to the requirements in Section 07 84 00 FIRESTOPPING.

### 3.5 SPREADERS

Provide spreaders between adjacent piping runs to prevent contact during seismic activity whenever pipe or insulated pipe surfaces are less than 4 inches apart. Apply spreaders at same interval as sway braces at an equal distance between the sway braces. If rack type hangers are used where the pipes are restrained from contact by mounting to the rack, spreaders are not required for pipes mounted in the rack. Apply spreaders to surface of bare pipe and over insulation on insulated pipes utilizing high-density inserts and pipe protection shields in accordance with the requirements of Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

### 3.6 SWAY BRACES FOR PIPING

Provide sway braces to prevent movement of the pipes under seismic loading. Provide braces in both the longitudinal and transverse directions, relative to the axis of the pipe. Provide sufficient braces for equipment to resist a horizontal force as specified in UFC 3-301-01, and UFC 3-301-02 without exceeding safe working stress of bracing components. Provide bracing that does not interfere with thermal expansion requirements for the pipes as described in other sections of these specifications. For seismic analysis of horizontal pipes, the equivalent static force should be considered to act concurrently with the full dead load of the pipe, including contents.

### 3.6.1 Transverse Sway Bracing

Provide transverse sway bracing for steel and copper pipe at intervals not to exceed those shown on the drawings. All runs (length of pipe between end joints) must have a minimum of transverse bracing at each end.

### 3.6.2 Longitudinal Sway Bracing

Provide longitudinal sway bracing at 40 foot intervals unless otherwise indicated. All runs (length of pipe between end joints) must have one longitudinal brace minimum. Construct sway braces in accordance with the drawings. Do not use branch lines, walls, or floors as sway braces.

### 3.6.3 Vertical Runs

Run is defined as length of pipe between end joints. Do not brace vertical runs of piping no more than 10 foot vertical intervals. Braces for vertical runs must be above the center of gravity of the segment being braced. Flexible couplings should be provided at the bottoms of risers for pipes larger than 3.5 in. (89 mm) in diameter. Flexible couplings and expansion joints should be braced laterally and longitudinally unless such bracing would interfere with the action of the couplings or joints. When pipes enter buildings, flexible couplings should be provided to allow for relative movement between the soil and building. Construct all sway braces in accordance with the drawings. Attach sway braces to the structural system. Do not connect to branch lines, walls, or floors.

### 3.6.4 Clamps and Hangers

Apply clamps or hangers on uninsulated pipes directly to pipe. Insulated piping must have clamps or hangers applied over insulation in accordance with Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

Hanger rod stiffener angle or strut bracing must be securely attached by a series of attachment clamps manufactured from a one piece metal stamping and must include all require attachment hardware and locking nuts. Attachment clamps made from aluminum or cast iron must not be used in seismic applications. Do not weld vertical braces to hanger rods.

## 3.7 SWAY BRACES FOR DUCTS

### 3.7.1 Braced Ducts

Provide bracing details and spacing for rectangular and round ducts in accordance with SMACNA 1981. However, the design seismic loadings for these items must not be less than loadings obtained using the procedures in UFC 3-301-01, and UFC 3-301-02. Bracing must not attach to duct joints. Use shortest screws possible when penetrating ductwork to minimize airflow noise inside duct.

### 3.7.2 Unbraced Ducts

Attach hangers for unbraced ducts to the duct within 2 inches of the top of the duct with a minimum of two #10 sheet metal screws in accordance with FEMA P-414. Use shortest screws possible when penetrating ductwork to minimize airflow noise inside duct. Install unbraced ducts with a 6 inch minimum clearance to vertical ceiling hanger wires.

### 3.8 EQUIPMENT

#### 3.8.1 General

Ensure housekeeping pads have adequate space to mount equipment and seismic restraint devices allowing adequate edge distance and embedment depth for restraint anchor bolts. Identify position of reinforcing steel and other embedded items prior to drilling holes for anchors. Do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength. Install neoprene grommet washers or fill the gap with epoxy on equipment anchor bolts where clearance between anchor and equipment support hole exceeds 0.125 inches.

#### 3.8.2 Controls

Ensure that controls for critical equipment that must remain operational after an earthquake are certified per paragraph 3.11 SPECIAL TESTING FOR SEISMIC-RESISTING EQUIPMENT and are served by emergency power as required.

### 3.9 ANCHOR BOLTS

#### 3.9.1 Cast-in-Place Anchor Bolts

Use templates to locate cast-in-place bolts accurately and securely in formwork. Anchor bolts must have an embedded straight length equal to at least 12 times nominal diameter of the bolt. Anchor bolts that exceed the normal depth of equipment foundation piers or pads must either extend into concrete floor or the foundation or be increased in depth to accommodate bolt lengths.

#### 3.9.2 Drilled-In Anchor Bolts

Drill holes with rotary impact hammer drills. Drill bits must be of diameters as specified by the anchor manufacturer. Unless otherwise shown on the Drawings, all holes must be drilled perpendicular to the concrete surface. Where anchors are permitted to be installed in cored holes, use core bits with matched tolerances as specified by the manufacturer. Properly clean cored hole per manufacturer's instructions. Identify position of reinforcing steel and other embedded items prior to drilling holes for anchors. Exercise care in coring or drilling to avoid damaging existing reinforcing or embedded items. Notify the COR if reinforcing steel or other embedded items are encountered during drilling. Take precautions as necessary to avoid damaging prestressing tendons, electrical and telecommunications conduit, and gas lines. Unless otherwise specified, do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength. Perform anchor installation in accordance with manufacturer instructions.

##### 3.9.2.1 Wedge Anchors, Heavy-Duty Sleeve Anchors, and Undercut Anchors

Protect threads from damage during anchor installation. Heavy-duty sleeve anchors must be installed with sleeve fully engaged in part to be fastened. Set anchors to manufacturer's recommended torque, using a torque wrench. Following attainment of 10% of the specified torque, 100% of the specified torque must be reached within 7 or fewer complete turns of the nut. If the specified torque is not achieved within the required number of turns, the anchor must be removed and replaced unless otherwise directed by the Engineer.

### 3.9.2.2 Cartridge Injection Adhesive Anchors

Where approved for seismic application, clean all holes per manufacturer instructions to remove loose material and drilling dust prior to installation of adhesive. Inject adhesive into holes proceeding from the bottom of the hole and progressing toward the surface in such a manner as to avoid introduction of air pockets in the adhesive. Follow manufacturer recommendations to ensure proper mixing of adhesive components. Sufficient adhesive must be injected in the hole to ensure that the annular gap is filled to the surface. Remove excess adhesive from the surface. Shim anchors with suitable device to center the anchor in the hole. Do not disturb or load anchors before manufacturer specified cure time has elapsed.

### 3.9.2.3 Capsule Anchors

Where approved for seismic application, perform drilling and setting operations in accordance with manufacturer instructions. Clean all holes to remove loose material and drilling dust prior to installation of adhesive. Remove water from drilled holes in such a manner as to achieve a surface dry condition. Capsule anchors must be installed with equipment conforming to manufacturer recommendations. Do not disturb or load anchors before manufacturer specified cure time has elapsed.

Observe manufacturer recommendations with respect to installation temperatures for cartridge injection adhesive anchors and capsule anchors.

## 3.10 ANCHOR BOLT TESTING

Test in place expansion and chemically bonded anchors not more than 24 hours after installation of the anchor, conducted by an independent testing agency; testing must be performed on random anchor bolts as described below.

### 3.10.1 Torque Wrench Testing

Perform torque wrench testing on not less than 50 percent of the total installed applied torque expansion anchors and at least one anchor for every piece of equipment containing more than two anchors. The test torque must equal the minimum required installation torque as required by the bolt manufacturer. Calibrate torque wrenches at the beginning of each day the torque tests are performed. Recalibrate torque wrenches for each bolt diameter whenever tests are run on bolts of various diameters. Apply torque between 20 and 100 percent of wrench capacity. Reach the test torque within one half turn of the nut, except for 3/8 inch sleeve anchors which must reach their torque by one quarter turn of the nut. If any anchor fails the test, test similar anchors not previously tested until 20 consecutive anchors pass. Failed anchors must be retightened and retested to the specified torque; if the anchor still fails the test it must be replaced.

### 3.10.2 Pullout Testing

Test expansion and chemically bonded anchors by applying a pullout load using a hydraulic ram attached to the anchor bolt. Testing must be in accordance with ASTM E488/E488M or ICC ES AC193. At least 10 percent of each type and size of anchors, but not less than 3 per day must be tested. Apply the load to the anchor without removing the nut; when that is not possible, the nut must be removed and a threaded coupler must be

installed of the same tightness as the original nut. Check the test setup to verify that the anchor is not restrained from withdrawing by the baseplate, the test fixture, or any other fixtures. The support for the testing apparatus must be at least 1.5 times the embedment length away from the bolt being tested. Load each tested anchor to 1 times the design tension value for the anchor. The anchor must have no observable movement at the test load. If any anchor fails the test, similar type and size anchors not previously tested must be tested until 10 percent of those type consecutive anchors pass. Remove and replace failed anchors. Fill empty anchor holes and patch failed anchor locations with high-strength non-shrink, nonmetallic grout.

### 3.11 SPECIAL TESTING FOR SEISMIC-RESISTING EQUIPMENT

Equipment and components (including controls) designated as Designated Seismic Systems required to remain operational after an earthquake will be seismic qualified by shake table testing conforming to ICC ES AC156 procedures. The manufacturer is to provide a certification by a fully qualified testing agency for the specific equipment and/or components. Prequalified certifications are acceptable unless noted otherwise. Seismic component qualification documentation for each piece of equipment must contain the information required in UFC 3-301-02, Section 2-17.2.5 Component Qualification Documentation.

Mechanical components that are required to be certified must bear permanent marking or nameplates constructed of a durable heat and water resistant material. Nameplates must be mechanically attached to such nonstructural components and placed on each component for clear identification. The nameplate must not be less than 5 inches x 7 inches with red letters 1 inch in height on a white background stating "Certified Equipment." The following statement must be on the nameplate: "This equipment/component is certified. No modifications are allowed unless authorized in advance and documented in the Equipment Certification Documentation file." The nameplate must also contain the component identification number in accordance with the drawings/specifications and the O&M manuals.

### 3.12 SPECIAL INSPECTION FOR SEISMIC-RESISTING SYSTEMS AND EQUIPMENT

Perform special inspections for seismic-resisting mechanical systems, equipment and components designated mechanical seismic systems and equipment per ICC IBC 1705.12.4; and plumbing and mechanical components per ICC IBC 1705.12.6. Periodic special inspections will be conducted on mechanical equipment as required by Section 1705.12 of the International Building Code and paragraph 2-5.4 of UFC 3-301-01. Provide a Statement of Special Inspections and Final Report in accordance with paragraph 2-2.4.3 of UFC 3-301-01.

-- End of Section --



SECTION 23 05 93

TESTING, ADJUSTING, AND BALANCING FOR HVAC  
**11/15**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

AMCA 203 (1990; R 2011) Field Performance  
Measurements of Fan Systems

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING  
ENGINEERS (ASHRAE)

ASHRAE 62.1 (2022) Ventilation for Acceptable Indoor  
Air Quality

ASSOCIATED AIR BALANCE COUNCIL (AABC)

AABC MN-1 (2002; 6th ed) National Standards for  
Total System Balance

AABC MN-4 (1996) Test and Balance Procedures

NATIONAL ENVIRONMENTAL BALANCING BUREAU (NEBB)

NEBB MASV (2006) Procedural Standards for  
Measurements and Assessment of Sound and  
Vibration

NEBB PROCEDURAL STANDARDS (2015) Procedural Standards for TAB  
(Testing, Adjusting and Balancing)  
Environmental Systems

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION  
(SMACNA)

SMACNA 1780 (2023) HVAC Systems - Testing, Adjusting  
and Balancing, 3rd Edition

SMACNA 1858 (2004) HVAC Sound And Vibration Manual -  
First Edition

1.2 DEFINITIONS

- a. AABC: Associated Air Balance Council
- b. COTR: Contracting Officer's Technical Representative

- e. HVAC: Heating, ventilating, and air conditioning; or heating, ventilating, and cooling
- f. NEBB: National Environmental Balancing Bureau
- g. Out-of-tolerance data: Pertains only to field acceptance testing of Final TAB report. When applied to TAB work this phase means "a measurement taken during TAB field acceptance testing which does not fall within the range of plus 5 to minus 5 percent of the original measurement reported on the TAB Report for a specific parameter."
- h. Season of maximum heating load: The time of year when the outdoor temperature at the project site remains within plus or minus 30 degrees Fahrenheit of the project site's winter outdoor design temperature, throughout the period of TAB data recording.
- i. Season of maximum cooling load: The time of year when the outdoor temperature at the project site remains within plus or minus 5 degrees Fahrenheit of the project site's summer outdoor design temperature, throughout the period of TAB data recording.
- j. Season 1, Season 2: Depending upon when the project HVAC is completed and ready for TAB, Season 1 is defined, thereby defining Season 2. Season 1 could be the season of maximum heating load, or the season of maximum cooling load.
- k. Sound measurements terminology: Defined in AABC MN-1, NEBB MASV, or SMACNA 1858 (TABB).
- l. TAB: Testing, adjusting, and balancing (of HVAC systems)
- m. TAB'd: HVAC Testing/Adjusting/Balancing procedures performed
- n. TAB Agency: TAB Firm
- o. TAB team field leader: TAB team field leader
- p. TAB team supervisor: TAB team engineer
- q. TAB team technicians: TAB team assistants
- r. TABB: Testing Adjusting and Balancing Bureau

#### 1.2.1 Similar Terms

In some instances, terminology differs between the Contract and the TAB Standard primarily because the intent of this Section is to use the industry standards specified, along with additional requirements listed herein to produce optimal results.

The following table of similar terms is provided for clarification only. Contract requirements take precedent over the corresponding AABC, NEBB, or TABB requirements where differences exist.

| SIMILAR TERMS           |                                                                                                 |                                                                                    |                                                           |
|-------------------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Contract Term           | AABC Term                                                                                       | NEBB Term                                                                          | TABB Term                                                 |
| TAB Standard            | National Standards for Testing and Balancing Heating, Ventilating, and Air Conditioning Systems | Procedural Standards for Testing, Adjusting and Balancing of Environmental Systems | International Standards for Environmental Systems Balance |
| TAB Specialist          | TAB Engineer                                                                                    | TAB Supervisor                                                                     | TAB Supervisor                                            |
| Systems Readiness Check | Construction Phase Inspection                                                                   | Field Readiness Check & Preliminary Field Procedures                               | Field Readiness Check & Prelim. Field Procedures          |

### 1.3 WORK DESCRIPTION

The work includes testing, adjusting, and balancing (TAB) of new heating, ventilating, and cooling (HVAC) air and water distribution systems including equipment and performance data, ducts, and piping which are located within, on, under, between, and adjacent to buildings

Perform TAB in accordance with the requirements of the TAB procedural standard recommended by the TAB trade association that approved the TAB Firm's qualifications. Comply with requirements of AABC MN-1, NEBB PROCEDURAL STANDARDS, or SMACNA 1780 (TABB) as supplemented and modified by this specification section. All recommendations and suggested practices contained in the TAB procedural standards are considered mandatory.

Conduct TAB of the indicated existing systems and equipment and submit the specified TAB reports for approval. Conduct TAB work in accordance with the requirements of this section.

#### 1.3.1 Air Distribution Systems

Test, adjust, and balance systems (TAB) in compliance with this section. Obtain Contracting Officer's written approval before applying insulation to exterior of air distribution systems as specified under Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

#### 1.3.2 Water Distribution Systems

TAB systems in compliance with this section. Obtain Contracting Officer's written approval before applying insulation to water distribution systems as specified under Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. At Contractor's option and with Contracting Officer's written approval, the piping systems may be insulated before systems are TAB'd.

Terminate piping insulation immediately adjacent to each flow control valve, automatic control valve, or device. Seal the ends of pipe insulation and the space between ends of pipe insulation and piping, with waterproof vapor barrier coating.

After completion of work under this section, insulate the flow control valves and devices as specified under Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

#### 1.3.3 TAB SCHEMATIC DRAWINGS

Show the following information on TAB Schematic Drawings:

1. A unique number or mark for each piece of equipment or terminal.
2. Air quantities at air terminals.
3. Air quantities and temperatures in air handling unit schedules.
4. Water quantities and temperatures in thermal energy transfer equipment schedules.
5. Water quantities and heads in pump schedules.
6. Water flow measurement fittings and balancing fittings.
7. Ductwork Construction and Leakage Testing Table that defines the DALT test requirements, including each applicable HVAC duct system ID or mark, duct pressure class, duct seal class, and duct leakage test pressure. This table is included in the file for Graphics for Unified Facilities Guide Specifications:  
<http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-gra>

The Testing, Adjusting, and Balancing (TAB) Specialist must review the Contract Plans and Specifications and advise the Contracting Officer of any deficiencies that would prevent the effective and accurate TAB of the system, including records of existing conditions, and systems readiness check. The TAB Specialist must provide a Design Review Report individually listing each deficiency and the corresponding proposed corrective action necessary for proper system operation.

Submit three copies of the TAB Schematic Drawings and Report Forms to the Contracting Officer, no later than 21 days prior to the start of TAB field measurements.

#### 1.3.4 Related Requirements

Requirements for construction scheduling related to HVAC TAB work are specified in Section 01 32 17.00 20 COST LOADED NETWORK ANALYSIS SCHEDULES (NAS).

#### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Records of Existing Conditions; G

Independent TAB Agency and Personnel Qualifications; G

TAB Design Review Report; G

SD-02 Shop Drawings

TAB Schematic Drawings and Report Forms; G

SD-03 Product Data

Equipment and Performance Data; G

TAB Related HVAC Submittals; G

A list of the TAB Related HVAC Submittals, no later than 7 days after the approval of the TAB team engineer .

SD-06 Test Reports

Prerequisite HVAC Work Checkout List For Proportional Balancing; G

Certified Final TAB Report for Proportional Balancing; G

Prerequisite HVAC Work Checkout List For Season 1; G

Certified Final TAB Report for Season 1; G

Prerequisite HVAC Work Checkout List For Season 2; G

Certified Final TAB Report for Season 2; G

SD-07 Certificates

Independent TAB Agency and Personnel Qualifications; G

TAB Submittal and Work Schedule; G

TAB Pre-Field Engineering Report; G

Instrument Calibration Certificates; G

TAB Procedures Summary; G

Advance Notice of TAB Field Work for Proportional Balancing; G

Advance Notice of TAB Field Work for Season 1; G

Advance Notice of TAB Field Work for Season 2 G

## 1.5 QUALITY ASSURANCE

### 1.5.1 Independent TAB Agency and Personnel Qualifications

To secure approval for the proposed agency, submit information certifying that the TAB agency is a first tier subcontractor who is not affiliated with any other company participating in work on this contract, including design, furnishing equipment, or construction. Further, submit the following, for the agency, to Contracting Officer for approval:

a. Independent AABC or NEBB or TABB TAB agency:

TAB agency: AABC registration number and expiration date of current certification; or NEBB certification number and expiration date of current certification; or TABB certification number and expiration date of current certification.

TAB team supervisor: Name and copy of AABC or NEBB or TABB TAB supervisor certificate and expiration date of current certification.

TAB team field leader: Name and documented evidence that the team field leader has satisfactorily performed full-time supervision of TAB work in the field for not less than 3 years immediately preceding this contract's bid opening date.

TAB team field technicians: Names and documented evidence that each field technician has satisfactorily assisted a TAB team field leader in performance of TAB work in the field for not less than one year immediately preceding this contract's bid opening date.

Current certificates: Registrations and certifications are current, and valid for the duration of this contract. Renew Certifications which expire prior to completion of the TAB work, in a timely manner so that there is no lapse in registration or certification. TAB agency or TAB team personnel without a current registration or current certification are not to perform TAB work on this contract.

b. TAB Team Members: TAB team approved to accomplish work on this contract are full-time employees of the TAB agency. No other personnel is allowed to do TAB work on this contract.

c. Replacement of TAB team members: Replacement of members may occur if each new member complies with the applicable personnel qualifications and each is approved by the Contracting Officer.

#### 1.5.1.1 TAB Standard

Perform TAB in accordance with the requirements of the standard under which the TAB Firm's qualifications are approved, i.e., AABC MN-1, NEBB PROCEDURAL STANDARDS, or SMACNA 1780 unless otherwise specified herein. All recommendations and suggested practices contained in the TAB Standard are considered mandatory. Use the provisions of the TAB Standard, including checklists, report forms, etc., as nearly as practical, to satisfy the Contract requirements. Use the TAB Standard for all aspects of TAB, including qualifications for the TAB Firm and Specialist and calibration of TAB instruments. Where the instrument manufacturer calibration recommendations are more stringent than those

listed in the TAB Standard, adhere to the manufacturer's recommendations.

All quality assurance provisions of the TAB Standard such as performance guarantees are part of this contract. For systems or system components not covered in the TAB Standard, TAB procedures must be developed by the TAB Specialist. Where new procedures, requirements, etc., applicable to the Contract requirements have been published or adopted by the body responsible for the TAB Standard used (AABC, NEBB, or TABB), the requirements and recommendations contained in these procedures and requirements are considered mandatory, including the latest requirements of ASHRAE 62.1.

#### 1.5.1.2 Qualifications

##### a. TAB Firm

The TAB Firm must be either a member of AABC or certified by the NEBB or the TABB and certified in all categories and functions where measurements or performance are specified on the plans and specifications, including TAB of environmental systems.

Certification must be maintained for the entire duration of duties specified herein. If, for any reason, the firm loses subject certification during this period, the Contractor must immediately notify the Contracting Officer and submit another TAB Firm for approval. Any firm that has been the subject of disciplinary action by either the AABC, the NEBB, or the TABB within the five years preceding Contract Award is not be eligible to perform any duties related to the HVAC systems, including TAB. All work specified in this Section and in other related Sections to be performed by the TAB Firm will be considered invalid if the TAB Firm loses its certification prior to Contract completion and must be performed by an approved successor.

These TAB services are to assist the prime Contractor in performing the quality oversight for which it is responsible. The TAB Firm must be a prime subcontractor of the Contractor and be financially and corporately independent of the mechanical subcontractor, reporting directly to and paid by the Contractor.

##### b. TAB Specialist

The TAB Specialist must be either a member of AABC, an experienced technician of the Firm certified by the NEBB, or a Supervisor certified by the TABB. The certification must be maintained for the entire duration of duties specified herein. If, for any reason, the Specialist loses subject certification during this period, immediately notify the Contracting Officer and submit another TAB Specialist for approval. Any individual that has been the subject of disciplinary action by either the AABC, the NEBB, or the TABB within the five years preceding Contract Award is not eligible to perform any duties related to the HVAC systems, including TAB. All work specified in this Section and in other related Sections performed by the TAB Specialist will be considered invalid if the TAB Specialist loses its certification prior to Contract completion and must be performed by the approved successor.

##### c. TAB Specialist Responsibilities

TAB Specialist responsibilities include all TAB work specified herein and in related sections under his direct guidance. The TAB specialist is required to be onsite on a daily basis to direct TAB efforts. The TAB Specialist must participate in the commissioning process.

#### 1.5.1.3 TAB Related HVAC Submittals

The TAB Specialist must prepare a list of the submittals from the Contract Submittal Register that relate to the successful accomplishment of all HVAC TAB. Accompany the submittals identified on this list with a letter of approval signed and dated by the TAB Specialist when submitted to the Government. Ensure that the location and details of ports, terminals, connections, etc., necessary to perform TAB are identified on the submittals.

#### 1.5.2 Responsibilities

The Contractor is responsible for ensuring compliance with the requirements of this section. The following delineation of specific work responsibilities is specified to facilitate TAB execution of the various work efforts by personnel from separate organizations. This breakdown of specific duties is specified to facilitate adherence to the schedule listed in the paragraph TAB SUBMITTAL AND WORK SCHEDULE.

##### 1.5.2.1 Contractor

- a. TAB personnel: Ensure that the DALT work and the TAB work is accomplished by a group meeting the requirements specified in the paragraph TAB PERSONNEL QUALIFICATION REQUIREMENTS.
- b. Pre-TAB meeting: Attend the meeting with the TAB Supervisor, and ensure that a representative is present for the sheetmetal contractor, mechanical contractor, electrical contractor, and automatic temperature controls contractor.
- c. HVAC documentation: Furnish one complete set of the following HVAC-related documentation to the TAB agency:
  - (1) Contract drawings and specifications
  - (2) Approved submittal data for equipment
  - (3) Construction work schedule
  - (4) Up-to-date revisions and change orders for the previously listed items
- d. Submittal and work schedules: Ensure that the schedule for submittals and work required by this section and specified in the paragraph TAB SUBMITTAL AND WORK SCHEDULE is met.
- e. Coordination of supporting personnel:

Provide the technical personnel, such as factory representatives or HVAC controls installer required by the TAB field team to support the DALT and the TAB field measurement work.

Provide equipment mechanics to operate HVAC equipment and ductwork mechanics to provide the field designated test ports to enable TAB



field team to accomplish the TAB field measurement work. Ensure these support personnel are present at the times required by the TAB team, and cause no delay in the TAB field work.

Conversely, ensure that the HVAC controls installer has required support from the TAB team field leader to complete the controls check out.

- f. Deficiencies: Ensure that the TAB Agency supervisor submits all Design/Construction deficiency notifications directly to the Contracting officer within 3 days after the deficiency is encountered. Further, ensure that all such notification submittals are complete with explanation, including documentation, detailing deficiencies.
- g. Prerequisite HVAC work: Complete check out and debugging of HVAC equipment, ducts, and controls prior to the TAB engineer arriving at the project site to begin the TAB work. Debugging includes searching for and eliminating malfunctioning elements in the HVAC system installations, and verifying all adjustable devices are functioning as designed. Include as prerequisite work items, the deficiencies pointed out by the TAB team supervisor in the design review report.
- h. Prior to the TAB field team's arrival, ensure completion of the applicable inspections and work items listed in the TAB team supervisor's pre-field engineering report. Do not allow the TAB team to commence TAB field work until all of the following are completed.
  - (1) HVAC system installations are fully complete.
  - (2) HVAC prerequisite checkout work lists specified in the paragraph PRE-FIELD TAB ENGINEERING REPORT are completed, submitted, and approved. Ensure that the TAB Agency gets a copy of the approved prerequisite HVAC work checklist.
  - (4) HVAC system filters are clean for both Season 1 and Season 2 TAB field work.
- i. Advance notice: Furnish to the Contracting Officer with advance written notice for the commencement of the TAB field work.
- j. Insulation work: Ensure that openings in duct and machinery insulation coverings for TAB test ports are marked, closed and sealed.

#### 1.5.2.2 TAB Agency

Provide the services of a TAB team which complies with the requirements of the paragraph INDEPENDENT TAB AGENCY PERSONNEL QUALIFICATIONS. The work to be performed by the TAB agency is limited to testing, adjusting, and balancing of HVAC air and water systems to satisfy the requirements of this specification section.

#### 1.5.2.3 TAB Team Supervisor

- a. Overall management: Supervise and manage the overall TAB team work effort, including preliminary and technical TAB procedures and TAB team field work.

- b. Pre-TAB meeting: Attend meeting with Contractor.
- c. Design review report: Review project specifications and accompanying drawings to verify that the air systems and water systems are designed in such a way that the TAB engineer can accomplish the work in compliance with the requirements of this section. Verify the presence and location of permanently installed test ports and other devices needed, including gauge cocks, thermometer wells, flow control devices, circuit setters, balancing valves, and manual volume dampers.
- d. Support required: Specify the technical support personnel required from the Contractor other than the TAB agency; such as factory representatives for temperature controls or for complex equipment. Inform the Contractor in writing of the support personnel needed and when they are needed. Furnish the notice as soon as the need is anticipated, either with the design review report, or the pre-field engineering report, the during the DALT or TAB field work.
- f. Pre-field engineering report: Utilizing the following HVAC-related documentation; contract drawings and specifications, approved submittal data for equipment, up-to-date revisions and change orders; prepare this report.
- g. Prerequisite HVAC work checklist: Ensure the Contractor gets a copy of this checklist at the same time as the pre-field engineering report is submitted.
- j. Technical Assistance for TAB Work: Provide immediate technical assistance to the TAB field team for the TAB work.
  - (1) TAB field visit: Near the end of the TAB field work effort, visit the contract site to inspect the HVAC installation and the progress of the TAB field work. Conduct site visit full-time for a minimum of one 8 hour workday duration. Review the TAB final report data and certify the TAB final report.
- k. Certified TAB report: Certify the TAB report. This certification includes the following work:
  - (1) Review: Review the TAB field data report. From this field report, prepare the certified TAB report.
  - (2) Verification: Verify adherence, by the TAB field team, to the TAB plan prescribed by the pre-field engineering report and verify adherence to the procedures specified in this section.
- l. Design/Construction deficiencies: Within 3 working days after the TAB Agency has encountered any design or construction deficiencies, the TAB Supervisor must submit written notification directly to the Contracting Officer, with a separate copy to the Contractor, of all such deficiencies. Provide in this submittal a complete explanation,

including supporting documentation, detailing deficiencies. Where deficiencies are encountered that are believed to adversely impact successful completion of TAB, the TAB Agency must issue notice and request direction in the notification submittal.

- m. TAB Field Check: The TAB team supervisor must attend and supervise Season 1 and Season 2 TAB field check.

#### 1.5.2.4 TAB Team Field Leader

- a. Field manager: Manage, in the field, the accomplishment of the work specified in Part 3, EXECUTION.
- b. Full time: Be present at the contract site when TAB field work is being performed by the TAB team; ensure day-to-day TAB team work accomplishments are in compliance with this section.
- c. Prerequisite HVAC work: Do not bring the TAB team to the contract site until a copy of the prerequisite HVAC Checklist, with all work items certified by the Contractor to be working as designed, reaches the office of the TAB Agency.

#### 1.5.3 Sequencing and Scheduling

##### 1.5.3.1 Projects with Phased Construction

This specification section is structured as though the HVAC construction, and thereby the TAB work, will be completed in a single phase. When the construction is completed in phases, the DALT work and TAB work must be planned, completed, and accepted for each construction phase.

##### a. Phasing of Work

This specification section is structured as though the HVAC construction, and thereby the TAB work, is going to be completed in a single phase in spite of the fact that there will be two seasons. All elements of the TAB work are addressed on this premise. When a contract is to be completed in construction phases, including the TAB work, and the DALT work, the TAB work and DALT work must be planned for, completed and approved by the Contracting Officer with each phase. An example of this case would be one contract that requires the rehabilitation of the HVAC in each of several separated buildings. At the completion of the final phase, compile all approved reports and submit as one document.

##### 1.5.3.2 TAB Submittal and Work Schedule

Comply with additional requirements specified in Appendix C: TAB SUBMITTAL AND WORK SCHEDULE included at the end of this section.

Submit this schedule, and TAB Schematic Drawings, adapted for this particular contract, to the Contracting Officer (CO) for review and approval. Include with the submittal the planned calendar dates for each submittal or work item. Resubmit an updated version for CO approval every 90 calendar days. Compliance with the following schedule is the Contractor's responsibility.

Qualify TAB Personnel: Within 45 calendar days after date of contract award, submit TAB agency and personnel qualifications.

Pre-TAB Meeting: Within 30 calendar days after the date of approval of the TAB agency and personnel, meet with the COTR.

Design Review Report: Within 60 calendar days after the date of the TAB agency personnel qualifications approval, submit design review report.

Pre-Field TAB Engineering Report: Within 15 calendar days after approval of the TAB agency Personnel Qualifications, submit the Pre-Field TAB Engineering Report.

Prerequisite HVAC Work Check Out List For Season 1 and Advanced Notice For Season 1 TAB Field Work: At a minimum of 115 calendar days prior to CCD, submit Season 1 prerequisite HVAC work check out list certified as complete, and submit advance notice of commencement of Season 1 TAB field work.

Season 1 TAB Field Work: At a minimum of 90 calendar days prior to CCD, and when the ambient temperature is within Season 1 limits, accomplish Season 1 TAB field work; submit Season 1 TAB report; and conduct Season 1 field check.

Complete Season 1 TAB Work: Prior to CCD, complete all TAB work except Season 2 TAB work.

Prerequisite HVAC Work Check Out List For Season 2 and Advanced Notice For Season 2 TAB Field Work: Within 150 calendar days after date of the commencement of the Season 1 TAB field work, submit the Season 2 prerequisite HVAC work check out list certified as complete and submit advance notice of commencement of Season 2 TAB field work.

Season 2 TAB Field Work: Within 180 calendar days after date of commencement of the Season 1 TAB field work and when the ambient temperature is within Season 2 limits, accomplish Season 2 TAB field work; submit Season 2 TAB report; and conduct Season 2 field check.

Complete Season 2 TAB Work: Within 15 calendar days after the completion of Season 2 field data check, complete TAB work.

a. TAB Design Review Report

Submit typed report describing omissions and deficiencies in the HVAC system's design that would preclude the TAB team from accomplishing the duct leakage testing work and the TAB work requirements of this

section. Provide a complete explanation including supporting documentation detailing the design deficiency. State that no deficiencies are evident if that is the case.

#### 1.5.3.3 TAB Pre-Field Engineering Report

Submit report containing the following information:

a. Step-by-step TAB procedure:

- (1) Strategy: Describe the method of approach to the TAB field work from start to finish. Include in this description a complete methodology for accomplishing each seasonal TAB field work session.
- (2) Air System Diagrams: Use the contract drawings and duct fabrication drawings if available to provide air system diagrams in the report showing the location of all terminal outlet supply, return, exhaust and transfer registers, grilles and diffusers. Use a key numbering system on the diagrams which identifies each outlet contained in the outlet airflow report sheets. Show intended locations of all traverses and static pressure readings.
- (3) Procedural steps: Delineate fully the intended procedural steps to be taken by the TAB field team to accomplish the required TAB work of each air distribution system and each water distribution system. Include intended procedural steps for TAB work for subsystems and system components.

b. Pre-field data: Submit AABC or NEBB or SMACNA 1780 data report forms with the following pre-field information filled in:

- (1) Design data obtained from system drawings, specifications, and approved submittals.
- (2) Notations detailing additional data to be obtained from the contract site by the TAB field team.
- (3) Designate the actual data to be measured in the TAB field work.
- (4) Provide a list of the types of instruments, and the measuring range of each, which are anticipated to be used for measuring in the TAB field work. By means of a keying scheme, specify on each TAB data report form submitted, which instruments will be used for measuring each item of TAB data. If the selection of which instrument to use, is to be made in the field, specify from which instruments the choice will be made. Place the instrument key number in the blank space where the measured data would be entered.

c. Prerequisite HVAC work checkout list: Provide a list of inspections and work items which are to be completed by the Contractor. This list must be acted upon and completed by the Contractor and then submitted and approved by the Contracting Officer prior to the TAB team coming to the contract site.

At a minimum, a list of the applicable inspections and work items

listed in the NEBB PROCEDURAL STANDARDS, Section III, "Preliminary TAB Procedures" under paragraphs titled, "Air Distribution System Inspection" and "Hydronic Distribution System Inspection" must be provided for each separate system to be TAB'd.

#### 1.5.4 Subcontractor Special Requirements

All contract requirements of this section must be accomplished directly by a first tier subcontractor. No work may be performed by a second tier subcontractor.

#### 1.5.5 Instrument Calibration Certificates

It is the responsibility of the TAB firm to provide instrumentation that meets the minimum requirements of the standard under which the TAB Firm's qualifications are approved for use on a project. Instrumentation must be in proper operating condition and must be applied in accordance with the instrumentation's manufacturer recommendations.

All instrumentation must bear a valid NIST traceable calibration certificate during field work and during government acceptance testing. All instrumentation must be calibrated within no later than one year of the date of TAB work or government acceptance testing field work.

### PART 2 PRODUCTS

Not used.

### PART 3 EXECUTION

#### 3.1 WORK DESCRIPTIONS OF PARTICIPANTS

Comply with requirements of this section as specified in Appendix A WORK DESCRIPTIONS OF PARTICIPANTS.

#### 3.2 PRE-TAB MEETING

Meet with the Contracting Officer's technical representative (COTR) to develop a mutual understanding relative to the details of the TAB work requirements. Ensure that the TAB supervisor is present at this meeting. Requirements to be discussed include required submittals, work schedule, and field quality control.

#### 3.3 TAB PROCEDURES

##### 3.3.1 TAB Field Work

Test, adjust, and balance the HVAC systems until measured flow rates (air and water flow) are within plus or minus 5 percent of the design flow rates as specified or indicated on the contract documents.

That is, comply with the the requirements of AABC MN-1 or SMACNA 1780 (TABB) and SMACNA 1858 (TABB), except as supplemented and modified by this section.

Provide instruments and consumables required to accomplish the TAB work. Calibrate and maintain instruments in accordance with manufacturer's written procedures.

Test, adjust, and balance the HVAC systems until measured flow rates (air and water flow) are within plus or minus 5 percent of the design flow rates as specified or indicated on the contract documents. Conduct TAB work, including measurement accuracy, and sound measurement work in conformance with the AABC MN-1 and AABC MN-4, or NEBB TABES and NEBB MASV, or SMACNA 1780 (used by TABB) and SMACNA 1858 sound measurement procedures, except as supplemented and modified by this section. The only water flow and air flow reporting which can be deferred until the Season 2 is that data which would be affected in terms of accuracy due to outside ambient conditions.

### 3.3.2 Preliminary Procedures

Use the approved pre-field engineering report as instructions and procedures for accomplishing TAB field work. TAB engineer is to locate, in the field, test ports required for testing. It is the responsibility of the sheet metal contractor to provide and install test ports as required by the TAB engineer.

### 3.3.3 TAB Air Distribution Systems

#### 3.3.3.1 Units With Coils

Report heating and cooling performance capacity tests for hot water, chilled water, DX and steam coils for the purpose of verifying that the coils meet the indicated design capacity. Submit the following data and calculations with the coil test reports:

- a. For air handlers with capacities greater than 7.5 tons (90,000 Btu) cooling, such as factory manufactured units, central built-up units and rooftop units, conduct capacity tests in accordance with AABC MN-4, procedure 3.5, "Coil Capacity Testing."

Do not determine entering and leaving wet and dry bulb temperatures by single point measurement, but by the average of multiple readings in compliance with paragraph 3.5-5, "Procedures", (in subparagraph d.) of AABC MN-4, Procedure 3.5, "Coil Capacity Testing."

Submit part-load coil performance data from the coil manufacturer converting test conditions to design conditions; use the data for the purpose of verifying that the coils meet the indicated design capacity in compliance with AABC MN-4, Procedure 3.5, "Coil Capacity Testing," paragraph 3.5.7, "Actual Capacity Vs. Design Capacity" (in subparagraph c.).

- b. For units with capacities of 7.5 tons (90,000 Btu) or less, such as fan coil units, duct mounted reheat coils associated with VAV terminal units, and unitary units, such as through-the-wall heat pumps:

Determine the apparent coil capacity by calculations using single point measurement of entering and leaving wet and dry bulb temperatures; submit the calculations with the coil reports.

#### 3.3.3.2 Exhaust Fans

Exhaust fan systems including fans, ducts, plenums, grilles, and hoods for exhaust air.

### 3.3.4 TAB Water Distribution Systems

#### 3.3.4.1 Condenser Water Make-up Water

Condenser water make-up water systems including cooling towers, pumps, system balance valves and flow measuring devices.

### 3.3.5 TAB Work on Performance Tests Without Seasonal Limitations

#### 3.3.5.1 Performance Tests

In addition to the TAB proportionate balancing work on the air distribution systems and the water distribution systems, accomplish TAB work on the HVAC systems which directly transfer thermal energy. TAB the operational performance of the heating systems and cooling systems.

#### 3.3.5.2 Ambient Temperatures

On each tab report form used for recording data, record the outdoor and indoor ambient dry bulb temperature range and the outdoor and indoor ambient wet bulb temperature range within which the report form's data was recorded. Record these temperatures at beginning and at the end of data taking.

### 3.3.6 TAB Work on Performance Tests With Seasonal Limitations

#### 3.3.6.1 Performance Tests

Accomplish proportional balancing TAB work on the air distribution systems and water distribution systems, in other words, accomplish adjusting and balancing of the air flows and water flows, any time during the duration of this contract, subject to the limitations specified elsewhere in this section. However, accomplish, within the following seasonal limitations, TAB work on HVAC systems which directly transfer thermal energy.

#### 3.3.6.2 Season Of Maximum Load

Visit the contract site for at least two TAB work sessions for Season 1 and Season 2 field measures. Visit the contract site during the season of maximum heating load and visit the contract site during the season of maximum cooling load, the goal being to TAB the operational performance of the heating systems and cooling systems under their respective maximum outdoor environment-caused loading. During the seasonal limitations, TAB the operational performance of the heating systems and cooling systems.

#### 3.3.6.3 Ambient Temperatures

On each tab report form used for recording data, record the outdoor and indoor ambient dry bulb temperature range and the outdoor and indoor ambient wet bulb temperature range within which the report form's data was recorded. Record these temperatures at beginning and at the end of data taking.

### 3.3.7 Workmanship

Conduct TAB work on the HVAC systems until measured flow rates are within plus or minus 5 percent of the design flow rates as specified or indicated



on the contract documents. This TAB work includes adjustment of balancing valves, balancing dampers, and sheaves. Further, this TAB work includes changing out fan sheaves and pump impellers if required to obtain air and water flow rates specified or indicated. If, with these adjustments and equipment changes, the specified or indicated design flow rates cannot be attained, contact the Contracting Officer for direction.

### 3.3.8 Deficiencies

Strive to meet the intent of this section to maximize the performance of the equipment as designed and installed. However, if deficiencies in equipment design or installation prevent TAB work from being accomplished within the range of design values specified in the paragraph WORKMANSHIP, provide written notice as soon as possible to the Contractor and the Contracting Officer describing the deficiency and recommended correction.

Responsibility for correction of installation deficiencies is the Contractor's. If a deficiency is in equipment design, call the TAB team supervisor for technical assistance. Responsibility for reporting design deficiencies to Contractor is the TAB team supervisor's.

### 3.3.9 TAB Reports

Additional requirements for TAB Reports are specified in Appendix B  
REPORTS - TAB

### 3.3.10 Quality Assurance - COTR TAB Field Acceptance Testing

#### 3.3.10.1 TAB Field Acceptance Testing

During the field acceptance testing, verify, in the presence of the COTR, random selections of data (water, air quantities, air motion, sound level readings) recorded in the TAB Report. Points and areas for field acceptance testing are to be selected by the COTR. Measurement and test procedures are the same as approved for TAB work for the TAB Report.

Field acceptance testing includes verification of TAB Report data recorded for the following equipment groups:

Group 4: 25 percent of the return grilles, return registers, exhaust grilles and exhaust registers.

Group 5: 25 percent of the supply fans, exhaust fans, and pumps.

Further, if any data on the TAB Report for Groups 2 through 5 is found not to fall within the range of plus 5 to minus 5 percent of the TAB Report data, additional group data verification is required in the presence of the COTR. Verify TAB Report data for one additional piece of equipment in that group. Continue this additional group data verification until out-of-tolerance data ceases to be found.

#### 3.3.10.2 Additional COTR TAB Field Acceptance Testing

If any of the acceptance testing measurements for a given equipment group is found not to fall within the range of plus 5 to minus 5 percent of the TAB Report data, terminate data verification for all affected data for that group. The affected data for the given group will be disapproved. Make the necessary corrections and prepare a revised TAB Report. Reschedule acceptance testing of the revised report data with the COTR.

#### 3.3.10.3 Prerequisite for Approval

Compliance with the field acceptance testing requirements of this section is a prerequisite for the final Contracting Officer approval of the TAB Report submitted.

### 3.4 MARKING OF SETTINGS

Upon the final TAB work approval, permanently mark the settings of HVAC adjustment devices including valves, gauges, splitters, and dampers so that adjustment can be restored if disturbed at any time. Provide permanent markings clearly indicating the settings on the adjustment devices which result in the data reported on the submitted TAB report.

### 3.5 MARKING OF TEST PORTS

The TAB team is to permanently and legibly mark and identify the location points of the duct test ports. If the ducts have exterior insulation, make these markings on the exterior side of the duct insulation. Show the location of test ports on the as-built mechanical drawings with dimensions given where the test port is covered by exterior insulation.

### 3.6 APPENDICES

Appendix A WORK DESCRIPTIONS OF PARTICIPANTS  
Appendix B REPORTS - TAB  
Appendix C TAB SUBMITTAL AND WORK SCHEDULE  
Appendix D REQUIREMENTS FOR DUCT AIR LEAK TESTING

## Appendix A

### WORK DESCRIPTIONS OF PARTICIPANTS

The Contractor is responsible for ensuring compliance with all requirements of this specification section. However, the following delineation of specific work items is provided to facilitate and co-ordinate execution of the various work efforts by personnel from separate organizations.

#### 1. Contractor

- a. HVAC documentation: Provide pertinent contract documentation to the TAB Firm, to include the following: the contract drawings and specifications; copies of the approved submittal data for all HVAC equipment, air distribution devices, and air/water measuring/balancing devices; the construction work schedule; and other applicable documents requested by the TAB Firm. Provide the TAB Firm copies of contract revisions and modifications as they occur.
- b. Schedules: Ensure the requirements specified under the paragraph "TAB Schedule" are met.
- c. Pre-TAB meeting: Arrange and conduct the TAB meeting. Ensure that a representative is present for the sheet metal contractor, the mechanical contractor, the electrical contractor, and the automatic temperature controls contractor.
- d. Coordinate Support: Provide and coordinate support personnel required by the TAB Firm in order to accomplish the TAB field work. Support personnel may include factory representatives, HVAC controls installers, HVAC equipment mechanics, sheet metal workers, pipe fitters, and insulators. Ensure support personnel are present at the work site at the times required.
- e. Correct Deficiencies: Ensure the notifications of Construction Deficiencies are provided as specified herein. Refer to the paragraph CONSTRUCTION DEFICIENCIES. Correct each deficiency as soon as practical with the Contracting Officer, and submit revised schedules and other required documentation.
- f. Pre-TAB Work Checklists: Complete check out and debugging of HVAC equipment, ducts, and controls prior to the TAB engineer arriving at the project site to begin the TAB work. Debugging includes searching for and eliminating malfunctioning elements in the HVAC system installations, and verifying all adjustable devices are functioning as designed. Include as pre-TAB work checklist items, the deficiencies pointed out by the TAB team supervisor in the design review report.

Prior to the TAB field team's arrival, ensure completion of the applicable inspections and work items listed in the TAB team supervisor's TAB Work Procedures Summary. Do not allow the TAB team to commence TAB field work until all of the following are completed.

- g. Give Notice of Testing: Submit advance notice of proportional balancing, Season 1, and Season 2 TAB field work accompanied by completed prerequisite HVAC Work List

- h. Insulation work: Ensure that no insulation is shall not be installed on ducts until work on the subject ducts is complete.

Ensure the duct and piping systems are properly insulated and vapor sealed upon the successful completion and acceptance of the TAB work.

2. TAB Team Supervisor

- a. Overall management: Supervise and manage the overall TAB team work effort, including preliminary and technical TAB procedures and TAB team field work.
- b. Schedule: Ensure the requirements specified under the paragraph "TAB Schedule" are met.
- c. Submittals: Provide the submittals specified herein.
- d. Pre-DALT/TAB meeting: Attend meeting with Contractor. Ensure TAB personnel that will be involved in the TAB work under this contract attend the meeting.
- e. Design Review Report: Submit typed report describing omissions and deficiencies in the HVAC system's design that would preclude the TAB team from accomplishing the duct leakage testing work and the TAB work requirements of this section. Provide a complete explanation including supporting documentation detailing the design deficiency. State that no deficiencies are evident if that is the case.
- f. Support required: Specify the technical support personnel required from the Contractor other than the TAB agency; such as factory representatives for temperature controls or for complex equipment. Inform the Contractor in writing of the support personnel needed and when they are needed. Furnish the notice as soon as the need is anticipated, either with the design review report, or the TAB Procedures Summary, the during the TAB field work.

Ensure the Contractor is properly notified and aware of all support personnel needed to perform the TAB work. Maintain communication with the Contractor regarding support personnel throughout the duration of the TAB field work, including the TAB field acceptance testing checking.

Ensure all inspections and verifications for the Pre-Final DALT and Pre-TAB Checklists are completely and successfully conducted before DALT and TAB field work is performed.

- g. Advance Notice: Monitor the completion of the duct system installations and provide the Advance Notice for field work as specified herein.
- h. Technical Assistance: Provide technical assistance to the TAB field work.
- i. Deficiencies Notification: Ensure the notifications of Construction Deficiencies are provided as specified herein. Comply with requirements of the paragraph CONSTRUCTION DEFICIENCIES. Resolve each deficiency as soon as practical and submit revised schedules and other required documentation.

- j. Procedures: Develop the required TAB procedures for systems or system components not covered in the TAB Standard.
3. TAB Team Field Leader
- a. Field manager: Manage, in the field, the accomplishment of the work specified in Part 3, EXECUTION.
  - b. Full time: Be present at the contract site when TAB field work is being performed by the TAB team; ensure day-to-day TAB team work accomplishments are in compliance with this section.
  - c. Prerequisite HVAC work: Do not bring the TAB team to the contract site until a copy of the prerequisite HVAC work list, with all work items certified by the Contractor to be working as designed, reaches the office of the TAB Agency.

## Appendix B

### REPORTS - TAB

All submitted documentation must be typed, neat, and organized. All reports must have a waterproof front and back cover, a title page, a certification page, sequentially numbered pages throughout, and a table of contents. Tables, lists, and diagrams must be titled. Generate and submit for approval the following documentation:

#### 1. TAB Work Execution Schedule

Submit a detailed schedule indicating the anticipated calendar date for each submittal and each portion of work required under this section. For each work entry, indicate the support personnel (such as controls provider, HVAC mechanic, etc.) that are needed to accomplish the work. Arrange schedule entries chronologically.

#### 2. TAB Procedures Summary

Submit a detailed narrative describing all aspects of the TAB field work to be performed. Include the following:

- a. A list of the intended procedural steps for the TAB field work from start to finish. Indicate how each type of data measurement will be obtained. Include what Contractor support personnel are required for each step, and the tasks they need to perform.
- b. A list of the project's submittals that are needed by the TAB Firm in order to meet this Contract's requirements.
- c. The schematic drawings to be used in the required reports, which may include building floor plans, mechanical room plans, duct system plans, and equipment elevations. Indicate intended TAB measurement locations, including where test ports need to be provided by the Contractor.
- d. The data presentation forms to be used in the report, with the preliminary information and initial design values filled in.
- e. A list of TAB instruments to be used, edited for this project, to include the instrument name and description, manufacturer, model number, scale range, published accuracy, most recent calibration date, and what the instrument will be used for on this project.
- f. A thorough checklist of the work items and inspections that need to be accomplished before field work can be performed. The Contractor must complete, submit, and receive approval of the Completed Pre-Final Work Checklist before field work can be accomplished.
- g. A thorough checklist of the work items and inspections that need to be accomplished before the Season 1 TAB field work can be performed. The Contractor must complete, submit, and receive approval of the Completed Season 1 Pre-TAB Work Checklist before the Season 1 TAB field work can be accomplished.
- h. A thorough checklist of the work items and inspections that need to be accomplished before the Season 2 TAB field work can be performed. The

Contractor must complete, submit, and receive approval of the Completed Season 2 Pre-TAB Work Checklist before the Season 2 TAB field work can be accomplished.

- i. The checklists specified above shall be individually developed and tailored specifically for the work under this contract. Refer to NEBB PROCEDURAL STANDARDS, Section III, "Preliminary TAB Procedures" under the paragraphs titled, "Air Distribution System Inspection" and "Hydronic Distribution System Inspection" for examples of items to include in the checklists.

### 3. Design Review Report

Submit report containing the following information:

- a. Review the contract specifications and drawings to verify that the TAB work can be successfully accomplished in compliance with the requirements of this section. Verify the presence and location of permanently installed test ports and other devices needed, including gauge cocks, thermometer wells, flow control devices, circuit setters, balancing valves, and manual volume dampers.
- b. Submit a typed report describing omissions and deficiencies in the HVAC system's design that would preclude the TAB team from accomplishing the TAB work requirements of this section. Provide a complete explanation including supporting documentation detailing the design deficiency. If no deficiencies are evident, state so in the report.

### 4. TAB Reports: Submit TAB Report for Proportional Balancing, Season 1, and Season 2 in the following manner:

- a. Procedure Summary: Submit a copy of the approved TAB Procedures Summary. When applicable, provide notations describing how actual field procedures differed from the procedures listed.
- b. Report format: Submit the completed data forms approved in the pre-field TAB Engineering Report completed by TAB field team, reviewed, approved and signed by the TAB supervisor. Bind the report with a waterproof front and back cover. Include a table of contents identifying by page number the location of each report. Report forms and report data shall be typewritten. Handwritten report forms or report data are not acceptable.
- c. Temperatures: On each TAB report form reporting TAB work accomplished on HVAC thermal energy transfer equipment, include the indoor and outdoor dry bulb temperature range and indoor and outdoor wet bulb temperature range within which the TAB data was recorded. Include in the TAB report continuous time versus temperature recording data of wet and dry bulb temperatures for the rooms, or zones, as designated in the following list:

Chemical Room, Pump Room and Maintenance Room

- (1) Data shall be measured and compiled on a continuous basis for the period in which TAB work affecting those rooms is being done.

- (2) Data shall be measured/recorded only after the HVAC systems installations are complete, the systems fully balanced and the HVAC systems controls operating in fully automatic mode. Provide a detailed explanation wherever a final measurement did not achieve the required value.
  - (3) Data may be compiled using direct digital controls trend logging where available. Otherwise, the Contractor shall temporarily install calibrated time versus temperature/humidity recorders for this purpose. The HVAC systems and controls shall have been fully operational a minimum of 24 hours in advance of commencing data compilation. The specified data shall be included in the Season I and Season 2 TAB Report.
- d. Air System Diagrams: Provided updated diagrams with final installed locations of all terminals and devices, any numbering changes, and actual test locations.
- e. Air Static Pressure Profiles: Report static pressure profiles for air duct systems including: EF-1, EF-2, EF-3 and SF-1. Report static pressure data for all supply, return, relief, exhaust and outside air ducts for the systems listed. The static pressure report data shall include, in addition to AABC or NEBB or TABB required data, the following:
- (1) Report supply fan, return fan, relief fan, and exhaust fan inlet and discharge static pressures.
  - (2) Report static pressure drop across chilled water coils, DX coils, hot water coils, steam coils, electric resistance heating coils and heat reclaim devices installed in unit cabinetry or the system ductwork.
  - (3) Report static pressure drop across outside air, return air, and supply air automatic control dampers, both proportional and two-position, installed in unit cabinetry.
  - (4) Report static pressure drop across air filters, acoustic silencers, moisture eliminators, air flow straighteners, air flow measuring stations or other pressure drop producing specialty items installed in unit cabinetry, or in the system ductwork. Examples of these specialty items are smoke detectors, white sound generators, RF shielding, wave guides, security bars, blast valves, small pipes passing through ductwork, and duct mounted humidifiers.  
  
Do not report static pressure drop across duct fittings provided for the sole purpose of conveying air, such as elbows, transitions, offsets, plenums, manual dampers, and branch takes-offs.
  - (5) Report static pressure drop across outside air and relief/exhaust air louvers.
  - (6) Report static pressure readings of supply air, return air, exhaust/relief air, and outside air in duct at the point where these ducts connect to each air moving unit.
- f. Duct Transverses: Report duct traverses for main supply, return,



exhaust, relief and outside air ducts. The TAB Agency shall evaluate and report findings on the duct traverses taken. Evaluate the suitability of the duct traverse measurement based on satisfying the qualifications for a pitot traverse plane as defined by AMCA 203, "Field Measurements", Section 8, paragraph 8.3, "Location of Traverse Plane".

- g. Instruments: List the types of instruments actually used to measure the tab data. Include in the listing each instrument's unique identification number, calibration date, and calibration expiration date.

Instrumentation, used for taking wet bulb temperature readings shall provide accuracy of plus or minus 5 percent at the measured face velocities. Submit instrument manufacturer's literature to document instrument accuracy performance is in compliance with that specified.

- h. Performance Curves: The TAB Supervisor shall include, in the TAB Reports, factory pump curves and fan curves for pumps and fans TAB'd on the job.
- i. Calibration Curves: The TAB Supervisor shall include, in the TAB Reports, a factory calibration curve for installed flow control balancing valves, flow venturis and flow orifices TAB'd on the job.
- j. Data From TAB Field Work: After completion of the TAB field work, prepare the TAB field data for TAB supervisor's review and approval signature, using the reporting forms approved in the pre-field engineering report. Data required by those approved data report forms shall be furnished by the TAB team. Except as approved otherwise in writing by the Contracting Officer, the TAB work and thereby the TAB report shall be considered incomplete until the TAB work is accomplished to within the accuracy range specified in the paragraph WORKMANSHIP.

## Appendix C

### TAB SUBMITTAL AND WORK SCHEDULE

Perform the following items of work in the order listed adhering to the dates schedule specified below. Include the major items listed in this schedule in the project network analysis schedule required by Section 01 32 17.00 20 COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS).

Submit TAB Agency and TAB Personnel Qualifications: Within 42 calendar days after date of contract award.

Submit the TAB Work Execution Schedule: within 14 days after receipt of the TAB agency and TAB personnel qualifications approval. Revise and re-submit this schedule 28 days prior to the commencement of TAB Season 1 work and TAB Season 2 work.

Submit the TAB Work Procedures Summary: within 14 days after receipt of the initial approved TAB Work Execution Schedule.

Meet with the COTR at the Pre-TAB Meeting: Within 28 calendar days after receipt of the approved initial TAB Execution Schedule.

Submit Design Review Report: Within 56 calendar days after the receipt of the approved initial TAB Work Execution Schedule.

Advance Notice of Season 1 TAB Field Work: At a minimum of 14calendar days prior to Season 1 TAB Field Work, submit advance notice of TAB field work accompanied by completed Season 1 Pre-TAB Work Checklist.

Season 1 TAB Field Work: At a minimum of 84 calendar days prior to CCD, and when the ambient temperature is within Season 1 limits, accomplish Season 1 TAB field work.

Submit Season 1 TAB Report: Within 14 calendar days after completion of Season 1 TAB field work, submit initial Season 1 TAB report.

Season 1 Quality Assurance - COTR TAB Field Check: 30 calendar days after initial Season 1 TAB report is approved by the Contracting Officer, conduct Season 1 field check.

Complete Season 1 TAB Work: Prior to CCD, complete all TAB work except Season 2 TAB work and submit final.

Receive the approved TAB report: Within 21 calendar days, receive the report from Contracting Officer approved TAB report.

Advance Notice of Season 2 TAB Field Work: At a minimum of 126 calendar days after CCD, submit advance notice of Season 2 TAB field work accompanied by completed Season 2 Pre-TAB Work Checklist.

Season 2 TAB Field Work: Within 14 calendar days after date of advance notice of Season 2 TAB field work and when the ambient temperature is within Season 2 limits, accomplish Season 2 TAB field work.

Submit Season 2 TAB Report: Within 14 calendar days after completion of Season 2 TAB field work, submit Season 2 TAB report.

Season 2 Quality Assurance - COTR TAB Field Checks: 28 calendar days after the Season 2 TAB report is approved by the Contracting Officer, conduct Season 2 field check.

Complete Season 2 TAB Work: Within 14 calendar days after the completion of Season 2 TAB field data check, complete all TAB work.

Receive the approved TAB report: Within calendar 21 days, receive the report from Contracting Officer.

| Appendix D                                                    |                 |         |   |   |   |
|---------------------------------------------------------------|-----------------|---------|---|---|---|
| REQUIREMENTS FOR DUCT AIR LEAK TESTING                        |                 |         |   |   |   |
|                                                               |                 | SYSTEMS |   |   |   |
|                                                               |                 |         |   |   |   |
| Duct System Static Pressure, in inches W.C.                   | for Supply      |         |   |   |   |
|                                                               | for Return      |         |   |   |   |
|                                                               | for Exhaust     |         |   |   |   |
|                                                               | for Outside Air |         |   |   |   |
| System Oval/Round Duct and Rectangular Duct SMACNA Seal Class | for Supply      | A       | A | A | A |
|                                                               | for Return      | A       | A | A | A |
|                                                               | for Exhaust     | A       | A | A | A |
|                                                               | for Outside Air | A       | A | A | A |

| Appendix D                                         |                    |         |  |  |  |
|----------------------------------------------------|--------------------|---------|--|--|--|
| REQUIREMENTS FOR DUCT AIR LEAK TESTING             |                    |         |  |  |  |
|                                                    |                    | SYSTEMS |  |  |  |
|                                                    |                    |         |  |  |  |
| System<br>Oval/Round<br>Duct SMACNA<br>Leak Class  | for Supply         |         |  |  |  |
|                                                    | for Return         |         |  |  |  |
|                                                    | for Exhaust        |         |  |  |  |
|                                                    | for Outside<br>Air |         |  |  |  |
| System<br>Rectangular<br>Duct SMACNA<br>Leak Class | for Supply         |         |  |  |  |
|                                                    | for Return         |         |  |  |  |
|                                                    | for Exhaust        |         |  |  |  |
|                                                    | for Outside<br>Air |         |  |  |  |
| Duct Test<br>Pressure, in<br>inches W.C.           | for Supply         |         |  |  |  |
|                                                    | for Return         |         |  |  |  |
|                                                    | for Exhaust        |         |  |  |  |
|                                                    | for Outside<br>Air |         |  |  |  |

| Appendix D                                                    |                 |         |   |     |     |
|---------------------------------------------------------------|-----------------|---------|---|-----|-----|
| REQUIREMENTS FOR DUCT AIR LEAK TESTING                        |                 |         |   |     |     |
|                                                               |                 | SYSTEMS |   |     |     |
|                                                               |                 |         |   |     |     |
| Duct System Static Pressure, in millimeters W.C.              | for Supply      |         |   |     | n/a |
|                                                               | for Return      |         |   |     | n/a |
|                                                               | for Exhaust     |         |   | n/a |     |
|                                                               | for Outside Air |         |   | n/a | n/a |
| System Oval/Round Duct and Rectangular Duct SMACNA Seal Class | for Supply      | A       | A | A   | A   |
|                                                               | for Return      | A       | A | A   | A   |
|                                                               | for Exhaust     | A       | A | A   | A   |
|                                                               | for Outside Air | A       | A | A   | A   |
| System Oval/Round Duct SMACNA Leak Class                      | for Supply      |         |   | 12  | n/a |
|                                                               | for Return      |         |   | 12  | n/a |
|                                                               | for Exhaust     |         |   | n/a |     |
|                                                               | for Outside Air |         |   | n/a | n/a |

| Appendix D                                |                 |         |  |     |     |
|-------------------------------------------|-----------------|---------|--|-----|-----|
| REQUIREMENTS FOR DUCT AIR LEAK TESTING    |                 |         |  |     |     |
|                                           |                 | SYSTEMS |  |     |     |
|                                           |                 |         |  |     |     |
| System Rectangular Duct SMACNA Leak Class | for Supply      |         |  | 24  | n/a |
|                                           | for Return      |         |  | 24  | n/a |
|                                           | for Exhaust     |         |  | n/a |     |
|                                           | for Outside Air |         |  | n/a | n/a |
| Duct Test Pressure, in inches W.C.        | for Supply      |         |  |     | n/a |
|                                           | for Return      |         |  |     | n/a |
|                                           | for Exhaust     |         |  | n/a |     |
|                                           | for Outside Air |         |  | n/a | n/a |

-- End of Section --

SECTION 23 07 00

THERMAL INSULATION FOR MECHANICAL SYSTEMS  
**08/24**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only. At the discretion of the Government, the manufacturer of any material supplied will be required to furnish test reports pertaining to any of the tests necessary to assure compliance with the standard or standards referenced in this specification.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING  
ENGINEERS (ASHRAE)

|                  |                                                                               |
|------------------|-------------------------------------------------------------------------------|
| ASHRAE 90.1 - IP | (2019) Energy Standard for Buildings<br>Except Low-Rise Residential Buildings |
| ASHRAE 90.1 - SI | (2019) Energy Standard for Buildings<br>Except Low-Rise Residential Buildings |

ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A240/A240M | (2024) Standard Specification for Chromium<br>and Chromium-Nickel Stainless Steel Plate,<br>Sheet, and Strip for Pressure Vessels and<br>for General Applications |
| ASTM A580/A580M | (2023) Standard Specification for<br>Stainless Steel Wire                                                                                                         |
| ASTM B209       | (2014) Standard Specification for Aluminum<br>and Aluminum-Alloy Sheet and Plate                                                                                  |
| ASTM C195       | (2007; R 2024) Standard Specification for<br>Mineral Fiber Thermal Insulating Cement                                                                              |
| ASTM C450       | (2008) Standard Practice for Fabrication<br>of Thermal Insulating Fitting Covers for<br>NPS Piping, and Vessel Lagging                                            |
| ASTM C533       | (2017; R 2023) Standard Specification for<br>Calcium Silicate Block and Pipe Thermal<br>Insulation                                                                |
| ASTM C534/C534M | (2023) Standard Specification for<br>Preformed Flexible Elastomeric Cellular<br>Thermal Insulation in Sheet and Tubular<br>Form                                   |
| ASTM C547       | (2022a) Standard Specification for Mineral<br>Fiber Pipe Insulation                                                                                               |
| ASTM C552       | (2022) Standard Specification for Cellular                                                                                                                        |

Glass Thermal Insulation

|            |                                                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------------------------------|
| ASTM C553  | (2013; R 2019) Standard Specification for Mineral Fiber Blanket Thermal Insulation for Commercial and Industrial Applications |
| ASTM C591  | (2022) Standard Specification for Unfaced Preformed Rigid Cellular Polyisocyanurate Thermal Insulation                        |
| ASTM C610  | (2017; R 2023) Standard Specification for Molded Expanded Perlite Block and Pipe Thermal Insulation                           |
| ASTM C612  | (2014; R 2019) Standard Specification for Mineral Fiber Block and Board Thermal Insulation                                    |
| ASTM C647  | (2008; R 2013) Properties and Tests of Mastics and Coating Finishes for Thermal Insulation                                    |
| ASTM C755  | (2019b) Standard Practice for Selection of Water Vapor Retarders for Thermal Insulation                                       |
| ASTM C795  | (2008; R 2023) Standard Specification for Thermal Insulation for Use in Contact with Austenitic Stainless Steel               |
| ASTM C916  | (2020) Standard Specification for Adhesives for Duct Thermal Insulation                                                       |
| ASTM C920  | (2018) Standard Specification for Elastomeric Joint Sealants                                                                  |
| ASTM C921  | (2010; R 2015) Standard Practice for Determining the Properties of Jacketing Materials for Thermal Insulation                 |
| ASTM C1126 | (2018) Standard Specification for Faced or Unfaced Rigid Cellular Phenolic Thermal Insulation                                 |
| ASTM C1136 | (2023) Standard Specification for Flexible, Low Permeance Vapor Retarders for Thermal Insulation                              |
| ASTM C1290 | (2016; R 2021) Standard Specification for Flexible Fibrous Glass Blanket Insulation Used to Externally Insulate HVAC Ducts    |
| ASTM C1710 | (2011) Standard Guide for Installation of Flexible Closed Cell Preformed Insulation in Tube and Sheet Form                    |
| ASTM D882  | (2012) Tensile Properties of Thin Plastic Sheeting                                                                            |



|               |                                                                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D2863    | (2019) Standard Test Method for Measuring the Minimum Oxygen Concentration to Support Candle-Like Combustion of Plastics (Oxygen Index)                               |
| ASTM D5590    | (2017; R 2021) Standard Test Method for Determining the Resistance of Paint Films and Related Coatings to Fungal Defacement by Accelerated Four-Week Agar Plate Assay |
| ASTM E84      | (2023) Standard Test Method for Surface Burning Characteristics of Building Materials                                                                                 |
| ASTM E96/E96M | (2024) Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials                                                              |
| ASTM E2231    | (2021) Standard Practice for Specimen Preparation and Mounting of Pipe and Duct Insulation Materials to Assess Surface Burning Characteristics                        |

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

|                    |                                                                                                                                                            |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CDPH SECTION 01350 | (2017; Version 1.2) Standard Method for the Testing and Evaluation of Volatile Organic Chemical Emissions from Indoor Sources using Environmental Chambers |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|

FM GLOBAL (FM)

|              |                                                                                                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------|
| FM APP GUIDE | (updated on-line) Approval Guide<br><a href="https://www.approvalguide.com/">https://www.approvalguide.com/</a> |
|--------------|-----------------------------------------------------------------------------------------------------------------|

GREEN SEAL (GS)

|       |                                     |
|-------|-------------------------------------|
| GS-36 | (2013) Adhesives for Commercial Use |
|-------|-------------------------------------|

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS  
INDUSTRY (MSS)

|           |                                                                                                                |
|-----------|----------------------------------------------------------------------------------------------------------------|
| MSS SP-58 | (2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation |
|-----------|----------------------------------------------------------------------------------------------------------------|

MIDWEST INSULATION CONTRACTORS ASSOCIATION (MICA)

|                      |                                                                |
|----------------------|----------------------------------------------------------------|
| MICA Insulation Stds | (8th Ed) National Commercial & Industrial Insulation Standards |
|----------------------|----------------------------------------------------------------|

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                                                       |
|----------|---------------------------------------------------------------------------------------|
| NFPA 90A | (2024) Standard for the Installation of Air Conditioning and Ventilating Systems      |
| NFPA 90B | (2024) Standard for the Installation of Warm Air Heating and Air Conditioning Systems |

SCIENTIFIC CERTIFICATION SYSTEMS (SCS)

SCS SCS Global Services (SCS) Indoor Advantage

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

SCAQMD Rule 1168 (2022) Adhesive and Sealant Applications

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

TAPPI T403 OM (2022) Bursting Strength of Paper

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-A-3316 (1987; Rev C; Am 2 1990) Adhesives,  
Fire-Resistant, Thermal Insulation

MIL-A-24179 (1969; Rev A; Am 2 1980; Notice 1 1987;  
Notice 2 2020) Adhesive, Flexible  
Unicellular-Plastic Thermal Insulation

MIL-PRF-19565 (1988; Rev C) Coating Compounds, Thermal  
Insulation, Fire- and Water-Resistant,  
Vapor-Barrier

UL SOLUTIONS (UL)

UL 94 (2023; Reprint Jan 2024) UL Standard for  
Safety Tests for Flammability of Plastic  
Materials for Parts in Devices and  
Appliances

UL 723 (2020) UL Standard for Safety Test for  
Surface Burning Characteristics of  
Building Materials

UL 2818 (2022) GREENGUARD Certification Program  
For Chemical Emissions For Building  
Materials, Finishes And Furnishings

1.2 SYSTEM DESCRIPTION

1.2.1 General

Provide field-applied insulation and accessories on mechanical systems as specified herein; factory-applied insulation is specified under the piping, duct or equipment to be insulated. Provide field applied insulation materials required for use on Government-furnished items as listed in the SPECIAL CONTRACT REQUIREMENTS.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

Submit the three SD types, SD-02 Shop Drawings, SD-03 Product Data, and

SD-08 Manufacturer's Instructions at the same time for each system.

SD-02 Shop Drawings

MICA Plates; G, AE

Pipe Insulation Systems and Associated Accessories; G, AE

Duct Insulation Systems and Associated Accessories; G, AE

Equipment Insulation Systems and Associated Accessories; G, AE

Recycled content for insulation materials; G, AE

SD-03 Product Data

Pipe Insulation Systems; G, AE

Duct Insulation Systems; G, AE

Equipment Insulation Systems; G, AE

SD-04 Samples

Thermal Insulation; G, AE

Display Samples; G, AE

SD-07 Certificates

Indoor air quality for adhesives; G, AE

SD-08 Manufacturer's Instructions

Pipe Insulation Systems; G, AE

Duct Insulation Systems; G, AE

Equipment Insulation Systems; G, AE

## 1.4 CERTIFICATIONS

### 1.4.1 Adhesives and Sealants

Provide products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification or validation by other third-party programs that products meet the requirements of this Section. Provide current product certification documentation from certification body. When product does not have certification, provide validation that product meets the indoor air quality product requirements cited herein.

## 1.5 QUALITY ASSURANCE

### 1.5.1 Installer Qualification

Qualified installers must have successfully completed three or more similar type jobs within the last 5 years.

## 1.6 DELIVERY, STORAGE, AND HANDLING

Deliver materials in the manufacturer's unopened containers. Protect materials delivered and placed in storage from weather, humidity, dirt, dust and other contaminants. The Contracting Officer may reject insulation material and supplies that become dirty, dusty, wet, or contaminated by some other means. Attach manufacturer's stamp or label giving the name of the manufacturer and brand, and a description of the material, date codes, and approximate shelf life (if applicable) to packages or standard containers of insulation, jacket material, cements, adhesives, and coatings delivered for use, and samples required for approval. Insulation packages and containers must be asbestos free.

## PART 2 PRODUCTS

### 2.1 STANDARD PRODUCTS

Provide materials which are the standard products of manufacturers regularly engaged in the manufacture of such products and that essentially duplicate items that have been in satisfactory use for at least 2 years prior to bid opening and is an acceptable product per this specification. Submit a complete list of materials, including manufacturer's descriptive technical literature, performance data, catalog cuts, and installation instructions. Include the product number, k-value, thickness and furnished accessories including adhesives, sealants and jackets for each mechanical system requiring insulation. The product data must be copyrighted, have an identifying or publication number, and have been published prior to the issuance date of this solicitation. Submit materials furnished under this section together in a booklet.

#### 2.1.1 Insulation System

Provide insulation systems in accordance with the approved MICA National Insulation Standards plates as supplemented by this specification. Provide field-applied insulation for heating, ventilating, and cooling (HVAC) air distribution systems and piping systems that are located within, on, under, and adjacent to buildings; and for plumbing systems. Provide CFC and HCFC free insulation.

#### 2.1.2 Surface Burning Characteristics

Unless otherwise specified, insulation must have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Determine flame spread, and smoke developed indexes, by ASTM E84 or UL 723. Test insulation in the same density and installed thickness as the material to be used in the actual construction. Prepare and mount test specimens according to ASTM E2231.

### 2.2 MATERIALS

Provide insulation that meets or exceeds the requirements of ASHRAE 90.1 - IP. Ensure insulation exterior is cleanable, grease resistant, non-flaking and non-peeling. Provide compatible materials that do not contribute to corrosion, soften, or otherwise attack surfaces to which applied in either wet or dry state. Use materials on stainless steel surfaces meeting ASTM C795 requirements. Do not use calcium silicate on chilled or cold water systems. Use asbestos free materials. Provide product recognized under UL 94 (if containing plastic) and listed

in FM APP GUIDE.

#### 2.2.1 Adhesives

Provide non-aerosol adhesive products used on the interior of the building (defined as inside of the weatherproofing system) that meet either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168 (HVAC duct sealants must meet limit requirements of "Other" category within SCAQMD Rule 1168 sealants table). Provide aerosol adhesives used on the interior of the building that meet either emissions requirements of CDPH SECTION 01350 (use the office or classroom requirements, regardless of space type) or VOC content requirements of GS-36. Provide certification or validation of indoor air quality for adhesives.

##### 2.2.1.1 Acoustical Lining Insulation Adhesive

Provide a nonflammable, fire-resistant adhesive conforming to ASTM C916, Type I.

##### 2.2.1.2 Mineral Fiber Insulation Cement

Provide cement in accordance with ASTM C195.

##### 2.2.1.3 Lagging Adhesive

Lagging is the material used for thermal insulation, especially around a cylindrical object. This may include the insulation as well as the cloth/material covering the insulation. To resist mold/mildew, use lagging adhesive meeting ASTM D5590 with 0 growth rating. Provide nonflammable and fire-resistant lagging adhesives that have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Ensure adhesive is MIL-A-3316, Class 1, pigmented white and suitable, as listed by the manufacturer, for bonding fibrous glass cloth to faced and unfaced fibrous glass insulation board; for bonding cotton brattice cloth to faced and unfaced fibrous glass insulation board; for sealing edges of and bonding glass tape to joints of fibrous glass board; for bonding lagging cloth to thermal insulation; or Class 2 for attaching fibrous glass insulation to metal surfaces. Apply lagging adhesives in strict accordance with the manufacturer's recommendations for pipe and duct insulation.

##### 2.2.1.4 Contact Adhesive

Adhesives may be any of, but not limited to, the neoprene based, rubber based, or elastomeric type that have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Ensure adhesive does not adversely affect, initially or in service, the insulation to which it is applied, nor cause any corrosive effect on metal to which it is applied. Ensure that any solvent dispersing medium or volatile component of the adhesive has no objectionable odor and does not contain any benzene or carbon tetrachloride. Ensure dried adhesive does not emit nauseous, irritating, or toxic volatile matters or aerosols when the adhesive is heated to any temperature up to 212 degrees F. The dried adhesive must be nonflammable and fire resistant. Flexible Elastomeric Adhesive: Comply with MIL-A-24179, Type II, Class I. Provide product listed in FM APP GUIDE.

#### 2.2.2 Caulking

ASTM C920, Type S, Grade NS, Class 25, Use A.

#### 2.2.3 Corner Angles

Nominal 0.016 inch aluminum 1 by 1 inch with factory applied kraft backing. Aluminum must be ASTM B209, Alloy 3003, 3105, or 5005.

#### 2.2.4 Fittings

Fabricated Fittings are the prefabricated fittings for flexible elastomeric pipe insulation systems in accordance with ASTM C1710. Together with the flexible elastomeric tubes, they provide complete system integrity for retarding heat gain and controlling condensation drip from chilled-water and refrigeration systems. Flexible elastomeric, fabricated fittings provide thermal protection (0.25 k) and condensation resistance (0.05 Water Vapor Transmission factor). For satisfactory performance, use properly installed protective vapor retarder/barriers and vapor stops on high relative humidity and below ambient temperature applications to reduce movement of moisture through or around the insulation to the colder interior surface.

#### 2.2.5 Finishing Cement

ASTM C450: Mineral fiber hydraulic-setting thermal insulating and finishing cement. All cements that may come in contact with Austenitic stainless steel must comply with ASTM C795.

#### 2.2.6 Fibrous Glass Cloth and Glass Tape

Provide fibrous glass cloth, with 20X20 maximum mesh size, and glass tape with maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Provide tape consisting of 4 inch wide rolls. Provide Class 3 tape that is 4.5 ounces/square yard. Elastomeric Foam Tape: Black vapor-retarder foam tape with acrylic adhesive containing an anti-microbial additive.

#### 2.2.7 Staples

Outward clinching type monel.

#### 2.2.8 Jackets

##### 2.2.8.1 Aluminum Jackets

Provide aluminum jackets consisting of corrugated, embossed or smooth sheet, 0.016 inch nominal thickness; ASTM B209, Temper H14, Temper H16, Alloy 3003, 5005, or 3105. Do not use corrugated aluminum jacket outdoors. Aluminum jacket securing bands must be Type 304 stainless steel, 0.015 inch thick, 1/2 inch wide for pipe under 12 inch diameter and 3/4 inch wide for pipe over 12 inch and larger diameter. Aluminum jacket circumferential seam bands must be 2 by 0.016 inch aluminum matching jacket material. Ensure bands for insulation below ground are 3/4 by 0.020 inch thick stainless steel, or fiberglass reinforced tape. The jacket may, at the option of the Contractor, be provided with a factory fabricated Pittsburgh or "Z" type longitudinal joint. When the "Z" joint is used, use bands at the circumferential joints that are designed by the manufacturer to seal the joints and hold the jacket in

place.

#### 2.2.8.2 Polyvinyl Chloride (PVC) Jackets

Polyvinyl chloride (PVC) jacket and fitting covers must have high impact strength, ultraviolet (UV) resistant rating or treatment and moderate chemical resistance. The minimum thickness for a Standard jacket is 0.020 inch and where high abuse resistance is required use Heavy Duty PVC jacket 0.030 inch. Standard PVC fittings will have an initial core thickness (before thermoforming) of 0.025 inch and heavy duty fittings will have an initial core thickness of 0.030 inch.

#### 2.2.8.3 Vapor Barrier/Vapor Retarder

Apply the following criteria to determine which system is required.

- a. On ducts, equip piping and equipment operating below 90 degrees F or located outside with a vapor barrier.
- b. Install ducts, pipes and equipment that are located inside and that always operate above 90 degrees F with a vapor retarder where required as stated in paragraph VAPOR RETARDER REQUIRED.

#### 2.2.9 Vapor Retarder Required

ASTM C921, Type I, minimum puncture resistance 50 Beach units on all surfaces except concealed ductwork, where a minimum puncture resistance of 25 Beach units is acceptable. Minimum tensile strength, 35 pounds/inch width. ASTM C921, Type II, minimum puncture resistance 25 Beach units, tensile strength minimum 20 pounds/inch width. Use jackets on insulation exposed in finished areas that have white finish suitable for painting without sizing. Based on the application, insulation materials that require manufacturer or fabricator applied pipe insulation jackets are cellular glass, when all joints are sealed with a vapor barrier mastic, and mineral fiber. Ensure all non-metallic jackets have a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Flexible elastomerics require (in addition to vapor barrier skin) vapor retarder jacketing for high relative humidity and below ambient temperature applications.

##### 2.2.9.1 White Vapor Retarder All Service Jacket (ASJ)

ASJ is for use on hot/cold pipes, ducts, or equipment indoors or outdoors if covered by a suitable protective jacket. Provide product which meets all physical property and performance requirements of ASTM C1136, Type I, except a minimum burst strength of 85 psi. ASTM D2863 Limited Oxygen Index (LOI) is a minimum of 31.

In addition, do not use paper or other moisture-sensitive material for the outer exposed surface or the inner-most surface contacting the insulation. Ensure the outer exposed surface is white and has an emittance no less than 0.80.

##### 2.2.9.2 Vapor Retarder/Vapor Barrier Mastic Coatings

###### 2.2.9.2.1 Vapor Barrier

The vapor barrier must be self adhesive (minimum 2 mils adhesive, 3 mils embossed) greater than 3 plies standard grade, silver, white, black and

embossed white jacket for use on hot/cold pipes. Ensure permeability is less than 0.02 when tested in accordance with ASTM E96/E96M. Provide products meeting UL 723 or ASTM E84 flame and smoke requirements and that are UV resistant.

#### 2.2.9.2.2 Vapor Retarder

Provide fire and water resistant vapor retarder coating appropriately selected for either outdoor or indoor service. Color must be white. Ensure the water vapor permeance of the compound is in accordance with ASTM C755. Provide nonflammable, fire resistant coating. To resist mold/mildew, provide coating meeting ASTM D5590 with 0 growth rating. Ensure coating meets MIL-PRF-19565 Type II (if selected for indoor service) and is Qualified Products Database listed. Determine all other application and service properties pursuant to ASTM C647.

#### 2.2.9.3 Laminated Film Vapor Retarder

ASTM C1136, Type I, maximum moisture vapor transmission 0.02 perms, minimum puncture resistance 50 Beach units on all surfaces except concealed ductwork; where Type II, maximum moisture vapor transmission 0.02 perms, a minimum puncture resistance of 25 Beach units is acceptable. Provide vapor retarder with a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84. Flexible Elastomeric exterior foam with factory applied UV Jacket. Construction of laminate designed to provide UV resistance, high puncture, tear resistance and an excellent WVT rate.

#### 2.2.9.4 Polyvinylidene Chloride (PVDC) Film Vapor Retarder

Provide PVDC film vapor retarder with a maximum moisture vapor transmission of 0.02 perms, minimum puncture resistance of 150 Beach units, a minimum tensile strength in any direction of 30 lb/inch when tested in accordance with ASTM D882, and a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84.

#### 2.2.9.5 Polyvinylidene Chloride Vapor Retarder Adhesive Tape

Requirements must meet the same as specified for Laminated Film Vapor Retarder above.

#### 2.2.9.6 Vapor Barrier/Weather Barrier

Ensure the vapor barrier is greater than 3 ply self adhesive laminate -white vapor barrier jacket- superior performance (permeability less than 0.02 when tested in accordance with ASTM E96/E96M). Provide vapor barrier meeting UL 723 or ASTM E84 25 flame and 50 smoke requirements; and UV resistant. Minimum burst strength 185 psi in accordance with TAPPI T403 OM. Tensile strength 68 lb/inch width (PSTC-1000). Provide tape as specified for laminated film vapor barrier above.

#### 2.2.10 Vapor Retarder Not Required

ASTM C921, Type II, Class D, minimum puncture resistance 50 Beach units on all surfaces except ductwork, where Type IV, maximum moisture vapor transmission 0.10, a minimum puncture resistance of 25 Beach units is acceptable. Provide jacket with a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84.



#### 2.2.11 Wire

Soft annealed ASTM A580/A580M Type 302, 304 or 316 stainless steel, 16 or 18 gauge.

#### 2.2.12 Insulation Bands

Provide 1/2 inch wide; 26 gauge stainless steel insulation bands.

#### 2.2.13 Sealants

Choose sealants from the butyl polymer type, the styrene-butadiene rubber type, or the butyl type of sealants. Provide sealants with a maximum permeance of 0.02 perms based on Procedure B for ASTM E96/E96M, and a maximum flame spread index of 25 and a maximum smoke developed index of 50 when tested in accordance with ASTM E84.

### 2.3 PIPE INSULATION SYSTEMS

Conform insulation materials to Table 1 and minimum insulation thickness as listed in Table 2 and meet or exceed the requirements of ASHRAE 90.1 - SIASHRAE 90.1 - IP. Limit pipe insulation materials to those listed herein and meeting the following requirements:

#### 2.3.1 Recycled Materials

Provide insulation materials containing the following minimum percentage of recycled material content by weight:

- Rock Wool: 75 percent slag of weight
- Fiberglass: 20 percent glass cullet
- Rigid Foam: 9 percent recovered material
- Phenolic Rigid Foam: 9 percent recovered material

Provide data identifying percentage of recycled content for insulation materials.

#### 2.3.2 Aboveground Cold Pipeline ( -30 to 60 deg. F)

Provide insulation for outdoor, indoor, exposed or concealed applications, as follows:

##### 2.3.2.1 Cellular Glass

ASTM C552, Type II, and Type III. Supply the insulation from the fabricator with (paragraph WHITE VAPOR RETARDER ALL SERVICE JACKET (ASJ)) ASJ vapor retarder and installed with all longitudinal overlaps sealed and all circumferential joints ASJ taped or supply the insulation unfaced from the fabricator and install with all longitudinal and circumferential joints sealed with vapor barrier mastic.

##### 2.3.2.2 Flexible Elastomeric Cellular Insulation

Closed-cell, foam- or expanded-rubber materials containing anti-microbial additive, complying with ASTM C534/C534M, Grade 1, Type I or II. Type I, Grade 1 for tubular materials. Type II, Grade 1, for sheet materials. Ensure Type I and II have vapor retarder/vapor barrier skin on one or both sides of the insulation, and require an additional exterior vapor retarder

covering for high relative humidity and below ambient temperature applications.

#### 2.3.2.3 Mineral Fiber Insulation with Integral Wicking Material (MFIWM)

ASTM C547. Install in accordance with manufacturer's instructions. Do not use in applications exposed to outdoor ambient conditions in climatic zones 1 through 4.

#### 2.3.2.4 Polyisocyanurate Insulation

ASTM C591, Type I. Polyisocyanurate insulation may only be used on systems operating at temperatures not to exceed 140 degrees F. Supply the insulation with a factory applied vapor retarder/barrier that complies with Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. The insulation and all covering must pass the flame spread index of 25 and the smoke developed index of 50 when tested in accordance with ASTM E84.

#### 2.3.3 Aboveground Hot Pipeline (Above 60 deg. F)

Provide insulation for outdoor, indoor, exposed or concealed applications meeting the following requirements. Supply the insulation with manufacturer's recommended factory-applied jacket/vapor barrier.

##### 2.3.3.1 Mineral Fiber

ASTM C547, Types I, II or III, supply the insulation with manufacturer's recommended factory-applied jacket.

##### 2.3.3.2 Calcium Silicate

ASTM C533, Type I indoor only, or outdoors above 250 degrees F pipe temperature. Supply insulation with the manufacturer's recommended factory-applied jacket/vapor barrier.

##### 2.3.3.3 Cellular Glass

ASTM C552, Type II and Type III. Supply the insulation with manufacturer's recommended factory-applied jacket.

##### 2.3.3.4 Flexible Elastomeric Cellular Insulation

Closed-cell, foam- or expanded-rubber materials containing anti-microbial additive, complying with ASTM C534/C534M, Grade 1, Type I or II to 220 degrees F service. Type I for tubular materials. Type II for sheet materials.

##### 2.3.3.5 Phenolic Insulation

ASTM C1126 Type III to 250 degrees F service must comply with ASTM C795. Supply the insulation with manufacturer's recommended factory-applied jacket/vapor barrier.

##### 2.3.3.6 Perlite Insulation

ASTM C610

#### 2.3.3.7 Polyisocyanurate Insulation

ASTM C591, Type I. Supply the insulation with a factory applied vapor retarder/barrier that complies with Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. The insulation and all covering must pass the flame spread index of 25 and the smoke developed index of 50 when tested in accordance with ASTM E84.

### 2.4 DUCT INSULATION SYSTEMS

#### 2.4.1 Factory Applied Insulation

Provide factory-applied ASTM C552, cellular glass thermal or ASTM C534/C534M Grade 1, Type II, flexible elastomeric closed cell insulation according to manufacturer's recommendations for insulation with insulation manufacturer's standard reinforced fire-retardant vapor barrier, with identification of installed thermal resistance (R) value and out-of-package R value.

#### 2.4.2 Field Applied Insulation

##### 2.4.2.1 General

Calculate the minimum thickness in accordance with ASHRAE 90.1 - IP.

#### 2.4.3 Rigid Insulation

Mineral-Fiber Board Insulation: Mineral or glass fibers bonded with a thermosetting resin. Comply with ASTM C612, Type IA or Type IB; 3.0 pcf. Provide insulation with factory applied ASJ and factory-applied FSK jacket. Minimum thickness in accordance with ASHRAE 90.1 - IP Tables 3 and 4

#### 2.4.4 Blanket Insulation

Mineral-Fiber Blanket Insulation: Mineral or glass fibers bonded with a thermosetting resin. Comply with ASTM C553, Type II and ASTM C1290, Type III with factory-applied FSK jacket; 3/4 pcf. Minimum thickness in accordance with ASHRAE 90.1 - IP Tables 3 and 4.

#### 2.4.5 Acoustical Duct Lining

##### 2.4.5.1 General

For ductwork indicated or specified in Section 23 30 00 HVAC AIR DISTRIBUTION to be acoustically lined, provide external insulation in accordance with this specification section and in addition to the acoustical duct lining if R-value would not be met as required in Table 3 and 4. The installation of internal acoustical duct insulation is prohibited.

##### 2.4.5.2 Duct Liner

Flexible Elastomeric Acoustical and Conformable Duct Liner Materials: Flexible Elastomeric Thermal, Acoustical and Conformable Insulation Compliance with ASTM C534/C534M Grade 1, Type II; and NFPA 90A or NFPA 90B as applicable.

## 2.4.6 Duct Insulation Jackets

### 2.4.6.1 All-Purpose Jacket

Provide insulation with insulation manufacturer's standard reinforced fire-retardant jacket with or without integral vapor barrier as required by the service. In exposed locations, provide jacket with a white surface.

### 2.4.6.2 Metal Jackets

#### 2.4.6.2.1 Aluminum Jackets

ASTM B209, Temper H14, minimum thickness of 27 gauge ( 0.016 inch), with factory-applied polyethylene and kraft paper moisture barrier on inside surface. Provide smooth surface jackets for jacket outside dimension 8 inches and larger. Provide corrugated surface jackets for jacket outside dimension 8 inches and larger. Provide stainless steel bands, minimum width of 1/2 inch.

#### 2.4.6.2.2 Stainless Steel Jackets

ASTM A240/A240M; Type 304, minimum thickness of 33 gauge ( 0.010 inch), smooth surface with factory-applied polyethylene and kraft paper moisture barrier on inside surface. Provide stainless steel bands, minimum width of 1/2 inch.

### 2.4.6.3 Vapor Barrier/Weatherproofing Jacket

Provide vapor barrier/weatherproofing jacket that is laminated self-adhesive (minimum 2 mils adhesive, 3 mils embossed) less than 0.02 permeability, (greater than 3 ply, standard grade, silver, white, black and embossed or greater than 8 ply (minimum 2.9 mils adhesive), heavy duty white or natural).

## 2.4.7 Weatherproof Duct Insulation

Provide ASTM C552, cellular glass thermal insulation or ASTM C534/C534M Grade 1, Type II, flexible elastomeric cellular insulation, and weatherproofing as specified in manufacturer's instruction. Multi-ply, Polymeric Blend Laminate Jacketing: Construction of laminate designed to provide UV resistance, high puncture, tear resistance and an excellent WVT rate.

## 2.5 EQUIPMENT INSULATION SYSTEMS

Insulate equipment and accessories as specified in Tables 5 and 6. In outside locations, provide insulation 1/2 inch thicker than specified. Increase the specified insulation thickness for equipment where necessary to equal the thickness of angles or other structural members to make a smooth, exterior surface. Submit a booklet containing manufacturer's published installation instructions for the insulation systems. The instructions must be copyrighted, have an identifying or publication number, and have been published prior to the issuance date of this solicitation. A booklet is also required by paragraphs titled: Pipe Insulation Systems and Duct Insulation Systems.

## PART 3 EXECUTION

### 3.1 APPLICATION - GENERAL

Apply insulation to unheated and uncooled piping and equipment. Do not compress flexible elastomeric cellular insulation at joists, studs, columns, ducts, and hangers. The insulation must not pull apart after a one hour period; replace any insulation found to pull apart after one hour.

#### 3.1.1 Display Samples

Submit and display, after approval of materials, actual sections of installed systems, properly insulated in accordance with the specification requirements. Such actual sections must remain accessible to inspection throughout the job and will be reviewed from time to time for controlling the quality of the work throughout the construction site. Identify each material used by indicating on an attached sheet the specification requirement for the material and the material by each manufacturer intended to meet the requirement. The Contracting Officer will inspect display sample sections at the jobsite. Keep approved display sample sections on display at the jobsite during the construction period. Upon completion of construction, the display sample sections will be closed and sealed.

##### 3.1.1.1 Pipe Insulation Display Sections

Include as a minimum an elbow or tee, a valve, dielectric waterways and flanges, a hanger with protection shield and insulation insert, or dowel as required, at support point, method of fastening and sealing insulation at longitudinal lap, circumferential lap, butt joints at fittings and on pipe runs, and terminating points for each type of pipe insulation used on the job, and for hot pipelines and cold pipelines, both interior and exterior, even when the same type of insulation is used for these services.

##### 3.1.1.2 Duct Insulation Display Sections

Display sample sections for rigid and flexible duct insulation used on the job. Use a temporary covering to enclose and protect display sections for duct insulation exposed to weather

#### 3.1.2 Installation

Except as otherwise specified, install material in accordance with the manufacturer's written instructions. Do not apply insulation materials until tests and heat tracing specified in other sections of this specification are completed. Remove material such as rust, scale, dirt and moisture from surfaces to receive insulation. Keep insulation clean and dry. Do not remove insulation from its shipping containers until the day it is ready to use and return to like containers or equally protect from dirt and moisture at the end of each workday. Thoroughly clean insulation that becomes dirty prior to use. If insulation becomes wet or if cleaning does not restore the surfaces to like new condition, reject the insulation, and immediately remove from the jobsite. Stagger joints on multi layer insulation. Mix mineral fiber thermal insulating cement with demineralized water when used on stainless steel surfaces. Install insulation, jacketing and accessories in accordance with MICA Insulation Stds plates except where modified herein or on the drawings.

### 3.1.3 Firestopping

Where pipes and ducts pass through fire walls, fire partitions, above grade floors, and fire rated chase walls, seal the penetration with fire stopping materials as specified in Section 07 84 00 FIRESTOPPING. The protection of ducts at point of passage through firewalls must be in accordance with NFPA 90A and NFPA 90B. All other penetrations, such as piping, conduit, and wiring, through firewalls must be protected with a material or system of the same hourly rating that is listed by UL, FM, or a NRTL.

### 3.1.4 Installation of Flexible Elastomeric Cellular Insulation

Install flexible elastomeric cellular insulation with seams and joints sealed with rubberized contact adhesive. Do not use flexible elastomeric cellular insulation on surfaces greater than 220 degrees F. Stagger seams when applying multiple layers of insulation. Protect insulation exposed to weather with metal jacketing as recommended by the manufacturer after the adhesive is dry and cured.

#### 3.1.4.1 Adhesive Application

Apply a brush coating of adhesive to both butt ends to be joined and to both slit surfaces to be sealed. Allow the adhesive to set until dry to touch but tacky under slight pressure before joining the surfaces. Ensure insulation seals at seams and joints are not capable of being pulled apart one hour after application. Replace insulation that can be pulled apart one hour after installation.

#### 3.1.4.2 Adhesive Safety Precautions

Use natural cross-ventilation, local (mechanical) pickup, or general area (mechanical) ventilation to prevent an accumulation of solvent vapors, keeping in mind the ventilation pattern must remove any heavier-than-air solvent vapors from lower levels of the workspaces. Gloves and spectacle-type safety glasses are recommended in accordance with safe installation practices.

### 3.1.5 Welding

Welding is not permitted on piping, duct or equipment without written approval of the Contracting Officer. The capacitor discharge welding process may be used for securing metal fasteners to duct.

### 3.1.6 Pipes/Ducts/Equipment That Require Insulation

Insulation is required on all pipes, ducts, or equipment, except for omitted items as specified.

## 3.2 PIPE INSULATION SYSTEMS INSTALLATION

### 3.2.1 Pipe Insulation

#### 3.2.1.1 General

Install pipe insulation on aboveground hot and cold pipeline systems as specified below to form a continuous thermal retarder/barrier, including straight runs, fittings and appurtenances unless specified otherwise.

Install full length units of insulation using a single cut piece to complete a run. Do not use cut pieces or scraps abutting each other. Omit pipe insulation on the following:

- a. Air chambers.
- b. Adjacent insulation.
- c. ASME stamps.
- d. Access plates of fan housings.

#### 3.2.1.2 Pipes Passing Through Walls, Roofs, and Floors

Provide continuous pipe insulation through the sleeve.

Provide an aluminum jacket or vapor barrier/weatherproofing self adhesive jacket (minimum 2 mils adhesive, 3 mils embossed) less than 0.0000 permeability, greater than 3 ply standard grade, silver, white, black and embossed with factory applied moisture retarder over the insulation wherever penetrations require sealing.

##### 3.2.1.2.1 Penetrate Interior Walls

Provide aluminum jacket or vapor barrier/weatherproofing - self adhesive jacket (minimum 2 mils adhesive, 3 mils embossed) less than 0.02 permeability, greater than 3 plies standard grade, silver, white, black and embossed which extends 2 inches beyond either side of the wall and secure on each end with a band.

##### 3.2.1.2.2 Penetrating Floors

Extend the aluminum jacket from a point below the backup material to a point 10 inches above the floor with one band at the floor and one not more than 1 inch from the end of the aluminum jacket.

##### 3.2.1.2.3 Penetrating Waterproofed Floors

Extend the aluminum jacket from below the backup material to a point 2 inches above the flashing with a band 1 inch from the end of the aluminum jacket.

##### 3.2.1.2.4 Penetrating Exterior Walls

Continue the aluminum jacket required for pipe exposed to weather through the sleeve to a point 2 inches beyond the interior surface of the wall.

##### 3.2.1.2.5 Penetrating Roofs

Insulate pipe as required for interior service to a point flush with the top of the flashing and sealed with flashing sealant. Tightly butt the insulation for exterior application to the top of flashing and interior insulation. Extend the exterior aluminum jacket 2 inches down beyond the end of the insulation to form a counter flashing. Seal the flashing and counter flashing underneath with metal jacketing/flashing sealant.

#### 3.2.1.3 Pipes Passing Through Hangers

Ensure insulation, whether hot or cold application, is continuous through

hangers. Support all horizontal pipes 2 inches and smaller on hangers with the addition of a Type 40 protection shield to protect the insulation in accordance with MSS SP-58. Whenever insulation shows signs of being compressed, or when the insulation or jacket shows visible signs of distortion at or near the support shield, install insulation inserts as specified below for piping larger than 2 inches, or factory insulated hangers (designed with a load bearing core) can be used.

#### 3.2.1.3.1 Horizontal Pipes Larger Than 2 Inches at 60 Degrees F and Above

Supported on hangers in accordance with MSS SP-58.

#### 3.2.1.3.2 Horizontal Pipes Larger Than 2 Inches and Below 60 Degrees F

Supported on hangers with the addition of a Type 40 protection shield in accordance with MSS SP-58. Install an insulation insert of cellular glass, prefabricated insulation pipe hangers, or perlite above 80 degrees F above each shield. Ensure insert covers no less than the bottom 180-degree arc of the pipe. Provide inserts that are the same thickness as the insulation, and extend 2 inches on each end beyond the protection shield. When insulation inserts are required in accordance with the above, and the insulation thickness is less than 1 inch, wooden or cork dowels or blocks may be installed between the pipe and the shield to prevent the weight of the pipe from crushing the insulation, as an option to installing insulation inserts. Ensure the insulation jacket is continuous over the wooden dowel, wooden block, or insulation insert.

#### 3.2.1.3.3 Vertical Pipes

Supported with either Type 8 or Type 42 riser clamps with the addition of two Type 40 protection shields in accordance with MSS SP-58 covering the 360-degree arc of the insulation. Install an insulation insert of cellular glass or calcium silicate between each shield and the pipe. Ensure the insert covers the 360-degree arc of the pipe. Provide inserts that are the same thickness as the insulation, and extend 2 inches on each end beyond the protection shield. When insulation inserts are required in accordance with the above, and the insulation thickness is less than 1 inch, wooden or cork dowels or blocks may be installed between the pipe and the shield to prevent the hanger from crushing the insulation, as an option instead of installing insulation inserts. Ensure the insulation jacket is continuous over the wooden dowel, wooden block, or insulation insert. Support the vertical weight of the pipe with hangers located in a horizontal section of the pipe. When the pipe riser is longer than 30 feet, support the weight of the pipe additionally with hangers in the vertical run of the pipe that are directly clamped to the pipe, penetrating the pipe insulation. Use insulated hangers and seal the insulation jacket as indicated herein for anchors in a similar service.

#### 3.2.1.3.4 Inserts

Covered with a jacket material of the same appearance and quality as the adjoining pipe insulation jacket, overlap the adjoining pipe jacket 1-1/2 inches, and seal as required for the pipe jacket. Use jacket material to cover inserts in flexible elastomeric cellular insulation conforming to ASTM C1136, Type 1, and is allowed to be of a different material than the adjoining insulation material.



#### 3.2.1.4 Flexible Elastomeric Cellular Pipe Insulation

Use tubular form flexible elastomeric cellular pipe insulation for pipe sizes 6 inches and less. Grade 1, Do not stretch Type II sheet insulation used on pipes larger than 6 inches around the pipe. On pipes larger than 12 inches, adhere the insulation directly to the pipe on the lower 1/3 of the pipe. Stagger seams when applying multiple layers of insulation. Insulate sweat fittings with miter-cut pieces the same size as on adjacent piping. Insulate screwed fittings with sleeved fitting covers fabricated from miter-cut pieces and overlap and seal to the adjacent pipe insulation. Type II requires an additional exterior vapor retarder/barrier covering for high relative humidity and below ambient temperature applications.

#### 3.2.1.5 Pipes in High Abuse Areas

In high abuse areas such as janitor closets and traffic areas in equipment rooms, kitchens, and mechanical rooms, utilize , aluminum or flexible laminate cladding (comprised of elastomeric, plastic or metal foil laminate) laminated self-adhesive (minimum 2 mils adhesive, 3 mils embossed) vapor barrier/weatherproofing jacket, - less than 0.02 permeability; (greater than 3 ply, standard grade, silver, white, black and embossed) aluminum jackets. Protect pipe insulation to the 6 foot level.

#### 3.2.1.6 Pipe Insulation Material and Thickness

Pipe insulation materials must be as listed in Table 1 and must meet or exceed the requirements of ASHRAE 90.1 - IP.

| TABLE 1                                                                |                                                                                                                                     |                 |      |       |             |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|-------|-------------|
| Insulation Material for Piping                                         |                                                                                                                                     |                 |      |       |             |
| Service                                                                |                                                                                                                                     |                 |      |       |             |
|                                                                        | Material                                                                                                                            | Specification   | Type | Class | VR/VB Req'd |
| Chilled Water (Supply & Return, Dual Temperature Piping, 40 F nominal) |                                                                                                                                     |                 |      |       |             |
|                                                                        | Cellular Glass                                                                                                                      | ASTM C552       | II   | 2     | Yes         |
|                                                                        | Flexible Elastomeric Cellular                                                                                                       | ASTM C534/C534M | I    |       | Yes         |
|                                                                        | Mineral Fiber with Wicking Material Do not use in applications exposed to outdoor ambient conditions in climatic zones 1 through 4. | ASTM C547       | I    |       | Yes         |
|                                                                        | Polyisocyanurate                                                                                                                    | ASTM C591       | I    | 2     | Yes         |
| Heating Hot Water Supply & Return, Heated Oil (Max 250 F)              |                                                                                                                                     |                 |      |       |             |
|                                                                        | Mineral Fiber                                                                                                                       | ASTM C547       | I    | 1     | No          |

| TABLE 1                                                                                            |                               |                 |      |       |             |
|----------------------------------------------------------------------------------------------------|-------------------------------|-----------------|------|-------|-------------|
| Insulation Material for Piping                                                                     |                               |                 |      |       |             |
| Service                                                                                            |                               |                 |      |       |             |
|                                                                                                    | Material                      | Specification   | Type | Class | VR/VB Req'd |
|                                                                                                    | Calcium Silicate              | ASTM C533       | I    |       | No          |
|                                                                                                    | Cellular Glass                | ASTM C552       | II   | 2     | No          |
|                                                                                                    | Faced Phenolic Foam           | ASTM C1126      | III  |       | Yes         |
|                                                                                                    | Flexible Elastomeric Cellular | ASTM C534/C534M | I    | 2     | No          |
|                                                                                                    | Polyisocyanurate<br>Max temp  | ASTM C591       | I    | 2     | Yes         |
| Cold Domestic Water Piping, Makeup Water, and Condenser Water Piping                               |                               |                 |      |       |             |
|                                                                                                    | Cellular Glass                | ASTM C552       | II   | 2     | No          |
|                                                                                                    | Flexible Elastomeric Cellular | ASTM C534/C534M | I    |       | No          |
| Hot Domestic Water Supply & Recirculating Piping (Max 200 F)                                       |                               |                 |      |       |             |
|                                                                                                    | Mineral Fiber                 | ASTM C547       | I    | 1     | No          |
|                                                                                                    | Cellular Glass                | ASTM C552       | II   | 2     | No          |
|                                                                                                    | Flexible Elastomeric Cellular | ASTM C534/C534M | I    |       | No          |
|                                                                                                    | Faced Phenolic Foam           | ASTM C1126      | III  |       | Yes         |
| Refrigerant Suction Piping (35 degrees F nominal)                                                  |                               |                 |      |       |             |
|                                                                                                    | Flexible Elastomeric Cellular | ASTM C534/C534M | I    |       | No          |
|                                                                                                    | Cellular Glass                | ASTM C552       | II   | 1     | Yes         |
| Compressed Air Discharge, Steam and Condensate Return (201 to 250 Degrees F                        |                               |                 |      |       |             |
|                                                                                                    | Cellular Glass                | ASTM C552       | II   |       | No          |
|                                                                                                    | Mineral Fiber                 | ASTM C547       | I    | 1     | No          |
|                                                                                                    | Calcium Silicate              | ASTM C533       | I    |       | No          |
|                                                                                                    | Faced Phenolic Foam           | ASTM C1126      | III  |       | Yes         |
|                                                                                                    | Perlite                       | ASTM C610       |      |       | No          |
|                                                                                                    | Flexible Elastomeric Cellular | ASTM C534/C534M | I    | 2     | No          |
| Exposed Lavatory Drains, Exposed Domestic Water Piping & Drains to Areas for Handicapped Personnel |                               |                 |      |       |             |
|                                                                                                    | Flexible Elastomeric Cellular | ASTM C534/C534M | I    |       | No          |
| Condensate Drain Located Inside Building                                                           |                               |                 |      |       |             |
|                                                                                                    | Cellular Glass                | ASTM C552       | II   | 2     | No          |

| TABLE 1                                                                   |                               |                 |         |       |             |
|---------------------------------------------------------------------------|-------------------------------|-----------------|---------|-------|-------------|
| Insulation Material for Piping                                            |                               |                 |         |       |             |
| Service                                                                   |                               |                 |         |       |             |
|                                                                           | Material                      | Specification   | Type    | Class | VR/VB Req'd |
|                                                                           | Flexible Elastomeric Cellular | ASTM C534/C534M | I       |       | No          |
| Medium Temperature Hot Water, Steam and Condensate (251 to 350 Degrees F) |                               |                 |         |       |             |
|                                                                           | Mineral Fiber                 | ASTM C547       | I       | 1     | No          |
|                                                                           | Calcium Silicate              | ASTM C533       | I       |       | No          |
|                                                                           | Cellular Glass                | ASTM C552       | I or II |       | No          |
| High Temperature Hot Water & Steam (351 to 700 Degrees F)                 |                               |                 |         |       |             |
|                                                                           | Mineral Fiber                 | ASTM C547       | I       | 2     | No          |
|                                                                           | Calcium Silicate              | ASTM C533       | I       |       | No          |
|                                                                           | Cellular Glass                | ASTM C552       |         |       | No          |
| Brine Systems Cryogenics (-30 to 0 Degrees F)                             |                               |                 |         |       |             |
|                                                                           | Cellular Glass                | ASTM C552       | II      | 2     | No          |
|                                                                           | Flexible Elastomeric Cellular | ASTM C534/C534M | I       |       | No          |
| Brine Systems Cryogenics (0 to 34 Degrees F)                              |                               |                 |         |       |             |
|                                                                           | Cellular Glass                | ASTM C552       | II      | 2     | No          |
|                                                                           | Flexible Elastomeric Cellular | ASTM C534/C534M | I       |       | No          |
| Note: VR/VB = Vapor Retarder/Vapor Barrier                                |                               |                 |         |       |             |

| TABLE 2                                                                                                                                                                       |                                     |                                                           |                           |        |        |      |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------------------------|---------------------------|--------|--------|------|-----------|
| Piping Insulation Thickness (inch)<br>Do not use integral wicking material in Chilled water applications exposed to outdoor ambient conditions in climatic zones 1 through 4. |                                     |                                                           |                           |        |        |      |           |
| Service                                                                                                                                                                       |                                     |                                                           |                           |        |        |      |           |
|                                                                                                                                                                               | Material                            | Conductivity<br>K-value<br><br>(Btu-in/<br>hr-ft^2-deg F) | Tube And Pipe Size (inch) |        |        |      |           |
|                                                                                                                                                                               |                                     |                                                           | <1                        | 1-<1.5 | 1.5-<4 | 4-<8 | > or = >8 |
| Chilled Water (Supply & Return, Dual Temperature Piping, 40 Degrees F nominal)                                                                                                |                                     |                                                           |                           |        |        |      |           |
|                                                                                                                                                                               | Cellular Glass                      | 0.30                                                      | 1.5                       | 1.5    | 1.5    | 1.5  | 2         |
|                                                                                                                                                                               | Flexible Elastomeric Cellular       | 0.235                                                     | 1                         | 1      | 1      | N/A  | N/A       |
|                                                                                                                                                                               | Mineral Fiber with Wicking Material | 0.22                                                      | 1                         | 1.5    | 1.5    | 2    | 2         |
|                                                                                                                                                                               | Polyisocyanurate                    | 0.18                                                      | 0.5                       | 1      | 1      | 1    | 1.5       |
| Heating Hot Water Supply & Return, Heated Oil (Max 250 F)                                                                                                                     |                                     |                                                           |                           |        |        |      |           |
|                                                                                                                                                                               | Mineral Fiber                       | 0.27                                                      | 2.5                       | 2.5    | 2.5    | 3    | 3         |
|                                                                                                                                                                               | Calcium Silicate                    | 0.37                                                      | 4                         | 4      | 4      | 4    | 4         |
|                                                                                                                                                                               | Cellular Glass                      |                                                           | 3.5                       | 3.5    | 3.5    | 4    | 4         |
|                                                                                                                                                                               | Faced Phenolic Foam                 | 0.24                                                      | 2.5                       | 2.5    | 2.5    | 3    | 3         |
|                                                                                                                                                                               | Polyisocyanurate<br>Max temp 140F   | 0.23                                                      | 1                         | 1      | 1.5    | 1.5  | 1.5       |
| Cold Domestic Water Piping, Makeup Water, and Condenser Water                                                                                                                 |                                     |                                                           |                           |        |        |      |           |
|                                                                                                                                                                               | Cellular Glass                      | 0.30                                                      | 1.5                       | 1.5    | 1.5    | 1.5  | 1.5       |

| TABLE 2                                                                                                                                                                       |                               |                                                           |                           |        |        |      |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------------------------|---------------------------|--------|--------|------|-----------|
| Piping Insulation Thickness (inch)<br>Do not use integral wicking material in Chilled water applications exposed to outdoor ambient conditions in climatic zones 1 through 4. |                               |                                                           |                           |        |        |      |           |
| Service                                                                                                                                                                       |                               |                                                           |                           |        |        |      |           |
|                                                                                                                                                                               | Material                      | Conductivity<br>K-value<br><br>(Btu-in/<br>hr-ft^2-deg F) | Tube And Pipe Size (inch) |        |        |      |           |
|                                                                                                                                                                               |                               |                                                           | <1                        | 1-<1.5 | 1.5-<4 | 4-<8 | > or = >8 |
|                                                                                                                                                                               | Flexible Elastomeric Cellular | 0.235                                                     | 1                         | 1      | 1      | N/A  | N/A       |
| Hot Domestic Water Supply & Recirculating Piping (Max 200 F)                                                                                                                  |                               |                                                           |                           |        |        |      |           |
|                                                                                                                                                                               | Mineral Fiber                 | 0.26                                                      | 1.5                       | 1.5    | 2      | 2    | 2         |
|                                                                                                                                                                               | Cellular Glass                | 0.34                                                      | 2                         | 2      | 2.5    | 2.5  | 2.5       |
|                                                                                                                                                                               | Flexible Elastomeric Cellular | 0.30                                                      | 1.5                       | 1.5    | 2      | N/A  | N/A       |
| Refrigerant Suction Piping (35 degrees F nominal)                                                                                                                             |                               |                                                           |                           |        |        |      |           |
|                                                                                                                                                                               | Flexible Elastomeric Cellular | 0.235                                                     | 1                         | 1      | 1      | N/A  | N/A       |
|                                                                                                                                                                               | Cellular Glass                | 0.29                                                      | 1.5                       | 1.5    | 1.5    | 1.5  | 1.5       |
| Compressed Air Discharge, Steam and Condensate Return (201 to 250 Degrees F                                                                                                   |                               |                                                           |                           |        |        |      |           |
|                                                                                                                                                                               | Mineral Fiber                 | 0.27                                                      | 2.5                       | 2.5    | 2.5    | 3    | 3         |
|                                                                                                                                                                               | Calcium Silicate              | 0.37                                                      | 2.5                       | 3      | 4      | 4    | 4.5       |
|                                                                                                                                                                               | Cellular Glass                | 0.35                                                      | 2                         | 2.5    | 3      | 3    | 3         |
| Exposed Lavatory Drains, Exposed Domestic Water Piping & Drains to Areas for Handicapped Personnel                                                                            |                               |                                                           |                           |        |        |      |           |
|                                                                                                                                                                               | Flexible Elastomeric Cellular |                                                           | 0.5                       | 0.5    | 0.5    | 0.5  | 0.5       |

| TABLE 2                                                                                                                                                                       |                               |                                                           |                           |        |        |      |           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------------------------|---------------------------|--------|--------|------|-----------|
| Piping Insulation Thickness (inch)<br>Do not use integral wicking material in Chilled water applications exposed to outdoor ambient conditions in climatic zones 1 through 4. |                               |                                                           |                           |        |        |      |           |
| Service                                                                                                                                                                       |                               |                                                           |                           |        |        |      |           |
|                                                                                                                                                                               | Material                      | Conductivity<br>K-value<br><br>(Btu-in/<br>hr-ft^2-deg F) | Tube And Pipe Size (inch) |        |        |      |           |
|                                                                                                                                                                               |                               |                                                           | <1                        | 1-<1.5 | 1.5-<4 | 4-<8 | > or = >8 |
| Condensate Drain Located Inside Building                                                                                                                                      |                               |                                                           |                           |        |        |      |           |
|                                                                                                                                                                               | Cellular Glass                | 0.30                                                      | 1.5                       | 1.5    | 1.5    | 1.5  | 1.5       |
|                                                                                                                                                                               | Flexible Elastomeric Cellular | 0.235                                                     | 1                         | 1      | 1      | N/A  | N/A       |
| Medium Temperature Hot Water, Steam and Condensate (251 to 350 Degrees F)                                                                                                     |                               |                                                           |                           |        |        |      |           |
|                                                                                                                                                                               | Mineral Fiber                 | 0.29                                                      | 2.5                       | 3.5    | 4      | 4.5  | 4.5       |
|                                                                                                                                                                               | Calcium Silicate              | 0.39                                                      | 5                         | 6.5    | 6.5    | 6.5  | 6.5       |
| High Temperature Hot Water & Steam (351 to 700 Degrees F)                                                                                                                     |                               |                                                           |                           |        |        |      |           |
|                                                                                                                                                                               | Mineral Fiber                 | 0.32                                                      | 4.5                       | 5      | 5      | 5    | 5         |
|                                                                                                                                                                               | Calcium Silicate              | 0.41                                                      | 4.5                       | 5      | 6      | 6    | 6         |
| Brine Systems Cryogenics (-30 to 0 Degrees F)                                                                                                                                 |                               |                                                           |                           |        |        |      |           |
|                                                                                                                                                                               | Cellular Glass                | 0.25                                                      | 2.5                       | 2.5    | 3      | 3    | 3.5       |
|                                                                                                                                                                               | Flexible Elastomeric Cellular | 0.24                                                      | 1                         | 1      | N/A    | N/A  | N/A       |
| Brine Systems Cryogenics (0 to 34 Degrees F)                                                                                                                                  |                               |                                                           |                           |        |        |      |           |

| TABLE 2                                                                                                                                                                       |                                                                       |                           |        |        |      |           |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------|--------|--------|------|-----------|--|
| Piping Insulation Thickness (inch)<br>Do not use integral wicking material in Chilled water applications exposed to outdoor ambient conditions in climatic zones 1 through 4. |                                                                       |                           |        |        |      |           |  |
| Service                                                                                                                                                                       |                                                                       |                           |        |        |      |           |  |
| Material                                                                                                                                                                      | Conductivity<br>K-value<br><br>(Btu-in/<br>hr-ft <sup>2</sup> -deg F) | Tube And Pipe Size (inch) |        |        |      |           |  |
|                                                                                                                                                                               |                                                                       | <1                        | 1-<1.5 | 1.5-<4 | 4-<8 | > or = >8 |  |
| Cellular Glass                                                                                                                                                                | 0.29                                                                  | 2                         | 2      | 2      | 2.5  | 3         |  |
| Flexible Elastomeric Cellular                                                                                                                                                 | 0.26                                                                  | 1                         | 1      | 1      | N/A  | N/A       |  |

### 3.2.2 Aboveground Cold Pipelines

Insulate the following cold pipelines for minus 30 to plus 60 degrees F in accordance with Table 2 except those piping listed in subparagraph Pipe Insulation in PART 3 as to be omitted. This includes but is not limited to the following:

- a. Make-up water.
- b. Horizontal and vertical portions of interior roof drains.
- c. Refrigerant suction lines.
- d. Chilled water.
- e. Dual temperature water, i.e. HVAC hot/chilled water.
- f. Air conditioner condensate drains.
- g. Brine system cryogenics.
- h. Condenser water piping where pipe temperature may fall below ambient dew point.
- i. Exposed lavatory drains and domestic water lines serving plumbing fixtures for handicap persons.
- j. Domestic cold and chilled drinking water.

### 3.2.2.1 Insulation Material and Thickness

Determine insulation thickness for cold pipelines using Table 2.

### 3.2.2.2 Factory or Field applied Jacket

Cover insulation with a factory applied vapor retarder jacket/vapor barrier or field applied seal welded PVC jacket or greater than 3 ply laminated self-adhesive (minimum 2 mils adhesive, 3 mils embossed) vapor barrier/weatherproofing jacket - less than 0.02 permeability, standard grade, silver, white, black and embossed for use with Mineral Fiber, Cellular Glass, and Phenolic Foam Insulated Pipe. For insulation inside the building, to be protected with an aluminum jacket, PVC jacket or fittings (installed over a vapor retarder jacket), or greater than 3 ply vapor barrier/weatherproofing self-adhesive (minimum 2 mils adhesive, 3 mils embossed) product, less than 0.02 permeability, standard grade, Embossed Silver, White & Black, install the insulation and vapor retarder jacket as specified herein. Install the aluminum jacket, PVC jacket or fittings (installed over a vapor retarder jacket), or greater than 3 ply vapor barrier/weatherproofing self-adhesive (minimum 2 mils adhesive, 3 mils embossed) product, less than 0.02 permeability, standard grade, embossed silver, White & Black, as specified for piping exposed to weather, except sealing of the laps of the aluminum jacket is not required. In high abuse areas such as janitor closets and traffic areas in equipment rooms, kitchens, and mechanical rooms, provide aluminum jackets, PVC jacket or fittings (installed over a vapor retarder jacket), or greater than 3 ply vapor barrier/weatherproofing self-adhesive (minimum 2 mils adhesive, 3 mils embossed) product, less than 0.02 permeability, standard grade, embossed silver, white & black, for pipe insulation to the 6 ft level.

### 3.2.2.3 Installing Insulation for Straight Runs Hot and Cold Pipe

Apply insulation to the pipe with tight butt joints. Seal all butted joints and ends with joint sealant and seal with a vapor retarder coating, greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape or PVDC adhesive tape.

#### 3.2.2.3.1 Longitudinal Laps of the Jacket Material

Overlap not less than 1-1/2 inches. Provide butt strips 3 inches wide for circumferential joints.

#### 3.2.2.3.2 Laps and Butt Strips

Secure with adhesive and staple on 4 inch centers if not factory self-sealing. If staples are used, seal in accordance with paragraph STAPLES below. Note that staples are not required with cellular glass systems.

#### 3.2.2.3.3 Factory Self-Sealing Lap Systems

May be used when the ambient temperature is between 40 and 120 degrees F during installation. Install the lap system in accordance with manufacturer's recommendations. Use a stapler only if specifically recommended by the manufacturer. Where gaps occur, replace the section or repair the gap by applying adhesive under the lap and then stapling.



#### 3.2.2.3.4 Staples

Coat all staples, including those used to repair factory self-seal lap systems, with a vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate jacket - 0.0000 perm adhesive tape. Coat all seams, except those on factory self-seal systems, with vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape.

#### 3.2.2.3.5 Breaks and Punctures in the Jacket Material

Patch by wrapping a strip of jacket material around the pipe and secure it with adhesive, staple, and coat with vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape. Extend the patch not less than 1-1/2 inches past the break.

#### 3.2.2.3.6 Penetrations Such as Thermometers

Fill the voids in the insulation and seal with vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape.

#### 3.2.2.3.7 Flexible Elastomeric Cellular Pipe Insulation

Install by slitting the tubular sections and applying them onto the piping or tubing. Alternately, whenever possible slide un-slit sections over the open ends of piping or tubing. Secure all seams and butt joints and seal with adhesive. When using self seal products, secure only the butt joints with adhesive. Push insulation on the pipe, never pulled. Stretching of insulation may result in open seams and joints. Clean cut all edges. Rough or jagged edges of the insulation are not be permitted. Use proper tools such as sharp knives. Do not stretch Grade 1, Type II sheet insulation around the pipe when used on pipe larger than 6 inches. On pipes larger than 12 inches, adhere sheet insulation directly to the pipe on the lower 1/3 of the pipe.

#### 3.2.2.4 Insulation for Fittings and Accessories

- a. Butt pipe insulation tightly to the insulation of the fittings and accessories. Seal the butted joints and ends with joint sealant and seal with a vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape.
- b. Place precut or preformed insulation around all fittings and accessories and conform to MICA plates except as modified herein: 5 for anchors; 10, 11, and 13 for fittings; 14 for valves; and 17 for flanges and unions. Insulation must be the same insulation as the pipe insulation, including same density, thickness, and thermal conductivity. Where precut/preformed is unavailable, rigid preformed pipe insulation sections may be segmented into the shape required. Use insulation of the same thickness and conductivity as the adjoining pipe insulation. If nesting size insulation is used, overlap the insulation 2 inches or one pipe diameter. Elbows insulated using segments must conform to MICA Tables 12.20 "Mitered Insulation Elbow". Submit a booklet containing completed MICA Insulation Stds plates detailing each insulating system for each pipe, duct, or equipment insulating system, after approval of materials and prior to applying insulation.

- (1) Ensure MICA plates detail the materials to be installed and the specific insulation application. Submit all MICA plates required showing the entire insulating system, including plates required to show insulation penetrations, vessel bottom and top heads, legs, and skirt insulation as applicable. Present all variations of insulation systems including locations, materials, vaporproofing, jackets and insulation accessories.
  - (2) If the Contractor elects to submit detailed drawings instead of edited MICA Plates, ensure the detail drawings are technically equivalent to the edited MICA Plate submittal.
- c. Upon completion of insulation installation on flanges, unions, valves, anchors, fittings and accessories, terminations, seams, joints and insulation not protected by factory vapor retarder jackets or PVC fitting covers must be protected with PVDC or greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape or two coats of vapor retarder coating with a minimum total thickness of 1/16 inch, applied with glass tape embedded between coats. Overlap tap seams 1 inch. Extend the coating out onto the adjoining pipe insulation 2 inches. Protect fabricated insulation with a factory vapor retarder jacket with either greater than 3 ply laminate jacket - less than 0.0000 perm adhesive tape, standard grade, silver, white, black and embossed or PVDC adhesive tape or two coats of vapor retarder coating with a minimum thickness of 1/16 inch and with a 2 inch wide glass tape embedded between coats. Where fitting insulation butts to pipe insulation, seal the joints with a vapor retarder coating and a 4 inch wide ASJ tape which matches the jacket of the pipe insulation.
  - d. Insulate anchors attached directly to the pipe for a sufficient distance to prevent condensation but no less than 6 inches from the insulation surface.
  - e. Mark insulation to show the location of unions, strainers, and check valves.

#### 3.2.2.5 Optional PVC Fitting Covers

At the option of the Contractor, premolded, one or two piece PVC fitting covers may be used in lieu of the vapor retarder and embedded glass tape. Factory precut insulation inserts or premolded insulation segments must be used under the fitting covers for elbows, tees and other miscellaneous configurations. Insulation inserts must be the same installed insulation thickness. One insert must be installed for each inch of pipe insulation. The installed and compressed insert must have the same thermal performance as the pipe insulation. Premolded insulation segments must be the same insulation as the pipe insulation including same density, thickness, and thermal conductivity. PVC fitting covers must be secured by PVC vapor retarder tape, adhesive, seal welding or with tacks made for securing PVC covers. For above ambient pipe temperatures, secure PVC covers using tacks. As an option circumferential joints can be covered with PVC vapor retarder tape. For below ambient conditions seal system using PVC vapor retarder tape, and vapor retarder mastic with vapor dams. Longitudinal seams may be sealed with vapor retarder mastic, PVC vapor retarder tape or solvent weld adhesive to obtain a continuous vapor seal. For severe service conditions install PVC jackets over the existing ASJ jacket and seal using a solvent weld adhesive.

### 3.2.3 Aboveground Hot Pipelines

#### 3.2.3.1 General Requirements

Insulate all hot pipe lines above 60 degrees F, except those piping listed in subparagraph Pipe Insulation in PART 3 as to be omitted, in accordance with Table 2. This includes but is not limited to the following:

- a. Domestic hot water supply & re-circulating system.
- b. Steam.
- c. Condensate & compressed air discharge.
- d. Hot water heating.
- e. Heated oil.
- f. Water defrost lines in refrigerated rooms.

Cover insulation, in accordance with manufacturer's recommendations, with a factory applied Type I jacket or field applied aluminum where required or seal welded PVC.

#### 3.2.3.2 Insulation for Fittings and Accessories

Butt pipe insulation tightly to the insulation of the fittings and accessories. Seal butted joints and ends with joint sealant. Mark insulation to show the location of unions, strainers, check valves and other components that would otherwise be hidden from view by the insulation.

##### 3.2.3.2.1 Precut or Preformed

Place precut insulation inserts or preformed insulation inserts around all fittings and accessories. Insulation inserts must be the same installed insulation thickness as the pipe insulation. One insert must be installed for each inch of pipe insulation thickness. The installed and compressed insert must have the same thermal performance as the pipe insulation. Preformed insulation inserts must be the same insulation as the pipe insulation including same density, thickness, and thermal conductivity.

##### 3.2.3.2.2 Rigid Preformed

Where precut/preformed is unavailable, rigid preformed pipe insulation sections may be segmented into the shape required. Use insulation that is the same thickness and conductivity as the adjoining pipe insulation. If nesting size insulation is used, do not overlap insulation 2 inches or one pipe diameter. Elbows insulated using segments must conform to MICA Tables 12.20 "Mitered Insulation Elbow".

### 3.2.4 Piping Exposed to Weather

Insulate and jacket piping exposed to weather as specified for the applicable service inside the building. After this procedure, apply a laminated self-adhesive (minimum 2 mils adhesive, 3 mils embossed) vapor barrier/weatherproofing jacket - less than 0.02 permeability (greater than 3 ply, standard grade, silver, white, black and embossed aluminum jacket, stainless steel or PVC jacket).

PVC jacketing requires no factory-applied jacket beneath it, however apply an all service jacket if factory applied jacketing is not furnished.

#### 3.2.4.1 Aluminum Jacket

The jacket for hot piping may be factory applied. Overlap the jacket no less than 2 inches at longitudinal and circumferential joints and secure with bands at no more than 12 inch centers. Overlap longitudinal joints down to shed water and locate at 4 or 8 o'clock positions. Seal joints on piping 60 degrees F and below with metal jacketing/flashings sealant while overlapping to prevent moisture penetration. Where jacketing on piping 60 degrees F and below abuts an un-insulated surface, caulk joints to prevent moisture penetration. Seal joints on piping above 60 degrees F with a moisture retarder.

#### 3.2.4.2 Insulation for Fittings

Insulate and finish flanges, unions, valves, fittings, and accessories as specified for the applicable service. Apply two coats of breather emulsion type weatherproof mastic (impermeable to water, permeable to air) recommended by the insulation manufacturer with glass tape embedded between coats. Overlap tap no less than 1 inch and the adjoining aluminum jacket no less than 2 inches. Factory preformed aluminum jackets may be used in lieu of the above. Provide molded PVC fitting covers when PVC jackets are used for straight runs of pipe. PVC fitting covers must be sealed based upon the pipe temperature in combination with ambient conditions and if the pipe insulation will be exposed to severe service conditions. For below ambient conditions, the pipe insulation and insulation fittings must be covered and sealed with a vapor barrier or weatherproofing jacket with a minimum 0.02 permeability as defined per ASTM C1136 before installing the PVC jacket and fitting cover. When in doubt as to the pipe temperature or service conditions, secure the PVC jacket using a weld adhesive using the manufacturer's recommended adhesive.

#### 3.2.4.3 PVC Jacket

Provide ultraviolet resistant PVC jacket that is adhesive welded weather tight with manufacturer's recommended adhesive. Include provision for thermal expansion.

#### 3.2.4.4 Stainless Steel Jackets

ASTM A240/A240M; Type 304, minimum thickness of 33 gauge (0.010 inch), smooth surface with factory-applied polyethylene and kraft paper moisture barrier on inside surface. Provide stainless steel bands, minimum width of 1/2 inch.

### 3.3 DUCT INSULATION SYSTEMS INSTALLATION

Except for oven hood exhaust duct insulation, install corner angles on external corners of insulation on ductwork in exposed finished spaces before covering with jacket. Omit duct insulation on exposed supply and return ducts in air conditioned spaces unless otherwise shown. Air conditioned spaces are defined as those spaces directly supplied with cooled conditioned air (or provided with a cooling device such as a fan-coil unit) and heated conditioned air (or provided with a heating

device such as a unit heater, radiator or convector).

### 3.3.1 Duct Insulation Minimum Thickness

Duct insulation minimum thickness in accordance with Table 4.

| Table 4 - Minimum Duct Insulation (inches) |     |
|--------------------------------------------|-----|
| Cold Air Ducts                             | 2.0 |
| Relief Ducts                               | 1.5 |
| Fresh Air Intake Ducts                     | 1.5 |
|                                            |     |
| Warm Air Ducts                             | 2.0 |
| Relief Ducts                               | 1.5 |
| Fresh Air Intake Ducts                     | 1.5 |

### 3.3.2 Insulation and Vapor Retarder/Vapor Barrier for Cold Air Duct

Provide insulation and vapor retarder/vapor barrier for the following cold air ducts and associated equipment.

- a. Supply ducts.
- b. Return air ducts.
- c. Relief ducts.
- d. Flexible run-outs (field-insulated).
- e. Plenums.
- f. Duct-mounted coil casings.
- g. Coil headers and return bends.
- h. Coil casings.
- i. Fresh air intake ducts.
- j. Filter boxes.
- k. Mixing boxes (field-insulated).
- l. Supply fans (field-insulated).
- m. Site-erected air conditioner casings.
- n. Ducts exposed to weather.
- o. Combustion air intake ducts.
- p. Exhaust ducts passing through concealed spaces, unconditioned spaces, or semi-heated spaces exhausting conditioned air.

Use insulation for rectangular ducts that is flexible type where concealed, minimum density 3/4 pcf, and rigid type where exposed, minimum density 3 pcf. Provide insulation for both concealed or exposed round/oval ducts that is flexible type, minimum density 3/4 pcf or a semi rigid board, minimum density 3 pcf, formed or fabricated to a tight fit, edges beveled and joints tightly butted and staggered. Provide insulation for all exposed ducts with either a white, paint-able, factory-applied Type I jacket or a field applied vapor retarder/vapor barrier jacket coating finish as specified. Ensure the total field applied dry film thickness is approximately 1/16 inch. Provide insulation on all concealed duct with a factory-applied Type I or II vapor retarder/vapor barrier jacket. Continue duct insulation through sleeves and prepare openings except firewall penetrations. Duct insulation terminating at fire dampers, must be continuous over the damper collar and retaining angle of fire dampers, which are exposed to unconditioned air and which may be prone to condensate formation. Provide duct insulation and vapor retarder/vapor barrier to cover the collar, neck, and un-insulated surfaces of diffusers, registers and grills. Apply vapor retarder/vapor barrier materials to form a complete unbroken vapor seal over the insulation. Seal sheet metal duct in accordance with Section 23 30 00 HVAC AIR DISTRIBUTION.

#### 3.3.2.1 Installation on Concealed Duct

- a. For rectangular, oval or round ducts, attach flexible insulation by applying adhesive around the entire perimeter of the duct in 6 inch wide strips on 12 inch centers.
- b. For rectangular and oval ducts, 24 inches and larger, additionally secure insulation to bottom of ducts using mechanical fasteners. Space fasteners on 16 inch centers and no more than 16 inches from duct corners.
- c. For rectangular, oval and round ducts, provide mechanical fasteners on sides of duct risers for all duct sizes. Space fasteners on 16 inch centers and no more than 16 inches from duct corners.
- d. Impale insulation on the mechanical fasteners (self stick pins) where used and press thoroughly into the adhesive. Take care to ensure vapor retarder/vapor barrier jacket joints overlap 2 inches. Do not compress insulation to a thickness less than that specified. Carry insulation over standing seams and trapeze-type duct hangers.
- e. Where mechanical fasteners are used, install self-locking washers and trim and bend the pin over.
- f. Secure jacket overlaps with staples and tape as necessary to ensure a secure seal. Coat staples, tape and seams with a brush coat of vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate (minimum 2 mils adhesive, 3 mils embossed) - less than 0.0000 perm adhesive tape.
- g. Cover breaks in the jacket material with patches of the same material as the vapor retarder jacket. Do not extend patches less than 2 inches beyond the break or penetration in all directions and secure with tape and staples. Seal staples and tape joints with a brush coat of vapor retarder coating or PVDC adhesive tape or greater than 3 ply laminate (minimum 2 mils adhesive, 3 mils embossed) - less than 0.0000 perm adhesive tape.

- h. At jacket penetrations such as hangers, thermometers, and damper operating rods, fill voids in the insulation and seal the penetration with a brush coat of vapor retarder coating or PVDC adhesive tape greater than 3 ply laminate (minimum 2 mils adhesive, 3 mils embossed) - less than 0.0000 perm adhesive tape.
- i. Seal insulation terminations and pin punctures and flash with a reinforced vapor retarder coating finish or tape with a brush coat of vapor retarder coating.. Ensure the coating overlaps the adjoining insulation and un-insulated surface 2 inches. Extend pin puncture coatings 2 inches from the puncture in all directions.
- j. Where insulation standoff brackets occur, extend insulation under the bracket and terminate the jacket at the bracket.

#### 3.3.2.2 Installation on Exposed Duct Work

- a. For rectangular ducts, secure rigid insulation to the duct by mechanical fasteners on all four sides of the duct, space no more than 12 inches apart and no more than 3 inches from the edges of the insulation joints. Provide a minimum of two rows of fasteners for each side of duct 12 inches and larger. Provide one row for each side of duct less than 12 inches. Provide mechanical fasteners that are corrosion resistant as G60 coated galvanized steel, and indefinitely sustain a 50 lb tensile dead load test perpendicular to the duct wall.
- b. Form duct insulation with minimum jacket seams. Fasten each piece of rigid insulation to the duct using mechanical fasteners. When the height of projections is less than the insulation thickness, bring insulation up to standing seams, reinforcing, and other vertical projections and do not carry over. Continue vapor retarder/barrier jacket across seams, reinforcing, and projections. When height of projections is greater than the insulation thickness, carry over insulation and jacket . Apply insulation with joints tightly butted. Neatly bevel insulation around name plates and access plates and doors.
- c. Impale insulation on the fasteners; install self-locking washers and trim and bend the pin over.
- d. Seal joints in the insulation jacket with a 4 inch wide strip of tape. Seal taped seams with a brush coat of vapor retarder coating.
- e. Cover breaks and ribs or standing seam penetrations in the jacket material with a patch of the same material as the jacket. Do not extend patches less than 2 inches beyond the break or penetration and secure with tape and staple. Seal staples and joints with a brush coat of vapor retarder coating.
- f. At jacket penetrations such as hangers, thermometers, and damper operating rods, fill the voids in the insulation and seal the penetrations with a flashing sealant.
- g. Seal and flash insulation terminations and pin punctures with a reinforced vapor retarder coating finish. Ensure coating overlaps the adjoining insulation and un-insulated surface 2 inches. Extend pin puncture coatings 2 inches from the puncture in all directions.
- h. Insulation for both concealed or exposed round or oval ducts must be

flexible type, minimum density 3/4 pcf or a semi rigid board, minimum density 3 pcf, formed or fabricated to a tight fit, edges beveled and joints tightly butted and staggered. Insulation for all exposed ducts must be provided with either a white, factory-applied Type I jacket or a field applied vapor retarder or vapor barrier jacket coating finish as specified, the total field applied dry film thickness must be approximately 1/16 inch.

- i. Insulate oval and round ducts, flexible type, with factory Type I jacket insulation with minimum density of 3/4 pcf, attach in accordance with MICA standards.

### 3.3.3 Insulation for Warm Air Duct

Provide insulation and vapor barrier for the following warm air ducts and associated equipment:.

- a. Supply ducts.
- b. Return air ducts.
- c. Relief air ducts
- d. Flexible run-outs (field insulated).
- e. Plenums.
- f. Duct-mounted coil casings.
- g. Coil-headers and return bends.
- h. Coil casings.
- i. Fresh air intake ducts.
- j. Filter boxes.
- k. Mixing boxes.
- l. Supply fans.
- m. Site-erected air conditioner casings.
- n. Ducts exposed to weather.
- o. Exhaust ducts passing through concealed spaces, unconditioned spaces, or semi-heated spaces exhausting conditioned air.

Provide insulation for rectangular ducts that is flexible type where concealed, and rigid type where exposed. Provide insulation on exposed ducts with a white, paint-able, factory-applied Type II jacket, or finish with adhesive finish. Use flexible type insulation for round ducts, with a factory-applied Type II jacket. Provide insulation on concealed duct with a factory-applied Type II jacket. Accomplish adhesive finish where indicated to be used by applying two coats of adhesive with a layer of glass cloth embedded between the coats. Ensure total dry film thickness is approximately 1/16 inch. Continue duct insulation through sleeves and prepare openings. Terminate duct insulation at fire dampers and flexible connections.



#### 3.3.3.1 Installation on Concealed Duct

- a. For rectangular, oval and round ducts, attach insulation by applying adhesive around the entire perimeter of the duct in 6 inch wide strips on 12 inch centers.
- b. For rectangular and oval ducts 24 inches and larger, secure insulation to the bottom of ducts using mechanical fasteners. Space fasteners on 18 inch centers and no more than 18 inches from duct corner.
- c. For rectangular, oval and round ducts, provide mechanical fasteners on sides of duct risers for all duct sizes. Space fasteners on 18 inch centers and no more than 18 inches from duct corners.
- d. Impale insulation on the mechanical fasteners where used. Do not compress insulation to a thickness less than that specified. Carry insulation over standing seams and trapeze-type hangers.
- e. Install self-locking washers where mechanical fasteners are used and trim and bend the pin over.
- f. Do not overlap insulation jacket less than 2 inches at joints and secure the lap and staple on 4 inch centers.

#### 3.3.3.2 Installation on Exposed Duct

- a. For rectangular ducts, secure the rigid insulation to the duct using mechanical fasteners on all four sides of the duct, space no more than 16 inches apart and no more than 6 inches from the edges of the insulation joints. Provide a minimum of two rows of fasteners for each side of duct 12 inches and larger and a minimum of one row for each side of duct less than 12 inches.
- b. Form duct insulation with factory-applied jacket with minimum jacket seams, and fasten each piece of rigid insulation to the duct using mechanical fasteners. When the height of projection is less than the insulation thickness, bring insulation up to standing seams, reinforcing, and other vertical projections and do not carry over the projection. Continue jacket across seams, reinforcing, and projections. Where the height of projections is greater than the insulation thickness, carry insulation and jacket over the projection.
- c. Impale insulation on the fasteners; install self-locking washers and trim and bend the pin over.
- d. Seal joints on jacketed insulation with a 4 inch wide strip of tape and brush with vapor retarder coating.
- e. Cover breaks and penetrations in the jacket material with a patch of the same material as the jacket. Extend patches no less than 2 inches beyond the break or penetration and secure with adhesive and staple.
- f. Seal insulation terminations and pin punctures with tape and brush with vapor retarder coating.
- g. Insulate oval and round ducts, flexible type, with factory Type I jacket insulation, minimum density of 3/4 pcf attach by staples spaced no more than 16 inches and no more than 6 inches from the degrees of

joints. Seal joints in accordance with item "d." above.

#### 3.3.4 Duct Test Holes

After duct systems have been tested, adjusted, and balanced, repair breaks in the insulation and jacket in accordance with the applicable section of this specification for the type of duct insulation to be repaired. After the insulation is repaired, identify on the outside of the insulation with marker, sticker, or other means "TAB test ports."

#### 3.3.5 Duct Exposed to Weather

##### 3.3.5.1 Installation

Insulate and finish ducts exposed to weather as specified for the applicable service for exposed duct inside the building. After the above is accomplished, further finish the insulation as detailed in the following subparagraphs.

##### 3.3.5.2 Round Duct

Laminated self-adhesive (minimum 2 mils adhesive, 3 mils embossed) vapor barrier/weatherproofing jacket - Less than 0.02 permeability, (greater than 3 ply, standard grade, silver, white, black and embossed or greater than 8 ply, heavy duty, white and natural) membrane must be applied overlapping material by 3 inches no bands or caulking needed - see manufacturer's recommended installation instructions. Aluminum jacket with factory applied moisture retarder must be applied with the joints lapped no less than 3 inches and secured with bands located at circumferential laps and at no more than 12 inch intervals throughout. Lap horizontal joints down to shed water and located at 4 or 8 o'clock position. Seal joints with metal jacketing sealant to prevent moisture penetration. Where jacketing abuts an un-insulated surface, seal joints with metal jacketing sealant.

##### 3.3.5.3 Fittings

Finish fittings and other irregular shapes as specified for rectangular ducts.

##### 3.3.5.4 Rectangular Ducts

Apply two coats of weather barrier mastic reinforced with fabric or mesh for outdoor application to the entire surface. Ensure each coat of weatherproof mastic has a minimum thickness of 1/16 inch. Ensure exterior is a metal jacketing applied for mechanical abuse and weather protection, and secure with screws or vapor barrier/weatherproofing jacket less than 0.02 permeability greater than 3 ply, standard grade, silver, white, black, and embossed or greater than 8 ply, heavy duty white and natural. Apply membrane overlapping material by 3 inches. No bands or caulking needed-see manufacturing recommend installation instructions.

#### 3.4 EQUIPMENT INSULATION SYSTEMS INSTALLATION

##### 3.4.1 General

Provide removable insulation sections to cover parts of equipment that must be opened periodically for maintenance including vessel covers, fasteners, flanges and accessories. Omit equipment insulation on the

following:

- a. Hand-holes.
- b. Boiler manholes.
- c. Cleanouts.
- d. ASME stamps.
- e. Manufacturer's nameplates.
- f. Duct Test/Balance Test Holes.

#### 3.4.2 Insulation for Cold Equipment

Cold equipment below 60 degrees F: Furnish insulation on equipment handling media below 60 degrees F including the following:

- a. Pumps.
- b. Refrigeration equipment parts that are not factory insulated.
- c. Drip pans under chilled equipment.
- d. Cold water storage tanks.
- e. Water softeners.
- f. Duct mounted coils.
- g. Cold and chilled water pumps.
- h. Pneumatic water tanks.
- i. Roof drain bodies.
- j. Air handling equipment parts that are not factory insulated.
- k. Expansion and air separation tanks.

##### 3.4.2.1 Insulation Type

Provide insulation suitable for the temperature encountered. Provide material and thicknesses as shown in Table 5:

| TABLE 5                                           |          |                    |
|---------------------------------------------------|----------|--------------------|
| Insulation Thickness for Cold Equipment (inches)  |          |                    |
| Equipment handling media at indicated temperature |          |                    |
|                                                   | Material | Thickness (inches) |
| 35 to 60 degrees F                                |          |                    |

| TABLE 5                                           |                               |                    |
|---------------------------------------------------|-------------------------------|--------------------|
| Insulation Thickness for Cold Equipment (inches)  |                               |                    |
| Equipment handling media at indicated temperature |                               |                    |
|                                                   | Material                      | Thickness (inches) |
|                                                   | Cellular Glass                | 1.5                |
|                                                   | Flexible Elastomeric Cellular | 1                  |
| 1 to 34 degrees F                                 |                               |                    |
|                                                   | Cellular Glass                | 3                  |
|                                                   | Flexible Elastomeric Cellular | 1.5                |
| Minus 30 to 0 degrees F                           |                               |                    |
|                                                   | Cellular Glass                | 3.5                |
|                                                   | Flexible Elastomeric Cellular | 1.75               |

#### 3.4.2.2 Pump Insulation

- a. Insulate pumps by forming a box around the pump housing. Construct the box by forming the bottom and sides using joints that do not leave raw ends of insulation exposed. Join joints between sides and between sides and bottom by adhesive with lap strips for rigid mineral fiber and contact adhesive for flexible elastomeric cellular insulation. Ensure box conforms to the requirements of MICA Insulation Stds plate No. 49 when using flexible elastomeric cellular insulation. Ensure joints between top cover and sides fit tightly forming a female shiplap joint on the side pieces and a male joint on the top cover, thus making the top cover removable.
- b. Protect exposed insulation corners with corner angles.
- c. Upon completion of installation of the insulation, including removable sections, apply two coats of vapor retarder coating with a layer of glass cloth embedded between the coats. The total dry thickness of the finish must be 1/16 inch. Provide a parting line between the box and the removable sections allowing the removable sections to be removed without disturbing the insulation coating. Apply flashing sealant to parting line, between equipment and removable section insulation, and at all penetrations.

#### 3.4.2.3 Other Equipment

- a. Form or fabricate insulation to fit the equipment. To ensure a tight fit on round equipment, bevel edges and tightly butt and stagger joints.
- b. Secure insulation in place with bands or wires at intervals as recommended by the manufacturer but no more than 12 inch centers except adhere flexible elastomeric cellular with contact adhesive.

Protect insulation corners under wires and bands with suitable corner angles.

- c. Install cellular glass in accordance with manufacturer's instructions. Seal joints and ends with joint sealant, and seal with a vapor retarder coating.
- d. Use removable insulation on heads of heat exchangers. Fabricate removable section joints using a male-female shiplap type joint. Finish the entire surface of the removable section by applying two coats of vapor retarder coating with a layer of glass cloth embedded between the coats. The total dry thickness of the finish must be 1/16 inch.
- e. Protect exposed insulation corners with corner angles.
- f. Apply insulation on equipment with ribs over 6 by 6 inches by 12 gauge welded wire fabric which has been cinched in place, or if approved by the Contracting Officer, spot weld to the equipment over the ribs. Secure insulation to the fabric with J-hooks and 2 by 2 inches washers or securely band or wire in place on 12 inch centers.

#### 3.4.2.4 Vapor Retarder/Vapor Barrier

Upon completion of installation of insulation, caulk penetrations. Apply two coats of vapor retarder coating or vapor barrier jacket over insulation, including removable sections, with a layer of open mesh synthetic fabric embedded between the coats. Ensure the total dry thickness of the finish is 1/16 inch. Apply flasing sealant or vapor barrier tape to parting line between equipment and removable section insulation.

#### 3.4.3 Insulation for Hot Equipment

Furnish insulation on equipment handling media above 60 degrees F including the following:

- a. Converters.
- b. Heat exchangers.
- c. Hot water generators.
- d. Water heaters.
- e. Pumps handling media above 130 degrees F.
- f. Fuel oil heaters.
- g. Hot water storage tanks.
- h. Air separation tanks.
- i. Surge tanks.
- j. Flash tanks.
- k. Feed-water heaters.

- l. Unjacketed boilers or parts of boilers.
- m. Boiler flue gas connection from boiler to stack (if inside).
- n. Induced draft fans.
- o. Fly ash and soot collectors.
- p. Condensate receivers.

#### 3.4.3.1 Insulation

Provide insulation suitable for the temperature encountered. Insulate shell and tube-type heat exchangers for the temperature of the shell medium.

Determine insulation thickness for hot equipment using Table 6:

| TABLE 6                                                                      |                                        |                    |
|------------------------------------------------------------------------------|----------------------------------------|--------------------|
| Insulation Thickness for Hot Equipment (inches)                              |                                        |                    |
| Equipment handling steam or media at indicated pressure or temperature limit |                                        |                    |
|                                                                              | Material                               | Thickness (inches) |
| 15 psig or 250 degrees F                                                     |                                        |                    |
|                                                                              | Rigid Mineral Fiber                    | 2                  |
|                                                                              | Flexible Mineral Fiber                 | 2                  |
|                                                                              | Calcium Silicate/Perlite               | 4                  |
|                                                                              | Cellular Glass                         | 3                  |
|                                                                              | Faced Phenolic Foam                    | 1.5                |
|                                                                              | Flexible Elastomeric Cellular (<200 F) | 1                  |
| 200psig or 400 degrees F                                                     |                                        |                    |
|                                                                              | Rigid Mineral Fiber                    | 3                  |
|                                                                              | Flexible Mineral Fiber                 | 3                  |
|                                                                              | Calcium Silicate/Perlite               | 4                  |
|                                                                              | Cellular Glass                         | 4                  |
| 600 degrees F                                                                |                                        |                    |
|                                                                              | Rigid Mineral Fiber                    | 5                  |
|                                                                              | Flexible Mineral Fiber                 | 6                  |
|                                                                              | Calcium Silicate/Perlite               | 6                  |

| TABLE 6                                                                                                                                                                                |                |                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------------------|
| Insulation Thickness for Hot Equipment (inches)                                                                                                                                        |                |                    |
| Equipment handling steam or media at indicated pressure or temperature limit                                                                                                           |                |                    |
|                                                                                                                                                                                        | Material       | Thickness (inches) |
|                                                                                                                                                                                        | Cellular Glass | 6                  |
| 600 degrees F: Thickness necessary to limit the external temperature of the insulation to 120 F. Submit heat transfer calculations to substantiate insulation and thickness selection. |                |                    |

#### 3.4.3.2 Insulation of Pumps

Insulate pumps by forming a box around the pump housing. Construct the box by forming the bottom and sides using joints that do not leave raw ends of insulation exposed. Band bottom and sides to form a rigid housing that does not rest on the pump. Ensure joints between top cover and sides fit tightly. The top cover must have a joint forming a female shiplap joint on the side pieces and a male joint on the top cover, making the top cover removable. Apply two coats of Class I adhesive over insulation, including removable sections, with a layer of glass cloth embedded between the coats. Provide a parting line between the box and the removable sections allowing the removable sections to be removed without disturbing the insulation coating. The total dry thickness of the finish must be 1/16 inch. Apply caulking to parting line of the removable sections and penetrations.

#### 3.4.3.3 Other Equipment

- Form or fabricate insulation to fit the equipment. To ensure a tight fit on round equipment, bevel edges and tightly butt and stagger joints.
- Secure insulation in place with bands or wires at intervals as recommended by the manufacturer but no greater than 12 inch centers except adhere flexible elastomeric cellular. Protect insulation corners under wires and bands with suitable corner angles.
- On high vibration equipment, set cellular glass insulation in a coating of bedding compound as recommended by the manufacturer, and seal joints with bedding compound. Fill mineral fiber joints with finishing cement.
- Provide removable insulation on heads of heat exchangers. Fabricate the removable section joint using a male-female shiplap type joint. Finish the entire surface of the removable section as specified.
- Protect exposed insulation corners with corner angles.
- On equipment with ribs, such as boiler flue gas connection, draft fans, and fly ash or soot collectors, apply insulation over 6 by 6 inch by 12 gauge welded wire fabric which has been cinched in place, or if

approved by the Contracting Officer, spot weld to the equipment over the ribs. Secure insulation to the fabric with J-hooks and 2 by 2 inch washers or securely band or wire in place on 12 inch (maximum) centers.

- g. On equipment handling media above 600 degrees F, apply insulation in two or more layers with staggered joints.
- h. Upon completion of installation of insulation, caulk penetrations. Apply two coats of adhesive over insulation, including removable sections, with a layer of glass cloth embedded between the coats. The total dry thickness of the finish must be 1/16 inch. Apply caulking to parting line between equipment and removable section insulation.

#### 3.4.4 Equipment Exposed to Weather

##### 3.4.4.1 Installation

Insulate equipment exposed to weather and finish in accordance with the requirements for ducts exposed to weather in paragraph DUCT INSULATION INSTALLATION.

##### 3.4.4.2 Optional Panels

At the option of the Contractor, prefabricated metal insulation panels may be used in lieu of the insulation and finish previously specified. Thermal performance must be equal to or better than that specified for field applied insulation. Provide panels that are the standard catalog product of a manufacturer of metal insulation panels. Provide fastenings, flashing, and support system conforming to published recommendations of the manufacturer for weatherproof installation and that prevent moisture from entering the insulation. Design panels to accommodate thermal expansion and to support a 250 pound walking load without permanent deformation or permanent damage to the insulation. Exterior metal cover sheet must be aluminum and exposed fastenings must be stainless steel or aluminum.

-- End of Section --



SECTION 23 08 00

COMMISSIONING OF MECHANICAL SYSTEMS

05/23, CHG 1: 08/24

PART 1 GENERAL

Building Commissioning is a systematic, quality-focused process for enhancing the delivery of a project that focuses on verifying and documenting that all of the commissioned systems and assemblies are planned, designed, installed, tested, operated, and maintained to meet the project requirements. The purpose is to reduce the cost and performance risks associated with delivering facilities projects, and to increase value to owners, occupants, and users. Comply with specification Section 01 91 00.15 BUILDING COMMISSIONING.

1.1 SEQUENCING AND SCHEDULING

Complete functional performance testing prior to performance verification testing required by Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC. Complete the following prior to starting Functional Performance Tests of mechanical systems:

- a. All equipment and systems completed, cleaned, flushed, disinfected, calibrated, tested, and operate in accordance with contract documents and construction plans and specifications
- b. Final DALT Report submitted and approved in accordance with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC
- c. Pre-final Testing, Adjusting, and Balancing Report submitted in accordance with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC
- e. The Certificate of Readiness submitted and approved in accordance with Section 01 91 00.15 BUILDING COMMISSIONING

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Test Equipment; G

SD-06 Test Reports

Pipe Flushing, Testing, And Water Treatment Reports; G

Completed Pre-Functional Checklists; G

Seasonal Test Report; G

Full-Load Test Report; G

Post-Construction Trend Log Report; G

SD-07 Certificates

Certificate Of Readiness; G

1.3 ACCESSIBILITY REQUIREMENTS

Equipment, systems, and devices for commissioned systems must be accessible. Make necessary modifications if systems and devices are not accessible for inspections and testing.

Assist commissioning team in testing by removing equipment covers, opening access panels, and other required activities that assist with visual oversight. Furnish ladders, flashlights, meters, gauges, or other inspection equipment as necessary.

1.4 COORDINATION

Refer to Section 01 91 00.15 BUILDING COMMISSIONING for requirements pertaining to coordination during the commissioning process. Coordinate with the Commissioning Provider in accordance with Section 01 91 00.15 and in accordance with the Commissioning Plan to schedule inspections as required to support the commissioning process. Furnish additional information requested by the Commissioning Provider. Coordinate scheduling of Functional Performance Testing with the commissioning team. Provide plans, reports, notes, and other documentation to the Commissioning Provider as specified in the commissioning plan, as it is completed.

1.5 PIPE FLUSHING, TESTING, AND WATER TREATMENT REPORTS

Test requirements are specified in Division 22 and 23 piping Sections. Prepare a pipe system cleaning, flushing, and hydrostatic testing log. Provide cleaning, flushing, testing, and water treatment log and final reports.

Include the following in the pipe system cleaning, flushing, and hydrostatic testing log:

- a. Minimum flushing water velocity.
- b. Water treatment reports.
- c. Tracking checklist for managing and ensuring that all pipe sections have been cleaned, flushed, hydrostatically tested, and chemically treated.

1.6 CERTIFICATE OF READINESS

Submit Certificate of Readiness documentation in accordance with Section 01 91 00.15 BUILDING COMMISSIONING for all equipment and systems including start-up reports; completed Pre-Functional Checklists; Testing, Adjusting, and Balancing (TAB) Report; Issues Log; HVAC Controls Start-Up Reports. Do not schedule Functional Performance Tests for the system until the Certificate of Readiness for that system receives approval by the Government. The CxC, Quality Control Manager, and the Mechanical, Electrical, Controls, and TAB subcontractor representatives must sign and date the Certificate of Readiness.

## PART 2 PRODUCTS

### 2.1 TEST EQUIPMENT

Provide all testing equipment required to perform testing for the systems to be commissioned, except for equipment specific to and used by TAB as required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC. Provide a sufficient quantity of two-way radios for each subcontractor. Submit list of Test Equipment and instrumentation to be used for testing including equipment/instrument identification number, equipment application or planned use, manufacturer, make, model, and serial number, and calibration history with certificates. Also list special equipment and proprietary tools specific to a piece of equipment required for testing.

#### 2.1.1 Proprietary Equipment

Provide manufacturer's proprietary test equipment and software required by any equipment manufacturer for programming and start-up, whether specified or not. Provide manufacturer test equipment, demonstrate its use, and assist in the commissioning process as needed. Provide data logging equipment and software required to test equipment.

#### 2.1.2 Calibration and Accuracy

Comply with equipment manufacturer's test equipment calibration procedures and intervals. Recalibrate test instruments immediately after instruments have been repaired resulting from being dropped or damaged. Affix calibration tags to test instruments. Furnish calibration records to Contracting Officer upon request.

Provide all testing equipment of sufficient quality and accuracy to test and measure system performance with the tolerances specified. Unless otherwise noted, the following minimum requirements apply: Provide temperature sensors and digital thermometers with a certified calibration within the past year to an accuracy of 0.5 degrees F and a resolution of plus or minus 0.1 degrees F. Provide pressure sensors with an accuracy of plus or minus 2.0 percent of the value range being measured (not full range of meter) and calibrated within the last year.

## PART 3 EXECUTION

### 3.1 MEETINGS

Attend all meetings in accordance with Section 01 91 00.15 BUILDING COMMISSIONING.

Provide timely updates on construction schedule changes to allow the Commissioning Provider to execute commissioning process efficiently. Notify Contracting Officer of anticipated construction delays to commissioning activities not yet performed or not yet scheduled.

### 3.2 COMMISSIONING CONSTRUCTION OBSERVATION CHECKLISTS

Commissioning construction observation checklists include Pre-Functional Checklists and Functional Performance Test Checklists. Provide commissioning construction observation checklists for the Interim and Final Construction Phase Commissioning Plan in accordance with Section

01 91 00.15 BUILDING COMMISSIONING.

### 3.2.1 Pre-Functional Checklists

The Pre-Functional Checklists must include items for physical inspection or testing that demonstrate that installation and start-up of equipment and systems is complete. Refer to paragraph PRE-FUNCTIONAL CHECKS. Pre-functional checklists must be tailored to verify the specific installation requirements and details of the construction documents and manufacturer's instructions.

Use the Pre-Functional Checklists prepared by the CxC for physical inspection or testing to demonstrate that installation and start-up of equipment and systems is complete. Refer to paragraph PRE-FUNCTIONAL CHECKS.

### 3.2.2 Functional Performance Test Checklists

Functional Performance Test Checklists must include procedures that explain, step-by-step, the actions and expected results that will demonstrate that the system performs in accordance with the contract. Refer to paragraph FUNCTIONAL PERFORMANCE TESTING. Include the following sections and details appropriate to the systems being tested in the Functional Performance Test Checklists:

- a. Notable system features including information about controls to facilitate understanding of system operation and maintenance
- b. Conclusions and recommendations. Conclusions must clearly indicate if system does or does not perform in accordance with contract requirements and specifications. Recommendation must clearly indicate that the system should or should not be approved by the Government.
- c. Test conditions including date, beginning and ending time, and beginning and ending outdoor air conditions
- d. Attendees
- e. Identification of the equipment involved in the test
- f. **System feature identification**
- g. Point-to-point observations including demonstrating system flow meters and sensors have been calibrated and are correctly displayed on the Operator work station
- h. Actuator operation observations demonstrating actuator responses to commands from the control system
- i. As-found condition of the system operation
- j. List of test items with step numbers along with the corresponding feature or control operation, intended test procedure, expected system response, and pass/fail indication.
- k. Space for comments for each test item.

Use the Functional Performance Test Checklists prepared by the CxC that list, step-by-step, the actions and expected results that will demonstrate that the system performs in accordance with the contract. Refer to paragraph FUNCTIONAL PERFORMANCE TESTING.

### 3.3 PRE-FUNCTIONAL CHECKS

Pre-Functional Checks are a type of Commissioning Inspection in accordance with Section 01 91 00.15 BUILDING COMMISSIONING. Complete one Pre-Functional Checklist for each individual item of equipment or system for each system required to be commissioned including, but not limited to, ductwork, piping, equipment, fixtures, and controls. Include manufacturer start-up checklists associated with equipment with the submission of the Pre-Functional Checklists. Provide manufacturer's installation manual for each type of unit. Indicate commissioning team member inspection and validation of each Pre-Functional Checklist item by initials. Validation of each Pre-Functional Checklist item by each team member indicates that item conforms to the contract documents and validated design in their area of responsibility. Commissioning Specialist validation of each Pre-Functional Checklist item indicates that each item has been installed correctly and in accordance with contract documents. Required commissioning team members for Pre-Functional Checks includes the CxC, Mechanical Commissioning Specialist, Quality Control Manager, sub-contractor representative for each trade responsible for construction/installation of the checklist item, and the TAB representative (for items impacting TAB). Submit the initialed and Completed Pre-Functional Checklists no later than 7 calendar days after completion of inspection of all checklist items for each system.

### 3.4 STARTUP AND INITIAL CHECKOUT

Document start-up and initial testing procedures, and include in the Completed Pre-Functional Checklists submittal, including:

- a. Startup tests and factory testing reports.
- b. Manufacturer's representative start-up, operating, troubleshooting and maintenance procedures.
- d. Perform and clearly document system operational checks and quality control checks as they are completed, and providing a copy to the commissioning team.
- e. Correct deficiencies and sign the Certificate of Readiness for each system before functional performance testing

### 3.5 TESTING, ADJUSTING, AND BALANCING REPORT REVIEW AND VERIFICATION

#### 3.5.1 Testing, Adjusting, and Balancing (TAB) Report Review

The Commissioning Specialist is responsible for reviewing the pre-final TAB Report required by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identifying any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel. All deficiencies must be resolved prior to TAB Report approval.

#### 3.5.2 Testing, Adjusting, and Balancing (TAB) Report Verification

The Commissioning Specialist is responsible for witnessing the TAB Field

Acceptance Testing specified by Section 23 05 93 TESTING, ADJUSTING, AND BALANCING FOR HVAC and identifying any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel. All deficiencies must be resolved prior to TAB Report approval.

### 3.6 HVAC CONTROLS START-UP AND PERFORMANCE VERIFICATION TEST REVIEW

The Commissioning Specialist is responsible for reviewing the Start-Up Testing Report, PVT Procedures and PVT Reports including endurance testing trend data required by Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC and identifying any deficiencies to the Contracting Officer's Representative and the Contractor's Quality Control Personnel. All deficiencies must be resolved prior to final acceptance.

### 3.7 FUNCTIONAL PERFORMANCE TESTING

Functional Performance Testing are a type of Commissioning Testing in accordance with Section 01 91 00.15 BUILDING COMMISSIONING. Conduct Functional Performance Testing in accordance with Section 01 91 00.15 BUILDING COMMISSIONING and requirements in this Section. Prior to Functional Performance Testing, complete all prerequisites in accordance with paragraph SEQUENCING AND SCHEDULING. Demonstrate that all system components have been installed, that each control device and item of equipment operates, and that the systems operate and perform, including interactive operation between systems, in accordance with contract documents.

#### 3.7.1 Test Scheduling and Coordination

Schedule and conduct Initial Functional Performance Tests as soon as all contract work is completed, regardless of the season. Develop and implement means of artificial loading to demonstrate, to a reasonable level of confidence, the ability of the HVAC systems to handle peak seasonal loads. Schedule Functional Performance Tests for each system only after the Certificate of Readiness has been approved by the Government for the system. Correct all deficiencies identified through any prior review, inspection, or test activity before the start of Functional Performance Tests.

Functional Performance Tests must be performed with the CxC present. Government reserves the right to witness all tests. Coordinate test schedule with Government representatives.

#### 3.7.2 Preparation

Put equipment and systems into operation and continue operation during each working day of functional performance testing, as required. Verify temperature and pressure taps in accordance with Contract Documents. Provide a pressure/temperature plug at each water sensor which is an input point to control system.

Perform minor adjustments to equipment and systems during Functional Performance Tests as deemed necessary by the commissioning team. Where calibrated DDC sensors cannot be used to record test data, provide measuring instruments, logging devices, and data acquisition equipment to record data for the complete range of test data for the required test period.

### 3.7.3 Testing Procedures

Provide all necessary materials and system modifications to produce the necessary flows, pressures, temperatures, and other conditions necessary to execute the test according to the specified conditions. At completion of the test, return the affected building equipment and systems to their pre-test condition.

Follow the Functional Performance Test from the approved Final Construction Phase Commissioning Plan. Perform Functional Performance Tests for each item of equipment and each system required to be commissioned. Verify all sensor calibrations, control responses, safeties, interlocks, operating modes, sequences of operation, capacities, and all other performance requirements comply with contract, regardless of the specific items listed within the checklists provided. In general, testing must progress from equipment or components to subsystems to systems to interlocks and connections between systems. Commissioning Specialists are responsible for determining the order of components and systems to be tested.

Indicate validation of each item of equipment and systems tested by signature of each commissioning team member for each test. The Quality Control Representative, Commissioning Specialists, and Contracting Officer's Representative, if present, must indicate validation after the equipment and systems are free of deficiencies.

### 3.7.4 Simulating Conditions

Functional performance testing is conducted by simulating conditions at control devices to initiate a control system response. Before testing, calibrate all sensors, transducers and devices. Over-writing control input values through the control system is not acceptable unless approved by the Contracting Officer. Perform each test under conditions that simulate actual conditions as close as is practically possible. Specific examples of simulating conditions are provided below. Do not simulate conditions when damage to the system or building may result.

- a. When varying static pressures inside ductwork cannot be simulated within the duct, and where a sensor signals the controls system to initiate sequences at various duct static pressures, it is acceptable to simulate the various pressures with a Pneumatic Squeeze-Bulb Type Signaling Device with gauge temporarily attached to the sensing tube leading to the transmitter. It is not acceptable to reset the various set-points, nor to simulate an electric analog signal (unless approved as noted above).
- b. Dirty filter pressure drops can be simulated by partially blocking filter face.
- c. Freeze-stat safeties can be simulated by packing portion of sensor with ice.
- d. High outside air temperatures can be simulated with a hair blower.
- e. Raising entering cooling coil temperatures by activating a heating/preheat coil can be used to simulate entering cooling coil conditions.
- f. Do not use signal generators to simulate sensor signals unless

approved by the Contracting Officer, as noted above, for special cases.

- g. Control set points can be altered. For example, to see the air conditioning compressor lockout work at an outside air temperature below 55 degrees F, when the outside air temperature is above 55 degrees F, temporarily change the lockout set point to be 0 degrees F above the current outside air temperature. Caution: Set points are not to be raised or lowered to a point to cause damage to the components, systems, or the building structure and contents.
- h. Test duct mounted smoke detectors in accordance with the manufacturer's recommendations. Perform the tests with air system at minimum airflow condition.
- i. Test current sensing relays used for fan and pump status signals to control system to indicate unit failure and run status by resetting the set point on the relay to simulate a lost belt or unit failure while the unit is running. Confirm that the failure alarm was generated and received at the control system. After the test is conducted, return the set point to its original set-point or a set-point as indicated by the Contracting Officer.

#### 3.7.5 Manufacturer's Representative

Provide a factory trained representative authorized by the equipment manufacturer to perform Functional Performance Testing for the following equipment:

Air Conditioners

Booster Pumps

Ensure the test representative reviews, approves, and signs the completed field test report. Include person's name with signatures.

#### 3.7.6 Sample Strategy

Perform Functional Performance Tests using the sample strategy identified herein. Test all central plant equipment, primary air handling units, and process cooling or heating equipment. Test all system-level equipment serving multiple zones. Twenty-five percent sample testing is allowed for large groups of identical equipment with identical controllers serving single zones such as air terminal units, fan coil units, unitary equipment, and plumbing fixtures. Sample size may be no less than three units. Complete a Functional Performance Test Checklist for each item of equipment or system to be tested.

##### 3.7.6.1 100 Percent Sample Procedures

Systems or equipment for which 100 percent sample size are tested fail if one or more of the test procedures results in discovery of a deficiency and the deficiency cannot be resolved within 5 minutes during the test.

Re-test to the extent necessary to confirm that the deficiencies have been corrected without negatively impacting the performance of the rest of the system.



#### 3.7.6.2 Less than 100 Percent Sample Procedures

Randomly test each sample group of identical equipment. If 10 percent of the units in the first sample fail the functional performance tests, test a second sample group, the same size as the first sample group. The second sample must not include any units from the first sample group.

If 10 percent of the units in the second sample fail, test all remaining units. If at any point frequent failures occur the CxC may stop the testing and require the contractor to perform and document a checkout of the remaining units prior to continuing functional testing.

#### 3.8 RETESTING REQUIREMENTS

Abort tests if any deficiency prevents successful completion of the test or if any required commissioning team member is not present for the test. Re-test only after all deficiencies identified during the original tests have been corrected.

Contracting Officer may withhold payment equivalent to lost time, re-testing, and aborted tests. These costs may include salary, travel costs, and per diem for Government team members. Correct deficiencies as identified by the commissioning team and retest the systems to be commissioned.

#### 3.9 SYSTEM ACCEPTANCE

Systems may be partially accepted prior to seasonal testing if they comply with all construction contract and accepted design requirements that can be tested during initial Functional Performance Tests. All test procedures must be successful completed prior to full systems acceptance.

#### 3.10 SEASONAL TESTS

Perform Initial Functional Performance Tests as soon as all contract work is completed, but prior to facility turnover, regardless of the season.

In addition to the Initial Functional Performance Tests, perform Functional Performance Tests of HVAC systems during peak heating and cooling seasons during outdoor air condition design extremes. Schedule Seasonal Functional Performance Tests in coordination with the Contracting Officer. Submit Seasonal Test Report within 14 days of test completion. Include seasonal test report in the Updated Final Commissioning Report as specified in Section 01 91 00.15 BUILDING COMMISSIONING

Execute seasonal functional performance testing, witnessed by the Contracting Officer. Correct deficiencies and make adjustments to O&M manuals and as-built drawings for applicable issues identified in any seasonal testing.

#### 3.11 FULL-LOAD TESTS

Perform Initial Functional Performance Tests as soon as all contract work is completed, but prior to facility turnover. In addition to the Initial Functional Performance Tests, perform Functional Performance Tests of HVAC systems under full-load conditions. Develop and implement means of artificial loading to demonstrate the ability of the process cooling systems to handle peak process loads. Schedule Full-Load Functional Performance Tests in coordination with the Contracting Officer. Submit

Full-Load Test Report within 14 days of test completion.

Execute full-load functional performance testing, witnessed by the Contracting Officer. Correct deficiencies and make adjustments to O&M manuals and as-built drawings for applicable issues identified in any full load testing.

### 3.12 TRAINING

Coordinate, schedule, and document all required training. At a minimum, include the following items in the training report for commissioned systems:

- a. Complete commissioning documentation
- b. Complete O&M data
- c. Complete Training
- d. Purpose of equipment.
- e. Principle of how the equipment works.
- f. Important parts and assemblies.
- g. How the equipment achieves its purpose and necessary operating conditions.
- h. Most likely failure modes, causes and corrections.
- i. On site demonstration.
- j. Provide updates to O&M manuals based on field modifications.
- k. Provide training of the post-occupancy operations and maintenance staff.

### 3.13 COMMISSIONING REPORT

Include all completed Pre-Functional Checklists and Functional Performance Checklists in the Commissioning Report as specified in Section 01 91 00.15 BUILDING COMMISSIONING. Include the approved TAB Report.

### 3.14 POST-CONSTRUCTION ENDURANCE TEST

Perform an Endurance Test in accordance with the paragraph ENDURANCE TESTING in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC once during the peak heating season and once during the peak cooling season during outdoor air condition extremes with the exception that network bandwidth usage measurement and recording is not required. Use the Temporary Trending Hardware, if necessary, in accordance with Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

The Mechanical System Commissioning Specialists must review the trend logs from the Endurance Tests to ensure that the systems have stable operation

and operate as required by the construction contract. The Commissioning Specialists must provide a Post-Construction Trend Log Report that identifies any deficiencies noted in operation, recommendations for correction, and includes a graphical representation of the trends. Provide one Trend Log Report for the peak cooling season and one Trend Log Report for the peak heating season. Submit the Post-Construction Trend Log Reports no later than 14 calendar days following receipt of the trend log data by the Commissioning Specialist. Provide a Post-Construction Trend Log Report meeting the same requirements as the Endurance Testing Results Format in Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC. Provide one Trend Log Report for the peak cooling season and one Trend Log Report for the peak heating season. Submit the Post-Construction Trend Log Reports no later than 14 calendar days following download of the trended data. Include the Post-Construction Trend Log Reports in the Updated Final Commissioning Report as specified in Section 01 91 00.15 BUILDING COMMISSIONING

-- End of Section --

SECTION 23 09 00

INSTRUMENTATION AND CONTROL FOR HVAC  
**08/24**

PART 1 GENERAL

1.1 SUMMARY

Provide a complete Direct Digital Control (DDC) system, suitable for the control of the heating, ventilating and air conditioning (HVAC) and other building-level systems as indicated on contract drawings.

1.1.1 System Requirements

Provide systems meeting the requirements this Section and other Sections referenced by this Section, and which have the following characteristics:

- a. The system implements the control sequences of operation shown in the Contract Drawings using DDC hardware to control mechanical and electrical equipment
- b. The system meet the requirements of this specification as a stand-alone system and does not require connection to any other system.
- c. Control sequences reside in DDC hardware in the building. The building control network is not dependent upon connection to a Utility Monitoring and Control System (UMCS) Front End or to any other system for performance of control sequences. To the greatest extent practical, the hardware performs control sequences without reliance on the building network.
- d. The hardware is installed such that individual control equipment can be replaced by similar control equipment from other equipment manufacturers with no loss of system functionality.
- e. All necessary documentation, configuration information, programming tools, programs, drivers, and other software are licensed to and otherwise remain with the Government such that the Government or their agents are able to perform repair, replacement, upgrades, and expansions of the system without subsequent or future dependence on the Contractor, Vendor or Manufacturer.
- f. Sufficient documentation and data, including rights to documentation and data, are provided such that the Government or their agents can execute work to perform repair, replacement, upgrades, and expansions of the system without subsequent or future dependence on the Contractor, Vendor or Manufacturer.
- g. Hardware is installed and configured such that the Government or their agents are able to perform repair, replacement, and upgrades of individual hardware without further interaction with the Contractor, Vendor or Manufacturer.

#### 1.1.2 End to End Accuracy

Select products, install and configure the system such that the maximum error of a measured value as read from the DDC Hardware over the network is less than the maximum allowable error specified for the sensor or instrumentation.

#### 1.1.3 Verification of Dimensions

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

#### 1.1.4 Drawings

The Government will not indicate all offsets, fittings, and accessories that may be required on the drawings. Carefully investigate the mechanical, electrical, and finish conditions that could affect the work to be performed, arrange such work accordingly, and provide all work necessary to meet such conditions.

### 1.2 RELATED SECTIONS

Related work specified elsewhere:

#### a. Section 01 91 00.15 BUILDING COMMISSIONING

### 1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 135 (2020; Interpretation 1-8 2021; Errata 1-2 2021; Addenda CD 2021; Addenda BV-CE 2022; Interpretation 9-12 2022; Interpretation 13-24 2023; Addenda BV-CF 2023; Errata 3 2023) BACnet—A Data Communication Protocol for Building Automation and Control Networks

ASHRAE FUN IP (2021) Fundamentals Handbook, I-P Edition

#### INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C62.41 (1991; R 1995) Recommended Practice on Surge Voltages in Low-Voltage AC Power Circuits

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250 (2020) Enclosures for Electrical Equipment  
(1000 Volts Maximum)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

NFPA 90A (2024) Standard for the Installation of  
Air Conditioning and Ventilating Systems

UNDERWRITERS LABORATORIES (UL)

UL 5085-3 (2006; Reprint Jan 2022) UL Standard for  
Safety Low Voltage Transformers - Part 3:  
Class 2 and Class 3 Transformers

#### 1.4 DEFINITIONS

The following list of definitions includes terms used in Sections referenced by this Section and are included here for completeness. The definitions contained in this Section may disagree with how terms are defined or used in other documents, including documents referenced by this Section. The definitions included here are the authoritative definitions for this Section and all Sections referenced by this Section.

After each term the protocol related to that term is included in parenthesis.

##### 1.4.1 Alarm Generation (All protocols)

Alarm Generation is the monitoring of a value, comparison of the value to alarm conditions and the creation of an alarm when the conditions set for the alarm are met. Note that this does NOT include delivery of the alarm to the final destination (such as a user interface).

##### 1.4.2 Building Automation and Control Network (BACnet) (BACnet)

The term BACnet is used in two ways. First meaning the BACnet Protocol Standard - the communication requirements as defined by ASHRAE 135 including all annexes and addenda. The second to refer to the overall technology related to the ASHRAE 135 protocol.

##### 1.4.3 BACnet Advanced Application Controller (B-AAC) (BACnet)

A hardware device BTL Listed as a B-AAC, which is required to support BACnet Interoperability Building Blocks (BIBBs) for scheduling and alarming, but is not required to support as many BIBBs as a B-BC.

##### 1.4.4 BACnet Application Specific Controller (B-ASC) (BACnet)

A hardware device BTL Listed as a B-ASC, with fewer BIBB requirements than a B-AAC. It is intended for use in a specific application.

##### 1.4.5 BACnet Building Controller (B-BC) (BACnet)

A hardware device BTL Listed as a B-BC. A general-purpose,

field-programmable device capable of carrying out a variety of building automation and control tasks including control and monitoring via direct digital control (DDC) of specific systems and data storage for trend information, time schedules, and alarm data. Like the other BTL Listed controller types (B-AAC, B-ASC etc.) a B-BC device is required to support the server ("B") side of the ReadProperty and WriteProperty services, but unlike the other controller types it is also required to support the client ("A") side of these services. Communication between controllers requires that one of them support the client side and the other support the server side, so a B-BC is often used when communication between controllers is needed.

#### 1.4.6 BACnet Broadcast Management Device (BBMD) (BACnet)

A communications device, typically combined with a BACnet router. A BBMD forwards BACnet broadcast messages to BACnet/IP devices and other BBMDs connected to the same BACnet/IP network. Each IP subnet that is part of a BACnet/IP network must have at least one BBMD. Note there are additional restrictions when multiple BBMDs share an IP subnet.

#### 1.4.7 BACnet/IP (BACnet)

An extension of BACnet, Annex J, defines the use of a reserved UDP socket to transmit BACnet messages over IP networks. A BACnet/IP network is a collection of one or more IP subnets that share the same BACnet network number. See also paragraph BACNET BROADCAST MANAGEMENT DEVICE.

#### 1.4.8 BACnet Internetwork (BACnet)

Two or more BACnet networks, connected with BACnet routers. In a BACnet Internetwork, there exists only one message path between devices.

#### 1.4.9 BACnet Interoperability Building Blocks (BIBBs) (BACnet)

A BIBB is a collection of one or more ASHRAE 135 Services intended to define a higher level of interoperability. BIBBs are combined to build the BACnet functional requirements for a device in a specification. Some BIBBs define additional requirements (beyond requiring support for specific services) in order to achieve a level of interoperability. For example, the BIBB DS-V-A (Data Sharing-View-A), which would typically be used by a front-end, not only requires the client to support the ReadProperty Service, but also provides a list of data types (Object / Properties) which the client must be able to interpret and display for the user.

In the BIBB shorthand notation, -A is the client side and -B is the server side.

|                                                                            |                                                  |
|----------------------------------------------------------------------------|--------------------------------------------------|
| The following is a list of some BIBBs used by this or referenced Sections: |                                                  |
| DS-COV-A                                                                   | Data Sharing-Change of Value (A side)            |
| DS-COV-B                                                                   | Data Sharing-Change of Value (B side)            |
| NM-RC-B                                                                    | Network Management-Router Configuration (B side) |

|                                                                            |                                                         |
|----------------------------------------------------------------------------|---------------------------------------------------------|
| The following is a list of some BIBBs used by this or referenced Sections: |                                                         |
| DS-RP-A                                                                    | Data Sharing-Read Property (A side)                     |
| DS-RP-B                                                                    | Data Sharing-Read Property (B side)                     |
| DS-RPM-A                                                                   | Data Sharing-Read Property Multiple (A Side)            |
| DS-RPM-B                                                                   | Data Sharing-Read Property Multiple (B Side)            |
| DS-WP-A                                                                    | Data Sharing-Write Property (A Side)                    |
| DM-TS-B                                                                    | Device Management-Time Synchronization (B Side)         |
| DM-UTC-B                                                                   | Device Management-UTC Time Synchronization (B Side)     |
| DS-WP-B                                                                    | Data Sharing-Write Property (B side)                    |
| SCHED-E-B                                                                  | Scheduling-External (B side)                            |
| DM-OCD-B                                                                   | Device Management-Object Creation and Deletion (B side) |
| AE-N-I-B                                                                   | Alarm and Event-Notification Internal (B Side)          |
| AE-N-E-B                                                                   | Alarm and Event-Notification External (B Side)          |
| T-VMT-I-B                                                                  | Trending-Viewing and Modifying Trends Internal (B Side) |
| T-VMT-E-B                                                                  | Trending-Viewing and Modifying Trends External (B Side) |

#### 1.4.10 BACnet Network (BACnet)

In BACnet, a portion of the control Internetwork consisting of one or more segments connected by repeaters. Networks are separated by routers.

#### 1.4.11 BACnet Operator Display (B-OD) (BACnet)

A basic operator interface with limited capabilities relative to a B-OWS. It is not intended to perform direct digital control. A B-OD profile could be used for LCD devices, displays affixed to BACnet devices, handheld terminals or other very simple user interfaces.

#### 1.4.12 BACnet Segment (BACnet)

One or more physical segments interconnected by repeaters (ASHRAE 135).

#### 1.4.13 BACnet Smart Actuator (B-SA) (BACnet)

A simple actuator device with limited resources intended for specific applications.

#### 1.4.14 BACnet Smart Sensor (B-SS) (BACnet)

A simple sensing device with limited resources.



#### 1.4.15 BACnet Testing Laboratories (BTL) (BACnet)

Established by BACnet International to support compliance testing and interoperability testing activities and consists of BTL Manager and the BTL Working Group (BTL-WG). BTL also publishes Implementation Guidelines.

#### 1.4.16 BACnet Testing Laboratories (BTL) Listed (BACnet)

A device that has been listed by BACnet Testing Laboratory. Devices may be certified to a specific device profile, in which case the listing indicates that the device supports the required capabilities for that profile, or may be listed as "other".

#### 1.4.17 Binary (All protocols)

A two-state system where an "ON" condition is represented by a high signal level and an "OFF" condition is represented by a low signal level. 'Digital' is sometimes used interchangeably with 'binary'.

#### 1.4.18 Broadcast (BACnet)

Unlike most messages, which are intended for a specific recipient device, a broadcast message is intended for all devices on the network.

#### 1.4.19 Building Control Network (BCN) (All protocols)

The network connecting all DDC Hardware within a building (or specific group of buildings).

#### 1.4.20 Building Point of Connection (BPOC) (All protocols)

A FPOC for a Building Control System. (This term is being phased out of use in preference for FPOC but is still used in some specifications and criteria. When it was used, it typically referred to a piece of control hardware. The current FPOC definition typically refers instead to IT hardware.)

#### 1.4.21 Commandable (All protocols)

See Overridable.

#### 1.4.22 Commandable Objects (BACnet)

Commandable Objects have a Commandable Property, Priority\_Array, and Relinquish\_Default Property as defined in ASHRAE 135, Clause 19.2, Command Prioritization.

#### 1.4.23 Configurable (All protocols)

A property, setting, or value is configurable if it can be changed via hardware settings on the device, via the use of engineering software or over the control network from the front end, and is retained through (after) loss of power.

- a. In a BACnet system, a property, setting, or value is configurable if it can be changed via one or more of:
  - (1) via BACnet services (including proprietary BACnet services)
  - (2) via hardware settings on the device

Note this is more stringent than the ASHRAE 135 definition.

#### 1.4.24 Control Logic Diagram (All protocols)

A graphical representation of control logic for multiple processes that make up a system.

#### 1.4.25 Device (BACnet)

A Digital Controller that contains a BACnet Device Object and uses BACnet to communicate with other devices.

#### 1.4.26 Device Object (BACnet)

Every BACnet device requires one Device Object, whose properties represent the network visible properties of that device. Every Device Object requires a unique Object Identifier number on the BACnet Internetwork. This number is often referred to as the device instance or device ID.

#### 1.4.27 Device Profile (BACnet)

A collection of BIBBs determining minimum BACnet capabilities of a device, defined in ASHRAE 135. Standard device profiles include BACnet Advanced Workstations (B-AWS), BACnet Building Controllers (B-BC), BACnet Advanced Application Controllers (B-AAC), BACnet Application Specific Controllers (B-ASC), BACnet Smart Actuator (B-SA), and BACnet Smart Sensor (B-SS).

#### 1.4.28 Digital Controller (All protocols)

An electronic controller, usually with internal programming logic and digital and analog input/output capability, which performs control functions.

#### 1.4.29 Direct Digital Control (DDC) (All protocols)

Digital controllers performing control logic. Usually the controller directly senses physical values, makes control decisions with internal programs, and outputs control signals to directly operate switches, valves, dampers, and motor controllers.

#### 1.4.30 Field Point of Connection (FPOC) (All protocols)

The FPOC is the point of connection between the UMCS IP Network and the field control network (either an IP network, a non-IP network, or a combination of both). The hardware at this location which provides the connection is generally an IT device such as a switch, IP router, or firewall.

In general, the term "FPOC Location" means the place where this connection occurs, and "FPOC Hardware" means the device that provides the connection. Sometimes the term "FPOC" is used to mean either and its actual meaning (i.e. location or hardware) is determined by the context in which it is used.

#### 1.4.31 Gateway (All protocols)

A device that translates from one protocol application data format to another. Devices that change only the transport mechanism of the protocol - "translating" from TP/FT-10 to Ethernet/IP or from BACnet MS/TP to

BACnet over IP for example - are not gateways as the underlying data format does not change. Gateways are also called Communications Bridges or Protocol Translators.

#### 1.4.32 IEEE 802.3 Ethernet (All protocols)

A family of local-area-network technologies providing high-speed networking features over various media, typically Cat 5, 5e or Cat 6 twisted pair copper or fiber optic cable.

#### 1.4.33 Internet Protocol (IP, TCP/IP, UDP/IP) (All protocols)

A communication method, the most common use is the World Wide Web. At the lowest level, it is based on Internet Protocol (IP), a method for conveying and routing packets of information over various LAN media. Two common protocols using IP are User Datagram Protocol (UDP) and Transmission Control Protocol (TCP). UDP conveys information to well-known "sockets" without confirmation of receipt. TCP establishes connections, also known as "sessions", which have end-to-end confirmation and guaranteed sequence of delivery.

#### 1.4.34 Input/Output (I/O) (All protocols)

Physical inputs and outputs to and from a device, although the term sometimes describes network or "virtual" inputs or outputs. See also "Points".

#### 1.4.35 I/O Expansion Unit (All protocols)

An I/O expansion unit provides additional point capacity to a digital controller

#### 1.4.36 IP subnet (All protocols)

A group of devices which share a defined range IP addresses. Devices on a common IP subnet can share data (including broadcasts) directly without the need for the traffic to traverse an IP router.

#### 1.4.37 Local-Area Network (LAN) (All protocols)

A communication network that spans a limited geographic area and uses the same basic communication technology throughout.

#### 1.4.38 Local Display Panels (LDPs) (All protocols)

A DDC Hardware with a display and navigation buttons, and must provide display and adjustment of points as shown on the Points Schedule and as indicated.

#### 1.4.39 MAC Address (All protocols)

Media Access Control address. The physical device address that identifies a device on a Local Area Network.

#### 1.4.40 Master-Slave/Token-Passing (MS/TP) (BACnet)

Data link protocol as defined by the BACnet standard. Multiple speeds

(data rates) are permitted by the BACnet MS/TP standard.

#### 1.4.41 Monitoring and Control (M&C) Software (All protocols)

The UMCS 'front end' software which performs supervisory functions such as alarm handling, scheduling and data logging and provides a user interface for monitoring the system and configuring these functions.

#### 1.4.42 Network Number (BACnet)

A site-specific number assigned to each network. This network number must be unique throughout the BACnet Internetwork.

#### 1.4.43 Object (BACnet)

An ASHRAE 135 Object. The concept of organizing BACnet information into standard components with various associated Properties. Examples include Analog Input objects and Binary Output objects.

#### 1.4.44 Object Identifier (BACnet)

A grouping of two Object properties: Object Type (e.g. Analog Value, Schedule, etc.) and Object Instance (in this case, a number). Object Identifiers must be unique within a device.

#### 1.4.45 Object Instance (BACnet)

See paragraph OBJECT IDENTIFIER

#### 1.4.46 Object Properties (BACnet)

Attributes of an object. Examples include present value and high limit properties of an analog input object. Properties are defined in ASHRAE 135; some are optional and some are required. Objects are controlled by reading from and writing to object properties.

#### 1.4.47 Operator Configurable (All protocols)

Operator configurable values are values that can be changed from a single common front end user interface across multiple vendor systems.

For non Niagara-based BACnet systems, a property, setting, or value in a device is Operator Configurable when it is Configurable and is either:

- a. a Writable Property of a Standard BACnet Object; or
- b. a Property of a Standard BACnet Object that is Writable when Out\_Of\_Service is TRUE and Out\_Of\_Service is Writable.

#### 1.4.48 Override (All protocols)

Changing the value of a point outside of the normal sequence of operation where the change has priority over the sequence and where there is a mechanism for releasing the change such that the point returns to the normal value. Overrides persist until released or overridden at the same or higher priority but are not required to persist through a loss of power. Overrides are often used by operators to change values, and generally originate at a user interface (workstation or local display panel).

#### 1.4.49 Packaged Equipment (All protocols)

Packaged equipment is a single piece of equipment provided by a manufacturer in a substantially complete and operable condition, where the controls (DDC Hardware) are factory installed, and the equipment is sold and shipped from the manufacturer as a single entity. Disassembly and reassembly of a large piece of equipment for shipping does not prevent it from being packaged equipment. Package units may require field installation of remote sensors. Packaged equipment is also called a "packaged unit".

Note industry may use the term "Packaged System" to mean a collection of equipment that is designed to work together where each piece of equipment is packaged equipment and there is a network that connects the equipment together. A "packaged system" of this type is NOT packaged equipment; it is a collection of packaged equipment, and each piece of equipment must individually meet specification requirements.

#### 1.4.50 Packaged Unit (All protocols)

See packaged equipment.

#### 1.4.51 Performance Verification Test (PVT) (All protocols)

The procedure for determining if the installed BAS meets design criteria prior to final acceptance. The PVT is performed after installation, testing, and balancing of mechanical systems. Typically the PVT is performed by the Contractor in the presence of the Government.

#### 1.4.52 Physical Segment (BACnet)

A single contiguous medium to which BACnet devices are attached (ASHRAE 135).

#### 1.4.53 Polling (All protocols)

A device periodically requesting data from another device.

#### 1.4.54 Points (All protocols)

Physical and virtual inputs and outputs. See also paragraph INPUT/OUTPUT (I/O).

#### 1.4.55 Proportional, Integral, and Derivative (PID) Control Loop (All protocols)

Three parameters used to control modulating equipment to maintain a setpoint. Derivative control is often not required for HVAC systems (leaving "PI" control).

#### 1.4.56 Proprietary (BACnet)

Within the context of BACnet, any extension of or addition to object types, properties, PrivateTransfer services, or enumerations specified in ASHRAE 135. Objects with Object\_Type values of 128 and above are Proprietary Objects. Properties with Property\_Identifier of 512 and above are proprietary Properties.

#### 1.4.57 Protocol Implementation Conformance Statement (PICS) (BACnet)

A document, created by the manufacturer of a device, which describes which portions of the BACnet standard may be implemented by a given device. ASHRAE 135 requires that all ASHRAE 135 devices have a PICS, and also defines a minimum set of information that must be in it. A device as installed for a specific project may not implement everything in its PICS.

#### 1.4.58 Repeater (All protocols)

A device that connects two control network segments and retransmits all information received on one side onto the other.

#### 1.4.59 Router (All protocols)

A device that connects two ASHRAE 135 networks and controls traffic between the two by retransmitting signals received from one side onto the other based on the signal destination. Routers are used to subdivide a BACnet internetwork and to limit network traffic.

#### 1.4.60 Segment (All protocols)

A 'single' section of a control network that contains no repeaters or routers. There is generally a limit on the number of devices on a segment, and this limit is dependent on the topology/media and device type.

#### 1.4.61 Standard BACnet Objects (BACnet)

Objects with Object\_Type values below 128 and specifically enumerated in Clause 21 of ASHRAE 135. Objects which are not proprietary. See paragraph PROPRIETARY.

#### 1.4.62 Standard BACnet Properties (BACnet)

Properties with Property\_Identifier values below 512 and specifically enumerated in Clause 21 of ASHRAE 135. Properties which are not proprietary. See Proprietary.

#### 1.4.63 Standard BACnet Services (BACnet)

ASHRAE 135 services other than ConfirmedPrivateTransfer or UnconfirmedPrivateTransfer. See paragraph PROPRIETARY.

#### 1.4.64 UMCS (All protocols)

UMCS stands for Utility Monitoring and Control System. The term refers to all components by which a project site monitors, manages, and controls real-time operation of HVAC and other building systems. These components include the UMCS "front-end" and all field building control systems connected to the front-end. The front-end consists of Monitoring and Control Software (user interface software), browser-based user interfaces and network infrastructure.

The network infrastructure (the "UMCS Network"), is an IP network connecting multiple building or facility control networks to the Monitoring and Control Software.

#### 1.4.65 UMCS Network (All protocols)

The UMCS Network connects multiple building or facility control networks to the Monitoring and Control Software.

#### 1.4.66 Writable Property (BACnet)

A Property is Writable when it can be changed through the use of one or more of the WriteProperty services defined in ASHRAE 135, Clause 15 regardless of the value of any other Property. Note that in the ASHRAE 135 standard, some Properties may be writable when the Out of Service Property is TRUE; for purposes of this Section, Properties that are only writable when the Out of Service Property is TRUE are not considered to be Writable.

### 1.5 PROJECT SEQUENCING

TABLE II: PROJECT SEQUENCING lists the sequencing of submittals as specified in paragraph SUBMITTALS (denoted by an 'S' in the 'TYPE' column) and activities as specified in PART 3 EXECUTION (denoted by an 'E' in the 'TYPE' column). TABLE II does not specify overall project milestone and completion dates; these dates are specified in the contract documents.

- a. Sequencing for Submittals: The sequencing specified for submittals is the deadline by which the submittal must be initially submitted to the Government. Following submission there will be a Government review period as specified in Section 01 33 00 SUBMITTAL PROCEDURES. If the submittal is not accepted by the Government, revise the submittal and resubmit it to the Government within 14 days of notification that the submittal has been rejected. Upon resubmittal there will be an additional Government review period. If the submittal is not accepted the process repeats until the submittal is accepted by the Government.
- b. Sequencing for Activities: The sequencing specified for activities indicates the earliest the activity may begin.
- c. Abbreviations: In TABLE II the abbreviation AAO is used for 'after approval of' and 'ACO' is used for 'after completion of'.

| TABLE II. PROJECT SEQUENCING |      |                                 |                                               |
|------------------------------|------|---------------------------------|-----------------------------------------------|
| ITEM #                       | TYPE | DESCRIPTION                     | SEQUENCING (START OF ACTIVITY OR DEADLINE FOR |
| 1                            | S    | Existing Conditions Report      |                                               |
| 2                            | S    | DDC Contractor Design Drawings  |                                               |
| 3                            | S    | Manufacturer's Product Data     |                                               |
| 4                            | S    | Pre-construction QC Checklist   |                                               |
| 5                            | E    | Install Building Control System | AAO #1 thru #4                                |
| 6                            | E    | Start-Up and Start-Up Testing   | ACO #5                                        |

| TABLE II. PROJECT SEQUENCING |      |                                                                      |                                                   |
|------------------------------|------|----------------------------------------------------------------------|---------------------------------------------------|
| ITEM #                       | TYPE | DESCRIPTION                                                          | SEQUENCING (START OF ACTIVITY OR DEADLINE FOR     |
| 7                            | S    | Post-Construction QC Checklist                                       | 14daysACO #6                                      |
| 8                            | S    | Programming Software<br>Configuration Software                       | 14daysACO #6                                      |
| 9                            | S    | Draft As-Built Drawings                                              | 14 daysACO #6                                     |
| 10                           | S    | Start-Up Testing Report                                              | 14 days ACO #6                                    |
| 11                           | S    | PVT Procedures                                                       | 14days before schedule start of #12 and AAO #10   |
| 12                           | S,E  | PVT Testing Activities                                               | As indicated in PART 3 of this Section            |
| 13                           | S    | PVT Report                                                           | As indicated in PART 3 of this Section            |
| 14                           | S    | Controller Application Programs<br>Controller Configuration Settings | 14days AAO #13                                    |
| 15                           | S    | Final As-Built Drawings                                              | 14 days AAO #13                                   |
| 16                           | S    | O&M Instructions                                                     | AAO #15                                           |
| 17                           | S    | Training Documentation                                               | AAO #10 and 14 days before scheduled start of #18 |
| 18                           | E    | Training                                                             | AAO #16 and #17                                   |



| TABLE II. PROJECT SEQUENCING |      |                       |                                               |
|------------------------------|------|-----------------------|-----------------------------------------------|
| ITEM #                       | TYPE | DESCRIPTION           | SEQUENCING (START OF ACTIVITY OR DEADLINE FOR |
| 19                           | S    | Closeout QC Checklist | ACO #18                                       |

#### 1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-02 Shop Drawings

DDC Contractor Design Drawings; G

Draft As-Built Drawings; G

Final As-Built Drawings; G

##### SD-03 Product Data

Programming Software; G

Controller Application Programs; G

Configuration Software; G

Controller Configuration Settings; G

Manufacturer's Product Data; G

##### SD-06 Test Reports

Pre-Construction Quality Control (QC) Checklist; G

Post-Construction Quality Control (QC) Checklist; G

Start-Up Testing Report; G

Control Contractor's Performance Verification Testing Plan; G

Equipment Supplier's Performance Verification Testing Plan; G

Endurance Testing Results; G

Performance Verification Test Report; G

#### SD-10 Operation and Maintenance Data

Operation and Maintenance (O&M) Instructions; G

Training Documentation; G

#### SD-11 Closeout Submittals

Enclosure Keys; G

Closeout Quality Control (QC) Checklist; G

### 1.7 DATA PACKAGE AND SUBMITTAL REQUIREMENTS

Technical data packages consisting of technical data and computer software (meaning technical data which relates to computer software) which are specifically identified in this project and which may be defined/required in other specifications must be delivered strictly in accordance with the CONTRACT CLAUSES and in accordance with the Contract Data Requirements List, DD Form 1423. Data delivered must be identified by reference to the particular specification paragraph against which it is furnished. All submittals not specified as technical data packages are considered 'shop drawings' under the Federal Acquisition Regulation Supplement (FARS) and must contain no proprietary information and be delivered with unrestricted rights.

### 1.8 SOFTWARE FOR DDC HARDWARE AND GATEWAYS

Provide all software related to the programming and configuration of DDC Hardware and Gateways as indicated. License all Software to the project site. The term "controller" as used in these requirements means both DDC Hardware and Gateways.

#### 1.8.1 Configuration Software

For each type of controller, provide the configuration tool software. Submit hard copies of the software user manuals for each software with the software submittal.

Submit Configuration Software on CD-ROM as a Technical Data Package. Submit USB flash drive hard copies of the software user manual for each piece of software.

#### 1.8.2 Controller Configuration Settings

For each controller, provide copies of the installed configuration settings as source code compatible with the configuration tool software for that controller.

Submit Controller Configuration Settings on CD-ROM as a Technical Data Package. Include on the CD-ROM a list or table of contents clearly indicating which files are associated with each device. Submit 2 USB flash drive copies of the Controller Configuration Settings CD-ROM.

#### 1.8.3 Programming Software

For each type of programmable controller, provide the programming software. Submit hard copies of software user manuals for each software with the software submittal.

Submit Programming Software on CD-ROM as a Technical Data Package. Submit USB flash drive hard copies of the software user manual for each piece of software.

#### 1.8.4 Controller Application Programs

For each programmable controller, provide copies of the application program as source code compatible with the programming software for that controller.

Submit Controller Application Programs on CD-ROM as a Technical Data Package. Include on the CD-ROM a list or table of contents clearly indicating which application program is associated with each device. Submit 2 copies of the Controller Application Programs CD-ROM.

### 1.9 QUALITY CONTROL CHECKLISTS

The QC Checklist for BACnet Systems in APPENDIX A of this Section must be completed by the Contractor's Chief Quality Control (QC) Representative and submitted as indicated.

The QC Representative must verify each item indicated and initial in the space provided to indicate that the requirement has been met. The QC Representative must sign and date the Checklist prior to submission to the Government.

#### 1.9.1 Pre-Construction Quality Control (QC) Checklist

Complete items indicated as Pre-Construction QC Checklist items in the QC Checklist. Submit four copies of the Pre-Construction QC Checklist.

#### 1.9.2 Post-Construction Quality Control (QC) Checklist

Complete items indicated as Post-Construction QC Checklist items in the QC Checklist. Submit four copies of the Post-Construction QC Checklist.

#### 1.9.3 Closeout Quality Control (QC) Checklist

Complete items indicated as Closeout QC Checklist items in the QC Checklist. Submit four copies of the Closeout QC Checklist.

## PART 2 PRODUCTS

Provide products meeting the requirements of other referenced Sections, and this Section.

### 2.1 GENERAL PRODUCT REQUIREMENTS

Units of the same type of equipment must be products of a single manufacturer. Each major component of equipment must have the manufacturer's name and address, and the model and serial number in a conspicuous place. Materials and equipment must be standard products of a manufacturer regularly engaged in the manufacturing of these and similar products. The standard products must have been in a satisfactory commercial or industrial use for two years prior to use on this project. The two year use must include applications of equipment and materials under similar circumstances and of similar size. DDC Hardware not meeting the two-year field service requirement is acceptable provided it has been successfully used by the Contractor in a minimum of two previous projects. The equipment items must be supported by a service organization. Items of the same type and purpose must be identical, including equipment, assemblies, parts and components.

### 2.2 PRODUCT DATA

Provide manufacturer's product data sheets documenting compliance with product specifications for each product provided under this Section. Provide product data for all products in a single indexed compendium, organized by product type.

For all BACnet hardware: for each manufacturer, model and version (revision) of DDC Hardware provide the Protocol Implementation Conformance Statement (PICS).

Submit Manufacturer's Product Data on CD-ROM.

### 2.3 OPERATION ENVIRONMENT

Unless otherwise specified, provide products rated for continuous operation under the following conditions:

- a. Pressure: Pressure conditions normally encountered in the installed location.
- b. Vibration: Vibration conditions normally encountered in the installed location.
- c. Temperature:
  - (1) Products installed indoors: Ambient temperatures in the range of 32 to 112 degrees F and temperature conditions outside this range normally encountered at the installed location.
- d. Humidity: 10 to 95 percent relative humidity, noncondensing and humidity conditions outside this range normally encountered at the installed location.

## 2.4 WIRELESS CAPABILITY

Wireless capabilities must comply with Section 25 05 11 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS

## 2.5 ENCLOSURES

Enclosures supplied as an integral (pre-packaged) part of another product are acceptable. Provide two Enclosure Keys for each lockable enclosure on a single ring per enclosure with a tag identifying the enclosure the keys operate. Provide enclosures meeting the following minimum requirements:

### 2.5.1 Outdoors

For enclosures located outdoors, provide enclosures meeting NEMA 250 Type 3 Type 4 requirements.

### 2.5.2 Mechanical and Electrical Rooms

For enclosures located in mechanical or electrical rooms, provide enclosures meeting NEMA 250 Type 2Type 4 requirements.

### 2.5.3 Other Locations

For enclosures in other locations including but not limited to occupied spaces, above ceilings, and in plenum returns, provide enclosures meeting NEMA 250 Type 1 requirements.

## 2.6 WIRE AND CABLE

Provide wire and cable meeting the requirements of NFPA 70 and NFPA 90A in addition to the requirements of this specification and referenced specifications.

### 2.6.1 Terminal Blocks

For terminal blocks which are not integral to other equipment, provide terminal blocks which are insulated, modular, feed-through, clamp style with recessed captive screw-type clamping mechanism, suitable for DIN rail mounting, and which have enclosed sides or end plates and partition plates for separation.

### 2.6.2 Control Wiring for Binary Signals

For Control Wiring for Binary Signals, provide 18 AWG copper or thicker wire rated for 300-volt service.

### 2.6.3 Control Wiring for Analog Signals

For Control Wiring for Analog Signals, provide 18 AWG or thicker, copper, single- or multiple-twisted wire meeting the following requirements:

- a. minimum 2 inch lay of twist
- b. 100 percent shielded pairs
- c. at least 300-volt insulation
- d. each pair has a 20 AWG tinned-copper drain wire and individual overall

pair insulation

- e. cables have an overall aluminum-polyester or tinned-copper cable-shield tape, overall 20 AWG tinned-copper cable drain wire, and overall cable insulation.

#### 2.6.4 Power Wiring for Control Devices

For 24-volt circuits, provide insulated copper 18 AWG or thicker wire rated for 300 VAC service. For 120-volt circuits, provide 14 AWG or thicker stranded copper wire rated for 600-volt service.

#### 2.6.5 Transformers

Provide UL 5085-3 approved transformers. Select transformers sized so that the connected load is no greater than 80 percent of the transformer rated capacity.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

Fully install and test the control system in accordance with this Section.

##### 3.1.1 Dielectric Isolation

Provide dielectric isolation where dissimilar metals are used for connection and support. Install control system in a manner that provides clearance for control system maintenance by maintaining access space required to calibrate, remove, repair, or replace control system devices. Install control system such that it does not interfere with the clearance requirements for mechanical and electrical system maintenance.

##### 3.1.2 Penetrations in Building Exterior

Make all penetrations through and mounting holes in the building exterior watertight.

##### 3.1.3 Device Mounting Criteria

Install devices in accordance with the manufacturer's recommendations and as indicated and shown. Provide a weathershield for all devices installed outdoors. Provide clearance for control system maintenance by maintaining access space required to calibrate, remove, repair, or replace control system devices. Provide clearance for mechanical and electrical system maintenance; do not not interfere with the clearance requirements for mechanical and electrical system maintenance.

##### 3.1.4 Labels and Tags

Key all labels and tags to the unique identifiers shown on the As-Built drawings. For labels exterior to protective enclosures provide engraved plastic labels mechanically attached to the enclosure or DDC Hardware. Labels inside protective enclosures may be attached using adhesive, but must not be hand written. For tags, provide plastic or metal tags mechanically attached directly to each device or attached by a metal chain or wire.

- a. Label all Enclosures and DDC Hardware.

- b. Tag Airflow measurement arrays (AFMA) with flow rate range for signal output range, duct size, and pitot tube AFMA flow coefficient.
- c. Tag duct static pressure taps at the location of the pressure tap

### 3.1.5 Surge Protection

#### 3.1.5.1 Power-Line Surge Protection

Protect equipment connected to AC circuits to withstand power-line surges in accordance with IEEE C62.41. Do not use fuses for surge protection.

#### 3.1.5.2 Surge Protection for Transmitter and Control Wiring

Protect DDC hardware against or provided DDC hardware capable of withstanding surges induced on control and transmitter wiring installed outdoors and as shown. Protect equipment against the following two waveforms:

- a. A waveform with a 10-microsecond rise time, a 1000-microsecond decay time and a peak current of 60 amps.
- b. A waveform with an 8-microsecond rise time, a 20-microsecond decay time and a peak current of 500 amperes.

### 3.2 DRAWINGS AND CALCULATIONS

Provide drawings in the form and arrangement indicated and shown. Use the same abbreviations, symbols, nomenclature and identifiers shown. Assign a unique identifier as shown to each control system element on a drawing. When packaging drawings, group schedules by system. When space allows, it is permissible to include multiple schedules for the same system on a single sheet. Except for drawings covering all systems, do not put information for different systems on the same sheet.

Submit hardcopy drawings on ISO A1 34 by 22 inches or A3 17 by 11 inches sheets, and electronic drawings in PDF and in AutoCAD Microstation Bentley BIM V8 Autodesk Revit 2013 format. In addition, submit electronic drawings in editable Excel format for all drawings that are tabular, including but not limited to the Point Schedule and Equipment Schedule.

- a. Submit DDC Contractor Design Drawings consisting of each drawing indicated with pre-construction information depicting the intended control system design and plans. Submit DDC Contractor Design Drawings as a single complete package: 2 hard copies and 2 copies on CD-ROM.
- b. Submit Draft As-Built Drawings consisting of each drawing indicated updated with as-built data for the system prior to PVT. Submit Draft As-Built Drawings as a single complete package: 2 hard copies and 2 copies on CD-ROM.
- c. Submit Final As-Built Drawings consisting of each drawing indicated updated with all final as-built data. Final As-Built Drawings as a single complete package: 2 hard copies and 2 copies on CD-ROM.

### 3.2.1 Sample Drawings

Sample drawings in electronic format are available at the Whole Building Design Guide page for this section:

<http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-23-09-00>

These drawings may prove useful in demonstrating expected drawing formatting and example content and are provided for illustrative purposes only. Note that these drawings do not meet the content requirements of this Section and must be completed to meet project requirements.

### 3.2.2 Drawing Index and Legend

Provide an HVAC Control System Drawing Index showing the name and number of the building, military site, State or other similar designation, and Country. In the Drawing Index, list all Contractor Design Drawings, including the drawing number, sheet number, drawing title, and computer filename when used. In the Design Drawing Legend, show and describe all symbols, abbreviations and acronyms used on the Design Drawings. Provide a single Index and Legend for the entire drawing package.

### 3.2.3 Thermostat and Occupancy Sensor Schedule

Provide a thermostat and occupancy sensor schedule containing each thermostat's unique identifier, room identifier and control features and functions as shown. Provide a single thermostat and occupancy sensor schedule for the entire project.

### 3.2.4 Valve Schedule

Provide a valve schedule containing each valve's unique identifier, size, flow coefficient Kv (Cv), pressure drop at specified flow rate, spring range, positive positioner range, actuator size, close-off pressure to torque data, dimensions, and access and clearance requirements data. In the valve schedule include actuator selection data supported by calculations of the force required to move and seal the valve, access and clearance requirements. Provide a single valve schedule for the entire project.

### 3.2.5 Damper Schedule

Provide a damper schedule containing each damper's unique identifier, type (opposed or parallel blade), nominal and actual sizes, orientation of axis and frame, direction of blade rotation, actuator size and spring ranges, operation rate, positive positioner range, location of actuators and damper end switches, arrangement of sections in multi-section dampers, and methods of connecting dampers, actuators, and linkages. Include the AMCA 511 maximum leakage rate at the operating static-pressure differential for each damper in the Damper Schedule. Provide a single damper schedule for the entire project.

### 3.2.6 Project Summary Equipment Schedule

Provide a project summary equipment schedule containing the manufacturer, model number, part number and descriptive name for each control device, hardware and component provided under this specification. Provide a single project equipment schedule for the entire project.



### 3.2.7 Equipment Schedule

Provide system equipment schedules containing the unique identifier, manufacturer, model number, part number and descriptive name for each control device, hardware and component provided under this specification. Provide a separate equipment schedule for each HVAC system.

### 3.2.8 Occupancy Schedule

Provide an occupancy schedule drawing containing the same fields as the occupancy schedule Contract Drawing with Contractor updated information. Provide a single occupancy schedule for the entire project.

### 3.2.9 DDC Hardware Schedule

Provide a single DDC Hardware Schedule for the entire project and including following information for each device.

#### 3.2.9.1 DDC Hardware Identifier

The Unique DDC Hardware Identifier for the device.

#### 3.2.9.2 HVAC System

The system "name" used to identify a specific system (the name used on the system schematic drawing for that system).

#### 3.2.9.3 BACnet Device Information

##### 3.2.9.3.1 Device Object Identifier

The Device Object Identifier: The Object\_Identifier of the Device Object

##### 3.2.9.3.2 Network Number

The Network Number for the device.

##### 3.2.9.3.3 MAC Address

The MAC Address for the device

##### 3.2.9.3.4 BTL Listing

The BTL Listing of the device. If the device is listed under multiple BTL Profiles, indicate the profile that matches the use and configuration of the device as installed.

##### 3.2.9.3.5 Proprietary Services Information

If the device uses non-standard ASHRAE 135 services, indicate that the device uses non-standard services and include a description of all non-standard services used. Describe usage and content such that a device from another vendor can interoperate with the device using the non-standard service. Provide descriptions with sufficient detail to allow a device from a different manufacturer to be programmed to both read and write the non-standard service request:

- a. read: interpret the data contained in the non-standard service and;

- b. write: given similar data, generate the appropriate non-standard service request.

#### 3.2.9.3.6 Alarming Information

Indicate whether the device is used for alarm generation, and which types of alarm generation the device implements: intrinsic, local algorithmic, remote algorithmic.

#### 3.2.9.3.7 Scheduling Information

Indicate whether the device is used for scheduling.

#### 3.2.9.3.8 Trending Information

Indicate whether the device is used for trending, and indicate if the device is used to trend local values, remote values, or both.

#### 3.2.10 Points Schedule

Provide a Points Schedule in tabular form for each HVAC system, with the indicated columns and with each row representing a hardware point, network point or configuration point in the system.

- a. When a Points Schedule was included in the Contract Drawing package, use the same fields as the Contract Drawing with updated information in addition to the indicated fields.
- b. When Point Schedules are included in the contract package, items requiring contractor verification or input have been shown in angle brackets ("<" and ">"), such as <\_\_\_> for a required entry or <value> for a value requiring confirmation. Complete all items in brackets as well as any blank cells. Do not modify values which are not in brackets without approval.

Points Schedule Columns must include:

##### 3.2.10.1 Point Name

The abbreviated name for the point using the indicated naming convention.

##### 3.2.10.2 Description

A brief functional description of the point such as "Supply Air Temperature".

##### 3.2.10.3 DDC Hardware Identifier

The Unique DDC Hardware Identifier shown on the DDC Hardware Schedule and used across all drawings for the DDC Hardware containing the point.

##### 3.2.10.4 Settings

The value and units of any setpoints, configured setpoints, configuration parameters, and settings related to each point.

#### 3.2.10.5 Range

The range of values, including units, associated with the point, including but not limited to a zone temperature setpoint adjustment range, a sensor measurement range, occupancy values for an occupancy input, or the status of a safety.

#### 3.2.10.6 Input or Output (I/O) Type

The type of input or output signal associated with the point. Use the following abbreviations for entries in this column:

- a. AI: The value comes from a hardware (physical) Analog Input
- b. AO: The value is output as a hardware (physical) Analog Output
- c. BI: The value comes from a hardware (physical) Binary Input
- d. BO: The value is output as a hardware (physical) Binary Output
- e. PULSE: The value comes from a hardware (physical) Pulse Accumulator Input
- f. NET-IN: The value is provided from the network (generally from another device). Use this entry only when the value is received from another device as part of scheduling or as part of a sequence of operation, not when the value is received on the network for supervisory functions such as trending, alarming, override or display at a user interface.
- g. NET-OUT: The value is provided to another controller over the network. Use this entry only when the value is transmitted to another device as part of scheduling or as part of a sequence of operation, not when the value is transmitted on the network for supervisory functions such as trending, alarming, override or display at a user interface.

#### 3.2.10.7 Object and Property Information

The Object Type and Instance Number for the Object associated with the point. If the value of the point is not in the Present\_Value Property, then also provide the Property ID for the Property containing the value of the point. Any point that is displayed at the front end or on an LDP, is trended, is used by another device on the network, or has an alarm condition must be documented here.

#### 3.2.10.8 Network Data Exchange Information (Gets Data From, Sends Data To)

Provide the DDC Hardware Identifier of other DDC Hardware the point is shared with.

#### 3.2.10.9 Override Information (Object Type and Instance Number)

For each point requiring an Override, indicate if the Object for the point is Commandable or, if the use of a separate Object was specifically approved by the Contracting Officer, provide the Object Type and Instance Number of the Object to be used in overriding the point.

#### 3.2.10.10 Trend Object Information

For each point requiring a trend, indicate if the trend is Local or Remote, the trend Object type and the trend Object instance number. For remote trends provide the DDC Hardware Identifier for the device containing the trend Object in the Points Schedule notes.

#### 3.2.10.11 Alarm Information

Indicate the Alarm Generation Type, Event Enrollment Object Instance Number, and Notification Class Object Instance Number for each point requiring an alarm. (Note that not all alarms will have Event Enrollment Objects.)

#### 3.2.10.12 Configuration Information

Indicate the means of configuration associated with each point.

- a. For Operator Configurable Points indicate BACnet Object and Property information (Name, Type, Identifiers) containing the configurable value. Indicate whether the property is writable always, or only when Out\_Of\_Service is TRUE.
- b. For Configurable Points indicate the BACnet Object and Property information as for Operator Configurable points, or identification of the configurable settings from within the engineering software for the device or identification of the hardware settings on the device.

#### 3.2.11 Riser Diagram

The Riser Diagram of the Building Control Network may be in tabular form, and must show all DDC Hardware and all Network Hardware, including network terminators. For each item, provide the unique identifier, common descriptive name, physical sequential order (previous and next device on the network), room identifier and location within room. A single riser diagram must be submitted for the entire system.

#### 3.2.12 Control System Schematics

Provide control system schematics in the same form as the control system schematic Contract Drawing with Contractor updated information. Provide a control system schematic for each HVAC system.

#### 3.2.13 Sequences of Operation Including Control Logic Diagrams

Provide HVAC control system sequence of operation and control logic diagrams in the same format as the Contract Drawings. Within these drawings, refer to devices by their unique identifiers. Submit sequences of operation and control logic diagrams for each HVAC system

#### 3.2.14 Controller, Motor Starter and Relay Wiring Diagram

Provide controller wiring diagrams as functional wiring diagrams which show the interconnection of conductors and cables to each controller and to the identified terminals of input and output devices, starters and package equipment. Show necessary jumpers and ground connections and the

labels of all conductors. Identify sources of power required for control systems and for packaged equipment control systems back to the panel board circuit breaker number, controller enclosures, magnetic starter, or packaged equipment control circuit. Show each power supply and transformer not integral to a controller, starter, or packaged equipment. Show the connected volt-ampere load and the power supply volt-ampere rating. Provide wiring diagrams for each HVAC system.

### 3.3 CONTROLLER TUNING

Tune each controller in a manner consistent with that described in the ASHRAE FUN IP and in the manufacturer's instruction manual. Tuning must consist of adjustment of the proportional, integral, and where applicable, the derivative (PID) settings to provide stable closed-loop control. Each loop must be tuned while the system or plant is operating at a high gain (worst case) condition, where high gain can generally be defined as a low-flow or low-load condition. Upon final adjustment of the PID settings, in response to a change in controller setpoint, the controlled variable must settle out at the new setpoint with no more than two (2) oscillations above and below setpoint. Upon settling out at the new setpoint the controller output must be steady. With the exception of naturally slow processes such as zone temperature control, the controller must settle out at the new setpoint within five (5) minutes. Set the controller to its correct setpoint and record and submit the final PID configuration settings with the O&M Instructions and on the associated Points Schedule.

### 3.4 START-UP

#### 3.4.1 Start-Up Test

Perform the following startup tests for each control system to ensure that the described control system components are installed and functioning per this specification.

Adjust, calibrate, measure, program, configure, set the time schedules, and otherwise perform all necessary actions to ensure that the systems function as indicated and shown in the sequence of operation and other contract documents.

##### 3.4.1.1 Systems Check

An item-by-item check must be performed for each HVAC system

##### 3.4.1.1.1 Step 1 - System Inspection

With the system in unoccupied mode and with fan hand-off-auto switches in the OFF position, verify that power and main air are available where required and that all output devices are in their failsafe and normal positions. Inspect each local display panel and each M&C Client to verify that all displays indicate shutdown conditions.

##### 3.4.1.1.2 Step 2 - Calibration Accuracy Check

Perform a two-point accuracy check of the calibration of each HVAC control system sensing element and transmitter by comparing the value from the test instrument to the network value provided by the DDC Hardware. Use digital indicating test instruments, such as digital thermometers, motor-driven psychrometers, and tachometers. Use test instruments with

accuracy at least twice as accurate as the specified sensor accuracy and with calibration traceable to National Institute of Standards and Technology standards. Check one the first check point in the bottom one-third of the sensor range, and the second in the top one-third of the sensor range. Verify that the sensing element-to-DDC readout accuracies at two points are within the specified product accuracy tolerances, and if not recalibrate or replace the device and repeat the calibration check.

#### 3.4.1.1.3 Step 3 - Actuator Range Check

With the system running, apply a signal to each actuator through the DDC Hardware controller. Verify proper operation of the actuators and positioners for all actuated devices and record the signal levels for the extreme positions of each device. Vary the signal over its full range, and verify that the actuators travel from zero stroke to full stroke within the signal range. Where applicable, verify that all sequenced actuators move from zero stroke to full stroke in the proper direction, and move the connected device in the proper direction from one extreme position to the other. For valve actuators and damper actuators, perform the actuator range check under normal system pressures.

#### 3.4.1.2 Weather Dependent Test

Perform weather dependent test procedures in the appropriate climatic season.

#### 3.4.2 Start-Up Testing Report

Submit 4 copies of the Start-Up Testing Report. The report may be submitted as a Technical Data Package documenting the results of the tests performed and certifying that the system is installed and functioning per this specification, and is ready for the Performance Verification Test (PVT).

### 3.5 PERFORMANCE VERIFICATION TESTING

#### 3.5.1 General

PVT testing must demonstrate compliance of controls work with contract document requirements and must be performed by the Controls Contractor and Equipment Suppliers. No less than 14 calendar days prior to start of controls system installation, meet with the Contracting Office's technical representative (COTR) and the designing engineer of the HVAC systems, the Contractor's QA representative, the Contractor's Controls Contractor representative, and the control system Owner to develop a mutual understanding relate to the details of the PVT work requirements, including required submittals, work schedule, and field quality control.

#### 3.5.2 Performance Verification Testing and Commissioning

PVT testing is a Government quality assurance function that includes systems trending and field tests. Commissioning is a quality control function that is the Commissioning Team's responsibility to the extent required by this contract.

#### 3.5.3 Performance Verification Testing of Equipment with Packaged Controls

Controls Contractor and Equipment Supplier(s) must share and coordinate PVT testing responsibilities for equipment provided with on-board factory

packaged controls such as boiler controllers, dedicated outside air systems (DOAS's), and packaged pumping systems.

#### 3.5.3.1 Controls Contractor Responsibilities

The Controls Contractor must provide a PVT Plan separate from Equipment Supplier's performance verification testing plan, perform endurance testing, and perform PVT testing concurrent with Equipment Suppliers' testing for equipment provided with on-board factory packaged controls to demonstrate the following:

- a. Equipment enabling and disabling.
- b. Equipment standard and optional control points necessary to accomplish functionality regardless if specified in contract documents or not.
- c. Equipment standard and optional alarms critical to safe operation regardless if specified in contract documents or not.
- d. All control points added by Controls Contractor in addition to onboard factory packaged controls regardless if specified in contract documents or not.

Refer to paragraphs titled "Performance Verification Test Plan" and "Endurance Testing" for additional information.

#### 3.5.3.2 Equipment Supplier Responsibilities

Each Equipment Supplier must provide PVT Plans separate from Controls Contractor's plans and perform PVT testing concurrent with Controls Contractor's testing for their equipment provided with on-board factory packaged controls to demonstrate the following:

- a. Equipment standard and optional control features necessary to accomplish functionality regardless if specified in contract documents or not.
- b. Equipment standard and optional operation modes necessary to accomplish functionality regardless if specified in contract documents or not.
- c. Equipment standard and optional alarm conditions for safe operation regardless if specified in contract documents or not.

Refer to all paragraphs under paragraph titled "Performance Verification Testing" except for section titled "Endurance Testing" for additional information.

#### 3.5.4 Sequencing of Performance Verification Testing Activities

PVT activities must be sequenced with major activities listed below for Test and Balance (TAB) Contractor, Equipment Suppliers, Commissioning Specialists, and others to demonstrate fully functioning systems. Refer to Section 01 32 17.00 20 COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS). Complete the items in TABLE III: SEQUENCING OF PVT TESTING ACTIVITIES as schedule activities or milestones.

| TABLE III: SEQUENCING OF PVT TESTING ACTIVITIES |                                                                                                                  |
|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| SEQUENCE                                        | ITEM                                                                                                             |
| 1                                               | Submission, review, and approval of Control Contractors PVT Plans.                                               |
| 2                                               | Submission, review, and approval of Equipment Suppliers PVT Plans.                                               |
| 3                                               | Submission, review, and approval of certified final Test and Balance Report.                                     |
| 4                                               | Conduct commissioning functional performance tests.                                                              |
| 5                                               | Submission, review, and approval of all of the Commissioning Specialists completed functional performance tests. |
| 6                                               | Request Contracting Officer to allow beginning of Government-witnessed PVT testing.                              |
| 7                                               | Contracting Officers approval to begin PVT testing.                                                              |
| 8                                               | Conduct PVT field work.                                                                                          |
| 9                                               | Governments verbal approval of PVT field work for all systems.                                                   |
| 10                                              | Conduct Test and Balance verification field work.                                                                |
| 11                                              | Governments written approval of Test and Balance verification field work.                                        |
| 12                                              | Submission, review, and approval of endurance testing.                                                           |
| 13                                              | Governments written approval of PVT field work for all systems.                                                  |
| 14                                              | Facility acceptance recommendation.                                                                              |
| 15                                              | Submission, review, and approval of Control Contractors PVT Report.                                              |
| 16                                              | Submission, review, and approval of Equipment Suppliers PVT Report.                                              |
| 17                                              | Conduct applicable re-testing and seasonal testing within 10 months of beneficial occupancy.                     |

#### 3.5.4.1 PVT Testing for Multi-Phase Construction

For air moving systems except outside air systems serving multiple phases, all major activities listed in TABLE III through Government's verbal approval of Test and Balance verification field work can be completed by phase if all ductwork construction is completed for that phase.

For primary systems such as chilled water systems, HVAC heating hot water systems, and outside air systems serving multiple phases, all major activities listed listed in TABLE III through Government's verbal approval of Test and Balance verification field work for all air moving systems served by that primary system for that phase must be completed prior to conducting PVT field work for that primary system.



### 3.5.5 Control Contractor's Performance Verification Testing Plan

Submit a detailed PVT Plan of the proposed control systems testing in this contract for approval prior to its use. Develop and use a single PVT Plan for each system with a unique control sequence. Systems sharing an identical control sequence can be tested using copies of the PVT Plan intended for these systems.

PVT Plans must include system-based, step-by-step test methods demonstrating system performs in accordance with contract document requirements. The Government may provide sample PVT Plans upon request. PVT Plans must include the following:

- a. Control sequences from contract documents segmented such that each control algorithm, operation mode, and alarm condition is immediately followed by numbered test methods required to initiate a response, expected response, space for comments, and "pass" or "fail" indication for each expected response.
- b. PVT Plans with control sequences from contract documents that are not segmented into parts will not be accepted.
- c. Indication where assisting personnel are required such as Mechanical Contractor.
- d. Signature and date lines for the Contractor's PVT administrator, Contractor's quality assurance representative, and Contracting Officer's representative acknowledging completion of testing.

### 3.5.6 Performance Verification Testing Sample Size

PVT testing sample sizes will be as follows:

- a. 100-Percent of the following systems:
  - (1) primary systems including, but not limited to, chilled water and HVAC heating hot water systems
  - (2) air handling unit systems including all associated fans except for remote exhaust air fans
  - (3) DOAS's including all associated fans except for remote exhaust air fans
- b. 20-Percent of each set of systems with a shared identical control sequence for systems such as:
  - (1) air terminal units
  - (2) exhaust air fans
  - (3) terminal equipment such as fan coil units and unit heaters

#### 3.5.6.1 Selection of Systems to Test

For sample sets less than 100-percent, the Government will choose which systems will be tested. The Government may require additional testing if previous testing results are inconsistent or demonstrate improper system control as follows:

- a. An additional 25-percent after five-percent failure rate of first sample set.
- b. 100-percent after any failures occurring in additional sample set.

### 3.5.7 Conducting Performance Verification Testing

At least 15 days prior to preferred test date, request the Contracting Officer to allow the beginning of Government-witnessed PVT testing. Provide an estimated time table required to perform testing of each system. Furnish personnel, equipment, instrumentation, and supplies necessary to perform all aspects of testing. Testing personnel must be regularly employed in the testing and calibration of control systems. After receipt of Contracting Officer's approval to begin testing, perform PVT testing using project's as-built (shop) control system drawings, project's design drawings, and approved PVT Plans.

During testing, identify deficiencies that do not meet contract document requirements. Deficiencies must be investigated, corrected with corrections documented, and re-tested at a later date following procedures for the initial PVT testing. The Government may require re-testing of any control system components affected by the original failed test.

### 3.5.8 Endurance Testing

#### 3.5.8.1 General

Conduct endurance testing in conjunction with the PVT to demonstrate control loop stability and accuracy. For all control loops tested, record trend data of the control variables over time, demonstrating that the control loop responds to a sudden change of the control variable set point without excessive overshoot or undershoot. Conduct endurance testing for each system subject to PVT testing. Systems must be operating as normally anticipated during occupancy throughout endurance testing.

Endurance testing results must clearly demonstrate control loop stability and accuracy. Controlled loop outputs must be stable and accurately maintain each setpoint.

#### 3.5.8.2 Hardware

Use hardware provided in this contract for testing. Use Government furnished hardware for testing if available when endurance testing begins. If unavailable, the Contractor must provide suitable hardware for required testing.

If insufficient buffer capacity exists to trend the entire endurance test, upload trend data during the course of endurance testing to ensure all trend data is retained. Lost trend data will require retesting of all control points for affected system(s).

#### 3.5.8.3 Endurance Testing Results Format

Submit endurance testing results for each tested system in a graphical format complete with clear indication of value(s) for y-axis, value for x-axis, and legend identifying each trended control point. The number of control points contained on a single graph must be such that all control

points can be clearly visible. Control points must be logically grouped such that related points appear on a single graph. In addition, submit a separate comma separated value (CSV) file of raw trend data for each trended system. Each trended control point in CSV file must be clearly identified.

For control points recorded based on change of value, change of value for recording data must be clearly identified for each control point.

#### 3.5.8.4 Endurance Testing Start, Duration, and Frequency

Trending of all control points for a given system must start at an identical date and time regardless of the basis of data collection. Duration of all endurance tests must be at least one-week.

Unless specified otherwise for control points recorded based on time, frequency of data collection must be 15-minutes. Frequency of data collection for specific types of control points is as follows:

##### 3.5.8.4.1 Points Trended at One Minute Intervals

- a. Temperature for supply air, return air, mixed air, supply water, and return water
- c. Flow for supply air, return air, outside air, chilled water, and HVAC heating hot water
- g. Command and status for control dampers and control valves
- h. Speed for fans and pumps
- i. Pressure for fans and pumps

##### 3.5.8.4.2 Points Trended at 15 Minute Intervals

- a. Temperature for zones
- b. Temperature for outside air not associated with energy recovery
- c. Command and status for equipment
- d. Pressure relative to the outside for facility

#### 3.5.9 Performance Verification Test Report

Submit a PVT Report after receiving Government's written approval of PVT field work that is intended to document test results and final control system sequences and settings prior to turnover. The PVT Report must contain the following:

- a. Executive summary that briefly discusses results of each system's endurance testing and PVT testing and conclusions for each system.
- b. Endurance testing for each system.

- c. Completed PVT Plan for each system used during testing that includes hand written field notes and participant signatures.
- d. Blank PVT Plan for each system approved prior to testing that is edited to reflect changes occurring during testing. Edits must be typed and must reflect changes to control sequences from contract documents, must reflect changes to numbered test methods required to initiate a response, and must reflect changes to expected response. Only one blank PVT Plan is required for each set of systems sharing an identical control sequence, such as air terminal units, exhaust air fans, fan coil units and unit heaters.
- e. Written certification that the installation and testing of all systems are complete and meet all contract document requirements.

### 3.6 OPERATION AND MAINTENANCE (O&M) INSTRUCTIONS

Provide HVAC control System Operation and Maintenance Instructions which include:

- a. "Data Package 3" as indicated in Section 01 78 23 OPERATION AND MAINTENANCE DATA for each piece of control equipment.
- b. "Data Package 4" as described in Section 01 78 23 OPERATION AND MAINTENANCE DATA for all air compressors.
- c. HVAC control system sequences of operation formatted as indicated.
- d. Procedures for the HVAC system start-up, operation and shut-down including the manufacturer's supplied procedures for each piece of equipment, and procedures for the overall HVAC system.
- e. As-built HVAC control system detail drawings formatted as indicated.
- f. Routine maintenance checklist. Provide the routine maintenance checklist arranged in a columnar format, where the first column lists all installed devices, the second column states the maintenance activity or that no maintenance required, the third column states the frequency of the maintenance activity, and the fourth column is used for additional comments or reference.
- g. Qualified service organization list, including at a minimum company name, contact name and phone number.
- h. Start-Up Testing Report.
- i. Performance Verification Test (PVT) Procedures and Report.

Submit 2 copies of the Operation and Maintenance Instructions, indexed and in booklet form. The Operation and Maintenance Instructions may be submitted as a Technical Data Package.

### L3.7 MAINTENANCE AND SERVICE

Provide services, materials and equipment as necessary to maintain the entire system in an operational state as indicated for a period of one year from the date of final acceptance of the project. Minimize impacts on

facility operations.

- a. The integration of the system specified in this section into a Utility Monitoring and Control System must not, of itself, void the warranty or otherwise alter the requirement for the one year maintenance and service period. Integration into a UMCS includes but is not limited to establishing communication between devices in the control system and the front end or devices in another system.
- b. The changing of configuration properties must not, of itself, void the warranty or otherwise alter the requirement for the one year maintenance and service period.

#### 3.7.1 Description of Work

Provide adjustment and repair of the system including the manufacturer's required sensor and actuator (including transducer) calibration, span and range adjustment.

#### 3.7.2 Personnel

Use only service personnel qualified to accomplish work promptly and satisfactorily. Advise the Government in writing of the name of the designated service representative, and of any changes in personnel.

#### 3.7.3 Scheduled Inspections

Perform two inspections at six-month intervals and provide work required. Perform inspections in June and December. During each inspection perform the indicated tasks:

- a. Perform visual checks and operational tests of equipment.
- b. Clean control system equipment including interior and exterior surfaces.
- c. Check and calibrate each field device. Check and calibrate 50 percent of the total analog inputs and outputs during the first inspection. Check and calibrate the remaining 50 percent of the analog inputs and outputs during the second major inspection. Certify analog test instrumentation accuracy to be twice the specified accuracy of the device being calibrated. Randomly check at least 25 percent of all binary inputs and outputs for proper operation during the first inspection. Randomly check at least 25 percent of the remaining binary inputs and outputs during the second inspection. If more than 20 percent of checked inputs or outputs failed the calibration check during any inspection, check and recalibrate all inputs and outputs during that inspection.
- d. Run system software diagnostics and correct diagnosed problems.
- e. Resolve any previous outstanding problems.

#### 3.7.4 Scheduled Work

This work must be performed during regular working hours, Monday through Friday, excluding Federal holidays.

### 3.7.5 Emergency Service

The Government will initiate service calls when the system is not functioning properly. Qualified personnel must be available to provide service to the system. A telephone number where the service supervisor can be reached at all times must be provided. Service personnel must be at the site within 24 hours after receiving a request for service. The control system must be restored to proper operating condition.

### 3.7.6 Operation

After performing scheduled adjustments and repairs, verify control system operation as demonstrated by the applicable tests of the performance verification test.

### 3.7.7 Records and Logs

Keep dated records and logs of each task, with cumulative records for each major component, and for the complete system chronologically. Maintain a continuous log for all devices, including initial analog span and zero calibration values and digital points. Keep complete logs and provide logs for inspection onsite, demonstrating that planned and systematic adjustments and repairs have been accomplished for the control system.

### 3.7.8 Work Requests

Record each service call request as received and include its location, date and time the call was received, nature of trouble, names of the service personnel assigned to the task, instructions describing what has to be done, the amount and nature of the materials to be used, the time and date work started, and the time and date of completion. Submit a record of the work performed within 5 days after work is accomplished.

### 3.7.9 System Modifications

Submit recommendations for system modification in writing. Do not make system modifications, including operating parameters and control settings, without prior approval of the Government.

## 3.8 TRAINING

Conduct a training course for 10 operating staff members designated by the Government in the maintenance and operation of the system, including specified hardware and software. Conduct 16 hours of training at the project site within 30 days after successful completion of the performance verification test. The Government reserves the right to make audio and visual recordings (using Government supplied equipment) of the training sessions for later use. Provide audiovisual equipment and other training materials and supplies required to conduct training. A training day is defined as 8 hours of classroom instruction, including two 15 minute breaks and excluding lunchtime, Monday through Friday, during the daytime shift in effect at the training facility.

### 3.8.1 Training Documentation

Prepare training documentation consisting of:

- a. Course Attendee List: Develop the list of course attendees in coordination with and signed by the ControlsHVACElectrical shop

supervisor.

- b. Training Manuals: Provide training manuals which include an agenda, defined objectives for each lesson, and a detailed description of the subject matter for each lesson. When presenting portions of the course material by audiovisuals, deliver copies of those audiovisuals as a part of the printed training manuals.

### 3.8.2 Training Course Content

For guidance in planning the required instruction, assume that attendees will have a high school education, and are familiar with HVAC systems. During the training course, cover all of the material contained in the Operating and Maintenance Instructions, the layout and location of each controller enclosure, the layout of one of each type of equipment and the locations of each, the location of each control device external to the panels, the location of the compressed air station, preventive maintenance, troubleshooting, diagnostics, calibration, adjustment, commissioning, tuning, and repair procedures. Typical systems and similar systems may be treated as a group, with instruction on the physical layout of one such system. Present the results of the performance verification test and the Start-Up Testing Report as benchmarks of HVAC control system performance by which to measure operation and maintenance effectiveness.

### 3.8.3 Training Documentation Submittal Requirements

Submit hardcopy training manuals and all training materials on CD-ROM. Provide one hardcopy manual for each trainee on the Course Attendee List and 2 additional copies for archive at the project site. Provide 2 copies of the Course Attendee List with the archival copies. Training Documentation may be submitted as a Technical Data Package.

**APPENDIX A**

| <b><u>QC CHECKLIST FOR BACNET SYSTEMS</u></b>                                                                                                                                                                                                  |                                                                                                                   |      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------|
| <p>This checklist is not all-inclusive of the requirements of this specification and should not be interpreted as such.</p> <p>Instructions: Initial each item in the space provided ( ____ ) verifying that the requirement has been met.</p> |                                                                                                                   |      |
| <p>This checklist is for (circle one:)</p> <p>Pre-Construction QC Checklist Submittal</p> <p>Post-Construction QC Checklist Submittal</p> <p>Close-out QC Checklist Submittal</p>                                                              |                                                                                                                   |      |
| Items verified for Pre-Construction, Post-Construction and Closeout QC Checklist Submittals:                                                                                                                                                   |                                                                                                                   |      |
| 1                                                                                                                                                                                                                                              | All DDC Hardware is numbered on Control System Schematic Drawings.                                                | ____ |
| 2                                                                                                                                                                                                                                              | Signal lines on Control System Schematic are labeled with the signal type.                                        | ____ |
| 3                                                                                                                                                                                                                                              | Local Display Panel (LDP) Locations are shown on Control System Schematic drawings.                               | ____ |
| Items verified for Post-Construction and Closeout QC Checklist Submittals:                                                                                                                                                                     |                                                                                                                   |      |
| 4                                                                                                                                                                                                                                              | All sequences are performed as specified using DDC Hardware.                                                      | ____ |
| 5                                                                                                                                                                                                                                              | Training schedule and course attendee list has been developed and coordinated with shops and submitted.           | ____ |
| Items verified for Closeout QC Checklist Submittal:                                                                                                                                                                                            |                                                                                                                   |      |
| 6                                                                                                                                                                                                                                              | Final As-built Drawings, including all Points Schedule drawings, accurately represent the final installed system. | ____ |
| 7                                                                                                                                                                                                                                              | Programming software has been submitted for all programmable controllers.                                         | ____ |
| 8                                                                                                                                                                                                                                              | All software has been licensed to the Government.                                                                 | ____ |
| 9                                                                                                                                                                                                                                              | O&M Instructions have been completed and submitted.                                                               | ____ |



| <b><u>QC CHECKLIST FOR BACNET SYSTEMS</u></b>                                  |                                                                                                                                                                                                                      |        |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 10                                                                             | Training course has been completed.                                                                                                                                                                                  | ____   |
| 11                                                                             | All DDC Hardware is installed on a BACnet ASHRAE 135 network using either MS/TP in accordance with Clause 9 or IP in accordance with Annex J.                                                                        | ____   |
| 12                                                                             | All DDC Hardware is BTL listed.                                                                                                                                                                                      | ____   |
| 13                                                                             | Communication between DDC Hardware is only via BACnet using standard services, except as specifically permitted by the specification. Non-standard services have been fully documented in the DDC Hardware Schedule. | ____   |
| 14                                                                             | Scheduling, Alarming, and Trending have been implemented using the standard BACnet Objects for these functions.                                                                                                      | ____   |
| 15                                                                             | All Properties indicated as required to be Writable are Writable and Overrides have been provided as indicated                                                                                                       | ____   |
| <div style="border-bottom: 1px solid black; height: 20px; width: 100%;"></div> |                                                                                                                                                                                                                      |        |
|                                                                                | (QC Representative Signature)                                                                                                                                                                                        | (Date) |
|                                                                                |                                                                                                                                                                                                                      |        |

-- End of Section --

**QC CHECKLIST: BACNET**

This checklist is not all-inclusive of the requirements of this specification and should not be interpreted as such.

This checklist is for (check one:)

|                                          |              |  |
|------------------------------------------|--------------|--|
| Pre-Construction QC Checklist Submittal  | (Items 1-3)  |  |
| Post-Construction QC Checklist Submittal | (Items 1-5)  |  |
| Close-out QC Checklist Submittal         | (Items 1-15) |  |

Initial and date each item in the spaces provided verifying that each requirement has been met.

| Items verified for Pre-Construction, Post-Construction and Closeout QC Checklists Submittal: |                                                                                     |         |      |
|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------|------|
|                                                                                              | Item                                                                                | Initial | Date |
| 1                                                                                            | All DDC Hardware is numbered on Control System Schematic Drawings.                  |         |      |
| 2                                                                                            | Signal lines on Control System Schematic are labeled with the signal type.          |         |      |
| 3                                                                                            | Local Display Panel (LDP) Locations are shown on Control System Schematic drawings. |         |      |

| Items verified for Post-Construction and Closeout QC Checklist Submittal: |                                                                                                         |         |      |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------|------|
|                                                                           | Item                                                                                                    | Initial | Date |
| 4                                                                         | All sequences are performed as specified using DDC Hardware.                                            |         |      |
| 5                                                                         | Training schedule and course attendee list has been developed and coordinated with shops and submitted. |         |      |

| Items verified for Closeout QC Checklists Submittal: |                                                                                                                                                                                                                      |         |      |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------|
|                                                      | Item                                                                                                                                                                                                                 | Initial | Date |
| 6                                                    | Final As-built Drawings, including all Points Schedule drawings, accurately represent the final installed system.                                                                                                    |         |      |
| 7                                                    | Programming software has been submitted for all programmable controllers.                                                                                                                                            |         |      |
| 8                                                    | All software has been licensed to the Government.                                                                                                                                                                    |         |      |
| 9                                                    | O&M Instructions have been completed and submitted.                                                                                                                                                                  |         |      |
| 10                                                   | Training course has been completed.                                                                                                                                                                                  |         |      |
| 11                                                   | All DDC Hardware is installed on a BACnet ASHRAE 135 network using either MS/TP in accordance with Clause 9 or IP in accordance with Annex J.                                                                        |         |      |
| 12                                                   | All DDC Hardware is BTL listed.                                                                                                                                                                                      |         |      |
| 13                                                   | Communication between DDC Hardware is only via BACnet using standard services, except as specifically permitted by the specification. Non-standard services have been fully documented in the DDC Hardware Schedule. |         |      |

APPENDIX A

| Items verified for Closeout QC Checklists Submittal: |                                                                                                                  |         |      |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------|------|
| Item                                                 |                                                                                                                  | Initial | Date |
| 14                                                   | Scheduling, Alarming, and Trending have been implemented using the standard BACnet Objects for these functions.  |         |      |
| 15                                                   | All Properties indicated as required to be Writeable are Writeable and Overrides have been provided as indicated |         |      |

---

(QC Representative Signature)

(Date)

SECTION 23 21 13

POLYPROPYLENE PRESSURE PIPING  
12/24

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

NSF INTERNATIONAL (NSF)

NSF/ANSI 14 (2023) Plastics Piping System Components and Related Materials

ASTM INTERNATIONAL (ASTM)

ASTM F2389 (2024a) Standard Specification for Pressure-rated Polypropylene (PP) Piping Systems

PART 2 PRODUCTS

2.1 COMPONENTS

2.1.1 Pipe

Polypropylene pipe shall meet ASTM F2389 and NSF/ANSI 14.

- a. Polypropylene Pipe, fittings, and valves shall contain no rework or recycled materials except that generated in the manufacturer's own plant from resin of the same specification from the same raw material.
- b. Pipe shall be manufactured from polypropylene (PP-RCT) thermoplastic, multi-layer fiber-composite, with heat-fused connections, and crystallinity-resistant, temperature-resistant, and naturally corrosion-resistant properties. Pipe shall be high-performance pressure, suitable for aggressive medium applications. Provide fiber-composite technology to reduce linear expansion. Pipe shall be made in an extrusion process. Polypropylene pipe standard dimension ratio (wall thickness) shall be SDR 11.
- c. Pipe shall be labeled every three feet to identify the size and type of pipe and the test standards which they meet.
- d. Manufacturer shall warranty pipe and fittings for 10 years to be free of defects in materials or manufacturing. Pipe shall be Aquatherm Blue Pipe MF.
- e. Pipe shall be manufactured by Aquatherm pipe located at 825 West 600 North Lindon, Utah 84042, or equal.

2.1.2 UV Protection

- a. Avoid exposing PP-RCT pipe and fittings to UV radiation during

transport, storage and construction. Store pipe in UV-resistant bags or wrap.

- b. For outdoor above ground installation, provide PP-RCT pipe with factory applied UV resistant coating layer. UV protected pipe shall have outer layer of black polyethylene, factory extruded over the top of the pipe. To fuse UV protected pipe, remove portions of protective layer per pipe manufacturer's instructions. All above ground pipe, fittings and joints without factory UV resistant coating shall be field protected with black elastomeric paint.
- c. Polypropylene fittings shall conform to dimensional requirements of SDR 11.
- d. Polypropylene piping that will be exposed to UV light shall be provided with a Factory applied UV resistant coating.
- e. Provide PP-RCT fittings with stainless steel threaded inserts for transition to metal threaded components.

#### 2.1.3 Joints

- a. PP-RCT joints shall be socket or butt type, with heat-fusion connections in accordance with ASTM F2389. Heat fusion joints shall maintain the same properties as the pipe itself. Parts shall be installed by the manufacturer's approved installer.
- b. Provide transitions where required, to ANSI flanges, NPT threads, and stainless steel piping, using flanges, adapters, and stainless steel threads.
- c. Provide flange joints with full-face gaskets. Faces of flanges must be clean, free of paint, and other imperfections. PP-RCT flanges must be installed per manufacturer guidelines utilizing proper flange adaptors and coated steel flanged backing ring when required.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

- a. Polypropylene pipe and fittings shall be joined by heat fusion methods or electrofusion by properly trained and qualified fusion equipment operators using well-maintained fusion equipment specifically designed for that purpose. The heat fusion procedure shall be conducted in accordance with ASTM F3722 and the pipe and/or fitting manufacturer's recommendations
- b. The installed pipe needs to be pressure tested at twice the working pressure or the pressure rating of the pipe, which ever is less. The pressure needs to remain the same for 2 hours without loss of pressure.
- c. Leaking fusion joints or mechanical connections shall be repaired in accordance with ASTM F3722 and the manufacturer's recommendation. Once the appropriate repairs are made, the system shall be re-tested for leaks using the same pressure testing procedure.

- d. Install hangers and supports at intervals specified in accordance with ANSI/MSS SP-58 or in the applicable code and as recommended by the manufacturer. Hangers and supports shall be located within one foot of any change in direction and within one foot of any fittings or valves.
- e. Contractor installers shall be trained by the manufacturer. When new contractor installers are implemented by the contractor due to staff change over, they shall be trained by the manufacturer prior to installation of pipe.

-- End of Section --

SECTION 23 21 23

HYDRONIC PUMPS

08/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1 (2024) Unified Inch Screw Threads (UN, UNR, and UNJ Thread Form)

ASTM INTERNATIONAL (ASTM)

ASTM A123/A123M (2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A307 (2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength

HYDRAULIC INSTITUTE (HI)

HI 1.1-1.2 (2014) Rotodynamic (Centrifugal) Pump for Nomenclature and Definitions

HI 1.3 (2013) Rotodynamic (Centrifugal) Pump Applications

HI ANSI/HI 2.1-2.2 (2014) Rotodynamic Vertical Pumps of Radial, Mixed, and Axial Flow Types for Nomenclature and Definitions

HI ANSI/HI 9.6.3 (2024) Rotodynamic Pumps - Guideline for Operating Regions - B120

HI ANSI/HI 14.6 (2011) Rotodynamic Pumps for Hydraulic Performance Acceptance Tests - A136

INTERNATIONAL CODE COUNCIL (ICC)

ICC IgCC (2018) International Green Construction Code

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 (2021) Motors and Generators

NEMA Z535.4 (2023) Product Safety Signs and Labels

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC Paint 21 (1982; E 2004) White or Colored Silicone Alkyd Paint (Type I, High Gloss and Type II, Medium Gloss)

SSPC Paint 25 (1997; E 2004) Zinc Oxide, Alkyd, Linseed Oil Primer for Use Over Hand Cleaned Steel, Type I and Type II

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.219 Mechanical Power Transmission Apparatus

UL SOLUTIONS (UL)

UL 778 (2016; Reprint Jul 2024) UL Standard for Safety Motor-Operated Water Pumps

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

System Coordination; G

SD-03 Product Data

Instructions; G

Equipment Data; G

Training Period; G

SD-06 Test Reports

Factory Tests

Field Quality Control

SD-07 Certificates

Manufacturer's Representative

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

Training; G



### 1.3 QUALITY ASSURANCE

#### 1.3.1 Manufacturer Services

Provide the services of a manufacturer's representative experienced in the installation, adjustment, and operation of the equipment specified. The representative must supervise the installation, adjustment, testing of the equipment, and conduct training.

Submit the names and qualifications of the manufacturer's representative and training engineers and written certification from the manufacturer that the representative and trainers are technically qualified.

#### 1.3.2 Standard Products

Provide material and equipment which are the standard products of a manufacturer regularly engaged in the manufacture of such products and that essentially duplicate equipment that has been in satisfactory HVAC operation at least 2 years prior to issuance of this solicitation. Support equipment with a service organization that is reasonably convenient to the jobsite. Pumps and motors of the same types must each be the product of one manufacturer.

#### 1.3.3 Conformance with Agency Requirements

Where materials or equipment are specified to be an approved type, attach the seal or label of approval from a nationally recognized testing agency, adequately equipped and competent to perform such services. A written certificate from the testing agency must accompany the materials or equipment and be submitted stating that the items have been tested and that they conform to the applicable requirements of the specifications and to the standards listed herein. The certificate must indicate the methods of testing used by the testing agency. In lieu of a certificate from a testing agency, published catalog specification data, accompanied by the manufacturer's certified statement to the effect that the items are in accordance with the applicable requirements of the specifications and the referenced standards, will be considered and may be acceptable as evidence that the items conform with agency requirements.

### 1.4 DELIVERY, STORAGE, AND HANDLING

Protect equipment, delivered and designated for storage, from the weather, humidity and temperature variations, dirt and dust, or other contaminants.

## PART 2 PRODUCTS

### 2.1 SYSTEM DESCRIPTION

Hydronic pumps used for heating and air conditioning applications are defined by the type of impeller, number of impellers, type of casing, method of connection to the driver, and mounting position. Provide centrifugal water pumps of the types indicated and specified. Use an electric motor driving unit for each pump as indicated and specified.

#### 2.1.1 Selection Criteria

Select pumps at a point within the maximum efficiency for a given impeller casing combination. Deviations within 3 percent of maximum efficiency are permissible, provided the lesser efficiency is not less than the scheduled

efficiency in the construction design documents. Pumps having impeller diameters larger or smaller than manufacturer's published maximum and minimum impeller diameters for a given impeller casing combination will be rejected. Pump performance data, as shown in performance curves, must be based on factory tests using precision instrumentation and exacting procedures as detailed in HI ANSI/HI 14.6.

#### 2.1.2 System Coordination

Submit drawings containing complete wiring and piping schematic diagrams and any other details required to demonstrate that the system has been coordinated and will properly function as a unit. Show the proposed layout and anchorage of equipment and appurtenances, and equipment relationship to other parts of the work including clearances for maintenance and operation. Provide a complete listing of equipment, materials and miscellaneous components including mechanical seals, bearings, and couplings.

#### 2.1.3 Safety Requirements

Fully enclose or guard couplings, projecting set-screws, keys, and other rotating parts, that pose an entangling hazards..

### 2.2 MATERIALS AND EQUIPMENT

#### 2.2.1 Nameplates

Securely affix a standard nameplate to pumps and motors in a conspicuous place showing the manufacturer's name, address, type or style, model, serial number, and catalog number. In addition, for each pump show the capacity in gpm at rated speed in rpm and total head in feet of water. For each electric motor show at least the minimum information required by NEMA MG 1. Show such other information as the manufacturer may consider necessary to complete identification on the nameplate. Pumps must be listed and labeled by UL, and comply with UL 778 for pumps not using universal motors rated more than 250 volts such as circulating pumps.

#### 2.2.2 Framed Instructions

Submit proposed diagrams, instructions, and other sheets, prior to posting. Post approved wiring and control diagrams showing the complete layout of the entire system, including equipment, piping valves, and control sequence, framed under glass or in approved laminated plastic, where directed. Provide condensed operating instructions explaining preventive maintenance procedures, methods of checking the system for normal safe operation, and procedures for safely starting and stopping the system, framed as specified above for the wiring and control diagrams, and posted beside the diagrams. Post the framed instructions before acceptance testing of the systems.

#### 2.2.3 Pump Characteristic

Construct hydronic water pumps in accordance with HI 1.1-1.2 and HI ANSI/HI 2.1-2.2. The pumps must be capable of discharging quantities at total discharge heads measured at the discharge flange, between the following limits:

Operate pumps at optimum efficiencies to produce the most economical pumping system under the conditions encountered . Pumps must furnish not

less than 150 percent of rated capacity at a total discharge head of not less than 65 percent of total rated head. The shutoff total head must not be greater than 120 percent of total rated head. Operate pumps at specified system fluid temperatures without vapor binding and cavitation. Operate pumps to HI ANSI/HI 9.6.3 standard for Preferred Operating Region (POR).

#### 2.2.4 Pump Drivers

Provide electric motors as indicated for each pump and in compliance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

#### 2.2.5 Equipment Data

Submit manufacturer's descriptive data and technical literature, performance charts and curves for all impeller sizes for a given casing, catalog cuts, and installation instructions. Provide spare parts data for each different item of material and equipment specified, after approval of the detail drawings and not later than three months prior to the date of beneficial occupancy. Include a complete list of parts and supplies, with current unit prices and local source of supply with contact information.

Submit catalog information, certified pumps curves, rated capacities, final impeller dimensions, and accessories provided for the product indicated. Indicate operating point of each pump on curves. Furnish pump curves for each pump and combination of pumps designed to operate in parallel. The pump curve must show as a minimum; bhp, flow, total dynamic head, efficiency, NPSH, impeller diameter and system curve (individually and in combination for each pump operating in a parallel application). Select pumps operating in parallel operation to cross the system curve when operating individually.

### 2.3 HYDRONIC PUMPS

Provide centrifugal, single-stage type, or multi-stage type, designed for HVAC service in the following configurations:

| Configuration        | Pump No.        |
|----------------------|-----------------|
|                      |                 |
| Vertical, Multistage | P9-A/B, P10-A/B |
|                      |                 |

#### 2.3.1 Vertical, Multistage

Type: Vertical, multistage as defined in HI 1.1-1.2 and HI 1.3 for in-line, multistage, separately coupled, overhung impeller, centrifugal pump.

2.3.1.1 Casing

Cast-iron or steel base and stainless-steel chamber.

2.3.1.2 Impeller

Closed, stainless steel; statically and dynamically balanced and keyed to shaft.

2.3.1.3 Shaft

Stainless steel.

2.3.1.4 Seal

Mechanical.

2.3.1.5 Bearing

Water-lubricated sleeve type.

2.3.1.6 Motor

Provide electric motor conforming to NEMA MG 1 and of the horsepower, and voltage indicated. Motor enclosure must be open drip proof; with heavy duty grease lubricated ball bearings completely adequate for the maximum load for which the motor is designed. Motor must be non-overloading at any point on the pump curve and premium efficiency. Provide motor efficiencies as shown in the ICC IgCC standard. Totally enclosed fan cooled motor efficiencies must be as shown in NEMA MG 1.

2.3.1.7 Control Panel

Factory installed and connected as an integral part of the booster pump; automatic for multiple-pumps, constant-speed operation, with load control and protection functions. Hand-off-automatic selector switch for each pump in cover of control panel. Duplex, automatic, alternating starter; switches lead pump to lag main pump and to two-pump operation.

2.4 ELECTRICAL WORK

Provide electrical motor driven equipment specified herein complete with motors, motor starters, and controls. Provide electric equipment and wiring in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Electrical characteristics must be as indicated. Provide motor starters complete with properly sized thermal overload protection in each phase and other appurtenances necessary for the motor control specified. Each motor must be of sufficient capacity to drive the equipment at the specified capacity without exceeding the nameplate rating of the motor when operating at proper electrical system voltage and frequency. Manual or automatic control and protective or signal devices required for the operation herein specified and any control wiring required for controls and devices but not indicated must be provided under this section of the specifications.

2.5 ELECTRICAL EQUIPMENT

Provide electrical equipment in conformance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide electrical motor driven equipment herein

specified complete with motors, motor starters, and controls. Motor controls, equipment, and wiring must be in accordance with NFPA 70.

#### 2.5.1 Electric Motors

Drive each electric motor-driven pump by a continuous-duty electric motor with enclosure type for specific service as defined in paragraph HYDRONIC PUMPS. Motor must have a 1.5 service factor. Provide squirrel-cage induction motors having normal-starting-torque and low-starting-current characteristics, and of sufficient size so that the nameplate horsepower rating will not be exceeded throughout the entire published pump characteristic curve. Integral size motors must be the premium efficiency type in accordance with NEMA MG 1. Pump electric motor efficiencies must meet or exceed the requirements of the ICC IgCC standard. Motor bearings must provide smooth operations under the conditions encountered for the life of the motor. Provide adequate thrust bearing in the motor to carry the weight of all rotating parts plus the hydraulic thrust and be capable of withstanding upthrust imposed during pump starting and under variable pumping head conditions specified. Motors must be rated 460 volts, 3 phase, 60 Hz and such rating must be stamped on the nameplate. Provide motors in conformance with NEMA MG 1.

#### 2.5.2 Control Equipment

Manually controlled pumps must have START-STOP pushbutton in cover. Automatically controlled pumps must have three-position "MANUAL-OFF-AUTOMATIC" selector switch in cover. Provide additional controls or protective devices as indicated.

#### 2.5.3 Variable Speed Control

The variable speed motor controllers must meet the requirements of UFGS 26 29 23 ADJUSTABLE SPEED DRIVE SYSTEMS UNDER 600 VOLTS.

### 2.6 EQUIPMENT APPURTENANCES

#### 2.6.1 Attachments

Furnish all necessary bolts, nuts, washers, bolt sleeves, and other types of attachments with the equipment for the installation of the equipment. Bolts conform to the requirements of ASTM A307 and hexagonal nuts of the same quality as the bolts used. Threads must be clean-cut and conform to ASME B1.1. Bolts, nuts, and washers specified to be galvanized or not otherwise indicated or specified, must be zinc coated after being threaded, by the hot-dip process conforming to ASTM A123/A123M as appropriate. Bolts, nuts, and washers specified or indicated to be stainless steel must be Type 316.

#### 2.6.2 Equipment Guards

Provide equipment driven by open shafts, belts, chains, or gears with all-metal guards enclosing the drive mechanism. Secure guards in position with steel braces or straps that permit easy removal for servicing the equipment. Coupler guards must comply with current national safety standards including 29 CFR 1910.219 and NEMA Z535.4. Provide guards with gaps no greater than 0.250 inches, safety orange in color, and have an NEMA Z535.4 compliant warning label.

### 2.6.3 Tools

Furnish a complete set of all special tools which may be necessary for the adjustment, operation, maintenance, and disassembly of all equipment. Special tools are considered to be those tools which because of their limited use are not normally available, but which are necessary for the particular equipment. Special tools must be high-grade, smooth, forged, alloy, tool steel. Furnish one pressure grease gun for each type of grease required. Deliver all tools at the same time as the equipment to which they pertain. Properly store and safeguard such tools until completion of the work, at which time deliver them to the Contracting Officer.

### 2.7 FINISHES

All motors, pump casings, and similar parts of equipment must be thoroughly cleaned, primed, and given two finish coats of paint at the factory in accordance with the recommendations of the manufacturer. Give ferrous surfaces not to be painted a shop coat of grease or other suitable rust-resistant coating.

### 2.8 FACTORY TESTS

Pumps must be tested by the manufacturer or a nationally recognized testing agency in compliance with HI 1.3. Submit certified test results.

## PART 3 EXECUTION

### 3.1 EXAMINATION

After becoming familiar with all details of the work, verify all dimensions in the field and advise the Contracting Officer of any discrepancy before performing the work.

### 3.2 INSTALLATION

Install each pump and motor in accordance with the written instructions of the manufacturer. Provide access space around the device for servicing no less than the minimum recommended by the manufacturer.

### 3.3 FIELD QUALITY CONTROL

After installation of the pumping units and appurtenances, including coupling guard, is complete, carry out operating tests to assure that the pumping installation operates properly. Make arrangements to have the manufacturer's representatives present when field equipment tests are made.

Give each pumping unit a running field test in the presence of the Contracting Officer for a minimum of 2 hours. Operate each pumping unit at its rated capacity or such other point on its head-capacity curve selected by the Contracting Officer. Provide an accurate and acceptable method of measuring the discharge flow. Tests must assure that the units and appurtenances have been installed correctly, that there is no objectionable heating, vibration, or noise from any parts, and that all manual and automatic controls function properly. If any deficiencies are revealed during any tests, correct such deficiencies and reconduct the tests.

Submit test reports in booklet form showing all field tests performed to adjust each component and all field tests performed to prove compliance

with the specified performance criteria, upon completion and testing of the installed system. Each test report must indicate the final position of controls.

### 3.4 FIELD PAINTING

Do not paint stainless steel, galvanized steel, and nonferrous surfaces.

#### 3.4.1 Touch-up painting

Factory painted items requiring touching up in the field must be thoroughly cleaned of all foreign material, and primed and topcoated with the manufacturer's standard factory finish.

#### 3.4.2 Exposed Ferrous Surfaces

Paint exposed ferrous surfaces with two coats of enamel paint conforming to SSPC Paint 21. Solvent clean factory primed surfaces before painting. Surfaces that have not been factory primed must be prepared and primed with one coat of SSPC Paint 25 or in accordance with the enamel paint manufacturer's recommendations.

### 3.5 CLOSEOUT ACTIVITIES

#### 3.5.1 Operation and Maintenance Manuals

Submit one complete set at the time the tests procedure is submitted; remaining sets before the contract is completed. Permanently bind each in a hard cover. Inscribe the following identification on the covers: the words "OPERATING AND MAINTENANCE INSTRUCTIONS," name and location of the building, name of the Contractor, and contract number. Place flysheets before instructions covering each subject. Use 8-1/2 by 11 inches paper for instruction sheets, with large sheets of drawings folded in.

Include, but do not limit to, the following in the Instructions:

- a. System layout showing piping, valves, and controls.
- b. Approved wiring and control diagrams including variable frequency drives.
- c. A control sequence describing startup, operation, and shutdown.
- d. Operating and maintenance instructions for each piece of equipment, including task list for routine maintenance, routine inspections, intermediate inspections, and annual inspections; lubrication instructions; and troubleshooting guide.
- e. Manufacturer's bulletins, cuts, and descriptive data; and parts list and recommended spare parts.

#### 3.5.2 Training

Upon completion of the work, and at a time designated by the Contracting Officer, provide the services of one or more competent engineers for a training period of not less than 8 hours to instruct a representative of the Government in the contents of the operation and maintenance manuals for the equipment furnished under these specifications. These field instructions must cover all the items contained in the bound

instructions. Submit the training course curriculum and training instructions 14 days prior to the start of training.

-- End of Section --



SECTION 23 23 00

REFRIGERANT PIPING  
**08/21**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

|               |                                                                                 |
|---------------|---------------------------------------------------------------------------------|
| AHRI 710 I-P  | (2009) Performance Rating of Liquid-Line Driers                                 |
| AHRI 720      | (2002) Refrigerant Access Valves and Hose Connectors                            |
| AHRI 750 I-P  | (2016) Performance Rating of Thermostatic Refrigerant Expansion Valves          |
| AHRI 760 I-P  | (2014) Performance Rating of Solenoid Valves for Use with Volatile Refrigerants |
| AHRI 1370 I-P | (2017) Performance Rating of Electronic Expansion Valves                        |

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

|                  |                                                                                                                                                                                |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASHRAE 15 & 34   | (2013) ASHRAE Standard 34-2016 Safety Standard for Refrigeration Systems/ASHRAE Standard 34-2016 Designation and Safety Classification of Refrigerants-ASHRAE Standard 34-2016 |
| ASHRAE 17        | (2022) Method of Testing Capacity of Thermostatic Refrigerant Expansion Valves                                                                                                 |
| ASHRAE 90.1 - IP | (2019) Energy Standard for Buildings Except Low-Rise Residential Buildings                                                                                                     |

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|              |                                                                                      |
|--------------|--------------------------------------------------------------------------------------|
| ASME B1.20.1 | (2013; R 2018) Pipe Threads, General Purpose (Inch)                                  |
| ASME B16.3   | (2021) Malleable Iron Threaded Fittings, Classes 150 and 300                         |
| ASME B16.5   | (2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard |
| ASME B16.9   | (2018) Factory-Made Wrought Butt welding Fittings                                    |

|                  |                                                                                |
|------------------|--------------------------------------------------------------------------------|
| ASME B16.11      | (2022) Forged Fittings, Socket-Welding and Threaded                            |
| ASME B16.21      | (2021) Nonmetallic Flat Gaskets for Pipe Flanges                               |
| ASME B16.22      | (2021) Wrought Copper and Copper Alloy Solder Joint Pressure Fittings          |
| ASME B16.26      | (2018) Standard for Cast Copper Alloy Fittings for Flared Copper Tubes         |
| ASME B31.1       | (2022) Power Piping                                                            |
| ASME B31.5       | (2022) Refrigeration Piping and Heat Transfer Components                       |
| ASME B31.9       | (2020) Building Services Piping                                                |
| ASME B40.100     | (2022) Pressure Gauges and Gauge Attachments                                   |
| ASME BPVC SEC IX | (2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications |

AMERICAN WELDING SOCIETY (AWS)

|                  |                                                                      |
|------------------|----------------------------------------------------------------------|
| AWS A5.8/A5.8M   | (2019) Specification for Filler Metals for Brazing and Braze Welding |
| AWS A5.31/A5.31M | (2012) Specification for Fluxes for Brazing and Braze Welding        |
| AWS BRH          | (2007; 5th Ed) Brazing Handbook                                      |
| AWS D1.1/D1.1M   | (2020; Errata 1 2021) Structural Welding Code - Steel                |
| AWS Z49.1        | (2021) Safety in Welding and Cutting and Allied Processes            |

ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                                         |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A53/A53M   | (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless                                                   |
| ASTM A193/A193M | (2024) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications |
| ASTM A334/A334M | (2004a; R 2021) Standard Specification for Seamless and Welded Carbon and Alloy-Steel Tubes for Low-Temperature Service                                 |
| ASTM A653/A653M | (2023) Standard Specification for Steel                                                                                                                 |

Sheet, Zinc-Coated (Galvanized) or  
Zinc-Iron Alloy-Coated (Galvannealed) by  
the Hot-Dip Process

|               |                                                                                                             |
|---------------|-------------------------------------------------------------------------------------------------------------|
| ASTM B32      | (2020) Standard Specification for Solder Metal                                                              |
| ASTM B62      | (2017) Standard Specification for Composition Bronze or Ounce Metal Castings                                |
| ASTM B75/B75M | (2020) Standard Specification for Seamless Copper Tube                                                      |
| ASTM B117     | (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus                                           |
| ASTM B280     | (2023) Standard Specification for Seamless Copper Tube for Air Conditioning and Refrigeration Field Service |
| ASTM B813     | (2024) Standard Specification for Liquid and Paste Fluxes for Soldering of Copper and Copper Alloy Tube     |
| ASTM D520     | (2000; R 2011) Zinc Dust Pigment                                                                            |
| ASTM D3308    | (2012; R 2017) Standard Specification for PTFE Resin Skived Tape                                            |
| ASTM E84      | (2023) Standard Test Method for Surface Burning Characteristics of Building Materials                       |

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS  
INDUSTRY (MSS)

|           |                                                                                                                |
|-----------|----------------------------------------------------------------------------------------------------------------|
| MSS SP-58 | (2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation |
|-----------|----------------------------------------------------------------------------------------------------------------|

U.S. DEPARTMENT OF DEFENSE (DOD)

|              |                                                    |
|--------------|----------------------------------------------------|
| UFC 3-301-01 | (2023; with Change 2, 2024) Structural Engineering |
|--------------|----------------------------------------------------|

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Refrigerant Piping System; G, AE

SD-03 Product Data

Refrigerant Piping System

Qualifications

Refrigerant Piping Tests

Verification of Dimensions

SD-06 Test Reports

Refrigerant Piping Tests

SD-07 Certificates

Service Organization

SD-10 Operation and Maintenance Data

Maintenance; G, AE

Operation and Maintenance Manuals; G, AE

Demonstrations; G, AE

### 1.3 QUALITY ASSURANCE

#### 1.3.1 Qualifications

Submit six copies of qualified procedures, and list of names and identification symbols of qualified welders and welding operators, prior to non-factory welding operations. Weld piping in accordance with the qualified procedures using performance qualified welders and welding operators. Procedures and welders must be qualified in accordance with ASME BPVC SEC IX. Welding procedures qualified by others, and welders and welding operators qualified by another employer may be accepted as permitted by ASME B31.1. Notify the Contracting Officer 24 hours in advance of tests to be performed at the work site, if practical. The welder or welding operator must apply the personally assigned symbol near each weld made, as a permanent record. Weld structural members in accordance with Section 05 12 00 STRUCTURAL STEEL. Welding and nondestructive testing procedures are specified in Section 40 17 26.00 20 WELDING PRESSURE PIPING.

#### 1.3.2 Contract Drawings

Because of the small scale of the drawings, it is not possible to indicate all offsets, fittings, and accessories that may be required. Carefully investigate the plumbing, fire protection, electrical, structural and finish conditions that would affect the work to be performed and arrange such work accordingly, furnishing required offsets, fittings, and accessories to meet such conditions.

### 1.4 DELIVERY, STORAGE, AND HANDLING

Protect stored items from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Proper protection and care of all material both before and during installation is the Contractor's responsibility. Replace any materials found to be damaged at

the Contractor's expense. During installation, cap piping and similar openings to keep out dirt and other foreign matter.

## 1.5 MAINTENANCE

### 1.5.1 General

Submit Data Package 2 plus operation and maintenance data complying with the requirements of Section 01 78 23 OPERATION AND MAINTENANCE DATA and as specified herein.

## PART 2 PRODUCTS

### 2.1 STANDARD COMMERCIAL PRODUCTS

- a. Provide materials and equipment which are standard products of a manufacturer regularly engaged in the manufacturing of such products, that are of a similar material, design and workmanship and that have been in satisfactory commercial or industrial use for 2 years prior to bid opening.
- b. The 2 year use must include applications of equipment and materials under similar circumstances and of similar size. The 2 years' experience must be satisfactorily completed by a product which has been sold or is offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures. Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation, for not less than 6000 hours exclusive of the manufacturer's factory tests, can be shown.
- c. Products must be supported by a service organization. System components must be environmentally suitable for the indicated locations. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. The service organizations must be reasonably convenient to the equipment installation and be able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.
- d. Exposed equipment moving parts, parts that produce high operating temperature, parts which may be electrically energized, and parts that may be a hazard to operating personnel must be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Install safety devices so that proper operation of equipment is not impaired. Welding and cutting safety requirements must be in accordance with AWS Z49.1.
- e. Provide the manufacturer's standard catalog data, at least 5 weeks prior to the purchase or installation of a particular component. Highlight the data to show information such as, but not limited to, material, size, options, performance charts, and curves in adequate detail to demonstrate compliance with contract requirements. Include the manufacturer's recommended installation instructions and procedures in the data provided. Provide data for the following components as a minimum:
  - (1) Piping and Fittings
  - (2) Valves
  - (3) Piping Accessories

(4) Pipe Hangers, Inserts, and Supports

2.2 ELECTRICAL WORK

Manual or automatic control and protective or signal devices required for the operation specified and any control wiring required for controls and devices specified, but not shown, must be provided.

2.3 REFRIGERANT PIPING SYSTEM

Provide refrigerant piping, valves, fittings, and accessories in accordance with ASHRAE 15 & 34 and ASME B31.5, except as specified herein. Refrigerant piping, valves, fittings, and accessories must be compatible with the fluids used and capable of withstanding the pressures and temperatures of the service. Refrigerant piping, valves, and accessories used for refrigerant service must be cleaned, dehydrated, and sealed (capped or plugged) prior to shipment from the manufacturer's plant. Submit drawings, at least 5 weeks prior to beginning construction, provided in adequate detail to demonstrate compliance with contract requirements. Drawings must consist of:

- a. Piping layouts which identify all valves and fittings.
- b. Plans and elevations which identify clearances required for maintenance and operation.

2.4 PIPE, FITTINGS AND END CONNECTIONS (JOINTS)

2.4.1 Steel Pipe

Steel pipe for refrigerant service must conform to ASTM A53/A53M, Schedule 40, Type E or S, Grades A or B. Do not use Type F pipe.

2.4.1.1 Welded Fittings and Connections

Butt-welded fittings must conform to ASME B16.9. Socket-welded fittings must conform to ASME B16.11. Identify welded fittings with the appropriate grade and marking symbol. Welded valves and pipe connections (both butt-welds and socket-welds types) must conform to ASME B31.9.

2.4.1.2 Threaded Fittings and Connections

Threaded fitting must conform to ASME B16.3. Threaded valves and pipe connections must conform to ASME B1.20.1.

2.4.1.3 Flanged Fittings and Connections

Flanges must conform to ASME B16.5, Class 150. Gaskets must be non-asbestos compressed material in accordance with ASME B16.21, 1/16 inch thickness, full face or self-centering flat ring type. Gaskets must contain aramid fibers bonded with styrene butadiene rubber (SBR) or nitrile butadiene rubber (NBR). Bolts, nuts, and bolt patterns must conform to ASME B16.5. Bolts must be high or intermediate strength material conforming to ASTM A193/A193M.

2.4.2 Steel Tubing

Tubing must be cold-rolled, electric-forged, welded-steel in accordance with ASTM A334/A334M, Grade 1. Joints and fittings must be socket type

provided by the steel tubing manufacturer.

#### 2.4.3 Copper Tubing

Provide copper tubing conforming to ASTM B280 annealed or hard drawn as required. Copper tubing must bear the product identification markings in accordance with ASTM B280, "ACR" must be present on copper tubing. Copper tubing must be soft annealed where bending is required and hard drawn where no bending is required. Soft annealed copper tubing must not be used in sizes larger than 1-3/8 inches. Joints must be brazed except that joints on lines 7/8 inch and smaller may be flared. Cast copper alloy fittings for flared copper tube must conform to ASME B16.26 and ASTM B62. Wrought copper and bronze solder-joint pressure fittings must conform to ASME B16.22 and ASTM B75/B75M. Joints and fittings for brazed joint must be wrought-copper or forged-brass sweat fittings. Cast sweat-type joints and fittings are not allowed for brazed joints. Brass or bronze adapters for brazed tubing may be used for connecting tubing to flanges and to threaded ends of valves and equipment.

#### 2.4.4 Solder

Solder must conform to ASTM B32, grade Sb5, tin-antimony alloy for service pressures up to 150 psig. Solder flux must be liquid or paste form, non-corrosive and conform to ASTM B813.

#### 2.4.5 Brazing Filler Metal

Filler metal must conform to AWS A5.8/A5.8M, Type BAg-5 with AWS Type FB3-A or Type FB3-C flux, except Type BCuP-3, BCuP-4, or BCuP-5 may be used for brazing copper-to-copper joints. BAlSi-4 with AWS Type FB1-A flux may be used when joining copper piping to aluminum components.

#### 2.4.6 Brazing Flux

Brazing flux must conform to AWS A5.31/A5.31M, Type FB3-A or Type FB3-C when using Type BAg-5 filler metal. Type FB1-A is to be used with Type BAlSi-4 filler metal.

#### 2.4.7 Press Fittings

Press fittings are not acceptable for use in refrigerant piping systems.

### 2.5 VALVES

Valves must be designed, manufactured, and tested specifically for refrigerant service. The valve material and all internal components must be compatible with the specific refrigerant and lubricant used. Valve bodies must be of brass, bronze, steel, or ductile iron construction. Valves 1 inch and smaller must have brazed or socket welded connections. Valves larger than 1 inch must have butt welded end connections. Do not use threaded end connections, except in pilot pressure or gauge lines where maintenance disassembly is required and welded flanges cannot be used. Internal parts must be removable for inspection or replacement without applying heat or breaking pipe connections. Valve stems exposed to the atmosphere must be stainless steel or corrosion resistant metal plated carbon steel. Direction of flow must be legibly and permanently indicated on the valve body. Control valve inlets must be fitted with integral or adapted strainer or filter where recommended or required by the manufacturer. Purge, charge and receiver valves must be of

manufacturer's standard configuration.

#### 2.5.1 Refrigerant Stop Valves

Valve must be the globe or full-port ball type with a back-seating stem especially packed for refrigerant service. Valve packing must be replaceable under line pressure. Provide valve with a handwheel or wrench operator and a seal cap. Valve must be the straight or angle pattern design as indicated.

#### 2.5.2 Check Valves

Valve must be the swing or lift type as required to provide positive shutoff at the differential pressure indicated. Valve must be provided with resilient seat.

#### 2.5.3 Liquid Solenoid Valves

Provide valves that comply with AHRI 760 I-P and are suitable for continuous duty with applied voltages 15 percent under and 5 percent over nominal rated voltage at maximum and minimum encountered pressure and temperature service conditions. Valves must be direct-acting or pilot-operating type, packless, except that packed stem, seal capped, manual lifting provisions must be furnished. Provide solenoid coils that are moisture-proof, UL approved, totally encapsulated or encapsulated and metal jacketed as required. Valves must have safe working pressure of 610 psi and a maximum operating pressure differential of at least 200 psi at 85 percent rated voltage. Valves must have an operating pressure differential suitable for the refrigerant used.

#### 2.5.4 Expansion Valves

Provide valve conforming to AHRI 750 I-P and ASHRAE 17. Valve must be the diaphragm and spring-loaded type with internal or external equalizers, and bulb and capillary tubing. Provide valve with an external superheat adjustment along with a seal cap. Internal equalizers may be utilized where flowing refrigerant pressure drop between outlet of the valve and inlet to the evaporator coil is negligible and pressure drop across the evaporator is less than the pressure difference corresponding to 2 degrees F of saturated suction temperature at evaporator conditions. Bulb charge must be determined by the manufacturer for the application and such that liquid will remain in the bulb at all operating conditions. Do not use gas limited liquid charged valves and other valve devices for limiting evaporator pressure without a distributor or discharge tube or effective means to prevent loss of control when bulb becomes warmer than valve body. Pilot-operated valves must have a characterized plug to provide required modulating control. A de-energized solenoid valve may be used in the pilot line to close the main valve in lieu of a solenoid valve in the main liquid line. Provide an isolatable pressure gauge in the pilot line, at the main valve. Automatic pressure reducing or constant pressure regulating expansion valves may be used only where indicated or for constant evaporator loads.

#### 2.5.5 Electronic Expansion Valves

Valve must conform to AHRI 1370 I-P and ASHRAE 17. The valve must prevent the return of liquid to the compressor in the event of power loss or low superheat.



#### 2.5.6 Safety Relief Valves

Valve must be the two-way type, unless indicated otherwise. Valve must bear the ASME code symbol. Valve capacity must be certified by the National Board of Boiler and Pressure Vessel Inspectors. Valve must be of an automatically reseating design after activation.

#### 2.5.7 Evaporator Pressure Regulators, Direct-Acting

Valve must include a diaphragm/spring assembly, external pressure adjustment with seal cap, and pressure gauge port. Valve must maintain a constant inlet pressure by balancing inlet pressure on diaphragm against an adjustable spring load. Pressure drop at system design load must not exceed the pressure difference corresponding to a 2 degrees F change in saturated refrigerant temperature at evaporator operating suction temperature. Spring must be selected for indicated maximum allowable suction pressure range.

#### 2.5.8 Refrigerant Access Valves

Provide refrigerant access valves and hose connections in accordance with AHRI 720.

### 2.6 PIPING ACCESSORIES

#### 2.6.1 Filter Driers

Driers must conform to AHRI 710 I-P. Sizes 5/8 inch and larger must be the full flow, replaceable core type. Sizes 1/2 inch and smaller must be the sealed type. Cores must be of suitable desiccant that will not plug, cake, dust, channel, or break down, and must remove water, acid, and foreign material from the refrigerant. Construct filter driers so that none of the desiccant will pass into the refrigerant lines. Minimum bursting pressure must be 1,500 psi.

#### 2.6.2 Sight Glass and Liquid Level Indicator

##### 2.6.2.1 Assembly and Components

Assembly must be pressure- and temperature-rated and constructed of materials suitable for the service. Glass must be borosilicate type. Ferrous components subject to condensation must be electro-galvanized.

##### 2.6.2.2 Gauge Glass

Gauge glass must include top and bottom isolation valves fitted with automatic checks, and packing followers; red-line or green-line gauge glass; elastomer or polymer packing to suit the service; and gauge glass guard.

##### 2.6.2.3 Bull's-Eye and Inline Sight Glass Reflex Lens

Provide bull's-eye and inline sight glass reflex lens for dead-end liquid service. For pipe line mounting, provide two plain lenses in one body suitable for backlighted viewing.

##### 2.6.2.4 Moisture Indicator

Indicator must be a self-reversible action, moisture reactive, color

changing media. Indicator must be furnished with full-color-printing tag containing color, moisture, and temperature criteria. Unless otherwise indicated, the moisture indicator must be an integral part of each corresponding sight glass.

#### 2.6.3 Vibration Dampeners

Dampeners must be of the all-metallic bellows and woven-wire type.

#### 2.6.4 Flexible Pipe Connectors

Connector must be a composite of interior corrugated phosphor bronze or Type 300 Series stainless steel, as required for fluid service, with exterior reinforcement of bronze, stainless steel or monel wire braid. Assembly must be constructed with a safety factor of not less than 4 at 300 degrees F. Unless otherwise indicated, the length of a flexible connector must be as recommended by the manufacturer for the service intended.

#### 2.6.5 Strainers

Strainers used in refrigerant service must have brass or cast-iron body, Y-or angle-pattern, cleanable, not less than 60-mesh noncorroding screen of an area to provide net free area not less than ten times the pipe diameter with pressure rating compatible with the refrigerant service. Screens must be stainless steel or monel and reinforced spring-loaded where necessary for bypass-proof construction.

#### 2.6.6 Pressure and Vacuum Gauges

Provide gauges conforming to ASME B40.100 with throttling type needle valve or a pulsation dampener and shut-off valve. Gauge must be a minimum of 3-1/2 inches in diameter with a range from 0 psig to approximately 1.5 times the maximum system working pressure. Select each gauge range so that at normal operating pressure, the needle is within the middle-third of the range.

#### 2.6.7 Temperature Gauges

Provide industrial duty type temperature gauges for the required temperature range. Gauges must have Fahrenheit scale in 2 degrees graduations scale (black numbers) on a white face. The pointer must be adjustable. Provide rigid stem type temperature gauges in thermowells located within 5 feet of the finished floor. Provide universal adjustable angle type or remote element type temperature gauges in thermowells located 5 to 7 feet above the finished floor. Provide remote element type temperature gauges in thermowells located 7 feet above the finished floor.

##### 2.6.7.1 Stem Cased-Glass

Provide stem cased-glass case composed of polished stainless steel or cast aluminum, 9 inches long, with clear acrylic lens, and non-mercury filled glass tube with indicating-fluid column.

##### 2.6.7.2 Bimetallic Dial

Provide bimetallic dial type case that is greater than 3-1/2 inches, stainless steel, and hermetically sealed with clear acrylic lens. Bimetallic element must be silicone dampened and unit fitted with external calibrator adjustment. Accuracy must be one percent of dial range.

#### 2.6.7.3 Liquid-, Solid-, and Vapor-Filled Dial

Provide liquid-, solid-, and vapor-filled dial type cases that are greater than 3-1/2 inches, stainless steel or cast aluminum with clear acrylic lens. Fill must be nonmercury, suitable for encountered cross-ambients, and connecting capillary tubing must be double-braided bronze.

#### 2.6.7.4 Thermowell

Thermowell must be identical size, 1/2 or 3/4 inch NPT connection, brass or stainless steel. Where test wells are indicated, provide captive plug-fitted type 1/2 inch NPT connection suitable for use with either engraved stem or standard separable socket thermometer or thermostat. Mercury must not be used in thermometers. Extended neck thermowells must be of sufficient length to clear insulation thickness by 1 inch.

#### 2.6.8 Pipe Hangers, Inserts, and Supports

Provide pipe hangers, inserts, guides, and supports conforming to MSS SP-58.

#### 2.6.9 Escutcheons

Escutcheons must be chromium-plated iron or chromium-plated brass, either one piece or split pattern, held in place by internal spring tension or set screws.

### 2.7 FABRICATION

#### 2.7.1 Factory Coating

Unless otherwise specified, equipment and component items, when fabricated from ferrous metal, must be factory finished with the manufacturer's standard finish, except that items located outside of buildings must have weather resistant finishes that will withstand 500 hours exposure to the salt spray test specified in ASTM B117 using a 5 percent sodium chloride solution. Immediately after completion of the test, the specimen must show no signs of blistering, wrinkling, cracking, or loss of adhesion and no sign of rust creepage beyond 1/8 inch on either side of the scratch mark. Cut edges of galvanized surfaces where hot-dip galvanized sheet steel is used must be coated with a zinc-rich coating conforming to ASTM D520, Type I.

#### 2.7.2 Factory Applied Insulation

Factory installed insulation must be in accordance with ASHRAE 90.1 - IP. Refrigerant suction lines between the cooler and each compressor and cold gas inlet connections to gas cooled motors must be insulated with not less than 1/2 inch thick unicellular plastic foam. Factory insulated items installed outdoors are not required to be fire-rated. As a minimum, factory insulated items installed indoors must have a flame spread index no higher than 25 and a smoke developed index no higher than 50. Factory insulated items (no jacket) installed indoors and which are located in air plenums, in ceiling spaces, and in attic spaces must have a flame spread index no higher than 25 and a smoke developed index no higher than 50. Flame spread and smoke developed indexes must be determined by ASTM E84. Test insulation in the same density and installed thickness as the material to be used in the actual construction. Test material supplied by a manufacturer with a jacket as a composite material. Provide jackets,

facings, and adhesives that have a flame spread index less than 25 and a smoke developed index less than 50 when tested in accordance with ASTM E84.

## PART 3 EXECUTION

### 3.1 EXAMINATION

After becoming familiar with all details of the work, perform a verification of dimensions in the field. Submit a letter, at least 2 weeks prior to beginning construction, including the date the site was visited, conformation of existing conditions, and any discrepancies found before performing any work.

### 3.2 INSTALLATION

Pipe and fitting installation must conform to the requirements of ASME B31.1. Cut pipe accurately to measurements established at the jobsite, and work into place without springing or forcing, completely clearing all windows, doors, and other openings. Cutting or other weakening of the building structure to facilitate piping installation is not permitted without written approval. Cut pipe or tubing square, remove by reaming, and permit free expansion and contraction without causing damage to the building structure, pipe, joints, or hangers.

#### 3.2.1 Directional Changes

Make changes in direction with fittings, except that bending of pipe 4 inches and smaller is permitted, provided a pipe bender is used and wide weep bends are formed. Mitering or notching pipe or other similar construction to form elbows or tees is not permitted. The centerline radius of bends must not be less than 6 diameters of the pipe. Bent pipe showing kinks, wrinkles, flattening, or other malformations will not be accepted.

#### 3.2.2 Functional Requirements

Install piping 1/2 inch/10 feet of pipe in the direction of flow to ensure adequate oil drainage. Properly cap or plug open ends of refrigerant lines or equipment during installation to keep moisture, dirt, or other foreign material out of the system. Piping must remain capped until installation. Equipment piping must be in accordance with the equipment manufacturer's recommendations and the contract drawings. Equipment and piping arrangements must fit into space allotted and allow adequate acceptable clearances for installation, replacement, entry, servicing, and maintenance.

#### 3.2.3 Fittings and End Connections

##### 3.2.3.1 Threaded Connections

Make threaded connections with tapered threads and make tight with PTFE tape complying with ASTM D3308 or equivalent thread-joint compound applied to the male threads only. Do not show more than three threads after the joint is made.

##### 3.2.3.2 Brazed Connections

Perform brazing in accordance with AWS BRH, except as modified herein. During brazing, fill the pipe and fittings with a pressure regulated inert

gas, such as nitrogen, to prevent the formation of scale. Before brazing copper joints, clean both the outside of the tube and the inside of the fitting with a wire fitting brush until the entire joint surface is bright and clean. Do not use brazing flux on copper-to-copper connections. Remove surplus brazing material at all joints. Make steel tubing joints in accordance with the manufacturer's recommendations. Paint joints in steel tubing with the same material as the baked-on coating within 8 hours after joints are made. Protect tubing against oxidation during brazing by continuous purging of the inside of the piping using nitrogen. Support piping prior to brazing and do not spring or force.

#### 3.2.3.3 Welded Connections

Fusion-weld joints in steel refrigerant piping. Make branch connections with welding tees or forged welding branch outlets. Thoroughly clean pipe of all scale and foreign matter before the piping is assembled. During welding, fill the pipe and fittings with an inert gas, such as nitrogen, to prevent the formation of scale. Beveling, alignment, heat treatment, and inspection of weld must conform to ASME B31.1. Remove and reweld weld defects at no additional cost to the Government. Store and dry electrodes in accordance with AWS D1.1/D1.1M or as recommended by the manufacturer. Do not use electrodes that have been wetted or that have lost any of their coating

#### 3.2.3.4 Flared Connections

When flared connections are used, use a suitable lubricant between the back of the flare and the nut in order to avoid tearing the flare while tightening the nut.

#### 3.2.3.5 Flanged Connections

When steel refrigerant piping is used, provide union or flange joints in each line immediately preceding the connection to each piece of equipment requiring maintenance, such as compressors, coils, chillers, control valves, and other similar items. Flanged joints must be assembled square end tight with matched flanges, gaskets, and bolts. Provide gaskets that are suitable for use with the refrigerants to be handled.

#### 3.2.4 Valves

##### 3.2.4.1 General

Install refrigerant stop valves on each side of each piece of equipment such as compressors condensers, evaporators, receivers, and other similar items in multiple-unit installation, to provide partial system isolation as required for maintenance or repair. Install stop valves with stems horizontal unless otherwise indicated. Install ball valves must be installed with stems positioned to facilitate operation and maintenance. Isolating valves for pressure gauges and switches must be external to thermal insulation. Safety switches must not be fitted with isolation valves. Filter dryers having access ports may be considered a point of isolation. Purge valves must be provided at all points of systems where accumulated non-condensable gases would prevent proper system operation. Valves must be furnished to match line size, unless otherwise indicated or approved.

#### 3.2.4.2 Expansion Valves

Install expansion valves with the thermostatic expansion valve bulb located on top of the suction line when the suction line is less than 2-1/8 inches in diameter and at the 4 o'clock or 8 o'clock position on lines larger than 2-1/8 inches. Fasten the bulb securely with two clamps. Insulate the bulb. Install the bulb in a horizontal portion of the suction line, if possible, with the pigtail on the bottom. If the bulb must be installed in a vertical line, the bulb tubing must be facing up.

#### 3.2.4.3 Valve Identification

Tag each system valve, including those which are part of a factory assembly. Tags must be in alphanumeric sequence, progressing in direction of fluid flow. Tags must be embossed, engraved, or stamped plastic or nonferrous metal of various shapes, sized approximately 1-3/8 inch diameter, or equivalent dimension, substantially attached to a component or immediately adjacent thereto. Attach tags with nonferrous, heavy duty, bead or link chain, 14 gauge annealed wire, nylon cable bands or as approved. Reference tag numbers in Operation and Maintenance Manuals and system diagrams.

#### 3.2.5 Vibration Dampers

Provide vibration damper in the suction and discharge lines on spring mounted compressors. Install vibration dampers parallel with the shaft of the compressor and anchor firmly at the upstream end on the suction line and the downstream end in the discharge line.

#### 3.2.6 Strainers

Provide strainers immediately ahead of solenoid valves and expansion devices. Strainers may be an integral part of an expansion valve.

#### 3.2.7 Filter Dryer

Provide a liquid line filter dryer on each refrigerant circuit located such that all liquid refrigerant passes through a filter dryer. Size dryers in accordance with the manufacturer's recommendations for the system in which it is installed. Install dryers such that it can be isolated from the system, the isolated portion of the system evacuated, and the filter dryer replaced. Install dryers in the horizontal position except replaceable core filter dryers may be installed in the vertical position with the access flange on the bottom.

#### 3.2.8 Sight Glass

Install a moisture indicating sight glass in all refrigerant circuits downstream of all filter dryers and where indicated. Provide full line size sight glasses.

#### 3.2.9 Discharge Line Oil Separator

Provide discharge line oil separator in the discharge line from each compressor. Connect the oil return line to the compressor as recommended by the compressor manufacturer.

### 3.2.10 Accumulator

Provide accumulators in the suction line to each compressor.

### 3.2.11 Flexible Pipe Connectors

Install connectors perpendicular to line of motion being isolated. Fit piping for equipment with bidirectional motion with two flexible connectors, in perpendicular planes. Install reinforced elastomer flexible connectors in accordance with manufacturer's instructions. Provide piping guides and restraints related to flexible connectors as required.

### 3.2.12 Temperature Gauges

Locate temperature gauges specifically on, but not limited to the following: the sensing element of each automatic temperature control device where a thermometer is not an integral part thereof, the liquid line leaving a receiver and the suction line at each evaporator or liquid cooler. Thermowells for insertion thermometers and thermostats must extend beyond thermal insulation surface not less than 1 inch.

### 3.2.13 Pipe Hangers, Inserts, and Supports

Pipe hangers, inserts, and supports must conform to MSS SP-58, except as modified herein. Do not use pipe hanger types 5, 12, and 26. Fabricate hangers used to support piping 2 inches and larger to permit adequate adjustment after erection while still supporting the load. Support piping subjected to vertical movement, when operating temperatures exceed ambient temperatures, by variable spring hangers and supports or by constant support hangers.

#### 3.2.13.1 Hangers

Do not use Type 3 on insulated piping. Type 24 may be used only on trapeze hanger systems or on fabricated frames.

#### 3.2.13.2 Inserts

Secure Type 18 inserts to concrete forms before concrete is placed. Continuous inserts which allow more adjustments may be used if they otherwise meet the requirements for Type 18 inserts.

#### 3.2.13.3 C-Clamps

Torque Type 19 and 23 C-clamps in accordance with MSS SP-58 and have both locknuts and retaining devices, furnished by the manufacturer. Field-fabricated C-clamp bodies or retaining devices are not acceptable.

#### 3.2.13.4 Angle Attachments

Furnish Type 20 attachments used on angles and channels with an added malleable-iron heel plate or adapter.

#### 3.2.13.5 Saddles and Shields

Where Type 39 saddle or Type 40 shield are permitted for a particular pipe attachment application, the Type 39 saddle, connected to the pipe, must be used on all pipe 4 inches and larger when the temperature of the medium is

60 degrees F or higher. Use Type 40 shields on all piping less than 4 inches and all piping 4 inches and larger carrying medium less than 60 degrees F. Use a high-density insulation insert of cellular glass under the Type 40 shield for piping 2 inches and larger.

#### 3.2.13.6 Horizontal Pipe Supports

Space horizontal pipe supports as specified in MSS SP-58 and install a support no more than 1 foot from the pipe fitting joint at each change in direction of the piping. Space pipe supports no more than 5 feet apart at valves.

#### 3.2.13.7 Vertical Pipe Supports

Support vertical pipe at each floor, except at slab-on-grade, and at intervals of not more than 15 feet not more than 8 feet from end of risers, and at vent terminations.

#### 3.2.13.8 Pipe Guides

Provide Type 35 guides using, steel, reinforced polytetrafluoroethylene (PTFE) or graphite slides where required to allow longitudinal pipe movement. Provide lateral restraints as required. Provide slide materials that are suitable for the system operating temperatures, atmospheric conditions, and bearing loads encountered.

#### 3.2.13.9 Steel Slides

Where steel slides do not require provisions for restraint of lateral movement, an alternate guide method may be used. On piping 4 inches and larger, use a Type 39 saddle. On piping under 4 inches, a Type 40 protection shield may be attached to the pipe or insulation and freely rest on a steel slide plate.

#### 3.2.13.10 High Temperature Guides with Cradles

Where there are high system temperatures and welding to piping is not desirable, the Type 35 guide must include a pipe cradle, welded to the guide structure and strapped securely to the pipe. Separate the pipe from the slide material by at least 4 inches, or by an amount adequate for the insulation, whichever is greater.

#### 3.2.13.11 Multiple Pipe Runs

In the support of multiple pipe runs on a common base member, use a clip or clamp where each pipe crosses the base support member. Spacing of the base support members must not exceed the hanger and support spacing required for an individual pipe in the multiple pipe run.

#### 3.2.13.12 Seismic Requirements

Support and brace piping and attached valves to resist seismic loads as specified under UFC 3-301-01 and 23 05 48.19 SEISMIC BRACING FOR HVAC. Provide structural steel required for reinforcement to properly support piping, headers, and equipment but not shown under this section. Specify material used for support under Section 05 12 00 STRUCTURAL STEEL.



### 3.2.13.13 Structural Attachments

Attachment to building structure concrete and masonry must be by cast-in concrete inserts, built-in anchors, or masonry anchor devices. Inserts and anchors must be applied with a safety factor not less than 5. Do not attach supports to metal decking. Construct masonry anchors for overhead applications of ferrous materials only. Provide structural steel brackets required to support piping, headers, and equipment, but not shown, under this section. Specify material used for support under Section 05 12 00 STRUCTURAL STEEL.

### 3.2.14 Pipe Alignment Guides

Provide pipe alignment guides where indicated for expansion loops, offsets, and bends and as recommended by the manufacturer for expansion joints, not to exceed 5 feet on each side of each expansion joint, and in lines 4 inches or smaller not more than 2 feet on each side of the joint.

### 3.2.15 Pipe Anchors

Provide anchors wherever necessary or indicated to localize expansion or to prevent undue strain on piping. Provide anchors consisting of heavy steel collars with lugs and bolts for clamping and attaching anchor braces, unless otherwise indicated. Install anchor braces in the most effective manner to secure the desired results using turnbuckles where required. Do not attach supports, anchors, or stays where they will injure the structure or adjacent construction during installation or by the weight of expansion of the pipeline. Where pipe and conduit penetrations of vapor barrier sealed surfaces occur, immediately anchor these items adjacent to each penetrated surface, to provide essentially zero movement within penetration seal. Submit detailed drawings of pipe anchors for approval before installation.

### 3.2.16 Building Surface Penetrations

Do not install sleeves in structural members except where indicated or approved. Provide galvanized sheet metal sleeves in non-load bearing surfaces conforming to ASTM A653/A653M, Coating Class G-90, 20 gauge. Provide uncoated carbon steel pipe sleeves in load bearing surfaces conforming to ASTM A53/A53M, Schedule 30. Apply sealants to moisture and oil-free surfaces and elastomers to not less than 1/2 inch depth. Do not install sleeves in structural members.

#### 3.2.16.1 Refrigerated Space

Fit refrigerated space building surface penetrations with sleeves fabricated from hand-lay-up or helically wound, fibrous glass reinforced polyester or epoxy resin with a minimum thickness equal to equivalent size Schedule 40 steel pipe. Construct sleeves with integral collar or fit cold side with a bonded slip-on flange or extended collar. In the case of masonry penetrations where sleeve is not cast-in, fill voids with latex mixed mortar cast to shape of sleeve and assemble flange/external collar type sleeve with butyl elastomer vapor barrier sealant through penetration to cold side surface vapor barrier overlap and fastened to surface with masonry anchors. Flash integral cast-in collar type sleeve as indicated. Normally seal noninsulated penetrating round surfaces to sleeve bore with mechanically expandable seals in vapor tight manner and insulate remaining warm and cold side sleeve depth with not less than 4 inches of foamed-in-place rigid polyurethane or foamed-in-place silicone elastomer.

Apply vapor barrier sealant to finish warm side insulation surface. Insulate warm side of penetrating surface beyond vapor barrier sealed sleeve insulation for a distance which prevents condensation. Seal wires in refrigerated space surface penetrating conduit with vapor barrier plugs or compound to prevent moisture migration through conduit and condensation therein.

### 3.2.16.2 General Service Areas

Extend each sleeve through its respective wall, floor, or roof, and cut flush with each surface. Provide pipes passing through concrete or masonry wall or concrete floors or roofs with pipe sleeves fitted into place at the time of construction. Provide sleeves that allow a minimum of 1/4 inch all-around clearance between bare pipe and sleeves or between jacketed-insulation and sleeves. Except in pipe chases or interior walls, seal the annular space between pipe and sleeve or between jacket over-insulation and sleeve in accordance with Section 07 92 00 JOINT SEALANTS.

### 3.2.16.3 Waterproof Penetrations

Install pipes passing through roof or floor waterproofing membrane through a 17 ounce copper sleeve, or a 0.032 inch thick aluminum sleeve, each within an integral skirt or flange. Form flashing sleeve, and extend skirt or flange greater than 8 inches from the pipe and set over the roof or floor membrane in a troweled coating of bituminous cement. Extend the flashing sleeve up the pipe a minimum of 2 inches above the roof or floor penetration. Seal the annular space between the flashing sleeve and the bare pipe or between the flashing sleeve and the metal-jacket-covered insulation as indicated. Seal penetrations by either one of the following methods.

#### 3.2.16.3.1 Waterproofing Clamping Flange

Pipes up to and including 10 inches in diameter passing through roof or floor waterproofing membrane may be installed through a cast iron sleeve with caulking recess, anchor lugs, flashing clamp device, and pressure ring with brass bolts. Clamp waterproofing membrane into place and place sealant in the caulking recess.

#### 3.2.16.3.2 Modular Mechanical Type Sealing Assembly

In lieu of a waterproofing clamping flange and caulking and sealing of annular space between pipe and sleeve or conduit and sleeve, a modular mechanical type sealing assembly may be installed. Provide seals consisting of interlocking synthetic rubber links shaped to continuously fill the annular space between the pipe/conduit and sleeve with corrosion protected carbon steel bolts, nuts, and pressure plates. Loosely assemble links with bolts to form a continuous rubber belt around the pipe with a pressure plate under each bolt head and each nut. After the seal assembly is properly positioned in the sleeve, tighten the bolt to cause the rubber sealing elements to expand and provide a watertight seal between the pipe/conduit and the sleeve. Size each seal assembly as recommended by the manufacturer to fit the pipe/conduit and sleeve involved. The Contractor electing to use the modular mechanical type seals must provide sleeves of the proper diameters.

#### 3.2.16.4 Fire-Rated Penetrations

Seal penetration of fire-rated walls, partitions, and floors as specified in Section 07 84 00 FIRESTOPPING.

#### 3.2.16.5 Escutcheons

Provide escutcheons for finished surfaces where exposed piping, bare or insulated, pass through floors, walls, or ceilings, except in boiler, utility, or equipment rooms. Where sleeves project slightly from floors, use special deep-type escutcheons. Secure escutcheon to pipe or pipe covering.

#### 3.2.17 Field Applied Insulation

Field installed insulation is specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS, except as defined differently herein.

#### 3.2.18 Field Painting

Painting required for surfaces not otherwise specified, and finish painting of items only primed at the factory are specified in Section 09 90 00 PAINTS AND COATINGS.

##### 3.2.18.1 Color Coding

Color coding for piping identification is specified in Section 09 90 00 PAINTS AND COATINGS.

#### 3.2.19 Identification Tags

Provide identification tags made of brass, engraved laminated plastic or engraved anodized aluminum indicating service and item number on all valves and dampers. Tags must be 1-3/8 inch minimum diameter and marking must be stamped or engraved. Indentations must be black for reading clarity. Attach tags to valves with No. 12 AWG copper wire, chrome-plated beaded chain or plastic straps designed for that purpose.

#### 3.3 CLEANING AND ADJUSTING

Clean uncontaminated system(s) by evacuation and purging procedures currently recommended by refrigerant and refrigerant equipment manufacturers, and as specified herein, to remove small amounts of air and moisture. Systems containing moderate amounts of air, moisture, contaminated refrigerant, or any foreign matter are considered contaminated systems. Restore contaminated systems to clean condition including disassembly, component replacement, evacuation, flushing, purging, and re-charging, using currently approved refrigerant and refrigeration manufacturer's procedures. Restore contaminated systems at no additional cost to the Government as determined by the Contracting Officer. Do not use water in any procedure or test.

#### 3.4 TRAINING COURSE

- a. Submit a schedule, at least 2 weeks prior to the date of the proposed training course, which identifies the date, time, and location for the training. Conduct a training course for 20 members of the operating staff as designated by the Contracting Officer. The training period must consist of a total 8 hours of normal working time and start after

the system is functionally completed but prior to final acceptance tests.

- b. Cover all of the items contained in the approved operation and maintenance manuals as well as demonstrations of routine maintenance operations in the field posted instructions..
- c. Submit 6 complete copies of an operation manual in bound 8 1/2 by 11 inch booklets listing step-by-step procedures required for system startup, operation, abnormal shutdown, emergency shutdown, and normal shutdown at least 4 weeks prior to the first training course. Include the manufacturer's name, model number, and parts list in the booklets. Include the manufacturer's name, model number, service manual, and a brief description of all equipment and their basic operating features in the manuals.
- d. Submit 6 complete copies of maintenance manual in bound 8 1/2 x 11 inch booklets listing routine maintenance procedures, possible breakdowns and repairs, and a trouble shooting guide. Include piping layouts and simplified wiring and control diagrams of the system as installed in the manuals.

### 3.5 REFRIGERANT PIPING TESTS

After all components of the refrigerant system have been installed and connected, subject the entire refrigeration system to pneumatic, evacuation, and startup tests as described herein. Submit a schedule, at least 2 weeks prior to the start of related testing, for each test. Identify the proposed date, time, and location for each test. Conduct tests in the presence of the Contracting Officer. Water and electricity required for the tests will be furnished by the Government. Provide all material, equipment, instruments, and personnel required for the test. Provide the services of a qualified technician, as required, to perform all tests and procedures indicated herein. Coordinate field tests with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC SYSTEMS. Submit 6 copies of the tests report in bound 8 1/2 by 11 inch booklets documenting all phases of the tests performed. Include initial test summaries, all repairs/adjustments made, and the final test results in the report.

#### 3.5.1 Preliminary Procedures

Prior to pneumatic testing, isolate equipment which has been factory tested and refrigerant charged as well as equipment which could be damaged or cause personnel injury by imposed test pressure, positive or negative, from the test pressure, or remove from the system. Remove safety relief valves and rupture discs that are not part of factory sealed systems, and cap or plug openings.

#### 3.5.2 Pneumatic Test

Provide pressure control and excess pressure protection at the source of test pressure. Valves must be wide open, except those leading to the atmosphere. Test gas must be dry nitrogen, with minus 70 degree F dewpoint and less than 5 ppm oil. Apply test pressure in two stages before any refrigerant pipe is insulated or covered. In accordance with ASME B31.5, a preliminary test not to exceed 25 psi must be applied as a means of locating major leaks. Every joint being tested must be coated with a thick soap or color indicating solution. The second stage test

pressure must be at least 110 percent of the design pressure, but cannot exceed 130 percent of the design pressure of any component in the system. For large systems that are not completely visible, the pressure in the system must be gradually increased to one-half of the test pressure after which the pressure must be increased in steps of one-tenth of the test pressure, until the required test pressure has been reached. The test pressure must be continuously maintained for at least 24 hours, after which it can be reduced to the leak test pressure. A correction factor of 0.3 psi will be allowed for each degree F change between test space initial and final ambient temperature, plus for increase and minus for a decrease. The leak test pressure must be the design pressure, or a pressure specified in the engineering design. To repair leaks, the joint must be taken apart, thoroughly cleaned, and reconstructed as a new joint. Joints repaired by caulking, re-melting, or back-welding/brazing are not acceptable. Following repair, the entire system must be retested using the pneumatic tests described above. Reassemble the entire system once the pneumatic tests are satisfactorily completed.

### 3.5.3 Evacuation Test

Following satisfactory completion of the pneumatic tests, relieve the pressure and evacuate the entire system to an absolute pressure of 300 micrometers. During evacuation of the system, the ambient temperature must be higher than 35 degrees F. Do not evacuate no more than one system at one time by one vacuum pump. Once the desired vacuum has been reached, close the vacuum line and allow the system to stand for 1 hour. If the pressure rises over 500 micrometers after the 1 hour period, evacuate the system again down to 300 micrometers and let set for another 1 hour period. Do not charge the system until a vacuum of at least 500 micrometers is maintained for a period of 1 hour without the assistance of a vacuum line. If during the testing the pressure rises above 500 micrometers, continue to repeat the evacuation procedures until all residual moisture has been removed. During evacuation, record pressures by a thermocouple-type, electronic-type, or a calibrated-micrometer type gauge.

### 3.5.4 System Charging and Startup Test

Following satisfactory completion of the evacuation tests, charge the system with the required amount of refrigerant by raising pressure to normal operating pressure and in accordance with manufacturer's procedures. Following charging, the system must operate with high-side and low-side pressures and corresponding refrigerant temperatures, at design or improved values. Test the entire system tested for leaks. Test fluorocarbon systems with halide torch or electronic leak detectors.

### 3.5.5 Refrigerant Leakage

If a refrigerant leak is discovered after the system has been charged, the leaking portion of the system must be immediately isolated from the remainder of the system and the refrigerant pumped into the system receiver or other suitable container. The refrigerant must not be discharged into the atmosphere.

### 3.5.6 Contractor's Responsibility

At all times during the installation and testing of the refrigeration system, take steps to prevent the release of refrigerants into the atmosphere. The steps must include, but not be limited to, procedures

which will minimize the release of refrigerants to the atmosphere and the use of refrigerant recovery devices to remove refrigerant from the system and store the refrigerant for reuse or reclaim. At no time will the allowable leak rate exceed the leak rates allowed in Section 608 of the Clean Air Act: 30 percent of the full charge per year for industrial refrigeration, 20 percent of the full charge per year for commercial refrigeration, and 10 percent of the full charge per year for comfort cooling. Any system leaks within the first year must be repaired in accordance with the requirements herein at no cost to the Government including material, labor, and refrigerant if the leak is the result of defective equipment, material, or installation.

-- End of Section --

SECTION 23 30 00

HVAC AIR DISTRIBUTION  
**05/20, CHG 1: 02/22**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR MOVEMENT AND CONTROL ASSOCIATION INTERNATIONAL, INC. (AMCA)

|          |                                                                              |
|----------|------------------------------------------------------------------------------|
| AMCA 201 | (2002; R 2011) Fans and Systems                                              |
| AMCA 210 | (2016) Laboratory Methods of Testing Fans for Aerodynamic Performance Rating |
| AMCA 300 | (2014) Reverberant Room Method for Sound Testing of Fans                     |
| AMCA 301 | (2014) Methods for Calculating Fan Sound Ratings from Laboratory Test Data   |

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

|                  |                                                                                                                |
|------------------|----------------------------------------------------------------------------------------------------------------|
| AHRI 350         | (2015) Sound Rating of Non-Ducted Indoor Air-Conditioning Equipment                                            |
| AHRI 410         | (2001; Addendum 1 2002; Addendum 2 2005; Addendum 3 2011) Forced-Circulation Air-Cooling and Air-Heating Coils |
| AHRI Guideline D | (1996) Application and Installation of Central Station Air-Handling Units                                      |

AMERICAN BEARING MANUFACTURERS ASSOCIATION (ABMA)

|         |                                                          |
|---------|----------------------------------------------------------|
| ABMA 9  | (2015) Load Ratings and Fatigue Life for Ball Bearings   |
| ABMA 11 | (2014) Load Ratings and Fatigue Life for Roller Bearings |

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

|             |                                                                                                           |
|-------------|-----------------------------------------------------------------------------------------------------------|
| ASHRAE 52.2 | (2017) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size |
| ASHRAE 62.1 | (2022) Ventilation for Acceptable Indoor Air Quality                                                      |
| ASHRAE 70   | (2023) Method of Testing the Performance of Air Outlets and Inlets                                        |

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A13.1 (2023) Scheme for the Identification of  
Piping Systems

ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M (2024) Standard Specification for Pipe,  
Steel, Black and Hot-Dipped, Zinc-Coated,  
Welded and Seamless

ASTM A123/A123M (2024) Standard Specification for Zinc  
(Hot-Dip Galvanized) Coatings on Iron and  
Steel Products

ASTM A167 (2011) Standard Specification for  
Stainless and Heat-Resisting  
Chromium-Nickel Steel Plate, Sheet, and  
Strip

ASTM A924/A924M (2022a) Standard Specification for General  
Requirements for Steel Sheet,  
Metallic-Coated by the Hot-Dip Process

ASTM B117 (2019) Standard Practice for Operating  
Salt Spray (Fog) Apparatus

ASTM B152/B152M (2019) Standard Specification for Copper  
Sheet, Strip, Plate, and Rolled Bar

ASTM B209 (2014) Standard Specification for Aluminum  
and Aluminum-Alloy Sheet and Plate

ASTM B766 (2023) Standard Specification for  
Electrodeposited Coatings of Cadmium

ASTM C553 (2013; R 2019) Standard Specification for  
Mineral Fiber Blanket Thermal Insulation  
for Commercial and Industrial Applications

ASTM D520 (2000; R 2011) Zinc Dust Pigment

ASTM D1654 (2008; R 2016; E 2017) Standard Test  
Method for Evaluation of Painted or Coated  
Specimens Subjected to Corrosive  
Environments

ASTM D3359 (2017) Standard Test Methods for Rating  
Adhesion by Tape Test

ASTM E2016 (2022) Standard Specification for  
Industrial Woven Wire Cloth

CALIFORNIA DEPARTMENT OF PUBLIC HEALTH (CDPH)

CDPH SECTION 01350 (2017; Version 1.2) Standard Method for  
the Testing and Evaluation of Volatile  
Organic Chemical Emissions from Indoor



Sources using Environmental Chambers

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

|            |                                                                                                                            |
|------------|----------------------------------------------------------------------------------------------------------------------------|
| NEMA MG 1  | (2021) Motors and Generators                                                                                               |
| NEMA MG 10 | (2017) Energy Management Guide for Selection and Use of Fixed Frequency Medium AC Squirrel-Cage Polyphase Induction Motors |
| NEMA MG 11 | (1977; R 2012) Energy Management Guide for Selection and Use of Single Phase Motors                                        |

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                                                                |
|----------|------------------------------------------------------------------------------------------------|
| NFPA 70  | (2023) National Electrical Code                                                                |
| NFPA 90A | (2024) Standard for the Installation of Air Conditioning and Ventilating Systems               |
| NFPA 701 | (2023; ERTA 1 2023) Standard Methods of Fire Tests for Flame Propagation of Textiles and Films |

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION (SMACNA)

|             |                                                                                          |
|-------------|------------------------------------------------------------------------------------------|
| SMACNA 1819 | (2002) Fire, Smoke and Radiation Damper Installation Guide for HVAC Systems, 5th Edition |
| SMACNA 1966 | (2020) HVAC Duct Construction Standards Metal and Flexible, 4th Edition                  |
| SMACNA 1981 | (2008) Seismic Restraint Manual Guidelines for Mechanical Systems, 3rd Edition           |

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT (SCAQMD)

|                  |                                          |
|------------------|------------------------------------------|
| SCAQMD Rule 1168 | (2022) Adhesive and Sealant Applications |
|------------------|------------------------------------------|

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|           |                                   |
|-----------|-----------------------------------|
| 40 CFR 82 | Protection of Stratospheric Ozone |
|-----------|-----------------------------------|

UL SOLUTIONS (UL)

|        |                                                                                           |
|--------|-------------------------------------------------------------------------------------------|
| UL 6   | (2022) UL Standard for Safety Electrical Rigid Metal Conduit-Steel                        |
| UL 181 | (2013; Reprint Dec 2021) UL Standard for Safety Factory-Made Air Ducts and Air Connectors |
| UL 555 | (2006; Reprint Aug 2016) UL Standard for Safety Fire Dampers                              |
| UL 586 | (2009; Reprint Sep 2022) UL Standard for                                                  |

Safety High-Efficiency Particulate, Air  
Filter Units

|                            |                                                                               |
|----------------------------|-------------------------------------------------------------------------------|
| UL 705                     | (2017; Reprint Sep 2024) UL Standard for Safety Power Ventilators             |
| UL 900                     | (2015; Reprint Aug 2022) UL Standard for Safety Standard for Air Filter Units |
| UL 1995                    | (2015; Reprint Aug 2022) UL Standard for Safety Heating and Cooling Equipment |
| UL Bld Mat Dir             | (updated continuously online) Building Materials Directory                    |
| UL Electrical Construction | (2012) Electrical Construction Equipment Directory                            |
| UL Fire Resistance         | (2014) Fire Resistance Directory                                              |

## 1.2 SYSTEM DESCRIPTION

Furnish ductwork, piping offsets, fittings, and accessories as required to provide a complete installation. Coordinate the work of the different trades to avoid interference between piping, equipment, structural, and electrical work. Provide complete, in place, all necessary offsets in piping and ductwork, and all fittings, and other components, required to install the work as indicated and specified.

### 1.2.1 Mechanical Equipment Identification

The number of charts and diagrams must be equal to or greater than the number of mechanical equipment rooms. Where more than one chart or diagram per space is required, mount these in edge pivoted, swinging leaf, extruded aluminum frame holders which open to 170 degrees.

#### 1.2.1.1 Charts

Provide chart listing of equipment by designation numbers and capacities such as flow rates, pressure and temperature differences, heating and cooling capacities, horsepower, pipe sizes, and voltage and current characteristics.

#### 1.2.1.2 Diagrams

Submit proposed diagrams, at least 2 weeks prior to start of related testing. provide neat mechanical drawings provided with extruded aluminum frame under 1/8-inch glass or laminated plastic, system diagrams that show the layout of equipment, piping, and ductwork, and typed condensed operation manuals explaining preventative maintenance procedures, methods of checking the system for normal, safe operation, and procedures for safely starting and stopping the system. After approval, post these items where directed.

### 1.2.2 Service Labeling

Label equipment, including fans, air handlers, terminal units, etc. with labels made of self-sticking, plastic film designed for permanent installation. Provide labels in accordance with the typical examples

below:

| SERVICE               | LABEL AND TAG DESIGNATION |
|-----------------------|---------------------------|
| Supply Fan Number     | SF-1                      |
|                       |                           |
| Exhaust Fan Number    | EF - 1, 2 and 3           |
| Duct Heater Number    | DH-1                      |
| Gravity Intake Number | GV-1 and 2                |
|                       |                           |
|                       |                           |

Identify similar services with different temperatures or pressures. Where pressures could exceed 125 pounds per square inch, gage, include the maximum system pressure in the label. Label and arrow piping in accordance with the following:

- a. Each point of entry and exit of pipe passing through walls.
- b. Each change in direction, i.e., elbows, tees.
- c. In congested or hidden areas and at all access panels at each point required to clarify service or indicated hazard.
- d. In long straight runs, locate labels at distances within eyesight of each other not to exceed 75 feet. All labels must be visible and legible from the primary service and operating area.

| For Bare or Insulated Pipes |            |
|-----------------------------|------------|
| for Outside Diameters of    | Lettering  |
| 1/2 thru 1-3/8 inch         | 1/2 inch   |
| 1-1/2 thru 2-3/8 inch       | 3/4 inch   |
| 2-1/2 inch and larger       | 1-1/4 inch |

### 1.2.3 Color Coding

Color coding of all piping systems must be in accordance with ASME A13.1.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance

with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Detail Drawings; G

SD-03 Product Data

Metallic Flexible Duct

Insulated Nonmetallic Flexible Duct Runouts

Duct Connectors

Duct Access Doors; G

Fire Dampers

Manual Balancing Dampers; G

Diffusers

Louvers

Air Vents, Penthouses, and Goosenecks

In-Line Centrifugal Fans

Axial Flow Fans

Centrifugal Type Power Roof Ventilators

Unit Ventilators

Indoor Air Quality for Duct Sealants; S

Diagrams; G

SD-06 Test Reports

Performance Tests; G

Damper Acceptance Test; G

Test Procedures; G

SD-07 Certificates

Ozone Depleting Substances Technician Certification

SD-08 Manufacturer's Instructions

Manufacturer's Installation Instructions

Operation and Maintenance Training

SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

Fire Dampers; G

Manual Balancing Dampers; G

In-Line Centrifugal Fans; G

Axial Flow Fans; G

Centrifugal Type Power Roof Ventilators; G

Unit Ventilators; G

SD-11 Closeout Submittals

Indoor Air Quality During Construction; S

#### 1.4 QUALITY ASSURANCE

Except as otherwise specified, approval of materials and equipment is based on manufacturer's published data.

- a. Where materials and equipment are specified to conform to the standards of the Underwriters Laboratories, the label of or listing

with reexamination in UL Bld Mat Dir, and UL 6 is acceptable as sufficient evidence that the items conform to Underwriters Laboratories requirements. In lieu of such label or listing, submit a written certificate from any nationally recognized testing agency, adequately equipped and competent to perform such services, stating that the items have been tested and that the units conform to the specified requirements. Outline methods of testing used by the specified agencies.

- b. Where materials or equipment are specified to be constructed or tested, or both, in accordance with the standards of the ASTM International (ASTM), the ASME International (ASME), or other standards, a manufacturer's certificate of compliance of each item is acceptable as proof of compliance.
- c. Conformance to such agency requirements does not relieve the item from compliance with other requirements of these specifications.
- d. Where products are specified to meet or exceed the specified energy efficiency requirement of FEMP-designated or ENERGY STAR covered product categories, equipment selected must have as a minimum the efficiency rating identified under "Energy-Efficient Products" at <http://femp.energy.gov/procurement>.

#### 1.4.1 Prevention of Corrosion

Protect metallic materials against corrosion. Provide rust-inhibiting treatment and standard finish for the equipment enclosures. Do not use aluminum in contact with earth, and where connected to dissimilar metal. Protect aluminum by approved fittings, barrier material, or treatment. Provide hot-dip galvanized ferrous parts such as anchors, bolts, braces, boxes, bodies, clamps, fittings, guards, nuts, pins, rods, shims, thimbles, washers, and miscellaneous parts not of corrosion-resistant steel or nonferrous materials in accordance with ASTM A123/A123M for exterior locations and cadmium-plated in conformance with ASTM B766 for interior locations.

#### 1.4.2 Asbestos Prohibition

Do not use asbestos and asbestos-containing products.

#### 1.4.3 Ozone Depleting Substances Technician Certification

All technicians working on equipment that contain ozone depleting refrigerants must be certified as a Section 608 Technician to meet requirements in 40 CFR 82, Subpart F. Provide copies of technician certifications to the Contracting Officer at least 14 calendar days prior to work on any equipment containing these refrigerants.

#### 1.4.4 Detail Drawings

Submit detail drawings showing equipment layout, including assembly and installation details and electrical connection diagrams; ductwork layout showing the location of all supports and hangers, typical hanger details, gauge reinforcement, reinforcement spacing rigidity classification, and static pressure and seal classifications. Include any information required to demonstrate that the system has been coordinated and functions properly as a unit on the drawings and show equipment relationship to other parts of the work, including clearances required for operation and

maintenance. Submit drawings showing bolt-setting information, and foundation bolts prior to concrete foundation construction for all equipment indicated or required to have concrete foundations. Submit function designation of the equipment and any other requirements specified throughout this Section with the shop drawings.

#### 1.4.5 Test Procedures

Conduct performance tests as required in Section 23 05 93 Testing, Adjusting and Balancing for HVAC and Section 23 09 00 Instrumentation and Control for HVAC.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

Protect stored equipment at the jobsite from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Additionally, cap or plug all pipes until installed.

### PART 2 PRODUCTS

#### 2.1 STANDARD PRODUCTS

Except for the fabricated duct, plenums and casings specified in paragraphs "Metal Ductwork" and "Plenums and Casings for Field-Fabricated Units", provide components and equipment that are standard products of manufacturers regularly engaged in the manufacturing of products that are of a similar material, design and workmanship. This requirement applies to all equipment, including diffusers, registers, fire dampers, and balancing dampers.

- a. Standard products are defined as components and equipment that have been in satisfactory commercial or industrial use in similar applications of similar size for at least two years before bid opening.
- b. Prior to this two year period, these standard products must have been sold on the commercial market using advertisements in manufacturers' catalogs or brochures. These manufacturers' catalogs, or brochures must have been copyrighted documents or have been identified with a manufacturer's document number.
- c. Provide equipment items that are supported by a service organization. In product categories covered by ENERGY STAR or the Federal Energy Management Program, provide equipment that is listed on the ENERGY STAR Qualified Products List or that meets or exceeds the FEMP-designated Efficiency Requirements.

#### 2.2 IDENTIFICATION PLATES

In addition to standard manufacturer's identification plates, provide engraved laminated phenolic identification plates for each piece of mechanical equipment. Identification plates are to designate the function of the equipment. Submit designation with the shop drawings. Provide identification plates that are layers, black-white-black, engraved to show white letters on black background. Letters must be upper case. Identification plates that are 1-1/2-inches high and smaller must be 1/16-inch thick, with engraved lettering 1/8-inch high; identification plates larger than 1-1/2-inches high must be 1/8-inch thick, with engraved lettering of suitable height. Identification plates 1-1/2-inches high and



larger must have beveled edges. Install identification plates using a compatible adhesive.

### 2.3 EQUIPMENT GUARDS AND ACCESS

Fully enclose or guard belts, pulleys, chains, gears, couplings, projecting setscrews, keys, and other rotating parts exposed to personnel contact according to OSHA requirements. Properly guard or cover with insulation of a type specified, high temperature equipment and piping exposed to contact by personnel or where it creates a potential fire hazard.

### 2.4 ELECTRICAL WORK

- a. Provide motors, controllers, integral disconnects, contactors, and controls with their respective pieces of equipment, except controllers indicated as part of motor control centers. Provide electrical equipment, including motors and wiring, as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide manual or automatic control and protective or signal devices required for the operation specified and control wiring required for controls and devices specified, but not shown. For packaged equipment, include manufacturer provided controllers with the required monitors and timed restart.
- b. For single-phase motors, provide high-efficiency type, fractional-horsepower alternating-current motors, including motors that are part of a system, in accordance with NEMA MG 11. Provide premium efficiency type integral size motors in accordance with NEMA MG 1.
- c. For polyphase motors, provide squirrel-cage medium induction motors, including motors that are part of a system, and that meet the efficiency ratings for premium efficiency motors in accordance with NEMA MG 1. Select premium efficiency polyphase motors in accordance with NEMA MG 10.
- d. Provide motors in accordance with NEMA MG 1 and of sufficient size to drive the load at the specified capacity without exceeding the nameplate rating of the motor. Provide motors rated for continuous duty with the enclosure specified. Provide motor duty that allows for maximum frequency start-stop operation and minimum encountered interval between start and stop. Provide motor torque capable of accelerating the connected load within 20 seconds with 80 percent of the rated voltage maintained at motor terminals during one starting period. Provide motor starters complete with thermal overload protection and other necessary appurtenances. Fit motor bearings with grease supply fittings and grease relief to outside of the enclosure.
- e. Where two-speed or variable-speed motors are indicated, solid-state variable-speed controllers are allowed to accomplish the same function. Use solid-state variable-speed controllers for motors rated 10 hp or less and adjustable frequency drives for larger motors. Provide variable frequency drives for motors as specified in Section 26 29 23 ADJUSTABLE SPEED DRIVE SYSTEMS UNDER 600 VOLTS.

### 2.5 ANCHOR BOLTS

Provide anchor bolts for equipment placed on concrete equipment pads or on

concrete slabs. Bolts to be of the size and number recommended by the equipment manufacturer and located by means of suitable templates. Installation of anchor bolts must not degrade the surrounding concrete.

## 2.6 SEISMIC ANCHORAGE

Anchor equipment in accordance with applicable seismic criteria for the area and as defined in SMACNA 1981

## 2.7 PAINTING

Paint equipment units in accordance with approved equipment manufacturer's standards unless specified otherwise. Field retouch only if approved. Otherwise, return equipment to the factory for refinishing.

## 2.8 INDOOR AIR QUALITY

Provide equipment and components that comply with the requirements of ASHRAE 62.1 unless more stringent requirements are specified herein.

## 2.9 DUCT SYSTEMS

### 2.9.1 Metal Ductwork

Provide metal ductwork construction, including all fittings and components, that complies with SMACNA 1966, as supplemented and modified by this specification .

- a. Construct ductwork meeting the requirements for the duct system static pressure specified in APPENDIX D of Section 23 05 93 TESTING, ADJUSTING AND BALANCING FOR HVAC.
- b. Provide radius type elbows with a centerline radius of 1.5 times the width or diameter of the duct where space permits. Otherwise, elbows having a minimum radius equal to the width or diameter of the duct or square elbows with factory fabricated turning vanes are allowed.
- d. Provide ductwork that meets the requirements of Seal Class A. Provide ductwork in VAV systems upstream of the VAV boxes that meets the requirements of Seal Class A.
- e. Provide sealants that conform to fire hazard classification specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS and are suitable for the range of air distribution and ambient temperatures to which it is exposed. Do not use pressure sensitive tape as a sealant. Provide duct sealant products that meet either emissions requirements of CDPH SECTION 01350 (limit requirements for either office or classroom spaces regardless of space type) or VOC content requirements of SCAQMD Rule 1168 (HVAC duct sealants are classified as "Other" within the SCAQMD Rule 1168 sealants table). Provide validation of indoor air quality for duct sealants.
- f. Make spiral lock seam duct, and flat oval with duct sealant and lock with not less than 3 equally spaced drive screws or other approved methods indicated in SMACNA 1966. Apply the sealant to the exposed male part of the fitting collar so that the sealer is on the inside of the joint and fully protected by the metal of the duct fitting. Apply one brush coat of the sealant over the outside of the joint to at least

2 inch band width covering all screw heads and joint gap. Dents in the male portion of the slip fitting collar are not acceptable.

- g. Fabricate outdoor air intake ducts and plenums with watertight soldered or brazed joints and seams.

#### 2.9.1.1 Metallic Flexible Duct

- a. Provide duct that conforms to UL 181 and NFPA 90A with factory-applied insulation, vapor barrier, and end connections. Provide duct assembly that does not exceed 25 for flame spread and 50 for smoke developed. Provide ducts designed for working pressures of 2 inches water gauge positive and 1.5 inches water gauge negative. Provide flexible round duct length that does not exceed 5 feet. Secure connections by applying adhesive for 2 inches over rigid duct, apply flexible duct 2 inches over rigid duct, apply metal clamp, and provide minimum of three No. 8 sheet metal screws through clamp and rigid duct.
- b. Inner duct core: Provide interlocking spiral or helically corrugated flexible core constructed of zinc-coated steel, aluminum, or stainless steel; or constructed of inner liner of continuous galvanized spring steel wire helix fused to continuous, fire-retardant, flexible vapor barrier film, inner duct core.
- c. Insulation: Provide inner duct core that is insulated with mineral fiber blanket type flexible insulation, minimum of 1 inch thick. Provide insulation covered on exterior with manufacturer's standard fire retardant vapor barrier jacket for flexible round duct.

#### 2.9.1.2 Insulated Nonmetallic Flexible Duct Runouts

Use flexible duct runouts only where indicated. Runout length is indicated on the drawings, and is not to exceed 5 feet. Provide runouts that are preinsulated, factory fabricated, and that comply with NFPA 90A and UL 181. Provide either field or factory applied vapor barrier. Provide not less than 20 ounce glass fabric duct connectors coated on both sides with neoprene. Where coil induction or high velocity units are supplied with vertical air inlets, use a streamlined, vaned and mitered elbow transition piece for connection to the flexible duct or hose. Provide a die-stamped elbow and not a flexible connector as the last elbow to these units other than the vertical air inlet type. Insulated flexible connectors are allowed as runouts. Provide insulated material and vapor barrier that conform to the requirements of Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. Do not expose the insulation material surface to the air stream.

#### 2.9.1.3 General Service Duct Connectors

Provide a flexible duct connector approximately 6 inches in width where sheet metal connections are made to fans or where ducts of dissimilar metals are connected. For round/oval ducts, secure the flexible material by stainless steel or zinc-coated, iron clinch-type draw bands. For rectangular ducts, install the flexible material locked to metal collars using normal duct construction methods. Provide a composite connector system that complies with NFPA 701 and is classified as "flame-retardant fabrics" in UL Bld Mat Dir.

#### 2.9.1.4 High Temperature Service Duct Connections

Provide material that is approximately 3/32 inch thick, 35 to 40-ounce per square yard weight, plain weave fibrous glass cloth with, nickel/chrome wire reinforcement for service in excess of 1200 degrees F.

#### 2.9.1.5 Aluminum Ducts

ASTM B209, alloy 3003-H14 for aluminum sheet and alloy 6061-T6 or equivalent strength for aluminum connectors and bar stock.

#### 2.9.1.6 Copper Sheets

ASTM B152/B152M, light cold rolled temper.

#### 2.9.1.7 Corrosion Resisting (Stainless) Steel Sheets

ASTM A167

#### 2.9.2 Duct Access Doors

Provide hinged access doors conforming to SMACNA 1966 in ductwork and plenums where indicated and at all air flow measuring primaries, automatic dampers, fire dampers, coils, thermostats, and other apparatus requiring service and inspection in the duct system. Provide access doors upstream and downstream of air flow measuring primaries and heating and cooling coils. Provide doors that are a minimum 15 by 18 inches, unless otherwise shown. Where duct size does not accommodate this size door, make the doors as large as practicable. Equip doors 24 by 24 inches or larger with fasteners operable from inside and outside the duct. Use insulated type doors in insulated ducts.

#### 2.9.3 Fire Dampers

Use 1.5 hour rated fire dampers unless otherwise indicated. Provide fire dampers that conform to the requirements of NFPA 90A and UL 555. Perform the fire damper test as outlined in NFPA 90A. If the ductwork connected to the fire damper is to be insulated then provide a factory installed pressure relief damper. Provide automatic operating fire dampers with a dynamic rating suitable for the maximum air velocity and pressure differential to which it is subjected. Provide fire dampers approved for the specific application, and install according to their listing. Equip fire dampers with a steel sleeve or adequately sized frame installed in such a manner that disruption of the attached ductwork, if any, does not impair the operation of the damper. Equip sleeves or frames with perimeter mounting angles attached on both sides of the wall or floor opening. Construct ductwork in fire-rated floor-ceiling or roof-ceiling assembly systems with air ducts that pierce the ceiling of the assemblies in conformance with UL Fire Resistance. Provide single blade type or multi-blade type fire dampers. Install dampers that do not reduce the duct or the air transfer opening cross-sectional area. Install dampers so that the centerline of the damper depth or thickness is located in the centerline of the wall, partition or floor slab depth or thickness. Unless otherwise indicated, comply with the installation details given in SMACNA 1819 and in manufacturer's instructions for fire dampers. Perform acceptance testing of fire dampers according to paragraph DAMPER ACCEPTANCE TEST and NFPA 90A.

#### 2.9.4 Manual Balancing Dampers

- a. Furnish manual balancing dampers with accessible operating mechanisms. Use chromium plated operators (with all exposed edges rounded) in finished portions of the building. Provide manual volume control dampers that are operated by locking-type quadrant operators.
- b. Unless otherwise indicated, provide opposed blade type multileaf dampers with maximum blade width of 12 inches. Provide access doors or panels for all concealed damper operators and locking setscrews. Provide access doors or panels in hard ceilings, partitions and walls for access to all concealed damper operators and damper locking setscrews. Coordinate location of doors or panels with other affected contractors.
- c. Provide stand-off mounting brackets, bases, or adapters not less than the thickness of the insulation when the locking-type quadrant operators for dampers are installed on ducts to be thermally insulated, to provide clearance between the duct surface and the operator. Provide stand-off mounting items that are integral with the operator or standard accessory of the damper manufacturer.

##### 2.9.4.1 Square or Rectangular Dampers

###### 2.9.4.1.1 Duct Height 12 inches and Less

###### 2.9.4.1.1.1 Frames

| Width               | Height            | Galvanized Steel Thickness | Length           |
|---------------------|-------------------|----------------------------|------------------|
| Maximum 19 inches   | Maximum 12 inches | Minimum 20 gauge           | Minimum 3 inches |
| More than 19 inches | Maximum 12 inches | Minimum 16 gauge           | Minimum 3 inches |

###### 2.9.4.1.1.2 Single Leaf Blades

| Width               | Height            | Galvanized Steel Thickness | Length           |
|---------------------|-------------------|----------------------------|------------------|
| Maximum 19 inches   | Maximum 12 inches | Minimum 20 gauge           | Minimum 3 inches |
| More than 19 inches | Maximum 12 inches | Minimum 16 gauge           | Minimum 3 inches |

###### 2.9.4.1.1.3 Blade Axles

To support the blades of round dampers, provide galvanized steel shafts supporting the blade the entire duct diameter frame-to-frame. Provide axle shafts that extend through standoff bracket and hand quadrant.

| Width               | Height            | Material         | Square Shaft     |
|---------------------|-------------------|------------------|------------------|
| Maximum 19 inches   | Maximum 12 inches | Galvanized Steel | Minimum 3/8 inch |
| More than 19 inches | Maximum 12 inches | Galvanized Steel | Minimum 1/2 inch |

#### 2.9.4.1.1.4 Axle Bearings

Support the shaft on each end at the frames with shaft bearings. Press fit shaft bearings configuration to provide a tight joint between blade shaft and damper frame.

| Width               | Height            | Material                                                            |
|---------------------|-------------------|---------------------------------------------------------------------|
| Maximum 19 inches   | Maximum 12 inches | solid nylon, or equivalent solid plastic, or oil-impregnated bronze |
| More than 19 inches | Maximum 12 inches | oil-impregnated bronze                                              |

#### 2.9.4.1.1.5 Control Shaft/Hand Quadrant

Provide dampers with accessible locking-type control shaft/hand quadrant operators.

Provide stand-off mounting brackets, bases, or adapters for the locking-type quadrant operators on dampers installed on ducts to be thermally insulated. Provide a minimum stand-off distance of 2 inches off the metal duct surface. Provide stand-off mounting items that are integral with the operator or standard accessory of the damper manufacturer.

#### 2.9.4.1.1.6 Finish

Mill Galvanized

#### 2.9.4.1.2 Duct Height Greater than 12 inches

##### 2.9.4.1.2.1 Dampers

Provide dampers with multi-leaf opposed-type blades.

##### 2.9.4.1.2.2 Frames

Maximum 48 inches in height; maximum 48 inches in width; minimum of 16 gauge galvanized steel, minimum of 5 inches long.

##### 2.9.4.1.2.3 Blades

Minimum of 16 gauge galvanized steel; 6 inch nominal width.

#### 2.9.4.1.2.4 Blade Axles

To support the blades of round dampers, provide galvanized square steel shafts supporting the blade the entire duct diameter frame-to-frame. Provide axle shafts that extend through standoff bracket and hand quadrant.

#### 2.9.4.1.2.5 Axle Bearings

Support the shaft on each end at the frames with shaft bearings constructed of oil-impregnated bronze, or solid nylon, or a solid plastic equivalent to nylon. Press fit shaft bearings configuration to provide a tight joint between blade shaft and damper frame.

#### 2.9.4.1.2.6 Blade Actuator

Minimum 1/2 inch diameter galvanized steel.

#### 2.9.4.1.2.7 Blade Actuator Linkage

Mill Galvanized steel bar and crank plate with stainless steel pivots.

#### 2.9.4.1.2.8 Control Shaft/Hand Quadrant

Provide dampers with accessible locking-type control shaft/hand quadrant operators.

Provide stand-off mounting brackets, bases, or adapters for the locking-type quadrant operators on dampers installed on ducts to be thermally insulated. Provide a minimum stand-off distance of 2 inches off the metal duct surface. Provide stand-off mounting items that are integral with the operator or standard accessory of the damper manufacturer.

#### 2.9.4.1.2.9 Finish

Mill Galvanized

### 2.9.4.2 Round Dampers

#### 2.9.4.2.1 Frames

| Size            | Galvanized Steel Thickness | Length           |
|-----------------|----------------------------|------------------|
| 4 to 20 inches  | Minimum 20 gauge           | Minimum 6 inches |
| 22 to 30 inches | Minimum 20 gauge           | Minimum 6 inches |
| 32 to 40 inches | Minimum 16 gauge           | Minimum 6 inches |

#### 2.9.4.2.2 Blades

| Size            | Galvanized Steel Thickness |
|-----------------|----------------------------|
| 4 to 20 inches  | Minimum 20 gauge           |
| 22 to 30 inches | Minimum 16 gauge           |
| 32 to 40 inches | Minimum 10 gauge           |

#### 2.9.4.2.3 Blade Axles

To support the blades of round dampers, provide galvanized steel shafts supporting the blade the entire duct diameter frame-to-frame. Provide axle shafts that extend through standoff bracket and hand quadrant.

| Size            | Shaft Size and Shape    |
|-----------------|-------------------------|
| 4 to 20 inches  | Minimum 3/8 inch square |
| 22 to 30 inches | Minimum 1/2 inch square |
| 32 to 40 inches | Minimum 3/4 inch square |

#### 2.9.4.2.4 Axle Bearings

Support the shaft on each end at the frames with shaft bearings constructed of oil-impregnated bronze, nylon, or a solid plastic equivalent to nylon. Axle bearings intended for low leakage at the damper frame must be neoprene, nitrile, or equivalent of 60 or greater durometer to reduce damper blade vibration. Press fit shaft bearings configuration to provide a tight joint between blade shaft and damper frame.

| Size            | Material                                                            |
|-----------------|---------------------------------------------------------------------|
| 4 to 20 inches  | solid nylon, or equivalent solid plastic, or oil-impregnated bronze |
| 22 to 30 inches | solid nylon, or equivalent solid plastic, or oil-impregnated bronze |
| 32 to 40 inches | oil-impregnated bronze, or stainless steel sleeve bearing           |

#### 2.9.4.2.5 Control Shaft/Hand Quadrant

Provide dampers with accessible locking-type control shaft/hand quadrant operators.

Provide stand-off mounting brackets, bases, or adapters for the locking-type quadrant operators on dampers installed on ducts to be



thermally insulated. Provide a minimum stand-off distance of 2 inches off the metal duct surface. Provide stand-off mounting items that are integral with the operator or standard accessory of the damper manufacturer.

#### 2.9.4.2.6 Finish

Mill Galvanized

### 2.9.5 Automatic Balancing Dampers

Provide dampers as specified in paragraph SUPPLEMENTAL COMPONENTS/SERVICES, subparagraph CONTROLS.

### 2.9.6 Diffusers, Registers, and Grilles

Provide factory-fabricated units of corrosion-resistant steel or aluminum that distribute the specified quantity of air evenly over space intended without causing noticeable drafts, air movement faster than 50 fpm in occupied zone, or dead spots anywhere in the conditioned area. Provide outlets for diffusion, spread, throw, and noise level as required for specified performance. Certify performance according to ASHRAE 70. Provide sound rated and certified inlets and outlets according to ASHRAE 70. Provide sound power level as indicated. Provide diffusers and registers with volume damper with accessible operator, unless otherwise indicated; or if standard with the manufacturer, an automatically controlled device is acceptable. Provide opposed blade type volume dampers for all diffusers and registers, except linear slot diffusers. Provide linear slot diffusers with round or elliptical balancing dampers. Where the inlet and outlet openings are located less than 7 feet above the floor, protect them by a grille or screen according to NFPA 90A.

#### 2.9.6.1 Diffusers

Provide diffuser types indicated. Furnish ceiling mounted units with anti-smudge devices, unless the diffuser unit minimizes ceiling smudging through design features. Provide diffusers with air deflectors of the type indicated. Provide air handling troffers or combination light and ceiling diffusers conforming to the requirements of UL Electrical Construction for the interchangeable use as cooled or heated air supply diffusers or return air units. Install ceiling mounted units with rims tight against ceiling. Provide sponge rubber gaskets between ceiling and surface mounted diffusers for air leakage control. Provide suitable trim for flush mounted diffusers. For connecting the duct to diffuser, provide duct collar that is airtight and does not interfere with volume controller. Provide return or exhaust units that are similar to supply diffusers.

#### 2.9.6.2 Registers

Double-deflection supply registers. Provide manufacturer-furnished volume dampers. Provide volume dampers of the group-operated, opposed-blade type and key adjustable by inserting key through face of register. Operating mechanism must not project through any part of the register face. Automatic volume control devices are acceptable. Provide exhaust and return registers as specified for supply registers, except provide exhaust and return registers that have a single set of nondirectional face bars or vanes having the same appearance as the supply registers. Set face bars or vanes at 90 degrees.

### 2.9.7 Louvers

Provide louvers for installation in exterior walls that are associated with the air supply and distribution system as specified in Section 07 60 00 FLASHING AND SHEET METAL and 08 91 00 METAL WALL AND DOOR LOUVERS.

### 2.9.8 Air Vents, Penthouses, and Goosenecks

Fabricate air vents, penthouses, and goosenecks from galvanized steel or aluminum sheets with galvanized or aluminum structural shapes. Provide sheet metal thickness, reinforcement, and fabrication that conform to SMACNA 1966. Accurately fit and secure louver blades to frames. Fold or bead edges of louver blades for rigidity and baffle these edges to exclude driving rain. Provide air vents, penthouses, and goosenecks with bird screen.

### 2.9.9 Bird Screens and Frames

Provide bird screens that conform to ASTM E2016, No. 2 mesh, aluminum or stainless steel. Provide "medium-light" rated aluminum screens. Provide "light" rated stainless steel screens. Provide removable type frames fabricated from either stainless steel or extruded aluminum.

## 2.10 AIR SYSTEMS EQUIPMENT

### 2.10.1 Fans

Test and rate fans according to AMCA 210. Calculate system effect on air moving devices in accordance with AMCA 201 where installed ductwork differs from that indicated on drawings. Install air moving devices to minimize fan system effect. Where system effect is unavoidable, determine the most effective way to accommodate the inefficiencies caused by system effect on the installed air moving device. The sound power level of the fans must not exceed 85 dBA when tested according to AMCA 300 and rated in accordance with AMCA 301. Provide all fans with an AMCA seal. Connect fans to the motors either directly or indirectly with V-belt drive. Use V-belt drives designed for not less than 150 percent of the connected driving capacity. Provide variable pitch motor sheaves for 15 hp and below, and fixed pitch as defined by AHRI Guideline D (A fixed-pitch sheave is provided on both the fan shaft and the motor shaft. This is a non-adjustable speed drive.). Select variable pitch sheaves to drive the fan at a speed which can produce the specified capacity when set at the approximate midpoint of the sheave adjustment. When fixed pitch sheaves are furnished, provide a replaceable sheave when needed to achieve system air balance. Provide motors for V-belt drives with adjustable rails or bases. Provide removable metal guards for all exposed V-belt drives, and provide speed-test openings at the center of all rotating shafts. Provide fans with personnel screens or guards on both suction and supply ends, except that the screens need not be provided, unless otherwise indicated, where ducts are connected to the fan. Provide fan and motor assemblies with vibration-isolation supports or mountings as indicated. Use vibration-isolation units that are standard products with published loading ratings. Select each fan to produce the capacity required at the fan static pressure indicated. Provide sound power level as indicated. Obtain the sound power level values according to AMCA 300. Provide standard AMCA arrangement, rotation, and discharge as indicated. Provide power ventilators that conform to UL 705 and have a UL label.

#### 2.10.1.1 In-Line Centrifugal Fans

Provide in-line fans with centrifugal backward inclined blades, stationary discharge conversion vanes, internal and external belt guards, and adjustable motor mounts. Mount fans in a welded tubular casing. Provide a fan that axially flows the air in and out. Streamline inlets with conversion vanes to eliminate turbulence and provide smooth discharge air flow. Enclose and isolate fan bearings and drive shafts from the air stream. Provide precision, self aligning ball or roller type fan bearings that are sealed against dust and dirt and are permanently lubricated. Provide L50 rated bearing life at not less than 200,000 hours as defined by ABMA 9 and ABMA 11. Provide motors with open drip proof enclosure. Provide magnetic motor starters across-the-line with general-purpose enclosures. Provide remote manual switch with pilot indicating light where indicated.

#### 2.10.1.2 Axial Flow Fans

Provide axial flow fans complete with drive components and belt guard, with steel housing, cast fan wheel, cast or welded steel diffusers, fan shaft, bearings, and mounting frame as a factory-assembled unit. Provide fan wheels that are dynamically balanced and keyed to the fan shaft, with radially projecting blades of airfoil cross-section. Enclose and isolate fan bearings and drive shafts from the air stream. Permanently lubricate fan bearings or provide them with accessible grease fittings. Provide precision self-aligning ball or roller type fan bearings that are sealed against dust and dirt. Provide fan bearings that have a L50 rated bearing life at not less than 200,000 hours of operation as defined by ABMA 9 and ABMA 11. Provide fan inlets with an aerodynamically shaped bell and an inlet cone. Install diffuser or straightening vanes at the fan discharge to minimize turbulence and provide smooth discharge air flow. Furnish fan unit with inlet and outlet flanges, inlet screen, duct equalizer section, and manual operation adjustable inlet vanes. Unless otherwise indicated, provide motors that do not exceed 1800 rpm and have open drip proof enclosure. Provide magnetic motor starters across-the-line with general-purpose enclosure. Provide remote manual switch with pilot indicating light where indicated.

#### 2.10.1.3 Centrifugal Type Power Roof Ventilators

Provide direct or V-belt driven centrifugal type fans with backward inclined, non-overloading wheel. Provide hinged or removable and weatherproof motor compartment housing, constructed of heavy gauge aluminum. Provide fans with birdscreen, disconnect switch, gravity dampers, roof curb, and extended base. Provide drip proof type motor enclosure. Use only lubricated bearings. If there is a conflict between NFPA 96 and UL 705 the most stringent wording must be adhered to.

#### 2.10.2 Coils

Provide fin-and-tube type coils constructed of seamless copper tubes and aluminum or copper fins mechanically bonded or soldered to the tubes. Provide copper tube wall thickness that is a minimum of 0.016 inches. Provide aluminum fins that are 0.0055 inch minimum thickness. Provide copper fins that are 0.0045 inch minimum thickness. Provide casing and tube support sheets that are not lighter than 16 gauge galvanized steel, formed to provide structural strength. When required, provide multiple tube supports to prevent tube sag. Mount coils for counterflow service. Rate and certify coils to meet the requirements of AHRI 410. Provide

factory applied phenolic, vinyl or epoxy/electrodeposition coating.

2.10.2.1 Electric Heating Coil

Provide an electric duct heater coil in accordance with UL 1995 and NFPA 70. Provide duct- or unit-mounted coil. Provide nickel chromium resistor, single stage, strip or stainless steel, fin tubular type coil. Provide coil with a built-in or surface-mounted high-limit thermostat interlocked electrically so that the coil cannot be energized unless the fan is energized. Provide galvanized steel or aluminum coil casing and support brackets. Mount coil to eliminate noise from expansion and contraction and for complete accessibility for service.

### 2.10.3 Air Filters

List air filters according to requirements of UL 900, except list high efficiency particulate air filters of 99.97 percent efficiency by the DOP Test method under the Label Service to meet the requirements of UL 586.

#### 2.10.3.1 Extended Surface Pleated Panel Filters

Provide 2 inch depth, sectional, disposable type filters of the size indicated with a MERV of 8 when tested according to ASHRAE 52.2. Provide initial resistance at 500 fpm that does not exceed 0.36 inches water gauge. Provide UL Class 2 filters, and nonwoven cotton and synthetic fiber mat media. Attach a wire support grid bonded to the media to a moisture resistant fiberboard frame. Bond all four edges of the filter media to the inside of the frame to prevent air bypass and increase rigidity.

#### 2.10.3.2 Holding Frames

Fabricate frames from not lighter than 16 gauge sheet steel with rust-inhibitor coating. Equip each holding frame with suitable filter holding devices. Provide gasketed holding frame seats. Make all joints airtight.

#### 2.10.3.3 Filter Gauges

Provide dial type filter gauges, diaphragm actuated draft for all filter stations, including those filters which are furnished as integral parts of factory fabricated air handling units. Provide gauges that are at least 3-7/8 inches in diameter, with white dials with black figures, and graduations with a minimum range of 1 inch of water beyond the specified final resistance for the filter bank on which each gauge is applied. Provide each gauge with a screw operated zero adjustment and two static pressure taps with integral compression fittings, two molded plastic vent valves, two 5 foot minimum lengths of 1/4 inch diameter aluminum tubing, and all hardware and accessories for gauge mounting.

## 2.11 TERMINAL UNITS

### 2.11.1 Unit Ventilators

Provide unit ventilators that include an enclosure, galvanized casing, coil assembly, resistance heating coil assembly, fan assembly, fan drive, motor, motor controller, dampers, damper operators, and sound power level as indicated. Obtain sound power level data or values for these units according to test procedures based on AHRI 350. Sound power values apply to units provided with factory fabricated cabinet enclosures and standard grilles, when handling standard flow for which the unit air capacity is

rated. Secure each unit to the building structure. Provide the unit ventilators with capacity indicated.

#### 2.11.1.1 Enclosures

Fabricate enclosures from not lighter than 16 gauge galvanized steel, reinforced and braced, or all welded framework with panels to provide equivalent strength. Provide casing that is acoustically and thermally insulated internally with not less than 1/2 inch thick dual density fibrous glass insulation. Make the exposed side a high density, erosion-proof material suitable for use in air streams with velocities up to 4500 fpm. Fasten the insulation with waterproof, fire-resistant adhesive. Design front panel for easy removal by one person. Provide discharge grilles that have adjustable grilles or grilles with adjustable vanes and properly distribute air throughout the conditioned space. Provide return grilles that are removable where front panel does not provide access to interior components. Plastic discharge or return grilles are not acceptable. Furnish removable panels or access doors for all piping and control compartments. Provide fan switch that is key operated or accessible through a locked access panel. Install gaskets at the back and bottom of the unit for effective air seal, as required.

#### 2.11.1.2 Electric Resistance Heating Elements

Provide electric resistance heating elements that are of the sheathed, finned, tubular type, or of the open resistance type designed for direct exposure to the air stream. Provide heating element electrical characteristics as indicated. Where fan motor or control voltage is lower than required for the electric-resistance heating element, install a fused factory mounted and wired transformer.

#### 2.11.1.3 Fans

Provide fans that meet the requirements as specified in paragraph AIR SYSTEMS EQUIPMENT. Provide galvanized steel or aluminum, multiblade, centrifugal type fans, dynamically and statically balanced. Equip fan housings with resilient mounted, self-aligning permanently lubricated ball bearings, sleeve bearings, or combination ball and sleeve bearings, capable of not less than 2000 hours of operation on one oiling. Provide direct-connected fans.

#### 2.11.1.4 Dampers

Provide an outside air proportioning damper on each unit. In addition, provide a vane to prevent excessive outside air from entering unit and to prevent blow-through of outside air through the return air grille under high wind pressures. Where outside air and recirculated air proportioning dampers are provided on the unit, an additional vane is not required. Provide face and bypass dampers for each unit to ensure constant air volume at all positions of the dampers. Furnish each unit with a factory installed control cam assembly, pneumatic motor, or electric motor to operate the face and bypass dampers and outside air damper or outside air and recirculated air dampers in the sequence as specified in paragraph SUPPLEMENTAL COMPONENTS/SERVICES, subparagraph CONTROLS.

#### 2.11.1.5 Motors

Provide permanent split-capacitor type motors with built-in thermal overload protection and automatic reset. Mount motor on a resilient

mounting, isolated from the casing and suitable for operation on electric service available. Provide a manually operated motor switch that provides for 2 or 3 speeds and off, mounted on an identified plate inside the unit below or behind an access door or adjacent to the room thermostats indicated. In lieu of speed control, provide a solid state variable speed controller having minimum speed reduction of 50 percent.

#### 2.11.1.6 Outside Air Intakes

Provide the manufacturer's standard design outside air intakes furnished with 1/2 inch mesh bird screen or louvers on 1/2 inch centers.

### 2.12 FACTORY PAINTING

Factory paint new equipment, which are not of galvanized construction. Paint with a corrosion resisting paint finish according to ASTM A123/A123M or ASTM A924/A924M. Clean, phosphatize and coat internal and external ferrous metal surfaces with a paint finish which has been tested according to ASTM B117, ASTM D1654, and ASTM D3359. Submit evidence of satisfactory paint performance for a minimum of 125 hours for units to be installed indoors and 500 hours for units to be installed outdoors. Provide rating of failure at the scribe mark that is not less than 6, average creepage not greater than 1/8 inch. Provide rating of the inscribed area that is not less than 10, no failure. On units constructed of galvanized steel that have been welded, provide a final shop docket of zinc-rich protective paint on exterior surfaces of welds or welds that have burned through from the interior according to ASTM D520 Type I.

Field paint factory painting that has been damaged prior to acceptance by the Contracting Officer in compliance with the requirements of paragraph FIELD PAINTING OF MECHANICAL EQUIPMENT.

### 2.13 SUPPLEMENTAL COMPONENTS/SERVICES

#### 2.13.1 Condensate Drain Lines

Provide and install condensate drainage for each item of equipment that generates condensate.

#### 2.13.2 Insulation

The requirements for shop and field applied insulation are specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

#### 2.13.3 Controls

The requirements for controls are specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC SYSTEMS and Section 23 09 00 INSTRUMENTATION AND CONTROL FOR HVAC.

## PART 3 EXECUTION

### 3.1 EXAMINATION

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing the work.

### 3.2 INSTALLATION

- a. Install materials and equipment in accordance with the requirements of the contract drawings and approved manufacturer's installation instructions. Accomplish installation by workers skilled in this type of work. Perform installation so that there is no degradation of the designed fire ratings of walls, partitions, ceilings, and floors.
- b. No installation is permitted to block or otherwise impede access to any existing machine or system. Install all hinged doors to swing open a minimum of 120 degrees. Provide an area in front of all access doors that clears a minimum of 3 feet. In front of all access doors to electrical circuits, clear the area the minimum distance to energized circuits as specified in OSHA Standards, part 1910.333 (Electrical-Safety Related work practices) and an additional 3 feet.
- c. Except as otherwise indicated, install emergency switches and alarms in conspicuous locations. Mount all indicators, to include gauges, meters, and alarms in order to be easily visible by people in the area.

#### 3.2.1 Condensate Drain Lines

Provide water seals in the condensate drain from all units. Provide a depth of each seal of 2 inches plus the number of inches, measured in water gauge, of the total static pressure rating of the unit to which the drain is connected. Provide water seals that are constructed of 2 tees and an appropriate U-bend with the open end of each tee plugged. Provide pipe cap or plug cleanouts where indicated. Connect drains indicated to connect to the sanitary waste system using an indirect waste fitting. Insulate air conditioner drain lines as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

#### 3.2.2 Equipment and Installation

Provide frames and supports for tanks, compressors, pumps, valves, air handling units, fans, coils, dampers, and other similar items requiring supports. Floor mount or ceiling hang air handling units as indicated. Anchor and fasten as detailed. Set floor-mounted equipment on not less than 6 inch concrete pads or curbs doweled in place unless otherwise indicated. Make concrete foundations heavy enough to minimize the intensity of the vibrations transmitted to the piping, duct work and the surrounding structure, as recommended in writing by the equipment manufacturer. In lieu of a concrete pad foundation, build a concrete pedestal block with isolators placed between the pedestal block and the floor. Make the concrete foundation or concrete pedestal block a mass not less than three times the weight of the components to be supported. Provide the lines connected to the pump mounted on pedestal blocks with flexible connectors. Submit foundation drawings as specified in paragraph DETAIL DRAWINGS. Provide concrete for foundations as specified in Section 03 30 00 CAST-IN-PLACE CONCRETE.

#### 3.2.3 Access Panels

Install access panels for concealed valves, vents, controls, dampers, and items requiring inspection or maintenance of sufficient size, and locate them so that the concealed items are easily serviced and maintained or completely removed and replaced.

#### 3.2.4 Flexible Duct

Install pre-insulated flexible duct in accordance with the latest printed instructions of the manufacturer to ensure a vapor tight joint. Provide hangers, when required to suspend the duct, of the type recommended by the duct manufacturer and set at the intervals recommended.

#### 3.2.5 Metal Ductwork

Install according to SMACNA 1966 unless otherwise indicated. Install duct supports for sheet metal ductwork according to SMACNA 1966, unless otherwise specified. Do not use friction beam clamps indicated in SMACNA 1966. Anchor risers on high velocity ducts in the center of the vertical run to allow ends of riser to move due to thermal expansion. Erect supports on the risers that allow free vertical movement of the duct. Attach supports only to structural framing members and concrete slabs. Do not anchor supports to metal decking unless a means is provided and approved for preventing the anchor from puncturing the metal decking. Where supports are required between structural framing members, provide suitable intermediate metal framing. Where C-clamps are used, provide retainer clips.

#### 3.2.6 Dust Control

To prevent the accumulation of dust, debris and foreign material during construction, perform temporary dust control protection. Protect the distribution system (supply and return) with temporary seal-offs at all inlets and outlets at the end of each day's work. Keep temporary protection in place until system is ready for startup.

#### 3.2.7 Insulation

Provide thickness and application of insulation materials for ductwork, piping, and equipment according to Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS. Externally insulate outdoor air intake ducts and plenums up to the point where the outdoor air reaches the conditioning unit or up to the point where the outdoor air mixes with the return air stream.

#### 3.2.8 Duct Test Holes

Provide holes with closures or threaded holes with plugs in ducts and plenums as indicated or where necessary for the use of pitot tube in balancing the air system. Plug insulated duct at the duct surface, patched over with insulation and then marked to indicate location of test hole if needed for future use.

#### 3.2.9 Power Roof Ventilator Mounting

Provide foamed 1/2 inch thick, closed-cell, flexible elastomer insulation to cover width of roof curb mounting flange. Where wood nailers are used, predrill holes for fasteners.

#### 3.2.10 Power Transmission Components Adjustment

Test V-belts and sheaves for proper alignment and tension prior to operation and after 72 hours of operation at final speed. Uniformly load belts on drive side to prevent bouncing. Make alignment of direct driven couplings to within 50 percent of manufacturer's maximum allowable range of misalignment.



### 3.3 EQUIPMENT PADS

Provide equipment pads to the dimensions shown or, if not shown, to conform to the shape of each piece of equipment served with a minimum 3-inch margin around the equipment and supports. Allow equipment bases and foundations, when constructed of concrete or grout, to cure a minimum of 28 calendar days before being loaded.

### 3.4 CUTTING AND PATCHING

Install work in such a manner and at such time that a minimum of cutting and patching of the building structure is required. Make holes in exposed locations, in or through existing floors, by drilling and smooth by sanding. Use of a jackhammer is permitted only where specifically approved. Make holes through masonry walls to accommodate sleeves with an iron pipe masonry core saw.

### 3.5 CLEANING

Thoroughly clean surfaces of piping and equipment that have become covered with dirt, plaster, or other material during handling and construction before such surfaces are prepared for final finish painting or are enclosed within the building structure. Before final acceptance, clean mechanical equipment, including piping, ducting, and fixtures, and free from dirt, grease, and finger marks. When the work area is in an occupied space such as office, laboratory or warehouse protect all furniture and equipment from dirt and debris. Incorporate housekeeping for field construction work which leaves all furniture and equipment in the affected area free of construction generated dust and debris; and, all floor surfaces vacuum-swept clean.

### 3.6 PENETRATIONS

Provide sleeves and prepared openings for duct mains, branches, and other penetrating items, and install during the construction of the surface to be penetrated. Cut sleeves flush with each surface. Place sleeves for round duct 15 inches and smaller. Build framed, prepared openings for round duct larger than 15 inches and square, rectangular or oval ducts. Sleeves and framed openings are also required where grilles, registers, and diffusers are installed at the openings. Provide one inch clearance between penetrating and penetrated surfaces except at grilles, registers, and diffusers. Pack spaces between sleeve or opening and duct or duct insulation with mineral fiber conforming with ASTM C553, Type 1, Class B-2.

#### 3.6.1 Sleeves

Fabricate sleeves, except as otherwise specified or indicated, from 20 gauge thick mill galvanized sheet metal. Where sleeves are installed in bearing walls or partitions, provide black steel pipe conforming with ASTM A53/A53M, Schedule 20.

#### 3.6.2 Framed Prepared Openings

Fabricate framed prepared openings from 20 gauge galvanized steel, unless otherwise indicated.

### 3.6.3 Insulation

Provide duct insulation in accordance with Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS continuous through sleeves and prepared openings except firewall penetrations. Terminate duct insulation at fire dampers and flexible connections. For duct handling air at or below 60 degrees F, provide insulation continuous over the damper collar and retaining angle of fire dampers, which are exposed to unconditioned air.

### 3.6.4 Closure Collars

Provide closure collars of a minimum 4 inches wide, unless otherwise indicated, for exposed ducts and items on each side of penetrated surface, except where equipment is installed. Install collar tight against the surface and fit snugly around the duct or insulation. Grind sharp edges smooth to prevent damage to penetrating surface. Fabricate collars for round ducts 15 inches in diameter or less from 20 gauge galvanized steel. Fabricate collars for square and rectangular ducts, or round ducts with minimum dimension over 15 inches from 18 gauge galvanized steel. Fabricate collars for square and rectangular ducts with a maximum side of 15 inches or less from 20 gauge galvanized steel. Install collars with fasteners a maximum of 6 inches on center. Attach to collars a minimum of 4 fasteners where the opening is 12 inches in diameter or less, and a minimum of 8 fasteners where the opening is 20 inches in diameter or less.

### 3.6.5 Firestopping

Where ducts pass through fire-rated walls, fire partitions, and fire rated chase walls, seal the penetration with fire stopping materials as specified in Section 07 84 00 FIRESTOPPING.

## 3.7 FIELD PAINTING OF MECHANICAL EQUIPMENT

Clean, pretreat, prime and paint metal surfaces; except aluminum surfaces need not be painted. Apply coatings to clean dry surfaces. Clean the surfaces to remove dust, dirt, rust, oil and grease by wire brushing and solvent degreasing prior to application of paint, except clean to bare metal on metal surfaces subject to temperatures in excess of 120 degrees F. Where more than one coat of paint is specified, apply the second coat after the preceding coat is thoroughly dry. Lightly sand damaged painting and retouch before applying the succeeding coat. Provide aluminum or light gray finish coat.

### 3.7.1 Temperatures less than 120 degrees F

Immediately after cleaning, apply one coat of pretreatment primer applied to a minimum dry film thickness of 0.3 mil, one coat of primer applied to a minimum dry film thickness of one mil; and two coats of enamel applied to a minimum dry film thickness of one mil per coat to metal surfaces subject to temperatures less than 120 degrees F.

### 3.7.2 Temperatures between 120 and 400 degrees F

Apply two coats of 400 degrees F heat-resisting enamel applied to a total minimum thickness of two mils to metal surfaces subject to temperatures between 120 and 400 degrees F.

### 3.7.3 Temperatures greater than 400 degrees F

Apply two coats of 315 degrees C 600 degrees F heat-resisting paint applied to a total minimum dry film thickness of two mils to metal surfaces subject to temperatures greater than 400 degrees F.

### 3.8 IDENTIFICATION SYSTEMS

Provide identification tags made of brass, engraved laminated plastic, or engraved anodized aluminum, indicating service and item number on all valves and dampers. Provide tags that are 1-3/8 inch minimum diameter with stamped or engraved markings. Make indentations black for reading clarity. Attach tags to valves with No. 12 AWG 0.0808-inch diameter corrosion-resistant steel wire, copper wire, chrome-plated beaded chain or plastic straps designed for that purpose.

### 3.9 DUCTWORK LEAK TESTS

The requirements for ductwork leak tests are specified in Section 23 05 93 TESTING, ADJUSTING AND BALANCING FOR HVAC.

### 3.10 DAMPER ACCEPTANCE TEST

Submit the proposed schedule, at least 2 weeks prior to the start of test. Operate all fire dampers and smoke dampers under normal operating conditions, prior to the occupancy of a building to determine that they function properly. Test each fire damper equipped with fusible link by having the fusible link cut in place. Test dynamic fire dampers with the air handling and distribution system running. Reset all fire dampers with the fusible links replaced after acceptance testing. To ensure optimum operation and performance, install the damper so it is square and free from racking.

### 3.11 TESTING, ADJUSTING, AND BALANCING

The requirements for testing, adjusting, and balancing are specified in Section 23 05 93 TESTING, ADJUSTING AND BALANCING FOR HVAC. Begin testing, adjusting, and balancing only when the air supply and distribution, including controls, has been completed, with the exception of performance tests.

### 3.12 PERFORMANCE TESTS

Conduct performance tests as required in Section 23 05 93 Testing, Adjusting and Balancing for HVAC and Section 23 09 00 Instrumentation and Control for HVAC.

### 3.13 CLEANING AND ADJUSTING

Provide a temporary bypass for water coils to prevent flushing water from passing through coils. Inside of air terminal units, unit ventilators, thoroughly clean ducts, plenums, and casing of debris and blow free of small particles of rubbish and dust and then vacuum clean before installing outlet faces. Wipe equipment clean, with no traces of oil, dust, dirt, or paint spots. Provide temporary filters prior to startup of all fans that are operated during construction, and provide new filters after all construction dirt has been removed from the building, and the ducts, plenums, casings, and other items specified have been vacuum cleaned. Perform and document that proper "Indoor Air Quality During

Construction" procedures have been followed; provide documentation showing that after construction ends, and prior to occupancy, new filters were provided and installed. Maintain system in this clean condition until final acceptance. Properly lubricate bearings with oil or grease as recommended by the manufacturer. Tighten belts to proper tension. Adjust control valves and other miscellaneous equipment requiring adjustment to setting indicated or directed. Adjust fans to the speed indicated by the manufacturer to meet specified conditions. Maintain all equipment installed under the contract until close out documentation is received, the project is completed and the building has been documented as beneficially occupied.

### 3.14 OPERATION AND MAINTENANCE

#### 3.14.1 Operation and Maintenance Manuals

Submit six manuals at least 2 weeks prior to field training. Submit data complying with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA. Submit Data Package 3 for the items/units listed under SD-10 Operation and Maintenance Data

#### 3.14.2 Operation And Maintenance Training

Conduct a training course for the members of the operating staff as designated by the Contracting Officer. Make the training period consist of a total of 8 hours of normal working time and start it after all work specified herein is functionally completed and the Performance Tests have been approved. Conduct field instruction that covers all of the items contained in the Operation and Maintenance Manuals as well as demonstrations of routine maintenance operations. Submit the proposed On-site Training schedule concurrently with the Operation and Maintenance Manuals and at least 14 days prior to conducting the training course.

-- End of Section --

SECTION 23 64 26

DESALINATION INTAKE SCREEN SYSTEM

08/09, CHG 7: 02/24

PART 1 GENERAL

1.1 SUMMARY

This Section includes intake screens, airburst backwash system, intake valves and valve actuators.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B31.9 (2020) Building Services Piping

ASME BPVC SEC IX (2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications

AMERICAN WELDING SOCIETY (AWS)

AWS Z49.1 (2021) Safety in Welding and Cutting and Allied Processes

1.3 SYSTEM DESCRIPTION

This specification includes desal intake system. Certain intake system components are a delegated design to be performed by the Contractor. Components and arrangements indicated are for reference only and are intended to show performance requirements. The intake screen, valves, valve extension and operators, airburst backwash system, controls, devices, and accessories must be designed by the system manufacturer. The intake design must include calculations showing pressure losses and flow velocities through the intake, screens, and the airburst system. Design must include calculations stamped and signed by a licensed Professional Engineer. The calculations must include a quantification of adequate volume and speed of the airburst system under the specified requirements. Commissioning program requirements to be set forth by delegated design.

1.4 BASIS OF DESIGN

The Basis of Design for the intake screens and airburst backwash system is Johnson Screens, New Brighton, MN, Hendrick Screen Design, Owensboro, KY, Elgin, Canal Winchester, OH, or approved equal. Products must meet requirements of the Buy American Act.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval.. When used, a code following the "G"

classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

The intake screens and airburst backwash system are long-lead items and submittals are critical submittals.

SD-01 Preconstruction Submittals

Valves; G, AE

Intake screens and airburst back wash system; G, AE

Intake Screen Flow Distribution; G, AE

SD-02 Shop Drawings

Detail Drawings; G, AE

SD-03 Product Data

Piping; G, AE

Valves; G, AE

Intake screens and airburst back wash system; G, AE

SD-05 Design Data; G

INTAKE SCREENS AND AIRBURST BACK WASH SYSTEM; G, AE

SD-06 Test Reports

Piping Welds NDE Report; G, AE

Intake Screen Flow Distribution; G, AE

INTAKE SCREENS AND AIRBURST BACK WASH SYSTEM; G, AE

SD-07 Certificates

Employer's Record Documents (For Welding); G

Welding Procedures and Qualifications; G

Certificates shall be submitted showing conformance with the referenced standards contained in this section.

INTAKE SCREENS AND AIRBURST BACK WASH SYSTEM; G, AE

SD-08 Manufacturer's Instructions

Lesson plan for the Instruction Course; G, AE

SD-10 Operation and Maintenance Data

Requirements for data packages are specified Section 01 78 23 OPERATION AND MAINTENANCE DATA, except as supplemented and modified by this specification section. Refer to Section

01 78 24.00 20 FACILITY ELECTRONIC OPERATION AND MAINTENANCE  
SUPPORT INFORMATION (eOMSI) for additional requirements.

Submit spare parts data for each different item of equipment specified, with operation and maintenance data packages. Include a complete list of parts and supplies, with current unit prices and source of supply, a recommended spare parts list for one year of operation, and a list of the parts recommended by the manufacturer to be replaced on a routine basis.

Submit a list of qualified permanent service organizations with operation and maintenance data packages. Include service organization addresses and service area or expertise. The service organizations shall be reasonably convenient to the equipment installation and be able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

Valves; G, AE

INTAKE SCREENS AND AIRBURST BACK WASH SYSTEM; G, AE

#### 1.6 SAFETY REQUIREMENTS

Exposed moving parts, parts that produce high operating temperature, parts which may be electrically energized, and parts that may be a hazard to operating personnel shall be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Safety devices shall be installed so that proper operation of equipment is not impaired.

#### 1.7 QUALITY CONTROL

##### 1.7.1 Detail Drawings

Submit drawings of sufficient size to be easily read, within 90 days of notice of Award of Contract. Submit information in the English language. Provide with English (IP) dimensions. Drawings must consist of complete designs of the intake and airburst backwash systems and other equipment, installation instructions, performance charts, brochures, and other information required to illustrate that the entire system has been coordinated and will function as a unit.

- a. Outline drawings of the system showing pertinent dimensions and weight of each component of the system. Prepare drawings to scale.
- b. Drawing showing details and dimensions of mounting design or layout including any embedded items.
- c. Cross-sectional drawings of the system showing each component. Show major or complicated sections of the system in detail. Indicate on each drawing an itemized list of components showing type, grade, and class of material used and make and model number of standard component used.
- d. Detail and assembly drawings required for manufacturing showing dimensions, tolerances, and clearances.
- e. Drawings covering erection and installation that are intended to be furnished to the erecting engineer.

## 1.8 DELIVERY, STORAGE, AND HANDLING

Protect stored items from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Proper protection and care of all material both before and during installation shall be the Contractor's responsibility. Any materials found to be damaged shall be replaced at the Contractor's expense. During installation, cap piping and similar openings to keep out dirt and other foreign matter. Any porous materials found to be contaminated with mold or mildew will be replaced at the Contractor's expense. Non-porous materials found to be contaminated with mold or mildew will be disinfected and cleaned prior to installation.

## 1.9 PROJECT/SITE CONDITIONS

### 1.9.1 Verification of Dimensions

The Contractor shall become familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy(ies) before performing any work.

### 1.9.2 Drawings

Because of the small scale of the drawings, it is not possible to indicate all offsets, fittings, and accessories that may be required. The Contractor shall carefully investigate the plumbing, fire protection, electrical, structural and finish conditions that would affect the work to be performed and shall arrange such work accordingly, furnishing required offsets, fittings, and accessories to meet such conditions.

### 1.9.3 Accessibility

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible.

## PART 2 PRODUCTS

### 2.1 STANDARD COMMERCIAL PRODUCTS

Materials and equipment must be standard products of a manufacturer regularly engaged in the manufacturing of such products, which are of a similar material, design, and workmanship. The standard products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The two-year use must include applications of equipment and materials under similar circumstances and of similar size. The 2 years' experience must be satisfactorily completed by a product which has been sold or is offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures.

The equipment items shall be supported by service organizations. These service organizations shall be reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

### 2.2 PIPING

Copper Nickel Pipe: 70/30 copper nickel pipe ASTM B411 and ASME SB466, IPS Class 200 wall.



Copper Nickel Flanges: 70/30 copper nickel ASTM B466 and ASME SB466, Class 150, coordinate bolt pattern with wet well bulkhead, valves, and intake screen

## 2.3 VALVES

### 2.3.1 Intake Screen Isolation Butterfly Valve

Butterfly valves for intake screens must be double offset design with sealing elements to provide zero leakage through non-rubbing 90-degree rotation. The body style shall be lug type and body and disk shall be constructed from super duplex material (ASTM A995 Gr 5A). Pressure rating and bolt pattern shall be class 150. The valve shall be manufactured by DelVal Flow Control LCC, or equal.

Valve must be suitable for sea water applications, The valve shall have the following characteristics:

| Item No | Part Name            | Material Specification |
|---------|----------------------|------------------------|
| 1       | Body                 | ASTM A 995 Gr. 5A      |
| 2       | Disc                 | ASTM A 995 Gr. 5A      |
| 3       | Seat                 | RPTFE                  |
| 4       | Stem                 | ASTM A479              |
| 5       | Bearing              | IGLIDUR-X              |
| 6       | Seat Retainer Ring   | ASTM A240              |
| 7       | Retainer Screw       | ASTM A1082             |
| 8       | Disc Spacer          | ASTM A479              |
| 9       | Wedge Key            | ASTM A479              |
| 10      | Stem Packing Support | ASTM 479               |
| 11      | Stem Packing         | PTFE                   |
| 12      | Gland                | ASTM A 479             |
| 13      | Snap Ring            | ASTM A313              |
| 14      | Gland Flange         | ASTM A240              |
| 15      | Belleville Washer    | ASTM A666              |
| 16      | Hex Nut              | ASTM A1082             |
| 17      | Bottom Plate Gasket  | PTFE                   |
| 18      | Bottom Plate         | ASTM A240              |
| 19      | Spring Washer        | ASTM A580              |
| 20      | Hex Bolt             | ASTM A1082             |

| Item No | Part Name              | Material Specification |
|---------|------------------------|------------------------|
| 21      | Square Nut Size: 2 in. | ASTM A1082             |

## 2.4 INTAKE SCREENS AND AIRBURST BACK WASH SYSTEM

### 2.4.1 General

- a. Provide delegated designs of intake screens and airburst backwash systems. The pump intake system components and equipment must be furnished as a complete integrated system by one manufacturer to ensure a properly designed and integrated intake system.
- b. System must be designed for hydrostatic load depth 30 feet and collapse rating of 13 psi.
- c. Intake Screens assembly must be horizontal orientation, of all-welded continuous slot construction and must be designed for air flushing of screen surface for debris removal. Equipment manufacturer must maintain an ongoing quality assurance program, preferably ISO-9000 certification. All welders must maintain certification to ASME Section IX. Copies of certification must be provided upon request.
- d. The single manufacturer supplying this equipment must be able to furnish proof of at least 5 installations and 10 years of manufacturing equipment of similar technology.
- e. Manufacturer must submit a complete and detailed pre-submittal package, including delegated designs, containing the below information:
  - (1) Detailed layout drawings.
  - (2) Supporting Intake Screen Flow Distribution data via a CFD (Computational Fluid Dynamics) analysis.
  - (3) Weld certifications.
  - (4) Evidence of a recognized ongoing quality assurance program.
  - (5) Detailed component specifications and catalog cuts as required.
  - (6) Detailed list of all variations required from the original design if an alternate screen is used, referencing appropriate Sections of the specifications and locations on the drawings.

### 2.4.2 Capacity and Strength

- a. Intake assembly capacity must be 1550 GPM (2.23 MGD) at a maximum through-slot velocity, as a result of water withdrawal, of 0.35 feet per second when screens are clean and 0.45 feet per second when the

screens are 20 percent blocked. The corresponding average through-slot velocity must be 80-90 percent of the maximum velocity. The hydraulic design of this system is based on this maximum headloss of 8 inches through a 20 percent blocked screen assembly. Screens that exceed this headloss are not acceptable. The manufacturer's clean screen assembly headloss must be stated in the delegated design documents.

- b. Evidence of the intake assembly capacity and flow distribution must be proven by a Computational Fluid Dynamic (CFD) analysis, supplied by the manufacturer. The CFD Analysis Method must be verified by calculations supported by previously completed physical flow distribution tests.
- c. The intake assembly shall be designed to a maximum of 4.33 psi negative pump pressure or differential head loss.

#### 2.4.3 Construction and Slot Opening Size

- a. Intake screen surface wire, support beam and stiffener structure must be an all-welded matrix designed to provide the specific strength with minimal interference with the through screen flow pattern.
- b. End plates must be a minimum of 0.105 inches thick. All structural butt welds must be full penetration fillet welds and must be the thickness of the thinner component.
- c. Screen slot size must be 0.0315 inch (0.8 mm). Slot size must be controlled and continuously monitored during manufacture.
- d. The mean slot size must be within +/- 0.002-inch with a standard deviation no greater than 0.002-inch throughout the assembly.
- e. The screen shall be provided with an end support connection to allow the screen to be supported.
- f. The screen shall have a maximum outside diameter of 36 inches and a maximum overall length of 70 inches.

#### 2.4.4 Materials

- a. Main outlet flange must mate with a flange pattern to mount directly to the pipe as shown on the drawings.
- b. Air connection must be 2.0-inch pipe size flange.
- c. Intake screen assembly must be manufactured of Z-Alloy or 70/30 copper-nickel material.
- d. The end of the screen shall be extended to allow support by a stanchion or provided with legs welded to structural supports of the screen that would allow stanchion supports to be welded to the legs.

#### 2.4.5 Airburst Backwash System

Intake screen supplier must provide, as part of the overall intake screen system, an airburst backwash system designed to remove debris from the screen surface by delivering a suitable volume of compressed air to the inside of the screen body. The exiting air must scour the screen surface to maintain adequate design flow and through slot velocity characteristics.

- a. Airburst Backwash System
  - (1) The airburst backwash system must consist of an integrated system of compressor, receiver tank, valves, and control panel. All components must be skid mounted and arrive on site ready to operate. Systems requiring field assembly and not ready to operate is not acceptable.
- b. Compressed air lines after the air compressor serving the intake structure must be:
  - (1) 70-30 copper nickel Class 700 with socket weld joints once at the docking seawall and in the seawater.
  - (2) Compressed air lines from building, to the 90 degree bend may be 316 stainless steel, polyethylene coated and wrapped.
- c. The air compressor must be a reciprocating type or rotary screw, pressure lubricated, and must be sized to recover from each backwash in 15 minutes. The compressor system shall have the following characteristics:
  - (1) 3/60/460V, TEFC 5HP Electric Motor.
  - (2) IP 54 Electrical Enclosure.
  - (3) Belt drive with automatic tensioning.
  - (4) Provided with a sound Enclosure to provide 63 dBa. rating or less.
  - (5) Ambient Temp (max/min) rating: 115F / 40F.
  - (6) Factory filled with Food grade oil.
    - (a) Provided with Moisture separator & condensate management.
- d. The receiver must be a 200 gallon (min) horizontal or vertical receiver, ASME coded for 200 psig,. The tanks shall be sized for the system piping and to displace three (3) screen volumes of air at the screen in 3-5 seconds during a backwash to provide suitable debris removal and cleaning. Alternatively, the tanks can be size per manufacturer recommendations such that the performance would be equal or better than as described above. The receiver must be equipped with a 4-inch isolated pressure gauge, safety relief valve, check valve, and automatic drain valve. The pipe supplied shall be of a size and thickness to be self-supporting of any gauges or equipment mounted to it.
- e. The system must include two fast-acting pneumatic valves rated at 200 psi with standard ANSI class 125/150 flanges. There must be one valve per screen and must be sized to match the tank flange and the backwash piping as part of the delegated design.
- f. Automatic Operation
  - (1) The above fast-acting valves must be provided with a spring return pneumatic actuator, single solenoid control with

position indicator, and position switches.

- (2) A stainless steel NEMA 4X control panel must be included that will contain as a minimum, the motor starter for the compressor, control power transformer (if not wired directly to 120V) and a programmable logic controller (PLC) with Human Machine Interface (MHI) touchscreen to perform the specified control functions. The panel will have the following features:
  - (a) All indicators will be LED type
  - (b) Lockable Disconnect with Type J Fusing
  - (c) 7-Day Programmable Time
  - (d) System On-Off Selector Switch
  - (e) Cycle Start Push Button
  - (f) Event Mode Selector Switch
  - (g) Digital pressure controller with readout
  - (h) System Power On Pilot Light
  - (i) Low Air Pressure Alarm Pilot Light
  - (j) Adequate Air Pressure Pilot Light
  - (k) Cycle in progress Pilot Light
  - (l) Relays
  - (m) Auxiliary Contacts
  - (n) Compression Type Terminal Blocks capable of 1 #10 wire or 2, #12 wires
  - (o) Isolated SCADA / PLC Output Contacts
    - 1. Cycle Start - Option for remote signal to initiate an air burst cycle.
    - 2. Valve Open
    - 3. Valve Closed
    - 4. Adequate Air
    - 5. Low Air Pressure
- (3) Controls must allow timed automatic, low level in the wetwell automatic clean, or manual initiated air cleaning cycles. A cycle consists of cleaning both intake system screens. The control system included must allow a full week duration of programmed airburst cycles.
- (4) Contractor must provide interconnecting piping and control wiring between the intake screen assembly and the airburst backwash

system.

- g. Onsite Commissioning and Training: A manufacturer's technician must be on the site for system start up and commissioning, and conduct a training course for system operators. Cost shall include all travel costs as well.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Cut pipe accurately to measurements established at the jobsite, and work into place without springing or forcing, completely clearing all windows, doors, and other openings. Cutting or other weakening of the building structure to facilitate piping installation is not permitted without written approval. Cut pipe or tubing square, remove burrs by reaming, and fashion to permit free expansion and contraction without causing damage to the building structure, pipe, joints, or hangers.

Notify the Contracting Officer in writing at least 15 calendar days prior to the date the connections are required. Obtain approval before interrupting service. Furnish materials required to make connections into existing systems and perform excavating, backfilling, compacting, and other incidental labor as required. Furnish labor and tools for making actual connections to existing systems.

#### 3.1.1 Welding

##### 3.1.1.1 Welding and Brazing:

Provide welding work specified in this Section for piping systems in conformance with ASME B31.9, as modified and supplemented by this specification Section and the accompanying drawings. The welding work includes: qualification of welding procedures, welders, welding operators, and nondestructive examination personnel; maintenance of welding records, and examination methods for welds.

##### 3.1.1.2 Employer's Record Documents (For Welding)

Submit for review and approval the following documentation. This documentation and the subject qualifications shall be in compliance with ASME B31.9.

- a. List of qualified welding procedures that is proposed to be used to provide the work specified in this specification section.
- b. List of qualified welders, brazers, welding operators, and brazing operators that are proposed to be used to provide the work specified in this specification section.
- c. List of qualified weld examination personnel that are proposed to be used to provide the work specified in this specification section.

##### 3.1.1.3 Welding Procedures and Qualifications

- a. Specifications and Test Results: Submit copies of the welding procedures specifications and procedure qualification test results for each type of welding required. Approval of any procedure does not relieve the Contractor of the responsibility for producing acceptable

welds. Submit this information on the forms printed in ASME BPVC SEC IX or their equivalent.

- b. Certification: Before assigning welders or welding operators to the work, submit a list of qualified welders, together with data and certification that each individual is performance qualified as specified. Do not start welding work prior to submitting welder, and welding operator qualifications. The certification shall state the type of welding and positions for which each is qualified, the code and procedure under which each is qualified, date qualified, and the firm and individual certifying the qualification tests.

#### 3.1.1.4 Examination of Piping Welds

Conduct non-destructive examinations (NDE) on piping welds and brazing and verify the work meets the acceptance criteria specified in ASME B31.9. NDE on piping welds covered by ASME B31.9 is visual inspection only. Submit a piping welds NDE report meeting the requirements specified in ASME B31.9.

#### 3.1.1.5 Welding Safety

Welding and cutting safety requirements shall be in accordance with AWS Z49.1.

#### 3.1.2 Fittings and End Connections

##### Welded Connections

- a. Branch connections must be made with welding tees or forged welding branch outlets. Pipe must be thoroughly cleaned of all scale and foreign matter before the piping is assembled. During welding, the pipe and fittings must be filled with an inert gas, such as nitrogen, to prevent the formation of scale. Beveling, alignment, heat treatment, and inspection of weld must conform to ASME B31.9. Weld defects must be removed and rewelded at no additional cost to the Government. Electrodes must be stored and dried in accordance with AWS D1.1/D1.1M or as recommended by the manufacturer.
- b. Electrodes that have been wetted or that have lost any of their coating must not be used.

#### 3.1.3 Flanges

Flanged joints must be provided in each line immediately preceding the connection to each piece of equipment or material requiring maintenance such as pumps, control valves, and other similar items. Flanged joints must be assembled square end tight with matched flanges, gaskets, and bolts. Gaskets must be suitable for the intended application.

#### 3.1.4 Pipe Hangers, Inserts, and Supports

Supports must be designed by and stamped by a Professional Engineer and submitted. This cost will be carried by the Contractor as part of their bid.

Material must be compatible with the environment the structural attachment it is placed in.

#### 3.1.4.1 Multiple Pipe Runs

In the support of multiple pipe runs on a common base member, a clip or clamp shall be used where each pipe crosses the base support member. Spacing of the base support members shall not exceed the hanger and support spacing required for an individual pipe in the multiple pipe run.

#### 3.1.4.2 Structural Attachments

Structural brackets required to support piping, headers, and equipment, but not shown, must be provided under this Section. Material must be compatible with the environment the structural attachment is placed in. Supports must be designed by and stamped by a Professional Engineer. This cost will be carried by the Contractor.

#### 3.1.5 Building Surface Penetrations

Cores or sleeves must not pass through structural members unless specifically detailed on the Drawings.

##### 3.1.5.1 Waterproof Penetrations

Penetrations must be sealed by either one of the following methods.

- a. Modular Mechanical Type Sealing Assembly: Modular mechanical type sealing assembly must be installed. Seals must consist of interlocking synthetic rubber links shaped to continuously fill the annular space between the pipe/conduit and sleeve with 316 stainless steel bolts, nuts, and pressure plates. Links must be loosely assembled with bolts to form a continuous rubber belt around the pipe with a pressure plate under each bolt head and each nut.
- b. After the seal assembly is properly positioned in the sleeve, tightening of the bolt must cause the rubber sealing elements to expand and provide a watertight seal between the pipe/conduit and the sleeve. Each seal assembly must be sized as recommended by the manufacturer to fit the pipe/conduit and sleeve involved. The Contractor must provide sleeves of the proper diameters.
- c. Existing Construction: Pipe penetration of the existing concrete can be done without sleeve. Seal concrete core hole with a mechanical link seal. Fill interior with silicone sealant, and fill exterior with non-shrink grout. Sleeves must be provided in holes through existing concrete where holes would not allow tight seal.

#### 3.2 INSPECTING AND CLEANING

The pipes must be flushed with cold water after finishing the installation. Flush the system until the water runs clear of debris and dirt. Inspect and test piping systems following procedures of authorities having jurisdiction and as specified by the piping system manufacturer.

Clean water distribution piping following procedures of the manufacturer and/or the Authority Having Jurisdiction.

#### 3.3 ELECTRICAL INSTALLATION

Install electrical equipment in accordance with NFPA 70 and manufacturers



instructions. Wiring must be installed in conduit.

### 3.4 FIELD TESTS

Field tests shall be conducted in the presence of the QC Manager or his designated representative to verify systems compliance with specifications. Any material, equipment, instruments, and personnel required for the test shall be provided by the Contractor.

### 3.5 INSTRUCTION TO GOVERNMENT PERSONNEL

Furnish the services of competent instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the desalination intake structure. Instructors shall be thoroughly familiar with all parts of the installation and shall be instructed in operating theory as well as practical operation and maintenance work. Submit a lesson plan for the instruction course for approval. The lesson plan and instruction course shall be based on the approved operation and maintenance data and maintenance manuals.

Conduct a training course for the operating staff and maintenance staff selected by the Contracting Officer. Give the instruction during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The number of man-days (8 hours per day) of instruction furnished shall be one man-day. Use approximately half of the time for classroom instruction and the other time for instruction at the location of equipment or system.

When significant changes or modifications in the equipment or system are made under the terms of the contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

Training Course Documentation: Training must be based on the operation and maintenance manuals and control system training manual. Deliver manuals for each trainee with two additional sets for archiving at the project site. Include an agenda, defined objectives, and a detailed description of subject matter for each lesson.

-- End of Section --

SECTION 23 81 00

DECENTRALIZED UNITARY HVAC EQUIPMENT  
05/24

PART 1 GENERAL

1.1 RELATED REQUIREMENTS

Section 23 03 00.00 20 BASIC MECHANICAL MATERIALS AND METHODS, applies to this section with the additions and modifications specified herein.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AIR-CONDITIONING, HEATING AND REFRIGERATION INSTITUTE (AHRI)

AHRI 700 (2016) Specifications for Fluorocarbon Refrigerants

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS (ASHRAE)

ASHRAE 15 & 34 (2013) ASHRAE Standard 34-2016 Safety Standard for Refrigeration Systems/ASHRAE Standard 34-2016 Designation and Safety Classification of Refrigerants-ASHRAE Standard 34-2016

ASHRAE 52.2 (2017) Method of Testing General Ventilation Air-Cleaning Devices for Removal Efficiency by Particle Size

ASHRAE 55 (2023) Thermal Environmental Conditions for Human Occupancy

ASHRAE 62.1 (2022) Ventilation for Acceptable Indoor Air Quality

ASHRAE 90.1 - IP (2019) Energy Standard for Buildings Except Low-Rise Residential Buildings

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME BPVC SEC IX (2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications

ASME BPVC SEC VIII D1 (2019) BPVC Section VIII-Rules for Construction of Pressure Vessels Division 1

AMERICAN WELDING SOCIETY (AWS)

AWS Z49.1 (2021) Safety in Welding and Cutting and Allied Processes

ASSOCIATION OF HOME APPLIANCE MANUFACTURERS (AHAM)

AHAM RAC-1 (1982; R2008) Directory of Certified Room Air Conditioners

ASTM INTERNATIONAL (ASTM)

ASTM B117 (2019) Standard Practice for Operating Salt Spray (Fog) Apparatus

ASTM C1071 (2019) Standard Specification for Fibrous Glass Duct Lining Insulation (Thermal and Sound Absorbing Material)

ASTM D520 (2000; R 2011) Zinc Dust Pigment

ASTM E84 (2023) Standard Test Method for Surface Burning Characteristics of Building Materials

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 (2021) Motors and Generators

NEMA MG 2 (2014) Safety Standard for Construction and Guide for Selection, Installation and Use of Electric Motors and Generators

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-DTL-5541 (2006; Rev F; Notice 2 2024) Chemical Conversion Coatings on Aluminum and Aluminum Alloys

UL SOLUTIONS (UL)

UL 484 (2014; Reprint Jun 2022) UL Standard for Room Air Conditioners

UL 586 (2009; Reprint Sep 2022) UL Standard for Safety High-Efficiency Particulate, Air Filter Units

UL 900 (2015; Reprint Aug 2022) UL Standard for Safety Standard for Air Filter Units

UL 1995 (2015; Reprint Aug 2022) UL Standard for Safety Heating and Cooling Equipment

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-03 Product Data

Spare Parts

Posted Instructions

Coil Corrosion Protection

System Performance Tests

Training; G

Inventory

Supplied Products

Manufacturer's Standard Catalog Data

#### SD-06 Test Reports

System Performance Tests; G

#### SD-07 Certificates

Service Organizations

#### SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

#### SD-11 Closeout Submittals

Ozone Depleting Substances; S

### 1.4 QUALITY ASSURANCE

Carefully investigate the plumbing, fire protection, electrical, structural and finish conditions that would affect the work to be performed and arrange such work accordingly, furnishing required offsets, fittings, and accessories to meet such conditions. Submit drawings consisting of:

- a. Equipment layouts which identify assembly and installation details.
- b. Plans and elevations which identify clearances required for maintenance and operation.

- c. Wiring diagrams which identify each component individually and interconnected or interlocked relationships between components.
- d. Foundation drawings, bolt-setting information, and foundation bolts prior to concrete foundation construction for equipment indicated or required to have concrete foundations.
- e. Details, if piping and equipment are to be supported other than as indicated, which include loadings and type of frames, brackets, stanchions, or other supports.
- f. Automatic temperature control diagrams and control sequences.
- g. Installation details which includes the amount of factory set superheat and corresponding refrigerant pressure/temperature.
- h. Equipment schedules

#### 1.5 DELIVERY, STORAGE, AND HANDLING

Protect stored items from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Properly protect and care for all material both before and during installation. Submit an inventory of all the stored items. Replace any materials found to be damaged, at no additional cost to the Government. During installation, cap piping and similar openings capped to keep out dirt and other foreign matter.

#### 1.6 ENVIRONMENTAL REQUIREMENTS

For proper Indoor Environmental Quality, maintain pressure within the building as indicated. Ventilation must meet or exceed ASHRAE 62.1 and all published addenda. Meet or exceed filter media efficiency as tested in accordance with ASHRAE 52.2. Thermal comfort must meet or exceed ASHRAE 55.

#### 1.7 WARRANTY

Provide equipment with the Manufacturer's Standard Warranty.

### PART 2 PRODUCTS

#### 2.1 ENERGY EFFICIENCY REQUIREMENTS

42 USC 8259b requires the procurement of energy efficient products in product categories covered by the Energy Star program or the Federal Energy Management Program for designated products. A list of covered product categories is available from the Federal Energy Management Web site at: <https://www.energy.gov/femp/search-energy-efficient-products>. A list of qualified light commercial products is available at: <http://www.energystar.gov/productfinder/product/certified-light-commercial-hvac/result>

Submit Material, Equipment, and Fixtures List of all supplied products within a covered product category, including manufacturer's catalog numbers, specification and drawing reference number, warranty information, fabrication site, and energy performance data. For product categories covered by the Energy Star program, submit documentation that the product is Energy Star-qualified. For product categories covered by the Federal

Energy Management Program, submit documentation that the product meets or exceeds FEMP-designated efficiency requirements.

#### 2.1.1 Room Air Conditioners

Selected room air conditioners are required to meet performance requirements specified by Energy Star. Information on the requirements can be found at:

[https://www.energystar.gov/products/room\\_air\\_conditioners/partners](https://www.energystar.gov/products/room_air_conditioners/partners).

### 2.2 MATERIALS

Provide Manufacturer's standard catalog data, at least 5 weeks prior to the purchase or installation of a particular component, highlighted to show material, size, options, performance charts and curves, etc. in adequate detail to demonstrate compliance with contract requirements. Data includes manufacturer's recommended installation instructions and procedures. If vibration isolation is specified for a unit, include vibration isolator literature containing catalog cuts and certification that the isolation characteristics of the isolators provided meet the manufacturer's recommendations. Submit data for each specified component. Minimum efficiency requirements must be in accordance with ASHRAE 90.1 - IP.

#### 2.2.1 Standard Products

Provide materials and equipment that are standard products of a manufacturer regularly engaged in the manufacturing of such products, which are of a similar material, design and workmanship. The standard products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2 year use includes applications of equipment and materials under similar circumstances and of similar size. The 2 years' experience must be satisfactorily completed by a product which has been sold or is offered for sale on the commercial market through advertisements, manufacturer's catalogs, or brochures. Products having less than a 2 year field service record will be acceptable if a certified record of satisfactory field operation, for not less than 6000 hours exclusive of the manufacturer's factory tests, can be shown. Products must be supported by a service organization. Ensure system components are environmentally suitable for the indicated geographic locations.

#### 2.2.2 Product Sustainability Criteria

##### 2.2.2.1 Energy Efficient Equipment

Provide equipment meeting the efficiency requirements as stated within this section.

##### 2.2.2.2 Electrical Equipment / Motors

Provide electrical equipment, motors, motor efficiencies, and wiring which are in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Electrical motor driven equipment specified must be provided complete with motors, motor starters, and controls. Electrical characteristics must be as shown, and unless otherwise indicated, all motors of 1 horsepower and above with open, dripproof, totally enclosed, or explosion proof fan cooled enclosures, must be the premium efficiency type in accordance with

NEMA MG 1. Field wiring must be in accordance with manufacturer's instructions. Each motor must conform to NEMA MG 1 and NEMA MG 2 and be of sufficient size to drive the equipment at the specified capacity without exceeding the nameplate rating of the motor. Motors must be continuous duty with the enclosure specified. Motor starters must be provided complete with thermal overload protection and other appurtenances necessary for the motor control indicated. Motors must be furnished with a magnetic across-the-line or reduced voltage type starter as required by the manufacturer. Motor duty requirements must allow for maximum frequency start-stop operation and minimum encountered interval between start and stop. Motors must be sized for the applicable loads. Motor torque must be capable of accelerating the connected load within 20 seconds with 80 percent of the rated voltage maintained at motor terminals during one starting period. Motor bearings must be fitted with grease supply fittings and grease relief to outside of enclosure. Manual or automatic control and protective or signal devices required for the operation specified and any control wiring required for controls and devices specified, but not shown, must be provided.

#### 2.2.2.3 Ozone Depleting Substances

Unitary air conditioning equipment must not use CFC-based refrigerants. Refrigerant may be an approved alternative refrigerant in accordance with EPA's Significant New Alternative Policy (SNAP) listing..

#### 2.2.2.4 Local/Regional Materials

Use materials or products extracted, harvested, or recovered, as well as manufactured, within a 500 mile radius from the project site, if available from a minimum of three sources.

#### 2.2.3 Nameplates

Major equipment including compressors, condensers, receivers, heat exchanges, fans, and motors must have the manufacturer's name, address, type or style, model or serial number, and catalog number on a plate secured to the item of equipment. Plates must be durable and legible throughout equipment life and made of stainless steel. Fix plates in prominent locations with nonferrous screws or bolts.

#### 2.2.4 Safety Devices

Exposed moving parts, parts that produce high operating temperature, parts which may be electrically energized, and parts that may be a hazard to operating personnel must be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Safety devices must be installed so that proper operation of equipment is not impaired. Welding and cutting safety requirements must be in accordance with AWS Z49.1.

### 2.3 EQUIPMENT

#### 2.3.1 Room Air Conditioner

Provide a through-the-wall mounted, appliance grade, factory assembled air conditioner unit in accordance with AHAM RAC-1 and UL 484. Units must include a self-contained, precharged, slide-in and removable chassis-mounted, air-cooled refrigeration system. Provide units removable from inside the building for servicing without removing the outside cabinet. Mount compressors on vibration isolators. Minimum cooling

capacity must be not less than that indicated. Provide units listed in the AHAM RAC-1. Cooling section must be equipped with a filter-drier on the suction line. Fan and condenser motors must have totally enclosed enclosures. Room Air Conditioners must have a minimum energy efficiency ratio (EER) of 10, and a minimum COP of 2.

#### 2.3.1.1 Primary/Supplemental Heat

Provide heating unit with internal thermal insulation having a fire hazard rating not to exceed 25 for flame spread and 50 for smoke developed as determined by ASTM E84.

##### 2.3.1.1.1 Electric Heating

Provide electric duct heater in accordance with UL 1995 and NFPA 70. Coil must be completely assembled, unit-mounted, and integral to the unit. Provide coil with nickel chromium elements and a maximum density of 40 watts per square inch. provide coil with automatic reset high limit control operating through heater backup contactors. Provide coil casing and support brackets of galvanized steel or aluminum. Mount coil to eliminate noise from expansion and contraction and be completely accessible for service. Electric resistance heating elements with high temperature-limit safety device, factory-mounted, and wired to chassis.

#### 2.3.1.2 Filters

Provide air filters of the throw-away or permanent washable type removable without the use of tools and arranged to filter both room and ventilating air. Filters must have a minimum efficiency reporting value (MERV) of 13 when tested in accordance with ASHRAE 52.2.

#### 2.3.1.3 Fans

Provide direct driven, statically and dynamically, centrifugal or propeller type fans. Design outdoor fan so that condensate evaporates without drip, splash, or spray on building exterior. Remove condensate by means of a drain or by evaporation and diffusion.

#### 2.3.1.4 Casing

Provide exterior casings for the specified room HVAC Units constructed of factory phosphatized and painted galvanized steel or aluminum sheet metal and galvanized or aluminum structural members. Fit casing with lifting provisions, access panels or doors, fan vibration isolators, electrical control panel, corrosion-resistant components, structural support members, insulated condensate drip pan and drain, and internal insulation in the cold section of the casing. Incorporate provisions to permit replacement of major unit components. Seal penetrations of cabinet surfaces, including the floor. Unit base must be watertight. Fit unit with a drain pan which extends under all areas where water may accumulate. Fabricate drain pan from Type 30X stainless steel, galvanized steel with protective coating as required, or an approved plastic material. Pan insulation must be water impervious. Extent and effectiveness of the insulation of unit air containment surfaces must prevent, within limits of the specified insulation, heat transfer between the unit exterior and ambient air, heat transfer between the two conditioned air streams, and condensation on surfaces. Insulation must conform to ASTM C1071.

Construct outside cabinets, including metal grilles to protect condenser



coils, of zinc-coated steel or aluminum. Steel and zinc-coated surfaces must receive at least one coat of primer and manufacturer's standard factory-applied finish. Insulate cabinets to prevent condensation and run off of moisture. Provide mounting hardware made of corrosion-resistant material or protected by a corrosion-resistant finish. Provide with metal or plastic mounting flanges on each side, top, and bottom of unit. For through-the-wall installations provide aluminum or shop painted zinc-coated steel flanged telescopic wall sleeves. Design wall sleeves to restrict driving rain. For window mounted units provide shop-painted metal mounting brackets, braces, and sill plates.

#### 2.3.1.5 Energy Efficiency

Minimum energy efficiency ratio (EER) must be in accordance with the paragraph EQUIPMENT EFFICIENCY. Room air conditioners must include the Energy Star label affixed to the equipment.

#### 2.3.1.6 Units for Operation on 115 Volts

Provide 3-wire cords of manufacturer's standard length. If not existing, provide a receptacle within reach of the standard length cord. Cords must have a 15- or 20-amp, 3-pole, 125-volt ground type plug to match receptacle.

#### 2.3.1.7 Units for Operation on 208 or 230 Volts

Provide 3-wire cords of manufacturer's standard length. If not existing, provide a receptacle within reach of the standard length cord. Cords must have a 15-, 20-, or 30-amp, 3-pole, 250-volt ground type plug to match receptacle.

#### 2.3.1.8 Controls

Provide units internally prewired by manufacturer with a 24 volt control circuit powered by an internal transformer. Terminal blocks must be provided for power wiring and external control wiring. Unit must be internally protected by fuses or a circuit breaker in accordance with UL 1995. Unit must be provided with microprocessor controls to provide all 24V control functions.

Mount controls in cabinet. Manual controls must permit operation of either the fan or the fan and refrigerating equipment. Fan control must provide two fan speed settings. Automatic controls must include a thermostat for controlling air temperature. Thermostat must have an adjustable range, including 72 to 80 degrees F and must automatically turn the refrigeration system on or off to maintain the preselected temperature within plus or minus 4 degrees F.

### 2.4 COMPONENTS

#### 2.4.1 Refrigerant and Oil

Refrigerant must be one of the fluorocarbon gases. Refrigerants must have number designations and safety classifications in accordance with ASHRAE 15 & 34. Refrigerants must meet the requirements of AHRI 700 as a minimum. Provide a complete charge of refrigerant for the installed system as recommended by the manufacturer. Lubricating oil must be of a type and grade recommended by the manufacturer for each compressor. Where

color leak indicator dye is incorporated, charge must be in accordance with manufacturer's recommendation.

#### 2.4.2 Fans

Fan wheel shafts must be supported by either maintenance-accessible lubricated antifriction block-type bearings, or permanently lubricated ball bearings. Unit fans must be selected to produce the cfm required at the fan total pressure. Motor starters, if applicable, must be magnetic across-the-line type with a opendriprooof enclosure. Thermal overload protection must be of the manual or automatic-reset type. Fan wheels or propellers must be constructed of aluminum or galvanized steel. Centrifugal fan wheel housings must be of galvanized steel, and both centrifugal and propeller fan casings must be constructed of aluminum or galvanized steel. Steel elements of fans, except fan shafts, must be hot-dipped galvanized after fabrication or fabricated of mill galvanized steel. Mill-galvanized steel surfaces and edges damaged or cut during fabrication by forming, punching, drilling, welding, or cutting must be recoated with an approved zinc-rich compound. Fan wheels or propellers must be statically and dynamically balanced. Forward curved fan wheels must be limited to 24 inches. Direct-drive fan motors must be of the multiple-speed variety. Belt-driven fans must have adjustable sheaves to provide not less than 100 percent fan-speed adjustment. The sheave size must be selected so that the fan speed at the approximate midpoint of the sheave adjustment will produce the specified air quantity. Centrifugal scroll-type fans must be provided with streamlined orifice inlet and V-belt drive. Each drive will be independent of any other drive. Propeller fans must be direct-drive drive type with adjustable pitch blades. V-belt driven fans must be mounted on a corrosion protected drive shaft supported by either maintenance-accessible lubricated antifriction block-type bearings, or permanently lubricated ball bearings. Each drive will be independent of any other drive. Drive bearings must be protected with water slingers or shields. V-belt drives must be fitted with guards where exposed to contact by personnel and fixed pitch sheaves.

#### 2.4.3 Primary/Supplemental Heating

##### 2.4.3.1 Electric Heating Coil

Coil must be an electric duct heater in accordance with UL 1995 and NFPA 70. Coil must be duct- or unit-mounted. Coil must be of the nickel chromium resistor, single stage, strip or stainless steel, fin tubular type. Coil must be provided with a built-in or surface-mounted high-limit thermostat interlocked electrically so that the coil cannot be energized unless the fan is energized. Coil casing and support brackets must be of galvanized steel or aluminum. Coil must be mounted to eliminate noise from expansion and contraction and be completely accessible for service. Supplemental Electric Resistance Heating controls must be provided to prevent operation when the heating load can be met by the primary source.

#### 2.4.4 Air Filters

Provide filters to filter outside air and return air and locate as indicated inside air conditioners inside filter box. Provide replaceable (throw-away) type. Filters must conform to UL 900, Class 1 or Class 2. Polyurethane filters cannot be used on units with multiframe filters.

Air filters must be listed in accordance with requirements of UL 900, except high efficiency particulate air filters of 99.97 percent efficiency

by the DOP Test Method must be as listed under the label service and must meet the requirements of UL 586.

#### 2.4.4.1 Extended Surface Pleated Panel Filters

Filters must be 2 inch depth sectional type of the size indicated and must have an average efficiency of 25 to 30 percent when tested in accordance with ASHRAE 52.2. Initial resistance at 500 feet/minute must not exceed 0.36 inches water gauge. Filters must be UL Class 2. Media must be nonwoven cotton and synthetic fiber mat. A wire support grid bonded to the media must be attached to a moisture resistant fiberboard frame. Four edges of the filter media must be bonded to the inside of the frame to prevent air bypass and increase rigidity.

#### 2.4.4.2 Manometers

Provide inclined-type manometers for filter stations of 2,000 cfm capacity or larger including filters furnished as integral parts of air-handling units and filters installed separately. Provide sufficient length to read at least one inch of water column with 10 major graduations, and equipped with spirit level. Equip manometers with overpressure safety traps to prevent loss of fluid, and two three-way vent valves for checking zero setting.

#### 2.4.5 Internal Dampers

Dampers must be parallel blade type with renewable blade seals and be integral to the unitary unit. Damper provisions must be provided for each outside air intake, exhaust, economizer, and mixing boxes. Dampers must have minimum position stops automatic modulation and operate as specified.

#### 2.4.6 Mixing Boxes

Mixing boxes must match the base unit in physical size and must include equally-sized flanged openings, each capable of full air flow. Arrangement must be as indicated.

#### 2.4.7 Cabinet Construction

Casings for the specified unitary equipment must be constructed of galvanized steel or aluminum sheet metal and galvanized or aluminum structural members. Minimum thickness of single wall exterior surfaces must be 18 gauge galvanized steel or 0.071 inch thick aluminum on units with a capacity above 20 tons and 20 gauge galvanized steel or 0.064 inch thick aluminum on units with a capacity less than 20 tons. Casing must be fitted with lifting provisions, access panels or doors, fan vibration isolators, electrical control panel, corrosion-resistant components, structural support members, insulated condensate drip pan and drain, and internal insulation in the cold section of the casing. Where double-wall insulated construction is proposed, minimum exterior galvanized sheet metal thickness must be 20 gauge. Provisions to permit replacement of major unit components must be incorporated. Penetrations of cabinet surfaces, including the floor, must be sealed. Unit must be fitted with a drain pan which extends under all areas where water may accumulate. Drain pan must be fabricated from Type 300 stainless steel, galvanized steel with protective coating as required, or an approved plastic material. Pan insulation must be water impervious. Extent and effectiveness of the insulation of unit air containment surfaces must prevent, within limits of the specified insulation, heat transfer between the unit exterior and

ambient air, heat transfer between the two conditioned air streams, and condensation on surfaces. Insulation must conform to ASTM C1071. Paint and finishes must comply with the requirements specified in paragraph FACTORY COATING.

#### 2.4.7.1 Indoor Cabinet

Indoor cabinets must be suitable for the specified indoor service and enclose all unit components.

#### 2.4.7.2 Outdoor Cabinet

Outdoor cabinets must be suitable for outdoor service with a weathertight, insulated and corrosion-protected structure. Cabinets constructed exclusively for indoor service which have been modified for outdoor service are not acceptable.

#### 2.4.8 Condensate Drain Piping

provide condensate drain piping in accordance with Section 23 05 15 COMMON PIPING FOR HVAC.

#### 2.4.9 Ductwork

Provide ductwork in accordance with Section 23 30 00 HVAC AIR DISTRIBUTION.

#### 2.4.10 Temperature Controls

Provide temperature controls fully coordinated with and integrated into the existing air-conditioning system.

### 2.5 UNITARY EQUIPMENT ACCESSORIES AND MISCELLANEOUS EQUIPMENT

#### 2.5.1 Air Economizer

Provide factory installed economizer with fully modulating 0-100 percent motor and dampers, barometric relief, minimum position setting and fixed dry bulb.

### 2.6 FINISHES

#### 2.6.1 Coil Corrosion Protection

Provide coil with a uniformly applied epoxy electrodeposition, phenolic, or vinyl type coating to all coil surface areas without material bridging between fins. Submit product data on the type coating selected, the coating thickness, the application process used, the estimated heat transfer loss of the coil, and verification of conformance with the salt spray test requirement. Coating must be applied at either the coil or coating manufacturer's factory. Coating process must ensure complete coil encapsulation. Coating must be capable of withstanding a minimum 1,000 hours exposure to the salt spray test specified in ASTM B117 using a 5 percent sodium chloride solution.

#### 2.6.2 Equipment and Components Factory Coating

Unless otherwise specified, equipment and component items, when fabricated from ferrous metal, must be factory finished with the manufacturer's standard finish, except that items located outside of buildings must have

weather resistant finishes that will withstand 500 hours exposure to the salt spray test specified in ASTM B117 using a 5 percent sodium chloride solution. Immediately after completion of the test, the specimen must show no signs of blistering, wrinkling, cracking, or loss of adhesion and no sign of rust creepage beyond 1/8 inch on either side of the scratch mark. Cut edges of galvanized surfaces where hot-dip galvanized sheet steel is used must be coated with a zinc-rich coating conforming to ASTM D520, Type I.

Where stipulated in equipment specifications of this section, coat finned tube coils of the affected equipment as specified below. Apply coating at the premises of a company specializing in such work. Degrease and prepare for coating in accordance with the coating applicator's procedures for the type of metals involved. Completed coating must show no evidence of softening, blistering, cracking, crazing, flaking, loss of adhesion, or "bridging" between the fins.

#### 2.6.2.1 Phenolic Coating

Provide a resin base thermosetting phenolic coating. Apply coating by immersion dipping of the entire coil. Provide a minimum of two coats. Bake or heat dry coils following immersions. After final immersion and prior to final baking, spray entire coil with particular emphasis given to building up coating on sheared edges. Total dry film thickness must be 2.5 to 3.0 mils.

#### 2.6.2.2 Chemical Conversion Coating with Polyelastomer Finish Coat

Dip coils in a chemical conversion solution to molecularly deposit a corrosion resistant coating by electrolysis action. Chemical conversion coatings must conform to MIL-DTL-5541, Class 1A. Cure conversion coating at a temperature of 110 to 140 degrees F for a minimum of 3 hours. Coat coil surfaces with a complex polymer primer with a dry film thickness of 1 mil. Cure primer coat for a minimum of 1 hour. Using dip tank method, provide three coats of a complex polyelastomer finish coat. After each of the first two finish coats, cure the coils for 1 hour. Following the third coat, spray a fog coat of an inert sealer on the coil surfaces. Total dry film thickness must be 2.5 to 3.0 mils. Cure finish coat for a minimum of 3 hours. Coating materials must have 300 percent flexibility, operate in temperatures of minus 50 to plus 220 degrees F, and protect against atmospheres of a pH range of 1 to 14.

#### 2.6.2.3 Vinyl Coating

Apply coating using an airless fog nozzle. For each coat, make at least two passes with the nozzle. Materials to be applied are as follows:

- a. Total dry film thickness, 6.5 mils maximum
- b. Vinyl Primer, 24 percent solids by volume: One coat 2 mils thick
- c. Vinyl Copolymer, 30 percent solids by volume: One coat 4.5 mils thick

#### 2.6.3 Factory Applied Insulation

Refrigeration equipment must be provided with factory installed insulation on surfaces subject to sweating including the suction line piping. Where motors are the gas-cooled type, factory installed insulation must be provided on the cold-gas inlet connection to the motor in accordance with

manufacturer's standard practice. Factory insulated items installed outdoors are not required to be fire-rated. As a minimum, factory insulated items installed indoors must have a flame spread index no higher than 75 and a smoke developed index no higher than 150. Factory insulated items (no jacket) installed indoors and which are located in air plenums, in ceiling spaces, and in attic spaces must have a flame spread index no higher than 25 and a smoke developed index no higher than 50. Flame spread and smoke developed indexes must be determined by ASTM E84. Insulation must be tested in the same density and installed thickness as the material to be used in the actual construction. Material supplied by a manufacturer with a jacket must be tested as a composite material. Jackets, facings, and adhesives must have a flame spread index no higher than 25 and a smoke developed index no higher than 50 when tested in accordance with ASTM E84.

## 2.7 TESTS, INSPECTIONS, AND VERIFICATIONS

All manufactured units must be inspected and tested, and documentation provided to demonstrate that each unit is in compliance with ANSI/AHRI and UL requirements and that the minimum efficiency requirements of ASHRAE 90.1 - IP have been met.

## PART 3 EXECUTION

### 3.1 EXAMINATION

After becoming familiar with all details of the work, perform Verification of Dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

### 3.2 INSTALLATION

Perform work in accordance with the manufacturer's published diagrams, recommendations, and equipment warranty requirements. Where equipment is specified to conform to the requirements of ASME BPVC SEC VIII D1 and ASME BPVC SEC IX, the design, fabrication, and installation of the system must conform to ASME BPVC SEC VIII D1 and ASME BPVC SEC IX.

#### 3.2.1 Equipment

Provide refrigeration equipment conforming to ASHRAE 15 & 34. Provide necessary supports for all equipment, appurtenances, and pipe as required, including frames or supports for compressors, pumps, cooling towers, condensers, and similar items. Isolate compressors from the building structure. If mechanical vibration isolators are not provided, provide vibration absorbing foundations. Each foundation must include isolation units consisting of machine and floor or foundation fastenings, together with intermediate isolation material. Other floor-mounted equipment must be set on not less than a 6 inch concrete pad doweled in place. Concrete foundations for floor mounted pumps must have a mass equivalent to three times the weight of the components, pump, base plate, and motor to be supported. In lieu of concrete pad foundation, concrete pedestal block with isolators placed between the pedestal block and the floor may be provided. Concrete pedestal block must be of mass not less than three times the combined pump, motor, and base weights. Isolators must be selected and sized based on load-bearing requirements and the lowest frequency of vibration to be isolated. Isolators must limit vibration to 20-40 percent at lowest equipment rpm. Provide lines connected to pumps mounted on pedestal blocks with flexible connectors. Provide foundation

drawings, bolt-setting information, and foundation bolts prior to concrete foundation construction for all equipment indicated or required to have concrete foundations. Concrete for foundations must be as specified in Section 03 30 00 CAST-IN-PLACE CONCRETE. Equipment must be properly leveled, aligned, and secured in place in accordance with manufacturer's instructions.

### 3.2.2 Mechanical Room Ventilation

Provide mechanical ventilation systems in accordance with Section 23 30 00 HVAC AIR DISTRIBUTION.

### 3.2.3 Field Applied Insulation

Apply field applied insulation as specified in Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS, except as defined differently herein.

### 3.2.4 Field Painting

Painting required for surfaces not otherwise specified, and finish painting of items only primed at the factory are specified in Section 09 90 00 PAINTS AND COATINGS.

## 3.3 CLEANING AND ADJUSTING

Equipment must be wiped clean, with all traces of oil, dust, dirt, or paint spots removed. Temporary filters must be provided for all fans that are operated during construction, and new filters must be installed after all construction dirt has been removed from the building. System must be maintained in this clean condition until final acceptance. Bearings must be properly lubricated with oil or grease as recommended by the manufacturer. Belts must be tightened to proper tension. Control valves and other miscellaneous equipment requiring adjustment must be adjusted to setting indicated or directed. Fans must be adjusted to the speed indicated by the manufacturer to meet specified conditions. Testing, adjusting, and balancing must be as specified in Section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC SYSTEMS.

## 3.4 TRAINING

Conduct a training course for the operating staff as designated by the Contracting Officer. The training period must consist of a total 8 hours of normal working time and start after the system is functionally completed but prior to final acceptance tests.

- a. Submit a schedule, at least 2 weeks prior to the date of the proposed training course, which identifies the date, time, and location for the training.
- b. Submit the field posted instructions, at least 2 weeks prior to construction completion, including equipment layout, wiring and control diagrams, piping, valves and control sequences, and typed condensed operation instructions. The condensed operation instructions must include preventative maintenance procedures, methods of checking the system for normal and safe operation, and procedures for safely starting and stopping the system. The posted instructions must be framed under glass or laminated plastic and be posted where indicated by the Contracting Officer.

- c. The posted instructions must cover all of the items contained in the approved operation and maintenance manuals as well as demonstrations of routine maintenance operations. Submit 6 complete copies of an operation manual in bound 8-1/2 by 11 inch booklets listing step-by-step procedures required for system startup, operation, abnormal shutdown, emergency shutdown, and normal shutdown at least 4 weeks prior to the first training course. The booklets must include the manufacturer's name, model number, and parts list. The manuals must include the manufacturer's name, model number, service manual, and a brief description of all equipment and their basic operating features.
- d. Submit 6 complete copies of maintenance manual in bound 8-1/2 by 11 inch booklets listing routine maintenance procedures, possible breakdowns and repairs, and a trouble shooting guide. The manuals must include piping and equipment layouts and simplified wiring and control diagrams of the system as installed.

### 3.5 SYSTEM PERFORMANCE TESTS

Before each refrigeration system is accepted, conduct tests to demonstrate the general operating characteristics of all equipment by a registered professional engineer or an approved manufacturer's start-up representative experienced in system start-up and testing, at such times as directed. Six copies of the report provided in bound 8-1/2 by 11 inch booklets. The report must document compliance with the specified performance criteria upon completion and testing of the system. The report must indicate the number of days covered by the tests and any conclusions as to the adequacy of the system.

For equipment providing heating and cooling the system performance tests must be performed during the heating and cooling seasons.

- a. Submit a schedule, at least 2 weeks prior to the start of related testing, for the system performance tests. The schedules must identify the proposed date, time, and location for each test. Tests must cover a period of not less than 48 hours for each system and must demonstrate that the entire system is functioning in accordance with the drawings and specifications.
- b. Make corrections and adjustments, as necessary, tests must be re-conducted to demonstrate that the entire system is functioning as specified. Prior to acceptance, install and tighten service valve seal caps and blanks over gauge points. Replace any refrigerant lost during the system startup.
- c. If tests do not demonstrate satisfactory system performance, correct deficiencies and retest the system. Conduct tests in the presence of the Contracting Officer. Water and electricity required for the tests will be furnished by the Government. Provide all material, equipment, instruments, and personnel required for the test.
- d. Coordinate field tests with Section 23 05 93 TESTING, ADJUSTING, AND BALANCING OF HVAC SYSTEMS. Submit 6 copies of the report provided in bound 8-1/2 by 11 inch booklets. The report must document compliance with the specified performance criteria upon completion and testing of the system. The report must indicate the number of days covered by the tests and any conclusions as to the adequacy of the system. Submit the report including the following information (where values



are taken at least three different times at outside dry-bulb temperatures that are at least 5 degrees F apart):

- (1) Date and outside weather conditions.
- (2) The load on the system based on the following:
  - (a) The refrigerant used in the system.
  - (b) Condensing temperature and pressure.
  - (c) Suction temperature and pressure.
  - (d) Ambient, condensing and coolant temperatures.
  - (e) Running current, voltage and proper phase sequence for each phase of all motors.
- (3) The actual on-site setting of operating and safety controls.
- (4) Thermostatic expansion valve superheat - value as determined by field test.
- (5) Subcooling.
- (6) High and low refrigerant temperature switch set-points
- (7) Low oil pressure switch set-point.
- (8) Defrost system timer and thermostat set-points.
- (9) Moisture content.
- (10) Capacity control set-points.
- (11) Field data and adjustments which affect unit performance and energy consumption.
- (12) Field adjustments and settings which were not permanently marked as an integral part of a device.

### 3.6 MAINTENANCE

#### 3.6.1 EXTRA MATERIALS

Submit spare parts data for each different item of equipment specified, after approval of detail drawings and not later than 2 months prior to the date of beneficial occupancy. Include in the data a complete list of parts and supplies, with current unit prices and source of supply, a recommended spare parts list for 1 year of operation, and a list of the parts recommended by the manufacturer to be replaced on a routine basis.

#### 3.6.2 Maintenance Service

Submit a certified list of qualified permanent service organizations, which includes their addresses and qualifications, for support of the equipment. The service organizations must be reasonably convenient to the equipment installation and be able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

-- End of Section --

SECTION 23 83 00.00 20

ELECTRIC SPACE HEATING EQUIPMENT  
**04/06**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 2 (2000; R 2020) Industrial Control and  
Systems Controllers, Contactors, and  
Overload Relays Rated 600 V

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

UL SOLUTIONS (UL)

UL 1025 (1980; R 1990, Bul. 1991) Electric Air  
Heaters

1.2 GENERAL REQUIREMENTS

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, applies to this section, with the additions and modifications specified herein.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Heater Installation Drawing

SD-03 Product Data

Electric Unit and Cabinet Heaters

Thermostat

SD-10 Operation and Maintenance Data

Electric Unit and Cabinet Heaters, Data Package 5

Submit in accordance with Section 01 78 23 OPERATION AND  
MAINTENANCE DATA.

## PART 2 PRODUCTS

### 2.1 ELECTRIC UNIT AND CABINET HEATERS

UL 1025; wattage, voltage, phase, number of steps, Btu/hr and CFM as indicated. Provide control-circuit terminals and single source of power supply. Heaters 5 Kw and larger shall be 3-phase, with load balanced on each of the three phases. Limit leaving air temperature to 140 degrees F with entering air of 60 degrees F.

#### 2.1.1 Enclosure

Minimum 20 gage steel.

#### 2.1.2 Heating Element

Nickel chromium heating wire element, free from expansion noise and 60 Hz hum. Embed element in magnesium-oxide insulating refractory. Seal element in high-mass steel or corrosion-resisting metallic sheath with fins. Enclose element ends in terminal box. Provide not more than six fins per inch. Limit fin surface temperature 550 degrees F at any point during normal operation.

#### 2.1.3 Controls

Include limit controls for overheat protection of heaters. For remote thermostatic operation, provide contactor rated for 100,000 duty cycles. Provide a control transformer to supply 120-volt thermostat control circuit for each heater.

#### 2.1.4 Wiring

Completely factory-prewired to terminal strips, ready to receive branch circuit and control connections for 140 degrees F copper wiring.

#### 2.1.5 Accessories

Provide fan switching devices to independently operate the fan motor for summer ventilation and winter heat recovery.

#### 2.1.6 Thermostat

Provide tamper resistant space thermostat, adjustable without requiring removal of heater components. Thermostat operating range shall be approximately 50 degrees F to a maximum of 85 degrees F with operating differential of 3 degrees F or less.

#### 2.1.7 Disconnect Means

Provide factory-installed safety disconnect switch in the housing or in an auxiliary matching control section with "off" position marking on the face plate.

#### 2.1.8 Outdoor Sensor

Provide outdoor sensor with sunlight-and-rain protection shield. The sensor shall provide a positive heater shut off when outdoor air temperature is 65 degrees F or higher.

#### 2.2 CONTACTORS

NEMA ICS 2, Enclosure Type 1.

#### 2.3 DISCONNECTS

Disconnect. UL listed. Enclosed fused switch, rated 480 volt, 3 phase, 4 wire, NEMA Type 1 enclosure. Disconnect shall be capable of being locked in the open position.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

Install in conformance with the approved heater installation drawing, NFPA 70, UL listing, and manufacturer's instructions, with necessary clearances for air circulation, maintenance, inspection, service testing and repair. Connect to electrical supply in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

##### 3.1.1 Unit Heaters

Mount units plumb, square and level with ceiling and walls.

##### 3.1.2 Cabinet Heaters

Where recessed mounting is indicated, seal entire recessed opening from exterior wall cavities, and provide a minimum 1/2-inch thick rigid fire resistant insulation on the wall behind the cabinet. Verify manufacturer's clearance requirements from electrical cords, drapes, and other furnishings.

##### 3.1.3 Remote Thermostat

Mount remote room space thermostats 4 feet 6 inches above finished floor on wall or as indicated. Connect remote thermostats with conduit and wiring to heaters as indicated.

#### 3.2 FIELD QUALITY CONTROL

Provide necessary personnel, instruments, and equipment to perform tests. Notify the Contracting Officer 5 working days prior to scheduled testings and locations.

##### 3.2.1 Field Inspection

Prior to initial operation, inspect installed equipment for conformance with drawings and specifications.

##### 3.2.2 Insulation Resistance Tests

Test 600-volt wiring to verify that no short circuits or grounds exist. Tests shall be made using an instrument which applies a voltage of

approximately 500 volts and provides a direct reading of resistance in ohms.

### 3.2.3 Operational Tests

Test equipment circuits and devices to demonstrate proper operation. Test each item of control equipment not less than 5 times.

-- End of Section --

SECTION 25 05 11  
CYBERSECURITY FOR DESALINATION SCADA SYSTEM  
08/24

PART 1 GENERAL

Many subparts in this Section contain text in curly braces ("{" and "}") indicating which cybersecurity control and control correlation identifier (CCI) the requirements of the subpart relate to. The text inside these curly braces is for Government reference only and enables coordination of the requirements of this Section with the RMF process throughout the design and construction process. Text in curly braces are not contractor requirements.

This Section refers to Security Requirements Guide (SRGs) and Security Technical Implementation Guide (STIGs). STIGs and SRGs are available online at the Information Assurance Support Environment (IASE) website at <https://public.cyber.mil/stigs/downloads/> and an SRG/STIG Applicability Guide and Collection Tool is available at <https://public.cyber.mil/stigs/SCAP/>. Not all control system components have applicable STIGs or SRGs. The "Control Systems SRG" does not apply to work performed under this Section; all requirements within this section to apply applicable SRGs DO NOT include the "Control Systems SRG".

1.1 CONTROL SYSTEM APPLICABILITY

There are multiple versions of this Section associated with this project. Different versions have requirements applicable to different control systems. This specific Section applies only to the Desalination SCADA System.

1.2 RELATED REQUIREMENTS

This section does not contain sufficient requirements to procure a control system and must be used in conjunction with other Sections which specify control systems. This Section adds cybersecurity requirements to the control systems specified in other Sections, and as these requirements are conditioned on the control system being provided, there may be requirements in this Section that will not apply to this project. All Sections containing facility-related control systems or control system components are related to the requirements of this Section. Review all specification sections to determine related requirements.

In cases where a requirement is specified in both this Section and in another Section, the more stringent requirement must be met. In cases where a requirement in this Section conflicts with the requirements of another Section such that both requirements cannot be met at the same time, request direction from the Contracting Officer (KO) to determine which requirement applies to the project.

Incorporate each of the requirements contained in this specification for facility-related control systems and equipment specified in the following UFGS Section(s):

- a. 26 29 23 ADJUSTABLE SPEED DRIVE (ASD) SYSTEMS UNDER 600 VOLTS
- b. 40 60 00 PROCESS CONTROL

### 1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

#### INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 802.1x (2010) Local and Metropolitan Area  
Networks - Port Based Network Access  
Control

#### INTERNET ENGINEERING TASK FORCE (IETF)

IETF RFC 2819 (2000) Remote Network Monitoring (RMON)  
Management Information Base (MIB)

#### NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST FIPS 140-2 (2001) Security Requirements for  
Cryptographic Modules

#### U.S. DEPARTMENT OF DEFENSE (DOD)

DODI 8551.01 (2014) Ports, Protocols, and Services  
Management (PPSM)

DTM 08-060 (2008) Policy on Use of Department of  
Defense (DoD) Information Systems -  
Standard Consent Banner and User Agreement

### 1.4 DEFINITIONS

#### 1.4.1 Administrator Account

An administrator account is an account with full permissions to a device, application, or operating system, including the ability to create and modify other user accounts.

Note that the operating system Administrator Account may be different than Administrator Accounts for applications hosted on that operating system. Also, most controllers will not have any support for accounts and will therefore not have an 'Administrator Account'.

#### 1.4.2 Controller

A device other than a computer or Ethernet switch.

#### 1.4.3 Mission Space

A device or media is in mission space if physical access to the device or media is controlled by the organization served by the device. For example, a control panel in an electrical room is in mission space if the organization (mission) served by that control panel controls access to the electrical room.

#### 1.4.4 Network

A network is a group of two or more devices that can communicate using a

network protocol. Network protocols must provide a method for addressing devices on the network; a communication method that does not provide an addressing scheme is not a networked form of communication. Devices that communicate using a method of communication that does not support device addressing are not using a network. Addresses may be other than IP addresses, and addressing may be at either Open Systems Interconnection (OSI) layer 2 or layer 3.

#### 1.4.5 Network Connected

A component is network connected (or "connected to a network") only when the device has a network transceiver which is directly connected to the network and implements the network protocol. A device lacking a network transceiver (and accompanying protocol implementation) can never be considered network connected. Note that (unlike many IT definitions of "Network Connected") a device connected to a non-IP network is still considered network connected (an IP connection or IP address is not required for a device to be network connected).

##### 1.4.5.1 Wireless Network Connected

Any device that supports wireless network communication is network connected to a wireless network, regardless of whether the device is communicating using wireless. Unless physically disabled, devices with wireless transceivers support wireless, it is not sufficient to disable the wireless in software.

#### 1.4.6 Network Media

The thing that provides the communication channel between the devices on a network. Typically wire, but might include wireless, fiber optic, or even power line (some network protocols allow sending network signals over power wiring).

#### 1.4.7 User Account Support Levels

The support for user accounts is categorized in this Section as one of three levels:

##### 1.4.7.1 FULLY Supported

Device supports configurable individual accounts. Accounts can be created, deleted, modified, etc. Privileges can be assigned to accounts. These devices support user-based (as opposed to role-based) authentication.

##### 1.4.7.2 MINIMALLY Supported

Device supports a small, fixed number of accounts (perhaps only one). Accounts cannot be modified. A device with only a "User" and an "Administrator" account would fit this category. Similarly, a device with two PINs for logon - one for restricted and one for unrestricted rights would fit here (in other words, the accounts do not have to be the traditional "username and password" structure). These devices typically only support role-based authentication.

Examples of devices which MINIMALLY support accounts are a) a variable frequency drive with a single account which requires a PIN for access to configuration; and b) a room lighting control touchpad interface that has a single account.



#### 1.4.7.3 NOT Supported

Device does not support any Access Enforcement therefore the whole concept of "account" is meaningless.

#### 1.4.8 Manual Local Input

Manual Local Inputs are system analog or binary inputs that are adjustable by a person but are, by intrinsic hardware design, very limited in potential capabilities. Manual Local Inputs do not have touch screens or full keyboards, but may have a few buttons or dials to allow input. Manual Local Inputs do not have full graphic screens or dot-matrix displays, but may have simple lights (LEDs) or 7-segment displays. Manual Local Inputs do not have any sort of menu structure, each button has a single well-defined function.

Examples of Manual Local Inputs are H-O-A switches, simple thermostats, and disconnect switches.

#### 1.4.9 Card Reader

A card reader is an input/output device whose primary function is to assist in two-factor authentication. A card reader must have an interface to read data from a card and may be able to write data to a card. A card reader may have a means (such as buttons, keypad, touchscreen, etc.) for a user to input a PIN or password, as well as a limited display.

#### 1.4.10 User Interface

A User Interface (UI) is something other than a Manual Local Input or Card Reader that allows a person to interact with the system or device. Note that while a Card Reader is not by itself a User Interface, a User Interface may contain a Card Reader in order for it to authenticate its user. Within control systems, there are a wide range of User Interfaces.

Two important distinctions are 1) whether the user interface is Local or Remote, and 2) the effective capabilities of the User Interface to alter data, which is the "privilege" of the user interface (where effective privilege available to a specific user at a specific user interface is the combination of the greatest privilege offered by the user interface and the specific account the user is logged into).

##### 1.4.10.1 Local User Interface

A Local User Interface is a user interface where the physical hardware the user interacts with (keyboard, buttons, display, etc.) is physically part of the device being affected. All of the relevant characteristics of the user interface are embodied within a single device.

Note that a Local UI may be able to access data in a different device, Local versus Remote in this context refers to the user interface itself; the capability to access data in a different device is covered under "Full User Interface".

##### 1.4.10.2 Remote User Interface

A Remote User Interface implements a Client/Server model where the physical hardware the user interacts with (Client) is physically distinct

from the device being affected (Server). Most or all of the security and functionality characteristics of the user interface are defined by the Server, not the Client. The Client and Server communicate via a network connection. A common example of a remote user interface is a web-based interface where the browser (client) is generally on different hardware than the web server (server). A Remote UI remains a Remote UI even if the user happens to be at a Client on the same hardware as the Server. What is important is that a) the Client may be on different hardware than the Server and b) the majority of the security and functional characteristics of the interface are defined at the Server.

Note that this definition of "remote" is consistent with that generally used in the control industry but is not aligned with the NIST 800-53 definition of "Remote", which refers to "outside the system". The term "Remote" here better aligns with the NIST 800-53 definition of "Network" (remote from within the system) Access.

#### 1.4.10.3 Types of User Interface (by capability)

User interfaces are also categorized by their capabilities as being Read Only, Limited, or Full.

##### 1.4.10.3.1 Read-Only User Interface

A Read Only User Interface (also referred to as a View-Only User Interface) is a user interface that only allows for reading data, it does not allow (have the capability to) modify data. A Read Only User Interface may be either Local or Remote. A User Interface that is configured to be Read Only (by some other means than the interface itself, such as using configuration software on a laptop) is a Read-Only Interface. Note a Read Only User Interface may have buttons (or touch screen, etc.) allowing the user to navigate through the presentation of data.

Examples of a Read Only User Interfaces are a) a publicly viewable "energy dashboard" showing weather data and energy usage within a building and b) digital wayfinding signage.

##### 1.4.10.3.2 Limited User Interface

A Limited User Interface is a user interface that - by design - can only alter information local to the user interface. Note that the determination of "alter" includes only direct interactions, it explicitly excludes interactions that might occur as secondary effects. For example, an interface changing the flow setpoint in a pump controller is a direct interaction, the subsequent change in flow (as well as any subsequent downstream changes in valve position) are not direct interactions.

Two examples of LIMITED UIs are: a) a variable speed drive has a Limited Local User Interface which allows the user to change properties within the drive, but does not allow affecting things outside the drive; and b) a typical home WiFi Router has a Limited Remote User Interface which allows configuration of the Router, but does not allow direct interaction with other devices.

##### 1.4.10.3.3 Full User Interface

A Full User Interface can alter information in devices outside the device with the user interface. For example, a typical Local Display Panel is a

Full Local User Interface while a browser-based front end is a Full Remote User Interface.

#### 1.4.10.3.4 View-Only User Interface

See Read-Only User Interface

#### 1.4.10.4 Other User Interface Terminology

In addition to defining whether a user interface is a Hardware Limited, Read-Only, Limited or Full, and whether it is Local or Remote, user interfaces are classified by whether they are writable or privileged.

##### 1.4.10.4.1 Writable User Interface

Any User Interface that is not Read-Only is Writable. (Limited User Interfaces and Full User Interfaces are both writable user interfaces (as they are capable of changing a value)).

##### 1.4.10.4.2 Privileged User Interface

A Privileged UI is a UI that has sufficient capabilities or functionality that it requires specific cybersecurity measures to be put in place to limit its unauthorized use. Ultimately, whether a specific user interface is considered a Privileged User Interface must be determined by usage. Unless otherwise specified, user interfaces can be determined to be privileged or not using the following:

- a. Read-Only User Interfaces are not privileged user interfaces.
- b. Full User Interfaces are privileged user interfaces.
- c. User interfaces that allow for configuration of auditing or allows for modification or deletion of audit logs are privileged user interface.
- d. User interfaces that allow for reprogramming a network connected device is a privileged user interface.
- e. Except as specified above, a Limited User Interface must be determined to be privileged or not based on the specific capabilities and use case of the user interface. In general however, user interfaces that do not offer significant capabilities above and beyond those available at that location via other means (e.g. such as a disconnect switch, breaker, or hand-off-auto switch, or physical attack) are not privileged.

#### 1.4.11 Wireless Network

Any network that communicates without using wires or fiber optics as the communication media. Wireless networks include: WiFi, Bluetooth, ZigBee, cellular, satellite, long and short wave radio, 2.4 GHz, free space optical, point-to-point laser, microwave, and IR.

#### 1.4.12 Wired Broadcast Network

Wired Broadcast Networks are any network, such as powerline carrier networks and modem (wired telephony), that use wire-based technologies where there is not a clearly defined boundary for signal propagation.

#### 1.4.13 Facility-Related Control System (FRCS)

A control system (a system of digital controllers, communication architecture, and user interfaces that monitor, or monitor and control, infrastructure and equipment) which controls equipment and infrastructure that is part of the facility (building, structure, or linear structure).

#### 1.4.14 Field Control System

A Building Control System, Utility Control System, Electronic Security System, etc. within the Facility and "downstream" of the Field Point of Connection (FPOC).

#### 1.4.15 Field Point of Connection (FPOC)

The FPOC is the point of connection between the Level 4 control system IP network and the Level 2 field control IP network. Note the Level 2 IP network here may consist entirely of a single patch cable from the FPOC to a Level 1-to-Level 2 device. The hardware which provides the connection at this location is an IT device such as a switch, IP router, or firewall.

#### 1.4.16 SCADA System

A SCADA system, or Supervisory Control and Data Acquisition system, is a system that monitors and controls water and wastewater equipment in real-time. SCADA systems use sensors, controllers, and software to collect data from various sources, such as pumps, tanks, and valves.

### 1.5 ADMINISTRATIVE REQUIREMENTS

#### 1.5.1 Points of Contact

When coordination is required, coordinate with the Contracting Officer (K0).

#### 1.5.2 Coordination

Coordinate the execution of this Section with the execution of all other Sections related to control systems as indicated in the paragraph RELATED REQUIREMENTS. Items that must be considered when coordinating project efforts include but are not limited to:

- a. If requesting permission for wireless or wired broadcast communication, the Wireless and Wired Broadcast Communication Request submittal must be approved prior to control system device selection and installation.
- b. If requesting permission for alternate account lock permissions, the Device Account Lock Exception Request must be approved prior to control system device selection and installation.
- c. If requesting permission for the use of a device with multiple physical connections to IP networks, the Multiple IP Connection Device Request must be approved prior to control system device selection and installation.
- d. Wireless testing may be required as part of the control system testing. See requirements for the Wireless Communication Test Report submittal.

- e. If the Device Audit Record Upload Software is to be installed on a computer not being provided as part of the control system, coordination is required to identify the computer on which to install the software.
- f. The Cybersecurity Interconnection Schedule must be coordinated with other work that will be interconnected to, and interconnections must be approved by the Government before relying on them for system functionality.
- g. Cybersecurity testing support must be coordinated across control systems and with the Government cybersecurity testing schedule.
- h. Passwords must be coordinated with the indicated contact for the project site.
- i. If applicable, HTTPS web server certificates must be obtained from the Contracting Officer (KO).
- j. Contractor Computer Cybersecurity Compliance Statements must be provided for each contractor using contractor owned computers.

#### 1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-01 Preconstruction Submittals

Device Account Lock Exception Request; G  
Multiple Ethernet Connection Device Request; G  
Contractor Computer Cybersecurity Compliance Statements; G  
Contractor Temporary Network Cybersecurity Compliance Statements; G  
Cybersecurity Interconnection Schedule ; G  
Proposed STIG and SRG Applicability Report; G  
Pre-Construction Control System Inventory Report ; G

##### SD-02 Shop Drawings

Network Communication, Ports, Protocols and Services Report; G  
Cybersecurity Network (Riser) Diagram ; G

##### SD-03 Product Data

Control System Cybersecurity Documentation; G

##### SD-06 Test Reports

Control System Cybersecurity Testing Procedures; G  
Control System Cybersecurity Testing Report; G

SD-07 Certificates

Software Licenses; G

SD-11 Closeout Submittals

Confidential Password Report; G

Enclosure Keys; G

Software and Configuration Backups; G

Auditing Front End Software; G

Device Audit Record Upload Software; G

System Maintenance Tool Software; G

Control System Scanning Tools; G

STIG, SRG and Vendor Guide Compliance Result Report; G

Final Control System Inventory Report; G

1.7 CYBERSECURITY DOCUMENTATION

{For Government Reference Only: This subpart (and its subparts) relates to PL-7; CCI-003071}

1.7.1 Proposed STIG and SRG Applicability Report

For each model of network connected or network infrastructure device, use the DISA SRG/STIG Applicability Guide and Collection Tool (available at <https://public.cyber.mil/stigs/SCAP/>) to identify applicable STIGs or SRGs and provide a report indicating applicable STIGs and SRGs for each model. Provide the Proposed STIG and SRG Applicability Report concurrently with the Pre-Construction Control System Inventory Report.

1.7.2 Cybersecurity Interconnection Schedule

{For Government Reference Only: This subpart relates to CA-3(b), PL-8, SC-7(9), SC-7(11); CCI-000258, CCI-003072, CCI-003073, CCI-003075, CCI-002398, CCI-002399, CCI-002401, CCI-002402, CCI-002403.}

Provide a completed Cybersecurity Interconnection Schedule documenting network connections between the installed system and other systems. Provide the following information for each device directly communicating between systems: Device Identifier, Device Manufacturer, Device Description, Transport layer Protocol, Network Address, Port (if applicable), MAC (Layer 2) address (if applicable), Media, Application Protocol, Service (if applicable), Descriptive Purpose of communication. For communication with other authorized systems also provide the Foreign Destination and POC for Destination. If other control system Sections used on this project include submittals documenting this information, provide copies of those submittals to meet this requirement.

In addition to the requirements of Section 01 33 00 SUBMITTAL PROCEDURES, provide the Cybersecurity Interconnection Schedule as an editable

Microsoft Excel file (a template Cybersecurity Interconnection Schedule in Excel format is available at <https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-25-05-11>.)

#### 1.7.3 Network Communication, Ports, Protocols and Services Report

{For Government Reference Only: This subpart (and its subparts) relates to CA-9, PL-8; CCI-002102, CCI-002103, CCI-002104, CCI-002105, CCI-003072, CCI-003073, CCI-003075 and also the submittal requirements associated with CM-6, CM-7, including CM-7(3), CCI-000388.}

Provide a Network Communication, Ports, Protocols and Services Report. For each networked device, document device identifier and the communication characteristics of the device including communication protocols, services used, encryption employed, and a general description of what information is communicated over the network. For each device using IP communication, document all TCP and UDP ports used. For each device using non-IP communication, document communication protocol and media used. If other control system Sections used on this project include submittals documenting this information, provide copies of those submittals to meet this requirement.

In addition to the requirements of Section 01 33 00 SUBMITTAL PROCEDURES, provide the Network Communication, Ports, Protocols and Services Report as an editable Microsoft Excel file.

#### 1.7.4 Control System Inventory Reports

{For Government Reference Only: This subpart (and its subparts) relates to CM-8(a), SI-17, IA-3; CCI-000389, CCI-000392, CCI-000398, CCI-002773, CCI-002774, CCI-002775, CCI-000777, CCI-000778, CCI-001958}

Provide a Pre-Construction Control System Inventory Report report and a Final Control System Inventory Report, using the Inventory Spreadsheet listed under this Section at <https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-25-05-11>. In the Control System Inventory Reports, document all devices with networking capabilities (both IP and non-IP), including network infrastructure devices and devices once capable of wireless communication. For each device provide all applicable information for which there is a field on the spreadsheet in accordance with the instructions on the spreadsheet.

In addition to the requirements of Section 01 33 00 SUBMITTAL PROCEDURES, provide the Control System Inventory Reports as editable Microsoft Excel files. Provide the Pre-Construction Control System Inventory Report concurrently with the Proposed STIG and SRG Applicability Report

#### 1.7.5 Software and Configuration Backups

{For Government Reference Only: This subpart (and its subparts) relates to CP-10; CCI-000550, CCI-000551, CCI-000552}

For all controllers, provide a backup of the controller configuration and the source code for all loaded application programs (all software that is not common to every controller of the same manufacturer and model).

If any or all of these are provided under another Section, provide documentation indicating this and referencing those submittals.

#### 1.7.6 Cybersecurity Network (Riser) Diagram

{For Government Reference Only: This subpart (and its subparts) relates to PL-2(a), PL-8; CCI-003051, CCI-003053, CCI-003072, CCI-003073, CCI-003075}

Provide a cybersecurity network (riser) diagram of the complete control system including all network and device hardware. For each device, include the device identifier, device type, and manufacturer. If the control system specifications require a riser diagram submittal, provide a copy of that submittal as the cybersecurity riser diagram. Otherwise, provide a riser diagram in one-line format.

#### 1.7.7 STIG, SRG and Vendor Guide Compliance Result Report

For every component (device or software) with an applicable STIG or SRG in the Proposed STIG and SRG Applicability Report, document compliance with the STIG or SRG requirements.

- a. For components which are scannable by the SCAP (security content automation protocol) tool (available online at <https://public.cyber.mil/stigs/scap>), include the SCAP report and raw scan results in addition to the final, manually reviewed and revised, documentation of compliance with STIG and SRG requirements. Checklist files should not contain any findings with a Not Reviewed (NR) status after manual reviews.
- b. For components which do not support automated scanning, a manual review using the General Purpose STIG option should be done. A completed Checklist file should not contain any findings with a Not Reviewed (NR) status after manual reviews

For every component (device or software) with manufacturer provided cybersecurity documentation, procedure, or method for secure configuration or installation, provide a report documenting how the component was configured and any deviation from the manufacturer instructions.

#### 1.7.8 Control System Cybersecurity Documentation

{For Government Reference Only: This subpart (and its subparts) relates to SA-5(a), SA-5(b), SA-5(c), SA-22(b); CCIs: CCI-003124, CCI-003125, CCI-003126, CCI-003127, CCI-003128, CCI-003129, CCI-003130, CCI-003131, CCI-003374}

Provide a Control System Cybersecurity Documentation submittal containing the indicated information for each device and software application.

##### 1.7.8.1 Desalination SCADA System

For Desalination SCADA System Devices, use the Default Requirements for Control System Devices.

##### 1.7.8.2 Default Requirements for Control System Devices

For control system devices where Control System Cybersecurity Documentation requirements are not otherwise indicated in this Section,



provide:

- a. Documentation that describes secure configuration of the device {For Government Reference Only: relates to CCI-003124}
- b. Documentation that describes secure installation of the device {For Government Reference Only: relates to CCI-003125}
- c. Documentation that describes secure operation of the device {For Government Reference Only: relates to CCI-003126}
- d. Documentation that describes effective use and maintenance of security functions or mechanisms for the device {For Government Reference Only: relates to CCI-003127}
- e. Documentation that describes known vulnerabilities regarding configuration and use of administrative (i.e. privileged) functions for the device {For Government Reference Only: relates to CCI-003128}
- f. Documentation that describes user-accessible security functions or mechanisms in the device and how to effectively use those security functions or mechanisms {For Government Reference Only: relates to CCI-003129}
- g. Documentation that describes methods for user interaction which enables individuals to use the device in a more secure manner {For Government Reference Only: relates to CCI-003130}
- h. Documentation that describes user responsibilities in maintaining the security of the device {For Government Reference Only: relates to CCI-003131}
- i. Documentation of the published last date of support by the manufacturer or indication that a published date is not available. {For Government Reference Only: relates to CCI-003374}

#### 1.8 SOFTWARE LICENSING

{For Government Reference Only: This subpart (and its subparts) relates to SI-2(a), SI-2(c), SI-7(14); CCI-001227, CCI-002605, CCI-002737}

For all software provided that has not already been licensed to the government or project site, provide a license to the project site for a period of no less than 5 years, and the license must also include the following software updates:

- a. Security and bug-fix patches issued by the software manufacturer.
- b. Security patches to address any vulnerability identified in the National Vulnerability Database at <http://nvd.nist.gov> with a Common Vulnerability Scoring System (CVSS) severity rating of MEDIUM or higher.

Provide a single Software Licenses submittal with documentation of the software licenses for all software provided

#### 1.9 CYBERSECURITY DURING CONSTRUCTION

{For Government Reference Only: This subpart (and its subparts) relates

to AC-18, CA-3; CCI-000258}

In addition to the control system cybersecurity requirements indicated in this section, meet following requirement throughout the construction process.

#### 1.9.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. Contractor computers connected to the control system, control system network, or a control system component at any point during construction must meet the following requirements:

##### 1.9.1.1 Operating System

The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

##### 1.9.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. Computers used on this project must perform a full antivirus scan at least once per day. Computers which have connected to any other network since the last full antivirus scan must perform a full antivirus scan prior to connection to the control system network or to the temporary contractor-installed IP network. Perform an antivirus scan on all removable media (e.g., external hard drives, CDs, DVDs, USB flash drives) prior to connecting to the control system environment.

##### 1.9.1.3 Passwords and Passphrases

The passwords and passphrases for computers, applications, and web-based applications supporting passwords must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

##### 1.9.1.4 User-Based Authentication

Each user must have a unique account; sharing of a single account between multiple users is prohibited.

##### 1.9.1.5 Demonstration of Compliance

The Government has the right to require demonstration of computer compliance with these requirements at any time during the project.

##### 1.9.1.6 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using contractor owned computers. Contractor Computer Cybersecurity Compliance Statements must use the template published at <https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-25-05-11>. Each Statement must

be signed by a cybersecurity representative for the relevant company.

#### 1.9.2 Temporary IP Networks

Temporary contractor-installed IP networks may be used during construction. When used, temporary contractor-installed IP networks connected to the control system, control system network, or a control system component at any point during construction must meet the following requirements:

##### 1.9.2.1 Network Boundaries and Connections

The network must not extend outside the project site and must not connect to any IP network other than those specifically provided or furnished for this project. Any and all access to the network from outside the project site is prohibited.

##### 1.9.3 Government Access to Network

Government personnel must be allowed to have complete and immediate access to the network at any time in order to verify compliance with this specification.

##### 1.9.4 Temporary Wireless IP Networks

In addition to the other requirements on temporary IP networks, temporary wireless IP (WiFi) networks, when permitted, must not interfere with existing wireless networks, must use WPA2 security and must not broadcast the network name (SSID). Network names (SSID) for wireless networks must be changed from their default values.

##### 1.9.5 Passwords and Passphrases

The passwords and passphrases for all network devices and network access must be changed from their default values. Passwords must be a minimum 8 characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

##### 1.9.6 Contractor Temporary Network Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Temporary Network Cybersecurity Compliance Statements for each company implementing a temporary IP network. Contractor Temporary Network Cybersecurity Compliance Statements must use the template published at <https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-25-05-11>. Each Statement must be signed by a cybersecurity representative for the relevant company. If no temporary IP networks will be used, provide a single copy of the Statement indicating this.

#### 1.10 CYBERSECURITY DURING WARRANTY PERIOD

All work performed on the control system after acceptance must be performed using Government Furnished Equipment or equipment specifically and individually approved by the Government.

## PART 2 PRODUCTS

All products used on this project must meet the indicated requirements, but not all products specified here will be required by every project.

## 2.1 ETHERNET SWITCH

Provide Open Systems Interconnection (OSI) Layer 2 Ethernet switches with the following capabilities, and with an interface to support switch configuration for these capabilities:

### 2.1.1 Required Functionality

Switches must:

- a. Copper Ethernet ports must auto negotiate for 10, 100 and 1000 megabits-per-second links.
- b. Be capable of implementing port level access control by MAC address and limit the number of MAC addresses to one MAC address per port.
- c. Be capable of implementing per-port access control lists (ACLs) where the list can be filtered by source and destination IP addresses, and by source and destination UDP or TCP ports.
- d. Support Remote Network Monitoring (RMON) Port Analysis in accordance with IETF RFC 2819
- e. Configure target port and analysis port such that switch clones all target port traffic to analysis port.
- f. Support authentication via RADIUS server (for management and 802.1x)
- g. Support IEEE 802.1x network login.

### 2.1.2 Configuration Requirements

Switches must:

- a. Support configuration save and restore.
- b. Support both manual IP address assignment and acquisition of a dynamic IP address via Dynamic Host Configuration Protocol (DHCP).
- c. Be capable of limiting access for configuration to one or more of: a web interface using HTTPS, a command line interface using SSH, or an SNMP connection using SNMP version 3 or later.
- d. Support the ability to lock configuration capability to a dedicated management port.

## 2.2 DAISY CHAIN IP CONTROLLERS

Controllers used as Daisy Chain IP Controllers must be IP controllers with exactly two Ethernet network connections and basic built-in switch capabilities to allow implementation of an Ethernet network in a daisy chain architecture. Switches incorporated by Daisy Chain IP Controllers are not required to meet the requirements for Ethernet Switches as defined in this Section.

## PART 3 EXECUTION

### 3.1 CYBERSECURITY HARDENING AND CONFIGURATION GUIDES

Install, configure, and harden all hardware and software furnished on this project in accordance with manufacturer provided documentation, procedures, or methods for secure configuration or installation. Do not implement specific hardening actions if that action would conflict with required functionality or another requirement of this Section.

### 3.2 NETWORK REQUIREMENTS

#### 3.2.1 Wireless and Wired Broadcast Communication

{For Government Reference Only: This subpart (and its subparts) relates to AC-18, AC-18(3); CCI-001438, CCI-001439, CCI-002323, CCI-001441, CCI-001449}

The use of wireless and wired broadcast communication for Desalination SCADA systems within a facility is prohibited. Wireless communication may be used to provide communication from the Desalination SCADA system at a facility to the central monitoring station. Communication between the Desalination SCADA system and the central monitoring station must be via FIPS 140-2 certified devices.

#### 3.2.2 Non-IP Control Networks

When control system specifications require particular communication protocols, use only those communication protocols and only as specified. Do not implement any other communication protocol.

When control system specifications do not indicate requirements for communication protocols, use only those protocols required for operation of the system as specified.

#### 3.2.3 IP Control Networks

{For Government Reference Only: This subpart relates to CM-6(a), CM-7(a), CM-7(b), CM-7(1)(b), SC-41; CCI-001588, CCI-000381, CCI-000380, CCI-000381, CCI-000382, CCI-001761, CCI-001762, CCI-002544, CCI-002545, CCI-002546.}

IP Networks must be Ethernet networks and must use switches which are Ethernet Switches or Daisy Chain IP Controllers as defined in this Section. Do not use nonsecure functions, ports, protocols and services as defined in DODI 8551.01 unless those ports, protocols and services are specifically required by the control system specifications or otherwise specifically authorized by the Government. Do not use ports, protocols and services that are not specified in the control system specifications or required for operation of the control system.

##### 3.2.3.1 IP Network Routers

Do not install any device that performs IP routing.

##### 3.2.3.2 IP Devices With Multiple Ethernet Connections

Except for Ethernet Switches and Daisy Chain IP Controllers, devices must not have more than one Ethernet connection to IP networks unless doing so

is required by the project specifications and the specific application is approved. If a device with Multiple Ethernet Connections to IP networks is required, provide a Multiple Ethernet Connection Device Request using the Multiple Ethernet Connection Device Request Template at <https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-25-05-11> to request approval for each device. If a device with Multiple Ethernet Connections to IP networks is not required, instead provide a document stating that no approval is being requested.

#### 3.2.4 Device Identification and Authentication

{For Government Reference Only: This subpart (and its subparts) relates to IA-3; CCI-000777, CCI-000778, CCI-001958.}

##### 3.2.4.1 Default Requirements for Control System Devices

For control system devices where Device Identification and Authentication requirements are not otherwise indicated in this Section: Devices using HTTP as a control protocol must use HTTPS instead.

#### 3.2.5 Cryptographic Module Authentication

{For Government Reference Only: This subpart (and its subparts) relates to IA-7; CCI-000803}

For devices (including but not limited to NIST FIPS 140-2 compliant radios) that have STIG/SRGs related to cryptographic module authentication (CCI-000803), comply with the requirements of those STIG/SRGs.

### 3.3 ACCESS CONTROL REQUIREMENTS

#### 3.3.1 User Accounts

{For Government Reference Only: This subpart (and its subparts) relate to AC-2(a), AC-3, AC-6(1), AC-6(10), AC-6(2), AC-6(9), CM-11(2), and IA-2; CCI-002110, CCI-000213, CCI-001558, CCI-002221, CCI-002222, CCI-002223, CCI-002235, CCI-000039, CCI-001419, CCI-002234, CCI-001812, and CCI-000764. }

Any user interface supporting user accounts (either FULLY or MINIMALLY) must limit access according to specified limitations for each account. Install and configure any device having a STIG or SRG in accordance with that STIG or SRG.

All user interfaces FULLY supporting accounts must implement user-based authentication where each account is uniquely assigned to a specific user. User interfaces FULLY supporting accounts must implement at least three (3) levels of user account privilege including: 1) an account with read-only permissions 2) an account with full permissions including account creation and modification and 3) an account with greater permissions than read-only but without account creation and modification. Disable any unnecessary or unused accounts. Disable any "guest-level" accounts that are created on the system by default.

##### 3.3.1.1 Controllers

For user interfaces provided by controllers, provide access control in accordance with the User Interface Requirements table for the applicable control system and user interface type.

- a. For table entries of "NA": NA means Not Applicable, there are no interfaces in this category.
- b. For table entries of "None Required": The user interface is not required to support user accounts.
- c. For table entries of "MINIMALLY": The user interface must at least MINIMALLY support user accounts.
- d. For table entries of "FULLY": The user interface must at FULLY support user accounts.
- e. For table entries of "KEY": The user interface must have physical security in the form of either a key lock on the interface itself or be furnished inside a lockable enclosure. Where this is required for a read only interface, the lockable enclosure must prevent viewing of data on the interface; for other interfaces, this lockable enclosure must prevent using the interface to alter data.
- f. For table entries of "Physical Security": For Local FULL interfaces, the interface must be located inside mission space. For Local Limited (not FULL) interfaces, the user interface must either a) be located within mission space or b) be protected by physical security at least as good as the control devices (and equipment controlled by the control devices) affected by the interface. For purposes of this requirement, 'affected' includes controllers with data that can be directly altered by the interface, as well as mechanical and/or electrical equipment directly controlled by those controllers, but does not include other interactions.
- g. Entries of the form "X and Y" must meet both the requirement indicated for X and the requirement indicated for Y. For example, an entry of "MINIMALLY and Physical Security" indicates the user interface must both MINIMALLY support accounts and have physical security.
- h. Entries of the form "X or Y" must meet either the requirement indicated for X or the requirement indicated for Y.

#### 3.3.1.1.1 Desalination SCADA System

For Desalination SCADA System Devices, use the Default Requirements for Control Systems.

#### 3.3.1.1.2 Default Requirements for Control Systems

For control system devices where User Interface Requirements are not otherwise indicated in this Section, use the Default User Interface Requirements tables.

| Default User Interface Requirements for Control Systems |                                   |
|---------------------------------------------------------|-----------------------------------|
| <u>User Interface Type</u>                              | <u>Access Control Requirement</u> |
| Local Read Only (see note 1)                            | None Required                     |

| Default User Interface Requirements for Control Systems                                                                      |                                   |
|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| <u>User Interface Type</u>                                                                                                   | <u>Access Control Requirement</u> |
| Local Limited, Non-privileged                                                                                                | MINIMALLY                         |
| Local Limited, Privileged                                                                                                    | MINIMALLY                         |
| Local Full                                                                                                                   | MINIMALLY                         |
| Remote Read Only                                                                                                             | None Required                     |
| Remote Limited, Non-Privileged                                                                                               | MINIMALLY                         |
| Remote Limited, Privileged<br>AND<br>Remote Full (see note 2)                                                                | FULLY                             |
| Notes:<br>1)Local Read Only User Interfaces are always Non-Privileged<br>2)Remote Full User Interfaces are always Privileged |                                   |

### 3.3.2 Unsuccessful Logon Attempts

{For Government Reference Only: This subpart (and its subparts) relate to AC-7 (a), AC-7 (b); CCI-000043, CCI-000044, CCI-001423, CCI-002236, CCI-002237, CCI-002238}

Except for high availability user interfaces indicated as exempt, devices must meet the indicated requirements for handling unsuccessful logon attempts. If a device cannot meet these requirements, document device capabilities to protect from subsequent logon attempts and propose alternate protections in a Device Account Lock Exception Request submittal. Do not implement alternate protection measures in lieu of the indicated requirements without explicit permission from the Government. If no Device Account Lock Exceptions are requested, provide a document stating that no approval is being requested as the Device Account Lock Exception Request.

#### 3.3.2.1 Devices MINIMALLY Supporting Accounts

Devices which MINIMALLY (but not FULLY) support accounts are not required to lock based on unsuccessful logon attempts.

#### 3.3.2.2 Devices FULLY Supporting Accounts

Devices which FULLY support accounts must meet the following requirements.

- a. It must lock the user account when three unsuccessful logon attempts occur within a 15 minute interval.
- b. Once an account is locked, the account must stay locked until unlocked by an administrator. If the account being locked is the sole administrator account on the device, the account must stay locked for one hour and then automatically unlock.



- c. Once the indicated number of unsuccessful logon attempts occurs, delay further logon prompts by five seconds.

### 3.3.2.3 High Availability Interfaces Exempt from Unsuccessful Logon Attempts Requirements

There are no high availability interfaces which are exempt from unsuccessful logon attempts requirements.

### 3.3.3 System Use Notification

{For Government Reference Only: This subpart (and its subparts) relates to AC-8; CCI-000048, CCI-002247, CCI-002243, CCI-002244, CCI-002245, CCI-002246, CCI-000050, CCI-002248}

#### 3.3.3.1 System Use Notification for Local User Interfaces

Devices which are connected to a network and have a local user interface must display a warning banner meeting the requirements of DTM 08-060 on the user interface screen if capable of doing so and must have a permanently affixed label with an approved banner from DTM 08-060 if unable to display the warning banner on the screen. Where it is impractical (perhaps due to device size) to affix the label to the device, affix the label to the device enclosure.

Labels must be machine printed or engraved, plastic or metal, designed for permanent installation, must use a font no smaller than 14 point, and must provide a high contrast between font and background colors.

Digital banners must be acknowledged before allowing users to take further action.

#### 3.3.4 Permitted Actions Without Identification or Authentication

{For Government Reference Only: This subpart (and its subparts) relates to AC-14; CCI-000061, CCI-000232}

The control system must require identification and authentication before allowing any actions, except read-only actions, by a user acting from a user interface which MINIMALLY or FULLY supports accounts.

#### 3.3.5 Enclosures

Prior to final acceptance of the system, lock all lockable enclosures. Submit an Enclosure Keys submittal with all copies of keys for all enclosures and a key inventory list documenting all keys. Label each key with the matching enclosure identifier.

### 3.4 USER IDENTIFICATION AND AUTHENTICATION

{For Government Reference Only: This subpart (and its subparts) relates to IA-2, IA-2(1), IA-2(12), IA-5 IA-5(b), IA-5(c), IA-5(e), IA-5(g), IA-5(1), IA-5(11); CCI-000764, CCI-000765, CCI-001953, CCI-001954, CCI-001544, CCI-001989, CCI-000182, CCI-001610, CCI-000192, CCI-000193, CCI-000194, CCI-000205, CCI-001619, CCI-001611, CCI-001612, CCI-001613, CCI-001614, CCI-000195, CCI-001615, CCI-000196, CCI-000197, CCI-000199, CCI-000198, CCI-001616, CCI-001617, CCI-000200, CCI-001618, CCI-002041, CCI-002002, CCI-002003. }

This subpart indicates requirements for specific methods of identification and authentication for users and user accounts. Where these requirements conflict apply the following order of precedence: 1) If present, Device Specific Requirements take precedence over any other requirements; and then 2) multifactor authentication requirements take precedence over password requirements.

#### 3.4.1 User Identification and Authentication for All System Types

Unless otherwise indicated, all user interfaces supporting accounts (either FULLY or MINIMALLY) must implement Identification and Authorization via passwords.

#### 3.4.2 User Identification and Authentication for Specific System Types

System specific requirements are in addition to and supersede those indicated for all system types. When no additional requirements are indicated for a specific system type the requirements for all systems still apply to that system type.

#### 3.4.3 User Identification and Authentication for Specific Devices

There are no additional device specific user interface requirements

#### 3.4.4 Implementation of Identification and Authorization Requirements

Identification and Authorization must be met by direct implementation in the user interface.

#### 3.4.5 Password-Based Authentication Requirements

##### 3.4.5.1 Passwords for Controllers FULLY Supporting Accounts

All controllers FULLY supporting accounts and supporting password-based authentication must enforce the following requirements:

- a. Minimum password length of 12 characters.
- b. Password must contain at least one uppercase character.
- c. Password must contain at least one lowercase character.
- d. Password must contain at least one numeric character.
- e. Password must contain at least one special character. The list of supported special characters must include at least four separate characters.
- f. Password must have a maximum lifetime of 60 days. When passwords expire, prompt users to change passwords. Do not lock accounts due to expired passwords.
- g. Password must differ from previous five passwords, where differ is defined as changing at least 50 percent of the characters.
- h. Passwords must be cryptographically protected during storage and transmission.

#### 3.4.5.2 Passwords for Devices Minimally Supporting Accounts

Devices MINIMALLY supporting accounts must support passwords with a minimum length of four characters.

#### 3.4.5.3 Password Configuration and Reporting

For all devices with a password, change the password from the default password. Coordinate selection of passwords with the Password Point of Contact. Do not use the same password for more than one device unless specifically instructed to do so. Provide a Confidential Password Report documenting the password for each device and describing the procedure to change the password for each device.

Do not provide the Confidential Password Report in electronic format. Provide two hardcopies of the Confidential Password Report, each copy in its own sealed envelope.

#### 3.4.6 Authenticator Feedback

{For Government Reference Only: This subpart relates to IA-6; CCI-000206}

Devices must never show authentication information, including passwords, on a display. Devices that momentarily display a character as it is entered, and then obscure the character, are acceptable. For devices that have STIGs or SRGs related to obscuring of authenticator feedback (CCI-000206), comply with the requirements of those STIGS/SRGs.

### 3.5 CYBERSECURITY AUDITING

#### 3.5.1 Audit Events, Content of Audit Records, and Audit Generation

{For Government Reference Only: This subpart (and its subparts) relates to AU-2(a), AU-2(c), AU-2(d), AU-3, AU-10, AU-12, AU-14(b), AU-14(1), AU-14(2), AU-14(3), CM-5(1), SC-7 (9); CCI-000123, CCI-001571, CCI-000125, CCI-001485, CCI-000130, CCI-000131, CCI-000132, CCI-001230, CCI-000133, CCI-000134, CCI-001487, CCI-000166, CCI-001899, CCI-000169, CCI-001459, CCI-000171, CCI-000172, CCI-001910, CCI-001914, CCI-001919, CCI-001464, CCI-001462, CCI-001920, CCI-001814, CCI-002400.}

For devices that have STIG/SRGs related to audit events, content of audit records or audit generation, comply with the requirements of those STIG/SRGs.

If auditing requirements can be met using existing control system alarm or event capabilities, those existing capabilities may be used to meet these requirements.

##### 3.5.1.1 Desalination SCADA System

For Desalination SCADA System Controllers, use the Default Requirements for Control System Controllers.

##### 3.5.1.2 Default Requirements for Control System Controllers

For control system controllers where Audit Events, Content of Audit Records, and Audit Generation are not otherwise indicated in this Section:

#### 3.5.1.2.1 Controllers Which FULLY Support Accounts

For each controller which FULLY supports accounts, provide the capability to select audited events and the content of audit logs. Configure controllers to audit the indicated events, and to record the indicated information for each auditable event.

##### 3.5.1.2.1.1 Audited Events

Configure each controller to audit the following events (where capable to the device):

- a. Successful and unsuccessful attempts to access, modify, or delete privileges, security objects, security levels, or categories of information (e.g. classification levels)
- b. Successful and unsuccessful logon attempts
- c. Successful logouts
- d. Concurrent logons from different workstations
- e. All account creations, modifications, disabling, and terminations.
- f. All kernel module load, unload, and restart

##### 3.5.1.2.1.2 Audit Event Information To Record

Configure each controller to record, for each auditable event, the following information (where applicable to the event):

- a. what type of event occurred
- b. when the event occurred
- c. where the event occurred
- d. the source of the event
- e. the outcome of the event
- f. the identity of any individuals or subjects associated with the event

#### 3.5.1.2.2 Controllers Which Do Not FULLY Support Accounts

For each controller which does not FULLY support accounts configure the controller to audit all controller shutdown and startup events and to record for each event the type of event and when the event occurred.

#### 3.5.2 Audit Time Stamps

{For Government Reference Only: This subpart (and its subparts) relates to AU-8; CCI-000159, CCI-001889, CCI-001890.}

Any device (computer or controller) generating audit records must have an internal clock capable of providing time with a resolution of one second. Clocks must not drift more than 10 seconds per day. Configure the system so that each device (computer or controller) generating audit records maintains accurate time to within one second. Note that if the control

system specifications include requirement for clocks, the most stringent requirement applies.

### 3.5.3 Auditing Front End Software

The project site currently has the following software to support control system auditing: TBD. If there is no existing auditing front end software or the software is not compatible with the provided control systems, provide Auditing Front End Software with audit log import and upload, export, notification, and analysis functionality. The Auditing Front End Software may be provided as a component of the control system front end or as a separate software package, and a single package may serve multiple control systems provided under the same projects if they are sharing a cybersecurity authorization.

When the Auditing Front End Software is neither existing nor installed under the requirements of another Section, furnish the Auditing Front End Software media and license the control system front end computer in TBD. Submit copies of Auditing Front End Software if this function is not part of the software provided with the control system to meet requirements of other Sections.

#### 3.5.3.1 Import and Upload Requirements

Auditing Front End Software must be capable of importing audit logs from the Device Audit Record Upload Software and of uploading audit logs over the network from all control system devices supporting network upload of audit logs.

#### 3.5.3.2 Export Requirements

Auditing Front End Software must be capable of exporting to a file format supported by Microsoft Excel.

### 3.5.4 Audit Storage Capacity and Audit Upload

{For Government Reference Only: This subpart (and its subparts) relates to AU-4; CCI-001848, CCI-001849}

The creation of audit records must never interfere with normal device operation. Devices must cease collection of auditing information if required to maintain normal operation.

- a. For devices that have STIG/SRGs related to audit storage capacity (CCI-001848 or CCI-001849) comply with the requirements of those STIG/SRGs.
- b. For controllers capable of generating audit records, provide 60 days worth of secure local storage, assuming 10 auditable events per day.

#### 3.5.4.1 Device Audit Record Upload Software

For each device required to audit events and for which audit logs cannot be uploaded over the network by the Auditing Front End Software, provide and license to the Government software implementing a secure mechanism of uploading audit records from the device and exporting them to the Auditing Front End Software. Where different devices use different software, provide software of each type required to upload audit logs from all devices.

When Device Audit Record Upload Software is capable of uploading audit logs over the network, install Device Audit Record Upload Software on the same computer as the Auditing Front End Software. Submit copies of device audit record upload software if this function is not part of the software provided with the control system to meet requirements of other Sections. If there are no devices requiring this software, provide a document stating this in lieu of this submittal.

### 3.5.5 Response to Audit Processing Failures

{For Government Reference Only: This subpart (and its subparts) relates to AU-5; CCI-000139, CCI-000140, CCI-001490.}

In case of an audit failure, if possible, continue to collect audit records by overwriting existing audit records.

### 3.6 REQUIREMENTS FOR LEAST FUNCTIONALITY

{For Government Reference Only: This subpart (and its subparts), along with the Network Communication, Ports, Protocols and Services Report submittal specified elsewhere in this section, relates to CM-6(a), CM-6(c), CM-7, CM-7(1)(b), SC-41; CCI-000363, CCI-000364, CCI-000365, CCI-001588, CCI-001755, CCI-000381, CCI-000380, CCI-000382, CCI-001761, CCI-001762, CCI-002544, CCI-002545, CCI-002546. }

For devices that have a STIG or SRG related to Requirements for Least Functionality (such as configuration settings and port and device I/O access for least functionality), install and configure the device in accordance with that STIG or SRGs.

#### 3.6.1 Device Capabilities

Do not provide devices with remote user interfaces or with full user interfaces where one was not required. Do not use a networked sensor or actuator where a non-networked sensor or actuator would suffice.

#### 3.6.2 Software

For software that has a STIG or SRG related to Requirements for Least Functionality (such as configuration settings and port access for least functionality), install and configure the software in accordance with that STIG or SRG.

Do not install software that is not specifically required to meet a contract requirement. Remove any previously installed that is not specifically required to meet a contract requirement. Do not implement functionality within software that is not specifically required to meet contract requirements.

### 3.7 SYSTEM AND COMMUNICATION PROTECTION

#### 3.7.1 Collaborative Computing

{For Government Reference Only: This subpart relates to SC-15(a), SC-15(b); CCI-001150, CCI-001152.}

Without explicit approval from the project site, control systems must not use collaborative computing technologies.

### 3.7.2 Denial of Service Protection

{For Government Reference Only: This subpart relates to SC-5, SC-12, SC-7(a); CCI-001093, CCI-002385, CCI-002386, CCI-002430, CCI-001097. }

To the greatest extent practical, implement control logic without reliance on the network. Except when required to meet the requirements of the control system Section (where the requirement can only be met using computer hardware), do not implement control logic in computers.

### 3.8 SAFE MODE AND FAIL SAFE OPERATION

{For Government Reference Only: This subpart (and its subparts) relates to CP-12, SI-10(3), SI-17; CCI-002855, CCI-002856, CCI-002857, CCI-002754, CCI-002773, CCI-002774, CCI-002775}

For all control system components with an applicable STIG or SRG, configure the component in accordance with all applicable STIGs and SRGs.

### 3.9 SYSTEM MAINTENANCE TOOL SOFTWARE

{For Government Reference Only: This subpart (and its subparts) relates to MA-3; CCI-000865.}

Submit and license to the Government all software required to operate, maintain and modify the control system such the Government or their agents are able to perform repair, replacement, upgrades, and expansions of the system without subsequent or future dependence on the Contractor, Vendor or Manufacturer. Submit hard copies of user manuals for each software with the software submittal. Provide any hardware keys or dongles, software keys, license numbers, and other information required to enable the Government to access or change the system.

For software provided and licensed to the Government under the requirements of another Section, submit a statement indicating the Section and Submittal under which the software was provided. For software provided to meet the requirements of this Section and not provided and licensed under another Section, submit software and software user manuals on DVD or CD as a Technical Data Package and submit one hard copy of the software user manual for each piece of software.

### 3.10 VULNERABILITY SCANNING

{For Government Reference Only: This subpart (and its subparts) relates to RA-5 RA-5(a), RA-5(b), RA-5(c), RA-5(d); CCI-001054, CCI-001055, CCI-000156, CCI-001641, CCI-001643, CCI-001057, CCI-001058, CCI-001059. }

All IP devices must be scannable, such that the device can be scanned by industry standard IP network scanning utilities without harm to the device, application, or functionality.

#### 3.10.1 Computers and Software Running on Computers

Computers and applications running on computers must meet relevant vulnerability scanning STIGs/SRGs and respond to approved DoD vulnerability scanning tools.

### 3.10.2 Controllers

Provide controllers that are scannable by standard control system discovery tools or control system browsers and return meaningful status information including the network inputs and outputs for the controller. This information must contain sufficient detail to detect vulnerabilities or exploits of the controller.

Provide all software needed to scan the control system as the Control System Scanning Tools submittal. If the software required to scan the system is already installed at the project site or is provided under a separate section instead provide a statement indicating this.

### 3.11 CONTROL SYSTEM CYBERSECURITY TESTING

#### 3.11.1 Control System Cybersecurity Testing Procedures

Prepare and submit Control System Cybersecurity Testing Procedures explaining step-by-step, the actions and expected results that will demonstrate that the control system meets the requirements of this Section. The Control System Cybersecurity Testing Procedures may be submitted as a Technical Data Package.

#### 3.11.2 Control System Cybersecurity Testing Execution

Using the Control System Cybersecurity Testing Procedures verify that the control system meets the requirements of this Section. UNLESS GOVERNMENT WITNESSING OF A TEST IS SPECIFICALLY WAIVED BY THE GOVERNMENT, PERFORM ALL TESTS WITH A GOVERNMENT WITNESS. If testing reveals deficiencies in the system, correct the deficiency and retest until successful.

#### 3.11.3 Control System Cybersecurity Testing Report

Prepare and submit a Control System Cybersecurity Testing Report documenting all tests performed and their results. Include all tests in the Control System Cybersecurity Testing Procedures and any additional tests performed during testing. Document test failures and repairs conducted with the test results. The Control System Cybersecurity Testing Report may be submitted as a Technical Data Package

### 3.12 FIELD QUALITY CONTROL, CYBERSECURITY VALIDATION SUPPORT

In addition to testing and testing support required by other Sections, provide a minimum of 40 hours of technical support for cybersecurity testing of control systems to support the DoD Risk Management Framework process Cybersecurity assessment of the control system. This support is independent of (and in addition to) the Control System Cybersecurity Testing specified in this section.

### 3.13 CYBERSECURITY TRAINING

Provide eight hours of classroom and hands-on training for Government personnel on the cybersecurity operation and maintenance of the control system provided. This training is in addition to and must be coordinated with control system training specified in other Sections.

The Government will provide the training location. Training must cover, at a minimum: (a) applying software and firmware updates, (b) user account creation, modification and deletion, (c) audit log upload



procedures and (d) identification of privileged user interfaces and system impact of those interfaces. Training session must include a question and answer period during which government staff questions about cybersecurity aspects of the control system are answered.

-- End of Section --

SECTION 26 05 13.00 10

MEDIUM-VOLTAGE CABLES

**11/23**

PART 1 GENERAL

UNDERGROUND ELECTRICAL DISTRIBUTION applies to work specified in this section.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASSOCIATION OF EDISON ILLUMINATING COMPANIES (AEIC)

- |          |                                                                                                |
|----------|------------------------------------------------------------------------------------------------|
| AEIC CS1 | (2012) Impregnated-Paper-Insulated,<br>Metallic Sheathed Cable, Solid Type                     |
| AEIC CS8 | (2013) Specification for Extruded<br>Dielectric Shielded Power Cables Rated 5<br>Through 46 kV |

ASTM INTERNATIONAL (ASTM)

- |           |                                                                                                               |
|-----------|---------------------------------------------------------------------------------------------------------------|
| ASTM B3   | (2013; R 2024) Standard Specification for<br>Soft or Annealed Copper Wire                                     |
| ASTM B8   | (2023) Standard Specification for<br>Concentric-Lay-Stranded Copper Conductors,<br>Hard, Medium-Hard, or Soft |
| ASTM D746 | (2014) Standard Test Method for<br>Brittleness Temperature of Plastics and<br>Elastomers by Impact            |

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

- |            |                                                                                                                                                                                                                                      |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IEEE 48    | (2020) Test Procedures and Requirements<br>for Alternating-Current Cable Terminations<br>Used on Shielded Cables Having Laminated<br>Insulation Rated 2.5 kV through 765 kV or<br>Extruded Insulation Rated 2.5 kV through<br>500 kV |
| IEEE 383   | (2023) Qualifying Electric Cables and<br>Splices for Nuclear Facilities                                                                                                                                                              |
| IEEE 386   | (2016) Separable Insulated Connector<br>Systems for Power Distribution Systems<br>Rated 2.5 kV through 35 kV                                                                                                                         |
| IEEE 400.2 | (2013) Guide for Field Testing of Shielded<br>Power Cable Systems Using Very Low<br>Frequency (VLF)                                                                                                                                  |

IEEE 404 (2012) Standard for Extruded and Laminated Dielectric Shielded Cable Joints Rated 2500 V to 500,000 V

IEEE 1202 (2006; R 2012; CORR 1 2012) Flame-Propagation Testing of Wire and Cable

IEEE C2 (2023) National Electrical Safety Code

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS (2021) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C119.1 (2023) Electric Connectors - Sealed Insulated Underground Connector Systems Rated 600 Volts

ANSI/NEMA WC 71/ICEA S-96-659 (2014; R 2022) Standard for Nonshielded Cables Rated 2001-5000 Volts for use in the Distribution of Electric Energy

NEMA WC 70 (2021) Power Cable Rated 2000 Volts or Less for the Distribution of Electrical Energy

NEMA WC 74/ICEA S-93-639 (2022) 5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

NFPA 70B (2023) Recommended Practice for Electrical Equipment Maintenance

NFPA 70E (2024) Standard for Electrical Safety in the Workplace

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FED-STD-228 (2014; Rev A; Notice 1) Test Methods for Cable and Wire, Insulated

1.2 DEFINITIONS

Medium-voltage power cables include all cables rated above 600 volts up to 35,000 volts.

1.3 ADMINISTRATIVE REQUIREMENTS

1.3.1 Pre-Installation Meetings

No later than 30 days of Contract Award, coordinate with the Contracting Officer to schedule a pre-installation meeting. Submit the following for

review and approval prior to the meeting:

- a. Pulling Plan including calculations of pulling tension and side wall pressure anticipated, and the maximum allowable pulling tension for each pull. Do not perform any pull until Government reviews and approves the pulling plan.
- b. Splicer/Terminator Certifications
- c. List of Splices and Terminations to be Installed by Splicer/Terminator
- d. Manufacturer's catalog data for all cables, cable supports and fittings, cable tags, fireproof tape, splice kits, terminations, and any other product data required to complete the work.
- e. Certificates showing that the cable manufacturer has made factory-conducted tests on each shipping length (reel) of cable. Include certified copies of test data showing conformance with the referenced standards and approval prior to delivery of cable.

#### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-01 Preconstruction Submittals

List of Splices and Terminations to be Installed by  
Splicer/Terminator; G

##### SD-02 Shop Drawings

Pulling Plan; G

##### SD-03 Product Data

Single-Conductor Shielded Cables; G

Cable Supports and Fittings; G

Polyethylene Cable Tags; G

Fireproof Tape; G

Splices; G

Terminations; G

Polyethylene Cable Tags; G

SD-06 Test Reports

Field Testing; G

Qualification Test Reports; G

Radiographic Tests; G

SD-07 Certificates

Splicer/Terminator Certifications; G

Cable Installer Qualifications; G

SD-08 Manufacturer's Instructions

Medium-Voltage Power Cables; G

Terminations; G

Splices; G

1.5 QUALITY CONTROL

1.5.1 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Ensure equipment, materials, installation, and workmanship are in accordance with the mandatory and advisory provisions of NFPA 70, IEEE C2 unless more stringent requirements are specified or indicated.

1.5.2 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Provide products which have been in satisfactory commercial or industrial use for 2 years prior to bid opening. Ensure the 2-year period includes applications of equipment and materials under similar circumstances and of similar size. Ensure the product has been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer.

1.5.2.1 Cable Installer Qualifications

Provide at least one onsite person in a supervisory position with a documentable level of competency and experience to supervise all cable pulling operations. Provide a resume showing the cable installers' experience in the last three years, including a list of references complete with points of contact, addresses and telephone numbers. Cable installer must demonstrate experience with a minimum of three medium voltage cable installations. The Contracting Officer reserves the right to require additional proof of competency or to reject the individual and call for an alternate qualified cable installer.

## Qualifications`

Verify personnel performing Medium Voltage (MV) splicing or terminations have 15 years minimum experience in cable splicing and terminations of the type used in this project. Submit splicer/terminator certifications issued by the cable splice and termination manufacturer who has examined and tested a test splice or termination of each type required by this contract for each cable splicer. Ensure the certification identifies which splices and terminations it applies to. Require each individual, certified or not, with the required medium voltage splicing and terminating experience, who is to perform cable splicing or terminating, to perform a minimum of one splice or termination of each type in the presence of the manufacturer's and Government's representative. Splicer/terminator certifications are required to be submitted no less than 45 days before splices or terminations are to be made in medium voltage cables. Supply all materials and tools required for the demonstration splices and terminations.

Once a splice or termination has been started by a splicer, ensure the same splicer completes that particular splice, and that each termination and splice is started and completed in one continuous work period.

Maintain and submit a list of splices and terminations to be installed by splicer/terminator. Ensure the list includes the following for each splice or termination completed.

- a. Name of splicer/terminator.
- b. Date splice or termination was performed.
- c. Location of splice or termination. For terminations at equipment indicate equipment number as required to completely define the location.
- d. Feeder number.

## 1.6 DELIVERY, STORAGE, AND HANDLING

Ship cables on reels in a way that protects the cable from mechanical injury. Hermetically seal end of each cable length using heat-shrinkable molded cable end caps to exclude moisture and securely attached to the reel..

Make the minimum reel drum diameter 14 times the overall diameter of the cable. Ensure that each cable length is installed with a pulling eye installed by the manufacturer, for installation in ducts, manholes, and utility tunnels.

## PART 2 PRODUCTS

### 2.1 SYSTEM DESCRIPTION

Provide medium-voltage power cables including multiple- and single-conductor cables rated 15,000 volts, ungrounded neutral, on 13,200-volt three-phase, 60-hertz, phase-to-phase, for grounded and ungrounded neutral systems.

Provide conductor cable assemblies consisting of:

- a. Conductor core with an extruded semiconductor shield over the conductors
- b. Insulation
- c. A polyethylene (PE) jacket.
- d. An extruded semiconductor insulation shield, a concentric neutral

## 2.2 EQUIPMENT

Ensure that ethylene-propylene rubberinsulated conductors are lead-free.

### 2.2.1 Single-Conductor Shielded Cables

#### 2.2.1.1 Ethylene-Propylene-Rubber-Insulated with PVC Jacket

Provide single-conductor 15 kV rated cable assemblies that consist of the following: Class B stranded copper conductors, an extruded semiconducting shield over the conductors, 220 mils of ethylene propylene rubber insulation, an extruded or other approved semiconducting shield, a 5-mil minimum copper tape shield wrapped helically with a minimum 12.5 percent overlap and a PVC jacket.

Provide single-conductor, ethylene-propylene-insulated, PVC-jacketed, shielded cable that conforms to NEMA WC 70, ANSI/NEMA WC 71/ICEA S-96-659, NEMA WC 74/ICEA S-93-639 and AEIC CS8.

### 2.2.2 Insulated Medium Voltage Connectors

Provide connector with a steel reinforced hook-stick eye, grounding eye, test point, and arc-quenching contact material per IEEE 386. Ensure connections are compatible with equipment bushings. Provide connectors as follows:

- a. 600 Ampere deadbreak connector ratings: Voltage: 15kV, 95kV BIL. Short time rating: 27,000 ampere rms, symmetrical for a time duration of 4.0 seconds.
- b. Provide connectors with a steel reinforced hook-stick eye, grounding eye, test point, and arc-quenching contact material per IEEE 386. Provide hot line voltage indicators on all connectors.

Ensure connections are compatible with equipment bushings.

### 2.2.3 Splices

Provide heat shrink splice kits which are the product of a single manufacturer. Ensure the power cable splice meets the requirements of IEEE 404 for a 15 kV rating, and must be rated by the manufacturer for use on 15 kV class cable systems. Ensure splices are rated for continuous operation at 221 degrees F, with an emergency overload temperature rating of 284 degrees F. Ensure the kit is capable of splicing cables with copper conductors sized as indicated in the contract drawings or accommodate a conductor size transitionProvide splices specifically

designed for the cable and grounding provisions.

Provide heat shrink splices which include but are not limited to the following:

- a. Inner heat shrink stress control tube with external end sealant, additional heat shrink tube over inner tube and inner tube end sealant.
- b. Heat shrink outer wraparound sleeve with heat sensitive indications on both the tube and rail/channel area to indicate proper torch heating, stress relief material, mastic, sealant, shielding mesh, and silicone grease.

#### 2.2.4 Terminations

Provide Class 1 terminations per IEEE 48.

#### 2.2.5 Cable Supports and Fittings

Provide cable supports, related fittings, and accessories for use in corrosive underground locations, such as manholes and utility tunnels, with a factory-applied coating of PVC of at least 20 mils thick. Provide PVC coated items that have a uniform thickness and are free of blisters, breaks, and holidays. Provide PVC compound that conforms to ASTM D746.

Provide cable racks, cable tray supports and related fittings that are UL-listed heavy-duty nonmetallic glass-reinforced nylon.

#### 2.2.6 Polyethylene Cable Tags

Provide tags of polyethylene that have an average tensile strength of 4500 pounds per square inch, and are 0.035-inch thick, non-corrosive non-conductive. Ensure tags are resistive to acids, alkalis, organic solvents, salt water, and are distortion resistant to 300 degrees F. Provide a one-piece nylon, self-locking tie at each end of the cable tag. Ensure ties have a minimum loop tensile strength of 175 pounds. Provide cable tags with block letters, numbers, and symbols 1 inch high on a yellow background. Ensure letters, numbers, and symbols do not fall off or change positions regardless of the cable tags orientation.

#### 2.2.7 Fireproof Tape

Provide fireproof tape approximately 30 mils thick by 3 inches wide, consisting of a flexible, unsupported elastomer that expands in fire to provide a thick char buildup between the flame and the cable. Ensure the tape does not give off a smoke when subjected to flames or support combustion. Also, ensure tape does not deteriorate when subjected to oil, water, gases, salt water, sewage and fungus.

### 2.3 MATERIALS

#### 2.3.1 Conductors

Ensure that conductors conform to the applicable requirements of NEMA WC 70, ANSI/NEMA WC 71/ICEA S-96-659, or NEMA WC 74/ICEA S-93-639.

Ensure that conductors are annealed copper core conforming to ASTM B3 and ASTM B8 and that they are bare, or tin-alloy-coated, according to the type of insulation used. Do not intermix copper and aluminum conductors in the



same raceway.

Provide Class B stranded conductors.

### 2.3.2 Insulation

Ensure the provided cables are rated for minimum 194 degrees F continuous conductor temperature and 266 degrees F emergency overload.

Provide cables with 133 percent insulation. Ensure insulation thickness is in accordance with the following:

| Voltage Rating (kV) | Insulation Level (%) | Typical Insulation Thickness |      |
|---------------------|----------------------|------------------------------|------|
|                     |                      | mm                           | mils |
| 5                   | 100                  | 2.29                         | 90   |
|                     | 137                  | 2.92                         | 115  |
|                     | 173                  | 3.56                         | 140  |
| 8                   | 100                  | 2.92                         | 115  |
|                     | 133                  | 3.56                         | 140  |
|                     | 173                  | 4.45                         | 175  |
| 15                  | 100                  | 4.45                         | 175  |
|                     | 133                  | 5.59                         | 220  |
|                     | 173                  | 6.6                          | 260  |
| 25                  | 100                  | 6.6                          | 260  |
|                     | 133                  | 8.13                         | 320  |
|                     | 173                  | 10.67                        | 420  |
| 35                  | 100                  | 8.76                         | 345  |
|                     | 133                  | 10.67                        | 420  |
|                     | 173                  | 14.73                        | 580  |

### 2.3.3 Cable Identification

Provide cables with printing on the outer jacket showing the cable type, name of the manufacturer, the year in which the cable was manufactured, sequential cable reel length markings and a unique number for identification purposes. Closely group the information on the tape at 6 foot maximum intervals to permit complete identification.

#### 2.3.4 Non-metallic Insulation Shield

Provide extruded insulation shield made of an extruded thermoset material compatible with the insulation and jacket. Ensure insulation shield is applied directly over and bonded to the insulation, and complies with AEIC CS8.

#### 2.3.5 Cable Shielding & bending Requirements

All cables shall be copper-tape shielded and installed with a minimum bending radius of 12 time the overall cable diameter. Copper-wire shielded cables shall only be permitted in locations where existing manholes are utilized and the minimum bending radius requirements for copper-tape shielded cables cannot be met. Cable bending radii shall conform to the guidelines specified in NEMA WC 74/ICEA S-93-639

#### 2.3.6 Jacket

Provide polyethylene (PE) jacketed cable extruded over the cable to a minimum thickness of 80 mils.

#### 2.3.7 Interlock Armored Cable

Provide a nonmetallic, corrosion-resistant jacket over interlock-armored cable that is black polyethylene conforming to NEMA WC 70, ANSI/NEMA WC 71/ICEA S-96-659 and NEMA WC 74/ICEA S-93-639.

### 2.4 TESTS, INSPECTIONS, AND VERIFICATIONS

#### 2.4.1 FACTORY TESTING

Submit certified evidence that the cable manufacturer has made factory-conducted tests on each shipping length (reel) of cable. Submit certified copies of test data in accordance with applicable provisions of the referenced standard. Include in tests on each length of cable to include:

- (1) Conductor Resistance
- (2) Accelerated Water Absorption Test
- (3) Water Immersion Test
- (4) Ionization
- (5) High-Voltage
- (6) Partial Discharge Test

Contracting Officer or designee has the option of witnessing required factory testing at no additional cost. Provide a schedule of manufacturing and testing in advance to permit such witnessing, if requested.

Submit certified qualification test reports in accordance with AEIC CS8 made in accordance with the applicable referenced standards. Ensure certified copies of test data show conformance to the requirements of referenced standards and submit for approval prior to shipment of the cable.

Prior to manufacturing, provide data regarding degradation of proposed insulating material and cable performance due to water immersion test as specified in this specification to the Contracting Officer or designee. Indicate in information AC breakdown stress in kV/mm or V/mil versus immersion time. Ensure a complete description and condition under which cable was tested accompanies the test information. Submit an accelerated water absorption test.

For cables not to be enclosed in metallic conduit, test for flammability in accordance with FED-STD-228, Method 5221 vertical, spark IEEE 383 and IEEE 1202, 70,000 Btu per hour per hour vertical tray flame test.

### PART 3 EXECUTION

#### 3.1 DEMOLITION OR CABLE CUTTING

Notify the Contracting Officer 14 working days prior to an outage for demolition or cable cutting of medium voltage electrical system.

The Government has established a mandatory inspection point prior to Contractor performing any medium voltage cable cuts or demolition. Notify the Contracting Officer 48 hours in advance of this mandatory inspection point.

As part of the mandatory inspection point, positively identify and label the medium voltage cable to be worked utilizing an electronic cable identifier. Ensure the process of identifying and labeling the cable to be worked is witnessed by the Government. Cable cutting and demolition of any medium voltage cable can occur only after approval by the Contracting Officer.

Cut medium voltage cables and conductors by indirect means using cable cutters specifically designed to be operated remotely only. Cutting of medium voltage cables and conductors by direct means is not permitted.

#### 3.2 INSTALLATION

Install medium-voltage cables in accordance with NFPA 70, NFPA 70E; and IEEE C2.

Refer to contract provisions for safety submittals and requirements associated with working in the vicinity of energized cables and equipment. The use of arc-flash and shock prevention equipment and personal protective equipment is mandatory.

Notify the Contracting Officer 14 working days prior to an outage that requires testing for phasing and phase rotation of medium voltage electrical systems. The Government will identify and tag the phasing of equipment and provide to the Contractor, in writing, the results of phasing and phase rotation tests. The Contractor is responsible for maintaining the phasing and phase rotation tests, and is responsible for maintaining the phasing, and matching the existing phase rotation and phasing when installing conductors in existing electrical systems.

Install the cables in the following locations:

Exterior:

- a. In underground duct banks
- b. In conduit above and below grade
- c. In manholes
- d. And by direct burial

Inside Buildings:

- a. By open wire method
- b. On insulator hooks
- c. On racks
- d. In wall and ceiling mounted cable trays

Secure cables with heavy-duty cable ties in existing or new trays mounted horizontally, where the cable rests on the tray bottom. Install cable ties at a minimum of 10 foot intervals.

Secure cables with non-metallic cable clamps, straps, hangers, or other approved supporting devices to tunnel walls, ceilings, and in new or existing cable trays mounted vertically, where the tray bottom is in a vertical plane.

When field cuts or other damage occurs to the PVC coating, apply a liquid PVC patch to maintain the integrity of the coating. After the installation is complete, perform an inspection to ensure that the coating has no voids, pinholes, or cuts.

Before installing new armored cable, ensure that cable trays are properly secured and supported. Add new permanent or temporary tray support devices as required to preclude cable tray failure during cable pulling or after cable is installed.

Installed cable or conductors of a primary distribution system will be rejected by the Government when placed:

- a. Openly in cable trays or openly racked along interior walls
- b. In the same raceway or conduit with AC/DC control circuits or AC power circuits operating at less than 600 volts
- c. In a manner allowing the cable to support its own weight

### 3.2.1 Duct Cleaning

Thoroughly clean ducts before installation of power cables. Pull a standard flexible mandrel through each duct to loosen particles of earth, sand, or foreign material in the line. Use a mandrel that is not less than 12 inches long with a diameter 1/2 inch less than the inside diameter of the duct. Then pull a brush with stiff bristles through each duct to remove the loosened particles. Use a brush with a diameter that is the same as or slightly larger than the diameter of the duct.

### 3.2.2 Pulling Cables in Ducts, Manholes and Utility Tunnels

Submit a Pulling Plan including calculations of pulling tension and side wall pressure anticipated, and the maximum allowable pulling tension for each pull. Do not perform any pull until Government reviews and approves the pulling plan.

Pull medium-voltage cables into ducts and utility tunnels with equipment designed for this purpose, including a power-driven winch, cable-feeding flexible tube guide, cable grips, and lubricants. Employ a sufficient number of trained personnel and equipment to ensure correct installation of the cable.

Set up the cable reel at the side of the manhole or tunnel hatch opening and above the duct or hatch level, allowing the cable to enter through the opening without reverse bending. Install a flexible tube guide through the opening in a manner that prevents the cable from rubbing against the edges of structural members.

Ensure that the pulling force for a cable grip on lead-sheathed cable does not exceed the force calculated in the pulling plan for the 1,500 pounds per square inch of sheath cross-sectional area. Use a dynamometer in the pulling line to ensure that the pulling force is not exceeded. Ensure that the pulling force for a nonmetallic-sheathed cable does not exceed the smaller of 1,000 pounds or a value computed from the following equation:

$$TM = 0.036 \times N \times CM$$

Where: TM = maximum allowable pulling tension in pounds

N = number of conductors in the cable

CM = cross-sectional area of each conductor in circular mils

#### 3.2.2.1 Allowable Sidewall Pressure

The allowable sidewall pressure is the smaller of 500 pounds per foot of bend radius or the cable manufacturer's recommended maximum value. Show in the pulling plan submittal the calculations for allowable tension and sidewall pressure as well as the anticipated tension and sidewall pressure for each pull in the project.

Unreel cable from the top of the reel, carefully controlling payout. Attach cable to be pulled through a swivel to the main pulling wire by means of a pulling eye installed by the factory or approved cable splicer.

Attach pulling eyes to the cable conductors of the 3-1/C circuit to prevent damage to the cable structure. Pull the entire 3-1/C circuit simultaneously.

#### 3.2.2.2 Minimum Bending Radius

Minimum bending radius during cable pulling operations is 30 inches. For permanent cable bending/racking the minimum bending radius is 12 times cable diameter.

#### 3.2.2.3 Coating of Cables

Liberally coat cables with a suitable cable-pulling lubricant as it enters the tube guide or duct. Do not use greaser and oil lubricants. Cover nonmetallic sheathed cables with wire-pulling compounds, when required, which have no deleterious effects on the cable. Use rollers, sheaves or tube guides, around which the cable is pulled, conforming to the 30 inches minimum bending radius of the cable during the pulling operations.

#### 3.2.2.4 Pulling Speed

Pull cables into ducts at a speed not to exceed 50 feet per minute and not in excess of maximum permissible pulling tension specified by the cable manufacturer. Cable pulling using a vehicle is not be permitted. Stop pulling operations immediately with any indication of binding or obstruction and do not resume until such difficulty is corrected. Provide sufficient slack for free movement of cable due to expansion or contraction.

#### 3.2.2.5 Cable Splice Support And Sealing

Firmly support cable splices made up in manholes on cable racks as indicated. Do not pull cable splices in ducts. Overlap cable ends at the ends of a section to provide sufficient undamaged cable for splicing. Overlap cables to be spliced in manholes to the centerline of the proposed joint by not less than 2 feet.

Immediately seal cut ends of cables cut in the field to prevent entrance of moisture with heat-shrinkable molded cable end caps.

#### 3.2.3 Splices and Terminations

Make splices in manholes or tunnels except where cable terminations are specifically indicated. Expedite splicing and terminating of cables in order to minimize exposure and cable deterioration.

Use only equipment and materials recommended by the splice manufacturer including calibrated cutting equipment to ensure consistent cut depths when preparing cable ends for the application of the splice kit. Connect the cable concentric neutral/shield wires across one side of the splice by split bundling the splice neutral wiring and connecting each bundle set to a continuous No. 4 AWG solid bare copper conductor via two compression conductors. Ensure the No. 4 AWG conductor extrudes from the cable splice jacket and connects to the manholes grounding system. Make all connections within the splice utilizing long barrel-type compression connectors and appropriate compression tools with proper size dies to ensure a satisfactory mechanical and electrical joint. Ensure bare connections of concentric neutral/shield wires are either contained within the splice kit or sealed via an additional outer covering, consisting of a heavy wall, heat-shrinkable tubing containing adhesive material (mastic) that melts as heat is applied and the outer tubing shrinks to form a moisture proof environmental seal. Provide outer tubing conforming to ANSI C119.1. Ensure splice meets the requirements of IEEE 404 for a 15 kV rating and is rated by the manufacturer for use on 15 kV class feeder cable systems. Take extra precautions to seal around the exit area of the bare copper jumpers with an additional mastic per the splice manufacturer's recommendations.

Terminate cables in approved cable terminations, rated Class 1 per IEEE 48.

Dry terminations with medium voltage pennants, preformed, and hand wrapped stress cones can be used for terminating cables. Provide terminations with adequate means for making external connections to the cable conductors of single-conductor cables (phase and concentric neutral), protecting the cable insulation against moisture, oil, or other contaminants. Take extra precautions in physically protecting and supporting cables, and maintaining the insulation level of the cable.

Ensure that installation includes built-up or prefabricated heat or cold shrink stress-relief cones at the terminals of all shielded cables and at the terminals of single-conductor lead-covered cables rated 15 kV and above, ungrounded.

Field-fabricate cable splices from splicing kits supplied by and in accordance with the cable manufacturer's recommendations for the type, size, and electrical characteristics of the cable specified. Locate cable splices in manholes midway between the cable racks on the walls of the manholes and supported with cable arms at approximately the same elevation as the enclosing duct.

If cable splices in the tunnel are not installed in cable trays, install the cable splices on cable racks or by other approved methods that minimize physical stress on the splice connections. Support splices at approximately the same elevation as the installed cable except where space limitations or existing cable length limitations make this method impractical or impossible.

Support all universal demountable splices in a manner that minimizes physical stress on the splice connections. Support each cable end termination using a pair of saddle supports under the cable end termination or cable with a minimum 12 inches and a maximum 30 inches separation between the supports. Secure the cable end termination and cable to the supports in a manner that prevents movement of termination or cable at the support. Install saddle supports on a galvanized steel framing channel that is anchored to the wall, securely fastened to the cable tray, or installed by other approved methods.

#### 3.2.4 Fireproofing

Provide fireproofing (Arc Proofing) for individual cable conductor in manholes, handholes and vaults which carry current at 2200 volts or more.

Tightly wrap strips of fireproofing tape around each cable spirally in half-lapped wrapping. Extend the tape 1 inch into the ducts. To prevent unraveling, random wrap the fireproofing tape the entire length of the fireproofing with pressure-sensitive glass cloth tape.

#### 3.2.5 Cable Tag Installation

Install cable tags in each manhole and at each termination as specified. Install cable tags over the fireproofing and position the tags so that they are clearly visible without disturbing any cabling or wiring in the manholes and equipment.

### 3.3 FIELD QUALITY CONTROL

After the installation of power cables has been completed, including splices, joints, and terminations, and before the cable is energized, subject each medium voltage cable to field testing in accordance with the

following requirements:

- a. Provide test equipment, labor, and trained technical personnel as necessary to perform the electrical acceptance tests.
- b. Record all tests on an approved medium voltage cable test form and submit completed forms to the Contracting Officer.
- c. Make arrangements to have tests witnessed and approved by the Contracting Officer.
- d. Isolate each power-cable installation completely from extraneous electrical connections at cable splices/terminations and joints. Observe all safety precautions.
- e. Ensure each power cable is first given an insulation resistance test using a meg-ohmmeter with a voltage output of at least 2,500-volts. Apply test for a long enough time to fully charge the cable (no less than one minute). Record readings as indicated on forms provided. The minimum reading is 5000 megohms at an ambient temperature of 68 degrees F. Correct readings taken at other than 68 degrees F ambient temperatures accordingly.
- f. Conform testing to NETA ATS, and NFPA 70B.

Upon successful completion of the insulation resistance test, subject the cable to a Very Low Frequency (VLF) AC high potential test. Adhere general VLF testing measures, parameters, considerations, and results to the following:

- a. Ensure test voltage duration is continuous duty for 30 minutes (non-interrupted)
- b. Provide the test equipment to test the cable capacity in microfarads.
- c. Provide the test equipment to generate the test voltages required for the 30 minute test duration and adhere to the following table:

| <u>VLF Test Voltage for Sinusoidal Waveform</u> |                                                |                                             |
|-------------------------------------------------|------------------------------------------------|---------------------------------------------|
| Cable Rating<br>phase to phase                  | Acceptance<br>(phase to ground test<br>values) | Acceptance<br>(phase to ground test values) |
|                                                 | (New cable)                                    | (Existing cable spliced to<br>new cable)    |
| (rms voltage, KV)                               | (peak voltage, KV)                             | (peak voltage, KV)                          |
| 5                                               | 14                                             | 10                                          |
| 15                                              | 28                                             | 22; (EPR TO EPR)                            |
| 15                                              | 15                                             | 15; (EPR TO NON-EPR)                        |

- d. Ensure the sinusoidal test frequency is 0.1 Hertz.



e. Do not perform test on cable attached to equipment.

Ensure Splices/terminations are clean, dry, and tested per IEEE 48 and IEEE 400.2.

Upon successful completion of the insulation resistance tests, subject the cable to a direct-current high-potential test for 5 minutes applying test voltages in accordance with AEIC CS1 and IEEE 400.2 for paper-impregnated, lead-covered cable; AEIC CS8 and IEEE 400.2 for cross-linked, polyethylene-insulated cable; and AEIC CS8 and IEEE 400.2 for ethylene propylene rubber-insulated cable.

Record leakage current readings every 30 seconds during the first 2 minutes and every minute thereafter for the remainder of the test. When the leakage current continues to increase after the first minute, immediately terminate the test and take steps to find and correct the fault. When a second test becomes necessary, repeat this test procedure.

Upon satisfactory completion of the high-potential test, give the cable a second insulation resistance test as before.

Provide results of the second insulation resistance test that agree with the first test and that indicate no evidence that the cable has been permanently injured by the high-potential test.

Record test data identifying the cable and location, megohm readings versus time, leakage current readings versus time, and cable temperature versus time.

Final acceptance depends upon the satisfactory performance of the cable under test. Do not energize cable until recorded test data has been approved by the Contracting Officer.

Perform radiographic tests on all potheads at the discretion of the Contracting Officer to determine if voids exist in the pothead. Rework unacceptable terminations at no additional expense to the Government.

### 3.4 CLOSEOUT ACTIVITIES

Provide manufacturer's instructions showing the recommended sequence and method of installation for medium-voltage power cables.

-- End of Section --

SECTION 26 05 19.00 10

INSULATED WIRE AND CABLE  
**11/22**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INSULATED CABLE ENGINEERS ASSOCIATION (ICEA)

ICEA S-58-679 (2014) Control, Instrumentation and Thermocouple Extension Conductor Identification

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA WC 71/ICEA S-96-659 (2014; R 2022) Standard for Nonshielded Cables Rated 2001-5000 Volts for use in the Distribution of Electric Energy

NEMA WC 26 (2008) Binational Wire and Cable Packaging Standard

NEMA WC 57 (2014) Standard for Control, Thermocouple Extension, and Instrumentation Cables

NEMA WC 70 (2021) Power Cable Rated 2000 Volts or Less for the Distribution of Electrical Energy

NEMA WC 74/ICEA S-93-639 (2022) 5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

UNDERWRITERS LABORATORIES (UL)

UL 44 (2018; Reprint May 2021) UL Standard for Safety Thermoset-Insulated Wires and Cables

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Cable Installation Plan; G

SD-03 Product Data

Wire and Cable; G

Conductors; G

Cable Manufacturing Data

SD-06 Test Reports

Test Report(s), Inspection Report(s), and Verification Report(s); G

### 1.3 DELIVERY, STORAGE, AND HANDLING

Furnish cables on reels or coils. Each cable and the outside of each reel or coil, must be plainly marked or tagged to indicate the cable length, voltage rating, conductor size, and manufacturer's lot number and reel number. Each coil or reel of cable must contain only one continuous cable without splices. Cables for exclusively dc applications, as specified in paragraph "High-Voltage Test Source," must be identified as such. Shielded cables rated 2,001 volts and above must be reeled and marked in accordance with NEMA WC 26, as applicable. Reels must remain the property of the Contractor..

## PART 2 PRODUCTS

### 2.1 MATERIALS

#### 2.1.1 Wire Table

Furnish wire and cable in accordance with the requirements of the wire table below, conforming to the detailed requirements specified herein.

#### 2.1.2 Rated Circuit Voltages

All power wire and cable must have minimum rated circuit voltages in accordance with NEMA WC 70, ANSI/NEMA WC 71/ICEA S-96-659, or NEMA WC 74/ICEA S-93-639 as applicable. Power wire and cable for circuit voltages rated 0-600 volts must be rated not less than 600 volts. Control wire and cable must have minimum rated circuit voltages in accordance with NEMA WC 57, but must be rated 600 volts if routed in raceway with other conductors that are rated 600 volts.

#### 2.1.3 Conductors

##### 2.1.3.1 Material for Conductors

A.Manufacturers: Subject to compliance with requirements, manufacturers offering products that may be incorporated into the Work include, but are not limited to the following:

1.Encore

2.General Cable

### 3.Southwire

B.Conductors shall be Underwriters Laboratories, Inc. (UL) listed, 600 volt rated, of insulation type specified below or elsewhere in the Specifications.

#### 1.Branch Circuits - Lighting and Power

a. #10 AWG and smaller shall be solid conductors.

b.#8 AWG and larger shall be stranded conductors.

c.Insulation shall be type THHN or THWN

#### 2.Feeders (60A and more): insulation shall be type XHHW-2.

3.For conductors installed in exposed conduit outside of Buildings, in underground conduit or in conduit within or just under roofing material, provide type XHHW.

4.Control Circuits: Use 600 volts U.L. type THWN or XHHW conductors except where subject to abnormally high temperatures such as on or near boilers.

5.Wiring through fluorescent fixtures in continuous rows shall be type THHN.

Conductors must conform to all the applicable requirements of NEMA WC 57, NEMA WC 70, ANSI/NEMA WC 71/ICEA S-96-659, or NEMA WC 74/ICEA S-93-639 as applicable. Copper conductors must be annealed copper material and they may be bare, or tin- or lead-alloy-coated, if required by the type of insulation used.

#### 2.1.3.2 Size

Minimum wire size must be as listed below.

a. No. 12 AWG for power and lighting circuits

b. No. 10 AWG for current transformer secondary circuits

c. No. 14 AWG for potential transformer, relaying, and control circuits

d. No. 16 AWG for annunciator circuits

e. No. 19 AWG for alarm circuits

Minimum wire sizes for rated circuit voltages of 2,001 volts and above must not be less than those listed for the applicable voltage in ANSI/NEMA WC 71/ICEA S-96-659 or NEMA WC 74/ICEA S-93-639, as applicable.

#### 2.1.3.3 Stranding

Conductor stranding classes cited herein must be as defined for control conductors in NEMA WC 57 or as defined for 0-2,000 volts power conductors in NEMA WC 70, as applicable. Lighting conductors No. 10 AWG and smaller must be solid or have Class B stranding. Any conductors used between stationary and moving devices, such as hinged doors or panels, must have Class H or K stranding. All other conductors must have Class B or C

stranding, except that conductors as shown, or in the schedule, as No. 12 AWG may be 19 strands of No. 25 AWG, and conductors shown as No. 10 AWG may be 19 strands of No. 22 AWG. Conductor stranding classes for circuit voltages 2,001 volts and above must be as defined in ANSI/NEMA WC 71/ICEA S-96-659 and NEMA WC 74/ICEA S-93-639, as applicable.

#### 2.1.3.4 Conductor Shielding

Use conductor shielding conforming to NEMA WC 57 for control wire and cable as applicable. Use conductor shielding conforming to ANSI/NEMA WC 71/ICEA S-96-659 or NEMA WC 74/ICEA S-93-639, as applicable, on power cables having a rated circuit voltage above 2,000 volts.

#### 2.1.3.5 Separator Tape

Where conductor shielding, strand filling, or other special conductor treatment is not required, a separator tape between conductor and insulation is permitted.

#### 2.1.4 Insulation

##### 2.1.4.1 Insulation Material

Unless specified otherwise or required by NFPA 70, wires in conduit, other than service entrance, must be 600-volt, conforming to Type HHW conforming to UL 44. Insulation for control wire and cable must meet the requirements of NEMA WC 57. Insulation requirements for wire and cable rated less than 2,000 volts must meet the requirements of NEMA WC 70. Insulation requirements for wire and cable rated 2,001-5,000 volts must meet the requirements of ANSI/NEMA WC 71/ICEA S-96-659. Insulation requirements for wire and cable rated 5,001 volts and greater must meet the requirements of NEMA WC 74/ICEA S-93-639.

For shielded cables of rated circuit voltages above 2,000 volts, the following provisions must also apply:

- a. XLPE, if used, must be tree-retardant.
- b. Insulation must be chemically bonded to conductor shielding.
- c. The insulation material and its manufacturing, handling, extrusion and vulcanizing processes must all be subject to strict procedures to prevent the inclusion of voids, contamination, or other irregularities on or in the insulation. Insulation material must be inspected for voids and contaminants.
- d. Cables with repaired insulation defects discovered during factory testing, or with splices or insulation joints, are prohibited unless specifically approved.

##### 2.1.4.2 Insulation Thickness

The insulation thickness for each conductor must be based on its rated circuit voltage.

##### 2.1.4.2.1 Power Cables, 2,000 Volts and Below

The insulation thickness for single-conductor and multiple-conductor power cables rated 2,000 volts and below must be as required by NEMA WC 70, as

applicable. Some thicknesses of NEMA WC 70 will be permitted only for single-conductor cross-linked thermosetting polyethylene insulated cables without a jacket. NEMA WC 70 ethylene-propylene rubber-insulated conductors must have a jacket.

#### 2.1.4.2.2 Single-Conductor and Multiple-Conductor Control Cables

The insulation thickness of control conductor sizes 22 AWG to 10 AWG used for control and related purposes must be as required by NEMA WC 57, as applicable. Control conductors larger than 10 AWG must be as required by NEMA WC 70.

#### 2.1.5 Metal-Clad Cable

##### 2.1.5.1 General

The metallic covering or sheath must be interlocked metal tape, conforming to the applicable requirements of NEMA WC 57, NEMA WC 70, ANSI/NEMA WC 71/ICEA S-96-659, or NEMA WC 74/ICEA S-93-639. The type of metal for the metallic covering must be galvanized steel. If the covering is of ferrous metal, it must be galvanized. Grounding conductor(s) conforming to NEMA WC 57, NEMA WC 70, ANSI/NEMA WC 71/ICEA S-96-659, or NEMA WC 74/ICEA S-93-639 as applicable must be furnished for each multiple-conductor metal-clad cable. Assembly and cabling must be as specified in paragraph "Cabling." The metallic covering must be applied over an inner jacket or filler tape. The cable must be assembled so that the metallic covering will be tightly bound over a firm core.

##### 2.1.5.2 Jackets

Metal-clad cables may have a jacket under the armor, and must have a jacket over the armor. Jackets must comply with the requirements of NEMA WC 57, NEMA WC 70, ANSI/NEMA WC 71/ICEA S-96-659, or NEMA WC 74/ICEA S-93-639 as applicable. The outer jacket for the metal-clad cable may be of polyvinyl chloride (PVC) only if specifically approved.

#### 2.1.6 Multiple-Conductor Cables

Grounding conductor(s) conforming to NEMA WC 57, NEMA WC 70, ANSI/NEMA WC 71/ICEA S-96-659, or NEMA WC 74/ICEA S-93-639 as applicable must be furnished for each multiple-conductor cable. Assembly and cabling must be as specified in paragraph CABLING.

## 2.2 CABLE IDENTIFICATION

### 2.2.1 Color-Coding

Insulation of individual conductors of multiple-conductor cables must be color-coded in accordance with ICEA S-58-679, except that colored braids will not be permitted. Only one color-code method must be used for each cable construction type. Control cable color-coding must be in accordance with ICEA S-58-679, Method as follows: Power cable color-coding must be as follows:

a. 208/120 volt, three-phase

(1) Phase A - black

- (2) Phase B - red
- (3) Phase C - blue
- (4) Grounded neutral - white
- (5) Insulated grounding conductor - green

b. 480/277 volt, three-phase

- (1) Phase A - brown
- (2) Phase B - orange
- (3) Phase C - yellow

c. 240/120 volt, single phase: Black and red

Other individual conductors must be color-coded as indicated, but such color-coding may be accomplished by applying colored plastic tapes or colored sleeves at terminations.

#### 2.2.2 Cabling

Individual conductors of multiple-conductor cables must be assembled with flame-and moisture-resistant fillers, binders, and a lay conforming to NEMA WC 57, NEMA WC 70, ANSI/NEMA WC 71/ICEA S-96-659, or NEMA WC 74/ICEA S-93-639. Flat twin cables are prohibited. Fillers must be used in the interstices of multiple-conductor round cables with a common covering where necessary to give the completed cable a substantially circular cross section. Fillers must be non-hygroscopic material, compatible with the cable insulation, jacket, and other components of the cable. The rubber-filled or other approved type of binding tape must consist of a material that is compatible with the other components of the cable and must be lapped at least 10 percent of its width.

#### 2.2.3 Dimensional Tolerance

The outside diameters of single-conductor cables and of multiple-conductor cables must not vary more than 5 percent and 10 percent, respectively, from the manufacturer's published catalog data.

### PART 3 EXECUTION

#### 3.1 INSTALLATION INSTRUCTIONS

Submit cable manufacturing data as requested. The following information must be provided by the cable manufacturer for each size, conductor quantity, and type of cable furnished:

- a. Minimum bending radius, in inches - For multiple-conductor cables, this information must be provided for both the individual conductors and the multiple-conductor cable.
- b. Pulling tension and sidewall pressure limits, in pounds.
- c. Instructions for stripping semiconducting insulation shields, if furnished, with minimum effort without damaging the insulation.

- d. Upon request, compatibility of cable materials and construction with specific materials and hardware manufactured by others must be stated. Also, if requested, recommendations must be provided for various cable operations, including installing, splicing, terminating, etc.

### 3.2 CABLE INSTALLATION PLAN

Provide a cable installation plan which includes the following for each raceway.

- a. Raceway name
  - b. Circuit names with wire sizes and quantities
  - c. Maximum tolerable pulling tension
  - d. Maximum tolerable sidewall pressure
  - e. Estimated maximum pulling tension
  - f. Estimated maximum sidewall pressure
  - g. Jam ratio
  - h. Lubricant and coefficient of friction
  - i. Pulling device(s)
  - j. Direction of pull

### 3.3 TEST REPORT(S), INSPECTION REPORT(S), AND VERIFICATION REPORT(S)

#### 3.3.1 Cable Data

Do not begin any wire and cable fabrication until materials are submitted and approved by the Contracting Officer. Submit cable data for approval including, but not limited to, dimensioned sketches showing cable construction and sufficient additional data to show that wire and cable meet the requirements of this Section.

#### 3.3.2 Inspection and Tests

Inspection and tests of wire and cable furnished under these specifications must be made by and at the plant of the manufacturer, and the manufacturer must provide certification and certification reports of completed inspections and completed tests. The Government may require or perform further tests before or after installation. Testing in general must comply with NEMA WC 57, NEMA WC 70, ANSI/NEMA WC 71/ICEA S-96-659, or NEMA WC 74/ICEA S-93-639 as applicable. Specific tests required for particular materials, components, and completed cables must be as specified in the sections of the above standards applicable to those materials, components, and cable types. Tests must also be performed in accordance with the additional requirements specified below. Submit certified copies of test reports.

- A. After installing conductors and cables and before electrical circuitry has been energized, test service entrance and feeder conductors for compliance with requirements.



B. Perform each of the following visual and electrical tests:

- 1) Inspect exposed sections of conductor and cable for physical damage and correct connection according to the single-line diagram.
- 2) Test bolted connections for high resistance using one of the following:
  - a. A low-resistance ohmmeter.
  - b. Calibrated torque wrench.
  - c. Thermographic survey.
- 3) Inspect compression-applied connectors for correct cable match and indentation.
- 4) Inspect for correct identification.
- 5) Inspect cable jacket and condition.
- 6) Insulation-resistance test on each conductor for ground and adjacent conductors. Apply a potential of 500-V dc for a one-minute duration.
- 7) Continuity test on each conductor and cable.
- 8) Uniform resistance of parallel conductors.

C. Cables will be considered defective if they do not pass tests and inspections.

- D. All conductors failing to comply with the manufacturers' minimum standard or recommendations shall be replaced at Contractor's expense.

E. Tests shall be made prior to final connection of equipment.

F. Prepare test and inspection reports to record the following:

- 1) Procedures used.
- 2) Results that comply with requirements.
- 3) Results that do not comply with requirements, and corrective action taken to achieve compliance with requirements.

3.3.2.1 Reports

Furnish results of tests. No wire or cable must be shipped until authorized. Lot number and reel or coil number of wire and cable tested must be indicated on the test reports.

[illegible]

-- End of Section --

SECTION 26 05 48

SEISMIC PROTECTION FOR ELECTRICAL EQUIPMENT  
**11/23**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

</REF>

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.21.1 (2009; R 2016) Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)

ASME B18.22M (1981; R 2017) Metric Plain Washers

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M (2019) Standard Specification for Carbon Structural Steel

ASTM A53/A53M (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A153/A153M (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM A307 (2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength

ASTM A500/A500M (2023) Standard Specification for Cold-Formed Welded and Seamless Carbon Steel Structural Tubing in Rounds and Shapes

ASTM A563 (2021; E 2022a) Standard Specification for Carbon and Alloy Steel Nuts

|                   |                                                                                                                                                                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A572/A572M   | (2021; E 2021) Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Structural Steel                                                                                                                                                   |
| ASTM A992/A992M   | (2022) Standard Specification for Structural Steel Shapes                                                                                                                                                                                               |
| ASTM B695         | (2021) Standard Specification for Coatings of Zinc Mechanically Deposited on Iron and Steel                                                                                                                                                             |
| ASTM F436         | (2011) Hardened Steel Washers                                                                                                                                                                                                                           |
| ASTM F959/F959M   | (2017a; R 2023) Standard Specification for Compressible-Washer-Type Direct Tension Indicators for Use with Structural Fasteners, Inch and Metric Series                                                                                                 |
| ASTM F1554        | (2020) Standard Specification for Anchor Bolts, Steel, 36, 55, and 105-ksi Yield Strength                                                                                                                                                               |
| ASTM F3125/F3125M | (2019) Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength |

ICC EVALUATION SERVICE, INC. (ICC-ES)

|              |                                                                                                         |
|--------------|---------------------------------------------------------------------------------------------------------|
| ICC ES AC156 | (2012) Acceptable Criteria for Seismic Certification by Shake-Table Testing of Nonstructural Components |
|--------------|---------------------------------------------------------------------------------------------------------|

INTERNATIONAL CODE COUNCIL (ICC)

|         |                                    |
|---------|------------------------------------|
| ICC IBC | (2021) International Building Code |
|---------|------------------------------------|

METAL FRAMING MANUFACTURERS ASSOCIATION (MFMA)

|        |                                            |
|--------|--------------------------------------------|
| MFMA-4 | (2004) Metal Framing Standards Publication |
|--------|--------------------------------------------|

U.S. DEPARTMENT OF DEFENSE (DOD)

|              |                                                                                       |
|--------------|---------------------------------------------------------------------------------------|
| MIL-STD-810  | (2022; Rev H; Change 1) Environmental Engineering Considerations and Laboratory Tests |
| UFC 3-301-01 | (2023; with Change 1, 2023) Structural Engineering                                    |
| UFC 3-301-02 | (2023) Design of Risk Category V Structures, National Strategic Military Assets       |

VIBRATION ISOLATION AND SEISMIC CONTROL MANUFACTURERS ASSOCIATION  
(VISCMA)

VISCMA 413

(2014) Installing Seismic Restraints for  
Electrical Equipment

## 1.2 SYSTEM DESCRIPTION

### 1.2.1 General Requirements

Design and provide seismic supports and attachments in accordance with UFC 3-301-01 and ASCE 7-16.

Provide seismic supports and attachments as indicated and in accordance with UFC 3-301-01 and ASCE 7-16.

Components, supports, and attachments must comply with following structural design criteria:

Risk Category: III.

Seismic Design Category: B.

Seismic Design Spectral Response Acceleration Parameter at Short Period (SDS): 0.23.

Seismic Design Spectral Response Acceleration Parameter at period of 1 second (SD1): 0.088.

Apply the seismic requirements described in this section and on the drawings to the electrical components listed in paragraph ELECTRICAL COMPONENTS.

Electrical components and their supports must be attached or anchored to structure.

### 1.2.2 Electrical Components

Provide seismic supports and attachments for the following electrical components in accordance with the requirements of this specification:

Components with Importance Factor ( $I_p$ ) = 1.0:

Control Panels

Light Fixtures

Motors

Switchboards

Transformers

Unit Substations

Distribution Systems Conduit, Cable Tray, Busway, and Raceway

Components with Importance Factor ( $I_p$ ) = 1.5 Designated Seismic Systems

Emergency and standby power systems

Fire detection and suppression systems

Emergency exit lighting

### 1.2.3 Contractor Designed Supports and Attachments

Provide seismic supports and attachments for electrical components.

Contractor must retain services of a Registered Professional Engineer to design supports and attachments for electrical components.

Submit copies of the Design Calculations and Design Drawings.  
Calculations and Drawings must be stamped and signed by Contractor's  
Registered Professional Engineer.

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

Equipment Requirements; G

#### SD-03 Product Data

Supports and Attachments; G

Equipment Requirements; G

Flexible Fittings; G

Anchors; G

#### SD-05 Design Data

Design Calculations; G

Design Drawings; G

#### SD-06 Test Reports

Anchors; G

#### SD-07 Certificates

ICC ES AC156 Shake Table Test; G

## PART 2 PRODUCTS

### 2.1 EQUIPMENT REQUIREMENTS

Equipment must be rugged enough to survive design seismic event.  
Equipment components designated as Designated Seismic Systems must remain operational as required in paragraph SPECIAL TESTING FOR SEISMIC-RESISTING EQUIPMENT.

Submit detail drawings of supports and attachments along with calculations, catalog cuts, templates, and erection and installation details, as appropriate, for the items listed in paragraph ELECTRICAL COMPONENTS. Indicate thickness, type, grade, class of metal, and dimensions; and show construction details, reinforcement, anchorage, and installation with relation to the building construction.

Submit calculations and drawings that are stamped and signed by Contractor's registered professional engineer. Design must be based on actual equipment and system layout. Design must include calculated loads and capacity of materials utilized for the connection of the equipment or system to the structure. Analysis must detail anchoring methods. Include drawing for Mission Critical Equipment and Designated Seismic System Equipment indicating the equipment location in the facility to be used for the installation. Equipment must be rigidly or flexibly mounted as indicated. Roof mounted equipment both vibration isolated and nonisolated, must have support members designed and anchored to building structure.

#### 2.1.1 Rigidly (Base and Suspended) Mounted Equipment

The following specific items of equipment must be constructed and assembled to withstand the seismic forces specified in UFC 3-301-01. Entirely locate each item of rigid electrical equipment and rigidly attach on one side only of a building expansion joint. Provide items such as electrical conduit, which cross the building expansion joint with flexible joints that are capable of accommodating displacements equal to the full width of the joint in each orthogonal direction.

Substations  
Transformers  
Switch Boards and Switch Gear

Equipment furnished under this Contract must be rigidly mounted using cast-in-place anchor bolts or post-installed anchors that are qualified for use in cracked concrete. Cast-in-place anchor bolts must conform to ASTM F1554. For any rigid equipment which is rigidly anchored, provide flexible joints for electrical conduit, cabletray, busway, and raceway, that are capable of accommodating displacements equal to the full width of the joint in both orthogonal directions. Mission critical base mounted and suspended equipment for Risk Category (RC) V, and Designated Seismic Systems (DSS) for RC IV buildings assigned to Seismic Design Category (SDC) C, D, E, or F and Risk Category IV components needed for continued operation after an earthquake must have two nuts provided on each bolt.

#### 2.2 SUPPORTS AND ATTACHMENTS

Material used for members listed in this section must conform with the following:

- a. Plates, rods, and rolled shapes, ASTM A36/A36M, ASTM A572/A572M Grade 50, or ASTM A992/A992M.
- b. Tubes, ASTM A500/A500M, Grade B.
- c. Pipes, ASTM A53/A53M, Grade B.
- d. Angles, ASTM A36/A36M.

- e. Channels (Struts) with in-turned lips and associated hardware for fastening to channels at discrete points conforming to MFMA-4.
- f. Fasteners:
  - (1) High-strength bolts, ASTM F3125/F3125M, Grade A325, heavy hex.
  - (2) Nuts, ASTM A563. Use heavy hex nuts for high strength bolts and anchor bolts.
  - (3) Washers, ASME B18.22M and ASME B18.21.1, except use ASTM F436 washers for high strength bolts.
  - (4) Galvanized coating, ASTM A153/A153M.
  - (5) Direct Tension Indicator Washers, ASTM F959/F959M. Provide ASTM B695, Class 55, Type 1 galvanizing. Submit product data for direct tension indicator washers.
  - (6) Standard bolts, ASTM A307.
- g. Anchor bolts:
  - (1) Cast in anchors: Refer to 05 50 13 MISCELLANEOUS METAL FABRICATIONS for requirements.
  - (2) Post-installed anchors: Refer to Section 05 05 20 POST-INSTALLED CONCRETE AND MASONRY ANCHORS for requirements.
- h. Welding: STRUCTURAL METAL FABRICATIONS

## 2.3 FLEXIBLE FITTINGS

Provide flexible fittings to allow conduit, cabletray, busway, and raceway systems to accommodate seismic movement between equipment and conduit, cabletray, busway, and raceway systems and supporting structure. Use specification grade steel fittings conforming to Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM MIL-STD-810. Provide suitable watertight expansion fittings that maintain electrical continuity by bonding jumper or other means. Flexible fittings must accommodate minimum of 4 inches seismic movement in each orthogonal directions.

## PART 3 EXECUTION

### 3.1 SUPPORTS AND ATTACHMENTS

Provide supports and attachments with continuous load path to distribute electrical component seismic loads to structure.

Provide supports and attachments for electrical components that conform to the arrangements shown. Install vertical diagonal braces at a 45-degree slope. Where interference is present, the slope may be minimum of 30 degrees or a maximum of 60 degrees per VISCMA 413.

Provide bolted and welded connections for supports and attachments in accordance with UFC 3-301-01.

Provide welding in accordance with AWS D1.1/D1.1M. Grind visible welds smooth in the finished installation.



### 3.2 BUILDING DRIFT

Do not attach electrical components to two dissimilar structural elements of a building that may respond differentially during an earthquake unless a flexible joint is provided. Electrical components, supports, and attachments must be capable of accommodating building story drifts, deflections, and relative displacements.

### 3.3 CONDUIT, CABLETRAY, BUSWAY, AND RACEWAY

Provide supports and attachments for conduit, cabletray, busway, and raceway conforming to the requirements of UFC 3-301-01.

### 3.4 ANCHORS

#### 3.4.1 General

Submit copies of test results to verify the adequacy of the specific anchor and application, as specified. Ensure housekeeping pads have adequate space to mount equipment and seismic restraint devices allowing adequate edge distance and embedment depth for restraint anchor bolts. Identify position of reinforcing steel and other embedded items prior to drilling holes for post-installed anchors. Do not drill holes in concrete or masonry until concrete, mortar, or grout has achieved full design strength. Fill annular gap with nonshrink grout at equipment anchor bolts where clearance between anchor and equipment support hole exceeds 0.125 inches.

#### 3.4.2 Cast-In-Place Anchors

Use templates to locate cast-in-place bolts accurately and securely in formwork. Provide anchor bolts with an embedded straight length with heavy hex nut and washer to provide required strength and ductility. Anchor bolts that exceed the normal depth of equipment foundation piers or pads must either extend into concrete floor or the foundation or increase depth of the piers or pads to accommodate bolt lengths.

#### 3.4.3 Post-Installed Anchors

Refer to Section 05 05 20 POST-INSTALLED CONCRETE AND MASONRY ANCHORS for requirements.

### 3.5 EQUIPMENT SUPPORT REQUIREMENTS

#### 3.5.1 Suspended Equipment

Provide supports and attachments for components supported from structural systems. Provide supports and attachments that consist of angles, rods, wire rope, bars, channels (struts) or pipes arranged as shown in bracing submittals and secured at both ends with not less than 1/2 inch bolts. Provide sufficient supports and attachments to resist seismic forces as specified in UFC 3-301-01 without exceeding capacity of structural components. Provide, for approval, specific force calculations in accordance with UFC 3-301-01 for the project.

Submit details of supports and attachments for acceptance. In lieu of bracing with vertical supports, these items may be supported with hangers inclined at 45 degrees directed up and radially away from equipment and

oriented symmetrically in 90-degree intervals on the horizontal plane, bisecting the angles of each corner of the equipment, provided that supporting members are properly sized to support operating weight when hangers are inclined. Where interference is present, the inclined hanger slope may be minimum of 30 degrees or a maximum of 60 degrees per VISCMA 413.

### 3.5.2 Floor or Pad Mounted Equipment

#### 3.5.2.1 Shear Resistance

Bolt components to floor or pads. Provide bolts to resist seismic forces in accordance with paragraph ANCHORS.

#### 3.5.2.2 Overturning Resistance

Use the ratio of the overturning moment from seismic forces to the resisting moment due to gravity loads to determine if overturning forces need to be considered in the sizing of anchor bolts. Provide calculations to verify the adequacy of the anchor bolts for combined shear and tension. Provide bolts to resist seismic forces in accordance with paragraph ANCHORS.

### 3.6 SPECIAL TESTING FOR SEISMIC-RESISTING COMPONENTS

Electrical components designated as Designated Seismic Systems required to remain operational after an earthquake must be seismic qualified by shake table testing conforming to ICC ES AC156 Shake Table Test procedures. The manufacturer is to provide a certification by a fully qualified testing agency for the specific components. Prequalified certifications are acceptable unless noted otherwise. Seismic qualification documentation for each component must contain the information required in UFC 3-301-02, Section 2-17.2.5 Component Qualification Documentation.

Components that are required to be certified must bear permanent marking or nameplates constructed of a durable heat and water resistant material.

Provide component identification nameplates in accordance with UFC 3-301-01.

Mechanically attach nameplates to electrical components.

### 3.7 SPECIAL INSPECTION FOR COMPONENTS, SUPPORTS, AND ATTACHMENTS

Perform special inspections for seismic-resisting systems, designated seismic systems, components, supports and attachments, and equipment per Section 01 45 35 SPECIAL INSPECTION attachment Schedule of Special Inspections and ICC IBC 1705.13.4; electrical components per ICC IBC 1705.13.6.

Special Inspector must examine Designated Seismic Systems requiring seismic qualification in accordance with ASCE 7-16 and verify the label, anchorage and mounting conform to the certificate of compliance.

Provide a Statement of Special Inspections and Final Special Inspection Report in accordance with UFC 3-301-01 .

-- End of Section --

SECTION 26 05 73

POWER SYSTEM STUDIES

08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

|             |                                                                                                                          |
|-------------|--------------------------------------------------------------------------------------------------------------------------|
| IEEE 1584   | (2018; E 2019) Guide for Performing Arc-Flash Hazard Calculations                                                        |
| IEEE 1584.1 | (2022) Guide for the Specification of Scope and Deliverable Requirements for an Arc-Flash Hazard Calculation Study       |
| IEEE 3002.2 | (2018) Recommended Practice for Conducting Load-Flow Studies and Analysis of Industrial and Commercial Power Systems     |
| IEEE 3002.3 | (2018) Recommended Practice for Conducting Short-Circuit Studies and Analysis of Industrial and Commercial Power Systems |

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                        |
|----------|--------------------------------------------------------|
| NFPA 70E | (2024) Standard for Electrical Safety in the Workplace |
|----------|--------------------------------------------------------|

U.S. DEPARTMENT OF DEFENSE (DOD)

|              |                                                                           |
|--------------|---------------------------------------------------------------------------|
| UFC 3-560-01 | (2017; with Change 2, 2019) Operations and Maintenance: Electrical Safety |
|--------------|---------------------------------------------------------------------------|

1.2 SYSTEM DESCRIPTION

The power system covered by this specification consists of: the new distribution system and the upstream over-current protective devices.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Field Examination Plan; G

Arc Flash Label Formats; G

SD-06 Test Reports

Field Examination

SD-07 Certificates

System Analyzer; G

SD-11 Closeout Submittals

Model Files; G

Load Flow Study; G

Fault Current Study; G

System Coordination Study; G

Arc Flash Hazard Study; G

#### 1.4 QUALITY ASSURANCE

##### 1.4.1 System Analyzer

The System Analyzer must perform the power system studies. The System Analyzer must be a registered professional electrical engineer with a minimum of 10 years of experience with power system studies. Include a list of three comparable jobs performed by the System Analyzer with specific names and telephone numbers for reference. Include the license number and state of the registered Professional Engineer.

##### 1.4.2 System Verifier

The System Verifier may inspect and record settings, markings, and otherwise document the existing equipment. The System Verifier may also place appropriate hazard labels on equipment. The System Verifier must not make any calibrations or adjustments or place equipment into service. When working with energized equipment, the system verifier must be a Qualified Person per NFPA 70E or working under the direct supervision of a Qualified Person.

#### PART 2 PRODUCTS

Not used.

#### PART 3 EXECUTION

##### 3.1 FIELD EXAMINATION

For each pre- and post-study inspection, submit a field examination plan identifying which facilities must be examined to complete the required work. Include a complete information verification procedure, where Qualified Personnel will be required, PPE requirements for "live" equipment examinations, and equipment to be used within the limited approach boundary of the equipment.

### 3.1.1 General

Perform field inspections of the site and equipment before the first study and after the last study to determine the state and settings of the equipment and to verify the final settings agree with the studies. Perform the post-study inspection(s) after changing settings, after performing acceptance tests, and before placing equipment in service. The post-study inspection may be performed by the testing organization approved in 26 08 00 APPARATUS INSPECTION AND TESTING.

Schedule the field examination by the System Verifier with the Contracting Officer at least 10 business days before conducting examination. Furnish all materials, labor, and equipment necessary to conduct the examination. Maintain a written record of the all equipment examined, equipment location, equipment ratings, settings, personnel involved, and the date the examination was performed. Submit the written record of each field examination.

### 3.1.2 Safety

Where examination of the equipment requires energized equipment to be opened, provide a Qualified Person to directly supervising any non-Qualified Person within the hazard area. Use PPE, protective barriers, danger signs and other safety devices to protect and warn personnel in the vicinity of "live" equipment being examined.

### 3.1.3 Application of Arc Flash Labels

Install arc flash warning labels using Qualified Personnel as necessary after the setting and inspection is complete. For new or modified equipment, install labels before the equipment is energized for the first time after installation/modification or setting changes. Schedule the label placement with the Contracting Officer at least 5 business days before label placement. Furnish all materials, labor, and equipment necessary to place the labels. Maintain a written record of the all equipment that received labels, personnel involved, and the date labels were placed.

## 3.2 POWER SYSTEM STUDIES

Perform power system studies to demonstrate that the equipment selected and system constructed meet the contract requirements for fault current and interrupt ratings, coordination, protection, and Arc Flash Hazard. Submit reports of the studies along with protective device equipment submittals. Apply Arc Flash Hazard labels to equipment after the studies are approved. Update and resubmit the studies after any changes to the equipment or systems which may affect the study results, and re-apply Arc Flash Hazard labels to equipment after the resubmitted studies are approved. The Government is not responsible for any changes to equipment, device ratings, settings, or additional labor for installation of equipment or devices ordered and/or procured before approval of the study.

Perform power system studies to demonstrate the system operation with regard to ratings, coordination, protection, and Arc Flash Hazard.

### 3.2.1 Scope of Studies

All new electric equipment shown on the one line diagram and extend through the downstream protective devices at the load's end and the existing upstream over current protective devices are included in the power study.

The "source bus" is the source of energy for system being analyzed. This may be the energy feed from a utility, the first bus upstream of the work, generators within the work or upstream of the work, or any other source capable of contributing significant energy into the system being analyzed.

In the systems model for the studies, incorporate all existing and new equipment within the scope of the studies. Incorporate any additional sources or load equipment necessary to accurately model the system's performance.

### 3.2.2 Determination of Facts

Determine and document the time-current characteristics, features, ratings, ampacities, and nameplate data for each existing protective device, electrical equipment, and feeder cables.

### 3.2.3 Single Line Diagram

Provide a single line diagram showing the electrical system buses, devices, transformation points, and all sources of load current and fault current, including generator and motor contributions. Provide a diagram from the system model. Each bus, device or transformation point must have a unique identifier. Show the location of switches, breakers, and circuit interrupting devices on the diagram together with available fault data, and the device interrupting rating.

The naming of existing components within the system model and single line diagram must match existing installed equipment names. The naming of new components within the system model and single line diagram must use unique identifiers and be coordinated with the Government.

### 3.2.4 Load Flow Study

Perform the load flow study in accordance with IEEE 3002.2 to identify initial steady-state conditions for the fault current study. For each operating scenario, provide load flow results on the diagram or in the report. Show or describe the loading factors and assumptions used in each operating mode.

### 3.2.5 Fault Current Study

Use the results of the load flow study to perform the fault current study in accordance with IEEE 3002.3. Provide balanced three-phase fault, bolted line-to-line fault, and single line-to-ground fault current values at each voltage transformation point and at each power distribution bus.

For each location, show in tabular form on the diagram or in the report the maximum and minimum available fault currents of all modes of operation for that location.

Where the available fault current at the source bus is not available, describe how the fault contribution from the source bus was determined and why this method is reasonable for the study.

### 3.2.6 System Coordination Study

Use the results of the load flow study and fault current study. For normal modes of operation, demonstrate that selectivity has been obtained between the devices within the scope of the project. Demonstrate the equipment, machines, and conductors are protected from damage from overloads and fault conditions. Include a description of the coordination of the protective devices in this project. Provide a written narrative describing which devices may operate in the event of a fault at each bus; the logic used to arrive at device ratings and settings; situations where system coordination is not achievable due to device limitations; coordination between upstream and downstream devices; and relay settings. Provide recommendations to improve or enhance system reliability, and detail where such changes would involve additions or modifications to the contract and cost damages (addition or reduction). Provide composite coordination plots on a log-log scale. Where recommendations are provided to improve or enhance system reliability, provide separate coordination plots on log-log scale showing the system coordination before and after the implementation of the recommendations. Where coordination has been adjusted to lower arc flash values, include a discussion of the change and the impact on the system within the report.

### 3.2.7 Arc Flash Hazard Study

Perform the arc flash hazard study in accordance with IEEE 1584.1. Utilize the data from the fault current study to determine the worst case incident energy per IEEE 1584 and OSHA 29 CFR 1910.269 Appendix E. Use identified modes of operation to determine the worst case arc flash energy. If not included in another study, include a description of the devices and device settings for the operating modes that provided the highest arc flash energy. The use of a "maintenance switch" to temporarily reduce the Arc Flash incident energy for maintenance work is acceptable. .

### 3.2.8 Study report(s)

- a. Include a narrative describing the studies performed; the bases and methods used; and the desired method of coordinated protection of the power system.
- b. Include descriptive and technical data for existing devices and new protective devices proposed. Include manufacturers published data, nameplate data, and definition of the fixed or adjustable features of the existing or new protective devices. For existing devices, included statements on the condition of the equipment based upon field inspections and owner's statements and reports.
- c. Document utility company data including system voltages, fault MVA,



system X/R ratio, time-current characteristic curves, current transformer ratios, and relay device numbers and settings; generator resistance and reactance values, and time constants; and existing power system data including time-current characteristic curves and protective device ratings and settings. Identify all assumptions about the protection devices, equipment, and system where data was not available.

- d. For each bus in the system, provide fully coordinated composite time-current characteristics (TCC) curves as required to ensure coordinated power system protection between protective devices and equipment. In a tabular format, provide existing and recommended ratings and settings of all protective devices.
- e. Provide an arc flash study report in accordance with IEEE 1584.1.
- f. Provide the calculations performed for the studies, including computer programs utilized. Provide the name of the software package, developer, and version number.
- f. The System Analyzer must stamp the study reports.

### 3.2.9 Arc Flash Labels

Provide arc flash warning labels on electrical equipment likely to require examination, servicing, or maintenance while energized. Typical types of equipment include pad-mounted transformers, switchgear, switchboards, panelboards, disconnect switches, industrial control panels, meter socket enclosures, and motor control centers that are in other than dwelling occupancies. The arc flash label naming must match the naming used in the system modeling and the single line diagram.

Comply with the label requirements specified in UFC 3-560-01. Obtain approval of arc flash label formats before printing.

- a. Provide a 3.5 inch x 5 inch to 4 inch x 6 inch thermal transfer type label of high adhesion polyester for each location device analyzed. The label must remain in place and be legible for at least 5 years in the installed environment.
- b. Labels must be machine printed with no field markings. Provide arc flash labels in the following manner. All labels must be based on implemented overcurrent devices and settings.
  - 1. Provide at least one arc flash label for each 480 volt and 208 volt panelboards.
  - 2. Provide at least one arc flash label for each low voltage (less than 1000 V) switchboard.
  - 3. Where the equipment construction has different hazards levels in different areas, provide a label for each compartment, cubicle, or section separated by barriers a single label representing the worst case hazard for all branch circuits and a separate unique label for the compartment containing the main protective device. On the worst case label, indicate which branch circuit compartments have the worst case hazard.
- c. Use the worst case hazard of all operating scenarios unless mitigation

procedures are used. If mitigation procedures are used, explain the procedures on the label.

### 3.3 MODELING

Develop a software model of the electrical system identified in the scope of the studies. Use the latest version of SKM PowerTools(TM). Develop the model with accurate, verified information. Model existing electrical equipment, machines, devices, and conductors directly connected to, altered by, or otherwise affected by the work. This includes, but is not limited to generators, transformers, switchgear, switchboards, panelboards, motor control centers, motors, voltage regulators, tap changers, protective relays, circuit breakers, switches, fuses, conductors, capacitors, reactors, grounding system, and control equipment.

Provide the final model files in their native editable formats for future use by the Government. Include device information for devices not in the software vendor's standard distribution.

-- End of Section --

SECTION 26 08 00

APPARATUS INSPECTION AND TESTING  
**11/22**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

|          |                                                                                                        |
|----------|--------------------------------------------------------------------------------------------------------|
| NETA ATS | (2021) Standard for Acceptance Testing<br>Specifications for Electrical Power<br>Equipment and Systems |
|----------|--------------------------------------------------------------------------------------------------------|

1.2 RELATED REQUIREMENTS

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM applies to this section with additions and modifications specified herein.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-06 Test Reports

Acceptance Tests and Inspections; G

SD-07 Certificates

Qualifications of Organization, and Lead Engineering Technician; G

Acceptance Test and Inspections Procedure; G

1.4 QUALITY ASSURANCE

1.4.1 Qualifications

Engage the services of a qualified testing organization to provide inspection, testing, calibration, and adjustment of the electrical distribution system and generation equipment listed in paragraph ACCEPTANCE TESTS AND INSPECTIONS herein. Organization must be independent of the supplier, manufacturer, and installer of the equipment. The organization must be a first tier subcontractor. No work required by this section of the specification may be performed by a second tier subcontractor.

- a. Submit name and qualifications of organization. Organization must have been regularly engaged in the testing of electrical materials,

devices, installations, and systems for a minimum of 5 years. The organization must have a calibration program, and test instruments used must be calibrated in accordance with NETA ATS.

- b. Submit name and qualifications of the lead engineering technician performing the required testing services. Include a list of three comparable jobs performed by the technician with specific names and telephone numbers for reference. Testing, inspection, calibration, and adjustments must be performed by an engineering technician, certified by NETA (Level III) or the National Institute for Certification in Engineering Technologies (NICET) with a minimum of 5 years' experience inspecting, testing, and calibrating electrical distribution and generation equipment, systems, and devices.

#### 1.4.2 Acceptance Tests and Inspections Reports

Submit certified copies of inspection reports and test reports. Include certification of compliance with specified requirements, identify deficiencies, and recommend corrective action when appropriate. Type and neatly bind test reports to form a part of the final record. Submit test reports documenting the results of each test not more than 10 days after test is completed.

#### 1.4.3 Acceptance Test and Inspections Procedure

Submit test procedure reports for each item of equipment to be field tested at least 45 days prior to planned testing date. Do not perform testing until after test procedure has been approved.

### PART 2 PRODUCTS

Not used.

### PART 3 EXECUTION

#### 3.1 ACCEPTANCE TESTS AND INSPECTIONS

Testing organization will perform acceptance tests and inspections. Test methods, procedures, and test values must be performed and evaluated in accordance with NETA ATS, the manufacturer's recommendations, and paragraph FIELD QUALITY CONTROL of each applicable specification section. Tests identified as optional in NETA ATS are not required unless otherwise specified. Place equipment in service only after completion of required tests and evaluation of the test results have been completed. Supply to the testing organization complete sets of shop drawings, settings of adjustable devices, and other information necessary for an accurate test and inspection of the system prior to the performance of any final testing. Notify Contracting Officer at least 14 days in advance of when tests will be conducted by the testing organization. Perform acceptance tests and inspections on applicable equipment and systems specified in the following sections:

g. Section 26 11 16 SECONDARY UNIT SUBSTATIONS

h.

k. Section 26 24 13 SWITCHBOARDS

### 3.2 SYSTEM ACCEPTANCE

Final acceptance of the system is contingent upon satisfactory completion of acceptance tests and inspections.

### 3.3 PLACING EQUIPMENT IN SERVICE

A representative of the approved testing organization must be present when equipment tested by the organization is initially energized and placed in service.

-- End of Section --

SECTION 26 11 16

SECONDARY UNIT SUBSTATIONS

11/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 318M (2014; ERTA 2015) Building Code  
Requirements for Structural Concrete &  
Commentary

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z540.1 (1994; R 2002) Calibration Laboratories  
and Measuring and Test Equipment - General  
Requirements

ASTM INTERNATIONAL (ASTM)

ASTM A123/A123M (2017) Standard Specification for Zinc  
(Hot-Dip Galvanized) Coatings on Iron and  
Steel Products

ASTM A153/A153M (2023) Standard Specification for Zinc  
Coating (Hot-Dip) on Iron and Steel  
Hardware

ASTM A653/A653M (2023) Standard Specification for Steel  
Sheet, Zinc-Coated (Galvanized) or  
Zinc-Iron Alloy-Coated (Galvannealed) by  
the Hot-Dip Process

ASTM A780/A780M (2020) Standard Practice for Repair of  
Damaged and Uncoated Areas of Hot-Dip  
Galvanized Coatings

ASTM C260/C260M (2010a; R 2016) Standard Specification for  
Air-Entraining Admixtures for Concrete

ASTM D709 (2017) Standard Specification for  
Laminated Thermosetting Materials

ASTM D1535 (2014; R 2018) Standard Practice for  
Specifying Color by the Munsell System

ASTM D6871 (2017) Standard Specification for Natural  
(Vegetable Oil) Ester Fluids Used in  
Electrical Apparatus

FM GLOBAL (FM)

FM APP GUIDE (updated on-line) Approval Guide  
<http://www.approvalguide.com/>

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 100 (2000; Archived) The Authoritative  
Dictionary of IEEE Standards Terms

IEEE C2 (2023) National Electrical Safety Code

IEEE C37.20.3 (2013) Standard for Metal-Enclosed  
Interrupter Switchgear

IEEE C37.121 (2012) American National Standard for  
Switchgear-Unit Substations - Requirements

IEEE C57.12.00 (2021) General Requirements for  
Liquid-Immersed Distribution, Power, and  
Regulating Transformers

IEEE C57.12.29 (2014) Standard for Pad-Mounted Equipment  
- Enclosure Integrity for Coastal  
Environments

IEEE C57.12.80 (2010) Standard Terminology for Power and  
Distribution Transformers

IEEE C57.12.90 (2021) Test Code for Liquid-Immersed  
Distribution, Power, and Regulating  
Transformers

IEEE C57.98 (2011) Guide for Transformer Impulse Tests

IEEE C62.11 (2020) Standard for Metal-Oxide Surge  
Arresters for Alternating Current Power  
Circuits (>1kV)

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS (2021) Standard for Acceptance Testing  
Specifications for Electrical Power  
Equipment and Systems

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO ISO/IEC 17025 (2017) General Requirements for the  
Competence of Testing and Calibration  
Laboratories

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 6 (1993; R 2016) Industrial Control and  
Systems: Enclosures

NEMA ST 20 (2014) Dry-Type Transformers for General  
Applications

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT (OECD)

OECD Test 203 (1992) Fish Acute Toxicity Test

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA 712-C-98-075 (1998) Fate, Transport and Transformation  
Test Guidelines - OPPTS 835.3100- "Aerobic  
Aquatic Biodegradation"

EPA 821-R-02-012 (2002) Methods for Measuring the Acute  
Toxicity of Effluents and Receiving Waters  
to Freshwater and Marine Organisms

UNDERWRITERS LABORATORIES (UL)

UL 467 (2022) UL Standard for Safety Grounding  
and Bonding Equipment

1.2 RELATED REQUIREMENTS

Section 26 08 00 APPARATUS INSPECTION AND TESTING, FRONT END AND  
INTEGRATION applies to this section, with the additions and modifications  
specified herein.

1.3 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms  
used in these specifications, and on the drawings, must be as defined in  
IEEE 100.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S"  
classification. Submittals not having a "G" or "S" classification are for  
information only. When used, a code following the "G" classification  
identifies the office that will review the submittal for the Government.  
Submit the following in accordance with Section 01 33 00 SUBMITTAL  
PROCEDURES:

In addition, submit in accordance with paragraph COORDINATED SUBMITTAL  
REVIEWS herein.

1.4.1 Coordinated Submittal Reviews

- a. Submit transformer submittals to Code DC44, Atlantic Division, Naval  
Facilities Engineering Systems Command for approval. In addition,  
submit one set of the remaining substation components for surveillance.
- b. Submit remaining substation component submittals to Engineer of Record  
for approval. In addition, submit one set of transformer submittals  
for surveillance and to insure alignment of equipment and coordination  
for interconnections.

SD-02 Shop Drawings

Unit Substation Drawings; G

Transformer Drawings (to Code DC44); G



Include wiring diagrams and installation details of equipment indicating proposed location, layout and arrangement, control panels, accessories, piping, ductwork, and other items that must be shown to ensure a coordinated installation. Wiring diagrams must identify circuit terminals and indicate the internal wiring for each item of equipment and the interconnection between each item of equipment. Drawings must indicate adequate clearance for operation, maintenance, and replacement of operating equipment devices. Submittals must include the nameplate data, size, and capacity. Submittals must also include applicable federal, military, industry, and technical society publication references.

SD-03 Product Data

Fuse Curves; G

Secondary Unit Substation Excluding Transformer Data; G

Unit Substation Transformer (Liquid-filled) (to Code DC44); G

SD-06 Test Reports

Acceptance Checks and Tests; G

SD-07 Certificates

Paint Coating System; G

Transformer Efficiencies; G

SD-09 Manufacturer's Field Reports

Load Interrupter Switch Production Tests; G

Unit Substation Transformer Design Tests (Liquid-filled) (to Code DC44); G

Unit Substation Transformer Routine and Other Tests (Liquid-filled) (to Code DC44); G

SD-10 Operation and Maintenance Data

Unit Substations, Data Package 5; G

SD-11 Closeout Submittals

Assembled Operation and Maintenance Manuals; G

Equipment Test Schedule (to Code DC44 for Liquid-filled Units); G

1.5 QUALITY ASSURANCE

1.5.1 Drawing Requirements

1.5.1.1 Unit Substation Drawings

Drawings must include, but are not limited to the following:

- a. An outline drawing, with dimensional plan view, elevation, foundation plan and side views showing incoming, transformer, and outgoing sections. as part of the total unit substation.
- b. One-line diagram showing all components and their ratings.
- c. Elementary diagrams and wiring diagrams with terminals identified, and indicating prewired interconnections between items of equipment and the interconnection between the items.
- d. Three-line diagram showing bus configuration, bus rating and overcurrent protective devices.
- f. Time-current characteristic fuse curves (on full size logarithmic paper) for the load interrupter switch fuse.

#### 1.5.1.2 Transformer Drawings

Drawings must include, but are not limited to the following:

- a. An outline drawing, with front, top, and side views.
- b. ANSI nameplate data.

#### 1.5.2 Paint Coating System

Submit IEEE C57.12.29 coating system performance requirement tests. When interrupter switchgear and transformer are provided by two different manufacturers, each one must provide certification.

#### 1.5.3 Transformer Efficiencies

Submit certification from the manufacturer indicating conformance with the paragraph SPECIFIED TRANSFORMER EFFICIENCIES".

#### 1.5.4 Substation Product Data

Submittal must include manufacturer's information for each component, device, and accessory provided with the equipment.

#### 1.5.5 Test Reports

Submit report of acceptance test results as specified by paragraph FIELD QUALITY CONTROL.

#### 1.5.6 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship must be in accordance with the mandatory and advisory provisions unless more stringent requirements are specified or indicated.

#### 1.5.7 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year period must include applications of equipment and materials under similar circumstances and of similar size. The product must have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

##### 1.5.7.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

##### 1.5.7.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site must not be used, unless specified otherwise.

#### 1.6 MAINTENANCE

##### 1.6.1 Assembled Operation and Maintenance Manuals

Manuals must be assembled in durable, hard covered, water resistant binders. The manual must be assembled and indexed in the order noted in a table of contents. The contents of the assembled operation and maintenance manuals must be as follows:

- a. Manufacturer's O&M information required by the paragraph, SD-10 OPERATION AND MAINTENANCE DATA.
- b. Catalog data required by the paragraph, SD-03 PRODUCT DATA.
- c. Drawing required by the paragraph, SD-02 SHOP DRAWINGS.
- d. Price for spare parts and supply list
- e. Routine and field acceptance test reports

##### 1.6.2 Operation and Maintenance Data

Submit operation and maintenance data in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA and as specified herein.

#### 1.7 WARRANTY

The equipment items must be supported by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

## PART 2 PRODUCTS

### 2.1 PRODUCT COORDINATION

Products and materials not considered to be secondary unit substations and related accessories are specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

### 2.2 SECONDARY UNIT SUBSTATION

Secondary Unit substations must comply with IEEE C37.121 regardless of the kVA rating specified. Substation must consist of primary incoming sections, one transformer section, and one outgoing section. Substation must be designed for outdoor service with ventilation openings and gasketing provided to ensure a weatherproof assembly under rain, snow, sleet, and hurricane conditions. Substations must be subassembled and coordinated by one manufacturer and must be shipped in complete sections ready for connection at the site. Where practicable, substation must be shipped as one unit. External doors must have provisions for padlocking. Bus bars and conductors must be copper.

#### 2.2.1 Incoming Section

The incoming section must consist of a dual metal-enclosed interrupter switchgear section for connecting the incoming circuits through a fused load interrupter switch to the transformer. Operation of the switchgear must not be affected by opening the low voltage compartment doors. If required for proper connection and alignment, include a transition section with the incoming section.

##### 2.2.1.1 Incoming Section Enclosure

The incoming section enclosure must be NEMA ICS 6 Type 3R, fabricated entirely of type 304L stainless steel. Bases, frames and channels of enclosure must be corrosion resistant and must be fabricated of 304L stainless steel. Base must include any part of enclosure that is within 3 inches of concrete pad. Galvanized steel must be ASTM A123/A123M, ASTM A653/A653M G90 coating, and ASTM A153/A153M, as applicable. Paint enclosure, including bases, ASTM D1535 light gray No. 61 or No. 49. Paint coating system must comply with IEEE C57.12.29.

##### 2.2.1.2 Cable Terminations

Provide medium voltage cable terminations as specified .

##### 2.2.1.3 Surge Arresters

IEEE C62.11, rated 12 kV. Arresters must be intermediate class. Arresters for use at elevations in excess of 6000 feet above mean sea level must be specifically rated for that purpose. Arresters must be equipped with mounting brackets suitable for the indicated installations.

##### 2.2.1.4 Load Interrupter Switch

IEEE C37.20.3. Provide a dual three-pole, single-throw, deadfront, metal-enclosed, load interrupter switch with manual stored energy operator. Switch must be fused, and designed for easy inspection and fuse replacement. The switch must be operated by a manually charged spring

stored energy mechanism which must simultaneously disconnect or connect ungrounded conductors. The moveable blade of the switch must be de-energized when in the open position. The mechanism must enable the switch to close against a fault equal to the momentary rating of the switch without affecting its continuous current carrying or load interrupting ability. A ground bus must extend the width of the switch enclosure and must be bolted directly thereto. Connect frame of unit to ground bus. The door must have an inspection window to allow full view of the position of the three switch blades through the closed door. Switch ratings must be:

- a. 13.8 kV, 95 kV BIL for service on a 15 kV system with a fault close rating of not less than 101 kA amperes asymmetrical.
- b. The switch must be capable of carrying continuously or interrupting 1200 A amperes with a momentary rating of 63 kA amperes at 13.8 kV.
- c. Switch must have provision for padlocking in the open and closed positions.
- d. Fuses must be current limiting type rated as shown on drawings amperes continuous, and 63 kA amperes interrupting capacity.

#### 2.2.1.5 Primary Protective Device Connection

Connections between the primary protective device and transformer must be cable mounted on porcelain insulators, and sized and braced to withstand the specified short-circuit and short-time currents.

#### 2.2.2 Transformer (Liquid-Filled) Sections

IEEE C57.12.29. Less-flammable, bio-degradable liquid-filled. Transformer base, including the tank, radiators, flanges, base, lifting provisions, and hardware, must be fabricated of type 316 stainless steel. Transformer base must include any part of the transformer that is within 3 inches of concrete pad. Paint coating system must comply with IEEE C57.12.29.

##### 2.2.2.1 Transformer Ratings

- a. Cooling Class: ONAN-Liquid-filled, self-cooled.
- b. Frequency: 60 Hz.
- c. Phases: Three phase.
- d. Rated Kilovolt Amperes: 1500 kVA
- e. Voltage Rating: 13,200 V - 480 V. For GrdY - GrdY transformers, provide transformer with five-legged core design for third harmonic suppression.
- f. Impedance: Minimum tested impedance must not be less than 5.75 percent at 85 degrees C.
- g. Insulation Level: 95 kV BIL
- h. Temperature Rise: 55/65 degree C average winding temperature rise

above a 30 degree ambient.

- i. Audible Sound Levels: Audible sound levels must comply with the following:

| <u>kVA Range</u> | <u>DECIBELS (MAX)</u> |
|------------------|-----------------------|
| 225-300          | 55                    |
| 301-500          | 56                    |
| 501-700          | 57                    |
| 701-1000         | 58                    |
| 1001-1500        | 60                    |
| 1501-2000        | 61                    |
| 2001-2500        | 62                    |
| 2501-3000        | 63                    |
| 3001-4000        | 64                    |

#### 2.2.2.2 Transformer Accessories

The transformer must have the following accessories:

- a. Four 2.5 percent full capacity taps, two above and two below rated primary voltage.
- b. Tap changer, with external, pad-lockable, manual type operating handle, for changing tap setting when transformer is de-energized.
- c. Ground pads.
- d. Liquid-level indicator.
- e. Pressure-vacuum gage.
- f. Liquid temperature indicator.
- g. Drain and filter valves.
- h. Pressure relief device, top mounted.
- i. Diagrammatic stainless steel or laser-etched anodized aluminum nameplate in accordance with IEEE C57.12.00 and as modified or supplemented by this section.
- j. Transformer base with provisions for jacking and for rolling in either direction.
- k. Lifting provisions.
- l. Bolted transformer top or welded top with bolted handhole access.

### 2.2.2.3 Specified Transformer Efficiencies

Provide transformer efficiency calculations utilizing the actual no-load and load loss values obtained during the routine tests performed on the actual transformer(s) prepared for this project. Reference no-load losses (NLL) at 20 degrees C. Reference load losses (LL) at 55 degrees C and at 50 percent of the nameplate load. The transformer is not acceptable if the calculated transformer efficiency is less than the efficiency indicated in the "KVA / Efficiency" table below. The table is based on requirements contained within 10 CFR 431, Subpart K. Submit certification, including supporting calculations, from the manufacturer indicating conformance.

| KVA        | EFFICIENCY (percent) |
|------------|----------------------|
| 15         | 98.65                |
| 30         | 98.83                |
| 45         | 98.92                |
| 75         | 99.03                |
| 112.5      | 99.11                |
| 150        | 99.16                |
| 225        | 99.23                |
| 300        | 99.27                |
| 500        | 99.35                |
| 750        | 99.40                |
| 1000       | 99.43                |
| 1500       | 99.48                |
| 2000       | 99.51                |
| 2500       | 99.53                |
| above 2500 | 99.54                |

### 2.2.2.4 Insulating Liquid

- a. Less-flammable transformer liquids: Must meet the requirements of ASTM D6871, and be approved by the FM APP GUIDE for Less or Non-Flammable Liquid Insulated Transformers. Provide identification of transformer as "non-PCB" and "manufacturer's name and type of fluid" on the nameplate.

Provide a fluid that is a biodegradable, electrical insulating, and

cooling liquid classified by UL and approved by FM as "less flammable" with the following properties:

- (1) Aquatic biodegradation: EPA 712-C-98-075, 99 percent.
- (2) Trout toxicity: The fluid must have passed OECD Test 203 following the methods of EPA 821-R-02-012 and be determined to be non-toxic.

#### 2.2.2.4.1 Liquid-Filled Transformer Nameplates

Provide nameplate information in accordance with IEEE C57.12.00 and as modified or supplemented by this section.

#### 2.2.3 Outgoing Section

The outgoing section must consist of a full height air terminal compartment for physical protection of and connection point for the secondary conductors between the transformer and the main switchgear located in building 72A pump room

The outgoing section must consist of a full height air terminal compartment. This compartment must be the connection point for the secondary conductors between the transformer and the switchboard located in the building. Provide one three point latching hinged door, either full height or on the upper half of the outgoing section to provide access to metering. The upper section must contain the current transformers and a watthour meter mounted to a dead front interior barrier as defined below. If using upper half section door only, the lower section must be bolt on type and contain bus bars and lugs to terminate the service entrance conductors. Provide insulated barriers between the upper and lower sections to permit the bus bars to pass between the sections. Provide locking access handle to eliminate unauthorized access.

##### 2.2.3.1 Outgoing Section Enclosure

Provide outgoing section enclosure in accordance with the requirements in paragraph INCOMING SECTION ENCLOSURE.

#### 2.2.4 Switchgear Configuration

##### 2.2.4.1 Duplex Load Interruptor Switches

- a. For 5 or 15 kV ratings, furnish a duplex load interruptor switch for the selection of two incoming sources in series with the load break MVS switch where shown on contract drawing. The duplex load interruptor switches shall have the same continuous current rating. The load interruptor switches shall be arranged and interlocked to permit make-before-break closed-transition operation between two sources. Interlocking logic shall be provided that:
  - (1) Closed transition transfer occurs only when both sources are within voltage, phase, and frequency limits
  - (2) The duration of parallel connection between sources during automatic operation shall be limited to a maximum of 100 milliseconds, unless otherwise approved



- (3) Both switches shall be rated for the sum of the available short circuit currents from both sources during parallel period.
  - (4) Automatic and manual transfer shall be possible without loss of load, subject to synchronism and protection permissive.
  - (5) Manual operation opening of second source to be performed in sequential manner, assumed 30 seconds or less.
- b. Provide mechanical and/or electrical interlocks to prevent closed transition operation unless all permissive conditions are met. The selector switch shall be positively locked in one of the two source positions when closed transition is not being performed.

#### 2.2.4.2 Duplex Automatic Transfer Control

- a. Duplex Automatic transfer control shall provide for a switchgear assembly with a common load bus and two normally energized medium-voltage power sources, designated "normal" and "standby," connected to the load bus. The fused switches connecting the two power sources to the load bus shall be controlled by a microprocessor-based automatic transfer control. Power for the transfer control shall be from the voltage sensing transformers.
- b. In the automatic mode, the load bus is connected to the normal power source. When the normal source fails, the control shall automatically open the normal power source and close the standby source circuit breaker.
- c. Sequence of operation:
  - (1) The default operation shall be with the normal source main breaker closed and the standby main breaker open. Upon detection of an undervoltage to the line side of the normal main breaker and after a field-adjustable time delay, that main breaker shall open and after an additional field-adjustable time delay, the standby breaker shall close restoring power to the facility.
  - (2) Upon restoration of voltage to the line side of the normal main breaker and after a field-adjustable time delay, the standby main breaker shall open and after a field-adjustable time delay, the normal main breaker shall close.
- d. Field-Adjustable Transfer Parameters:
  - (1) Delay the transfer from the normal power source to the standby power source and from the standby power source to the normal source. The time delay is to allow the load voltage to decay before reconnecting to another power source. Delay range: 0 seconds to 30 minutes.
  - (2) Delay the initiation of the transfer sequence. The time delay is recommended to override a momentary power outage or voltage fluctuation. Delay range: 0 seconds to 120 seconds.
  - (3) Delay the transfer from the standby power source to the normal power source: (0 seconds to 30 minutes).

- (4) A relay with contact that changes state when the power is available on the normal source and a relay with contact that changes state when the power is available on the standby source.
- e. Controls and Indicators: In addition to the delay setting controls, provide the following:
  - (1) Interlocks or relay control to prevent transfer when either of the two controlled circuit breakers tripped due to overcurrent or ground fault.
  - (2) Three-position selector switch to select the normal source: Source 1, Source 2, or none.
  - (3) Transfer-control automatic and manual selector.
    - (a) Interlock shall prevent paralleling of the two power sources in manual mode.
    - (b) Open-close control switch for manual electrical operation of each controlled circuit breaker.
    - (c) Selector to place control into programming mode.
    - (d) Circuit breaker control switch for each of the normal and standby source breakers, providing open and close operation.
    - (e) Push button to initiate manual retransfer to the normal source when the transfer controller is in the automatic mode.
  - (4) Meters and display to show the following:
    - (a) Voltage and frequency of both sources.
    - (b) A multi-line display showing the following:
      - 1. Set points of timers, and voltage pickup and dropout set points.
      - 2. Date, time, and reason for at least the last 10 transfers. The display may show the information for one transfer at a time using a scrolling control, with the others held in memory.
      - 3. When the control system is in the transferring process, the display shall show delay countdown in seconds.
      - 4. LED lights to show the following: Normal source available, Standby source available, Normal source connected, Standby source connected, Load bus energized.
      - 5. Voltage Transformers: Primary and secondary protection and disconnecting means for sensing functions and control power.
      - 6. Voltage Sensing Relays: Microprocessor-based IEEE device number 27/47 voltage detection relays for three-phase undervoltage protection and negative sequence voltage protection.

- f. Motor operators, where indicated on the drawings, shall have the following features:
- (1) All motor-operated switches shall consist of a standard manually-operated switch in combination with an electric motor driven linear actuator, which charges the spring. Connection between the linear actuator and switch mechanism shall be by reliable rigid metal-to-metal linkages; not chains or cables. The linear actuator and all associated low voltage wiring shall be located in a low voltage compartment or barriered to separate it from the high voltage.
  - (2) Operating voltage shall be 120 volts, 60 Hz available from an external source as shown on the drawings. The switch shall be capable of manual operation should a loss of control power be encountered.
  - (3) The linear actuator shall be a highly repetitively manufactured item, completely sealed and weather protected, and designed for rugged industrial application. No lubrication or adjustments shall be necessary for its normal operating life. The motor shall be equipped with an automatically reset thermal overload protector.
  - (4) Motor operator shall be easily removable for maintenance purpose.

#### 2.2.5 Control Power Transformers

Transformer must conform to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

#### 2.2.6 Heaters

Provide 120-volt heaters in incoming section and outgoing section. Heaters must be of sufficient capacity to control moisture condensation in the compartments, must be 250 watts minimum, and must be controlled by a thermostat and humidistat located in each section. Thermostat must be industrial type, high limit, to maintain compartments within the range of 60 to 90 degrees F. Humidistat must have a range of 30 to 60 percent relative humidity. If heater voltage is different than substation equipment voltage, provide transformer rated to carry 125 percent of heater full load rating. Transformer must have 220 degrees C insulation system with a temperature rise not exceeding 115 degrees C and must conform to NEMA ST 20. Energize electric heaters while the equipment is in storage or in place prior to being placed in service. Provide method for easy connection of heater to external power source.

#### 2.2.7 Terminal Boards

Provide with engraved plastic terminal strips and screw type terminals for external wiring between components and for internal wiring between removable assemblies. Terminal boards associated with current transformers must be short-circuiting type. Terminate conductors for current transformers with ring-tongue lugs. Terminal board identification must be identical in similar units. External wiring must be color coded consistently for similar terminal boards.

#### 2.2.8 Wire Marking

Mark control and metering conductors at each end. Provide factory-installed, white, plastic tubing, heat stamped with black block type letters on factory-installed wiring. On field-installed wiring, provide white, preprinted, polyvinyl chloride (PVC) sleeves, heat stamped with black block type letters. Each sleeve must be elliptically shaped to securely grip the wire, and must be keyed in such a manner to ensure alignment with adjacent sleeves. Provide specific wire markings using the appropriate combination of individual sleeves. Each wire marker must indicate the device or equipment, including specific terminal number to which the remote end of the wire is attached.

#### 2.2.9 Grounding and Bonding

Provide a continuous ground bus bar. It shall run the length of the switchgear, the ground bus bar shall be accessible from the back of the switchgear. The ground bus bar must be hard-drawn copper of 98 percent minimum conductivity, minimum size 1/4 by 2 inches.

#### 2.2.10 Padlocks

Padlocks must be provided for secondary unit substation equipment and for each fence gate. Padlocks must be keyed alike.

#### 2.2.11 Cast-in-Place Concrete

Concrete associated with electrical work for other than encasement of underground ducts must be 4000 psi minimum 28-day compressive strength unless specified otherwise. All concrete must conform to the requirements of Section 03 30 00 CAST-IN-PLACE CONCRETE.

Must be composed of fine aggregate, coarse aggregate, portland cement, and water so proportioned and mixed as to produce a plastic, workable mixture. Fine aggregate must be of hard, dense, durable, clean, and uncoated sand. The coarse aggregate must be reasonably well graded from 3/16 inch to one inch. The fine and coarse aggregates must be free from injurious amounts of dirt, vegetable matter, soft fragments or other deleterious substances. Water must be fresh, clean, and free from salts, alkali, organic matter, and other impurities. Concrete associated with electrical work for other than encasement of underground ducts must be 4000 psi minimum 28-day compressive strength unless specified otherwise. Slump must not exceed 4 inches. Retempering of concrete will not be permitted. Exposed, unformed concrete surfaces must be given a smooth, wood float finish. Concrete must be cured for a period of not less than 7 days, and concrete made with high early strength portland cement must be repaired by patching honeycombed or otherwise defective areas with cement mortar as directed by the Contracting Officer. Air entrain concrete exposed to weather using an air-entraining admixture conforming to ASTM C260/C260M. Air content must be between 4 and 6 percent.

#### 2.3 MANUFACTURER'S NAMEPLATES

Each item of equipment must have a nameplate bearing, as a minimum, the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable. Include additional information as applicable to fully identify the equipment. Nameplates must be made of noncorrosive metal. Equipment containing liquid dielectric must include the type of

dielectric on the nameplate. Sectionalizer switch nameplates must have a schematic with all switch positions shown and labeled. As a minimum, provide nameplates for transformers, circuit breakers, meters, switches, and switchgear.

## 2.4 FIELD FABRICATED NAMEPLATES

ASTM D709. Provide laminated plastic nameplates for each secondary unit substation, equipment enclosure, relay, switch, and device; as specified in this section or as indicated on the drawings. Each nameplate inscription must identify the function and, when applicable, the position. Nameplates must be melamine plastic, 0.125 inch thick, white with black center core. Provide red laminated plastic label with white center core where indicated. Surface must be matte finish. Corners must be square. Accurately align lettering and engrave into the core. Minimum size of nameplates must be one by 2.5 inches. Lettering must be a minimum of 0.25 inch high normal block style.

## 2.5 WARNING SIGNS

Provide warning signs for the enclosures of secondary unit substations having a nominal rating exceeding 600 volts.

- a. When the enclosure integrity of such equipment is specified to be in accordance with IEEE C57.12.29, such as for secondary unit substations, provide self-adhesive warning signs on the outside of the high voltage compartment door(s). Sign must be a decal and must have nominal dimensions of 7 by 10 inches with the legend "DANGER HIGH VOLTAGE" printed in two lines of nominal 2 inch high letters. The word "DANGER" must be in white letters on a red background and the words "HIGH VOLTAGE" must be in black letters on a white background.
- b. When such equipment is guarded by a fence, mount signs on the fence. Provide metal signs having nominal dimensions of 14 by 10 inches with the legend "DANGER HIGH VOLTAGE KEEP OUT" printed in three lines of nominal 3 inch high white letters on a red and black field.

## 2.6 SOURCE QUALITY CONTROL

### 2.6.1 Equipment Test Schedule

The Government reserves the right to witness tests. Provide equipment test schedules for tests to be performed at the manufacturer's test facility. Submit required test schedule and location, and notify the Contracting Officer 30 calendar days before scheduled test date. Notify Contracting Officer 15 calendar days in advance of changes to scheduled date.

#### Test Instrument Calibration

- (1) The manufacturer must have a calibration program which assures that all applicable test instruments are maintained within rated accuracy.
- (2) The accuracy must be directly traceable to the National Institute of Standards and Technology. Test equipment must qualify for the UL standard of Scope of Accreditation ISO ISO/IEC 17025 and ANSI Z540.1.

- (3) Instrument calibration frequency schedule must not exceed 12 months for both test floor instruments and leased specialty equipment.
- (4) Dated calibration labels must be visible on all test equipment.
- (5) Calibrating standard must be of higher accuracy than that of the instrument tested.
- (6) Keep up-to-date records that indicate dates and test results of instruments calibrated or tested. For instruments calibrated by the manufacturer on a routine basis, in lieu of third party calibration, include the following:
  - (a) Maintain up-to-date instrument calibration instructions and procedures for each test instrument.
  - (b) Identify the third party/laboratory calibrated instrument to verify that calibrating standard is met.

#### 2.6.2 Load Interrupter Switch Production Tests

IEEE C37.20.3. Furnish reports of production tests performed on the actual equipment for this project. Required tests must be as follows:

##### Production Tests

- (1) Dielectric
- (2) Mechanical operation
- (3) Grounding of instrument transformer case
- (4) Electrical operation and control wiring

#### 2.6.3 Transformer Design Tests (Liquid-Filled)

In accordance with IEEE C57.12.00 and IEEE C57.12.90. Additionally, IEEE C57.12.80 section 5.1.2 states that "design tests are made only on representative apparatus of basically the same design." Submit design test reports (complete with test data, explanations, formulas, and results), in the same submittal package as the product data and shop drawings for each of the specified transformers. Design tests must have been performed prior to the award of this contract.

- a. Tests must be certified and signed by a registered professional engineer.
- b. Temperature rise: "Basically the same design" for the temperature rise test means a unit-substation transformer with the same coil construction (such as wire wound primary and sheet wound secondary), the same kVA, the same cooling type (ONAN), the same temperature rise rating, and the same insulating liquid as the transformer specified.
- c. Lightning impulse: "Basically the same design" for the lightning impulse dielectric test means a unit-substation transformer with the same BIL, the same coil construction (such as wire wound primary and sheet wound secondary), and a tap changer, if specified. Design lightning impulse tests must include both the primary and secondary

windings of that transformer.

- (1) IEEE C57.12.90 paragraph 10.3 entitled "Lightning Impulse Test Procedures," and IEEE C57.98.
  - (2) State test voltage levels.
  - (3) Provide photographs of oscilloscope display waveforms or plots of digitized waveforms with test report.
- d. Lifting and moving devices: "Basically the same design" for the lifting and moving devices test means a transformer in the same weight range as the transformer specified.
- e. Pressure: "Basically the same design" for the pressure test means a unit-substation transformer with a tank volume within 30 percent of the tank volume of the transformer specified.

#### 2.6.4 Transformer Routine and Other Tests (Liquid-Filled)

In accordance with IEEE C57.12.00 and IEEE C57.12.90. Routine and other tests must be performed by the manufacturer on each of the actual transformers prepared for this project to ensure that the design performance is maintained in production. Submit test reports, by serial number and receive approval before delivery of equipment to the project site. Required tests and testing sequence must be as follows:

- a. Cold resistance measurements (provide reference temperature)
- b. Phase relation
- c. Ratio
- d. Insulation power-factor by manufacturer's recommended test method
- e. No-load losses (NLL) and excitation current
- f. Load losses (LL) and impedance voltage
- g. Dielectric
  - (1) Impulse: Per IEEE C57.12.90 paragraph 10.3 entitled "Lightning Impulse Test Procedures," and IEEE C57.98. Test the primary winding only.
    - (a) State test voltage levels
    - (b) Provide photographs of oscilloscope display waveforms or plots of digitized waveforms with test reports. As an alternative, photographs of oscilloscope display waveforms or plots of digitized waveforms may be hand-delivered at the factory witness test.
  - (2) Applied voltage
  - (3) Induced voltage
- h. Leak

i. Sample insulating liquid. Sample must be tested for:

- (1) Dielectric breakdown voltage
- (2) Acid neutralization number
- (3) Specific gravity
- (4) Interfacial tension
- (5) Color
- (6) Visual condition
- (7) Water in insulating liquid
- (8) Measure dissipation factor or power factor

j. Perform dissolved gas analysis (DGA)

### PART 3 EXECUTION

#### 3.1 INSTALLATION

Electrical installations must conform to IEEE C2, and to the requirements specified herein.

#### 3.2 GROUNDING

IEEE C2, except that grounds and grounding systems must have a resistance to solid earth ground not exceeding 5 ohms.

##### 3.2.1 Grounding Electrodes

Provide driven ground rods as shown on drawings. Connect ground conductors to the upper end of the ground rods by exothermic weld or compression connector. Provide compression connectors at equipment end of ground conductors.

##### 3.2.2 Substation Grounding

Provide bare copper cable not smaller than No. 4/0 AWG not less than 24 inches below grade interconnecting the indicated ground rods. Surge arrester and neutrals must be bonded directly to the transformer enclosure and then to the grounding electrode system with bare copper conductors, sized as shown. Lead lengths must be kept as short as practicable with no kinks or sharp bends. Substation transformer neutral connections must not be smaller than No. 1/0 AWG. When work in addition to that indicated or specified is directed to obtain the specified ground resistance, the provision of the contract covering "Changes" must apply. Fence and equipment connections must not be smaller than No. 4 AWG. Ground fence at each gate post and cornerpost and at intervals not exceeding 10 feet. Bond each gate section to the fence post through a 1/8 by one inch flexible braided copper strap and clamps.

##### 3.2.3 Connections

Make joints in grounding conductors and loops by exothermic weld or compression connector.



### 3.2.4 Grounding and Bonding Equipment

UL 467, except as indicated or specified otherwise.

## 3.3 INSTALLATION OF EQUIPMENT AND ASSEMBLIES

Install and connect unit substations furnished under this section as indicated on project drawings, the approved shop drawings, and as specified herein.

### 3.3.1 Interrupter Switchgear

IEEE C37.20.3.

### 3.3.2 Field Applied Painting

Where field applied painting of enclosures is required to correct damage to the manufacturer's factory applied coatings, provide manufacturer's recommended coatings and apply in accordance with manufacturer's instructions.

### 3.3.3 Field Fabricated Nameplate Mounting

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

### 3.3.4 Warning Sign Mounting

Provide the number of signs required to be readable from each accessible side, but space the signs a maximum of 30 feet apart.

### 3.3.5 Galvanizing Repair

Repair damage to galvanized coatings using ASTM A780/A780M, zinc rich paint, for galvanizing damaged by handling, transporting, cutting, welding, or bolting. Do not heat surfaces that repair paint has been applied to.

## 3.4 FOUNDATION FOR EQUIPMENT AND ASSEMBLIES

### 3.4.1 Exterior Location

Mount unit substation on concrete slab. Unless otherwise indicated, the slab must be at least 8 inches thick, reinforced with a 6 by 6 - W2.9 by W2.9 mesh placed uniformly 4 inches from the top of the slab. Slab must be placed on a 6 inch thick, well-compacted gravel base. Top of concrete slab must be approximately 4 inches above the finished grade. Edges above grade must have 1/2 inch chamfer. The slab must be of adequate size to project at least 8 inches beyond the equipment. Provide conduit turnups and cable entrance space required by the equipment to be mounted. Seal voids around conduit openings in slab with water- and oil-resistant caulking or sealant. Seals must be of sufficient strength and durability to protect all energized live parts of the equipment from rodents, insects, or other foreign matter. Cut off and bush conduits 3 inches above slab surface.

### 3.4.2 Cast-in-Place Concrete

Cast-in-place concrete work must conform to the requirements of Section 03 30 00 CAST-IN-PLACE CONCRETE ACI 318M.

### 3.5 PADLOCKS

Provide padlocks for secondary unit substation equipment and for each fence gate.

### 3.6 FIELD QUALITY CONTROL

#### 3.6.1 Performance of Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations and include the following visual and mechanical inspections and electrical tests, performed in accordance with NETA ATS.

##### 3.6.1.1 Transformers (Liquid-Filled)

###### a. Visual and mechanical inspection

- (1) Compare equipment nameplate data with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition. Check for damaged or cracked insulators and leaks.
- (3) Verify that cooling fans and pumps operate correctly and that fan and pump motors have correct overcurrent protection.
- (4) Verify operation of all alarm, control, and trip circuits from temperature and level indicators, pressure relief device, and fault pressure relay.
- (5) Inspect all bolted electrical connections for high resistance using low-resistance ohmmeter, verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey.
- (6) Verify correct liquid level in transformer tank.
- (7) Perform specific inspections and mechanical tests as recommended by manufacturer.
- (8) Verify correct equipment grounding.
- (9) Verify the presence of transformer surge arresters.
- (10) Verify that positive pressure is maintained on gas blanketed transformers.

###### b. Electrical Tests

- (1) Perform resistance measurements through all bolted connections with low-resistance ohmmeter, if applicable.
- (2) Perform dissolved gas analysis (DGA).

- (3) Verify that the tap-changer is set at specified ratio.
- (4) Verify proper secondary voltage phase-to-phase and phase-to-neutral after energization and prior to loading.

#### 3.6.1.2 Current Transformers

##### a. Visual and Mechanical Inspection

- (1) Compare equipment nameplate data with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Verify correct connection.
- (4) Verify that adequate clearances exist between primary and secondary circuit.
- (5) Inspect all bolted electrical connections for high resistance using low-resistance ohmmeter, verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey.
- (6) Verify that all required grounding and shorting connections provide good contact.

##### b. Electrical Tests

- (1) Perform resistance measurements through all bolted connections with low-resistance ohmmeter, if applicable.
- (2) Perform insulation-resistance tests.
- (3) Perform polarity tests.
- (4) Perform ratio-verification tests.

#### 3.6.1.3 Metering and Instrumentation

##### a. Visual and Mechanical Inspection

- (1) Compare equipment nameplate data with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Verify tightness of electrical connections.

##### b. Electrical Tests

- (1) Verify accuracy of meters at 25, 50, 75, and 100 percent of full scale.
- (2) Calibrate watthour meters according to manufacturer's published data.
- (3) Verify all instrument multipliers.

- (4) Verify that current transformer and voltage transformer secondary circuits are intact.

#### 3.6.1.4 Grounding System

##### a. Visual and Mechanical Inspection

- (1) Inspect ground system for compliance with contract plans and specifications.

##### b. Electrical Tests

- (1) Perform ground-impedance measurements utilizing the fall-of-potential method. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of a single ground rod perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground testing megger in accordance with manufacturer's instructions to test each ground or group of grounds. The instrument must be equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test.
- (2) Submit the measured ground resistance of each ground rod or grounding system, indicating the location of the rod or grounding system. Include the test method and test setup (i.e., pin location) used to determine ground resistance and soil conditions at the time the measurements were made.

#### 3.6.2 Pre-Energization Services

Calibration, testing, adjustment, and placing into service of the installation must be accomplished by a manufacturer's product field service engineer or independent testing company with a minimum of 2 years of current product experience. The following services must be performed subsequent to testing but prior to the initial energization. The equipment must be inspected to ensure that installation is in compliance with the recommendations of the manufacturer and as shown on the detail drawings. Terminations of conductors at major equipment must be inspected to ensure the adequacy of connections. Bare and insulated conductors between such terminations must be inspected to detect possible damage during installation. If factory tests were not performed on completed assemblies, tests must be performed after the installation of completed assemblies. Components must be inspected for damage caused during installation or shipment to ensure packaging materials have been removed. Components capable of being both manually and electrically operated must be operated manually prior to the first electrical operation. Components capable of being calibrated, adjusted, and tested must be calibrated, adjusted, and tested in accordance with the instructions of the equipment manufacturer.

#### 3.6.3 Follow-Up Verification

Upon completion of acceptance checks, settings, and tests, the Contractor must show by demonstration in service that circuits and devices are in

good operating condition and properly performing the intended function. Test must require each item to perform its function not less than three times. As an exception to requirements stated elsewhere in the contract, the Contracting Officer must be given 5 working days' advance notice of the dates and times for checks, settings, and tests.

-- End of Section --

SECTION 26 20 00

INTERIOR DISTRIBUTION SYSTEM

08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM B1 (2013) Standard Specification for  
Hard-Drawn Copper Wire

ASTM D709 (2017) Standard Specification for  
Laminated Thermosetting Materials

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 81 (2012) Guide for Measuring Earth  
Resistivity, Ground Impedance, and Earth  
Surface Potentials of a Ground System

IEEE Stds Dictionary (2009) IEEE Standards Dictionary: Glossary  
of Terms & Definitions

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS (2021) Standard for Acceptance Testing  
Specifications for Electrical Power  
Equipment and Systems

NATIONAL ELECTRICAL CONTRACTORS ASSOCIATION (NECA)

NECA NEIS 1 (2015) Standard for Good Workmanship in  
Electrical Construction

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C80.1 (2020) American National Standard for  
Electrical Rigid Steel Conduit (ERSC)

ANSI C80.3 (2020) American National Standard for  
Electrical Metallic Tubing (EMT)

NEMA 250 (2020) Enclosures for Electrical Equipment  
(1000 Volts Maximum)

NEMA FU 1 (2012) Low Voltage Cartridge Fuses

NEMA ICS 1 (2022) Standard for Industrial Control and  
Systems: General Requirements

NEMA ICS 2 (2000; R 2020) Industrial Control and

Systems Controllers, Contactors, and  
Overload Relays Rated 600 V

|            |                                                                                                                         |
|------------|-------------------------------------------------------------------------------------------------------------------------|
| NEMA ICS 4 | (2015) Application Guideline for Terminal Blocks                                                                        |
| NEMA ICS 6 | (1993; R 2016) Industrial Control and Systems: Enclosures                                                               |
| NEMA KS 1  | (2013) Enclosed and Miscellaneous Distribution Equipment Switches (600 V Maximum)                                       |
| NEMA RN 1  | (2005; R 2013) Polyvinyl-Chloride (PVC) Externally Coated Galvanized Rigid Steel Conduit and Intermediate Metal Conduit |
| NEMA ST 20 | (2014) Dry-Type Transformers for General Applications                                                                   |
| NEMA TC 2  | (2020) Standard for Electrical Polyvinyl Chloride (PVC) Conduit                                                         |
| NEMA TC 3  | (2021) Polyvinyl Chloride (PVC) Fittings for Use With Rigid PVC Conduit and Tubing                                      |
| NEMA TC 14 | (2002) Standard for Reinforced Thermosetting Resin Conduit (RTRC) and Fittings                                          |
| NEMA VE 2  | (2018; ERTA 1-2 2018) Cable Tray Installation Guidelines                                                                |
| NEMA WD 1  | (1999; R 2020) Standard for General Color Requirements for Wiring Devices                                               |
| NEMA WD 6  | (2021) Wiring Devices Dimensions Specifications                                                                         |

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|          |                                                        |
|----------|--------------------------------------------------------|
| NFPA 70  | (2023) National Electrical Code                        |
| NFPA 70E | (2024) Standard for Electrical Safety in the Workplace |

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

|         |                                                                                           |
|---------|-------------------------------------------------------------------------------------------|
| TIA-607 | (2019d) Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises |
|---------|-------------------------------------------------------------------------------------------|

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|            |                                                                           |
|------------|---------------------------------------------------------------------------|
| 10 CFR 431 | Energy Efficiency Program for Certain Commercial and Industrial Equipment |
|------------|---------------------------------------------------------------------------|

UNDERWRITERS LABORATORIES (UL)

|              |                                                                                                                           |
|--------------|---------------------------------------------------------------------------------------------------------------------------|
| UL 1         | (2005; Reprint Jan 2022) UL Standard for Safety Flexible Metal Conduit                                                    |
| UL 6         | (2022) UL Standard for Safety Electrical Rigid Metal Conduit-Steel                                                        |
| UL 20        | (2018; Reprint May 2023) UL Standard for Safety General-Use Snap Switches                                                 |
| UL 44        | (2018; Reprint May 2021) UL Standard for Safety Thermoset-Insulated Wires and Cables                                      |
| UL 50        | (2015) UL Standard for Safety Enclosures for Electrical Equipment, Non-Environmental Considerations                       |
| UL 67        | (2018; Reprint May 2023) UL Standard for Safety Panelboards                                                               |
| UL 83        | (2017; Reprint Mar 2020) UL Standard for Safety Thermoplastic-Insulated Wires and Cables                                  |
| UL 248-4     | (2010; Reprint Apr 2019) Low-Voltage Fuses - Part 4: Class CC Fuses                                                       |
| UL 248-8     | (2011; Reprint Aug 2020) Low-Voltage Fuses - Part 8: Class J Fuses                                                        |
| UL 248-10    | (2011; Reprint Aug 2020) Low-Voltage Fuses - Part 10: Class L Fuses                                                       |
| UL 248-12    | (2011; Reprint Aug 2020) Low Voltage Fuses - Part 12: Class R Fuses                                                       |
| UL 248-15    | (2018) Low-Voltage Fuses - Part 15: Class T Fuses                                                                         |
| UL 360       | (2013; Reprint Apr 2023) UL Standard for Safety Liquid-Tight Flexible Metal Conduit                                       |
| UL 467       | (2022) UL Standard for Safety Grounding and Bonding Equipment                                                             |
| UL 486A-486B | (2018; Reprint May 2021) UL Standard for Safety Wire Connectors                                                           |
| UL 486C      | (2018; Reprint May 2021) UL Standard for Safety Splicing Wire Connectors                                                  |
| UL 489       | (2016; Rev 2019) UL Standard for Safety Molded-Case Circuit Breakers, Molded-Case Switches and Circuit-Breaker Enclosures |
| UL 498       | (2017; Reprint May 2023) UL Standard for Safety Attachment Plugs and Receptacles                                          |



|         |                                                                                                               |
|---------|---------------------------------------------------------------------------------------------------------------|
| UL 506  | (2017; Reprint Jan 2022) UL Standard for Safety Specialty Transformers                                        |
| UL 508  | (2018; Reprint Jul 2021) UL Standard for Safety Industrial Control Equipment                                  |
| UL 510  | (2020; Dec 2022) UL Standard for Safety Polyvinyl Chloride, Polyethylene and Rubber Insulating Tape           |
| UL 514A | (2013; Reprint Jun 2022) UL Standard for Safety Metallic Outlet Boxes                                         |
| UL 514B | (2012; Reprint May 2020) Conduit, Tubing and Cable Fittings                                                   |
| UL 514C | (2014; Reprint Feb 2020) UL Standard for Safety Nonmetallic Outlet Boxes, Flush-Device Boxes, and Covers      |
| UL 651  | (2011; Reprint May 2022) UL Standard for Safety Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings |
| UL 719  | (2015; Reprint Jan 2023) UL Standard for Safety Nonmetallic-Sheathed Cables                                   |
| UL 797  | (2007; Reprint Apr 2023) UL Standard for Safety Electrical Metallic Tubing -- Steel                           |
| UL 817  | (2015; Reprint May 2023) UL Standard for Safety Cord Sets and Power-Supply Cords                              |
| UL 854  | (2020; Reprint Jan 2022) Standard for Service-Entrance Cables                                                 |
| UL 869A | (2006; Reprint Jun 2020) Reference Standard for Service Equipment                                             |
| UL 870  | (2016; Reprint Mar 2019) UL Standard for Safety Wireways, Auxiliary Gutters, and Associated Fittings          |
| UL 943  | (2016; Reprint Feb 2018) UL Standard for Safety Ground-Fault Circuit-Interrupters                             |
| UL 1063 | (2017; Reprint Jun 2022) UL Standard for Safety Machine-Tool Wires and Cables                                 |
| UL 1242 | (2006; Reprint Apr 2022) UL Standard for Safety Electrical Intermediate Metal Conduit -- Steel                |
| UL 1283 | (2017) UL Standard for Safety Electromagnetic Interference Filters                                            |
| UL 1449 | (2021; Reprint Dec 2022) UL Standard for Safety Surge Protective Devices                                      |

|            |                                                                                |
|------------|--------------------------------------------------------------------------------|
| UL 1561    | (2011; Reprint Jun 2015) Dry-Type General Purpose and Power Transformers       |
| UL 1569    | (2018) UL Standard for Safety Metal-Clad Cables                                |
| UL 1660    | (2019; Reprint Jan 2022) Liquid-Tight Flexible Nonmetallic Conduit             |
| UL 4248-1  | (2022) UL Standard for Safety Fuseholders - Part 1: General Requirements       |
| UL 4248-12 | (2018; Reprint Feb 2022) UL Standard for Safety Fuseholders - Part 12: Class R |

## 1.2 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined in IEEE Stds Dictionary.

## 1.3 RELATED REQUIREMENTS

Section 25 05 11 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS applies to this section, with the additions and modifications specified herein.

## 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

Panelboards; G

Transformers; G

Cable Trays; G

Wireways; G

Marking Strips Drawings; G

### SD-03 Product Data

Receptacles; G

Circuit Breakers; G

Switches; G

Transformers; G

Enclosed Circuit Breakers; G

Motor Controllers; G

Manual Motor Starters; G

Secondary Bonding Busbar; G

Surge Protective Devices; G

Cable Trays; G

SD-05 Design Data

Cable Tray Design; G

SD-06 Test Reports

600-volt Wiring Test; G

Grounding System Test; G

Transformer Tests; G

Ground-fault Receptacle Test; G

SD-07 Certificates

Fuses; G

SD-09 Manufacturer's Field Reports

Transformer Factory Tests

SD-10 Operation and Maintenance Data

Electrical Systems, Data Package 5; G

## 1.5 QUALITY ASSURANCE

### 1.5.1 Fuses

Submit coordination data as specified in paragraph, FUSES of this section.

### 1.5.2 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Provide equipment, materials, installation, and workmanship in accordance with NFPA 70 unless more stringent requirements are specified or indicated. NECA NEIS 1 shall be considered the minimum standard for workmanship.

### 1.5.3 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship and:

- a. Have been in satisfactory commercial or industrial use for 2 years prior to bid opening including applications of equipment and materials under similar circumstances and of similar size.
- b. Have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.
- c. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

The contractor is responsible for implementing protective device settings as recommended by the System Coordination Study 26 05 73 POWER SYSTEM STUDIES. Where protective devices require a programming tool or cord the contractor is responsible for providing such equipment.

#### 1.5.3.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

#### 1.5.3.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site are not acceptable.

## 1.6 MAINTENANCE

### 1.6.1 Electrical Systems

Submit operation and maintenance data in accordance with Section 01 78 23, OPERATION AND MAINTENANCE DATA and as specified herein. Submit operation

and maintenance manuals for electrical systems that provide basic data relating to the design, operation, and maintenance of the electrical distribution system for the building. Include the following:

- a. Single line diagram of the "as-built" building electrical system.
- b. Schematic diagram of electrical control system (other than HVAC, covered elsewhere).
- c. Manufacturers' operating and maintenance manuals on active electrical equipment.

#### 1.7 WARRANTY

Provide equipment items supported by service organizations that are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

#### 1.8 SEISMIC REQUIREMENTS

Provide seismic details conforming to SEISMIC PROTECTION FOR ELECTRICAL EQUIPMENT.

### PART 2 PRODUCTS

#### 2.1 MATERIALS AND EQUIPMENT

As a minimum, meet requirements of UL, where UL standards are established for those items, and requirements of NFPA 70 for all materials, equipment, and devices.

#### 2.2 CONDUIT AND FITTINGS

Conform to the following:

##### 2.2.1 Rigid Metallic Conduit

###### 2.2.1.1 Rigid, Threaded Zinc-Coated Steel Conduit

ANSI C80.1, UL 6.

##### 2.2.2 Rigid Nonmetallic Conduit

PVC Type EPC-40 and EPC-80 in accordance with NEMA TC 2, UL 651, or fiberglass conduit, in accordance with NEMA TC 14.

##### 2.2.3 Intermediate Metal Conduit (IMC)

UL 1242, zinc-coated steel only.

##### 2.2.4 Electrical, Zinc-Coated Steel Metallic Tubing (EMT)

UL 797, ANSI C80.3.

##### 2.2.5 Plastic-Coated Rigid Steel and IMC Conduit

NEMA RN 1, Type 40( 40 mils thick).

#### 2.2.6 Flexible Metal Conduit

UL 1, limited to 6 feet.

##### 2.2.6.1 Liquid-Tight Flexible Metal Conduit, Steel

UL 360, limited to 6 feet.

#### 2.2.7 Fittings for Metal Conduit, EMT, and Flexible Metal Conduit

UL 514B. Ferrous fittings: cadmium- or zinc-coated in accordance with UL 514B.

##### 2.2.7.1 Fittings for Rigid Metal Conduit and IMC

Threaded-type. Split couplings unacceptable.

##### 2.2.7.2 Fittings for EMT

Steel compression type.

#### 2.2.8 Fittings for Rigid Nonmetallic Conduit

NEMA TC 3 for PVC and NEMA TC 14 for fiberglass, and UL 514B.

#### 2.2.9 Liquid-Tight Flexible Nonmetallic Conduit

UL 1660.

### 2.3 CABLE TRAYS

NEMA VE 2. Provide the following:

Submit cable tray design, including dimensional layout, load and seismic calculations, and fill calculations. Dimensional layout includes cable spacings, cable tray splices, and supports. Fill calculations include an index of cables for each section and identification of the lb/ft, cross sectional area, and insulation voltage class for each cable.

- a. Cable trays: form a wireway system, with a nominal depth as indicated. Cable tray is to have a span/load class per NEMA VE 2.
- b. Cable trays: constructed of fiberglass.
- c. Cable trays: include splice and end plates, dropouts, and miscellaneous hardware.
- d. Edges, fittings, and hardware: finished free from burrs and sharp edges.
- e. Fittings: ensure not less than load-carrying ability of straight tray sections and have manufacturer's minimum standard radius.
- f. Radius of bends: as indicated.

##### 2.3.1 Trough-Type Cable Trays

Provide size as indicated. Cable tray must be suitable for use as an equipment grounding conductor.

### 2.3.2 Ladder-Type Cable Trays

Provide size as indicated Cable tray must be suitable for use as an equipment grounding conductor.

### 2.3.3 Channel-Type Cable Trays

Provide size as indicated Cable tray must be suitable for use as an equipment grounding conductor.

### 2.3.4 Solid Bottom-Type Cable Trays

Provide size as indicated Provide solid covers.

## 2.4 OUTLET BOXES AND COVERS

UL 514A, cadmium- or zinc-coated, if ferrous metal. UL 514C, if nonmetallic.

### 2.4.1 Floor Outlet Boxes

Provide the following:

- a. Boxes: adjustable and concrete tight.
- b. Each outlet: consisting of cast-metal body with threaded openings, for conduits,, and cover plate with 3/4 inch threaded plug.
- d. Receptacle outlets: consisting of surface-mounted aluminum or stainless steel housing with duplex-type receptacle as specified herein.
- e. Provide gaskets where necessary to ensure watertight installation.
- f. Provide plugs with installation instructions to the Contracting Officer for 5 percent of outlet boxes for the capping of outlets upon removal of service fittings.

## 2.5 CABINETS, JUNCTION BOXES, AND PULL BOXES

UL 50; volume greater than 100 cubic inches, NEMA Type 1 enclosure; sheet steel, hot-dip, zinc-coated. Where exposed to wet, damp, or corrosive environments, NEMA Type 4X.

## 2.6 WIRES AND CABLES

Provide wires and cables in accordance applicable requirements of NFPA 70 and UL for type of insulation, jacket, and conductor specified or indicated. Do not use wires and cables manufactured more than 24 months prior to date of delivery to site.

### 2.6.1 Conductors

Provide the following:

- a. Conductor sizes and capacities shown are based on copper, unless indicated otherwise.

- b. Conductors No. 8 AWG and larger diameter: stranded.
- c. Conductors No. 10 AWG and smaller diameter: solid.
- d. Conductors for remote control, alarm, and signal circuits, classes 1, 2, and 3: stranded unless specifically indicated otherwise.
- e. All conductors: copper.

#### 2.6.1.1 Equipment Manufacturer Requirements

When manufacturer's equipment requires copper conductors at the terminations or requires copper conductors to be provided between components of equipment, provide copper conductors or splices, splice boxes, and other work required to satisfy manufacturer's requirements.

#### 2.6.2 Color Coding

Provide color coding for service, feeder, branch, control, and signaling circuit conductors.

##### 2.6.2.1 Ground and Neutral Conductors

Provide color coding of ground and neutral conductors as follows:

- a. Grounding conductors: Green.
- b. Neutral conductors: White.
- c. Exception, where neutrals of more than one system are installed in same raceway or box, other neutrals color coding: white with a different colored (not green) stripe for each.

##### 2.6.2.2 Ungrounded Conductors

Provide color coding of ungrounded conductors in different voltage systems as follows:

- a. 208/120 volt, three-phase
  - (1) Phase A - black
  - (2) Phase B - red
  - (3) Phase C - blue
- b. 480/277 volt, three-phase
  - (1) Phase A - brown
  - (2) Phase B - orange
  - (3) Phase C - yellow
- c. 120/240 volt, single phase: Black and red
- d. On three-phase, four-wire delta system, high leg: orange, as required by NFPA 70.



### 2.6.3 Insulation

Unless specified or indicated otherwise or required by NFPA 70, provide power and lighting branch wires rated for 600-volts, Type THWN/THHN conforming to UL 83 feeders 60 amps and greater Type XHHW conforming to UL 44, except that grounding wire may be type TW conforming to UL 83; remote-control and signal circuits: Type TW or TF, conforming to UL 83. Where equipment or devices require 90-degree Centigrade (C) conductors, provide only conductors with 90-degree C insulation or better.

### 2.6.4 Bonding Conductors

ASTM B1, solid bare copper wire for sizes No. 8 AWG and smaller diameter; , Class B, stranded bare copper wire for sizes No. 6 AWG and larger diameter.

### 2.6.5 Service Entrance Cables

Service Entrance (SE) and Underground Service Entrance (USE) Cables, UL 854.

### 2.6.6 Nonmetallic Sheathed Cable

UL 719, Type NM or NMC.

### 2.6.7 Wire and Cable for 400 Hertz (Hz) Circuits

Insulated copper conductors.

### 2.6.8 Metal-Clad Cable

UL 1569; NFPA 70, Type MC cable.

### 2.6.9 Cord Sets and Power-Supply Cords

UL 817.

## 2.7 SPLICES AND TERMINATION COMPONENTS

UL 486A-486B for wire connectors and UL 510 for insulating tapes. Connectors for No. 10 AWG and smaller diameter wires: insulated, pressure-type in accordance with UL 486A-486B or UL 486C (twist-on splicing connector). Provide solderless terminal lugs on stranded conductors.

## 2.8 DEVICE PLATES

Provide the following:

- a. UL listed, one-piece device plates for outlets to suit the devices installed.
- b. For metal outlet boxes, plates on unfinished walls: zinc-coated sheet steel or cast metal having round or beveled edges.
- c. Screws: machine-type with countersunk heads in color to match finish of plate.
- d. Sectional type device plates are not be permitted.

- e. Plates installed in wet locations: gasketed and UL listed for "wet locations."

## 2.9 SWITCHES

### 2.9.1 Toggle Switches

NEMA WD 1, UL 20, single pole mounting strap with grounding screw. Include the following:

- a. Handles: gray thermoplastic.
- b. Wiring terminals: screw-type, side-wired or of the solderless pressure type having suitable conductor-release arrangement.
- c. Contacts: silver-cadmium and contact arm - one-piece copper alloy.
- d. Switches: rated quiet-type ac only, 120/277 volts, with current rating and number of poles indicated.

### 2.9.2 Switch with Red Pilot Handle

NEMA WD 1. Provide the following:

- a. Pilot lights that are integrally constructed as a part of the switch's handle.
- b. Pilot light color: red and illuminate whenever the switch is closed or "on".
- c. Pilot lighted switch: rated 20 amps and 120 volts or 277 volts as indicated.
- d. The circuit's neutral conductor to each switch with a pilot light.

### 2.9.3 Breakers Used as Switches

For 120- and 277-Volt fluorescent fixtures, mark breakers "SWD" in accordance with UL 489.

### 2.9.4 Disconnect Switches

NEMA KS 1. Provide heavy duty-type switches where indicated, where switches are rated higher than 240 volts, and for double-throw switches. Utilize Class R fuseholders and fuses for fused switches, unless indicated otherwise. Provide horsepower rated for switches serving as the motor-disconnect means. Provide switches in NEMA 4X enclosure as indicated per NEMA ICS 6.

## 2.10 FUSES

NEMA FU 1. Provide complete set of fuses for each fusible switch. Coordinate time-current characteristics curves of fuses serving motors or connected in series with circuit breakers or other circuit protective devices for proper operation. Submit coordination data for approval. Provide fuses with a voltage rating not less than circuit voltage.

#### 2.10.1 Fuseholders

Provide in accordance with UL 4248-1.

#### 2.10.2 Cartridge Fuses, Current Limiting Type (Class R)

UL 248-12, Class RK-5 time-delay type. Provide only Class R associated fuseholders in accordance with UL 4248-12.

#### 2.10.3 Cartridge Fuses, High-Interrupting Capacity, Current Limiting Type (Classes J, L, and CC)

UL 248-8, UL 248-10, UL 248-4, Class J for zero to 600 amperes, Class L for 601 to 6,000 amperes, and Class CC for zero to 30 amperes.

#### 2.10.4 Cartridge Fuses, Current Limiting Type (Class T)

UL 248-15, Class T for zero to 1,200 amperes, 300 volts; and zero to 800 amperes, 600 volts.

### 2.11 RECEPTACLES

Provide the following:

- a. UL 498, general purpose specification grade, grounding-type. Residential grade receptacles are not acceptable.
- b. Ratings and configurations: as indicated.
- c. Bodies: gray as per NEMA WD 1.
- d. Face and body: thermoplastic supported on a metal mounting strap.
- e. Dimensional requirements: per NEMA WD 6.
- f. Screw-type, side-wired wiring terminals or of the solderless pressure type having suitable conductor-release arrangement.
- g. Grounding pole connected to mounting strap.
- h. The receptacle: containing triple-wipe power contacts and double or triple-wipe ground contacts.
- i. Controlled receptacles: as required per ASHRAE 90.1. Provide marking for controlled receptacle per NFPA 70.

#### 2.11.1 Weatherproof Receptacles

Provide receptacles, UL listed for use in "wet locations" with integral GFCI protection. Include cast metal box with gasketed, hinged, lockable and weatherproof while-in-use, polycarbonate, UV resistant/stabilized extra-duty rated hood.

#### 2.11.2 Ground-Fault Circuit Interrupter Receptacles

UL 943, duplex type for mounting in standard outlet box. Provide device capable of detecting current leak when the current to ground is 6 milliamperes or higher, and tripping per requirements of UL 943 for Class A ground-fault circuit interrupter devices. Provide screw-type,

side-wired wiring terminals or pre-wired (pigtail) leads.

### 2.11.3 Special Purpose Receptacles

Receptacles serving are special purpose. Provide in ratings indicated. Furnish one matching plug with each receptacle.

## 2.12 PANELBOARDS

Provide panelboards in accordance with the following:

- a. UL 67 and UL 50 having a short-circuit current rating as indicated.
- b. Panelboards for use as service disconnecting means: additionally conform to UL 869A.
- c. Panelboards: circuit breaker-equipped.
- d. Designed such that individual breakers can be removed without disturbing adjacent units or without loosening or removing supplemental insulation supplied as means of obtaining clearances as required by UL.
- e. "Specific breaker placement" is required in panelboards to match the breaker placement indicated in the panelboard schedule on the design drawings. If it is not possible to match "specific breaker placement" during construction, obtain Government approval prior to device installation.
- f. Use of "Subfeed Breakers" is not acceptable.
- g. Main breaker: "separately" mounted "above" branch breakers.
- h. Where "space only" is indicated, make provisions for future installation of breakers.
- i. Directories: indicate load served by each circuit in panelboard.
- j. Directories: indicate source of service to panelboard (e.g., Panel PA served from Panel MDP).
- k. Provide new directories for existing panels modified by this project as indicated.
- l. Type directories and mount in holder behind transparent protective covering.
- m. Panelboards: listed and labeled for their intended use.
- n. Panelboard nameplates: provided in accordance with paragraph FIELD FABRICATED NAMEPLATES.

### 2.12.1 Enclosure

Provide panelboard enclosure in accordance with the following:

- a. UL 50.
- b. Cabinets mounted outdoors surface on unistrut .

- c. Cabinets: painted in accordance with paragraph FIELD APPLIED PAINTING.
- d. Outdoor cabinets: NEMA 4X a removable steel plate 1/4 inch thick in the bottom for field drilling for conduit connections.
- e. Front edges of cabinets: form-flanged or fitted with structural shapes welded or riveted to the sheet steel, for supporting the panelboard front.
- f. All cabinets: fabricated such that no part of any surface on the finished cabinet deviates from a true plane by more than 1/8 inch.
- g. Holes: provided in the back of indoor surface-mounted cabinets, with outside spacers and inside stiffeners, for mounting the cabinets with a 1/2 inch clear space between the back of the cabinet and the wall surface.
- h. Flush doors: mounted on hinges that expose only the hinge roll to view when the door is closed.
- i. Each door: fitted with a combined catch and lock latch.
- j. Keys: two provided with each lock, with all locks keyed alike.
- k. Finished-head cap screws: provided for mounting the panelboard fronts on the cabinets.

#### 2.12.2 Panelboard Buses

Support bus bars on bases independent of circuit breakers. Design main buses and back pans so that breakers may be changed without machining, drilling, or tapping. Provide isolated neutral bus in each panel for connection of circuit neutral conductors. Provide separate ground bus identified as equipment grounding bus per UL 67 for connecting grounding conductors; bond to steel cabinet.

#### 2.12.3 Circuit Breakers

UL 489, solid state-type having a minimum short-circuit current rating equal to the short-circuit current rating of the panelboard in which the circuit breaker will be mounted. Breaker terminals: UL listed as suitable for type of conductor provided. Where indicated on the drawings, provide circuit breakers with shunt trip devices. Series rated circuit breakers and plug-in circuit breakers are unacceptable.

##### 2.12.3.1 Multipole Breakers

Provide common trip-type with single operating handle. Design breaker such that overload in one pole automatically causes all poles to open. Maintain phase sequence throughout each panel so that any three adjacent breaker poles are connected to Phases A, B, and C, respectively.

##### 2.12.3.2 Circuit Breaker With Ground-Fault Circuit Interrupter

UL 943 and NFPA 70. Provide with auto-monitoring (self-test) and lockout features, "push-to-test" button, visible indication of tripped condition, and ability to detect and trip when current imbalance is 6 milliamperes or higher per requirements of UL 943 for Class A ground-fault circuit

interrupter devices.

## 2.13 ENCLOSED CIRCUIT BREAKERS

UL 489. Individual molded case circuit breakers with voltage and continuous current ratings, number of poles, overload trip setting, and short circuit current interrupting rating as indicated. Enclosure type as indicated. Provide solid neutral.

## 2.14 TRANSFORMERS

Provide transformers in accordance with the following:

- a. NEMA ST 20, general purpose, dry-type, self-cooled, ventilated.
- b. Provide transformers in NEMA4X enclosure.
- c. Taps for transformers 15 kVA and larger: Two 2.5 percent taps Full Capacity Above Nominal (FCAN) and two 2.5 percent taps Full Capacity Below Nominal (FCBN).
- d. Transformer insulation system:
  - (1) 220 degrees C insulation system for transformers 15 kVA and greater, with temperature rise not exceeding 115 degrees C under full-rated load in maximum ambient of 40 degrees C.
- e. Transformer of 115 degrees C temperature rise: capable of carrying continuously 115 percent of nameplate kVA without exceeding insulation rating.
- f. Transformers: quiet type with maximum sound level at least 3 decibels less than NEMA standard level for transformer ratings indicated.

### 2.14.1 Specified Transformer Efficiency

Transformers, indicated and specified with: 480V primary, 80 degrees C or 115 degrees C temperature rise, kVA ratings of 37.5 to 100 for single phase or 30 to 500 for three phase, energy efficient type. The transformer is not acceptable if the calculated transformer efficiency is less than the efficiency indicated in 10 CFR 431, Subpart K.

### 2.14.2 K-Rated Transformers

Provide K-rated transformers for non-linear loads in accordance with the following:

- a. Transformer insulation: UL recognized 220 degrees C system. Neither the primary nor the secondary temperature is allowed to exceed 220 degrees C at any point in the coils while carrying their full rating of non-sinusoidal load.

- b. Transformers are to be UL listed and labeled for K-13 K-Factor rating as indicated in accordance with UL 1561.
- c. Transformers evaluated by the UL K-Factor evaluation: listed for 115 degrees C average temperature rise only.
- d. Transformers with K-Factor ratings with temperature rise of 150 degrees C rise are not acceptable.
- e. K-Factor rated transformers impedance: allowed range of 3 percent to 5 percent, with a minimum reactance of 2 percent to prevent excessive neutral current when supplying loads with large amounts of third harmonic.

## 2.15 MOTOR CONTROLLERS

Provide motor controllers in accordance with the following:

- a. UL 508, NEMA ICS 1, and NEMA ICS 2.
- b. Provide controllers with thermal overload protection in each phase, and one spare normally open auxiliary contact, and one spare normally closed auxiliary contact.
- c. Provide controllers for motors rated 1-hp 746 watts and above with electronic phase-voltage monitors designed to protect motors from phase-loss, undervoltage, and overvoltage.
- d. Provide protection for motors from immediate restart by a time adjustable restart relay.
- e. When used with pressure, float, or similar automatic-type or maintained-contact switch, provide a hand/off/automatic selector switch with the controller.
- f. Connections to selector switch: wired such that only normal automatic regulatory control devices are bypassed when switch is in "hand" position.
- g. Safety control devices, such as low and high pressure cutouts, high temperature cutouts, and motor overload protective devices: connected in motor control circuit in "hand" and "automatic" positions.
- h. Control circuit connections to hand/off/automatic selector switch or to more than one automatic regulatory control device: made in accordance with indicated or manufacturer's approved wiring diagram.
- i. Provide selector switch with the means for locking in any position.
- j. Provide a disconnecting means, capable of being locked in the open position, for the motor that is located in sight from the motor location and the driven machinery location. As an alternative, provide a motor controller disconnect, capable of being locked in the open position, to serve as the disconnecting means for the motor if it is in sight from the motor location and the driven machinery location.
- k. Overload protective devices: provide adequate protection to motor windings; be thermal inverse-time-limit type; and include manual reset-type pushbutton on outside of motor controller case.

- l. Cover of combination motor controller and manual switch or circuit breaker: interlocked with operating handle of switch or circuit breaker so that cover cannot be opened unless handle of switch or circuit breaker is in "off" position.
- m. Minimum short circuit withstand rating of combination motor controller: 65 KAIC rms symmetrical amperes.
- n. Provide controllers in hazardous locations with classifications as indicated.

#### 2.15.1 Control Wiring

Provide control wiring in accordance with the following:

- a. All control wire: stranded tinned copper switchboard wire with 600-volt flame-retardant insulation Type SIS meeting UL 44, or Type MTW meeting UL 1063, and passing the VW-1 flame tests included in those standards.
- b. Hinge wire: Class K stranding.
- c. Current transformer secondary leads: not smaller than No. 10 AWG.
- d. Control wire minimum size: No. 14 AWG.
- e. Power wiring for 480-volt circuits and below: the same type as control wiring with No. 12 AWG minimum size.
- f. Provide wiring and terminal arrangement on the terminal blocks to permit the individual conductors of each external cable to be terminated on adjacent terminal points.

#### 2.15.2 Control Circuit Terminal Blocks

Provide control circuit terminal blocks in accordance with the following:

- a. NEMA ICS 4.
- b. Control circuit terminal blocks for control wiring: molded or fabricated type with barriers, rated not less than 600 volts.
- c. Provide terminals with removable binding, fillister or washer head screw type, or of the stud type with contact and locking nuts.
- d. Terminals: not less than No. 10 in size with sufficient length and space for connecting at least two indented terminals for 10 AWG conductors to each terminal.
- e. Terminal arrangement: subject to the approval of the Contracting Officer with not less than four spare terminals or 10 percent, whichever is greater, provided on each block or group of blocks.
- f. Modular, pull apart, terminal blocks are acceptable provided they are of the channel or rail-mounted type.
- g. Submit data showing that any proposed alternate will accommodate the specified number of wires, are of adequate current-carrying capacity,



and are constructed to assure positive contact between current-carrying parts.

#### 2.15.2.1 Types of Terminal Blocks

- a. Short-Circuiting Type: Short-circuiting type terminal blocks: furnished for all current transformer secondary leads with provision for shorting together all leads from each current transformer without first opening any circuit. Terminal blocks: comply with the requirements of paragraph CONTROL CIRCUIT TERMINAL BLOCKS above.
- b. Load Type: Load terminal blocks rated not less than 600 volts and of adequate capacity: provided for the conductors for NEMA Size 3 and smaller motor controllers and for other power circuits, except those for feeder tap units. Provide terminals of either the stud type with contact nuts and locking nuts or of the removable screw type, having length and space for at least two indented terminals of the size required on the conductors to be terminated. For conductors rated more than 50 amperes, provide screws with hexagonal heads. Conducting parts between connected terminals must have adequate contact surface and cross-section to operate without overheating. Provide each connected terminal with the circuit designation or wire number placed on or near the terminal in permanent contrasting color.

#### 2.15.3 Control Circuits

Control circuits: maximum voltage of 120 volts derived from control transformer in same enclosure. Transformers: conform to UL 506, as applicable. Transformers, other than transformers in bridge circuits: provide primaries wound for voltage available and secondaries wound for correct control circuit voltage. Size transformers so that 80 percent of rated capacity equals connected load. Provide disconnect switch on primary side. Provide fuses in each ungrounded primary feeder. Provide one fused secondary lead with the other lead grounded.

#### 2.15.4 Enclosures for Motor Controllers

NEMA ICS 6.

#### 2.15.5 Multiple-Speed Motor Controllers and Reversible Motor Controllers

Across-the-line-type, electrically and mechanically interlocked. Multiple-speed controllers: include compelling relays and multiple-button, station-type with pilot lights for each speed.

#### 2.15.6 Pushbutton Stations

Provide with "start/stop" momentary contacts having one normally open and one normally closed set of contacts, and red lights to indicate when motor is running. Stations: heavy duty, oil-tight design.

#### 2.15.7 Pilot and Indicating Lights

Provide LED cluster lamps.

#### 2.16 MANUAL MOTOR STARTERS (MOTOR RATED SWITCHES)

Single pole designed for surface mounting with overload protection and pilot lights.

### 2.16.1 Pilot Lights

Provide yoke-mounted, seven element LED cluster light module. Color: green in accordance with NEMA ICS 2.

## 2.17 ELECTRICAL HEATING

This section includes material, equipment, and installation requirements for a complete electric heat trace system, including multi-loop control panel, heat trace cables, branch circuit wiring, temperature sensors, and all associated componenets.

A. Submit shop drawings for approval prior to ordering equipment. Include the following:

- a. Panel layout, internal wiring diagram, and circuit breaker schedule.
- b. Heat trace cable routing diagram with tag numbers and lengths
- c. RTD sensor locations and integration method
- d. Panel dimensions, mounting details, and nameplate schedule
- e. Commissioning checklist and test procedure.

B. Provide products listed and labeled by UL or other recognized testing agency.

C. Installation shall conform to NEC, UFC, and local authority requirements.

D. Provide manufacturer's standard warranty for all supplied heat trace system components.

### 2.17.1 Heat Trace Control Panel & Cables

A. Rating: 480/277V, 3-phase, 4-wire, 70A minimum main input.

B. Minimum twelve (12) branch circuits, each with individual protection

C. Features shall include:

- a. SCR control, PID mode or duty-cycle based
- b. Integrated RTD input, ambient or pipe sensing.
- c. Local touchscreen HMI interface.
- d. Ethernet or Modbus RTU communication.
- e. NEMA 4X enclosure.
- f. Built-in GFCI monitoring and alarming.

D. Provide self-regulating or constant wattage heat trace cable as approved by manufacturer.

E. Total of twelve (12) individual cable runs anticipated.

F. Cable shall be UL-listed and suitable for application environment.

G. Provide lighted end seal kits for each heat trace circuit. Kits shall visibly illuminate when the associated heat trace cable is energized. Light indicators shall be suitable for outdoor use and comply with the cable manufacturer's recommendations.

## 2.17.2 Installation & Commissioning

- A. Install panel, wiring, and cables per manufacturer's instructions and NEC Article 427.
- B. Provide separate branch circuit conduit and wiring from panel to each heat trace cable.
- C. Secure cables with fiberglass or manufacturer-approved tape only - zip ties not permitted.
- D. Install junction boxes, RTDs, and labels at all required locations
- E. Perform the following tests and verifications prior to acceptance
  - a. Insulation resistance testing (pre- and post-insulation).
  - b. Verify correct operation of SCR, alarms, GFCI, and setpoints.
  - c. Configure touchscreen HMI with labels and modes
  - d. Submit startup report including circuit loads, insulation tests, setpoints, and alarm testing.
- F. Notify Engineer five (5) business days prior to startup.
- G. Commissioning shall not be considered complete until all circuits function as intended and documentation is accepted by the Engineer.

## 2.18 GROUNDING AND BONDING EQUIPMENT

### 2.18.1 Ground Rods

UL 467. Ground rods: cone pointed copper-clad steel, with minimum diameter of 3/4 inch and minimum length 10 feet. Sectional type rods may be used for rods 20 feet or longer.

### 2.18.2 Ground Bus

Copper ground bus: provided in the electrical equipment rooms as indicated.

### 2.18.3 Secondary Bonding Busbar

Provide corrosion-resistant grounding busbar suitable for indoor installation in accordance with TIA-607. Busbars: plated for reduced contact resistance. If not plated, clean the busbar prior to fastening the conductors to the busbar and apply an anti-oxidant to the contact area to control corrosion and reduce contact resistance. . The Primary bonding busbar (PBB) and the Secondary bonding busbar (SBB): sized in accordance with the immediate application requirements and with consideration of future growth. Provide Secondary bonding busbars with the following:

- a. Predrilled copper busbar provided with holes for use with standard sized lugs,

- b. Minimum dimensions of 0.25 in thick by 4 in wide for the PBB and 2 in wide for SBBs with length as indicated;
- c. Listed by a nationally recognized testing laboratory.

#### 2.19 MANUFACTURER'S NAMEPLATE

Provide on each item of equipment a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

#### 2.20 FIELD FABRICATED NAMEPLATES

Provide field fabricated nameplates in accordance with the following:

- a. ASTM D709.
- b. Provide laminated plastic nameplates for each equipment enclosure, relay, switch, and device; as specified or as indicated on the drawings.
- c. Each nameplate inscription: identify the function and, when applicable, the position.
- d. Nameplates: melamine plastic, 0.125 inch thick, white with black center core.
- e. Provide red laminated plastic label with white center core where indicated.
- f. Surface: matte finish. Corners: square. Accurately align lettering and engrave into the core.
- g. Minimum size of nameplates: one by 2.5 inches.
- h. Lettering size and style: a minimum of 0.25 inch high normal block style.

#### 2.21 WARNING SIGNS

Provide warning signs for flash protection in accordance with NFPA 70E for switchboards, panelboards, industrial control panels, and motor control centers that are in other than dwelling occupancies and are likely to require examination, adjustment, servicing, or maintenance while energized. Provide field installed signs to warn qualified persons of potential electric arc flash hazards when warning signs are not provided by the manufacturer. Provide marking that is clearly visible to qualified persons before examination, adjustment, servicing, or maintenance of the equipment.

#### 2.22 FIRESTOPPING MATERIALS

Provide firestopping around electrical penetrations in accordance with Section 07 84 00 FIRESTOPPING.

#### 2.23 WIREWAYS

UL 870. Material: steel epoxy painted 16 gauge for heights and depths up

to 6 by 6 inches, and 14 gauge for heights and depths up to 12 by 12 inches. Provide in length indicated required for the application with hinged-screw-cover NEMA 4X enclosure per NEMA ICS 6.

## 2.24 SURGE PROTECTIVE DEVICES

Provide parallel type surge protective devices (SPD) which comply with UL 1449 at the , panelboards. Provide surge protectors in a NEMA 4X enclosure per NEMA ICS 6. Do not install SPD inside a panelboard or switchboard enclosure. However, SPD may be installed in a separate compartment of a switchgear provided that it is supplied by a dedicated circuit breaker. SPD must have the same short-circuit current rating as the protected equipment and must not be installed at a point of system where the available fault current is in excess of that rating. Use Type 1 or Type 2 SPD and connect on the load side of a dedicated circuit breaker. Submit performance and characteristic curves.

Provide the following modes of protection:

### FOR SINGLE PHASE AND THREE PHASE WYE CONNECTED SYSTEMS-

- Phase to phase ( L-L )
- Each phase to neutral ( L-N )
- Neutral to ground ( N-G )
- Phase to ground ( L-G )

SPDs at the service entrance: provide with a minimum surge current rating of 80,000 amperes for L-L mode minimum and 40,000 amperes for other modes (L-N, L-G, and N-G) and downstream SPDs rated 40,000 amperes for L-L mode minimum and 20,000 amperes for other modes (L-N, L-G, and N-G).

The minimum MCOV (Maximum Continuous Operating Voltage) rating for L-N and L-G modes of operation: 120 percent of nominal voltage for 240 volts and below; 115 percent of nominal voltage above 240 volts to 480 volts.

Provide EMI/RFI filtering per UL 1283 for each mode with the capability to attenuate high frequency noise. Minimum attenuation: 20db.

## 2.25 FACTORY APPLIED FINISH

Provide factory-applied finish on electrical equipment in accordance with the following:

- a. NEMA 250 corrosion-resistance test and the additional requirements as specified herein.
- b. Interior and exterior steel surfaces of equipment enclosures: thoroughly cleaned followed by a rust-inhibitive phosphatizing or equivalent treatment prior to painting.
- c. Exterior surfaces: free from holes, seams, dents, weld marks, loose scale or other imperfections.
- d. Interior surfaces: receive not less than one coat of corrosion-resisting paint in accordance with the manufacturer's standard practice.
- e. Exterior surfaces: primed, filled where necessary, and given not less

than two coats baked enamel with semigloss finish.

- f. Equipment located indoors: ANSI Light Gray, and equipment located outdoors: ANSI Light Gray.
- g. Provide manufacturer's coatings for touch-up work and as specified in paragraph FIELD APPLIED PAINTING.

## 2.26 SOURCE QUALITY CONTROL

### 2.26.1 Transformer Factory Tests

Submittal: include routine NEMA ST 20 transformer test results on each transformer and also provide the results of NEMA "design" and "prototype" tests that were made on transformers electrically and mechanically equal to those specified.

## 2.27 COORDINATED POWER SYSTEM PROTECTION

Prepare analyses as specified in Section 26 05 73 POWER SYSTEM STUDIES.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Electrical installations, including weatherproof and hazardous locations and ducts, plenums and other air-handling spaces: conform to requirements of NFPA 70 and to requirements specified herein.

#### 3.1.1 Underground Service

Underground service conductors and associated conduit: continuous from service entrance equipment to outdoor power system connection.

#### 3.1.2 Hazardous Locations

Perform work in hazardous locations, as defined by NFPA 70, in strict accordance with NFPA 70 for particular "Class," "Division," and "Group" of hazardous locations involved. Provide conduit and cable seals where required by NFPA 70. Provide conduit with tapered threads.

#### 3.1.3 Service Entrance Identification

Service entrance disconnect devices, switches, and enclosures: labeled and identified as such.

##### 3.1.3.1 Labels

Wherever work results in service entrance disconnect devices in more than one enclosure, as permitted by NFPA 70, label each enclosure, new and existing, as one of several enclosures containing service entrance disconnect devices. Label, at minimum: indicate number of service disconnect devices housed by enclosure and indicate total number of enclosures that contain service disconnect devices. Provide laminated plastic labels conforming to paragraph FIELD FABRICATED NAMEPLATES. Use lettering of at least 0.25 inch in height, and engrave on black-on-white matte finish. Service entrance disconnect devices in more than one enclosure: provided only as permitted by NFPA 70.

#### 3.1.4 Wiring Methods

Provide insulated conductors installed in rigid steel conduit, IMC, rigid nonmetallic conduit, or EMT, except where specifically indicated or specified otherwise or required by NFPA 70 to be installed otherwise. Grounding conductor: separate from electrical system neutral conductor. Provide insulated green equipment grounding conductor for circuit(s) installed in conduit and raceways. Shared neutral, or multi-wire branch circuits, are not permitted with arc-fault circuit interrupters. Minimum conduit size: 1/2 inch in diameter for low voltage lighting and power circuits. Vertical distribution in multiple story buildings: made with metal conduit in fire-rated shafts, with metal conduit extending through shafts for minimum distance of 6 inches. Firestop conduit which penetrates fire-rated walls, fire-rated partitions, or fire-rated floors in accordance with Section 07 84 00 FIRESTOPPING.

##### 3.1.4.1 Pull Wire

Install pull wires in empty conduits. Pull wire: plastic having minimum 200-pound force tensile strength. Leave minimum 36 inches of slack at each end of pull wire.

##### 3.1.4.2 Metal-Clad Cable

Install in accordance with NFPA 70, Type MC cable.

#### 3.1.5 Conduit Installation

Unless indicated otherwise, conceal conduit under floor slabs and within finished walls, ceilings, and floors. Keep conduit minimum 6 inches away from parallel runs of flues and steam or hot water pipes. Install conduit parallel with or at right angles to ceilings, walls, and structural members where located above accessible ceilings and where conduit will be visible after completion of project. Run conduits in crawl space under floor slab as if exposed.

##### 3.1.5.1 Restrictions Applicable to Aluminum Conduit

- a. Do not install underground or encase in concrete or masonry.
- b. Do not use brass or bronze fittings.
- c. Do not use when the enclosed conductors must be shielded from the effects of High-altitude Electromagnetic Pulse (HEMP).

##### 3.1.5.2 Restrictions Applicable to EMT

- a. Do not install underground.
- b. Do not encase in concrete, mortar, grout, or other cementitious materials.
- c. Do not use in areas subject to physical damage including but not limited to equipment rooms where moving or replacing equipment could physically damage the EMT.
- d. Do not use in hazardous areas.
- e. Do not use outdoors.

- f. Do not use in fire pump rooms.
- g. Do not use when the enclosed conductors must be shielded from the effects of High-altitude Electromagnetic Pulse (HEMP).

#### 3.1.5.3 Restrictions Applicable to Nonmetallic Conduit

##### a. PVC Schedule 40.

- (1) Do not use where subject to physical damage, including but not limited to, mechanical equipment rooms, electrical equipment rooms, fire pump rooms, and where restrictions are applying to both PVC Schedule 40 and PVC Schedule 80.
- (2) Do not use above grade, except where allowed in this section for rising through floor slab or indicated otherwise.

##### b. PVC Schedule 80.

- (1) Do not use where subject to physical damage, including but not limited to, hospitals, power plant, missile magazines, and other such areas.
- (2) Do not use in hazardous (classified) areas.
- (3) Do not use in penetrating fire-rated walls or partitions, or fire-rated floors.

#### 3.1.5.4 Restrictions Applicable to Flexible Conduit

Use only as specified in paragraph FLEXIBLE CONNECTIONS. Do not use when the enclosed conductors must be shielded from the effects of High-altitude Electromagnetic Pulse (HEMP).

#### 3.1.5.5 Underground Conduit

Plastic-coated rigid steel; plastic-coated steel IMC; PVC, Type EPC-40; or fiberglass. Convert nonmetallic conduit, to plastic-coated rigid, steel conduit before rising through floor slab. Plastic coating: extend minimum 6 inches above floor.

#### 3.1.5.6 Conduit for Circuits Rated Greater Than 600 Volts

Rigid metal conduit or IMC only.

#### 3.1.5.7 Stub-Ups

Provide conduits stubbed up through concrete floor for connection to free-standing equipment with adjustable top or coupling threaded inside for plugs, set flush with finished floor. Extend conductors to equipment in rigid steel conduit, except that flexible metal conduit may be used 6 inches above floor. Where no equipment connections are made, install screwdriver-operated threaded flush plugs in conduit end.

#### 3.1.5.8 Conduit Support

Support conduit by pipe straps, wall brackets, threaded rod conduit hangers, or ceiling trapeze. Plastic cable ties are not acceptable.



Fasten by wood screws to wood; by toggle bolts on hollow masonry units; by concrete inserts or expansion bolts on concrete or brick; and by machine screws, welded threaded studs, or spring-tension clamps on steel work. Threaded C-clamps may be used on rigid steel conduit only. Do not weld conduits or pipe straps to steel structures. Do not exceed one-fourth proof test load for load applied to fasteners. Provide vibration resistant and shock-resistant fasteners attached to concrete ceiling. Do not cut main reinforcing bars for any holes cut to depth of more than 1 1/2 inches in reinforced concrete beams or to depth of more than 3/4 inch in concrete joints. Fill unused holes. In partitions of light steel construction, use sheet metal screws. In suspended-ceiling construction, run conduit above ceiling. Do not support conduit by ceiling support system. Conduit and box systems: supported independently of both (a) tie wires supporting ceiling grid system, and (b) ceiling grid system into which ceiling panels are placed. Do not share supporting means between electrical raceways and mechanical piping or ducts. Identify independent conduit support in both fire and non-fire rated assemblies per NFPA 70. Coordinate installation with above-ceiling mechanical systems to assure maximum accessibility to all systems. Spring-steel fasteners may be used for lighting branch circuit conduit supports in suspended ceilings in dry locations. Support exposed risers in wire shafts of multistory buildings by U-clamp hangers at each floor level and at 10 foot maximum intervals. Where conduit crosses building expansion joints, provide suitable watertight expansion fitting that maintains conduit electrical continuity by bonding jumpers or other means. For conduits greater than 2 1/2 inches inside diameter, provide supports to resist forces of 0.5 times the equipment weight in any direction and 1.5 times the equipment weight in the downward direction.

#### 3.1.5.9 Directional Changes in Conduit Runs

Make changes in direction of runs with symmetrical bends or cast-metal fittings. Make field-made bends and offsets with hickey or conduit-bending machine. Do not install crushed or deformed conduits. Avoid trapped conduits. Prevent plaster, dirt, or trash from lodging in conduits, boxes, fittings, and equipment during construction. Free clogged conduits of obstructions.

#### 3.1.5.10 Locknuts and Bushings

Fasten conduits to sheet metal boxes and cabinets with two locknuts where required by NFPA 70, where insulated bushings are used, and where bushings cannot be brought into firm contact with the box; otherwise, use at least minimum single locknut and bushing. Provide locknuts with sharp edges for digging into wall of metal enclosures. Install bushings on ends of conduits, and provide insulating type where required by NFPA 70.

#### 3.1.5.11 Flexible Connections

Provide flexible steel conduit between 3 and 6 feet in length; for equipment subject to vibration, noise transmission, or movement; and for motors. Install flexible conduit to allow 20 percent slack. Minimum flexible steel conduit size: 1/2 inch diameter. Provide liquid tight flexible nonmetallic conduit in wet and damp locations for equipment subject to vibration, noise transmission, movement or motors. Provide separate ground conductor across flexible connections. Plastic cable ties are not acceptable as a support method.

### 3.1.6 Cable Tray Installation

Install and ground in accordance with NFPA 70. Install cable trays parallel with or at right angles to ceilings, walls, and structural members. Cable tray and tray supports must not partially nor completely obstruct access to the room. Support in accordance with NEMA VE 2 and with manufacturer recommendations. Coat contact surfaces of aluminum connections with an antioxidant compound prior to assembly. Adjacent cable tray sections: bonded together by connector plates of an identical type as the cable tray sections. For grounding of cable tray system provide No. 2 AWG bare copper wire throughout cable tray system, and bond to each section, except use No. 1/0 aluminum wire if cable tray is aluminum. Terminate cable trays 10 inches from both sides of smoke and fire partitions. Install conductors run through smoke and fire partitions in 4 inch rigid steel conduits with grounding bushings, extending 12 inches beyond each side of partitions. Seal conduit on both ends to maintain smoke and fire ratings of partitions. Firestop penetrations in accordance with Section 07 84 00, FIRESTOPPING. Provide supports to resist forces of 0.5 times the equipment weight in any direction and 1.5 times the equipment weight in the downward direction.

### 3.1.7 Boxes, Outlets, and Supports

Provide boxes in wiring and raceway systems wherever required for pulling of wires, making connections, and mounting of devices or fixtures. Boxes for metallic raceways: cast-metal, hub-type when located in wet locations, when surface mounted on outside of exterior surfaces, when surface mounted on interior walls exposed up to 7 feet above floors and walkways, or when installed in hazardous areas and when specifically indicated. Boxes in other locations: sheet steel, except that aluminum boxes may be used with aluminum conduit, and nonmetallic boxes may be used with nonmetallic sheathed cable conduit system. Provide each box with volume required by NFPA 70 for number of conductors enclosed in box. Boxes for mounting lighting fixtures: minimum 4 inches square, or octagonal, except that smaller boxes may be installed as required by fixture configurations, as approved. Boxes for use in masonry-block or tile walls: square-cornered, tile-type, or standard boxes having square-cornered, tile-type covers. Provide gaskets for cast-metal boxes installed in wet locations and boxes installed flush with outside of exterior surfaces. Provide separate boxes for flush or recessed fixtures when required by fixture terminal operating temperature; provide readily removable fixtures for access to boxes unless ceiling access panels are provided. Support boxes and pendants for surface-mounted fixtures on suspended ceilings independently of ceiling supports. Fasten boxes and supports with wood screws on wood, with bolts and expansion shields on concrete or brick, with toggle bolts on hollow masonry units, and with machine screws or welded studs on steel. Threaded studs driven in by powder charge and provided with lock washers and nuts or nail-type nylon anchors may be used in lieu of wood screws, expansion shields, or machine screws. In open overhead spaces, cast boxes threaded to raceways need not be separately supported except where used for fixture support; support sheet metal boxes directly from building structure or by bar hangers. Where bar hangers are used, attach bar to raceways on opposite sides of box, and support raceway with approved-type fastener maximum 24 inches from box. When penetrating reinforced concrete members, avoid cutting reinforcing steel.

#### 3.1.7.1 Boxes

Boxes for use with raceway systems: minimum 1 1/2 inches deep, except where shallower boxes required by structural conditions are approved. Boxes for other than lighting fixture outlets: minimum 4 inches square, except that 4 by 2 inch boxes may be used where only one raceway enters outlet.

#### 3.1.7.2 Pull Boxes

Construct of at least minimum size required by NFPA 70 of code-gauge galvanized sheet steel, and compatible with nonmetallic raceway systems, except where cast-metal boxes are required in locations specified herein. Provide boxes with screw-fastened covers. Where several feeders pass through common pull box, tag feeders to indicate clearly electrical characteristics, circuit number, and panel designation.

#### 3.1.7.3 Extension Rings

Extension rings are not permitted for new construction. Use only on existing boxes in concealed conduit systems where wall is furred out for new finish.

#### 3.1.8 Mounting Heights

Mount panelboards, enclosed circuit breakers, motor controller and disconnecting switches so height of center of grip of the operating handle of the switch or circuit breaker at its highest position is maximum 79 inches above floor or working platform or as allowed in Section 404.8 per NFPA 70. Mount lighting switches 48 inches above finished floor. Mount receptacles and telecommunications outlets 18 inches above finished floor, unless otherwise indicated.. Mount other devices as indicated.. Measure mounting heights of receptacle outlet boxes in the to the bottom of the outlet box.

#### 3.1.9 Conductor Identification

Provide conductor identification within each enclosure where tap, splice, or termination is made. For conductors No. 6 AWG and smaller diameter, provide color coding by factory-applied, color-impregnated insulation. For conductors No. 4 AWG and larger diameter, provide color coding by plastic-coated, self-sticking markers; colored nylon cable ties and plates; or heat shrink-type sleeves. manufacturer's recommendations.

##### 3.1.9.1 Marking Strips

Provide marking strips for identification of power distribution, control, data, and communications cables in accordance with the following:

- a. Provide white or other light-colored plastic marking strips, fastened by screws to each terminal block, for wire designations.
- b. Use permanent ink for the wire numbers
- c. Provide reversible marking strips to permit marking both sides, or provide two marking strips with each block.
- d. Size marking strips to accommodate the two sets of wire numbers.

- e. Assign a device designation in accordance with NEMA ICS 1 to each device to which a connection is made. Mark each device terminal to which a connection is made with a distinct terminal marking corresponding to the wire designation used on the Contractor's schematic and connection diagrams.
- f. The wire (terminal point) designations used on the Contractor's wiring diagrams and printed on terminal block marking strips may be according to the Contractor's standard practice; however, provide additional wire and cable designations for identification of remote (external) circuits for the Government's wire designations.
- g. Prints of the marking strips drawings submitted for approval will be so marked and returned to the Contractor for addition of the designations to the terminal strips and tracings, along with any rearrangement of points required.

#### 3.1.10 Splices

Make splices in accessible locations. Make splices in conductors No. 10 AWG and smaller diameter with insulated, pressure-type connector. Make splices in conductors No. 8 AWG and larger diameter with solderless connector, and cover with insulation material equivalent to conductor insulation.

#### 3.1.11 Terminating Aluminum Conductors

#### 3.1.12 Covers and Device Plates

Install with edges in continuous contact with finished wall surfaces without use of mats or similar devices. Plaster fillings are not permitted. Install plates with alignment tolerance of 1/16 inch. Use of sectional-type device plates are not permitted. Provide gasket for plates installed in wet locations.

#### 3.1.13 Electrical Penetrations

Seal openings around electrical penetrations through fire resistance-rated walls, partitions, floors, or ceilings in accordance with Section 07 84 00 FIRESTOPPING.

#### 3.1.14 Grounding and Bonding

Provide in accordance with NFPA 70. Ground exposed, non-current-carrying metallic parts of electrical equipment, metallic raceway systems, grounding conductor in metallic and nonmetallic raceways, system grounds, grounding conductor of nonmetallic sheathed cables, and neutral conductor of wiring systems. Make ground connection at main service equipment, and extend grounding conductor to point of entrance of metallic water service. Make connection to water pipe by suitable ground clamp or lug connection to plugged tee. If flanged pipes are encountered, make connection with lug bolted to street side of flanged connection. Supplement metallic water service grounding system with additional made electrode in compliance with NFPA 70. Make ground connection to driven ground rods on exterior of building. Bond additional driven rods together with a minimum of 4 AWG soft bare copper wire buried to a depth of at least 12 inches. Interconnect all grounding media in or on the structure to provide a common ground potential. This includes lightning protection, electrical service, telecommunications system grounds, as well as underground

metallic piping systems. Make interconnection to the gas line on the customer's side of the meter. Use main size lightning conductors for interconnecting these grounding systems to the lightning protection system.

In addition to the requirements specified herein, provide telecommunications grounding. Where ground fault protection is employed, ensure that connection of ground and neutral does not interfere with correct operation of fault protection.

#### 3.1.14.1 Ground Rods

Provide ground rods and measure the resistance to ground using the fall-of-potential method described in IEEE 81. Do not exceed 25 ohms under normally dry conditions for the maximum resistance of a driven ground. If this resistance cannot be obtained with a single rod, additional rods, spaced on center. Spacing for additional rods must be a minimum of 10 feet, or if sectional type rods are used, additional sections may be coupled and driven with the first rod. In high-ground-resistance, UL listed chemically charged ground rods may be used. If the resultant resistance exceeds 25 ohms measured not less than 48 hours after rainfall, notify the Contracting Officer who will decide on the number of ground rods to add.

#### 3.1.14.2 Grounding Connections

Make grounding connections which are buried or otherwise normally inaccessible, excepting specifically those connections for which access for periodic testing is required, by exothermic weld or high compression connector.

- a. Make exothermic welds strictly in accordance with the weld manufacturer's written recommendations. Welds which are "puffed up" or which show convex surfaces indicating improper cleaning are not acceptable. Mechanical connectors are not required at exothermic welds.
- b. Make high compression connections using a hydraulic or electric compression tool to provide the correct circumferential pressure. Provide tools and dies as recommended by the manufacturer. Use an embossing die code or other standard method to provide visible indication that a connector has been adequately compressed on the ground wire.

#### 3.1.14.3 Ground Bus

Provide a copper ground bus in the electrical equipment rooms as indicated. Noncurrent-carrying metal parts of transformer neutrals and other electrical equipment: effectively grounded by bonding to the ground bus. Bond the ground bus to both the entrance ground, and to a ground rod or rods as specified above having the upper ends terminating approximately 4 inches above the floor. Make connections and splices of the brazed, welded, bolted, or pressure-connector type, except use pressure connectors or bolted connections for connections to removable equipment. For raised floor equipment rooms in computer and data processing centers, provide a minimum of four, one at each corner, ground buses connected to the building grounding system. Use bolted connections in lieu of thermoweld, so they can be changed as required by additions and alterations.

#### 3.1.14.4 Resistance

Maximum resistance-to-ground of grounding system: do not exceed 5 ohms under dry conditions. Where resistance obtained exceeds 25 ohms, contact Contracting Officer for further instructions.

#### 3.1.15 Equipment Connections

Provide power wiring for the connection of motors and control equipment under this section of the specification. Except as otherwise specifically noted or specified, automatic control wiring, control devices, and protective devices within the control circuitry are not included in this section of the specifications and are provided under the section specifying the associated equipment.

#### 3.1.16 Government-Furnished Equipment

Contractor rough-in for Government-furnished equipment make connections to Government-furnished equipment to make equipment operate as intended, including providing miscellaneous items such as plugs, receptacles, wire, cable, conduit, flexible conduit, and outlet boxes or fittings.

#### 3.1.17 Repair of Existing Work

Perform repair of existing work, demolition, and modification of existing electrical distribution systems as follows:

##### 3.1.17.1 Workmanship

Lay out work in advance. Exercise care where cutting, channeling, chasing, or drilling of floors, walls, partitions, ceilings, or other surfaces is necessary for proper installation, support, or anchorage of conduit, raceways, or other electrical work. Repair damage to buildings, piping, and equipment using skilled craftsmen of trades involved.

##### 3.1.17.2 Existing Concealed Wiring to be Removed

Disconnect existing concealed wiring to be removed from its source. Remove conductors; cut conduit flush with floor, underside of floor, and through walls; and seal openings.

##### 3.1.17.3 Removal of Existing Electrical Distribution System

Removal of existing electrical distribution system equipment includes equipment's associated wiring, including conductors, cables, exposed conduit, surface metal raceways, boxes, and fittings, back to equipment's power source as indicated.

##### 3.1.17.4 Continuation of Service

Maintain continuity of existing circuits of equipment to remain. Maintain existing circuits of equipment energized. Restore circuits wiring and power which are to remain but were disturbed during demolition back to original condition.

#### 3.1.18 Surge Protective Devices

Connect the surge protective devices in parallel to the power source, keeping the conductors as short and straight as practically possible.

Maximum allowed lead length is 3 feet avoiding 90 degree bends. Do not locate surge protective devices inside a panelboard or switchboard enclosure.

### 3.2 FIELD FABRICATED NAMEPLATE MOUNTING

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets. Provide nameplate on all equipment in access controlled spaces and areas.

### 3.3 WARNING SIGN MOUNTING

Provide the number of signs required to be readable from each accessible side. Space the signs in accordance with NFPA 70E.

### 3.4 FIELD APPLIED PAINTING

Paint electrical equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria. Where field painting of enclosures for panelboards, load centers or the like is specified to match adjacent surfaces, to correct damage to the manufacturer's factory applied coatings, or to meet the indicated or specified safety criteria, provide manufacturer's recommended coatings and apply in accordance to manufacturer's instructions.

### 3.5 FIELD QUALITY CONTROL

Furnish test equipment and personnel and submit written copies of test results. Give Contracting Officer 14 working days notice prior to each test. Where applicable, test electrical equipment in accordance with NETA ATS.

#### 3.5.1 Devices Subject to Manual Operation

Operate each device subject to manual operation at least five times, demonstrating satisfactory operation each time.

#### 3.5.2 600-Volt Wiring Test

Test wiring rated 600 volt and less to verify that no short circuits or accidental grounds exist. Perform insulation resistance tests on wiring No. 6 AWG and larger diameter using instrument which applies voltage of 1,000 volts DC for 600 volt rated wiring and 500 volts DC for 300 volt rated wiring per NETA ATS to provide direct reading of resistance. All existing wiring to be reused must also be tested.

#### 3.5.3 Transformer Tests

Perform the standard, not optional, tests in accordance with the Inspection and Test Procedures for transformers, dry type, air-cooled, 600 volt and below; as specified in NETA ATS. Measure primary and secondary voltages for proper tap settings. Tests need not be performed by a recognized independent testing firm or independent electrical consulting firm.

#### 3.5.4 Ground-Fault Receptacle Test

Test ground-fault receptacles with a "load" (such as a plug in light) to

verify that the "line" and "load" leads are not reversed. Press the TEST button and then the RESET button to verify by LED status that the device is a self-test model as specified in UL 943.

#### 3.5.5 Grounding System Test

Test grounding system to ensure continuity, and that resistance to ground is not excessive. Test each ground rod for resistance to ground before making connections to rod; tie grounding system together and test for resistance to ground. Make resistance measurements in dry weather, not earlier than 48 hours after rainfall. Submit written results of each test to Contracting Officer, and indicate location of rods as well as resistance and soil conditions at time measurements were made.

#### 3.5.6 Phase Rotation Test

Perform phase rotation test to ensure proper rotation of service power prior to operation of new or reinstalled equipment using a phase rotation meter. Follow the meter manual directions performing the test.

-- End of Section --



SECTION 26 24 13

SWITCHBOARDS

08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C12.1 (2014; Errata 2016) Electric Meters - Code for Electricity Metering

ASTM INTERNATIONAL (ASTM)

ASTM A123/A123M (2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A153/A153M (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM A240/A240M (2023) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

ASTM A653/A653M (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM A780/A780M (2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings

ASTM D709 (2017) Standard Specification for Laminated Thermosetting Materials

ASTM D1535 (2014; R 2018) Standard Practice for Specifying Color by the Munsell System

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 81 (2012) Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System

IEEE 100 (2000; Archived) The Authoritative Dictionary of IEEE Standards Terms

IEEE C2 (2023) National Electrical Safety Code

IEEE C37.90.1 (2013) Standard for Surge Withstand Capability (SWC) Tests for Relays and Relay Systems Associated with Electric Power Apparatus

IEEE C57.12.29 (2014) Standard for Pad-Mounted Equipment - Enclosure Integrity for Coastal Environments

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS (2021) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA PB 2.1 (2013) General Instructions for Proper Handling, Installation, Operation and Maintenance of Deadfront Distribution Switchboards Rated 600 V or Less

NEMA ICS 6 (1993; R 2016) Industrial Control and Systems: Enclosures

NEMA PB 2 (2011) Deadfront Distribution Switchboards

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

UNDERWRITERS LABORATORIES (UL)

UL 467 (2022) UL Standard for Safety Grounding and Bonding Equipment

UL 489 (2016; Rev 2019) UL Standard for Safety Molded-Case Circuit Breakers, Molded-Case Switches and Circuit-Breaker Enclosures

UL 891 (2019) UL Standard for Safety Switchboards

1.2 RELATED REQUIREMENTS

Section 26 08 00 APPARATUS INSPECTION AND TESTING applies to this section, with the additions and modifications specified herein.

1.3 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined in IEEE 100.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification

identifies the office that will review the submittal for the Government.  
Submit the following in accordance with Section 01 33 00 SUBMITTAL  
PROCEDURES:

SD-02 Shop Drawings

Switchboard Drawings; G

SD-03 Product Data

Switchboard; G

SD-06 Test Reports

Switchboard Design Tests; G

Switchboard Production Tests; G

Acceptance Checks and Tests; G

SD-07 Certificates

Cybersecurity Installation Certification; G

Submit certification indicating conformance with the paragraph  
CYBERSECURITY INSTALLATION CERTIFICATION.

SD-10 Operation and Maintenance Data

Switchboard Operation and Maintenance, Data Package 5; G

SD-11 Closeout Submittals

Assembled Operation and Maintenance Manuals; G

Equipment Test Schedule; G

Request for Settings; G

1.5 QUALITY ASSURANCE

1.5.1 Product Data

Include manufacturer's information on each submittal for each component, device and accessory provided with the switchboard including:

- a. Circuit breaker type, interrupting rating, and trip devices, including available settings.
- b. Manufacturer's instruction manuals and published time-current curves (in electronic format) of the main secondary breaker and largest secondary feeder device.

1.5.2 Switchboard Drawings

Include wiring diagrams and installation details of equipment indicating proposed location, layout and arrangement, control panels, accessories, piping, ductwork, and other items that must be shown to ensure a

coordinated installation. Identify circuit terminals on wiring diagrams and indicate the internal wiring for each item of equipment and the interconnection between each item of equipment. Indicate on the drawings adequate clearance for operation, maintenance, and replacement of operating equipment devices. Include the nameplate data, size, and capacity on submittal. Also include applicable federal, military, industry, and technical society publication references on submittals. Include the following:

- a. One-line diagram including breakers, fuses, current transformers, and meters.
- b. Outline drawings including front elevation, section views, footprint, and overall dimensions.
- c. Bus configuration including dimensions and ampere ratings of bus bars.
- d. Markings and NEMA nameplate data, including fuse information (manufacturer's name, catalog number, and ratings).
- e. Circuit breaker type, interrupting rating, and trip devices, including available settings.
- f. Wiring diagrams and elementary diagrams with terminals identified, and indicating prewired interconnections between items of equipment and the interconnection between the items.
- g. Manufacturer's instruction manuals and published time-current curves (in electronic format) of the main secondary breaker and largest secondary feeder device. Use this information (designer of record) to provide breaker settings that ensures protection and coordination are achieved. For Navy installations, provide electronic format curves using SKM's Power Tools for Windows device library electronic format or EasyPower device library format depending on installation modeling software requirements.
- h. Provisions for future expansion by adding switchboard sections.

#### 1.5.3 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" or "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Provide equipment, materials, installation, and workmanship in accordance with the mandatory and advisory provisions unless more stringent requirements are specified or indicated.

#### 1.5.4 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship, and:

- a. Have been in satisfactory commercial or industrial use for 2 years prior to bid opening including applications of equipment and materials under similar circumstances and of similar size.

- b. Have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.
- c. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

#### 1.5.4.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

#### 1.5.4.2 Material and Equipment Manufacturing Date

Products manufactured more than one year prior to date of delivery to site are not acceptable.

### 1.6 MAINTENANCE

#### 1.6.1 Switchboard Operation and Maintenance Data

Submit Operation and Maintenance Manuals in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

#### 1.6.2 Assembled Operation and Maintenance Manuals

Assemble and securely bind manuals in durable, hard covered, water resistant binders. Assemble and index the manuals in the following order with a table of contents:

- a. Manufacturer's O&M information required by the paragraph SD-10, OPERATION AND MAINTENANCE DATA.
- b. Catalog data required by the paragraph SD-03, PRODUCT DATA.
- c. Drawings required by the paragraph SD-02, SHOP DRAWINGS.
- d. Prices for spare parts and supply list.
- e. Information on metering.
- f. Design test reports.
- g. Production test reports.

#### 1.6.3 Spare Parts

Provide spare parts as specified below. Provide spare parts that are of the same material and workmanship, meet the same requirements, and are interchangeable with the corresponding original parts furnished.

- a. Quantity 2 - Fuses of each type and size.

### 1.7 WARRANTY

Provide equipment items that are supported by service organizations

reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

## PART 2 PRODUCTS

### 2.1 PRODUCT COORDINATION

Products and materials not considered to be switchboards and related accessories are specified in the drawings and Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

### 2.2 SWITCHBOARD

NEMA PB 2 and UL 891.

#### 2.2.1 Ratings

Provide equipment with the following ratings:

- a. Voltage rating: 480Y/277 volts AC, 4-wire as indicated.
- b. Continuous current rating of the main bus: as indicated.
- c. as indicated.
- d. UL listed and labeled as service entrance equipment.

#### 2.2.2 Construction

Provide the following:

- a. Switchboard: consisting of one or more vertical sections bolted together to form a rigid assembly and aligned as indicated.
- b. All circuit breakers: front accessible.
- c. Rear aligned switchboards: front accessible load connections.
- d. Where indicated, "space for future" or "space" means to include a vertical bus provided behind a blank front cover. Where indicated, "provision for future" means full hardware provided to mount a breaker suitable for the location.
- e. Completely factory engineered and assembled, including protective devices and equipment indicated with necessary interconnections, instrumentation, and control wiring.

##### 2.2.2.1 Enclosure

Provide the following:

- a. Enclosure: NEMA ICS 6 Type 4X as indicated fabricated entirely of 12 gauge ASTM A240/A240M 304L stainless steel.
- b. Enclosure: bolted together with removable bolt-on side and hinged rear covers, and sloping roof downward toward rear.
- c. Front doors: provided with stainless steel padlockable vault handles

with a three point catch.

- d. Bases, frames and channels of enclosure: corrosion resistant and fabricated of ASTM A240/A240M type 304 or 304L stainless steel. Separate sections using vertical steel barriers.
- e. Base: includes any part of enclosure that is within 3 inches of concrete pad.
- f. Galvanized steel: ASTM A123/A123M, ASTM A653/A653M G90 coating, and ASTM A153/A153M, as applicable. Galvanize after fabrication where practicable.
- g. Paint color: ASTM D1535 light gray No. 61 over rust inhibitor.
- h. Paint coating system: comply with IEEE C57.12.29 for stainless steel.

#### 2.2.2.2 Bus Bars

Provide the following:

- a. Bus bars: copper with silver-plated contact surfaces.
  - (1) Phase bus bars: uninsulated
  - (2) Neutral bus: rated 100 percent of the main bus continuous current rating.
- b. Make bus connections and joints with hardened steel bolts.
- c. Main-bus : rated at the full ampacity of the main throughout the switchboard. Vertical rating equals the horizontal bus rating for single section, for multiple sections the vertical equals 50% of horizontal bus.
- d. Minimum one-quarter by 2 inch copper ground bus secured to each vertical section along the entire length of the switchboard.

#### 2.2.2.3 Main Section

Provide the main section for the main an individually mounted fixed insulated-case circuit breaker.

#### 2.2.2.4 Distribution Sections

Provide the distribution sections consisting of individually mounted insulated-case circuit breakers as indicated.

#### 2.2.3 Protective Device

Provide main and branch protective devices as indicated.

##### 2.2.3.1 Insulated-Case Breaker

Provide the following:

- a. UL 489. UL listed and labeled, 100 percent rated main breaker manually operated with electronic adjustable trip LSI features, low voltage, insulated-case circuit breaker, with a short-circuit current rating as

indicated at 480V volts.

- b. Breaker frame size: as required to accommodate the number of conductors per phase.
- c. Series rated circuit breakers are unacceptable.

#### 2.2.3.2 Molded-Case Circuit Breaker

Provide the following:

- a. UL 489. UL listed and labeled, standard rated branch breakers, manually operated with electronic adjustable trip LSI feature for all circuit breakers 150A and greater and LI features for all circuit breakers 125A and below, low voltage molded-case circuit breaker, with a short-circuit current rating as indicated at 480V volts.
- b. Breaker frame size: as required to accommodate the number of conductors per phase.
- c. Series rated circuit breakers are unacceptable.

#### 2.2.4 Electronic Trip Units

Equip main and distribution breakers as indicated with a solid-state tripping system consisting of three current sensors and a microprocessor-based trip unit that provides true rms sensing adjustable time-current circuit protection. Include the following:

- a. Current sensors ampere rating: the same as the breaker frame rating.
- b. Trip unit ampere rating: as indicated.
- c. Ground fault protection: as indicated.
- d. Electronic trip units: provide additional features as indicated:
  - (1) Breakers: include long delay pick-up and time settings, and LED indication of cause of circuit breaker trip. Include short delay pick-up and time settings, instantaneous settings, and ground fault settings as indicated.
  - (2) Breakers: include provisions for communication via a network twisted pair cable for remote monitoring and control. Provide the following communications protocol: Modbus, IEC 61850.
  - (3) For electronic trip units that are rated for or can be adjusted to 1,200 amperes or higher, provide arc energy reduction capability with an energy-reducing maintenance switch with local status indicator.

#### 2.2.5 Metering

##### 2.2.5.1 Digital Meters

IEEE C37.90.1 for surge withstand. Provide true rms, plus/minus one percent accuracy, programmable, microprocessor-based meter enclosed in a sealed case with the following features.



a. Display capability:

- (1) Multi-Function Meter: Display a selected phase to neutral voltage, phase to phase voltage, percent phase to neutral voltage THD, percent phase to phase voltage THD; a selected phase current, neutral current, percent phase current THD, percent neutral current; selected total PF, kW, KVA, kVAR, FREQ, kVAh, kWh. Detected alarm conditions include over/under current, over/under voltage, over/under KVA, over/under frequency, over/under selected PF/kVAR, voltage phase reversal, voltage imbalance, reverse power, over percent THD.
- b. Design meters to accept input from standard 5A secondary instrument transformers and direct voltage monitoring range to 600 volts, phase to phase.
- c. Provide programming via a front panel display and a communication interface accessible by a computer.
- d. Provide password secured programming stored in non-volatile EEPROM memory.
- e. Provide digital communications in a Modbus RTU protocol via a RS485 serial port.
- f. Provide meter that calculates and stores average max/min demand values with time and date for all readings based on a user selectable sliding window averaging period.
- g. Provide meter with programmable hi/low set limits with two Form C dry contact relays when exceeding alarm conditions.
- h. Provide meter with a display of Total Harmonic Distortion (THD) measurement to a minimum of the thirty-first order.
- i. Include historical trend logging capability with the ability to store up to 100,000 data points with intervals of 1 second to 180 minutes. Provide a unit that can store and time stamp up to 1000 programmable triggered conditions.

#### 2.2.6 Terminal Boards

Provide with engraved plastic terminal strips and screw type terminals for external wiring between components and for internal wiring between removable assemblies. Provide short-circuiting type terminal boards associated with current transformer. Terminate conductors for current transformers with ring-tongue lugs. Provide terminal board identification that is identical in similar units. Provide color coded external wiring that is color coded consistently for similar terminal boards.

#### 2.2.7 Wire Marking

Mark control and metering conductors at each end. Provide factory installed, white, plastic tubing, heat stamped with black block type letters on factory-installed wiring. On field-installed wiring, provide white, preprinted, polyvinyl chloride (PVC) sleeves, heat stamped with black block type letters. Provide a single letter or number on each sleeve, elliptically shaped to securely grip the wire, and keyed in such a manner to ensure alignment with adjacent sleeves. Provide specific wire

markings using the appropriate combination of individual sleeves. Indicate on each wire marker the device or equipment, including specific terminal number to which the remote end of the wire is attached.

### 2.3 MANUFACTURER'S NAMEPLATE

Provide a nameplate on each item of equipment bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent is not acceptable. This nameplate and method of attachment may be the manufacturer's standard if it contains the required information.

### 2.4 FIELD FABRICATED NAMEPLATES

ASTM D709. Provide laminated plastic nameplates for each switchboard, equipment enclosure, relay, switch, and device; as specified in this section or as indicated on the drawings. Identify on each nameplate inscription the function and, when applicable, the position. Provide nameplates of melamine plastic, 0.125 inch thick, white with black center core. Provide matte finish surface. Provide square corners. Accurately align lettering and engrave into the core. Provide nameplates with minimum size of one by 2.5 inches. Provide lettering that is a minimum of 0.25 inch high normal block style.

### 2.5 SOURCE QUALITY CONTROL

#### 2.5.1 Equipment Test Schedule

The Government reserves the right to witness tests. Provide equipment test schedules for tests to be performed at the manufacturer's test facility. Submit required test schedule and location, and notify the Contracting Officer 30 calendar days before scheduled test date. Notify Contracting Officer 15 calendar days in advance of changes to scheduled date.

Provide the following as part of test equipment calibration:

- a. Provide a calibration program which assures that all applicable test instruments are maintained within rated accuracy.
- b. Accuracy: Traceable to the National Institute of Standards and Technology.
- c. Instrument calibration frequency schedule: less than or equal to 12 months for both test floor instruments and leased specialty equipment.
- d. Dated calibration labels: visible on all test equipment.
- e. Calibrating standard: higher accuracy than that of the instrument tested.
- f. Keep up-to-date records that indicate dates and test results of instruments calibrated or tested. For instruments calibrated by the manufacturer on a routine basis, in lieu of third party calibration, include the following:
  - (1) Maintain up-to-date instrument calibration instructions and procedures for each test instrument.

- (2) Identify the third party/laboratory calibrated instrument to verify that calibrating standard is met.

## 2.5.2 Switchboard Design Tests

NEMA PB 2 and UL 891.

### 2.5.2.1 Design Tests

Furnish documentation showing the results of design tests on a product of the same series and rating as that provided by this specification.

- a. Short-circuit current test.
- b. Enclosure tests.
- c. Dielectric test.

### 2.5.2.2 Additional Design Tests

In addition to normal design tests, perform the following tests on the actual equipment. Furnish reports which include results of design tests performed on the actual equipment.

- a. Temperature rise tests.
- b. Continuous current.

## 2.5.3 Switchboard Production Tests

NEMA PB 2 and UL 891. Furnish reports which include results of production tests performed on the actual equipment for this project. These tests include:

- a. 60-hertz dielectric tests.
- b. Mechanical operation tests.
- c. Electrical operation and control wiring tests.
- d. Ground fault sensing equipment test.

## 2.6 COORDINATED POWER SYSTEM PROTECTION

Provide a power system study as specified in Section 26 05 73 POWER SYSTEM STUDIES.

## 2.7 ARC FLASH WARNING LABEL

Provide warning label for switchboards. Locate this self-adhesive warning label on the outside of the enclosure warning of potential electrical arc flash hazards and appropriate PPE required. Provide label format as indicated.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Conform to IEEE C2, and to the requirements specified herein. Provide new

equipment and materials unless indicated or specified otherwise.

### 3.2 GROUNDING

NFPA 70 and IEEE C2, except that grounds and grounding systems with a resistance to solid earth ground not exceeding 15 ohms.

#### 3.2.1 Grounding Electrodes

Provide driven ground rods as shown on the drawings. Connect ground conductors to the upper end of the ground rods by exothermic weld or compression connector. Provide compression connectors at equipment end of ground conductors.

#### 3.2.2 Equipment Grounding

Provide bare copper cable not smaller than No. 4/0 AWG not less than 24 inches below grade connecting to the indicated ground rods. When work in addition to that indicated or specified is directed to obtain the specified ground resistance, the provision of the contract covering "Changes" applies.

#### 3.2.3 Connections

Make joints in grounding conductors and loops by exothermic weld or compression connector.

#### 3.2.4 Grounding and Bonding Equipment

UL 467, except as indicated or specified otherwise.

### 3.3 INSTALLATION OF EQUIPMENT AND ASSEMBLIES

Install and connect equipment furnished under this section as indicated on project drawings, the approved shop drawings, and as specified herein.

#### 3.3.1 Switchboard

ANSI/NEMA PB 2.1.

#### 3.3.2 Meters and Instrument Transformers

ANSI C12.1.

#### 3.3.3 Field Applied Painting

Where field painting of enclosures is required to correct damage to the manufacturer's factory applied coatings, provide manufacturer's recommended coatings and apply in accordance with manufacturer's instructions.

#### 3.3.4 Galvanizing Repair

Repair damage to galvanized coatings using ASTM A780/A780M, zinc rich paint, for galvanizing damaged by handling, transporting, cutting, welding, or bolting. Do not heat surfaces that repair paint has been applied to.

### 3.3.5 Field Fabricated Nameplate Mounting

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

## 3.4 FOUNDATION FOR EQUIPMENT AND ASSEMBLIES

### 3.4.1 Interior Location

Mount switchboard on concrete slab as follows:

- a. Unless otherwise indicated, provide the slab with dimensions at least 4 inches thick.
- b. Install slab such that the top of the concrete slab is approximately 4 inches above the finished grade.
- c. Provide edges above grade 1/2 inch chamfer.
- d. Provide slab of adequate size to project at least 8 inches beyond the equipment.
- e. Provide conduit turnups and cable entrance space required by the equipment to be mounted.
- f. Seal voids around conduit openings in slab with water- and oil-resistant caulking or sealant.
- g. Cut off and bush conduits 3 inches above slab surface.

## 3.5 FIELD QUALITY CONTROL

Submit request for settings of breakers to the Contracting Officer after approval of switchboard and at least 30 days in advance of their requirement.

### 3.5.1 Performance of Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations and include the following visual and mechanical inspections and electrical tests, performed in accordance with NETA ATS.

#### 3.5.1.1 Switchboard Assemblies

- a. Visual and Mechanical Inspection
  - (1) Compare equipment nameplate data with specifications and approved shop drawings.
  - (2) Inspect physical, electrical, and mechanical condition.
  - (3) Verify appropriate anchorage, required area clearances, and correct alignment.
  - (4) Clean switchboard and verify shipping bracing, loose parts, and documentation shipped inside cubicles have been removed.
  - (5) Inspect all doors, panels, and sections for paint, dents,

scratches, fit, and missing hardware.

- (6) Verify that fuse and circuit breaker sizes and types correspond to approved shop drawings as well as to the circuit breaker's address for microprocessor-communication packages.
- (7) Verify that current transformer ratios correspond to approved shop drawings.
- (8) Inspect all bolted electrical connections for high resistance using low-resistance ohmmeter, verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey.
- (9) Confirm correct operation and sequencing of electrical and mechanical interlock systems.
- (10) Confirm correct application of manufacturer's recommended lubricants.
- (11) Inspect insulators for evidence of physical damage or contaminated surfaces.
- (12) Verify correct barrier installation and operation.
- (13) Exercise all active components.
- (14) Inspect all mechanical indicating devices for correct operation.
- (15) Verify that filters are in place and vents are clear.
- (16) Test operation, alignment, and penetration of instrument transformer withdrawal disconnects.
- (17) Inspect control power transformers.

b. Electrical Tests

- (1) Perform insulation-resistance tests on each bus section.
- (2) Perform dielectric withstand voltage tests.
- (3) Perform insulation-resistance test on control wiring; Do not perform this test on wiring connected to solid-state components.
- (4) Perform control wiring performance test.
- (5) Perform primary current injection tests on the entire current circuit in each section of assembly.
- (6) Perform phasing check on double-ended switchboard to ensure correct bus phasing from each source.
- (7) Verify operation of switchboard heaters.

3.5.1.2 Circuit Breakers

Low Voltage - Insulated-Case and Low Voltage Molded Case with Solid State Trips

a. Visual and Mechanical Inspection

- (1) Compare nameplate data with specifications and approved shop drawings.
- (2) Inspect circuit breaker for correct mounting.
- (3) Operate circuit breaker to ensure smooth operation.
- (4) Inspect case for cracks or other defects.
- (5) Inspect all bolted electrical connections for high resistance using low resistance ohmmeter, verifying tightness of accessible bolted connections and/or cable connections by calibrated torque-wrench method, or performing thermographic survey.
- (6) Inspect mechanism contacts and arc chutes in unsealed units.

b. Electrical Tests

- (1) Perform contact-resistance tests.
- (2) Perform insulation-resistance tests.
- (3) Perform Breaker adjustments for final settings in accordance with Government provided settings.
- (4) Perform long-time delay time-current characteristic tests
- (5) Determine short-time pickup and delay by primary current injection.
- (6) Determine ground-fault pickup and time delay by primary current injection.
- (7) Determine instantaneous pickup current by primary injection.
- (8) Verify correct operation of any auxiliary features such as trip and pickup indicators, zone interlocking, electrical close and trip operation, trip-free, and anti-pump function.

3.5.1.3 Current Transformers

a. Visual and Mechanical Inspection

- (1) Compare equipment nameplate data with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Verify correct connection.
- (4) Verify that adequate clearances exist between primary and secondary circuit.
- (5) Inspect all bolted electrical connections for high resistance using low-resistance ohmmeter, verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey.

- (6) Verify that all required grounding and shorting connections provide good contact.

b. Electrical Tests

- (1) Perform resistance measurements through all bolted connections with low-resistance ohmmeter, if applicable.
- (2) Perform insulation-resistance tests.
- (3) Perform polarity tests.
- (4) Perform ratio-verification tests.

3.5.1.4 Metering and Instrumentation

a. Visual and Mechanical Inspection

- (1) Compare equipment nameplate data with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Verify tightness of electrical connections.

b. Electrical Tests

- (1) Determine accuracy of meters at 25, 50, 75, and 100 percent of full scale.
- (2) Calibrate watthour meters according to manufacturer's published data.
- (3) Verify all instrument multipliers.
- (4) Electrically confirm that current transformer and voltage transformer secondary circuits are intact.

3.5.1.5 Grounding System

a. Visual and Mechanical Inspection

- (1) Inspect ground system for compliance with contract plans and specifications.

b. Electrical Tests

- (1) IEEE 81. Perform ground-impedance measurements utilizing the fall-of-potential method. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of a single ground rod perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground resistance tester in accordance with manufacturer's instructions to test each ground or group of grounds. Use an instrument equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test.



- (2) Submit the measured ground resistance of each ground rod and grounding system, indicating the location of the rod and grounding system. Include the test method and test setup (i.e., pin location) used to determine ground resistance and soil conditions at the time the measurements were made.

#### 3.5.1.6 Cybersecurity Installation Certification

Furnish a certification that control systems are installed in accordance with DoD Instruction 8500.01, DoD Instruction 8510.01, and as required by individual Service Implementation Policy.

#### 3.5.2 Follow-Up Verification

Upon completion of acceptance checks, settings, and tests, show by demonstration in service that circuits and devices are in good operating condition and properly performing the intended function. Trip circuit breakers by operation of each protective device. Test each item to perform its function not less than three times. As an exception to requirements stated elsewhere in the contract, provide the Contracting Officer five working days advance notice of the dates and times for checks, settings, and tests.

-- End of Section --

SECTION 26 29 23

ADJUSTABLE SPEED DRIVE (ASD) SYSTEMS UNDER 600 VOLTS

02/20, CHG 1: 05/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

EUROPEAN COMMITTEE FOR STANDARDIZATION (CEN/CENELEC)

EN 61800-3 (2017) Requirements for the Control of  
Electromagnetic Interference  
Characteristics of Subsystems and Equipment

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 519 (2022) Standard for Harmonic Control in  
Electrical Power Systems

IEEE C62.41.1 (2002; R 2008) Guide on the Surges  
Environment in Low-Voltage (1000 V and  
Less) AC Power Circuits

IEEE C62.41.2 (2002) Recommended Practice on  
Characterization of Surges in Low-Voltage  
(1000 V and Less) AC Power Circuits

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 61000-3-12 (2012) Electromagnetic Compatibility (EMC)  
- Part 3-12: Limits - Limits for harmonic  
currents produced by equipment connected  
to public low-voltage systems with input  
current >16 A and ≤75 A per phase

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250 (2020) Enclosures for Electrical Equipment  
(1000 Volts Maximum)

NEMA ICS 1 (2022) Standard for Industrial Control and  
Systems: General Requirements

NEMA ICS 3.1 (2019) Guide for the Application,  
Handling, Storage, Installation and  
Maintenance of Medium-Voltage AC  
Contactors, Controllers and Control Centers

NEMA ICS 6 (1993; R 2016) Industrial Control and  
Systems: Enclosures

NEMA ICS 7 (2020) Adjustable-Speed Drives

|                                                          |                                                                                                                                                                           |
|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NEMA ICS 7.2                                             | (2015) Application Guide for AC Adjustable Speed Drive Systems                                                                                                            |
| NEMA ICS 61800-2                                         | (2005) Adjustable Speed Electrical Power Drive Systems Part 2: General Requirements - Rating Specifications for Low Voltage Adjustable Frequency A.C. Power Drive Systems |
| NEMA MG 1                                                | (2021) Motors and Generators                                                                                                                                              |
| NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)              |                                                                                                                                                                           |
| NFPA 70                                                  | (2023) National Electrical Code                                                                                                                                           |
| U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA) |                                                                                                                                                                           |
| 47 CFR 15                                                | Radio Frequency Devices                                                                                                                                                   |
| UNDERWRITERS LABORATORIES (UL)                           |                                                                                                                                                                           |
| UL 489                                                   | (2016; Rev 2019) UL Standard for Safety Molded-Case Circuit Breakers, Molded-Case Switches and Circuit-Breaker Enclosures                                                 |
| UL 61800-5-1                                             | (2016) Adjustable Speed Electrical Power Drive Systems - Part 5-1: Safety Requirements - Electrical, Thermal and Energy                                                   |

## 1.2 RELATED REQUIREMENTS

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM applies to this section with additions and modifications specified herein.

## 1.3 SYSTEM DESCRIPTION

### 1.3.1 Performance Requirements

#### 1.3.1.1 Electromagnetic Interference Suppression

Computing devices, as defined by 47 CFR 15 and EN 61800-3 rules and regulations, must be certified to comply with the requirements for class A computing devices and labeled.

#### 1.3.1.2 Electromechanical and Electrical Components

Ensure electrical and electromechanical components of the Adjustable Speed Drive (ASD) do not cause electromagnetic interference to adjacent electrical or electromechanical equipment while in operation.

### 1.3.2 Electrical Requirements

#### 1.3.2.1 Power Line Surge Protection

IEEE C62.41.1 and IEEE C62.41.2, IEEE 519, IEC 61000-3-12 Control panel must have surge protection, included within the panel to protect the unit from damaging transient voltage surges. Surge protective device must be mounted near the incoming power source and properly wired to all three

phases and ground. Fuses must not be used for surge protection.

#### 1.3.2.2 Sensor and Control Wiring Surge Protection

I/O functions as specified must be protected against surges induced on control and sensor wiring installed outdoors and as shown. Test the inputs and outputs in both normal mode and common mode using the following two waveforms:

- a. A 10 microsecond by 1000 microsecond waveform with a peak voltage of 1500 volts and a peak current of 60 amperes.
- b. An 8 microsecond by 20 microsecond waveform with a peak voltage of 1000 volts and a peak current of 500 amperes.

#### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-02 Shop Drawings

Schematic Diagrams; G  
Interconnecting Diagrams; G  
Installation Drawings; G  
As-Built Drawings; G

##### SD-03 Product Data

Adjustable Speed Drives; G  
Wires and Cables  
Equipment Schedule

##### SD-06 Test Reports

ASD Test  
Performance Verification Tests  
Endurance Test

##### SD-07 Certificates

Testing Agency's Field Supervisor NETA Certificate; G

##### SD-08 Manufacturer's Instructions

Installation instructions

##### SD-09 Manufacturer's Field Reports

ASD Test Plan; G

Standard Products

SD-10 Operation and Maintenance Data

Adjustable Speed Drives, Data Package 4

## 1.5 QUALITY ASSURANCE

### 1.5.1 Schematic Diagrams

Submit diagrams showing circuits and device elements for each replaceable module. Schematic diagrams of printed circuit boards are permitted to group functional assemblies as devices, provided that sufficient information is provided for government maintenance personnel to verify proper operation of the functional assemblies.

### 1.5.2 Interconnecting Diagrams

Show interconnections between equipment assemblies, and external interfaces, including power and signal conductors. Include for enclosures and external devices.

### 1.5.3 Installation Drawings

Show floor plan of each site, with ASD's and motors indicated. Indicate ventilation requirements, adequate clearances, and cable routes. Submit drawings for government approval prior to equipment construction or integration. Immediately record modifications to original drawings made during installation for inclusion into the as-built drawings.

### 1.5.4 Equipment Schedule

Provide schedule of equipment supplied. Schedule must provide a cross reference between manufacturer data and identifiers indicated in shop drawings. Schedule must include the total quantity of each item of equipment supplied and data indicating compatibility with motors being driven. For complete assemblies, such as ASD's, provide the serial numbers of each assembly, and a sub-schedule of components within the assembly. Provide recommended spare parts listing for each assembly or component.

### 1.5.5 Installation Instructions

Provide installation instructions issued by the manufacturer of the equipment, including notes and recommendations, prior to shipment to the site. Provide operation instructions prior to acceptance testing.

### 1.5.6 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship and:

- a. Have been in satisfactory commercial or industrial use for 2 years prior to bid opening including applications of equipment and materials under similar circumstances and of similar size.

- b. Have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.
- c. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

#### 1.6 DELIVERY AND STORAGE

Store delivered equipment to protect from the weather, humidity and temperature variations, dirt and dust, or other contaminants.

#### 1.7 WARRANTY

The complete system must be warranted by the manufacturer for a period of five years. Repair or replace any component failing to perform its function as specified and documented at no additional cost to the Government. Items repaired or replaced must be warranted for an additional period of at least one year from the date that it becomes functional again, as specified in FAR 52.246-21 Warranty of Construction.

#### 1.8 MAINTENANCE

##### 1.8.1 Spare Parts

Manufacturers provide spare parts in accordance with recommended spare parts list.

- a. Furnish extra materials that match products installed and that are packaged with protective covering for storage and identified with labels describing contents.
  - (1) Power Fuses: Equal to 10 percent of quantity installed for each size and type, but no fewer than 3 of each size and type.
  - (2) Control Power Fuses: Equal to 10 percent of quantity installed for each size and type, but no fewer than 3 of each size and type.
  - (3) Indicating Lights: 4 of each type and color installed.
  - (4) Auxiliary Contacts: Furnish 2 spare(s) for each size and type of magnetic controller installed.
  - (5) Power Contacts: Furnish 3 spare(s) for each size and type of magnetic contactor installed.

##### 1.8.2 Operation and Maintenance Data

Provide in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA. Provide service and maintenance information including preventive maintenance, assembly, and disassembly procedures. Include electrical drawings from electrical general sections. Provide additional information necessary to provide complete operation, repair, and maintenance information, detailed to the smallest replaceable unit. Include copies of as-built submittals. Provide routine preventative maintenance instructions, and equipment required. Provide instructions on how to modify program settings, and modify the control program. Provide

instructions on drive adjustment, trouble-shooting, and configuration.  
Provide instructions on process tuning and system calibration.

#### 1.8.3 Maintenance Support

During the warranty period, provide on-site, on-call maintenance services by drive manufacturer's personnel on the following basis: The service must be on a per-call basis with 36 hour response. Contractor is responsible for the maintenance of all hardware and software of the system during the warranty period. Various personnel of different expertise must be sent on-site depending on the nature of the maintenance service required. Costs must include travel, local transportation, living expenses, and labor rates of the service personnel while responding to the service request. The provisions of this Section are not in lieu of, nor relieve the Contractor of, warranty responsibilities covered in this specification. Should the result of the service request be the uncovering of a system defect covered under the warranty provisions, all costs for the call, including the labor necessary to identify the defect, must be borne by the Contractor.

#### 1.8.4 Technical Support

Provide the ASDs with manufacturer's technical telephone support in English, readily available during normal working hours.

### PART 2 PRODUCTS

#### 2.1 ADJUSTABLE SPEED DRIVES (ASD)

Provide adjustable speed drive to control the speed of induction motor(s). The ASD must include the following minimum functions, features and ratings.

- a. Input circuit breaker per UL 489 with a minimum of 65,000 amps symmetrical interrupting capacity and door interlocked external operator.
- b. A converter stage per UL 61800-5-1 must change fixed voltage, fixed frequency, ac line power to a fixed dc voltage. The converter must utilize a full wave bridge design incorporating diode rectifiers. Silicon Controlled Rectifiers (SCR) are not acceptable. The converter must be insensitive to three phase rotation of the ac line and must not cause displacement power factor of less than .95 lagging under any speed and load condition.
- c. An inverter stage must change fixed dc voltage to variable frequency, variable ac voltage for application to a standard NEMA MG 1 Part 30 motor designed for use with adjustable frequency power supplies. Switch the inverter to produce a sine coded pulse width modulated (PWM) output waveform.
- d. The ASD shall be capable of supplying 110 percent of rated full load current for one minute at maximum ambient temperature.
- e. The ASD must be designed to operate from a 480V volt, plus or minus 10 percent, three phase, 60 Hz supply, and control motors with a corresponding voltage rating.
- f. Acceleration and deceleration time must be independently adjustable

from one second to 60 seconds.

Adjust decelerating time by providing an external dynamic braking resistor designed to meet NEMA ICS 61800-2 to be capable of decelerating six times the motor inertia with no more than 150 percent of rated current with the motor at its base speed. Required deceleration time may be achieved using not only dynamic braking resistor but with other methods described in NEMA ICS 7.2-2015 paragraph 5.2.5.

- g. Adjustable full-time current limiting must limit the current to a preset value which must not exceed 110 percent of the controller rated current. The current limiting action must maintain the V/Hz ratio constant so that variable torque can be maintained. Short time starting override must allow starting current to reach 175 percent of controller rated current to maximum starting torque.
- h. The controllers must be capable of producing an output frequency over the range of 3 Hz to 60 Hz (20 to one speed range), without low speed cogging. Over frequency protection must be included such that a failure in the controller electronic circuitry must not cause frequency to exceed 110 percent of the maximum controller output frequency selected.
- i. Minimum and maximum output frequency must be adjustable over the following ranges: 1) Minimum frequency 3 Hz to 50 percent of maximum selected frequency; 2) Maximum frequency 40 Hz to 60 Hz.
- j. The controller efficiency at any speed must not be less than 96 percent.
- k. The controllers must be capable of being restarted into a motor coasting in the forward direction without tripping.
- l. Protection of power semiconductor components must be accomplished without the use of fast acting semiconductor output fuses. Subjecting the controllers to any of the following conditions must not result in component failure or the need for fuse replacement:
  - (1) Short circuit at controller output
  - (2) Ground fault at controller output
  - (3) Open circuit at controller output
  - (4) Input undervoltage
  - (5) Input overvoltage
  - (6) Loss of input phase
  - (7) AC line switching transients
  - (8) Instantaneous overload
  - (9) Sustained overload exceeding 115 percent of controller rated current
  - (10) Over temperature



(11) Phase reversal

- m. Solid state motor overload protection must be included such that current exceeding an adjustable threshold must activate a 60 second timing circuit. Should current remain above the threshold continuously for the timing period, the controller will automatically shut down. Havesensor in each phase, Class 10/20 selectable tripping characteristic selected to protect motor against voltage and current unbalance and single phasing, Class II ground-fault protection, with start and run delays to prevent nuisance trip on starting, analog communication module, NC/NO isolated overload alarm contact, external overload, reset push button.
- n. Include slip compensation circuit that will sense changing motor load conditions and adjust output frequency to provide speed regulation of NEMA MG 1 Part 30 designed for use with adjustable frequency power supplies motors to within plus or minus 0.5 percent of maximum speed without the necessity of a tachometer generator.
- o. The controller shall not create any feedback noise on the input line that will adversely affect electronic or microprocessor based equipment (such as computers or electron microscopes), and the controller shall not impress voltage or current spikes on modes the system. The minimum requirements shall conform to IEEE Standard 519, Special Applications for Line Notching and Distortion. The manufacturer shall provide at no additional cost any equipment to meet this requirement; i.e., A.C. line filters of the RLC type and/or isolation transformer, or both as required to meet full compliance with IEEE 519, if controller does not meet all standards.
- p. The ASD must include external fault reset capability. All the necessary logic to accept an external fault reset contact must be included.
- q. Provide critical speed lockout circuitry to prevent operating at frequencies with critical harmonics that cause resonant vibrations. The ASD must have a minimum of three user selectable bandwidths.
- r. General Requirements
  - (1) Provide a factory-assembled bypass system consisting of IEC-rated bypass and isolation contactors, configured to enable motor operation in the event of ASD (adjustable speed drive) failure, or to allow for safe isolation during maintenance.
  - (2) The bypass system shall support manual and automatic transfer of the motor between VFD output and bypass power, using a field-selectable control mode.
  - (3) Provide mechanical and electrical interlocks between bypass and isolation contactors to prevent simultaneous energization.
  - (4) Include a selector switch, transfer delay timer, and status indication lights (e.g., "Drive," "Bypass," and "Fault") mounted on the enclosure door.
  - (5) Bypass systems shall maintain motor overload and short-circuit protection while operating in bypass mode.

s. Bypass Type Based on Motor Size

(1) Motors less than 50 HP:

- (a) Provide an Across-the-Line (ATL) Bypass.
- (b) ATL bypass shall use full-voltage contactors rated for motor duty, with short-circuit protection coordinated with the motor branch circuit.
- (c) System shall support automatic restart on VFD failure, with adjustable time delay.

(2) Motors 50 HP and above:

- (a) Provide a Soft Starter Bypass.
- (b) Soft starter shall be solid-state, three-phase, with adjustable start and stop ramps, and integrated overload protection.
- (c) Soft starter shall be configured for full-voltage start with reduced current to minimize mechanical stress and power system voltage dip.
- (d) Provide control logic to automatically transfer to soft start mode upon VFD fault or loss of signal, with a manual override switch.

t. Additional Requirements for All Bypass Systems

- (1) All bypass components shall be UL listed and suitable for the voltage and motor ratings specified.
- (2) Enclosures shall be NEMA 1 or NEMA 12 as required, with adequate ventilation and labeled wiring terminals.
- (3) Provide remote bypass status indication and fault outputs for interface with BAS/SCADA systems, where applicable.
- (4) Provide all wiring, relays, and terminal blocks necessary to fully integrate the bypass system with the drive and motor controls.

u. LINE CONDITIONING AND FILTERING

- (1) Input Line Conditioning: Based on the manufacturer's harmonic analysis study and report, provide input filtering, as required, to limit total demand (harmonic current) distortion and total harmonic voltage demand at the defined point of common coupling to meet IEEE 519 recommendations.
- (2) The variable frequency drive (VFD) shall include as a minimum an integral 3% line reactors to reduce line noise generated by the inverter and 5% DC reactor to limit harmonics. The VFD manufacturer shall prepare and submit calculations confirming that the line harmonics and notching generated by the VFD do not exceed the levels for a general system as stated in the Institute of Electrical and Electronics Engineers Standard 519 "Guide for

Harmonic Control of Static Power Converters" to a maximum of 5% Total Voltage Harmonic Distortion (VTHD), and 8% TDD for current at Point of Common Coupling (PCC). The POC shall be assumed as the Service switchboard for VTHD and TDD calculations.

- (3) Line notch area shall not exceed 22,800 (V/S) and 20% notch depth.
  - (4) The manufacturer shall provide additional line reactors or isolation transformers to ensure that this standard is met, as necessary.
  - (5) Independent timed linear acceleration and deceleration functions will be provided, adjustable from 4 to 20 seconds.
  - (6) Output Filtering: Motor dv/dt filter shall be provided for motor cable runs exceeding 100 feet for motors with a peak voltage insulation rating less than 1600 Vac. The dv/dt filter shall be located at the AFD and shall reduce the dv/dt clamp any voltage overshoots of the AFD output. It will return the energy in the voltage overshoots to the AFD Dc bus. A power dissipating resistance device is not acceptable.
  - (7) The adjustable speed drive shall be designed to meet the guidelines regarding emissions of Radio Frequency and Electro-magnetic Interference (RFI/EMI) set by FCC Class A guidelines through the use of a tuned line filter, adjuste
- v. Minimum Operating Conditions. Designed and constructed ASD's to operate within the following service conditions:
- (1) Ambient Temperature Rating: 0 to 120 degrees F.
  - (2) Non-condensing relative humidity rating: less than 95 percent.
  - (3) Ambient rating: Not exceed 3,300 feet.
- w. Adjustable frequency drives for motors 50 hp and above shall be 18 pulse drives to mitigate the impact of harmonics on the building electric system.

#### 18 PULSE VFD REQUIREMENTS

- (1) The VFD shall provide microprocessor-based control for three-phase induction motors. The controller's full load output current rating shall be based on 50 degree C ambient at 250HP and below and 40 degree ambient above 250HP VT and no less than a 3.6 kHz switching frequency to reduce motor noise and avoid increased motor losses. Drive shall have been tested to and UL listed as conforming to the requirements of UL508C at rated load currents and ambient temperature per this specification. Drive shall have a UL listed interrupting rating of 100kAIC.
- (2) The VFD shall be of the Pulse Width Modulated (PWM) design converting the utility input voltage and frequency to a variable voltage and frequency output via a two-step operation. Adjustable Current Source VFD is not acceptable. Insulated Gate Bipolar Transistors (IGBT's) shall be used in the inverter section. Bipolar Junction Transistors, GTO's or SCR's are not acceptable. The VFD shall run at the above listed switching frequency.

- (3) The VFD shall have an efficiency at full load and speed that exceeds 95%. The efficiency shall exceed 90% at 50% speed and load. The VFDs shall maintain the line side displacement power factor at no less than 0.96, regardless of speed and load.
- (4) 18 pulse drive shall include the following protective features, as a minimum: Input line fuses, Metal Oxide Varistor (MOV), and Lineside choke rated 7.5% of rated line current.
- (5) The VFDs shall have a one (1) minute overload current rating of 110% for variable torque loads 150% for constant torque loads.
- (6) The AFDs shall be capable of operating of operating any NEMA design B squirrel cage induction motor, regardless of manufacturer, with a horsepower and current rating within the capacity of the VFD.
- (7) For systems with a standby generator, the harmonic distortion shall meet the distortion levels shown in table 10.3 of IEEE519-1992 for ISC/IL<20. The contractor shall provide the following data for the stand

#### 2.1.1 ASD for Industrial Application

Provide the following operator control and monitoring devices mounted on the front panel of the ASD:

- a. Manual speed potentiometer.
- b. Hand-Off-Auto (HOA) switch.
- c. Power on light.
- d. Drive run power light.
- e. Local display capable of including ASD status, frequency, motor RPM, phase current, fault diagnostic in descriptive text, and all programmed parameters.

#### 2.1.2 ASD for Pump Application

ASDs must have the following features:

- a. A local operator control providing the following functions:
  - (1) Remote/Local operator selection with password access.
  - (2) Run/Stop and manual speed commands.
  - (3) All programming functions.
  - (4) Scrolling through all display functions.
- b. A local operator control panel with the following data displayed:
  - (1) ASD status.
  - (2) Frequency.

- (3) Motor RPM.
- (4) Phase current.
- (5) Scrolling through all display functions.
- (6) Fault diagnostics in descriptive text.
- (7) All programmed parameters.
- c. Standard PI loop controller with input terminal for controlled variable and parameter settings.
- d. User interface terminals for remote control of ASD speed, speed feedback, and an isolated form C SPDT relay, which energizes on a drive fault condition.
- e. An isolated form C SPDT auxiliary relay which energizes on a run command.
- f. An adjustable carrier frequency with 16 KHz minimum upper limit.
- g. A built-in or external line reactor with 3 percent minimum impedance to protect the DC bus capacitors and rectifier section diodes, reduce power line transient voltage, line notching, DC bus over-voltage tripping and improve the inverter over-current and over-voltage conditions.
- h. Historical logging information and displays:
  - (1) Real-time clock with current time and date.
  - (2) Running log of total power versus time.
  - (3) Total run time.
  - (4) Fault log, maintaining last four faults with time and data stamp for each.
- i. The ASD must be capable of automatic control by a remote 4-20 mA, 0 to 10 VDC signal, by BACnet network command, or manually by the ASD control panel.
- j. ASDs must include the following operator programmable parameters:
  - (1) Upper and lower limit frequency.
  - (2) Acceleration and deceleration rate.
  - (3) Variable torque volts per Hertz curve.
  - (4) Starting voltage level.
  - (5) Starting frequency level.
  - (6) Display speed scaling.
  - (7) Enable/disable soft stall feature.

- (8) Motor overload level.
  - (9) Motor stall level.
  - (10) Jump frequency and hysteresis band.
  - (11) PWM carrier frequency.
- k. ASD must have the following protective features:
- (1) An electronic adjustable inverse time current limit with consideration for additional heating of the motor at frequencies below 45Hz, for the protection of the motor.
  - (2) An electronic adjustable soft stall feature, allowing the ASD to lower the frequency to a point where the motor will not exceed the full-load amperage when an overload ASD will automatically return to the requested frequency when load conditions permit.
  - (3) A separate electronic stall at 110 percent ASD rated current, and a separate hardware trip at 190 percent current.
  - (4) The ability to shut down if inadvertently started into a rotating load without damaging the ASD or the motor.
  - (5) The ability to keep a log of a minimum of four previous fault conditions, indicating the fault type and time of occurrence in descriptive text.
  - (6) The ability to sustain 110 percent rated current for 60 seconds.
  - (7) The ability to shutdown safely or protect against and record the following fault conditions:
    - (a) Over current (and an indication if the over current was during acceleration, deceleration, or running).
    - (b) Over current internal to the drive.
    - (c) Motor overload at start-up.
    - (d) Over voltage from utility power.
    - (e) Motor running overload.
    - (f) Over voltage during deceleration.
    - (g) ASD over heat.
    - (h) Load and ground fault.
    - (h) Abnormal parameters or data in ASD EEPROM.

## 2.2 ENCLOSURES

Provide equipment enclosures conforming to NEMA 250, Type 4X stainless steel NEMA ICS 7, and NEMA ICS 6, with a heater if located outdoors. An HMCP device shall provide the disconnecting means. The operating handle

shall protrude through the door, but the disconnect shall not be mounted on the door. The handle shall indicate ON, OFF, and tripped conditions. The handle shall have provisions to accommodate a minimum of three padlocks in the OFF position. Interlocks shall prevent unauthorized opening or closing of the ASD door with the disconnect handle in the ON position. The door handle interlock should have provisions to be defeated by qualified maintenance personnel.

## 2.3 WIRES AND CABLES

All wires and cables must conform to NEMA 250, NEMA ICS 7, NFPA 70.

## 2.4 NAMEPLATES

Nameplates external to NEMA enclosures must conform with the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide manufacturer's standard, permanent nameplates for internal areas of enclosures.

## 2.5 SOURCE QUALITY CONTROL

### 2.5.1 ASD Test Plan

To ensure quality, each ASD must be subject to a series of in-plant quality control inspections before approval for shipment from the manufacturer's facilities. Provide test plans.

### 2.5.2 ASD Test Report

To ensure quality, each ASD must be subject to a series of in-plant quality control inspections before approval for shipment from the manufacturer's facilities. Provide test reports.

## 2.6 ACCESSORIES

- a. General Requirements for Control-Circuit and Pilot Devices: NEMA ICS 5; factory installed in VFC enclosure cover unless otherwise indicated.
  - (1) Push Buttons: Lockable.
  - (2) Pilot Lights: Push to test.
  - (3) Selector Switches: Rotary type.
  - (4) Stop and Lockout Push-Button Station: Momentary-break, push-button station with a factory-applied hasp arranged so padlock can be used to lock push button in depressed position with control circuit open.
- b. NC/NO bypass contactor auxiliary contact(s).
- c. Control Relays: Auxiliary and adjustable solid-state time-delay relays.
- d. Phase-Failure, Phase-Reversal, and Undervoltage and Overvoltage Relays: Solid-state sensing circuit with isolated output contacts for hard-wired connections. Provide adjustable undervoltage, overvoltage, and time-delay settings.
  - (1) Current Transformers: Continuous current rating, basic impulse insulating level (BIL) rating, burden, and accuracy class suitable

for connected circuitry. Comply with IEEE C57.13.

e. Supplemental Digital Meters:

- (1) Elapsed-time meter.
- (2) Kilowatt meter.
- (3) Kilowatt-hour meter.

- f. Breather and drain assemblies, to maintain interior pressure and release condensation in NEMA 250, Type 4, Type 4X, Type 12 enclosures installed outdoors or in unconditioned interior spaces subject to humidity and temperature swings.
- g. Space heaters, with NC auxiliary contacts, to mitigate condensation in NEMA 250, Type 3R and Type 4X enclosures installed outdoors or in unconditioned interior spaces subject to humidity and temperature swings.
- h. Cooling Fan and Exhaust System: For NEMA 250, Type 1 or Type 12; UL 508 component recognized: Supply fan, with composite intake and exhaust grills and filters; 120-V ac; obtained from integral CPT.
- i. Sun shields installed on fronts, sides, and tops of enclosures installed outdoors and subject to direct and extended sun exposure.
- j. Spare control-wiring terminal blocks wired.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

Per NEMA ICS 3.1, install equipment in accordance with the approved manufacturer's printed installation drawings, instructions, wiring diagrams, and as indicated on project drawings and the approved shop drawings. A field representative of the drive manufacturer must supervise the installation of all equipment, and wiring.

#### 3.2 GROUNDING

Per NEMA ICS 7.2, ASD must be solidly grounded to the main distribution.

a. General

Grounding shall comply with NEC Article 250, NEMA ICS 7.2, and Section 26 20 00 - Grounding and Bonding for Electrical Systems.

b. VFD Enclosure Grounding

- (1) VFD enclosures shall be solidly grounded to the building electrical service ground.
- (2) Provide copper equipment grounding conductors (EGCs) in all raceways feeding the VFDs, sized per NEC Table 250.122.
- (3) Bond all metallic conduit, cable tray, and enclosures entering or exiting the VFD with grounding bushings or locknuts.

c. Motor Grounding



- (1) Ground all pump motor frames to the facility equipment ground system using UL-listed grounding lugs.
- (2) For outdoor or remote pump installations, provide a supplemental grounding electrode system consisting of two ground rods spaced a minimum of 6 feet apart and bonded with a minimum #6 AWG bare copper conductor. Ground rods shall comply with NEC 250.52(A)(5) and 250.53(G).

d. VFD Output Cable Shielding and EMI Mitigation

- (1) VFD output conductors shall be shielded or use VFD-rated armored cable.
- (2) Terminate shielding at the VFD end only unless manufacturer recommends otherwise.
- (3) Maintain separation between VFD output cables and low-voltage signal/control wiring to reduce electromagnetic interference.

e. Testing and Verification

- (1) Verify and document equipment ground continuity between VFD and motor.
- (2) Measure and record grounding system resistance; ensure resistance to earth does not exceed 5 ohms unless otherwise noted.

### 3.3 FIELD QUALITY CONTROL

Specified products must be tested as a system for conformance to specification requirements prior to scheduling the acceptance tests. Conduct performance verification tests in the presence of Government representative, observing and documenting complete compliance of the system to the specifications. Submit a signed copy of the test results, certifying proper system operation before scheduling tests.

#### 3.3.1 ASD Test

A proposed test plan must be submitted to the contracting officer at least 28 calendar days prior to proposed testing for approval. The tests must conform to NEMA ICS 1, NEMA ICS 7, and all manufacturer's safety regulations. The Government reserves the right to witness all tests and review any documentation. Inform the Government at least 14 working days prior to the dates of testing. Perform the ASD test engaging a qualified testing agency's field supervisor currently certified by NETA to supervise on-site testing.

a. Acceptance Testing Preparation:

- (1) Test insulation resistance for each VFC element, bus, component, connecting supply, feeder, and control circuit.
- (2) Test continuity of each circuit.

b. Tests and Inspections:

- (1) Inspect VFC, wiring, components, connections, and equipment installation. Test and adjust controllers, components, and equipment.
  - (2) Test insulation resistance for each VFC element, component, connecting motor supply, feeder, and control circuits.
  - (3) Test continuity of each circuit.
  - (4) Verify that voltages at VFC locations are within 10 percent of motor nameplate rated voltages. If outside this range for any motor, notify Construction Manager before starting the motor(s).
  - (5) Test each motor for proper phase rotation.
  - (6) Perform tests according to the Inspection and Test Procedures for Adjustable Speed Drives stated in NETA Acceptance Testing Specification. Certify compliance with test parameters.
  - (7) Correct malfunctioning units on-site, where possible, and retest to demonstrate compliance; otherwise, replace with new units and retest.
  - (8) Test and adjust controls, remote monitoring, and safeties. Replace damaged and malfunctioning controls and equipment.
- c. VFCs will be considered defective if they do not pass tests and inspections.
- d. Prepare test and inspection reports, including a certified report that identifies the VFC and describes scanning results. Include notation of deficiencies detected, remedial action taken, and observations made after remedial action.

### 3.3.2 Performance Verification Tests

"Performance Verification Test" plan must provide the step by step procedure required to establish formal verification of the performance of the ASD. Compliance with the specification requirements must be verified by inspections, review of critical data, demonstrations, and tests. The Government reserves the right to witness all tests, review data, and request other such additional inspections and repeat tests as necessary to ensure that the system and provided services conform to the stated requirements. Inform the Government 14 calendar days prior to the date the test is to be conducted.

### 3.3.3 Endurance Test

Immediately upon completion of the performance verification test, the endurance test must commence. The system must be operated at varying rates for not less than 192 consecutive hours, at an average effectiveness level of 0.9998, to demonstrate proper functioning of the complete PCS. Continue the test on a day-to-day basis until performance standard is met. The contractor is not allowed in the building during the endurance test. The system must respond as designed.

### 3.4 DEMONSTRATION

#### 3.4.1 Training

Coordinate training requirements with the Contracting Officer. Provide video tapes, if available, of all training provided to the Government for subsequent use in training new personnel. Provide all training aids, texts, and expendable support material for a self-sufficient presentation shall be provided, the amount of which to be determined by the contracting officer.

##### 3.4.1.1 Instructions to Government Personnel

Provide the services of competent instructors with minimum two-year field experience with the operation and maintenance of similar ASDs who will give full instruction to designated personnel in operation, maintenance, calibration, configuration, and programming of the complete control system. Orient the training specifically to the system installed. Instructors must be thoroughly familiar with the subject matter they are to teach. The number of training days of instruction furnished must be as specified. A training day is defined as eight hours of instruction, including two 15-minute breaks and excluding lunch time; Monday through Friday. Provide a training manual for each student at each training phase which describes in detail the material included in each training program. Provide one additional copy for archiving. Provide equipment and materials required for classroom training. Provide a list of additional related courses, and offers, noting any courses recommended. List each training course individually by name, including duration, approximate cost per person, and location of course. Unused copies of training manuals must be turned over to the Government at the end of last training session.

##### 3.4.1.2 Operating Personnel Training Program

Provide one 2-hour training session at the site at a time and place mutually agreeable between the Contractor and the Government. Provide session to train 4 operation personnel in the functional operations of the system and the procedures that personnel will follow in system operation. This training shall include:

- a. System overview
- b. General theory of operation
- c. System operation
- d. Alarm formats
- e. Failure recovery procedures
- f. Troubleshooting

##### 3.4.1.3 Engineering/Maintenance Personnel Training

Accomplish the training program as specified. Training must be conducted on site at a location designated by the Government. Provide a one-day training session to train four engineering personnel in the functional operations of the system. This training must include:

- a. System overview

- b. General theory of operation
- c. System operation
- d. System configuration
- e. Alarm formats
- f. Failure recovery procedures
- g. Troubleshooting and repair
- h. Maintenance and calibration
- i. System programming and configuration

-- End of Section --

SECTION 26 51 00

INTERIOR LIGHTING  
05/20, CHG 2: 11/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

|                   |                                                                                                                                                                                                     |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A580/A580M   | (2023) Standard Specification for Stainless Steel Wire                                                                                                                                              |
| ASTM A641/A641M   | (2019) Standard Specification for Zinc-Coated (Galvanized) Carbon Steel Wire                                                                                                                        |
| ASTM A653/A653M   | (2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process                                                             |
| ASTM A1008/A1008M | (2024) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable |
| ASTM B164         | (2003; R 2014) Standard Specification for Nickel-Copper Alloy Rod, Bar, and Wire                                                                                                                    |
| ASTM B633         | (2023) Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel                                                                                                               |

EUROPEAN UNION (EU)

|                      |                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------------|
| Directive 2011/65/EU | (2011) Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment |
|----------------------|------------------------------------------------------------------------------------------------------|

ILLUMINATING ENGINEERING SOCIETY (IES)

|                |                                                                                                           |
|----------------|-----------------------------------------------------------------------------------------------------------|
| ANSI/IES LM-79 | (2019) Approved Method: Electrical and Photometric Measurements of Solid State Lighting Products          |
| ANSI/IES LM-80 | (2020) Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays and Modules |
| ANSI/IES LS-1  | (2020) Lighting Science: Nomenclature and Definitions for Illuminating Engineering                        |

|                                                          |                                                                                                                       |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| ANSI/IES TM-21                                           | (2021) Technical Memorandum: Projecting Long-Term Luminous, Photon, and Radiant Flux Maintenance of LED Light Sources |
| ANSI/IES TM-30                                           | (2020) Technical Memorandum: IES Method for Evaluating Light Source Color Rendition                                   |
| IES Lighting Library                                     | IES Lighting Library                                                                                                  |
| INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE) |                                                                                                                       |
| IEEE 100                                                 | (2000; Archived) The Authoritative Dictionary of IEEE Standards Terms                                                 |
| IEEE C2                                                  | (2023) National Electrical Safety Code                                                                                |
| NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)     |                                                                                                                       |
| NEMA 250                                                 | (2020) Enclosures for Electrical Equipment (1000 Volts Maximum)                                                       |
| NEMA ANSLG C78.377                                       | (2017) Electric Lamps— Specifications for the Chromaticity of Solid State Lighting Products                           |
| NEMA C82.77-10                                           | (2020) Harmonic Emission Limits - Related Power Quality Requirements                                                  |
| NEMA SSL 1                                               | (2016) Electronic Drivers for LED Devices, Arrays, or Systems                                                         |
| NEMA SSL 3                                               | (2011) High-Power White LED Binning for General Illumination                                                          |
| NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)              |                                                                                                                       |
| NFPA 70                                                  | (2023) National Electrical Code                                                                                       |
| NFPA 101                                                 | (2024) Life Safety Code                                                                                               |
| NFPA 110                                                 | (2025) Standard for Emergency and Standby Power Systems                                                               |
| U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA) |                                                                                                                       |
| 47 CFR 15                                                | Radio Frequency Devices                                                                                               |
| UNDERWRITERS LABORATORIES (UL)                           |                                                                                                                       |
| UL 924                                                   | (2016; Reprint Dec 2022) UL Standard for Safety Emergency Lighting and Power Equipment                                |
| UL 1598                                                  | (2021; Reprint Jan 2024) UL Standard for Safety Luminaires                                                            |
| UL 1598C                                                 | (2014) Standard for Light-Emitting Diode (LED) Retrofit Luminaire Conversion Kits                                     |

UL 8750

(2020; Reprint Jul 2024) UL Standard for  
Safety Light Emitting Diode (LED)  
Equipment for Use in Lighting Products

## 1.2 RELATED REQUIREMENTS

Materials not considered to be luminaires, luminaire accessories, or lighting equipment are specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

## 1.3 DEFINITIONS

- a. Unless otherwise specified or indicated, electrical and electronics terms used in these specifications and on the drawings, must be as defined in IEEE 100 and ANSI/IES LS-1.
- b. For LED luminaire light sources, "Useful Life" is the operating hours before reaching 70 percent of the initial rated lumen output (L70) with no catastrophic failures under normal operating conditions. This is also known as 70 percent "Rated Lumen Maintenance Life" as defined in ANSI/IES LM-80.
- c. For LED luminaires, "Luminaire Efficacy" (LE) is the appropriate measure of energy efficiency, measured in lumens/watt. This is gathered from LM-79 data for the luminaire, in which absolute photometry is used to measure the lumen output of the luminaire as one entity, not the source separately and then the source and housing together.
- d. Total harmonic distortion (THD) is the root mean square (RMS) of all the harmonic components divided by the total fundamental current.

## 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

Luminaire Drawings; G

### SD-03 Product Data

Luminaires; G

Light Sources; G

LED Drivers; G

Luminaire Warranty; G

Mini Inverters; G

Exit Signs; G

Emergency Drivers; G

#### SD-04 Samples

Luminaire Samples; G

#### SD-05 Design Data

Luminaire Design Data; G

Photometric Plan; G

#### SD-06 Test Reports

ANSI/IES LM-79 Test Report; G

ANSI/IES LM-80 Test Report; G

ANSI/IES TM-21 Test Report; G

ANSI/IES TM-30 Test Report; G

#### SD-07 Certificates

LED Driver and Dimming Switch Compatibility Certificate; G

#### SD-10 Operation and Maintenance Data

Lighting System, Data Package 5; G

Maintenance Staff Training Plan; G

End-User Training Plan; G

### 1.5 QUALITY ASSURANCE

Data, drawings, and reports must employ the terminology, classifications and methods prescribed by the IES Lighting Library as applicable, for the lighting system specified.

#### 1.5.1 Luminaire Drawings

Include dimensions, accessories installation details, and construction details. Photometric data, including CRI, CCT, LED driver type, zonal lumen data, and candlepower distribution data must accompany shop drawings.

#### 1.5.2 Luminaire Design Data

- a. Provide safety certification and file number for the luminaire family



that must be listed, labeled, or identified in accordance with the NFPA 70. Applicable testing bodies are determined by the US Occupational Safety Health Administration (OSHA) as Nationally Recognized Testing Laboratories (NRTL) and include: CSA (Canadian Standards Association), ETL (Edison Testing Laboratory), and UL (Underwriters Laboratories).

- b. Provide long term lumen maintenance projections for each LED luminaire in accordance with ANSI/IES TM-21. Data used for projections must be obtained from testing in accordance with ANSI/IES LM-80.

#### 1.5.3 ANSI/IES LM-79 Test Report

Submit test report on manufacturer's standard production model of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data in IES format as outlined under "14.0 Test Report" in ANSI/IES LM-79.

#### 1.5.4 ANSI/IES LM-80 Test Report

Submit report on manufacturer's standard production LED light source (package, array, or module) of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data as outlined under "8.0 Test Report" in ANSI/IES LM-80.

#### 1.5.5 ANSI/IES TM-21 Test Report

Submit test report on manufacturer's standard production LED light source (package, array, or module) of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data, as well as required interpolation information as outlined under "7.0 Report" in ANSI/IES TM-21.

#### 1.5.6 ANSI/IES TM-30 Test Report

Submit color vector graphic in accordance with ANSI/IES TM-30 on manufacturer's standard production LED light source (package, array, or module) of specified luminaire. Include spectral distribution of test LED light source.

#### 1.5.7 LED Driver and Dimming Switch Compatibility Certificate

Submit certification from the luminaire, driver, or dimmer switch manufacturer that ensures compatibility and operability between devices without flickering and to specified dimming levels.

#### 1.5.8 Photometric Plan

##### 1.5.8.1 Computer-generated Photometric Plans

Computer-generated photometric plans for each space are required to verify proposed luminaires and locations meet the required performance criteria of the design using the applicable light loss factor (LLF).

Target illumination levels are provided for each Interior Application. Depending on the application and the recommendations provided by the IES, values are given as one of the following:

- a. Minimum: No values anywhere on the calculation grid may be less than this value, within a 10 percent margin of error.
- b. Minimum Average: An average, taken over the entire task area for the application, may not be less than this value, within a 10 percent margin of error.
- c. Maximum: No values anywhere on the calculation grid may be greater than this value, within a 10 percent margin of error.
- d. Maximum Average: An average, taken over the entire task area for the application, may not be greater than this value, within a 10 percent margin of error.
- e. Uniformity: Unless otherwise noted, uniformity is calculated as a ratio of the average calculated illuminance over the minimum calculated illuminance of the calculation grid.

#### 1.5.8.2 Schematic Photometric Plan Calculations

Schematic photometric plan calculations must include:

- a. Horizontal illuminance measurements at workplane or other designated height above finished floor, taken at a maximum of every one foot across the task area.
- b. Average maintained illuminance level.
- c. Minimum and maximum maintained illuminance levels.
- d. Lighting power density (Watts per square foot).
- e. LLF. Recommended LLF is 0.81 for LED luminaires but LLF varies based on environment and application.

#### 1.5.8.3 Final Photometric Plan Calculations

Final photometric plan calculations must include:

- a. Horizontal illuminance measurements at workplane or other designated height above finished floor, taken at a maximum of every one foot across the task area.
- b. Where applicable, vertical illuminance measurements at designated surface, taken at a maximum of every one foot across task area.
- c. Minimum and maximum maintained illuminance levels.
- d. Average maintained illuminance level.
- e. Average to minimum and maximum to minimum ratios for horizontal illuminance.
- f. Lighting power density (Watts per square foot).
- g. LLF. Recommended LLF is 0.81 for LED luminaires but LLF varies based on environment and application.

#### 1.5.9 Test Laboratories

Test laboratories for the ANSI/IES LM-79 and ANSI/IES LM-80 test reports must be one of the following:

- a. National Voluntary Laboratory Accreditation Program (NVLAP) accredited for solid-state lighting testing as part of the Energy-Efficient Lighting Products laboratory accreditation program for both LM-79 and LM-80 testing.
- b. One of the qualified labs listed on the Department of Energy - LED Lighting Facts Approved Testing Laboratories List for LM-79 testing.
- c. One of the EPA-Recognized Laboratories listed for LM-80 testing.

#### 1.5.10 Regulatory Requirements

Equipment, materials, installation, and workmanship must be in accordance with the mandatory and advisory provisions of NFPA 70, unless more stringent requirements are specified or indicated. Provide luminaires and assembled components that are approved by and bear the label of UL for the applicable location and conditions unless otherwise specified.

#### 1.5.11 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design, and workmanship. Products must have been in satisfactory commercial or industrial use for six months prior to bid opening. The six-month period must include applications of equipment and materials under similar circumstances and of similar size. The product must have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the six-month period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

##### 1.5.11.1 Alternative Qualifications

Products having less than a six-month field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

##### 1.5.11.2 Material and Equipment Manufacturing Date

Do not use products manufactured more than six months prior to date of delivery to site, unless specified otherwise.

#### 1.6 WARRANTY

Support all equipment items by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

#### 1.6.1 Luminaire Warranty

Provide and transfer to the government the original LED luminaire manufacturers standard commercial warranty for each different luminaire manufacturer used in the project.

- a. Provide a written five year minimum replacement warranty for material, luminaire finish, and workmanship. Provide written warranty document that contains all warranty processing information needed, including customer service point of contact, whether or not a return authorization number is required, return shipping information, and closest return location to the luminaire location.

- (1) Finish warranty must include failure and substantial deterioration such as blistering, cracking, peeling, chalking, or fading.

- (2) Material warranty must include:

- (a) All LED drivers and integral control equipment.

- (b) Replacement when more than 15 percent of LED sources in any lightbar or subassembly(s) are defective, non-starting, or operating below 70 percent of specified lumen output.

- (c) Replacement when more than 15 percent of LED sources in any lightbar or subassembly(s) show a color shift greater than 0.003 delta u'v' from the zero hour measurement stated in the ANSI/IES LM-79 Test Report.

- b. Warranty period must begin in accordance with the manufacturer's standard warranty starting date.

- c. Provide replacements that are promptly shipped, without charge, to the using Government facility point of contact and that are identical to or an improvement upon the original equipment. All replacements must include testing of new components and assembly.

#### 1.7 OPERATION AND MAINTENANCE MANUALS

##### 1.7.1 Lighting System

Provide operation and maintenance manuals for the lighting system in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA that provide basic data relating to the design, operation, and maintenance of the lighting system for the building. Additional requirements for the Navy are provided in Section 01 78 24.00 20 FACILITY DATA WORKBOOK (FDW). Include the following:

- a. Manufacturers' operating and maintenance manuals.
- b. Luminaire shop drawings for modified and custom luminaires.
- c. Luminaire Manufacturers' standard commercial warranty information as specified in paragraph LUMINAIRE WARRANTY.

PART 2 PRODUCTS

2.1 PRODUCT COORDINATION

2.2 LUMINAIRES

UL 1598, NEMA C82.77-10. Provide luminaires as indicated in the luminaire schedule and NL plates or details on project plans, complete with light source, wattage, and lumen output indicated. All luminaires of the same type must be provided by the same manufacturer. Luminaires must be specifically designed for use with the driver and light source provided.

2.2.1 Luminaire Samples

Submit one sample of each luminaire type , complete with light source, LED driver rated for 120 V operation, and 6 feet pigtail with 3-prong Edison plug. Sample will be returned to the Contractor for installation in the project work.

2.2.2 Luminaires

UL 8750, ANSI/IES LM-79, ANSI/IES LM-80. For all luminaires, provide:

- a. Complete system with LED drivers and light sources.
- b. Housings constructed of non-corrosive materials. All new aluminum housings must be anodized or powder-coated. All new steel housings must be treated to be corrosion resistant.
- c. ANSI/IES TM-21, ANSI/IES LM-80. Minimum L70 lumen maintenance value of 50,000 hours unless otherwise indicated in the luminaire schedule. Luminaire drive current value must be identical to that provided by test data for luminaire in question.
- d. Minimum efficacy as specified in the luminaire schedule. Theoretical models of initial lamp lumens per watt are not acceptable. If efficacy values are not listed in the luminaire schedule, provide luminaires that meet the following minimum values:

| Luminaire Style                                       | Minimum Luminaire Efficacy |
|-------------------------------------------------------|----------------------------|
|                                                       |                            |
|                                                       |                            |
|                                                       |                            |
| Linear, Ambient (indirect wall mount, linear pendent) | 100 LPW                    |
| High Bay, Low Bay, and Industrial Locations           | 100 LPW                    |
|                                                       |                            |
|                                                       |                            |

| Luminaire Style | Minimum Luminaire Efficacy |
|-----------------|----------------------------|
|                 |                            |
|                 |                            |
|                 |                            |

- e. UL listed for dry or damp location typical of interior installations. Any luminaire mounted on the exterior of the building must be UL listed for wet location typical of exterior installations.
- f. LED driver and light source package, array, or module are accessible for service or replacement without removal or destruction of luminaire.
- g. Lenses constructed of heat tempered borosilicate glass, UV-resistant acrylic, or silicone.

#### 2.2.2.1 Luminaire Conversion Kits

Provide luminaire conversion kits that meet UL 1598C Standard for Light-Emitting Diode (LED) Retrofit Luminaire Conversion Kits.

### 2.3 LIGHT SOURCES

NEMA ANSLG C78.377, NEMA SSL 3. Provide type, delivered lumen output, and wattage as indicated in the luminaire schedule on project plans.

#### 2.3.1 LED Light Sources

Provide LED light sources that meet the following requirements:

- a. NEMA ANSLG C78.377. Emit white light and have a nominal CCT of 3500 and 4000 Kelvin.
- b. Minimum Color Rendering Index (CRI) of 90.
- c. Directive 2011/65/EU. Restriction of Hazardous Substances (RoHS) compliant.
- d. Light source color consistency by utilizing a binning tolerance within a 3-step McAdam ellipse.
- e. Color maintenance value of no greater than 0.003 ( $\Delta u'v'$ ) at 6000 hours as listed in ANSI/IES LM-79 Test Report.

### 2.4 LED DRIVERS

NEMA SSL 1, UL 8750. Provide LED drivers that are electronic, UL Class 1 or Class 2, constant-current type and that comply with the following requirements:

- a. The combined driver and LED light source system does not exceed the minimum luminaire efficacy values as listed in the luminaire schedule provided.

- b. Operates at a voltage of 120-277 volts at 50/60 hertz, with input voltage fluctuations of plus/minus 10 percent.
- c. Power Factor (PF) greater than or equal to 0.90 at full input power and across specified dimming range.
- d. Maximum Total Harmonic Distortion (THD) less than 20 percent at full input power and across specified dimming range.
- e. Operates for at least 50,000 hours at maximum case temperature and 90 percent non-condensing relative humidity.
- f. Withstands Category A surges of 2 kV without impairment of performance. Provide surge protection that is integral to the driver.
- g. Integral thermal protection that reduces the output power to protect the driver and light source from damage if the case temperature approaches or exceeds the driver's maximum operating temperature.
- h. 47 CFR 15. Complies with the requirements of the Federal Communications Commission (FCC) rules and regulations, Non-Consumer (Class A) for EMI/RFI (conducted and radiated).
- i. Class A sound rating.
- j. Directive 2011/65/EU. Restriction of Hazardous Substances (RoHS) compliant.
- k. Provide dimming capability as indicated in the luminaire schedule on project plans. Dimmable drivers must dim down to 10 percent. Dimmable drivers must be controlled by a Class 2 low voltage 0-10VDC controller

## 2.5 LIGHTING CONTROLS

## 2.6 EXIT AND EMERGENCY LIGHTING EQUIPMENT

### 2.6.1 Exit Signs

UL 924, NFPA 101. Provide wattage as indicated in the luminaire schedule on project plans. Provide LED Exit Signs that meet the following criteria:

- a. Housing constructed of .Edge-lit type with clear acrylic, edge-lit face and aluminum trim having .
- b. UL listed for damp location.
- c. Configured for universal mounting.
- d. 6 inch high, 3/4 inch stroke red lettering on face of sign with chevrons on either side of lettering to indicate direction.
- e. Single or double face as indicated in project plans and luminaire schedule.

#### 2.6.1.1 Exit Signs with Battery Backup

Equip with automatic power failure device, test switch, and pilot light, and fully automatic high/low trickle charger in a self-contained power

pack. Battery must be sealed, maintenance free nickel-cadmium type, and must operate unattended for a period of not less than five years. Emergency run time must be a minimum of 1-1/2 hours. LEDs must have a minimum rated life of 10 years. Provide self-diagnostic circuitry integral to emergency LED driver.

#### 2.6.2 LED Emergency Drivers

UL 924, NFPA 101. Provide LED emergency driver with automatic power failure detection, test switch and LED indicator (or combination switch/indicator) located on luminaire exterior, and fully-automatic solid-state charger, battery and inverter integral to a self-contained housing. Provide self-diagnostic function integral to emergency driver. Integral nickel-cadmium battery is required to supply a minimum of 90 minutes of emergency power at 10 watts, constant output. Driver must be RoHS compliant, rated for installation in plenum-rated spaces and damp locations, and be warranted for a minimum of five years.

#### 2.6.3 Self-Diagnostic Circuitry for LED Drivers

UL 924, NFPA 101. Provide emergency lighting unit with fully-automatic, integral self-testing/diagnostic electronic circuitry. Circuitry must provide for a one minute diagnostic test every 28 days, and a 30 minute diagnostic test every six months, minimum. Any malfunction of the unit must be indicated by LED(s) visible from the exterior of the luminaire. A manual test switch must also be provided to perform a diagnostic test at any given time.

#### 2.6.4 Mini Inverters

UL 924, NFPA 101. Provide mini inverters that are designed to provide power to emergency luminaires. Provide mini inverters that are suitable for damp installations, operate at a voltage of 120-277 volts at 50/60 hertz, and are capable of operating 0-10V dimming override. Provide mini inverters that supply a minimum of 90 minutes of emergency power.

#### 2.6.5 Central Emergency Lighting System

UL 924, NFPA 101, NFPA 110 level 1, NFPA 70. Provide a central power system providing emergency power at 120 volts, 60 hertz, for a minimum period of 90 minutes. Design the system to handle surges during loss and recovery of the voltage, and to deliver its full rated output to the designated lamp load. Provide backup ac source for power.

##### 2.6.5.1 Operation

Provide system such that when the lighting system loses normal supply voltage, it automatically disengages itself from the normal input line, and switches to a self-contained inverter with built-in protection when the output is shorted or overloaded. Ensure that, when normal line voltage resumes, the emergency system automatically switches back to normal operation. Size the transfer switch for this function to handle 125 percent of full load. Provide the battery system with self-contained inverters with overload protection.

##### 2.6.5.2 Charger

Provide a completely automatic battery charger that maintains the batteries in a fully charged condition and recharges the batteries to full



capacity within 24 hours after full discharge in accordance with UL 924.

#### 2.6.5.3 Batteries

Provide sealed nickel-cadmium batteries, maintenance-free for a period of not less than 0 years under normal operating conditions.

#### 2.6.5.4 Accessories

Provide visual indicators to indicate normal power, inverter power, and battery-charger operation. Provide a low-voltage test switch to simulate power failure by interrupting the input line, voltage meter, electrolyte level detector to automatically disable the charging circuit in the event of a fault, and low-voltage cutoff to prevent extreme battery power dissipation.

#### 2.6.5.5 Enclosure

Provide a free-standing cabinet with floor stand and constructed of 12-gage sheet steel with baked-on enamel finish and a locking latch.

### 2.7 LUMINAIRE MOUNTING ACCESSORIES

#### 2.7.1 Suspended Luminaires

- a. Provide hangers capable of supporting twice the combined weight of luminaires supported by hangers.
- b. Hangers must allow luminaires to swing within an angle of 45 degrees. Brace pendants 4 feet or longer to limit swinging. Provide with swivel hangers to ensure a plumb installation for rigid stem pendants. Provide cadmium-plated steel with a swivel-ball tapped for the conduit size indicated.
- c. Single-unit suspended luminaires must have twin-stem hangers. Multiple-unit or continuous row luminaires with a separate power supply cord must have a tubing or stem for wiring at one point and a tubing or rod suspension provided for each unit length of chassis, including one at each end.
- d. Provide all linear pendent and surface mounted luminaires with two supports per four-foot section or three per eight-foot section unless otherwise recommended by manufacturer.
- e. Provide rods in minimum 0.18 inch diameter.

#### 2.7.2 Recess and Surface Mounted Luminaires

Provide access to light source and LED driver from bottom of luminaire. Provide trim and lenses for the exposed surface of flush-mounted luminaires as indicated on project drawings and specifications. Luminaires recessed in ceilings which have a fire resistive rating of one hour or more must be enclosed in a box which has a fire resistive rating equal to that of the ceiling. For surface mounted luminaires with brackets, provide flanged metal stem attached to outlet box, with threaded end suitable for supporting the luminaire rigidly in design position. Flanged part of luminaire stud must be of broad base type, secured to outlet box at not fewer than three points.

### 2.7.3 Luminaire Support Hardware

#### 2.7.3.1 Wire

ASTM A641/A641M. Galvanized, soft tempered steel, minimum 0.11 inches in diameter, or galvanized, braided steel, minimum 0.08 inches in diameter.

#### 2.7.3.2 Wire for Humid Spaces

ASTM A580/A580M. Composition 302 or 304, annealed stainless steel, minimum 0.11 inches in diameter.

ASTM B164. UNS N04400, annealed nickel-copper alloy, minimum 0.11 inches in diameter.

#### 2.7.3.3 Threaded Rods

Threaded steel rods, 3/16 inch diameter, zinc or cadmium coated.

#### 2.7.3.4 Straps

Galvanized steel, one by 3/16 inch, conforming to ASTM A653/A653M, with a light commercial zinc coating or ASTM A1008/A1008M with an electrodeposited zinc coating conforming to ASTM B633, Type RS.

### 2.8 EQUIPMENT IDENTIFICATION

#### 2.8.1 Manufacturer's Nameplate

Each item of equipment must have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

#### 2.8.2 Labels

UL 1598. All luminaires must be clearly marked for operation of specific light sources and LED drivers. The labels must be easy to read when standing next to the equipment, and durable to match the life of the equipment to which they are attached. Note the following light source characteristics in the format "Use Only \_\_\_\_\_":

- a. Correlated Color Temperature (CCT) and Color Rendering Index (CRI) for all luminaires.
- b. Driver and dimming protocol.

All markings related to light source type must be clear and located to be readily visible to service personnel, but unseen from normal viewing angles when light sources are in place. LED drivers must have clear markings indicating dimming type and indicate proper terminals for the various outputs.

### 2.9 FACTORY APPLIED FINISH

NEMA 250. Provide all luminaires and lighting equipment with factory-applied painting system that as a minimum, meets requirements of corrosion-resistance testing.

## PART 3 EXECUTION

### 3.1 INSTALLATION

IEEE C2, NFPA 70.

#### 3.1.1 Light Sources

When light sources are not provided as an integral part of the luminaire, deliver light sources of the type, wattage, lumen output, color temperature (CCT), color rendering index (CRI), and voltage rating indicated to the project site and install just prior to project completion, if not already installed in the luminaires from the factory.

#### 3.1.2 Luminaires

Set luminaires plumb, square, and level with ceiling and walls, in alignment with adjacent luminaires and secure in accordance with manufacturers' directions and approved drawings. Provide accessories as required for ceiling construction type indicated on Finish Schedule. Luminaire catalog numbers do not necessarily denote specific mounting accessories for type of ceiling in which a luminaire may be installed. Provide wires, straps, or rods for luminaire support in this section. Install luminaires with vent holes free of air blocking obstacles.

##### 3.1.2.1 Suspended Luminaires

Measure mounting heights from the bottom of the luminaire for ceiling-mounted luminaires and to center of luminaire for wall-mounted luminaires. Obtain architect approval of the exact mounting height on the job before commencing installation and, where applicable, after coordinating with the type, style, and pattern of the ceiling being installed. Support suspended luminaires from structural framework of ceiling or from inserts cast into slab.

- a. Provide suspended luminaires with 45 degree swivel hangers so that they hang plumb and level.
- b. Locate so that there are no obstructions within the 45 degree range in all directions.
- c. The stem, canopy and luminaire must be capable of 45 degree swing.
- d. Rigid pendent stem, aircraft cable, rods, or chains 4 feet or longer excluding luminaire must be braced to prevent swaying using three cables at 120 degree separation.
- e. Suspended luminaires in continuous rows must have internal wireway systems for end to end wiring and must be properly aligned to provide a straight and continuous row without bends, gaps, light leaks or filler pieces.
- f. Utilize aligning splines on extruded aluminum luminaires to assure minimal hairline joints.
- g. Support steel luminaires to prevent "oil-canning" effects.
- h. Match supporting pendants with supported luminaire. Aircraft cable must be stainless steel.

- i. Match finish of canopies to match the ceiling, and provide low profile canopies unless otherwise shown.
- j. Maximum distance between suspension points must be 10 feet or as recommended by the manufacturer, whichever is less.

#### 3.1.2.2 Recessed and Semi-Recessed Luminaires

- a. Support recessed and semi-recessed luminaires independently from the building structure by a minimum of two wires, straps or rods per luminaire and located near opposite corners of the luminaire. Secure horizontal movement with clips provided by manufacturer. Ceiling grid clips are not allowed as an alternative to independently supported luminaires.
- b. Support round luminaires or luminaires smaller in size than the ceiling grid independently from the building structure by a minimum of four wires, straps or rods per luminaire, spaced approximately equidistant around.
- c. Do not support luminaires by acoustical tile ceiling panels.
- d. Where luminaires of sizes less than the ceiling grid are indicated to be centered in the acoustical panel, support each independently and provide at least two 3/4 inch metal channels spanning, and secured to, the ceiling tees for centering and aligning the luminaire.
- e. Adjust aperture rings on all applicable ceiling recessed luminaires to accommodate various ceiling material thickness. Coordinate cut-out size in ceiling to ensure aperture covers cut-out entirely. Install aperture rings such that the bottom of the ring is flush with finished ceiling or not more than 1/16 inch above. Do not install luminaires such that the aperture ring extends below the finished ceiling surface.
- f. For luminaire recessed in plaster ceilings, provide plaster frames for setting. Install setting such that the bottom of the frame is flush with finished ceiling. Support luminaires with plaster frames utilizing yokes or leveling lugs. Do not mount luminaires or support elements to ducts or pipes. Yokes must support a luminaire by no fewer than two bolts each.

#### 3.1.2.3 Field Applied Painting

Provide field applied painting for all luminaire type . Paint lighting equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria. Provide painting as specified in Section 09 90 00 PAINTS AND COATINGS.

#### 3.1.3 LED Drivers

Provide LED drivers integral to luminaire as constructed by the manufacturer.

#### 3.1.4 Exit Signs

NFPA 101. Wire exit signs and emergency lighting units ahead of the local

switch, to the normal lighting circuit located in the same room or area.

Connect exit signs on separate circuits and serve from a separate circuit breaker. . Paint source of control red and provide lockout capability.

### 3.2 FIELD QUALITY CONTROL

#### 3.2.1 Tests

##### 3.2.1.1 Lighting Control Verification Tests

Verify lighting control system and devices operate according to approved sequence of operations. Verification tests are to be completed after commissioning.

- a. Verify wall box dimmers and scene wallstations operate as described in sequence of operations.

##### 3.2.1.2 Emergency Lighting Test

Interrupt power supply to demonstrate proper operation of emergency lighting. If adjustments are made to the lighting system, re-test system to show compliance with standards.

### 3.3 CLOSEOUT ACTIVITIES

#### 3.3.1 Commissioning

NFPA 101. Factory Trained Field Service Technician is responsible for calibration and programming sequences for input devices and systems in accordance with the requirements described in the sequence of operation.

#### 3.3.2 Training

##### 3.3.2.1 Maintenance Staff Training

Submit a Maintenance Staff Training Plan at least 30 calendar days prior to training session that describes training procedures for Owner's personnel in the operation and maintenance of lighting and lighting control system. Provide on-site training which demonstrates full system functionality, assigning schedules, calibration adjustments for light levels and sensor sensitivity, integration procedures for connecting to third-party devices, and manual override including information on appropriate use. Provide protocols for troubleshooting, maintenance, repair, and replacement, and literature on available system updates and process for implementing updates.

##### 3.3.2.2 End-User Training

Submit an End-User Training Plan at least 30 calendar days prior to training session that describes training procedures for end-users on the lighting control system. Provide users with a list of control devices located within user-occupied spaces, such as photosensors and occupancy and vacancy sensors, including information on the proper operation and schedule for each device. Provide demonstration for each type of interface. Provide users with the building schedule as currently commissioned, including conditional programming based on astronomic time

clock functionality. Provide users with the correct contact information for maintenance personnel who will be available to address any lighting control issues.

Provide laminated instructions to the user at each scene wallstation. Provide only instructions relevant to the functionality of the specific scene wallstation. Provide a description of each labeled scene control button. If the room utilizes occupancy/vacancy sensors or photosensors, include a description of this functionality on the instruction sheet.

-- End of Section --

SECTION 27 10 00

BUILDING TELECOMMUNICATIONS CABLING SYSTEM  
**08/11**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D709 (2017) Standard Specification for  
Laminated Thermosetting Materials

ELECTRONIC COMPONENTS INDUSTRY ASSOCIATION (ECIA)

ECIA EIA/ECA 310-E (2005) Cabinets, Racks, Panels, and  
Associated Equipment

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 100 (2000; Archived) The Authoritative  
Dictionary of IEEE Standards Terms

INSULATED CABLE ENGINEERS ASSOCIATION (ICEA)

ICEA S-83-596 (2021) Indoor Optical Cable

ICEA S-90-661 (2021) Category 3 and 5E Individually  
Unshielded Twisted Pairs, Indoor Cables  
(With or Without an Overall Shield) for  
Use in General Purpose and LAN  
Communications Wiring Systems

NATIONAL ELECTRICAL CONTRACTORS ASSOCIATION (NECA)

NECA/BICSI 568 (2006) Standard for Installing Building  
Telecommunications Cabling

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA WC 66 (2019) Performance Standard for Category 6  
and Category 7 100 Ohm Shielded and  
Unshielded Twisted Pairs

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-455-21 (1988a; R 2012) FOTP-21 - Mating  
Durability of Fiber Optic Interconnecting  
Devices

|                |                                                                                                           |
|----------------|-----------------------------------------------------------------------------------------------------------|
| TIA-526-14     | (2023d) OFSTP-14A Optical Power Loss Measurements of Installed Multimode Fiber Cable Plant                |
| TIA-568.0      | (2020e) Generic Telecommunications Cabling for Customer Premises                                          |
| TIA-568.1      | (2020e) Commercial Building Telecommunications Infrastructure Standard                                    |
| TIA-568.2      | (2024e) Balanced Twisted-Pair Telecommunications Cabling and Components Standards                         |
| TIA-568.3      | (2022e) Optical Fiber Cabling Components Standard                                                         |
| TIA-569        | (2019e; Add 1 2022) Telecommunications Pathways and Spaces                                                |
| TIA-606        | (2021d) Administration Standard for Telecommunications Infrastructure                                     |
| TIA-607        | (2019d) Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises                 |
| TIA-1152       | (2016; R 2021) Requirements for Field Test Instruments and Measurements for Balanced Twisted-Pair Cabling |
| TIA/EIA-598    | (2014D; Add 2 2018) Optical Fiber Cable Color Coding                                                      |
| TIA/EIA-604-10 | (2002a) FOCIS 10 Fiber Optic Connector Intermateability Standard - Type LC                                |

U.S. FEDERAL COMMUNICATIONS COMMISSION (FCC)

|             |                                                                       |
|-------------|-----------------------------------------------------------------------|
| FCC Part 68 | Connection of Terminal Equipment to the Telephone Network (47 CFR 68) |
|-------------|-----------------------------------------------------------------------|

UL SOLUTIONS (UL)

|         |                                                                                                          |
|---------|----------------------------------------------------------------------------------------------------------|
| UL 50   | (2024) UL Standard for Safety Enclosures for Electrical Equipment, Non-Environmental Considerations      |
| UL 444  | (2017; Reprint Jun 2021) UL Standard for Safety Communications Cables                                    |
| UL 467  | (2022) UL Standard for Safety Grounding and Bonding Equipment                                            |
| UL 514C | (2014; Reprint Apr 2024) UL Standard for Safety Nonmetallic Outlet Boxes, Flush-Device Boxes, and Covers |



|         |                                                                                   |
|---------|-----------------------------------------------------------------------------------|
| UL 969  | (2017; Reprint May 2023) UL Standard for Safety Marking and Labeling Systems      |
| UL 1286 | (2022; Reprint Mar 2024) UL Standard for Safety Office Furnishings                |
| UL 1863 | (2004; Reprint Oct 2019) UL Standard for Safety Communication Circuit Accessories |

## 1.2 RELATED REQUIREMENTS

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM , apply to this section with additions and modifications specified herein.

## 1.3 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in this specification shall be as defined in TIA-568.1, TIA-568.2, TIA-568.3, TIA-569, TIA-606 and IEEE 100 and herein.

### 1.3.1 Campus Distributor (CD)

A distributor from which the campus backbone cabling emanates. (International expression for main cross-connect (MC).)

### 1.3.2 Building Distributor (BD)

A distributor in which the building backbone cables terminate and at which connections to the campus backbone cables may be made. (International expression for intermediate cross-connect (IC).)

### 1.3.3 Floor Distributor (FD)

A distributor used to connect horizontal cable and cabling subsystems or equipment. (International expression for horizontal cross-connect (HC).)

### 1.3.4 Telecommunications Room (TR)

An enclosed space for housing telecommunications equipment, cable, terminations, and cross-connects. The room is the recognized cross-connect between the backbone cable and the horizontal cabling.

### 1.3.5 Entrance Facility (EF) (Telecommunications)

An entrance to the building for both private and public network service cables (including wireless) including the entrance point at the building wall and continuing to the equipment room.

### 1.3.6 Equipment Room (ER) (Telecommunications)

An environmentally controlled centralized space for telecommunications equipment that serves the occupants of a building. Equipment housed therein is considered distinct from a telecommunications room because of the nature of its complexity.

### 1.3.7 Open Cable

Cabling that is not run in a raceway as defined by NFPA 70. This refers to cabling that is "open" to the space in which the cable has been

installed and is therefore exposed to the environmental conditions associated with that space.

#### 1.3.8 Open Office

A floor space division provided by furniture, moveable partitions, or other means instead of by building walls.

#### 1.3.9 Pathway

A physical infrastructure utilized for the placement and routing of telecommunications cable.

### 1.4 SYSTEM DESCRIPTION

The building telecommunications cabling and pathway system shall include permanently installed backbone and horizontal cabling, horizontal and backbone pathways, service entrance facilities, work area pathways, telecommunications outlet assemblies, conduit, raceway, and hardware for splicing, terminating, and interconnecting cabling necessary to transport telephone and data (including LAN) between equipment items in a building. The horizontal system shall be wired in a star topology from the telecommunications work area to the floor distributor or campus distributor at the center or hub of the star. The backbone cabling and pathway system includes intrabuilding and interbuilding interconnecting cabling, pathway, and terminal hardware. The intrabuilding backbone provides connectivity from the floor distributors to the building distributors or to the campus distributor and from the building distributors to the campus distributor as required. The backbone system shall be wired in a star topology with the campus distributor at the center or hub of the star. The interbuilding backbone system provides connectivity between the campus distributors and is existing. Provide telecommunications pathway systems referenced herein as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. The telecommunications contractor must coordinate with the NMCI/COSC/NGEN contractor concerning access to and configuration of telecommunications spaces. The telecommunications contractor may be required to coordinate work effort within the telecommunications spaces with the NMCI/COSC/NGEN contractor.

### 1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-02 Shop Drawings

Telecommunications Drawings; G

Telecommunications Space Drawings; G

In addition to Section 01 33 00 SUBMITTAL PROCEDURES, provide shop drawings in accordance with paragraph SHOP DRAWINGS.

#### SD-03 Product Data

Telecommunications Cabling (backbone and horizontal); G

Patch Panels; G

Telecommunications Outlet/Connector Assemblies; G

Equipment Support Frame; G

Connector Blocks; G

Spare Parts; G

Submittals shall include the manufacturer's name, trade name, place of manufacture, and catalog model or number. Include performance and characteristic curves. Submittals shall also include applicable federal, military, industry, and technical society publication references. Should manufacturer's data require supplemental information for clarification, the supplemental information shall be submitted as specified in paragraph REGULATORY REQUIREMENTS and as required in Section 01 33 00 SUBMITTAL PROCEDURES.

#### SD-06 Test Reports

Telecommunications Cabling Testing; G

#### SD-07 Certificates

Telecommunications Contractor Qualifications; G

Key Personnel Qualifications; G

Manufacturer Qualifications; G

Test Plan; G

#### SD-09 Manufacturer's Field Reports

Factory Reel Tests; G

#### SD-10 Operation and Maintenance Data

Telecommunications Cabling and Pathway System Data Package 5; G

#### SD-11 Closeout Submittals

Record Documentation; G

### 1.6 QUALITY ASSURANCE

#### 1.6.1 Shop Drawings

In exception to Section 01 33 00 SUBMITTAL PROCEDURES, submitted plan drawings shall be a minimum of 11 by 17 inches in size using a minimum scale of 1/8 inch per foot. Include wiring diagrams and installation details of equipment indicating proposed location, layout and arrangement, control panels, accessories, piping, ductwork, and other items that must be shown to ensure a coordinated installation. Wiring diagrams shall identify circuit terminals and indicate the internal wiring for each item of equipment and the interconnection between each item of equipment.

Drawings shall indicate adequate clearance for operation, maintenance, and replacement of operating equipment devices. Submittals shall include the nameplate data, size, and capacity. Submittals shall also include applicable federal, military, industry, and technical society publication references.

#### 1.6.1.1 Telecommunications Drawings

Provide drawings in accordance with TIA-606. The identifier for each termination and cable shall appear on the drawings. Drawings shall depict final telecommunications installed wiring system infrastructure in accordance with TIA-606. The drawings should provide details required to prove that the distribution system shall properly support connectivity from the EF telecommunications and ER telecommunications, CD's, and FD's to the telecommunications work area outlets. Provide a plastic laminated schematic of the as-installed telecommunications cable system showing cabling, CD's, BD's, FD's, and the EF and ER for telecommunications keyed to floor plans by room number. Mount the laminated schematic in the EF telecommunications space as directed by the Contracting Officer. The following drawings shall be provided as a minimum:

- a. T1 - Layout of complete building per floor - Building Area/Serving Zone Boundaries, Backbone Systems, and Horizontal Pathways. Layout of complete building per floor. The drawing indicates location of building areas, serving zones, vertical backbone diagrams, telecommunications rooms, access points, pathways, grounding system, and other systems that need to be viewed from the complete building perspective.
- b. T2 - Serving Zones/Building Area Drawings - Drop Locations and Cable Identification (ID'S). Shows a building area or serving zone. These drawings show drop locations, telecommunications rooms, access points and detail call outs for common equipment rooms and other congested areas.
- c. T4 - Typical Detail Drawings - Faceplate Labeling, Firestopping, Americans with Disabilities Act (ADA), Safety, Department of Transportation (DOT). Detailed drawings of symbols and typicals such as faceplate labeling, faceplate types, faceplate population installation procedures, detail racking, and raceways.

#### 1.6.1.2 Telecommunications Space Drawings

Provide T3 drawings in accordance with TIA-606 that include telecommunications rooms plan views, pathway layout (cable tray, racks, ladder-racks, etc.), mechanical/electrical layout, and cabinet, rack and wall elevations. Drawings shall show layout of applicable equipment including incoming cable stub or connector blocks, building protector assembly, outgoing cable connector blocks, patch panels and equipment spaces and cabinet/racks. Drawings shall include a complete list of equipment and material, equipment rack details, proposed layout and anchorage of equipment and appurtenances, and equipment relationship to other parts of the work including clearance for maintenance and operation. Drawings may also be an enlargement of a congested area of T1 or T2 drawings.

#### 1.6.2 Telecommunications Qualifications

Work under this section shall be performed by and the equipment shall be

provided by the approved telecommunications contractor and key personnel. Qualifications shall be provided for: the telecommunications system contractor, the telecommunications system installer, and the supervisor (if different from the installer). A minimum of 30 days prior to installation, submit documentation of the experience of the telecommunications contractor and of the key personnel.

#### 1.6.2.1 Telecommunications Contractor

The telecommunications contractor shall be a firm which is regularly and professionally engaged in the business of the applications, installation, and testing of the specified telecommunications systems and equipment. The telecommunications contractor shall demonstrate experience in providing successful telecommunications systems within the past 3 years of similar scope and size. Submit documentation for a minimum of three and a maximum of five successful telecommunication system installations for the telecommunications contractor.

#### 1.6.2.2 Key Personnel

Provide key personnel who are regularly and professionally engaged in the business of the application, installation and testing of the specified telecommunications systems and equipment. There may be one key person or more key persons proposed for this solicitation depending upon how many of the key roles each has successfully provided. Each of the key personnel shall demonstrate experience in providing successful telecommunications systems within the past 3 years.

Supervisors and installers assigned to the installation of this system or any of its components shall be Building Industry Consulting Services International (BICSI) Registered Cabling Installers, Technician Level. Submit documentation of current BICSI certification for each of the key personnel.

In lieu of BICSI certification, supervisors and installers assigned to the installation of this system or any of its components shall have a minimum of 3 years experience in the installation of the specified copper and fiber optic cable and components. They shall have factory or factory approved certification from each equipment manufacturer indicating that they are qualified to install and test the provided products. Submit documentation for a minimum of three and a maximum of five successful telecommunication system installations for each of the key personnel. Documentation for each key person shall include at least two successful system installations provided that are equivalent in system size and in construction complexity to the telecommunications system proposed for this solicitation. Include specific experience in installing and testing telecommunications systems and provide the names and locations of at least two project installations successfully completed using optical fiber and copper telecommunications cabling systems. All of the existing telecommunications system installations offered by the key persons as successful experience shall have been in successful full-time service for at least 18 months prior to the issuance date for this solicitation. Provide the name and role of the key person, the title, location, and completed installation date of the referenced project, the referenced project owner point of contact information including name, organization, title, and telephone number, and generally, the referenced project description including system size and construction complexity.

Indicate that all key persons are currently employed by the

telecommunications contractor, or have a commitment to the telecommunications contractor to work on this project. All key persons shall be employed by the telecommunications contractor at the date of issuance of this solicitation, or if not, have a commitment to the telecommunications contractor to work on this project by the date that the bid was due to the Contracting Officer.

Note that only the key personnel approved by the Contracting Officer in the successful proposal shall do work on this solicitation's telecommunications system. Key personnel shall function in the same roles in this contract, as they functioned in the offered successful experience. Any substitutions for the telecommunications contractor's key personnel requires approval from The Contracting Officer.

#### 1.6.2.3 Minimum Manufacturer Qualifications

Cabling, equipment and hardware manufacturers shall have a minimum of 3 years experience in the manufacturing, assembly, and factory testing of components which comply with TIA-568.1, TIA-568.2 and TIA-568.3.

#### 1.6.3 Test Plan

Provide a complete and detailed test plan for the telecommunications cabling system including a complete list of test equipment for the components and accessories for each cable type specified, 60 days prior to the proposed test date. Include procedures for certification, validation, and testing.

#### 1.6.4 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship shall be in accordance with the mandatory and advisory provisions of NFPA 70 unless more stringent requirements are specified or indicated.

#### 1.6.5 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products shall have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year period shall include applications of equipment and materials under similar circumstances and of similar size. The product shall have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Where two or more items of the same class of equipment are required, these items shall be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

##### 1.6.5.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests,

is furnished.

#### 1.6.5.2 Material and Equipment Manufacturing Date

Products manufactured more than 1 year prior to date of delivery to site shall not be used, unless specified otherwise.

#### 1.7 DELIVERY AND STORAGE

Provide protection from weather, moisture, extreme heat and cold, dirt, dust, and other contaminants for telecommunications cabling and equipment placed in storage.

#### 1.8 ENVIRONMENTAL REQUIREMENTS

Connecting hardware shall be rated for operation under ambient conditions of 32 to 140 degrees F and in the range of 0 to 95 percent relative humidity, noncondensing.

#### 1.9 WARRANTY

The equipment items shall be supported by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

#### 1.10 MAINTENANCE

##### 1.10.1 Operation and Maintenance Manuals

Commercial off the shelf manuals shall be furnished for operation, installation, configuration, and maintenance of products provided as a part of the telecommunications cabling and pathway system, Data Package 5. Submit operations and maintenance data in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA and as specified herein not later than 2 months prior to the date of beneficial occupancy. In addition to requirements of Data Package 5, include the requirements of paragraphs TELECOMMUNICATIONS DRAWINGS, TELECOMMUNICATIONS SPACE DRAWINGS, and RECORD DOCUMENTATION. Ensure that these drawings and documents depict the as-built configuration.

##### 1.10.2 Record Documentation

Provide T5 drawings including documentation on cables and termination hardware in accordance with TIA-606. T5 drawings shall include schedules to show information for cut-overs and cable plant management, patch panel layouts and cover plate assignments, cross-connect information and connecting terminal layout as a minimum. T5 drawings shall be provided on electronic media using Windows based computer cable management software. Provide the following T5 drawing documentation as a minimum:

- a. Cables - A record of installed cable shall be provided in accordance with TIA-606. The cable records shall include the required data fields for each cable and complete end-to-end circuit report for each complete circuit from the assigned outlet to the entry facility in accordance with TIA-606. Include manufacture date of cable with submittal.
- b. Termination Hardware - A record of installed patch panels,

cross-connect points, distribution frames, terminating block arrangements and type, and outlets shall be provided in accordance with TIA-606. Documentation shall include the required data fields as a minimum in accordance with TIA-606.

### 1.10.3 Spare Parts

In addition to the requirements of Section 01 78 23 OPERATION AND MAINTENANCE DATA, provide a complete list of parts and supplies, with current unit prices and source of supply, and a list of spare parts recommended for stocking.

## PART 2 PRODUCTS

### 2.1 COMPONENTS

Components shall be UL or third party certified. Where equipment or materials are specified to conform to industry and technical society reference standards of the organizations, submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. The certificate shall state that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard. Provide a complete system of telecommunications cabling and pathway components using star topology. Provide support structures and pathways, complete with outlets, cables, connecting hardware and telecommunications cabinets/racks. Cabling and interconnecting hardware and components for telecommunications systems shall be UL listed or third party independent testing laboratory certified, and shall comply with NFPA 70 and conform to the requirements specified herein.

### 2.2 TELECOMMUNICATIONS PATHWAY

Provide telecommunications pathways in accordance with TIA-569 and as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide system furniture pathways in accordance with UL 1286.

### 2.3 TELECOMMUNICATIONS CABLING

Cabling shall be UL listed for the application and shall comply with TIA-568.0, TIA-568.1, TIA-568.2, TIA-568.3 and NFPA 70. Provide a labeling system for cabling as required by TIA-606 and UL 969. Ship cable on reels or in boxes bearing manufacture date for for unshielded twisted pair (UTP) in accordance with ICEA S-90-661 and optical fiber cables in accordance with ICEA S-83-596 for all cable used on this project. Cabling manufactured more than 12 months prior to date of installation shall not be used.

#### 2.3.1 Horizontal Cabling

Provide horizontal cable in compliance with NFPA 70 and performance characteristics in accordance with TIA-568.1.

##### 2.3.1.1 Horizontal Copper

Provide horizontal copper cable, UTP, 100 ohm in accordance with TIA-568.2,



UL 444, ANSI/NEMA WC 66, ICEA S-90-661 . Provide four each individually twisted pair, minimum size 24 AWG conductors, Category 6, with a blue thermoplastic jacket. Cable shall be imprinted with manufacturers name or identifier, flammability rating, gauge of conductor, transmission performance rating (category designation) and length marking at regular intervals in accordance with ICEA S-90-661. Provide plenum (CMP), riser (CMR), or general purpose (CM or CMG) communications rated cabling in accordance with NFPA 70. Substitution of a higher rated cable shall be permitted in accordance with NFPA 70. Cables installed in conduit within and under slabs shall be UL listed and labeled for wet locations in accordance with NFPA 70.

#### 2.3.1.2 Horizontal Optical Fiber

Provide optical fiber horizontal cable in accordance with ICEA S-83-596 and TIA-568.3. Cable shall be tight buffered, multimode, 50/125-um diameter, OM2. Cable shall be imprinted with manufacturer, flammability rating and fiber count at regular intervals not to exceed 40 inches.

Provide plenum (OFNP), riser (OFNR), or general purpose (OFN or OFNG) rated non-conductive, fiber optic cable in accordance with NFPA 70. Substitution of a higher rated cable shall be permitted in accordance with NFPA 70. Cables installed in conduit within and under slabs be UL listed and labeled for wet locations in accordance with NFPA 70. The cable jacket shall be of single jacket construction with color coding of cordage jacket, fiber, unit, and group in accordance with TIA/EIA-598.

#### 2.3.2 Work Area Cabling

##### 2.3.2.1 Work Area Copper

Provide work area copper cable in accordance with TIA-568.2, with a blue, thermoplastic jacket.

##### 2.3.2.2 Work Area Optical Fiber

Provide optical work area cable in accordance with TIA-568.3.

#### 2.4 TELECOMMUNICATIONS SPACES

Provide connecting hardware and termination equipment in the telecommunications entrance facility and telecommunication equipment room to facilitate installation as shown on design drawings for terminating and cross-connecting permanent cabling. Provide telecommunications interconnecting hardware color coding in accordance with TIA-606.

##### 2.4.1 Equipment Support Frame

Provide in accordance with ECIA EIA/ECA 310-E and UL 50.

- a. Cabinets, wall-mounted modular type, 16 gauge steel or 11 gauge aluminum construction, minimum, treated to resist corrosion. Cabinet shall have lockable front door, louvered side panels, 250 CFM roof mounted fan, ground lug, and top and bottom cable access. Cabinet shall be compatible with 19 inches panel mounting. All cabinets shall be keyed alike. A duplex AC outletsurge protected power strip with 6 duplex 20 amp receptacles shall be provided within the cabinet.

#### 2.4.2 Connector Blocks

Provide insulation displacement connector (IDC) Type 110 for Category 6 systems. Provide blocks for the number of horizontal and backbone cables terminated on the block plus 25 percent spare.

#### 2.4.3 Cable Guides

Provide cable guides specifically manufactured for the purpose of routing cables, wires and patch cords horizontally and vertically on 19 inches equipment cabinets. Cable guides of ring or bracket type devices mounted on cabinet panels for horizontal cable management and individually mounted for vertical cable management. Mount cable guides with screws, or nuts and lockwashers.

#### 2.4.4 Patch Panels

Provide ports for the number of horizontal and backbone cables terminated on the panel plus 25 percent spare. Provide pre-connectorized optical fiber and copper patch cords for patch panels. Provide patch cords, as complete assemblies, with matching connectors as specified. Provide fiber optic patch cables with crossover orientation in accordance with TIA-568.3. Patch cords shall meet minimum performance requirements specified in TIA-568.1, TIA-568.2 and TIA-568.3 for cables, cable length and hardware specified.

##### 2.4.4.1 Modular to 110 Block Patch Panel

Provide in accordance with TIA-568.1 and TIA-568.2. Panels shall be third party verified and shall comply with EIA/TIA Category 6 requirements. Panel shall be constructed of 0.09 inches minimum aluminum and shall be wall mounted and compatible with an ECIA EIA/ECA 310-E 19 inches equipment cabinet. Panel shall provide 48 non-keyed, 8-pin modular ports, wired to T568A. Patch panels shall terminate the building cabling on Type 110 IDCs and shall utilize a printed circuit board interface. The rear of each panel shall have incoming cable strain-relief and routing guides. Panels shall have each port factory numbered and be equipped with laminated plastic nameplates above each port.

##### 2.4.4.2 Fiber Optic Patch Panel

Provide panel for maintenance and cross-connecting of optical fiber cables. Panel shall be constructed of 16 gauge steel or 11 gauge aluminum minimum and shall be wall mounted and compatible with a ECIA EIA/ECA 310-E 19 inches equipment rack. Each panel shall provide 12 adapters as duplex LC in accordance with TIA/EIA-604-10 with zirconia ceramic alignment sleeves, alignment sleeves. Provide dust cover for unused adapters. The rear of each panel shall have a cable management tray a minimum of 8 inches deep with removable cover, incoming cable strain-relief and routing guides. Panels shall have each adapter factory numbered and be equipped with laminated plastic nameplates above each adapter.

#### 2.5 TELECOMMUNICATIONS OUTLET/CONNECTOR ASSEMBLIES

##### 2.5.1 Outlet/Connector Copper

Outlet/connectors shall comply with FCC Part 68, TIA-568.1, and TIA-568.2. UTP outlet/connectors shall be UL 1863 listed, non-keyed, 8-pin modular,

constructed of high impact rated thermoplastic housing and shall be third party verified and shall comply with TIA-568.2 Category 6 requirements. Outlet/connectors provided for UTP cabling shall meet or exceed the requirements for the cable provided. Outlet/connectors shall be terminated using a Type 110 IDC PC board connector, color-coded for both T568A and T568B wiring. Each outlet/connector shall be wired T568A. UTP outlet/connectors shall comply with TIA-568.2 for 200 mating cycles. UTP outlet/connectors installed in outdoor or marine environments shall be jell-filled type containing an anti-corrosive, memory retaining compound.

#### 2.5.2 Optical Fiber Adapters(Couplers)

Provide optical fiber adapters suitable for duplex LC in accordance with TIA/EIA-604-10 with zirconia ceramic alignment sleeves, as indicated. Provide dust cover for adapters. Optical fiber adapters shall comply with TIA-455-21 for 500 mating cycles.

#### 2.5.3 Optical Fiber Connectors

Provide in accordance with TIA-455-21. Optical fiber connectors shall be duplex LC in accordance with TIA/EIA-604-10 with zirconia ceramic alignment sleeves, ferrule, epoxyless crimp style compatible with 50/125 multimode fiber. The connectors shall provide a maximum attenuation of 0.3 dB at 850 nm with less than a 0.2 dB change after 500 mating cycles.

#### 2.5.4 Cover Plates

Telecommunications cover plates shall comply with UL 514C, and TIA-568.1, TIA-568.2, TIA-568.3; flush or oversized design constructed of high impact thermoplastic material ivory in color. Provide labeling in accordance with the paragraph LABELING in this section.

### 2.6 GROUNDING AND BONDING PRODUCTS

Provide in accordance with UL 467, TIA-607, and NFPA 70. Components shall be identified as required by TIA-606. Provide ground rods, bonding conductors, and grounding busbars as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

### 2.7 FIRESTOPPING MATERIAL

Provide as specified in Section 07 84 00 FIRESTOPPING.

### 2.8 MANUFACTURER'S NAMEPLATE

Each item of equipment shall have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

### 2.9 FIELD FABRICATED NAMEPLATES

ASTM D709. Provide laminated plastic nameplates for each equipment enclosure, relay, switch, and device; as specified or as indicated on the drawings. Each nameplate inscription shall identify the function and, when applicable, the position. Nameplates shall be melamine plastic, 0.125 inches thick, white with black center core. Surface shall be matte finish. Corners shall be square. Accurately align lettering and engrave into the core. Minimum size of nameplates shall be one by 2.5 inches.

Lettering shall be a minimum of 0.25 inches high normal block style.

## 2.10 TESTS, INSPECTIONS, AND VERIFICATIONS

### 2.10.1 Factory Reel Tests

Provide documentation of the testing and verification actions taken by manufacturer to confirm compliance with TIA-568.1, TIA-568.2, TIA-568.3, and TIA-526-14 for multimode optical fiber cables.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Install telecommunications cabling and pathway systems, including the horizontal and backbone cable, pathway systems, telecommunications outlet/connector assemblies, and associated hardware in accordance with NECA/BICSI 568, TIA-568.1, TIA-568.2, TIA-568.3, TIA-569, NFPA 70, and UL standards as applicable. Provide cabling in a star topology network. Pathways and outlet boxes shall be installed as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Install telecommunications cabling with copper media in accordance with the following criteria to avoid potential electromagnetic interference between power and telecommunications equipment. The interference ceiling shall not exceed 3.0 volts per meter measured over the usable bandwidth of the telecommunications cabling. Cabling shall be run with horizontal and vertical cable guides in telecommunications spaces with terminating hardware and interconnection equipment.

#### 3.1.1 Cabling

Install UTP, and optical fiber telecommunications cabling system as detailed in TIA-568.1, TIA-568.2, TIA-568.3. Screw terminals shall not be used except where specifically indicated on plans. Use an approved insulation displacement connection (IDC) tool kit for copper cable terminations. Do not exceed manufacturers' cable pull tensions for copper and optical fiber cables. Provide a device to monitor cable pull tensions. Do not exceed 25 pounds pull tension for four pair copper cables. Do not chafe or damage outer jacket materials. Use only lubricants approved by cable manufacturer. Do not over cinch cables, or crush cables with staples. For UTP cable, bend radii shall not be less than four times the cable diameter. Cables shall be terminated; no cable shall contain unterminated elements. Cables shall not be spliced. Label cabling in accordance with paragraph LABELING in this section.

##### 3.1.1.1 Open Cable

Use only where specifically indicated on plans for use in cable trays, or below raised floors. Install in accordance with TIA-568.1, TIA-568.2 and TIA-568.3. Do not exceed cable pull tensions recommended by the manufacturer. Copper cable not in a wireway or pathway shall be suspended a minimum of 8 inches above ceilings by cable supports no greater than 60 inches apart. Cable shall not be run through structural members or in contact with pipes, ducts, or other potentially damaging items. Placement of cable parallel to power conductors shall be avoided, if possible; a minimum separation of 12 inches shall be maintained when such placement cannot be avoided.

Plenum cable shall be used where open cables are routed through plenum

areas. Cable routed exposed under raised floors shall be plenum rated. Plenum cables shall comply with flammability plenum requirements of NFPA 70. Install cabling after the flooring system has been installed in raised floor areas. Cable 6 feet long shall be neatly coiled not less than 12 inches in diameter below each feed point in raised floor areas.

#### 3.1.1.2 Backbone Cable

- a. Copper Backbone Cable. Install intrabuilding backbone copper cable, in indicated pathways, between the campus distributor, located in the telecommunications entrance facility or room, the building distributors and the floor distributors located in telecommunications rooms and telecommunications equipment rooms as indicated on drawings.
- b. Optical fiber Backbone Cable. Install intrabuilding backbone optical fiber in indicated pathways. Do not exceed manufacturer's recommended bending radii and pull tension. Prepare cable for pulling by cutting outer jacket 10 inches leaving strength members exposed for approximately 10 inches. Twist strength members together and attach to pulling eye. Vertical cable support intervals shall be in accordance with manufacturer's recommendations.

#### 3.1.1.3 Horizontal Cabling

Install horizontal cabling as indicated on drawings. Do not untwist Category 6 UTP cables more than one half inch from the point of termination to maintain cable geometry. Provide slack cable in the form of a figure eight (not a service loop) on each end of the cable, 10 feet in the telecommunications room, and 12 inches in the work area outlet.

#### 3.1.2 Pathway Installations

Provide in accordance with TIA-569 and NFPA 70. Provide building pathway as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

#### 3.1.3 Cable Tray Installation

Install cable tray as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Only CMP and OFNP type cable shall be installed in a plenum.

#### 3.1.4 Work Area Outlets

##### 3.1.4.1 Terminations

Terminate UTP cable in accordance with TIA-568.1, TIA-568.2 and wiring configuration as specified. Terminate fiber optic cables in accordance with TIA-568.3.

##### 3.1.4.2 Cover Plates

As a minimum, each outlet/connector shall be labeled as to its function and a unique number to identify cable link in accordance with the paragraph LABELING in this section.

##### 3.1.4.3 Cables

Unshielded twisted pair and fiber optic cables shall have a minimum of 12 inches of slack cable loosely coiled into the telecommunications outlet boxes. Minimum manufacturer's bend radius for each type of cable shall

not be exceeded.

#### 3.1.4.4 Pull Cords

Pull cords shall be installed in conduit serving telecommunications outlets that do not have cable installed.

#### 3.1.5 Telecommunications Space Termination

Install termination hardware required for Category 6 and optical fiber system. An insulation displacement tool shall be used for terminating copper cable to insulation displacement connectors.

##### 3.1.5.1 Connector Blocks

Connector blocks shall be cabinet mounted in orderly rows and columns. Adequate vertical and horizontal wire routing areas shall be provided between groups of blocks. Install in accordance with industry standard wire routing guides in accordance with TIA-569.

##### 3.1.5.2 Patch Panels

Patch panels shall be mounted in equipment cabinets with sufficient ports to accommodate the installed cable plant plus 25 percent spares.

- a. Fiber Optic Patch Panel. Fiber optic cable loop shall be provided as recommended by the manufacturer. The outer jacket of each cable entering a patch panel shall be secured to the panel to prevent movement of the fibers within the panel, using clamps or brackets specifically manufactured for that purpose.

##### 3.1.5.3 Equipment Support Frames

Install in accordance with TIA-569:

- a. Cabinets, wall-mounted modular type. Mount cabinet to plywood backboard in accordance with manufacturer's recommendations. Mount cabinet so height of highest panel does not exceed 78 inches above floor.

#### 3.1.6 Electrical Penetrations

Seal openings around electrical penetrations through fire resistance-rated wall, partitions, floors, or ceilings as specified in Section 07 84 00 FIRESTOPPING.

#### 3.1.7 Grounding and Bonding

Provide in accordance with TIA-607, NFPA 70 and as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

### 3.2 LABELING

#### 3.2.1 Labels

Provide labeling in accordance with TIA-606. Handwritten labeling is unacceptable. Stenciled lettering for voice and data circuits shall be provided using thermal ink transfer process .

### 3.2.2 Cable

Cables shall be labeled using color labels on both ends with identifiers in accordance with TIA-606.

### 3.2.3 Termination Hardware

Workstation outlets and patch panel connections shall be labeled using color coded labels with identifiers in accordance with TIA-606.

## 3.3 FIELD APPLIED PAINTING

Paint electrical equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria. Painting shall be as specified in Section 09 90 00 PAINTS AND COATINGS.

### 3.3.1 Painting Backboards

If backboards are required to be painted, then the manufactured fire retardant backboard must be painted with fire retardant paint, so as not to increase flame spread and smoke density and must be appropriately labeled. Label and fire rating stamp must be unpainted.

## 3.4 FIELD FABRICATED NAMEPLATE MOUNTING

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

## 3.5 TESTING

### 3.5.1 Telecommunications Cabling Testing

Perform telecommunications cabling inspection, verification, and performance tests in accordance with TIA-568.1, TIA-568.2, TIA-568.3. Test equipment shall conform to TIA-1152. Perform optical fiber field inspection tests via attenuation measurements on factory reels and provide results along with manufacturer certification for factory reel tests. Remove failed cable reels from project site upon attenuation test failure.

#### 3.5.1.1 Inspection

Visually inspect UTP and optical fiber jacket materials for UL or third party certification markings. Inspect cabling terminations in telecommunications rooms and at workstations to confirm color code for T568A or T568B pin assignments, and inspect cabling connections to confirm compliance with TIA-568.1, TIA-568.2, TIA-568.3, . Visually confirm Category 6, marking of outlets, cover plates, outlet/connectors, and patch panels.

#### 3.5.1.2 Verification Tests

UTP backbone copper cabling shall be tested for DC loop resistance, shorts, opens, intermittent faults, and polarity between conductors, and between conductors and shield, if cable has overall shield. Test operation of shorting bars in connection blocks. Test cables after termination but prior to being cross-connected.

For multimode optical fiber, perform optical fiber end-to-end attenuation tests in accordance with TIA-568.3 and TIA-526-14 using Method A, Optical Power Meter and Light Source for multimode optical fiber. Perform verification acceptance tests.

#### 3.5.1.3 Performance Tests

Perform testing for each outlet and MUTOA as follows:

- a. Perform Category 6 link tests in accordance with TIA-568.1 and TIA-568.2. Tests shall include wire map, length, insertion loss, NEXT, PSNEXT, ELFEXT, PSELFEXT, return loss, propagation delay, and delay skew.
- b. Optical fiber Links. Perform optical fiber end-to-end link tests in accordance with TIA-568.3.

#### 3.5.1.4 Final Verification Tests

Perform verification tests for UTP and optical fiber systems after the complete telecommunications cabling and workstation outlet/connectors are installed.

- a. Data Tests. These tests assume the Information Technology Staff has a network installed and are available to assist with testing. Connect to the network interface device at the demarcation point. Log onto the network to ensure proper connection to the network.

-- End of Section --



SECTION 28 31 60

INTERIOR FIRE ALARM SYSTEM, NON-ADDRESSABLE  
**08/20**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

FM GLOBAL (FM)

FM APP GUIDE (updated on-line) Approval Guide  
<https://www.approvalguide.com/>

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 4 (2018) Standard for Integrated Fire Protection and Life Safety System Testing

NFPA 70 (2023; ERTA 1 2024; TIA 24-1) National Electrical Code

NFPA 72 (2022; ERTA 22-1) National Fire Alarm and Signaling Code

NFPA 170 (2024; ERTA 1 2023) Standard for Fire Safety and Emergency Symbols

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 4-010-06 (2023) Cybersecurity of Facility-Related Control Systems

UL SOLUTIONS (UL)

UL 464 (2023) UL Standard for Safety Audible Signaling Devices for Fire Alarm and Signaling Systems, Including Accessories

UL 1638 (2023) UL Standard for Safety Visible Signaling Devices for Fire Alarm and Signaling Systems, Including Accessories

UL 1971 (2002; Reprint Feb 2024) UL Standard for Safety Signaling Devices for the Deaf and Hard of Hearing

UL Fire Prot Dir UL Product IQ (updated online) at  
<https://productiq.ulpropsector.com/en>

1.2 RELATED SECTIONS

Section 25 05 11 CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS, applies to this section, with the additions and modifications specified

herein. In addition, refer to the following sections for related work and coordination:

Section 21 13 13 WET PIPE SPRINKLER SYSTEM, FIRE PROTECTION

Section 07 84 00 FIRESTOPPING for additional work related to firestopping.

### 1.3 SUMMARY

#### 1.3.1 Scope

- a. This work includes designing and modifying the existing fire alarm system as described herein and on the contract drawing Building 72. Include system wiring, raceways, pull boxes, terminal cabinets, outlet and mounting boxes, control equipment, initiating devices, notification appliances, supervising station fire alarm transmitters, and other accessories and miscellaneous items required for a complete operational system even though each item is not specifically mentioned or described. Provide system complete and ready for operation. Existing interior fire alarm system was manufactured by King-Fisher. Design and installation must comply with UFGS 25 05 11 and UFC 4-010-06.
- b. Provide equipment, materials, installation, workmanship, inspection, and testing in strict accordance with NFPA 72, except as modified herein. The system layout on the drawings show the intent of coverage and suggested locations. Final quantity, system layout, and coordination are the responsibility of the Contractor.
- c. The fire alarm system must be independent of the building security, building management, and energy/utility monitoring systems other than for control functions.

#### 1.3.2 Qualified Fire Protection Engineer (QFPE)

The QFPE in this section is the same as FPQC in Section 01 45 00.00 22. Wherever Qualified Fire Protection Engineer (QFPE) or QFPE appears in this section, substitute with Fire Protection Qualify Control Specialist (FPQC) Services of the QFPE must include:

- a. Reviewing SD-02, SD-03, and SD-05 submittal packages for completeness and compliance with the provisions of this specification. Construction (shop) drawings and calculations must be prepared and signed by the fire alarm designer.
- b. Providing a letter documenting that the SD-02, SD-03, and SD-05 submittal package has been reviewed and noting any outstanding comments. The QFPE must affix their professional engineering stamp with signature to this letter, indicating approval prior to submitting to the DFPE.
- c. Performing in-progress construction surveillance prior to installation of ceilings (rough-in inspection).
- d. Witnessing pre-Government and final Government functional performance testing and performing a final installation review.

- e. Signing applicable certificates under SD-07.

#### 1.4 DEFINITIONS

Wherever mentioned in this specification or on the drawings, the equipment, devices, and functions must be defined as follows:

##### 1.4.1 Terminal Cabinet

A steel cabinet with locking, hinge-mounted door where terminal strips are securely mounted inside the cabinet.

##### 1.4.2 Designated Fire Protection Engineer (DFPE)

The DoD fire protection engineer that oversees that Area of Responsibility for that project. This is sometimes referred to as the "cognizant" fire protection engineer. Interpret reference to "authority having jurisdiction" and/or AHJ in referenced standards to mean the Designated Fire Protection Engineer (DFPE). The DFPE may be responsible for review of the contractor submittals having a "G" designation, and for witnessing final inspection and testing.

##### 1.4.3 Qualified Fire Protection Engineer (QFPE)

A QFPE is an individual who is a licensed professional engineer (P.E.), who has passed the fire protection engineering written examination administered by the National Council of Examiners for Engineering and Surveying (NCEES) and has relevant fire protection engineering experience.

#### 1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval.

Shop drawings (SD-02), product data (SD-03) and calculations (SD-05) must be prepared by the fire alarm designer and combined and submitted as one complete package. The QFPE must review the SD-02/SD-03/SD-05 submittal package for completeness and compliance with the Contract provisions prior to submission to the Government. The QFPE must provide a Letter of Confirmation that they have reviewed the submittal package for compliance with the contract provisions. This letter must include their registered professional engineer stamp and signature. Partial submittals and submittals not reviewed by the QFPE will be returned by the Government disapproved without review.

Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-01 Preconstruction Submittals

Qualified Fire Protection Engineer (QFPE); G

Fire alarm system designer; G

Technician; G

Installer; G

Test Technician; G

SD-02 Shop Drawings

Wiring Diagrams; G

System Layout; G

Notification Appliances; G

Initiating devices; G

Battery Power; G

Voltage Drop Calculations; G

SD-03 Product Data

Manual Stations; G

Notification Appliances; G

Alarm Wiring; G

Back Boxes and Conduit; G

Terminal Cabinets; G

Environmental Enclosures or Guards; G

SD-05 Design Data

SD-06 Test Reports

Test Procedures; G

SD-07 Certificates

Verification of Compliant Installation; G

Request for Government Final Test; G

SD-10 Operation and Maintenance Data

Operation and Maintenance (O&M) Instructions; G

SD-11 Closeout Submittals

As-Built Drawings; G

## 1.6 SYSTEM OPERATION

Fire alarm system components requiring power, except for the FACU(s) power supply, must operate on 24 volts DC unless noted otherwise in this section.

Provide a fire alarm system that is a complete, supervised fire alarm reporting system. Activate the system into the alarm mode by actuation of any alarm initiating device. Remain in the alarm mode until the initiating device is reset and the fire alarm control unit is reset and restored to normal. Audible and visual appliances and systems must comply with NFPA 72. Operate fire alarm system components requiring power, except for the control unit power supply, on 24 Volts dc.

### 1.6.1 Alarm Initiating Devices and Notification Appliances (Visual, Audible)

- a. Connect alarm initiating devices to initiating device circuits (IDC) Class "B" and installed in accordance with NFPA 72.
- b. Connect notification appliances to notification appliance circuits (NAC) Class "B".

### 1.6.2 Functions and Operating Features

The sequence of operation for the existing system will not be modified. All manual pull stations water flow devices will be set as an alarm function. All monitor modules connected to tamper switches will be set as supervisory functions.

## 1.7 EXISTING EQUIPMENT

- a. Equipment and devices must be compatible and operable with the existing fire alarm system and must not impair reliability or operational functions of existing supervising station fire alarm system..

## 1.8 QUALITY ASSURANCE

### 1.8.1 Submittal Documents

#### 1.8.1.1 Preconstruction Submittals

Within 36 days of contract award but not less than 14 days prior to commencing any work on site, the Contractor must submit the following for review and approval. SD-02, SD-03 and SD-05 submittals received prior to the review and approval of the qualifications of the fire alarm subcontractor and QFPE must be returned disapproved without review. All resultant delays must be the sole responsibility of the Contractor.

#### 1.8.1.2 Shop Drawings

Shop drawings must not be smaller than the Contract Drawings. Drawings must comply with the requirements of NFPA 72 and NFPA 170. Minimum scale for floor plans must be 1/8"=1'.

#### 1.8.1.3 Wiring Diagrams

Electronic copies of point-to-point wiring diagrams showing the points of connection and terminals used for electrical field connections in the system, including interconnections between the equipment or systems that are supervised or controlled by the system. Diagrams must show connections from field devices to the FACU and remote FACU, initiating circuits, switches, relays and terminals, including pathway diagrams between the control unit and shared communications equipment within the protected premises. Point-to-point wiring diagrams must be job specific and must not indicate connections or circuits not being utilized. Provide complete riser diagrams indicating the wiring sequence of all devices and their connections to the control equipment. Include a color-code schedule for the wiring.

#### 1.8.1.4 System Layout

Electronic copies of plan view drawing showing device locations, terminal cabinet locations, junction boxes, other related equipment, conduit routing, conduit sizes, wire counts, conduit fill calculations, wire color-coding, circuit identification in each conduit, and circuit layouts for all floors. Indicate candela rating of each visual notification appliance. Indicate the addresses of all devices, modules, relays, and similar. Indicate if the environment for the FACU is within its environmental listing (e.g. temperature/humidity).

Provide a complete description of the system operation in matrix format similar to the "Typical Input/Output Matrix" included in the Annex of NFPA 72.

#### 1.8.1.5 Notification Appliances

Calculations and supporting data on each circuit to indicate that there is at least 25 percent spare capacity for notification appliances. Annotate data for each circuit within the project on the drawings.

#### 1.8.1.6 Initiating Devices

Calculations and supporting data on each circuit to indicate that there is at least 25 percent spare capacity for initiating devices. Annotate data for each circuit within the project on the drawings.

#### 1.8.1.7 Battery Power

Calculations and supporting data as required in paragraph Battery Power Calculations for alarm, alert, and supervisory power requirements. Calculations including ampere-hour requirements for each system component and each control unit component, and the battery recharging period, must be included on the drawings.

#### 1.8.1.8 Voltage Drop Calculations

Voltage drop calculations for each notification circuit indicating that sufficient voltage is available for proper operation of the system and all components, at a minimum rated voltage of the system operating on batteries. Include the calculations on the system layout drawings. Due to the lack of existing as-builts, this calculation will be a rough estimation using the distance between the control panel and the end of the circuit.

#### 1.8.1.9 Product Data

Electronic copies of annotated descriptive data to show the specific model, type, and size of each item. Catalog cuts must also indicate the NRTL listing. The data must be highlighted to show model, size, and options that are intended for consideration. Data must be adequate to demonstrate compliance with all contract requirements. Product data for all equipment must be combined into a single submittal.

Provide an equipment list identifying the type, quantity, make, and model number of each piece of equipment to be provided under this submittal. The equipment list must include the type, quantity, make and model of spare equipment. Types and quantities of equipment submitted must coincide with the types and quantities of equipment used in the battery calculations and those shown on the shop drawings.

#### 1.8.1.10 As-Built Drawings

The drawings must show the system as installed, including deviations from both the project drawings and the approved shop drawings. These drawings must be submitted within two weeks after the final Government test of the system. At least one set of the as-built (marked-up) drawings must be provided at the time of, or prior to the final Government test.

### 1.8.2 Qualifications

#### 1.8.2.1 Fire Alarm System Designer

The fire alarm system designer must be certified as a Level III(minimum) Technician by National Institute for Certification in Engineering Technologies (NICET) in the Fire Alarm Systems subfield of Fire Protection Engineering Technology or meet the qualifications for a QFPE.

#### 1.8.2.2 Technician

Fire alarm technicians with a minimum of four years of experience must be utilized to install and terminate fire alarm devices, cabinets and control units. The fire alarm technicians installing the equipment must be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings.

#### 1.8.2.3 Installer

Fire alarm installer with a minimum of two years of experience utilized to assist in the installation of fire alarm devices, cabinets and control units. A licensed electrician must be allowed to install wire, cable, conduit and backboxes for the fire alarm system. The fire alarm installer must be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings.

#### 1.8.2.4 Test Technician

Fire alarm technicians with a minimum of eight years of experience and NICET Level III utilized in testing and certification of the installation of the fire alarm devices, cabinets and control units. The fire alarm technicians testing the equipment must be factory trained in the installation, adjustment, testing, and operation of the equipment installed as part of this project.

#### 1.8.2.5 Manufacturer

Components must be of current design and must be in regular and recurrent production at the time of installation. Provide design, materials, and devices for a protected premises fire alarm system, complete, conforming to NFPA 72, except as specified herein.

#### 1.8.3 Regulatory Requirements

Equipment and material must be listed or approved. Listed or approved, as used in this Section, means listed, labeled or approved by a Nationally Recognized Testing Laboratory (NRTL) such as UL Fire Prot Dir or FM APP GUIDE. The omission of these terms under the description of any item of equipment described must not be construed as waiving this requirement. All listings or approvals by testing laboratories must be from an existing ANSI or UL published standard. The recommended practices stated in the manufacturer's literature or documentation must be considered as mandatory requirements.

#### 1.9 DELIVERY, STORAGE, AND HANDLING

Protect equipment delivered and placed in storage from the weather, humidity, and temperature variation, dirt and dust, and other contaminants.

#### 1.10 MAINTENANCE

### PART 2 PRODUCTS

#### 2.1 GENERAL PRODUCT REQUIREMENT

All fire alarm equipment must be listed for use under the applicable reference standards.



## 2.2 MATERIALS AND EQUIPMENT

### 2.2.1 Standard Products

Provide materials, equipment, and devices that have been tested by a nationally recognized testing laboratory and listed for fire protection service when so required by NFPA 72 or this specification. Select material from one manufacturer, where possible, and not a combination of manufacturers, for any particular classification of materials. Material and equipment must be the standard products of a manufacturer regularly engaged in the manufacture of the products for at least 2 years prior to bid opening.

### 2.3 MANUAL STATIONS

Provide metal or plastic, surface mounted, double-action, manual stations, that are not subject to operation by jarring or vibration. Stations must be equipped with screw terminals for each conductor. Stations that require the replacement of any portion of the device after activation are not permitted. Stations must be finished in red with molded raised lettering operating instructions of contrasting color. The use of a key must be required to reset the station.

### 2.4 Notification Appliances

#### 2.4.1 Audible Notification Appliances

Audible appliances must conform to the applicable requirements of UL 464. Appliances must be connected into notification appliance circuits. Surface mounted audible appliances must be painted red. Recessed audible appliances must be installed with a grill that is painted red.

##### 2.4.1.1 Horns

Horns must be surface-mounted, with the matching mounting backbox surface mounted vibrating type suitable for use in an electrically supervised circuit. Horns must produce a sound rating of at least 85 dBA at 10 feet. Horns used in exterior locations must be specifically listed or approved for outdoor use and be provided with metal housing and protective grilles. Horns must be explosionproof in the corrosive storage room.

#### 2.4.2 Visual Notification Appliances

Visual notification appliances must conform to the applicable requirements of UL 1638, UL 1971 and conform to the Architectural Barriers Act (ABA). Visual Notification Appliances must have clear high intensity optic lens, xenon flash tubes, or light emitting diode (LED) and be marked "Alert" in letters of contrasting color. The light pattern must be disbursed so that it is visible above and below the strobe and from a 90 degree angle on both sides of the strobe. Strobe flash rate must be 1 flash per second and a minimum of 15 candela based on the UL 1971 test. Strobe must be surface mounted.

### 2.5 WIRING

Provide wiring materials under this section as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM with the additions and modifications specified herein.

### 2.5.1 Alarm Wiring

IDC wiring must be solid copper cable in accordance with the manufacturers requirements. Copper initiating device circuit field wiring must be No. 16 AWG size conductors at a minimum. Visual notification appliance circuit conductors, that contain audible alarm appliances, must be copper No. 14 AWG size conductors at a minimum. Wire size must be sufficient to prevent voltage drop problems. Circuits operating at 24 VDC must not operate at less than the listed voltages for the detectors and/or appliances. Power wiring, operating at 120 VAC minimum, must be a minimum No. 12 AWG solid copper having similar insulation. Acceptable power-limited cables are FPL, FPLR or FPLP as appropriate with red colored covering. Nonpower-limited cables must comply with NFPA 70.

## 2.6 SYSTEM MONITORING

### 2.6.1 Valves

Each valve affecting the proper operation of a fire protection system, including automatic sprinkler control valves, sprinkler service entrance valve, isolating valves for pressure type waterflow or supervision switches, and valves at backflow preventers, whether supplied under this contract or existing, must be electrically monitored to ensure its proper position. Provide each tamper switch with a separate zone, unless they are within the same room, then a maximum of five can use the same zone.

## 2.7 ENVIRONMENTAL ENCLOSURES OR GUARDS

Environmental enclosures must be provided to permit fire alarm components to be used in areas that exceed the environmental limits of the listing. The enclosure must be listed for the device or appliance as either a manufactured part number or as a listed compatible accessory for the component is currently listed. Guards required to deter mechanical damage must be either a listed manufactured part or a listed accessory for the category of the initiating device or notification appliance.

## PART 3 EXECUTION

### 3.1 VERIFYING ACTUAL FIELD CONDITIONS

Before commencing work, examine all adjoining work on which the contractor's work is in any way dependent for perfect workmanship according to the intent of this specification section, and report to the Contracting Officer's Representative any condition which prevents performance of first class work. No "waiver of responsibility" for incomplete, inadequate or defective adjoining work will be considered unless notice has been filed before submittal of a proposal.

### 3.2 INSTALLATION

#### 3.2.1 Manual Stations

Locate manual stations as required by NFPA 72 and as indicated on the drawings. Mount stations so they are located no farther than 5 feet from the exit door they serve, measured horizontally. Manual stations must be mounted at 44 inches measured to the operating handle.

### 3.2.2 Notification Appliances

- a. Locate notification appliance devices as required by NFPA 72. Where more than two visual notification appliances are located in the same room or corridor or field of view, provide synchronized operation. Devices must use screw terminals for all field wiring.
- b. Audible and visual notification appliances mounted on the exterior of the building, within unconditioned spaces, or in the vicinity of showers must be listed weatherproof appliances installed on weatherproof backboxes.
- c. Speakers must not be located in close proximity to the FACU or LOC so as to cause feedback when the microphone is in use.

### 3.3 SYSTEM FIELD WIRING

#### 3.3.1 Wiring within Cabinets, Enclosures, and Boxes

Provide wiring installed in a neat and professional manner and installed parallel with or at right angles to the sides and back of any box, enclosure, or cabinet. Conductors that are terminated, spliced, or otherwise interrupted in any enclosure, cabinet, mounting, or junction box must be connected to screw-type terminal blocks. Mark each terminal in accordance with the wiring diagrams of the system. The use of wire nuts or similar devices is prohibited. Wiring to conform with NFPA 70.

Indicate the following in the wiring diagrams:

- a. Point-to-point wiring diagrams showing the points of connection and terminals used for electrical field connections in the system, including interconnections between the equipment or systems that are supervised or controlled by the system. Diagrams must show connections from field devices to the FACU and remote fire alarm control units, initiating circuits, switches, relays and terminals.
- b. Complete riser diagrams indicating the wiring sequence of devices and their connections to the control equipment. Include a color code schedule for the wiring. Include floor plans showing the locations of devices and equipment.

#### 3.3.2 Terminal Cabinets

Provide a terminal cabinet at the base of any circuit riser, on each floor at each riser, and where indicated on the drawings. Terminal size must be appropriate for the size of the wiring to be connected. Conductor terminations must be labeled and a drawing containing conductors, their labels, their circuits, and their interconnection must be permanently mounted in the terminal cabinet. Minimum size is 8 inches by 8 inches. Only screw-type terminals are permitted. Provide an identification label, that displays "FIRE ALARM TERMINAL CABINET" with 2-inch lettering, on the front of the terminal cabinet.

#### 3.3.3 Alarm Wiring

- a. Voltages must not be mixed in any junction box, housing or device, except those containing power supplies and control relays.
- b. Utilize shielded wiring where recommended by the manufacturer. For

shielded wiring, ground the shield at only one point, in or adjacent to the FACU.

- c. Pigtail or T-tap connections to initiating device circuits, supervisory alarm circuits, and notification appliance circuits are prohibited.
- d. Color coding is required for circuits and must be maintained throughout the circuit. Conductors used for the same functions must be similarly color coded. Conform wiring to NFPA 70.
- e. Pull all conductors splice free. The use of wire nuts, crimped connectors, or twisting of conductors is prohibited. Where splices are unavoidable, the location of the junction box or pull box where they occur must be identified on the as-built drawings. The number and location of splices must be subject to approval by the Designated Fire Protection Engineer (DFPE).

#### 3.3.4 Back Boxes and Conduit

In addition to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, provide all wiring in rigid metal conduit or intermediate metal conduit unless specifically indicated otherwise. Minimum conduit size must be 3/4-inch in diameter. Do not use electrical non-metallic tubing (ENT) or flexible non-metallic tubing and associated fittings.

- a. Galvanized rigid steel (GRS) conduit must be utilized where exposed to weather, where subject to physical damage, and where exposed on exterior of buildings. Intermediate metal conduit (IMC) may be used in lieu of GRS as allowed by NFPA 70.
- b. Electrical metallic tubing (EMT) is permitted above suspended ceilings or exposed where not subject to physical damage. Do not use EMT underground, encased in concrete, mortar, or grout, in hazardous locations, where exposed to physical damage, outdoors or in fire pump rooms. Use die-cast compression connectors.
- c. For rigid metallic conduit (RMC), only threaded type fitting are permitted for wet or damp locations.
- d. Flexible metal conduit is permitted for initiating device circuits 6 feet in length or less. Flexible metal conduit is prohibited for notification appliance circuits and signaling line circuits. Use liquid tight flexible metal conduit in damp and wet locations.
- e. Schedule 40 (minimum) polyvinyl chloride (PVC) is permitted where conduit is routed underground or underground below floor slabs. Convert non-metallic conduit, other than PVC Schedule 40 or 80, to plastic-coated rigid, or IMC, steel conduit before turning up through floor slab.
- f. Exterior wall penetrations must be weathertight. Conduit must be sealed to prevent the infiltration of moisture.

#### 3.3.5 Conductor Terminations

Labeling of conductors at terminal blocks in terminal cabinets, FACU, and

remote FACU must be provided at each conductor connection. Each conductor or cable must have a shrink-wrap label to provide a unique and specific designation. Each terminal cabinet, FACU, and remote FACU must contain a laminated drawing that indicates each conductor, its label, circuit, and terminal. The laminated drawing must be neat, using 12 point lettering minimum size, and mounted within each cabinet, control unit, or unit so that it does not interfere with the wiring or terminals. Maintain existing color code scheme where connecting to existing equipment.

### 3.4 FIRESTOPPING

Provide firestopping for holes at conduit penetrations through floor slabs, fire-rated walls, partitions with fire-rated doors, corridor walls, and vertical service shafts in accordance with Section 07 84 00 FIRESTOPPING.

### 3.5 PAINTING

- a. In unfinished areas (including areas above drop ceilings), paint all exposed electrical conduit (serving fire alarm equipment), fire alarm conduit, surface metal raceway, junction boxes and covers red. In lieu of painting conduit, the contractor may utilize red conduit with a factory applied finish.
- b. In finished areas, paint exposed electrical conduit (serving fire alarm equipment), fire alarm conduit, surface metal raceways, junction boxes, and electrical boxes to match adjacent finishes. The inside cover of the junction box must be identified as "Fire Alarm" and the conduit must have painted red bands 3/4-inch wide at 10-foot centers and at each side of a floor, wall, or ceiling penetration.
- c. Painting must comply with Section 09 90 00 PAINTS AND COATINGS.

### 3.6 FIELD QUALITY CONTROL

#### 3.6.1 Test Procedures

Submit detailed test procedures, prepared and signed by the NICET Level III Fire Alarm Technician, and the representative of the installing company, and reviewed by the QFPE 5 days prior to performing system tests. Detailed test procedures must list all components of the installed system such as initiating devices and circuits, notification appliances and circuits, control devices/equipment, batteries, transmitting and receiving equipment, power sources/supply, annunciators, special hazard equipment, emergency communication equipment, interface equipment, and surge protective devices. Test procedures must include sequence of testing, time estimate for each test, and sample test data forms. The test data forms must be in a check-off format (pass/fail with space to add applicable test data; similar to the forma in NFPA 72 and NFPA 4.) The test procedures and accompanying test data forms must be used for the pre-Government testing and the Government testing. The test data forms must record the test results and must:

- a. Identify the NFPA Class of all Initiating Device Circuits (IDC), and Notification Appliance Circuits (NAC).
- b. Identify each test required by NFPA 72 Test Methods and required test herein to be performed on each component, and describe how these tests must be performed.

- c. Identify each component and circuit as to type, location within the facility, and unique identity within the installed system. Provide necessary floor plan sheets showing each component location, test location, and alphanumeric identity.
- d. Identify all test equipment and personnel required to perform each test (including equipment necessary for smoke detector testing. The use of magnets is not permitted.
- e. Provide space to identify the date and time of each test. Provide space to identify the names and signatures of the individuals conducting and witnessing each test.

### 3.6.2 Pre-Government Testing

#### 3.6.2.1 Verification of Compliant Installation

Conduct inspections and tests to ensure that devices and circuits are functioning properly. Tests must meet the requirements of paragraph entitled "Minimum System Tests" as required by NFPA 72. The contractor and an authorized representative from each supplier of equipment must be in attendance at the pre-Government testing to make necessary adjustments. After inspection and testing is complete, provide a signed Verification of Compliant Installation letter by the QFPE that the installation is complete, compliant with the specification and fully operable. The letter must include the names and titles of the witnesses to the pre-Government tests. Provide all completion documentation as required by NFPA 72 including all referenced annex sections and the test reports noted below.

- a. NFPA 72 Record of Completion.
- b. NFPA 72 Record of Inspection and Testing.
- c. Fire Alarm and Emergency Communication System Inspection and Testing Form.
- d. Audibility test results with marked-up test floor plans.
- e. Documentation that all tests identified in the paragraph "Minimum System Tests" are complete.

#### 3.6.2.2 Request for Government Final Test

When the verification of compliant installation has been completed, submit a formal request for Government final test to the Designated Fire Protection Engineer (DFPE). Government final testing will not be scheduled until the DFPE has received copies of the request for Government final testing and Verification of Compliant Installation letter with all required reports. Government final testing will not be performed until after the connections to the installation-wide fire reporting system has been completed and tested to confirm communications are fully functional. Submit request for test at least 5 calendar days prior to the requested test date.

### 3.6.3 Correction of Deficiencies

If equipment was found to be defective or non-compliant with contract requirements, perform corrective actions and repeat the tests. Tests must be conducted and repeated if necessary until the system has been demonstrated to comply with all contract requirements.

### 3.6.4 Government Final Tests

The tests must be performed in accordance with the approved test procedures in the presence of the DFPE. Furnish instruments and personnel required for the tests. The following must be provided at the job site for Government Final Testing:

- a. The manufacturer's technical representative.
- b. The contractor's Qualified Fire Protection Engineer (QFPE).
- c. Marked-up red line drawings of the system as actually installed.
- d. Copy of pre-Government Test Certificate, test procedures and completed test data forms.
- e. Audibility test results with marked-up floor plans.

Government Final Tests will be witnessed by the Designated Fire Protection Engineer and Qualified Fire Protection Engineer (QFPE). At this time, any and all required tests noted in the paragraph "Minimum System Tests" must be repeated at their discretion.

## 3.7 MINIMUM SYSTEM TESTS

### 3.7.1 System Tests

Test the system in accordance with the procedures outlined in NFPA 72. The required tests are as follows:

- a. Verify the absence of unwanted voltages between circuit conductors and ground. The tests must be accomplished at the pre-Government test with results available at the final system test.
- b. Verify that the control unit is in the normal condition as detailed in the manufacturer's O&M manual.
- c. Test each initiating device and notification appliance and circuit for proper operation and response at the control unit. Smoke detectors must be tested in accordance with manufacturer's recommended calibrated test method. Use of magnets is prohibited. Testing of duct smoke detectors must comply with the requirements of NFPA 72 except disconnect at least 20 percent of devices. If there is a failure at these devices, then supervision must be tested at each device.
- d. Test the system for specified functions in accordance with the contract drawings and specifications and the manufacturer's O&M manual.

- e. Test both primary power and secondary power. Verify, by test, the secondary power system is capable of operating the system for the time period and in the manner specified.
- f. Determine that the system is operable under trouble conditions as specified.
- g. Visually inspect wiring.
- h. Test the battery charger and batteries.
- i. Verify that red-line drawings are accurate.
- j. Measure the current in circuits to ensure there is the calculated spare capacity for the circuits.
- k. Measure voltage readings for circuits to ensure that voltage drop is not excessive.
- l. Measure the voltage drop at the most remote appliance (based on wire length) on each notification appliance circuit.
- m. Verify the documentation cabinet is installed and contains all as-built shop drawings, product data sheets, design calculations, site-specific software data package, and all documentation required by paragraph titled "Test Reports".

### 3.7.2 Audibility Tests

Sound pressure levels from audible notification appliances must be a minimum of 15 dBa over ambient with a maximum of 110 dBa in any occupiable area. The provisions for audible notification (audibility) must be met with doors, fire shutters, movable partitions, and similar devices closed.

## 3.8 SYSTEM ACCEPTANCE

Following acceptance of the system, as-built drawings and O&M manuals must be delivered to the Contracting Officer for review and acceptance. The drawings must show the system as installed, including deviations from both the project drawings and the approved shop drawings. These drawings must be submitted within two weeks after the final Government test of the system. At least one set of as-built (marked-up) drawings must be provided at the time of, or prior to the Final Government Test.

- a. The drawings must be prepared electronically and sized no less than the contract drawings. Furnish one set of CDs or DVDs containing software back-up and CAD based drawings in latest version of AutoCAD and DXF and portable document formats of as-built drawings and schematics.
- b. Include complete wiring diagrams showing connections between devices and equipment, both factory and field wired.
- c. Include a riser diagram and drawings showing the as-built location of devices and equipment.
- d. Provide Operation and Maintenance (O&M) Instructions.

In existing buildings, the transfer of devices from the existing system to



the new system and the permission to begin demolition of the old fire alarm system will not be permitted until the as-built drawings and O&M manuals are received.

-- End of Section --

SECTION 31 00 00.00 22

EARTHWORK  
08/2024

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                                                           |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C33/C33M   | (2018) Standard Specification for Concrete Aggregates                                                                                                                     |
| ASTM C150/C150M | (2024) Standard Specification for Portland Cement                                                                                                                         |
| ASTM C260/C260M | (2010a; R 2016) Standard Specification for Air-Entraining Admixtures for Concrete                                                                                         |
| ASTM C618       | (2023; E 2023) Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete                                                           |
| ASTM C989/C989M | (2022) Standard Specification for Slag Cement for Use in Concrete and Mortars                                                                                             |
| ASTM D1140      | (2017) Standard Test Methods for Determining the Amount of Material Finer than 75- $\mu$ m (No. 200) Sieve in Soils by Washing                                            |
| ASTM D1557      | (2012; E 2015) Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft <sup>3</sup> ) (2700 kN-m/m <sup>3</sup> ) |
| ASTM D2487      | (2017; E 2020) Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)                                                |
| ASTM D4318      | (2017; E 2018) Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils                                                                       |
| ASTM D4832      | (2016; E 2018) Standard Test Method for Preparation and Testing of Controlled Low Strength Material (CLSM) Test Cylinders                                                 |
| ASTM D6023      | (2016) Standard Test Method for Density (Unit Weight), Yield, Cement Content, and Air Content (Gravimetric) of Controlled Low-Strength Material (CLSM)                    |

ASTM D6938 (2017a) Standard Test Method for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2024) Safety -- Safety and Occupational Health (SOH) Requirements

40 CFR 260 Hazardous Waste Management Systems

40 CFR 273 Standards for Universal Waste Management

40 CFR 279 Standards for the Management of Used Oil

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

STATE OF MAINE (SHS)

SHS MEDOT (2014) State of Maine Department of Transportation Standard Specifications, including revisions through award of this contract (See <http://www.maine.gov/mdot/publications/>)

Maine DEP Hazardous Waste Regulation  
Chapter 50-588

## 1.2 DEFINITIONS

### 1.2.1 Structural Fill

Soil material placed to support buildings, walls, pads, pavements, and other similar facilities.

### 1.2.2 Utility Bedding Material

Structural Fill or Sand placed to directly support pipes, conduits, cables, and appurtenant structures. Bedding may also be used to provide a cushion between utilities and bedrock, obstacles, obstructions and other unyielding materials.

### 1.2.3 Flowable Fill

Fill placed in a plastic or liquid form that flows to near its final placement location with limited assistance and subsequently cures or solidifies to provide a stable or impermeable barrier.

### 1.2.4 Satisfactory Materials

Satisfactory materials for fill, backfill, and/or any in-situ soils to remain in place comprise any materials classified by ASTM D2487 as GW, GP, GM, GP-GM, GW-GM, SW, SP, SM, or SW-SM and shall be free of debris, roots, wood, scrap material, vegetation, refuse, soft unsound particles, and frozen, deleterious, or objectionable materials. Maximum particle size to be no greater than one-half of the allowable lift thickness.

#### 1.2.5 Unsatisfactory Materials

Materials which do not comply with the requirements for satisfactory materials are unsatisfactory. Unsatisfactory materials also include man-made fills; trash; refuse; backfills from previous construction; roots and other organic matter or frozen material; and suspected contaminated materials (excavated material with any of the following characteristics: significant petroleum and/or chemical odor; an oily sheen; and/or material with staining or significant change of color). Rock, boulders or stones with a particle diameter greater than one-half the lift thickness at the intended location, asphalt, or concrete materials shall be separated from any excess soils requiring disposal and shall be transported & disposed of by the Contractor. Notify the Contracting Officer when encountering any contaminated materials.

#### 1.2.6 Hard/Unyielding Materials

Hard/Unyielding materials comprise weathered rock, dense consolidated deposits, or conglomerate materials which are not included in the definition of "rock" with stones greater than 6 inches in any dimension or as defined by the pipe manufacturer, whichever is smaller. These materials usually require the use of heavy excavation equipment, ripper teeth, or jack hammers for removal.

#### 1.2.7 Unstable Material

Unstable materials are too weak to adequately support the utility pipe, conduit, equipment, appurtenant structure, or pavement. Satisfactory material may become unstable due to ineffective drainage, dewatering, becoming frozen, excessive loading.

#### 1.2.8 Rock

Solid homogeneous interlocking crystalline material with firmly cemented, laminated, or foliated masses or conglomerate deposits, neither of which can be removed without systematic drilling, drilling and the use of expansion jacks or feather wedges, or the use of backhoe-mounted pneumatic hole punchers or rock breakers; also large boulders, buried masonry, or concrete other than pavement exceeding 1 cubic yard in volume. Removal of hard material will not be considered rock excavation because of intermittent drilling that is performed merely to increase production.

SPECIAL NOTE: BLASTING IS NOT PERMITTED ON PORTSMOUTH NAVAL SHIPYARD.

#### 1.2.9 Capillary Water Barrier

A layer of clean, poorly graded crushed rock, stone, or natural sand or gravel having a high porosity which is placed beneath a building slab with or without a vapor barrier to cut off the capillary flow of pore water to the area immediately below a slab.

#### 1.2.10 Non- Hazardous Soil

Non-hazardous soils shall include all soils which, upon generation, do not meet the definition of a hazardous waste as defined by the EPA Hazardous Waste Regulations 40 CFR 260 through 40 CFR 268, 40 CFR 273, and 40 CFR 279 and the State of Maine DEP Hazardous Waste Regulations Chapter 50-855.

#### 1.2.11 Hazardous Soils

Hazardous soils shall include all soils which, upon generation, do meet the definition of a hazardous waste as defined by the EPA Hazardous Waste Regulations 40 CFR 260, 40 CFR 268, 40 CFR 273, and 40 CFR 279 and the State of Maine DEP Hazardous Waste Regulations Chapter 850-855.

#### 1.2.12 Installation Restoration (IR) Sites

A known, demarcated area impacted by past waste disposal operations and/or hazardous substance spills currently regulated by the Navy and outside Federal and State agencies.

#### 1.2.13 Degree of Compaction (Proctor)

Degree of compaction required, except as noted in the second sentence, is expressed as a percentage of the maximum density obtained by the test procedure presented in ASTM D1557 abbreviated as a percent of laboratory maximum dry density (MDD). Since ASTM D1557 applies only to soils that have 30 percent or less by weight of their particles retained on the 3/4 inch sieve, express the degree of compaction for material having more than 30 percent by weight of their particles retained on the 3/4 inch sieve as a percentage of the maximum density in accordance with ASTM D1557.

#### 1.2.14 Borrow

Soil brought to the project site from an external location for the purposes of project construction.

#### 1.2.15 Subgrade

Earth materials exposed in the bottom of the excavation for utilities, building foundation slabs, and pavement areas including shoulders.

### 1.3 SUBSURFACE DATA

Subsurface soil boring logs are shown in project plans. These data represent available subsurface information; however, variations may exist between boring locations.

### 1.4 CRITERIA FOR BIDDING

Base bids on the following criteria:

#### 1.4.1 General Criteria

- a. Surface elevations are as indicated
- b. Ground water elevation is estimated to be at or near mean high water level of the Piscataqua River. Approximate groundwater elevations are indicated on the boring logs. The Contractor shall include all costs associated with all dewatering efforts. Failure by the Contractor to plan for groundwater conditions shall not be the basis for any claim nor equitable price or contract time adjustments.
- c. Testing results used for geotechnical analysis or collection of any soil waste characterization samples cannot also be used for waste characterization unless Code 106.3 reviews and approves the sampling plan and has oversight of the sampling process.

- d. Removal of soils and rock beyond the lines and grades indicated will not be grounds for claims for additional payment.
- e. Blasting will not be permitted on the Shipyard. Contractors means and methods may include hoe ramming or other approved methods. Provide protection of adjacent facilities as necessary during prosecution of work.

#### 1.4.2 Soil Management Criteria

The project will involve the excavation and handling of soil in areas within and outside the limits of one or more IR sites, which are also referred to as Operable Units (OUs). The limits of work and the IR site boundaries are shown in the project plans. The management of soil in IR and non-IR areas of the project shall conform to the soil management criteria described herein.

IR site where non-hazardous soil is assumed to be present.

This section applies to IR sites where non-hazardous soil is assumed to present (previously remediated) but is still regulated under a Land Use Control (LUC). This section specifically applies to OU1, OU2 (with the exception of an area beneath a 2-foot soil cover around Building 310), OU7, and OU9 (with the exception of an area beneath the Building 62 Annex).

- a. Coordination with PWD ME EV IR Coordinator must occur at least three weeks prior to the start of work to allow adequate time for PWD ME EV notification to EPA and MEDEP. The notification must include a site plan identifying IR site boundaries and the proposed activities therein.
- b. The Government is responsible for soil testing, transport, and disposal.
- c. The Contractor is responsible for soil excavation, stockpiling, onsite management, and loading (into Government furnished trucks and/or containers).
- d. A site specific HASP is not required as the IR site has been remediated to below acceptable Construction Worker exposure levels. Contractor must stop work and notify the PWD ME Construction Manager (CM) if suspect soil contamination is encountered (e.g. - presence of staining, odor, debris).
- e. Reuse of excavated soil is prohibited.
- f. Soil stockpiles must be located within the IR site boundary.
- g. Soil stockpiles must be placed on and covered with 6 mil poly sheeting and surrounded with erosion and sediment controls to prevent runoff. Soil stockpiles must follow MEDEP Erosion and Sediment Control Best Management Practices.
- h. Soil stockpiles must remain covered unless being actively worked or sampled.
- i. Contractor must coordinate directly with Code 106.3 to schedule soil testing (at least seven days prior to desired date).
- j. Soil stockpiles must not be added to once Government soil sampling

activities are complete.

- k. Contractor must clearly placard stockpiles with "Pending Characterization" while awaiting Government soil characterization test results and either "Non-Hazardous" or "Hazardous" based on those results. Signs must be clearly legible and designed to withstand weathering for the duration of the project.
- l. Contractor must coordinate directly with Code 106.3 to schedule Government furnished trucks and/or containers.

Excavation outside IR site where non-hazardous soil is assumed present.

- a. The Contractor is responsible for all aspects of soil management from generation to disposal; including, excavation, stockpiling, onsite management, testing, loading, transport, and disposal. Refer to Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME).
- b. The Contractor is responsible for waste characterization testing of excess soil. The Contractor must submit an Excess Soil Disposal & Sampling Plan to PWD ME EV and Code 106.3 for review and approval prior to performing sampling and analysis.
- c. A site specific HASP is not required. Contractor must stop work and notify the PWD ME CM if suspect soil contamination is encountered (e.g. - presence of staining, odor, debris).
- d. Reuse of excavated soil as backfill is allowed given project specifications are met.
- e. Soil stockpiles must follow MEDEP Erosion and Sediment Control Best Management Practices. Refer to Section 01 57 19.00 22 TEMPORARY ENVIRONMENTAL CONTROLS - PORTSMOUTH NAVAL SHIPYARD (PWD ME).
- f. Contractor must use an experienced environmental professional for the collection of soil samples for disposal characterization.
- g. Code 106.3 must be present onsite to witness Contractor soil sampling events, which must be scheduled at least seven days prior.
- h. Contractor must use a laboratory that has Maine state certification and NELAP/NELAC accreditation for all test methods required for soil analysis (see requirements in 1.5.4).
- i. Soil stockpiles must not be added to once Contractor sample collection is complete.
- j. Contractor must provide Code 106.3 with all documentation pertaining to the sampling event and results. Documentation includes but is not limited to: analytical results, maps and diagrams, copy of chain of custody and Contractor sampling field log book entry.
- k. Contractor must clearly placard stockpiles with "Pending Characterization" while awaiting Contractor soil characterization test results and either "Non-Hazardous" or "Hazardous" based on those results. Signs must be clearly legible and designed to withstand weathering for the duration of the project.

- l. Contractor must not ship any soil without the written approval of the PWD ME CM and Code 106.3.
- m. Once loaded into trucks and/or containers, the Contractor must transport the soil to Building 357 for processing (e.g. - weigh load, manifest paperwork) prior to leaving the Shipyard.
- n. Soil transported off the Shipyard must be disposed of at a licensed non-municipal landfill disposal facility, preapproved by Code 106.3.
- o. The Contractor must provide documentation to the PWD ME CM and Code 106.3 that soil removed from the Shipyard was disposed of at the approved landfill disposal facility.
- p. Sampled soil stockpiles must be placed on and covered with 6 mil poly sheeting and surrounded with erosion and sediment controls to prevent runoff.
- q. Sampled soil stockpiles must follow MEDEP Erosion and Sediment Control Best Management Practices' and 'Sampled soil stockpiles must remain covered unless being actively worked or sampled.

Excavation outside an IR site where hazardous or potential hazardous soil is discovered.

- a. The Government is responsible for soil testing, transport, and disposal.
- b. The Contractor is responsible for soil excavation, stockpiling, onsite management, and loading (into Government furnished trucks and/or containers).
- c. A site specific HASP describing proper HAZWOPER site controls must be developed, approved by the Contracting Officer, and implemented for worker protection.
- d. Reuse of excavated soil is prohibited.
- e. Soil stockpiles must be located within the project footprint, as close to the excavation as possible.
- f. Soil stockpiles must be placed on and covered with 6 mil poly sheeting and surrounded with erosion and sediment controls to prevent runoff. Soil stockpiles must follow MEDEP Erosion and Sediment Control Best Management Practices.
- g. Soil stockpiles must remain covered unless being actively worked or sampled.
- h. Contractor must segregate soil stockpiles based on evidence of contamination (e.g. - presence of staining, odor, debris) from other materials excavated.
- i. Contractor must coordinate directly with Code 106.3 to schedule soil testing (at least seven days prior to desired date).
- j. Soil stockpiles must not be added to once Government soil sample collection activities are complete.



- k. Contractor must clearly placard stockpiles with "Pending Characterization" while awaiting Government soil characterization test results and either "Non-Hazardous" or "Hazardous" based on those results. Signs must be clearly legible and designed to withstand weathering for the duration of the project.
- l. Contractor must coordinate directly with Code 106.3 to schedule Government furnished trucks and/or containers.

#### 1.5 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government. The following shall be submitted in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-01 Preconstruction Submittals

Shoring and Sheet piling Plan; G  
Excavation and Trenching Plan; G  
Dewatering Work Plan; G  
Archeologists Qualifications; G  
Preconstruction Meeting; G  
Excess Soil Disposal & Sampling Plan; G  
Submit 15 days prior to starting work.

##### SD-03 Product Data

Flowable Fill Mix Design; G  
Geotextiles

##### SD-06 Test Reports

Dewatering Performance Records; G  
Material Test Report; G  
Contractor Sampling Field Log Book Entry; G  
Laboratory gradation test; G  
Laboratory compaction tests; G  
Pipe Inspection Report; G  
Geotechnical Evaluation Report; G

Submit copies of all laboratory and field test reports within 24 hours of the completion of the test. If applicable, submit unit price item rock quantity surveys at least 24 hours in advance of removal.

## 1.6 QUALITY CONTROL

### 1.6.1 Geotechnical Engineer

The Contractor shall hire a Professional Geotechnical Engineer to provide inspection of excavations and soil/groundwater conditions throughout construction. The Geotechnical Engineer is responsible for performing pre-construction and periodic site visits throughout construction to assess site conditions. The Geotechnical Engineer is responsible for preparing and updating the Excavation and Trenching Plan and Dewatering Work Plan as construction progresses to reflect changing conditions and submit an updated plan if necessary. Submit a monthly Geotechnical Evaluation report, informing the Contractor and Contracting Officer of the status of the plan and an accounting of the Contractor's adherence to the plan addressing any present or potential problems. The Contractor is responsible for arranging meetings with the Geotechnical Engineer and Contracting Officer throughout the contract duration.

The Geotechnical Engineer shall be available to meet with the Contracting Officer at any time throughout the contract duration to review the performance of the dewatering system and temporary support of excavation systems.

### 1.6.2 Qualified Technician

Provide a Qualified Technician to inspect, monitor, sample, and perform field testing. The technician qualifications need to be one of the following: a current National Institute for Certification in Engineering Technologies (NICET) Level II minimum certification in Construction Materials Testing Soils; a Geologist-in-Training with minimum one-year experience; an Engineer-in-Training with minimum one-year experience; a Registered Geologist; or a Professional Engineer.

### 1.6.3 Lab Validation

Perform testing by a Corps validated commercial testing laboratory or Contractor established testing laboratory meeting the requirements of Section 01 45 00.00 22 QUALITY CONTROL and approved by the Contracting Officer. Submit testing laboratory validation for the testing to be performed. Do not permit work requiring testing until testing facilities have been inspected, Corps validated and approved by the Contracting Officer.

### 1.6.4 Preconstruction Meeting

Conduct a preconstruction meeting at the jobsite at least five business days prior to the start of earthwork operations on the project. The preconstruction meeting is to be arranged by the Contractor and is to follow the written agenda submitted prior to the meeting. The purpose of this meeting is to review the requirements of this specification and the associated plans. The following individuals must be in attendance at this meeting: Contractor's Project Manager and Project Superintendent, earthwork subcontractor's Project Manager and Site Foreman, Contractor's Geotechnical Engineer and Testing Agency, Government Geotechnical Engineer and Civil Engineer, and Government Construction Manager and Engineering Technician.

The minutes of this meeting are to be recorded by the Contractor and published via email within 48 hours to all attendees. The minutes must be

re-published within 48 hours via email pending any subsequent comments from the attendees.

#### 1.7 DELIVERY, STORAGE, and HANDLING

Perform in a manner to prevent contamination or segregation of material.

#### 1.8 GENERAL SOIL MANAGEMENT REQUIREMENTS

##### 1.8.1 Soil Transported Off Base

Any soil transported off base from the Shipyard must be disposed of at a licensed non-municipal landfill disposal facility, which has been preapproved by Code 106.3.

##### 1.8.2 Bulk Materials in Contact With Soil

Bulk materials which are in contact with soil (e.g. - undisturbed bedrock, asphalt, concrete) may be transported off base for disposal or recycling, given there are no suspected hazardous constituents associated with the materials (e.g., PCB paint) and soil is completely removed prior to being loaded for transport (field determination by PWD ME Construction Manager (CM) or Engineering Tech (ET)). Information regarding the destination of these materials must be submitted to the PWD ME CM.

#### 1.9 SOIL SAMPLING FOR WASTE CHARACTERIZATION

Testing parameters must meet the requirements of the selected landfill disposal facility including the type, number, and frequency of tests. Chemical analysis of the samples must be performed by a laboratory that has Maine state certification and National Environmental Laboratory Accreditation Program (NELAP/NELAC) accreditation for all test methods required for soil analysis and a Quality Systems Manual that conforms to the standards of ISO 17025.

#### 1.10 SITE WASTE REMOVAL (SWR) MEETING

A SWR meeting must be held with the PWD ME CM and Code 106.3 prior to the removal of excavated soil from a project site to ensure all pertinent requirements of this PNSY Standard Operating Procedure for Site Waste Removal (SWR) Process (see Attachment B) have been met.

No transport of soil will be allowed until full concurrence is provided by the PWD ME CM and Code 106.3 following the satisfactory completion of the SWR.

#### 1.11 REQUIREMENTS FOR TRANSPORTING BACKFILL ON BASE

Aggregates containing less than 10% fines (material passing the number 4 sieve) transported to the Shipyard from off base to be used as backfill do not require testing.

All other backfill materials which do not meet the above conditions must be tested for the following contaminants:

- Total Petroleum Hydrocarbons (TPH) (GC/FID)
- Total 8 RCRA Metals (7060, 7740, etc.)
- Total Volatiles (8260)
- Total Semi-Volatiles (8270)

- Total Pesticides (8081)
- Total Herbicides (8151)
- Total PCB's (8082)
- Ignitability/flash (1010-liquids, 1030-solids)
- Corrosivity/pH (9045)
- Reactive Sulfide (7.3.4.1)

Off base soil must not contain any contaminant concentration which exceeds the most current "Residential Scenario" values listed in Appendix 1 of the MEDEP Remedial Action Guidelines found at <https://www.maine.gov/dep/spills/publications/guidance/>. TPH concentrations must not exceed 100 mg/kg. If the total concentration of any Toxicity Characteristic Leaching Procedure (TCLP) regulated contaminant listed in 40 CFR 261.24 is greater than or equal to twenty times its regulatory threshold, TCLP analysis will be required. The off base soil must not contain any TCLP concentration which exceeds the current EPA values listed in 40 CFR 261.24, Table 1 Maximum Concentration of Contaminants for Toxicity Characteristic EPA limits.

Representative sampling and testing of the backfill must be conducted at a rate of 1 sample for every 1,000 tons, with a minimum of 1 sample per borrow source. Contractor must use a licensed, Maine state-certified and NELAP accredited laboratory for the chemical analysis. Backfill must not be transported on base until all laboratory test results have been approved by PWD ME EV.

Borrow pits must meet all appropriate state certification guidelines. Written confirmation that the purchased backfill material did not come from a chemically impacted area must be provided by borrow pit supplier. Proof of certification and written confirmation must be provided to the PWD ME EV prior to backfill being brought on base.

NOTE: Should any clean backfill be removed for excess soil disposal, testing is required even if the project profile amount covers the amount being disposed of.

#### 1.12 EXCESS SOIL DISPOSAL & SAMPLING PLAN

The Contractor shall submit an Excess Soil Disposal & Soil Sampling Plan indicating the licensed facility that will be used for off site disposal of excess non-hazardous soil material. The Excess Soil Disposal & Soil Sampling Plan shall include all information required in Attachment A "Environmental Sampling Plan Expectations for Non-Shipyard Personnel" and the following:

1. Introduction - Provide a location and description of the excavation work required to complete the work under the contract. A site plan showing the limits of excavation and planned stockpile locations for soils to be reused and unsuitable or excess soils requiring disposal.
2. Excavation - Provide quantities of excavated materials. Provide an estimated quantity of soils to be reused and soils requiring off site disposal.
3. Soil Characterization - The Contractor shall collect representative soil samples to confirm the material meets the requirements of the Contractor's selected licensed disposal facility. The soil samples shall be sent to a laboratory that has Maine state certification and National Environmental Laboratory Accreditation Program (NELAP/NELAC)

accreditation for all test methods required for soil analysis and Quality Systems Manual that conforms to the standards of ISO 17025. The quantity & location of soil samples and laboratory tests shall be based on the disposal facility requirements.

The Contractor shall specify the sample collection methods and procedures. Soil sampling can be conducted post-excavation from stockpiles or pre-excavation in-situ using test pits or soil borings. The Contractor's Excess Soil Disposal and Sampling Plan shall identify the preferred method. For in-situ sampling, the plan must specify the quantity, locations, and depths of test pits/borings. All sampling plans and procedures require approval from Code 106.3.

4. Soil Sampling - The Contractor shall provide the analytical sampling requirements based on the licensed facility that will be accepting the soils for disposal. The soil shall be tested using a laboratory that has Maine state certification and National Environmental Laboratory Accreditation Program (NELAP) accreditation for all test methods required for soil analysis and Quality Systems Manual that conforms to the standards of ISO 17025.
5. Stockpiling - Provide description on how the soils (soils to be reused & excess soils requiring disposal) will be stockpiled within the limits of work and in accordance with the requirements provided in the plans & contract documents. The location of the proposed stockpile areas shall be shown on a site plan.
6. Transportation on base - The excess soils shall be transported directly to Building 357 from the project site to be inspected, weighed & manifested by the Government prior to leaving the Shipyard. B357 is open Monday-Friday from 8 am to 2 pm. The contractor shall coordinate deliveries with the Contracting Officer & PNSY 106.3 at least 14 calendar days prior to making any shipments to B357. The licensed facility must be approved by PWD ME Environmental & PNSY Code 106.3 prior to any soil leaving the Shipyard. The Contractor shall contact Code 106.3 for information regarding the appropriate disposal facilities and what is CURRENTLY approved.
7. Transportation off base - Provide a description of how the soils will be transported to the approved disposal facility. The Contractor shall provide documentation to the Contracting Officer that soil removed from the Shipyard was disposed of at the approved disposal facility.

## 1.13 QUALITY ASSURANCE

### 1.13.1 Excavation and Trenching Plan

Provide Manufacturers Tabulated Data for trench boxes and other pre-manufactured support systems to be used on the project. See EM 385-1-1, Section 25, Excavation and Trenching for requirements.

Provide shop drawings and calculations for site specific support systems including, but not limited to, soldier pile and lagging and timber sheeting. See EM 385-1-1, Section 25, Excavation and Trenching for requirements.

### 1.13.2 Dewatering Work Plan

The Contractor shall submit procedures for accomplishing dewatering work

to the Contracting Officer for approval. The Dewatering Work Plan submittal shall be prepared by a Professional Geotechnical Engineer licensed in the State of Maine with specialized experience in geotechnical engineering. The requirements of the plan are specified in Paragraph entitled DEWATERING herein.

#### 1.13.3 Utilities

Movement of construction machinery and equipment over pipes and utilities during construction shall be at the Contractor's risk. Perform work adjacent to non-Government utilities as indicated in accordance with procedures outlined by utility company. Excavation made with power-driven equipment is not permitted within three (3) feet of known Government-owned utility or subsurface construction. For work immediately adjacent to or for excavations exposing a utility or other buried obstruction, excavate by hand. Start hand excavation on each side of the indicated obstruction and continue until the obstruction is uncovered or until clearance for the new grade is assured. Support uncovered lines or other existing work affected by the contract excavation until approval for backfill is granted by the Contracting Officer. Report damage to utility lines or subsurface construction immediately to the Contracting Officer.

#### 1.13.4 Post- Review Archaeological Discoveries

In the event that a potential archaeological resource is encountered during ground disturbing activities, stop work within the surrounding area immediately and notify the CM.

If suspected human remains (including but not limited to bone, bone fragments, teeth, mummified flesh, burials, and cremations) are encountered, stop work in the surrounding area immediately and notify the CM. The remains shall be treated with respect and not be disturbed or photographed. In coordination with the CM, provide the materials and labor necessary to secure the area to prevent damage and protect the discovery.

Work may not resume until direction is provided by the CM. All artifacts remain property of the GOV.

#### 1.13.5 Archaeological Monitoring

If archaeological monitoring is required for the project, ensure all earthwork activities are conducted in coordination with 01 44 00.22. If archaeological resources are determined to be significant, the KTR may be required, in coordination with the COR, to provide a qualified archaeologist meeting the Secretary of the Interior's Professional Qualification Standards and on the State of Maine's Level II-approved list to monitor excavations. Submit archeologists qualifications.

### PART 2 PRODUCTS

#### 2.1 SOIL MATERIALS

##### 2.1.1 Structural Fill

Satisfactory Material consisting of well-graded sand and gravel mixture free of roots, topsoil, loam, where indicated, organic material, and any other deleterious materials, as well as clods of silt or clay, and meet the following gradation requirements or meet the SHS MEDOT, Section 703.20

Gravel Borrow specification:

| Screen or Sieve Size | Percent Passing |
|----------------------|-----------------|
| 6 inches             | 100             |
| 3 inches             | 70 - 100        |
| No. 4                | 35 - 70         |
| No. 40               | 5 - 35          |
| No. 200              | 0 - 10          |

2.1.2 Ordinary Fill

Satisfactory Material consisting of a mixture of sand and gravel meeting the SHS MEDOT, Section 703.19 Granular Borrow with 20 percent or less passing the No. 200 sieve and no stone greater than one-half the loose lift thickness, or 6 inches, whichever is smaller.

2.1.3 Sand

Clean, coarse-grained sand classified as SP-SM, material and gradation requirements for Maine DOT 703.22: Underdrain Backfill Material, Type C.

2.1.4 Crushed stone

Clean, quarried material meeting the material and gradation requirements for SHS MEDOT 703.13: Crushed Stone  $\frac{3}{4}$  inch.

2.1.5 Gravel Subbase

Clean, quarried material meeting the material and gradation requirements for SHS MEDOT 703.10: Aggregate for Untreated Surface Course and Leveling Course, Type A.

2.2 FLOWABLE FILL

Design and submit flowable fill mix design to consist of Portland cement, fly ash, and/or slag cement and fine aggregate. Include the dry weights of cementitious material(s); quality and gradation of aggregates in the saturated surface-dry weights along with gradation tests; quantities, types, and names of admixtures; and quantity of water per cubic yard. The maximum unconfined compressive strength to be 100 psi at 28 days in accordance with ASTM D4832. The aggregates in accordance with ASTM C33/C33M Fine Aggregates. Air-entrain fill in accordance with ASTM C260/C260M. The air content to be between 8 and 15 percent in accordance with ASTM D6023. Portland cement to be Type I or II in accordance with ASTM C150/C150M. Fly ash to be Class C in accordance with ASTM C618. Provide slag cement in Grade 100 or 120 in accordance with ASTM C989/C989M.

2.3 BURIED WARNING AND IDENTIFICATION MARKERS

Provide polyethylene plastic and metallic core or metallic-faced, acid- and alkali-resistant, polyethylene plastic warning tape manufactured specifically for warning and identification of buried utility lines as indicated on the drawings. Provide tape on rolls, 3 inches minimum width, color coded as specified below for the intended utility with warning and identification imprinted in bold black letters continuously over the entire tape length. Warning and identification to read, "CAUTION, BURIED

(intended service) LINE BELOW" or similar wording. Provide permanent color and printing, unaffected by moisture or soil.

| Warning Tape Color Codes |                                                 |
|--------------------------|-------------------------------------------------|
| Red                      | Electric                                        |
| Yellow                   | Gas, Oil; Steam, Air, Other Dangerous Materials |
| Orange                   | CCTV, Telephone and Other Communications        |
| Blue                     | Water Systems                                   |
| Green                    | Sewer Systems                                   |
| Purple                   | Reclaimed Water                                 |
| White                    | Proposed Excavation                             |
| Pink                     | Temporary Survey Markings                       |
|                          |                                                 |

#### 2.3.1 Warning Tape for Metallic Piping

Provide acid and alkali-resistant polyethylene plastic tape conforming to the width, color, and printing requirements specified above, with a minimum thickness of 0.004 inch and a minimum strength of 1500 psi lengthwise, and 1250 psi crosswise, with a maximum 350 percent elongation.

#### 2.3.2 Detectable Warning Tape for Non-Metallic Piping

Provide polyethylene plastic tape conforming to the width, color, and printing requirements specified above, with a minimum thickness of 0.004 inch, and a minimum strength of 1500 psi lengthwise and 1250 psi crosswise. Manufacture tape with integral wires, foil backing, or other means of enabling detection by a metal detector when tape is buried up to 3 feet deep. Encase metallic element of the tape in a protective jacket or provide with other means of corrosion protection.

#### 2.3.3 Detection Wire for Non-Metallic Piping

Insulate a single strand, solid copper detection wire with a minimum of 12 AWG.

#### 2.4 BORROW

Obtain borrow materials required in excess of those furnished from excavations from sources outside of Government property. Borrow materials shall meet the requirements for Soil Materials for intended use as defined herein.

#### 2.5 GEOTEXTILE

Provide Geotextile meeting the requirements of AASHTO M288 - geotextile for separation. Submit material product data for all Geotextiles utilized.



## PART 3 EXECUTION

### 3.1 PROTECTION

Perform all work specified in accordance with applicable requirements of the Corps of Engineers publication EM 385-1-1 Safety and Health Requirements Manual. Provide a Geotechnical Engineer to monitor construction activities and to prepare necessary work plans and reports; see paragraph QUALITY CONTROL.

Use equipment of type and size appropriate for the site conditions (soil character and moisture content). Maintenance of exposed subgrades and fills is the responsibility of the Contractor. The Contractor is required to prevent damage by ineffective drainage, dewatering, and heavy loads and equipment by implementing precautionary measures. Repair or replace any defects or damage.

#### 3.1.1 Shoring and Sheeting

Submit a Shoring and Sheeting Plan where needed to complete and provide shoring bracing, cribbing, trench boxes, underpinning, and sheeting where the Work is shown. Submit drawings and calculations, certified by a registered professional engineer, describing the methods for shoring and sheeting of excavations. Drawings to include material sizes and types, arrangement of members, and the sequence and method of installation and removal. Calculations are to include data and references used.

In addition to requirements detailed in EM 385-1-1, the Contractor shall include provisions in the Shoring and Sheeting Plan that will accomplish the following:

- a. Stabilize features, prevent undermining or unintended horizontal and vertical movement of pavements, foundations, utility pipelines, structures, slabs, and other adjacent structures.
- b. Prevent slippage or movement in banks or slopes adjacent to the excavation.
- c. All shoring materials installed permanently below grade shall consist of material not susceptible to rot.

#### 3.1.2 Machinery and Equipment

Movement of construction machinery and equipment over pipes during construction shall be at the Contractor's risk. Repair, or remove and provide new pipe for existing or newly installed pipe that has been displaced or damaged.

#### 3.1.3 Underground Utilities

The location of the existing utilities shown on the plans are approximate. The contractor shall physically field verify the location and elevation of the existing utilities indicated prior to starting construction. The Contractor shall submit a PWD Maine Dig Safe Utility Request Form and contact DIG SAFE (1-888-344-7233). The Contractor shall confirm the location of all existing underground utilities prior to commencing any excavation work.

Perform work adjacent to utilities as indicated. Excavation made with

power-driven equipment is not permitted within 3 feet of known utility or subsurface construction. For work immediately adjacent to or for excavations exposing a utility or other buried obstruction, excavate by hand. Start hand excavation on each side of the indicated obstruction and continue until the obstruction is uncovered or until clearance for the new grade is assured. Support uncovered lines or other existing work affected by the contract excavation until approval for backfill is granted by the Contracting Officer. Report damage to utility lines or subsurface construction immediately to the Contracting Officer.

#### 3.1.4 Drainage and Dewatering

Provide for the collection and disposal of surface and subsurface water encountered during construction. Prepare a Dewatering Plan that complies with State and Federal regulations, and the requirements specified herein.

##### 3.1.4.1 Drainage

Provide for the collection and disposal of surface and subsurface water encountered during construction. Construct storm drainage features (ponds/basins) at the earliest stages of site development, and throughout construction grade the construction area to provide positive surface water runoff away from the construction activity and provide temporary ditches, swales, and other drainage features and equipment as required to keep soils from becoming unstable, prevent erosion, or undermining of structures and utilities. Remove unstable material from working platforms for equipment operation and soil support for subsequent construction features and provide new material as specified herein. It is the responsibility of the Contractor to assess the site conditions to employ necessary measures to permit construction to proceed. Excavated slopes and backfill surfaces shall be protected to prevent erosion and sloughing. Excavation shall be performed so that the site, the area immediately surrounding the site, and the area affecting operations at the site shall be continually and effectively drained.

##### 3.1.4.2 Dewatering

Control groundwater flowing toward or into excavations to prevent sloughing of excavation slopes and walls, boils, uplift and heave in the excavation and to eliminate interference with orderly progress of construction. French drains, sumps, ditches or trenches are not allowed within 3 feet of the foundation of any structure, except with specific written approval, and after specific contractual provisions for restoration of the foundation area have been made. Perform control measures by the time the excavation reaches the water level in order to maintain the integrity of the in-situ material. While the excavation is open, maintain the water level continuously, at least 2 feet below the working level. Submit a Dewatering Work Plan outlining procedures for accomplishing dewatering work. Operate dewatering system continuously until construction work below existing water levels is complete. Measure and record performance of dewatering system at same time each day by use of observation wells or piezometers installed in conjunction with the dewatering system. Submit dewatering performance records weekly.

- a. The majority of the Portsmouth Naval Shipyard experiences shallow groundwater areas. Dewatering work and other aspects of the project must be planned under the assumption groundwater may be encountered up to the elevation of the highest tide. Any groundwater elevations shown on drawings accompanying the specifications are elevations which

existed at the time of subsurface investigations, and are for reference only.

- b. The Contractor shall hire a Professional Geotechnical Engineer licensed in the State of Maine with specialized experience in geotechnical engineering aspects of design and construction to prepare a project specific dewatering plan.

The dewatering plan shall at a minimum address anticipated dewatering discharge volumes, pumping rates required to maintain construction in the dry, demonstrate adequate capacity to treat and dispose the discharge volume, and illustrate on a site plan the location proposed for dewatering discharge treatment systems and disposal locations. The plan shall describe implementation of construction dewatering practices in accordance with Maine Erosion and Sediment Control Best Management Practices (Maine DEPLW0588, latest edition). At no point shall construction dewatering be discharged to storm sewers. The Geotechnical Engineer shall stamp and date the dewatering plan prior to submission to the Contracting Officer.

The Geotechnical Engineer shall be responsible for performing pre-construction site inspections and periodic site visits throughout construction to assess site conditions. The Geotechnical Engineer shall update the excavation, sheeting and dewatering plans as construction progresses to reflect changing conditions and shall submit an updated plan to the Contracting Officer. The Geotechnical Engineer shall submit a written report to the Contractor's Quality Control Manager and Contracting Officer, at least weekly, addressing the performance of the dewatering efforts, the Contractor's adherence to the plan and addressing any present or potential problems and recommendations.

The Geotechnical Engineer shall be available to meet with the Contracting Officer at any time throughout the contract duration to review the performance of the dewatering system.

- c. The Contractor's dewatering efforts shall control groundwater flowing toward or into excavations to prevent sloughing of excavation slopes and walls, boils, uplift and heave in the excavation and to eliminate interference with orderly progress of the construction. Dewatering Pits, french drains, sumps, ditches or trenches shall not be permitted within three (3) feet of the foundation of any structure. Control measures shall be taken by the time the excavation reaches the water level in order to maintain the integrity of the in situ material. The water level shall be maintained continuously as recommended by the Contractor's Geotechnical Engineer.
- d. The Contractor shall collect system performance records weekly. Relieve hydrostatic head in pervious zones below subgrade elevation in layered soils to prevent uplift.
- e. All water removed by the dewatering system shall be discharged back into the subsurface using infiltration pits (width of excavation must exceed the depth) or other means defined by the Contractor's Geotechnical Engineer and approved by the Contracting Officer. The dewatering system shall include temporary above ground storage tanks (ie. Frac Tanks) to manage the water based on soil infiltration rates. The size and number of storage tanks shall be defined by the Contractor's Geotechnical Engineer based on the required dewatering

pump rates, soil infiltration rates, planned work schedule and shall be presented in the dewatering plan.

- f. If contaminated groundwater is encountered while dewatering, dewatering operations shall cease and the contractor shall notify the Contracting Officer to request disposal requirements, issued by PWD-ME Environmental Division and PNSY Environmental Division. Discharge of hazardous substances (e.g. petroleum) shall not be permitted under any circumstances.

Dewatering in an IR Site or an area where hazardous soil is present:

If dewatering is being performed within an IR site, the pumped water shall be discharged into infiltration pits located within the same IR boundary. All groundwater discharges generated within the boundaries of an IR Site shall be infiltrated into the subsurface within the boundaries of the same IR site from which the discharge was generated. Under no circumstances shall water be released to the storm drain system, sanitary sewer, a wetland resource, or river.

Dewatering outside of an IR site where non-hazardous soil is assumed to be present:

If dewatering pumps are located outside of an IR site, the pumped water shall be discharged into infiltration pits located within the same area outside of the IR boundary. Under no circumstances shall water be released to the storm drain system, sanitary sewer, a wetland resource, or the river.

- g. The water in the storage tanks shall be discharged into infiltration pits (width of excavation must exceed the depth) within the limits of work. Multiple infiltration pits may be required. The locations of the infiltration pits shall be identified by the Contractors' Geotechnical Engineer and approved by the Contracting Officer. Under no circumstances shall water be released to the storm drain system, sanitary sewer, a wetland resource, or the river, unless specifically approved by the Contracting Officer.
- h. The Contractor shall include in the bid all necessary controls required to manage all dewatering activities required to complete the work within the specified contract time and in accordance with the plans and specifications. Failure by the Contractor to plan for groundwater conditions and dewatering requirements necessary to complete the work shall not be the basis for any claim nor an equitable price or contract time adjustment.

### 3.1.5 Protection of Graded Surfaces

Protect newly backfilled, graded, and topsoiled areas from traffic, erosion, and settlements that may occur. Repair or reestablish damaged grades, elevations, or slopes.

### 3.2 BORROW

Select borrow material to meet the requirements and conditions of the fill or embankment for which it is to be used. Obtain borrow material from approved private sources. Unless otherwise provided in the contract, the Contractor is responsible for obtaining the right to procure material, pay royalties and other charges involved, and bear the expense of developing

the sources, including rights-of-way for hauling from the owners. Unless specifically provided, do not obtain borrow within the limits of the project site without prior written approval.

#### 3.2.1 Contractor Furnished Borrow Area(s)

Obtain approved borrow materials from approved offsite sources. If a borrow source is selected that is not a commercial entity from which soil material is directly purchased, submit a Borrow Plan that includes the borrow source location, geotechnical test results showing the fill material meets the Contract requirements, environmental test results in accordance with paragraph REQUIREMENTS FOR TRANSPORTING BACKFILL ON BASE, and any Federal, State, and local permits required for excavation and reclamation of the borrow area.

### 3.3 SURFACE PREPARATION

#### 3.3.1 Clearing and Grubbing

Remove pavement, concrete, and other items that would interfere with construction operations within lines 5 feet outside of each building and structure lines.

#### 3.3.2 Stripping

Strip suitable soil from the site where excavation or grading is indicated and stockpile separately from other excavated material. Material unsuitable for use as fill shall be wasted, except that material meeting the requirements for backfill may be used for backfilling.

#### 3.3.3 Stockpiling Operations

Place and grade stockpiles of satisfactory, unsatisfactory, and wasted materials as specified. Keep stockpiles in a neat and well drained condition, giving due consideration to drainage at all times. Clear, grub, and seal by rubber-tired equipment, the ground surface at stockpile locations; separately stockpile excavated satisfactory and unsatisfactory materials. Protect stockpiles of satisfactory materials from contamination which may destroy the quality and fitness of the stockpiled material. Do not create stockpiles that could obstruct the flow of any stream, endanger a partly finished structure, impair the efficiency or appearance of any structure, or be detrimental to the completed work in any way. If the Contractor fails to protect the stockpiles, and any material becomes unsatisfactory, remove and replace such material with satisfactory material from approved sources.

Separately stockpile soil from IR and non-IR areas. Coordinate stockpiling requirements with Soil Management Criteria for IR and non-IR areas.

### 3.4 EXCAVATION

Excavate to contours, elevation, and dimensions indicated. Keep excavations free from water. Excavate soil disturbed or weakened by Contractor's operations, and soils softened or made unstable for subsequent construction due to exposure to weather. Use material removed from excavations meeting the specified requirements in the construction of fills, embankments, subgrades, shoulders, bedding (as backfill), and for similar purposes to minimize surplus material and to minimize additional material brought on site. Do not excavate below indicated depths

and/or limits identified in SHS MEDOT Section 203, Excavation and Embankments and SHS MEDOT Section 206, except to remove unstable material as determined by the Geotechnical Engineer and confirmed by the Contracting Officer. Remove and replace excavations below the grades shown with appropriate materials as directed by the Contracting Officer. Structural Excavation will not be permitted except to remove unsatisfactory material, and shall be in accordance with paragraph OVER EXCAVATION.

If at any time during excavation, the Contractor encounters material that may be classified as rock or as hard/unyielding material, uncover such material, and notify the Contracting Officer. Do not proceed with the excavation of this material until the Contracting Officer has classified the materials as common excavation or rock excavation. Failure on the part of the Contractor to uncover such material, notify the Contracting Officer, and allow sufficient time for classification and delineation of the undisturbed surface of such material will cause the forfeiture of the Contractor's right of claim to any classification or volume of material to be paid for other than that allowed by the Contracting Officer for the areas of work in which such deposits occur.

#### 3.4.1 Trench Excavation Requirements

Excavate trenches as indicated on the Drawings. Slope trench walls below the top of the pipe, or make vertical, and of such width as indicated on the Drawings. Do not exceed the trench width below the top of pipe of 24 inches plus pipe outside diameter (O.D.) for pipes of less than 24 inches inside diameter, and do not exceed 36 inches plus pipe outside diameter for pipe sizes larger than 24 inches inside diameter. Where recommended trench widths are exceeded, provide redesign, stronger pipe, or special installation procedures. The Contractor is responsible for the cost of redesign, stronger pipe, or special installation procedures without any additional cost to the Government.

##### 3.4.1.1 Bottom Preparation

Grade the bottoms of trenches accurately to provide uniform bearing and support for the bottom quadrant of each section of the pipe. Excavate bell holes to the necessary size at each joint or coupling to eliminate point bearing. Remove stones of 6 inch or greater in any dimension, or as recommended by the pipe manufacturer, whichever is smaller, to avoid point bearing.

##### 3.4.1.2 Removal of Unyielding Material

Where bedrock or unyielding material is encountered in the bottom of the trench, notify the Contracting Officer. Following approval, remove such material 6 inch below the required trench grade and replaced with crushed stone.

##### 3.4.1.3 Removal of Unstable Material

Where unstable material is encountered in the bottom of the trench, remove such material to the depth directed and replace it to the proper grade with crushed stone wrapped in geotextile or Structural Fill. When removal of unstable material is required due to the Contractor's fault or neglect in performing the work, the Contractor is responsible for excavating the resulting material and replacing it without additional cost to the Government.

#### 3.4.1.4 Excavation for Appurtenances

Provide excavation for manholes, vaults, catch-basins, inlets, or similar structures sufficient size to permit the placement of structures, and placement and compaction of backfill

#### 3.4.2 OVER Excavation

Excavation below the grades shown will not be permitted except to remove unsatisfactory material. Unsatisfactory material extending below the grades shown shall be classified by Contracting Officer and removed as directed. Refill with crushed stone wrapped in geotextile or structural fill material, depending on location with respect to structures and compact as required in herein. Satisfactory material removed below the depths indicated, without specific direction of the Contracting Officer, shall be replaced with satisfactory materials to the indicated excavation grade; except as specified for spread footings. Determination of elevations and measurements of approved over depth excavation of unsatisfactory material below grades indicated shall be done under the direction of the Contracting Officer. No extra payment will be made for removal of unsuitable soils and replacement with satisfactory material or select fills as may be directed by the Contracting Officer. Make excavations to the lines, grades, and elevations shown, or as directed. Provide trenches and foundation pits of sufficient size to permit the placement and removal of forms for the full length and width of structure footings and foundations as shown. Clean rock or other hard foundation material of loose debris and cut to a firm, level, stepped, or serrated surface. Remove loose disintegrated rock and thin strata.

Concrete placement is not allowed until footing subgrades have been inspected and approved by the Contracting Officer.

#### 3.4.3 Pile Cap Excavation

Stop the excavation at an elevation 6 to 12 inches above the bottom of the footing before installing the micropiles. After pile installation has been completed, complete the remainder of the excavation to the elevations shown.

#### 3.4.4 Rock Excavation

Excavate rock encountered in the cut section the excavation depth indicated on the Drawings and replace with . Roughen level surfaces and cut sloped surfaces into benches for bond with concrete. Protect shale from conditions causing decomposition along joints or cleavage planes and other types of erosion. BLASTING IS NOT PERMITTED ON PORTSMOUTH NAVAL SHIPYARD

#### 3.5 CONTROL OF EXCAVATIONS AND EXCAVATED MATERIALS

All Excavation work shall conform with all applicable OSHA and EM385-1-1 requirements. All excavation activities shall conform with Shipyard requirements. All excavations shall be closed as soon as work within the excavation is complete. Excavated materials shall not be allowed to spread in an uncontrolled manner nor become contaminated with other debris. All excavations shall be properly barricaded with barricades and suitable warning devices in a good state of repair. The Government will not provide barricades for Contractor use. The following controls also apply:

### 3.5.1 Unnatural Fill Materials/Industrial Debris

Upon discovery of Unnatural Fill Materials / Industrial Debris, notify the Contracting Officer, who will arrange for the PWD ME Environmental Division to inspect the materials.

### 3.5.2 Excavated Materials to be Reused as Backfill

Excavated non-hazardous soil materials from excavations which will remain open for LESS THAN TWO WEEKS may remain at the site of excavation.

## 3.6 SUBGRADE PREPARATION

### 3.6.1 General Requirements

Shape subgrade to line, grade, and cross section as indicated. Remove unsatisfactory and unstable material in surfaces to receive fill or in excavated areas, as determined by proof rolling, and replaced with satisfactory materials as directed by the Contracting Officer. Do not place material on surfaces that are muddy, frozen, contain frost, or otherwise containing unstable material. Scarify the surface to a depth of 6 inches prior to placing fill. Step, plow, break up or bench sloped surfaces steeper than 1 vertical to 4 horizontal prior to scarifying. When subgrades are less than the specified density, the ground surface shall be broken up to a minimum depth of 6 inches, pulverized, and compacted to the specified density. Place 6 inches of loose fill and blend with scarified material. When subgrade is part fill and part excavation or natural ground, scarify to a depth of 12 inches.

#### 3.6.1.1 Proof Rolling and Compaction

After stripping, excavating, and rough grading to the planned elevation, proof roll the exposed subgrade of all building and pavement locations with six passes of a loaded 10 yard tandem axle dump truck or 15 ton, pneumatic-tired or smooth drum roller. Operate the roller or truck in a systematic manner to ensure the number of passes over all areas, and at speeds between 2.5 to 3.5 miles per hour. When proof rolling under buildings, the building subgrade is considered to extend 5 feet beyond the building lines, and make one-half of the passes with the roller in a direction perpendicular to the other passes.

After excavating and rough grading trenches to the planned elevation, proof compact exposed soil subgrades with six passes of a vibratory plate compactor.

Notify the Contracting Officer a minimum of 3 days prior to proof rolling and compacting. Proof rolling and compacting shall be performed in the presence of the Contracting Officer. Subgrade materials that exhibit excessive deflection and/or rutting during proof rolling and compaction need to be undercut as directed by the Contracting Officer and replaced with fill and backfill material.

Perform proof rolling and compaction on exposed subgrade that is unfrozen and free of surface water.

### 3.6.2 Subgrade for Structures, Spread Footings, and Concrete Slabs

Do not excavate below depth shown for structures, spread footings, and



concrete slabs. If over excavation occurs, notify the Contracting Officer and remove, replace, and compact as directed. After final rolling, the surface of the subgrade for buildings and pavements must not show deviations greater than 0.05 foot when tested with a 12-foot straightedge applied both parallel and at right angles to the centerline of the area.

### 3.6.3 Subgrade for Pavements and Shoulders

Compact top 12 inches of subgrade for pavements and shoulders to at least 95 percent of ASTM D1557.

## 3.7 FILLING AND COMPACTION

Prepare subgrade on which backfill is to be placed and provide compaction requirements for backfill materials in conformance with the applicable portions of paragraphs for SUBGRADE PREPARATION. Do not place material on surfaces that are muddy, frozen, or contain frost. Finish compaction by pneumatic-tired rollers, steel-wheeled rollers, or other approved equipment well suited to the soil being compacted. Moisten material as necessary to plus or minus 2 percent of optimum moisture. Compact and test each lift before placing overlaying lift.

Compaction using water jetting will not be permitted.

### 3.7.1 Trench Backfill

Backfill trenches to the grade shown. Do not backfill the trench until all specified tests are performed.

Backfill as rapidly as construction, testing, and acceptance of work permits. Place and compact trench backfill in 6 inch lifts to top of trench and in 6 inch lifts to one foot over pipe outside structures and paved areas.

#### 3.7.1.1 Replacement of Unyielding Material

Replace unyielding material removed from the bottom of the trench with satisfactory material or initial backfill material.

#### 3.7.1.2 Replacement of Unstable Material

Replace unstable material removed from the bottom of the trench or excavation with satisfactory material placed in layers not exceeding 6 inches loose thickness.

#### 3.7.1.3 Bedding and Initial Backfill

Provide bedding of the type and thickness shown. Place initial backfill material and compact it with approved tampers to a height of at least one foot above the utility pipe or conduit. Bring up the backfill evenly on both sides of the pipe for the full length of the pipe. Take care to ensure thorough compaction of the fill under the haunches of the pipe. Except where shown or when specified otherwise in the individual piping section, provide bedding for buried piping in accordance with PART 2 paragraph UTILITY BEDDING MATERIAL. Compact backfill to top of pipe to 85 percent of ASTM D1557. Provide plastic piping with bedding to spring line of pipe.

#### 3.7.1.4 Final Backfill

Do not begin backfill until construction below finish grade has been approved, underground utilities systems have been inspected, tested and approved, forms removed, and the excavation cleaned of trash and debris. Bring backfill to indicated finish grade. Where pipe is coated or wrapped for protection against corrosion, the backfill material up to an elevation 2 feet above sewer lines and one foot above other utility lines need to be free from stones larger than one inch in any dimension. Heavy equipment for spreading and compacting backfill are not to be operated closer to foundation or retaining walls than a distance equal to the height of backfill above the top of footing; compact remaining area in layers not more than 4 inches in compacted thickness with power-driven hand tampers suitable for the material being compacted. Place backfill carefully around pipes or tanks to avoid damage to coatings, wrappings, or tanks. Do not place backfill against foundation walls prior to 7 days after completion of the walls. As far as practicable, bring backfill up evenly on each side of the wall and sloped to drain away from the wall.

Fill the remainder of the trench, except for special materials for buildings and pavements with satisfactory material. Place backfill material and compact as follows:

##### 3.7.1.4.1 Trenches, Buildings, Railroads, and Pavements

Place backfill up to the required elevation as specified. Do not permit water flooding or jetting methods of compaction. Compact as specified for Structural Fill.

##### 3.7.1.4.2 Turfed or Seeded Areas and Miscellaneous Areas

Deposit backfill in layers of a maximum of 12 inches loose thickness, and compact it to 85 percent maximum density for cohesive soils and 90 percent maximum density for cohesionless soils.

#### 3.7.1.5 Electrical Distribution System

Provide a minimum cover of 24 inches from the finished grade to direct burial cable and conduit or duct line, unless otherwise indicated.

#### 3.7.1.6 Displacement of Features

After other required tests have been performed and the trench backfill compacted to the finished grade surface, inspect the pipe to determine whether unexpected or damaging displacement has occurred. Conduct walk-through inspection of pipe sizes larger than 48 inches. Inspect pipes smaller than 48 inches using remote methods using closed circuit television, sonar, or hybrid that can provide a 360-degree inspection of the pipe. Prepare and submit a pipe inspection report consisting of digital video or photos. If, in the judgment of the Contracting Officer, the interior of the pipe shows poor alignment or any other defects that would cause improper functioning of the system, replace or repair the defects as directed at no additional cost to the Government.

#### 3.7.1.7 Buried Tape And Detection Wire

##### 3.7.1.7.1 Buried Warning and Identification Tape

Provide buried utility lines with utility identification tape. Bury tape

12 inches below finished grade; under pavements and slabs, bury tape 6 inches below top of subgrade.

#### 3.7.1.7.2 Buried Detection Wire

Bury detection wire directly above non-metallic piping at a distance not to exceed 12 inches above the top of pipe. Extend the wire continuously and unbroken, from manhole to manhole. Terminate the ends of the wire inside the manholes at each end of the pipe, with a minimum of 3 feet of wire, coiled, remaining accessible in each manhole. Furnish insulated wire over its entire length. Install wires at manholes between the top of the corbel and the frame, and extend up through the chimney seal between the frame and the chimney seal. For force mains, terminate the wire in the valve pit at the pump station end of the pipe.

#### 3.7.2 Structural Fill Placement

Place fill and backfill beneath and adjacent to structures in successive horizontal layers of loose material not more than 6 inches in depth, or in loose layers not more than 6 inches in depth when using hand-operated compaction equipment. Do not place over wet or frozen materials. Compact to at least 95 percent of laboratory maximum density per ASTM D1557 except as otherwise specified. Perform compaction in such a manner as to prevent wedging action or eccentric loading upon or other damage to the structure.

Moisture condition fill and backfill material to within range of plus 2 or minus 2 percent of optimum moisture content at the time of compaction.

#### 3.7.3 Backfill for Appurtenances

After the manhole, catchbasin, inlet, or similar structure has been constructed and the concrete has been allowed to cure for 2 days, place backfill in such a manner that the structure is not be damaged by the shock of falling earth. Deposit the backfill material, compact it as specified for final backfill, and bring up the backfill evenly on all sides of the structure to prevent eccentric loading and excessive stress.

#### 3.7.4 Flowable Fill

Place fill in a manner to completely fill voids in the location indicated. Do not place when atmospheric temperatures are expected to be below 33 degrees F at any time during the 3 day period following placement.

### 3.8 FINISHING/FINISH OPERATIONS

During construction, keep embankments and excavations shaped and drained. Maintain ditches and drains along subgrade to drain effectively at all times. Do not disturb the finished subgrade by traffic or other operation. Protect and maintain the finished subgrade in a satisfactory condition until ballast, subbase, base, or pavement is placed. Do not permit the storage or stockpiling of materials on finished subgrade. Do not lay subbase, base course, ballast, or pavement until the subgrade has been checked and approved, and in no case place subbase, base, surfacing, pavement, or ballast on a muddy, spongy, frozen or otherwise unstable subgrade.

Finish the surface of excavations, embankments, and subgrades to a smooth and compact surface in accordance with the lines, grades, and cross

sections or elevations shown. Provide the degree of finish for graded areas within 0.1 foot of the grades and elevations indicated except as indicated for subgrades specified in paragraph SUBGRADE PREPARATION. Finish gutters and ditches in a manner that will result in effective drainage. Finish the surface of areas to be turfed to a smoothness suitable for the application of turfing materials. Repair graded, topsoiled, or backfilled areas prior to acceptance of the work, and re-established grades to the required elevations and slopes.

#### 3.8.1 Capillary Water Barrier

Place a capillary water barrier under concrete floor and area-way slabs grade directly on the subgrade and compact with a minimum of two passes of a hand-operated plate-type vibratory compactor.

#### 3.8.2 Grading Around Structures

Construct areas within 5 feet outside of each building and structure line true-to-grade, shape to drain, and maintain free of trash and debris until final inspection has been completed and the work has been accepted.

#### 3.8.3 Grading

Finish grades as indicated within one-tenth of one foot. Grade areas to drain water away from structures. Maintain areas free of trash and debris. For existing grades that will remain but which were disturbed by Contractor's operations, grade as directed.

### 3.9 FIELD QUALITY CONTROL

#### 3.9.1 Sampling

Take the number and size of samples required to perform the following tests.

#### 3.9.2 Testing

Perform one of each of the following tests for each material used. Provide additional tests for each source change. Laboratory testing shall have been completed within 12 month.

##### 3.9.2.1 Soil Material Testing

Test soil materials defined in Section 2.1 in accordance with ASTM C136 for conformance to ASTM D2487 gradation limits; ASTM D1140 for material finer than the No. 200 sieve; ASTM D4318 for liquid limit and for plastic limit; ASTM D1557 for moisture density relations, as applicable.

Submit Laboratory gradation test and Laboratory compaction tests results for each soil material proposed for use as fill and backfill.

### 3.10 TESTING

Test density in accordance with ASTM D6938. Perform density test at the start of the job and as follows:

- a. Density testing per the following will be required at the expense of the Contractor.
- b. Bedding and backfill in trenches: One test per 50 linear feet in each lift.
- c. Other site areas as follows:

| Material Type   | Location of Material  | Test Frequency                                                                                                | Degree of Compaction |
|-----------------|-----------------------|---------------------------------------------------------------------------------------------------------------|----------------------|
| Structural Fill | Below Structures      | One test per lift per each independent foundation or one every per 100 sq ft (whichever results in more tests | 95%                  |
| Structural Fill | Adjacent to Structure | One test per lift per 200 square feet of each lift                                                            | 95%                  |
| Ordinary Fill   | Below pavement        | One test per lift per 100 linear feet of trench placed                                                        | 95%                  |

Perform testing as indicated in the Table above. Submit Material Test Reports within 24 hours of tests being completed.

-- End of Section --

## ATTACHMENT 1

### Table 1: Soil Analytical Data

Table 1: Soil Analytical Data  
Portsmouth Naval Shipyard  
Kittery, ME

|                                                         |       | Location ID                              |       | P1112-B-101-0106 |        | P1112-B-102B-0106 |       | P1112-B-103-0106 |       | P1112-B-104-0106 |       | P1112-B-105-0106 |  |
|---------------------------------------------------------|-------|------------------------------------------|-------|------------------|--------|-------------------|-------|------------------|-------|------------------|-------|------------------|--|
|                                                         |       | Sample Date                              |       | 12/11/2023       |        | 12/12/2023        |       | 12/12/2023       |       | 12/12/2023       |       | 12/11/2023       |  |
|                                                         |       | Sample Type                              |       | Soil             |        | Soil              |       | Soil             |       | Soil             |       | Soil             |  |
| Analyte                                                 | Unit  | MEDEP Soil RAGS;<br>Construction Workers | Value | Qualifier        | Value  | Qualifier         | Value | Qualifier        | Value | Qualifier        | Value | Qualifier        |  |
| General Chemistry - EPA Method SW3540C                  |       |                                          |       |                  |        |                   |       |                  |       |                  |       |                  |  |
| TOTAL SOLIDS                                            | %     | ~                                        | 90    |                  | 97     |                   | 79.2  |                  | 88.1  |                  | 80.3  |                  |  |
| RCRA Metals - EPA Method SW6010C                        |       |                                          |       |                  |        |                   |       |                  |       |                  |       |                  |  |
| ARSENIC                                                 | mg/kg | 41                                       | 10.9  |                  | 38.9   | N                 | 16.2  |                  | 22.2  |                  | 28.2  |                  |  |
| BARIUM                                                  | mg/kg | 100,000                                  | 67.8  |                  | 24.3   |                   | 131   |                  | 143   |                  | 59.4  |                  |  |
| CADMIUM                                                 | mg/kg | 140                                      | 0.58  | U                | 0.52   | U                 | 0.34  | J                | 0.67  | U                | 0.68  | U                |  |
| CHROMIUM                                                | mg/kg | 100,000                                  | 34.6  |                  | 10.4   |                   | 47    |                  | 39.5  |                  | 62    |                  |  |
| LEAD                                                    | mg/kg | 440                                      | 534   |                  | 9.45   |                   | 1650  |                  | 1030  |                  | 134   |                  |  |
| SELENIUM                                                | mg/kg | 8,000                                    | 1.4   | U                | 1.2    | U                 | 1.4   | U                | 1.6   | U                | 1.6   | U                |  |
| SILVER                                                  | mg/kg | 8,000                                    | 0.78  | U                | 0.7    | U                 | 0.78  | U                | 0.89  | U                | 0.9   | U                |  |
| MERCURY (SW7471B)                                       | mg/kg | 3.1                                      | 2.17  |                  | 0.0074 | J                 | 16.1  |                  | 0.874 |                  | 0.323 | N                |  |
| Total Petroleum Hydrocarbons (TPH) - EPA Method SW8015D |       |                                          |       |                  |        |                   |       |                  |       |                  |       |                  |  |
| Extractable TPH C9-C36                                  | mg/kg | ~                                        | 780   |                  | 46     | B                 | 170   |                  | 250   |                  | 96    |                  |  |
| Pesticides - EPA Method SW8081B                         |       |                                          |       |                  |        |                   |       |                  |       |                  |       |                  |  |
| 4,4'-DDD                                                | ug/kg | 130,000                                  | 17    | U                | 1.6    | U                 | 18    | U                | 19    | U                | 2     | U                |  |
| 4,4'-DDE                                                | ug/kg | 130,000                                  | 17    | U                | 1.6    | U                 | 18    | U                | 19    | U                | 2     | U                |  |
| 4,4'-DDT                                                | ug/kg | 120,000                                  | 17    | U                | 1.6    | U                 | 18    | U                | 19    | U                | 2     | U                |  |
| ALDRIN                                                  | ug/kg | 2,500                                    | 9     | U                | 0.84   | U                 | 9.5   | U                | 9.6   | U                | 1     | U                |  |
| ALPHA BHC                                               | ug/kg | 5,000                                    | 9     | U                | 0.84   | U                 | 9.5   | U                | 9.6   | U                | 1     | U                |  |
| ALPHA-CHLORDANE                                         | ug/kg | 110,000                                  | 9     | U                | 0.84   | U                 | 9.5   | U                | 9.6   | U                | 1     | U                |  |
| BETA BHC                                                | ug/kg | 17,000                                   | 9     | U                | 0.84   | U                 | 9.5   | U                | 9.6   | U                | 1     | U                |  |
| DELTA BHC                                               | ug/kg | ~                                        | 9     | U                | 0.84   | U                 | 9.5   | U                | 9.6   | U                | 1     | U                |  |
| DIELDRIN                                                | ug/kg | 2,000                                    | 17    | U                | 1.6    | U                 | 18    | U                | 19    | U                | 2     | U                |  |
| ENDOSULFAN I                                            | ug/kg | 9,600,000                                | 9     | U                | 0.84   | U                 | 9.5   | U                | 9.6   | U                | 1     | U                |  |
| ENDOSULFAN II                                           | ug/kg | 9,600,000                                | 17    | U                | 1.6    | U                 | 18    | U                | 19    | U                | 2     | U                |  |
| ENDOSULFAN SULFATE                                      | ug/kg | 9,600,000                                | 17    | U                | 1.6    | U                 | 18    | U                | 19    | U                | 2     | U                |  |
| ENDRIN                                                  | ug/kg | 340,000                                  | 26    | U                | 2.5    | U                 | 28    | U                | 28    | U                | 3     | U                |  |
| ENDRIN ALDEHYDE                                         | ug/kg | 340,000                                  | 17    | U                | 1.6    | U                 | 18    | U                | 19    | U                | 2     | U                |  |
| ENDRIN KETONE                                           | ug/kg | 340,000                                  | 17    | U                | 1.6    | U                 | 18    | U                | 19    | U                | 2     | U                |  |
| GAMMA BHC                                               | ug/kg | 35,000                                   | 9     | U                | 0.84   | U                 | 9.5   | U                | 9.6   | U                | 1     | U                |  |
| GAMMA-CHLORDANE                                         | ug/kg | 110,000                                  | 9     | U                | 0.84   | U                 | 9.5   | U                | 9.6   | U                | 1     | U                |  |
| HEPTACHLOR                                              | ug/kg | 8,700                                    | 9     | U                | 0.84   | U                 | 9.5   | U                | 9.6   | U                | 1     | U                |  |
| HEPTACHLOR EPOXIDE                                      | ug/kg | 4,500                                    | 9     | U                | 0.84   | U                 | 9.5   | U                | 9.6   | U                | 1     | U                |  |
| METHOXYCHLOR                                            | ug/kg | 5,600,000                                | 90    | U                | 8.4    | U                 | 95    | U                | 96    | U                | 10    | U                |  |
| TOXAPHENE                                               | ug/kg | ~                                        | 170   | U                | 16     | U                 | 180   | U                | 180   | U                | 19    | U                |  |

Table 1: Soil Analytical Data  
Portsmouth Naval Shipyard  
Kittery, ME

|                                                        |       | Location ID                              | P1112-B-101-0106 |           | P1112-B-102B-0106 |           | P1112-B-103-0106 |           | P1112-B-104-0106 |           | P1112-B-105-0106 |           |
|--------------------------------------------------------|-------|------------------------------------------|------------------|-----------|-------------------|-----------|------------------|-----------|------------------|-----------|------------------|-----------|
|                                                        |       | Sample Date                              | 12/11/2023       |           | 12/12/2023        |           | 12/12/2023       |           | 12/12/2023       |           | 12/11/2023       |           |
|                                                        |       | Sample Type                              | Soil             |           | Soil              |           | Soil             |           | Soil             |           | Soil             |           |
| Analyte                                                | Unit  | MEDEP Soil RAGS;<br>Construction Workers | Value            | Qualifier | Value             | Qualifier | Value            | Qualifier | Value            | Qualifier | Value            | Qualifier |
| Polychlorinated biphenyls (PCBs) - EPA Method SW8082A  |       |                                          |                  |           |                   |           |                  |           |                  |           |                  |           |
| AROCLOR-1016                                           | ug/kg | 13,000                                   | 45 U             |           | 8.4 U             |           | 47 U             |           | 48 U             |           | 52 U             |           |
| AROCLOR-1221                                           | ug/kg | 13,000                                   | 45 U             |           | 8.4 U             |           | 47 U             |           | 48 U             |           | 52 U             |           |
| AROCLOR-1232                                           | ug/kg | 13,000                                   | 53 U             |           | 9.9 U             |           | 56 U             |           | 57 U             |           | 61 U             |           |
| AROCLOR-1242                                           | ug/kg | 13,000                                   | 45 U             |           | 8.4 U             |           | 47 U             |           | 48 U             |           | 52 U             |           |
| AROCLOR-1248                                           | ug/kg | 13,000                                   | 45 U             |           | 8.4 U             |           | 47 U             |           | 48 U             |           | 52 U             |           |
| AROCLOR-1254                                           | ug/kg | 13,000                                   | 45 U             |           | 8.4 U             |           | 47 U             |           | 48 U             |           | 52 U             |           |
| AROCLOR-1260                                           | ug/kg | 13,000                                   | 45 U             |           | 8.4 U             |           | 47 U             |           | 48 U             |           | 52 U             |           |
| Herbicides - EPA Method SW8151A                        |       |                                          |                  |           |                   |           |                  |           |                  |           |                  |           |
| 2,4,5-T                                                | ug/kg | 9,000,000                                | 49 U             |           | 45 U              |           | 59 U             |           | 50 U             |           | 55 U             |           |
| 2,4-D                                                  | ug/kg | ~                                        | 73 U             |           | 68 U              |           | 88 U             |           | 75 U             |           | 82 U             |           |
| 2,4-DB                                                 | ug/kg | ~                                        | 49 U             |           | 45 U              |           | 59 U             |           | 50 U             |           | 55 U             |           |
| DALAPON                                                | ug/kg | 34,000,000                               | 83 U             |           | 77 U              |           | 100 U            |           | 85 U             |           | 93 U             |           |
| DICAMBA                                                | ug/kg | ~                                        | 49 U             |           | 45 U              |           | 59 U             |           | 50 U             |           | 55 U             |           |
| DICHLOROPROP                                           | ug/kg | ~                                        | 49 U             |           | 45 U              |           | 59 U             |           | 50 U             |           | 55 U             |           |
| DINOSEB                                                | ug/kg | 1,100,000                                | 49 U             |           | 45 U              |           | 59 U             |           | 50 U             |           | 55 U             |           |
| MCPA                                                   | ug/kg | 560,000                                  | 4900 UL          |           | 4500 UL           |           | 5900 UL          |           | 5000 UL          |           | 5500 UL          |           |
| MCPP                                                   | ug/kg | 1,100,000                                | 4900 U           |           | 4500 U            |           | 5900 U           |           | 5000 U           |           | 5500 U           |           |
| PENTACHLOROPHENOL                                      | ug/kg | 54,000                                   | 49 U             |           | 45 U              |           | 59 U             |           | 50 U             |           | 55 U             |           |
| SILVEX                                                 | ug/kg | 9,000,000                                | 49 U             |           | 45 U              |           | 59 U             |           | 50 U             |           | 55 U             |           |
| Volatile Organic Compounds (VOCs) - EPA Method SW8260C |       |                                          |                  |           |                   |           |                  |           |                  |           |                  |           |
| BENZENE                                                | ug/kg | 75,000                                   | 2.3 U            |           | 2.4 U             |           | 3.2 U            |           | 2.9 U            |           | 2.4 U            |           |
| ETHYLBENZENE                                           | ug/kg | 380,000                                  | 2.3 U            |           | 2.4 U             |           | 3.2 U            |           | 2.9 U            |           | 2.4 U            |           |
| M-,P-XYLENE                                            | ug/kg | 260,000                                  | 4.6 U            |           | 4.7 U             |           | 6.3 U            |           | 5.8 U            |           | 4.7 U            |           |
| O-XYLENE                                               | ug/kg | 260,000                                  | 2.7 U            |           | 2.8 U             |           | 3.8 U            |           | 3.5 U            |           | 2.8 U            |           |
| TOLUENE                                                | ug/kg | 810,000                                  | 2.7 U            |           | 6.7               |           | 3.8 U            |           | 3.5 U            |           | 2.8 U            |           |
| XYLENES (TOTAL)                                        | ug/kg | 260,000                                  | 6.8 U            |           | 7.1 U             |           | 9.5 U            |           | 8.7 U            |           | 7.1 U            |           |



Table 1: Soil Analytical Data  
Portsmouth Naval Shipyard  
Kittery, ME

|                                                              |       | Location ID                              | P1112-B-101-0106 |           | P1112-B-102B-0106 |           | P1112-B-103-0106 |           | P1112-B-104-0106 |           | P1112-B-105-0106 |           |
|--------------------------------------------------------------|-------|------------------------------------------|------------------|-----------|-------------------|-----------|------------------|-----------|------------------|-----------|------------------|-----------|
|                                                              |       | Sample Date                              | 12/11/2023       |           | 12/12/2023        |           | 12/12/2023       |           | 12/12/2023       |           | 12/11/2023       |           |
|                                                              |       | Sample Type                              | Soil             |           | Soil              |           | Soil             |           | Soil             |           | Soil             |           |
| Analyte                                                      | Unit  | MEDEP Soil RAGS;<br>Construction Workers | Value            | Qualifier | Value             | Qualifier | Value            | Qualifier | Value            | Qualifier | Value            | Qualifier |
| Semi-Volatile Organic Compounds (SVOCs) - EPA Method SW8270D |       |                                          |                  |           |                   |           |                  |           |                  |           |                  |           |
| 2-METHYLNAPHTHALENE                                          | ug/kg | 4,100,000                                | 270 U            |           | 250 U             |           | 280 U            |           | 270 U            |           | 280 U            |           |
| ACENAPHTHENE                                                 | ug/kg | 62,000,000                               | 270 U            |           | 250 U             |           | 280 U            |           | 270 U            |           | 280 U            |           |
| ACENAPHTHYLENE                                               | ug/kg | 45,000,000                               | 270 U            |           | 250 U             |           | 280 U            |           | 270 U            |           | 280 U            |           |
| ANTHRACENE                                                   | ug/kg | 100,000,000                              | 160 J            |           | 250 U             |           | 250 J            |           | 270 U            |           | 280 U            |           |
| BENZO(A)ANTHRACENE                                           | ug/kg | 280,000                                  | 1000             |           | 250 U             |           | 1800             |           | 830              |           | 280 U            |           |
| BENZO(A)PYRENE                                               | ug/kg | 29,000                                   | 1000             |           | 250 U             |           | 1300             |           | 880              |           | 280 U            |           |
| BENZO(B)FLUORANTHENE                                         | ug/kg | 290,000                                  | 1200             |           | 380 U             |           | 1800             |           | 1000             |           | 430 U            |           |
| BENZO(K)FLUORANTHENE                                         | ug/kg | 2,900,000                                | 440              |           | 250 U             |           | 760              |           | 420              |           | 280 U            |           |
| BENZO[G,H,I]PERYLENE                                         | ug/kg | 23,000,000                               | 980              |           | 250 U             |           | 1400             |           | 1000             |           | 150 J            |           |
| CHRYSENE                                                     | ug/kg | 29,000,000                               | 1100             |           | 250 U             |           | 1800             |           | 1000             |           | 280 U            |           |
| DIBENZO(A,H)ANTHRACENE                                       | ug/kg | 29,000                                   | 300 J            |           | 380 U             |           | 460 J            |           | 260 J            |           | 430 U            |           |
| FLUORANTHENE                                                 | ug/kg | 41,000,000                               | 1200             |           | 250 U             |           | 2700             |           | 920              |           | 280 U            |           |
| FLUORENE                                                     | ug/kg | 41,000,000                               | 270 U            |           | 250 U             |           | 280 U            |           | 270 U            |           | 280 U            |           |
| INDENO(1,2,3-CD)PYRENE                                       | ug/kg | 290,000                                  | 960              |           | 250 U             |           | 1300             |           | 920              |           | 280 U            |           |
| NAPHTHALENE                                                  | ug/kg | 120,000                                  | 270 U            |           | 250 U             |           | 280 U            |           | 270 U            |           | 280 U            |           |
| PHENANTHRENE                                                 | ug/kg | 23,000,000                               | 1100             |           | 250 U             |           | 2400             |           | 590              |           | 280 U            |           |
| PYRENE                                                       | ug/kg | 31,000,000                               | 2400             |           | 250 U             |           | 3400             |           | 2000             |           | 290 J            |           |
| TCLP RCRA 8 Metals - EPA Method SW6010C                      |       |                                          |                  |           |                   |           |                  |           |                  |           |                  |           |
| ARSENIC                                                      | ug/L  | ~                                        | 25 U             |           | 25 U              |           | 25 U             |           | 25 U             |           | 25 U             |           |
| BARIUM                                                       | ug/L  | ~                                        | 607              |           | 198               |           | 1020             |           | 720              |           | 224              |           |
| CADMIUM                                                      | ug/L  | ~                                        | 15 U             |           | 15 U              |           | 8 J              |           | 15 U             |           | 15 U             |           |
| CHROMIUM                                                     | ug/L  | ~                                        | 20 U             |           | 20 U              |           | 20 U             |           | 20 U             |           | 20 U             |           |
| LEAD                                                         | ug/L  | ~                                        | 1330             |           | 30                |           | 4400             |           | 2220             |           | 39               |           |
| SELENIUM                                                     | ug/L  | ~                                        | 35 U             |           | 35 U              |           | 35 U             |           | 35 U             |           | 35 U             |           |
| SILVER                                                       | ug/L  | ~                                        | 20 U             |           | 20 U              |           | 20 U             |           | 20 U             |           | 20 U             |           |
| MERCURY (SW7470A)                                            | ug/L  | ~                                        | 0.1 U            |           | 0.1 U             |           | 0.1 U            |           | 0.1 U            |           | 0.054 J          |           |

MEDEP Remedial Action Guidelines (RAGS) for Soil Exposure Pathway to Construction Worker used.

**Bold** = Detected

Exceeds Soil RAG

~ = No RAG Available

U= Indicates the compound was analyzed for but not detected above the specified level.

J = Estimated value

B = Indicates the analyte was detected in the laboratory method blank analyzed concurrently with the sample.

SECTION 31 63 33

DRILLED MICROPILES

02/25

PART 1 GENERAL

1.1 GENERAL

- a. This work shall consist of designing and constructing drilled micropiles (micropiles) as shown on the Contract Drawings and approved working drawings prepared as specified herein. The Contractor is responsible for furnishing all design, materials, products, accessories, tools, equipment, services, transportation, labor and supervision, and manufacturing techniques required for design, installation, and testing of drilled micropiles.
- b. The Contractor shall select the micropile type, size, pile top attachment, installation means and methods, estimate the ground-grout bond value and determine the required grout bond length and final micropile diameter. The Contractor shall design and install micropile that will develop the load capacities indicated on the Contract Drawings. The micropile load capacities shall be confirmed by verification and proof load testing as required and must meet the test acceptance criteria specified herein. The Contractor's micropile design shall conform to requirements set forth in this specification and any micropile design minimums and/or maximums as shown on the Contract Drawings.
- c. Existing utilities within 10 feet of the proposed drilled micropiles shall be exposed prior to drilled micropile installation.
- d. Micropile testing shall include a verification test within the equipment pad; and proof tests on at least one production pile within the equipment pad and on at least one production pile below the break tank pad.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN PETROLEUM INSTITUTE (API)

|              |                                                                                      |
|--------------|--------------------------------------------------------------------------------------|
| API RP 13B-1 | (2009; R 2016) Recommended Practice for<br>Field Testing Water-Based Drilling Fluids |
|--------------|--------------------------------------------------------------------------------------|

ASTM INTERNATIONAL (ASTM)

|           |                                                                           |
|-----------|---------------------------------------------------------------------------|
| ASTM A252 | (2010) Standard Specification for Welded<br>and Seamless Steel Pipe Piles |
|-----------|---------------------------------------------------------------------------|

|                 |                                                                                                         |
|-----------------|---------------------------------------------------------------------------------------------------------|
| ASTM A615/A615M | (2024) Standard Specification for Deformed<br>and Plain Carbon-Steel Bars for Concrete<br>Reinforcement |
|-----------------|---------------------------------------------------------------------------------------------------------|

|                 |                                 |
|-----------------|---------------------------------|
| ASTM C109/C109M | (2023) Standard Test Method for |
|-----------------|---------------------------------|

Compressive Strength of Hydraulic Cement  
Mortars (Using 2-in. or (50-mm) Cube  
Specimens)

|                   |                                                                                                   |
|-------------------|---------------------------------------------------------------------------------------------------|
| ASTM C150/C150M   | (2024) Standard Specification for Portland Cement                                                 |
| ASTM D1143/D1143M | (2007; R 2013) Piles Under Static Axial Compressive Load                                          |
| ASTM D3689/D3689M | (2007; R 2013; E 2013) Standard Test Methods for Deep Foundations Under Static Axial Tensile Load |

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-01 Preconstruction Submittals

Contractor Experience; G  
Foreman Operators; G  
Drill Rig Operators; G  
Project Reference List; G  
Personnel List; G  
Professional Engineer; G

#### SD-02 Shop Drawings

Working Drawings; G  
Micropile Structures; G  
Pile Load Test; G

#### SD-05 Design Data

Micropile Structures; G  
Steel Pipe; G  
Reports; G  
Design Criteria and Notes; G  
Design Calculations; G  
Plans; G  
Computations; G

Safety and Strength Factors; G

SD-06 Test Reports

Verification Load Test Report; G

Proof Load Test Reports; G

SD-08 Manufacturer's Instructions

Micropile Structures; G

Installation; G

1.4 ADMINISTRATIVE REQUIREMENTS

Work shall not begin until the construction submittals have been received, reviewed, and accepted in writing by the Contracting Officer. Any submittals that are found to be unacceptable by the engineer shall be revised, resubmitted and accepted prior to commencing work.

Submit the following:

1.4.1 Contractor Experience Requirements

- a. The micropile Contractor shall have Contractor Experience in the construction and load testing of drilled micropiles and have successfully constructed at least 5 projects in the last 5 years involving construction totaling at least 100 micropiles of similar capacity to those required in these Contract Drawings and specifications.
- b. The Contractor Experience shall include previous drilling and grouting in soil and rock similar to project conditions. The Contractor shall submit construction details, structural details and load test results for at least three previous successful micropile load tests from different projects of similar scope to this project.
- c. The Contractor shall assign a Professional Engineer, licensed in the State of Maine, to supervise the work. This engineer shall have experience on at least 10 projects of similar scope to this project completed over the past 5 years. The Contractor shall not use manufacturers representatives to satisfy the supervising engineer requirements of this section. The contractor may use a single independent consultant for this purpose, provided the consultant has specific experience as listed above, and operates their business specifically for the purpose of transferring technology and skills in micropiling to contractors.
- d. The on-site Foreman Operators and Drill Rig Operators shall also have experience on at least 10 projects over the past 5 years installing drilled micropiles of equal or greater capacity than required in these Contract Drawings and specifications.
- e. The Contractor shall assign a Professional Engineer, licensed in the State of Maine, to design the micropiles. This engineer shall also be responsible for design, supervision and reporting of the verification and Product Proof Test. This engineer shall have experience in the design of at least 3 successfully completed drilled micropile projects

over the past 5 years, with micropiles of similar capacity to those required in these Contract Drawings and specifications in soil and rock similar to the project conditions.

- f. At least 45 calendar days before the planned start of micropile construction, the Contractor shall submit 5 copies of the completed Project Reference List and a Personnel List. The project reference list shall include a brief project description with the owner's name and current phone number and load test reports. The personnel list shall identify the supervising project engineer, drill rig operators, and on-site foremen to be assigned to the project. The personnel list shall contain a summary of each individual's experience and be complete enough for the Engineer to determine whether each individual satisfies the required qualifications.
- g. Work shall not start, nor materials ordered, until the Contracting Officers written approval of the Contractors experience qualification is given. The Contracting Officer may suspend work if the Contractor uses non-approved personnel. If work is suspended, the Contractor shall be fully liable for all resulting cost and no adjustment in contract time will result from the suspension.

#### 1.4.2 Micropile Design Requirements

- a. The micropiles shall be designed to meet the specific loading conditions, as shown on the Contract Drawings and approved working drawings. The micropile design shall conform to all required minimum and/or maximum properties shown on the Contract Drawings and the Service Load Design (SLD) procedures contained in the FHWA Micropile Design and Construction Reference Manual , FHWA NHI-05-039.
- b. Micropiles shall be designed to derive their support entirely by side resistance within bedrock. Side friction in the sand and fill, and end bearing resistance shall be ignored. Micropiles shall have a minimum bond length of 5 feet in bedrock.
- c. Steel pipe used for micropile permanent casing shall incorporate an additional 1/16 inch thickness of sacrificial steel for corrosion protection.
- d. The Contractor shall prepare and submit to the Contracting Officer Working Drawings. The working drawings shall include all information required for the design, plans, construction and quality control of the piling. This information should include the following, but not necessarily be limited to:

##### (1) Design Computations

- (a) A written summary report which describes the overall micropile design.
- (b) Applicable code requirements and design references.
- (c) Micropile Structures critical design cross-section(s) geometry including soil/rock strata and piezometric levels and location, magnitude and direction of applied loadings.
- (d) Design Criteria and Notes including, soil/rock shear strengths (friction angle and cohesion), unit weights, and

grout-to-ground bond values and micropile drillhole diameter assumptions for each soil/rock strata.

- (e) Safety and Strength Factors used in the design of the ground-grout bond values, surcharges, soil/rock and material unit weights, steel, grout and concrete materials.
- (f) Design Calculations - sheets with the project number, micropile structure location, designation, date of preparation, initials of designer and checker, and page number at the top of each page. Provide an index page with the design calculations.
- (g) Design Criteria and Notes including an explanation of any symbols and computer program used in the design.
- (h) Pile to footing connection calculations.

#### 1.4.3 Plans

- a. A plan view of the Micropile Structures identifying:
  - (1) A reference elevation datum.
  - (2) Right-of-way and permanent or temporary construction easement limits, location of all known active and abandoned utilities, adjacent structures or other potential interference. The centerline of any buried utility (including, but not limited to, drainage structures, drainage pipes, and electrical duct bank) within 10 feet of the proposed micropile location.
  - (3) Subsurface exploration locations shown on the plan view of the proposed Micropiles with appropriate reference baselines to fix the locations of the exploration relative to the micropile structure.
- b. An elevation view of the micropile structure(s) identifying:
  - (1) Micropile locations and elevations; vertical and horizontal spacing; batter and alignment and the location of existing subsurface features.
  - (2) Existing and finish grade profiles both behind and in front of the micropile structure.
- c. Design parameters and applicable codes.
- d. General notes for constructing the Micropile Structures including construction sequencing or other special construction requirements.
- e. A listing of the summary of quantities on the elevation drawing of each micropile structure showing pay item estimated quantities.
- f. Micropile typical sections including micropile spacing and inclination; minimum drillhole diameter; pipe casing and reinforcing bar size and details; splice type and locations; centralizers and spacers; grout bond zone and casing plunge length (if used); corrosion protection details; and connection details to the substructure footing, anchorage, plates, etc.

- g. A typical detail of verification and production proof test micropiles defining the micropile length, minimum drillhole diameter, inclination, and load test bonded and unbonded test lengths.
- h. Details, dimensions and schedules for all micropiles, casing and reinforcing steel, including reinforcing bar bending details.
- i. Details for constructing micropile structures around drainage facilities (if applicable).

#### 1.4.4 Construction Procedures

- a. Detailed step-by-step description of the proposed micropile construction procedure, including personnel, testing and equipment to assure quality control. This step-by-step procedure shall be shown in sufficient detail to allow the Contracting Officer to monitor the construction and quality of the micropiles.
- b. Proposed start date and time schedule and micropile Installation schedule providing the following:
  - (1) Micropile number.
  - (2) Micropile design load.
  - (3) Type and size of rebar.
  - (4) Minimum total bond length.
  - (5) Total micropile length.
  - (6) Micropile top footing attachment.
- c. If welding of casing is proposed, submit the welding procedure. All welding shall be done in accordance with the current AWS Structural Welding Code.
- d. Information on space requirements for Installation equipment that verify the proposed equipment can perform at the site.
- e. Plans describing how surface water, drill flush, and excess waste grout will be controlled and disposed.
- f. Proposed Grouting Plans. The grouting plan shall include complete descriptions, and details for the following:
  - (1) Grout mix design and type of materials to be used in the grout including certified test data and trial batch reports. The Contractor shall also provide specific gravity of the wet mix design.
  - (2) Methods and equipment for accurately monitoring and recording the grout depth and grout volume as the grout is being placed.
  - (3) Estimated curing time for grout to achieve specified strength. Previous test results for the proposed grout mix completed within one year of the start of grouting may be submitted for initial verification and acceptance and start of production work.

(4) Procedure and equipment for Contractor monitoring of grout quality. At a minimum, the Contractor shall be required to use a Baroid Mud Balance (per API RP-13B-1) to check the specific gravity of the mixed grout prior to placement of the grout into each drilled micropile.

- g. Computations showing that the proposed equipment used for flushing the micropile during installation (i.e. pumps for water flushing and compressors for air flushing) will maintain the required up-hole (flushing) velocities.

#### 1.4.5 Pile Load Test

- a. Detailed Plans for the proposed micropile load testing method. This shall include all drawings, details, and structural design calculations necessary to clearly describe the proposed test method, reaction load system capacity and equipment setup, types and accuracy of apparatus to be used for applying and measuring the test loads and pile top movements in accordance with this specification.
- b. Plan showing the proposed location of the verification test pile.
- c. Calibration Reports and data for each test jack, pressure gauge and master pressure gauge and electronic load cell to be used. The calibration tests shall have been performed by an independent testing laboratory, and tests shall have been performed within 90 calendar days of the date submitted. Testing shall not commence until the Contracting Officer has reviewed and accepted the jack, pressure gauge, master pressure gauge and electronic load cell calibration data.

#### 1.4.6 Reports

- a. Micropile Installation Records
- (1) The micropile Contractor shall prepare and submit to the Contracting Officer full-length installation records for each micropile installed. The records shall be submitted within one work shift after that pile installation is completed. The data shall be recorded on a micropile installation log. A separate log shall be provided for each micropile.
- b. Upon completion of the micropile verification load test, the Contractor shall prepare and submit a professional engineer stamped report of the test results for review and acceptance by the Contracting Officer. This report shall include the acceptance criteria for the micropile verification test, design skin friction in bedrock, and written confirmation of the micropiles capacity.
- c. Upon completion of each micropile proof load test, the Contractor shall prepare and submit a professional engineer stamped report of the test results for review and acceptance by the Contracting Officer. This report shall include the acceptance criteria for the micropile proof test and written confirmation of the micropiles capacity.

#### 1.5 QUALITY CONTROL

- a. Codes, Regulations, Reference Standards and Specifications:



- (1) Codes and regulations of the jurisdictional authorities.
- (2) American Society of State Highway and Transportation Officials (AASHTO), latest edition.
- (3) FHWA Micropile Design and Construction Reference Manual, FHWA NHI-05-039, 2005

- b. The work shall be performed and supervised by personnel experienced in the installation and testing of micropiles of similar size and soil conditions. Personnel shall be subject to the approval of the Contracting Officer.

## PART 2 PRODUCTS

### 2.1 MATERIALS

Furnish materials new and without defects. Materials for micropiles shall consist of the following:

#### 2.1.1 Admixtures for Grout

Accelerators are not permitted. Expansive admixtures and admixtures containing chlorides are not permitted.

#### 2.1.2 Cement

All cement shall be Portland cement conforming to ASTM C150/C150M, Types II, III or V.

#### 2.1.3 Centralizers and Spacers

Centralizers and spacers shall be fabricated from schedule 40 PVC pipe or tube, steel, or material non-detrimental to the reinforcing steel. Wood shall not be used. Centralizers and spacers shall be securely attached to the reinforcement; sized to position the reinforcement within 0.25 inch of plan location from center of pile; sized to allow grout tremie pipe insertion to the bottom of the drillhole; and sized to allow grout to freely flow up the drillhole and casing and between adjacent reinforcing bars.

#### 2.1.4 Grout

Neat cement or fine aggregate/cement mixture. The designer is responsible for indicating the 3 day and 28 day compressive strengths. The grout shall conform to the specification ASTM C109/C109M and to any minimum and/or maximum properties shown on the Contract Drawings.

#### 2.1.5 Permanent Casing Pipe

Permanent steel casing/pipe shall conform to required minimum and/or maximum properties shown on the Contract Drawings. The permanent steel casing/pipe shall be designed to withstand the design service loadings shown on the Contract Drawings and the proof/verification tests loading described in this specification. The steel casing/pipe shall conform to one or more of the following specifications ASTM A252. The designer will be responsible for indicating the applicable material specification(s) and any welding or fabrication conditions that apply.

#### 2.1.6 Reinforcing Bars

Reinforcing steel shall be deformed bars in accordance with ASTM A615/A615M. The grade, thickness and number of bars shall be indicated by the designer and shall conform to any minimum and/or maximum properties shown on the Contract Drawings. Continuous spiral deformations (i.e. continuous threadbars) shall be used. Bar tendon couplers, if required, shall develop the ultimate tensile strength of the bars without evidence of any failure.

### PART 3 EXECUTION

#### 3.1 PRE-CONSTRUCTION MEETING

A pre-construction meeting will be scheduled by the Contracting Officer and held prior to the start of micropile construction. The Contracting Officer, prime Contractor, micropile specialty Contractor and micropile design engineer shall attend the meeting. Attendance is mandatory. The pre-construction meeting will be conducted to clarify the construction requirements for the work, to coordinate the construction schedule and activities, and to identify contractual relationships and delineation of responsibilities amongst the prime Contractor and the various Subcontractors; specifically those pertaining to excavation for micropile structures, installation of temporary sheeting, anticipated subsurface conditions, micropile installation and testing, micropile structure survey control and site drainage control.

#### 3.2 SITE DRAINAGE CONTROL

- a. The Contractor shall control and properly dispose of drill flush and construction related waste, including excess grout, in accordance with any related specifications within the contract documents and all applicable codes and regulations. Drill flush shall be conveyed by pipe, hose or conduit away from the location where the micropile is being drilled and away from any adjacent structure or facility. The acceptable distance required to convey the drill flush away from the micropile location shall be determined by the Contracting Officer. Provide positive control and discharge of all surface water that will affect construction of the micropile installation. Maintain all pipes or conduits used to control surface water during construction. Repair damage caused by surface water at no additional cost. Upon substantial completion of the work, remove surface water control pipes or conduits from the site. Alternatively, with the approval of the Contracting Officer, pipes or conduits may be left in place if fully grouted.
- b. Immediately contact the Contracting Officer if unanticipated existing subsurface drainage structures or other utilities are discovered during excavation or drilling. Suspend work in these areas until remedial measures meeting the Contracting Officer's approval are implemented.

#### 3.3 MICROPILE ALLOWABLE CONSTRUCTION TOLERANCES

- a. Centerline of piling shall not be more than 3 inches from indicated plan location. Centerline of reinforcing steel shall not be more than 0.5 inches from the centerline of the pile.
- b. Pile shall be plumb or battered within 2 percent of total-length plan alignment.

- c. Top elevation of pile shall be plus 1 inch or minus 1 inch maximum from vertical elevation indicated.

#### 3.4 MICROPILE INSTALLATION

- a. The micropile Contractor shall select the drilling method, the grouting procedure and the grouting pressure used for installation of the micropiles. The micropile Contractor shall also determine the micropile casing size, final drillhole diameter and bond length, and central tendon reinforcement steel sizing necessary to develop the specified load capacities and load testing requirements. All micropile material properties and dimensions shall conform to minimum/maximum properties/dimensions as shown in the contract drawings. The micropile Contractor is also responsible for estimating the grout take. There will be no extra payment for grout overruns.
- b. The permanent and/or temporary casing shall be drilled a minimum 6 inches into bedrock or to a depth within the ledge so as to prevent subsidence of overburden into the bond zone portion of the drill hole (i.e. the rock socket). The Contract Drawings include boring logs and record drawings with information on bedrock depth. Any difference in the depth to bedrock and corresponding required length of permanent casing installed and accepted by the Contracting Officer from the information shown on the Contract Drawings shall be measured for payment and/or credit. There will be no payment for differences in required length of temporary casing.
- c. The drilling equipment and methods shall be suitable for drilling through the conditions to be encountered, without causing damage to the overburden, any overlying or adjacent structures, buried structures or utilities, or services. If called for in the drilling method description, or by the nature of the stratum to be drilled through, the micropile Contractor shall furnish an overburden casing of the type and thickness, which can be installed without distortion. Casings that fail, fracture, or otherwise distort during drilling or after drilling shall, unless otherwise directed, be withdrawn or replaced at the micropile Contractor's expense. The drillhole must be open along its full length to at least the design minimum drillhole diameter prior to placing grout and reinforcement.
- d. The up-hole (flushing) velocity shall be maintained within the following ranges:
  - (1) For air flush or air/water mist flush the minimum velocity is 5,000 ft/sec and the maximum velocity is 6,800 ft/sec.
  - (2) For water flush the minimum velocity is 40 ft/sec and the maximum is 390 ft/sec.
- e. Temporary casing or other approved method of pile drillhole support will be required in caving or unstable ground to permit the pile shaft to be formed the minimum design drillhole diameter. The Contractor's proposed method(s) to provide drillhole support and to prevent detrimental ground movements shall be reviewed by the Contracting Officer. Detrimental ground movement is defined as movement which requires remedial repair measures, in order to maintain site conditions as determined by the Contracting Officer. Use of drilling fluid containing bentonite or any other non-reverting drilling fluid is not allowed.

- f. During construction, the Contractor shall observe the ground conditions in the vicinity of the micropile construction site on a daily basis for signs of ground heave or subsidence. Immediately notify the Contracting Officer if signs of movements are observed. The micropile Contractor shall immediately suspend or modify drilling or grouting operations if ground heave or subsidence is observed, if the micropile structure is adversely affected, or if adjacent structures are damaged from the drilling or grouting. If the Contracting Officer determines that the movements require corrective action, the micropile Contractor shall take corrective actions necessary to stop the movement or perform repairs.
- g. Reinforcement may be placed prior to grouting the drillhole. Reinforcement surface shall be free of deleterious substances such as soil, mud, grease or oil that might contaminate the grout or coat the reinforcement and impair bond. Pile reinforcement groups, if used, shall be sufficiently strong to withstand the installation and grouting process without damage or disturbance.
- h. The micropile Contractor shall check pile top elevations and adjust all installed micropiles to the planned elevations.
- i. Centralizers and spacers shall be provided at 10ft centers maximum spacing. The upper and lower most centralizer shall be located a maximum of 3ft from the top and bottom of the micropile. Centralizers and spacers shall permit the free flow of grout without misalignment of the reinforcing bar(s) and permanent casing. The reinforcing steel shall be inserted into the drill hole to the desired depth without difficulty. Partially inserted reinforcing bars shall not be driven or forced into the hole. The micropile Contractor shall re-drill and reinsert reinforcing steel when necessary to facilitate insertion.
- j. Lengths of casing and reinforcing bars to be spliced shall be secured in proper alignment and in a manner to avoid eccentricity or angle between the axes of the two lengths to be spliced. Splices and threaded joints shall meet the requirements of the rebar material. Threaded pipe casing joints shall be located at least two casing diameters (OD) from a splice in any reinforcing bar. When multiple bars are used, bar splices shall be staggered at least 1 foot.
- k. Micropiles shall be grouted the same day the load transfer bond length is drilled. The grouting equipment used shall be a colloidal grout plant, and shall produce a grout free of lumps and undispersed cement. Paddle type mixers are not acceptable. The micropile Contractor shall have means and methods of measuring the grout quantity and pumping pressures during the grouting operations. The grout pump shall be equipped with a pressure gauge to monitor grout pressure. A second pressure gauge shall be placed at the point of injection into the pile top. The pressure gauge shall be capable of measuring pressures of at least 145 psi or twice the actual grout pressure used, whichever is greater. The grout shall be kept in agitation prior to placing. Grout shall be placed within one hour of mixing. The grouting equipment shall be sized to enable each pile to be grouted in one continuous operation. The grout shall be injected from the lowest point of the drill hole and injection shall continue until uncontaminated grout flows from the top of the pile. The grout may be pumped through grout tubes, casing, or drill rods. Temporary casing, if used, shall be extracted in stages ensuring that, after each length of casing is

removed the grout level is brought back up to the ground level before the next length is removed. The tremie pipe or casing shall always extend below the level of the existing grout in the drillhole. The grout takes shall be controlled to prevent excessive heave or fracturing of rock or soil formations. Upon completion of grouting, the grout tube may remain in the hole, but must be filled with grout.

- l. If the Contractor elects to use a post-grouting system, Working Drawings and details shall be submitted to the Contracting Officer for review.
- m. Grout within the micropile verification and proof test piles shall attain the minimum required 3-day compressive strength prior to load testing. During production, micropile grout shall be tested by the Contractor for compressive strength in accordance with ASTM C109/C109M at a frequency of no less than one set of three 2 inch grout cubes, or 3 cylinders, from each grout plant each day of operation or per every 10 piles, whichever occurs more frequently. The compressive strength shall be the average of the 3 cubes tested.
- n. Grout consistency as measured by grout density shall be determined by the micropile Contractor per API RP-13B-1 at a frequency of at least one test per pile, conducted just prior to start of pile grouting. The Baroid Mud Balance used in accordance with API RP 13B-1 is an approved device for determining the grout density of neat cement grout.
- o. Provide grout cube/cylinder compressive strength and grout density test results to the Contracting Officer within 24 hours of testing.

### 3.5 OBSTRUCTION REMOVAL DURING MICROPILE INSTALLATION

- a. Naturally occurring cobbles, boulders, bedrock, claystones, sand and gravel layers, glacial deposits, and other dense, well bounded in-situ soils shall not be considered as obstructions. The Contractor is also advised as to the presence of existing timber piles large rip-rap boulders, which also may be encountered during micropile installation. Encountering such conditions is to be anticipated and is consistent with the general nature of the subsurface conditions. The cost associated with these conditions shall be incidental to and included in the micropile bid prices.
- b. The Contractor shall notify the Contracting Officer immediately upon encountering any potential obstruction which unavoidably and completely stops the advancement of the micropile drill progress, despite the Contractor's most diligent and practical efforts. The Contractor's notification may be oral, but if oral, shall also be confirmed by the Contractor in writing within 24 hours. The notification shall include all pertinent information relating to the nature, depth, plan location coordinates, expected extent of the potential obstruction. The Contractor shall also provide physical evidence, acceptable to the Contracting Officer, of the material comprising the potential obstruction and evidence that the potential obstruction is not bedrock.
- c. The Contractor shall remove or penetrate the obstruction by the most diligent, practical means and shall comply with noise and vibration limitations and other specified requirements, and shall maintain excavation support and protection of nearby structures and utilities at all times.

- d. At all times during the removal of obstructions, the Contractor shall diligently pursue the removal or penetration of the obstructions using all the necessary and appropriate means and methods. The Contractor shall have on hand at all times, readily available, equipment, tools, materials, and labor appropriate for the effective removal of all types of potential obstructions. No payment shall be made for idle time due to the contractor's failure to have readily available such equipment, tools, materials, or labor.
- e. At any time during the course of obstruction removal, the Contracting Officer may provide alternative direction.
- f. If in the judgment of the Contracting Officer, obstructions make it impossible to install certain micropiles at locations to the required depths shown on the Working Drawings after resorting to all usual methods, the Contracting Officer may order that additional micropile or micropiles be installed or that other remedial actions be taken, for which the Contractor will be paid in accordance with the unit prices established in this Contract.
- g. Fill with grout any drill holes which are abandoned at the direction of the Contracting Officer because of unmovable obstructions.

### 3.6 VERIFICATION AND PROOF TESTS

- a. Perform verification of piles at the locations indicated on the approved submittal. Perform proof testing on piles selected by Contracting Officer. Perform compression load testing in accord with ASTM D1143/D1143M and tension load testing in accord with ASTM D3689/D3689M, except as modified herein.
- b. Perform one(1) pre-production verification pile load test to verify the design of the pile system and the construction methods proposed prior to installing any production piles. Sacrificial verification test pile shall be constructed in conformance with the approved Working Drawings. Verification test pile(s) shall be installed within the equipment pad at a location(s) approved by the Contracting Officer.
- c. Proof load tests shall be performed on production piles to verify that the Contractor installed micropiles will meet the compression and/or tensile load capacities and load test acceptance criteria and to verify the length of the micropile load transfer bond zone is adequate. The micropile verification load test results must verify the Contractor's design and installation methods. Two (2) proof load test shall be performed: one on a pile supporting the equipment pad and one on a pile supporting the break tank pad.
- d. The drilling method, grouting method, permanent casing length, micropile diameter (cased and uncased) and bond zone length for the verification test pile shall be identical to those specified for the production piles at the given locations. The verification test micropile structural steel sections shall be sized to safely resist the maximum test load.
- e. The maximum verification and proof test loads applied to the micropile shall not exceed 80 percent of the structural capacity of the micropile structural elements; including steel yield in tension, steel yield or buckling in compression, or grout crushing in compression.

Any required increase in strength of the verification and proof test pile elements above the strength required for the production piles shall be provided for in the Contractor's bid price.

- f. Testing equipment shall include dial gauges, dial gauge independent reference frame, load jack and pressure gauge, electronic load cell (with readout device), and a reaction frame. The load cell is required only for the creep test portion of the verification test. The contractor shall provide a description of test setup and jack, pressure gauge and load cell calibration curves in accordance with the Submittals Section.
- g. Design the testing reaction frame to be sufficiently rigid and of adequate dimensions such that excessive deformation of the testing equipment does not occur. Align the jack, bearing plates, load cell, and stressing anchorage such that unloading and repositioning of the equipment will not be required during the test.
- h. Apply and measure the test load with a hydraulic jack and pressure gauge. The pressure gauge shall be graduated in 100psi increments or less. The jack and pressure gauge shall have a pressure range not exceeding twice the anticipated maximum test pressure. Jack ram travel shall be sufficient to allow the test to be done without resetting the equipment. Monitor the creep test load hold during verification tests with both the pressure gauge and the electronic load cell. Use the load cell to accurately maintain a constant load hold during the creep test load hold increment of the verification test.
- i. Measure the pile top movement with dial gauges capable of measuring to 0.001 inches. The dial gauge shall have a travel sufficient to allow the test to be done without having to reset the gauge. Visually align the gauge to be parallel with the axis of the micropile and support the gauge independently from the jack, pile or reaction frame. Use a minimum of two dial gauges when the test setup requires reaction against the ground or single reaction piles on each side of the test pile.
- j. Test verification piles as follows:
  - (1) Test verification piles to the following loads; Alignment Load (AL) and the Maximum Design Load (DL). The DL loads are provided on the contract drawings. The AL is the minimum load applied to the micropile during testing needed to keep the testing equipment correctly positioned. The AL shall not exceed 5% of the DL. The verification pile load tests shall be made by incrementally loading the micropile in accordance with the following cyclic load schedule for both compression and tension loading:

| Verification Load Test Table |          |              |                  |
|------------------------------|----------|--------------|------------------|
| Step                         | Loading  | Applied Load | Hold Time (min.) |
| 1                            | Apply AL | 0.05 DL      | 2.5              |

| Verification Load Test Table                                                                                                                                                                                                                                                                                  |                                                             |                                                                                                       |                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 2                                                                                                                                                                                                                                                                                                             | Cycle 1                                                     | 0.15 DL<br>0.30 DL<br>0.45 DL<br>AL                                                                   | 2.5<br>2.5<br>2.5<br>1.0                                            |
| 3                                                                                                                                                                                                                                                                                                             | Cycle 2                                                     | 0.15 DL<br>0.45 DL<br>0.60 DL<br>0.75 DL<br>0.90 DL<br>1.00 DL<br>AL                                  | 1.0<br>1.0<br>2.5<br>2.5<br>2.5<br>2.5<br>1.0                       |
| 4                                                                                                                                                                                                                                                                                                             | Cycle 3                                                     | 0.15 DL<br>1.00 DL<br>1.15 DL<br>1.30 DL                                                              | 1.0<br>1.0<br>2.5<br>Creep Test - See Below                         |
| Hold the load for the Creep Test at least 10 minutes while recording movement at 1, 2, 3, 4, 5, 6, and 10-minute intervals. If the total movement measuring during the load hold exceeds the specified maximum value, the load should be extended for an additional 50 minutes (i.e., 1 hr. total hold time). |                                                             |                                                                                                       |                                                                     |
| 5                                                                                                                                                                                                                                                                                                             | Cycle 3 cont'd                                              | 1.45 DL<br>AL                                                                                         | 2.5<br>1.0                                                          |
| 6                                                                                                                                                                                                                                                                                                             | Cycle 4                                                     | 0.15 DL<br>1.45 DL<br>1.60 DL<br>1.75 DL<br>1.90 DL<br>2.00 DL<br>1.50 DL<br>1.00 DL<br>0.50 DL<br>AL | 1.0<br>1.0<br>1.0<br>2.5<br>2.5<br>10.0<br>5.0<br>5.0<br>5.0<br>5.0 |
| 7                                                                                                                                                                                                                                                                                                             | Remove the load and compare results to acceptable criteria. |                                                                                                       |                                                                     |

- (2) Pile top movement shall be measured at each load increment. The load-hold period shall start as soon as each test load increment is applied. Dial gauges shall be reset to zero after the initial AL is applied.
- (3) Acceptance criteria for micropile verification load test is:
  - (a) The total vertical movement at the micropile head for the 1.0 DL load increment should not exceed 0.5-inch.
  - (b) The total vertical movement at the end of the 1.3 DL load increment should not exceed 0.04-inch for the hold time between 1 and 10 minutes and should not exceed 0.08-inch between 6 and 60 minutes if held longer.



- (c) The failure load is defined as the load where the slope of the load versus micropile head deflection first exceeds 0.025-in./kip.
  - (4) If a verification tested micropile fails to meet the acceptance criteria, the Contractor shall modify the design, the construction procedure, or both. These modifications may include modifying the installation methods, increasing the bond length, or changing the micropile type. Any modification that necessitates changes to the structure shall be submitted as a revision to the working drawings and require the Contracting Officer's review and acceptance. Any modifications of design or construction procedures or cost of additional verification test piles and load testing shall be at the Contractor's expense. At the completion of verification testing, test piles shall be removed down to the elevation specified by the Contracting Officer.
- k. Perform proof tests as follows:
- (1) Perform proof load tests on production piles selected by the Contracting Officer. A minimum of 2 micropiles shall be proof tested. Upon completion of each test, the Contractor shall prepare and submit a professional engineer stamped report of the test results for review and acceptance by the Contracting Officer.
  - (2) Test proof test piles to a maximum test load of 1.60 times the Maximum Design Load (DL). The DL load is provided on the contract drawings. Proof tests shall be made by incrementally loading the micropile in accordance with the following schedule:
    - (a) Increase load by 15% DL up to 130% DL (creep test load). Each increment shall be held for 2.5 minutes except for the creep test load which shall be held for 10 minutes. If creep exceeds 0.04-inch during the creep test hold, hold for an additional 50 minutes.
    - (b) Increase load by 15% DL up to 160% DL. Each increment shall be held for 2.5 minutes except for the 160% DL which shall be held for 10 minutes.
    - (c) Decrease the load in 30% increments. Each increment shall be held for 4 minutes.
    - (d) Measure rebound at zero load for 30 minutes.
  - (3) The acceptance criteria for micropile proof load tests are:
    - (a) At the end of the 1.3 DL test load increment, test piles shall have a creep rate not exceeding 0.04 inch/log cycle time (1 to 10 minutes) or 0.08 inch/log cycle time (6 to 60 minutes). The creep rate shall be linear or decreasing throughout the creep load hold period.
    - (b) Failure does not occur at the 1.67 DL maximum test load. Failure is defined as the load at which attempts to further increase the test load simply result in continued pile movement.

- (4) If a proof-tested micropile fails to meet the acceptance criteria, the Contractor shall immediately proof test another micropile within that footing. For failed piles and further construction of other piles, the Contractor shall modify the design, the construction procedure, or both. These modifications may include installing replacement micropiles, incorporating piles at not more than 50% of the maximum load attained, post-grouting the tested pile and re-proof testing the pile, modifying installation methods, increasing the bond length, or changing the micropile type. Any modification that necessitates changes to the structure design shall require the Contracting Officer's prior review and acceptance. Any modifications of design or construction procedures, or cost of additional verification test piles and verification and/or proof load testing, or replacement production micropiles, shall be at the Contractor's expense.

l. Load test reports:

- (1) Within 3 days after completion of the load tests, submit the Verification Load Test Report and Proof Load Test Reports. The load test reports shall include.
  - (a) Test pile installation log(s)
  - (b) Information on the load test including, at a minimum, the load testing sequence, rates and durations of load increments, equipment calibrations, graphical and tabular measurements of deflection, and drawings indicating the test pile and reaction pile locations.
  - (c) Comparison of the load test data to the Acceptance Criteria provided herein and statement regarding the load carrying capacity of the pile relative to the design loads.
  - (d) The submittals shall be prepared by, signed and sealed by the Contractor's Professional Engineer.
  - (e) Production pile installation shall not begin prior to approval of the verification load test report.

-- End of Section --

SECTION 32 11 26

HOT-MIX BITUMINOUS BASE COURSE FOR ROADS AND STREETS  
05/20

PART 1 GENERAL

1.1 UNIT PRICES

1.1.1 Measurement

The amount paid for will be the number of 2000-pound tons of bituminous mixture used in the accepted work. Weigh bituminous mixture after mixing. No payment will be made for defective areas until corrected.

1.1.2 Basis for Payment

The quantities of bituminous base course will be paid for at the respective contract unit prices in the bid schedule. Payment will constitute full compensation for preparing and reconditioning the underlying layer; for furnishing all material, equipment, plant, and tools; and for all labor and other incidentals necessary to complete the work required by this section.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS  
(AASHTO)

AASHTO M 156 (2013; R 2017) Standard Specification for  
Requirements for Mixing Plants for  
Hot-Mixed, Hot-Laid Bituminous Paving  
Mixtures

ASPHALT INSTITUTE (AI)

AI MS-2 (2015) Asphalt Mix Design Methods

ASTM INTERNATIONAL (ASTM)

ASTM C29/C29M (2023) Standard Test Method for Bulk  
Density ("Unit Weight") and Voids in  
Aggregate

ASTM C88 (2018) Standard Test Method for Soundness  
of Aggregates by Use of Sodium Sulfate or  
Magnesium Sulfate

ASTM C127 (2015) Standard Test Method for Density,  
Relative Density (Specific Gravity), and  
Absorption of Coarse Aggregate

ASTM C128 (2022) Standard Test Method for Density,  
Relative Density (Specific Gravity), and

Absorption of Fine Aggregate

|                   |                                                                                                                                            |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C131/C131M   | (2020) Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine |
| ASTM C136/C136M   | (2019) Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates                                                               |
| ASTM C183/C183M   | (2022) Standard Practice for Sampling and the Amount of Testing of Hydraulic Cement                                                        |
| ASTM D75/D75M     | (2019) Standard Practice for Sampling Aggregates                                                                                           |
| ASTM D140/D140M   | (2016) Standard Practice for Sampling Asphalt Materials                                                                                    |
| ASTM D242/D242M   | (2009; R 2014) Mineral Filler for Bituminous Paving Mixtures                                                                               |
| ASTM D1856        | (2009; R 2015) Recovery of Asphalt from Solution by Abson Method                                                                           |
| ASTM D2041/D2041M | (2011) Theoretical Maximum Specific Gravity and Density of Bituminous Paving Mixtures                                                      |
| ASTM D2172/D2172M | (2017; E 2018) Standard Test Methods for Quantitative Extraction of Asphalt Binder from Asphalt Mixtures                                   |
| ASTM D2216        | (2019) Standard Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass                             |
| ASTM D2726/D2726M | (2019) Standard Test Method for Bulk Specific Gravity and Density of Non-Absorptive Compacted Bituminous Mixtures                          |
| ASTM D3381/D3381M | (2018) Standard Specification for Viscosity-Graded Asphalt Binder for Use in Pavement Construction                                         |
| ASTM D3665        | (2012; R 2017) Standard Practice for Random Sampling of Construction Materials                                                             |
| ASTM D3666        | (2016) Standard Specification for Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials                       |
| ASTM D4318        | (2017; E 2018) Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils                                        |
| ASTM D4791        | (2019) Flat Particles, Elongated Particles, or Flat and Elongated Particles                                                                |

in Coarse Aggregate

|                   |                                                                                                                                                                       |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D4867/D4867M | (2009; R 2014) Effect of Moisture on Asphalt Concrete Paving Mixtures                                                                                                 |
| ASTM D5821        | (2013; R 2017) Standard Test Method for Determining the Percentage of Fractured Particles in Coarse Aggregate                                                         |
| ASTM D6373        | (2016) Standard Specification for Performance Graded Asphalt Binder                                                                                                   |
| ASTM D6925        | (2014) Standard Test Method for Preparation and Determination of the Relative Density of Hot Mix Asphalt (HMA) Specimens by Means of the Superpave Gyrotory Compactor |

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

#### SD-03 Product Data

Sources of Aggregates

Job Mix Formula; G

#### SD-06 Test Reports

Sources of Aggregates

Bituminous Materials

Test Section; G

Service Record; G

#### SD-09 Manufacturer's Field Reports

Batch Tickets

### 1.4 QUALITY CONTROL

#### 1.4.1 Qualifications

Perform sampling and testing using an approved commercial testing laboratory or on-site facilities. Submit accreditation of the commercial laboratory by an independent evaluation authority, indicating conformance to ASTM D3666, including all applicable test procedures. Do not start work requiring testing until the facilities have been inspected and approved. Schedule and provide payment for laboratory inspections. Additional payment or a time extension due to failure to acquire the required laboratory validation is not allowed. Maintain this certification for the duration of the project.

#### 1.4.2 Test Results

Verify that materials comply with the specification. When a material source is changed, test the new material for compliance. When deficiencies are found, repeat the initial analysis and retest the material already placed to determine the extent of unacceptable material. Replace or repair all in-place unacceptable material to conform to the contract requirements. Submit copies of field tests results within 24 hours after the tests are performed and certified copies of tests results for approval not less than 30 days before material is required for the work.

#### 1.4.3 Batch Tickets

Provide batch tickets in accordance with AASHTO M 156.

#### 1.4.4 Aggregates

Select sources of aggregates and submit a plan for operation of a new source of aggregates at least 10 days in advance of starting production. If a previously developed source is selected, submit test results with evidence that central plant hot-mix bituminous pavements constructed with the aggregates have had a satisfactory service record of at least 5 years under similar climatic conditions. Include in the service record a tabulation of aggregate gradation and quality test results, typical hot-mix asphalt mix design using the aggregate, and a list of representative paving projects using the aggregate. Make such tests and other investigations as necessary to determine whether or not aggregates meeting the requirements specified can be produced from the proposed sources. Sample aggregates in accordance with ASTM D75/D75M and test them at the start of production.

#### 1.4.5 Mineral Filler

Sample mineral filler in accordance with ASTM C183/C183M.

#### 1.4.6 Bituminous Materials

Select sources where bituminous materials are obtained in advance of time when materials will be required in the work. Sample bituminous materials in accordance with ASTM D140/D140M. Submit test results not less than 10 days before such material is required for use in the work.

### 1.5 ENVIRONMENTAL REQUIREMENTS

Do not construct bituminous courses when the underlying course contains free surface water, or when temperature of the surface of the underlying course is below 40 degrees F, unless otherwise directed.

### 1.6 ACCEPTANCE

#### 1.6.1 Tolerances

Acceptance of bituminous base course is based on compliance with the tolerances presented in Table 1. Remove and replace bituminous base course represented by the failing tests or submit repair plan for approval.

| TABLE 1                            |                             |
|------------------------------------|-----------------------------|
| Attribute                          | Measurement                 |
| Plant Mixture                      |                             |
| Delivery to Laydown Machine        | Minimum 250 deg F           |
| Laboratory Air Voids               | 3 to 5 percent              |
| Finished Mat                       |                             |
| Mat Density (avg of 4 cores/lot)   | Minimum 92 percent of TMD   |
| Joint Density (avg of 4 cores/lot) | Minimum 90.5 percent of TMD |
| Grade                              | plus/minus 0.05 foot        |
| Smoothness                         | plus/minus 3/8 inch         |
| Longitudinal Joint Offset          | Minimum 1 foot              |
| Transverse Joint Offset            | Minimum                     |

#### 1.6.2 Test Section

At the start of plant operation, prepare a quantity of the mixture sufficient to construct a test section at least 100 feet long and two spreader widths wide. Place, spread, and compact the mixture with equipment to be used in the project and in accordance with requirements specified herein. Construct a cold joint between spreader widths. Test and evaluate the test section and conform to all specified requirements. If tests indicate that the pavement does not conform to the tolerances of Table 1, remove and construct additional test sections and sample for conformance to specification requirements. Do not start production of the bituminous base course mixture without approval.

### PART 2 PRODUCTS

#### 2.1 AGGREGATES

Provide aggregates consisting of crushed stone, crushed slag, crushed gravel screenings, sand, and mineral filler, as required. The portion of these materials retained on the No. 4 sieve is classified as coarse aggregate; the portion passing the No. 4 sieve and retained on the No. 200 sieve, as fine aggregate; and the portion passing the No. 200 sieve, as mineral filler.

##### 2.1.1 Coarse Aggregates

Provide coarse aggregates consisting of clean, sound, durable fragments of crushed stone, crushed slag, or crushed gravel meeting the following requirements:

#### 2.1.1.1 Aggregate Wear

Percentage of wear not exceeding 40 after 500 revolutions, as determined in accordance with ASTM C131/C131M.

#### 2.1.1.2 Aggregate Loss

Percentage of loss not exceeding 18 after five cycles performed in accordance with ASTM C88, using magnesium sulfate.

#### 2.1.1.3 Fractured Faces

At least 75 percent by weight of coarse aggregate containing two or more fractured faces produced by crushing when tested in accordance with ASTM D5821.

#### 2.1.1.4 Flat and Elongated Pieces

Particle shape essentially cubical and containing not more than 20 percent, by weight, of flat particles and elongated particles (3:1 ratio of maximum to minimum) when tested in accordance with ASTM D4791.

#### 2.1.1.5 Dry Weight of Crushed Slag

Dry weight of crushed slag not less than 75 pcf as determined in accordance with ASTM C29/C29M.

#### 2.1.2 Fine Aggregates

Provide fine aggregates consisting of clean, durable natural sands; manufactured sands prepared by crushing stone, slag, or gravel, or any combination of natural and manufactured sands. Natural sands consist of grains of clean, hard, durable rock. Limit the quantity of uncrushed material to a maximum of 25 percent by weight of total aggregate.

#### 2.1.3 Mineral Filler

Mineral filler conforming to ASTM D242/D242M.

#### 2.1.4 Liquid Limit and Plasticity Index

Measure liquid limit and plasticity index on the portion of the aggregate passing the No. 40 sieve in accordance with ASTM D4318. Requirements apply to the individual aggregate fractions and the combined blend in the completed base course. Provide aggregates classified as either nonplastic or having a liquid limit not greater than 25 and a plasticity index not greater than 5.

### 2.2 BITUMINOUS MATERIALS

#### 2.2.1 Asphalt Cement

Provide asphalt cement binder conforming to ASTM D6373 Performance Grade (PG). As an alternate, provide ASTM D3381/D3381M Viscosity Grade asphalt cement.

### 2.3 AGGREGATE GRADATION

Provide mineral aggregate of such size that percentage composition by



weight, as determined by ASTM C136/C136M, conforms to the gradation specified in TABLE 2, and does not vary from the low limit on one sieve to the high limit on the adjacent sieve or vice versa, but grade uniformly from coarse to fine.

| TABLE 2. AGGREGATE GRADATION |             |             |             |
|------------------------------|-------------|-------------|-------------|
| Percent Passing              |             |             |             |
| Sieve Size                   | Gradation 1 | Gradation 2 | Gradation 3 |
| 1 inch                       | 100         | ---         | ---         |
| 3/4 inch                     | 90-100      | 100         | ---         |
| 1/2 inch                     | 68-88       | 90-100      | 100         |
| 3/8 inch                     | 60-82       | 72-88       | 90-100      |
| No. 4                        | 45-67       | 53-73       | 58-78       |
| No. 8                        | 32-54       | 38-60       | 40-60       |
| No. 16                       | 22-44       | 26-48       | 28-48       |
| No. 30                       | 15-35       | 18-38       | 18-38       |
| No. 50                       | 9-25        | 11-27       | 11-27       |
| No. 100                      | 6-18        | 6-18        | 6-18        |
| No. 200                      | 3-6         | 3-6         | 3-6         |

## 2.4 COMPOSITION OF MIXTURE

### 2.4.1 Job-Mix Formula (JMF)

#### 2.4.1.1 Develop the JMF

Provide an asphalt mix composed of a mixture of well-graded aggregate, mineral filler if required, and asphalt binder. Size the aggregate fractions, handle in separate size groups, and combine in such proportions that the resulting mixture meets the grading requirements of Table 2. Submit proposed JMF; do not produce hot-mix asphalt for payment until a JMF has been approved. Design the hot-mix asphalt in accordance with Marshall or Superpave procedures and the criteria shown in Table 3. Use the hand-held hammer to compact the specimens for Marshall mix design. Design Superpave mixes with the number of gyrations specified in Table 3, unless the DOT option is chosen. If the Tensile Strength Ratio (TSR) of the composite mixture, as determined by ASTM D4867/D4867M, is less than 75, reject the aggregates or treat the asphalt mixture with an approved anti-stripping agent. Add the amount of anti-stripping agent sufficient to produce a TSR of not less than 75. Provide an antistrip agent, if required, at no additional cost. Provide sufficient materials to produce 200 pound of blended mixture for verification of mix design at least 14

days prior to construction of test section.

#### 2.4.1.2 Option

A currently used DOT Superpave hot mix may be used in lieu of developing a Marshall hot mix design as described herein. Design the Superpave volumetric mix in accordance with AI MS-2 and ASTM D6925. Provide a nominal maximum aggregate size (NMAS) of 3/4 inch. Other DOT hot mix design methods may be suitable, as approved.

#### 2.4.2 JMF Requirements

Submit in writing the job mix formula for approval at least 10 days prior to the start of the test section including as a minimum:

- a. Percent passing each sieve size.
- b. Percent of asphalt binder.
- c. Percent of each aggregate and mineral filler to be used.
- d. Asphalt performance grade viscosity grade.
- e. Number of blows of hand-held hammer per side of molded specimen. (NA for Superpave)
- f. Number of gyrations of Superpave gyratory compactor, (NA for Marshall mix design)
- g. Laboratory mixing temperature.
- h. Lab compaction temperature.
- i. Temperature-viscosity relationship of the asphalt cement.
- j. Plot of the combined gradation on the 0.45 power gradation chart, stating the nominal maximum size.
- k. Graphical plots of stability (NA for Superpave), flow (NA for Superpave), air voids, voids in the mineral aggregate, and unit weight versus asphalt content as shown in AI MS-2.
- l. Specific gravity and absorption of each aggregate.
- m. Percent natural sand.
- n. Percent particles with 2 or more fractured faces (in coarse aggregate).
- o. Fine aggregate angularity.
- p. Percent flat or elongated particles (in coarse aggregate).
- q. Tensile Strength Ratio(TSR).
- r. Antistrip agent (if required) and amount.
- s. List of all modifiers and amount.

- t. Correlation of hand-held hammer with mechanical hammer (NA for Superpave).
- u. Percentage and properties (asphalt content, binder properties, and aggregate properties) of reclaimed asphalt pavement (RAP) in accordance with paragraph RECYCLED HOT-MIX ASPHALT, if RAP is used.

| Table 3. Mix Design Criteria                                                                                           |                           |                           |
|------------------------------------------------------------------------------------------------------------------------|---------------------------|---------------------------|
| Test Property                                                                                                          | 50 Blows or Mix Gyrations | 75 Blows or Mix Gyrations |
| Stability, pounds, minimum (NA for Superpave)                                                                          | *1350                     | *1800                     |
| Flow, 0.01 inch, (NA for Superpave)                                                                                    | 8-18                      | 8-16                      |
| Air voids, percent                                                                                                     | 3-5                       | 3-5                       |
| Percent Voids in mineral aggregate (VMA), (minimum)                                                                    |                           |                           |
| Gradation 1                                                                                                            | 14.0                      | 14.0                      |
| Gradation 2                                                                                                            | 15.0                      | 15.0                      |
| Gradation 3                                                                                                            | 16.0                      | 16.0                      |
| TSR, minimum percent                                                                                                   | 75                        | 75                        |
| * This is a minimum requirement.                                                                                       |                           |                           |
| ** Calculate VMA in accordance with AI MS-2, based on ASTM C127 and ASTM C128 bulk specific gravity for the aggregate. |                           |                           |

#### 2.4.2.1 Adjustment to JMF

The JMF for each mixture is in effect until a new formula is approved in writing. Should a change in sources of any materials be made, perform a new mix design and obtain approval before the new material is used. Make minor adjustments within the specification limits to the JMF to optimize mix volumetric properties. Adjustments to the original JMF are limited to plus or minus 4 percent on the No. 4 and coarser sieves; plus or minus 3 percent on the No. 8 to No. 50 sieves; and plus or minus 1 percent on the No. 100 sieve. Adjustments to the JMF are limited to plus or minus 1.0 percent on the No. 200 sieve. Asphalt content adjustments are limited to plus or minus 0.40 from the original JMF. If adjustments are needed that exceed these limits, develop a new mix design.

#### 2.5 RECYCLED ASPHALT PAVEMENT

RAP is not permitted. 2.6 EQUIPMENT, TOOLS, AND MACHINES

#### 2.6.1 Bituminous Plant

Provide a bituminous plant of such capacity to produce the quantities of bituminous mixtures required for the project within the completion time of the contract. Provide hauling equipment, paving machines, rollers, miscellaneous equipment, and tools in sufficient numbers and capacity and in proper working condition to place the bituminous paving mixtures at a rate equal to the plant output. Provide a sufficient number of adequately trained personnel during paving operations to produce a pavement meeting the requirements in this specification.

#### 2.6.2 Mixing Plants

Provide mixing plants in accordance with AASHTO M 156 which are automatic or semiautomatic controlled, commercially manufactured units designed, coordinated, and operated to consistently produce a mixture within the job-mix formula (JMF). Prequalify drum or batch mixers at the production rate to be used during actual mix production. The prequalification tests include extraction in accordance with ASTM D2172/D2172M and recovery of the asphalt binder in accordance with ASTM D1856.

#### 2.6.3 Asphalt Paver

Provide asphalt pavers which are self-propelled, with an activated screed, heated as necessary, and capable of spreading and finishing courses of hot-mix asphalt which will meet the specified thickness, smoothness, and grade, with sufficient power to propel itself and the hauling equipment without adversely affecting the finished surface. Provide a receiving hopper of sufficient capacity to permit a uniform spreading operation and equipped with a distribution system to place the mixture uniformly in front of the screed without segregation and produce a finished surface of the required evenness and texture without tearing, shoving, or gouging the mixture. If screed extensions are used to increase the paving width, provide auger extensions to distribute the hot mix along the additional screed length. Equip the paver with a control system capable of automatically maintaining the specified screed elevation. Automatically actuate the control system from either a reference line and/or through a system of mechanical sensors or sensor-directed mechanisms or devices which will maintain the paver screed at a predetermined transverse slope and at the proper elevation to obtain the required surface.

#### 2.6.4 Hauling Equipment

Provide trucks for hauling hot-mix asphalt having tight, clean, and smooth metal beds. To prevent the mixture from adhering to them, lightly coat the truck beds with a release agent specifically designed for use with hot mix asphalt. Provide each truck with a suitable cover to protect the mixture from adverse weather. When necessary to maintain the mixture at the specified temperature, insulate or heat truck beds and securely fasten covers (tarps).

#### 2.6.5 Rollers

Provide the number, type, and weight of rollers sufficient to compact the mixture to the required density while it is still in a workable condition. Do not use equipment which causes excessive crushing of the aggregate or displacement of the asphalt mixture.

### 2.6.6 Straightedge

Furnish and maintain at the site, in good condition, one 12 foot straightedge for each bituminous paver for use in testing the finished surface. Construct straightedges of aluminum with blades of box or box-girder cross section and a flat bottom reinforced to insure rigidity and accuracy. Provide handles to facilitate movement on pavement.

## PART 3 EXECUTION

### 3.1 CONDITIONING OF UNDERLYING COURSE

Prior to placing the bituminous base course, clean the underlying surface of foreign or objectionable matter. Apply a prime coat in accordance with 32 12 13 BITUMINOUS TACK AND PRIME COATS.

### 3.2 MIXING

#### 3.2.1 Preparation of Mineral Aggregates

Place and maintain each aggregate stockpile in such a manner to prevent segregation. Regulate rates of feed of aggregates so that the moisture content and temperature of aggregates will be within tolerances specified herein. Provide dry storage for mineral filler.

#### 3.2.2 Preparation of Bituminous Mixtures

Convey aggregates, mineral filler, and bitumen into the mixer in proportionate quantities required to meet the JMF. Set the mixing time as required to obtain a uniform coating of the aggregate with the bituminous material. Limit the temperature of bitumen at time of mixing not to exceed 300 degrees F. Maintain the temperature of aggregate and mineral filler in the mixer within the range of 293 to 329 degrees F when bitumen is added. Overheated and carbonized mixtures or mixtures that foam will be rejected.

#### 3.2.3 Water Content of Aggregates

Perform drying operations to reduce the water content of mixture to less than 0.75 percent. Conduct the water content test in accordance with ASTM D2216. If the water content is determined on individual hot bin samples, calculate the water content as a weighted average based on composition of blend.

#### 3.2.4 Storage of Bituminous Paving Mixture

Store the mixture according to the requirements of AASHTO M 156. Empty uninsulated surge bins at the end of each working day. If excessive heat loss, segregation, or oxidation of the stored asphalt mixture is observed, discontinue the use of the surge bin.

### 3.3 TRANSPORTATION OF BITUMINOUS MIXTURE

Transport the bituminous mixture from the paving plant to the site in trucks having tight, clean, smooth beds lightly coated with an approved release agent to prevent adhesion of mixture to truck bodies. Drain excessive release agent prior to loading. Cover each load with canvas or other approved material of ample size to protect mixture from weather and

prevent loss of heat. Reject loads that have crusts of cold, unworkable material or have become wet by rain. Do not haul over freshly placed material.

### 3.4 PLACING

Do not place bituminous mixtures without ample time to complete placement and compaction during daylight hours, unless artificial lighting is provided.

#### 3.4.1 Tack Coat

Spray contact surfaces of previously constructed pavement, curbs, manholes, and similar structures with a tack coat conforming to the requirements of Section 32 12 13 BITUMINOUS TACK AND PRIME COATS.

#### 3.4.2 Offsetting Joints in Bituminous Base Course

Place the bituminous base course so that longitudinal joints are offset from joints in the underlying course by at least 1 foot. Offset transverse joints by at least 2 feet from transverse joints in the underlying course.

#### 3.4.3 Use of Laydown Machine

Reject mixtures having temperatures less than 250 degrees F when delivered to the laydown machine. Adjust the laydown machine and regulate the speed so that the surface of the course being laid will be smooth and continuous without tears and pulls, and of such depth that, when compacted, the surface conforms to the cross section, grade, and contour indicated. Begin placement of the mixture along the centerline of a crowned section or on the high side of areas with a one-way slope. Place the mixture as nearly continuous as possible, and adjust the speed of placing to permit proper compaction. When segregation occurs in the mixture during placing, suspend the laydown operation until the cause is determined and corrected. Correct irregularities in alignment of the course left by the laydown machine by trimming directly behind machine. Immediately after trimming, thoroughly compact the edges of the course by tamping laterally with a lute. Do not permit distortion of the course during tamping.

#### 3.4.4 Placing Strips Succeeding Initial Strips

In placing each succeeding strip after the initial strip has been spread and compacted as specified below, overlap the screed of the laydown machine 1/2 to 1 inch over the previously placed strip and sufficiently high so that compaction will produce a smooth, dense joint. Use a lute to push back the mixture placed on the edge of the previously placed strip to the edge of the strip being placed. Do not broadcast material onto the mat. Remove and waste excess mixture.

#### 3.4.5 Hand Spreading in Lieu of Machine Spreading

In areas where the use of machine spreading is impractical, spread the mixture by hand. Prevent segregation during spreading. Do not broadcast material onto the mat. Remove and waste excess mixture. Maintain grade and smoothness tolerances presented in Table 1.

### 3.5 COMPACTION OF MIXTURE

Begin compaction as soon after placing as the mixture will bear roller without undue displacement. Do not permit delays in compacting the freshly placed mixture. After the initial rolling, perform preliminary tests of the crown, grade, and smoothness. Correct deficiencies so that the finished course will conform to requirements for the grade and smoothness specified in subpart: ACCEPTANCE. After meeting crown, grade, and smoothness requirements, continue rolling until a mat density of at least 92 percent of the theoretical maximum density (TMD) determined in accordance with ASTM D2041/D2041M is obtained. Roll the joints until until a joint density of at least 90.5 percent of the theoretical maximum density (TMD) determined in accordance with ASTM D2041/D2041M is obtained. Thoroughly compact areas inaccessible to rollers with hot hand tampers.

#### 3.5.1 Correcting Deficient Areas

Remove mixtures that become contaminated or are defective. Do not permit skin patching of an area that has been rolled. Cut holes the full thickness of the base course so that the sides are perpendicular and parallel to the direction of traffic and the edges are vertical. Spray sides with tack coat conforming to requirements of Section 32 12 13 BITUMINOUS TACK AND PRIME COATS. Place hot mix asphalt in the holes in sufficient quantity so that the finished surface will conform to grade, smoothness, and density requirements.

### 3.6 JOINTS

#### 3.6.1 General

Carefully construct joints between old and new pavements or between successive day's work or joints that have become cold to establish a continuous bond between old and new sections of the course. Construct joints having the same texture, density, and smoothness as other sections of the course. Clean contact surfaces of previously constructed pavements that have become coated with dust, sand, or other objectionable material by brushing or cut back with approved power saw, as approved. Spray the surface against which new material is placed with a thin, uniform coat of tack coat conforming to requirements of Section 32 12 13 BITUMINOUS TACK AND PRIME COATS. Apply the material far enough in advance of placement of the fresh mixture to insure adequate curing. Take care to prevent damage or contamination of sprayed surface.

#### 3.6.2 Transverse Joints

Pass the roller over the unprotected end of freshly placed mixture only when placing of the course is discontinued or when delivery of the mixture is interrupted to the extent that the unrolled material may become cold. In all cases, cut back the edge of the previously placed course a minimum of 2 inches to expose an even, straight, vertical surface for the full thickness of the course. In continuing placement of the strip, position the mechanical spreader on the transverse joint so that sufficient hot mixture will be spread to obtain a joint after rolling that conforms to the required density and smoothness specified herein.

#### 3.6.3 Longitudinal Joints

Cut back edges of a previously placed strip that have cooled or are

irregular, honeycombed, poorly compacted, damaged, or otherwise defective. In all cases, cut back the edge of the previously placed course a minimum of 2 inches to expose an even, straight, vertical surface for the full thickness of the course.

### 3.7 EDGES OF PAVEMENT

Neatly trim outside edges adjacent to shoulders.

### 3.8 QUALITY CONTROL

Perform tests in sufficient numbers and at the locations and times directed to ensure that materials, mixtures and compaction meet specified requirements. Obtain samples of finished pavement, including samples that span the longitudinal joint. Sample bituminous materials during construction when shipments of bituminous materials are received or when necessary to assure that some condition of handling or storage has not been detrimental to the bituminous material.

#### 3.8.1 Sampling

Obtain plant mix and in-place samples on a lot and subplot basis. Each full day's production or a maximum of 1000 tons is considered a lot. Divide the lot into four (4) equal sublots and obtain random samples in accordance with ASTM D3665 within each subplot. Obtain plant mix samples from the haul truck or from behind the paver. Test for grade and smoothness on a total lot basis.

#### 3.8.2 In-Place Density

Take one random core (4 inches or larger in diameter) from the mat (interior of the lane) of each subplot, and one random core from the joint (immediately over joint) of each subplot, with each random core the full thickness of the layer being placed. When the random core is less than 1 inch thick, do not include in the analysis. In this case, take another random core. After air drying to a constant weight, determine the density of each core in accordance with ASTM D2726/D2726M. Determine percent compaction using the TMD. Evaluate for acceptance in accordance with subpart: ACCEPTANCE. Remove and replace unacceptable lots.

#### 3.8.3 Laboratory Air Voids and Theoretical Maximum Density

Calculate laboratory air voids by determining the bulk density of each lab compacted specimen using the laboratory-prepared, thoroughly dry method of ASTM D2726/D2726M and determining the theoretical maximum density of each subplot sample using ASTM D2041/D2041M. Use the latest theoretical maximum density value to calculate the laboratory air voids for each subplot. Evaluate for acceptance in accordance with subpart: ACCEPTANCE. Complete and report all laboratory air void tests within 24 hours after completion of construction of each lot.

#### 3.8.4 Plan Grade

Provide finished surfaces conforming, within tolerances specified, to the lines, grades, and cross sections indicated. Do not permit finished surfaces to vary more than the tolerances provided in subpart: ACCEPTANCE from the plan gradeline and elevation established and approved at the site. Maintain finished surfaces flush with finished surfaces of abutting pavements. Do not permit deviations from the plan gradeline and elevation



in areas of pavements where closer conformance with plan grade and elevation is required for the proper functioning of drainage and other appurtenant structures involved.

#### 3.8.5 Surface Smoothness

Provide finished surfaces not deviating from the testing edge of a straightedge more than the tolerances of subpart: ACCEPTANCE in any direction.

#### 3.8.6 Temperatures

Check temperatures at least four times per lot, at necessary locations, to determine the temperature at the dryer, the asphalt cement in the storage tank, the asphalt mixture at the plant, and the asphalt mixture at the job site.

#### 3.9 PROTECTION OF PAVEMENT

After final rolling of the pavement, do not permit vehicular traffic of any kind until the pavement has cooled to ambient temperature.

-- End of Section --

SECTION 32 12 13

BITUMINOUS TACK AND PRIME COATS  
05/17

PART 1 GENERAL

1.1 UNIT PRICES

1.1.1 Measurement

The bituminous material paid for will be the measured quantities of residual bituminous material used in the accepted work, provided that the measured quantities are not 10 percent over the specified quantities. Any amount of bituminous material more than 10 percent over the specified quantity will be deducted from the measured quantities. Express measured quantities in 2000 pound tons at 60 degrees F. Correct volumes measured at temperatures other than 60 degrees F in accordance with ASTM D1250 using a coefficient of expansion of 0.00025 per degree F for asphalt emulsion.

1.1.2 Payment

The quantities of bituminous material, determined as specified above, will be paid for at the respective contract unit prices. Payment will constitute full compensation for all operations necessary to complete the work as specified herein.

1.1.3 Waybills and Delivery Tickets

Submit waybills and delivery tickets, during progress of the work. Before the final statement is allowed, file with the Contracting Officer certified waybills and certified delivery tickets for all bituminous materials used in the construction of the pavement covered by the contract. These submittals are required for Unit Pricing bid only. Do not remove bituminous material from storage until the initial outage and temperature measurements have been taken. The delivery or storage units will not be released until the final outage has been taken.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS  
(AASHTO)

AASHTO T 102 (2009; R 2013) Standard Method of Test for  
Spot Test of Asphaltic Materials

ASTM INTERNATIONAL (ASTM)

ASTM D140/D140M (2016) Standard Practice for Sampling  
Asphalt Materials

ASTM D946/D946M (2020) Standard Specification for  
Penetration-Graded Asphalt Cement for Use

in Pavement Construction

|                   |                                                                                                                                                                                                                          |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D977         | (2019a; E 2019) Standard Specification for Emulsified Asphalt                                                                                                                                                            |
| ASTM D1250        | (2019; E 2020) Standard Guide for Use of the Joint API and ASTM Adjunct for Temperature and Pressure Volume Correction Factors for Generalized Crude Oils, Refined Products, and Lubricating Oils: API MPMS Chapter 11.1 |
| ASTM D2026/D2026M | (2015) Cutback Asphalt (Slow-Curing Type)                                                                                                                                                                                |
| ASTM D2027/D2027M | (2019) Cutback Asphalt (Medium-Curing Type)                                                                                                                                                                              |
| ASTM D2028/D2028M | (2015) Cutback Asphalt (Rapid-Curing Type)                                                                                                                                                                               |
| ASTM D2995        | (1999; R 2009) Determining Application Rate of Bituminous Distributors                                                                                                                                                   |

### 1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval.. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Waybills and Delivery Tickets

Local/Regional Materials

SD-06 Test Reports

Sampling and Testing

### 1.4 QUALITY ASSURANCE

Certificates of compliance for asphalt materials delivered will be obtained and checked to ensure that specification requirements are met. Quantities of applied material will be determined. Payment will be for amount of residual asphalt applied. Tack coat materials will not be diluted. Prime coat materials when emulsions are used can be diluted on site with potable water up to 1 part emulsion to 1 part water.

### 1.5 DELIVERY, STORAGE, AND HANDLING

Inspect the materials delivered to the site for contamination and damage. Unload and store the materials with a minimum of handling.

### 1.6 EQUIPMENT, TOOLS AND MACHINES

#### 1.6.1 General Requirements

Equipment, tools and machines used in the work are subject to approval.

Maintain in a satisfactory working condition at all times. Calibrate equipment such as asphalt distributors, scales, batching equipment, spreaders and similar equipment within 12 months of their use. If the calibration expires during project, recalibrate the equipment before work can continue.

#### 1.6.2 Bituminous Distributor

Provide a self propelled distributor with pneumatic tires of such size and number to prevent rutting, shoving or otherwise damaging the surface being sprayed. Calibrate the distributor in accordance with ASTM D2995. Design and equip the distributor to spray the bituminous material in a uniform coverage at the specified temperature, at readily determined and controlled total liquid rates from 0.03 to 1.0 gallons per square yard, with a pressure range of 25 to 75 psi and with an allowable variation from the specified rate of not more than plus or minus 5 percent, and at variable widths. Include with the distributor equipment a separate power unit for the bitumen pump, full-circulation spray bars, tachometer, pressure gauges, volume-measuring devices, adequate heaters for heating of materials to the proper application temperature, a thermometer for reading the temperature of tank contents, and a hand hose attachment suitable for applying bituminous material manually to areas inaccessible to the distributor. The distributor will be capable of circulating and agitating the bituminous material during the heating process.

#### 1.6.3 Heating Equipment for Storage Tanks

Use steam, electric, or hot oil heaters for heating the bituminous material. Provide steam heaters consisting of steam coils and equipment for producing steam, so designed that the steam cannot come in contact with the bituminous material. Fix an armored thermometer to the tank with a temperature range from 40 to 400 degrees F so that the temperature of the bituminous material may be determined at all times.

#### 1.6.4 Power Brooms and Power Blowers

Use power brooms and power blowers suitable for cleaning the surfaces to which the bituminous coat is to be applied.

### 1.7 ENVIRONMENTAL REQUIREMENTS

Apply bituminous coat only when the surface to receive the bituminous coat is dry. A limited amount of moisture (approximately 0.03 gallon/square yard) can be sprayed on the surface of unbound material when prime coat is used to improve coverage and penetration of asphalt material. Apply bituminous coat only when the atmospheric temperature in the shade is 50 degrees F or above and when the temperature has not been below 35 degrees F for the 12 hours prior to application, unless otherwise directed.

## PART 2 PRODUCTS

### 2.1 PRIME COAT

Provide asphalt conforming to one of the following grades:

#### 2.1.1 Cutback Asphalt

Provide cutback asphalt conforming to ASTM D2026/D2026M, Grade SC-70  
ASTM D2027/D2027M, Grade MC-30.

### 2.1.2 Emulsified Asphalt

Provide emulsified asphalt conforming to ASTM D977, Type SS-1. Asphalt emulsion can be diluted up to 1 part water to 1 part emulsion for prime coat use. Do not dilute asphalt emulsion for tack coat use.

## 2.2 TACK COAT

### 2.2.1 Asphalt Cement

Provide asphalt cement conforming to ASTM D946/D946M.

### 2.2.2 Cutback Asphalt

Provide cutback asphalt conforming to ASTM D2028/D2028M.

### 2.2.3 Emulsified Asphalt

Provide emulsified asphalt conforming to ASTM D977, Type SS-1. For prime coats the emulsified asphalt can be diluted with up to 1 part emulsion to 1 part water. No dilution is allowed for tack coat applications. The base asphalt used to manufacture the emulsion is required to show a negative spot when tested in accordance with AASHTO T 102 using standard naphtha.

### 2.2.4 Local/Regional Materials

Use Local/Regional Materials or products extracted, harvested, or recovered, as well as manufactured, within a 500 mile radius from the project site, if available from a minimum of three sources.

## PART 3 EXECUTION

### 3.1 PREPARATION OF SURFACE

Immediately before applying the bituminous coat, remove all loose material, dirt, clay, or other objectionable material from the surface to be treated by means of a power broom or blower supplemented with hand brooms. Apply treatment only when the surface is dry and clean.

### 3.2 APPLICATION RATE

The exact quantities within the range specified, which may be varied to suit field conditions, will be determined by the Contracting Officer.

#### 3.2.1 Tack Coat

Apply bituminous material for the tack coat in quantities of not less than 0.03 gallons nor more than 0.10 gallons per square yard of residual asphalt onto the pavement surface as approved by the Contracting Officer. Do not dilute asphalt emulsion when used as a tack coat.

#### 3.2.2 Prime Coat

Apply bituminous material for the prime coat in quantities of not less than 0.05 gallons nor more than 0.12 gallons per square yard of residual asphalt for asphalt emulsion up to a 1 to 1 dilution rate or for residual asphalt for cutback asphalt.

### 3.3 APPLICATION TEMPERATURE

#### 3.3.1 Viscosity Relationship

Apply asphalt at a temperature that will provide a viscosity between 10 and 60 seconds, Saybolt Furol, or between 20 and 120 centistokes, kinematic. Furnish the temperature viscosity relation to the Contracting Officer.

#### 3.3.2 Temperature Ranges

The viscosity requirements determine the application temperature to be used. The following is a normal range of application temperatures:

| Cutback Asphalts          |                   |
|---------------------------|-------------------|
| MC-30                     | 85-190 degrees F  |
| SC-70, MC-70,<br>RC-70    | 120-225 degrees F |
| SC-250, MC-250,<br>RC-250 | 165-270 degrees F |
| Asphalt Emulsion          |                   |
| All Grades                | 70-160 degrees F  |
| Asphalt Cement            |                   |
| All Grades                | 275-350 degrees F |

Some of these temperatures for rapid cure cutbacks are above the flash point of the material and care should be taken in their heating.

### 3.4 APPLICATION

#### 3.4.1 General

Following preparation and subsequent inspection of the surface, apply the bituminous prime or tack coat with the bituminous distributor at the specified rate with uniform distribution over the surface to be treated. Properly treat all areas and spots, not capable of being sprayed with the distributor, with the hand spray. Until the succeeding layer of pavement is placed, maintain the surface by protecting the surface against damage and by repairing deficient areas at no additional cost to the Government. If required, spread clean dry sand to effectively blot up any excess bituminous material. No smoking, fires, or flames other than those from the heaters that are a part of the equipment are permitted within 25 feet of heating, distributing, and transferring operations of cutback materials. Prevent all traffic, except for paving equipment used in constructing the surfacing, from using the underlying material, whether primed or not, until the surfacing is completed. The bituminous coat requirements are described herein.

#### 3.4.2 Prime Coat

Apply a prime coat at locations shown on the Drawings. The prime coat is

required if it will be at least 7 days before the asphalt mixture is constructed on the underlying (base course, etc.) compacted material. The type of liquid asphalt and application rate will be as specified herein. Protect the underlying layer from any damage (water, traffic, etc.) until the surfacing is placed. If the Contractor places the surfacing within seven days, the choice of protection measures or actions to be taken is at the Contractor's option. Repair (recompact or replace) damage to the underlying material caused by lack of, or inadequate, protection by approved methods at no additional cost to the Government. If the Contractor opts to use the prime coat, apply as soon as possible after consolidation of the underlying material. Apply the bituminous material uniformly over the surface to be treated at a pressure range of 25 to 75 psi; the rate will be as specified above in paragraph APPLICATION RATE. To obtain uniform application of the prime coat on the surface treated at the junction of previous and subsequent applications, spread building paper on the surface for a sufficient distance back from the ends of each application to start and stop the prime coat on the paper and to ensure that all sprayers will operate at full force on the surface to be treated. Immediately after application remove and destroy the building paper.

#### 3.4.3 Tack Coat

Apply tack coat at the locations shown on the drawings. A tack coat should be applied to every bound surface (asphalt or concrete pavement) that is being overlaid with asphalt mixture and at transverse and longitudinal joints. Apply the tack coat when the surface to be treated is clean and dry. Immediately following the preparation of the surface for treatment, apply the bituminous material by means of the bituminous distributor, within the limits of temperature specified herein and at a rate as specified above in paragraph APPLICATION RATE. Apply the bituminous material so that uniform distribution is obtained over the entire surface to be treated. Treat lightly coated areas and spots missed by the distributor by spraying with a hand wand or using other approved method. Following the application of bituminous material, allow the surface to cure without being disturbed for period of time necessary to permit setting of the tack coat. Apply the bituminous tack coat only as far in advance of the placing of the overlying layer as required for that day's operation. Maintain and protect the treated surface from damage until the succeeding course of pavement is placed.

#### 3.5 CURING PERIOD

Following application of the bituminous material and prior to application of the succeeding layer of asphalt mixture allow the bituminous coat to cure and water or volatiles to evaporate prior to overlaying. Maintain the tacked surface in good condition until the succeeding layer of pavement is placed, by protecting the surface against damage and by repairing and recoating deficient areas. Allow the prime coat to cure without being disturbed for a period of at least 48 hours or longer, as may be necessary to attain penetration into the treated course. Furnish and spread enough sand to effectively blot up excess bituminous material.

#### 3.6 FIELD QUALITY CONTROL

Obtain certificates of compliance for all asphalt material delivered to the project. Obtain samples of the bituminous material under the supervision of the Contracting Officer. The sample may be retained and tested by the Government at no cost to the Contractor.

### 3.7 SAMPLING AND TESTING

Furnish certified copies of the manufacturer's test reports indicating temperature viscosity relationship for cutback asphalt or asphalt cement, compliance with applicable specified requirements, not less than 5 days before the material is required in the work.

#### 3.7.1 Sampling

Unless otherwise specified, sample bituminous material in accordance with ASTM D140/D140M.

#### 3.7.2 Calibration Test

Furnish all equipment, materials, and labor necessary to calibrate the bituminous distributor. Calibrate using the approved job material and prior to applying the bituminous coat material to the prepared surface. Calibrate the bituminous distributor in accordance with ASTM D2995.

#### 3.7.3 Trial Applications

Before applying the spray application of tack or prime coat, apply three lengths of at least 100 feet for the full width of the distributor bar to evaluate the amount of bituminous material that can be satisfactorily applied.

##### 3.7.3.1 Tack Coat Trial Application Rate

Unless otherwise authorized, apply the trial application rate of bituminous tack coat materials in the amount of 0.05 gallons per square yard. Make other trial applications using various amounts of material as may be deemed necessary.

##### 3.7.3.2 Prime Coat Trial Application Rate

Unless otherwise authorized, apply the trial application rate of bituminous materials in the amount of 0.15 gallon per square yard. Make other trial applications using various amounts of material as may be deemed necessary.

#### 3.7.4 Sampling and Testing During Construction

Perform quality control sampling and testing as required in paragraph FIELD QUALITY CONTROL.

### 3.8 TRAFFIC CONTROLS

Keep traffic off surfaces freshly treated with bituminous material. Provide sufficient warning signs and barricades so that traffic will not travel over freshly treated surfaces.

-- End of Section --



SECTION 32 31 13

CHAIN LINK FENCES AND GATES  
**11/21**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                           |
|-----------------|---------------------------------------------------------------------------------------------------------------------------|
| ASTM A392       | (2011; R 2022a) Standard Specification for Zinc-Coated Steel Chain-Link Fence Fabric                                      |
| ASTM A780/A780M | (2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings                          |
| ASTM A824       | (2001; R 2022) Standard Specification for Metallic-Coated Steel Marcellled Tension Wire for Use With Chain Link Fence     |
| ASTM C94/C94M   | (2023) Standard Specification for Ready-Mixed Concrete                                                                    |
| ASTM F567       | (2023) Standard Practice for Installation of Chain Link Fence                                                             |
| ASTM F626       | (2014; R 2023) Standard Specification for Fence Fittings                                                                  |
| ASTM F668       | (2017; R 2022) Standard Specification for Polyvinyl Chloride (PVC) and Other Polymer-Coated Steel Chain Link Fence Fabric |
| ASTM F934       | (2016; R 2022) Standard Specification for Standard Colors for Polymer-Coated Chain Link Fence Materials                   |

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

|               |                                                                                            |
|---------------|--------------------------------------------------------------------------------------------|
| FS RR-F-191/1 | (Rev F) Fencing, Wire and Post, Metal (Chain-Link Fence Fabric)                            |
| FS RR-F-191/2 | (Rev E) Fencing, Wire and Post, Metal (Chain-Link Fence Gates)                             |
| FS RR-F-191/3 | (2023; Rev F) Fencing, Wire and Post, Metal (Chain-Link Fence Posts, Top Rails and Braces) |
| FS RR-F-191/4 | (Rev F) Fencing, Wire and Post, Metal (Chain-Link Fence Accessories)                       |

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

- Fence Assembly; G
- Location of Gate, Corner, End, and Pull Posts; G
- Gate Assembly; G
- Gate Hardware and Accessories; G
- Erection/Installation Drawings; G

### SD-03 Product Data

- Fence Assembly; G
- Gate Assembly; G
- Gate Hardware and Accessories; G
- Zinc Coating; G
- PVC Coating; G
- Aluminum Alloy Coating; G
- Fabric; G
- Stretcher Bars; G
- Concrete; G

### SD-04 Samples

- Fabric; G
- Posts; G
- Braces; G
- Line Posts; G
- Sleeves; G
- Top Rail; G
- Bottom Rail; G
- Tension Wire; G

Stretcher Bars; G

Gate Posts; G

Gate Hardware and Accessories; G

Padlocks; G

Wire Ties; G

SD-07 Certificates

Certificates of Compliance

SD-08 Manufacturer's Instructions

Fence Assembly

Gate Assembly

Hardware Assembly

Accessories

### 1.3 QUALITY CONTROL

#### 1.3.1 Certificates of Compliance

Submit certificates of compliance in accordance with the applicable reference standards and descriptions of this section for the following:

- a. Zinc coating
- b. Fabric
- c. Stretcher bars
- d. Gate hardware and accessories
- e. Concrete

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Deliver materials to site in an undamaged condition. Store materials off the ground to provide protection against oxidation caused by ground contact.

## PART 2 PRODUCTS

### 2.1 SYSTEM DESCRIPTION

Submit reports of listing chain-link fencing and accessories regarding weight in ounces for zinc coating, and chemical composition and thickness of aluminum alloy coating.

Submit manufacturer's catalog data for complete fence assembly, gate assembly, hardware assembly and accessories.

## 2.2 FENCES

### 2.2.1 Fabric

FS RR-F-191/1; Type II, aluminum-coated steel, 9 gauge, conforming to ASTM A491, 9-gauge core wire size, conforming to ASTM F668. Color to be black complying with ASTM F934.I, zinc-coated steel, 9 gauge conforming to ASTM A392, with 2.0 ounces per square foot of coated surface. Provide selva ge knuckled at both selvages.

Height of fabric, as indicated.

Provide fabric consisting of wires woven into a 1-3/4 inch diamond mesh. Provide one-piece fabric widths for fence heights up to 12 feet.

#### 2.2.1.1 Top and Bottom Selvages

Provide knuckled selvages at top and bottom for fabric with 2 inch mesh and up to 60 inches high, and if over 60 inches high, provide twisted and barbed top selva ge and knuckled bottom selva ge.

Knuckle top and bottom selvages for 1-3/4 inch and 1 inch mesh fabric.

### 2.2.2 Posts

#### 2.2.2.1 Metal

##### 2.2.2.1.1 Line Posts

Provide line posts complying with FS RR-F-191/3. Provide Class 1, steel pipe, Grade A Class 3, formed steel sections, in minimum sizes listed in FS RR-F-191/3 for each class and grade.

##### 2.2.2.1.2 End, Corner and Pull Posts

Provide end, corner, and pull posts in minimum sizes listed in FS RR-F-191/3 for each class and grade. Provide Class 1, steel pipe, Grade A Regular Strength.

##### 2.2.2.1.3 Steel Pipe, Class 1, Grade B Test Requirements

Steel pipe, Class 1, Grade B meeting the following performance criteria when subjected to salt spray testing in accordance with ASTM B117:

- a. Exterior: 1,000 hours with maximum 5 percent red rust.
- b. Interior: 650 hours with maximum 5 percent red rust.

#### 2.2.2.2 Post Tops

Provide steel, wrought iron, or malleable ironaluminum tops with PVC coating of minimum thickness of 0.006 in and designed as a weathertight closure cap. Post top to have finish and coating matching rails and posts. Steel type to be pressed steel galvanized after fabrication having a minimum zinc coating of 1.20 ounces per square foot. Provide resin reinforced tops for resin reinforced posts. Provide one cap for each post, unless equal protection is provided by a combination post-cap and wire supporting arm. Provide caps with an opening to permit through passage of the top rail.

### 2.2.3 Braces and Rails

Class 1, steel pipe, Grade A Regular Strength, in minimum sizes listed in FS RR-F-191/3 for each class and grade.

#### 2.2.3.1 Top Rail

Provide top rail conforming to minimum sizes specified in FS RR-F-191/3 for each class and grade unless members are to be oversized. Provide expansion couplings 6 inches long at each joint in top rails.

#### 2.2.3.2 Center Rails Between Line Posts

Provide center rail conforming to minimum sizes specified in FS RR-F-191/3 for each class and grade unless members are to be oversized.

#### 2.2.3.3 Bottom Rail

Provide bottom rail conforming to minimum sizes specified in FS RR-F-191/3 for each class and grade unless members are to be oversized.

#### 2.2.3.4 Post-Brace Assembly

Provide bracing conforming to minimum sizes specified in FS RR-F-191/3 for each class and grade, and 3/8 inch adjustable truss rods and turnbuckles.

### 2.2.4 Wire Ties

Provide 9-gauge wire for tying fabric to line posts, spaced 12 inches on center. For tying fabric to rails and braces, space wire ties 24 inches on center. For tying fabric to tension wire, space 0.105-inch hog rings 24 inches on center. Manufacturer's standard procedure will be accepted if of equal strength and durability.

FS RR-F-191/4. Provide wire ties constructed of the same material and coating as the fencing fabric.

### 2.2.5 Sleeves

Provide sleeves for setting into concrete construction of the same material as post sections, sized 1 inch greater than the diameter or dimension of the post. Weld flat plates to each sleeve base to provide anchorage and prevent intrusion of concrete.

### 2.2.6 Stretcher Bars

Provide bars that have one-piece lengths equal to the full height of the fabric with a minimum cross section of 3/16 by 3/4 inch, in accordance with ASTM F626.

### 2.2.7 Stretcher Bar Bands

Provide bar bands for securing stretcher bars to posts that are steel, wrought iron, or malleable iron spaced not over 15 inches on center. Bands may also be used in conjunction with special fittings for securing rails to posts. Provide bands with projecting edges chamfered or eased.

#### 2.2.8 Tension Wire

Provide metallic coated steel marcelled tension wire, ( No. 7-gauge) complying with ASTM A824. Provide aluminum-coated (aluminized) steel wire with coating that weighs not less than 0.4 ounces per square foot. Provide zinc-coated steel wire with zinc coating that weighs not less than 1.2 ounces per square foot.

#### 2.2.9 Miscellaneous Hardware

Provide miscellaneous hot-dip galvanized hardware as required.

### 2.3 GATES

FS RR-F-191/2; Type I, single swing III, single cantilever sliding, wheel sliding gate. Shape and size of gate frame as shown on drawings. Framing and bracing members, round of steel alloy. Steel member finish, zinc-coated. Provide gate frames and braces of minimum sizes listed in FS RR-F-191/3 for each Class and Grade, except that steel pipe frames are a minimum of 1.90 inches o.d., 0.120 inches minimum wall thickness and aluminum pipe frames and intermediate braces are 1.869 inches o.d. minimum, 0.940 lb/ft of length. Provide intermediate members as necessary for gate leaves more than 8 feet wide, to provide rigid construction, free from sag or twist. Provide truss rods or intermediate braces for gate leaves less than 8 feet wide.

#### 2.3.1 Gate Posts

Provide gate posts for supporting each gate leaf in minimum sizes listed in FS RR-F-191/3 for each material class and grade. Gate post material class, grade and finish to match other fence posts.

#### 2.3.2 Gate Fabric

Gate fabric, is as specified for fencing fabric. Attach gate fabric to gate frame in accordance with manufacturer's standards, except that welding is not permitted.

#### 2.3.3 Gate Frame

Provide gate frame assembly that is welded or assembled with special malleable or pressed-steel fittings and rivets to provide rigid connections. Install fabric with stretcher bars at vertical edges; stretcher bars may also be used at top and bottom edges. Attach stretcher bars and fabric to gate frames on all sides at intervals not exceeding 15 inches. Attach hardware with rivets or by other means which provides equal security against breakage or removal.

#### 2.3.4 Gate Bracing

Provide diagonal cross-bracing, consisting of 3/8 inch diameter adjustable-length truss rods on welded gate frames, where necessary to obtain frame rigidity without sag or twist. Provide nonwelded gate frames with diagonal bracing.

#### 2.3.5 Padlocks

Provide padlocks conforming to ASTM F883, with chain.

### 2.3.6 Gate Hardware and Accessories

Provide gate hardware and accessories that conforms to ASTM A392 and ASTM F626, and as specified. Coating for steel latches, stops, hinges, keepers, and accessories, is galvanized

- a. Provide forged steel hinges to suit gate size, non-lift-off type, offset to permit 180-degree opening. Provide hinge with stainless steel pin.
- b. Provide latch that permits accessibility and operation from either side of the gate regardless of the latching arrangement, and with a padlock eye provided as an integral part of the latch. Provide fork type gate latches.
- c. Provide stops and holders of malleable iron for vehicular gates. Provide stops that automatically engage the gate and hold it in the open position until manually released.
- e. Provide double gates with a cane bolt and ground-set keeper, with latch or locking device and padlock eye designed as an integral part.
- f. Provide manufacturer's standard heavy-duty track ball bearing hanger sheaves, overhead framing and supports, guides, stays, bracing, and accessories as required for easy operation of manual sliding gates.

### 2.4 MATERIALS

#### 2.4.1 Zinc Coating

Provide zinc-coated ferrous metal components and accessories that are factory coated after fabrication, except as otherwise specified.

For galvanizing field repairs, provide material that is cold-applied zinc-rich coating conforming to ASTM A780/A780M.

#### 2.4.2 Concrete

Provide concrete conforming to ASTM C94/C94M, and obtaining a minimum 28-day compressive strength of 3,000 psi.

#### 2.4.3 Grout

Provide grout of proportions one part portland cement to three parts clean, well-graded sand and a minimum amount of water to produce a workable mix.

### PART 3 EXECUTION

Submit manufacturer's erection/installation drawings and instructions that detail proper assembly and materials in the design for fence, gate, hardware and accessories.

Provide complete installation conforming to ASTM F567.

### 3.1 PREPARATION

Ensure final grading and established elevations are complete prior to commencing fence installation.

### 3.2 INSTALLATION

#### 3.2.1 Security

Install new chain link fencing, remove existing fencing, and perform related work to provide continuous security for facility. Schedule and fully coordinate work with Contracting Officer and cognizant Security Officer.

#### 3.2.2 Fence Installation

Install fence on prepared surfaces to line and grade indicated. Secure fastening and hinge hardware in place to fence framework by peening or welding. Allow for proper operation of components. Coat peened or welded areas with a repair coating matching original coating. Install fence in accordance with fence manufacturer's written installation instructions except as modified herein.

##### 3.2.2.1 Post Spacing

Provide line posts spaced equidistantly apart, not exceeding 10 feet on center. Provide gate posts spaced as necessary for size of gate openings. Do not exceed 500 feet on straight runs between braced posts. Provide corner or pull posts, with bracing in both directions, for changes in direction of 15 degrees or more, or for abrupt changes in grade. Submit drawings showing location of gate, corner, end, and pull posts.

##### 3.2.2.2 Top and Bottom Tension Wire

Install tension wires before installing chain-link fabric, and pull wires taut. Place top and bottom tension wires within 8 inches of respective fabric line.

#### 3.2.3 Excavation

Provide excavations for post footings which are drilled holes in concrete slab as indicated. Space footings for line posts 10 feet on center maximum and at closer intervals when indicated, with bottoms of the holes approximately 3 inches below the bottoms of the posts.

#### 3.2.4 Setting Posts

Remove loose and foreign materials from holes and moisten the soil prior to placing concrete. Provide tops of footings that are trowel finished and sloped or domed to shed water away from posts. Set hold-open devices, sleeves, and other accessories in concrete.

Keep exposed concrete moist for at least 7 calendar days after placement or cured with a membrane curing material, as approved. Grout all posts set into sleeved holes in concrete with an approved grouting material. Maintain vertical alignment of posts in concrete construction until concrete has set.



#### 3.2.4.1 Earth and Bedrock

Provide concrete bases of dimensions indicated on the manufactures installation drawings. Compact concrete to eliminate voids, and finish to a dome shape.

#### 3.2.4.2 Concrete Slabs and Walls

Set posts into zinc-coated sleeves, set in concrete slab or wall, to a minimum depth of 12 inches. Fill sleeve joint with nonshrink grout or other approved material. Set posts for support of removable fence sections into sleeves that provide a tight sliding joint and hold posts aligned and plumb without use of setting material.

#### 3.2.4.3 Bracing

Brace gate, corner, end, and pull posts to nearest post with a horizontal brace used as a compression member, placed at least 12 inches below top of fence, and a diagonal tension rod.

#### 3.2.4.4 Tolerances

Provide posts that are straight and plumb within a vertical tolerance of 1/4 inch after the fabric has been stretched. Provide fencing and gates that are true to line with no more than 1/2 inch deviation from the established centerline between line posts. Repair defects as directed.

#### 3.2.5 Concrete Strength

Provide concrete that has attained at least 75 percent of its minimum 28-day compressive strength, but in no case sooner than 7 calendar days after placement, before rails, tension wire, or fabric are installed. Do not stretch fabric and wires or hang gates until the concrete has attained its full design strength.

#### 3.2.6 Top Rails

Provide top rails that run continuously through post caps or extension arms, bending to radius for curved runs. Provide expansion couplings as recommended by the fencing manufacturer.

#### 3.2.7 Center Rails

Provide single piece center rails between posts set flush with posts on the fabric side, using special offset fittings where necessary.

#### 3.2.8 Brace Assembly

Provide bracing assemblies at end and gate posts and at both sides of corner and pull posts, with the horizontal brace located at midheight of the fabric.

Install brace assemblies so posts are plumb when the diagonal rod is under proper tension. Provide two complete brace assemblies at corner and pull posts where required for stiffness and as indicated.

### 3.2.9 Tension Wire Installation

Install tension wire by weaving them through the fabric and tying them to each post with not less than 7-gauge galvanized wire or by securing the wire to the fabric with 10-gauge ties or clips spaced 24 inches on center.

### 3.2.10 Fabric Installation

Provide fabric in single lengths between stretch bars with bottom barbs placed approximately 1-1/2 inches above the ground line. Pull fabric taut and tied to posts, rails, and tension wire with wire ties and bands.

Install fabric on the security side of fence, unless otherwise directed. Ensure fabric remains under tension after the pulling force is released.

### 3.2.11 Stretcher Bar Installation

Thread stretcher bars through or clamped to fabric 4 inches on center and secured to posts with metal bands spaced 15 inches on center.

### 3.2.12 Gate Installation

Install gates plumb, level, and secure, with full opening without interference. Install ground set items in concrete for anchorage as recommended by the fence manufacturer. Adjust hardware for smooth operation and lubricated where necessary.

### 3.2.13 Tie Wires

Provide tie wires that are U-shaped to the pipe diameters to which attached. Twist ends of tie wires not less than two full turns and bent so as not to present a hazard.

### 3.2.14 Fasteners

Install nuts for tension bands and hardware on the side of the fence opposite the fabric side. Peen ends of bolts to prevent removal of nuts.

### 3.2.15 Zinc-Coating Repair

Clean and repair galvanized surfaces damaged by welding or abrasion, and cut ends of fabric, or other cut sections with specified galvanizing repair material applied in strict conformance with the manufacturer's printed instructions.

### 3.2.16 Accessories Installation

#### 3.2.16.1 Post Caps

Install post caps as recommended by the manufacturer.

#### 3.2.16.2 Padlocks

Provide padlocks for gate openings and provide chains that are securely attached to gate or gate posts. Provide padlocks keyed alike, and provide two keys for each padlock.

### 3.2.17 Grounding

Ground fencing as specified.

Ground all fences crossed by overhead power lines in excess of 600 volts, and all electrical equipment attached to the fence. Ground fences on each side of all gates, at each corner, at the closest approach to each building located within 50 feet of the fence, and where the fence alignment changes more than 15 degrees. Grounding locations can not exceed 650 feet. Bond each gate panel with a flexible bond strap to its gate post. Ground fences crossed by power lines of 600 volts or more at or near the point of crossing and at distances not exceeding 150 feet on each side of crossing. Provide ground conductor consisting of No. 6 AWG solid copper wire. Provide solid copper rod grounding electrodes 3/4 inch by 10 foot long. Drive electrodes into the earth so that the top of the electrode is at least 6 inches below the grade. Where driving is impracticable, bury electrodes a minimum of 12 inches deep and radially from the fence, with top of the electrode not less than 2 feet or more than 8 feet from the fence. Clamp ground conductor to the fence and electrodes with bronze grounding clamps to create electrical continuity between fence posts, fence fabric, and ground rods. Total resistance of the fence to ground cannot exceed 25 ohms.

### 3.3 CLOSEOUT ACTIVITIES

Remove waste fencing materials and other debris from the work site.

Submit manufacturer's data indicating percentage of recycled material content in protective fence materials, including chain link fence, fabric, and gates to verify affirmative procurement compliance.

-- End of Section --

SECTION 33 11 13.28

HIGH DENSITY POLYETHYLENE PIPE

07/24

PART 1 GENERAL

1.1 WORK INCLUDED

This Section covers furnishing, handling, laying, joining and installation of HDPE piping, fittings and appurtenances.

The Contractor shall furnish and install the various pipelines and appurtenant work as indicated on the Contract Drawings and as specified herein, or as reasonably required to produce a complete, proper, and functional installation in accordance with the intent of these Contract Documents.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

|            |                                                        |
|------------|--------------------------------------------------------|
| ASME SB466 | Specification for Seamless Copper Nickel Pipe and Tube |
|------------|--------------------------------------------------------|

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|            |                                                        |
|------------|--------------------------------------------------------|
| ASME SB466 | Specification for Seamless Copper Nickel Pipe and Tube |
|------------|--------------------------------------------------------|

ASTM INTERNATIONAL (ASTM)

|           |                                                                        |
|-----------|------------------------------------------------------------------------|
| ASTM B466 | (2018) Standard Specification for Seamless Copper-Nickel Pipe and Tube |
|-----------|------------------------------------------------------------------------|

|            |                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D3261 | (2016) Standard Specification for Butt Heat Fusion Polyethylene (PE) Plastic Fittings for Polyethylene (PE) Plastic Pipe and Tubing |
|------------|-------------------------------------------------------------------------------------------------------------------------------------|

1.3 QUALITY ASSURANCE

All pipe and fittings shall be inspected and tested at the factory as required by the standard specifications to which the material is manufactured. The Contractor shall furnish in duplicate to the Engineer sworn certificates providing evidence of such tests.

The Government reserves the right to have any or all pipe, fittings, and special castings inspected and/or tested by an independent service at either the manufacturer's plant or elsewhere. Such inspection and/or tests shall be at the Owner's expense.

Deflections in horizontal alignment will not be permitted at joints without written consent of the Engineer. If approved, deflections shall

not exceed one-half the manufacturer's recommendation.

When requested by the Engineer, the Contractor shall ensure that a qualified representative of the manufacturer shall be present at the jobsite for the first day of pipe laying, to assure that proper procedures are followed.

The Engineer shall be notified in advance when the location of an existing pipeline conflicts with the proposed location of the Work.

All piping shall be of the type and size shown on the drawings and described in this section of the Specifications.

1.4 SUBMITTALS: IN ACCORDANCE WITH REQUIREMENTS OF GENERAL SPECIFICATIONS, SUBMIT THE FOLLOWING:

Manufacturer's literature on the materials of this Section.

Manufacturer's certification that the product was manufactured, tested, and supplied in accordance with this specification.

1.5 DELIVERY, STORAGE, AND HANDLING

Pipe shall be packaged to withstand shipment without damage and handled carefully on the jobsite.

PART 2 PRODUCTS

2.1 HIGH DENSITY POLYETHYLENE (HDPE) PIPE AND FITTINGS:

2.1.1 Design Requirements

HDPE pipe and fittings shall be manufactured from PE 3408 resin meeting the cell classification of 345434C per ASTM 1248 and 3350.

All polyethylene fittings shall have a pressure rating equal to or higher than the pressure rating of the pipe.

HDPE pipe shall be furnished in standard lengths.

All HDPE connections shall have ANSI Class 150. Gaskets for flanged connections shall be full faced EPDM and hardware shall be 70/30 Copper / Nickel, Titanium or similar material rated for seawater.

All pipe and fittings shall be IDR 11 rated at 160 psi.

All piping and valves shall be supported by the size and style supports shown in the drawings, or an approved equal. The support shall be designed and stamped by a professional engineer. The cost shall be borne by the Contractor.

Pipe shall be homogeneous throughout; free from voids, cracks, and other defects; as uniform as commercially practicable in color, density, and other physical properties.

Pipe surfaces shall be free from nicks, scratches, and other blemishes. The joining surfaces of pipe shall be free from gouges and other imperfections that could cause leakage at joints.

For fused saddles, transition couplings for HDPE to PVC shall be submitted for approval. Pipe shall be blue for water irrigation or blue strip meeting NSF or AWWA.

## 2.2 MANUFACTURERS

Pipe and fittings shall be manufactured by Ipex, Inc.; Performance Pipe, Division of Chevron Chemical Co.; JM Eagle Pipe Co.; Pipeline Plastics, LLC; E.J. Prescott or approved equal.

## 2.3 COPPER NICKEL PIPE MATERIAL ALTERNATIVE

Copper Nickel Pipe may be used in place of HDPE, as approved by the Engineer. Copper nickel pipe material must be 70/30 copper nickel pipe ASTM B411 and ASME SB466, IPS Class 200 wall.

### 2.3.1 Copper Nick Flanges

70/30 copper nickel ASTM B466 and ASME SB466 Class 150.

## PART 3 EXECUTION

### 3.1 INSPECTION BEFORE INSTALLATION

Each length of pipe and each fitting shall be carefully inspected prior to being lowered into the pipe. All materials not meeting the requirements of these specifications, or otherwise found defective or unsatisfactory by the Engineer, shall be rejected and immediately marked and removed from the jobsite by the Contractor.

### 3.2 PIPE INSTALLATION

Except as modified herein, installation of HDPE pipe installation shall be in accordance with ASTM D2321 and manufacturer standards.

All pipe and fittings shall be installed according to the manufacturer's instructions. Pipe interiors, fitting interiors, and joint surfaces shall be thoroughly cleaned prior to installation. Pipes and fittings shall be maintained clean. For HDPE pipe, a clean cotton cloth shall be employed for cleaning; polyester-type materials shall not be used as they may melt during fusion procedures.

Branches and fittings shall be laid by the Contractor as indicated on the drawings, and/or as required by the Engineer. Open ends of pipe and branches shall be closed with PVC caps secured in place with premolded gasket joints or as required by the Engineer.

Care shall be taken to prevent earth and other materials from entering the pipe, and when pipe laying operations are suspended, the Contractor shall maintain a suitable stopper in the end of the pipe.

If any defective pipe is discovered after being placed, removal and replacement with sound pipe will be required at no additional cost to the Owner.

### 3.3 HDPE PIPE JOINING

HDPE pipe should be joined by butt-fusion methods, having a completely uniform and monolithic pipe interior according to the fusion joining

procedures as instructed by the manufacturer meeting standards of ASTM D3261.

Each individual performing fusion joining shall have had at least one year of experience in the use of the fusion procedure.

Inspection of joints shall be performed by a person qualified by training or experience to evaluate the acceptability of HDPE joints made under the applicable joining procedures.

The pipe sections shall be joined at ground level to a length recommended by the manufacturer, such that when pulling the pipe into position, maximum allowable stress is not exceeded. Use appropriate materials and equipment, as recommended by the HDPE pipe manufacturer, when pulling butt-fused pipe sections in the 48" RCP, to prevent pipe damage.

#### 3.4 LEAKAGE TESTING

On completion of a section of pipe is installed, the Contractor shall install suitable bulkheads as required, dewater and test the pipe for leakage.

Unless otherwise approved, the section shall be tested in accordance with manufacturer recommended procedure. Pressure testing shall use water as test media and not air.

#### 3.5 TESTING

Prior to the pressure and leakage tests, the piping shall be thoroughly flushed clean of all dirt, dust, oil, grease and other foreign material. This work shall be done with care to avoid damage to linings and coatings.

The installed pipe shall be pressure tested and leakage tested in accordance with AWWA Standard C600.

#### 3.6 PRESSURE AND LEAKAGE TESTS

Unless otherwise approved, all pipelines shall be given a combined pressure and leakage tests between line valves. The Contractor shall furnish and install suitable temporary testing plugs or caps; all necessary pressure pumps, pipe connections, meters, gates, and other necessary equipment; and all labor required. The Owner or Engineer shall have the privilege of using their own gages.

Subject to approval and provided that the tests are made within a reasonable time considering the progress of the project as a whole, and the need to put the section into service, the Contractor may make the tests when it desires.

Unless it has already been done, the section of pipe to be tested shall be filled with water of approved quality, and all air shall be expelled from the pipe. The Contractor shall follow established procedures for filling the pipe and expelling trapped air to avoid exposing the piping system to water hammer. If blowoffs are not available at high points for releasing air, the Contractor shall install the necessary taps. If the Contractor changes the grade of pipe installation, it will be responsible for locating the taps at the correct location in the system for testing. After completion of the test, if so, required by the Engineer, the Contractor shall remove corporations used for testing and plug the holes.

The section under test shall be maintained full of water for a period of 24 hours prior to the combined pressure and leakage test being applied.

The pressure test shall consist of first raising the water pressure (based on the elevation of the lowest point of the section under test corrected to the gage location) to a pressure in pounds per square inch numerically equal to the pressure rating of the pipe (160 psi, unless otherwise noted). If the Contractor cannot achieve the specified pressure and maintain it for a period of one hour, the section shall be considered as having failed to pass the pressure test.

If the pressure test fails, the Contractor shall make a leakage test by metering the flow of water into the pipe while maintaining in the section being tested a pressure equal to the pressure rating of the pipe. If the average leakage during a two hour period exceeds a rate of 11.6 gallons per inch of diameter per 24 hours per mile of pipeline, the section shall be considered as having failed the leakage test. For example, if 1,000 feet of 12 inch pipe is to be tested, the allowable leakage is 2.2 gallons over a 2-hour period, calculated as follows:

$$L = (11.6 \text{ gal}) \times (12") \times (2 \text{ hr.}) \times (1000') = 2.2 \text{ gal} \\ (1") \times (24 \text{ hr.}) \times (5280')$$

If the section fails to pass the pressure and leakage test, the Contractor shall do everything necessary to locate, uncover, and repair or replace the defective pipe, fitting, or joint, all at its own expense and without extension of time for completion of the work. Additional tests and repairs shall be made until the section passes the specified test.

-- End of Section --



SECTION 33 21 00.19

VERTICAL MULTI-STAGE CENTRIFUGAL PUMPS  
**11/24**

PART 1 GENERAL

1.1 WORK INCLUDED

This Section covers the furnishing and installation of vertical multi-stage centrifugal pumps, motors, and appurtenances complete for pumping BWRO permeate water.

1.2 RELATED WORK

- a. 03 30 00 CAST-IN-PLACE CONCRETE
- b. Division 40 - Instrumentation
- c. Division 26 - Electrical

1.3 PUMP SCHEDULE

- a. Adequate tolerances in the listed capacity, head and efficiency values have been included. No deviation below the listed parameters will be permitted.
- b. All listed parameters shall be considered design points.
- c. Pump Schedule

Table 1.3

| Pump                   | Description                             | Design Capacity (gpm) | Design Total Dynamic Head (feet) | Rated Motor HP | RPM                 | Voltage | Starter Type | Make / Model            |
|------------------------|-----------------------------------------|-----------------------|----------------------------------|----------------|---------------------|---------|--------------|-------------------------|
| 06-01-001<br>06-01-002 | Permeate supply to condensate tank      | 300                   | 60                               | 10             | 3600                | 480     | VFD          | Xylem / 92SVX221NNAXX0  |
| 06-01-003<br>06-01-004 | Permeate supply to South cooling towers | 100                   | 93.23                            | 25             | 1891,<br>(3600 max) | 480     | VFD          | Xylem / HVX2D6L-46SVX30 |
| 06-01-005<br>06-01-006 | Permeate supply to North cooling towers | 200                   | 207.53                           | 25             | 2993,<br>(3600 max) | 480     | VFD          | Xylem / HVX2D6L-46SVX30 |

#### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-02 Shop Drawings

PUMP CONSTRUCTION; G, AE

Show complete details of construction and installation of all equipment, including appurtenances; include all dimensions.

Include all specifics of this installation.

Delete all references not applicable to this installation.

Outline specific areas where the proposed equipment differs from that specified.

Include complete wiring diagrams required for installation.

Prior to shipment, the certified pump curves shall be approved by the Engineer and shall show the actual performance of the pumps under factory testing. Pump certification shall include testing for head, capacity, efficiency and brake horsepower, at 50%, 65%, 85% and 100% for variable speed driven pumps, and at the rated speed for constant speed pumps.

Complete test results on each motor from the routine test as defined in the NEMA standard for a motor of its class and rating to determine that it is free from electrical and mechanical defects and to provide assurance that it meets the design specifications.

The manufacturer's complete initial test for each motor model design tested for efficiency by the following means: The full load Dynamometer test shall be based on IEEE 112 Method B, per NEMA MG1 12.53.

##### SD-06 Test Reports

FIELD AND ACCEPTANCE TESTS, G, AE

Upon completion of installation, the results of the acceptance tests as specified under this section of the specification shall be submitted to the Engineer.

##### SD-07 Certificates

Installation; G, AE

Furnish written certification from the manufacturer's representative of the proper installation of each component.

##### SD-09 Manufacturer's Field Reports

FIELD AND ACCEPTANCE TESTS; G

Upon completion of installation, the results of the field tests as specified under this section of the specification shall be submitted to the Engineer.

SD-10 Operation and Maintenance Data

Operations and Maintenance Manuals; G

Complete operation and maintenance information for this specific equipment.

The manuals will be reviewed by the Engineer for completeness; those that are deemed inadequate will be returned for correction.

Complete parts list including the manufacturer's reference and ordering numbers.

A complete list of the manufacturer's name, address and phone number, the local representative's name, address and phone number, and the model and serial numbers of all equipment supplied.

Spare Parts; G

1.5 QUALITY ASSURANCE

a. General

- (1) This specification directs special attention to certain features but does not purport to cover all details of the design, manufacture or installation of the pumping units. The final responsibility for supplying and installing pumping equipment which functions as specified herein rests with the Contractor.
- (2) Where more than one pump is being supplied, they should all be supplied by the same Pump Manufacturer.
- (3) Workmanship and the method and materials of construction shall conform to the best practice and highest standards applicable for the design use as specified.
- (4) Pumping units furnished shall be complete in all particulars and ready for final assembly, installation and operation.
- (5) Any structural, mechanical, civil, architectural, HVAC, plumbing or electrical changes required by the use of an approved equal product shall be the responsibility of the Contractor.

b. Manufacturers

- (1) Pump manufacturers shall certify proof of successful operating experience during the last 10 years of five installations of equipment comparable to that specified herein.
- (2) The electrical variable speed drive equipment specified in Division 26 (including a/c drive induction motor specified herein) shall be designed and sized by the pump supplier who

shall have successfully manufactured, installed and started up at least ten systems similar to this installation in the past five years. The pump and motor supplier shall assume responsibility for the correct operation of the entire a/c drive system.

- (3) Pumps shall be Goulds Xylem, or Engineer approved equal. See Table 1.3 in Section 1.3 for approved manufacturer and model numbers.

## 1.6 STORAGE

- a. Pump shafts and columns stored on site shall have covered and taped ends for protection. Pump equipment damaged or bent during shipping or storage shall be replaced at no cost to the Government.
- b. All pump motors shall be stored on site according to motor manufacturer's recommendations until pumps are operational and accepted by the Engineer. Pumps with motor space heaters shall be energized during storage; the Contractor shall be responsible for such arrangements.

## 1.7 WARRANTY

- a. The pump manufacturer and pump motor manufacturer shall each individually and separately warranty that the equipment they supplied under this Section fully meets the criteria specified herein and shall further warranty that the equipment is free from all defects in materials and workmanship.
- b. The manufacturer's warranties from defects shall contain a provision that the manufacturer shall repair or replace any defects, to the satisfaction of and at no additional cost to the Owner, for a period of twenty-four (24) months for the pump and motor from the date of Substantial Completion of the project.

## PART 2 PRODUCTS

### 2.1 PUMP CONSTRUCTION

#### 2.1.1 Pump Casing

- a. The pump casing shall be of deep drawn, laser welded AISI 304L or 316L stainless steel and shall be capable of withstanding maximum working pressures of 360 psi. Piping connections shall be in-line and shall be compatible with ANSI raised face flanges.
- b. Wear ring composed of PPS shall be provided with each stage. Wear rings must be self centering and easily replaceable.
- c. The interior of the pump head and base shall be smooth and free from surface defects.
- d. Pump casings shall be designed for 175 psi working pressure and shall be hydrostatically tested at 150 percent of working pressure.
- e. The seal housing shall be of concave design and shall hold the seal faces below the topmost part of the pump casing.

#### 2.1.2 Pump Impellers

- a. The impellers shall of enclosed design and constructed of AISI 316L or AISI 304L stainless steel. Impellers shall provide internal thrust balance in each stage.
- b. Each stage shall have a bowl with attached diffuser and be constructed of AISI 304L or 316L stainless steel.
- c. The mechanical seal shall have a carbon rotating face, silicon carbide graphite filled stationary face, viton elastomers, and 316 SS metal spring.

#### 2.1.3 Inverter Duty Motors

- a. The pumps shall be driven over the specified speed range by a/c motors sized to operate in conjunction with variable voltage/variable frequency rectified power and provide 100 percent motor torque and horsepower rating at 1.0 service factor.
- b. The motor base speed shall be selected so that the maximum operating speed of the centrifugal pump and motor are compatible.
- c. Motors shall not exceed 3,600 rpm.
- d. The a/c motors for variable frequency applications shall be 3 Phase, 60 Hertz, 460 Volts. They shall be rated for operation in 40NC ambient at 1.15 service factor for constant speed operation and 1.0 service factor for operation on variable frequency, variable voltage. The motor shall have a KVA per horsepower rating of NEMA code letter G or better.
- e. The motors shall be totally enclosed fan cooled enclosure, premium efficiency design, cast iron construction, mill and chemical duty, NEMA design B, Class B insulation with the following minimum features:
  - (1) The motor shall develop sufficient torque to start the pump and bring it up to full speed when the pump is full of water and operate the pump at any point of its characteristic curve without entering into the service factor.
- f. Each motor shall be designed for the highest efficiency standardly available in the marketplace.
- g. The motors shall be tested as specified in this section.

#### 2.2 PUMP TESTS

All pumps shall be tested in the shop of the manufacturer for head, capacity, efficiency, and brake horsepower at 50%, 65%, 85% and 100% of the rated speed. Six certified copies of the results in the form of pump characteristic curves shall be furnished to the Engineer for review prior to shipment.

### PART 3 EXECUTION

#### 3.1 INSTALLATION

All pumps shall be assembled and installed in accordance with the

manufacturer's recommendations and as approved by the Engineer.

Pump checkout and testing shall be as described in Section 01 75 13, EQUIPMENT CHECKOUT AND TESTING, and as described below.

### 3.2 FIELD AND ACCEPTANCE TESTS

- a. After all pumps have been installed and connected and after inspection, operation, and adjustment has been completed by the manufacturer's representative, the pumping equipment shall be field tested in the presence of the Engineer, for overall wire to water efficiency and for general performance and fitness for the service specified. Results of the tests shall be submitted to the Engineer.
- b. The quantity of water discharged by the pumps shall be measured by the equipment installed.
- c. Pumping heads shall be measured by pressure gauges, especially calibrated, and electrical input by using suitable instruments. Duration of the tests shall be as determined by the Engineer.
- d. If the capacity of a pumping unit under the design pumping heads fails to deliver the design capacity, or if the wire to water efficiency under the designed head fails to meet the efficiency stated in the specification, the Contractor shall, at its own expense, upon the written request of the Engineer, replace the motor, impeller, or other parts, or otherwise improve the unit until the specified capacity and efficiency are fulfilled.

### 3.3 TECHNICAL DIRECTION

- a. The service of a factory trained manufacturer's representative shall be provided as specified herein.
- b. Service to be provided:
  - (1) The service representative shall be responsible for complete component inspection on site after delivery.
  - (2) Start up services and supervision, including coordinating the tuning of the variable frequency drive system with instrumentation.
  - (3) Complete instruction of the Owner's operating personnel. The Owner reserves the right to videotape the instruction of the operating personnel.
- c. Service shall be provided for the following minimum periods of time:
  - (1) Pumps and motors Two (2) eight hour days.
- d. The minimum period of time herein specified does not relieve the manufacturer from providing sufficient time to satisfactorily complete the required service functions.
- e. A manufacturer's representative shall certify in writing that the pumps have been properly installed.

#### 3.4 SPARE PARTS

All other spare parts recommended in the manufacturer's standard operations and maintenance information.

#### 3.5 OPERATION AND MAINTENANCE MANUALS

Provide standard operation and maintenance manual for the pumps provided  
-- End of Section --

SECTION 33 61 13.19

VALVES, PIPING, AND EQUIPMENT IN VALVE MANHOLES  
**02/16**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ALUMINUM ASSOCIATION (AA)

AA H35.1/35.1M (2017) American National Standard Alloy and Temper Designation Systems for Aluminum

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.20.1 (2013; R 2018) Pipe Threads, General Purpose (Inch)

ASME B16.5 (2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard

ASME B16.9 (2018) Factory-Made Wrought Butt welding Fittings

ASME B16.11 (2022) Forged Fittings, Socket-Welding and Threaded

ASME B16.20 (2017) Metallic Gaskets for Pipe Flanges

ASME B16.21 (2021) Nonmetallic Flat Gaskets for Pipe Flanges

ASME B16.34 (2021) Valves - Flanged, Threaded and Welding End

ASME B31.1 (2022) Power Piping

ASME B40.100 (2022) Pressure Gauges and Gauge Attachments

ASME BPVC SEC IX (2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications

ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A106/A106M (2019a) Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service



|                                                                                |                                                                                                                                                         |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A123/A123M                                                                | (2024) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products                                                         |
| ASTM A193/A193M                                                                | (2024) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications |
| ASTM A194/A194M                                                                | (2023) Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both  |
| ASTM A733                                                                      | (2016; R 2022) Standard Specification for Welded and Seamless Carbon Steel and Austenitic Stainless Steel Pipe Nipples                                  |
| ASTM B209                                                                      | (2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate                                                                           |
| ASTM D2822/D2822M                                                              | (2005; R 2011; E 2011) Standard Specification for Asphalt Roof Cement, Asbestos-Containing                                                              |
| ASTM D3359                                                                     | (2017) Standard Test Methods for Rating Adhesion by Tape Test                                                                                           |
| ASTM F1139                                                                     | (1988; R 2019) Steam Traps and Drains                                                                                                                   |
| MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS) |                                                                                                                                                         |
| MSS SP-25                                                                      | (2018) Standard Marking System for Valves, Fittings, Flanges and Unions                                                                                 |
| MSS SP-45                                                                      | (2020) Bypass and Drain Connections                                                                                                                     |
| MSS SP-58                                                                      | (2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation                                          |
| MSS SP-72                                                                      | (2010a) Ball Valves with Flanged or Butt-Welding Ends for General Service                                                                               |
| MSS SP-80                                                                      | (2019) Bronze Gate, Globe, Angle and Check Valves                                                                                                       |
| MSS SP-83                                                                      | (2014) Class 3000 Steel Pipe Unions Socket Welding and Threaded                                                                                         |
| MSS SP-110                                                                     | (2010) Ball Valves Threaded, Socket-Welding, Solder Joint, Grooved and Flared Ends                                                                      |

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC Paint 16 (2006; R 2015; E 2015) Coal Tar  
Epoxy-Polyamide Black (or Dark Red) Paint

SSPC Paint 29 (2002; E 2004) Zinc Dust Sacrificial  
Primer, Performance-Based

SSPC SP 10/NACE No. 2 (2015) Near-White Blast Cleaning

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Detail Drawings; G, AE

SD-03 Product Data

Support of the Equipment; G, AE

Piping and Fittings; G, AE

Valves; G, AE

Insulating Flanges; G, AE

Expansion Joints; G, AE

SD-04 Samples

Insulated Sections; G; G, AE

SD-10 Operation and Maintenance Data

Valve Manholes and Accessories; G; G, AE

Data Package 2; G; G, AE

1.3 QUALITY ASSURANCE

1.3.1 Detail Drawings

Submit detail drawings 30 days after notice to proceed for valve manholes and the piping and equipment in the valve manholes, such as steam traps, valves, sump pumps, pressure gauges, thermometers and insulation, including a complete list of equipment and materials, manufacturer's descriptive and technical literature, performance charts and curves, catalog cuts, installation instructions, and complete wiring and schematic diagrams. Show on the drawings pipe anchors and guides, and layout and anchorage of equipment and appurtenances in valve manholes, and equipment

relationship to other parts of the work including clearances for maintenance and operation.

#### 1.3.2 Insulated Sections

Submit sample sections, 30 days after notice to proceed, for insulation of pipe, elbow, tee, valve, support point, and terminating points. After approval of materials and prior to insulation of piping, prepare a display of insulated sections showing compliance with specifications and showing fastening, sealing, jacketing, straps, waterproofing, supports, hangers, anchors, and saddles. Display approved display sample sections at the jobsite during the construction period until no longer needed by Contracting Officer, then removed.

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Protect all materials and equipment delivered and placed in storage from the weather, excessive humidity, and excessive temperature variation; dirt, dust, or other contaminants.

### PART 2 PRODUCTS

#### 2.1 STANDARD PRODUCTS

Provide materials and equipment which are the standard products of a manufacturer regularly engaged in the manufacture of such products and that essentially duplicate items that have been in satisfactory use for at least 2 years prior to bid opening.

Equipment items must be supported by service organizations. Submit a certified list of qualified permanent service organizations for support of the equipment which includes their addresses and qualifications. These service organizations must be reasonably convenient to the equipment installation and able to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

**Products provided here must be compatible with seawater or deionized water. Material like titanium, copper-nickel, ASTM B62 Bronze, ASTM 148 Bronze, etc. shall be materials of construction when in contact with this water. Where materials are called for in the drawings and specifications that are not compatible with seawater, they shall be provided with sacrificial anodes.**

#### 2.2 NAMEPLATES

Supply each major item of equipment such as sump pump, motor, steam trap, and pressure reducing valve with the manufacturer's name, address, type or style, model or serial number, and catalog number on a plate secured to the item of equipment.

#### 2.3 ASBESTOS PROHIBITION

Asbestos and asbestos-containing products are not allowed.

#### 2.4 ELECTRICAL WORK

Provide motors, manual or automatic motor control equipment, and protective or signal devices required for the operation specified under this section in accordance with NFPA 70 and contract documents.

## 2.5 PIPING AND FITTINGS

### 2.5.1 General Requirements

Provide piping, fittings and piping accessories inside the valve manholes suitable for the working pressure and temperature requirements of the system conforming to ASME B31.1. To the greatest extent possible, match the piping and fittings inside the valve manholes to the piping and fittings located on the outside of the valve manhole. Provide steel piping in valve manholes with joints welded except that joints 3/4 inch and smaller may be threaded. When threaded joints are used on High Temperature Water Systems, seal weld (continuous fillet weld) the interface area where the pipe threads meet the threaded fittings to preclude any water leakage. Do not attach supports, anchors, or stays to any piping system in places where either the installation of or the movement of the pipe and its contents will cause damage to the construction.

### 2.5.2 Steel Pipe

Provide black steel, seamless or electric-resistance welded, conforming to the requirements of ASTM A53/A53M, Grade B or ASTM A106/A106M, Grade B. Provide schedule 40 type for pipe up to and including 10 inches in diameter. Provide 0.375 inch nominal wall thickness for pipe 12 inches in diameter and greater. Provide schedule 80 type for gauge piping and piping 3/4 inch in diameter and smaller.

#### 2.5.2.1 Nipples

Provide nipples that conform to ASTM A733 as required to match adjacent piping.

#### 2.5.2.2 Pipe Threads

Provide pipe threads that conform to ASME B1.20.1. Use pipe threads only on pipe 3/4 inch or smaller.

### 2.5.3 Fittings

Provide fittings, valves, flanges and unions with the manufacturer's trademark affixed in accordance with MSS SP-25 so as to permanently identify the manufacturer.

#### 2.5.3.1 Welded Fittings

Provide welded fittings to match connecting pipes with butt welded fittings, conforming to ASME B16.9, and socket welded fittings, conforming to ASME B16.11.

#### 2.5.3.2 Unions

Provide unions that conform to MSS SP-83 as required to match adjacent piping.

#### 2.5.3.3 Ball Valves

Provide ball valves having flanged or butt welded end connections conforming to MSS SP-72; provide ball valves having threaded end

connections conforming to MSS SP-110.

#### 2.5.4 Insulating Flanges and Dielectric Waterways

##### 2.5.4.1 Insulating Flanges

For systems in which cathodic protection is used, provide insulating flanges or flange gasket kits in the valve manhole at the pipe connection to or from the heat distribution system, at the interface of dissimilar metals, and when the carrier pipe and appurtenances are supported in such a way as to electrically ground or alter the cathodic protection system voltages or currents. Provide a kit that consists of flanges, a flange gasket, nuts and bolts, bolt sleeves, and one insulating washer and one steel washer for both ends of each bolt. Provide manufacturer certified gasket kits capable of electrically isolating the pipe at the 175 psig pressure and 100 degrees F temperature of the heating medium at the point of application. Submit evidence of satisfactory installations operating not less than 2 years, in accordance with paragraph SUBMITTALS, before materials are delivered. Ensure that these kits are provided and properly installed according to manufacturer's published instructions. Provide bolts torqued to the correct tightness and in the correct bolt pattern as recommended by the manufacturer's published instructions. Provide steel flanges that conform to ASME B16.5 Class 150 300 and that match valves or flanged fittings on which used. Provide flat faced steel flanges. Provide non-asbestos compressed material gaskets in accordance with ASME B16.21. Provide bolts that conform to the requirements of ASTM A193/A193M, Grade B7. Provide bolt heads marked to identify the manufacturer and the standard to which the bolt complies. Extend bolt lengths to no less than 2 full threads beyond the nut at the required tension with the washer seated. Provide nuts that conform to the requirements of ASTM A194/A194M, Grade 7.

##### 2.5.4.2 Dielectric Waterways

Provide dielectric waterways that have temperature and pressure rating equal to or greater than that specified for the connecting piping and used for joining dissimilar metals on 3/4 inch and smaller threaded pipe. Provide waterways that have metal connections on both ends suited to match connecting piping. Provide dielectric waterways that are internally lined with an insulator specifically designed to prevent current flow between dissimilar metals. Provide dielectric flanges that meet the performance requirements described herein for dielectric waterways.

##### 2.5.4.3 Gaskets Non-Insulating

Provide spiral wound, non-asbestos gasket with centering ring that conform to ASME B16.20.

#### 2.6 VALVES

Provide valves that conform to the material, fabrication, and operating requirements of ASME B31.1, unless otherwise specified. Provide valves suitable for the service temperatures and pressures utilized. Provide valves for that conform to ASME B31.1 Class 150, as suitable for service temperatures and pressures utilized. Valves 3/4 inch and smaller may be bronze where seal welding is not required. Provide valves 6 inches and larger with a 1 inch minimum gate or globe bypass valve sized in conformance with MSS SP-45.

### 2.6.1 Steel Valves

Provide steel globe, gate, angle, and check valves that conform to the requirements of ASME B16.34 and ASME B31.1 for the service temperatures and pressures utilized. Provide gate valves 2-1/2 inches and smaller with a rising stem. Provide gate valves 3 inches and larger with an outside screw and yoke.

### 2.6.2 Bronze Valves

#### 2.6.2.1 Globe, Gate, and Angle Valves

Provide bronze globe, gate, and angle valves that conform to MSS SP-80, union bonnet type.

#### 2.6.2.2 Check Valves

Provide bronze check valves that conform to MSS SP-80.

### 2.6.3 Packing

Provide asbestos free valve packing. Provide die-formed, ring type specifically designated valve stem packing suitable for service temperatures and pressures utilized. Provide polytetrafluoroethylene packing that has a with minimum 50 percent graphite filament. Provide valves 1-1/2 inches and smaller with four or five packing rings and provide valves 2 inches and larger with at least six packing rings. Spiral or continuous packing will not be acceptable. Provide a metal insert having proper clearance around the valve stem at the bottom of the stuffing box and acting as a base for the packing material. Provide one piece construction with provisions for not less than two bolts for packing adjustment, with a liner of noncorrosive material for packing glands.

## 2.7 STEAM TRAPS

Provide fail open traps with trap bodies suitable for a working pressure of not less than 1.5 times the steam supply pressure, but not less than 200 psi.

### 2.7.1 Bucket Traps

Provide inverted-bucket type bucket traps with automatic air discharge conforming to ASTM F1139.

### 2.7.2 Thermostatic Traps

Provide thermostatic traps that have a bimetallic element with automatic air discharge conforming to ASTM F1139.

## 2.8 PRESSURE GAUGES

Provide pressure gauges that conform to ASME B40.100 with a minimum dial size of 4-1/4 inches. Provide each gauge with a throttling type needle valve or a pulsation dampener and shut-off valve.

## 2.9 DIAL THERMOMETERS

Provide dial type thermometers 3-1/2 inches in diameter with stainless steel case, remote-type bulb or direct-type bulb as required. Provide

thermometers that have an accuracy of plus or minus 2 degrees F. Provide thermometer wells of the separable socket type for each thermometer with a direct-type bulb. Provide thermometer with a white face with black digits graduated in 2 degrees F increments.

## 2.10 COATINGS

Coat steel manhole piping with an organic zinc undercoat that conforms to SSPC Paint 29 Type II followed by a thermal barrier coating having a manufacturer's documented minimum thermal conductivity of 0.058 Btu/hr•ft•°F. Provide the undercoat and thermal barrier coating with a continuous use service temperature rating that exceeds the nominal system operating temperature by a minimum of 50 degrees F.

## 2.11 CONCRETE VALVE MANHOLES AND ACCESSORIES

### 2.11.1 Wall and Floor Construction

Provide manhole in accordance with Section 03 42 13.00 10 PLANT-PRECAST CONCRETE PRODUCTS FOR BELOW GRADE CONSTRUCTION. Construct walls and floors of reinforced concrete not less than 8 inches thick. Construct walls using one monolithic pour. Extend walls flush with finished grade. Provide floor with an internal sump; slope the floor in all directions to the sump to allow water collection. Provide construction joints with water stops. Waterproof manhole exterior in accordance with Section 07 13 53 ELASTOMERIC SHEET WATERPROOFING.

### 2.11.2 Manhole Supported Cover(s)

Provide an 5/16 inch thick checker pattern, aluminum solid plate cover that conforms to AA H35.1/35.1M cover that is supported by and is flush with the top of the manhole walls. Construct cover(s) to be removable and sectionalized as indicated. Provide hot-dipped galvanized structural steel supports, anchor bolts, nuts, and washers. Provide a cover and support system that can support a load up to 150 psf. Install an 5/16 inch thick checker pattern, aluminum solid plate cover that conforms to AA H35.1/35.1M on top of the open grating. Attach the checkered plate to the grating with removable, galvanized steel fasteners.

### 2.11.3 Raised Frame Cover(s)

Provide a raised support structure constructed out of hot-dipped galvanized steel that is designed to sit on top of the manhole walls. Provide an 5/16 inch thick checker pattern, aluminum solid plate cover that conforms to AA H35.1/35.1M. Construct cover(s) to be removable and sectionalized as indicated. Provide ventilation openings as indicated around the entire perimeter below the raised top. Provide hot-dipped galvanized steel lifting lugs on the cover.

### 2.11.4 Concrete Cover

Provide a 6 inches thick cast concrete cover designed to support loads up to 150 psf. Provide a 4 by 4 foot aluminum access door in the concrete top. Provide two 6 inch goosenecks; terminate one gooseneck inside the manhole within 2 feet of the manhole's floor; terminate the other gooseneck inside the manhole just below the manhole top.

#### 2.11.5 Ladders

Provide steel valve manhole ladders, with nonslip surfaces, and consisting of uprights with steps or rungs. fabricate ladders with two stringers a minimum 3/8 inch thick and 2-1/2 inches wide, and rungs not be less than 16 inches in width, 3/4 inch diameter, spaced 12 inches apart. Anchor the ladders to the wall by means of steel inserts spaced not more than 6 feet apart vertically, and install to provide at least 6 inches of space between the wall and rungs. Galvanize ladders and inserts after fabrication in conformance with ASTM A123/A123M.

#### 2.11.6 Pipe Sleeves

Provide zinc-coated steel pipe, conforming to ASTM A53/A53M, Schedule 40 or standard weight. Install so there is no electrical continuity between the pipe sleeve and the pipe casing.

##### 2.11.6.1 Pipe Sleeves Through Valve Manhole Cover

Provide insulation continuously through sleeves and provide aluminum jacket over the insulation. Provide smooth sheet 0.016 inch nominal thickness aluminum jacket conforming to ASTM B209. Where penetrations in valve manhole tops are required, insulate piping and seal with waterproof coating up to a point flush with the top of the flashing and the end of the insulation. Butt insulation exposed to the weather tightly against the flashing and valve manhole insulation, and extend the aluminum jacket required for piping exposed to the weather 2 inches beyond the insulation to form a counterflashing. Flash and counterflash valve manhole penetrations and apply waterproof coating conforming to ASTM D2822/D2822M, Type I.

##### 2.11.6.2 Pipe Sleeves for Conduit Penetrations

Provide a modular mechanical type sealing assembly between the valve manhole pipe sleeve and the conduit casing . The mechanical seal consists of interlocking elastomeric links shaped to continuously fill the annular space between the casing and sleeve. The link material is a synthetic elastomeric capable of withstanding long term exposure at 400 degrees F without deterioration. Attach the links to each other with corrosion resistant steel bolts, nuts and pressure plates. The link, bolts, nuts and pressure plates must be the product of single manufacturer and furnished as the product of single manufacturer as a package or kit.

#### 2.11.7 Pipe Supports

Provide pipe supports in accordance with MSS SP-58. Galvanize all pipe supports, including structural cross support members, Chains, straps, or single point supports are not allowed.

#### 2.12 EXPANSION JOINTS

Submit manufacturer's descriptive data and technical literature, performance charts, catalog cuts and installation instructions.

##### 2.12.1 Guided Slip Tube

Internally-externally guided type, injected semiplastic type packing, with service outlets. Construct joints for minimum working pressure of ASME Class 150. Provide single or double slip tube type as indicated. Provide



flanged or butt welding end connections as indicated.

#### 2.12.2 Flexible Ball

Provide chromium plated steel balls capable of 360-degree rotation plus 15-degree angular flex movement. Provide pressure molded composition gaskets designed for continuous operation temperature of 525 degrees F. Construct joints for minimum working pressure of ASME Class 150. Provide flanged or butt welding end connections as indicated.

#### 2.12.3 Bellows-Type

Type 304 stainless steel corrugated bellows, reinforced with rings, internal sleeves, and external protective covers, designed to withstand 10,000 cycles over a 20 year period and a minimum working pressure of ASME Class 150. Provide limit stops to limit total movement in both directions. Cold set the joints to compensate for temperature at time of installation. Provide single or double bellows expansion joint as indicated. Provide first pipe alignment guide no more than four pipe diameters from the expansion joint; provide second pipe alignment guide no more than 14 pipe diameters from the first guide. Provide flanged or butt welding end connections as indicated.

#### 2.13 MISCELLANEOUS METAL

Hot-dip galvanize miscellaneous metal bolted together, shop welded, or assembled in the field, and pipe supports, including structural cross support members and anchors

### PART 3 EXECUTION

#### 3.1 EXAMINATION

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing the work.

#### 3.2 SITE WORK

##### 3.2.1 Excavation, Trenching, and Backfilling

Excavate, trench, and backfill the valve manholes as indicated and in accordance with Section 31 00 00.00 02 EARTHWORK.

##### 3.2.2 Painting

Clean the heat affected zone of field welded galvanized surfaces and other galvanized surfaces damaged during installation in compliance with SSPC SP 10/NACE No. 2, and paint in accordance with Section 09 90 00 PAINTS AND COATINGS. Clean steel and iron appurtenances, piping, and supports in compliance with SSPC SP 10/NACE No. 2, and paint in accordance with SSPC Paint 16.

#### 3.3 PIPING

##### 3.3.1 General

Use steel piping and insulate in valve manholes. Protect insulation with a galvanized steel jacket. Cut pipe to measurements established at the site and work into place without springing or forcing. Clear all openings and equipment, and avoid cutting or other weakening of structural members

to facilitate piping installation. Remove burrs from ends of pipe by reaming. Install to permit free expansion and contraction without damage to joints or hangers and in accordance with ASME B31.1. Do not attach supports, anchors, or stays where either expansion or the weight of the pipe could cause damage to permanent construction. The method of attaching supports must not interfere with the operation of the cathodic protection system.

### 3.3.2 Welded Joints

Weld all pipe joints for piping in valve manholes, except joints at traps, strainers, and at valves and piping 3/4 inch and smaller which may be threaded. Conform welding to the requirements specified in paragraph WELDING.

### 3.3.3 Flanged and Threaded Joints

#### 3.3.3.1 Flanged Joints

Flanged joints are permitted for dielectric isolation only. Construct flanged joints to be faced true, provided with gaskets, and made perfectly square and tight. Use flanged joints only for electrical isolation and in other special cases where connected equipment is available with only flanged joints, or when specifically indicated. Provide electrically isolated flange joints at all connections to or from the heat distribution system and between dissimilar metals.

#### 3.3.3.2 Threaded Joints

Apply graphite or inert filler and oil, graphite compound, or polytetrafluoroethylene tape to the male threads only. Provide unions at all screwed valves, strainers and connections to equipment 3/4 inch and smaller. Use dielectric unions at connections of dissimilar metals in 3/4 inch and smaller piping. When used on High Temperature Water Systems, seal weld threaded joints.

### 3.3.4 Reducing Fittings

#### 3.3.4.1 Horizontal Water Heating Lines

Provide eccentric reducers for all pipe size changes. Provide eccentric type reducing fittings to maintain the tops of adjoining pipes at the same level.

### 3.3.5 Branch Connections

Branch off top of mains as indicated providing unrestricted circulation, elimination of air pockets, and permitting the complete drainage of the system. Branch connections may be made with either welding tees or forged branch outlet fittings. If branch outlet fittings are used, provide forged fittings no larger than two nominal pipe sizes smaller than the main run. Reinforce branch outlet fittings to withstand external strains and designed to withstand full pipe bursting strength.

### 3.3.6 Pipe Supports in Valve Manholes

Securely support horizontal and vertical runs of pipe in valve manholes.

### 3.4 WELDING

Weld pipe in accordance with qualified procedures, using performance qualified welders and welding operators. Procedures and welders must be qualified in accordance with ASME BPVC SEC IX. Welding procedures qualified by others, and welders and welding operators qualified by another employer may be accepted as permitted by ASME B31.1. Notify the Contracting Officer 24 hours in advance of tests and perform the tests at the work site. The welder or welding operator must apply his assigned symbol near each weld he makes as a permanent record. Perform welding and nondestructive testing procedures for piping as specified in Section 40 17 26.00 20 WELDING PROCESS PIPING. Weld structural members in accordance with Section 05 05 23.16 STRUCTURAL WELDING.

### 3.5 COATINGS

Prepare the steel piping surface by abrasive blasting to the near-white metal grade in conformance with SSPC SP 10/NACE No. 2. Within eight hours of blasting, or prior to any condensation of moisture or other surface deterioration whichever occurs first, coat all surfaces with an organic zinc primer conforming to SSPC Paint 29 Type II. Spray apply the primer to a minimum thickness at any point of 4 mils. Allow the primer to cure according to the manufacturer's recommendations prior to overcoating with the thermal barrier coating. Provide thermal barrier coating having film forming properties, an adhesion value of 5 when tested according to ASTM D3359 and a minimum thermal conductivity of 0.058 Btu/hr•ft•°F. Spray apply the thermal barrier coating in accordance with manufacturers recommendations to a minimum thickness at any point of 50 mils.

### 3.6 INSULATION

Install insulation so that it is not damaged by pipe expansion or contraction. Keep insulation dry before, during, and after installation. Groove insulation installed over welds to assure a snug fit. Hold insulation in place with stainless steel straps. Install a minimum of 2 bands on each individual length of insulation, with maximum spacing not exceeding 18 inch centers.

#### 3.6.1 Installation

Install material in accordance with published installation instructions of the manufacturer. Do not apply insulation materials until piping tests are complete. Prior to application, thoroughly clean surfaces of moisture, grease, dirt, rust, and scale; paint where required.

#### 3.6.2 Insulation on Pipes Passing Through Sleeves

Provide continuous insulation, as required by paragraph PIPE SLEEVES THROUGH VALVE MANHOLE COVER. Provide aluminum jackets over the insulation. When penetrating valve manhole walls, extend aluminum jacket not less than 2 inches beyond the sleeve on each side of the wall and secure with an aluminum band on each side of the wall. Where flashing is provided, secure the jacket with not less than one band located not more than 1 inch from the end of the jacket. When penetrating valve manhole tops, insulate pipe as required for valve manhole service.

#### 3.6.3 Covering of Insulation in Valve Manholes

Cover insulation for pipe, flanges, valves, and fittings with aluminum

jackets.

#### 3.6.4 Insulation of Piping Accessories in Valve Manholes

Insulate flanges, couplings, unions, valves, fittings, and other pipe accessories, unless otherwise indicated, with removable and reusable factory premolded, prefabricated, or field fabricated insulation. Provide aluminum sheet over insulation manholes and neatly terminate for accessories that are not to be insulated.

#### 3.6.5 Insulation Sealing for Chilled Water Systems

Seal the ends of insulation with vapor barrier, caulk penetrations and apply caulking to parting line between equipment and removable section insulation. Upon completion of installation of the insulation, including removable sections, apply two coats of vapor barrier coating with a layer of glass cloth embedded between the coats, providing a total dry thickness of the finish of 1/16 inch while maintaining removability of the sections as designed. Apply coating to flanges, unions, valves, anchors, fittings and accessories, all terminations, and all insulation not protected by factory vapor barrier jackets or PVC fitting covers. Overlap tape seams 1 inch. Extend the coating out onto the adjoining pipe insulation 2 inches. Taper insulation terminations to unions at a 45-degree angle.

#### 3.6.6 Insulation Thickness

Provide the minimum thickness of insulation for shall be 2 inches.

### 3.7 VALVE MANHOLES AND ACCESSORIES

#### 3.7.1 Piping and Equipment in Valve Manholes

Provide easy access in valve manholes without stepping on piping or equipment, and allow sufficient working area for maintenance work. Refer to drawings of piping and equipment in valve manholes for installation and support details. Install all globe, angle and gate valves with the stems horizontal or above.

Submit Data Package 2 as related to all equipment provided for the project in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA. Detail in the operation manuals the step-by-step procedures required for equipment startup, operation, and shutdown. Include in the operation manuals the manufacturer's name, model number, parts list, and brief description of all equipment and their basic operating features. List in the maintenance manuals routine maintenance procedures, possible breakdowns and repairs, and troubleshooting guides. Include in the maintenance manuals piping and equipment layout and simplified wiring and control diagrams indicating location of electrical components with terminals designated for wiring, as installed.

### 3.8 TESTS

Perform tests of piping in the valve manholes as part of the testing of the direct buried conduit system. Include the piping in the valve manhole in these tests and perform in accordance with the system supplier's Approved Brochure or the contract specifications.

-- End of Section --



SECTION 33 71 02

UNDERGROUND ELECTRICAL DISTRIBUTION  
**08/21**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS  
(AASHTO)

AASHTO HB-17 (2002; Errata 2003; Errata 2005, 17th Edition) Standard Specifications for Highway Bridges

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 318M (2014; ERTA 2015) Building Code Requirements for Structural Concrete & Commentary

ACI SP-66 (2004) ACI Detailing Manual

ASSOCIATION OF EDISON ILLUMINATING COMPANIES (AEIC)

AEIC CS8 (2013) Specification for Extruded Dielectric Shielded Power Cables Rated 5 Through 46 kV

ASTM INTERNATIONAL (ASTM)

ASTM B1 (2013) Standard Specification for Hard-Drawn Copper Wire

ASTM B3 (2013) Standard Specification for Soft or Annealed Copper Wire

ASTM B8 (2011; R 2017) Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft

ASTM B496 (2016; R 2021) Standard Specification for Compact Round Concentric-Lay-Stranded Copper Conductors

ASTM C32 (2023) Standard Specification for Sewer and Manhole Brick (Made from Clay or Shale)

ASTM C309 (2019) Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete

|           |                                                                                                                                                |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM C478 | (2018) Standard Specification for Circular Precast Reinforced Concrete Manhole Sections                                                        |
| ASTM C857 | (2016) Standard Practice for Minimum Structural Design Loading for Underground Precast Concrete Utility Structures                             |
| ASTM C990 | (2009; R 2019) Standard Specification for Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants |

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

|                      |                                                                                                                                                                                                                       |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IEEE 48              | (2020) Test Procedures and Requirements for Alternating-Current Cable Terminations Used on Shielded Cables Having Laminated Insulation Rated 2.5 kV through 765 kV or Extruded Insulation Rated 2.5 kV through 500 kV |
| IEEE 81              | (2012) Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System                                                                                                       |
| IEEE 386             | (2016) Separable Insulated Connector Systems for Power Distribution Systems Rated 2.5 kV through 35 kV                                                                                                                |
| IEEE 400.2           | (2013) Guide for Field Testing of Shielded Power Cable Systems Using Very Low Frequency (VLF)                                                                                                                         |
| IEEE 404             | (2012) Standard for Extruded and Laminated Dielectric Shielded Cable Joints Rated 2500 V to 500,000 V                                                                                                                 |
| IEEE C2              | (2023) National Electrical Safety Code                                                                                                                                                                                |
| IEEE C37.20.3        | (2013) Standard for Metal-Enclosed Interrupter Switchgear                                                                                                                                                             |
| IEEE Stds Dictionary | (2009) IEEE Standards Dictionary: Glossary of Terms & Definitions                                                                                                                                                     |

INSULATED CABLE ENGINEERS ASSOCIATION (ICEA)

|               |                                                        |
|---------------|--------------------------------------------------------|
| ICEA S-94-649 | (2021) Concentric Neutral Cables Rated 5 Through 46 KV |
|---------------|--------------------------------------------------------|

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

|          |                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------|
| NETA ATS | (2021) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems |
|----------|--------------------------------------------------------------------------------------------------|

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

|                          |                                                                                                                                                                                                                                                              |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ANSI C119.1              | (2023) Electric Connectors - Sealed Insulated Underground Connector Systems Rated 600 Volts                                                                                                                                                                  |
| NEMA C119.4              | (2011) Electric Connectors - Connectors for Use Between Aluminum-to-Aluminum or Aluminum-to-Copper Conductors Designed for Normal Operation at or Below 93 Degrees C and Copper-to-Copper Conductors Designed for Normal Operation at or Below 100 Degrees C |
| NEMA RN 1                | (2005; R 2013) Polyvinyl-Chloride (PVC) Externally Coated Galvanized Rigid Steel Conduit and Intermediate Metal Conduit                                                                                                                                      |
| NEMA TC 2                | (2020) Standard for Electrical Polyvinyl Chloride (PVC) Conduit                                                                                                                                                                                              |
| NEMA TC 3                | (2021) Polyvinyl Chloride (PVC) Fittings for Use With Rigid PVC Conduit and Tubing                                                                                                                                                                           |
| NEMA TC 9                | (2020) Standard for Fittings for Polyvinyl Chloride (PVC) Plastic Utilities Duct for Underground Installation                                                                                                                                                |
| NEMA WC 74/ICEA S-93-639 | (2022) 5-46 kV Shielded Power Cable for Use in the Transmission and Distribution of Electric Energy                                                                                                                                                          |

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|         |                                              |
|---------|----------------------------------------------|
| NFPA 70 | (2023; ERTA 4 2023) National Electrical Code |
|---------|----------------------------------------------|

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

|         |                                                                                 |
|---------|---------------------------------------------------------------------------------|
| TIA-758 | (2012b) Customer-Owned Outside Plant Telecommunications Infrastructure Standard |
|---------|---------------------------------------------------------------------------------|

U.S. DEPARTMENT OF AGRICULTURE (USDA)

|                    |                                       |
|--------------------|---------------------------------------|
| RUS Bull 1751F-644 | (2002) Underground Plant Construction |
|--------------------|---------------------------------------|

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

|               |                                                                                  |
|---------------|----------------------------------------------------------------------------------|
| CID A-A-60005 | (Basic; Notice 2) Frames, Covers, Gratings, Steps, Sump And Catch Basin, Manhole |
|---------------|----------------------------------------------------------------------------------|

UNDERWRITERS LABORATORIES (UL)

|       |                                                                                      |
|-------|--------------------------------------------------------------------------------------|
| UL 6  | (2022) UL Standard for Safety Electrical Rigid Metal Conduit-Steel                   |
| UL 44 | (2018; Reprint May 2021) UL Standard for Safety Thermoset-Insulated Wires and Cables |



|              |                                                                                                                                 |
|--------------|---------------------------------------------------------------------------------------------------------------------------------|
| UL 94        | (2023; Reprint May 2023) UL Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances |
| UL 467       | (2022) UL Standard for Safety Grounding and Bonding Equipment                                                                   |
| UL 486A-486B | (2018; Reprint Jul 2023) UL Standard for Safety Wire Connectors                                                                 |
| UL 510       | (2020; Dec 2022) UL Standard for Safety Polyvinyl Chloride, Polyethylene and Rubber Insulating Tape                             |
| UL 514A      | (2013; Reprint Jun 2022) UL Standard for Safety Metallic Outlet Boxes                                                           |
| UL 514B      | (2012; Reprint May 2020) Conduit, Tubing and Cable Fittings                                                                     |
| UL 651       | (2011; Reprint May 2022) UL Standard for Safety Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings                   |
| UL 854       | (2020; Reprint Jan 2022) Standard for Service-Entrance Cables                                                                   |
| UL 1072      | (2006; Reprint Apr 2020) Medium-Voltage Power Cables                                                                            |
| UL 1242      | (2006; Reprint Apr 2022) UL Standard for Safety Electrical Intermediate Metal Conduit -- Steel                                  |

## 1.2 RELATED REQUIREMENTS

Section 26 08 00 APPARATUS INSPECTION AND TESTING applies to this section, with the additions and modifications specified herein.

## 1.3 DEFINITIONS

- a. Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined in IEEE Stds Dictionary.
- b. In the text of this section, the words conduit and duct are used interchangeably and have the same meaning.
- c. In the text of this section, "medium voltage cable splices," and "medium voltage cable joints" are used interchangeably and have the same meaning.

## 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Precast Underground Structures; G

SD-03 Product Data

Medium Voltage Cable; G

Medium Voltage Cable Joints; G

Medium Voltage Cable Terminations; G

Precast Concrete Structures; G

Sealing Material

Pulling-In Irons

Manhole Frames and Covers; G

Handhole Frames and Covers; G

Composite/Fiberglass Handholes; G

Cable Supports (racks, arms and insulators); G

SD-06 Test Reports

Medium Voltage Cable Qualification and Production Tests; G

Existing Cable Assessment Tests; G

Field Acceptance Checks and Tests; G

Arc-proofing Test for cable fireproofing tape; G

Cable Installation Plan and Procedure; G

Six copies of the information described below in 8-1/2 by 11 inch binders having a minimum of three rings from which material may readily be removed and replaced, including a separate section for each cable pull. Separate sections by heavy plastic dividers with tabs, with all data sheets signed and dated by the person supervising the pull.

- a. Site layout drawing with cable pulls numerically identified.
- b. A list of equipment used, with calibration certifications. The manufacturer and quantity of lubricant used on pull.
- c. The cable manufacturer and type of cable.
- d. The dates of cable pulls, time of day, and ambient temperature.
- e. The length of cable pull and calculated cable pulling tensions.

f. The actual cable pulling tensions encountered during pull.

#### SD-07 Certificates

Cable splicer/terminator; G

Cable Installer Qualifications; G

### 1.5 QUALITY ASSURANCE

#### 1.5.1 Precast Underground Structures

Submittal required for each type used. Provide calculations and drawings for precast manholes and handholes bearing the seal of a registered professional engineer including:

- a. Material description (i.e.,  $f'_c$  and  $F_y$ )
- b. Manufacturer's printed assembly and installation instructions
- c. Design calculations
- d. Reinforcing shop drawings in accordance with ACI SP-66
- e. Plans and elevations showing opening and pulling-in iron locations and details

#### 1.5.2 Certificate of Competency for Cable Splicer/Terminator

Submit certification of the qualification of the cable splicer/terminator for approval, 30 days before splices or terminations are to be made in medium voltage (5 kV to 35 kV) cables. Include the training, and experience of the individual on the specific type and classification of cable to be provided under this contract. Indicate that the individual has had three or more years recent experience splicing and terminating medium voltage cables. List a minimum of three splices/terminations that have been in operation for more than one year. In addition, the individual may be required to perform a dummy or practice splice/termination in the presence of the Contracting Officer, before being approved as a qualified cable splicer. If that additional requirement is imposed, the Contractor must provide short sections of the approved types of cables along with the approved type of splice/termination kit, and detailed manufacturer's instructions for the cable to be spliced. The Contracting Officer reserves the right to require additional proof of competency or to reject the individual and call for certification of an alternate cable splicer.

#### 1.5.3 Cable Installer Qualifications

Provide at least one onsite person in a supervisory position with a documentable level of competency and experience to supervise all cable pulling operations. Provide a resume showing the cable installers' experience in the last three years, including a list of references complete with points of contact, addresses and telephone numbers. Cable installer must demonstrate experience with a minimum of three medium voltage cable installations. The Contracting Officer reserves the right to require additional proof of competency or to reject the individual and call for an alternate qualified cable installer.

#### 1.5.4 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship must be in accordance with the mandatory and advisory provisions of IEEE C2 and NFPA 70 unless more stringent requirements are specified or indicated.

#### 1.5.5 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year period must include applications of equipment and materials under similar circumstances and of similar size. The product must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

##### 1.5.5.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

##### 1.5.5.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site are not acceptable, unless specified otherwise.

## PART 2 PRODUCTS

### 2.1 CONDUIT, DUCTS, AND FITTINGS

#### 2.1.1 Rigid Metal Conduit

UL 6.

##### 2.1.1.1 Rigid Metallic Conduit, PVC Coated

NEMA RN 1, Type A40, except that hardness must be nominal 85 Shore A durometer, dielectric strength must be minimum 400 volts per mil at 60 Hz, and tensile strength must be minimum 3500 psi.

#### 2.1.2 Intermediate Metal Conduit

UL 1242.

##### 2.1.2.1 Intermediate Metal Conduit, PVC Coated

NEMA RN 1, Type A40, except that hardness must be nominal 85 Shore A

durometer, dielectric strength must be minimum 400 volts per mil at 60 Hz, and tensile strength must be minimum 3500 psi.

#### 2.1.3 Plastic Conduit for Direct Burial and Riser Applications

UL 651 and NEMA TC 2, EPC-40 or as indicated.

#### 2.1.4 Plastic Duct for Concrete Encasement

Provide Type EPC-40 per UL 651 and NEMA TC 2, as indicated.

#### 2.1.5 Duct Sealant

UL 94, Class HBF. Provide high-expansion urethane foam duct sealant that expands and hardens to form a closed, chemically and water resistant, rigid structure. Sealant must be compatible with common cable and wire jackets and capable of adhering to metals, plastics and concrete. Sealant must be capable of curing in temperature ranges of 35 degrees F to 95 degrees F. Cured sealant must withstand temperature ranges of -20 degrees F to 200 degrees F without loss of function.

#### 2.1.6 Fittings

##### 2.1.6.1 Metal Fittings

UL 514B.

##### 2.1.6.2 PVC Conduit Fittings

UL 514B, UL 651, and NEMA TC 3.

##### 2.1.6.3 PVC Duct Fittings

NEMA TC 9.

##### 2.1.6.4 Outlet Boxes for Steel Conduit

Outlet boxes for use with rigid or flexible steel conduit must be cast-metal cadmium or zinc-coated if of ferrous metal with gasketed closures and must conform to UL 514A.

### 2.2 LOW VOLTAGE INSULATED CONDUCTORS AND CABLES

Insulated conductors must be rated 600 volts and conform to the requirements of NFPA 70, including listing requirements. Wires and cables manufactured more than 12 months prior to date of delivery to the site are not acceptable. Service entrance conductors must conform to UL 854, type USE.

#### 2.2.1 Conductor Types

Cable and duct sizes indicated are for copper conductors and THHN/THWN unless otherwise noted. Conductors No. 10 AWG and smaller must be solid. Conductors No. 8 AWG and larger must be stranded. All conductors must be copper.

#### 2.2.2 Conductor Material

Unless specified or indicated otherwise or required by NFPA 70, wires in

conduit, other than service entrance, must be 600-volt, Type XHHW conforming to UL 44. Copper conductors must be annealed copper complying with ASTM B3 and ASTM B8.

#### 2.2.3 Jackets

Provide cables with an overall PVC outer jacket.

#### 2.2.4 Cable Marking

Insulated conductors must have the date of manufacture and other identification imprinted on the outer surface of each cable at regular intervals throughout the cable length.

Identify each cable by means of a fiber, laminated plastic, or non-ferrous metal tags in each manhole, handhole, junction box, and each terminal. Each tag must contain the following information; cable type, conductor size, circuit number, circuit voltage, cable destination and phase identification.

Color code conductors. Provide conductor identification within each enclosure where a tap, splice, or termination is made. Conductor identification must be by color-coded insulated conductors, plastic-coated self-sticking printed markers, colored nylon cable ties and plates, heat shrink type sleeves, or colored electrical tape. Properly identify control circuit terminations. Color must be green for grounding conductors and white for neutrals; except where neutrals of more than one system are installed in same raceway or box, other neutrals may be white with a different colored (not green) stripe for each. Color of ungrounded conductors in different voltage systems are as follows:

- a. 208/120 volt, three-phase
  - (1) Phase A - black
  - (2) Phase B - red
  - (3) Phase C - blue
- b. 480/277 volt, three-phase
  - (1) Phase A - brown
  - (2) Phase B - orange
  - (3) Phase C - yellow
- c. 120/240 volt, single phase: Black and red

#### 2.3 LOW VOLTAGE WIRE CONNECTORS AND TERMINALS

Provide a uniform compression over the entire conductor contact surface. Use solderless terminal lugs on stranded conductors.

- a. For use with copper conductors: UL 486A-486B.

#### 2.4 LOW VOLTAGE SPLICES

Provide splices in conductors with a compression connector on the

conductor and by insulating and waterproofing using one of the following methods which are suitable for continuous submersion in water and comply with ANSI C119.1.

#### 2.4.1 Heat Shrinkable Splice

Provide heat shrinkable splice insulation by means of a thermoplastic adhesive sealant material applied in accordance with the manufacturer's written instructions.

#### 2.4.2 Cold Shrink Rubber Splice

Provide a cold-shrink rubber splice which consists of EPDM rubber tube which has been factory stretched onto a spiraled core which is removed during splice installation. The installation must not require heat or flame, or any additional materials such as covering or adhesive. It must be designed for use with inline compression type connectors, or indoor, outdoor, direct-burial or submerged locations.

### 2.5 MEDIUM VOLTAGE CABLE

Cable (conductor) sizes are designated by American Wire Gauge (AWG) and Thousand Circular Mils (Kcmil). Conductor and conduit sizes indicated are for copper conductors unless otherwise noted. Insulated conductors must have the date of manufacture and other identification imprinted on the outer surface of each cable at regular intervals throughout cable length. Wires and cables manufactured more than 12 months prior to date of delivery to the site are not acceptable. Provide single conductor type cables unless otherwise indicated.

#### 2.5.1 Cable Configuration

Provide Type MV cable, conforming to NEMA WC 74/ICEA S-93-639 and UL 1072. Provide cables manufactured for use in duct applications as indicated. Cable must be rated 15 kV or 35 kV as indicated with 133 percent insulation level.

#### 2.5.2 Conductor Material

Provide concentric-lay-stranded, Class B conductors. Provide soft drawn copper cables complying with ASTM B3 and ASTM B8 for regular concentric and compressed stranding or ASTM B496 for compact stranding.

#### 2.5.3 Insulation

Provide ethylene-propylene-rubber (EPR) insulation conforming to the requirements of ANSI/NEMA WC 74/ICEA S-93-639 and ICEA S-94-649.

#### 2.5.4 Shielding

Cables rated for 2 kV and above must have a semiconducting conductor shield, a semiconducting insulation shield, and an overall copper tape shield for each phase.

#### 2.5.5 Neutrals

Neutral conductors must be copper, employing the same insulation and jacket materials as phase conductors.

#### 2.5.6 Jackets

Provide PVC jackets with a separator that prevents contact with underlying semiconducting insulating shield.

### 2.6 MEDIUM VOLTAGE CABLE TERMINATIONS

IEEE 48 Class 1; of the molded elastomer, prestretched elastomer, or heat-shrinkable elastomer. Acceptable elastomers are track-resistant silicone rubber or track-resistant ethylene propylene compounds, such as ethylene propylene rubber or ethylene propylene diene monomer. Separable insulated connectors may be used for apparatus terminations, when such apparatus is provided with suitable bushings. Provide terminations, where required, with mounting brackets suitable for the intended installation and with grounding provisions for the cable shielding, metallic sheath, or armor. Provide terminations in a kit, including: skirts, stress control terminator, ground clamp, connectors, lugs, and complete instructions for assembly and installation. Terminations must be the product of one manufacturer, suitable for the type, diameter, insulation class and level, and materials of the cable terminated. Do not use separate parts of copper or copper alloy in contact with aluminum alloy parts in the construction or installation of the terminator.

#### 2.6.1 Cold-Shrink Type

Terminator must be a one-piece design, utilizing the manufacturer's latest technology, where high-dielectric constant (capacitive) stress control is integrated within a skirted insulator made of silicone rubber. Termination must not require heat or flame for installation. Termination kit must contain all necessary materials (except for the lugs). Design termination for installation in low or highly contaminated indoor and outdoor locations and must resist ultraviolet rays and oxidative decomposition.

#### 2.6.2 Heat Shrinkable Type

Terminator must consist of a uniform cross section heat shrinkable polymeric construction stress relief tubing and environmentally sealed outer covering that is nontracking, resists heavy atmospheric contaminants, ultra violet rays and oxidative decomposition. Provide heat shrinkable sheds or skirts of the same material. Design termination for installation in low or highly contaminated indoor or outdoor locations.

#### 2.6.3 Separable Insulated Connector Type

IEEE 386. Provide connector with steel reinforced hook-stick eye, grounding eye, test point, and arc-quenching contact material. Provide connectors of the loadbreak or deadbreak type as indicated, of suitable construction for the application and the type of cable connected, and that include cable shield adaptors. Provide external clamping points and test points. Do not use separable connectors in manholes/handholes.

- a. 200 Ampere loadbreak connector ratings: Voltage: 15 kV, 95 kV BIL. Short time rating: 10,000 rms symmetrical amperes.
- b. 600 Ampere deadbreak connector ratings: Voltage: 15 kV, 95 kV BIL. Short time rating: 25,000 rms symmetrical amperes.



## 2.7 MEDIUM VOLTAGE CABLE JOINTS

Provide joints (splices) in accordance with IEEE 404 suitable for the rated voltage, insulation level, insulation type, and construction of the cable. Joints must be certified by the manufacturer for waterproof, submersible applications. Upon request, supply manufacturer's design qualification test report in accordance with IEEE 404. Connectors for joint must be tin-plated electrolytic copper, having ends tapered and having center stops to equalize cable insertion.

### 2.7.1 Heat-Shrinkable Joint

Consists of a uniform cross-section heat-shrinkable polymeric construction with a linear stress relief system, a high dielectric strength insulating material, and an integrally bonded outer conductor layer for shielding. Replace original cable jacket with a heavy-wall heat-shrinkable sleeve with hot-melt adhesive coating.

### 2.7.2 Cold-Shrink Rubber-Type Joint

Joint must be of a cold shrink design that does not require any heat source for its installation. Splice insulation and jacket must be of a one-piece factory formed cold shrink sleeve made of black EPDM rubber. Splice should be packaged three splices per kit, including complete installation instructions.

## 2.8 TELECOMMUNICATIONS CABLING

Provide telecommunications cabling in accordance with Section 33 82 00 TELECOMMUNICATIONS OUTSIDE PLANT (OSP).

### 2.9 LIVE END CAPS

Provide live end caps using a "kit" including a heat-shrinkable tube and a high dielectric strength, polymeric plug overlapping the conductor. Conform to applicable portions of IEEE 48.

### 2.10 TAPE

#### 2.10.1 Insulating Tape

UL 510, plastic insulating tape, capable of performing in a continuous temperature environment of 80 degrees C.

#### 2.10.2 Buried Warning and Identification Tape

Provide detectable tape in accordance with Section 31 00 00 EARTHWORK.

#### 2.10.3 Fireproofing Tape

Provide tape composed of a flexible, conformable, unsupported intumescent elastomer. Tape must be not less than .030 inch thick, noncorrosive to cable sheath, self-extinguishing, noncombustible, adhesive-free, and must not deteriorate when subjected to oil, water, gases, salt water, sewage, and fungus.

### 2.11 PULL ROPE

Plastic or flat pull line (bull line) having a minimum tensile strength of

200 pounds.

## 2.12 GROUNDING AND BONDING

### 2.12.1 Driven Ground Rods

Provide copper-clad steel ground rods conforming to UL 467 not less than 3/4 inch in diameter by 10 feet in length. Sectional type rods may be used for rods 20 feet or longer.

### 2.12.2 Grounding Conductors

Stranded-bare copper conductors must conform to ASTM B8, Class B, soft-drawn unless otherwise indicated. Solid-bare copper conductors must conform to ASTM B1 for sizes No. 8 and smaller. Insulated conductors must be of the same material as phase conductors and green color-coded, except that conductors must be rated no more than 600 volts. Aluminum is not acceptable.

## 2.13 CAST-IN-PLACE CONCRETE

Provide concrete in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE. In addition, provide concrete for encasement of underground ducts with 3000 psi minimum 28-day compressive strength. Concrete associated with electrical work for other than encasement of underground ducts must be 4000 psi minimum 28-day compressive strength unless specified otherwise.

## 2.14 UNDERGROUND STRUCTURES

Provide precast concrete underground structures or standard type cast-in-place manhole types as indicated, conforming to ASTM C857 and ASTM C478. Top, walls, and bottom must consist of reinforced concrete. Walls and bottom must be of monolithic concrete construction. Locate duct entrances and windows near the corners of structures to facilitate cable racking. Covers must fit the frames without undue play. Form steel and iron to shape and size with sharp lines and angles. Castings must be free from warp and blow holes that may impair strength or appearance. Exposed metal must have a smooth finish and sharp lines and arises. Provide necessary lugs, rabbets, and brackets. Set pulling-in irons and other built-in items in place before depositing concrete. Install a pulling-in iron in the wall opposite each duct line entrance. Cable racks, including rack arms and insulators, must be adequate to accommodate the cable.

### 2.14.1 Cast-In-Place Concrete Structures

Concrete must conform to Section 03 30 00 CAST-IN-PLACE CONCRETE. Construct walls on a footing of cast-in-place concrete except that precast concrete base sections may be used for precast concrete manhole risers.

### 2.14.2 Precast Concrete Structures, Risers and Tops

Precast concrete underground structures shall be provided in lieu of cast-in-place where practical and subject to the requirements specified below. Precast units must be the product of a manufacturer regularly engaged in the manufacture of precast concrete products, including precast manholes.

#### 2.14.2.1 General

Precast concrete structures must have the same accessories and facilities as required for cast-in-place structures. Likewise, precast structures must have plan area and clear heights not less than those of cast-in-place structures. Concrete materials and methods of construction must be the same as for cast-in-place concrete construction, as modified herein. Slope in floor may be omitted provided precast sections are poured in reinforced steel forms. Concrete for precast work must have a 28-day compressive strength of not less than 4000 psi. Structures may be precast to the design and details indicated for cast-in-place construction, precast monolithically and placed as a unit, or structures may be assembled sections, designed and produced by the manufacturer in accordance with the requirements specified. Structures must be identified with the manufacturer's name embedded in or otherwise permanently attached to an interior wall face.

#### 2.14.2.2 Design for Precast Structures

ACI 318Mand as shown on manhole structural drawings. In the absence of detailed on-site soil information, design for the following soil parameters/site conditions:

- a. Angle of Internal Friction ( $\phi$ ) = 30 degrees
- b. Unit Weight of Soil (Dry) = 120 pcf, (Saturated) = 130 pcf
- c. Coefficient of Lateral Earth Pressure ( $K_a$ ) = 0.33
- d. Ground Water Level = at ground elevation
- e. Vertical design loads must include full dead, superimposed dead, and live loads including a 30 percent magnification factor for impact. Live loads must consider all types and magnitudes of vehicular (automotive, industrial, or aircraft) traffic to be encountered. The minimum design vertical load must be for H20 highway loading per AASHTO HB-17.
- f. Horizontal design loads must include full geostatic and hydrostatic pressures for the soil parameters, water table, and depth of installation to be encountered. Also, horizontal loads imposed by adjacent structure foundations, and horizontal load components of vertical design loads, including impact, must be considered, along with a pulling-in iron design load of 6000 pounds.
- g. Each structural component must be designed for the load combination and positioning resulting in the maximum shear and moment for that particular component.
- h. Design must also consider the live loads induced in the handling, installation, and backfilling of the manholes. Provide lifting devices to ensure structural integrity during handling and installation.

#### 2.14.2.3 Construction

Provide a uniform thickness for structure top, bottom, and wall not less than 6 inches. Thin-walled knock-out panels for designed or future duct

bank entrances are not permitted. Provide quantity, size, and location of duct bank entrance windows as directed, and cast completely open by the precaster. Size of windows must exceed the nominal duct bank envelope dimensions by at least 12 inches vertically and horizontally to preclude in-field window modifications made necessary by duct bank misalignment. However, the sides of precast windows must be a minimum of 6 inches from the inside surface of adjacent walls, floors, or ceilings. Form the perimeter of precast window openings to have a keyed or inward flared surface to provide a positive interlock with the mating duct bank envelope. Provide welded wire fabric reinforcing through window openings for in-field cutting and flaring into duct bank envelopes. Provide additional reinforcing steel comprised of at least two No. 4 bars around window openings. Provide drain sumps a minimum of 12 inches in diameter and 4 inches deep for precast structures.

#### 2.14.2.4 Joints

Provide tongue-and-groove joints on mating edges of precast components. Shiplap joints are not allowed. Design joints to firmly interlock adjoining components and to provide waterproof junctions and adequate shear transfer. Seal joints watertight using preformed plastic strip conforming to ASTM C990. Install sealing material in strict accordance with the sealant manufacturer's printed instructions. Provide waterproofing at conduit/duct entrances into structures, and where access frame meets the top slab, provide continuous grout seal.

#### 2.14.3 Manhole Frames and Covers

Provide cast iron frames and covers for manholes conforming to CID A-A-60005. Cast the words "ELECTRIC" or "TELECOMMUNICATIONS" in the top face of power and telecommunications manhole covers, respectively, and as required in drawings.

#### 2.14.4 Handhole Frames and Covers

Frames and covers of steel must be welded by qualified welders in accordance with standard commercial practice. Provide rolled-steel floor plate covers having an approved antislip surface. Hinges must be of stainless steel with bronze hinge pin, 5 by 5 inches by approximately 3/16 inch thick, without screw holes, and must be for full surface application by fillet welding. Hinges must have nonremovable pins and five knuckles. The surfaces of plates under hinges must be true after the removal of raised antislip surface, by grinding or other approved method.

#### 2.14.5 Manhole Riser and Collars

Provide precast reinforced concrete grade rings, precast concrete risers, and sewer and manhole brick conforming to ASTM C32, Grade MS as required on drawings

### 2.15 CABLE SUPPORTS (RACKS, ARMS, AND INSULATORS)

Anchors shall be sleeve type and constructed of stainless steel.

#### 2.15.1 Cable Rack Stanchions

The wall bracket or stanchion must be 4 inches by approximately 1 inch glass-reinforced nylon with recessed bolt mounting holes, 48 inches long (minimum) in manholes. Space slots for mounting cable rack arms at 8 inch

intervals.

#### 2.15.2 Rack Arms

Cable rack arms must be glass reinforced nylon and must be of the removable type. Rack arm length must be a minimum of 8 inches and a maximum of 21 inches.

#### 2.15.3 Insulators

Insulators are not required for nylon arms.

#### 2.16 CABLE TAGS IN MANHOLES

Provide polyethylene tags for each power cable located in manholes. Do not provide handwritten letters. The first position on the power cable tag denotes the voltage. The second through sixth positions on the tag identifies the circuit. The next to last position denotes the phase of the circuit and include the Greek "phi" symbol. The last position denotes the cable size. As an example, a tag could have the following designation: "11.5 NAS 1-8(Phase A)500," denoting that the tagged cable is on the 11.5kV system circuit number NAS 1-8, underground, Phase A, sized at 500 kcmil.

##### 2.16.1 Polyethylene Cable Tags

Provide tags of polyethylene having an average tensile strength of 3250 pounds per square inch; and that are 0.08 inch thick (minimum), non-corrosive non-conductive; resistive to acids, alkalis, organic solvents, and salt water; and distortion resistant to 170 degrees F. Provide 0.05 inch (minimum) thick black polyethylene tag holder. Provide a one-piece nylon, self-locking tie at each end of the cable tag, having a minimum loop tensile strength of 175 pounds and black block letters, numbers, and symbols one inch high on a yellow background. Letters, numbers, and symbols must not fall off or change positions regardless of the cable tags' orientation.

#### 2.17 MEDIUM VOLTAGE ABOVE GROUND CABLE TERMINATING CABINETS

Cable terminating cabinets must be hook-stick operable, deadfront construction conforming to the requirements of IEEE C37.20.3, Category A. Provide cabinets with 600 A. dead-break junctions and elbow-type separable dead-break connectors, cable parking stands, and grounding lugs. Provide cable terminating equipment in conformance with IEEE 386.

Ratings at 60 Hz must be:

|                                                      |              |
|------------------------------------------------------|--------------|
| Nominal voltage (kV)                                 | As indicated |
| Rated maximum voltage (kV)                           | 15 or 35     |
| Rated continuous current (A)                         | 600          |
| One-second short-time current-carrying capacity (kA) | As indicated |

|          |              |
|----------|--------------|
| BIL (kV) | As indicated |
|----------|--------------|

## 2.18 LOW VOLTAGE ABOVE GROUND TERMINATION PEDESTAL

Provide copolymer polypropylene, low voltage above ground termination pedestal manufactured through an injection molding process. Pedestals must resist fertilizers, salt air environments and ultra-violet radiation. Pedestal top must be imprinted with a "WARNING" and "ELECTRIC" identification. Pedestal must contain three lay-in six port connectors, NEMA C119.4, Class "A", dual rated for aluminum or copper, and capable of terminating conductors ranging from 10 AWG to 500 kcmil. Protect each connector with a clear, hard lexan (plastic) cover. Provide pedestal with rust-free material and stainless steel hardware that is lockable.

## 2.19 SOURCE QUALITY CONTROL

### 2.19.1 Arc-Proofing Test for Cable Fireproofing Tape

Manufacturer must test one sample assembly consisting of a straight lead tube 12 inches long with a 2 1/2 inch outside diameter, and a 1/8 inch thick wall, and covered with one-half lap layer of arc and fireproofing tape per manufacturer's instructions. The arc and fireproofing tape must withstand extreme temperature of a high-current fault arc 13,000 degrees K for 70 cycles as determined by using an argon directed plasma jet capable of constantly producing and maintaining an arc temperature of 13,000 degrees K. Temperature (13,000 degrees K) of the ignited arc between the cathode and anode must be obtained from a dc power source of 305 (plus or minus 5) amperes and 20 (plus or minus 1) volts. Direct the arc toward the sample assembly accurately positioned 5 (plus or minus 1) millimeters downstream in the plasma from the anode orifice by fixed flow rate of argon gas (0.18 g per second). Test each sample assembly at three unrelated points. Start time for tests must be taken from recorded peak current when the specimen is exposed to the full test temperature. Surface heat on the specimen prior to that time must be minimal. The end point is established when the plasma or conductive arc penetrates the protective tape and strikes the lead tube. Submittals for arc-proofing tape must indicate that the test has been performed and passed by the manufacturer.

### 2.19.2 Medium Voltage Cable Qualification and Production Tests

Results of AEIC CS8 qualification and production tests as applicable for each type of medium voltage cable.

## 2.20 INNERDUCT

Innerduct shall be constructed of corrugated high density polyethylene. Product shall come in 5-unit assembly, 1.25-inch diameter each unit, with each unit of varying color. Pull tape shall be provided in each innerduct.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Install equipment and devices in accordance with the manufacturer's published instructions and with the requirements and recommendations of NFPA 70 and IEEE C2 as applicable. In addition to these requirements,

install telecommunications in accordance with TIA-758 and  
RUS Bull 1751F-644.

### 3.2 CABLE INSPECTION

Inspect each cable reel for correct storage positions, signs of physical damage, and broken end seals prior to installation. If end seal is broken, remove moisture from cable prior to installation in accordance with the cable manufacturer's recommendations.

### 3.3 CABLE INSTALLATION PLAN AND PROCEDURE

Obtain from the manufacturer an installation manual or set of instructions which addresses such aspects as cable construction, insulation type, cable diameter, bending radius, cable temperature limits for installation, lubricants, coefficient of friction, conduit cleaning, storage procedures, moisture seals, testing for and purging moisture, maximum allowable pulling tension, and maximum allowable sidewall bearing pressure. Perform pulling calculations and prepare a pulling plan and submit along with the manufacturer's instructions in accordance with SUBMITTALS. Install cable strictly in accordance with the cable manufacturer's recommendations and the approved installation plan.

Calculations and pulling plan must include:

- a. Site layout drawing with cable pulls identified in numeric order of expected pulling sequence and direction of cable pull.
- b. List of cable installation equipment.
- c. Lubricant manufacturer's application instructions.
- d. Procedure for resealing cable ends to prevent moisture from entering cable.
- e. Cable pulling tension calculations of all cable pulls.
- f. Cable percentage conduit fill.
- g. Cable sidewall bearing pressure.
- h. Cable minimum bend radius and minimum diameter of pulling wheels used.
- i. Cable jam ratio.
- j. Maximum allowable pulling tension on each different type and size of conductor.
- k. Maximum allowable pulling tension on pulling device.

### 3.4 UNDERGROUND STRUCTURE CONSTRUCTION

Provide standard type cast-in-place construction as specified herein and as indicated, or precast construction as specified herein. Horizontal concrete surfaces of floors must have a smooth trowel finish. Cure concrete by applying two coats of white pigmented membrane forming-curing compound in strict accordance with the manufacturer's printed instructions, except that precast concrete may be steam cured. Curing compound must conform to ASTM C309. Locate duct entrances and windows in

the center of end walls (shorter) and near the corners of sidewalls (longer) to facilitate cable racking and splicing. Covers for underground structures must fit the frames without undue play. Form steel and iron to shape and size with sharp lines and angles. Castings must be free from warp and blow holes that may impair strength or appearance. Exposed metal must have a smooth finish and sharp lines and arises. Provide necessary lugs, rabbets, and brackets. Set pulling-in irons and other built-in items in place before depositing concrete. Manhole locations, as indicated, are approximate. Coordinate exact manhole locations with other utilities and finished grading and paving.

#### 3.4.1 Cast-In-Place Concrete Structures

Construct walls on a footing of cast-in-place concrete except that precast concrete base sections may be used for precast concrete manhole risers.

#### 3.4.2 Precast Concrete Construction

Set commercial precast structures on 6 inches of level, 90 percent compacted granular fill, 3/4 inch to 1 inch size, extending 12 inches beyond the structure on each side. Compact granular fill by a minimum of four passes with a plate type vibrator. Installation must additionally conform to the manufacturer's instructions.

#### 3.4.3 Pulling-In Irons

Provide steel bars bent as indicated, and cast in the walls and floors. Alternatively, pipe sleeves may be precast into the walls and floors where required to accept U-bolts or other types of pulling-in devices possessing the strengths and clearances stated herein. The final installation of pulling-in devices must be made permanent. Cover and seal exterior projections of thru-wall type pulling-in devices with an appropriate protective coating. In the floor, locate the irons a minimum of 6 inches from the edge of the sump, and in the walls, locate the irons within 6 inches of the projected center of the duct bank pattern or precast window in the opposite wall. However, the pulling-in iron must not be located within 6 inches of an adjacent interior surface, or duct or precast window located within the same wall as the iron. If a pulling-in iron cannot be located directly opposite the corresponding duct bank or precast window due to this clearance limitation, locate the iron directly above or below the projected center of the duct bank pattern or precast window the minimum distance required to preserve the 6 inch clearance previously stated. In the case of directly opposing precast windows, pulling-in irons consisting of a 3 foot length of No. 5 reinforcing bar, formed into a hairpin, may be cast-in-place within the precast windows simultaneously with the end of the corresponding duct bank envelope. Irons installed in this manner must be positioned directly in line with, or when not possible, directly above or below the projected center of the duct bank pattern entering the opposite wall, while maintaining a minimum clear distance of 3 inches from any edge of the cast-in-place duct bank envelope or any individual duct. Pulling-in irons must have a clear projection into the structure of approximately 4 inches and must be designed to withstand a minimum pulling-in load of 6000 pounds. Hot-dip galvanize irons after fabrication.

#### 3.4.4 Cable Racks, Arms and Insulators

Cable racks, arms and insulators must be sufficient to accommodate the cables. Space racks in power manholes not more than 3 feet apart, and



provide each manhole wall with a minimum of two racks. Space racks in signal manholes not more than 16 1/2 inches apart with the end rack being no further than 12 inches from the adjacent wall. Methods of anchoring cable racks are as follows:

- a. Provide a 5/8 inch diameter by 5 inch long anchor bolt with 3 inch foot cast in structure wall with 2 inch protrusion of threaded portion of bolt into structure. Provide 5/8 inch steel square head nut on each anchor bolt. Coat threads of anchor bolts with suitable coating immediately prior to installing nuts.
- b. Provide concrete channel insert with a minimum load rating of 800 pounds per foot. Insert channel must be steel of the same length as "vertical rack channel;" and cast flush in structure wall. Provide 5/8 inch steel nuts in channel insert to receive 5/8 inch diameter by 3 inch long steel, square head anchor bolts.
- c. Provide concrete "spot insert" at each anchor bolt location, cast flush in structure wall. Each insert must have minimum 800 pound load rating. Provide 5/8 inch diameter by 3 inch long steel, square head anchor bolt at each anchor point. Coat threads of anchor bolts with suitable coating immediately prior to installing bolts.

#### 3.4.5 Field Painting

Clean cast-iron frames and covers not buried in concrete or masonry of mortar, rust, grease, dirt and other deleterious materials, and coat with bituminous paint.

### 3.5 UNDERGROUND CONDUIT AND DUCT SYSTEMS

#### 3.5.1 Requirements

Run conduit in straight lines except where a change of direction is necessary. Provide numbers and sizes of ducts as indicated. Provide a 4/0 AWG bare copper grounding conductor above medium-voltage distribution duct banks. Bond bare copper grounding conductor to ground rings (loops) in all manholes and to ground rings (loops) at all equipment slabs (pads). Route grounding conductor into manholes with the duct bank (sleeving is not required). Ducts must have a continuous slope downward toward underground structures and away from buildings, laid with a minimum slope of 3 inches per 100 feet. Depending on the contour of the finished grade, the high-point may be at a terminal, a manhole, a handhole, or between manholes or handholes. Terminate all PVC conduit end points in utility holes, switching cabinets, transform handholes and buildings with end bells. The bell end of the conduits that enter manholes and handholes must be flush with the wall.

Perform changes in ductbank direction as follows:

- a. Short-radius manufactured 90-degree duct bends may be used only for pole or equipment risers, unless specifically indicated as acceptable.
- b. The minimum manufactured bend radius must be 18 inches for ducts of less than 3 inch diameter, and 36 inches for ducts 3 inches or greater in diameter.
- c. As an exception to the bend radius required above, provide field manufactured longsweep bends having a minimum radius of 25 feet for a

change of direction of more than 5 degrees, either horizontally or vertically, using a combination of curved and straight sections. Maximum manufactured curved sections allowed for use in field manufactured longsweep bend: 30 degrees.

### 3.5.2 Treatment

Keep ducts clean of concrete, dirt, or foreign substances during construction. Make field cuts requiring tapers with proper tools and match factory tapers. Use a coupling recommended by the duct manufacturer whenever an existing duct is connected to a duct of different material or shape. Store ducts to avoid warping and deterioration with ends sufficiently plugged to prevent entry of any water or solid substances. Thoroughly clean ducts before being laid. Store plastic ducts on a flat surface and protected from the direct rays of the sun.

### 3.5.3 Conduit Cleaning

As each conduit run is completed, for conduit sizes 3 inches and larger, draw a flexible testing mandrel approximately 12 inches long with a diameter less than the inside diameter of the conduit through the conduit. After which, draw a stiff bristle brush through until conduit is clear of particles of earth, sand and gravel; then immediately install conduit plugs. For conduit sizes less than 3 inches, draw a stiff bristle brush through until conduit is clear of particles of earth, sand and gravel; then immediately install conduit plugs.

### 3.5.4 Multiple Conduits

Separate multiple conduits by a minimum distance of 3 inches, except that light and power conduits must be separated from control, signal, and telephone conduits by a minimum distance of 12 inches. Stagger the joints of the conduits by rows (horizontally) and layers (vertically) to strengthen the conduit assembly. Provide plastic duct spacers that interlock vertically and horizontally. Spacer assembly must consist of base spacers, intermediate spacers, ties, and locking device on top to provide a completely enclosed and locked-in conduit assembly. Install spacers per manufacturer's instructions, but provide a minimum of two spacer assemblies per 10 feet of conduit assembly.

### 3.5.5 Conduit Plugs and Pull Rope

Provide new conduit indicated as being unused or empty with plugs on each end. Plugs must contain a weephole or screen to allow water drainage. Provide a plastic pull rope having 3 feet of slack at each end of unused or empty conduits.

### 3.5.6 Conduit and Duct Without Concrete Encasement

Depths to top of the conduit must be not less than 24 inches below finished grade. Provide not less than 3 inches clearance from the conduit to each side of the trench. Grade bottom of trench smooth; where rock, soft spots, or sharp-edged materials are encountered, excavate the bottom for an additional 3 inches, fill and tamp level with original bottom with sand or earth free from particles, that would be retained on a 1/4 inch sieve. The first 6 inch layer of backfill cover must be sand compacted as previously specified. The rest of the excavation must be backfilled and compacted in 3 to 6 inch layers. Provide color, type and depth of warning tape as specified in Section 31 00 00 EARTHWORK.

#### 3.5.6.1 Encasement Under Roads and Structures

Under roads, paved areas, and railroad tracks, install conduits in concrete encasement of rectangular cross-section providing a minimum of 3 inch concrete cover around ducts. Extend concrete encasement at least 5 feet beyond the edges of paved areas and roads, and 12 feet beyond the rails on each side of railroad tracks. Depths to top of the concrete envelope must be not less than 24 inches below finished grade, and under railroad tracks not less than 50 inches below the top of the rails.

#### 3.5.7 Duct Encased in Concrete

Construct underground duct lines of individual conduits encased in concrete. Depths to top of the concrete envelope must be not less than 18 inches below finished grade, except under roads and pavement, concrete envelope must be not less than 24 inches below finished grade, and under railroad tracks not less than 50 inches below the top of the rails. Do not mix different kinds of conduit in any one duct bank. Concrete encasement surrounding the bank must be rectangular in cross-section and provide at least 3 inches of concrete cover for ducts. Separate conduits by a minimum concrete thickness of 3 inches. Before pouring concrete, anchor duct bank assemblies, prevent floating during concrete pouring by driving reinforcing rods adjacent to duct spacer assemblies and attaching the rods to the spacer assembly. Provide steel reinforcing in the concrete envelope as indicated. Provide color, type and depth of warning tape as specified in Section 31 00 00 EARTHWORK.

##### 3.5.7.1 Connections to Manholes

Duct bank envelopes connecting to underground structures must be flared to have enlarged cross-section at the manhole entrance to provide additional shear strength. Dimensions of the flared cross-section must be larger than the corresponding manhole opening dimensions by no less than 12 inches in each direction. Perimeter of the duct bank opening in the underground structure must be flared toward the inside or keyed to provide a positive interlock between the duct bank and the wall of the structure. Use vibrators when this portion of the encasement is poured to assure a seal between the envelope and the wall of the structure.

##### 3.5.7.2 Connections to Existing Underground Structures

For duct bank connections to existing structures, break the structure wall out to the dimensions required and preserve steel in the structure wall. Cut steel and bend out to tie into the reinforcing of the duct bank envelope. Chip the perimeter surface of the duct bank opening to form a key or flared surface, providing a positive connection with the duct bank envelope.

##### 3.5.7.3 Connections to Existing Concrete Pads

For duct bank connections to concrete pads, break an opening in the pad out to the dimensions required and preserve steel in pad. Cut the steel and bend out to tie into the reinforcing of the duct bank envelope. Chip out the opening in the pad to form a key for the duct bank envelope.

##### 3.5.7.4 Connections to Existing Ducts

Where connections to existing duct banks are indicated, excavate the banks

to the maximum depth necessary. Cut off the banks and remove loose concrete from the conduits before new concrete-encased ducts are installed. Provide a reinforced concrete collar, poured monolithically with the new duct bank, to take the shear at the joint of the duct banks. Abandon in place those no longer used ducts and cables which do not interfere with the work.

#### 3.5.7.5 Partially Completed Duct Banks

During construction wherever a construction joint is necessary in a duct bank, prevent debris such as mud, and, and dirt from entering ducts by providing suitable conduit plugs. Fit concrete envelope of a partially completed duct bank with reinforcing steel extending a minimum of 2 feet back into the envelope and a minimum of 2 feet beyond the end of the envelope. Provide one No. 4 bar in each corner, 3 inches from the edge of the envelope. Secure corner bars with two No. 3 ties, spaced approximately one foot apart. Restrain reinforcing assembly from moving during concrete pouring.

#### 3.5.8 Duct Sealing

Seal all electrical penetrations for radon mitigation, maintaining integrity of the vapor barrier, and to prevent infiltration of air, insects, and vermin.

### 3.6 CABLE PULLING

Test existing duct lines with a mandrel and thoroughly swab out to remove foreign material before pulling cables. Pull cables down grade with the feed-in point at the manhole or buildings of the highest elevation. Use flexible cable feeds to convey cables through manhole opening and into duct runs. Do not exceed the specified cable bending radii when installing cable under any conditions, including turnups into switches, transformers, switchgear, switchboards, and other enclosures. Cable with tape shield must have a bending radius not less than 12 times the overall diameter of the completed cable. If basket-grip type cable-pulling devices are used to pull cable in place, cut off the section of cable under the grip before splicing and terminating.

#### 3.6.1 Cable Lubricants

Use lubricants that are specifically recommended by the cable manufacturer for assisting in pulling jacketed cables.

### 3.7 CABLES IN UNDERGROUND STRUCTURES

Do not install cables utilizing the shortest path between penetrations, but route along those walls providing the longest route and the maximum spare cable lengths. Form cables to closely parallel walls, not to interfere with duct entrances, and support on brackets and cable insulators. Support cable splices in underground structures by racks on each side of the splice. Locate splices to prevent cyclic bending in the spliced sheath. Install cables at middle and bottom of cable racks, leaving top space open for future cables, except as otherwise indicated for existing installations. Provide one spare three-insulator rack arm for each cable rack in each underground structure.

### 3.7.1 Cable Tag Installation

Install cable tags in each manhole as specified, including each splice. Tag wire and cable provided by this contract. Install cable tags over the fireproofing, if any, and locate the tags so that they are clearly visible without disturbing any cabling or wiring in the manholes.

### 3.8 CONDUCTORS INSTALLED IN PARALLEL

Group conductors such that each conduit of a parallel run contains one Phase A conductor, one Phase B conductor, one Phase C conductor, and one neutral conductor.

### 3.9 LOW VOLTAGE CABLE SPLICING AND TERMINATING

Make terminations and splices with materials and methods as indicated or specified herein and as designated by the written instructions of the manufacturer. Do not allow the cables to be moved until after the splicing material has completely set. Make splices in underground distribution systems only in accessible locations such as basements, vaults, manholes, handholes, or aboveground termination pedestals.

### 3.10 MEDIUM VOLTAGE CABLE TERMINATIONS

Make terminations in accordance with the written instruction of the termination kit manufacturer.

### 3.11 MEDIUM VOLTAGE CABLE JOINTS

Provide power cable joints (splices) suitable for continuous immersion in water. Make joints only in accessible locations in manholes or handholes by using materials and methods in accordance with the written instructions of the joint kit manufacturer.

#### 3.11.1 Joints in Shielded Cables

Cover the joined area with metallic tape, or material like the original cable shield and connect it to the cable shield on each side of the splice. Provide a bare copper ground connection brought out in a watertight manner and grounded to the manhole grounding loop as part of the splice installation. Ground conductors, connections, and rods must be as specified elsewhere in this section. Wire must be trained to the sides of the enclosure to prevent interference with the working area.

### 3.12 CABLE END CAPS

Cable ends must be sealed at all times with coated heat shrinkable end caps. Cables ends must be sealed when the cable is delivered to the job site, while the cable is stored and during installation of the cable. The caps must remain in place until the cable is spliced or terminated. Sealing compounds and tape are not acceptable substitutes for heat shrinkable end caps. Cable which is not sealed in the specified manner at all times will be rejected.

### 3.13 LIVE END CAPS

Provide live end caps for single conductor medium voltage cables where indicated.

### 3.14 FIREPROOFING OF CABLES IN UNDERGROUND STRUCTURES

Fireproof (arc proof) wire and cables which will carry current at 2200 volts or more in underground structures.

#### 3.14.1 Fireproofing Tape

Tightly wrap strips of fireproofing tape around each cable spirally in half-lapped wrapping. Install tape in accordance with manufacturer's instructions.

#### 3.14.2 Tape-Wrap

Tape-wrap metallic-sheathed or metallic armored cables without a nonmetallic protective covering over the sheath or armor prior to application of fireproofing. Wrap must be in the form of two tightly applied half-lapped layers of a pressure-sensitive 10 mil thick plastic tape, and must extend not less than one inch into the duct. Even out irregularities of the cable, such as at splices, with insulation putty before applying tape.

### 3.15 GROUNDING SYSTEMS

NFPA 70 and IEEE C2, except provide grounding systems with a resistance to solid earth ground not exceeding 5 ohms.

#### 3.15.1 Grounding Electrodes

Provide cone pointed driven ground rods driven full depth plus 6 inches, installed to provide an earth ground of the appropriate value for the particular equipment being grounded.

If the specified ground resistance is not met, provide an additional ground rod in accordance with the requirements of NFPA 70 (placed not less than 6 feet from the first rod). Should the resultant (combined) resistance exceed the specified resistance, measured not less than 48 hours after rainfall, notify the Contracting Officer immediately.

#### 3.15.2 Grounding Connections

Make grounding connections which are buried or otherwise normally inaccessible, by exothermic weld or compression connector.

- a. Make exothermic welds strictly in accordance with the weld manufacturer's written recommendations. Welds which are "puffed up" or which show convex surfaces indicating improper cleaning are not acceptable. Mechanical connectors are not required at exothermic welds.
- b. Make compression connections using a hydraulic compression tool to provide the correct circumferential pressure. Tools and dies must be as recommended by the manufacturer. An embossing die code or other standard method must provide visible indication that a connector has been adequately compressed on the ground wire.

#### 3.15.3 Grounding Conductors

Provide bare grounding conductors, except where installed in conduit with associated phase conductors. Ground cable sheaths, cable shields, conduit, and equipment with No. 6 AWG. Ground other noncurrent-carrying

metal parts and equipment frames of metal-enclosed equipment. Ground metallic frames and covers of handholes and pull boxes with a braided, copper ground strap with equivalent ampacity of No. 6 AWG. Provide direct connections to the grounding conductor with 600 v insulated, full-size conductor for each grounded neutral of each feeder circuit, which is spliced within the manhole.

#### 3.15.4 Ground Cable Crossing Expansion Joints

Protect ground cables crossing expansion joints or similar separations in structures and pavements by use of approved devices or methods of installation which provide the necessary slack in the cable across the joint to permit movement. Use stranded or other approved flexible copper cable across such separations.

#### 3.15.5 Manhole Grounding

Loop a 4/0 AWG grounding conductor around the interior perimeter, approximately 12 inches above finished floor. Secure the conductor to the manhole walls at intervals not exceeding 36 inches. Connect the conductor to the manhole grounding electrode with 4/0 AWG conductor. Connect all incoming 4/0 grounding conductors to the ground loop adjacent to the point of entry into the manhole. Bond the ground loop to all cable shields, metal cable racks, and other metal equipment with a minimum 6 AWG conductor.

#### 3.15.6 Fence Grounding

Provide grounding for fences as indicated. Drive ground rods until the top is 12 inches below grade. Attach a No. 4 AWG copper conductor, by exothermic weld to the ground rods and extend underground to the immediate vicinity of fence post. Lase the conductor vertically into 12 inches of fence mesh and fasten by two approved bronze compression fittings, one to bond wire to post and the other to bond wire to fence. Bond each gate section to its gatepost by a 1/8 by one inch flexible braided copper strap and ground post clamps. Clamps must be of the anti-electrolysis type.

### 3.16 EXCAVATING, BACKFILLING, AND COMPACTING

Provide in accordance with NFPA 70 and Section 31 00 00 EARTHWORK.

#### 3.16.1 Reconditioning of Surfaces

##### 3.16.1.1 Unpaved Surfaces

Restore to their original elevation and condition unpaved surfaces disturbed during installation of duct. Preserve sod and topsoil removed during excavation and reinstall after backfilling is completed. Replace sod that is damaged by sod of quality equal to that removed. When the surface is disturbed in a newly seeded area, re-seed the restored surface with the same quantity and formula of seed as that used in the original seeding, and provide topsoiling, fertilizing, liming, seeding, sodding, sprigging, or mulching. Provide work in accordance with Section 32 92 19 SEEDING and Section 32 93 00 EXTERIOR PLANTS.

##### 3.16.1.2 Paving Repairs

Where trenches, pits, or other excavations are made in existing roadways and other areas of pavement where surface treatment of any kind exists ,

restore such surface treatment or pavement the same thickness and in the same kind as previously existed, except as otherwise specified or indicated in details, and to match and tie into the adjacent and surrounding existing surfaces. Make repairs as specified in Section 32 13 13.06 PORTLAND CEMENT CONCRETE PAVEMENT FOR ROADS AND SITE FACILITIES.

### 3.17 CAST-IN-PLACE CONCRETE

Provide concrete in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE.

#### 3.17.1 Concrete Slabs (Pads) for Equipment

Unless otherwise indicated, the slab must be at least 8 inches thick, reinforced with a 6 by 6 - W2.9 by W2.9 mesh, placed uniformly 4 inches from the top of the slab. Place slab on a 6 inch thick, well-compacted gravel base. Top of concrete slab must be approximately 4 inches above finished grade with gradual slope for drainage. Edges above grade must have 1/2 inch chamfer. Slab must be of adequate size to project at least 8 inches beyond the equipment.

Stub up conduits, with bushings, 2 inches into cable wells in the concrete pad. Coordinate dimensions of cable wells with transformer cable training areas.

#### 3.17.2 Sealing

When the installation is complete, seal all conduit and other entries into the equipment enclosure with an approved sealing compound. Seals must be of sufficient strength and durability to protect all energized live parts of the equipment from rodents, insects, or other foreign matter.

### 3.18 FIELD QUALITY CONTROL

#### 3.18.1 Performance of Field Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations, and include the following visual and mechanical inspections and electrical tests, performed in accordance with NETA ATS.

##### 3.18.1.1 Medium Voltage Cables

Perform tests after installation of cable, splices, and terminators and before terminating to equipment or splicing to existing circuits.

##### a. Visual and Mechanical Inspection

- (1) Inspect exposed cable sections for physical damage.
- (2) Verify that cable is supplied and connected in accordance with contract plans and specifications.
- (3) Inspect for proper shield grounding, cable support, and cable termination.
- (4) Verify that cable bends are not less than ICEA or manufacturer's minimum allowable bending radius.
- (5) Inspect for proper fireproofing.



(6) Visually inspect jacket and insulation condition.

(7) Inspect for proper phase identification and arrangement.

b. Electrical Tests

(1) Perform a shield continuity test on each power cable by ohmmeter method. Record ohmic value, resistance values in excess of 10 ohms per 1000 feet of cable must be investigated and justified.

(2) Perform acceptance test on new cables before the new cables are connected to existing cables and placed into service, including terminations and joints. Submit results per submittals for Existing Cable Assessment Tests.

(3) Perform maintenance test on complete cable system after the new cables are new cables connected to existing cables and placed into service, including existing cable, terminations, and joints. Tests must be very low frequency (VLF) alternating voltage withstand tests in accordance with IEEE 400.2. VLF test frequency must be 0.05 Hz minimum for a duration of 60 minutes using a sinusoidal waveform. Test voltages must be as follows:

| CABLE RATING AC TEST VOLTAGE for<br>ACCEPTANCE TESTING |                |
|--------------------------------------------------------|----------------|
| 5 kV                                                   | 10kV rms(peak) |
| 8 kV                                                   | 13kV rms(peak) |
| 15 kV                                                  | 20kV rms(peak) |
| 25 kV                                                  | 31kV rms(peak) |
| 35 kV                                                  | 44kV rms(peak) |

| CABLE RATING AC TEST VOLTAGE for<br>MAINTENANCE TESTING |                |
|---------------------------------------------------------|----------------|
| 5 kV                                                    | 7kV rms(peak)  |
| 8 kV                                                    | 10kV rms(peak) |
| 15 kV                                                   | 16kV rms(peak) |
| 25 kV                                                   | 23kV rms(peak) |
| 35 kV                                                   | 33kV rms(peak) |

3.18.1.2 Low Voltage Cables, 600-Volt

Perform tests after installation of cable, splices and terminations and before terminating to equipment or splicing to existing circuits.

a. Visual and Mechanical Inspection

- (1) Inspect exposed cable sections for physical damage.
- (2) Verify that cable is supplied and connected in accordance with contract plans and specifications.
- (3) Verify tightness of accessible bolted electrical connections.
- (4) Inspect compression-applied connectors for correct cable match and indentation.
- (5) Visually inspect jacket and insulation condition.
- (6) Inspect for proper phase identification and arrangement.

b. Electrical Tests

- (1) Perform insulation resistance tests on wiring No. 6 AWG and larger diameter using instrument which applies voltage of approximately 1000 volts dc for one minute.
- (2) Perform continuity tests to ensure correct cable connection.

3.18.1.3 Grounding System

a. Visual and mechanical inspection

Inspect ground system for compliance with contract plans and specifications.

b. Electrical tests

Perform ground-impedance measurements utilizing the fall-of-potential method in accordance with IEEE 81. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of a single ground rod perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground resistance tester in accordance with manufacturer's instructions to test each ground or group of grounds. The instrument must be equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test. Provide site diagram indicating location of test probes with associated distances, and provide a plot of resistance vs. distance.

3.18.2 Follow-Up Verification

Upon completion of acceptance checks and tests, show by demonstration in service that circuits and devices are in good operating condition and properly performing the intended function. As an exception to requirements stated elsewhere in the contract, the Contracting Officer must be given 5 working days advance notice of the dates and times of checking and testing.

.... -- End of Section --

SECTION 35 45 01

VERTICAL PUMPS, MIXED-FLOW IMPELLER-TYPE  
**02/21, CHG 3: 08/23**

PART 1 GENERAL

1.1 PRICES

1.1.1 Vertical Pumps, Mixed-Flow Impeller-Type

1.1.1.1 Payment

Payment will be made for costs associated with furnishing and installing the vertical pumps, and mixed-flow impeller-type, as specified.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN PETROLEUM INSTITUTE (API)

API RP 686 (2009) Recommended Practice for Machinery Installation and Installation Design

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B17.1 (1967; R 2017) Keys and Keyseats

ASME B31.1 (2022) Power Piping

ASME B46.1 (2020) Surface Texture, Surface Roughness, Waviness and Lay

ASME BPVC SEC IX (2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C200 (2012) Steel Water Pipe - 6 In. (150 mm) and Larger

AWWA C203 (2020) Coal-Tar Protective Coatings and Linings for Steel Water Pipelines - Enamel and Tape - Hot-Applied

AWWA C207 (2018) Standard for Steel Pipe Flanges for Waterworks Service, Sizes 4 in. through 144 in. (100 mm through 3600 mm)

AWWA C208 (2022) Dimensions for Fabricated Steel Water Pipe Fittings

AWWA M11 (2016) Steel Pipe: A Guide for Design and Installation

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding  
Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM A27/A27M (2020) Standard Specification for Steel  
Castings, Carbon, for General Application

ASTM A36/A36M (2019) Standard Specification for Carbon  
Structural Steel

ASTM A48/A48M (2022) Standard Specification for Gray  
Iron Castings

ASTM A108 (2024) Standard Specification for Steel  
Bar, Carbon and Alloy, Cold-Finished

ASTM A217/A217M (2022) Standard Specification for Steel  
Castings, Martensitic Stainless and Alloy,  
for Pressure-Containing Parts, Suitable  
for High-Temperature Service

ASTM A242/A242M (2013; R 2018) Standard Specification for  
High-Strength Low-Alloy Structural Steel

ASTM A276/A276M (2024) Standard Specification for  
Stainless Steel Bars and Shapes

ASTM A285/A285M (2017) Standard Specification for Pressure  
Vessel Plates, Carbon Steel, Low- and  
Intermediate-Tensile Strength

ASTM A312/A312M (2022a) Standard Specification for  
Seamless, Welded, and Heavily Cold Worked  
Austenitic Stainless Steel Pipes

ASTM A351/A351M (2018) Standard Specification for  
Castings, Austenitic, for  
Pressure-Containing Parts

ASTM A352/A352M (2021) Standard Specification for Steel  
Castings, Ferritic and Martensitic, for  
Pressure-Containing Parts, Suitable for  
Low-Temperature Service

ASTM A516/A516M (2017) Standard Specification for Pressure  
Vessel Plates, Carbon Steel, for Moderate-  
and Lower-Temperature Service

ASTM A576 (2023) Standard Specification for Steel  
Bars, Carbon, Hot-Wrought, Special Quality

ASTM A668/A668M (2023) Standard Specification for Steel  
Forgings, Carbon and Alloy, for General  
Industrial Use

|                 |                                                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------|
| ASTM A743/A743M | (2021) Standard Specification for Castings, Iron-Chromium, Iron-Chromium-Nickel, Corrosion Resistant, for General Application |
| ASTM B148       | (2014) Standard Specification for Aluminum-Bronze Sand Castings                                                               |
| ASTM B584       | (2022) Standard Specification for Copper Alloy Sand Castings for General Applications                                         |
| ASTM D2000      | (2018) Standard Classification System for Rubber Products in Automotive Applications                                          |
| ASTM E165/E165M | (2023) Standard Practice for Liquid Penetrant Examination for General Industry                                                |
| ASTM F1476      | (2007; R 2019) Standard Specification for Performance of Gasketed Mechanical Couplings for Use in Piping Applications         |

HYDRAULIC INSTITUTE (HI)

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| HI 9.1-9.5      | (2021) Pumps General Guidelines for Materials, Sound Testing, and Decontamination - B117 |
| HI 9.6.4        | (2009) Rotodynamic Pumps for Vibration Analysis and Allowable Values                     |
| HI ANSI/HI 14.6 | (2011) Rotodynamic Pumps for Hydraulic Performance Acceptance Tests - A136               |

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

|          |                                                                                                  |
|----------|--------------------------------------------------------------------------------------------------|
| NETA ATS | (2021) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems |
|----------|--------------------------------------------------------------------------------------------------|

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

|             |                                                                                                                                                                                                                                                                                  |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ISO 1940-1  | (2003; R 2008) Mechanical Vibration - Balance Quality Requirements for Rotors in a Constant (Rigid) State - Part 1: Specification and Verification of Balance Tolerances                                                                                                         |
| ISO 10816-3 | (2017, AMD 1) Mechanical Vibration - Evaluation of Machine Vibration by Measurements on Non-rotating Parts - Part 3: Industrial Machines with Nominal Power above 15 kW and Nominal Speeds Between 120 r/min and 15 000 r/min when Measured in situ AMENDMENT 1 - Second Edition |

INTERNATIONAL SOCIETY OF AUTOMATION (ISA)

|           |                         |
|-----------|-------------------------|
| ISA RP2.1 | (1978) Manometer Tables |
|-----------|-------------------------|

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA 800-R-11-002

(2011) Environmentally Acceptable  
Lubricants

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Contractor Selected Manufacturers; G, AE

SD-02 Shop Drawings

Detail Drawings; G, AE

SD-03 Product Data

Materials; G, AE

Shipping Bills

Pump Curves; G, AE

Installation and Erection Instructions Manual; G, AE

Field Tests; G, AE

In Water Bearing Lubrication; G, AE

Grout; G, AE

SD-04 Samples

Materials; G, AE

SD-05 Design Data

Stress-Relieving Procedure; G, AE

Baseplates and the Anchoring Bolts; G, AE

SD-06 Test Reports

Factory Test; G, AE

Balancing; G, AE

Field Test Results; G, AE

SD-07 Certificates

Qualified Welders; G

Examination Procedure And Qualification Of The Examiner; G

SD-10 Operation and Maintenance Data

Operation and Maintenance Instructions Manual; G, AE

1.4 QUALITY ASSURANCE

1.4.1 Detail Drawings

Submit drawings of sufficient size to be easily read, within 90 days of notice of award of contract. Submit information in the English language. Provide with English (IP) dimensions. Drawings must consist of complete designs of the pump, pump installation instructions, performance charts, brochures, and other information required to illustrate that the entire pumping system (including the pump, motor has been coordinated and will function as a unit.

- a. Outline drawings of the pump showing pertinent dimensions and weight of each component of the pump. Prepare drawings to scale.
- b. Drawing showing details and dimensions of pump mounting design or layout including any embedded items.
- c. Cross-sectional drawings of the pump showing each component. Show major or complicated sections of the pump in detail. Indicate on each drawing an itemized list of components showing type, grade, and class of material used and make and model number of standard component used.
- d. Detail and assembly drawings of entire pump. Include all dimensions required to manufacture pump.
- e. Drawings covering erection and installation, that are intended to be furnished to the erecting engineer.
- f. Drawings of the pump; base plate showing its dimensions.

1.4.2 Welding

Weld structural members in accordance with Section 05 05 23.16 STRUCTURAL WELDING. For all other welding, procedures and welders must be qualified in accordance with ASME BPVC SEC IX. Welding procedures qualified by others, and welders and welding operators qualified by a previously qualified employer may be accepted as permitted by ASME B31.1. Perform welder qualification tests for each welder whose qualifications are not in compliance with the referenced standards. Notify the Contracting Officer 24 hours in advance of qualification tests. Perform the qualification tests at the work site if practical. The welder or welding operator must apply their assigned symbol near each weld made as a permanent record.

The names of all qualified welders, their identifying symbols, and the qualifying procedures for each welder including support data such as test procedures used, and standards tested to.

1.5 DELIVERY, STORAGE, AND HANDLING

1.5.1 General

Outfit major pump components with lifting lugs or eye bolts to facilitate handling. Design and arrange lugs or bolts to allow safe handling of pump

components singly or collectively as required during shipping, installation, and maintenance. Submit copies of certified shipping bills to the Contracting Officer or memorandums of all shipments of finished pieces or members to designated site, giving designation mark and weight of each piece, number of pieces, total weight, and if shipped by rail in carload lots, car initial and number.

#### 1.5.2 Processing for Storage

Prepare pumps and spare parts for storage indoors. Indoor storage consists of a permanent building that has a leak-proof roof, full walls to contain stored equipment, and a concrete floor or temporary trailers. A temporary structure may also be built at the job site for equipment storage that will contain features of the permanent building above except that provision for ventilation must be provided and floor may be crushed rock. Provide a vapor barrier below the crushed rock. Use cribbing below equipment stored on crushed rock so that equipment does not come in contact with crushed rock. Provide a plastic barrier between equipment and wood cribbing. Submit a list of equipment and materials requiring to the Contracting Officer no later than 30 days prior to shipment of pumping units. Store long term (greater than 6 months) in accordance with Contractor selected pump specifications.

### 1.6 PROJECT/SITE CONDITIONS

#### 1.6.1 Static Head

Static head is the difference, in feet, between water surface elevation in the immediately inside trash rack and water surface elevation of river. Total head includes static head, friction losses outside of equipment being furnished, plus velocity head loss.

### 1.7 MAINTENANCE

#### 1.7.1 Special Tools

Include one set of special tools required to completely assemble, disassemble, or maintain the pumps. Special tools refer to oversized or specially dimensioned tools, special attachments or fixtures, or any similar items. If required, provide a device for temporarily supporting the pump shaft and impeller during assembly, disassembly, and reassembly of the motor when the thrust bearing is not in place.

### 1.8 WARRANTY

Provide a warranty for all equipment included under this section against defective workmanship, materials, design, and performance for a period of 12 months from the date the equipment is accepted. If the equipment or part does not conform to these warranties, and the Government notifies the Contractor within a reasonable time after its discovery, the Contractor must promptly correct such nonconformity by repair or replacement. Coordinate the down time and repair for the equipment with the Contracting Officer. The Contractor is liable during the warranty period for the direct cost of removal of the equipment from the installed location, transportation for repair, and reinstallation on site. The expense of removing adjacent apparatus, installing spare equipment, costs of supplying temporary service, is not included in this warranty provision.



## PART 2 PRODUCTS

### 2.1 SYSTEM DESCRIPTION

Design and provide identical, vertical, mixed-flow, single stage impeller-type pumps, as indicated. The pumping systems include pumps and electric motor pump drives.

#### 2.1.1 Design Requirements

- a. Pumps are for the purpose of pumping seawater from B242 into to B72. Water pumped will not exceed 68 degrees F, will be relatively turbid, and may contain sand and silt capable of passing through the intake screen. An intake screen will have 0.0315 inch clear openings.
- b. Design pumps so that no major modifications, alterations, or additions will be required to the pumping station to accommodate them. Design pumps so that pump parts will fit within the limiting horizontal and vertical dimensions shown and that installation and maintenance can be accomplished by the interior; overhead traveling crane. Pumps must be capable of being lowered through floor openings shown with a minimum of 1 inch clearance around each side.
- c. Design documents must be signed and sealed by a professional engineer.

#### 2.1.2 Capacities

Each pump must:

- a. The guarantee point for the pump is 780 gal/min with a differential total dynamic head pressure of 64 feet with water surface in sump at Elevation -0.25 ft (NAV 88) feet. Note high water level is 4.18 ft and low water elevation is -4.71 ft. With both pumps in operation, the system shall achieve 1,550 gal/min with a differential total dynamic head pressure of 72 feet. 1

#### 2.1.3 Pump Curves

Indicate on the pump curves for the submitted pumps that the pumps are capable of operating over the entire total head corresponding to the static head range.

### 2.2 MATERIALS

Material selection not specified is guided by HI 9.1-9.5 for corrosion, erosion, and abrasion resistance. Submit copies of purchase orders, deviations from the specified materials, mill orders, shop orders for materials, and work orders, including orders placed or extended by each supplier. Furnish a list designating materials to be used for each item at time of submittal of drawings. Within 60 days of notice of award submit names of Contractor selected manufacturers of machinery and other equipment contemplated to be incorporated in the work, together with performance capacities and other relevant information pertaining to the equipment. Submit samples of materials as directed. Equipment, materials, and articles installed or used without the approval of the Contracting Officer are at risk of subsequent rejection.

- a. Identify each pump by means of a separate nameplate permanently affixed in a conspicuous location. The plate must bear the

manufacturer's name, model designation, serial number if applicable, and other pertinent information e.g. horsepower, speed, capacity, type, direction of rotation. Make the plate of corrosion-resisting metal with raised or depressed lettering and a contrasting background.

- b. Equip each pump with suitably located instruction plates, including any warnings and cautions, describing any special and important procedures to be followed in starting, operating, and servicing the equipment. Make the plates of corrosion-resisting metal with raised or depressed lettering and contrasting background.
- c. Provide safety guards and/or covers wherever necessary to protect the operators from accidental contact with moving parts. Make guards and covers of sheet steel, expanded metal, or another acceptable material and removable for disassembly of the pump.

## 2.3 METALWORK FABRICATION

### 2.3.1 Designated Materials

Designated materials must conform to the following specifications, grades, and classifications.

| MATERIALS                                           | SPECIFICATION   | GRADE, CLASS                                      |
|-----------------------------------------------------|-----------------|---------------------------------------------------|
| Aluminum-Bronze                                     | ASTM B148       | Alloy No. C95500 Castings                         |
| Cast Iron                                           | ASTM A48/A48M   | Class Nos. 150A 150B, and 150C; 30A, 30B, and 30C |
| Cast Steel                                          | ASTM A27/A27M   | Grade 65-35, annealed                             |
| Cast Stainless Steel                                | ASTM A743/A743M | CF8                                               |
| Coal Tar Protective Coatings                        | AWWA C203       |                                                   |
| Cold-Rolled Steel Bars                              | ASTM A108       | min, Wt. Strm 65,000 psi                          |
| Copper Alloy Castings                               | ASTM B584       | Alloy No. C93700                                  |
| Corrosion-Resistant Alloy Casting                   | ASTM A217/A217M | Grade CA15                                        |
|                                                     | ASTM A352/A352M | CA6NM                                             |
|                                                     | ASTM A351/A351M | CF8M                                              |
| Dimensions for Steel Water Piping Fittings          | AWWA C208       |                                                   |
| Hot-Rolled Stainless                                | ASTM A576       | Graded G10200 and G11410                          |
| Ring Flanges                                        | AWWA C207       | Class B                                           |
| Rubber Products in Automotive Applications          | ASTM D2000      |                                                   |
| Seamless and Welded Austenitic Stainless Steel Pipe | ASTM A312/A312M |                                                   |
| Stainless Bars and Shapes                           | ASTM A276/A276M | Grades S30400 and S41000                          |
| Steel Forging                                       | ASTM A668/A668M | Class F                                           |
| Cast Stainless Steel                                | ASTM A743/A743M | CF8                                               |

| MATERIALS                                              | SPECIFICATION   | GRADE, CLASS |
|--------------------------------------------------------|-----------------|--------------|
| Steel Pipe<br>6 inch and Larger                        | AWWA C200       |              |
| Steel Plates, Pressure Vessel                          | ASTM A516/A516M | Grade 55     |
| Steel Plates, Pressure Vessel                          | ASTM A242/A242M |              |
| Steel Plate, Structural                                | ASTM A285/A285M | Grade B      |
| Structural Steel                                       | ASTM A36/A36M   |              |
| Surface Texture (Surface Roughness, Waviness, and Lay) | ASME B46.1      |              |

## 2.3.2 Bolted Connections

### 2.3.2.1 Bolts, Nuts, and Washers

Bolts, nuts, and washers must conform to requirements of paragraphs under METALWORK FABRICATION, and paragraphs under VERTICAL PUMPS for types required. Use beveled washers where bearing faces have a slope of more than 1:20 with respect to a plane normal to bolt axis.

### 2.3.2.2 Materials Not Specifically Described

Materials not specifically described must conform to latest ASTM specification or to other listed commercial specifications covering class or kinds of materials to be used.

## 2.3.3 Metalwork

### 2.3.3.1 Flame Cutting of Material

Flame cutting of material other than steel is subject to approval of the Contracting Officer. Shear accurately, and neatly finish all portions of work. Steel may be cut by mechanically guided or hand-guided torches, provided an accurate profile with a smooth surface free from cracks and notches is secured. Prepare surfaces and edges to be welded in accordance with AWS D1.1/D1.1M. Do not chip or grind except where specified and as necessary to remove slag and sharp edges of mechanically guided or hand-guided cuts not exposed to view. Visible or exposed hand-guided cuts must be chipped, ground, or machined to metal free of voids, discontinuities, and foreign materials.

### 2.3.3.2 Alignment of Wetted Surfaces

Exercise care to assure that correct alignment of wetted surfaces being joined by a flanged joint is being obtained. Where plates of the water passage change thickness, transition on the outer surface, leaving inner surface properly aligned. When welding has been completed and welds have been cleaned, but prior to stress relieving, carefully check joining of plates in the presence of the Contracting Officer for misalignment of adjoining parts. Localized misalignment between inside or wetted surfaces of an adjoining flange-connected section of pump cannot exceed the amount shown in Column 4 of Table 1 for the respective radius or normal distance from the theoretical flow centerline. Correct misalignments greater than the allowable amount by grinding away offending metal, providing the maximum depth to which metal is to be removed does not exceed amount shown in Column 5 of Table 1. Do not remove metal until assuring the Contracting Officer that no excessive stresses will occur in the remaining material and that excessive local vibration will not result from removal of the material. Where required correction is greater than the amount in Column 5 of Table 1, reject pipe for use. Proposed procedure for all corrective work, other than minor grinding, must be approved by the Contracting Officer prior to start of corrective work. Finish corrective work by grinding corrected surface to a smooth taper. Length of the taper along each flow line element must be 10 times the depth of the offset error at the flow line. Wetted surface irregularities that might have existed in an approved model cannot be reason for accepting comparable surface irregularities in the prototype pump.

| TABLE 1                |                                  |                         |                         |                              |
|------------------------|----------------------------------|-------------------------|-------------------------|------------------------------|
| (1)                    | (2)                              | (3)                     | (4)                     | (5)                          |
| Pipe Diameter (inches) | Pipe Radius or Distance (inches) | Pipe Thickness (inches) | Maximum Offset (inches) | Grind-Not More Than (inches) |
| 24                     | 12                               | 3/8                     | 1/16                    | 3/32                         |
| 30                     | 15                               | 3/8                     | 1/16                    | 3/32                         |
| 36                     | 18                               | 3/8                     | 3/32                    | 3/32                         |
| 42                     | 21                               | 1/2                     | 3/32                    | 1/8                          |
| 48                     | 24                               | 1/2                     | 1/8                     | 1/8                          |
| 54                     | 27                               | 1/2                     | 1/8                     | 1/8                          |
| 60                     | 30                               | 3/4                     | 5/32                    | 5/32                         |
| 72                     | 36                               | 1                       | 5/32                    | 5/32                         |
| 84                     | 42                               | 1-1/8                   | 3/16                    | 1/4                          |

### 2.3.3.3 Stress-Relieving Procedure

After all fabrication welding is completed, and prior to any machining, stress-relieve the bell and the impeller (if it is fabricated) by heat

treatment. Submit proposed stress-relieving procedure.

#### 2.3.4 Examination of Castings

Clean and carefully examine all castings for surface defects. Examine all discovered defects by nondestructive means. Examination personnel must be qualified/certified in accordance with applicable ASTM requirements. Submit the examination procedure and qualification of the examiner. Conduct examination tests in the presence of the Contracting Officer. Choose the examination procedure best suited for the application.

##### 2.3.4.1 Examination Procedures

###### 2.3.4.1.1 Vibration

The pump, motor, and base shall operate outside of the natural resonant frequencies.

###### 2.3.4.1.2 Liquid Penetrant

Conform inspection to the applicable provisions of ASTM E165/E165M.

#### 2.4 VERTICAL PUMPS

##### 2.4.1 Speed

Rotative speed of pump must be no greater than 1,800 rpm.

##### 2.4.2 Efficiency

Pump efficiency, as defined in HI ANSI/HI 14.6, must include losses from the suction bell to the discharge elbow outlet and must not be less than 76 percent at the head-capacity condition(s) specified in paragraph CAPACITIES.

##### 2.4.3 Suction Bell

Make each suction bell of stainless steel plate. Provide a flanged connection for mating with the impeller bowl with a rabbet fit or four equally spaced dowels installed in the vertical position for initial alignment purposes and to maintain concentric alignment of the pump. Support each suction bell entirely by the pump casing. Supports from sump floor will not be acceptable.

##### 2.4.4 Impeller Bowl

Make each impeller bowl of nickel aluminum bronze (ASTM B148 Alloy 958). Heat-treat and stress-relieve welds before final machining. Provide flanges for mating with the suction bell and the impeller bowl or two-piece construction of the impeller. Provide flanged connections with the suction bell and the diffuser or split construction with a rabbet fit or four equally spaced dowels installed in the vertical position for initial alignment purposes and to maintain concentric alignment of the pump. Machine finish the impeller-swept area in the impeller bowl to at least 125 microinch rms and concentric with the impeller axis. For mixed-flow impellers, the angle in the impeller bowl must equal the outside angle of the impeller blade tips. Tolerance for concentricity of the impeller with the impeller axis is not greater than 20 percent of the operating clearance between the impeller and the impeller bowl.

#### 2.4.5 Diffuser Bowl

Make each diffuser bowl of nickel aluminum bronze (ASTM B148 Alloy 958). The diffuser must contain support for the upper impeller shaft bearing and have vanes to guide the pumped flow. Equip the diffuser bowl with a bypass drain, if required, to outside of the pump from the diffuser cavity located between the enclosing tube connection and the impeller. Furnish throttle bushing located in the cavity immediately above the impeller. Design bypass drain and throttle bushing to reduce water pressure on the lower seal. Impeller back-wear rings can also be used to reduce water pressure on the lower seal.

#### 2.4.6 Pump Column and Discharge Elbow

##### 2.4.6.1 Column and Discharge Elbow

Make each discharge elbow of duplex stainless steel plate (ASTM A240), Pipe (ASTM A790), UNS S32205. If turning vanes made of welded steel plate are used, provide an access door in the discharge elbow to allow personnel to inspect turning vanes and remove trash if necessary. Design column and discharge elbow to withstand internal pressures and external loadings associated with various conditions of pump operation. Provide flanges for mating individual segments together and for mating the pump column to the diffuser bowl. Flanges must have rabbeted fits or four equally spaced dowels installed in flanges for initial alignment purposes and to maintain concentric alignment. Terminate the elbow in a plain-end circular section. . Provide diameter of discharge end of elbow as indicated and allow standard diameter flexible couplings to be used. Use adjustable thrust rods and thrust lugs to transfer the load by bridging the coupling between the pump discharge elbow and discharge piping. Determine the size and number of thrust rods needed for the expected loading. Use a minimum of four thrust rods. Maintain complete access to the discharge piping until the discharge pipe installation is complete, inspected, and approved.

##### 2.4.6.2 Column and Discharge Elbow Support

Make each pump pipe column of fiber glass (D2996 Gr.1) material and the column size shall be a minimum of 8-inches outside diameter to reduce friction losses and velocity through the column. Column pipe thickness shall be Schedule 40 and the flange bolt material shall be monel bolts (ASTM F467 N04400) and nuts (ASTM F46 N0440).

Design pump column and discharge elbow for suspension from a baseplate assembly specified in paragraph BASEPLATE AND SUPPORTS and located at operating floor level.

Design each pumping unit for installation as indicated. Install the baseplate for supporting the electric motor. Design the support system to maintain proper alignment of the pumping unit and propeller blade setting. Include the support system in the dynamic analysis.

##### 2.4.6.3 Flanges

Machine flanges and drill bolt holes concentric with pump shaft vertical centerline, having a tolerance of plus or minus one fourth of the clearance between the bolt and bolt hole. When fabricated from steel plate, flanges must not be less than 1 inch thick after machining. Flange thickness after machining cannot vary more than 10 percent of greatest

flange thickness. Provide external stiffeners, if needed. Construct fabricated flanges, as a minimum, to the dimensions of AWWA C207, Class B. Design flanges on major components of pump casing (suction bell, impeller bowl, diffuser bowl, and column and elbow piping) such that blind holes necessitating use of cap screws or stud bolts are not used. Design flanges for connection to column pipe by at least two continuous fillet welds. Connect the inside diameter of the flange to pump column with one weld, and connect the outside diameter of the pump column to the flange with the other weld. Final design of welds rests with the manufacturer, and specified welds are the minimum requirement. Parallel machine, when provided on each end of the same component, and mount parallel to a plane that is normal to pump shaft centerline. Flanges on each end of the same component must have parallel tolerance of 0.002 inch. Finish machine mating surface on flange to 125 microinch finish or better. Provide flanges with a minimum of three jacking bolts to aid in disassembly of the pump.

#### 2.4.6.4 Flanged Joints

Design flanged joints to be air-and water-tight, against positive and negative operating pressures that will be experienced. Provide mating flanges, unless of the male-female rabbet type, with not less than four tapered dowels equally spaced around each flange. If rabbeted fit is not used, then provide the method used to determine concentricity of connected pieces. Gaskets are permitted to be used between column sections, gasket material must be EPDM.

#### 2.4.6.5 Nuts and Bolts

Monel material shall be used for bolts used in assembling the pump and its supporting members, including anchor bolts and dowels. Use only monel nuts and hexagonal bolts and nuts. Also use monel washers.

#### 2.4.6.6 Harnessed Coupling

Provide a flexible mechanical coupling or split-sleeve type coupling that either conforms to ASTM F1476, Type II, Class 3, stainless steel or ASTM F1476, Type 1, to connect the pump discharge elbow to the discharge piping. Install a minimum of four harness bolts (sized by the pump manufacturer) at each coupling.

#### 2.4.7 Impeller

Make the impeller of nickel aluminum bronze.

##### 2.4.7.1 Removal and Prior To Finishing

After removal from mold, and prior to finishing of surface imperfections, the Contracting officer will inspect castings. Fill and grind minor surface imperfections as necessary to preserve correct contour and outline of impeller and to restore surface imperfections to the same degree of finish as surrounding surfaces. Correct surface pits, depressions, projections, or overlaps showing greater than 1/16 inch variation from the general contour for that section. Castings that exhibit surface imperfections (as defined above) covering an area of more than 10 percent of blade surface will be rejected.

#### 2.4.7.2 Balancing

Balance each impeller by the two-plane balancing technique. Balance each impeller at rated operating speed. Check the balance at 110 percent of balance speed, and make needed corrections. Amount of allowable unbalance is in accordance with grade G6.3 of ISO 1940-1. Securely fasten weights needed to obtain the required level of balance to the inside cavity of the impeller hub. In no case can portions of the impeller be removed or weights be added to the outside of the hub, vanes, or water passages. Provide documentation of completed balance test with certification statement.

#### 2.4.8 Shafting

##### 2.4.8.1 Shaft

Each impeller shaft must be Monel K-500 and intermediate shaft(s) must be same material as the impeller shaft without interfering with shaft alignment. Each impeller shaft shall be provided with tungsten carbide bearing journal hardfacing. Also provide for removal of the impeller from below without disassembly of the pump above the impeller bowl. If the pump is multi-staged, design to permit the lower bowls and impeller to be easily removed for in-place inspections of upper propeller and bowl. Design shafts for two different design cases. The first uses a factor of safety of 5 based on ultimate tensile strength of the shaft material and rated horsepower of the motor. The second uses 75 percent of the yield strength of the shaft material and the locked rotor torque of the motor .

##### 2.4.8.2 Couplings

Couple the pump and motor shafts together using rigid flanged coupling capable of transmitting the forces and torques involved. Bolt coupling halves together and maintain concentric with each other, by means of a rabbet fit, to within 0.002 inch. Retain a shaft coupling nut, if used, by fitted bolts, and comply with all tolerances specified for the coupling. Finish machine the flange and bore in one setup to insure that the flange of the coupling is true to the bore. Each flange must be perpendicular to the bore, and parallel to the opposite end and mating flanges to within 0.002 inch. Each flange must be concentric to the centerline of the shaft to within 0.002 inch. Join together pump shaft sections with sleeve-type couplings capable of taking rotation in either direction. Threads, except on fasteners, cannot be employed in construction of sleeve-type couplings. Construct couplings, including keys and fasteners, of Monel K-500 (ASTM B865 K-500 UNS N05500) materials. All keys and keyseats (keyways) must meet the requirements of ASME B17.1. The finished shaft assembly must be concentric about the shaft centerline to within 0.004 inch. Shop assemble couplings and the pump shaft, and inspect for compliance with contract requirements. After inspection, matchmark parts, including fitted bolts, to their mating pieces.

##### 2.4.8.3 Journals

Finish sleeves at all bearings and packing gland locations to at least 32 rms and finish sleeves at seal locations to 16 rms. Securely fasten sleeves to the shaft to prevent movement. Make keys and fasteners, if used, from corrosion resisting steel; fastening by adhesive or welding is not acceptable. All keys and keyseats (keyways) must meet the requirements of ASME B17.1. The surface hardness of the sleeves at the bearing and packing gland locations must be as recommended by the pump



manufacturer.

Finish the shaft journal at all guide bearing and packing gland locations to at least 32 rms and finish the shaft at seal journal locations to 16 rms. The option exists to install replaceable stainless steel one-piece sleeves at each bearing, packing gland, and seal locations with the finishes stated above. Securely fasten sleeves to the shaft to prevent movement. Make keys and fasteners, if used, from corrosion resisting steel; fastening by adhesive or welding is not acceptable. All keys and keyseats (keyways) must meet the requirements of ASME B17.1. The surface hardness at the seal locations must be as recommended by the seal manufacturer.

#### 2.4.8.4 Circumferential Line

Inscribe or etch a circumferential line on the shaft above the stuffing box and mount an adjustable pointer opposite this line in order to indicate a change in vertical position of shaft and to permit realignment after motor removal.

#### 2.4.9 Shaft Enclosure

Provide a shaft enclosure to cover the intermediate shaft and coupling. It must be rigid enough to be self-supporting. Design each enclosure to be watertight and for easy assembly and disassembly in the field. Split each enclosure longitudinally to permit easy removal without removing or disassembling the pump shaft. Construct enclosing tubes, constructed with screw type joints and using tension in the tube to hold alignment, to prohibit the tension tube from unscrewing when the packing gland adjustments are made.

#### 2.4.10 Guide Bearings and Seals

##### 2.4.10.1 Guide Bearings

All in water bearing lubrication must conform EPA 800-R-11-002 for environmental acceptability. Submit in water bearing lubrication product data to show bearings meet environmental, performance, and comparability requirements for the application.

Bearings shall be Thordon GM22401 composite materials. Ensure the bearings are submerged in water when the pump is in operation. Accomplish lubrication by pump discharge water (product water) free-flowing through perforations in the shaft enclosure tube.

Provide each pump with bearings that are grooved and continuous. Machine the bearings to marine clearances for strained water lubrication and of sufficient length and diameter to keep bearing pressure to the allowable design. Select bearing thicknesses and marine clearances to allow for material swell from water submergence. Shaft bearings may be located as the design dictates with the following exceptions. Provide no fewer than three bearings in the pump with a maximum bearing spacing of 5 feet. Locate bearings at the top of the diffuser section, above the impeller, and near the baseplate. Support the line shaft bearing by the diffuser inner cone fabricated assembly. Secure tubing for bearing lubrication water to and routed on the outside of the casing for each bearing. The tubing must be continuously welded to a diffuser vane or bearing support to cross from the outside of the casing to the bearing mounting assembly.

Bearings must be easily removable for servicing in the field.

#### 2.4.10.2 Product Lubricated Pump Bearings

Submerged pump bearing must be product lubricated and meet the following requirements:

- a. Fabricated from a Thordon GM2401 composite material and not require petroleum lubricants for operation.
- b. Operate in brackish water that may contain silt.
- c. Does not require service or replacement for 50,000 operating hours.

#### 2.4.11 Thrust Bearing

Provide a thrust bearing in the motor to carry total thrust load occurring across pump operation.

#### 2.4.12 Packing Gland

Provide water-lubricated packing gland split longitudinally to facilitate removal or renewal. Arrange it to permit inspection, repair, removal, or replacement of packing without entering pump from below operating room floor. Provide eye bolts and tapped holes in each half of the split gland if halves weigh over 30 pounds each.

#### 2.4.13 Siphon Breaker Valve

Provide a siphon breaker valve assembly for each of the vertical axial-flow pumps. Provide mechanical, self-actuating valves. Install each assembly at the top (summit) of the discharge pipe and must vent air from the discharge pipe when the pump is started. The assembly must also permit air to enter the discharge pipe through the siphon breaker valve to prevent reverse siphoning of water when the pump is stopped. Size the valves based on the pumping conditions (e.g. flow, head, discharge pipe diameter) of the specific system. If a mechanically-actuated valve is chosen, size the valve to allow air to relieve at a maximum of 300 feet per second. Provide a means to operate the valve manually to stop back-siphoning through the pump in case the normal operator of the siphon breaker valve fails.

### 2.5 LUBRICATION SYSTEM

All in water bearing lubrication must conform EPA 800-R-11-002 for environmental acceptability. Submit in water bearing lubrication product data to show bearings meet environmental, performance, and comparability requirements for the application.

Provide lubrication for the bearings from the water being pumped (i.e., product water) by means of perforations in the shaft enclosure. Design the perforations such that the shaft tube is constantly filled with water during pump operation, thereby continually covering the contact surfaces of the bearings. If upper bearings will not receive sufficient pumped water for lubrication, use another means to provide sufficient water to these bearings using either product water or water from the intake canal. Provide a means to indicate visually that the upper bearing(s) and packing gland are obtaining sufficient lubrication water.

### 2.5.1 Supplemental Lubrication Water

If it is determined that the pumped product water cannot be lifted sufficiently to lubricate the upper bearing(s), it is the Contractor's responsibility to design and provide a complete water lubrication system. Provide a separate water lubrication system for each pump. The water cannot come from the potable/well water system. This water lubrication system must include a means for providing sufficient lubrication water to the upper bearing(s). The supplemental lubrication water system must include an alarm and shutdown in case of low-flow or no-flow to the upper bearing(s), along with electrical input to the engine control panel and SCADA/PLC systems as needed. Provide any necessary filtering, flow switches, time delays, to ensure that the upper bearings are satisfactorily lubricated. Coordinate any additional electrical power and controls requirements with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM and other electrical sections. The supplemental lubrication water system must be fully maintainable. If the supplemental lubrication water system is needed, provide shop drawings, operations and maintenance data, and as-built drawings as required.

### 2.6 PAINTING

Perform painting in accordance with Section 09 90 00 PAINTING.

### 2.7 FACTORY ASSEMBLY

The pump must be assembled at the Contractor's selected manufacturer's plant to assure proper fitting and alignment of all parts. Tolerances cannot exceed those specified or shown on the the manufacturing drawings. Check rotating elements for binding. The suction bell, impeller housing, diffuser, and the discharge elbow must be properly match marked and have their centerlines clearly marked on the outside of all flanges to facilitate erection and alignment in the field. Notify the Contracting Officer sufficiently in advance to permit a representative of the Contracting Officer to inspect and witness the pump assembly. Matchmark all parts disassembled for shipment.

### 2.8 BASEPLATE AND SUPPORTS

Design and proportion the soleplate and baseplate to support the entire pump assembly, the motor and the loads (including the results of the dynamic analysis) to which it may be subjected during operation. Furnish lifting lugs or eye bolts, special slings, strongbacks, or other devices necessary to handle the pump during loading, unloading, erection, installation, and subsequent disassembly and assembly. Provide a sole plate under the baseplate. Install, level and grout the sole plate in accordance with API RP 686, Chapter 5 - Mounting Plate Grouting. Provide leveling nuts for leveling the baseplate assembly. Provide an anchor bolt layout to aid in placement of anchor bolts. The pedestal supporting the motor must contain a 1-inch lip to contain water leakage from the shaft packing. Provide a sole plate under the baseplate. Provide the calculations used in the design of the baseplates and the anchoring bolts to ensure that the proper forces (e.g. shear, torsion, bending) have been considered. Submit product data for grout used as pump supports. The baseplates must be structural steel plate of adequate thickness to support the weight of the pump plus the maximum hydraulic thrust of the pump. Provide plates of the length, width, and thickness as determined by the Contractor.

## 2.9 TESTS, INSPECTIONS, AND VERIFICATIONS

### 2.9.1 Pump Testing

Test specified pumps in accordance with HI ANSI/HI 14.6. For all pumps include Performance, hydrostatic, NPSH, and mechanical testing as described in HI ANSI/HI 14.6.

### 2.9.2 Critical Speeds

The assembled pumping unit, consisting of the motor and pump must be free from critical speeds or harmful torsional vibrations at all speeds encountered within the operating range.

Before the pump and motor are released for manufacture, the pump/motor structure must be analyzed by the Contractor for harmful natural frequencies in the lateral and torsional directions. A natural frequency that occurs within 25 percent above or below normal operating speed is unacceptable. Construct a dynamic analysis model using a commercially available program such as Ansys, Cosmos/M, or equivalent, which utilize finite element methods. Incorporate effects of column pipes, cover pipes, shafts, bearings, mass concentrations, and other such features as necessary to accurately model the pump structure. Analyze the structure in the run (wet) condition and consider the effect of water mass in the column and damping effect of water in the sump (vertical units only) at highest and lowest sump water levels. Incorporate Reed critical frequency and mass elastic diagram information for the Contractor's selected motors. If the Contractor cannot demonstrate to the satisfaction of the Contracting Officer (based on impact tests of similar units) that the Reed critical frequency value is accurate, the Contractor must conduct a dynamic analysis using finite element methods as described to determine motor Reed critical frequency for use with the Contractor selected pumps. Submit the complete detailed dynamic analysis report including the following information:

- a. Computer program used.
- b. Schematic diagram of the model depicting nodes and elements.
- c. Input data consisting of node coordinates, element types, material properties, element characteristics, element connectivities, and specified displacements.
- d. Motor mass elastic and Reed critical information (or dynamic analysis, if required).
- e. Analysis results, including significant natural frequencies.
- f. Interpretation of results.

Impact the test motor provided before shipment to determine actual Reed critical frequency of motor. Include results of impact tests included in motor test data to be submitted. The Contractor must address any discrepancy between calculated and actual motor Reed critical frequency values to determine whether design changes are required to prevent harmful natural frequencies in the pump/motor structure. Incorporate any design changes which are required to correct these discrepancies.

#### 2.9.2.1 Torsional Analysis

Before the pump, gear drive, and engine are released for manufacture, the Contractor must analyze the system for harmful torsional natural frequencies using mass elastic information for the selected pump(s) and gear drive(s). A natural frequency that occurs within 25 percent above or below normal operating speed is considered to be unacceptable.

#### 2.9.3 Lubricating System Tests

Test the complete lubricating system for each pumping unit, as deemed necessary by the Contracting Officer, to determine that the system meets the operational requirements specified. Test at least one valve of each size furnished with the lubrication line removed from its bearing and fitted with a pressure relief valve and pressure gage. Adjust the pressure relief valve to discharge it at the operating pressure specified and operate the system through one or more cycles as required to obtain an accurate measurement of the quantity of lubricant delivered. This must be within plus or minus 20 percent of the theoretical delivery of the respective valve. Replace, reinstall, and retest any component parts that are damaged as the result of these tests or that fail to meet the requirements of the specification.

#### 2.9.4 Factory Test

##### 2.9.4.1 General

Performance of the pump to be furnished will be accepted on the basis of the factory test. The factory test shall be performed in accordance with Standard Performance Test per ANSI/HI 14.6 Grade 1U. Conduct this test using the first pump produced for this contract.

##### 2.9.4.2 Instrumentation and Procedures

In the test report describe each instrument to be used in the tests in detail, giving all data applicable, such as manufacturer's name, type, model number, certified accuracy, coefficient, ratios, specific gravity of manometer fluid to be used, and smallest scale division. Provide calibration data on each of the instruments used. When necessary for clarity, include a sketch of the instrument or instrument arrangement. Include fully detailed narrative description of each proposed method of instrumentation, procedures to be used, and a sample set of computations. State the lowest equivalent static head that is obtainable with the testing when operating along the head-capacity curve of the proposed pump. Perform test procedures, except as specified, in accordance with applicable provisions of HI ANSI/HI 14.6.

##### 2.9.4.2.1 Head Measurements

Make head measurements using either a direct reading water column, mercury-air, mercury-water, a Meriam fluid manometer, or a pressure transducer. Measure vacuums with either a mercury-air manometer, a mercury-water manometer, or a pressure transducer. Dampen fluctuations sufficiently to permit column gages or a differential pressure transducer to be read to either closest 0.01 foot of water or Meriam fluid or 0.1 inch of mercury. Use manometers as indicated by ISA RP2.1. When pressure transducers are used, check their accuracy with a manometer.

#### 2.9.4.2.2 Capacity

Determine capacity by a calibrated flowmeter or long-radius ASME flow nozzle. Do not use orifice plates. Connect the flowmeter or nozzle taps to the column gages equipped with dampening devices that permits differential head to be determined to either the closest 0.01 foot of water or Meriam fluid or 0.1 inch of mercury. Magnetic flowmeters and flowmeters utilizing ultrasonic flow measurements will be acceptable if calibration of flowmeter has been completed within the last 6 months.

#### 2.9.4.2.3 Rotational Speed of Pump

Measure the rotational speed of the pump in accordance with "Method of Rotary Speed Movement" in HI ANSI/HI 14.6, except that revolution counters cannot be used. Non-contacting hand-held electronic tachometers are acceptable. The device used must permit the speed to be determined to 1 rpm.

#### 2.9.4.2.4 Power Input

Measure power input to the pump in accordance with "Power Measurements" in HI ANSI/HI 14.6. Use a method to permit pump brake horsepower to be determined to the closest 0.5 horsepower.

#### 2.9.4.2.5 Mechanical Tests

Take vibration measurements of the assembled pumping units per HI ANSI/HI 14.6.

#### 2.9.4.3 Pump Test

Demonstrate that the proposed pump complies with the specified performance. The pump must be capable of operation without instability over the entire range of heads specified in paragraph CAPACITIES. Tolerances must be in accordance with HI requirements. Instability is defined, for this specification, as when one or more of the following conditions occur:

- a. the pump has two or more flow rates at the same total head;
- b. The head-capacity curve has a dip (region on curve where change in flow rate produces an abnormally low head);
- c. When any point in the usable range of head-capacity curve cannot be repeated within 3 percent.

Rerun the test if this occurs. Compliance with specifications will be determined from curves required by paragraph TEST RESULTS. Test procedures, except as specified, must be in accordance with applicable provisions of HI ANSI/HI 14.6. The acceptance grade is 1U as described in HI ANSI/HI 14.6. Pumps are acceptable if they achieve the guarantee point requirements stated in paragraph CAPACITIES. Use water at approximately the same temperature for all tests run and record the temperature during test runs.

#### 2.9.4.4 Motor, Cables, and Controller Test

Test the pump motor per NETA ATS. For induction motors use NETA ATS 7.15.1.

#### 2.9.4.5 Test Procedure

##### 2.9.4.5.1 Performance of The Pump

Determine the performance of the pump by a series of test points sufficient in number to develop a constant-speed curve over the range of total heads corresponding to the static heads in paragraph CAPACITIES. The performance/test range must include additional testing at total heads 2 feet higher than the total head determined in paragraph CAPACITIES. The lowest total head for testing must be, as a minimum, the total head determined from paragraph CAPACITIES. If the test setup permits testing at lower total heads, extend the range of total heads 2 feet lower. Conduct tests using prototype total heads. Head differentials between adjacent test points cannot exceed 3 feet, but in no case may fewer than 10 points be plotted in the pumping range. If the plot of the data indicates a possibility of instability or dip in the head-versus-capacity curve, a sufficient number of additional points on either side of instability must be made to clearly define the head-capacity characteristics.

##### 2.9.4.5.2 Sump Elevations

Conduct tests at two different sump elevations (approximately a 5 foot differential) to determine the effect of test sump geometry on the performance of the pump. Should the test results indicate that the performance is not the same in all respects for both sump conditions, take whatever corrective action is necessary to produce congruent results. Use the sump elevations specified in paragraph CAPACITIES. The test results with this sump elevation must meet all specified conditions of capacity, head, and brake horsepower. Submit curves indicating test results.

##### 2.9.4.5.3 Tests Results

Plot results of tests to show total head, static , brake horsepower and efficiency as ordinates; all plotted against pump discharge as the abscissa. Plot curves showing prototype performance to a scale that will permit reading head directly to 0.5 foot, capacity to 100 gpm, , efficiency to 1 percent, and power input to 10 horsepower.

##### 2.9.4.5.4 Demonstration

Demonstrate to the Contracting Officer that the blade templates fit the tested pump. Perform the demonstration immediately after testing is completed. Retain all templates for the tested pump, and provide them to the Government upon request of the Contracting Officer, to permit the Government to verify geometric similarity with the Contractor's pump. In addition to providing templates, submit dimensioned drawings of the impeller that contain all dimensions needed to manufacture it. Stamp the tested impeller with identification marks. Provide necessary facilities and instruments needed to permit the Government to verify that pumps are in complete geometric similarity with the tested pump.

#### 2.10 PUMP DRAINAGE

Provide drain holes for all parts of the pump to eliminate trapped water. These drain provisions must be self-draining without any requirement to enter the sump.

## 2.11 CERTIFIED TEST REPORT

Submit Contractor's pump manufacturer certified completed test report stating the specified pumps meet HI ANSI/HI 14.6 after testing is complete.

## 2.12 TEST REPORT

Submit, in accordance with HI ANSI/HI 14.6, and within 30 days of receipt of approval of the witness test, to Government digital copies of a report covering completely the test setup and performance and cavitation tests. Include, as a minimum, the following in each test report:

- a. Provide a statement of the purpose of the test, the name of the project, contract number, and design conditions. Also provide where guaranteed values differ from specified values.
- b. A resume of preliminary studies, if such studies were made.
- c. A description of the test pump and motor, including serial numbers, if available. Information required under "b" may be included here.
- d. A description of the test procedure used, including dates, test personnel, any retest events, and witness test data.
- e. A list of all test instruments with model numbers and serial numbers.
- f. Sample computations (complete).
- g. A discussion of test results.
- h. Conclusions.
- i. Copies of instrument calibration.
- j. Copies of all recorded test data.
- k. Curves required by paragraph TESTS RESULTS.
- l. Curves showing the performance of the test pump.
- m. Drawings of the test setup showing all pertinent dimensions and elevations and a detailed dimensioned cross section of the pump.
- n. Drawings including cross sections of water passages that must be incorporated in the construction contract.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Install the soleplate and baseplate in accordance with API RP 686. Provide leveling nuts for leveling the baseplate assembly. Provide an anchor bolt layout to aid in placement of anchor bolts. The pedestal supporting the motor must contain a 1-inch lip to contain water leakage from the shaft packing. Provide a threaded drain to the sump.



### 3.1.1 Equipment

- a. Install the equipment specified under this section and related drive machinery specified under other sections of this specification in accordance with the approved Installation and Erection Instructions Manual; no later than time of pump delivery, submit digital manual describing procedures to be followed by the erecting engineer in erecting, assembling, installing, and dry-and wet-testing the pump. To the extent necessary or desirable, coordinate and consolidate description of the pump with similar descriptions specified for the gear reducer and diesel enginemotor.
  - (1) The description must be a complete, orderly, step-by-step explanation of operations required, and also include such things as alignment procedures, bolt torque values, permissible blade/bowl clearances; permissible bowl out-of-roundness; permissible shaft misalignment; recommended instrument setups; recommended gages and instruments; bearing clearances; and similar details.
  - (2) Complement and supplement the description with drawings, sketches, photos, and similar materials to whatever extent necessary or desirable, resulting in a description that may be comprehended by an engineer or mechanic without extensive experience in erecting or installing pumps of this type.
- b. The Contractor's erection engineer(s), familiar with the equipment to be installed, must supervise the handling, installation, start-up and testing of the equipment as required by paragraph QUALITY ASSURANCE.
- c. Submit digital copies of Operation and Maintenance Instructions Manual containing complete information on operation, lubrication, adjustment, routine and special maintenance, disassembly, repair, reassembly, and trouble diagnostics for the pump and auxiliary units. Provide the operation and maintenance manual and both parts lists on flash drive, formatted to print on ANSI size A 8-1/2 by 11-inch paper. Drawings incorporated in manual or parts lists, may be reduced to page size provided they are clear and legible. Photographs or catalog cuts of components may be included for identification. Submit operation and maintenance data in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

### 3.1.2 Pipes and Joints

Install pipes and joints in accordance with AWWA M11.

### 3.2 FIELD TESTS

After the pumping unit is installed and prior to start-up, completely clean the sump area of any accumulated construction debris. Once final cleaning of the sump area is completed, the area will be area reviewed by a representative of the Government. Correct any damage to the pumping units or related equipment during initial start-up due to foreign objects left in the sump areas.

Prior to proceeding with construction of the model, but not later than 90 days after the date of notice to proceed, submit a description of the proposed model and test procedure. Include dimensioned drawings and cross-sectional views of the model pump showing with the location of all

instruments and the point of their connection to the model.

#### 3.2.1 Wet Tests

Test each pump unit under load, at or near normal operating conditions, for at least 2 hours or as directed by the Contracting Officer; the test will be witnessed by the Government. Provide all supplies and equipment required to conduct the test. During the test observe, measure and record the operation of the pumps during the test for vibration and bearing temperatures in accordance with HI 9.6.4ISO 10816-3. Submit field test results. Without additional costs to the Government, make all changes and correct any errors. The Contracting Officer may waive or postpone the test if sufficient water is not available.

-- End of Section --

SECTION 40 05 13

PIPELINES, LIQUID PROCESS PIPING  
10/07, CHG 2: 02/20

PART 1 GENERAL

1.1 UNIT PRICES

Measurement and payment will be based on completed work performed in accordance with the drawings, specifications and the contract payment schedules. No payment will be made under this section for excavation, trenching, or backfilling. Payment for such work will be made under Section 31 00 00.00 22 EARTHWORK.

1.1.1 Measurement

Determine the length of pipelines, for which payment will be made, by measuring along the centerlines of the various piping systems and sizes as furnished and installed. Measure pipe from the center of fitting to center of fitting and from center of main header to end of pipe. Make no deduction for the space occupied by valves or fittings.

1.1.2 Payment

Payment will be made at the price per linear foot listed in the bid form for the various types and sizes of piping, and will be full compensation for all pipes, joints, fittings and specialties, complete in place. Payment for valves and other appurtenances will be made at the respective contract unit price for each item complete in place. Payment will include the furnishing of all testing, plant, labor, and material and incidentals necessary to complete the work, as specified and as shown in contract documents.

1.1.2.1 Connections to Existing Piping

Connections to existing piping systems where new fittings in the existing line are required will be paid for according to the contract prices for such connection. The price will be considered as full compensation for material and labor required for the removal and replacement of the existing pipe as necessary.

1.1.2.2 Connections to Existing Equipment

Connections to existing equipment where new fittings for the existing equipment are required will be paid for according to the contract prices for such connection. The price will be considered as full compensation for material and labor required for the installation of new fittings or the removal and replacement of existing fittings, as necessary.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|              |                                                                                         |
|--------------|-----------------------------------------------------------------------------------------|
| ASME B1.1    | (2024) Unified Inch Screw Threads<br>(UN, UNR, and UNJ Thread Forms)                    |
| ASME B1.20.1 | (2013; R 2018) Pipe Threads, General<br>Purpose (Inch)                                  |
| ASME B16.1   | (2020) Gray Iron Pipe Flanges and Flanged<br>Fittings Classes 25, 125, and 250          |
| ASME B16.5   | (2020) Pipe Flanges and Flanged Fittings<br>NPS 1/2 Through NPS 24 Metric/Inch Standard |
| ASME B16.9   | (2018) Factory-Made Wrought Buttwelding<br>Fittings                                     |
| ASME B16.11  | (2022) Forged Fittings, Socket-Welding and<br>Threaded                                  |
| ASME B16.15  | (2018) Cast Copper Alloy Threaded Fittings<br>Classes 125 and 250                       |
| ASME B16.18  | (2021) Cast Copper Alloy Solder Joint<br>Pressure Fittings                              |
| ASME B16.21  | (2021) Nonmetallic Flat Gaskets for Pipe<br>Flanges                                     |
| ASME B16.22  | (2021) Wrought Copper and Copper Alloy<br>Solder Joint Pressure Fittings                |
| ASME B16.26  | (2018) Standard for Cast Copper Alloy<br>Fittings for Flared Copper Tubes               |
| ASME B16.34  | (2021) Valves - Flanged, Threaded and<br>Welding End                                    |
| ASME B31.1   | (2022) Power Piping                                                                     |
| ASME B31.3   | (2022; Errata 2023) Process Piping                                                      |
| ASME B36.10M | (2022) Welded and Seamless Wrought Steel<br>Pipe                                        |
| ASME B36.19M | (2022; Errata 2023) Welded and Seamless<br>Wrought Stainless Steel Pipe                 |

AMERICAN SOCIETY OF SANITARY ENGINEERING (ASSE)

|           |                                                                         |
|-----------|-------------------------------------------------------------------------|
| ASSE 1001 | (2021) Performance Requirements for<br>Atmospheric Type Vacuum Breakers |
|-----------|-------------------------------------------------------------------------|

AMERICAN WATER WORKS ASSOCIATION (AWWA)

|                |                                                                                                          |
|----------------|----------------------------------------------------------------------------------------------------------|
| ANSI/AWWA C541 | (2016; R 2021) Hydraulic and Pneumatic<br>Cylinder and Vane-Type Actuators for<br>Valves and Slide Gates |
|----------------|----------------------------------------------------------------------------------------------------------|

|                  |                                                                                  |
|------------------|----------------------------------------------------------------------------------|
| ANSI/AWWA C542   | (2016; R 2021) Electric Motor Actuators for Valves and Slide Gates               |
| AWWA C104/A21.4  | (2022) Cement-Mortar Lining for Ductile-Iron Pipe and Fittings for Water         |
| AWWA C110/A21.10 | (2021) Ductile-Iron and Gray-Iron Fittings                                       |
| AWWA C111/A21.11 | (2023) Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings          |
| AWWA C115/A21.15 | (2020) Flanged Ductile-Iron Pipe With Ductile-Iron or Gray-Iron Threaded Flanges |
| AWWA C153/A21.53 | (2019) Ductile-Iron Compact Fittings for Water Service                           |
| AWWA C500        | (2019) Metal-Seated Gate Valves for Water Supply Service                         |
| AWWA C510        | (2017; R 2021) Double Check Valve Backflow Prevention Assembly                   |
| AWWA C511        | (2017; R 2021) Reduced-Pressure Principle Backflow Prevention Assembly           |
| AWWA C550        | (2017) Protective Interior Coatings for Valves and Hydrants                      |
| AWWA C606        | (2022) Grooved and Shouldered Joints                                             |
| AWWA C651        | (2014) Standard for Disinfecting Water Mains                                     |

AMERICAN WELDING SOCIETY (AWS)

|                |                                                                      |
|----------------|----------------------------------------------------------------------|
| AWS A5.8/A5.8M | (2019) Specification for Filler Metals for Brazing and Braze Welding |
| AWS D1.1/D1.1M | (2020; Errata 1 2021) Structural Welding Code - Steel                |

ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------|
| ASTM A36/A36M   | (2019) Standard Specification for Carbon Structural Steel                                             |
| ASTM A47/A47M   | (1999; R 2022; E 2022) Standard Specification for Ferritic Malleable Iron Castings                    |
| ASTM A53/A53M   | (2024) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless |
| ASTM A105/A105M | (2023) Standard Specification for Carbon Steel Forgings for Piping Applications                       |
| ASTM A106/A106M | (2019a) Standard Specification for                                                                    |

Seamless Carbon Steel Pipe for  
High-Temperature Service

|                 |                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A108       | (2024) Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished                                                                                  |
| ASTM A126       | (2004; R 2023) Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings                                                           |
| ASTM A153/A153M | (2023) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware                                                                           |
| ASTM A167       | (2011) Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip                                                  |
| ASTM A182/A182M | (2023) Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service |
| ASTM A193/A193M | (2024) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications       |
| ASTM A194/A194M | (2023) Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both        |
| ASTM A269/A269M | (2022) Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service                                                   |
| ASTM A307       | (2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength                                                     |
| ASTM A312/A312M | (2022a) Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes                                                 |
| ASTM A334/A334M | (2004a; R 2021) Standard Specification for Seamless and Welded Carbon and Alloy-Steel Tubes for Low-Temperature Service                                       |
| ASTM A351/A351M | (2018) Standard Specification for Castings, Austenitic, for Pressure-Containing Parts                                                                         |
| ASTM A395/A395M | (1999; R 2022) Standard Specification for Ferritic Ductile Iron Pressure-Retaining Castings for Use at Elevated Temperatures                                  |
| ASTM A403/A403M | (2022b) Standard Specification for Wrought                                                                                                                    |

Austenitic Stainless Steel Piping Fittings

|                 |                                                                                                                                                 |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A423/A423M | (2019) Standard Specification for Seamless and Electric-Welded Low-Alloy Steel Tubes                                                            |
| ASTM A479/A479M | (2023a) Standard Specification for Stainless Steel Bars and Shapes for Use in Boilers and Other Pressure Vessels                                |
| ASTM A513/A513M | (2020a) Standard Specification for Electric-Resistance-Welded Carbon and Alloy Steel Mechanical Tubing                                          |
| ASTM A576       | (2023) Standard Specification for Steel Bars, Carbon, Hot-Wrought, Special Quality                                                              |
| ASTM A632       | (2004; R 2014) Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing (Small-Diameter) for General Service            |
| ASTM A727/A727M | (2014; R2019) Standard Specification for Carbon Steel Forgings for Piping Components with Inherent Notch Toughness                              |
| ASTM A780/A780M | (2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings                                                |
| ASTM A813/A813M | (2014; R 2019) Standard Specification for Single- or Double-Welded Austenitic Stainless Steel Pipe                                              |
| ASTM A814/A814M | (2015; R 2019) Standard Specification for Cold-Worked Welded Austenitic Stainless Steel Pipe                                                    |
| ASTM A858/A858M | (2019) Standard Specification for Heat-Treated Carbon Steel Fittings for Low-Temperature and Corrosive Service                                  |
| ASTM A865/A865M | (2023) Standard Specification for Threaded Couplings, Steel, Black or Zinc-Coated (Galvanized) Welded or Seamless, for Use in Steel Pipe Joints |
| ASTM B32        | (2020) Standard Specification for Solder Metal                                                                                                  |
| ASTM B42        | (2020) Standard Specification for Seamless Copper Pipe, Standard Sizes                                                                          |
| ASTM B62        | (2017) Standard Specification for Composition Bronze or Ounce Metal Castings                                                                    |
| ASTM B88        | (2022) Standard Specification for Seamless Copper Water Tube                                                                                    |
| ASTM B98/B98M   | (2013) Standard Specification for                                                                                                               |

Copper-Silicon Alloy Rod, Bar, and Shapes

|                 |                                                                                                                                                                                                                 |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM B124/B124M | (2020) Standard Specification for Copper and Copper Alloy Forging Rod, Bar, and Shapes                                                                                                                          |
| ASTM C552       | (2022) Standard Specification for Cellular Glass Thermal Insulation                                                                                                                                             |
| ASTM D1784      | (2020) Standard Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds                                                                              |
| ASTM D1785      | (2021) Standard Specification for Poly(Vinyl Chloride) (PVC), Plastic Pipe, Schedules 40, 80, and 120                                                                                                           |
| ASTM D2464      | (2015) Standard Specification for Threaded Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80                                                                                                        |
| ASTM D2466      | (2023) Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 40                                                                                                                 |
| ASTM D2467      | (2015) Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80                                                                                                                 |
| ASTM D2564      | (2020) Standard Specification for Solvent Cements for Poly(Vinyl Chloride) (PVC) Plastic Piping Systems                                                                                                         |
| ASTM D2609      | (2015) Standard Specification for Plastic Insert Fittings for Polyethylene (PE) Plastic Pipe                                                                                                                    |
| ASTM D2657      | (2007; R 2015) Heat Fusion Joining Polyolefin Pipe and Fittings                                                                                                                                                 |
| ASTM D2683      | (2020) Standard Specification for Socket-Type Polyethylene Fittings for Outside Diameter-Controlled Polyethylene Pipe and Tubing                                                                                |
| ASTM D2774      | (2021) Underground Installation of Thermoplastic Pressure Piping                                                                                                                                                |
| ASTM D2855      | (2020) Standard Practice for the Two-Step (Primer and Solvent Cement) Method of Joining Poly (Vinyl Chloride) (PVC) or Chlorinated Poly (Vinyl Chloride) (CPVC) Pipe and Piping Components with Tapered Sockets |
| ASTM D3222      | (2018a) Standard Specification for Unmodified Poly(Vinylidene Fluoride) (PVDF) Molding Extrusion and Coating                                                                                                    |



Materials

|                 |                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM D3261      | (2016) Standard Specification for Butt Heat Fusion Polyethylene (PE) Plastic Fittings for Polyethylene (PE) Plastic Pipe and Tubing                                                 |
| ASTM D3308      | (2012; R 2017) Standard Specification for PTFE Resin Skived Tape                                                                                                                    |
| ASTM D3350      | (2021) Polyethylene Plastics Pipe and Fittings Materials                                                                                                                            |
| ASTM D3892      | (2015) Standard Practice for Packaging/Packing of Plastics                                                                                                                          |
| ASTM E814       | (2024) Standard Test Method for Fire Tests of Penetration Firestop Systems                                                                                                          |
| ASTM F402       | (2018) Standard Practice for Safe Handling of Solvent Cements, Primers, and Cleaners Used for Joining Thermoplastic Pipe and Fittings                                               |
| ASTM F437       | (2021) Standard Specification for Threaded Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 80                                                               |
| ASTM F438       | (2017) Standard Specification for Socket-Type Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 40                                                            |
| ASTM F439       | (2019) Standard Specification for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 80                                                                        |
| ASTM F441/F441M | (2023) Standard Specification for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe, Schedules 40 and 80                                                                         |
| ASTM F493       | (2022) Standard Specification for Solvent Cements for Chlorinated Poly (Vinyl Chloride) (CPVC) Plastic Pipe and Fittings                                                            |
| ASTM F656       | (2021) Standard Specification for Primers for Use in Solvent Cement Joints of Poly(Vinyl Chloride) (PVC) Plastic Pipe and Fittings                                                  |
| ASTM F876       | (2023a) Standard Specification for Crosslinked Polyethylene (PEX) Tubing                                                                                                            |
| ASTM F1055      | (2016; R 2022a) Standard Specification for Electrofusion Type Polyethylene Fittings for Outside Diameter Controlled Polyethylene and Crosslinked Polyethylene (PEX) Pipe and Tubing |

ASTM F1056 (2018; R 2022) Standard Specification for Socket Fusion Tools for Use in Socket Fusion Joining Polyethylene Pipe or Tubing and Fittings

ASTM F1199 (2021) Standard Specification for Cast (All Temperatures and Pressures) and Welded Pipe Line Strainers (150 psig and 150 degrees F Maximum)

ASTM F1200 (2021a) Standard Specification for Fabricated (Welded) Pipe Line Strainers (Above 150 psig and 150 degrees F (1Mpa and 65 degrees C))

ASTM F1290 (2019) Standard Practice for Electrofusion Joining Polyolefin Pipe and Fittings

DUCTILE IRON PIPE RESEARCH ASSOCIATION (DIPRA)

DIPRA TRD (2016) Thrust Restraint Design for Ductile Iron Pipe

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-25 (2018) Standard Marking System for Valves, Fittings, Flanges and Unions

MSS SP-58 (2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250 (2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 704 (2022) Standard System for the Identification of the Hazards of Materials for Emergency Response

PLASTICS PIPE INSTITUTE (PPI)

PPI TR-21 (2001) Thermal Expansion and Contraction in Plastic Piping Systems

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC SP 6/NACE No.3 (2007) Commercial Blast Cleaning

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01 (2023; with Change 2, 2024) Structural Engineering

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910

Occupational Safety and Health Standards

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings; G, AE

Materials and Equipment

SD-03 Product Data

Qualifications; G, AE

Welders; G, AE

Waste Water Disposal; G, AE

Manufacturer's Field Services; G, AE

Delivery, Storage and Handling; G, AE

Materials and Equipment; G, AE

Installation; G, AE

Valve Schedule; G, AE

Operator Schedule; G, AE

SD-06 Test Reports

Double Containment Piping Leak Detection System; G, AE

Pipe Leakage Tests; G, AE

Hydrostatic Tests; G, AE

Pneumatic Tests; G, AE

Valve Testing; G, AE

Disinfection; G, AE

SD-07 Certificates

Plastic Piping System; G, AE

Plastic Pipe Installation; G, AE

SD-10 Operation and Maintenance Data

Piping and Appurtenances; G, AE

#### 1.4 QUALIFICATIONS

##### 1.4.1 Experience

Submit a statement certifying that the Contractor has the specified experience. Ensure Contractor has successfully completed at least three projects of the same scope and size or larger within the last six years. Demonstrate specific experience in regard to the system installation to be performed.

##### 1.4.2 Double Containment Piping System Manufacturer

The double containment piping system manufacturer will have at least ten years of installation experience with leak detection/location sensor cable technology.

##### 1.4.3 Welders

Submit the names of all qualified welders, their identifying symbols, and the qualifying procedures for each welder including support data such as test procedures used, standards tested to, etc. Weld pressure piping systems in accordance with qualifying procedures using performance qualified welders and operators. Use procedures and welders in accordance with 40 17 26.00 20 WELDING PRESSURE PIPING. Weld structural members in accordance with Section 05 05 23.16 STRUCTURAL WELDING.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

Materials delivered and placed in storage must be stored with protection from the weather, excessive humidity variation, excessive temperature variation, dirt, dust and/or other contaminants. Proper protection and care of material before, during and after installation is the Contractor's responsibility. Any material found to be damaged must be replaced at the Contractor's expense. During installation, piping must be capped to keep out dirt and other foreign matter. A Safety Data Sheet in conformance with 29 CFR 1910 Section 1200(g) must accompany each chemical delivered for use in pipe installation. At a minimum, this includes all solvents, solvent cements, glues and other materials that may contain hazardous compounds. Handling must be in accordance with ASTM F402. Storage facilities must be classified and marked in accordance with NFPA 704. Materials must be stored with protection from puncture, dirt, grease, moisture, mechanical abrasions, excessive heat, ultraviolet (UV) radiation damage, or other damage. Pipe and fittings must be handled and stored in accordance with the manufacturer's recommendation. Plastic pipe must be packed, packaged and marked in accordance with ASTM D3892.

#### 1.6 PROJECT/SITE CONDITIONS

##### 1.6.1 Environmental Requirements

**The project is handling salt water and dionized water. Material of construction shall be compatable with liquid it is carrying or environment placed in. Where materials are called out in the drawings that may not be compatable with the environment, cathodic protection shall be provided.** Should the drawings miss calling out this proction, the general contract should assume this protection shall be provided. .

##### 1.6.2 Existing Conditions

Verify existing piping and penetrations. Prior to ordering materials,

expose all existing pipes which are to be connected to new pipelines. Verify the size, material, joint types, elevation, horizontal location, and pipe service of existing pipes, and inspect size and location of structure penetrations to verify adequacy of wall sleeves, and other openings before installing connecting pipes.

## 1.7 SEQUENCING AND SCHEDULING

For slab, floor, wall, and roof penetrations, keep on site pertinent wall pipes and sleeves before they are required for placement in concrete forms. Verify and coordinate the size and location of building and structure pipe penetrations before forming and placing concrete.

## 1.8 MAINTENANCE

### 1.8.1 Service

Provide services for automatic valvedouble containment leak detection monitoring systemdouble containment leak sensor cable, systems by a manufacturer's representative who is experienced in the installation, adjustment and operation of the equipment specified. Inspect the installation, and supervise the adjustment and testing of the equipment.

### 1.8.2 Extra Materials

Submit the manufacturer's installation recommendations or instructions for each material or procedure to be utilized, including materials preparation. Concurrent with delivery and Installation of the specified piping systems and appurtenances, furnish spare parts for each different item of material and equipment specified that is recommended by the manufacturer to be replaced any time up to 3 years of service. For each type and size of valve, provide the following extra materials: lubricator, lubricant (with appropriate temperature rating), lubricator/isolating valve; galvanized operating wrench, 4.1 feet long, for T-handled operators; galvanized operating key for cross handled valves. Extra materials include 2 of the following spare parts for each type and size of valve: gaskets; all elastomer parts; stem packing; seat rings.

## PART 2 PRODUCTS

### 2.1 SYSTEM REQUIREMENTS

This specification covers the requirements for above and below grade liquid process pipe, pipe supports, fittings, equipment and accessories located both inside and outside of treatment plants.

#### 2.1.1 Design Requirements

Select and design support systems in accordance with MSS SP-58 within the specified spans and component requirements. The absence of pipe supports and details on the contract drawings does not relieve the Contractor of responsibility for sizing and providing supports throughout facility. The structural design, selection, fabrication and erection of piping support system components must satisfy the seismic requirements in accordance with UFC 3-301-01 a

#### 2.1.2 Performance Requirements

The pressure ratings and materials specified represent minimum acceptable

standards for piping systems. Provide piping systems suitable for the services specified and intended. Coordinate each piping system to function as a unit. Provide flanges, valves, fittings and appurtenances with a pressure rating no less than that required for the system in which they are installed.

#### 2.1.2.1 Above Grade Piping Systems

Provide piping systems suitable for design conditions, considering the piping both with and without internal pressure, and installation factors such as insulation, support spans, and ambient temperatures. Give consideration to all operating and service conditions both internal and external to the piping systems.

### 2.2 MATERIALS AND EQUIPMENT

Submit manufacturer's descriptive and technical literature for each piping system, including design recommendations; pressure and temperature ratings; dimensions, type, grade and strength of pipe and fittings; thermal characteristics (coefficient of expansion and thermal conductivity); and chemical resistance to each chemical and chemical mixture in the liquid stream. Provide piping materials and appurtenances as specified and as shown on the drawings, and suitable for the service intended. Ensure piping materials, appurtenances, and equipment supplied as part of this contract are of equal material and ratings as the connecting pipe, new and unused except for testing equipment. Ensure components that serve the same function and are the same size are identical products of the same manufacturer. Provide general materials to be used for the piping systems in accordance with TABLE I and indicate by service in the contract drawings. Submit a list of piping systems, pressure ratings and source of supply for each piping system broken out by material, size and application as indicated on the contract drawings. A list of any special tools necessary for each piping system and appurtenances furnished for adjustment, operation, maintenance and disassembly of the system. Provide pipe fittings compatible with the applicable pipe materials.

#### 2.2.1 Standard Products

Provide material and equipment which are the standard products of a manufacturer regularly engaged in the manufacturing of the products and that essentially duplicate items that have been in satisfactory use for at least 2 years prior to bid opening. Submit the following: Equipment shop drawings and support system detail drawings showing piping systems and appurtenances, such as mechanical joints, valves, local indicators and hangers, including a complete list of equipment and materials. As-built drawings showing pipe anchors and guides, and layout of piping systems relative to other parts of the work including clearances for maintenance and operation. As-built piping and instrumentation diagrams (P&IDs) identifying and labeling equipment, instrumentation, valves, vents, drains, and all other inline devices; if the contract drawings contained P&IDs, revise the P&IDs found in the contract drawings to reflect the constructed process system, as directed by the Contracting Officer. Use nominal sizes for standardized products. Provide pipe, valves, fittings and appurtenances supported by a service organization that is, in the opinion of the Contracting Officer, reasonably convenient to the site.

### 2.2.2 Identification and Tagging

Each piece of pipe must bear the ASTM designation and all other markings required for that designation. Mark valves in accordance with MSS SP-25 and securely attach an identification tag using stainless steel wire. Identification tags must be 1.375 inch minimum diameter, made of engraved laminated plastic engraved anodized aluminum stamped stainless steel. Display service, valve identification number shown on the contract drawings, the manufacturer's name, and the valve model number.

## 2.3 DUCTILE IRON PIPING SYSTEM

### 2.3.1 Ductile Iron Pipe

Provide ductile iron pipe for pressure service with a design and wall thickness conforming to AWWA C115/A21.15. Provide ductile iron pipe with a double thickness cement lining conforming to AWWA C104/A21.4. The ductile iron shall be glass lined and class 52 thickness when not in a buried condition and class 53 when buried.

### 2.3.2 Ductile Iron Joints

Provide joints with a working pressure rating for liquids equal to the pressure rating of the connected pipe. Provide dielectric fittings or isolation joints between all dissimilar metals.

#### 2.3.2.1 Restrained Joints

Provide restrained joints conforming to the requirements of AWWA C111/A21.11, and designed for a working pressure equal to connected pipe rating. When using ductile iron pipe with restrained joints, supply field cuts with a lock ring complete with retainer, retainer lock and roll-pin, as required by manufacturer's recommendations, procedures and/or installation instructions.

#### 2.3.2.2 Flanged Joints

Provide flanged joints conforming to AWWA C110/A21.10. Provide gaskets, bolts and nuts with flanged joints in sufficient quantity for the complete assembly of each joint. Gaskets must be vulcanized synthetic rubber, reclaimed rubber is not acceptable.

### 2.3.3 Ductile Iron Fittings

Fittings must be ductile iron AWWA C153/A21.53. Up to 12 inches inclusive, the fittings must be 250 psig rated. Pipe and fittings must be glass lined. Provide flanges and flanged fittings conforming to ASME B16.1 and rated for 150 psig service. Provide ductile iron materials. For tie-in to existing flanges, field check existing flanges for nonstandard bolt hole configurations and design as required to assure new pipe and flange mate properly. Bolts and nuts must be carbon steel conforming to ASTM A307, Grade B. Provide bolts with washers of the same material as the bolts. Gaskets must be rubber full face, maximum 0.125 inch thick. Sacrificial zinc anode bolts from Mars Company shall be provided for each bolt or studs (both sides).

### 2.3.4 Corrosion Control

Ductile iron piping must be hot-dipped galvanized in accordance with

ASTM A153/A153M. Coat and wrap buried pipe per manufacturer standards.,  
2.4 CARBON STEEL PIPING SYSTEM

#### 2.4.1 Carbon Steel Pipe

##### 2.4.1.1 General Service

Provide carbon steel pipe meeting the requirements of ASTM A53/A53M seamless , Grade A, Schedule 80 and , hot-dipped galvanized. Provide Schedule 80 buried carbon steel piping and fittings.

##### 2.4.1.2 High Temperature Service

Seamless carbon steel pipe for high temperature service must conform to ASTM A106/A106M Grade A , , Schedule 80 with dimensions conforming to ASME B36.10M.

#### 2.4.2 Carbon Steel Tubing

Provide tubing meeting the requirements of ASTM A334/A334M, seamless Grade 1 carbon steel or ASTM A423/A423M, seamless , Grade 1 low-alloy carbon steel with nominal size and wall thickness in accordance with Pipe Schedule.

#### 2.4.3 Carbon Steel Joints

Join carbon steel piping by flanges or mechanical joints for grooved ends meeting the requirements of AWWA C606. Provide dielectric fittings or isolation joints between all dissimilar metals.

#### 2.4.4 Carbon Steel Fittings

Fittings must be heat-treated low-carbon steel. Where cast fittings are not available, fabricate segmental welded steel fittings, ASTM A53/A53M, Grade B, meeting the requirements of manufacturer's recommended wall thicknesses.

##### 2.4.4.1 Threaded Fittings

Threaded fittings must be Class 150 300, low carbon steel, ASTM A858/A858M, conforming to ASME B16.11, and threaded in accordance with ASME B1.20.1. Threaded, rigid couplings must be seamless, Type I hot-dipped galvanized Type II electrogalvanized carbon steel in accordance with ASTM A865/A865M and threaded in accordance with ASME B1.20.1. Use polytetrafluoroethylene (PTFE) pipe-thread tape conforming to ASTM D3308 for lubricant/sealant.

##### 2.4.4.2 Welding Fittings

Provide butt-welding or socket-welding welding fittings. Provide welding fittings consisting of forged steel, ASTM A105/A105M Class 150 .

##### 2.4.4.3 Flanged Fittings

Ensure the internal diameter bores of flanges and flanged fittings are the same as that of the associated pipe. The flanges must be welding neck or threaded type. Flanges and flanged fittings must be forged steel, ASTM A105/A105M or ASTM A727/A727M , faced and drilled to ASME B16.5 Class 150 with a flat face. For tie-in to existing flanges, field check existing flanges for non-standard bolt hole configurations and design as



required to assure new pipe and flange mate properly. Provide bolting consisting of alloy-steel ASTM A193/A193M Grade B5 hex head bolts and ASTM A194/A194M Grade 5 hex head nuts . When mating flange on valves or equipment is cast iron, use ASTM A193/A193M Grade B8 Class 1 bolts and ASTM A194/A194M Grade 8 heavy hex head nuts . Provide bolts with washers of the same material as the bolts. Provide gaskets meeting the requirements of ASME B16.5. Provide nonmetallic gaskets conforming to ASME B16.21 that are a maximum 0.125inch thick chloroprene rubber, durometer hardness No.80, 1,500 psi minimum tensile strength, 125 percent minimum elongation, flat ring type for use with raised face flanges and full face type for use with flat face flanges.

#### 2.4.4.4 Compression Fittings for Tubing

Provide compression fittings consisting of carbon steel ASTM A108 or ASTM A576 nuts, ferrules and bodies rated for pressure of application. Provide straight threads conforming to ASME B1.1.

#### 2.4.5 Carbon Steel Coatings

Coat carbon steel piping components with corrosion resistant materials. Ensure coatings and finishes are 100 percent holiday free.

##### 2.4.5.1 Silicone Coating

Prepare carbon steel piping surfaces in accordance with SSPC SP 6/NACE No.3. Provide surfaces with an alkyd primer of 2.5 mils dry film thickness followed by two alkyd modified silicone final coats.

##### 2.4.5.2 Zinc Coating

Galvanizing must be hot-dip applied and meet the requirements of ASTM A153/A153M; electroplated zinc or cadmium plating is unacceptable. Stainless steel components may be substituted where galvanizing is specified.

##### 2.4.5.3 Thermoplastic Resin Coating System

Carbon steel piping surfaces must have a minimum of 4 coats of phenolic type coatings applied at a minimum dry film thickness of 1.6 mils per coat. Bake each coat at 300 degrees F for 10 minutes. Cure the full coating system in an oven at 375 degrees F for 30 minutes.

#### 2.4.6 Carbon Steel Cathodic Protection

Buried ferrous piping must have cathodic protection.

### 2.5 STAINLESS STEEL PIPING SYSTEM

#### 2.5.1 Austenitic Piping

##### 2.5.1.1 Stainless Steel Pipe

Stainless steel pipe intended for general corrosive service must meet the requirements of ASTM A312/A312M, seamless ASTM A813/A813M for fit-up and alignment quality, ASTM A814/A814M for flanged and cold-bending quality, C, Grade TP316L , Schedule in accordance with Pipe Schedule with dimensions conforming to ASME B36.19M.

#### 2.5.1.2 Stainless Steel Tubing

Provide stainless steel tubing meeting the requirements of ASTM A269/A269M or ASTM A632, seamless, Grade TP316L with nominal size and wall thickness in accordance with Pipe Schedule.

#### 2.5.1.3 Stainless Steel Joints

Join stainless steel piping by welded fittings or flanges. Provide dielectric fittings or isolation joints between all dissimilar metals.

#### 2.5.1.4 Stainless Steel Welding Fittings

Provide butt-welding or socket-welding welding fittings. Provide welding fittings consisting of forged austenitic stainless steel, ASTM A403/A403M Grade TP316L, .

#### 2.5.1.5 Stainless Steel Flanged Fittings

Provide internal diameter bores of flanges and flanged fittings that are the same as that of the associated pipe. Ensure flanges are welding neck slip-on or threaded type. Provide flanges and flanged fittings consisting of forged austenitic stainless steel, ASTM A182/A182M Grade TP316L, Class 150300, drilled to ASME B16.5 with a flat face. For tie-in to existing flanges, field check existing flanges for non-standard bolt hole configurations and design as required to assure new pipe and flange mate properly. Bolting must be alloy-steel ASTM A193/A193M Grade B8 hex head bolts and ASTM A194/A194M Grade 8 8C hex head nuts. When mating flange on valves or equipment is cast iron, use ASTM A193/A193M Grade B8 Class 1 bolts and ASTM A194/A194M Grade 8 heavy hex head nuts. Provide bolts with washers of the same material as the bolts. Provide gaskets meeting the requirements of ASME B16.5. Provide nonmetallic gaskets conforming to ASME B16.21 and that are a maximum 1/8 inch thick chloroprene rubber, durometer hardness No.80, 1,500 psi minimum tensile strength, 125 percent minimum elongation, flat ring type for use with raised face flanges and full face type for use with flat face flanges.

#### 2.5.1.6 Stainless Steel Crimping Fittings

Crimping fittings must be cold drawn, TP316L austenitic stainless steel. O-ring seals must be fluoro-elastomeric.

#### 2.5.1.7 Compression Fittings for Tubing

Provide compression fittings consisting of ASTM A479/A479M stainless steel, Grade TP316, nuts, ferrules and bodies rated to pressure of the application. Provide straight threads conforming to ASME B1.1.

### 2.6 COPPER PIPING SYSTEM

#### 2.6.1 Copper Pipe

Seamless C12200 copper alloy pipe, must be a 061 annealed Temper with extra strong standard dimensions conforming to ASTM B42

#### 2.6.2 Copper Tubing

Provide seamless copper alloy tubing conforming to ASTM B88 alloy C12200, Type K, with a 060 (annealed) . .

### 2.6.3 Copper Joints

Join pipe using threaded soldered or brazed fittings and flanged connections to equipment. Join tubing using solder flared or compression fittings. Provide dielectric fittings or isolation joints between all dissimilar metals.

### 2.6.4 Copper Fittings

Provide component castings of flanges and fittings consisting of copper alloy C83600 (also known as alloy 85-5-5-5), Temper in accordance with ASTM B62. Provide solder joint fittings conforming to ASME B16.22 and ASME B16.18. Provide fittings for flared copper tube conforming to ASME B16.26. Provide cast bronze threaded fittings conforming to ASME B16.15 and are threaded in accordance with ASME B1.20.1. Ensure flanges and flanged fittings are faced and drilled Class 150 in accordance with ASME B16.26. Provide copper and bronze press fittings conforming to material requirements of ASME B16.18 or ASME B16.22 and performance criteria of IAPMO PS 117. Sealing elements must be of EPDM and factory installed or an alternative supplied by fitting manufacturer. For tie-in to existing flanges, field check existing flanges for non-standard bolt hole configurations and design as required to assure new pipe and flange mate properly. Tubing compression fittings must be forged brass alloy C37700, conforming to ASTM B124/B124M, nuts, ferrules and bodies rated for pressure of the application with straight threads conforming to ASME B1.1.

#### 2.6.4.1 Bolting For Copper Piping

Provide bolting materials that meet the requirements of ASME B31.1 and consist of ASTM B98/B98M alloy C65100 or C66100 materials. Provide bolts with washers of the same material as the bolts.

#### 2.6.4.2 Gaskets For Copper Piping

Gaskets listed in ASME B16.5, Annex E, Group 1a may be used with any flange rating class and bolting. Provide nonmetallic, full faced gaskets used with low strength or non-ferrous bolting with a seating pressure less than 1,600 psi. Ensure gasket dimensions conform to ASME B16.21.

### 2.6.5 Solder For Copper Piping

Use solder and flux conforming to ASTM B32 and AWS A5.8/A5.8M. .

### 2.6.6 Copper Piping Supports

Support piping system using padded steel piping supports that conform to MSS SP-58. Do not use conventional steel and galvanized pipe hangers for copper piping systems. Ensure all valves, instruments and other equipment attached to the piping system are individually supported.

## 2.7 PLASTIC PIPING SYSTEM

Submit documentation certifying that the manufacturer of each thermoplastic piping system is listed with the Plastic Pipe Institute as meeting the recipe and mixing requirements of the resin manufacturer for the resin used to manufacture each of the respective thermoplastic pipe systems.

#### 2.7.1 PVC Pipe

PVC, ASTM D1784, minimum cell classification 12545-C, pipe must be Schedule 80 conforming to ASTM D1785 .

#### 2.7.2 PVC Tubing

Provide flexible and clear tubing with nominal size and wall thickness in accordance with Pipe Schedule and reinforcement.

#### 2.7.3 PVC Joints

Join piping system by socket-weld or flanged connections except where connecting to unions, valves, and equipment with threaded connections that may require future disassembly. Ensure connections at those points are threaded and back-welded. Tubing connections must use compression fittings.

#### 2.7.4 PVC Fittings

Ensure the schedule rating for the fittings is not less than that for the associated pipe. Fittings must be ASTM D1784, , PVC conforming to the requirements of ASTM D2464, threaded in accordance with ASME B1.20.1 or ASTM D2466, socket type or ASTM D2467, socket type.

##### 2.7.4.1 Flanged Fittings

Provide flanges and flanged fittings that are Class 125, one piece, molded hub type, flat faced, and conform to ASME B16.1 drilling. Provide flanges complete with one-piece, molded PVC stub ends. Ensure flanged connections have the same pressure rating as the pipe or greater. Bolting must be stainless steel, ASTM A193/A193M, Grade B8 hex head bolts and ASTM A194/A194M, Grade 8 hex head nuts. Provide bolts with washers of the same material as the bolts. Gaskets must be full-faced, maximum 1/8 inch thick, fabricated from polytetrafluoroethylene (PTFE) in accordance with ASME B16.21. When the mating flange has a raised face, use a flat ring gasket and provide a filler gasket between outer diameter of the raised face and the flange outer diameter to protect the PVC flange from bolting moment.

##### 2.7.4.2 Tubing Fittings

Provide compression type fittings comprised of polyvinylidene fluoride (PVDF) nuts and bodies, with elastomeric O-ring seals , rated to pressure of application with straight threads conforming to ASME B1.1.

#### 2.7.5 PVC Solvent Cement

Join socket connections with PVC solvent cement conforming to ASTM D2564. Ensure manufacture and viscosity is as recommended by the pipe and fitting manufacturer to assure compatibility. Prepare joints with primers conforming to ASTM F656 prior to cementing and assembly.

### 2.8 CHLORINATED POLYVINYL CHLORIDE (CPVC)

#### 2.8.1 CPVC Pipe

CPVC, ASTM D1784, minimum cell classification 23447, pipe must be Schedule

80 conforming to ASTM F441/F441M .

#### 2.8.2 CPVC Joints

Join piping system by socket-weld or flanged connections except where connecting to unions, valves, and equipment with threaded connections that may require future disassembly. Ensure connections at those points are threaded and back-welded.

#### 2.8.3 CPVC Fittings

Ensure the schedule rating for the fittings is not less than that for the associated pipe. Fittings must be ASTM D1784, cell classification 23447 , CPVC conforming to the requirements of ASTM F437, threaded in accordance with ASME B1.20.1 , ASTM F438, socket type or ASTM F439, socket type.

##### 2.8.3.1 Flanged Fittings

Provide flanges and flanged fittings that are Class 125, one piece, molded hub type, flat faced, and conforming to ASME B16.1 drilling. Provide flanges complete with one-piece, molded CPVC stub ends. Ensure flanged connections have the same pressure rating as the pipe or greater. Bolting must be stainless steel, ASTM A193/A193M, Grade B8 B8M hex head bolts and ASTM A194/A194M, Grade 8 hex head nuts. Provide bolts with washers of the same material as the bolts. Gaskets must be full-faced, maximum 0.125 inch thick, fabricated from polytetrafluoroethylene (PTFE) in accordance with ASME B16.21. When the mating flange has a raised face, use a flat ring gasket and provide a filler gasket between outer diameter of the raised face and the flange outer diameter to protect the CPVC flange from the bolting moment.

#### 2.8.4 Solvent Cement

Join socket connections with PVC solvent cement conforming to ASTM F493. Ensure manufacture and viscosity is as recommended by the pipe and fitting manufacturer to assure compatibility. For sodium hypochlorite, use Weld On 724 CPVC cement or equal.

### 2.9 POLYVINYLIDENE FLUORIDE (PVDF)

#### 2.9.1 PVDF Pipe

PVDF, conforming to ASTM D3222, pipe must be 80.

#### 2.9.2 PVDF Tubing

Provide flexible tubing with nominal size and wall thickness in accordance with Pipe Schedule.

#### 2.9.3 PVDF Joints

Join PVDF pipe by thermal butt-fusion , except where connecting to valves and equipment that may require future disassembly, provide flanged joints. Tubing connections must use compression fittings.

#### 2.9.4 PVDF Fittings

Provide molded PVDF fittings. Provide fittings with the same or higher pressure rating as the pipe when installed in accordance with the latest

technical specifications. Provide flanges and flanged fittings that are Class 125, one piece, molded hub type, flat faced, and conforming to ASME B16.1 drilling. Provide flanges complete with one-piece, molded PVDF stub ends. Ensure flanged connections have the same pressure rating as the pipe or greater. Bolting must be stainless steel, ASTM A193/A193M, Grade B8 B8M hex head bolts and ASTM A194/A194M, Grade 8 hex head nuts. Provide bolts with washers of the same material as the bolts. Gaskets must be full-faced, maximum 1/8 inch thick, fabricated from polytetrafluoroethylene (PTFE) in accordance with ASME B16.21. When the mating flange has a raised face, use a flat ring gasket and provide a filler gasket between outer diameter of the raised face and the flange outer diameter to protect the PVDF flange from bolting moment. Provide compression type tubing fittings comprised of polyvinylidene fluoride (PVDF) nuts and bodies, with elastomeric O-ring seals, rated for the pressure of the application with straight threads conforming to ASME B1.1.

2.10 POLYETHYLENE (PE)

#### 2.10.1 PE Pipe

Ensure pipe is extruded from PE, ASTM D3350 with a minimum cell classification of 324433-C. The PE pipe must be Schedule 80.

#### 2.10.2 PE Tubing

Tubing must be flexible crosslinked PE conforming to ASTM D3350, minimum cell classification 35400, and dimensioned in accordance with ASTM F876 with nominal size in accordance with Pipe Schedule.

#### 2.10.3 PE Joints

Join PE pipe by thermal butt-fusion socket heat fusion and/or socket electrofusion, except where connecting to valves and equipment that may require future disassembly, use flanged joints.

#### 2.10.4 PE Fittings

Provide PE fittings with the same or higher pressure rating as the pipe when installed in accordance with the latest technical specifications. Provide molded PE fittings. Provide butt-fusion fittings conforming to ASTM D3261. Provide socket-fusion fittings conforming to ASTM D2683 with tools meeting the requirements of ASTM F1056. Provide insert fittings conforming to ASTM D2609.

##### 2.10.4.1 Couplings

Join couplings and saddle joints by electrofusion in accordance with ASTM F1055.

##### 2.10.4.2 Flanged Fittings

Provide flanges and flanged fittings that are Class 125, ASME B16.1 ASME B16.5 drilling. Provide flanges complete with one-piece, molded PE stub ends. Ensure flanged connections have the same pressure rating as the pipe or greater. Bolting must be stainless steel, ASTM A193/A193M, Grade B8 hex head bolts and ASTM A194/A194M, Grade 8 hex head nuts. Provide bolts with washers of the same material as the bolts. Gaskets must be full-faced, maximum 1/8 inch thick, fabricated from polytetrafluoroethylene (PTFE) in accordance with ASME B16.21.

#### 2.10.4.3 Tubing Fittings

Provide compression type fittings comprised of forged brass alloy C37700, conforming to ASTM B124/B124M, nuts, polyvinylidene fluoride (PVDF) nuts and bodies, with elastomeric O-ring seals rated for pressure of application with straight threads conforming to ASME B1.1.

### 2.11 ISOLATION JOINTS AND COUPLINGS

#### 2.11.1 Metallic Piping Couplings

Provide thrust ties where shown on the contract drawings and where required to restrain the force developed by 1.5 times the maximum allowable operating pressures specified. For metallic pipe other than ductile iron, attach thrust ties with fabricated lugs. For ductile iron pipe, attach thrust ties with socket clamps against a grooved joint coupling or flange. For exposed installations, use zinc-plated nuts and bolts. However, high-strength, low-alloy steel, in accordance with AWWA C111/A21.11, may be substituted for use on cast iron and ductile iron couplings. For buried and submerged installations, provide 316 stainless steel bolts and nuts. Use steel middle rings and followers that are fusion bonded epoxy-lined and coated in accordance with Section 09 90 00 PAINTS AND COATINGS and pressure tested beyond yield point.

##### 2.11.1.1 Sleeve-Type Couplings

Use sleeve-type couplings for joining plain end pipe sections in a flexible manner with a diameter to properly fit the pipe. A coupling consists of one stainless steel or ductile iron middle ring, two stainless steel or ductile iron followers, two elastomeric wedge section gaskets and elliptic-neck, track-head steel bolts designed to properly compress the gaskets. For pipe sizes between 0.5 through 1.5 inch, ensure the followers are ductile iron, and the middle ring is in accordance with ASTM A513/A513M or ASTM A395/A395M with AWWA C111/A21.11 bolting. For pipe sizes 2 inch and larger, ensure the followers are ASTM A395/A395M, and the middle ring is ASTM A513/A513M or ASTM A395/A395M with AWWA C111/A21.11 bolting. Gaskets must be fluoro-elastomeric. Split sleeve-type couplings may be used in aboveground installations under special situations and when approved in advance by the Contracting Officer.

##### 2.11.1.2 Transition Couplings

Transitional couplings may be used to connect two pipes of the same material that have small differences in outside diameter. Size a fully assembled transitional coupling to properly fit pipe diameters. The coupling must consist of one stainless steel or ductile iron middle ring, two stainless steel or ductile iron followers, two elastomeric wedge section gaskets and elliptic-neck, track-head steel bolts designed to properly compress the gaskets. The coupling must use fluoro-elastomeric, wedge gaskets. Size the coupling to match the associated piping.

##### 2.11.1.3 Flanged Coupling Adapters

Provide flanged coupling adapters consisting of fully assembled units manufactured to meet ASTM A126 Class B, cast iron. The flanges must mate with ASME B16.1 ASME B16.5 Class 150 flanges of the same nominal size. Apply a factory applied corrosion resistant coating. The coupling must use fluoro-elastomeric, wedge gaskets. Where pipe movement out of the adaptor may occur, provide proper anchorage of the pipe and furnish

couplings with lock pins. Size the coupling to match the associated piping.

## 2.11.2 Couplings for Nonmetallic Piping

### 2.11.2.1 Bellows Coupling

Provide a bellows coupling with a minimum of two polytetrafluoroethylene (PTFE) convolutions unless otherwise shown, with ductile iron flanged, faced and drilled to ASME B16.1/ASME B16.5 Class 150 end connections, and metal reinforcing bands. The maximum allowable working pressure is rate for the application. Limit bolting to restrain the force developed by 1.5 times the specified maximum allowable operating pressure. Size the coupling to match the associated piping.

### 2.11.2.2 Compression Coupling

A compression coupling consists of one stainless steel middle section, two stainless steel mechanical nuts, two elastomeric gaskets and two machined steel lock rings. The coupling must use fluoro-elastomeric, wedge gaskets. The rated pressure must work with application.  
B. Size the coupling to match the associated piping.

## 2.12 VALVE BOXES SERVICE BOXES, VALVE MANHOLES AND VALVE PITS

The box length must adapt to the depth of cover required over the pipe at the valve location without full extension. Boxes must be cast iron or concrete, except that concrete boxes may be installed only in locations not subjected to vehicular traffic. Provide boxes with housings of sufficient size to completely cover the valve or service stop and complete with covers.

### 2.12.1 Valve Boxes

Provide cast-iron valve boxes with minimum metal thickness of 3/16 inch and of the extension type with slide-type adjustment and flared base. Provide concrete boxes that are the standard product of a manufacturer of precast concrete equipment.

### 2.12.2 Service Boxes

Provide extension service boxes with either screw or slide-type adjustment.

### 2.12.3 Valve Manholes or Pits

Construct valve manholes or pits for automatic valves and meters installed below grade in accordance with Section 33 61 13.19 VALVES, PIPING, AND EQUIPMENT IN VALVE MANHOLES.

## 2.13 VALVES

### 2.13.1 General Requirements For Valves

Valves include operator, actuator, handwheel, chain wheel, extension stem, floor stand, worm and gear operator, operating nut, chain, wrench, and all other accessories required for a complete operation. Ensure valves are suitable for the intended service. Renewable parts are not to be of a lower quality than those specified. Ensure valve ends are compatible with adjacent piping system. Size an operator to operate the associated



valve for the full range of pressures and velocities. Valves will open by turning counterclockwise. Provide factory mounted operators, actuators, and accessories.

#### 2.13.2 Valve Schedule

Submit a list of valve materials, pressure ratings, valve operator's materials, air supply pressure, electrical service, location, source of supply, and reference identification as indicated in the contract drawings. Provide a list of any special tools necessary for each valve type and appurtenances furnished for adjustment, operation, maintenance and disassembly. Requirements relative to this paragraph are shown on the located in the contract drawings .

#### 2.13.3 Factory Finishing

Provide valves with an epoxy lining and coating in accordance with AWWA C550 unless otherwise specified. Ensure the epoxy is either a two-part liquid material or a heat-activated fusion material except that only a heat-activated material applies if a valve coating is specified as "fusion" or "fusion bonded" epoxy. Provide epoxy lining and coating with a minimum 7.0 mils dry film thickness except where it is limited by valve operating tolerances. Finish exposed valves in accordance with Section 09 90 00 PAINTS AND COATINGS.

#### 2.13.4 Check Valves

##### 2.13.4.1 Thermoplastic Check Valve

Provide thermoplastic check valves, 0.25 inch through 16 inch, of a ball-check design, manufactured of chlorinated polyvinyl chloride (CPVC) with flanged end connections. Use valves that are rated for 150 psig service. Provide valves with fluoro-elastomeric O-ring seals and seats.

##### 2.13.4.2 Double Disc Swing Check Valve

Double disc swing check valves, 2 inch through 52 inch, must be wafer style, spring loaded swing check valve, with a Duplex Stainless Steel or Marine Bronze body, a Duplex Stainless Steel or marine grade bronze disc, resilient seats, stainless steel hinge pin, and a stainless steel stop pin spring. Ensure valves 2 inch through 12 inch are rated for 200 psig service at 140 degrees F and valves 14 inch through 52 inch are rated for 150 psig service at 140 degrees F.

##### 2.13.4.3 Silent Check Valve

Provide silent check valves conforming to the following:

- a. Silent check valves, 2 through 10 inch, center guided valve with a bronze body, bronzetrim, bronze springs. Ensure valves are rated for 200 psig service at 140 degrees F.
- b. Silent check valves, 2.5 through 42 inch, must be globe style, center guided valve with ASME B16.1 Class 125 flanged end connections, a bronze body, bronze trim, butadiene acrylonitrile seat, and bronze spring. Ensure valves are rated for 150 psig service.

## 2.13.5 Ball Valves

### 2.13.5.1 General Purpose Ball Valves

Provide general purpose ball valves conforming to the following:

- a. Ball valves, 2 inch and smaller, must be end entry type with bronze bodies and threaded, in accordance with ASME B1.20.1, regular ports. Provide valves with polytetrafluoroethylene (PTFE) seats and packing, brassballs and hand lever or pneumatically actuated, orelectrically actuated operators as shown in the drawings. Ensure valves are rated for 400 psig psig service at 150 degrees F and conform to ASME B16.34 . Install a union adjacent to the valves to provide access to the seat.
- b. Ball valves, 2.5 inch and larger, must be end entry type with bronze bodies and ASME B16.1 flanged ends. Provide valves with polytetrafluoroethylene (PTFE)packing and seats, a brass ball, full bore ports, and hand wheel,pneumatically actuated, or electrically actuated operators as shown in the contract drawings. Ensure valves are rated for 400 psig psig service at 150 degrees F and conform to ASME B16.34 Class.
- c. Provide ball valves, 2 to 12 inch, conforming to ASME B16.34, and with a bronzebody, stainless steel ball and stem, polytetrafluoroethylene (PTFE) packing and gasket, and flangedends, full port. Ensure valves are rated for 200 psig service, and have hand lever, pneumatically actuatedor electrically actuatedoperators as shown in contract drawings..

### 2.13.5.2 Multiple Piece Body Ball Valves

Provide multiple piece body ball valves, 1.5 to 6 inch, with three piece bodies constructed of cast steel ASTM A351/A351M Grade CF8Mstainless steel. Valves must have a TP316 stainless steel ball, and ASME B16.5 flangedend connections. Ensure valves are rated for 1000 psig service and conform to ASME B16.34 . Provide valves with reinforced polytetrafluoroethylene (PTFE) seats and stem packing, that are fullbore, and equipped with hand lever, , pneumatically actuated, or electrically actuated operators as shown on the contract drawings.

### 2.13.5.3 Thermoplastic Ball Valve

Provide thermoplastic ball valves, 6 inch and smaller, rated for 150 psig service at 120 degrees F, and have ASTM D1784, chlorinated polyvinyl chloride (CPVC) bodies, balls, and stems. Provide valves of end entry, double union design, with flangedends connections, a polytetrafluoroethylene (PTFE) O-ring stem seals. Provide valves with operator as shown in the contract drawings.

## 2.13.6 Gate Valves

### 2.13.6.1 General Service Gate Valves

Provide general service gate valves conforming to the following:

- a. Provide gate valves, 2 inch and smaller, with bronze bodies and stems, screwed or bolted bronze bonnets, single solid wedge bronze discs, and non-rising stems. Provide valves rated for 175 psig service and conforming to ASME B16.34 Class 150 . End connections

must be ASME B16.1 flanged. Equip valves operators as specified in the contract documents.

- b. Provide gate valves, 2.5 inch and larger, with marine grade bronze bodies with bronze trim. Provide valves that have Class 125 flanged or mechanical joint (buried only end connections. Provide a NRS Bolted type bonnet. Discs must be wedge type of bronze construction, and have nonrising stems . Include a by-pass of the same materials as the gate valve for each gate valve, 16 inch and larger. Provide bypass meeting the requirements of AWWA C500. Use valves rated for 200 psig service. Equip valves with operators as required in the contract documents.

#### 2.13.6.2 Thermoplastic Gate Valve

Provide thermoplastic gate valves, 1/2 inch and larger, with ASTM D1784 chlorinated polyvinyl chloride (CPVC), , bolted bonnets, single polypropylene wedge discs, non-rising stems, and flanged end connections. Provide valves rated for 150 psig psig service at 140 degrees F. Equip valves with operators as called out in contract drawings.

#### 2.13.7 Globe Valves

##### 2.13.7.1 General Requirements For Globe Valves

Provide globe valves, 3 inch and smaller, that are globe style valve with bronzebodies, with bronze trim, and bronzebonnets. Provide valves conforming to ASME B16.34 Class 150, and with ASME B16.11 threaded or ASME B16.5 flanged or ASME B16.1 flanged end connections. Include threaded bonnets, inside screws, rising stems, plug discs constructed of bronze, and bronze rings. Ensure valves are rated for 200 psig service. Equip valves with operators as required in contract documents.

##### 2.13.7.2 Needle Valve

Needle valves, 1 inch and smaller, must be of a straightpattern and must have TP316 stainless steel bodies and trim. Provide valves conforming to ASME B16.34 Class 150 with ASME B16.11 threaded, in accordance with ASME B1.20.1, or if used on tubing, have tubing compression fittings, that match associated tubing fittings, end connections. Ensure valves include integral bonnets, TP316 stainless steel stems, plug stem tips constructed of polytetrafluoroethylene (PTFE)bronze, fluoro-elastomer packing and O-ring stem seals. Ensure valves are rated for 300 psig service. Equip valves with operators as required in contract documents.

##### 2.13.7.3 Hose Valve

Hose valves, 0.75 through 3 inch, must be globe style angle pattern hose valves with cast iron ductile iron carbon steel bronzebodies, bronze trim, inside screws, rising stems, and polytetrafluoroethylene (PTFE)disc. The outlet ports must be cast brass in accordance with ASME B1.20.1 pipe threads, male by male, nipple adapter with hexagonal wrench feature and brass cap with chain. Provide valves rated for 125 psig service.

#### 2.13.8 Plug Valves

##### 2.13.8.1 Eccentric Valve

Provide nonlubricated type eccentric valves, 3 inch and smaller, rated

for 175 psig service at 140 degrees F. Valves must have drip-tight shutoff with pressure from either direction, and bronze bodies, in accordance with ASME B16.5 flanged or ASME B16.1 flanged or ASME B16.11 threaded end connections as applicable for application, all metal, matching body plugs with round ports, stainless steel seats, Monel stem bearings, and fluoro-elastomer filled polytetrafluoroethylene (PTFE) seals. Provide valves conforming to ASME B16.34 Class 150. Equip valves with operators as defined in contract documents.

#### 2.13.8.2 Lined Eccentric Valve

Provide nonlubricated type eccentric valves, 3 through 54 inch, rated for 175 psig service at 140 degrees F. Provide valves with drip-tight shutoff with pressure from either direction, and bronze bodies in accordance with ASME B16.1 flanged end connections. Provide cast iron plugs with round or rectangular ports of no less than 80 percent of the connecting pipe area and coated with fluoro-elastomer. Provide valves with stainless steel seats, self-lubricating reinforced polytetrafluoroethylene (PTFE) stem bearings, and nitrile rubber grit seals on the stems. Equip valves with operators as required in contract drawings. Ensure valves conform to ASME B16.34 Class 150.

#### 2.13.9 Butterfly Valves

##### 2.13.9.1 Standard Service Butterfly Valve

Provide butterfly valves, 2 inch and larger, with marine rated bronze or super duplex stainless steel bodies, with ASME B16.5 flanged or ASME B16.1 flanged end connections. Provide valves conforming to Class 150 or ASME B16.34 Class 150 as required for application. Discs must be contoured duplex stainless steel. Provide duplex stainless steel valve shafts with self-lubricating, corrosion-resistant sleeve type bearings. Provide valve seats, constructed of chloroprene, for 24 inch and smaller valves and attach to either the valve body or the disc. Provide valves with operators as required in the contract documents.

##### 2.13.9.2 Thermoplastic Butterfly Valves

Provide thermoplastic butterfly valves, 1.5 inch and larger, with lugged style bodies constructed of polyvinyl chloride (PVC) coated ductile iron. Provide valves with polyvinyl chloride (PVC) discs, fluoro-elastomeric seats, polytetrafluoroethylene (PTFE) seals, and operators per the contract documents. Ensure valves are rated for 150 psig service at 140 degrees F.

#### 2.13.10 Diaphragm Valves

##### 2.13.10.1 Standard Service Diaphragm Valve

Provide diaphragm valves, 1/2 inch and larger, with marine grade bronze or super duplex stainless steel bodies, ASME B16.1 or ASME B16.5 Class 150 flanged end connections and fluoro-elastomeric seals, and are actuated as indicated on contract drawings. Provide position indicators to indicate diaphragm position. Valves must be rated for 150 psig service at 140 degrees F.

##### 2.13.10.2 Thermoplastic Diaphragm Valve

Provide thermoplastic diaphragm valves, 1/2 inch and larger, with polyvinyl chloride (PVC), ASME B16.1 or ASME B16.5 Class 150 flanged end

connections and polytetrafluoroethylene (PTFE) seals, and that are actuated per the contract drawings. Pneumatically operated valves must be fail-closed. Provide position indicators to indicate diaphragm position. Valves must be rated for 150 psig service at 68 degrees F.

#### 2.13.11 Self-Contained Automatic Valves

##### 2.13.11.1 Pressure-Reducing Valve

Pressure-reducing valves, 1/2 inch and larger, must be direct operated, diaphragm actuated, spring controlled globe valves with bronze bodies. End connections must be ASME B16.1 or ASME B16.5 Class 150 flanged as for application. Provide stainless steel trim and stem. Valves must have externally mounted strainers with cocks. Use valve sizes and ratings as shown in the Valve Schedule on the contract drawings.

Pressure-reducing valves, for the R.O. membrane system and ABW system shall be constructed of duplex stainless steel rated for seawater application. The purpose of these valves is to shut when there is no flow (e.g. act like a check valve when pump stops) as well as control the pressure such that common pressure in the discharge line to the outfall does not exceed 25 psig.

The controls shall consist of an electric actuator including the electric motor, reduction gearing, valve stem drive nut/bushing, position limit switches, mechanical overload torque switches, and automatic declutch-able handwheel. In addition, the valve will close upon power failure.

These valves shall be supplied with a non-intrusive electric actuator capable of being controlled based on a 4-20mA analog signal for valve position.

The electric actuator shall be powered from a 230/460 VAC, 3 Phase, 60 Hz. Power source and are to incorporate motor, integral reversing starter, local control facilities and terminals for remote control and indication connections housed within a self-contained, sealed enclosure.

All programming shall be done digitally. Mechanical dials used to program limits is not acceptable.

Actuators shall be suitable for indoor and outdoor use. The actuator shall be capable of functioning in an ambient temperature ranging from °F(-10 °C) to °F(110 °C).

##### Motor

1. The electric motor shall be Class F insulated, with a time rating of at least 15 minutes at 104°F(40°C) or twice the valve stroking time, whichever is the longer, at an average load of at least 33% of maximum valve torque.

2. Electrical and mechanical disconnection of the motor should be possible without draining the lubricant from the actuator gearcase.

##### Hand operation

1. A handwheel shall be provided for emergency operation, engaged when the motor is declutched by a lever or similar means, the drive being restored to power automatically by starting the motor. The handwheel or selection lever shall not move on restoration of motor drive.

2. Provision shall be made for the hand/auto selection lever to be

locked in both hand and auto positions. It should be possible to select hand operation while the actuator is running or start the actuator motor while the hand/auto selection lever is locked in hand without damage to the drive train.

3. The handwheel drive must be mechanically independent of the motor drive and any handwheel gearing should be such as to permit emergency manual operation in a reasonable time.

4. Clockwise operation of the handwheel shall give closing movement of the valve unless otherwise stated in the job specification.

Remote valve position/actuator status indication.

1. Four contacts shall be provided which can be selected to indicate any position of the valve; Provision shall be made for the selection of a normally closed or open contact form.

2. Contacts shall maintain and update position indication during handwheel operation when all external power to the actuator is isolated.

3. The contacts shall be rated at 5A, 250V AC, 30V DC.

Local position indication

1. The actuator shall include a digital position indicator with a display from fully open to fully closed in 1% increments. Red, green and yellow lights corresponding to Open, Closed and Intermediate position shall be included on the indicator.

2. End of travel indication colors shall be reversible.

Integral starter and transformer

1. The reversing starter, control transformer and local controls shall be integral with the valve actuator suitably housed to prevent breathing and condensation.

2. The starter shall be suitable for 60 starts per hour and of rating appropriate to motor size.

3. The controls supply transformer shall be fed from two of the incoming three phases and incorporate overload protection.

Local Controls

1. The actuator shall incorporate local controls for Open, Close and Stop and a Local/Stop/Remote mode selector switch lockable in any one of the following three positions: local control only, stop (no electrical operation), remote control plus local stop only. It shall be possible to select maintained or non-maintained local control.

2. The local controls shall be arranged so that the direction of valve travel can be reversed without the necessity of stopping the actuator. Provision shall be made to orientate the local controls through increments of 90°.

The actuator shall be factory installed on the valve prior to shipment to the job site.

Pump Control Electrical Panel

A. The valve shall be supplied with an electrical control panel, designed to control the valve.

B. The control panel shall be NEMA 4X Rated. The panel shall not exceed 16" Wide x 16" Tall x 6" Deep.

C. The control panel shall house the following components required to perform the necessary logic functions.

1. Relays

2. Timers
3. Dry Contact (Input) for "Pump Start" command
4. Dry Contact (Output) for "Valve Fail" annunciation

D. The following components shall be mounted on the door of the control panel. All lights, switches, and pushbuttons shall be supplied with independent labels.

1. HAND-OFF-AUTOMATIC selector switch
2. PUMP RUNNING indicating light
3. VALVE FAIL indicating light
4. ALARM RESET pushbutton

E. All wiring within the control panel shall be neat and tidy. Each wire shall be numbered on both ends and should collate with the wire numbers indicated on the electrical schematic provided.

These PRVs will be manufactured by Ross, Cal-Val or equal.

#### 2.13.12 Operators

##### 2.13.12.1 Operator Schedule

Requirements relative to this paragraph are shown on the Operator Schedule located in the contract drawings.

##### 2.13.12.2 Manual Operator

The force in a manual operator is not allowed to exceed 39.3 pound under any operating condition, including initial breakaway. Equip the operator with gear reduction when force exceeds 39.3 pound. Provide a self-locking type manual operator or equip with a self-locking device. Supply a position indicator on quarter-turn valves. Provide worm and gear operators consisting of a one-piece design with worm-gears of gear bronze material. Worm must be hardened alloy steel with the thread ground and polished. Provide traveling nut type operators with threader steel reach rods with an internally threaded bronze or ductile iron nut.

##### 2.13.12.2.1 Exposed Operators

Provide exposed operators with galvanized and painted handwheels. Lever operators are allowed on quarter-turn valves 8 inch and smaller. Supply cranks on gear type operators. If located off of the operator floor, provide chain wheel operator with tiebacks, extension stem, floor stands, and other accessories to permit operation from normal operation level. Ensure valve handles are capable of padlocking, and provide wheels that are lockable with a chain and padlock.

##### 2.13.12.2.2 Underground Operators

Buried service operators on valves larger than 2.5 inch must have a 2 inch operating nut. Buried operators on valves 2 inch and smaller must have a cross handle for operation by a forked key. Enclose the moving parts of valve and operator in housing to prevent contact with the soil. Design buried service operators for quarter-turn valves to withstand an input

torque of 450 foot-pound of input torque at the fully open or fully closed positions, and grease pack and gasket to withstand a submersion in water to 10.2 psig. Provide buried valves with extension stems, bonnets, and valve boxes.

#### 2.13.12.3 Pneumatic Operator

Provide pneumatic operators complete with actuators, air sets, exhaust mufflers, speed controls, pilot solenoids, safety vented isolation valves, and accessories. Ensure pneumatic operators are suitable for full operation range of valve at air supply pressure indicated. Use actuators to return the valve to the closed position upon loss of signal unless otherwise indicated. Use springs to return valve to this failed position. Furnish pneumatic operators with features noted on the valve Schedule in the contract drawings. Provide limit switches on all actuators.

##### 2.13.12.3.1 Cylinder Actuators

Provide cylinder actuators conforming to ANSI/AWWA C541 and ANSI/AWWA C542, and operate with an air supply pressure of 80 psig. Totally enclose the nonswivel type with travel stops and position indicator, and factory lubricate and seal, requiring no additional lubrication. The double acting type must be nonmetallic for operation on nonlubricated air and must have a manual handwheel override independent of the cylinder. Locate the manual override locally next to valve.

##### 2.13.12.3.2 Diaphragm Actuators

Provide diaphragm actuators with a spring return with a aluminum diaphragm case and spring barrel, steel spring and actuator stem, and fabric-reinforced chloroprene diaphragm. Use actuators on quarter-turn valves which include a totally enclosed valve actuating mechanism with adjustable travel stops and valve position indicator with manual override if indicated. The actuating mechanism must factory lubricated and sealed. Size and configure diaphragm actuators for the service indicated and an air supply pressure of 35 psig.

##### 2.13.12.3.3 Air Sets

Include a pressure regulator with internal relief, filter, outlet pressure gauge, and adjustable reduced pressure range as required by the valve actuator. The air set must have an aluminum body and handwheel, safety vented lockout isolation valve, and gauge range 1.33 to 2 times maximum operating pressure.

##### 2.13.12.3.4 Limit Switches

Provide single-pole, double-throw (SPDT) type limit switches, rated 10 amps at 120 volts ac, housed in a NEMA 250 Type 4 X enclosure, and adjustable for open and closed valve positions.

##### 2.13.12.3.5 Positioners

The positioners for modulating actuators must be pneumatic force balance instruments to control valve positions as a function of the input signals. The positioners must accomplish positive positioning of valve by a mechanical feedback connection from the valve actuating mechanism. Provide position feedback through a characterized linear cam to allow adjustment of valve positioning and input signal. Provide positioner



suitable for either a double acting or spring return actuator. Provide positioner with zero and span adjustment and field reversible for direct or reverse action. Include gauges for supply and output pressure and for input signal pressure. Operate modulating valve positioners on a 3 to 15 psig pneumatic 4 to 20 mA or 1 to 5 v dc electric input signal unless otherwise indicated. A positioner for dc input signal with transducers must convert the electrical signal to the appropriate pneumatic signal. Ensure transducer is integral with the positioner or a separate component. If separate, factory mount the transducer on the pneumatic operator. Line electric power is not required for transducer operation. Corrosion-resistant enclosures for positioners and transducers must be splash- and moisture-proof with gasketed covers.

#### 2.13.12.3.6 Solenoid Valve

Use a solenoid valve to pilot the control actuator in the appropriate configuration for the type of actuator being controlled. Provide a pilot operated diaphragm type solenoid valve consisting of a brass body and resilient seat and operate with minimum operating pressure differential no greater than 10.2 psi and maximum operating pressure differential no less than 150 psi . Provide corrosion-resistant internal parts. Provide solenoid valve with Class F molded coils for operation on 120 volts, 60-Hz, ac, unless otherwise indicated. Provide solenoid enclosure conforming to NEMA 250 Type 4X. Solenoids on double acting cylinders for open-close and throttling valves must be four-way with dual coils. Solenoids on spring return cylinders for open-close and throttling valves must be three-way, spring return. Furnish an air exhaust muffler in the exhaust port of all actuator pilot solenoid valves.

#### 2.13.12.4 Electric Operator

Provide electric operators complete with actuators, speed controls and accessories. Operate actuators on 120 VAC, 60 Hz with a 75 percent duty cycle and equip with an AC thermal overload protector with automatic rest, reversing (bi-directional) operation for use with quarter-turn valves, or rotating equipment to full rotation. Gearing must be a two-stage planetary, permanently lubricated self-locking gear train with self-lubricating bearings; connections via male output shaft. Use start-up torque of 120 foot-pound. Use stall torque of 150 foot-pound. Mount two travel stop limit switches with cams, internal, independent, adjustable, and actuated by cams, on the drive shaft. Provide a side mounted hand turn wheel for a manual override. Ensure actuators have a NEMA 250 Type 4 X enclosure with a corrosion resistant, baked epoxy finish as standard. Operate actuator in a temperature range of -40 to plus 150 degrees F. Actuators must fail in last position unless otherwise indicated. Furnish electric operators with features noted on the Valve Schedule in the contract drawings. Provide limit switches on all actuators.

##### 2.13.12.4.1 Limit Switches

Limit switches must be single-pole, double-throw (SPDT) type, rated 10 amps at 120 volts ac, housed in a NEMA 250 Type 4 X enclosure, and adjustable for open and closed valve positions.

##### 2.13.12.4.2 Positioners

Provide positioners for modulating actuators to control valve positions as a function of the input signals. Operate positioner on 120 VAC, 60 Hz.

voltage. The mode of operation must be direct acting reverse acting. Operate modulating valve positioners on a either 4 to 20 mA or 1 to 5 v DC input signal as indicate in the contract drawings unless otherwise indicated. Provide splash-and moisture-proof corrosion-resistant enclosures with gasketed covers for positioners.

#### 2.13.13 Valve Accessories

##### 2.13.13.1 Extension Bonnet for Valve Operator

Provide all extension bonnets as necessary, complete with stem and accessories applicable to the specific valve and operator.

##### 2.13.13.2 Floor Stand and Extension Stem

Provide the nonrising, indicating type floor stand and extension stem, complete with stem, coupling, handwheel, stem guide brackets, and yoke attachment. Space stem guide such that stem L/R ratio does not exceed 200. Supply anchors as required.

##### 2.13.13.3 Floor Box and Stem

Provide the plain type floor box and stem for support of nonrising type stem; complete with stem, operating nut, and stem guide brackets. Space stem guide such that stem L/R ratio does not exceed 200. Supply anchors as required.

##### 2.13.13.4 Chain Wheel and Guide

Provide the handwheel direct-mount type chain wheel and guide, complete with galvanized or cadmium-plated chain.

#### 2.14 DRAINS

Valved drains may not be shown on the detailed drawings for individual pipelines; their absence will not relieve the Contractor of the responsibility for providing and installing them as indicated in the piping and instrumentation diagrams to complete the piping system for the use intended.

##### 2.14.1 Locations

Locate drains as indicated on the contract drawings. Drain all pipeline low points.

##### 2.14.2 Sizes

For pipelines 2.5 inch and larger, provide 0.75 inch drains equipped with ball valves. For pipelines 2 inch and smaller, provide 1/2 inch drains equipped with ball valves.

#### 2.15 SAMPLE PORTS

Sample ports, shown on the flow diagrams and piping and instrument diagrams of the contract drawings, may not be shown on the detailed drawings of the individual pipelines; their absence does not relieve the Contractor of the responsibility for providing them. Provide sample ports as indicated in the piping and instrument diagrams to complete the piping systems for the use intended. Locate sample ports in easily accessible

locations, and avoid potential stagnant points and/or areas where material could collect. S Provide valve body consisting of stainless steel Class 150 with a male ASME B1.20.1 pipe threads inlet connection and female ASME B1.20.1 pipe threads outlet connection. Provide sampling ports comprised of pipe fittings, pipe, and ball valves which comply with material, temperature, and pressure requirements of the associated piping system as specified elsewhere in this Section.

## 2.16 MISCELLANEOUS PIPING COMPONENTS

### 2.16.1 Air Release and Vacuum Breakers

Locate air release vents, and vent such that a hazardous atmosphere will not be created upon operation.

#### 2.16.1.1 Locations

Locate vacuum breakers as indicated on the contract drawings. Ensure all pipeline high points have vacuum breakers. Provide vacuum breakers on all tanks and process equipment.

#### 2.16.1.2 Vacuum Breakers

Provide vacuum breakers 2 inch and smaller of an angle type with all bronze bodies and bonnets, and install at least 6 inch above the flood line of associated equipment and conforming to ASSE 1001 for pipe applied units.

#### 2.16.1.3 Air and Vacuum Valve Suitable for Corrosive Service

Provide air and vacuum valve conforming to ASSE 1001, and automatically exhaust air during the filling of a system while allowing air to re-enter during draining or when vacuum occurs. Ensure valve is rated for 150 psig working pressure and built with a standard elongated body. Provide valve consisting of a bronze body and cover, with duplex stainless steel float and trim. Provide end connections as follows: for 1/2 through 3 inch ASME B1.20.1 pipe threaded inlet and outlet, for 4 inch and larger ASME B16.5 or ASME B16.1 Class 150 flanged inlet with outlet as required to be compatible with application. Fit air and vacuum valve with blowoff valve, quick disconnect couplings, and a minimum 6.6 feet of hose in order to permit back flushing after installation without dismantling the valve.

#### 2.16.1.4 Air Release Valve Suitable for Corrosive Service

Use air release valve to automatically exhaust entrained air that accumulates in a system. Ensure valve is rated for 150 psig working pressure and built with a standard elongated body. Provide valve consisting of a bronze body and cover, with duplex stainless steel float and trim. Ensure valve end connections are ASME B16.5 or ASME B16.1 Class 150 flanged a compatible for application used. Fit air and vacuum valve with blowoff valve, quick disconnect couplings, and a minimum 6.6 feet of hose in order to permit back flushing after installation without dismantling the valve.

#### 2.16.1.5 Combination Air Valve Suitable for Corrosive Service

The valve combines the operating functions of both an air and vacuum valve and an air release valve. The air and vacuum portion must automatically exhaust air during filling of a piping system and allow air to re-enter during draining or when a vacuum occurs. Air release portion must

automatically exhaust entrained air that accumulates in the piping system. Provide valve consisting of a an individual air and vacuum valve and an air relief valve mounted on a common header. Ensure valve is rated for 150psig working pressure and built with a standard elongated body. Provide valve with a bronze body and cover, with duplex stainless steel float and trim. Ensure valve end connections are ASME B16.5or ASME B16.1 Class 150 flangedsuitable for application. he air and vacuum valve to be fitted with a blowoff valve, quick disconnect couplings, and a minimum 6.6 feet of hose in order to permit back flushing after installation without dismantling the valve.

#### 2.16.2 Backflow Preventer

The backflow preventer must be identical in size to pipe. Do not allow total head loss through the complete backflow assembly to exceed 10.1 psi at rated flow.

##### 2.16.2.1 Double Check Valve Assembly

Provide backflow preventer consisting of two independently operating, spring loaded, "Y" check valves rated for 175 psig service at 140 degrees F, with one isolation gate valve, one differential relief valve, and testing cocks. Use port size of 1inch and ASME B1.20.1 threaded . Ensure check valve assembly is rated for 150 psig working pressure at 150 degrees F. Provide assembly meeting the requirements of AWWA C510.

##### 2.16.2.2 Reduced Pressure Backflow Preventer

Provide assembly body consisting of two independent bronze body check valves rated at 150 psig at 140 degrees F, with an intermediate relief valve, and isolation gate valve, as testing cocks. Ensure all internal parts are serviceable in-line. Use port sizes of 1 inch and ASME B1.20.1 threaded. Ensure reduced pressure backflow prevention assembly is rated for 150 working pressure at 150 degrees F. Provide assembly body in accordance with AWWA C511.

##### 2.16.2.3 Backflow Preventer with Intermediate Vent

Provide assembly body consisting of two independent bronze body check valves rated at 175psig at 140 degrees F, with an intermediate atmospheric vent, and isolation gate valve, as testing cocks. Use port sizes of 1 inch and ASME B1.20.1 threaded . Ensure backflow prevention assembly is rated for 150 psig working pressure at 150 degrees F. Provide assembly body in accordance with AWWA C511.

#### 2.16.3 Strainers

Provide simplex strainers with a Y-pattern body. Use port sizes of 1inch and ASME B1.20.1 threaded. Ensure strainers are rated for 150psig working pressure at 150 degrees F and conform to ASTM F1199or ASTM F1200. Provide body consisting of cast bronze with a screwed bronze cap. The screen must be heavy-gauge Monel, 30 mesh and equipped with a ASME B1.20.1 pipe threaded blowoff hole.

#### 2.16.4 Indicating Devices

##### 2.16.4.1 Pressure and Vacuum Gauges

Pressure and vacuum gauges must be stem orpanel mounted, with brass cases

equipped with safety pressure blowout backs aglycerine -filled dials. If stem mounted, the thickness of pipe must be sufficient size and thickness to support the gauge. The gauge sensors must be diaphragm actuated and constructed of Monel or Inconel or beryllium-copper. Equip gauges with Monel or TP316L stainless steel threaded connections that supports the weight of the gauge. The dials of the gauges must be 4.5 inch in diameter or larger with scale readings in psig and inches of mercury ranging from zero to approximately twice the anticipated process operating or equipment pressure. Provide a slotted adjustable pointer . Provide a lever handled gauge cock and filter type snubber. Install snubber between the pipeline and the gauge. Isolate the gauges from the process fluids using remote corrosion resistant diaphragm seals. Construct the housing of the corrosion resistant seals of Monel polytetrafluoroethylene (PTFE) or compatible material for application. Compose seals of Monel, Hastelloy polytetrafluoroethylene (PTFE) or compatible material for application .

#### 2.16.4.2 Thermometers

Provide bi-metal actuated thermometers, with 5 inch dished anti-parallax dials that have external calibration adjustment and stainless steel cases. Do not use mercury in thermometers. Use thermometers with stainless steel stems, adjustable angle type for the correct viewing angle. Include union connections with associated thermowells. Scale must be 25 to 125 degrees F with accuracy within one scale division.

#### 2.16.4.3 Thermowells

Provide P316 stainless steel thermowells with a diameter of 1 inch. Use length as shown on the contract drawings and coordinate with the associated temperature element. Construct process connections of stainless steel with flanges, faced and drilled to ASME B16.5 Class 150 . Use thermowells, equipped with terminal connection heads rated NEMA 250 Type 4 X, with thermocouples or RTDs.

#### 2.16.5 Static Mixer

Design static mixer to disperse the design flow, as shown in the hydraulic profile of added chemicals in a process flow stream . The maximum allowable pressure drop for the static mixer is 5 feet of water column in accordance with the requirements of the process stream pumping system. Size diameter of the mixer housing identical to the process piping. Use length in accordance with the number of mixing elements required. Provide housing materials compatible with chemicals being injected, providing chemical resistance to both the chemical additives and process stream. Ensure coatings on coated components are factory spark-tested to verify that the coating is free from pinholes. Ensure end configurations are , flanged faced and drilled to ASME B16.5 or ASME B16.1 Class 150 and compatible with the piping system. Provide injection ports of the same materials as the mixer housing in the number, dimensions, and positions shown on drawings, and with connections compatible with the chemical feed piping system. Supply each housing with a name plate which at a minimum provides the manufacturer's name and address, part model number, and direction of flow. Construct mixing elements providing resistance to both the chemical additives and process stream. Install elements consecutively, with the number required designed to provide mixing with a homogeneity of the final mix of less than or equal to 0.05 by the end of the static mixer or within 10 pipe diameters downstream.

#### 2.16.6 Expansion Joints

Provide all structural work and equipment required to control expansion and contraction of piping. Verify that the anchors, guides, and expansion joints provided, adequately protect the piping systems.

##### 2.16.6.1 Expansion Joint for Metallic Pipe

The expansion joint must be a bellows type with 316L stainless steel wetted materials of construction. Size expansion joint to match the associated piping. The maximum allowable working pressure is 150 psig at 120 degrees F. Size expansion joint for a maximum axial compressing and expanding for a temperature range of and an angular rotation of 15 degrees. End connections must be ASME B16.5 or ASME B16.1 Class 150 flanged. Provide required accessories for a complete assembly including: limit stops, internal flow liners, and control rods, .

##### 2.16.6.2 Expansion Joint for Nonmetallic Piping

Provide a bellows expansion joint with a minimum convolutions to accommodate an axial deflection for a temperature range of 0°F to 120°F flanged, faced and drilled to ASME B16.1 or ASME B16.5 Class 150 end connections, and metal reinforcing bands. The maximum allowable working pressure is 140 psig at 120 degrees F. Limit bolting to restrain the force developed by 1.5 times the specified maximum allowable operating pressure. Size expansion joint to match the associated piping. Provide required accessories for a complete assembly including: limit stops, internal flow liners, and control rods.

#### 2.16.7 Pressure Relief Devices

Provide pressure relief devices conforming to the requirements of ASME B31.3.

##### 2.16.7.1 Pressure-Relief Valve

Provide pressure-relief valves conforming to the following:

- a. Pressure-relief valves, 2 inch and smaller, must be a direct diaphragm, spring controlled type with bronze bodies and spring cases. Trim must be bronze and seats, nitrile. Diaphragms must be elastomeric, chloroprene. Ensure miscellaneous parts such as the valve stems, nuts, and springs are stainless steel. Ensure end connections are flanged, faced and drilled to ASME B16.1 or ASME B16.5 Class 15. The valves must open when the upstream pressure reaches a maximum set point. Sizes and ratings as shown in the Valve Schedule in the contract drawings.
- b. Pressure relief valves, 2.5 inch and larger, must be hydraulically operated, diaphragm actuated, pilot controlled globe or angle valves with externally mounted strainers and test cocks. Provide bodies consisting of bronze and trim consisting of bronze. Stem must be stainless steel. Ensure end connections are flanged, faced and drilled to ASME B16.1 or ASME B16.5 Class 150. The valves must open when the upstream pressure reaches a maximum set point. Sizes and ratings as shown in the Valve Schedule in the contract drawings..

## 2.17 PIPE SUPPORTS AND PENETRATIONS

Provide auxiliary steel where the support of piping systems and equipment is required between building structural elements. Provide light gauge and structural steel shapes conforming to the requirements of ASTM A36/A36M. The Contractor has the option to use pre-engineered support systems of electrogalvanized steel products. However, a mixture of support system manufacturers products is not permitted. Where auxiliary steel is indicated as stainless steel, provide stainless steel conforming to ASTM A167, No. 1 Finish or as noted on the contract drawings.

### 2.17.1 Pipe Supports

Provide pipe supports conforming to the requirements of MSS SP-58. Where pipe supports contact bare piping or in-line devices, provide supports of compatible material so that neither has a deteriorating action on the other.

#### 2.17.1.1 Beam Clamps

For upper attachments on structural steel, provide beam clamps of ASTM A36/A36M carbon steel and MSS SP-58 Types 19 through 23, 25 or 27 through 30. Holes drilled in structural steel for hanger support rods will not be permitted. Provide clamps with hardened steel cup-point set screws and lock-nuts for anchoring in place. Base clamp size selection only on the support of the required load.

#### 2.17.1.2 Riser Clamps

Support vertical runs of piping at each floor, or closer where required, with ASTM A36/A36M carbon steel clamps bolted around pipes and attach to the building construction. Provide copper plated clamps for copper tubing support. Use two bolt-type clamps designed for installation under insulation on insulated pipe runs.

#### 2.17.1.3 Brackets

Where piping is run adjacent to walls or steel columns, provide welded ASTM A36/A36M steel brackets, pre-punched with a minimum of two fastener holes.

#### 2.17.1.4 Offset Pipe Clamp

Where pipes are indicated as offset from wall surfaces, supply a double-leg design two-piece pipe clamp.

#### 2.17.1.5 Racks

Fabricate multiple pipe racks or trapeze hangers from ASTM A36/A36M steel, and design to suit the conditions at the points of installation. Keep pipes in their relative positions to each other by the use of clamps or clips. Pipelines subject to thermal expansion must be free to slide or roll.

#### 2.17.1.6 Hangers

Fabricate hangers of malleable iron, ASTM A47/A47M or ASTM A36/A36M carbon steel. Ensure all hangers are of a uniform type and material for a given pipe run and application. Use coated or plated hangers to isolate steel

hangers from dissimilar metal tube or pipe. Ensure hangers for pipe sizes 2.5 inch or larger incorporate a means of vertical adjustment after erection while supporting the load. For piping systems with operating temperatures from 122 to 446 degrees F, use the following: MSS SP-58 Type 1 or 3 through 12 hangers with overhead support and appropriate saddle of MSS SP-58 Type 40 for insulated pipe; MSS SP-58 Types 41 or 43 through 46 hangers or supports with roller support and appropriate saddle of MSS SP-58 Type 39 on insulated pipe; MSS SP-58 Types 35 through 38 for sliding support. For piping systems with liquid temperatures up to 122 degrees F, use the following: MSS SP-58 Types 1, 3 through 12, Types 24 and 26 with overhead support, or Types 35 through 38 with support from below.

#### 2.17.1.7 Hanger Rods

Provide carbon steel hanger rods conforming to ASTM A576. The diameter of the rods for piping system support must conform to ASME B31.1.

#### 2.17.2 Pipe Guides

##### 2.17.2.1 Intermediate Guides

For piping 6 inch and smaller, provide a pipe clamp with an oversize pipe sleeve for a minimum 0.16 inch clearance. For piping 8 inch and larger, use U-bolts with double nuts that are manufactured for the purpose to provide a minimum 0.28 inch clearance around pipe. The stock sizes for the U-bolts are as follows: for a 8 inch pipe use a 0.625 inch U-bolt; for a 10 inch pipe, use a 3/4 inch U-bolt; for a 12 inch to 16 inch pipe, use a 0.875 inch U-bolt; and for 18 inch to 30 inch pipes use 1 inch U-bolts.

##### 2.17.2.2 Alignment Guides

For piping, 8 inch and smaller, ensure alignment guides are galvanized steel, spider or sleeve type. For piping, 10 inch and larger, ensure alignment guides are galvanized steel, roller type guides unless shown otherwise in contract drawings.

#### 2.17.3 Flashing Sleeves

Install galvanized steel flashing sleeves wherever piping passes through concrete roof structures. Where piping penetrates roofs, provide 4 lb. lead flashing. Extend flashing 8 inches from the pipe in all directions, extend up the pipe, and fit with double-threaded flashing for pipes 3 inches and smaller. Turn down flashing inside the pipe for 4 inch and larger pipes.

#### 2.17.4 Wall Penetrations

##### 2.17.4.1 Above Grade Wall Penetrations

Provide piping which passes through fire-rated or smoke-rated walls, floors, or ceilings with insulated and encased pipe sleeves. Seal penetrations through an existing fire or fire barrier wall with a fire stop system that has an "F" rating no less than the required fire resistance rating of the penetrated wall. The fire stopping sealant for metal piping systems must be vibration resistant, polysiloxane (also known as silicone) based, nonslumping, premixed sealant with intumescent properties, that is rated for 3 hours pursuant to ASTM E814 and UL requirements. The fire stopping sealant for plastic and insulated piping



systems must be a polysiloxane (also known as silicone) based, nonslumping, premixed sealant with intumescent properties , that is vibration and moisture resistant, and is rated for 3 hours pursuant to ASTM E814 and UL requirements with metal collars. Fit vented plastic pipe penetrations with galvanized steel collars that have intumescent inlays.

#### 2.17.4.2 Below Grade Wall Penetrations

Provide below-grade wall penetrations with hydrostatic seals designed to seal opening between pipe or conduit and a through-structure opening. The seals must be modular mechanical type consisting of interlocking synthetic rubber links shaped to continuously fill the annular space between the pipe and wall opening .

#### 2.17.4.3 Galvanizing

Galvanizing must be zinc or cadmium plating. Stainless steel components may be substituted where galvanizing is specified.

### 2.18 MISCELLANEOUS MATERIALS

#### 2.18.1 Pipe Insulation Material

Provide insulation for pipes, valves, instrumentation and controls, and other equipment in accordance with Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS consisting of one inch ASTM C552 cellular glass and integral moisture barriers.

#### 2.18.2 Heat Trace

Heat trace must be electrical with materials selected for compatibility with transported liquids and ambient environment. The heat trace must be capable of maintaining the liquid process design temperature at 50 degrees F maximum when subjected to an exterior temperature of -20 degrees F. Ensure electrical work is in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, and UL listed, and includes all terminations, junction boxes, and automatic controls. Perform work in conformance to hazard classifications indicated on the drawings.

- a. Provide UL listed parallel conduction type heat tape, with adjustable thermostat for outdoor aboveground winterized piping. Ensure tape is not affected by direct sunlight, ambient temperature, operating temperature, rain, or salt laden atmosphere. Provide flexible, parallel circuit construction consisting of a continuous self-limiting resistance, conductive inner core material between two parallel copper bus wires, designed for cut-to-length at the job site and for wrapping around valves and complex fittings. Self-regulation must prevent overheating and burnouts even where the cable overlaps itself.
- b. Provide end seals for ends of circuits. Wire at the ends of circuits are not to be tied together.
- c. Provide sufficient cable, as recommended by the manufacturer, to keep the pipe surface at 34 degrees F minimum during winter outdoor design temperature as indicated, but not less than the following: 3 inch pipe and smaller with one inch thick insulation, 4 watts/feet; and 4 inch pipe and larger 1.5 inch thick insulation, 8 watts/feet of pipe.

## PART 3 EXECUTION

### 3.1 EXAMINATION

After becoming familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing the work.

### 3.2 PREPARATION

#### 3.2.1 Protection

Close pipe and equipment openings with caps or plugs during installation. Protect equipment from dirt, water, and chemical or mechanical damage.

#### 3.2.2 System Preparation

##### 3.2.2.1 Pipe and Fittings

Inspect pipe and fittings before exposed piping is installed or buried piping is lowered into the trench. Clean the ends of pipes thoroughly, remove foreign matter and dirt from inside of pipes, and keep piping clean during and after laying.

##### 3.2.2.2 Damaged Coatings

Repair damaged coating areas in the field with material equal to the original coating, except promptly remove damaged glass-lined pipe from the site. Do not install damaged piping materials. Field repair damaged and uncoated areas of galvanized piping conforming to ASTM A780/A780M.

##### 3.2.2.3 Field Fabrication

Notify the Contracting Officer at least 2 weeks prior to the field fabrication of pipe or fittings and at least 3 days prior to the start of any surface preparation or coating application work. Perform field welding in accordance with Section 40 17 26.00 20 WELDING PRESSURE PIPING. Provide welding electrodes in accordance with Table 3.1 of AWS D1.1/D1.1M as required for the applicable base metals and welding process. Fabricate fittings in accordance with the manufacturer's instructions.

### 3.3 EXPOSED PIPING INSTALLATION

Run exposed piping as straight as practical along the alignment shown on the contract drawings and with a minimum of joints. Install piping and appurtenances in conformance with reviewed shop drawings, manufacturer's instructions and ASME B31.3. Install piping without springing or forcing the pipe.

#### 3.3.1 Anchors and Fasteners

Impact expansion (hammer and explosive charge drive-type) anchors and fastener systems are not acceptable. Lead shields, plastic or fiber inserts, and drilled-in plastic sleeve/nail drive systems are also not acceptable.

##### 3.3.1.1 Drilled-In Expansion Anchors and Fasteners

Design anchors to accept both machine bolts and/or threaded rods. Such

anchors must consist of an expansion shield and expander nut contained inside the shield. Fabricate and design the expander nut to climb the bolt or rod thread and simultaneously expand the shield as soon as the threaded item, while being tightened, reaches, and bears against the shield bottom. Provide shield body consisting of four legs, taper the inside of each toward shield bottom (or nut end). Elongate the end of one leg and turn across shield bottom. Rib the outer surface of shield body for grip-action. The expander nut must be of square design with sides tapered inward from bottom to top. The anchor materials of construction must be zinc plated steel of 43,541 psi minimum tensile strength (unless shown otherwise in contract drawings). Provide fasteners consisting of machine bolts for use with above anchors; provide nuts and washers conforming to ASTM A194/A194M. Ensure the anchor length, diameter, and embedment depth meet the manufacturer's requirements for the maximum allowable working load of the application.

#### 3.3.1.2 Drilled-In Adhesive Anchors

Compose anchors of an anchor rod assembly and an anchor rod adhesive cartridge. Provide anchor rod assembly composed of a chamfered and threaded stud rod of zinc plated ASTM A36/A36M steel with a nut and washer of ASTM A194/A194M alloy-steel. Use anchor length, diameter, and embedment depth meeting the manufacturer's requirements for the maximum allowable working load of the application. Ensure the adhesive cartridge is a sealed capsule containing premeasured amounts of resin, quartz sand aggregate, and a hardener contained in a separate vial within the capsule. Activate the capsule ingredients by the insertion procedure of the anchor rod assembly.

#### 3.3.2 Piping Expansion and Contraction Provisions

Install piping to allow for thermal expansion and contraction resulting from the difference between installation and operating temperatures. Design for installation of plastic pipe exposed to ambient conditions or in which the temperature variation of the contents is substantial must have provisions for movement due to thermal expansion and contraction documented to be in accordance with PPI TR-21. Install anchors as shown in the contract drawings to withstand expansion thrust loads and to direct and control thermal expansion. Install an intermediate pipe guide for every pipe at each metal channel framing support not carrying an anchor or alignment guide. Where pipe expansion joints are required, install pipe alignment guides adjacent to the expansion device and within four pipe diameters. Install expansion devices in accordance with the manufacturer's instructions and at the locations shown in the contract drawings.

#### 3.3.3 Piping Flexibility Provisions

Provide thrust protection as required. Install flexible couplings and expansion joints at connections to equipment, and where shown on the contract drawings. Provide additional pipe anchors and flexible couplings, beyond those shown on the contract drawings, to facilitate piping installation, in accordance with reviewed shop drawings.

#### 3.3.4 Couplings, Adapters and Service Saddles

Thoroughly clean pipes of oil, scale, rust, and dirt in order to provide a clean seat for gaskets. Wipe gaskets clean prior to installation. Lubricate flexible couplings and flanged coupling adapter gaskets with

soapy water or the manufacturer's standard lubricant before installation on the pipe ends. Install couplings, service saddles, and anchor studs in accordance with manufacturer's instructions. Tighten bolts progressively, drawing up bolts on opposite sides a little at a time until all bolts have a uniform tightness. Use torque-limiting wrenches to tighten bolts.

### 3.3.5 Piping Equipment/Component Installation

Install piping components and indicators in accordance with manufacturer's instructions. Provide required upstream and downstream clearances, isolation valves, and miscellaneous devices for an operable installation.

Use upstream and downstream lengths of undisturbed piping in accordance with flow indicator manufacturer's recommendations.

#### 3.3.5.1 Backflow Preventers

Install backflow preventers with nameplate and test cocks accessible from front of unit, and with a minimum clearance of 12.2 inches between the port and grade. Install assemblies in accordance with local codes and discharge to an open drain with an air gap; vertical installation is prohibited.

#### 3.3.5.2 Local Indicators

Install all direct-reading indicator devices, thermometers, and pressure gauges so that they can be easily read from floor level, and are readily accessible for maintenance and service. Coat all temperature sensing bulbs with a silver base heat transfer grease prior to insertion into the thermowell. Install pressure gauges and thermometers where indicated in the contract drawings. Field calibrate all indicators at time of installation to ensure measuring and reading accuracy. Install differential pressure gauges across the process equipment indicated in the contract drawings, in accordance with the manufacturer's recommendations, and arranged for easy observation.

#### 3.3.6 Pipe Flanges

Pipe flanges must be set level, plumb, and aligned. Install flanged fittings true and perpendicular to the axis of the pipe. Ensure bolt holes are concentric to the centerline of the pipe and straddle the vertical centerline of the pipe.

#### 3.3.7 Valve Locations

Locate valves in accordance with the contract drawings where actuators are shown. Where actuators are not shown, locate and orient valves to permit easy access to the valve operator, and to avoid interferences.

#### 3.3.8 Pipe Tap Connections

Taps to pipe barrels are unacceptable. Make taps to ductile iron piping only with a service saddle or at a tapping boss of a fitting, valve body, or equipment casting. Make taps to steel piping only with a welded threadolet connection.

#### 3.3.9 Plastic Pipe Installation

Submit a statement signed by the plastic pipe manufacturer's representative certifying that the Contractor's personnel are capable of

properly installing the piping system on the project. Cut, make up, and install all plastic pipe in accordance with the pipe manufacturer's recommendations.

- a. Perform heat joining in accordance with ASTM D2657. Perform electrofusion joining in accordance with ASTM F1290. Do not thread Schedule 40 pipe. Use Schedule 80 threaded nipples where necessary to connect to threaded valves or fittings. Use strap wrenches for tightening threaded plastic joints, and do not over tighten these fittings.
- b. Do not lay pipe when the temperature is below 40.1 degrees F, nor above 90 degrees F when exposed to direct sunlight. Ensure any plastic pipe installed above grade and outdoors is ultraviolet (UV) protected or UV resistant.
- c. Shield pipe ends that are to be joined from direct sunlight prior to and during the laying operation. Provide adequate ventilation when working with pipe joint solvent cement and handle solvent cements, primers and cleaners in accordance with ASTM F402.
- d. Provide and install supports and hangers in accordance with the manufacturer's recommendations and which indicated as specified and indicated. Where plastic pipe is subjected to severe temperature fluctuations, provisions for expansion and contraction must be provided. Accomplish this with the use of expansion joints and offset piping arrangements. Hydrostatically test all lines at 1.5 times the pressures listed in the Pipe Schedule shown on the contract drawings or maximum operating pressure, whichever is smaller.

Submit 6 copies each of operation and maintenance manuals in indexed booklet form. Detail in the Operation Manuals the step-by-step procedures required for specialized startup, operation and shutdown of piping systems, and include the manufacturer's name, model number, parts list and brief description of piping equipment such as valves and other appurtenances and their basic operating features. List in the Maintenance Manuals routine maintenance procedures and troubleshooting guides for the equipment, and include piping layout and valve locations.

#### 3.3.9.1 PVC Piping

Construct solvent-cemented joints in accordance with ASTM D2855.

#### 3.3.10 Double Containment Piping Installation

Factory trained field representatives of the piping supplier must provide technical field support during critical periods of piping and leak detection system installation including final check out of the leak detection/location system, and end user training.

#### 3.3.11 Insulation

Install insulation on piping as indicated in the contract drawings in accordance with the provisions of Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

### 3.4 BURIED PIPE PLACEMENT

Install thermoplastic piping systems underground in accordance with ASTM D2774.

#### 3.4.1 Excavation and Backfilling

Perform earthwork as specified in Section 31 00 00.00 22 EARTHWORK. Accomplish backfilling after inspection by the Contracting Officer. Exercise care when lowering pipe into the trench to prevent damage or twisting of the pipe.

#### 3.4.2 Fittings

Make press connections in accordance with manufacturer's installation instructions using tools approved by the manufacturer. Insert tubing fully into the fitting and then mark at the shoulder of the fitting. Check fitting alignment against the mark on the tubing to ensure the tubing is fully inserted before the joint is pressed. At valves and connections, dig out the trench bottom with sufficient length, width, and depth to ensure clearance between the undisturbed trench bottom and the valves and such connections.

#### 3.4.3 Thrust Restraint

Thrust restraint devices are generally not shown in the contract drawings; their absence will not relieve Contractor of the responsibility for providing them as required to provide complete systems for the use intended. Provide thrust blocks and ties where required, whether or not shown on the contract drawings. At a minimum, provide thrust restraint at pipeline tees, plugs, caps, bends, and other locations where unbalanced forces exist.

##### 3.4.3.1 Thrust Blocks

Thrust blocking must be concrete of a mix not leaner than 1 cement, 2.5 sand and 5 gravel, and have a compressive strength of not less than 2000 psi after 28days. Place blocking between solid ground and the fitting to be anchored. Unless otherwise indicated or directed, pour the base and thrust bearing sides of the thrust blocks against undisturbed earth. The sides of thrust blocks not subject to thrusts may be poured against forms. Provide area of bearing as shown or directed. Place blocking so that fitting joints are accessible for repair. Use steel rods and clamps, protected by galvanizing or a coating of bituminous paint, to anchor vertical down bends into gravity thrust blocks.

##### 3.4.3.2 Restrained Joints

The restrained pipe according to manufacturer recommendations. For ductile iron pipe, design restrained joints in accordance with DIPRA TRD.

#### 3.4.4 Marking Tape

Provide and install pipe marking tape in accordance with the requirements of Section 31 00 00.00 22 EARTHWORK.

#### 3.4.5 Plastic Pipe Installation

Cut, fabricate, and install plastic pipe in strict conformance with the

pipe manufacturer's recommendations. Provide offset loops from the trench centerline as recommended by the manufacturer for the maximum temperature variation between the pipe temperature at the time of solvent welding and operating temperature. Design for installation of plastic pipe exposed to ambient conditions or in which the temperature variation of the contents is substantial must have provisions for movement due to thermal expansion and contraction documented to be in accordance with PPI TR-21. Support flexible plastic pipe connected to heavy fittings, manholes, and rigid structures in such a manner that no subsequent relative movement between the plastic pipe at the flanged joint and the rigid structures is possible. Do not use thrust blocking for flexible plastic piping. Design and install piping to withstand the compression and expansion forces imposed by the trench conditions. .

### 3.5 CONNECTING DISSIMILAR PIPE

Install flexible transition couplings, dielectric fittings and isolation joints in accordance with the manufacturer's instructions.

### 3.6 EXTERNAL CORROSION PROTECTION

Protect all pipe and piping accessories from corrosion and adverse environmental conditions.

#### 3.6.1 Underground Metallic Piping

Protect buried metallic piping from corrosion using protective coatings and cathodic protection. Where dissimilar metals are joined underground, use gas-tight isolation joints. Provide insulating joint material where shown to control galvanic or electrical action.

#### 3.6.2 Above Grade Metallic Piping

Do not paint nonferrous and stainless steel piping except for aluminum alloy piping. Where dissimilar metals are joined, use isolation joints.

##### 3.6.2.1 Ferrous Piping

Touch up shop primed surfaces with ferrous metal primer. Solvent clean surfaces that have not been shop primed. Clean surfaces that contain loose rust, mill scale or other foreign substances mechanically by power wire brushing or where requested by Owner/Engineer commercial sand blasting conforming to SSPC SP 6/NACE No.3 and prime with a ferrous metal primer. Finish primed surfaces with two coats of exterior vinyl paint in accordance with Section 09 90 00 PAINTS AND COATINGS.

### 3.7 DOUBLE CONTAINMENT PIPING LEAK DETECTION SYSTEM

- a. Install the system in accordance with the manufacturer's recommended installation procedures. Follow all local, state and federal codes and requirements. Install system by properly trained personnel. Provide a location map with the system by the Contractor indicating the "as built" system configuration and sensing string layout. Provide markings along the cable length as references to locate leaks. Base markings upon calibration points. Take and record calibration points along the sensing string in accordance with the manufacturer's recommended procedures. Provide cable not in containment piping with cable tags every 49.2 feet.

- b. Perform tests to demonstrate the ability of the system to detect and locate breaks, shorts and probes on the sensor string. Perform leak testing pursuant to the following procedure in order to verify operation and the ability to work with condensation pools or other static moisture. The double containment piping system leak detection field test procedures are as follows:

- (1) Wet the sensor cable near the start of the sensor string and acknowledge the detection/location alarm and remap the system.
- (2) Wet the sensor cable near the end of the sensor string with the first location still wetted and acknowledge the detection/location alarm and remap the system.
- (3) Wet the sensor cable in three additional locations between the first and second leak locations with each detection/location alarm being acknowledged and with all prior leak locations still wetted.
- (4) Prepare and submit a report verifying each leak location and detection accuracy. Furnish a hard copy report of the test results.

### 3.8 FLEXIBLE JOINTS AT CONCRETE STRUCTURES

Provide flexible joints at the face of all structures, whether or not shown on the contract drawings. Rubber ring joints, mechanical joints, flexible couplings, and proprietary restrained ductile iron pipe joints are considered flexible joints; welded pipe joints are not. Joints may be flush with the structure face or may be located up to 1 pipe diameter away from face, but not further than 17.7 inches away from face. For pipelines larger than 18 inch in diameter the first joint must be within 1 pipe diameter.

### 3.9 CLOSURES

Install closure pieces as necessary to end pipe runs and conforming to ASME B16.9 or ASME B16.11. Elastomer sleeves bonded to pipe ends are not acceptable. Provide pressure piping with closures of butt-welded caps blind flanges, threaded plugs, plain end pieces as required matching existing piping system, with thickness matching the nominal wall thickness of the associated pipe, mounted on double flexible couplings, unless otherwise shown on contract drawings or approved by the Contracting Officer. Ensure pipes with restrained joints have pipe closures installed with thrust tie-rod assemblies.

### 3.10 PENETRATIONS

Steel pipe sleeves must be hot-dipped galvanized after fabrication for above grade applications in nonsubmerged areas. For below grade, or in submerged and damp environments, line and coat steel pipe sleeves as specified in Section 09 90 00 PAINTS AND COATINGS. Isolate embedded metallic piping from concrete reinforcement using coated pipe penetrations. Coatings must be as specified in Section 09 90 00 PAINTS AND COATINGS. Support wall pipes securely by form work to prevent contact with reinforcing steel and tie-wires. Ensure joints are sealed with a wall penetration seal. For existing concrete walls, rotary drilled holes may be provided in lieu of sleeves.



### 3.11 VALVE INSTALLATION

Install flanged valve bolt holes so as to straddle the vertical centerline of pipe. Clean flanged faces prior to inserting the gasket and bolts, and then tighten the nuts progressively and uniformly. Clean threads of threaded ends by wire brushing or swabbing prior to installation.

#### 3.11.1 Valve Orientation

Install operating stem of a manual valve in a vertical position when the valve is installed in horizontal runs of pipe having centerline elevations 4.5 feet or less above finished floor, unless otherwise shown on contract drawings. Install operating stem of a manual valve in a horizontal position in horizontal runs of pipe having centerline elevations between 4.5 feet and 6.75 feet above finish floor, unless otherwise shown on contract drawings. Install automatic valves in accordance with the manufacturer's instructions and approved drawings.

##### 3.11.1.1 Butterfly Valves

Orientation of butterfly valves must take into account changes in pipe direction. Orient valve shafts so that unbalanced flows caused by pipe direction changes or other disturbances are equally divided to each half of the disc.

##### 3.11.1.2 Plug Valves

If a plug valve seat position is not shown in the contract drawings, locate the seat position as follows: for horizontal flow, the flow must produce an "unseating" pressure, and the plug must open into the top half of valve; and for vertical flow, install the seat in the highest portion of the valve.

#### 3.11.2 Line Size Ball Valves

Install a line size ball valve and union upstream of each solenoid valve, in-line flow switch, or other in-line electrical device, excluding magnetic flowmeters, for isolation during maintenance.

#### 3.11.3 Isolation Valve

Install safety isolation valves on compressed air supplies. Locate valve to provide accessibility for control and maintenance. If necessary, install access doors in finished walls and plaster ceilings for valve access.

#### 3.11.4 Operator Extension Stems

Where the depth of the valve is such that its centerline is more than 3 feet below grade, furnish an operator extension stem with a 2 inch operating nut to bring the operating nut to a point 5.9 inches below the surface of the ground and/or box cover. Locate operating nut in a floor box.

#### 3.11.5 Torque Tube

Where the operator for quarter-turn valve is located on a floor stand, furnish an extension stem torque tube, properly sized for the maximum torque capacity of the valve.

### 3.11.6 Chain Wheel and Guide

Install chain wheel and guide assemblies or chain lever assemblies on manually operated valves located over 6.73 feet above finished floor elevation. Where chains hang in normally traveled areas, use appropriate "L" type tie-back anchors.

### 3.12 AIR RELEASE, DRAINS AND SAMPLE PORTS

Provide sample ports where indicated on the contract drawings. Install specified vents at piping high points for entrapped air release and install drains in the low points of pipelines regardless of whether shown on contract drawings.

### 3.13 PIPING SUPPORT SYSTEMS INSTALLATION

The absence of pipe supports and details on the contract drawings does not relieve the Contractor of responsibility for sizing and providing supports throughout plant.

#### 3.13.1 General Support Requirements

Provide pipe support systems meeting the requirements of MSS SP-58. Install Contractor-designed and selected support systems in accordance with MSS SP-58, and as specified herein. Support piping connections to equipment by pipe supports and not off the equipment. Support large or heavy valves, fittings, and/or equipment independently of associated piping. Do not support pipes off other pipes. Provide supports at piping changes in direction or in elevation, adjacent to flexible joints and couplings, and where otherwise shown on the contract drawings. Do not install pipe supports and hangers in equipment access areas or bridge crane runs. Brace hanging pipes against horizontal movement by both longitudinal and lateral sway bracing. At each channel type support, provide every pipe with an intermediate pipe guide, except where pipe anchors are required. Existing support systems may be used to support additional new piping only if the Contractor can demonstrate that the existing support systems are adequate for the additional loads, or if the existing systems are strengthened to support the additional loads. Provide pedestal type pipe supports under base flanges adjacent to rotating equipment and where required to isolate vibration. Brace piping 2.5 inch in diameter and larger for seismic forces. Install lateral supports for seismic loads at all changes in direction.

#### 3.13.2 Support of Insulated Piping

Install oversized supports to fit the insulation inserts. Provide supports with galvanized or stainless steel protection shields and oversized rollers.

#### 3.13.3 Dielectric Barriers

Install dielectric barriers between supports and copper or stainless steel piping, and between stainless steel supports and non-stainless steel ferrous piping.

#### 3.13.4 Support Spacing

Provide a support and associated designed by a professional engineer.

### 3.13.5 Support Methods

Provide piping support as specified and as shown in the contract drawings. Support single horizontal suspended piping by adjustable swivel-ring, split-ring, or clevis hangers. Support multiple horizontal suspended piping by trapeze hangers with channel type supports. Provide horizontal pedestal mounted piping with saddle type supports. Provide horizontal wall mounted piping with wall brackets. Support vertical piping by wall brackets, base elbows, or riser clamps on floor penetrations.

### 3.13.6 Supports and Hangers for Stainless Steel Piping

Provide all hanger-pipe contact surfaces with a dielectric barrier consisting of chloroprene rubber wrapping or plastic coated hangers. The load rating of universal concrete inserts is not allowed to be less than that of the hanger rods they support.

### 3.14 PIPE IDENTIFICATION, PAINTING AND COLOR CODING

Ensure color, coating, and lettering requirements for exposed piping are in accordance with Section 09 90 00 PAINTS AND COATINGS. Except where piping is required to be completely painted in its code color, piping or its insulation covering may be banded either with plastic adhesive tapes or painted stripes around pipe designating piping contents in accordance with following options and requirements. Provide a single individual band, of plastic adhesive tape or paint, designating pipe contents with sufficient length to permit the stenciling of pipe contents in letters. Provide identification at branch connections, inlets and outlets of equipment, every 19.7 feet of straight run, upstream of valves, and within 3.3 feet of entrance to or exit from wall curtains, or other similar type barrier.

### 3.15 FIELD QUALITY CONTROL

#### 3.15.1 Hydrostatic Tests

Where any section of a pipeline is provided with concrete thrust blocking for fitting, do not make the hydrostatic tests until at least 5 days after the installation of the concrete thrust blocking, unless otherwise approved by the Contracting Officer.

##### 3.15.1.1 Buried Piping

After the pipe is laid, the joints completed and the trench partially backfilled leaving the joints exposed for examination, subject the newly laid piping or any valved section of piping, unless otherwise specified, for 1 hour to a hydrostatic test pressure as listed in the Pipe Schedule in the contract drawings. Open and close each valve several times during the test. Carefully examine exposed pipe, joints, fittings, and valves during the partially open trench test. Replace joints showing visible leakage as necessary. Remove defective pipe, joints, fittings, and valves found during the pressure test and replace with new material, and repeat the test until the test results are satisfactory. The requirement for the joints to remain exposed for the hydrostatic tests may be waived by the Contracting Officer when one or more of the following conditions are encountered: (1) wet or unstable soil conditions in the trench; (2) compliance would require maintaining barricades and walkways around and across an open trench in a heavily used area that would require continuous surveillance to assure safe conditions; or (3) maintaining the trench in

an open condition would delay completion of the Contract. The Contractor may request a waiver, setting forth in writing the reasons for the request and stating the alternative procedure proposed to comply with the hydrostatic tests. Place backfill placed prior to the tests in accordance with the requirements of Section 31 00 00.00 22 EARTHWORK.

#### 3.15.1.2 Exposed Piping

Conduct hydrostatic testing in accordance with ASME B31.3. Test piping systems under normal service conditions (as indicated in the Pipe Schedule in the contract drawings) to demonstrate compliance. Test pressure less than 1.5 times the design pressure is prohibited. Use water as the hydrostatic test fluid. Provide clean test water of such quality to prevent corrosion of the piping system materials. Open air release vents at all high points of the piping system in order to purge air pockets while the piping system is filling.

##### 3.15.1.2.1 Rigid Piping

For rigid piping hydrostatic testing, calculate the maximum test pressure according to ASME B31.3, but do not exceed the yield strength of the piping system. The maximum velocity during filling is 0.25 fps applied over full area of pipe. Venting during filling may also be provided by loosening flanges with a minimum of four bolts or by the use of equipment vents. Test all parts of the piping system. Maintain hydrostatic test pressure continuously for 30 minutes minimum and for such additional time as necessary to conduct examinations for leakage. Examine all joints and connections for leakage. Ensure piping system, exclusive of possible localized instances at pump or valve packing, shows no visual evidence of leaking. Correct visible leakage and retest. Unless otherwise directed by the Contracting Officer, leave the piping system full of water after leaks are repaired.

##### 3.15.1.2.2 Non-Rigid, Non-Metallic Piping

For non-rigid, non-metallic piping hydrostatic testing, calculate the maximum test pressure according to ASME B31.3, but do not exceed 1.5 times the maximum pressure rating of the lowest rated component in the piping system. The maximum velocity during filling is 0.25 fps applied over full area of pipe. Pressurize the system initially to 50 percent of the normal service conditions and inspect. Repair any leaks. Then pressurize the system to the test pressure. Add small amounts of water as required on a hourly basis for a maximum of 3 hours in order to maintain the test pressure. After 4 hours, lower the test pressure by 10.2 psi. If the hydrostatic pressure remains steady for 1 hour, then no leakage is indicated. Inspect for leaks, repair and retest if necessary. Allow piping system to relax for 8 hours before retesting.

##### 3.15.1.3 Double Containment

Test the primary piping of a double containment piping system in accordance with Subparagraph Exposed Piping of this paragraph. Pneumatically pressure test secondary containment piping of a double containment piping system in accordance with Subparagraph Double Containment Secondary Piping. Pressure test at the maximum test pressure of 5 psi or manufacturer's recommended maximum, 1 times the maximum pressure rating of the lowest rated component in the piping system. The maximum velocity during filling is 0.25 fps applied over full area of pipe. Pressurize the system initially to 50 percent of the normal service

conditions and inspect. Repair any leaks. Then pressurize the system to the test pressure. Add small amounts of water as required on a hourly basis for a maximum of 3 hours in order to maintain the test pressure. After 4 hours, lower the test pressure by 10.2 psi. If the hydrostatic pressure remains steady for 1 hour, then no leakage is indicated. Inspect for leaks, repair and retest if necessary. Allow piping system to relax for 8 hours before retesting.

#### 3.15.1.4 Time for Making Test

Except for joint material setting or where concrete thrust blocks necessitate a delay, underground piping jointed with rubber gaskets, mechanical or push-on joints, or couplings may be subjected to hydrostatic pressure, inspected, and tested for leakage at any time after partial completion of backfill. Conduct tests for above ground pressure piping after the piping has been completely installed, including all supports, hangers, and anchors, and inspected for proper installation but prior to installation of insulation.

#### 3.15.2 Pneumatic Tests

Prepare for and conduct pneumatic testing in accordance with the requirements of ASME B31.3. Care must be taken to minimize the chance of a brittle fracture or failure during a pneumatic leak test. Use only non-toxic, nonflammable, inert gases or air.

##### 3.15.2.1 Pressure Relief Device

During pneumatic testing, provide a pressure relief device for each piping section being tested. The device must have a set pressure no higher than the test pressure plus the lesser of 10 percent of the test pressure or 50.8 psi.

##### 3.15.2.2 Pneumatic Testing Procedures

The test fluid must be air and the test pressure must be 110 percent of the design pressure. Increase test pressure incrementally until the gage pressure reaches the lesser of 50 percent of the test pressure or 25 psig. Examine piping joints for leakage. If no leakage is occurring, continue to increase the pressure incrementally, while maintaining each incremental increase long enough to equalize pipe strains, until the test pressure is reached. Then reduce the test pressure to the design pressure and maintain for 10 minutes without additional energy expenditure. If the pneumatic pressure remains steady, then no leakage is indicated. Inspect for and repair leaks, and retest if necessary.

#### 3.15.3 Pipe Leakage Tests

Unless approved by the Contracting Officer, conduct leakage testing after the pressure tests have been satisfactorily completed. The duration of each leakage test is at least 2 hours, and during the test, subject the piping to no less than 200 psig pressure. Leakage is defined as the quantity of the test liquid, water, that is supplied to the piping system, or any valved or approved section thereof, in order to maintain pressure within 5 psi of the specified leakage test pressure after the piping has been filled with the test liquid and all air is expelled. No piping installation will be accepted if leakage exceeds the allowable leakage determined by the following formula:

|                                            |                                                 |                  |
|--------------------------------------------|-------------------------------------------------|------------------|
| $L = C_f \times N \times D \times P^{0.5}$ |                                                 |                  |
| Cf                                         | conversion factor                               | 0.0001351        |
| L                                          | allowable leakage                               | gallons per hour |
| N                                          | number of joints in the length of piping tested |                  |
| D                                          | nominal pipe diameter                           | inches           |
| P                                          | average test pressure during the test           | psig             |

Should any test disclose leakage greater than that allowed, locate the leaks and repair until the leakage is within the specified allowance, without additional cost.

#### 3.15.4 Testing New to Existing Connections

Test new piping connected to existing pipe, existing equipment, existing treatment systems, or tanks and treatment systems furnished under other Sections. Isolate the new piping with pipe caps, spectacle blinds, or blind flanges. Test the joint between new piping and existing piping by methods that do not place the entire existing system under the test load. Proceed. then, with the testing of new piping systems as specified herein.

#### 3.15.5 Valve Testing

Submit copies of all field test reports within 24 hours of the completion of the test. Valves may either be tested while testing pipelines, or as a separate step. Demonstrate that valves open and close smoothly with operating pressure on one side and atmospheric pressure on the other, and in both directions for two-way valve applications. Count and record the number of turns required to open and close each valve, and account for any discrepancies with manufacturer's data. Examine air and vacuum relief valves as the associated pipe is being filled to verify venting and seating is fully functional. Set, verify, and record set pressures for all relief and regulating valves. Test self-contained automatic valves at both maximum and minimum operating ranges, and reset upon completion of test to the design value.

#### 3.16 FINAL CLEANING

##### 3.16.1 Interim Cleaning

Prevent the accumulation of weld rod, weld spatter, pipe cuttings and filings, gravel, cleaning rags, and other foreign material within piping sections during fabrication. Examine the piping to assure removal of these and other foreign objects prior to assembly and installation.

##### 3.16.2 Flushing

Following assembly and testing, and prior to final acceptance, flush piping systems with water to remove accumulated construction debris and other foreign matter. Flush piping until all foreign matter is removed from the pipeline. Provide all hoses, temporary pipes, ditches, and other items as required to properly dispose of flushing water without damage to adjacent properties. The minimum flushing velocity is 2.5 fps . For

large diameter pipe where it is impractical to flush the pipe at the minimum flushing velocity, clean the pipeline in-place from the inside by brushing and sweeping, then flushing the pipeline at a lower velocity. Install cone strainers in the flushing connections of attached equipment and leave in place until cleaning is completed. Remove accumulated debris through drains, or by removing spools or valves.

### 3.16.3 Disinfection

. Before acceptance of (water) piping system operation, disinfect each section of completed pipeline in accordance with AWWA C651. After pressure tests have been made, thoroughly flush the piping section to be disinfected with water until all entrained dirt and mud have been removed before introducing the chlorinating material. The chlorinating material must be liquid chlorine, calcium hypochlorite, or sodium hypochlorite. Ensure the chlorinating material provides a dosage of no less than 50 ppm and introduce into the piping in an approved manner. Chlorinate PVC pipe lines using only the above specified chlorinating material in solution. Do not introduce the agent into the line in a dry solid state. Retain the treated water in the pipe long enough to destroy all non-spore-forming bacteria. Except where a shorter period is approved, the retention time is at least 24 hours and produce no less than 25 ppm of free chlorine residual throughout the line at the end of the retention period. Open and close all valves on the lines being disinfected several times during the contact period. Then flush the line with clean water until the residual chlorine is reduced to less than 1.0 ppm. During the flushing period, open and close each outlet on the line several times. From several points in the pipeline section, the Contracting Officer will take samples of water in sterilized containers for bacterial examination. Repeat the disinfection until the piping system passes the bacterial examination for 2 consecutive days. The piping system will not be accepted until satisfactory bacteriological results have been obtained.

### 3.17 WASTE WATER DISPOSAL

Submit the method proposed for disposal of waste water from hydrostatic tests and disinfection, and all required permits, prior to performing hydrostatic tests. Dispose of water used for testing, cleaning, flushing and/or disinfection in accordance with all applicable regulations. Disposal is solely the responsibility of the Contractor. Provide the method proposed for disposal of waste water to the Contracting Officer for approval prior to performing any testing, cleaning, flushing and disinfection activities.

### 3.18 MANUFACTURER'S FIELD SERVICES

Submit a signed statement certifying that the installation is satisfactory and in accordance with the contract drawings and specifications and the manufacturer's prescribed procedures and techniques, upon completion of the project and before final acceptance. Obtain manufacturer's technical assistance for Contractor training, installation inspection, start up, and owner operating and maintenance training. Follow manufacturer's instructions for installation.

-- End of Section --

SECTION 40 05 13.96

WELDING PROCESS PIPING  
**05/10**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)

|                         |                                                                                                                           |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------|
| ASNT SNT-TC-1A          | (2020) Recommended Practice for Personnel Qualification and Certification in Nondestructive Testing                       |
| ASNT SNT-TC-1A Q&A Bk A | (2010) Supplement to Recommended Practice No. SNT-TC-1A (Q&A Book): Radiographic Testing Method                           |
| ASNT SNT-TC-1A Q&A Bk B | (2007) Supplement to Recommended Practice SNT-TC-1A (Q&A Book): Magnetic Particle Method                                  |
| ASNT SNT-TC-1A Q&A Bk C | (2011; Text Correction 2011) Supplement to Recommended Practice No. SNT-TC-1A (Q&A Book): Ultrasonic Testing Method       |
| ASNT SNT-TC-1A Q&A Bk D | (2011; Text Correction 2011) Supplement to Recommended Practice No. SNT-TC-1A (Q&A Book): Liquid Penetrant Testing Method |

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|                    |                                                                                                      |
|--------------------|------------------------------------------------------------------------------------------------------|
| ASME B31.1         | (2022) Power Piping                                                                                  |
| ASME B31.3         | (2022; Errata 2023) Process Piping                                                                   |
| ASME B31.5         | (2022) Refrigeration Piping and Heat Transfer Components                                             |
| ASME B31.8         | (2022; Supplement 2022) Gas Transmission and Distribution Piping Systems                             |
| ASME BPVC SEC I    | (2017) BPVC Section I-Rules for Construction of Power Boilers                                        |
| ASME BPVC SEC II-C | (2017) BPVC Section II-Materials Part C-Specifications for Welding Rods Electrodes and Filler Metals |
| ASME BPVC SEC IX   | (2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications                       |



ASME BPVC SEC V

(2017) BPVC Section V-Nondestructive  
Examination

AMERICAN WELDING SOCIETY (AWS)

|                |                                                                                |
|----------------|--------------------------------------------------------------------------------|
| AWS A2.4       | (2012) Standard Symbols for Welding,<br>Brazing and Nondestructive Examination |
| AWS A3.0M/A3.0 | (2020) Standard Welding Terms and<br>Definitions                               |
| AWS B2.1/B2.1M | (2021) Specification for Welding Procedure<br>and Performance Qualification    |
| AWS QC1        | (2016) Specification for AWS Certification<br>of Welding Inspectors            |
| AWS Z49.1      | (2021) Safety in Welding and Cutting and<br>Allied Processes                   |

1.2 DEFINITIONS AND SYMBOLS

Definitions are in accordance with AWS A3.0M/A3.0. Symbols are in accordance with AWS A2.4.

1.3 PERFORMANCE REQUIREMENTS

Quality of all joint preparation, welding, and examination is the Contractor's responsibility for. Clearly identify and record all materials used in the welding operations. The inspection and testing defined in this specification are minimum requirements. Additional inspection and testing is the responsibility of the Contractor when it is deemed necessary to achieve the quality required.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Pressure Piping; G

SD-03 Product Data

Welding Operations

SD-07 Certificates

Qualifications

1.5 QUALIFICATIONS

Welding procedures, welders, and welding operators previously qualified by test may be accepted for the work without requalification, provided that all of the following conditions are fulfilled:

- a. Copies of the welding procedures, the procedure qualification test records, and the welder and welding operator performance qualification test records are submitted and approved in accordance with paragraph SUBMITTALS.
- b. Testing was performed by an approved testing laboratory or technical consultant or by the Contractor's approved quality assurance organization.
- c. The welding procedures, welders, and welding operators were qualified in accordance with ASME BPVC SEC IX, or AWS B2.1/B2.1M, AR-2 level; and base materials, filler materials, electrodes, equipment, and processes conformed to the applicable requirements of this specification.
- d. The requirements of paragraph "Renewal of Qualification" below are met and records showing name of employer and period of employment using the process for which qualified are submitted as evidence of conformance.

#### 1.5.1 Welding Procedures Qualification

Record in detail and qualify the Welding Procedure Specifications for every proposed welding procedure. Qualification for each welding procedure must conform to the requirements of ASME B31.1, ASME B31.3, ASME B31.5, ASME B31.8, and to this specification. Specify end preparation for butt welds including cleaning, alignment, and root openings. Provide preheat, interpass temperature control, and postheat treatment of welds as required by approved welding procedures, unless otherwise indicated or specified. Describe the type of backing rings or consumable inserts, if used, and if they are to be removed, describe the removal process. Submit copies of the welding procedure specifications and procedure qualification test results for each type of welding required in accordance with paragraph SUBMITTALS. Approval of any procedure does not relieve the Contractor of the sole responsibility for producing acceptable welds. Identify welding procedures individually and reference on the detail drawings or key to the contract drawings.

#### 1.5.2 Welder and Welding Operator Performance

Provide qualified welder and welding operator assigned to work in accordance with ASME B31.1, ASME B31.3, ASME B31.5, ASME B31.8.

##### 1.5.2.1 Certification

Before assigning welders or welding operators to the work, provide the Contracting Officer with their names together with certification that each individual is performance-qualified as specified. State the type of welding and positions for which each is qualified, the code and procedure under which each is qualified, date qualified, and the firm and individual certifying the qualification tests.

##### 1.5.2.2 Identification

Identify each particular weld with the personal number, letter, or symbol assigned to each welder or welding operator. To identify welds, submit written records indicating the location of welds made by each welder or welding operator, and each welder or welding operator must apply the personal mark adjacent to the welds using a rubber stamp or felt-tipped

marker with permanent, weatherproof ink or other methods approved by the Contracting Officer that do not deform the metal. For seam welds, place identification marks adjacent to the welds at 3 foot intervals. Identification by die stamps or electric etchers will not be allowed.

#### 1.5.2.3 Renewal of Qualification

Requalification of a welder or welding operator is required under any of the following conditions:

- a. When a welder or welding operator has not used the specific welding process for a period of 3 months; the period may be extended to 6 months if the welder or welding operator has been employed on some other welding process.
- b. When a welder or welding operator has not welded with any process during a period of 3 months, all the personal qualifications are considered expired, including any extended by virtue of a., above.
- c. There is specific reason to question the person's ability to make welds that will meet the requirements of the specifications.
- d. The welder or welding operator was qualified by an employer, other than those firms performing work under this contract, and a qualification test has not been taken within the preceding 12 months.
- e. Renewal of qualification for a specific welding process under conditions a., b., and d., above, needs to be made on only a single test joint or pipe of any thickness, position, or material to reestablish the welder's or welding operator's qualification for any thickness, position, or material covered under previous qualification.

#### 1.5.3 Inspection and NDE Personnel

All inspection and NDE personnel must be qualified in accordance with the following requirements.

##### 1.5.3.1 Inspector Certification

Welding inspectors must be qualified in accordance with AWS QC1.

##### 1.5.3.2 NDE Personnel

Furnish certified NDE personnel, and establish a written procedure for the control and administration of NDE personnel training, examination, and certification. Base the procedures on appropriate specific and general guidelines of training and experience recommended by ASNT SNT-TC-1A, ASNT SNT-TC-1A Q&A Bk A, ASNT SNT-TC-1A Q&A Bk B, ASNT SNT-TC-1A Q&A Bk C and ASNT SNT-TC-1A Q&A Bk D.

#### 1.6 REGULATORY REQUIREMENTS

This section covers the welding of pressure piping systems. Submit detail drawings showing location, length, and type of welds; and indicating postweld heat treatment and NDE as required. Deviations from applicable codes, approved procedures, and approved detail drawings will not be permitted without prior written approval. Materials or components with welds made offsite will not be accepted if the welding does not conform to the requirements of this specification, unless otherwise specified.

Develop procedures for welding all metals included in the work. Do not start welding until welding procedures, welders, and welding operators have been qualified. Perform qualification testing by an approved testing laboratory, or by the Contractor if approved by the Contracting Officer. Notify the Contracting Officer at least 24 hours in advance of the time and place of the tests. When practicable, perform the qualification tests at or near the worksite. Maintain current records of the test results obtained in the welding procedure, welding operator, welder performance qualifications, and nondestructive examination (NDE) procedures readily available at the site for examination by the Contracting Officer. Qualify the procedures for making transition welds between different materials or between plates or pipes of different wall thicknesses. ASME B31.1, ASME B31.3, ASME B31.5, ASME B31.8 requirements for branch connections may be used in lieu of detailed designs. Unless otherwise specified, the choice of welding process is the responsibility of the Contractor. Safety precautions must conform to AWS Z49.1.

## 1.7 DELIVERY, STORAGE, AND HANDLING

Deliver all filler metals, electrodes, fluxes, and other welding materials to the site in manufacturers' original packages and stored in a dry space until used. Properly label and design packages to give maximum protection from moisture and to insure safe handling.

### 1.7.1 Material Control

Store materials in a controlled access and clean, dry area that is weathertight and maintained at a temperature recommended by the manufacturer. The materials must not be in contact with the floor and store on wooden pallets or cribbing.

#### 1.7.1.1 Damaged Containers

Store low-hydrogen steel electrodes in their sealed shipping container. If the seal is damaged during shipment or storage, and the damage is not immediately detected, rebake the covered electrodes in that container in accordance with the manufacturer's instructions prior to issuance or discard. If a container is damaged in storage and the damage is witnessed, immediately place the electrodes from that container in a storage oven. Provide storage oven temperature as recommended by the manufacturer or the welding material specification.

#### 1.7.1.2 Partial Issues

When a container of covered electrodes is opened and only a portion of the content is issued, place the remaining portion, within 1/2 hour, in a storage oven.

### 1.7.2 Damaged Materials

Discard damaged materials. Discard covered electrodes which are oil or water-soaked, dirty, or on which the flux has separated from the wire.

## PART 2 PRODUCTS

### 2.1 MATERIALS

Provide welding materials which comply with ASME BPVC SEC II-C. Provide welding equipment, electrodes, welding wire, and fluxes capable of

producing satisfactory welds when used by a qualified welder or welding operator using qualified welding procedures.

## PART 3 EXECUTION

### 3.1 WELDING OPERATIONS

Perform welding in accordance with qualified procedures using qualified welders and welding operators. Submit detailed procedures which define methods of compliance to contract drawings and specifications. Inspection and material procurement records. System and material testing and certification records. Written records and drawings indicating location of welds made by each welder or welding operator.

Do not perform welding when the quality of the completed weld could be impaired by the prevailing working or weather conditions. The Contracting Officer will determine when weather or working conditions are unsuitable for welding. Welding of hangers, supports, and plates to structural members must conform to Section 05 05 23.16 STRUCTURAL WELDING.

#### 3.1.1 Base Metal Preparation

Do not use oxy-fuel cutting on austenitic stainless steel or nonferrous materials.

#### 3.1.2 Weld Joint Fit-Up

Fit, align, and retain parts that are to be joined by welding in position during the welding operation by the use of bars, jacks, clamps, or other mechanical fixtures. Do not use welded temporary attachments except when it is impractical to use mechanical fixtures. When temporary attachments are used, they must be the same material as the base metal. Completely remove by grinding or thermal cutting after the welding operation is completed. If thermal cutting is used, cut the attachment no less than 1/4 inch from the member and remove the balance by grinding. After the temporary attachment has been removed, visually examine the area.

#### 3.1.3 Preheat and Interpass Temperatures

Provide preheat temperatures meeting the requirements specified by ASME B31.1, ASME B31.3, ASME B31.5, ASME B31.8. However, preheat below 50 degrees F for ferritic steel or austenitic stainless steel, or 32 degrees F for nonferrous alloys is not acceptable. Do not exceed maximum interpass temperatures of 300 degrees F for austenitic stainless steels, nickel alloys, and copper alloys; and 500 degrees F for carbon steels. Use preheat techniques to ensure that the full thickness of the weld joint preparation and/or adjacent base material, at least 3 inches in all directions, is at the specified temperature. Preheating by induction or resistance methods is preferred. When flame heating is used, employ only a neutral flame. Do not use oxy-fuel heating on austenitic stainless steel or nickel-alloy materials; however, air-fuel heating is acceptable if controlled to insure that the surface temperature does not exceed 150 degrees F. Check interpass temperatures on the surface of the component within 1 inch of the weld groove and at the starting location of the next weld pass, and for a distance of about 6 inches ahead of the weld, but not on the area to be welded.

#### 3.1.4 Production Welding Instructions

- a. Do not perform welding when the ambient temperature is lower than 0 degree F.
- b. Welding is not permitted on surfaces that are wet or covered with ice, when snow or rain is falling on the surfaces to be welded, or during periods of high winds, unless the welders and the work are properly protected.
- c. Provide welding grade gases for purging and shielding with a dew point of minus 40 degrees F or lower.
- d. Back purges are required for austenitic stainless steels and nonferrous alloys welded from one side and set up such that the flow of gas from the inlet to the outlet orifice passes across the area to be welded. The oxygen content of the gas exiting from the purge vent must be less than 2 percent prior to welding.
- e. Maintain the purge on groove welds for at least three layers or 3/16 inch.
- f. Furnish removable purge dam materials made of expandable or flexible plugs, such as plexiglass, plywood (which must be dry when used), etc. Provide wood dams of kiln-dried quality. Provide nonremovable purge dams and purge dam adhesives made of water soluble materials. Purge dams made of polyvinyl alcohol are not permitted.
- g. Do not perform any welding process which requires the use of external gas shielding in a draft or wind unless the weld area is protected by a shelter. This shelter must be of material and shape appropriate to reduce wind velocity in the vicinity of the weld to a maximum of 5 mph (440 fpm).
- h. Welding of low-alloy and hardenable high-alloy steels may be interrupted provided a minimum of at least 3/8 inch thickness of weld deposit or 25 percent of the weld groove is filled, whichever is greater, and the preheat temperature is maintained during the time that welding is interrupted. If the temperature falls below the minimum preheat temperature before all welding has been completed on a joint, or, where required, before post weld heat treatment, perform a liquid penetrant or magnetic particle examination to insure sound deposited metal before reheating. Welding of other materials may be interrupted without restriction provided a visual inspection is performed before welding is resumed.
- i. Taper the ends of tack welds to be incorporated in the final welds by grinding or welding technique. Remove tack welds that are cracked or defective and retack the groove prior to welding. Remove temporary tack welds, ground the surface smooth, and visually inspect. For low-alloy and hardenable high-alloy steels, the area must be magnetic particle examination inspected.
- j. When joining ferritic steel pressure piping components to austenitic stainless steel pressure piping components and postweld heat treatment is required, the following requirements apply:
  - (1) Butter the weld-end preps of ferritic steel components, which are to be welded to austenitic stainless steel, with one of the

following weld filler metals conforming to the specified requirements: ASME BPVC SEC II-C, SFA 5.14, Classification ERNiCr-3; or ASME BPVC SEC II-C, SFA 5.11, Classification ENiCrFe-2.

- (2) Butter the ferritic steel weld-end prep, apply a postweld heat treatment as required by ASME B31.1, ASME B31.3, ASME B31.5, ASME B31.8 and then machine with the applicable weld-end preparation. After machining, the buttered layer must be a minimum of 1/4 inch thick.
- (3) Complete pressure piping transition joints using ERNiCr-3 or ENiCrFe-2 weld filler metals. Do not perform further postweld heat treatment.

- k. When joining ferritic steel pressure piping components to austenitic stainless steel pressure piping components and postweld heat treatment is not required, prepare and weld the joint using either ERNiCr-3 or ENiCrFe-2 filler metals. For service temperatures of 200 degrees F or less, stainless filler metal 309 ASME BPVC SEC II-C, SFA 5.4 or 5.9 is permissible in lieu of the nickel-based alloys.
- l. Grinding of completed welds is to be performed only to the extent required for NDE, including any inservice examination, and to provide weld reinforcement within the requirements of ASME B31.1, ASME B31.3, ASME B31.5, ASME B31.8. If the surface of the weld requires grinding, avoid reducing the weld or base material below the minimum required thickness. Provide minimum weld external reinforcement flush between external surfaces.

### 3.1.5 Postweld Heat Treatment

- a. Perform postweld heat treatment in accordance with ASME B31.1, ASME B31.3, ASME B31.5, ASME B31.8. Measure temperatures for local postweld heat treatment continuously by thermocouples in contact with the weldment.
- b. Perform postweld heat treatment of low-alloy steels, when required, immediately upon completion of welding and prior to the temperature of the weld falling below the preheat temperature. However, postweld heat treatment may be postponed after the completion of the weld, if, immediately after the weld is completed, it is maintained at a minimum temperature of 300 degrees F or the preheat temperature, whichever is greater, for 2 hours per inch of weld thickness.
- c. For low-alloy steels, use cooling rates that avoid temper embrittlement.

### 3.2 EXAMINATIONS, INSPECTIONS, AND TESTS

Visual and NDE must be performed by the Contractor to detect surface and internal discontinuities in completed welds. Employ the services of a qualified commercial inspection or testing laboratory or technical consultant approved by the Contracting Officer. Visually inspect all tack welds, weld passes, and completed welds. In addition, perform magnetic particle or liquid penetrant examination on root passes. Radiographic, Liquid penetrant, Magnetic particle or Ultrasonic examination is required as indicated in TABLE I. When inspection and testing indicates defects in a weld joint, repair the weld by a qualified welder in accordance with

paragraph CORRECTIONS AND REPAIRS.

| TABLE I. MANDATORY MINIMUM NONDESTRUCTIVE EXAMINATIONS                                                                 |                                                             |                                                                                                              |                                      |
|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------|
| Type Weld                                                                                                              | Piping Service Conditions and Nondestructive Test           |                                                                                                              |                                      |
|                                                                                                                        | Temperatures over 750 degrees F and at all pressures        | Temperatures between 350 degrees F and 750 degrees F inclusive and at pressures above 1025 psig              | All others                           |
| Butt Welds (Girth and Longitudinal)                                                                                    | RT for NPS over 2 inches MT or PT for NPS 2 inches and less | RT for over 2 inches NPS with thickness over 3/4 inch. Visual for all sizes with thickness 3/4 inch or less. | Visual for all sizes and thicknesses |
| Welded Branch Connections (Size indicated is branch size) (See Note 7)                                                 | RT for NPS over 4 inches MT or PT for NPS 4 inches and less | RT for over 4 inches NPS with thickness over 3/4 inch. Visual for all sizes with thickness 3/4 inch or less. | Visual for all sizes and thicknesses |
| Fillet, Socket Attachment and Seal Welds                                                                               | PT or MT for all sizes and thicknesses                      | Visual for all sizes and thicknesses                                                                         | Visual for all sizes and thicknesses |
| NOTES TO TABLE I                                                                                                       |                                                             |                                                                                                              |                                      |
| (1) All welds must be given a visual examination in addition to type of specific nondestructive examination specified. |                                                             |                                                                                                              |                                      |
| (2) NPS - nominal pipe size.                                                                                           |                                                             |                                                                                                              |                                      |
| (3) RT - Radiographic examination; MT - magnetic particle examination; PT - liquid penetrant examination.              |                                                             |                                                                                                              |                                      |
| (4) RT of branch welds must be performed before any nonintegral reinforcing material is applied.                       |                                                             |                                                                                                              |                                      |
| (5) The thickness of butt welds is defined as the thicker of the two abutting ends after end preparation.              |                                                             |                                                                                                              |                                      |
| (6) Temperatures and pressures shown are design.                                                                       |                                                             |                                                                                                              |                                      |



| TABLE I. MANDATORY MINIMUM NONDESTRUCTIVE EXAMINATIONS                                                                                                                                                                                                                                                    |                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Type Weld                                                                                                                                                                                                                                                                                                 | Piping Service Conditions and Nondestructive Test |
| (7) In lieu of radiography of welded branch connections when required above, liquid penetrant or magnetic particle examination is acceptable and, when used, must be performed at the lesser of one half of the weld thickness or each 1/2 inch of weld thickness and all accessible final weld surfaces. |                                                   |
| (8) For nondestructive examination of the pressure retaining component, refer to the standards listed in applicable code or the manufacturing specifications.                                                                                                                                             |                                                   |
| (9) Fillet welds not exceeding 1/4 inch throat thickness which are used for the permanent attachment of nonpressure retaining parts are exempt from the PT or MT requirements of the above table.                                                                                                         |                                                   |

### 3.2.1 Random NDE Testing

When random radiographic, liquid penetrant, magnetic particle or ultrasonic examination is required, test a minimum of 10 percent of the total length or number of piping welds. Randomly select welds for inspection, but include an examination of welds made by each welding operator or welder in the selection. If the random testing reveals that any welds fail to meet minimum quality requirements, inspect an additional 10 percent of the welds in that same group. If all of the additional welds inspected meet the quality requirements, accept the entire group of welds represented and repair the defective welds. If any of the additional welds inspected also fail to meet the quality requirements, reject that entire group of welds. Remove and reweld the rejected welds, or 100 percent inspect the rejected welds and remove and reweld all defective weld areas.

### 3.2.2 Visual Inspection

Visually inspect weld joints as follows:

#### 3.2.2.1 Before Welding

For compliance with requirements for joint preparation, placement of backing rings or consumable inserts, alignment and fit-up, and cleanliness.

#### 3.2.2.2 During Welding

For cracks and conformance to the qualified welding procedure.

#### 3.2.2.3 After Welding

For cracks, contour and finish, bead reinforcement, undercutting, overlap, and size of fillet welds.

### 3.2.3 NDE Testing

Perform NDE in accordance with written procedures. Procedures for radiographic, liquid penetrant, magnetic particle or ultrasonic tests and methods must conform to ASME BPVC SEC V. Demonstrate the approved procedure to the satisfaction of the Contracting Officer. In addition to the information required in ASME BPVC SEC V, include the timing of the NDE in relation to the welding operations and safety precautions.

### 3.2.4 Inspection and Tests by the Government

The Government will perform inspection and supplemental nondestructive or destructive tests as deemed necessary. The cost of supplemental NDE will be borne by the Government. Correct and repair defects and reexamine weld repairs at no additional cost to the Government. Inspection and tests will be performed as required for visual inspection and NDE, except that destructive tests may be required also. When destructive tests are ordered by the Contracting Officer and performed by the Contractor, and the specimens or other supplemental examinations indicate that the materials and workmanship do not conform to the contract requirements, the cost of the tests, corrections, and repairs are borne by the Contractor. When the specimens or other supplemental examinations of destructive tests indicate that materials or workmanship do conform to the specification requirements, the cost of the tests and repairs will be borne by the Government. When destructive tests are made, repairs must be made by qualified welders or welding operators using welding procedures which will develop the full strength of the members cut. Welding is subject to inspection and tests in the mill, shop, and field. When materials or workmanship do not conform to the specification requirements, the work may be rejected at any time before final acceptance of the system containing the weldment.

### 3.3 ACCEPTANCE STANDARDS

#### 3.3.1 Visual

The following indications are unacceptable:

- a. Cracks.
- b. Undercut on surface which is greater than 1/32 inch deep.
- c. Weld reinforcement greater than 3/16 inch.
- d. Lack of fusion on surface.
- e. Incomplete penetration (applies only when inside surface is readily accessible).
- f. Convexity of fillet weld surface greater than 10 percent of longest leg plus 0.03 inch.
- g. Concavity in groove welds.
- h. Concavity in fillet welds greater than 1/16 inch.
- i. Fillet weld size less than indicated or greater than 1.25 times the minimum indicated fillet leg length.

#### 3.3.2 Magnetic Particle Examination

The following relevant indications are unacceptable:

- a. Any cracks and linear indications.
- b. Rounded indications with dimensions greater than 3/16 inch.

- c. Four or more rounded indications in a line separated by 1/16 inch or less edge-to-edge.
- d. Ten or more rounded indications in any 6 square inches of surface with the major dimension of this area not to exceed 6 inches with the area taken in the most unfavorable location relative to the indications being evaluated.

### 3.3.3 Liquid Penetrant Examination

Indications with major dimensions greater than 1/16 of an inch are considered relevant. The following relevant indications are unacceptable:

- a. Any cracks or linear indications.
- b. Rounded indications with dimensions greater than 3/16 inch.
- c. Four or more rounded indications in a line separated by 1/16 inch or less edge-to-edge.
- d. Ten or more rounded indications in any 6 square inches of surface with the major dimension of this area not to exceed 6 inches with the area taken in the most unfavorable location relative to the indications being evaluated.

### 3.3.4 Radiography

Welds that are shown by radiography to have any of the following discontinuities are unacceptable:

- a. Porosity in excess of that shown as acceptable in ASME BPVC SEC I, Appendix A-250.
- b. Any type of crack or zone of incomplete fusion or penetration.
- c. Any other elongated indication which has a length greater than:
  - (1) 1/4 inch for "t" up to 3/4 inch inclusive. Where "t", here and below, pertains to the thickness of the weld being examined; if a weld joins two members having different thickness at the weld, "t" is the thinner of these two thicknesses.
  - (2) 1/3 "t" for "t" from 3/4 inch to 2-1/4 inch, inclusive.
  - (3) 3/4 inch for "t" over 2-1/4 inch.
- d. Any group of indications in line that have an aggregate length greater than "t" in a length of "12t", except where the distance between the successive indications exceeds 6L where L is the longest indication in the group.

### 3.3.5 Ultrasonic Examination

Where discontinuities are interpreted to be cracks, lack of fusion, and incomplete penetration, they are unacceptable regardless of length. Linear-type discontinuities are unacceptable if the amplitude exceeds the reference level and discontinuities have lengths which exceed the following:

- a. 1/4 inch for "t" up to 3/4 inch. Where "t", here and below, is the thickness of the weld being examined; if the weld joins two members having different thicknesses at the weld, "t" is the thinner of these two thicknesses.
- b. 1/3 inch for "t" from 3/4 to 2-1/4 inch.
- c. 3/4 inch for "t" over 2-1/4 inch.

#### 3.4 CORRECTIONS AND REPAIRS

Remove and repair defects as specified in ASME B31.1, ASME B31.3, ASME B31.5, ASME B31.8 unless otherwise specified. Repair disqualifying defects discovered between weld passes before additional weld material is deposited. Wherever a defect is removed, and repair by welding is not required, blend the affected area into the surrounding surface eliminating sharp notches, crevices, or corners. After defect removal is complete and before rewelding, examine the area by the same test method which first revealed the defect to ensure that the defect has been eliminated. After rewelding, reexamine the repaired area by the same test method originally used for that area. Regard any indication of a defect as a defect unless reevaluation by NDE or by surface conditioning shows that no disqualifying defects are present. The use of any foreign material to mask, fill in, seal, or disguise welding defects will not be permitted.

-- End of Section --

SECTION 40 17 26.00 20

WELDING PRESSURE PIPING  
04/06

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN PETROLEUM INSTITUTE (API)

API Std 1104 (2013; Errata 1-3 2014; Addendum 1 2014; Errata 4 2015; Addendum 2 2016) Welding of Pipeline and Related Facilities

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)

ASNT SNT-TC-1A (2020) Recommended Practice for Personnel Qualification and Certification in Nondestructive Testing

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B31.1 (2022) Power Piping  
ASME B31.3 (2022; Errata 2023) Process Piping  
ASME B31.4 (2022) Pipeline Transportation Systems for Liquid Hydrocarbons and Other Liquid  
ASME B31.5 (2022) Refrigeration Piping and Heat Transfer Components  
ASME B31.9 (2020) Building Services Piping  
ASME BPVC SEC I (2017) BPVC Section I-Rules for Construction of Power Boilers  
ASME BPVC SEC II-C (2017) BPVC Section II-Materials Part C-Specifications for Welding Rods Electrodes and Filler Metals  
ASME BPVC SEC IX (2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications  
ASME BPVC SEC V (2017) BPVC Section V-Nondestructive Examination

AMERICAN WELDING SOCIETY (AWS)

AWS A2.4 (2012) Standard Symbols for Welding, Brazing and Nondestructive Examination

|                |                                                                          |
|----------------|--------------------------------------------------------------------------|
| AWS A3.0M/A3.0 | (2020) Standard Welding Terms and Definitions                            |
| AWS B2.1/B2.1M | (2021) Specification for Welding Procedure and Performance Qualification |
| AWS D1.1/D1.1M | (2020; Errata 1 2021) Structural Welding Code - Steel                    |
| AWS QC1        | (2016) Specification for AWS Certification of Welding Inspectors         |
| AWS Z49.1      | (2021) Safety in Welding and Cutting and Allied Processes                |

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

|             |                                                |
|-------------|------------------------------------------------|
| 29 CFR 1910 | Occupational Safety and Health Standards       |
| 29 CFR 1926 | Safety and Health Regulations for Construction |

1.2 RELATED REQUIREMENTS

Section 23 03 00.00 20 BASIC MECHANICAL MATERIALS AND METHODS applies to this section with the additions and modifications specified herein.

1.3 DEFINITIONS

AWS A3.0M/A3.0 and applicable ANSI piping documents.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Welding Pressure Piping

SD-07 Certificates

Welding Procedures Qualification

Nondestructive Examination (NDE) Procedures

NDE personnel Certification Procedures

Inspector Certification

Submit inspector certification and NDE personnel certification for record.

SD-11 Closeout Submittals

Weld Identifications

## 1.5 QUALITY ASSURANCE

### 1.5.1 Welding Pressure Piping

Show location, length, and type of welds, and indicate postweld heat treatment and nondestructive testing as required.

### 1.5.2 Procedures

Develop and qualify procedures for welding metals included in the work. Do not start welding until welding procedures, welders, and welding operators have been qualified. Perform qualification testing by an approved testing laboratory, or by the Contractor if approved by the Contracting Officer in accordance with the qualified procedures. Notify the Contracting Officer at least 24 hours in advance of the time and place of the tests. When practicable, perform the qualification tests at or near the work site. Maintain current records of the test results obtained in welding procedure, welding operator/welder performance qualifications, and nondestructive examination (NDE) procedures. These records must be readily available at the site for examination by the Contracting Officer. Qualify the procedures for making transition welds between different materials or between plates or pipes of different wall thicknesses. ANSI Piping requirements for branch connections may be used in lieu of detailed designs. Unless otherwise specified, the choice of welding process must be the responsibility of the Contractor.

#### 1.5.2.1 Previous Qualifications

Welding procedures, welders, and welding operators previously qualified by test may be accepted for the work without requalification provided that the following conditions are fulfilled:

- a. Copies of welding procedures, procedure qualification test records, and welder and welding operator performance qualification test records are submitted and approved in accordance with the paragraph SUBMITTALS.
- b. Testing was performed by an approved testing laboratory or technical consultant or by the Contractor's approved quality control organization.
- c. The welding procedures, welders, and welding operators were qualified in accordance with ASME BPVC SEC IX or AWS B2.1/B2.1M, AR-2 level; and base materials, filler materials, electrodes, equipment, and processes conformed to the applicable requirements of this specification.
- d. The requirements of paragraph WELDER AND WELDING OPERATOR PERFORMANCE QUALIFICATION for renewal of qualification were met, and records showing name of employer and period of employment using the process for which qualified are submitted as evidence of conformance.

#### 1.5.2.2 Performance

The Contractor must be responsible for the quality of joint preparation, welding, and examination. Clearly identify and record materials used in the welding operations. The examination and testing defined in this specification are minimum requirements. Provide additional examination and testing as necessary to achieve the quality required.

### 1.5.3 Welding Procedures Qualification

Qualification of the welding procedures for each group of materials to be welded is required as indicated in ASME BPVC SEC IX. Record in detail and qualify the "Welding Procedure Specifications" for every welding procedure proposed. Qualification for each welding procedure must conform to the requirements of ANSI Standards and to this specification. The welding procedures must specify end preparation for welds, including cleaning, alignments, and root openings. Preheat, interpass temperature control, and postheat treatment of welds must be as required by ANSI Piping documents, unless otherwise indicated or specified. Describe the type of backing rings or consumable inserts, if used, and, if they are to be removed, the removal process. Welding procedure qualifications must be identified individually and referenced on the shop drawings or suitably keyed to the contract drawings.

### 1.5.4 Welder and Welding Operator Performance Qualification

Qualify each welder and welding operator assigned to work covered by this specification by performance tests using equipment, positions, procedures, base metals, and electrodes or bare filler wires from the same specification, classification, or group number that will be encountered on his assignment. Welders or welding operators who make acceptable procedure qualification tests will be considered performance-qualified for the welding procedure used. Determine performance qualification in accordance with ASME BPVC SEC IX, ANSI Piping Standards and as specified.

### 1.5.5 Renewal of Qualification

Requalification of a welder or welding operator must be required under one or any combination of the following conditions:

- a. When a welder or welding operator has not used the specific welding process for a period of 3 months. The period may be extended to 6 months if the welder has been employed on another welding process.
- b. There is specific reason to question the welder's ability to make welds that will meet the requirements of the specifications.
- c. The welder or welding operator was qualified by an employer other than those firms performing work under this contract and a qualification test has not been taken within the preceding 12 months. Renewal of qualification under this condition need be made on only a single test joint or pipe of any thickness, position, or material to reestablish qualification for any thickness, position, or material for which the welder or welding operator had qualified previously.

### 1.5.6 Qualification of Inspection and (NDE) Personnel

Qualification of Inspection and Nondestructive Examination (NDE) Personnel: Qualify inspection and nondestructive examination personnel in accordance with the following requirements:

#### 1.5.6.1 Inspector Certification

Qualify welding inspectors in accordance with AWS QC1.



#### 1.5.6.2 NDE Personnel Certification Procedures

Certify NDE personnel and establish a written procedure for the control and administration of NDE personnel training, examination, and certification. Base procedures on appropriate specific and general guidelines of training and experience recommended by ASNT SNT-TC-1A, Supplement A-Radiographic, Supplement B-Magnetic particle, Supplement C-Ultrasonic and Supplement D-Liquid Penetrant.

#### 1.5.6.3 Welding Procedures and Qualifications

- a. Specifications and Test Results: Submit copies of the welding procedure specifications and procedure qualification test results for each type of welding required. Approval of any procedure does not relieve the Contractor of the responsibility for producing acceptable welds. Submit this information on the forms printed in ASME BPVC SEC IX or their equivalent.
- b. Certification: Before assigning welders or welding operators to the work, submit their names, together with certification that each individual is performance qualified as specified. Do not start welding work prior to procedure qualification. The certification must state the type of welding and positions for which each is qualified, the code and procedure under which each is qualified, date qualified, and the firm and individual certifying the qualification tests.

#### 1.5.7 Symbols

Conform to AWS A2.4.

##### 1.5.7.1 Weld Identifications

Submit a list of the welders' names and symbol for each welder. To identify welds, submit written records indicating the location of welds made by each welder or welding operator.

#### 1.5.8 Safety

Conform to AWS Z49.1, 29 CFR 1910-SUBPART Q, "Welding, Cutting, and Brazing," 29 CFR 1926-SUBPART J, "Welding and Cutting."

#### 1.6 ENVIRONMENTAL

Do not perform welding when the quality of the completed weld could be impaired by the prevailing working or weather conditions. The Contracting Officer will determine when weather or working conditions are unsuitable for welding.

#### 1.7 DELIVERY AND STORAGE

Deliver filler metals, electrodes, fluxes and other welding materials to the site in manufacturers' original packages and store in a dry space until used. Label and design packages properly to give maximum protection from moisture and to assure safe handling.

## PART 2 PRODUCTS

### 2.1 WELDING MATERIALS

Comply with ASME BPVC SEC II-C. Welding equipment, electrodes, welding wire, and fluxes must be capable of producing satisfactory welds when used by a qualified welder or welding operator using qualified welding procedures.

## PART 3 EXECUTION

### 3.1 WELDING

Do not deviate from applicable codes, approved procedures and approved shop drawings without prior written approval from the Contracting Officer. Materials or components with welds made off the site will not be accepted if the welding does not conform to the requirements of this specification unless otherwise specified. Assign each welder or welding operator an identifying number, letter, or symbol that must be used to identify his welds. Each welder or welding operator must apply his mark adjacent to his weld using an approved rubber stamp or felt-tipped marker with permanent, weatherproof ink or other approved methods that do not deform the metal. For seam welds, place identification marks adjacent to the welds at 3 foot intervals. Confine identification by die stamps or electric etchers to the weld reinforcing crown, preferably in the finished crater.

### 3.2 WELDING OPERATORS

Perform welding in accordance with qualified procedures using qualified welders and welding operators.

### 3.3 SUPPORTS

Welding of hangers, supports, and plates to structural members must conform to AWS D1.1/D1.1M.

### 3.4 EXAMINATIONS AND TESTS

Visual and nondestructive examinations must be performed by the Contractor to detect surface and internal discontinuities in completed welds. Employ the services of a qualified commercial inspection or testing laboratory or technical consultant approved by the Contracting Officer. Visually examine welds Radiographic, Liquid penetrant, Magnetic particle, or Ultrasonic examination must be required as indicated in Tables IV and V attached to this section. Random NDE testing applies to ASME B31.3 and ASME B31.4 piping unless specified otherwise. When examination and testing indicates defects in a weld joint, a qualified welder must repair the weld in accordance with the paragraph CORRECTIONS AND REPAIRS of this section.

#### 3.4.1 Random NDE Testing

When radiographic, liquid penetrant, magnetic particle, or ultrasonic examination is required, test a minimum of 10 percent of the total length or number of piping welds. Randomly select the welds examined, but include an examination of welds made by each welding operator or welder. If random testing reveals that a weld fails to meet minimum quality requirements, examine an additional 10 percent of the welds in that same

group. If the additional welds examined meet the quality requirements, the entire group of welds represented must be accepted and the defective welds must be repaired. If any of the additional welds examined also fail to meet the quality requirements, that entire group of welds must be rejected. Remove and reweld rejected welds or examine rejected welds 100 percent and remove and reweld defects.

#### 3.4.2 Visual Examination

Visually examine welds as follows:

- a. Before welding -- for compliance with requirements for joint preparation, placement of backing rings or consumable inserts, alignment and fit-up, and cleanliness.
- b. During welding -- for conformance to the qualified welding procedure.
- c. After welding -- for cracks, contour and finish, bead reinforcement, undercutting, overlap, and size of fillet welds.

#### 3.4.3 Nondestructive Examination

NDE must be in accordance with written procedures. Procedures for radiographic, liquid penetrant, magnetic particle, or ultrasonic tests and methods must conform to ASME BPVC SEC V. The approved procedure must be demonstrated to the satisfaction of the Contracting Officer's QA personnel.

In addition to the information required in ASME BPVC SEC V, the written procedures must include:

- a. Timing of the nondestructive examination in relation to the welding operations.
- b. Safety precautions.

#### 3.4.4 Examinations and Tests by the Government

Examinations and tests will conform to paragraphs "Visual Examination" and "Nondestructive Examination" of this section, except that destructive tests may be required also. When destructive tests are made, qualified welders or welding operators must make repairs using welding procedures which will develop the full strength of the members cut. Welding must be subject to examination and tests in the mill, shop, and field.

#### 3.4.5 Piping Subject to 100 Percent NDE

ASME B31.4 and ASME B31.3 Piping Subject to 100 Percent NDE: 100 percent of each day's girth welds installed in the following locations must be nondestructively examined 100 percent by radiographic, magnetic particle, or liquid penetrant examination unless impracticable, in which case at least 90 percent must be examined. Nondestructive examination must be impracticable for each girth weld not examined.

- a. At onshore locations where a loss of hazardous liquid (petroleum, petroleum products, or anhydrous ammonia) could reasonably be expected to pollute stream, river, lake, reservoir, or other body of water, and any offshore area;
- b. Within railroad or public road rights-of-way;

- c. At overhead road crossings and within tunnels;
- d. Within the limits of any incorporated subdivision of a State government; and
- e. Within populated areas, including, but not limited to, residential subdivisions, shopping centers, schools, designated commercial areas, industrial facilities, public institutions, and places of public assembly.

### 3.5 ACCEPTANCE STANDARDS

#### 3.5.1 Visual

The following indications are unacceptable:

- a. Cracks--external surface.
- b. Undercut on surface which is greater than 1/32 inch deep or 25 percent for ASME B31.3 and 12.5 percent for ASME B31.4 and ASME B31.9 of the wall thickness, whichever is less, provided that the remaining wall thickness is not less than the minimum design thickness. For ASME B31.4 and in accordance with API Std 1104, undercuts over 1/64 inch through 1/32 inch or over 6 to 12.5 percent of the pipe wall thickness, whichever is smaller, must not exceed 2 inches in a continuous weld length of 12 inches or 1/6 the length of the weld, whichever is smaller; and undercuts 1/64 inch or 6 percent of the wall thickness, whichever is smaller, are acceptable regardless of length.
- c. Weld reinforcement:
  - (1) ASME B31.1, conform to Table I.

| TABLE I REINFORCEMENT OF GIRTH AND LONGITUDINAL BUTT WELDS |                                                           |                               |                         |
|------------------------------------------------------------|-----------------------------------------------------------|-------------------------------|-------------------------|
|                                                            | Maximum Thickness of Reinforcement for Design Temperature |                               |                         |
| Thickness of Base Metal, inches                            | Greater than 750 degrees F                                | 350 degrees F - 750 degrees F | Less than 350 degrees C |
|                                                            | inch                                                      | inch                          | inch                    |
| Up to 1/8, incl.                                           | 1/16                                                      | 3/32                          | 3/16                    |
| Over 1/8 to 3/16, incl.                                    | 1/16                                                      | 1/8                           | 3/16                    |
| Over 3/16 to 1/2, incl.                                    | 1/16                                                      | 5/32                          | 3/16                    |
| Over 1/2 to 1, incl.                                       | 3/32                                                      | 3/16                          | 3/16                    |

| TABLE I REINFORCEMENT OF GIRTH AND LONGITUDINAL BUTT WELDS                                                                                             |                                                           |                                                                       |                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------|-------------------------|
|                                                                                                                                                        | Maximum Thickness of Reinforcement for Design Temperature |                                                                       |                         |
| Thickness of Base Metal, inches                                                                                                                        | Greater than 750 degrees F                                | 350 degrees F - 750 degrees F                                         | Less than 350 degrees C |
|                                                                                                                                                        | inch                                                      | inch                                                                  | inch                    |
| Over 1 to 2, incl.                                                                                                                                     | 1/8                                                       | 1/4                                                                   | 1/4                     |
| Over 2                                                                                                                                                 | 5/32                                                      | The greater of 1/4 inch or 1/8 times the width of the weld in inches. |                         |
| NOTES:                                                                                                                                                 |                                                           |                                                                       |                         |
| 1. For double welded butt joints, this limitation on reinforcement given above must apply separately to both inside and outside surfaces of the joint. |                                                           |                                                                       |                         |
| 2. For single welded butt joints, the reinforcement limits given above must apply to the outside surface of the joint only.                            |                                                           |                                                                       |                         |
| 3. The thickness of weld reinforcement must be based on the thickness of the thinner of the materials being joined.                                    |                                                           |                                                                       |                         |
| 4. The weld reinforcement thicknesses must be determined from the higher of the abutting surfaces involved.                                            |                                                           |                                                                       |                         |
| 5. Weld reinforcement may be removed if so desired.                                                                                                    |                                                           |                                                                       |                         |

(2) ASME B31.3, conform to Table II.

| TABLE II                                       |                            |
|------------------------------------------------|----------------------------|
| Wall Thickness, inches                         | Height, inches             |
| Less than or equal to 1/4                      | Less than or equal to 1/16 |
| Greater than 1/4, is less than or equal to 1/2 | Less than or equal to 1/8  |
| Greater than 1/2, is less than or equal to 1   | Less than or equal to 5/32 |
| Greater than 1                                 | Less than or equal to 3/16 |
| NOTES:                                         |                            |

| TABLE II                                                                                                                                                                                                                                                                                                                                                                  |                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Wall Thickness, inches                                                                                                                                                                                                                                                                                                                                                    | Height, inches |
| 1. Wall thickness is the nominal wall thickness of the thinner of components joined by butt weld.                                                                                                                                                                                                                                                                         |                |
| 2. Height: For "Normal Service" and "Severe Cyclic" conditions, use the listed value. For "Category D Fluid Service," use twice the listed value. Measure from surfaces of adjacent components. The lesser of the two measurements, in any plane through the weld, must not exceed the applicable value at right. Weld metal must merge smoothly into component surfaces. |                |

(3) ASME B31.4, conform to Table I for under 350 degrees F.

(4) ASME B31.5, conform to Table III.

| TABLE III                   |                                 |
|-----------------------------|---------------------------------|
| Pipe Wall Thickness, inches | Reinforcement Thickness, inches |
| 1/4 and under               | 1/16                            |
| Over 1/4 through 1/2        | 3/32                            |
| Over 1/2, through 1         | 1/8                             |
| Over 1                      | 3/16                            |

(5) ASME B31.9: Thickness of weld reinforcement must not exceed 3/16 inch.

- d. Lack of fusion on surface.
- e. Incomplete penetration (applies only when inside surface is readily accessible).
- f. Convexity of fillet weld surface greater than 10 percent of longest leg plus 0.03 inch.
- g. Concavity in groove welds.
- h. Concavity in fillet welds greater than 1/16 inch.
- i. Fillet weld size less than indicated or greater than 1 1/4 times the minimum specified fillet leg length.

### 3.5.2 Magnetic Particle Examination

The following relevant indications are unacceptable:

- a. Any cracks and linear indications.

- b. Rounded indications with dimensions greater than 3/16 inch.
- c. Four or more rounded indications in a line separated by 1/16 inch or less edge-to-edge.
- d. Ten or more rounded indications in any 6 square inches of surface, with the major dimension of this area not to exceed 6 inches, with the area taken in the most unfavorable location relative to the indications being evaluated.

#### 3.5.3 Liquid Penetrant Examination

Indications whose major dimensions are greater than 1/16 inch must be considered relevant. The following relevant indications are unacceptable:

- a. Any cracks or linear indications.
- b. Rounded indications with dimensions greater than 3/16 inch.
- c. Four or more rounded indications in a line separated by 1/16 inch or less edge-to-edge.
- d. Ten or more rounded indications in any 6 square inches of surface, with the major dimension of this area not to exceed 6 inches, with the area taken in the most unfavorable location relative to the indications being evaluated.

#### 3.5.4 Radiography

Welds that are shown by radiography to have any of the following discontinuities are unacceptable:

- a. Any type of crack or zone of incomplete fusion or penetration.
  - b. Any other elongated indication which has a length greater than:
    - (1) 1/4 inch for t up to 3/4 inch, inclusive;
    - (2) 1/3 t for t from 3/4 inch to 2 1/4 inches, inclusive;
    - (3) 3/4 inch for t over 2 1/4 inches where t is the thickness of the thinner portion of the weld.
- ("t" pertains to the thickness of the weld being examined. If a weld joins two members having different thickness at the weld, "t" is the thinner of these two thicknesses.)
- c. Any group of indications in line that have an aggregate length greater than t in a length of 12t, except where the distance between the successive indications exceeds 6L where L is the longest indication in the group.
  - d. Porosity in excess of that shown acceptable in Appendix A-250, Acceptance Standard for Radiographically Determined Rounded Indications in Welds, ASME BPVC SEC I.

#### 3.5.5 Ultrasonic Examination

Permitted for ASME B31.3 and ASME B31.4 piping only. Linear type

discontinuities are unacceptable if the amplitude exceeds the reference level and discontinuities have lengths which exceed the following:

- a.  $1/4$  inch for  $t$  up to  $3/4$  inch
- b.  $1/3 t$  for  $t$  from  $3/4$  inch to  $2\ 1/4$  inches
- c.  $3/4$  inch for  $t$  over  $2\ 1/4$  inches

("t" is the thickness of the weld being examined. If the weld joins two members having different thickness at the weld, "t" is the thinner of these two thicknesses. Discontinuities are interpreted to be cracks, lack of fusion, and incomplete penetration are unacceptable regardless of length.)

### 3.6 CORRECTIONS AND REPAIRS

Remove defects and replace welds as specified in ANSI Piping Standards, unless otherwise specified. Repair defects discovered between weld passes before additional weld material is deposited. Wherever a defect is removed, and repair by welding is not required, the affected area must be blended into the surrounding surface eliminating sharp notches, crevices, or corners. After defect removal is complete and before rewelding, reexamine the area by the same test methods which first revealed the defect to ensure that the defect has been eliminated. After rewelding, reexamine the repaired area by the same test methods originally used for that area. For repairs to base material, the minimum examination must be the same as required for butt welds. Indication of a defect must be regarded as a defect unless reevaluation by NDE or by surface conditioning shows that no unacceptable indications are present. The use of foreign material to mask, fill in, seal, or disguise welding defects will not be permitted.



| TABLE IV - EXAMINATIONS AND TESTS FOR VARIOUS MATERIALS AND SERVICES                               |        |                                                                                    |                                       |                        |
|----------------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------------|---------------------------------------|------------------------|
| <u>Examinations or Tests Required</u>                                                              |        |                                                                                    |                                       |                        |
| Material or Application                                                                            | Visual | Radiographic                                                                       | Magnetic Particle or Liquid Penetrant | Ultrasonic             |
| High-alloy austenitic or nickel steels or nickel alloys for cryogenic service and vacuum service   |        |                                                                                    |                                       |                        |
| a. Tack welds                                                                                      | Yes    | No                                                                                 | No No                                 |                        |
| b. Root passes                                                                                     | Yes    | No                                                                                 | Yes No                                |                        |
| c. Intermediate passes                                                                             | Yes    | No                                                                                 | No No                                 |                        |
| d. Completed weld                                                                                  | Yes    | 100 percent for NPS over 1-1/4 inches 60 percent for NAPA PS 1-1/4 inches and less | Yes(PT only); 1/2 inch and over       | Yes for wall thickness |
| High-alloy austenitic or nickel steels or nickel alloys for other than cryogenic or vacuum service |        |                                                                                    |                                       |                        |
| a. Tack welds                                                                                      | Yes    | No                                                                                 | No No                                 |                        |
| b. Root passes                                                                                     | Yes    | No                                                                                 | NoYes                                 | No                     |
| c. Intermediate passes                                                                             | Yes    | No                                                                                 | No No                                 |                        |
| d. Completed weld                                                                                  | Yes    | 100 percent                                                                        | Yes (PT only)                         | No                     |
| Stainless steel to carbon steel                                                                    |        |                                                                                    |                                       |                        |

| TABLE IV - EXAMINATIONS AND TESTS FOR VARIOUS MATERIALS AND SERVICES |        |              |                                       |            |
|----------------------------------------------------------------------|--------|--------------|---------------------------------------|------------|
| <u>Examinations or Tests Required</u>                                |        |              |                                       |            |
| Material or Application                                              | Visual | Radiographic | Magnetic Particle or Liquid Penetrant | Ultrasonic |
| a. Completed weld                                                    | Yes    | No           | Yes (PT only)                         | No         |
| Carbon steel piping systems                                          |        |              |                                       |            |
| a. Tack welds                                                        | Yes    | No           | No                                    | No         |
| b. Root passes                                                       | Yes    | No           | No                                    | No         |
| c. Intermediate passes                                               | Yes    | No           | No                                    | No         |
| d. Completed weld                                                    | Yes    | Random       | No                                    | No         |

| TABLE V - MANDATORY MINIMUM NONDESTRUCTIVE EXAMINATIONS FOR ASME B31.1 PIPING                                         |                                                              |                                                                                                                                   |                                       |
|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
|                                                                                                                       | Temperatures over 750 degrees F and at all pressures.        | Temperatures between 350 degrees F and 750 degrees F inclusive and at all pressures over 1052 psig gage                           | All others                            |
| Buttwelds (Girth and Longitudinal)                                                                                    | RT for NPS over 2 inches MT or PT for NPS 2 inches and less. | RT for over 2 inch NPS with thickness over 3/4 inch. Visual for all sizes with thickness 3/4 inch or less.                        | Visual for all sizes and thicknesses. |
| Welded Branch Connections (Size indicated is Branch Size)                                                             | RT for NPS over 2 inch MT or PT for NPS 2 inch and less.     | RT for branch over 4 inch NPS and thickness of branch over 3/4 inch. Visual for all sizes with branch thickness 3/4 inch or less. | Visual for all sizes and thicknesses. |
| Fillet, Socket Welds                                                                                                  | PT or MT for all sizes and thicknesses.                      | Visual for all sizes and thicknesses.                                                                                             | Visual for all sizes and thicknesses. |
| NOTES:                                                                                                                |                                                              |                                                                                                                                   |                                       |
| 1. Thickness refers to pressure boundary wall thickness (such as pipe wall, fitting wall, or nozzle wall thickness).  |                                                              |                                                                                                                                   |                                       |
| 2. All welds must be given a visual examination in addition to type of specific nondestructive examination specified. |                                                              |                                                                                                                                   |                                       |
| 3. NPS-Nominal Pipe Size.                                                                                             |                                                              |                                                                                                                                   |                                       |
| 4. RT-Radiographic examination; MT-magnetic particle examination; PT-liquid penetrant examination.                    |                                                              |                                                                                                                                   |                                       |

| TABLE V - MANDATORY MINIMUM NONDESTRUCTIVE EXAMINATIONS FOR ASME B31.1 PIPING                                                                                                                                                                                                                            |                                                       |                                                                                                         |            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------------|
|                                                                                                                                                                                                                                                                                                          | Temperatures over 750 degrees F and at all pressures. | Temperatures between 350 degrees F and 750 degrees F inclusive and at all pressures over 1052 psig gage | All others |
| 5. RT of branch welds must be performed before any nonintegral reinforcing material is applied.                                                                                                                                                                                                          |                                                       |                                                                                                         |            |
| 6. The thickness of butt welds is defined as the thicker of the two abutting ends after end preparation.                                                                                                                                                                                                 |                                                       |                                                                                                         |            |
| 7. Temperatures and pressures shown are design.                                                                                                                                                                                                                                                          |                                                       |                                                                                                         |            |
| 8. In lieu of radiography of welded branch connections when required above, liquid penetrant or magnetic particle examination is acceptable and, when used, must be performed at the lesser of one-half of the weld thickness or each 1/2 inch of weld thickness and all accessible final weld surfaces. |                                                       |                                                                                                         |            |
| 9. For nondestructive examination of the pressure retaining component, refer to the standards listed in applicable code or the manufacturing specifications.                                                                                                                                             |                                                       |                                                                                                         |            |

-- End of Section --

SECTION 40 60 00

PROCESS CONTROL  
05/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C12.1 (2014; Errata 2016) Electric Meters - Code for Electricity Metering

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 142 (2007; Errata 2014) Recommended Practice for Grounding of Industrial and Commercial Power Systems - IEEE Green Book

IEEE C37.90.1 (2013) Standard for Surge Withstand Capability (SWC) Tests for Relays and Relay Systems Associated with Electric Power Apparatus

IEEE C62.41.1 (2002; R 2008) Guide on the Surges Environment in Low-Voltage (1000 V and Less) AC Power Circuits

IEEE C62.41.2 (2002) Recommended Practice on Characterization of Surges in Low-Voltage (1000 V and Less) AC Power Circuits

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 61131-3 (2013) Programmable Controllers - Part 3: Programming Languages

INTERNATIONAL SOCIETY OF AUTOMATION (ISA)

ISA 101.01 (2015) Human Machine Interfaces for Process Automation Systems

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250 (2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

NEMA ICS 1 (2022) Standard for Industrial Control and Systems: General Requirements

NEMA ICS 2 (2000; R 2020) Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 V

NEMA ICS 3 (2005; R 2010) Medium-Voltage Controllers  
Rated 2001 to 7200 V AC

NEMA ICS 4 (2015) Application Guideline for Terminal  
Blocks

NEMA ICS 5 (2017) Industrial Control and Systems:  
Control Circuit and Pilot Devices

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

NFPA 79 (2015) Electrical Standard for Industrial  
Machinery

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST SP 250 (1991) Calibration Services Users Guide

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

47 CFR 15 Radio Frequency Devices

UNDERWRITERS LABORATORIES (UL)

UL 508A (2018; Reprint Jul 2022) UL Standard for  
Safety Industrial Control Panels

UL 1059 (2019; Reprint Jul 2022) UL Standard for  
Safety Terminal Blocks

## 1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

### SD-02 Shop Drawings

Contractor Design Drawings; G, AE

Draft As-Built Drawings; G, AE

### SD-03 Product Data

Control Drawings; G

Sensors; G

Performance Verification Test (PVT); G

Factory Test Procedure; G

### SD-06 Test Reports

Factory Test Report

Testing, Adjusting and Commissioning

Performance Verification Test(PVT)

Endurance Test

#### SD-07 Certificates

Control and Sensor Wiring

Ground Rods

Wiring

Installation

#### SD-10 Operation and Maintenance Data

Training Manual; G, AE

Control System; G, AE

#### SD-11 Closeout Submittals

Final As-Built Drawings; G, AE

### 1.3 SITE ENVIRONMENTAL CONDITIONS

The expected site environmental conditions are a minimum of 14 degrees F and a maximum of 81 degrees F.

### 1.4 SEQUENCING

TABLE I: PROJECT SEQUENCING specifies the sequencing of submittals as specified in paragraph SUBMITTALS (denoted by an 'S' in the 'TYPE' column) and activities as specified in PART 3 EXECUTION (denoted by an 'E' in the 'TYPE' column).

#### 1.5.1 Sequencing for Submittals

The sequencing specified for submittals is the deadline by which the submittal must be initially submitted to the Government. Following submission there will be a Government review period as specified in Section 01 33 00 SUBMITTAL PROCEDURES. If the submittal is not accepted by the Government, revise the submittal and resubmit it to the Government within 14 days of notification that the submittal has been rejected. Upon re-submittal there will be an additional Government review period. If the submittal is not accepted the process repeats until the submittal is accepted by the Government.

#### 1.5.2 Sequencing for Activities

The sequencing specified for activities indicates the earliest the activity may begin.

#### 1.5.3 Abbreviations

In TABLE I the abbreviation AA0 is used for 'after approval of' and 'AC0' is used for 'after completion of'.

## PART 2 PRODUCTS

### 2.1 SYSTEM DESCRIPTION

The process control system must be used to monitor and control the operation of process equipment as specified and in accordance with the sequence of operation and control schematics shown on the drawings. The process control system must provide for operator interaction, overall process control system supervision, and process equipment control and monitoring. The system must adhere to Section 25 05 11 . Provide hardware configured and sized to support expansion as specified and shown on the drawings.

The process control system must be complete including sensors, field preamplifiers, signal conditioners, offset and span adjustments, amplifiers, transducers, transmitters, control devices, engineering units conversions and algorithms for the applications; and must maintain the specified end-to-end process control loop accuracy from the sensor to display and final control element. Connecting conductors must be suitable for installed controls. Enclosures must be rated for NEMA 4X.

#### 2.1.1 Operation

The process control system provided under this specification must operate using a combination of sequential function charts, function block diagrams, structured text, instruction, and ladder logic type as defined in IEC 61131-3 and supervisory control to provide the required sequences of operation. Input data to the controller must be obtained by using instruments and controls interfaced to mechanical, electrical, utility systems and other systems as shown and specified. All required setpoints, settings, alarm limits, and sequences of operation must be as identified in the database/ settings tables .

#### 2.1.2 Points

Provide inputs to and outputs from the process control system in accordance with the Input/Output (I/O) Summary Table indicated. Each connected analog output (AO), analog input (AI), binary output (BO), binary input (BI), pulse accumulator (PA) input and other input or output device connected to the control system must represent a "point" where referred to in this specification.

#### 2.1.3 Building Telecommunications Cabling (BTC)

Provide data transmission systems for communication between PLCs and between PLCs and the central station and server as shown on the drawings and as indicated.

#### 2.1.4 System Reliability

The system must be designed for maximum reliability, safety and integrity while maintaining an availability of 99% or better.

### 2.2 MATERIALS AND EQUIPMENT

#### 2.2.1 Product Certifications

Computing devices, as defined in FCC Part 15, supplied as part of the



process control system must be certified to comply with the requirements of Class B computing devices.

#### 2.2.2 Standard Products

Materials and equipment must be standard unmodified products of a manufacturer regularly engaged in the manufacturing of such products. Units of the same type of equipment must be products of a single manufacturer. Items of the same type and purpose must be identical and supplied by the same manufacturer, unless replaced by a new version approved by the Government.

#### 2.2.3 Nameplates

Each major component of equipment must have the manufacturer's name and address, and the model and serial number in a conspicuous place. Laminated plastic nameplates must be provided for equipment devices and panels furnished. Each nameplate must identify the device, such as pump "P-1" or valve "VLV-402". Labels must be coordinated with the schedules and the process and instrumentation drawings. Laminated plastic must be 1/8 inch thick, white with black center core. Nameplates must be a minimum of 1 by 3 inches with minimum 1/4 inch high engraved block lettering. Nameplates for devices smaller than 1 by 3 inches must be attached by a nonferrous metal chain. All other nameplates must be attached to the device.

### 2.3 GENERAL REQUIREMENTS

Equipment located outdoors, not provided with climate controlled enclosure, must be capable of operating in the ambient temperature range. Electrical equipment will conform to Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Equipment and wiring must be in accordance with NFPA 70, with proper consideration given to environmental conditions such as moisture, dirt, corrosive agents, and hazardous area classification.

### 2.4 SENSORS

#### 2.4.1 Transmitter

Unless indicated otherwise, each sensor must be provided with a transmitter, selected to match the sensor. Except where specifically indicated otherwise on the drawings, the transmitter must be provided with a four digit or analog visual display of the measured parameter and with a 4 to 20 mAdc output signal proportional to the level of the measured parameter. Accuracy must be plus or minus 2 percent of full scale reading with output error not exceeding plus or minus 0.5 percent of full scale. Transmitter must be located where indicated, mounted integrally with the sensor, pipe mounted, wall mounted or installed in the control panel. The distance between the sensor and transmitter must not exceed the manufacturer's recommendation. Field preamplifiers and signal conditioners must be included when necessary to maintain the accuracy from sensor to the programmable logic controller or recorder.

#### 2.4.2 Liquid Service

Sensors and meters in liquid service must be rated for continuous duty service at fluid approach velocities from 2.5 ft/s to 33 ft/s with correspondingly higher constriction velocities over a fluid temperature range from 32 degrees F to 105 degrees F at pressures up to 150 psi gage.

### 2.4.3 Flow Rate Sensors and Meters

Liquid flow indication must be provided in gpm. Vapor flow indication must be provided in cubic feet per minute. Pressure taps must incorporate appropriate snubbers. Unless indicated otherwise, the flow transmitter must produce a signal that is proportional to the volumetric flow rate, compensated for fluid temperature, and must have an accuracy of plus or minus 1 percent of full flow. Flow transmitter must be located within 15 feet of the flow element. The flow transmitter must include a digital readout of the volumetric flow rate to 3 significant figures. The controller must be provided with a minimum of three sets of dry contacts rated in accordance with NEMA ICS 1. The first set of contacts must close when the lower (warning) detection level has been exceeded. The second set of contacts must close when the upper (alarm) detection level has been exceeded. The third set of contacts must close when a controller malfunction has occurred, including loss of power or loss of sensor input.

The alarm levels must be individually adjustable. The controller must be provided with an audible warning horn that sounds when the upper detection level has been exceeded, and a warning horn silence button. The controller must provide a 4-20 mA<sub>dc</sub> output signal to the programmable logic controller, proportional to the measured parameter. The controller must be provided with an internal battery to maintain operation for a minimum of 12 hours if power is lost. Flow rate must be controlled to within plus or minus 5 percent of the design flow.

#### 2.4.3.1 Magnetic Flowmeter

Magnetic flowmeter must be non-intrusive, DC pulse type and must measure fluid flow through the use of a self generated magnetic field. The meter must have automatic zeroing circuitry. The magnetic flow element must be encapsulated in anodized aluminum. Flowmeter must be capable of measuring up to a maximum flow velocity of 33 fps. The metering tube must be constructed of anodized aluminum. The meter must be rated for a process temperature range of 32 to 212 F and 0 to 149F ambient. The maximum pressure drop across the meter and appurtenances must be 5 psi at the maximum flow rate.

#### 2.4.4 Level Instrumentation

Pressure taps must incorporate appropriate snubbers. Relays and housing must be intrinsically safe or explosion proof as required by the NFPA hazard rating for compatibility with the contents of the tank or sump. The controller must be provided with a minimum of three sets of dry contacts rated in accordance with NEMA ICS 1. The first set of contacts must close when the lower (warning) detection level has been exceeded. The second set of contacts must close when the upper (alarm) detection level has been exceeded. The third set of contacts must close when a controller malfunction has occurred, including loss of power or loss of sensor input.

The alarm levels must be individually adjustable. The controller must be provided with an audible warning horn that sounds when the upper detection level has been exceeded, and a warning horn silence button. The controller must provide a 4-20 mA<sub>dc</sub> output signal to the programmable logic controller, proportional to the measured parameter. The controller must be provided with an internal battery to maintain operation for a minimum of 12 hours if power is lost.

#### 2.4.4.1 Non-Contact Ultrasonic Level Sensor

The sensor must be microprocessor based and must provide continuous, non-contact level measurement of liquids and solids utilizing microwave pulsed time of flight measurement method. Sensor must have pushbutton calibration and LCD display. The sensor must operate in a frequency band approved for industrial use. The sensor must be capable of measuring in a range of 0 to 20 feet with an accuracy of plus or minus .25 percent of full scale and repeatability  $\pm 0.125$ ". The sensor must be capable of distinguishing between real echoes, reflections and background noise. The sensor must automatically compensate for temperature changes. The sensor must be capable of operating in a temperature range from minus 15 degrees F to 120 degrees F. Assembly must be flange mounted of sufficient size to eliminate echoing and suitable for the installed environment indicated. Mounting assembly must be suitable for service without requiring entry or drainage of the vessel where level is being measured.

#### 2.4.4.2 Leak Detection

Double walled containment system leak detectors must use electrodes mounted in the interstices of double walled containment systems with a minimum time delay of 0.5 seconds. Leak detectors for open systems must be mounted at slab or floor level with either a minimum time delay of 0.5 seconds or a minimum built-in-vertical adjustment of 1/8 inch to prevent activation due to high humidity. Detector must have a contact rating of 1.0 amps resistive or 200 mA inductive at 28 VDC. Leak detector panel must indicate the location and detector causing the alarmed state. The indicator must be manual reset type. A framed, non-fading half-size as-built location map in laminated plastic must be provided for the cable leak detection system in double containment piping systems indicating the as installed system configuration; sensing string layout must be furnished. Marks in meters feet along the length of pipeline interstitial cable must be provided as references to locate leaks.

#### 2.4.5 Pressure Instrumentation

Pressure taps must incorporate appropriate snubbers.

##### 2.4.5.1 Pressure Sensor and Transducer

The sensing element must be either diaphragm type. The pressure transducer must withstand up to 150 percent of rated pressure, with an accuracy of plus or minus 1.0 percent of full scale selected to put the design range of the measured pressure in the middle third of the transducer's range. Pressure must be measured in psi gage with a range, plus or minus 10 percent of design range and must be furnished with display to the nearest 0.145 psi. The transmitter output error must not exceed 0.1 percent of calibrated span.

##### 2.4.5.2 Pressure Switch

Sensors must be diaphragm and must be constructed of brass. Pressure switch must have a repetitive accuracy of plus or minus 5.0 percent of the operating range and must withstand up to 150 percent of rated pressure.

Switch actuation set point must be adjustable over the operating pressure range with a differential adjustment span of 20 to 40 percent of the range of the switch. The switch must have Form C snap-action contacts rated in

accordance with NEMA ICS 1.

#### 2.4.6 Process Analytical Instrumentation

Probes must be easily removable without interrupting service. Sampling pumps must be included where necessary or applicable to the sensing device. For sensors integral to the electronic controller the sample may be drawn directly into the sensor or may be drawn through a sample tube. For sensors remotely located the sample may be drawn through a sample tube. Outdoor sample tubes must be heat traced. Sensor and controller construction must be suitable for operation in the monitored medium. Systems requiring automated zero and calibration gas or reagents must be provided with 7 days supply of calibration gas or reagent. The controller must be provided with a minimum of three sets of dry contacts rated in accordance with NEMA ICS 1. The first set of contacts must close when the lower (warning) detection level has been exceeded. The second set of contacts must close when the upper (alarm) detection level has been exceeded. The third set of contacts must close when a controller malfunction has occurred, including loss of power or loss of sensor input. The alarm levels must be individually adjustable. The controller must be provided with an audible warning horn that sounds when the upper detection level has been exceeded, and a warning horn silence button. The controller must provide a 4-20 mA<sub>dc</sub> output signal to the programmable logic controller, proportional to the measured parameter. The controller must be provided with an internal battery to maintain operation for a minimum of 12 hours if power is lost.

##### 2.4.6.1 Chlorine in Liquid

All parts of the chlorine measurement system, including the sensors, transmitters, controllers and peripheral devices, that may come in contact with chlorine or a chlorine-filled environment must be constructed of materials suitable for this application. Residual chlorine sensor must continuously monitor the chlorine residual. The sensor must be capable of detecting chlorine in the range of 0 to 10 mg/L with an accuracy of plus or minus 1 percent of full scale reading. The sensor response time must be 90 percent in a maximum of 60 seconds.

##### 2.4.6.2 pH Monitoring

The sensor must be flow-through type. Sensor must be a differential type. Sensor must have a range of 1 . pH units to 14 . pH units and must have an accuracy of plus or minus 0.1 . pH unit. The sensor must automatically compensate for temperature over the temperature range. The sensor body must be PVC, CPVC or epoxy.

##### 2.4.6.3 Total Dissolved Solids (TDS)

The TDS sensor must measure the specific conductance using a conductivity sensor, displaying the total dissolved solids value in microohms (microSiemens) per centimeter (uS/cm) and transmitting an analog signal for remote processing. System must be industrial grade, torodial and suitable for measurement of conductivity in a solution by insertion of the sensing element into the pipeline, using a hot tap assembly . Sensor assembly must be suitable for periodic removal for adjustment and cleaning without requiring shut down of the process. Sensor must be suitable for range of 0 to 10,000 . Range must be field verified for the application and adjusted as required. Sensing element must be constructed of 316 stainless steel and glass, including temperature element, and be capable

of continuous operation. Sensing element must be unaffected by color in the fluid, pressure, and rate of flow. Sensor must have automatic temperature compensation and must require no normal maintenance or periodic recalibration.

#### 2.4.6.4 Water Turbidity

System must be complete and include indicating meter, sensing element and a transmitter. System must be industrial grade and suitable for measurement of turbidity by insertion of the sensing element into the pipeline, using a hot tap assembly. Sensor assembly must be suitable for periodic removal for adjustment and cleaning without requiring shutdown of the process. Sensor must be suitable for range from 0 to 200 Nephelometric turbidity units (NTU). The accuracy must be plus or minus 2 percent of full scale reading. Range must be field verified for the application and adjusted as required. Sensing element must be constructed of 316 stainless steel and glass. Sensing element must be unaffected by color in the fluid, pressure, temperature and rate of flow. Sensor must have automatic zeroing and must require no normal maintenance or periodic recalibration. Sensor must have active and passive anti-fouling mechanisms.

#### 2.4.7 Limit Switch

Limit switch must be rotary/lever type, contacts must be heavy duty and rated A600 or N600. Enclosure must be metal. The actuating head must vary based upon the application.

#### 2.4.8 Emergency Stop

Emergency stop pushbutton must have red mushroom actuator and yellow background. Actuator must be self-latching and manually reset. Provide contacts that are NEMA ICS A600 and directly opened. Pushbutton must meet NFPA 79 SECTION 10.7.

### 2.5 PROGRAMMABLE LOGIC CONTROLLER (PLC)

#### 2.5.1 PLC General Requirements

PLCs must be micro-processor based, capable of receiving binary and analog inputs and, through programming, must be able to control binary and analog output functions, perform data handling operations and communicate with external devices. PLCs must meet the requirements of Class A computing devices, and must be labeled as set forth in 47 CFR 15 and must be able to withstand conducted susceptibility test as outlined in NEMA ICS 1, NEMA ICS 2, NEMA ICS 3, or IEEE C37.90.1. PLCs must function properly at temperatures between 32 and 122 degrees F at 5 to 95 percent relative humidity non-condensing and must tolerate storage temperatures between minus 40 and plus 140 degrees F at 5 to 95 percent relative humidity non-condensing.

#### 2.5.2 Modular PLC

PLCs must be based on a modular, field expandable design allowing the system to be tailored to the process control application. The system must be expandable through the use of additional hardware and/or user software. As a minimum, the PLC must include a mounting backplane, power supply module, central processing unit (CPU) module, communications module, and input/output (I/O) module. The modules must be grouped

together in a mounting rack or cabinet. The mounting rack backplane must provide the communications mechanism to fully integrate the individual modules located within the rack. Modules other than I/O modules must plug directly into the backplane. The use of wire connectors between modules will not be allowed except for expansion of the system to include multiple backplanes. The rack size must be as needed to hold the equipment necessary while performing the required control functions.

#### 2.5.2.1 Central Processing Unit (CPU) Module

The CPU module must be a self contained, microprocessor based unit that provides time of day, scanning, application program execution, storage of application programs, storage of numerical values related to the application process and logic, I/O bus traffic control, peripheral and external device communications and self diagnostics. The scan time must be 250 milliseconds or better including spare I/O channels.

#### 2.5.2.2 Communications Module

The communications module must allow peer-to-peer communication with other PLCs and must allow the PLC to communicate with the central station, or workstation. The communication module must utilize the manufacturer's standard communication architecture and protocol, ethernet architecture and protocol or a combination of these. The communication module must allow programming of the PLC to be done locally through the use of a laptop computer or from the central station or remote workstation.

#### 2.5.2.3 Power Supply Module

One or more power supply modules must be provided as necessary to power other modules installed in the same cabinet. Power supply modules must plug directly into the backplane. Auxiliary power supplies may be used to supply power to remote cabinets or modules.

- a. Power supply modules must use AC power with a nominal voltage of 120 VAC plus or minus 5 percent. The power supply module must monitor the incoming line voltage level and must provide over current and over voltage protection. If the voltage level is detected as being out of range the power supply module must continue to provide power for an adequate amount of time to allow for a safe and orderly shutdown. Power supply modules must be capable of withstanding a power loss for a minimum of 20 milliseconds while still remaining in operation and providing adequate power to all connected modules.
- b. Each power supply module must be provided with an on-off switch integral to the module. If the manufacturer's standard power supply module is not provided with an on-off switch, a miniature toggle type switch must be installed near the PLC and must be clearly labeled as to its function..
- c. Provide power supply modules with an indicating light which must be lit when the module is operating properly.

#### 2.5.2.4 Input/Output (I/O) Modules

Modules must be self contained, microprocessor based units that provide an interface to field devices. The modules must be located in the same cabinet as the other PLC components. Each module must contain visual indication to display the on-off status of individual inputs or outputs. Each I/O

must be protected against reversal of polarity of the signal. Analog inputs and analog outputs must have 'open, short and out of range circuit' detection. It must be configurable per channel.

### 2.5.3 Program Storage/Memory Requirements

The CPU must utilize the manufacturer's standard non-volatile memory for the operating system. The controller must have electronically battery backup volatile memory. Must be possible to change battery with power on for storage of user programs. The user programs must be loaded through the controller keypad, central station or through the use of a laptop computer. The CPU memory capacity must be based on the system's control requirements. The memory capacity must be sized such that, when the system is completely programmed and functional, no more than 50 percent of the memory allocated for these purposes is used.

### 2.5.4 Input/Output Characteristics

Each controller must allow for analog input, analog output, binary input and binary output. The number and type of inputs and outputs for the system must be as shown on the drawings and must comply with the sequence of control. The system capacity must include a minimum of 20 percent spare input and output points (no less than two points) for each point type provided. During normal operation, a malfunction in any input/output channel must affect the operation of that channel only and must not affect the operation of the CPU or any other channel. All input circuits must have a minimum optical isolation of 1500 VRMS and must be filtered to guard against high voltage transients from the externally connected devices. All output circuits must have a minimum optical isolation of 1500 VRMS and must be filtered to guard against high voltage transients from the externally connected devices.

#### 2.5.4.1 Analog Inputs

Analog input circuits must be available in 4-20 mA.

#### 2.5.4.2 Binary Inputs

Binary input circuits must be available in  
79-132 VAC.

#### 2.5.4.3 Analog Outputs

Analog output circuits must be available in 4-20 mA.

#### 2.5.4.4 Binary Outputs

Binary output circuits must be available in 79-132 VAC.

#### 2.5.4.5 Pulse Inputs

Pulse inputs must be able to detect a pulse of 100 milliseconds or less.

### 2.5.5 Wiring Connections

Wiring connections must be heavy duty, self lifting, pressure type screw terminals to provide easy wire insertion and secure connections. The terminals must accept two #14 AWG wires. A hinged protective cover must be provided over the wiring connections. The cover must have write-on

areas for identification of the external circuits.

#### 2.5.6 On-Off Switch

Each controller must be provided with an integral on-off power switch. If the controller is not provided with a manufacturer's standard on-off switch, a miniature toggle type switch must be installed in the control panel near the controller and must be clearly labeled as to its function.

#### 2.5.7 Diagnostics

Each PLC must have diagnostic routines implemented in firmware. The CPU must continuously perform self-diagnostic routines that will provide information on the configuration and status of the CPU, memory, communications and input/output. The diagnostic routines must be regularly performed during normal system operation. A portion of the scan time of the controller must be dedicated to performing these housekeeping functions. In addition, a more extensive diagnostic routine must be performed at power up and during normal system shutdown. The CPU must log input/output and system faults in fault tables which must be accessible for display. When a fault affects input/output or communications modules the CPU must shut down only the hardware affected and continue operation by utilizing the healthy system components. All faults must be annunciated at the PLC and the central station. Diagnostic software must be useable in conjunction with the portable tester. The following diagnostics must be performed:

- a. Analog Inputs: Sensor out of range, open or shorted loop, analog-to-digital converter check
- b. Analog Outputs: Open or shorted loop
- c. Configuration: Check compatibility and availability of selected I/O hardware and software
- d. Memory: Checksum, parity check End-to End CPU memory

#### 2.5.8 Accuracy

Provide controllers with an accuracy of plus or minus 0.25 percent of input span.

### 2.6 PLC SOFTWARE

Furnish all PLC software described in this specification as part of the complete control system.

#### 2.6.1 Operating System

Each PLC must be provided with the manufacturer's standard operating system software package. The PLC must maintain a point database in its memory that includes all parameters, constraints and the latest value or status of all points connected to the PLC. Execution of the PLC application programs must use the data in memory resident files. The operating system must support a full compliment of process control functions. It must be possible to define these functions using a mix of ladder logic diagrams, function blocks, sequential function charts and text programming. Programming methods and interactions must be based on IEC 61131-3. A combination of the programming methods must be possible



within a single controller. The operating system must allow loading of control logic locally or from the central station in which case it must require a password to do so and data files from the portable tester. It must also support data entry and diagnostics using an operator interface panel attached directly to the PLC. Each PLC must be capable of operating in stand alone mode.

#### 2.6.1.1 Startup

The PLC must have startup software that causes automatic commencement of operation without human intervention, including startup of all connected I/O functions. A PLC restart program based on detection of power failure at the PLC must be included in the PLC software. The restart program must include start time delays between successive commands to prevent demand surges or overload trips.

#### 2.6.1.2 Failure Mode

Upon failure for any reason, each PLC must perform an orderly shutdown. Systems which are not Primary/Secondary must force all PLC outputs to a predetermined (failure mode) state, consistent with the failure modes shown and the associated control device. Primary/Secondary systems must transfer I/O scan and control to the PLC not currently failed.

#### 2.6.2 Functions

The controller operating system must be able to scan inputs, control outputs, and read and write to its internal memory in order to perform the required control as indicated in the sequence of control on the drawings. The controller must periodically perform self diagnostics to verify that it is functioning properly. If the system is set up as a Primary/Secondary system the system must attempt to switch to the other PLC upon sensing a fault in the currently controlling PLC.

##### 2.6.2.1 Analog Monitoring

Measure and transmit all analog values including calculated analog points.

##### 2.6.2.2 Logic (Virtual)

Logic (virtual) points must be software points entered in the point database which are not directly associated with a physical I/O function. Logic (virtual) points must be analog or binary points created by calculation from any combination of binary and analog points, or other data having all the properties of real points, including alarms, without the associated hardware. Logic (virtual) points must be defined or calculated and entered into the database. The calculated analog point must have point identification in the same format as any other analog point.

##### 2.6.2.3 State Variables

If an analog point represents more than two (up to 8) specific states, each state must be nameable. For example, a level sensor must be displayed at its measured engineering units plus a state variable with named states usable in programs or for display such as low alarm/low/normal/high/high alarm.

#### 2.6.2.4 Analog Totalization

Any analog point must be operator assignable to the totalization program. Up to eight analog values must be totalized within a selectable time period.

#### 2.6.3 Alarm Processing

Each PLC must have alarm processing software for AI, DI, and PA alarms for all real and virtual points connected to that PLC.

##### 2.6.3.1 Binary Alarms

Binary alarms are those abnormal conditions indicated by BIs as specified and shown. The system must automatically suppress analog alarm reporting associated with a binary point when that point is turned off.

##### 2.6.3.2 Analog Alarms

Analog alarms are those conditions higher or lower than a defined value, as measured by an AI. Analog readings must be compared to predefined high and low limits, and alarmed each time a value enters or returns from a limit condition. Unique high and low limits must be assigned to each analog point in the system. In control point adjustment (CPA) applications, key the limit to a finite deviation traveling with the setpoint. The system must automatically suppress analog alarm reporting associated with an analog point when that analog point is turned off.

##### 2.6.3.3 Pulse Accumulator (PA) Alarms

Pulse accumulator alarms are those conditions calculated from totalized values of accumulator inputs or PA input rates that are outside defined limits as specified and shown. PA totalized values must be compared to predefined limits and alarmed each time a value enters a limit condition. Unique limits must be assigned to each PA point in the system.

#### 2.6.4 Constraints

##### 2.6.4.1 Equipment Constraints Definitions

Each control point in the database must have PLC resident constraints defined and entered by the Contractor, including as applicable: maximum starts (cycles) per hour; minimum off time; minimum on time; high limit (value in engineering units); and low limit (value in engineering units).

##### 2.6.4.2 Constraints Checks

All control devices connected to the system must have the PLC constraints checked and passed before each command is issued. Each command point must have unique constraints assigned. High and low "reasonableness" values or one differential "rate-of-change" value must be assigned to each AI. Each individual point must be capable of being selectively disabled by the operator from the central station.

#### 2.6.5 Control Sequences and Control Loops

Specific functions to be implemented are defined in individual system control sequences and database tables shown on the drawings, and must include, as applicable, the following functions: PI control must provide

proportional control and proportional plus integral control; two position control must provide control for a two state device by comparing a set point against a process variable and an established dead band; floating point control must exercise control when an error signal exceeds a selected dead band, and must maintain control until the error is within the dead band limits; signal selection must allow the selection of the highest or lowest analog value from a group of analog values as the basis of control and must include the ability to cascade analog values so that large numbers of inputs can be reduced to one or two outputs; signal averaging must allow the mathematical calculation of the average analog value from a group of analog values as the basis of control and must include the ability to "weight" the individual analog values so that the function output can be biased as necessary to achieve proper control; reset function must develop an AO based on up to two AIs and one operator specified reset schedule.

#### 2.6.6 Command Priorities

A scheme of priority levels must be provided to prevent interaction of a command of low priority with a command of higher priority. Override commands entered by the operator must have higher priority than those emanating from applications programs.

#### 2.6.7 Resident Application Software

Provide resident applications programs developed in accordance with paragraph GRAPHICAL PROGRAMMING to achieve the sequences of operation, parameters, constraints, and interlocks necessary to provide control of the process systems connected to the process control system. All application programs must be resident in the PLC and must execute in the PLC, and must coordinate with each other, to ensure that no conflicts or contentions remain unresolved.

##### 2.6.7.1 Program Inputs and Outputs

Use program inputs listed for each application program to calculate the required program outputs. Where specific program inputs are not available, a "default" value or virtual point appropriate for the equipment being controlled and the proposed sequence of operation must be provided to replace the missing input, thus allowing the application program to operate.

##### 2.6.7.2 Failure Mode

In the event of a PLC failure, the controlled equipment must continue to function in the failure mode shown on the drawings. Systems that are Primary/Secondary must transfer control to the non-failed system.

### 2.7 CONTROL PANELS

#### 2.7.1 Components

##### 2.7.1.1 Enclosures

The enclosure for each control panel must conform to the requirements of NEMA 250 for the types specified. Finish color must be the manufacturer's standard, unless otherwise indicated. Enclosures for installation in mechanical equipment rooms must be Type 4 ; those for installation in clean, dry indoor occupied space may be Type 1; other locations must be as

otherwise specified or shown. Enclosures for equipment installed outdoors must be Type 4 or as shown. Enclosures for installation in a corrosive environment must be Type 4X and must be constructed of stainless steel . Painted steel must not be allowed for use in a corrosive environment. Enclosure must be provided with a single, continuously hinged exterior door with print pocket, 3-point latching mechanism and key lock and a single, continuously hinged interior door.

#### 2.7.1.2 Controllers

Provide controllers in accordance with paragraph Programmable Logic Controller (PLC).

#### 2.7.1.3 Standard Indicator Light

Indicator lights showing on, off, stand-by, automatic, manual depending on the application must comply with NEMA ICS 1, NEMA ICS 2 and UL 508A. Lights must be heavy duty, round and must mount in a 0.875 inch mounting hole. Indicator lights must be LED type and must operate at 120 VAC or 24 VDC. Long life bulbs must be used. Indicator light must be provided with a legend plate labeled as shown on the drawings. Lens color must be as indicated on the drawings. Lights must be push to test (lamp) type.

#### 2.7.1.4 Selector Switches

Selector switches must comply with NEMA ICS 1, NEMA ICS 2 and UL 508A. Selector switches must be heavy duty, round and must mount in a 0.875 inch mounting hole. The number of positions must be as indicated on the drawings. Switches must be illuminated . Switches must be rated for 600 volts, 10 amperes continuous. Selector switches must be provided with a legend plate labeled as shown on the drawings. Where indicated or required, dual auxiliary contacts must be provided for the automatic position to provide position sensing at the central station or workstation. Auxiliary contacts must be rated for 120 VAC, 1A as a minimum. Where indicated on the drawings, switches must be key operated. All keys must be identical.

#### 2.7.1.5 Push Buttons

Push buttons must comply with NEMA ICS 1, NEMA ICS 2 and UL 508A. Push buttons must be heavy duty, round and must mount in a 22.5 mm 0.875 inch mounting hole. The number and type of contacts must be as indicated on the drawings or required by the Sequence of Control. Push buttons must be rated for 600 volts, 10 amperes continuous. Push buttons must be provided with a legend plate labeled as shown on the drawings.

#### 2.7.1.6 Relays

Relays must comply with NEMA ICS 5 and derated for altitude above 1,500 m. Relays must be as required by the Sequence of Control. Relay coil must be 120 VAC and must be provided with matching mounting socket. Power consumption must not be greater than 3 watts. Coils must have a minimum current rating of 10 amps and minimum voltage rating of 120 volts. Contacts must have a minimum current rating of 10 amps and minimum voltage rating of 240 volts.

#### 2.7.1.7 Terminal Blocks

Terminal blocks must comply with NEMA ICS 4 and UL 1059. Terminal blocks

for conductors exiting control panels must be two-way type with double terminals, one for internal wiring connections and the other for external wiring connections. Terminal blocks must be made of bakelite or other suitable insulating material with full deep barriers between each pair of terminals. A terminal identification strip must form part of the terminal block and each terminal must be identified by a number in accordance with the numbering scheme on the approved wiring diagrams.

#### 2.7.1.8 Alarm Horns

Alarm horns must be provided where indicated on the drawings. Horns must be vibrating type and must comply with UL 508A. Horns must provide 100 dB at 10 feet. Exterior mounted horns must be weather proof by design or must be mounted in a weather proof enclosure that does not reduce the effectiveness of the horn.

#### 2.7.2 Panel Assembly

Control panels must be factory assembled and shipped to the jobsite as a single unit. Panels must be fabricated as indicated and devices must be mounted as shown or required. Each panel must be fabricated as a bottom-entry connection point for process control system electrical power, process control system main air source, process control system wiring, communications system wiring to other control panels and operators workstation.

#### 2.7.3 Electrical Requirements

Each panel must be powered by a dedicated 120 volts ac circuit, with a fuse, sized as recommended by the equipment manufacturer, and a disconnect switch located inside the panel. Wiring must terminate inside the panel on terminal blocks. Electrical work must be as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM and as shown on the drawings.

#### 2.7.4 Power Line Conditioner

Each control panel must be provided with a power line conditioner to provide both voltage regulation and noise rejection. The power line conditioner must be of the ferro-resonant design, with no moving parts and no tap switching, while electrically isolating the secondary from the power line side. The power line conditioner must be sized for 125 percent of the actual connected kva load. Characteristics of the power line conditioner must be as follows:

##### 2.7.4.1 85 Percent Load

At 85 percent load, the output voltage must not deviate by more than plus or minus 1 percent of nominal voltage when the input voltage fluctuates between minus 20 percent to plus 10 percent of nominal voltage.

##### 2.7.4.2 Load Changes

During load changes of zero to full load, the output voltage must not deviate by more than plus or minus 3 percent of nominal voltage. Full correction of load switching disturbances must be accomplished within 5 cycles, and 95 percent correction must be accomplished within 2 cycles of the onset of the disturbance.

#### 2.7.5 Grounding

Control panel enclosures must be equipped with a solid copper ground bus or equivalent. The ground bus must be securely anchored to the enclosure so as to effectively ground the entire structure. Clamp-type terminals sized large enough to carry the maximum expected current must be provided on the ground bus for grounding cables. Where a definite circuit ground is required, a single wire not less than #10 AWG must run independently to the panel ground bus and must be fastened to the ground bus with a bolted terminal lug. Cases of instruments, relays and other devices must be effectively grounded through the enclosures steel structure unless otherwise indicated. Insulated wiring having a continuous rated current of not less than the circuit fuse rating must be used for grounding. Grounding terminals of power receptacles must be solidly grounded to the panel enclosure.

#### 2.7.6 Convenience Outlet

A 120 volt ac, 20 amp, ground fault interruption (GFI) type duplex convenience outlet must be provided inside the panel. The outlet circuit must be separate from the panel power circuit.

#### 2.7.7 Panel Interior Light

Each control panel must be provided with a 15 watt LED light. The light must be operated by a manual on-off switch mounted on the interior door of the enclosure. The light must be powered by the same circuit as the convenience outlet.

#### 2.7.8 Ventilation System

Each control panel must be provided with two single phase, 120 volt ac ventilation fans. Each fan must supply a minimum of 100 cfm of ventilation air through the enclosure. Each fan must be provided with a line voltage thermostat. Thermostat setpoints must be adjustable in a range of 70 to 140 degrees F as a minimum. Each supply and exhaust grille must contain a filter that is easily removed for cleaning or replacement.

#### 2.7.9 Heating System

Each control panel(s) must be provided with a thermostatically controlled electric heater capable of maintaining an enclosure temperature of 35 degrees F when continuously exposed to an ambient temperature of 0 degrees F.

#### 2.7.10 Uninterruptible Power Supply

Each control panel must be provided with an Uninterruptible Power Supply in accordance with Computer Hardware, UPS specification.

#### 2.7.11 Air Conditioning System

Where indicated, control panel must be provided with a mechanical refrigeration air conditioning system. The system must be capable of maintaining a temperature of 100 degrees F inside the enclosure with all equipment in the panel operating and while continuously exposed to full sunlight and an ambient air temperature of 95 degrees F. The compressor and condenser must be located outside the control panel enclosure. Provisions must be made to remove condensate from the control panel and to

protect all devices within the enclosure from condensate.

## 2.8 COMPUTER HARDWARE

For computer hardware furnished under this specification provide standard products of a single manufacturer which advertises service in all 48 contiguous states, and provide only model currently in production. Except for PCI-E cards installed into expansion slots provided in a desktop or server computer in order to meet the requirements of this specification, do not modify computer hardware from the manufacturer configuration.

### 2.8.1 Server Hardware

Computer Server Hardware (server) must be an industrial panel PC or small form factor PC meeting the following minimum requirements::

#### 2.8.1.1 Processor

Quad-core processor designed for server applications. Processor speed must be at least 50 percent of the speed of the fastest Intel server processor commercially available.

#### 2.8.1.2 Random Access Memory (RAM)

300 percent of the recommended requirements of the software to be installed on the server.

#### 2.8.1.3 Communications Ports

Four USB 3.0 3.1 ports.

#### 2.8.1.4 Hard Drives

##### 2.8.1.4.1 Internal Hard Drives

Solid State Hard drives with SATA-3 Controller providing at least 2TB usable disk space. Hard drives must use RAID (Redundant Array of Inexpensive Disks) at levels 1 or 5 (RAID-1 or RAID-5).

#### 2.8.1.5 Optical Drive

Not required.

#### 2.8.1.6 Video Output

32-bit color at a minimum resolution of 1920 by 1080 at a minimum refresh rate of 70 Hz and a DVI or display port output.

#### 2.8.1.7 Network Interface

One integrated 1000Base-T Ethernet with RJ45 connector.

#### 2.8.1.8 Monitor

Must be an Industrial panel or monitor suitable for use in a NEMA4 environment, mounted in the control panel. The monitor may be a combined Panel PC if the server requirements are met.

#### 2.8.1.9 Keyboard

Provide a 101 key wired USB keyboard having a minimum 64 character standard ASCII character set based on ANSI INCITS 154. Primary key input must be by touchscreen.

#### 2.8.1.10 Mouse

Provide a 2-button wired USB optical scroll mouse with a minimum resolution of 400 dots per inch. Primary input must be by touchscreen.

#### 2.8.1.11 Power Supplies

Power supply must be provided by the control panel the server is mounted within. Provide in accordance with Control Panel Components specification.

#### 2.8.2 Uninterruptible Power Supply (UPS)

A self contained UPS suitable for installation and operation at the Server and Workstation must be provided. The units must be sized to provide a minimum of 10 minutes of operation of the central station computer. The UPS must incorporate surge suppression, noise filtering (normal and common mode) short circuit protection and voltage regulation (brownout and overvoltage protection). UPS must be complete with all necessary power supplies, transformers, batteries, and accessories and must include visual indication of normal power operation, UPS operation, abnormal operation and visual and audible indication of low battery power. The UPS must comply with the Federal Communications Commission Standard 15J part A for radio noise emissions.

#### 2.9 TOUCHSCREEN HMI

Human Machine Interface (HMI) display screens must be supplied to provide an interface to the PLC. The touch screens must be compatible with the specific PLC being furnished. Locate touch screens in the cabinets as shown in the Contract Drawings. The touch screens must provide maintenance information and indication of the systems. The touch screens must be 19" panel mount with a minimum resolution of 1024x768. Provide operator interface as shown on the Contract Drawings or approved equal. If a panel PC is provided, no additional HMI is required.

##### 2.9.1 TOUCHSCREEN HMI Display Layout

Develop HMI control and supervision display layouts complying with the minimum requirements of the Control Narrative in the Contract Drawings. Submit for approval prior to implementation.

#### 2.10 MONITORING AND CONTROL SOFTWARE

Provide a single software package which implements the Scheduling, Alarming, Trending, Graphical System Display, and System Display Editor functionality. Other specified M&C functionality may be implemented in the same software package or in additional software packages. The monitoring and control software must provide the communication, programming and control capabilities necessary to support all specified points and functions, plus a minimum expansion of 50 percent of the current number of points, complete with their point database. The monitoring and control station must be online at all times and must perform all required functions as specified. Where multiple modules are



used the modules must be capable of sharing data and operating together seamlessly. Software must be a client-server software package with a graphical user interface (GUI) using web-browser based clients. The system must support multiple user operation with multiple tasks for each user and must support operation and management of all peripheral devices.

#### 2.10.1 M&C Software Update Licensing

In addition to all other licensing requirements, provide M&C Software licensing which includes licensing of the following software updates for a period of no less than 5 years:

- a. Security and bug-fix patches issued by the M&C Software manufacturer.
- b. Security patches to address any vulnerability identified in the National Vulnerability Database at <http://nvd.nist.gov> with a Common Vulnerability Scoring System (CVSS) severity rating of MEDIUM or higher.

#### 2.10.2 Graphical Operations

##### 2.10.2.1 Graphical User Interface

The central station must be provided with a mouse driven, graphical user interface.

##### 2.10.2.2 Display Information

The central station must display information necessary to support all requirements specified, including: operator commands; alarm notification; reports; system graphics as specified and as shown, incorporating dynamic data; and curve plotting.

##### 2.10.2.3 System Graphics Implementation

System graphics displays must be hierarchical displays which integrate dynamic data into the display. System graphics must reflect actual system configuration. Each system schematic must be included as a separate display. Different colors, textures, and use of inverted video must be used for various components and dynamic data. The displays must include standard and/or custom symbols. A library of callable display symbols containing symbols for all necessary equipment and control devices must be furnished. Symbols must conform to ISA 101.01 where applicable. Data associated with a display must be updated within 5 seconds of the binary status change or the analog change in excess of the analog change differential. Any dynamic data which is not current, due to PLC communications failure, PLC failure, or point out of service, must be highlighted or flagged.

##### 2.10.2.4 Display Editor

The display editor must enable the user to create, modify, save and delete displays and symbols. Within the display there must be dynamic fields with the function of linking the dynamic fields with the database and must be executed automatically as the last step of the database generation and modification procedure.

#### 2.10.2.5 Graphical Programming

The system must include a graphical programming function which must be used to create all control sequences utilized in the control panels. This function must reside in the central station to create, modify, and test software for control panel resident programs. The graphical programming function must provide programming elements to be connected together to create a logic diagram. The diagram must be compilable to produce executable code for the control panel. The graphical programming function must include elements necessary to create logic diagrams that represent sequences of operation. Program elements must be able to be combined into a custom template which can then be used as a standard function. Program checkout and debug facilities must include display of dynamic and/or simulated system variables and points on the programming screens. The user must be able to fix or force values of variables to enable program checkout during debugging. The programming must allow for the use of the portable tester for loading files directly into the control panel, uploading of existing control panel programming and database information and downloading of control panel programming and database information.

#### 2.10.2.6 System Menus and Displays

The user must be able to call up the following displays by dedicated function key, pull down menu or by icon and must be able to page forward and backward on linked multiple page displays. The system menu and index displays must also contain icons which can be used to call up subsequent displays.

- a. System Menu (list of all graphics and menus).
- b. Index (list of all PLCs).
- c. Alarm Summary (list of all uncleared alarms).
- d. Abnormal Summary (list of all devices not in normal state; keeps track of alarm conditions which have been cleared).
- e. Data Communications Summary (listing of availability for each communication channel, by statistically processing the number of transmission errors, outages, and other abnormal conditions for each channel).

#### 2.10.2.7 Hard-Copy Screen Request

The central station must be able to obtain a hard copy of the monitor display being viewed. This must be an exact "snapshot" of the data and device symbols shown on the selected monitor.

#### 2.10.3 Command Software

The software must provide for defining and selecting points, parameters, graphics, report generation, and all other functions associated with operation. The operator commands must be usable from central station computer and workstation keyboards with individual operator passwords as specified.

##### 2.10.3.1 Command Input

Command menus must utilize full words and acronyms selected to allow

operators to use the system without extensive training or data processing backgrounds. The system must prompt the operator.

#### 2.10.3.2 Command Input Errors

The system must supervise operator inputs to ensure they are correct for proper execution. Provide operator input assistance whenever a command cannot be executed because of operator input errors.

#### 2.10.3.3 Special Functions

The system must support the following special functions by using a mouse, in addition to all other commands specified:

##### 2.10.3.3.1 Help

Produce a display of all commands available to the operator. The help command, followed by a specific command, must produce context sensitive listing with a short explanation of the purpose, use, and system reaction to that command.

##### 2.10.3.3.2 Start/Enable

Manually start equipment and enable monitoring and control of points.

##### 2.10.3.3.3 Stop/Disable

Manually stop equipment and disable monitoring and control components.

##### 2.10.3.3.4 Display Diagram

Display diagrams of specific utility systems or other systems.

##### 2.10.3.3.5 Diagram Development

Facilitate development of diagrams of specific utility systems or other systems.

##### 2.10.3.3.6 Auto/Override

Override automatic operation of a point or return a point to automatic operation.

##### 2.10.3.3.7 Print Report

Allow the operator to print reports.

##### 2.10.3.3.8 Confirm Action

Allow the operator to confirm that the desired command sequence has been correctly entered and is to be executed.

##### 2.10.3.3.9 Cancel Action

Perform the opposite function of the confirm action, at any time prior to executing confirm action.

#### 2.10.3.3.10 Memo Pad

Allow the operator to create, store and retrieve pop-up notes.

#### 2.10.3.4 Operator's Commands

The operator's commands must provide the means for entry of control and monitoring commands, and for retrieval of information. The operator's commands must perform such tasks as requesting a display of any binary, analog, or accumulator point, or any group of related points, startup and shutdown selected systems or devices, modifying

#### 2.10.3.5 Level of Addressing

Provide four levels of addressing for identification as follows:

##### 2.10.3.5.1 Point

The individual sensor or control device within a unit.

##### 2.10.3.5.2 Unit

The unit that a point is associated with, such as a blower.

##### 2.10.3.5.3 Sub-System

The sub-system that a point is located in or near.

##### 2.10.3.5.4 System

The system that a sub-system is located in or near.

#### 2.10.3.6 System Access Control

A minimum of 5 passwords must be usable with the process control system software. The system must maintain an ASCII disk file logging all operators logged onto the system, alarm acknowledgments, commands issued and all database modifications for each password. Each password must be definable as to the functions that the operator can perform.

#### 2.10.4 Alarms

Provide M&C Software meeting the following minimum requirements for alarms:

- a. The M&C software must be capable of generating alarms by comparing the value of any point from any connected system to user-configurable limits and must notify an operator of the occurrence of an alarm condition. The process control system alarm history must be stored and must be recallable by the operator using the report generator. Alarm messages must take precedence over other functions. A minimum of the most recent 25 system alarms must be directly available at the central station computer. Operator acknowledgment of one alarm must not be considered as acknowledgment of any other alarm nor must it inhibit reporting of subsequent alarms. Alarm data to be displayed and stored must include: identification of the alarm; date and time to the nearest second of occurrence; device or sensor type; limit exceeded (if analog); engineering units; current value or status; alarm class; and alarm messages. All alarms recorded by the M&C system must be recommunicated to the existing control system as described in

contract documents.

- b. The M&C software must support at least two alarm priority levels: critical and informational. Critical alarms must remain in alarm until acknowledged by an operator and the alarm condition no longer exists; informational alarms must remain in alarm until the alarm condition no longer exists or until the alarm is acknowledged.
- c. The creation, modification, and handling (routing) of alarms must be fully accessible and fully adjustable from the graphical user interface.
- d. Alarm Data. Alarm data to be displayed and stored must include:
  - 1) Identification of alarm including building, system (or sub-system), and device name.
  - 2) Date and time to the nearest second of occurrence.
  - 3) Alarm type:
  - 4) Unreliable: Indicates that the source device has failed due to the sensing device or alarm parameter being out-of-range or bad data.
  - 5) High Alarm.
  - 6) Low Alarm.
  - 7) Current value or status of the alarm point, including engineering units
  - 8) Alarm limits, including engineering units.
  - 9) Alarm priority.
  - 10) Alarm Message: A unique message with a field of at least 60 characters. Assignment of messages to an alarm must be an operator editable function.
  - 11) Acknowledgement status of the alarm including the time, date and user of acknowledgement.

#### 2.10.4.1 Binary Alarms

Binary alarms must be subject to immediate reporting, within the alarm response time, at the central station.

#### 2.10.4.2 Analog Alarms

These alarms must be subject to immediate reporting, within the alarm response time, at the central station. The control panel analog readings must be compared to predefined high and low limits, and alarmed to the central station each time a value enters or returns from a limit condition. The program must automatically change the high or low limits, or both, of any analog point, based on time scheduled operations as specified, allowing for a time interval before the new alarm limit becomes effective. For those applications where setpoint adjustments are made, the alarm limit must be keyed to a finite deviation traveling with the setpoint.

#### 2.10.4.3 Alarm Messages

A unique message with a field of 60 characters must be provided for each alarm. Assignment of messages to a point must be an operator editable function. Secondary messages must be assignable by the operator for printing to provide further information, such as telephone lists or maintenance functions, and must be editable by the operator.

#### 2.10.4.4 Alarm Classes

Classes of alarms, which will be identified for each item, include class 1 and class 2 alarm conditions. Class 1 (Critical) must include display, print, and audible alarm at occurrence and at return-to-normal. Acknowledgment of class 1 alarms by the operator must be required at occurrence and at return-to-normal. Class 2 (Informational) must include display, print, and audible alarm at occurrence and at return-to-normal. No acknowledgment of class 2 alarms is required unless otherwise shown.

#### 2.10.5 Pop-up Note Function

A pop-up note function must be included with the workstation, providing the operator a capability of noting any data which may be associated with alarms or with any other event. A note created by an operator must be automatically called up when any other workstation calls up the associated point, alarm, or alarm summary. The pop-up note function must also support free form entry of data which can be used by any workstation operators as general reminders or instructions.

#### 2.10.6 Real Time Clock Synchronization

The system must synchronize each central station computer, real time clock, within one second and at least once per day automatically, without operator intervention and without requiring system shutdown. The central station computer must automatically initiate a call once per day to the NIST clock to obtain the correct time and date and update the real time clock. The central station computer must generate a report showing the time difference.

#### 2.10.7 System Reaction

Under system normal heavy load, no more than 10 seconds must lapse from the time a binary status alarm or analog alarm occurs at a PLC until the change is displayed at the central station and operator's workstation. The total system response time from initiation of a control action command to display of the resulting status change must not exceed 20 seconds under system normal heavy load conditions, assuming a zero response time for operation of the PLC's control device. The alarm printer must continue to print out all occurrences, including time of occurrence, to the nearest second. All system normal heavy load conditions must be introduced to the system via AIs and DIs.

##### 2.10.7.1 Occurrence

System normal heavy load conditions are defined as the occurrence throughout the system of a total of three status changes, three binary alarms, three analog high or low limit alarms, and three analog quantity changes within the high and low limits during a single 1-second interval. This number of similar occurrences must repeat on a continuous basis during successive 1-second intervals for a period of 2 minutes.

#### 2.10.7.2 Location

System normal heavy load conditions, as specified, must have 50 percent of the changes and alarms, including no less than one of each type, occurring at a single PLC with the remaining changes and alarms distributed among the remaining PLCs.

#### 2.10.8 Report Generator

Software must be provided to generate and format standard and custom reports for displaying, printing, and storing on disk. Reports must use database values and parameters, values calculated using the real time static database or historical data base; with the reports subsequently stored on hard disk or zip drive. Dynamic operation of the system must not be interrupted to generate a report. The report must contain the time and date when the sample was taken, and the time and date when the report was printed. The software must allow for automatic and manual generation of reports. For automatic reports an operator must be able to specify the time the initial report is to be generated, the time interval between reports, end of period, and the output format for the report. Manual report generation must allow for the operator to request at any time the output of any report.

##### 2.10.8.1 Periodic Automatic Report

The system must allow for specifying, modifying, or inhibiting the report to be generated, the time the initial report is to be generated, the time interval between reports, end of period, and the output peripheral.

##### 2.10.8.2 Request Report Mode

The system must allow for the operator to request, at any time, an immediate printout of any report.

#### 2.10.9 Data Interchange

Software must be provided to format and store on a government approved removable hard drive the data, trends, profiles, reports and logs as specified in a defined, standard format such as ASCII text or DIF for export and further processing by other software and/or computer systems.

#### 2.10.10 Control Panel and DTS Circuit Alarms

The system must supervise each control panel, I/O function and DTS circuit for alarm reporting, including: control panel not responding; control panel responding (return to normal); control panel to central station DTS circuit high error rate; control panel to control panel DTS circuit high error rate; control panel/central station real time clock error more than 15 seconds (adjustable); control panel intrusion alarm; control panel offline; control panel online (return to normal); control panel failure (self-diagnostics); point not responding to command; and point change of state without command.

#### 2.10.11 Central Station Database

The central station database must be stored on disk and in memory. The static database must be downloadable as required to control panels in the system.

#### 2.10.11.1 Database Definition Process

Software must be provided to define and modify each point in the database using operator commands. The definition must include all physical parameters and constraints associated with each point. Each database item must be callable for display or printing, including EEPROM, ROM and RAM resident data. Each point must be defined and entered into central station database.

#### 2.10.11.2 Dynamic Database

The dynamic database includes those variables which change with time or conditions including all DIIs, AIs, PAs, and virtual (logic) points.

#### 2.10.11.3 Dynamic Database Update

Update the dynamic database from the field, allowing the operator to select update times from 0.1 seconds to 2.0 seconds.

#### 2.10.11.4 Static Database

The static database includes those fixed parameters and constraints from all PLCs which define the characteristics of the system and I/O functions such as alarm limits, start/stop times, point names, PLC channel addresses, and sensor spans.

#### 2.10.11.5 Central Station Static Database Update

A copy of each control panels static database must be updated automatically once per day, each time an authorized change is submitted or upon demand from the central station database.

#### 2.10.11.6 Workstation Access to Dynamic Data

Any workstation with proper access password and connected to the central station via the BTS must have access to the central station's dynamic data. Display of data must commence within 5 seconds.

#### 2.10.12 Historical Data Storage and Retrieval

A historical data storage and retrieval function must be provided at the central station to collect and store dynamic data. This function must be in addition to other data storage requirements. The function must have the capability to collect and store alarm status changes, point values, events and operator commands, and system responses. The storage function must also have the capability to collect and store multiple sets of analog data at pre-specified sampling rates. This function must have the capability to retain historical data on hard disk for pre-specified time periods, up to forty five days using last day roll over, for short-term analysis, and then output the data to a solid state drive in a SQL database for long-term retention. The operator must also be able to selectively recall short-term data stored on hard disk. Retrieval and printing of the contents of any selected historical data file must be available using the data retrieval and report generation program. The output of the report generation program must be capable of being viewed on the screen, printed in a report, or stored.



#### 2.10.13 Trending

Provide M&C software capable of creating, modifying, uploading and archiving Point properties and performing real-time display of point properties with a minimum trending rate of 100 points per second.

- a. The M&C Software must include a graphical display for trend configuration, creation and deletion accessible through the graphical user interface. Each trend must be user-configurable for:
  - (1) Point to trend.
  - (2) Sampling interval: adjustable between 1 second and 1 hour.
  - (3) Start and Stop Time of Trend: Start and stop times determined by one or more of the following methods:
    - (a) Start time and stop time
    - (b) Start time and duration
    - (c) Start time and number of samples
- b. The M&C software must be capable of displaying and printing a graphical representation of each trend, and of multiple trended points on the same graph. The software must be capable of saving trend logs to a file. If the file format is not plain ASCII text in a Comma-Separated-Value (CSV) format, provide a means to export or convert the file to plain ASCII text in a CSV format.

#### 2.10.14 Analog Monitoring

The system must measure, transmit, and display analog values, including calculated analog points. Differential measurements must be displayed as positive or negative values with respect to their reference points shown. An analog change in value is defined as a change exceeding a preset differential value as specified. Each analog change in value must be operator selectable and settable to provide for a minimum reporting change in value of one-half the specified end-to-end accuracy of the measured variable. Displays and reports must express analog values in proper engineering units with sign. Provide 128 different sets of engineering unit conversions. Each engineering conversion unit must include range, span, and conversion equation.

#### 2.10.15 Analog Totalization

Any analog or calculated point must be operator assignable to the totalization program. The analog totalization time period must be defined uniquely for each point. At the end of the period, totals must be stored on disk for future reference. Totalization must then restart from zero for the next time period. The program must keep track of the peak and total value measured during the current period and for the previous period. The operator must be able to initiate a summary of totalization information on a point, unit, sub-system or system. The operator must be able to set or reset each totalized value individually. The operator must be able to define, modify, or delete the time period online.

#### 2.10.16 LAN Software

A network operating system must be supplied as part of the LAN software. The network operating system must support network device access to the central station. The system must provide workstation access to the central station as a virtual terminal. The network must provide network access to shared peripherals. The LAN software must provide for transparent communication with any node on the system. The LAN software must support the following:

##### 2.10.16.1 Access Control

Access control to the central station computer and workstations. Operators must be able to perform all specified functions, given the proper passwords, including database definition/modification, graphic creation/modification, and trending.

##### 2.10.16.2 Other Functions and Configurations

#### 2.10.17 Virus Protection Software

Virus Protection Software will be furnished by the Government.

#### 2.10.18 Disk Imaging (Backup) Software

Provide Disk imaging (backup) software consisting of the project site's standard disk imaging software.

### 2.11 DATA COMMUNICATION REQUIREMENTS

Process control system data communications must support the specified functions and process control system configuration shown on the drawings.

#### 2.11.1 Central Station/Workstation

Each workstation must be able to communicate with the central station as a virtual terminal. The workstation must be able to initiate uploads or downloads of programs and resident data, including parameters of connected systems PLCs and devices, constraints and programs in the central station.

#### 2.11.2 Central Station/PLC

The central station must be able to initiate an upload or download of PLC data programs.

#### 2.11.3 Communication

Communication with other computer systems must be accomplished using an Ethernet communications port. The central station or workstation must be able to initiate upload or download of data files.

#### 2.11.4 Error Detection and Retransmission

Error detection and retransmission must comply with current Ethernet standards.

### 2.12 FACTORY TEST

The process control system must be tested at the factory prior to

shipment. Written notification of planned testing must be given to the Government at least 21 days prior to testing, and in no case must notice be given until after the Contractor has received written Government approval of the test procedures.

#### 2.12.1 Factory Test Setup

Assemble and integrate the factory test setup as specified to prove that performance of the system satisfies all requirements of this project, including system communications requirements in accordance with the approved test procedures. The factory test must take place during regular daytime working hours on weekdays. Equipment used must be the same equipment that is to be delivered to the site. The factory test setup must include the following:

| Factory Test               |                                                                                                                                          |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| central station equipment  | one each of the components                                                                                                               |
| workstation                | one of each type                                                                                                                         |
| control panel              | not less than two control panels: at least one of each type used in the system plus at least one per DTS type                            |
| test set                   | one of each type                                                                                                                         |
| portable tester            | one of each type                                                                                                                         |
| communications circuits    | one of each type and speed to be utilized in the proposed system including bridges, modems, encoder/decoders, transceivers and repeaters |
| surge protection equipment | for power, communications, I/O functions and networks                                                                                    |
| I/O functions              | sufficient to demonstrate the I/O capability and system normal operation                                                                 |
| software                   | software required for proper operation of the proposed system including application programs and sequences of operation                  |

#### 2.12.2 Factory Test Procedure

Test procedures must define the tests required to ensure that the system meets technical, operational, and performance requirements. The test procedures must define location of tests, milestones for the tests, and identify simulation programs, equipment, personnel, facilities, and supplies required. Provide for testing all process control system capabilities and functions specified and shown. Cover actual equipment and sequences to be used for the specified project and include detailed instructions for test setup, execution, and evaluation of test results. The test reports must document results of the tests. Surge testing need not be conducted acceptable documented proof can be provided that such

testing has been satisfactorily demonstrated to the Government with identical surge protection applied. The procedures must include the following:

|                                                                                                              |                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                              | Test Procedure                                                                                                                         |
| equipment                                                                                                    | block diagram                                                                                                                          |
| hardware and software                                                                                        | descriptions                                                                                                                           |
| commands                                                                                                     | operator commands                                                                                                                      |
| I/O functions                                                                                                | test database points with failure modes                                                                                                |
| passwords                                                                                                    | required for each operator access level                                                                                                |
| each type of digital and analog point in the test database                                                   | description                                                                                                                            |
| test equipment                                                                                               | list                                                                                                                                   |
| surge protection                                                                                             | circuit diagrams                                                                                                                       |
| inputs required (I/O point values and status) and corresponding expected results of each set of input values | for each application program                                                                                                           |
| default values                                                                                               | for the application program inputs not implemented or provided for in the contract documents for the application programs to be tested |

### 2.12.3 Factory Test Report

Submit original copies of data produced during the factory test, including results of each demonstration procedure within 7 days after completion of each test. Arrange the report so that commands, responses, and data acquired are correlated to allow logical interpretation of the data.

## PART 3 EXECUTION

### 3.1 FACTORY TEST

Perform factory testing of the System as specified. The Contractor is responsible for providing personnel, equipment, instrumentation, and supplies necessary to perform required testing. Provide written notification of planned testing to the Government at least 21 days prior to testing, and do not give this notice until after receiving written Government approval of the specific Factory Test Procedures. Provide Factory Test Procedures which define the tests required to ensure that the system meets technical, operational, and performance specifications. Within the Procedures define location of tests, milestones for the tests, and identify simulation programs, equipment, personnel, facilities, and supplies required. Provide procedures which test all capabilities and functions specified and indicated. Perform the Factory Test using equipment and software of the same manufacturer, model and revision as

will be used for the specified project. Include detailed instructions for test setup, execution, and evaluation of test results in the Procedures. Upon completion of the test, prepare a Factory Test Report, documenting the results of the Test, and submit it as specified. This report must be approved before any equipment is shipped.

Perform the Factory Test and provide Factory Test Submittals as shown in TABLE II. FACTORY TEST SEQUENCING.

TABLE II FACTORY TEST SEQUENCING

| ITEM # | DESCRIPTION                      | SEQUENCING                                    |
|--------|----------------------------------|-----------------------------------------------|
|        |                                  | (START of ACTIVITY or DEADLINE FOR SUBMITTAL) |
| 1      | Submit Factory Test Procedure to | 7 days after notice Proceed                   |
| 2      | Perform Factory Test             | After Approval of #1                          |
| 3      | Submit Factory Test Report       | 5 days After Completion of #2                 |

### 3.2 EQUIPMENT INSTALLATION REQUIREMENTS

#### 3.2.1 Installation

Install system components and appurtenances in accordance with the manufacturer's instructions and provide necessary interconnections, services, and adjustments required for a complete and operable system. Adjust or replace devices not conforming to the required accuracies. Replace factory sealed devices, rather than adjusting. Installation, adjustment, and operation of the equipment specified must be supervised by a manufacturer's representative experienced in the installing, adjusting, and testing of the equipment.

- a. Install instrumentation and communication equipment and cable grounding as necessary to preclude ground loops, noise, and surges from adversely affecting system operation.
- b. Install wiring in exposed areas, including low voltage wiring, as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Wiring in air plenum areas installed without conduit must be plenum-rated in accordance with NFPA 70.
- c. Submit detail drawings containing complete piping, wiring, schematic, flow diagrams and any other details required to demonstrate that the system has been coordinated and will properly function as a unit. Piping and Instrumentation (P&ID) drawings (prepared using industry recognized device symbols, clearly defined and describing piping designations to define the service and materials of individual pipe segments and instrument tags employing Instrument Society of America suggested identifiers). Include in the Drawings, as appropriate: product specific catalog cuts; a drawing index; a list of symbols; a series of drawings for each process control system using abbreviations, symbols, nomenclature and identifiers as shown; valve schedules; compressed instrument air station schematics and ASME air storage tank certificates for each type and make of compressed instrument air station.

#### 3.2.1.1 Isolation, Penetrations and Clearance from Equipment

Dielectric isolation must be provided where dissimilar metals are used for connection and support. Penetrations through and mounting holes in the building exteriors must be made watertight. Holes in concrete, brick, steel and wood walls must be drilled or core drilled with proper equipment; conduits installed through openings must be sealed with materials which are compatible with existing materials. Openings must be sealed with materials which meet the requirements of NFPA 70 and Section 07 84 00 FIRESTOPPING. Installation must provide clearance for control-system maintenance. Process control system installation must not interfere with the clearance requirements for mechanical and electrical system maintenance.

#### 3.2.1.2 Device Mounting

Devices must be installed in accordance with manufacturers' recommendations and as shown. Control devices to be installed in piping must be provided with required gaskets, flanges, thermal compounds, insulation, piping, fittings, and manual valves for shutoff, equalization, purging, and calibration. Any deviations must be documented and submitted to the Government for approval prior to mounting. Damaged insulation must be replaced or repaired after devices are installed to match existing work. Damaged galvanized surfaces must be repaired by touching up with zinc paint.

#### 3.2.1.3 Pneumatic Tubing

Tubing must be concealed in finished areas. Tubing may be run exposed in unfinished areas, such as mechanical equipment rooms. For tubing to be enclosed in concrete, rigid metal or intermediate metal conduit must be provided. Tubing must be installed parallel or perpendicular to building walls throughout. Maximum spacing between tubing supports must be 1.5 m 5 feet. Each tubing system must be tested pneumatically at 1.5 times the working pressure for 24 hours, with a maximum pressure drop of 2 psig with compressed air supply turned off. Joint leaks must be corrected by remaking the joint. Caulking of joints will not be permitted. Tubing and two insulated copper phone wires for installation checkout may be run in the same conduit. Tubing and electrical power conductors must not be run in the same conduit; however, control circuit conductors may be run in the same conduit as polyethylene tubing.

#### 3.2.2 Sequences of Operation

Study the operation and sequence of local equipment controls, as a part of the conditions report, and note any deviations from the described sequences of operation on the contract drawings. Perform necessary adjustments to make the equipment operate in an optimum manner and must fully document changes made.

#### 3.3 INSTALLATION OF EQUIPMENT

Install equipment as specified, as shown and as required in the manufacturer's instructions for a complete and fully operational control system.

### 3.3.1 Control Panels

Control panels must be located as indicated on the drawings. Devices located in the control panels must be as shown on the drawings or as needed to provide the indicated control sequences.

### 3.3.2 Flow Measuring Device

Fluid flow instruments must be installed in accordance with manufacturer's recommendations, unless otherwise indicated in the specification. The minimum straight unobstructed piping for the flowmeter installation must be 10.0 pipe diameters upstream and 5.0 pipe diameters downstream. Meters for gases and vapors must be installed in vertical piping, and meters for liquids must be installed in horizontal piping, unless otherwise recommended by the manufacturer or indicated in the specifications.

#### 3.3.2.1 Flow Switch

Flow switches must be installed in such a manner as to minimize disturbance of the flow of fluid while maintaining reliable operation of the switch.

#### 3.3.2.2 Magnetic Flowmeter

Meter must be installed in vertical piping so that the flow tube remains full of the process fluid under all operating conditions. A minimum of ten pipe diameters straight run upstream of the flowmeter and five pipe diameters straight run downstream of the flowmeter must be provided. The flowmeter and piping system must be grounded to earth ground.

### 3.3.3 Level Instruments

#### 3.3.3.1 Float Switches

Switches must be mounted in accordance with manufacturer's published instructions. Procedures must be those used for equipment in hazardous locations.

#### 3.3.3.2 Ultrasonic Level Sensor

Sensor must be installed vertically in the top of the tank and in accordance with the manufacturer's instructions. Switches must be accessible for maintenance and calibration. In applications where switches cannot be directly mounted to a tank by the threaded or flanged connection, a mounting bracket must be provided for connection to the inside tank wall. Sensor must be positioned to maximize the return echo signal and minimize vessel obstructions in the sensors line of sight. The minimum recommended distance from the tank fill opening and from the side of the tank must be maintained.

### 3.3.4 Pressure Instruments

Pressure sensors and pressure transducers must be verified by calibration. All pressure taps must incorporate appropriate snubbers. Pressure sensors and pressure switches must have valves for isolation, venting, and taps for calibration. Pressure switches and pressure transducers installed on liquid or steam lines must have drains. Pressure transducers, differential pressure sensors and differential pressure switches must have nulling valves. Pressure switches must be adjusted to

the proper setpoint and must be verified by calibration. Switch contact ratings and duty must be selected for the application.

### 3.3.5 Temperature Instrument Installation

#### 3.3.5.1 RTD

When the RTD is installed in pipe or is susceptible to corrosion or vibration, the RTD must be installed in a thermowell. Thermowells must be filled with conductive heat transfer fluid prior to installation of the RTD in the thermowell. RTDs used for space temperature sensing must include a housing suitable for wall mounting. RTDs used for outside air sensing must have an instrument shelter or sun shield as shown to minimize solar effects, and must be mounted to minimize building effects. RTD assemblies must be readily accessible and installed to allow easy replacement.

#### 3.3.5.2 Temperature Switches

Temperature switches must be installed as specified for RTDs. Temperature switches must be adjusted to the proper setpoint and must be verified by calibration. Switch contact ratings and duty must be selected for the application.

### 3.3.6 Process Analytical Instrumentation

#### 3.3.6.1 Ammonia Monitor

The controller must be located as shown on the drawings. The ammonia sensor must be mounted as recommended by the manufacturer. The location of the sensor must be representative of the area to be monitored and must allow access to the sensor for periodic calibration. The sensor must be located in a dry area or must be protected from moisture without restricting the flow of ammonia gas to the sensor.

#### 3.3.6.2 Chlorine in Water

The controller must be located in the control panel or other location as shown on the drawings. The chlorine sensor must be immersed in the fluid being monitored using an assembly that will allow removal of the sensor from the water. The sensors must be located in an area of continuous flow.

#### 3.3.6.3 Dissolved Oxygen

The dissolved oxygen sensor must be immersed in the fluid to be monitored using manufacturer's mounting assembly. The sensor must be located in an area of continuous fluid flow. The transmitter must be located remote from the sensor. The transmitter and wiring connections must be in a weathertight enclosure. The transmitter must be mounted to allow the digital readout to be easily viewed.

#### 3.3.6.4 PH and ORP Sensor

Pipe mounted flow sensor must be located in a threaded tee or fitting to allow removal from the pipe. Submersible sensor must be completely immersed in the fluid being monitored using an ensemble that will allow for removal of the sensor from the fluid for replacement. The sensor must be located in an area of continuous flow. The transmitter must be located remote from the sensor. The transmitter must be mounted to allow



the digital readout to be easily viewed.

#### 3.3.6.5 Total Dissolved Solids

The sensor must be pipe type as indicated on the drawings. Pipe mounted sensor must be mounted in a threaded tee or fitting to allow removal of the sensor. The transmitter must be located remote from the sensor. The transmitter and wiring connections must be located in a weathertight enclosure.

#### 3.3.7 Instrument Shelters

Instrument shelters must be installed in the location shown with the bottom 4.0 feet above the supporting surface using legs and secured rigidly to minimize vibrations from winds. Instrument shelters must be oriented with door facing North. Instruments located in shelters must be mounted in the 3-dimensional center of the open space of the shelter.

#### 3.3.8 Output Devices

Output devices (transducers, relays, contactors, or other devices) which are not an integral part of the control panel, must be mounted in an enclosure mounted adjacent to the control panel, unless otherwise shown. Where H-O-A and/or override switches on the drawings or required by the control sequence, the switches must be installed so that the process control system controls the function through the automatic position and other controls work through the hand position.

#### 3.3.9 Enclosures

All enclosure penetrations must be from the bottom of the enclosure, and must be sealed to preclude entry of water using a silicone rubber sealant.

#### 3.3.10 Transformers

Transformers for control voltages below 120 VAC must be fed from the nearest power panel or motor control center, using circuits provided for the purpose. Provide a disconnect switch on the primary side and a fuse on the secondary side. Transformers must be enclosed in a steel cabinet with conduit connections.

### 3.4 WIRE, CABLE AND CONNECTING HARDWARE

#### 3.4.1 LAN Cables and Connecting Hardware

LAN cables and connecting hardware must be installed in accordance with Section 27 10 00 BUILDING TELECOMMUNICATIONS CABLING SYSTEM Modify the existing B72 fiber loop to connect the control system specified herein to the plant network. Provide new ethernet and fiber optic cable per Section 27 10 00.

#### 3.4.2 Metering and Sensor Wiring

Metering and sensor wiring must be installed in accordance with the requirements of ANSI C12.1, NFPA 70,

##### 3.4.2.1 Power Line Surge Protection

Control panels must be protected from power line surges. Protection must

meet the requirements of IEEE C62.41.1 and IEEE C62.41.2. Fuses must not be used for surge protection.

#### 3.4.2.2 Sensor and Control Wiring Surge Protection

Digital and analog inputs must be protected against surges induced on control and sensor wiring. Protect binary and analog outputs against surges induced on control and sensor wiring installed outdoors and as shown. Fuses must not be used for surge protection. Test the inputs and outputs in both the normal and common mode using the following two waveforms: The first waveform must be 10 microseconds by 1000 microseconds with a peak voltage of 1500 volts and a peak current of 60 amperes. The second waveform must be 8 microseconds by 20 microseconds with a peak voltage of 1000 volts and a peak current of 500 amperes. Submit certified test results for surge protection.

### 3.5 SOFTWARE INSTALLATION

Load software required for an operational process control system, including databases (for points specified and shown), operational parameters, and system, command, and application programs. Adjust, tune, debug, and commission all software and parameters for controlled systems to assure proper operation in accordance with the sequences of operation and database tables.

### 3.6 CONTROL DRAWINGS

#### 3.6.1 Control

Control drawings, reproducible, with corresponding CADD files, must be provided for equipment furnished and for interfaces to equipment at each respective equipment location. Condensed operating instructions explaining preventive maintenance procedures, methods of checking the system for normal safe operation and procedures for safely starting and stopping the system manually must be prepared in typed form, reproducible, with corresponding word processor files and posted beside the diagrams. Diagrams and instructions must be submitted prior to posting. The framed instructions must be posted before acceptance testing of the system.

#### 3.6.2 Contractor Design Drawings

Contractor Design Drawings as a single complete package: 2 hard copies and 2 copies in electronic form. As a minimum they must include wiring, logic, and layout. Submit hardcopy drawings on A3 420 by 297 mm 17 by 11 inches sheets, and electronic drawings in PDF and in AutoCAD format. In addition, submit electronic drawings in editable Excel format for all drawings that are tabular, including but not limited to the Point Schedule and Equipment Schedule. Contractor Design Drawings must be approved prior to any fabrication.

##### 3.6.2.1 Draft As-Built

Draft As-Built Drawings as a single complete package: 2 hard copies and 2 copies in electronic form. Submit hardcopy drawings on A3 420 by 297 mm 17 by 11 inches sheets, and electronic drawings in PDF and in AutoCAD format. In addition, submit electronic drawings in editable Excel format for all drawings that are tabular, including but not limited to the Point Schedule and Equipment Schedule.

#### 3.6.2.2 Final As-Built

Final As-Built Drawings as a single complete package: 2 hard copies and 2 copies in electronic form. Submit hardcopy drawings on A3 420 by 297 mm 17 by 11 inches sheets, and electronic drawings in PDF and in AutoCAD format. In addition, submit electronic drawings in editable Excel format for all drawings that are tabular, including but not limited to the Point Schedule and Equipment Schedule.

#### 3.6.3 Points Schedule

Provide a Points Schedule in tabular form for each system, with the indicated columns and with each row representing a hardware point, network point or configuration point in the system.

- a. When a Points Schedule was included in the Contract Drawing package, use the same fields as the Contract Drawing with updated information in addition to the indicated fields.
- b. When Point Schedules are included in the contract package, items requiring contractor verification or input have been shown in angle brackets (" $<$ " and " $>$ "), such as  $< \_\_\_ >$  for a required entry or  $< \text{value} >$  for a value requiring confirmation. Complete all items in brackets as well as any blank cells. Do not modify values which are not in brackets without approval. Points Schedule Columns must include:

##### 3.6.3.1 Point Name

The abbreviated name for the point using the indicated naming convention.

##### 3.6.3.2 Description

A brief functional description of the point such as "Supply Air Temperature".

##### 3.6.3.3 DDC Hardware Identifier

The Unique DDC Hardware Identifier shown on the DDC Hardware Schedule and used across all drawings for the DDC Hardware containing the point.

##### 3.6.3.4 Settings

The value and units of any setpoints, configured setpoints, configuration parameters, and settings related to each point.

##### 3.6.3.5 Range

The range of values, including units, associated with the point, including but not limited to setpoint adjustment range, a sensor measurement range, or the status of a safety.

##### 3.6.3.6 Input or Output (I/O) Type

The type of input or output signal associated with the point. Use the following abbreviations for entries in this column:

- a. AI: The value comes from a hardware (physical) Analog Input
- b. AO: The value is output as a hardware (physical) Analog Output

- c. BI: The value comes from a hardware (physical) Binary Input
- d. BO: The value is output as a hardware (physical) Binary Output
- e. PULSE: The value comes from a hardware (physical) Pulse Accumulator Input
- f. NET-IN: The value is provided from the network (generally from another device). Use this entry only when the value is received from another device as part of scheduling or as part of a sequence of operation, not when the value is received on the network for supervisory functions such as trending, alarming, override or display at a user interface.
- g. NET-OUT: The value is provided to another controller over the network. Use this entry only when the value is transmitted to another device as part of scheduling or as part of a sequence of operation, not when the value is transmitted on the network for supervisory functions such as trending, alarming, override or display at a user interface.

#### 3.6.3.7 Network Data Exchange Information

(Gets Data From, Sends Data To) Provide the DDC Hardware Identifier of other DDC Hardware the point is shared with.

#### 3.6.3.8 Override Information

For each point requiring an Override, indicate if the Object for the point is Commandable.

#### 3.6.3.9 Trend Object Information

For each point requiring a trend, indicate if the trend is Local or Remote. For remote trends provide the DDC Hardware Identifier for the device performing the trend.

#### 3.6.3.10 Alarm Information

Indicate the Alarm Generation Type.

### 3.7 FIELD TESTING AND ADJUSTING EQUIPMENT

Provide personnel, equipment, instrumentation, and supplies necessary to perform site testing. The Government will witness the PVT, and written permission must be obtained from the Government before proceeding with the testing. Original copies of data produced, including results of each test procedure, during PVT must be turned over to the Government at the conclusion of each phase of testing prior to Government approval of the test. The test procedures must cover actual equipment and functions specified for the project.

#### 3.7.1 Testing, Adjusting and Commissioning

After successful completion of the factory test as specified, the Contractor will be authorized to proceed with the installation of the system equipment, hardware, and software. Once the installation has been completed, tested, adjusted, and commissioned each control loop and system

in accordance with NIST SP 250 and must verify proper operation of each item in the sequences of operation, including hardware and software. Calibrate field equipment, including control devices, adjust control parameters and logic (virtual) points including control loop setpoints, gain constants, constraints, and verify data communications before the system is placed online. Test installed ground rods as specified in IEEE 142 and submit certification stating that the test was performed in accordance with IEEE 142. Calibrate each instrumentation device connected to the process control system control network by making a comparison between the reading at the device and the display at the workstation, using a standard at least twice as accurate as the device to be calibrated. Check each control point within the process control system control network by making a comparison between the control command at the central station and field-controlled device. Deliver trend logs/graphs of all points showing to the Government that stable control has been achieved. Points on common systems must be trended simultaneously. One log must be provided showing concurrent samples taken once a minute for a total of 4 hours. One log must be provided showing concurrent samples taken once every 30 minutes, for a total of 24 hours. Verify operation of systems in the specified failure modes upon Process control system network failure or loss of power, and verify that systems return to process control system control automatically upon a resumption of process control system network operation or return of power. Deliver a report describing results of functional tests, diagnostics, calibrations and commissioning procedures including written certification to the Government that the installed complete system has been calibrated, tested, adjusted and commissioned and is ready to begin the PVT. The report must also include a copy of the approved PVT procedure.

### 3.7.2 Performance Verification Test (PVT)

Submit test procedures for the PVT. The test procedure must describe all tests to be performed and other pertinent information such as specialized test equipment required and the length of the PVT. The test procedures must explain, in detail, step-by-step actions and the expected results, to demonstrate compliance with all the requirements of the drawings and this specification. The test procedure must be site specific and based on the inputs and outputs, required calculated points and the sequence of control. Refer to the actions and expected results to demonstrate that the process control system performs in accordance with the sequence of control. Include a list of the equipment to be used during the testing plus manufacturer's name, model number, equipment function, the date of the latest calibration and the results of the latest calibration.

Demonstrate that the completed Process control system complies with the contract requirements. All physical and functional requirements of the project including communication requirements must be demonstrated and shown. Demonstrate that each system operates as required in the sequence of operation. The PVT as specified must not be started until after receipt of written permission by the Government, based on the written report including certification of successful completion of testing, adjusting and commissioning as specified, and upon successful completion of training as specified. Upon successful completion of the PVT, furnish test reports and other documentation.

### 3.7.3 Endurance Test

Use the endurance test to demonstrate the overall system reliability of the completed system. The endurance test must be conducted in phases.

The endurance test must not be started until the Government notifies the Contractor in writing that the PVT is satisfactorily completed, training as specified has been completed, outstanding deficiencies have been satisfactorily corrected, and that the Contractor has permission to start the endurance test. Provide an operator to man the system during Phase I endurance testing, in addition to any Government personnel that may be made available. The Government may terminate testing at any time when the system fails to perform as specified. Upon termination of testing by the Government or by the Contractor, commence an assessment period as described for Phase II. Upon successful completion of the endurance test, deliver test reports and other documentation, as specified, to the Government prior to acceptance of the system.

#### 3.7.3.1 Phase I (Testing)

The test must be conducted 24 hours per day, 7 days per week, for 30 consecutive calendar days, including holidays, and the system must operate as specified. Make no repairs during this phase of testing unless authorized by the Government in writing.

#### 3.7.3.2 Phase II (Assessment)

After the conclusion of Phase I, identify failures, determine causes of failures, repair failures, and deliver a written report to the Government. The report must explain in detail the nature of each failure, corrective action taken, results of tests performed, and must recommend the point at which testing should be resumed. After delivering the written report, convene a test review meeting at the job site to present the results and recommendations to the Government. The meeting must not be scheduled earlier than 5 business days after receipt of the report by the Government. As a part of this test review meeting, demonstrate that failures have been corrected by performing appropriate portions of the performance verification test. The Government reserves the right to cancel the test review meeting if no failures or deficiencies occur during the Phase I testing. If the Government chooses to do so, the Contractor will be notified in writing. Based on the Contractor's report and the test review meeting, the Government will determine if retesting is necessary and the restart point. The Government reserves the right to require that the Phase I test be totally or partially rerun. Do not commence any required retesting until after receipt of written notification by the Government. After the conclusion of any retesting which the Government may require, the Phase II assessment must be repeated as if Phase I had just been completed.

#### 3.7.3.3 Exclusions

The Contractor will not be held responsible for failures resulting from the following: Outage of the main power supply in excess of the capability of any backup power source, provided that the automatic initiation of all backup sources was accomplished and that automatic shutdown and restart of the process control system performed as specified. Failure of a Government furnished communications link, provided that the PLC automatically and correctly operates in the stand-alone mode as specified, and that the failure was not due to Contractor furnished equipment, installation, or software. Failure of existing Government owned equipment, provided that the failure was not due to Contractor furnished equipment, installation, or software.

### 3.8 FIELD TRAINING

Field training oriented to the specific system must be provided for designated personnel. Furnish a copy of the training manual for each trainee plus two additional copies. Manuals must include an agenda, the defined objectives for each lesson, and a detailed description of the subject matter for each lesson. Furnish audiovisual equipment and other training supplies and materials. Copies of the audiovisuals must be delivered with the printed training manuals. The Government reserves the right to videotape training sessions for later use. A training day is defined as 8 hours of classroom instruction, excluding lunchtime, Monday through Friday, during the daytime shift in effect at the training facility. Submit the training manual and schedule to receive approval from the Government at least 30 days before the training.

#### 3.8.1 Preliminary Operator Training

Prior to the start of field testing, preliminary operator training must be taught at the project site for 2 consecutive training days. Upon completion of this course, each student, using appropriate documentation, should be able to perform elementary operations with guidance and describe the general hardware architecture and functionality of the system. This course must include: general system architecture; functional operation of the system, including workstations; operator commands; application programs, control sequences, and control loops; database entry and modification; reports generation; alarm reporting; diagnostics; and historical files.

#### 3.8.2 Additional Operator Training

Following the field testing, additional classroom training for operators must be taught for 2 consecutive training days; individual instruction sessions of 4 -hour periods in the morning (or afternoon) of the same weekday for 2 consecutive weeks and an additional 1 day classroom session for answering operator questions. Individual instruction must consist of "hands-on" training under the constant monitoring of the instructor. Classroom training must include instruction on the specific hardware configuration of the installed process control system and specific instructions for operating the installed system. Schedule activities during this period so that the specified amount of time on the equipment will be available for each student. The final session will address specific topics that the students need to discuss and to answer questions concerning the operation of the system. Upon completion of the course, the students should be fully proficient in system operation and have no unanswered questions regarding operation of the installed process control system. Each student should be able to start the system, operate the system, recover the system after a failure and describe the specific hardware architecture and operation of the system and be fully proficient in all system operations. Report the skill level of each student at the end of this course.

#### 3.8.3 Maintenance Training

Following the endurance test a minimum period of one training days must be provided by a factory representative or a qualified Contractor trainer for five designated personnel on maintenance of the equipment. The training must include: physical layout of each piece of hardware, calibration procedures, preventive maintenance procedures, schedules, troubleshooting, diagnostic procedures and repair instructions.

#### 3.8.4 Specialized Training

Following the maintenance training, a minimum period of five , total training day(s) must be provided by a factory representative or a qualified Contractor trainer for ten people on the input devices.

##### 3.8.4.1 Flow Meter Training

Each type of flow meter, to include calibration, maintenance and testing of flow elements and transducers.

##### 3.8.4.2 Specialized Sensor Training

Provide training on each type of specialized sensor such as chlorine, turbidity, pH, to include calibration, maintenance and testing of sensing elements and transducers for five people.

#### 3.9 OPERATION AND MAINTENANCE DATA REQUIREMENTS

Outline the step-by-step procedures required for system startup, operation and shutdown. Include in the instructions layout, wiring and control diagrams of the system as installed, the manufacturer's name, model number, service manual, parts list and a brief description of all equipment and their basic operating features. List routine maintenance procedures, possible breakdowns and repairs and troubleshooting guides.

-- End of Section --



SECTION 41 22 13.14

BRIDGE CRANES, OVERHEAD ELECTRIC, TOP RUNNING  
**11/19, CHG 1: 02/21**

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

- |                  |                                                                                                                                                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AGMA 908         | (1989B; R 1999) Information Sheet:<br>Geometry Factors for Determining the<br>Pitting Resistance and Bending Strength of<br>Spur, Helical and Herringbone Gear Teeth |
| ANSI/AGMA 2001   | (2004D; R 2010) Fundamental Rating Factors<br>and Calculation Methods for Involute Spur<br>and Helical Gear Teeth                                                    |
| ANSI/AGMA 2015-1 | (2001A; R 2014) Accuracy Classification<br>System - Tangential Measurements for<br>Cylindrical Gears                                                                 |

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

- |          |                                                        |
|----------|--------------------------------------------------------|
| AISC 360 | (2016) Specification for Structural Steel<br>Buildings |
|----------|--------------------------------------------------------|

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

- |             |                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------|
| ASME B30.2  | (2022) Overhead and Gantry Cranes (Top<br>Running Bridge, Single or Multiple Girder,<br>Top Running Trolley Hoist) |
| ASME B30.10 | (2024) Hooks                                                                                                       |

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

- |           |                             |
|-----------|-----------------------------|
| ASSP Z359 | (2013) Fall Protection Code |
|-----------|-----------------------------|

AMERICAN WELDING SOCIETY (AWS)

- |                  |                                                                                                            |
|------------------|------------------------------------------------------------------------------------------------------------|
| AWS D1.1/D1.1M   | (2020; Errata 1 2021) Structural Welding<br>Code - Steel                                                   |
| AWS D14.1/D14.1M | (2019) Specification for Welding of<br>Industrial and Mill Cranes and Other<br>Material Handling Equipment |

ASTM INTERNATIONAL (ASTM)

- |                 |                                                                                 |
|-----------------|---------------------------------------------------------------------------------|
| ASTM A275/A275M | (2023) Standard Practice for Magnetic<br>Particle Examination of Steel Forgings |
|-----------------|---------------------------------------------------------------------------------|

|                   |                                                                                                                                                                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A668/A668M   | (2023) Standard Specification for Steel Forgings, Carbon and Alloy, for General Industrial Use                                                                                                                                                          |
| ASTM A931         | (2008; R 2013) Standard Test Method for Tension Testing of Wire Ropes and Strand                                                                                                                                                                        |
| ASTM A1023/A1023M | (2021) Standard Specification for Stranded Carbon Steel Wire Ropes for General Purposes                                                                                                                                                                 |
| ASTM E543         | (2021) Standard Specification for Agencies Performing Non-Destructive Testing                                                                                                                                                                           |
| ASTM E1417/E1417M | (2016) Standard Practice for Liquid Penetrant Testing                                                                                                                                                                                                   |
| ASTM F3125/F3125M | (2019) Standard Specification for High Strength Structural Bolts and Assemblies, Steel and Alloy Steel, Heat Treated, Inch Dimensions 120 ksi and 150 ksi Minimum Tensile Strength, and Metric Dimensions 830 MPa and 1040 MPa Minimum Tensile Strength |

CRANE MANUFACTURERS ASSOCIATION OF AMERICA (CMAA)

|         |                                                                                                                |
|---------|----------------------------------------------------------------------------------------------------------------|
| CMAA 70 | (2020) Specification for Top Running Bridge and Gantry Type Multiple Girder Electric Overhead Traveling Cranes |
|---------|----------------------------------------------------------------------------------------------------------------|

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

|            |                                                                                                        |
|------------|--------------------------------------------------------------------------------------------------------|
| NEMA 250   | (2020) Enclosures for Electrical Equipment (1000 Volts Maximum)                                        |
| NEMA ICS 2 | (2000; R 2020) Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 V |
| NEMA ICS 5 | (2017) Industrial Control and Systems: Control Circuit and Pilot Devices                               |
| NEMA ICS 6 | (1993; R 2016) Industrial Control and Systems: Enclosures                                              |
| NEMA ICS 8 | (2011) Crane and Hoist Controllers                                                                     |

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|         |                                                        |
|---------|--------------------------------------------------------|
| NFPA 70 | (2023; ERTA 1 2024; TIA 24-1) National Electrical Code |
|---------|--------------------------------------------------------|

RESEARCH COUNCIL ON STRUCTURAL CONNECTIONS (RCSC)

|           |                                                                           |
|-----------|---------------------------------------------------------------------------|
| RCSC A348 | (2020) RCSC Specification for Structural Joints Using High-strength Bolts |
|-----------|---------------------------------------------------------------------------|

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC SP 6/NACE No.3 (2007) Commercial Blast Cleaning

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910 Occupational Safety and Health Standards

29 CFR 1910.147 The Control of Hazardous Energy (Lock Out/Tag Out)

29 CFR 1910.179 Overhead and Gantry Cranes

29 CFR 1910.306 Specific Purpose Equipment and Installations

UL SOLUTIONS (UL)

UL 50 (2024) UL Standard for Safety Enclosures for Electrical Equipment, Non-Environmental Considerations

UL 489 (2025) UL Standard for Safety Molded-Case Circuit Breakers, Molded-Case Switches and Circuit-Breaker Enclosures

UL 943 (2016; Reprint Sep 2023) UL Standard for Safety Ground-Fault Circuit-Interrupters

UL 1004-1 (2012; Reprint Nov 2020) UL Standard for Safety Rotating Electrical Machines - General Requirements

UL 1449 (2021; Reprint Dec 2022) UL Standard for Safety Surge Protective Devices

1.2 DEFINITIONS

- a. Bridge Crane: That part of an overhead crane system consisting of girder(s), end trucks, end ties, walkway, and drive mechanism which carries the trolley(s) and travels along the runway rails parallel to the runway.
- b. Crane Runway: The track system along which the crane operates horizontally, including track hangar rods, track connection devices, and runway structural supports.
- c. Dead Loads: The loads on a structure which remain in a fixed position relative to the structure.
- d. Girder: The principal horizontal beam of the crane bridge. It is supported by the crane end trucks.
- e. Lifted Load: The load consisting of the rated load and the weight of lifting devices attached to the crane such as the load block, bucket, or other supplemental devices.
- f. Pendant: A control for a hoist and a crane. The pendant hangs from the hoist or the crane by a cable at a height that is easy for the

operator to reach.

- g. Rated Load: The maximum working load suspended under the load hook.
- h. Standard Commercial Cataloged Product: A product which is currently being sold, or previously has been sold, in substantial quantities to the general public, industry or Government in the course of normal business operations. Models, samples, prototypes or experimental units do not meet this definition. The term "cataloged" as specified in this section is defined as "appearing" on the manufacturer's published product data sheets. These data sheets must have been published or copyrighted prior to the issue date of this solicitation and have a document identification number or bulletin number.
- i. Top Running Crane: An electric overhead traveling crane that runs on rails on top of support girders.
- j. Trolley Load: Weight of the trolley and its associated equipment carried by the trolley wheels.
- k. Operating Environments:
  - (1) General Purpose Service: This applies to most cranes and are, in large measure, the manufacturers' standard designs. Cranes should be classified as General Purpose Service if they are operating in routine environments.

### 1.3 SYSTEM DESCRIPTION

The overhead crane system shall be a top-running, single-girder bridge crane designed and constructed in accordance with CMAA Class C minimum service requirements, and the hoist shall conform to ASME HST-4 Duty Class H4 minimum. The system shall include a 7.5-ton wire rope hoist with a motor-driven trolley, dual bridge drives, and variable frequency drive (VFD) controls to provide smooth acceleration and deceleration of bridge and trolley motions. The hoist shall provide true vertical lift through a double-reeved configuration and include both electro-mechanical and mechanical load brakes for safe load handling. The hoist headroom shall not exceed 24 inches, measured from the hook saddle to the underside of the bridge girder. All motors shall have a 60-minute duty rating, and the bridge control panel shall include a keyed bypass for the hoist brake circuit. The crane shall be suitable for indoor service in an industrial environment typical of Navy maintenance facilities. Requirements for the crane runway system and supporting structures are specified in Section 05 12 00 - STRUCTURAL STEEL and shall conform to AISC 360.

#### 1.3.1 Crane Design Criteria

Cranes will operate in the given spaces and match the runway dimensions and rails indicated. Hook coverage, hook vertical travel, clear hook height, lifting capacity, and load test weight must not be less than that indicated.

##### 1.3.1.1 Classification

Provide top running overhead electric traveling (OET) single girder cranes conforming to CMAA 70 service class C for operation in an indoor environment, general purposeservice.

The crane must be pendant controlled and radio controlled and operate in the spaces and within the loading conditions indicated. The pendant controller must be mounted on a separate festooned cable system from the trolley power supply. The crane must operate on 480-volts AC, 60 Hz, three phase power source. Maximum crane wheel loads (without impact) due to dead, trolley, and lifted loads, with the trolley in any position, must not cause a more severe loading condition in the runway support structure than that produced by the design wheel loads and spacing indicated.

- a. The 7.5-ton top running crane shall consist of a single girder and a structural runway with a festoon system. The crane shall have a maximized lifting height as shown on the Drawings (minimum of 11' 3/4" from the hook to the top of the curb). Top Running girder and structural runway dimensions shall be based on the limits identified on the Contract Documents.
- b. The crane manufacturer shall be responsible for furnishing and installing the gantry beam, hoist, and runway rail system on top of horizontal structural beams provided by the General Contractor, as shown on the structural drawings, to deliver a complete and operable system. The crane manufacturer shall coordinate with the General Contractor and the structural engineer to confirm final runway elevation, anchorage details, and interface tolerances prior to fabrication and installation. The gantry crane shall be sized to span approximately 12' - 9 3/4", as indicated on the structural plans, and shall provide a minimum lifting height of 15 feet. The crane manufacturer shall ensure the crane rails are installed in accordance with applicable Crane Manufacturers Association of America (CMAA) tolerances. The Contractor shall furnish a complete, coordinated, and fully operable crane system in accordance with the crane manufacturer's requirements for rated load and duty cycle. The Contractor shall also submit a design certification letter, stamped by a licensed structural engineer, verifying that the crane girders, rails, and all related components have been designed, detailed, and installed in accordance with CMAA requirements, applicable building codes and standards, and are capable of safely supporting the crane's rated capacity under all operating conditions.
- c. Hoist to include 7.5-ton capacity R&M wire rope hoist, two speed (19/3.2 FPM) motor.
- d. Trolley Drive to provide approximately 65 FPM travel speed via VFD motor.
- e. Bridge drive to provide approximately 100 FPM travel speed via VFD motor; controls to be push-button, festooned on a separate track.
- f. Deflection of the rail shall be limited to a maximum of L/600 (vertical) and L/400 (horizontal) of the span.
- g. All crane supporting members, connections, and attachment details shall be fabricated of structural steel, plate or forgings only. No castings will be permitted. System shall be suspended using components supplied by the crane manufacturer or approved equal.
- h. The rated load of the crane shall be the maximum load for which the crane is designed and built by the manufacturer. In determining the rated load, all handling devices such as buckets, magnets, grabs, etc., shall be included as part of the load to be handled.

#### 1.3.1.2 Rated Capacity and Speeds

Provide crane with rated capacity of 7.5 tons. Lower load block or assembly of hook, swivel bearing sheaves, pins, and frame suspended by the hoisting ropes are not considered part of the rated capacity.

Rated (maximum) speeds plus or minus 10 percent (feet/min) for the main hoist, bridge, and trolley at the rated load are specified in the table below. The minimum speed must not exceed the values listed. Values in the table are for a fully loaded crane. Using overspeed, the hoist function must be capable of approximately double when not loaded.

| Rated Speeds<br>feet/min |         |         |
|--------------------------|---------|---------|
| Description              | Minimum | Maximum |
| Main Hoist               | 10      | 20      |
| Auxiliary Hoist          | NA      | NA      |
| Trolley                  | 15      | 65      |
| Bridge                   | 20      | 100     |

#### 1.4 VERIFICATION OF DIMENSIONS

The Contractor is responsible for the coordination and proper relation of their work to the building structure and to the work of all trades. Verify all dimensions of the building that relate to fabrication of the crane and notify the Contracting Officer of any discrepancy before finalizing the crane order.

#### 1.5 SUBMITTALS

Government approval is required for submittals with a "G" classification.

Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00

SUBMITTAL PROCEDURES. Submittals shall include complete data demonstrating conformance with this specification and all referenced standards. Drawings and data shall clearly indicate equipment arrangement, performance ratings, electrical characteristics, and all interfaces with structural and electrical systems. Shop drawings shall show the runway beams, rail alignment, gantry assembly, and all connections to the supporting structure furnished by others. Submittals shall provide sufficient detail to verify compliance with CMAA and NAVFAC requirements.

##### SD-02 Shop Drawings

Overhead Electric Traveling Crane; G

Complete Schematic Wiring Diagram; G

Control System and Network Drawings; G

Runway Beams and Gantry Assembly (rails, beam connections, and end truck interface details only; supporting columns by others); G

##### SD-03 Product Data

Gear Reducers; G

Hoist Brakes; G

Travel Brakes; G

Couplings; G

Load Blocks and Hooks; G

Wheels; G  
Hoists; G  
Sheaves; G  
Commercial Hoist and Trolley Units; G  
End Trucks; G  
Bridge Rails; G  
End Stops; G  
Bumpers; G  
Variable Frequency Drives; G  
Motors; G  
Runway Conductor System; G  
Bridge Conductor System; G  
Limit Switches; G  
Radio Control System; G  
Pendant Pushbutton Station; G  
Pendant Conductor System; G  
Controls; G  
Control Parameter Settings; G  
Capacity Overload Protective Device; G  
Painting System; G  
Control System and Network; G

SD-05 Design Data

Load and Sizing Calculations; G

SD-06 Test Reports

Hook Proof Test; G  
Hook Non-destructive Test (NDT); G  
Post-erection Inspection; G  
Operational Tests; G  
Hook Tram Measurement; G



Load Tests; G

Crane Rail Survey

#### SD-07 Certificates

Wire Rope including Breaking Strength Test Report; G

Crane Runway System; G

Hazardous Material; G

Loss of Power Test; G

Coupling Alignment Verification Record; G

Overload Test; G

Brake Adjustment Record; G

Compliance with Listed Standards; G

Public Domain Software; G

Software and Services; G

#### SD-10 Operation and Maintenance Data

Operation and Maintenance Manuals; G

#### SD-11 Closeout Submittals

Disabled Ports, Connectors, and Interfaces; G

Network-Capable Control Devices; G

Control System Access Control; G

Control System Account Management; G

Patch Management and Updates; G

Malware Detection and Protection; G

Wireless Technology Provisions; G

Control System Inventory; G

Evaluation Status of Hardware and Software; G

### 1.6 QUALITY ASSURANCE

#### 1.6.1 Manufacturer Qualification

Overhead Electric Traveling Crane must be designed and manufactured by a company with a minimum of 10 years of specialized experience in designing and manufacturing the type of overhead crane required to meet requirements of the Contract Documents. Crane design must be accomplished by, or directly supervised by, a registered professional engineer (PE). PE

licensing must be by a board or agency authorized to license and register professional engineers. The PE may be a Contractor's regular employee or a consultant. The PE's review and attestation of specification compliance and professional responsibility must be signified by his or her PE original seal and dated signature on the final drawings. The professional engineers must only undertake and perform work under this contract in the branch(s) of engineering in which they are licensed.

#### 1.6.2 Pre-Delivery Inspections

Contractor is responsible for performance of quality control inspections, testing, and documentation. Submit all crane test data recorded on appropriate test record forms suitable for retention for the life of the crane.

##### 1.6.2.1 Inspection of Steel Castings

Visually inspect load-carrying steel castings. All load bearing components, couplings, shafts, and gears, in the hoist drive train must be rolled or forged steel, except brake drums which may be ductile iron. Methods of repairing the discontinuities is subject to review by the Contracting Officer.

##### 1.6.2.2 Inspection of Hook Assembly

Inspect hook by a magnetic-particle type inspection prior to delivery. Furnish documentation of hook inspection to Contracting Officer prior to field operational testing. As part of the acceptance standard, linear indications are not allowed. Welding repairs of hook are not permitted. A hook showing linear indications, damage or deformation is not acceptable and must be replaced immediately.

##### 1.6.2.2.1 Hook Non-Destructive Test (NDT)

Magnetic-particle inspect the hook over the entire area in accordance with ASTM A275/A275M. Acceptance standard is no defects. A defect is defined as a linear indication that is greater than 1/8 inch long. For hooks of non-magnetic material, NDT will be liquid penetrant (PT) method in accordance with ASTM E1417/E1417M. For PT testing of hooks containing stainless steels, titanium, or nickel based alloys, total halogens and Sulphur used in the NDT process must be controlled as specified in T9074-AS-GIB-010/271.

Inspect each hook and shank over the entire surface area by magnetic particle inspection.

- a. Procedure: Conduct magnetic particle inspection in accordance with ASTM A275/A275M with the following restrictions: Do not use DC yokes (including switchable AC/DC yokes used in the DC mode) or permanent magnet yokes. Do not use automatic powder blowers or any other form of forced air other than from a hand-held bulb for the application or removal of dry magnetic particles. Remove arc strikes. Equipment ammeters must have an accuracy of plus or minus 5 percent of full scale (equipment ammeter accuracy other than that stated is acceptable provided the MT procedure states that a magnetic field indicator is used to establish and verify adequate field strength for all aspects of the inspection.)
- b. Acceptance Criteria: Defects found on the hook will result in

rejection of defective items for use on furnished hoist. For this inspection, a defect is defined as a linear indication for which the largest dimension is greater than 1/8 inch.

- c. Test Report: Submit a test report of the magnetic particle inspection of each hook provided the Contracting Officer for approval prior to final acceptance of hoist installation. Certify test reports by the testing organization. The performing organization must provide a written statement of certification to ASTM E543, current within one year of the date the NDT was performed. The NDT procedures including technique sheets specific to the types, shapes, and size of the parts being examined must adequately describe the orientation of the hooks within the magnetizing equipment. The performing organization must have the NDT procedures and its technique sheet used for testing of the hook reviewed and approved by an independent Level III examiner. Submit the (Level III examiner) approved procedures, technique sheet, and certification to the Contracting Officer with the test report.

#### 1.6.3 Certificates

All certifications must be dated and bear the original signature (above the printed name) of the authorized representative of the Contractor or the manufacturer of the items or equipment being certified. Each certification will clearly identify the crane, the drives, components, and location (as applicable) to which it applies:

- a. Submit a Wire Rope Certification with the wire rope manufacturer's certification that the rope meets the published breaking strength or the actual breaking strength of a sample taken from the reel and tested. Certification is to be traceable to the hoist, crane, and reel.
- b. Submit a Crane Runway System Certificate stating that the new crane will operate properly on the runway; if the crane(s) cannot operate without restriction, the Contractor must indicate crane limitations.
- c. Submit a Hazardous Material Certificate that the crane does not contain hazardous material including asbestos, lead, cadmium, chromium, PCBs, or elemental mercury. Products required for the designing and manufacturing of cranes must not contain the prohibited materials.
- d. Submit a Loss of Power Test Certificate stating that a test may be performed in which power is removed from the crane while the hoist, bridge, and trolley are in operation.
- e. Submit a Certificate of the Coupling Alignment Verification Record.
- f. Submit an Overload Test Certificate stating that the crane can be periodically load tested to 125 percent (plus 0 minus 5 percent) of rated load.
- g. Submit a Certificate of the Brake Adjustment Record. Provide a brake adjustment record and installation/maintenance manuals for each brake on the crane. Each brake measurement must have a tolerance traceable to the associated brake manual or documentation provided by the brake manufacturer, location of measurements, and the actual brake setting. Changes made to settings of the brake, at any time, will void the record.

- h. Submit a Certificate of Compliance with Listed Standards
  - i. Provide a Contractor Hazardous Environment Certificate stating that the new crane and all associated components excluding the hoist are designed for operation in the hazardous environment specified in the Classification section.
  - j. The Contractor must provide a Public Domain Software Certificate declaring that public domain software (e.g., freeware, shareware) is not used in the system.
  - k. The Contractor must provide a certificate stating that all Software and Services that are not required for operation and/or maintenance of the product have been removed. The software/services to be removed are identified in paragraph SOFTWARE AND SERVICES.
- 1.6.4 Drawings: Overhead Electric Traveling Crane

- a. Submit drawings showing the general arrangement of all components in plan, elevation, and end views. Show all major features of the crane including: hook approaches on all four sides, clearances and principal dimensions, assemblies of hoist, trolley and bridge drives, motor nameplate data, overcurrent protective device ratings, and electrical schematic drawings. Include weights and centers of gravity of major components. Provide maximum wheel loads (without impact) and spacing imparted to the crane runway system track beams. Indicate the crane speeds along the runway, the trolley speeds along the bridge girder, and the hoist lifting speeds; all speeds indicated are speeds with hoist loaded with rated crane capacity load.
- b. Submit shop drawings of all fabricated components. Shop drawing quality must be equivalent to the contract drawings accompanying this solicitation. Drawings must be reviewed, signed, and sealed by a registered professional engineer.
- c. Provide integral schedule of crane components on each drawing. Drawings should also include fastener torques (installation/mounting fasteners) and specific lubrication (Greases/oils) as applicable. The schedule must provide a cross reference between manufacturer data and shop drawings. Components listed on the schedule of crane components must include total quantity, description, original manufacturer, and part number. Distributing agents will not be acceptable in lieu of the original manufacturer.
- d. Provide control system and network drawings. Network diagram must show equipment locations, names, models, and IP addresses on network communications schematic for all Programmable Logic Controllers (PLCs), Remote Terminal Unit (RTU), Supervisory Controller, and Other Network-Capable Devices. In addition, the drawings must consist of all software block, flow, and ladder diagrams.

1.6.5 Design Data: Load and Sizing Calculations

Submit complete list of equipment and materials, including manufacturer's descriptive data and technical literature, performance charts and curves, catalog cuts, and installation instructions. Submit calculations reviewed, signed, and sealed by a registered professional engineer verifying the load cases, sizing of the bridge girders, end trucks, travel

drives, brake selections, and overcurrent protection for motors, controllers, and branch circuits. Provide a list of all codes and standards, design assumptions, equations, specified efficiencies, limits, factors of safety, component ratings, and sources of values used. Include free body diagrams or sketches of each load case.

#### 1.6.6 Welding Qualifications and Procedures

Welding must be in accordance with qualified procedures using AWS D14.1/D14.1M as modified. Written welding procedures must specify the Contractor's standard dimensional tolerances for deviation from camber and sweep and not exceed those specified in AWS D14.1/D14.1M and CMAA 70. Welders and welding operators must be qualified in accordance with AWS D1.1/D1.1M or AWS D14.1/D14.1M.

#### 1.7 CRANE SAFETY

Comply with the mandatory and advisory safety requirements of ASME B30.10, ASME B30.2, CMAA 70, 29 CFR 1910.147, 29 CFR 1910.179, 29 CFR 1910.306, and all applicable provisions of 29 CFR 1910 and NFPA 70. Where personal fall arrest anchorages are provided, design anchorages in accordance with ASSP Z359.

### PART 2 PRODUCTS

#### 2.1 MATERIALS

##### 2.1.1 General

Provide materials and equipment which are standard products of manufacturers regularly engaged in the fabrication of complete and totally functional cranes including necessary ancillary equipment. Material will be free from defects and imperfections that might affect the serviceability and appearance of the finished product. All material must be new and unused.

##### 2.1.2 Nameplates

Secure nameplates to each major component of equipment with the manufacturer's name, address, type or style, model or catalog number, and serial number. Provide two bridge identification plates, one for each side of the bridge. Provide noncorrosive metal identification plates with letters "BC-1" such that they are easily read from the floor.

##### 2.1.3 Capacity Marking

Mark the rated capacity in tons on each side of the crane on the bridge girders and both sides of the hook block. Capacity marks must be large enough to be clearly visible from the floor. The markings must be positioned to be visible at the operator's position after the crane has been installed.

##### 2.1.4 Safety Warnings

Affix labels in a readable position to each lift block or control station in accordance with ASME B30.2. Submit safety warnings, diagrams and other instructions suitably framed and protected for display as indicated by the Contracting Officer as follows:

Design and locate the word "WARNING" or other legend to bring the label to the attention of the operator. Provide durable type warning labels and display the following information concerning safe-operating procedures: Cautionary language against lifting more than the rated load; operating the hoist when the hook is not centered under the hoist; operating hoist with twisted, kinked or damaged rope; operating damaged or malfunctioning hoist; operating a rope hoist with a rope that is not properly seated in its hoist drum groove; lifting people; lifting loads over people; and removing or obscuring the warning label.

## 2.2 STRUCTURAL REQUIREMENTS

Structural requirements must be in accordance with CMAA 70, Section 3. Structural steel materials must conform to the standards permitted in CMAA 70 and AISC 360.

### 2.2.1 Structural Connections

High-strength bolted structural connections must be designed and installed in accordance with RCSC A348. Bolts must be of ASTM F3125/F3125M Grade A325/A325M or Grade A490/A490M material. Galvanized bolts are not acceptable.

Welded connections must be performed in accordance with AWS D14.1/D14.1M. Allowable stress values must comply with CMAA 70.

### 2.2.2 Bridge Girder or Girders

Provide wide flange beam, standard I-Beam, or section fabricated from rolled plates and shapes bridge girders. If the ends of bridge girders are notched to fit over the end trucks, the notches must be reinforced with vertical diaphragms and horizontal stiffeners.

### 2.2.3 Bridge Rails

Provide bridge rails, crane girders and other sections that are straight and true. Make all rail joints flush and true without misalignment of running tread and design to minimize vibration. The gap between adjacent rail ends and the vertical misalignment of running treads must not exceed 1/32 inch. Solid stock (e.g. square bar, roundstock) is not permitted as bridge rail. Center bridge rail on top flange or position bridge rail over girder web for torsion box girders. Fasten rail to girder with welded clips. Position rail clips in pairs and at not more than 36 inches on center. Bolt bridge rail joints using standard joint bars. Stagger and position rail joints directly over girder diaphragms. Provide a positive stop at bridge rail ends to prevent creep.

### 2.2.4 End Ties and Bridge Girder End Connections

If equalizing end trucks are used, provide rigid end ties between girders to form a frame that is rigid about the vertical and horizontal axes. If compensating end trucks are used, provide end ties which are rigid about the vertical axis but relatively flexible about the horizontal axis to permit partial rocking motion for wheel load compensation. Provide full depth diaphragms at girder connections and jacking points. Provide horizontal gusset plates at the elevation of top and bottom end tie flanges for connection to girder ends. Make end connections with high-strength bolts in accordance with the Structural Connections section of this specification. Use tapered alignment pins to maintain original

shop alignment between bridge girders and end ties/trucks.

#### 2.2.5 End Trucks

Provide fixed axle type end trucks fabricated from structural steel plate to provide a rigid box section structure. Center wheels between the webs of the box section. Configure bridge and trolley trucks with a feature that limits load movement to 1 inch in the event of wheel or shaft failure. Provide jacking pads for removal of wheel assemblies. Wheel axle bearing seats must be designed such that wheel and axle bearing assembly can be removed with not more than 3 inches of jacking. Provide tapered alignment pins between bridge girders and end truck flanges to ensure accurate field alignment of components, in accordance with NAVFAC 11450.2A paragraph 2-1.3.1.4(c).

#### 2.2.6 Trolley Frame

Provide trolley frame as a one-piece structural steel weldment. Provide pads for the use of jacks or wedges when changing truck wheels. Make all trolley yokes and load bars of drop forged, cast or rolled steel.

#### 2.2.7 End Stops and Bumpers

Fit bridge girders with structural steel end stops. Locate stops to permit maximum trolley travel. Fit bridge end trucks and trolley frames with shock-absorbing bumpers capable of decelerating and stopping the bridge and trolley within the limits stated by 29 CFR 1910 and CMAA 70. Bumpers must fully engage end stops. Mount bumpers so that there is no direct shear on mounting bolts (if any) upon impact. Bumpers must provide adequate clearance between the crane and surrounding structure when compressed to preclude damaging equipment. Ensure bridge and trolley bumper retention in accordance with ASME B30.2. When more than one crane is located and operated on the same runway, bumpers must be provided on their adjacent ends or on one end of one crane. Fit the other end of the end-truck with a structural steel stop to engage the bumpers of the adjacent crane. Ensure bridge bumpers are properly aligned with runway end stops.

### 2.3 MECHANICAL REQUIREMENTS

- a. Provide steel shafts, gears, keys, and couplings. Cast iron and aluminum used to support components of the hoist power transmission train must be ductile.
- b. All bearings, except those subject only to small rocker motion, must be anti-friction type. All connections subject only to small rocking motion are to be fitted with bushings or thrust washers in the pivot pin bore, as applicable. Bronze bushings must have provisions for grease lubrication.
- c. All mechanical components must be accurately aligned and positively secured to maintain the alignment. Parts must not be forced into position to obtain apparent alignment.

#### 2.3.1 Hoists

Provide hoist conforming to ASME B30.2 and CMAA 70 service class C or better, double reeved, except as modified and supplemented in this section. Hoist shall conform to ASME HST-4 Duty Class H4 minimum.

## 2.3.2 Drives

### 2.3.2.1 Bridge Drives

Provide A-4 bridge drive arrangement as specified in CMAA 70 consisting of a single electric motor mechanically connected through gear reduction and drive shafts to the drive wheels or separate drive motors at each end of bridge.

Acceleration and deceleration must meet the requirements specified in this section. Gears must conform to applicable AGMA standards. Provide oil tight fully enclosed gear reducers with pressure or splash type lubrication. Bridge travel limit switches are optional.

### 2.3.2.2 Trolley Drives

Provide complete trolley drive arrangement with a minimum of two wheels driven by an integral electric motor. Drive mechanism must run in totally enclosed oil bath. Limit switches are optional for drive mechanism. Acceleration and deceleration controls must meet requirements specified in this section.

### 2.3.3 Load Blocks and Hooks

The load block must be forged carbon steel-unpainted/unplated hook-minimum throat dimensions of 2.5 inches and shall be designed to prevent steel-to-steel contact of moving parts. . The block must be fully enclosed, concealing the sheaves and wire ropes, except for wire rope slots and drain holes. The block must be clearly marked with the capacity in pounds on both sides. Standard commercial blocks may be used at their published ratings when their published design factors are 5.0 or greater.

Provide an unpainted single barbed forged steel hook complying with ASTM A668/A668M. Fit hook with a safety latch designed to preclude inadvertent displacement of slings from the hook saddle. The hook and hook nut must be removable without unreeving of the hoist. Provide hook nut with a removable type set screw or other similar fastener, installed in a plane parallel to the longitudinal axis of the hook shank. Do not weld hook nut. Uniquely mark the hook in a permanent fashion that is traceable to the NDT certification. The hook nut must be of non-sparking materials.

Hook must be free to rotate through 360 degrees when supporting the test load up to 125 percent of the rated capacity. Provide only hooks which are designed and commercially rated in accordance with CMAA and conforming to ASME B30.10 and CMAA 70.

### 2.3.4 Wire Ropes

- a. Wire ropes must conform to ASTM A1023/A1023M and be tested as required by ASTM A931. The wire rope must be in a double reeved configuration and equalized with a sheave. Provide wire rope with a minimum design factor of 5 to 1 based on the load experienced at rated capacity and minimum breaking strength of the wire rope.
- b. Provide hoisting ropes with improved plow steel, extra improved plow steel, or extra-extra improved plow steel, regular lay, bright, and uncoated with an independent wire rope, wire strand, or otherwise,



steel core. Hot-dipped galvanized wire rope is not permitted.

- c. Maximum hoisting rope fleet angles must be 4 degrees for drums and 4.75 degrees for sheaves.

For hoists with external wire rope terminations, rope end connections (other than drum connections) must be speltered sockets with forged steel terminals or swaged fittings installed to provide 100 percent of the wire rope's breaking strength.

For hoists utilizing a true vertical lift configuration with equalizer sheaves, rope dead ends are located at the drum only, and this requirement is not applicable.

Provide proof of wire rope Breaking Strength Test Report. Wedge sockets or aluminum swages are not permitted on wire rope end connections.

#### 2.3.5 Sheaves

Provide steel sheaves. Minimum pitch diameters must be not less than 18 times the rope diameter for running sheaves, and not less than 12 times the rope diameter for equalizer sheaves. Sheave surfaces that contact wire rope must not be painted. Sheaves must be heat-treated to a minimum hardness of 320 Brinell Hardness Number (BHN) in the wire rope contact area, have a groove depth not less than 1.5 times the hoisting rope diameter, and a throat angle of 30 to 40 degrees.

#### 2.3.6 Hoist Drum

Provide drum made of steel. Design the drum such that all hoisting rope is wound in a single layer and so that not less than two dead wraps of hoisting rope remain on each anchorage when the hook is in its extreme low position. Drum grooving must be machined right and left hand beginning at the ends and grooving toward the center of the drum. Minimum drum groove depth must be 0.375 times the rope diameter. Minimum drum groove pitch must be either 1.14 times the rope diameter or the rope diameter plus 1/8 inch, whichever is smaller. Do not paint, coat, or galvanize the surface of the drum which comes in contact with wire rope.

For wire rope drums installed directly onto the output shaft of the hoist speed reducer without an intermediate flexible coupling, the drum to shaft connection must be a barrel coupling.

#### 2.3.7 Gearing

Provide gearing of the enclosed gear reducers type. Provide steel spur, helical, or herringbone type gears and pinions only. Gearing must conform to ANSI/AGMA 2001 and AGMA 908. Internal and external gear dimensional tolerances must conform to the applicable AGMA standard for tooth geometry and tolerances. Gearing within hoist, trolley, and bridge reducers shall meet these requirements. Open-type gearing is not acceptable, except for final drives.

##### 2.3.7.1 Gear Reducers

Gear reducers must be integral components of the hoist, trolley, and bridge drive assemblies furnished with the crane package and shall be the standard products of manufacturers regularly engaged in the design and manufacture of gear reducers for CMAA Class C cranes. Gear reducers must

be designed, manufactured, and rated in accordance with ANSI/AGMA 6113 (ANSI/AGMA 6013 for hoist and bridge drives), fully enclosed and oil-tight except for final reduction, and operate in an oil bath with smooth, quiet performance. Reducers shall have a minimum service factor of 1.25 for bridge and trolley drives and 1.5 for hoist drives, with bearings providing a minimum L10 life of 10,000 hours at rated load and duty. Enclosed gearing shall be selected for Industrial Duty and designed in accordance with CMAA 70 and NAVFAC 11450.2A requirements.

#### 2.3.7.2 Open Gearing

Provide all gears and pinions with adequate strength and durability for the crane service class and manufactured to ANSI/AGMA 2015-1 Accuracy Grade A8 or better. Open gears must be enclosed with safety guards provided with openings with covers for inspection and access for grease lubrication.

#### 2.3.8 Wheels

Provide double flanged, straight tread trolley and bridge travel wheels of rolled-to-shape or roll-forged steel. The rim, flanges, and wheel tread must be hardened to not less than 320 Brinell Hardness Number (BHN). Wheel sizing and flange-to-rail head clearances must be in accordance with CMAA 70 recommendations.

#### 2.3.9 Bridge and Trolley Travel Brakes

Provide bridge and trolley drives with electro-mechanical brakes or non-freecoasting mechanical drive capable of stopping the motion of the bridge or trolley within a distance in feet equal to 10 percent of the full load speed in feet per minute when traveling at full speed with a full load. Mechanical bridge or trolley brakes are not required when the VFD drive provides dynamic braking and controlled deceleration under all operating conditions. Brakes must have an externally accessible means to manually release the brake.

#### 2.3.10 Hoist Brakes

- a. Equip hoist with a minimum of two electro-mechanical holding brakes, each with a minimum torque rating of 125 percent of the rated load hoisting torque. Designate one brake as the primary brake (mounted closest to the motor) and one as the secondary brake.
- b. The primary brake shall be non-time-delayed and shall set immediately upon removal of power to the hoist motor.  
The secondary brake shall have an adjustable setting time delay of 1 to 3 seconds after the primary brake sets.  
Do not use an uninterruptible power supply (UPS) or other stored-energy device to create the time delay.
- c. Each brake shall be adjustable to 50 percent of its rated capacity, have externally accessible manual release means, and be designed for inspection and adjustment without disassembly.  
Brakes shall have manual or automatic return-to-ON feature upon release. Torque adjustments shall not be required where brakes are used solely as holding brakes.
- d. Brakes shall be electrically and mechanically interlocked so that loss of power to one brake does not prevent operation of the other.

Brake components shall comply with NAVFAC 11450.2A paragraphs 2-4.2.1 and 2-4.2.2.

- e. Provide hoist load-limit switch factory-set to actuate at 100 percent of rated load per ASME B30.2 § 3-1.3.

#### 2.3.11 Couplings

Provide flexible, grid, or gear-type couplings between motor and gearbox shafts, and between gearbox and drum shafts, where applicable. Couplings shall be designed for continuous duty and capable of transmitting 150 percent of rated motor torque. Chain and continuous sleeve type couplings are not permitted. Spline couplings are acceptable when provided as part of C-face or D-face motor assemblies. Conventional couplings must not be loaded in the radial direction. Brake wheel or brake disc couplings (if used) must be compatible with the required coupling type. Flexible couplings must not be relied upon to compensate for poor alignment. Ends of coupled shafts must be aligned within the manufacturer's recommended tolerances. Provide coupling guards in accordance with OSHA 29 CFR 1910.219.

#### 2.3.12 Drip Pans

- a. The crane must be designed to preclude leakage of lubricants onto the lifted loads or the floor. Equipment or components, which cannot be made leak-proof, must be fitted with unpainted corrosion resistant steel drip pans or must have the foundations seal welded to create a dam. Drip pans that utilize liquid sealant to prevent leakage of lubricants are not permitted.
- b. The drip pans must be sized to hold the entire gear case fluid capacity, installed under all drive machinery, designed to permit easy removal of collected lubricant. A trolley floor designed to contain any lubricant drips may be used as fluid containment for any equipment that is mounted on it.

#### 2.3.13 Shims

Provide shims in accordance with NAVCRANECENINST 11450.2A § 2-2.15.3. Shims shall be corrosion-resistant for all indoor and outdoor applications. Shim stacks shall not exceed four shims, and commercially available shim kits shall be used to ensure consistent, pre-cut shims sized for standard fasteners. All new equipment shall be installed with shims under each foot, with a minimum shim thickness of 0.100 inch. Soft foot shall be limited to 0.001 inch and shall not exceed 0.003 inch after installation.

### 2.4 ELECTRICAL REQUIREMENTS

#### 2.4.1 Motors

Motors must meet all applicable requirements of NEMA MG 1 and UL 1004-1. All motors must have a minimum of a 60-minute duty rating and be Totally Enclosed Non-Ventilated (TENV), Totally Enclosed Fan Cooled (TEFC), or Totally Enclosed Blower Cooled (TEBC). Provide inverter-duty motors for open-loop variable frequency drives (VFDs). Provide vector-duty motors for closed-loop VFDs. Provide single-speed AC squirrel-cage induction type motors for the bridge and trolley drives. Provide vector-duty AC squirrel-cage induction type motor with encoder for the

hoist. Provide motors with a minimum of Class F insulation and Class B temperature rise at rated load. Provide motor overload protection utilizing thermal sensors embedded in the windings and include space/anti-condensation heaters where applicable. Coordinate motor ratings with the drive manufacturer to ensure compatibility with the specified VFD system.

#### 2.4.2 Controls

- a. Provide static reversing, variable frequency drives (VFD) for the bridge and trolley electric controls. Provide static reversing, VFD, speed regulated, closed loop, flux vector electric controls for the hoist. For feedback, provide hoist motors with encoders. The hoist controller must enable the drive motor to develop full torque continuously at zero speed. The hoist secondary brake must be controlled separate from the primary and connected to different output (within the drive) from the primary brake. VFD controllers must meet NEMA ICS 8, Part 8 and at a minimum, provide under-voltage protection, electronic instantaneous over current protection, DC bus over voltage protection, and be able to withstand output line to line shorts without component failure. Select bridge and trolley drives such that the continuous rating of the controller is not less than the calculated motor full load current based on CMAA 70 5.2.9.1.1.1 and NEC Table 430.250. Select hoist drives such that the continuous rating of the controller is not less than 130 percent of the calculated motor full load current based on CMAA 70 5.2.9.1.1.1 and NEC Table 430.250. All hoist drives must have a motor over-torque limit to lock out the hoist and prevent gross overload of the associated hoist. Provide dynamic braking for each electric drive that is sized per VFD manufacturer's requirements. Submit VFD Control Parameter Settings. Crane controls to be designed such that operator will be safely away from lifting load. Emergency safety stops to be provided locally at the crane motor and via remote control.
- b. Push button controls and cable assembly shall be suspended from an independent festoon track. The pushbutton station shall have a NEMA4 rating with the following features: bright yellow polypropylene enclosure for easy visibility, 8-button enclosure, momentary on / off, six each 2-speed buttons, double break silver-plated switch contacts, multi-grip cord grip accepts eight through 24 conductor round cable, strain relief cable and fittings, easy to read legend plates, replaceable neoprene boots, neoprene brushing, and warning tag.
- c. Push button festooning shall carry control wiring from the independently suspended push button station to the control panel. Push button festooning shall be a rugged steel C track system with trolleys. NEMA12 junction box, flat cable, and all other components required to control the 2-step bridge control panel, the 2-speed hoist, and the 2-stop trolley.
- d. The hoist, trolley, and bridge brakes must set after the associated controller decelerates the drive motor to a controlled stop. The hoist, trolley, and bridge controllers must be sized to provide sufficient starting torque to initiate motion of that crane drive mechanism from standstill with 0 to 125 percent of rated load on the hook. The hoist controller must prove torque before release of the brakes and enable the drive motor to develop full torque continuously at zero speed. Motors must operate smoothly at all speeds without torque pulsations, and must only be energized within the frequency

range of 50-60 Hz at rated speed. Radio controls shall have safety features that include receiver self-diagnosing, transmitter self-diagnosing, transmitter low voltage detection and warning, transmitter and receiver MAIN deactivation when the transmitter is not in use, during transmitter low voltage, and when the system is in sleep mode. The encoder / decoder system shall utilize advanced microprocessor control. If the receiver MAIN relay is defective (fails to open or close during operation), the safety relay shall signal the system to shut-down immediately to avoid the possibility of any accidents occurring. Synthesized PLL receiving module shall accept signals from 30 available channels. Transmitter and receiver enclosures shall be fully waterproofed and also be impervious to dust, water, oil, grease, acid, alkaline, heat and sunlight.

- e. Provide two-speed magnetic controls for the bridge drive, trolley drive, and hoist drive. Controllers must meet the requirements of NEMA ICS 8. Ensure that an energized drive motor initially rotates only in the direction selected by the operator by activating the corresponding direction; i.e., is not overhauled. For AC squirrel cage motor controllers, the requirements of NEMA ICS 2, Part 2, for general-purpose controllers, must be met. Remote controller unit shall operate via rechargeable battery. Equipment supplier to provide the following:
  - (1) Battery backups to cover power requirements for three full shifts of operation.
  - (2) Battery charger unit and / or cords.
- f. Provide the bridge and trolley motor control systems with a drift point between OFF and the first speed control point in each direction. Crane mounted controls shall be enclosed in NEMA 4 enclosures, pre-wired and tested prior to shipment, unless otherwise noted.
- g. The use of definite purpose contactors is prohibited. If IEC contactors are used, the application cannot exceed the contactor manufacturer's AC3 ratings for the contactor at a minimum. Provide electrical transformer, sized to system, to step down voltages for the required controls and accessories.
- h. On hoist function roll-up must be less than 1/8 inch measured at the hook block and roll-back must not occur over the entire load range. Disconnect and power feeds shall be in accordance with Division 26 Electrical and so shown on the Electrical Drawings.
- i. Use of Uninterruptable Power Supplies (UPS) is prohibited. Feed control circuits from a single phase, air cooled, double wound transformer with a grounded metal screen between the primary and secondary windings of the transformer. Crane manufacturer shall provide final power and control wiring diagrams, load data, and conduit requirements to the electrical contractor.
- j. Provide a main line contactor. Energization of the main line contactor must be controlled by the POWER-OFF/POWER-ON switch/pushbutton on all controllers. Upon actuation of the POWER-OFF pushbutton; power to all drive motors, brakes, and controls must be removed. The mainline contactor must not be able to be energize while the POWER-OFF pushbutton is actuated. The POWER-OFF pushbutton circuitry must be independent of all controls or any other electronic

devices.

- k. Provide anti-condensation heaters in all control panels and electrical enclosures to prevent moisture accumulation and corrosion resulting from humidity or temperature fluctuations.

#### 2.4.3 Protection

Protection must not be less than that required by NEMA ICS 8, CMAA 70, NFPA 70, UL 1004-1, UL 1449, UL 489, UL 50, UL 943, 29 CFR 1910.147, 29 CFR 1910.179, 29 CFR 1910.306 and all applicable provisions of 29 CFR 1910. Provide an On/Off button that removes power from the motors, brakes and control circuit on all operator control stations. Provide for lockout/tagout of all hazardous energy sources. The main crane disconnect will be furnished and installed under Division 26 work.

#### 2.4.4 Resistors

Provide resistors with natural convection cooling sized as recommended by the VFD OEM and fabricated of corrosion resistant metal; the use of "wire wound" type resistors is prohibited for segments of 8 ohms or less. Mount resistors in substantial, ventilated enclosures constructed entirely of non-combustible materials. Provide resistors with terminals fitted in the coolest position in the enclosure.

#### 2.4.5 Transients and Harmonics Protection

- a. Provide contactors and relays with appropriate Metal Oxide Varistors (MOV) or resistor-capacitor (R-C) surge absorbers installed across the respective coil.
- b. Provide transient protection for electronic drive controllers that is either internal to the drive or via an MOV connected line-to-ground close to the line terminals of the drive.
- c. Provide line reactors rated for continuous duty operation based upon the motor nameplate amperes. With motors of 50 horsepower or greater, harmonics protection must be provided by an isolations transformer or as recommended by the VFD OEM. For a drive motor branch circuit that exceeds 150 feet in length, a reactor must also be connected in series with the controller load (output) terminals to provide standing wave protection or as otherwise recommended by the VFD or motor OEM.

#### 2.4.6 Limit Switches

- a. Provide primary upper and lower geared limit switches. Geared limits must allow reversing direction to back out of the limit without resetting. The lower limit switch must be set such that there are a minimum of two wraps of rope on the hoist drum.
- b. Provide a backup mechanical hook block activated upper limit switch wired independent of the directional controllers and the primary upper limit switch that removes power from the hoist motor, hoist brake and hoist controls conforming to NEMA ICS 5. The backup limit must require hoist resetting prior to operation of the hoist in any direction.
- c. Travel limit switches must be provided to slow the crane to 25 percent of its rated speed 1 foot before the bridge end stops and 5 feet from

the trolley end stops. Limit switches must be mounted rigidly in a manner so as to protect the switch from misalignment or damage. The target/trip arm must be large enough to provide interception given a misalignment were to occur.

#### 2.4.7 Operator Controls

- a. Provide crane equipped with a pendant pushbutton station and radio control system.
- b. Provide a selector switch to allow the use of only one of the two available control stations on the pendant controller.

##### 2.4.7.1 Pendant Pushbutton Station

The cranes must be controlled from a pendant pushbutton station suspended from an independent festooned messenger track system, operating the length of the bridge. Provide multiconductor flexible cords for pendant pushbutton stations with No. 16 AWG minimum conductors. Provide a method of strain relief to protect the electrical conductors from damage. Locate the pendant pushbutton station 4 feet above the finished floor. Pushbutton pendant station must have its elements legibly marked and arranged vertically, in order, in accordance with CMAA 70. Provide two speed pendant pushbuttons for control of the hoist bridge and trolley. Provide pendant pushbuttons for control that spring return to the OFF position. Voltage in the pendant pushbutton station must not exceed 150 Volts AC or 300 Volts DC. Provide a maintained two-position selector switch for slow speed selection.

##### 2.4.7.1.1 Pendant Conductor System

Provide a festoon type pendant conductor system. The festoon cables must be flat cables suspended from carriers riding on an I-beam or C-track. The pendant controller must be capable of traveling the entire length of the bridge and move independently of the trolley. Festoon loops must not extend below the high hook position.

##### 2.4.7.2 Radio Control System

Provide each system with a handheld type portable transmitter unit. Only one transmitter at a time can control the crane and there must be no interference from one crane's controller affecting operation of the other cranes in the building. Each transmitter must include: individual push button controls for each hoist, trolley, and bridge; a maintained contact, keyed switch, marked ON-OFF, for portable transmitter unit power; indication of Battery Power, and indication of Transmitting Status; a red emergency STOP mushroom pushbutton. The transmitters and all controls must each be clearly and permanently labeled with functionality and direction. Directions for controllers must be in accordance with CMAA 70 recommendations. The remote radio control system must be designed to meet the requirements of NEMA ICS 8, Part 9 and ECMA 15. Each radio remote control lever must be in the OFF position before the associated crane function can begin. The system frequency must be within the unlicensed FCC Part 15 range. Each control unit must maintain a continuous status signal to the associated receiver during operation. There must be no significant loss in systems efficiency and function at the end of eight hours of continuous battery use.

## 2.4.8 Electrification Systems

### 2.4.8.1 Runway Conductor System

- a. Provide a Festoon System for the runway conductor system utilizing cables suspended from carriers riding on an I-beam or C-track for the crane, including all necessary cables and hardware to the crane from a wall or column mounted disconnect switch. Provide electrification system with three power conductors and an equipment grounding conductor. Conductors must be fabricated from copper. The crane is required to be grounded through this conductor system. Provide conductors sized for simultaneous motions of the hoist, bridge, trolley mechanisms and any ancillary loads. Festooned cable loops must not extend low enough to come into contact with any obstructions.

### 2.4.8.2 Bridge Conductor System

- a. Provide Festoon System for the bridge conductor system utilizing cables suspended from carriers riding on an I-beam or C-track. Conductors must be fabricated from copper. A minimum of 20 percent of the festoon control circuit conductors for each electrification system must be spares at the time of crane acceptance. The trolley is required to be grounded through this conductor system. Festooned cable loops must not extend low enough to come into contact with any obstructions.

### 2.4.9 Capacity Overload Protective Device

- a. Provide a capacity overload protective device for all hoist systems using VFD drive capacity overload protection (separate from torque limiting feature of the VFD) . Set hoist capacity overload protection at 110 percent. Provide a keyed override or other means to disable the hoist capacity overload protection when performing a load test.
- b. Initially, set the torque limiting capability of the VFD (that is separate from the capacity overload protective device) to 150 percent of the motor torque (amperage) necessary to hoist 100 percent load. It may be adjusted up only to avoid nuisance trips and adjusted down if possible while still avoiding nuisance trips.

### 2.4.10 Enclosures

- a. Provide enclosures for control panels, controls, and brakes in accordance with NEMA 250 and NEMA ICS 6, Classification Type 1 indoor, general purpose12 indoor without knockouts, general purpose. There must not be any condensation inside the control panels. Provide anti-condensation heaters that remain energized when the main line contractor is energized.
- b. Provide a non-resettable hour meter, connected across the main line contactor, readable from the exterior of the main control panel, to indicate the elapsed number of hours the crane is energized.
- c. Gaskets of enclosures and fixtures, and joints and contact surfaces of hazardous/explosive enclosures must be kept free of any paint to prevent damage during removal and reinstallation of gaskets of enclosures.



#### 2.4.11 Warning Devices

Provide a warning horn that is operable from a push button at the radio control station. Provide a warning rotating beacon that is illuminated at all times during movement of the hoist, trolley, or bridge function.

#### 2.4.12 Indicator Lights

Provide fault light, slow speed light, and hoist brake fault, as applicable. Provide Indicator Lights mounted in an enclosure on the bottom of the bridge with lights sized and positioned to be visible from the ground. The lights must be the dual-lamp type. Provide a white light to indicate that power is available on the load side of the crane disconnect and a blue light to indicate that the main contactor is energized. Light voltage must be 115 VAC. Provide nameplates that are legible from ground level. The nameplates must read, in their respective order, "POWER AVAILABLE" and "CRANE ENERGIZED". The POWER AVAILABLE light must be supplied by a separate, fused transformer for its energization.

#### 2.4.13 Cyber Security of Control Systems

- a. Provide the following for PLC, RTU, Supervisory Controller, or other network-capable (whether networked or not upon delivery) control devices as applicable:
  - (1) Hardware list (Hardware list must include the following for each device):
    - (a) Manufacturer
    - (b) Model
    - (c) Location
    - (d) Key technical ratings (e.g. memory)
    - (e) Serial number
    - (f) MAC addresses
    - (g) IP addresses
  - (2) Software List (Software list must include the following for each device):
    - (a) Manufacturer
    - (b) Version/subversion
    - (c) Location/device
    - (d) Used network ports/protocols/services
  - (3) List and discussion of all security features of Contractor hardware and software.
- b. For every PLC, RTU, Supervisory Controller, or other network-capable control devices (whether networked or not upon delivery), deliver the following on CD/DVD:
  - (1) Original firmware
  - (2) Original firmware hash
  - (3) SOP for application of firmware updates/patches
  - (4) POC or website for firmware updates/patches
  - (5) Count of interfaces and types
  - (6) Protocols in use, per interface
  - (7) Configuration file
  - (8) SOP for configuration

#### 2.4.13.1 Control System and Network

- a. Provide a rugged laptop type workstation (computer) complete with all compatible software (including software licenses), redundant physical back-up copies on CD/DVD of the installed software, and all necessary cables and special connectors to allow crane software to be troubleshot, checked and upgraded, and for the data recorder to be accessed and information retrieved. Equip the workstation with a CD/DVD drive and the associated CD/DVD burning software. The workstation must also be equipped with USB ports (2.0 and 3.0), an Ethernet port, and a serial port. Delivering the software on a USB (flash drive) device is prohibited.
- b. The laptops must be designed for an industrial environment and must be shock resistant and weatherproof as a minimum. Provide the laptop with a built-in CD/DVD reader with the capability to burn CDs and DVDs including associated software to burn CDs and DVDs.
- c. The Contractor must provide all equipment, including software and hardware, necessary for testing, installation, and communicating/troubleshooting all systems provided with the crane (e.g. engine/generator, control system, LID, etc.). The Contractor must provide all crane specific operational software files (e.g. ladder logic, functional block programming, etc.) for their associated systems (e.g. control systems, LID, engine generator, etc.).
- d. A single common networked design must not be used for the control systems. A network for an individual function may be used as long as a failure of the network does not affect any other function/network except as defined for specific safety interlocks (e.g. LMI system). A common crane network may be used in a monitoring mode for recording faults and trending and is encouraged. Failure of the monitoring system must not affect crane functions.
- e. All provided hardware and software must be currently marketed products, not currently scheduled for end of life or obsolescence, to ensure system sustainability.
- f. Ensure there is no remote access capability enabled as remote access capabilities are prohibited. Physically disable or remove all modem/network devices not required for operational purposes.

#### 2.4.13.2 Software and Services

- a. Remove all Software and Services not required for operation and/or maintenance of the product. If removal is not technically feasible, then disable software not required for the operation and/or maintenance of the product. Configure the product to allow the ability to re-enable ports and/or services if they are disabled by software. The removal of software or services may not impede the primary function of the product. If software that is not required cannot be removed or disabled, document a specific explanation and provide risk mitigating recommendations and/or specific technical justification. The software/service to be removed and/or disabled includes, but is not limited to:
  - (1) Cameras
  - (2) Games

- (3) Device drivers for product components not procured/delivered
  - (4) Messaging services (e.g., email, instant messenger, peer-to-peer file sharing)
  - (5) Source code
  - (6) Software compilers in user workstations and servers
  - (7) Software compilers for programming languages that are not used in the control system
  - (8) Unused networking and communications protocols
  - (9) Unused administrative utilities, diagnostics, network management, and system management functions
  - (10) Backups of files, databases, and programs used only during system development
  - (11) All unused data and configuration files
  - (12) Remove and/or disable, through software, physical disconnection, or engineered barriers, all services and/or ports in the procured product not required for normal operation, emergency operations, or troubleshooting. This includes communication ports and physical input/output ports (e.g., USB docking ports, video ports, and serial ports).
- b. Provide documentation showing all disabled ports, connectors, and interfaces for all network-capable devices. In addition, the documentation must provide summary documentation of the procured product's security features and security-focused instructions on product maintenance, support, and reconfiguration of default settings.
  - c. For the evaluation status of hardware and software, the Contractor must provide information on Common Criteria or National Information Assurance Partnership (NIAP) or Federal Information Processing Standards (FIPS) evaluation status of hardware and software.

#### 2.4.13.3 Access Control

- a. The Contractor must configure each component of the procured product to operate using the principle of least privilege. This includes operating system permissions, file access, user accounts, application-to-application communications, and energy delivery system services.
- b. Provide user accounts with configurable access and permissions associated with one or more organizationally defined user role(s), where roles are used.
- c. Provide a system administration mechanism for changing user(s') role (e.g., group) associations.
- d. The Contractor must document control system access control options by defining access and security permissions, user accounts, and applications with associated roles.

- e. Provide recommended methods for the Acquirer to prevent unauthorized changes to the Basic Input/Output System (BIOS) and other firmware. If it is not technically feasible to protect the BIOS to reduce the risk of unauthorized changes, the Contractor must document this case and provide mitigation recommendations.

#### 2.4.13.4 Control System Account Management

The Contractor must document all accounts (including, but not limited to, generic and/or default) that need to be active for proper operation of the procured product.

Remove or disable any accounts that are not needed for normal or maintenance operations, emergency, or troubleshooting of the energy delivery system.

#### 2.4.13.5 Session Management

The Contractor may not allow multiple concurrent logins using the same authentication credentials, allow applications to retain login information between sessions, provide any auto-fill functionality during login, or allow anonymous logins.

Provide account-based and group-based configurable session-based logout and timeout settings (e.g., alarms and human-machine interfaces).

#### 2.4.13.6 Authentication/Password Policy and Management

Provide a configurable account password management system that allows for, but is not limited to, the following:

- a. Changes to passwords (including default passwords)
- b. Selection of password length
- c. Frequency of change
- d. Setting of required password complexity
- e. Number of login attempts prior to lockout
- f. Inactive session logout
- g. Screen lock by application
- h. Comparison to a library of forbidden strings
- i. Derivative use of the user name
- j. Denial of repeated or recycled use of the same password

The Contractor must time stamp log files.

#### 2.4.13.7 Logging and Auditing

Provide logging capabilities that cover the following events, at a minimum (as appropriate to their function):

- a. Information requests and server responses
- b. Successful and unsuccessful authentication and access attempts
- c. Account changes
- d. Privileged use
- e. Application start-up and shutdown
- f. Application failures
- g. Major application configuration changes

#### 2.4.13.8 Heartbeat Signals

The Contractor must identify heartbeat signals or protocols and recommend which should be included in network monitoring. At a minimum, a last gasp report from a dying component or equivalent must be included in network monitoring.

The Supplier must provide packet definitions of the heartbeat signals and examples of the heartbeat traffic if the signals are included in network monitoring.

#### 2.4.13.9 Patch Management and Updates

The Contractor must verify that procured products (including third-party hardware, software, firmware, and services) have appropriate updates and patches installed prior to delivery.

Provide documentation of the patch management program and update process (including third-party hardware, software, and firmware). This documentation must include resources and technical capabilities to sustain this program and process. Provide the Contractor's method or a recommendation for how the integrity of the patch is validated by the Acquirer as well as the Supplier's approach and capability to remediate newly reported zero-day vulnerabilities.

#### 2.4.13.10 Malware Detection and Protection

- a. The Contractor is required to implement at least one of the following:
  - (1) Provide a host-based malware detection capability that quarantines (instead of automatically deleting) suspected infected files. Provide an updating scheme for malware signatures. The Contractor must test and confirm compatibility of malware detection application patches and upgrades.
  - (2) If the Contractor is not providing the host-based malware detection capability, the Contractor must suggest malware detection products to be used and provide guidance on malware detection and configuration settings that will work with Contractor products.
- b. The Contractor must validate that cybersecurity services running on the procured product (e.g., virus checking and malware detection) do not conflict with other such services running on the procured product.

- c. For malware detection and protection, the Contractor must provide, or specify how to implement, the capability to automatically scan any removable media that is introduced to the product being acquired.

#### 2.4.13.11 Physical Security

Provide lockable or locking enclosures or rooms for energy delivery systems and system components (e.g., servers, clients, and networking hardware) and for the systems used to manage and control physical access (e.g., servers, lock controllers, and alarm control panels). Provide a method for tamper detection on lockable or locking enclosures. If a physical security and monitoring system is used, tamper detection must be compatible. The Contractor must ensure that physical security features do not hamper the crane system operations. Provide the tools and instructions for making changes to locks, locking codes, keycards, and any other keyed entrances.

#### 2.4.13.12 Wireless Technology

For wireless technology provisions, the Contractor must document:

- a. Specific protocols and other detailed information required for wireless devices to communicate with the control network, including other wireless equipment that can communicate with the Contractor-supplied devices.
- b. Use, capabilities, and limits for the wireless devices.
- c. Power and frequency requirements of the wireless devices (e.g., microwave devices meet the frequency requirements of Generic Requirements (GR)-63 Network Equipment Building System (NEBS) and GR-1089).
- d. Range of the wireless devices and verify that the range of communications is minimized to both meet the needs of the Acquirer's proposed deployment and reduce the possibility of signal interception from outside the designated security perimeter.
- e. Wireless technology and associated devices compliance with standard operational and security requirements specified in applicable wireless standard(s) or specification(s) (e.g., applicable IEEE standards, such as 802.11).
- f. Configuration control options that enable varying of the security level of the devices.

#### 2.4.13.13 Control System Inventory

Provide the complete control system inventory. The Control System Inventory must include the following attributes, in tabular format, as applicable:

| General Information        | Location Information                   | Hardware Details                                     | Operating System and Platform | Network Information (Actual Function, not potential function) |
|----------------------------|----------------------------------------|------------------------------------------------------|-------------------------------|---------------------------------------------------------------|
| Unique ID                  | Facility Name                          | Device Type                                          | Embedded OS (Yes/No)          | MAC Address(es)                                               |
| Barcode or Identifier      | NFAID                                  | Device Sub-Type                                      | OS Contractor                 | IP Address(es)                                                |
| Region                     | Commodity                              | Device Function                                      | Operating System (O/S)        | Upstream Device                                               |
| Installation               | Floor                                  | Manufacturer                                         | O/S Version                   | Protocols In Use                                              |
| Special Area (Option DNA1) | Room                                   | Product Line                                         | Platform Contractor           | Host Name                                                     |
|                            | Location                               | Model No.                                            | Platform Product Line         |                                                               |
|                            | System Type                            | Serial No.                                           | Platform                      |                                                               |
|                            | Functional System or Equipment Control | Remote Connectivity: (Wired / Wireless / None)       | Platform Version              |                                                               |
|                            |                                        | Network Type Used: (Serial / Ethernet / Both / None) |                               |                                                               |

## 2.5 PAINTING SYSTEM

- a. Remove all grease, oil, and surface debris by solvent wiping or detergent/water scrubbing, prior to blast cleaning. Prepare surfaces to be coated by abrasive blasting to SSPC SP 6/NACE No.3, Commercial Blast Cleaning, or in accordance with the coating manufacturer's requirements, whichever is more stringent.
- b. Use a painting system appropriate for the conditions provided in the Crane Design Criteria section. Paint exposed portions of the crane using a three-coat system as follows: zinc-rich primer consisting of a minimum of 85 percent zinc by weight in the dry film, an anticorrosive epoxy intermediate coat, and an aliphatic polyurethane top coat . All paint products must be supplied by a single manufacturer and free of chromates, lead, and mercury. Apply each coat in accordance with manufacturer's instructions and requirements. Ensure each coat is smooth, even, and free of runs, sags, orange peel, and other defects. Desired color of finish coat is brilliant yellow. Submit product data for painting system.
- c. Coat faying surfaces of bolted connections per RCSC A348, but do not apply finish paint.
- d. Paint the load block brilliant yellow with black diagonal striping. Paint, coatings, or galvanizing on the following items or areas is not

acceptable: hoist wire ropes, hooks, hook nuts, running bearing surfaces (including sheaves and wheel treads), grease fittings, or other items not normally painted.

- e. Factory paint electrical and mechanical equipment in accordance with the manufacturer's best standard practice (for the specified environment), except that electrical equipment doors, which expose current-carrying electrical conductors when opened, must be orange.

## 2.6 IDENTIFICATION PLATES

Furnish and install identification plates. Provide non-corrosive metal identification plates with clearly legible permanent lettering giving the manufacturer's name, model number, serial number, capacity in pound units printed in different colors, and other essential information or identification.

### 2.6.1 Markings on Crane, Trolley, and Hook

To avoid operation of the crane in the wrong direction, affix the appropriate directions (NORTH, SOUTH, EAST, and WEST) with arrows on the bottom of the bridge girder where they can be easily seen by the operator and from the loading point. Provide labels on the controls with corresponding directional (NORTH, SOUTH, EAST, and WEST) markings. Markings must agree with the markings on controller. Do not indicate directional arrows on controller.

Mark the hook rated capacity in pounds on both sides of the hoist load block.

## PART 3 EXECUTION

### 3.1 EXAMINATION

After becoming familiar with all details of the work, and before performing any work, verify all dimensions in the field. The Contractor is responsible for the coordination and proper relation of the contracted work to the building structure and to the work of all trades. Verify all dimensions of the building that relate to fabrication of the crane and notify the Contracting Officer of any discrepancy before finalizing the crane order.

### 3.2 SHOP ASSEMBLY AND TESTS

Shop assemble major components as completely as possible, except for reeving of drums and sheaves. Functionally test the crane system at the construction facility prior to shipment. The Government reserves the right to inspect the crane for compliance with this specification and to witness the functionality tests. Notify the Contracting Officer 14 days prior to starting testing operations.

### 3.3 ERECTION AND INSTALLATION

Perform the entire crane erection in accordance with manufacturer's instructions under the full-time supervision of the manufacturer's representative.



### 3.3.1 Mechanical Alignment

Align motors, couplings, brakes, gear boxes and drive components in accordance with manufacturer's instructions. Complete the Coupling Alignment Verification Record.

### 3.3.2 Electrical Adjustments

Adjust control system in accordance with manufacturer's instructions. Store a copy of all Control Parameter Settings (PLC, VFD). Provide the final alignment data on the Complete Schematic Wiring Diagram, including but not limited to, timer settings, resistor tap settings, potentiometer settings, test-point voltages, supply voltages, motor voltages, motor currents. Provide the test conditions such as ambient temperature, motor load, date performed and person performing the alignment as part of the Operational Tests report.

### 3.3.3 Field Welding

Perform welding indoors, where possible. Surface of parts to be welded must be free from rust, scale, paint, grease or other foreign matter. Minimum preheat and interpass temperatures must conform to the requirements of AWS D14.1/D14.1M. Perform welding of girders and beams conforming to AWS D14.1/D14.1M.

### 3.3.4 Field Painting

Perform painting indoors, where possible. Field painting (including touch-up) must conform to the requirements of the coating manufacturer and as specified in PAINTING SYSTEM.

## 3.4 FIELD QUALITY CONTROL

### 3.4.1 Post-Erection Inspection

After erection, the Contractor, the Activity Crane Inspector/Test Director, and the Contracting Officer must jointly inspect the crane bridge and hoist systems and components to determine compliance with specifications and approved submittals. Notify the Contracting Officer 14 days before the inspection. Provide a report of the inspection indicating the crane is considered ready for operational tests.

### 3.4.2 Operational Tests

Check the clearance envelope of the entire crane prior to picking or traversing any load to ensure there are no obstructions. Test the systems in service to determine that each component of the system operates as specified, is properly installed and adjusted, and is free from defects in material, manufacture, installation, and workmanship. Rectify all deficiencies disclosed by testing and retest the system or component to prove the crane is operational.

#### 3.4.2.1 No-Load Test

Raise and lower each hook through the full range of normal travel at rated speed for three complete cycles. Raise and lower each hook, testing other speeds of the crane. Verify proper operation of hoist limit switches. Operate the bridge and trolley in each direction the full distance between end stops. Operate through the entire speed range and verify proper brake

operation. Verify correct operation of all indication and ancillary devices.

#### 3.4.3 Test Data

Record test data on appropriate test record forms suitable for retention for the life of the crane. Record operating and startup current measurements for hoist, trolley, and bridge motors using appropriate instrumentation (i.e., clamp-on ammeters). Compare recorded values with design specifications or manufacturer's recommended values; abnormal differences (i.e., greater than 10 percent from manufacturer's or design values) must be justified or appropriate adjustments performed. In addition, note, investigate, and correct any high temperatures or abnormal operation of any equipment or machinery. Record hoist, trolley, and bridge speeds during each test cycle.

#### 3.4.4 Hook Tram Measurement

Establish a throat dimension base measurement by installing two tram points and measuring the distance between these tram points (plus or minus 1/64 inch). Record this base dimension. Measure the distance between tram points before and after load test. An increase in the throat opening from the base measurement is cause for rejection.

#### 3.4.5 Load Tests

Perform the following tests for each hoist, as specified below.

Test loads used in this section are defined as the following:

Wire rope run-in load: 25 - 50 percent of rated load.

Rated load test: 100 percent (plus 0 minus 10) of rated load.

Overload test: 125 percent (plus 0 minus 5) of rated load.

##### 3.4.5.1 Wire Rope Run-In

The primary purpose of this procedure is to exercise the newly installed wire rope.

Place the load on the hook. Start at ground level and hoist up to one foot below upper limit at slow speed. Hoist down to lower limit at slow speed. Repeat hoisting and lowering of the load for approximately 10 hoisting cycles, increasing the speed for each cycle. During this test, the capacity overload lockout should not activate.

##### 3.4.5.2 Rated Load Test

###### 3.4.5.2.1 Hoist

- a. Static Load Test: With the trolley in the center of the bridge span, raise the test load approximately one foot. Hold the load for 10 minutes. Rotate the load and hook 360 degrees to check bearing operation with no binding. Observe lowering that may occur which indicates a weakness in the structure or malfunction of hoisting components or brakes. Verify that maximum beam and girder deflections do not exceed CMAA 70 design limits.

- b. For hoists with primary and secondary holding brakes, raise the test load and release the secondary holding brake while testing the primary holding brake. Hold for 10 minutes. Observe for lowering of the load, which may indicate malfunction of hoisting components or brakes. Re-engage secondary holding brake and release the primary holding brake. Hold for 10 minutes. Observe for lowering of the load, which may indicate malfunction of hoisting components or brakes. Re-engage the primary holding brake. Recheck proper operation of time delay and ensure smooth positive stopping.
- c. Hoist Mechanical Load Brake (if present): Raise test load approximately 5 feet. With the hoist controller in the neutral position, release (by hand) the holding brake. Document the method used to release the holding brake. The load brake must hold the test load. Again with the holding brake in the released position start the test load down at slow speed and return the controller to the "off" position as the test load lowers. The load brake must stop and hold the test load.
- d. Raise and lower test load through the full lift range and visually observe smooth control and acceleration between points. Completely stop the machinery at least once in each direction to ensure proper brake operation.
- e. Hoist Loss of Power Test: Raise the test load to approximately 8 feet. While slowly lowering the test load, disconnect the crane's power source. Verify that the test load does not lower and that the brake is set.

#### 3.4.5.2.2 Trolley

Operate the trolley (if space is available) the full distance of the bridge rails in each direction with a test load on the hook. Check proper functioning through the range of speeds. Verify proper brake action.

#### 3.4.5.2.3 Bridge

With a test load on the hook, operate the bridge for the full length of the runway (if space is available) in one direction with the trolley at the far end of the bridge, and in the opposite direction with the trolley at the opposite end of the bridge. Use extreme caution. Check proper functioning through the range of speeds. Check for any binding of the bridge end trucks and verify proper brake action. Record deficiencies. Secure from testing if deficiencies are found.

#### 3.4.5.2.4 Trolley Loss of Power Test

With a test load of 100 percent of rated load, raise the test load approximately midway between the trolley and any permanent obstruction on the operating floor. Starting at a safe distance from walls or other obstructions, attain a slow speed of trolley travel. While maintaining a safe distance from obstructions, disconnect the main power source at the wall mounted safety switch (disconnect) to simulate a power failure. Verify that the trolley stops and that the brake sets properly. Measure the distance required for the trolley to stop.

#### 3.4.5.2.5 Bridge Loss of Power Test

With a test load of 100 percent of rated load, raise the test load

approximately midway between the trolley and any permanent obstruction on the operating floor. Starting at a safe distance from walls or other obstructions, attain a slow speed of bridge travel. While maintaining a safe distance from obstructions, disconnect the main power source at the wall mounted safety switch (disconnect) to simulate a power failure. Verify that the bridge stops and that the brake sets properly. Measure the distance required for the bridge to stop.

#### 3.4.5.3 Overload Test

##### 3.4.5.3.1 Hoist

Disconnect or adjust the overload limit device to allow the hoist to lift the test load. Verify proper operation of the overload limit device after it is reconnected.

- a. Static Load Test: With the trolley in the center of the bridge span, raise the test load approximately one foot. Hold the load for 10 minutes. Rotate the load and hook 360 degrees to check bearing operation with no binding. Observe lowering that may occur which indicates a weakness in the structure or malfunction of hoisting components or brakes.

For hoists with primary and secondary holding brakes, raise the test load and release the secondary holding brake while testing the primary holding brake. Hold for 10 minutes. Observe for lowering of the load, which may indicate malfunction of hoisting components or brakes. Re-engage secondary holding brake and release the primary holding brake. Hold for 10 minutes. Observe for lowering of the load, which may indicate malfunction of hoisting components or brakes. Re-engage the primary holding brake. Recheck proper operation of time delay and ensure smooth positive stopping.

- b. Raise and lower test load and visually observe smooth control. Stop the load during raising and lowering to verify that the brakes holds the load.
- c. Hoist Load Brake (if present): Raise test load approximately 5 feet. With the hoist controller in the neutral position, release (by hand) the holding brake. Document the method used to release the holding brake. The load brake must hold the test load. Again with the holding brake in the released position start the test load down at slow speed and return the controller to the "off" position as the test load lowers. The load brake must stop and hold the test load.
- d. Hoist Loss of Power Test: Raise the test load to approximately 8 feet. While slowly lowering the test load, disconnect the crane's power source. Verify that the test load does not lower and that the brake is set.

##### 3.4.5.3.2 Trolley

Operate the trolley the full distance of the bridge rails in each direction with a test load on the hook (one cycle) through the range of speeds. Verify proper brake action.

##### 3.4.5.3.3 Bridge

With a test load on the hook, operate the bridge for the full length of

the runway in one direction with the trolley at the extreme end of the bridge, and in the opposite direction with the trolley at the opposite extreme end of the bridge (one cycle). Check proper functioning through the range of speeds. Check for any binding of the bridge end trucks and verify proper brake action. Record deficiencies. Secure from testing if deficiencies are found.

### 3.5 MANUFACTURER'S FIELD SERVICE REPRESENTATIVE

Furnish a qualified experienced manufacturer's field service representative to supervise the crane installation, assist in the performance of the on-site testing, and instruct personnel in the operational and maintenance features of the equipment.

### 3.6 OPERATION AND MAINTENANCE MANUALS

Provide two hard copies of operation and two hard copies of maintenance manuals for the equipment furnished along with an electronic copy (PDF) of each on a Compact Disc. Provide one complete set prior to performance testing and final copies upon acceptance. Provide operation manuals that detail the step-by-step procedures required for system startup, operation and shutdown. Include the manufacturer's name, model number, parts list, and brief description of all equipment and basic operating features. List in the maintenance manuals routine maintenance procedures, including weekly, monthly, semi-annual, and annual required maintenance items, possible breakdowns and repairs, and troubleshooting guides. Also include as-built drawings, piping and equipment layout, design calculations, Control Parameter Settings and printouts of any software, and simplified wiring and control diagrams of the system as installed. Secure approval of operation and maintenance manuals prior to the field training course.

### 3.7 FIELD TRAINING

Conduct a training course for 8 operating and maintenance staff and provide a copy of the training material to each participant. Provide a training period consisting of a total of hours of normal working time and starting after the system is functionally completed but prior to final acceptance. Cover all pertinent points involved in operating, starting, stopping, and servicing the equipment, including all major elements of the Operation and Maintenance Manuals. Demonstrate in course instructions all routine maintenance operations such as lubrication, general inspection, and critical maintenance.

### 3.8 FINAL ACCEPTANCE

Final acceptance of crane system will not be given until Contractor has successfully completed all testing operations, corrected all material and equipment defects, made all proper operation adjustments, and removed paint or overspray on wire rope, hook, and electrical collector bars.

-- End of Section --

SECTION 43 21 13.16

HORIZONTAL END SUCTION PUMPS  
09/24

PART 1 GENERAL

1.1 WORK INCLUDED

This Section covers the furnishing and installation of pump motor and variable frequency motor drive as required for complete operation of the pumps.

1.2 RELATED WORK

- A. Equipment Checkout and Testing is covered by Section 01 75 13 .
- B. Finish paint: all items of equipment such as motors, pumps, instrumentation panels, electrical switchgear, and similar items, that have been given shop coats, paint filler, enamel or other treatment customary with the manufacturer, shall have, after installation, all scratches and blemishes touch up prior to application of the first field coat. Factory prefinished items not to be field painted shall be touched up with matching paint to repair any areas damaged during installation. Shop coats shall be fully compatible with the finish coat.
- C. Power to all equipment specified in this section shall be coordinated with Division 26 ELECTRICAL . Coordinate the requirements as outlined in Section 26 29 23 ADJUSTABLE SPEED DRIVES (ASD) SYSTEMS UNDER 600 VOLTS and the requirements listed in sections 2.02 and 2.03 below.
- D. Discharge piping and valves for all pumping equipment shall be as specified in Section 40 05 13 PIPELINES, LIQUID PROCESS PIPING.

1.3 DESIGN CRITERIA

- A. Adequate tolerances in the listed capacity, head and efficiency values have been included. No deviation below the listed parameters will be permitted.
- B. All listed parameters shall be considered design points.
- C. End Suction Pump Schedule

| Pump Tag               | Pump                          | Description                                                      | Max.<br>Capacity<br>(gpm) | TDH<br>(PSI) | Motor<br>Horsepower | Voltage | Starter<br>Type |
|------------------------|-------------------------------|------------------------------------------------------------------|---------------------------|--------------|---------------------|---------|-----------------|
| 04-01-001<br>04-01-002 | P4A and P4B<br>(Duty/Standby) | Transfer flow<br>UF break tank<br>to SWRO<br>Cartridge<br>Filter | 625                       | 50           | 30                  | 480     | VFD             |

| Pump Tag  | Pump                                | Description                                                | Max.<br>Capacity<br>(gpm) | TDH<br>(PSI) | Motor<br>Horsepower | Voltage | Starter<br>Type |
|-----------|-------------------------------------|------------------------------------------------------------|---------------------------|--------------|---------------------|---------|-----------------|
| 03-01-004 | SWRO High Pressure Pump             | From Cartridge Filter to SWRO                              | 685                       | 2320         | 350                 | 480     | VFD             |
| 04-01-003 | BWRO High Pressure Pump             | From SWRO to BWRO                                          | 450                       | 220          | 100                 | 480     | VFD             |
|           | ABW Residuals Pumps                 | Transfer ABW from SWRO to outfall                          | 280                       | 17.3         | 5                   | 480     | VFD             |
|           | Drain Pumps (Duty/Standby)          | Drainage backwash waste                                    | 558                       | 30           | 3                   | 480     | VFD             |
| 04-01-007 | SWRO CIP Pump                       | Transfer from UF CIP Tank to SWRO (via camlock connection) | 300                       | 26           | 40                  | 480     | VFD             |
|           | CIP Pump (P6A & P6B) (Duty/Standby) | Permeate supply to CIP Tank                                | 150                       | 20           | 7                   | 480     | VFD             |

#### 1.4 SUBMITTALS

##### SUBMITTAL PROCEDURES:

##### A. Shop Drawings (Six Copies)

1. Show complete details of construction and installation of all equipment, including appurtenances; include all dimensions.
2. Include all specifics of this installation.
3. Delete all references not applicable to this installation.
4. Outline specific areas where the proposed equipment differs from that specified.
5. Include complete wiring diagrams required for installation.
6. Certified pump curves showing the actual performance of the pumps to be supplied under factory testing. Pump certification shall include testing for head, capacity, efficiency and brake horsepower, at 50%, 65%, 85% and 100% for variable speed driven pumps; and at the rated speed for constant speed pumps. Six certified copies of the results in the form of pump characteristic curves shall be furnished to the Engineer for review prior to shipment.
7. Complete test results on each motor for the tests as defined in the NEMA standard for a motor of its class and rating, to determine that it is free from electrical and mechanical defects and to provide assurance that it meets the design specifications.

8. The manufacturer's complete test results for each motor model design tested for efficiency by the following means:
  - (a) The full load Dynamometer test shall be based on IEEE 112 Method B, per NEMA MG1 12.53 A & B including six (6) load point testing and stray load loss segregation and correction steps.
9. A complete, easily readable functional description of the proposed equipment.

B. Operations and Maintenance Manuals (Four Sets)

1. Complete operation and maintenance information for this specific equipment.
2. The manuals will be reviewed by the Engineer for completeness; those that are deemed inadequate will be returned for correction.
3. Complete parts list including the manufacturer's reference and ordering numbers.
4. A complete list of the manufacturer's name, address and phone number, the local representative's name, address and phone number, and the model and serial numbers of all equipment supplied.
5. Recommended Spare Parts List.

C. Installation Submittals (Four Sets)

Upon completion of installation, the results of the field and acceptance tests as specified under this section of the specification shall be submitted to the Engineer.

Upon completion of installation, the results of the field and acceptance tests as specified under this section of the specification shall be submitted to the Engineer.

1.5 QUALITY ASSURANCE

A. General

1. This Section directs special attention to certain features but does not purport to cover all details of the design, manufacture or installation of the pumping units. The final responsibility for supplying and installing pumping equipment which functions as specified herein rests with the Contractor.
2. Workmanship and the method and materials of construction shall conform to the best practice and highest standards applicable for the design use as specified.
3. Pumping units furnished shall be complete in all particulars and ready for final assembly, installation and operation.
4. Any structural, mechanical, civil, architectural, HVAC, plumbing or electrical changes required by the use of an approved equal product shall be the responsibility of the Contractor.



B. Manufacturer Requirements

1. Pump manufacturers shall certify proof of successful operating experience during the last 10 years of five installations of equipment comparable to that specified herein.
2. The electrical drive equipment specified herein (including a/c drive induction motor) shall be designed and sized by the supplier, who shall assume responsibility for correct operation of the system.
3. The manufacturer of the a/c drive induction motor specified herein shall have a factory trained service engineer in residence for at least 50 major cities within the United States.

1.6 STORAGE

- A. Pump shafts stored on site shall have covered and taped ends for protection. Pump equipment damaged or bent during shipping or storage shall be replaced.
- B. All pump motors shall be stored on site according to motor manufacturer's recommendations until pumps are operational and accepted by the Engineer. Pumps with motor space heaters shall be energized during storage; the Contractor shall be responsible for such arrangements.

1.7 WARRANTY

- A. The pump manufacturer and the pump motor manufacturer shall each individually and separately warrant that the equipment they supplied under this Section fully meets the criteria specified herein and shall further warrant that the equipment is free from all defects in materials and workmanship.
- B. The manufacturer's warranties from defects shall contain a provision that the manufacturer shall repair or replace any defects, to the satisfaction of and at no additional cost to the Owner, for a period of twenty-four (24) months for the pump and motor from the date of Substantial Completion of the project.
- C. The variable frequency drive shall be warranted for a period of 24 months after date of shipment. Warranty shall include parts and labor by the manufacturer.

PART 2 PRODUCTS

2.1 HORIZONTAL END SUCTION PUMPS

A. Pump Manufacturer

The pump supplier shall assume complete responsibility for each pump, motor, variable frequency drive and controls. The pump supplier shall have successfully manufactured pumping units for water for a minimum of five years and shall manufacture the major components of the pumping units.

Horizontal end suction pumps shall be manufactured by Goulds

Pumps, Seneca Falls, NY, or approved equal.

B. Horizontal Pump Casings

1. The pump casings shall be of cast alloy, high alloy/stainless steel, conforming to ASTM A890/A995, horizontally end suction.
2. The casings shall have tapped and plugged holes for priming, vent and drain and shall be equipped \_\_\_\_\_.
3. The interior of the casings shall be smooth and free from surface defects.
4. Suction and discharge flanges for pump shall be cast solid with the casing. Drilling and dimensions of flanges shall conform to the 125 pound American Standard.
5. Pump casings shall be designed for 290 psi maximum allowable working pressure and shall be hydrostatically tested at 150 percent of working pressure.

C. Horizontal Pump Impellers

1. The impellers shall be carefully designed to give the characteristics desired, and shall be both hydraulically and dynamically balanced.
2. The impellers shall be of the semi-open type, cast in one piece and meeting current lead standards, of cast alloy, high alloy/stainless steel, conforming to ASTM A890/A995.
3. The impellers shall be finished all over, the exterior being turned and polished and the interior finished smooth by hand to match templates.

D. Horizontal Pump Shaft Sleeves

1. Sleeves shall be straight slip on type, o ring sealed and secured in place with threaded sleeve nuts located external to the stuffing box.
2. Each sleeve shall be driven by an impeller key.
3. Sleeve material shall be 316 stainless steel. The sleeve shall extend through and beyond the packing gland.

E. Horizontal Pump Mechanical Seal

1. The A backplate with deflector vanes constructed shall be provided, including a single mechanical seal shall be John Crane (Morton Grove, IL) model 5615Q (Silicone Carbide versus Silicon Carbide with EPR) or better. The design shall allow for continuous operation without the need for external flush water or venting.

F. Horizontal Pump Bearings

1. Bearings shall be designed and sized for at least 100,000 hours minimum rated bearing life.

2. The bearings shall be protected from water seepage and dust.
3. Bearing lubrication shall be oil bath equipped with oil slinger.

G. Horizontal Pump Shafts

1. The pump shafts shall be of series SAE 4140 alloy steel, ground to accurate dimensions to run true, finished and polished on all sections.
2. The shafts shall be of ample strength and rigidity, with the shortest practicable distance between bearings to keep deflection and vibration to a minimum.
3. All parts of the shaft in contact with water or passing through stuffing boxes shall be protected by easily removable locked on stainless steel sleeves.
4. The maximum allowable deflection of the shaft shall be 0.002-inch at any point of operation on the pump curve.

H. Horizontal Pump Baseplates

1. Suitable baseplates, each with an opening for grout, shall be furnished with the pumping units. Baseplates shall be solidly imbedded in the concrete pad for vibration free operation.
2. The baseplate shall be designed to support both pump and close-coupled motor.
3. Planed surfaces or bearing pads shall be provided for the pump and motor.
4. The baseplates shall be provided with a suitable raised edge or drip pan arrangement for catching oil and water drops and shall have tapped connections for drainage.
5. Baseplates shall be of Galvanized steel to ANSI 1991. SWRO distribution pump skid P4, P10, and P11 shall be of stainless steel or corrosion resistant material.

I. Miscellaneous Fittings

1. Drain connections shall be provided at the pump volute, the drip pocket below the stuffing box (if applicable) and at all other possible drainage points.
2. The volute shall be provided with air vents so that the pump can be freed from air pockets.

2.2 A/C INDUCTION DRIVE MOTORS FOR VARIABLE SPEED APPLICATIONS

- A. The pumps shall be driven over the specified speed range by a/c motors sized to operate in conjunction with variable voltage/variable frequency rectified power and provide 100 percent motor torque and horsepower rating at 1.0 service factor. Motors used for the a/c variable speed drives shall be Reliance Duty Master, High Efficient XE Design, Premium Efficiency by U.S. Motors, or approved equal.

- B. The motor base speed shall be selected so that the maximum operating speed of the centrifugal pump and motor are compatible.
- C. Motors shall not exceed 3600 rpm.
- D. The a/c motors for variable frequency applications shall be 3 Phase, 60 Hertz, 208 Volts. They shall be rated for operation in 40°C ambient at 1.15 service factor for constant speed operation and 1.0 service factor for operation on variable frequency, variable voltage. The motor shall have a KVA per horsepower rating of NEMA code letter G or better.
- E. The motors shall be open protected enclosure, premium high efficiency design (minimum 94% nominal efficiency at full load), cast iron construction, NEMA design B, Class B insulation with the following minimum features:
  - 1. Cast iron conduit boxes, diagonally split neoprene gasketed, rotatable 360° in 90° increments.
  - 2. Grounding clamp in box.
  - 3. Removable lifting lug.
  - 4. Permanently numbered non-wicking leads.
  - 5. Lead separator motor frame and conduit box.
  - 6. Zinc-plated hardware.
  - 7. Stainless steel nameplate.
  - 8. Stator windings shall be copper construction.
  - 9. Stator and rotor completely epoxy coated for corrosion protection.
  - 10. The motors shall have a non reverse ratchet shall be furnished with the motor of the ball or pin type.
  - 11. The motor shall develop sufficient torque to start the pump and bring it up to full speed when the pump is full of water and operate the pump at any point of its characteristic curve without entering into the service factor.

## 2.3 A/C VARIABLE FREQUENCY CONTROLLER

### A. General

- 1. The alternating current (a/c) drive systems shall provide variable speed to the booster pumps. The Contractor shall provide a new VFD to control each pump. The drive shall be sized to meet the amperage conditions of each individual booster pump.
- 2. All pumps specified herein shall be placed on duplex pump skids and operate lead/lag. Pumps shall have hand/off/automatic control capabilities.
- 3. The Variable Frequency Controllers and all associated appurtenances (e.g. harmonic filters, etc.) shall be as specified

in 26 29 23 ADJUSTABLE SPEED DRIVE (ASD) SYSTEMS UNDER 600 VOLTS.

## 2.4 PAINTING

- A. Ferrous surfaces obviously not to be painted shall be given a shop coat of grease or other suitable rust resistant coating.

Painting shall be done in accordance with 09 90 00 PAINTS AND COATINGS.

## PART 3 EXECUTION

### 3.1 FIELD ACCEPTANCE TESTS

- A. After the pump has been installed and connected, checked out as required by Section 01 75 00 and after inspection, operation, and adjustment have been completed by the manufacturer's representative, the equipment shall be field tested in the presence of the Engineer, for overall wire to water efficiency and for general performance and fitness for the service specified. Results of these tests shall be submitted to the Engineer.

### 3.2 MANUFACTURER'S SERVICES:

- A. The services of a factory trained manufacturer's representative shall be provided as described in Section 01 75 00 START UP AND TESTING and as specified herein.

- B. Services to be Provided:

- 1. For the pumps, the service representative shall be responsible for complete component inspection on site after delivery and shall assist in the correct assembly of the components for a minimum period of one (1) eight hour day.

- (a) For inspection and check out of the erected equipment.

- (b) For start up services and supervision.

- (c) For complete instruction of the Owner's operating personnel.

- C. The minimum period of time herein specified does not relieve the manufacturer from providing sufficient time to satisfactorily complete the required service functions.

- D. The manufacturer's representative shall certify in writing that the pump has been properly installed.

- E. The Owner reserves the right to video tape the instruction of the operating personnel for future use in training.

### 3.3 SPARE PARTS

- A. Provide all other spare parts as recommended in the manufacturer's standard operations and maintenance information.

- B. Provide minimum two (2) mechanical seals for each pump.

-- End of Section --

SECTION 43 41 45

FIBERGLASS REINFORCED PLASTIC CHEMICAL STORAGE TANKS

01/25

PART 1 GENERAL

1.1 REQUIREMENTS

The Contractor shall furnish, deliver, install, test and place in satisfactory operation fiberglass reinforced plastic storage tanks, complete with all necessary accessories, as shown on the Drawings and as specified herein.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

01 42 00 SOURCES FOR REFERENCE PUBLICATIONS

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.5 (2020) Pipe Flanges and Flanged Fittings  
NPS 1/2 Through NPS 24 Metric/Inch Standard

ASTM INTERNATIONAL (ASTM)

ASTM C581 (2020) Standard Practice for Determining  
Chemical Resistance of Thermosetting  
Resins Used in Glass-Fiber-Reinforced  
Structures, Intended for Liquid Service

ASTM INTERNATIONAL (ASTM)

ASTM INTERNATIONAL (ASTM)

ASTM INTERNATIONAL (ASTM)

ASTM INTERNATIONAL (ASTM)

ASTM E1067/E1067M (2018) Standard Practice for Acoustic  
Emission Examination of Fiberglass  
Reinforced Plastic Resin (FRP)  
Tanks/Vessels

ASTM INTERNATIONAL (ASTM)

ASTM INTERNATIONAL (ASTM)

All reference specifications, codes, and standards shall be the current version available at the time of Bid.

### 1.3 CONDITIONS OF SERVICE / STORAGE TANK SCHEDULE

| Parameter                                                           | R0 Permeate Tank                           |
|---------------------------------------------------------------------|--------------------------------------------|
| Number of tanks                                                     | One (1)                                    |
| Tank Identification No.                                             | 05-07-002                                  |
| Max Solution Concentration                                          | 100% Deionized Water                       |
| Specific Gravity                                                    | 1.10                                       |
| Freezing Point                                                      | 32 degrees F                               |
| Design Temperature                                                  | -10 to 100 degrees F                       |
| Design Pressure                                                     | Atmospheric / As Required for Flanged Vent |
| pH                                                                  | 3 to 10                                    |
| Type                                                                | Vertical Cylindrical                       |
| Bottom Configuration                                                | Flat Bottom                                |
| Top Configuration                                                   | Dome Top                                   |
| Useable Capacity (to invert of overflow)                            | 30,000 gallons                             |
| Maximum Diameter                                                    | 13' - 0"                                   |
| Maximum Overall Height                                              | 33'- 0" w/dome cover                       |
| Connection Openings*                                                |                                            |
| a) Fill                                                             | 8"                                         |
| b) Spare                                                            | 8"                                         |
| c) Outlet 1                                                         | 8"                                         |
| d) Outlet 2                                                         | Not Required                               |
| e) Overflow                                                         | 12"                                        |
| f) Drain                                                            | Not Required                               |
| g) Vent                                                             | 12"                                        |
| h) Vent Type                                                        | Piped Vent                                 |
| i) Emergency Vent                                                   | As Required by Tank Manufacturer           |
| j) Manway Diameter                                                  | 36"                                        |
| k) Continuous Level Instrument                                      | 3" flanged, 1" flanged backup              |
| Heating Panels and Insulation<br>Design Temp / Minimum Ambient Temp | Yes                                        |

| Parameter                                                                 | R0 Permeate Tank     |
|---------------------------------------------------------------------------|----------------------|
| Materials of Construction of Metal Components                             | Not applicable       |
| Materials of Construction of Metal Components above Containment Area      | 316L Stainless Steel |
| Materials of Construction of Elastomers                                   | EPDM                 |
| Ladder                                                                    | Yes                  |
| *Refer to Drawings for Location Orientation, and Elevation of Connections |                      |

#### 1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

##### SD-02 Shop Drawings

CONDITIONS OF SERVICE / STORAGE TANK SCHEDULE; G, AE

Drawings and details for tanks and ladder as recommended by Manufacturer and conforming with OSHA standards. Submittal shall also include weight of the tanks, resin specification and complete manufacturer's recommendation, calculation for tank design signed by a professional engineer including restraint system, and details of the anchoring system.

##### SD-03 Product Data

Installation; G, AE

List of at least five similar installations of the tank type, size, chemical service, and location conditions being proposed, including date installed, contact name, address and phone number.

Dimensions; G, AE

Dimensions of tank and dimensions, location, and orientation of openings, fittings, accessories, attachments, restraints and



supports, anchor bolts, manways, and flexible connections.

#### SD-06 Test Reports

INSPECTION AND TESTING; G, AE

#### SD-08 Manufacturer's Instructions

Instructions

Detailed instructions for pipe connections and bolt torque values.

Post Curing Requirements

Manufacturer's recommendation regarding post cure requirements

#### SD-10 Operation and Maintenance Data

Operation and Maintenance Manual; G, AE

#### SD-11 Closeout Submittals

Warranty; G, AE

### 1.5 WARRANTY

- A. Warranty and Guarantee shall be a minimum of two (2) years.

## PART 2 PRODUCTS

### 2.1 ACCEPTABLE MANUFACTURERS

- a. The fiberglass reinforced plastic storage tanks shall be manufactured by Plas-Tanks Industries or Premier Production Equipment.
- b. The fiberglass reinforced plastic tank manufacturer shall specialize in manufacture, assembly, and field service of FRP chemical storage tanks with a minimum of five years of experience.

### 2.2 GENERAL

- a. The Contractor is responsible for the coordination and selection of corrosion resistant materials for the chemical solutions specified below. The fiberglass reinforced plastic storage tank manufacturer shall become familiar with the characteristics of the specified chemical solutions and guarantee the suitability of the materials used in manufacturing of the tank and appurtenances. The Contractor and manufacturer shall include all features as necessary for satisfactory operation of the tank system for all specified chemical solution concentrations and temperatures.
- b. All tank capacities (volumes) specified shall include only that volume

in the straight shell below the maximum tank fill level (generally 6" below the overflow pipe invert elevation) and above the top of the outlet pipe. At least four inches of freeboard shall be provided between the top of the overflow pipe and the top of the straight shell.

- c. The tank manufacturer shall be fully responsible for the structural design and integrity and watertightness of all tanks including all connections.
- d. Tanks shall be anchored to the concrete base by the Contractor in accordance with the Drawings and manufacturer recommendations.

## 2.3 MATERIALS AND CONSTRUCTION

### a. Construction

(1) Tank construction shall be as follows:

| Parameter           | RO Permeate Tank                                |
|---------------------|-------------------------------------------------|
| Resin               | CorVE8301 or Derakane 411 resin with nexus Veil |
| Resin Cure System   | MEKP                                            |
| Liner Reinforcement | Double C-glass veil                             |
| Structural Layer    |                                                 |
| Resin Cure System   | MEKP                                            |

### b. Resin

- (1) The resin for fiberglass reinforced plastic storage tanks shall be a commercial-grade, corrosion-resistant thermoset that has either been evaluated in a laminate by testing in accordance with ASTM C581 or that has been determined by previous documented service to be acceptable for the service conditions.
- (2) The resin used shall not contain any fillers, pigments, dyes, or colorants, which may interfere with visual inspection of laminate quality, except as required for viscosity control. The limit of filler shall be 5 percent by weight. No fillers or bulking agents shall be used in the exterior structural layer to decrease the glass loading ratio. Resin pastes used to fill crevices before overlay are permitted.
- (3) The initiators used will be of the type, manufacturing origin, and amounts specified by the resin manufacturer.
- (4) Tank shall be provided with UV inhibitor in the resin mix to prevent degradation of the tank.

### c. Reinforcement

- (1) The reinforcing material shall be a commercial grade glass fiber

having a coupling agent which shall provide a suitable bond between the glass reinforcement and the resin and shall be suitable for the fabrication method used. The reinforcing material shall be comparable to that used to generate corrosion resistance.

- (2) Chopped Strand Mat - Chopped strand mat shall be constructed from chopped commercial-grade E-type glass strands bonded together using a binder. The strands should be treated with a sizing that is chemically compatible with the resin system used.
- (3) Continuous Roving - Continuous roving shall be a commercial-grade E-type glass fiber with a sizing that is chemically compatible with the resin system used.
- (4) Woven Roving - Woven roving shall be in accordance with referenced ASTM Specifications.
- (5) Surface Mat - The reinforcement used for the inner surface shall be a commercial-grade chemical resistant glass surface mat as recommended by the resin manufacturer for the intended service.

d. Laminate Construction

- (1) The laminate comprising the structural tank (bottom, cylindrical shell, top head) shall consist of a corrosion-resistant barrier comprised of an inner surface, interior layer, and a structural layer.
- (2) The inner surface exposed to the chemical environment shall be resin-rich, not less than 0.01 inch thick, and reinforced with chemically resistant surfacing material. The surface shall be smooth, glossy, and free of pits. The inner surface of the tanks shall have one or two layers of 'C' glass veil or a polyester synthetic veil depending on the chemical environment. Material used as reinforcing on the surface exposed to chemical attack shall be a commercial grade chemical resistant glass fiber having a coupling agent.
- (3) The interior layer shall be not less than 0.1 inch thick and composed of resin, reinforced only with noncontinuous glass strands applied in a minimum of two plies of chopped strand mat or in a minimum of two passes by the spray up process. Glass strands shall not be shorter than 1.0 inch or longer than 2.0 inches. Glass content of the inner liner and interior layer combined shall be 27% +/- 5% by weight.
- (4) Before the reinforcement of the exterior layer is applied, the interior layer shall be allowed to cure completely so that the thickness of the corrosion barrier, consisting of the inner surface and interior layer, will not be reduced. Cure systems for corrosion barrier and structural layers shall be as specified above. The degree of cure of the laminate, after post cure, shall be such as to exhibit a Barcol hardness on the inner surface of at least 90% of the resin manufacturer's minimum specified hardness for the cured laminate. Post cure of the tanks shall be provided when recommended by the resin manufacturer for a particular service; post cure shall be provided in accordance with the resin manufacturer's guidelines.

- (5) The structural layer shall provide additional strength necessary to meet the tensile and flexural requirements. The reinforcement shall be filament wound, contact molded or a combination of both and may consist of continuous roving, woven roving, chopped strand mat or chopped strands. Where separate layers of reinforcement are used, all layers shall be lapped a minimum of 1.0 inch. Laps shall be staggered as much as possible. If woven roving or cloth is used in successive layers, it shall be alternated with a layer of chopped strand glass.
- (6) Glass content of the filament-wound structural layer shall be 50 to 80 percent by weight. The thickness of the filament wound portion of the tank shell may vary with tank height, provided that all stress and other requirements are met at any height level. All reinforcement used shall be resistant to corrosion by the particular chemical stored in the tank.
- (7) The outer surface shall consist of chopped strands or surfacing mat, or both, over which shall be applied a resin rich coating. The outer surface shall not be pigmented, painted or dyed except to prevent ultraviolet degradation of the tank contents, as applicable. This surface shall be at least 0.02 inch thick.
- (8) The tank top shall be domed with openings and connections as shown and specified. The tank top shall be able to support a 250pound load on a 4 inch by 4 inch area.
- (9) Tank bottom attachment to tank wall shall be per ASTM D 3299 and D 4097.

#### 2.4 CONNECTIONS AND ACCESSORIES

- a. All flanged nozzles shall be of hand lay-up construction with pipe stub molded integrally with the pipe flange. All connections/openings shall be flanged in accordance with ASME B16.5 150 pounds and provided with flanged gasket. Flanged connections, nozzles, and openings shall be FRP gusseted and flat face. All pipe supports, hardware, accessories, etc., shall be provided. All piping connected to the tanks shall be perpendicular or parallel to the straight shell of the tanks. All piping into the tanks shall be supported such that no weight is placed on the tank and its connections. Piping supports requiring holes through the side wall of the tanks shall not be allowed.
- b. Tank outlet connections shall be full bottom drain connections to allow for the tank to be completely drained. If recommended by tank manufacturer, each tank outlet connection shall be provided with a flexible connector, supplied by the tank manufacturer, resistant to the specified chemical to allow for expansion and contraction of the tank and to isolate the tank from vibration. Each tank drain line shall be provided with an isolation valve by the Contractor as indicated on the Drawings.
- c. Vent lines shall be top mounted. Each vent shall be extended to the atmosphere, extend 24-inches above the roof, and shall have a 180 degree return and a fiberglass vent insect screen. Vent lines shall be supplied and furnished by the Contractor as required or as directed by the Engineer.

- d. Each tank shall be provided with an overflow pipe as specified and indicated on the Drawings. Contractor shall provide the overflow pipe, but tank manufacturer shall provide pipe supports. The Contractor shall provide a Flange Insert Check Valves (FIV) for the overflow.
- e. Each tank shall be provided the mounting and connecting requirements, including mounting flange diameter, required clearance between mounting flange and tank wall, and height above liquid level, shall be coordinated with the Instrument Supplier.
- f. Each tank shall be provided with flat faced flanged manway(s) with gaskets, blind flanges, and bolts that are constructed of materials resistant to corrosion by the specified chemicals as shown in the Tank Schedule. Manway covers shall be hinged.
- g. Each storage tank shall be equipped with an exterior access ladder for access to the manway. Ladders shall comply with specification 05 51 33 Metal Ladders. Ladders must be secured to the tank and secured to the concrete to allow for tank expansion/contraction due to temperature and loading changes. Use proper chemical resistant materials when anchoring to tank dome or sidewall. Ladders shall be designed to meet applicable OSHA standards.
- h. Each tank shall be provided with a minimum of four lifting lugs. Lifting lugs shall be capable of withstanding the weight of an empty tank with a safety factor of 3 to 1.
- i. Each tank shall be provided with a minimum of four tie down lugs and all necessary anchor bolts. The tank restraints shall withstand wind load or seismic load, whichever is greater, calculated in accordance with current UBC code.
  - (1) The anchor lugs shall be designed and supplied by the tank manufacturer.
  - (2) The tank manufacturer shall submit calculations, sealed by a Professional Engineer, to verify that tie-down lugs can withstand buoyancy, seismic activity, and wind load.
  - (3) The tank manufacturer shall submit calculations, sealed by a Professional Engineer, to verify that tie-down lugs can withstand buoyancy, seismic activity, and wind load.
  - (4) The Contractor shall submit calculations, sealed by a Professional Engineer, to verify that anchor bolts can withstand buoyancy, seismic activity, and wind load.
- j. J. Each tank shall be provided with a harness lug to meet the OSHA 5000-lb rating on top of the tank for safety tie-off. This lug shall be adjacent to and reachable from the ladder.
- k. The tank shall be provided with a permanently attached label providing the following information:
  - (1) Name, concentration, and specific gravity of material stored.
  - (2) Resin, cure system, and liner reinforcement for corrosion barrier layer.

- (3) Resin and cure system for structural layer.
- (4) Tank dimensions and capacity.
- (5) Maximum temperature rating of tank.
- (6) Manufacturer.
- (7) Date of manufacture.
- l. All metallic parts, fasteners, brackets, mounting hardware, and accessories provided by the tank manufacturer shall be constructed of corrosion resistant metals as specified in the Tank Schedule.
- m. Tank shall be provided with 2 inches of insulation and heating system by Chloramax Heating and Control. The insulation shall be glass fiber (ASTM C527) and designed to meet a minimum temperature of 36 degrees F in the water in the tank with an outside temperature of -10 degrees F. Wind will be 20 mph and a safety factor of 25% will be implemented.

## 2.5 PIPING SUPPORT

All piping into the tanks shall be supported such that no weight is placed on the tank or its connections.

## PART 3 EXECUTION

### 3.1 MANUFACTURER'S FIELD SERVICES

The services of a qualified manufacturer's technical representative shall be provided and shall include the following site visits for each series of tanks:

| Service                     | Number of Trips                            | Number of Days / Trips |
|-----------------------------|--------------------------------------------|------------------------|
| Installation and Inspection | 1                                          | 1                      |
| Annual Inspection           | As required for warranty.<br>Minimum of 1. | 1                      |

### 3.2 INSTALLATION

- a. The Contractor shall furnish and install the Fiberglass Reinforced Plastic storage tanks, and related items in accordance with the manufacturers' recommendations and in accordance these specifications.
- b. A manufacturer's field representative shall be on site when each tank is installed to observe installation and verify that each tank has been installed according to the manufacturer's recommendations. The manufacturer shall provide a report certifying that each tank has been installed properly.
- c. All piping, valves, fittings, conduit, wiring, etc., required to interconnect system components shall be furnished and installed by the Contractor.
- d. All metallic fasteners, brackets, mounting hardware, and accessories

located in chemical storage and feed areas shall be constructed of corrosion-resistant metals as specified in the Tank Schedule.

- e. All tanks shall be mounted on concrete pads only when demonstrated to be fully cured. Tank pad notches are required for the full drain outlet connections.
- f. The finished surface of each tank bottom shall be sufficiently flat, smooth, and free of irregularities, to prevent the possibility of tank failure from point loads or other sources.
- g. The Contractor shall install a 3/8" EPDM mat between each concrete pad and storage tank. The tanks shall be installed on level pads.

### 3.3 INSPECTION AND TESTING

- a. A 3-stage inspection process shall be performed on the tank during various stages in the fabrication process. The first inspection shall be performed at the completion of the corrosion barrier and before structural winding occurs. The second inspection will be performed after the tank is removed from the mandrel and before any nozzles are attached. The third inspection shall be a final inspection prior to shipment of the tank and shall be performed by an inspector with at least 5 years of experience with FRP vessels.
- b. Upon completion of installation of tank and prior to connecting piping, the Contractor shall provide blind flanges or other suitable plugs for all openings in the side wall of the tank, fill tank with potable water from a source approved by the Engineer and conduct a leakage test. Tank shall be filled up to the top of the straight shell and left to sit over a 5-consecutive day test period. There shall be no leakage over the test period. Leakage tests shall be repeated for any replaced or repaired tank. Upon satisfactory completion of leakage test, Contractor shall drain, thoroughly clean, and dry the tank and dispose of water in a suitable manner.
- c. Degree of surface cure shall be tested per the referenced ASTM specifications.
- d. Quality control for visual defects shall be per the referenced ASTM specifications.
- e. After the tank has completed a successful field hydrostatic test, the tank shall undergo a mechanical integrity test using Acoustical Emission Test (AE Test) in accordance with the latest version of ASTM E1067/E1067M. The Contractor shall secure the services of Non-Destructive Evaluation International (Indian Trail, NC) or an equally qualified firm to perform the AE testing. Firms shall be considered equally qualified by demonstrating the following:
  - (1) The firm shall specialize in AE testing.
  - (2) The firm shall have all the necessary equipment to perform AE testing.
  - (3) The firm shall have a minimum of five (5) years of experience in AE testing and shall provide a list of AE tests performed to demonstrate experience.

(4) The firm's testing staff shall have a minimum of five (5) years of experience with AE testing.

- f. Contractor shall provide all labor and materials necessary for completing AE testing including water to fill tanks with during testing. After testing is completed, a final report shall be submitted to the Engineer. In the event the AE test results indicate repairs are required; the tank manufacturer shall promptly repair all faulty areas of the tank as identified by the test. The tank shall then be re-tested using the hydrostatic test and the AE test. The second AE test report shall be submitted to the Engineer. Any tank failing the second hydrostatic test, or the AE test shall be removed from the project site and replaced with a new tank at no cost to the Owner. The replacement tank shall undergo hydrostatic and AE testing and shall be subject to the same acceptance criteria as the initial tank. This process shall be repeated until all tanks provided to the project pass the AE test.

#### 3.4 IOperation and Maintenance Manual

Provide manufacturer recommended operation and maintenance manual  
-- End of Section --



SECTION 46 30 00

WATER AND WASTEWATER CHEMICAL FEED SYSTEMS

08/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z400.1/Z129.1 (2010) Hazardous Workplace Chemicals - Hazard Evaluation and Safety Data Sheet and Precautionary Labeling Preparation

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME A13.1 (2023) Scheme for the Identification of Piping Systems

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA B300 (2018) Hypochlorites

ASTM INTERNATIONAL (ASTM)

ASTM D1785 (2021) Standard Specification for Poly(Vinyl Chloride) (PVC), Plastic Pipe, Schedules 40, 80, and 120

ASTM E1067/E1067M (2018) Standard Practice for Acoustic Emission Examination of Fiberglass Reinforced Plastic Resin (FRP) Tanks/Vessels

HYDRAULIC INSTITUTE (HI)

HI 9.1-9.5 (2021) Pumps General Guidelines for Materials, Sound Testing, and Decontamination - B117

HI ANSI/HI 7.1-7.5 (2023) Controlled-Volume Metering Pumps American National Standard for Nomenclature, Definitions, Application, and Operation - B115

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250 (2020) Enclosures for Electrical Equipment (1000 Volts Maximum)

NEMA ICS 1 (2022) Standard for Industrial Control and Systems: General Requirements

NEMA ICS 2 (2000; R 2020) Industrial Control and

Systems Controllers, Contactors, and  
Overload Relays Rated 600 V

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

PLUMBING-HEATING-COOLING CONTRACTORS ASSOCIATION (PHCC)

NAPHCC NSPC (2015) National Standard Plumbing Code  
Illustrated

UNDERWRITERS LABORATORIES (UL)

UL 50 (2024) UL Standard for Safety Enclosures  
for Electrical Equipment,  
Non-Environmental Considerations

1.2 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Detail Drawings; G, AE

SD-03 Product Data

Chemical Feed System; G, AE

Safety Data Sheets; G, AE

Framed Instructions; G, AE

Auxiliary Equipment and Spare Parts; G, AE

Factory Test Report; G, AE

SD-06 Test Reports

Operating Tests; G, AE

Tank Testing; G, AE

Controlled Volume Pumps - Operational Tests; G, AE

System Pressure Tests; G, AE

Flow Tests; G, AE

Synchronization Tests; G, AE

SD-07 Certificates

Supplied Chemical; G, AE

Service Organization; G, AE

Certificates; G, AE

#### SD-10 Operation and Maintenance Data

Operating Instructions; G, AE

Maintenance Instructions; G, AE

#### SD-11 Closeout Submittals

Extended Warranty; G, AE

### 1.3 MAINTENANCE MATERIAL SUBMITTALS

Concurrent with delivery and installation of the specified equipment, furnish auxiliary equipment and spare parts as follows:

- a. Spare parts for each different item of material and equipment specified including all of the parts recommended by the manufacturer to be replaced after 1 year service.
- b. For each machine: one extra of each part used that is made from glass, hard rubber, or clear plastic; one extra set of solution-hose connections; one extra set of diaphragms, two filler plug seal washers; two ball checks; two seats; two complete sets of all gaskets; one spare diaphragm for each back pressure regulator; one hydraulic plunger assembly for each different size metering pump; one of each type of material back pressure regulator, with three spare springs and fluorocarbon resin diaphragms for each; one spare diaphragm and air valve for pulsation dampener.
- c. For each different size of direct current motor one SCR circuit board with 12 drive motor fuses and 12 SCR controller fuses; overload replacement elements for each size and type of motor.
- d. One set of special tools for each type of equipment including calibration devices, and instruments required for adjustment, calibration, disassembly, operation, and maintenance of the equipment.
- e. Two pairs of safety goggles and/or face shields, two chemical resistant aprons, and two pairs of chemical resistant gloves in one or more wall mounted steel cases.
- f. One assembly tool for tubular diaphragm.
- g. One lever type grease gun or other lubricating device for each type of grease required.
- h. One or more steel tool cases mounted on the wall in a convenient location complete with flat key locks, two keys, and clips or hooks to hold each special tool.

#### 1.4 DELIVERY, STORAGE, AND HANDLING

Protect material and equipment delivered and placed in storage from the weather, excessive humidity and excessive temperature variation, dirt, dust, or other contaminants.

### PART 2 PRODUCTS

#### 2.1 SYSTEM DESCRIPTION

Provide a chemical feed system consisting of a chemical supply storage tank from which the chemical solution is pumped through piping or tubing, as appropriate, to the point of application. Include with each chemical feed system controlled volume pumps, tanks, mixers, gauges, back pressure regulators, strainers, pressure relief valves, sight glasses and flow metering devices, check valves, and hand valves.

##### 2.1.1 System Requirements

Select and fabricate the pumps in accordance with HI ANSI/HI 7.1-7.5 and HI 9.1-9.5 except as modified herein. Provide a pump simplex or duplex skid mounted system as noted below and in Contract Drawings. Provide pump stands and platforms adequate to support the pumping and skid system.

##### 2.1.2 Performance Requirements

Capacity and features of the chemical feed systems and accessories must be suitable for 24-hour full load service in ambient, non-freezing conditions.

##### 2.1.3 Submittal Data for Chemical Feed System

Submit manufacturer's performance charts, and pump curves. List of materials, list of equipment, including a complete list of parts and supplies with current unit prices and source of supply. List of special tools for each type of equipment furnished including special tools necessary for adjustment, operation, maintenance, and disassembly.

##### 2.1.4 Extended Warranty

provide Manufacturer's Extended Warranty to cover system components for 3 years.

#### 2.2 EQUIPMENT

##### 2.2.1 Standard Products

Provide materials and equipment which are the standard products of a manufacturer regularly engaged in the manufacture of the products and that essentially duplicate items that have been in satisfactory use for at least 2 years prior to bid opening. Equipment must be supported by a service organization that is, in the opinion of the Contracting Officer, reasonably convenient to the site. Submit Service Organization Certificates outlining their location and qualifications to the Contracting Office.

##### 2.2.2 Nameplates

Secure a plate to each major item of equipment containing the manufacturer's name, address, type or style, model or serial number, and

catalog number.

### 2.2.3 Controlled Volume Pumps

Provide the chemical feed systems in Table 1.

Table 1: Chemical Feed Pumps

### 2.2.4 Controls

Provide the chemical metering equipment with the appurtenances and accessories, as required, for flow capacity adjustment. Provide manual range adjustment on all systems.

#### 2.2.4.1 Automatic Control

Provide automatic control with the capability to vary feed rate based on signals from a flow or pH meter or controller. The pump controller shall be supplied by the manufacturer, but must be compatible with CTI Dynamic LLC control station out of Attleborough, MA for the control of the pump in automatic/hand/off modes.

#### 2.2.4.2 Manual Control

Provide non-automatic control with the capability for starting or stopping the chemical metering equipment and adjustment of the solution feed rate by the operator.

##### 2.2.4.2.1 Spare Pumps

Provide one spare pump for each model listed in Table 1. Where the pump head or internals are not compatible with the chemical, provide a spare pump head without the pump motor. The intention is to provide spare pump and if necessary interchangeable heads/internals to be compatible with the chemicals being implemented in the project, so project has redundancy without providing a spare pump for each chemical.

- a. Prominent model No. GMXA1604PVT2M000UDC0300EN.
- b. Finish Thompson model No. DB9V-E-FF-M219.
- c. Prominent model No. S2CBH07220PVTS070UD11001EN.

### 2.2.5 Drives for Controlled Volume Pumps

Provide with and drive the metering pumps by alternating current electric motor drives for chemical pumps.

#### 2.2.5.1 Electric Motor Drive

Electric motor must be of sufficient capacity to operate the chemical metering equipment under all operating conditions without exceeding their rated nameplate current or power, or their specified temperature limits. Provide motors having starting characteristics and ruggedness necessary under the actual conditions of operations or clean-up procedures used in the areas where they are located. Alternating current motors with power rating of 1/3 hp or less must be 115 volts, single-phase, 60-Hz service; motors with power rating in excess of 1/3 hp must be 460 volts, three-phase, 60-Hz service. Electrical features of direct current motors, including the ratings of the motors, must be compatible with the

capabilities and ratings of the rectifier controllers with which they are used.

#### 2.2.6 Calibration Standpipes

Provide chemical metering equipment with a calibration standpipe for measuring pump output. Note that chemical transfer pumps do not require calibration standpipes. See the Contract Drawings for more detail. The standpipe must allow convenient observation of the change of fluid level for at least 1/2 minute at full stroke and maximum speed settings, and of Schedule 80, clear PVC pipe conforming to ASTM D1785 with Schedule 80 fittings equipped with a flanged connection to the pump manifold and an end cap fitted with a PVC vacuum breaker and ball valve for air venting. The standpipe must have a clear, observable length of at least 12 inches and be permanently calibrated to allow for gph or lph readings, to allow reading of the fluid contents with an accuracy of 1 percent.

#### 2.2.7 Valves

##### 2.2.7.1 Metering Pump Valves

Equip the metering pump with adjustable internal vacuum and pressure relief valve, hydraulic oil refill valve (if required by model), and automatic air bleed valve. The relief valve must be adjustable over the full pressure range of the pump and preset at the factory.

##### 2.2.7.2 Suction and Discharge Valves

Fit suction and discharge valve cartridges with ball checks that open to full pipe diameter.

##### 2.2.7.3 Back Pressure Valve

Install back pressure regulating valve on the pump discharge and factory adjust to crack open at the indicated pressure. Provide spring opposed diaphragms with loading pressures adjustable by means of a screw in the top works. Locate back pressure valve as close to the injection point as possible.

##### 2.2.7.4 Pulsation Dampeners

Size pulsation dampeners for the displacement of each pump. Pulsation dampeners must have a diaphragm separating the upper chamber from the lower chamber. Charge the upper chamber with compressed air to 50 percent of the expected line pressure. Provide the diaphragm of molded construction and prevent the air charge from being dissolved in the process fluid. The lower chamber must be plastic or lined with inert plastic material to prevent corrosion by the process fluid. Equip the upper chamber with a tire valve type charging valve and air pressure gauge. Provide an air line and air hose with pressure regulator and hand-operated, lever-type valve suitable for charging the pulsation dampers.

#### 2.2.8 Solution Tanks

##### 2.2.8.1 Polyethylene Storage Tanks

- a. Service: Chemical storage tanks shall be suited for the following operating conditions:

- b. High Density Cross-linked Polyethylene resin used in the tank manufacture shall be Poly CL™ and shall contain ultraviolet stabilizer as recommended by resin manufacturer. Where black tanks are indicated, the resin shall have a carbon black compounded into it. The tank material shall be rotationally molded and be a resin that is commercially available at the time of tank manufacture.
- c. For sodium hypochlorite and sulfuric acid, tank resin shall include an antioxidant polyethylene system (OR-1000) with four times the antioxidant properties of a standard polyethylene bonded to the interior surface during the manufacturing process.
- d. Wall thickness for a given hoop stress is to be calculated in accordance with ASTM D 1998. In NO case shall the tank thickness be less than design requirements per ASTM D 1998.

- (1) The wall thickness of any cylindrical portion at any fluid level shall be determined by the following equation:

$$T = P \times OD/2SD \text{ or } 0.433 \times SG \times H \times OD/2SD$$

Where:

T = wall thickness, in

P = pressure, psi

SG = specific gravity, gm/cc

H = fluid head, ft

OD = outside diameter, ft

SD = hydrostatic design stress

- (a) The minimum wall thickness shall be sufficient to support its own weight in an upright position without external support but shall not be less than 0.187" thick.
- (b) **Specific Gravity for design in all tanks shall be 1.9.**
- (2) On closed top tanks the top head shall be integrally molded with the cylindrical wall. Its minimum thickness shall be equal to the thickness of the top of the straight sidewall. In most cases, flat areas shall be provided for attachment of large fittings on the dome of the tank.
- (3) The bottom head shall be integrally molded with the cylindrical wall. Knuckle radius shall be:

| Tank Diameter, ft.      | Min Knuckle Radius, in. |
|-------------------------|-------------------------|
| less than or equal to 6 | 1                       |
| greater than 6          | 1 - 1/2                 |

- (4) Tanks with 3000 gal capacity or larger shall have at least 3 lifting lugs. Lugs shall be designed for lifting the tank when empty.
- (a) Unless otherwise indicated by Contract drawings, for indoor pneumatic fill, manways shall be 24-in diameter or greater and equipped with an emergency pressure relief device or SAFE- Surge™ Manway with pressure relief at 6" water column to prevent over-pressurization. The SAFE-Surge manway shall be chemically compatible with the chemical being stored. Gaskets shall be closed cell, cross-linked polyethylene foam, Viton, or EPDM materials.
  - (b) Unless otherwise indicated by Contract drawings, for outdoor pneumatic fill, manways shall be 24-in diameter or greater and equipped with Poly Processing Company's F.S.2650® combined manway and vent to prevent over pressurization of tank. Manway must be capable of relieving a volume flow rate of up to 2650 ACFM. Gaskets shall be closed cell, cross-linked polyethylene foam, Viton, or EPDM materials.
  - (c) Unless otherwise indicated, tanks less than 2000 gallons in non-pneumatic applications shall have a manway cover 17-in or smaller of Polyethylene material with a coarse thread. The 405 gallon tanks shown on the drawings are allowed to have a 7 inch diameter, threaded cap for access. Gaskets, when required for sealing, shall be closed cell, cross-linked polyethylene foam, viton or EPDM materials compatible with the chemical being implemented.
  - (d) NOTE: Tanks must be vented to allow for performance at atmospheric pressure, in accordance with the Venting Requirements for Polyethylene Tanks matrix.

e. Tank color shall be natural.

#### 2.2.8.2 Tank Accessories

##### a. Ladder

- (1) Ladders shall comply with specification 05 51 33 Metal Ladders.
- (2) Ladder will only be provided when climb height is greater than 20 ft. Ladders must be secured to the tank and secured to the concrete to allow for tank expansion/contraction due to temperature and loading changes. Use proper chemical resistant materials when anchoring to tank dome or sidewall.
- (3) All ladders shall be designed to meet applicable OSHA standards.

##### b. Restraint System (See Tables 2 and 3):

- (1) Metal components to be stainless steel and tension ring with stainless steel cables and clamps.
- (2) Tank restraint system shall be supplied and the design of same certified by a Structural Engineer registered in the State of tank installation. Design shall conform to the most recent edition of the IBC code for seismic and wind load. Anchor bolts



as required by the calculations shall be supplied by the general contractor.

c. Heat Tracing (See Tables 2 and 3)

(1) Heat tracing system for temperature maintenance shall be SilcoPad® tank heating systems, designed to maintain a desired product temperature, not to exceed 100 degrees F. Each system shall include tank heating pads and a temperature controller. The quantity and type of SilcoPad® is determined by the size of the tank, the desired temperature maintenance and environmental conditions. Systems shall be able to provide a delta of 60 degrees F. Tanks are supplied with the heating panels and a controller installed by Poly Processing Company. Power supply to be the only field installation required.

(a) Pads to operate on 120 vac single phase with a maximum power density of 0.5 watts / sq. inch.

(b) Silicone pad heaters must fully comply with Article 427-23(b) of the National Electric Code.

(c) Tank shall be supplied with a 2XTC Microprocessor based tank heating controller. Controller shall have high-limit protection, High temperature, Low temperature, and low current alarms, and a 4-20mA retransmission output. Controller shall have LED's on the face for alarm indications.

d. Insulation: The tanks shall be insulated to maintain the desired temperature. See Tables 2 and 3 for those tanks that require insulation.

(1) Insulation used shall be polyurethane foam with a density of 2.5 lb/ft<sup>3</sup> with a minimum an "R" value of 6.3/in. The foam shall be applied with a nominal thickness of 2" to the external tank surfaces except the tank bottom.

(2) Upon completion of application and curing of the insulation, 2 full coverage coats of latex mastic coating shall be applied to the surface of the insulation in such manner as to seal the insulation from the outside environment.

2.2.8.3 Tanks

a. See Tables 2 and 3 per the following specifications.

Note 1: Approximate overall height is measured along the straight cylindrical portion of the tank and includes the dome top.

b. Fittings

(1) Tank fittings shall be according to the fitting schedule. Threaded fittings shall use American Standard Pipe Threads. If tanks are insulated, fittings shall be installed at the factory prior to application of the insulation.

(2) Bolted flange fittings shall be constructed of one 150 lb. flange with ANSI bolt pattern, one flange gasket and stud bolts with

gaskets. Stud bolts to have chemical resistant polyethylene injection molded heads and gaskets to provide a sealing surface between the bolt head and the interior tank wall. Stud bolt heads are to be color coded for visual ease of identifying the bolt material by onsite operators. Green- 316 Stainless Steel, Black- Titanium, Red- Alloy C-276, Blue- Monel. All materials shall be compatible with chemical service and as indicated in the fitting schedule above. EPDM or Viton GF gaskets shall be supplied as required based on the chemical.

- (3) For sodium hypochlorite and sulfuric acid storage, Bolted One-Piece Sure Seal (B.O.S.S.), double flange fittings constructed of virgin polyethylene shall be supplied. Bolts will be welded to a common backing ring and encapsulated with polyethylene preventing fluid contact with the metal material. Flange will have one full face gasket to provide a sealing surface against inside tank wall. All materials shall be compatible with chemical service and as indicated in the fitting schedule above.
- (4) Down Pipes and Fill Pipes: Down pipes and fill pipes shall be supported at 6-ft max intervals. Down pipes and fill pipes shall be PVC or material compatible with the chemical stored.
- (5) U-Vents: Each tank must be vented for the material and flow and withdrawal rates expected. Vents should comply with OSHA 1910.106(F)(iii)(2)(IV)(9). U-vents shall be sized by the tank manufacturer and be furnished complete with insect screen if required in accordance with the venting schedule listed above.
- (6) On dual wall tank(s) greater than 1000 gallons, bottom fitting(s) must be designed to maintain 110% secondary containment integrity. Bottom containment fitting must include PTFE expansion joint designed to accommodate movement of primary tank in design accordance with ASTM-D 1998 tolerances. All secondary containment fittings and parts shall be resistant to chemical fume corrosion. Fitting shall include the option to connect a secondary containment pipe over primary pipe.
- (7) All fittings on the 1/3 lower sidewall of tanks with capacities > 1000 gallons shall have 100% virgin PTFE Flexijoint® expansion joint. Expansion joint to have 3 convolutions, stainless steel limit cables, FRP composite flanges and meet the following minimum performance specifications. Galvanized parts will not be accepted.

Expansion joints to meet the following minimum performance requirements:

Axial Compression greater than or equal to 0.67"

Axial Extension greater than or equal to 0.67"

Lateral Deflection greater than or equal to 0.51"

Angular Deflection greater than or equal to 14"

Torsional Rotation greater than or equal to 4"

#### 2.2.8.4 Factory Testing

##### a. Material Testing

- (1) Perform gel and low temperature impact tests in accordance with ASTM D 1998 on condition samples cut from each polyethylene chemical storage tank.
- (2) Degree of Crosslinking. Use Method C of ASTM D 1998- Section 11.4 to determine the ortho-xylene insoluble fraction of cross-linked polyethylene gel test. Samples shall test at no less than 60 percent.

##### b. Tank Testing

- (1) Dimensions: Take exterior dimensions with the tank empty, in the vertical position. Outside diameter tolerance, including out-of-roundness, shall be per ASTM D 1998. Fitting placement tolerance shall be +/- 1/2-in vertical and +/- 1 degree radial.
- (2) Visual: Inspect for foreign inclusions, air bubbles, pimples, crazing, cracking, and delamination.
- (3) Hydrostatic test: Following fabrication, the bottom tanks, including inlet and outlet fittings, shall be hydraulically tested with water by filling to the top sidewall for a minimum of 1 hour and inspected for leaks. Following successful testing, the tank shall be emptied and cleaned prior to shipment.

Table 2: Chemical Tanks

Table 3: Tank Accessories

#### 2.2.9 Pressure Gauges

Provide diaphragm type gauges with Bourdon tube and diaphragm compartments filled completely with oil, and made of materials suitable for the application. Install diaphragm seals at each gauge connection to isolate gauges from corrosion, sludge or other hazards of the process fluid. Provide seal material that is compatible with the oil in the gauge and the process fluid.

#### 2.2.10 Mixers and Injectors

##### 2.2.10.1 Static Mixer

Provide a tab style static mixer manufactured by Westfall Manufacturing. Mixers shall be manufactured from fiberglass material and rated for the pressure of the application.

##### 2.2.10.2 Injectors

Introduce injectors for chemical solution into the pipeline mains by means of a CPVC plastic injection nozzle, or by means of a suitable diffuser tube inserted through a corporation cock. Construct the device for introducing the solution into a pressure main in such a way that accidental breakage of discharge hose or tubing does not cause water to escape from the pipeline, and allows disassembling of the unit without leakage. Injectors shall be manufactured by Saf-T-Flo and rated for the

pressure of the application.

#### 2.2.10.2.1 Sample Waste Pump and Tank

Provide sample waste pump model 730EnN drive with 720 R pumphead and controls for pumping analyzer waste to the sanitary sewer drain. Provide 19 mm diameter hose. Pumps capacity to be 30 psig. Pump shall be provided with a control panel to control the speed of the pump based on the level of water in the associated waste tank. Waste tank to be 10 gallons size and provided with a connection to supply water to the waste pump and inlet collection line.

#### 2.2.11 Piping

##### 2.2.11.1 Backflow Preventor

Provide backflow prevention devices or air gaps on tank fill lines in accordance with NAPHCC NSPC.

##### 2.2.11.2 Chemical Solution Piping

Incorporate provisions to allow solution piping to be conveniently and safely bleed off trapped air and minimize infiltration of air bubbles. Provide chemical solution piping in accordance with Section 40 05 13 PIPELINES, LIQUID PROCESS PIPING.

##### 2.2.11.2.1 Smaller than 1 inch Diameter

Chemical solution piping smaller than 1 diameter must be braided PVC tubing rated for 250 psig by Kuri-tec. Make joints for tubing using a clamp-type mechanical coupling.

##### 2.2.11.2.2 Piping 1 inch Diameter or Greater

Chemical solution piping of 1 inch diameter or larger must be CPVC pipe conforming to ASTM F441/F441M, sch. 80

#### 2.2.12 Electrical Work

Provide electric motor-driven equipment complete with motor, motor starter, and controls. Provide electrical equipment and wiring in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Implement hazard classifications in accordance with NFPA 70.

##### 2.2.12.1 Motor Starters

Provide motor starters complete with thermal overload protection and other appurtenances necessary for the motor control specified.

##### 2.2.12.2 Control and Protective Devices

Provide manual or automatic control and protective or signal devices required for the operation and any control wiring required for controls and devices. Provide motor controls conforming to NEMA ICS 1 or NEMA ICS 2. Prewire equipment to the maximum practicable extent. Control cabinets must conform to the requirements of UL 50, NEMA 250, Type 4X.

#### 2.2.13 Equipment Appurtenances

Provide PTFE coated steel, cadmium plated or Type 316 stainless steel bolts, nuts, anchors, washers and all other types of supports necessary for the installation of the equipment.

#### 2.2.14 Factory Painting

Factory painting must conform to manufacturer's standard factory finish, provided it does not discolor in the presence of hydrogen sulfide fumes, high water vapor atmosphere, alkaline water vapor, and concentrated chlorine (oxidizing) conditions. Coating must be at least 1.75 mils thick.

#### 2.2.15 Factory Test Report

Factory examine fiberglass tanks in accordance with ASTM E1067/E1067M prior to shipping. Furnish a copy of the corresponding test report with each tank.

### 2.3 MATERIALS

Submit Safety Data Sheets in conformance with ANSI Z400.1/Z129.1 for each chemical. Provide a 90 day supply at the maximum pumping rate for each feeder or pair of duplexed feeders. Submit two copies of certification stating that each supplied chemical meets the following requirements.

#### 2.3.1 Acids

Sulphuric (sulfuric acid).

Sulfuric acid for ultrafilter cleaning procedures, citric acid for ultrafilter cleaning procedures, Kleen MTC 113 or similar for RO CIP procedures.

#### 2.3.2 Bases

AWWA B501 caustic soda for ultrafilter cleaning procedures, AWWA B501 for continuous injection, Kleen MTC405 or similar for RO CIP procedures.

#### 2.3.3 Disinfecting Agents

AWWA B300 hypochlorites. 2.3.4 Miscellaneous

Sodium bisulfite for continuous injection, antiscalant Hypersperse MDC714 or similar for continuous injection, sodium bisulfite for ultrafilter cleaning procedures.

## PART 3 EXECUTION

### 3.1 EXAMINATION

After becoming familiar with details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy before performing the work.

### 3.2 INSTALLATION

Submit detail drawings containing complete piping, wiring, schematic, flow diagrams, and any other details required to demonstrate that the system

has been coordinated and properly functions as a unit. On the drawings show proposed layout and anchorage of equipment and appurtenances, and equipment relationship to other parts of the work including clearances for installation, maintenance and operation.

### 3.2.1 Chemical Feeding Equipment

Install controlled volume pumps, equipment, and appurtenances to provide a complete and integrated system in accordance with the instruction of the manufacturer and under the direct supervision of the manufacturer's representative.

### 3.2.2 Field Painting

Thoroughly clean, prime and top-coat factory painted items requiring touching up in the field with the manufacturer's standard factory finish provided it does not discolor in the presence high water vapor atmosphere, alkaline water vapor, and concentrated chlorine (oxidizing) conditions. Paint the equipment which did not receive a factory finish as specified in Section 09 90 00 PAINTS AND COATINGS. Coating must be at least 1.75 mils thick. Provide piping identification as specified in Section 09 90 00 PAINTS AND COATINGS. Mark pipe carrying materials not listed in Section 09 90 00 PAINTS AND COATINGS in accordance with ASME A13.1.

### 3.2.3 Framed Instructions

Submit framed instructions for approval prior to postings. Final size must be easy to read by operators 5 feet from instructions. Limit frame size to 11" x 17" size provide multiple frames if needed. Post framed instructions, containing wiring and control diagrams, where directed. Post condensed operating instructions as outlined in paragraph 3.4.2 Operating Instructions. Post the framed instructions before acceptance testing of the systems.

## 3.3 FIELD QUALITY CONTROL

### 3.3.1 Testing

After installation of each controlled volume pump, carry out operating tests as specified below to assure that the chemical metering installation operates properly. If any deficiencies are revealed during any tests, correct such deficiencies and reconduct the tests. Submit reports of all tests in booklet form prior to final acceptance of the installation. Show all field tests performed to adjust each component and all field tests performed to prove compliance with the specified performance criteria, upon completion and testing of the installed system. Indicate in each test report the final position of controls.

#### 3.3.1.1 Tank Testing

Clean tanks of loose debris and dry prior to testing. Field test tanks for leaks or damage in shipment. Hydrostatically test tanks to detect large leaks and then with the specified chemical to detect small leaks. Test each tank with its solution for a period of 24 hours at which time no visible leakage is evident. Supply all pipes, hoses, pumps, water, power and other equipment required to convey the test liquids and to carry out the tests. Repair damage or leaks in tanks or replace tanks. Replace damaged ceramic tanks.

#### 3.3.1.2 Controlled Volume Pumps - Operational Tests

Test pumps to demonstrate that the pumps are capable of operating without vibration or leakage. Perform testing at the pump's maximum flow rate and at half the flow rate. Demonstrate testing while controlled and operated in all feasible modes with the pumps operated singly and in unison. Plot the response of each pump on curves for the various operating pressures encountered and the results compare to the curves shown on the manufacturer's published pump data. If control characteristic curves are not available at the time of testing, the pump manufacturer's service engineer must generate such curves for each pump; graphically depicting the pump displacement at 25, 50, 75, and 100 percent of motor speed for SCR equipped pumps, and at 25, 50, 75, and 100 percent of maximum stroke position for all pumps. Generate curves only for the specified back pressure. Chemical can be recycled back to the bulk tank. If chemical cannot be safely handled, the general contractor and request from the Engineer to use water instead of the chemical of interest.

#### 3.3.1.3 System Pressure Tests

Carry out tests at system pressure. Manually control back pressure valves for this testing, and reset as necessary after testing. The time to deliver a given quantity of chemical at a given stroke and speed setting must be the same at all pressures.

#### 3.3.1.4 Flow Tests

Test pumps to demonstrate zero gpm flow at a zero stroke or speed setting. Failure to meet this test is cause for rejection. Test pumps through full range of performance. Submit test results to prove pump functions within specified system parameters.

#### 3.3.1.5 Synchronization Tests

Operate the pumps for a period of 4 hours to demonstrate that the double diaphragm systems do not lose their synchronization. Loss of synchronization is also cause for rejection; repair or replace the pump as necessary to achieve synchronization. Fully retest repaired or replaced equipment.

#### 3.3.2 Chemical Waste

Neutralize chemicals wasted during testing procedures to achieve a pH value between 6.5 and 9.5 and a chlorine concentration of not more than 1 percent (10,000 mg/L). Route all chemicals wasted during testing procedures to the sanitary sewer at a rate that the process can assimilate without upset.

#### 3.3.3 Manufacturer Field Service

Provide the services of a manufacturer's representative who is certified by manufacturer in the installation, adjustment, and operation of the equipment specified. The representative must supervise the installation, adjustment, and testing of the equipment. Submit certificates to Contracting Officer for approval.

### 3.4 CLOSEOUT ACTIVITIES

#### 3.4.1 FIELD TRAINING

Conduct a field training course for designated operating, maintenance and supervisory staff members. Provide training for a total period of 8 hours of normal working time and start after the system is functionally complete but prior to final acceptance tests. Cover all of the items contained in the Operating and Maintenance Instructions during field training.

#### 3.4.2 Operating Instructions

Submit complete copies of operating instructions outlining the step-by-step procedures required for system startup, operation and shutdown. Include in the instructions the manufacturer's name, model number, service manual, parts list, and brief description of all equipment and their basic operating features. Also include in the instructions as-built drawings of the piping layout, equipment layout, simplified wiring and control diagrams of the system as installed, and flow diagrams.

#### 3.4.3 Maintenance Instructions

Submit complete copies of maintenance instructions listing routine maintenance procedures, possible breakdowns and repairs, and trouble-shooting guides.

-- End of Section --



Attachment #1: Table 1 Chemical Feed Pumps

| Tag       | Name                                                          | Concentration | Control                   | Pump Manufacturer | Model No                   | Flow     | Head     | HP    | Speed | PH | HZ | Volt |
|-----------|---------------------------------------------------------------|---------------|---------------------------|-------------------|----------------------------|----------|----------|-------|-------|----|----|------|
| 02-01-001 | Bleach UF Break Tank Feed - Biocontrol, Pump No. 1            | 12.5%         | Analogue Or Pulse control | Prominent         | GMXA1604PVT2M000UDC0300EN  | 0.95 gph | 232 psig | 30W   | N/A   | 1  | 60 | 120  |
| 02-01-025 | Bleach Pre - UF Feed - Biocontrol, Pump No. 2                 | 12.5%         | Analogue Or Pulse control | Prominent         | GMXA1604PVT2M000UDC0300EN  | 0.95 gph | 232 psig | 30W   | N/A   | 1  | 60 | 120  |
| 02-01-003 | Bleach CIP, Pump No. 1                                        | 12.5%         | On/off                    | Finish Thompson   | DB9V-E-FF-M219             | 17.1 gpm | 25 psig  | 1.5   | 3450  | 3  | 60 | 460  |
|           |                                                               |               |                           |                   |                            |          |          |       |       |    |    |      |
| 02-01-005 | Citric Acid for CIP, Pump No. 1                               | 50%           | On/off                    | Finish Thompson   | DB9V-E-FF-M219             | 17.1 gpm | 25 psig  | 1.5   | 3450  | 3  | 60 | 460  |
|           |                                                               |               |                           |                   |                            |          |          |       |       |    |    |      |
| 02-07-002 | Sulfuric Acid for UF CIP and waste neutralization, Pump No. 1 | 93%-98%       | Analogue Or Pulse control | Prominent         | S2CBH07220PVT5070UD11001EN | 71.6 gpm | 101 psig | 110 W | N/A   | 1  | 60 | 120  |
|           |                                                               |               |                           |                   |                            |          |          |       |       |    |    |      |
| 02-01-009 | Bisulfite for chlorine neutralization prior to RO, Pump No. 1 | 38%           | Analogue Or Pulse control | Prominent         | GMXA1604PVT2M000UDC0300EN  | 0.95 gph | 232 psig | 30W   | N/A   | 1  | 60 | 120  |
|           |                                                               |               |                           |                   |                            |          |          |       |       |    |    |      |
| 02-01-011 | Bisulfite for CIP, Pump No. 1                                 | 38%           | On/off                    | Finish Thompson   | DB9V-E-FF-M219             | 17.1 gpm | 25 psig  | 1.5   | 3450  | 3  | 60 | 460  |
|           |                                                               |               |                           |                   |                            |          |          |       |       |    |    |      |
| 02-01-013 | Anti-scalent prior to SWRO, Pump No. 1                        | 100%          | Analogue Or Pulse control | Prominent         | GMXA1604PVT2M000UDC0300EN  | 0.95 gph | 232 psig | 30W   | N/A   | 1  | 60 | 120  |
|           |                                                               |               |                           |                   |                            |          |          |       |       |    |    |      |
| 02-01-015 | RO Cleaning Chemical Kleen MCT 113, Pump No. 1                | 100%          | On/off                    | Finish Thompson   | DB9V-E-FF-M219             | 17.1 gpm | 25 psig  | 1.5   | 3450  | 3  | 60 | 460  |
|           |                                                               |               |                           |                   |                            |          |          |       |       |    |    |      |
| 02-01-017 | RO Cleaning Chemical Kleen MCT 405, Pump No. 1                | 100%          | On/off                    | Finish Thompson   | DB9V-E-FF-M219             | 17.1 gpm | 25 psig  | 1.5   | 3450  | 3  | 60 | 460  |
|           |                                                               |               |                           |                   |                            |          |          |       |       |    |    |      |
| 02-01-019 | pH adjustment post RO, Pump No. 1                             | 25%           | Analogue Or Pulse control | Prominent         | GMXA1604PVT2M000UDC0300EN  | 0.95 gph | 232 psig | 30W   | N/A   | 1  | 60 | 120  |
| 02-01-021 | pH adjustment CIP, Pump No. 1                                 | 25%           | On/off                    | Finish Thompson   | DB9V-E-FF-M219             | 17.1 gpm | 25 psig  | 1.5   | 3450  | 3  | 60 | 460  |
|           |                                                               |               |                           |                   |                            |          |          |       |       |    |    |      |

## Attachment #2: Venting Requirements for Polyethylene Tanks

| Venting Requirements For Polyethylene Tanks                             |                                                      |                    |                   |                                                                                 |                    |                   |                                                                                                                         |                    |                   |
|-------------------------------------------------------------------------|------------------------------------------------------|--------------------|-------------------|---------------------------------------------------------------------------------|--------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------|
| Mechanical Pump Fill                                                    | Pneumatic Fill                                       |                    |                   |                                                                                 |                    |                   |                                                                                                                         |                    |                   |
| IF ≤ 1000 gallons                                                       | IF - Vent length ≤ 3 feet                            |                    |                   | IF - Vent length > 3' and ≤ 30'                                                 |                    |                   | IF - Scrubber Application                                                                                               |                    |                   |
| Vent size should equal size of largest fill or discharge fitting        | AND - Vent screen mesh size ≥ 1/4" or no screen used |                    |                   | AND - 3 or less 90° elbows with no other restrictions or reduction in pipe size |                    |                   | Vent pipe size throughout scrubber system <b>CANNOT be reduced!</b>                                                     |                    |                   |
|                                                                         |                                                      |                    |                   |                                                                                 |                    |                   | Centerline of dispersion pipe not to be submersed > 6 inches                                                            |                    |                   |
| IF > 1000 gallons                                                       | Emergency Pressure Relief Cover Required             |                    |                   | Emergency Pressure Relief Cover Required                                        |                    |                   | Perforated dispersion pipe must be same diameter or larger, as vent. Sum of perforations ≥ cross sectional area of pipe |                    |                   |
| Vent size should exceed the largest fill or discharge fitting by 1 inch | Tanker Discharge                                     | Inlet/Fitting Size | Minimum Vent Size | Tanker Discharge                                                                | Inlet/Fitting Size | Minimum Vent Size | Tanker Discharge                                                                                                        | Inlet/Fitting Size | Minimum Vent Size |
|                                                                         | 2"                                                   | 2"                 | 4"                | 2"                                                                              | 2"                 | 6"                | 2"                                                                                                                      | 2"                 | 6"                |
|                                                                         | 3"                                                   | 2"                 | 6"                | 3"                                                                              | 2"                 | 6"                | 3"                                                                                                                      | 2"                 | 8"                |
|                                                                         | 3"                                                   | 3"                 | 6"                | 3"                                                                              | 3"                 | 8"                | 3"                                                                                                                      | 3"                 | 10"               |

(2) 2 inch vents **DO NOT EQUAL** 4 inch venting capacity

rev. Nov 2006

For detailed venting guidelines, please visit our Technical Resources at [www.polyprocessing.com](http://www.polyprocessing.com)

Attachment #3: Table 2 Chemical Tanks

| Tag       | Name                           | Concentration                                                                                                                                                                                                                                                                                                                                                                                                             | Style         | Size, gal | Dia, ft | H, ft   | Ladder | Linear PE Liner | Manway, in | Vent, in | Level Element opening, in      | PRV Return Line, in                   |
|-----------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------|---------|---------|--------|-----------------|------------|----------|--------------------------------|---------------------------------------|
| 05-05-002 | UF Break Tank                  | Seawater - 33 ppth                                                                                                                                                                                                                                                                                                                                                                                                        | Single Wall   | 20,000    | 13'-9"  | 20'-3"  | Yes    | No              | 24         | 8        | 3 -Flanged<br>1-Flanged backup | N/A                                   |
| 05-07-003 | CIP Tank                       | All Chemicals below except Anti-scalent. Max. concentrations in the tank will be as follows (not concurrently):<br>1. 500 mg/L as sodium hypochlorite<br>2. 2 g/L citric acid adjust down to pH 2 with Sulfuric Acid<br>3. 1.4 g/L as sodium bisulfite<br>4. Caustic Sode to neutralize citric acid.<br>5. High and low pH cleaner for RO (MCT 405 and MCT 113).<br>Roughly 40 gallons into 2,700 gals of permeate water. | Double Walled | 5,000     | 8'-6"   | 17'-11" | No     | No              | 24         | 8        | 3 -Flanged<br>1-Flanged backup | -<br>3 inch<br>Permeate<br>water fill |
| 02-07-001 | Citric Acid                    | 50% dry wt/wet wt                                                                                                                                                                                                                                                                                                                                                                                                         | Double Walled | 1,000     | 6'-5"   | 7'-1"   | No     | No              | 17         | 6        | 3 -Flanged<br>1-Flanged backup | 1                                     |
| 02-07-002 | Sulfuric Acid                  | 93% dry wt/wet wt                                                                                                                                                                                                                                                                                                                                                                                                         | Double Walled | 405       | 4'-0"   | 6'-1"   | No     | Yes             | 7 threaded | 4        | 3 -Flanged<br>1-Flanged backup | 1                                     |
| 02-07-003 | Sodium Bisulfite               | 38% dry wt/wet wt                                                                                                                                                                                                                                                                                                                                                                                                         | Double Walled | 1,000     | 6'-5"   | 7'-1"   | No     | No              | 17         | 6        | 3 -Flanged<br>1-Flanged backup | 1                                     |
| 02-07-004 | Caustic Soda                   | 25% dry wt/wet wt                                                                                                                                                                                                                                                                                                                                                                                                         | Double Walled | 1,000     | 6'-5"   | 7'-1"   | No     | No              | 17         | 6        | 3 -Flanged<br>1-Flanged backup | 1                                     |
| 02-07-005 | Antiscalent HYPERSPERSE MDC714 | 100% wet weight                                                                                                                                                                                                                                                                                                                                                                                                           | Double Walled | 1,000     | 6'-5"   | 7'-1"   | No     | No              | 17         | 6        | 3 -Flanged<br>1-Flanged backup | 1                                     |
| 02-07-006 | Kleen MCT 405                  | These are all wet wt./wet wt.<br>1-3% Benzenesulfonic acid<br>3-10% Sodium alkyl benzene sulphonate<br>3-10% Sodium carbonate<br>3-10% N-hydroxyethylenediamine triacetic acid trisodium salt                                                                                                                                                                                                                             | Double Walled | 405       | 4'-0"   | 6'-1"   | No     | No              | 7 threaded | 4        | 3 -Flanged<br>1-Flanged backup | 1                                     |
| 02-07-007 | Sodium Hypochlorite            | 12.5% trade weight                                                                                                                                                                                                                                                                                                                                                                                                        | Double Walled | 1,000     | 6'-5"   | 7'-1"   | No     | Yes             | 17         | 6        | 3 -Flanged<br>1-Flanged backup | 1                                     |
| 02-07-008 | Kleen MCT 113                  | These are all wet wt./wet wt.<br>10-20% Citric Acid<br>3-10% Formic Acid<br>3-10% Hydroxyacetic acid<br>3-10% N-hydroxyethylenediamine triacetic acid trisodium salt                                                                                                                                                                                                                                                      | Double Walled | 405       | 4'-0"   | 6'-1"   | No     | No              | 7 threaded | 4        | 3 -Flanged<br>1-Flanged backup | 1                                     |

Attachment #4: Table 3 Tank Accessories

| Tag       | Name                                 | Overflow Tank and Size, gal** | Outlet, in                        | Overflow, in | Inlet, in                         | Fill, in                             | Gaskets | Metals   | Manufacturer or equal   | Insulation and Heating |
|-----------|--------------------------------------|-------------------------------|-----------------------------------|--------------|-----------------------------------|--------------------------------------|---------|----------|-------------------------|------------------------|
| 05-05-002 | UF Break Tank - Seawater             | No                            | 8 - Bolted double flange fittings | 8            | 8 - Bolted double flange fittings | 3 - BWRO reject return               | EPDM    | Titanium | Syndar Industries       | Yes                    |
| 05-07-003 | CIP Tank                             | Yes (100 gal)                 | 3 - Bellow                        | 3            | 3 - Sidewall                      | Two- 3-inch openings at top of tank* | EPDM    | Titanium | Syndar Industries       | Yes                    |
| 02-07-001 | Citric Acid                          | Yes (50 gal)                  | 1 - bolted flange fitting         | 2            | N/A                               | 1                                    | EPDM    | 316SS    | Polyprocess Safety Tank | No                     |
| 02-07-002 | Sulfuric Acid                        | Yes (50 gal)                  | 1 - bolted flange fitting         | 2            | N/A                               | 1                                    | Viton   | C-276    | Polyprocess Safety Tank | No                     |
| 02-07-003 | Sodium Bisulfite                     | Yes (75 gal)                  | 2 - bolted flange fitting         | 3            | N/A                               | 2                                    | EPDM    | 316SS    | Polyprocess Safety Tank | No                     |
| 02-07-004 | Caustic Soda                         | Yes (75 gal)                  | 2 - bolted flange fitting         | 3            | N/A                               | 2                                    | EPDM    | 316SS    | Polyprocess Safety Tank | No                     |
| 02-07-005 | Anti-scalent<br>HYPERPERSE<br>MDC714 | Yes (50 gal)                  | 1 - bolted flange fitting         | 2            | N/A                               | 1                                    | EPDM    | 316SS    | Polyprocess Safety Tank | No                     |
| 02-07-006 | Kleen MCT 405                        | Yes (50 gal)                  | 1 - bolted flange fitting         | 2            | N/A                               | 1                                    | EPDM    | 316SS    | Polyprocess Safety Tank | No                     |
| 02-07-007 | Sodium Hypochlorite                  | Yes (75 gal)                  | 2 - bolted flange fitting         | 3            | N/A                               | 2                                    | EPDM    | Titanium | Polyprocess Safety Tank | No                     |
| 02-07-008 | Kleen MCT 113                        | Yes (50 gal)                  | 1 - bolted flange fitting         | 2            | N/A                               | 1                                    | EPDM    | 316SS    | Polyprocess Safety Tank | No                     |

Notes

- \* Two 3-inch openings for the following source waters: 1.) heating recirculation, and 2.) neutralization recirculation.
- \*\* Overflow tank will receive the overflow from the bulk tank have a vent and be provided with a 1-inch flanged connection to accept a level and accessories. The float in the tank will alarm the facility. In the case of the CIP tank, it will shut down the CIP loop and/or the plant.

SECTION 46 64 30

TEMPORARY ASW PUMP STATION  
**11/24**

PART 1 GENERAL

1.1 SUMMARY

This Section specifies the temporary ASW pump station equipment and components including associated testing.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

|              |                                                                                      |
|--------------|--------------------------------------------------------------------------------------|
| ASME B1.20.1 | (2013; R 2018) Pipe Threads, General Purpose (Inch)                                  |
| ASME B16.1   | (2020) Gray Iron Pipe Flanges and Flanged Fittings Classes 25, 125, and 250          |
| ASME B16.5   | (2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard |
| ASME B31.3   | (2022; Errata 2023) Process Piping                                                   |
| ASME B40.100 | (2022) Pressure Gauges and Gauge Attachments                                         |
| ASME B16.5   | (2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard |

AMERICAN WATER WORKS ASSOCIATION (AWWA)

|                  |                                                                          |
|------------------|--------------------------------------------------------------------------|
| AWWA C104/A21.4  | (2022) Cement-Mortar Lining for Ductile-Iron Pipe and Fittings for Water |
| AWWA C110/A21.10 | (2021) Ductile-Iron and Gray-Iron Fittings                               |
| AWWA C111/A21.11 | (2023) Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings  |
| AWWA C150/A21.50 | (2021; R 2023) Thickness Design of Ductile-Iron Pipe                     |
| AWWA C151/A21.51 | (2023) Ductile-Iron Pipe, Centrifugally Cast                             |
| AWWA C542        | (2016) Electric Motor Actuators for Valves and Slide Gates               |

ASTM INTERNATIONAL (ASTM)

|                 |                                                                                                                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASTM A48/A48M   | (2022) Standard Specification for Gray Iron Castings                                                                                                          |
| ASTM A126       | (2004; R 2023) Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings                                                           |
| ASTM A182/A182M | (2024) Standard Specification for Forged or Rolled Alloy and Stainless Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service |
| ASTM A240/A240M | (2025) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications      |
| ASTM A312/A312M | (2022a) Standard Specification for Seamless, Welded, and Heavily Cold Worked Austenitic Stainless Steel Pipes                                                 |
| ASTM A351/A351M | (2024; E 2025) Standard Specification for Castings, Austenitic, for Pressure-Containing Parts                                                                 |
| ASTM A479/A479M | (2024) Standard Specification for Stainless Steel Bars and Shapes for Use in Boilers and Other Pressure Vessels                                               |
| ASTM A494       | (2018) Standard Specification for Castings, Nickel and Nickel Alloy                                                                                           |
| ASTM B16        | Standard Specification for Free-Cutting Brass Rod, Bar and Shapes for Use in Screw Machines                                                                   |
| ASTM B61        | (2015; R 2021) Standard Specification for Steam or Valve Bronze Castings                                                                                      |
| ASTM B164       | (2003; R 2014) Standard Specification for Nickel-Copper Alloy Rod, Bar, and Wire                                                                              |
| ASTM B494       | (2020) Standard Specification for Primary Zirconium                                                                                                           |
| ASTM B505/B505M | (2023) Standard Specification for Copper Alloy Continuous Castings                                                                                            |
| ASTM F1199      | (2021) Standard Specification for Cast (All Temperatures and Pressures) and Welded Pipe Line Strainers (150 psig and 150 degrees F Maximum)                   |
| ASTM F2389      | (2024a) Standard Specification for Pressure-rated Polypropylene (PP) Piping Systems                                                                           |

FEDERAL SPECIFICATIONS

|              |                                                                                            |
|--------------|--------------------------------------------------------------------------------------------|
| FS HH-I-551  | Insulation Block and Boards, Thermal<br>(Cellular Glass)                                   |
| FS WW-P-460B | (4) Pipe Fittings; Brass or Bronze<br>(Thread) 125- and 250-pound                          |
| FS WW-U-516B | Unions, Brass or Bronze, Threaded Pipe<br>Connections and Solder-Joint Tube<br>Connections |
| FS WW-U-531E | Unions, Pipe, Steel or Malleable Iron;<br>Threaded Connection, 150 Lbs. and 250 Lbs.       |

HYDRAULIC INSTITUTE (HI)

|                  |                                                                      |
|------------------|----------------------------------------------------------------------|
| HI ANSI/HI 9.6.3 | (2024) Rotodynamic Pumps - Guideline for<br>Operating Regions - B120 |
|------------------|----------------------------------------------------------------------|

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

|           |                              |
|-----------|------------------------------|
| NEMA MG 1 | (2021) Motors and Generators |
|-----------|------------------------------|

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

|         |                                                           |
|---------|-----------------------------------------------------------|
| NFPA 70 | (2023; ERTA 1 2024; TIA 24-1) National<br>Electrical Code |
|---------|-----------------------------------------------------------|

MILITARY SPECIFICATIONS (MIL SPEC)

|             |                                                                     |
|-------------|---------------------------------------------------------------------|
| MIL-A-17472 | Gaskets                                                             |
| MIL-R-1149D | Gasket Materials, Synthetic Rubber, 50 and<br>65 Durometer Hardness |

NSF INTERNATIONAL (NSF)

|             |                                                                   |
|-------------|-------------------------------------------------------------------|
| NSF/ANSI 14 | (2023) Plastics Piping System Components<br>and Related Materials |
|-------------|-------------------------------------------------------------------|

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Sequence of Construction; G, AE

SD-02 Shop Drawings

System Coordination; G, AE

General Arrangement and Dimensional Drawings for Pumps, Piping Systems, Cooling Water Intake Structure, and associated equipment and components; G

SD-03 Product Data

TEMPORARY ASW SYSTEM; G, AE

Submittal Package to be inclusive of the following items:

- (a) Cooling Water Intake Structure
- (b) Airburst Backwash System
- (c) Pumps
- (d) Y-Strainer
- (e) Motors
- (f) Gate Valves
- (g) Butterfly Valves
- (h) Check Valves
- (i) Air and Vacuum Valves
- (j) Pressure Relief Valve
- (k) Pressure Gauges
- (l) Magnetic Flow Meter
- (m) Pressure Transmitter
- (n) Pressure Switches
- (o) Stem Covers
- (p) Discharge Piping
- (q) Hard Piping
- (r) Fittings and Flanges
- (s) Bolts, Nuts, Washers and Fasteners
- (t) Gaskets
- (u) Unions
- (v) Pipe Hangers, Inserts, and Supports
- (w) Insulation and Heat Tracing
- (x) Electrical Equipment
- (y) Controls
- (z) Field Testing and Commissioning Procedures



SD-06 Test Reports

Temporary ASW System Performance; G, AE

Field Quality Control and Commissioning; G, AE

SD-07 Certificates

Manufacturer's Representative; G, AE

Pumps; G, AE

Gate Valves; G, AE

Butterfly Valves; G, AE

Check Valves; G, AE

Air and Vacuum Valves; G, AE

Pressure Relief Valve; G, AE

Magnetic Flow Meter; G, AE

Pressure Transmitter; G, AE

Piping; G, AE

Fittings and Flanges; G, AE

Bolts, Nuts, Washers and Fasteners; G, AE

Gaskets; G, AE

Union; G, AE

1.4 QUALITY ASSURANCE

1.4.1 Manufacturer Services

Provide the services of a manufacturer's representative experienced in the installation, adjustment, and operation of the equipment specified. The representative must supervise the installation, adjustment, testing of the equipment, and conduct training.

Submit the names and qualifications of the manufacturer's representative and training engineers and written certification from the manufacturer that the representative and trainers are technically qualified.

1.4.2 Standard Products

Provide material and equipment which are the standard products of a manufacturer regularly engaged in the manufacture of such products and that essentially duplicate equipment that has been in satisfactory operation at least 2 years prior to issuance of this solicitation. Support equipment with a service organization that is reasonably convenient to the jobsite. Pumps and motors of the same types must each be the product of one manufacturer.

#### 1.4.3 Conformance with Agency Requirements

Where materials or equipment are specified to be an approved type, attach the seal or label of approval from a nationally recognized testing agency, adequately equipped and competent to perform such services. A written certificate from the testing agency must accompany the materials or equipment and be submitted stating that the items have been tested and that they conform to the applicable requirements of the specifications and to the standards listed herein. The certificate must indicate the methods of testing used by the testing agency. In lieu of a certificate from a testing agency, published catalog specification data, accompanied by the manufacturer's certified statement to the effect that the items are in accordance with the applicable requirements of the specifications and the referenced standards, will be considered and may be acceptable as evidence that the items conform with agency requirements.

#### 1.5 DELIVERY, STORAGE, AND HANDLING

Protect equipment, delivered and designated for storage, from the weather, humidity and temperature variations, dirt and dust, or other contaminants.

### PART 2 PRODUCTS

#### 2.1 SYSTEM DESCRIPTION

A Temporary ASW Pumping System will be required to bypass the existing ASW and CWIS system, and will be considered as a single, standalone package, encompassing all components required meet NDPES permit requirements and performance requirements. The delegated design, selection, installation, performance testing, training, maintenance, monitoring, repair and subsequent demolition of the Temporary ASW System will be the responsibility of the Contractor. Operation of the Temporary ASW System will be provided by NAVSEA Code 984 and Shop 99DD. The contractor must provide initial training to NAVSEA personnel for system operation, and maintain operational capability of the Temporary ASW System at all times between decommissioning and recommissioning of the permanent ASW system unless there is an approved outage. Any outage associated with maintenance or repair of the Temporary ASW System must be coordinated with Code 984 and follow the outage request procedures outlined in paragraph 1.4 of Section 01 14 00.00 22 Work Restrictions.

##### 2.1.1 System Coordination

Submit drawings containing complete wiring and piping schematic diagrams and any other details required to demonstrate that the system has been coordinated and will properly function as a unit. Show the proposed layout and anchorage of equipment and appurtenances, and equipment relationship to other parts of the work including clearances for maintenance and operation. Provide a complete listing of equipment, materials and miscellaneous components including mechanical seals, bearings, and couplings.

##### 2.1.2 Safety Requirements

Fully enclose or guard couplings, projecting set-screws, keys, and other rotating parts, that pose an entangling hazards.

### 2.1.3 General

All pumps must conform to the standards of the Hydraulic Institute with respect to nomenclature, composition, construction, dimensions, tolerances, safety, operating characteristics, performance, quality, rating, testing, and related items. Drives and intermediate shafting and appurtenances must be provided by the pump manufacturer.

## 2.2 MATERIALS AND EQUIPMENT

### 2.2.1 Nameplates

Securely affix a standard nameplate to pumps and motors in a conspicuous place showing the manufacturer's name, address, type or style, model, serial number, and catalog number. In addition, for each pump show the capacity in gpm at rated speed in rpm and total head in feet of water. For each electric motor show at least the minimum information required by NEMA MG 1. Show such other information as the manufacturer may consider necessary to complete identification on the nameplate. Pumps must be listed and labeled by UL.

### 2.2.2 Pump Characteristic

Operate pumps at optimum efficiencies to produce the most economical pumping system under the conditions encountered and size to make optimum match with the system head curve as shown. Pumps must furnish not less than 150 percent of rated capacity at a total discharge head of not less than 65 percent of total rated head. The shutoff total head must not be greater than 120 percent of total rated head. Operate pumps at specified system fluid temperatures without vapor binding and cavitation. Operate pumps to HI ANSI/HI 9.6.3 standard for Preferred Operating Region (POR).

### 2.2.3 Equipment Data

Submit catalog information, certified pumps curves, rated capacities, final impeller dimensions, and accessories provided for the product indicated. Indicate operating point of each pump on curves. Furnish pump curves for each pump and combination of pumps designed to operate in parallel. The pump curve must show as a minimum; bhp, flow, total dynamic head, efficiency, NPSH, impeller diameter and system curve (individually and in combination for each pump operating in a parallel application). Select pumps operating in parallel operation to cross the system curve when operating individually.

## 2.3 TEMPORARY ASW SYSTEM

### 2.3.1 Cooling Water Intake Structure

The temporary intake system components and equipment shall be furnished as a complete integrated system by one manufacturer to ensure a properly designed and integrated intake system.

System shall be designed for hydrostatic load depth of 30 feet and collapse rate of 13 psi. Install at an elevation of -11.5 feet (NAVD88) or lower, to meet all low tide conditions.

Intake Screens assembly shall be horizontal orientation, of all-welded continuous slot construction and shall be designed for air flushing of screen surface for debris removal. Equipment manufacturer shall maintain

an ongoing quality assurance program, including ISO-9000 certification. All welders shall maintain certification to ASME Section IX. Copies of certification shall be provided upon request. The single manufacturer supplying this equipment shall be able to furnish proof of at least 5 installations and 10 years of manufacturing equipment of similar technology.

Manufacturer shall submit a complete and detailed pre-submittal package containing the below information:

- a. Detailed layout drawings.
- b. Supporting flow distribution data via a CFD (Computational Fluid Dynamics) analysis.
- c. Weld certifications.
- d. Evidence of a recognized ongoing quality assurance program.
- e. Detailed component specifications and catalog cuts as required.
- f. Detailed list of all variations required from the original design if an alternate screen is used, referencing appropriate sections of the specifications and locations on the drawings.

#### 2.3.1.1 Capacity and Strength

Intake assembly capacity shall be 950 GPM at a maximum through-slot velocity, as a result of water withdrawal, of 0.35 feet per second. The corresponding average through-slot velocity shall be 80-90 percent of the maximum velocity. The hydraulic design of this system is based on this maximum headloss - screens that exceed this headloss are not acceptable. The manufacturer's clean screen assembly headloss shall be stated in the bid documents.

Total intake assembly of 1900 GPM shall be handled by two horizontal screens of intake drum screen assembly.

Evidence of the intake assembly capacity and flow distribution shall be proven by a Computational Fluid Dynamic (CFD) analysis, supplied by the manufacturer. The CFD Analysis Method must be verified by actual physical testing.

Intake assembly shall be designed to a maximum 4.33 psi negative pump pressure or differential headloss. Design stress used for determining strength of the assembly shall be no more than 90 percent of the published yield strength of the material used. Strength calculations verifying compliance with these criteria shall be provided upon request.

#### 2.3.1.2 Construction and Slot Opening Size

Intake screen surface wire #69, support beam and stiffener structure shall be an all-welded matrix designed to provide the specific strength with minimal interference with the through screen flow pattern.

End plates and tee body shall be a minimum of 0.105 inches thick. All structural butt welds shall be full penetration fillet welds and shall be the thickness of the thinner component.

Screen slot size shall be 0.8 mm. The open area for this slot opening shall be 30.73 percent. Slot size shall be controlled and continuously monitored during manufacture.

For slot openings of 0.040-inch through 0.100-inch the mean slot size shall be within +/- 0.002-inch with a standard deviation no greater than 0.002-inch throughout the assembly.

For slot openings greater than 0.100-inch the mean slot size shall be within +/- 0.003-inch with a standard deviation no greater than 0.003-inch throughout the entire assembly.

Intake Screen shall be manufactured of 70-30 copper nickel material.

#### 2.3.1.3 Intake Screen Butterfly Valves

Butterfly valves for intake screens shall be triple offset design with sealing elements to provide zero leakage through non-rubbing 90-degree rotation. Valve shall be suitable for sea water applications, designed to ISO 5752, API 609 and ASME B34. Flanges shall be per ASME B16.5, ASME B16.47. Pressure rating Class 150, temperature range minus 76 degrees F to 842 degrees F. Metal seal ring in Duplex and highly wear-resistant seat shall ensure prolonged superior tightness. Asymmetric design and disparate torque shall ensure tightness in both directions.

Valves must be electrically actuated but shall also include manual top-side handwheels.

##### 2.3.1.3.1 Butterfly Valve Materials

- a. Body: Double flanged, Super Duplex stainless steel.
- b. Disc: Stainless steel ASTM A351/A351M.
- c. Weld surfacing on seat: Stellite Grade 21.
- d. Shaft: ASTM A479/A479M Type XM19.
- e. Packing: Graphite
- f. Bolts and Nuts: ISO 3506 A4 (AISI 316).
- g. Packing and Bottom Flanges: ASTM A240/A240M Type 316.
- h. Bracket: Carbon steel.

##### 2.3.1.3.2 Electric Actuator

Provide quarter-turn electric actuators for intake screens butterfly valve control. The actuator shall be compact and reliable in operation, with high torque output. Actuators shall utilize double reduction worm gear drives - bronze worm gears and alloy steel worms for self-locking with minimum backlash. Motor shall be with Class F insulation, and built-in thermals for motor protection. Actuator shall have 7-10 watt space heater to prevent condensation. Operators shall be AWWA C542 complaint. Mount valve actuator on a floor stand.

#### 2.3.1.3.3 Enclosure

Enclosure shall be waterproof, IP67, NEMA 4, suitable for submersible and hazardous operation. Materials shall be high grade aluminum alloy with corrosion resistant polyester coating, and internally and externally anodized casing. The main housing cover shall be spigot tapper, fitted with captive bolts for removal. Inside the enclosure electrical and mechanical design shall be color-coded for easy maintenance. Limit switch settings can be adjusted without disturbing other switch settings.

#### 2.3.1.3.4 Controls

Local Control Units (LCU) shall provide a way to check actuator status and control operation. Controls shall have two selector switches: Open/Close and Local/Off/Remote. Provide continuous mechanical position indicator with large window display. LED display shall indicate valve positions: Open, Close, Remote, Fault, Power On. Supply with control transformer 110V/230V.

Proportional Control Unit (PCU) shall have control (input) and feedback (output) of valve position for modulating service. Provide manual handwheel with declutching mechanism for emergency manual operation.

#### 2.3.1.3.5 Rigid Rod System

Rigid rod system shall be used for remote valve operation. Gear boxes or swivel gear joints shall be provided to accommodate bends. Materials shall be high grade bronze and stainless steel, suitable for sea water applications. System components shall be low maintenance, gear boxes shall be sealed for life.

#### 2.3.1.3.6 Delegated Design

Contractor is responsible for design and selection of the remote valve rigid rod components based on the valve size, operating speed, operating torque, transmission system losses, and as recommended by the system manufacturer.

#### 2.3.1.4 Emergency Bypass

The intake screen shall include an emergency bypass with a minimum 1 inch by 2 inch trash rack.

#### 2.3.1.5 Structural Support

The intake screen must be structurally independent of seawall. Structure must be supported by floating rack and may be secured to existing bollards on Berth. No modifications to the seawall are permitted.

The screen manufacturer is responsible for the design the intake support system.

#### 2.3.2 Airburst Backwash System

Intake screen supplier shall provide, as part of the overall temporary intake screen system, an air backwash system designed to remove debris from the screen surface by delivering a suitable volume of compressed air to the inside of the screen body. The exiting air shall scour the screen surface to maintain adequate design flow and through slot velocity

characteristics.

The backwash system shall consist of an integrated system of compressor, receiver tank, valves and control panel. All components shall be skid mounted and arrive on site ready to operate. Any system provided requiring field assembly and not ready to operate shall not be acceptable.

Compressed air lines after the air compressor serving the intake structure shall be 70-30 copper nickel Class 700 with socket weld joints. The compressor shall be a reciprocating type, 5 HP pressure lubricated, and shall be sized to recover from each backwash in 15 minutes. The receiver shall be a 120 gallon horizontal receiver, ASME coded for 200 psig, sized for the system piping and to displace three screen volumes of air at the screen in 3-5 seconds during a backwash to provide suitable debris removal and cleaning. The receiver shall be equipped with a 4-inch isolated pressure gauge, safety valve and automatic drain valve. The system shall include one butterfly valve rated at 200 psi with standard ANSI class 125/150 flanges. There shall be one valve per screen and shall be sized to match the tank flange and the backwash piping.

The above butterfly valve must be provided with a pneumatic actuator with direct mounted solenoid valves, position indicator and position switches. The system must include a suitably size air receiver interconnected and charged by the main receiver to allow valve operations.

A NEMA 4X control panel shall be included that will contain as a minimum, the motor starter for the compressor, control power transformer and a relay logic to perform the specified control functions.

Controls shall allow timed automatic or manual initiated air cleaning cycles. A cycle consists of cleaning all intake system screens. The control system included shall allow a full week duration of programmed Hydroburst cycles.

Contractor shall provide all interconnecting piping and control wiring between the intake screen assembly and the Hydroburst air backwash system.

#### 2.3.3 Pumps

Provide (2) temporary trailer-mounted suction lift pumps capable of pumping a minimum of 950 GPM at 260 ft Total Dynamic Head in lead-standby operation, for seawater intake application.

Pumps should be self-priming and all design, installation, operation, and maintenance of the system is the responsibility of the contractor.

#### 2.3.4 Y-Strainer

Strainer shall meet ASTM F1199, and shall be the cleanable "Y" type, the same size as the pipeline. Strainer bodies shall be fabricated of cast or forged 316 stainless steel with bottoms drilled, and tapped. Provide blowoff outlet with pipe nipple, gate valve, and discharge pipe nipple. The bodies shall have arrows clearly cast on the sides indicating the direction of flow. Y-strainers to be blown off weekly to prevent clogging.

Provide strainer with removable cover and sediment screen. The screen shall be made of minimum 22 gauge corrosion-resistant steel, with small perforations numbering not less than 400 per square inch to provide a net free area through the basket of at least 3.30 times that of the entering

pipe. The flow shall be into the screen and out through the perforations.

#### 2.3.5 Gate Valves

All gate valves 12 inches and smaller must be manufacturer's standard pattern flanged unless otherwise indicated, resilient seated, outside screw and yoke, rated at 225 psi working pressure at temperatures between 0 to 66 deg C (32 to 150 deg F). The valve body, bonnet, bonnet yoke, gland, and wedge must all be cast bronze meeting ASTM B61. The gasket must be constructed of synthetic rubber, packing of Teflon, and seat ring of cast bronze conforming to ASTM B505/B505M. The gate must be fully supported throughout travel by an integrally cast tongue and groove fit between it and the valve body. All gate valves must open by turning counterclockwise and be handwheel operated.

#### 2.3.6 Butterfly Valves

All butterfly valves 2 inches and larger, must be manufacturer's standard pattern, lugged to typical ANSI Class 150 flanges, and resiliently seated. The valves must be rated for a working pressure of 200 psi at temperatures between -29 to 121 deg C (-20 to 250 deg F). The valves must be designed for use in salt water. The body must be constructed of bronze conforming to ASTM B61. The shaft and disc must be constructed of Monel conforming to ASTM B164 Class A and ASTM A494 M30C respectively. The seat and seals must be constructed of Buna N. Butterfly valves must also be operated by manual locking hand levers.

All lugged butterfly valves must be equipped with custom fabricated flange spacers constructed of 1.125 inch thick Type 316 stainless steel plate conforming to ASTM A240/A240M. The fabricated flanged spacers must be drilled to conform with an ASME B16.5 bolt pattern. One spacer must be added to each side of the lugged butterfly valve to allow the internal disc to open properly when used with polypropylene piping.

#### 2.3.7 Check Valves

The check valves on the ASW Pump System discharge piping must be of swing check type designed to allow flow forward and downstream of the pump, but disallow flow reversal. The valves must be Class 150 flanged and rated for a 225 psi WOG.

The check valves must be designed for use in salt water and to take the full thrust from the Vertical Turbine Pump at shut-off head. The body, cap, and lever must be bronze conforming to ASTM B61.

The disc must be a nickel copper alloy conforming to ASTM B494 GR M30-C. The lever plug, pin, disc nut, washer, and cotter pin must be a nickel copper alloy conforming to ASTM B164. The gasket must be of synthetic rubber type conforming to MIL-R-1149D.

The manufacturer of the check valves must have been regularly engaged in the design and manufacture of the swing check valves for at least five (5) years and submit at least five (5) similar installations in service for a minimum of five (5) years.

#### 2.3.8 Air and Vacuum Valves

The air valves for the ASW Pump System must vent large quantities of air thru the orifice when the pump starts, close tight when liquid enters, and



permit large quantities of air to re-enter thru the orifice when the pump stops to prevent a vacuum from forming in the pump discharge column.

The body and cover must be fabricated of cast iron, conforming to ASTM A126, Gr. B. The baffle must be cast iron, conforming to ASTM A48/A48M and must shield the float from direct impact of air and water to prevent premature float closure. The seat must be Buna-N and slip fit into the baffle or cover and lock in place without any distortion, but be easily removable. The float must be Type 316 stainless steel center guided for positive seating and rated for 1000 psi non-shock service. The entire float and baffle assembly must be shrouded with a brass water diffuser, conforming to ASTM B16, to prevent water from slamming the float shut. The outlet of the valve must be flanged and piped as shown on the Drawings. The interior of the valves must be protected by a factory applied epoxy coating resistant to salt water with a dry film thickness of 10 mils. The exterior of the valves must be factory primed with two coats of phenolic red oxide primer, compatible with the finish painting as specified under 09 90 00 PAINTS AND COATINGS.

#### 2.3.9 Pressure Relief Valve

A 3-inch Pressure Relief Valve must maintain constant upstream pressure by by-passing or relieving excess pressure and must maintain close pressure limits without causing surges. The valve must be of the control globe type and must be capable of operating at a pressure of 120 psi with a differential pressure of 106 psi and must be adjustable. The valve body must be stainless steel with 150 lb flanged ends, single seat, unbalanced with cage retained seat, full port, trim must be stainless steel and equal %, EPDM packing, and PEEK bearings. The valve must be equipped with an electric actuator with a control signal of 4-20 mA CL, 115 VAC, IP 67 with heater, and a capacitive fail safe mode to be able to open with a loss of power or signal. The inlet and outlet of the valve must be flanged and piped as shown on the Drawings.

#### 2.3.10 Pressure Gauges

Each pressure gauge must be liquid filled with open front design and must be suitable for bottom stem mounting. The Bourdon tube must be phosphor bronze with brass socket. Each gauge must be 3-1/2 inches in diameter, must have a phenolic or nylon gauge case and must have graduated markings for every 1 or 2 psi. Gauge accuracy must be +/- 1.0 percent of span, ASME B40.100, Grade 1A.

- a. Each gauge must be furnished and installed with a diaphragm seal. The seal's diaphragm and bottom section must be constructed from 316 stainless steel. Each diaphragm seal must have a 1/4 inch NPT flushing connection with brass gauge cock.
- b. Each gauge and its diaphragm seal must be filled with silicone oil.
- c. Each gauge on the side of a pump must be a compound gauge.
- d. Each pressure gauge assembly must have bronze ball valve between the diaphragm seal and the pumped fluid.
- e. Each gauge must have a piston type pressure snubber between it and its diaphragm seal.

Gauges must be furnished in accordance with the following list:

- a. Vertical Turbine Pumps:  
Suction: 3 at 0-30 psi  
Discharge: 3 at 0-125 psi
- b. 8-inch Simplex Strainers at Suction & Discharge of Each Strainer  
Suction: 3 at 0-150 psi  
Discharge: 3 at 0-150 psi
- c. Downstream of Flow Meter at Pressure Transducer  
Discharge: 1 at 0-150 psi

#### 2.3.11 Magnetic Flow Meter

One (1) 8-inch magnetic flow meter, with a flow range of 30 gpm to 5975 gpm, must be supplied for flange mounting and installed as shown on the Drawings. The meter must be of the low frequency electromagnetic induction type and must produce a DC pulse signal directly proportional and linear to the liquid flow rate. The body must be of the flanged design and must be suitable for mounting between ANSI Class 150 flanges, by using the appropriate mounting studs and nuts supplied. Pressure limitations must be of maximum pressure dictated by flange rating. Flange material must be steel ASTM A105. The liner must be of hard rubber and the electrodes must be Hastalloy-C22. The maximum fluid temperature must be 80 deg C (176 deg F). The meter must be designed for operation on 100-230VAC and 60 Hz. The flow converter must be a compact model with a local display and control unit. The electronic standard output must be 4 to 20 mA dc. The current output must be basic I/O HART active/passive. The flow converter housing must be aluminum.

#### 2.3.12 Pressure Transmitter

One (1) 0-150 PSI (adjustable), titanium saltwater-rated pressure transmitter with 1/2-inch MS male threads and surge suppressor must be supplied and installed as shown on the Drawings. There must be a 4-20mA signal output (directly proportional to the adjusted span) from the transmitter to both the System Control Panel and at the Johnson/Yokogawa system. I/I Booster(s) must be installed in the loop as needed. Span limit of the transmitter must be 0.54-in H<sub>2</sub>O to 2320 psi, and its turn down ratio must be up to 100:1. The unit must have a 5-year stability and a base accuracy of  $\pm 0.075$  percent. The upper range and lower range must be 348 psi and 0.15 psi, respectively; and a minimum span of 3.5 psi. The insulation resistance must be 100M at 1000VDC (terminals to earth). The transmitter must have an operating temperature range of -20 to +70 deg C (-4 to +158 deg F) and operating humidity of up to 100 percent annual average. The filling used in the transmitter must be silicone oil; and must meet ANSI/ISA-S 82.03 hydrostatic test requirements and SAMA PMC 27.1. Vibration and shock resistance levels of the transmitter must meet IEC 60068-2-26 and 60068-2-27, respectively; and be dust and sand tight and protected against immersion effects (NEMA 4X). An integral, LCD, 7-digit, alphanumeric display/meter must be part of the transmitter and a factory calibration certificate must be supplied with the transmitter.

One (1) 200 PSI (adjustable) pressure ranged 304 stainless steel transmitter with 1/4-inch MNPT threaded connection must be supplied and installed as shown on the Drawings. There must be a 4-20mA signal output from the transmitter to the system control panel. The unit must have a base accuracy of plus or minus 0.5 percent. The upper and lower range

must be 200 psi and 0 psi. The insulation resistance must be 100M at 100VDC. The transmitter must have an operating temperature range of -28 to 82 deg C (-20 to + 180 deg F) and operating humidity of up to 95 percent annual average. The transmitter's response time must be less than 5 ms. Vibration effect must be less than plus or minus 0.1% F.S. effect for 0 to 2000 Hz at 20 g's in any axis. Shock effect must be less than plus or minus 0.05% F.S. effect for 100 g's and 20 ms shock in any axis. The transmitter must be dust and sand tight and protected against immersion effects (NEMA 4X).

#### 2.3.13 Pressure Switches

Three (3) 15-150 PSI (adjustable), 316 stainless steel port, Teflon coated polyamide diaphragm and Buna-N O-ring spring pressure switches must be installed. Electrical Form must be 15 amp at 125 or 250 VAC, 1/8 hp at 125 VAC, 1/4 hp at 250 VAC, 0.5 A resistive, 0.04 A inductive at 125 VDC. Each switch also must be protected against immersion effects (NEMA 4X).

#### 2.3.14 Stem Covers

Stem covers for gate valves must be clear plastic manufactured of strong butyrate plastic. Stem indication must be provided by clear mylar tape to indicate "open" and "closed" positions. A cast aluminum adaptor must be used to mount the cover to the gate valve bonnet.

#### 2.3.15 Discharge Piping

Provide all suction hoses, discharge piping, fittings, etc. needed to meet requirements per design drawings. Discharge piping should be EPDM hoses with one main discharge pipe and a full length spare on site as backup supply.

The Temporary system shall be hard piped in accordance with Section 2.3.16 Hard Piping - from pump discharge through the flow meter.

#### 2.3.16 Hard Piping

##### 2.3.16.1 Polypropylene-Random (PP-RCT)

All piping 3-inch and above must be polypropylene random copolymer with modified crystallinity and temperature resistance (PP-RCT) and must be SDR11 MF RP and conform to ASTM F2389, ASME B31.3, and NSF/ANSI 14. Piping must be rated to a minimum of 160 psi at 73 deg F. Fittings and pipe must be joined by heat welding with a proprietary device. Horizontally laid pipe must be secured in such a manner that temperature-induced expansion and contraction are accommodated. Manufacturer shall warrant pipe and fitting for 10 years to be free of defects in materials or manufacturing. Pipe shall be Aquatherm Blue Pipe MF RP or approved equal.

##### 2.3.16.2 Ductile Piping

All 8-inch ductile iron piping located just upstream and downstream of the flow meter must be Pressure Class 350 conforming to AWWA C150/A21.50 and AWWA C151/A21.51. Flanged pipe must have factory assembled, long hub screw on flanges and conform to ASME B16.1 for 125-pound bolting. All ductile iron pipe must be double cement mortar lined conforming to AWWA C104/A21.4 with seal coat inside. All ductile-iron piping to be painted must have the factory primed with two coats of epoxy primer and be coated in accordance to 09 90 00 PAINTS AND COATINGS. Thickness

of ductile iron piping must conform to AWWA C150/A21.50. Restrained joints must be locked type joints manufactured by the pipe and fittings manufacturer that utilize restraint independent of the joint gasket. Restrained joints must be for the specified 150 psig test pressure. Mechanical joint retainer glands must be as manufactured by EBAA Iron Inc., Clow Corporation, U.S. Pipe and Foundry Company, or equal.

#### 2.3.16.3 Stainless Steel Tubing

Fittings on corrosion-resistant lines must be grade F316L, 150 pound conforming to ASTM A182/A182M. Piping and fittings 1/2-inch and smaller shall have threaded connections conforming to ASME B1.20.1. The manufacturer of each fitting shall be permanently identified on the body of the fitting in accordance with MSS SP-25. Bronze screwed fittings may be used for small connections such as gage assembler or small leakage lines and must conform to FS WW-P-460B (4).

#### 2.3.17 Fitting and Flanges

##### 2.3.17.1 Polypropylene-Random (PP-RCT) Fittings

Polypropylene random copolymer with modified crystallinity and temperature resistance (PP-RCT) fittings must be provided with PP-RCT piping that is socket, butt, fusion outlet, or electrofusion heat welded and must be installed in accordance with the manufacturer's published installation instructions as well as conforming to ASTM F2389. Flange fittings must be 150 pound conforming to ASME B16.5. Pipe shall be labeled every three feet to identify the size and type of pipe and the test standards which they meet. PP-RCT flanges must be installed per manufacturer guidelines utilizing proper flange adapters and stainless steel 316L flange backing rings.

##### 2.3.17.2 Ductile Iron Fittings

Ductile iron fittings must be provided with ductile iron piping and must be flanged, push-on, or restrained mechanical joints where indicated, conforming to AWWA C110/A21.10 and AWWA C111/A21.11, and double cement mortar lining conforming to AWWA C104/A21.4 with seal coat inside. Ductile iron fittings must match Pressure Class 350 ductile iron pipe conforming in thickness to AWWA C150/A21.50. Flanged fittings must be 125 pound conforming to ASME B16.1. All ductile iron fittings to be painted must be factory primed with two coats of epoxy primer and be finish coated in accordance with 09 90 00 PAINTS AND COATINGS.

##### 2.3.17.3 Stainless Steel Tube Fittings

All piping 2-inch and smaller shall be corrosion resistant, stainless steel Type 316, Schedule 40, seamless, per ASTM A312/A312M

#### 2.3.18 Insulation and Heat Tracing

Provide insulation with jacketing for Temporary ASW piping equipment installed outside.

All ASW piping installed outside shall also be heat traced.

The insulation around the pipe must be aluminum jacketed cellular glass rigid urethane foam, specific weight of 1.8-2.2 pcf pipe insulation conforming to FS HH-I-551, Type II, Class 2 and 3 combined,

except as specified hereinafter. The insulation must be a minimum of 2 inches thick furnished in curved segments with beveled edges and ends to fit the curvature of the pipe, including the pipe fittings. Fibrated cutback asphalt coating together with a glass fiber mesh must be applied around the insulation prior to placement of the aluminum jacketing. The cutback asphalt must be Pittseal III, manufactured by Lion Oil Company; 65-07, manufactured by Benjamin Foster or approved equal. Nonhardening mastic tape must be at least 1 inch wide and 1/16-inch thick, of a nondrying nature, containing 100 percent solids and no volatile oils, must withstand temperatures from minus 80 degrees F. to plus 300 degrees F., must be nonstaining, nonbleeding, and nontoxic to the skin. Fitting covers must be made by an approved experienced fabricator. The pipe must be dry and clean before insulation is applied. The insulation must be shaped to the curvature of the pipe including the pipe coating. The insulation must be tightly butted and must be staggered. Sections of insulation must be covered with banks of nonmetallic tape of 1-inch width wrapped twice around the insulation, spaced 9 inches on center. The insulation joints must be sealed with the above-mentioned joint sealer. The insulation must be coated with the fibrated cutback asphalt. All insulation must be covered with 26 gauge (0.016 inch wall thickness) aluminum jacketing. Longitudinal and transverse joints must be lapped 3 inches. The joints must be sealed with nonhardening mastic tape compressed between the laps. The jacketing must be banded with stainless steel bands spaced 9 inches on center starting 1-1/2 inches from the end of any one section. Bands must not be less than 3/4 inches wide by 0.02 inches thick and must be fastened with crimped stainless steel seal clips. The aluminum jacketing material must be alloy 6061-T6 and must be finished using Alcoa's Allumilite finish No. 215, or an approved equal finish. The insulation and covering must extend along the pipe line as shown on the Drawings, and be fitted for the class of Ductile Iron Pipe specified. The aluminum jacketing must be given 2 field coats of coal tar protective coating.

#### 2.3.19 Electrical Work

The Temporary ASW System shall be powered by the 400a electric pedestal on Berth 5 for industrial power. Provide all cable routing and protection to the laydown area(s) as indicated in the design drawings. Supply and install temporary Automatic Transfer Switch, protection devices for two (2) temporary pumps as selected by Contractor, and one (1) backup diesel generator.

Provide electrical motor driven equipment specified herein complete with motors, pump control panels, and controls. Electrical characteristics are as indicated. Provide pump control panels complete with properly sized thermal overload protection in each phase and other appurtenances necessary for the motor control specified. Each motor must be of sufficient capacity to drive the equipment at the specified capacity without exceeding the nameplate rating of the motor when operating at proper electrical system voltage and frequency. Manual or automatic control and protective or signal devices required for the operation herein specified and any control wiring required for controls and devices but not indicated must be provided under this Section of the specifications.

#### 2.3.20 Electrical Equipment

Provide electrical motor driven equipment herein specified complete with motors, pump control panels, and controls. Motor controls, equipment, and wiring must be in accordance with NFPA 70.

#### 2.3.21 Controls

Follow the same control procedures as specified in 40 60 00 PROCESS CONTROL.

#### 2.3.22 Performance Testing

The Temporary ASW System may not be placed into active service until a two-week performance test has been passed and approved, meeting all pumping and operation requirements. The two-week period shall test and verify operation of both pumps, along with all valves, strainers, and appurtenances. The primary discharge hose shall be utilized and tested during the two-week period.

#### 2.3.23 Bolts, Nuts, Washers and Fasteners

All bolts, nuts and washers must be stainless steel conforming to Type 316, unless otherwise indicated. Pump anchor bolts and anchor bolts for bearings and other devices anchored to floors must be stainless steel, Type 316. Bolts for fastening all flanged or mechanical joint piping, fittings and valves must be stainless steel, Type 316. All bolts on valves, fittings, hangers and supports, etc. must be stainless steel, Type 316. Existing nuts, bolts, and washers must not be reused.

#### 2.3.24 Gaskets

Gaskets must conform to AWWA C111/A21.111 or MIL-A-174722, full face as appropriate to the connecting fittings. Surface to be gasketed must be thoroughly cleaned before assembly, removing paint, dirt, spatter, and similar items.

#### 2.3.25 Unions

Unions must conform to FS WW-U-516B or FS WW-U-531E to match connecting piping or tubing.

#### 2.3.26 Pipe Hangers, Inserts, and Supports

All piping, shafting, extension stems, intermediate bearings, and similar items must be effectively supported. Stainless steel tubing must be carried on corrosion resistant hangers. All bolts and nuts must be Type 316 stainless steel.

Pipe hangers, inserts, guides, and supports: to MSS SP-58 and MSS SP-69. Supports shall be stainless steel with rubber lining. Pipe support spacing for PP-RCT pipe shall be as recommended by the pipe manufacturer, based on the temperature of the operating fluid and the ambient temperature.

##### 2.3.26.1 Hangers and Clamps

Provide rubber-lined stainless steel clamps. Use tape to pad the space between the plastic pipe and metal on non-clamping hangers. Do not install metal in direct contact with the PP-RCT pipe. In high humidity areas provide non-crushable pipe shields.

#### 2.3.26.2 Fixed Points (Anchor)

Fixed points shall be measured and installed to accommodate forces of expansion in the pipe and probable additional loads. Swinging clamps can't be used as fixed points.

#### 2.3.26.3 Sliding Points (Guide)

Sliding points shall allow axial pipe movement, without damaging the pipe. When positioning a sliding point, movement of the pipe shall not be blocked by walls, fittings, or equipment.

### PART 3 EXECUTION

#### 3.1 EXAMINATION

After becoming familiar with all details of the work, verify all dimensions in the field and advise the Contracting Officer of any discrepancy before performing the work.

#### 3.2 INSTALLATION

Install each pump and motor in accordance with the written instructions of the manufacturer and under the direct supervision of the manufacturer's representative. Provide access space around the device for servicing no less than the minimum recommended by the manufacturer.

Pumps, shafting, piping, and other equipment and appurtenances must be installed in accordance with the Sequence of Construction, as indicated, in strict accordance with the manufacturer's approved shop drawings and printed installation instructions and the Standards of the Hydraulic Institute. All supports necessary whether indicated or not must be provided.

Pumps less than 30 feet in length shall be shipped fully assembled with motor and shaft seal unmounted. Discharge flange shall be protected with a wooden flange cover. The skidded pump and related equipment shall be unloaded, stored, and installed per the manufacturer's operation and installation manuals. The pump manufacturer shall include two days of start-up field services to supervise the start-up and instructions of proper maintenance and operations.

##### 3.2.1 Flanged Joints

Flanged joints must be made up with full-face or ring gaskets as appropriate. Faces of flanges must be clean, free of paint and other imperfections. PP-RCT flanges must be installed per manufacturer guidelines utilizing proper flange adapters and stainless steel 316L flange backing rings. Provide 316 stainless steel UNC nuts, torque per PP-RCT manufacturer's recommendations.

##### 3.2.2 Tubing Joints

Flared or flareless joints must be made up tight with accurately assembled parts. If solder joints are used, they must be made with high-temperature solder (over 1000 degrees F melting point).

### 3.2.3 Unions and Flanges

Unions or flanges must be placed where indicated and as necessary to permit easy disassembly of piping and apparatus. Each connection having a screwed-end valve must have a union.

### 3.2.4 Valves

Valves must be installed as indicated. Installation must be with the stem vertical up unless the valve is installed in the vertical position, or otherwise indicated or directed by the Contracting Officer. All valves must be installed in such a manner as to be accessible and not interfering with adjacent or nearby equipment.

### 3.2.5 Polypropylene-Random (PP-RCT) Piping

When installed in systems with pumps in excess of 7.5 HP, piping shall be protected from excessive heat generated by operating the pump at shut-off conditions. Where the possibility exists that the pump will operate with no flow, the protection method shall be a temperature relief valve or comparable level of protection, set to a maximum temperature of 185°F.

If heat tracing or freeze protection is specified for the piping, it should be installed on the pipe interior or exterior. It must be suitable for use with plastic piping and be self-regulating to ensure that the surface temperature of the pipe and fittings will not exceed 158°F.

Pipe alignment guides shall be provided where indicated for expansion loops, offsets, and bends and as recommended by the manufacturer for expansion joints, not to exceed 5 feet on each side of each expansion joint.

### 3.2.6 Expansion and Contraction

Provide expansion and contraction controls, guides and anchors to take into account the expansion and contraction of the pipe. Provide expansion loops or offsets as required and as indicated in the manufacturer's literature.

## 3.3 FIELD QUALITY CONTROL AND COMMISSIONING

After installation of the pumping units and appurtenances, including coupling guard, is complete, carry out operating tests to assure that the pumping installation operates properly. Make arrangements to have the manufacturer's representatives present when field equipment tests are made. Give each pumping unit a running field test in the presence of the Contracting Officer for a minimum of 2 hours. Operate each pumping unit at its rated capacity or such other point on its head-capacity curve selected by the Contracting Officer. Provide an accurate and acceptable method of measuring the discharge flow. Tests must assure that the units and appurtenances have been installed correctly, that there is no objectionable heating, vibration, or noise from any parts, and that all manual and automatic controls function properly. If any deficiencies are revealed during any tests, correct such deficiencies and reconduct the tests.

Submit test reports in booklet form showing all field tests performed to adjust each component and all field tests performed to prove compliance with the specified performance criteria, upon completion and testing of



the installed system. Each test report must indicate the final position of controls.

### 3.3.1 General

Submit Field Testing and Commissioning Procedures for approval before initiating quality control testing and commissioning.

Pump test data as observed, recorded and submitted to the Contractor quality control representative and to the Contracting Officer must comply with the latest edition of the Standards of the Hydraulic Institute. All instruments must be calibrated as specified in the paragraph Calibration of Instruments from the Test Code.

### 3.3.2 Formal Inspection and Tests

The Contracting Officer will witness formal tests and approve all systems before they are accepted. Submit the request for formal inspection at least 15 days prior to the date the inspection is to take place. An experienced technician regularly employed by the pump installer must be present during the inspection. Tests must simulate pump performance throughout the entire operating curve for conformance with manufacturer's characteristic curves. At this inspection repeat any or all of the required tests as directed. Correct defects in the work provided by the Contractor, and make additional tests until it has been demonstrated that the system complies with all Contract requirements. Manufacturer's certified shop test characteristic curves for each pump being tested must be furnished by the Contractor at the time of the pump acceptance test. All temporary piping, pressure gauges, personnel, and equipment required for field tests must be furnished by the Contractor. The Contractor must also furnish all required instruments and personnel for pump motor tests, such as tachometer and wattmeter. A recording watt meter must be used. All test instruments (including test pressure gauges, tachometer, recording watt meter/electrical analyzer, etc.) must have been calibrated within the prior one year period. All piping must be cleaned and flushed after completion of installation and prior to testing. The Government will furnish water and electricity for the tests.

-- End of Section --

SECTION 46 80 00

BRACKISH WATER REVERSE OSMOSIS UNIT  
09/24

PART 1 GENERAL

1.1 SUMMARY

- a. All equipment listed below shall be furnished by a single manufacturer and shall be products of manufacturer regularly engaged in the production of said equipment:
  - (1) Water and Wastewater Chemical Feed Systems - Refer to Division 46 Specification 46 30 00 WATER AND WASTEWATER CHEMICAL FEED SYSTEMS.
  - (2) Seawater Reverse Osmosis Unit - Refer to Division 46 Specification 46 81 00 SEAWATER WATER REVERSE OSMOSIS UNIT.
  - (3) Immersed Membrane Filtration System and Ancillary Equipment - Refer to Division 46 Specification 46 82 00 IMMERSED MEMBRANE FILTRATION SYSTEM AND ANCILLARY EQUIPMENT
- b. The manufacturer shall have the sole responsibility for proper functioning of the complete system
- c. Any structural, mechanical, civil, architectural, HVAC, plumbing, electrical or instrumentation changes required by the use of any product (named vendor or approved equal product) other than the design product or an alternate layout shall be the responsibility of the Contractor and shall be provided at no additional cost to Owner.
- d. List of Desalination System Manufacturers"
  - (1) Veolia Water Technologies (Basis of Design)
  - (2) Ovivo Inc.
  - (3) WesTech Engineering, LLC
  - (4) Approved equal

1.2 WORK INCLUDED

This Section covers the furnishing and installation of one (1) brackish water reverse osmosis (BWRO) unit and ancillary equipment. All products under the PART 2 PRODUCTS section of the specification herein shall be provided by one (1) manufacturer. The manufacturer shall be the same for the UF, BWRO, and SWRO systems.

All installed equipment shall be suitable for seawater application and should be designed and constructed to withstand the harsh conditions of seawater. The contractor shall be responsible for supplying all MCCs, SCADA systems, motor starters, and VFDs. Any items not provided by the manufacturer will be supplied by the general contractor.

#### 1.2.1 BWR0 System

The manufacturer's scope of supply for the BWR0 system is described in the following table.

- a. RO membrane element vessels
- b. Process skid including:
  - (1) three (3) RO feed cartridge filters
  - (2) one (1) post cartridge filter pressure transmitter
  - (3) one (1) RO feed pH analyzer
  - (4) one (1) RO feed pressure switch low
  - (5) one (1) RO feed pressure transmitter
  - (6) one (1) high pressure pump
  - (7) one (1) RO permeate conductivity analyzer
  - (8) one (1) RO permeate flow meter
  - (9) one (1) RO permeate pressure switch high
  - (10) one (1) RO concentrate pressure transmitter
  - (11) one (1) RO concentrate flow transmitter
  - (12) one (1) RO concentrate pressure switch high
  - (13) one (1) RO concentrate control valve
  - (14) one (1) lot of pressure gauges
  - (15) one (1) lot of process valves (both manual and automatic valves)
  - (16) one (1) lot of on-skid piping

#### 1.3 DESIGN CRITERIA

Adequate tolerances in the listed capacity, head and efficiency values have been included. No deviation below the listed parameters will be permitted.

#### 1.4 QUALITY ASSURANCE

- a. General
  - (1) This Section directs special attention to certain features but does not purport to cover all details of the design, manufacture or installation of the pumping units. The final responsibility for supplying and installing pumping equipment which functions as specified herein rests with the Contractor.
  - (2) Seismic rating:  $S_s = 1.24$ . Equipment importance factor:  $I_p = 1.25$ . Seismic response coefficient:  $SR = 0.358$ . Seismic

requirements are applicable. Seismic load parameters for the building are provided in the structural general notes sheet.

- (3) Paint System Shall be the following or similar: Two Coat. Primer: Tnemec Series 20 Pota-Pox or Tnemec Series 20 Pota-Pox FC, 3-5 mil thick. Topcoat: Tnemec Endura Shield II Series 1075 Epoxoline, 4-6 mil thick. Recommend for indoor and non-coastal applications. Recommended manufacturers paints and finishing shall be used.
- (4) Sling lift with overhead clearing and spreader bars with three slings. OR use forklift with roller dolly's. Forklift from cartridge filter end. To be lifted by qualified lifters.

#### 1.5 SUBMITTALS

Submittals including shop drawings, O&M manuals, and installation documents shall be submitted in electronic format in accordance with the requirements of Section 01 33 00. Additionally, four hard copies of the final O&M manual shall be provided. In accordance with requirements of general specifications, submit the following:

##### SD-02 Shop Drawings

Equipment Construction Details; G

Equipment Installation; G

- a. Show complete details of construction and installation of all equipment, including appurtenances; include all dimensions.
- b. Include all specifics of this installation.
- c. Delete all references not applicable to this installation.
- d. Outline specific areas where the proposed equipment differs from that specified.

Wiring Diagram; G

- a. Include complete wiring diagrams required for installation.

##### SD-03 Product Data

Pump Curves; G

Pump Certification; G

- a. Certified pump curves showing the actual performance of the feed pump to be supplied under factory testing. Pump certification shall include testing for head, capacity, efficiency and brake horsepower, at 50%, 65%, 85% and 100% for variable speed driven pumps; and at the rated speed for constant speed pumps. Six certified copies of the results in the form of pump characteristic curves shall be furnished to the Engineer for review prior to shipment.

Product Description; G

- a. A complete, easily readable functional description of the proposed

equipment.

Product Data; G

- a. A complete list of the manufacturer's name, address and phone number, the local representative's name, address and phone number, and the model and serial numbers of all equipment supplied.

SD-06 Test Reports

Feed Pump Motor Test Results; G

- a. Complete test results on the feed pump motor for the tests as defined in the NEMA standard for a motor of its class and rating, to determine that it is free from electrical and mechanical defects and to provide assurance that it meets the design specifications.

Motor Test Results; G

- a. The manufacturer's complete test results for each motor model design tested for efficiency by the following means:
  - (1) The full load Dynamometer test shall be based on IEEE 112 Method B, per NEMA MG1 12.53 A & B including six (6) load point testing and stray load loss segregation and correction steps.

Factory Acceptance Test; G

- a. Results and reporting of the factory acceptance test described under PART 3 EXECUTION.

Field Test Report; G

- a. Upon completion of installation, the results of the field and acceptance tests as specified under this section of the specification shall be submitted to the Engineer.

SD-07 Certificates

Parts List; G

- a. Complete parts list including the manufacturer's reference and ordering numbers.

Spare Parts List; G

- a. Recommended Spare Parts List.

Installation Certification; G

- a. Furnish written certification from the manufacturer's representative of the proper installation of each component.

SD-10 Operation and Maintenance Data

Operating And Maintenance Manual; G

- a. Complete operation and maintenance information for this specific equipment.

- b. The manuals will be reviewed by the Engineer for completeness; those that are deemed inadequate will be returned for correction.

SD-11 Closeout Submittals

Warranty; G

1.6 STORAGE

- a. Equipment damaged or bent during shipping or storage shall be replaced.
- b. Feed pump motor shall be stored on site according to motor manufacturer's recommendations until pumps are operational and accepted by the Engineer. Contractor shall be responsible for such arrangements.
- c. Disassembled equipment shall be kept clean and protected against rodents and pests until equipment is reassembled, operational and accepted by the Engineer.

1.7 WARRANTY

- a. The pump manufacturer and the pump motor manufacturer shall each individually and separately warranty that the equipment they supplied under this Section fully meets the criteria specified herein, and shall further warranty that the equipment is free from all defects in materials and workmanship.
- b. The manufacturer's warranty from defects shall contain a provision that the manufacturer shall repair or replace any defects, to the satisfaction of and at no additional cost to the Owner, for a period of twenty-four (24) months for the pump and motor from the date of Substantial Completion of the project.
- c. The variable frequency drive shall be warranted for a period of 24 months after date of shipment. Warranty shall include parts and labor by the manufacturer.

PART 2 PRODUCTS

2.1 GENERAL

A. The overall BWRO unit and products specified Herein shall be provided by one manufacturer. The Basis of Design for the BWRO system is RO-PROflex, 72-2, FRP, 50HP, AB, SG by Veolia Water Technologies, Trevose, PA or similar.

B. Performance Data

|                       |           |
|-----------------------|-----------|
| Design Permeate Flow  | 278 gpm   |
| Feed Flow             | 312 gpm   |
| Concentrate Flow Rate | 34.42 gpm |

|                           |           |
|---------------------------|-----------|
| Recovery                  | 89.0%     |
| Design Temperature        | 1 C       |
| Design pH                 | 5.5 - 8.5 |
| System Inlet Pressure     | 30 psi    |
| System Operating Pressure | 235 psi   |

## 2.2 PIPING

Refer to the pipe schedule in the mechanical process bid drawings for specified pipe materials. All fittings, flanges, connections, fasteners, nuts, washers, nipples, joints, and seals shall be suitable for the intended application.

## 2.3 WIRING

The following table applies to all wiring offered for the BWRO unit.

| Wiring                 | Size         | Description                                           |
|------------------------|--------------|-------------------------------------------------------|
| Control system wiring  | 110/240VAC   | 18AWG minimum, TFFN insulation                        |
| Power circuit wiring   | 110/240VAC   | 14AWG minimum, TFFN or THWN insulation                |
| Instrumentation wiring | 4-20mA loops | 22AWG minimum                                         |
| Motor wiring           | All sizes    | Dependent on motor size (usually outside scope of R0) |

## 2.4 VALVES

The material of construction shall be corrosion-resistant. Duplex 2205 or approved equal shall be used for high-pressure applications, while PVC can be used for low-pressure application.

The following table applies to all valves offered for the BWRO unit.

| Manufacturer | Description |
|--------------|-------------|
|--------------|-------------|

### BALL

|                          |                                                                                                                                                                                                                              |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Flow-Tek, Star, or equal | Duplex 2205 stainless steel body, threaded ends (drains, vents), Victaulic ends (concentrate backpressure), and socket ends (process piping). MSS-SP72 compliant. Gear Operators included for pressure reducing Ball Valves. |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|              |                                                                                |
|--------------|--------------------------------------------------------------------------------|
| SMC or equal | 150 lb. class, PVC Body, threaded and hose-barb ends (drains and sample ports) |
|--------------|--------------------------------------------------------------------------------|

#### BUTTERFLY

|               |                                                                                                    |
|---------------|----------------------------------------------------------------------------------------------------|
| Bray or equal | 150 lb. class, ASME B16.34, Nylon coated ductile iron disc, EPDM seat, cast iron body, 316 SS stem |
|---------------|----------------------------------------------------------------------------------------------------|

#### CHECK

|                       |                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------|
| Techno-check or equal | 150 lb. class, EPDM seat, Duplex 2205 stainless steel disc, with Victaulic grooved ends |
| SMC or equal          | Polypropylene body with EPDM seal (used on pH sensor slipstream)                        |

#### PRESSURE REGULATING VALVE

|                     |                                                                                                                       |
|---------------------|-----------------------------------------------------------------------------------------------------------------------|
| Pneumatics or equal | 5 Micron Element, NPTF thread 3/8 Port Size Gauge, Metal Bowl with sight glass<br>Standard output pressure 0-125 PSIG |
|---------------------|-----------------------------------------------------------------------------------------------------------------------|

#### RELIEF

|                  |                                                                                                                                                    |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Farris, or equal | 300 lb. class, internals of Duplex 2205 stainless steel, body and bonnet of carbon steel (used when pump pressure exceeds housing pressure rating) |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|

### 2.5 INSTRUMENTATION

The following table applies to all instrumentation offered for the BWRO unit. Named instrumentation or equal shall be provided per Engineer approval.

#### Instrument List

| Instrument           | Display | Manufacturer               | Locations                   |
|----------------------|---------|----------------------------|-----------------------------|
| Instrument Center    |         | Thornton                   | Thornton M800               |
| Flow Meters          | HMI     | GF Signet 2537             | Permeate, concentrate       |
| Pressure Switch      | HMI     | United Electric Spectra 10 | Feed, permeate, concentrate |
| Pressure Transmitter | HMI     | E+H PMC11                  | Primary, Final              |



|                          |                  |                     |                                                                                            |
|--------------------------|------------------|---------------------|--------------------------------------------------------------------------------------------|
| Pressure Gauge           | Local indication | Ashcroft or equal   | Pre-filter, post-filter, primary, final, interstage, permeate, concentrate, pump discharge |
| Conductivity/Temperature | HMI              | GF Signet 2850      | Permeate, Feed                                                                             |
| pH Meter                 | HMI              | GF Signet 2724/2750 | Feed                                                                                       |
| ORP                      |                  | N/A                 | Feed                                                                                       |

Accessories shipped loose or Mounted on Control Panel

#### Chlorine Monitor

| Instrument                                                                        | Display                         | Manufacturer    | Locations |
|-----------------------------------------------------------------------------------|---------------------------------|-----------------|-----------|
| Pressure Switch (feed, permeate, concentrate)                                     | Model 10 Series pressure switch | United Electric |           |
| Devices will be identified with tags on the P&ID and adhesive labels near devices |                                 |                 |           |

#### Instrument - Local Indication

| Instrument                                          | Display | Manufacturer             | Locations |
|-----------------------------------------------------|---------|--------------------------|-----------|
| Pressure Gauges - 2.5" gauges with 1/4" connections |         | Ashcroft, WIKA, or equal |           |

## 2.6 EQUIPMENT LIST

The following table details the main process equipment to be supplied in this proposal. Named suppliers or equal shall be provided per Engineer approval.

#### Rotating Equipment

| Equipment               | Details                                                                      | Supplier          |
|-------------------------|------------------------------------------------------------------------------|-------------------|
| R0 High Pressure Pump   | Horizontal Multi-stage centrifugal, model AS30010 or equal, quantity one (1) | Tonkaflo or equal |
| Inlet/discharge housing | Duplex 2205 stainless steel                                                  | Tonkaflo or equal |

|             |                                                                                          |                            |
|-------------|------------------------------------------------------------------------------------------|----------------------------|
| Shell       | Duplex 2205 stainless steel                                                              | Tonkaflo or equal          |
| Impeller    | SS Series = Noryl; AS Series = 304 Stainless Steel.                                      | Tonkaflo or equal          |
| Shaft       | Duplex 2205 stainless steel                                                              | Tonkaflo or equal          |
| Lubrication | Flooded oil                                                                              | Tonkaflo or equal          |
| Seal type   | Mechanical Crane type 21, Type 1, or 2B                                                  | Tonkaflo or equal          |
| Coupling    | Flex rubber spacer type, with 1.15 service factor. Coupling guard meeting OSHA standards | Tonkaflo or equal          |
| Motor       | VAC, 3-phase, 60 HZ, 75 HP                                                               | Marathon, Balder, or equal |

#### Cartridge Filter Housing

| Equipment                          | Details                                                                                            | Supplier        |
|------------------------------------|----------------------------------------------------------------------------------------------------|-----------------|
| RO Cartridge Filter housings       | FRP Housings, 100 psi max pressure, non code stamped.                                              | ROPV, or equal  |
| RO Cartridge Filter housing OPTION | SS Housings, 100 psi max pressure, non code stamped.                                               | VEOLIA or equal |
| Cartridge Filter                   | FLT,CTG,DEPTH,HF.Zs<br>01-40-FSB,Melt-blown Polypropylene. 1- micron, 40 in, quantity of three (3) | VEOLIA or equal |
| Valves                             | Pre-filter manual isolation, post filter auto isolation, vent and drain                            | BRAY or Equal   |

#### Membrane Element Housing

| Equipment           | Details                                                                                                                                                                                                                                        | Supplier           |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| RO Element housings | FRP Housings, model 450P-8 or equal, 450 psi max pressure, 4 port style , thin film composite membrane element (or other options to choose). ASME B & PV Section X code Stamp. Quantity of 12, 6' length, 8" diameter, 8-4 banking arrangement | Codeline, or equal |

|                                      |                                                                                                                                                                                       |                 |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Membrane Element                     | Model AK-400H or equal,<br>quantity 72, FRP Wrapped                                                                                                                                   | VEOLIA or equal |
| Installation/Utility<br>Requirements | Inlet- 6" flange<br>Permeate- 4" Flange<br>Concentrate- 3" Flange<br>Inlet (CIP)- 6" Flange<br>Permeate (CIP)- 4" Flange<br>Concentrate (CIP)- 3" Flange<br>Compressed air- 0.25" FPT |                 |

## 2.7 ELECTRICAL AND CONTROL SYSTEM

1. This section herein details the electrical and control system to be supplied in this proposal. Named suppliers or equal shall be provided per Engineer approval.
2. A control system is provided to allow the plant operators to easily operate, monitor and control the equipment supplied by Membrane Manufacturer.
3. The control system is mounted on a corrosion resistant mounting panel. It incorporates optional motor controls, solenoid valves, slipstream instrumentation fully wired and tubed.
4. Control panel is cUL/ UL 508 Standard for Industrial Control Equipment.
5. Membrane System Alarms required:
  - a. Low Inlet Pressure
  - b. High/Low pH
  - c. High Temperature
  - d. High Permeate Pressure
  - e. High Concentrate Pressure
6. Control System: Refer to specification 46 82 00 section 2.13 Control System.

Main Control Panel containing PLC and HMI - NEMA 4 painted carbon steel (or NEMA 4X stainless steel option)

PLC and HMI

Allen-Bradley CompactLogix 1769 and PanelView Plus Standard

- (a) Processor: PLC - AB CompactLogix 1769
- (b) Operator Interface: PanelView Plus® 700
- (c) Communications: Ethernet

Control Screens Included:

- (a) Security access screen
- (b) System overview
- (c) System data display
- (d) Alarm History
- (e) Individual component display of operational status, 4-20 mA instruments and control
- (f) Alarm status and indication
- (g) Alarm description
- (h) Set point screens

#### NOTES

- a. The control panel will be fully assembled and wired and simulated control tests will be carried out to prove correct operation of controls, sequencing of valves, etc.
- b. Maximum ambient air temperature for the control panels is 35°C
- c. Provide UPS. Panel must operate at site atmospheric conditions.
- d. 120 VAC control power.
- e. Components are UL listed or recognized.

#### Electrical and Instrumentation Assembly Components

|         |                            |                                             |
|---------|----------------------------|---------------------------------------------|
| 1 lot   | Wiring                     | Per BWRO Manufacturer standards             |
| 1 lot   | Terminal Blocks            | AB, or equal                                |
| 1 lot   | Air Solenoid Valves        | Valve ASCO Pneumatics manufacture standard  |
| 1 lot   | Cable Tray                 | Painted steel                               |
| 1 lot   | Air Tubing and Fittings    | Polyethylene Tubing / PPL - ¼"              |
| 1 lot   | Sample Tubing and Fittings | Polyethylene Tubing / PPL - ¼"              |
| 1 piece | Sample Solenoids           | ¼" Parker Skinner stainless steel, or equal |

#### NOTES

- a. Solenoids for valves will be gang-mounted in on corrosion resistant panel.
- b. Limit switches are not provided.
- c. Pneumatic tubing will be run and supported within skid limits, from solenoid to valve actuators, as required.

## Motor Control Components

Motors shall be VFD controlled.

Local disconnect switches shall be included with motors.

## PART 3 EXECUTION

### 3.1 INSTALLATION

A. Contractor shall be responsible for disassembly and reassembly of all equipment in order to complete installation as shown in the Contract Documents. Required measures to inspect onsite conditions, as well as coordination with manufacturer and planning to ensure proper installation of all equipment and membranes as recommended by the manufacturer, shall be the sole responsibility of the Contractor.

B. Manufacturer will provide necessary oversight to ensure any disassembly/reassembly is properly done without any adverse effects on the equipment. Manufacturer's on-site representative will confirm the final equipment has been installed as per manufacturer requirements and provide written confirmation that all warranties remain in effect.

### 3.2 FIELD AND FACTORY ACCEPTANCE TEST

A. After the pump has been installed and connected, and after inspection, operation, and adjustment have been completed by the manufacturer's representative, the equipment shall be field tested in the presence of the Engineer, for overall wire to water efficiency and for general performance and fitness for the service specified. Results of these tests shall be submitted to the Engineer.

### 3.3 MANUFACTURER'S SERVICES

A. As applicable, all components as defined within are factory tested as per manufacturer standards before leaving the manufacturer's facility. The ITP (Inspection and Test Plan) includes a test and / or inspection as defined in the ITP procedure for platform skids, prefabricated pipe (all assembled pipe, not loose supplied pipe), machinery (pumps, blowers, etc.), tanks, valves, instruments, electrical panels / junction boxes, electrical instruments, loose materials (cables, cable trays, etc.), and VFDs, as applicable. A FAT report (Factory Acceptance Test) is provided to the customer. It includes certificates, test reports, and CE declaration of conformity, as applicable. The Contractor is responsible for obtaining the FAT report from the Manufacturer.

B. Services to be provided:

1. For the pumps, the service representative shall be responsible for complete component inspection on site after delivery and shall assist in the correct assembly of the components for a minimum period of one (1) eight hour day.

a. For inspection and check out of the erected equipment.

b. For start up services and supervision.

c. For complete instruction of the Owner's operating personnel.

C. The minimum period of time herein specified does not relieve the manufacturer from providing sufficient time to satisfactorily complete the required service functions.

D. The manufacturer's representative shall certify in writing that the pump has been properly installed. Contractor shall coordinate with Manufacturer to remedy any necessary/recommended corrections.

E. The Owner reserves the right to video tape the instruction of the operating personnel for future use in training.

F. For services related to the BWRO system, refer to Section 46 82 00 IMMERSSED MEMBRANE FILTRATION SYSTEM AND ANCILLARY EQUIPMENT 3.2.1 - Services.

#### 3.4 SPARE PARTS

A. The Contractor will provide all other spare parts as recommended in the manufacturer's standard operations and maintenance information.

B. Provide minimum two (2) mechanical seals for feed pump.

-- End of Section --

SECTION 46 81 00

SEAWATER WATER REVERSE OSMOSIS UNIT  
09/24

PART 1 GENERAL

1.1 SUMMARY

- a. All equipment listed below shall be furnished by a single manufacturer and shall be products of manufacturer regularly engaged in the production of said equipment:
  - (1) Water and Wastewater Chemical Feed Systems - Refer to Specification 46 30 00 WATER AND WASTEWATER CHEMICAL FEED SYSTEMS.
  - (2) Brackish Water Reverse Osmosis Unit - Refer to Specification 46 80 00 BRACKISH WATER REVERSE OSMOSIS UNIT.
  - (3) Immersed Membrane Filtration System and Ancillary Equipment - Refer to Specification 46 82 00 IMMERSED MEMBRANE FILTRATION SYSTEM AND ANCILLARY EQUIPMENT.
- b. The manufacturer shall have the sole responsibility for proper functioning of the complete system.
- c. Any structural, mechanical, civil, architectural, HVAC, plumbing, electrical or instrumentation changes required by the use of any product (named vendor or approved equal product) other than the design product or an alternate layout shall be the responsibility of the Contractor and shall be provided at no additional cost to Owner.
- d. List of Desalination System Manufacturers:.
  - (1) Veolia Water Technologies (Basis of Design)
  - (2) Ovivo Inc.
  - (3) WesTech Engineering, LLC
  - (4) Approved Equal

1.2 WORK INCLUDED

A. This Section covers the furnishing and installation of one (1) seawater water reverse osmosis (SWRO) unit and ancillary equipment. All products under the PART 2 PRODUCTS section of the specification herein shall be provided by one (1) manufacturer. The following shall be included as part of the SWRO unit:

- (1) High pressure pump skid
- (2) Seawater SWRO Membrane skid
- (3) Seawater SWRO cartridge filter skid

B. The manufacture shall be the same for the UF, BWRO, and SWRO systems.

All installed equipment shall be suitable for seawater application and should be designed and constructed to withstand the harsh conditions of seawater

All products under the PART 2 PRODUCTS section of the specification herein shall be provided by one (1) manufacturer.

The manufacturer's scope of supply for the SWRO system is described in the following table.

a. Description

(1) RO membrane element vessels

(2) Process skid including:

(a) eleven (11) RO feed cartridge filters

(b) one (1) RO feed ORP analyzer

(c) one (1) RO feed pressure switch low

(d) one (1) high pressure pump

(e) one (1) high pressure booster

(f) one (1) high pressure booster auxiliary bypass control valve

(g) one (1) RO feed pressure transmitter

(h) one (1) RO permeate temperature and conductivity analyzer

(i) one (1) RO permeate flow meter

(j) one (1) RO permeate pressure switch high

(k) one (1) RO concentrate pressure transmitter

(l) one (1) RO concentrate pressure switch high

(m) one (1) lot of pressure gauges

(n) one (1) lot of process valves (both manual and automatic valves)

(o) one (1) lot of pressure safety valves on the RO permeate and concentrate piping

(p) one (1) lot of on-skid piping

b. The contractor shall be responsible for supplying all MCCs, SCADA systems, motor starters, and VFDs. Any items not provided by the manufacturers will be supplied by the general contractor.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for



information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

Submittals including shop drawings, O&M manuals, and installation documents shall be submitted in electronic format in accordance with the requirements of Section 01 33 00. Additionally, four hard copies of the final O&M manual shall be provided.

SD-02 Shop Drawings

Equipment Construction Details; G

Equipment Installation; G

- a. Show complete details of construction and installation of all equipment, including appurtenances; include all dimensions.
- b. Include all specifics of this installation.
- c. Delete all references not applicable to this installation.
- d. Outline specific areas where the proposed equipment differs from that specified.

Wiring Diagram; G

- a. Include complete wiring diagrams required for installation.

SD-03 Product Data

Pump Curves; G

Pump Certification; G

- a. Certified pump curves showing the actual performance of the feed pump to be supplied under factory testing. Pump certification shall include testing for head, capacity, efficiency and brake horsepower, at 50%, 65%, 85% and 100% for variable speed driven pumps; and at the rated speed for constant speed pumps. Six certified copies of the results in the form of pump characteristic curves shall be furnished to the Engineer for review prior to shipment.

Product Description; G

- a. A complete, easily readable functional description of the proposed equipment.

Product Data; G

- a. A complete list of the manufacturer's name, address and phone number, the local representative's name, address and phone number, and the model and serial numbers of all equipment supplied.

Feed Pump Motor Test Results; G

- a. Complete test results on the feed pump motor for the tests as defined in the NEMA standard for a motor of its class and rating, to determine

that it is free from electrical and mechanical defects and to provide assurance that it meets the design specifications.

Motor Test Results; G

- a. The manufacturer's complete test results for each motor model design tested for efficiency by the following means:
  - (1) The full load Dynamometer test shall be based on IEEE 112 Method B, per NEMA MG1 12.53 A & B including six (6) load point testing and stray load loss segregation and correction steps.

Factory Acceptance Test; G

- a. Results and reporting of the factory acceptance test described under PART 3 EXECUTION.

Field Test Report; G

- a. Upon completion of installation, the results of the field and acceptance tests as specified under this section of the specification shall be submitted to the Engineer.

SD-07 Certificates

Parts List; G

- a. Complete parts list including the manufacturer's reference and ordering numbers.

Spare Parts List; G

- a. Recommended Spare Parts List.

Installation Certification; G

- a. Furnish written certification from the manufacturer's representative of the proper installation of each component.

SD-10 Operation and Maintenance Data

Operating And Maintenance Manuals; G

- a. Complete operation and maintenance information for this specific equipment.
- b. The manuals will be reviewed by the Engineer for completeness; those that are deemed inadequate will be returned for correction.

## 1.4 QUALITY ASSURANCE

### 1.4.1 General

1. This Section directs special attention to certain features but does not purport to cover all details of the design, manufacture or installation of the pumping units. The final responsibility for supplying and installing pumping equipment which functions as specified herein rests with the Contractor.

2. Seismic rating:  $S_s = 1.24$ . Equipment importance factor:  $I_p = 1.25$ . Seismic response coefficient:  $SR = 0.358$ . Seismic requirements are applicable. Seismic load parameters for the building are provided in the structural general notes sheet.

3. A paint system shall be designed that ensures maximum protection and extends the service life of equipment exposed to seawater and marine environments.

4. Sling lift with overhead clearing and spreader bars with three slings. OR use forklift with roller dolly's. Forklift from cartridge filter end. To be lifted by qualified lifters.

#### 1.5 STORAGE

A. Equipment damaged or bent during shipping or storage shall be replaced.

B. Feed pump motor shall be stored on site according to motor manufacturer's recommendations until pumps are operational and accepted by the Engineer. Contractor shall be responsible for such arrangements.

C. Disassembled equipment shall be kept clean and protected against rodents and pests until equipment is reassembled, operational and accepted by the Engineer.

#### 1.6 WARRANTY

A. The pump manufacturer and the pump motor manufacturer shall each individually and separately warranty that the equipment they supplied under this Section fully meets the criteria specified herein, and shall further warranty that the equipment is free from all defects in materials and workmanship.

B. The manufacturer's warranties from defects shall contain a provision that the manufacturer shall repair or replace any defects, to the satisfaction of and at no additional cost to the Owner, for a period of twenty-four (24) months for the pump and motor from the date of Substantial Completion of the project.

### PART 2 PRODUCTS

#### 2.1 GENERAL

A. The overall SWRO unit and products specified Herein shall be provided by one manufacturer. The Basis of Design for the SWRO system is RO, SeaPRO-126, 35K, 480, PX, AB, PRV by Veolia Water Technologies, Trevose, PA.

##### B. Performance Data

|                    |            |
|--------------------|------------|
| Feed Flow          | 659 gpm    |
| Design TDS         | 33,000 ppm |
| Design Temperature | 1 degree C |

|                       |           |
|-----------------------|-----------|
| System Inlet Pressure | 30 PSI    |
| Design Permeate Flow  | 312 gpm   |
| Design Permeate TDS   | 21.5 mg/L |
| Concentrate Flow Rate | 347 gpm   |
| SWRO Recovery         | 47.4%     |
| Nominal Rejection %   | >97%      |

## 2.2 PIPING

Pipe material shall be compatible with seawater. Refer to the pipe schedule in the mechanical process bid drawings for specified pipe materials. All fittings, flanges, connections, fasteners, nuts, washers, nipples, joints, and seals shall be suitable for the intended application and compatible with the seawater environment.

## 2.3 WIRING

The following table applies to all wiring offered for the SWRO unit.

| Wiring                 | Size         | Description                                           |
|------------------------|--------------|-------------------------------------------------------|
| Control system wiring  | 110/240VAC   | 18AWG minimum, TFFN insulation                        |
| Power circuit wiring   | 110/240VAC   | 14AWG minimum, TFFN or THWN insulation                |
| Instrumentation wiring | 4-20mA loops | 22AWG minimum                                         |
| Motor wiring           | All sizes    | Dependent on motor size (usually outside scope of R0) |

## 2.4 VALVES

The material of construction shall be corrosion-resistant for seawater applications. Duplex 2205 or approved equal shall be used for high-pressure applications, while PVC can be used for low-pressure application. The following table applies to all valves offered for the SWRO unit.

| Manufacturer | Description |
|--------------|-------------|
|--------------|-------------|

### BALL

|                          |                                                                                                                                                                                                                              |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Flow-Tek, Star, or equal | Duplex 2205 stainless steel body, threaded ends (drains, vents), Victaulic ends (concentrate backpressure), and socket ends (process piping). MSS-SP72 compliant. Gear Operators included for pressure reducing Ball Valves. |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|              |                                                                                |
|--------------|--------------------------------------------------------------------------------|
| SMC or equal | 150 lb. class, PVC Body, threaded and hose-barb ends (drains and sample ports) |
|--------------|--------------------------------------------------------------------------------|

#### BUTTERFLY

|               |                                                                                            |
|---------------|--------------------------------------------------------------------------------------------|
| Bray or equal | 150 lb. class, ASME B16.34, Nylon coated ductile iron disc, EPDM seat, Duplex 2205 SS stem |
|---------------|--------------------------------------------------------------------------------------------|

Note: Lug style bodies on 2.0 inch and greater and wafer style bodies on 1.5 inch

#### CHECK

|                       |                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------|
| Techno-check or equal | 150 lb. class, EPDM seat, Duplex 2205 stainless steel disc, with Victaulic grooved ends |
| SMC or equal          | Polypropylene body with EPDM seal (used on pH sensor slipstream)                        |

#### PRESSURE REGULATING VALVE

|                     |                                                                                |
|---------------------|--------------------------------------------------------------------------------|
| Pneumatics or equal | 5 Micron Element, NPTF thread 3/8 Port Size Gauge, Metal Bowl with sight glass |
|---------------------|--------------------------------------------------------------------------------|

#### RELIEF

|                  |                                                                                                                                            |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Farris, or equal | 300 lb. class, internals of 316 stainless steel, body and bonnet of carbon steel (used when pump pressure exceeds housing pressure rating) |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------|

### 2.5 INSTRUMENTATION

1. The following table applies to all instrumentation offered for the SWRO unit. Named instrumentation or equal shall be provided per Engineer approval.

2. The following table applies to all instrumentation offered for the BWRO unit. Named instrumentation or equal shall be provided per Engineer approval.

#### Instrument List

| Instrument        | Display | Manufacturer      | Locations     |
|-------------------|---------|-------------------|---------------|
| Instrument Center |         | Thornton or equal | Thornton M800 |

|                          |                  |                                   |                                                                                                            |
|--------------------------|------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------|
| Flow Meters              | Transmitter      | GF Signet 51530-P0 or equal       | Permeate, concentrate                                                                                      |
| ERD Flow Meters          | Transmitter      | GE Panametrics AT868 or equal     | HP ERD outlet, LP ERD inlet                                                                                |
| Pressure Switch          | Discrete         | United Electric or equal          | High pressure pump feed, boost pump feed, permeate, concentrate                                            |
| Pressure Transmitter     | Transmitters     | E+H or equal                      | Feed inlet to membrane pressure vessels, concentrate outlet from membrane pressure vessels.                |
| Pressure Gauge           | Local Indication | Wika, Ashcroft or equal           | Pre-filter, post-filter, feed inlet to membrane pressure vessels, boost pump outlet, permeate, concentrate |
| Conductivity/Temperature | Transmitter      | Thornton M800 58-031-409 or equal | Permeate                                                                                                   |
| pH Meter                 | Transmitter      | Thornton M800 52-005-381 or equal | LP ERD Inlet (Feed)                                                                                        |
| Temperature              | Transmitter      | Thornton or equal                 | Permeate                                                                                                   |

Accessories, shipped loose or Mounted on Control Panel

#### Chlorine Monitor

|                                                                                   |                                          |                          |
|-----------------------------------------------------------------------------------|------------------------------------------|--------------------------|
| Level Transmitter (CIP tank level control for permeate flush)                     | PX3228 or equal                          |                          |
| Pressure Switch (feed, permeate, concentrate)                                     | Model 10 Series pressure switch or equal | United Electric          |
| Devices will be identified with tags on the P&ID and adhesive labels near devices |                                          |                          |
| Instrument - Local Indication                                                     |                                          |                          |
| Pressure Gauges - 2.5" gauges with 1/4" connections                               |                                          | Ashcroft, WIKA, or equal |

## 2.6 EQUIPMENT LIST

The following table details the main process equipment to be supplied in

this proposal. Named supplier or equal shall be provided per Engineer approval. The material of construction shall be corrosion-resistant for seawater applications. Duplex 2205 or approved equal shall be used for high-pressure applications.

| Equipment | Type | Supplier |
|-----------|------|----------|
|-----------|------|----------|

#### Rotating Equipment

|                         |                                                                                                                                                                                               |                            |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| RO High Pressure Pump   | Horizontal Multi-stage                                                                                                                                                                        | FEDCO or equal             |
| Inlet/discharge housing | Duplex 2205 stainless steel                                                                                                                                                                   | FEDCO or equal             |
| Shell                   | Duplex 2205 stainless steel                                                                                                                                                                   | FEDCO or equal             |
| Impeller                | SS Series = Noryl; AS Series = Duplex 2205 Stainless Steel.                                                                                                                                   | FEDCO or equal             |
| Shaft                   | Duplex 2205 stainless steel                                                                                                                                                                   | FEDCO or equal             |
| Lubrication             | Flooded oil                                                                                                                                                                                   | FEDCO or equal             |
| Seal Type               | Mechanical Crane type 21, Type 1, or 2B                                                                                                                                                       | FEDCO or equal             |
| Motor & VFD             | 300 HP motor, VFD model # Danfoss VLT Aqua FC202 or GE AF-650 GP                                                                                                                              | Danfoss, GE or equal       |
| Turbocharger            | Model HPB-130 or equal                                                                                                                                                                        | FEDCO or equal             |
| Coupling                | Flex rubber spacer type, with 1.15 service factor. Coupling guard meeting OSHA standards                                                                                                      | Tonkaflo or equal          |
| Motor                   | Ultra Nema Premium Extra Severe Duty (IP55) Motor. TEFC, NEMA B Standard, EISAct2007 Compliant, Class H Insulation. Maximum Speed: 3,600 rpm, Service Factor: 1.15. Inverter Duty Nema MG1-31 | Marathon, Baldor, or equal |

#### Cartridge Filter Housings

|                             |                                                                              |                 |
|-----------------------------|------------------------------------------------------------------------------|-----------------|
| RO Cartridge Filter housing | SS Housings, 100 psi max pressure, non code stamped. Model HFRP1-60 or equal | VEOLIA or equal |
|-----------------------------|------------------------------------------------------------------------------|-----------------|

|                  |                                                                                                |                 |
|------------------|------------------------------------------------------------------------------------------------|-----------------|
| Cartridge Filter | 1-micron nominal, model HF.R0.Zs 01-60-TSE or equal melt-blown polypropylene, quantity six (6) | VEOLIA or equal |
|------------------|------------------------------------------------------------------------------------------------|-----------------|

#### Membrane Element Housing

|                                   |                                                                                                                                           |                 |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| RO Element housings               | R8040B1200S model or equal, ring and plate style, quantity 18. Each housing holds 7 elements                                              | ROPV or equal   |
| Membrane Element                  | Model AE-400 or equal, quantity 126, polyamide membranes                                                                                  | VEOLIA or equal |
| Installation/Utility Requirements | Inlet- 6" Flange<br>Permeate- 4" Flange<br>Concentrate- 6" Flange<br>Power: 460VAC, 3-phase, 60Hz<br>Control Circuit: 220V, 1-phase, 50Hz |                 |

#### 2.7 ELECTRIC AND CONTROL SYSTEM

1. This section herein details the electrical and control system to be supplied in this proposal. Named suppliers or equal shall be provided per Engineer approval.
2. A control system is provided to allow the plant operators to easily operate, monitor and control the equipment supplied by Membrane Manufacturer.
3. The control system is mounted on a corrosion resistant mounting panel. It incorporates optional motor controls, solenoid valves, slipstream instrumentation fully wired and tubed.
4. Control panel is cUL/ UL 508 Standard for Industrial Control Equipment.
5. Membrane System Alarms required:
  - (a) Membrane System Alarms required:
  - (b) High permeate conductivity
  - (c) High permeate temperature
  - (d) VFD fault
  - (e) High permeate pressure
  - (f) High concentrate pressure
  - (g) High / Low inlet pH



(h) High / low concentrate flow

(i) High pressure inlet / outlet of membrane pressure valves

6. Control System: Refer to specification 46 82 00 section 2.13

Control System

Description

Control Panels and Remote I/O Boxes

Main Control Panel containing PLC and HMI - NEMA 4 painted carbon steel (or NEMA 4X stainless steel option)

Notes:

- a. The control panel will be fully assembled and wired and simulated control tests will be carried out to prove correct operation of controls, sequencing of valves, etc.
- b. Maximum ambient air temperature for the control panels is 35°C.
- c. Provide UPS. Panel must operate at site atmospheric conditions.
- d. 120 VAC control power.
- e. Components are UL listed or recognized.

Electrical and Instrumentation Assembly Components

|         |                            |                                            |
|---------|----------------------------|--------------------------------------------|
| 1 lot   | Wiring                     | Per SWRO Manufacturer standards            |
| 1 lot   | Terminal Blocks            | AB, or equal                               |
| 1 lot   | Air Solenoid Valves        | Valve ASCO Pneumatics manufacture standard |
| 1 lot   | Cable Tray                 | Painted steel                              |
| 1 lot   | Air Tubing and Fittings    | Polyethylene Tubing / PPL - "              |
| 1 lot   | Sample Tubing and Fittings | Polyethylene Tubing / PPL - "              |
| 1 Piece | Sample Solenoids           | " Parker Skinner stainless steel, or equal |

Notes:

- a. Solenoids for valves will be gang-mounted in on corrosion resistant panel
- b. Limit switches are not provided.
- c. Pneumatic tubing will be run and supported within skid limits, from solenoid to valve actuators, as required.

Motor Control Components

Motors shall be VFD controlled.

Local disconnect switches shall be included with motors.

## PART 3 EXECUTION

### 3.1 INSTALLATION

Manufacturer's on-site representative will confirm the final equipment has been installed as per manufacturer requirements and provide written confirmation that all warranties remain in effect.

### 3.2 FIELD AND FACTORY ACCEPTANCE TESTS

A. After the pump has been installed and connected, checked out as required by Section 01 75 00 and after inspection, operation, and adjustment have been completed by the manufacturer's representative, the equipment shall be field tested in the presence of the Engineer, for overall wire to water efficiency and for general performance and fitness for the service specified. Results of these tests shall be submitted to the Engineer.

### 3.3 MANUFACTURER'S SERVICES

A. The services of a factory trained manufacturer's representative shall be provided as described in Section 01 75 00 START UP AND TESTING and as specified herein.

B. As applicable, all components as defined within are factory tested as per manufacturer standards before leaving the manufacturer's facility. The ITP (Inspection and Test Plan) includes a test and / or inspection as defined in the ITP procedure for platform skids, prefabricated pipe (all assembled pipe, not loose supplied pipe), machinery (pumps, blowers, etc.), tanks, valves, instruments, electrical panels / junction boxes, electrical instruments, loose materials (cables, cable trays, etc.), and VFDs, as applicable. A FAT report (Factory Acceptance Test) is provided to the customer. It includes certificates, test reports, and CE declaration of conformity, as applicable. The Contractor is responsible for obtaining the FAT report from the Manufacturer.

C. Services to be provided:

1. For the pumps, the service representative shall be responsible for complete component inspection on site after delivery and shall assist in the correct assembly of the components for a minimum period of one (1) eight hour day.

(a) For inspection and check out of the erected equipment.

(b) For start up services and supervision.

(c) For complete instruction of the Owner's operating personnel.

D. The minimum period of time herein specified does not relieve the manufacturer from providing sufficient time to satisfactorily complete the required service functions.

E. The manufacturer's representative shall certify in writing that

the pump has been properly installed. Contractor shall coordinate with Manufacturer to remedy any necessary/recommended corrections.

F. The Owner reserves the right to video tape the instruction of the operating personnel for future use in training.

G. For services related to the SWRO system, refer to Section 46 82 00 IMMERSSED MEMBRANE FILTRATION SYSTEM AND ANCILLARY EQUIPMENT 3.2 Startup Commissioning, operator training.

#### 3.4 SPARE PARTS

A. Provide all other spare parts as recommended in the manufacturer's standard operations and maintenance information. Provide a list of recommended spare parts for the cartridge filters and the RO membrane skid.

B. Provide minimum two (2) mechanical seals for feed pump.

-- End of Section --

SECTION 46 82 00

IMMERSED MEMBRANE FILTRATION SYSTEM AND ANCILLARY EQUIPMENT  
**09/24**

PART 1 GENERAL

1.1 SUMMARY

- a. This Section includes the automatic strainer, immersed membrane filtration system, permeate and backwash rotary pumps, drain pumps, and ancillary equipment. All installed equipment shall be suitable for seawater application and should be designed and constructed to withstand the harsh conditions of seawater.
- b. All equipment listed below shall be furnished by a single manufacturer and shall be products of manufacturer regularly engaged in the production of said equipment:
  - (1) Water and Wastewater Chemical Feed Systems - Refer to Division 46 specification 46 30 00 WATER AND WASTEWATER CHEMICAL FEED SYSTEMS.
  - (2) Brackish Water Reverse Osmosis Unit - Refer to Division 46 Specification 46 80 00 BRACKISH WATER REVERSE OSMOSIS UNIT
  - (3) Seawater Reverse Osmosis Unit - Refer to Division 46 specification 46 81 00 SEAWATER WATER REVERSE OSMOSIS UNIT
- c. The manufacturer shall have the sole responsibility for proper functioning of the complete system.
- d. Any structural, mechanical, civil, architectural, HVAC, plumbing, electrical or instrumentation changes required by the use of any product (named vendor or approved equal product) other than the design product or an alternate layout shall be the responsibility of the Contractor and shall be provided at no additional cost to Owner.
- e. List of Desalination System Manufacturers:
  - (1) Veolia Water Technologies (Basis of Design)
  - (2) Ovivo Inc.
  - (3) WesTech Engineering, LLC
  - (4) Approved equal

1.1.1 Membrane Filtration System

This Section outlines the furnishing of a complete and operable Membrane Filtration System and provides the Scope of Work that shall be performed by the Membrane Filtration manufacturer (named herein the VENDOR). The Membrane Filtration System (MFS) shall be based on a low pressure submerged ultrafiltration membrane configuration using a vacuum to extract the membrane filtrate. The system shall include, but not be limited to, the following major equipment:

- (1) Membranes.
- (2) Permeate & Backwash Rotary Pumps.
- (3) Turbidity meters, analyzer, and flow measurement.
- (4) Low pressure blowers.
- (5) Clean-in-place (CIP) system.
- (6) Membrane support structures and frames.
- (7) MFS electrical and control (E&IC) system.
- (8) Monitoring equipment.
- (9) Piping and valves internal to and directly above the membrane tanks with termination points at or near the edge of the tanks
- (10) Valves and Instrumentation devices.
- (11) Drain Pumps.

The Manufacturer's scope of supply for the UF system is described in the following table.

| Quantity                      | Description                                                                                                                                                            |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2                             | 2 Feed Strainers with two (2) pressure gauges, one (1) pressure differential switch, and one (1) automatic backwash valve, one (1) automatic drain valve per strainer. |
| 2                             | Strainer backwash                                                                                                                                                      |
| 2                             | Strainer backwash pumps suction isolation valves (one (1) per pump)                                                                                                    |
| 2                             | Sets of sample / drain valves surrounding strainer backwash pumps (two (2) valves per set)                                                                             |
| 1                             | Strainer backwash pumps suction pressure gauge with required isolation valve.                                                                                          |
| 2                             | Strainer backwash pumps discharge pressure gauges with required isolation valves (one (1) per pump)                                                                    |
| 1                             | Strainer backwash pump flowmeter                                                                                                                                       |
| Positive Displacement Blowers |                                                                                                                                                                        |
| 2                             | Membrane air scour blower package (one (1) duty and (1) standby) supplied loose, complete with required motor and appurtenances. VFDs required - by others.            |
| 2                             | Membrane air scour blower isolation valves, manual                                                                                                                     |

| Quantity                                     | Description                                                                                                       |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 3                                            | Membrane air scour blower isolation valves, automatic air relief valves                                           |
| 1                                            | Membrane air scour blower low-flow switches                                                                       |
| UF Membranes and Membrane Tank Appurtenances |                                                                                                                   |
| 3                                            | Sets of membrane cassette support brackets                                                                        |
| 3                                            | Permeate pressure transmitters                                                                                    |
| 3                                            | Membrane tank level transmitters with required isolation valves                                                   |
| 3                                            | Membrane tank inlet flow control valves                                                                           |
| 3                                            | Membrane tank inlet automatic isolation valves                                                                    |
| 3                                            | Membrane tank drain automatic isolation valves                                                                    |
| 3                                            | Membrane tank inlet sample valve                                                                                  |
| 3                                            | Cleaning solution automatic isolation valve, to / from CIP pump                                                   |
| 3                                            | Ejector assemblies, complete with required isolation valves                                                       |
| 3                                            | Ejector air supply assemblies, complete with required isolation valves                                            |
| 2                                            | UF submersible Drain pumps (one (1) duty and one (1) standby)                                                     |
| 2                                            | UF Drain pumps suction isolation valves (one (1) per pump)                                                        |
| 3                                            | Mud valves (one (1)) UF drain valve per Tank                                                                      |
| 1                                            | UF Drain pumps suction pressure gauge with required isolation valve                                               |
| 2                                            | UF Drain pumps discharge pressure gauges with required isolation valve (one (1) per pump)                         |
| 2                                            | UF Drain pumps discharge check and isolation valves (one(1) set per pump)                                         |
| 1                                            | UF Drain pump flowmeter.                                                                                          |
| UF Permeate / BW Pumps                       |                                                                                                                   |
| 3                                            | Process pumps (VFD's required - by others), supplied loose, complete with required motor (one (1) duty per train) |
| 6                                            | Pressure switch high on permeate / backpulse line (two (2) per pump)                                              |
| 3                                            | Permeate / Backpulse isolation valve (one (1) per pump)                                                           |

| Quantity                             | Description                                                                                                                         |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 3                                    | Sets of sample / drain valves surrounding process pump (two (2) valves per set) (one (1) per pump)                                  |
| 6                                    | Process pumps suction and discharge isolation valves (two (2) per pump)                                                             |
| 3                                    | Suction pressure gauges with required isolation valves (one (1) per pump)                                                           |
| 3                                    | Discharge pressure gauges with required isolation valves (one (1) per pump)                                                         |
| 3                                    | Permeate / backpulse flowmeter (one (1) per pump)                                                                                   |
| 3                                    | Permeate turbidimeter panels.<br>Panel is supplied loose with instrument, isolation valve, and needle valve installed on the panel. |
| UF Break Tank & Transfer Pump System |                                                                                                                                     |
| 1                                    | Break tank level transmitter and associated isolation valve                                                                         |
| 1                                    | Permeate temperature transmitter                                                                                                    |
| 1                                    | UF Break tank level switches                                                                                                        |
| 1                                    | Chlorine Analyzer                                                                                                                   |
| 2                                    | Low pressure RO transfer pumps (one (1) duty + one (1) standby), complete with motors (VFD by Others)                               |
| 2                                    | Low pressure RO transfer pumps suction isolation valves (one (1) per pump)                                                          |
| 2                                    | Sets of sample / drain valves surrounding Low pressure RO transfer pumps (two (2) valves per set)                                   |
| 1                                    | Low pressure RO transfer pumps suction pressure gauge with required isolation valve                                                 |
| 2                                    | Low pressure RO transfer pumps discharge pressure gauges with required isolation valves (one (1) per pump)                          |
| 2                                    | Low pressure RO transfer pumps discharge check and isolation valves (one (1) per pump)                                              |
| 1                                    | Low pressure RO transfer pumps flowmeter                                                                                            |
| 1                                    | Low pressure RO transfer pumps isolation valve                                                                                      |
| 1                                    | Set of antiscalant dosing valves, supplied loose                                                                                    |
| 1                                    | Set of sodium bisulfite valves supplied loose.                                                                                      |

| Quantity                                                    | Description                                                                                                                                                           |
|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UF Membrane Cleaning / Neutralization System                |                                                                                                                                                                       |
| 1                                                           | CIP tank level transmitter and associated isolation valve                                                                                                             |
| 1                                                           | CIP tank automatic feed isolation valve                                                                                                                               |
| 1                                                           | CIP tank drain valve                                                                                                                                                  |
| 1                                                           | CIP pump feed automatic isolation valve                                                                                                                               |
| 1                                                           | CIP pump recirculation automatic isolation valve                                                                                                                      |
| 1                                                           | CIP neutralized solution disposal automatic isolation valve                                                                                                           |
| 1                                                           | CIP membrane tank drain automatic isolation valve                                                                                                                     |
|                                                             | CIP membrane tank fill automatic isolation valve                                                                                                                      |
| 1                                                           | CIP pumps (one (1) duty (VFDs required - by others)                                                                                                                   |
| 1                                                           | CIP pump suction pressure gauge with required isolation valve                                                                                                         |
| 1                                                           | CIP pump suction isolation hand valves                                                                                                                                |
| 1                                                           | Sets of sample / drain valves surrounding CIP pump                                                                                                                    |
| 1                                                           | CIP pump discharge pressure gauge with required isolation valve                                                                                                       |
| 1                                                           | CIP pump discharge check valve                                                                                                                                        |
| 1                                                           | CIP pump discharge isolation hand valve                                                                                                                               |
| 1                                                           | CIP flowmeter                                                                                                                                                         |
| 1                                                           | CIP temperature transmitter                                                                                                                                           |
| 1                                                           | CIP in-line electric heater with local control panel, flow switch and temperature switch                                                                              |
| 1                                                           | CIP in-line electric heater feed automatic isolation valve                                                                                                            |
| 1                                                           | CIP in-line electric heater drain automatic isolation valve                                                                                                           |
| 1                                                           | Neutralized cleaning solution check valve                                                                                                                             |
| 1                                                           | Chlorine and pH analyzer panel, supplied loose, complete with isolation valves, check valve and needle valve installed on the panel.                                  |
| 5                                                           | Set of chemical dosing control valves, one set per chemical, to include manual isolation valves, automatic isolation valves, drain / sample valves, and check valves. |
| UF Membrane Cleaning / Neutralization Chemical Feed Systems |                                                                                                                                                                       |



| Quantity | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | <p>Refer to Drawings and Specification Section 46 30.00 Chemical Feed System. The Manufacturer shall provide a complete skid-mounted system. The Contractor is responsible for providing all components not included on the skid system. Each metering pump shall have one spare. For metering pumps of the same size, only one spare unit per size is required. Spare pumps must be suitable for retrofitting into the chemical metering system for the corresponding chemical.</p> |

#### 1.1.2 Equipment and Services

- a. Equipment and Services to be provided by the general contractor, herein referred to as the CONTRACTOR, separate from VENDOR responsibility:
  - (1) Application for all required permits and permissions including but not limited to: civil works, mechanical and electrical works, environmental permits and any and all other permits required for membrane system operation.
  - (2) Design, supply and installation of CIP Tank, UF break tank and permeate Water storage tank.
  - (3) Design, supply and installation of interconnecting piping to and from the VENDOR supplied equipment and/or skid packages including but not limited to:
    - (a) Temporary piping required for the start-up and commissioning of the VENDOR equipment.
  - (4) Design, supply and installation of all pipe supports including pipe hangers for all piping supplied by the VENDOR.
  - (5) Design, supply and installation of pneumatic lines supplying air to the VENDOR supplied pneumatic actuators.
  - (6) Design and supply of all processes and chemical piping.
  - (7) Design, supply and installation of membrane tank and equipment access platforms, walkways, stairs, safety tie-off points, handrails, grating, ladders, etc, as required.

- (8) Design, supply and installation of membrane tank grating and their support over the membrane tanks.
  - (9) Any on-site touch-up primer or painting of equipment supplied by the VENDOR.
  - (10) Design, supply and installation of an overhead traveling bridge crane and pulley/hoist above the membrane tanks for installation and removal of the membrane cassettes. Refer to Bid Option No. 12 New Bridge Crane.
  - (11) Design, supply and installation of all floor drains.
  - (12) Design, supply and installation of bulk chemical storage facilities.
  - (13) Supply and installation of any spacer rings that may be required for VENDOR supplied valves.
  - (14) Design, supply and installation of panel boards, transformers, and other equipment as necessary to provide power distribution and control for VENDOR supplied equipment.
  - (15) Design, supply and installation of the Motor Control Center complete with motor starters, variable frequency drives, motor disconnects and Uninterruptible Power Supply required for the VENDOR supplied equipment.
  - (16) Design, supply and integration of the plant SCADA, including SCADA configuration, testing and interfacing with the VENDOR supplied control system.
  - (17) Design, supply and installation of all power distribution and instrumentation interconnecting wiring, optical fibers, conduit and appurtenances, as follows:
    - (a) Power connections as needed from the electrical power source to the VENDOR control panels, the MCC (by Electrical CONTRACTOR) and from the MCC to the VENDOR supplied equipment.
    - (b) Connections as needed between the various field-mounted instruments and valve actuators to the VENDOR control panels.
    - (c) Includes all cabling trays, conduits, fittings and supports as necessary.
  - (18) Control of equipment and systems not supplied by the VENDOR.
  - (19) Unloading, unpacking, storage (according to the VENDOR's recommendation), installation, assembly and field erection of the VENDOR's supplied equipment.
  - (20) Any items not provided by the vendor will be supplied by the general contractor.
- b. Equipment installation includes:
- (1) Installation of all VENDOR supplied loose-ship equipment.

- (2) Alignment and coupling of all pumps and other rotating equipment by a qualified person.
- (3) Installation of VENDOR supplied control panels.
- (4) Supply one month of storage of all chemicals (pre-treatment chemicals and chemicals used for the cleaning of the membrane modules and plant operation) during start-up, commissioning and acceptance testing of the membrane system. The existing chemical storage tanks are designed to provide one month of chemical storage required for the operation and maintenance of the system.
- (5) Supply of all required oil and lubricants for equipment start-up and initial operation per the manufacturer's specifications for the VENDOR's equipment.
- (6) Replacement of lubricants in all drives and intermediate drives of mechanical equipment after initial break-in of the VENDOR's equipment.
- (7) Flushing of all piping and membrane tanks and verification of removal of all residual debris from construction.
- (8) Make available laboratory services, operating and maintenance personnel during VENDOR equipment start-up, commissioning and acceptance testing.
- (9) CONTRACTOR to coordinate with VENDOR regarding equipment installation, operation, startup and commissioning.
- (10) VENDOR to provide assistance and recommendations to the CONTRACTOR and Engineer during equipment installation, operation startup and commissioning.
- (11) Flushing and disposal of membrane preservative.
- (12) CIP/neutralization tank, concrete construction.

## 1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

### ASTM INTERNATIONAL (ASTM)

|               |                                                                 |
|---------------|-----------------------------------------------------------------|
| ASTM A48/A48M | (2022) Standard Specification for Gray Iron Castings            |
| ASTM B148     | (2024) Standard Specification for Aluminum-Bronze Sand Castings |

## 1.3 BASIS OF DESIGN

The Basis of Design for the intake screens and airburst backwash system is ZeeWeed Immersed 500D three-cassette model, Veolia Water Technologies and Solutions, Trevose, Pennsylvania, or approved equal. Products must meet requirements of the Buy American Act.

A. General

- (1) The UF Membrane process shall utilize a hollow fiber, reinforced, porous media, composed of a polymeric material.
- (2) Three (3) UF membranes are to be immersed in three (30 separate tanks, in direct contact with feed water. A vacuum shall draw water through the UF membranes producing filtrate water. A vacuum is to be applied through the use of a permeate (or filtrate) pump. Intermittent airflow shall be introduced beneath the membrane modules producing turbulence that scours the external surface of the hollow fibers. Filtered water is to be systematically reversed through the membranes while the raw water in the tank is evacuated, effecting a backwash operation.
- (3) The UF system should provide a flexible mode of aeration to automatically respond to challenging feed water quality events.
- (4) Membranes are to be cleaned daily utilizing a low strength oxidant and/or acid, and monthly using a high strength oxidant and/or acid.
- (5) Pre-treatment of the raw water, including chlorine injection, shall be accomplished prior to the membrane system. This pre-treatment, and the clarification pre-treatment process are not within the scope of the VENDOR.
- (6) Screen / strainer shall be 50 micron and supplied by UF vendor.
- (7) Pipe material shall be compatible with seawater. Refer to the pipe schedule in the mechanical process bid drawings for specified pipe materials. All fittings, flanges, connections, fasteners, nuts, washers, nipples, joints, and seals shall be suitable for the intended application and compatible with the seawater environment.

B. Raw Water Quality

The pretreatment system shall include a chlorine injection upstream of the UF. The membrane feed water shall be raw water from the Source Water of the quality described below:

| PARAMETER      | UNITS                     | AVERAGE | MINIMUM | MAXIMUM |
|----------------|---------------------------|---------|---------|---------|
| Alkalinity     | mg/L as CaCO <sub>3</sub> | 70      | 50      | 120     |
| Salinity       | PSS                       | 28.01   | 16      | 33      |
| Total Hardness | mg/L as CaCO <sub>3</sub> | 300     | 2000    | 6364    |
| pH             | -                         | 7.88    | 7.24    | 8.2     |
| Temperature    | C                         | 10.48   | 1       | 20      |

|                                  |      |        |        |        |
|----------------------------------|------|--------|--------|--------|
| Total Organic Carbon (TOC)       | mg/L | 2      | 1      | 3      |
| Dissolved Organic Carbon (DOC)   | mg/L | 1.5    | 0.5    | 2      |
| Apparent Color                   | TCU  | 20     | 10     | 45     |
| Turbidity                        | NTU  | 2      | 1      | 20     |
| Total Suspended Solids           | mg/L | 16.2   | 1.8    | 49     |
| Total Dissolved Solids (TDS)     | mg/L | 23,500 | 18,329 | 35,225 |
| Calcium                          | mg/L | 200    | 100    | 408    |
| Magnesium                        | mg/L | 700    | 500    | 1300   |
| Barium                           | mg/L | 1      | 0      | 4      |
| Strontium                        | mg/L | 6      | 0      | 8.2    |
| Dissolved Iron                   | mg/L | 0.5    | 0      | 1      |
| Sulfate                          | mg/L | 20     | 15     | 25     |
| Dissolved Aluminum               | mg/L | 0.05   | 0      | 0.12   |
| Dissolved Manganese              | mg/L | 0      | 0      | 0.02   |
| Total Silica                     | mg/L | 2.1    | 0.5    | 4.3    |
| Langelier Saturation Index (LSI) | -    | -0.97  | -0.05  | 1.06   |

### C. Permeate Water Quality

Permeate from the membrane system shall be guaranteed to meet the following requirements:

#### Guaranteed Permeate Water Quality

| Parameter                  | Value         |
|----------------------------|---------------|
| pH                         | 7.5 - 9.5     |
| Conductivity               | 0 - 100 umhos |
| Hardness (mg / L as CaCO3) | <0.1 ppm      |

| Parameter  | Value                                          |
|------------|------------------------------------------------|
| Alkalinity | <150 ppm                                       |
| Iron       | <0.1 ppm                                       |
| Turbidity  | Less than or equal to 0.1 NTU 95% of the time. |

Note 1: All guarantees are contingent upon proper maintenance, calibration and service of instruments and other related equipment as per the VENDOR's and original equipment manufacturer's instruction.

#### 1.4 System Design Conditions and Criteria

A. The membrane system shall meet the following design conditions and performance requirements:

##### Design Conditions

| Parameter                                                      | Quantity / Units             |
|----------------------------------------------------------------|------------------------------|
| Number of Trains                                               | 3                            |
| Number of Cassettes Per Train                                  | 1                            |
| Design Minimum Temperature                                     | 1 degree C                   |
| Design Capacity (Net) With All Trains in Service at 1 degree C | 777 gallons per minute (GPM) |
| Recovery at Design Capacity                                    | 90%                          |

R0 Design Capacity (Net) With All UF Trains In Service at 1°C: 0.4 MGD

#### 1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

The ultrafiltration system and ancillary equipment are long-lead items and submittals are critical submittals.

##### SD-01 Preconstruction Submittals

MEMBRANES; G, AE

PERMEATE AND BACKWASH ROTARY PUMPS; G, AE

ASSOCIATED COMPRESSED AIR EQUIPMENT; G, AE

LOW PRESSURE BLOWERS; G, AE

CLEAN-IN-PLACE CIP SYSTEM; G, AE

MEMBRANE SUPPORT STRUCTURE AND FRAMES; G, AE

MFS ELECTRICAL AND CONTROL SYSTEM; G, AE

MONITORING EQUIPMENT; G, AE

PIPING AND VALVES INTERNAL TO AND DIRECTLY ABOVE THE MEMBRANE  
TANKS WITH TERMINATION POINTS AT OR NEAR THE EDGE OF THE TANKS; G,  
AE

VALVES AND INSTRUMENTATION DEVICES; G, AE

DRAIN PUMPS; G, AE

EQUIPMENT LIST; G, AE

1. Major Equipment Mechanical Bill of Material for the project.
2. Includes all major equipment including pumps, tanks, heater, modules, blowers etc.

MECHANICAL BILL OF MATERIALS; G, AE

1. Includes quantity, tag numbers, P&ID reference number, and item description including manufacturer and manufacturer part number, materials of construction, connection sizes and types, and VENDOR part number.

I/O LIST; G, AE

1. A complete listing of the analog and digital I/O on the system.
2. Shows where the I/O is being sent to/from and which panel it is wired to.
3. It identifies the exact location of each IO point on the control modules of every PLC component of the system.  
Issued for information only.

#### SD-02 Shop Drawings

DETAIL DRAWINGS; G, AE

PIPING AND INSTRUMENTATION DIAGRAMS (P&IDS); G, AE

1. For VENDOR-manufactured/supplied major components.
2. Includes scope of supply, pipe and equipment sizing, and materials of construction.
3. Issued for approval.

PROCESS FLOW DIAGRAM (PFD); G, AE

1. Shows the relationship between the major components in

the system.

2. It must show all the unit operations in the process but does not need to show every piece of equipment.
3. Issued for approval.

#### GENERAL ARRANGEMENT DRAWINGS; G, AE

1. General Arrangement and dimensions of individual membrane tanks and all membrane tank trains together in plan, elevation and section views, and including maximum and minimum water levels.

#### NETWORK ARCHITECTURE DIAGRAM; G, AE

1. Provides definition of the control system architecture at a high level.
2. Indicates all PLCs and Remote I/O Panels including PLC make/model.
3. Shows all software communication protocols planned for the system and where employed including the use of any "smart" or networked components.
4. Indicates communication strategy with other entities (SCADA, other controls, etc.).

#### ELECTRICAL DRAWINGS; G, AE

1. Includes the following components: Electrical schematic diagrams, and electrical bill of material.
2. Issued for approval.

#### PLC AND CONTROL PANEL LAYOUT DRAWINGS; G, AE

1. A physical layout of the main control panel and remote I/O panels.
2. Includes the panel front elevation view for both the interior and exterior, the terminal block layouts, and the bill of material.

#### CONTROL LOGIC AND SEQUENCE CHARTS; G, AE

1. An Excel Workbook consisting of several worksheets which describe the required process automation in detail including system modes, equipment modes, interlocks, permissives, sequences, and alarms.
2. The two (2) most significant worksheets are the Sequence Chart (SC) and the Control Logic Chart (CLC).
3. Issued for approval.

SD-03 Product Data



AUTOMATIC STRAINERS; G, AE

MEMBRANES; G, AE

PERMEATE PUMPS AND BACKWASH ROTARY PUMPS; G, AE

ASSOCIATED COMPRESSED AIR EQUIPMENT; G, AE

LOW PRESSURE BLOWERS; G, AE

CLEAN-IN-PLACE CIP SYSTEM; G, AE

MEMBRANE SUPPORT STRUCTURES AND FRAMES; G, AE

MFS ELECTRICAL AND CONTROL SYSTEM; G, AE

MONITORING EQUIPMENT; G, AE

PIPING AND VALVES INTERNAL TO AND DIRECTLY ABOVE THE MEMBRANE  
TANKS WITH TERMINATION POINTS AT OR NEAR THE EDGE OF THE TANKS; G,  
AE

VALVES AND INSTRUMENTATION DEVICES; G, AE

DRAIN PUMPS; G, AE

#### SD-05 Design Data

AUTOMATIC STRAINERS; G, AE

MEMBRANES; G, AE

PERMEATE PUMPS AND BACKWASH ROTARY PUMPS; G, AE

ASSOCIATED COMPRESSED AIR EQUIPMENT; G, AE

LOW PRESSURE BLOWERS; G, AE

CLEAN-IN-PLACE CIP SYSTEM; G, AE

MEMBRANE SUPPORT STRUCTURES AND FRAMES; G, AE

MFS ELECTRICAL AND CONTROL SYSTEM; G, AE

MONITORING EQUIPMENT; G, AE

PIPING AND VALVES INTERNAL TO AND DIRECTLY ABOVE THE MEMBRANE  
TANKS WITH TERMINATION POINTS AT OR NEAR THE EDGE OF THE TANKS; G,  
AE

VALVES AND INSTRUMENTATION DEVICE; G, AE

DRAIN PUMPS; G, AE

#### SD-06 Test Reports

MEMBRANE AND HOUSING TESTS; G, AE

Prior to delivery, all the membranes and housings shall be tested  
and permanently marked with factory serial number.

- (1) A test report shall be submitted to the OWNER through the Engineer prior to membrane delivery.
- (2) The test report shall provide the following:
  - (a) Membrane Serial Number.
  - (b) Date of test.
  - (c) Test conditions.
  - (d) Test results.
- (3) Prior to delivery, submit Factory Test reports as described in Section 40 60 00.

SD-07 Certificates

EMPLOYER'S RECORD DOCUMENTS (FOR WELDING) ; G, AE

WELDING PROCEDURES AND QUALIFICATIONS; G, AE

CERTIFICATES MUST BE SUBMITTED SHOWING CONFORMANCE WITH THE REFERENCED STANDARDS CONTAINED IN THIS SECTION.; G, AE

SD-08 Manufacturer's Instructions

LESSON PLAN FOR THE INSTRUCTION COURSE; G, AE

SD-10 Operation and Maintenance Data

REQUIREMENTS FOR DATA PACKAGES ARE SPECIFIED SECTION 01 78 23 OPERATION AND MAINTENANCE DATA, EXCEPT AS SUPPLEMENTED AND MODIFIED BY THIS SPECIFICATION SECTION. REFER TO SECTION 01 78 24.00 22 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (EOMSI) FOR ADDITIONAL REQUIREMENTS.; G, AE

SUBMIT SPARE PARTS DATA FOR EACH DIFFERENT ITEM OF EQUIPMENT SPECIFIED, WITH OPERATION AND MAINTENANCE DATA PACKAGES. ; G, AE

1. Include a complete list of parts and supplies, with current unit prices and source of supply, a recommended spare parts list for 1 year of operation, and a list of the parts recommended by the manufacturer to be replaced on a routine basis.

AUTOMATIC STRAINERS; G, AE

MEMBRANES; G, AE

PERMEATE AND BACKWASH ROTARY PUMPS; G, AE

ASSOCIATED COMPRESSED AIR EQUIPMENT; G, AE

LOW PRESSURE BLOWERS; G, AE

CLEAN-IN-PLACE CIP SYSTEM; G, AE

MEMBRANE SUPPORT STRUCTURES AND FRAMES; G, AE

MFS ELECTRICAL AND CONTROL SYSTEM; G, AE

MONITORING EQUIPMENT; G, AE

PIPING AND VALVES INTERNAL TO AND DIRECTLY ABOVE THE MEMBRANE  
TANKS WITH TERMINATION POINTS AT OR NEAR THE EDGE OF THE TANKS; G,  
AE

VALVES AND INSTRUMENTATION DEVICES; G, AE

DRAIN PUMPS; G, AE

## 1.6 SAFETY REQUIREMENTS

Exposed moving parts, parts that produce high operating temperature, parts which may be electrically energized, and parts that may be a hazard to operating personnel must be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Safety devices must be installed so that proper operation of equipment is not impaired.

## 1.7 QUALITY CONTROL

### 1.7.1 Detail Drawings

Submit drawings of sufficient size to be easily read, within 90 days of notice of Award of Contract. Submit information in the English language. Provide with English (IP) dimensions. Drawings must consist of complete designs of the manufacturer's systems and other equipment, installation instructions, performance charts, brochures, and other information required to illustrate that the entire system has been coordinated and shall function as a unit.

- a. Outline drawings of the system showing pertinent dimensions and weight of each component of the system. Prepare drawings to scale.
- b. Drawing showing details and dimensions of mounting design or layout including any embedded items.
- c. Cross-sectional drawings of the system showing each component. Show major or complicated sections of the system in detail. Indicate on each drawing an itemized list of components showing type, grade, and class of material used and make and model number of standard component used.
- d. Detail and assembly drawings required for manufacturing showing dimensions, tolerances, and clearances.
- e. Drawings covering erection and installation that are intended to be furnished to the erecting engineer.

### 1.7.2 Quality Assurance

- a. The equipment and material to be furnished under this contract shall be free from defects in materials and workmanship. The complete membrane filtration system shall be provided by one Manufacturer having complete responsibility for the design of the system, including all necessary ancillary equipment, to provide a complete operating

system.

- b. The materials and equipment furnished by the VENDOR shall be installed by a CONTRACTOR. The VENDOR shall be required to provide factory trained technical representatives to monitor installation and to provide technical assistance for start-up, commissioning and testing of the VENDOR's equipment. The VENDOR shall be responsible for the membrane system's ability to meet the specifications and performance guarantees.

#### 1.8 DELIVERY, STORAGE, AND HANDLING

Protect stored items from the weather, humidity and temperature variations, dirt and dust, or other contaminants. Proper protection and care of all material both before and during installation must be the CONTRACTOR's responsibility. Any materials found to be damaged must be replaced at the CONTRACTOR's expense. During installation, cap piping and similar openings to keep out dirt and other foreign matter. Any porous materials found to be contaminated with mold or mildew shall be replaced at the CONTRACTOR's expense. Non-porous materials found to be contaminated with mold or mildew must be disinfected and cleaned prior to installation.

#### 1.9 PROJECT/SITE CONDITIONS

##### 1.9.1 Verification of Dimensions

The CONTRACTOR must become familiar with all details of the work, verify all dimensions in the field, and advise the Contracting Officer of any discrepancy(ies) before performing any work.

##### 1.9.2 Drawings

Because of the small scale of the drawings, it is not possible to indicate all offsets, fittings, and accessories that may be required. The CONTRACTOR must carefully investigate the conditions that would affect the work to be performed and must arrange such work, accordingly, furnishing required offsets, fittings, and accessories to meet such conditions.

##### 1.9.3 Accessibility

Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible.

#### PART 2 PRODUCTS

All installed equipment shall be suitable for seawater application and should be designed and constructed to withstand the harsh conditions of seawater.

##### 2.1 AUTOMATIC STRAINERS

Two automatic self-cleaning strainers shall be installed, configured as shown in the bid drawing, with one serving as duty and the other as standby. The strainers shall meet the following design criteria:

- a. Strainer Manufacturer/Model: Boll model Aquaboll, or an approved equal.
- b. Material of Construction: Cast iron with epoxy lining.
- c. Screen Type & Size: 50 micron mesh.

- d. Strainer Motor: 480 Watt, 120 VAC/60 Hz.
- e. Certification: The strainers shall be certified for seawater application to ensure their suitability and durability in marine environments.

## 2.2 MEMBRANE TANK

General-Each membrane train shall be in a separate, isolatable concrete tank constructed by the CONTRACTOR.

Provide tank level transmitter (one per train), a feed turbidimeter (one common) with instrument, instrument auto-cleaner, isolation valve, and needle valve installed on the panel. Other valves and instruments as required for a complete system.

### 2.2.1 Membrane Modules

- a. Three (3) ultrafiltration modules shall be of hollow fiber construction and configured for a normal filtration flow direction from outside the fiber through to the inside (lumen) of the fiber. The Basis of Design utilizes a total of 204 modules, with 68 modules per cassette, one cassette per train, and three trains in total.
- b. A module is a unit assembly of UF hollow fibers and is the base serviceable element of the membrane filtration process.
- c. The modules are to be constructed such that the fibers are arranged vertically, and permanently bonded in position at both the top and bottom of the module.
- d. The internal lumens of the fibers shall terminate within a channel area at both ends of each module that facilitates permeate collection and flow.
- e. Membrane modules are to be constructed of chemically resistant materials matching or bettering the chemical resistance of the membranes.
- f. The filtration area of each module is to be based on the wetted external surface area of the fibers.
- g. The membranes shall be of reinforced hollow fiber construction and have a nominal pore size of no greater than 0.05  $\mu\text{m}$  and an absolute pore size no greater than 0.1  $\mu\text{m}$ .
- h. Membranes shall be constructed of PVDF.
- i. Membranes are to be capable of being cleaned with sodium hypochlorite, caustic, or citric acid at temperatures up to 30°C.
- j. Cleaning pH can vary between 2 and 10.5. Membranes are tolerant to chlorine to at least 30,000,000 mg-min/L of free chlorine, for cleans up to 1,000 mg/L and a pH of up to 10.5 for up to 10 hours at a time.

### 2.2.2 Certification

Membranes shall be certified for seawater application to ensure their

suitability and durability in marine environments.

#### 2.2.3 Membrane Racks/Cassettes

- a. Membrane modules shall be assembled into units called racks or cassettes.
- b. Each rack/cassette shall be supported within a frame manufactured of material suitable for the intended application and compatible with the seawater environment.

#### 2.2.4 Piping

Provide all piping within the internal limits of the Membrane Tank. Refer to the Membrane System Process and Instrumentation Drawings which delineate the scope of supply included with the MFS. The contractor should supply all piping embedded in concrete. Vendor scope would be limited to the air supply downcomer

The VENDOR shall ensure that all pipes are sized to minimize pipe losses.

### 2.3 PERMEATE (FILTRATE) SYSTEM

#### 2.3.1 General

Provide a pumped permeate collection system including pumps, controls, isolation valves, check valves, flow meters, and pressure gauges. All equipment is furnished loose by the VENDOR for installation and wiring by the CONTRACTOR. Automated valves shall have solenoids mounted on the valve actuator. Solenoids shall be as per Section 2.11 Valves of this specification.

#### 2.3.2 Permeate (Filtrate) and Backwash Pumps

Provide individual permeate pumping system for each of the three (3) trains. The permeate pumps shall be rotary lobe type to provide vacuum pressure, achieve backwashing when operated in the reverse direction, seawater corrosion resistance material, and meet the following design criteria:

- a. Pump Manufacturers/model - Boerger, Model FL 518, or equal
- b. Pump Type - Rotary Lobe Type
- c. Flanges - ANSI #150
- d. Casing - Bronze (e.g. ASTM B148 C95800 Nickel-Aluminum Bronze, Super duplex stainless steel (e.g. SAF 2507) or approved equal
- e. Impeller - Super duplex stainless steel or approved equal
- f. Shaft Sleeve - 316 Stainless Steel
- g. Control: VFD
- h. Motor: 460 Volt, three-phase electric motor, totally enclosed fan cooled (TEFC), class B temperature rise, class F insulation, normal starting torque. Motors shall be Siemens, GE USEM, Brooks Hansen,

Reliance, Westinghouse, Toshiba, or approved equal.

### 2.3.3 Permeate System Valves and Instrumentation

Supplied loose by the VENDOR for installation and wiring by the CONTRACTOR:

- a. Permeate pressure transmitter per train on the permeate line.
- b. Isolation hand valves on suction and discharge of each pump.
- c. Compound pressure gauges on suction and discharge of each pump, with isolation valves.
- d. Drain valves on suction and discharge of each pump.
- e. Sample valves on discharge of each pump, to feed turbidimeter panel.
- f. A dedicated turbidimeter shall be supplied, by VENDOR, per membrane train, to measure the turbidity of the treated water before it is discharged to the T2 UF Break Tank. See section 2.12 Instrumentation for specifications for all metering equipment related to ultrafiltration.

### 2.4 BACKWASH SYSTEM

Components to be provided by the VENDOR, installation to be performed by the CONTRACTOR. The CONTRACTOR shall coordinate with the VENDOR, and the VENDOR shall provide guidance, requirements and recommendations to the CONTRACTOR.

- a. Under normal operation, the membrane trains shall automatically cycle between permeation mode and backwash mode. The treated water used for the backwash procedure is collected in the UF Break Tank. A Backpulse Pump Skid is used to draw the water from the Break Tank and pump it to the inside of the membrane fibers ("inside-out" flow path). Permeate pumps (loose shipped) are used for backwash.
- b. Furnish equipment to create a complete, integral, and automatic backwash system designed to maintain and restore membrane performance. All equipment is furnished loose by the VENDOR for installation and wiring by the CONTRACTOR.
- c. Design the backwash system to clean the membranes in place without requiring their disconnection from the membrane train. The backwash system shall be connected to all membrane trains and shall be designed to backwash one membrane train at a time while the other membrane trains remain in operation.

#### 2.4.1 UF Break Tank

Contractor to provide one (1) 18,800 gallon Tank and interconnecting piping.

- a. The Tank shall have sufficient ports installed in number and size to fill with permeate, Backwash Pump inlet, overflow, discharge to SWRO distribution pump, and drain. Each port shall be fitted with an ANSI #150, gasket, and bolt style mechanical flanges.
- b. The Tank shall be equipped with a level transmitter with local display

and isolation valve.

- c. Provide isolation hand valves on the tank outlet, and on the tank drain line.
- d. The Tank shall be constructed of HDPE or FRP.
- e. Tank shall be configured as per contract drawings.

#### Drain Pumps

Two submersible pumps will be installed inside the wet well to manage the backwash drainage from the UF tanks and to handle any overflow scenarios. The pumps will operate based on level control and will be driven by variable frequency drives (VFDs). The basis of design is the Flygt NP 3127 HT 3 Adaptive 488 model, equipped with a 10 HP motor rated at 3-phase, 480 volts. Each pump is designed to deliver 558 GPM at 18 PSI.

- a. Pump Manufacturers - Flygt Model NP 3127 H3 or approved equal
- b. Impeller Material - Material shall be certified for seawater application.
- c. The pumps shall be certified for seawater application to ensure their suitability and durability in marine environments
- d. Refer to specification 22 13 29 for submersible centrifugal pumps

#### 2.5 CIP/NEUTRALIZATION SYSTEM

- a. Furnish equipment to create a complete, integral, and automatic CIP system designed to maintain and restore membrane performance by reducing or controlling the transmembrane pressure. The CIP system should also be capable of neutralization of the spent chemical solution post clean. All equipment is furnished loose by the VENDOR for installation and wiring by the CONTRACTOR. Automated valves shall have solenoids mounted on the valve actuator per Section 2.11 Valves of this specification.
- b. Chemical cleaning shall be fully automatic, specifically:
  - (1) No operator intervention shall be required to initiate the cleaning process;
  - (2) No operator intervention shall be required during the chemical cleaning process;
  - (3) Upon completion of the cleaning process, the system shall automatically return to normal operation; and
  - (4) Initiation of the chemical cleaning process shall be by either a scheduled event from the PLC/SCADA, or by an operator-initiated command at the OIT / SCADA system.
- c. Design the CIP system to clean the membranes in place without requiring their disconnection from the membrane train. The CIP system shall be connected to all membrane trains and shall be designed to clean one membrane train at a time while the other membrane train



remains in operation.

#### 2.5.1 CIP Pump

A single UF CIP (Clean-in-Place) pump shall be provided for chemical cleaning of the UF membranes.

- a. Manufacturer - Boerger, Netzsch, or approved equal
- b. Pump Type Rotary Lobe Type Reversible Pump
- c. Design Point: 250 GPM at 30 feet TDH
- d. Maximum priming lift: 18 ft
- e. Flange - ANSI #150
- f. Casing/Impeller/ Shaft Sleeve: Pump material shall be certified for highly corrosive materials such as Citric Acid (500 mg / L) and sodium Hypochlorite (500 mg / L).
- g. Control: VFD
- h. Motor: 460 Volt, three-phase electric motor, totally enclosed fan cooled (TEFC), class B temperature rise, class F insulation, normal starting torque. Motors shall be Siemens, GE USEM, Brooks Hansen, Reliance, Westinghouse, or Toshiba.

A single RO CIP pump shall be provided for chemical cleaning of the RO membranes.

- a. Manufacturer: Goulds; Grundfos; or approved equal
- b. Pump Type: Centrifugal operated with VFD
- c. Basis of Design Goulds GL03SH
- d. Application: Highly acidic or alkaline chemical solution
- e. Design Points: 320 GPM at 150 ft TDH; 720 GPM at 150 TDH
- f. Casing/Impeller/Shaft Sleeve: Stainless steel 316 L
- g. Motor: 460 Volt, three-phase electric motor, 60 Hz, 40 hp.

#### 2.5.2 CIP Heater

The CIP system shall include at minimum one (1) three-phase electrical heater suitable for the heating of the chemical cleaning solution to the required heated cleaning temperature (40°C), at minimum one (1) degree Celsius raw water temperature. The heater shall provide numerous protections to avoid overheating the solution and severely damaging piping, membranes, or tanks. These must include:

- a. A dedicated heater electrical panel, with a controller to maintain the desired heater element temperature and hard-wired interlocks as described below.
- b. Low flow switch with hard wired interlock to ensure water is always

flowing across the heater.

- c. High Temperature switch with hard wired interlock.
- d. Heater must be installed in a S10 316SS pipe (part of interconnecting piping supplied by the CONTRACTOR).
- e. If heater is in contact with cleaning solution, it must be of a titanium alloy construction, to avoid corrosion. Ideally, the heater shall not come in contact with cleaning solution, in which case it's acceptable for the heater to be of 316SS construction.
- f. Heater system must include an automated drain valve to ensure heater is dry whenever not in operation in order to prevent corrosion.
- g. Anticipated electrical Requirements: 250 kW, 480 Volt, 3 Phase
- h. Basis of design: Hubbell electric water Model CR250SST4-E-9

#### 2.5.3 CIP Tank

The tank(s) are to be assumed to be of HDPE construction, to be in the scope of the CONTRACTOR. VENDOR to provide a size and quantity estimate for this tank in proposal, 5,000 gallons storage capacity. VENDOR to provide:

- a. Level transmitter(s) with local display and isolation valve.
- b. Provide isolation hand valves on the tank(s) outlet, and on the tank drain line.

#### 2.5.4 CIP System Valves and Instrumentation

Supplied loose by the VENDOR for installation and wiring by the CONTRACTOR, at minimum the below:

- a. One (1) CIP temperature transmitter.
- b. One (1) CIP flow meter.
- c. Isolation hand valves on the suction and discharge of pumps.
- d. Drain valves on suction and discharge of pumps.
- e. One (1) check valve on discharge of pumps.
- f. Pressure gauges on suction and discharge of pumps, with isolation valves.
- g. CIP tank filled from the BWRO permeate pipe (valve by others). Back up fill line from city water.
- h. Automatic valving that allows CIP solution to flow to the CIP heater, returned to the CIP tank, or waste to drain as VENDOR requires.

#### 2.6 pH/CHLORINE ANALYZER PANEL

A pH/Chlorine Analyzer Panel shall be used to ensure the neutralized cleaning solution is at an acceptable pH and a low enough residual free

chlorine concentration to permit discharge to sewer. The pH/Chlorine Analyzer Panel includes the following equipment installed and pre-wired on the panel with interconnecting tubing between the various pieces of instrumentation and valves:

- a. One (1) in-line chlorine sensor, complete with display, capable of measuring 5 ppm or less of free chlorine, and tolerant of up to 200 ppm of chlorine.
- b. One (1) in-line pH sensor meter to measure pH of cleaning solution during neutralization of cleaning solution waste.
- c. Flush line to ensure both sensors are in contact with water during CIP downtime.
- d. Isolation, dosing, and needle valves.

## 2.7 MAINTENANCE CLEAN AND RECOVERY CLEAN DESCRIPTIONS

- a. Table below provides a summary for typical cleaning type and frequencies:

|                                 |                                                                                                                                                                                                      |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maintenance Clean (MC) Protocol | 2 / day / train<br><br>12 times per week using 250 mg/L NaOCl<br><br>1 time per week using 500 mg/L citric acid with mineral acid for pH adjustment to 2.1<br><br>Maintenance cleans are not heated. |
| Recovery Clean (RC) Protocol    | 24 / year / train<br><br>500 mg/L NaOCl followed by 12 / year / train 2 g/L citric acid with mineral acid for pH adjustment to 2.1<br><br>Recovery clean solutions are heated prior to each clean.   |

- b. A Recovery Clean (RC) is utilized to recover lost capacity of the membranes due to particulate fouling which cannot be removed through the processes of air scouring, back pulsing, and maintenance cleaning. The RC operation shall be facilitated by the same CIP system and chemical dosing equipment supplied for performance of the maintenance clean and shall be a similar operation of longer duration. A recovery clean is heated.
- c. The cleaning procedure shall be initiated automatically through the PLC and shall be a fully automated procedure.
- d. The Contractor shall provide two (2) sets of 6-inch diameter, 25-foot long flexible hose connections suitable for use with highly corrosive chemicals during SWRO membrane cleaning, which is performed four (4) times per year.

## 2.8 CHEMICAL CLEAN DOSING EQUIPMENT:

Adhere to the following specifications as well as Section 46 30 00 WATER AND WASTEWATER CHEMICAL FEED EQUIPMENT of the contract documents.

- a. Chemical Dosing Pumps: Chemical dosing systems shall be supplied for all cleaning and neutralization chemicals required by the membrane system. The cleaning and neutralization chemicals requiring chemical dosing systems are:
  - (1) 12.5% sodium hypochlorite
  - (2) 50% liquid citric acid
  - (3) 38% sodium bisulfite
  - (4) 25% sodium hydroxide
  - (5) 50% sulfuric acid
- b. Chemicals with larger flows shall be pumped using a mag drive pump system. The mag drive pump system includes the following (all supplied loose for installation by the CONTRACTOR):
  - (1) One (1) strainer
  - (2) One (1) chemical pump suction isolation hand valve per pump
  - (3) One (1) chemical pump discharge isolation hand valve per pump
  - (4) One (1) duty mag drive chemical pump and one (1) standby
  - (5) One (1) pump discharge check valve per pump
  - (6) Two (2) sample/drain valves per pump
  - (7) One (1) chemical pump suction pressure gauge with isolation valve per pump
  - (8) One (1) chemical pump discharge pressure gauge with isolation valve per pump
  - (9) One (1) chemical pump flowmeter
  - (10) One (1) hand control valve
  - (11) Two (2) isolation valves.
  - (12) One (1) pressure relief valve (for sodium hypochlorite service only)
- c. Chemicals with smaller flows may be pumped from a panel-mounted chemical pump system. The Chemical Pump Panel includes the following equipment installed and pre-wired on the panel with interconnecting piping between the various pieces of instrumentation and valves:
  - (1) One (1) duty and one (1) standby chemical dosing pump
  - (2) One (1) calibration column

- (3) One (1) pressure relief valve per pump
- (4) One (1) pressure gauge with diaphragm seal
- (5) Lot isolation valves
- (6) PVC or PVDF piping and/or tubing, selected to be chemically compatible with the cleaning chemical
- (7) Electrical junction box for any solenoids, level switches, motor or other device requiring an electrical connection.

## 2.9 BLOWER SYSTEM

Furnish equipment to create a complete, integral, and automatic blower system designed to air scour the membranes during backwash and sized for the maximum air demand with all module spaces on a train filled. All equipment is furnished loose by the VENDOR for installation and wiring by the CONTRACTOR.

- a. One (1) duty and one (1) standby blower shall be supplied. The Blower shall conform to the following:
  - (1) Blower Type - Positive Displacement Rotary Lobe - Oil Free
  - (2) Manufacturer - Aerzen, Kaeser, or approved equal
  - (3) Model - Aerzen GM Series (Basis of Design)
  - (4) Connections - Flexible Sleeve Type Connection
  - (5) Casing & Head Plates: cast iron EN-GJL-200, equivalent to ASTM A48/A48M class 30
  - (6) Shaft - C45N, equivalent to AISI 1043
  - (7) Rotors - ductile iron or forge steel C45N, equivalent to AISI 1043
  - (8) Drive - V-Belt Drive, automatic tension
  - (9) Guard - plate OSHA-rated V-belt drive guard
  - (10) Control: VFD
  - (11) Motor - TEFC 1,800 or 3,600 rpm, inverter-rated
  - (12) Accessories: one (1) air inlet filter, one (1) air intake/silencer, one (1) air discharge silencer, one (1) pressure relief valve, one (1) discharge pressure gauge with isolation valve, and one (1) check valve, and one (1) acoustic enclosure.
  - (13) Blower System Valves and Instrumentation: Supplied loose by the VENDOR for installation and wiring by the CONTRACTOR at minimum the below:
    - (a) One (1) Blower isolation hand valve.
    - (b) One (1) membrane air scour flow switch.

## 2.10 VALVES

Valves in contact with raw water or permeate shall contain wetted components of stainless steel, PVC, or food grade elastomers.

### 2.10.1 Automated Valves

- a. Automated on/off valves that cycle regularly shall be butterfly type and incorporate a pneumatic actuator controlled by a 2-way solenoid valve. Butterfly valves shall be lug-style suitable for installation between ANSI # 150 flanges.
- b. Pneumatic actuators shall be double acting cylinder - rack and pinion type. Actuators shall operate at a drive air pressure of 80 psig or greater.
- c. Air to valve actuators must be controlled by 2-way electrical solenoid valves that receive electrical signals from the control system. Solenoid valve is mounted on the actuator. Manual actuation capability is available so that the system's normally automatic valves can be manually controlled in the event that the main system controller fails.
- d. Electrically operated valves can be provided at the location indicated on the bid drawings.

### 2.10.2 Manual Valves

Manual valves for on/off service shall be ball or butterfly type. For throttling service, manual valves shall lock in place to prevent position changes without manual intervention.

### 2.10.3 General System Valves

- a. Butterfly valves shall be cast iron, epoxy coated, lug style body with Nylon coated ductile iron disc or stainless-steel disc (dependent on application), EPDM seat (NSF 61 certified), and 316 stainless steel shaft.
  - (1) Manufacturer/Model: Bray Series 31 or 3L
- b. Chemical feed ball valves (< 3") shall be Tru-Union style with PVC body, plug, and shaft, with EPDM or Viton seals (as required for chemical service).
  - (1) Manufacturer/Model: George Fischer Type 546 or equal
- c. Instrument isolation ball valves shall be 316 stainless steel body valves with PTFE seats and socket weld or threaded end connections.
  - (1) Manufacturer/Model: Apollo model 76F or equal
- d. General Service water check valves shall be wafer swing check valves suitable for installation between ANSI #150 flanges, with stainless steel body and disc, and EPDM seat.
  - (1) Manufacturer: Ritepro or equal
- e. Check valves on chemical lines are ball-style with PVC body and disc.

(1) Manufacturer: George Fischer or equal

- f. Two-way isolation solenoid valves for air or water service shall be F-NPT style with 316 stainless steel or brass body and EPDM seal.

(1) Manufacturer: Asco, Burkert or equal

- g. Needle valves shall have 316 stainless steel body and seat materials.

(1) Manufacturer: Whitey or equal

#### 2.10.4 General System Pneumatic Actuators

- a. Pneumatic actuators on general service butterfly valves shall be double acting cylinder - rack and pinion type. Actuators shall operate at a drive air pressure of 80 psig or greater. Actuators shall have body material of extruded aluminum alloy, anodized. Include travel stops  $\pm 5^\circ$  at  $0^\circ$  and  $90^\circ$  positions. End caps and pistons of die cast aluminum alloy. Output shaft of carbon steel (zinc plated). Acetal shaft bearings. Stainless steel fasteners. Buna-N O-rings.

(1) Manufacturer/Model: Bray Series 92

- b. Pneumatic actuators on PVC valves shall be double acting. Actuators shall operate at a drive air pressure of 80 psig or greater. Actuators shall have body material and end caps of PP, fiberglass reinforced.

(1) Manufacturer: George Fischer

#### 2.11 INSTRUMENTATION

General: Unless otherwise indicated, VENDOR shall supply, calibrate, configure, test and start-up all instruments and equipment related to the VENDOR provided packaged system.

##### 2.11.1 Level Transmitters

Level transmitters shall be supplied for monitoring the liquid levels in the Membrane Tanks, Backwash, and CIP Tank(s). The transmitter shall be hydrostatic type, immersion entry with Stainless Steel and PPS NEMA 4X housing. Acceptable manufacturers and specifications are:

- a. Manufacturer/Model: E&H FMX21
- b. Application - Salt Water, Immersion
- c. Mounting - Suspension clamp, cable mounting screws
- d. Output - 4-20 mA
- e. Voltage - 11.5 to 30 VDC
- f. Housing - PPS/Polyolefin & 316L

##### 2.11.2 Pressure Transmitters

Pressure transmitters shall be supplied to monitor the positive pressure

on both the inside and outside of the membrane in order to measure TMP, and should be located on the permeate line. Acceptable manufacturer and specifications for the pressure transmitters are:

- a. Manufacturer/Model: E&H Cerabar PMC11 or equal
- b. Mounting - 0.5" NPT
- c. Output Signal - 4-20 mA
- d. Voltage - 11.5 to 30 VDC
- e. Diaphragm - Ceramic
- f. Housing - 316L NEMA 4X
- g. Integral LCD - Yes

#### 2.11.3 Electromagnetic Flowmeters

All flow meters must be installed according to the manufacturer's recommendations including the following considerations:

- a. Minimum upstream and downstream pipe diameters.
- b. Installed vertically (with flow going up) or horizontally in flooded pipe.
- c. Grounding rings installed, supplied with instrument.
- d. Upstream of any chemical injection or given additional downstream distance for dampening of noise caused by chemical pulses.
- e. Manufacturer/Model: E&H Promag W400 or equal
  - (1) Process Connections ANSI #150 flanged
  - (2) Housing 316L Stainless Steel NEMA 4X
  - (3) Wetted Materials:
    - (a) Electrodes: Hastelloy / C22
    - (b) Liner PTFE
  - (4) Accuracy +/- 0.5%
  - (5) Operator Interface Local
  - (6) Output 4-20 mA DC

#### 2.11.4 Turbidity Meter and Controller

All turbidity meters and controllers must be installed according to manufacturer's recommendations. In addition, they must be supplied with the following:

- a. Sensor and controller mounted on a panel with minimal or no installation required by the OWNER.



- b. Automatic isolation valve, so equipment is not drained when not in production.
- c. For the Permeate Turbidimeter, include a bubble trap, large enough to remove any air bubbles introduced to the permeate stream during normal operation. It is not acceptable to ignore high turbidity alarm conditions caused by air bubbles beyond what is deemed acceptable by the State/Provincial Regulator/Authorities.
- d. Manufacturer/Model: Hach TU5300sc low range laser turbidimeter
  - (1) Ranges as VENDOR requires
  - (2) Output Signal - 4-20 mA
  - (3) Power - 120V/1ph/60 Hz
  - (4) Display - Hach SC 4500
  - (5) Accessories: Include Auto Cleaner

#### 2.11.5 pH/Residual Chlorine Sensor and Transmitter

All pH/chlorine meters and transmitters must be installed according to manufacturer's recommendations. In addition, they must be supplied with the following:

- a. Sensor(s) and controller mounted on a panel with minimal or no installation required by the OWNER.
- b. Automatic isolation valve, so equipment is not drained when not in CIP, and not exposed to high chemical concentrations that may damage the equipment.
- c. Automatic isolation valve for flush water to clean and maintain chlorine probe between CIP operations.
- d. Tolerant of up to 500 ppm of chlorine, but accurate to 0.1 ppm to confirm neutralization of cleaning solution.
- e. pH range 0-12, accuracy of +/- 0.01 units.
- f. Chlorine Sensor Manufacturer/Model: E&H Memosens CCS51D
- g. pH Sensor Manufacturer/Model: E&H Memosens CPS31D
- h. Chlorine/pH Transmitter Manufacturer/Model: E&H Liquiline CM444

#### 2.11.6 Temperature Transmitters

Temperature sensors shall be RTD assemblies. These shall be installed according to the manufacturer's recommendations. Transmitters supplied shall be:

- a. Acceptable manufacturer and specifications for the temperature transmitter(s) are:
  - (1) Manufacturer/Model: E&H TH 11 or 13 RTD assembly

- (2) Mounting - 0.5" NPT process connection
- (3) Power Supply - 24 V DC
- (4) Output Signal - 4-20 mA
- (5) Ambient Temperature Range: -40 to 85°C

#### 2.11.7 Pressure Gauges

Pressure gauges shall be included for the various pumps included with the membrane system. Pressure gauges shall be liquid filled with 316SS wetted parts. Pressure gauges shall be supplied complete with an isolation ball valve / gauge cock between the pressure gauge and the process line.

- a. Manufacturer/Model: Ashcroft Type 1008S
- b. Air Flow Switches
- c. An air flow switch shall be included to provide flow confirmation on the blower line.
- d. Manufacturer/Model: Kobald KAL-L

#### 2.11.8 Pressure Switches

Refer to Division 40 specifications.

#### 2.11.9 Liquid Flow Switches

Refer to Division 40 specifications.

#### 2.11.10 Temperature Switches

Refer to Division 40 specifications.

### 2.12 SPECIAL TOOLS

- a. Furnish all special tools required for routine operation of the membrane filtration system. Special tools include but are not limited to the following:
  - (1) One (1) membrane repair kit
  - (2) Hand tools and hoisting rings used for cassette and module removal (if required).

### 2.13 CONTROL SYSTEM

- a. A single programmable logic controller (PLC) with hard wired I/O shall be provided. The PLC CPU and Ethernet communication modules shall be located in the UF Control Panel fully assembled and tested, to be mounted indoors in a non temperature controlled environment. The control panel shall have a dead front panel.
- b. A Human-Machine Interface (HMI) shall be located on the interior door of the UFCP. Ethernet connections to the VENDOR's control panel and other communication nodes is by the CONTRACTOR. The UFCP shall also

provide terminations for all non-train equipment, such as chemical equipment, air compressors, blowers, CIP, and Backwash equipment, and any other loose shipped equipment supplied by the VENDOR.

- c. There shall be one (1) remote I/O panel per membrane train. This panel shall provide terminations for all train specific equipment, such as the permeate pump, pneumatic and solenoid valves, mag flow meter, pressure transmitters and switches, turbidity transmitter, temperature transmitter, etc.
- d. Control System: The control system shall be designed based on a distributed I/O architecture. The control system shall be based around Allen Bradley Control Logix platform of Programmable Logic Controllers (PLC). The control system shall include the following equipment:
  - (1) One (1) Control Panel (UFCP), rated NEMA 4X with dead front. Shall include PLC, HMI and all required control system equipment.
  - (2) The PLC shall include a minimum of 20% spare I/O for future use. Power integrity shall be furnished via Uninterruptible Power Supply (UPS), supplied by the CONTRACTOR.
  - (3) The UFCP shall include a color touch screen HMI mounted in the door of the enclosure. The HMI shall provide a graphical interface to the membrane filtration system process. The terminal shall communicate via an Ethernet connection to the PLC. The terminal shall be an Allen Bradley PanelView Plus or approved equal.
  - (4) The UFCP shall include a five port industrial Ethernet switch to facilitate communication between the PLC, HMI, and to the plant-wide control network as required. The switch shall be DIN rail mounted and be 24VDC powered.
  - (5) To protect against lightning and transient voltage surges, each control panel shall be protected at the incoming power by a transient surge suppressor located in the control panel.
  - (6) All Digital I/O shall be provided with interposing relays.
  - (7) All Analog I/O shall be provided with surge protection.
  - (8) Dry relay contacts shall be provided for all alarms.
  - (9) Field wiring of all the I/O to and from the control panels shall be by the CONTRACTOR in accordance with wiring diagrams provided by the VENDOR. Unless otherwise specified, control panels shall be designed primarily to accommodate field wiring entering from the bottom of the enclosure.
  - (10) MCP should include programming for all equipment within the UF, SWRO and BWRO systems. Local remote I/O panels shall be rated NEMA 4 and be provided the UF system (One RIO panel for the UF system). The local I/O shall be Allen-Bradley Flex I/O. The remote I/O shall communicate via Ethernet.
- e. The membrane system PLC shall be compatible for communication and integration with the Plant PLC/SCADA system.

- f. Base the control architecture on a PLC located in a MCP with distributed I/O to control the membrane trains. Provide controls for all elements scope supplied. Provide all instrumentation required to control and monitor the operation of the membrane filtration system. Provide all PLC and HMI programming for all supplied equipment. Supply all software updates to control, monitor and operate the membrane filtration system.
- g. To provide warranty support, VENDOR shall provide field service personnel as required by the warranty conditions

#### 2.14 CONTROL OPERATION

- a. All valves and control devices shall be interlocked through the PLC to allow smooth and continuous automatic operation. Valves shall open, close and/or modulate, depending on signals from the PLC. These signals shall be predetermined through PLC programming and allow the system to operate at optimal conditions. Variable speed pumps shall also be controlled by the PLC and vary their vacuum/flow output based on level signals from the process tanks.
- b. All operating parameters shall be continuously monitored by the PLC. If an alarm or emergency condition occurs, the PLC program shall instruct the various components to change operation conditions and/or shut down the system and alert the operator. The system control logic shall be designed with the ability to shut down the system in the event of an alarm condition that could be detrimental to the equipment.

#### 2.15 HMI SYSTEM

- a. The HMI shall be an Allen-Bradley PanelView Plus. It shall be installed on the door of the Master Control Panel. It shall be programmed to provide the following features:
  - (1) Monitoring of the system and of individual devices.
  - (2) Control of the system and of individual devices.
  - (3) Adjustment of operating parameters.
  - (4) Operator and supervisor level passwords.
  - (5) Current alarm summary.
  - (6) Alarm history.
- b. All monitoring, control, and alarm logic shall reside in the PLC CPU. Data points defined in the CPU shall be communicated over the Ethernet network to provide the operator monitoring and control functionality at the HMI. These data points shall be available to any device on the Ethernet network, such as the Plant SCADA system (by the CONTRACTOR), to provide some or all of the operator functionality that is found on the HMI. VENDOR recommends that the Plant SCADA system incorporate full Operator functionality to act as a backup in the unlikely event of an HMI failure and to provide a more convenient location for operator interaction.
- c. Since the Plant SCADA system shall have access to the data in the VENDOR's PLC, features such as alarm annunciation and logging, data

logging and trending, and report generation can be easily implemented on the Plant SCADA system.

## 2.16 HMI SOFTWARE STANDARDS

- a. HMI applications shall be developed using VENDOR's standards as described below. In general, these standards apply equally to HMI and Plant SCADA applications as the intent is to maintain a common "look and feel" regardless of platform. HMI applications shall be developed using Rockwell FactoryTalk View Studio to run on an Allen-Bradley PanelView Plus HMI.
- b. All screens shall have the following basic features:
  - (1) The top left corner of the screen shall have the OWNER's logo. For convenience, the OWNER's logo shall also function as a previous screen pushbutton. if the OWNER's logo is not available then it shall be replaced by the VENDOR's logo.
  - (2) The top right corner shall have the VENDOR's logo.
  - (3) The date and time shall be displayed on two lines immediately to the left of the VENDOR's logo.
  - (4) The screen title shall be displayed centered at the top of the screen.
  - (5) Common navigation and control buttons shall be located at the bottom of the screen.
  - (6) Navigation arrows shall provide a convenient means to change from one screen to another to follow the process.
  - (7) A "screen select" pushbutton shall list all the main screens that are available and shall provide a convenient means to change to any screen from the current screen.
  - (8) When a device on the screen is pressed, a popup screen shall appear providing control of that device.
  - (9) An alarm banner shall pop-up when a new alarm occurs.
- c. Alarms triggered by the PLC shall be displayed and recorded with a date and time stamp. The Active Alarm screen shall list all alarms that are currently active. The Alarm History page shall list all alarms that have occurred. All alarm messages shall be configured with red text on a black background. The system shall be set-up to automatically delete daily log files older than two months.
- d. The HMI shall be configured for various security levels to ensure that users are permitted the monitoring and control functionality that they have been granted. Unless otherwise specified, there shall be four (4) security levels: GUEST, OPER, SUPER, and VENDOR's security level, each level having an increasing level of privileges. The system shall be configured to log out the current user automatically after a pre-determined amount of time or inactivity.

## 2.17 ELECTRICAL

### 2.17.1 Junction Boxes

- a. Junction boxes to facilitate field wiring shall only be supplied for skid or panel-mounted equipment where it is possible to group numerous electrical connections together in order to minimize field installation.
- b. Design, supply, and installation of all other junction boxes shall be by the CONTRACTOR.

### 2.17.2 Electrical Panel Wiring

All wiring within the control panel shall conform to the following standards, at a minimum:

- a. All conductors shall be copper and have 600 Volt rated insulation.
- b. Control circuit and instrument wires shall be a minimum of 18AWG (max 7 Amps) Type MTW construction B, or TFFN for UL or type TEWN for CSA.
- c. Power circuits (branch and feeder circuits) and bonding conductors shall be a minimum of 14 AWG (max 15 Amps) Type THWN-2 or THHN/THWN for UL or type T90 Nylon for CSA.
- d. Wiring of all the I/O to and from the control panels to devices shall be by CONTRACTOR.

## 2.18 SPARE PARTS

An itemized spare parts list shall be generated by the VENDOR after the final scope of supply by the VENDOR has been determined (after Final Shop Drawing submittal). This list shall include pricing. The CONTRACTOR will provide all other spare parts as recommended in the manufacturer's standard operations and maintenance information. The Contractor will provide minimum two (2) mechanical seals for feed pump.

## PART 3 EXECUTION

### 3.1 INSTALLATION

The CONSTRUCTION CONTRACTOR shall install the MFS in strict accordance with VENDOR's instructions and recommendations.

VENDOR's field service representative (FSR) shall instruct the CONTRACTOR on installation, and inspect and oversee the proper placement and installation of membrane modules and ancillary equipment furnished by the VENDOR. VENDOR shall recommend any necessary adjustments to CONTRACTOR after VENDOR's inspection to remedy any membrane modules or ancillary equipment not properly installed.

VENDOR shall provide oversight of all work to install components provided to ensure that MFS warrantee remains intact. This includes any required disassembly/reassembly of membrane skids, pump skids, etc. needed to deliver the components into the final locations within the building.

CONTRACTOR shall be responsible for making any recommended adjustments to the MFS as a result of VENDOR's inspection and other inspections performed by the OWNER or Engineer.

### 3.2 START UP, COMMISSIONING, OPERATOR TRAINING

#### 3.2.1 Services (Base services)

The following services shall be provided by the Vendor, coordinated by the CONTRACTOR:

- a. VENDOR shall provide a minimum of start up, commissioning and training services as specified below, allocated across the specific number of trips to site included. Additional trips and time on site can be discussed further with the OWNER to suit project and site considerations, at additional cost.
- b. The VENDOR shall furnish qualified field personnel (full time employee of the membrane manufacturer). The field personnel shall demonstrate experience with installation, start-up, and operation of MFS's of comparable size, application, and complexity and shall be acceptable to the OWNER. The VENDOR's FSR shall be a qualified individual who has previously provided onsite services for the installation, testing, and startup of a membrane system at a minimum of three immersed UF or MBR's of comparable size.
- c. The VENDOR shall furnish qualified field personnel for the RO systems. The field personnel shall demonstrate experience with installation, start-up, and operation of RO systems.

#### Startup, Commissioning, and Operator Training Requirements

| # Trips | Minimum Days On-Site | General Scope of Work / Description                                                                                                                                                                                                                       |
|---------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | 2                    | Pre-start-up Inspection Visit: VENDOR shall provide time to inspect installation work, address questions, assist CONTRACTOR in development of punch list of completion items necessary prior to return visit, and provide guidance on module installation |
| 2       | 6                    | For assistance installing the system with times and activities to be determined jointly by VENDOR and CONTRACTOR.                                                                                                                                         |
| 5       | 25                   | System Commissioning including: <ul style="list-style-type: none"><li>- Electrical and mechanical checkouts</li><li>- Membrane installation verification</li><li>- Controls testing including integration</li><li>- Process start-up</li></ul>            |
| 1       | 5                    | Assistance during Performance Testing                                                                                                                                                                                                                     |

| # Trips     | Minimum Days On-Site | General Scope of Work / Description                                                                                                                                                                                                                                                                                                                                                         |
|-------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1           | 10                   | Operator Training: VENDOR shall provide formal training that shall take place prior to the Performance Test. Trainer shall instruct OWNER's operation and maintenance personnel on proper operation, installation, sampling, cleaning, maintenance, and troubleshooting for the membrane system. OWNER shall be allowed to video tape training for future use.                              |
| Total<br>10 | Total<br>48          | <p>VENDOR shall additionally include for travel time to and from the job site for VENDOR's Field Service personnel. Travel / living expenses shall be also included by the VENDOR.</p> <p>VENDOR shall assume an 8-hr workday and no weekends or holidays are required.</p> <p>The OWNER shall provide the VENDOR a minimum of two weeks' advance notice to schedule all trips to site.</p> |

### 3.2.2 VENDOR's Responsibilities Following Substantial Completion

Following Substantial Completion, the CONTRACTOR shall coordinate with the VENDOR to ensure the following services are provided: (Unless specifically noted below, the costs for these services are included in the lump sum price for the membrane system.)

- a. Emergency Support Services: Telephone Assistance - Provide 24-hour per day, seven-day per week support for two (2) years for troubleshooting assistance by technical support specialists at a toll-free number.
- b. Ten (10) service visits (each service visit being two (2) days) during the first five (5) years of operation, to be scheduled at the agreement of the OWNER and VENDOR.

### 3.2.3 Vendor's Service Visit

The service visit shall include the following activities:

#### 3.2.3.1 Process Monitoring

- a. Inspect and confirm proper operation of the membrane system in accordance with the manufacturer operation & maintenance manual.
- b. Review operating logs, analytical tests and InSight data (if available) with the operator(s). Discuss operator concerns and manufacturer issues emerging from this review.
- c. Discuss imminent seasonal shifts. Plan and implement forward looking adjustments.
- d. Advise the operations manager of technical updates as they become available from manufacturer.



#### 3.2.3.2 Membrane Integrity

- a. Assess pressure decay test or bubble test data as it correlates to the membrane condition and/or permeate water quality. Assist plant operators to repair membrane fibers as required.
- b. Assess the effectiveness of on-going membrane cleaning procedures (recovery cleaning, maintenance cleaning, backwashing) and provide recommendations to the plant operators as necessary.

#### 3.2.3.3 Controls

- a. Review system alarm history, discuss any related issues with operator(s) and recommend appropriate actions to be taken.
- b. Perform limited PLC code modifications as planned in advance, secured by proper documentation, dial-in capabilities and file backup precautions.
- c. Verify operation of all safety interlock/controllers, pressure switches and temperature switches.

#### 3.2.3.4 Verification of Instrument Calibration

- a. Review set points, verify the condition of all control instruments, sensors, probes, and transmitters, including switching action and output. Assist operators with re-calibration, as necessary.
- b. Collaborate with the plant operator in maintaining a log of calibration activities.

#### 3.2.3.5 Preventative Maintenance Planning

- a. Develop a preventive maintenance plan with the plant operator.
- b. Review the spare parts provisions with the plant operator and identify any additional parts to provide the desired level of security.

#### 3.2.3.6 Training

- a. During scheduled site visits, the manufacturers service representative can provide operators with informal training on any areas of concern; to explain the operation, process, maintenance or troubleshooting activities and, in general, to enhance operator ability and confidence.

#### 3.2.3.7 Reporting

- a. Manufacturers shall provide a report to record membrane condition, tasks accomplished during the visit and identify key operating and maintenance issues.

#### 3.2.3.8 Spares

- a. The service representative shall review the spares provisions with the plant operator to identify any additional spare parts which should be brought into inventory to provide the desired level of security to the plant.

#### 3.2.4 Bid Option No. 7

Post construction completion follow-on services: Price to provide continuing operation of the desalination plant after project completion and commissioning for one year. Under this bid option, the manufacturer shall provide a full-time Lead Plant Operator (40 hours per week for one year) with experience operating similar equipment and facilities. The Lead Plant Operator will be responsible for overseeing plant operations and training the team in equipment operation, process control, record-keeping, reporting, troubleshooting, and maintenance.

#### 3.3 INSPECTING AND CLEANING

The pipes must be flushed with cold water after finishing the installation. Flush the system until the water runs clear of debris and dirt. Inspect and test piping systems following procedures of authorities having jurisdiction and as specified by the piping system manufacturer.

Clean water distribution piping following procedures of the manufacturer and/or the Authority Having Jurisdiction.

#### 3.4 ELECTRICAL INSTALLATION

Install electrical equipment in accordance with NFPA 70 and manufacturer's instructions. Wiring must be in conduit. Electrical installation by others.

#### 3.5 FIELD TESTS

Field tests must be conducted in the presence of the QC Manager or their designated representative to verify systems compliance with specifications. Material, equipment, instruments, and personnel required for the test must be provided by the CONTRACTOR.

#### 3.6 INSTRUCTION TO GOVERNMENT PERSONNEL

- a. Furnish the services of competent instructors to give full instruction to the designated Government personnel in the adjustment, operation, and maintenance, including pertinent safety requirements, of the desalination system (Strainer, UF, SWRO, BWRO, CIP cleaning process. Instructors must be thoroughly familiar with all parts of the installation and must be instructed in operating theory as well as practical operation and maintenance work. Submit a lesson plan for the instruction course for approval. The lesson plan and instruction course must be based on the approved operation and maintenance data and maintenance manuals.
- b. Conduct a training course for the operating staff and maintenance staff selected by the Contracting Officer. Give the instruction during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. The number of days (8 hours per day) of instruction furnished must be ten (10) business days. Use approximately half of the time for classroom instruction and the other time for instruction at the location of equipment or system.
- c. When significant changes or modifications in the equipment or system

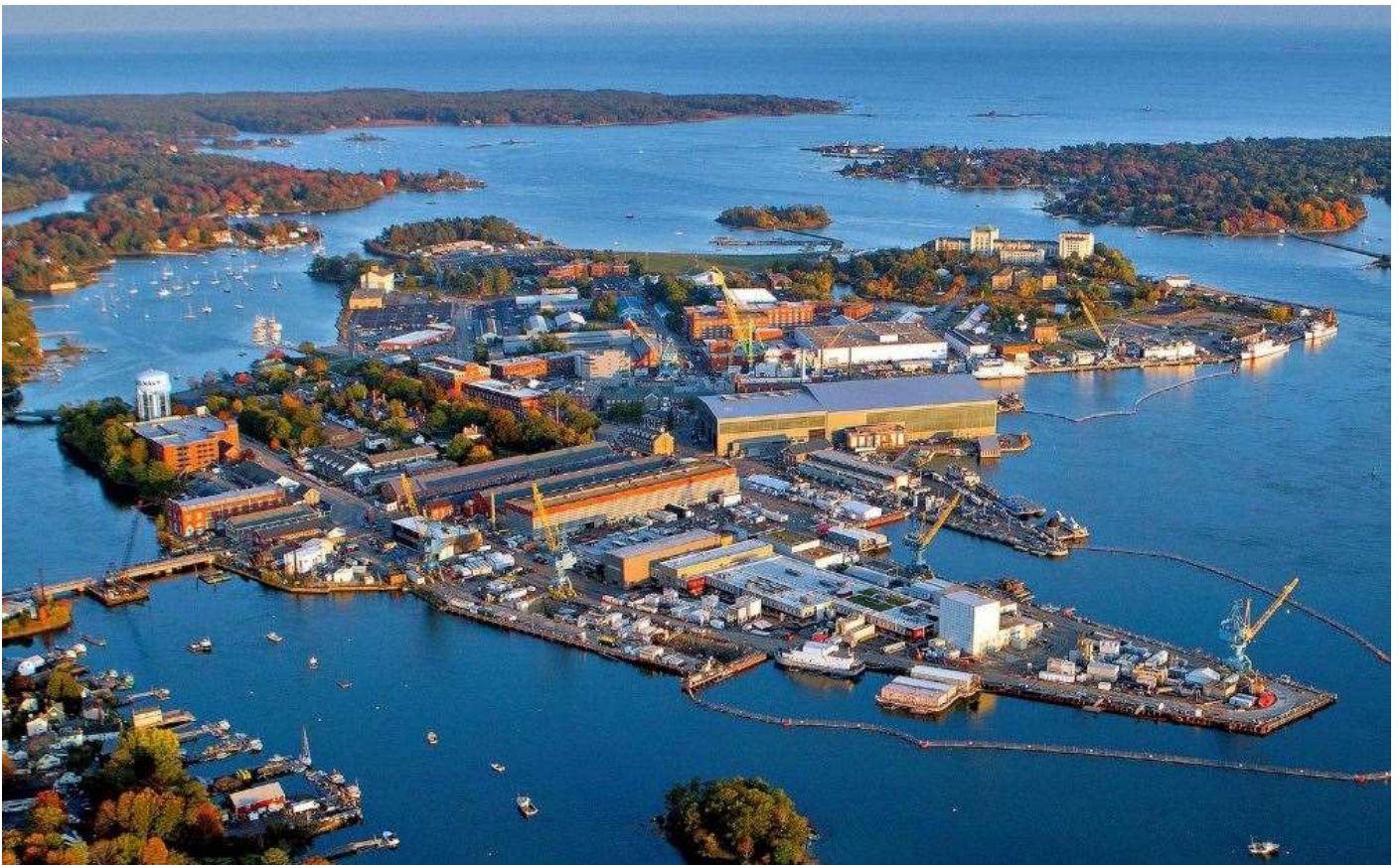
are made under the terms of the Contract, provide additional instruction to acquaint the operating personnel with the changes or modifications.

- d. Training Course Documentation: Training must be based on the operation and maintenance manuals and control system training manual. Deliver manuals for each trainee with two additional sets for archiving at the project site. Include an agenda, defined objectives, and a detailed description of subject matter for each lesson.

-- End of Section --

# ***Basis of Design Desalination Plant Sequence of Operation & Control***

***Attachment to Section 46 82 00***



**August 8, 2025**

## 1.0 INTRODUCTION

### 1.1 REPORT ORGANIZATION

This basis of design Desalination Plant sequence of operation and Control is organized into the following sections:

- Section 1: Overall Process Description
- Section 2: Sequence of Operation & Control (SOC)
- Section 3: Chemical Cleaning (CIP) Sequences

This document can serve as a guide to how the control systems (PLC, SCADA) should operate. However, during commissioning and operation, the equipment manufacturers recommended procedures take precedence.

### 1.2 OVERALL PROCESS DESCRIPTION

The desalination unit processes consist of the following major process treatment components:

1. **Raw Water Intake Pumps (2):** Initially, introduces seawater from the river into the desalination system.
2. **Strainer:** A space-efficient, automatically backwashing strainer will serve as pretreatment upstream of the UF system.
3. **Ultrafiltration (UF) as Pre-Treatment:** The UF system consists of three separate trains , each equipped with a single cassette and associated equipment. This configuration allows two-thirds of the design flow to continue during backwashing or cleaning.
4. **Reverse Osmosis (RO1) - Pass 1:** After ultrafiltration, the first pass involves a single-stage seawater reverse osmosis (SWRO) unit. This includes a cartridge filtration skid, pump skid, membrane skid, high-pressure pump, booster pump, and energy recovery equipment to manage osmotic pressures and remove dissolved solids.
5. **Reverse Osmosis (RO2) - Pass 2:** The second pass involves a two-stage brackish water reverse osmosis (BWRO) unit, fed at lower pressures than the RO1 unit to further reduce total dissolved solids. The RO2 feed pump will require a minimum pressure of 25-35 psi, but it will reach the required 200 psi net driving pressure without booster pumps.
6. **Storage Tanks and Associated Pumps:** The facility will feature multiple storage tanks, including an 18,800-gallon UF break tank, a 30,000-gallon permeate tank, and several chemical storage tanks and associated pumps.
7. **CIP Cleaning System:** The CIP cleaning process consists of equipment required to perform maintenance and recovery cleaning of the UF system, as well as periodic cleaning of the RO membranes.
8. **Duplex Submerged UF Drain Pump Station:** designed to transfer UF backwash flow and/or any overflow to the outfall.

**Figure 1** depicts the overall process flow schematic for the desalination facility.



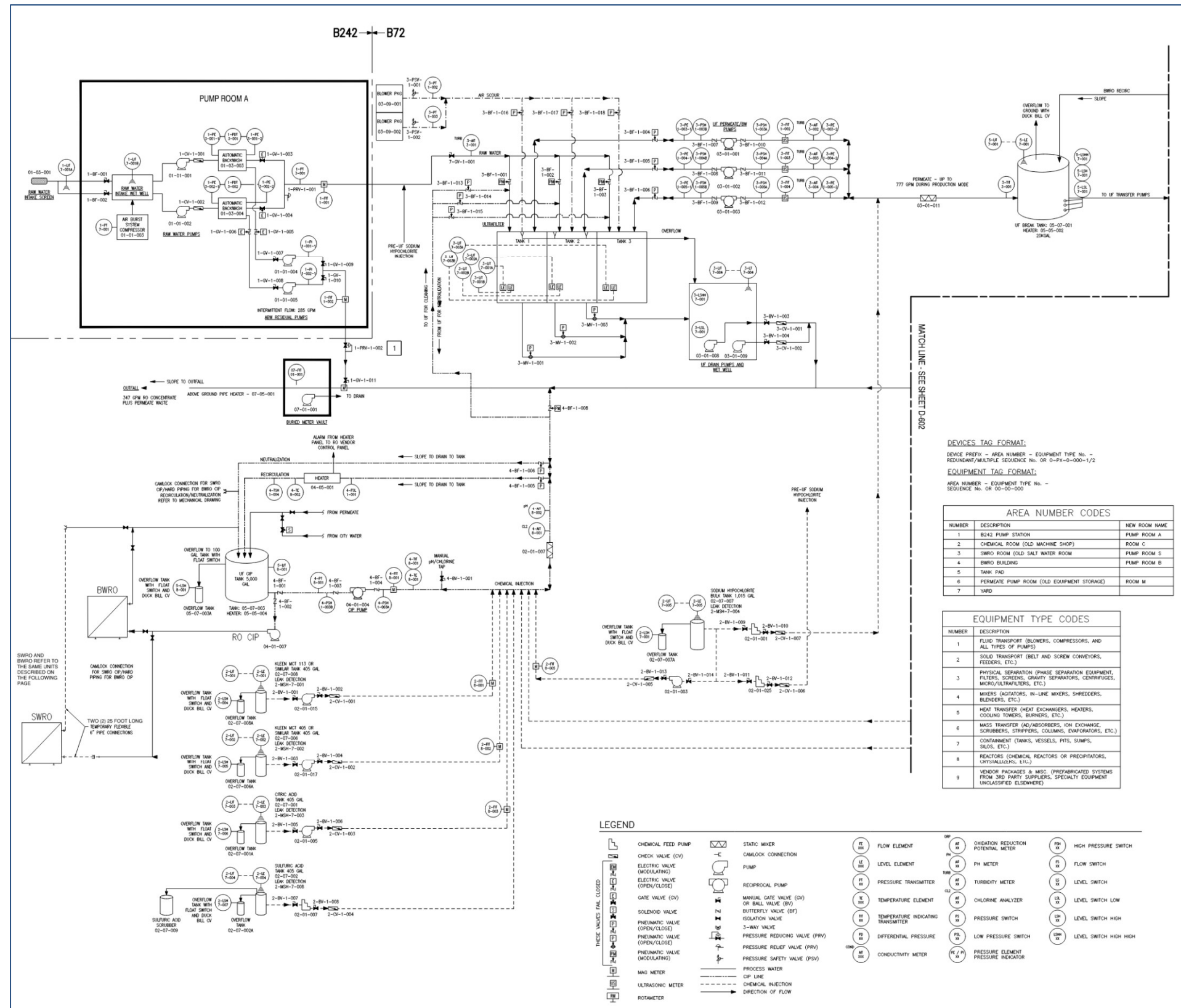


Figure 1 Overall Process flow schematic for the desalination facility.

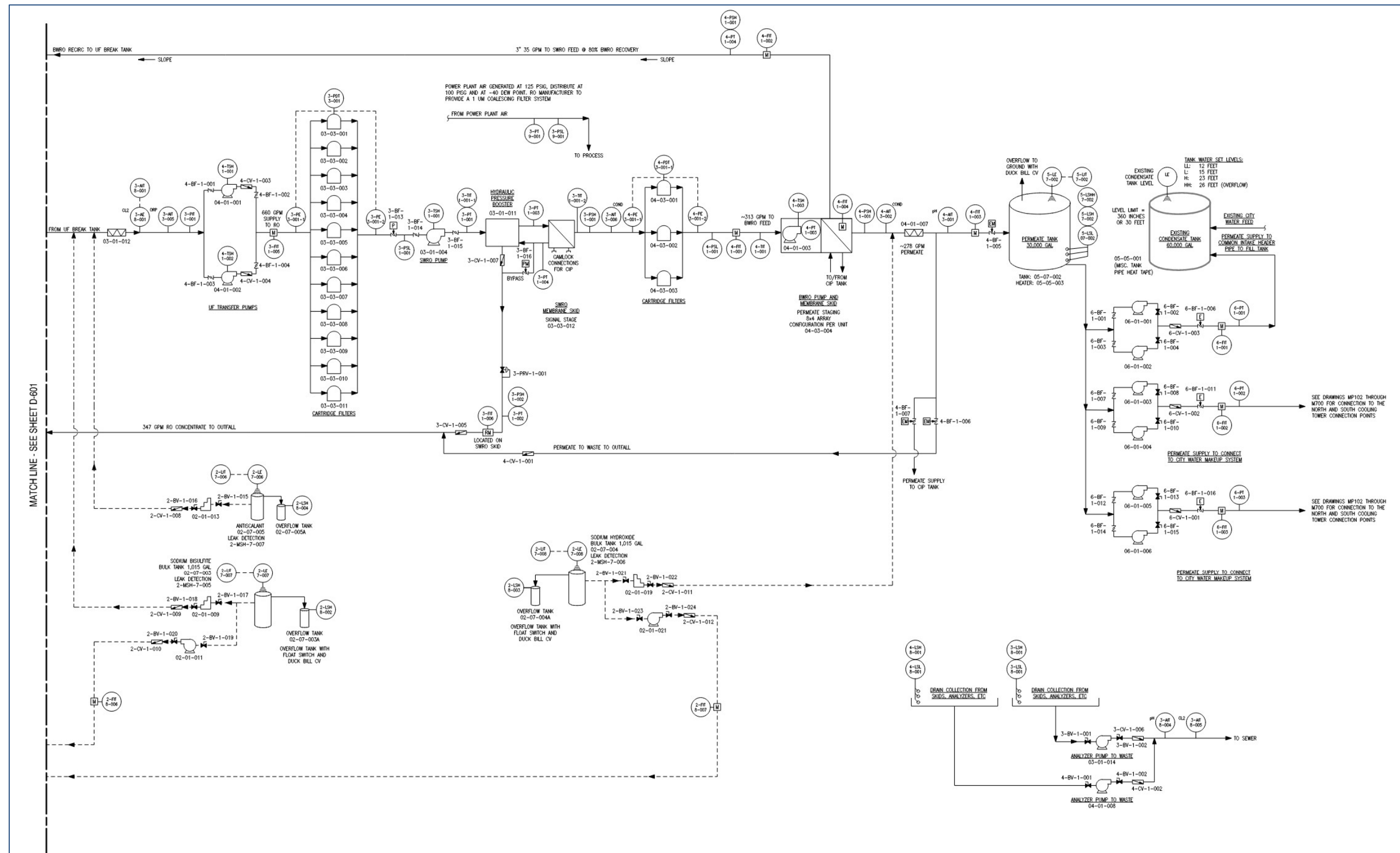


Figure 1 Overall Process flow schematic for the Desalination Facility (Continued)

## 2.0 SEQUENCE OF OPERATION & CONTROL (SOC)

### 2.1 SYSTEM STARTUP PREREQUISITES

1. The Power supply is available and all safety interlocks (pressure, level, leak detection) have been verified.
2. The operator must confirm that all manual valves are set in the in Open positions.
3. The Plant local/remote by the (DCS/PLC) control confirmed.
4. Process tanks (e.g., permeate, CIP) at operational levels.

### 2.2 INTAKE PUMP OPERATION

- **Start Condition:**
  - 2 Intake Pumps Lead / lag arrangement
  - Automatic start command initiated.
- **Control Logic:**
  - Normal operation one pump starts (duty); second on standby or alternates based on runtime.
  - When any UF unit is in backwash or cleaning mode, operate with a single intake pump at approximately 70% capacity.
  - Once the UF unit returns to service, both intake pumps (duty / duty) and divert more flow to the recently cleaned unit to expedite filling. When the water level in the unit reaches its normal operating level, return to standard operation with one pump running at approximately 777 GPM.
  - Monitor Flow and pressure at the SCADA and PLC
- **Shutdown Condition:**
  - Emergency stops or faults will shut down the entire system. A manual restart is required to begin operation again.

### 2.3 STRAINER OPERATION

- **Start Condition:**
  - Inline continuous operation with intake pump start.
- **Control Logic:**
  - Monitors differential pressure (DP).
  - Auto backflush on high DP which is set during startup and can be adjusted by the operator.
  - Alarm logic if high DP persists.
  - Differential setpoints and alarms shall be configured during startup and testing by the manufacturer's field technician. The manufacturer shall provide the setpoints.

### 2.4 SUBMERGED ULTRAFILTRATION (UF) SYSTEM

- **Start Condition:**



- Flow confirmed from intake/strainer.
- UF tank at minimum operating level to set during startup and could be adjusted by the operator.

- **Control Logic:**

- Permeate production is based on level in break tank and number of cassettes in operation.
- Periodic backwash/air scour based on timer, DP, or percent recovery.
- Automatic periodic backwash, alarms, and air scour settings shall be configured during startup and testing by the manufacturer's field technician. The manufacturer shall provide the setpoints.

- **Shutdown Condition:**

- UF tank low-low level.
- High turbidity or fouling alarm.
- UF module failure.

## 2.5 BREAK TANK (18,000 GAL)

- **Control Points:**

- Level transmitters monitor tank level.
- High/low level alarms.
- Transfer pump interlocked with tank level.

### Water Level Control and Shutdown Setpoints

The preferred water level operating range should be maintained between **3 feet above the bottom** of the tank and **3 feet below the top** of the tank.

The initial setpoints are configured as follows; however, the operator shall have the option to adjust these setpoints automatically:

- **High Water Depth:** 18 feet from the bottom of the tank. An alarm shall be initiated and displayed on the SCADA/PLC system. At this level and above, the intake pump speed shall be reduced to approximately 70%. The corresponding change in flow rate shall be indicated on the SCADA/PLC interface.
- **Low Water Depth:** 3 feet from the bottom of the tank. An alarm shall be initiated and displayed on the SCADA/PLC system.
- **High-High Water Depth (19 feet):** Automatically shuts down upstream processes, including the intake pumps and UF permeate/backwash pumps. An alarm shall be initiated and displayed on the SCADA/PLC system.
- **Low-Low Water Depth:** 2 feet from the bottom of the tank. Automatically shut down downstream processes, including the transfer pumps and RO pumps. An alarm shall be initiated and displayed on the SCADA/PLC system.
- Automatic resume normal operations once the water level returns to within the preferred operating range.

- The plant operator should have the option to update these level settings at any time via the SCADA system.

## 2.6 TRANSFER PUMP

- **Start Condition:**
  - Break tank - A minimum low level of 3 feet from the bottom of the tank shall be set during startup. This level can be adjusted by the operator as needed.
  - SWRO demands signal.
- **Control Logic:**
  - VFD controlled stable feed flow indicated at the SCADA / PLC
  - Positive Flow/pressure feedback from booster inlet.
- **Shutdown Condition:**
  - Low-Low Water Depth (2 feet) in the break tank: Automatically shuts down downstream processes, including the transfer pumps and RO pumps. An alarm shall be initiated and displayed on the SCADA/PLC system.
  - Pump fault or no flow an alarm shall be initiated and displayed on the SCADA/PLC system.

## 2.7 BOOSTER PUMP (RO)

- **Start Condition:**
  - Transfer pump running.
  - SWRO skid ready.
- **Control Logic:**
  - VFD control for stable SWRO pressure.
- **Shutdown Condition:**
  - No feed pressure. An alarm shall be initiated and displayed on the SCADA/PLC system.
  - High pressure or overcurrent alarm. An alarm shall be initiated and displayed on the SCADA/PLC system.

## 2.8 SWRO SKID

- **Start Condition:**
  - Feed pressure and flow within setpoints.
- **Control Logic:**
  - Operate based on permeate tank demand.
  - Monitors permeate conductivity, flow, and pressure. Indicated at the SCADA / PLC

- CIP on scheduled maintenance.
- **Shutdown Condition:**
  - High feed pressure.
  - High permeate conductivity.
  - Low feed pressure or flow.
  - An alarm shall be initiated and displayed on the SCADA/PLC system.

## 2.9 BWRO SKID

- Start Condition:
  - SWRO permeate available.
- Control Logic:
  - Same control logic as SWRO.
- Shutdown Condition:
  - High DP, conductivity, or fault. An alarm shall be initiated and displayed on the SCADA/PLC system.

## 2.10 PERMEATE TANK (30,000 GAL)

- **Control Points:**

### Water Level Control and Shutdown Setpoints

The preferred water level operating range should be maintained between **3 feet above the bottom** of the tank and **3 feet below the top** of the tank.

The initial setpoints are configured as follows; however, the operator shall have the option to adjust these setpoints automatically:

- **High Water Depth:** 30 feet from the bottom of the tank. An alarm shall be initiated and displayed on the SCADA/PLC system.
- 
- **Low Water Depth:** 3 feet from the bottom of the tank. An alarm shall be initiated and displayed on the SCADA/PLC system.
- **High-High Water Depth (31 feet):** Automatically shuts down upstream processes, an alarm shall be initiated and displayed on the SCADA/PLC system.
- **Low-Low Water Depth:** 2 feet from the bottom of the tank. Automatically shut down downstream permeate supply pumps and switch to City water makeup system. An alarm shall be initiated and displayed on the SCADA/PLC system.
- Automatic resume normal operations once the water level returns to within the preferred operating range.
  - The plant operator should have the option to update these level settings at any time via the SCADA system.

## 2.11 PERMEATE SUPPLY PUMPS

- **Equipment Involved**

- Duplex Pump Set – South Cooling Tower (1 duty/1 standby)
- Duplex Pump Set – North Cooling Tower (1 duty/1 standby)
- Duplex Pump Set – Condensate Tank (1 duty/1 standby)

- **Control Points:**

The start and stop of the permeate pumps shall be controlled based on water demand from the South Cooling Tower, North Cooling Tower, and the water level in the condensate tank, respectively. However, If Permeate Tank Level in the permeate tank reaches Low-Low (2 foot above bottom of tank)

- Do NOT start/stop permeate pumps.
- Automatically switch water supply to City Water Backup.
- Trigger alarm/notification to operations. An alarm shall be initiated and displayed on the SCADA/PLC system.
- Continue using city water until the permeate tank level returns to the preferred operating level.

### Condensate Tank Level Setpoints and Pump Control

The level setpoints for the condensate tank are already established and integrated into the existing B72 control system (refer to Process Flow Diagram – Figure 1). The contractor shall ensure the following level setpoints are configured:

- **High-High:** 26 feet
- **High:** 23 feet
- **Low:** 15 feet
- **Low-Low:** 12 feet

### Control Logic:

- At **High Level (23 feet)** – the permeate feed pump dedicated to the condensate tank shall stop.
- At **Low Level (15 feet)** – the permeate feed pump shall start.

## 2.12 GENERAL CONTROLS AND ALARMS

- All critical equipment should have:
  - Local and remote start/stop. The remote stop command shall be initiated through the SCADA/PLC interface by an operator or automatically by system logic (e.g., based on tank levels, alarms, or interlocks).
  - Flow, pressure, and status feedback at SCADA / PLC.

- Interlocks to protect downstream units.
- Fault alarms and trips for:
  - High/low pressure
  - No flow
  - Overload

The plant operator should have the option to update these settings at any time via the SCADA system.

All alarms and warnings shall be indicated at SCADA & PLC's

## 2.13 UF DRAIN DUPLEX PUMP STATION SEQUENCE OF OPERATION

The UF drain pump station operates as a typical duplex pump system, with pump operation controlled by water level. A level sensor will be installed for primary control, along with a backup float switch for redundancy.

### **Operational Sequence:**

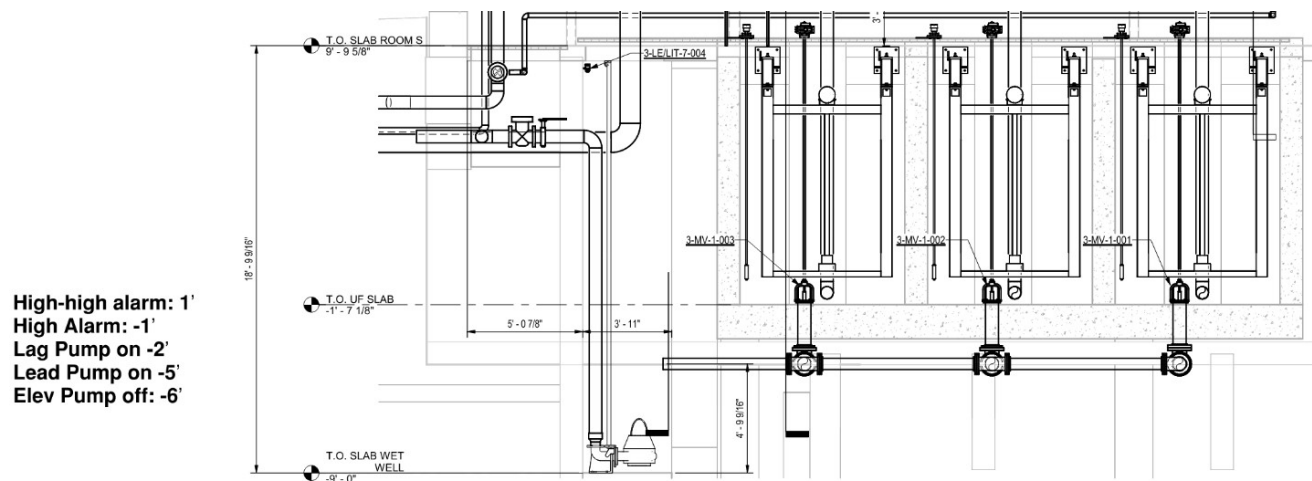
- After each UF unit backwash, the corresponding UF tank drain valve automatically opens, allowing backwash water to flow by gravity into the pump station.
- As the inflow increases, the wet well water level begins to rise.

### **Pump Control Logic:**

1. Lead Pump Start:
  - When the water level reaches the lead pump start point (detected by the level sensor), the lead pump is activated.
  - The lead pump begins discharging water from the wet well.
2. Lag Pump Start:
  - If inflow continues and exceeds the capacity of the lead pump, the water level will rise to the lag pump starting point.
  - The lag pump is then activated.
  - Both pumps operate together at full capacity to prevent overflow.
3. Lag Pump Stop:
  - When the water level drops below the lag pump stop level, the lag pump turns off.
  - The lead pump remains in operation.
4. Lead Pump Stop:
  - When the water level falls below the lead pump stop point, the pump turns off.
  - The system returns to standby mode.
5. Pump Alternation:
  - PLC logic to be used to alternate the lead pump designation between Pump 1 and Pump 2 with each operating cycle.
6. High-Level Alarm:
  - If both pumps are unable to keep up with the inflow, and the water level reaches the high-level alarm point:
7. All alarms shall be indicated at SCADA / PLC.

### **Initial setting Point:**

See Figure below.



The plant operator should have the option to update these settings at any time via the SCADA system. However, it is required to maintain a water level above the pump motor to keep it submerged when the pumps are off.

## 2.14 POWER OUTAGE

Post- Outage Restart Procedure:

### 1. Login and Initial Checks

- Log into the SCADA system using authorized credentials.
- Check system status overview for alarms, offline devices, or fault indications.
- Verify that main power or backup power is stable.
- Confirm that communication links with all PLCs and remote devices are re-established

### 2. Safety and System Readiness

- Ensure field inspections have been completed:
  - No leaks, electrical hazards, or mechanical faults.
  - Emergency stop buttons are disengaged.
- Check that emergency shutdown interlocks are reset.
- Acknowledge and clear non-critical alarms on SCADA.

### 3. Manual Restart Sequence (via SCADA)

Restart systems in logical process order:

- Intake System
  - Open intake valves.
  - Start intake pumps manually via SCADA.
  - Confirm stable flow rate and pressure at intake.
- Pre-Treatment System
  - Start strainer /UF filters and backwash system
  - Start chemical dosing systems
  - Verify dosing pump status and flow rates.

- Reverse Osmosis (RO) Unit
  - Start booster pumps
  - Confirm feed water pressure is within safe range.
  - Start RO trains.
  - Monitor key parameters:
    - Pressure differentials
    - Permeate water conductivity
    - Temperature and flow
- Permeate Water
  - Confirm permeate water quality meets standards
  - Confirm water level in the permeate tank and condensate tank
  - Start permeate water transfer pumps to condensate or cooling towers.

### 3.0 CHEMICAL CLEANING (CIP) SEQUENCES

#### 3.1 UF CIP CLEANING SEQUENCE OF OPERATION

To chemically clean the UF membranes by recirculating a cleaning solution, removing organic/inorganic foulants, and restoring flux/permeate quality. The table below provides a summary for typical cleaning type and frequency:

|                                 |                                                                                                                                                                                                  |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maintenance Clean (MC) Protocol | 2/day/train<br>12 times per week using 250 mg/L NaOCl<br>1 time per week using 500 mg/L citric acid<br>with mineral acid for pH adjustment to 2.1<br>maintenance cleans are not heated           |
| Recovery Clean (RC) Protocol    | 24/year/train<br>500 mg/L NaOCl<br>followed by<br>12/year/train<br>2 g/L citric acid<br>with mineral acid for pH adjustment to 2.1<br>recovery clean solutions are<br>heated prior to each clean |

#### Notes:

1. Preconditions Before CIP Start:
  - UF skid in standby mode (not producing permeate).
  - Operator shall Verify that all manual valves are fully opened prior to operation.
  - The operator shall confirm the UF tank is isolated from raw water intake.
  - CIP tank filled with appropriate CIP chemical solution.
  - Proper CIP chemical selected.
  - The operator shall confirm the CIP pump, heating system (for UF RC only), and verified operational valves.
  - The operator shall confirm the Safety interlocks active: temperature, pressure, level sensors OK.
  - Operator input or auto-trigger based on scheduled maintenance.



2. The UF cleaning process is intended to be performed automatically without operator intervention.
3. The operator is responsible for monitoring equipment performance and system alarms. The operator shall have the ability to adjust system setpoints as needed. However, any adjustments to the setpoints must be confirmed with the equipment manufacturer prior to implementation.
4. Additionally, the operator shall be able to schedule maintenance cleaning automatically through the control system.
5. The operator shall have the ability to either automatically schedule or automatically initiate the cleaning process for each cassette unit via the SCADA system. The cleaning cycle shall apply only to the selected cassette unit. To maintain continuous plant operation, only one cassette shall undergo cleaning at a time, while the remaining two cassettes remain in service.
6. It is also recommended to start the cleaning process when the break tank is full to ensure sufficient water availability.

The automatic UF CIP cycle phases are as follows:

1. Flush / Drain (Optional)
  - Purpose: Remove the remaining feed water and loose particles from membrane housing.
  - Steps:
    - Open drain valve from UF feed line and drain by gravity to UF drain pump station.
    - Start UF drain pump automatic based on level.
    - Duration: ~2–5 minutes
2. Fill CIP Tank
  - Purpose: Prepare and mix the cleaning chemical.
  - Steps:
    - Fill CIP tank with permeate.
    - Heat to required temperature (e.g., 35–40°C) for UF recovery cleaning only.
    - Automatic dosing into CIP tank.
    - Add chemicals (NaOH, citric acid, etc.).
    - Mix solution using CIP circulation pump.
    - Confirm pH/temperature setpoints.
    - Interlocks: pH in range, level in CIP tank OK.
3. Circulation Phase
  - Purpose: Recirculate cleaning solution through UF membranes.
  - Steps:
    - Automatic close normal process valves (feed/permeate).
    - Open CIP inlet/outlet valves.
    - Start CIP pump (A bidirectional rotary lobe pump will be used for the recirculation process).
    - Flow cleaning solution through UF membranes (inlet → membrane → outlet → return to CIP tank).
    - Maintain target flow and pressure (low pressure to avoid membrane damage).
    - Monitor temperature and pH throughout.
    - Duration: 20–60 minutes per solution.
4. Discharge UF CIP wastewater to River
  - Drain CIP solution to CIP tank for neutralization prior to discharge to river.
  - Fill UF unit with water from break tank.



- Drain rinse water to CIP tank.
- Confirm the CIP wastewater pH has been neutralized to acceptable environmental standards (typically 6.5–8.5).
- Test for and ensure residual chlorine is fully removed or below regulatory discharge limits.
- Record test results in the daily log.
- Open the valve to the CIP discharge line leading to the river.
- CIP wastewater begins to discharge at a controlled flow rate of approximately 200 GPM.
- The discharged CIP waste will be combined with the concentrate prior to being released to the river through the proposed 12" HDPE outfall pipe. This blending process will reduce contaminant concentrations at the outfall.

#### 5. Return to Operation

- Reset valves to normal process position.
- Reopen feed water inlet, start back normal filtration.
- Monitor initial permeate flow, turbidity, and DP for confirmation of cleaning effectiveness.
- Log CIP operation: date, chemical used, parameters, results.

### 3.2 SWRO & BWRO MEMBRANE CLEANING (CIP) – SEQUENCE OF OPERATION

To remove foulants (e.g., scaling, biofilm, organics) from RO membranes using chemical cleaning in a controlled loop.

- The SWRO CIP process is intended to occur four (4) times a year, with a high pH cleaner and a low pH cleaner.
- The BWRO CIP process is intended to occur four (4) times a year, with a high pH cleaner and a low pH cleaner.
- The cleaning process is intended to be performed Manually but the chemical dosing is conducted automatically.
- Preconditions Before Starting Cleaning (Operator is to confirm the following):
  - RO system in off mode.
  - CIP tank filled with prepared cleaning solutions (acidic or alkaline).
  - Verify that all manual valves are fully opened prior to operation.
  - CIP pump and valves operational.
  - Temporary flexible piping is connected.
  - System interlocks functional (pressure, flow, level, temp).

The CIP cycle phases are as follows:

#### 1. Flush & Drain

- Purpose: Remove residual feed water from pressure vessels.
- Actions:
  - Open drain valves on RO skid.
  - Use permeate or clean water to flush out concentrate.
  - Close drain valves after flushing.

#### 2. Chemical Preparation

- Fill CIP tank with RO or permeate water.
- Add appropriate cleaning chemicals.

- Alkaline (NaOH) for organics/biofouling (pH 10–11).
- Acid (citric) for scaling (pH 2–4).
- Mix with CIP pump recirculation.
- Confirm pH and level within range.

### 3. Recirculation (Cleaning Loop)

- Close process valves; open CIP inlet/outlet valves.
- Start CIP pump.
- Loop cleaning solution through RO elements (feed → concentrate → return to CIP tank).
- Maintain recommended flow and pressure.
- Monitor  $\Delta P$ , flow, pH, temp.
- Typical duration: 30–60 min.

### 4. Soak Phase (Optional)

- Stop CIP pump.
- Let solution soak for 30–60 minutes.
- May alternate soak and circulation phases for tough fouling.

### 5. Rinse Phase

- Drain used CIP solution to CIP tank for neutralization or waste tank.
- Fill CIP tank with clean RO/permeate water.
- Recirculate rinse water through membranes.
- Continue until:
  - pH is neutral.
  - Conductivity stabilizes.
  - No foam or chemical odor
- Drain rinse water.

### 6. Discharge CIP wastewater to River

- Confirm the CIP wastewater pH has been neutralized to acceptable environmental standards (typically 6.5–8.5).
- Test for and ensure residual chlorine is fully removed or below regulatory discharge limits.
- Record test results in the daily log.
- Open the valve to the CIP discharge line leading to the river.
- Begin CIP wastewater discharge at a controlled flow rate of approximately 100 GPM.

To reduce contaminant concentrations at the outfall it's recommended the operator perform the following:

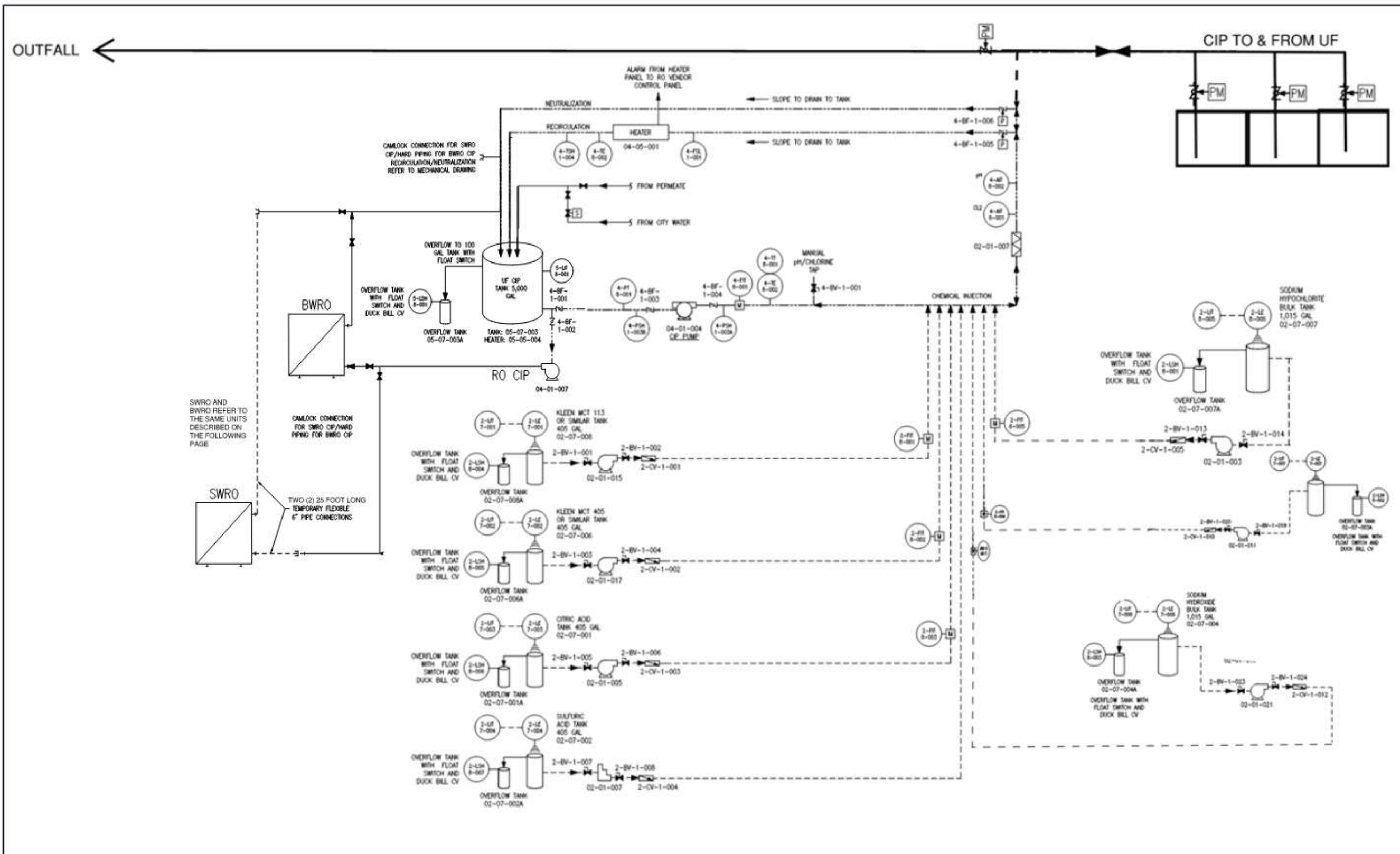
- Open all UF drain valves.
- Simultaneously, start the intake pump to draw raw river water at approximately 800 GPM into the UF tank. From the UF tank, the water will drain into the UF drain pump station and then be returned to the river. The UF drain flow will be combined with the CIP waste, helping to reduce the overall discharge concentration to the river.
- Do NOT start the UF permeate pumps during the SWRO and BWRO CIP cleaning/discharge operation.

### 7. Return to Operation

- Reset valve positions to normal feed/concentrate/permeate routing.
- Start feed pump and RO unit in slow ramp-up mode.
- Discard initials permeate (first 15–30 min).
- Monitor:
  - $\Delta P$
  - Permeate flow and conductivity.
  - Recovery rate

Note that Plant operators should follow the manufacturer's recommendations for the UF and RO CIP cleaning procedure.

The figure below shows the CIP cleaning flow diagram.



CIP CLEANING PROCESS FLOW DIAGRAM

